SGA 3: Group Schemes

Avertissement and Introduction

Avertissement

We present here a slightly revised re-edition of the original Séminaire, whose aim and content are indicated in the Introduction. The revision consisted essentially in the correction of typographical errors, the addition (in footnotes) of some supplementary remarks or references, the present subdivision into three volumes each equipped with a detailed table of contents and an index of notation, and the adjunction of a terminological index at the end of Volume 3. In addition, Exposé VI_B by J.-E. Bertin has been partially rewritten by him, in particular §§ 5 and 10, so that some references to that Exposé differ from references to the original Exposé. The reader will find a list of the Séminaire's Exposés at the beginning of the present volume.

Since the first edition of the present Séminaire appeared, the *Éléments de Géométrie Algébrique*, Chap. IV, have appeared in their entirety, which renders certain passages of the Séminaire unnecessary; we have occasionally indicated in footnotes the relevant references to EGA IV that allow such passages to be short-circuited.

For another exposition on algebraic groups using systematically the language of schemes, we point to the book by M. Demazure and P. Gabriel, *Groupes algébriques* (North-Holland & Masson et Cie). Unlike the present Séminaire, this book assumes no knowledge of algebraic geometry, but contains all the necessary preliminaries from the theory of schemes, and its reading can therefore serve as an introduction to the study of our Séminaire. (It also contains themes not covered in the Séminaire, such as the Dieudonné-style structure theory of commutative affine algebraic groups, in *loc. cit.* Chap. V.)

Bures-sur-Yvette, March 1970

Introduction

1. The aim of the present seminar is twofold.

On the one hand, we aim to give convenient foundations for the theory of group schemes in general. Exposés I to IV will provide for this purpose the indispensable preliminary exercises in schematic and categorical syntax. To obtain a language that "fits" geometric intuition effortlessly, and to avoid eventually unbearable circumlocutions, we always identify a prescheme X over another S with the functor $(Sch)^\circ/S \to (Ens)$ that it represents,[1] and it is necessary to give many definitions in such a way that they apply to arbitrary functors, representable or not. Moreover,

[1] Such a viewpoint seems to have been envisaged for the first time eight or nine years ago, in connection with the theory of formal groups, by P. Cartier, who unfortunately did not take the trouble to make it precise and systematic as it deserved.

nearly all the functors we shall have to use will be "sheaves" (for the "faithfully flat quasi-compact topology"); Exposé IV, which treats groups only incidentally, gives a sketch of the language of "localization" and sheaves, which turns out to be very convenient for questions of representability of functors as well. Above all, this Exposé will provide us, for questions of passage to the quotient, with the most convenient framework for what follows.

Exposé V gives some general results on the existence of quotients, taken up in Exposé VI_A in the case of the quotient of an algebraic group over a field (or more generally, over an Artinian ring) by a subgroup.[2] The latter Exposé and Exposé VI_B which follows it also contain various elementary results specific to algebraic groups over a field, used routinely thereafter.

Exposé VII studies certain facts linked to the characteristic of the base field, and in particular develops with appropriate generality the correspondence between radicial group schemes of height 1 and restricted p-Lie algebras.

Finally, Exposé XVIII contains the generalization, in the theory of schemes, of Weil's theorem on the "birational" definition of algebraic groups.

On the other hand, we propose to generalize to groups over an arbitrary base prescheme the Borel–Chevalley structure theory of affine algebraic groups. It has moreover become apparent in the course of writing up the seminar notes that the affine hypothesis is unnecessary for many results of the theory. The most complete results are obtained, evidently, in the case of "semisimple group schemes", or more generally "reductive" ones, with which we shall be exclusively concerned from Exposé XIX onwards. Chevalley himself had already given the construction of the "Tôhoku" groups over the ring of integers, a construction that will be taken up in the present seminar. The principal uniqueness theorem[3] gives a simple characterization of the "twisted" variants of these Tôhoku groups over a base prescheme S: they are the affine and smooth groups over S whose geometric fibers are connected semisimple groups in the usual sense.[4]

2. As in the case of the theory known over an algebraically closed field, a crucial role is played by the sub-tori of the group schemes under consideration. The preliminary study of tori — and, more generally, of "group schemes of multiplicative type" (both from the intrinsic viewpoint and from the viewpoint of multiplicative-

[2]For a more thorough study of passage to the quotient, in particular by reductive groups, see the important study by D. Mumford, *Geometric Invariant Theory*, Ergebnisse der Mathematik, Bd. 34, Springer, 1965. Let us observe that, on an important point, the terminology of this book does not agree with ours: over a field of characteristic $p > 0$, the groups that Mumford calls "reductive"[^N.D.E-I-intro-1] are the smooth groups of multiplicative type in the sense of the seminar (cf. Exp. IX). One may presumably consider that Mumford's acceptation of the word "reductive", which loses its meaning over a base that is not a field, was adopted by him on a provisional basis and as a kind of stopgap (and this is also roughly what Mumford explains, for other motives, from the second paragraph of his preface!).

[3]*N.D.E.* This refers to Corollary XXIII.5.6, which is easily deduced from the uniqueness theorem for split reductive groups (cf. XXIII, Th. 4.1 and Cor. 5.2), given that every semisimple S-group is a "twisted form" (for the étale topology) of a "Tôhoku" group (cf. XXII, Cor. 2.3).

[4]This is the essential result of M. Demazure's thesis (*Schémas en groupes réductifs*, Bull. Soc. Math. France **93** (1965), 369–413).

type subgroups of a given group) — therefore takes up a fairly large place in this seminar (Exposés VIII to XII). Their remarkable rigidity (in certain respects greater even than that of abelian schemes, or of semisimple group schemes) makes them very effective working instruments for the study of certain more general groups.

3. From Exposé XII onwards (with the exception of Exposé XVIII already mentioned) we shall routinely use the theory of affine algebraic groups over an algebraically closed field, which the reader will find in the *Séminaire Chevalley* 1956, in particular in Exposés IV to IX of that seminar. We shall also use, but to a lesser extent, the later Exposés of the *Séminaire Chevalley*, devoted to the structure of semisimple algebraic groups. Indeed, we shall take up Chevalley's theory directly in the framework of schemes: it will be seen that in this way (even over a base field) the exposition gains in simplicity and in precision.

4. The principal object of the present seminar is evidently to develop techniques that apply to the study of group schemes over an arbitrary base, i.e. essentially to the study of families of algebraic groups. As such, the infinitesimal properties of such families, and in particular the case of an Artinian base scheme, play an important role. These properties intervene even for the study

of algebraic groups over a field k, in the case where the latter is not perfect, in order in particular to apply the technique of descent in the non-Galois case. Among the new results obtained in this case, let us mention the existence of maximal tori and Cartan subgroups in any smooth algebraic group, the rationality of the variety of maximal tori, and various related results (Exp. XIV[5]), or the correspondence between the "forms" of a semisimple group and the principal homogeneous bundles under a suitable (in general non-connected) semisimple algebraic group (Exp. XXIV, 1.16–1.20). In general, one may say that the methods required to work over a non-perfect base field are essentially those used for arbitrary base preschemes, and thereby fall outside the framework of classical algebraic geometry.

5. It has not seemed useful to indicate at the head of the written-up Exposés the date or dates of the corresponding oral presentations at the Séminaire. Let us merely say that the order of the mimeographed Exposés (from I to XXVI) does correspond to the order of the oral presentations. On the other hand, the writing-up of the final text is sometimes notably later than the oral presentation, and often differs from it fairly substantially, the written text being generally more detailed and more complete (such as Exp. IV and VII_B), or indeed substantially more general (such as Exp. XII or VII_B) than the oral presentation. Other written Exposés correspond to no oral presentation (VI_B, VII_A, XV, XVI, XVII, and essentially XXVI), and were written up and inserted into the mimeographed seminar either to provide convenient references for various other Exposés (this is notably the case of VI_B), or because they constitute a natural extension of the notions and techniques already developed. One should note as a consequence that the reading of Exposés VII_A, VII_B, XV, XVI, XVII is not necessary for the study of the rest of the seminar, and in particular for the part of this seminar devoted to reductive group schemes.

[5] *N.D.E.* In particular, Theorems 1.1 and 6.1.

6. From the theory of schemes, we shall mainly use the general language of schemes, expounded in EGA I; the notions of flat morphism, étale morphism, smooth morphism, expounded in SGA 1, I to V; and finally the theory of faithfully flat descent of SGA 1, VIII. We have as far as possible avoided formulating unnecessary noetherian hypotheses, which has required us in return to replace the usual hypothesis "of finite type" by the hypothesis "of finite presentation". For the notion of morphism of finite presentation, the reader is referred to EGA IV, 1.4 and 1.6. The results of SGA 1, I to IV, most often stated in the noetherian context, will be developed in the general case in EGA IV,[6] where standard methods for reducing certain types of statements (involving hypotheses of finite presentation) to the noetherian case will also be developed in detail (EGA IV, §§ 8, 9, 11). The reader who does not wish to admit these results from EGA IV can simplify certain statements or their proofs by assuming the base prescheme to be locally noetherian. He will, however, expose himself to difficulties in cases where the proof

proceeds by descent from Â to A, where Â is the completion of a noetherian local ring A, since this method leads to the introduction of the (in general non-noetherian) ring $\hat{A} \otimes_A \hat{A}$.

7. References will be made following the usual decimal system: the reference 5.7.11 refers to the proposition (or lemma, definition, etc.) of that name in the same Exposé; in the reference XVII 7.8 the Roman numeral indicates the number of the Exposé. We shall use the following abbreviations for our standard references:

Abbreviation	Expansion
Bible	Séminaire Chevalley, *Groupes de Lie algébriques*, 1956/58
EGA X, x.y.z	J. Dieudonné and A. Grothendieck, *Éléments de Géométrie Algébrique*, Chap. X, statement x.y.z (or sub-paragraph x.y, etc.)
SGA n, X y.z	*Séminaire de Géométrie Algébrique du Bois-Marie*, year n, statement y.z of Exposé X.
TDTE	A. Grothendieck, *Techniques de descente et théorèmes d'existence en Géométrie Algébrique*, Exposés in the *Séminaire Bourbaki* between 1959 and 1962.

Bibliography

[CTS79] J.-L. Colliot-Thélène and J.-J. Sansuc, *Fibrés quadratiques et composantes connexes réelles*, Math. Ann. **244** (1979), 105–134.

[Ha75] W. J. Haboush, *Reductive groups are geometrically reductive*, Ann. of Math. **102** (1975), no. 1, 67–83.

[6] Since the writing of this introduction, the four parts (§§ 1 to 21) of EGA IV have appeared.

[KM97] S. Keel and S. Mori, *Quotient by groupoids*, Ann. of Math. **145** (1997), no. 1, 193–213.

[Ko97] J. Kollár, *Quotient spaces modulo algebraic groups*, Ann. of Math. **145** (1997), no. 1, 33–79.

[MF82] D. Mumford and J. Fogarty, *Geometric invariant theory*, 2nd ed., Springer-Verlag, 1982; (resp. 3rd ed., with F. Kirwan, 1994).

[Na64] M. Nagata, *Invariants of a group in an affine ring*, J. Math. Kyoto Univ. **3** (1964), no. 3, 369–377.

[NM64] M. Nagata and T. Miyata, *Note on semi-reductive groups*, J. Math. Kyoto Univ. **3** (1964), no. 3, 379–382.

[Se77] C. S. Seshadri, *Geometric reductivity over an arbitrary base*, Adv. Math. **26** (1977), no. 3, 225–274.

Errata — Tome I

Misprints and errors in the new edition of SGA 3, Tome 1.

13 October 2024.

1. SGA 3, Tome 1

Page numbers refer to the version published in 2011. The new compilation compressed the pages to remove the final (odd-numbered) page of Exposés 3 and 6A, so the pagination of the 2024 version is shifted by −2 (resp. −4) starting from Exposé 4 (resp. Exposé 6B).

1.1. Exposé I

- p. 18, in the added Proposition 3.1.2, the proof that O is a generator is incorrect. This has been corrected in the 13/10/2024 version.
- p. 29, N.D.E. (40): $C\dot{f}VI_B$ should be: Cf. `VI_B`.

1.2. Exposé II

- p. 82: the 2nd sentence of N.D.E. 71 should appear in N.D.E. 72.

1.3. Exposé IV

- p. 182, proof of Prop. 1.8: in the diagram, on the upper-left arrow, `vu` should be `uv`, and likewise in the 1st line below the diagram. This has been corrected in the 13/10/2024 version.

1.4. Exposé VI_A

- p. 302, the proof of Lemma 2.1.3 contained an unwarranted simplification; one must revert to the original proof. This has been corrected in the 13/10/2024 version.
- p. 329, in the proof of Lemma 6.1, the word "à droite" (on the right) is missing after "resp.".

1.5. Exposé VI_B

- p. 359, end of Cor. 4.4. There was a slip in N.D.E. (31): keeping the same notation, one must consider the subgroup $G = \operatorname{Spec} A[X]/(\delta X)$ of the additive group over S, which to every A-algebra R associates the subgroup of $r \in R$ such that $\delta r = 0$. This has been corrected in the 13/10/2024 version.
- p. 374, Th. 6.2.3: in the proof, line 2 of the 3rd paragraph, replace "free module" by "projective module".

1.6. Exposé VII_B

- p. 587, line 6 of 5.0: $\Delta(A)$ should be $\Delta(a)$.

Preface to Tome II

A note from the translator.

Tome II of the 2011 SMF re-edition of SGA 3 contains Exposés VIII through XVIII, with no separate front matter, errata sheet, or notation index of its own — the editors chose to consolidate the volume-level apparatus into Tomes I and III only. Reading-order navigation for the Tome II Exposés can be found in the README; references to the Avertissement, Introduction, and reference convention all live with Tome I's front matter; and the translation glossary applies uniformly across the three Tomes. The Tome I errata sheet flags Exposé II only (§ 1.2), and the Tome III errata sheet does not concern this Tome.

Topically, Tome II spans three threads. Exposés VIII–XII (Grothendieck) develop the theory of groups of multiplicative type — diagonalizable groups (VIII), homomorphisms between groups of multiplicative type (IX), characterization and classification (X), representability criteria (XI), and the foundational structure theorems on maximal tori, the Weyl group, Cartan subgroups, and the reductive center (XII), which together provide the language used throughout Tome III. Exposés XIII–XV (Grothendieck, with Serre's appendix in XIV, and Raynaud in XV) develop the theory of regular elements in algebraic groups and Lie algebras and apply it to the construction and parametrization of maximal tori, in particular in smooth group schemes over a non-perfect field. Exposés XVI–XVIII close the Tome with Raynaud's treatment of groups of zero unipotent rank and of unipotent algebraic groups and their extensions (XVI–XVII), and Artin's writing-up of Weil's theorem on the construction of a group from a rational law (XVIII).

Several of the Exposés in this Tome were inserted into the mimeographed Séminaire without a corresponding oral presentation, as Grothendieck notes in §5 of the Introduction: VII_A and VII_B from Tome I, and XV, XVI, XVII from this Tome, may be read as natural extensions of the techniques developed in the surrounding material rather than as steps in the main line; the reader heading for the reductive-group material of Tome III need not pause on them.

The Exposés in this Tome are listed below; for the full reading order across all three Tomes, see the README.

- Exposé VIII — Diagonalizable groups (Grothendieck)
- Exposé IX — Groups of multiplicative type: homomorphisms (Grothendieck)
- Exposé X — Characterization and classification of groups of multiplicative type (Grothendieck)
- Exposé XI — Representability criteria (Grothendieck)
- Exposé XII — Maximal tori, the Weyl group, Cartan subgroups, the reductive center (Grothendieck)
- Exposé XIII — Regular elements: algebraic groups and Lie algebras (Grothendieck)
- Exposé XIV — Regular elements: continuation, applications (Grothendieck, app. Serre)
- Exposé XV — Complements on sub-tori. Application to smooth groups (Raynaud)
- Exposé XVI — Groups of zero unipotent rank (Raynaud)
- Exposé XVII — Unipotent algebraic groups; extensions (Raynaud)
- Exposé XVIII — Weil's theorem on the construction of a group from a rational law (Artin)

Errata — Tome III

Misprints and errors in the new edition of SGA 3, Tome 3.

13 October 2024.

1. SGA 3, Tome 3

1.1. Exposé XIX

- p. 15, line 4: after "take T split", add: "and use the proof of Th. 2.5".
- p. 23, line −4 of 5.13: $t \neq 0$ should be $b \neq 0$.
- p. 24, at the level of page 33 of the original: add "of r copies" before "of $nG =$". And, 6 lines below, replace $nG_{m,S'}$ by $(nG_{m,S'})^r$.

1.2. Exposé XX

- p. 29, 2 lines below original page 39: x should be X.
- p. 37, 3 lines above B), $G(0) = 1$ should be $E(0) = 1$.

- p. 38, line above C), r should be α.
- p. 38, 2 lines below C), $N(S')$ should be $N'(S')$.
- p. 40, 1st formula in D): $E(x)$ should be $E(x)^{-1}$. (This formula was not written in the original.)
- p. 40, paragraph E): in the 3 first formulas with $\alpha(H)$ on the left, replace on the right $(1+bx)$ by $(1+ax)$. The proof is unchanged.
- p. 45, Th. 3.1 (v): the 1st formula is incorrect and Y is undefined. (This error appears in the original.) One should understand that $Y = zX$, with $z \in G_m(S')$; then by (iv) one has $w_\alpha(Y) = w_\alpha(X)\alpha^\vee(z^{-1})$, hence $w_\alpha(X)^2 = \alpha^\vee(-1)$ gives

  $$w_\alpha(X)\ w_\alpha(Y) = \alpha^\vee(-z^{-1}).$$

  Finally, X^{-1} and Y^{-1} are defined in Corollary 2.6: one has $XX^{-1} = 1$ and $Y^{-1} = z^{-1}X^{-1}$, hence indeed $-z^{-1} = -XY^{-1}$.
- p. 51, Prop. 3.12: replace $Q(S)$ by $N^\times(S)$.

1.3. Exposé XXI

- p. 85, N.D.E. (19): a closing parenthesis is missing after $w_{k-1}(\rho)$.

1.4. Exposé XXII

- p. 122, last line of 4.1.12: $\exp_\alpha(Y)$ should be $\exp_{-\alpha}(Y)$.

1.5. Exposé XXIII

- p. 210, line above 6.2: replace $-X_{-\alpha}^{-1}$ by $-X_\alpha^{-1}$.

1.6. Exposé XXIV

- p. 245, line 8 of the proof of 4.5.1: $o\ \sim$ should be où.

1.7. Exposé XXVI

- p. 311, diagram of 4.5.3: extra parenthesis before the last O_f.

Exposé I. Algebraic structures. Group cohomology

by M. Demazure

[7] This Exposé consists of two parts; the first gathers a certain number of general definitions and sets up notations that will often be used in the sequel, while the second treats group cohomology and culminates in Theorem 5.3.3 (vanishing of the cohomology of diagonalizable groups).

[7]Version of 13/10/2024.

We choose once and for all a Universe.[8] All the definitions stated and all the constructions carried out will be relative to this Universe. We shall systematically allow ourselves the following abuse of language: in order to define a functor $f : C \to C'$, we shall content ourselves with defining the object $f(S)$ of C' for every object S of C, each time that there is no ambiguity about the way to define $f(h)$ for an arrow h of C. In practice, we shall say: let $f : C \to C'$ be the functor defined by $f(S) = \cdots$.

1. Generalities

1.1.

Let C be a category. We shall denote by Ĉ the category $\mathrm{Hom}(C^\circ, (Ens))$ of contravariant functors from C to the category (Ens) of sets.[9] There exists a canonical functor $h : C \to \hat{C}$ which associates to every $X \in Ob(C)$ the functor hX such that

$$hX(S) = \mathrm{Hom}(S, X).$$

For every functor $F \in Ob(\hat{C})$, one defines (cf. for example EGA 0_III, 8.1.4) a bijection

$$\mathrm{Hom}(hX, F) \xrightarrow{\sim} F(X).$$

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In particular, for every pair X, Y of objects of C, the following canonical map is bijective:

Hom_C(X, Y) Hom_Ĉ(hX, hY);

i.e. the functor h is fully faithful. It thus defines an isomorphism of C onto a full subcategory of Ĉ, and an equivalence of C with the full subcategory of Ĉ formed by the representable functors (i.e. those isomorphic to a functor of the form hX). In the sequel, we shall often identify X and hX. The following numbers aim to show that this identification can be made without danger.

Remark 1.1.1.[11] *We sometimes need the following variant. Let D be a full subcategory of C and let $X, Y \in Ob(D)$; denote by h'_X and h'_Y the restrictions to D of hX and hY. Then one has*

Hom_C(X, Y) = Hom_D(X, Y) = Hom_D̂(h′_X, h′_Y)

and therefore: to give a morphism $X \to Y$ "is the same thing" as to give, functorially in T, a map $\phi(T) : \mathrm{Hom}(T, X) \to \mathrm{Hom}(T, Y)$, for every $T \in Ob(D)$.

[8]N.D.E.: Cf. SGA 4, Exp. I, § 0 and Appendix; see also the discussion in [DG70], p. xxvi.
[9]N.D.E.: One calls it the category of presheaves on C, cf. IV 4.3.1.
[10]N.D.E.: This result is often called the "Yoneda Lemma"; we shall use this terminology in other N.D.E.
[11]N.D.E.: This remark has been added.

1.2.

[12] We shall say that F is a *subobject* (or a *subfunctor*) of G if $F(S)$ is a subset of $G(S)$ for each S.

In $\hat{C}$, “arbitrary” inverse limits exist and are computed by:

$$(\varprojlim_i F_i)(S) = \varprojlim_i F_i(S).$$

[13] In particular, fiber products are defined by:

$$(F \times_G F')(S) = F(S) \times_{G(S)} F'(S).$$

[14]

We shall choose as final object of $\hat{C}$ the functor e such that $e(S) = \emptyset$[15]. Every $F \in Ob(\hat{C})$ has a unique morphism into e, and one sets

$$F \times F' = F \times_e F'.$$

The functor h commutes with inverse limits; in particular, for $X \times X'$ to exist ($X, X' \in Ob(C)$), resp. for C to admit a final object e, it is necessary and sufficient that $hX \times hX'$ be representable, resp. that e be representable, and one has

$$hX \times hX' \simeq h_{X \times X'} \quad \text{and} \quad he \simeq e.$$

A monomorphism of $\hat{C}$ is nothing other than a morphism $F \to G$ such that, for every $S \in Ob(C)$, the corresponding map of sets $F(S) \to G(S)$ is injective.[16]

The functor Γ. For every $F \in Ob(\hat{C})$, one sets

$$\Gamma(F) = \mathrm{Hom}(e, F);$$

an element of $\Gamma(F)$ is therefore a family $(\gamma_S)_{S \in Ob(C)}$, $\gamma_S \in F(S)$, such that for every arrow $f : S' \to S''$ of C one has $F(f)(\gamma_{S''}) = \gamma_{S'}$.

One sets $\Gamma(X) = \Gamma(hX)$ for $X \in Ob(C)$. If C has a final object e, one therefore has an isomorphism $\Gamma(X) \simeq \mathrm{Hom}(e, X)$.

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[12] N.D.E.: The order has been modified, in order to introduce fiber products before monomorphisms, cf. N.D.E. (9).

[13] N.D.E.: Likewise, “arbitrary” direct limits exist and are computed “argument by argument”, i.e. $(\lim \to_i F_i)(S) = \lim \to_i F_i(S)$; but in general the functor h does not commute with direct limits.

[14] N.D.E.: In particular, the kernel of a pair of morphisms $u, v : F \rightrightarrows G$ is the subfunctor $Ker(u, v)$ of F defined by $Ker(u, v)(S) = x \in F(S) | u(x) = v(x)$.

[15] N.D.E.: $\emptyset$ (the set of subsets of the empty set) denotes the one-element set.

[16] N.D.E.: If $F(S) \to G(S)$ is injective for every S, it is clear that $F \to G$ is a monomorphism; the converse is seen by considering the diagram $F \times_G F \rightrightarrows F \to G$. One thus obtains that: “$F \to G$ is a monomorphism if and only if the diagonal morphism $F \to F \times_G F$ is an isomorphism” (cf. EGA I, 5.3.8). Likewise, it is clear that if $F(S) \to G(S)$ is surjective for every S, then $F \to G$ is an epimorphism, and the converse is seen by considering the amalgamated sum $G \sqcup_F G$, cf. the proof of Lemma 4.4.4 in Exp. IV.

1.3.

Let $S \in Ob(C)$. We denote by C/S the category of objects of C over S, i.e. the category whose objects are the arrows $f : T \to S$ of C, with $\mathrm{Hom}(f, f')$ being the subset of $\mathrm{Hom}(T, T')$ formed by those u such that $f = f' \circ u$. If C has a final object e, then C/e is isomorphic to C. The category C/S has a final object: the identity arrow $S \to S$.

If $f : T \to S$ is an object of C/S, then one can form the category $(C/S)/f$, which by abuse of language one denotes $(C/S)/T$, and one has a canonical isomorphism

$$C/T \simeq (C/S)/T.$$

This construction also applies to the category $\hat{C}$; one defines in particular the category $\hat{C}/hS$. On the other hand, one can form the category $\hat{C}_{/S}$.

If $f : T \to S$ is an object of C/S, then $\Gamma(f)$ is identified with the set $\Gamma(T/S)$ of sections of T over S, that is, of arrows $S \to T$ which are right inverses of f. Note that $hf : hT \to hS$ is then an object of $\hat{C}/hS$, and one has:

$$\Gamma(hf) \simeq \Gamma(hT/hS) \simeq \Gamma(T/S) \simeq \Gamma(f).$$

1.4.

We now propose to define an equivalence of categories $\hat{C}_{/S}$ and $\hat{C}/hS$, that is, to prove that "to give a functor on the category of objects of C over S is the same thing as to give a functor on C endowed with a morphism into hS".

(i) *Construction of $\alpha S : \hat{C}/hS \to \hat{C}_{/S}$.*

Let first $H : F \to hS$ be an object of $\hat{C}/hS$. We must define a functor $\alpha S(H)$ on C/S. Let first $f : T \to S$ be an object of C/S; we define $\alpha S(H)(f)$ as the inverse image of $f \in hS(T)$ by the map $H(T) : F(T) \to hS(T)$.[17]

Next, let $u : f \to f'$ be an arrow of C/S; then $F(u) : F(T') \to F(T)$ induces a map from $\alpha S(H)(f')$ into $\alpha S(H)(f)$, which we denote $\alpha S(H)(u)$. One verifies at once that the maps

$$f \mapsto \alpha S(H)(f) \quad \text{and} \quad u \mapsto \alpha S(H)(u)$$

do define a functor on C/S, hence an object $\alpha S(H)$ of $\hat{C}_{/S}$.

Finally, let $H : F \to hS$ and $H' : F' \to hS$ be two objects of $\hat{C}/hS$ and $U : H \to H'$ a morphism of $\hat{C}/hS$:

```
        U
  F ────────→ F′
   \         /
  H \       / H′
```

[17] N.D.E.: For example, if $F = hX$ then H corresponds to a morphism $h : X \to S$ and $H(T) : \mathrm{Hom}_C(T, X) \to \mathrm{Hom}_C(T, S)$ is the map $g \mapsto h \circ g$, whence $\alpha S(hX) = \mathrm{Hom}_{C/S}(-, X)$.

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\   /
 hS
```

Then for every $f : T \to S$, the map $U(T) : F(T) \to F'(T)$ induces a map

$$\alpha S(U)(f) : \alpha S(H)(f) \to \alpha S(H')(f),$$

which defines a morphism of functors

$$\alpha S(U) : \alpha S(H) \to \alpha S(H').$$

One verifies easily that the maps

`H ↦ αS(H)     and    U ↦ αS(U)`

do define a functor $\alpha S : \hat{C}/hS \to \hat{C}_{/S}$.

(ii) Proposition 1.4.1. *The functor $\alpha S : \hat{C}/hS \to \hat{C}_{/S}$ is an equivalence of categories.*

We only indicate the principle of the construction of a quasi-inverse functor $\beta S : \hat{C}_{/S} \to \hat{C}/hS$. Let G be a functor on C/S; for every object T of C, one sets

`βS(G)(T) = sum of the sets G(f) for f ∈ Hom(T, S) = hS(T),`

which defines a functor $\beta S(G)$ on C, equipped with an obvious projection onto `hS`.

1.5.

The equivalence αS commutes with the functors Γ. In other words, if $H : F \to hS$ is an object of $\hat{C}/hS$ and $\alpha S(H)$ the corresponding object of $\hat{C}_{/S}$, one has

$$\Gamma(\alpha S(H)) \simeq \Gamma(H) \simeq \Gamma(F/hS).$$

The equivalence αS commutes with the functors h, i.e. if $f : T \to S$ is an object of C/S, then $hf : hT \to hS$ is an object of $\hat{C}/hS$ whose transform by αS is nothing other than $h_{C/S}(f)$, where

$$h_{C/S} : C/S \to \hat{C}_{/S}$$

is the canonical functor.[18] As a consequence:

Proposition 1.5.1. *Let $H : F \to hS$ be an object of $\hat{C}/hS$. For $\alpha S(H) : (C/S)^\circ \to (Ens)$ to be representable, it is necessary and sufficient that $F : C^\circ \to (Ens)$ be representable; if $F \simeq hT$, then $\alpha S(H)$ is representable by the object $T \to S$ of C/S.*

The equivalence αS is transitive in S: if $f : T \to S$ is an object of C/S, one has a commutative diagram of equivalences

[18] N.D.E.: Cf. N.D.E. (10).

$$\begin{tikzcd}
(\hat{C}/hS)/hT \arrow[r, "\alpha_{S/hT}"] \arrow[dr, "\simeq"'] & (\hat{C}_{/S})/h_T \arrow[r, "\alpha_f"] & \hat{C}_{(C/S)/T} \\
& \hat{C}/hT \arrow[r, "\alpha_T"'] & \hat{C}_{/T} \arrow[u, "\simeq"']
\end{tikzcd}$$

where $\alpha_{S/hT}$ denotes (provisionally) the restriction (cf. 1.6) of the functor αS to objects above hT.

1.6. Base change in a functor

For every $S \in Ob(C)$, one has a canonical functor

$$iS : C/S \to C$$

defined by $iS(f) = T$ if f is the arrow $T \to S$. If $f : T \to S$ is an object of C/S, one denotes by $i_{T/S} = i_f$ the functor:

$$i_{T/S} : (C/S)/T \to C/S,$$

and one has the commutative diagram:

$$\begin{tikzcd}
(C/S)/T \arrow[r, "i_{T/S}"] \arrow[dr, "\simeq"'] & C/S \arrow[r, "iS"] & C \\
& C/T \arrow[ur, "iT"'] &
\end{tikzcd}$$

that is, identifying $(C/S)/T$ with C/T as we shall do henceforth,

$$iS \circ i_{T/S} = iT.$$

In the same way, if one identifies C and C/e when C has a final object e, then $i_{S/e} : C/S \to C/e$ is identified with iS.

For $X \in Ob(C)$ (resp. $Y \in Ob(C/S)$), let $pS(X)$ (resp. $p_{T/S}(Y)$) denote the object of C/S (resp. of C/T), when it exists, defined by $X \times S$ (resp. $Y \times_S T$) equipped with its second projection:

$$\begin{tikzcd}
X \times S \arrow[d, "pS(X)"'] \\
S
\end{tikzcd}
\quad \text{resp.} \quad
\begin{tikzcd}
Y \times_S T \arrow[d, "p_{T/S}(Y)"'] \\
T
\end{tikzcd}$$

The (partially defined) functor pS (resp. $p_{T/S}$) is called the *base change functor*. It is by definition of the product (resp. of the fiber product) the right adjoint of the functor iS (resp. $i_{T/S}$).[19] One also denotes

pS(X) = XS and p_{T/S}(Y) = YT.

The functor iS defines a functor (restriction)

$$i*_S : \hat{C} \to \hat{C}_{/S};$$

one denotes $FS = i*_S(F) = F \circ iS$. One has obviously

$$i*_{T/S} \circ i*_S = i*_T,$$

that is, for every functor $F \in Ob(\hat{C})$,

(FS)T = FT.

The notation requires a justification, which is the following:

Proposition 1.6.1. *For the functor $(hX)S : (C/S)^{\circ} \to (Ens)$ to be representable, it is necessary and sufficient that the product $X \times S$ exist. One then has*

$$(hX)S \simeq h_{XS}.$$

This shows that FS has two interpretations: restriction of the functor F to C/S, and functor obtained by base change $e \leftarrow S$. This leads to the following notation:

```
 F          FS             FT

 |           |              |
 e ←———————  S ←——————————  T
```

which renders both of the preceding interpretations. Note that one has

Γ(FS) ≃ Hom(hS, F) ≃ F(S),

in particular

$$\Gamma(XS) \simeq \mathrm{Hom}(S, X).$$

[19] N.D.E.: I.e. if $g : U \to S$ (resp. $h : V \to T$) is an object of C/S (resp. C/T) then $iS(g) = U$ and $i_{T/S}(h)$ is the object $f \circ h : V \to S$ of C/S, and one has:

Hom_{C/S}(U, X × S) ≃ Hom_C(U, X) resp. Hom_{C/T}(V, X ×_S T) ≃ Hom_{C/S}(V, X).

1.7.0.

[20] Let E be an object of $\hat{C}$. Consider the category C_E of objects of C above E: its objects are the pairs (V, ρ) formed by an object V of C and a $\hat{C}$-morphism $\rho : hV \to E$, i.e. $\rho \in E(V)$; a morphism from (V, ρ) to (V', ρ') is

the datum of a C-morphism $f : V \to V'$ such that $\rho' \circ f = \rho$ (i.e. $E(f)(\rho') = \rho$). Denote by L the functor

$$\lim \to_{(V,\rho)\in C_E} hV,$$

i.e. for every $S \in Ob(C)$, $L(S) = \lim \to_{(V,\rho)} hV(S)$ is the set of equivalence classes of triples (V, ρ, v), where $v : S \to V$ is a C-morphism, and where one identifies (V, ρ, v) with $(V', \rho', f \circ v)$ for every C-morphism $f : V \to V'$ such that $\rho' \circ f = \rho$.

Then the map $\phi_E(S)$ which to the class of (V, ρ, v) associates the element $\rho \circ v$ of $E(S)$ is well defined, and defines a morphism of functors

φ_E : lim→_{(V,ρ)∈C_E} hV → E.

Lemma. *ϕ_E is an isomorphism.*

Indeed, let $S \in Ob(C)$. Every $x \in E(S)$ is the image under $\phi_E(S)$ of the triple (S, x, id_S); this shows that $\phi_E(S)$ is surjective. On the other hand, let $\ell_1 = (V_1, \rho_1, v_1)$ and $\ell_2 = (V_2, \rho_2, v_2)$ be two elements of $L(S)$ having the same image in $E(S)$; set $\gamma = \rho_1 \circ v_1 = \rho_2 \circ v_2$. Then ℓ_1 and ℓ_2 are both equal, in $L(S)$, to the class of the triple (S, γ, id_S). This shows that $\phi_E(S)$ is injective.

Corollary. *For every object F of $\hat{C}$, one has*

Hom(E, F) = lim←_{(V,ρ)∈C_E} F(V).

1.7. Objects Hom, Isom, etc.

Let F and G be two objects of $\hat{C}$. We shall define another object of $\hat{C}$ in the following way:

Hom(F, G)(S) = Hom_{Ĉ_{/S}}(FS, GS) ≃ Hom_{Ĉ/hS}(F × hS, G × hS) ≃ Hom_Ĉ(F × hS, G).

The object $\mathrm{Hom}(F, G)$ defined above has the following properties:

(i) $\mathrm{Hom}(e, G) \simeq G$.[21]

(ii) *The formation of* Hom *commutes with base extension:*

Hom(FS, GS) ≃ Hom(F, G)_S.

(iii) $(F, G) \mapsto \mathrm{Hom}(F, G)$ *is a bifunctor, contravariant in F and covariant in G.*

These three properties are obvious from the definitions.

[20] N.D.E.: The following lemma has been added (cf. SGA 4, I.3.4); it is used in the proof of 1.7.1 and will be useful several times in the sequel.

[21] N.D.E.: And, if E is a third object of $\hat{C}$, one has Hom(E, F × G) ≅ Hom(E, F) × Hom(E, G).

We shall show that, for every object E of $\hat{C}$, one has

$$\mathrm{Hom}(E, \mathrm{Hom}(F, G)) \cong \mathrm{Hom}(E \times F, G).$$

Let $\phi : E \times F \to G$; we must associate to it a morphism of E into $\mathrm{Hom}(F, G)$. So let $S' \to S$ be an arrow of C. One has maps

$$E(S) \times F(S') \to E(S') \times F(S') \xrightarrow{\phi(S')} G(S').$$

Every element e of $E(S)$ therefore defines, for every $S' \to S$, a map $F(S') \to G(S')$ functorial in S', i.e. an element $\theta_\phi(e)$ of $\mathrm{Hom}(F, G)(S)$. One has thus obtained a map

$$\phi \mapsto \theta_\phi, \quad \mathrm{Hom}(E \times F, G) \to \mathrm{Hom}(E, \mathrm{Hom}(F, G)),$$

which is "functorial in E".

Proposition 1.7.1.[22] *Let $E, F, G \in Ob(\hat{C})$.*

(a) The map $\phi \mapsto \theta_\phi$ is a bijection:

$$\mathrm{Hom}(E \times F, G) \xrightarrow{\sim} \mathrm{Hom}(E, \mathrm{Hom}(F, G)).$$

(b) Moreover, one has an isomorphism of functors:

$$\mathrm{Hom}(E, \mathrm{Hom}(F, G)) \simeq \mathrm{Hom}(E \times F, G).$$

(a) Consider both members as functors in E. The asserted result is true if $E = hX$; indeed, in that case it is nothing other than the definition of the functor $\mathrm{Hom}(F, G)$. On the other hand, both members as functors in E transform direct limits into inverse limits. Finally, by Lemma 1.7.0, every object E of $\hat{C}$ is isomorphic to the direct limit of the hX, where X runs over the category C_E. This proves (a).

[23] Let us sketch a direct proof of (a). To every $\theta \in \mathrm{Hom}(E, \mathrm{Hom}(F, G))$, one associates the element ϕ_θ of $\mathrm{Hom}(E \times F, G)$ defined as follows. For every $S \in Ob(C)$, one has a map

$$\theta(S) : E(S) \to \mathrm{Hom}(F \times S, G),$$

functorial in S. If $(e, f) \in E(S) \times F(S)$, then f is a morphism $S \to F$, hence $f \times id_S$ is a morphism $S \to F \times S$; on the other hand, $\theta(S)(e)$ is a morphism $F \times S \to G$, so by composition one obtains a morphism:

$$\theta(S)(e) \circ (f \times id_S) : S \to G,$$

i.e. an element $\phi_\theta(S)(e, f)$ of $G(S)$. One verifies easily that the correspondence $S \mapsto \phi_\theta(S)$ is functorial in S, hence defines a morphism ϕ_θ from $E \times F$ to G. We leave to the reader the verification that the maps $\theta \mapsto \phi_\theta$ and $\phi \mapsto \theta_\phi$ are mutually inverse bijections.

Let us prove (b). If $S \in Ob(C)$, one has, by 1.7 (ii) and (a) applied to C/S:

---

[22] N.D.E.: Point (b) has been added, which will be useful in II.1 and II.3.11. On the other hand, a second, more direct, proof of (a) has been sketched.

[23] (continuation of the proof of Proposition 1.7.1, sketching a direct argument)

```
Hom(E, Hom(F, G))(S) ≃ Hom_S(ES, Hom_S(FS, GS))
                     ≅ Hom_S(ES ×_S FS, GS)
                     ≅ Hom(E × F × S, G)
                     ≅ Hom(E × F, G)(S)
```

and these isomorphisms are functorial in S.

Corollary 1.7.2. *One has:*

```
Hom(E, Hom(F, G)) ≃ Hom(F, Hom(E, G)),
Hom(E, Hom(F, G)) ≃ Hom(F, Hom(E, G)).
```

In particular, taking $E = e$, and taking into account $\mathrm{Hom}(e, G) \simeq G$, one has

```
Γ(Hom(F, G)) ≃ Hom(F, G).
```

Note that the composition of `Hom`'s provides functorial morphisms

```
Hom(F, G) × Hom(G, H) → Hom(F, H).
```

If F and G are two objects of Ĉ, one denotes by $Isom(F,G)$ the subset of $\mathrm{Hom}(F,G)$ formed by the isomorphisms of F onto G. One then defines a subobject $Isom(F,G)$ of $\mathrm{Hom}(F,G)$ by:

```
Isom(F, G)(S) = Isom(FS, GS).
```

One then has isomorphisms

```
Γ(Isom(F, G)) ≃ Isom(F, G),
Isom(F, G) ≃ Isom(G, F).
```

In the particular case where $F = G$, one sets

```
End(F) = Hom(F, F),    End(F) = Hom(F, F) ≃ Γ(End(F)),
Aut(F) = Isom(F, F),    Aut(F) = Isom(F, F) ≃ Γ(Aut(F)).
```

The formation of the objects `Hom, Isom, Aut, End` commutes with base change.

Remark 1.7.3.[24] One can construct an object isomorphic to $Isom(F,G)$ in the following manner: one has a morphism

```
Hom(F, G) × Hom(G, F) → End(F);
```

permuting F and G, one deduces a morphism

```
Hom(F, G) × Hom(G, F) → End(F) × End(G).
```

On the other hand, the identity morphism of F is an element of $\mathrm{End}(F)$ and therefore defines a morphism $e \to \mathrm{End}(F)$. Doing the same with G and forming the product, one finds a morphism

$$e \to \mathrm{End}(F) \times \mathrm{End}(G).$$

[24]N.D.E.: The numbering 1.7.3 has been added, for subsequent references.

It is then immediate that the fiber product of e and of $\mathrm{Hom}(F,G)\times\mathrm{Hom}(G,F)$ over $\mathrm{End}(F)\times\mathrm{End}(G)$ is isomorphic to $Isom(F,G)$.

All these definitions apply in particular to the case where $F = hX, G = hY$. In the case where $\mathrm{Hom}(hX,hY)$ is representable by an object of C, one denotes this object by $\mathrm{Hom}(X,Y)$. It has the following property: if $Z\times X$ exists, then

```
Hom(Z, Hom(X, Y)) ≃ Hom(Z × X, Y).
```

This property characterizes it when the products exist in C.

One defines similarly (when they happen to exist) objects

```
Isom(X, Y),    End(X),    Aut(X);
```

we simply note that, by the construction given above, $Isom(X,Y)$ exists whenever fiber products exist in C and `Hom(X, Y)`, `Hom(Y, X)`, `End(X)` and $\mathrm{End}(Y)$ exist.

All that precedes applies equally to categories of the form C/S. The corresponding objects will be denoted as explicitly as possible by appropriate symbols: for example, if T and T' are two objects of C above S, one will denote by $\mathrm{Hom}_S(T,T')$ the object $\mathrm{Hom}_{C/S}(T/S,T'/S)$.

1.8. Constant objects

Let C be a category in which direct sums and fiber products exist, and in which direct sums commute with base change (for example, the category of schemes[25]). For every set E and every object S of C, we set

```
(1.8.1)    ES =  the direct sum of a family (S_i)_{i∈E}
                 of objects of C all isomorphic to S.
```

This object is characterized by the formula:

```
(1.8.2)    Hom_C(ES, T) = Hom(E, Hom_C(S, T)),
```

for every $T\in Ob(C)$, where the second `Hom` is taken in the category of sets.

The object `ES` is equipped with a canonical projection onto S, in such a way that $E\mapsto ES$ is in fact a functor from `(Ens)` to C/S.

[26] If $S'\to S$ is an arrow of C, one has, since direct sums commute with base change,

$$E_{S'} = (ES)_{S'}.$$

In particular, if C has a final object e, one has

$$ES = (Ee)_S.$$

[25]N.D.E.: The former terminology “préschémas/schémas” has been replaced by the current terminology “schémas/schémas séparés”.

[26]N.D.E.: In the three paragraphs that follow, the order of the sentences has been modified and some clarifications added concerning the role of the hypothesis (∗) below.

The functor $E \mapsto ES/S$, from (Ens) to C/S, commutes with finite products. For this, it suffices to see that

$$(\times) \qquad ES \times_S FS = (E \times F)_S.$$

Now, by the results of 1.7 applied to C/S, one has, for every $T \in Ob(C/S)$, natural isomorphisms (all unspecified Hom's being taken in C/S):

$$\mathrm{Hom}((E \times F)_S, T) \cong \mathrm{Hom}_{(\mathrm{Ens})}(E \times F, \mathrm{Hom}(S, T)) \cong \mathrm{Hom}_{(\mathrm{Ens})}(E, \mathrm{Hom}_{(\mathrm{Ens})}(F, \mathrm{Hom}(S, T))) \cong \mathrm{Hom}_{(\mathrm{Ens})}(E, \mathrm{Hom}(FS, T))$$

and

$$\mathrm{Hom}(ES \times_S FS, T) \cong \mathrm{Hom}(ES, \mathrm{Hom}(FS, T)) \cong \mathrm{Hom}_{(\mathrm{Ens})}(E, \mathrm{Hom}(S, \mathrm{Hom}(FS, T))).$$

Now $\mathrm{Hom}(S, \mathrm{Hom}(FS, T)) \cong \mathrm{Hom}(FS, T)$, hence $(\times)$.

Suppose that, with $\emptyset$ denoting an initial object of C, the diagram

$$\begin{array}{ccc} \emptyset & \longrightarrow & S \\ \downarrow & & \downarrow \\ S & \longrightarrow & S \sqcup S \end{array}$$

$(*)$ *is cartesian.*

(This is the case for the category of schemes.)[27] Then the functor $E \mapsto ES$ commutes with finite inverse limits.

Indeed, taking $(\times)$ into account, it suffices to see that $E \mapsto ES$ commutes with fiber products. Let $u : E \to G$ and $v : F \to G$ be two maps of sets. Since in C direct sums commute with base change, one has

$$(1) \qquad FS \times_{GS} ES \cong \coprod_{f\in F} S_f \times_{GS} ES \cong \coprod_{f\in F,\, x\in E} S_f \times_{GS} S_x.$$

If $v(f) \neq u(x)$, there exists in C/S a morphism

$$S_f \times_{GS} S_x \to S_f \times_{S_{v(f)} \sqcup S_{u(x)}} S_x;$$

now by hypothesis $(*)$ the right-hand term is $\emptyset$. Consequently,

$$(2) \qquad S_f \times_{GS} S_x \cong \emptyset \qquad \text{if } v(f) \neq u(x).$$

On the other hand, if $v(f) = u(x)$, there exists in C/S a morphism

$$S \to S_f \times_{GS} S_x;$$

since $S \xrightarrow{id} S$ is the final object of C/S, it follows that

$$(3) \qquad S_f \times_{GS} S_x \cong S \qquad \text{if } v(f) = u(x).$$

Combining (1), (2), and (3) one obtains an isomorphism functorial in S

$$FS \times_{GS} ES \cong \coprod_{F \times_G E} S = (F \times_G E)_S.$$

[27] N.D.E.: $(*)$ is not verified if C is the category whose arrows are $A \to B$ and id_A, id_B; in this case $B \sqcup B = B$ and $B \times_B B = B \neq A$.

An object of the form ES will be called a *constant object*. Note that one has a morphism functorial in E:

$$E \to \Gamma(ES/S)$$

which associates to each $i \in E$ the section of ES over S defined by the isomorphism of S onto S_i. Suppose condition $(*)$ is satisfied for every object S of C; then the morphism $E \to \Gamma(ES/S)$ is a monomorphism for every $S \neq \varnothing$.

If C is the category of schemes, then $\Gamma(ES/S)$ is identified with the locally constant maps from the topological space S to the set E, the preceding map associating to each element of E the corresponding constant map. Note that, by what was just said, ES may also be defined as representing the functor which to every S' over S associates the set of locally constant functions from the topological space S' to the set E.[28]

2. Algebraic structures

Given a species of algebraic structure in the category of sets, we propose to extend it to the category C. Let us first treat one example: the case of groups.

2.1. Group structures

We retain the notations of the preceding paragraph.

Definition 2.1.1. *Let $G \in Ob(\hat{C})$. By a* $\hat{C}$-group structure *on G, we mean the datum, for every $S \in Ob(C)$, of a group structure on the set $G(S)$, in such a way that for every arrow $f : S' \to S''$ of C, the map $G(f) : G(S'') \to G(S')$ is a group homomorphism. If G and H are two $\hat{C}$-groups, by a* morphism of $\hat{C}$-groups *from G into H we mean any morphism $u \in \mathrm{Hom}(G,H)$ such that for every $S \in Ob(C)$, the map of sets $u(S) : G(S) \to H(S)$ is a group homomorphism.*

One denotes by $\mathrm{Hom}_{\hat{C}-Gr.}(G,H)$ the set of morphisms of $\hat{C}$-groups from G into H, and by $(\hat{C} - Gr.)$ the category of $\hat{C}$-groups.

Examples. Let $E \in Ob(\hat{C})$; the object $\mathrm{Aut}(E)$ is endowed in an obvious manner with a $\hat{C}$-group structure. The final object e has a unique $\hat{C}$-group structure which makes it a final object of $(\hat{C} - Gr.)$.

For every $S \in Ob(C)$, let $e_G(S)$ be the identity element of $G(S)$. The family of the $e_G(S)$

defines an element $e_G \in \Gamma(G) = \mathrm{Hom}(e,G)$ which is a morphism of $\hat{C}$-groups $e \to G$, and which is called the *unit section* of G.

Note that to give a $\hat{C}$-group structure on G amounts to giving a composition law on G, i.e. a $\hat{C}$-morphism

[28] N.D.E.: Note also that the diagonal morphism $ES \to ES \times_S ES = (E \times E)_S$ is a closed immersion, i.e. ES is separated over S.

$\pi_G : G \times G \to G$

such that, for every $S \in Ob(C)$, $\pi_G(S)$ endows $G(S)$ with a group structure.

In the same way, $f : G \to H$ is a morphism of Ĉ-groups if and only if the following diagram is commutative:

$$\begin{tikzcd} G \times G \arrow[r, "\pi_G"] \arrow[d, "{(f,f)}"] & G \arrow[d, "f"] \\ H \times H \arrow[r, "\pi_H"] & H \end{tikzcd}$$

A subobject H of G such that, for every $S \in Ob(C)$, $H(S)$ is a subgroup of $G(S)$, obviously has a Ĉ-group structure induced by that of G: it is the only one for which the monomorphism $H \to G$ is a morphism of Ĉ-groups. The Ĉ-group H endowed with this structure is called a *sub-$\hat{C}$-group of G*.

If G and H are two Ĉ-groups, the product $G \times H$ is endowed with an obvious Ĉ-group structure: for every $S \in Ob(C)$, one endows $G(S) \times H(S)$ with the product group structure of the group structures given on $G(S)$ and $H(S)$. The Ĉ-group $G \times H$ endowed with this structure will be called the *product $\hat{C}$-group* of G and H (it is indeed the product in the category of Ĉ-groups).

If G is a Ĉ-group, then for every $S \in Ob(C)$, GS is a $\hat{C}_{/S}$-group. If G and H are two Ĉ-groups, one defines the object $\mathrm{Hom}_{\hat{C}-Gr.}(G, H)$ of Ĉ by:

$\mathrm{Hom}_{\hat{C}\text{-Gr.}}(G, H)(S) = \mathrm{Hom}_{\hat{C}_{/S}\text{-Gr.}}(GS, HS)$

(Note: $\mathrm{Hom}_{\hat{C}-Gr.}$ is not in general a Ĉ-group, nor a fortiori the Hom object in

the category $(\hat{C} - Gr.)$).

One defines similarly the objects

$\mathrm{Isom}_{\hat{C}\text{-Gr.}}(G, H), \quad \mathrm{End}_{\hat{C}\text{-Gr.}}(G), \quad \mathrm{Aut}_{\hat{C}\text{-Gr.}}(G).$

Definition 2.1.2. *Let $G \in Ob(C)$. By a C-*group structure *on G, we mean a $\hat{C}$-group structure on $hG \in Ob(\hat{C})$. By a* morphism of the C-group G into the C-group H, *we mean an element $u \in \mathrm{Hom}(G, H) \simeq \mathrm{Hom}(hG, hH)$ which defines a morphism of $\hat{C}$-groups from hG into hH.*

One denotes by (C-Gr.) the category of C-groups. Note that there exists in (Cat) a cartesian square

$$\begin{tikzcd} (\text{C-Gr.}) \arrow[r] \arrow[d] & (\hat{\text{C}}\text{-Gr.}) \arrow[d] \\ C \arrow[r, "h"] & \hat{C} \end{tikzcd}$$

All the preceding definitions and constructions therefore transport at once to (C-Gr.) whenever the functors they involve (products, Hom objects, etc.) are representable. They also apply to the categories C/S. In that case, we shall write $\mathrm{Hom}_{S-Gr.}$ for $\mathrm{Hom}_{\hat{C}/S-Gr.}$, etc.

2.2.

More generally, if (T) is a species of structure on n base sets defined by finite inverse limits (for example, by commutativities of diagrams constructed with cartesian products: structures of monoid, group, set with operators, module over a ring, Lie algebra over a ring, etc.), the preceding construction permits one to define the notion of "structure of species (T) on n objects $F_1, \cdots, F_n$ of $\hat{C}$":

such a structure will be the datum, for each S of C, of a structure of species (T) on the sets $F_1(S), \cdots, F_n(S)$ in such a way that for every arrow $S' \to S''$ of C, the family of maps $(F_i(S'')) \to (F_i(S'))$ is a polyhomomorphism for the species of structure (T). One defines similarly the morphisms of the species of structure (T), whence a category $(\hat{C} \times \hat{C} \cdots \times \hat{C})^{(T)}$. The fully faithful functor $(h \times h \times \cdots \times h)$ then allows one to define by inverse image the category $(C \times C \times \cdots \times C)^{(T)}$, and then, as it commutes with inverse limits, to transport to it all the properties, notions, and notations of a functorial kind introduced in $\hat{C}$. Suppose now that in C fiber products exist, and let (T) be a species of algebraic structure defined by the datum of certain morphisms between cartesian products satisfying axioms consisting of certain commutativities of diagrams constructed using the preceding arrows. A structure of species (T) on a family of objects of C will therefore be defined by certain morphisms between cartesian products satisfying certain commutation conditions. It follows that if C and C' are two categories possessing products and $f : C \to C'$ is a functor commuting with products, then for every family of objects (F_i) of C endowed with a structure of species (T), the family $(f(F_i))$ of objects of C' will thereby also be endowed with a structure of species (T). Every C-group will be transformed into a C'-group, every pair (C-ring, C-module over this C-ring) into an analogous pair in C', etc.

In particular, let C be a category satisfying the conditions of 1.8;[29] the functor $E \mapsto ES$ defined in *loc. cit.* commutes with finite inverse limits; it therefore transforms group into S-group (i.e. C/S-group), ring into S-ring, etc.

Remark. It is good to note that the preceding construction procedure applied to the category $\hat{C}$ does give back the notions already defined there; in other words, it amounts to the same thing to give on an object of $\hat{C}$ a structure of species (T) when one considers this object as a functor on C, or to give a structure of species (T) on the representable functor on $\hat{C}$ defined by this object.[30]

We shall still treat two particular cases of the preceding construction, the case of structures with operator groups and the case of modules.

[29]N.D.E.: Including the condition (∗).

[30]N.D.E.: For example, for group structures: let $G \in Ob(\hat{C})$; if the functor $\hat{C} \to (Ens)$, $F \mapsto \mathrm{Hom}_{\hat{C}}(F, G)$ is endowed with a group structure, the same is true of its restriction to C, $X \mapsto G(X)$. Conversely, if G is a $\hat{C}$-group, then the "multiplication" morphism $\pi_G : G \times G \to G$ induces for every $F \in Ob(\hat{C})$ a group structure on $\mathrm{Hom}_{\hat{C}}(F, G)$, functorial in F.

2.3. Structures with operator groups

Definition 2.3.1. *Let $E \in Ob(\hat{C})$ and $G \in Ob(\hat{C} - Gr.)$. A structure of* object with $\hat{C}$-operator group G (*or of G-object*) *on E is the datum, on $E(S)$, for every $S \in Ob(C)$, of a structure of set with operator group $G(S)$ in such a way that, for every arrow $S' \to S''$ of C, the map of sets $E(S'') \to E(S')$ is compatible with the operator homomorphism $G(S'') \to G(S')$.*

As usual, it amounts to the same thing to give a morphism

$$\mu : G \times E \to E$$

which for each S endows $E(S)$ with a structure of set with operators $G(S)$. But $\mathrm{Hom}(G \times E, E) \simeq \mathrm{Hom}(G, \mathrm{End}(E))$, so μ defines a morphism $G \to \mathrm{End}(E)$, and it is immediate that this maps G into $\mathrm{Aut}(E)$ and is a morphism of $\hat{C}$-groups. Consequently: to give on E a structure of object with $\hat{C}$-operator group G is equivalent to giving a morphism of $\hat{C}$-groups

$$\rho : G \to \mathrm{Aut}(E).$$

In particular, every element $g \in G(S)$ defines an automorphism $\rho(g)$ of the functor E_S, that is, an automorphism of $E \times hS$ commuting with the projection $E \times hS \to hS$, and in particular an automorphism of the set $E(S')$ for every $S' \to S$.

Definition 2.3.2. *One denotes by E^G the subobject of E defined as follows:*

$$E^G(S) = \{x \in E(S) \mid x_{S'} \text{ invariant under } G(S') \text{ for every } S' \to S\},$$

where $x_{S'}$ denotes the image of x by $E(S) \to E(S')$.

Then E^G (*"subobject of invariants of G"*)

is the largest subobject of E on which G operates trivially.

Definition 2.3.3. *Let F be a subobject of E. One denotes by $Norm_G F$ and $Centr_G F$ the sub-$\hat{C}$-groups of G defined by*

[31]

$$\begin{aligned}(Norm_G F)(S) &= \{g \in G(S) \mid \rho(g)\, F_S = F_S\}\\ &= \{g \in G(S) \mid \rho(g)\, F(S') = F(S'), \text{ for every } S' \to S\},\end{aligned}$$

$$\begin{aligned}(Centr_G F)(S) &= \{g \in G(S) \mid \rho(g)|F_S = \text{identity}\}\\ &= \{g \in G(S) \mid \rho(g)|F(S') = \text{identity, for every } S' \to S\},\end{aligned}$$

where the vertical bar after $\rho(g)$ denotes restriction.

[31] N.D.E.: We have corrected the original, replacing the inclusion $\rho(g)F_S \subset F_S$ by an equality, in order to ensure that $Norm_G(F)$ is indeed a group (see VI_B 6.4 for conditions under which the "transporter" coincides with the "strict transporter").

Scholium 2.3.3.1.[32] *In particular, let $x \in \Gamma(E)$, i.e. (cf. 1.2) a collection of elements $x_S \in E(S)$, $S \in Ob(C)$, such that for every arrow $f : S' \to S$ one has $E(f)(x_S) = x_{S'}$ (if C has a final object* `S_0` *one has $\Gamma(E) = E(S_0)$). Then x defines a subfunctor of E, which we shall denote $\bar{x}$, and one has $Norm_G \bar{x} = Centr_G \bar{x}$. We shall denote by $Stab_G(x)$ and call* stabilizer of x *this functor; for every $S \in Ob(C)$ one therefore has:*

`Stab_G(x)(S) = {g ∈ G(S) | ρ(g) x_S = x_S}.`

Suppose that fiber products exist in C; if $G = hG$ (resp. $E = hE$), *where G is a C-group (resp. $E \in Ob(C)$), and if C has a final object* `S_0`*, so that x is a morphism $S_0 \to E$, then $Stab_G(x)$ is representable by the fiber product $G \times_E S_0$, where $G \to E$ is the composite of $id_G \times x : G = G \times S_0 \to G \times E$ and of $\mu : G \times E \to E$.*

Remark 2.3.3.2.[33] *The formation of E^G, $Norm_G F$ and $Centr_G F$ commutes with base change, i.e. for every $S \in Ob(C)$ one has*

`(E^G)_S = (ES)^{GS},    (Norm_G F)_S ≃ Norm_{GS} F_S,    (Centr_G F)_S ≃ Centr_{GS} F_S.`

Definition 2.3.4. *If G is a C-group and E an object of $\hat{C}$* (resp. E an object of C) *a structure of G-object on E* (resp. on E) *is a structure of* `hG`*-object on E* (resp. `hE`).

In view of this definition, all the notions and notations defined above transport to C when they involve only representable functors: for example, if $Norm_{hG}(hF)$ is representable, then there exists one and only one subobject of G that represents it, and which is then a sub-C-group of G; one denotes it $Norm_G(F)$, etc.

Definition 2.3.5. *a) One says that the $\hat{C}$-group G* operates *on the $\hat{C}$-group H if H is endowed with a structure of G-object such that, for every $g \in G(S)$, the automorphism of $H(S)$ defined by g is a group automorphism.*

It amounts to the same thing to say that for every $g \in G(S)$, the automorphism $\rho(g)$ of `HS`

is an automorphism of $\hat{C}_{/S}$-groups, or that the morphism of Ĉ-groups $G \to \mathrm{Aut}(H)$ maps G into $\mathrm{Aut}_{\hat{C}-Gr.}(H)$.

b) In the situation above, there exists on the product $H \times G$ a unique $\hat{C}$-group structure such that, for every S, $(H \times G)(S)$ is the semidirect product of the groups $H(S)$ and $G(S)$ relative to the given operation of $G(S)$ on $H(S)$. We shall denote this $\hat{C}$-group $H \cdot G$ and call it the semidirect product *of H by G. One therefore has by definition*

`(H · G)(S) = H(S) · G(S).`

Let G be a Ĉ-group. For every arrow $S' \to S$ of C and every $g \in G(S)$, let $Int(g)$ be the automorphism of $G(S')$ defined by $Int(g)h = ghg^{-1}$. This definition extends into that of a morphism of Ĉ-groups

`Int : G → Aut_{Ĉ-Gr.}(G) ⊂ Aut(G).`

[32] N.D.E.: Scholium 2.3.3.1 and Remark 2.3.3.2 have been added.
[33] N.D.E.: Scholium 2.3.3.1 and Remark 2.3.3.2 have been added.

Definition 2.3.3 therefore applies, and one has sub-$\hat{C}$-groups of G

Norm_G(E) and Centr_G(E)

for every subobject E of G.

Definition 2.3.6. *One calls* center of G *and one denotes by* $Centr(G)$ *the sub-*$\hat{C}$*-group* $Centr_G(G)$ *of* G. *One says that* G *is* commutative *if* $Centr(G) = G$, *or, what amounts to the same thing, if* $G(S)$ *is commutative for every* S.

One says that the sub-$\hat{C}$*-group* H *of* G *is* invariant *in* G *(or* normal*) if $Norm_G(H) = G$, or, what amounts to the same thing, if $H(S)$ is invariant in $G(S)$ for every S.*[34]

Definition 2.3.6.1.[35] *Let* $f : G \to G'$ *be a morphism of* $\hat{C}$*-groups. One calls* kernel of f, *and one denotes by* Ker f, *the sub-*$\hat{C}$*-group of* G *defined by*

(Ker f)(S) = {x ∈ G(S) | f(S)(x) = 1} = Ker f(S)

for every $S \in Ob(C)$; *it is an invariant sub-*$\hat{C}$*-group.*

Note that if $G = hG$ *and* $G' = hG'$, *if* C *has a final object* S_0 *and if fiber products exist in* C, *then* $Ker(f)$ *is representable by* $S_0 \times_{G'} G$.

Definition 2.3.6.2.[36] *Let* $E \in \hat{C}$ *and* G *a* $\hat{C}$*-group operating on* E. *One says that the operation of* G *on* E *is* faithful *if the kernel of the morphism* $G \to \mathrm{Aut}(E)$ *is trivial, i.e. if for every* $S \in Ob(C)$ *and* $g \in G(S)$, *the condition* $g_{S'} \cdot x = x$ *for every* $S' \to S$ *and* $x \in E(S')$ *entails* $g = 1$.

Many definitions and propositions of the elementary theory of groups transpose easily. Let us simply point out the following, which will be useful to us:

Proposition 2.3.7. *Let* $f : W \to G$ *be a morphism of* $\hat{C}$*-groups. Set* $H(S) = Ker f(S)$. *Let* $u : G \to W$ *be a morphism of* $\hat{C}$*-groups which is a section of* f

(and which is then necessarily a monomorphism). Then W *is identified with the semidirect product of* H *by* G *for the operation of* G *on* H *defined by* $(g,h) \mapsto Int(u(g))h$ *for* $g \in G(S)$, $h \in H(S)$, $S \in Ob(C)$.

The set of these definitions and propositions transports as usual to C. One defines in particular the semidirect product of two C-groups H and G (with G operating on H) when the cartesian product $H \times G$ exists, and one has the analogue of Proposition 2.3.7 in the following form:

Proposition 2.3.8. *Let* $H \xrightarrow{i} W \xrightarrow{f} G$ *be a sequence of morphisms of* C*-groups such that for every* $S \in Ob(C)$, (H(S), i(S)) *is a kernel of* $f(S) : W(S) \to G(S)$. *Let* $u : G \to W$ *be a morphism of* C*-groups which is a section of* f. *Then* W *is identified with the semidirect product of* H *by* G *for the operation of* G *on* H *such that if* $S \in Ob(C)$, $g \in G(S)$ *and* $h \in H(S)$, *one has* $Int(u(g))i(h) = i({}^g h)$.

[34]N.D.E.: Moreover, one says that H is *central* in G if $Centr_G(H) = G$, or, what amounts to the same thing, if $H(S)$ is central in $G(S)$ for every S.
[35]N.D.E.: Definitions 2.3.6.1 and 2.3.6.2 have been added.
[36]N.D.E.: Definitions 2.3.6.1 and 2.3.6.2 have been added.

3. The category of O-modules, the category of G-O-modules

Definition 3.1. *Let O and F be two objects of $\hat{C}$. One says that F is a* $\hat{C}$-module *over the $\hat{C}$-ring O, or in abbreviated form an O*-module*, if, for every $S \in Ob(C)$, one has endowed $O(S)$ with a ring structure and $F(S)$ with a structure of module over this ring in such a way that, for every arrow $S' \to S''$ of C, $O(S'') \to O(S')$ is a ring homomorphism and $F(S'') \to F(S')$ is a homomorphism of abelian groups compatible with this ring homomorphism.*

If O is fixed, one defines in the usual way a morphism of the O-modules F and F', whence the commutative group $\mathrm{Hom}_O(F, F')$, and the category of O-modules, denoted `(O-Mod.)`.

Lemma 3.1.1.[37] *`(O-Mod.)` is endowed with a structure of abelian category, defined "argument by argument". Moreover, `(O-Mod.)` verifies the axiom (AB 5) (cf. [Gr57, 1.5]), i.e. arbitrary direct sums exist, and if M is an O-module, N a submodule, and $(F_i)_{i \in I}$ a directed family of submodules of M, then*

`□_{i∈I} (F_i ∩ N) = (□_{i∈I} F_i) ∩ N.`

Indeed, let $f : F \to F'$ be a morphism of O-modules. One defines the O-modules `Ker f` (resp. `Im f` and `Coker f`) by setting, for every $S \in Ob(C)$, $(Kerf)(S) = Kerf(S)$ (resp. □). Then `Ker f` (resp. `Coker f`) is a kernel (resp. cokernel) of f, and one has an isomorphism of O-modules $F/Kerf \xrightarrow{\sim} Imf$. This proves that `(O-Mod.)` is an abelian category.

Arbitrary direct sums exist and are defined "argument by argument". Finally, if M is an O-module, N a submodule, and $(F_i)_{i \in I}$ a directed family of submodules of M, then the inclusion

`□_{i∈I} (F_i ∩ N) ⊂ (□_{i∈I} F_i) ∩ N`

is an equality: indeed, if $S \in Ob(C)$ and $x \in N(S) \cap \bigcup_i F_i(S)$, there exists $i \in I$ such that $x \in N(S) \cap F_i(S)$.

Proposition 3.1.2.[38] *Suppose the category C is small, i.e. that $Ob(C)$ is a set. Then `(O-Mod.)` has a generator, and therefore enough injective objects.*

Proof. For every object S of C, one defines the O-module `O_S` as follows. For every object S', $O_S(S')$ is a direct sum of copies of $O(S')$ indexed by $\mathrm{Hom}(S', S)$, i.e.

`O_S(S′) = ⊕_{g : S′ → S} O(S′) g.`

For every morphism $f : S'' \to S'$, denoting $f*$ the ring homomorphism $O(S') \to O(S'')$, the morphism $O_S(f) : O_S(S') \to O_S(S'')$ sends each element `a g` of $O_S(S')$ (where

[37]N.D.E.: Statements 3.1.1 and 3.1.2 have been added to make explicit that the category `(O-Mod.)` is abelian and verifies the axiom (AB 5), and moreover has enough injective objects if the category C is small. An error in 3.1.2, reported in 2016 by Linyuan Liu, has been corrected here.

[38]N.D.E.: Statements 3.1.1 and 3.1.2 have been added to make explicit that the category `(O-Mod.)` is abelian and verifies the axiom (AB 5), and moreover has enough injective objects if the category C is small. An error in 3.1.2, reported in 2016 by Linyuan Liu, has been corrected here.

$a \in O(S')$ and $g : S' \to S$) to the element $f * (a) g \circ f$ of $O_S(S'')$. One verifies easily that this does define a functor in O-modules.

Since, by hypothesis, $Ob(C)$ is a set, one can consider the direct sum $G = \bigoplus_{S \in Ob(C)} O_S$.

Lemma 3.1.2.1. *Let F be an O-module. For every $S \in Ob(C)$, every element x of $F(S)$ defines a unique morphism of O-modules $\tilde{x} : O_S \to F$ such that $\tilde{x}(1id_S) = x$.*

Proof. Let ϕ be a morphism of O-modules $O_S \to F$ such that $\phi(1id_S) = x$. Let $g : S' \to S$ and $a \in O(S')$. The $O(S')$-linearity and the commutativity of the diagram

$$\begin{tikzcd} O_S(S) \arrow[r, "\varphi"] \arrow[d, "O_S(g)"'] & F(S) \arrow[d, "F(g)"] \\ O_S(S') \arrow[r, "\varphi"] & F(S') \end{tikzcd}$$

entail that $\phi(ag) = a\phi(1g) = aF(g)(\phi(1id_S)) = aF(g)(x)$. So ϕ, if it exists, is determined by the previous equality. Conversely, if one defines ϕ thus, then for every $f : S'' \to S'$ one has

$$\varphi \circ O_S(f)(ag) = \varphi(f^*(a)\, g \circ f) = f^*(a)\, F(g \circ f)(x) = F(f)(a\, F(g)(x)) = F(f) \circ \varphi(ag)$$

i.e. the diagram

$$\begin{tikzcd} O_S(S') \arrow[r, "\varphi"] \arrow[d, "O_S(f)"'] & F(S') \arrow[d, "F(f)"] \\ O_S(S'') \arrow[r, "\varphi"] & F(S'') \end{tikzcd}$$

is commutative. This proves the lemma.

Now, let F be an O-module. For every $S \in Ob(C)$, let $F_0(S)$ be a system of generators of the $O(S)$-module $F(S)$, and let L_S be the direct sum indexed by $F_0(S)$ of copies of O_S. By the lemma, one obtains a morphism of O-modules $L_S \to F$, such that the morphism $L_S(S) \to F(S)$ is surjective, hence an epimorphism in the category of $O(S)$-modules.

Set $I = \coprod_{S \in Ob(C)} F_0(S)$. Then one obtains a morphism of O-modules

$$G^{\oplus I} \longrightarrow \bigoplus_{S \in Ob(C)} L_S \longrightarrow F$$

which is "argument by argument" an epimorphism, hence an epimorphism of (O-Mod.). This shows that G is a generator of (O-Mod.) (cf. [Gr57, 1.9.1]). Since (O-Mod.) verifies (AB 5), it then follows from [Gr57, 1.10.1] that (O-Mod.) has enough injective objects.

Remark 3.1.3. *If O_0 is the $\hat{C}$-ring defined by $O_0(S) = \mathbb{Z}$ (which should not be confused with the functor associated to the constant object $\mathbb{Z}$), then the category* *of O_0-modules is isomorphic to the category of commutative $\hat{C}$-groups.*

Definition 3.1.4. *Note that, if F is an O-module, then for every $S \in Ob(C)$,* FS *is an O_S-module. One can therefore define a $\hat{C}$-abelian group* $\mathrm{Hom}_O(F, F')$ *by*

Hom_O(F, F′)(S) = Hom_{O_S}(FS, F′_S).

One defines similarly the objects

Isom_O(F, F′),   End_O(F)   and   Aut_O(F),

the last being a $\hat{C}$-group for the structure induced by the composition of automorphisms.

Definition 3.2. *Let O be a $\hat{C}$-ring, F an O-module and G a $\hat{C}$-group. By a* structure of G-O-module *on F we mean a structure of G-object such that for every $S \in Ob(C)$ and every $g \in G(S)$, the automorphism of $F(S)$ defined by g is an automorphism of its $O(S)$-module structure.*

It amounts to the same thing to say that the morphism of $\hat{C}$-groups

$$\rho : G \to \mathrm{Aut}(F)$$

defined in 2.3 maps G into the sub-$\hat{C}$-group $\mathrm{Aut}_O(F)$ of $\mathrm{Aut}(F)$. To give a structure of G-O-module on the O-module F is therefore to give a morphism of $\hat{C}$-groups

$$\rho : G \to \mathrm{Aut}_O(F).$$

One defines in a natural way the abelian group $\mathrm{Hom}_{G-O}(F, F')$, hence the additive category of G-O-modules denoted (G-O-Mod.).

Remark 3.2.1. *The reader will note that (G-O-Mod.) may also be defined as the category of O[G]-modules, where O[G] is the algebra of the $\hat{C}$-group G over the $\hat{C}$-ring O, whose definition is clear.*[39] *Consequently, by 3.1.1 and 3.1.2, (G-O-Mod.) is an abelian category verifying the axiom (AB 5); moreover, if C is small, (G-O-Mod.) has enough injective objects.*

All the constructions of this paragraph specialize at once to the case where G (or O, or both) are representable by objects of C, which are thereby endowed with the corresponding algebraic structures.

We have treated succinctly the case of the principal algebraic structures encountered in the rest of this seminar. For the others (structure of O-Lie algebra for example), we believe that the reader will have had enough examples in this paragraph to make the general mechanism sketched in 2.2 work himself in each particular case.

We shall now apply what we have just done to the category of schemes, denoted (Sch), and more generally to the categories $(Sch)/S$ (also denoted (Sch/S)).

[39] N.D.E.: The following sentence has been added; this will be used in section 5.

4. Algebraic structures in the category of schemes

We shall allow ourselves, whenever there is no ambiguity, the following abuses of language: one will designate by T the object $T \xrightarrow{f} S$ of C/S, the datum of f ("structural morphism of T") being understood, and one will identify C with a subcategory of $\hat{C}$. Given the compatibilities stated in the preceding paragraphs, these identifications can be made without danger.

We shall further simplify the terminology along the following model: a (Sch)-group will also be called a *group scheme*, a $(Sch)/S$-group a *group scheme over S*, or *S-group scheme*, or *S-group*, or *A-group* when S is the spectrum of the ring A.

4.1. Constant schemes

The category of schemes satisfies the conditions required in 1.8. One can therefore define constant objects in it; for every set E, one has a scheme $E_{\mathbb{Z}}$, and for every scheme S, an S-scheme $ES = (E_{\mathbb{Z}})_S$. Recall (cf. *loc. cit.*) that for every S-scheme T, $\mathrm{Hom}_S(T, ES)$ is the set of locally constant maps

from the topological space T to E.

The functor $E \mapsto ES$ commutes with finite inverse limits. In particular if G is a group, GS is an S-group scheme; if O is a ring, OS is an S-ring scheme, etc.

4.2. Affine S-groups

Let us recall a certain number of things about affine S-schemes (EGA II, § 1). One says that the S-scheme T is *affine over S* if the inverse image of every affine open subset of S is affine. The OS-algebra $f_*(O_T)$, which one denotes $A(T)$, is then quasi-coherent (f denotes the structural morphism of T). Conversely, to every quasi-coherent OS-algebra A, one can associate an S-scheme affine over S, denoted $\mathrm{Spec}(A)$. These functors $T \mapsto A(T)$ and $A \mapsto \mathrm{Spec}(A)$ are quasi-inverse equivalences between the category of S-schemes affine over S and the category opposite to that of quasi-coherent OS-algebras.

It follows that to give an algebraic structure on an S-scheme T affine over S is equivalent to giving the corresponding structure on $A(T)$ in the category opposite to that of quasi-coherent OS-algebras. In particular, if G is an S-group affine over S, $A(G)$ is endowed with a structure of augmented OS-bialgebra, that is, one has two morphisms of OS-algebras

$$\Delta : A(G) \to A(G) \otimes_{OS} A(G) \quad \text{and} \quad \varepsilon : A(G) \to OS$$

corresponding to the morphisms of S-schemes

$$\pi : G \times G \to G \quad \text{and} \quad e_G : S \to G.$$

The maps Δ and ϵ satisfy the following conditions (which express that G is an S-monoid):[40]

[40] N.D.E.: And, of course, the inversion morphism $G \to G$ induces a morphism of OS-algebras $\tau : A(G) \to$

(HA 1) Δ *is co-associative: the following diagram is commutative*

$$\begin{tikzcd}
& A(G) \otimes_{OS} A(G) \arrow[dr, "id \otimes \Delta"] & \\
A(G) \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & & A(G) \otimes_{OS} A(G) \otimes_{OS} A(G) \\
& A(G) \otimes_{OS} A(G) \arrow[ur, "\Delta \otimes id"'] &
\end{tikzcd}$$

(HA 2) Δ *is compatible with* ϵ*: the two following composites are the identity*

$$A(G) \xrightarrow{\Delta} A(G) \otimes_{OS} A(G) \xrightarrow{id \otimes \varepsilon} A(G) \otimes_{OS} OS \xrightarrow{\sim} A(G),$$

$$A(G) \xrightarrow{\Delta} A(G) \otimes_{OS} A(G) \xrightarrow{\varepsilon \otimes id} OS \otimes_{OS} A(G) \xrightarrow{\sim} A(G).$$

Let us take this opportunity to note once more that it follows from the definition of an S-group structure that to give such a structure on an S-scheme G affine over S, it is not necessary to verify anything on $A(G)$, but simply to endow each $G(S')$ for S' above S with a group structure functorial in S'.

This remark applies *mutatis mutandis* to morphisms.

4.3. The groups G_a and G_m. The ring O

4.3.1. Let G_a be the functor from $(Sch)^\circ$ to (Ens) defined by

$$G_a(S) = \Gamma(S, OS)$$

endowed with the $\hat{Sch}$-group structure defined by the additive group structure of the ring $\Gamma(S, OS)$. It is representable by an affine scheme, which we shall denote G_a, which is therefore a group scheme

$$G_a = \operatorname{Spec} \mathbb{Z}[T].$$

Indeed, $G_a(S) = \mathrm{Hom}(S, G_a) = \mathrm{Hom}_{Alg.}(\mathbb{Z}[T], \Gamma(S, OS)) \simeq \Gamma(S, OS)$.

For every scheme S, one therefore has an S-group affine over S, denoted $G_{a,S}$, which corresponds to the OS-bialgebra OS[T], with the diagonal map and the augmentation defined by:

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \quad \varepsilon(T) = 0.$$

$A(G)$ which, together with Δ and ϵ, makes $A(G)$ an 0S-Hopf algebra.

4.3.2. Let G_m be the functor from $(Sch)^{\circ}$ to `(Ens)` defined by

$$G_m(S) = \Gamma(S, OS)^{\times},$$

where $\Gamma(S, OS)^{\times}$ denotes the multiplicative group of invertible elements of the ring $\Gamma(S, OS)$, endowed with its natural $\hat{Sch}$-group structure. It is representable by an affine scheme denoted G_m

`G_m = Spec ℤ[T, T^{-1}] = Spec ℤ[ℤ],`

where $\mathbb{Z}[\mathbb{Z}]$ denotes the algebra of the group $\mathbb{Z}$ over the ring $\mathbb{Z}$. Indeed,

`G_m(S) = Hom_{Alg.}(ℤ[T, T^{-1}], Γ(S, OS)) ≃ Γ(S, OS)^×.`

For every scheme S, one therefore has an S-group affine over S denoted $G_{m,S}$, which corresponds to the `OS`-algebra $OS[\mathbb{Z}]$, with the diagonal map and augmentation defined by:

`Δ(x) = x ⊗ x    and    ε(x) = 1,    for x ∈ ℤ.`

4.3.3. Let O be the functor G_a endowed with its $\hat{Sch}$-ring structure. It is represented by the scheme $\operatorname{Spec} \mathbb{Z}[T]$, which we shall denote O when considering it as endowed with its ring scheme structure.

For every scheme S, `OS = S ×_{Spec ℤ} Spec ℤ[T] = Spec(OS[T])` is therefore an S-ring scheme, affine over S. (Note: in EGA II 1.7.13, `OS` is denoted `S[T]`).

4.3.3.1. [41] For every object X of $\hat{Sch}$, $O(X) = \mathrm{Hom}(X, O)$ is endowed with a ring structure, functorial in X. In particular, if X' is a scheme and one is given morphisms $x : X' \to X$ and $f : X \to O$ (i.e. $f \in O(X)$), then $f(x) = f \circ x$ is an element of $O(X') = \Gamma(X', O_{X'})$.

Definition. *Let $\pi : M \to X$ be a morphism of $\hat{Sch}$, and let $OX = O \times X$. One says that M is an OX-module if one has given, for every X-scheme X', a structure of $O(X')$-module on* $\mathrm{Hom}_X(X', M)$*, functorial in X'.*

This amounts to giving a law of X-abelian group $\mu : M \times_X M \to M$ and an "external law"

`O × M = OX ×_X M → M,    (f, m) ↦ f · m`

which is an X-morphism (i.e. $\pi(f \cdot m) = \pi(m)$) and which, for every $x \in X(S)$, endows $M(x) = m \in M(S) | \pi(m) = x$ with a structure of $O(S)$-module.

In this case, for every $Y \in Ob\hat{Sch}_{/X}$ (not necessarily representable), $\mathrm{Hom}_X(Y, M) = \Gamma(M_Y/Y)$ is an $O(Y)$-module, functorially in Y.

[41] N.D.E.: This paragraph has been added, which will be useful later (cf. II 1.3).

4.4. Diagonalizable groups

4.4.1. The construction of G_m generalizes as follows: let M be a commutative group and $M_{\mathbb{Z}}$ the group scheme associated to it by the procedure of 4.1. Consider the functor defined by

`D(M)(S) = Hom_{groups}(M, G_m(S)) ≃ Hom_{S-Gr.}(MS, G_{m,S}).`

It is a commutative $\hat{Sch}$-group representable by a group scheme that one will denote $D(M)$; one will therefore have by definition:

$$D(M) \simeq \mathrm{Hom}_{(Sch)-Gr.}(M_{\mathbb{Z}}, G_m).$$

Set in fact

$$D(M) = \operatorname{Spec} \mathbb{Z}[M],$$

where $\mathbb{Z}[M]$ is the algebra of the group M over the ring $\mathbb{Z}$; one has

`D(M)(S) = Hom_{Alg.}(ℤ[M], Γ(S, OS)) ≃ Hom_{groups}(M, Γ(S, OS)^×)`

by the very definition of the algebra $\mathbb{Z}[M]$.

4.4.2. For every scheme S one therefore has an S-group scheme affine over S

`D_S(M) = D(M)_S = Hom_{(Sch)-Gr.}(M_ℤ, G_m)_S = Hom_{S-Gr.}(MS, G_{m,S}).`

It is associated to the `OS`-bialgebra `OS[M]`, endowed with the diagonal map and augmentation defined by

`Δ(x) = x ⊗ x    and    ε(x) = 1,    for x ∈ M.`

4.4.3. If $f : M \to N$ is a homomorphism of commutative groups, one defines in an obvious manner a morphism of S-groups

$$D_S(f) : D_S(N) \to D_S(M),$$

whence a functor

$$D_S : M \mapsto D_S(M)$$

from the category of abelian groups to the category of S-groups affine over S, which one may also define as the composite of the functor $M \mapsto MS$ and of the functor $MS \mapsto \mathrm{Hom}_{S-Gr.}(MS, G_{m,S})$. This functor commutes with extensions of the base.

An S-group isomorphic to a group of the form $D_S(M)$ is said to be *diagonalizable*. Note that the elements of M are interpreted as certain characters of $D_S(M)$, that is, certain elements of $\mathrm{Hom}_{S-Gr.}(D_S(M), G_{m,S})$. (Confer VIII 1).

4.4.4. Let us give some examples of diagonalizable groups. One has first

```
D(ℤ) = G_m    and    D(ℤ^r) = (G_m)^r.
```

One sets

$$\mu_n = D(\mathbb{Z}/n\mathbb{Z}),$$

and calls it the *group of n-th roots of unity*. Indeed, one has

```
μ_n(S) = Hom_{groups}(ℤ/nℤ, Γ(S, OS)^×) = {f ∈ Γ(S, OS) | f^n = 1}.
```

The S-group $\mu_{n,S}$ corresponds to the OS-algebra $OS[T]/(T^n - 1)$. Suppose in particular that S is the spectrum of a field k of characteristic $p = n$. Setting $T - 1 = s$, one finds

$$k[T]/(T^p - 1) = k[s]/(s^p),$$

which shows that the underlying topological space of $\mu_{p,k}$ reduces to a point, the local ring of this point being the artinian k-algebra $k[s]/(s^p)$. (In the same vein, let us note that the S-schemes $G_{a,S}, G_{m,S}, OS$ are smooth over S, that $D_S(M)$ is

flat over S and that it is formally smooth (resp. smooth) over S if and only if no residue characteristic of S divides the torsion of M (resp. and if moreover M is of finite type), cf. Exp. VIII, 2.1).

4.5. Other examples of groups

The preceding procedure applies to the "classical groups" (linear groups GL_n, symplectic Sp_n, orthogonal O_n, etc.). One defines for example GL_n as representing the functor GL_n such that:

```
GL_n(S) = GL(n, Γ(S, OS)) = Aut_{OS}(OS^n).
```

One can construct it for example as the open subscheme of Spec $\mathbb{Z}[T_{ij}]$ $(1 \leqslant i, j \leqslant n)$ defined by the function $f = det((T_{ij})_{i,j=1}^n)$, that is, Spec $\mathbb{Z}[T_{ij}, f^{-1}]$.

4.6. Functor-modules in the category of schemes

We propose to associate to every OS-module on the scheme S an OS-module (where OS denotes the functor-ring introduced in 4.3.3). This can be done in two different ways. Precisely:

Definition 4.6.1. *Let S be a scheme. For every OS-module F one denotes by $V(F)$ and $W(F)$ the contravariant functors on $(Sch)/S$ defined by:*

```
V(F)(S′) = Hom_{O_{S′}}(F ⊗_{OS} O_{S′}, O_{S′}),
W(F)(S′) = Γ(S′, F ⊗_{OS} O_{S′})
```

(where $F \bigotimes_{OS} O_{S'}$ denotes the inverse image on S' of the OS-module F).

Then $V(F)$ and $W(F)$ are endowed in an obvious manner with a structure of OS-modules (recall that $OS(S') = \Gamma(S', O_{S'}) = W(OS)(S')$), in such a way that one has in fact defined two functors V and W from the category of OS-modules to the category of OS-modules, with V contravariant and W covariant.

We restrict ourselves in the rest of this paragraph to the case where the OS-modules in question are quasi-coherent, that is, we consider V and W as

functors from the category (OS-Mod.q.c.) of quasi-coherent OS-modules to the category of OS-modules

$$V : (OS - Mod.q.c.)^{\circ} \to (OS - Mod.), W : (OS - Mod.q.c.) \to (OS - Mod.).$$

Remark 4.6.1.1.[42] *The reader will note that, in what follows, all the propositions involving only the functor W are valid for arbitrary OS-modules, not necessarily quasi-coherent.*

Proposition 4.6.2. *(i) The functors V and W commute with base extension: if S' is above S and if F is a quasi-coherent OS-module, one has*

$$V(F \otimes O_{S'}) \simeq V(F)_{S'} \quad \text{and} \quad W(F \otimes O_{S'}) \simeq W(F)_{S'}.$$

(ii) The functors V and W are fully faithful: the canonical maps

$$\mathrm{Hom}_{OS}(F, F') \to \mathrm{Hom}_{OS}(V(F'), V(F))$$
$$\mathrm{Hom}_{OS}(F, F') \to \mathrm{Hom}_{OS}(W(F), W(F'))$$

are bijective.

(iii) The functors V and W are additive:

$$V(F \oplus F') \simeq V(F) \times_S V(F') \quad \text{and} \quad W(F \oplus F') \simeq W(F) \times_S W(F').$$

Parts (i) and (iii) are obvious from the definitions. For (ii), one takes for S' open subschemes of S. We leave the proof to the reader (for V, use EGA II, 1.7.14).

Recall (cf. 3.1.4) that if F, F' are OS-modules, $\mathrm{Hom}_{OS}(F, F')$ denotes the S-functor (in abelian groups) which to every $S' \to S$ associates $\mathrm{Hom}_{O_{S'}}(F_{S'}, F'_{S'})$.

Proposition 4.6.3.[43] *One has canonical morphisms in (OS-Mod.):*

$$\begin{tikzcd} & W(\mathrm{Hom}_{OS}(F, F')) \arrow[dl] \arrow[dr] & \\ \mathrm{Hom}_{OS}(W(F), W(F')) & & \mathrm{Hom}_{OS}(V(F'), V(F)) \end{tikzcd}$$

This follows immediately from 4.6.2 (i) and (ii).

[42]N.D.E.: The numbering 4.6.1.1 has been added, for subsequent references.
[43]N.D.E.: The isomorphism $\mathrm{Hom}_{OS}(W(F), W(F')) \cong \mathrm{Hom}_{OS}(V(F'), V(F))$ has been added.

Notation 4.6.3.1. *Let F be a quasi-coherent 0S-module. One knows (EGA II, 1.7.8) that the S-functor $V(F)$ is representable by an S-scheme affine over S which one denotes $V(F)$ and calls the* vector fibration[44] *defined by F:*

$$V(F) = \mathrm{Spec}(S(F)),$$

where $S(F)$ denotes the symmetric algebra of the 0S-module F.[45]

Proposition 4.6.4. *Let F and F' be two quasi-coherent 0S-modules, A a quasi-coherent 0S-algebra. One has a functorial isomorphism:*

Hom_S(Spec(A), Hom_{OS}(W(F′), W(F))) ▯ Hom_{OS}(F′, F ⊗_{OS} A).

Indeed, denoting $X = \mathrm{Spec}(A)$, the first member is canonically isomorphic to $\mathrm{Hom}_{OS}(W(F'), W(F))(X)$, that is, by definition, to

Hom_{OX}(W(F′)_X, W(F)_X) ≃ Hom_{OX}(W(F′ ⊗ O_X), W(F ⊗ O_X))

(cf. 4.6.2 (i)), which by 4.6.2 (ii) can also be written

Hom_{OX}(F′ ⊗ O_X, F ⊗ O_X) = Hom_{OS}(F′, π_*(π*(F))),

where one denotes $\pi : X \to S$ the structural morphism. But, by EGA II, 1.4.7, one has $\pi_*(\pi * (F)) \simeq F \bigotimes A$, which completes the proof.

Corollary 4.6.4.1. *One has a canonical isomorphism*

W(F ⊗ A) ≃ Hom_S(Spec(A), W(F)).

Indeed,[46] let $f : S' \to S$ be an S-scheme and $X' = X \times_S S'$; one has a cartesian square

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d, "\pi'"'] & X \arrow[d, "\pi"] \\ S' \arrow[r, "f"] & S \end{tikzcd}$$

and by EGA II, 1.5.2, X' is affine over S' and $\pi'_*(O_{X'}) = f * (A)$. One has therefore

Hom_S(Spec(A), W(F))(S′) = Hom_{S′}(Spec(f*(A)), W(f*(F)))

and by 4.6.4 applied to $f * (F)$, $F' = O_{S'}$ and $f * (A)$, this equals

Γ(S′, f*(F) ⊗ f*(A)) = Γ(S′, f*(F ⊗ A)) = W(F ⊗ A)(S′).

[44] N.D.E.: "vector bundle" has been replaced by "vector fibration"; current usage being to call a vector fibration that is locally trivial of rank r a "vector bundle of rank r", i.e. one whose sheaf of sections is locally isomorphic to $OS^{\oplus r}$.

[45] N.D.E.: Let us point out here the articles [Ni04] (resp. [Ni02]), which show that if S is noetherian and F is a coherent 0S-module, then $W(F)$ (resp. the S-group which to every $T \to S$ associates $\mathrm{Aut}_{OT}(F \bigotimes OT)$) is representable if and only if F is locally free.

[46] N.D.E.: The original has been detailed in what follows.

Proposition 4.6.5. *If F and F' are two OS-modules locally free of finite type, the morphisms of 4.6.3 are isomorphisms.*

Indeed, for every $S' \to S$, one has

W(Hom_{OS}(F, F′))(S′) = Γ(S′, Hom_{OS}(F, F′) ⊗ O_{S′}) = Hom_{O_{S′}}(F ⊗ O_{S′}, F′ ⊗ O_{S′}).

But the second member is indeed isomorphic to $\mathrm{Hom}_{OS}(W(F), W(F'))(S')$ and to $\mathrm{Hom}_{OS}(V(F'), V(F))(S')$, by 4.6.2 (i) and (ii).

Corollary 4.6.5.1. *Let F be an OS-module locally free of finite type. Set $F^\vee = \mathrm{Hom}_{OS}(F, OS)$. One has canonical isomorphisms:*

W(F^v) ≃ Hom_{OS}(W(F), OS) ≃ V(F),
V(F^v) ≃ Hom_{OS}(V(F), OS) ≃ W(F).

One has finally the following proposition:

Proposition 4.6.6. *Let $f : F \to F'$ be a morphism of OS-modules locally free of finite rank. For $W(f) : W(F) \to W(F')$ to be a monomorphism, it is necessary and sufficient that f identify F locally with a direct factor of F'.*

The direct proposition is essentially contained in EGA 0_I, 5.5.5.[47] Conversely, if F is locally a direct factor of F', then for every $\pi : S' \to S$, $\pi * F$ is a submodule of $\pi * F'$ (because locally a direct factor), so $W(F)(S') = \Gamma(S', \pi * F)$ is a submodule of $W(F')(S') = \Gamma(S', \pi * F')$.

4.7. The category of G-OS-modules

Let G be an S-group and F an OS-module; $W(F)$ is endowed with a structure of OS-module.

Definition 4.7.1. *By a* structure of G-OS-module on F *we mean a structure of hG-OS-module on $W(F)$ (cf. 3.2). A morphism of G-OS-modules is by definition a morphism of the associated hG-OS-modules. One thus obtains the category (G-OS-Mod.), and one denotes (G-OS-Mod.q.c.) the full subcategory formed by the G-OS-modules that are quasi-coherent as OS-modules.*

To give a structure of G-OS-module on F is therefore, by 3.2, to give a morphism of $S\hat{c}h_{/S}$-groups

$$\rho : hG \to \mathrm{Aut}_{OS}(W(F)).$$

Remark 4.7.1.0.[48] *Since by 4.6.2 one has an anti-isomorphism of S-functors in groups*

(†) Aut_{OS}(W(F)) ≃ Aut_{OS}(V(F)),

[47] N.D.E.: The following sentence has been detailed.
[48] N.D.E.: This remark has been added, taken from [DG], II, § 2, 1.1.

one sees that it is equivalent to give a structure of hG-OS*-module on* $W(F)$ *or on* $V(F)$. Indeed, let $\rho : hG \to \mathrm{Aut}_{OS}(W(F))$ be as above. For every $T \to S$ and $g \in G(T)$, denote by $\rho*(g)$ the image of $\rho(g)$ by the anti-isomorphism (†); one has therefore $\rho*(gh) = \rho*(h) \circ \rho*(g)$, i.e. $\rho*$ defines a structure of hG-OS-module "on the right" on $V(F)$. Setting $\rho^\vee(g) = \rho*(g^{-1})$, one obtains indeed a structure of hG-OS-module on $V(F)$, the datum of which is equivalent to that of ρ.

Remark 4.7.1.1. One can say that one has constructed the categories one has just defined by the cartesian squares:

$$\begin{array}{ccccc} (\texttt{G-OS-Mod.q.c.}) & \longrightarrow & (\texttt{G-OS-Mod.}) & \longrightarrow & (\texttt{hG-OS-Mod.}) \\ & & & & \big\downarrow \text{forget} \\ (\texttt{OS-Mod.q.c.}) & \longrightarrow & (\texttt{OS-Mod.}) & \xrightarrow{W} & (\texttt{OS-Mod.}). \end{array}$$

[49] The categories (OS-Mod.) and (OS-Mod.) are abelian, but one should be careful that in general the functor W is neither left nor right exact.

Remark 4.7.1.2.[50] Let F be a G-OS-module. The subsheaf of invariants F^G is defined as follows: for every open subset U of S,

$$F^G(U) = W(F)^G(U) = \{x \in F(U) \mid g \cdot x_{S'} = x_{S'} \text{ for every } S' \xrightarrow{f} U,\ g \in G(S')\},$$

where $x_{S'}$ denotes the image of x in $\Gamma(S', f*(F)) = \Gamma(U, f_* f*(F))$.

One should be careful that the natural morphism $W(F^G) \to W(F)^G$ is not in general an isomorphism. For example, if $S = \mathrm{Spec}(\mathbb{Z})$ and G is the constant group $\mathbb{Z}/2\mathbb{Z} = 1, \tau$ acting on $F = OS$ by $\tau \cdot 1 = -1$, one has $F^G = 0$ but, if R is an $\mathbb{F}_2$-algebra, $W(F)^G(\mathrm{Spec}(R)) = R$.

4.7.2. We suppose from now until the end of n° 4.7 that G is affine over S.[51] Then, by virtue of 4.6.4, the datum of a morphism of S-functors

$$\rho : hG \to \mathrm{End}_{OS}(W(F))$$

is equivalent to that of a morphism of OS-modules

$$\mu : F \to F \otimes_{OS} A(G).$$

To say that ρ is a morphism of $\hat{Sch}_{/S}$-groups is then equivalent to saying that μ satisfies the following axioms:

(CM 1) *the following diagram is commutative*

[49] N.D.E.: The original has been corrected by suppressing the assertion that the category (G-OS-Mod.) is abelian, see 4.7.2.1 below.

[50] N.D.E.: This remark has been added.

[51] N.D.E.: Cf. VI_B, §§ 11.1–11.6 for the extension of the results of 4.7.2 to the case where G is not necessarily affine, but where G and F are assumed to be flat over S.

$$\begin{array}{ccc} F & \xrightarrow{\mu} & F \otimes_{O_S} A(G) \\ \Big\downarrow{\scriptstyle \mu} & & \Big\downarrow{\scriptstyle id \otimes \Delta} \\ F \otimes_{O_S} A(G) & \xrightarrow{\mu \otimes id} & F \otimes_{O_S} A(G) \otimes_{O_S} A(G). \end{array}$$

(CM 2) *the following composite is the identity*

$$F \xrightarrow{\mu} F \otimes A(G) \xrightarrow{id \otimes \varepsilon} F \otimes O_S \xrightarrow{\sim} F.$$

These axioms (CM 1) and (CM 2) are those of a (right) comodule structure[52] over the bialgebra $A(G)$.

Set $A = A(G)$. If F and F' are A-comodules, a morphism of comodules $f : F \to F'$ is a morphism of O_S-modules such that the following diagram is

commutative:

$$\begin{array}{ccc} F & \xrightarrow{f} & F' \\ \Big\downarrow{\scriptstyle \mu_F} & & \Big\downarrow{\scriptstyle \mu_{F'}} \\ F \otimes A & \xrightarrow{f \otimes id} & F' \otimes A. \end{array}$$

One thus obtains the category (A-Comod.), and one will denote by (A-Comod.q.c.) the full subcategory formed by the A-comodules that are quasi-coherent as O_S-modules. One has thus obtained:

Proposition 4.7.2. *Let G be an S-group affine over S. One has equivalences of categories:*

$$(G - OS - Mod.) \cong (A(G) - Comod.)(G - OS - Mod.q.c.) \cong (A(G) - Comod.q.c.)$$

[53] *If moreover $S = \mathrm{Spec}(\Lambda)$ is affine and if one denotes $\Lambda[G] = \Gamma(S, A(G))$, one has an equivalence of categories*

$$(A(G) - Comod.q.c.) \cong (\Lambda[G] - Comod.).$$

[54] Suppose moreover that $A = A(G)$ is a flat O_S-module. Let E be an A-comodule and F a sub-O_S-module of E. Since A is flat over O_S, one can identify $F \otimes A$ (resp. $F \otimes A \otimes A$) with a sub-O_S-module of $E \otimes A$ (resp. $E \otimes A \otimes A$). Suppose that μ_E maps F into $F \otimes A$; then its restriction $\mu_F : F \to F \otimes A$ endows F with a structure

[52] N.D.E.: Left G-O_S-modules correspond in a natural way to right $A(G)$-comodules.
[53] N.D.E.: The following sentence has been added.
[54] N.D.E.: The following paragraph has been added, taken from [Se68, § 1.3].

of A-comodule; one says that F is a *subcomodule* of E. By passage to the quotient, μ_E defines a morphism of OS-modules $E/F \to E/F \otimes A$, which endows E/F with a structure of A-comodule. If $f : E \to E'$ is a morphism of A-comodules, Ker f (resp. Im f) is a sub-A-comodule of E (resp. E'), and f induces an isomorphism of A-comodules: $E/Kerf \xrightarrow{\sim} Imf$. Moreover, if E and E' are quasi-coherent OS-modules, so are Ker f and Im f. Consequently, (A-Comod.) and (A-Comod.q.c.) are abelian categories.

Corollary 4.7.2.1. *Suppose that G is affine and flat over S. Then the category (G-OS-Mod.q.c.) (resp. (G-OS-Mod.)), equivalent to the category of $A(G)$-comodules quasi-coherent over OS (resp. $A(G)$-comodules), is abelian.*

4.7.3. Suppose now that G is a diagonalizable group, that is, that $A(G)$ is the algebra of a commutative group M over the sheaf of rings OS. If F is an OS-module, one has

F ⊗ A(G) = ∐_{m ∈ M} F ⊗ m OS.

To give a morphism of OS-modules

μ : F → F ⊗ A(G)

is therefore equivalent to giving OS-endomorphisms $(\mu_m)_{m\in M}$ of F, such that for every section x of F on an open subset of S, $(\mu_m(x))$ is a section of the direct sum $\coprod_{m\in M} F$ (this means that on every sufficiently small open subset, there are only finitely many restrictions of the $\mu_m(x)$ that are nonzero).

For μ defined by

μ(x) = ∑_{m ∈ M} μ_m(x) ⊗ m

to satisfy (CM 1) (resp. (CM 2)) it is necessary and sufficient that one have

μ_m ∘ μ_{m′} = δ_{mm′} μ_m,   (resp. ∑_{m ∈ M} μ_m = Id_F),

which means that the μ_m are pairwise orthogonal projectors with sum the identity. One has thus proved:

Proposition 4.7.3. *If $G = D_S(M)$ is a diagonalizable S-group, the category of quasi-coherent G-OS-modules (resp. of G-OS-modules) is equivalent to the category of quasi-coherent OS-modules (resp. of OS-modules) graded of type M.*

Remark. If F is endowed with a structure of OS-algebra preserved by the operations of G, then the gradation of F is a gradation of algebra. More precisely:

Corollary 4.7.3.1. *The functor $A \mapsto \operatorname{Spec} A$ induces an equivalence between the category of quasi-coherent OS-algebras graded of type M and the category opposite to that of S-schemes affine over S with S-operator group $G = D_S(M)$.*

Proposition 4.7.4. *Let G be a diagonalizable S-group. If $0 \to F_1 \to F_2 \to F_3 \to 0$ is an exact sequence of quasi-coherent G-OS-modules that splits as a sequence of OS-modules, then it also splits as a sequence of G-OS-modules.*

Indeed, if $G = D_S(M)$, each F_i is graded by the $(F_i)_m$, and for each $m \in M$ the sequence

$$0 \to (F_1)_m \to (F_2)_m \to (F_3)_m \to 0$$

of OS-modules is split. The preceding proposition then entails the result.

5. Group cohomology

5.1. The standard complex

[55] Let C be a category, G a $\hat{C}$-group, O a $\hat{C}$-ring, and F a G-O-module. One sets, for $n \geqslant 0$,

```
C^n(G, F) = Hom(G^n, F)    and    C^n(G, F) = Hom(G^n, F),
```

where G^0 is the final object e. Then $C^n(G,F)$ (resp. $C^n(G,F)$) is endowed in an obvious manner with a structure of O-module (resp. of $\Gamma(O)$-module), and one has

```
C^n(G, F) ≅ Γ(C^n(G, F))    and    C^n(G, F)(S) = C^n(GS, FS).
```

To give an element of $C^n(G,F)$ is to give for each $S \in Ob(C)$ an n-cochain of $G(S)$ in $F(S)$, functorially in S. The boundary operator

```
∂ : C^n(G(S), F(S)) → C^{n+1}(G(S), F(S)),
```

which, recall, is given by the formula

```
∂f(g_1, …, g_{n+1}) = g_1 f(g_2, …, g_{n+1})
                    + ∑_{i=1}^{n} (-1)^i f(g_1, …, g_i g_{i+1}, …, g_{n+1})
                    + (-1)^{n+1} f(g_1, …, g_n),
```

is functorial in S and therefore defines a homomorphism:

```
∂ : C^n(G, F) → C^{n+1}(G, F)
```

such that $\partial \circ \partial = 0$. One has thus defined a complex of abelian groups (and even of $\Gamma(O)$-modules) denoted $C*(G,F)$. One defines in the same way the complex of O-modules $C*(G,F)$, and one has:

```
C*(G, F) = Γ(C*(G, F)).
```

One denotes by $H^n(G,F)$ (resp. $H^n(G,F)$) the groups (resp. the $\hat{C}$-groups) of cohomology of the complex $C*(G,F)$ (resp. $C*(G,F)$).

One has in particular

```
H^0(G, F) = F^G    and    H^0(G, F) = Γ(F^G).
```

[55] N.D.E.: This complex is often called the “Hochschild complex”; see for example § II.3 of [DG70].

Remark 5.1.1.[56] *The "set-theoretic" description of ∂ given above is convenient for verifying that $\partial \circ \partial = 0$. One can also define ∂ in terms of the multiplication $m : G \times G \to G$ and the action $\mu : G \times F \to F$ as follows: for every $f \in C^n(G,F)$,*

```
∂f = μ ∘ (id_G × f) + ∑_{i=1}^{n} (-1)^i f ∘ (id_{G^{i-1}} × m × id_{G^{n-i}}) + (-1)^{n+1} f ∘ pr_{[
```

where $pr_{[1,n]}$ denotes the projection of $G^{n+1} = G^n \times G$ onto G^n. Similarly, for every $S \in Ob(C)$ and $f \in C^n(G,F)(S) = C^n(GS,FS)$, one has

```
∂f = μ_S ∘ (id_{GS} × f) + ∑_{i=1}^{n} (-1)^i f ∘ (id_{GS^{i-1}} × m_S × id_{GS^{n-i}}) + (-1)^{n+1}
```

where m_S and μ_S are deduced from m and μ by base change.

5.2.

[57] Recall (cf. § 3) that `(G-O-Mod.)` is endowed with a structure of abelian category, defined "argument by argument"; thus,

$$0 \to F' \to F \to F'' \to 0$$

is an exact sequence of G-O-modules if and only if the sequence of abelian groups

$$0 \to F'(S) \to F(S) \to F''(S) \to 0$$

is exact, for every $S \in Ob(C)$.

[58] Suppose C small; then, by 3.2.1, `(G-O-Mod.)` has enough injective objects, so that the derived functors of the left exact functors H^0 and H^0 are defined. We now propose to show that the functors H^n (resp. H^n) are indeed the derived functors of H^0 (resp. H^0).

Definition 5.2.0.[59] *For every O-module P, one denotes by $E(P)$ the object* $\mathrm{Hom}(G,P)$ *of $\hat{C}$ endowed with the structure of G-O-module defined as follows: for every $S \in Ob(C)$ one has* $\mathrm{Hom}(G,P)(S) = \mathrm{Hom}_S(GS,PS)$*, and one makes $g \in G(S)$ and $a \in O(S)$ operate on $\phi \in \mathrm{Hom}_S(GS,PS)$ by the formulas*

```
(g · φ)(h) = φ(hg)    and    (a · φ)(h) = a φ(h),
```

for every $h \in G(S')$, $S' \to S$. Moreover, for every $\phi \in \mathrm{Hom}_S(GS,PS)$ one sets

$$\epsilon(\phi) = \phi(1) \in P(S)$$

(where 1 denotes the unit element of $G(S)$).

---

[56] N.D.E.: This remark has been added.
[57] N.D.E.: The following recollection has been added.
[58] N.D.E.: The following sentence has been added.
[59] N.D.E.: The original has been modified, in order to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

This defines a functor $E : (O-Mod.) \to (G-O-Mod.)$ and a natural transformation $\epsilon : E \to Id$, where Id denotes the identity functor of (O-Mod.).

Remark 5.2.0.1.[60] *In what follows, let us denote by G_1 and G_2 two copies of G. Then the morphism*

$$G_1 \times E(P) \to E(P), \qquad (g_1, \varphi) \mapsto (g_2 \mapsto \varphi(g_2 g_1))$$

corresponds via the isomorphisms

$$\begin{aligned} \mathrm{Hom}(G_1 \times E(P), E(P)) &\simeq \mathrm{Hom}(E(P), \mathrm{Hom}(G_1, \mathrm{Hom}(G_2, P))) \\ &\simeq \mathrm{Hom}(E(P), \mathrm{Hom}(G_2 \times G_1, P)) \end{aligned}$$

to the morphism $\phi \mapsto ((g_2, g_1) \mapsto \phi(g_2 g_1))$, i.e. to the morphism

$$\mathrm{Hom}(G, P) \to \mathrm{Hom}(G_2 \times G_1, P)$$

induced by the multiplication $\mu_G : G \times G \to G$, $(g_2, g_1) \mapsto g_2 g_1$.

Lemma 5.2.0.2.[61] *(i) The functor E is right adjoint to the forgetful functor $(G-O-Mod.) \to (O-Mod.)$; more precisely, $\epsilon : E \to Id$ induces for every $M \in (G-O-Mod.)$ and $P \in Ob(O-Mod.)$ a bijection*

$$\mathrm{Hom}_{G\text{-}O\text{-}Mod.}(M, E(P)) \cong \mathrm{Hom}_{O\text{-}Mod.}(M, P)$$

functorial in M and P.

(ii) Consequently, if I is an injective object of (O-Mod.) then $E(I)$ is an injective object of (G-O-Mod.).

Proof. To every O-morphism $f : M \to P$, one associates the element ϕ_f of $\mathrm{Hom}_O(M, E(P))$ defined as follows. For every $S \in Ob(C)$ and $m \in M(S)$, $\phi_f(m)$ is the element of $\mathrm{Hom}_S(GS, PS)$ defined by: for every $g \in G(S')$, $S' \to S$,

$$\varphi_f(m)(g) = f(g\, m) \in P(S').$$

Then, for every $h \in G(S)$, one has $\phi_f(hm) = h \cdot f(m)$, i.e. ϕ_f is an element of

$$\mathrm{Hom}_{G-O-Mod.}(M, E(P)).$$

If $\phi \in \mathrm{Hom}_{G-O-Mod.}(M, E(P))$ and if one denotes, for every $m \in M(S)$, $f(m) = \phi(m)(1)$, then

$$\varphi_f(m)(g) = f(g\, m) = \varphi(g\, m)(1) = (g \cdot \varphi(m))(1) = \varphi(m)(g),$$

i.e. $\phi_f = \phi$. Conversely, it is clear that $\phi_f(m)(1) = f(m)$. This proves (i), and (ii) follows at once.

Definition 5.2.0.3. *Let M be a G-O-module; the identity map of M (considered as an O-module) corresponds by adjunction to the morphism of G-O-modules*

[60] N.D.E.: The original has been modified, in order to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

[61] N.D.E.: The original has been modified, in order to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

$$\mu_M : M \to E(M)$$

such that for every $S \in Ob(C)$ and $m \in M(S)$, $\mu_M(m)$ is the morphism $GS \to MS$ defined by: for every $S' \to S$ and $g \in G(S')$, $\mu_M(m)(g) = g \cdot m_{S'} \in M(S')$.

Note that μ_M is a monomorphism. Indeed, $\epsilon_M : E(M) \to M$ is a morphism of O-modules such that $\epsilon_M \circ \mu_M = id_M$; consequently M is a direct factor, as an O-module, of $E(M)$.

Proposition 5.2.1. *Suppose that C is small, that finite products exist in it, and that G is representable. Then, the functors $H^n(G,-)$ (resp. $H^n(G,-)$) are the derived functors of the left exact functor $H^0(G,-)$ (resp. $H^0(G,-)$) on the category of G-O-modules.*

By virtue of the well-known general results,[62] it suffices to verify that the $H^n(G,-)$ (resp. $H^n(G,-)$) form an effaceable cohomological functor in degrees > 0.

Let

$$0 \to F' \to F \to F'' \to 0$$

be an exact sequence of G-O-modules, and let $S \in Ob(C)$. By hypothesis, G is representable by an object $G \in Ob(C)$, and finite products exist in C; in particular C has a final element e. Hence each $G^n \times hS$ is representable by $G^n \times S$ (with $G^0 = e$), and the sequence

0 → F′(G^n × S) → F(G^n × S) → F″(G^n × S) → 0

is exact. This shows that the sequence of O-modules

0 → C^n(hG, F′) → C^n(hG, F) → C^n(hG, F″) → 0

is exact. It follows that $C*(G,-)$, considered as a functor on (G-O-Mod.) with values in the category of complexes of (O-Mod.), is exact. This shows that the

$H^n(G,-)$ indeed form a cohomological functor. Since the functor Γ is exact, the same is true for the $H^n(G,-)$. It will now suffice to prove:

Lemma 5.2.2. *For every $P \in Ob(O-Mod.)$, one has:*

H^n(G, Hom(G, P)) = 0   and   H^n(G, Hom(G, P)) = 0, for n > 0.

It suffices to prove that $C*(G, \mathrm{Hom}(G,P))$ and $C*(G, \mathrm{Hom}(G,P))$ are homotopically trivial in degrees > 0. It suffices even to do this for the second, the corresponding result for the first being deduced from it by base change.[63] Now, one defines for every $n \geqslant 0$ a morphism

σ : C^{n+1}(G, Hom(G, P)) → C^n(G, Hom(G, P))

[62] N.D.E.: Cf. [Gr57], 2.2.1 and 2.3. Moreover, the original has been detailed in what follows.
[63] N.D.E.: The original has been detailed in what follows.

as follows. Let $f \in C^{n+1}(G, \mathrm{Hom}(G,P))$; for every $S \in Ob(C)$ and $g_1, \cdots, g_n \in G(S)$, $\sigma(f)(g_1, \cdots, g_n)$ is the element of $\mathrm{Hom}_S(GS, PS)$ defined by: for every $S' \to S$ and $x \in G(S')$,

$$\sigma(f)(g_1, \dots, g_n)(x) = f(x, g_1, \dots, g_n)(e) \in P(S')$$

(where e denotes the unit element of $G(S')$). Then, σ is a homotopy operator in degrees > 0. Indeed, for every $g_1, \cdots, g_{n+1} \in G(S)$ and $x \in G(S')$ one has, on the one hand:

$$\begin{aligned}\partial\sigma(f)(g_1, \dots, g_{n+1})(x) &= f(x g_1, g_2, \dots, g_{n+1})(e) \\ &+ \sum_{i=1}^{n} (-1)^i f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1})(e) \\ &+ (-1)^{n+1} f(x, g_1, \dots, g_n)(e),\end{aligned}$$

and on the other hand:

$$\begin{aligned}\sigma(\partial f)(g_1, \dots, g_{n+1})(x) &= (x f(g_1, g_2, \dots, g_{n+1}))(e) - f(x g_1, g_2, \dots, g_{n+1})(e) \\ &+ \sum_{i=1}^{n} (-1)^{i+1} f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ &+ (-1)^{n+2} f(x, g_1, \dots, g_n)(e),\end{aligned}$$

whence

$$(\partial\sigma(f) + \sigma(\partial f))(g_1, \dots, g_{n+1})(x) = f(g_1, \dots, g_{n+1})(x),$$

i.e. $\partial\sigma + \sigma\partial$ is the identity map of $C^{n+1}(G, \mathrm{Hom}(G,P))$, for every $n \geqslant 0$.

Remark 5.2.3.[64] *The hypothesis "C small" is used only to ensure the existence of the derived functors $R^n H^0$ and $R^n H^0$. In any case, the foregoing shows that the functors $H^n(G,-)$ (resp. $H^n(G,-)$) form a cohomological functor, effaceable in degrees > 0, hence they are the (right) satellite functors of the left exact functor $H^0(G,-)$ (resp. $H^0(G,-)$), in the sense of [Gr57, 2.2]; if (G-O-Mod.) has enough injective objects (which is the case if C is small), they coincide with the derived functors (*loc. cit.* 2.3).*

5.3. Cohomology of G-OS-modules

Let S be a scheme, G an S-group, and F a quasi-coherent G-OS-module. One defines the cohomology groups of G with values in F by

$$H^n(G, F) = H^n(hG, W(F)).$$

(for the notations, cf. 4.6).

Suppose G is affine over S. Then, in view of Proposition 4.6.4, this cohomology is computed as follows: $H^n(G,F)$ is the n-th homology group of the complex $C*(G,F)$ whose n-th term is:

$$C^n(G, F) = \Gamma(S, F \otimes \underbrace{A(G) \otimes \cdots \otimes A(G)}_{n \text{ times}})$$

If f (resp. a_i) is a section of F (resp. of $A(G)$) on an open subset of S, one has

[64] N.D.E.: This remark has been added.

$$
\begin{aligned}
\partial(f \otimes a_1 \otimes \cdots \otimes a_n) = {} & \mu_F(f) \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \\
& + \sum_{i=1}^{n} (-1)^i f \otimes a_1 \otimes \cdots \otimes \Delta a_i \otimes \cdots \otimes a_n \\
& + (-1)^{n+1} f \otimes a_1 \otimes a_2 \cdots \otimes a_n \otimes 1,
\end{aligned}
$$

where $\Delta : A(G) \to A(G) \bigotimes A(G)$ and $\mu_F : F \to F \bigotimes A(G)$ describe the coalgebra structure of $A(G)$ and the comodule structure of F. Note in passing that the cohomology of G with values in F therefore depends only on the comodule structure of F, and in particular only on the S-monoid structure of G.

One has in particular

$$H^0(G, F) = \Gamma(S, F^G),$$

where F^G, the *sheaf of invariants of* F, is defined as the sheaf whose sections on the open subset U of S are the sections of F on U whose inverse image in any S' over U is invariant under $G(S')$ (cf. 4.7.1.2).

Theorem 5.3.1. *Let S be an affine scheme, G an S-group affine and flat over S. The functors $H^n(G, -)$ are the derived functors of $H^0(G, -)$ on the category of quasi-coherent G-OS-modules.*

Proof.[65] Since G is affine and flat over S, by 4.7.2.1, the category `(G-OS-Mod.q.c.)` is equivalent to the category `(A(G)-Comod.q.c.)` of $A(G)$-comodules quasi-coherent over `OS`, and is therefore abelian. On the other hand, $A(G)$ being a flat `OS`-module, each functor $F \mapsto F \bigotimes_{OS} A(G)^{\bigotimes n}$ is exact; since moreover S is affine, one obtains that $C*(G, -)$ is an exact functor on `(G-OS-Mod.q.c.)`.

Denote by Δ (resp. η) the comultiplication (resp. the augmentation) of $A(G)$. For every quasi-coherent `OS`-module P, one denotes $Ind(P) = P \bigotimes_{OS} A(G)$ endowed with the structure of $A(G)$-comodule defined by

$$id_P \otimes \Delta : P \otimes_{OS} A(G) \to P \otimes_{OS} A(G) \otimes_{OS} A(G);$$

this defines a functor `Ind : (OS-Mod.q.c.) → (G-OS-Mod.q.c.)`.

It follows from 4.6.4.1, 5.2.0, and 5.2.0.1 that one has an isomorphism of G-`OS`-modules:

$$(*) \qquad W(Ind(P)) \simeq E(W(P)) = Hom(G, W(P)).$$

Via this identification, the morphism $\epsilon : E(W(P)) \to W(P)$ corresponds to the morphism of `OS`-modules $id_P \bigotimes \eta : Ind(P) \to P$.

One has already used that the functor $W : (OS{-}Mod.) \to (OS{-}Mod.)$ is fully faithful; the same is true, by Definition 4.7.1, of its restriction to `(G-OS-Mod.)`, i.e. if M, M' are G-`OS`-modules, one has a functorial isomorphism

$$Hom_{G\text{-}OS\text{-}Mod.}(M, M') \simeq Hom_{G\text{-}OS\text{-}Mod.}(W(M), W(M')).$$

[65] N.D.E.: The original has been modified to show that the category `(G-OS-Mod.q.c.)` is abelian and has enough injective objects. One may compare with [Ja03], Part I, 3.3-3.4, 3.9, 4.2 and 4.14-4.16 (where one should be aware that "k-group scheme" means "affine k-group scheme", cf. *loc. cit.*, 2.1).

Consequently, one deduces from Lemma 5.2.0.2 the following:

Corollary 5.3.1.1. *(i) The functor Ind is right adjoint to the forgetful functor $(G-OS-Mod.q.c.) \to (OS-Mod.q.c.)$. More precisely, the map $id_P \otimes \eta : Ind(P) \to P$ induces for every object M of (G-OS-Mod.q.c.) a bijection*

$$\mathrm{Hom}_{G\text{-}OS\text{-}Mod.}(M, Ind(P)) \cong \mathrm{Hom}_{OS}(M, P).$$

(ii) Consequently, if I is an injective object of (OS-Mod.q.c.) then $Ind(I)$ is an injective object of (G-OS-Mod.q.c.).

Let F be a G-OS-module and $\mu_F : F \to Ind(F)$ the map defining the structure of $A(G)$-comodule. It follows from 5.2.0.3 (or from axioms (CM 1) and (CM 2) of 4.7.2) that μ_F is a morphism of G-OS-modules, and that $(id_F \otimes \eta) \circ \mu_F = id_F$, hence that F is a direct factor of $Ind(F)$ as OS-module; in particular, μ_F is a monomorphism. Since, by $(*)$ and 5.2.2,

$$H^n(G, W(Ind(F))) \simeq H^n(G, \mathrm{Hom}_S(G, W(F))) = 0 \quad \text{for } n > 0,$$

one therefore obtains that $H^n(G, -)$ is effaceable for $n > 0$.

Finally, S being affine, (OS-Mod.q.c.) has enough injective objects. So let $F \hookrightarrow I$ be a monomorphism of OS-modules, where I is an injective object of (OS-Mod.q.c.); then, $A(G)$ being flat over OS, $Ind(F)$ is a sub-G-OS-module of $Ind(I)$, whence:

Corollary 5.3.1.2. *The abelian category (G-OS-Mod.q.c.) has enough injective objects.*

Taking [Gr57, 2.2.1 and 2.3] into account (already used in the proof of 5.2.1), this completes the proof of Theorem 5.3.1.

Remark 5.3.1.3. *One can also prove 5.3.1.1 by the following computation. To every morphism of G-OS-modules $\phi : M \to P \otimes_{OS} A(G)$ one associates the OS-morphism $(id_P \otimes \eta) \circ \phi : M \to P$. Conversely, to every OS-morphism $f : M \to P$ one associates the morphism of G-OS-modules $(f \otimes id_{A(G)}) \circ \mu_M : M \to P \otimes_{OS} A(G)$. One sees at once that*

$$(id_P \otimes \eta) \circ (f \otimes id_{A(G)}) \circ \mu_M = (f \otimes id_{OS}) \circ (id_P \otimes \eta) \circ \mu_M = f.$$

On the other hand, for every ϕ the following diagram is commutative:

$$\begin{tikzcd}
M \arrow[r, "\varphi"] \arrow[d, "\mu_M"'] & P \otimes_{OS} A(G) \arrow[d, "id_P \otimes \Delta"] \\
M \otimes_{OS} A(G) \arrow[r, "\varphi \otimes id_{A(G)}"] & P \otimes_{OS} A(G) \otimes_{OS} A(G).
\end{tikzcd}$$

It follows that

$$\begin{aligned}
((id_P \otimes \eta) \circ \varphi \otimes id_{A(G)}) \circ \mu_M &= (id_P \otimes \eta \otimes id_{A(G)}) \circ (\varphi \otimes id_{A(G)}) \circ \mu_M \\
&= (id_P \otimes \eta \otimes id_{A(G)}) \circ (id_P \otimes \Delta) \circ \varphi = \varphi.
\end{aligned}$$

This proves 5.3.1.1 (i) (and (ii) follows).

Let F be a G-$\mathcal{O}S$-module; one has seen above that axiom (CM 2) of 4.7.2 shows that, considered as $\mathcal{O}S$-module, F is a direct factor of $E(F)$. This entails:

Proposition 5.3.2. *Let S be an affine scheme and G an S-group affine and flat; suppose that every exact sequence $0 \to F_1 \to F_2 \to F_3 \to 0$ of quasi-coherent G-$\mathcal{O}S$-modules that splits as a sequence of $\mathcal{O}S$-modules also splits as a sequence of G-$\mathcal{O}S$-modules.*

Then, the functors $H^n(G, -)$ are zero for $n > 0$ (or what amounts to the same, the functor $H^0(G, -)$ is exact).

Indeed, by hypothesis, the sequence of G-$\mathcal{O}S$-modules

$$0 \to F \to E(F) \to E(F)/F \to 0$$

is split; F is therefore a direct factor, as G-$\mathcal{O}S$-module, of $E(F)$, whose cohomology is zero.

One immediately deduces from this and from Proposition 4.7.4:

Theorem 5.3.3. *Let S be an affine scheme and G a diagonalizable S-group. For every quasi-coherent G-$\mathcal{O}S$-module F, one has $H^n(G, F) = 0$, for $n > 0$.*

Remark. *Proposition 5.3.2 remains valid when G is not necessarily flat over S; the proof then appeals to relative cohomology.*[66]

6. G-equivariant objects and modules

[67] Let C be a category having a final object and in which fiber products exist. Let G be a $\hat{C}$-group, $\pi : M \to X$ a morphism of $\hat{C}$, and $\lambda = \lambda_X : G \times X \to X$ an action of G on X. In the sequel, one will denote $Y \times_f M$ the fiber product of $\pi : M \to X$ and an X-functor $f : Y \to X$.

For every $U \in Ob(C)$ and $x \in X(U)$, one will denote $M_x = U \times_x M$, i.e. for every $\phi : U' \to U$, one has

```
M_x(U′) = {m ∈ M(U′) | π(m) = x_{U′} = φ*(x)}.
```

Finally, if $g \in G(U)$ one will also denote gx the element $\lambda(g, x)$ of $X(U)$.

Definition 6.1. *a) One says that M is a G-*equivariant X-object *if one has given an action $\Lambda : G \times M \to M$ of G on M lifting λ, i.e. such that the square below is commutative:*

```
                    Λ
    G × M ──────────────→ M
    |                     |
```

[66]N.D.E.: The editors did not seek to develop this remark.
[67]N.D.E.: This section has been added.

$$\begin{array}{ccc} & id_G \times \pi & & \pi \\ & \downarrow & \lambda & \downarrow \\ & G \times X & \longrightarrow & X \end{array}$$

This amounts to giving for every morphism $(g, x) : U \to G \times X$ maps

$$\Lambda_x^U(g) : M_x(U) \to M_{gx}(U), \quad m \mapsto g \cdot m$$

verifying $1 \cdot m = m$ and $g \cdot (h \cdot m) = (gh) \cdot m$ and functorial in the $(G \times X)$-object U. This still amounts to giving morphisms of U-objects:

$$\Lambda_x(g) : M_x \to M_{gx}$$

verifying $\Lambda_x(1) = id$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$.

*b) Let now O be a $\hat{C}$-ring and let $OX = O \times X$. Under the conditions of (a), one says that M is a G-*equivariant OX-module *if it is an OX-module (cf. Definition 4.3.3.1, valid for any ring functor on a category C) and if the action Λ is compatible with the OX-module structure of M, i.e. if for every $(g, x) \in G(U) \times X(U)$ ($U \in Ob(C)$), the map $\Lambda_x(g) : M_x \to M_{gx}$, $m \mapsto g \cdot m$ is a morphism of OU-modules.*

Remark 6.2. *(i) In (a) above, the conditions $\Lambda_x(1) = id$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ evidently entail that each $\Lambda_x(g)$ is an isomorphism, with inverse $\Lambda_{gx}(g^{-1})$. Conversely, if one assumes that each $\Lambda_x(g)$ is an isomorphism, the condition $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ applied to $h = 1$ gives $\Lambda_x(1) = id$.*

(ii) Let M be an OX-module. First, one sees that giving a morphism $\Lambda : G \times M \to M$ making the diagram of 6.1 commutative and such that each morphism

$\Lambda_x(g) : M_x \to M_{gx}$, $m \mapsto g \cdot m$, is an isomorphism of OU-modules, is equivalent to giving an isomorphism of $O_{G \times X}$-modules:

$$\theta : G \times M = (G \times X) \times_{pr_X} M \xrightarrow{\sim} (G \times X) \times_\lambda M$$
$$(g, x, m) \mapsto (g, x, g \cdot m).$$

Since one has assumed that each $\Lambda_x(g)$ is an isomorphism, the equality $\Lambda_x(1) = id$ will be a consequence of the equality $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ applied to $h = 1$. One therefore obtains that Λ is an action of G on M (i.e. $g \cdot (h \cdot m) = (gh) \cdot m$) if and only if the diagram of $(G \times G \times X)$-isomorphisms below is commutative (where one denotes μ the multiplication of G and $f*(\theta)$ the isomorphism deduced from θ by a base change $f : G \times G \times X \to G \times X$):

$$\begin{array}{ccc} (G \times G \times X) \times_{pr_X \circ pr_{2,3}} M & \xrightarrow{pr^*_{2,3}(\theta)} & (G \times G \times X) \times_{\lambda \circ pr_{2,3}} M \\ (\mu \times id_X)^*(\theta) \downarrow & & \downarrow (id_G \times \lambda)^*(\theta) \\ (G \times G \times X) \times_{\lambda \circ (\mu \times id_X)} M & & (G \times G \times X) \times_{\lambda \circ (id_G \times \lambda)} M. \end{array}$$

One sees therefore that giving a G-equivariant OX-module structure on M is equivalent to giving an isomorphism θ of $O_{G\times X}$-modules as above, such that the diagram above is commutative.

(iii) All that precedes extends to the case where G is only a $\hat{C}$-monoid: in that case, giving an action $\Lambda : G \times M \to M$ lifting λ and such that each $\Lambda_x(g) : M_x \to M_{gx}$ is a morphism of OU-modules is equivalent to giving a morphism θ of $O_{G\times X}$-modules as in (ii), such that the diagram above (without the ∼ under the arrows) is commutative, and such that $p_M \circ \theta \circ (\epsilon_G \times id_M) = id_M$, where ϵ_G denotes the unit section of G and p_M the projection onto M.

6.3. G-equivariant morphisms

Let Y be a second object of Ĉ, endowed with an action $\lambda_Y : G \times Y \to Y$ of G, and let N be a second G-equivariant OX-module. One says that a Ĉ-morphism $f : Y \to X$ (resp. a morphism of OX-modules $\phi : M \to N$) is *G-equivariant* if it commutes with the action of G, i.e. if one has set-theoretically $f(g \cdot y) = g \cdot f(y)$ (resp. $\phi(g \cdot m) = g \cdot \phi(m)$), which is equivalent to saying that $f \circ \lambda_Y = \lambda_X \circ (id_G \times f)$ (resp. $\phi \circ \Lambda_M = \Lambda_N \circ (id_G \times \phi)$).

One then obtains at once the following lemma:

Lemma 6.3.1. *Let $f : Y \to X$ be a G-equivariant morphism and M a G-equivariant OX-module. Then the inverse image $f*(M) = Y \times_f M$ is a G-equivariant OY-module.*

On the other hand, G acts on $\mathrm{Hom}_{\hat{C}}(Y, X)$. Indeed, let $T \in Ob(C)$, $g \in G(T)$ and ϕ a T-morphism $Y_T \to X_T$; then g defines automorphisms $\lambda_Y(g)$ and $\lambda_X(g)$ of Y_T and X_T, and one will denote $g \cdot \phi$ (or also $g\phi g^{-1}$) the morphism $\lambda_X(g) \circ \phi \circ \lambda_Y(g^{-1})$. This defines an action of $G(T)$ on $\mathrm{Hom}_T(Y_T, X_T)$, functorial in T.

Definition 6.3.2. *If $\phi : Y \to X$ is an arbitrary $\hat{C}$-morphism, one can therefore consider its stabilizer $Stab_G(\phi)$ (cf. 2.3.3.1): for every $T \in Ob(C)$, $Stab_G(\phi)(T)$ is the subgroup $G(T)$ formed by the g such that $g \circ \phi_T = \phi_T \circ g$, i.e. such that the diagram*

$$\begin{array}{ccc} Y_T & \xrightarrow{\phi_T} & X_T \\ \downarrow g & & \downarrow g \\ Y_T & \xrightarrow{\phi_T} & X_T \end{array}$$

commutes. Then, the morphism $\phi : Y \to X$ is equivariant under $Stab_G(\phi)$.

6.4. Global sections

Let M be a G-equivariant OX-module. Denote by S_0 the final object of C, and (cf. Exp. II, 1.1) by $\prod_{X/S_0} M$ the "functor of sections of M over X": it is the functor which to every $T \in Ob(C)$ associates

$$\mathrm{Hom}_X(X_T, M) = \mathrm{Hom}_{X_T}(X_T, M_T) = \Gamma(M_T/X_T).$$

Recall, on the other hand, that every morphism $g : Z \to Y$ of $\hat{C}$-objects above X induces a morphism of abelian groups $M(g) : M(Y) \to M(Z)$, which is compatible with the ring morphism $g* : O(Y) \to O(Z)$. In particular, when $Z = Y$ (with g then an X-endomorphism of Y), one obtains a morphism of abelian groups

$$M(g) : M(Y) \to M(Y)$$

which is not in general a morphism of $O(Y)$-modules, but which verifies, for every $m \in M(Y)$ and $\alpha \in O(Y)$:

$$M(g)(\alpha \cdot m) = g^*(\alpha) \cdot M(g)(m).$$

This said, one will write simply, in the sequel, $g*$ instead of $M(g)$.

Let $T \in Ob(C)$ and let $X_T = X \times T$ and pr_X the projection $X_T \to X$. For every $\alpha \in O(X_T)$ and $g \in G(T)$, set $g(\alpha) = (g^{-1}) * (\alpha)$: it is the element of $O(X_T)$ defined set-theoretically by: for every $S \in Ob(C)$ and $x \in X(S)$, $t \in T(S)$,

$$g(\alpha)(x, t) = \alpha(g^{-1} x, t).$$

One thus obtains a (left) action of $G(T)$ by ring automorphisms on $O(X_T)$, functorial in T, and such that $g(\alpha) = \alpha$ if α is the image in $O(X_T)$ of an element of $O(T)$.

Now denote by ϕ the identity morphism of X_T (cf. 6.3 and the generalization further below to a morphism $\phi : Y \to X$) and designate $\mathrm{Hom}_X(X_T, M)$ by $M(\phi g^{-1})$ resp. $M(\phi)$, according as X_T is regarded as an X-object via $pr_X \circ \lambda(g^{-1})_T$, resp. pr_X.

Then $\lambda(g^{-1})_T$ is an X-morphism between these two X-objects, hence, by the preceding, one obtains a morphism of abelian groups

$$(g^{-1})^* : M(\varphi) \to M(\varphi g^{-1}), \qquad m \mapsto m \circ \lambda(g^{-1})_T$$

which verifies $(g^{-1})*(\alpha \cdot m) = (g\alpha)\cdot(g^{-1})*(m)$ for every $m \in M(\phi)$ and $\alpha \in O(X_T)$. (If m is a section of M_T on X_T then $(g^{-1}) * (m)$ is the section defined set-theoretically by $(x, t) \mapsto m(g^{-1}x, t)$.) In particular, $(g^{-1})*$ is a morphism of $O(T)$-modules.

By the functoriality of the morphisms of $O(X_T)$-modules $\Lambda_x(g)$, one obtains a commutative diagram:

$$\begin{tikzcd}
M(\varphi) \arrow[r, "(g^{-1})^*"] \arrow[d, "\Lambda_\varphi(g)"'] & M(\varphi g^{-1}) \arrow[d, "\Lambda_{\varphi g^{-1}}(g)"] \\
M(g \varphi) \arrow[r, "(g^{-1})^*"] & M(g \varphi g^{-1})
\end{tikzcd}$$

and $g\phi g^{-1} = \phi$, since ϕ is the identity map. Setting $A(g) = (g^{-1}) * \circ \Lambda_\phi(g)$, one therefore obtains a morphism of abelian groups

$$A(g) : M(\phi) \to M(\phi)$$

which is “compatible with the action of $G(T)$ on $O(X_T)$”, i.e. which verifies

`A(g)(α · m) = (g α) · A(g)(m).`

Finally, if h is a second element of $G(T)$, it follows from the functoriality of M and of the morphisms $\Lambda_x(g)$ that one has a commutative diagram

```
        M(φ)
          \
         Λ_φ(g)
            \                    Λ_φ(hg)
             \
   Λ_φ(h)     \                              \
               ↓                              ↘
           M(g φ) ——————→ M(hg φ) —————→ M(φ)
           (g^{-1})*   Λ_φ(h)    (hg)^{-1}*

               (g^{-1})*                ((hg)^{-1})*

           M(φ) —————→ M(h φ) ——————→ M(φ)
                  Λ_φ(h)         (h^{-1})*
```

whence $A(h) \circ A(g) = A(hg)$. Consequently, one has obtained the following proposition:

Proposition 6.4.1. *For every T, the $O(X_T)$-module $(\prod_{X/S_0} M)(T) = \Gamma(M_T/X_T)$ is endowed, in a way functorial in T, with an action of $G(T)$ “compatible with the action of $G(T)$ on $O(X_T)$”, i.e. which verifies*

`A(g)(α · m) = g(α) · A(g)(m).`

Since $g(\alpha) = \alpha$ for $\alpha \in O(T)$, this gives in particular that $\prod_{X/S_0} M$ is a G-O_{S_0}-module.

More generally, let, as in 6.3, Y be a second G-equivariant $\hat{C}$-object, $\phi : Y \to X$ a $\hat{C}$-morphism and $H = Stab_G(\phi)$. Then the fiber product $M_\phi = Y \times_\phi M$ is an H-equivariant OY-module. One therefore obtains:

Corollary 6.4.2. *The functor $\prod_{Y/S_0} M_\phi$ is a $Stab_G(\phi)$-O_{S_0}-module.*

6.5. G-equivariant OX-modules

Let us apply what precedes in the following situation. Let S be a scheme, G an S-group scheme acting on an S-scheme X, and F an OX-module (not necessarily quasi-coherent).

Definition 6.5.1. *One says that F is a G-*equivariant OX-module *if the OX-module $M = W(F)$ is G-equivariant, i.e. if one has given, for every morphism $(g, x) : U \to G \times_S X$, isomorphisms of OU-modules*

$$\Lambda_x(g) : W(x*(F)) \xrightarrow{\sim} W((gx)*(F)),$$

functorial in U and verifying $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$. Since the functor W is fully faithful (cf. 4.6.1.1 and 4.6.2), one therefore obtains isomorphisms of OU-modules

$$\Lambda_x(g) : x*(F) \xrightarrow{\sim} (gx)*(F),$$

where one recalls that gx denotes the morphism $\lambda \circ (g, x) : U \to X$. In particular, applying this to the identity morphism of $G \times_S X$, one obtains an isomorphism of $O_{G\times_S X}$-modules

$$(\star)\theta : pr *_X (F) \xrightarrow{\sim} \lambda * (F)$$

such that the diagram $(\star\star)$ of morphisms of sheaves on $G \times_S G \times_S X$ below is commutative:

$$\begin{array}{ccccc}
pr^*_{\{2,3\}} \circ pr^*_X(F) & \xrightarrow{pr^*_{\{2,3\}}(\theta)} & pr^*_{\{2,3\}} \circ \lambda^*(F) & & (id_G \times \lambda)^* \circ pr^*_X(F) \\
 & & (\star\star) & & \square\ (id_G \times \lambda)^*(\theta) \\
(\mu \times id_X)^* \circ pr^*_X(F) & \longrightarrow & (\mu \times id_X)^* \circ \lambda^*(F) & & (id_G \times \lambda)^* \circ \lambda^*(F).
\end{array}$$

Moreover, if E is a second G-equivariant OX-module, one says that a morphism of OX-modules $\phi : E \to F$ is *G-equivariant* if the morphism $W(\phi) : W(E) \to W(F)$ is. With the above notations, this is equivalent to saying that $\theta_F \circ pr*_X(\phi) = \lambda*(\phi) \circ \theta_E$.

Remark 6.5.2. *If $f : Y \to X$ is a G-equivariant morphism, it follows from 6.3.1 that $f*(F)$ is a G-equivariant OY-module.*

Remark 6.5.3. *Suppose moreover F quasi-coherent. Then it follows from what precedes that giving a G-equivariant OX-module structure on $M = W(F)$ is equivalent to giving such a structure on $V(F)$, which is itself equivalent to giving an action of G on the vector fibration $V(F)$, compatible with the action on X and "linear" on the fibers.*

Indeed, denote by ϕ the isomorphism $\lambda*(F) \xrightarrow{\sim} pr *_X (F)$ inverse of θ. For every morphism $(g, x) : U \to G \times_S X$, one has an isomorphism of OU-modules $\phi_x(g) = \Lambda_x(g)^{-1} = \Lambda_{gx}(g^{-1})$ from $(gx)*(F)$ to $x*(F)$; it induces an isomorphism of OU-modules

$$V((gx)*(F)) \xrightarrow{\sim} V(x*(F))$$

which one will denote $t\phi_x(g)$. One then has

$$t\,\varphi_{gx}(h) \circ t\,\varphi_x(g) = t\,(\Lambda_{gx}(h) \circ \varphi_x(g))^{-1} = \Lambda_x(hg)^{-1} = t\,\varphi_x(hg).$$

Since, for every X-scheme $f : Y \to X$, one has $V(F) \times_f Y \simeq V(f*(F))$ (cf. 4.6.2), one therefore obtains that the isomorphism

```
t φ : G ×_S V(F) = (G ×_S X) ×_{pr_X} V(F) = V(pr*_X(F))
                                          V(λ*(F)) = (G ×_S X) ×_λ V(F)
```

endows $V(F)$ with a G-equivariant 0X-module structure. Finally, identifying each functor $V(-)$ with the vector fibration that represents it, and composing $t\phi$ with the projection onto $V(F)$, one obtains an action of G on the vector fibration $V(F)$, compatible with the action on X and "linear" on the fibers.

One thus recovers the definition given, for example, in [GIT, Chap. 1, § 3], up to the fact that in *loc. cit.* Mumford considers a locally free 0X-module of finite rank E, and defines an action of G on $V(E) = W(E^\vee)$. Indeed, the diagram ($\star\star$) above, with θ replaced by ϕ and the direction of the arrows reversed, is exactly the one one finds in *loc. cit.*, Def. 1.6, and the isomorphism $t\phi$ above coincides with the isomorphism Φ of *loc. cit.*, p. 31.

Remark 6.5.4. *Consider in particular the case where $X = S$, endowed with the trivial action of G. In this case, a G-equivariant 0S-module F is the same thing as a G-0S-module (cf. 4.7.1). Moreover, if one denotes f the morphism $G \to S$ (equal here to pr_X and to λ), then the isomorphism*

$$(\star)\theta : f*(F) \xrightarrow{\sim} f*(F)$$

is an element of

```
Hom_{OG}(f*(F), f*(F)) = Hom_{OG}(W(f*(F)), W(f*(F))) = End_{OS}(W(F))(G)
```

which is nothing other than the morphism $\rho : G \to \mathrm{End}_{OS}(W(F))$ defining the operation of G on $W(F)$. Moreover, θ corresponds by adjunction to the morphism of 0S-modules $f_*(\theta) \circ \tau : F \to f_* f*(F)$, where $\tau : F \to f_* f*(F)$ is the "unit" morphism of the adjunction. (This will be used in VI_B, 11.10.bis.)

6.6. The functors $\prod_{X/S} W(F)$ and $W(p^*(F))$

Let S be a scheme, G an S-group scheme acting on S-schemes X and Y via the morphisms $\lambda : G \times_S X \to X$ and $\mu : G \times_S Y \to Y$, and let $p : X \to Y$ be a G-equivariant morphism. Suppose that the morphisms $p : X \to Y$ and $\nu_Y : Y \to S$ are quasi-compact and quasi-separated, and that $\pi : G \to S$ is flat.

Then the projection $pr_Y : G \times_S Y \to Y$ is flat, as is μ (since μ is the composite of pr_Y and of the automorphism $(g, y) \mapsto (g, gy)$). Finally, for a variable S-scheme $f : T \to$, one will denote by $p_T : X_T \to T$ and $f_X : X_T \to X$ the morphisms deduced from p and f by base change.

Let F be a quasi-coherent and G-equivariant 0X-module, and let θ be the isomorphism $pr*_X(F) \xrightarrow{\sim} \lambda*(F)$ of 6.5.1 ($\star$). Since p is quasi-compact and quasi-separated, $p_*(F)$ is quasi-coherent (cf. EGA I, 9.2.1).

Lemma 6.6.1. *The quasi-coherent $\mathcal{O}_Y$-module $p_*(F)$ is G-equivariant.*

Indeed, one has the two cartesian squares below:

$$\begin{array}{ccccc} G \times_S X & \xrightarrow{pr_X} & X & \xleftarrow{\lambda} & G \times_S X \\ \downarrow q & & \downarrow p & & \downarrow q \\ G \times_S Y & \xrightarrow{pr_Y} & Y & \xleftarrow{\mu} & G \times_S Y. \end{array}$$

Since p is quasi-compact and quasi-separated and pr_Y and μ are flat, it follows from EGA III, 1.4.15 (completed by EGA IV_1, 1.7.21) that one has $q_* pr *_X (F) = pr *_Y p_*(F)$ and $q_* \lambda * (F) = \mu * p_*(F)$. Consequently, θ induces an isomorphism:

$$\theta_Y : pr^*_Y p_*(F) \cong \mu^* p_*(F),$$

and one obtains similarly that θ_Y verifies the "cocycle condition" 6.5.1 ($\star\star$) (since the morphisms $G \times_S G \times_S Y \to G \times_S Y$ involved are flat). This proves the lemma.

In particular, take $Y = S$ endowed with the trivial action of G. Taking Remark 6.5.4 into account, one then obtains that $p_*(F)$ is a G-$\mathcal{O}_S$-module. If moreover G is affine over S and if one denotes $A(G) = \pi_*(O_G)$, then $p_*(F)$ is therefore an $A(G)$-comodule, by 4.7.2.

On the other hand, by 6.4.1, the functor $\prod_{X/S} W(F)$, which to every $f : T \to S$ associates

$$W(f_X^*(F))(X_T) = \Gamma(X_T, f_X^*(F)) = \Gamma(T, p_{T*} f_X^*(F))$$

is a G-$\mathcal{O}_S$-module. Moreover, one has a canonical morphism $\tau : W(p_*(F)) \to \prod_{X/S} W(F)$, which is given for every $f : T \to S$ by the canonical morphism:

$$\Gamma(T, f^* p_*(F)) \to \Gamma(T, p_{T*} f_X^*(F))$$

and which is an isomorphism when restricted to the full subcategory of schemes T flat over S, and one verifies without difficulty that τ is a morphism of G-$\mathcal{O}_S$-modules. Consequently, one has obtained the following proposition (for point (ii), compare with [GIT], p. 32).

Proposition 6.6.2. *Let S be a scheme, G an S-group scheme acting on an S-scheme X, and F a quasi-coherent and G-equivariant $\mathcal{O}_X$-module. Suppose that $\pi : G \to S$ is flat and that $p : X \to S$ is quasi-compact and quasi-separated.*

(i) Then $p_*(F)$ is a quasi-coherent G-$\mathcal{O}_S$-module. Moreover, the canonical morphism $W(p_*(F)) \to \prod_{X/S} W(F)$ is a morphism of G-$\mathcal{O}_S$-modules, and these two functors coincide on the category of flat S-schemes.

(ii) If moreover G is affine over S and if one denotes $A(G) = \pi_*(O_G)$, then $p_*(F)$ is endowed with a structure of $A(G)$-comodule.[68]

[68] N.D.E.: The same is true if G is flat, quasi-compact and quasi-separated over S and if $p_*(F)$ is a flat $\mathcal{O}_S$-module, cf. Exp. VI_B, 11.6.1 (ii).

6.7. Stabilizers

Let S be a scheme, G an S-group scheme acting on an S-scheme X, and let F be a quasi-coherent G-equivariant OX-module. Let Y be a second S-scheme endowed with an action of G (possibly trivial), let $\phi : Y \to X$ be an S-morphism, not necessarily G-equivariant, and let $Stab_G(\phi)$ be the stabilizer in G of ϕ (cf. 6.3.2).

Let us point out at once (see Exp. VI_B, § 6) that $Stab_G(\phi)$ is representable by a closed sub-group scheme H of G if X is separated over S and if Y is essentially free over S (cf. *loc. cit.*, Déf. 6.2.1). Indeed, consider the morphism $r : G \times_S Y \to X \times_S X$ given set-theoretically by $r(g, y) = (\phi(y), g\phi(g^{-1}y))$, and let $P = G \times_S Y$ and P' the inverse image by r of the diagonal $\Delta_{X/S}$. Then one has (cf. *loc. cit.*, 6.2.4 (a))

$$Stab_G(\phi) = \prod_{P/G} P'$$

and therefore, by *loc. cit.*, $Stab_G(\phi)$ is representable by a closed sub-group scheme H of G if X is separated over S and if Y is essentially free over S; this second condition being automatically verified if S is the spectrum of a field, or if $Y = S$. Under these hypotheses, $\phi * (F)$ is then a quasi-coherent H-equivariant OY-module (cf. 6.5.2).

Hence, if moreover $\pi : G \to S$ and $p : Y \to S$ are quasi-compact and quasi-separated over S, and π flat, then $p_*\phi * (F)$ is an H-OS-module, by 6.6.2. In particular, one obtains:

Corollary 6.7.1. *Let S be a scheme, G an S-group scheme acting on an S-scheme X, and F a quasi-coherent G-equivariant OX-module. Suppose that $\pi : G \to S$ is flat and that $X \to S$ is quasi-compact and separated. Let $\tau : S \to X$ be a section of X over S.*

(i) The stabilizer $H = Stab_G(\tau)$ is a closed sub-group scheme of G, and $\tau * (F)$ is a quasi-coherent H-OS-module.

(ii) If moreover H is affine over S (for example, if G is) and if one denotes $A(H) = \pi_*(O_H)$, then $\tau * (F)$ is an $A(H)$-comodule.

6.8. G-equivariant sheaves on G

To conclude, let us point out two results (6.8.1 and 6.8.6 below) that will be used in Exposés II and III (cf. in particular III, 4.25).

Proposition 6.8.1. *Let S be a scheme, $\pi : G \to S$ an S-group scheme, $\epsilon : S \to G$ the unit section. Consider the action by left translations of G on itself. Then the functors $E \mapsto \pi * (E)$ and $F \mapsto \epsilon * (F)$ induce equivalences, quasi-inverse to one another, between the category of quasi-coherent OS-modules and that of quasi-coherent G-equivariant OG-modules.*

Proof. Denote by μ the multiplication of G and by pr_2 the second projection $G \times_S G \to G$. Since $\pi \circ \mu = \pi \circ pr_2$, then, for every quasi-coherent OS-module E, one has a canonical isomorphism $\mu * \pi * (E) = pr *_2 \pi * (E)$, and one verifies easily that this isomorphism satisfies the "cocycle condition" 6.5.1 ($\star\star$), i.e. $\pi * (E)$ is a G-equivariant 0G-module. Since $\epsilon * \pi * (E) = E$, the functor $E \mapsto \pi * (E)$ is

fully faithful; it remains therefore to see that for every G-equivariant 0G-module F, one has $F \simeq \pi * \epsilon * (F)$. By hypothesis, one has an isomorphism $\theta : pr *_2 (F) \xrightarrow{\sim} \mu * (F)$; taking the inverse image of θ by the morphism $\tau : G \to G \times_S G$ of components $(id_G, \epsilon \circ \pi)$, one obtains an isomorphism $F \xrightarrow{\sim} \pi * \epsilon * (F)$.

Remarks 6.8.2. *(a) Consider the action of $G \times_S G$ on G defined by $(g_1, g_2) \cdot g = g_1 g g_2^{-1}$; then the stabilizer of the unit section $\epsilon : S \to G$ is the diagonal subgroup H of $G \times_S G$. Consequently, if F is a quasi-coherent $(G \times_S G)$-equivariant 0G-module then, by 6.5.2, $E = \epsilon * (F)$ is endowed with a structure of H-0S-module.*

(b) One can show that $F \mapsto \epsilon * (F)$ is an equivalence of categories, between the category of quasi-coherent $(G \times_S G)$-equivariant 0G-modules and that of quasi-coherent H-0S-modules. This is a particular case of more general "descent" results (cf. Exp. IV, § 2 and SGA 1, VIII), see for example [Th87], 1.2–1.3.

Remark 6.8.3. *One retains the notations of 6.8.1. For every quasi-coherent 0S-module E, denote by $\pi_*^G \pi * (E)$ the sub-0S-module of $\pi_* \pi * (E)$ whose sections on every open subset V of S are the $\gamma \in \Gamma(\pi^{-1}(V), \pi * (E))$ such that $g \cdot \gamma_{S'} = \gamma_{S'}$ for every $S' \to V$ and $g \in G(S')$. Then the natural morphism $E \to \pi_*^G \pi * (E)$ is an isomorphism: this is immediate if $\pi_* \pi * (E) = E \otimes_{OS} \pi_*(O_G)$ (for example if $G \to S$ is affine, or if $G \to S$ is quasi-compact and quasi-separated and E flat), and it is verified without difficulty in the general case.*

Remark 6.8.4. *Let S be a scheme, H an S-group scheme acting on an S-scheme X, F an 0X-module. Suppose H flat over S and denote by $W_P(F)$ the restriction of the functor $W(F)$ to the full subcategory formed by the S-schemes that are flat. Since, by 6.5.1, endowing F with a structure of H-equivariant module is equivalent to giving an isomorphism $\theta : pr *_X (F) \to \lambda * (F)$ of sheaves on $G \times_S X$, verifying the "cocycle condition" ($\star\star$), one sees that to give an H-equivariant module structure on F, it suffices to give such a structure on $W_P(F)$.*

Recollection 6.8.5. *Let X be an S-scheme and Y a sub-S-scheme of X. One denotes by $N_{Y/X}$ the conormal sheaf of the immersion $i : Y \hookrightarrow X$ (cf. EGA IV_4, 16.1.2). If $S' \to S$ is a flat morphism and if one denotes $i' : Y' \hookrightarrow X'$ the immersion deduced from i by base change, then by* loc. cit.*, 16.2.2 (iii), one has $N_{Y/X} \otimes_{OY} O_{Y'} = N_{Y'/X'}$.*

Proposition 6.8.6. *Let S be a scheme, X an S-group scheme, Y a sub-S-group scheme of X. Suppose Y flat over S. Then the conormal sheaf $N_{Y/X}$ is a $(Y \times_S Y)$-equivariant 0Y-module.*

Indeed, $H = Y \times_S Y$ is flat over S, so by Remark 6.8.4, it suffices to endow $W_P(N_{Y/X})$ with an H-equivariant module structure. Let S' be a flat S-scheme, let $Y' \hookrightarrow X'$ be the immersion obtained by base change, and let $N' = N_{X/Y} \otimes_{OY} O_{Y'}$. By 6.8.5, one has $N' = N_{Y'/X'}$.

Every $h \in H(S')$ induces an automorphism of Y', and one therefore obtains, for every $y \in Y'(S')$, isomorphisms

$$\Gamma(S', y^*(N')) \xrightarrow{\sim} \Gamma(S', y^* h^*(N'))$$

which endow $W_P(N)$ with a structure of H-equivariant module (cf. 6.1).

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[69]

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Footnotes

Exposé II. Tangent bundles, Lie algebras

by M. Demazure

[70] We propose in this Exposé to construct the analogue, in the theory of schemes, of the tangent bundles and Lie algebras of the classical theory. It will however be useful not to restrict ourselves to schemes properly speaking, but also to consider certain functors on the category of schemes which are not necessarily representable (for example the functors Hom, Norm, etc.). As announced in the preceding Exposé (cf. I 1.1), we shall identify a scheme with the functor associated to it.

On the other hand, the constructions presented below go beyond the framework of the theory of schemes. They are equally valid, for example, in the theory of analytic spaces with nilpotent elements, modulo a few modifications of detail.

[69] N.D.E.: additional references cited in this Exposé.
[70] Version of 14/10/2024.

Before beginning this construction, we must lay down a few general definitions that complement those of I 1.7.

1. The functors $\mathrm{Hom}_{Z/S}(X, Y)$

Let us resume the notations of I 1.1. We identify the category C with a full subcategory of $\hat{C} = \mathrm{Hom}(C^{\circ}, (Ens))$ (in particular we suppress the underlines[71] that allowed us to distinguish graphically an object of $\hat{C}$ from an object of C).

Consider the following situation: four objects of $\hat{C}$, denoted S, X, Y, Z, the first being in fact an object of C, with X and Y above Z, and Z above S:

$$\begin{tikzcd} X \arrow[dr, "pX"'] & & Y \arrow[dl, "pY"] \\ & Z \arrow[d] & \\ & S. & \end{tikzcd}$$

Definition 1.1. *We define an object of $\hat{C}/S$, denoted $\mathrm{Hom}_{Z/S}(X, Y)$, by:*

$$(1) \quad \mathrm{Hom}_{Z/S}(X, Y)(S') = \mathrm{Hom}_{Z_{S'}}(X_{S'}, Y_{S'}) = \mathrm{Hom}_Z(X \times_S S', Y),$$

for every object S' of C/S. One sees at once that $\mathrm{Hom}_{Z/S}(X, Y)$ is nothing other than the subobject of $\mathrm{Hom}_S(X, Y)$ formed by the morphisms compatible with pX and pY, that is to say, the kernel of the pair of morphisms

$$\mathrm{Hom}_S(X, Y) \rightrightarrows \mathrm{Hom}_S(X, Z)$$

defined: the first by composition with pY; the second as the constant morphism whose "image" is pX.

On the other hand, one sees as in I 1.7 that, for every object T of $\hat{C}$ above S, one has a natural bijection:[72]

$$(2) \quad \mathrm{Hom}_S(T, \mathrm{Hom}_{Z/S}(X, Y)) \simeq \mathrm{Hom}_Z(X \times_S T, Y).$$

Moreover, by I 1.7.1, if E, F are objects of $\hat{C}$ above Z, one has:

$$\mathrm{Hom}_Z(E, \mathrm{Hom}_Z(F, Y)) \simeq \mathrm{Hom}_Z(E \times_Z F, Y) \simeq \mathrm{Hom}_Z(F, \mathrm{Hom}_Z(E, Y)).$$

Applying this with $E = X$ and $F = Z \times_S T$, one obtains natural bijections, for every object T of $\hat{C}/S$:

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[71] Rendered in this edition by boldface characters $\mathbf{F}$, $\mathbf{G}$, $\mathbf{V}$, $\mathbf{W}$, cf. Exposé I. However, for functors such as `Norm` and `Centr` (cf. 5.2) the underlines have been kept in the original, and they have been added for the functor `Lie`, this to distinguish the functor $Lie(G/S)$ from the $\Gamma(S, O_S)$-Lie algebra $Lie(G/S) = Lie(G/S)(S)$, used for example in Exposé VII_A.

[72] One has detailed the points (2), (3), (4); in particular, (4) will be used in 3.11.

$$\mathrm{Hom}_S(T, \mathrm{Hom}_{Z/S}(X, Y)) \simeq \mathrm{Hom}_Z(X \times_S T, Y) \simeq \mathrm{Hom}_Z(Z \times_S T, \mathrm{Hom}_Z(X, Y)) \tag{3}$$
$$\simeq \mathrm{Hom}_Z(X, \mathrm{Hom}_Z(Z \times_S T, Y)).$$

Moreover, these bijections are functorial in T, so one obtains isomorphisms of S-functors:

$$\mathrm{Hom}_S(T, \mathrm{Hom}_{Z/S}(X, Y)) \simeq \mathrm{Hom}_{Z/S}(X, \mathrm{Hom}_Z(Z \times_S T, Y)) \tag{4}$$
$$\simeq \mathrm{Hom}_{Z/S}(X \times_S T, Y).$$

Let us point out two particular cases of the definition. If $Z = S$, one has:

$$\mathrm{Hom}_{S/S}(X, Y) = \mathrm{Hom}_S(X, Y).$$

On the other hand, when $X = Z$, one sets

$$\prod_{Z/S} Y = \mathrm{Hom}_{Z/S}(Z, Y), \tag{5}$$

so that by definition

$$\Big(\prod_{Z/S} Y\Big)(S') = \mathrm{Hom}_Z(Z \times_S S', Y) \simeq \Gamma(Y_{S'}/Z_{S'}).$$

The functor $\prod_{Z/S} : \hat{C}/Z \to \hat{C}/S$ is right adjoint to the base-change functor from S to Z: for every S-functor U one has

$$\mathrm{Hom}_S\Big(U, \prod_{Z/S} Y\Big) = \mathrm{Hom}_Z(U \times_S Z, Y).$$

(If $C = (Sch)$ and if Z is an S-scheme, the functor $\prod_{Z/S}$ is called *"Weil restriction of scalars"*.)[73] Let us note also that one has an isomorphism:

$$\mathrm{Hom}_{Z/S}(X, Y) \simeq \mathrm{Hom}_{X/S}(X, Y \times_Z X) = \prod_{X/S}(Y \times_Z X), \tag{6}$$

which gives in particular, for $Z = S$, an isomorphism:

$$\mathrm{Hom}_S(X, Y) \simeq \prod_{X/S} Y_X. \tag{7}$$

Remark 1.2. The functor $Y \mapsto \mathrm{Hom}_{Z/S}(X,Y)$ commutes with products in the following sense: one has a functorial isomorphism

$$\mathrm{Hom}_{Z/S}(X, Y \times_Z Y') \simeq \mathrm{Hom}_{Z/S}(X, Y) \times_S \mathrm{Hom}_{Z/S}(X, Y'). \tag{$*$}$$

It follows that if Y is a Z-group, resp. a Z-ring, etc., then $\mathrm{Hom}_{Z/S}(X,Y)$ is an S-group, resp. an S-ring, etc.

Remark 1.3.[74] Moreover, let $\pi : M \to Y$ be a Y-functor of $\mathcal{O}_Y$-modules (cf. I, 4.3.3.1). Set $H = \mathrm{Hom}_{Z/S}(X,Y)$. Then $\mathrm{Hom}_{Z/S}(X,M)$ is equipped with a natural structure of $\mathcal{O}_H$-module; more precisely, for every H' above H, $\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X,M))$ is equipped with a natural structure of $O(H' \times_S X)$-module.

Indeed, denote by $m : M \times_Y M \to M$ and $\lambda : O_Y \times_Y M \to M$ the morphisms defining the structures of Y-(abelian) group and of $\mathcal{O}_Y$-module. Let H' be an S-scheme above $H = \mathrm{Hom}_{Z/S}(X,Y)$, i.e. one has been given a Z-morphism $f : X \times_S H' \to Y$, which therefore makes $X \times_S H'$ a Y-object. Then,

[73] One has added the two preceding sentences.
[74] One has added this remark, useful for Corollary 3.11.1.

$$\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, M))$$

is the set of Z-morphisms $\phi : X \times_S H' \to M$ such that $\pi \circ \phi = f$, i.e. of Y-morphisms $X \times_S H' \to M$.

If ϕ, ψ are two such morphisms, one defines $\phi + \psi$ as the Y-morphism composite

$$X \times_S H' \xrightarrow{\phi \times \psi} M \times_Y M \xrightarrow{m} M$$

and one verifies that this equips $\mathrm{Hom}_{Z/S}(X, M)$ with a structure of abelian group above $H = \mathrm{Hom}_{Z/S}(X, Y)$.[75]

Likewise, if a is an element of $O(X \times_S H')$, i.e. an S-morphism $a : X \times_S H' \to O_S$, one defines $a\phi$ as the composite $\lambda \circ (a \times \phi)$, where $a \times \phi$ denotes the Y-morphism from $X \times_S H'$ to $O_Y \times_Y M \simeq O_S \times_S M$ with components a and ϕ; one verifies that this equips $\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, M))$ with a structure of $O(X \times_S H')$-module, functorial in the H-object H'.

2. The schemes $I_S(M)$

Definition 2.1. *Let S be a scheme and M a quasi-coherent 0_S-module. We denote by $D_{O_S}(M)$ the quasi-coherent 0_S-algebra $O_S \oplus M$ (where M is considered as an ideal of square zero). We denote by $I_S(M)$ the S-scheme* $\mathrm{Spec}\, D_{O_S}(M)$.[76]

In particular we set $D_{O_S} = D_{O_S}(O_S)$, $I_S = I_S(O_S)$, and we call them respectively the algebra of dual numbers over S and the scheme of dual numbers over S.

Then $M \mapsto I_S(M)$ is a contravariant functor from the category of quasi-coherent 0_S-modules to that of S-schemes. In particular the morphisms $0 \to M$ and $M \to 0$ define respectively the structural morphism $\rho : I_S(M) \to I_S(0) = S$ and a section ϵ_M of it, which we call the *zero section*.[77]

2.1.1.[78] Since $M \mapsto I_S(M)$ is a contravariant functor, every $a \in \mathrm{End}_{O_S}(M)$ defines an S-endomorphism a^* of $I_S(M)$, and one has $1^* = id$, $(ab)^* = b^* \circ a^*$, $0^* = \epsilon_M \circ \rho$ and $a^* \circ \epsilon_M = \epsilon_M$. Consequently, the S-scheme $I_S(M)$ is equipped with a right action of the multiplicative monoid of $\mathrm{End}_{O_S}(M)$, which commutes with the S-morphisms $I_S(M) \to I_S(M')$ arising from morphisms $M' \to M$; in particular, the operators a^* preserve the zero section of $I_S(M)$.

For every $a \in \mathrm{End}_{O_S}(M)$ and $f : S' \to S$ and $m \in I_S(M)(S')$, we shall write $m \cdot a = a^*(m)$; then $m \cdot 1 = m$, $(m \cdot a) \cdot b = m \cdot (ab)$, $m \cdot 0 = \epsilon_M(\rho(m))$ and, if $m = \epsilon_M(f)$, then $m \cdot a = m$.[79]

[75] Of course, if M is a Y-group (not necessarily abelian) and if one sets $\phi \cdot \psi = m \circ (\phi \times \psi)$, this makes $\mathrm{Hom}_{Z/S}(X, M)$ a group above $\mathrm{Hom}_{Z/S}(X, Y)$.

[76] Note that $I_S(M)$ has the same underlying space as S.

[77] Compare with 3.1.1.

[78] One has detailed the original in what follows, and one has added the numbering 2.1.1 to 2.1.3.

[79] In the sequel, one will be principally interested in the case where $a \in O(S)$, acting by homotheties on M. For example, if M is a free 0_S-module of rank r, then, for every $S' \to S$, $\mathrm{Hom}_S(S', I_S(M))$ is identified with the set of r-tuples $(\epsilon_1, \cdots, \epsilon_r) \in O(S')$ such that $\epsilon_i \epsilon_j = 0$ for all i, j, and one has

Remark 2.1.2. *The formation of the $I_S(M)$ commutes with base extension: one has canonical isomorphisms*

$$I_S(M)_{S'} \simeq I_{S'}(M \otimes_{O_S} O_{S'}).$$

To simplify, we shall write $I_{S'}(M) = I_S(M)_{S'}$; more generally, if X is an S-functor (not necessarily representable), we shall write $I_X(M) = I_S(M) \times_S X$.

2.1.3.[80] By what precedes, the multiplicative monoid of $O(S')$ acts on the S'-scheme $I_{S'}(M)$, functorially in M, i.e. the S-scheme $I_S(M)$ is equipped with a structure of object with monoid of operators O_S, this structure being functorial in M. One thus has a morphism of S-schemes

$$\lambda : I_S(M) \times_S O_S \to I_S(M),$$

verifying obvious conditions. For every S-functor X, one obtains by base change a morphism of X-functors:

$$\lambda_X : I_X(M) \times_S O_S \to I_X(M)$$

which makes the S-functor $I_X(M)$ into an object with monoid of operators $O(X)$: every element a of $O(X) = \mathrm{Hom}_S(X, O_S)$ defines an X-endomorphism $a^* = \lambda_X \circ (id_{I_X(M)} \times (a \circ pr_X))$ of $I_X(M)$; explicitly, if $x \in X(S')$ and $m \in I_S(M)(S') = I_{S'}(M)(S')$, then $a(x) = a \circ x$ belongs to $O(S')$ and one has:

$$(m, x) \cdot a = (m \cdot a(x), x).$$

This operation is functorial in M and preserves the zero section $\epsilon_M : X \to I_X(M)$, i.e. $a^* \circ \epsilon_M = \epsilon_M$ for every $a \in O(X)$.

Moreover, this operation is "functorial in X" in the following sense: if $\pi : Y \to X$ is a morphism of S-functors and $u : O(X) \to O(Y)$ the corresponding ring morphism (i.e. $u(a) = a \circ \pi \in O(Y)$ for every $a \in O(X) = \mathrm{Hom}_S(X, O_S)$), then the following diagram is commutative:

$$\begin{tikzcd}
I_X(M) \arrow[r, "a^*"] & I_X(M) \\
I_Y(M) \arrow[u, "\pi"] \arrow[r, "u(a)^*"] & I_Y(M) \arrow[u, "\pi"] .
\end{tikzcd}$$

2.2. Let now M and N be two quasi-coherent O_S-modules. The commutative diagram

$$\begin{tikzcd}
 & M \oplus N \arrow[dr] \arrow[dd, shift left] & \\
M \arrow[ur] \arrow[dr] & & N \arrow[dl] \\
 & {} \arrow[uu, shift left] &
\end{tikzcd}$$

$(\epsilon_1, \cdots, \epsilon_r) \cdot a = (a\epsilon_1, \cdots, a\epsilon_r)$ for every $a \in O(S)$.

[80]One has added this paragraph, which will be useful in 3.4.2 and 4.6.2.

```
           ↘    |    ↙
                0
```

defines a commutative diagram of S-schemes

```
               I_S(M ⊕ N)
           ↗      ↑       ↖
         ↗        |         ↖
       ↗       ε_{M⊕N}        ↖
    I_S(M)          |          I_S(N)
(∗)     ↖         |           ↗
         ↖        |         ↗
          ε_M     |      ε_N
              ↘   |   ↙
                  S .
```

Proposition 2.2. *For every S-scheme X, the diagram of functors over S obtained by applying the functor $\mathrm{Hom}_S(-, X)$ to diagram (∗) is cartesian:*

```
               Hom_S(I_S(M ⊕ N), X)
           ↙                       ↘
    Hom_S(I_S(M), X)        Hom_S(I_S(N), X)
           ↘                       ↙
               Hom_S(S, X) = X .
```

It must be verified that for every $S' \to S$, the diagram of sets obtained by taking the value of the functors on S' is cartesian. Since the formation of $I_S(P)$ commutes with base extension in the sense explained above, it suffices to do it for $S' = S$, hence to verify that the following diagram of sets is cartesian:

```
               X(I_S(M ⊕ N))
           ↙               ↘
    X(I_S(M))        X(I_S(N))
           ↘     X(ε_{M⊕N})↙
          X(ε_M)  ↓  X(ε_N)
                X(S) .
```

Now, if $x \in X(S)$, it follows from SGA 1, III 5.1[81], that $X(\epsilon_M)^{-1}(x)$ is isomorphic, functorially in M, to

$$\mathrm{Hom}_{O_S}(x^*(\Omega^1_{X/S}), M),$$

where $\Omega^1_{X/S}$ denotes the sheaf of relative differentials of X with respect to S. Now this latter functor (in M) evidently transforms a direct sum of O_S-modules into the product of the corresponding sets, whence the result.

[81] See also the addition 0.1.8 in Exp. III.

Corollary 2.2.1. *Let X be an S-scheme and M a free 0_S-module of finite type. The functor $\mathrm{Hom}_S(I_S(M), X)$ is isomorphic (as functor above X) to a finite product (above X) of copies of $\mathrm{Hom}_S(I_S, X)$.*

Nota 2.2.2. *It follows from the proof of the proposition that $\mathrm{Hom}_S(I_S, X)$ is isomorphic as X-functor to $V(\Omega^1_{X/S})$ (I 4.6.1), hence representable by the vector fibration*[82] *$V(\Omega^1_{X/S})$.*

3. The tangent bundle, condition (E)

In this section, unless otherwise stated, the letters M, M', N, etc., will always denote free 0_S-modules of finite type (that is to say, isomorphic to a finite direct sum of copies of 0_S).

We shall systematically use the identifications justified in Exposé I; thus we shall say "functor above S" to designate indifferently a functor equipped with a morphism into S (= h_S) or a functor on the category of objects above S. Likewise we shall say "group-functor above S", etc.

Definition 3.1. *Let S be a scheme and M a free 0_S-module of finite type. Let X be a functor above S. One calls* **tangent bundle to X above S relative to the 0_S-module** *M and denotes by $T_{X/S}(M)$ the S-functor*

$$T_{X/S}(M) = \mathrm{Hom}_S(I_S(M), X).$$

In particular, one calls **tangent bundle to X above** *S and denotes by $T_{X/S}$ the functor*[83]

$$T_{X/S} = T_{X/S}(0_S) = \mathrm{Hom}_S(I_S, X).$$

Then $M \mapsto T_{X/S}(M)$ is a covariant functor from the category of free 0_S-modules of finite type to the category of S-functors. In particular the morphisms[84] $M \to 0$ and $0 \to M$ define respectively an S-morphism

$$\pi_M : T_{X/S}(M) \longrightarrow T_{X/S}(0) \simeq X$$

and a section τ_0 of it, called the *zero section* (or *null section*).

Remark 3.1.1.[85] One should note that the projection $\pi_M : T_{X/S}(M) \to X$ is induced by the zero section $\epsilon_M : S \to I_S(M)$, while the null section $\tau_0 : X \to T_{X/S}(M)$ is induced by the structural morphism $\rho : I_S(M) \to S$; i.e., for every point $t \in T_{X/S}(M)(S')$ (resp. $x \in X(S')$), corresponding to an S-morphism $f : S' \times_S I_S(M) \to X$ (resp. $g : S' \to X$), one has $\pi(t) = f \circ (id_{S'} \times \epsilon_M)$ (resp. $\tau_0(x) = g \circ (id_{S'} \times \rho)$).

[82] Cf. N.D.E. (33) of Exp. I.
[83] When $S = \mathrm{Spec}(k)$ and X is a k-scheme, one has $T_{X/S}(S') = \mathrm{Hom}_S(S' \otimes_k k[\epsilon], X) = X(S' \otimes_k k[\epsilon])$; one thus recovers one of the usual definitions of the tangent bundle.
[84] One has corrected "$0 \to M$ and $M \to 0$" to: "$M \to 0$ and $0 \to M$".
[85] One has added the paragraphs 3.1.1 and 3.1.2.

It follows from 3.1 that $M \mapsto T_{X/S}(M)$ is a covariant functor from the category of free 0_S-modules of finite type to that of functors above X. In particular $O(S)$ is a monoid of operators of the X-functor $T_{X/S}(M)$ which respects "functoriality in M".

Scholie 3.1.2.[86] What precedes means, in particular, the following. For every S-morphism $\phi : X' \to X$, set

$$\Sigma(X', M) = \mathrm{Hom}_X(X', T_{X/S}(M)).$$

One has an action of the multiplicative monoid $\mathrm{End}_{O_S}(M)$ on $\Sigma(X', M)$, denoted $(\lambda, x) \mapsto \lambda * x$, such that $\lambda * (\mu * x) = (\lambda\mu) * x$, $1 * x = x$, and $0 * x = \tau_0 \circ \phi$, where τ_0 is the null section $X \to T_{X/S}(M)$. One has likewise an action of $\mathrm{End}_{O_S}(M \oplus M)$ on $\Sigma(X', M \oplus M)$.[87]

Moreover, let $m : M \oplus M \to M$ (resp. $\delta : M \to M \oplus M$) be the addition (resp. the diagonal map) of M, and denote by $m_{X'} : \Sigma(X', M \oplus M) \to \Sigma(X', M)$ and $\delta_{X'} : \Sigma(X', M) \to \Sigma(X', M \oplus M)$ the morphisms induced by m and δ. For $\lambda, \mu \in O(S)$, denote by ℓ_λ (resp. $\ell_{\lambda,\mu}$) multiplication by λ in M (resp. by (λ, μ) in $M \oplus M$). Since $m \circ \ell_{\lambda,\lambda} = \ell_\lambda \circ m$ and $m \circ \ell_{\lambda,\mu} \circ \delta = \ell_{\lambda+\mu}$, one has, for every $z \in \Sigma(X', M \oplus M)$ and $x \in \Sigma(X', M)$:

$$(\dagger) \quad \lambda * m(z) = m((\lambda, \lambda) * z), \qquad m((\lambda, \mu) * \delta(x)) = (\lambda + \mu) * x.$$

Definition 3.2. *Let $u \in X(S) = \mathrm{Hom}_S(S, X) = \Gamma(X/S)$. One calls* **tangent space to X above S at the point u relative to** *M, and denotes by $L^u_{X/S}(M)$, the S-functor obtained from the X-functor $T_{X/S}(M)$ by pullback along the morphism $u : S \to X$:*

$$\begin{tikzcd} L^u_{X/S}(M) \arrow[r] \arrow[d] & T_{X/S}(M) \arrow[d, "\pi"] \\ S \arrow[r, "u"] & X . \end{tikzcd}$$

In particular $L^u_{X/S}(O_S)$ is denoted $L^u_{X/S}$. It is the **tangent space to X above S at the point u**.

Remark 3.2.1.[88] It follows from 3.1.1 that, for every $t : S' \to S$, $L^u_{X/S}(M)(S')$ is the set of S-morphisms $f : I_{S'}(M) \to X$ such that $f \circ \epsilon_M = u \circ t$, where $\epsilon_M : S' \to I_{S'}(M)$ is the zero section.

Let us note immediately the

Proposition 3.3. *If X is representable by an S-scheme denoted $\tilde{X}$, then $T_{X/S}(M)$ and $L^u_{X/S}(M)$ are representable. In particular $T_{X/S}$ and $L^u_{X/S}$ are representable by vector fibrations on $\tilde{X}$ and on S, which are respectively $V(\Omega^1_{\tilde{X}/S})$ and $V(u^*(\Omega^1_{\tilde{X}/S}))$.*

It clearly suffices to prove the proposition for $T_{X/S}(M)$; the analogous results for $L^u_{X/S}(M)$ follow by pullback. By Corollary 2.2.1, it even suffices to do it for $T_{X/S}$, and

[86] One has added the paragraphs 3.1.1 and 3.1.2.

[87] In fact, one is interested only in the action of $O(S)$ (resp. $O(S) \times O(S)$) on $\Sigma(X', M)$ (resp. $\Sigma(X', M \oplus M)$), cf. below and the proof of 3.6.

[88] One has added this remark.

in that case, the proposition is nothing other than the remark noted in 2.2.2.

Remark 3.3.1. From this proposition follows a particularly simple description of the vector fibration representing $L^u_{X/S}$: the image of the section u of X over S is locally closed[89], hence defined by a quasi-coherent ideal m of a scheme induced on an open subset of X. The quotient m/m^2 can be considered as a quasi-coherent module on S. It is this module that defines the sought-after vector fibration.

Take for example X an algebraic scheme over a field k and u a k-rational point of X. Let m be the maximal ideal of the local ring $O_{X,u}$ and let t be the k-vector space dual to m/m^2; it is the Zariski tangent space of $O_{X,u}$ at the point u. Then, with the notations of I 4.6.5.1, one has:

$$L^u_{X/k} = V(m/m^2) = W(t).$$

Having closed this parenthesis, let us return to the general situation. Let us remark first that $L^u_{X/S}(M)$ is a covariant functor from the category of free O_S-modules of finite type to that of functors above S. In particular $O(S)$ is a set of operators of the S-functor $L^u_{X/S}(M)$ which respects functoriality in M.[90]

Proposition 3.4. *The formation of $T_{X/S}(M)$ and $L_{X/S}(M)$ commutes with base extension: for every S-scheme S', one has isomorphisms functorial in M*

$$T_{X_{S'}/S'}(M \otimes O_{S'}) \simeq T_{X/S}(M)_{S'},$$
$$L^{u'}_{X_{S'}/S'}(M \otimes O_{S'}) \simeq L^u_{X/S}(M)_{S'}, \quad \text{where } u' = u_{S'}.$$

This results immediately from the fact that Hom commutes with base extension.

Corollary 3.4.1. *The X-functor $T_{X/S}(M)$ (resp. the S-functor $L^u_{X/S}(M)$) is naturally equipped with a structure of object with operators O_X (resp. O_S), this structure being functorial in M.*

Let us show this first for $L^u_{X/S}(M)$. For each S' above S, $O(S')$ acts on $M \otimes O_{S'}$ hence on $L^{u'}_{X_{S'}/S'}(M \otimes O_{S'}) = L^u_{X/S}(M)_{S'}$; now one verifies that this action is functorial in S'. It thus equips $L^u_{X/S}(M)$, as announced, with a structure of functor with operators O_S.

For $T_{X/S}(M)$ it is a little more complicated. For each X' above X, set $T_{X/S}(M)_{X'} = T_{X/S}(M) \times_X X'$; one must equip $T_{X/S}(M)_{X'}(X') = \mathrm{Hom}_X(X', T_{X/S}(M))$ with a structure of set with monoid of operators $O(X')$ in a manner functorial in X'. For this one constructs the following diagram, where $X_{X'}$ denotes $X \times_S X'$ and f' denotes the section of $X_{X'}$ over X' defined by $f : X' \to X$:

$$\begin{array}{ccc} & T_{X_{X'}/X'}(M) & \\ \swarrow & & \searrow \\ T_{X/S}(M) & \longleftarrow & T_{X/S}(M)_{X'} \\ \downarrow & & \downarrow \end{array}$$

[89] Cf. EGA I, 5.3.11.
[90] Cf. 3.1.2.

```
                X_{X′}
             ↙        ↘ f′
           ↙             ↘
X ←———   f                 X′
             ↘           ↙
                S .
```

[91] This diagram, joined with 3.2.1, shows that $T_{X/S}(M)_{X'}(X')$ is identified with

$$L^{f'}_{X_{X'}/X'}(M)(X') = \{ X'\text{-morphisms } \psi : I_{X'}(M) \to X_{X'} \text{ such that } \psi \circ \varepsilon_M = f' \},$$

on which every $a \in O(X')$ acts via its action on $I_{X'}(M)$, i.e., with the notations of 2.1.1, one has: $a\psi = \psi \circ a^*$, i.e. for every $X'' \to X'$ and $x \in I_{X'}(M)(X'')$, $(a\psi)(x) = \psi(x \cdot a)$. One then verifies easily that this construction is functorial in X'.

The isomorphisms of Proposition 3.4 are then, by construction, isomorphisms for the structures of $O_{X_{S'}}$-objects, resp. $O_{S'}$-objects.

Remark 3.4.2.[92] The action of O_X on $T_{X/S}(M)$ can be seen, more simply, as follows. For every $f : X' \to X$, one has

$$\mathrm{Hom}_X(X', T_{X/S}(M)) = \{ \varphi \in \mathrm{Hom}_S(I_{X'}(M), X) \mid \varphi \circ \varepsilon_M = f \},$$

and one saw in 2.1.3 that $I_{X'}(M)$, considered as S-functor, is equipped with an action of the monoid $O(X')$ which preserves the zero section $\epsilon_M : X' \to I_{X'}(M)$. Consequently, if one denotes by a^* the X'-endomorphism of $I_{X'}(M)$ defined by $a \in O(X')$, one has: $a\phi = \phi \circ a^*$, i.e., for every $S' \to S$ and $(m, x') \in \mathrm{Hom}_S(S', I_S(M) \times_S X')$,

$$(a\varphi)(m, x') = \varphi(m \cdot a(x'), x')$$

(note that $a^* \circ \epsilon_M = \epsilon_M$, whence $(a\phi) \circ \epsilon_M = \phi \circ \epsilon_M = f$).

Likewise, the action of O_S on $L^u_{X/S}(M)$ can be described as follows. For every $t : S' \to S$, $L^u_{X/S}(M)(S')$ is the set of S-morphisms $\phi : I_{S'}(M) \to X$ such that $\phi \circ \epsilon_M = u \circ t$; for such a ϕ and $a \in O(S')$, one has: $a\phi = \phi \circ a^*$.

Remark 3.4.3.[93] The observations of 3.1.2 hold equally for the action of O_S on $L^u_{X/S}(M)$ and that of O_X on $T_{X/S}(M)$.

Definition 3.5. *Let S be a scheme and X an S-functor. One says that X satisfies* **condition (E)** *relative to S if, for every S' above S and all free $O_{S'}$-modules M and N of finite type, the diagram of sets*

```
                X(I_{S′}(M ⊕ N))
             ↙                  ↘
    X(I_{S′}(M))              X(I_{S′}(N))
             ↘                  ↙
                X(S′) ,
```

[91] One has detailed the original in what follows.
[92] One has added Remarks 3.4.2 and 3.4.3.
[93] One has added Remarks 3.4.2 and 3.4.3.

obtained by applying X to the diagram (∗) defined in 2.1, is cartesian.[94]

3.5.1.[95] It comes to the same thing to say that the functor $M \mapsto T_{X/S}(M)$ transforms direct sums of free O_S-modules of finite type into products of X-functors; in this case, the same holds for the functor $M \mapsto L^u_{X/S}(M) = S \times_X T_{X/S}(M)$, for every $u \in \Gamma(X/S)$.

Proposition 2.2 shows that every representable functor satisfies condition (E).

Abbreviation: instead of saying "X satisfies condition (E) with respect to S", we shall sometimes say "X/S satisfies condition (E)".

If X/S satisfies condition (E), the functor $M \mapsto T_{X/S}(M)$ commutes with products and therefore transforms groups into groups. In particular $T_{X/S}(M)$ is a commutative X-group. For the same reason, $L^u_{X/S}(M)$ is a commutative S-group.

Proposition 3.6. *If X/S satisfies (E), the abelian group structure on $T_{X/S}(M)$ (resp. $L^u_{X/S}(M)$) and the action of O_X (resp. O_S) equip $T_{X/S}(M)$ (resp. $L^u_{X/S}(M)$) with a structure of O_X-module (resp. O_S-module).*

The action of O_X (resp. O_S) is functorial in M; it therefore respects the abelian group structure that is deduced by functoriality from that of M.[96] Indeed, let us resume the notations of 3.1.2. The structure of (abelian) X-group on $T_{X/S}(M)$ is defined by the composite:

$$T_{X/S}(M) \times_X T_{X/S}(M) \simeq T_{X/S}(M \oplus M) \xrightarrow{m} T_{X/S}(M),$$

and on the other hand the morphism

$$T_{X/S}(M) \xrightarrow{\delta} T_{X/S}(M \oplus M) \simeq T_{X/S}(M) \times_X T_{X/S}(M)$$

is the diagonal morphism. Taking into account Remark 3.4.3, one deduces from the equalities (†) of 3.1.2 that

$$\lambda(x + y) = \lambda x + \lambda y, \qquad (\lambda + \mu)x = \lambda x + \mu x,$$

for every $f : X' \to X$, $x, y \in \mathrm{Hom}_X(X', T_{X/S}(M))$ and $\lambda, \mu \in O(X')$.

Remark 3.6.1. *If X is representable, in which case it satisfies (E) and $T_{X/S}$ and $L^u_{X/S}$ are representable by vector fibrations, the preceding laws are the same as those deduced from the vector fibration structures (cf. I 4.6).*[97]

Proposition 3.4 bis. *If X/S satisfies (E), then $X_{S'}/S'$ satisfies (E) and the isomorphisms of 3.4 respect the structures of $O_{X_{S'}}$-modules, resp. of $O_{S'}$-modules.*

Without comment.

[94] For examples of S-functors and S-groups not satisfying (E), see 6.2 and 6.3.

[95] One has added what follows.

[96] One has detailed what follows.

[97] I.e., for every S-morphism $f : X' \to X$, the action of $O(X')$ on $\mathrm{Hom}_X(X', T_{X/S}(M))$ corresponds, via the identification $\mathrm{Hom}_X(X', T_{X/S}(M)) \simeq \mathrm{Hom}_{O_{X'}}(f^*(\Omega^1_{X/S}), M \otimes_{O_S} O_{X'})$, to the natural action of $O(X')$ on the right-hand term; this follows from the proof of SGA 1, III 5.1 (see also the addition 0.1.8 in Exp. III).

Proposition 3.7. *The functors $T_{X/S}(M)$ and $L^u_{X/S}(M)$ are functorial in X, i.e., if $f : X \to X'$ is an S-morphism, one has commutative diagrams:*

$$\begin{tikzcd} T_{X/S}(M) \arrow[r, "T(f)"] \arrow[d] & T_{X'/S}(M) \arrow[d] \\ X \arrow[r] & X' \end{tikzcd}, \qquad \begin{tikzcd} L^u_{X/S}(M) \arrow[rr, "L(f)"] \arrow[dr] & & L^{f\circ u}_{X'/S}(M) \arrow[dl] \\ & S & \end{tikzcd}.$$

Moreover,[98] *if f is a monomorphism, so are $T(f)$ and $L(f)$.*

The existence of $T(f)$ and $L(f)$, as well as the last assertion, follow immediately from the definitions. The commutativity of the diagrams then follows from the functoriality of these morphisms with respect to M and from the fact that $T_{X/S}(0) = X$.

Remark 3.7.1.[99] Suppose X and X' representable and let r be the rank of the free $\mathcal{O}_S$-module M. Then, by 2.2.2, $T_{X/S}(M)$ is isomorphic to the product over X of r copies of $V(\Omega^1_{X/S})$, and likewise for $T_{X'/S}(M)$. Consequently, the above square is cartesian when f is an open immersion, more generally when $f^*(\Omega^1_{X'/S}) = \Omega^1_{X/S}$, for example if f is étale; under these conditions, one has an isomorphism of S-functors

$$L^u_{X/S}(M) \xrightarrow{\sim} L^{f\circ u}_{X'/S}(M).$$

In general, the cartesian square of 3.7 defines a morphism of X-functors:

$$\begin{tikzcd} T_{X/S}(M) \arrow[rr] \arrow[dr] & & T_{X'/S}(M) \times_{X'} X \arrow[dl] \\ & X & \end{tikzcd}.$$

Proposition 3.7 bis. *Let $f : X \to X'$ be an S-morphism; if X and X' satisfy (E) with respect to S, then*

$$T_{X/S}(M) \xrightarrow{T(f)} T_{X'/S}(M)_X \qquad \text{resp.} \qquad L^u_{X/S}(M) \xrightarrow{L(f)} L^{f\circ u}_{X'/S}(M)$$

is a morphism of $\mathcal{O}_X$-modules (resp. of $\mathcal{O}_S$-modules).

This follows from Proposition 3.7 by functoriality in M.

Proposition 3.8. *Let X and Y be two functors above S. One has isomorphisms functorial in M:*

$$T_{X/S}(M) \times_S T_{Y/S}(M) \xrightarrow{\sim} T_{(X\times_S Y)/S}(M), \tag{3.8.1}$$
$$L^u_{X/S}(M) \times_S L^v_{Y/S}(M) \xrightarrow{\sim} L^{(u,v)}_{(X\times_S Y)/S}(M). \tag{3.8.2}$$

[100] The first isomorphism follows from 1.2 $(*)$; the second is deduced from it by the base change $(u,v) : S \to X \times_S Y$.

[98] One has added the sentence that follows.
[99] One has added the hypothesis that X and X' be representable, and one has detailed what follows.
[100] One has detailed what follows.

Remark 3.8.0. Note that (3.8.1) can also be interpreted as an isomorphism of $X \times_S Y$-functors

$$T_{X/S}(M) \times_X (X \times_S Y) \underset{(X\times_S Y)}{\times} T_{Y/S}(M) \times_Y (X \times_S Y) \xrightarrow{\sim} T_{(X\times_S Y)/S}(M).$$

Corollary 3.8.1. *If X/S is equipped with an algebraic structure defined by finite cartesian products, then $T_{X/S}(M)$ is equipped with a structure of the same kind and the projection $T_{X/S}(M) \to X$ is a morphism of this kind of structure.*

Proposition 3.8 bis. *If X/S and Y/S satisfy (E), then $(X \times_S Y)/S$ satisfies (E) and (3.8.1) (resp. (3.8.2)) is an isomorphism of $\mathcal{O}_{X\times_S Y}$-modules (resp. $\mathcal{O}_S$-modules).*

Proof.[101] Suppose that X/S and Y/S satisfy (E). Then, by 3.5.1 and (3.8.1), so does $(X \times_S Y)/S$. Let us show that (3.8.1) is an isomorphism of $\mathcal{O}_{X\times_S Y}$-modules.

Let $(x, y) : Z \to X \times_S Y$ be an S-morphism; taking into account 3.4.2, it suffices to see that the map

$$\{ \varphi \in \mathrm{Hom}_S(I_Z(M), X) \mid \varphi \circ \varepsilon_M = x \} \times \{ \psi \in \mathrm{Hom}_S(I_Z(M), Y) \mid \psi \circ \varepsilon_M = y \}$$
$$\xrightarrow{\sim} \{ \theta \in \mathrm{Hom}_S(I_Z(M), X \times_S Y) \mid \theta \circ \varepsilon_M = (x, y) \}$$

which to (ϕ, ψ) associates $\phi \times \psi$ is a morphism of $\mathcal{O}(Z)$-modules. But this is clear, because if $a \in \mathcal{O}(Z)$ then $a \cdot (\phi, \psi) = (\phi \circ a^*, \psi \circ a^*)$ is sent to

$$(\varphi \circ a^*) \times (\psi \circ a^*) = (\varphi \times \psi) \circ a^* = a \cdot (\varphi \times \psi).$$

Likewise, using 3.2.1, one shows that (3.8.2) is an isomorphism of $\mathcal{O}_S$-modules.

3.9.0.[102] If X is an S-group and if $e : S \to X$ denotes its unit section, one writes:

$$Lie(X/S, M) = L^e_{X/S}(M),$$

i.e., $Lie(X/S, M)$ is defined by the cartesian square:

$$\begin{tikzcd}
Lie(X/S, M) \arrow[r, "i"] \arrow[d] & T_{X/S}(M) \arrow[d, "p"] \\
S \arrow[r, "e"] & X .
\end{tikzcd}$$

By Corollary 3.8.1, the projection $p : T_{X/S}(M) \to X$ is a morphism of S-groups, and it follows that $Lie(X/S, M)$ is equipped with a structure of S-group, and is isomorphic via i to the kernel of p.

If, moreover, X/S satisfies condition (E), we shall see in Proposition 3.9 that the S-group structure on $Lie(X/S, M)$, induced by that of X, coincides with the abelian group structure induced by functoriality in M (cf. 3.5.1). In fact, this result is valid

[101] One has added this proof.
[102] One has added this paragraph, in order to explain the introduction of the notion of S-H-functor.

under the weaker hypothesis that X be an S-functor of monoids or, more generally, an S-functor of H-sets (cf. the definition below).

Definitions 3.9.0.1. *a) Let us introduce the following terminology:*[103] *an H-**set** is a set X equipped with a composition law with two-sided unit, denoted e_X or simply e. If $f : X \to Y$ is a morphism of H-sets, its kernel $\mathrm{Ker}\ f$ is $f^{-1}(e_Y)$; it is a sub-H-set of X.*

*b) An H-**object** in a category C is defined in the usual manner: it is therefore an object X of C, equipped with a morphism $X \times X \to X$ such that there exists a section of X (above the final object) possessing the properties of a two-sided unit. Every C-monoid, in particular every C-group, is therefore a C-H-object. In particular, an H-object of the category of functors above the scheme S will be called an **S-H-functor**.*

c) If X is an S-H-functor (for example, an S-group), and if $e : S \to X$ denotes the unit section of X, one writes:

$$\mathrm{Lie}(X/S, M) = L^e_{X/S}(M) \quad \text{and} \quad \mathrm{Lie}(X/S) = \mathrm{Lie}(X/S, O_S).$$

Let us make explicit the following particular case of 3.8.1.[104]

Corollary 3.9.0.2. *If X is an S-H-functor (resp. an S-group), then $T_{X/S}(M)$ and $Lie(X/S, M)$ are also S-H-functors (resp. S-groups), and one has morphisms of S-H-functors (resp. of S-groups):*

$$\mathrm{Lie}(X/S, M) \xrightarrow{\ i\ } T_{X/S}(M) \underset{s}{\overset{p}{\rightleftarrows}} X,$$

where i is an isomorphism of $Lie(X/S)(M)$ onto $\mathrm{Ker}\ p$ and s is a section of p.

Proposition 3.9. *Let X be an S-H-functor satisfying (E) with respect to S. The S-H-functor structure of $Lie(X/S, M)$ coming from that of X coincides with the S-group structure defined in 3.5.1.*

It follows from what was said above that $Lie(X/S, M)$ is an H-object in the category of O_S-modules. The proposition will then follow from the following lemma:[105]

Lemma 3.10. *Let C be a category. Let G be an H-object in the category of C-H-objects; G is therefore a C-H-object (whose composition law we shall denote $f : G \times G \to G$) equipped with a morphism of C-H-objects $h : G \times G \to G$.*[106] *Then $f = h$ and f is commutative.*

By taking the values of the functors on a variable argument, one reduces, in the usual manner, to verifying the lemma when C is the category of sets. One thus has a set G and two maps $f, h : G \times G \to G$ such that

[103] This is inspired by the notion of H-space in topology.
[104] One has added Corollary 3.9.0.2, which will be useful in 4.1.
[105] Inspired by the standard proof showing that the fundamental group of an H-space is abelian.
[106] $G \times G$ is equipped with the law $(G \times G) \times (G \times G) \to G \times G$, $((x, z), (y, t)) \mapsto (f(x, y), f(z, t))$.

```
h(f(x, y), f(z, t)) = f(h(x, z), h(y, t)).
```

One has on the other hand two elements of G, namely e and u, with $f(e,x) = f(x,e) = x$, $h(u,x) = h(x,u) = x$. One sees first that

```
h(f(u, y), f(x, u)) = f(x, y) = h(f(x, u), f(u, y)).
```

In particular, for $y = e$, resp. $x = e$, one obtains, respectively,

```
x = f(x, e) = h(f(u, e), f(x, u)) = h(u, f(x, u)) = f(x, u),
y = f(e, y) = h(f(e, u), f(u, y)) = h(u, f(u, y)) = f(u, y),
```

whence, on substituting in the original equality,

```
h(y, x) = f(x, y) = h(x, y).
```

This proves the lemma, and thus Proposition 3.9. From 3.9 one then deduces the following corollaries.

Corollary 3.9.1. *If X is an S-H-functor satisfying (E) with respect to S, every element of $X(I_S(M))$ which projects onto the unit element of $X(S)$ is invertible.*

Corollary 3.9.2. *If X is an S-monoid satisfying (E) with respect to S, an element of $X(I_S(M))$ is invertible if and only if its image in $X(S)$ is.*

Corollary 3.9.3. *If X is an S-group satisfying (E) with respect to S, the two S-group laws on $Lie(X/S,M)$ coincide.*

Corollary 3.9.4.[107] *Let G be an S-group satisfying (E) with respect to S. For $n \in \mathbb{Z}$, let $n_G : G \to G$ be the morphism of S-functors defined by $g \mapsto g^n$. Then the derived morphism $L(n_G) : Lie(G/S) \to Lie(G/S)$ is "multiplication by n", i.e. the map that to every $x \in Lie(G/S)(S')$ associates $n\ x$.*

Let us first remark that n_G is not in general a morphism of groups, but it preserves the unit section $e : S \to G$, so the derived morphism $L(n_G)$ indeed sends $Lie(G/S) = L^e_{X/S}$ into itself. If one denotes by i the inclusion $Lie(G/S) \hookrightarrow T_{G/S}$, then $L(n_G)$ is defined by the equality $i(L(n_G)(x)) = i(x)^n$, for every $S' \to S$ and $x \in Lie(G/S)(S')$. Now, by 3.9, the two group laws on $Lie(G/S)$ (coming from condition (E) and coming from the law of G) coincide, i.e. one has $i(x)^n = i(nx)$, whence $L(n_G)(x) = nx$.

Before drawing other consequences of Proposition 3.9, let us prove another functoriality result:

Proposition 3.11. *In the situation of Section 1, one has an isomorphism functorial in M*

$$T_{\mathrm{Hom}_{Z/S}(X,Y)/S}(M) \xrightarrow{\sim} \mathrm{Hom}_{Z/S}(X, T_{Y/Z}(M)).$$

Indeed, by definition (cf. 3.1):

```
T_{Hom_{Z/S}(X,Y)/S}(M) = Hom_S(I_S(M), Hom_{Z/S}(X, Y)).
```

[107] One has added this corollary, which amplifies Remark 4.1.1.2 below.

By the isomorphism (4) of 1.1, applied to $T = I_S(M)$, one has:

$$\mathrm{Hom}_S(I_S(M), \mathrm{Hom}_{Z/S}(X, Y)) \simeq \mathrm{Hom}_{Z/S}(X, \mathrm{Hom}_Z(Z \times_S I_S(M), Y)).$$

Taking into account the isomorphism $Z \times_S I_S(M) \simeq I_Z(M)$, this gives

$$T_{\mathrm{Hom}_{Z/S}(X,Y)/S}(M) \simeq \mathrm{Hom}_{Z/S}(X, \mathrm{Hom}_Z(I_Z(M), Y)) = \mathrm{Hom}_{Z/S}(X, T_{Y/Z}(M)).$$

Corollary 3.11.1. *If Y/Z satisfies (E), then $\mathrm{Hom}_{Z/S}(X,Y)/S$ satisfies (E) and the isomorphism of 3.11 respects the O-module structures above $\mathrm{Hom}_{Z/S}(X,Y)$.*[108]

Proof. Let M, N be two free O_S-modules of finite type. If Y/Z satisfies (E), then

$$T_{Y/Z}(M \oplus N) \simeq T_{Y/Z}(M) \times_Y T_{Y/Z}(N).$$

The right-hand term is a subfunctor of $T_{Y/Z}(M) \times_S T_{Y/Z}(N)$ and via the isomorphism of 1.2 (∗), one obtains an isomorphism

$$\mathrm{Hom}_{Z/S}(X, T_{Y/Z}(M \oplus N)) \simeq \mathrm{Hom}_{Z/S}(X, T_{Y/Z}(M)) \times_{\mathrm{Hom}_{Z/S}(X,Y)} \mathrm{Hom}_{Z/S}(X, T_{Y/Z}(N)).$$

Combined with 3.11, this implies:

$$T_{\mathrm{Hom}_{Z/S}(X,Y)/S}(M \oplus N) \simeq T_{\mathrm{Hom}_{Z/S}(X,Y)/S}(M) \times_{\mathrm{Hom}_{Z/S}(X,Y)} T_{\mathrm{Hom}_{Z/S}(X,Y)/S}(N),$$

so $\mathrm{Hom}_{Z/S}(X,Y)$ satisfies (E) with respect to S. This proves the first assertion of the corollary.

Let us see the second. Denote by $H = \mathrm{Hom}_{Z/S}(X,Y)$ and give ourselves an S-morphism $\Delta : H' \to \mathrm{Hom}_{Z/S}(X,Y)$, i.e., a Z-morphism $\delta : H' \times_S X \to Y$, which therefore makes $H' \times_S X$ a Y-object.

On the one hand, one has a commutative diagram:

$$\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, T_{Y/Z}(M))) \longrightarrow \mathrm{Hom}_S(H', \mathrm{Hom}_{Z/S}(X, T_{Y/Z}(M)))$$

$$\mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(M)) \longrightarrow \mathrm{Hom}_Z(H' \times_S X, T_{Y/Z}(M))$$

$$\{ \psi \in \mathrm{Hom}_Z(I_{H' \times_S X}(M), Y) \mid \psi \circ \varepsilon_M = \delta \} \longrightarrow \mathrm{Hom}_Z(I_{H' \times_S X}(M), Y) .$$

By 1.3, the action of $\alpha \in O(H' \times_S X)$ on $\Psi \in \mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(M))$ is given by: for every $U \to S$ and $(h,x) \in \mathrm{Hom}_S(U, H' \times_S X)$ (U being then above Y via $\delta \circ (h,x)$),

$$(\alpha\Psi)(h, x) = \alpha(h, x)\, \Psi(h, x),$$

where $\alpha(h,x) \in O(U)$ acts on $\Psi(h,x) \in T_{Y/Z}(M)(U)$ via the structure of O_Y-module of $T_{Y/Z}(M)$. By 3.4.2, this last is given, via the identification

$$\mathrm{Hom}_Y(H' \times_S X, T_{Y/Z}(M)) = \{ \psi \in \mathrm{Hom}_Z(I_{H' \times_S X}(M), Y) \mid \psi \circ \varepsilon_M = \delta \},$$

by: for every $(m,h,x) \in \mathrm{Hom}_S(U, I_S(M) \times_S H' \times_S X)$,

[108] Setting $H = \mathrm{Hom}_{Z/S}(X,Y)$, this implies in particular that $T_{H/S}(M)$ is equipped with a natural structure of module over $\prod_{X \times_S H/H} O_H$. This result has seemed somewhat surprising to the editors; for this reason one has detailed the proof.

$$(1)\qquad (\alpha\psi)(m, h, x) = \psi(m \cdot \alpha(h, x), h, x).$$

On the other hand, consider the tangent space $T_{H/S}(M) = \mathrm{Hom}_S(I_S(M), H)$; one has a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Hom}_H(H', T_{H/S}(M)) & \longrightarrow & \mathrm{Hom}_S(H', T_{H/S}(M)) \\
\{\Phi \in \mathrm{Hom}_S(I_{H'}(M), H) \mid \Phi \circ \varepsilon_M = \Delta\} & \longrightarrow & \mathrm{Hom}_S(I_{H'}(M), H) \\
(*) & & \\
\{\varphi \in \mathrm{Hom}_Z(I_{H' \times_S X}(M), Y) \mid \varphi \circ \varepsilon_M = \delta\} & \longrightarrow & \mathrm{Hom}_Z(I_{H' \times_S X}(M), Y)\ ,
\end{array}$$

where the bijection $(*)$ is given as follows (cf. 1.1 (2) and I 1.7.1): for every $U \to S$ and $(m, h, x) \in \mathrm{Hom}_S(U, I_S(M) \times_S H' \times_S X)$ (so that U is above Z via $U \xrightarrow{x} X \to Z$), one has $\Phi(m, h) \in \mathrm{Hom}_Z(X \times_S U, Y)$ and

$$(\dagger)\qquad \varphi(m, h, x) = \Phi(m, h) \circ (x \times \mathrm{id}_U) \in \mathrm{Hom}_Z(U, Y).$$

By 3.4.2 (where one replaces X by $\mathrm{Hom}_{Z/S}(X, Y)$ and X' by H'), the action of $a \in O(H')$ on $\Phi \in \mathrm{Hom}_S(I_{H'}(M), H)$ is given by: for every $U \to S$ and $(m, h) \in \mathrm{Hom}_S(U, I_S(M) \times_S H')$,

$$(a\Phi)(m, h) = \Phi(m \cdot a(h), h).$$

Consequently, if ϕ (resp. $a\phi$) is the element of $\mathrm{Hom}_Z(I_{H' \times_S X}(M), Y)$ associated with Φ (resp. $a\Phi$), one has, by $(\dagger)$,

$$(2)\qquad (a\varphi)(m, h, x) = \Phi(m \cdot a(h), h) \circ (x \times \mathrm{id}_U) = \varphi(m \cdot a(h), h, x).$$

Combined with (1), this shows that the isomorphism $T_{H/S}(M) \xrightarrow{\sim} \mathrm{Hom}_{Z/S}(X, T_{Y/Z}(M))$ of 3.11.1 is an isomorphism of $O(H)$-modules; moreover, for every $H' \to H$, the structure of $O(H')$-module on $\mathrm{Hom}_H(H', T_{H/S}(M))$ extends, in a manner functorial in H', to a structure of $O(H' \times_S X)$-module.

In particular, for $Z = S$, one obtains the following corollary.

Corollary 3.11.2. *One has an isomorphism functorial in M*

$$T_{\mathrm{Hom}_S(X,Y)/S}(M) \xrightarrow{\sim} \mathrm{Hom}_S(X, T_{Y/S}(M)).$$

Moreover, if Y/S satisfies (E), then $\mathrm{Hom}_S(X, Y)/S$ satisfies (E) and the preceding isomorphism respects the O-module structures above $\mathrm{Hom}_S(X, Y)$.

[109] Let $u : X \to Y$ be an S-morphism; one identifies it with the constant morphism $u : S \to \mathrm{Hom}_S(X, Y)$ such that $u(f) = u$ for every $f : S' \to S$. One sees at once that the fiber product of u and of $\mathrm{Hom}_S(X, T_{Y/S}(M)) \to \mathrm{Hom}_S(X, Y)$ is identified with $\mathrm{Hom}_{Y/S}(X, T_{Y/S}(M))$, where X is above Y via u. Consequently, from the definition of $L^u_{\mathrm{Hom}_S(X,Y)/S}(M)$ and the preceding corollary one deduces:

Corollary 3.11.3. *Let $u : X \to Y$ be an S-morphism. One has an isomorphism functorial in M (where in the right-hand term X is above Y via u):*

[109] One has added the sentences that follow.

$$L^u_{\mathrm{Hom}_S(X,Y)/S}(M) \xrightarrow{\sim} \mathrm{Hom}_{Y/S}(X, T_{Y/S}(M)).$$

[110] *It is an isomorphism of 0_S-modules if Y/S satisfies (E).*

In particular, for $Y = X$, $\mathrm{End}_S(X)$ is an S-functor of monoids, hence a fortiori an S-H-functor; recalling that $Lie(\mathrm{End}_S(X)/S, M)$ denotes $L^e_{\mathrm{End}_S(X)/S}(M)$, where e is the unit section (cf. 3.9.0.1), one obtains:

Corollary 3.11.4. *One has an isomorphism functorial in M*

```
Lie(End_S(X)/S, M) □ ∏_{X/S} T_{X/S}(M);
```

[111] *it is an isomorphism of 0_S-modules if X/S satisfies (E).*

Remark 3.11.5.[112] Suppose that X/S satisfies (E). Then $\prod_{X/S} T_{X/S}(M) = \mathrm{Hom}_{X/S}(X, T_{X/S}(M))$ is equipped with a structure of $(\prod_{X/S} O_X)$-module, i.e. for every $S' \to S$,

```
Hom_{X/S}(X, T_{X/S}(M))(S′) = { ψ ∈ Hom_X(I_{S′}(M) ×_S X, X) | ψ ∘ (ε_M × id_X) = pr_X }
```

is equipped with a structure of $O(X \times_S S')$-module, functorial in S'. This follows, at one's choice, from 3.6 and the properties of the functor $\prod_{X/S}$ (cf. 1.2), or from the proof of 3.11.1.

We shall now interpret geometrically the definition of the tangent bundle.

[113] Let U be an S-functor; by I 1.7.2, one has isomorphisms functorial in M

```
T_{X/S}(M)(U) = Hom_S(U, Hom_S(I_S(M), X)) ≃ Hom_S(I_S(M), Hom_S(U, X))
              ≃ Hom_{I_S(M)}(U_{I_S(M)}, X_{I_S(M)}).
```

In particular, the morphism $M \to 0$ gives a commutative diagram:

```
Hom_S(U, T_{X/S}(M))  □  Hom_{I_S(M)}(U_{I_S(M)}, X_{I_S(M)})
          |                                 |
          | π_M                             | identity
          ↓                                 ↓
Hom_S(U, X)  ──────────────────────→  Hom_S(U, X) ,
```

where the second vertical arrow is obtained by the base change $\epsilon_M : S \to I_S(M)$.[114]

In consequence:

Proposition 3.12. *Let $h_0 : U \to X$ be an S-morphism. Then $\mathrm{Hom}_X(U, T_{X/S}(M))$ is identified with the set of $I_S(M)$-morphisms from $U_{I_S(M)}$ to $X_{I_S(M)}$ which restrict to h_0 on U (viewed as a subobject of $U \times_S I_S(M)$ via $id_U \times_S \epsilon_M$).*

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[110] One has added the sentence that follows.
[111] One has added the sentence that follows.
[112] One has added this remark.
[113] One has simplified the original in what follows.
[114] Via the isomorphism $\mathrm{Hom}_{I_S(M)}(U_{I_S(M)}, X_{I_S(M)}) \simeq \mathrm{Hom}_S(U \times_S I_S(M), X)$, this is equivalent to Remark 3.1.1. Likewise, 3.12 is equivalent to the first part of Remark 3.4.2.

[115] In particular, for $U = X$ and $h_0 = id_X$, one obtains the

Corollary 3.12.1. *The set $\Gamma(T_{X/S}(M)/X)$ is identified with the set of $I_S(M)$-endomorphisms ϕ of $X_{I_S(M)}$ which induce the identity on X, i.e. such that the following diagram be commutative:*[116]

$$\begin{tikzcd} I_X(M) \arrow[r, "\varphi"] & I_X(M) \\ X \arrow[u, "\varepsilon_M"] \arrow[r, "id"] & X \arrow[u, "\varepsilon_M"'] \end{tikzcd} .$$

[117] On the other hand, by 3.11.2, $\Gamma(T_{X/S}(M)/X) \simeq Lie(\mathrm{End}_S(X)/S, M)(S)$; if moreover X/S satisfies (E), then $\mathrm{End}_S(X)/S$ satisfies (E), so $Lie(\mathrm{End}_S(X)/S, M)$ is an O_S-module by 3.6 (and in fact a $(\prod_{X/S} O_X)$-module, by 3.11.5). Applying 3.9, one deduces the

Proposition 3.13. *If X/S satisfies (E), the abelian group $\Gamma(T_{X/S}(M)/X)$ is identified with the set of $I_S(M)$-endomorphisms of $X_{I_S(M)}$ inducing the identity on X. Consequently, every $I_S(M)$-endomorphism of $X_{I_S(M)}$ which induces the identity on X is an automorphism.*

Corollary 3.13.1. *Let $u : X \to Y$ be an S-isomorphism, Y/S satisfying (E). Every $I_S(M)$-morphism from $X_{I_S(M)}$ into $Y_{I_S(M)}$ which extends u is an isomorphism.*

Corollary 3.13.2. *If Y/S satisfies (E), then the monomorphism $Isom_S(X,Y) \to \mathrm{Hom}_S(X,Y)$ induces, for every $u \in Isom_S(X,Y)$, an isomorphism*

$$L^u_{Isom_S(X,Y)/S}(M) \xrightarrow{\sim} L^u_{\mathrm{Hom}_S(X,Y)/S}(M).$$

Proof.[118] One must see that $L^u_{Isom_S(X,Y)/S}(M)(S') \to L^u_{\mathrm{Hom}_S(X,Y)/S}(M)(S')$ is a bijection, for every $S' \to S$. By base change (cf. 3.4), it suffices to do it for $S' = S$. In this case, $L^u_{\mathrm{Hom}_S(X,Y)/S}(M)(S)$ (resp. $L^u_{Isom_S(X,Y)/S}(M)(S)$) is the set of $I_S(M)$-morphisms (resp. automorphisms) $X_{I_S(M)} \to Y_{I_S(M)}$ which extend u, and one applies the preceding corollary.

Corollary 3.13.3. *If X/S satisfies (E), the monomorphism $\mathrm{Aut}_S(X) \to \mathrm{End}_S(X)$ induces, for every $u \in \mathrm{Aut}_S(X)$, an isomorphism $L^u_{\mathrm{Aut}_S(X)/S}(M) \xrightarrow{\sim} L^u_{\mathrm{End}_S(X)/S}(M)$. In particular, one has*

$$Lie(\mathrm{Aut}_S(X)/S, M) \simeq Lie(\mathrm{End}_S(X)/S, M) \simeq \prod_{X/S} T_{X/S}(M),$$

[119] *so that $Lie(\mathrm{Aut}_S(X)/S, M)$ is equipped with a structure of $(\prod_{X/S} O_X)$-module.*

---

[115] One has detailed the original in what follows.
[116] When $M = O_X$, these are called "infinitesimal endomorphisms" of X; see 6.2 at the end of this Exposé.
[117] One has detailed the original in what follows.
[118] One has added the proof that follows.
[119] One has added what follows.

3.14. Suppose to conclude that X is representable.[120] In this case, one saw in 2.2.2 that the X-functor $T_{X/S}$ is representable by $V(\Omega^1_{X/S})$, whence bijections:

$$\Gamma(T_{X/S}/X) \simeq \mathrm{Hom}_X(\Omega^1_{X/S}, O_X) \simeq \mathrm{D\acute{e}r}_{O_S}(O_X).$$

This is also deduced from what precedes, as follows. By 3.13, $\Gamma(T_{X/S}/X)$ is identified with the set of infinitesimal endomorphisms of X (i.e. of I_S-endomorphisms of X_{I_S} inducing the identity on X). Now X and X_{I_S} have the same underlying topological space, the corresponding rings of sheaves being O_X and $D_{O_X} = O_X \oplus M$, where $M = O_X$ is considered as ideal of square zero. Denoting by $\pi : D_{O_X} \to O_X$ the morphism of O_X-algebras that vanishes on M, one deduces that to give an infinitesimal endomorphism of X is equivalent to giving a morphism of O_S-algebras $\phi : O_X \to D_{O_X}$ such that $\pi \circ \phi = id_{O_X}$, which is equivalent to giving an O_S-derivation of the sheaf of rings O_X.

Moreover, one sees easily that if $D, D' \in D\acute{e}r_{O_S}(O_X)$ and if one denotes by ϕ_D the infinitesimal endomorphism corresponding to D, then

$$\phi_{D+D'} = \phi_D \circ \phi_{D'}.$$

This shows that the identification

$$\{ \text{infinitesimal endomorphisms of } X \} \simeq \mathrm{D\acute{e}r}_{O_S}(O_X)$$

is an isomorphism of abelian groups. Taking into account 3.13 (and 3.11.5), one has therefore constructed an isomorphism of abelian groups (and even of $O(X)$-modules)

$$\Gamma(T_{X/S}/X) \xrightarrow{\sim} D\acute{e}r_{O_S}(O_X),$$

which recovers the classical interpretation of tangent vector fields in terms of derivations of the structure sheaf.[121] Let us remark in passing that $\Gamma(T_{X/S}/X)$ is equal to $H^0(X, g_{X/S})$, where $g_{X/S}$ is the dual of $\Omega^1_{X/S}$.

4. Tangent space to a group — Lie algebras

4.1. Let G be a group-functor above S. By 3.9.0.2, $T_{G/S}(M)$ and $Lie(G/S, M)$ are equipped with structures of groups above S and one has morphisms of groups

$$\begin{tikzcd} \mathrm{Lie}(G/S, M) \arrow[r, "i"] & T_{G/S}(M) \arrow[r, shift left, "p"] & G \arrow[l, shift left, "s"] \end{tikzcd};$$

[120]One has added the reminder of 2.2.2, and detailed the sequel.

[121]Moreover, this isomorphism is functorial in S: if $S' \to S$, setting $X' = X_{S'}$, one has $Lie(\mathrm{Aut}_S(X)/S)(S') \simeq \Gamma(T_{X'/S'}/X') \simeq D\acute{e}r_{O_{S'}}(O_{X'})$.

by definition i is an isomorphism of $Lie(G/S)(M)$ onto the kernel of p and s is a section of p. It then follows from I 2.3.7 that this sequence of morphisms allows one to identify $T_{G/S}(M)$ with the semi-direct product of G by $Lie(G/S, M)$.

Definition 4.1.A.[122] *The corresponding action of G on $Lie(G/S, M)$ is denoted*

$$\mathrm{Ad} : G \to \mathrm{Aut}_{S\text{-gr.}}(Lie(G/S, M))$$

and called the **adjoint representation** *(relative to M) of G; one has therefore by definition, for $x \in G(S')$ and $X \in Lie(G/S, M)(S')$:*

$$\mathrm{Ad}(x)\, X = i^{-1}(s(x)\, i(X)\, s(x)^{-1}).$$

If G is commutative, then $T_{G/S}(M)$ is also commutative and $Ad(x)X = X$.

Definition 4.1.B.[123] *If G and H are two group-functors above S and if $f : G \to H$ is a morphism of groups, by functoriality there is deduced a morphism of exact sequences compatible with the canonical sections:*

$$\begin{tikzcd}
1 \arrow[r] & Lie(G/S, M) \arrow[r] \arrow[d, "L(f)"] & T_{G/S}(M) \arrow[r] \arrow[d, "T(f)"] & G \arrow[r] \arrow[d, "f"] & 1 \\
1 \arrow[r] & Lie(H/S, M) \arrow[r] & T_{H/S}(M) \arrow[r] & H \arrow[r] & 1 ;
\end{tikzcd}$$

$L(f)$, which we shall also write $Lie(f)$, is the **derived morphism** *of f.*[124]

Remark 4.1.C.[125] *If G/S and H/S satisfy (E), then $L(f)$ respects the O_S-module structures deduced from "functoriality in M" (cf. 3.6).*

Proposition 4.1.1. *Let $g \in G(S)$. Then $Ad(g) : Lie(G/S, M) \to Lie(G/S, M)$ is the derived morphism of $Int(g) : G \to G$.*

Indeed $Ad(g)X = i^{-1}(Int(g)i(X))$, which is nothing other than $L(Int(g))(X)$ by the very definition of the derived morphism.

Suppose that G/S satisfies (E). Then, by Proposition 3.9, the group structure on $Lie(G/S, M)$ defined as above is nothing other than the structure induced by its O_S-module structure (defined thanks to (E)). One then deduces from the preceding proposition and the functoriality of the action of O_S (cf. 3.6) the corollary:

Corollary 4.1.1.1. *Suppose that G/S satisfies (E). Then Ad sends G into the subgroup*

$$\mathrm{Aut}_{O_S\text{-mod.}}(Lie(G/S, M))$$

of $\mathrm{Aut}_{S-gr.}(Lie(G/S, M))$*, i.e., for every $g \in G(S')$, $Ad(g)$ respects the $O_{S'}$-module structure of $Lie(G'/S', M)$ (where $G' = G \times_S S'$). In other words, Ad is a* **linear representation** *(cf. I, 3.2) of G in the O_S-module $Lie(G/S, M)$.*

[122] One has added the numbering 4.1.A and 4.1.B to highlight these definitions.
[123] One has added the numbering 4.1.A and 4.1.B to highlight these definitions.
[124] If f is a monomorphism, so are $L(f)$ and $T(f)$, cf. 3.7.
[125] One has added this remark.

Remarks 4.1.1.2. *a) For G/S to satisfy (E), it is necessary and sufficient that for every pair (M,N) of free $\mathcal{O}_S$-modules of finite type, the diagram*

$$\begin{tikzcd}
 & Lie(G/S, M \oplus N) \arrow[dl] \arrow[dr] & \\
Lie(G/S, M) \arrow[dr] & & Lie(G/S, N) \arrow[dl] \\
 & Lie(G/S, 0) = S ,&
\end{tikzcd}$$

obtained by applying the functor $Lie(G/S, -)$ to the diagram $(*)$ of 2.1, be cartesian.

b) Suppose that G/S satisfies (E). Then the derived morphism of the group law $\pi : G \times_S G \to G$ is nothing other than the addition law in $Lie(G/S, M)$. (N.B. π is not a morphism of groups, but $\pi(e,e) = e$, so the derived morphism $L(\pi)$ sends $T^{(e,e)}_{(G\times_S G)/S}(M) = Lie(G/S,M) \times_S Lie(G/S,M)$ into $Lie(G/S,M)$, cf. 3.7 and 3.8.) For every $n \in \mathbb{Z}$, one shows likewise that if one denotes by $n_G : G \to G$ the morphism of S-functors defined by $g \mapsto g^n$, then the derived morphism $L(n_G)$ is multiplication by n on $Lie(G/S)$, cf. 3.9.4.

4.1.2.0.[126] Consider now the S-functor $\mathrm{Hom}_{G/S}(G, T_{G/S}(M))$; for every $S' \to S$, one has $T_{G/S}(M)_{S'} \simeq T_{G_{S'}/S'}(M)$ (cf. 3.4) and therefore:

$$\mathrm{Hom}_{G/S}(G, T_{G/S}(M))(S') \simeq \mathrm{Hom}_{G_{S'}}(G_{S'}, T_{G_{S'}/S'}(M)) = \Gamma(T_{G_{S'}/S'}(M)/G_{S'}).$$

Note first that one has an isomorphism, functorial in S',

$$(*)\quad \mathrm{Hom}_{S'}(G_{S'}, Lie(G_{S'}/S', M)) \xrightarrow{\sim} \Gamma(T_{G_{S'}/S'}(M)/G_{S'})$$

which to every $f : G_{S'} \to Lie(G_{S'}/S', M)$ associates the section $s_f : G_{S'} \to T_{G_{S'}/S'}(M)$ such that, for every $S'' \to S'$ and $g \in G(S'')$:

$$s_f(g) = i(f(g))s(g).$$

Let h be an automorphism of the functor $G_{S'}$ above S', not necessarily respecting the group structure. To every section τ of $T_{G_{S'}/S'}(M)$, one can associate $h(\tau)$ defined by transport of structure: it is, for example, the only section of $T_{G_{S'}/S'}(M)$ making commutative the diagram

$$\begin{tikzcd}
G_{S'} \arrow[r, "\tau"] \arrow[d, "h"'] & T_{G_{S'}/S'}(M) \arrow[d, "T(h)"] \\
G_{S'} \arrow[r, "h(\tau)"'] & T_{G_{S'}/S'}(M) .
\end{tikzcd}$$

In particular, take for h the right translation t_x by an element x of $G(S')$, i.e., $h(g) = t_x(g) = g \cdot x$, for every $g \in G(S'')$, $S'' \to S'$. Then one has immediately

[126] One has detailed the original in what follows, in order to show the action of G on the functors $\mathrm{Hom}_{G/S}(G, T_{G/S}(M))$ and $\mathrm{Hom}_S(G, Lie(G/S, M))$.

$$t_x(s_f) = s_{t_x(f)}$$

where $t_x(f)$ denotes the morphism from $G_{S'}$ into $Lie(G_{S'}/S', M)$ defined by

$$t_x(f)(g) = f(g \cdot x^{-1}),$$

for every $g \in G(S'')$, $S'' \to S'$.

It follows that if one makes G act by right translations on

$\mathrm{Hom}_{G/S}(G, T_{G/S}(M))$ and $\mathrm{Hom}_S(G, Lie(G/S, M))$

in the following way: for every $S' \to S$, $x \in G(S')$, $\sigma \in \Gamma(T_{G_{S'}/S'}(M)/G_{S'})$ and $f \in \mathrm{Hom}_{S'}(G_{S'}, Lie(G_{S'}/S', M))$,

$(\sigma \cdot x)(g) = \sigma(g \cdot x^{-1}) \cdot s(x)$ and $(f \cdot x)(g) = f(g \cdot x^{-1})$,

for every $g \in G(S'')$, $S'' \to S'$, then the isomorphism $(*)$ above respects the actions of G.

In particular, by this isomorphism, the elements of $\mathrm{Hom}_{G/S}(G, T_{G/S}(M))^G(S')$ correspond to the constant morphisms from $G_{S'}$ into $Lie(G_{S'}/S', M)$ (i.e. factoring through the projection $G_{S'} \to S'$) or also to the elements of $Lie(G_{S'}/S', M)(S') = Lie(G/S, M)(S')$.

Terminology. The elements of $\mathrm{Hom}_{G/S}(G, T_{G/S}(M))^G(S')$ will be called "sections of $T_{G_{S'}/S'}(M)$ invariant under right translation".

One then obtains the:

Proposition 4.1.2.[127] *The map $Lie(G/S, M)(S) \to \Gamma(T_{G/S}(M)/G)$ which to $X \in Lie(G/S, M)(S)$ associates the section $x \mapsto X \cdot x$ is a bijection of $Lie(G/S, M)(S)$ onto the part of $\Gamma(T_{G/S}(M)/G)$ formed by the sections invariant under right translation.*

Likewise, one makes G act on the right on $\mathrm{End}_{I_S(M)/S}(G_{I_S(M)})$ as follows: for every $S' \to S$, $x \in G(S')$ and $u \in \mathrm{End}_{I_S(M)/S}(G_{I_S(M)})(S') = \mathrm{End}_{I_{S'}(M)}(G_{I_{S'}(M)})$,

$(u \cdot x)(g) = u(g \cdot x^{-1}) \cdot x$,

for every $g \in G(S'')$, $S'' \to I_{S'}(M)$. Then the morphism of 3.12.1

$$\mathrm{Hom}_{G/S}(G, T_{G/S}(M)) \longrightarrow \mathrm{End}_{I_S(M)/S}(G_{I_S(M)})$$

respects the actions of G and therefore induces, for every $S' \to S$, a bijection between $\Gamma(T_{G_{S'}/S'}(M)/G_{S'})$ and the set of $I_{S'}(M)$-endomorphisms u of $G_{I_{S'}(M)}$ which induce the identity on G and "commute with right translations", i.e. which verify $u_{S''} \cdot x = u_{S''}$ for every $S'' \to S'$ and $x \in G(S'')$. One obtains therefore:

[127] The statements 4.1.2, 4.1.3 and 4.1.4 are obtained more simply by remarking that the automorphisms of G invariant by right translations are the left translations.[^N.D.E-II-53]

Proposition 4.1.3.[128] *There exists a bijection functorial in G between the set $Lie(G/S, M)(S)$ and the set of $I_S(M)$-endomorphisms of $G_{I_S(M)}$ inducing the identity on G and commuting with the right translations of G (in the sense indicated above).*

Taking now into account 3.13:

Theorem 4.1.4.[129] *Let G be an S-functor of groups; suppose that G/S satisfies (E). Then the group $Lie(G/S, M)(S)$ is identified, functorially in G, with the group of $I_S(M)$-automorphisms of $G_{I_S(M)}$ inducing the identity on G and commuting with the right translations of G (in the sense indicated above).*

One thus recovers (in the case $M = O_S$) one of the classical definitions of the Lie algebra of a group.

4.2.0.[130] Before going further, let us establish new corollaries of 3.11. Let X, Y be above Z, with Z above S, as in Section 1. As one saw in 3.11, the isomorphisms 1.1 (4):

$$(1)\quad \mathrm{Hom}_S(I_S(M), \mathrm{Hom}_{Z/S}(X, Y)) \xrightarrow{\sim} \mathrm{Hom}_{Z/S}(X, \mathrm{Hom}_Z(I_Z(M), Y)) \simeq \mathrm{Hom}_{Z/S}(X \times_S I_S(M), Y)$$

induce the isomorphism θ below:

$$(2)\quad T_{\mathrm{Hom}_{Z/S}(X,Y)}(M) \xrightarrow{\sim} \mathrm{Hom}_{Z/S}(X, T_{Y/Z}(M)) \simeq \mathrm{Hom}_{Z/S}(X \times_S I_S(M), Y)\ .$$

By 1.3, if Y is a Z-group, so is $\mathrm{Hom}_Z(V, Y)$ for every $V \to Z$ (in particular for $V = I_Z(M)$); explicitly, if $Z'' \to Z' \to Z$ and $\phi, \psi \in \mathrm{Hom}_Z(V_{Z'}, Y)$, then $\phi \cdot \psi$ is defined by $(\phi \cdot \psi)(v) = \phi(v)\psi(v)$, for every $v \in V_{Z'}(Z'')$.

Suppose now that X and Y are Z-groups and let us pose the following definition.

Definition 4.2.0.1. *Let $\mathrm{Hom}_{(Z/S)-gr.}(X, Y)$ be the subfunctor of $\mathrm{Hom}_{Z/S}(X, Y)$ defined by: for every $S' \to S$,*

$$(3)\quad \mathrm{Hom}_{(Z/S)\text{-gr.}}(X, Y)(S') = \mathrm{Hom}_{Z_{S'}\text{-gr.}}(X_{S'}, Y_{S'}).$$

This definition applies equally when one replaces Y by the Z-group $T_{Y/Z}(M)$.

One then sees easily that $T_{\mathrm{Hom}_{(Z/S)-gr.}(X,Y)/S}(M)(S')$ corresponds, in the preceding isomorphisms (2), to the $Z_{S'}$-morphisms

$$\varphi : X_{S'} \times_{S'} I_{S'}(M) \to Y_{S'}$$

which are "multiplicative in X", i.e. which verify $\phi(x_1x_2, m) = \phi(x_1, m)\phi(x_2, m)$, and these correspond to the morphisms of $Z_{S'}$-groups $X_{S'} \to T_{Y/Z}(M)_{S'}$. One has thus obtained:

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[128] The statements 4.1.2, 4.1.3 and 4.1.4 are obtained more simply by remarking that the automorphisms of G invariant by right translations are the left translations.[^N.D.E-II-53]

[129] The statements 4.1.2, 4.1.3 and 4.1.4 are obtained more simply by remarking that the automorphisms of G invariant by right translations are the left translations.[^N.D.E-II-53]

[130] One has added paragraph 4.2.0, whose results are used implicitly in 4.2 and 4.7 of the original.

Proposition 4.2.0.2. *Let X, Y be Z-groups, with Z above S. One has isomorphisms of S-functors, functorial in M:*

$$T_{\mathrm{Hom}_{(Z/S)-gr.}(X,Y)}(M) \xrightarrow{\sim} \mathrm{Hom}_{(Z/S)-gr.}(X, T_{Y/Z}(M)).$$

In particular, for $Z = S$, one obtains the following corollary. Before stating it, let us remark that if Y is a commutative S-group, so is $T_{Y/S}(M)$, then $H = \mathrm{Hom}_{S-gr.}(X,Y)$ and $\mathrm{Hom}_{S-gr.}(X, T_{Y/S}(M))$, and finally $T_{H/S}(M)$.

Corollary 4.2.0.3. *Let X, Y be S-groups. One has isomorphisms of S-functors, functorial in M:*

$$T_{\mathrm{Hom}_{S-gr.}(X,Y)/S}(M) \xrightarrow{\sim} \mathrm{Hom}_{S-gr.}(X, T_{Y/S}(M)).$$

If Y is commutative, these are moreover isomorphisms of abelian S-groups.

Definition 4.2.0.4.[131] *If Y is an 0_S-module, the functor $T_{Y/S}(M)$ (resp. $Lie(Y/S, M)$) is equipped with an 0_S-module structure deduced from that of Y. Equipped with this structure, it will be denoted $T'_{Y/S}(M)$ (resp. $Lie'(Y/S, M)$).*

Consequently, if X, Y are 0_S-modules, then $T'_{Y/S}(M) = \mathrm{Hom}_S(I_S(M), Y)$ and $H = \mathrm{Hom}_{O_S-mod.}(X,Y)$, then $\mathrm{Hom}_{O_S-mod.}(X, T'_{Y/S}(M))$ and $T'_{H/S}(M)$, are equipped with an 0_S-module structure, and one obtains the:

Corollary 4.2.0.5. *If X, Y are 0_S-modules, one has isomorphisms of 0_S-modules, functorial in M:*

$$T'_{\mathrm{Hom}_{O_S\text{-mod.}}(X,Y)/S}(M) \xrightarrow{\sim} \mathrm{Hom}_{O_S\text{-mod.}}(X, T'_{Y/S}(M)).$$

Definition 4.2.A.[132] *Let X, L be S-groups, with X acting on L by group automorphisms (cf. I 2.3.5). One defines the subfunctor $Z^1_S(X, L)$ of $\mathrm{Hom}_S(X, L)$ as follows: for every $S' \to S$,*

$$Z^1_S(X, L)(S') = \{ \varphi \in \mathrm{Hom}_{S'}(X_{S'}, L_{S'}) \mid \varphi(x_1 x_2) = \varphi(x_1)\,(x_1 \cdot \varphi(x_2)), \text{ for all } x_1, x_2 \in X(S''),\ S'' \to S' \}.$$

It is called the "functor of **crossed homomorphisms** *from X to L".*

Remark 4.2.B.[133] *a) If L' is a second S-group on which X acts by group automorphisms, one has*

$$Z^1_S(X, L \times_S L') \simeq Z^1_S(X, L) \times_S Z^1_S(X, L').$$

b) If L is a G-0_S-module, $Z^1_S(X, L)$ coincides with the kernel of the differential $\partial : \mathrm{Hom}_S(X, L) \to \mathrm{Hom}_S(X^2, L)$ defined in I 5.1; in particular, $Z^1_S(X, L)$ is in this case an 0_S-module.

[131] One has inserted here this definition, which in the original appeared in 4.3.

[132] One has detailed the original in what follows; in particular, one has added Definition 4.2.A and Remark 4.2.B.

[133] One has detailed the original in what follows; in particular, one has added Definition 4.2.A and Remark 4.2.B.

4.2. Let $u : X \to Y$ be a morphism of S-groups; then $L^u_{\mathrm{Hom}_{S-gr.}(X,Y)/S}(M)$ is described as follows. First, one saw in 3.11.3 that one has an isomorphism of S-functors, functorial in M:

$$(\dagger) \quad L^u_{\mathrm{Hom}_S(X,Y)/S}(M) \cong \mathrm{Hom}_{Y/S}(X, T_{Y/S}(M)).$$

On the other hand, since Y is an S-group, one has

$$T_{Y/S}(M) = \mathrm{Lie}(Y/S, M) \cdot Y = \mathrm{Lie}(Y/S, M)_Y;$$

there follows an isomorphism of S-functors, functorial in M:

$$L^u_{\mathrm{Hom}_S(X,Y)/S}(M) \cong \mathrm{Hom}_S(X, \mathrm{Lie}(Y/S, M)).$$

For every $S' \to S$, let $u' : X' \to Y'$ denote the morphism deduced from u by base change. Consider the S-functor defined as follows:[134] for every $S' \to S$,

$$\begin{aligned}\mathrm{Hom}_{(Y/S)\text{-}gr.}(X, \mathrm{Lie}(Y/S, M) \cdot Y)(S') &= \mathrm{Hom}_{Y'\text{-}gr.}(X', (\mathrm{Lie}(Y/S, M) \cdot Y)_{S'}) \\ &= \mathrm{Hom}_{Y'\text{-}gr.}(X', \mathrm{Lie}(Y'/S', M) \cdot Y').\end{aligned}$$

Then one sees easily that the isomorphism $(\dagger)$ induces an isomorphism

$$(\dagger') \quad L^u_{\mathrm{Hom}_{S\text{-}gr.}(X,Y)/S}(M) \cong \mathrm{Hom}_{(Y/S)\text{-}gr.}(X, \mathrm{Lie}(Y/S, M) \cdot Y).$$

On the other hand, the morphism

$$X \xrightarrow{\ u\ } Y \xrightarrow{\ \mathrm{Ad}\ } \mathrm{Aut}_{S\text{-}gr.}(\mathrm{Lie}(Y/S, M))$$

defines an action of X on $L = Lie(Y/S, M)$ by group automorphisms.

If $\Phi \in \mathrm{Hom}_{Y/S}(X, Lie(Y/S, M) \cdot Y)$, then, for every $S'' \to S' \to S$ and $x \in X(S'')$, one can write in a unique manner

$$\Phi(S')(x) = \varphi(S')(x) \cdot u'(x), \qquad \text{where } \varphi(S')(x) \in \mathrm{Lie}(Y'/S', M)(S'');$$

this determines an element ϕ of $\mathrm{Hom}_S(X, Lie(Y/S, M))$. Then $\Phi(S')$ is a morphism of groups if, and only if, for all $x_1, x_2 \in X(S'')$ one has:

$$\begin{aligned}\varphi(S')(x_1 x_2) &= \varphi(S')(x_1)\,\big(u(x_1)\,\varphi(S')(x_2)\,u(x_1)^{-1}\big) \\ &= \varphi(S')(x_1)\,(x_1 \cdot \varphi(S')(x_2)),\end{aligned}$$

i.e., if and only if $\phi \in Z^1_S(X, Lie(Y/S, M))$. One has thus obtained the:

Proposition 4.2. *Let $u : X \to Y$ be a morphism of S-groups. One has an isomorphism of S-functors, functorial in M:*

$$L^u_{\mathrm{Hom}_{S\text{-}gr.}(X,Y)/S}(M) \cong Z^1_S(X, \mathrm{Lie}(Y/S, M)).$$

[135] Suppose further that Y/S satisfies (E). Then it follows from 4.2.0.3, exactly as in the proof of 3.11.1, that $\mathrm{Hom}_{S-gr.}(X,Y)/S$ satisfies (E). So one has (cf. 3.5.1):

$$L^u_{\mathrm{Hom}_{S\text{-}gr.}(X,Y)/S}(M \oplus N) \simeq L^u_{\mathrm{Hom}_{S\text{-}gr.}(X,Y)/S}(M) \times_S L^u_{\mathrm{Hom}_{S\text{-}gr.}(X,Y)/S}(N).$$

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[134] This is Definition 4.2.0.1 applied to $Z = Y$ and to the Y-groups $u : X \to Y$ and $T_{Y/S}(M)$.

[135] One has added the sentences that follow and Proposition 4.2 bis, implicit in the original.

(This also follows from 4.2 and 4.2.B a).) Consequently, $L^u_{\mathrm{Hom}_{S-gr.}(X,Y)/S}(M)$ is equipped, like $Z^1_S(X, Lie(Y/S, M))$ (cf. 4.2.B b)), with an 0_S-module structure, deduced from functoriality in M. One deduces that the isomorphism of 4.2 is, in this case, an isomorphism of 0_S-modules:

Proposition 4.2 bis.[136] *Let $u : X \to Y$ be a morphism of S-groups; suppose that Y/S satisfies (E). One has an isomorphism of 0_S-modules, functorial in M:*

$$L^u_{\mathrm{Hom}_{S\text{-}gr.}(X,Y)/S}(M) \xrightarrow{\sim} Z^1_S(X, Lie(Y/S, M)).$$

Moreover, when Y/S satisfies (E), one deduces from 3.13.1, exactly as in the proof of 3.13.2, that for every $u \in Isom_{S-gr.}(X,Y)$ one has an isomorphism functorial in M:

$$(*) L^u_{Isom_{S-gr.}(X,Y)/S}(M) \xrightarrow{\sim} L^u_{\mathrm{Hom}_{S-gr.}(X,Y)/S}(M).$$

One deduces from this the following two corollaries.

Corollary 4.2.1. *Let $u : X \to Y$ be a morphism of S-groups; if Y/S satisfies (E), one has an isomorphism of 0_S-modules, functorial in M:*

$$L^u_{Isom_{S\text{-}gr.}(X,Y)/S}(M) \xrightarrow{\sim} Z^1_S(X, Lie(Y/S, M)).$$

Corollary 4.2.2. *Let X be an S-group; if X/S satisfies (E), one has an isomorphism of 0_S-modules, functorial in M:*

$$Lie(Aut_{S\text{-}gr.}(X)/S, M) \xrightarrow{\sim} Z^1_S(X, Lie(X/S, M)).$$

[137] Moreover, if Y is commutative, the adjoint action of Y on $L = Lie(X/S, M)$ is trivial, whence $Z^1_S(X, L) = \mathrm{Hom}_{S-gr.}(X, L)$. Therefore:

Corollary 4.2.3. *Let Y be a commutative S-group; one has an isomorphism of S-functors, functorial in M:*

$$L^u_{\mathrm{Hom}_{S\text{-}gr.}(X,Y)/S}(M) \xrightarrow{\sim} \mathrm{Hom}_{S\text{-}gr.}(X, Lie(Y/S, M)).$$

4.3. Consider now the case where X and Y are 0_S-modules. Recall (cf. 4.2.0.4) that one writes $T'_{Y/S}(M)$ (resp. $Lie'(Y/S, M)$) for the functor $T_{Y/S}(M)$ (resp. $Lie(Y/S, M)$) equipped with the 0_S-module structure deduced from that of Y.

When Y/S satisfies (E), we shall always write $Lie(Y/S, M)$ for the functor $Lie(Y/S, M)$ equipped with the 0_S-module structure defined for every functor satisfying (E). In this case, we know (cf. 3.9) that the abelian-group structures of $Lie(Y/S, M)$ and $Lie'(Y/S, M)$ coincide, but the same is not a priori true for those of module (see a counterexample in §6.3). For every $S' \to S$ and $a \in O(S')$, we shall write $a \cdot' m$ (resp. $a \cdot m$) for the action of a on $m \in Lie'(Y/S, M)(S')$ (resp. on $m \in Lie(Y/S, M)(S')$), and similarly for the action of a on $T'_{Y/S}(M)$ and $T_{Y/S}(M)$.

[136] One has added the sentences that follow and Proposition 4.2 bis, implicit in the original.
[137] One has added the sentence that follows.

One has $T'_{Y/S}(M) \simeq Lie'(Y/S, M) \bigoplus Y$ as 0_S-modules; consequently one obtains, exactly as for Propositions 4.2 and 4.2 bis, the:

Proposition 4.3. *Let $u : X \to Y$ be a morphism of 0_S-modules. One has an isomorphism of S-functors, functorial in M:*

$$(*) \quad L^u_{\mathrm{Hom}_{O_S\text{-mod.}}(X,Y)/S}(M) \cong \mathrm{Hom}_{O_S\text{-mod.}}(X, Lie'(Y/S, M)).$$

[138] *If Y/S satisfies (E), then $\mathrm{Hom}_{O_S-mod.}(X,Y)/S$ satisfies (E) and $(*)$ is an isomorphism of 0_S-modules when one equips both sides with the 0_S-module structure deduced from functoriality in M.*[139]

Remark 4.3 bis.[140] Let $u : X \to Y$ be a morphism of 0_S-modules; denote by τ_u the map which to every morphism of 0_S-modules $\phi : X \to Lie'(Y/S, M)$ associates the morphism

$$u \oplus \varphi : X \to T'_{Y/S}(M) = Y \oplus Lie'(Y/S, M).$$

Then the isomorphism of 4.3 fits into the following commutative diagram, functorial in M:

$$\begin{tikzcd}
L^u_{\mathrm{Hom}_{O_S\text{-mod.}}(X,Y)/S}(M) \arrow[r, "\cong"] \arrow[d] & \mathrm{Hom}_{O_S\text{-mod.}}(X, Lie'(Y/S, M)) \arrow[d, "\tau_u"] \\
T_{\mathrm{Hom}_{O_S\text{-mod.}}(X,Y)/S}(M) \arrow[r, "\cong"] & \mathrm{Hom}_{O_S\text{-mod.}}(X, T'_{Y/S}(M)).
\end{tikzcd}$$

Moreover, when Y/S satisfies (E), one deduces from 3.13.1, exactly as in the proof of 3.13.2, that for every $u \in Isom_{O_S-mod.}(X, Y)$, one has

$$(*) \quad L^u_{\mathrm{Isom}_{O_S\text{-mod.}}(X,Y)/S}(M) = L^u_{\mathrm{Hom}_{O_S\text{-mod.}}(X,Y)/S}(M).$$

Corollary 4.3.1. *Let X be an 0_S-module satisfying (E) with respect to S. One has an isomorphism functorial in M:*

$$\mathrm{Lie}(\mathrm{Aut}_{O_S\text{-mod.}}(X)/S, M) \cong \mathrm{Hom}_{O_S\text{-mod.}}(X, Lie'(X/S, M))$$

which respects the 0_S-module structures deduced from functoriality in M.[141] *In particular, $\mathrm{Aut}_{O_S-mod.}(X)/S$ satisfies (E).*

Proof. The first assertion follows from $(*)$ and 4.3; let us prove the second. As X/S satisfies (E), one has isomorphisms of 0_S-modules $Lie'(X/S, M \oplus N) \simeq Lie'(X/S, M) \times_S Lie'(X/S, N)$, and thus:

$$\begin{aligned}&\mathrm{Lie}(\mathrm{Aut}_{O_S\text{-mod.}}(X)/S, M \oplus N) \simeq \\ &\quad \mathrm{Lie}(\mathrm{Aut}_{O_S\text{-mod.}}(X)/S, M) \times_S \mathrm{Lie}(\mathrm{Aut}_{O_S\text{-mod.}}(X)/S, N).\end{aligned}$$

Taking 4.1.1.2 a) into account, this proves that $\mathrm{Aut}_{O_S-mod.}(X)/S$ satisfies (E).

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[138] One has added the sentence that follows.
[139] Note that on the right-hand term, this is the structure defined by $(a\phi)(x) = a \cdot \phi(x)$, where $a \cdot \phi(x)$ denotes the action of a on $\phi(x) \in Lie(Y/S, M)$. This differs from the action $(a \cdot' \phi)(x) = a \cdot' \phi(x) = \phi(ax)$, but coincides with it if $Lie(Y/S, M) = Lie'(Y/S, M)$.
[140] One has added this remark, which will be useful in 4.5.1.
[141] Cf. N.D.E. (61). Moreover, one has added the sentence that follows, as well as its proof.

4.3.2. Before continuing in this direction, let us examine more closely the relations between Y, $Lie(Y/S)$ and $Lie'(Y/S)$. Let us remark first that

(1) $\quad Lie(O_S/S, M) = Lie'(O_S/S, M) = W(M)$

(where $W(M)$ is defined in I 4.6) and that one therefore has a canonical isomorphism

$$(2) d : O_S \xrightarrow{\sim} Lie(O_S/S).$$

Let now F be an O_S-module. For every $S_2 \to S_1 \to S$[142], one has a dihomomorphism

(3) $\quad F(S_1) \to F(S_2)$
$\qquad O(S_1) \to O(S_2)$,

whence a morphism of $O(S_2)$-modules

$F(S_1) \otimes_{O(S_1)} O(S_2) \to F(S_2)$.

In particular, setting $S_1 = S'$ and $S_2 = I_{S'}(M)$, one deduces morphisms of $O(S')$-modules, functorial in M,

$F(S') \otimes_{O(S')} T_{O_S/S}(M)(S') \to T'_{F/S}(M)(S')$;

letting S' vary, one obtains morphisms of O_S-modules, functorial in M,

(4) $\quad F \otimes_{O_S} T_{O_S/S}(M) \to T'_{F/S}(M)$.

These morphisms are functorial in M, hence compatible with the projections of the tangent bundles onto their bases; they therefore define morphisms of O_S-modules

(5) $\quad F \otimes_{O_S} Lie(O_S/S, M) \to Lie'(F/S, M)$

such that one has the commutative diagram:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & F \otimes_{O_S} Lie(O_S/S, M) & \longrightarrow & F \otimes_{O_S} T_{O_S/S}(M) & \longrightarrow & F & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & Lie'(F/S, M) & \longrightarrow & T'_{F/S}(M) & \longrightarrow & F & \longrightarrow & 0 \end{array} .$$

One may consider the morphisms (5) as morphisms of abelian S-groups

(6) $\quad F \otimes_{O_S} Lie(O_S/S, M) \to Lie(F/S, M)$;

tensoring F with the isomorphism $d : O_S \xrightarrow{\sim} Lie(O_S/S)$, one deduces (for $M = O_S$) a morphism of abelian S-groups

(7) $\quad F \to F \otimes_{O_S} Lie(O_S/S) \to Lie(F/S)$

also denoted $d : F \to Lie(F/S)$.

[142] One has changed $S' \to S$ to $S_2 \to S_1 \to S$, because one must let S_2 and S_1 vary (cf. below). Moreover, one has detailed the original in what follows.

Remark 4.3.3.[143] When F/S satisfies (E), the morphisms (6) and (7) are not necessarily morphisms of O_S-modules, when one equips both sides with the module structures deduced from that of M thanks to condition (E).

For example, let k be a field of characteristic $p > 0$, $S = \mathrm{Spec}(k)$, and let F be the O_S-module which to every S-scheme T associates $F(T) = \Gamma(T, O_T)$ equipped with the $O(T)$-module structure obtained by making scalars act through the p-th power, i.e., $r \cdot f = r^p f$, for $r \in O(T)$, $f \in F(T)$. As S-functor of groups, F is isomorphic to $G_{a,S}$. Therefore F satisfies (E) and $Lie(F/S)$ is identified with $Lie(G_{a,S}/S) \cong O_S$. Then, the canonical morphism $d : F \to Lie(F/S)$ is, for every $T \to S$, the identity map $F(T) \to O(T)$: it indeed respects the abelian-group structure but not the O_S-module structure.

Remark 4.3.4.[144] One can make the morphisms (4) and (5) explicit as follows. The morphism $\Theta : F \otimes_{O_S} T_{O_S/S}(M) \to T'_{F/S}(M) = \mathrm{Hom}_S(I_S(M), F)$ is defined by: for every $S' \to S$, $\alpha \in O(I_{S'}(M))$, and $f : S' \to F$,

$$\Theta(f \otimes \alpha) = \alpha \cdot (\tau_0 \circ f) = \alpha \cdot (f \circ \rho),$$

where τ_0 is the null section $F \to T'_{F/S}(M)$ and ρ the structural morphism $I_{S'}(M) \to S'$. Then Θ induces a morphism $\theta : F \otimes_{O_S} Lie(O_S/S, M) \to Lie'(F/S, M)$; this follows from the "functoriality in M" already mentioned after (4), and can be seen explicitly as follows. On the one hand, one has

$$Lie'(F/S, M)(S') = \{ \varphi \in \mathrm{Hom}_S(I_{S'}(M), F) \mid \varphi \circ \varepsilon_M = e \},$$

where e denotes the unit section $S' \to F$. On the other hand, $Lie(O_S/S, M)(S') = \Gamma(S', M)$ is the kernel of the augmentation $\eta : O(I_{S'}(M)) \to O(S')$, and it is therefore a question of seeing that if $f \in F(S')$ and $\alpha \in \Gamma(S', M)$, then $\alpha \cdot (f \circ \rho) \circ \epsilon_M = e$.

Consider the dihomomorphism (3), in the case where $S_2 \to S_1$ is the S-morphism $\epsilon_M : S' \to I_{S'}(M)$; one has then a commutative diagram

$$\begin{array}{ccc} F(I_{S'}(M)) & \xrightarrow{F(\varepsilon_M)} & F(S') \\ \downarrow \alpha & & \downarrow \eta(\alpha) \\ F(I_{S'}(M)) & \xrightarrow{F(\varepsilon_M)} & F(S') \end{array}.$$

For every $\phi : I_{S'}(M) \to F$, one therefore has $(\alpha \cdot \phi) \circ \epsilon_M = \eta(\alpha) \cdot (\phi \circ \epsilon_M)$, whence $(\alpha \cdot \phi) \circ \epsilon_M = e$ if $\eta(\alpha) = 0$.

In particular, taking $M = tO_S$, one has $Lie(O_S/S) = tO_S$ and the morphism $F \to Lie'(F/S)$ is given by $f \mapsto t \cdot (f \circ \rho)$.

Remark 4.3.5.[145] Let F be an O_S-module, set $E = \mathrm{End}_{O_S-mod.}(F)$ and denote by

[143] One found in the original the assertion that when F/S satisfies (E), the morphisms (6) (and hence (7)) are morphisms of O_S-modules, assertion contradicted by a counterexample given in 6.3. One has deleted this assertion, inserted here the example just mentioned, and modified Definition 4.4 accordingly. This is without consequence for the sequel.
[144] One has added this remark, which will be useful in 4.6.2.
[145] One has added this remark, which will be useful in 4.5 and 4.7.

d_F and d_E the morphisms of $\mathcal{O}_S$-modules given by 4.3.2 (5):

$$d_F : F \otimes_{\mathcal{O}_S} \mathrm{Lie}(\mathcal{O}_S/S, M) \to \mathrm{Lie}'(F/S, M),$$
$$d_E : E \otimes_{\mathcal{O}_S} \mathrm{Lie}(\mathcal{O}_S/S, M) \to \mathrm{Lie}'(E/S, M).$$

One deduces from 4.3.4 the following commutative diagram of morphisms of $\mathcal{O}_S$-modules:

$$\begin{tikzcd}
E \otimes_{\mathcal{O}_S} \mathrm{Lie}(\mathcal{O}_S/S, M) \arrow[r, "d_E"] & \mathrm{Lie}'(E/S, M) \\
\mathrm{Hom}_{\mathcal{O}_S}(F, F \otimes_{\mathcal{O}_S} \mathrm{Lie}(\mathcal{O}_S/S, M)) \arrow[r, "d_F"] & \mathrm{Hom}_{\mathcal{O}_S}(F, \mathrm{Lie}'(F/S, M)) \arrow[u, "\simeq\ (*)"']
\end{tikzcd}$$

where the right vertical arrow is the isomorphism $(*)$ of 4.3. Therefore (loc. cit.), if F/S satisfies (E), then E/S satisfies (E) and $(*)$ is also an isomorphism of $\mathcal{O}_S$-modules when one equips the right-hand terms with the $\mathcal{O}_S$-module structure deduced from (E).

Remark 4.4.0.[146] In 4.3.2, the morphisms (4) are isomorphisms if and only if the morphisms (5) are. Moreover, if these conditions are verified, then F/S satisfies (E). Indeed, it suffices to verify that $\mathrm{Lie}'(F/S, M \oplus N) \simeq \mathrm{Lie}'(F/S, M) \times_S \mathrm{Lie}'(F/S, N)$. Now one has the commutative diagram below, where by hypothesis the horizontal arrows are isomorphisms:

$$\begin{tikzcd}
F \otimes_{\mathcal{O}_S} \mathrm{Lie}(\mathcal{O}_S/S, M \oplus N) \arrow[r] \arrow[d, "\simeq"] & \mathrm{Lie}'(F/S, M \oplus N) \arrow[d] \\
F \otimes_{\mathcal{O}_S} (\mathrm{Lie}(\mathcal{O}_S/S, M) \times_S \mathrm{Lie}(\mathcal{O}_S/S, N)) \arrow[r] & \mathrm{Lie}'(F/S, M) \times_S \mathrm{Lie}'(F/S, N);
\end{tikzcd}$$

the second vertical arrow is therefore also an isomorphism, i.e. F/S satisfies (E).

Definition 4.4. *One says that F is a* **good $\mathcal{O}_S$-module** *if the morphisms*

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(M) \to T_{F/S}(M)$$
or, equivalently,
$$F \otimes_{\mathcal{O}_S} \mathrm{Lie}(\mathcal{O}_S/S, M) \to \mathrm{Lie}(F/S, M),$$

are isomorphisms of abelian S-groups (so that F/S satisfies (E)) and if moreover they respect the $\mathcal{O}_S$-module structures deduced from condition (E).

Corollary 4.4.1.[147] *Let F be an $\mathcal{O}_S$-module. Consider the following conditions:*

(i) F is a good $\mathcal{O}_S$-module.

(ii) F/S satisfies (E) and $d : F \to Lie(F/S)$ is an isomorphism of $\mathcal{O}_S$-modules.

(iii) $Lie(F/S, M) = Lie'(F/S, M)$.

Then one has (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii).

[146] Taking into account the modifications in Definition 4.4 (cf. N.D.E. (65)), one has inserted here this remark, which in the original figured in the proof of 4.4.1.

[147] The original showed that (i) implies (ii) and (iii); one has added the implication (ii) $\Rightarrow$ (i).

Proof. (i) ⇒ (ii) follows from the definition. To prove (ii) ⇒ (i), one must show that the morphisms of abelian S-groups (functorial in M)

$$F \otimes_{O_S} \mathrm{Lie}(O_S/S, M) \to \mathrm{Lie}(F/S, M)$$

are isomorphisms of O_S-modules. Since F/S satisfies (E), both sides transform finite direct sums of copies of O_S into finite products of abelian S-groups. This reduces us to the case $M = O_S$, which follows from the hypothesis.

Finally, (i) ⇒ (iii) follows from the definition and from the fact that the isomorphisms

$$F \otimes_{O_S} \mathrm{Lie}(O_S/S, M) \to \mathrm{Lie}'(F/S, M)$$

of 4.3.2 (5) are morphisms of O_S-modules.

Examples 4.4.2. *For every quasi-coherent O_S-Module E, the O_S-modules $V(E)$ and $W(E)$ defined in I 4.6 are good.*

[148] Indeed, for every $f : S' \to S$, the morphisms

$$V(E)(S') \otimes_{O(S')} O(I_{S'}(M)) \to T_{V(E)/S}(M)(S')$$
$$W(E)(S') \otimes_{O(S')} O(I_{S'}(M)) \to T_{W(E)/S}(M)(S')$$

correspond, respectively, to the morphisms:

$$\mathrm{Hom}_{O_{S'}\text{-mod.}}(f^*(E), O_{S'}) \otimes_{O(S')} \Gamma(S', D_{O_{S'}}(M)) \to \mathrm{Hom}_{O_{S'}\text{-mod.}}(f^*(E), D_{O_{S'}}(M))$$
$$\Gamma(S', f^*(E)) \otimes_{O(S')} \Gamma(S', D_{O_{S'}}(M)) \to \Gamma(S', f^*(E) \otimes_{O_{S'}} D_{O_{S'}}(M));$$

these are isomorphisms, since $D_{O_{S'}}(M)$ is isomorphic, as $O_{S'}$-module, to a finite direct sum of copies of $O_{S'}$.

Proposition 4.5. *Let F be a good O_S-module. Then:*

(i) $\mathrm{Aut}_{O_S-mod.}(F)/S$ *satisfies (E) and one has a functorial isomorphism*

$$\mathrm{Lie}(\mathrm{Aut}_{O_S\text{-mod.}}(F)/S, M) \to \mathrm{Hom}_{O_S\text{-mod.}}(F, \mathrm{Lie}(F/S, M))$$

which respects the O_S-module structures deduced from condition (E). In particular, one has an isomorphism of O_S-modules

$$\mathrm{Lie}(\mathrm{Aut}_{O_S\text{-mod.}}(F)/S) \to \mathrm{End}_{O_S\text{-mod.}}(F).$$

(ii)[149] *Moreover,* $\mathrm{End}_{O_S-mod.}(F)$ *is a good O_S-module.*

Indeed, by 4.4.1, F/S satisfies (E) and

$$(1) \quad \mathrm{Lie}(F/S, M) = \mathrm{Lie}'(F/S, M) \simeq F \otimes_{O_S} \mathrm{Lie}(O_S/S, M).$$

Assertion (i) then follows from 4.3.1. Set $E = \mathrm{End}_{O_S-mod.}(F)$. By (1) and Remark 4.3.5, one has the following commutative diagram of morphisms of abelian S-groups:

[148] One has added what follows.

[149] One has added the point (ii), immediate consequence of what precedes.

$$\begin{tikzcd}
\mathrm{End}_{\mathcal{O}_S\text{-mod.}}(F) \otimes_{\mathcal{O}_S} Lie(\mathcal{O}_S/S, M) \arrow[r, "d_E"] & Lie(\mathrm{End}_{\mathcal{O}_S\text{-mod.}}(F)/S, M) \\
\mathrm{Hom}_{\mathcal{O}_S}(F, F \otimes_{\mathcal{O}_S} Lie(\mathcal{O}_S/S, M)) \arrow[r, "d_F"] & \mathrm{Hom}_{\mathcal{O}_S}(F, Lie(F/S, M)) \arrow[u, "\simeq\ (*)"']
\end{tikzcd}$$

where d_F and $(*)$ are isomorphisms of $\mathcal{O}_S$-modules; consequently, so is d_E. This proves (ii).

Scholie 4.5.1.[150] Set (cf. 2.1) $O_{I_S} = O_S \bigoplus tO_S$ (with $t^2 = 0$), and let F be a good $\mathcal{O}_S$-module. Then, for every $S' \to S$, the morphism

$$F(S') \oplus t\,F(S') = F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}) \xrightarrow{\sim} F(I_{S'}) = F(S') \oplus Lie(F/S)(S')$$

(which is the identity on $F(S')$) induces an isomorphism of $O(S')$-modules $tF(S') \simeq Lie(F/S)(S')$. Letting S' vary, one obtains an isomorphism which one may write $Lie(F/S) \simeq tF$.

For every $S' \to S$, one has therefore, by 4.5, a commutative diagram:

$$\begin{tikzcd}
\mathrm{End}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}) \arrow[r, "\sim"] \arrow[dr] & \mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}, t\,F_{S'}) \arrow[r, "\sim"] \arrow[d] & Lie(\mathrm{Aut}_{\mathcal{O}_S\text{-mod.}}(F)/S)(S') \arrow[d] \\
& \mathrm{Aut}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}}) & T_{\mathrm{Aut}_{\mathcal{O}_S\text{-mod.}}(F)/S}(S')
\end{tikzcd}$$

and one deduces from 4.3 bis that every $X \in \mathrm{End}_{O_{S'}-mod.}(F_{S'})$ corresponds to the element $id + tX$ of $\mathrm{Aut}_{O_{I_{S'}}-mod.}(F_{I_{S'}})$.

Definition 4.6. *One says that the S-functor of groups G is* **good** *if G/S satisfies condition (E) and if $Lie(G/S)$ is a good $\mathcal{O}_S$-module.*

[151] Note that if F is a good $\mathcal{O}_S$-module, it is a good S-group; indeed F/S satisfies (E) and $Lie(F/S) \simeq F$ is a good $\mathcal{O}_S$-module.

Example 4.6.1. *If G is representable, it is good. Indeed, G/S satisfies (E) and $Lie(G/S)$ is of the form $V(E)$, hence good, by 4.4.2.*

Lemma 4.6.2.[152] *Let G be an S-functor of groups such that G/S satisfies (E), and let $F = Lie(G/S)$. Then F satisfies (E) and the morphism of abelian groups $d : F \to Lie(F/S)$ respects the $\mathcal{O}_S$-module structures.*

Consequently, G is good if and only if $F \to Lie(F/S)$ is bijective.

Proof. Suppose that G/S satisfies (E). Let M, N be free $\mathcal{O}_S$-modules of finite rank. Write $F(N) = Lie(G/S, N)$ and e the unit section of G.

For every $S' \to S$, one has $F(N)(S') = g \in \mathrm{Hom}_S(I_{S'}(N), G) | g \circ \epsilon_N = e$ and $Lie(F(N)/S, M)(S') = \mathrm{Hom}_S(I_{S'}(M), F(N))$ is identified with

[150] One has added this scholie, implicit in the original, which will be useful in 4.7.1. (Here and in the sequel, one writes t, t', etc. for variables of square zero, the letter ϵ being reserved for the unit section of groups.)

[151] One has added the sentence that follows.

[152] One has added this lemma.

$$\{ \Phi \in \mathrm{Hom}_S(I_{S'}(N) \times_{S'} I_{S'}(M), G) \mid \Phi \circ (\varepsilon_N \times \mathrm{id}_{I_{S'}(M)}) = e \in G(I_{S'}(M)), \ \Phi \circ (\mathrm{id}_{I_{S'}(N)} \times \varepsilon_M) = e \in G(I_{S'}(N)) \}.$$

This shows that

$$(1) \quad Lie(F(N)/S, M) \simeq Lie(F(M)/S, N).$$

As G/S satisfies (E), one deduces:

$$Lie(F(N)/S, M_1 \oplus M_2) \simeq Lie(F(M_1 \oplus M_2)/S, N) \simeq Lie((F(M_1) \times_S F(M_2))/S, N).$$

By 3.8, the right-hand term is isomorphic to

$$Lie(F(M_1)/S, N) \times_S Lie(F(M_2)/S, N) \simeq Lie(F(N)/S, M_1) \times_S Lie(F(N)/S, M_2).$$

It follows that

$$(2) \quad Lie(F(N)/S, M_1 \oplus M_2) \simeq Lie(F(N)/S, M_1) \times_S Lie(F(N)/S, M_2),$$

so $F(N)/S$ satisfies (E), by Remark 4.1.1.2 a).

Let us now show that the morphism of abelian groups $d : F(N) \to Lie(F(N)/S)$ respects the O_S-module structures. Consider the free O_S-module $M = tO_S$, so that

$$O(I_{S'}(M)) = O(S')[t]/(t^2) = O(S') \oplus t\, O(S')$$

and $Lie(O_S/S) \simeq tO_S$. We denote by ρ_t the structural morphism $I_S(M) \to S$, which corresponds to the injection $u_t : O_S \hookrightarrow D_{O_S}(M) = O_S \bigoplus tO_S$. Recall (cf. 3.4.2) that, for every S-functor X, $F(N)(X) = Lie(G/S, N)(X)$ is the set of S-morphisms $\phi : I_X(N) \to G$ such that $\phi \circ \epsilon_N = e$, and that the action of $a \in O(X)$ is given by $a \cdot \phi = \phi \circ a^*$, where a^* is the endomorphism of $I_X(N)$ associated with a, cf. 2.1.3.

Consequently, by 4.3.4, the morphism $d : F(N) \xrightarrow{\sim} F(N) \bigotimes_{O_S} tO_S \to Lie(F(N)/S)$ is given by: for every $S' \to S$ and $f \in \mathrm{Hom}_S(S', F(N))$,

$$f \mapsto f \otimes t \mapsto t \cdot (f \circ \rho_t) = f \circ \rho_t \circ t^*,$$

where, in the last term, $f \circ \rho_t \in F(N)(I_{S'}(M))$ is considered as an S-morphism $\phi : I_{I_{S'}(M)}(N) \to G$ (such that $\phi \circ \epsilon_N = e$). It is a question of seeing that d is a morphism of O_S-modules, i.e., that $d(a \cdot f) = u_t(a) \cdot d(f)$ for every $a \in O(S')$.

Considering $f \in F(N)(S')$ as a morphism $I_{S'}(N) \to G$, one has likewise $a \cdot f = f \circ a^*$. On the other hand, by the functoriality in X of the action of $O(X)$ on $I_X(N)$ (cf. 2.1.3), one has the commutative diagram below:

$$\begin{tikzcd} I_{S'}(N) \arrow[r, "a^*"] & I_{S'}(N) \\ I_{I_{S'}(M)}(N) \arrow[u, "\rho_t"] \arrow[r, "u_t(a)^*"] & I_{I_{S'}(M)}(N) \arrow[u, "\rho_t"] \end{tikzcd}$$

One has therefore

$$d(a \cdot f) = f \circ a^* \circ \rho_t \circ t^* = f \circ \rho_t \circ u_t(a)^* \circ t^* = f \circ \rho_t \circ t^* \circ u_t(a)^* = u_t(a) \cdot d(f)$$

(the second-to-last equality resulting from the fact that O is commutative). This completes the proof of Lemma 4.6.2.

Theorem 4.7. *If F is a good O_S-module, the S-group $\mathrm{Aut}_{O_S-mod.}(F)$ is good.*

[153] Indeed, by 4.5, $\mathrm{Aut}_{O_S-mod.}(F)/S$ satisfies (E) and $\mathrm{Lie}(\mathrm{Aut}_{O_S\text{-mod.}}(F)/S) \simeq \mathrm{End}_{O_S\text{-mod.}}(F)$ is a good O_S-module.

4.7.1.[154] Let now G be an S-group and F a good O_S-module. Suppose given a linear representation of G in F, that is to say (I 4.7.1), a morphism of S-groups

$$\rho : G \longrightarrow \mathrm{Aut}_{O_S-mod.}(F).$$

If G/S satisfies (E), one deduces from 4.1.C and 4.5 a morphism of O_S-modules, denoted ρ' or $d\rho$:

$$\mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(\mathrm{Aut}_{O_S\text{-mod.}}(F)/S) \simeq \mathrm{End}_{O_S\text{-mod.}}(F).$$

[155] Moreover, setting $O_{I_S} = O_S \oplus tO_S$ (with $t^2 = 0$), one deduces from 4.5.1 that, if $S' \to S$ and $X \in Lie(G/S)(S') \subset G(I_{S'})$, then one has in $\mathrm{Aut}_{O_{I_{S'}}-mod.}(F_{I_{S'}})$ the following equality:

$$(*) \qquad \rho(X) = \mathrm{id} + t\,\rho'(X),$$

i.e. for every $S'' \to I_{S'}$ and $f \in F(S'')$, one has in $F(S'')$ the equality $\rho(X)(f) = f + t\rho'(X)(f)$.

Definition 4.7.2. *Let G be a good S-group. Then $Lie(G/S)$ is a good O_S-module, and one has a morphism of S-groups $\mathrm{Ad} : G \to \mathrm{Aut}_{O_S\text{-mod.}}(\mathrm{Lie}(G/S))$. One deduces from 4.7.1 a morphism of O_S-modules*

$$\mathrm{ad} : \mathrm{Lie}(G/S) \longrightarrow \mathrm{End}_{O_S\text{-mod.}}(\mathrm{Lie}(G/S)),$$

or, what amounts to the same, an O_S-bilinear morphism:

$$\mathrm{Lie}(G/S) \times_S \mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(G/S), \qquad (x, y) \mapsto [x, y] = \mathrm{ad}(x) \cdot y$$

(where x and y denote two arbitrary elements of $Lie(G/S)(S') = Lie(G_{S'}/S')(S')$). If G is commutative, then $[x, y] = 0$.

4.7.3. One can give an equivalent definition of the bracket as follows: let us remark first that it suffices to do it for $x, y \in Lie(G/S)(S)$. Let us remark next that there is a canonical isomorphism $I_S \times_S I_S \simeq I_{I_S}$; to avoid confusions, let us denote by I and I' two copies of I_S and set $O_I = O_S[t]$, $O_{I'} = O_S[t']$, where $t^2 = 0 = t'^2$. One has then a commutative diagram

[153] One has simplified the original, using the additions made in 4.3.1 and 4.5.
[154] One has added the numbering 4.7.1, 4.7.2, 4.7.3.
[155] One has added the sentence that follows.

$$
\begin{tikzcd}
I \times I' \arrow[r] \arrow[d] & I' \arrow[d] \\
I \arrow[r] & S ,
\end{tikzcd}
$$

the two arrows starting from $I \times I'$ identifying it with the scheme of dual numbers over I or over I'. There follows a commutative diagram of groups (where one writes $L = Lie(G/S)$):

$$
(1)\qquad
\begin{tikzcd}
 & & 1 \arrow[d] & 1 \arrow[d] & \\
 & & L(I) \arrow[r] \arrow[d] & L(S) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & L(I') \arrow[r] \arrow[d] & G(I \times I') \arrow[r] \arrow[d] & G(I') \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & L(S) \arrow[r] & G(I) \arrow[r] \arrow[d] & G(S) \arrow[r] \arrow[d] & 1 \\
 & & 1 & 1 . &
\end{tikzcd}
$$

The ninth piece of the puzzle is nothing other than $Lie(L/S)(S)$. If G is good, this is $L(S)$ and one has therefore the following commutative diagram, where the lines and columns are exact sequences of groups, the five $L(-)$ are commutative,[156] and where, taking into account the identification $L(I) = L(S) \bigoplus tL(S)$ (resp. $L(I') = L(S) \bigoplus t'L(S)$), the injection $L(S) \hookrightarrow L(I)$ (resp. $L(S) \hookrightarrow L(I')$) is given by $u \mapsto tu$ (resp. $u \mapsto t'u$):

$$
\begin{tikzcd}
z \in L(S) \arrow[r, "t"] \arrow[d, "t'"] & L(I) \arrow[r] & L(S) \ni x \arrow[d] \\
L(I') \arrow[r] \arrow[d] & G(I \times I') \arrow[r] & G(I') \arrow[d] \\
y \in L(S) \arrow[r] & G(I) \arrow[r] & G(S) .
\end{tikzcd}
$$

Now in such a diagram, if one takes two elements x and y as marked, and one lifts arbitrarily x resp. y to an element $\tilde{x} \in L(I)$ resp. $y' \in L(I')$, the commutator $\tilde{x}y'\tilde{x}^{-1}y'^{-1}$ in $G(I \times I')$ does not depend on the chosen lifts and is the image of an element z as marked. The reader will verify that one has $z = [x, y]$.[157]

Indeed, if one still denotes by x the image of x by the canonical section $L(S) \to L(I)$ (and likewise for y), then $\tilde{x} = xu$ and $y' = yv$, with $u, v \in L(S) = L(I) \cap L(I')$, and

[156] One has detailed the original in what follows.

[157] The original indicated in a footnote: "($*$) The author acknowledges, at Gabriel's request, that the exercise is not immediate; this is moreover the reason why it is not in the text". One has given the details, taking into account the addition made in 4.7.1; see also [DG70], § II.4, 4.2.

since $L(I)$ and $L(I')$ are commutative one has

$$\begin{aligned}\tilde{x}\, y'\, \tilde{x}^{-1}\, y'^{-1} &= x\, u\, y\, v\, u^{-1}\, x^{-1}\, v^{-1}\, y^{-1}\\ &= x\, u\, y\, u^{-1}\, v\, x^{-1}\, v^{-1}\, y^{-1}\\ &= x\, y\, x^{-1}\, y^{-1}.\end{aligned}$$

Moreover, this element is sent to the unit element of $G(I)$ and of $G(I')$, hence comes from a (unique) z as indicated. Finally, considering y (resp. x) as an element of $L_{I'}(I')$ (resp. of $L(S) \subset G(I')$), one has by 4.7.1 (∗):

$$x\, y\, x^{-1} = Ad(x)(y) = (id + t'\, ad(x))(y) = y + t'\, [x, y],$$

so the element $xyx^{-1}y^{-1}$ of $L(I')$ is the image of the element $z = [x, y]$ of $L(S)$.

On this construction the following two properties appear:

(i) the bracket is "functorial in G": precisely, $G \mapsto Lie(G/S)$ is a functor from the category of good S-groups to the category of good O_S-modules equipped with an O_S-bilinear composition law.

(ii) One has $[x, y] + [y, x] = 0$: indeed the diagram is symmetric with respect to the first diagonal.[158]

Proposition 4.8. *Let F be a good O_S-module. Via the identification*

$$Lie(Aut_{O_S\text{-mod.}}(F)/S) = End_{O_S\text{-mod.}}(F)$$

one has

$$Ad(g) \cdot Y = g \circ Y \circ g^{-1} \quad \text{and} \quad [X, Y] = X \circ Y - Y \circ X,$$

for every $S' \to S$, $g \in \mathrm{Aut}_{O_{S'}-mod.}(F_{S'})$, and $X,\ Y \in Lie(Aut_{O_S\text{-mod.}}(F)/S)(S') = End_{O_{S'}\text{-mod.}}(F_{S'})$.

Proof.[159] By base change, one reduces to $S' = S$, which permits one to lighten the notation. Set $I = I_S$ and $O_I = O_S[t]$ (with $t^2 = 0$). Recall (cf. 4.5.1) that the inclusion $i : End_{O_S\text{-mod.}}(F) \hookrightarrow Aut_{O_I\text{-mod.}}(F_I)$ sends Y to $id + tY$. Then, by definition of $Ad(g)$ (cf. 4.1.A), one has:

$$id + t\, Ad(g)(Y) = g \circ (id + t\, Y) \circ g^{-1} = id + t\, (g \circ Y \circ g^{-1}),$$

whence $Ad(g)(Y) = g \circ Y \circ g^{-1}$.

Let I' be a second copy of I_S, set $O_{I'} = O_S[t']$ (with $t'^2 = 0$). Apply the results of 4.7.3 to $G = \mathrm{Aut}_{O_S-mod.}(F)$ and $L = Lie(G/S) = Aut_{O_S\text{-mod.}}(F)$. One identifies X with its image under the canonical section $L(S) \hookrightarrow L(I)$; its image in $G(I \times I')$ is then $id + t'X$, whose inverse is $id - t'X$. Likewise, Y lifts to $id + tY$, whose inverse is $id - tY$. Then, the commutator

[158] I.e. x and y play symmetric roles: the image in $G(I \times I')$ of $[y, x]$ equals the commutator $y'\tilde{x}y'^{-1}\tilde{x}^{-1}$, which is the inverse of $\tilde{x}y'\tilde{x}^{-1}y'^{-1} = [x, y]$. On the other hand (cf. 4.10 below), one does not know whether one necessarily has $[x, x] = 0$: if one considers x as an element $\tilde{x} \in L(I)$ (resp. $x' \in L(I')$) it is not a priori clear that $\tilde{x}$ and x' commute. . .

[159] One has detailed and simplified the original in what follows.

$$(id + t' X) \circ (id + t Y) \circ (id - t' X) \circ (id - t Y) = id + t t' (X \circ Y - Y \circ X)$$

is the image in $G(I \times I')$ of the element $Z = X \circ Y - Y \circ X$ of $L(S)$ (indeed, Z is sent to $tZ \in L(I)$, then to $id + t'tZ \in G(I \times I')$). By 4.7.3, this shows that $[X, Y] = X \circ Y - Y \circ X$.

Corollary 4.8.1. *Let G be a good S-group and $x, y, z \in Lie(G/S)(S')$. One has:*

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

[160] Indeed, since G is good, $Lie(G/S)$ is a good O_S-module and therefore, by 4.7, $\mathrm{Aut}_{O_S-mod.}(Lie(G/S))$ is a good S-group. Then, the morphism of S-groups

$$Ad : G \to \mathrm{Aut}_{O_S\text{-mod.}}(Lie(G/S))$$

gives, by the functoriality 4.7.3 (i): $ad[x, y] = [adx, ady]$. Combined with 4.8, this gives:

$$ad[x, y] = [ad\, x, ad\, y] = ad\, x \circ ad\, y - ad\, y \circ ad\, x,$$

which, applied to an element z, gives the Jacobi relation.

Corollary 4.8.2. *Let G be a good S-group acting linearly on a good O_S-module F (i.e. let F be a G-O_S-module, with G and F good). Then the linear map $\rho' : Lie(G/S) \to \mathrm{End}_{O_S\text{-mod.}}(F)$ is a representation, i.e. one has*

$$\rho'([x, y]) = \rho'(x) \circ \rho'(y) - \rho'(y) \circ \rho'(x).$$

Scholie 4.9. *To every good S-group (for example representable), one has associated a good O_S-module $Lie(G/S)$ functorially equipped with a bilinear map satisfying*

$$[x, y] + [y, x] = 0, \qquad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

We shall call $Lie(G/S)$ equipped with this structure the "Lie algebra" of G over S (the quotation marks being justified by the fact that one does not know whether $Lie(G/S)$ is, strictly speaking, an O_S-Lie algebra[161]). To every linear representation of G in a good O_S-module F is associated a representation of its "Lie algebra". In particular, to the adjoint representation of G is associated the adjoint representation of its "Lie algebra".

Definition 4.10. *A group-functor G above S is said to be* **very good** *if it is good and if $Lie(G/S)$ is an O_S-Lie algebra (i.e. if one has identically $[x, x] = 0$).*

Examples 4.10.1. *The following S-groups are very good: $\mathrm{Aut}_{O_S-mod.}(F)$ for every good O_S-module F (cf. 4.7 and 4.8), every representable group (see below), every good S-group admitting a monomorphism into a very good S-group, for example every good subgroup-functor of a representable group, or every good S-group admitting a faithful linear representation in a good O_S-module, for example every good S-group such that Ad is a monomorphism . . .*

---

[160] One has detailed the original in what follows.

[161] Because one does not know whether $[x, x] = 0$, see 4.10 which follows.

4.11. Suppose now that G is a group scheme over S. By 4.1.4, $Lie(G/S)(S)$ is identified with the group of infinitesimal automorphisms of G/S right-invariant, that is to say, by 3.14, with the group of derivations of O_G over O_S invariant by right translation. Moreover this identification respects the module structure and is an[162] anti-isomorphism of Lie algebras, as one sees by reasoning as in 4.7.3: set $O_I = O_S[t]$ and $O_{I'} = O_S[t']$ and let $x \in L(I)$ and $y \in L(I')$. The left translation λ_x (resp. λ_y) is an S-automorphism of $G_{I\times I'}$ which induces the identity on $G_{I'}$ (resp. G_I) and which corresponds to an O_S-automorphism

```
u = id + t d_x    resp.    v = id + t′ d_y
```

of $O_{G_{I\times I'}} = O_G[t,t']/(t^2,t'^2)$, where d_x, d_y are O_S-derivations of O_G invariant by right translation. Since the correspondence between S-automorphisms of $G_{I\times I'}$ and O_S-automorphisms of $O_{G_{I\times I'}}$ is contravariant, $\lambda_x\lambda_y\lambda_x^{-1}\lambda_y^{-1}$ corresponds to $v^{-1}u^{-1}vu = id + tt'(d_y d_x - d_x d_y)$. One deduces, by 4.7.3, that the map $x \mapsto -d_x$ is an isomorphism of Lie algebras (for more details, see [DG70], § II.4, 4.4 and 4.6). What precedes is valid for $Lie(G/S)(S') = Lie(G_{S'}/S')(S')$ for every $S' \to S$. One thus recovers the classical definition:[163]

Scholie 4.11.1. *Via the isomorphism $x \mapsto -d_x$, $Lie(G/S)$ is identified with the functor that to every S' above S associates the $O(S')$-Lie algebra of derivations of $G_{S'}$ with respect to S' invariant by right translation.*

Since one knows already, by 4.6.1, that every representable group is good, it follows from what precedes:

Corollary 4.11.2. *Every representable group is very good.*

Let $\epsilon : S \to G$ be the unit section of G. Set $\omega^1_{G/S} = \epsilon^*(\Omega^1_{G/S})$ and recall (cf. 3.3) that $Lie(G/S)$ is representable by the vector fibration $V(\omega^1_{G/S})$.

Scholie 4.11.3. *One has therefore associated functorially to every S-group scheme G a vector fibration $Lie(G/S) = V(\omega^1_{G/S})$ over S, which represents the functor $Lie(G/S)$, hence is equipped with a structure of S-scheme of O_S-Lie algebras. Moreover (cf. 3.4 and 3.8), this construction commutes with base extension and finite products.*

Remarks 4.11.4.[164] Denote by π the morphism $G \to S$.

a) The O_G-module $\Omega^1_{G/S}$ is evidently $(G \times_S G)$-equivariant (cf. I, § 6) and therefore, by I 6.8.1, one has $\Omega^1_{G/S} \simeq \pi^*(\omega^1_{G/S})$. It follows for example that $\Omega^1_{G/S}$ is locally free (resp. locally free of finite type) if $\omega^1_{G/S}$ is, which is in particular the case if S is the spectrum of a field (resp. if S is the spectrum of a field and G locally of finite type over S).

[162] One has corrected a sign error in the original, cf. [DG70], § II.4, 4.4 and 4.6.
[163] One has detailed the original in what follows; in particular, one has added, for later references, the numbering 4.11.1 to 4.11.8.
[164] One has detailed what follows, taking into account the additions made in Exp. I, § 6.8.

b) Moreover, by I 6.8.2, $\omega^1_{G/S}$ is equipped with a canonical structure of G-0_S-module, which induces on $V(\omega^1_{G/S}) = Lie(G/S)$ the adjoint action.[165]

c) On the other hand (cf. EGA I, 5.3.11 and 5.4.6), ϵ is an immersion, and is a closed immersion if G is separated over S. Therefore $\omega^1_{G/S}$ is identified with I/I^2, where I is the quasi-coherent ideal defining $\epsilon(S)$ in an open subset U of G in which $\epsilon(S)$ is closed (if $G \to S$ is separated, one may take $U = G$, and if $G = \operatorname{Spec} A(G)$ is affine over S, I is nothing other than the augmentation ideal of $A(G)$, i.e. the kernel of $\epsilon^{?} : A(G) \to O_S$, cf. I 4.2).[166]

Remark 4.11.5. One can deduce from the isomorphism $\Omega^1_{G/S} \simeq \pi^*(\omega^1_{G/S})$ that the 0_S-module $\omega^1_{G/S}$ is identified with the sheaf $\pi_*(\Omega^1_{G/S})^G$ of "differentials of G with respect to S invariant on the right", that is to say, with the sheaf whose sections on an open subset U of S are the sections of $\Omega^1_{G/S}$ on $\pi^{-1}(U)$ invariant by right translation (cf. I, 6.8.3, compare also with VII_A, 2.4).[167]

Notation 4.11.6. *One denotes by $Lie(G/S)$ the sheaf of sections of the vector fibration $Lie(G/S) \to S$; it is the 0_S-module $(\omega^1_{G/S})^\vee = \mathrm{Hom}_{O_S}(\omega^1_{G/S}, O_S)$ dual to $\omega^1_{G/S}$ (cf. EGA II 1.7.9). It is equipped with a structure of 0_S-Lie algebra.*

Since this construction does not commute with base extension (in general), the Lie algebra structure on this module does not allow one to reconstruct the structure of S-scheme of 0_S-Lie algebras on $Lie(G/S)$.[168]

However one has:

Lemma 4.11.7. *Suppose $\omega^1_{G/S}$ locally free of finite type (which is the case in particular if G is smooth over S (cf. EGA IV_4, 17.2.4), or if S is the spectrum of a field and G locally of finite type over S). Then $Lie(G/S)^\vee \cong (\omega^1_{G/S})^{\vee\vee} \cong \omega^1_{G/S}$ and therefore*

Lie(G/S) = V(ω¹_{G/S}) = V(Lie(G/S)^v) = W(Lie(G/S))

(the last equality following from I 4.6.5.1).

Finally, let $G \to H$ be a monomorphism of group-functors. Then $Lie(G/S) \to Lie(H/S)$ is also a monomorphism (cf. 3.7). Since every vector monomorphism of vector fibrations is a closed immersion[169] one obtains:

[165]For this reason, the linear action of G on $\omega^1_{G/S}$ could be called the "pre-adjoint action"; in fact, by abuse of language, one will still speak of "the adjoint action" on $\omega^1_{G/S}$. Let us point out here a slightly more general construction, which will be used in Exp. III (cf. in particular III 0.8). Suppose given, for every $Y \to S$, an 0_Y-module $N(Y)$, and for every S-morphism $\phi : Z \to Y$, a morphism of 0_Z-modules $N(\phi) : \phi^* N(Y) \to N(Z)$, which is functorial in ϕ (this is the case, for example, if J is a quasi-coherent ideal of 0_S and if one sets $N(Y) = JO_Y$, cf. Exp. III); then the action of G on $\omega^1_{G/S}$ induces a (generalized) "adjoint action" of G on the S-functor of abelian groups that to every $f : Y \to S$ associates $\mathrm{Hom}_{O_Y}(f^*(\omega^1_{G/S}), N(Y))$.

[166]One has added the preceding description of $\omega^1_{G/S}$, which will be useful later (for example, in VII_A, 5.5).

[167]This description of $\omega^1_{G/S}$ in terms of invariant differentials will not be used in the sequel.

[168]Because $Lie(G/S) \cong (\omega^1_{G/S})^\vee$ does not necessarily determine $\omega^1_{G/S}$. For example, if $S = \mathbb{A}^1_k$ (k a field) and if G is the S-group whose fiber at $s = 0$ is $G_{a,k}$ and the other fibers are the unit group, then $Lie(G/S) = 0$ but $Lie(G_s/k)(k) = k$.

[169]Let $f : M \to N$ be a morphism of 0_S-modules and $P = Coker(f)$. If $V(N) \to V(M)$ is a monomorphism, the surjective morphism $Sym(N) \to Sym(P)$ factors through 0_S, so $P = 0$.

Corollary 4.11.8. *Let $G \hookrightarrow H$ be a monomorphism of S-group schemes.*

(i) $Lie(G/S) \to Lie(H/S)$ is a closed immersion and therefore $\omega^1_{H/S} \to \omega^1_{G/S}$ is an epimorphism.

(ii) If $\omega^1_{G/S}$ is locally free of finite type, then the corresponding morphism $Lie(G/S) \to Lie(H/S)$ is an isomorphism of $Lie(G/S)$ onto a sub-module of $Lie(H/S)$ locally direct factor.

5. Computation of some Lie algebras

5.1. Examples of Lie algebras: diagonalizable groups

Let $G = D_S(M)$ be a diagonalizable group over S (I 4.4). Since the formation of $Lie(G/S)$ commutes with base extension, it suffices to make the construction for $G = D(M)$. One has then:

$$G(I_S) = \mathrm{Hom}_{gr.}(M, \Gamma(I_S, O_{I_S})^\times) = \mathrm{Hom}_{gr.}(M, \Gamma(S, D_{O_S})^\times).$$

Now one has a split exact sequence

$$1 \to \Gamma(S, O_S) \to \Gamma(S, D_{O_S})^\times \to \Gamma(S, O_S)^\times \to 1,$$

which gives that $Lie(G)(S)$ is identified with $\mathrm{Hom}_{gr.}(M, O(S))$ equipped with its obvious $O(S)$-module structure. One thus obtains, after base change:

Proposition 5.1. *One has isomorphisms*

$$\mathrm{Hom}_{S\text{-}gr.}(M_S, O_S) \xrightarrow{\sim} Lie(D_S(M)/S) \quad \text{and} \quad \mathrm{Hom}_{gr.}(\tilde{M}_S, O_S) \xrightarrow{\sim} Lie(D_S(M)/S),$$

(where, in the second isomorphism, $\tilde{M}_S$ denotes the constant sheaf of groups on S defined by M, and $\mathrm{Hom}_{gr.}$ *the sheaf of homomorphisms of sheaves of groups).*

Corollary 5.1.1. *If M is free of finite type (or, as we shall say later, if $D_S(M)$ is a split torus), then (see I, 4.6.5 for the definition of W)*

$$W(Lie(D_S(M)/S)) \xrightarrow{\sim} Lie(D_S(M)/S),$$
$$M^\vee \otimes_{\mathbb{Z}} O_S \xrightarrow{\sim} Lie(D_S(M)/S),$$

where $M^\vee$ denotes the dual of the abelian group M. In particular

$$O_S \xrightarrow{\sim} Lie(G_{m,S}/S) \quad \text{and} \quad O_S \xrightarrow{\sim} Lie(G_{m,S}/S).$$

5.2. Normalizers and centralizers

Let us first prove a few lemmas. Recall (cf. I 3.1.1) that a sequence $0 \to F' \to F \to F'' \to 0$ of O_S-modules is said to be exact if for every $S' \to S$ the sequence $0 \to F'(S') \to F(S') \to F''(S') \to 0$ of $O(S')$-modules is exact.

Likewise, a sequence $1 \to G' \to G \to G'' \to 1$ of S-groups is said to be exact if for every $S' \to S$ the sequence of groups $1 \to G'(S') \to G(S') \to G''(S') \to 1$ is exact.

Lemma 5.2.1.[170] *Let $1 \to G' \to G \to G'' \to 1$ be an exact sequence of S-groups.*

[170] One has detailed the statement of the lemma (and its proof); this will be useful in 5.3.1.

(i) The sequences $1 \to T_{G'/S}(M) \to T_{G/S}(M) \to T_{G''/S}(M) \to 1$ *and*

$$1 \to \mathrm{Lie}(G'/S, M) \to \mathrm{Lie}(G/S, M) \to \mathrm{Lie}(G''/S, M) \to 1$$

are then exact.

(ii) Let $1 \to H' \to H \to H'' \to 1$ *be a second sequence of groups; it is exact if and only if the sequence below is exact:*

$$1 \to G' \times_S H' \to G \times_S H \to G'' \times_S H'' \to 1.$$

(iii) If two of the S*-groups* G', G, G'' *satisfy (E), the third also satisfies (E).*

(iv) If $0 \to F' \to F \to F'' \to 0$ *is an exact sequence of* $\mathcal{O}_S$*-modules and if two of the modules* F', F, F'' *are good, the third is also.*

(v) If two of the S*-groups* G', G, G'' *are good, the third is also.*

[171] The first part of (i) is immediate, and the second part follows. Likewise, (ii) is immediate. Let us prove (iii). To abbreviate, write $L(M) = Lie(G, M)$, $L'(M) = Lie(G', M)$, etc. Then one has the following commutative diagram

$$\begin{tikzcd}
1 \arrow[r] & L'(M \oplus N) \arrow[r] \arrow[d] & L(M \oplus N) \arrow[r] \arrow[d] & L''(M \oplus N) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & L'(M) \times_S L'(N) \arrow[r] & L(M) \times_S L(N) \arrow[r] & L''(M) \times_S L''(N) \arrow[r] & 1
\end{tikzcd}$$

in which the first (resp. the second) row is exact by (i) (resp. (i) and (ii)). Assertion (iii) follows.

Let us prove (iv). One has a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & F' \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(M) \arrow[r] \arrow[d] & F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(M) \arrow[r] \arrow[d] & F'' \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(M) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & T_{F'/S}(M) \arrow[r] & T_{F/S}(M) \arrow[r] & T_{F''/S}(M) \arrow[r] & 0
\end{tikzcd}$$

with exact rows (the first, because $T_{\mathcal{O}_S/S}(M)$ is a free $\mathcal{O}_S$-module hence flat, the second by (i)). It follows that if two of the modules F', F, F'' are good, the third is also. Finally, (v) follows from (iii) and (iv).

Lemma 5.2.2. *Let* G *be an* S*-group,* E*,* H *two* G*-objects,* F *a* G*-*$\mathcal{O}_S$*-module.*

(i) The canonical homomorphism $E^G \times_S H^G \to (E \times_S H)^G$ *is an isomorphism.*

(ii) If F *is a good* $\mathcal{O}_S$*-module, so is* F^G*.*

[172] The first assertion is immediate; let us prove the second. For every $S' \to S$, one has a commutative diagram

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[171] One has added the proof of points (iii) and (v).

[172] One has detailed the original in what follows.

$$\begin{array}{ccc} F^G(I_{S'}(M)) & \longrightarrow & F(I_{S'}(M)) \\ \uparrow \varphi & & \uparrow \simeq \varphi_1 \\ F^G(S') \otimes_{O(S')} O(I_{S'}(M)) & \longrightarrow & F(S') \otimes_{O(S')} O(I_{S'}(M)) \end{array}$$

and one must show that ϕ is bijective; now it is evidently injective; let us show that it is surjective. Let $(t_0, \cdots, t_n)$ be a basis of the free $O(S')$-module $O(I_{S'}(M))$ and let

$$u = \sum_{i=0}^n f_i \otimes t_i$$

be an element of $F(S') \bigotimes_{O(S')} O(I_{S'}(M))$ such that $\phi_1(u)$ belongs to $F^G(I_{S'}(M))$. Let us show that the f_i belong to $F^G(S')$.

Let $S'' \to S'$; one can consider S'' as above $I_{S'}(M)$ via the zero section ϵ_M. Then, for every $g \in G(S'')$, one has:

$$u_{S''} = g \cdot u_{S''} = \sum_{i=0}^n g \cdot (f_i)_{S''} \otimes (t_i)_{S''}.$$

Since the $(t_i)_{S''}$ form a basis of $O(I_{S''}(M)) \simeq O(I_{S'}(M)) \bigotimes_{O(S')} O(S'')$ over $O(S'')$, it follows that $g \cdot (f_i)_{S''} = (f_i)_{S''}$, whence $f_i \in F^G(S')$ for every i.

Notations.[173] If E is an S-group and F a sub-S-group of E, one denotes by E/F the S-functor that to every $S' \to S$ associates the set $E(S')/F(S')$ of classes $\bar{x} = xF(S')$, $x \in E(S')$. If E is a commutative S-group then E/F is a commutative S-group.

Let now G be an S-group and K a sub-S-group of G; set $E = Lie(G/S, M)$ and $F = Lie(K/S, M)$. The adjoint action of K on E stabilizes F, hence induces an action of K on the S-functor E/F. Then, for every $S' \to S$, one has:

$$(E/F)^K(S') = \{ \bar{x} \in E(S')/F(S') \mid \text{for every } f : S'' \to S' \text{ and } k \in K(S''), \; f^*(\bar{x}^{-1})\, \mathrm{Ad}(k)(f^*(\bar{x})) \in F(S'') \},$$

where $f^*(\bar{x})$ denotes the image of $\bar{x}$ in $E(S'')$.

Theorem 5.2.3. *Let G be an S-group, K a sub-S-group of G. Write (I 2.3.3)*

$$N = \mathrm{Norm}_G(K), \quad Z = \mathrm{Centr}_G(K).$$

Make K act on $Lie(G/S, M)$ through the adjoint representation of G.

(i) If the group law of $Lie(G/S, M)$ is commutative[174] [175]*, then*

$$Lie(N/S, M) / Lie(K/S, M) = (Lie(G/S, M) / Lie(K/S, M))^K.$$

(ii) If the group law of $Lie(G/S, M)$ is commutative[176]*, then*

$$Lie(Z/S, M) = Lie(G/S, M)^K.$$

(iii) If G satisfies (E) (resp. if G and K satisfy (E)), then Z satisfies (E) (resp. N satisfies (E)).

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[173] One has detailed these notations.
[174] Condition automatically verified if G satisfies (E) (cf. 3.9), for example if G is representable.
[175] For an example where the S-group $Lie(G/S)$ is not commutative, see 6.3.
[176] Condition automatically verified if G satisfies (E) (cf. 3.9), for example if G is representable.

(iv) Suppose G is good; then Z is good; if moreover G is very good, then Z is very good.

(v) Suppose G and K are good; then N is good; if moreover G is very good, then N is very good.

To prove (i) and (ii)[177] we shall use the following lemma, which follows from the diagram of exact sequences considered in 4.1.B (with G and H interchanged).

Lemma 5.2.3.0. *Let G be an S-group, H a sub-S-group of G, and M a free O_S-module of finite type. Then $T_{H/S}(M)$ and $Lie(G/S, M)$ are sub-S-groups of $T_{G/S}(M)$ and one has:*

$$T_{H/S}(M) \cap_T Lie(G/S, M) = Lie(H/S, M),$$

where one has set $T_{H/S}(M) \cap_T Lie(G/S, M) :=_{def} T_{H/S}(M) \times_{T_{G/S}(M)} Lie(G/S, M)$.

Since the functors considered in (i) and (ii) commute with base extension, it suffices to show the equalities of S-points.

Set $H = N$ (resp. $= Z$) and let $\alpha = \pm 1$. By the preceding lemma and the definition of N and Z (cf. I 2.3.3), one has: $Lie(H/S, M)(S) =$

$$\{ X \in Lie(G/S, M)(S) \subset G(I_S(M)) \mid \text{for every } f : S' \to I_S(M) \text{ and } u \in K(S'), \; f^*(X^\alpha) \cdot u \cdot f^*(X^{-\alpha}) \cdot u^{-1} \in K(S'), \text{ resp. } f^*(X) \cdot u \cdot f^*(X^{-1}) \cdot u^{-1} = 1 \quad (*) \},$$

where $f^*(X)$ denotes the image of X in $G(S')$.

To simplify the writing, let us write

$$\mathfrak{g} = Lie(G/S, M), \quad \mathfrak{k} = Lie(K/S, M), \quad \mathfrak{n} = Lie(N/S, M), \quad \mathfrak{z} = Lie(Z/S, M).$$

If $X \in Lie(H/S, M)(S)$, the equalities $(*)$ are valid for every $f : S' \to S$, since $f = \rho \circ \epsilon_M \circ f$ factors through $I_S(M)$ (where $\rho : I_S(M) \to S$ is the structural morphism and $\epsilon_M : S \to I_S(M)$ the zero section). One deduces that

$$\mathfrak{n}(S)/\mathfrak{k}(S) \subset (\mathfrak{g}/\mathfrak{k})^K(S) \quad \text{and} \quad \mathfrak{z}(S) \subset \mathfrak{g}^K(S).$$

To prove the reverse inclusions, suppose henceforth $g = Lie(G/S, M)$ commutative; then g^K and $(g/k)^K$ are commutative S-groups. Let $X \in g(S)$ and $\bar{X}$ its image in $(g/k)(S)$, suppose that $\bar{X} \in (g/k)^K(S)$ (resp. $X \in g^K(S)$) and let us show that $X \in n(S)$ (resp. $X \in z(S)$).

Let $f : S' \to I_S(M)$; let us show that the preceding condition $(*)$ is verified for every $u \in K(S')$. Consider the cartesian square

$$\begin{array}{ccc} I_{S'}(M) & \xrightarrow{p} & I_S(M) \\ \Big\downarrow{\scriptstyle \rho'} & & \Big\downarrow{\scriptstyle \rho} \end{array}$$

[177] One has detailed the proof, adding in particular Lemma 5.2.3.0, implicit in the original.

$$\begin{tikzcd} {} \arrow[d] & {} \arrow[d] \\ S' \arrow[r, "\rho \circ f"] & S \end{tikzcd}$$

and let h be the section of ρ' defined by f. It suffices to show that, for every $v \in K(I_{S'}(M))$, one has

$$(**) \quad \begin{aligned} & p^*(X^\alpha) \cdot v \cdot p^*(X^{-\alpha}) \cdot v^{-1} \in K(I_{S'}(M)), \\ & \text{resp. } p^*(X) \cdot v \cdot p^*(X^{-1}) \cdot v^{-1} = 1. \end{aligned}$$

Indeed, taking $v = \rho'^*(u)$ and applying h^* to $(**)$, one obtains $(*)$, since $\rho' \circ h = id_{S'}$ and $p \circ h = f$.

Let us now show $(**)$; for simplicity, we shall write X instead of $p^*(X)$. Every $v \in K(I_{S'}(M))$ is written in a unique manner $Y \cdot k$ where $Y \in Lie(K/S, M)(S')$ and $k \in K(S')$. The expression $X^\alpha \cdot u \cdot X^{-\alpha} \cdot u^{-1}$ then becomes $X^\alpha \cdot Y \cdot (k \cdot X^{-\alpha} \cdot k^{-1}) \cdot Y^{-1}$ which, since $Lie(G/S, M)$ is commutative, is written $X^\alpha Ad(k)(X^{-\alpha})$. Now this is a priori in $Lie(G/S, M)(S')$; taking into account Lemma 5.2.3.0, the condition $(**)$ for X therefore becomes: for every $k \in K(S')$,

$$\begin{cases} Ad(k)(X) = X, & \text{if } H = Z; \\ X^\alpha Ad(k)(X^{-\alpha}) \in Lie(K/S, M)(S'), & \text{if } H = N. \end{cases}$$

When $H = Z$, this condition is indeed a consequence of the hypothesis $X \in g^K(S)$. When $H = N$, the condition is also written:

$$(*') \quad Ad(k)(X) = X \quad \text{and} \quad Ad(k)(X^{-1}) = X^{-1},$$

now the second condition of $(*')$ is a consequence of the first, since the action of K on g/k respects the group structure of this latter. Therefore, when $H = N$, the condition $(**)$ for X is indeed a consequence of the hypothesis $\bar{X} \in (g/k)^K(S)$. This proves (i) and (ii).

To prove (iii)–(v), let us write $g(M) = Lie(G/S, M)$ and define similarly $k(M)$, $z(M)$ and $n(M)$. If G/S satisfies (E), then $g(M \bigoplus N) \simeq g(M) \times_S g(N)$ and therefore, by 5.2.2 (i),

$$g(M \oplus N)^K \simeq g(M)^K \times_S g(N)^K,$$

whence $z(M \bigoplus N) \simeq z(M) \times_S z(N)$, so Z satisfies (E). If moreover K/S satisfies (E), one obtains successively isomorphisms:

$$\begin{aligned} (g/k)(M \oplus N) &\simeq (g/k)(M) \times_S (g/k)(N), \\ (g/k)^K(M \oplus N) &\simeq (g/k)(M)^K \times_S (g/k)(N)^K, \end{aligned}$$

then a commutative diagram:

$$\begin{tikzcd} 0 \arrow[r] & k(M \oplus N) \arrow[r] \arrow[d, "\wr"] & n(M \oplus N) \arrow[r] \arrow[d] & (n/k)(M \oplus N) \arrow[r] \arrow[d, "\wr"] & 0 \\ 0 \arrow[r] & k(M) \times_S k(N) \arrow[r] & n(M) \times_S n(N) \arrow[r] & (n/k)(M) \times_S (n/k)(N) \arrow[r] & 0 \end{tikzcd}$$

from which it follows that $n(M \bigoplus N) \simeq n(M) \times_S n(N)$, so N satisfies (E).

Henceforth, write $g = g(O_S) = Lie(G/S)$ and define similarly k, z, n. If G is good, g is a good 0_S-module so, by 5.2.2 (ii), $z \simeq g^K$ is also, so Z (which satisfies (E) by (iii)) is good. If moreover K is good, then k is a good 0_S-module so, by 5.2.1 (iv) and 5.2.2 (ii), so are g/k and $(g/k)^K$. Taking into account the exact sequence

$$0 \longrightarrow k \longrightarrow n \longrightarrow (g/k)^K \longrightarrow 0,$$

one obtains, by 5.2.1 (iv) again, that n is good. Finally, if in addition to the preceding conditions, G is very good, i.e., if one has identically $[x, x] = 0$ for every $x \in g$, it is clear that Z and N are very good. This proves (iii), (iv) and (v).

Corollary 5.2.3.1. *One has Lie(Centr(G)/S) = Lie(G/S)^G if the group law of $Lie(G/S)$ is commutative.*

Corollary 5.2.3.2. *If the group law of $Lie(G/S)$ is commutative and if K is a normal subgroup of G, then*

(Lie(G/S) / Lie(K/S))^K = Lie(G/S) / Lie(K/S).

5.3. Linear representations

Let G be a good S-group acting linearly on a good 0_S-module F via

$$\rho : G \longrightarrow \mathrm{Aut}_{O_S-mod.}(F).$$

One has defined (4.7.1 and 4.8.2) a corresponding linear representation

ρ′ : Lie(G/S) □ End_{0_S-mod.}(F).

The sub-S-groups $Norm_G(E)$ and $Centr_G(E)$ are defined for every part E of F, for example for every sub-0_S-module E of F.

Definition 5.3.0. *One will pose analogously: for every $S' \to S$,*

Norm_{Lie(G/S)}(E)(S′) = { X ∈ Lie(G/S)(S′) | ρ′(X) E_{S′} ⊂ E_{S′} };
Centr_{Lie(G/S)}(E)(S′) = { X ∈ Lie(G/S)(S′) | ρ′(X) E_{S′} = 0 }.

(Note that this construction can be made for any linear representation of an 0_S-“Lie algebra” (in the sense of 4.9) and that the two subobjects constructed are sub-0_S-modules stable under the bracket.)

Theorem 5.3.1. *Let G be a good S-group acting linearly on a good 0_S-module F and let E be a sub-0_S-module of F.*

(i) One has Lie(Centr_G(E)/S) = Centr_{Lie(G/S)}(E) and $Centr_G(E)$ is a good S-group; it is very good if G is.

(ii) Suppose that E is a good 0_S-module. Then[178] Lie(Norm_G(E)/S) = Norm_{Lie(G/S)}(E) and $Norm_G(E)$ is a good S-group; it is very good if G is.

[178] One has corrected the original, which stated the equality that follows without hypotheses on E; see 6.5.1 for counterexamples.

The proof is left to the reader.

5.3.2. Let G be a good S-group; what precedes applies in particular to the case where one takes for ρ the adjoint representation of G. Let E be a good[179] submodule of $Lie(G/S)$; to it one therefore associates two subgroups of G, its centralizer and its normalizer. By 5.3.1, their Lie algebras are respectively the centralizer and the normalizer of E in $Lie(G/S)$ computed as usual by means of the bracket:

```
Centr_{Lie(G/S)}(E)(S′) = { X ∈ Lie(G/S)(S′) | [X, E_{S′}] = 0 },
Norm_{Lie(G/S)}(E)(S′)  = { X ∈ Lie(G/S)(S′) | [X, E_{S′}] ⊂ E_{S′} }.
```

5.3.3. Let K be a sub-S-group of G, then $Lie(K/S)$ is a sub-0_S-module of $Lie(G/S)$; suppose that $Lie(K/S)$ is a good 0_S-module[180] (which is the case if K is a good S-group). One evidently has

```
Centr_G(K) ⊂ Centr_G(Lie(K/S)),
Norm_G(K) ⊂ Norm_G(Lie(K/S)),
```

whence, by 5.3.1,

```
Lie(Centr_G(K)/S) ⊂ Centr_{Lie(G/S)}(Lie(K/S)),
Lie(Norm_G(K)/S) ⊂ Norm_{Lie(G/S)}(Lie(K/S)),
```

but none of these four inclusions is a priori an equality; we shall see by the sequel many examples.

It follows in particular from these inclusions that if K is a normal subgroup of G, then $Lie(K/S)$ is an ideal of $Lie(G/S)$.

6. Miscellaneous remarks

6.1. One can define the bracket of two infinitesimal automorphisms for an S-functor X which is not necessarily a group. It suffices to apply the results of this Exposé to the group $\mathrm{Aut}_S(X)$. To arrive at an agreeable formalism, one is led to suppose X good, i.e. to suppose that the 0_X-module $T_{X/S}$ is good (if X is an S-group, this definition evidently coincides with Definition 4.6).

6.2. There exist functors possessing infinitesimal endomorphisms that are not automorphisms, hence a fortiori not satisfying condition (E).[181] For every pointed set (E, x_0), let $MA(E)$ be the free abelian monoid generated by E and let $MAP(E, x_0)$ be the abelian monoid obtained by quotienting $MA(E)$ by the equivalence relation generated by the relation $m \sim x_0 + m$. Then $(E, x_0) \mapsto MAP(E, x_0)$ is the left adjoint of the forgetful functor from the category of abelian monoids to that of pointed sets; one will say that $MAP(E, x_0)$ is the "free abelian monoid on the pointed set (E, x_0)".

[179] One has added this hypothesis; cf. 6.5.1.
[180] One has added this hypothesis; cf. 6.5.1.
[181] One has detailed the original in what follows. In particular, the correct definition of the "free abelian monoid on a pointed set" was pointed out to us by O. Gabber.

Take then for X the functor that to every scheme S associates the free abelian monoid on the set $O(S)$, pointed by the zero element. Every morphism $f : S \to I_{\mathbb{Z}} = \mathrm{Spec}(\mathbb{Z}[\epsilon])$ corresponds to an element of square zero u_f of $O(S)$, and hence defines an endomorphism of $X(S)$ by $x \mapsto x + u_f$ (sum in $MAP(O(S), 0)$). One thus obtains an endomorphism ϕ of $X_{I_{\mathbb{Z}}} = X \times_{\mathbb{Z}} I_{\mathbb{Z}}$, defined as follows. For every $f \in I_{\mathbb{Z}}(S)$ and $x \in X(S)$,

```
φ((x, f)) = (x + u_f, f).
```

If $f_0 : S \to I_{\mathbb{Z}}$ is the composite of the structural morphism $S \to \mathrm{Spec}(\mathbb{Z})$ and the zero section of $I_{\mathbb{Z}}$, the corresponding element is $u_{f_0} = 0$, and so $\phi((x, f_0)) = (x, f_0)$, since $x+0 = x$ in $MAP(O(S), 0)$. This shows that ϕ induces the identity on X; it is therefore an infinitesimal endomorphism of X which is evidently not an automorphism.

6.3. There exist modules that are not good. On the one hand, one can modify the preceding counterexample slightly:[182] take for $F(S)$ the free $O(S)$-module with basis the elements of $O(S)$; then F does not satisfy condition (E) with respect to $\mathrm{Spec}(\mathbb{Z})$; moreover, let G be the $\mathbb{Z}$-group defined by $G(S) = GL_n(R(S))$, where $n \geqslant 2$ and $R(S)$ is the $O(S)$-algebra of the abelian group $O(S)$; then the $\mathbb{Z}$-group $Lie(G/\mathbb{Z})$ is not commutative.

[183] On the other hand, one can give the following counterexamples. Let $S = \mathrm{Spec}(k)$, with k a field of characteristic $p > 0$.

a) Let F be the `O_S`-module which to every S-scheme T associates $F(T) = \Gamma(T, O_T)$ equipped with the $O(T)$-module structure obtained by making scalars act through the p-th power, that is to say, $r \cdot f = r^p f$, for $r \in O(T)$, $f \in F(T)$. As S-functor of groups, F is isomorphic to $G_{a,S}$. Therefore F/S satisfies (E) and $Lie(F/S)$ is identified with $Lie(G_{a,S}/S) \cong O_S$. Then, the canonical morphism $F \to Lie(F/S)$ is,

for every $T \to S$, the identity map $F(T) \to O(T)$: it respects the abelian-group structure but not the module structure. Therefore F is not good (cf. 4.4.1).

b) Let $\alpha_{p,k}$ be the k-functor of groups which to every S-scheme T associates

```
α_{p,k}(T) = { x ∈ O(T) | x^p = 0 };
```

it is represented by $\mathrm{Spec}(k[X]/(X^p))$ so it is a very good S-group; it is also equipped with an `O_S`-module structure, but it is not a good `O_S`-module, because the canonical morphism $\alpha_{p,k} \to Lie(\alpha_{p,k}/k) = G_{a,k}$ is not bijective.

6.4. Let G be a group-functor on S. One has by definition the following implications

```
(G/S satisfies (E)) ⇐ (G is good) ⇐ (G is very good).
```

[184] It would be interesting to prove or counter-exemplify the implications in the reverse direction.

[182] One has added what follows.
[183] In what follows, one has detailed example a) and added example b).
[184] And G is very good if it is representable (4.11). For representability criteria, see for example [MO67]. On the other hand, for the automorphisms of an affine algebraic group (over an algebraically closed field of characteristic 0), let us cite [HM69].

6.5.[185] Let Nil be the $\mathbb{Z}$-functor of groups defined as follows: for every scheme S, $Nil(S)$ is the ideal of $O(S)$ formed by the nilpotent elements, i.e.

$$Nil(S) = \{ x \in O(S) \mid \text{there exists } n \in \mathbb{N} \text{ such that } x^n = 0 \}.$$

(Nil is very good but is not representable.) Let Nil^2, $O_{réd}$ and F be the $\mathbb{Z}$-functors of groups that to every scheme S associate, respectively, the ideal $Nil(S)^2$ and

$$O_{réd}(S) = O(S) / Nil(S), \qquad F(S) = O(S) / Nil(S)^2.$$

One sees easily that $Lie(O_{réd}/\mathbb{Z}) = 0$, so the $O_{\mathbb{Z}}$-module $O_{réd}$ is not good (although it is a good $\mathbb{Z}$-group). If M, N are $\mathbb{Z}$-modules free of finite rank, one has

$$Nil^2(I_S(M \oplus N)) \simeq Nil^2(S) \oplus Nil(S) \otimes_{\mathbb{Z}} M \oplus Nil(S) \otimes_{\mathbb{Z}} N$$

and therefore

$$F(I_S(M \oplus N)) \simeq F(S) \oplus O_{réd}(S) \otimes_{\mathbb{Z}} M \oplus O_{réd}(S) \otimes_{\mathbb{Z}} N.$$

One deduces, on the one hand, that the $\mathbb{Z}$-functor of groups F satisfies condition (E) and, on the other hand, that $Lie(F/\mathbb{Z}) = O_{réd}$; since this latter is not a good $O_{\mathbb{Z}}$-module, this shows that F is an example of a $\mathbb{Z}$-group that satisfies (E) but is not good.

6.5.1. Let us also give the counterexamples mentioned in 5.3.1–5.3.3. Let S be a scheme, F the good O_S-module $O_S^{\oplus 2}$ equipped with the natural action of the good S-group $G = GL_{2,S}$, and E the sub-O_S-module of F formed by the pairs (x_1, x_2) such that x_2 is nilpotent. Set $N = Norm_G(E)$. Then $Lie(N/S) = Lie(G/S)$ while, for every $S' \to S$, one has

$$Norm_{Lie(G/S)}(E)(S') = \left\{ \begin{bmatrix} a & b \\ x & c \end{bmatrix} \mid a, b, c, x \in O(S'),\ x \text{ nilpotent} \right\},$$

so $Lie(Norm_G(E)/S) \neq Norm_{Lie(G/S)}(E)$.

By considering the semi-direct product $G_1 = F \cdot G$, one obtains an analogous counterexample where E is a sub-O_S-module of $Lie(G_1/S)$; moreover, with the notations introduced above, $E = Lie(K/S)$ where K is the subgroup $O_S \bigoplus Nil^2$ of F (i.e., for every $S' \to S$, $K(S')$ is formed by the pairs (x_1, x_2) such that $x_2 \in Nil(S')^2$).

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[186]

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[185] One has added this paragraph.
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Footnotes

Exposé III. Infinitesimal extensions

by M. Demazure

[^N.D.E-III-0] In this Exposé, we place ourselves in the following general situation. We have a scheme S and a coherent nilpotent ideal I on S. We denote by S_n the closed subscheme of S defined by the ideal I^{n+1} ($n \geq 0$). In particular S_0 is defined by I. Since I is nilpotent, S_n is equal to S for n large enough, and the `Sᵢ` have the same underlying topological space. A typical example of this situation is the following: S is the spectrum of a local Artinian ring A, I is the ideal defined by the radical of A, so that S_0 is the spectrum of the residue field of A.

In the preceding situation, one is given a certain number of data above S_0, and one looks above S for data which lift them, that is to say, which give them back by base change from S to S_0. This is done step by step, by way of the S_n. At each step, we propose to define the obstructions encountered, and to classify, when they exist, the solutions obtained.

The passage from S_n to S_{n+1} can be generalized as follows: one has a scheme S, two ideals I and J with $I \supset J$, and $I{\cdot}J = 0$ (in the preceding case S, I and J are respectively S_{n+1}, I/I^{n+2}, I^{n+1}/I^{n+2}). We denote by S_0 (resp. `S_J`) the closed subscheme of S defined by I (resp. J), and we pose a problem of extension from `S_J` to S.

In (SGA 1, III), problems of extension of morphisms of schemes and of extension of schemes were treated. We pose here the problems of extension of morphisms of groups, of extension of group structures, and of extension of subgroups.

We have gathered in a § 0 the results of (SGA 1, III) which will be useful to us, in order to put them in the most convenient form for our purpose, and to spare the reader from having to refer constantly to (SGA 1, III).[^N.D.E-III-1] § 1 collects some computations of group cohomology which will be useful in what follows and which have nothing to do with scheme theory. §§ 2 and 3 treat respectively of the extension of group morphisms and the extension of group structures. In § 4, we have briefly recalled the proof of a result stated in (TDTE IV) concerning the extension of subschemes, and applied this result to the problem of extension of subgroups. For the rest of the Seminar, only the result of § 2, concerning the extension of morphisms of groups, will be indispensable.[^N.D.E-III-2]

The idea of reducing infinitesimal extension problems to the usual computations of cohomology in extensions of groups was suggested by J. Giraud at the oral exposé (whose computations were notably more complicated and less transparent). Unfortunately, it seems that this method only applies well to the first two problems studied, and we have been unable to escape rather painful computations in the case of extensions of subgroups.

To simplify the language, we shall call a Y-functor, resp. Y-scheme, etc., a functor, resp. scheme, etc., equipped with a morphism into the functor Y, thus extending the definitions of Exposé I (which concerned only the case of a representable Y).

0. Reminders from SGA 1 III and various remarks

Let us first state a general definition.

Definition 0.1. *Let C be a category, X an object of $\hat{C}$, G a $\hat{C}$-group acting on X. We say that X is* ***formally principal homogeneous***[^N.D.E-III-3] *under G if the following equivalent conditions are satisfied:*

(i) for each object S of C, the set $X(S)$ is empty or principal homogeneous under $G(S)$;

(ii) the morphism of functors $G \times X \to X \times X$ defined set-theoretically by $(g, x) \mapsto (gx, x)$ is an isomorphism.

This being so, we shall put the results of (SGA 1, III, § 5)[^N.D.E-III-4] in the form which will be most useful to us. We shall employ the following general notations throughout this section. We have a scheme S and on S two quasi-coherent ideals I and J such that

I ⊃ J   and   I · J = 0.

In particular we shall therefore have $J^2 = 0$. We shall denote by S_0 (resp. S_J) the closed subscheme of S defined by the ideal I (resp. J). For every S-functor X, we shall systematically denote by X_0 and X_J the functors obtained by base change from S to S_0 and S_J. Same notation for a morphism.

Definition 0.1.1.[^N.D.E-III-5] *Let X be an S-functor. Define a functor X^+ above S by the formula:*

Hom_S(Y, X⁺) = Hom_{S_J}(Y_J, X_J) = Hom_S(Y_J, X)

for a variable S-scheme Y. In the notations of (Exp. II, 1), one has

$$X^+ \simeq \prod_{S_J/S} X_J.$$

The identity morphism of X_J defines by construction an S-morphism

$$p_X : X \to X^+.$$

[^N.D.E-III-6] Explicitly, for every S-scheme Y, the map

p_X(Y) : Hom_S(Y, X) → Hom_S(Y, X⁺) = Hom_S(Y_J, X) is the map induced by the morphism $Y_J \to Y$.

Remark 0.1.2. *Let us now observe that if X is an S-group functor, then X_J is an S_J-group functor; then the formula defining X^+ endows it with a structure of S-group functor, and the description of p_X above shows that $p_X : X \to X^+$ is a morphism of S-group functors.*

Remark 0.1.3. *On the other hand, for every S-group functor Y, one has:*

$$\mathrm{Hom}_{S\text{-gr.}}(Y, X^+) = \mathrm{Hom}_{S_J\text{-gr.}}(Y_J, X_J).$$

Indeed, let $f \in \mathrm{Hom}_S(Y, X^+)$, corresponding to $f_J \in \mathrm{Hom}_{S_J}(Y_J, X_J)$; the condition that $f \in \mathrm{Hom}_{S-gr.}(Y, X^+)$ is that, for every $T \to S$ and $y, y' \in Y(T)$, one has $f(y \cdot y') = f(y) \cdot f(y')$, and this is equivalent to

$$f_J(y_J) \cdot f_J(y_J') = f_J((y \cdot y')_J);$$

since $(y \cdot y')_J = y_J \cdot y_J'$ (because $Y(T) \to Y(T_J) = Y_J(T_J)$ is a morphism of groups), this is the condition for f_J to be a morphism of groups. Applying this to $Y = X$, we recover that p_X, which corresponds to id_{X_J}, is a morphism of S-group functors.

Let us now return to the general case, but assume that X is an S-scheme. Since an S-morphism of a variable S-scheme Y into X^+ is by definition an S_J-morphism g_J of Y_J into X_J, we shall define an X^+-functor in abelian groups L_X as follows.

Scholie 0.1.4.[^N.D.E-III-7] *If $\pi : Y \to S$ is a morphism of schemes and M an O_S-module, we denote by $M \bigotimes_{O_S} O_Y$ the inverse image $\pi * (M)$. If J is an ideal of O_S, we denote by $J \cdot O_Y$ the ideal of O_Y image of the morphism*

$$\pi^*(J) = J \otimes_{O_S} O_Y \to O_Y .$$

Note that, for every morphism of S-schemes $f : Z \to Y$, one has an epimorphism of O_Z-modules:

$$f^*(J \cdot O_Y) \to J \cdot O_Z , \qquad (0.1.4)$$

as follows from the commutative diagram below:

$$\begin{tikzcd} J \otimes_{O_S} O_Y \otimes_{O_Y} O_Z \arrow[r] \arrow[d] & J \otimes_{O_S} O_Z \arrow[d] \\ f^*(J \cdot O_Y) = (J \cdot O_Y) \otimes_{O_Y} O_Z \arrow[r] & J \cdot O_Z . \end{tikzcd}$$

Definition 0.1.5.[^N.D.E-III-8] *Let X be an S-scheme. For every X^+-scheme Y, given by a morphism $g_J : Y_J \to X_J$, we set:*

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{O_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), J \cdot O_Y),$$

where $\Omega^1_{X_0/S_0}$ denotes the module of relative differentials of X_0 with respect to S_0 (cf. SGA 1, I.1 or EGA IV₄, 16.3), and where $J \cdot O_Y$ is regarded as an O_{Y_0}-module thanks to the fact that it is annihilated by I.

Then L_X is an X^+-functor in abelian groups.[^N.D.E-III-9] Indeed, for every X^+-morphism $f : Z \to Y$, the functor f_0* and the morphism $f_0*(J \cdot O_Y) \to J \cdot O_Z$ of (0.1.4) induce a natural morphism of abelian groups $L_X(f)$:

$$\begin{aligned} &\mathrm{Hom}_{O_{Y_0}}(g_0*(\Omega^1_{X_0/S_0}), J \cdot O_Y) \\ &\quad\to \mathrm{Hom}_{O_{Z_0}}(f_0* g_0*(\Omega^1_{X_0/S_0}), f_0*(J \cdot O_Y)) \\ &\quad\to \mathrm{Hom}_{O_{Z_0}}(f_0* g_0*(\Omega^1_{X_0/S_0}), J \cdot O_Z). \end{aligned}$$

Let us finally note that $L_X(f)$ is described locally as follows. Note first that Y and Y_J have the same underlying topological space, and similarly for V and V_J if V is an open subset of Y. Let then $U = \mathrm{Spec}(A)$ be an affine open of X above an affine open $\mathrm{Spec}(\Lambda)$ of S, $V = \mathrm{Spec}(B)$ an affine open of Y such that $g_J(V_J) \subset U$, and $W = \mathrm{Spec}(C)$ an affine open of $f^{-1}(V)$. Let J and I be the ideals of Λ corresponding to J and I. Then f (resp. g_J) induces a morphism of Λ-algebras $\theta : B \to C$ (resp. $\phi : A \to B/JB$), and one obviously has $\theta(JB) \subset JC$. On the other hand, $m \in \mathrm{Hom}_{O_{Y_0}}(g_0*(\Omega^1_{X/S}), J \cdot O_Y)$ induces an element D of

$$\mathrm{Hom}_{O_{V_0}}(g_0*(\Omega^1_{U/S}), J \cdot O_V) = \mathrm{Hom}_{B/IB}(\Omega^1_{A/\Lambda} \otimes_A B/IB, JB) = \mathrm{Der}_\Lambda(A, JB),$$

and the image of $L_X(f)(m)$ in

$$\mathrm{Hom}_{O_{W_0}}(f_0* g_0*(\Omega^1_{U/S}), J \cdot O_Z) = \mathrm{Hom}_{C/IC}(\Omega^1_{A/\Lambda} \otimes_A C/IC, JC) = \mathrm{Der}_\Lambda(A, JC)$$

is none other than $\theta \circ D$.

Remark 0.1.6.[^N.D.E-III-10] *Let $f : X \to W$ be an S-morphism. It induces an S-morphism $f^+ : X^+ \to W^+$ defined as follows: if g is an element of $\mathrm{Hom}_S(Y, X^+)$, corresponding to an S-morphism $g_J : Y_J \to X$, then $f^+(g)$ is the element $f \circ g_J$ of $\mathrm{Hom}_S(Y, W^+)$. It is clear that the following diagram is commutative:*

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "p_X"'] & W \arrow[d, "p_W"] \\ X^+ \arrow[r, "f^+"] & W^+ \end{tikzcd} .$$

Reminders 0.1.7.[^N.D.E-III-11] *In this paragraph, given an S-scheme X, we "recall" certain functorial properties of the module of differentials $\Omega^1_{X/S}$ and of the first infinitesimal neighborhood of the diagonal $\Delta^{(1)}_{X/S}$, properties which follow easily from (EGA IV₄, §§ 16.1–16.4), but which do not figure there explicitly.*

a) Let us begin by recalling the following facts (cf. EGA II, §§ 1.2–1.5). Let $g : Y \to X$ be a morphism of schemes, $\pi : X' \to X$ an affine X-scheme, B the quasi-coherent O_X-algebra $\pi_*(O_{X'})$; then the Y-scheme $Y \times_X X'$ is affine and corresponds to the quasi-coherent O_Y-algebra $g*(B)$, and one has a commutative diagram of bijections:

$$\begin{tikzcd} \mathrm{Hom}_X(Y, X') \arrow[r, "\sim"] \arrow[d] & \mathrm{Hom}_Y(Y, Y \times_X X') \arrow[d] \\ \square & \square \end{tikzcd}$$

$$\mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(\mathcal{B}, g_*(\mathcal{O}_Y)) \xrightarrow{\ \sim\ } \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^*(\mathcal{B}), \mathcal{O}_Y).$$

Moreover, these bijections are functorial in the pair (X, X'), i.e., if W' is an affine W-scheme, corresponding to the quasi-coherent $\mathcal{O}_W$-algebra $\mathcal{A}$, if one has a commutative diagram of morphisms of schemes

$$\begin{tikzcd}
 & X' \arrow[r, "f'"] & W' \arrow[d, dash] \\
Y \arrow[ur, "g'"] \arrow[r, "g"] & X \arrow[r, "f"] & W ,
\end{tikzcd}$$

and if we denote by $\phi : \mathcal{A} \to f_*(\mathcal{B})$ and $\phi^\sharp : f^*(\mathcal{A}) \to \mathcal{B}$ (resp. $\theta : \mathcal{B} \to g_*(\mathcal{O}_Y)$ and $\theta^\sharp : g^*(\mathcal{B}) \to \mathcal{O}_Y$) the algebra morphisms associated to f' (resp. to a variable X-morphism $g' : Y \to X'$), then one has the following commutative diagram (where Y is viewed as a W-scheme via $f \circ g$):

$$\begin{tikzcd}
\mathrm{Hom}_X(Y, X') \arrow[r, "g' \mapsto f' \circ g'"] & \mathrm{Hom}_W(Y, W') \arrow[d, "\simeq"] \\
 & \mathrm{Hom}_{\mathcal{O}_W\text{-alg.}}(\mathcal{A}, f_* g_*(\mathcal{O}_Y)) \arrow[d, "\simeq"] \\
\mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(\mathcal{B}, g_*(\mathcal{O}_Y)) \arrow[r, "\theta \mapsto \theta \circ \varphi^\sharp"] \arrow[d, "\simeq"] & \mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(f^*(\mathcal{A}), g_*(\mathcal{O}_Y)) \arrow[d, "\simeq"] \\
\mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^*(\mathcal{B}), \mathcal{O}_Y) \arrow[r, "\psi \mapsto \psi \circ g^*(\varphi^\sharp)"] & \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^* f^*(\mathcal{A}), \mathcal{O}_Y).
\end{tikzcd}$$

b) Let now X be an S-scheme. Let $\Omega^1_{X/S}$ be the module of differentials of X over S, and $\Delta^{(1)}_{X/S}$ the first infinitesimal neighborhood of the diagonal immersion $\delta_X : X \to X \times_S X$; it is a subscheme of $X \times_S X$, of which the diagonal $\Delta_{X/S}$ is a closed subscheme. We denote by pr^i_X ($i = 1, 2$) the two projections $X \times_S X \to X$, and by π_X the restriction of pr^1_X to $\Delta^{(1)}_{X/S}$.

On the one hand, every morphism $f : X \to W$ of S-schemes induces an S-morphism $\Delta^{(1)} f : \Delta^{(1)}_{X/S} \to \Delta^{(1)}_{W/S}$ such that the following diagram is commutative:

$$\begin{tikzcd}
X \arrow[r, "\delta_X"] \arrow[d, "f"'] & \Delta^{(1)}_{X/S} \arrow[r] \arrow[d, "\Delta^{(1)} f"] & X \times_S X \arrow[r, "pr_X"] \arrow[d, "f \times f"] & X \arrow[d, "f"] \\
W \arrow[r, "\delta_W"] & \Delta^{(1)}_{W/S} \arrow[r] & W \times_S W \arrow[r, "pr_W"] & W .
\end{tikzcd}$$

On the other hand, $\Delta^{(1)}_{X/S}$ is, via the projection π_X, an affine X-scheme, spectrum of the augmented quasi-coherent $\mathcal{O}_X$-algebra

$$P^1_{X/S} = \mathcal{O}_X \bigoplus \Omega^1_{X/S},$$

where $\Omega^1_{X/S}$ is an ideal of square zero; the augmentation is the morphism of $\mathcal{O}_X$-algebras $\epsilon_X : P^1_{X/S} \to \mathcal{O}_X$ which vanishes on $\Omega^1_{X/S}$ and which corresponds to the closed immersion $\delta_X : X \hookrightarrow \Delta^{(1)}_{X/S}$. Then, every morphism of S-schemes $f : X \to W$ induces a morphism of augmented $\mathcal{O}_X$-algebras

$$f^*(P^1_{W/S}) = \mathcal{O}_X \oplus f^*(\Omega^1_{W/S}) \to P^1_{X/S} = \mathcal{O}_X \oplus \Omega^1_{X/S}$$

that is, equivalently, a morphism of $\mathcal{O}_X$-modules

$$f_{X/W/S} : f*(\Omega^1_{W/S}) \to \Omega^1_{X/S},$$

cf. (EGA IV_4, 16.4.3.6) (and (16.4.18.2) for the notation $f_{X/W/S}$).

Since $\pi_X : \Delta^{(1)}_{X/S} \to X$ is affine, then, by a), $\Delta^{(1)}f$ is entirely determined by $f_{X/W/S}$ and, for every X-scheme $g : Y \to X$, the set

$$\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \simeq \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(\mathcal{O}_Y \oplus g^*(\Omega^1_{X/S}), \mathcal{O}_Y)$$

is identified with a subset of $\mathrm{Hom}_{\mathcal{O}_Y}(g*(\Omega^1_{X/S}), \mathcal{O}_Y)$, namely the subset

$$\tilde{\mathrm{Hom}}_{\mathcal{O}_Y}(g*(\Omega^1_{X/S}), \mathcal{O}_Y)$$

formed by the $\mathcal{O}_Y$-morphisms $\psi : g*(\Omega^1_{X/S}) \to \mathcal{O}_Y$ such that $Im(\psi)$ is an ideal of $\mathcal{O}_Y$ of square zero.[^N.D.E-III-12]

Consequently, applying a) to the diagram

$$\begin{tikzcd}
\Delta^{(1)}_{X/S} \arrow[r, "\Delta^{(1)} f"] & \Delta^{(1)}_{W/S} & \\
Y \arrow[u, "g'"] \arrow[r, "g"] & X \arrow[r, "f"] & W
\end{tikzcd}$$

and taking into account that $\Delta^{(1)}f$ is the restriction to $\Delta^{(1)}_{X/S}$ of $f \times f$, one obtains the following commutative diagram, functorial in the X-scheme $Y \xrightarrow{g} X$:

$$\begin{tikzcd}
\mathrm{Hom}_X(Y, X \times_S X) \arrow[r, "(g, g') \mapsto (f\circ g, f\circ g')"] & \mathrm{Hom}_W(Y, \Delta^{(1)}_{W/S}) \\
\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \arrow[u] \arrow[r, "g' \mapsto \Delta^{(1)}f \circ g'"] & \mathrm{Hom}_W(Y, \Delta^{(1)}_{W/S}) \arrow[u]
\end{tikzcd} \qquad (0.1.7\ (*))$$

$$\begin{tikzcd}[column sep=huge]
{} \arrow[d, "\simeq"] & {} \arrow[d, "\simeq"] \\
\widetilde{\mathrm{Hom}}_{O_Y}(g^*(\Omega^1_{X/S}), O_Y) \arrow[r, "\psi \mapsto \psi \circ g^*(f_{X/W/S})"] & \widetilde{\mathrm{Hom}}_{O_Y}(g^* f^*(\Omega^1_{W/S}), O_Y).
\end{tikzcd}$$

Remark 0.1.7.1. *Let us end this paragraph with the following remark, which will be useful later (cf. 0.1.10). If we denote by $\tilde{L}_X$ the X-functor which to every X-scheme $g : Y \to X$ associates $\mathrm{Hom}_{O_Y}(g*(\Omega^1_{X/S}), O_Y)$, and $\tilde{L}_f : \tilde{L}_X \to \tilde{L}_W$ the morphism of functors defined above (which to every $\psi \in \tilde{L}_X(Y)$ associates $\psi \circ g*(f_{X/W/S})$), what precedes shows that we have a commutative diagram of functors:*

$$\begin{tikzcd}
X \tilde{\times}_X \tilde{L}_X \arrow[d, "f \tilde{\times} f"'] & \Delta^{(1)}_{X/S} \arrow[l, "\sim"'] \arrow[r] \arrow[d, "\Delta^{(1)} f"'] & X \times_S X \arrow[d, "f \times f"'] \\
W \tilde{\times}_W \tilde{L}_W & \Delta^{(1)}_{W/S} \arrow[l, "\sim"'] \arrow[r] & W \times_S W.
\end{tikzcd}$$

Theorem 0.1.8. *(SGA 1, III 5.1)[^N.D.E-III-13] Let Y, X be two S-schemes, J a quasi-coherent ideal of O_Y of square zero, Y_J the closed subscheme of Y defined by J, and $g_J : Y_J \to X$ an S-morphism.*

a) The set $P(g_J)$ of S-morphisms $g : Y \to X$ which extend g_J is either empty, or principal homogeneous under the abelian group

$$\mathrm{Hom}_{O_{Y_J}}(g_J * (\Omega^1_{X/S}), J).$$

b) If $i : Y_0 \hookrightarrow Y_J$ is the closed immersion defined by a quasi-coherent ideal $I \supset J$ such that $I \cdot J = 0$, and if $g_0 = g_J \circ i$, the preceding abelian group is isomorphic to

$$\mathrm{Hom}_{O_{Y_0}}(g_0 * (\Omega^1_{X/S}), J).$$

Proof. (b) follows at once from (a). Indeed, J, being annihilated by I, can be considered as an O_{Y_0}-module, whence, by adjunction:

$$\mathrm{Hom}_{O_{Y_J}}(g_J^*(\Omega^1_{X/S}), J) = \mathrm{Hom}_{O_{Y_0}}(i^* g_J^*(\Omega^1_{X/S}), J).$$

To prove (a), we may assume $P(g_J) \neq \emptyset$, i.e. that there exists an S-morphism $g : Y \to X$ extending g_J. Let us denote by j the immersion $Y_J \hookrightarrow Y$. Then $P(g_J)$ is the set of S-morphisms $g' : Y \to X$ such that $g' \circ j = g_J$. The datum of such a g' is equivalent to the datum of an S-morphism

$$h : Y \to X \times_S X$$

such that $pr_1 \circ h = g$ and $h_J = \delta \circ g_J$, where $h_J = h \circ j$ and δ is the diagonal immersion $X \hookrightarrow X \times_S X$:

$$\begin{tikzcd}[column sep=huge]
X \times_S X & Y_J \arrow[l, "h_J = \delta \circ g_J"'] \arrow[d, dash] \\
{} \arrow[u, "h"', dash] & {}
\end{tikzcd}$$

```
pr₁|   \                      j |
   ▼    \                       ▼
   X  ◄─g──                     Y .
```

Since h_J factors through δ and Y is in the first infinitesimal neighborhood of the immersion $j : Y_J \to Y$ (since $J^2 = 0$), then, by functoriality (cf. EGA IV$_4$, 16.2.2 (i)), the h's sought factor uniquely through $\Delta^{(1)}_{X/S}$ (cf. 0.1.7). Set

$$Y' = \Delta^{(1)}_{X/S} \times_X Y \quad \text{and} \quad Y_J' = Y' \times_Y Y_J = \Delta^{(1)}_{X/S} \times_X Y_J .$$

Then the h's sought are in bijection with the sections u of $Y' \to Y$ which extend the section $u_J = (\delta \circ g_J, id)$ of $Y_J' \to Y_J$. On the other hand, Y' (resp. Y_J') is an affine scheme over Y (resp. Y_J), corresponding to the quasi-coherent algebra

$$A = O_Y \oplus g^*(\Omega^1_{X/S}), \quad \text{resp.} \quad A_J = A \otimes_{O_Y} O_{Y_J} = O_{Y_J} \oplus g_J^*(\Omega^1_{X/S}).$$

Let us denote by $\epsilon : A \to O_Y$ the canonical augmentation of A (i.e. the morphism of O_Y-algebras $A \to O_Y$ which vanishes on $g*(\Omega^1_{X/S})$), and likewise define $\epsilon_J : A_J \to O_{Y_J}$. Then,

$$\Gamma(Y'/Y) \simeq \mathrm{Hom}_{O_Y\text{-alg.}}(A, O_Y), \quad \Gamma(Y_J'/Y_J) \simeq \mathrm{Hom}_{O_{Y_J}\text{-alg.}}(A_J, O_{Y_J})$$

and, via these isomorphisms, the section $u = (\delta \circ g, id)$ (resp. u_J) corresponds to ϵ (resp. ϵ_J). Consequently, $P(g_J)$ is in bijection with the set of algebra morphisms $A \to O_Y$ which reduce to ϵ_J, and via this bijection, g corresponds to ϵ.

Set $M = g * (\Omega^1_{X/S})$. Then $\mathrm{Hom}_{O_Y-alg.}(A, O_Y)$ is identified with the set of O_Y-morphisms $\psi : M \to O_Y$ such that $Im(\psi)$ is an ideal of square zero, and we are interested in those which induce the null morphism $M \to O_{Y_J} = O_Y/J$, i.e. which send M into J. Conversely, since $J^2 = 0$, every O_Y-morphism $\phi : M \to J$ comes from a (unique) algebra morphism $A \to O_Y$, reducing to ϵ_J. Finally, we have $\mathrm{Hom}_{O_Y}(g * (\Omega^1_{X/S}), J) = \mathrm{Hom}_{O_{Y_J}}(g_J * (\Omega^1_{X/S}), J)$ since $J^2 = 0$ (cf. the proof of (b) already seen). One thus obtains a bijection

$$P(g_J) \simeq \mathrm{Hom}_{O_{Y_J}}(g_J * (\Omega^1_{X/S}), J)$$

by which g corresponds to the null morphism.

For every $m \in \mathrm{Hom}_{O_{Y_J}}(g_J * (\Omega^1_{X/S}), J)$, denote by $m \cdot g$ the element of $P(g_J)$ associated to g and m by the preceding bijection. We have already seen that $0 \cdot g = g$; it remains to see that

$$m' \cdot (m \cdot g) = (m + m') \cdot g. \qquad (0.1.8\ (*))$$

This is verified locally.[^N.D.E-III-14] Indeed, the two preceding morphisms $Y \to X$ induce the same continuous map as g between the underlying topological spaces; it therefore suffices to verify that for every affine open $U = \mathrm{Spec}(A)$ of X above an affine open $\mathrm{Spec}(\Lambda)$ of S, and every affine open $V = \mathrm{Spec}(B)$ of $g^{-1}(U)$, they induce the same morphism of Λ-algebras $A \to B$.

Let $J = \Gamma(V, \mathcal{J})$ and let ϕ, ψ and η be the morphisms $A \to B$ induced by g, $m \cdot g$ and $m' \cdot (m \cdot g)$ respectively; they coincide modulo J. One can uniquely write $\psi = \phi + D$ (resp. $\eta = \psi + D'$), where D (resp. D') is an element of

$$\mathrm{Der}_\varphi(A, J) = \{\delta \in \mathrm{Hom}_\Lambda(A, J) \mid \delta(ab) = \varphi(a)\,\delta(b) + \varphi(b)\,\delta(a)\}$$

(resp. $\mathrm{Der}_\psi(A, J)$). But $\mathrm{Der}_\phi(A, J) = \mathrm{Der}_\psi(A, J)$ since $J^2 = 0$, and both are identified with

$$\mathrm{Hom}_{B/J}(\Omega^1_{A/\Lambda} \otimes_A B/J, J),$$

and via this identification D corresponds to m and D' to m'. Then $\eta = \phi + D + D'$ and $D + D'$ corresponds to $m + m'$, whence the equality $(*)$.

Corollary 0.1.9.[^N.D.E-III-15] *Let X be an S-scheme; resume the notations of 0.1.5. Then X is endowed with a (left) action of the X^+-abelian group L_X, which makes X into a formally principal homogeneous object under L_X above X^+, i.e. one has an isomorphism of X^+-functors:*

$$L_X \times_{X^+} X \xrightarrow{\sim} X \times_{X^+} X$$

(defined set-theoretically by $(m, x) \mapsto (x, m \cdot x)$).

Proof. Let i_0 be the immersion $X_0 \hookrightarrow X$. Note first that, since $X_0 = X \times_S S_0$, one has $i_0 * (\Omega^1_{X/S}) \simeq \Omega^1_{X_0/S_0}$ (cf. EGA IV, 16.4.5).

Let Y be an X^+-scheme, given by an S-morphism $g_J : Y_J \to X$, and let $g_0 : Y_0 \to X_0$ be the morphism obtained by base change. By 0.1.8, if $\mathrm{Hom}_{X^+}(Y, X)$ is non-empty, it is a principal homogeneous set under the group

$$\mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^* i_0^*(\Omega^1_{X/S}), \mathcal{J} \cdot \mathcal{O}_Y),$$

which is identified with $\mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0 * (\Omega^1_{X_0/S_0}), \mathcal{J} \cdot \mathcal{O}_Y) = L_X(Y)$. One therefore has a bijection

$$L_X(Y) \times \mathrm{Hom}_{X^+}(Y, X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(Y, X \times_{X^+} X)$$

given by $(m, g) \mapsto (g, m \cdot g)$. Let us show that this is "functorial in Y".

Let $f : Z \to Y$ be a morphism of S-schemes. It is a question of showing that the diagram below is commutative:

$$\begin{tikzcd}
L_X(Y) \times \mathrm{Hom}_{X^+}(Y, X) \arrow[r, "\sim"] \arrow[d, "L_X(f) \times f"'] & \mathrm{Hom}_{X^+}(Y, X \times_{X^+} X) \arrow[d, "f \times f"] \\
L_X(Z) \times \mathrm{Hom}_{X^+}(Z, X) \arrow[r, "\sim"] & \mathrm{Hom}_{X^+}(Z, X \times_{X^+} X).
\end{tikzcd}$$

If $\mathrm{Hom}_{X^+}(Y, X) = \emptyset$, there is nothing to show. It therefore suffices to see that, for every S-morphism $g : Y \to X$ extending g_J and every $m \in L_X(Y)$, one has:

$$(m \cdot g) \circ f = L_X(f)(m) \cdot (g \circ f). \qquad (0.1.9\ (*))$$

These two S-morphisms $Z \to X$ coincide on Z_J with $g_J \circ f_J$; in particular, they induce the same continuous map as $g \circ f$ between the underlying topological spaces. Consequently, it suffices to see that, if $z \in Z$, $y = f(z)$, $x = g(y)$, and if A, B, C denote respectively the local rings $O_{X,x}$, $O_{Y,y}$, $O_{Z,z}$, then the morphisms $A \to C$ induced by $(m \cdot g) \circ f$ and $L_X(f)(m) \cdot (g \circ f)$ coincide. Denote by s the image of x in S, $\Lambda = O_{S,s}$, J and I the ideals of Λ corresponding to J and I, and let $\phi, \psi : A \to B$ and $\theta : B \to C$ be the morphisms of Λ-algebras induced by g, $m \cdot g$, and f. Then m induces an element D of

$$\mathrm{Hom}_{B/IB}(\Omega^1_{A/\Lambda} \otimes_A B/IB, JB) = \mathrm{Der}_\Lambda(A, JB)$$

and one has $\psi = \phi + D$; hence $(m \cdot g) \circ f$ corresponds to $\theta \circ \psi = \theta \circ \phi + \theta \circ D$. Now, we have seen in 0.1.5 that $\theta \circ D$ is the image of $L_X(f)(m)$ in

$$\mathrm{Hom}_{C/IC}(\Omega^1_{A/\Lambda} \otimes_A C/IC, JC) = \mathrm{Der}_\Lambda(A, JC);$$

consequently, $\theta \circ \phi + \theta \circ D$ is the image of $L_X(f)(m) \cdot (g \circ f)$. This proves the equality (0.1.9 (∗)).

Corollary 0.1.10.[^N.D.E-III-16] *a)* L_X *depends functorially on* X*: for every* S*-morphism* $f : X \to W$*, there exists an* S*-morphism* $L_f : L_X \to L_W$ *which is a morphism of abelian groups "above* f^+*", i.e., the diagram*

$$\begin{tikzcd} L_X \arrow[r, "L_f"] \arrow[d] & L_W \arrow[d] \\ X^+ \arrow[r, "f^+"] & W^+ \end{tikzcd}$$

is commutative, and for every $Y \to X^+$*,*

$$L_f(Y) : \mathrm{Hom}_{X^+}(Y, L_X) \to \mathrm{Hom}_{W^+}(Y, L_W)$$

(where Y *is above* W^+ *via* f^+*) is a morphism of abelian groups.*

b) Moreover, the following diagram is commutative:

$$\begin{tikzcd} L_X \times X \arrow[r, "\sim"] \arrow[d, "L_f \times f"] & X \times_{X^+} X \arrow[d, "f \times f"] \\ L_W \times W \arrow[r, "\sim"] & W \times_{W^+} W . \end{tikzcd}$$

Proof. a) L_f is induced by the morphism of O_{X_0}-modules $f_{X_0/W_0/S_0} : f_0*(\Omega^1_{W_0/S_0}) \to \Omega^1_{X_0/S_0}$ (cf. 0.1.7 b)): for every X^+-scheme Y, given by an S-morphism $g_J : Y_J \to X$, one has a commutative diagram, functorial in Y:

$$\begin{tikzcd} \mathrm{Hom}_{O_{Y_0}}(g_0*(\Omega^1_{X_0/S_0}), J \cdot O_Y) \arrow[r, "L_f(Y)"] \arrow[d] & \mathrm{Hom}_{O_{Y_0}}(g_0* f_0*(\Omega^1_{W_0/S_0}), J \cdot O_Y) \arrow[d] \\ \{g_J\} & \{f_J \circ g_J\} \end{tikzcd}$$

where $L_f(Y)$ is the map $\psi \mapsto \psi \circ g_0 * (f_{X_0/W_0/S_0})$, which is indeed a morphism of abelian groups.[^N.D.E-III-17]

Let us prove (b). Let Y be an X^+-scheme; if $\mathrm{Hom}_{X^+}(Y, X) = \emptyset$ there is nothing to show. So let $g \in \mathrm{Hom}_{X^+}(Y, X)$; it must be seen that for every $m \in L_X(Y)$, one has:

$$f \circ (m \cdot g) = L_f(Y)(m) \cdot (f \circ g). \qquad (0.1.10\ (*))$$

Now, g being fixed, $\mathrm{Hom}_X(Y, X \times_{X^+} X)$ is a subset of $\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S})$ and

$$L_X(Y) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), J \cdot \mathcal{O}_Y) = \mathrm{Hom}_{\mathcal{O}_Y}(g^*(\Omega^1_{X/S}), J \cdot \mathcal{O}_Y)$$

a subset of $\tilde{L}_X(Y)$ (cf. 0.1.7); finally, $L_f(Y)$ is the restriction to $L_X(Y)$ of the map $\tilde{L}_f(Y)$. Moreover, the bijection

$$L_X(Y) \xrightarrow{\sim} \mathrm{Hom}_X(Y, X \times_{X^+} X), \qquad m \mapsto (g, m \cdot g)$$

is (the inverse of) the restriction to $L_X(Y) \subset \tilde{L}_X(Y)$ of the bijection $\mathrm{Hom}_X(Y, \Delta^{(1)}_{X/S}) \xrightarrow{\sim} \{g\} \times \tilde{L}_X(Y)$ considered in 0.1.7.1. Consequently, the equality (0.1.10 $(*)$) results from (0.1.7 $(*)$); indeed, if we denote by g' the X-morphism $Y \to \Delta^{(1)}_{X/S}$ defined by $(g, m \cdot g)$, then the element of $\tilde{L}_W(Y)$ corresponding to $(f \circ g, f \circ g')$ is $\tilde{L}_f(Y)(m) = L_f(Y)(m)$, i.e., one indeed has

$$L_f(m) \cdot (f \circ g) = f \circ (m \cdot g).$$

Lemma 0.1.11. *Let X, X' be two S-schemes. One has a commutative diagram:*

$$\begin{array}{ccc} L_X \times_S L_{X'} & \xrightarrow{\sim} & L_{X \times_S X'} \\ \downarrow & & \downarrow \\ X^+ \times_S X'^+ & \xrightarrow{\sim} & (X \times_S X')^+ . \end{array}$$

[^N.D.E-III-18] *Proof.* First, for every S-scheme Y, $\mathrm{Hom}_S(Y, X^+ \times_S X'^+)$ equals $\mathrm{Hom}_S(Y, X^+) \times \mathrm{Hom}_S(Y, X'^+)$ and this is isomorphic to

$$\mathrm{Hom}_S(Y_J, X) \times \mathrm{Hom}_S(Y_J, X') = \mathrm{Hom}_S(Y, (X \times_S X')^+);$$

this proves that $X^+ \times_S X'^+ \simeq (X \times_S X')^+$.

Next, let Y be a scheme above $X^+ \times_S X'^+$ via a morphism $h : Y_J \to X \times_S X'$; set $f = p \circ h$ and $g = q \circ h$, where we have denoted by p, q the projections of $X \times_S X'$ to X and X'. Since $\Omega^1_{(X_0 \times_{S_0} X'_0)/S_0} \cong p_0 * (\Omega^1_{X_0/S_0}) \bigoplus q_0 * (\Omega^1_{X'_0/S_0})$ (cf. EGA IV₄, 16.4.23), one obtains a natural isomorphism:

$$\mathrm{Hom}_{\mathcal{O}_{Y_0}}(f_0^*(\Omega^1_{X_0/S_0}), J \cdot \mathcal{O}_Y) \times \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega^1_{X'_0/S_0}), J \cdot \mathcal{O}_Y)$$
$$\simeq \mathrm{Hom}_{\mathcal{O}_{Y_0}}(h_0^*(\Omega^1_{(X_0 \times_{S_0} X'_0)/S_0}), J \cdot \mathcal{O}_Y)$$

i.e., $L_X(Y) \times L_{X'}(Y) \simeq L_{X \times_S X'}(Y)$.

Remark 0.1.12.[^N.D.E-III-19] *Let C be a category stable under fibered products, S an object of C, T_1, T_2 two objects above S and, for $i = 1, 2$, L_i and X_i two objects above T_i:*

$$\begin{tikzcd}
L_1 \arrow[r] & T_1 & X_1 \arrow[l] \arrow[dr] & & L_2 \arrow[r] & T_2 & X_2 \arrow[l] \arrow[dll] \\
& & & S & & & .
\end{tikzcd}$$

Then one has a natural isomorphism:

$$(L_1 \times_{T_1} X_1) \times_S (L_2 \times_{T_2} X_2) \simeq (L_1 \times_S L_2) \times_{T_1 \times_S T_2} (X_1 \times_S X_2).$$

Consequently, from the preceding lemma one deduces the:

Corollary 0.1.13. *Let X_1, X_2 be two S-schemes. One has a commutative diagram of isomorphisms:*

$$\begin{tikzcd}
L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) \arrow[d, "(0.1.11)\ \simeq"'] \arrow[dr, "\simeq"] & \\
(L_{X_1} \times_S L_{X_2}) \times_{(X_1^+ \times_S X_2^+)} (X_1 \times_S X_2) \arrow[r, "\sim"] & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_{X_2^+} X_2) .
\end{tikzcd}$$

(0.1.12)

We can now state:

Proposition 0.2. *For every S-scheme X, one can define a (left) action of the X^+-abelian group L_X on the X^+-object X, such that:*

(i) this action makes X into a formally principal homogeneous object under L_X above X^+, i.e. the morphism

$$L_X \times_{X^+} X \longrightarrow X \times_{X^+} X$$

is an isomorphism of X^+-functors;

(ii) this action is functorial in the S-scheme X, i.e., for every S-morphism $f : X \to W$, the following diagram is commutative:

$$\begin{tikzcd}
L_X \times_{X^+} X \arrow[r] \arrow[d, "L_f \times f"'] & X \arrow[d, "f"] \\
L_W \times_{W^+} W \arrow[r] & W ;
\end{tikzcd}$$

(iii) this action "commutes with fibered product", i.e. for all S-schemes X_1 and X_2, the following diagram is commutative:

$$\begin{tikzcd}
L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) \arrow[r] \arrow[d, "\simeq"] & X_1 \times_S X_2 \\
(L_{X_1} \times_S L_{X_2}) \times_{(X_1^+ \times_S X_2^+)} (X_1 \times_S X_2) \arrow[r, "\sim"] & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_ \arrow[u]
\end{tikzcd}$$

Proof.[^N.D.E-III-20] (i) and (ii) follow respectively from Corollaries 0.1.9 and 0.1.10. To prove (iii), denote $P(X) = L_X \times_{X^+} X$, for every S-scheme X. Then, by (ii) applied to the projections $p_i : X_1 \times_S X_2 \to X_i$, one obtains commutative squares

$$\begin{tikzcd}
P(X_1 \times_S X_2) \arrow[r] \arrow[d, "L_{p_i} \times p_i"'] & X_1 \times_S X_2 \arrow[d, "p_i"] \\
P(X_i) \arrow[r] & X_i
\end{tikzcd}$$

for $i = 1, 2$, and hence a commutative square:

$$\begin{tikzcd}
P(X_1 \times_S X_2) \arrow[r] \arrow[d] & X_1 \times_S X_2 \arrow[d, equal] \\
P(X_1) \times_S P(X_2) \arrow[r] & X_1 \times_S X_2 .
\end{tikzcd}$$

Combining this with Corollary 0.1.13, one obtains that the vertical arrow is an isomorphism, and that one has the commutative diagram announced in (iii).

Remark 0.3. *Suppose the X^+-scheme Y flat over S (cf. SGA 1, IV). Then one can write*

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{O_{Y_0}}(g_0^*(\Omega^1_{X_0/S_0}), J \otimes_{O_{S_0}} O_{Y_0}).$$

Remark 0.4. *Denote by $\pi_0 : X_0 \to S_0$ the structural morphism and suppose there exists an O_{S_0}-module $\omega^1_{X_0/S_0}$ such that $\Omega^1_{X_0/S_0} \simeq \pi_0 * (\omega^1_{X_0/S_0})$ (the case will arise in particular when X_0 is an S_0-group, cf. II, 4.11). If one defines a functor L'_X above S by the formula*

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{O_{Y_0}}(\omega^1_{X_0/S_0} \otimes_{O_{S_0}} O_{Y_0}, J \cdot O_Y), \tag{0.4.1}$$

one then has $\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X)$ for every X^+-scheme Y, that is to say

$$L_X = L'_X \times_S X^+ .$$

[^N.D.E-III-21] Then, since $L_X \times_{X^+} X = L'_X \times_S X$, the action of L_X on X induces an action of L'_X on X, and this action respects the morphism $p_X : X \to X^+$; indeed, if Y is an S-scheme, $h : Y \to X$ an S-morphism and m an element of $L'_X(Y)$, then h and $m \cdot h$ have the same restriction to Y_J, i.e. $p_X(m \cdot h) = p_X(h)$.

Remark 0.5. *Keep the hypotheses and notations of 0.4 and suppose moreover that Y is an S-scheme flat over S. Then we have*

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{S_0}(Y_0, L^0_X),$$

where the S_0-functor in abelian groups L^0_X is defined by the following identity (with respect to the variable S_0-scheme T):

$$\mathrm{Hom}_{S_0}(T, L^0_X) = \mathrm{Hom}_{O_T}(\omega^1_{X_0/S_0} \otimes_{O_{S_0}} O_T, J \otimes_{O_{S_0}} O_T). \tag{0.5.1}$$

In the notations of (II, 1), we have therefore shown that the functors L'_X and $\prod_{S_0/S} L^0_X$ have the same restriction to the full subcategory of $(Sch)/S$ whose objects are the S-schemes Y flat over S.

Remark 0.6. *Keep the hypotheses and notations of 0.5[^N.D.E-III-22] and suppose moreover that there exists a section ϵ_0 of $\pi_0 : X_0 \to S_0$; one then has $\omega^1_{X_0/S_0} \simeq \epsilon_0 * (\Omega^1_{X_0/S_0})$.*

First, one has (independently of the preceding hypothesis):

$$\mathrm{Hom}_{S_0}(T, L^0_X) = \Gamma(T, \mathrm{Hom}_{O_T}(\omega^1_{X_0/S_0} \otimes_{O_{S_0}} O_T, J \otimes_{O_{S_0}} O_T)).$$

Now suppose that $\omega^1_{X_0/S_0}$ admits a finite presentation (cf. EGA 0_I, 5.2.5), which will in particular be the case if X_0 is locally of finite presentation over S_0 (cf. EGA IV_4, 16.4.22). Then, if T is flat over S_0, it follows from (EGA 0_I, 6.7.6) that

$$\mathrm{Hom}_{O_T}(\omega^1_{X_0/S_0} \otimes_{O_{S_0}} O_T, J \otimes_{O_{S_0}} O_T) \simeq \mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J) \otimes_{O_{S_0}} O_T ,$$

whence

$$\mathrm{Hom}_{S_0}(T, L^0_X) = \Gamma(T, \mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J) \otimes_{O_{S_0}} O_T).$$

Introducing the notation $W(\cdot)$ of (I, 4.6.1), we have therefore proved that for every S_0-scheme T flat over S_0, one has

$$\mathrm{Hom}_{S_0}(T, L^0_X) = \mathrm{Hom}_{S_0}(T, W(\mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J))).$$

In summary, if $\omega^1_{X_0/S_0}$ admits a finite presentation, and if one restricts to the category of S-schemes flat over S, one has

$$L'_X = \prod_{S_0/S} W(\mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J)), \qquad (0.6.1)$$

and $\mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J)$ is a quasi-coherent O_{S_0}-module, by EGA I, 9.1.1.

Note finally that if $\omega^1_{X_0/S_0}$ is moreover locally free (of finite rank), for example if X_0 is smooth over S_0 (in which case it is automatically locally of finite presentation over S_0), one has

$$\mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J) \simeq \mathrm{Lie}(X_0/S_0) \otimes_{O_{S_0}} J , \qquad (0.6.2)$$

where, by abuse of language (X_0 not being necessarily an S_0-group), we denote by $Lie(X_0/S_0)$ the dual of the O_{S_0}-module $\omega^1_{X_0/S_0}$.[^N.D.E-III-23]

Proposition 0.2 (and its proof) has two important corollaries.[^N.D.E-III-24]

Corollary 0.7. *Let X be an S-scheme.*

a) Every S-endomorphism of X inducing the identity on X_J is an automorphism.

b) One has an exact sequence of groups:

$$0 \longrightarrow \mathrm{Hom}_{O_{X_0}}(\Omega^1_{X_0/S_0}, J \cdot O_X) \xrightarrow{\ i\ } \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_J}(X_J) .$$

c) Moreover, if one makes $\mathrm{Aut}_S(X)$ act on the first group by transport of structure, one has, for all $u \in \mathrm{Aut}_S(X)$ and $m \in \mathrm{Hom}_{O_{X_0}}(\Omega^1_{X_0/S_0}, J \cdot O_X)$:

$$i(u \cdot m) = u \cdot i(m) \cdot u^{-1}.$$

Proof. By 0.2 (i), $\mathrm{Hom}_{X^+}(X, X)$ is a principal homogeneous set under $\mathrm{Hom}_{X^+}(X, L_X)$, since it is certainly non-empty: it contains a marked point, namely the identity automorphism of X.[^N.D.E-III-25] Consequently, the map $m \mapsto m \cdot id_X$ induces a bijection

$$\mathrm{Hom}_{\mathcal{O}_{X_0}}(\Omega^1_{X_0/S_0}, J \cdot \mathcal{O}_X) = L_X(X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, X).$$

Let $m \in L_X(X)$ and let $f = m' \cdot id_X$ be an element of $\mathrm{Hom}_{X^+}(X, X)$. Applying 0.2 (ii) to f, one obtains:

$$f \circ (m \cdot id_X) = L_f(X)(m) \cdot f = L_f(X)(m) \cdot (m' \cdot id_X).$$

On the other hand, since f is an X^+-endomorphism of X, one has $f_J = id_{X_J}$ and therefore $f_0 = id_{X_0}$; since L_f depends only on f_0 (cf. N.D.E. (17) in 0.1.10), one therefore has $L_f(X)(m) = m$. Consequently, the equality above rewrites as:

$$(m' \cdot id_X) \circ (m \cdot id_X) = m \cdot (m' \cdot id_X) = (m + m') \cdot id_X\ .$$

This shows that the bijection $m \mapsto m \cdot id_X$ transforms the group law of $\mathrm{Hom}_{X^+}(X, L_X)$ into the composition law of X^+-endomorphisms of X.

It follows first that every element of $\mathrm{Hom}_{X^+}(X, X)$ is invertible, which is the first assertion of the statement, and then that one has an exact sequence

$$0 \longrightarrow \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_J}(X_J)\ ,$$

which is the second.

Let us now note that the morphism i defined above is functorial in X for isomorphisms, because it is defined in structural terms from the action of L_X on X above X^+, itself functorial in X by assertion (ii) of Proposition 0.2.[^N.D.E-III-26] Hence every automorphism u of X above S induces by transport of structure isomorphisms

$$h : \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, L_X)$$

and $f : \mathrm{Aut}_S(X) \xrightarrow{\sim} \mathrm{Aut}_S(X)$ such that the following diagram is commutative:

$$\begin{tikzcd}
\mathrm{Hom}_{X^+}(X, L_X) \arrow[r, "i"] \arrow[d, "h"] & \mathrm{Aut}_S(X) \arrow[d, "f"] \\
\mathrm{Hom}_{X^+}(X, L_X) \arrow[r, "i"] & \mathrm{Aut}_S(X)
\end{tikzcd}$$

i.e. such that $f \circ i = i \circ h$. On the other hand, f is given by the commutative diagram

$$\begin{tikzcd}
X \arrow[r, "a"] \arrow[d, "u"'] & X \arrow[d, "u"] \\
X \arrow[r, "f(a)"] & X\ ,
\end{tikzcd}$$

that is, $f(a) = u \circ a \circ u^{-1}$ for every $a \in \mathrm{Aut}_S(X)$. Writing $i(h(m)) = f(i(m))$, one finds the desired formula.

Corollary 0.7.bis. *Let X be an S-scheme such that X_J is an S_J-monoid. Then L_X is endowed with a structure of S-monoid, one has a split exact sequence of S-monoids:*

$$1 \longrightarrow L'_X \xrightarrow{\ i\ } L_X \underset{s}{\overset{p}{\rightleftarrows}} X^+ \longrightarrow 1$$

and the monoid law induced on L'_X coincides with its abelian group structure. In particular, if X_J is an S_J-group, then L_X is an S-group and is the semidirect product of X^+ and L'_X.

Proof. Indeed, since X_J is an S_J-monoid, then $X^+ = \prod_{S_J/S} X_J$ is an S-monoid (indeed, one has $X^+(Y) = X_J(Y_J)$ for every $Y \to S$). For every S-scheme Y, denote by $\tilde{Y}_J$ the Y_J-affine scheme corresponding to the quasi-coherent O_{Y_J}-algebra $O_{Y_J} \bigoplus J \cdot O_Y$ (i.e. the graded algebra associated to the filtration $O_Y \supset J \cdot O_Y$). Then $L_X(Y)$ is identified with $X_J(\tilde{Y}_J)$ and $L'_X(Y)$ with the kernel of the morphism $p : X_J(\tilde{Y}_J) \to X_J(Y_J)$ induced by the "zero section" $Y_J \to \tilde{Y}_J$ (i.e. by the morphism of O_{Y_J}-algebras $O_{\tilde{Y}_J} \to O_{Y_J}$ vanishing on the ideal $J \cdot O_Y$). One has therefore, for every $Y \to S$, a split exact sequence of monoids, functorial in Y:

$$1 \longrightarrow L'_X(Y) \xrightarrow{\ i\ } L_X(Y) \underset{s}{\overset{p}{\rightleftarrows}} X^+(Y) \longrightarrow 1\ .$$

It remains to see that the monoid law induced on L'_X coincides with its abelian group structure. Denote by μ the monoid law of L_X and e its unit section; one must show that for all $m, m' \in L'_X(Y)$, one has

$$\mu(m \cdot e,\ m' \cdot e) = (m + m') \cdot e\ .$$

This can be seen in either of the following ways. On the one hand, one can revisit the proof of equality (0.1.10 (∗)) by replacing the morphism $f : X \to W$ appearing there with the morphism $\mu : L_X \times_S L_X \to L_X$. Identifying $X^+(Y) = X_J(Y_J)$ with its image by s in $L_X(Y) = X_J(\tilde{Y}_J)$, one obtains that, for all $g, g' \in X_J(Y_J)$ and $m, m' \in L'_X(Y)$, one has

$$\mu(m \cdot g,\ m' \cdot g') = L_\mu^{(g,\ g')}(m,\ m') \cdot \mu(g,\ g'), \qquad (*)$$

where $L_\mu^{(g,\ g')}$ denotes the morphism derived from μ at the point (g, g') (i.e. $\tilde{Y}_J$ is above $L_X \times_S L_X$ via (g, g')). In particular, one has $\mu(m \cdot e,\ m' \cdot e) = L_\mu^{(e,\ e)}(m,\ m') \cdot e$; now $L_\mu^{(e,\ e)}(m,\ m') = L_{\ell_e}(m') + L_{r_e}(m)$, where ℓ_e (resp. r_e) denotes left (resp. right) translation by e, which is the identity map of X_J, whence $L_\mu^{(e,\ e)}(m,\ m') = m + m'$.

Alternatively, one can proceed as follows (cf. the proof of [DG70], § II.4, Th. 3.5). By Lemma 0.1.11, the formation of X^+ and of L_X "commutes with products", and hence the same holds for L'_X; it follows that the morphism $\mu' : L'_X \times L'_X \to$

L'_X induced by μ is a homomorphism for the abelian-group structure. One then deduces from Lemma 3.10 of Exp. II that μ' coincides with the abelian-group law.

0.8.[^N.D.E-III-27] Let now X be an S-scheme such that X_J is an S_J-group. Suppose there exists an S-morphism

$$P : X \underset{S}{\times} X \longrightarrow X$$

such that the morphism obtained by base change

$$P_J : X_J \underset{S_J}{\times} X_J \longrightarrow X_J$$

is the group law of X_J. (An important particular case of the preceding situation will be the case where X is an S-group and one takes for P its group law.) From this one deduces a morphism

$$L_P : L_X \underset{S}{\times} L_X \simeq L_{X \times_S X} \longrightarrow L_X$$

which, in fact, does not depend on P, because it is computed by means of the group law P_J of X_J, as we shall now see.[^N.D.E-III-28] Indeed, by (ii) and (iii) of 0.2, for every $Y \to S$ and $x, x' \in X(Y)$, $m, m' \in L'_X(Y)$, one has

$$P(m \cdot x,\ m' \cdot x') = L_P^{(x, x')}(m, m') \cdot (x, x') = L_P^{(x, x')}(m, m') \cdot \mu(g, g')$$

where g (resp. g') is the image of x (resp. x') in $X^+(Y)$. Moreover (cf. the proof of 0.10), $L_P^{(x,x')}$ equals $L_\mu^{(g, g')}$ and, by 0.7.bis (), this is the element of $L'_X(Y)$ defined by the following equality in $L_X(Y)$:

$$L_\mu^{(g, g')}(m, m') \cdot \mu(g, g') = \mu(m \cdot g,\ m' \cdot g'),$$

that is, if we denote by $\times$ (instead of μ) the group law of L_X and Ad the "adjoint action" of X^+ on L'_X (which factors through X_0 and is induced by the adjoint action of X_0 on $\omega^1_{X_0/S_0}$), one obtains that

$$L_\mu^{(g, g')}(m, m') \times g \times g' = m \times g \times m' \times g' = (m \times \mathrm{Ad}(g)(m')) \times g \times g'$$

hence finally $L_P^{(x,x')}(m, m') = m \times Ad(g)(m')$. We therefore obtain:

Proposition 0.8. *Let $P : X \times_S X \to X$ be an S-morphism such that P_J endows X_J with a structure of S_J-group. Denote by $\times$ the group law of L'_X and by $(m, x) \mapsto m \cdot x$ the morphism $L'_X \times_S X \to X$ defining the action of L'_X on X, and let $Ad : X^+ \to \mathrm{Aut}_{S-gr.}(L'_X)$ be the "adjoint action" of X^+ on L'_X (which is induced by the adjoint action of X_0 on $\omega^1_{X_0/S_0}$). Then, for every $S' \to S$ and $x, x' \in X(S')$, $m, m' \in L'_X(S')$, one has:*

$$P(m \cdot x,\ m' \cdot x') = (m \times \mathrm{Ad}(p_X(x))(m')) \cdot P(x, x'). \qquad (0.8.1)$$

If X is an S-group, we shall denote by $*$ its law, e its unit section, and i the S-morphism defined by:

$$i(m) = m \cdot e,$$

for every $S' \to S$ *and* $m \in L'_X(S')$*.*

Corollary 0.9. *Let* X *be an* S*-group. Then* X^+ *is naturally endowed with a structure of* S*-group, and* p_X *is a morphism of* S*-groups. Moreover, the* S*-morphism*

$$i : L'_X \longrightarrow X, \quad m \mapsto m \cdot e$$

is an isomorphism of S*-groups from* L'_X *onto* $Ker(p_X)$*, and one has, for all* $S' \to S$*,* $x' \in X(S')$*,* $m \in L'_X(S')$*:*

$$m \cdot x' = (m \cdot e) * x' = i(m) * x' . \tag{0.9.1}$$

The first two assertions have already been proved in 0.1.2. Since X is formally principal homogeneous over X^+ under $L_X = L'_X \times_S X^+$, the morphism i is indeed an isomorphism of S-functors from L'_X onto the kernel of p_X. The fact that i is a morphism of groups and the formula (0.9.1) follow from formula (0.8.1) applied respectively to $x = x' = e$, and to $x = e$, $m' = 1$.

Corollary 0.10. *Let* X *be an* S*-group. With the preceding notations, for every* $S' \to S$ *and all* $x \in X(S')$ *and* $m' \in L'_X(S')$*, one has*

$$x * i(m') * x^{-1} = i(\mathrm{Ad}(p_X(x))(m')) . \tag{0.10.1}$$

This follows from the equality $i(m') * x^{-1} = m' \cdot x^{-1}$ and from (0.8.1) applied to $m = 1$ and $x' = x^{-1}$.

When X is an S-group, we have therefore explicitly determined the kernel of $X \to X^+$ and the action of the inner automorphisms of X on this kernel. We shall now see that one can do the same for certain S-group functors not necessarily representable. One case will be useful to us, namely that of the $\mathtt{Aut}$ functors (I, 1.7). Let us state at once:

Proposition 0.11. *Let* E *be an* S*-scheme. Denote* $X = \mathrm{Aut}_S(E)$*. The kernel of the morphism of* S*-group functors*

$$p_X : X \longrightarrow X^+$$

is canonically identified with the S*-functor in commutative groups* L'_X *defined by*

$$\mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{O_{E_0 \times_{S_0} Y_0}}(\Omega^1_{E_0/S_0} \otimes_{O_{S_0}} O_{Y_0}, J \cdot O_{E \times_S Y}),$$

where Y *denotes a variable* S*-scheme.*

Indeed, if Y is a variable S-scheme, one has $\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$, and

$$\mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_S(Y_J, X) = \mathrm{Aut}_{Y_J}(E \times_S Y_J) = \mathrm{Aut}_{Y_J}((E \times_S Y) \times_Y Y_J).$$

Applying 0.7 b) to the Y-scheme $E \times_S Y$, one obtains an isomorphism of groups:

$$\mathrm{Hom}_S(Y, L'_X) \simeq \mathrm{Ker}(\mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+)),$$

an isomorphism that one verifies easily to be functorial in the S-scheme Y. One thus obtains an isomorphism of S-groups

$$L'_X \simeq \mathrm{Ker}(X \longrightarrow X^+),$$

which completes the proof of Proposition 0.11.

Corollary 0.12.[^N.D.E-III-29] *We keep the notations of 0.11: E is an S-scheme and $X = \mathrm{Aut}_S(E)$. One has a natural action f of X on L'_X defined as follows. For every S-scheme Y, one has*

$$\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$$
$$\text{and} \quad \mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathcal{O}_{E_0 \times_{S_0} Y_0}}(\Omega^1_{E_0 \times_{S_0} Y_0 / Y_0}, J \cdot \mathcal{O}_{E \times_S Y})$$

(N. B. $\Omega^1_{E_0/S_0} \bigotimes_{S_0} \mathcal{O}_{Y_0} \simeq \Omega^1_{E_0 \times_{S_0} Y_0/Y_0}$, cf. EGA IV₄, 16.4.5); the first group acts on the second by transport of structure and this action is indeed functorial in Y. One then has the formula:

$$x \cdot i(m) \cdot x^{-1} = i(f(x)\, m), \qquad (0.12.1)$$

for every $Y \to S$ and all $x \in \mathrm{Hom}_S(Y, X)$, $m \in \mathrm{Hom}_S(Y, L'_X)$.

Indeed, this follows from 0.7 c) applied to the Y-scheme $E \times_S Y$.

Reminder 0.13. *The direct image of a quasi-coherent module by a morphism of finite presentation is quasi-coherent. Under the same conditions, the formation of the direct image commutes with flat base change: in the situation*

$$\begin{array}{ccc} T & \xleftarrow{g'} & T' = T \times_S S' \\ \downarrow f & & \downarrow f' \\ S & \xleftarrow{g} & S' \end{array},$$

if one supposes f (and therefore f') of finite presentation and g (and therefore g') flat, one has for every quasi-coherent $\mathcal{O}_T$-module F

$$f_*(F) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} = f'_*(F \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$

where, more elegantly,

$$g^*(f_*(F)) = f'_*(g'^*(F)).$$

These two facts hold more generally for a quasi-compact and quasi-separated morphism f, cf. (EGA I, 9.2.1) and (EGA III₁, 1.4.15) in the quasi-compact separated case (taking into account EGA III₂, Err_III 25) and (EGA IV₁, 1.7.4 and 1.7.21).

Remark 0.14.[^N.D.E-III-30] *Resume the notations of 0.11: let E be an S-scheme, $X = \mathrm{Aut}_S(E)$ and L'_X the S-functor in commutative groups defined by:*

$$\begin{aligned} L'_X(Y) &= \mathrm{Hom}_{\mathcal{O}_{E_0 \times_{S_0} Y_0}}(\Omega^1_{E_0/S_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, J \cdot \mathcal{O}_{E \times_S Y}) \\ &= \mathrm{Hom}_{\mathcal{O}_{E \times_S Y}}(\Omega^1_{E/S} \otimes_{\mathcal{O}_S} \mathcal{O}_Y, J \cdot \mathcal{O}_{E \times_S Y}) \\ &= \Gamma(E \times_S Y, \mathcal{H}om_{\mathcal{O}_{E \times_S Y}}(\Omega^1_{E/S} \otimes_{\mathcal{O}_S} \mathcal{O}_Y, J \cdot \mathcal{O}_{E \times_S Y})). \end{aligned}$$

Suppose Y flat over S; then one has isomorphisms:

$$J \cdot O_{E \times_S Y} \xleftarrow{\sim} (J \cdot O_E) \otimes_{O_S} O_Y \xrightarrow{\sim} (J \cdot O_E) \otimes_{O_{S_0}} O_{Y_0} .$$

Suppose moreover E of finite presentation over S; then $\Omega^1_{E/S}$ is an O_E-module of finite presentation (cf. EGA IV$_4$, 16.4.22), and hence, by (EGA 0_I, 6.7.6),

$$\mathrm{Hom}_{O_{E \times_S Y}}(\Omega^1_{E/S} \otimes_{O_S} O_Y, (J \cdot O_E) \otimes_{O_S} O_Y) \simeq \mathrm{Hom}_{O_E}(\Omega^1_{E/S}, J \cdot O_E) \otimes_{O_S}$$

Denote by $\pi : E \to S$ and $g : Y \to S$ the structural morphisms; applying 0.13 to the diagram

$$\begin{tikzcd} E \arrow[d, "\pi"] & E \times_S Y \arrow[l, "g'"'] \arrow[d, "\pi'"] \\ S & Y , \arrow[l, "g"] \end{tikzcd}$$

and to the O_E-module $F = \mathrm{Hom}_{O_E}(\Omega^1_{E/S}, J \cdot O_E)$, one obtains

$$\Gamma(E \times_S Y, g'^* F) = \Gamma(Y, \pi'_* g'^* F) = \Gamma(Y, g^* \pi_* F) = W(\pi_* F)(Y).$$

We have therefore shown that, if E is of finite presentation over S, one has

$$L'_X = W(\pi_* \mathrm{Hom}_{O_E}(\Omega^1_{E/S}, J \cdot O_E)) \qquad (0.14.1)$$

on the category of S-schemes flat over S. Note moreover that the module of which we take the W is quasi-coherent, by (EGA I, 9.1.1 and 9.2.1).

[^N.D.E-III-31] Denote by L_0 the S_0-functor

$$W(\pi_{0*} \mathrm{Hom}_{O_{E_0}}(\Omega^1_{E_0/S_0}, J \cdot O_E)).$$

Then, returning to the definition of $L'_X(Y)$ and taking into account the isomorphism

$$J \cdot O_{E \times_S Y} \simeq (J \cdot O_E) \otimes_{O_{S_0}} O_{Y_0},$$

one obtains, by arguing as above, that

$$L'_X(Y) = L_0(Y_0) = L_0(Y \times_S S_0) = \textstyle\prod_{S_0/S} L_0(Y).$$

Hence, on the category of S-schemes flat over S, one has:

$$L'_X = \textstyle\prod_{S_0/S} W(\pi_{0*} \mathrm{Hom}_{O_{E_0}}(\Omega^1_{E_0/S_0}, J \cdot O_E)).$$

It is not obvious that the action of X on L'_X defined in 0.12 comes from an action of $X_0 = \mathrm{Aut}_{S_0}(E_0)$ on L_0; this is however the case when, moreover, E is flat over S.

Indeed, in this case one has canonical isomorphisms:

$$J \cdot O_E \simeq J \otimes_{O_S} O_E \simeq J \otimes_{O_{S_0}} O_{E_0}.$$

$$L_0 \simeq W(\pi_{0*} \mathrm{Hom}_{O_{E_0}}(\Omega^1_{E_0/S_0}, J \otimes_{O_{S_0}} O_{E_0})),$$

$$L'_X = \textstyle\prod_{S_0/S} W(\pi_{0*} \mathrm{Hom}_{O_{E_0}}(\Omega^1_{E_0/S_0}, J \otimes_{O_{S_0}} O_{E_0})). \qquad (0.14.2)$$

Then, for every S_0-scheme T, one has

$L_0(T) \simeq \mathrm{Hom}_{O_{E_0 \times_{S_0} T}}(\Omega^1_{E_0 \times_{S_0} T / T}, J \otimes_{O_{S_0}} O_{E_0 \times_{S_0} T})$

and $\mathrm{Hom}_{S_0}(T, X_0) = \mathrm{Aut}_T(E_0 \times_{S_0} T)$ acts by transport of structure on $L_0(T)$, functorially in T; finally, for every S-scheme Y flat over S, the action by transport of structure of $\mathrm{Hom}_S(Y, X) = \mathrm{Aut}_Y(E \times_S Y)$ on $L'_X(Y) = L_0(Y_0)$ factors through $\mathrm{Aut}_{Y_0}(E_0 \times_{S_0} Y_0)$.

Let us finally extract from (SGA 1, III) the following two propositions.

Proposition 0.15. *(SGA 1, III, 6.8)[^N.D.E-III-32] For every S_J-scheme Y smooth over S_J and affine, there exists an S-scheme X smooth over S such that $X \times_S S_J \simeq Y$, and such an X is unique up to (non-unique) isomorphism.*

Proposition 0.16. *(SGA 1, III, 5.5)[^N.D.E-III-33] Let X be an S-scheme smooth over S. For every affine S-scheme Y, the canonical map*

$$p_X(Y) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_J}(Y_J, X_J)$$

is surjective.

Corollary 0.17. *Let E be an S-scheme smooth over S and affine; denote $X = \mathrm{Aut}_S(E)$. For every affine S-scheme Y, the canonical map*

$$\mathrm{Aut}_Y(E \times_S Y) = \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Aut}_{Y_J}(E_J \times_{S_J} Y_J)$$

is surjective.

Indeed, $Y \times_S E$ is affine over Y, itself affine, so affine. Applying 0.16, one deduces that every S_J-morphism $Y_J \times_{S_J} E_J \to E_J$ extends to an S-morphism $Y \times_S E \to E$.[^N.D.E-III-34] In other words, every Y_J-endomorphism of $Y_J \times_{S_J} E_J$ lifts to a Y-endomorphism of $Y \times_S E$. Then, 0.7 a) shows that every Y_J-automorphism of $Y_J \times_{S_J} E_J$ lifts to a Y-automorphism of $Y \times_S E$, which is the announced property.

1. Extensions and cohomology

1.1.

Let C be a category stable under fibered products.[^N.D.E-III-35] Let S be an object of C, G an (representable) S-group, and F an S-functor in commutative groups on which G acts. We defined in (I, 5.1) the cohomology groups $H^n(G, F)$. Recall that these are the homology groups of a complex denoted $C*(G, F)$ where, denoting $(G/S)^n = G \times_S \cdots \times_S G$ (n factors),

$$C^n(G, F) = \mathrm{Hom}_S((G/S)^n, F).$$

Since G, and hence the $(G/S)^n$, are representable, one has also

$$C^n(G, F) = F((G/S)^n);$$

from this, and from the definition of the boundary operator, one sees that the complex $C*(G, F)$ depends only on the restriction of F to the full subcategory of C/S whose objects are the cartesian powers of G over S. Consequently, one has the:

Lemma 1.1.1. *Let C be a category stable under fibered products,[^N.D.E-III-35] S an object of C, G a representable S-group. Denote by $C(G)$ the full subcategory of C/S whose objects are the cartesian powers of G over S. Let F and F' be two S-functors in commutative groups on which G acts. If F and F' have the same restriction to $C(G)$, one has a canonical isomorphism*

$$H^*(G, F) \xrightarrow{\sim} H^*(G, F').$$

1.1.2. Cohomology and restriction of scalars.

[^N.D.E-III-36] Let us state another comparison result. Let now $T \to S$ be a morphism in C. If F is a T-functor in commutative groups, then the functor obtained by "restriction of scalars" (cf. Exp. II, 1)

$$F_1 = \prod_{T/S} F$$

is an S-functor in commutative groups and one has a morphism of S-group functors

$$u : \prod_{T/S} \mathrm{Aut}_{T\text{-gr.}}(F) \longrightarrow \mathrm{Aut}_{S\text{-gr.}}(F_1).$$[^N.D.E-III-37]

Let now G be an S-group functor and let

$$G_T \longrightarrow \mathrm{Aut}_{T\text{-gr.}}(F)$$

be an action of G_T on F. By definition of the functor $\prod_{T/S}$, one deduces a morphism of S-group functors

$$G \longrightarrow \prod_{T/S} \mathrm{Aut}_{T\text{-gr.}}(F)$$

whence, by composition with u, an action of G on $F_1 = \prod_{T/S} F$.[^N.D.E-III-38]

Lemma 1.1.2. *Under the preceding conditions, one has a canonical isomorphism*

$$H^*(G, \prod_{T/S} F) \simeq H^*(G_T, F).$$

Indeed, by the definition of cohomology, the standard complexes are canonically isomorphic.

1.2. Lifting of group morphisms.

[^N.D.E-III-39] Following the general principles, we lay down the following definition:

Definition 1.2.1. *Let $1 \to M \xrightarrow{u} E \xrightarrow{v} G$ be a sequence of morphisms of $\hat{C}$-groups. We say that it is **exact** if the following equivalent conditions are verified:*

(i) for every $S \in ObC$, the following sequence of ordinary groups is exact:

$$1 \longrightarrow M(S) \xrightarrow{u(S)} E(S) \xrightarrow{v(S)} G(S)$$

(ii) for every object H of $\hat{C}$, the following sequence of ordinary groups is exact:

$$1 \longrightarrow \mathrm{Hom}(H, M) \xrightarrow{u(H)} \mathrm{Hom}(H, E) \xrightarrow{v(H)} \mathrm{Hom}(H, G)$$

Taking in particular $H = G$ in (ii), one sees that the set of sections of v (not respecting *a priori* the group structures) is empty or principal homogeneous under $\mathrm{Hom}(G, M)$. Suppose it is non-empty; so let

$$s : G \longrightarrow E$$

be a section of v. Then for every $S \in ObC$ and every $x \in G(S)$, the element $s(x)$ of $E(S)$ defines an inner automorphism of E_S which normalizes M_S (more correctly, the image of M_S by u_S), hence an automorphism of M_S.

Scholie 1.2.1.1.[^N.D.E-III-40] *If M is commutative, one sees "set-theoretically" that this automorphism does not depend on the chosen section, but only on x, and depends multiplicatively on it. In summary, to every exact sequence*

$$(E) \qquad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

such that M is commutative and that v admits a section, is associated a morphism of $\hat{C}$-groups

$$G \longrightarrow \mathrm{Aut}_{\hat{C}\text{-gr.}}(M)$$

*which is called the **action of G on M defined by the extension (E)**.*

Definition 1.2.1.2. *We saw in (I, 2.3.7) that v admits a section which is a morphism of $\hat{C}$-groups if and only if the extension (E) is isomorphic ("as an extension") to the semidirect product of M by G relative to the preceding action. Such a section of v will be called a **section of the extension (E)**, or simply a **section of (E)**.*

If s is a section of (E) and if $m \in \Gamma(M) \simeq Ker(\Gamma(E) \to \Gamma(G))$ (for the definition of Γ, see I, 1.2), then the morphism $G \to E$ defined by[^N.D.E-III-41]

$$x \mapsto u(m)\, s(x)\, u(m)^{-1}$$

*is also a section of (E), said to be **deduced from s by the inner automorphism defined by** m (or by $u(m)$).*

Lemma 1.2.2. *Let $(E) : 1 \to M \xrightarrow{u} E \xrightarrow{v} G$ be an exact sequence of $\hat{C}$-groups such that M is commutative and v admits a section. Let G act on M in the manner defined by (E).*

(i) The extension (E) canonically defines a class $c(E) \in H^2(G, M)$ whose vanishing is necessary and sufficient for the existence of a section of (E).

(ii) If $c(E) = 0$, the set of sections of (E) is principal homogeneous under the group $Z^1(G, M)$, and the set of sections of (E) modulo the action of the inner automorphisms defined by the elements of $\Gamma(M)$ is principal homogeneous under the group $H^1(G, M)$.

(iii)[^N.D.E-III-42] Let s be a section of (E); the set of conjugates of s by the inner automorphisms defined by $\Gamma(M)$ is in bijection with $\Gamma(M)/H^0(G, M)$.

Proof. It proceeds exactly as in the case of ordinary groups, the fact that one starts from a section of v ensuring the functoriality of the set-theoretic computations. Let us briefly indicate the principal steps.

a) To each section s of v, one associates the morphism

$$Ds : G \times G \longrightarrow M,$$

defined set-theoretically by

$$u(Ds(x, y)) = s(xy)\, s(y)^{-1}\, s(x)^{-1}.$$

One shows that Ds is a 2-cocycle by the following computation.[^N.D.E-III-43] By the definition of the differential of the standard complex (I, 5.1), one has:

$$(\partial^2 Ds)(x, y, z) = (s(x)\, Ds(y, z)\, s(x)^{-1}) \cdot Ds(x, y)^{-1} \cdot Ds(xy, z)^{-1} \cdot Ds(x, yz);$$

it suffices to substitute the definition of Ds into this formula to find (without using any commutativity) $Ds \in Z^2(G, M)$.

b) If s and s' are two sections of v, there exists $f : G \to M$ such that $s(x) = f(x)s'(x)$. One then has

$$\begin{aligned} Ds'(x, y) &= f^{-1}(xy)\, Ds(x, y)\, s(x)\, f(y)\, s(x)^{-1}\, f(x), \\ &= Ds(x, y) \cdot f^{-1}(xy) \cdot (s(x)\, f(y)\, s(x)^{-1}) \cdot f(x), \end{aligned}$$

i.e.

$$Ds' = Ds \cdot \partial^1 f.$$

[^N.D.E-III-44] This shows that the class of Ds in $H^2(G, M)$ does not depend on the chosen section s of v; it is the class $c(E)$ of the extension (E).

c) Let s and s' be two sections of v and let $m \in \Gamma(M)$. Then, the equality $s(x) = m^{-1}s'(x)m$ (for every $x \in G(S)$, $S \in ObC$) is equivalent to

$$s(x) = m^{-1}\, s'(x)\, m\, s'(x)^{-1}\, s'(x), \quad \text{i.e.} \quad s = \partial^0 m \cdot s' .$$

In particular, the stabilizer of s in $\Gamma(M)$ is the subgroup of $m \in \Gamma(M)$ such that $\partial^0 m = e_M$, i.e. the subgroup $H^0(G, M)$. This already proves (iii).

d) The reasoning is now habitual.[^N.D.E-III-45] Let s_0 be an arbitrary section of v; there exists a section s, necessarily of the form $s = f \cdot s_0$, which is a morphism of groups, i.e. which satisfies $Ds = 0$, if and only if $(Ds_0)^{-1} = \partial^1 f$, i.e., if and only if the class $c(E)$ is zero. This proves (i).

In this case, the set of sections of (E) consists of the sections $s' = h \cdot s$, where $h : G \to M$ satisfies $\partial^1 h = 0$, i.e. $h \in Z^1(G, M)$. Moreover, by point c), two such sections $h_1 \cdot s$ and $h_2 \cdot s$ are conjugate under $\Gamma(M)$ if and only if h_1 and h_2 have the same image in $H^1(G, M)$. This proves (ii).

Let still

$$(E) \qquad 1 \longrightarrow M \xrightarrow{\ u\ } E \xrightarrow{\ v\ } G$$

be an exact sequence of $\hat{C}$-groups with M commutative. Let

$$f : H \longrightarrow G$$

be a morphism of $\hat{C}$-groups. Consider $E_f = H \times_G E$; it is a $\hat{C}$-group and the projection $v_f : E_f \to H$ is a morphism of $\hat{C}$-groups. Likewise for $e_f : E_f \to E$. On the other hand, if one sends M into E via u and into H via the unit morphism, one defines a morphism of $\hat{C}$-groups $u_f : M \to E_f$. We have thus constructed a commutative diagram of $\hat{C}$-groups:

$$\begin{tikzcd}
(E) & 1 \arrow[r] & M \arrow[r, "u"] & E \arrow[r, "v"] & G \\
(E_f) & 1 \arrow[r] & M \arrow[u, equal, "id"] \arrow[r, "u_f"] & E_f \arrow[u, "e_f"] \arrow[r, "v_f"] & H . \arrow[u, "f"]
\end{tikzcd}$$

One has immediately:

Lemma 1.2.3. *(i) The sequence (E_f) is exact.*

(ii) The map $s \mapsto e_f \circ s = f'$ realizes a bijective correspondence between the morphisms

$$s : H \longrightarrow E_f$$

such that $v_f \circ s = id$ (that is, the sections of v_f) and the morphisms

$$f' : H \longrightarrow E$$

such that $v \circ f' = f$ (that is, the morphisms f' “lifting” f).

(iii) In the preceding correspondence, sections of (E_f) and morphisms of groups f' lifting f correspond.

Applying Lemma 1.2.2 to the extension (E_f) and taking into account 1.2.3, one obtains the following proposition (which formally contains 1.2.2):

Proposition 1.2.4. *Let $(E) : 1 \to M \to E \xrightarrow{v} G$ be an exact sequence of $\hat{C}$-groups with M commutative. Let*

$$f : H \longrightarrow G$$

be a morphism of $\hat{C}$-groups; suppose it lifts to a morphism (not necessarily of groups) $f' : H \to E$. Let H act on M by the composite morphism (multiplicative and independent of the choice of f'),

$$H \xrightarrow{f'} E \xrightarrow{int} \mathrm{Aut}_{\hat{C}\text{-gr.}}(M).$$

(i) The morphism f canonically defines a class $c(f) \in H^2(H, M)$ whose vanishing is necessary and sufficient for the existence of a morphism of $\hat{C}$-groups

$$f' : H \longrightarrow E$$

lifting f.

(ii) If $c(f) = 0$, the set of morphisms of $\hat{C}$-groups f' lifting f, modulo the action of the inner automorphisms defined by elements of $\Gamma(M)$ (i.e. by elements m of $\Gamma(E)$ such that $v(m) = e$), is principal homogeneous under $H^1(H, M)$.

(iii) If $f' : H \to E$ is a morphism of groups lifting f, the set of transforms of f' by the inner automorphisms defined by the elements of $\Gamma(M)$ is isomorphic to $\Gamma(M)/\Gamma(M^H) = \Gamma(M)/H^0(H, M)$.

1.3. Extensions of group laws.

Consider the following situation: one has a morphism of $\hat{C}$

(†) $\quad p : X \longrightarrow Y$

and a commutative $\hat{C}$-group M acting on X, such that X is formally principal homogeneous above Y under M_Y.

If $g : Y \to Z$ is an arbitrary morphism of $\hat{C}$, then $g \circ p : X \to Z$ is invariant under M: for each $S \in ObC$, $(g \circ p)(S)$ is invariant under the action of $M(S)$ acting on $X(S)$. Conversely, we shall assume the following condition verified for $n = 1, 2, 3, 4$.

$(+)_n$: Every morphism from X^n to M, invariant under the action of M^n on X^n, factors uniquely through $p^n : X^n \to Y^n$ (where the powers n denote cartesian powers).

Lemma 1.3.1. *(i) If h is a morphism from Y to M, the automorphism u_h of X defined set-theoretically by $x \mapsto h(p(x)) \cdot x$ preserves the fibers of p and commutes with the actions of M on X,[^N.D.E-III-46] i.e. for all $S \in ObC$ and $x \in X(S)$, $m \in M(S)$, one has*

$$p(h(p(x)) \cdot x) = p(x), \qquad m \cdot (h(p(x)) \cdot x) = h(p(m \cdot x)) \cdot (m \cdot x) .$$

(ii) This construction realizes a bijective correspondence between morphisms from Y to M and automorphisms of X preserving the fibers of p and commuting with the actions of M.

The first part is clear, since $p(m \cdot x) = p(x)$ and M is commutative.

Conversely, an automorphism u of X preserving the fibers of p is written set-theoretically $x \mapsto g(x) \cdot x$, where g is some morphism from X to M. If u commutes with the actions of M, the morphism g is invariant under M[^N.D.E-III-47] and one concludes by condition $(+)_1$.

We now suppose given in addition a group law on Y and an action of Y on M, that is, a morphism of $\hat{C}$-groups:

(‡) $\quad f : Y \longrightarrow \mathrm{Aut}_{\hat{C}\text{-gr.}}(M).$

Definition 1.3.2. *A composition law on X*

$$P : X \times X \longrightarrow X$$

is said to be ***admissible*** *if it verifies the following two conditions:*

(i) P lifts the group law of Y, i.e. the diagram

$$\begin{tikzcd} X \times X \arrow[r, "P"] \arrow[d, "(p,p)"'] & X \arrow[d, "p"] \\ Y \times Y \arrow[r] & Y \end{tikzcd}$$

is commutative.

(ii) For every $S \in ObC$ and all $x, y \in X(S)$, $m, n \in M(S)$, one has the following relation:

$$(++) \qquad P(m \cdot x, n \cdot y) = m \cdot f(p(x))(n) \cdot P(x, y).$$

Proposition 1.3.3. *For a group law $*$ on X to be admissible, it is necessary and sufficient that the following four conditions be satisfied:*

(i) $p : X \to Y$ is a morphism of groups.

(ii) The morphism $i : M \to X$ defined by $i(m) = m \cdot e_X$ is an isomorphism of groups from M onto $Ker(p)$, that is to say: one has set-theoretically $(m \cdot e_X) * (n \cdot e_X) = (mn) \cdot e_X$.

(iii) One has $m \cdot x = (m \cdot e_X) * x = i(m) * x$ for each $m \in M(S)$, $x \in X(S)$.

(iv) The inner automorphisms of X act on $Ker(p)$ by the set-theoretic formula

$$x * i(m) * x^{-1} = i(f(p(x))\, m).$$

The proof is immediate.

Lemma 1.3.4.[^N.D.E-III-48] *Let h be a morphism $Y \to M$ and u_h the automorphism $x \mapsto h(p(x)) \cdot x$ of X (cf. 1.3.1). Let P be an admissible composition law (resp. an admissible group law) on X and let P' be the composition law on X deduced from P by means of u_h, i.e. $P'(x, y) = u_h^{-1}(P(u_h(x), u_h(y)))$. Then:*

(i) P' is an admissible composition law (resp. an admissible group law).

(ii) For every $x, y \in X(S)$ ($S \in ObC$), setting $v = p(x)$ and $w = p(y)$, one has

$$P'(x, y) = (h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w))) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

Proof. One has $u_h^{-1} = u_{h^\vee}$, where $h^\vee : Y \to M$ is defined by $h^\vee(y) = h(y)^{-1}$. By 1.3.2 (i) and (ii), one has $P(h(v) \cdot x, h(w) \cdot y) = h(v) \cdot f(v)(h(w)) \cdot P(x, y)$ and $p(P(x, y)) = vw$, whence

$$P'(x, y) = (h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w))) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

It is then immediate that P' verifies conditions (i) and (ii) of 1.3.2.

Definition 1.3.5. *Two admissible composition laws deduced from one another by the procedure of 1.3.4 are said to be* ***equivalent****.[^N.D.E-III-49]*

Proposition 1.3.6. *Suppose there exists an admissible composition law on X. Then:*

(i) There exists a class $c \in H^3(Y, M)$ (canonically determined), whose vanishing is necessary and sufficient for the existence of an admissible ***associative*** *composition law on X.*

(ii) If $c = 0$, the set of admissible and associative composition laws (resp. of equivalence classes of admissible and associative composition laws) on X is principal homogeneous under $Z^2(Y, M)$ (resp. $H^2(Y, M)$).

The proof proceeds in several steps.

a) Let P be an admissible composition law on X. Since P lifts the composition law of Y which is associative, there exists a unique morphism $a : X^3 \to M$ such that

$$(*) \qquad P(x, P(y, z)) = a(x, y, z) \cdot P(P(x, y), z).$$

By applying conditions 1.3.2 (i) and (ii), one sees at once that a is invariant under the action of M^3 on X^3,[^N.D.E-III-50] whence, by applying hypothesis $(+)_3$, the following result:

(1) There exists a unique morphism $DP : Y^3 \to M$ such that

$$P(x, P(y, z)) = DP(p(x), p(y), p(z)) \cdot P(P(x, y), z),$$

and P is associative if and only if $DP = 0$.

b) Compute step by step $P(P(P(x,y),z),t)$ by means of the preceding formula. Setting $p(x) = u$, $p(y) = v$, $p(z) = w$, $p(t) = h$, one obtains[^N.D.E-III-51] the following pentagonal diagram, where an arrow $a \xrightarrow{m} b$ means that $b = m \cdot a$:

$$\begin{tikzcd}[column sep=small]
 & P(x, P(y, P(z, t))) \arrow[dl, "DP(u,v,wh)"'] \arrow[dr, "f(u)(DP(v,w,h))"] & \\
P(P(x, y), P(z, t)) \arrow[d, "DP(uv,w,h)"'] & & P(x, P(P(y, z), t)) \arrow[d, "DP(u,vw,h)"] \\
P(P(P(x, y), z), t) & & P(P(x, P(y, z)), t) \arrow[ll, "DP(u,v,w)"']
\end{tikzcd}$$

so one finds

$$DP(u, v, w) \cdot DP(u, vw, h) \cdot f(u)(DP(v, w, h)) \cdot DP(u, v, wh)^{-1} \cdot DP(uv, w, h)^{-1} = e_M$$

i.e., $\partial^3 DP(u,v,w,h) = e_M$. As moreover the first member of the preceding formula can be written, by means of P and a, as the expression in (x,y,z,t) of a certain morphism $X^4 \to M$, it follows from the uniqueness hypothesis in $(+)_4$ that $\partial^3 DP$ and e_M, which factor through the same morphism, are equal, hence

(2) DP is a cocycle, i.e. one has $DP \in Z^3(Y, M)$.

c) If P and P' are two admissible composition laws on X, there exists a unique morphism

$$b : X^2 \longrightarrow M$$

such that $P'(x,y) = b(x,y) \cdot P(x,y)$. Applying 1.3.2 (ii) to P and P', one sees that b is invariant under M^2, whence, by $(+)_2$:

(3) For every pair of admissible composition laws (P,P'), there exists a unique $d(P,P') : Y^2 \to M$ such that

$$P'(x, y) = d(P, P')(p(x), p(y)) \cdot P(x, y),$$

and the set of admissible composition laws becomes in this way principal homogeneous under $\mathrm{Hom}(Y^2, M) = C^2(Y, M)$.

d) Under the preceding conditions, one has the formula:

$$(4) \qquad DP' - DP = \partial^2 d(P, P').$$

e) P and P' are equivalent if and only if there exists a morphism $h \in C^1(Y,M) = \mathrm{Hom}(Y,M)$ such that $d(P,P') = \partial^1 h$; this follows from the definition of equivalence and from 1.3.4 (ii).

f) It now only remains to conclude: one seeks a P' that is associative, i.e. such that $DP' = e_M$. Now DP is a cocycle whose class in $H^3(Y,M)$ does not depend on the chosen admissible composition law P (by (3) and (4)). This class is the desired obstruction c. One will be able to choose a P' answering the conditions if and only if $c = 0$; indeed, choosing an arbitrary P, one will have to solve, by (1):

$$0 = DP' = DP + \partial^2 d(P, P'),$$

which is possible by (3) and (4) if and only if $c = 0$. The set of associative P' is principal homogeneous under $Z^2(Y,M)$, again by (3) and (4). The set of associative P' up to equivalence is principal homogeneous under $H^2(Y,M)$ by (e).

2. Infinitesimal extensions of a morphism of group schemes

Resume the notations of § 0. Let Y and X be two S-group functors. Let M be the kernel of the morphism of groups $p_X : X \to X^+$. One thus has an exact sequence of S-group functors

$$1 \longrightarrow M \longrightarrow X \xrightarrow{p_X} X^+ .$$

By definition of X^+, one has isomorphisms

$$\begin{aligned} \mathrm{Hom}_S(Y, X^+) &\xrightarrow{\sim} \mathrm{Hom}_{S_J}(Y_J, X_J) \\ \mathrm{Hom}_{S\text{-gr.}}(Y, X^+) &\xrightarrow{\sim} \mathrm{Hom}_{S_J\text{-gr.}}(Y_J, X_J), \end{aligned}$$

and the morphism

$$\mathrm{Hom}_S(Y, p_X) : \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+)$$

associates to an S-morphism $f : Y \to X$ the S-morphism $f^+ : Y \to X^+$ corresponding by the preceding isomorphisms to the S_J-morphism $f_J : Y_J \to X_J$ obtained by base change from f. If M is commutative, one can apply to this situation Proposition 1.2.4.

2.0.

[^N.D.E-III-52] In what follows, we shall be interested in the following case: Y is flat over S, and X is an S-group functor of one of the following two species:

a) X is an S-group scheme,

b) $X = \mathrm{Aut}_S(E)$ where E is an S-scheme, of finite presentation over S.

Denote by $(Flat)/S$ the category of S-schemes flat over S. In case (a) (resp. (b)), the S-group functor $M = Ker(X \to X^+)$, its restriction L to $(Flat)/S$, and the actions of the inner automorphisms of X on M, have been computed in 0.9, 0.5, and 0.10 (resp. 0.11, 0.14, and 0.12). That is to say, in case (a), let L_0 be the S_0-functor in commutative groups defined by: for every S_0-scheme T_0,

$$\mathrm{Hom}_{S_0}(T_0, L_0) = \mathrm{Hom}_{O_{T_0}}(\omega^1_{X_0/S_0} \otimes_{O_{S_0}} O_{T_0}, J \otimes_{O_{S_0}} O_{T_0}),$$

on which X_0 acts via its adjoint representation in $\omega^1_{X_0/S_0}$; then $L = \prod_{S_0/S} L_0$, i.e. for every S-scheme T, one has $L(T) = L_0(T \times_S S_0)$.

In case (b), denote by π the structural morphism $E \to S$; then L is the functor in abelian groups on $(Flat)/S$ defined by

$$\mathrm{Hom}_S(T, L) = \Gamma(T, \pi_*(\mathrm{Hom}_{O_E}(\Omega^1_{E/S}, J \cdot O_E)) \otimes_{O_S} O_T),$$

on which X, considered as a functor on $(Flat)/S$, acts as we saw in 0.12.

Then, one has an exact sequence of functors in groups on $(Flat)/S$:

$$(E) \qquad 1 \longrightarrow L \longrightarrow X \longrightarrow X^+ .$$

On the other hand, Y being supposed flat over S, the groups $H^i(Y, M)$ depend, by 1.1.1, only on the restriction L of M to $(Flat)/S$. Since $L = \prod_{S_0/S} L_0$ in case (a), then by 1.1.2, one has in this case isomorphisms $H^i(Y, L) \simeq H^i(Y_0, L_0)$.

Then, taking the preceding into account, one deduces from Proposition 1.2.4 the:[^N.D.E-III-53]

Theorem 2.1. *Let S be a scheme, I and J two quasi-coherent ideals such that $I \supset J$ and $I \cdot J = 0$, defining the closed subschemes S_0 and S_J, and let:*

— X an S-group functor of type (a) or (b), and L_0, L as above;

— Y an S-group scheme flat over S and $f_J : Y_J \to X_J$ a morphism of S_J-groups.

Then:

(i) For f_J to lift to a morphism of S-groups $Y \to X$, it is necessary and sufficient that the following two conditions be satisfied:

(i_1) f_J lifts to a morphism of S-functors $Y \to X$ (by 1.2.4, this defines an action of Y on L, which does not depend on the chosen lift; moreover, in case (a), the action thus obtained of Y_0 on L_0 comes from the morphism $f_0 : Y_0 \to X_0$ and from "the adjoint action" of X_0 on L_0);

(i_2) A certain obstruction $c(f_J)$, defined canonically by f_J, vanishes, where $c(f_J)$ is a class in $H^2(Y, L)$ ($\simeq H^2(Y_0, L_0)$ in case (a)).

(ii) If the conditions of (i) are satisfied, the set E of morphisms of S-group functors $Y \to X$ extending f_J is principal homogeneous under $Z^1(Y, L)$ ($\simeq Z^1(Y_0, L_0)$ in case (a)), and E modulo the action of the inner automorphisms of X defined by the sections of X over S inducing the unit section of X_J over S_J, is principal homogeneous under $H^1(Y, L)$ ($\simeq H^1(Y_0, L_0)$ in case (a)).

(iii) If $f : Y \to X$ is a morphism of S-group functors extending f_J, the set of morphisms $Y \to X$ transforms of f by the inner automorphisms defined by the sections of X over S inducing the unit section of X_J over S_J is isomorphic to $\Gamma(L)/H^0(Y, L)$ ($\simeq \Gamma(L_0)/H^0(Y_0, L_0)$ in case (a)).

Remark 2.1.1.[^N.D.E-III-54] *If $f, f' : Y \to X$ are morphisms of S-group functors extending f_J, one therefore obtains a cocycle $d(f, f') \in Z^1(Y, L)$ ($\simeq Z^1(Y_0, L_0)$ in case (a)), such that*

```
(∗)      f′ = d(f, f′) · f .[^N.D.E-III-55]
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We shall denote by $\bar{d}(f, f')$ the image of $d(f, f')$ in $H^1(Y, L)$ ($\simeq H^1(Y_0, L_0)$ in case (a)).

Remark 2.2. *We keep the preceding notations; in particular, Y is flat over S. In case (b), L is, by (0.14.1), the restriction to $(Flat)/S$ of the functor*

$$W(\pi_*(\mathrm{Hom}_{O_E}(\Omega^1_{E/S}, J \cdot O_E))),$$

where $\pi : E \to S$ is the structural morphism. In case (a), suppose moreover that X is locally of finite presentation over S; then by (0.6.1), L is the restriction to $(Flat)/S$ of the functor

$$\prod_{S_0/S} W(\mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J)).$$

In both cases, the module of which we take the W is quasi-coherent, by (EGA I, 9.1.1). Suppose moreover Y affine over S.[^N.D.E-III-56] Then, by (I, 5.3), one obtains:

a) $H^i(Y, L) = H^i(Y_0, L_0) = H^i(Y_0, \mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J))$,

b) $H^i(Y, L) = H^i(Y, \pi_*(\mathrm{Hom}_{O_E}(\Omega^1_{E/S}, J \cdot O_E)))$.

Remark 2.3. *1) By 0.16 and 0.17, condition (i_1) is automatically verified when Y is an affine scheme and*

```
(∗)      {  in case (a), X is smooth over S;
            in case (b), E is smooth and affine over S.
```

2) Moreover, under these conditions (Y always being supposed flat over S, cf. 2.0), one can write in case (a), by 2.2 a) and (0.6.2),

$$H^i(Y, L) = H^i(Y_0, L_0) = H^i(Y_0, \mathrm{Lie}(X_0/S_0) \otimes_{O_{S_0}} J),$$

[^N.D.E-III-57] and in case (b), by (0.14.2), 1.1.2 and (I, 5.3),

$$H^i(Y, L) = H^i(Y_0, \pi_{0*} \mathrm{Hom}_{O_{E_0}}(\Omega^1_{E_0/S_0}, J \otimes_{O_{S_0}} O_{E_0})).$$

Let us now state a certain number of corollaries concerning the case where Y is an S-diagonalizable group (I, 4.4); one knows then (loc. cit. 5.3.3) that if S is affine, $H^n(Y,F) = 0$ for $n > 0$ and every quasi-coherent O_S-module F. First, a particular case:

Corollary 2.4. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal. Let Y be a diagonalizable S-group and let:*

a) X an S-group locally of finite presentation over S,

b) $X = \mathrm{Aut}_S(E)$ where E is an S-scheme locally of finite presentation.

Let $f : Y \to X$ be a morphism of S-groups such that the morphism $f_0 : Y_0 \to X_0$ obtained by base change is the unit morphism. Then f is the unit morphism.

Indeed, the question is local on S and (in (b)) on E. We may therefore suppose S affine and (in (b)) E of finite presentation over S. Now introducing the closed subschemes S_n of S defined by the powers of the ideal defining S_0, one is reduced to the case where S_0 is defined by an ideal of square zero, and in that case the asserted statement follows from the theorem, via 2.2.

In the case where one does not necessarily suppose that f_0 is the unit morphism, one has:

Corollary 2.5. *Let S and S_0 be as in 2.4. Suppose moreover S affine. Let Y be a diagonalizable S-group, X an S-group functor, and $f_0 : Y_0 \to X_0$ a morphism of S_0-group functors.*

(i)[^N.D.E-III-58]

(ii) Suppose that one of the following two properties holds:

(a) X is an S-group smooth over S;

(b) $X = \mathrm{Aut}_S(E)$ where E is smooth and affine over S.

Then f_0 extends to a morphism of S-groups $Y \to X$; two such extensions are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0.

Introduce the S_n as above.[^N.D.E-III-59] For (ii), note first that a scheme smooth over S is necessarily locally of finite presentation over S; hence, in case (b), E being smooth and affine over S is necessarily of finite presentation over S, i.e. we are indeed under hypothesis (b) of 2.0.

Then, under the hypotheses of (ii), condition (i_1) of 2.1 is automatically verified by 0.16 and 0.17; moreover every section of X_{S_n} over S_n lifts to a section of $X_{S_{n+1}}$ over S_{n+1}, by the definition of "smooth over S" in case (a), and by 0.17 in case (b). Consequently, if f and f' are two lifts of f_0, one can suppose step by step that $f_n = f'_n$ by lifting the inner automorphism whose existence is asserted by part (ii) of the theorem, which completes the proof.

By reasoning likewise, taking into account Remark 2.3, one obtains:

Corollary 2.6. *Let S be a scheme, I a nilpotent ideal defining the closed subscheme S_0, Y an S-group flat over S and affine, X an S-group smooth over S.*

(i) If, for every $n \geq 0$, one has $H^2(Y_0, Lie(X_0/S_0) \otimes_{O_{S_0}} I^{n+1}/I^{n+2}) = 0$, every morphism of S_0-groups $f_0 : Y_0 \to X_0$ lifts to a morphism of S-groups $f : Y \to X$.

(ii) If, for every $n \geq 0$, one has $H^1(Y_0, Lie(X_0/S_0) \otimes_{O_{S_0}} I^{n+1}/I^{n+2}) = 0$, two such lifts are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0.

Now one has the following lemma:

Lemma 2.7. *Let S be an affine scheme, G an affine S-group, F a quasi-coherent 0_S-module, L a locally free 0_S-module. Suppose one has an action of G on F in the sense of Exposé I, which defines an action of G on $F \otimes_{O_S} L$[^N.D.E-III-60]. Denote by Λ the ring of S, L the Λ-module defining L (which is therefore a projective module). One has a canonical isomorphism*

H*(G, F ⊗_{O_S} L) ≃ H*(G, F) ⊗_Λ L.

[^N.D.E-III-61] Indeed, denote by A the 0_S-algebra $A(G)$ and consider the complex C of quasi-coherent 0_S-modules:

0 ⟶ F ⟶ F ⊗_{O_S} A ⟶ F ⊗_{O_S} A ⊗_{O_S} A ⟶ · · ·

By (I, 5.3), $H*(G, F)$ (resp. $H*(G, F \otimes_{O_S} L)$) is the cohomology of the complex $\Gamma(S, C)$ (resp. $\Gamma(S, C \otimes_{O_S} L)$). Now, since S is affine, one has (cf. EGA I, 1.3.12)

Γ(S, C ⊗_{O_S} L) ≃ Γ(S, C) ⊗_Λ L.

Since L is a projective Λ-module (hence flat), one has also $H*(\Gamma(S, C) \otimes_\Lambda L) \simeq H*(\Gamma(S, C)) \otimes_\Lambda L$, whence the announced result.

By using the lemma, one transforms 2.6 into:

Corollary 2.8. *Let S be an affine scheme, I a nilpotent ideal on S defining the closed subscheme S_0. Suppose the I^{n+1}/I^{n+2} locally free on S_0. Let Y be an S-group flat over S and affine, X an S-group smooth over S, and $f_0 : Y_0 \to X_0$ a morphism of S-groups.*

(i) If $H^2(Y_0, Lie(X_0/S_0)) = 0$, f_0 lifts to a morphism of S-groups $Y \to X$.

(ii) If $H^1(Y_0, Lie(X_0/S_0)) = 0$, two such lifts are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0.

In particular, taking $Y = X$:

Corollary 2.9. *Let S and S_0 be as above. Let X be an S-group smooth over S and affine.*

(i) If $H^1(X_0, Lie(X_0/S_0)) = 0$, every endomorphism of X over S inducing the identity on X_0 is the inner automorphism defined by a section of X over S inducing the unit section of X_0 over S_0.

(ii) If $H^2(X_0, Lie(X_0/S_0)) = 0$, every S_0-automorphism of X_0 extends to an S-automorphism of X.[^N.D.E-III-62]

Remark 2.10. *The assertions concerning H^1 have converses by the theorem. Let us signal as an example the following: if $S = IS_0$ is the scheme of dual numbers over S_0 (II, 2.1) and if X is a flat S-group such that every automorphism of X over S inducing the identity on S_0 is the inner automorphism defined by a section of X over S inducing the unit section of X_0 over S_0, then $H^1(X_0, Lie(X_0/S_0)) = 0$.[^N.D.E-III-63]*

Corollary 2.11. *Let S, I and J be as in 2.1. Let Y be an S-group scheme flat over S, X an S-group scheme, $f : Y \to X$ a morphism of S-groups. The set of morphisms from Y to X deduced from f by conjugation by elements $x \in X(S)$ inducing the unit of $X(S_J)$ is isomorphic to the quotient*

$$E = \mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J) / \mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J)^{ad(Y_0)} ,$$

where the second group consists of the O_{S_0}-morphisms $\omega^1_{X_0/S_0} \to J$ which by every base change $S' \to S_0$ give morphisms $\omega^1_{X_0/S_0} \otimes_{O_{S_0}} O_{S'} \to J \otimes_{O_{S_0}} O_{S'}$ invariant under the action of $Y_0(S')$ on the first factor.

By 2.1 (iii), one knows that the set sought is isomorphic to $\Gamma(L_0)/H^0(Y_0, L_0)$. Now $\Gamma(L_0) = \mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J)$ and $H^0(Y_0, L_0)$ is evidently none other than $\Gamma(L_0)^{ad(Y_0)}$ in the sense of the preceding statement.

Corollary 2.12. *Under the conditions of 2.11, suppose moreover $\omega^1_{X_0/S_0}$ locally free of finite rank. Then*

$$E \simeq \Gamma(S_0, Lie(X_0/S_0) \otimes_{O_{S_0}} J) / H^0(Y_0, Lie(X_0/S_0) \otimes_{O_{S_0}} J).$$

[^N.D.E-III-64] Indeed, if $\omega^1_{X_0/S_0}$ is locally free of finite rank, one has $\mathrm{Hom}_{O_{S_0}}(\omega^1_{X_0/S_0}, J) \simeq Lie(X_0/S_0) \otimes_{O_{S_0}} J$.

Corollary 2.13. *Suppose moreover Y_0 diagonalizable. Then*

$$E \simeq \Gamma(S_0, Lie(X_0/S_0) \otimes_{O_{S_0}} J) / \Gamma(S_0, Lie(X_0/S_0)^{ad(Y_0)} \otimes_{O_{S_0}} J)$$

where $Lie(X_0/S_0)^{ad(Y_0)}$ can be constructed as the factor of the decomposition of (I, 4.7.3) corresponding to the null character of Y_0.

Indeed, if $Y_0 \simeq D_{S_0}(M)$, one has by loc. cit. a decomposition into direct sum:

$$Lie(X_0/S_0) = Lie(X_0/S_0)_0 \oplus \bigoplus_{m \in M,\, m \neq 0} Lie(X_0/S_0)_m .$$

By tensoring with J, one finds an analogous decomposition for $Lie(X_0/S_0) \bigotimes_{O_{S_0}} J$, whence the relation

$$H^0(Y_0, Lie(X_0/S_0) \otimes J) \simeq \Gamma(S_0, Lie(X_0/S_0)_0 \otimes_{O_{S_0}} J).$$

Corollary 2.14. *Suppose moreover S_0 affine. Then*

$$E \simeq \Gamma(S_0, [Lie(X_0/S_0) / Lie(X_0/S_0)^{ad(Y_0)}] \otimes_{O_{S_0}} J).$$

3. Infinitesimal extensions of a group scheme

Still in the notation of n° 0 (S, I, J, etc.), let us give ourselves an S-scheme X and suppose X_J endowed with a group structure. We propose to find the S-group structures on X inducing on X_J the given structure.

From now on, we assume X flat over S. Let C be the category of S-schemes flat over S. We have therefore $X \in ObC$. We shall denote by Y, resp. M, the functor on C defined by X^+, resp. L'_X. The canonical morphism $p_X : X \to X^+$ defines a morphism of $\hat{C}$

$$p : X \longrightarrow Y$$

and the action of L'_X on X in $(Sch)/S$ defines an action of M on X in $\hat{C}$. One verifies at once that X thus becomes formally principal homogeneous under M_Y above Y (cf. 0.2 (i) and 0.4).

The action of X^+ on L'_X defined in 0.8 (denoted Ad in loc. cit.) defines an action denoted f of Y on M. One knows, on the other hand (0.5), that

$$Hom_{\hat{C}}(Z, M) \simeq Hom_{S_0}(Z_0, L_0), \qquad Z \in Ob\, C,$$

where L_0 is the functor defined in 0.5.

Lemma 3.1. *(i) Condition $(+)_n$ of 1.3 is satisfied for every positive integer n.*

(ii) If one makes the S_0-group X_0 act on the S_0-functor L_0 through its adjoint representation, one has a canonical isomorphism

$$H^*(X_0, L_0) \simeq H^*(Y, M),$$

(the first cohomology being computed in $(Sch)/S_0$, the second in C).

Both parts of the lemma follow from the relation:

$$\begin{aligned} Hom_{\hat{C}}(Y, M) &\simeq Hom_{(Sch)/S_0}(X^+ \times_S S_0, L_0) \\ &\simeq Hom_{S_0}(X_0, L_0) \\ &\simeq Hom_{\hat{C}}(X, M), \end{aligned}$$

which arises at once from the definition of M as a $\prod_{S_0/S}$. This relation being more generally satisfied on replacing X, Y by X_n, Y_n, one deduces that every morphism $X_n \to M$ factors in a unique manner through Y_n, which entails $(+)_n$. One also deduces from it the relation $C*(Y,M) = C*(X_0,L_0)$, which entails (ii).

We may therefore apply the constructions of 1.3. In particular:

Lemma 3.2. *Let $P : X \times_S X \to X$ be a morphism. In order for P to induce the group law of X_J, it is necessary and sufficient that P be an admissible composition law (cf. 1.3.2) on X.*

Indeed, in order for P to induce the group law of X_J, it is necessary and sufficient that P lift the group law of X^+, or equivalently that of Y. It therefore only remains to show that every morphism P lifting the group law of X_J satisfies the identity (++) of 1.3.2 (ii), and this is exactly what was seen in 0.8.

Proposition 3.3. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal. Let X be a flat S-scheme, quasi-compact or locally of finite presentation over S. Let $P : X \times_S X \to X$ be a composition law on X. In order for P to be a group law, it is necessary and sufficient that the two following conditions be satisfied:*

(i) P is associative.

(ii) P induces on $X_0 = X \times_X S_0$ a group law.

These conditions are obviously necessary. Let us show that they are sufficient. Suppose first that $X \to S$ has a section. Since $X(S')$ is then non-empty for each $S' \to S$, it suffices[187] to show that, for every $x \in X(S')$, the left and right translations by x are isomorphisms of $X_{S'}$.[188]

One may evidently suppose $S' = S$; the translation in question t induces on X_0 a translation t_0 of X_0, which is therefore an automorphism since X_0 is a group. One concludes by flatness (SGA 1 III 4.2).[189]

No longer supposing now that X has a section over S, suppose that there exists an $S' \to S$ such that $X_{S'}$ has a section over S'. Then $X_{S'}$ is an S'-group according to what we have just seen; consider its unit section e'. The inverse image of e' by $pr_i : S' \times_S S' \to S'$ $(i = 1, 2)$ is the unit section of $X_{S''}$ for the group law inverse image of $P_{S'}$ by pr_i. But since P is "defined over S", these two group laws coincide, and therefore so do their unit sections. One has therefore $pr_1 * (e') = pr_2 * (e')$.

[187] N.D.E.: We have corrected the original by suppressing the inadequate reference to an exercise of Bourbaki on semigroups (cf. [BAlg], § I.2, Exercises 9–13) and by indicating the role of left and right translations; see the following N.D.E.

[188] N.D.E.: Let E be a non-empty set equipped with an associative composition law, such that every left translation ℓ_x is bijective; fix $x_0 \in E$. There exists a unique $e \in E$ such that $x_0 \cdot e = x_0$; then $x_0 \cdot e \cdot x = x_0 \cdot x$ entails $e \cdot x = x$, for every $x \in E$. On the other hand, for every x there exists a unique x' such that $x \cdot x' = e$. Suppose moreover that there exists $b \in E$ such that the right translation r_b is injective. Then, for every x, the equality $x \cdot e \cdot b = x \cdot b$ gives $x \cdot e = x$ (i.e. e is a neutral element), and $x \cdot x' \cdot x = x = x \cdot e$ entails $x' \cdot x = e$, i.e. x' is the inverse of x from the left and the right, hence E is a group. Note that the hypothesis "r_b injective" is necessary: on every set E one can define a composition law by $x \cdot y = y$, for all $x, y \in E$; then every left translation is the identity (whence the associativity of the law), but for every y one has $r_y(E) = y$, hence E is not a group if $|E| > 1$.

[189] N.D.E.: Since X and X_0 have the same underlying topological space and t_0 is an automorphism, t is a homeomorphism, hence an affine morphism, cf. Exp. VI_B, 2.9.1 or EGA IV₄, 18.12.7.1. It thus suffices to see that if J is a nilpotent ideal of a ring Λ, and $\phi : A \to B$ a morphism of Λ-algebras, with B flat over Λ, such that $\phi \otimes_\Lambda (\Lambda/J)$ is bijective, then ϕ is bijective. By the "nilpotent Nakayama lemma", ϕ is surjective; moreover, B being flat over Λ, one also has $Ker(\phi) \otimes_\Lambda (\Lambda/J) = 0$, whence $Ker(\phi) = 0$, hence ϕ is bijective.

If $S' \to S$ is a descent morphism (cf. Exp. IV n° 2), there will exist a section of X giving e' by base extension, and we shall be done. Since X_X has a section over X (the diagonal section), one sees that it now suffices to prove that $X \to S$ is a descent morphism. Now it is flat and surjective, and quasi-compact or locally of finite presentation, hence covering for (fpqc), hence a descent morphism (Exp. IV, n° 6).

Remark. *In fact the hypothesis $X \to S$ quasi-compact or locally of finite presentation is superfluous, by virtue of the following result which the reader will prove as an exercise on Exposé IV:*

Under the conditions of the text on S and S_0, if $X \to S$ is a flat morphism and $X_0 \to S_0$ a morphism covering for (fpqc), then $X \to S$ is a descent morphism.

Lemma 3.4. *In order for two admissible composition laws on X to be equivalent (cf. 1.3.5), it is necessary and sufficient that they be deduced from one another by an automorphism of X over S inducing the identity on X_J.*

Indeed, the morphisms constructed in 1.3.1 are exactly those of the preceding statement (by 0.7).[190]

Taking all the preceding results into account, Proposition 1.3.6 gives:

Theorem 3.5. *Let S be a scheme, I and J two ideals on S such that $I \supset J$, $I \cdot J = 0$, S_0 and S_J the closed subschemes of S which they define. Let X be an S-scheme flat over S (and locally of finite presentation or quasi-compact over S), X_0 and X_J the schemes obtained by base change. Suppose X_J endowed with an S_J-group structure and denote by L_0 the S_0-functor in commutative groups defined by the formula*

$$\mathrm{Hom}_{S_0}(T, L_0) = \mathrm{Hom}_{O_T}(\omega^1_{X_0/S_0} \otimes_{O_{S_0}} O_T, J \otimes_{O_{S_0}} O_T)$$

on which X_0 acts through its adjoint representation.

(i) For there to exist an S-group structure on X inducing the given structure on X_J, it is necessary and sufficient that the following conditions be satisfied:

(i$_1$) There exists a morphism of S-schemes $X \times_S X \to X$ inducing the group law of X_J.

(i$_2$) A certain obstruction class belonging to $H^3(X_0, L_0)$ (defined canonically by the data of X and the group law of X_J) is zero.

(ii) If the conditions of (i) are satisfied, the set E of group laws on X inducing the given law of X_J is a principal homogeneous set under $Z^2(X_0, L_0)$, and E modulo the S-automorphisms of X inducing the identity on X_J, is a principal homogeneous set under $H^2(X_0, L_0)$.

[190] N.D.E.: Indeed, by the proof of 0.7, the S-endomorphisms of X inducing the identity on X_J are the automorphisms $m \cdot id_X$, for m ranging over $M(X) = \mathrm{Hom}_S(X, M)$ (for every $S' \to S$ and $x \in X(S')$, one has $(m \cdot id_X)(x) = m(x) \cdot x$). Now, by the proof of 3.1, each $m : X \to M$ factors in a unique manner through a morphism h from $Y = X^+$ to M, and therefore $m \cdot id_X$ is the automorphism u_h introduced in 1.3.1. The lemma then follows from the definition of equivalence, cf. 1.3.4 and 1.3.5.

[191] Indeed, every morphism of S-schemes $f : X \times_S X \to X$ inducing the group law of X_J is, by 3.2, an admissible composition law on X; then, by 1.3.6 (i), the existence of an associative admissible composition law $P : X \times_S X \to X$ is equivalent to the vanishing of a certain class $c(f) \in H^3(X_0, L_0)$, and in this case, by 3.3, P is a group law. This proves (i), and (ii) then follows from 3.3 and 1.3.6 (ii).

Remark 3.5.1.[192] *If μ, μ' are group laws on X inducing the given law of X_J, one therefore obtains a cocycle $\delta(\mu,\mu') \in Z^2(X_0, L_0)$, the sign convention chosen being that $\mu' = \delta(\mu,\mu') \cdot \mu$, that is, for every $S' \to S$ and $x, y \in X(S')$,*

$$\mu'(x, y) = \delta(\mu, \mu')(x_0, y_0) \cdot \mu(x, y).$$

[193]

We shall denote by $\bar{\delta}(\mu,\mu')$ the image of $\delta(\mu,\mu')$ in $H^2(X_0, L_0)$. Finally, if X endowed with the group law μ (resp. μ') is designated simply by X (resp. X'), one will write $\delta(X, X')$ instead of $\delta(\mu,\mu')$, and likewise for $\bar{\delta}(X, X')$.

Remark 3.6. *Let X_J be an S_J-scheme smooth over S_J and affine. By 0.15, there exists up to isomorphism a unique S-scheme X, smooth over S, reducing to X_J. If X_J is endowed with an S_J-group structure, it follows from 0.16 that condition (i_1) is automatically satisfied. Moreover, by 0.6 the definition of L_0 simplifies and one obtains:*

Corollary 3.7. *Let S, I and J be as in 3.1. Let X_J be an S_J-group smooth over S_J and affine.*

(i) The set of S-groups smooth over S and reducing to X_J, up to isomorphism (inducing the identity on X_J), is empty or principal homogeneous under the group

$$H^2(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} J).$$

(ii) There exists an S-group smooth over S reducing to X_J if and only if a certain obstruction in

$$H^3(X_0, \mathrm{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} J)$$

is zero.

One deduces as usual the following corollaries:

Corollary 3.8. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal I. Let X_0 be an S_0-group smooth over S and affine.*

(i) If $H^2(X_0, Lie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} I^{n+1}/I^{n+2}) = 0$ for every $n \geq 0$, two S-groups smooth over S reducing to X_0 are isomorphic (by an isomorphism inducing the identity on X_0).

[191] N.D.E.: We have added what follows.

[192] N.D.E.: We have added this remark, analogue of 4.5.1, to introduce the notation $\delta(\mu,\mu')$ (or $\delta(X,X')$), used in 4.38; consequently, we have also added in 3.5 (ii) above the part concerning E itself.

[193] N.D.E.: We have conformed to the sign conventions of the original, in order to have in 4.38 (5) the equality $\partial^1 \bar{d}(X,X') = \bar{\delta}(X,X')$ (see also N.D.E. (54)).

(ii) If $H^3(X_0, Lie(X_0/S_0) \otimes_{O_{S_0}} I^{n+1}/I^{n+2}) = 0$ for every $n \geq 0$, there exists an S-group smooth over S, reducing to X_0.

Corollary 3.9. *Let S be an affine scheme and S_0 a closed subscheme defined by a nilpotent ideal I. Suppose the I^{n+1}/I^{n+2} locally free on S_0. Let X_0 be an S_0-group smooth and affine over S_0.*

(i) If $H^2(X_0, Lie(X_0/S_0)) = 0$, two S-groups smooth over S reducing to X_0 are isomorphic.

(ii) If $H^3(X_0, Lie(X_0/S_0)) = 0$, there exists an S-group smooth over S reducing to X_0.

Corollary 3.10. *Let S_0 be a scheme and $S = IS_0$ the scheme of dual numbers over S_0. Let X_0 be an S_0-group smooth over S_0. In order for every S-group Y, smooth over S, such that Y_0 be S_0-isomorphic to X_0, to be S-isomorphic to $X = X_0 \times_{S_0} S$, it is necessary and sufficient that $H^2(X_0, Lie(X_0/S_0)) = 0$.*[194]

Indeed, by virtue of 3.5 the set of classes, up to an S-group isomorphism "inducing the identity on X_0", of such groups Y, is in bijection with $H^2(X_0, Lie(X_0/S_0))$; hence the set of classes, up to an arbitrary S-group isomorphism, is in bijection with

$$H^2(X_0, Lie(X_0/S_0))/\Gamma_0,$$

where

$$\Gamma_0 = \mathrm{Aut}_{S_0-gr.}(X_0)$$

(which acts in the evident manner on the H^2). The conclusion follows at once.[195]

4. Infinitesimal extensions of closed subgroups

Let us first state a result valid in an arbitrary abelian category.

Lemma 4.1. *Let $0 \to A' \to^i A \to^p A'' \to 0$ be an exact sequence, $\phi : A' \to Q$ a morphism and $\pi : A'' \to P$ an epimorphism with kernel C. Let E be the set (up to isomorphism) of quadruples (B, f, g, h) such that the sequence*

$$0 \longrightarrow Q \xrightarrow{f} B \xrightarrow{g} P \longrightarrow 0$$

be exact and the diagram below commutative:

$$\begin{tikzcd}
0 \arrow[r] & A' \arrow[r, "i"] \arrow[d, "\varphi"] & A \arrow[r, "p"] \arrow[d, "h"] & A'' \arrow[r] \arrow[d, "\pi"] & 0 \\
0 \arrow[r] & Q \arrow[r, "f"] & B \arrow[r, "g"] & P \arrow[r] & 0.
\end{tikzcd}$$

[194] N.D.E.: This is used in XXIV, 1.13.

[195] N.D.E.: Indeed, $\mathrm{Aut}_{S_0-gr.}(X_0)$ acts by group automorphisms on the abelian group $H^2(X_0, Lie(X_0/S_0))$, hence the orbit of 0 is the singleton $\{0\}$; consequently the quotient set is a singleton if and only if $H^2(X_0, Lie(X_0/S_0)) = 0$.

(i) For E to be non-empty, it is necessary and sufficient that the image in $Ext^1(C,Q)$ of the element A of $Ext^1(A'',A')$ be zero.

(ii) Under these conditions, E is a principal homogeneous set under the abelian group $\mathrm{Hom}(C,Q)$.

Introduce the amalgamated sum $B' = A \sqcup_{A'} Q$. One then has a commutative diagram with exact rows:[196]

$$\begin{tikzcd}
0 \arrow[r] & A' \arrow[r, "i"] \arrow[d, "\varphi"] & A \arrow[r, "p"] \arrow[d, "j"] & A'' \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & Q \arrow[r] & B' \arrow[r] & A'' \arrow[r] & 0,
\end{tikzcd}$$

and it is clear that the category of solutions of the problem posed is canonically isomorphic to the category of solutions of the corresponding problem for the sequence

$$0 \longrightarrow Q \longrightarrow B' \longrightarrow A'' \longrightarrow 0$$

and the morphisms id_Q and $\pi : A'' \to P$.[197] In this case, the set E is in bijection with the set of subobjects N of B' such that $B' \to A''$ induces an isomorphism of N with the kernel C of $A'' \to P$, that is to say, the set of morphisms $e : C \to B'$ lifting the canonical morphism $C \to A''$. The abelian group $G = \mathrm{Hom}(C,Q)$ acts on E by $g \cdot e = g + e$ (addition in $\mathrm{Hom}(C,B')$), and if $E \neq \emptyset$ this makes E into a principal homogeneous set under G.

One deduces from this:

Proposition 4.2.[198] *Let S be a scheme, S_J the closed subscheme defined by a quasi-coherent ideal J of square zero, X an S-scheme, F an O_X-module, $X_J = X \times_S S_J$, $F_J = F \bigotimes_{O_S} O_{S_J}$, and $G_J = F_J/H_J$ a quotient module of F_J. Suppose given a morphism of O_{X_J}-modules*

$$f : J \otimes_{O_{S_J}} G_J \to Q.$$

Let E be the sheaf of sets on X defined as follows: for every open U of X, $E(U)$ is the set of quotient modules G of $F|_U$, such that $G/JG = G_J|_U$ and there exists an isomorphism

$$h : JG \xrightarrow{\sim} Q|_U$$

making the diagram

[196] N.D.E.: One has $Coker(j) = B' \sqcup_Q 0 = A \sqcup_{A'} 0 = A''$, and one sees that $Ker(j) \simeq Ker(i) = 0$ by reasoning "as if C were a category of modules"; for a proof solely in terms of arrows, see for example [Fr64], Th. 2.5.4 (∗).

[197] N.D.E.: In what follows, we have replaced A by B' and detailed the end of the argument.

[198] N.D.E.: We have rewritten the statement to be exactly in the setting of the application made of it in 4.3; moreover, we have detailed the proof, following the indications given by M. Demazure.

$$\begin{tikzcd} J \otimes_{O_{S_J}} (G_J|_U) \arrow[r, "f|_U"] \arrow[d, "\text{can.}"] & Q|_U \\ JG \arrow[ur, "h \simeq"'] & \end{tikzcd}$$

commutative (h is then unique, since $J \bigotimes_{O_{S_J}} (G_J|_U) \to JG$ is an epimorphism). Then E is a sheaf formally principal homogeneous under the sheaf in commutative groups

$$A = \mathrm{Hom}_{O_X}(H_J, Q) = \mathrm{Hom}_{O_{X_J}}(H_J, Q).$$

Proof. If $E(U) = \emptyset$ there is nothing to prove; one may therefore suppose that $E(U)$ contains an element $\tilde{G}$. Then, in the diagram below, h is an isomorphism and all the arrows are epimorphisms:

$$\begin{tikzcd} J \otimes_{O_{S_J}} (F_J|_U) \arrow[r] \arrow[d, "\text{can.}"] & J \otimes_{O_{S_J}} (G_J|_U) \arrow[r, "f|_U"] \arrow[d, "\text{can.}"] & Q|_U \\ JF|_U \arrow[r] & J\tilde{G} \arrow[ur, "h \simeq"'] & \end{tikzcd}$$

Therefore, the morphism $J \bigotimes_{O_{S_J}} (F_J|_U) \to Q|_U$ induces an epimorphism (necessarily unique) $\phi : JF|_U \to Q|_U$, and if G is an O_U-module such that $G/JG = G_J|_U$ and one has a commutative diagram with exact rows:

$$\begin{tikzcd} 0 \arrow[r] & JF|_U \arrow[r] \arrow[d, "\varphi"] & F|_U \arrow[r] \arrow[d] & F_J|_U \arrow[r] \arrow[d, "\pi"] & 0 \\ 0 \arrow[r] & Q|_U \arrow[r, "p_J"] & G \arrow[r] & G_J|_U \arrow[r] & 0 \end{tikzcd}$$

(where p_J is the projection $G \to G/JG = G_J|_U$, so that $Q|_U = Ker(p_J) = JG$), then one can identify G with a quotient module of $F|_U$. Consequently, by 4.1 (ii), the set $E(U)$ is principal homogeneous under the abelian group

$$\mathrm{Hom}_{O_X}(H_J, Q)(U) = \mathrm{Hom}_{O_{X_J}}(H_J, Q)(U).$$

Proposition 4.3. *(TDTE IV 5.1) Let S be a scheme, S_J the closed subscheme defined by a quasi-coherent ideal J of square zero, X an S-scheme, F a quasi-coherent O_X-module, $X_J = X \times_S S_J$, $F_J = F \bigotimes_{O_S} O_{S_J}$. Let $G_J = F_J/H_J$ be a quasi-coherent quotient module of F_J, flat over S_J.*

For every open U of X, let $E(U)$ be the set of quasi-coherent[199] quotient modules G of $F|_U$, flat over S, and such that $G/JG \simeq G_J|_U$. Then the $E(U)$ form a sheaf of sets

[199]N.D.E.: To lighten the statement, we have added here the hypothesis that G be quasi-coherent, and deferred to the proof the remark that this hypothesis is automatically satisfied; we have detailed the proof accordingly.

E on X, which is formally principal homogeneous under the sheaf in commutative groups

$$A = \mathrm{Hom}_{O_{X_J}}(H_J, J \otimes_{O_{S_J}} G_J).$$

Proof. Denote by $\pi : X \to S$ the structural morphism. Let U be an open of X and G an O_U-module flat over S and such that $G/JG \simeq G_J|_U$. Then, for every $x \in U$, G_x is a flat module over the local ring $O_{S,s}$ (where $s = \pi(x)$), and therefore the morphism

$$J_s \otimes_{O_{S,s}} (G/JG)_x = J_s \otimes_{O_{S,s}} \overline{G}_x \to (JG)_x$$

is bijective; one has therefore an exact sequence

$$0 \to J \otimes_{O_S} (G_J|_U) \to G \to G_J|_U \to 0$$

and since $J \otimes_{O_S} (G_J|_U)$ and $G_J|_U$ are quasi-coherent O_U-modules, so is G (cf. EGA III, 1.4.17).

Conversely, since one has supposed G_J flat over S_J, if G is a quasi-coherent O_U-module such that $G/JG \simeq G_J$ and such that the morphism $J \otimes_{O_{S_J}} G_J \to JG$ is bijective, then G is flat over S, by the "fundamental criterion of flatness" (cf. SGA 1 IV, 5.5[200]).

Consequently, the set $E(U)$ considered here coincides with the set considered in 4.2, taking for f the identity morphism of $J \otimes_{O_{S_J}} G_J$, and the conclusion follows therefore from 4.2. QED.

[201] Let us preserve the preceding notation. Let Y_J be a closed subscheme of X_J, defined by a quasi-coherent ideal I_{Y_J}. We assume Y_J flat over S_J. Then, applying 4.3 to $F = O_X$ and $G_J = O_{Y_J} = O_{X_J}/I_{Y_J}$, one obtains the following corollary.

Corollary 4.3.1. *Let S, S_J, J, X, X_J, Y_J and I_{Y_J} be as above; one assumes Y_J flat over S_J. Denote by A_J the sheaf in commutative groups*

$$\mathrm{Hom}_{O_{X_J}}(I_{Y_J}, J \otimes_{O_{S_J}} O_{Y_J})$$

on X_J and $A = i_*(A_J)$, where i is the immersion $X_J \hookrightarrow X$.

For every open U of X, let $E(U)$ be the set of closed subschemes Y of U, flat over S, such that $Y \times_S S_J = Y_J \cap U$. Then E is an A-pseudo-torsor.

If moreover a Y exists locally (that is, if every $x \in X$ has an open neighborhood U such that $E(U) \neq \emptyset$), then E is an A-torsor. Now one knows (see for example EGA IV_4, 16.5.15) that the A-torsors on X are parametrized by the group $H^1(X, A) = H^1(X_J, A_J)$, and that E has a global section (i.e. $E(X) \neq \emptyset$) if and only if the cohomology class corresponding to E is zero. One thus obtains:

Corollary 4.4. *Let S, S_J, J, X, X_J, Y_J and I_{Y_J} be as above; one assumes Y_J flat over S_J. Let E be the set of closed subschemes Y of X, flat over S, such that $Y \times_S S_J = Y_J$.*

(i) The set E is empty or principal homogeneous under the abelian group

[200] N.D.E.: See also [BAC], § III.5, th. 1.

[201] N.D.E.: We have detailed what follows and added Corollary 4.3.1. Recall that "pseudo-torsor" is synonymous with "formally principal homogeneous".

$$H^0(X, A) = H^0(X_J, A_J) = \mathrm{Hom}_{O_{X_J}}(I_{Y_J}, J \otimes_{O_{S_J}} O_{Y_J}).$$

(ii) For E to be non-empty, it is necessary and sufficient that the two following conditions be satisfied:

(a) There exists locally on X a solution of the problem.

(b) A certain obstruction is zero, lying in

$$H^1(X_J, \mathrm{Hom}_{O_{X_J}}(I_{Y_J}, J \otimes_{O_{S_J}} O_{Y_J})).$$

Complement 4.4.1.[202] *Let us keep the notation of 4.4 and suppose that E contains an element Y. Denote by I_Y the ideal of O_X defining Y, and I_{Y_J} its image in O_{X_J}. Then, as was seen in the proof of 4.2, one has a commutative diagram*

$$\begin{tikzcd}
J \otimes_{O_{S_J}} O_{X_J} \arrow[r] \arrow[d] & JO_X \arrow[d] \\
J \otimes_{O_{S_J}} O_{Y_J} \arrow[r, "\simeq"] & JO_Y
\end{tikzcd}$$

hence an epimorphism of O_X-modules $\phi : JO_X \to J \bigotimes_{O_{S_J}} O_{Y_J}$; denote by K its kernel. Then, for every element Y' of E, the morphism $O_X \to O_{Y'}$ factors through O_X/K (which is the amalgamated sum B' of the proof of Lemma 4.1) and, denoting by $I_{Y'}$ the ideal of Y' in O_X, one has a commutative diagram:

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & & \\
 & & I_{Y'}/K \arrow[r, "\simeq"] \arrow[d] & I_{Y_J} \arrow[d] & & \\
0 \arrow[r] & (JO_X)/K \arrow[r] \arrow[d] & O_X/K \arrow[r] \arrow[d] & O_{X_J} \arrow[r] \arrow[d] & 0 & \\
0 \arrow[r] & J \otimes_{O_{S_J}} O_{Y_J} \arrow[r] & O_{Y'} \arrow[r] \arrow[d] & O_{Y_J} \arrow[r] \arrow[d] & 0 & \\
 & & 0 & 0. & &
\end{tikzcd}$$

Therefore, replacing X by the closed subscheme defined by K, one reduces to $K = 0$. Then, the datum of Y' is equivalent to that of the sub-O_X-module $I_{Y'}$ of O_X, sending bijectively onto I_{Y_J} by the projection $p : O_X \to O_{X_J}$; denote by $f' : I_{Y_J} \xrightarrow{\sim} I_{Y'}$ (resp. $f : I_{Y_J} \xrightarrow{\sim} I_Y$) the inverse isomorphism. Then $f' - f$ is an element of

$$\mathrm{Hom}_{O_{X_J}}(I_{Y_J}, J \otimes_{O_{S_J}} O_{Y_J}) = \mathrm{Hom}_{O_{X_J}}(I_{Y_J}, JO_Y)$$

which we shall denote $d(Y', Y)$. (Note that $d(Y, Y') = -d(Y', Y)$.)

For our fixed Y and variable Y', consider the morphism:

[202] N.D.E.: We have added this complement, useful to prove point (ii) of Proposition 4.8.

$$I_{Y'} \to O_X \to O_Y = O_X/I_Y;$$

since the composition with $O_Y \to O_{Y_J}$ is zero, one knows that it takes values in $JO_Y = J \bigotimes_{O_{S_J}} O_{Y_J}$. More precisely, if V is an open of X, x' a section of $I_{Y'}$ on V and x_J its image in $\Gamma(V, I_{Y_J})$, then

$$x' = f'(x_J) = f(x_J) + (f' - f)(x_J) = f(x_J) + d(Y', Y)(x_J).$$

Consequently: the morphism $I_{Y'} \to JO_Y$ is given by $d(Y', Y)$.

4.5.0.[203] Let us keep the notation of 4.3.1 and 4.4 and carry out a certain number of transformations: $I_{Y_J}/I_{Y_J}^2$ is a quasi-coherent O_{X_J}-module annihilated by I_{Y_J}, hence is the direct image of a quasi-coherent O_{Y_J}-module denoted N_{Y_J/X_J}, called the *conormal sheaf* to Y_J in X_J.[204] Since $J \bigotimes_{O_{S_J}} O_{Y_J}$ is annihilated by I_{Y_J}, the sheaf in commutative groups A_J of 4.3.1 identifies with:

$$\mathrm{Hom}_{O_{Y_J}}(I_{Y_J}/I_{Y_J}^2, J \otimes_{O_{S_J}} O_{Y_J}) = \mathrm{Hom}_{O_{Y_J}}(N_{Y_J/X_J}, J \otimes_{O_{S_J}} O$$

whence, for every $i \geq 0$:

$$H^i(X_J, A_J) = H^i(Y_J, \mathrm{Hom}_{O_{Y_J}}(N_{Y_J/X_J}, J \otimes_{O_{S_J}} O_{Y_J})).$$

[205] One can then suppress the hypothesis "Y closed", as follows. Let us first note that every open U_J of X_J comes by base change from the open subscheme U of X having the same underlying topological space as U_J. Let now Y_J be a closed subscheme of U_J, flat over S_J, and I_{Y_J} the quasi-coherent ideal of O_{U_J} defining Y_J. If Y_J lifts to a subscheme Y of X, then Y, having the same underlying topological space as Y_J, is a closed subscheme of U; consequently, the obstruction to lifting Y_J to a subscheme, flat over S, of X or of U is "the same", it resides in

$$H^1(Y_J, \mathrm{Hom}_{O_{Y_J}}(N_{Y_J/X_J}, J \otimes_{O_{S_J}} O_{Y_J})).$$

Finally, let us return to the notation of n° 0: let I be a quasi-coherent ideal of O_S such that $J \subset I$ and $IJ = 0$, and let S_0 be the closed subscheme of S_J defined by I. For every S-scheme Z, one denotes $Z_J = Z \times_S S_J$ and $Z_0 = Z \times_S S_0$. Then, since J is annihilated by I, one has, with the notation of 4.4:

$$J \otimes_{O_{S_J}} O_{Y_J} = J \otimes_{O_{S_0}} O_{Y_0}$$
$$\mathrm{Hom}_{O_{Y_J}}(N_{Y_J/X_J}, J \otimes_{O_{S_J}} O_{Y_J}) = \mathrm{Hom}_{O_{Y_0}}(N_{Y_J/X_J} \otimes_{O_{Y_J}} O_{Y_0}, J \otimes$$

etc. One thus obtains:

Proposition 4.5. *Let S be a scheme, S_0 and S_J the closed subschemes defined by the quasi-coherent ideals I and J, such that $I \supset J$ and $I \cdot J = 0$. Let X be an S-scheme and Y_J a subscheme of X_J, flat over S_J. Let A_0 be the O_{Y_0}-module defined by*

$$A_0 = \mathrm{Hom}_{O_{Y_0}}(N_{Y_J/X_J} \otimes_{O_{Y_J}} O_{Y_0}, J \otimes_{O_{S_0}} O_{Y_0}).$$

(i) For there to exist a subscheme Y of X, reducing to Y_J, flat over S, it is necessary and sufficient that the following conditions be satisfied:

[203] N.D.E.: We have added the numbering 4.5.0 to mark the return to the original.
[204] N.D.E.: We have corrected the following sentence.
[205] N.D.E.: We have detailed what follows.

(a) Such a Y exists locally on X.

(b) A certain obstruction in $R^1\Gamma(Y_0, A_0)$ is zero.[206]

(ii) Under these conditions, the set of Y satisfying the required conditions is principal homogeneous under the commutative group $\Gamma(Y_0, A_0)$.

Remark 4.5.1.[207] *It follows from 4.5 (ii) that for every pair (Y, Y') of subschemes*[208] *of X, flat over S and reducing to Y_J, one has a "deviation"*

$$d(Y', Y) \in \Gamma(Y_0, A_0) = \mathrm{Hom}_{O_{Y_0}}(N_{Y_J/X_J} \otimes_{O_{Y_J}} O_{Y_0}, J \otimes_{O_{S_0}} O_{Y_0});$$

the sign convention adopted in 4.4.1 being that $d(Y', Y)$ corresponds to the morphism of O_X-modules

$$I_{Y'} \hookrightarrow O_X \longrightarrow O_Y$$

(which takes values in $JO_Y \simeq J \otimes_{O_{S_0}} O_{Y_0}$ and factors through $I_{Y'_J} = I_{Y_J}$ and then through N_{Y_J/X_J}).

Remark 4.6.[209] *If X is flat over S and if Y_J is locally complete intersection in X_J, then condition (a) is always satisfied and every Y flat over S lifting Y_J is then locally complete intersection in X. If moreover Y_0 is affine, condition (b) is also satisfied.*

Definition 4.6.1. *(cf. SGA 6, VII 1.1) Let B be a commutative ring, $f : E \to B$ a B-linear morphism, where E is a free B-module of finite rank d, and I the ideal $f(E)$ (if one chooses a basis of E, f is given by a d-tuple $(f_1, \dots, f_d)$ of elements of B, and I is the ideal generated by the f_i). The* **Koszul complex** *$K\bullet(f)$ is the graded B-module $\bigwedge\bullet E$, equipped with the differential (of degree -1):*

$$x_1 \wedge \cdots \wedge x_i \mapsto \sum_{j=1}^{i} (-1)^{j-1} f(x_j)\, x_1 \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_i.$$

One has therefore an augmented chain complex (B/I being in degree -1):

$$\cdots \to \wedge^2 E \to E \xrightarrow{f} B \to B/I \to 0$$

which by definition is exact in degree 0, since $I = f(E)$. One says that f is **regular** if $K\bullet(f)$ is acyclic in degrees > 0, that is, if the augmented complex above is a resolution of $C = B/I$.

In this case, the proof of SGA 6, VII 1.2 b) shows that the C-modules I^n/I^{n+1} ($n \in \mathbb{N}$) are free, I/I^2 being of rank d.

[206] N.D.E.: Here, we have denoted $R^1\Gamma(Y_0, A_0)$ the "coherent" cohomology group $H^1(Y_0, A_0)$ of the O_{Y_0}-module A_0, in order to distinguish it from the "Hochschild" cohomology groups $H^i(Y_0, M_0)$ (Y_0 an S_0-group, M_0 an O_{S_0}-module) which will be considered starting from 4.16.

[207] N.D.E.: We have placed here this remark, which replaces Remark 4.7 of the original.

[208] N.D.E.: We have corrected "closed subschemes" to "subschemes".

[209] N.D.E.: We have kept, for the record, Remark 4.6 of the original, in which the definition of "locally complete intersection" does not appear. We have added next the "good" definition, drawn from SGA 6, VII 1.4 (which replaces that of EGA IV_4, 16.9.2), and the proof of the three results stated in the remark.

Definition 4.6.2. *(cf. SGA 6, VII 1.4) Let X be a scheme, Y a subscheme, U an open of X such that Y is a closed subscheme of U, defined by the quasi-coherent ideal $\mathcal{I}_Y$.*

One says that Y is **locally complete intersection** *in X if $Y \hookrightarrow X$ is a regular immersion in the sense of SGA 6, VII 1.4, that is, if for every $y \in Y$ there exists an affine open neighborhood V of y in U, a finite free $\mathcal{O}_V$-module $\mathcal{E}$, and a regular morphism $f : \mathcal{E} \to \mathcal{O}_V$ of image $\mathcal{I}_Y|_V$, i.e. such that $K_\bullet(f)$ be a resolution of $\mathcal{O}_{Y\cap V}$.*

This implies that the immersion $Y \hookrightarrow X$ is locally of finite presentation, and, by 4.6.1, that the conormal sheaf $\mathcal{N}_{Y/X} = \mathcal{I}_Y/\mathcal{I}_Y^2$ is a finite locally free $\mathcal{O}_Y$-module.

Lemma 4.6.3.[210] *Let A be a ring, J an ideal of A of square zero, $\bar{A} = A/J$, B a flat A-algebra, E a free B-module of finite rank, $f : E \to B$ a morphism of B-modules. One supposes that the morphism $\bar{g} : \bar{E} = E \otimes_A \bar{A} \to \bar{B} = B \otimes_A \bar{A}$ induced by f is regular and that $\bar{B}/\bar{g}(\bar{E})$ is flat over $\bar{A}$.*

Then f is regular and $B/f(E)$ is flat over A.

Proof. Set $C = B/f(E)$ and $\bar{C} = C \otimes_A \bar{A} = \bar{B}/\bar{g}(\bar{E})$. First, the $\bigwedge_B^i(E)$ are free B-modules, hence flat A-modules, since B is flat over A. As $\bigwedge^\bullet_B E \otimes_A \bar{A} \simeq \bigwedge^\bullet_{\bar{B}} \bar{E}$, one obtains therefore an exact sequence of complexes:

$$0 \to J \otimes_A \textstyle\bigwedge^\bullet_B E \to \bigwedge^\bullet_B E \to \bigwedge^\bullet_{\bar{B}} \bar{E} \to 0.$$

Moreover, since $J^2 = 0$, one has $J \otimes_A M = J \otimes_A \bar{A} \otimes_A M$ for every A-module M. Denoting by dashed arrows the augmentation morphisms, and by d the rank of E, one therefore obtains the bicomplex that follows, where the rows are exact:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & J \otimes_A \bigwedge^d_{\bar{B}} \bar{E} \arrow[r] \arrow[d] & \bigwedge^d_B E \arrow[r] \arrow[d] & \bigwedge^d_{\bar{B}} \bar{E} \arrow[r] \arrow[d] & 0 \\
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
0 \arrow[r] & J \otimes_A \bar{E} \arrow[r] \arrow[d, "\mathrm{id}\otimes\bar{g}"] & E \arrow[r] \arrow[d, "f"] & \bar{E} \arrow[r] \arrow[d, "\bar{g}"] & 0 \\
0 \arrow[r] & J \otimes_A \bar{B} \arrow[r] \arrow[d, dashed] & B \arrow[r] \arrow[d, dashed] & \bar{B} \arrow[r] \arrow[d, dashed] & 0 \\
 & J \otimes_A \bar{C} \arrow[r] \arrow[d] & C \arrow[r] \arrow[d] & \bar{C} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Moreover, the right and left columns are exact, since $K_\bullet(\bar{g})$ is a resolution of $\bar{C}$ and the latter is flat over $\bar{A}$. Hence, considering the long exact homology sequence associated with the exact sequence of unaugmented complexes, one obtains that $K_\bullet(f)$ is acyclic in degrees > 0, and that one has in degree 0 an exact sequence:

[210] N.D.E.: In order to prove the results stated in Remark 4.6, we have added Lemmas 4.6.3, 4.6.4 and Proposition 4.6.5, as well as Remark 4.6.6.

$$0 \to J \otimes_A \bar{C} \to C \to \bar{C} \to 0.$$

Hence C is flat over A, by the "fundamental criterion of flatness" (cf. [BAC], § III.5, th. 1).

Lemma 4.6.4.[211] *Let A be a commutative ring, J a nilpotent ideal, $N \subset M$ two A-modules such that M/N is flat over A. If $x_1, \ldots, x_n$ are elements of N whose images generate the image $\bar{N}$ of N in M/JM, then they generate N.*

Indeed, denote by N' the submodule of N generated by the x_i, and $Q = N/N'$. Then the morphism $N' \otimes (A/J) \to \bar{N}$ is surjective. On the other hand, since M/N is flat over A, the morphism $N \otimes (A/J) \to \bar{N}$ is bijective. One thus obtains that $Q \otimes (A/J) = 0$, whence $Q = 0$ by the "nilpotent Nakayama lemma" (one has $Q = JQ = J^2Q = \cdots = 0$).

One can now prove:

Proposition 4.6.5.[212] *Let S, I, J and X, Y_J be as in 4.5. Suppose moreover X flat over S and Y_J locally complete intersection in X_J.*

a) Then condition (a) of 4.5 (i) is satisfied; moreover, every Y flat over S lifting Y_J is locally complete intersection in X.

b) If moreover Y_0 is affine, condition (b) of loc. cit. is likewise satisfied.

Proof. The first assertion of (a) follows from Lemma 4.6.3; the second then results from Lemma 4.6.4. On the other hand, the hypothesis entails (cf. 4.6.2) that N_{Y_J/X_J} is a finite locally free O_{Y_J}-module, hence the O_{Y_0}-module

$$A_0 = \mathrm{Hom}_{O_{Y_0}}(N_{Y_J/X_J} \otimes_{O_{Y_J}} O_{Y_0}, J \otimes_{O_{S_0}} O_{Y_0}).$$

is quasi-coherent (cf. EGA I, 1.3.12), whence $R^1\Gamma(Y_0, A_0) = 0$ if Y_0 is affine.

Remark 4.6.6.[213] *Let us conclude this paragraph by the following example, which shows that, under the hypotheses of Lemma 4.6.3, if $(\bar{g}_1, \bar{g}_2)$ is a regular sequence generating the ideal $\bar{I} = \bar{g}(\bar{E})$, it does not necessarily lift to a regular sequence in B.*

Let k be a field, $\bar{A} = k[X, Y]$, denote by $k\epsilon$ the $\bar{A}$-module $\bar{A}/(X, Y)$ (i.e. $P \cdot \epsilon = P(0, 0)\epsilon$ for every $P \in \bar{A}$), and let $A = \bar{A} \bigoplus k\epsilon$, where $J = k\epsilon$ is an ideal of square zero. One has $A/J = \bar{A}$.

The algebra $B = A \bigotimes_k k[Z, T]$ is free over A, hence flat; one has $\bar{B} = k[X, Y, Z, T]$. Set $\bar{g}_1 = XZ - YT$ and $\bar{g}_2 = XZ - 1$. Since the polynomial $\bar{g}_1$ is irreducible, $\bar{B}/(\bar{g}_1)$ is integral, and therefore $(\bar{g}_1, \bar{g}_2)$ is a regular sequence in $\bar{B}$, generating the ideal $\bar{I} = (XZ - 1, YT - 1)$. Hence

[211] N.D.E.: In order to prove the results stated in Remark 4.6, we have added Lemmas 4.6.3, 4.6.4 and Proposition 4.6.5, as well as Remark 4.6.6.

[212] N.D.E.: In order to prove the results stated in Remark 4.6, we have added Lemmas 4.6.3, 4.6.4 and Proposition 4.6.5, as well as Remark 4.6.6.

[213] N.D.E.: In order to prove the results stated in Remark 4.6, we have added Lemmas 4.6.3, 4.6.4 and Proposition 4.6.5, as well as Remark 4.6.6.

$$\bar{C} = \bar{B}/\bar{I} = k[X, Y, X^{-1}, Y^{-1}] = \bar{A}[X^{-1}, Y^{-1}]$$

is a flat $\bar{A}$-algebra (and also a flat A-algebra). But every lift in B of $\bar{g}_1$ is of the form $XY - ZT + \lambda\epsilon$, where $\lambda \in k[Z,T]$, hence annihilates ϵ.

4.7. One has suppressed here Remark 4.7, placed in 4.5.1.

Remark 4.8.0.[214] *Let S be a scheme, S' a closed subscheme, X an S-scheme, Y a sub-S-scheme of X, and $X' = X \times_S S'$, $Y' = Y \times_S S'$. Then, one has a surjective morphism of $\mathcal{O}_{Y'}$-modules*

$$\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\text{surj}} \mathcal{N}_{Y'/X'}.$$

Indeed, up to replacing X by a certain open, one may suppose that Y is closed, defined by an ideal $\mathcal{I}_Y$ of $\mathcal{O}_X$; then the image of $\mathcal{I}_Y$ in $\mathcal{O}_{X'}$ is the ideal $\mathcal{I}_{Y'}$ defining Y', and one has a surjective morphism of $\mathcal{O}_{Y'}$-modules

$$\pi : (\mathcal{I}_Y/\mathcal{I}_Y^2) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\text{surj}} \mathcal{I}_{Y'}/\mathcal{I}_{Y'}^2.$$

Suppose moreover that $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{I}_Y$ is flat over $\mathcal{O}_S$; then the natural morphism

$$\mathcal{I}_Y \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \xrightarrow{\sim} \mathcal{I}_{Y'}$$

is bijective (cf. EGA IV_2, 2.1.8). One then has the following commutative diagram with exact rows:

$$\begin{tikzcd}
\mathcal{I}_Y^2 \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \arrow[r] \arrow[d, "\text{surj.}"'] & \mathcal{I}_Y \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \arrow[r] \arrow[d, "\sim"] & (\mathcal{I}_Y/\mathcal{I}_Y^2) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \arrow[r] \arrow[d, "\pi \text{ surj.}"] & 0 \\
0 \arrow[r] \mathcal{I}_{Y'}^2 \arrow[r] & \mathcal{I}_{Y'} \arrow[r] & \mathcal{I}_{Y'}/\mathcal{I}_{Y'}^2 \arrow[r] & 0
\end{tikzcd}$$

whence one deduces, by the snake lemma:[215]

$$\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\sim} \mathcal{N}_{Y'/X'} \qquad \text{if } Y \text{ is flat over } S. \tag{4.8.0}$$

Proposition 4.8. *Let S, S_0, S_J and $\mathcal{I}$, $\mathcal{J}$ be as in 4.5.*[216] *Let X be an S-scheme, Y a subscheme of X, and i the immersion $Y \hookrightarrow X$.*

(i) For every S-morphism $f : T \to X$ such that $f_J : T_J \to X_J$ factors through Y_J, one can define an obstruction

$$(*) \qquad c(X, Y, f) \in \mathrm{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T)$$

whose vanishing is equivalent to the existence of a factorization of f through Y.

(ii) Let Y' be a second subscheme of X. Suppose that $Y'_J = Y_J$ and that Y, Y' are flat over S. One then has isomorphisms (cf. 4.8.0):

[214] N.D.E.: We have inserted here this remark, used in the following proposition; it appeared in 4.10 of the original.

[215] N.D.E.: In the original, this was indicated in Remark 4.10, under the additional hypothesis that Y' was locally complete intersection in X'. This hypothesis figured also, consequently, in statements 4.12–4.14; it seems in fact superfluous, and we have suppressed it from the above-mentioned statements.

[216] N.D.E.: We have suppressed the hypothesis that $\mathcal{I}$ be nilpotent, which appears superfluous (cf. the proof).

$$JO_Y \simeq J \otimes_{O_{S_0}} O_{Y_0} \simeq JO_{Y'} \quad \text{and} \quad N_{Y/X} \otimes_{O_Y} O_{Y_J} ⥲ N_{Y_J/X_J}$$

whence an isomorphism:

$$u : \mathrm{Hom}_{O_{Y_0}}(N_{Y_J/X_J} \otimes_{O_{Y_J}} O_{Y_0}, JO_Y) ⥲ \mathrm{Hom}_{O_{Y_0}}(N_{Y/X} \otimes_{O_Y} O_{Y_0}, JO_{Y'}$$

Denoting by $i' : Y' \to X$ the canonical immersion and $d(Y, Y')$ the deviation of 4.5.1, one has:[217]

$$(**) \qquad c(X, Y, i') = u(d(Y, Y')).$$

(iii) The canonical morphism $N_{Y/X} \xrightarrow{D} i^*(\Omega^1_{X/S})$ *(cf. SGA 1 II, formula 4.3)*[218] *induces a morphism:*

$$D_0 : N_{Y/X} \otimes_{O_Y} O_{Y_0} ⥲ \Omega^1_{X_0/S_0} \otimes_{O_{X_0}} O_{Y_0}$$

and hence, for every S-morphism $f : T \to X$ as in (i), a morphism:

$$v_{f_0} : \mathrm{Hom}_{O_{T_0}}(f_0^*(\Omega^1_{X_0/S_0}), JO_T) \to \mathrm{Hom}_{O_{T_0}}(f_0^*(N_{Y/X} \otimes_{O_Y} O_{Y_0}), JO_T),$$
$$a \mapsto a \circ f_0^*(D_0)$$

where above the first group is $\mathrm{Hom}_{X^+}(T, L_X)$*, cf. 0.1.5. For $a \in$* $\mathrm{Hom}_{X^+}(T, L_X)$*, one has:*

$$(***) \qquad c(X, Y, a \cdot f) - c(X, Y, f) = v_{f_0}(a),$$

where $a \cdot f$ denotes the composite morphism $T \xrightarrow{a \times f} L_X \times_{X^+} X \to X$.

We shall prove part (i) of the proposition, leaving the reader to (not) verify assertions (ii) and (iii); this verification is done by reduction to the affine case, then by comparison of explicit definitions.[219]

Let us therefore prove (i). The morphism $f : T \to X$ defines a morphism of sheaves of rings $\phi : O_X \to f_*(O_T)$.[220] Let U be an open subscheme of X in which Y is closed; since T (resp. Y_J) has the same underlying space as T_J (resp. Y), the continuous map underlying f sends T into U, and since U is an open of X, ϕ induces a morphism of sheaves of rings $O_U = O_X|_U \to f_*(O_T)$, i.e. f factors through U.

Therefore, one may restrict to the case where Y is closed, hence defined by a sheaf of ideals I_Y. For f to factor through Y, it is necessary and sufficient that the composite map $I_Y \to O_X \to f_*(O_T)$ be zero. Since f_J factors through Y_J, the composite map $I_{Y_J} \to O_{X_J} \to f_*(O_{T_J})$ is zero. Considering the commutative diagram, where the first row is exact:

$$\begin{array}{ccccccc}
0 & \longrightarrow & f_*(JO_T) & \longrightarrow & f_*(O_T) & \longrightarrow & f_*(O_{T_J}) \\
 & & \nwarrow & & \uparrow & & \uparrow \\
 & & \nwarrow & & \varphi & & \varphi_J
\end{array}$$

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[217] N.D.E.: See also 4.27 further on.

[218] N.D.E.: See also EGA IV₄, 16.4.21. Recall that if U is an affine open of X such that $Y \cap U$ is defined by the ideal I of $A = O_X(U)$, if one denotes by d the differential $A \to \Gamma(U, \Omega^1_{X/S})$, and if $x \in I$, then $D(x + I^2)$ is the element $d(x) \otimes 1$ of $\Gamma(U, \Omega^1_{X/S}) \otimes_A (A/I)$.

[219] N.D.E.: We have done these verifications below.

[220] N.D.E.: On the one hand, we have suppressed the hypothesis that I be nilpotent, i.e. that X_0 have the same underlying topological space as X; on the other hand, we have detailed the following sentence.

$$\begin{tikzcd} & \mathcal{O}_X \arrow[r] & \mathcal{O}_{X_J} \\ & I_Y \arrow[u] \arrow[r] \arrow[ul, "\varphi"] & I_{Y_J} \arrow[u] \end{tikzcd}$$

one deduces that ϕ sends I_Y into $f_*(J\mathcal{O}_T)$.[221] Since $J^2 = 0$, it follows that $f_*(J\mathcal{O}_T)$, viewed as $\mathcal{O}_X$-module via ϕ, is annihilated by I_Y; consequently, ϕ induces a morphism of $\mathcal{O}_X$-modules

$$h : i_*(N_{Y/X}) = I_Y/I_Y^2 \to f_*(J\mathcal{O}_T).$$

On the other hand, one has cartesian squares:

$$\begin{tikzcd} T_0 \arrow[r, "f_0"] \arrow[d, "\tau_{T_0}"] & X_0 \arrow[d, "\tau_{X_0}"] & Y_0 \arrow[l, "i_0"'] \arrow[d, "\tau_{Y_0}"] \\ T \arrow[r, "f"] & X & Y. \arrow[l, "i"'] \end{tikzcd}$$

where iT_0 etc. are the closed immersions deduced by base change from $S_0 \hookrightarrow S$. Since $J\mathcal{O}_T$ is a quasi-coherent $\mathcal{O}_T$-module annihilated by I, one has an isomorphism

$$J\mathcal{O}_T \simeq (\tau_{T_0})_*\tau_{T_0} * (J\mathcal{O}_T),$$

whence $f_*(J\mathcal{O}_T) \simeq (\tau_{X_0})_*(f_0)_*\tau_{T_0} * (J\mathcal{O}_T)$. Therefore h corresponds, by adjunction, to a morphism of $\mathcal{O}_{T_0}$-modules

$$h_0 : f_0^*\tau_{X_0}^* i_*(N_{Y/X}) \to i_{T_0}^*(J\mathcal{O}_T).$$

Now, $\tau_{X_0} * i_*(N_{Y/X}) \simeq (i_0)_*\tau_{Y_0} * (N_{Y/X}) = N_{Y/X} \bigotimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}$. Hence, returning to the abuse of notation $i_{T_0} * (J\mathcal{O}_T) = J\mathcal{O}_T$ constantly used, h_0 identifies with a morphism of $\mathcal{O}_{T_0}$-modules

$$h_0 : f_0^*(N_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}) \to J\mathcal{O}_T$$

which is the obstruction $c(X, Y, f)$ sought. This proves (i).

When f is the immersion $i' : Y' \hookrightarrow X$, one sees that $c(X, Y, i')$ comes from the morphism $I_Y \hookrightarrow \mathcal{O}_X \to \mathcal{O}_{Y'}$ hence corresponds, by 4.4.1 and 4.5.1, to the class $d(Y, Y')$. This proves (ii).

Let us prove (iii). First, $D : N_{Y/X} \to i_*(\Omega^1_{X/S})$ induces a morphism

$$D_0 : \tau_{Y_0}^*(N_{Y/X}) \to \tau_{Y_0}^* i_*(\Omega^1_{X/S}) = i_0^* \tau_{X_0}^*(\Omega^1_{X/S})$$

and, since $X_0 = X \times_S S_0$, one has $\tau_{X_0} * (\Omega^1_{X/S}) \simeq \Omega^1_{X_0/S_0}$ (cf. EGA IV₄, 16.4.5). One thus obtains the announced morphism

$$D_0 : N_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} \to \Omega^1_{X_0/S_0} \otimes_{\mathcal{O}_{X_0}} \mathcal{O}_{Y_0}.$$

Finally, we shall verify equality $(***)$ after the remark below.

[221] N.D.E.: We have detailed what follows.

Remark 4.9. *The obstruction $c(X,Y,f)$ is computed locally on T. Let $U = \mathrm{Spec}(C)$ be an affine open of T above an affine open $\mathrm{Spec}(A)$ of X, itself above an affine open $\mathrm{Spec}(\Lambda)$ of S; let $J \subset I \subset \Lambda$ (resp. $I_Y \subset A$) be the ideals corresponding to $J \subset I$ (resp. to I_Y), let $B = A/I_Y$ and let $\phi : A \to C$ be the morphism of Λ-algebras corresponding to $f : T \to X$; since $f(T_J) \subset Y_J$ one has $\phi(I_Y) \subset JC$ and therefore ϕ induces a morphism of Λ-algebras $B \to C/JC \to C_0 = C/IC$. Then the obstruction $c = c(X,Y,f)$ is computed by the following commutative diagram:*

$$\begin{tikzcd}
I_Y \arrow[rr] & & A \arrow[r, "\varphi"] & C \\
I_Y/I_Y^2 \arrow[r] & I_Y/I_Y^2 \otimes_B C_0 \arrow[rr] & & JC, \arrow[u, "c"']
\end{tikzcd}$$

that is, it is defined, above the open U, as the unique element of

$$\mathrm{Hom}_{C_0}(I_Y/I_Y^2 \otimes_B C_0, JC) = \mathrm{Hom}_{B_0}(I_Y/I_Y^2 \otimes_B B_0, JC)$$

such that, with the evident notation, one has $c(x \otimes_B 1) = \phi(x)$, for every $x \in I_Y$.

[222] One can now complete the proof of 4.8 (iii). The equality $(***)$ is verified locally on T, so one is reduced to the affine situation described above. Let us denote by $d_{A/\Lambda}$ the differential $A \to \Omega^1_{A/\Lambda}$. Then a corresponds, above U, to an element a_U of

$$\mathrm{Hom}_{C_0}(\Omega^1_{A_0/\Lambda_0} \otimes_{A_0} C_0, JC) \simeq \mathrm{Hom}_{B_0}(\Omega^1_{A/\Lambda} \otimes_A B_0, JC) \simeq \mathrm{Hom}_A(\Omega^1_{A/\Lambda}, JC).$$

Then, on the one hand, $v_{f_0}(a)$ corresponds above U to the element $a_U \circ \bar{D}_0$, where $\bar{D}_0$ is the morphism of B-modules[223]

$$I_Y/I_Y^2 \to \Omega^1_{A/\Lambda} \otimes_A B_0, \qquad x + I_Y^2 \mapsto d_{A/\Lambda}(x) \otimes 1.$$

On the other hand (cf. the proofs of 0.1.8 and 0.1.9), the morphism of Λ-algebras $\phi' : A \to C$ corresponding to $a \cdot f$ differs from ϕ by the Λ-derivation $A \to JC$ associated with a_U, i.e. one has:

$$\varphi' = \varphi + a_U \circ d_{A/\Lambda} = \varphi + a_U \circ (d_{A/\Lambda} \otimes 1).$$

Consequently, denoting $c' = c(X,Y,a \cdot f)$, one has for every $x \in I_Y$, denoting by $\bar{x}$ its image in I_Y/I_Y^2:

$$(c' - c)(\bar{x} \otimes 1) = a_U(d_{A/\Lambda}(x) \otimes 1) = (a_U \circ \bar{D}_0)(\bar{x}) = v_{f_0}(a)(\bar{x}).$$

This shows that $c' - c = v_{f_0}(a)$.

4.10. One has suppressed Remark 4.10 of the original, made obsolete by the addition of Remark 4.8.0.

4.11. We now propose to study the following situation. Let S, S_J and S_0 be as in 4.8; one has three S-schemes X, X', T, a subscheme Y of X (resp. Y' of X'), and morphisms $f : T \to X'$ and $g : X' \to X$.

[222] N.D.E.: We have added what follows.
[223] N.D.E.: Cf. N.D.E. (93).

$$\begin{tikzcd} & & Y' \arrow[d, hook, "i'"] & Y \arrow[d, hook, "i"] \\ T \arrow[r, "f"] & X' \arrow[r, "g"] & X. & \end{tikzcd}$$

One supposes that by reduction modulo J, this diagram completes into a commutative diagram

$$\begin{tikzcd} & & Y'_J \arrow[r] \arrow[d, hook, "i'_J"] & Y_J \arrow[d, hook, "i_J"] \\ T_J \arrow[r, "f_J"] \arrow[urr] & X'_J \arrow[r, "g_J"] & X_J. & \end{tikzcd}$$

One has therefore by 4.8 obstructions:

$$c(X, Y, g \circ i') \in \mathrm{Hom}_{O_{Y'_0}}(i_0'^*g_0^*(N_{Y/X} \otimes_{O_Y} O_{Y_0}), JO_{Y'}),$$
$$c(X', Y', f) \in \mathrm{Hom}_{O_{T_0}}(f_0^*(N_{Y'/X'} \otimes_{O_{Y'}} O_{Y'_0}), JO_T),$$
$$c(X, Y, g \circ f) \in \mathrm{Hom}_{O_{T_0}}(f_0^*g_0^*(N_{Y/X} \otimes_{O_Y} O_{Y_0}), JO_T),$$

whose relations one seeks to compute.[224]

Lemma 4.12. *Suppose Y' flat over S, so that $J \bigotimes_{O_{S_0}} O_{Y'_0} = JO_{Y'}$.*

(i) One has a natural morphism

$$b_{f_0} : \mathrm{Hom}_{O_{Y'_0}}(i_0'^*g_0^*(N_{Y/X} \otimes O_{Y_0}), JO_{Y'}) \to \mathrm{Hom}_{O_{T_0}}(f_0^*g_0^*(N_{Y/X} \otimes O_{Y_0}),$$

(ii) One has also a natural morphism, functorial in T,[225]

$$a_{g_0}(f_0) : \mathrm{Hom}_{O_{T_0}}(f_0^*(N_{Y'/X'} \otimes O_{Y'_0}), JO_T) \to \mathrm{Hom}_{O_{T_0}}(f_0^*g_0^*(N_{Y/X} \otimes O_{Y_0}), JO$$

Proof.[226] Let us first note that, X, X', Y, Y' being fixed, to give a T as above is equivalent to giving a morphism $(f, f_J) : T \to X' \times_{X'^+} Y'^+$. Set $Z = X' \times_{X'^+} Y'^+$ and denote by M and M' the Z-functors defined by: for every $f : T \to Z$,

$$M(T) = \mathrm{Hom}_{O_{T_0}}(f_0^*g_0^*(N_{Y/X} \otimes O_{Y_0}), JO_T)$$
$$M'(T) = \mathrm{Hom}_{O_{T_0}}(f_0^*(N_{Y'/X'} \otimes O_{Y'_0}), JO_T).$$

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One has in any case a commutative diagram:

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[224] N.D.E.: From 4.17 on, we shall apply this to the case where X is an S-group, $g : X \times_S X \to X$ the multiplication, Y a subscheme of X such that Y_J is a subgroup of X_J, $Y' = Y \times_S Y$, and to the two morphisms $Y^3 \to X^2$ which send (y_1, y_2, y_3) to (y_1y_2, y_3), resp. (y_1, y_2y_3). In this case, the comparison of the above obstructions will show that the obstruction to Y being a subgroup of X resides in a certain cohomology group (Hochschild) $H^2(Y_0, N_0)$.

[225] N.D.E.: We have suppressed the hypothesis "Y'_0 locally complete intersection in X'_0", superfluous by 4.8.0; on the other hand, we have added that $a_{g_0}(f_0)$ is "functorial in T", this playing a crucial role in the proof of 4.17.

[226] N.D.E.: We have detailed the proof, to make visible the "functoriality in T" of a_{g_0}.

[227] N.D.E.: The situation will simplify from 4.16 on: one will restrict to schemes flat over S, Y will be a flat S-group and $Y' = Y \times_S Y$; one will then obtain S_0-functors N_0 and N'_0.

$$\begin{tikzcd}
f_0^*(J \otimes_{O_{S_0}} O_{Y_0'}) \arrow[r, dash] \arrow[d] & J \otimes_{O_{S_0}} O_{T_0} \arrow[d] \\
f_0^*(JO_{Y'}) \arrow[r, dashed] & JO_T
\end{tikzcd}$$

and since Y' is flat over S, the left arrow is an isomorphism, hence one obtains a morphism of O_{T_0}-modules $f_0 * (JO_{Y'}) \to JO_T$. The latter induces a morphism of abelian groups

$$\mathrm{Hom}_{O_{T_0}}(f_0^* g_0^*(N_{Y/X} \otimes O_{Y_0}), f_0^*(JO_{Y'})) \to M(T)$$

and, composing with the morphism

$$M(Y') \to \mathrm{Hom}_{O_{T_0}}(f_0^* g_0^*(N_{Y/X} \otimes O_{Y_0}), f_0^*(JO_{Y'})),$$

induced by f_0*, one obtains the morphism $b_{f_0} : M(Y') \to M(T)$.

Likewise, one has in any case a diagram

$$\begin{tikzcd}
g_0^*(N_{Y/X} \otimes_{O_Y} O_{Y_0}) \arrow[r, dashed] \arrow[d] & N_{Y'/X'} \otimes_{O_{Y'}} O_{Y_0'} \arrow[d] \\
g_0^*(N_{Y_0/X_0}) \arrow[r] & N_{Y_0'/X_0'}
\end{tikzcd}$$

and since Y' is flat over S, the second vertical arrow is an isomorphism by 4.8.0. One thus obtains an $O_{Y_0'}$-morphism

$$i_0'^* g_0^*(N_{Y/X} \otimes_{O_Y} O_{Y_0}) \to N_{Y'/X'} \otimes_{O_{Y'}} O_{Y_0'}$$

which induces a morphism $a_{g_0}(id_{Y'}) : M'(Y') \to M(Y')$ and, for every $f : T \to Z$, a morphism $a_{g_0}(f) : M'(T) \to M(T)$ such that one has a commutative diagram

$$\begin{tikzcd}
M'(Y') \arrow[r, "a_{g_0}(id_{Y_0'})"] \arrow[d, "b'_{f_0}"'] & M(Y') \arrow[d, "b_{f_0}"] \\
M'(T) \arrow[r, "a_{g_0}(f_0)"] & M(T)
\end{tikzcd}$$

(where b'_{f_0} is defined like b_{f_0}). QED.

Remark 4.12.1.[228] *Denote by M_0 and M_0' the Y_0'-functors defined by: for every $f : T_0 \to Y_0'$,*

$$M_0(T) = \mathrm{Hom}_{O_{T_0}}(f_0^* g_0^*(N_{Y/X} \otimes O_{Y_0}), J \otimes_{O_{S_0}} O_{T_0})$$
$$M_0'(T) = \mathrm{Hom}_{O_{T_0}}(f_0^*(N_{Y'/X'} \otimes O_{Y_0'}), J \otimes_{O_{S_0}} O_{T_0}).$$

Note immediately that $Z_0 = Y_0'$ and that on the category of Z-schemes T which are flat over S, M and M' coincide, respectively, with the functors $\prod_{S_0/S} M_0$ and $\prod_{S_0/S} M_0'$. In this case, b_{f_0} is simply the morphism

$$f_0* : M_0(Y_0') \longrightarrow M(T_0)$$

[228] N.D.E.: We have added this remark, used in the proof of 4.17.

induced by f_0, and for every morphism $u : U \to T$, setting $h = f \circ u$, one has a commutative diagram

$$\begin{array}{ccc}
M_0'(T_0) & \xrightarrow{a_{g_0}(f_0)} & M_0(T_0) \\
\downarrow{\scriptstyle u_0^*} & & \downarrow{\scriptstyle u_0^*} \\
M_0'(U_0) & \xrightarrow{a_{g_0}(h_0)} & M_0(U_0)
\end{array}$$

i.e. a_{g_0} becomes a morphism of functors $\prod_{S_0/S} M_0' \to \prod_{S_0/S} M_0$.

Proposition 4.13. *Suppose Y' flat over S. One has then the formula:*

$$c(X, Y, g \circ f) = a_{g_0}(c(X', Y', f)) + b_{f_0}(c(X, Y, g \circ i')).$$

Since the definition of the different obstructions and of the morphisms a_{g_0} and b_{f_0} is local, one easily sees that it suffices to verify the given formula when the different schemes in play are affine. Let us thus denote $S = \mathrm{Spec}(\Lambda)$, $S_J = \mathrm{Spec}(\Lambda/J)$, $S_0 = \mathrm{Spec}(\Lambda/I)$, $T = \mathrm{Spec}(C)$, $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(A/I_Y) = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(A')$, $Y' = \mathrm{Spec}(A'/I_{Y'}) = \mathrm{Spec}(B')$.

One has therefore a diagram of rings and ideals[229]

$$\begin{array}{ccccc}
 & & B' & & B \\
 & & \uparrow{\scriptstyle \pi'} & & \uparrow{\scriptstyle \pi} \\
C & \xleftarrow{f} & A' & \xleftarrow{g} & A \\
 & & \uparrow & & \uparrow \\
 & & I_{Y'} & & I_Y.
\end{array}$$

Let us study the different terms of the formula to be proved. In what follows, if $x \in I_Y$ (resp. $u \in I_{Y'}$), we denote by $\bar{x}$ (resp. $\bar{u}$) its image in I_Y/I_Y^2 (resp. $I_{Y'}/I_{Y'}^2$); on the other hand, if m belongs to a Λ-module M, we denote by $\bar{m}_0$ its image in $M_0 = M/IM$.

One has seen that $c = c(X,Y,g \circ f)$ is the unique C_0-morphism $I_Y/I_Y^2 \otimes_B C_0 \to JC$ such that $c(\bar{x} \otimes 1) = f(g(x))$, for every $x \in I_Y$.

Fix $x \in I_Y$; one has $g(x) \in I_{Y'} + JA'$ since $g_J(Y_J') \subset Y_J$. Write $g(x) = x' + \Sigma \lambda_i a_i'$, with $x' \in I_{Y'}$, $\lambda_i \in J$, $a_i' \in A'$. One therefore has

$$(1) \qquad c(X, Y, g \circ f)(\bar{x} \otimes 1) = f(g(x)) = f(x') + \Sigma \lambda_i f(a_i').$$

Now consider $a_{g_0}(c(X', Y', f))$. According to the definitions laid down, it is defined by the diagram

[229]N.D.E.: We have kept the notation of the original, denoting $f : A' \to C$ and $g : A \to A'$ the morphisms of rings corresponding to $f : T \to X'$ and $g : X' \to X$. This explains the formula $c(X,Y,g \circ f)(\bar{x} \otimes 1) = f(g(x))$, for $x \in I_Y$.

$$\begin{tikzcd}
 & I_{Y'} \arrow[r, "f"] & C & \\
I_{Y_0'}/I_{Y_0'}^2 \otimes_{B_0'} C_0 & & I_{Y'}/I_{Y'}^2 \otimes_{B'} C_0 \arrow[ll, "\simeq"'] \arrow[u, "c(X',Y',f)"'] \arrow[r] & JC \\
I_{Y_0}/I_{Y_0}^2 \otimes_{B_0} C_0 \arrow[u, "g_0"] & I_Y/I_Y^2 \otimes_B C_0 \arrow[l] \arrow[urr, "a_{g_0}(c(X',Y',f))"'] & &
\end{tikzcd}$$

One has therefore $a_{g_0}(c(X',Y',f))(\bar{x}\otimes 1) = f(u)$, where u is an element of $I_{Y'}$ whose image $\bar{u}$ in $I_{Y_0'}/I^2_{Y_0'}$ satisfies $\bar{u}_0 \otimes 1 = \bar{g}_0(\bar{x}_0) \otimes 1 = \bar{g}_0(\bar{x}_0) \otimes 1$. One can therefore take $u = x'$ and one finds

$$(2) \qquad a_{g_0}(c(X', Y', f))(\bar{x} \otimes 1) = f(x').$$

Consider finally $b_{f_0}(c(X,Y,g \circ i'))$. By hypothesis, the morphism of Λ_0-algebras $f_0 : A_0' \to C_0$ factors through B_0', and therefore, since $J \otimes_{\Lambda_0} B_0' \xrightarrow{\sim} JB'$ (B' being flat over Λ), one obtains a morphism of B_0'-modules $\psi : JB' \to JC$ such that one has a commutative diagram:

$$\begin{tikzcd}
J \otimes_{\Lambda_0} A_0' \arrow[r, "id\otimes\pi'"] \arrow[d] & J \otimes_{\Lambda_0} B_0' \arrow[r, "id\otimes f_0"] \arrow[d, "\wr"] & J \otimes_{\Lambda_0} C_0 \\
JA' \arrow[r, "\pi'"] & JB' \arrow[r, "\psi"] & JC.
\end{tikzcd}$$

Denote $\phi : JB' \otimes_{B_0'} C_0 \to JC$ the morphism of C_0-modules deduced from ψ; then one has, for every $a' \in A'$, $\lambda \in J$,

$$(\dagger) \qquad \varphi(\lambda\pi'(a') \otimes 1) = \lambda f(a').$$

Then, $b_{f_0}(c(X,Y,g \circ i'))$ is defined by the commutative diagram:

$$\begin{tikzcd}
I_Y \arrow[r, "\pi' \circ g"] & B' & \\
I_Y/I_Y^2 \otimes_B B_0' \arrow[r] \arrow[d] & JB' \arrow[u, "c(X,Y,g\circ i')"'] \arrow[d] & \\
I_Y/I_Y^2 \otimes_B C_0 \arrow[r] \arrow[drr, "b_{f_0}(c(X,Y,g\circ i'))"'] & JB' \otimes_{B_0'} C_0 \arrow[dr, "\varphi"] & \\
 & & JC.
\end{tikzcd}$$

One has therefore at once

$$b_{f_0}(c(X, Y, g \circ i'))(\bar{x} \otimes 1) = \varphi\left(\sum \lambda_i \pi'(a'_i) \otimes 1\right) = \sum \lambda_i f(a'_i), \tag{3}$$

the last equality following from (†) above. The comparison of the three explicit results (1), (2), (3) gives the formula sought.

Corollary 4.14. *Let Y, Y' be two flat subschemes of X, reducing to Y_J; suppose Y_0 locally complete intersection in X_0. If $f : T \to X$ is an S-morphism such that f_J factors through $Y_J \to X_J$, one has the formula*

$$c(X, Y, f) - c(X, Y', f) = b_{f_0}(d(Y, Y')).$$

Indeed, applying the preceding formula to the diagram

$$\begin{tikzcd} & Y' \arrow[d, hook, "i'"] & Y \arrow[d, hook, "i"] \\ T \arrow[r, "f"] & X \arrow[r, "\mathrm{id}"] & X \end{tikzcd}$$

one finds $c(X, Y, f) - c(X, Y', f) = b_{f_0}(c(X, Y, i'))$. Moreover, by 4.8 (ii), one has $c(X, Y, i') = d(Y, Y')$.

Proposition 4.15. *Let X be an S-group smooth over S and Y a sub-S-group flat and locally of finite presentation over S. Then Y is locally complete intersection (cf. 4.6.2) in X.*

Proof.[230] We shall show that the immersion $Y \to X$ is regular in the sense of EGA IV₄, 16.9.2, which implies that it is also regular in the sense of 4.6.2, by EGA IV₄, 19.5.1 (moreover, by loc. cit., the two definitions are equivalent if S is locally noetherian). Therefore, in what follows, we take "regular immersion" in the sense of EGA IV₄, 16.9.2. Since X and Y are flat and locally of finite presentation over S, then, by EGA IV₄, 19.2.4, it suffices to show that, for every $s \in S$, $Y_s \to X_s$ is a regular immersion. By EGA IV₄, 19.1.5 (ii), one is reduced to verifying the assertion on the geometric fibers of S, that is, when S is the spectrum of an algebraically closed field k.

Then, by VI_A, 3.2, the quotient X/Y exists and is smooth, the morphism $\pi : X \to X/Y$ is flat, and one has a cartesian square

$$\begin{tikzcd} Y \arrow[r, "f"] \arrow[d, "i"] & X \arrow[d, "\pi"] \end{tikzcd}$$

[230] N.D.E.: We have added in the statement the hypothesis that Y be locally of finite presentation over S, and have given the following proof, more direct than the one sketched in the original. To be complete, let us also detail the latter. As in the proof given above, one reduces first to the case where $S = \mathrm{Spec}(k)$, k being an algebraically closed field. By EGA IV₄, 16.9.10 and 19.3.2, it suffices to see that, for every $y \in Y$, the completion of the local ring $\mathcal{O}_{Y,y}$ is the quotient of a complete noetherian local ring by a regular sequence. By loc. cit., 19.3.3, the set of $y \in Y$ satisfying this property is an open U of Y; since Y is of finite type over k, it suffices to show that U contains every closed point. Since Y is a k-group it suffices, by a translation argument, to show that the property is true for the completion of $\mathcal{O}_{Y,e}$, that is, for the "formal group" $\hat{Y}$ corresponding to Y (cf. Exp. VII_B). Now, since X is smooth, the affine algebra $A(\hat{X})$ is an algebra of formal power series $k[[X_1, \ldots, X_n]]$, and one concludes with the help of the Dieudonné structure theorem which shows that $A(\hat{Y})$ is isomorphic to a quotient $k[[X_1, \ldots, X_{r+s}]]/(X_1^{p^{n_1}}, \ldots, X_r^{p^{n_r}})$, where p is the characteristic exponent of k and $r + s \leq n$, cf. VII_B, Remark 5.5.2 (b).

$$\begin{array}{ccc} \downarrow & & \downarrow \\ e & \longrightarrow & X/Y \end{array}$$

(where e is the image in X/Y of the unit point of X). Therefore, by flat base change (cf. EGA IV_4, 19.1.5 (ii)), it suffices to see that i is a regular immersion, which is immediate since the noetherian local ring $O_{X/Y,e}$ is smooth, hence its maximal ideal is generated by a regular sequence.

4.16.[231] Let X be an S-group smooth over S, denote $\mu : X \times_S X \to X$ its group law. Suppose given a sub-S_J-group Y_J of X_J, flat and locally of finite presentation over S_J. By 4.15, Y_J is locally complete intersection in X.

Hence, by 4.6.5, every flat S-scheme[232] Y lifting Y_J is locally complete intersection in X. For such a Y one has, by 4.8.0,

(4.16.1) $\quad N_{Y/X} \otimes_{O_Y} O_{Y_0} = N_{Y_0/X_0} = N_{Y_J/X_J} \otimes_{O_{Y_J}} O_{Y_0}.$

On the other hand, denote by $\epsilon_0 : S_0 \to Y_0$ the unit section of Y_0 and n_{Y_0/X_0} the quasi-coherent O_{S_0}-module:

$$n_{Y_0/X_0} = \epsilon_0 * (N_{Y_0/X_0}).$$

Since Y_0 and X_0 are S_0-groups, one sees easily that N_{Y_0/X_0} is invariant under the (say left) translations of Y_0, hence[233] is the inverse image by $Y_0 \to S_0$ of n_{Y_0/X_0}, i.e. one has

(4.16.2) $\quad N_{Y_0/X_0} = n_{Y_0/X_0} \otimes_{O_{S_0}} O_{Y_0}.$

Taking (4.16.1) and (4.16.2) into account, one deduces on the one hand from 4.5 that the set of sub-S-schemes Y of X, flat over S, lifting Y_J, is empty or principal homogeneous under

(4.16.3) $\quad \mathrm{Hom}_{O_{Y_0}}(n_{Y_0/X_0} \otimes_{O_{S_0}} O_{Y_0}, J \otimes_{O_{S_0}} O_{Y_0}),$

and one deduces on the other hand from 4.8 (i) that, for every such Y and every S-morphism $f : T \to X$ such that $f_J : T_J \to X_J$ factors through Y_J, the obstruction $c(X, Y, f)$ to f factoring through Y is an element of

$\mathrm{Hom}_{O_{T_0}}(n_{Y_0/X_0} \otimes_{O_{S_0}} O_{T_0}, JO_T);$

if moreover T is flat over S, this last group equals

$\mathrm{Hom}_{O_{T_0}}(n_{Y_0/X_0} \otimes_{O_{S_0}} O_{T_0}, J \otimes_{O_{S_0}} O_{T_0}).$

This leads to introducing the group functor N_0 below:

Definition 4.16.1. *Let N_0 be the S_0-functor in commutative groups defined by: for every $Z \in Ob(Sch)/S_0$,*

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[231] N.D.E.: We have reorganized 4.16 by regrouping there, on the one hand, the hypotheses stated at the end of 4.15 and, on the other hand, the definition of the obstruction DY.

[232] N.D.E.: We have corrected the original by adding "flat".

[233] N.D.E.: See 4.25 further on.

$$(*) \qquad \mathrm{Hom}_{S_0}(Z, N_0) = \mathrm{Hom}_{\mathcal{O}_Z}(n_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_Z, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_Z).$$

Then, the set of sub-S-schemes Y of X, flat over S, lifting Y_J, is empty or principal homogeneous under

$$\mathrm{Hom}_{S_0}(Y_0, N_0) = C^1(Y_0, N_0).$$

For each such Y, consider the following diagram:

$$\begin{tikzcd} Y \times_S Y \arrow[d, hook, "{(i,i)}"] & & Y \arrow[d, hook, "i"] \\ X \times_S X \arrow[rr, "\mu"] & & X \end{tikzcd}$$

and denote $DY = c(X, Y, \mu \circ (i,i))$ the obstruction to $\mu \circ (i,i)$ factoring through Y, i.e. to Y being stable under the group law of X; by what precedes, DY is an element of

$$N_0(Y_0 \times_{S_0} Y_0) = C^2(Y_0, N_0).$$

Lemma 4.17.[234] *Let X be an S-group smooth over S and Y_J a sub-S_J-group of X_J, flat and locally of finite presentation over S_J. For each subscheme Y of X, flat over S and lifting Y_J, consider the obstruction defined in 4.16.1:*

$$DY \in \mathrm{Hom}_{S_0}(Y_0 \times_{S_0} Y_0, N_0) = C^2(Y_0, N_0)$$

(i) For Y to be a sub-S-group of X, it is necessary and sufficient that $DY = 0$.

(ii) If one makes Y_0 act on N_0 by functoriality from the inner automorphisms of Y_0, then $DY \in Z^2(Y_0, N_0)$.

(iii) If Y and Y' are two subschemes of X, flat over S, lifting Y_J (so that the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ is defined, cf. 4.5.1), one has $DY' - DY = \partial^1 d(Y, Y')$.[235]

Let us successively prove these various assertions.

4.18. *Proof of 4.17 (i).* By definition, one has $DY = 0$ if and only if Y is stable under the group law of X. Hence $DY = 0$ if Y is a subgroup of X. Conversely, if $DY = 0$, Y is equipped with the induced law μ_Y, which is associative and reduces modulo J to the group law on X_J; since Y is flat and locally of finite presentation over S, it follows from 3.3 that μ_Y is a group law.

4.19. *Proof of 4.17 (ii).* This is done by comparing the two values of $u = c(X, Y, \mu^2 \circ (i,i,i))$ computed in the two following diagrams (D_j), $j = 1, 2$:

$$(D_j) \qquad \begin{tikzcd} Y \times_S Y \times_S Y \arrow[d, hook, "{(i,i,i)}"] \arrow[r, "f_j"] & Y \times_S Y \arrow[d, hook, "{(i,i)}"] & Y \arrow[d, hook, "i"] \\ {} & {} \arrow[r, "\mu"] & {} \end{tikzcd}$$

[234] N.D.E.: We have modified 4.17 and 4.18 taking into account the additions made in 4.16.

[235] N.D.E.: In the original, one finds $DY' - DY = -\partial d(Y, Y')$, but their ∂ is the opposite of the differential ∂^1 defined in I, 5.1.

$$X \times_S X \times_S X \longrightarrow X \times_S X \longrightarrow X$$

where $f_1 = (1, \pi)$, $f_2 = (\pi, 1)$, and where one denotes by μ^2 the morphism

$$\mu \circ f_1 = \mu \circ f_2 : X \times_S X \times_S X \to X.$$

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Set $\mu_Y = \mu \circ (i,i)$, $f_{j,Y} = f_j \circ (i,i,i)$ and $\mu_{2,Y} = \mu^2 \circ (i,i,i)$. For $j = 1,2$, denote by a_j and b_j the morphisms

$$a_j = a_{\mu_0}((f_{j,Y})_0) \quad \text{and} \quad b_j = b_{(f_{j,Y})_0},$$

associated with the pair of morphisms $(f_{j,Y}, \mu)$ by Lemma 4.12; one has therefore:

$$(\dagger) \quad \mathrm{Hom}_{O_{Y_0^3}}((f_{j,Y})_0^*(N_{Y_0 \times Y_0/X_0 \times X_0}), JO_{Y_0^3}) \xrightarrow{a_j} \mathrm{Hom}_{O_{Y_0^3}}((\mu_{2,Y})_0^*(N_{Y$$

$$\mathrm{Hom}_{O_{Y_0^2}}((\mu_Y)_0^*(N_{Y_0/X_0}), JO_{Y_0^2}) \xrightarrow{b_j} \mathrm{Hom}_{O_{Y_0^3}}((\mu_{2,Y})_0^*(N_{Y_0/X_0}), J$$

Since $N_{Y_0 \times Y_0/X_0 \times X_0} \simeq pr_1 * N_{Y_0/X_0} \bigoplus pr_2 * N_{Y_0/X_0}$ (since X_0 and Y_0 are flat over S_0), and $N_{Y_0/X_0} \simeq n_{Y_0/X_0} \bigotimes_{O_{S_0}} O_{Y_0}$, then:

$$(f_{j,Y})_0^*(N_{Y_0 \times Y_0/X_0 \times X_0}) \simeq (n_{Y_0/X_0} \oplus n_{Y_0/X_0}) \otimes O_{Y_0^3}$$

and, likewise,

$$(\mu_{2,Y})_0^*(N_{Y_0/X_0}) \simeq n_{Y_0/X_0} \otimes O_{Y_0^3} \quad \text{and} \quad (\mu_Y)_0^*(N_{Y_0/X_0}) \simeq n_{Y_0/X_0} \otimes O_{Y_0^2}.$$

Moreover, since Y_0^2 and Y_0^3 are flat over S_0, then $(\dagger)$ rewrites in the following form:

$$(\ddagger) \quad \begin{cases} a_j : \mathrm{Hom}_{S_0}(Y_0^3, N_0 \oplus N_0) \to \mathrm{Hom}_{S_0}(Y_0^3, N_0) \\ b_j : \mathrm{Hom}_{S_0}(Y_0^2, N_0) \to \mathrm{Hom}_{S_0}(Y_0^3, N_0). \end{cases}$$

Applying 4.13 twice to $c(X, Y, \mu_{2,Y}) = u$, one obtains:

$$a_1(c(X^2, Y^2, f_1)) + b_1(c(X, Y, \mu_Y)) = u = a_2(c(X^2, Y^2, f_2)) + b_2(c(X, Y, \mu_Y)).$$

Now, $c(X, Y, \mu_Y) = DY$ and, since $f_1 = (1, \mu)$ and $f_2 = (\mu, 1)$, one has, with evident notations:

$$c(X^2, Y^2, f_1) = (0, DY) \quad \text{and} \quad c(X^2, Y^2, f_2) = (DY, 0).$$

Hence, one obtains:

$$u = a_1((0, DY)) + b_1(DY) = a_2((DY, 0)) + b_2(DY).$$

The first thing one notes is that b_j is nothing other than $\mathrm{Hom}_{S_0}((f_{j,Y})_0, N_0)$, that is to say, the morphism deduced from $(f_{j,Y})_0$ by functoriality.

The identity above therefore becomes:

$$a_1((0, DY)) - \mathrm{Hom}((\mu, 1), N_0)(DY) + \mathrm{Hom}((1, \mu), N_0)(DY) - a_2((DY, 0)) = 0.$$

[236] N.D.E.: We have slightly modified the notations, and detailed the beginning of the argument.

One recognizes the two middle terms: they are the parts "$DY(xy,z)$" and "$DY(x,yz)$" of the 2-coboundary formula. It only remains, then, to identify the two other terms.

We must first compute the map a_j. Now it comes, by inverse image by $(f_{j,Y})_0$, from the morphism of $O_{Y_0^2}$-modules

$$P : n_{Y_0/X_0} \otimes O_{Y_0^2} \to (n_{Y_0/X_0} \oplus n_{Y_0/X_0}) \otimes O_{Y_0^2}$$

induced by the product in Y_0. Now this morphism is described in the following way: consider the vector bundle $V = V(n_{Y_0/X_0})$; P gives by duality a morphism

$$V(P) : V \times_{S_0} V \times_{S_0} Y_0 \times_{S_0} Y_0 \to V \times_{S_0} Y_0 \times_{S_0} Y_0$$

which is expressed set-theoretically by[237]

$$V(P)(u, v, a, b) = (u + \mathrm{Ad}(a)v, ab, b).$$

This is proved exactly like the corresponding fact on Lie algebras, that is, on the module $\omega^1_{Y_0/S_0}$. One first notes that V is endowed by functoriality in Y_0 with a group structure in the category of vector bundles on S_0; by virtue of the lemma already used for Lie algebras (Exposé II, 3.10), this structure coincides with the group structure underlying its O_S-module structure. One then sees that $V(n_{Y_0/X_0} \bigotimes_{O_{S_0}} O_{Y_0}) = V(N_{Y_0/X_0})$ is also endowed with a structure of S_0-group which is none other than the semi-direct product of that of V by that of Y_0. It only remains to identify the operations of Y_0 on V to establish the desired formula.

Let us now compute the two remaining terms. Consider first $a_1((0,DY))$. One computes it by the diagram (where n denotes n_{Y_0/X_0}):

$$\begin{tikzcd}
n \otimes O_{Y_0^2} \arrow[r, "P"] \arrow[d, "(f_{1,Y})_0^*"] & (n + n) \otimes O_{Y_0^2} \arrow[d, "(f_{1,Y})_0^*"] \\
n \otimes O_{Y_0^3} \arrow[dr, "a_1((0,DY))"'] & (n + n) \otimes O_{Y_0^3} \arrow[d, "(0,\, DY)"] \\
& J \otimes O_{Y_0^3}.
\end{tikzcd}$$

Considering now the vector bundles defined by these different modules as so many schemes over S_0 and taking points with values in anything, one has, denoting (u,x,y,z) a point of $V(J) \times Y_0^3$;

$$\begin{tikzcd}
(\mathrm{Ad}(x)DY_{y,z}(u),\, x,\, yz) & (0 + DY_{y,z}(u),\, x,\, yz) \arrow[l] \\
\arrow[u] & (0 + DY_{y,z}(u),\, x,\, y,\, z) \arrow[u] \\
& \arrow[u]
\end{tikzcd}$$

[237]N.D.E.: We have replaced a, b by ab, b to make visible that $V(P)$ comes by inverse image on Y_0^2 from the multiplication morphism $V_{Y_0} \times_{S_0} V_{Y_0} \to V_{Y_0}$.

$$(Ad(x)DY_{y,z}(u), x, y, z) \longleftarrow (u, x, y, z).$$

One has thus obtained $a_1((0, DY))(x,y,z) = Ad(x)DY(y,z)$, which is indeed the first term of the coboundary. One would have likewise $a_2((DY,0))(x,y,z) = DY(x,y)$, whence[238]

$$0 = Ad(x)DY(y, z) - DY(xy, z) + DY(x, yz) - DY(x, y) = (\partial^2 DY)(x, y, z).$$

4.20. *Proof of 4.17 (iii).*[239] This is done by comparing the two values of $v = c(X, Y, \mu \circ (i', i'))$ computed in the two following diagrams

$$(*)\qquad \begin{tikzcd} & Y' \arrow[d, hook, "i'"] & Y \arrow[d, hook, "i"] \\ Y' \times_S Y' \arrow[r, "\mu\circ(i',i')"] & X & X \end{tikzcd}$$

$$(\dagger)\qquad \begin{tikzcd} & Y \times_S Y \arrow[d, hook, "(i,i)"] & Y \arrow[d, hook, "i"] \\ Y' \times_S Y' \arrow[r, "(i',i')"] & X \times_S X \arrow[r, "\mu"] & X. \end{tikzcd}$$

Denote $f = \mu \circ (i', i')$; then $(*)$ gives

(1) $\quad v = DY' + f_0{}^*(c(X, Y, i')).$

Now $Y_0' = Y_0$ and f_0 is the multiplication $Y_0^2 \to Y_0$; one deduces from this that

(2) $\quad f_0{}^*(c(X, Y, i'))(x_0, y_0) = c(X, Y, i')(x_0 y_0).$

Set $c = c(X, Y, i')$; via the identification $N_0' \simeq N_0 \oplus N_0$, $c(X \times_S X, Y \times_S Y, (i', i'))$ identifies with the pair (c, c). Then, denoting $h = (i', i')$, $(\dagger)$ gives

(3) $\quad v = h_0{}^*(DY) + a_{\mu_0}(c, c).$

Now h_0 is the identity map of Y_0^2, whence $h_0 * (DY) = DY$. Finally, by the computation of a_{μ_0} done previously, one has for every $S' \to S$ and $x_0, y_0 \in Y_0(S_0')$,

(4) $\quad a_{\mu_0}(c, c)(x_0, y_0) = c(x_0) + Ad(x_0)(c(y_0)).$

One thus obtains:

$$\begin{aligned}(DY' - DY)(x_0, y_0) &= Ad(x_0)c(X, Y, i')(y_0) - c(X, Y, i')(x_0 y_0) + c(X, Y, i')(x_0) \\ &= (\partial^1 c(X, Y, i'))(x_0, y_0).\end{aligned}$$

Since $c(X, Y, i') = d(Y, Y')$ (cf. 4.8 (ii)), this shows that $DY' - DY = \partial^1 d(Y, Y')$.

Theorem 4.21. *Let S be a scheme, I and J two ideals*[240] *on S such that $I \supset J$ and $I \cdot J = 0$. Let X be an S-group smooth over S and Y_J a sub-S_J-group of X_J, flat*

[238] N.D.E.: We have changed the signs to make them compatible with I 5.1.
[239] N.D.E.: We have detailed the original in what follows.
[240] N.D.E.: We have suppressed the hypothesis that I be nilpotent, which appears superfluous.

and locally of finite presentation over S_J. Consider the S_0-functor in commutative groups N_0 defined by

$$\mathrm{Hom}_{S_0}(T, N_0) = \mathrm{Hom}_{O_T}(n_{Y_0/X_0} \otimes_{O_{S_0}} O_T, J \otimes_{O_{S_0}} O_T), \qquad T \in \mathrm{Ob}\,(\mathrm{Sch})/S_0,$$

on which Y_0 acts via the inner automorphisms of X_0.

(i) For there to exist a sub-S-group of X, flat over S, which reduces to Y_J, it is necessary and sufficient that the two following conditions be verified:

(i_1) There exists a subscheme Y of X, flat over S, lifting Y_J (condition automatically satisfied if Y_0 is affine, cf. 4.6.5).

(i_2) A certain canonical obstruction, element of $H^2(Y_0, N_0)$, is zero.

(ii) If the conditions of (i) are satisfied, the set of sub-S-groups Y of X, flat over S and reducing to Y_J is a principal homogeneous set under the group $Z^1(Y_0, N_0)$.[241]

Indeed, condition (i_1) is necessary. Suppose it satisfied and let Y be flat over S lifting Y_J. We must seek a Y' flat over S lifting Y_J as well such that $DY' = 0$,[242] cf. 4.17 (i). By 4.17 (iii), this amounts to seeking a $d(Y', Y) \in C^1(Y_0, N_0)$ such that $DY = \partial^1 d(Y', Y)$.[243]

Let $c \in H^2(Y_0, N_0)$ be the image class of DY, which is a cocycle by 4.17 (ii). It does not depend on the choice of Y by 4.17 (iii), and its vanishing is necessary and sufficient for the existence of a $d(Y', Y)$ satisfying the preceding equation. This proves (i).

If one has now chosen Y such that $DY = 0$, the equation to solve becomes $\partial^1 d(Y', Y) = 0$, which proves (ii).

Remark 4.22. *Let us keep the notation of 4.21. By 4.15, Y_0 is locally complete intersection in X_0, hence N_{Y_0/X_0} is a finite locally free O_{Y_0}-module, and consequently $n_{Y_0/X_0} = \epsilon_0 * (N_{Y_0/X_0})$ is a finite locally free O_{S_0}-module. Hence, denoting $n^\vee_{Y_0/X_0} =$ $\mathrm{Hom}_{O_{Y_0}}(n_{Y_0/X_0}, O_{Y_0})$, one has*

$$\mathrm{Hom}_{O_T}(n_{Y_0/X_0} \otimes_{O_{S_0}} O_T, J \otimes_{O_{S_0}} O_T) \simeq n_{Y_0/X_0}^\vee \otimes_{O_{S_0}} J \otimes_{O_{S_0}} O_T.$$

for every $T \to S_0$.[244] *Consequently, the S_0-functor N_0 is isomorphic to the functor*

$$W(n_{Y_0/X_0}^\vee \otimes_{O_{S_0}} J) \simeq W(\mathrm{Hom}_{O_{S_0}}(n_{Y_0/X_0}, J)).$$

It results in isomorphisms:[245]

$$H^2(Y_0, N_0) \simeq H^2(Y_0, \mathrm{Hom}_{O_{S_0}}(n_{Y_0/X_0}, J)) \simeq H^2(Y_0, n_{Y_0/X_0}^\vee \otimes_{O_{S_0}} J),$$
$$Z^1(Y_0, N_0) \simeq Z^1(Y_0, \mathrm{Hom}_{O_{S_0}}(n_{Y_0/X_0}, J)) \simeq Z^1(Y_0, n_{Y_0/X_0}^\vee \otimes_{O_{S_0}} J).$$

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[241] N.D.E.: The question of whether the preceding set, modulo conjugation by the $x \in X(S)$ inducing the unit of $X(S_J)$, is principal homogeneous under $H^1(Y_0, N_0)$, occupies n°s 4.23 to 4.36.

[242] N.D.E.: We have corrected $\partial DY'$ to DY'.

[243] N.D.E.: Cf. N.D.E. (110).

[244] N.D.E.: We have detailed what precedes; this shows that the following isomorphism is valid without flatness hypothesis; on the other hand, since 4.16, we have restricted ourselves to S-schemes $f : T \to S$ flat over S to ensure that the group $\mathrm{Hom}_{O_T}(n_{Y_0/X_0} \otimes_{O_{S_0}} O_{T_0}, JO_T)$, in which the obstruction $c(X, Y, f)$ resides, coincides with $N_0(T_0)$ (cf. the end of 4.16).

[245] N.D.E.: With the notation of I 5.3, assuming Y_0 affine over S_0.

4.23. Still under the hypotheses of 4.21, we are now going to study how the set of Y lifting Y_J behaves with respect to conjugation by sections of X. If x is a section of X over S inducing the unit section of X_J, the inner automorphism $Int(x)$ defined by x transforms flat subgroups of X lifting Y_J into flat subgroups of X lifting Y_J. Now, under the conditions of 4.21 (ii), the set of these subgroups is principal homogeneous under $Z^1(Y_0, N_0)$; we shall see that there then exists a subgroup Γ of $B^1(Y_0, N_0)$[246] such that two subgroups of X, flat over S, and lifting Y_J, are conjugate (by $x \in X(S)$ inducing the unit of $X(S_J)$) if and only if their "difference" in $Z^1(Y_0, N_0)$ is an element of Γ. In the best cases, we shall show that Γ equals $B^1(Y_0, N_0)$, hence that the set of flat subgroups of X lifting Y_J, modulo conjugation by $x \in X(S)$ inducing the unit of $X(S_J)$, is empty or principal homogeneous under $H^1(Y_0, N_0)$ (cf. 4.29 and 4.36).

4.24. We keep the notation of 4.21. Let Y be a flat subgroup of X, reducing to Y_J. Recall that we introduced in 0.5 the functor L_0^X (resp. L_0^Y) defined by the identity with respect to the variable S_0-scheme T:

$$\mathrm{Hom}_{S_0}(T, L_0^X) = \mathrm{Hom}_{O_T}(\omega^1_{X_0/S_0} \otimes_{O_{S_0}} O_T, J \otimes_{O_{S_0}} O_T)$$

(resp. similarly on replacing X by Y), as well as the functor $L'_X = \prod_{S_0/S} L_0^X$.

Now one has:

Lemma 4.25. *There exists a canonical exact sequence of Y_0-O_{S_0}-modules*

$$(+) \qquad n_{Y_0/X_0} \xrightarrow{\ d\ } \omega^1_{X_0/S_0} \longrightarrow \omega^1_{Y_0/S_0} \longrightarrow 0$$

possessing the following properties:

(i) By inverse image on Y_0, d gives the morphism $\bar{D}_0$ of 4.8 (iii).

(ii) If X_0 and Y_0 are smooth over S_0, then d is injective. Since the two ω^1 are then locally free of finite type, so is n_{Y_0/X_0} and the sequence is locally split.

Proof.[247] Denote by π_0 the morphism $Y_0 \to S_0$. By SGA 1 II, formula (4.3) (see also EGA IV₄, 16.4.21), one has a canonical exact sequence of O_{Y_0}-modules

$$(\dagger) \qquad N_{Y_0/X_0} \xrightarrow{\ \bar{D}_0\ } \Omega^1_{X_0/S_0} \otimes_{O_{X_0}} O_{Y_0} \longrightarrow \Omega^1_{Y_0/S_0} \longrightarrow 0.$$

Since this sequence is formed of $(Y_0 \times_S Y_0)$-equivariant modules and morphisms, its inverse image $(+)$ by $\epsilon_0{*}$ is an exact sequence of Y_0-O_{S_0}-modules, and $(\dagger)$ is the inverse image of $(+)$ by $\pi_0{*}$ (cf. Exp. I, § 6.8). This proves (i).

Suppose moreover X_0 and Y_0 smooth over S_0. Then, by SGA 1 II 4.10 (see also EGA IV₄, 17.2.3 (i) and 17.2.5), D is injective and the sequence $(\dagger)$ is formed of O_{Y_0}-modules locally free of finite type (hence is locally split). By the equivalence of categories I, 6.8.1, d is also injective, and therefore the sequence $(+)$ has the indicated properties.

[246] N.D.E.: We have replaced $Z^1(Y_0, N_0)$ by $B^1(Y_0, N_0)$, since the proof shows that Γ is a subgroup of $B^1(Y_0, N_0)$, cf. 4.27–4.29.

[247] N.D.E.: We have detailed the proof, taking into account the additions made in Exp. I, § 6.8.

4.26.[248] For every S_0-scheme $f : T \to S_0$, (+) gives an exact sequence of $Y_0(T)$-$O(T)$-modules

$$0 \to \mathrm{Hom}_{O_T}(f^*(\omega^1_{Y_0/S_0}), f^*(J)) \to \mathrm{Hom}_{O_T}(f^*(\omega^1_{X_0/S_0}), f^*(J)) \to \mathrm{Hom}_{O_T}(f^*(n_{Y_0/X_0}), f^*$$

hence one has an exact sequence of Y_0-O_{S_0}-modules:

$$0 \to L_0^Y \to L_0^X \xrightarrow{d} N_0. \tag{4.26.1}$$

From this one deduces an exact sequence of complexes of abelian groups:

$$0 \to C^*(Y_0, L_0^Y) \to C^*(Y_0, L_0^X) \xrightarrow{d^*} C^*(Y_0, N_0),$$

and in particular, a commutative diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & C^0(Y_0, L_0^Y) \arrow[r] \arrow[d, "\partial"] & C^0(Y_0, L_0^X) \arrow[r, "d^0"] \arrow[d, "\partial"] & C^0(Y_0, N_0) \arrow[d, "\partial"] \\
0 \arrow[r] & C^1(Y_0, L_0^Y) \arrow[r] & C^1(Y_0, L_0^X) \arrow[r, "d^1"] & C^1(Y_0, N_0).
\end{tikzcd}$$

Note that $C^0(Y_0, L_0^Y)$ (resp. $C^0(Y_0, L_0^X)$) is none other than $\mathrm{Hom}_{S_0}(S_0, L_0^Y) = \mathrm{Hom}_S(S, L'_Y)$ (resp. $\cdots$), i.e. (cf. 0.9) the group of sections of Y (resp. X) over S inducing the unit section of X_J. Note also that d^1 is none other than the morphism $v_{i_{Y_0}}$ of 4.8 (iii), where $i_{Y_0} : Y_0 \to X_0$ is the canonical immersion.[249]

Lemma 4.27. *Under the conditions of 4.21 for S, I, J and X, let Y be a subgroup of X, flat over S and lifting Y_J. Denote $i : Y \hookrightarrow X$ the canonical immersion.*[250]

(i) Let $i' : Y \to X$ be a morphism of S-schemes lifting i_0 (so that i' is also an immersion), let $Y' = i'(Y)$ and let $d(i,i')$ be the element of $C^1(Y_0, L_0^X)$ such that $i' = d(i,i')\cdot i$ (cf. 1.2.4). Then the deviation $d(Y,Y') \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by the formula:

$$d(Y, Y') = d^1(d(i, i')).$$

(ii) Let $x \in C^0(Y_0, L_0^X)$ be a section of X over S inducing the unit section of X_J over S_J. Then the deviation $d(Y, Int(x)Y) \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by the formula:

$$-d(Y, \mathrm{Int}(x)Y) = d^1\partial x = \partial\, d^0 x.$$

Indeed, Y' is the image of Y by the composite morphism:[251]

$$Y \xrightarrow{(d(i,i'),\, i)} L'_X \times_S X \to X,$$

which is denoted $d(i,i')\cdot i$ in 4.8 (iii); by loc. cit. and the equality $v_{i_0} = d^1$, one has therefore:

$$c(X, Y', d(i, i')\cdot i) - c(X, Y', i) = v_{i_0}(d(i, i')) = d^1(d(i, i')).$$

[248] N.D.E.: We have detailed the original, to make visible that one has an exact sequence of Y_0-O_{S_0}-modules.

[249] N.D.E.: This results from the definition of $d : n_{Y_0/X_0} \to \Omega^1_{X_0/S_0}$ (cf. 4.25) and from that of $v_{i_{Y_0}}$ (cf. 4.8).

[250] N.D.E.: We have added point (i) below, which will be useful in 4.35.1 and then in 4.38 (4) and (5).

[251] N.D.E.: Recall that $L'_X = \prod_{S_0/S} L_0^X$.

But $d(i,i') \cdot i = i'$ factors through Y' by definition, hence the first term is zero; moreover, by 4.8 (ii), one has $c(X,Y',i) = d(Y',Y) = -d(Y,Y')$. Hence $d(Y,Y') = d^1(d(i,i'))$, which proves (i).

Let now x be as in (ii). By the formula

$$xyx^{-1} = xyx^{-1}y^{-1}y = (x - \mathrm{Ad}(y)x)y = (-\partial x)(y) \cdot y,$$

one sees that Y' is the image of Y by the immersion $i' = (-\partial x) \cdot i_Y$. Hence, by (i) one obtains

$$-d(Y, \mathrm{Int}(x)Y) = d^1\partial x = \partial\, d^0 x.$$

Corollary 4.28. *For two subgroups Y and Y' of G, flat over S and lifting Y_J, to be conjugate by a section of X over S inducing the unit section of X_J, it is necessary and sufficient that $d(Y,Y') \in \partial d^0 C^0(Y_0, L_0^X) \subset \partial C^0(Y_0, N_0) = B^1(Y_0, N_0)$.*

Corollary 4.29. *If d^0 is surjective, Y and Y' as above are conjugate by a section of X over S inducing the unit section of X_J if and only if $d(Y,Y') \in B^1(Y_0, N_0)$.*

Corollary 4.30. *Let Y be as in 4.27; the set of conjugates of Y by sections of X over S inducing the unit section of X_J is isomorphic to:*

$$d^1\partial(C^0(Y_0, L_0^X)) = C^0(Y_0, L_0^X) / \mathrm{Ker}\, d^1\partial.$$

Note now that $C^0(Y_0, L_0^X)/Ker d^1\partial$ is computed solely with the help of the left square of the commutative diagram of 4.26. It follows in particular that one can also compute it in any diagram of the same type having the same left square. Consider in particular the functor L_0^X/L_0^Y above S_0 defined by

$$\mathrm{Hom}_{S_0}(T, L_0^X/L_0^Y) = \mathrm{Hom}_{S_0}(T, L_0^X) / \mathrm{Hom}_{S_0}(T, L_0^Y).$$

One has a commutative diagram

$$\begin{array}{ccccccccc}
0 & \to & C^0(Y_0, L_0^Y) & \to & C^0(Y_0, L_0^X) & \to & C^0(Y_0, L_0^X/L_0^Y) & \to & 0 \\
 & & \downarrow{\scriptstyle\partial} & & \downarrow{\scriptstyle\partial} & & \downarrow{\scriptstyle\partial} & & \\
0 & \to & C^1(Y_0, L_0^Y) & \to & C^1(Y_0, L_0^X) & \to & C^1(Y_0, L_0^X/L_0^Y) & \to & 0,
\end{array}$$

whence by the preceding remark:

Corollary 4.31. *Let Y be as in 4.27; the set of conjugates of Y by sections of X over S inducing the unit section of X_J is isomorphic to*

$$E = \partial(C^0(Y_0, L_0^X/L_0^Y)) = C^0(Y_0, L_0^X/L_0^Y) / H^0(Y_0, L_0^X/L_0^Y).$$

Corollary 4.32. *Suppose moreover S_0 affine and $\omega^1_{Y_0/S_0}$ finite locally free.*[252] *If one denotes $F_0 = [Lie(X_0/S_0)/Lie(Y_0/S_0)] \otimes_{\mathcal{O}_{S_0}} J$, one has $E = \Gamma(S_0, F_0)/H^0(Y_0, F_0)$.*

[253] Indeed, since $\omega^1_{Y_0/S_0}$ is finite locally free, as is $\omega^1_{X_0/S_0}$ (since X is supposed smooth over S), one has, by 0.6:

[252] N.D.E.: We have corrected $Lie(Y_0/S_0)$ to $\omega^1_{Y_0/S_0}$.
[253] N.D.E.: We have detailed the original in what follows.

$$L_0^Y = W(Lie(Y_0/S_0) \otimes_{O_{S_0}} J) \quad \text{and} \quad L_0^X = W(Lie(X_0/S_0) \otimes_{O_{S_0}} J).$$

On the other hand, by 4.25, one has an exact sequence of Y_0-O_{S_0}-modules:

$$0 \to K \to \omega^1_{X_0/S_0} \xrightarrow{\phi} \omega^1_{Y_0/S_0} \to 0$$

(where $K = Ker(\phi)$). Since $\omega^1_{Y_0/S_0}$ and $\omega^1_{X_0/S_0}$ are finite locally free, one has a locally split exact sequence:

$$0 \to Lie(Y_0/S_0) \otimes_{O_{S_0}} J \to Lie(X_0/S_0) \otimes_{O_{S_0}} J \to F_0 \to 0.$$

It follows that one has an exact sequence of Y_0-O_{S_0}-modules:

$$0 \longrightarrow L_0^Y \longrightarrow L_0^X \longrightarrow W(F_0).$$

By the reasoning that served us to prove 4.31, we can compute E as the image of the composite map

$$C^0(Y_0, L_0^X) \xrightarrow{\pi} C^0(Y_0, W(F_0)) \xrightarrow{\partial} C^1(Y_0, W(F_0)).$$

Now the map π above is the map $\Gamma(S_0, Lie(X_0/S_0) \bigotimes_{O_{S_0}} J) \to \Gamma(S_0, F_0)$. Hence, S_0 being affine, π is surjective and one finds indeed the announced result.

Corollary 4.33. *Let S, I, J and X be as in 4.21, and let Y be a diagonalizable subgroup of X. Suppose $\omega^1_{Y_0/S_0}$ finite locally free and S_0 affine.*[254] *The set of subgroups of X conjugate to Y by a section of X over S inducing the unit section of X_J is isomorphic to*

$$E = \Gamma(S_0, [Lie(X_0/S_0) / Lie(X_0/S_0)^{ad(Y_0)}] \otimes_{\Gamma(S_0, O_{S_0})} \Gamma(S_0, J))$$

[255] *that is, isomorphic to $L_0^X(Y_0)/H^0(Y_0, L_0^X)$.*

Indeed, one writes by I 4.7.3 (cf. 2.13):

$$Lie(X_0/S_0) = Lie(X_0/S_0)^{ad(Y_0)} \bigoplus R.$$

Since Y_0 is commutative one has $Lie(Y_0/S_0) \subset Lie(X_0/S_0)^{ad(Y_0)}$, hence

$$F_0 = [Lie(X_0/S_0)^{ad(Y_0)} / Lie(Y_0/S_0)] \otimes J \oplus R \otimes J,$$
$$F_0^{ad(Y_0)} = [Lie(X_0/S_0)^{ad(Y_0)} / Lie(Y_0/S_0)] \otimes J.$$

By 4.32, one has therefore $E \simeq \Gamma(S_0, R \bigotimes J)$. Returning to the definition of R, one is done.

Corollary 4.34. *Let S, I, J and X be as in 4.21, and let Y be a diagonalizable subgroup of X. Suppose $\omega^1_{Y_0/S_0}$ finite locally free and S_0 affine.*[256] *If $x \in X(S)$ induces the unit section of X_J and normalizes Y, then it centralizes Y.*

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[254] N.D.E.: We have added the hypothesis on $\omega^1_{Y_0/S_0}$ and replaced the hypothesis "S affine" by "S_0 affine".
[255] N.D.E.: We have added what follows, cf. 4.34.
[256] N.D.E.: Cf. N.D.E. (129).

This results immediately from comparison of the preceding corollary and 2.14. Indeed, 4.33 shows that the elements of $C^0(Y_0, L_0^X)$ which globally preserve Y are the elements of $H^0(Y_0, L_0^X)$, and one has seen in 2.14 that these are precisely those which act trivially on the canonical immersion $Y \to X$.

4.35. Let us return to the general situation of 4.21 and suppose Y_J smooth over S_J. Then, by 4.25 (ii), one has an exact sequence of Y_0-O_{S_0}-modules:

$$(*) 0 \longrightarrow Lie(Y_0/S_0) \longrightarrow Lie(X_0/S_0) \longrightarrow n^{\vee}_{Y_0/X_0} \longrightarrow 0$$

and they are finite locally free O_{S_0}-modules.

On the other hand, by SGA 1, II 4.10, every subscheme Y of X lifting Y_J and flat over S will be smooth over S.[257] Suppose moreover S_0 and Y_J affine. Then, since $n^{\vee}_{Y_0/X_0}$ is a locally free O_{S_0}-module, the sequence $(*)$ remains exact when one applies $\otimes_{O_{S_0}} J$ to it, and then takes the inverse image on Y_0^n, and as the Y_0^n are affine, one therefore obtains an exact sequence of complexes of abelian groups:

$$0 \to C^*(Y_0, L_0^Y) \to C^*(Y_0, L_0^X) \xrightarrow{d^*} C^*(Y_0, N_0) \to 0$$

and in particular, a commutative diagram with exact rows

$$\begin{array}{ccccccccc}
0 & \to & C^1(Y_0, L_0^Y) & \to & C^1(Y_0, L_0^X) & \xrightarrow{d^1} & C^1(Y_0, N_0) & \to & 0 \\
 & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial & & \\
0 & \to & C^2(Y_0, L_0^Y) & \to & C^2(Y_0, L_0^X) & \xrightarrow{d^2} & C^2(Y_0, N_0) & \to & 0.
\end{array}$$

Let now Y, Y' be two subgroups of X lifting Y_J and flat, hence smooth, over S. As Y_J is affine, then, by 0.15, Y and Y' are isomorphic as schemes extending Y_J, i.e. there exists an isomorphism of S-schemes

$$f : Y \xrightarrow{\sim} Y'$$

inducing the identity on Y_J. On the one hand, by 1.2.4, f defines an element a of $C^1(Y_0, L_0^X)$ such that $f(y) = a(y_0)y$, for every $y \in Y(S')$, $S' \to S$, and by 4.27 (i), one has

$$d^1(a) = d(Y, Y').$$

Moreover, since Y, Y' are subgroups of X, the above element belongs to $Z^1(Y_0, N_0)$ (cf. 4.21). Then ∂a is an element of $Z^2(Y_0, L_0^Y)$ whose image $\bar{\partial} a$ in $H^2(Y_0, L_0^Y)$ depends only on the class $\bar{d}(Y, Y') \in H^1(Y_0, N_0)$; this being the definition of the connecting map $\partial^1 : H^1(Y_0, N_0) \to H^2(Y_0, L_0^Y)$, one has therefore:

$$\partial^1(\bar{d}(Y, Y')) = \bar{\partial} a.$$

[257]N.D.E.: We have added what follows and Proposition 4.35.1, implicit in the original, cf. 4.38 (5).

On the other hand, let us transport by f the group structure of Y' and let Y_1 be the group obtained (which thus has Y as underlying scheme), that is, the group law μ_1 of Y_1 is defined by: for every $S' \to S$ and $x, y \in Y(S')$,

$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$

By 3.5.1, Y_1 defines a cocycle $\delta(Y, Y_1) \in Z^2(Y_0, Lie(Y_0/S_0))$ such that, for every $S' \to S$ and $x, y \in Y(S')$, one has

$$\delta(Y, Y_1)(x_0, y_0)\, xy = \mu_1(x, y) = f^{-1}(f(x)f(y)).$$

Set $b = \delta(Y, Y_1)$. For every $S' \to S$ and $x, y \in Y(S')$, one has $(b(x_0, y_0)xy)_0 = x_0y_0$ and therefore one obtains that $f(b(x_0, y_0)xy)$ equals, on the one hand, $a(x_0y_0)b(x_0, y_0)xy$ and, on the other hand,

$$f(x)f(y) = a(x_0)x\, a(y_0)y = a(x_0)\, \mathrm{Ad}(x_0)(a(y_0))\, xy.$$

Comparing the two expressions, one obtains that $b(x_0, y_0)$ equals

$$a(x_0y_0)^{-1}\, a(x_0)\, \mathrm{Ad}(x_0)(a(y_0)) = \mathrm{Ad}(x_0)(a(y_0)) - a(x_0y_0) + a(x_0) = (\partial a)(x_0, y_0),$$

i.e. $\delta(Y, Y_1) = \partial a$. We have thus obtained:

Proposition 4.35.1.[258] *Under the hypotheses of 4.21, suppose moreover S_0 affine and Y_J smooth over S_J and affine. Let Y, Y' be two subgroups of X lifting Y_J and flat (hence smooth) over S, let $f : Y \xrightarrow{\sim} Y'$ be an isomorphism of S-schemes inducing the identity on Y_J, denote by Y_1 the group obtained by transporting via f the group structure of Y'. Then one has*

$$\partial^1(\bar{d}(Y, Y')) = \delta(Y, Y_1).$$

Proposition 4.36. *Under the hypotheses of 4.21, suppose moreover Y_J smooth over S_J and S_0 affine. The set of sub-S-groups Y of X flat (or smooth) over S, reducing to Y_J, modulo conjugation by sections of X over S inducing the unit section of X_J, is either empty, or a principal homogeneous set under the group*

$$H^1(Y_0, [Lie(X_0/S_0)/Lie(Y_0/S_0)] \otimes_{\mathcal{O}_{S_0}} J).$$

It suffices for us to verify that Corollary 4.29 applies, that is, that

$$d^0 : \mathrm{Hom}_{\mathcal{O}_{S_0}}(\omega^1_{X_0/S_0}, J) \to \mathrm{Hom}_{\mathcal{O}_{S_0}}(n_{Y_0/X_0}, J)$$

is surjective. Now this follows from the fact that the sequence (+) of 4.25 (ii) is split, S_0 being affine.[259]

Let us finally state a corollary common to 4.21 and 4.36, which will in fact be the only form under which we shall use in what follows the general results of this section.[260]

[258] N.D.E.: We have added what follows and Proposition 4.35.1, implicit in the original, cf. 4.38 (5).
[259] N.D.E.: This also follows from the proof of 4.32.
[260] N.D.E.: For example, 4.37 is used in Exposé IX to prove statements 3.2 bis and 3.6 bis.

Corollary 4.37. *Let S be a scheme and S_0 the closed subscheme defined by a nilpotent ideal I. Let X be an S-group smooth over S, and Y_0 a sub-S_0-group of X_0, flat over S_0.*

(i) If S_0 is affine, Y_0 smooth over S_0, and if

$$H^1(Y_0, [Lie(X_0/S_0)/Lie(Y_0/S_0)] \otimes_{O_{S_0}} I^{n+1}/I^{n+2}) = 0$$

for every $n \geq 0$, two sub-S-groups of X, flat (or smooth) over S, reducing to Y_0, are conjugate by a section of X over S inducing the unit section of X_0.

(ii) If Y_0 is affine and of finite presentation and if[261]

$$H^2(Y_0, n_{Y_0/X_0}^\vee \otimes_{O_{S_0}} I^{n+1}/I^{n+2}) = 0$$

for every $n \geq 0$, there exists a sub-S-group of X, flat over S, reducing to Y_0.

4.38. It remains to relate the three constructions which we have made in this Exposé. To avoid inessential complications, we shall place ourselves in the following situation: S_0 is the spectrum of a field k, S is the spectrum of the dual numbers $D(k)$, G is an S-group smooth over S, K a sub-S-group, smooth over S and affine.

[262] Denote $g_0 = LieG_0$ (which here equals $\Gamma(S_0, LieG_0) = Lie(G_0/S_0)(S_0)$) and $k_0 = LieK_0$. One has an exact sequence of k-vector spaces[263]:

$$0 \to k_0 \xrightarrow{i} g_0 \xrightarrow{d} n_{K_0/G_0}^\vee \to 0,$$

giving rise to an exact cohomology sequence:

$$\begin{aligned} 0 \to H^0(K_0, k_0) &\xrightarrow{i^0} H^0(K_0, g_0) \xrightarrow{d^0} H^0(K_0, g_0/k_0) \\ &\xrightarrow{\partial^0} H^1(K_0, k_0) \xrightarrow{i^1} H^1(K_0, g_0) \xrightarrow{d^1} H^1(K_0, n_{K_0/G_0}^\vee) \xrightarrow{\partial^1} H^2(K_0, k_0). \end{aligned}$$

Now these various groups all have a geometric meaning.

a) $H^0(K_0, k_0) = LieCentr(K_0)$[264] by II 5.2.3.

b) $H^0(K_0, g_0) = LieCentr_{G_0}(K_0)$[265] (idem).

c) $H^0(K_0, g_0/k_0) = LieNorm_{G_0}(K_0)/k_0$[266] (idem).

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[261] N.D.E.: We have replaced $\mathrm{Hom}_{O_{S_0}}(n_{Y_0/X_0}, I^{n+1}/I^{n+2})$ by $n_{Y_0/X_0}^\vee \otimes_{O_{S_0}} I^{n+1}/I^{n+2}$, in accordance with Remark 4.22.

[262] N.D.E.: We have slightly modified the original in what follows. In particular, we have replaced X by G and Y by K, and we have denoted g_0 and k_0 their Lie algebras. On the other hand, we have written explicitly $H^i(K_0, \cdot)$ instead of the abbreviation $H^i(\cdot)$ of the original.

[263] N.D.E.: Equipped with the adjoint action of K_0.

[264] N.D.E.: Since the formation of centralizers and normalizers commutes with base change (cf. I 2.3.3.1), we have written $Centr(K_0)$ instead of $Centr(K)_0$ in the original, and similarly $Centr_{G_0}(K_0)$ and $Norm_{G_0}(K_0)$ instead of $Centr_G(K)_0$ and $Norm_G(K)_0$.

[265] N.D.E.: Since the formation of centralizers and normalizers commutes with base change (cf. I 2.3.3.1), we have written $Centr(K_0)$ instead of $Centr(K)_0$ in the original, and similarly $Centr_{G_0}(K_0)$ and $Norm_{G_0}(K_0)$ instead of $Centr_G(K)_0$ and $Norm_G(K)_0$.

[266] N.D.E.: Since the formation of centralizers and normalizers commutes with base change (cf. I 2.3.3.1), we have written $Centr(K_0)$ instead of $Centr(K)_0$ in the original, and similarly $Centr_{G_0}(K_0)$ and $Norm_{G_0}(K_0)$ instead of $Centr_G(K)_0$ and $Norm_G(K)_0$.

d) $H^1(K_0, k_0) = Lie\,\mathrm{Aut}_{S_0-gr.}(K_0)/Im(k_0)$, where $Im(k_0)$ denotes the image of k_0 by the morphism $Lie(Int_0)$ deduced from $Int_0 : K_0 \to \mathrm{Aut}_{S_0-gr.}(K_0)$. Indeed, it follows from 2.1 (ii), applied to $Y = X = K$ and $f_0 = id_{K_0}$, that $Z^1(K_0, k_0)$ is the group of infinitesimal automorphisms of the S_0-group K_0, and that $H^1(K_0, k_0)$ is obtained by quotienting by inner infinitesimal automorphisms, i.e. by the image of k_0. Moreover, by II 4.2.2, one also has $Z^1(K_0, k_0) = Lie\,\mathrm{Aut}_{S_0-gr.}(K_0)$[267].

e) $H^1(K_0, g_0)$ is, by 2.1 (ii), the group of deviations between homomorphisms $K \to G$ extending the canonical immersion $i_0 : K_0 \to G_0$, modulo the deviations obtained by the action of the inner automorphisms of G defined by elements of $G(S)$ giving the unit of $G(S_0)$ (that is, elements of g_0).

f) $H^1(K_0, n^\vee_{K_0/G_0})$ is, by 4.36, the group of deviations between subgroups K' of G extending K_0 and flat over S (hence smooth over S, cf. SGA 1, II 4.10), modulo the deviations obtained by the action of the inner automorphisms of G constructed as previously.

g) $H^2(K_0, k_0)$ is, by 3.5 (ii), the group of deviations between group structures on K extending that of K_0, modulo the S-automorphisms of K inducing the identity on K_0.

We now propose to show how one can make explicit the six morphisms of the preceding exact sequence in the geometric interpretation we have just given.

1. i^0 and d^0 are nothing other than the morphisms obtained by passage to the Lie algebra (then by passage to the quotient for d^0), starting from the canonical monomorphisms:

$\mathrm{Centr}(K_0) \hookrightarrow \mathrm{Centr}_{G_0}(K_0) \hookrightarrow \mathrm{Norm}_{G_0}(K_0)$.

This indeed results immediately from the definition of the identifications (a), (b), and (c).

1. One constructs ∂^0 as follows. Let $\bar{x} \in LieNorm_{G_0}(K_0)/k_0$. Lift it to $x \in \mathrm{Lie}\,\mathrm{Norm}_{G_0}(K_0) \subset \mathrm{Norm}_G(K)(S)$. Then $Int(x)$ defines an automorphism of K inducing the identity on K_0, hence an element of $Lie\,\mathrm{Aut}_{S_0-gr.}(K_0)$. Denote $Int(\bar{x})$ the image of this element in $Lie\,\mathrm{Aut}_{S_0-gr.}(K_0)/Im(k_0)$. Then one has:

$$(*)\partial^0(\bar{x}) = -Int(\bar{x}) = Int(\bar{x}^{-1}).$$

Indeed, let us compute the element of $Lie\,\mathrm{Aut}_{S_0-gr.}(K_0)$ defined by $Int(x)$. It will correspond by definition to an element a of $Z^1(K_0, k_0)$ such that

$x\,y\,x^{-1} = a(y_0)\,y, \quad$ for every $y \in K(S'),\ S' \to S.$

But this can also be written $a(y_0) = xyx^{-1}y^{-1} = x - Ad(y)x = -\partial(x)(y_0)$, whence $a = -\partial(x)$.

[267]N.D.E.: And this is the Lie algebra $\mathrm{Der}_k(k_0)$ of derivations of k_0; hence $H^1(K_0, k_0)$ is the quotient of $\mathrm{Der}_k(k_0)$ by inner derivations (i.e. by the image of $ad : k_0 \to \mathrm{Der}_k(k_0)$).

[268] On the other hand, the image of $x \in LieNorm_{G_0}(K_0) \subset g_0$ by ∂ is an element of $Z^1(K_0, k_0)$, whose image $\bar{\partial}(x)$ in $H^1(K_0, k_0)$ depends only on $\bar{x}$, and by definition of the connecting map ∂^0, one has

$$\partial^0(\bar{x}) = \bar{\partial}(x);$$

combined with the equality $a = -\partial(x)$, this proves $(*)$.

3)[269] Denote $i : K \to G$ the canonical immersion. Let ū be an element of $H^1(K_0, k_0)$, image of a

```
u ∈ Lie Aut_{S₀-gr.}(K₀) ⊂ Aut_{S-gr.}(K).
```

Then one has:

```
(∗∗)    i¹(ū) = d̄(i, i ∘ u),
```

where $\bar{d}(i, i \circ u)$ is the class defined in 2.1.1.

Indeed, u is the image of an element $v \in Z^1(K_0, k_0)$ such that $u(y) = v(y_0)y$, and $i^1(\bar{u})$ is the image in $H^1(K_0, g_0)$ of the cocycle $i \circ v \in Z^1(K_0, g_0)$.

Now, since i is a morphism of groups, the equality $u(y) = v(y_0)y$ entails $iu(y) = iv(y_0)i(y)$. It follows that $i \circ v = d(i, i \circ u)$, whence $(**)$.

1. Let $i' : K \to G$ be a morphism of groups lifting i_0, let $d(i, i')$ be the class defined in 2.1.1, and let $d(K, i'(K)) \in C^1(K_0, n_{K_0/X_0})$ be the deviation defined in 4.5.1; by 4.21, $d(K, i'(K))$ belongs to $Z^1(K_0, n_{K_0/X_0})$. Denote $\bar{d}(K, i'(K))$ its image in $H^1(K_0, n_{K_0/X_0})$. Then, by 4.27 (i), one has:

```
(†)    d¹(d̄(i, i′)) = d̄(K, i′(K)).
```

1. Finally, let K' be a subgroup of G lifting K_0 and flat, hence smooth, over S. We have supposed that K_0 is affine. Then one knows that K and K' are isomorphic as schemes extending K_0 (cf. 0.15), hence there exists an isomorphism of S-schemes

$$f : K \xrightarrow{\sim} K'$$

inducing the identity on K_0. Let us transport by f the group structure of K' and let K_1 be the group obtained (which thus has K as underlying scheme), that is, the group law μ_1 of K_1 is defined by: for every $S' \to S$ and $x, y \in K(S')$,

$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$

[268]N.D.E.: We have added the following sentence.
[269]N.D.E.: We have detailed the original in what follows, and in $(**)$ we have corrected $u \circ i$ to $i \circ u$.

[270] By 3.5.1, K_1 defines a cocycle $\delta(K, K_1) \in Z^2(K_0, k_0)$ such that, for every $S' \to S$ and $x, y \in K(S')$, one has

$$\delta(K, K_1)(x_0, y_0)\, xy = \mu_1(x, y) = f^{-1}(f(x)f(y)).$$

Then, by 4.35.1, one has:

$$(\ddagger) \qquad \partial^1(\bar{d}(K, K')) = \delta(K, K_1).$$

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Exposé IV. Topologies and sheaves

by M. Demazure[272]

[273]

This Exposé is intended to acquaint the reader with the essentials of the language of topologies and sheaves (without cohomology), particularly convenient in questions of passage to the quotient (among others).

The first three sections develop the language of passage to the quotient. The fourth, which is the central part, is the exposition of the theory of sheaves, oriented principally towards the application to questions of quotients; the fifth is an application to passage to the quotient in groups and to principal homogeneous fibered objects. The last section concerns more specifically the category of schemes, and defines various useful topologies on this category.

The reader will profitably refer to [AS], [MA], [D], and SGA 4; [D] in particular as regards the applications of topologies to the theory of descent, and SGA 4 for questions of universes (particularly mistreated in this Exposé).

[270] N.D.E.: We have modified the original in what follows, taking into account the additions made in 3.5.1 and 4.35.1.

[271] N.D.E.: Additional references cited in this Exposé.

[272] This text develops the substance of two oral expositions of A. Grothendieck, completing the latter on several important points which had been passed over in silence or scarcely touched on.

[273] N.D.E.: Version of 13/10/2024.

1. Universal effective epimorphisms

In what follows in this Exposé, we suppose fixed a category C.

Definition 1.1. *A morphism $u : T \to S$ is called an* epimorphism *if, for every object X, the corresponding map*

$$X(S) = \mathrm{Hom}(S, X) \to X(T) = \mathrm{Hom}(T, X)$$

is injective.[274] *One says that u is a* universal epimorphism *if for every morphism $S' \to S$, the fiber product $T' = T \times_S S'$ exists, and $u' : T' \to S'$ is an epimorphism.*

Definition 1.2. *A diagram*

$$A \xrightarrow{u} B \underset{v_2}{\overset{v_1}{\rightrightarrows}} C$$

of maps of sets is said to be exact *if u is injective and if its image is formed by the elements b of B such that $v_1(b) = v_2(b)$. A diagram of the same type in C is said to be* exact *if for every object X of C, the corresponding diagram of sets*

$$A(X) \to B(X) \rightrightarrows C(X)$$

is exact; one then also says that u makes A into a kernel *of the pair of arrows (v_1, v_2). Dually, a diagram*

$$C \underset{v_2}{\overset{v_1}{\rightrightarrows}} B \xrightarrow{u} A$$

in C is said to be exact if it is exact as a diagram in the opposite category C°, i.e. if for every object X of C, the corresponding diagram of sets

$$X(A) \to X(B) \rightrightarrows X(C)$$

is exact.[275] *One also says that u makes A into a* cokernel *of the pair of arrows (v_1, v_2).*

Definition 1.3. *A morphism $u : T \to S$ is called an* effective epimorphism *if the fiber square $T \times_S T$ exists, and if the diagram*

$$T \times_S T \underset{pr_2}{\overset{pr_1}{\rightrightarrows}} T \xrightarrow{u} S$$

is exact, i.e. if u makes S into a cokernel of (pr_1, pr_2). One says that u is a universal effective epimorphism *if for every morphism $S' \to S$, the fiber product $T' = T \times_S S'$ exists, and the morphism $u' : T' \to S'$ is an effective epimorphism.*

One evidently has the implications:

$$\begin{array}{ccc} \text{universal effective epimorphism} & \Rightarrow & \text{effective epimorphism} \\ \Downarrow & & \Downarrow \\ \text{universal epimorphism} & \Rightarrow & \text{epimorphism,} \end{array}$$

[274] N.D.E.: that is, if u is right-cancellable.

[275] N.D.E.: This implies, in particular, that u be an epimorphism.

but in general no other implication holds.[276]

Definition 1.4.0.[277] *We “recall” that a morphism $u : T \to S$ is said to be* squarable *if for every morphism $S' \to S$, the fiber product $T \times_S S'$ exists.*

Lemma 1.4. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then*

a) u, v epimorphisms $\Longrightarrow$ uv epimorphism $\Longrightarrow$ u epimorphism,

b) u, v universal epimorphisms $\Longrightarrow$ uv universal epimorphism, and u squarable $\Longrightarrow$ u universal epimorphism.

Lemma 1.4 is trivial from the definitions. From it we conclude:

Corollary 1.5. *Let $u : X \to Y$ and $u' : X' \to Y'$ be universal epimorphisms, such that $Y \times Y'$ exists; then $X \times X'$ exists and $u \times u' : X \times X' \to Y \times Y'$ is a universal epimorphism.*

Let us also note:

Definition 1.6.0.[278] *One says that an object S of C is* squarable *if its product with every object of C exists. (If C has a final object e, this is equivalent to saying that the morphism $S \to e$ is squarable, cf. 1.4.0.)*

Lemma 1.6. *Let $u : X \to Y$ be a morphism in C/S; for it to be an epimorphism (resp. universal epimorphism, resp. effective epimorphism, resp. universal effective epimorphism), it suffices that the corresponding morphism in C be such, and this is also necessary if one supposes that S is a squarable object of C.*

Immediate proof left to the reader. One uses the hypothesis “S squarable” in order to interpret the C-morphisms from an object Y of C/S into an object Z of C as the C/S-morphisms from Y into $Z \times S$.

Lemma 1.7. *With the notation of 1.4: u, v effective epimorphisms and v universal epimorphism $\Longrightarrow$ uv effective epimorphism.*

To see this, one considers the diagram

```
S ⇇ T ⇇ T ×_S T
  ↑    ↑v     ↑v×_S v
     U ⇇ U ×_S U
          ↖
        U ×_T U
```

One notes that by hypothesis, row 1 and column 1 are exact, and that by virtue of 1.5 and 1.6, $v \times_S v$ is an epimorphism (v being a universal epimorphism). The

[276]N.D.E.: For example, if $C = (Sch)$ is the category of schemes, one sees easily that every universal epimorphism is surjective. Let $T = \coprod_{pprime} \mathrm{Spec}(F_p)$ and $S = \mathrm{Spec}(Z)$; then the morphism $u : T \to S$ is an epimorphism that is not universal. On the other hand, one sees that $T \times_S T$ is identified with T, so that the two projections $T \times_S T \rightrightarrows T$ coincide; since id_T does not descend to a morphism $S \to T$, this shows that u is not an effective epimorphism.
[277]N.D.E.: The numbering 1.4.0 has been added for later references.
[278]N.D.E.: The numbering 1.6.0 has been added for later references.

conclusion follows by an evident diagram-chase: if an element of $X(U)$ has the same images in $X(U \times_S U)$, it has *a fortiori* the same images in $X(U \times_T U)$, hence comes from an element of $X(T)$ since column 1 is exact. As row 1 is exact, it suffices to verify that the element under consideration has the same images in $X(T \times_S T)$, and since $v \times_S v$ is an epimorphism, it suffices to verify that the images in $X(U \times_S U)$ are the same, which is indeed the case.

Proposition 1.8. *Consider morphisms $U \xrightarrow{v} T \xrightarrow{u} S$. Then u, v universal effective epimorphisms $\Longrightarrow$ uv universal effective epimorphism, and u squarable $\Longrightarrow$ u universal effective epimorphism.*

The first implication follows at once from 1.7. For the second, one looks at the diagram (of "bisimplicial" type):

```
S ⇇ T ⇇ T ×_S T
↑uv   ↑    ↑
U ⇇ U ×_S T ⇇ U ×_S T ×_S T
↑     ↑              ↑
U ×_S U ⇇ U ×_S U ×_S T ⇇ U ×_S U ×_S T ×_S T.
```

Columns 1, 2, 3 are exact by virtue of the hypothesis "uv universal effective epimorphism", row 2 is exact, since $U \times_S T \to U$ is an effective epimorphism (because it has a section over U), and the same holds for row 3 (same reason). An evident diagram-chase then shows that row 1 is exact, i.e. u is an effective epimorphism. As the hypotheses made are invariant under any change of base $S' \to S$, it follows that u is in fact a universal effective epimorphism.

Corollary 1.9. *Let $u : X \to Y$ and $u' : X' \to Y'$ be universal effective epimorphisms, such that $Y \times Y'$ exists; then $X \times X'$ exists and $u \times u' : X \times X' \to Y \times Y'$ is a universal effective epimorphism.*

Proof as for 1.5 by the diagram

```
Y′ ⇇ X′
↑     ↑u′
X ⇇ X × Y′ ⇇ X × X′
↑u                ↘ u×u′
Y ⇇ Y × Y′.
```

Corollary 1.10. *Consider a squarable morphism $u : T \to S$, and a change-of-base morphism $S' \to S$ which is a universal effective epimorphism. For u to be a universal effective epimorphism, it is necessary and sufficient that $u' : T' = T \times_S S' \to S'$ be one:*

```
T ⇇v′ T′
↓u     ↓u′
S ⇇v  S′.
```

Only the "it suffices" requires a proof. Now if u' is a universal effective epimorphism, so is vu' by 1.8, and since $vu' = uv'$, one concludes by 1.8 that u is a universal effective epimorphism.

Remark 1.11. *The same reasoning shows that in 1.10 one can replace "universal effective epimorphism" by "universal epimorphism" or "universal and effective epimorphism", or simply by "epimorphism" (and in this last case, the hypothesis "u squarable" is evidently unnecessary).*

In the proof of 1.8 we used the following result, which deserves to be made explicit:

Proposition 1.12. *Let $u : T \to S$ be a morphism that admits a section. Then u is an epimorphism, and if $T \times_S T$ exists, it is an effective epimorphism, and a universal effective epimorphism if moreover u is squarable.*

The first assertion is contained in 1.4 a), and the third will follow at once from the second, which it will therefore suffice to establish. In fact we have a stronger conclusion: for every functor $F : C^{\circ} \to (Ens)$ (not necessarily representable), the diagram of sets

$$F(S) \to F(T) \rightrightarrows F(T \times_S T)$$

is exact. This may be considered as a particular case of the formalism of Čech cohomology (in dimension 0!), which we content ourselves with recalling here. Suppose simply that $T \times_S T$ exists; one then sets

$$\check{H}^0(T/S, F) = \operatorname{Ker}(F(T) \rightrightarrows F(T \times_S T)).$$

One may evidently regard $\check{H}^0(T/S, F)$ as a contravariant functor in the argument T varying in C/S, every S-morphism $T' \to T$ defining a map

$$(+) \qquad \check{H}^0(T/S, F) \to \check{H}^0(T'/S, F).$$

Fix T and T' in C/S. A well-known calculation shows that if there exists an S-morphism from T' into T, the corresponding map (+) is in fact independent of the choice of this morphism,[279] so that $\check{H}^0(T/S, F)$ may be regarded as a functor on the category associated to the set ObC/S preordered by the relation of "domination" (T' dominates T if there exists an S-morphism from T' into T). In particular, if T and T' are isomorphic in this latter category, i.e. if each dominates the other, then (+) is an isomorphism of sets. This applies in particular to the case where T' is the final object of C/S, i.e. essentially S itself; in any case T dominates $T' = S$, and the converse is true precisely if T/S has a section. This establishes 1.12 in the strengthened form announced.

Remark 1.13. *For various applications, the notions introduced in the present Exposé, and the results stated, must be developed more generally relative to a family of morphisms $u_i : T_i \to S$ with the same target (instead of a single morphism*

[279]N.D.E.: This is the following argument, communicated by M. Demazure. Let $f, g : T' \to T$, and let $\phi : T' \to T \times_S T$ be the morphism with components f and g, whence $p_1 \circ \phi = f$ and $p_2 \circ \phi = g$. Then $F(\phi) : F(T \times_S T) \to F(T')$ satisfies $F(\phi) \circ F(p_1) = F(f)$ and $F(\phi) \circ F(p_2) = F(g)$. Now, for every $x \in \check{H}^0(T/S, F)$, one has $F(p_1)(x) = F(p_2)(x)$. Hence, applying $F(\phi)$ to both sides, one obtains $F(f)(x) = F(g)(x)$, which shows that f and g induce the same morphism.

$u : T \to S$). Thus, such a family will be said to be epimorphic *if for every object X of C, the corresponding map*

$$X(S) \to \prod_i X(T_i)$$

is injective, and one introduces in the same way the notion of an effective epimorphic *family and the "universal" variants of these notions. We shall admit, if need be, in what follows, that the results of the present Exposé extend to this more general situation.*

Remark 1.14. *Every morphism that is at once a monomorphism and an effective epimorphism is an isomorphism.*

Indeed, in the notation of 1.3, the fact that $T \to S$ be a monomorphism entails that the two morphisms

$$pr_1, pr_2 : T \times_S T \rightrightarrows T$$

are equal (and are isomorphisms). Now a diagram

$$C \rightrightarrows_{v,v} B \xrightarrow{u} A$$

is exact only if u is an isomorphism, as follows immediately from the definition.[280]

2. Descent morphisms

Let us recall the following definitions:

Definition 2.1. *Let $f : S' \to S$ be a morphism such that $S'' = S' \times_S S'$ exists, and let $u' : X' \to S'$ be an object over S'. One calls* gluing datum *on X'/S', relative to f, an S''-isomorphism*

$$c : X''_1 \xrightarrow{\sim} X''_2$$

where X''_i ($i = 1, 2$) denotes the inverse image (supposed to exist) of X'/S' under the projection $pr_i : S'' \to S'$. One says that the gluing datum c is a descent datum *if it satisfies the "cocycle condition"*

$$pr*_{3,1}(c) = pr*_{3,2}(c) pr*_{2,1}(c)$$

where $pr_{i,j}$ ($1 \le j < i \le 3$) are the canonical projections from $S''' = S' \times_S S' \times_S S'$ into S'' (N.B. one now supposes that S''' also exists), where $pr*_{i,j}(c)$ is the inverse image of c, considered as an S'''-morphism from X'''_j into X'''_i, and where for every

[280] N.D.E.: One will note that a monomorphism which is an epimorphism is not necessarily an isomorphism. For example, in $C = (Sch)$, the morphism $\mathrm{Spec}(\mathbb{F}_p) \sqcup \mathrm{Spec}(\mathbb{Z}[1/p]) \to \mathrm{Spec}(\mathbb{Z})$ is a monomorphism and a surjective epimorphism, but is not an isomorphism.

integer k between 1 and 3, X'''_k denotes the inverse image (supposed to exist) of X'/S' under the projection of index k, $q_k : S''' \to S'$.

In the second part of the definition, we have therefore used identifications and abuses of writing in current use,[281] which experience proves to be harmless, but which it is evidently fitting to avoid in a rigorous exposition of the theory of descent (which must precisely justify to a certain extent these common abuses of language). Such a formal exposition ([D]) has been written by Giraud (with the aim of justifying and making precise SGA 1, VII, which was never written). For a detailed exposition of the results of faithfully flat descent of which constant use will be made in the present Séminaire, one may consult SGA 1, VIII.

Let $f : S' \to S$ still be a morphism such that $S'' = S' \times_S S'$ exists, and let X be an object over S such that $X' = X \times_S S'$ and $X'' = X \times_S S''$ exist; then the inverse images of X' under pr_i ($i = 1, 2$) exist and are canonically isomorphic, and consequently X'/S' is endowed with a canonical gluing datum relative to f. When S''' and $X''' = X \times_S S'''$ exist, this is even a descent datum. If Y is another object over S, satisfying the same conditions as X/S, then for every S-morphism $X \to Y$, the corresponding S'-morphism $X' \to Y'$ is "compatible with the canonical gluing data" on X', Y'. If in particular $S' \to S$ is a squarable morphism, then

$$X \mapsto X' = X \times_S S'$$

is a functor from the category C/S into the "category of objects over S' equipped with a descent datum relative to f" — a category whose definition is left to the reader, and which is a full subcategory of the "category of objects over S' equipped with a gluing datum relative to f".

This being so:

Definition 2.2. *One says that a morphism $f : S' \to S$ is a* descent morphism *(resp. an* effective descent morphism*) if f is squarable (i.e. for every $X \to S$, the fiber product $X \times_S S'$ exists), and if the preceding functor $X \mapsto X' = X \times_S S'$ from the category C/S of objects over S, into the category of objects over S' equipped with a descent datum relative to f, is fully faithful (resp. an equivalence of categories).*

One notes that the first of these two notions can be expressed using only the notion of gluing datum (hence without involving the triple fiber product S'''), f being a descent morphism if f is squarable and $X \mapsto X'$ is a fully faithful functor from the category C/S into the category of objects over S' equipped with a gluing datum relative to f. When one makes this definition explicit, one finds that it means that for two objects X, Y over S, the following diagram of sets

$$(\times) \qquad \mathrm{Hom}_S(X, Y) \to \mathrm{Hom}_{S'}(X', Y') \rightrightarrows_{p_1, p_2} \mathrm{Hom}_{S''}(X'', Y'')$$

is exact, where $p_i(u')$ denotes the inverse image of $u' \in \mathrm{Hom}_{S'}(X', Y')$ under the projection $pr_i : S'' \to S'$, for $i = 1, 2$; indeed, the kernel of the pair (p_1, p_2) is by

[281] N.D.E.: for example, one has identified, on the one hand, $pr*_{2,1}(X''_1) = pr*_{3,1}(X''_1)$ and, on the other hand, $pr*_{2,1}(X''_2) = pr*_{3,2}(X''_2)$.

definition none other than the set of S'-morphisms $X' \to Y'$ compatible with the canonical gluing data.

Note that, by definition of the inverse images Y', Y'', one has canonical bijections

$$\mathrm{Hom}_{S'}(X', Y') \simeq \mathrm{Hom}_S(X', Y) \quad \text{and} \quad \mathrm{Hom}_{S''}(X'', Y'') \simeq \mathrm{Hom}_S(X'', Y),$$

so that the exactness of the diagram (×) is equivalent to that of

$$(\times\times) \qquad \mathrm{Hom}_S(X, Y) \to \mathrm{Hom}_S(X', Y) \rightrightarrows \mathrm{Hom}_S(X'', Y),$$

obtained by applying $\mathrm{Hom}_S(-, Y)$ to the diagram in C/S:

$$(\times\times\times) \qquad X'' \rightrightarrows X' \to X$$

which is deduced from

$$S'' \rightrightarrows S' \to S$$

by the base change $X \to S$. This proves, taking account of 1.6, the first part of the following proposition:

Proposition 2.3. *Let $f : S' \to S$ be a morphism. If it is a universal effective epimorphism, it is a descent morphism, and the converse is true if S is a squarable object of C (i.e. its product with every object of C exists).*

It remains to prove that if f is a descent morphism, it is a universal effective epimorphism, that is, that for every morphism $X \to S$, the diagram (×××) is exact, i.e. for every object Z of C, the transform of this diagram by $\mathrm{Hom}(-, Z)$ is an exact diagram of sets. Now by hypothesis $Z \times S$ exists; let Y be the object of C/S that it defines; then the transform of (×××) by $\mathrm{Hom}(-, Z)$ is isomorphic to the transform by $\mathrm{Hom}_S(-, Y)$, and this last is exact by the hypothesis on f.

One may therefore apply to universal effective epimorphisms the results on descent morphisms, such as the following:

Proposition 2.4. *Let $f : S' \to S$ be a descent morphism (for example a universal effective epimorphism). Then:*

a) For every S-morphism $u : X \to Y$, u is an isomorphism (resp. a monomorphism) if and only if $u' : X' \to Y'$ is.

b) Let X, Y be two subobjects of S, X' and Y' the subobjects of S' inverse images of the preceding ones. For X to be contained in Y (resp. to be equal to Y), it is necessary and sufficient that the same hold for X', Y'.

For (a), it follows from the definition that if u' is an isomorphism in the category of objects with gluing datum, then u is an isomorphism; now one notes at once that every isomorphism of objects over S' compatible with gluing data is an isomorphism of objects with gluing datum, i.e. its inverse is also compatible with the gluing data. For b), one is reduced to proving that if X' is contained in Y', i.e. if there exists an S'-morphism $X' \to Y'$, then the same holds for X, Y over S. Now since $Y' \to S'$, and hence also $Y'' \to S''$, is a monomorphism, one sees that $X' \to Y'$ is automatically compatible with the gluing data, hence comes from an S-morphism

$X \to Y$. Note that the proof holds more generally when one has two objects X, Y over S, with $Y \to S$ a monomorphism, and one asks whether the morphism $X \to S$ factors through Y: it suffices that $X' \to S'$ factor through Y'.

Corollary 2.5. *Let $f : S' \to S$ be a universal effective epimorphism and $g : S \to T$ a morphism such that $S \times_T S$ exists. Suppose that $S'' = S' \times_S S'$ is also a fiber product of S' with itself over T, i.e. $S' \times_S S' \xrightarrow{\sim} S' \times_T S'$. Then $g : S \to T$ is a monomorphism (and conversely, of course).*

Indeed, consider the cartesian diagram

$$\begin{array}{ccc} S' \times_S S' & \square & S' \times_T S' \\ \downarrow & & \downarrow f \\ S & \to & S \times_T S, \end{array}$$

where the second horizontal arrow is the diagonal morphism. By virtue of 1.9 the second vertical arrow f is a universal effective epimorphism, by hypothesis the first horizontal arrow is an isomorphism, hence by virtue of 2.4 a) or b) at one's choice,[282] the same holds for $S \to S \times_T S$, which means precisely that $g : S \to T$ is a monomorphism.

Remark 2.6. *The notions introduced in 2.1, in terms of morphisms between certain inverse limits, become explicit in an obvious way in terms of the contravariant functors defined by the objects S, S', X' under consideration: subject to the existence of the fiber products envisaged in 2.1, there is a one-to-one correspondence between gluing data (resp. descent data) on X'/S' relative to $S' \to S$, and gluing data (resp. descent data) for the corresponding objects in $\hat{C} = \mathrm{Hom}(C^\circ, (Ens))$. This allows one, if one wishes, to extend these notions to the case where one makes no hypothesis of existence of fiber products in C.*

Remark 2.7. *The notions introduced in this number generalize to the case of families of morphisms. They can on the other hand be presented in a more intrinsic manner using the notion of sieve (4.1). For these questions, the reader is referred to [D].*

3. Universal effective equivalence relations

3.1. Equivalence relations: definitions

Definition 3.1.1. *One calls a C-*equivalence relation *in $X \in ObC$ a representable subfunctor R of the functor $X \times X$, such that for every $S \in ObC$, $R(S)$ is the graph of an equivalence relation in $X(S)$.*

This definition applies in particular to the category Ĉ. If one considers X as an object of Ĉ, one then sees that a Ĉ-equivalence relation in X is nothing other than a subfunctor R of $X \times X$ such that $R(S)$ is the graph of an equivalence relation in $X(S)$ for every object S of C.[283]

[282] N.D.E.: applied to $f : S' \times_T S' \to S \times_T S = Y$.

[283] N.D.E.: The condition is evidently necessary. Conversely, if for every $S \in ObC$, $R(S)$ is the graph of

If R is a C-equivalence relation in X, one denotes by p_i the morphism from R into X induced by the projection $pr_i : X \times X \to X$. One thus has a diagram

$$p_1, p_2 : R \rightrightarrows X.$$

Definition 3.1.2. *A morphism $u : X \to Z$ is said to be* compatible *with R if $up_1 = up_2$. An object of C that is a cokernel of the pair (p_1, p_2) is also called a* quotient object *of X by R and denoted X/R.*

One thus has an exact diagram

$$R \underset{p_2}{\overset{p_1}{\rightrightarrows}} X \xrightarrow{p} X/R$$

and X/R represents the covariant functor

$$\mathrm{Hom}_C(X/R, Z) = \{\text{morphisms from } X \text{ into } Z \text{ compatible with } R\}.$$

Since quotient objects have been chosen in C, the quotient X/R is unique (when it exists).

These definitions generalize at once to the case of a $\hat{C}$-equivalence relation in X, but one will notice that the identification of objects of C with their images in $\hat{C}$ does not commute with the formation of quotients, that is, the quotient X/R of X by R in C is not *a priori* a quotient of X by R in $\hat{C}$. One should therefore guard against rashly identifying C with its image in $\hat{C}$ in questions involving passages to the quotient.[284]

In what follows, we shall say simply *equivalence relation* for *$\hat{C}$-equivalence relation*; we shall specify, when appropriate, whether we are dealing with C-equivalence relations.[285]

Definition 3.1.3. *If X is an object of C over S, one calls an* equivalence relation in X over S *an equivalence relation R in X such that the structural morphism $X \to S$ is compatible with R.*

The canonical monomorphism $R \to X \times X$ then factors through the monomorphism

$$X \times_S X \hookrightarrow X \times X$$

an equivalence relation, then this equivalence relation extends to $R(F)$ for every $F \in Ob\hat{C}$, by declaring that two morphisms $\phi, \psi : F \to R$ are equivalent if, for every $S \in ObC$ and $x \in F(S)$, $\phi(x)$ and $\psi(x)$ are equivalent in $X(S)$.

[284] N.D.E.: Let us illustrate this by giving an outline of the sequel of this Exposé. Let G be a C-group, H a sub-C-group, R the equivalence relation in G defined by $G \times H \to G \times G$, $(g, h) \mapsto (g, gh)$ (cf. 3.2). The functor Q defined by $Q(S) = G(S)/H(S)$ is a quotient in $\hat{C}$ (according to 4.4.9 applied to the least fine topology, cf. 4.4.2), but it is not in general the quotient that one wishes. For example, for $C = (Sch)$, one has an exact sequence of (affine) group schemes:

$$1 \to \mu_2 \to G_m \xrightarrow{p} G_m \to 1$$

which identifies G_m with the quotient G_m/μ_2. Moreover, since p is a finite and locally free morphism, then G_m is the sheaf-quotient of G_m by μ_2 in the larger category of sheaves for the (fppf) topology, cf. 4.6.6 (ii) and 6.3.2. By contrast, the quotient Q in $\hat{C}$ is not isomorphic to G_m since, for example, $Q(Z) = 1$ while $G_m(Z) = \pm 1$. Hence Q is not an (fppf) sheaf, and *a fortiori* Q is not representable.

[285] N.D.E.: Note that, even for a $\hat{C}$-equivalence relation in X, one is interested in the existence of a quotient in C.

and defines an equivalence relation in the object $X \to S$ of C/S. When the quotient X/R exists, it is endowed with a canonical morphism into S and the corresponding object of C/S is a quotient of $X \in ObC/S$ by the preceding equivalence relation.

Conversely, if S is a squarable object of C and if $Y \to S$ is a quotient of X by this equivalence relation (in C/S), then Y is a quotient of X by R in C. In any case, we shall never have to consider quotients in C/S that are not already quotients in C.

Definition 3.1.4. *If X (resp. X') is an object of C equipped with an equivalence relation R (resp. R'), a morphism*

$$u : X \to X'$$

is said to be compatible with R and R' *if the following equivalent conditions are satisfied:*

(i) for every $S \in ObC$, two points of $X(S)$ congruent modulo $R(S)$ are transformed by u into two points of $X'(S)$ congruent modulo $R'(S)$;

(ii) there exists a morphism $R \to R'$ (necessarily unique) making commutative the diagram

$$\begin{tikzcd} R \arrow[r] \arrow[d] & R' \arrow[d] \\ X \times X \arrow[r, "(u \times u)"'] & X' \times X'. \end{tikzcd}$$

By the universal property of X/R, there then exists (when the quotients X/R and X'/R' exist) a unique morphism v making commutative the diagram

$$X \xrightarrow{p} X/R \downarrow u \downarrow v X' \xrightarrow{p'} X'/R'.$$

Definition 3.1.5. *A subobject (= a representable subfunctor) Y of X is said to be* stable under the equivalence relation R *if the following equivalent conditions are satisfied:*

(i) For every $S \in ObC$, the subset $Y(S)$ of $X(S)$ is stable under $R(S)$.

(ii) The two subobjects of R inverse images of Y under p_1 and p_2 are identical.

An important particular case of stable subobject of X is the following: the quotient X/R exists and Y is the inverse image on X of a subobject of X/R.

Definition 3.1.6. *Let R be an equivalence relation in X and $X' \to X$ a morphism. The equivalence relation R' in X' obtained by the cartesian diagram*

$$\begin{tikzcd} R' \arrow[r] \arrow[d] & R \arrow[d] \\ X' \times X' \arrow[r] & X \times X \end{tikzcd}$$

is said to be the equivalence relation in X' inverse image *of the equivalence relation R in X. In particular, if X' is a subobject of X, one says it is the equivalence relation* induced in X' by R, *and one denotes it $R_{X'}$.*

The morphism $X' \to X$ is compatible with R' and R; one thus has, when the quotients exist, a morphism $X'/R' \to X/R$ (3.1.4). If X' is a subobject of X, we shall see later that in certain cases one can prove that $X'/R' \to X/R$ is a monomorphism, hence identifies X'/R' with a subobject of X/R. When this is so, the inverse image of this subobject in X will be a subobject of X containing X' and stable under R: the *saturation* of X' for the equivalence relation R.

Proposition 3.1.7. *If the subobject Y of X is stable under R, one has two cartesian squares, for $i = 1, 2$:*

$$R_Y \to R \downarrow p_i \downarrow p_i Y \to X.$$

Immediate proof.

3.2. Equivalence relation defined by a group acting freely

Definition 3.2.1. *Let X be an object of C and H a C-group acting on X (that is, equipped with a morphism of $\hat{C}$-groups*

`H → Aut(X)).`

One says that H acts freely *on X if the following equivalent conditions are satisfied:*

(i) For every $S \in ObC$, the group $H(S)$ acts freely on $X(S)$;

(ii) The morphism of functors

`H × X → X × X`

defined set-theoretically by `(h, x) ↦ (hx, x)` *is a monomorphism.*

(In the preceding definition, one supposed that the group H acted "on the left" on X. One evidently has an analogous notion in the case where the group acts "on the right", that is, when one is given a morphism of groups from the group H° opposite to H into Aut(X)).

If H acts freely on X, the image of $H \times X$ under the morphism of (ii) is an equivalence relation in X called the *equivalence relation defined by the action of H on X*. When the quotient of X by this equivalence relation exists, it is denoted $H \backslash X$

(X/H when H acts on the right). It represents the following covariant functor: if Z is an object of C, one has

`Hom(H\backslash X, Z) = {morphisms from X into Z invariant under H}`

where the morphism $f : X \to Z$ is said to be *invariant under H* if for every $S \in ObC$, the corresponding morphism $X(S) \to Z(S)$ is invariant under the group $H(S)$.

Lemma 3.2.2. *Under the conditions of 3.2.1, let Y be a subobject of X. The following conditions are equivalent:*

(i) Y is stable under the equivalence relation defined by H (3.1.5);

(ii) For every $S \in ObC$, the subset $Y(S)$ of $X(S)$ is stable under $H(S)$;

(iii) There exists a morphism f, necessarily unique, making commutative the diagram

$$\begin{tikzcd} H \times Y \arrow[r, "f"] \arrow[d] & Y \arrow[d] \\ H \times X \arrow[r] & X. \end{tikzcd}$$

Under these conditions, f defines a morphism of $\hat{C}$-groups

$$H \longrightarrow \mathrm{Aut}(Y)$$

and the equivalence relation defined in Y by this action of H is none other than the equivalence relation induced in Y by the equivalence relation defined in X by the action of H.

Immediate proof. One evidently has an analogous statement for a "right action". The action of H on Y defined above will be called the *action induced in Y by the given action of H on X.*

Let us now consider the following situation: H and G are two C-groups and one is given a morphism of groups

$$u : H \longrightarrow G.$$

Then H acts on G by translations (one sets set-theoretically $hg = u(h)g$) and acts freely there if and only if u is a monomorphism. The quotient of G by this action of H is denoted, when it exists, $H\backslash G$. One defines similarly a right action of H on G and a quotient G/H. These quotients are functorial with respect to the groups in question; more precisely, one has the following lemma, stated for right quotients:

Lemma 3.2.3. *Let $u : H \to G$ and $u' : H' \to G'$ be two monomorphisms of C-groups. Suppose given a morphism of C-groups*

$$f : G \longrightarrow G'.$$

The following conditions are equivalent:

(i) f is compatible with the equivalence relations defined in G and G' by H and H'.

(ii) For every $S \in ObC$, one has $fu(H(S)) \subset u'(H'(S))$.

(iii) There exists a morphism $g : H \to H'$, necessarily unique and multiplicative, such that the following diagram is commutative

$$H \xrightarrow{g} H' \downarrow u \downarrow u' G \xrightarrow{f} G'.$$

Under these conditions, if the quotients G/H and G'/H' exist, there exists a unique morphism $\bar{f}$ making commutative the diagram

$$G \xrightarrow{f} G' \downarrow p \downarrow p' G/H \xrightarrow{\bar{f}} G'/H'.$$

The first part is proved by reduction to the set-theoretic case. The second follows at once from (i).

One could translate to the present situation the notions introduced above for general equivalence relations. Let us simply signal the following lemma, whose proof is immediate by reduction to the set-theoretic case:

Lemma 3.2.4. *Let $u : H \to G$ be a monomorphism of C-groups and G' a sub-C-group of G. For the subobject G' of G to be stable under the equivalence relation defined by H, it is necessary and sufficient that u factor through the canonical monomorphism $G' \to G$, and under this condition the action of H on G' induced by the action of H on G defined by u is none other than the action deduced from the monomorphism $H \to G'$ that factors u.*

3.3. Universal effective equivalence relations

Definition 3.3.1. *Let $f : X \to Y$ be a morphism. One calls the* equivalence relation defined by f in X, *and denotes by $R(f)$, the $\hat{C}$-equivalence relation in X image of the canonical monomorphism*

$X \times_Y X \to X \times X.$

Definition 3.3.2. *Let R be an equivalence relation in X. One says that R is* effective *if*

(i) R is representable (i.e. is a C-equivalence relation);

(ii) the quotient $Y = X/R$ exists in C;[286]

(iii) the diagram

$R \rightrightarrows_{p_1, p_2} X \xrightarrow{p} Y$

makes R the fiber square of X over Y, that is, R is the equivalence relation defined by p.

Scholie 3.3.2.1.[287] *If R is an effective equivalence relation in X, then p is an effective epimorphism (1.3). If $f : X \to Y$ is an effective epimorphism, then $R(f)$ is an effective equivalence relation in X a quotient of which is Y. There is therefore*

[286]N.D.E.: "in C" has been added.

[287]N.D.E.: The numbering 3.3.2.1 (resp. 3.3.3.1) has been added for later references. On the other hand, Remark 3.3.3.2 has been added.

a "Galois" one-to-one correspondence between effective equivalence relations in X and effective quotients of X (i.e. equivalence classes of effective epimorphisms with source X).

Definition 3.3.3. *One says that the equivalence relation R in X is* universally effective *if the quotient $Y = X/R$ exists, and if, for every $Y' \to Y$, the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist and R' is a fiber square of X' over Y'. It amounts to the same to say that R is effective and that $p : X \to X/R$ is a universal effective epimorphism.*

Scholie 3.3.3.1.[288] *There is therefore, as above, a one-to-one correspondence between universal effective equivalence relations in X and universal effective quotients of X.*

Remark 3.3.3.2.[289] *Suppose that C is the category of S-schemes and let $\mathbb{A}^1$ denote the affine space of dimension 1 over S. Let $R \subset X \times_S X$ be a universal effective equivalence relation and $p : X \to Y$ the quotient. Then, for every open U of Y, $O(U) = \mathrm{Hom}_S(U, \mathbb{A}^1_S)$ is the set of elements ϕ of $O(p^{-1}(U)) = \mathrm{Hom}_S(p^{-1}(U), \mathbb{A}^1_S)$ such that $\phi \circ pr_1 = \phi \circ pr_2$. In particular, if R is given by the action of a group H acting freely on the right on X (cf. 3.2.1), then $O(U)$ is the set of $\phi \in O(p^{-1}(U))$ such that $\phi(xh) = \phi(x)$, for every $S' \to S$ and $x \in X(S')$, $h \in H(S')$.*

Proposition 3.3.4. *Let R be a universal effective equivalence relation in X. Let $f : X \to Z$ be a morphism compatible with R, hence factoring through $g : X/R \to Z$. The following conditions are equivalent:*

(i) g is a monomorphism;

(ii) R is the equivalence relation defined by f.

Indeed, (i) trivially entails (ii); the converse is none other than 2.5.

Definition 3.3.5. *Let H be a C-group acting freely on X. One says that H acts* effectively, *or that the action of H on X is* effective *(resp.* universally effective*), if the equivalence relation defined in X by the action of H is effective (resp. universally effective).*

3.4. (M)-effectivity

In practice, it is most often difficult to characterize universal effective epimorphisms. One often disposes, nevertheless, of a certain number of morphisms of this type, for example in scheme theory, of faithfully flat quasi-compact morphisms. This leads to the developments below.

3.4.1. Let us first state a certain number of conditions bearing on a family `(M)` of morphisms of C:

[288] N.D.E.: The numbering 3.3.2.1 (resp. 3.3.3.1) has been added for later references. On the other hand, Remark 3.3.3.2 has been added.

[289] N.D.E.: The numbering 3.3.2.1 (resp. 3.3.3.1) has been added for later references. On the other hand, Remark 3.3.3.2 has been added.

(a) (M) is stable under extension of the base, i.e., every $u : T \to S$ element of (M) is squarable (cf. 1.4.0) and for every $S' \to S$, $u' : T \times_S S' \to S'$ is an element of (M).

(b) The composite of two elements of (M) is in (M).

(c) Every isomorphism is an element of (M).

(d) Every element of (M) is an effective epimorphism.

Note that (a) and (b) entail:

(a′) The cartesian product of two elements of (M) is in (M): let $u : X \to Y$ and $u' : X' \to Y'$ be two S-morphisms, elements of (M). If $Y \times_S Y'$ exists, then $X \times_S X'$ exists and $u \times_S u'$ is an element of (M).

This follows from the diagram

```
Y′      ⇇      X′
↑              ↑u′
X  ⇇  X ×_S Y′  ⇇  X ×_S X′
↑u                     ↘ u×_S u′
Y  ⇇  Y ×_S Y′.
```

Likewise (a) and (d) entail:

(d′) Every element of (M) is a universal effective epimorphism.

3.4.2. The family (M_0) of universal effective epimorphisms satisfies the conditions (a) through (d) of 3.4.1. Indeed, (a), (c) and (d) are satisfied by definition, (b) follows from 1.8. In what follows, we shall suppose given a family (M) of morphisms of C satisfying these conditions: our results will therefore apply in particular to the family (M_0).

Definition 3.4.3. *One says that the equivalence relation R in X is of* type (M) *if it is representable and if $p_1 \in (M)$ (which by (b) and (c) entails that $p_2 \in (M)$).*

One says that R is (M)-effective *if it is effective and if the canonical morphism $X \to X/R$ is an element of (M).*

One says that the quotient Y of X is (M)-effective *if the canonical morphism $X \to Y$ is an element of (M).*

The following consequences result from this definition:[290]

Proposition 3.4.3.1. *(i) An (M)-effective equivalence relation is of type (M) and universally effective.*

(ii) An (M)-effective quotient is universally effective (cf. 3.3.3).

(iii) The maps $R \mapsto X/R$ and $p \mapsto R(p)$ realize a one-to-one correspondence between (M)-effective equivalence relations in X and (M)-effective quotients of X.

[290] N.D.E.: The numbering 3.4.3.1 has been added for later references, and the proof of point (i) has been detailed.

(iv) (M_0)*-effective is equivalent to universally effective.*

Let us prove point (i). Since R is (M)-effective, one has a cartesian square

$$R \xrightarrow{p_2} X \downarrow p_1 \downarrow p X \xrightarrow{p} X/R,$$

and $p \in (M)$. Then, by 3.4.1 (a), p_1 and p_2 belong to (M), hence R is of type (M).

Set $Y = X/R$ and let $Y' \to Y$ be an arbitrary morphism. By 3.4.1 (a), the fiber products $X' = X \times_Y Y'$ and $R' = R \times_Y Y'$ exist and the morphisms $X' \to Y'$ and $p_i' : R' \to X'$ belong to (M). Finally, since $R = X \times_Y X$, one obtains, by associativity of the fiber product:

$$R' = X \times_Y X \times_Y Y' = X' \times_{Y'} X'.$$

This shows that R' is (M)-effective; hence in particular, R is universally effective. This proves (i), and also (iv). Points (ii) and (iii) follow from this, taking account of Definition 3.3.2.

3.4.4. Let H be an S-group whose structural morphism is an element of (M). Then if H acts freely on the S-object X, it defines an equivalence relation of type (M). Indeed, by (a) the fiber product $H \times_S X$ exists and $pr_2 : H \times_S X \to X$ is an element of (M). One says that the action of H is (M)-effective if the equivalence relation defined in X by this action is (M)-effective.

Proposition 3.4.5 ((M)-effectivity and base change). *Let R be an (M)-effective equivalence relation in X over S. Set $Y = X/R$. Let $S' \to S$ be a change of base such that $Y' = Y \times_S S'$ exists. Then $X' = X \times_S S'$ exists, $R' = R \times_S S'$ is an (M)-effective equivalence relation in X' over S' and $X'/R' \simeq (X/R)'$.*

Indeed, the canonical morphisms $X \to Y$ and $R \to Y$ are elements of (M), hence by (a′) X' and R' are representable. By associativity of the product, R' is the equivalence relation defined in X' by the canonical morphism $X' \to Y'$ which is an element of (M), whence the conclusion.

Proposition 3.4.6 ((M)-effectivity and cartesian products). *Let R (resp. R') be an (M)-effective equivalence relation in X (resp. X') over S. If $(X/R) \times_S (X'/R')$ exists, then $X \times_S X'$ exists, $R \times_S R'$ is an (M)-effective equivalence relation in $X \times_S X'$ over S and*

$$(X \times_S X')/(R \times_S R') \simeq (X/R) \times_S (X'/R').$$

Set $Y = X/R$, $Y' = X'/R'$. By (a′), the fiber product $X \times_S X'$ exists and the canonical morphism

$$q : X \times_S X' \to Y \times_S Y'$$

is an element of (M). Now the formula

$$(X \times_S X') \times_{(Y \times_S Y')} (X \times_S X') \simeq (X \times_Y X) \times_S (X' \times_{Y'} X')$$

(all products without subscript are taken over S) shows that $R \times_S R'$ is the equivalence relation defined by q in $X \times_S X'$, which completes the proof.

3.4.7.[291] Suppose that C has a final object e and let $f : G \to G'$ be a morphism of C-groups, such that $f \in (M)$. Then, by 3.4.1 (a), the kernel $Ker(f)$ is representable by $e \times_{G'} G$, and the morphism $Ker(f) \to e$ belongs to (M).

On the other hand, the equivalence relation defined by f is the same as that defined by the action of $Ker(f)$ (say, on the right) on G, i.e., it is the image of the morphism $G \times Ker(f) \to G \times G$, defined set-theoretically by $(g, h) \mapsto (g, gh)$. Consequently, one deduces from 3.3.2.1 the following corollary:

Corollary 3.4.7.1. *Suppose that C has a final object e and let $f : G \to G'$ be a morphism of C-groups, such that $f \in (M)$. Then the action of $Ker(f)$ on G is (M)-effective and G' is the quotient $G/Ker(f)$.*

3.5. Construction of quotients by descent

It frequently happens that one does not know how to construct a quotient directly, but that one does know how to do so after a suitable change of base. The present number gives a criterion useful in this situation.

We have seen in §2.1 the definition of a descent datum on an object X' over S' relative to a morphism $S' \to S$.

Definition 3.5.1. *One says that a descent datum on X' relative to $S' \to S$ is* effective *if X' equipped with this descent datum is isomorphic to the inverse image on S' of an object X over S, equipped with its canonical descent datum.*

If $S' \to S$ is a descent morphism (2.2), then the X of the definition is unique up to unique isomorphism. To say that $S' \to S$ is an effective descent morphism (2.2), is to say that it is a descent morphism and that every descent datum relative to this morphism is effective.

Consider now an equivalence relation R in an object X over S. Let X' (resp. X'', resp. X''') be the inverse images of X on S', $S'' = S' \times_S S'$ and $S''' = S' \times_S S' \times_S S'$ and let R', R'', R''' be the equivalence relations deduced from R by inverse image. Suppose that the equivalence relation R' in X' is (M)-effective, and consider the quotient $Y' = X'/R'$, which is an object over S'. Its two inverse images on S'' are isomorphic to X''/R'' by 3.4.5. The S'-object Y' is therefore endowed with a canonical gluing datum. Using likewise the uniqueness of X'''/R''', one sees that this is even a descent datum. (Remark: it has been implicitly supposed in this proof that all the fiber products written existed, which is the case in particular if $S' \to S$ is squarable, for example a descent morphism).

Proposition 3.5.2. *Let R be an equivalence relation in the object X over S. Let $S' \to S$ be a universal effective epimorphism. Suppose that every S-morphism*

[291] N.D.E.: This paragraph has been added.

whose inverse image on S' is in (M) *is itself in* (M)*. The following conditions are equivalent:*

(i) R is (M)*-effective in X;*

(ii) R' is (M)*-effective in X' and the canonical descent datum on X'/R' is effective.*

If this is so, the "descended" object of X'/R' is canonically isomorphic to X/R.

The fact that (i) entails (ii) is none other than the translation, in the language of descent, of 3.4.4. If one shows the converse, the last assertion of the proposition will be a consequence of the fact that a universal effective epimorphism is a descent morphism (2.3), hence that the "descended object" is unique (up to unique isomorphism).

So let us prove (ii) ⇒ (i). Let Y' be the quotient X'/R' and Y the descended object. As the canonical morphism $p' : X' \to X'/R' = Y'$ is by construction compatible with the descent data (its inverse images on S'' coincide with the canonical morphism $X'' \to X''/R''$), it comes by inverse image on S' from an S-morphism $p : X \to Y$. Since p' is an element of (M), it follows from the hypothesis made on the morphism $S' \to S$ that p is also an element of (M). As p' is compatible with the equivalence relation R', p is compatible with R, again because a universal effective epimorphism is a descent morphism. One thus has a morphism

$$R \to X \times_Y X.$$

To prove that R is (M)-effective and that Y is isomorphic to X/R, it suffices to prove that this morphism is an isomorphism. Now it becomes one by extension of the base from S to S', since R' is effective; it is therefore an isomorphism for the same reason as before (2.4).

Note that the hypothesis of the text is satisfied if one takes for (M) the family (M_0) of universal effective epimorphisms and if C has fiber products (1.10). One deduces the

Corollary 3.5.3. *Suppose that C has fiber products (over S would suffice). Let R be an equivalence relation in X over S and $S' \to S$ a universal effective epimorphism. The following conditions are equivalent:*

(i) R is universally effective in X,

(ii) R' is universally effective in X' and the canonical descent datum on X'/R' is effective.

If this is so, the "descended" object of X'/R' is canonically isomorphic to X/R.

4. Topologies and sheaves

The notion of sieve, and the presentation of the notion of topology (4.2.1) adopted here (more intrinsic and in many respects more convenient than the one by covering families of [MA]), are due to J. Giraud [AS].

4.1. Sieves

Definition 4.1.1. *One calls a* sieve *of the category C a subfunctor C of the final functor $e : C^\circ \to (Ens)$.*

To every sieve C of C one associates the set $E(C)$ of objects X of C such that $C(X) \neq \emptyset$, that is, such that the structural morphism $X \to e$ factors through C. One thus has the equivalences

$$(+) \quad X \in E(C) \Leftrightarrow C(X) = e(X) = \{\emptyset\}.$$
$$X \notin E(C) \Leftrightarrow C(X) = \emptyset.$$

The set $E = E(C)$ enjoys the following property:

$$(++) \quad \text{If } X \in E \text{ and if } \mathrm{Hom}(Y, X) \neq \emptyset, \text{ then } Y \in E.$$

(Note that if one equips the set $\mathrm{Ob}\ C$ with its natural preorder structure (Y dominating X if there exists an arrow from Y into X), the sets E satisfying (++) are the subsets of $\mathrm{Ob}\ C$ that contain every dominator[292] of one of their elements.)

Conversely, if E is a subset of $\mathrm{Ob}\ C$ enjoying property (++), then E is written in a unique way in the form $E(C)$ and C is defined by the formulas (+). There is therefore a one-to-one correspondence between sieves of C and subsets of $\mathrm{Ob}\ C$ satisfying condition (++). By abuse of language, we shall sometimes say that the set $E(C)$ is a sieve of C.

[293] Let C and C' be two sieves of C; since they are two subfunctors of the final functor e, it amounts to the same to say that C dominates C' (for the relation of domination in $Ob\hat{C}$), or that C is a subfunctor of C', or yet that $E(C) \subset E(C')$; one then says that C is *finer* than C'. One sees that this gives an order structure on the set of sieves of C. Furthermore, one has $E(C) \cap E(C') = E(C \times C')$ and therefore the set of sieves of C is filtered for the order relation "being finer".

Every subset E of $Ob\hat{C}$, for example a subset of $\mathrm{Ob}\ C$, defines a sieve $C(E)$: the set of $X \in ObC$ such that $F(X) \neq \emptyset$ for at least one $F \in E$ satisfies condition (++) and defines the sought sieve.

This sieve can also be defined as the image of the family of morphisms $F \to e, F \in E$ in the sense of the following definition:

Definition 4.1.2. *Let $F_i \to F$ be a family of morphisms of $\hat{C}$ with the same target F. One calls* image *of this family the subfunctor of F defined by*

$$S \mapsto \bigcup_i \mathrm{Im}\ F_i(S) \subset F(S).$$

Proposition 4.1.3. *The formation of the image commutes with base extension: in the preceding notation, denote by I the image of the family $F_i \to F$; for every*

[292] N.D.E.: Here, "dominator" is taken in the sense of the preorder relation mentioned above, i.e., Y dominates X if there exists an arrow $Y \to X$. On the other hand, if X, Y are two subobjects of an object Z, one says (cf. 2.4) that Y contains X if $X \subset Y$. To avoid any ambiguity between these two terminologies, "majorant" has been replaced in the sequel by "dominating" in the first case, and by "containing", in the second.

[293] N.D.E.: The sentence that follows has been detailed.

morphism $G \to F$ of $\hat{C}$, the image of the family of morphisms $F_i \times_F G \to G$ is the subfunctor $I \times_F G$ of G.

Definition 4.1.4.0.[294] *Let C be a sieve of C. If E is a subset of $Ob\ C$ such that $C(E) = C$, one says that E is a* base *of C. Every sieve C has a base, for example the set $E(C)$.*

We propose to describe the set $\mathrm{Hom}(C, F)$, where C is a sieve of C and F an object of $\hat{C}$, using a base S_i of C. For each pair (i, j), one has a diagram in $\hat{C}$:

$$S_i \times S_j \to S_i \downdownarrows S_j \to e,$$

whence a diagram of sets

$$\Gamma(F) = \mathrm{Hom}(e, F) \xrightarrow{\sigma} \prod_i \mathrm{Hom}(S_i, F) \rightrightarrows_{\tau_1, \tau_2} \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)$$

such that $\tau_1\sigma = \tau_2\sigma$. One thus has a morphism

$$\mathrm{Hom}(e, F) \to \mathrm{Ker}(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)).$$

One verifies immediately:

Proposition 4.1.4. *One has an isomorphism, functorial in F,*

$$\mathrm{Hom}(C, F) \simeq \mathrm{Ker}(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)),$$

such that the diagram

$$\begin{array}{ccc} \mathrm{Hom}(e, F) & \to & \mathrm{Ker}(\prod_i \mathrm{Hom}(S_i, F) \rightrightarrows \prod_{i,j} \mathrm{Hom}(S_i \times S_j, F)) \\ & & \uparrow \simeq \\ \mathrm{Hom}(e, F) & \to & \mathrm{Hom}(C, F), \end{array}$$

where the last line is induced by the canonical morphism $C \to e$, is commutative.

Corollary 4.1.5. *Suppose that the fiber products $S_i \times S_j$ are representable, for example*

that the S_i are squarable. One then has, for every $F \in Ob\hat{C}$, an isomorphism

$$\mathrm{Hom}(C, F) \simeq \mathrm{Ker}(\prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times S_j)).$$

Remark 4.1.6. *Let R be a sieve of C; denote by $\bar{R}$ the full subcategory of C whose set of objects is $E(R)$ and by*

$$i_R : \bar{R} \longrightarrow C$$

the inclusion functor. One has an isomorphism, functorial in $F \in Ob\hat{C}$,

$$\mathrm{Hom}(R, F) \simeq \Gamma(F \circ i_R)$$

such that the diagram

[294] N.D.E.: The numbering 4.1.4.0 has been added in order to highlight this definition.

$$\begin{array}{ccc} \mathrm{Hom}(e, F) & \to & \mathrm{Hom}(R, F) \\ \updownarrow & & \updownarrow \\ \Gamma(F) & \to & \Gamma(F \circ i_R), \end{array}$$

where the second line is induced by the functor i_R, is commutative.

Definition 4.1.7. *Let C be a category. One calls a* sieve of the object S of C *a sieve of the category C/S.*

A sieve of S is therefore a sub-$\hat{C}$-object of S. To it corresponds canonically a subset of ObC/S containing the source of every arrow whose target it contains. By abuse of language, such a set will also be called a sieve of S.

4.2. Topologies: definitions

Definition 4.2.1. *Let C be a category. One calls a* topology *on C the datum, for each S of C, of a set $J(S)$ of sieves of S, called* covering sieves *or* refinements *of S,*

the datum satisfying the following axioms:

(T 1) For every refinement R of S and every morphism $T \to S$, the sieve $R \times_S T$ of T is covering ("stability under base change").

(T 2) If R, C are two sieves of S, if R is covering, and if for every $T \in ObC$ and every morphism $T \to R$, the sieve $C \times_S T$ of T is covering, then C is a refinement of S.[295]

(T 3) If $C \supset R$ are two sieves of S and if R is covering, then C is covering.

(T 4) For every S, S is a refinement of S.

One can reformulate these axioms in the following way. Suppose given a topology $S \mapsto J(S)$ on C and, for every object F of $\hat{C}$, denote by $J(F)$ the set of subfunctors R of F such that for every morphism $T \to F$ of $\hat{C}$ where T is representable, $R \times_F T$, which is a sieve of T, is covering. By virtue of (T 1), this notation is compatible with the preceding one. One will also say that $R \in J(F)$ is a *refinement* of F. One verifies immediately that the preceding axioms entail the following properties:

(T′ 0) If $F \supset G$ are two objects of $\hat{C}$, and if for every $S \in ObC$ and every morphism $S \to F$, $G \times_F S \in J(S)$, then $G \in J(F)$.

(T′ 1) If $G \in J(F)$, and if $H \to F$ is a morphism of $\hat{C}$, then $G \times_F H \in J(H)$.

(T′ 2) If $F \supset G \supset H$ are three objects of $\hat{C}$, if $G \in J(F)$ and $H \in J(G)$, then $H \in J(F)$.

(T′ 3) If $F \supset G \supset H$ are three objects of $\hat{C}$ and if $H \in J(F)$, then $G \in J(F)$.

(T′ 4) For every $F \in Ob\hat{C}$, $F \in J(F)$.

Conversely, if one gives oneself, for every $F \in Ob\hat{C}$, a set $J(F)$ of subobjects of F satisfying the properties (T′ 0) through (T′ 4), the map $S \mapsto J(S)$ defines a topology on C and the two preceding constructions are inverse to each other.

[295]N.D.E.: that is: if C is covering "locally with respect to the covering sieve R", then C is covering.

From (T′ 1), (T′ 2) and (T′ 3)[296] results the following property:

(T′ 5) If G and H are two subobjects of F and if $G, H \in J(F)$, then $G \cap H \in J(F)$.

The set $J(F)$, ordered by the relation $\supset$, is therefore filtered; this remark will be useful later.

4.2.2. One says that the topology defined by J is *finer* than the topology defined by J' if for every $S \in ObC$, $J(S) \supset J'(S)$ (it amounts to the same to say that for every $F \in Ob\hat{C}$, $J(F) \supset J'(F)$).

Every set of topologies on C has a greatest lower bound: let I be an indexing set, and for each $i \in I$, let $S \mapsto J_i(S)$ be a topology on C. Set $J(S) = \bigcap_{i \in I} J_i(S)$; it is immediate that one has thus defined a topology on C, and that this is indeed the greatest lower bound of the given set.

In particular, let us give ourselves, for each $S \in ObC$, a set $E(S)$ of sieves of S. One calls the *topology generated by these sets* the least fine topology for which the elements of $E(S)$ are refinements of S for every S.

Definition 4.2.3. *Let $F_i \to F$ be a family of morphisms of $\hat{C}$. Let $G \subset F$ be the image (4.1.2) of this family. The family is said to be* covering *if $G \in J(F)$. A morphism is said to be covering if the family reduced to this morphism is covering.*

This definition applies in particular to an inclusion: a sieve C of S is covering if and only if the canonical morphism $C \to S$ is covering.

The axioms (T′ 0) through (T′ 5) entail for covering families the following properties:

(C 0) Let $F_i \to F$ be a family of morphisms of $\hat{C}$. If for every representable base change $S \to F$, the family $F_i \times_F S \to S$ is covering, then so is the initial family.

(C 1) For every covering family $F_i \to F$ and every morphism $G \to F$, the family $F_i \times_F G \to G$ is covering ("stability under base change").

(C 2) If $F_i \to F$ is a covering family and if, for each i, $F_{ij} \to F_i$ is a covering family, then the composite family $F_{ij} \to F$ is covering ("stability under composition").

(C 3) If $G_j \to F$ is a covering family, and if $F_i \to F$ is a family of morphisms with target F such that for each j there exists an i such that $G_j \to F$ factors through $F_i \to F$, then $F_i \to F$ is covering ("saturation").

(C 4) Every family reduced to an isomorphism is covering.

Note that (C 2) and (C 3) also entail:

(C 5) If $F_i \to F$ is a family of morphisms with target F such that there exists a covering family $G_j \to F$ such that for every j the family $F_i \times_F G_j \to G_j$ is covering, then the family $F_i \to F$ is covering ("a locally covering family is covering").

4.2.4. Conversely, let C be a category having fiber products and let us give ourselves, for each $S \in ObC$, a set of families of morphisms of C with target S, said to

[296] N.D.E.: (T′ 1) and (T′ 2) suffice: $G \cap H = G \times_F H$ belongs to $J(H)$, by (T′ 1), hence to $J(F)$, by (T′ 2).

be *covering families*, the datum satisfying axioms (C 1) through (C 4) (hence also (C 5) which is a consequence). For every $S \in ObC$, let $J(S)$ be the set

of sieves of S having a covering base (or, what amounts to the same by (C 3), all of whose bases are covering). Then $S \mapsto J(S)$ defines a topology on C. The two preceding constructions are inverse to each other.

In fact, in applications, it is impractical to consider all covering families, since one sometimes has fairly simple descriptions of a "sufficient" number of such families. This leads to posing the following definitions.

Definition 4.2.5.0.[297] *Let C be a category. Suppose given, for each $S \in ObC$, a set $P(S)$ of families of morphisms of C with target S. One calls the* topology generated by *P the least fine topology for which the given families are covering.*

Definition 4.2.5. *Let C be a category. One calls a* pretopology *on C the datum for each $S \in ObC$ of a set $R(S)$ of families of morphisms $S_i \to S$ with target S, said to be* covering for the pretopology under consideration, *satisfying the following axioms:*

(P 1) For every family $S_i \to S \in R(S)$ and every morphism $T \to S$, the fiber products $S_i \times_S T$ exist and $S_i \times_S T \to T \in R(T)$.

(P 2) If $S_i \to S \in R(S)$ and if for each i, $T_{ij} \to S_i \in R(S_i)$, then the composite family $T_{ij} \to S$ belongs to $R(S)$.

(P 3) Every family reduced to an isomorphism is covering.

Proposition 4.2.6. *For every S, let $J(S)$ be the set of sieves of S covering for the topology generated by the pretopology R. Let $J_R(S)$ be the part of $J(S)$ formed by the sieves defined by the families of $R(S)$. Then $J_R(S)$ is cofinal in $J(S)$: every refinement of S contains a sieve defined by a family of $R(S)$.*

For every S, let $J'(S)$ be the set of sieves of S containing a sieve of $J_R(S)$. One evidently has $J'(S) \subset J(S)$. To show that $J(S) = J'(S)$, it suffices to show that the $J'(S)$ make a topology on C, that is, that they satisfy axioms (T 1) through (T 4). Now (T 1), (T 3), (T 4) are evidently satisfied. It remains to verify (T 2).[298]

So let U be an element of $J'(S)$ and C a sieve of S; one supposes that for every $T \to U$, the sieve $C \times_S T$ is in $J'(T)$ and one must prove that $C \in J'(S)$. By definition of J', U contains a refinement U' defined by a family $S_i \to S \in R(S)$. Since one has verified (T 3), it suffices to prove that $U' \cap C \in J'(S)$, so one may suppose that $U = U'$. By hypothesis, for every i, $C \times_S S_i \in J'(S_i)$; there therefore exists, for each i, a covering family $T_{ij} \to S_i \in R(S_i)$ such that $T_{ij} \to S_i$ factors through $C \times_S S_i \to S_i$. The morphism $T_{ij} \to S$ therefore factors through $C \to S$, which shows that C contains the sieve defined by the composite family $T_{ij} \to S$, and one has finished by (P 2).

The axioms (P 1) through (P 3) are those of [MA]. Given the practical interest of pretopologies, we shall interpret each important result with the aid of a pretopology

[297] N.D.E.: This definition has been placed here (placed in the original after 4.2.5), since it will be used in 6.2.1 in a slightly more general setting than that of 4.2.5.

[298] N.D.E.: in what follows, typographical errors of the original have been corrected.

defining the given topology.

Remark 4.2.7. *One can introduce a somewhat more general notion: one gives, for each S, a set of covering families satisfying (P 1), (P 3) and Proposition 4.2.6. This presents itself in particular when the given families satisfy (P 1), (P 3) and (C 5). The reader may consult [D].*

Definition 4.2.8. *Let C be equipped with a topology, and let S be an object of C. Let $P(S')$ be a relation involving an argument $S' \in ObC/S$. Suppose that* $\mathrm{Hom}(S'', S') \neq \emptyset$ *entails $P(S') \Rightarrow P(S'')$. One says that P is true* locally on *S for the topology under consideration, if the following equivalent conditions are satisfied:*

(i) The set of $S' \to S$ such that $P(S')$ is true is a refinement of S.

(ii) There exists a refinement of S such that $P(S')$ is true for every S' of this refinement.

(iii) (If the given topology is defined by a pretopology). There exists a covering family for this pretopology such that $P(S')$ is true for every S' of this family.

Example 4.2.9. *Let $f : X \to Y$ be an S-morphism. One will say that f is* locally an isomorphism *if there exists a covering family $S_i \to S$ such that for every i, $f \times_S S_i$*

is an isomorphism. It amounts to the same to require that there exist a refinement R of S such that for every $T \to R$, $X(T) \to Y(T)$ is an isomorphism.

One will see in the sequel many other examples of "local" language.

4.3. Presheaves, sheaves, sheaf associated to a presheaf

Definition 4.3.1. *Let C be a category. One calls a* presheaf of sets *on C any contravariant functor from C into the category of sets. The category $\hat{C}$ =* $\mathrm{Hom}(C^\circ, (Ens))$ *is called the* category of presheaves *on C. If C is equipped with a topology, one says that the presheaf P is* separated *(resp.* is a sheaf*) if for every $S \in ObC$ and every $R \in J(S)$, the canonical map*

`(+)    P(S) = Hom(S, P) → Hom(R, P)`

is injective (resp. bijective). One calls category of sheaves, *and denotes by $\tilde{C}$, the full subcategory of $\hat{C}$ whose objects are the sheaves.*[299]

Proposition 4.3.2. *Let P be a separated presheaf (resp. a sheaf). For every functor $H \in Ob\hat{C}$ and every $R \in J(H)$, the canonical map*

`(+)    Hom(H, P) → Hom(R, P)`

is injective (resp. bijective).

[299]N.D.E.: One will note that if Q is a subpresheaf of a separated presheaf P, then Q is separated. Indeed, for every sieve R of S, the composite map $Q(S) \hookrightarrow P(S) \hookrightarrow P(R)$ is injective and factors through $Q(S) \to Q(R)$.

Indeed, let P be a separated presheaf, H a presheaf, $R \in J(H)$, and $u, v : H \to P$ such that $uj = vj$. For every $f : S \to H$, $S \in ObC$, $R \times_H S$ is a refinement of S and $ufj_S = vfj_S$:

$$\begin{tikzcd}
R \times_H S \arrow[r, "j_S"] \arrow[d, "f_R"] & S \arrow[d, "f"] & \\
R \arrow[r, "j"] & H \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & P.
\end{tikzcd}$$

Since P is separated, one deduces $uf = vf$. This being true for every representable S, one has $u = v$.

Suppose now that P is a sheaf. Let $g : R \to P$; let us show that it factors through H. For every $f : S \to H$, $S \in ObC$, $g \circ f_R : R \times_H S \to P$ factors

in a unique way through S, hence defines a morphism $h : S \to P$, which is evidently functorial in f, by uniqueness:

$$\begin{tikzcd}
R \times_H S \arrow[r, "f_R"] \arrow[d, "j_S"] & R \arrow[r, "g"] \arrow[d, "j"] & P \\
S \arrow[r, "f"] & H. \arrow[ur, "h"] &
\end{tikzcd}$$

One has thus defined for every S a map from $H(S)$ into $P(S)$ functorial in S, hence a morphism from H into P that answers the required conditions.

Corollary 4.3.2.1.[300] *Let R, F be two sheaves. If R is a refinement of F, then $R = F$.*

Indeed, suppose that R is a refinement of F and denote by j the inclusion $R \hookrightarrow F$. By 4.3.2, one has $\mathrm{Hom}(F, R) = \mathrm{End}(R)$, hence there exists $\pi : F \to R$ such that $\pi \circ j = id_R$. One has likewise $\mathrm{End}(F) = \mathrm{Hom}(R, F)$, and the equality $j \circ \pi \circ j = j$ entails $j \circ \pi = id_F$, hence j is an isomorphism.

Proposition 4.3.3 ([AS], 1.3). *Let C be a category. Let P be a presheaf on C; for every $S \in ObC$, denote by $J(S)$ the set of sieves R of S such that for every $T \to S$, the map*

$$(+) \qquad \mathrm{Hom}(T, P) \to \mathrm{Hom}(R \times_S T, P)$$

is injective (resp. bijective). Then the $J(S)$ define a topology on C, i.e. satisfy axioms (T 1) through (T 4).

Corollary 4.3.4. *Let, for every $S \in ObC$, $K(S)$ be a family of sieves satisfying (T 1). Let P be a presheaf on C. For it to be separated (resp. a sheaf) for the topology generated by the $K(S)$, it is necessary and sufficient that for every $S \in ObC$ and every $R \in K(S)$, the canonical map*

$$(+) \qquad \mathrm{Hom}(S, P) \to \mathrm{Hom}(R, P)$$

be injective (resp. bijective).

Corollary 4.3.5. *Let, for each $S \in ObC$, $R(S)$ be a set of families of morphisms of C with target S, satisfying (P 1) (for example defining a pretopology). Let P be a*

[300] N.D.E.: This corollary has been added.

presheaf on C. For P to be separated (resp. a sheaf) for the topology generated by R, it is necessary and sufficient that for every $S \in ObC$ and every family $S_i \to S \in R(S)$, the map

$$P(S) \longrightarrow \prod_i P(S_i)$$

be injective, (resp. the diagram

$$P(S) \to \prod_i P(S_i) \rightrightarrows \prod_{i,j} P(S_i \times_S S_j)$$

be exact).

Definition 4.3.6. *Let C be a category. One calls the* canonical topology *on C the finest topology for which all representable functors are sheaves.*

Corollary 4.3.7. *For a sieve R of S to be a refinement for the canonical topology, it is necessary and sufficient that for every morphism $T \to S$ of C and every $X \in ObC$, the canonical map*

$$\mathrm{Hom}(T, X) \to \mathrm{Hom}(R \times_S T, X)$$

be bijective.

Definition 4.3.8. *A sieve covering for the canonical topology will be called a* universal effective epimorphic sieve.

Corollary 4.3.9. *A universal effective epimorphic family defines a universal effective epimorphic sieve. Conversely, every squarable family defining a universal effective epimorphic sieve is universal effective epimorphic.*

Let us return to the case where C is equipped with an arbitrary topology and pass to the construction of the sheaf associated to a presheaf P. Let S be an object of C. If $R \supset R'$ are

two refinements of S, one has a diagram

$$\begin{array}{ccc} \mathrm{Hom}(S, P) & \to & \mathrm{Hom}(R, P) \\ & & \searrow \\ & & \mathrm{Hom}(R', P). \end{array}$$

The ordered set $J(S)$ is filtered, as has already been remarked. Since S is an element of $J(S)$, one has an evident morphism

$$\mathrm{Hom}(S, P) \to \varinjlim_{R \in J(S)} \mathrm{Hom}(R, P).$$

Definition 4.3.10.0.[301] *One sets $\check{H}^0(S,P) = \lim \to_{R\in J(S)} \mathrm{Hom}(R,P)$. One verifies that $\check{H}^0(S,P)$ depends functorially on S, hence defines a functor LP by*

$$\text{(++)} \quad \mathrm{Hom}(S, LP) = \check{H}^0(S, P) = \varinjlim_{R \in J(S)} \mathrm{Hom}(R, P).$$

[301] N.D.E.: The numbering 4.3.10.0 has been added for later references. On the other hand, it follows from the definition that if $Q \to P$ is a monomorphism, the same holds for $LP \to LQ$; hence L "preserves monomorphisms" (see also 4.3.16 for a more general result: L "commutes with finite inverse limits").

One has by construction morphisms

```
□_P : P □ LP
z_R : Hom(R, P) → Hom(S, LP).
```

Lemma 4.3.10. *(i) For every refinement R of S and every $u : R \to P$, the diagram*

$$P \xrightarrow{\ell_P} LP \uparrow u \uparrow z_R(u) R \xrightarrow{i_R} S$$

is commutative.

(ii) For every morphism $v : S \to LP$, there exists a refinement R of S and a morphism $u : R \to P$ with $v = z_R(u)$.

(iii) Let Q be a functor and $u, v : Q \to P$ such that $\ell_P u = \ell_P v$. Then the kernel of the pair (u, v) is a refinement of Q.

(iv) Let $u : R \to P$ and $u' : R' \to P$; for $z_R(u) = z_{R'}(u')$, it is necessary and sufficient that there exist a refinement $R'' \subset R \cap R'$ of S such that u and u' coincide on R''.

Proof. (i): One must verify that $z_R(u) i_R = \ell_P u$. For this, it suffices to verify that the composites of these two morphisms with every morphism $g : T \to R$, where T is representable, are equal. Now consider $f = i_R g$ and the fiber product $R' = R \times_S T$:[302]

```
P  —□_P→  LP
↑u          ↑z_R(u)
R  —i_R→  S
↑p          ↑f
R′  —~→   T (via i_{R′}, g)
```

By definition of ℓ_P, $\ell_P u g = z_R(u) f$ (this is the particular case of what one is trying to prove in which i_R is an isomorphism), and now $z_R(u) f = z_R(u) i_R g$.

(ii) and (iv) merely translate the definition of $\mathrm{Hom}(S, LP)$ as a direct limit.

(iii): If K denotes the kernel of the pair (u, v), then for each morphism $f : S \to Q$ where S is representable, $K \times_Q S$ is a subfunctor of the kernel of the pair of arrows $uf, vf : S \to P$. One is therefore reduced, by (T′ 0), to proving the assertion in the case where $Q = S$ is representable. But in this case, it follows from (ii) and (iv) that K contains a refinement of S hence is a refinement of S.

One verifies finally that `P □ LP` defines a functor

$$L : \hat{C} \longrightarrow \hat{C}$$

and `P □ □_P` a morphism of functors

[302] N.D.E.: Denote by p the projection $R' \to R$. Since i_R is a monomorphism, one has $p = g i_{R'}$. Let g' be the section of $i_{R'}$ defined by g; then $pg' = g$ and therefore $pg' i_{R'} = g i_{R'} = p$; this entails $g' i_{R'} = id_{R'}$ and therefore $i_{R'} : R' \to T$ is an isomorphism, with inverse g'.

$$\ell : Id_{\hat{C}} \longrightarrow L.$$

Let us now state the essential result:

Proposition 4.3.11. *(i) If P is any presheaf, LP is separated and $\ell_P : P \to LP$ is covering (4.2.3).*

(ii) If P is a sheaf, $P \to LP$ is an isomorphism.

(iii) For every presheaf P and every separated presheaf (resp. sheaf) F, the map

$\mathrm{Hom}(\ell_P, F) : \mathrm{Hom}(LP, F) \to \mathrm{Hom}(P, F)$

is injective (resp. bijective).

(iv) If P is separated, $\ell_P : P \to LP$ is a covering monomorphism (hence P is a refinement of LP), and LP is a sheaf.

Proof. (i) First, $P \to LP$ is covering; indeed, for every morphism $v : S \to LP$, there exists, by 4.3.10 (i) and (ii), a refinement R of S such that one has a commutative diagram

$P \xrightarrow{\ell_P} LP$
$\uparrow v' \qquad\qquad \uparrow v$
$P \times_{LP} S \leftarrow S$
$\uparrow \qquad\qquad\qquad \uparrow i_R$
R

hence $P \times_{LP} S \to S$ is covering. It follows, by (C 0), that $P \to LP$ is covering.

If on the other hand two morphisms $v_1, v_2 : S \to LP$ induce the same morphism on a refinement R of S, let us show that they are equal. There exist refinements R_i, $i = 1, 2$, and morphisms $u_i : R_i \to P$ such that $z_{R_i}(u_i) = v_i$. Taking R small enough, one may suppose $R_1 = R_2 = R$. It then follows from the commutative diagram of 4.3.10

(i) that $\ell_P u_1 = \ell_P u_2$. By loc. cit. (iii), u_1 and u_2 therefore coincide on a refinement of R, hence a refinement of S, which entails that $z_R(u_1) = z_R(u_2)$, by loc. cit. (iv).

(ii) is clear, for if P is a sheaf, $\mathrm{Hom}(S, P) \to \mathrm{Hom}(R, P)$ is already an isomorphism for every refinement R of S.

(iii) Let u and v be two morphisms $LP \to F$ such that $u\ell_P = v\ell_P$. To show that $u = v$, it suffices to see that $uf = vf$ for every $f : S \to LP$ where S is representable. Now there exists a refinement R of S and a morphism $g : R \to P$ with $f = z_R(g)$. Then uf and vf coincide on R with $u\ell_P g = v\ell_P g$, hence coincide on R. If F is separated, one therefore has $uf = vf$. Suppose now that F is a sheaf; one then has the commutative diagram

$$\begin{tikzcd} P \arrow[r, "\ell_P"] \arrow[dr] & LP \arrow[d] \\ F \arrow[r] & LF \end{tikzcd}$$

which shows that $\mathrm{Hom}(\ell_P, F)$ is surjective.

(iv) Let us show that if P is separated, $P \to LP$ is a monomorphism. For this, it suffices to see that for every pair of morphisms $u, v : S \to P$ (where S is representable) such that $\ell_P u = \ell_P v$ one has $u = v$. Now loc. cit. (iii) shows that u and v coincide on a refinement of S, hence coincide because P is separated. This shows that $P \to LP$ is a monomorphism; as it is covering by (i), one obtains that P is a refinement of LP.

Let us finally show that LP is a sheaf. As we already know by (i) that it is a separated presheaf, it suffices to see that for every $S \in Ob C$, every refinement R of S and every morphism $h : R \to LP$, there exists a morphism $u : S \to LP$ with $ui_R = h$. Now $R' = P \times_{LP} R$ is a refinement of R, since P is a refinement of LP, hence R' is

a refinement of S. Set $u = z_{R'}(h')$:

$$\begin{tikzcd} P \arrow[rr, "\ell_P"] & & LP \\ R' \arrow[u, "h'"] \arrow[r, "j"] & R \arrow[r, "i_R"] & S. \arrow[u, "u \text{ (via } h)"'] \end{tikzcd}$$

One has $ui_{R'} = \ell_P h' = hj$, whence $ui_R j = hj$. Since R' is a refinement of R and since LP is separated, 4.3.2 shows that $ui_R = h$.

Corollary 4.3.12.[303] *Let F be a sheaf and R a sub-$\hat{C}$-object of F. Then R is a separated presheaf, $\ell_R : R \to LR$ is a covering monomorphism, and one has a commutative diagram*

$$\begin{tikzcd} R \arrow[rr, "i"] \arrow[dr, "\ell_R"'] & & F \\ & LR. \arrow[ur, "j"'] & \end{tikzcd}$$

Consequently, R is a refinement of F if and only if j is an isomorphism.

We have already noted that R is separated and that j is a monomorphism, cf. N.D.E. (24) and (26). By 4.3.11 (iv), ℓ_R is a covering monomorphism. Therefore, if j is an isomorphism, R is a refinement of F. Conversely, if i is covering, so is j, hence it is an isomorphism by 4.3.2.1.

Remark 4.3.13. *If $J'(S)$ is a cofinal subset of $J(S)$, one has*

$$\mathrm{Hom}(S, LP) = \varinjlim_{R \in J'(S)} \mathrm{Hom}(R, P).$$

In particular, let $S \mapsto R(S)$ be a pretopology generating the given topology. The functor L can be described using the covering families that are elements of $R(S)$. Making the formula above explicit, one recovers the construction of [MA].

[303] N.D.E.: The statement of the corollary and its proof have been detailed.

Denote by i the inclusion functor $\tilde{C} \to \hat{C}$. From Proposition 4.3.11 results the following theorem:

Theorem 4.3.14. *There exists a unique functor $a : \hat{C} \to \tilde{C}$ such that the following diagram is commutative*

$$\hat{C} \xrightarrow{L} \hat{C} \downarrow a \downarrow L\tilde{C} \xrightarrow{i} \hat{C},$$

i.e., for every presheaf P, $L(L(P))$ is a sheaf. The functors i and a are adjoint to one another: for every presheaf P and every sheaf F one has an isomorphism, functorial in P and F,

$$\mathrm{Hom}_{\hat{C}}(P, i(F)) \simeq \mathrm{Hom}_{\tilde{C}}(a(P), F),$$

that is,

$$\mathrm{Hom}(P, F) \simeq \mathrm{Hom}(a(P), F).$$

Definition 4.3.15. *The sheaf $a(P)$ is said to be* associated *to the presheaf P.*

Remark 4.3.16. *As the functor L is constructed using inverse limits and filtered direct limits, it commutes with finite inverse limits.*[304]

Moreover, if one identifies $L(P \times P)$ with $LP \times LP$, the morphism $\ell_{P\times P}$ is identified with $\ell_P \times \ell_P$. It follows for example that if P is a presheaf of groups, LP is also canonically equipped with a structure of presheaf of groups and the canonical morphism $P \to LP$ is a morphism of groups. The same holds for the functor a, which shows that if P is a presheaf of groups and F a sheaf of groups, one has an isomorphism

$$\mathrm{Hom}_{\hat{C}\text{-gr.}}(P, i(F)) \simeq \mathrm{Hom}_{\tilde{C}\text{-gr.}}(a(P), F).$$

See [D] for more details.

4.3.17. If V is any category, one calls a *presheaf on C with values in V* a contravariant functor from C into V. To define sheaves with values in V, we must first recall the definition of the inverse limit of a functor. If R and V are two categories, and

$$F : R^\circ \longrightarrow V$$

a contravariant functor from R to V, one denotes by $\varprojlim F$ the object of $\hat{V}$ defined as follows:

$$\varprojlim F(X) = \mathrm{Hom}_{\hat{V}}(X, \varprojlim F) = \varprojlim_{U \in \mathrm{Ob}\, R} \mathrm{Hom}_V(F(U), X) = \mathrm{Hom}(c_X, F),$$

where X is a variable object of V, where c_X denotes the contravariant functor from R to V that sends each object of R to X and each arrow of R to id_X, and where the last Hom is taken in the category $\mathrm{Hom}(R^\circ, V)$. If R has a final object e_R, one has

[304] N.D.E.: In particular, if K is the kernel of a pair of morphisms of presheaves $u, v : Q \rightrightarrows P$, then LK is the kernel of $Lu, Lv : LQ \rightrightarrows LP$ (this will be used in 4.4.5).

$\lim \leftarrow F = F(e_R)$. If V is the category of sets, the functor lim ← is identified with the functor Γ.

If S is an object of C and R a sieve of S, denote by $\bar{R}$ the full subcategory of C/S whose set of objects is $E(R)$ and $i_R : \bar{R} \to C/S$ the canonical functor. If P is a presheaf on C with values in V, it defines by restriction a functor $P_S : (C/S)^{\circ} \to V$. The functor i_R induces a morphism of $\hat{V}$:

`P(S) = P_S(S) = lim← P_S □ lim←(P_S ∘ i_R).`

One denotes by $\lim \leftarrow_R P$ the object $\lim \leftarrow (P_S \circ i_R)$ of $\hat{V}$. By virtue of 4.1.6, Definition 4.3.1 generalizes to the

Definition 4.3.18. *The presheaf P on C with values in V is said to be* separated *(resp.* a sheaf*), if for every $S \in ObC$ and every $R \in J(S)$, the canonical morphism of $\hat{V}$*

$$P(S) \longrightarrow \lim \leftarrow_R P$$

is a monomorphism (resp. an isomorphism).

In the case where V is the category `(Gr.)` of groups (or any other category of sets equipped with algebraic structures defined by finite inverse limits), one can see (cf. [D]) that there is equivalence between the following notions: a presheaf on C with values in `(Gr.)` whose underlying presheaf of sets is a sheaf, and a group in the category of sheaves of sets. Taking account of these identifications, we shall always consider sheaves with values in a category of sets equipped with algebraic structures defined by finite inverse limits

as sheaves of sets, equipped in the category $\tilde{C}$ with the corresponding algebraic structure.

4.4. Exactness properties of the category of sheaves

Theorem 4.4.1. *(i) Arbitrary inverse limits exist in $\tilde{C}$; "they are computed in $\hat{C}$", i.e. the inclusion functor $i : \tilde{C} \to \hat{C}$ commutes with inverse limits: if (X_α) is an inverse system of sheaves, the presheaf*

`lim← i(X_α) : S □ lim← X_α(S)`

is a sheaf and one has `i(lim← X_α) = lim← i(X_α)`.

(ii) Arbitrary direct limits exist in $\tilde{C}$: if (X_α) is a direct system of sheaves, one has

`lim→ X_α = a(lim→ i(X_α))`

where $\lim \to i(X_\alpha)$ *is the presheaf direct limit of the $i(X_\alpha)$:*

`lim→ i(X_α) : S □ lim→ X_α(S).`

(iii) The functor $a : \hat{C} \to \tilde{C}$ commutes with arbitrary direct limits and with finite inverse limits.

Assertions (i) and (ii) follow formally from the adjunction formula (4.3.14), and[305] the first assertion of (iii) follows from (ii). Finally, the second assertion of (iii) has already been pointed out in 4.3.16.

Scholie 4.4.2. *This theorem allows one to use the following method to prove in $\tilde{C}$ an assertion bearing simultaneously on arbitrary direct limits and finite inverse limits (for example: "every epimorphism is universally effective", cf. below). One begins by proving the corresponding assertion in the category of sets, then one extends it "argument by argument" to the category*

of presheaves. Next, one uses the preceding theorem to pass from the category of presheaves to the category of sheaves. One will see in the sequel many examples of this method (4.4.3, 4.4.6, 4.4.9, etc.).

Let us finally remark that the assertions relative to the category of presheaves are formally corollaries of the assertions relative to the category of sheaves. It suffices in fact to take as topology the least fine topology ("chaotic"), that is, the topology defined by $J(S) = S$ for every $S \in ObC$; every functor is indeed a sheaf for this topology.

Proposition 4.4.3. *Let $F = F_i \to F$ be a family of morphisms of sheaves. The following conditions are equivalent:*

(i) F is an epimorphic family.

(ii) F is a universal effective epimorphic family (1.13).

(iii) F is covering (4.2.3).

(iv) The sheaf image of F (that is, the sheaf associated to the presheaf image of F (4.1.2)) is F.

The equivalence of (iii) and (iv) follows from 4.3.12. The other equivalences will follow from the following lemmas.

Lemma 4.4.4. *Let $f : X \to Y$ be a monomorphism of sheaves which is an epimorphism. Then f is an isomorphism.*

The lemma is first clear in the category of sets. Let us prove it next in the category of presheaves. Consider the presheaf

$V : S \mapsto Y(S) \amalg_{X(S)} Y(S);$

it is the amalgamated sum of Y and Y under X in the category of presheaves.[306] Denote by i_1 and i_2 the two "coordinate" morphisms $Y \to V$. If $X \to Y$ is an epimorphism in the category of presheaves, then $i_1 = i_2$. In this case, for each S, the map $X(S) \to Y(S)$ is surjective; as it is also injective, it is a bijection, and therefore $X \to Y$ is an isomorphism.

[305] N.D.E.: The original has been modified here.
[306] N.D.E.: The continuation of the proof has been slightly modified.

Let us place ourselves finally in the category $\tilde{C}$ of sheaves. By 4.4.1 (ii), the amalgamated sum in $\tilde{C}$ of the sheaves Y and Y under X is the sheaf Z associated to the presheaf V. Consider the diagram of morphisms:

$$X \xrightarrow{f} Y \rightrightarrows_{i_1, i_2} V \xrightarrow{\tau} Z = a(V).$$

One has $i_1 f = i_2 f$, whence $\tau i_1 f = \tau i_2 f$, and therefore $\tau i_1 = \tau i_2$, since f is an epimorphism in $\tilde{C}$. By point (iii) of the lemma below, the presheaf V is separated, hence τ is a monomorphism (4.3.11 (iv)). Therefore $i_1 = i_2$, and we have seen above that this

entails that $f : X \to Y$ is an isomorphism. This proves Lemma 4.4.4, once one has verified that V is separated.

[307] Let $R_\emptyset$ be the presheaf that to every $S \in ObC$ associates the empty set; it is an initial object of $\hat{C}$.

Lemma 4.4.5. *(i) Suppose that $R_\emptyset \notin J(S)$, for every $S \in ObC$. If (X_i) is a family of separated presheaves, the direct sum presheaf $\coprod_i X_i$ is separated.*[308]

(ii) Consider an equivalence relation in the category of presheaves:

$$X \rightrightarrows_{u, v} Y$$

and let $w : Y \to Z$ be the quotient. If X is a sheaf and Y separated, then Z is separated.

(iii) Consider an amalgamated sum in the category of presheaves, where u and u' are monomorphisms:

$$\begin{tikzcd} X \arrow[r, "u"] \arrow[d, "u'"'] & Y \arrow[d] \\ Y' \arrow[r] & V. \end{tikzcd}$$

If Y and Y' are separated, and X a sheaf, then V is separated.

(i) Set $X = \coprod_i X_i$. Let $S \in ObC$ and $\tau : R \hookrightarrow S$ a refinement of S, and let x_1, x_2 be two elements of $X(S)$ such that $x_1 \circ \tau = x_2 \circ \tau$; there exist indices i, j such that $x_1 \in X_i(S)$ and $x_2 \in X_j(S)$. Since $R \neq R_\emptyset$, there exists a morphism $\phi : T \to R$, with $T \in ObC$. Then, $x_1 \circ \tau \circ \phi = x_2 \circ \tau \circ \phi$, and since $X(T)$ is the disjoint union of the $X_k(T)$, this entails $i = j$. Then, since X_i is separated and a subobject of X, the map $X_i(S) \to X_i(R) \to X(R)$ is injective, and therefore $x_1 = x_2$. This proves that X is separated.

[307] N.D.E.: Point (i) of Lemma 4.4.5 has been corrected, and the proof of the three points detailed.

[308] N.D.E.: In general, the direct sum of two sheaves F, G is not a sheaf. Indeed, let $S_1, S_2 \in ObC$; suppose that the direct sum $S = S_1 \coprod S_2$ exists in C and that the fiber product $S_1 \times_S S_2$ is an initial object $\emptyset$ of C (cf. I, 1.8). Let R be the sieve of S with base S_1, S_2; then $(F \coprod G)(R)$ is the disjoint union of $F(S) \coprod G(S)$ and of $F(S_i) \times G(S_j)$ for $i \neq j$, hence $F \coprod G$ is not a sheaf in general. On the other hand, if C is the category with a single object S and id_S as sole morphism, equipped with the topology defined by $J(S) = R_\emptyset, S$, then the only separated presheaves are $R_\emptyset$ and $X = h_S$ (which is a sheaf), and $X \coprod X$ is not separated.

Let us prove (iii). Consider the morphisms $i : Y \to V$ and $j : Y' \to V$, and let K be the kernel of:

$Y \times Y' \rightrightarrows_{p,\, q} V,$

where $p = i \circ pr_1$ and $q = j \circ pr_2$.

Let $S \in Ob(C)$. Since u and u' are monomorphisms, one can identify $X(S)$ with its image in $Y(S)$ (resp. $Y'(S)$); denote by $Z(S)$ (resp. $Z'(S)$) the complement. Then

$V(S)$ is identified with the disjoint union of $Z(S)$, $Z'(S)$ and $X(S)$, and one easily sees that the maps $i(S) : Y(S) \to V(S)$ and $j(S) : Y'(S) \to V(S)$ are injective, and that the map

$(u \times u')(S) : X(S) \to K(S)$

is bijective. Consequently, i and j are monomorphisms, and $u \times u' : X \to K$ is an isomorphism.

Set $U = Y \coprod Y'$, and let $\tau : R \hookrightarrow S$ be a refinement of S; one has a commutative diagram:

$$U(S) \xrightarrow{U(\tau)} U(R) \downarrow\downarrow V(S) \xrightarrow{V(\tau)} V(R).$$

Let $v_1, v_2 \in V(S)$ whose images in $V(R)$ coincide. By the definition of V, v_1, v_2 lift to elements y_1, y_2 of $U(S)$; denote by z_1, z_2 their images in $U(R)$. Then z_1, z_2 have the same image in $V(R)$.

Since $i : Y \to V$ and $j : Y' \to V$ are monomorphisms, the maps $Y(R) \to V(R)$ and $Y'(R) \to V(R)$ are injective. Therefore, since Y and Y' are separated, if y_1 and y_2 both belong to $Y(S)$ or to $Y'(S)$, then $y_1 = y_2$. Otherwise, one may suppose that $y_1 \in Y(S)$ and $y_2 \in Y'(S)$, whence $z_1 \in Y(R)$ and $z_2 \in Y'(R)$. But then, since $i \circ z_1 = j \circ z_2$, the morphism $z_1 \boxtimes z_2 : R \to Y \times Y'$ factors through $K = X$. Moreover, since $X(R) = X(S)$ (because X is a sheaf), there exists $x \in X(S)$ such that $u(x) \circ \tau = z_1 = y_1 \circ \tau$ and $u'(x) \circ \tau = z_2 = y_2 \circ \tau$, whence, since Y and Y' are separated, $u(x) = y_1$ and $u'(x) = y_2$, and therefore $v_1 = v_2$. This proves that V is separated.

Let us prove (ii). Let us first remark that the morphism of presheaves

$$X \xrightarrow{u \boxtimes v} K,$$

where K denotes the kernel of the pair of morphisms $w \circ pr_i : Y \times Y \to Z$, is an isomorphism. Indeed, for every $T \in ObC$, $X(T)$ is an equivalence relation in $Y(T)$, so that the diagram

$X(T) \xrightarrow{u \boxtimes v} Y(T) \times Y(T) \rightrightarrows_{w \circ pr_1,\, w \circ pr_2} Z(T)$

is exact in the category of sets.

Let now $S \in ObC$ and $g_1, g_2 : S \to Z$ be two morphisms that coincide on a refinement $\tau : R \hookrightarrow S$ of S. Since $S \in ObC$, one has $Z(S) = Y(S)/X(S)$,

by construction of Z, and therefore there exist morphisms $f_1, f_2 : S \to Y$ such that $wf_i = g_i$.

Then, $wf_1\tau = wf_2\tau$ and therefore, by what precedes, there exists a morphism $\phi : R \to X$ such that $u\phi = f_1\tau$ and $v\phi = f_2\tau$. Since X is a sheaf, there exists $\psi : S \to X$ such that $\phi = \psi\tau$, and therefore one has in $Y(R)$ the equalities:

$$u\,\psi\,\tau = f_1\,\tau, \qquad v\,\psi\,\tau = f_2\,\tau.$$

Since $Y(S) \to Y(R)$ is injective (Y being separated), this entails $u\psi = f_1$ and $v\psi = f_2$, whence it follows that $g_1 = g_2$. This proves that Z is separated.

Lemma 4.4.6.[309] *(i) Let $F_i \xrightarrow{f_i} F$ be a family of morphisms of presheaves, and let $G \subset F$ be the presheaf image. Then the following diagram in $\hat{C}$ is exact:*

$$\coprod_i F_i \times_G \coprod_j F_j \rightrightarrows_{\{pr_1,\, pr_2\}} \coprod_i F_i \to G$$

i.e., for every presheaf H, the following diagram of sets is exact:

$$(*) \qquad \mathrm{Hom}(G, H) \to \prod_i \mathrm{Hom}(F_i, H) \rightrightarrows_{\{p,\, q\}} \prod_{\{i,j\}} \mathrm{Hom}(F_i \times_G F_j, H).$$

(ii) Every covering family of morphisms of sheaves is universally effective epimorphic.

(i) Let H be a presheaf. The map $f*$ which to a morphism $\phi : G \to H$ associates the family of morphisms $\phi \circ f_i : F_i \to H$ is injective, since for every $S \in ObC$, $\phi(S)$ is determined by the $(\phi \circ f_i)(S)$, since the family $f_i(S) : F_i(S) \to G(S)$ is surjective. It is clear that the image of $f*$ is contained in $Ker(p, q)$. Conversely, let $\phi_i : F_i \to H$ be a family of morphisms such that, for every i, j, the diagram below is commutative:

$$\begin{tikzcd} F_i \times_G F_j \arrow[r] \arrow[d] & F_j \arrow[d, "\varphi_j"] \\ F_i \arrow[r, "\varphi_i"] & H. \end{tikzcd}$$

Then, for every S, the map $\coprod_i F_i(S) \to H(S)$ factors uniquely as a map $\phi(S) : G(S) \to H(S)$, and this defines a morphism $\phi : G \to H$ such that $f*(\phi) = (\phi_i)$. This proves the exactness of the sequence $(*)$, and point (i).

Let us prove (ii). Since the notion of covering family is stable under base extension, it suffices to show that every covering family is effective epimorphic. So let $F_i \to F$ be a covering family of morphisms of sheaves, and let G be the presheaf image of this family. Since the family is covering, so is the monomorphism $G \hookrightarrow F$, hence, by 4.3.12, one has $a(G) = F$.

On the other hand, since $G \hookrightarrow F$ is a monomorphism, the fiber products $F_i \times_G F_j$ and $F_i \times_F F_j$ are the same. Hence, by (i), the following diagram of sets is exact, for every presheaf H:

[309] N.D.E.: The statement of the lemma and its proof have been detailed.

(∗∗) $\mathrm{Hom}(G, H) \to \prod_i \mathrm{Hom}(F_i, H) \rightrightarrows_{p,q} \prod_{i,j} \mathrm{Hom}(F_i \times_F F_j, H)$.

If moreover H is a sheaf, one has

$\mathrm{Hom}(G, H) = \mathrm{Hom}(a(G), H) = \mathrm{Hom}(F, H)$,

and then (∗∗) shows that $F_i \to F$ is indeed an effective epimorphic family in the category $\tilde{C}$ of sheaves.

Lemma 4.4.7. *Every family of morphisms of sheaves $F_i \to F$ factors as a covering family $F_i \to G$ and a monomorphism $G \to F$.*

It suffices in fact to take for G the sheaf image of the given family.

Proof of Proposition 4.4.3: one has seen in 4.4.6 that (iii) ⇒ (ii), and one evidently has (ii) ⇒ (i). Let finally $F_i \to F$ be an epimorphic family; by Lemma 4.4.7, it factors as a covering family followed by a monomorphism. But the latter, being dominated[310] by an epimorphic family, is an epimorphism, hence an isomorphism by 4.4.4.

Remark 4.4.8.[311] *As the presheaf image of the family F is separated, the construction of the associated sheaf shows that the conditions of Proposition 4.4.3 are also equivalent to the following:*

(v) For every $S \in ObC$, every $f \in F(S)$ is locally in the image of F, that is:

(vi) For every $S \in ObC$ and every $f \in F(S)$, the set of $S' \to S$ such that the image of f in $F(S')$ is in the image of one of the $F_i(S')$ is a refinement of S.

(vii) (If the topology is defined by a pretopology R). For every $S \in ObC$ and every $f \in F(S)$, there exists a family $S_j \to S \in R(S)$ such that for every j the image f_j of f in $F(S_j)$ is in the image of one of the $F_i(S_j)$.

Remark 4.4.8.bis. *If the sheaf F is representable, the preceding conditions are also equivalent to:*

(viii) The set of $T \to F$ ($T \in ObC$), such that there exist an i and a commutative diagram

$$\begin{tikzcd} F_i \arrow[r] & F \\ T \arrow[u] \arrow[ur] & \end{tikzcd}$$

is a refinement of F.

Indeed, if (viii) is satisfied, the presheaf image of the F_i is dominated by[312] a refinement of F, which entails that the family is covering. Conversely, one applies (vi) to $id_F \in F(F)$.

[310] N.D.E.: Recall (cf. N.D.E. (17)) that one says that a morphism $g : G \to H$ is *dominated* by a family of morphisms $F_i \to H$ if this family factors through g.
[311] N.D.E.: In order to be in accord with later references, the numbering of the original, which contained two nos. 4.4.7, has been corrected.
[312] N.D.E.: "contains" has been replaced by: "is dominated by", cf. N.D.E. (17).

This condition is expressed in pictorial language in the following way: *locally on F, there exists an i such that $F_i \to F$ has a section*. In particular, a morphism

$G \to F$, where G is a sheaf and F a representable sheaf, will be covering if and only if it has *locally (on F) a section*.

[313] The following lemmas will be useful in VI_B, 8.1 and 8.2.

Lemma 4.4.8.1. *Let $Q \subset P$ be presheaves of groups on C, Q being normal in P. Suppose that P is separated, and that Q is a sheaf. Then the presheaf of groups P/Q is separated. Consequently, one has*

$$a(P/Q) = L(P/Q) = \varinjlim_{R \in J(S)} (P/Q)(R).$$

It suffices to show the first assertion, since the second follows from it, by 4.3.11 (iv) and (ii). Let $S \in ObC$ and R a sieve of S, and let $y \in P(S)/Q(S)$ whose image in $(P/Q)(R)$ is the identity. One must show that $y = 1$. Now, by hypothesis, the morphism $P(S) \to P(R)$ (resp. $Q(S) \to Q(R)$) is injective (resp. an isomorphism), and in the commutative diagram below, the top row is exact:

$$\begin{tikzcd} 1 \arrow[r] & Q(S) \arrow[r] \arrow[d, "\wr"] & P(S) \arrow[r] \arrow[d] & P(S)/Q(S) \arrow[r] \arrow[d] & 1 \\ 1 \arrow[r] & Q(R) \arrow[r] & P(R) \arrow[r] & (P/Q)(R). \end{tikzcd}$$

The result will follow from this, if one shows that the bottom row is exact. Let $f : R \to P$ whose image in $(P/Q)(R)$ is the identity, and let $\phi : T \to R$ with $T \in ObC$. Then $f \circ \phi$ is the identity of $(P/Q)(T) = P(T)/Q(T)$, i.e. $f \circ \phi \in Q(T)$. Therefore, $\varphi \mapsto f \circ \varphi$ is a functorial map $R(T) \to Q(T)$, hence defines a morphism of functors $\pi : R \to Q$ such that $\pi(id_R) = f$, whence $f \in Q(R)$. This proves the exactness of the bottom row, and the lemma is proved.

Lemma 4.4.8.2. *Let $H \subset G$ be presheaves of groups on C.*

(i) If H is normal in G, then $L(H)$ is normal in $L(G)$.

(ii) If H is central in G, then $L(H)$ is central in $L(G)$.

Let $S \in ObC$; it must be shown (cf. I 2.3.6) that $L(H)(S)$ is normal (resp. central) in $L(G)(S)$. Let $h \in L(H)(S)$ and $g \in L(G)(S)$; there exist a sieve R of S and elements $h' \in H(R)$, $g' \in G(R)$, such that $h = z_R(h')$ and $g = z_R(g')$ (notation of 4.3.10). Since z_R is a morphism of groups, one has $ghg^{-1}h^{-1} = z_R(g'h'g'^{-1}h'^{-1})$.

In case (i), one has $g'h'g'^{-1}h'^{-1} \in H(R)$, whence $ghg^{-1}h^{-1} \in LH(S)$; in case (ii), $g'h'g'^{-1}h'^{-1} = 1$ and therefore $ghg^{-1}h^{-1} = 1$.

Proposition 4.4.9. *Every equivalence relation in $\tilde{C}$ is universally effective (3.3.3): let R be a $\tilde{C}$-equivalence relation in the sheaf X; then the sheaf associated to the separated presheaf*

$$i(X)/i(R) : S \mapsto X(S)/R(S)$$

[313] N.D.E.: Lemmas 4.4.8.1 and 4.4.8.2 have been added.

is a universal effective quotient of X by R.

Let X/R be the sheaf quotient of X by R, which exists by 4.4.1 (ii): $X/R = a(i(X)/i(R))$. We must show that $X \to X/R$ is a universal effective epimorphism, and that the morphism $f : R \to X \times_{X/R} X$ is an isomorphism. The first assertion has already been proved (4.4.3). As for f, it comes by application of the functor a from the morphism $i(R) \to i(X) \times_{i(X/R)} i(X)$ or, since $i(X)/i(R)$ is separated (4.4.5 (ii)) so that $i(X)/i(R) \to i(X/R)$ is a monomorphism, from the canonical morphism $i(R) \to i(X) \times_{i(X)/i(R)} i(X)$.

One is therefore reduced to proving the same assertion in the category of presheaves. But $i(X)/i(R)$ is the presheaf $S \mapsto X(S)/R(S)$ and one is reduced to proving the analogous assertion in the category of sets, where it is immediate.

Proposition 4.4.10. *Under the conditions of 4.4.9, let Y be a subsheaf of X. Denote by R_Y the equivalence relation induced in Y by R. Then the canonical morphism (3.1.6)*

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism: it identifies Y/R_Y with a subsheaf of X/R, which is the sheaf image *of the composite morphism*

$$Y \longrightarrow X \longrightarrow X/R.$$

The morphism of presheaves

$$i(Y)/i(R_Y) = i(Y)/i(R)_{i(Y)} \longrightarrow i(X)/i(R)$$

is a monomorphism. Since the functor a is left exact (4.3.16), it transforms monomorphisms into monomorphisms, and therefore

$$Y/R_Y \longrightarrow X/R$$

is a monomorphism. The last assertion follows from the commutative diagram

$$Y \to X \downarrow\downarrow Y/R_Y \to X/R,$$

and the fact that $Y \to Y/R_Y$ is covering.

By virtue of this proposition, we shall always identify Y/R_Y with a subsheaf of X/R.

Proposition 4.4.11. *Let R be a $\tilde{C}$-equivalence relation in the sheaf X. For every subsheaf Y of X stable under R, denote by Y' the quotient Y/R_Y considered as a subsheaf of $X' = X/R$. Then $Y = Y' \times_{X'} X$, and the maps $Y \mapsto Y/R_Y$ and $Y' \mapsto Y'$*

×_{X′} *X realize a bijective correspondence between the set of subsheaves Y of X stable under R and the set of subsheaves Y' of X'.*

If Y' is a subsheaf of X', then $Y' \times_{X'} X$ is a subsheaf of X stable

under R.[314] If Y' is obtained by passage to the quotient from a subsheaf Y of X, then Y is a subobject of $Y' \times_{X'} X$. It therefore suffices to show that if one has two subsheaves Y and Y_1 of X, stable under R, Y_1 containing Y, and if the quotients Y/R_Y and Y_1/R_{Y_1} are identical, then $Y = Y_1$. One is evidently reduced to proving the same assertion in the case where $Y_1 = X$. Denoting then by P (resp. Q) the presheaf $i(X)/i(R)$ (resp. $i(Y)/i(R_Y)$), the diagram

$$X \to P \uparrow\uparrow Y \to Q$$

is cartesian. Since one has a commutative diagram

$$P \to a(P) \uparrow\uparrow Q \to a(Q),$$

and since $Q \hookrightarrow a(Q)$ is covering (4.3.11), the monomorphism $Q \hookrightarrow P$ is covering, hence Q is a refinement of P. By base change, Y is a refinement of X. Since X and Y are sheaves, this entails (4.3.12) $Y = X$.

4.4.12. In particular, if Y is a subsheaf of X, and if $Y' = Y/R_Y$, then the preceding correspondence defines a subsheaf $\bar{Y}$ of X, stable under R, containing Y and minimum for these properties, which one calls the *saturation* of Y for the equivalence relation R.

Footnotes (chunk-A)

4.5. The case of a topology coarser than the canonical topology

By 4.3.6 and 4.3.8, the following conditions are equivalent for a topology T on C:

(i) T is coarser than the canonical topology of C.

(ii) Every representable presheaf is a sheaf for T.

(iii) Every covering sieve for T is universal effective epimorphic.

If T is defined by a pretopology $S \mapsto R(S)$, these conditions are further equivalent to

(iv) Every family belonging to $R(S)$ is universal effective epimorphic.

In the case where these conditions are satisfied, the canonical functor $C \to \hat{C}$ factors through a functor $j_C = j : C \to \tilde{C}$ (we shall also write $j(S) = \tilde{S}$[315]).

[314] N.D.E.: and one has $(Y' \times_{X'} X)/R = Y'$.
[315] N.D.E.: and we shall also write $h_S = \hat{S}$, cf. the first commutative diagram of 4.5.4.

Proposition 4.5.1. *The functor $j : C \to \tilde{C}$ is fully faithful and commutes with arbitrary inverse limits. It is in particular left exact and therefore preserves the algebraic structures defined by finite inverse limits.*

This follows at once from consideration of the commutative diagram

$$\begin{tikzcd} & C \arrow[dl, "j"'] \arrow[dr, "\text{(canonical)}"] & \\ \tilde{C} \arrow[rr, "i"] & & \hat{C} \end{tikzcd}$$

and from 4.4.1 (i).

Before exhibiting other properties of the functor j, we need to define the topology induced on a category C/S. No longer assuming the given topology necessarily coarser than the canonical one, this is done as follows: if C is a sieve of T in C and one has a morphism $T \to S$, then C defines naturally a sieve of T in C/S, denoted $J(T)$ (since the definition of a sieve of T depends only on the category $C/T = (C/S)/T$). If, for example, C is defined by the family $T_i \to T$, then its image in C/S is defined by the same family considered as a family of morphisms of C/S. This being said, the map $T \mapsto J(T)$ defines a topology on C/S called the *topology induced* by the given topology. With the definitions of [AS], 2.3, it is the coarsest of the topologies on C/S for which the canonical functor

$$i_S : C/S \longrightarrow C$$

is a comorphism[316]. One will note that the identifications

$$(C/S)/T = C/T$$

respect the topologies by definition.

Proposition 4.5.2. *Let S be a representable sheaf on C and $F \to S$ a morphism of $\hat{C}$. In order for $S' \mapsto \mathrm{Hom}_S(S', F)$ to be a separated presheaf (resp. a sheaf) on C/S, it is necessary and sufficient that F be a separated presheaf (resp. a sheaf) on C.*

For every functor P, one has (I 1.4.1)

$$\mathrm{Hom}(P, F) = \coprod_{f \in \mathrm{Hom}(P, S)} \mathrm{Hom}_f(P, F).$$

[317]

Let $S' \in ObC$ and $\eta : C' \to S'$ a covering sieve. As S is a sheaf, the map $f \mapsto f \circ \eta$ establishes a bijection $\mathrm{Hom}(S', S) \xrightarrow{\sim} \mathrm{Hom}(C', S)$. Consequently, the map

$$\mathrm{Hom}_{\hat{C}}(S', F) \to \mathrm{Hom}_{\hat{C}}(C', F)$$

[316]N.D.E.: i.e., such that for every object $T \to S$ of C/S, every covering sieve R' of $i_S(T \to S) = T$, considered as a sieve of $(T \to S) \in ObC/S$, is covering.

[317]N.D.E.: We have expanded the original in what follows.

decomposes as a "disjoint union" of the maps

$$\mathrm{Hom}_f(S', F) \xrightarrow{\sim} \mathrm{Hom}_{f \circ \eta}(C', F).$$

The proposition follows.

Corollary 4.5.3. *The topology induced on C/S by a topology on C coarser than the canonical topology of C, is coarser than the canonical topology of C/S.*

Corollary 4.5.4. *Suppose the given topology on C is coarser than the canonical topology. For every $S \in Ob C$, one has an equivalence of categories*

$$\tilde{C}/\tilde{S} \xrightarrow{\sim} \tilde{C}/S.$$

The following diagrams are commutative up to isomorphism (all unlabelled arrows being equivalences):

$$\begin{tikzcd}
C/S \arrow[r, "(j_C)/S"] \arrow[d, equal] & \tilde{C}/\tilde{S} \arrow[r, "(i_C)/\tilde{S}"] \arrow[d] & \hat{C}/\hat{S} \arrow[r, "(a_C)/\hat{S}"] \arrow[d] & \tilde{C}/\tilde{S} \arrow[d] \\
C/S \arrow[r, "j_{C/S}"] & \tilde{C}/S \arrow[r, "i_{C/S}"] & \hat{C}/S \arrow[r, "a_{C/S}"] & \tilde{C}/S
\end{tikzcd}$$

$$\begin{tikzcd}
 & \tilde{T} \arrow[dl] \arrow[dr] & \\
\tilde{C}/\tilde{S} \arrow[dr, "\simeq"] & & \tilde{C}/S/\tilde{T} \\
 & \tilde{C}/\tilde{T} \arrow[r] \arrow[ur] & \tilde{C}/T
\end{tikzcd}$$

The commutativity of the first two squares results from the definition of the equivalence $\tilde{C}/\tilde{S} \to \tilde{C}/S$. To prove the commutativity of the last, one must see that the following square is commutative:

$$\begin{tikzcd}
\hat{C}/\hat{S} \arrow[r, "L'"] \arrow[d] & \hat{C}/\hat{S} \arrow[d] \\
\hat{C}/S \arrow[r, "L''"] & \hat{C}/S
\end{tikzcd},$$

where L' is the restriction of the functor L_C to $\hat{C}/\hat{S}$ and L'' the functor $L_{C/S}$; this is easily seen by going back to the definition of the functors L (given after 4.3.9). As for the second diagram, this is none other than the restriction to the sheaf categories of the corresponding diagram on the categories of presheaves (Exposé I, n° 1), which is commutative.

Scholie 4.5.5. *The various assertions of this number show that, in the case where the given topology is coarser than the canonical topology, one may identify C with a full subcategory of $\tilde{C}$, itself a full subcategory of $\hat{C}$, and that under this identification one may indulge in the usual abuses of language regarding $C \to \hat{C}$, justified by the commutativities above. Let us explicitly remark that the first diagram of 4.5.4 shows that one may use the functor a without special precaution.*

We shall see in the next number that the identification of C with a full subcategory of $\tilde{C}$ (contrary to what was the case for $\hat{C}$) commutes with the formation of certain direct limits, and we shall then say how to use this fact.

From now on, and unless explicitly stated otherwise, we shall assume the given topology coarser than the canonical topology and we shall systematically make the identifications described above.

Proposition 4.5.6. *Let F and G be two sheaves over S and $f : F \to G$ an S-morphism. The following conditions on f are equivalent:*

(i) f is a monomorphism (resp. an epimorphism, resp. an isomorphism) in $\tilde{C}$.

(ii) f is a monomorphism (resp. an epimorphism, resp. an isomorphism) in $\tilde{C}/S = \tilde{C}/S$.

For monomorphism and isomorphism, this is evident (it is a question of presheaves). For epimorphism, it follows from the description of epimorphisms as covering morphisms (4.4.3) and from the fact that, by definition of the induced topology, these are the same in $\tilde{C}$ and $\tilde{C}/S$.

Proposition 4.5.7. *Let $f : F \to G$ be a morphism of sheaves. The following conditions are equivalent:*

(i) f is a monomorphism (resp. an epimorphism, resp. an isomorphism).

(ii) For each $S \in ObC$, $f_S : F_S \to G_S$ is locally a monomorphism (resp. an epimorphism, resp. an isomorphism), that is:

(iii) For each $S \in ObC$, the set of $T \to S$ such that $F_T \to G_T$ is a monomorphism (resp. an epimorphism, resp. an isomorphism) is a refinement of S.

If the given topology is defined by a pretopology R, these conditions are further equivalent to the following:

(iv) For each $S \in ObC$, there exists a covering family $S_i \to S \in R(S)$ such that for every i, $F_{S_i} \to G_{S_i}$ is a monomorphism (resp. an epimorphism, resp. an isomorphism).

If the category C has a final object e, one may content oneself with taking $S = e$ in conditions (ii), (iii), and (iv).

One obviously has (ii) ⇔ (iii) ⇔ (iv). To prove the equivalence of (i) and (ii) as well as the supplement concerning the final object, one must show that (ii) ⇒ (i) and that the notions in question are stable under base extension. Let us first prove this last point. For monomorphism and isomorphism, this is evident (it is a question of presheaves). For epimorphism, it follows from the fact that every epimorphism of sheaves is universal (4.4.3).

Let us finally show that (ii) entails (i). Suppose first that $f_S : F_S \to G_S$ is locally a monomorphism (resp. an isomorphism). There then exists a covering sieve C of S such that for every $T \to C$, f_T is a monomorphism (resp. an isomorphism). As an inverse limit of monomorphisms (resp. isomorphisms) is one, $\mathrm{Hom}(C, f)$ will be a monomorphism (resp. an isomorphism) (cf. 4.1.4). The commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(C, F) \arrow[r, "\mathrm{Hom}(C, f)"] & \mathrm{Hom}(C, G) \\ F(S) \arrow[u] \arrow[r, "f(S)"] & G(S) \arrow[u] \end{tikzcd}$$

then shows that $f(S)$ is injective (resp. bijective). Suppose finally that $f : F \to G$ is locally an epimorphism and let $H \subset G$ be its image. For each $S \to G$, $H \times_G S$ is the image of $f \times_G S : F \times_G S \to S$. To show that f is an epimorphism, one must show that H is a refinement of G, that is, that $H \times_G S$ is a refinement of S for each S. But since this is so after every base change $T \to S$ of a refinement of S (since $f \times_G S$ is locally covering), $H \times_G S$ is indeed a refinement of S (Axiom (T 2)).

Corollary 4.5.8. *Let F and G be two sheaves over S and $f : F \to G$ an S-morphism. In order for f to be a monomorphism, resp. an epimorphism, resp. an isomorphism, it is necessary and sufficient that it be so locally on S.*

Remark 4.5.9. *The proof of the proposition shows that it remains valid, for the part concerning monomorphisms (resp. isomorphisms), when one only assumes that F is a separated presheaf (resp. a sheaf) and G an arbitrary presheaf (resp. a separated presheaf).*

Let us provisionally return to the case of an arbitrary topology and lay down a definition.

Definition 4.5.10. *Let $G \to F$ be a morphism of $\hat{C}$. We say that G is a* relative sheaf above F *if for every F-functor H and every refinement R of H, the canonical map*

$$(+) \qquad \mathrm{Hom}_F(H, G) \to \mathrm{Hom}_F(R, G)$$

is bijective.

Proposition 4.5.2 immediately generalizes:

Proposition 4.5.11. *If F is a sheaf, G is a relative sheaf above F if and only if it is a sheaf.*

Lemma 4.5.12. *In the situation $X \to T \to S$ (where X, T, S are three objects of $\hat{C}$), if X is a relative sheaf above T, then $U = \prod_{T/S} X$ is a relative sheaf above S.*

Indeed, one has for every S-functor Y

$$\mathrm{Hom}_S(Y, U) = \mathrm{Hom}_T(T \times_S Y, X).$$

If C is a sieve of Y, then $T \times_S C$ is a sieve of $T \times_S Y$; one concludes at once.

Corollary 4.5.13. *The presheaves* $\mathrm{Hom}_{T/S}(X,Y)$*,* $Isom_S(X,Y)$*, etc., are sheaves when the arguments appearing in them are also.*

Indeed, all these presheaves are constructed by means of fibered products and presheaves $\prod_{T/S} X$ (I 1.7 and II 1). It therefore suffices to verify the result for a presheaf $\prod_{T/S} X$; in this case, the assertion follows from 4.5.11 and 4.5.12.

4.6. Description of the quotient of a sheaf by an equivalence relation

Recall that we are assuming the topology T given to be coarser than the canonical topology.

Proposition 4.6.1. *Let $R \rightrightarrows X$ (with morphisms p_1, p_2) be a $\tilde{C}$-equivalence relation in the sheaf X. Let $F \in Ob\hat{C}$ be defined as follows: for each S of C,*

```
F(S) = { sub-S-sheaves Z of X_S stable under R × S[^N.D.E-IV-43], whose quotient by R_Z is S,
         i.e. such that the diagram R_Z ⇉ Z → S is exact }.
```

Then for every sheaf Y, $\mathrm{Hom}(Y,F)$ *is identified with the set:*

```
{ sub-Y-sheaves of X × Y stable under R × Y and whose quotient is Y }.
```

In particular the sub-sheaf R of $X \times X$ corresponds to an element p of $\mathrm{Hom}(X,F)$ *and the diagram*

```
R ⇉ X →^p F
```

is exact, hence identifies F with the sheaf-quotient X/R.

Indeed, set $Q = X/R$. For every sheaf Y and every morphism $f \in \mathrm{Hom}(Y,Q)$ corresponding to a section $s : Y \hookrightarrow Q \times Y$, consider the diagram

```
R × Y ⇉ X × Y ⟶ Q × Y
                 ↑     ↑
(∗)              |     | s
                 |     |
                 Z ⟶  Y
```

where the square is cartesian. It is immediate by 4.4.11 that Z is a sub-Y-sheaf of $X \times Y$, stable under $R \times Y$, whose quotient is Y, and that, conversely, every Z of this type comes from a unique section of $Q \times Y$ over Y. Taking first Y representable, one extracts an isomorphism $Q \simeq F$. Taking next Y arbitrary, one extracts the announced form of $\mathrm{Hom}(Y,F)$. Considering finally the canonical morphism $X \to Q$, one sees at once that it corresponds to the sub-X-sheaf R of $X \times X$, which completes the proof.

Corollary 4.6.2. *Let G be an arbitrary subfunctor of F such that* $\mathrm{Hom}(X,G) \subset \mathrm{Hom}(X,F)$ *contains R. Then the canonical morphism $p : X \to F$ factors through G.*

Since p is covering (4.4.9 and 4.4.3), it follows that G is a refinement of F. In particular, every sub-sheaf G of F satisfying the preceding condition is equal to F (4.3.12).

4.6.3. We are now going to interest ourselves in the case where X and R are representable. Let us first introduce some terminology. Besides the conditions (a) to (d) introduced in 3.4.1, we shall use other conditions on a family $(\mathtt{M})$ of morphisms of C, which we state below, recalling the conditions (a) to (c) already given, for completeness.

(a) $(\mathtt{M})$ is stable under base extension.

(b) The composite of two elements of $(\mathtt{M})$ is in $(\mathtt{M})$.

(c) Every isomorphism is an element of $(\mathtt{M})$.

(d_T) Every element of $(\mathtt{M})$ is covering.[318]

(e_T) Let $f : X \to Y$ be a morphism of C. If there exists a refinement R of Y such that for every $Y' \to R$, $X \times_Y Y' \to Y'$ is an element of $(\mathtt{M})$, then f is an element of $(\mathtt{M})$.

Recall that (a) and (b) entail

(a′) The cartesian product of two elements of $(\mathtt{M})$ is an element of $(\mathtt{M})$.

On the other hand, (a) and (d_T) entail by 4.3.9:

(d′) Every element of $(\mathtt{M})$ is a universal effective epimorphism.

4.6.4. The preceding conditions are verified by the family of covering morphisms, denoted $(\mathtt{M\_T})$, when C has fibered products. Indeed (cf. 4.2.3), (a) follows from (C 1), (b) from (C 2), (c) from (C 4), (d_T) from the definition, (e_T) from (C 5). The results we shall establish for a family satisfying these conditions will apply in particular to the family $(\mathtt{M\_T})$. In particular, one may take for T the canonical topology and for $(\mathtt{M})$ the family of universal effective epimorphisms.

Lemma 4.6.5. *Let $(\mathtt{M})$ be a family of morphisms satisfying the properties (a) to (e_T) above. Let R be a C-equivalence relation in $X \in Ob C$, of type $(\mathtt{M})$. Let $\tilde{X}$ be the sheaf defined by X, $\tilde{R}$ the $\tilde{C}$-equivalence relation in $\tilde{X}$ defined by R and $\tilde{X}/\tilde{R}$ the sheaf-quotient. In order for R to be $(\mathtt{M})$-effective, it is necessary and sufficient that $\tilde{X}/\tilde{R}$ be representable. If this is so, $\tilde{X}/\tilde{R}$ is represented by the quotient X/R.*

Suppose first that R is $(\mathtt{M})$-effective and denote $Y = X/R$. The canonical morphism $p : X \to Y$ is an element of $(\mathtt{M})$, hence covering by (d_T). The corresponding morphism

$$\tilde{p} : \tilde{X} \longrightarrow \tilde{X}/\tilde{R}$$

[318] N.D.E.: Recall that T denotes the given topology on C, coarser than the canonical topology.

is therefore a universal effective epimorphism of $\tilde{C}$ (4.4.3), hence identifies $\tilde{X}/\tilde{R}$ with the quotient of $\tilde{X}$ by the equivalence relation R' defined in $\tilde{X}$ by $\tilde{p}$. As the canonical functor $C \to \tilde{C}$ commutes with fibered products, R' is none other than $\tilde{R}$, since R is the equivalence relation defined by p.

Conversely, suppose $\tilde{X}/\tilde{R}$ representable by an object Y of C. Let $p : X \to Y$ be the morphism deduced from the canonical morphism $\tilde{X} \to \tilde{X}/\tilde{R}$; it is a covering morphism by 4.4.3. It is clear as before that R is the equivalence relation defined by p. It only remains to show that $p \in (M)$. Now the cartesian square

$$\begin{tikzcd} R \simeq X \times_Y X \arrow[r] \arrow[d, "pr_2"'] & X \arrow[d, "p"] \\ X \arrow[r] & Y \end{tikzcd}$$

shows that p becomes p_2, which is an element of (M), after base change by the covering morphism p. One concludes by (e_T).

Corollary 4.6.5.1.[319] *Let (M) satisfy the properties (a) to (e_T) above and let $f : G \to G'$ be a morphism of C-groups, such that $f \in (M)$. Suppose $Ker(f)$ representable (which is the case if C has a final object e). Then the equivalence relation in G defined by $H = Ker(f)$ is (M)-effective and G' represents the sheaf-quotient $\tilde{G}/\tilde{H}$ for the topology T.*

This follows from 4.6.5 and 3.4.7.1.

We are now in a position to state the main result of this number.

Theorem 4.6.6. *Let (M) be a family of morphisms verifying the axioms (a) to (e_T) of 4.6.3. Let R be a C-equivalence relation of type (M) (cf. 3.4.3) in the object X of C. Consider the functor $F \in Ob\hat{C}$ defined as follows:*

$$F(S) = \{ \text{sub-}S\text{-sheaves } Z \text{ of } X_S \text{ stable under } R \times S \text{ whose quotient by } R_Z \text{ is } S \}.$$

Let F_0 be the subfunctor of F defined as follows: $F_0(S)$ consists of the $Z \in F(S)$ that are representable, that is:

$$\begin{aligned} F_0(S) = \{ & \text{sub-}C/S\text{-objects } Z \text{ of } X_S \text{ stable under } R \times S, \text{ such that} \\ & R_Z \text{ is (M)-effective and has quotient } S \text{ (i.e. such that} \\ & Z \to S \text{ belongs to (M) and } R_Z \simeq Z \times_S Z) \}. \end{aligned}$$

Then:

(i) The morphism $p \in \mathrm{Hom}(X, F) = F(X)$ defined by the sub-object R of $X \times X$ identifies F with the sheaf-quotient of X by R.

(ii) The following conditions are equivalent:

- *a) F is representable.*
- *b) F_0 is representable.*

[319] N.D.E.: We have added this corollary.

- *c) R is (M)-effective.*

Under these conditions, $F = F_0 = X/R$.

(iii) Let (N) be a family of morphisms stable under base change, such that for every covering family $S_i \to S$ and every family $T_i \to S_i$ of morphisms of (N), every descent datum on the T_i relative to $S_i \to S$ is effective. Suppose X squarable (cf. 1.6.0) and the morphism $R \to X \times X$ an element of (N). Then $F_0 = F$.

Proof. (i) has already been proved (4.6.1).

(ii) We have seen the equivalence of a) and c) as well as the equality $F = X/R$. It remains to prove that b) or c) implies $F_0 = F$. Let us first remark, as is moreover affirmed in the statement, that F_0 is indeed a subfunctor of F; for every $S \in ObC$ and every $Z \in F_0(S)$, the morphism $Z \to S$ is squarable, hence $Z \times_S S'$ is an element of $F_0(S')$ for every $S' \to S$. As $R \in F(X)$ belongs to $F_0(X)$, 4.6.2 shows that b) implies $F_0 = F$.

Now suppose c) verified and let Q be an object of C representing X/R. Then the morphism $X \to Q$ is an element of (M) and, for every $S \in ObC$ and every $Z \in F(S)$, the diagram $(*)$ of 4.6.1 shows that $Z = S \times_{Q\times S} X \times S$ is representable, and $Z \to S$ belongs to (M), hence $Z \in F_0(S)$.

(iii) Let $f \in \mathrm{Hom}(S, F)$ correspond to $Z \in F(S)$. We must show that f factors through F_0, that is, that Z is representable. This is clear first if f factors through X, by virtue of:

Lemma 4.6.7. *Let $x_0 \in X(S)$. The image of x_0 in $F(S)$ corresponds to the sub-sheaf Z of X_S defined by the two cartesian squares*

$$\begin{tikzcd}
X_S \arrow[r, "\mathrm{id}_{X_S} \times x_0"] & X_S \times_S X_S \arrow[r] & X \times X \\
Z \arrow[u] \arrow[r] & R_S \arrow[u] \arrow[r] & R \arrow[u]
\end{tikzcd}$$

This lemma follows at once from the description of the morphism $X \to F$.

Let us return to the proof of the theorem. If f factors through X, then Z is representable and, as $R \to X \times X$ is an element of (N), the same holds for $Z \to X_S$.

In general, f does not necessarily factor through X; but since $X \to F$ is covering (4.4.3), there exists by 4.4.8 (vii) a covering family $S_i \to S$ and for each i a morphism $S_i \to X$ making commutative the diagram

$$\begin{tikzcd}
X \arrow[r] & F \\
S_i \arrow[u] \arrow[r] & S \arrow[u, "f"'] .
\end{tikzcd}$$

By what precedes, the morphism $f_i : S_i \to F$ defined by the preceding diagram belongs to $\mathrm{Hom}(S_i, F_0)$ and corresponds to the sub-sheaf $Z \times_S S_i$ of X_{S_i}. The morphism

$Z \times_S S_i \to X_{S_i}$ is an element of (N) and the family $X_{S_i} \to X_S$ covering. It therefore only remains to establish:

Proposition 4.6.8. *Let $S_i \to S$ be a covering family and Z a sheaf above S. Suppose that for each i, the S_i-functor $Z \times_S S_i$ is representable by an object T_i. Then the family of T_i is equipped with a canonical descent datum relative to $S_i \to S$. In order for Z to be representable, it is necessary and sufficient that this datum be effective; if this is so, the "descended" object represents Z.*

Let us first remark that by 4.4.3, $S_i \to S$ is a universal effective epimorphic family in $\tilde{C}$, hence a descent family in $\tilde{C}$ (2.3). If Z is representable by the object T, then $T \times_S S_i$ (considered as a sheaf) is isomorphic to $Z \times_S S_i$, hence the descent datum on the T_i is effective and the (unique) descended object is isomorphic to Z. Conversely, suppose that the canonical descent datum on the T_i is effective and let T be the descended object. Since the family $S_i \to S$ is a descent family in $\tilde{C}$, there exists an S-morphism $T \to Z$ which by base extension to each S_i recovers the canonical morphism $T_i \to Z \times_S S_i$. This morphism is locally an isomorphism; as T and Z are sheaves, it follows from 4.5.8 that it is an isomorphism.

Corollary 4.6.9. *Let R be an (M)-effective equivalence relation in X. For every sheaf F, the map*

$\mathrm{Hom}(X/R, F) \to \mathrm{Hom}(X, F)$

identifies the first set with the part of the second consisting of the morphisms compatible with R.

Corollary 4.6.10. *Let T' be a topology coarser than T, for which the morphisms of (M) are covering. Under the conditions of 4.6.6 (iii), X/R is also the sheaf-quotient of X by R in every intermediate topology between T' and the canonical topology.*

Remark 4.6.11. *If in the statement of 4.6.6 (iii), one furthermore assumes that, under the hypotheses of the text, if one denotes T the descended object, the morphism $T \to S$ is an element of (N), then the inclusion morphisms $Z \hookrightarrow X_S$ are also elements of (N), as follows at once from the construction of Z by descent.*

Remark 4.6.12. *The implications c) ⇒ b) ⇒ a) and c) ⇒ [F_0 = F = X/R] have been established without recourse to the "if" part of Lemma 4.6.5, which is the only place where condition (e_T) is used. They therefore remain valid if (M) satisfies only conditions (a) to (d_T). An example of such a family (M) is that of squarable covering morphisms (compare with 4.6.4). In the case of the canonical topology, these are none other than the universal effective epimorphisms. One therefore has:*

Corollary 4.6.13. *Let R be a universal effective equivalence relation in X. Then the object X/R of C is the sheaf-quotient of X by R for the canonical topology. It represents the following functor: $(X/R)(S)$ is the set of sub-C/S-objects Z of X_S stable under $R \times S$ and such that the induced equivalence relation is universal effective and has S as quotient.*

Similarly, for an arbitrary topology:

Corollary 4.6.14. *Let* (M) *be the family of squarable covering morphisms. If R is an* (M)*-effective equivalence relation in X, then the object X/R of C is the sheaf-quotient of X by R and represents the functor* F_0 *of 4.6.6.*

Scholie 4.6.15. *We can now bring the following precisions to 4.5.5.* Whereas in questions involving exclusively inverse limits (fibered products, algebraic structures, etc.), one may, by the results of Exposé I and 4.5.5, identify C indifferently with a full subcategory of $\tilde{C}$ or of $\hat{C}$, it is not the same in those that mix inverse and direct limits. In all questions involving both inverse and direct limits, in particular passages to the quotient (example: group structure on the quotient of a group by an invariant subgroup), we shall consider the given category as embedded in the category of sheaves; thus if R is a C-equivalence relation in the object X of C, X/R will denote the sheaf-quotient of X by R (previously denoted $j(X)/j(R)$), hence in the case where this sheaf is representable, the object it represents. The preceding results show that in the most important cases, a quotient in C will also be a quotient in the category of sheaves; in any case, we forbid ourselves the use of the notation X/R for a quotient in C that does not coincide with the quotient in $\tilde{C}$ (for example one that is not universal), thus modifying the definitions of n° 3.

To study a problem of the type above, one therefore places oneself first in the category of sheaves, where all the usual results are valid (cf. n° 4.4), then one specializes the results obtained to the original category, using the results of the present number and, when one has them, descent effectivity criteria. We shall see examples of this method in the following numbers.

4.7. Use of effectivity criteria: isomorphism theorem

In this number, we give an example of the use of effectivity criteria. The data of departure are a topology T on C (always coarser than the canonical topology), a family (M) of morphisms of C verifying the axioms (a) to (e_T) of 4.6.3, and a family (N) of morphisms of C liable to verify the following axioms:

(a) (N) is stable under base extension.

(f_T) "the morphisms of (N) descend by the given topology"; that is: for every $S \in ObC$, every covering family $S_i \to S$ and every family $T_i \to S_i$ of morphisms of (N), every descent datum on the T_i relative to $S_i \to S$ is effective, and if one denotes T the descended object, the morphism $T \to S$ is an element of (N).

Since every element of (M) is covering (condition 4.6.3 (d_T)), (f_T) entails the following axiom:[320]

(f_M) If $Y \to X$ is an element of (N) and $X \to X_1$ an element of (M), every descent datum on Y relative to $X \to X_1$ is effective; if one denotes Y_1 the descended object, $Y_1 \to X_1$ is an element of (N).

Let us at once signal an example of this situation, which will be treated later: C is the category of schemes, T the faithfully flat quasi-compact topology; (M) the family

[320] N.D.E.: We have placed the axiom (f_M) here (which figured before Proposition 4.7.2).

of faithfully flat quasi-compact morphisms, (N) the family of closed immersions, or that of quasi-compact immersions.[321]

Let us recall the principal result of 4.6.6 (taking into account 4.6.11):

Proposition 4.7.1. *If X is a squarable object of C, R an equivalence relation of type (M) in X, such that $R \to X \times X$ is an element of (N), with (N) verifying (a) and (f_T), then the sheaf-quotient X/R is defined by*

```
(X/R)(S) = { sub-S-objects Z of X_S, stable under R × S, such that Z → X_S
             belongs to (N), Z → S is covering (or an element of (M)),
             and R_Z ≃ Z ×_S Z }.
```

Moreover, one has:

Proposition 4.7.2. *Let $X \in ObC$ and R an (M)-effective equivalence relation in X. Let (N) be a family of morphisms verifying (a) and (f_M).*

For every sub-object Y of X, stable under R and such that $Y \to X$ belongs to (N), the equivalence relation induced in Y by R is (M)-effective and the quotient $Y/R_Y = Y'$ is a sub-object of $X' = X/R$ such that $Y' \to X'$ belongs to (N).

The map $Y \mapsto Y'$ is a bijection between the set of sub-objects Y of X, stable under R, such that $Y \to X$ belongs to (N), and the set of sub-objects Y' of X' such that $Y' \to X'$ belongs to (N). The inverse map is $Y' \mapsto Y' \times_{X'} X$.

Proof. As R is (M)-effective, the morphism $X \to X'$ belongs to (M). Let Y' be a sub-object of X' such that the canonical morphism $Y' \to X'$ belongs to (N). Then, the sub-object $Y = Y' \times_{X'} X$ of X is stable under R, and the morphism $Y \to X$ (resp. $Y \to Y'$) belongs to (N) (resp. (M)) since (N) and (M) are stable under base change. Let R_Y denote the equivalence relation induced in Y by R. By 4.4.11, the sheaf quotient Y/R_Y is represented by Y' and therefore, by 4.6.5, R_Y is (M)-effective.

Conversely, let us show that every sub-object Y of X, stable under R, such that the structural morphism $Y \to X$ belongs to (N), is obtained in this way. Indeed, if Y is stable under R, its two images in $R = X \times_{X'} X$ are identical and Y is equipped with a descent datum relative to $X \to X'$; the desired result follows, since the family (N) verifies the axiom (f_M).

Corollary 4.7.3. *Let $X \in ObC$ and R an (M)-effective equivalence relation in X; assume moreover that $R \to X \times X$ belongs to (N), where (N) verifies (a) and (f_T). Then, for every Y as in 4.7.2, $R_Y \to Y \times Y$ also belongs to (N) and therefore, by 4.7.1, one has:*

```
(Y/R_Y)(S) = { sub-S-objects Z of Y_S, stable under R_Y × S, such that Z → Y_S
               belongs to (N) (then Z → X_S also belongs to it), Z → S is covering,
               and R_Z ≃ Z ×_S Z }.
```

[321] N.D.E.: cf. § 6.4, see also VI_A, 5.3.1.

5. Passage to the quotient and algebraic structures

5.1. Principal homogeneous bundles

Definition 5.1.0.[322] *We recall (III 0.1) that an object X with (right) operator group H is said to be* formally principal homogeneous[323] *under H if the canonical morphism (of functors)*

$X \times H \to X \times X$

defined by $(x, h) \mapsto (x, xh)$ is an isomorphism. It amounts to the same thing to say (cf. loc. cit.) that for every $S \in Ob C$, $X(S)$ is formally principal homogeneous under $H(S)$, that is, empty or principal homogeneous under $H(S)$. In particular, if H is made to operate on itself by (right) translations, H becomes formally principal homogeneous under itself.

Definition 5.1.1. *The object X with operator group H is said to be* trivial *if it is isomorphic (as object with operator group H) to H on which H operates by translations.*

Proposition 5.1.2. *Let X be formally principal homogeneous under H. One has an isomorphism*

$$\Gamma(X) \xrightarrow{\sim} Isom_{H-obj.}(H, X)$$

of principal homogeneous sets under $\Gamma(H)$.

To every section x of X one associates the morphism from H to X defined setwise by $h \mapsto xh$. The stated assertion is immediate, by reduction to the set-theoretic case.

Corollary 5.1.3. *One has an isomorphism of objects with operators H*

$$X \xrightarrow{\sim} Isom_{H-obj.}(H, X).$$

Corollary 5.1.4. *In order for an object with operator group to be trivial, it is necessary and sufficient that it be formally principal homogeneous and possess a section.*

Definition 5.1.5. *Let C be a category equipped with a topology. We say that the S-object X with S-operator group H is a* principal homogeneous bundle under H *if it is locally trivial, that is, if the following equivalent conditions are satisfied:*[324]

(i) The set of $T \to S$ such that (the functor) $X \times_S T$ is trivial under $H \times_S T$ is a refinement of S.

[322]N.D.E.: We have introduced the numbering 5.1.0, to refer to it later.
[323]N.D.E.: One also says "pseudo-torsor", cf. EGA IV_4, 16.5.15. On the other hand, the more general notion of formally homogeneous object (not necessarily principal homogeneous), is defined in the addendum 6.7.1 at the end of this Exposé.
[324]N.D.E.: In this case, one also says that X is an H-torsor.

(ii) There exists a covering family (for a pretopology defining the given topology) $S_i \to S$ such that for each i, the S_i-functor $X \times_S S_i$ with S_i-functor-group of operators $H \times_S S_i$ is trivial (= has a section over S_i).

Proposition 5.1.6. *Let C be a category equipped with a topology T. Let (M) be a family of morphisms of C verifying the axioms (a) to (e_T) of 4.6.3. Let H be an S-group such that the structural morphism $H \to S$ is an element of (M) and P an S-object with S-operator group H. The following conditions are equivalent:*

(i) P is a principal homogeneous bundle under H (Definition 5.1.5).

(ii) P is formally principal homogeneous under H and the structural morphism $P \to S$ is an element of (M).

(iii) There exists a morphism $S' \to S$ element of (M) such that by base extension from S to S', P becomes trivial, that is, $P \times_S S'$ is trivial under $H \times_S S'$.

(iv) H operates freely on P, in an (M)-effective manner, and the quotient P/H is isomorphic to S.

Let us first remark that (ii) and (iv) are equivalent, taking into account that, in either case, $P \to S$ is an element of (M), hence squarable, which ensures the representability of the fibered products $H \times_S P$ and $P \times_S P$. It is clear that (ii) entails (iii), as one can take P itself as S', the hypothesis that P is formally principal homogeneous entailing that $P \times_S P$ is trivial under $H \times_S P$ (5.1.4), since it has a section (the diagonal section). It is clear that (iii) entails (i), since $S' \to S$ is a covering family, by axiom (d_T). It therefore remains to show that (i) entails (ii). The morphism of sheaves $P \times_S H \to P \times_S P$ is locally an isomorphism, hence an isomorphism (4.5.8); P is therefore formally principal homogeneous. The structural morphism $P \to S$ is locally isomorphic to the structural morphism $H \to S$ which is an element of (M). It is therefore itself an element of (M) by (e_T).

The equivalence between (i) and (iv) generalizes:

Proposition 5.1.7. *Under the same hypotheses on C and (M), let H be an S-group and X an S-object on which H operates (on the right). Suppose the structural morphism $H \to S$ is an element of (M). The following conditions are equivalent:*

(i) H operates freely on X and in an (M)-effective manner.

(ii) There exists an S-morphism $p : X \to Y$ compatible with the equivalence relation defined in X by the action of H and such that the operation of $H \times_S Y$ on X above Y thus deduced makes X a principal homogeneous bundle under H_Y above Y.

Under these conditions p identifies Y with the quotient X/H.

If $p : X \to Y$ is a morphism compatible with the action of H, then the operation of $H \times_S Y$ on X above Y thus deduced defines in X the same equivalence relation as the action of H, by virtue of the formula

H_Y ×_Y X H ×_S X.

The proposition follows from this remark and from the equivalence (iv) ⇔ (i) above.

Corollary 5.1.7.1.[325] *Let C be a category having a final object, stable under fibered products, and equipped with a topology T coarser than the canonical topology. Let $f : G \to H$ be a morphism of C-groups, and $K = Ker(f)$. Assume f covering for the topology T.*

Then H represents the sheaf quotient G/K, and f is a `K_H`-torsor. (N.B. One will also say that: "G is a K-torsor above H".)

Indeed, as f is covering, it is a universal effective epimorphism (4.4.3), hence by 3.3.3.1, H is the quotient of G by the equivalence relation $R(f) = G \times_H G$. On the other hand, the morphism $G \times K \to G \times_H G$, $(g,k) \mapsto (g, gk)$ is an isomorphism of objects with operator group $K_G = G \times_H K_H$ (its inverse being given by $(g,g') \mapsto (g, g^{-1}g')$). Hence, on the one hand, $R(f)$ is the equivalence relation defined by K; on the other hand, since the morphism $f : G \to H$ is covering, f is a `K_H`-torsor, by 5.1.6 (ii) (or directly by definition 5.1.5 (ii)).

We can now make Theorem 4.6.6 more precise in the case of passage to the quotient by an operator group:

Proposition 5.1.8. *Under the hypotheses of 5.1.7, let `F_0` denote the functor over S defined as follows: for each $S' \to S$, $F_0(S')$ is the set of representable sub-S'-functors Z of $X \times_S S'$, stable under $H \times_S S'$ and being principal homogeneous bundles under this S'-group for the induced action (3.2.2).*

(i) The following conditions are equivalent:

- *a) The operation of H on X is `(M)`-effective and free.*[326]
- *b) `F_0` is representable.*

Under these conditions, one has $F_0 = X/H$.

(ii) Let `(N)` be a family of morphisms, stable under base change, such that for every covering family $S'_i \to S'$ and every family $T_i \to S'_i$ of morphisms of `(N)`, every descent datum on the T_i relative to $S'_i \to S'$ is effective. Assume the morphism $X \times_S H \to X \times_S X$ is an element of `(N)` and X squarable. Then the element p of $\mathrm{Hom}(X, F_0)$ *corresponding to the sub-object $X \times_S H$ of $X \times_S X$ identifies `F_0` with the sheaf-quotient X/H.*

5.2. Group structures and passage to the quotient

In this number we interest ourselves in the algebraic structures one can place on the quotient G/H of a group by a subgroup. We shall first place ourselves in the category of sheaves on C for an arbitrary topology. By taking the canonical topology and using 4.5.12, we shall obtain results for the universal effective passage to the quotient in C.

[325] N.D.E.: We have added this particular case as a corollary, which will be used several times in the following Exposés.
[326] N.D.E.: We have added "and free".

Proposition 5.2.1. *Let $u : H \to G$ be a monomorphism of sheaves of groups. There exists on the sheaf-quotient G/H a unique structure of object with operator group G such that the canonical morphism*

$$p : G \longrightarrow G/H$$

is a morphism of objects with operator group G. This structure is functorial with respect to the pair (G, H): if one has a commutative diagram

$$\begin{tikzcd} H \arrow[r] \arrow[d] & G \arrow[d, "f"] \\ H' \arrow[r] & G' \end{tikzcd},$$

the morphism $f : G/H \to G'/H'$ (3.2.3) is compatible with the morphism f on the operator groups.

Indeed, the sheaf G/H is the sheaf associated with the presheaf

$$i(G)/i(H) : S \mapsto G(S)/H(S);$$

as the functor a is left exact, it transforms objects with operator groups into objects with operator group. As the presheaf $i(G)/i(H)$ is equipped with a structure of object with operator groups $i(G)$, then $G/H = a(i(G)/i(H))$ is equipped with a structure of object with operators $a(i(G)) = G$. This structure obviously enjoys all the stated properties.

Corollary 5.2.2. *Let $u : H \to G$ be a monomorphism of C-groups. Suppose that the operation of H on G is universal effective. There exists on the object-quotient $G/H \in ObC$ a unique structure of object with operator group G such that $p : G \to G/H$ is a morphism of objects with operators. This structure is functorial in the pair (H, G) (with H operating in a universal effective manner in G), in the preceding sense.*

Proposition 5.2.3. *Let $u : H \to G$ be a monomorphism of sheaves of groups identifying H with an invariant sub-sheaf of groups of G. There exists on the sheaf-quotient G/H a unique structure of sheaf of groups such that the canonical morphism $p : G \to G/H$ is a morphism of groups. This structure is functorial in the pair (H, G) (with H invariant).*

The proof is similar to that of 5.2.1.

Corollary 5.2.4. *Let $u : H \to G$ be a monomorphism of C-groups identifying H with an invariant subgroup of G. Suppose that the action of H on G is universal effective. There exists on the object-quotient $G/H \in ObC$ a unique group structure*

such that the canonical morphism $G \to G/H$ is a morphism of groups. This structure is functorial with respect to the pair (H, G) (H invariant, H operating in a universal effective manner).

One can characterize the group structure of G/H in a more telling manner:

Proposition 5.2.5. *Under the conditions of 5.2.4, let K be a C-group and $f : G \to K$ a morphism. The following conditions are equivalent:*

(i) f is a morphism of groups compatible with the equivalence relation defined by H.

(ii) f is a morphism of groups inducing the trivial morphism $H \to K$.

(iii) f factors as a morphism of groups $G/H \to K$.

In particular, one has an isomorphism, functorial in the group K

$$\mathrm{Hom}_{C\text{-gr.}}(G/H, K) \simeq \{f \in \mathrm{Hom}_{C\text{-gr.}}(G, K) \mid f \circ u = e\}.$$

The equivalence of (i) and (ii) is proved set-theoretically. One obviously has (iii) ⇒ (ii). The equivalence of (iii) and (ii) follows from the formula

$$\mathrm{Hom}(G/H, K) \simeq \mathrm{Hom}(i(G)/i(H), K)$$

and from the definition of the group structure of G/H.

Remark 5.2.6. *In the preceding situation, if the kernel of f is exactly H, the morphism $G/H \to K$ that factors f is a monomorphism. This follows at once from 3.3.4.*

In the case of sheaves of groups, one can make 4.4.11 more precise by means of the

Proposition 5.2.7. *Let G be a sheaf of groups, H an invariant sub-sheaf of groups. For every sub-sheaf of groups K of G containing H, let K' be the quotient group K/H considered as a subgroup of $G' = G/H$.*

One has $K = K' \times_{G'} G$, and the maps $K \mapsto K/H$ and $K' \mapsto K' \times_{G'} G$ realize a bijection between the set of sub-sheaves of groups of G containing H and the set of sub-sheaves of groups of G'. In this correspondence, the invariant sub-sheaves of groups of G and of G' correspond.

The first part follows easily from 4.4.11 and 3.2.4. It remains to see that K is invariant in G if and only if K' is invariant in G'. If K is invariant in G, then the presheaf $i(K)/i(H)$ is invariant in $i(G)/i(H)$. The same holds for the associated sheaves, by virtue of the usual argument. If conversely K' is invariant in G', then the fibered product $K \times_{G'} G$ is invariant in G, as one sees immediately.

If now L is an arbitrary sub-sheaf of groups of G, let $\bar{L}$ be the saturation of L for the equivalence relation defined by H; we shall also write $\bar{L} = L \cdot H$.

Proposition 5.2.8. *Under the preceding conditions, $L \cdot H$ is a sub-sheaf of groups of G containing H and the image of L in G/H is identified with*

$$(L \cdot H)/H \simeq L/(H \cap L).$$

Indeed, let L' denote the sheaf image of L in G/H. It is a sub-sheaf of groups of G/H corresponding to $L \cdot H$ in the correspondence of the preceding proposition. As the morphism $L \to L'$ is covering, hence a universal effective epimorphism of sheaves, it follows from 4.4.9 that L' is identified with the quotient of L by the kernel of $L \to L'$ which is obviously none other than $H \cap L$.

Let us finally consider the following situation: we have a sheaf of groups G, a sub-sheaf of groups K and a sub-sheaf of groups H of K, invariant in K. Let us first define a (right) operation of the sheaf of groups $H \backslash K$ $(= K/H)$ on G/H. The group K operates by right translations on G. As H is invariant in K, this operation is compatible with the equivalence relation defined by the action of H and therefore defines an operation of K on G/H, that is, a morphism from the opposite group K° to K into $\mathrm{Aut}(G/H)$. As the latter is a sheaf (4.5.13) and this morphism is trivial on H, it factors through K/H and defines the sought operation. As the right and left operations of G on itself commute, the operations of G and K/H on G/H commute.

Proposition 5.2.9. *Under the preceding conditions, K/H operates freely (on the right) on G/H and one has a canonical isomorphism of sheaves with operator group G*

$$(G/H)/(K/H) \simeq G/K.$$

When K is invariant in G, in which case K/H is invariant in G/H (5.2.7), this isomorphism respects the group structures of the two sides.

One has an isomorphism of presheaves

$$(i(G)/i(H))/(i(K)/i(H)) \xrightarrow{\sim} i(G)/i(K),$$

which respects the structures of objects with operator group $i(G)$. The announced result is obtained by applying the functor a to this relation.

Corollary 5.2.10. *Let G be a C-group, K a sub-C-group of G, H an invariant sub-C-group of K. Let (M) be a family of morphisms of C verifying the axioms (a) to (e_T). Suppose the operation of H on G (resp. K) on the right is (M)-effective. Then K/H operates in a natural manner freely on the right on G/H; this operation commutes with that of G. The following conditions are equivalent:*

(i) The operation of K on G is (M)-effective.

(ii) The operation of K/H on G/H is (M)-effective.

Under these conditions, one has an isomorphism of objects[327] *with operator group G:*

$$(G/H)/(K/H) \simeq G/K.$$

[327]N.D.E.: of C.

5.3. Use of effectivity criteria: Noether's theorem[328]

Let C, T and (M) be as usual. Let (N) be a family of morphisms verifying the axioms (a) and (f_M) of 4.7. Putting together 5.2.7 and 4.7.2, one obtains:

Proposition 5.3.1. *Let G be a C-group. Let H be a sub-C-group of G, invariant and operating in an (M)-effective manner in G.*

For every sub-C-group K of G containing H and such that the morphism $K \to G$ belongs to (N), H operates in K in an (M)-effective manner and the quotient $K/H = K'$ is a sub-C-group of $G/H = G'$ such that the morphism $K' \to G'$ belongs to (N).

The map $K \mapsto K'$ is a bijection between the set of sub-C-groups K of G, containing H and such that $K \to G$ belongs to (N), and the set of sub-C-groups K' of G' such that $K' \to G'$ belongs to (N). The inverse map is $K' \mapsto K \times_{G'} G$. In this correspondence, the invariant subgroups of G and G' correspond.

Corollary 5.3.2. *If $H \to G$ is an element of (N), then C has a final object e and the unit section $e \to G/H$ is an element of (N).*[329]

6. Topologies in the category of schemes

6.1. The Zariski topology

This is the topology generated by the following pretopology: a family of morphisms $S_i \to S$ is covering if each morphism is an open immersion and the union of the images of the S_i is all of S. It is denoted (Zar).

Definition 6.1.1. *A sheaf for the Zariski topology is also called* a functor of local nature*: it is a contravariant functor from (Sch) to (Ens) such that for every scheme S and every covering of S by opens S_i, one has an exact diagram:*

F(S) → ∏_i F(S_i) ⇒ ∏_{i,j} F(S_i ∩ S_j).

In particular, a functor of local nature transforms direct sums into products. As every representable functor is a sheaf, this topology is coarser than the canonical topology.

From the terminological point of view, whenever we say "local", "locally", without precision, it will be with reference to the Zariski topology, hence in the usual sense.

6.2. A procedure for constructing topologies

Proposition 6.2.1. *Let C be a category, C' a full subcategory, P a set of families of morphisms of C with the same target, stable under base change and under composition (i.e. verifying the axioms (P 1) and (P 2) of 4.2.5), P' a set of families of morphisms of C' containing the families reduced to an identity isomorphism.*

[328] N.D.E.: In fact, the isomorphisms established in 5.2.8 to 5.2.10 would deserve to be called "Noether isomorphism theorems".

[329] N.D.E.: This follows from the proposition applied to $K = H$.

Equip C with the topology generated by P and P' (cf. 4.2.5.0) and suppose the three conditions below are verified:

(a) If $S_i \to S \in P'$ (hence $S_i, S \in Ob C'$) and if $T \to S$ is a morphism of C', then the fibered products $S_i \times_S T$ (in C) exist and the family $S_i \times_S T \to T$ belongs to P' (hence $S_i \times_S T \in Ob C'$). (Remark: this condition entails that P' is stable under base change in C', but is not equivalent to it, since it further supposes that the inclusion functor from C' to C commutes with certain fibered products).

(b) For every $S \in Ob C$, there exists $S_i \to S \in P$ with $S_i \in Ob C'$ for each i.

(c) In the following situation:

```
              S_{ijk}
                │ (P′)
                ↓
        (P)  S_{ij}
   S_i ←──── │
        (P′) │
             ↓
             S ,
```

where $S, S_i, S_{ij}, S_{ijk} \in Ob C'$; $S_i \to S \in P'$; $S_{ij} \to S_i \in P$ for each i; $S_{ijk} \to S_{ij} \in P'$ for each ij, there exists a family $T_n \to S \in P'$ and for each n a multi-index (ijk) and a commutative diagram

```
S_{ijk} ←──── T_n
       \     /
        \   /
         \ /
          S .
```

Then, in order for a sieve R of $S \in Ob C$ to be covering, it is necessary and sufficient that there exist a composite family

```
R ←┈┈┈┈ S_{ij}
        │ (P′)
        ↓
   S ⟵ S_i
        (P)
```

where $S_i, S_{ij} \in Ob C'$, $S_i \to S \in P$, $S_{ij} \to S_i \in P'$ for each i, and the morphisms $S_{ij} \to S$ so obtained factor through R (in other words, the sieve generated by this composite family is contained in R).

Proof. As the families elements of P and of P' are covering, a family composed of such families will be also (C 2), hence a sieve of the indicated form will be covering, since it contains a covering sieve.

Conversely, it suffices to see that the sieves of this form do indeed form a topology, i.e. it suffices to verify the axioms (T 1) to (T 4) of 4.2.1.

Axiom (T 4). Let $S \in ObC$. There exists by (b) a family $S_i \to S \in P$ with $S_i \in ObC'$. The families $S_i \to^{id_{S_i}} S_i$ are elements of P' by hypothesis. The sieve S of S is therefore of the desired form:

```
S_i ───→ S
 │  (P′) id    id_S
 ↓  (P)
S_i ───→ S .
```

Axiom (T 3). Evident.

Axiom (T 2). Let R be a sieve of S of the desired form and let C be a sieve[330] of S such that, for every $T \in ObC$ and every morphism $T \to S$ factoring through R, the sieve $C \times_S T$ of T is of the desired form. Then, as $S_{ij} \to S$ factors through R, the sieve $C_{ij} = C \times_S S_{ij}$ of S_{ij}:

```
        S_{ij} ←─── C_{ij}
       /  (P′)
      /
    S_i
     │ (P)
     ↓
     S ←──── C
     ↑
     R
```

is of the desired form; hence, for each `ij`, one has a diagram of the form:

```
S_{ijkl}
   │ (P′)
   ↓
S_{ijk}
   │
   │ (P)
   ↓
S_{ij} ←──── C_{ij} .
```

One has therefore proved that there exists a composite family

```
S_{ijkl} ──(P′)⟶ S_{ijk} ──(P)⟶ S_{ij} ──(P′)⟶ S_i ──(P)⟶ S
```

belonging to $P \circ P' \circ P \circ P'$, factoring through C, and where all objects other than S are in C'. Applying condition (c) to each family $S_{ijkl} \to S_i$, one deduces that for each i there exists a family $T_{in} \to S_i \in P'$, such that $T_{in} \to S$ factors through one of the S_{ijkl}, hence through C:

```
T_{in} ──→ S_{ijkl}
   │ (P′)
   ↓
```

[330] N.D.E.: We have corrected the original here.

```
S_i
│ (P)
↓
S ←──── C .
```

The sieve C of S is therefore of the desired form, which completes the verification.

Axiom (T 1). Let R be a sieve of S of the given form and let $T \to S$ be a morphism of C. Let us show that the sieve $R \times_S T$ of T is of the desired form.

```
      S_{ij} ←──────── U_{ikj}
         │ (P′)             (P′)
         ↓                  ↓
S_i ←─────────── T_i ←─(P)── U_{ik}
         │                     /
         │ (P)          (P)   / (P)
         ↓                   /
R ──→ S ←──────── T ←───────
```

Let $T_i = S_i \times_S T$. The family $T_i \to T$ belongs to P (by (P 1)). Applying (b), one constructs $U_{ik} \to T_i \in P$, with the $U_{ik} \in ObC'$. By hypothesis (condition (P 2) on P), one has $U_{ik} \to T \in P$. By (a), $U_{ik} \times_{S_i} S_{ij} = U_{ikj}$ is an object of C' and for each ik, $U_{ikj} \to U_{ik} \in P'$. Then, the commutative diagram below

```
R ←──────── U_{ikj}
              │ (P′)
              ↓
              U_{ik}
              │ (P)
              ↓
S ←──── T ←──
```

shows that the morphisms $U_{ijk} \to T$ factor through the sieve $T \times_S R$ of T, which is therefore of the desired form, which completes the proof.

Corollary 6.2.2. *If $S \in ObC'$ and if R is a sieve of S, R is covering if and only if there exists a family $T_i \to S \in P'$, factoring through R.*

Indeed, such a sieve is covering. On the other hand, it suffices to apply (c) by taking the family $S_i \to S$ reduced to the identity isomorphism of S to deduce from the proposition that a covering sieve is of the indicated form.

Corollary 6.2.3. *In order for a presheaf F on C to be separated (resp. a sheaf), it is necessary and sufficient that the morphism*

$$F(S) \longrightarrow \prod_i F(S_i)$$

be injective (resp. that the diagram

$$F(S) \to \prod_i F(S_i) \rightrightarrows \prod_{i,j} F(S_i \times_S S_j)$$

be exact) in the two following cases:

(i) $S_i \to S \in P$*,*

(ii) $S, S_i \in Ob C'$*;* $S_i \to S \in P'$*.*

Indeed, the conditions are necessary, since the families in question are covering. If C is the sieve of S image of a family of morphisms $S_{ij} \to^{(P')} S_i \to^{(P)} S$, a straightforward diagram chase shows that the conditions of the corollary entail that $\mathrm{Hom}(S, F) \to^{g} \mathrm{Hom}(C, F)$ is injective (resp. bijective). But every refinement R of S contains a sieve C of the above type and one has a commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}(S, F) \arrow[rr, "f"] \arrow[dr, "g"'] & & \mathrm{Hom}(R, F) \arrow[dl, "h"] \\
& \mathrm{Hom}(C, F) . &
\end{tikzcd}$$

One knows that g is injective, hence so is f. Therefore F is separated. But R is a refinement of C, hence h is also injective. If g is bijective, then f is also, hence F is a sheaf.

Remark 6.2.4. *The preceding corollary does not follow from 4.3.5, because* P' *is not stable under base extension.*

Remark 6.2.5. *Condition (c) is verified in particular in the case where*

(i) P' *is stable under composition.*

(ii) If $S_j \to S$ *is a family of morphisms of* C'*, element of* P*, there exists a subfamily element of* P'*.*

6.3. Application to the category of schemes

One takes for C the category of schemes, for C' the full subcategory formed by affine schemes, for P the set of surjective families of open immersions. One will consider several sets P':

P'_1 : finite surjective families, composed of flat morphisms.

P'_2 : finite surjective families, composed of flat morphisms of finite presentation and quasi-finite.[331]

[331] N.D.E.: Let P''_2 denote the set of finite surjective families, composed of flat morphisms of C' of finite presentation. According to Proposition 6.3.1 below, the topology T_2 generated by P and P'_2 coincides with the topology generated by P and P''_2. This follows from the results of EGA IV_4, § 17.16 on quasi-sections; see the proof of 6.3.1. Let us cite here the following particular case of EGA IV_4, 17.16.2: let S be affine and $f : X \to S$ a surjective flat morphism locally of finite presentation; then there exists a morphism $S' \to S$ faithfully flat, of finite presentation, quasi-finite, with S' affine, and an S-morphism $S' \to X$.

P'_3 : finite surjective families, composed of étale morphisms.

P'_4 : finite surjective families, composed of étale and finite morphisms.

For each of these sets P'_i, except P'_4, the conditions of Proposition 6.2.1 are verified ((c) thanks to 6.2.5, since an affine scheme being quasi-compact, every family of morphisms of C', element of P, contains a finite subfamily that is also in P, hence in P'_i for $i = 1, 2, 3$). The topology T_i generated by P and P'_i is denoted and called in the following way:

```
T_1 = (fpqc) = faithfully flat quasi-compact topology.
T_2 = (fppf) = faithfully flat (locally) of finite presentation topology.
T_3 = (ét)  = étale topology.
T_4 = (étf) = étale finite topology.
```

As $P'_1 \supset P'_2 \supset P'_3 \supset P'_4$, one has

```
(fpqc) ≥ (fppf) ≥ (ét) ≥ (étf) ≥ (Zar).
```

Proposition 6.3.1. *(i) In order for the sieve R of S to be covering for T_i, $1 \leq i \leq 3$, it is necessary and sufficient that there exist a covering (S_p) of S by affine opens and for each p a family $S_{pq} \to S_p$ element of P'_i, the S_{pq} being affine, such that each morphism $S_{pq} \to S$ factors through R.*[332]

(ii) In order for a presheaf F on (Sch) to be a sheaf for (fpqc) (resp. (fppf), (ét), (étf)), it is necessary and sufficient that

- *a) F be a sheaf for (Zar), i.e. a functor of local nature.*
- *b) For every faithfully flat (resp. faithfully flat of finite presentation and quasi-finite, resp. surjective étale, resp. surjective étale finite) morphism $T \to S$, where T and S are affine, one has an exact diagram:*

```
F(S)  F(T) ⇒ F(T ×_S T).
```

(iii) The topologies T_i, $i = 1, 2, 3, 4$, are coarser than the canonical topology.

(iv) Every surjective family formed of flat and open morphisms (resp. flat and locally of finite presentation, resp. étale, resp. étale and finite) is covering for (fpqc) (resp. (fppf), resp. (ét), resp. (étf)).

(v) Every finite surjective family formed of flat and quasi-compact morphisms is covering for (fpqc). In particular, every faithfully flat and quasi-compact morphism is covering for (fpqc).

Proof. (i) follows from 6.2.1, (ii) from 6.2.3, taking into account the fact that a sheaf for the Zariski topology transforms direct sums into products. Every representable functor being a sheaf for (Zar) and verifying condition (b) of (ii) by SGA 1, VIII 5.3, T_1 is coarser than the canonical topology, which proves (iii).

[332] N.D.E.: By hypothesis, each family $S_{pq} \to S \in P'_i$ is finite, hence $S'_p = \sqcup_q S_{pq}$ is affine and the family can therefore be replaced by the morphism $S'_p \to S_p$, which still belongs to P'_i.

Let us prove (iv). Let $S_i \to S$ be a family of morphisms as in the statement. Considering a covering of S by affine opens, one reduces immediately to the case where S is affine.[333]

Let us first treat the case where the morphisms $S_i \to S$ are flat and open (resp. étale). Let S_{ij} be a covering of S_i by affine opens. As the morphisms in question are open, the images T_{ij} of the S_{ij} form an open covering of S. As S is affine, hence quasi-compact, it is covered by a finite number of opens T_{ij}, for (i, j) running through a finite set F. Then $S' = \bigsqcup_F S_{ij}$ is affine, and the morphism $S' \to S$ belongs to P'_1, resp. P'_3, hence is covering. As it factors through the given family, the latter is covering.

In the case (étf), each S_i is finite over S hence is affine; in the preceding argument, one can then take the covering S_i of S_i, and one obtains that $S' \to S$ belongs to P'_4.

Let us now consider the case where the morphisms $f_i : S_i \to S$ are flat and locally of finite presentation. For every $s \in S$, there exists, by (the proof of) EGA IV_4, 17.16.2, an affine subscheme $X(s)$ of a certain S_i, such that $s \in f_i(X(s))$ and that the morphism $g_i : X(s) \to S$, restriction of f_i, is flat, of finite presentation, and quasi-finite. Then, $g_i(X(s))$ is an open neighborhood $U(s)$ of s (EGA IV_2, 2.4.6), and, S being affine, it is covered by a finite number of such opens $U(s_j)$, j = 1, ..., n. Consequently, $X' = \bigsqcup_j X(s_j)$ is affine, and the morphism $X' \to S$ is surjective, flat, of finite presentation, and quasi-finite, hence belongs to P'_2.[334] This completes the proof of (iv).

Let us prove (v). Let $S_i \to S$ be a finite faithfully flat and quasi-compact family. Let T_j be a covering of S by affine opens. The $S_{ij} = T_j \times_S S_i$ are quasi-compact and therefore have finite affine open coverings T_{ijk}. Each morphism $T_{ijk} \to T_j$ is flat, and the family $T_{ijk} \to T_j$ is finite and surjective, hence covering for T_1. The family $T_{ijk} \to S$ is therefore also, by composition. It factors through the given family which is therefore also:

```
S_i ←──── S_{ij} ←──── T_{ijk}
                          /
                         /
                        /
S ←──── T_j ←──────────
```

Corollary 6.3.2. *Let (M_i) be the following family of morphisms:*

- *(M_1) : faithfully flat and quasi-compact morphisms.*
- *(M_2) : faithfully flat morphisms locally of finite presentation.*
- *(M_3) : surjective étale morphisms.*

[333] N.D.E.: We have simplified what follows, taking advantage of the fact that S is henceforth assumed affine.

[334] N.D.E.: This shows that, if one denotes P''_2 the set of finite surjective families of morphisms of C' flat of finite presentation, the topology generated by P and P''_2 equals T_2. On the other hand, with the notations at the start of the proof of (iv), if one takes a covering of S_i by affine opens, of finite presentation over S, one obtains that $S' \to S$ belongs to P''_2.

- *(M_4) : surjective étale finite morphisms.*[335]

The family (M_i) verifies the axioms (a), (b), (c), (d_{T_i}) and (e_{T_i}) of 4.6.3.

Indeed, for (a), (b), (c), it is classical (EGA and SGA 1, passim.).[336] By 6.3.1, (iv) and (v), (M_i) verifies (d_{T_i}). It remains to see that (M_i) verifies (e_{T_i}); for this, it suffices to see that (M_i) verifies (e_{T_1}), which entails the others. This follows from SGA 1, VIII (nos 4 and 5).

Corollary 6.3.3. *If X is a scheme and R an equivalence relation in X of type (M_i), R is (M_i)-effective if and only if the sheaf-quotient of X by R for T_i is representable and in this case it is represented by the quotient X/R.*

Indeed, this is 4.6.5.

6.4. Effectivity conditions

We now seek families `(N)` of morphisms verifying axiom (f_T) of 4.7. Let us first remark that (f_{T_1}) entails (f_{T_i}), so that we may restrict ourselves to the case of the topology (fpqc).

Lemma 6.4.1. *The following families of morphisms verify axiom (f_{T_1}) of 4.7, that is, "descend by (fpqc)":*

- *(N) : open immersions.*
- *(N′) : closed immersions.*
- *(N″) : quasi-compact immersions.*

By virtue of 6.3.1 (ii), it suffices to verify that the given families descend by the Zariski topology and by a faithfully flat quasi-compact morphism. The first assertion is clear; let us verify the second. For (N), this is SGA 1, VIII 4.4; for (N′), this is loc. cit., 1.9. For (N″) one argues as in loc. cit., 5.5, using the two preceding results.

Corollary 6.4.2. *The same result holds for quasi-compact open immersions.*

These results allow one to apply to the present situation the general results of 4.7.1, 4.7.2, 5.1.8, 5.3.1, etc. Let us state one as an example, the first.

Corollary 6.4.3. *(= 4.7.1 + 4.6.10). Let X be a scheme and R an equivalence relation in X. Suppose that $R \to X$ is faithfully flat and quasi-compact and that $R \to X \times X$ is a closed immersion (resp. open, resp. quasi-compact, resp. quasi-compact open). Then the sheaf-quotient X/R is the same for the topology (fpqc) and for the canonical topology, and for each scheme S, one has*

```
(X/R)(S) = { closed (resp. open, resp. retrocompact[^N.D.E-IV-63], resp. open retrocompact)
             subschemes Z of X_S, stable under R × S, such that Z → S is
             faithfully flat quasi-compact and the diagram R_Z ⇒ Z → S is exact }.
```

[335] N.D.E.: We have corrected the original by adding the surjectivity hypothesis for `(M_3)` and `(M_4)`, which is automatically satisfied in the other cases.

[336] N.D.E.: cf. EGA I, 6.6.4 for "quasi-compact", EGA II, 6.1.5 for "finite", EGA IV_1, 1.6.2 for "locally of finite presentation", EGA IV_2, 2.2.13 for "faithfully flat", and EGA IV_4, 17.3.3 for "étale".

6.5. Principal homogeneous bundles

Let us simply indicate the terminology:

topology	principal homogeneous bundles			
(fpqc)	"	"	"	(tout court)
(ét)	"	"	"	quasi-isotrivial
(étf)	"	"	"	locally isotrivial
(Zar)	"	"	"	locally trivial.

6.6. Other topologies

One sometimes uses other topologies on the category of schemes. Let us indicate one: the *global étale finite* topology (étfg), generated by the pretopology whose covering families are the surjective families formed of étale finite morphisms. It is not finer than the Zariski topology. The corresponding principal homogeneous bundles are called "isotrivial".

```
    (canonical)
         |
       (fpqc)
         |
       (fppf)
         |
        (ét)
         |
       (étf)
       /   \
      /     \
  (Zar)    (étfg)
      \     /
       \   /
     (chaotic)[^N.D.E-IV-64]
```

6.7. Homogeneous spaces[337]

Let G be an S-group scheme, X an S-scheme with (left) operator group G, and

$$\Phi : G \times_S X \to X \times_S X$$

the morphism of S-schemes defined setwise by $(g,x) \mapsto (gx,x)$. Let us recall (cf. 5.1.0 and III.0.1) that one says that X is a *formally principal homogeneous space* under G if the following equivalent conditions are satisfied:

(i) for every $T \to S$, the set $X(T)$ is empty or principal homogeneous under $G(T)$,

(ii) Φ is an isomorphism of S-functors,

(iii) Φ is an isomorphism of S-schemes.

[337] N.D.E.: We have added the numbers that follow.

(The equivalence (i) ⇔ (ii) is clear, and one has (ii) ⇔ (iii) since $C = (Sch/S)$ is a full subcategory of $\hat{C}$.)

The definition of formally homogeneous space (not necessarily principal homogeneous) is obtained by requiring that Φ be an epimorphism in the category of sheaves for an appropriate topology T. Indeed, the condition that Φ be an epimorphism of S-functors amounts to the requirement that, for every $T \to S$, the set $X(T)$ be empty or homogeneous (not necessarily principal homogeneous) under $G(T)$, but this condition is too restrictive, as the following simple example shows. Let $S = \operatorname{Spec} R$, $G = G_{m,R}$ and $X = G_{m,R}$ on which G acts via $t \cdot x = t^2 x$. Then the morphism Φ is étale, finite, and surjective, hence an epimorphism in the category of sheaves for the topology (étf) (a fortiori, an epimorphism of S-schemes); on the other hand, the points 1 and -1 of $X(R)$ are not conjugate by an element of $G(R)$, so that the morphism $G(R) \times X(R) \to X(R) \times X(R)$ is not surjective.[338] One is therefore led to lay down the following definition:

Definition 6.7.1. *Let G be an S-group, X an S-scheme with operator group G, and T a topology on (Sch/S), coarser than the canonical topology. We say that X is a* formally homogeneous space under G (relative to the topology T) *if the following equivalent conditions are satisfied:*

(i) the morphism $\Phi : G \times_S X \to X \times_S X$ is an epimorphism in the category of sheaves for the topology T,

(ii) for every $T \to S$, and $x, y \in X(T)$, there exists a morphism $T' \to T$ covering for the topology T, and $g \in G(T')$, such that $y_{T'} = g \cdot x_{T'}$.

Remark 6.7.2. *Condition (i) implies, in particular, that Φ is a universal effective epimorphism in (Sch/S) (cf. 4.4.3). This entails, as one easily sees, that Φ is surjective (cf. 1.3, N.D.E. (3)).*

Proposition and Definition 6.7.3.[339] *Let G be an S-group, X an S-scheme with operator group G, and T a topology on (Sch/S), coarser than the canonical topology. The following conditions are equivalent:*

(i) X verifies the two hypotheses below:

- *(1) the morphism $\Phi : G \times_S X \to X \times_S X$ is covering, i.e. X is a G-formally homogeneous space,*
- *(2) the morphism $X \to S$ is also covering, i.e. locally for the topology T, it has a section (cf. RefIV.4.4.8bis.bis).*

(ii) "Locally on S for the topology T", X is isomorphic, as a scheme with operator group G, to the sheaf quotient (for T) of G by a sub-group-scheme H, i.e. there

[338] N.D.E.: Evidently, this difficulty arises from the fact that if C' is a full subcategory of $\hat{C}$ containing C, for example, the category $\tilde{C}_T$ of sheaves on C for a topology T coarser than the canonical topology, and if $f : X \to Y$ is a morphism in C, then the implications:

$$f \text{ epimorphism of } \hat{C} \Rightarrow f \text{ epimorphism of } C' \Rightarrow f \text{ epimorphism of } C$$

are in general strict.

[339] N.D.E.: cf. [Ray70], Def. VI.1.1.

exists a covering family $S_i \to S$ such that each $X \times_S S_i$ represents the sheaf quotient of $G \times_S S_i$ by a certain sub-group-scheme H_i.

Under these conditions, one says that X is a G-homogeneous space* (relative to the topology T).*

Proof. Suppose (ii) is verified. Set $G_i = G \times_S S_i$ and $X_i = X \times_S S_i$. Then, X_i has a section over S_i, namely the composite of the unit section $\epsilon_i : S_i \to G_i$ and the projection $\pi_i : G_i \to X_i = G_i/H_i$. Hence $X \to S$ is covering.

On the other hand, π_i is covering, hence $\pi_i \times \pi_i$ is also (cf. 4.2.3 (C 1) and (C 2)), and one has a commutative diagram:

$$\begin{tikzcd}
G_i \times_{S_i} X_i \arrow[r, "\Phi_i"] & G_i \times_{S_i} X_i \\
G_i \times_{S_i} G_i \arrow[r, "\Psi_i", "\sim"'] \arrow[u, "\mathrm{id} \times \pi_i"] & G_i \times_{S_i} G_i \arrow[u, "\pi_i \times \pi_i"']
\end{tikzcd}$$

where Φ_i is deduced from Φ by the base change $S_i \to S$ and Ψ_i is the isomorphism defined setwise by $(g, g') \mapsto (gg', g)$. Then $(\pi_i \times \pi_i) \circ \Psi_i$ is covering, hence Φ_i is also (4.2.3 (C 3)). This shows that Φ is "locally covering", hence is covering (4.2.3 (C 5)). This proves that (ii) ⇒ (i).

Conversely, suppose (i) is verified, and suppose moreover that the structural morphism $X \to S$ has a section σ. By EGA I, 5.3.13, σ is an immersion. Let us define $H = G \times_X S$ by the diagram below, in which the two squares are cartesian:

$$\begin{tikzcd}
H \arrow[r] \arrow[d, "\sigma"] & G \arrow[r, "\mathrm{id}_G \boxtimes \sigma"] \arrow[d, "\pi"] & G \times_S X \arrow[d, "\Phi"] \\
S \arrow[r] & X \arrow[r, "\mathrm{id}_X \boxtimes \sigma"] & X \times_S X
\end{tikzcd}$$

where π, $id_G \boxtimes \sigma$ and $id_X \boxtimes \sigma$ denote the morphisms defined setwise, for $T \to S$ and $g \in G(T)$, $x \in X(T)$, by:

$$\pi(g) = g \cdot \sigma_T, \quad (\mathrm{id}_G \boxtimes \sigma)(g) = (g, \sigma_T), \quad (\mathrm{id}_X \boxtimes \sigma)(x) = (x, \sigma_T).$$

Then, π is covering, and H is a sub-group-scheme of G, representing the stabilizer $Stab_G(\sigma)$ of σ (cf. I, 2.3.3), i.e., for every $T \to S$, one has:

$$H(T) = \{ g \in G(T) \mid g \cdot \sigma_T = \sigma_T \}.$$

Let us denote G/H the presheaf $T \mapsto G(T)/H(T)$, and $a(G/H)$ the associated sheaf, for the topology T. By what precedes, one obtains a commutative diagram of morphisms of presheaves with operator group G:

$$\begin{tikzcd} G \arrow[r, "\pi"] \arrow[d] & X \\ G/H \arrow[ur, "\bar{\pi}"'] & \end{tikzcd}$$

where $\bar{\pi}$ is a monomorphism. As π is covering, $\bar{\pi}$ is also and therefore, by 4.3.12, $\bar{\pi}$ induces an isomorphism $a(G/H) \xrightarrow{\sim} X$. One has therefore proved that: if X is a G-homogeneous space such that $X \to S$ admits a section σ, then X represents the sheaf quotient G/H, where $H = G \times_X S$ is the stabilizer of σ.

In the general case, there exists by hypothesis a covering family $S_i \to S$ such that each morphism $X_i = X \times_S S_i \to S_i$ has a section σ_i. Set $G_i = G \times_S S_i$; then the morphism $\Phi_i : G_i \times_{S_i} X_i \to X_i \times_{S_i} X_i$ deduced from Φ by the base change $S_i \to S$ is again covering. Hence, by what precedes, $X_i \cong G_i/H_i$, where H_i is the stabilizer in G_i of σ_i. This completes the proof of the implication (i) ⇒ (ii).

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Footnotes

Exposé V. Construction of quotient schemes

by P. Gabriel

The aim of this Exposé is to prove the theorems stated in TDTE III.[341] If X and T

[340] N.D.E.: We have added this reference.

[341] N.D.E.: namely, theorems 5.1, 5.3, 6.1, 6.2 and 7.2 of TDTE III. The first two (resp. the next two) correspond to theorem 4.1 (resp. theorems 7.1 and 8.1) of this Exposé. Theorem 7.2 of TDTE III is

are two objects of a category C, we write $X(T)$ instead of $\mathrm{Hom}_C(T, X)$. Similarly, if $\phi : Y \to X$ is an arrow (resp. an object T) of C, then $\phi(T)$ denotes the map $g \mapsto \phi \circ g$ from $Y(T)$ to $X(T)$:

$$\begin{tikzcd} & T \arrow[dl, "g"'] \arrow[dr, "\varphi \circ g"] & \\ Y \arrow[rr, "\varphi"] & & X, \end{tikzcd}$$

and $T(\phi)$ denotes the map $g \mapsto g \circ \phi$ from $T(X)$ to $T(Y)$:

$$\begin{tikzcd} Y \arrow[rr, "\varphi"] \arrow[dr, "g \circ \varphi"'] & & X \arrow[dl, "g"] \\ & T. & \end{tikzcd}$$

Finally, if P is a scheme, we write P for the underlying set of P.

Exceptionally, in the present Exposé we do not follow the convention stated in IV 4.6.15 on the notation for quotients (loc. cit., top of page 227 of the original), since we wish to give here a construction of quotients which also applies to "pre-equivalence relations"[342] that are not equivalence relations.

1. C-groupoids

a) Let C be a category in which products and fiber products exist. Recall first that a diagram

$$X_1 \underset{d_0}{\overset{d_1}{\rightrightarrows}} X_0 \xrightarrow{p} Y$$

in C is said to be *exact* if $pd_0 = pd_1$ and if, for every $T \in C$, $T(p)$ is a bijection of $T(Y)$ onto the subset of $T(X_0)$ consisting of arrows $f : X_0 \to T$ such that $fd_0 = fd_1$. One also says that (Y, p) is the *cokernel* of (d_0, d_1) and writes

$$(Y, p) = \mathrm{Coker}(d_0, d_1).$$

b) Let, for example, C be the category `(Esp.An)` of ringed spaces. In this case, there always exists a cokernel (Y, p), which can be described as follows: the underlying topological space of Y is obtained from X_0 by identifying the points $d_0(x)$ and $d_1(x)$ and endowing Y with the quotient topology. The canonical map $\pi : X_0 \to Y$ together with d_0, d_1 then induces a double arrow of sheaves of rings on Y:

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proved in Exp. VI_A, 3.2 and 3.3.

[342] N.D.E.: that is, groupoids with base X, cf. the terminology at the end of section 1. When C is the category of schemes, the quotient $p : X \to Y$ of a groupoid $X*$ with base X exists under certain hypotheses (cf. 4.1, 6.1, 7.1); if, moreover, $X*$ is an equivalence relation, then p is, under the same hypotheses, faithfully flat and quasi-compact, hence a universal epimorphism, cf. loc. cit.

$$\pi_*(O_0) \underset{\delta_1}{\overset{\delta_0}{\rightrightarrows}} \pi_*(d_{0*} O_1) = \pi_*(d_{1*} O_1),$$

where O_i is the structure sheaf of X_i. We choose for the sheaf of rings on Y the subsheaf of $\pi_*(O_0)$ whose sections s satisfy $\delta_0(s) = \delta_1(s)$. The arrow p is defined in the evident way.

Let[343] $X_1 \rightrightarrows X_0$ be a diagram (with arrows d_0, d_1) in (Esp.An) and let (Y, p) be its cokernel. We say that an open set U of X_0 is *saturated* if $d_0^{-1}(U) = d_1^{-1}(U)$, which is equivalent to saying that $U = p^{-1}(p(U))$. In this case, since Y is endowed with the quotient topology, $p(U)$ is an open subset of Y.

Lemma 1.1. *Let U be a saturated open set of X and $V = p(U)$. If we denote by U_1 the open set $d_0^{-1}(U) = d_1^{-1}(U)$ of X_1, and by $\tilde{d}_0$, $\tilde{d}_1$, and $\tilde{p}$ the restrictions of d_0, d_1 to U_1, and of p to U, then $(V, \tilde{p})$ is a cokernel in (Esp.An) of:*[344]

$$U_1 \underset{\tilde{d}_0}{\overset{\tilde{d}_1}{\rightrightarrows}} U \xrightarrow{\tilde{p}} V.$$

The verification is straightforward.

Lemma 1.2. *Let $X_1 \rightrightarrows X_0$ be a diagram in (Sch) (with arrows d_0, d_1) and let (Y, p) be its cokernel in (Esp.An).*

(i) If Y is a scheme and p a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in (Sch).

(ii) Suppose that every point of X_0 possesses a saturated open neighborhood U such that, denoting by $\tilde{d}_0$ and $\tilde{d}_1$ the restrictions of d_0 and d_1 to $d_0^{-1}(U) = d_1^{-1}(U)$, and by (Q, q) the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in (Esp.An), the space Q is a scheme and q a morphism of schemes. Then (Y, p) is a cokernel of (d_0, d_1) in (Sch).

(i) is proved in § 4.c); since the proof is short, let us repeat it here. Let $f : X_0 \to Z$ be a morphism of schemes such that $fd_0 = fd_1$. By hypothesis, there is a unique morphism of ringed spaces $r : Y \to Z$ such that $f = rp$. It remains to show that, for every $y \in Y$, the homomorphism $O_{r(y)} \to O_y$ induced by r is local. This follows from the fact that p is surjective, so that y is of the form $p(x)$, and from the fact that the homomorphism $O_{f(x)} \to O_x$ induced by f is local.

(ii) follows from (i) and the preceding lemma.

[343] N.D.E.: Lemmas 1.1 and 1.2 have been added; they are used several times in sections 5 to 9.

[344] N.D.E.: This is not the case in the category of schemes. Take, for example, $S = \mathrm{Spec}(C)$, $X_0 = A_S^2 = \mathrm{Spec}(C[x_1, x_2])$, let $d_1 : G_{m,S} \times_S A_S^2 \to A_S^2$ be the action of $G_{m,S}$ by homotheties on A_S^2, let d_0 be the projection onto the second factor, and $U = A_S^2 - m$, where m is the point (0, 0). Then projective space P_S^1 is the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in (Esp. An) and in (Sch), and the cokernel Y of (d_0, d_1) in (Esp. An) is the union of P_S^1 and the point $y_0 = p(m)$; the only open set containing y_0 is Y, and one has $\Gamma(Y, O_Y) = C$. If $f : A_S^2 \to T$ is a morphism of S-schemes such that $fd_0 = fd_1$ and if $V = \mathrm{Spec}(A)$ is an affine open of T containing the point $t_0 = f(y_0)$, then $f^{-1}(V) = A^2$ and the ring morphism $A \to C[x_1, x_2]$ factors through C; this shows that $S = \mathrm{Spec}(C)$ is the cokernel of (d_0, d_1) in the category (Sch/S).

c) In this Exposé we study the existence of $Coker(d_0, d_1)$ when the double arrow (d_0, d_1) is inserted in a richer context; more precisely, let $X_2 = X_1 \times_{d_1,d_0} X_1$ denote the fiber product of the diagram

$$X_1 \downarrow d_1(*)X_0 \uparrow d_0 X_1,$$

and let d'_0 and d'_2 be the two canonical projections of X_2 onto X_1; one then has by definition a Cartesian square

$$\begin{tikzcd} X_2 \arrow[r, "d'_0"] \arrow[d, "d'_2"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_0"'] & X_0. \end{tikzcd}$$

(0)

Moreover, let us give ourselves a third arrow $d'_1 : X_2 \to X_1$; we say that $(d_0, d_1 : X_1 \rightrightarrows X_0, d'_1)$ is a C-*groupoid* if for every object T of C, $X_1(T)$ is the set of arrows of a groupoid $X * (T)$ whose set of objects is $X_0(T)$, with source map $d_1(T)$, target map $d_0(T)$, and composition map $d'_1(T)$ (one identifies, as usual, $(X_1 \times_{d_1,d_0} X_1)(T)$ with $X_1(T) \times_{d_1(T),d_0(T)} X_1(T)$; we also recall that a groupoid is a category in which every arrow is invertible).[345]

If ϕ is an arrow of the groupoid $X * (T)$, the map $f \mapsto \phi \circ f$ is a bijection from the set of arrows f whose target coincides with the source of ϕ onto the set of arrows having the same target as ϕ. One sees easily that this fact can be translated by saying that the square

$$\begin{tikzcd} X_2 \arrow[r, "d'_1"] \arrow[d, "d'_0"'] & X_1 \arrow[d, "d_0"] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd}$$

(1)

is Cartesian.

Similarly, the map $g \mapsto g \circ \phi$ is a bijection from the set of arrows g of $X * (T)$ having source equal to the target of ϕ onto the set of arrows having the same source as ϕ. This fact can again be translated by saying that the square

[345] N.D.E.: Hence, in this case, $X_2(T)$ is the set of pairs (f_2, f_1) of composable arrows, that is, such that $d_0(f_1) = d_1(f_2)$, and d'_0, d'_1, d'_2 send (f_2, f_1) to f_2, $f_2 \circ f_1$, f_1 respectively.

$$\begin{tikzcd}
X_2 \arrow[r, "d'_1"] \arrow[d, "d'_2"'] & X_1 \arrow[d, "d_1"] \\
X_1 \arrow[r, "d_1"'] & X_0
\end{tikzcd} \tag{2}$$

is Cartesian.

On the other hand, let $s : X_0 \to X_1$ be the unique arrow of C such that, for every T, $s(T) : X_0(T) \to X_1(T)$ associates to every object of $X*(T)$ the identity arrow of that object.[346] The arrow s satisfies the equalities

$$d_1\, s = \mathrm{id}_{X_0}, \tag{3}$$
$$d_0\, s = \mathrm{id}_{X_0}. \tag{3 bis}$$

Finally, the associativity of the composition maps $d'_1(T)$ translates into the commutativity of the diagram

$$\begin{tikzcd}
X_1 \times_{d_1, d_0} X_1 \times_{d_1, d_0} X_1 \arrow[r, "d'_1 \times X_1"] \arrow[d, "X_1 \times d'_1"'] & X_1 \times_{d_1, d_0} X_1 \arrow[d, "d'_1"] \\
X_1 \times_{d_1, d_0} X_1 \arrow[r, "d'_1"'] & X_1 .
\end{tikzcd} \tag{4}$$

Conversely, the conditions (1), (2), and (4) together with the existence of an arrow s satisfying (3) imply that $(X_1 \rightrightarrows X_0, d'_1)$ is a C-groupoid. The condition (3) is harmless; it merely ensures that the map $d_1(T) : X_1(T) \to X_0(T)$ is surjective for every $T \in C$. In what follows we shall mostly make use of the Cartesian squares (0), (1) and (2), which we summarize in the diagram

$$\begin{tikzcd}
X_2 \arrow[r, shift left, "d'_1"] \arrow[r, shift right, "d'_0"'] \arrow[d, "d'_2"'] & X_1 \arrow[r, "d_0"] \arrow[d, "d_1"] & X_0 \\
X_1 \arrow[r, "d_0"'] & X_0 &
\end{tikzcd} \tag{0,1,2}$$

In this diagram the two left-hand squares (i.e. the squares (0) and (2)) are Cartesian; the first row is exact, and X_2 is identified with the fiber product $X_1 \times_{d_0,d_0} X_1$.

[346] N.D.E.: $T \mapsto s(T)$ defines an element of $\mathrm{Hom}(h_{X_0}, h_{X_1})$, and the latter equals $\mathrm{Hom}(X_0, X_1)$ by the Yoneda lemma.

We use associativity only indirectly, for instance to ensure the existence of an arrow s satisfying (3) and (3 bis), or else to ensure the existence of an arrow

$$(\dagger) \quad \sigma : X_1 \to X_1 \quad \text{such that} \quad d_0\,\sigma = d_1 \quad \text{and} \quad d_1\,\sigma = d_0$$

(one chooses σ so that $\sigma(T) : X_1(T) \to X_1(T)$ sends every arrow of $X*(T)$ to its inverse).[347]

By abuse of language, we shall sometimes call a C-*groupoid* a diagram

$$X_2 \overset{d'_0,\, d'_1,\, d'_2}{\rightrightarrows} X_1 \overset{d_0,\, d_1}{\rightrightarrows} X_0$$

such that (0), (1) and (2) are Cartesian, (4) is commutative, and there exists s satisfying (3). The object X_2 will therefore be allowed to be "a" fiber product of $(*)$ without being "the" fiber product of $(*)$.[348]

Terminology. Instead of C-groupoid $X*$, we shall also speak of the *groupoid $X*$ with base X_0*, or of the *pre-equivalence relation $X*$ in X_0*.

2. Examples of C-groupoids

a) Let X be an object of C and G a C-group acting on the left on X. We denote by $d_0 : G \times X \to X$ the arrow defining the action of G on X, by $d_1 : G \times X \to X$ the projection of the product onto the second factor, by $\mu : G \times G \to G$ the arrow defining the C-group structure of G, and finally by $pr_{2,3}$ the projection of $G \times G \times X = G \times (G \times X)$ onto the second factor. Then

$$G \times G \times X \ \overset{pr_{2,3},\ \mu\times X,\ G\times d_0}{\rightrightarrows}\ G \times X \ \overset{d_1,\ d_0}{\rightrightarrows}\ X$$

is a C-groupoid.

b) Let $d_0, d_1 : X_1 \to X_0$ be an *equivalence pair*, i.e., if $d_0 \boxtimes d_1 : X_1 \to X_0 \times X_0$ is the arrow with components d_0 and d_1, we suppose that $(d_0 \boxtimes d_1)(T)$ is, for every object T of C, a bijection of $X_1(T)$ onto the graph of an equivalence relation on $X_0(T)$. The set $X_1(T)$ therefore identifies with the set of pairs (x, y) of elements of $X_0(T)$ such that $x \sim y$; similarly, the set $X_2(T) = (X_1 \times_{d_1, d_0} X_1)(T)$ identifies with the set of triples (x, y, z) of elements of $X_0(T)$ such that $x \sim y$ and $y \sim z$. There is therefore one and only one arrow $d'_1 : X_2 \to X_1$ making the squares (1) and (2) commute: $d'_1(T)$ must send $(x, y, z) \in X_2(T)$ to $(x, z) \in X_1(T)$. For this choice of d'_1, $(d_0, d_1 : X_1 \rightrightarrows X_0, d'_1)$ is a C-groupoid.

Conversely, consider a C-groupoid $X*$ such that $d_0 \boxtimes d_1 : X_1 \to X_0 \times X_0$ is a monomorphism. Then (d_0, d_1) is an equivalence pair and $X*$ can be reconstructed from

[347] N.D.E.: It follows from the Yoneda lemma that σ is an involutive automorphism of X_1; this will be used, for example, in 3.e) and in theorem 4.1.

[348] N.D.E.: see example 2.a) below.

(d_0, d_1) as explained a few lines above.[349]

c) If $p : X \to Y$ is any arrow of C and if pr_1 and pr_2 are the two projections of $X \times_{p,p} X$ onto X, then $(pr_1, pr_2) : X \times_{p,p} X \rightrightarrows X$ is an equivalence pair. One says that p is an *effective epimorphism* if the diagram

$$X \times_{p,p} X \underset{pr_2}{\overset{pr_1}{\rightrightarrows}} X \xrightarrow{\ p\ } Y$$

is exact, that is, if $(Y, p) = Coker(pr_1, pr_2)$.

Let, for example, S be a Noetherian scheme and let C be the category of schemes finite over S. Let us show that an epimorphism in C is not necessarily effective: take S equal to $\operatorname{Spec} k[T^3, T^5]$, where k is a commutative field, Y equal to S, and X equal to $\operatorname{Spec} k[T]$. If i is the inclusion of $B = k[T^3, T^5]$ into $A = k[T]$, take p equal to $\operatorname{Spec} i$. In this case $X \times_{p,p} X$ identifies with $\operatorname{Spec}(A \otimes_B A)$ and $Coker(pr_1, pr_2)$ with $\operatorname{Spec} B'$, where B' is the subring of A consisting of a such that $a \otimes_B 1 = 1 \otimes_B a$. Now

$$T^7 \otimes_B 1 = (T^2 T^5) \otimes_B 1 = T^2 \otimes_B T^5 = T^2 \otimes_B (T^3 T^2) = T^5 \otimes_B T^2 = 1 \otimes_B T^7.$$

So T^7 belongs to B', does not belong to B, and $\operatorname{Spec} B'$ is distinct from $\operatorname{Spec} B$, which yields the counterexample.[350]

3. Some sorites on C-groupoids

Here, in no particular order, are some remarks used in what follows:

a) Let

$$X_2 \overset{d'_0,\, d'_1,\, d'_2}{\Rrightarrow} X_1 \overset{d_0,\, d_1}{\rightrightarrows} X_0$$

be a C-groupoid and $f_0 : Y_0 \to X_0$ an arrow of C. We shall define a C-groupoid with base Y_0

$$Y_2 \overset{e'_0,\, e'_1,\, e'_2}{\Rrightarrow} Y_1 \overset{e_0,\, e_1}{\rightrightarrows} Y_0$$

which we shall call *induced by $X*$ and f_0*. One also says that $Y*$ is the *inverse image of $X*$ under the base change f_0*.

We choose for Y_1 the fiber product of the diagram

$$\begin{array}{ccc} Y_1 & \xrightarrow{\ f_1\ } & X_1 \\ | & & | \end{array}$$

[349] N.D.E.: In particular, if G is a C-group acting on the left on an object X of C and if $X*$ is the C-groupoid defined in a), then (d_0, d_1) is an equivalence pair if and only if G acts freely on X, cf. Exp. III, 3.2.1.

[350] N.D.E.: The same argument applies for $B = k[T^3, T^4]$ and $T^5 \otimes_B 1$; more generally, for $B = k[T^n, T^{n+r}]$ and the element $T^{n+2r} \otimes_B 1$, provided that n does not divide $2r$.

$$\begin{tikzcd} {} \arrow[d] & {} \arrow[d, "d_0 \boxtimes d_1"] \\ Y_0 \times Y_0 \arrow[r, "f_0 \times f_0"'] & X_0 \times X_0, \end{tikzcd}$$

and for e_0 and e_1 the arrows obtained by composing the canonical arrow $Y_1 \to Y_0 \times Y_0$ with the first and second projections of $Y_0 \times Y_0$. The morphism $Y_1 \to Y_0 \times Y_0$ is then $e_0 \boxtimes e_1$, and one has $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, where we have written f_1 for the projection of Y_1 onto X_1.

We set $Y_2 = Y_1 \times_{e_0,e_1} Y_1$, cf. 1.c). One can say that the pair (e_0, e_1) is defined in such a way that, for every $T \in C$ and every pair (y, x) of elements of $Y_0(T)$, there is a certain one-to-one correspondence $\psi \mapsto {}_y\psi_x$ between the arrows ψ of $X*(T)$ with source $f_0(x)$ and target $f_0(y)$ and the arrows ${}_y\psi_x$ of $Y*(T)$ with source x and target y. One therefore determines $e_1' : Y_2 \to Y_1$ by defining, for every $T \in C$, the composition of arrows of $Y*(T)$ by the formula

$${}_z\psi_y \circ {}_y\varphi_x = {}_z(\psi \circ \varphi)_x.$$

It is clear that this definition makes each $Y*(T)$ into a groupoid.

b) Knowing the C-groupoid $X*$ and the base change $f_0 : Y_0 \to X_0$, one can reconstruct the pair $(e_0, e_1) : Y_1 \rightrightarrows Y_0$ in another way:[351] construct $Y_0 \times_{X_0} X_1$, pr_1 and pr_2 so that the square

$$\begin{tikzcd} Y_0 \times_{X_0} X_1 \arrow[r, "pr_2"] \arrow[d, "pr_1"'] & X_1 \arrow[d, "d_0"] \\ Y_0 \arrow[r, "f_0"'] & X_0 \end{tikzcd}$$

is Cartesian. One then verifies without difficulty, by reduction to the set-theoretic case, that one has the Cartesian square

$$\begin{tikzcd} Y_1 \arrow[r, "e_0 \boxtimes f_1"] \arrow[d, "e_1"'] & Y_0 \times_{X_0} X_1 \arrow[d, "d_1 \circ pr_2"] \\ Y_0 \arrow[r, "f_0"'] & X_0, \end{tikzcd}$$

where f_1 denotes the canonical projection of $Y_1 = (Y_0 \times Y_0) \times_{(X_0 \times X_0)} X_1$ onto X_1.

c) We shall give two examples of inverse images of a C-groupoid. Take Y_0 equal to X_1, f_0 equal to d_0. For every object T of C, $Y_1(T)$ then identifies with the set of diagrams of the form

[351] N.D.E.: this second viewpoint will be used in 3.f) and in the proof of 6.1.

$$\begin{tikzcd} b \arrow[r, "\varphi"] & d \\ a \arrow[u, "f"] & c \arrow[u, "g"'] \end{tikzcd}$$

of $X * (T)$. The source of such a diagram is the arrow f, the target is the arrow g. These diagrams compose in the evident way.

Now put $Y'_0 = X_1$, $f'_0 = d_1$ (we add the primes[352] to avoid any confusion with the preceding example). In this case, $Y'_1(T)$ identifies, for every $T \in C$, with the set of diagrams of the form

$$\begin{tikzcd} b & d \\ a \arrow[u, "f"] \arrow[r, "\psi"] & c \arrow[u, "g"'] \end{tikzcd}$$

of the groupoid $X * (T)$. The source of such a diagram is f, the target is g; the composition of these diagrams is evident.

This being so, it is clear that the identity map of $Y_0(T)$ and the map

$$\begin{tikzcd} b \arrow[r, "\varphi"] & d & & b & d \\ a \arrow[u, "f"] & c \arrow[u, "g"'] & \mapsto & a \arrow[u, "f"] \arrow[r, "g^{-1}\varphi f"] & c \arrow[u, "g"'] \end{tikzcd}$$

from $Y_1(T)$ to $Y'_1(T)$ define an isomorphism of the groupoid $Y * (T)$ onto $Y' * (T)$. Moreover, this isomorphism depends functorially on T, so that the C-groupoids $Y*$ and $Y'*$ are isomorphic.[353]

d)

Proposition 3.1. *We keep the notations of a) and assume that f_0 is an effective and universal epimorphism. Then $Coker(d_0, d_1)$ exists if and only if $Coker(e_0, e_1)$ exists.*[354] *Moreover, in that case, f_0 induces an isomorphism*

$$\mathrm{Coker}(d_0, d_1) \xrightarrow{\sim} \mathrm{Coker}(e_0, e_1).$$

Let us first recall that an epimorphism $f_0 : Y_0 \to X_0$ is said to be *universal* if, for every Cartesian square

$$\begin{tikzcd} Y' \arrow[r] \arrow[d, no head] & Y_0 \arrow[d, no head] \\ {} & {} \end{tikzcd}$$

[352] N.D.E.: “accents” in the original.

[353] N.D.E.: This will play a crucial role in the proof of lemma 6.1.

[354] N.D.E.: The original has been modified to make explicit the isomorphism below.

$$\begin{tikzcd}
{} \arrow[d, "f'"'] & {} \arrow[d, "f_0"] \\
X' \arrow[r] & X_0,
\end{tikzcd}$$

f' is an epimorphism. This being so, let us denote by $C(d_0, d_1)$ the covariant functor from C to sets which associates to every $T \in C$ the kernel of the pair $T(d_0), T(d_1) : T(X_0) \rightrightarrows T(X_1)$. We define $C(e_0, e_1)$ similarly. For every $T \in C$, one therefore has a commutative diagram

$$\begin{tikzcd}
C(d_0, d_1)(T) \arrow[r] \arrow[d, "T(f)"'] & T(X_0) \arrow[r, shift left, "T(d_1)"] \arrow[r, shift right, "T(d_0)"'] \arrow[d, "T(f_0)"'] & T(X_1) \arrow[d, "T(f_1)"] \\
C(e_0, e_1)(T) \arrow[r] & T(Y_0) \arrow[r, shift left, "T(e_1)"] \arrow[r, shift right, "T(e_0)"'] & T(Y_1),
\end{tikzcd}$$

where $T(f)$ is the injection induced by the injection $T(f_0)$. If we show that $T(f)$ is a surjection for every T, we shall have a functorial isomorphism $f : C(d_0, d_1) \xrightarrow{\sim} C(e_0, e_1)$, so that the representability of one of these functors will be equivalent to that of the other; this will prove our proposition.

To prove the surjectivity of $T(f)$, consider the diagram

$$\begin{tikzcd}
 & Y_1 \arrow[rr, "f_1"] \arrow[d, shift right, "e_0"'] \arrow[d, shift left, "e_1"] & & X_1 \arrow[d, shift right, "d_0"'] \arrow[d, shift left, "d_1"] \\
Y_0 \times_{X_0} Y_0 \arrow[ur, "\Delta"] \arrow[r, shift left, "pr_2"] \arrow[r, shift right, "pr_1"'] & Y_0 \arrow[rr, "f_0"'] & & X_0,
\end{tikzcd}$$

where Δ is the section of $Y_1 \to Y_0 \times Y_0$ defined by the morphism $s \circ f_0 \circ pr_1 : Y_0 \times Y_0 \to X_1$, with $s : X_0 \to X_1$ the arrow satisfying equalities (3) and (3 bis) of section 1.

If the arrow $g : Y_0 \to T$ is such that $g \circ e_0 = g \circ e_1$, then $g \circ e_0 \circ \Delta = g \circ e_1 \circ \Delta$, so $g \circ pr_1 = g \circ pr_2$. Since f_0 is an effective epimorphism, g factors through f_0 and an arrow $h : X_0 \to T$, that is to say $g = T(f_0)(h)$. It remains to show that h belongs to $C(d_0, d_1)(T)$, i.e. satisfies $hd_0 = hd_1$; now one has

$$h\, d_0\, f_1 = h\, f_0\, e_0 = g\, e_0 = g\, e_1 = h\, f_0\, e_1 = h\, d_1\, f_1,$$

whence the desired equality, since f_1 is an epimorphism (because f_0 is a universal epimorphism).

e) Consider now a scheme S and choose C equal to (Sch/S). The data of a C-groupoid

$$X_2 \overset{d'_0,\, d'_1,\, d'_2}{\Rrightarrow} X_1 \overset{d_0,\, d_1}{\rightrightarrows} X_0$$

allows one to define an equivalence relation on the set X_0 underlying the scheme X_0: if $x, y \in X_0$, one writes $x \sim y$ when there exists $z \in X_1$ such that $x = d_1 z$ and $y = d_0 z$. The reflexivity and symmetry of this relation are evident;[355] let us prove transitivity: if $x \sim y$ and $y \sim z$, there exist $u, v \in X_1$ with $x = d_1 u$, $y = d_0 u$, $y = d_1 v$, $z = d_0 v$. It follows that (v, u) belongs to the set-theoretic fiber product $X_1 \times_{d_1, d_0} X_1$. Since the canonical map

$$X_1 \times_{d_1, d_0} X_1 \to X_1 \times_{d_1, d_0} X_1$$

from the set underlying the fiber product into the fiber product of the underlying sets is surjective, (v, u) is the image of some $w \in X_2$. One then has $x = d_1 d'_1 w$ and $z = d_0 d'_1 w$, whence $x \sim z$.

f) We keep the notations of a) and b), C still being (Sch/S). If x, y are points of Y_0, we shall see that one has $x \sim y$ if and only if $f_0(x) \sim f_0(y)$ (the inverse image of the equivalence relation defined by a groupoid is the equivalence relation defined by the inverse image of the groupoid).

Indeed, suppose $x \sim y$. There exists therefore $z \in Y_1$ such that $x = e_1(z)$ and $y = e_0(z)$. Since $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, one then has $f_0(x) = d_1 f_1(z)$ and $f_0(y) = d_0 f_1(z)$, whence $f_0(x) \sim f_0(y)$.

Conversely, suppose $f_0(x) \sim f_0(y)$ and let $z \in X_1$ be such that $f_0(y) = d_1(z)$ and $f_0(x) = d_0(z)$. Using the construction and notations of b), there is then a point t of $Y_0 \times_{X_0} X_1$ such that $pr_1(t) = x$ and $pr_2(t) = z$. Similarly, since $f_0(y) = d_1 pr_2(t)$, there is $s \in Y_1$ such that $y = e_1(s)$ and $(e_0 \boxtimes f_1)(s) = t$. One then has $e_0(s) = pr_1(e_0 \boxtimes f_1)(s) = pr_1(t) = x$. Whence $x \sim y$.

4. Passage to the quotient by a finite and flat groupoid (proof of a particular case)

Theorem 4.1. *Consider a (Sch/S)-groupoid*

$$X_2 \overset{d'_0,\, d'_1,\, d'_2}{\Rrightarrow} X_1 \overset{d_0,\, d_1}{\rightrightarrows} X_0.$$

Suppose the following conditions are satisfied:[356]

a) d_1 is finite locally free;

b) for every $x \in X_0$, the set $d_0 d_1^{-1}(x)$ is contained in an affine open of X_0.[357]

[355] N.D.E.: Reflexivity follows from the existence of $s : X_0 \to X_1$ which is a section of both d_0 and d_1; symmetry follows from the existence of the involution σ of X_1 which "exchanges d_0 and d_1", that is, satisfies $d_0 \sigma = d_1$ and $d_1 \sigma = d_0$, cf. § 1, (3), (3 bis) and (†).

[356] N.D.E.: Since $d_0 = d_1 \sigma$, where σ is an involutive automorphism of X_1, these two conditions are symmetric in d_1 and d_0; moreover, one has $d_0 d_1^{-1}(x) = d_1 d_0^{-1}(x)$.

[357] N.D.E.: One cannot omit hypothesis b). Indeed, H. Hironaka has given an example of an action of the finite group $G = \mathbb{Z}/2\mathbb{Z}$ on a proper C-variety X_0 such that the quotient X_0/G is an algebraic space

Then:

(i) There exists a cokernel (Y, p) of (d_0, d_1) in (Sch/S); moreover, such a (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.

(ii) p is integral and open, and Y is affine if X_0 is affine.[358]

(iii) The morphism $X_1 \to X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.

(iv) If (d_0, d_1) is an equivalence pair, then $X_1 \to X_0 \times_Y X_0$ is an isomorphism[359] *and $p : X_0 \to Y$ is finite locally free.*[360] *Moreover, (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the (fppf) topology and, for every base change $Y' \to Y$, Y' is the cokernel of the groupoid $X * \times_Y Y'$ obtained from $X*$ by the base change $X_0 \times_Y Y' \to X_0$.*

In particular, for every base change $S' \to S$, $Y' = Y \times_S S'$ is the cokernel of the S'-groupoid $X'* = X * \times_S S'$. So, in this case, "the formation of the quotient commutes with base change".

It evidently follows from (i) that the topological space underlying Y is the quotient of the topological space underlying X_0 by the equivalence relation defined by the (Sch/S)-groupoid $X*$.

We shall first prove this theorem when X_0 is affine and d_1 is locally free of constant rank n. We shall see next how to reduce to this particular case.

In the case in which we have placed ourselves, X_0, X_1 and X_2 are affine. We can therefore suppose that

$$X_i = \operatorname{Spec} A_i, \quad d_j = \operatorname{Spec} \delta_j, \quad d'_k = \operatorname{Spec} \delta'_k,$$

the A_i being commutative rings and the δ_j, δ'_k ring homomorphisms. One can then replace the diagram (0, 1, 2) by the following

$$\begin{array}{llll}
 & \qquad\quad \delta'_1 & & \\
 & A_2 \leftleftarrows A_1 \leftleftarrows A_0 & & \\
 & \qquad\quad \delta'_0 & & \qquad \delta_0 \\
(0,1,2)* & \quad \delta'_2 & \qquad \delta_1 & \\
 & \qquad\quad \delta_1 & & \\
 & A_1 \leftleftarrows A_0, & & \\
 & \qquad\quad \delta_0 & &
\end{array}$$

where the two left-hand squares are cocartesian.

Let B denote the subring of A_0 consisting of those $a \in A_0$ such that $\delta_0(a) = \delta_1(a)$.

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which is not a scheme ([Hi62], see also [Mum65], Chap. 4, § 3).

[358] N.D.E.: We have added that p is open, by taking up the analogous proof given in 6.1.

[359] N.D.E.: Note that, in this case, $X_1 \to X_0 \times_S X_0$ is therefore an immersion (EGA I, 5.3.10); see also VI_B, 9.2.1. On the other hand, for the existence of the quotient (in the category of schemes or that of algebraic spaces) under the weaker hypothesis that d_0 and d_1 are finite (but not necessarily flat), see [An73], § 1.1, [Fe03], [Ko08]...

[360] N.D.E.: We have made explicit the consequences which follow; see [Ray67a], th. 1 (iii) and the proof given further on, at the end of section 5.

a) Let us show that A_0 is integral over B. If a belongs to A_0, let

$$P_{\delta_1}(T, \delta_0(a)) = T^n - \sigma_1 T^{n-1} + \cdots + (-1)^n \sigma_n$$

be the characteristic polynomial of $\delta_0(a)$ when A_1 is regarded as an algebra over A_0 via the homomorphism δ_1 (cf. Bourbaki, Alg. VIII, § 12 and Alg. comm. II, § 5, exercise 9). Since the left-hand squares of $(0, 1, 2)*$ are cocartesian, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_0 \delta_0(a))$$

and

$$\delta_1(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_1 \delta_0(a)).$$

Since $\delta'_0\delta_0 = \delta'_1\delta_0$, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = \delta_1(P_{\delta_1}(T, \delta_0(a))),$$

that is, $\delta_0(\sigma_i) = \delta_1(\sigma_i)$ for every i. Hamilton–Cayley moreover tells us that

$$\delta_0(a)^n - \delta_1(\sigma_1)\, \delta_0(a)^{n-1} + \cdots + (-1)^n \delta_1(\sigma_n) = 0.$$

Since $\delta_1(\sigma_i) = \delta_0(\sigma_i)$, one also has

$$\delta_0(a)^n - \delta_0(\sigma_1)\, \delta_0(a)^{n-1} + \cdots + (-1)^n \delta_0(\sigma_n) = 0,$$

whence

$$a^n - \sigma_1 a^{n-1} + \cdots + (-1)^n \sigma_n = 0,$$

since there exists a homomorphism $\tau : A_1 \to A_0$ such that $\tau\delta_0 = id_{A_0}$, hence δ_0 is injective. It follows that A_0 is integral over B.

b) Consider now two prime ideals x and y of A_0. We shall show that the equality $x \cap B = y \cap B$ entails the existence of a prime ideal z of A_1 such that $x = d_0(z)$ and $y = d_1(z)$.

Indeed, if the assertion were not true, x would be distinct from $\delta_0^{-1}(t)$ for every prime ideal t of A_1 such that $\delta_1^{-1}(t) = y$. For such a t one would have $\delta_0^{-1}(t) \cap B = \delta_1^{-1}(t) \cap B = y \cap B = x \cap B$, whence by Cohen–Seidenberg (cf. Bourbaki, Alg. comm. V, § 2, cor. 1 of prop. 1) x would be contained in no $\delta_0^{-1}(t)$.[361] Now there are at most n prime ideals t of A_1 such that $\delta_1^{-1}(t) = y$ (cf. loc. cit., prop. 3), so, by the "Prime Avoidance Lemma" (loc. cit., II, § 1, prop. 3), there would exist $a \in x$ belonging to no $\delta_0^{-1}(t)$. Consequently, $\delta_0(a)$ would belong to none of these ideals t, and so, by the lemma below, the norm $N_{\delta_1}(\delta_0(a))$ would not belong to y (one computes this norm by regarding A_1 as an algebra over A_0 via the homomorphism δ_1; one has $N_{\delta_1}(\delta_0(a)) = \sigma_n$ with the notations of a)). But, since $(-1)^{n-1}\sigma_n = a^n + \sum_{i=1}^{n-1}(-1)^i \sigma_i a^{n-i}$, this norm belongs to $B \cap x = B \cap y$, whence the contradiction.

Lemma 4.1.1. *Let $A \to A'$ be a morphism of commutative rings making A' into a projective A-module of rank n. Let $p \in \mathrm{Spec}(A)$, $q_1, \cdots, q_r$ the elements of* $\mathrm{Spec}(A')$

[361] N.D.E.: We have expanded on the original in what follows; in particular, we have added lemma 4.1.1, taken from [DG70], III, § 2.4, Lemma 4.3.

above p, and $a \in A'$. Then a belongs to $q_1 \cup \cdots \cup q_r$ if and only if its norm $N(a)$ belongs to p.

Indeed, replacing A and A' by the localizations A_p and A'_p, we reduce to the case where (A, p) is local and A' is semilocal, with $\mathrm{Spec}(A') = q_1, \cdots, q_r$. In this case, A' is a free A-module of rank n (cf. Bourbaki, Alg. comm. II, § 3.2, cor. 2 of prop. 5), and $N(a)$ is the determinant of the endomorphism $\ell_a : a' \mapsto aa'$ of A', so one has the equivalences

$$N(a) \notin p \Longleftrightarrow N(a) \text{ invertible} \Longleftrightarrow \ell_a \text{ invertible} \Longleftrightarrow a \notin q_1 \cup \cdots \cup q_r.$$

c) Proof of (i):

Set $Y = \mathrm{Spec}\, B$ and $p = \mathrm{Spec}\, i$, where i is the inclusion of B into A_0. By a), the morphism $p : X_0 \to Y$ is surjective. Let us first show that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces: it follows indeed from b) that the set underlying $\mathrm{Spec}\, B$ is obtained from the set underlying X_0 by identifying the points x and y such that there exists $z \in X_1$ with $d_1 z = y$, $d_0 z = x$. Moreover, since i is integral, $p = \mathrm{Spec}\, i$ is closed, so Y is endowed with the quotient topology of that of X_0. It follows that p is open. Indeed, let U' be any open of X_0; since d_1 is surjective and finite locally free, hence faithfully flat and of finite presentation, and therefore open, the saturation $U = d_1(d_0^{-1}(U'))$ of U' for the equivalence relation defined by $X*$ is open. Then $p(U') = p(U)$ is open, since Y is endowed with the quotient topology.

It follows finally from the choice of B and from the fact that p, d_0 and d_1 are affine that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_{X_0}) \underset{p_*(\delta_0)}{\overset{p_*(\delta_1)}{\rightrightarrows}} p_*(d_{0*}(\mathcal{O}_{X_1})) = p_*(d_{1*}(\mathcal{O}_{X_1}))$$

is exact.

It remains to show that (Y, p) is also the cokernel of (d_0, d_1) in the category of schemes (more generally, in the category of ringed spaces in local rings). Let then $q : X_0 \to Z$ be a morphism of schemes such that $qd_0 = qd_1$. By what precedes, there is a unique morphism of ringed spaces $r : Y \to Z$ such that $q = rp$. It remains to show that, for every $y \in Y$, the homomorphism $\mathcal{O}_{r(y)} \to \mathcal{O}_y$ induced by r is local. This follows from the fact that p is surjective, so that y is of the form $p(x)$, and from the fact that the homomorphism $\mathcal{O}_{q(x)} \to \mathcal{O}_x$ induced by q is local.

d) Proof of (ii): Follows from a) and c).

e) Proof of (iii):

Recall that one denotes by P the set underlying a scheme P, and by $d : P \to Q$ the map induced by a morphism $d : P \to Q$.

Lemma 4.1.2.[362] *Let (A, m) be a local ring, k its residue field, and K an extension*

[362] N.D.E.: We have inserted this lemma, which is used several times in this Exposé and in subsequent

of the field k. Then there exists a local and flat A-algebra B such that B/mB is k-isomorphic to K; moreover, one can choose B finite and free over A if K is of finite degree over k.

This is proved in EGA 0_III, 10.3.1, where it is moreover shown that one can choose B Noetherian if A is. For the reader's convenience, let us indicate the proof.

Put $A' = A[T]$, where T is an indeterminate. If $K = k(T)$, let $p = mA'$ and $B = A'_p$. Then $B/mB \cong k[T]_{(0)} = k(T)$, and B is flat over A', which is a free A-module, so B is flat over A.

If $K = k(t) = k[t]$, where t is algebraic over k, set $B = A'/(F)$, where $F \in A'$ is a monic polynomial whose image in k[T] is the minimal polynomial f of t over k. Then B is a free A-module of finite rank $deg(F) = deg(f)$. In particular, B is integral over A, hence every maximal ideal of B contains m. Since $B/mB \cong k[T]/(f) \cong K$, it follows that B is local, with maximal ideal mB. This already shows that if $[K : k] < \infty$, one can choose B finite and free over A.

In the general case, let $(t_i)_{i \in I}$ be a system of generators of K over k, and endow I with a well-ordering (i.e., a total order $\leqslant$ such that every non-empty subset of I has a least element). For every $i \in I$, let k_i (resp. $k_{<i}$) denote the subfield of K generated by the t_j for $j \leqslant i$ (resp. $j < i$). Adding one element if necessary, we may suppose that I has a greatest element ξ, so that $K = k_\xi$. Consider the subset J of I consisting of indices i such that there exists an inductive system $(A_j)_{j \leqslant i}$ of local and flat A-algebras such that $A_j/mA_j \cong k_j$ and A_j is flat over A_ℓ for every $\ell < j$. Suppose $I - J$ non-empty; let i be its least element and let $A' = \lim_{j<i} A_j$. Since tensor product commutes with direct limits, A' is flat over A and over each A_j for $j < i$, and one has $A'/mA' \cong A' \otimes_A (A/m) \cong k_{<i}$. Moreover, A' is local, with maximal ideal mA'. Indeed, if $x = f_j(x_j)$ is non-invertible, then x_j is not invertible, hence belongs to the maximal ideal mA_j of A_j, whence $x \in mA'$. It then follows from the monogenic case treated above that there exists a local and flat A'-algebra A_i such that $A_i/mA_i \cong k_{<i}(t_i) = k_i$; then A_i is flat over each A_j for $j < i$, and so $i \in J$, contrary to hypothesis. This contradiction shows that $J = I$, and so A_ξ answers the question. Lemma 4.1.2 is proved.

Let us now prove 4.1 (iii). Recall that one denotes by P the set underlying a scheme P, and by $d : P \to Q$ the map induced by a morphism $d : P \to Q$. One can then translate b) by saying that the map

$$d_0 \boxtimes d_1 : X_1 \to X_0 \times_Y X_0$$

with components d_0 and d_1 is surjective; now this map factors as follows

$$X_1 \xrightarrow{d_0 \boxtimes d_1} X_0 \times X_0 \xrightarrow{q} X_0 \times_Y X_0,$$
(set-theoretic Y-product)

q being the canonical map; the image of $d_0 \boxtimes d_1$ therefore contains all points v of

Exposés (VI_A, VI_B). It appeared as Lemma VI_B, 4.5.1 in the original 1965 edition of SGAD.

$X_0 \times_Y X_0$ such that $v = q^{-1}(q(v))$. This last condition[363] will be realized in particular if v is rational over Y, that is to say, if the residue field $\kappa(v)$ of v identifies with the residue field $\kappa(w)$ of the image w of v in Y.

If $v \in X_0 \times_Y X_0$ is not rational over Y, let w again be the image of v in Y. By lemma 4.1.2, there exists a local ring C of radical m and a local and flat homomorphism $f : O_w \to C$ such that C/m is isomorphic to $\kappa(v)$ as a $\kappa(w)$-algebra. If one sets $Y' = \operatorname{Spec} C$ and if $\pi : Y' \to Y$ is the morphism induced by f, it is clear that the canonical projection of $(X_0 \times_Y X_0) \times_Y Y'$ to $X_0 \times_Y X_0$ sends to v a point v' of $(X_0 \times_Y X_0) \times_Y Y'$ which is rational over Y'. Since

$$(X_0 \times_Y X_0) \times_Y Y' \cong (X_0 \times_Y Y') \times_{Y'} (X_0 \times_Y Y'),$$

and since the hypotheses of theorem 4.1 and the previous results, in particular point b), remain valid after the base change $\pi : Y' \to Y$, then v' is the image of an element $u' \in X_1 \times_Y Y'$ by the morphism deduced from $d_0 \boxtimes d_1$ by base change. If u is the image of u' in X_1, one indeed has $v = (d_0 \boxtimes d_1)(u)$.

f) Proof of (iv):[364]

Lemma 4.1.3. *If a monomorphism of schemes $f : T \to Z$ is a finite morphism, it is a closed immersion.*

Indeed, covering Z by affine opens Z_i and replacing f by the induced morphisms $f^{-1}(Z_i) \to Z_i$, we reduce (f being finite, hence affine) to the case where $Z = \operatorname{Spec} B$ and $T = \operatorname{Spec} A$. Since f is a monomorphism, the diagonal morphism $T \to T \times_Z T$ is an isomorphism (EGA I, 5.3.8), i.e., $A \otimes_B A \to A$ is an isomorphism. Consequently, for every maximal ideal m of B, one has an isomorphism

$$(A/mA) \otimes_k (A/mA) \cong (A/mA),$$

where we have set $k = B/m$. Since A is finite over B, A/mA is a k-vector space of finite dimension d, and the above isomorphism entails $d = 0$ or 1, so that the morphism $k = B/m \to A/mA$ is surjective. Hence, by Nakayama's lemma (A being finite over B), the morphism $B_m \to A_m$ is surjective. It follows that the morphism of B-modules $B \to A$ is surjective (since its cokernel C satisfies $C_m = 0$ for every m, so is zero). This proves the lemma.

Let us now prove (iv). By hypothesis, $X_0 = \operatorname{Spec} A_0$, $X_1 = \operatorname{Spec} A_1$, and, for $i = 0, 1$, the morphism $\delta_i : A_0 \to A_1$ makes A_1 a finitely generated A_0-module; thus, a fortiori, the morphism $A_0 \otimes_B A_0 \to A_1$ is finite.

One assumes in addition that $d = d_0 \boxtimes d_1 : X_1 \to X_0 \times_Y X_0$ is a monomorphism; hence, by the preceding lemma, the morphism $A_0 \otimes_B A_0 \to A_1$ is surjective.

[363] N.D.E.: Note the permutation of pages in Lecture Notes 151; the real order is 265-266-268-269-267-270-271.

[364] N.D.E.: We have added the following lemma, taken from [DG70], I, § 5, Prop. 1.5 (see also the proof of EGA IV_3, 8.11.5), used implicitly in the original, and explicitly in [DG70], III, § 2, 4.6. It is moreover useful in th. 7.1 further on.

We shall show that it is an isomorphism (we shall prove along the way that $p : X_0 \to Y$ is finite and locally free). It suffices to show that, for every prime ideal p of B, the homomorphism $(A_0)_p \otimes_{B_p} (A_0)_p \to (A_1)_p$ with components δ_{0p} and δ_{1p} is bijective. In other words, one may suppose B local. It then follows from b) that $(A_0)_p$ is semilocal; indeed, if m is a maximal ideal of $(A_0)_p$, the other maximal ideals are of the form $\delta_0^{-1}(n)$, where n runs over the prime ideals of A_1 such that $\delta_1^{-1}(n) = m$; the assertion follows from the fact that there are at most $n = [A_1 : A_0]$ such prime ideals n. Possibly performing a faithfully flat base change,[365] one can also suppose that the residue field of B is infinite, so that one can use the following lemma:

Lemma 4.2. *Let B be a local ring with infinite residue field, A a semilocal ring, and $i : B \to A$ a homomorphism sending the maximal ideal n of B into the radical r of A. Let M be a free A-module of rank n and N a B-submodule of M that generates M as an A-module. Then N contains a basis of M over A.*

Recall indeed that a sequence $m_1, \cdots, m_n$ of elements of M is an A-basis of M if and only if the canonical images of $m_1, \cdots, m_n$ in M/rM form a basis of M/rM over A/r. One can therefore replace M by M/rM, N by $N/(N \cap rM)$, A by A/r and B by B/n. In this case the lemma is easy (if A is a product of fields $K_1 \times \cdots \times K_r$, one can identify M with the module $K_1^n \times \cdots \times K_r^n$; if x_j is then an element of N whose j-th component in $K_1^n \times \cdots \times K_r^n$ is non-zero, show that a certain linear combination x of the x_j with coefficients in B has all components non-zero; then replace M by M/Ax and proceed by induction on n).

We apply the preceding lemma in the following situation: $B = B$, $A = A_0$, i is the inclusion of B in A_0, $M = A_1$ regarded as an A_0-module via the homomorphism δ_1, $N = \delta_0(A_0)$. Indeed, since $d_0 \boxtimes d_1 : X_1 \to X_0 \times_Y X_0$ is a closed immersion, the homomorphism $A_0 \otimes_B A_0 \to A_1$ with components δ_0 and δ_1 is surjective; this means precisely that $\delta_0(A_0)$ generates the A_0-module A_1.

Let then $a_1, \cdots, a_n$ be elements of A_0 such that $\delta_0(a_1), \cdots, \delta_0(a_n)$ form a basis of A_1 over A_0. If we show that $a_1, \cdots, a_n$ is a basis of A_0 over B, it will follow that the homomorphism $A_0 \otimes_B A_0 \to A_1$ sends the basis $(1 \otimes a_i)_{1 \leqslant i \leqslant n}$ to the basis $(\delta_0(a_i))_{1 \leqslant i \leqslant n}$, hence is bijective. Consequently, if $\epsilon : \mathbb{Z}^n \to A_0$ is the morphism of abelian groups sending the natural basis of $\mathbb{Z}^n$ to $a_1, \cdots, a_n$, it suffices to prove that the map $B \otimes_{\mathbb{Z}} \mathbb{Z}^n \to A_0$ with components i and ϵ is bijective.

Now the diagram $(0, 1, 2)*$ considered at the beginning of this proof induces the following commutative diagram:

δ'_1 $\qquad\qquad\qquad$ δ_0

$A_2 \leftleftarrows A_1 \leftleftarrows A_0$

δ'_0

$u_2 \downarrow \qquad u_1 \downarrow \cong \qquad u_0 \downarrow$

$\delta_1 \otimes \mathbb{Z}^n \qquad i \otimes \mathbb{Z}^n$

[365] N.D.E.: cf. Lemma 4.1.2.

$$A_1 \otimes_{\mathbb{Z}} \mathbb{Z}^n \leftleftarrows A_0 \otimes_{\mathbb{Z}} \mathbb{Z}^n \leftleftarrows B \otimes_{\mathbb{Z}} \mathbb{Z}^n,$$
$$\delta_0 \otimes \mathbb{Z}^n$$

where u_0, u_1 and u_2 have respectively as components i and ϵ, δ_1 and $\delta_0\epsilon$, δ_2' and $\delta_0'\delta_0\epsilon$. We know that u_1 is an isomorphism. Since the two left-hand squares of $(0,1,2)*$ are cocartesian, u_2 is bijective. But the two horizontal rows of our diagram are exact, so u_0 is bijective.[366] This shows that A_0 is a B-module locally free of rank n, and, by the previous reductions, this entails that $\delta_0 \otimes \delta_1 : A_0 \otimes_B A_0 \to A_1$ is an isomorphism. This completes the proof of theorem 4.1 in the particular case considered (X_0 affine and d_1 locally free of constant rank n).

5. Passage to the quotient by a finite and flat groupoid (general case)

a) Let $U^{(n)}$ be the largest open subset of X_0 above which d_1 is finite locally free of rank n. One knows that X_0 is the direct sum of the $U^{(n)}$. It follows on the other hand from the two Cartesian squares

$$\begin{tikzcd} X_2 \arrow[r, "d_0'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_0"'] & X_0 \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} X_2 \arrow[r, "d_1'"] \arrow[d, "d_2'"'] & X_1 \arrow[d, "d_1"] \\ X_1 \arrow[r, "d_1"'] & X_0 \end{tikzcd}$$

that the inverse images of $U^{(n)}$ under d_0 and d_1 both coincide with the largest open subset of X_1 above which d_2' is locally free of rank n;[367] one therefore has $d_0^{-1}(U^{(n)}) = d_1^{-1}(U^{(n)})$, so that the groupoid $X*$ is the direct sum of the groupoids $X*^{(n)}$ induced by $X*$ on the open-and-closed subsets $U^{(n)}$. Consequently, as one sees easily, it suffices to prove theorem 4.1 for each of the $X*^{(n)}$: one is reduced to the case where d_1 is finite locally free of rank n.

b) We are now in a position to prove our theorem in the general case.

By a) one may suppose d_1 locally free of rank n. Let then (Y, p) be a cokernel of (d_0, d_1) in the category of all ringed spaces. The argument at the end of paragraph 4.c) shows that to prove 4.1 (i) it suffices to prove that Y is a scheme and $p : X_0 \to Y$ a morphism of schemes. By lemma 1.2, the question is local on Y: let $y \in Y$ and let $x \in X_0$ with $p(x) = y$; if x possesses a saturated affine open neighborhood U, then $p(U)$ will be an affine open of Y by § 4, and $p|U$ will be a morphism of schemes. It therefore suffices to prove that every $x \in X_0$ possesses a saturated affine open neighborhood U. Here is how one proceeds (the proof is taken from SGA 1, VIII, cor. 7.6).

[366] N.D.E.: We have added what follows.

[367] N.D.E.: indeed, since d_0 (resp. d_1) is surjective, flat and finite, hence faithfully flat and affine, then d_2' is of rank n above a neighborhood of a point x of X_1 if and only if d_1 is of rank n above a neighborhood of $d_0(x)$ (resp. $d_1(x)$).

```
d₁(d₀⁻¹(x)) ⊂ U = (V_f)′ ⊂ V_f ⊂ V′ ⊂ V ⊂ X₀

    ↑               ↑           ↑
affine open     special      affine
special of V    affine open  open
                of V
                    ↑           ↑
                largest     largest
                saturated   saturated
                open in V_f open in V
```

By condition b) of 4.1, there exists an affine open V of X_0 containing $d_1(d_0^{-1}(x))$;[368] if $F = X_0 - V$, then $d_1(d_0^{-1}(F))$ is closed since d_1 is integral, and $V' = X_0 - d_1(d_0^{-1}(F))$ is the largest saturated open contained in V. Since V' is a neighborhood of the finite set $d_1(d_0^{-1}(x))$, there exists a section f of the structure sheaf of V vanishing on $V - V'$ and such that $d_1(d_0^{-1}(x))$ is contained in the open V_f of V consisting of points where f does not vanish. We shall show that the largest saturated open $(V_f)'$ of V_f is affine, and therefore answers the question.

Indeed, let $Z(f) = V' - V_f$. Then $d_0^{-1}(Z(f))$ is the set of points of $d_0^{-1}(V') = d_1^{-1}(V')$ where the image $d_0^*(f)$ of f under the map induced by d_0 vanishes. On the other hand, since d_1 induces a locally free morphism of rank n from $d_0^{-1}(V') = d_1^{-1}(V')$ onto V',[369] then, by lemma 4.1.1, $d_1(d_0^{-1}(Z(f)))$ is the set of points where the norm N of $d_0^*(f)$ for the morphism d_1 vanishes. It follows that $(V_f)' = V' - d_1(d_0^{-1}(Z(f)))$ is the set of points of V_f where N does not vanish; consequently, $(V_f)'$ is affine.

This proves 4.1 (i); assertions (ii), (iii), and the first part of (iv) are then clear. Let us finally show the consequences indicated at the end of point (iv) (cf. [Ray67a], th. 1 (iii)).

By hypothesis, the groupoid $X*$ comes from an equivalence relation $i : R \to X_0 \times X_0$ (i being therefore an immersion, cf. N.D.E. 19), and one has established that R is effective (cf. Exp. IV, 3.3.2) and that $p : X_0 \to Y = X_0/R$ is a surjective and finite locally free morphism, hence in particular faithfully flat and of finite presentation.

Consequently, denoting by `(M)` the family of faithfully flat morphisms locally of finite presentation, R is `(M)`-effective. Therefore, by Exp. IV, 6.3.3, (Y, p) represents the quotient sheaf of X_0 by R for the (fppf) topology, and the assertions concerning base change follow from IV, 3.4.3.1.

Remark 5.1.[370] *We keep the hypotheses and notations of 4.1, and suppose in addition that S is locally Noetherian and $\pi_0 : X_0 \to S$ is quasi-projective. Let us then show that $\pi : Y \to S$ is quasi-projective.*

The above hypotheses imply that $Y \to S$ is of finite type, see the proof of 6.1 (ii). Let A be an invertible O_{X_0}-module that is ample for π_0. By EGA II, 6.1.12, $p_*(A)$ is an

[368] N.D.E.: one has $d_1(d_0^{-1}(x)) = d_0(d_1^{-1}(x))$, cf. N.D.E. 16 in theorem 4.1.
[369] N.D.E.: We have added the reference to lemma 4.1.1, cf. [DG70], III, § 5.2, p. 313.
[370] N.D.E.: We have added this paragraph.

invertible $p_*(O_{X_0})$-module. There therefore exists a covering $(V_i)_{i\in I}$ of Y by affine opens such that A is trivial above each of the saturated affine opens $U_i = p^{-1}(V_i)$.

For each index i, write $A_{i,0} = O_{X_0}(U_i)$, $A_{i,1}$ the ring of the affine open $d_0^{-1}(U_i) = d_1^{-1}(U_i)$ of X_1, $\delta_{i,0}$ (resp. $\delta_{i,1}$) the morphism $A_{i,0} \to A_{i,1}$ induced by d_0 (resp. d_1), and $B_i = O_Y(V_i) = b \in A_{i,0} | \delta_{i,0}(b) = \delta_{i,1}(b)$.

Following EGA II, § 6.5, consider the invertible O_{X_0}-module $N_{d_1}(d_0^*(A))$, the norm relative to the finite locally free morphism $d_1 : X_1 \to X_0$ of the invertible O_{X_1}-module $d_0^*(A)$. If A is given, relative to the covering $(U_i)_{i\in I}$, by transition functions $c_{ij} \in O_{X_0}(U_i \cap U_j)^\times$, then $N_{\delta_1}(d_0^*(A))$ is given by the transition functions $N_{d_1}(\delta_0(c_{ij})) \in O_{X_0}(U_i \cap U_j)^\times$; since, by paragraph 4.a), these elements belong to $O_Y(V_i \cap V_j)^\times$, they define an invertible O_Y-module L, such that $p^*(L) = N_{d_1}(d_0^*(A))$. Moreover, note that for every $n \in \mathbb{N}^*$, one has $p^*(L^n) = N_{d_1}(d_0^*(A^n))$, cf. loc. cit., (6.5.2.1).

Let us show that L is ample for the morphism $\pi : Y \to S$. For this, replacing S by an affine open, we may suppose S affine. Let then $y \in Y$, $x \in X_0$ with $p(x) = y$, V an affine open of Y containing y, and $U = p^{-1}(V)$. Since A is π_0-ample, there exists $n \in \mathbb{N}^*$ and a section $s \in \Gamma(X_0, A^n)$ such that the open $(X_0)_s$ satisfies $x \in (X_0)_s \subset U$. With the preceding notations, s is given by sections $a_i \in A_{i,0} = O_{X_0}(U_i)$ such that $a_i = c_{ij} a_j$ on $U_i \cap U_j$, and $(X_0)_s$ is the union of the opens $U_i' = p \in \mathrm{Spec}(A_{i,0}) | a_i \notin p$.

For each index i, put $N(a_i) = N_{\delta_1}(\delta_0(a_i)) \in B_i$. By 4.1 (i) and lemma 4.1.1, one has:

```
p(U′_i) = p d₁(d₀⁻¹(U′_i)) = p d₁({q ∈ Spec(A_{i,1}) | δ_{i,0}(a_i) ∉ q})
```

and $d_1(q \in \mathrm{Spec}(A_{i,1}) | \delta_{i,0}(a_i) \notin q) = p \in \mathrm{Spec}(A_{i,0}) | N_{\delta_1}(\delta_{i,0}(a_i)) \notin p$, whence

```
p(U′_i) = {p ∈ Spec(B_i) | N(a_i) ∉ p}.
```

It follows that $p((X_0)_s)$ equals $Y_{N(s)}$, where we have written $N(s)$ for the section of L^n over Y defined by the sections $N(a_i) \in O_Y(V_i)$. One thus has

```
(*)   y ∈ p((X₀)_s) = Y_{N(s)} ⊂ p(U) = V.
```

This shows that L is ample for $\pi : Y \to S$, which finishes showing that $\pi : Y \to S$ is quasi-projective.

6. Passage to the quotient when a quasi-section exists

We shall now prove a lemma of technical character which will be useful in the proof of the two theorems we have in view. Let S be a scheme and

```
        d′₀, d′₁, d′₂       d₀, d₁
X₂ ⇶ X₁ ⇉ X₀
```

a (Sch/S)-groupoid. We shall call a *quasi-section* of the groupoid $X*$ any subscheme U of X_0 such that (1) and (2) hold:

(1) The restriction v of d_1 to $d_0^{-1}(U)$ is a finite, locally free, and surjective morphism from $d_0^{-1}(U)$ onto X_0.

(2) Every subset E of U consisting of points pairwise equivalent for the equivalence relation defined by $X*$ (§ 3.e)) is contained in an affine open of U.[371]

If U is a quasi-section of $X*$, the (Sch/S)-groupoid

$$U_2 \overset{u'_0,\, u'_1,\, u'_2}{\Rrightarrow} U_1 \overset{u_0,\, u_1}{\rightrightarrows} U$$

induced by $X*$ and the inclusion of U into X_0 (cf. § 3.a)) satisfies the hypotheses of theorem 4.1. Set indeed $V = d_0^{-1}(U)$ and let u and v be the morphisms with source V induced respectively by d_0 and d_1:

$$X_0 \xleftarrow{v} V \xrightarrow{u} U.$$

By paragraph 3.b), one has a Cartesian square

$$\begin{tikzcd} U_1 \arrow[r] \arrow[d, "u_1"'] & V \arrow[d, "v"] \\ U \arrow[r, "\text{inclusion}"] & X_0, \end{tikzcd}$$

so u_1 is surjective and finite locally free by (1). With (2), condition (1) therefore ensures that the groupoid $U*$ satisfies the hypotheses of theorem 4.1. In particular $Coker(u_0, u_1)$ exists in (Sch/S). Moreover, d_0 has a section, so that u is a universal effective epimorphism (cf. III 1.12); it follows, by proposition 3.1, that $Coker(u_0, u_1)$ coincides with the cokernel $Coker(v_0, v_1)$ of the groupoid $V*$:

$$V_2 \overset{v'_0,\, v'_1,\, v'_2}{\Rrightarrow} V_1 \overset{v_0,\, v_1}{\rightrightarrows} V,$$

inverse image of $U*$ under the base change $u : V \to U$, that is also the inverse image of $X*$ under the base change:

$$V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_0} X_0.$$

By paragraph 3.c), $V*$ is isomorphic to the groupoid $V'*$, the inverse image of $X*$ under the base change:

$$v : \ V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_1} X_0,$$

and so $V'*$ admits a cokernel in (Sch/S). Now, being flat, surjective and finite, $v : V \to X_0$ is faithfully flat and quasi-compact, hence a universal effective epimorphism by III 6.3.2. Consequently, by proposition 3.1, the groupoid $X*$ admits a cokernel $Coker(d_0, d_1)$ in (Sch/S). We have thus proved the first assertion of point (i) of the following lemma:[372]

[371] N.D.E.: If $x, y \in E$, there exists $z \in X_1$ such that $d_1(x) = x$ and $d_0(z) = y$, that is, z belongs to the set $v^{-1}(x)$, which is finite by (1). Hence E is contained in the finite set $d_0(v^{-1}(x)) = d_0 d_1^{-1}(x) \cap U$.

[372] N.D.E.: We have slightly modified what follows; in particular, in lemma 6.1, the additional hypothesis that d_0 be flat has been moved to (iv), and (ii) has been separated into (i′) + (ii), and the second assertion of (i′) added.

Lemma 6.1. *Suppose that the (Sch/S)-groupoid $X*$ possesses a quasi-section. Then:*

(i) There exists a cokernel (Y,p) of (d_0,d_1) in (Sch/S); moreover, such a (Y,p) is a cokernel of (d_0,d_1) in the category of all ringed spaces.

(i′) p is surjective, and is open (resp. universally closed) if d_0 is.

(ii) Suppose S locally Noetherian and X_0 locally of finite type (resp. of finite type) over S. Then p and $Y \to S$ are locally of finite presentation (resp. of finite presentation).

(iii) The morphism $X_1 \to X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.

(iv) If (d_0,d_1) is an equivalence pair, then $X_1 \to X_0 \times_Y X_0$ is an isomorphism. Moreover, if $d_0 : X_1 \to X_0$ is flat, p is faithfully flat.

Before proving the second assertion of (i), we shall demonstrate (i′), (ii) and (iii).

a) Proof of (i′) and (ii):

We have just seen that (Y,p) identifies with $Coker(v_0,v_1)$ and $Coker(u_0,u_1)$. Let then q and r be the canonical epimorphisms from U and V into Y:

$$\begin{tikzcd} X_0 & V \arrow[l, "v"'] \arrow[r, "u"] & U \arrow[dl, "q"] \\ & Y. \arrow[from=1-1, "r"'] & \end{tikzcd}$$

By hypothesis, v is surjective and finite locally free, hence open. On the other hand, if $d_0 : X_1 \to X_0$ is open (resp. universally closed), then u, which is obtained from it by base change, is also.

Since, by theorem 4.1, q is surjective, integral, and open, it follows that r is surjective, and open (resp. universally closed) if d_0 is. The same therefore holds for p, since v is surjective. This proves (i′).

Suppose now S locally Noetherian and X_0 locally of finite type over S, so that X_0 is locally Noetherian.

Let us show that Y is locally of finite presentation over S. Let $S' = \operatorname{Spec} R$ be an affine open of S, $Y' = \operatorname{Spec} B$ an affine open of Y projecting into S', and $U' = \operatorname{Spec} A$ the inverse image of Y' in U. Since R is Noetherian, it suffices to show that B is a finitely generated R-algebra; but, by paragraphs 4 and 5, B is contained in A, which is a finitely generated R-algebra; the assertion therefore follows from the fact that R is Noetherian and A is integral over B.

Finally, since $X_0 \to S$ is locally of finite type, so is p (EGA I, 6.6.6), hence p is locally of finite presentation since Y is locally Noetherian.

It remains to show the last assertion of (ii). Suppose in addition X_0 of finite type over S. Then, since p is surjective, Y is also quasi-compact over S, hence of finite type over S. Since S is locally Noetherian, then $X_0 \to S$ and $Y \to S$ are of finite presentation, and so $p : X_0 \to Y$ is also (EGA IV_1, 1.6.2 (v)).

b) Proof of (iii):

Since the groupoid $V*$ with base V is isomorphic both to the inverse image of $U*$ under the base change u and to the inverse image of $X*$ under the base change v, one has a double Cartesian square

$$\begin{tikzcd}
X_1 \arrow[d, "d_0 \boxtimes d_1"'] & V_1 \arrow[l] \arrow[r] \arrow[d, "v_0 \boxtimes v_1"'] & U_1 \arrow[d, "u_0 \boxtimes u_1"] \\
X_0 \times_Y X_0 & V \times_Y V \arrow[l, "v \times v"] \arrow[r, "u \times u"'] & U \times_Y U.
\end{tikzcd}$$

Since $u_0 \boxtimes u_1$ is surjective, so is $v_0 \boxtimes v_1$. Since $v \times v$ is surjective, so is the composite morphism $V_1 \to X_0 \times_Y X_0$, and therefore so is $d_0 \boxtimes d_1$.

c) Proof of (i):

It remains to prove that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. We first show that Y is obtained from X_0 by identifying the points x and y such that there exists $z \in X_1$ with $d_0(z) = x$ and $d_1(z) = y$. Indeed, p is surjective and one has $pd_0 = pd_1$; moreover, if $p(x) = p(y)$, there is a point z' of $X_0 \times_Y X_0$ whose first projection is x and second projection is y. If z is a point of X_1 such that $(d_0 \boxtimes d_1)(z) = z'$, one indeed has $d_0(z) = x$ and $d_1(z) = y$.

On the other hand, if W is a saturated open of X_0, then $W \cap U$ is a saturated open of U; by 4.1, $q(W \cap U)$ is an open of Y. Since $q(W \cap U)$ is none other than $p(W)$, one sees that Y is endowed with the quotient topology of that of X_0.

It remains to show that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \to p_*(\mathcal{O}_{X_0}) \rightrightarrows p_* d_{0*}(\mathcal{O}_{X_1}) = p_* d_{1*}(\mathcal{O}_{X_1})$$

is exact.

Let then Y' be an open of Y and put $U' = q^{-1}(Y')$, $X'_0 = p^{-1}(Y')$, etc.[373] Then U' is an open of U saturated for the equivalence relation defined by the groupoid $U*$, and it follows from lemmas 1.1 and 1.2 that Y' is the cokernel, in (Sch/S) and in (Esp.An), of the groupoid induced by $U*$ on U'. Similarly, X'_0 is an open of X_0 saturated for the equivalence relation defined by $X*$, and one has the following commutative diagram, where the two squares are Cartesian:

$$\begin{tikzcd}
X'_0 \arrow[d, dash] & V' = d_0^{-1}(U') \arrow[l, "\tilde{d}_1"'] \arrow[r, "\tilde{d}_0"] \arrow[d, dash] & U' \arrow[d, dash] \\
{} & {} & {}
\end{tikzcd}$$

[373] N.D.E.: We have expanded what follows, in particular the fact that U' is a quasi-section of the groupoid induced on X'_0.

$$X_0 \xleftarrow{d_1} V = d_0^{-1}(U) \xrightarrow{d_0} U'.$$

Then $\tilde{d}_1$ is surjective, and finite locally free. On the other hand, let $x \in U'$. Since U is a quasi-section, the set $E := d_0 d_1^{-1}(x) \cap U$ is finite and contained in an affine open W of U. Then $E' = E \cap U'$ is a finite set, contained in the quasi-affine open $W \cap U'$. Consequently, there exists an affine open W' of $W \cap U'$ containing E'. This shows that U' is a quasi-section of the groupoid $X'*$ induced by $X*$ on X'_0. The first assertion of (i), applied to $X'*$ and U', then shows that Y' is the cokernel in (Sch/S) of $X'*$.

In particular, for every S-scheme T, one has the exact sequence

$$T(Y') \xrightarrow{T(p|_{X'_0})} T(X'_0) \underset{T(d_0|_{X'_1})}{\overset{T(d_1|_{X'_1})}{\rightrightarrows}} T(X'_1).$$

Now, if T is the "affine line" $G_{a,S}$ (I 4.3), this sequence identifies with the sequence

$$\Gamma(Y', \mathcal{O}_Y) \to \Gamma(p^{-1}(Y'), \mathcal{O}_{X_0}) \underset{\delta_0}{\overset{\delta_1}{\rightrightarrows}} \Gamma(d_0^{-1} p^{-1}(Y'), \mathcal{O}_{X_1}) = \Gamma(d_1^{-1} p^{-1}(Y'), \mathcal{O}_{X_1})$$

which is therefore exact for every open Y'. This completes the proof of 6.1 (i).

d) Proof of (iv):

If (d_0, d_1) is an equivalence pair, the same holds for (u_0, u_1). It follows that $u_0 \boxtimes u_1 : U_1 \to U \times_Y U$ is an isomorphism (theorem 4.1), hence so is $v_0 \boxtimes v_1$ (confer the Cartesian squares of b)); since $v \times v$ is faithfully flat and quasi-compact, $d_0 \boxtimes d_1$ is an isomorphism (SGA 1, VIII 5.4).

Moreover, if d_0 is flat, so is u. Now q is flat, by theorem 4.1, so r also is. Since v is faithfully flat, then p is flat, and therefore faithfully flat since surjective.

7. Quotient by a proper and flat groupoid

Theorem 7.1.[374] *Let S be a locally Noetherian scheme and*

$$X_2 \overset{d'_0,\, d'_1,\, d'_2}{\rightrightarrows} X_1 \overset{d_0,\, d_1}{\rightrightarrows} X_0$$

[374] N.D.E.: Let us mention here the article of S. Keel and S. Mori ([KM97]), where the following theorem is established. Let X be an algebraic space of finite type over a locally Noetherian base S, and $j : R \to X \times_S X$ a flat groupoid whose stabilizer $j^{-1}(\Delta_X)$ is finite over X; there then exists an algebraic space which is a geometric quotient of X by R and a uniform categorical quotient; moreover, if j is separated, this quotient is separated. In particular, if a flat S-group scheme G acts properly on X, with finite stabilizer (i.e., the morphism $G \times_S X \to X \times_S X$, $(g, x) \mapsto (x, g \cdot x)$, is proper and the stabilizer of the diagonal is finite over X), then there exists a geometric quotient $X \to X/G$. In the case of a reductive S-group scheme G, this is a result of J. Kollár ([Ko97]).

a (Sch/S)-groupoid such that d_1 is proper and flat, X_0 is quasi-projective over S,[375] *and the morphism $d : X_1 \to X_0 \times_S X_0$ with components d_0 and d_1 is quasi-finite. Then:*

(i) There exists a cokernel (Y, p) of (d_0, d_1) in (Sch/S); moreover, such a (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.

(ii) p is surjective, open, proper, of finite presentation, and $Y \to S$ is of finite presentation and separated.[376]

(iii) The morphism $X_1 \to X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.

(iv) If (d_0, d_1) is an equivalence pair, then $X_1 \to X_0 \times_Y X_0$ is an isomorphism and p is faithfully flat.[377] *Moreover, (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the (fppf) topology and, for every base change $Y' \to Y$, Y' is the cokernel of the groupoid $X * \times_Y Y'$ obtained from $X*$ by the base change $X_0 \times_Y Y' \to X_0$.*

In particular, for every base change $S' \to S$, $Y' = Y \times_S S'$ is the cokernel of the S'-groupoid $X'* = X * \times_S S'$. So, in this case, "the formation of the quotient commutes with base change".

Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Lemma 1.2 shows that, to prove (i), it suffices to show that every point z of X_0 possesses a saturated open neighborhood U_z such that, denoting by $\tilde{d}_0$ and $\tilde{d}_1$ the restrictions of d_0 and d_1 to $d_0^{-1}(U_z) = d_1^{-1}(U_z)$, and by (Q, q) the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in `(Esp.An)`, Q is a scheme and q a morphism of schemes.

By lemma 6.1 (i), it therefore suffices to show that every point z of X_0 possesses a saturated open neighborhood U_z such that the groupoid induced on U_z by $X*$ possesses a quasi-section. One can even suppose that z is closed in its fiber over S (we shall say that z is *closed relative to S*).[378] The existence of U_z then follows from the lemmas below:

Lemma 7.2. *Let T be an affine Noetherian scheme, X, Y, and Z T-schemes of finite type, with X quasi-projective over T, and*

$$\begin{tikzcd} Y \arrow[r, "u"] \arrow[d, "v"'] & X \arrow[d] \\ Z \arrow[r, dotted] & T \end{tikzcd}$$

[375] N.D.E.: This hypothesis on X_0 is necessary, cf. N.D.E. 17 in Th. 4.1.

[376] N.D.E.: In TDTE III, Th. 6.1, it is indicated that $Y \to S$ is quasi-projective if S is Noetherian. The editors have not seen how to deduce this from the local existence of quasi-sections.

[377] N.D.E.: We have made explicit the consequences which follow; see [Ray67a], th. 1 (iii) and the end of the proof of the theorem. Let us also mention that another proof of th. 7.1 in the case of an equivalence relation, based on the existence of Hilbert schemes, is given in [AK80], Th. 2.9, see also [BLR90], § 8.2, Th. 12; it is moreover shown there, in this case, that $Y \to S$ is quasi-projective.

[378] N.D.E.: Indeed, if one has constructed such an open neighborhood U_z for every point z closed relative to S, then the union of these U_z covers X_0, since each fiber over S of the closed complement is a Noetherian scheme without closed points, hence empty.

a diagram in (Sch/T). Let z be a point of $v(Y)$ that is closed relative to T and such that v is flat at the points of $v^{-1}(z)$. Then there exists a closed subscheme F of X such that $u(u^{-1}(F) \cap v^{-1}(z))$ is finite and non-empty, and such that the restriction of v to $u^{-1}(F)$ is flat at the points of $v^{-1}(z)$.

Let $T = \operatorname{Spec} A$. One may suppose X of the form $\operatorname{Proj} S$, where S is the symmetric algebra of a finitely generated A-module E.

If $u(v^{-1}(z))$ is finite, one can choose F equal to X. Otherwise, we denote by $y_1, \cdots, y_n$

the points of the fiber $v^{-1}(z)$ associated with the structure sheaf $O_{v^{-1}(z)}$ of $v^{-1}(z)$ (the y_i are such that, if O_i denotes the local ring of $v^{-1}(z)$ at y_i, the maximal ideal of O_i consists of zero divisors). If t is the image of z in T, $u(v^{-1}(z))$ is an infinite constructible subset of the fiber of t in X. There therefore exists a point x closed in this fiber, belonging to $u(v^{-1}z)$ and distinct from $u(y_1), \cdots, u(y_n)$. Then $X - x$ is an open neighborhood of $u(y_1), \cdots, u(y_n)$, hence contains an open neighborhood of the form $D_+(f)$, where f is a homogeneous element of degree d of S (the notations are those of EGA II, § 2.3).

Consequently, the closed subscheme $X_1 = V_+(f)$ defined by f contains x and avoids the points $u(y_1), \cdots, u(y_n)$. It follows of course that the inverse image $Y_1 = u^{-1}(V_+(f))$ of this subscheme is distinct from Y and meets $v^{-1}(z)$. We shall further show that the restriction v_1 of v to Y_1 is flat at the points of $v^{-1}(z)$; if $u(v_1^{-1}(z))$ is finite, we shall therefore only need to choose F equal to X_1; otherwise, we shall repeat the argument we have just developed, replacing Y by Y_1, v by v_1, u by the morphism u_1 induced on Y_1 by u; in this way we shall obtain a decreasing sequence $X, X_1, \cdots$ of closed subschemes of X; since such a sequence terminates, $u(u^{-1}(X_n) \cap v^{-1}(z))$ will be finite and non-empty for some n, and one will choose F equal to X_n.

It remains then to show that v_1 is flat at the points of $v^{-1}(z)$; let y be a point of Y_1 above z, O_y the local ring of y in Y, $\bar{O}_y$ the local ring of y in $v^{-1}(z)$, $O_{v(y)}$ the local ring of $v(y)$ in Z. If $g \in S_1$ is such that $D_+(g)$ is a neighborhood of $u(y)$ in X, let ϕ be the image of f/g^d in O_y and $\bar{\phi}$ the image of f/g^d in $\bar{O}_y$. It then follows from the construction of f that $\bar{\phi}$ is not a zero divisor in $\bar{O}_y$; since

O_y is flat over O_z, ϕ is not a zero divisor in O_y and $O_y/O_y\phi$ is flat over O_z (SGA 1, IV 5.7). But $O_y/O_y\phi$ is precisely the local ring of y in Y_1.

Lemma 7.3. *We keep the notations and hypotheses of 7.1. Every point z of X_0 closed relative to S therefore possesses a saturated open neighborhood U_z such that the groupoid induced by $X*$ on U_z possesses a quasi-section.*

The statement being local on S, one may suppose S affine Noetherian and apply the previous lemma to the diagram

$$\begin{tikzcd} X_1 \arrow[r, "d_0"] \arrow[d, "d_1"'] & X_0 \arrow[d] \\ {} & {} \end{tikzcd}$$

$$\begin{tikzcd} \arrow[d] & \arrow[d] \\ X_0 \arrow[r, dotted] & S \end{tikzcd}$$

of (Sch/S). Let then F be a closed subscheme of X_0 such that $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite and non-empty, and such that the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $d_1^{-1}(z)$.[379]

Denote by F_1 and F_2 the inverse images of F under d_0 and under $d_0 d_0' = d_0 d_1'$, and denote by $\tilde{d}_0$, $\tilde{d}_1$, etc., the morphisms induced by d_0, d_1, etc. One thus has a commutative diagram

$$\begin{tikzcd} F_2 \arrow[r, shift left, "\tilde{d}'_1"] \arrow[r, shift right, "\tilde{d}'_0"'] \arrow[d, "\tilde{d}'_2"'] & F_1 \arrow[r, "\tilde{d}_0"] \arrow[d, "\tilde{d}_1"] & F \arrow[d, dotted, "\tilde{q}"] \\ X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0 \arrow[r, dotted, "q"'] & S, \end{tikzcd}$$

where the two left-hand squares are Cartesian and the first row is exact (confer (0,1,2), § 1), and where q and $\tilde{q}$ denote the structure morphisms.

Let us first show that there are only finitely many points of F_1 above z.[380] Indeed, let s be the image of z in S; since F is of finite type over S, the fiber $\tilde{q}^{-1}(s)$ is a Noetherian scheme. On the other hand, since $\tilde{d}_0$ is proper, $\tilde{d}_0(\tilde{d}_1^{-1}(z))$ is a closed subscheme of $\tilde{q}^{-1}(s)$, consisting of finitely many points. Consequently (cf. EGA I, 6.2.2), the points of this set are closed in $\tilde{q}^{-1}(s)$, and also (since F is closed in X_0) in the fiber $q^{-1}(s)$ of s in X_0. Let y be one of these points; since the fiber $q^{-1}(s)$ is of finite type over $\kappa(s)$, it contains affine open neighborhoods $\operatorname{Spec} B$ and $\operatorname{Spec} C$ of y and z, respectively, where B and C are finitely generated $\kappa(s)$-algebras. Then y and z correspond to maximal ideals $p \subset B$ and $q \subset C$, the fields B/p and C/q are of finite degree over $\kappa(s)$, and so $(B/p) \otimes_{\kappa(s)} (C/q)$ is a $\kappa(s)$-algebra of finite dimension, whose maximal ideals correspond exactly to the points of $X_0 \times_S X_0$ whose second (resp. first) projection is z (resp. y). There are therefore only finitely many points u of $X_0 \times_S X_0$ whose

second projection is z and whose first projection belongs to $\tilde{d}_0(\tilde{d}_1^{-1}(z))$. Finally, since $X_1 \to X_0 \times_S X_0$ has finite fibers, such a point u comes from finitely many points of X_1, whence the desired assertion.

The morphism $\tilde{d}_1$ is therefore quasi-finite and flat at the points of F_1 above z. Since $\tilde{d}_1$ is of finite type, it follows from SGA 1, IV 6.10 and EGA III, 4.4.10,[381] that the

[379] N.D.E.: We have added details, and made explicit the role of the hypothesis that d_0 and d_1 are proper in theorem 7.1. (One can compare with the statement and proof of theorem 8.1, where this properness hypothesis is omitted.)

[380] N.D.E.: We have added details, and made explicit the role of the hypothesis that d_0 and d_1 are proper in theorem 7.1. (One can compare with the statement and proof of theorem 8.1, where this properness hypothesis is omitted.)

[381] N.D.E.: If $f : X \to Y$ is a morphism locally of finite type, the set of $x \in X$ isolated in their fiber

set Φ of points of F_1 where $\tilde{d}_1$ is not simultaneously flat and quasi-finite is closed in F_1, hence in X_1 (since F_1 is closed in X_1). Since d_1 is proper, $\tilde{d}_1(\Phi)$ is closed, and does not contain z by what precedes. Put $W = \tilde{d}_1(F_1) - \tilde{d}_1(\Phi)$. Then the restriction of $\tilde{d}_1$ to $\tilde{d}_1^{-1}(W)$ is[382] of finite presentation (in view of the Noetherian hypotheses), flat, proper and quasi-finite, hence finite, locally free, and open, by EGA III, 4.4.2, and EGA IV_2, 2.1.12 and 2.4.6. Consequently, $\tilde{d}_1(F_1)$ is a neighborhood of z, and W is the largest open of X_0 contained in $\tilde{d}_1(F_1)$ above which $\tilde{d}_1$ is simultaneously flat and quasi-finite.

We shall see in lemma 7.4 that the inverse images of Φ by $\tilde{d}'_1$ and $\tilde{d}'_0$ both identify with the set of points of F_2 where $\tilde{d}'_2$ is not simultaneously flat and quasi-finite. It follows that $d_0^{-1}(W) = \tilde{d}'_2(F_2) - \tilde{d}'_2(\tilde{d}'_0\Phi)$ coincides with $d_1^{-1}(W) = \tilde{d}'_2(F_2) - \tilde{d}'_2(\tilde{d}'_1\Phi)$, that is, W is saturated. Consequently, setting $W_1 = \tilde{d}_1^{-1}(W)$, the equality $d_0^{-1}(W) = d_1^{-1}(W)$ entails $\tilde{d}'_2\tilde{d}'^{-1}_0(W) = \tilde{d}'_2\tilde{d}'^{-1}_1(W)$, that is, $\tilde{d}'^{-1}_0(W_1) = \tilde{d}'^{-1}_1(W_1)$. Since $\tilde{d}_0$ is faithfully flat and quasi-compact (because d_0 is, like d_1, surjective, proper and flat), and the square

$$\begin{tikzcd} F_2 \arrow[r, "\tilde{d}'_1"] \arrow[d, "\tilde{d}'_0"'] & F_1 \arrow[d, "\tilde{d}_0"] \\ F_1 \arrow[r, "\tilde{d}_0"'] & F \end{tikzcd}$$

is Cartesian, it follows that W_1 is of the form $\tilde{d}_0^{-1}(U)$, where U is an open of F

(SGA 1, VIII 4.4). This open U of F is a quasi-section for the groupoid with base W induced by $X*$. One can therefore choose U_z equal to W.

It remains for us therefore to state lemma 7.4, whose proof is classical:

Lemma 7.4. *Consider a Cartesian square of schemes*

$$\begin{tikzcd} F_2 \arrow[r, "v"] \arrow[d, "d'"'] & F_1 \arrow[d, "d"] \\ X_1 \arrow[r, "u"'] & X_0 \end{tikzcd}$$

and let x be a point of F_2.

(i) If u is flat, d' is flat at x if and only if d is flat at $v(x)$.

(ii) If d is locally of finite type, d' is quasi-finite at x if and only if d is quasi-finite

$f^{-1}(f(x))$ is open in X: in EGA III, 4.4.10, this is deduced, for Y locally Noetherian, from Zariski's "Main Theorem"; on the other hand, for arbitrary Y, this follows from Chevalley's semi-continuity theorem (EGA IV_3, 13.1.3 and 13.1.4). Consequently, f is quasi-finite at x if and only if f is of finite type at x and x is isolated in $f^{-1}(f(x))$; this will be used further on, cf. N.D.E. 42.

[382] N.D.E.: We have expanded on the original in what follows.

at $v(x)$.[383]

We have thus proved that there exists a covering of X_0 by saturated opens W such that the groupoid $W*$ induced by $X*$ on W possesses a quasi-section.[384]

By lemma 6.1 and the reductions stated after theorem 7.1, this implies assertions (i) and (iii) of theorem 7.1, and the fact that p is surjective and open, and that p and $Y \to S$ are locally of finite presentation. Moreover, since $X_0 \to S$ is quasi-projective, hence separated and of finite type, then p is separated, and the proof of point (ii) of lemma 6.1 shows that p and $Y \to S$ are of finite presentation.

To show that p is proper, it remains therefore to show that it is universally closed. As the assertion is local on Y, one may place oneself on a saturated open W such that the groupoid $W*$ induced by $X*$ on W possesses a quasi-section U (since X_0 is covered by such opens). Taking up the notations of 6.a), one has a commutative diagram

$$\begin{tikzcd} W \arrow[dr, "r"'] & V \arrow[l, "v"'] \arrow[r, "u"] & U \arrow[dl, "q"] \\ & Z, & \end{tikzcd}$$

where Z is an open of Y, all the arrows are surjective, and q is integral. Moreover, by hypothesis, $d_0 : X_1 \to X_0$ is proper, so u, obtained from it by base change, is also. Consequently, r is universally closed, and so is p, since v is surjective.

Finally, p being surjective and universally closed, and X_0 quasi-projective hence separated, the diagonal $\Delta_{Y/S}(Y)$ is closed in $Y \times_S Y$, being the image under $p \times p$ of the diagonal $\Delta_{X_0/S}(X_0)$. So Y is separated over S. This completes the proof of 7.1 (ii).

The assertions to prove in 7.1 (iv) are local on Y; since X_0 is covered by the saturated opens U_z, it suffices to verify these assertions by replacing X and Y by U_z and $V = p(U_z)$. As one has already seen at the beginning of the proof of 7.1, it follows from lemmas 1.1, 1.2, and 6.1 (i), that $(V_z, p|_{U_z})$ is the cokernel in (Sch) and in (Esp.An) of the groupoid induced by $X*$ on U_z. Now the hypothesis that

$d = (d_0, d_1)$ is a monomorphism is preserved by the base change $U_z \to X_0$. Consequently, the first two assertions of 7.1 (iv) follow from 6.1 (iv).

Let us finally show the consequences indicated at the end of point (iv) (cf. [Ray67a], th. 1 (iii)). By hypothesis, the groupoid $X*$ comes from an equivalence relation

[383]N.D.E.: The conditions are sufficient, by base change (cf. EGA II, 6.2.4 (iii) and EGA IV_2, 2.1.4). Conversely, put $y = d'(x)$ and $z = u(y) = d(v(x))$, and suppose d' flat at x and u (hence also v) flat. Then $\mathcal{O}_{v(x)} \to \mathcal{O}_x$ is faithfully flat, as is $\mathcal{O}_z \to \mathcal{O}_y \to \mathcal{O}_x$. Consequently, $\mathcal{O}_z \to \mathcal{O}_{v(x)}$ is faithfully flat (cf. EGA IV_2, 2.2.11 (iv)). Finally, suppose d locally of finite type and d' quasi-finite at x. Then $v(x)$ is isolated in its fiber $d^{-1}(z)$, since x is in its fiber $d'^{-1}(y) = d^{-1}(z) \otimes_{\kappa(z)} \kappa(y)$. Hence, by Chevalley's semi-continuity theorem, there exists an open neighborhood of $v(x)$ every point of which is isolated in its fiber (EGA IV_3, 13.1.3 and 13.1.4), so that d is quasi-finite at $v(x)$.

[384]N.D.E.: We have modified the sequel, taking advantage of the additions made in lemma 6.1.

$R \to X_0 \times X_0$, and one has established that R is effective (cf. Exp. IV, 3.3.2) and that $p : X_0 \to Y = X_0/R$ is faithfully flat and of finite presentation. Consequently, denoting by `(M)` the family of faithfully flat morphisms locally of finite presentation, R is `(M)`-effective. Therefore, by Exp. IV, 6.3.3, (Y, p) represents the quotient sheaf of X_0 by R for the (fppf) topology, and the assertions concerning base change follow from IV, 3.4.3.1.

8. Passage to the quotient by a flat, not necessarily proper, groupoid

Theorem 8.1.[385] *Let S be a Noetherian scheme and*

$$X_2 \overset{d'_0,\, d'_1,\, d'_2}{\rightrightarrows} X_1 \overset{d_0,\, d_1}{\rightrightarrows} X_0$$

a (Sch/S)-groupoid such that d_1 is flat and of finite type, X_0 is of finite type over S, and the morphism $X_1 \to X_0 \times_S X_0$ with components d_0 and d_1 is quasi-finite.

There then exists an open W of X_0 which is dense, saturated, and satisfies the following properties:

(i) If $W_2 \rightrightarrows W_1 \rightrightarrows W$ (with arrows w'_i, w_j) is the groupoid induced by $X*$ on W, then (w_0, w_1) admits a cokernel (V, p) in (Sch/S); moreover, (V, p) is a cokernel of (w_0, w_1) in the category of all ringed spaces.

(ii) p is surjective and open.

(ii′) p and $V \to S$ are of finite presentation.

(iii) The morphism $W_1 \to W \times_V W$ with components w_0 and w_1 is surjective.

[385] N.D.E.: There exists a largest open W of X_0 satisfying the conclusions of the theorem. Indeed, let W be an open as in the theorem and $W^{\square}$ a dense saturated open contained in W. Since p is open, $V^{\square} = p(W^{\square})$ is an open of V, and $W^{\square} = p^{-1}(V^{\square})$, since $W^{\square}$ is saturated. By lemma 1.1, $V^{\square}$ is a cokernel for the groupoid induced on $W^{\square}$. Thus one can glue along their intersection $W^{\square}$ two opens W and W' satisfying the conclusions of the theorem, and the conditions (i), (ii), (iii), (iv), as well as the fact that p and $V \to S$ are locally of finite presentation, are preserved. The conclusion (ii′) follows, as in the proof of 6.1 (ii), from the hypothesis that X_0 is of finite type over S Noetherian. Moreover, lemmas 1.1 and 1.2 also show that the union of all saturated opens U of X_0 such that the open $p(U)$ of Y is a scheme and $p|_U : U \to p(U)$ is a morphism of schemes is the largest saturated open Ω of X_0 satisfying condition (i) of 8.1. Theorem 8.1 shows that Ω contains a dense open W, but it is not immediate that Ω satisfies properties (ii) to (iv). On this subject, the reader may consult [Ray67a], [Ray67b], and Appendix I of [An73], which give more precise results, and study the question of the representability of the quotient S-sheaf (fppf) $\tilde{X}/R$ (where one has denoted by R the groupoid $X*$ with base $X = X_0$), all this under weaker hypotheses (S an arbitrary scheme, X a scheme locally of finite type over S, and R an S-groupoid with base X such that d_0 (and therefore d_1) is flat and of finite presentation). Let us mention in particular the following results. If $\tilde{X}/R$ is representable by an S-scheme Y, then Y is also the cokernel in the category `(Esp. An)`. The converse is in general false (cf. example 0.4 of [Mum65], Chap. 0, § 3, cited in [Ray67a], Rem. 1), but is true if $d = (d_0, d_1)$ is an immersion. Under this hypothesis, the morphism $p : \Omega \to Z := \Omega/R_\Omega$ is faithfully flat and of finite presentation; if moreover S is locally Noetherian, then a point x of codimension 1 in X belongs to Ω if and only if the graph of the groupoid induced on $\mathrm{Spec}(O_{X,x})$ is closed. For all this, see [Ray67a], Prop. 1, [Ray67b], Prop. 1 and Theorems 2, 1 and 4, and [An73], Theorems 5 and 6 pages 66–67, and Prop. 3.3.1 page 49. (See also, in the case of an action of an algebraic group on an algebraically closed field k, the article [DR81].)

(iv) If (d_0, d_1) is an equivalence pair, $W_1 \to W \times_V W$ is an isomorphism and p is faithfully flat.

We shall show that one can choose W in such a way that the (Sch/S)-groupoid $W*$ induced by $X*$ possesses a quasi-section (confer § 7). Theorem 8.1 will then follow from lemma 6.1.

Suppose provisionally that, for every point $z \in X_0$ closed relative to S (confer § 7), there exists a saturated open W_z which possesses a quasi-section and meets all the irreducible components of X_0 passing through z. Then the exterior $X_0 - W_z$ of W_z in X_0 is saturated (since the saturation $d_1(d_0^{-1}(X_0 - W_z))$ of this exterior is open and does not meet W_z). If this exterior is non-empty, one can choose in it a point z' closed relative to S and associate to z' an open $W_{z'}$ as above; one may moreover suppose $W_{z'}$ contained in $X_0 - W_z$; then W_z and $W_{z'}$ are disjoint and the groupoid induced by $X*$ on $W_z \cup W_{z'}$ possesses a quasi-section. The process must stop, because X_0 has only finitely many irreducible components. It therefore remains to construct W_z.

For this, one may suppose S affine; in this case, let y be a point of X_1

such that $d_1(y) = z$, X an affine open of X_0 containing $d_0(y)$, Y the inverse image of X in X_1 under d_0, and finally $u : Y \to X$ and $v : Y \to X_0$ the morphisms induced by d_0 and d_1. Since X is affine, hence quasi-projective, one can apply lemma 7.2: there is therefore a closed subscheme F of X_0 such that $d_0^{-1}(F) \cap d_1^{-1}(z)$ is non-empty, $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite, and the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $v^{-1}(z)$. This allows us to take up the notations of lemma 7.3, denoting by F_1 and F_2 the inverse images of F in X_1 and X_2, etc.

$$\begin{tikzcd} F_2 \arrow[r, shift left, "\tilde{d}'_1"] \arrow[r, shift right, "\tilde{d}'_0"'] \arrow[d, "\tilde{d}'_2"'] & F_1 \arrow[r, "\tilde{d}_0"] \arrow[d, "\tilde{d}_1"] & F \\ X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X_0. & \end{tikzcd}$$

One then shows as in 7.3 that $\tilde{d}_1$ is quasi-finite at the points of $\tilde{d}_1^{-1}(z)$, so that it is natural to consider the open F_1' of F_1 consisting of points where $\tilde{d}_1$ is simultaneously flat and quasi-finite. By 7.4, the two inverse images of F_1' under $\tilde{d}_1'$ and $\tilde{d}_0'$ consist of the points of F_2 where $\tilde{d}_2'$ is flat and quasi-finite, so these two inverse images coincide, and F_1' is of the form $\tilde{d}_0^{-1}(F')$, where F' is an open of F

(SGA 1, VIII 4.4). Possibly replacing F by F', one may therefore suppose that $\tilde{d}_1$ is quasi-finite and flat. In this case, we denote by W_z the largest open of $\tilde{d}_1(F_1)$ above which $\tilde{d}_1$ is finite and flat.

This open W_z does not necessarily contain z, but it contains the generic points of the irreducible components of X_0 passing through z.[386] Since d_0 (resp. d_1) is faithfully flat and of finite presentation (hence open), it then follows from SGA 1, VIII 5.7, that $d_0^{-1}(W_z)$ and $d_1^{-1}(W_z)$ both coincide with the largest open of $\tilde{d}_2'(F_2)$ above which $\tilde{d}_2'$ is finite and flat. One sees consequently as in 7.3 that the two inverse images of $\tilde{d}_1^{-1}(W_z)$ under $\tilde{d}_0'$ and $\tilde{d}_1'$ coincide, so that $\tilde{d}_1^{-1}(W_z)$ is of the form $\tilde{d}_0^{-1}(U)$ where U is an open of F which is a quasi-section for the groupoid induced by $X*$ on W_z.

9. Elimination of the Noetherian hypotheses in theorem 7.1

a) We take up the notations and hypotheses of lemma 6.1 and let $\pi : S' \to S$ be an arbitrary base change. Denote by $f' : X' \to Y'$ the morphism of S'-schemes deduced by extension via π of the base from a morphism of S-schemes $f : X \to Y$. With this convention, $p' : X_0' \to Y'$ is surjective, as is the morphism $X_1' \to X_0' \times_{Y'} X_0'$ with components d_0' and d_1'. The set underlying Y' therefore identifies with the quotient of the set underlying X_0' by the equivalence relation defined in X_0' by the S'-groupoid $X'*$. Moreover, $q' : U' \to Y'$ is integral and surjective, so that the topology of Y' is the quotient topology of that of U', hence also of that of X_0' (confer the proof in § 6.c).

On the other hand, it is clear that U' is a quasi-section of the S'-groupoid $X'*$, to which one can therefore apply lemma 6.1. In particular, $X'*$ possesses a cokernel (Y_1, p_1) and the topological space underlying Y_1 is obtained from the topological space underlying X_0' by identifying the points equivalent under the relation defined by $X'*$. It follows that the canonical morphism $Y_1 \to Y'$ is a homeomorphism; I claim that $Y_1 \to Y'$ is even a universal homeomorphism: indeed, if S'' is above S', let Y_2 be the cokernel of $(d_0 \times_S S'', d_1 \times_S S'')$. By what precedes, applied to the base changes $S'' \to S'$ and $S'' \to S$,

$$Y_2 \longrightarrow Y_1 \times_{S'} S'' \quad \text{and} \quad Y_2 \longrightarrow Y \times_S S'' \simeq Y' \times_{S'} S''$$

are homeomorphisms, so the same holds for $Y_1 \times_{S'} S'' \to Y' \times_{S'} S''$.

b) Analogous remarks evidently apply to the case where the groupoid $X*$ "locally" possesses quasi-sections (confer the proof of theorem 7.1).[387] For example, one has the following theorem:

Theorem 9.0. *Let S be an arbitrary scheme and $X_2 \rightrightarrows X_1 \rightrightarrows X_0$ a (Sch/S)-groupoid (with arrows d_i' and d_j) such that: X_0 is of finite presentation and quasi-projective over S, d_1 is of finite presentation, proper and flat, the morphism $d_0 \boxtimes d_1 : X_1 \to X_0 \times_S X_0$ is quasi-finite. Then:*

(1) Every point x of X_0 has an open neighborhood W that is saturated and such that the groupoid induced by $X*$ on W possesses a quasi-section.

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[386] N.D.E.: Indeed, let η be such a generic point. The hypotheses imply that $O_{X_0,\eta}$ is an Artinian local ring, and $O_{F_1,\eta}$ a finitely generated $O_{X_0,\eta}$-module. Therefore, by SGA 1, VIII 6.5, there exists an open neighborhood of η above which $\tilde{d}_1$ is finite.

[387] N.D.E.: We have expanded what follows, to highlight theorem 9.0 below.

(2) Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Then Y is a scheme, p a morphism of schemes, and (Y, p) is a cokernel of (d_0, d_1) in (Sch/S).

(3) p is surjective, open and universally closed.

(4) The morphism $d : X_1 \to X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.

(5) If (d_0, d_1) is an equivalence pair, then:

(a) $d : X_1 \to X_0 \times_Y X_0$ is an isomorphism and p is faithfully flat.

(b) p and $Y \to S$ are of finite presentation, and (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the (fppf) topology.

Proof. For (1), the question is local on S, so one may suppose $S = \operatorname{Spec} B$ affine. There then exists a ring A of finite type over $\mathbb{Z}$, a morphism $S \to T = \operatorname{Spec} A$ and a (Sch/T)-groupoid $Z*$ such that $X*$ identifies with $Z* \times_T S$ (cf. EGA IV_3, 8.8.3, applied to $S_0 = \operatorname{Spec} \mathbb{Z}$ and $S_i = \operatorname{Spec} A_i$, with the A_i running over the finitely generated $\mathbb{Z}$-subalgebras of B). Moreover, one may suppose that $Z*$ satisfies the hypotheses of theorem 7.1 (cf. EGA IV_3, 8.10.5). Consequently, $Z*$ "locally" possesses quasi-sections.

The same therefore holds for $X*$, by a), and assertions (2), (3), (4) and (5) (a) follow from 6.1, as in the proof of 7.1.

c) Let us show that $Y \to S$ is of finite presentation.[388] By hypothesis, (d_0^{X*}, d_1^{X*}) is an equivalence pair, that is, $d^{X*} : X_1 \to X_0 \times_S X_0$ is a monomorphism. By EGA IV_3, 8.10.5, one may suppose, possibly enlarging A, that $d^{Z*} : Z_1 \to Z_0 \times_T Z_0$ is a monomorphism. Since $T = \operatorname{Spec} A$, with A Noetherian, it then follows from theorem 7.1 that the groupoid $Z*$ possesses a cokernel (Q, q) in (Sch/T), that q and $Q \to T$ are of finite presentation, and moreover that $q : Z_0 \to Q$ is faithfully flat and that d^{Z*} induces an isomorphism $Z_1 \xrightarrow{\sim} Z_0 \times_Q Z_0$. Put $Q_S = Q \times_T S$.

Since $X_i \cong Z_i \times_T S$, one therefore obtains an isomorphism:

$$d^{Z*} \times_T S : X_1 \xrightarrow{\sim} X_0 \times_{Q_S} X_0.$$

Denote its inverse by ϕ, and let π be the canonical morphism

$$X_0 \times_Y X_0 \to X_0 \times_{Q_S} X_0.$$

Then $\phi \circ \pi$ is the inverse of $d_0 \boxtimes d_1 : X_1 \xrightarrow{\sim} X_0 \times_Y X_0$. It follows that the equivalence relation defined by $X*$, that is, the monomorphism

$$X_1 \xrightarrow[\cong]{d_0 \boxtimes d_1} X_0 \times_Y X_0 \longrightarrow X_0 \times_S X_0,$$

identifies with the equivalence relation $R(q_S)$ defined by the morphism $q_S : X_0 \to Q_S$. Since the latter is faithfully flat and of finite presentation, hence a universal effective epimorphism, $R(q_S)$ has quotient Q_S (cf. IV 3.3.2). Consequently, $Y \cong$

[388] N.D.E.: The original states that this follows from proposition 9.1 below. We were not able to reconstruct that argument. The proof that follows was indicated to us by O. Gabber.

$Q \times_T S$, so $Y \to S$ and p are of finite presentation. Moreover, by IV 6.3.3, (Y, p) is also a cokernel of (d_0, d_1) in the category of sheaves for the (fppf) topology.

Proposition 9.1. *Consider morphisms of schemes*

$$X_0 \xrightarrow{p} Y \xrightarrow{q} S$$

such that qp *is of finite type (resp. of finite presentation) and* p *is faithfully flat of finite presentation. Then* q *is of finite type (resp. of finite presentation)*[389].

Since p is surjective and qp quasi-compact, q is quasi-compact. So one may suppose S, Y and X_0 affine, with rings A, B, C. One has $B = \lim B_i$, where the B_i run over the finitely generated A-subalgebras of B. Since C is of finite presentation over B, there exists an index i_0, a B_{i_0}-algebra of finite presentation C_{i_0}, and an isomorphism $C \simeq C_{i_0} \otimes_{B_{i_0}} B$; if we put $C_i = C_{i_0} \otimes_{B_{i_0}} B_i$ for $i \geqslant i_0$, we therefore have $C \simeq C_i \otimes_{B_i} B$.

$$\begin{array}{ccc} B & \longrightarrow & C \\ \uparrow & & \uparrow \\ B_i & \longrightarrow & C_i \\ \uparrow & & \\ A & & \end{array}$$

Since C is faithfully flat over B, one extracts from EGA IV_3, 11.2.6 and 8.10.5 (vi) the existence of an $i_1 \geqslant i_0$ such that C_{i_1} is faithfully flat over B_{i_1}; consequently C_i is faithfully flat over B_i for $i \geqslant i_1$. For $i \geqslant i_1$, the canonical map $C_i \to C$ is then injective, since deduced from $B_i \to B$ by faithfully flat extension of the base.

If C is of finite type over A, it follows that there exists an index $j \geqslant i_1$ such that $C_j = C$, whence $B_j = B$, since C_j is faithfully flat over B_j. Consequently, B is of finite type over A.

Suppose now C of finite presentation over A. By what precedes, B is of finite type over A, hence of the form $\bar{B}/I$ where $\bar{B}$ is a polynomial algebra over A in a finite number of indeterminates, and I an ideal of $\bar{B}$. Then I is the union of its finitely generated subideals I_α; whence the equality $B = \lim B_\alpha$ with $B_\alpha = \bar{B}/I_\alpha$. Proceeding as above, there exists an index α_0, a B_{α_0}-algebra of finite presentation C_{α_0}, and an isomorphism $C \simeq C_{\alpha_0} \otimes_{B_{\alpha_0}} B$. For $\alpha \geqslant \alpha_0$, one again sets $C_\alpha = C_{\alpha_0} \otimes_{B_{\alpha_0}} B_\alpha$ so that one has $C \simeq C_\alpha \otimes_{B_\alpha} B$ for $\alpha \geqslant \alpha_0$. Again by EGA IV_3, 11.2.6 and 8.10.5 (vi), one concludes as above that C_α is faithfully flat over B_α for α large enough. In this case, the kernel of the map $C_\alpha \to C$ (resp. $C_\alpha \to C_\beta$ for $\beta \geqslant \alpha$) identifies with $C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ (resp. with $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha)$).

Since C_α and C are of finite presentation over A and $C_\alpha \to C$ is surjective, $C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ is a finitely generated ideal[390] and is the union of the ideals $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha)$. One therefore has $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) = C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ for β large enough, whence also $I_\beta = I$ (since C_α is faithfully flat over B_α); so B is of finite presentation over A.

[389] Cf. EGA IV_4, 17.7.5 for a more general result.
[390] N.D.E.: cf. EGA IV_1, 1.4.4.

10. Complement: quotients by a group scheme

The following §§ 10.2–10.4, written following indications of M. Raynaud, aim to apply the preceding theorems to the case of an action of a group scheme. For the reader's convenience, we have begun by reproducing, in § 10.1, statements 2.1 to 2.3 of Exp. XVI.

10.1. Representability theorems for quotients.

"Recall" first the following result:

Theorem 10.1.1. *Let S be a scheme, X and Y two S-schemes, $f : X \to Y$ an S-morphism. Suppose that one is in one of the following two cases:*

α) The morphism f is locally of finite presentation.

β) The scheme S is locally Noetherian and X is locally of finite type over S.

Then the following conditions are equivalent:

(i) There exists an S-scheme X' and a factorization of f:

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S-morphism locally of finite presentation and f'' is a monomorphism.

(ii) The (first) projection:

$$p_1 : X \times_Y X \to X$$

is a flat morphism.

Moreover, if the preceding conditions are realized, (X', f') is a quotient of X by the equivalence relation defined by f (for the (fppf) topology), so that the factorization $f = f'' \circ f'$ of i) is unique up to isomorphism.

The case Y locally Noetherian, X of finite type over Y, is treated in [Mur65], cor. 2 of th. 2. We shall see that one can reduce to this case.

Let us first make a few remarks:

a) The implication (i) ⇒ (ii) is trivial. Indeed, the first projection

$$p'_1 : X \times_{X'} X \to X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism, since f'' is a monomorphism, and p'_1 is flat, since f' is flat, so p_1 is flat.

b) The assertions of 10.1.1 are local on Y (hence local on S); they are also local on X, as follows easily from the fact that a flat morphism locally of finite presentation is open (EGA IV_3, 11.3.1).

c) Under the hypotheses of 10.1.1 α), in view of what precedes, we are reduced to the case where X and Y are affine and f of finite presentation. Possibly replacing S by Y, one may suppose X and Y of finite presentation over S. One then reduces to the case S Noetherian thanks to EGA IV_3, 11.2.6.

d) Under the hypotheses of 10.1.1 β), one may suppose S, X, Y affine, S Noetherian and X of finite type over S. Consider Y as filtered inverse limit of affine schemes Y_i of finite type over S. The schemes $X \times_{Y_i} X$ form a filtered decreasing family of closed subschemes of $X \times_S X$, whose inverse limit is $X \times_Y X$. Since $X \times_S X$ is Noetherian, one has $X \times_{Y_i} X = X \times_Y X$ for i large enough, so that $f_i : X \to Y \to Y_i$ satisfies the hypotheses of 10.1.1 ii) if f does. Since the equivalence relation defined by f on X coincides with that defined by f_i, it is clear that it suffices to prove ii) ⇒ i) for f_i, which reduces us to the case where Y is of finite type over S.

Application to group schemes. Let S be a scheme, G an S-group scheme locally of finite presentation over S, acting (on the left) on an S-scheme X. If $X \to S$ possesses a section ξ, recall that the stabilizer $Stab_G(\xi)$ is representable by a subgroup scheme of G (cf. I, 2.3.3).

Theorem 10.1.2. *Let S be a scheme, G an S-group scheme locally of finite presentation over S, acting on an S-scheme X.*

One assumes that $X \to S$ possesses a section ξ, such that the stabilizer H of ξ in G is flat over S. If one of the following hypotheses is satisfied:

a) X is locally of finite type over S,

b) S is locally Noetherian,

then the quotient (fppf) sheaf G/H is representable by an S-scheme, locally of finite presentation over S, and the S-morphism:

$$f : G \to X, \quad g \mapsto g \cdot \xi$$

factors as:

$$\begin{tikzcd} G \arrow[d, "p"'] \arrow[dr, "f"] & \\ G/H \arrow[r, "i"] & X, \end{tikzcd}$$

where p is the canonical projection, which is a faithfully flat morphism locally of finite presentation, and i is a monomorphism.

Proof. The morphism f makes G an X-scheme. By definition of the stabilizer of ξ, the morphism:

$$G \times_S H \to G \times_X G, \quad (g, h) \mapsto (g, gh)$$

is an isomorphism. Since H is flat over S, $G \times_S H$ is flat over G, so the first projection $p_1 : G \times_S G \to G$ is a flat morphism. Moreover, if X is locally of finite type over S, f is locally of finite presentation (EGA IV_1, 1.4.3 (v)), and otherwise S is assumed locally Noetherian. It then suffices to apply 10.1.1 to the morphism f. It remains to see that G/H is locally of finite presentation over S, but this follows immediately from 9.1.

Corollary 10.1.3. *Let S be a scheme, $u : G \to H$ a morphism of S-group schemes. Suppose G locally of finite presentation over S and that either H is locally of finite type over S, or S is locally Noetherian.*

Then, if $K = Ker(u)$ is flat over S, the quotient group G/K is representable by an S-group scheme locally of finite presentation over S, and u factors as:

$$\begin{tikzcd} G \arrow[r, "u"] \arrow[d, "p"'] & H \\ G/K \arrow[ur, "i"'] & \end{tikzcd}$$

where p is the canonical projection and i a monomorphism.

Proof. One applies 10.1.2 taking $X = H$ and for ξ the unit section of H.

10.2. Stabilizer of the diagonal.

Let S be a Noetherian scheme, X an S-scheme of finite type, and G a flat S-group scheme of finite type acting on the left on X, i.e., one has an S-action $d_0 : G \times_S X \to X$. Denote by $d_1 : G \times_S X \to X$ the projection onto the second factor. Following § 2.a), one has the groupoid

$$\begin{tikzcd} G \times_S G \times_S X \arrow[r, shift left=2, "pr_{2,3}"] \arrow[r, "\mu \times X" description] \arrow[r, shift right=2, "G \times d_0"'] & G \times_S X \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X \end{tikzcd}$$

whose cokernel, if it exists, is denoted $G\backslash X$.

Definition 10.2.1. *We denote by $F \subset G \times_S X$ the* stabilizer of the diagonal section, *i.e. the X-scheme defined by the Cartesian product*

$$\begin{tikzcd} F \arrow[r] \arrow[d] & X \arrow[d, "\Delta"] \\ G \times_S X \arrow[r, "(d_0, d_1)"] & X \times_S X. \end{tikzcd}$$

Then F is an X-subgroup scheme of $G \times_S X$. Since $G \times_S X$ is of finite type over S Noetherian, hence Noetherian, F is of finite type over S and over X (EGA I, 6.3.5 and 6.3.6). Moreover, if $X \to S$ is separated, F is a closed X-subgroup scheme of $G \times_S X$.

Recall that one says that G *acts freely* on X if the morphism

$$G \times_S X \xrightarrow{(d_0, d_1)} X \times_S X$$

is a monomorphism (cf. Exp. III, 3.2.1). This amounts to saying that F is the trivial group scheme with base X.

10.3. Case where F is quasi-finite over X.

Since F is of finite type over X, it is quasi-finite over X if and only if the fixators of the geometric points of X are finite.

Theorem 10.3.1.[391] *Under the hypotheses of 10.2, suppose that F is quasi-finite over X. Then there exists an open U of X, dense and G-saturated, satisfying the following properties:*

(i) In (Sch/S), the cokernel $V = G\backslash U$ exists; moreover, the scheme V is a quotient in the category of ringed spaces.

(ii) $p : U \to V$ is surjective, open, and of finite presentation.

(iii) V is of finite presentation over S.

(iv) The morphism $G \times_S U \to U \times_V U$, $(g, x) \mapsto (gx, x)$, is surjective.

(v) Suppose in addition that G acts freely on X. Then $U \to V$ is a (left) G-torsor locally trivial for the (fppf) topology. In particular, $U \to V$ is faithfully flat.[392]

Proof. It is assumed that the morphism $G \times_S X \to X \times_S X$, $(g, x) \mapsto (gx, x)$, is quasi-finite. Theorem 8.1 therefore applies to the groupoid defined by (X, G). Thus there exists a dense saturated open $U \subset X$ such that the quotient $G\backslash U$ exists; it satisfies properties (i), (ii), (iii).

To establish (iv), recall that G acts freely on X if and only if (d_0, d_1) is an equivalence pair (III 3.2.1). In this case, theorem 8.1 (iv) shows that the morphism $G \times_S U \to U \times_V U$ is an isomorphism and that p is faithfully flat and of finite presentation. Thus U is a G-torsor with base V, locally trivial for the (fppf) topology.

10.4. Case where F is flat over X.

We denote

$$d = (d_0, d_1) : G \times_S X \longrightarrow X \times_S X$$

the morphism $d(g, x) = (gx, x)$. Recall that the sheaf-theoretic graph $\tilde{\Gamma}$ of the equivalence relation associated with (X, G) is the (fppf) S-subsheaf of $X \times_S X$ image of (d_0, d_1). It is the (fppf) sheaf associated to the graph functor:

$$T \mapsto \Gamma(T) = \{(x_0, x_1) \in X(T) \times X(T) \mid x_0 \in G(T)\, x_1\}.$$

Set $G_X = G \times_S X$. For every S-scheme T, one has a surjective map

[391] N.D.E.: Here too, there exists a largest open U of X satisfying the conclusions of the theorem, cf. N.D.E. 44.

[392] N.D.E.: If one assumes in addition that G is a reductive S-group scheme and that the (free) action of G on X is linearizable, then it is known that $G\backslash X$ is representable and that $X \to G\backslash X$ is a (left) G-torsor. This follows from results of Raynaud and Seshadri and is found in the article [CTS79] (proposition 6.11).

$$G_X(T) \to \Gamma(T), \quad (g, x) \mapsto (gx, x),$$

which induces a bijective map

$$\phi(T) : G_X(T)/F(T) \xrightarrow{\sim} \Gamma(T);$$

indeed, if $(g, x), (g', x') \in G_X(T)$ satisfy $(gx, x) = (g'x', x')$, then $x' = x$ and $g^{-1}g'x = x$, so $(g^{-1}g'x, x) \in F(T)$ and (g, x) and (g', x) have the same image in $G_X(T)/F_X(T)$.

By definition (cf. IV, 4.4.1 (ii) or proof of 5.2.1), the quotient sheaf G_X/F is the (fppf) sheaf associated to the presheaf

$$T \mapsto G_X(T)/F(T) \cong \Gamma(T).$$

One therefore has an isomorphism of sheaves $\phi : G_X/F \to \tilde{\Gamma}$.

Theorem 10.4.1.[393] *Under the hypotheses of 10.2, one has:*

a) $\tilde{\Gamma}$ is representable if and only if F is flat over X.

b) Suppose F flat over X. Then the morphisms induced by d_1 and d_0:

$$\begin{tikzcd} G_X/F \arrow[r, shift left, "d_0"] \arrow[r, shift right, "d_1"'] & X \end{tikzcd}$$

are faithfully flat and of finite presentation.

Proof of a): Suppose the (fppf) sheaf G_X/F representable by an X-scheme Y. Then, by IV 6.3.3, $p : G_X \to Y$ is faithfully flat and locally of finite presentation, and the second square of the diagram below is Cartesian:

$$\begin{tikzcd} F \arrow[r] \arrow[d] & F \times_X G_X \arrow[r] & G_X \arrow[d, "p"] \\ X \arrow[r, "e_X"] & G_X \arrow[r] & Y, \end{tikzcd}$$

the first square being obtained by base change along the unit section $e_X : X \to G_X$. Since p is faithfully flat and locally of finite presentation, so is $F \to X$.

Conversely, suppose F flat over X. Put $X_2 = X \times_S X$. The morphism $d : G_X \to X_2$ allows one to form the fiber product:

$$\begin{tikzcd} G_X \times_{X_2} G_X \arrow[r] \arrow[d] & G_X \arrow[d] \\ G_X \arrow[r] & X_2. \end{tikzcd}$$

Then the morphism $G_X \times_{X_2} G_X \to X_2$ is an $F \times_X X_2$-torsor over X_2, and is therefore flat and of finite type (since F is). By theorem 10.1.1, the morphism d factors uniquely:

[393] N.D.E.: This is point (2) of theorem 3 of [Ray67b]. In this Note another proof of th. 10.1.1 is sketched.

$$G_X \xrightarrow{\psi} Y \xrightarrow{\tau} X \times_S X,$$

where ψ is faithfully flat (of finite type) and τ is a monomorphism of schemes.

Consequently, the morphism of sheaves $\psi : G_X \to Y$ is therefore F-invariant, and there comes a morphism of sheaves $\bar{\psi} : G_X/F \to Y$. Moreover, since ψ is faithfully flat (of finite type), the monomorphism of sheaves τ factors through the sheaf image of d, that is $\tilde{\Gamma}$. The isomorphism of sheaves $G_X/F \cong \tilde{\Gamma}$ therefore factors through the monomorphism $Y \to \tilde{\Gamma}$. One concludes that Y represents G_X/F.

Proof of b): Suppose F flat over X. Then, by a) and its proof, G_X/F is representable, and the morphism $p : G_X \to G_X/F$ is faithfully flat and of finite presentation. On the other hand, the morphisms $d_i : G_X \to X$ ($i = 0, 1$) are faithfully flat and of finite presentation by hypothesis. Since $d_i = \bar{d}_i \circ p$, it follows from EGA IV_2, 2.2.13 (iii) and EGA IV_3, 11.3.16, that $\bar{d}_i$ is faithfully flat and of finite presentation.

Theorem 10.4.2.[394] *Under the hypotheses of 10.2, suppose F flat over X. Then there exists a dense saturated open U of X such that the (fppf) quotient $V = G\backslash U$ is an S-scheme of finite type and $U \to V$ is faithfully flat and of finite presentation.*

Proof. Theorem 10.4.1 shows that $G_X/F \cong \tilde{\Gamma}$ is representable. Then the (fppf) sheaf $G\backslash X$ identifies with the quotient sheaf of

$$G_X/F \overset{\bar{d}_1}{\underset{\bar{d}_0}{\rightrightarrows}} X.$$

By what precedes, $\bar{d}_i : G_X/F \to X$ is faithfully flat and of finite presentation ($i = 0, 1$), and the morphism

$$G_X/F \xrightarrow{\cong} \tilde{\Gamma} \longrightarrow X \times_S X$$

is a monomorphism, that is, $(\bar{d}_0, \bar{d}_1)$ is an equivalence pair. Consequently, theorem 8.1 applies. There therefore exists an open U of X, dense and saturated, such that the (fppf) quotient $V = G\backslash U$ is an S-scheme of finite type, and $U \to V$ is faithfully flat and of finite presentation.

Taking into account the generic flatness theorem (EGA IV_2, 6.9.3), one obtains the

Corollary 10.4.3. *Under the hypotheses of 10.2, suppose X reduced. Then there exists a dense saturated open U of X such that the (fppf) quotient $G\backslash U$ is an S-scheme of finite type and $U \to G\backslash U$ is faithfully flat and of finite presentation.*

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[394] N.D.E.: Here too, there exists a largest open U of X satisfying the conclusions of the theorem; moreover, a point $x \in X$ of codimension 1 in X belongs to U if and only if the morphism $(G_X/F) \times_X \mathrm{Spec}(O_{X,x}) \to \mathrm{Spec}(O_{X,x}) \times_S \mathrm{Spec}(O_{X,x})$ is a closed immersion, cf. N.D.E. 44.
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Exposé VI_A. Generalities on algebraic groups

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Throughout this entire chapter, A will denote an Artinian local ring with residue field k. A group scheme G over $\operatorname{Spec} A$ will be called simply an *A-group*. This A-group is said to be *locally of finite type* if the underlying scheme is locally of finite type over A; it is said to be *algebraic* if the underlying scheme is of finite type over A.

0. Preliminary remarks

0.1.

Let us first consider a group scheme G over an arbitrary scheme S. We call *multiplication* the structural morphism $\mu : G \times_S G \to G$, and *inversion* the morphism $c : G \to G$ defined by the equalities $c(T)(x) = x^{-1}$ (with T a scheme over S and x an element of $G(T)$). If U and V are subsets of the underlying set of G, we write $U \cdot V$ for the image under the multiplication morphism of the part of $G \times_S G$ consisting of points whose first projection lies in U and second projection in V. Likewise, the notations U^{-1} and $c(U)$ are equivalent.

Let pr_1 denote the projection of $G \times_S G$ onto the first factor and $\sigma : G \times_S G \to G \times_S G$ the morphism with components pr_1 and μ. For every S-scheme T, $\sigma(T)$ is the map $(x, y) \mapsto (x, xy)$; it follows that σ is an automorphism. The composition of this automorphism with the projection pr_2 of $G \times_S G$ onto the second factor is the multiplication morphism. When G is flat over S, pr_2 and hence μ are flat morphisms; when G is smooth over S, pr_2 and hence μ are smooth morphisms, etc.

0.2.

We assume from now on that S is the spectrum of an Artinian local ring A with residue field k. We denote by $(Sch/k)_{red}$ the category of reduced schemes over k. For every scheme X over A, the reduced scheme X_{red} is an object of $(Sch/k)_{red}$, and the functor $X \mapsto X_{red}$ is right adjoint to the inclusion of $(Sch/k)_{red}$ into (Sch/A). It follows that, for every A-group G, G_{red} is a group in the category $(Sch/k)_{red}$,[397] i.e., for every reduced k-scheme T, $G_{red}(T)$ is equipped with a group structure, functorial in T. One should beware that G_{red} is not necessarily a k-group, since the "multiplication" is only a morphism from $(G_{red} \times_k G_{red})_{red}$ into G_{red}.[398]

[399] However, if k is a perfect field, the inclusion of $(Sch/k)_{red}$ into (Sch/k) commutes with products, so that groups in the category $(Sch/k)_{red}$ are identified with group schemes over k whose underlying scheme is reduced. In this case, if G is a k-group, G_{red} is a group subscheme of G; this subgroup is not in general normal in G.

[396] N.D.E.: Version of 13/10/2024.
[397] N.D.E.: The sentences that follow have been added.
[398] N.D.E.: For examples of group schemes G over a non-perfect field k such that G_{red} is not a group scheme over k, see 1.3.2 below.
[399] N.D.E.: We have slightly modified what follows of 0.2, as well as 0.3.

For example, if k is a field of characteristic 3, the constant group $(\mathbb{Z}/2\mathbb{Z})_k$ acts non-trivially on the diagonalizable group $D_k(\mathbb{Z}/3\mathbb{Z})$ (cf. Exp. I, 4.1 and 4.4); if G denotes the semidirect product of $D_k(\mathbb{Z}/3\mathbb{Z})$ by $(\mathbb{Z}/2\mathbb{Z})_k$ defined by this action, G_{red} is identified with $(\mathbb{Z}/2\mathbb{Z})_k$ and is not normal in G.[400]

Let k be an arbitrary field, $k^{p^{-\infty}}$ its perfect closure, and H a group in the category $(Sch/k)_{red}$. Then $(H \otimes_k k^{p^{-\infty}})_{red}$ is a group scheme over the perfect field $k^{p^{-\infty}}$. Since $H \otimes_k k^{p^{-\infty}}$ and $(H \otimes_k k^{p^{-\infty}})_{red}$ have the same underlying topological space, one sees that the groups of $(Sch/k)_{red}$ share with k-groups certain topological properties invariant under extension of the base field: for example, it will follow from 0.3 and the remarks just made that every group of $(Sch/k)_{red}$ is separated.

We shall encounter groups of $(Sch/k)_{red}$ in what follows whenever we deal with a non-empty, locally closed subset U of an A-group G such that $U \cdot U = U$ and $U^{-1} = U$: indeed, the reduced subscheme of G defined by U is then a group of $(Sch/k)_{red}$.

0.3.

An A-group G is always separated, since the unit section $e : \operatorname{Spec} A \to G$ is a closed immersion. Indeed, let x be the unique point of $\operatorname{Spec} A$ and η the structural morphism $G \to \operatorname{Spec} A$. Since $\eta \circ e = id_{\operatorname{Spec} A}$, for every affine open $U = \operatorname{Spec} B$ of G containing $e(x)$, the morphism $B \to A$ has a section, and so is surjective. It follows that e is a closed immersion.[401] Now the diagonal of $G \times_A G$ is identified with the functor from $(Sch/A)^{\circ}$ with values in `(Ens)` that associates to every scheme S over A the inverse image of the unit element of $G(S)$ under the map $\phi(S) : (x, y) \mapsto x \cdot y^{-1}$ from $G(S) \times G(S)$ to $G(S)$. We therefore have the cartesian square below, so that the diagonal morphism, being obtained from a closed immersion by base change, is itself a closed immersion:

$$\begin{tikzcd} G \times_S G \arrow[r, "\varphi"] & G \\ G \arrow[u, "\text{diagonal}"'] \arrow[r] & \operatorname{Spec} A. \arrow[u, "\text{unit section}"'] \end{tikzcd}$$

0.4.

[402] Let G be an A-scheme. We shall say that a point g of G is *strictly rational over* A if there exists an A-morphism $s : \operatorname{Spec} A \to G$ sending the unique point of $\operatorname{Spec} A$ to g, i.e. if the morphism $A \to O_{G,g}$ admits a retraction. Note that one then has $\kappa(g) = k$,

[400] N.D.E.: To have an analogous example with G connected, one may consider, for $char(k) = p > 0$, the semidirect product of $\alpha_{p,k}$ and $G_{m,k}$.

[401] N.D.E.: This argument is valid for every local ring A of dimension zero, and shows that if S is a discrete scheme, every S-group is separated, cf. VI_B, 5.2; on the other hand (cf. VI_B, 5.6.4), if S contains a closed point s that is not isolated, the S-scheme G obtained by gluing two copies of S along the open subset $S - s$ is not separated over S, but is equipped with an S-group structure.

[402] N.D.E.: We have detailed the original in what follows, in particular we have added Remark 0.4.1.

and so g is a closed point of G (if B is the ring of an affine open neighborhood of g and $\mathfrak{p}$ the prime ideal of B corresponding to g, then $B/\mathfrak{p} \subset k$ is a finite A-algebra, hence an integral Artinian ring, hence a field).

Suppose henceforth that G is an A-group; then such a section $s : \operatorname{Spec} A \to G$ defines an automorphism r_s of the scheme G over A, which we call *right translation by* s: for every morphism $\pi : S \to \operatorname{Spec} A$, $r_s(\pi)$ is the automorphism of $G(S)$ defined by $x \mapsto x \cdot G(\pi)(s)$, for every $x \in G(S)$. Similarly, we write ℓ_s for the *left translation by* s, i.e. the automorphism of G defined by the equalities $\ell_s(\pi)(x) = G(\pi)(s) \cdot x$, for every $x \in G(S)$.

Since $G \otimes_A k$ and G have the same underlying topological space G, since $G \otimes_A k$ is a k-group, and since $s \otimes_A k$ depends only on g and not on s, one sees that the automorphisms of G induced by r_s and ℓ_s (or by $r_{s\otimes k}$ and $\ell_{s\otimes k}$) depend only on g and not on s; when P is a subset of G, we may therefore write $r_g(P)$ or $P \cdot g$ (resp. $\ell_g(P)$ or $g \cdot P$) instead of $r_s(P)$ (resp. $\ell_s(P)$), in agreement with 0.1.

Remark 0.4.1. *If g is a strictly rational point of G and if $A \to A'$ is a morphism of Artinian local rings, then $G' = G \otimes_A A'$ has a unique point g' above g, and g' is strictly rational over A'; moreover, if one writes P' for the inverse image of P in G', then $P' \cdot g'$ is the inverse image of $P \cdot g$ (cf. EGA I, 3.4.8).*

Proposition 0.5. [403] *Let G be an A-group and U, V two dense open subsets of G. Then $U \cdot V$ (i.e. the image of $U \times_A V$ under multiplication) is equal to the whole underlying space of G.*

Proof. Since G and $G \otimes_A k$ have the same underlying space, we may, possibly replacing A by k and G by $G \otimes_A k$, assume that $A = k$. Let $g \in G$. Set $K = \kappa(g)$; then left translation ℓ_g is an automorphism of G_K. Since the projection $G_K \to G$ is open, U_K and V_K are dense open subsets of G_K, as is the image of V_K under λ_g. There exists therefore $v \in V_K$ such that $u = \ell_g(v)$ belongs to U_K. Let L be an extension of K containing $\kappa(v)$ (and so $\kappa(u)$), and let g_L and v_L be the L-points of G_L deduced from g and v. Then $g_L \cdot v_L = u'$ is a point of G_L above u, and so $g_L = u' \cdot v_L^{-1}$ lies above $U \cdot V$, whence $g \in U \cdot V$, which proves the proposition.

Corollary 0.5.1. [404] *If G is an irreducible A-group, then G is quasi-compact.*

Proof. Let U be a non-empty affine open subset of G; then U is dense in G, so by 0.5 the morphism $\mu : U \times_A U \to G$ is surjective, and therefore G is quasi-compact since $U \times_A U$ is.

Corollary 0.5.2. [405] *Let G be an A-group scheme, and H a group subscheme of G over A. Then H is closed.*

Proof. Let k' be the perfect closure of the residue field k of A. Since the underlying

[403] N.D.E.: The original stated this result under the hypothesis that G is locally of finite type over A. Since it is useful to have it available in the general case, and since the proof is essentially the same, we have stated and proved the result in the general case. This will be used several times in what follows.
[404] N.D.E.: We have inserted this corollary here, cf. 2.4 and 2.6.2 below.
[405] N.D.E.: We have added this corollary, used implicitly in VIII, 6.7; see also VI_B 6.2.5.

topological spaces of G and H are unchanged under the base change $A \to k \to k'$, we may suppose that $A = k$ is a perfect field. We may then suppose that G and H are reduced, hence geometrically reduced.

Let $\bar{H}$ be the closure of H; then $\mu^{-1}(\bar{H})$ is a closed subset of $G \times G$ containing $H \times H$. Since the morphism $H \to \operatorname{Spec} k$ (resp. $\bar{H} \to \operatorname{Spec} k$) is universally open, and since H is dense in $\bar{H}$, then $H \times H$ is dense in $H \times \bar{H}$ and $H \times \bar{H}$ is dense in $\bar{H} \times \bar{H}$, so $H \times H$ is dense in $\bar{H} \times \bar{H}$. Therefore $\mu(\bar{H} \times \bar{H}) \subset \bar{H}$, and so, since $\bar{H} \times \bar{H}$ is reduced, μ induces a morphism $\mu' : \bar{H} \times \bar{H} \to \bar{H}$.

Let then $g \in \bar{H}$, and set $K = \kappa(g)$. Since the projection $\bar{H}_K \to \bar{H}$ is open, H_K and $\ell_g(H_K)$ are two dense open subsets of $\bar{H}_K$, so there exist $u, v \in H_K$ such that $\ell(g)(v) = u$. One concludes, as in the proof of 0.5, that g belongs to $H \cdot H = H$, whence $\bar{H} = H$.

1. Local properties of an A-group locally of finite type

We shall first see that, if G is locally of finite type and flat over A, one can "make strictly rational any closed point of G" by means of a finite, flat extension of the base.

1.1.

Unless explicit mention is made to the contrary, we assume from now on that G is an A-group locally of finite type. When A is a field k, we shall obtain in Exposé VII_B very precise results on the local rings of G.[406] We content ourselves here with a few elementary results:

Proposition 1.1.1. *Let x be a point of an A-group G locally of finite type and flat over A. Then the local ring $O_{G,x}$ is Cohen–Macaulay, and there exists a system $a_1, \cdots, a_n$ of parameters of $O_{G,x}$ such that $O_{G,x}/(a_1, \cdots, a_n)$ is a finite and flat (hence finite and free) A-module.*

We first assume A equal to its residue field k; it then suffices to prove that $O_{G,x}$ is Cohen–Macaulay and one may limit oneself to the case where x is a closed point (cf. EGA 0_IV, 16.5.13). By Lemma 1.1.2 below, G contains a closed point y such that $O_{G,y}$ is Cohen–Macaulay. By SGA 1, I § 9, this amounts to saying that, for every finite extension K of the base field k and every point $\bar{y}$ of $\bar{G} = G \otimes_k K$ above y, $O_{\bar{G},\bar{y}}$ is Cohen–Macaulay. If the extension K has been chosen large enough — i.e. if K contains a normal extension of k containing the residue fields $\kappa(y)$ and $\kappa(x)$ — then $\bar{y}$ is (strictly) rational over K, as is every point $\bar{x}$ of $\bar{G}$ above x.[407] Since the automorphism $r_{\bar{x}} \circ (r_{\bar{y}})^{-1}$ sends $\bar{y}$ to $\bar{x}$, it follows that $O_{\bar{G},\bar{x}}$, and hence $O_{G,x}$ (SGA 1, I § 9), are Cohen–Macaulay.

[406] N.D.E.: For example, they are always complete intersections, cf. VII_B, 5.5.1. Moreover, if $char(k) = 0$ then G is smooth (VI_B, 1.6.1; see also VII_B, 3.3.1).

[407] N.D.E.: Indeed, the hypothesis on K entails that, for every extension L of K, every k-morphism $\kappa(x) \to L$ (resp. $\kappa(y) \to L$) factors through K; consequently, all points of $G \otimes_k K$ above x or y have K as residue field, i.e. are (strictly) rational over K.

When A is again assumed arbitrary, the preceding argument applies to $k \otimes_A G$, so that $k \otimes_A O_{G,x}$ is Cohen–Macaulay. If $a_1, \cdots, a_n$ is a sequence of elements of $O_{G,x}$ whose image in $k \otimes_A O_{G,x}$ is a system of parameters, it follows from SGA 1, IV 5.7 or from EGA 0_IV, 15.1.16, that $a_1, \cdots, a_n$ is an $O_{G,x}$-regular sequence and that $O_{G,x}/(a_1, \cdots, a_n)$ is finite and flat (hence finite and free) over A.

Lemma 1.1.2. *Every non-empty scheme X, locally of finite type over an Artinian ring A, contains a closed point x whose local ring is Cohen–Macaulay.*

We may of course assume X affine with algebra B, and argue by induction on $\dim X$ (the assertion is clear if X is discrete, since all local rings are then Artinian). Since B is of finite type over A, if $\dim B > 0$, B contains an element a that is non-invertible and not a zero-divisor.[408] The closed subscheme $X' = \operatorname{Spec} B/(a)$ of X is then of dimension strictly less than $\dim X$, and by induction contains a closed point x such that $O_{X',x}$ is Cohen–Macaulay. Since $O_{X',x} = O_{X,x}/(a)$ and a is non-invertible and not a zero-divisor in $O_{X,x}$, then $O_{X,x}$ is Cohen–Macaulay (see also EGA IV_2, 6.11.3).

Proposition 1.2. *Let A be an Artinian local ring, G an A-group locally of finite type and flat over A, and x a closed point of G. There exists an A-algebra A' that is local, finite and free over A, such that every point of $G \otimes_A A'$ above x is strictly rational over A'.*

[409] Indeed, let k_1 be a normal extension of finite degree of k containing the residue field $\kappa(x)$ of x. By Lemma V 4.1.2, there exists an A-algebra A_1 that is local, finite and free over A, with residue field k_1. In this case (cf. N.D.E. (11)) all the points $g_1, \cdots, g_n$ of $G \otimes_A A_1$ above $x \in G$ have k_1 as residue field (i.e. $g_1, \cdots, g_n$ are rational over A_1 in the sense of Exp. V, § 4.e)).

Let then $B_1, \cdots, B_n$ be the local rings of $g_1, \cdots, g_n$. By 1.1.1, $B_1, \cdots, B_n$ have quotients $B_1', \cdots, B_n'$ that are Artinian and finite and free over A_1. Set $A' = B_1' \otimes_{A_1} \cdots \otimes_{A_1} B_n'$. Then A' is local, finite and free over A_1 and, for each $i = 1, \cdots, n$, we have surjective homomorphisms

B_i ⊗_{A_1} A′ ↠ B′_i ⊗_{A_1} A′ ↠ A′,

the second induced by the multiplication map $B_i' \otimes_{A_1} B_i' \twoheadrightarrow B_i'$. Consequently, A' answers the question.

1.3.

Let e denote the unit element (or origin) of G, i.e. the image of the unique point of $\operatorname{Spec} A$ by the unit section $\operatorname{Spec} A \to G$. By definition itself, e is strictly rational over A.

Proposition 1.3.1. [410] *Let G be a group locally of finite type and flat over an*

[408] N.D.E.: Indeed, B is a noetherian Jacobson ring (cf. [BAC], V § 3.4). If every non-invertible element is a zero-divisor, then every prime ideal is an associated prime ideal of B, so in particular B has only a finite number of maximal ideals $\mathfrak{m}_1, \cdots, \mathfrak{m}_n$. Since B is a Jacobson ring, the intersection of the $\mathfrak{m}_i$ is the nilradical of B, and it follows that each $\mathfrak{m}_i$ is a minimal prime ideal of B, so that $\dim B = 0$.
[409] N.D.E.: We have detailed the original in what follows.
[410] N.D.E.: We have added point (1), useful in the examples 1.3.2 that follow.

Artinian ring A, and K (resp. $\bar{k}$) the perfect closure (resp. an algebraic closure) of the residue field k of A.

(1) For every closed point x of $\bar{G} = G \otimes_k \bar{k}$, the local rings $O_{\bar{G},\bar{e}}$ and $O_{\bar{G},x}$ are isomorphic. In particular, the tangent spaces $T_e G$ and $T_x G$ have the same dimension.

(2) The following assertions are equivalent:

(i) $G \otimes_A K$ is reduced. (i bis) $O_{G,e} \otimes_A K$ is reduced. (ii) G is smooth over A. (ii bis) G is smooth over A at the origin.

Proof. (1) Let x be a closed point of $\bar{G}$; there is exactly one $\bar{k}$-morphism $s : \operatorname{Spec} \bar{k} \to \bar{G}$ whose image is x; right translation r_s then induces an isomorphism from $O_{\bar{G},\bar{e}} = O_{G,e} \otimes_k \bar{k}$ onto $O_{\bar{G},x}$, whence assertion (1).

Let us prove assertion (2). By SGA 1, II.2.1, one reduces immediately to the case where A is a field ($A = k$). The implications (i) ⇒ (i bis), (ii) ⇒ (ii bis), (ii) ⇒ (i) and (ii bis) ⇒ (i bis) are obvious, so it suffices to prove that (i bis) implies (ii).

Now, it follows from (i bis) that $O_{G,e} \otimes_k \bar{k}$ is reduced. Hence, by (1), $O_{\bar{G},x}$ is reduced for every closed point x of $\bar{G}$, so that $\bar{G}$ is reduced. Then, since $\bar{G}$ is locally of finite type over $\bar{k}$, there exists at least one closed point y such that $O_{\bar{G},y}$ is regular. Since, by (1), the local rings of the closed points of $\bar{G}$ are all isomorphic to $O_{\bar{G},\bar{e}}$, one sees that all these local rings are regular, so that $\bar{G}$ is smooth over $\bar{k}$, and hence G is smooth over k.

[411] One may now give the examples below, signalled by M. Raynaud, of group schemes G over a non-perfect field k such that G_{red} is not a group scheme over k.

Examples 1.3.2. *Let k be a non-perfect field of characteristic $p > 0$, $t \in k - k^p$, $\bar{k}$ an algebraic closure of k, and $\alpha \in \bar{k}$ such that $\alpha^p = t$.*

(1) Consider the additive group $G_{a,k} = \operatorname{Spec} k[X]$, and let G be the group subscheme, finite over k, defined by the additive polynomial $X^{p^2} - tX^p$. Then

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G_red = Spec k[X]/X(X^{p(p−1)} − t)
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is étale at the origin. If it were a group scheme over k, it would be smooth (by 1.3.1); but G_{red} is not geometrically reduced, so it is not a group scheme over k.

(2) Consider $G_{a,k}^4 = \operatorname{Spec} k[X,Y,U,V]$ and let G be the group subscheme defined by the ideal I generated by the additive polynomials $P = X^p - tY^p$ and $Q = U^p - tV^p$. Then G is of dimension 2 and irreducible, since $(G_{\bar{k}})_{red} \cong \operatorname{Spec} \bar{k}[Y,V]$ is.

Let $A = k[X,Y,U,V]$ and $\mathfrak{m}$ its augmentation ideal. Denote by `x, y, u, v` the images of `dX, dY, dU, dV` in $\Omega^1_{A/k} \otimes_A (A/\mathfrak{m})$, viewed as linear forms on the tangent space $k^4 = T_0 G_{a,k}^4$. Let us show that the subspace $E = T_0 G_{red}$ equals k^4. Otherwise, there would exist a linear form $f = ax + by + a'u + b'v$, with $a, b, a', b' \in k$ not all zero, vanishing on E. Recall that the formation of $\Omega^1_{A/k}$ (and hence of tangent spaces) commutes with base change (cf. EGA IV_4, 16.4.5), and identify f with its image in $(\bar{k}^4)*$. Since $(G_{\bar{k}})_{red} \subset (G_{red})_{\bar{k}}$, then f vanishes on the subspace $T_0(G_{\bar{k}})_{red}$ of $\bar{k}^4$,

[411] N.D.E.: We have added the sentence that follows and the examples 1.3.2.

which is defined by the equations $g_1 = x - \alpha y$ and $g_2 = u - \alpha v$, and so $f = \lambda g_1 + \mu g_2$, with $\lambda, \mu \in \bar{k}$. Now $\lambda g_1 + \mu g_2$ belongs to k^4 only if $\lambda = \mu = 0$! This contradiction shows that $E = k^4$, and so $T_0(G_{red})_{\bar{k}} = \bar{k}^4$.

On the other hand, $R = XV - YU$ belongs to $\sqrt{I}$, since $R^p = (X^p - tY^p)V^p - Y^p(U^p - tV^p)$. Consequently, the tangent space F at the point $(\alpha, 1, \alpha, 1)$ of $(G_{red})_{\bar{k}}$ is contained in the hyperplane H of $\bar{k}^4$ with equation $\alpha dV + dX - dU - \alpha dY = 0$, hence is of dimension ≤ 3.[412] Therefore, by point (1) of 1.3.1, G_{red} is not a group scheme over k.

2. Connected components of an A-group locally of finite type

2.1.

Let us first consider an arbitrary A-group G, and let G' be the connected component of the origin e of G. This connected component is evidently closed, so that we may identify it with the reduced closed subscheme of G having G' as underlying space.[413]

Proposition 2.1.1. *For every extension K of the residue field k of A, $G' \otimes_A K$ has as underlying space the connected component of the origin in the K-group $G \otimes_A K$ (i.e. G' is geometrically connected).*

Indeed, let $(G \otimes_A K)'$ be the connected component of the origin in $G \otimes_A K$. Since the image of $(G \otimes_A K)'$ in G is connected and contains the unit element of G, this image is contained in G', so that $(G \otimes_A K)'$ is contained in the inverse image

$G' \otimes_A K$ of G' in $G \otimes_A K$. The proposition therefore follows from the connectedness of $G' \otimes_A K$, which is proved in Lemma 2.1.2.

Lemma 2.1.2. *Let X and Y be two connected schemes over a field k. If X contains a rational point, then $X \times_k Y$ is connected.*

We give below a direct proof of this result from EGA IV_2 (4.5.8 and 4.5.14).

Suppose first Y non-empty, connected and affine, with algebra B. In this case, $X \times_k Y$ is the spectrum of the quasi-coherent $\mathcal{O}_X$-algebra $\mathcal{B} = \mathcal{O}_X \otimes_k B$. We want to show that every subset U of $X \times_k Y$ that is open, closed, and non-empty coincides with $X \times_k Y$. Now U is affine over X and its affine $\mathcal{O}_X$-algebra is a direct factor of $\mathcal{B}$. It therefore follows from Lemma 2.1.3 below that the image of U in X is open and closed,

i.e. coincides with all of X. This image contains in particular a rational point x of X, so that U meets the inverse image of x in $X \times_k Y$. Since this inverse image is isomorphic to Y, hence connected, U contains this inverse image. The same would hold for the complement of U in $X \times_k Y$ if U were distinct from $X \times_k Y$, which would be absurd.

[412] N.D.E.: In fact one sees without difficulty that $\sqrt{I}$ is generated by P, Q, R at every point $\neq 0$, so $F = H$.

[413] N.D.E.: We shall see below (2.2) that G' is a group in the category $(Sch/k)_{red}$.

If Y is now an arbitrary k-scheme, what precedes shows that the fibers of the canonical projection $X \times_k Y \to Y$ are connected. If x is a rational point of X, these fibers all meet the subscheme $x \times_k Y$, which is itself connected, whence the proposition.

Lemma 2.1.3. *Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$-algebra which is locally*[414] *a direct factor of a free $\mathcal{O}_X$-module. The image of* $\operatorname{Spec} \mathcal{A}$ *in X is then open and closed.*

Let V be this image. It is clear that V is contained in the support of $\mathcal{A}$. Conversely, if x belongs to the support of $\mathcal{A}$, then $\mathcal{A}_x$ is non-zero and is a free $\mathcal{O}_{X,x}$-module, since by Kaplansky every projective module over a local ring is free. Consequently the fiber of $\operatorname{Spec} \mathcal{A}$ at x, which is affine with algebra $\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$, is non-empty. One thus sees that the image of $\operatorname{Spec} \mathcal{A}$ coincides with the support of $\mathcal{A}$.

If s is the unit section of $\mathcal{A}$, the equality $s_x = 0$ implies that s, and hence $\mathcal{A}$, are zero in a neighborhood of x. So the support of $\mathcal{A}$ is closed. On the other hand, one may assume that $\mathcal{A}$ is a direct factor of the direct sum L of a family $(\mathcal{O}_X^\alpha)$ of copies of $\mathcal{O}_X$.[415] For every β, write ϕ_β for the restriction to $\mathcal{A}$ of the canonical projection of $L = \bigoplus_\alpha \mathcal{O}_X^\alpha$ onto $\mathcal{O}_X^\beta$. For $x \in X$, denote by $\mathcal{A}_x^*$ the dual module $\operatorname{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x})$. Since $\mathcal{A}_x$ is a direct factor of L_x, the restriction map

$$\operatorname{Hom}_{\mathcal{O}_{X,x}}(L_x, \mathcal{O}_{X,x}) \to \operatorname{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}) = \mathcal{A}_x^*$$

is surjective. If $\mathcal{A}_x \neq 0$ then, since $\mathcal{A}_x$ is free and non-zero, there exist $a \in \mathcal{A}_x$ and a linear form $\phi \in \mathcal{A}_x^*$ such that $\phi(a) = 1$. There exists therefore a family (ξ^α) of elements of $\mathcal{O}_{X,x}$ such that, for every $u \in \mathcal{A}_{X,x}$, one has $\phi(u) = \sum_\alpha \xi^\alpha \phi^\alpha(u)$ (this sum being finite since $\phi^\alpha(u) = 0$ except for finitely many α). Applying this to $u = a$, one obtains that there exist $\alpha_1, \cdots, \alpha_n$ such that

$$1 = \phi(a) = \xi^{\alpha_1} \phi^{\alpha_1}(a) + \cdots + \xi^{\alpha_n} \phi^{\alpha_n}(a).$$

There exists then an open neighborhood U of x such that a and the ξ^{α_i} come from sections $\tilde{a} \in \mathcal{A}(U)$ and $\tilde{\xi}^{\alpha_i} \in \mathcal{O}_X(U)$, and the equality $\sum_i \tilde{\xi}^{\alpha_i} \phi^\alpha(\tilde{a}) = 1$ on U shows that $\tilde{a}_y \neq 0$ for every $y \in U$. ("The support of a projective module is open.")

2.2.

The notations being still those of 2.1, it is clear that G' is a reduced k-scheme. Lemma 2.1.2 shows that $G' \times_k G'$ is connected, so that $(G' \times_k G')_{red}$ is the reduced subscheme of $G \times_A G$ having as underlying space the connected component of the origin. In particular, the multiplication morphism $\mu : G \times_A G \to G$ induces a morphism $\mu' : (G' \times_k G')_{red} \to G'$ that makes G' a group in $(Sch/k)_{red}$.

[414] N.D.E.: We have added the word "locally". Moreover, the reference to Kaplansky is: *Projective modules*, Ann. of Maths. **68** (1958), 372–377; see also [BAC], § II.3, Ex. 3.
[415] N.D.E.: We have detailed the original in what follows.

2.2.bis.

[416] One recalls (cf. Exp. V) that, if P is a scheme, one writes P for the underlying topological space of P. Now one defines a sub-A-functor G^0 of G by setting, for every A-scheme S,

$G^0(S) = \{u \in G(S) \mid u(S) \subset G'\}$.

Let $c : G \to G$ be the inversion morphism; since $c(G') = G'$, we have $c \circ u \in G^0(S)$ for every $u \in G^0(S)$. On the other hand, if $u, v \in G^0(S)$, then $u \boxtimes v$ sends S into the subspace of $G \times_A G$ consisting of points whose two projections belong to G'; this subspace is identified with the underlying space of $G' \times_A G'$, which is connected by Lemma 2.1.2. Consequently, $\mu \circ (u \boxtimes v)$ sends S into G'. This shows that G^0 is a sub-A-functor in groups of G.

If the connected component of e is open in G, then the subfunctor G^0 is representable by the subscheme induced by G on this open set, which is therefore a group subscheme of G; we shall also write G^0 for it. In this case, with the notations of 2.1, one has $G' = (G^0)_{red}$ and the topological spaces G' and G^0 coincide.[417]

2.3.

In accordance with our conventions of 1.1, we again assume from now on that G is locally of finite type over A. Then G is locally noetherian, hence locally connected,[418] hence: every connected component of G is open.

We then write G^0 for the subscheme induced by G on the connected component G' of the unit element. By 2.2.bis, G^0 is a group subscheme of G, which we shall call the *neutral component* of G;

for every A-scheme S we therefore have:

$G^0(S) = \{u \in G(S) \mid u(S) \subset G^0 = G'\}$.

Let G^α be an arbitrary connected component of G and $\nu^\alpha : G^\alpha \times_A G^0 \to G$ the morphism defined by the equalities

$$\nu^\alpha(S)(g, \gamma) = g\gamma g^{-1},$$

for every $S \in (Sch/A)$, $g \in G^\alpha(S)$, $\gamma \in G^0(S)$.

If e is the origin of G, the restriction of ν^α to $G^\alpha \times_A e$ is the trivial morphism; since $G^\alpha \times_A G^0$ is connected by 2.1.2, one sees that ν^α factors through G^0. Therefore, for

[416]N.D.E.: We have added this paragraph, in order to make the link with VI_B, § 3.
[417]N.D.E.: In this Exposé, the notation G^0 is reserved for the case where the connected component of e is open; in VI_B, § 3, this connected component will be denoted G^0 in all cases. This is a slightly abusive notation, but one that is compatible with what precedes when the connected component is the underlying topological space of an open group subscheme G^0 of G.
[418]N.D.E.: Indeed, in a noetherian space, the connected components are finite in number, hence each one is open; see also EGA I, 6.1.9.

every A-scheme S, $G^0(S)$ is a normal subgroup of $G(S)$. We have thus obtained the following proposition:[419]

Proposition 2.3.1. *Let G be an A-group locally of finite type. Then the neutral component G^0 is an open normal group subscheme of G.*

Proposition 2.4. *Let G be an A-group locally of finite type.*

(i) G^0 is irreducible, and $G^0 \otimes_A k$ is geometrically irreducible over k.

(ii) G^0 is quasi-compact, hence of finite type over A.

[420] *Proof.* (i) Since G^0 and $G^0 \otimes_A k$ have the same underlying topological space, it suffices to show the second assertion. Let $\bar{k}$ be an algebraic closure of k. By 2.2, $(G \otimes_A \bar{k})_{red}$ is a $\bar{k}$-group locally of finite type and reduced, hence smooth over $\bar{k}$ (1.3.1). A fortiori the local rings of $(G \otimes_A \bar{k})_{red}$ are integral domains, so,[421] since $G \otimes_A \bar{k}$ is locally noetherian, the connected components of $G \otimes_A \bar{k}$ are irreducible (cf. EGA I, 6.1.10). In particular, the connected component $G^0 \otimes_A \bar{k}$ (cf. 2.1.1) is irreducible.

(ii) Let us now show that G^0 is of finite type over A. Since G^0 is locally of finite type over A, it suffices to prove that G^0 is quasi-compact. As G^0 is irreducible, this follows from 0.5.1.

Corollary 2.4.1. *Every connected component of G is irreducible,[422] of finite type over A,[423] and of the same dimension as G^0.*

One may indeed assume A equal to its residue field k. Let then C be a connected component of G, x a closed point of C, $\kappa(x)$ the residue field of x, and k' a normal extension of k containing $\kappa(x)$ and of finite degree over k. The canonical projection $\pi : C \otimes_k k' \to C$ is open and closed;[424] consequently, if C' is a connected component of $C \otimes_k k'$, the projection $C' \to C$ is surjective, so C' contains a point $y \in \pi^{-1}(x)$, and such a point is rational over k' (cf. the proof of 1.2), so that C' is the image by translation r_y of $G^0 \otimes_k k'$. Now $G^0 \otimes_k k'$ is of finite type over k' by 2.4, and $\pi^{-1}(x)$ is finite (of cardinality $\leq [k' : k]$), so $C \otimes_k k'$ is of finite type over k', and hence C is of finite type over k.

On the other hand, since $G^0 \otimes_k k'$ is irreducible by 2.4, the same holds for C',[425] and hence also for C, since the projection $C' \to C$ is surjective.

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[419] N.D.E.: We have added the numbering 2.3.1 to make this statement explicit. Note moreover that G^0 is even a characteristic subgroup of G, cf. 2.6.5 (ii).

[420] N.D.E.: In the statement, we have replaced "G^0 is geometrically irreducible" by "$G^0 \otimes_A k$ is geometrically irreducible over k", and we have detailed the proof.

[421] N.D.E.: We have detailed the original by adding the reference to EGA I, 6.1.10.

[422] N.D.E.: One should beware that a non-neutral connected component is not in general geometrically connected. For example, if $k = \mathbb{R}$, the group μ_{3,ℝ}, represented by $\mathbb{R}[X]/(X^3 - 1)$, has two connected components: $e = \operatorname{Spec} \mathbb{R}$ and $C = \operatorname{Spec} \mathbb{R}[X]/(X^2 + X + 1)$, and $C \otimes_{\mathbb{R}} \mathbb{C}$ has two components.

[423] N.D.E.: We have added the assertion that follows, cf. VI_B, 1.5.

[424] N.D.E.: We have detailed the original in what follows.

[425] N.D.E.: We have simplified the original here.

Finally, we have seen above that $C \otimes_k k'$ is a disjoint union of finitely many translates of $G^0 \otimes_k k'$. Since dimension is invariant under extension of the base field (cf. EGA IV_2, 4.1.4), it follows that C has the same dimension as G^0. (Moreover, by EGA IV_2, 5.2.1, one has $\dim_g G = \dim G^0$ for every point $g \in G$.)

2.5.

This paragraph has been added. The results that follow appear in Exp. VI_B, but could (or should) have figured in VI_A, and it is useful to have them available from now on, in order to make precise Theorem 3.2 below.

Lemma 2.5.1. *Let $(A, \mathfrak{m})$ be an Artinian local ring, and $k = A/\mathfrak{m}$ its residue field.*

(i) If X is an A-scheme such that $X \otimes_A k$ is locally of finite type (resp. of finite type) over k, then the same holds for X over A.

(ii) Let $u : X \to Y$ be a morphism of A-schemes. If $u \otimes_A k$ is an immersion (resp. a closed immersion), then so is u.

Proof. (i) Suppose $X \otimes_A k$ locally of finite type over k. Let $U = \operatorname{Spec} B$ be an affine open of X. By hypothesis, there exist elements $x_1, \cdots, x_n$ of B whose images generate $B/\mathfrak{m}B$ as a k-algebra, and it follows from the "nilpotent Nakayama lemma" that the x_i generate B as an A-algebra. This proves that X is locally of finite type over A. If, in addition, $X \otimes_A k$ is quasi-compact, then so is X (which has the same underlying topological space), and hence X is of finite type over A. This proves (i).

Let us prove (ii). Suppose $u \otimes_A k$ is an immersion (resp. a closed immersion). Then u is a homeomorphism of X onto a locally closed (resp. closed) part of Y, and for every $x \in X$, the ring morphism $\phi_x : O_{Y,u(x)} \to O_{X,x}$ is such that $\phi_x \otimes_A k$ is surjective. By the nilpotent Nakayama lemma, it follows that ϕ_x is surjective, so u is an immersion (resp. a closed immersion).

Proposition 2.5.2. *Let A be an Artinian local ring with residue field k, and let $u : G \to H$ be a quasi-compact morphism between A-group schemes locally of finite type.*

(a) The set $u(G)$ is closed in H, and its connected components are irreducible and all of the same dimension.

(b) One has `dim G = dim u(G) + dim Ker(u)`.

(c) If u is a monomorphism, it is a closed immersion.

Proof. By the preceding lemma, it suffices to prove the proposition in the case where $A = k$. Moreover, since the properties under consideration are stable by

(fpqc) descent, and since dimension is invariant under extension of the base field, one may assume k algebraically closed.

Let us prove (a). Denote by C the reduced subscheme of H whose underlying topological space is $u(G)$. Since $u(G)$ is stable under the inversion morphism of H, so is C. On the other hand, $u : G \to C$ is quasi-compact and dominant, so by EGA IV_2, 2.3.7 the same holds for $u \times_k id_G$ and $id_H \times_k u$, and hence for their composition $u \times_k u : G \times_k G \to C \times_k C$. Consequently, the multiplication of H sends $C \times_k C$ into C, and so C is a group subscheme of H.

So, replacing H by C, we reduce to the case where u is dominant. Then $G(k)$ is dense in H, hence meets every connected component of H, and hence acts transitively on the set of these connected components. It therefore suffices to show that $u(G)$ contains H^0. Replacing G by $u^{-1}(H^0)$, we may suppose $H = H^0$; in this case, by 2.4, H is irreducible and of finite type over k, hence noetherian. On the other hand, u is locally of finite type (cf. EGA I, 6.6.6) and quasi-compact, hence of finite type. Consequently, by Chevalley's constructibility theorem (cf. EGA IV_1, 1.8.5), $u(G)$ is a constructible (and dense) part of $H = u(G)$, hence contains a dense open U of H (cf. EGA 0_III, 9.2.2). Then, by 0.5, one has $H = U \cdot U \subset u(G)$, whence $u(G) = H$. Taking 2.4.1 into account, this proves assertion (a).

Let us prove (b). Recall first that the functor $Ker(u)$ (cf. I, 2.3.6.1) is representable by $u^{-1}(e)$, where e denotes the unit element of H. Since u is of finite type, $Ker(u)$ is of finite type over k. On the other hand, replacing H by the reduced closed subscheme $u(G)$, we may assume u surjective. Denote by u^0 the restriction of u to G^0. Since G and $Ker(u)$ are equidimensional, and since $Ker(u)^0 \subset Ker(u^0)$, we reduce to the case where G, and hence also H, are irreducible.

Then, by EGA IV_3, 9.2.6.2 and 10.6.1 (ii), the set of $y \in H$ such that $\dim u^{-1}(y) = \dim G - \dim H$ contains a non-empty open set V. Since u is surjective, $U = u^{-1}(V)$ is then a non-empty open subset of G, hence contains a closed point x of G, since G is a Jacobson scheme (cf. EGA IV_3, 10.4.8). Then right translation r_x is an isomorphism of $Ker(u)$ onto $u^{-1}(u(x))$, whence:

$$\dim Ker(u) = \dim u^{-1}(u(x)) = \dim G - \dim H.$$

Let us prove (c), following [DG70], I, § 3.4. (Another proof is given in Exp. VI_B, 1.4.2.) Assume u a monomorphism. If C is a connected component of G, there exists a closed point $x \in G$ such that $C = r_x(G^0)$, and if one denotes by u_C (resp. u^0) the restriction of u to C (resp. to G^0), one has $u_C = r_{u(x)} \circ u^0 \circ r_x^{-1}$, so it suffices to show that u^0 is a closed immersion. We may therefore suppose $G = G^0$, so that G is irreducible and of finite type over k.

Let ξ be the generic point of G; then $O_{G,\xi}$ is an Artinian local ring; denote by $\mathfrak{m}$ its maximal ideal. On the other hand, let $h = u(\xi)$, $\mathfrak{n}$ the maximal ideal of $O_{H,h}$, and $A = O_{G,\xi}/\mathfrak{n}O_{G,\xi}$. Since u is a monomorphism, so is the morphism $u_h : \mathrm{Spec}(A) \to \mathrm{Spec}(\kappa(h))$ obtained by base change, so the multiplication morphism $A \otimes_{\kappa(h)} A \to A$ is an isomorphism (cf. EGA I, 5.3.8), whence $A = \kappa(h)$. By Nakayama's lemma (since $\mathfrak{n}O_{G,\xi}$ is contained in $\mathfrak{m}$, hence nilpotent), it follows that the morphism $O_{H,h} \to O_{G,\xi}$ is surjective.

Let then V be an affine open of H containing h, U a non-empty affine open of G contained in $u^{-1}(V)$, $\phi : O_H(V) \to O_G(U)$ the morphism of k-algebras induced by u, $\mathfrak{p}$ the prime ideal of $O_G(U)$ corresponding to ξ, and $\mathfrak{q} = \phi^{-1}(\mathfrak{p})$. Since G is of finite type over k, $O_G(U)$ is generated as a k-algebra by a finite number of elements

$a_1, \cdots, a_n$. By what precedes, there exist elements $b_1, \cdots, b_n$ and s in $O_H(V)$ such that $s \notin \mathfrak{q}$ and such that one has in $O_G(U)_\mathfrak{p}$ the equalities $a_i/1 = \phi(b_i)/\phi(s)$. There exist therefore elements $t_1, \cdots, t_n$ of $O_G(U) - \mathfrak{p}$ such that $t_i(a_i\phi(s) - \phi(b_i)) = 0$. Then, setting $t = t_1 \cdots t_n \phi(s) \in O_G(U) - \mathfrak{p}$, the equalities $a_i/1 = \phi(b_i)/\phi(s)$ already hold in $O_G(U)_t$ and, since $t \in O_G(U) = k[a_1, \cdots, a_n]$, there exists $b \in O_H(V)$ such that $t/1 = \phi(b)/\phi(s)^r$, for some $r \in \mathbb{N}$. Consequently, ϕ induces a surjection of $O_H(V)_{sb}$ onto $O_G(U)_t$, and so u is a local immersion at the point ξ.

The open subset W of G consisting of the points at which u is a local immersion is therefore non-empty. Since G is a Jacobson scheme, W contains a closed point y, and to show $W = G$, it suffices to show that every closed point of G belongs to W. Now every closed point x is the image of y by translation $r_x \circ r_y^{-1}$, hence belongs to W, whence $W = G$. This proves that u is a local immersion.

Since G is irreducible, it follows that u is an immersion. Indeed, for every $x \in G$, let U_x and V_x be open subsets of G and H such that $x \in U_x$ and such that u induces a closed immersion of U_x into V_x. Since U_x is dense in G, $u(U_x)$ is dense in $u(G) \cap V_x$, and since $u(U_x)$ is closed in V_x, one has $u(U_x) = V_x$. Moreover, since u is injective, one has $U_x = u^{-1}(V_x)$, and it follows that u induces a closed immersion of G into the open subscheme of H covered by the V_x. So $u : G \to H$ is an immersion. But we have already seen that $u(G)$ is closed in H, hence u is a closed immersion.

Lemma 2.5.3. [426] *Let A be an Artinian local ring, k its residue field, G a flat A-group, X an A-scheme equipped with a left action $\mu : G \times_A X \to X$ of G and with a section $s_0 : \operatorname{Spec} A \to X$. (This is the case, for instance, if G' is a second A-group and one is given an A-group morphism $G \to G'$.)*

Let ϕ be the morphism $\mu \circ (id_G \times s_0)$ from $G = G \times_A A$ to X. If ϕ is flat at a point g of G, then ϕ is flat.

Proof. Since G is flat over A, the flatness-by-fibers criterion (EGA IV_3, 11.3.10.2) shows that it suffices to show that $\phi \otimes_A k$ is flat; hence we may suppose $A = k$. In this case, the datum of s_0 is equivalent to that of a k-point $x_0 \in X(k)$, and ϕ is the morphism $h \mapsto hx_0$.

Let then $h \in G$; let us show that ϕ is flat at the point h. Let K be an extension of k containing a copy of $\kappa(g)$ and of $\kappa(h)$; one has a cartesian square

$$\begin{tikzcd} G_K \arrow[r, "\phi_K"] \arrow[d] & X_K \arrow[d] \\ G \arrow[r, "\phi"] & X \end{tikzcd}$$

in which the two vertical arrows are faithfully flat. So, by V 7.4 (i), ϕ_K is flat at every point $g' \in G_K$ above g, and to show that ϕ is flat at h, it suffices to show that

[426] N.D.E.: We make no finiteness hypotheses in this statement; this will be useful later (cf. 6.2 and VI_B, § 12).

ϕ_K is flat at a point h' above h. We are thus reduced to the case where g and h are rational. Let then $u = hg^{-1}$ and ℓ_u (resp. μ_u) the left translation of G (resp. of X) defined by u; since $\varphi \circ \ell_u = \mu_u \circ \varphi$, one obtains a commutative square

$$\begin{tikzcd}
O_{G,h} \arrow[r, "\sim"] & O_{G,g} \\
O_{X,hx_0} \arrow[u] \arrow[r, "\sim"] & O_{X,gx_0} \arrow[u]
\end{tikzcd}$$

in which the horizontal arrows are isomorphisms. Since the morphism $O_{X,gx_0} \to O_{G,g}$ is flat, the morphism $O_{X,hx_0} \to O_{G,h}$ is also flat.

Proposition 2.5.4. *Let A be an Artinian local ring, k its residue field, G an A-group locally of finite type, $X \neq \emptyset$ an A-scheme locally of finite type equipped with a left action of G. Assume that the morphism $\phi : G \times_A X \to X \times_A X$ defined set-theoretically by $(g, x) \mapsto (gx, x)$ is surjective. Then:*

(i) The connected components of X are of finite type, irreducible, and all of the same dimension.

(ii) More precisely, let $\bar{k}$ be an algebraic closure of k and x a closed point of $X \otimes_A \bar{k}$; its stabilizer F is a closed group subscheme of $G \otimes_A \bar{k}$, and the dimension of the irreducible components of X is $\dim G - \dim F$.

Taking Lemma 2.5.1 into account, we may suppose $A = k$. Suppose first k algebraically closed. Then G_{red} is a k-group locally of finite type, and hence, replacing G by G_{red} and X by X_{red}, we may suppose G and X reduced.

Since $G \times_k X$ is locally of finite type over k, ϕ is locally of finite type (cf. EGA I, 6.6.6 (v)), hence locally of finite presentation since $X \times_k X$ is locally noetherian. Let x be a rational point of X; then the morphism $\phi_x : G \to X$ obtained from ϕ by base change is surjective and locally of finite presentation. If η is a maximal point of X, then $O_{X,\eta}$ is a field (since X is reduced), so ϕ_x is flat at every point of G above η. So, by Lemma 2.5.3, ϕ_x is flat. Consequently, $\phi_x : G \to X$ is faithfully flat and locally of finite presentation, hence open (cf. EGA IV_2, 2.4.6). Since G^0 is open in G, irreducible and quasi-compact (by 2.4), each orbit $G^0 x = \phi_x(G^0)$, for x running through the rational points of X, is an open subset of X, irreducible and quasi-compact, hence of finite type over k (since X is locally of finite type over k).

Since every non-empty open subset of X contains a rational point, it follows that X is covered by these open sets. Moreover, two such open sets are either disjoint or equal. Indeed, if $\phi_x(G^0) \cap \phi_y(G^0)$ is non-empty, it contains a rational point z, and there exist therefore two rational points $g, h \in G^0$ such that $gx = z = hy$, whence $x = g^{-1}z$ and $y = h^{-1}z$, and so $\phi_x(G^0) = \phi_z(G^0) = \phi_y(G^0)$. It follows that the orbits $\phi_x(G^0)$ are also closed, and so are at once the connected components and the irreducible components of X.

Finally, let x, y be two rational points of X. Since ϕ_x is surjective, there exists a rational point $g \in G$ such that $y = gx$, and since G^0 is a normal subgroup of G, the orbit $G^0 y$ is the image of G^0 under translation ℓ_g of X, so that $G^0 y$ and $G^0 x$ have the same dimension.

Moreover, by I, 2.3.3.1, the stabilizer of x is represented by the closed subscheme F of G defined by the cartesian square below:

$$\begin{tikzcd} F \arrow[r] \arrow[d] & G \arrow[d, "\varphi_x"] \\ \operatorname{Spec} k \arrow[r] & X. \end{tikzcd}$$

Then F is a k-group locally of finite type, $F \cap G^0$ is a k-group of finite type containing F^0, and by 2.4.1, F and $F \cap G^0$ are equidimensional, of the same dimension as F^0. Let $C = \phi_x(G^0)$ be the irreducible component of X containing x. Proceeding as in the proof of point (b) of 2.5.2, one obtains that $\dim C = \dim G^0 - \dim F^0 = \dim G - \dim F$.

In the general case (i.e. for k an arbitrary field), let $\bar{k}$ be an algebraic closure of k. Let C be a connected component of X and C' a connected component of $C \otimes_k \bar{k}$; then C' is a connected component of $X' = X \otimes_k \bar{k}$. The morphism $\pi : X' \to X$ is open (cf. EGA IV_2, 2.4.10), and since it is integral, it is also closed; consequently $\pi(C') = C$. Since C' is irreducible and quasi-compact, C is irreducible and quasi-compact, hence of finite type over k (since X is locally of finite type over k).

Finally, since dimension is invariant under extension of the base field (cf. EGA IV_2, 4.1.4), $\dim C = \dim C'$, and since all the irreducible components of X' have the same dimension, the same holds for those of X.

2.6. Complements.

This paragraph has been added, drawn from [Per75] II, §§ 1–2, with complements due to O. Gabber.[427] This shows that the preceding results are valid for any group scheme G over a field k. (This will be used in sections 5, 6 and 7 of Exp. VI_B.)

We fix a field k. Let us begin with the following lemma (loc. cit., II 2.1.1), which does not appear explicitly in EGA IV_2, § 4.4 (although it can perhaps be read between the lines at the beginning of loc. cit., § 4.4.1).

Lemma 2.6.0. *Let X be an irreducible k-scheme, K an extension of k, X' an irreducible component of X_K. The projection $X' \to X$ is surjective.*

Indeed, let B be a transcendence basis of K over k and $L = k(B) \subset K$. By EGA IV_2, 4.3.2 and 4.4.1, X_L is irreducible and X' dominates X_L. The morphism $X' \to X_L$ is therefore integral and dominant, hence surjective. Since $X_L \to X$ is surjective (loc. cit., 4.4.1), so is $X' \to X$.

[427] N.D.E.: These results were communicated to us by O. Gabber, in particular 2.6.6, which plays an important role in section 5 of VI_B.

For the rest of 2.6, we fix a k-group scheme G and an action `μ : G × X → X` of G on a k-scheme X satisfying the following condition:

`(⋆)    the morphism Φ : G × X → X × X, (g, x) ↦ (gx, x) is surjective`

(this is the case in particular for G acting on itself by left translations). We shall then say, for short: "Let X be a G-scheme satisfying (□)". Finally, we write k' for the perfect closure of k.

Proposition 2.6.1. *Let X be a G-scheme satisfying* ($\star$)*.*

(i) X is geometrically pointwise irreducible over k, i.e. for every extension K of k, each $x \in X_K$ belongs to a unique irreducible component of X_K*.*

(ii) Each local ring of $(X_{k'})_{red}$ is normal.

(iii) Let η be a maximal point of $(X_{k'})_{red}$, $C = \eta^-$, and L the algebraic closure of k in $\kappa(\eta)$. Then C is an L-scheme, and is geometrically irreducible over L (i.e. $C \otimes_L K$ is irreducible for every extension K of L).

(iv) In particular, if x is a rational point of X, then the irreducible component of X containing x is geometrically irreducible over k.

Proof. (i) Since hypothesis ($\star$) is preserved by every base change $k \to K$, it suffices to show that each $x \in X$ belongs to a unique irreducible component of X. Since the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is a universal homeomorphism, we may moreover assume k perfect. We may then assume G and X reduced. Let η be a maximal point of X and z an arbitrary point of $Z = \eta^-$. Since X is reduced, the local ring $O_{X,\eta}$ equals $\kappa(\eta)$, and since k is perfect, for every extension K of k, $\kappa(\eta) \otimes_k K$ is normal (cf. EGA IV_2, 6.14.2), so every point of X_K above η is normal.

Since Φ is surjective, there exists a point γ of $G \times X$ such that $\Phi(\gamma)$ has projections z and η. Let $K = \kappa(\gamma)$; there exist then rational points g and η' of G_K and X_K such that η' lies above η and $z' = g\eta'$ lies above z. By what precedes, η' is a normal point of X_K, hence so is z'. Since $\pi : X_K \to X$ is flat, it follows that $z = \pi(z')$ is a normal point of X (cf. EGA IV_2, 2.1.13). This proves (ii) and (i).

The first assertion of (iii) then follows from (ii). Then, since L is algebraically closed in $\kappa(\eta)$, C is geometrically irreducible over L, by EGA IV_2, 4.5.9.

Finally, if X has a rational point x, it follows from (iii) that $L = k$, and so X is geometrically irreducible over k. This can also be seen directly as follows (cf. [Per75], II 2.1): let C be the irreducible component of X containing x and let K be an extension of k; X_K has a unique point e_K above e, and, by (i), e_K belongs to a unique irreducible component C' of X_K; on the other hand, by 2.6.0, every irreducible component of C_K contains e_K, hence equals C'.

Notation. Let us write provisionally C^0 for the reduced closed subscheme of G whose underlying space is the unique irreducible component of G containing the unit element e.

Corollary 2.6.2. *C^0 is geometrically irreducible over k and is set-theoretically stable under the group law, i.e. $(C^0_{k'})_{red}$ is a subgroup of $(G_{k'})_{red}$. Consequently, C^0 is quasi-compact.*

Indeed, by 2.6.1, C^0 is geometrically irreducible over k, so $C^0 \times C^0$ is irreducible, so $\nu(C^0 \times C^0) \subset C^0$, where ν denotes the morphism $(g,h) \mapsto gh^{-1}$. Since $\operatorname{Spec} k' \to \operatorname{Spec} k$ is a universal homeomorphism, the same conclusion holds for $C^0_{k'}$, and then for $H = (C^0_{k'})_{red}$, and so, since $H \times_{k'} H$ is reduced, ν induces a morphism $H \times_{k'} H \to H$, i.e. H is a subgroup of $(G_{k'})_{red}$. Consequently, by 0.5.1, H (and hence also C^0) is quasi-compact.

Recollection 2.6.3. *Let Y be a scheme. Recall (EGA 0_III, 9.1.1) that a subset E of Y is said to be* retrocompact *if the inclusion $E \hookrightarrow Y$ is quasi-compact, and that, by EGA IV_1, 1.9.5 (v) and 1.10.1, if U is a retrocompact open subset of Y, then the closure U^- of U is the union of the closures y^- of the points $y \in U$, and of course it is enough to take y running through the maximal points of U, i.e. the maximal points of Y contained in U.*

Consequently, if Y is quasi-separated and if η is a maximal point of Y, then the intersection of the closed neighborhoods of η equals η^-: indeed, if $y \in Y - \eta^-$, then y is contained in an affine open V not containing η; since Y is quasi-separated, V is retrocompact, so V^- is the union of the ξ^-, for ξ running through the maximal points of Y belonging to V, and hence $\eta \notin V^-$, i.e. $Y - V^-$ is a closed neighborhood of η not containing y.

Proposition 2.6.4. *Let X be a G-scheme satisfying $(\star)$ and let U be an open subset of X.*

(i) C^0U is an open subset of X, equal to the union of the irreducible components of X whose generic point belongs to U.

(i′) Consequently, if X is irreducible, it is quasi-compact.

(ii) If moreover U is retrocompact in X, then C^0U equals U^-, hence is an open-and-closed subset of X.

Proof. (i) First, C^0U is open, since it is the union, for $g \in C^0$, of the projections of the open sets $g \cdot U_{\kappa(g)} \subset X_{\kappa(g)}$, and each projection $X_{\kappa(g)} \to X$ is open.

To prove the second assertion of (i), we may replace k by k' (since $\operatorname{Spec} k' \to \operatorname{Spec} k$ is a universal homeomorphism), and so assume k perfect. We may then assume G and X reduced, hence geometrically reduced.

Let η be a maximal point of X contained in U and let Z be its closure. Consider the morphism $\mu' : C^0 \times Z \to X$. Since C^0 is geometrically irreducible, $C^0 \times Z$ is irreducible; write γ for its generic point. Since μ' sends the point $\epsilon(\eta)$ (where ϵ denotes the unit section of C^0) to η, then γ is sent to a generization of η, hence to η. So μ' sends the underlying space of $C^0 \times Z$ into Z, and so, since $C^0 \times Z$ is reduced, μ' factors through Z.

Let now $z \in Z$. Set $K = \kappa(z)$. Then the morphism μ_z : G_K → X_K, $h \mapsto h \cdot z$ is surjective; let α be a maximal point of X_K; the local ring $O_{X_K,\alpha}$ is

a field, since X_K is reduced, so μ_z is flat at every point of G_K above α, so μ_z is flat, by Lemma 2.5.3.

On the other hand, μ_z sends the generic point ω of C^0_K to a point $t \in Z_K$. Let β be a maximal point of Z_K such that $t \in \beta^-$; since μ_z is flat, there exists a generization ξ of ω such that μ_z(ξ) = β, and since ω is a maximal point of G_K, one necessarily has $\xi = \omega$, and so μ_z(ω) equals β, which lies above η (since $Z_K \to Z$ is flat).

Set $L = \kappa(\omega)$ and let ω_L and z_L be the L-points deduced from ω and z; then $\omega_L \cdot z_L = \beta'$ is a point of Z_L above β, and so $z_L = \omega_L^{-1} \cdot \beta' \in C^0_L \cdot U_L$, whence $z \in C^0 \cdot U$. This proves (i).

Since C^0 is quasi-compact, by 2.6.2, point (i′) follows from this: if X is irreducible and if U is a non-empty affine open, then X equals C^0U, i.e. is the image of the morphism $C^0 \times U \to X$, hence is quasi-compact. Finally, if U is retrocompact in X, then (cf. 2.6.3) U^- is the union of the η^- for η running through the maximal points of X contained in U, hence equals C^0U. This proves (ii).

One then obtains the following result ([Per75] II Th. 2.4, see also [Per76], Prop. 4.1.1):

Theorem 2.6.5. *Let k be a field, G a k-group scheme.*

(i) There exists a unique group subscheme G^0 of G, called the neutral component *of G, such that:*

(a) The underlying space of G^0 is the irreducible component of the unit element.
(b) $G^0 \to G$ is a flat closed immersion, i.e. $O_{G,g} = O_{G^0,g}$ for every $g \in G^0$.

(ii) Moreover, G^0 is quasi-compact, geometrically irreducible, and is a characteristic subgroup of G.

(iii) If G is connected, then $G = G^0$.

Proof. (i) Recall first that G is separated (0.3), hence a fortiori quasi-separated. Let U be an affine open of G containing the generic point ω of C^0. By 2.6.3 and 2.6.4, $U^- = C^0U$ is at once open and closed, and $C^0 = \omega^-$ is set-theoretically the intersection of these open-and-closed parts.

For U running through the affine opens containing ω, one obtains a projective system of G-schemes U^-, whose transition morphisms are affine (since they are closed immersions). One may therefore form the projective limit G^0 (cf. EGA IV_3, 8.2.2), i.e. for every affine open V of G, $G^0 \cap V$ is the spectrum of the algebra

```
lim O_G(V ∩ U) = O_G(V) / ∑_U I_U(V),
→
```

where $I_U(V)$ denotes the kernel of $O_G(V) \to O_G(V \cap U)$. It follows that G^0 has C^0 as underlying space, and that $G^0 \to G$ is a closed immersion. Moreover, for every $g \in G^0$, $O_{G^0,g}$ is the inductive limit, for V running through the affine opens of G

containing g, of the k-algebras $O_{G^0}(V \cap G^0) = \lim_U O_G(V \cap U)$, and this double inductive limit is

identified with

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lim lim O_G(V) = O_{G,g}
→   →
 U   V
g∈V⊂U
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i.e. one has $O_{G^0,g} = O_{G,g}$ (see also EGA IV_2, 5.13.3 (ii)). So $i : G^0 \to G$ is a flat closed immersion. Conversely, this condition implies that $i^*(O_G) = O_{G^0}$, and so G^0 is uniquely determined by conditions (a) and (b). This proves (i).

The first two assertions of (ii) follow from 2.6.2. Finally, let S be a k-scheme and ϕ an automorphism of the S-group G_S. For every $s \in S$, ϕ_s sends $G^0_{\kappa(s)}$ into itself, so $\phi(G^0_S) \subset G^0_S$. Moreover, the closed immersion $i_S : G^0_S \hookrightarrow G_S$ deduced from i by base change is flat, so one has $O_{G_S,\phi(z)} = O_{G^0_S,\phi(z)}$ for every $z \in G^0_S$, and so $\phi \circ i_S$ factors through G^0_S. This proves that G^0 is a characteristic subgroup of G, whence (ii). Finally, (iii) is a particular case of point (i) of the following proposition.[428]

Proposition 2.6.6. *Let k be a field, G a k-group acting on a k-scheme X in such a way that the morphism $G \times X \to X \times X$, $(g,x) \mapsto (gx,x)$ is surjective. Suppose X quasi-separated. Then:*

(i) Every connected component C of X is irreducible.

(ii) Let η be the generic point of C and L the algebraic closure of k in $\kappa(\eta)$. Then C is an L-scheme, geometrically irreducible over L, and the morphism

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G⁰_L ×_L C → C ×_L C
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is surjective.

(iii) In particular, if C contains a rational point x, then C is geometrically irreducible over k and the morphism $\phi : G^0 \to C$, $g \mapsto gx$ is surjective.

Proof. (i) Let C be a connected component of X, Y an irreducible component of X contained in C, η the generic point of Y, and U an affine open of X containing η. Since X is quasi-separated, U is retrocompact in X, so by 2.6.4, $U^- = G^0U$ is an open-and-closed part of X meeting C, hence containing C. Now, by 2.6.3, the intersection of the U^-, for U running through a fundamental system of affine open neighborhoods of η, equals η^-. It follows that $C = \eta^-$. This proves (i).

The first assertion of (ii) (and also of (iii)) follows from 2.6.1. Let us start by showing the second assertion of (iii). Denote by ω (resp. η) the generic point of G^0 (resp. C). Let $z \in C$ and $K = \kappa(z)$. Since G^0 (resp. C) is geometrically irreducible over k, G^0_K (resp. C_K) is irreducible; let ξ (resp. β) denote its generic point. We saw in

[428]N.D.E.: This proposition (as well as the preceding results) was communicated to us by O. Gabber; it will be used to correct the proof of Theorem 5.3 of VI_B.

the proof of 2.6.4 that the morphism $\mu_z : G_K \to C_K$, $h \mapsto h \cdot z$ sends ξ to β, and similarly one has $\phi(\omega) = \eta$.

Let $L = \kappa(\xi)$ and let ξ_L, z_L be the L-points deduced from ξ and z; then $\xi_L \cdot z_L = \beta'$ is a point of C_L above $\beta \in C_K$, hence also above $\eta \in C$. Consider the cartesian square:

$$\begin{tikzcd} G^0_L \arrow[r, "\phi_L"] \arrow[d, "\pi_{G^0}"'] & C_L \arrow[d, "\pi_C"] \\ G^0 \arrow[r, "\phi"] & C ; \end{tikzcd}$$

since $\phi_L^{-1}(\pi_{G^0}^{-1}(\omega)) = \pi_C^{-1}(\eta)$ (cf. EGA I, 3.4.8), there exists $g \in G^0_L$ such that $\phi_L(g) = \beta'$. One therefore has $z_L = \xi_L^{-1} \cdot \phi_L(g) = \phi_L(\xi_L^{-1} g)$, whence $\phi(\pi_{G^0}(\xi_L^{-1} g)) = \pi_C(z_L) = z$. This proves that ϕ is surjective.

Now let us prove the second assertion of (ii). It suffices to show that, for every $z \in C$, the morphism $\mu_z : G^0_L \otimes_L \kappa(z) \to C \otimes_L \kappa(z)$ is surjective, but this follows from (iii), since z is a rational point of $C \otimes_L \kappa(z)$.

Remark 2.6.7. *Under the hypotheses of 2.6.6, if $C(k) = \emptyset$, the morphism $G^0 \times_k C \to C \times_k C$ is not necessarily surjective. For example, for $k = \mathbb{R}$ and $G = \pm 1_{\mathbb{R}}$, the G-torsor $X = \operatorname{Spec} \mathbb{R}[X]/(X^2+1)$ is connected, but the morphism $G^0 \times_{\mathbb{R}} X \to X \times_{\mathbb{R}} X$ is not surjective. (But one has $L = \mathbb{C}$ and the morphism $G^0 \times_{\mathbb{R}} X \to X \times_{\mathbb{C}} X$ is an isomorphism.)*

3. Construction of quotients $F\backslash G$ (for G, F of finite type)

3.1.

Let A be an Artinian local ring and $u : F \to G$ a homomorphism of A-groups. If $\mu : F \times_A F \to F$ and $\nu : G \times_A G \to G$ denote the multiplication morphisms and λ the composite morphism

$$F \times_A G \xrightarrow{u \times G} G \times_A G \xrightarrow{\nu} G,$$

we recall that the *left quotient* $F\backslash G$ of G by F is the cokernel of the (Sch/A)-groupoid G_* described below:

$$\begin{tikzcd} F \times_A F \times_A G \arrow[r, shift left=2, "F \times \lambda"] \arrow[r, "\mu \times G" description] \arrow[r, shift right=2, "pr_{2,3}"'] & F \times_A G \arrow[r, shift left, "\lambda"] \arrow[r, shift right, "pr_2"'] & G \end{tikzcd}$$

(pr_2 and $pr_{2,3}$ are the projections of $F \times_A G$ and $F \times_A (F \times_A G)$ onto the second factors). We shall say that G_* is the *groupoid with base G defined by u* (cf. Exp. V, § 2.a; as in Exposé V, we do not follow in this Exposé the convention of IV, 4.6.15).

Since the unique A-morphism $F \to \operatorname{Spec} A$ is universally open (EGA IV_2, 2.4.9), pr_2 is an open morphism; the same therefore holds for λ, which is the composition of pr_2 and the automorphism σ of $F \times_A G$ defined by the formulas: $\sigma(S)(x, y) = (x, u(S)(x) \cdot y)$, where S is a variable A-scheme, and x and y belong to $F(S)$ and $G(S)$. One sees in the same way that pr_2 and λ are flat when F is flat over A.

Let us also note, to conclude these preliminaries, that every A-morphism s : $\operatorname{Spec} A \to G$ defines an automorphism of the groupoid G_* which induces on G, $F \times_A G$ and $F \times_A F \times_A G$ the automorphisms r_s, $id_F \times_A r_s$ and $id_F \times_A id_F \times_A r_s$ respectively. We shall again write r_s for this automorphism of G_*, and we shall say that r_s is the *right translation defined by* s (cf. 0.4).

3.2. Theorem.

Theorem 3.2. *Let F and G be flat groups locally of finite type over an Artinian local ring A. Let $u : F \to G$ be a quasi-compact A-group homomorphism with kernel finite over A. Then:*[429]

(i) The left quotient $F\backslash G$ of G by F exists in (Sch/A), and the sequence

$$F \times_A G \underset{pr_2}{\overset{\lambda}{\rightrightarrows}} G \xrightarrow{p} F\backslash G$$

is exact in the category of all ringed spaces.

(ii) The canonical morphism $p : G \to F\backslash G$ is surjective and open, and $F\backslash G$ is a direct sum of schemes of finite type over A.

(ii′) More precisely, $X = F\backslash G$ is equipped with a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$; consequently, the connected components of X are of finite type over A, irreducible, and all of dimension $\dim G - \dim F$.

(iii) The canonical morphism $F \times_A G \to G \times_{F\backslash G} G$ is surjective.

(iv) If u is a monomorphism,[430] *then:*

(a) $F \times_A G \xrightarrow{(\lambda, pr_2)} G \times_{F\backslash G} G$ is an isomorphism, and $G \to F\backslash G$ is faithfully flat and locally of finite presentation. (a′) $F\backslash G$ represents the (fppf) quotient sheaf $F\tilde{\backslash}G$, and $G \to F\backslash G$ is a locally trivial F-torsor for the (fppf) topology. (b) $F\backslash G$ is flat over A, and is smooth over A if G is so. (c) $u : F \to G$ is a closed immersion, and $F\backslash G$ is separated. (d) If, in addition, F is a normal subgroup of G, there exists on $F\backslash G$ one and only one A-group structure such that $p : G \to F\backslash G$ is a morphism of A-groups.

In the proof of this theorem, A' will denote an A-algebra that is local, finite and free over A. If R is a relation involving A', we shall say that "$R(A')$ is true when A'

[429] N.D.E.: We have added (ii′) and detailed point (iv), taking into account the additions made in 2.5.2, 2.5.4, and in Exp. V, 6.1.

[430] N.D.E.: In this case, the hypothesis that G is flat may be removed, cf. subsection 3.3.

is large enough" if there exists an A-algebra A_1 that is local, finite and free over A such that the relation $R(A')$ is satisfied for each A-algebra A' that is local, finite and free over A_1.

We shall first prove the theorem when F and G are of finite type over A.

3.2.1. Suppose for a moment that every point of G has an open saturated neighborhood W such that the groupoid induced by G_* on W possesses a quasi-section (cf. V § 6). Then, by V 6.1, one has assertions (i), (ii), (iii) and (iv)(a), and $F\backslash G$ is of finite type over k. Moreover, under the hypothesis of (iv), since $G \to F\backslash G$ is faithfully flat and locally of finite presentation, assertion (b) follows from EGA IV, 2.2.14 and 17.7.7. On the other hand, assertion (iv)(a′) follows from (iv)(a), by Exp. IV, 3.4.3.1, 5.2.2 and 5.1.6. Finally, (iv)(c) will be proved in 3.2.5, and (ii′) and (iv)(d) will be proved in section 5.

Let us now prove the following assertion:

(†) every finite subset of $F\backslash G$ is then contained in an affine open.

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If U is a quasi-section of the groupoid induced by G_* on a saturated open W of G, then $U \otimes_A A'$ is a quasi-section of the (Sch/A')-groupoid induced by $G_* \otimes_A A'$ on $W \otimes_A A'$. Moreover, if U_* is the (Sch/A)-groupoid induced by G_* on U, then $U_* \otimes_A A'$ is identified with the (Sch/A')-groupoid induced by $G_* \otimes_A A'$ on $U \otimes_A A'$.[432]

It follows from the proofs of Exposé V that the construction of the quotient $X = F\backslash G$ commutes with the extension $A \to A'$ of the base of the type considered here.[433]

Let then $x_1, \cdots, x_n$ be points of $X = F\backslash G$, which we may assume closed,[434] and $g_1, \cdots, g_n$ closed points of G projecting onto $x_1, \cdots, x_n$. Let V be an everywhere dense affine open of X,[435] and let U be the inverse image of V in G. By 1.2, there exists an A-algebra A' that is local, finite and free over A, such that the points $g'_1, \cdots, g'_p$ of $G' = G \otimes_A A'$ above $g_1, \cdots, g_n$ are strictly rational over A'.[436] Since the morphisms $G' \to G$ and $G \to X$ are open, $U' = U \otimes_A A'$ is dense in G', so the open set $\bigcap_{i=1}^{p} (U')^{-1} \cdot g'_i$ is non-empty, hence contains a closed point x. So, by 1.2 (and 0.4.1), one may

[431] N.D.E.: Let us point out here that if $A = k$ is a field, then every quasi-compact open of $F\backslash G$ is quasi-projective (a result due to Chow for smooth algebraic groups), cf. [Ray70], VI 2.6. By contrast, over the Artinian local ring $A = \mathbb{C}[\epsilon]/(\epsilon^2)$, there exist abelian A-schemes G that are not projective (loc. cit., XII 4.2).

[432] N.D.E.: What precedes is valid for every base change $A \to A'$.

[433] N.D.E.: This is detailed in 4.6 below: it is a matter of seeing that the formation of the direct image by the morphisms p, λ and pr_2 commutes with flat base changes $A \to A'$. Since F and G are of finite type over the Artinian ring A, the morphisms f in question are all quasi-compact and quasi-separated, and the equality $f_*(O_X) \otimes_A A' = f'_*(O_{X'})$ (with obvious notations) follows from EGA IV_1, 1.7.21.

[434] N.D.E.: Indeed, let $y_1, \cdots, y_n$ be arbitrary points of X; since X is of finite type over A, each y_i has in its closure a closed point x_i, and every open subset containing x_i contains y_i.

[435] N.D.E.: Such an open exists, since X is of finite type over k: X has a finite number of irreducible components $C_1, \cdots, C_p$, and it suffices to take, for each i, a non-empty affine open contained in $C_i - \bigcup_{j \neq i} C_j$. (Here one knows moreover, by (ii′), that the C_i are disjoint…)

[436] N.D.E.: We have detailed the original in what follows.

suppose, by possibly enlarging A', that x is strictly rational over A'. Then, since $x \in (U')^{-1} \cdot g'_i$, one has $g'_i \in U' \cdot x$.

Denote by V' the inverse image of V in $X' = X \otimes_A A'$; this is an affine open of X', and is also the image of U' under the projection $G' \to X'$. Since right translation r_x is an automorphism of the groupoid $G_* \otimes_A A'$, it induces an automorphism, again denoted r_x, of the quotient X'. Consequently, the image $V' \cdot x = r_x(V')$ of $U' \cdot x$ in X' is

an affine open of X' containing the images $x'_1, \cdots, x'_p$ of $g'_1, \cdots, g'_p$.

Consider now the equivalence relation on $X' = X \otimes_A A'$ defined by the projection $X \otimes_A A' \to X$:

$$X \otimes_A A' \otimes_A A' \underset{d_0}{\overset{d_1}{\rightrightarrows}} X \otimes_A A' \longrightarrow X,$$

where d_0 and d_1 are induced by the two canonical injections of A' into $A' \otimes_A A'$. Since A' is a finite, free A-algebra, say of rank n, then d_0 and d_1 are finite and locally free of rank n; consequently, one may apply the reasoning of Exp. V, 5.b (drawn from the proof of SGA 1, VIII.7.6). One thus obtains that $x'_1, \cdots, x'_p$ are contained in an affine saturated open W' contained in the affine open $V' \cdot x$. The image of W' in X then contains $x_1, \cdots, x_n$ and is an affine open of X, by V, 4.1 (ii).

3.2.2. For every A-algebra A' that is local, finite and free over A, let us now write $U(A')$ for the set of points of $G \otimes_A A'$ having an open saturated neighborhood W such that the groupoid induced by $G_* \otimes_A A'$ on W has a quasi-section. It is clear that $U(A')$ is saturated for the action of $G(\operatorname{Spec} A')$ on $G \otimes_A A'$. We shall see that, when A' is large enough, $U(A')$ is equal to $G \otimes_A A'$.

By Theorem V 8.1, $U(A)$ is non-empty, hence contains a closed point y. The proof then proceeds by induction on $\dim(G - U(A))$. Let $g_1, \cdots, g_n$ be closed points belonging to the various irreducible components of $G - U(A)$. By 1.2, there exists A' local, finite and free over A, such that the points $g'_1, \cdots, g'_p$ (resp. $x = x_1, \cdots, x_r$) of $G' = G \otimes_A A'$ projecting onto $g_1, \cdots, g_n$ (resp. onto y) are strictly rational over A'. Then $U(A')$ contains $(U(A) \otimes_A A') \cdot x^{-1} g'_i$ for every i; so $U(A')$ contains

$g'_1, \cdots, g'_p$, and one has

$$\dim(G' - U(A')) < \dim(G - U(A)).$$

The induction hypothesis then implies the existence of an algebra A'' local, finite and free over A' such that one has $U(A'') = G' \otimes_{A'} A'' = G \otimes_A A''$.

3.2.3. We are now in a position to prove the existence of $F \backslash G$ when F and G are of finite type over A. Let A' be large enough over A so that $U(A')$ coincides with $G \otimes_A A'$ (cf. 3.2.2). We shall set $A'' = A' \otimes_A A'$ and, for every A-scheme X, we shall denote by X' and X'' the fibered products $X \bigotimes_A A'$ and $X \bigotimes_A A''$. By 3.2.1 and 3.2.2, the quotients $F'\backslash G'$ and $F''\backslash G''$ exist, and one has the following commutative diagram, in which the first two lines and columns are exact:

$$\begin{tikzcd}
F'' \times_{A''} G'' \arrow[r, shift left, "pr''_2"] \arrow[r, shift right, "\lambda''"'] \arrow[d, shift right, "w_1"'] \arrow[d, shift left, "w_2"] & G'' \arrow[r, "p''"] \arrow[d, shift right, "v_1"'] \arrow[d, shift left, "v_2"] & F'' \backslash G'' \arrow[d, shift right, "u_1"'] \arrow[d, shift left, "u_2"] \\
(*) \quad F' \times_{A'} G' \arrow[r, shift left, "pr'_2"] \arrow[r, shift right, "\lambda'"'] \arrow[d, "h"] & G' \arrow[r, "p'"] \arrow[d, "g"] & F' \backslash G' \\
F \times_A G \arrow[r, shift left, "pr_2"] \arrow[r, shift right, "\lambda"'] & G &
\end{tikzcd} .$$

In this diagram, pr'_2 and λ' (resp. pr''_2 and λ'') are obtained from pr_2 and λ by obvious base changes; the morphisms g and h are induced by the canonical injection $A \to A'$. One denotes by p' and p'' the canonical morphisms; the morphisms v_1, v_2 and w_1, w_2 are induced by the two canonical injections of A' into A''. Finally, since the construction of the quotient $F'\backslash G'$ commutes with the two base changes

$f_1, f_2 : \operatorname{Spec} A'' \rightrightarrows \operatorname{Spec} A'$, one has, writing $\pi' : F'\backslash G' \to \operatorname{Spec} A'$ for the structural morphism, canonical isomorphisms, for $i = 1, 2$:

$$\tau_i : F''\backslash G'' \xrightarrow{\sim} (F'\backslash G') \times_{\pi', f_i} \operatorname{Spec} A'',$$

and the morphism u_i is the composite of τ_i and the projection $(F'\backslash G') \times_{\pi', f_i} \operatorname{Spec} A'' \to F'\backslash G'$.

Now, when one has a diagram of type $(*)$ with the first two rows and columns exact, one verifies easily that $Coker(pr_2, \lambda)$ exists if and only if $Coker(u_1, u_2)$ exists, and these two cokernels are identified. The existence of $F\backslash G$ will therefore follow from that of $Coker(u_1, u_2)$.

Now it follows from the compatibility of the formation of $F\backslash G$ with the base extensions considered here (cf. N.D.E. (37) in 3.2.1, and 4.6 below) that the composite morphism

$$(F'\backslash G') \times_{\pi', f_1} \operatorname{Spec} A'' \xrightarrow{\tau_1^{-1}} F''\backslash G'' \xrightarrow{\tau_2} (F'\backslash G') \times_{\pi', f_2} \mathrm{Spe}$$

is a descent datum on $F'\backslash G'$ relative to $f : \operatorname{Spec} A' \to \operatorname{Spec} A$. By 3.2.1 (†) and SGA 1, VIII 7.6, this descent datum is effective, that is, $Coker(u_1, u_2)$ exists (one could also use directly Theorem 4.1 of Exp. V).

3.2.4. To complete the proof of assertions (i), (ii), (iii) and (iv)(a) of 3.2 in the case where F and G are of finite type over A, it remains to study the quotient $F\backslash G$. By V 6.1, assertions (ii), (iii) and (iv)(a) "become true" after the base change $f : \operatorname{Spec} A' \to \operatorname{Spec} A$; by EGA IV_2, 2.6.1, 2.6.2 and 2.7.1, these assertions were therefore true before the base change. Finally, to prove the second assertion of (i), i.e. that $F\backslash G$ is the cokernel of (pr_2, λ) in the category of all ringed spaces, one need only refer to V § 6.c).

3.2.5. [437] Let us now prove assertion (iv)(c) of 3.2, by reproducing the proof of VI_B, 9.2.1. Write $X = F\backslash G$, and d for the morphism $F \times_A G \to G \times_A G$ with components λ and pr_2.

Since u is a closed immersion by 2.5.2, and since $d = \sigma \circ (u \times id_G)$, where σ is the automorphism of $G \times_A G$ defined by $\sigma(x, y) = (xy, y)$, d is a closed immersion. On the other hand, by (iv)(a), one has the cartesian square

$$\begin{tikzcd} F \times_A G \arrow[r, "d"] \arrow[d] & G \times_A G \arrow[d, "p \times p"] \\ X \arrow[r, "\Delta_X"] & X \times_A X \end{tikzcd}$$

and p, hence also $p \times p$, is faithfully flat and locally of finite presentation. So, by (fppf) descent, since d is a closed immersion, the same holds for Δ_X, i.e. X is separated.

3.3.

[438] In Theorem 3.2, the hypothesis that G be flat can be removed when $u : F \to G$ is a monomorphism. This generalization is mentioned in Remark 9.3 b) of Exp. VI_B, and also in [Ray67a], Example a) i), p. 82. The proof, found in Theorem 4 of [An73], follows from Theorem 3.2 and from the following theorem of Grothendieck (mentioned in [Ray67a], Th. 1 ii) and proved in [DG70], § III.2, 7.1). If X is a scheme and R an equivalence relation in X, we shall write $\tilde{X}/R$ for the (fppf) quotient sheaf of X by R (cf. IV 4.4.9).

Theorem 3.3.1 (Grothendieck). *Let A be a ring, X an A-scheme, and $d_0, d_1 : R \to X$ an A-equivalence relation in X, such that d_i is faithfully flat, of finite presentation. Let X_0 be a saturated closed subscheme of X, defined by a nilpotent ideal, and let R_0 be the equivalence relation induced by R on X_0. Then, if the (fppf) quotient sheaf $\tilde{X}_0/R_0$ is representable by an A-scheme, so is $\tilde{X}/R$.*

For the proof, we refer to [DG70], § III.2, 7.1. Let us now return to the case where A is an Artinian local ring. Let $u : F \to G$ be a quasi-compact morphism between A-

[437] N.D.E.: We have added this paragraph.
[438] N.D.E.: We have added this subsection.

groups locally of finite type, and suppose further that u is a monomorphism. Then, by 2.5.2, u is a closed immersion.

One may now state the following variant of Theorem 3.2.

Theorem 3.3.2. *Let A be an Artinian local ring, G an A-group locally of finite type, F a closed subgroup of G, flat over A.*[439] *Then:*

(i) The (fppf) quotient sheaf $\tilde{F}\backslash G$ is representable by an A-scheme $F\backslash G$ that is separated and locally of finite type; moreover, the sequence

$$F \times_A G \overset{//}{\longrightarrow} G \xrightarrow{p} F\backslash G$$

is exact in the category of all ringed spaces.

(ii) $F \times_A G \xrightarrow{(\lambda, pr_2)} G \times_{F\backslash G} G$ is an isomorphism, and $p : G \to F\backslash G$ is faithfully flat and locally of finite presentation, so that p is a locally trivial F-torsor for the (fppf) topology.

(iii) If G is flat (resp. of finite type, resp. smooth) over A, then so is $F\backslash G$.

(iv) $X = F\backslash G$ is equipped with a right action of G, such that $p(e) \cdot g = p(g)$ for every $g \in G$; consequently, the connected components of X are of finite type over A, irreducible, and all of dimension $\dim G - \dim F$.

(v) If, in addition, F is a normal subgroup of G, there exists on $F\backslash G$ one and only one A-group structure such that $p : G \to F\backslash G$ is a morphism of A-groups.

Assertions (i) and (ii) follow from 3.2 and 3.3.1, and since $G \to F\backslash G$ is faithfully flat and locally of finite presentation, assertion (iii) follows from EGA IV, 2.2.14, 2.7.1 and 17.7.7. We shall prove assertions (iv) and (v) in section 5. Let us note immediately the following corollary.

Corollary 3.3.3. *Let A be an Artinian local ring, G an A-group locally of finite type, H a closed subgroup of G, flat over A. Write p for the morphism $G \to G/H$ and λ (resp. pr_1) for the morphism $G \times H \to G$ defined by $\lambda(g, h) = gh$ (resp. the projection $G \times H \to G$). Then, for every open subset U of G/H, one has*

$$O(U) = \{\varphi \in O(p^{-1}(U)) \mid \varphi \circ \lambda = \varphi \circ pr_1\}$$

i.e. $O(U)$ is the set of $\phi \in O(p^{-1}(U))$ such that $\phi(gh) = \phi(g)$ for every A-scheme S and $g \in G(S)$, $h \in H(S)$.

Indeed, since $p : G \to G/H$ is faithfully flat and locally of finite presentation, hence covering for the (fppf) topology, this follows from IV, 3.3.3.2.

4. Construction of quotients $F\backslash G$ (general case)

We now assume the hypotheses of Theorem 3.2 to be satisfied, F and G not necessarily being of finite type over A.

[439] N.D.E.: For an example where F is not flat and $\tilde{F}\backslash G$ not representable, see [DG70], § III.3, n° 3.3.

4.1.

Let us first consider a connected component G^α of G and show that the saturation $S(G^\alpha)$[440] of G^α for the equivalence relation defined by the groupoid G_* is an open-and-closed subset of G (in other words, is the union of certain connected components of G).

This saturation is the image of $F \times_A G^\alpha$ under λ, hence is open in G (cf. § 3.1). If k is the residue field of A and $\bar{k}$ an algebraic closure of k, it remains to show that the image of $(F \times_A G^\alpha) \otimes_A \bar{k}$ by $\lambda \otimes_A \bar{k}$ is closed in $G \otimes_A \bar{k}$, or equivalently, by SGA 1, VIII.4.4, that the image of $(F \times_A G^\alpha) \otimes_A \bar{k}$ by $\lambda \otimes_A \bar{k}$ is closed. Since $G^\alpha \otimes_A \bar{k}$ is the union of a finite number of connected components of $G \otimes_A \bar{k}$, we are reduced to the case where A is an algebraically closed field, which we shall assume. In this case, $S(G^\alpha)$ is the union of the images of G^α under the left translations $\ell_{u(x)}$, where

x runs through the closed points of F; the assertion follows therefore from the fact that these images are connected components of G.

4.2.

Let us in particular take G^α to be the connected component G^0 of the origin of G. Then $S(G^0)$ obviously contains the image of F under u, which is none other than the equivalence class of the origin. On the other hand, if F^β is a connected component of F, $F^\beta \times_A G^0$ is connected (2.1.2), so that the image of $F^\beta \times_A G^0$ by λ is contained in the connected component of $u(F^\beta)$ in G. In other words, $S(G^0)$ is the union of the connected components which meet the image of F.

One will also note that the open subscheme of G having $S(G^0)$ as underlying space is a subgroup of G (which we still denote $S(G^0)$):

indeed, the inversion morphism of G preserves the image of F and permutes the connected components of G meeting this image; it therefore suffices to show that $\nu : G \times_A G \to G$ sends $S(G^0) \times_A S(G^0)$ into $S(G^0)$, and for this one may suppose that A is an algebraically closed field (with the notations of 4.1, $S(G^0) \otimes_A \bar{k}$ is indeed identified with the saturation of $(G \otimes_A \bar{k})^0$ by the equivalence relation defined by the homomorphism $u \otimes_A \bar{k}$); if G^γ and G^δ are then connected components of $S(G^0)$, $G^\gamma \times_A G^\delta$ is connected and its image under ν meets the image of F; consequently, $u(G^\gamma \times_A G^\delta)$ is contained in a connected component of G meeting $u(F)$.

4.3.

It follows from what precedes that the groupoid G_* with base G defined by u is the direct sum of the groupoids $S(G^\alpha)_*$ induced by G_* on the various open-and-closed parts of G of the form $S(G^\alpha)$. The cokernel of G_* is therefore the direct sum of the cokernels of these groupoids $S(G^\alpha)_*$, which one is led to study separately.

Let us first consider the groupoid $S(G^0)_*$ induced by G_* on $S(G^0)$. It is clear that $S(G^0)_*$ is the groupoid with base $S(G^0)$ defined by the homomorphism from F into

[440] N.D.E.: We have written $S(G^\alpha)$ instead of $\bar{G}^\alpha$ for the saturation of G^α.

$S(G^0)$ induced by u (§ 3.1). The cokernel whose existence we wish to prove is therefore identified with $F\backslash S(G^0)$. Consider on the other hand the groupoid

```
                 □′_2
                 ────→       □_1
                 □′_1            ────→
G⁰_2             ────→     G⁰_1  ────→   G⁰_0 = G⁰
                                 □_0
                 □′_0
```

induced by $S(G^0)_*$ on G^0. If one refers to the construction explicit in V § 3.b), the object then denoted $Y_0 \times_{X_0} X_1$ is none other than

$F \times_A G^0$, so that G_1^0 is the inverse image of G^0 under the morphism $F \times_A G^0 \to S(G^0)$ induced by λ.

I claim that this inverse image is $F^0 \times_A G^0$, where F^0 denotes the inverse image of G^0 under u. Indeed, if F^β is a connected component of F^0, $F^\beta \times_A G^0$ is connected (2.1.2) and $\lambda(F^\beta \times_A G^0)$ is contained in G^0; conversely, if F^β is a connected component of F not contained in F^0, the image of $F^\beta \times_A G^0$ is again connected and contains $u(F^\beta)$; if $u(F^\beta)$ is not contained in G^0, $\lambda(F^\beta \times_A G^0)$ does not meet G^0.

It follows from what precedes that the groupoid G_*^0 induced by G_* on G^0 is the groupoid with base G^0 defined by the homomorphism $F^0 \to G^0$ induced by u. Since G^0, and hence F^0, are of finite type over A, then, by paragraph 4, G_*^0 has a cokernel which is none other than $F^0\backslash G^0$.

I now claim that $F^0\backslash G^0$ is identified with $F\backslash S(G^0)$. Indeed, the proof is analogous to that of the first part of assertion (i) of Lemma V § 6.1; consider the diagram:

```
                 v                              pr_2
S(G⁰) ←────────────  F ×_A G⁰  ───────────→  G⁰,
```

where v is the morphism induced by λ. Since pr_2 has a section, pr_2 is a universal effective epimorphism, so that $F^0\backslash G^0$ coincides with $Coker(v_0, v_1)$, where

```
                 v′_2
                 ────→
                 v′_1            v_1
                                 ────→
V_2              ────→     V_1   ────→   V = F ×_A G⁰
                                 v_0
                 v′_0
```

is the inverse image by pr_2 of the groupoid G_*^0 (cf. V § 3.a), that is, also the inverse image of $S(G^0)_*$ by the composite morphism

```
           inclusion                  pr_2
F ×_A G⁰ ──────────→ F ×_A S(G⁰) ──────────→ S(G⁰).
```

Similarly, since v is faithfully flat and quasi-compact, $F\backslash S(G^0)$ coincides with the cokernel of the inverse image of $S(G^0)_*$ by the base change v. Now this inverse image is isomorphic to V_* by Exp. V, § 3.c; it follows that the canonical inclusion of G^0_* into $S(G^0)_*$ induces an isomorphism of $F^0\backslash G^0$ onto $F\backslash S(G^0)$.

We finally note that: the construction of $F\backslash S(G^0)$ commutes with finite locally free base changes, because the same holds for $F^0\backslash G^0$ (cf. N.D.E. (37) and 4.6 below).

4.4.

It remains to construct the cokernel of the groupoid $S(G^\alpha)_*$ when G^α is an arbitrary connected component of G. If A' is a large enough local, finite, free A-algebra (cf. 3.2), $G^\alpha \otimes_A A'$ is the union of a finite number of connected components $C_1, \cdots, C_n$ of $G \otimes_A A'$, all of which have a strictly rational point. For every i, there exists therefore a right translation r_i of $G \otimes_A A'$ sending $G^0 \otimes_A A'$ onto C_i; this translation induces an isomorphism of the groupoid $S(G^0_*) \otimes_A A'$ onto $S(C^i_*)$, so that the groupoid induced by $G_* \otimes_A A'$ on the saturation of C_i has a cokernel.

Since $S(G^\alpha_*) \otimes_A A'$ is the direct sum of some of the $S(C^i_*)$, then $S(G^\alpha_*) \otimes_A A'$ has a cokernel; this cokernel is the direct sum of a certain number of copies of $(F^0 \otimes_A A')\backslash(G^0 \otimes_A A')$, so that every finite subset of this cokernel is contained in an affine open; moreover, the construction of this cokernel commutes with finite locally free extensions

of the base (cf. N.D.E. (37) and 4.6 below). One thus sees, as in 3.2.3, that this cokernel is of the form $Y \otimes_A A'$, where Y is a cokernel of $S(G^\alpha_*)$.

4.5.

We have therefore constructed $F\backslash G$ and shown that it is a direct sum of schemes of finite type over A. The other assertions of Theorem 3.2 reduce directly to assertions concerning the groupoids $S(G^\alpha_*)$. As in V § 6, the second assertion of (i) follows from the first and from (ii) and (iii), so it suffices to prove (ii), (iii) and (iv)(a). Since A' is a local, finite, free A-algebra, the morphism $A \to A'$ is faithfully flat and of finite presentation, so, by SGA 1, VIII (3.1, 4.6, 5.4), it suffices to verify the corresponding assertions in the case of the groupoid $S(G^\alpha_*) \otimes_A A'$. Now the latter is isomorphic to the direct sum of a finite number of copies of $S(G^0_*) \otimes_A A'$ (cf. 4.4), so that one is reduced to the groupoid $S(G^0_*)$.

For this last one continues to mimic the proof established in V § 6, as one began to do in 4.3.

4.6.

Let us add to conclude this paragraph some remarks concerning Lemma 6.1 and § 9.a of Exposé V: with the hypotheses and notations of V § 9.a, we seek a condition under which the construction of the cokernel of the (Sch/S)-groupoid X_* commutes with an extension $\pi : S' \to S$ of the base. Since the cokernels of X_* and X'_* are identified with the cokernels of the groupoids U_* and U'_* induced by X_* and X'_* on the quasi-sections U and U', one is reduced to the case of a (Sch/S)-groupoid satisfying the hypotheses of Theorem V 4.1.

[441] If one denotes by Y the cokernel of U_*, $Y' = Y \times_S S'$, and Y_1 the cokernel of U'_*, one saw in V § 9.a that the canonical morphism $Y_1 \to Y'$ is a homeomorphism (and even a universal homeomorphism); one may therefore identify Y_1 and Y' as topological spaces. If $p : U \to Y$ is the canonical morphism and if $p' : U' \to Y'$ is obtained from it by base change, we then want the sequence of $O_{Y'}$-modules

$$(*)\quad O_{Y'} \xrightarrow{p'_*(O_{U'})} \rightrightarrows p'_* u'_{1*}(O_{U'_1}) = p'_* u'_{0*}(O_{U'_1})$$

to be exact.[442] Since we are under the hypotheses of V 4.1, u_0 and u_1 are finite and locally free; and, by V.4.1 (ii), p is integral. Then, p and $p \circ u_i$ are affine, hence separated and quasi-compact.

Consequently, if S' is flat over S, it follows from EGA III_1, 1.4.15 (taking into account the correction Err_III 25 in EGA III_2) that the sequence $(*)$ is identified with the inverse image of the sequence

$$(**)\quad O_Y \longrightarrow \overset{p_*(O_U)}{\rightrightarrows} p_* u_{1*}(O_{U_1}) = p_* u_{0*}(O_{U_1}),$$

which is an exact sequence.[443] An analogous argument applies when the groupoid X_* has "locally" quasi-sections (cf. the proof of Theorem V 7.1). One thus obtains the:

Proposition 4.6.1. *The construction of the cokernel of X_* commutes with flat extensions of the base when X_* has locally quasi-sections.*

4.7.

Let us now consider the case of the groupoid G_* of Theorem 3.2 when one provisionally assumes F and G of finite type over A.

By 3.2.2, there exists an A-algebra A' local, finite and free over A such that the groupoid $G_* \bigotimes_A A'$ has "locally" quasi-sections. For every extension $T \to \operatorname{Spec} A$ of the base, the sequence

$$(F''\backslash G'') \times_{\operatorname{Spec} A} T \overset{//}{\longrightarrow} (F'\backslash G') \times_{\operatorname{Spec} A} T \longrightarrow (F\backslash G) \times_{\{S$$

deduced from the diagram $(*)$ of 3.2.3 is exact. If one supposes in addition T flat over $\operatorname{Spec} A$, then $(F''\backslash G'') \times_{\operatorname{Spec} A} T$ and $(F'\backslash G') \times_{\operatorname{Spec} A} T$ are identified respectively, by 4.6, with the cokernels of the groupoids

$$(G_* \otimes_A A'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (G_* \otimes_A A') \times_{\operatorname{Spec} A} T.$$

[441] N.D.E.: We have added the sentence that follows.
[442] N.D.E.: In what follows, we have modified the original; the supplementary hypotheses made on X_* were superfluous.
[443] N.D.E.: We have added on the one hand the following sentence and, on the other hand, the numbering 4.6.1 below, to make the stated result explicit.

The diagram deduced from 3.2.3 (∗) by the base change $T \to \operatorname{Spec} A$ then shows that $(F\backslash G) \times_{\operatorname{Spec} A} T$ is identified with the cokernel of $G_* \times_{\operatorname{Spec} A} T$. An analogous argument is valid in the general case (i.e. when G and F are locally of finite type over A). One therefore obtains:[444]

Proposition 4.7.1. *Under the hypotheses of Theorem 3.2, for every flat A-scheme T, $(F\backslash G) \times_{\operatorname{Spec} A} T$ is identified with the left quotient of $G \times_{\operatorname{Spec} A} T$ by $F \times_{\operatorname{Spec} A} T$.*

5. Connections with Exposé IV and consequences

5.1.

[445] We resume the notations of § 3 and the hypotheses of Theorem 3.2; one then has the following commutative diagram

$$\begin{tikzcd}
F \times_A G \times_A G \arrow[r, "F \times \nu"] \arrow[d, shift right, "pr_2 \times G"'] \arrow[d, shift left, "\lambda \times G"] & F \times_A G \arrow[d, shift right, "pr_2"'] \arrow[d, shift left, "\lambda"] \\
G \times_A G \arrow[r, "\nu"] \arrow[d, "p \times G"'] & G \arrow[d, "p"] \\
(F\backslash G) \times_A G \arrow[r, dashed, "\rho"] & F\backslash G,
\end{tikzcd}$$

which satisfies the equalities $pr_2 \circ (F \times \nu) = \nu \circ (pr_2 \times G)$ and $\lambda \circ (F \times \nu) = \nu \circ (\lambda \times G)$. Moreover, since G is assumed flat over A, the left vertical sequence is exact by 4.7, so that ν induces a morphism of A-schemes:

$$\rho : (F\backslash G) \times_A G \to F\backslash G.$$

This morphism ρ makes G act on the right on $F\backslash G$, as one verifies immediately; moreover, the canonical morphism $G \to F\backslash G$ commutes with the right actions of G on G and on $F\backslash G$.

[446] This proves the first assertion of point (ii′) of 3.2. By 2.5.4, one then obtains that the connected components of $X = F\backslash G$ are of finite type, irreducible, and all of the same dimension. To evaluate this dimension, one may suppose $A = k$ and k algebraically closed. By I, 2.3.3.1, the stabilizer of the k-point $p(e)$ is represented by the fiber $H = p^{-1}(p(e))$, and since $F\backslash G$ is the quotient of G by F in the category of ringed spaces, this fiber has $u(F)$ as underlying space, and since $Ker(u)$ is finite, one therefore has $\dim H = \dim u(F) = \dim F$. By 2.5.4 (ii), one obtains therefore that $\dim X = \dim G - \dim F$. This proves point (ii′) of Theorem 3.2 (and hence also point (iv) of 3.3.2).

[444]N.D.E.: We have added the numbering 4.7.1 below to make the stated result explicit.
[445]N.D.E.: We have changed the title of this section (called "Compléments" in the original).
[446]N.D.E.: We have added what follows.

5.2.

When the homomorphism of A-groups $u : F \to G$ is a monomorphism, one can recover 5.1 by using the results of Exposé IV. Indeed, the canonical morphism $p : G \to F\backslash G$ is faithfully flat and open by 3.2; it is therefore covering for the (fpqc) topology (IV 6.3.1), and one may apply corollaries IV.5.2.2 and IV.5.2.4.

In particular, if we assume, in addition to the hypotheses of 3.2, that u is the inclusion in G of a normal subgroup F, there exists on $F\backslash G$ one and only one A-group structure such that the canonical morphism $p : G \to F\backslash G$ is a homomorphism of A-groups.[447] This proves point (v) of 3.3.2.

5.3.

We shall now review some statements from Exposé IV.

5.3.1. Statements IV 5.2.7 and IV 5.3.1 translate as follows. Let F and G be two groups locally of finite type and flat over A, F being a closed normal subgroup of G. The maps $H \mapsto F\backslash H$ and $H' \mapsto H' \times_{(F\backslash G)} G$ define a bijective correspondence between flat A-subgroups of G containing F and flat A-subgroups of $F\backslash G$. In this bijection, closed (resp. normal) subgroups of G containing F correspond to closed (resp. normal) subgroups of $F\backslash G$.[448]

5.3.2. Proposition IV 5.2.9 implies the following result. Let F, H and G be groups locally of finite type and flat over A; assume $F \subset H \subset G$, with F closed in G and normal in H. Under these conditions, $F\backslash H$ acts freely on the left on $F\backslash G$, the quotient scheme (F\backslash H)\(F\backslash G) exists, and there is a canonical isomorphism of schemes with operator group G:

$$(F\backslash H)\ (F\backslash G) = H\backslash G.$$

5.3.3. From IV 5.2.8, finally, follows the assertion below. Let F, H and G be groups locally of finite type and flat over A; assume that F is contained in, closed and normal

in G, that H is contained in G, and that $F \cap H$ is flat over A. Let $F \times^{\tau}_{A} H$ denote the A-group having the product $F \times_A H$ as underlying scheme, with multiplication defined by the morphism $((x,h),(y,h')) \mapsto (xhyh^{-1}, hh')$; similarly, let $u : H \cap F \to F \times^{\tau}_{A} H$ be the monomorphism $x \mapsto (x^{-1}, x)$, and let $F \cdot H$ be the quotient (F ∩ H)\(F ×^τ_A H). Under these conditions, there is a canonical isomorphism

F\(F · H) = (F ∩ H)\backslash H.

[447] N.D.E.: We have added the sentence that follows.

[448] N.D.E.: In addition to the aforementioned statements of Exp. IV, one uses the fact that, since $G \to F\backslash G$ is faithfully flat, an A-subgroup H of G is flat over A if and only if $F\backslash H$ is.

5.4.

[449] Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type, such that the kernel N of u is flat over A. In this case, by 3.3.2 and 5.2, the quotient A-group $C = N\backslash G$ exists and the morphism $p : G \to C$ is faithfully flat and locally of finite presentation. On the other hand, by IV 5.2.6, u induces a monomorphism $v : C \to H$, which is quasi-compact (because u is, and $G \to C$ is surjective; cf. EGA IV_1, 1.1.3), hence is a closed immersion by 2.5.2. We have therefore obtained the following proposition:

Proposition 5.4.1. *Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type, such that $N = Ker u$ is flat over A. Then one has the factorization:*

$$\begin{tikzcd} G \arrow[rr, "u"] \arrow[dr, "p"'] & & H \\ & N\backslash G \arrow[ur, "i"'] & \end{tikzcd}$$

where p is faithfully flat, locally of finite presentation, and i a closed immersion.

Suppose in addition G flat over A. Then, by 3.3.2, $C = N\backslash G$ is flat over A, and so the quotient $X = C\backslash H$ exists in (Sch/A) and represents the (fppf) quotient sheaf $\tilde{C}\backslash H$, and $q : H \to X$ is a C-torsor. Consequently, writing $e : \operatorname{Spec} A \to G$ for the unit section of G, v induces an isomorphism of (fppf) sheaves between $\tilde{C}$ and the fibered product of q and of $q \circ e : \operatorname{Spec} A \to X$, which is represented by a closed subscheme of H. Consequently, v is an isomorphism of C onto a closed group subscheme K of G (equal to the stabilizer of the A-point $q \circ e$ of X). (This gives another proof of the fact that every quasi-compact monomorphism $v : C \to H$ between A-groups locally of finite type is a closed immersion, cf. 2.5.2 and VI_B 1.4.2.)

Suppose in addition that C is a normal subgroup of H; in this case, the A-group $\bar{H} = C\backslash H$ is the cokernel in the category of A-groups of the morphism $u : G \to H$, and K is the kernel of the morphism $H \to \bar{H}$. When G and H are abelian A-groups, K is the image of u in the category of abelian A-groups, while $C = (Keru)\backslash G$ is the coimage of u. Taking into account the isomorphism $C \xrightarrow{\sim} K$ just established, one obtains:[450]

Theorem 5.4.2. *Let k be a field. The category of commutative algebraic k-groups is abelian.*

Indeed, when k is a field, $Ker(u)$ is flat over k whatever u.

[451] Let us note that the full subcategory of affine commutative algebraic k-groups

[449] N.D.E.: We have detailed the original in what follows; in particular, we have added Proposition 5.4.1.
[450] N.D.E.: We have added the number 5.4.2 to this theorem.
[451] N.D.E.: We have added what follows.

is *thick*. Indeed, consider an exact sequence of commutative algebraic k-groups:

$$1 \longrightarrow N \longrightarrow G \longrightarrow G/N \longrightarrow 1.$$

If G is affine, it is clear that N is so, and G/N is also so by a theorem of Chevalley, cf. VI_B, 11.17. Conversely, if N and G/N are affine, then so is G, by VI_B, 9.2 (viii). One therefore obtains:

Corollary 5.4.3. *Let k be a field. The category of commutative affine algebraic k-groups is abelian.*

Let us further point out that the category of all commutative affine k-groups (not necessarily of finite type) is abelian; this is deduced from VI_B, 11.17 and 11.18.2 (cf. [DG70], § III.3, 7.4), see also VII_B, 2.4.2 for a proof using formal groups.

5.5.

Let G be a group locally of finite type and flat over an Artinian local ring A. We know (2.3) that the connected component of the origin G^0 is a normal open group subscheme of G, hence also flat over A. Then, by 3.2 and 5.2, $G^0\backslash G$ is an A-group scheme, flat over A. Moreover, since each connected component G^α of G is saturated for the equivalence relation defined by G^0, then $G^0\backslash G$ is the direct sum of the $G^0\backslash G^\alpha$ (cf. 4.3). In particular, the connected component of the origin in $G^0\backslash G$ is none other than $G^0\backslash G^0 \cong \operatorname{Spec} A$, and so $G^0\backslash G \to \operatorname{Spec} A$ is a local isomorphism at the origin. Consequently, $G^0\backslash G$ is étale over $\operatorname{Spec} A$, by VI_B, 1.3.[452] One therefore obtains the following proposition (for point (ii), one refers to [DG70], § II.5, 1.7–1.10):

Proposition 5.5.1. *Let A be an Artinian local ring and G an A-group locally of finite type and flat.*

(i) $G^0\backslash G$ is an étale A-group.

(ii) Consequently, if $A = k$ is an algebraically closed field, $G^0\backslash G$ is a constant k-group, acting in a simply transitive way on the set of connected components of G; hence if G is algebraic, $G^0\backslash G$ is finite.

5.6.

Let k be a perfect field and G a k-group locally of finite type. We have seen (0.2) that G_{red} is then a group subscheme of G. Moreover, the equivalence class of the origin of G under the left action of G_{red} on G is the whole underlying space of G. So, by Theorem 3.2, one obtains:

Proposition 5.6.1. *Let k be a perfect field and G a k-group locally of finite type. Then the k-scheme $G_{red}\backslash G$ is the spectrum of a finite, local k-algebra with residue field k.*

[452] N.D.E.: We have detailed the original in what precedes and have added Proposition 5.5.1 to make this result explicit.

[453] Indeed, by 3.2, $G_{red}\backslash G$ has a single point, with residue field k, and is a k-scheme of finite type; it is therefore the spectrum of a local k-algebra of finite dimension (cf. EGA I, 6.4.4).

Proposition 5.6.2. *Let $u : F \to G$ be a morphism between groups locally of finite type over a perfect field k. The following assertions are equivalent:*

(i) u is flat.

(ii) $u^0 : F^0 \to G^0$ is dominant and the morphism $v : F_{red}\backslash F \to G_{red}\backslash G$ induced by u is flat.

[454] Indeed, consider the following commutative diagram:

$$\begin{tikzcd} F \arrow[r, "p"] \arrow[d, "u"'] & F_{red}\backslash F \arrow[d, "v"] \\ G \arrow[r, "q"] & G_{red}\backslash G, \end{tikzcd}$$

where p and q denote the canonical projections. By 3.2 (iv), p and q are faithfully flat; consequently, if u is flat, then $q \circ u = v \circ p$ is flat, hence so is v.

Conversely, suppose v flat and u^0 dominant. Since u^0 is quasi-compact (F^0 being of finite type over k by 2.4, hence noetherian), it sends the generic point ξ of F^0 to the generic point η of G^0. Let R be the finite local k-algebra of which $G_{red}\backslash G$ is the spectrum, and $\mathfrak{m}$ its maximal ideal. One has local morphisms of local rings: $R \to O_{G,\eta} \to O_{F,\xi}$. Note that one has a cartesian square:

$$\begin{tikzcd} G_{red} \arrow[r] & G \arrow[d, "q"] \\ \mathrm{Spec}(R/\mathfrak{m}) \arrow[r] & \mathrm{Spec}(R) \end{tikzcd}$$

and so $O_{G,\eta}/\mathfrak{m}O_{G,\eta} \cong O_{G_{red},\eta} = \kappa(\eta)$, so that $O_{F,\xi}/\mathfrak{m}O_{F,\xi}$ is flat over $O_{G,\eta}/\mathfrak{m}O_{G,\eta}$.

On the other hand, since q and $v \circ p$ are flat, G and F are flat over R. Consequently, by the local flatness criterion (cf. EGA IV_3, 11.3.10.2), $O_{F,\xi}$ is flat over $O_{G,\eta}$, i.e. u is flat at the point ξ. So, by 2.5.3, u is flat.

6. Complements on k-groups not necessarily of finite type

[455] Let us further point out the following results, which will be useful in the addendum VI_B, § 12. We fix a base field k.

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[453] N.D.E.: We have added the numbering 5.6.1, as well as the proof that follows.

[454] N.D.E.: We have detailed the proof of (ii) ⇒ (i), and have simplified the diagram below.

[455] N.D.E.: We have added the results that follow, taken from [Per75]. Note that Lemma 6.1 can be expressed, in Weil's language, by saying that "every point of G is the product of two generic points".

Lemma 6.1. *Let G be a k-group. For every $x \in G$, there exists a point $u \in G \times G$ such that $\mu(u) = x$ and that the two projections $p_1(u)$ and $p_2(u)$ are maximal points of G.*

Proof. Set $K = \kappa(x)$. Since the projection $G_K \to G$ sends maximal points to maximal points, one is reduced to the case where x is rational. Then left translation λ_x (resp. right translation ρ_x) gives us a morphism $G \to G \times G$, $g \mapsto (\lambda_x(g^{-1}), g)$ (resp. $g \mapsto (g, \rho_x(g^{-1}))$) which induces an isomorphism of G onto $\mu^{-1}(x)$, inverse of p_2 (resp. p_1). So, if u is a maximal point of $\mu^{-1}(x)$, then $p_1(u)$ and $p_2(u)$ are maximal points of G, and so u is suitable.

Corollary 6.2. *Let $f : G \to H$ be a quasi-compact, dominant morphism of k-groups.*

(i) f is surjective.

(ii) If H is reduced, f is faithfully flat.

Proof. Write μ_H (resp. μ_G) for the multiplication of H (resp. G). Let $h \in H$. By 6.1, there exists $u \in H \times H$ such that $\mu_H(u) = h$ and that $\alpha = p_1(u)$ and $\beta = p_2(u)$ are maximal points of H. Since f is quasi-compact and dominant, $f^{-1}(\alpha)$ and $f^{-1}(\beta)$ are non-empty (cf. EGA IV_1, 1.1.5), and so there exists $v \in G \times G$ such that $(f \times f)(v) = u$ (cf. EGA I, 3.5.2). Then $g = \mu_G(v)$ satisfies $f(g) = h$. This shows that f is surjective.

Suppose in addition H reduced. Then $O_{H,\alpha}$ is a field, and we have seen above that $f^{-1}(\alpha) \neq \emptyset$, so f is flat at every point ξ of $f^{-1}(\alpha)$, so f is flat by Lemma 2.5.3.

Recollection 6.3. *Recall (cf. EGA IV_3, 11.10.1) that a morphism of schemes $f : X \to Y$ is said to be* schematically dominant *if it satisfies the following condition: for every open U of Y, if Z is a closed subscheme of U such that the morphism $f^{-1}(U) \to U$ factors through Z, then $Z = U$. When f is quasi-compact and quasi-separated, this is equivalent to saying that the closed image of X by f is Y (cf. loc. cit., 11.10.3 (iv) and EGA I, 9.5.8).*

Proposition 6.4. *Let $f : H \to G$ be a quasi-compact morphism of k-groups. Then the closed image of f is a group subscheme H' of G, and f factors as:*

$$\begin{tikzcd} H \arrow[rr, "f"] \arrow[dr, "f'"'] & & G \\ & H' \arrow[ur, "i"'] & \end{tikzcd}$$

where f' is schematically dominant, quasi-compact and surjective.

Proof. Since H is separated (0.3), f is quasi-compact and separated, so $f_*(O_H)$ is a quasi-coherent O_G-module, and the closed image H' of f exists and is the closed subscheme of G defined by the quasi-coherent ideal $I = Ker(O_G \to f_*(O_H))$ (cf. EGA I, § 9.5).

Write c_G and μ_G (resp. c_H and μ_H) for the inversion and multiplication morphisms of G (resp. H). Then $c_G \circ f = f \circ c_H$ factors through the closed subscheme $c_G(H')$, whence $H' \subset c_G(H')$ and so $H' = c_G(H')$ (since $c_G^2 = id_G$). Similarly, since $f \circ \pi_H = \pi_G \circ (f \times f)$ factors through H', then $f \times f$ factors through the closed subscheme $\pi_G^{-1}(H')$ of $G \times G$. On the other hand, since the formation of the closed image commutes with flat base changes (EGA III 1.4.15 and IV_1 1.7.21), the closed image of $f \times id_H$ (resp. $id_{H'} \times f$) is $H' \times H$ (resp. $H' \times H'$). So, by "transitivity of closed images" (EGA I, 9.5.5), the closed image of $f \times f$ is $H' \times H'$, which is therefore contained in $\pi_G^{-1}(H')$, i.e. the restriction of π_G to $H' \times H'$ factors through H'. This shows that H' is a closed group subscheme of G. Write i for the inclusion $H' \hookrightarrow G$.

Then f equals $i \circ f'$, where $f' : H \to H'$ is schematically dominant and quasi-compact (since f is quasi-compact and i is separated). So, by 6.2, f' is surjective. This proves 6.4.

One can now state the following theorem ([Per75], V 3.1 & 3.2, see also [Per76], 0.0 & 0.1).

Theorem 6.5 (D. Perrin). *Let G be a quasi-compact k-group. Then*

(i) G is the projective limit of a filtered system (G_i) of k-groups of finite type (whose transition morphisms $u_{ij} : G_j \to G_i$ are affine for i large enough), and the morphisms $G \to G_i$ are faithfully flat (and affine for i large enough).

(ii) Let H be a closed k-subgroup of G. Then the (fpqc) quotient sheaf $\tilde{G}/H$ is a k-scheme in the following two cases:

(1) The immersion $H \to G$ is of finite presentation; in this case, G/H is of finite type over k. (2) H is normal in G.

For the proof of this theorem (which rests on several intermediate theorems), we refer to [Per75]. For the reader's convenience, let us however prove the two corollaries below, cf. [Per75] V 3.3 to 3.4 or [Per76] 4.2.3 to 4.2.5.

Corollary 6.6. *Let $f : G \to H$ be a quasi-compact morphism of k-groups.*

(i) If f is schematically dominant, it is faithfully flat.

(ii) This is the case, in particular, if H is affine and the morphism $f^{\sharp} : O(H) \to O(G)$ is injective.

Proof. Suppose f schematically dominant. Then, by 6.2 (i), f is surjective, so, by 2.5.3, it suffices to show that f is flat at the unit element of G. Since $O_{H,e} = O_{H^0,e}$ (cf. 2.6.5), one may replace H by H^0, and hence assume H irreducible. Then H is quasi-compact (loc. cit.), and since f is quasi-compact, so is G. By 6.5 (i), $H = \lim_i H_i$, where each H_i is an algebraic k-group. Denote by N_i the kernel of $G \to H_i$; this is a closed normal k-subgroup of G. Moreover, since the unit section $e \to H_i$ is of finite presentation, the immersion $N_i \to G$ is

also so; so, by 6.5 (ii), the (fpqc) quotient G/N_i is an algebraic k-group. One has then a commutative diagram

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[d, "q_i"'] & H \arrow[d, "p_i"] \\ G/N_i \arrow[r, "f_i"] & H_i \end{tikzcd}$$

where f is schematically dominant, p_i, q_i are faithfully flat (hence schematically dominant). Then $f_i \circ q_i = p_i \circ f$ is schematically dominant, and so f_i is also so. On the other hand, f_i is a monomorphism, hence a closed immersion, since G/N_i and H_i are algebraic (2.5.2). It follows that f_i is an isomorphism, and so $G \to H_i$ is faithfully flat. Then, by [BAC] I § 2.7, Prop. 9, the morphism $G \to H$ is flat; on the other hand, it is surjective by 6.2 (i), so it is faithfully flat. This proves point (i); and point (ii) follows, since if H is affine and $f^{\sharp}$ injective, then the closed image of f equals H, so f is schematically dominant (cf. 6.4).

Corollary 6.7. *Let $u : G \to H$ be a morphism of k-groups and $N = Ker(u)$. Suppose u quasi-compact.*

(i) The (fpqc) quotient sheaf $\tilde{G}/N$ is represented by a k-group scheme G/N, and u factors as:

$$\begin{tikzcd} G \arrow[rr, "u"] \arrow[dr, "p"'] & & H \\ & G/N \arrow[ur, "i"'] & \end{tikzcd}$$

where p is faithfully flat and i a closed immersion.

(ii) In particular, if u is a monomorphism, u is a closed immersion, and if u is schematically dominant, u is faithfully flat.

Proof. (i) By 6.4 and 6.6, the closed image G' of u is a closed group subscheme of G, and $p : G \to G'$ is faithfully flat and quasi-compact. One evidently has $Ker(p) = Ker(u) = N$, and so by Exp. IV, 3.3.2.1 and 5.1.7, G' represents the (fpqc) quotient sheaf $\tilde{G}/N$.

(ii) The second assertion is contained in 6.6; let us show the first. If u is a monomorphism, so is p; then p is at once a monomorphism and an effective epimorphism, hence an isomorphism (cf. IV, 1.14). This proves 6.7.

Let us finally point out the following corollaries (cf. [Per76], 4.2.6 to 4.2.8).

Corollary 6.8. *The category of commutative quasi-compact k-group schemes is abelian.*

Taking 6.7 into account, the proof is analogous to that of 5.4.2.

Corollary 6.9. *If $char(k) = 0$, every k-group scheme G is geometrically reduced.*

Indeed, if $g \in G$ and if K is an algebraic closure of $\kappa(g)$, one has $\mathcal{O}_{G,e} \otimes_k K \simeq \mathcal{O}_{G_K, g_K}$, so it suffices to show that $\mathcal{O}_{G,e} = \mathcal{O}_{G^0,e}$ is geometrically reduced. One is thus reduced to the case where G is connected, hence quasi-compact (2.6.5). Then the result follows from 6.5 and from Cartier's theorem for algebraic groups (cf. VI_B, 1.6.1 or [DG70] § II.6, Th. 1.1).

Corollary 6.10. *Let G be a quasi-compact k-group. Assume k algebraically closed.*

(i) Let $f : G \to H$ be a faithfully flat morphism of k-groups. Then the induced map $G(k) \to H(k)$ is surjective.

(ii) The set of rational points is dense in G.

For the proof, we refer to [Per75], V 3.7 & 3.9.

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Footnotes

Exposé VI_B. Generalities on group schemes

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[^N.D.E-VI_B-0] Version of 13/10/2024.

[456] N.D.E.: Additional references cited in this Exposé.

This Exposé, which corresponds to no oral lecture of the seminar, is intended to bring together a number of technical results, commonly used, concerning group schemes.[457]

1. Morphisms of groups locally of finite type over a field

1.1.

Let A be an Artinian local ring, G and H two A-groups[458], and $u : G \to H$ a morphism of A-groups. Then u induces a morphism of groups $u(A) : G(A) \to H(A)$.[459] Since $H(A)$ acts on H by right translation, $u(A)$ defines, by restriction, an action of $G(A)$ on H. This action is compatible with the morphism u and with the action of $G(A)$ on G defined by right translation. Since $G(A)$ acts transitively on the strictly rational points of G (see (VI_A, 0.4) for the definition), one sees that these points "all behave in the same way with respect to u"; from this spring the following properties.

Proposition 1.2. *Let $u : G \to H$ be a quasi-compact morphism between A-groups locally of finite type over A. Then the set $u(G)$ is closed in H and one has* $\dim G = \dim u(G) + \dim \mathrm{Ker}\, u$.[460]

Since u commutes with the inversion morphisms of G and H, the image $u(G)$ is invariant under the inversion morphism of H; the same is therefore true of its closure $u(G)^-$ in H. On the other hand, let L denote the set of points of $H \times_A H$ whose two projections both lie in $u(G)$; it is clear that L is the image of the morphism $u \times_A u : G \times_A G \to H \times_A H$. Hence the multiplication morphism of H sends L into $u(G)$, in other words $u(G) \cdot u(G) = u(G)$. On the other hand, Lemma 1.2.1 below shows that L^-, the closure of L in $H \times_A H$, is the set of points of $H \times_A H$ whose two projections lie in $u(G)^-$; hence $u(G)^- \cdot u(G)^- = u(G)^-$, so that the reduced subscheme of H whose underlying space is $u(G)^-$ is naturally equipped with a group structure in the category $(Sch/k)_{red}$, where k is the residue field of A (cf. (VI_A, 0.2)).

Let us prove the first assertion of 1.2. Replacing A by the algebraic closure of its residue field k, we may assume that A is an algebraically closed field k (cf. EGA IV_2, 2.3.12). Replacing u by $u_{red} : G_{red} \to H_{red}$, we may assume G and H reduced; in this case, as we have just seen, $u(G)$ is the underlying space of a reduced group subscheme of G; we may therefore assume u dominant. Then $G(k)$ acts transitively on the set of connected components of H, and it suffices to show that $u(G) \cap H^0$ is closed: we are reduced to the case where H is connected, hence irreducible and of finite type (VI_A, 2.4). Then u is of finite type, since it is quasi-compact and locally of finite type; since H is noetherian, $u(G)$ is constructible (EGA IV_1, 1.8.5), so it contains an open subset V of H (EGA 0_III, 9.2.2), and then, by (VI_A, 0.5), we have $H = V \cdot V \subset u(G) \cdot u(G) = u(G)$.

[457] This Exposé has been seriously reworked since its mimeographed edition; in particular, §§ 10 and 11 have been entirely rewritten.[^N.D.E-VI_B-1]

[458] Recall the convention adopted at the beginning of VI_A: an A-group is an A-group scheme (and such a scheme is separated, cf. (VI_A, 0.3)).

[459] We have replaced "homomorphism" by "morphism".

[460] This statement also appears in (VI_A, 2.5.2).

Let us prove the second assertion. Recall first that the functor `Ker u` (cf. I, 2.3.6.1) is representable by $u^{-1}(e)$, where e denotes the unit element of H. When u is locally of finite type, `Ker u` is therefore locally of finite type over A. We reduce as before, this time using EGA IV$_2$, 4.1.4, to the case where A is an algebraically closed field k.

We may further assume G and H irreducible and of finite type and u dominant: indeed, k being algebraically closed, it is clear that the connected components of G, on the set of which $G(k)$ acts transitively, all have the same dimension, and that if u^0 denotes the restriction of u to G^0, then $(Keru)^0 \subset Keru^0$, and $\dim \operatorname{Ker} u^0 = \dim \operatorname{Ker} u$. One sees likewise that u is then of finite type over k. If η denotes the generic point of H, one has $\dim u^{-1}(\eta) = \dim G - \dim H$ (EGA IV$_3$, 10.6.1 (ii)). By EGA IV$_3$, 9.2.3 and 9.2.6, the set of $y \in H$ such that $\dim u^{-1}(y) = \dim u^{-1}(\eta)$ contains a non-empty open subset V. Since u is dominant, $U = u^{-1}(V)$ is then a non-empty open subset of G and contains a closed point x of G, since G is a Jacobson scheme (EGA IV$_3$, 10.4.7). Then right translation r_x is an isomorphism of `Ker u` onto $u^{-1}(u(x))$, so that:

$$\dim \operatorname{Ker} u = \dim u^{-1}(u(x)) = \dim u^{-1}(\eta) = \dim G - \dim H.$$

Lemma 1.2.1. *Let $f : X' \to X$ and $g : Y' \to Y$ be two quasi-compact, dominant morphisms of schemes over an Artinian local ring A. Then $f \times_A g : X' \times_A Y' \to X \times_A Y$ is dominant (and quasi-compact).*

Indeed, one has $f \times_A g = (id_X \times_A g) \circ (f \times_A id_{Y'})$. It therefore suffices to show that $f \times_A id_{Y'}$ and $id_X \times_A g$ are dominant. For this one may replace A by its residue field k. In this case, X and Y' are flat over $A = k$, and since $f \times_A id_{Y'}$ (resp. $id_X \times_A g$) is deduced from f (resp. g) by the flat base change $Y' \to A$ (resp. $X \to A$), it is dominant (and quasi-compact), by EGA IV$_2$, 2.3.7.

Counterexample 1.2.2. *Let k be a field of characteristic 0, G the constant k-group $\mathbb{Z}$, and H the additive k-group $G_{a,k}$. Let $u : G \to H$ be a morphism of k-groups. If $u \neq 0$, then $u(G)$ is not closed in H.*

Proposition 1.3. [461] *Let A be an Artinian local ring, k its residue field, G an A-group locally of finite type and flat, $u : G \to H$ a morphism of A-groups. The following assertions are equivalent:*

(i) u is flat (resp. quasi-finite, resp. unramified, resp. smooth, resp. étale) at some point of G.[462]

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[461] In the original, 1.3, 1.3.1, and 1.3.1.1 are stated for A a field. For completeness, we have treated the case of an Artinian local ring; to do so, we have added Lemma 1.3.0.

[462] Recall that a morphism of schemes $f : X \to Y$ is said to be *of finite type* (resp. *of finite presentation*, resp. *quasi-finite*) *at* x if there exist affine opens V containing $f(x)$ and U containing x such that $f(U) \subset V$ and $O_X(U)$ is an $O_Y(V)$-algebra of finite type (resp. of finite presentation, resp. and moreover the fiber $f^{-1}(f(x'))$ is finite for all $x' \in U$; see also N.D.E. (40) of Exp. V). One says that f is *locally quasi-finite* (resp. *of finite type*, *of finite presentation*) if it has this property at every point x. On the other hand, one says that f is *smooth* (resp. *unramified*, resp. *étale*) *at* x if there exists an open neighborhood U of x such that the morphism $f|_U : U \to Y$ is smooth (resp. unramified, resp. étale). In view of these definitions, it is clear that the set of points where f is of finite presentation, resp. of finite type, quasi-finite, smooth, unramified, étale, is open in X. Moreover, since the flat locus of a

(ii) u is flat (resp. quasi-finite, resp. unramified, resp. smooth, resp. étale).

Proof. It suffices to show that (i) implies (ii). First, we have the following lemma.

Lemma 1.3.0. *Let $A \to B \to C$ be morphisms of commutative rings and let n be a nilpotent ideal of A. Suppose that C/nC is a (B/nB)-algebra of finite type.*

(i) Then C is a B-algebra of finite type.

(ii) Moreover, if C is flat over A and if C/nC is a (B/nB)-algebra of finite presentation, then C is a B-algebra of finite presentation.

Indeed, let $x_1, \cdots, x_n$ be elements of C whose images generate C/nC as a (B/n)-algebra. By the nilpotent Nakayama lemma, the x_i generate C as a B-algebra. This proves (i). Let ϕ be the resulting surjective morphism $B[X_1, \cdots, X_n] \to C$, and let $I = Ker(\phi)$.

Suppose now that C is flat over A and that C/nC is of finite presentation over B/nB. Then, on the one hand, I/nI is identified with the kernel of $\bar{\phi} = \phi \otimes_A (A/n)$. On the other hand, by EGA IV$_1$, 1.4.4, the kernel of $\bar{\phi}$ is an ideal of finite type. Let then $P_1, \cdots, P_s$ be elements of I whose images generate I/nI as an ideal; by the nilpotent Nakayama lemma, they generate I. This proves (ii).

Let us return to the proof of 1.3. Let x be an arbitrary point of G. Since G is flat over A, the fiber-wise flatness criterion in the form of EGA IV$_3$, 11.3.10.2, shows that u is flat at x if $u \otimes_A k$ is. Likewise, by the preceding lemma, one sees that u is of finite type (resp. of finite presentation) at x if $u \otimes_A k$ is. Since the other properties are then verified on fibers (cf. EGA IV$_4$, 17.4.1, 17.5.1, and 17.6.1, for unramified, smooth, and étale), we are reduced to the case $A = k$.

Let now x be a point of G where one of the conditions of 1.3 (i) holds. Since the properties under consideration are preserved by (fpqc) descent (cf. EGA IV, 2.5.1, 2.7.1, and 17.7.1), one reduces, by replacing k by an algebraic closure of $\kappa(x)$, to the case where k is algebraically closed and $x \in G(k)$.

Since G is a Jacobson scheme (cf. EGA IV$_3$, 10.4.7) and since the set W of points of G where u is flat (resp. quasi-finite, unramified, smooth, or étale) is stable under generization[463] (resp. open), it suffices to show that every point $y \in G(k)$ belongs to W. Now, for such a point y, the translation $r_y \circ r_x^{-1}$ sends x to y, hence u has the desired property at y, i.e. $y \in W$. ∎

Corollary 1.3.1. *Let A be an Artinian local ring, k its residue field, G a flat A-group. The following assertions are equivalent:*[464]

morphism locally of finite presentation is open (EGA IV$_3$, 11.3.1), the set of points of X where f is of finite presentation and flat is also open in X. All of this will be used repeatedly in what follows.

[463]The original stated that W_{flat} is open, which does not seem a priori obvious…

[464]Note that it does not suffice to assume that G is flat over A at one point: for instance, suppose $A \neq k$, let H be the constant A-group $(\mathbb{Z}/2\mathbb{Z})_A$ and G the closed A-subgroup of H whose non-neutral component is reduced; then the structural morphism $G \to \operatorname{Spec} A$ is a local isomorphism at the unit point e, but is not flat at the point $g \neq e$.

(i) G is locally quasi-finite (resp. unramified, resp. smooth, resp. étale) over A at some point.

(ii) G is locally quasi-finite (resp. unramified, resp. smooth, resp. étale) over A.

[465] *Proof.* Indeed, if G satisfies one of the conditions of (i) at a point x, there exists an open neighborhood U of x which is of finite type over A. Consequently, it suffices to apply 1.3 in the case where H is the unit A-group, taking into account the following lemma. ∎

Lemma 1.3.1.1. *Let A be an Artinian local ring and G an A-group. If there exists a non-empty open subset of G of finite type over A, then G is locally of finite type over A.*

By Lemma 1.3.0, we may assume A equal to its residue field k. Moreover, by (fpqc) descent, we may assume k algebraically closed (cf. EGA IV_2, 2.7.1). Let V be the open subset of G formed by the points where G is of finite type over k; by hypothesis, $V \neq \emptyset$. Since G is a Jacobson scheme, V contains a closed point x and, to show that $V = G$, it suffices to show that every closed point y of G belongs to V. Now, for such a point y, the translation $r_y \circ r_x^{-1}$ sends x to y, whence $y \in V$. ∎

Corollary 1.3.2. *Let A be an Artinian local ring, $u : G \to H$ a morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) u is universally open,

(ii) u is open,

(iii) u is open at some point of G,

(iv) the morphism $u^0 : G^0 \to H^0$ deduced from u is dominant,

(iv′) u^0 is surjective,

(v) there exists a connected component C of G such that, if D denotes the connected component of H containing $u(C)$, the morphism $u' : C \to D$ deduced from u is dominant.

Proof. The implications (i) ⇒ (ii) ⇒ (iii) and (iv′) ⇒ (iv) ⇒ (v) are clear. Since G^0 is of finite type over A (VI_A, 2.4), and hence noetherian, u^0 is quasi-compact, so $u^0(G^0)$ is closed in H^0 by 1.2; hence (iv) ⇒ (iv′). On the other hand, since G^0 (resp. C) is open in G (VI_A, 2.3) and H^0 (resp. D) is irreducible (VI_A, 2.4.1), one sees that (ii) implies (iv) (resp. (iii) implies (v)). It remains to show that (v) implies (i).

The open subset C (resp. D) of G (resp. H) will be endowed with its induced scheme structure, and u' will denote the morphism $C \to D$ deduced from u. Let k be the residue field of A.[466] Since the base change $\operatorname{Spec} k \to \operatorname{Spec} A$ is a universal homeomorphism, we may assume $A = k$. By hypothesis, u' is dominant and, since C is of finite type over k (VI_A, 2.4.1) and hence noetherian, u' is quasi-compact. By

[465] We have added the following sentence, and simplified the proof of Lemma 1.3.1.1, by invoking EGA IV, 2.7.1 instead of loc. cit., 17.7.5.

[466] We have expanded the original in what follows.

EGA IV$_2$, 2.3.7, $u' \otimes_k \bar{k}$ is again quasi-compact and dominant, where $\bar{k}$ denotes an algebraic closure of k. Then, since $C \otimes_k \bar{k}$ is a union of connected components of $G \otimes_k \bar{k}$, the morphism $u \otimes_k \bar{k} : G \otimes_k \bar{k} \to H \otimes_k \bar{k}$ satisfies assertion (v). We are thus reduced to the case where $A = k$ is an algebraically closed field, taking into account EGA IV$_2$, 2.6.4.

In this case, we may further replace u by u_{red}, and we are reduced to the case where H is reduced. Let then ξ (resp. η) be the generic point of C (resp. D). Since u' is quasi-compact and dominant, $u'(\xi) = \eta$ (cf. EGA IV$_1$, 1.1.5). On the other hand, since H is reduced, the local ring $O_{H,\eta}$ is a field, hence u' is flat at the point ξ.[467] Hence, by 1.3, u is flat; moreover, since u is locally of finite type and H is locally noetherian, u is locally of finite presentation. Therefore, by EGA IV$_2$, 2.4.6, u is universally open. ∎

Proposition 1.4. *Let A be an Artinian local ring, and $u : G \to H$ a quasi-compact morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) u is proper,

(ii) there exists $h \in H$ such that the fiber $u^{-1}(h)$ is non-empty and proper over $\kappa(h)$,

(iii) $\mathrm{Ker}\ u$ is proper over A.

Proof. It is clear that (i) implies (iii), and that (iii) implies (ii). On the other hand, it follows from the hypotheses that u is of finite type and, since G is separated (VI_A, 0.3), so is u (EGA I, 5.5.1). It therefore remains to show that assertion (ii) implies that u is universally closed, so that we may assume A equal to its residue field k.[468] Let k' be an algebraic closure of $\kappa(h)$, $u' : G' \to H'$ the morphism deduced from u by base change, and h' a point of H' above h; then the fiber $u'^{-1}(h') = u^{-1}(h) \times_{\kappa(h)} k'$ is non-empty and proper, and it suffices to show that u' is proper (EGA IV$_2$, 2.6.4). We may therefore assume that k is algebraically closed and $h \in H(k)$.

We have seen (1.2) that $u(G)$ is then the underlying set of a closed reduced group subscheme of H; since every closed immersion is proper (EGA II, 5.4.2), we may assume that u is surjective and that H is reduced. Since u is surjective, the group $G(k)$ acts transitively on the set of closed points of H; whatever the closed point y of H, $u^{-1}(y)$ is therefore proper over $\kappa(y)$. By EGA IV$_3$, 9.6.1, the set of $y \in H$ such that $u^{-1}(y)$ is not proper over $\kappa(y)$ is locally constructible; since it contains no closed point, it is empty (cf. EGA IV$_3$, 10.3.1 and 10.4.7).

[469] Consider now the generic point η of H^0; by what precedes, the fiber $u^{-1}(\eta) = G \times_H \operatorname{Spec} \kappa(\eta)$ is proper over $\kappa(\eta)$. On the other hand, since H is reduced, $\kappa(\eta)$ equals $O_{H,\eta}$. Since $O_{H,\eta}$ is the direct limit of the rings $O_H(V)$, as V runs through the non-empty open subsets of H^0, it follows from EGA IV$_3$, 8.1.2 a) and 8.10.5 (xii), that there exists a non-empty open subset V of H^0 such that the restriction of u

[467] The original invoked the generic flatness theorem (EGA IV$_2$, 6.9.1), which is not necessary here.
[468] We have added the following sentence.
[469] We have expanded the original in what follows.

over V is proper. It is then clear that the $g \cdot V$, for $g \in G(k)$, form an open cover of H such that, for every $g \in G(k)$, the restriction of u over the open $g \cdot V$ is proper; one deduces that u is proper (cf. EGA II, 5.4.1). ∎

Corollary 1.4.1. *Let A be an Artinian local ring, and $u : G \to H$ a morphism between A-groups locally of finite type. The following assertions are equivalent:*

(i) u is locally quasi-finite,

(ii) u is quasi-finite at some point,

(iii) $\mathrm{Ker}\ u$ is discrete,

(iv) the restriction of u to each connected component of G is finite.

If in addition u is quasi-compact, these assertions are equivalent to the following:

(v) u is finite.

Proof. It is clear that (iv) implies (iii), that (iii) implies (ii) (EGA I, 6.4.4), and that, in the case where u is quasi-compact, assertions (iv) and (v) are equivalent. We have already seen in 1.3 that (i) and (ii) are equivalent.

Let us show finally that (i) implies (iv). Let C be a connected component of G; since C is of finite type over A (VI_A, 2.4.1) and G and H are separated (VI_A, 0.2), by EGA I, 5.5.1 and 6.3.4, the restriction u' of u to C is separated and of finite type.[470] Since the fibers of u' are discrete, it follows that u' is quasi-finite (cf. EGA II, 6.2.2). Since every quasi-finite proper morphism is finite (cf. EGA III_1, 4.4.2), it therefore suffices to show that u' is universally closed.

For this, we may assume that A is equal to its residue field k. Then, by (fpqc) descent (cf. EGA IV_2, 2.6.4), it suffices to show that $u' \otimes_k \bar{k}$ is universally closed, where $\bar{k}$ denotes an algebraic closure of k. Moreover, since C is of finite type over k, $C \otimes_k \bar{k}$ is the sum of finitely many connected components $C'_1, \cdots, C'_n$ of $G \otimes_k \bar{k}$, and it suffices to show the assertion for each C'_i. One is thus reduced to the case where k is algebraically closed.

Let then g be a closed point of C. If $u^0 : G^0 \to H$ is the restriction of u to G^0, one has $u' = r_{u(g)} \circ u^0 \circ r_g^{-1}$, where r_g denotes right translation by g. Therefore, to show that u' is proper, it suffices to show that u^0 is. By hypothesis, u is locally quasi-finite, so the fiber $\mathrm{Ker}\ u$ is discrete (and non-empty); we have seen that u^0 is of finite type, so the fiber $Keru^0$ is finite (cf. EGA II, 6.2.2), hence proper and non-empty. Therefore u^0 is proper by 1.4. ∎

Corollary 1.4.2. *Let A be an Artinian local ring, and $u : G \to H$ a quasi-compact morphism between A-groups locally of finite type. The following assertions are equivalent:*[471]

[470] We have expanded the original in what follows.

[471] The hypothesis that G and H are locally of finite type may be removed, since according to [Per76, 4.2.4]: every quasi-compact monomorphism $u : G \to H$ between group schemes over a field k is a closed immersion; see also [DG70, III.3.7.2 b)] for the case where G and H are affine (in which case every morphism $G \to H$ is affine (EGA II, 1.6.2 (v)), hence quasi-compact).

(i) u is a closed immersion,

(ii) u is a monomorphism,

(iii) `Ker u` *is trivial, i.e. isomorphic to the unit k-group.*

In particular, every group subscheme[472] *of H is closed.*

Proof. It is clear that (i) implies (ii), and if one considers the functors represented respectively by G and H, it is immediate that conditions (ii) and (iii) are equivalent. Finally, if `Ker u` is the unit k-group, `Ker u` is a proper non-empty fiber, so u is a proper monomorphism by 1.4, of finite presentation since H is locally noetherian (EGA IV_1, 1.6.1), and hence a closed immersion (EGA IV_3, 8.11.5).

The last assertion follows from the fact that, since H is locally noetherian, every immersion $G \to H$ is quasi-compact (EGA I, 6.6.4). ∎

Counterexample 1.4.3. *Let k be a field of characteristic 0, G the constant k-group $\mathbb{Z}$, and H the k-group $G_{a,k}$. Let $u : G \to H$ be a morphism of k-groups. If $u \neq 0$, then $Ker u = 0$, but u is not a closed immersion (cf. 1.2.2).*

We shall use later the two following results, which should have appeared in Exposé VI_A:

Lemma 1.5. *Let k be a field and G a k-group locally of finite type. Then:*

(i) All irreducible components of G have the same dimension.

(ii) For every $g \in G$, one has $\dim_g G = \dim G$.

[473] Assertion (i) has already been proved in (VI_A, 2.4.1), and assertion (ii) follows from it. Indeed, let g be a point of G and C the connected component of G containing g. By definition (EGA 0_IV, 14.1.2), $\dim_g G$ is the infimum of the integers $\dim U$, as U runs through the open neighborhoods of g; one therefore has $\dim_g G = \dim U_0$ for some `U_0`, which one may assume contained in C (since $\dim V \leq \dim U$ if $V \subset U$). Then, since C is irreducible and of finite type over k (VI_A, 2.4.1), one has `dim U_0 = dim C = tr.deg_k κ(ξ)`, where ξ is the generic point of C, by EGA IV_2, 5.2.1. Hence `dim_g G = dim C = dim G`. ∎

Proposition 1.6. [474] *Let S be a scheme of characteristic zero and G an S-group scheme, locally of finite presentation over S at every point of the unit section $\epsilon(S)$. For G to be smooth over S at every point of the unit section, it is necessary and sufficient that the* `O_S`*-module $\omega_{G/S} = \epsilon * (\Omega^1_{G/S})$ (called the conormal module to the unit section of G) be locally free.*

[472] In this particular case, see also (VI_A, 0.5.2), valid without finiteness hypotheses.

[473] The implication (ii) ⇒ (i) is a general fact (cf. EGA 0_IV, 14.1.6), and (i) ⇒ (ii) follows from the fact that if X is an irreducible scheme of finite type over a field, one has $\dim X = \dim U$ for every non-empty open subset U of X; the essential point here is therefore to establish assertion (i), which has already been done in an addition to (VI_A, 2.4.1). We have modified the statement and proof of the lemma accordingly.

[474] In the statement, we have replaced "along the unit section" by "at every point of the unit section"; on the other hand, at the end of the proof, we have made explicit the results of EGA IV_4 cited in reference.

Recall that a scheme S is said to be *of characteristic zero* if for every closed point x of S, the field $\kappa(x)$ has characteristic zero.

Recall also (II 4.11) that, if π denotes the structural morphism $G \to S$, one has $\Omega^1_{G/S} = \pi * (\omega_{G/S})$, so that it comes to the same thing to say that the O_S-module $\omega_{G/S}$ is locally free, or that the O_G-module $\Omega^1_{G/S}$ is locally free.

If there exists an open neighborhood U of $\epsilon(S)$ which is smooth over S, then, by EGA IV$_4$, 17.2.3, $\Omega^1_{U/S}$ is locally free of finite type, as well as $\omega_{G/S} = \epsilon * (\Omega^1_{U/S})$.

Conversely, if $\omega_{G/S}$ is locally free, the same holds for $\Omega^1_{G/S} = \pi * (\omega_{G/S})$. Since S is of characteristic 0, the Jacobian criterion (EGA IV$_4$, 16.12.2) therefore implies that G is differentially smooth over S. Then it follows from EGA IV$_4$, 17.12.5, that G is smooth over S at every point of the unit section. ∎

Corollary 1.6.1 (Cartier). *Given a field k of characteristic zero, every k-group locally of finite type over k is smooth over k.*

Indeed, it is then clear that the k-module $\omega_{G/k}$ is locally free, so, by 1.6, G is smooth over k at the unit point e, and hence smooth over k by 1.3.1. ∎

2. "Open properties" of groups and group morphisms locally of finite presentation

2.0.

In all that follows, S will denote an arbitrary scheme; an S-group scheme will be called an *S-group*. Given an S-group G, we shall denote by ϵ the unit section, by $c : G \to G$ the inversion morphism, and by μ the multiplication morphism $G \times_S G \to G$. For every S-scheme X, we shall denote by π or π_X the structural morphism $X \to S$.

Given a property $P(u)$ of a morphism of S-schemes $u : X \to Y$, we shall say that $P(u)$ is *stable under base change* if, whenever u satisfies $P(u)$, so does the morphism $u_{(Y')}$, for any S-morphism $Y' \to Y$. We say that $P(u)$ is *local in nature for the topology* T (cf. Exp. IV, §§ 4 and 6) if $P(u)$ satisfies the following two conditions:

a) $P(u)$ is stable under base change,

b) whenever there exists a covering family of S-morphisms $Y_i \to Y$ for the topology T such that each morphism $u_{(Y_i)}$ satisfies $P(u)$, then u satisfies $P(u)$.

Proposition 2.1. *Let $P(u)$ be a property of a morphism of S-schemes, local in nature for the (fpqc) topology. Let $u : G \to H$ be a morphism of S-groups. Assume G flat and universally open over S.*

Let W be the largest open subset of H above which u satisfies the property $P(u)$, and let $V = u^{-1}(W)$. Then $U = \pi_G(V)$ is open in S and V is an open group subscheme of $G|_U$.

The existence of a largest open subset W of H above which u satisfies $P(u)$ follows from the fact that $P(u)$ is local in nature for the Zariski topology. Since π_G is universally open, $\pi_G(V)$ is an open subset U of S. It suffices to show that V is a group subscheme of $G|_U$. We may therefore assume $U = S$.

Set then $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$, $W' = W \times_S V$, and $u' = u_{(V)}$; let W_1' be the largest open subset of H' above which u' satisfies $P(u)$; since V is flat and universally open over S, so is H' over H, and Lemma 2.1.1 below shows that $W_1' = W'$. Now consider the automorphism of V-schemes a (resp. b) of G' (resp. H'), namely right translation by the inverse of the diagonal section δ (resp. by the inverse of $u(\delta)$), defined by

$$a(g, v) = (g \cdot v^{-1}, v), \qquad (\text{resp. } b(h, v) = (h \cdot u(v^{-1}), v)),$$

for any morphism $T \to S$, $g \in G(T)$, $v \in V(T)$, and $h \in H(T)$. It is clear that $u' \circ a = b \circ u'$, which shows that W' is stable under b, hence V' is stable under a, so that V is a group subscheme of G. $\square$

Lemma 2.1.1. *Let $P(u)$ be a property of an S-morphism u, local in nature for the (fpqc) topology. Consider a cartesian square of morphisms of S-schemes:*

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d] & Y' \arrow[d, "g"] \\ X \arrow[r, "f"'] & Y, \end{tikzcd}$$

where g is flat and open. Let W (resp. W_1') be the largest open subset of Y (resp. Y') above which f (resp. $f' = f_{(Y')}$) satisfies $P(u)$. Then $W_1' = W \times_Y Y'$.

Set $W' = W \times_Y Y'$; since $P(u)$ is stable under base change, one has $W' \subset W_1'$. Since g is open, $W_1 = g(W_1')$ is an open subset of Y. Set $V_1 = f^{-1}(W_1)$ and $V_1' = V_1 \times_{W_1} W_1'$; it is clear that $V_1' = f'^{-1}(W_1')$. Since g is flat and open, the morphism $W_1' \to W_1 = g(W_1')$ deduced from g is faithfully flat and open, hence covering for the (fpqc) topology (cf. IV 6.3.1 (iv)). Since the morphism $V_1' \to W_1'$ deduced from f' satisfies $P(u)$, the same holds for the morphism $V_1 \to W_1$ deduced from f; hence $W_1 \subset W$, and $W_1' \subset g^{-1}(W_1) \subset g^{-1}(W) = W'$; therefore $W' = W_1'$. $\square$

Remark 2.1.2. *A great many properties of a morphism are local in nature for the (fpqc) topology; let us mention the properties of being flat, (universally) open, (locally) of finite type, of finite presentation, quasi-finite (cf. EGA IV_2, 2.5.1, 2.6.1 and 2.7.1), smooth, étale, unramified (EGA IV_4, 17.7.3).*

The proof of 2.1 in fact uses only base changes by flat morphisms; the proposition therefore applies to a property satisfying condition b) of 2.0 relative to the (fpqc) topology, and stable under base change by flat morphisms (for example, that of being quasi-compact and dominant).

Of course, one can state an analogous proposition concerning properties local in nature for a topology T finer than the Zariski topology, the condition to verify on G being then that π_G be universally open and covering for the topology T.

In particular, if G is flat and locally of finite presentation over S, one has an analogous statement for properties stable under base change by flat morphisms locally of finite presentation, satisfying condition b) of 2.0 relative to the (fpqc) topology (e.g., the properties of being regular, reduced, Cohen–Macaulay, etc. (EGA IV_2, 6.8)).

Proposition 2.2. *Let G and H be two S-groups and $u : G \to H$ a morphism of S-groups. Then:*[475]

(i) Suppose G or H flat over S, and G or H locally of finite presentation over S, and let V be the largest open subset of G such that the restriction of u to V is flat and locally of finite presentation (resp. smooth, resp. étale). Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.

(ii) Suppose G or H universally open over S, and let V be the largest open subset of G such that the restriction of u to V is universally open. Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.

Proof. We first show (i). Let us show that the restriction π_V of π_G to V is flat and locally of finite presentation.

a) If π_G is flat (resp. locally of finite presentation), so is π_V.

b) If π_H is flat (resp. locally of finite presentation), so is π_V, as the composition of the restriction of u to V and π_H.

So in the four cases envisaged in the statement, π_V is flat and locally of finite presentation, hence universally open (EGA IV_2, 2.4.6). Set $U = \pi_V(V)$; U is therefore open in S. It then suffices to show that V is an open group subscheme of $G|_U$; we may therefore assume $U = S$.

Set then $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$ and $u' = u_{(V)}$. Then, since V is flat and locally of finite presentation over S, so is H' over H. By EGA IV_4, 17.7.4, V' is then the largest open subset of G' such that the restriction of u' to V' is flat and locally of finite presentation (resp. smooth, resp. étale). With the automorphisms a and b defined as in the proof of 2.1, it is then clear that V' is stable under a, hence V is a group subscheme of G.

Let us show (ii). The restriction π_V of π_G to V is a universally open morphism, either because so is π_G, or as the composition of the restriction of u to V and π_H in the case where π_H is universally open. Set $U = \pi_V(V)$; U is then open in S. It suffices to show that V is an open group subscheme of $G|_U$. We may therefore assume $U = S$.

Set as before $G' = G \times_S V$, $H' = H \times_S V$, $V' = V \times_S V$ and $u' = u_{(V)}$. Then $\pi_V : V \to S$ is surjective and universally open, the same is true of $G' \to G$, so that V' is the largest open subset of G' such that the restriction of u' to V' is universally open, by virtue

[475] In case (i), the open subset V is formed by all the points of G at which u is smooth, resp. étale, resp. of finite presentation and flat, cf. N.D.E. (6). On the other hand, in case (ii), V is the largest open subset contained in the set E of points of G at which u is universally open, but E is not necessarily open, as shown by the following example (EGA IV_3, 14.1.3 (i)): let k be a field, $H = S = \operatorname{Spec} A$ with $A = k[x]$, and G the S-group $\operatorname{Spec} A[y]/(xy)$; then E equals the unit section of G, which is not open.

of (EGA IV$_3$, 14.3.4 (i) and (ii)). With the automorphisms a and b defined as before, it is then clear that V' is stable under a, hence V is a group subscheme of G. □

Corollary 2.3. *Let G be an S-group and V the largest open subset of G which is flat and locally of finite presentation (resp. smooth, étale, universally open) over S. Then $U = \pi_V(V)$ is open in S, and V is an open group subscheme of $G|_U$.*

It suffices to apply 2.2 in the case where H is the unit S-group and u is the unique morphism of S-groups $G \to H$, since then π_H is an isomorphism and $\pi_G = \pi_H \circ u$. □

Corollary 2.4. *Let G be an S-group; if there exists a neighborhood X of the unit section having one (or several) of the following properties:*

X is flat and locally of finite presentation (resp. smooth, étale, universally open) over S,

then there exists an open group subscheme of G having the same properties.

It suffices to apply 2.3, remarking that here, with the notations of 2.2, one has $\epsilon(S) \subset V$, hence $U = S$. □

Proposition 2.5. [476] *Let $u : G \to H$ be a morphism of S-groups.*

(i) Suppose that G (resp. H) is of finite presentation and flat (resp. of finite type) over S at the points of its unit section. Then the sets:

$$U_{flat} \supset U_{smooth} \supset U_{\acute{e}}t,$$

formed of the points $s \in S$ such that u_s is flat (resp. smooth, étale), are open in S.

If moreover G (resp. H) is locally of finite presentation and flat (resp. locally of finite type) over S, then the set V_{flat} (resp. V_{smooth}, $V_{\acute{e}}t$) of points of G where u is flat (resp. smooth, resp. étale) is the inverse image under π_G of U_{flat} (resp. U_{smooth}, $U_{\acute{e}}t$).

(ii) Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$ (a condition satisfied if G is of finite type over S at the points of the unit section (1.3.1.1)), and that u is locally of finite type (resp. locally of finite presentation) at the points of the unit section of G. Then the sets:

$$U_l.q.f. \supset U_n.r.$$

formed of the $s \in S$ such that u_s is locally quasi-finite (resp. unramified) are open in S.

If moreover u is locally of finite type (resp. locally of finite presentation), then the set $V_l.q.f.$ (resp. $V_n.r.$) of points of G where u is quasi-finite (resp. unramified) is the inverse image under π_G of $U_l.q.f.$ (resp. $U_n.r.$).

[476] We have expanded the statement and the proof, and in (i) we have weakened the hypothesis on H by replacing "of finite presentation" by "of finite type".

Proof. Let us show (i). First note (1.3.1.1) that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Let Y be the open subset of H formed by the points where π_H is of finite type, and let X_1 be the open subset of $u^{-1}(Y)$ formed by the points where π_G is of finite presentation. By EGA IV$_3$, 11.3.1, the set X of points of X_1 where π_G is of finite presentation and flat is open in X_1, hence in G.

Let $u_X : X \to Y$ be the morphism deduced from u. Since Y (resp. X) contains the unit section of H (resp. G), the restriction π_X of π_G to X is surjective, and the morphism $S \to X$ deduced from ϵ_G is an S-section of X.

Consider the sets $W_{flat} \supset W_{smooth} \supset W_{\acute{e}}t$, formed by the points of X where u_X is flat (resp. smooth, resp. étale). It is clear that W_{smooth} and $W_{\acute{e}}t$ are open in X, cf. N.D.E. (6). On the other hand, since π_Y is locally of finite type and $\pi_X = \pi_Y \circ u_X$ is locally of finite presentation, by EGA IV$_1$, 1.4.3 (v), u_X is locally of finite presentation. Consequently, the flat locus W_X of u_X is open in X (EGA IV$_3$, 11.3.1).

Let $x \in X$ and set $s = \pi_X(x)$; then by EGA IV$_2$, 11.3.10 (combined with EGA IV$_4$, 17.5.1, resp. 17.6.1), x belongs to W_{flat} (resp. W_{smooth}, $W_{\acute{e}}t$) if and only if u_s is flat (resp. smooth, étale) at the point x, or, what comes to the same by 1.3, if and only if u_s is flat (resp. smooth, étale).

Consequently, for "♭ = flat, smooth or étale", one has $U_{▯} = \epsilon_G^{-1}(W_{▯})$ and $W_{▯} = \pi_X^{-1}(U_{▯})$. Since $W_{▯}$ is open in X, hence in G, the first equality shows that $U_{▯}$ is open in S.

The second assertion of (i) follows from what has just been seen, since then $Y = H$, $X = G$, and $V_{▯} = W_{▯} = \pi_G^{-1}(U_{▯})$.

Let us show assertion (ii). Let $V_l.t.f. \supset V_l.q.f.$ be the open subsets of G formed by the points at which u is of finite type (resp. quasi-finite). Set $X = V_l.t.f.$ and denote by u_X the restriction of u to X. By hypothesis, X contains $\epsilon(S)$. Let $x \in X$ and set $s = \pi_X(x)$.

If u is quasi-finite at x, then, by base change, u_s is quasi-finite at x, and so, since G_s is assumed locally of finite type, u_s is locally quasi-finite by 1.3.1.

Conversely, if u_s is locally quasi-finite, then $u_X^{-1}(u_X(x)) \subset u_s^{-1}(u_s(x))$ is a finite set. Since $u_X : X \to H$ is locally of finite type, it follows from Chevalley's semicontinuity theorem (EGA IV$_3$, 13.1.3) that there exists an open neighborhood W of x in X such that the fiber $u_X^{-1}(u_X(x'))$ is discrete for every $x' \in W$. Hence, by EGA II, 6.2.2 (and EGA III$_2$, Err_III 20), u_X is quasi-finite at x.

Consequently, one has $U_l.q.f. = \epsilon_G^{-1}(V_l.q.f.)$ and $V_l.q.f. = \pi_X^{-1}(U_l.q.f.)$, and the first equality shows that $U_l.q.f.$ is open in S.

If moreover G is locally of finite type over S, then $X = G$, and so $V_l.q.f.$ is the inverse image under π_G of $U_l.q.f.$. This proves the first half of (ii).

Consider now the open subsets $V_l.p.f. \supset V_n.r.$, formed by the points where u is of finite presentation (resp. unramified), and suppose that $Z = V_l.p.f.$ contains the unit section of G. Let $z \in Z$; set $s = \pi_Z(z)$ and $h = u_Z(z)$.

If u is unramified at z, then, by base change, u_s is unramified at z, and so, since G_s is assumed locally of finite type, u_s is unramified by 1.3.1.

Conversely, if u_s is unramified at the point z, then the fiber $u_s^{-1}(h) = u^{-1}(h)$ is unramified over $\kappa(h)$ at the point z. Since Z is an open subset of G, the fiber $u_Z^{-1}(h) = u_s^{-1}(h) \cap Z$ is unramified over $\kappa(h)$ at the point z. Since u_Z is locally of finite presentation, this entails, by EGA IV$_4$, 17.4.1, that u_Z is unramified at the point z.

One therefore has $U_n.r. = \epsilon_G^{-1}(V_n.r.)$ and $V_n.r. = \pi_Z^{-1}(U_n.r.)$, and the first equality shows that $U_n.r.$ is open in S.

If moreover G is locally of finite presentation over S, then $Z = G$, and so $V_n.r.$ is the inverse image under π_G of $U_n.r.$. This completes the proof of assertion (ii). ∎

Corollary 2.6. *Let $u : G \to H$ be a morphism of S-groups which is a radicial morphism (which is the case if u is a monomorphism (EGA I, 3.5.4)). Suppose G (resp. H) locally of finite presentation and flat (resp. locally of finite type) over S. Then the set U of $s \in S$ such that u_s is an open immersion is open in S, and the restriction of u over U is an open immersion.*

By 2.5 (i), the set U' of points $s \in S$ such that u_s is étale is open in S. Since u is radicial, so is u_s, for every $s \in S$, hence by EGA IV$_4$, 17.9.1, one has $U = U'$, which shows that U is open. Finally, by 2.5 (i), the restriction of u over U is étale; since u is radicial, this restriction is an open immersion (EGA IV$_4$, 17.9.1). ∎

Proposition 2.7. *Let G be an S-group. The following conditions are equivalent:*

(i) G is unramified over S at the points of the unit section,

(ii) the unit section is an open immersion,

(iii) G is of finite presentation over S at the points of the unit section, and for every $s \in S$, G_s is unramified over $\kappa(s)$.

If, moreover, one assumes G locally of finite presentation over S, then each of the three preceding conditions is equivalent to the following:

(iv) G is unramified over S.

The equivalence of assertions (i) and (ii) follows from the more general Lemma 2.7.1 below. Note (1.3.1.1) that either of conditions (i) or (iii) implies that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Then (EGA IV$_4$, 17.4.1), assertion (i) is equivalent to the fact that, for every $s \in S$, G_s is unramified over $\kappa(s)$ at the point e_s, unit of G_s, or again (1.3.1), to the fact that G_s is unramified over $\kappa(s)$, so assertions (i) and (iii) are equivalent. Finally, if G is locally of finite presentation over S, assertions (iii) and (iv) are equivalent (EGA IV$_4$, 17.4.1). ∎

Lemma 2.7.1. *Let G be an S-scheme equipped with a section ϵ. In order that G be unramified over S at the points of this section, it is necessary and sufficient that ϵ be an open immersion.*

The condition is necessary, by EGA IV₄, 17.4.1 a) ⇒ b″). Conversely, if ϵ is an open immersion, then the restriction to $\epsilon(S)$ of the structural morphism $G \to S$ is an isomorphism, hence G is unramified over S at the points of $\epsilon(S)$. ∎

Corollary 2.8. *Let $u : G \to H$ be a morphism of S-groups. Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$.*[477]

a) If u is locally of finite type, the following conditions are equivalent:

(i) u is locally quasi-finite,

(ii) for every $s \in S$, $u_s : G_s \to H_s$ is locally quasi-finite,

(iii) $\mathrm{Ker}\ u$ is locally quasi-finite over S,

(iv) the fibers of $\mathrm{Ker}\ u$ are discrete.

b) If u is locally of finite presentation, the following conditions are equivalent:

(v) u is unramified,

(vi) for every $s \in S$, $u_s : G_s \to H_s$ is unramified,

(vii) $\mathrm{Ker}\ u$ is unramified,

(viii) the unit section $S \to Ker u$ is an open immersion.

The equivalences (i) ⇔ (ii) and (v) ⇔ (vi) follow from 2.5 (ii), and Reminder 2.8.1 below shows that (i) ⇒ (iii) and (v) ⇒ (vii).

For every $s \in S$, denote by e_s the unit element of H_s. Then (iii) (resp. (vii)) implies that, for every $s \in S$,

$$(\mathrm{Ker}\ u)_s = \mathrm{Ker}\ u_s = u_s^{-1}(e_s)$$

is locally quasi-finite (resp. unramified) over $\kappa(s) = \kappa(e_s)$, hence u_s is quasi-finite (resp. unramified) at the unit point of G_s, hence, by 1.3, u_s is locally quasi-finite (resp. unramified). So (iii) ⇒ (ii) and (vii) ⇒ (vi). Finally, (ii) ⇔ (iv) by 1.4.1, and (vii) ⇔ (viii) by 2.7. ∎

Reminder 2.8.1. *Recall (cf. I 2.3.6.1) that, given a morphism $u : G \to H$ of S-groups, the* kernel *of u, denoted $\mathrm{Ker}\ u$, is the group subfunctor of G defined by setting, for any morphism $f : T \to S$,*

$$(\mathrm{Ker}\ u)(T) = \{a \in G(T) \mid u \circ a = \varepsilon_H \circ f\}.$$

By loc. cit. (or EGA I, 4.4.1), this functor is representable by the S-group $G \times_H S = u^{-1}(\epsilon_H(S))$, simply denoted $\mathrm{Ker}\ u$. In particular, the structural morphism $Ker u \to S$ is deduced from u by base change.

Lemma 2.9. *Let $\pi : X \to S$ be a morphism admitting an S-section ϵ.*

(i) If π is injective, it is integral.[478]

[477] We have modified the presentation, separating the assertions relative to the "quasi-finite" case from those relative to the "unramified" case.

[478] This is also a particular case of EGA IV₄, 18.12.11, since π is evidently a universal homeomorphism.

(ii) If π is locally of finite type, and if, for every $s \in S$, π_s is an isomorphism, then π is an isomorphism.[479]

Let us first note that, by Lemma 2.9.1 below, π, being a homeomorphism, is an affine morphism.

If π is injective, ϵ is surjective. Since ϵ is a surjective immersion, $\epsilon(S)$ is isomorphic to the closed subscheme of X defined by a nil-ideal I of 0_X. Since ϵ is an S-section of the morphism π, one has a direct sum decomposition $O_X = O_S \bigoplus I$ of 0_S-modules. Since I is a nil-ideal of 0_X, I is evidently integral over 0_S, hence 0_X is integral over 0_S, and π is integral.

Suppose now that π is locally of finite type. Since $\pi \circ \epsilon = id_S$, ϵ is locally of finite presentation (cf. EGA IV₁, 1.4.3 (v)), hence I is an ideal of finite type of 0_X (EGA IV₁, 1.4.1). For every $s \in S$, one has $O_{X_s} = \kappa(s) \bigoplus I \otimes_{O_S} \kappa(s)$. By hypothesis, ϵ_s is an isomorphism, hence $I \otimes_{O_S} \kappa(s) = 0$ for every $s \in S$, hence a fortiori $I \otimes_{O_X} \kappa(x) = 0$ for every $x \in X$, which implies, by Nakayama's lemma, that $I = 0$, hence π is an isomorphism. ∎

Lemma 2.9.1. *Let $f : X \to Y$ be a morphism of schemes which is a homeomorphism; then f is an affine morphism.*[480]

It suffices to show that every point $y \in Y$ has an open neighborhood W such that the restriction of f over W is an affine morphism. So let $y \in Y$, and let V be an affine open neighborhood of y in Y. Let $V' = f^{-1}(V)$. Then V' is an open neighborhood of $x = f^{-1}(y)$ in X. There exists an open affine neighborhood W' of x in X contained in V'. Set then $W = f(W')$. Then W is an open neighborhood of y in Y contained in the affine scheme V, hence W is separated. Since W' is an affine scheme, the restriction of f over W is then an affine morphism (EGA II, 1.2.3). ∎

Corollary 2.10. *Let G be an S-group locally of finite type. Suppose that, for every $s \in S$, G_s is the unit $\kappa(s)$-group; then G is the unit S-group.*

More generally:

Corollary 2.11. *Let $f : X \to Y$ be an S-morphism locally of finite type. In order that f be a monomorphism, it is necessary and sufficient that f_s be a monomorphism for every $s \in S$.*

It is clear that the condition is necessary; let us show that it is sufficient. If, for every $s \in S$, f_s is a monomorphism, then a fortiori for every $y \in Y$, f_y is a monomorphism; we may therefore assume $Y = S$.

By EGA I, 5.3.8, to show that f is a monomorphism, it suffices to show that $\Delta_f : X \to X \times_S X$ is an isomorphism, or, what comes to the same, that the first projection $p : X \times_S X \to X$ is an isomorphism. But, if f_s is a monomorphism, it likewise follows from EGA I, 5.3.8 that the first projection $X_s \times_{\kappa(s)} X_s \to X_s$ (which is identified with

[479] This is also an immediate consequence of EGA IV₄, 18.12.6.
[480] Cf. EGA IV₄, 18.12.7.1 for a slightly more general result, provable by the same proof.

p_s) is an isomorphism. Now p has the S-section Δ_f, hence Lemma 2.9 affirms that if for every $s \in S$, p_s is an isomorphism, then so is p. □

Corollary 2.12. *Let G be an S-scheme having an S-section ϵ and such that the structural morphism $\pi : G \to S$ is closed. Let $s \in S$ be such that π is of finite presentation at the point $\epsilon(s)$ and that $\epsilon_s : \mathrm{Spec}(\kappa(s)) \to G_s$ is an isomorphism (or, what comes to the same, that $\pi_s : G_s \to \mathrm{Spec}\,\kappa(s)$ is an isomorphism). Then there exists an open neighborhood U of s in S such that $\epsilon|_U : U \to G \times_S U$ is an isomorphism.*

Let V be the set of points of G where π is unramified; it is known that V is open (cf. N.D.E. (6)) and contains $\epsilon(s)$. Hence $W = \epsilon^{-1}(V)$ is an open subset of S containing s, such that for every $t \in W$, π is unramified at $\epsilon(t)$. Since π is closed, so is $\pi|_W$, hence we may assume $W = S$.

Then, by 2.7.1, $X = G - \epsilon(S)$ is a closed subset of G not meeting G_s, hence, since π is closed, $F = \pi(X)$ is a closed subset of S not containing s; set $U = S - F$; then U is an open subset of S such that $\epsilon|_U$ is an isomorphism of U onto $G|_U$. □

3. Neutral component of a group locally of finite presentation

3.0.

Given a part A (resp. B) of an S-scheme X (resp. Y), by abuse of notation, $A \times_S B$ will denote the part of $X \times_S Y$ formed by the points whose first projection lies in A and second in B.

Given a part A of an S-group G, we shall say that A is *stable under the group law of G* if $c(A) \subset A$ and $\mu(A \times_S A) \subset A$.

Definition 3.1. *Let G be an S-functor in groups satisfying the following condition:*

(+) for every $s \in S$, the functor $G_s = G \otimes_S \kappa(s)$ is representable.

One then denotes by G_s^0 the connected component of the unit element of the $\kappa(s)$-group G_s.[481] *One defines a sub-S-functor in groups of G, called the* neutral component *of G, denoted G^0, by setting, for every morphism $T \to S$:*

$$G^0(T) = \{u \in G(T) \mid \forall\, s \in S,\ u_s(T_s) \subset G^0_s\}.$$

One has thus defined the functor $G \mapsto G^0$ from (Sch/S)-gr. to (Sch/S)-gr.

Remark 3.2. *(i) Let G be an S-functor in groups satisfying condition (+); then $Lie(G^0/S) = Lie(G/S)$, by virtue of Exposé II.*[482] *Indeed, for every S-scheme T, denote by I_T the T-scheme of dual numbers over T and by $\tau : T \to I_T$ the zero section (cf. II, 2.1). By definition, one has*

$$Lie(G/S)(T) = \{u \in G(I_T) \mid u \circ \tau = e\},$$

[481] This is a slightly abusive notation, but one which is compatible with the notations of (VI_A, 2.3) when this connected component is the underlying topological space of an open group subscheme G^0 of G, cf. (VI_A, 2.2.bis).
[482] We have added what follows.

where $e : T \to G$ denotes the composition of $T \to S$ and the unit section $S \to G$, and likewise

$$\mathrm{Lie}(G^0/S)(T) = \{u \in G^0(I_T) \mid u \circ \tau = e\}$$
$$= \{u \in G(I_T) \mid u \circ \tau = e \text{ and } u_s((I_T)_s) \subset G^0_s,\ \forall\, s \in S\}.$$

Now, for every $s \in S$, T_s and $(I_T)_s = I_{T_s}$ have the same underlying set, hence if $u \in Lie(G/S)(T)$, one has $u_s(I_{T_s}) = e_s$, where e_s denotes the unit point of G_s, whence $u \in Lie(G^0/S)(T)$. So the inclusion $Lie(G^0/S)(T) \subset Lie(G/S)(T)$ is an equality for every T, whence $Lie(G^0/S) = Lie(G/S)$.

(ii) Let G and G' be two S-functors in groups satisfying (+); then:

a) if $G \subset G'$, then $G^0 \subset G'^0$,

b) if $G \subset G'$ and $G'^0 \subset G$, then $G^0 = G'^0$,

c) if for every $s \in S$, G_s is locally of finite type over $\kappa(s)$, then G^0 satisfies property (+), by (VI_A, 2.3), and one has $(G^0)^0 = G^0$.

Proposition 3.3. *Let G be an S-functor in groups satisfying condition (+) and let S' be an S-scheme. Then $(G \times_S S')^0 = G^0 \times_S S'$; in other words, the functor $G \mapsto G^0$ commutes with base change, i.e. the following diagram is commutative:*

$$\begin{tikzcd}
(\mathrm{Sch}/S)\text{-gr.} \arrow[r, "G \mapsto G^0"] \arrow[d] & (\mathrm{Sch}/S)\text{-gr.} \arrow[d] \\
(\mathrm{Sch}/S')\text{-gr.} \arrow[r] & (\mathrm{Sch}/S')\text{-gr.}
\end{tikzcd}$$

It suffices indeed to check that, for every $s' \in S'$ with image s in S, $((G \times_S S') \otimes_{S'} \kappa(s'))^0 = (G_s \otimes_{\kappa(s)} \kappa(s'))^0$ equals $G_s^0 \otimes_{\kappa(s)} \kappa(s')$; this follows from (VI_A, 2.1.2). Note, for later use in 4.2, that we have not used the group structure of G_s, only the fact that G_s^0 has a rational point, namely $\epsilon(s)$, hence is geometrically connected (see also EGA IV_2, 4.5.14). ∎

Special case 3.4. *Let G be an S-group scheme; denote by G^0 the subset of G equal to the union of the G_s^0 as s runs through S. Then G^0 is a part of G stable under the group law of G (cf. 3.0), and for any morphism $S' \to S$ one has:*

$$G^0(S') = \{u \in G(S') \mid u(S') \subset G^0\}.$$

When G^0 is an open part of G, it is clear that G^0 is representable by the subscheme of G induced on the open G^0.

Proposition 3.5. *Let S be a quasi-compact and quasi-separated scheme, and G an S-group with fibers locally of finite type. Then there exists a quasi-compact open subset U of G containing G^0.*

The unit section ϵ being an immersion, $\epsilon(S)$ is a quasi-compact subspace of G, so there exists a quasi-compact open subset V of G containing $\epsilon(S)$. Since S is quasi-separated and V quasi-compact, V is quasi-compact over S (EGA IV_1, 1.2.4), so $V \times_S V$ is quasi-compact over S, hence quasi-compact. Then $V \cdot V = \mu(V \times_S V)$ is quasi-compact. Set $V_s = V \cap G_s$ and $V_s^0 = V \cap G_s^0$. Then V_s^0 is an open subset of G_s^0, dense in G_s^0 since G_s^0 is irreducible (VI_A, 2.4), so $V_s^0 \cdot V_s^0 = G_s^0$ (VI_A, 0.5), which shows that $V_s \cdot V_s \supset G_s^0$, hence $V \cdot V \supset G^0$. Finally, since $V \cdot V$ is quasi-compact, there exists a quasi-compact open subset U of G containing $V \cdot V$ and a fortiori G^0. ∎

Corollary 3.6. *Let G be an S-group with fibers locally of finite type and connected. Then G is quasi-compact over S.*

Our assertion being local on S (EGA I, 6.6.1), one reduces to the case where S is affine. By 3.5, there then exists a quasi-compact open U of G containing $G^0 = G$, hence G is quasi-compact, hence quasi-compact over the affine scheme S (EGA I, 6.6.4 (v)). ∎

Proposition 3.7. [483] *Let G be an S-group locally of finite presentation. Then:*

(i) G^0 is ind-constructible in G.

(ii) If moreover G is quasi-separated over S, and S quasi-compact and quasi-separated, then G^0 is constructible.

(iii) Consequently, if G is quasi-separated over S, G^0 is locally constructible.

Proof. Let us first show the first assertion. Since $\pi : G \to S$ is locally of finite presentation, given $s \in S$, there exists an open subset U of G containing $\epsilon(s)$ and an open subset V of S containing s such that $\pi(U) \subset V$ and such that the morphism $\pi' : U \to V$ deduced from π is of finite presentation. Then $T = \epsilon^{-1}(U)$ is an open subset of S and if we let $W = \pi'^{-1}(T)$ and $\pi'' = \pi'|_W$, then $\pi'' : W \to T$ is of finite presentation, and admits as section the morphism $\epsilon'' : T \to W$ deduced from ϵ.

For every $t \in T$, since G_t^0 is irreducible (VI_A, 2.4), $W \cap G_t^0$ is dense in G_t^0, hence irreducible, hence connected: it is therefore the connected component of $\pi''^{-1}(t)$ containing $\epsilon''(t)$. It then follows from EGA IV$_3$, 9.7.12 that the union W^0 of the $W \cap G_t^0$, for $t \in T$, is locally constructible in W. On the other hand, it follows from (VI_A, 0.5) that $G^0 \cap \pi^{-1}(T) = W^0 \cdot W^0$, i.e. $G^0 \cap \pi^{-1}(T)$ is the image of $W^0 \times_T W^0$ under the morphism $\mu'' : W \times_T W \to \pi^{-1}(T)$ deduced from μ.[484] Since $W \times_T W$ (resp. $\pi^{-1}(T)$) is of finite presentation (resp. locally of finite presentation) over T, μ'' is locally of finite presentation and quasi-separated, by EGA IV$_1$, 1.4.3 and 1.2.2; if moreover $\pi^{-1}(T)$ is quasi-separated over T, then μ'' is quasi-compact (loc. cit. 1.2.4), hence of finite presentation. Since $W^0 \times_T W^0$ is locally constructible in $W \times_T W$ (since W^0 is in W), it follows from Chevalley's constructibility theorem

[483] Recall the following definitions and results (cf. EGA 0_III, § 9.1 and EGA IV$_1$, §§ 1.8 and 1.9). Let X be a topological space. (i) An open U of X is called *retrocompact* if the inclusion $U \hookrightarrow X$ is quasi-compact, i.e. if $U \cap V$ is quasi-compact for every quasi-compact open $V \subset X$. (ii) A part C of X is called *constructible* if it is a finite union of parts of the form $U - V$, where U and V are retrocompact opens in X. (iii) A part L of X is called *locally constructible* if for every $x \in X$ there exists an open neighborhood U of x in X such that $L \cap U$ is constructible in U. (N.B. If X is quasi-compact and quasi-separated, L is then constructible.) (iv) A part I of X is called *ind-constructible* if for every $x \in X$, there exists an open neighborhood U of x in X such that $I \cap U$ is a union of parts locally constructible in U. Let now $f : X \to Y$ be a morphism of schemes. By Chevalley's constructibility theorem (cf. EGA IV$_1$, 1.8.4 and 1.9.5 (viii)), if f is of finite presentation (resp. locally of finite presentation), then the image under f of any locally constructible part is locally constructible (resp. ind-constructible).

[484] We have expanded the original in what follows.

(loc. cit., 1.8.4 and 1.9.5 (viii)) that $G^0 \cap \pi^{-1}(T)$ is ind-constructible in $\pi^{-1}(T)$, and is locally constructible in $\pi^{-1}(T)$ if G (and hence $\pi^{-1}(T)$) is quasi-separated over T. This proves assertions (i) and (iii).

Suppose now G quasi-separated over S, and S quasi-compact and quasi-separated. Then, by 3.5, there exists a quasi-compact open U of G containing G^0. Since G is quasi-separated over S, G is quasi-separated, so the open U is retrocompact (EGA IV$_1$, 1.2.7), and it suffices to show that G^0 is constructible in U (EGA 0_III, 9.1.8). Moreover, U being quasi-compact, hence quasi-compact over S (EGA IV$_1$, 1.2.4), and quasi-separated over S, the restriction of π to U is of finite presentation, so by EGA IV$_3$, 9.7.12, G^0 is locally constructible in U, hence constructible in U, since U is quasi-compact and quasi-separated (EGA IV$_1$, 1.8.1). This proves (ii), and it follows that for every quasi-compact and quasi-separated open T of S (e.g., for every affine open of S), $G^0 \cap \pi^{-1}(T)$ is constructible. □

Corollary 3.8. *Let `S_0` be a quasi-compact and quasi-separated scheme, I a filtered increasing preordered set, $(A_i)_{i\in I}$ a direct system of commutative quasi-coherent O_{S_0}-algebras, $A = \lim A_i$ (direct limit), $S_i = \operatorname{Spec} A_i$ for $i \in I$, and $S = \operatorname{Spec} A$ (cf. EGA II, 1.3.1).*

Let G be an `S_0`-group scheme locally of finite presentation. Then the canonical map $\lim G^0(S_i) \to G^0(S)$ is bijective.

Since G is locally of finite presentation over `S_0`, the canonical map $\lim G(S_i) \to G(S)$ is bijective, by EGA IV$_2$, 8.14.2 c). It follows immediately that the canonical map $\lim G^0(S_i) \to G^0(S)$ is injective. Let us show that it is surjective. Let $g \in G^0(S) \subset G(S)$. There exists $i \in I$ such that g factors through $g_i \in G(S_i)$ via $S \to S_i$; by hypothesis, $g^{-1}(G^0) = S$. But, by 3.7, G^0 is ind-constructible in G, so $g_i^{-1}(G^0)$ is ind-constructible in S_i. It then follows from EGA IV$_2$, 8.3.4, that there exists an index $j \geq i$ such that $g_j^{-1}(G^0) = S_j$, where g_j is the map deduced from g_i by the base change $S_j \to S_i$. This shows that $g_j \in G^0(S_j)$, hence that g comes from an element of $\lim G^0(S_i)$. □

Proposition 3.9. *Let G be an S-group locally of finite presentation. Suppose that G^0 is representable; then the canonical morphism $i : G^0 \to G$ is an open immersion; moreover, G^0 is quasi-compact over S.*

Since G^0 is a subfunctor of the functor G, the morphism i is a monomorphism, hence radicial. By the definition of the functor G^0, one verifies immediately from the definition (EGA IV$_4$, 17.1.1) that i is a formally étale morphism (noting that G^0 is the image of i in G). Finally, it follows from the characterization (EGA IV$_3$, 8.14.2 c)) of S-schemes locally of finite presentation via the functor they represent, and from 3.8, that, since G is locally of finite presentation over S, so is G^0. Hence i is locally of finite presentation (cf. EGA IV$_1$, 1.4.3); it is therefore a radicial étale morphism; hence an open immersion (EGA IV$_4$, 17.9.1).

Finally, the last assertion follows from 3.6. □

Theorem 3.10. *Let G be an S-group. The following conditions are equivalent:*

(i) G is smooth over S at the points of the unit section.

(ii) G is flat and locally of finite presentation over S at the points of the unit section, and for every $s \in S$, G_s is smooth over $\kappa(s)$.

(iii) There exists an open group subscheme G' of G, smooth over S.

(iv) G^0 is representable by an open subscheme of G, smooth over S.

It is clear that (iv) ⇒ (iii) ⇒ (i) and, by 1.3.1 and 2.4, (i) implies (ii) and (iii). Moreover, (ii) ⇒ (i) by EGA IV_4, 17.5.1.

Let us show finally that (iii) implies (iv). Lemma 3.10.1 below shows that G' contains G^0, and that $G'^0 = G^0$. It therefore suffices to show that G^0 is open in G, since we have already seen (3.4) that G^0 will then be representable by the smooth group subscheme induced by G' on the open $G'^0 = G^0$. We may therefore assume $G' = G$.

To show that G^0 is open, it suffices to show that every $s \in S$ has a neighborhood T in S such that $G^0 \cap \pi^{-1}(T)$ is open in $\pi^{-1}(T)$. Let $s \in S$. Since $G = G'$, π is locally of finite presentation, so one can construct, as in the proof of 3.7, an open subset T of S containing s, and an open subset W of G containing $\epsilon(s)$, such that the morphism $\pi'' : W \to T$ deduced from π is of finite presentation and admits as section the morphism $\epsilon'' : T \to W$ deduced from ϵ. For every $t \in T$, $W \cap G_t^0$ is then the connected component of $\pi''^{-1}(t)$ containing $\epsilon''(t)$. Since π is smooth, so is π'', which is therefore smooth and of finite presentation. Then, by EGA IV_3, 15.6.5, the union W^0 of the $W \cap G_t^0$ for $t \in T$ is open in W.

On the other hand, by (VI_A, 0.5), one has $W^0 \cdot W^0 = G^0 \cap \pi^{-1}(T)$, and one must show that this is open in $\pi^{-1}(T)$. We may therefore now assume $T = S$; it remains to show that $W^0 \cdot W^0$ is open in G. Since π is universally open, so is μ (VI_A, 0.1). Hence, since W^0 is open in G, so is $W^0 \cdot W^0 = \mu(W^0 \times_S W^0)$.

This result will be improved in 4.4. ∎

Lemma 3.10.1. *Let G be an S-group with fibers locally of finite type. Then every open group subscheme H of G contains G^0, and satisfies $G^0 = H^0$.*

Let $s \in S$; set $G'_s = H_s \cap G_s^0$; then G'_s is an open subset of G_s^0, which is dense in G_s^0 since G_s^0 is irreducible (VI_A, 2.4), hence $G'_s \cdot G'_s = G_s^0$ (VI_A, 0.5), which shows that $G_s^0 = G'_s \cdot G'_s \subset H_s \cdot H_s = H_s$. One thus has $G_s^0 = H_s^0$ for every s, whence $G^0 \subset H$ and $G^0 = H^0$. ∎

Proposition 3.11. *Let $u : G \to H$ be a morphism between S-groups locally of finite presentation. If u is flat, the map $u^0 : G^0 \to H^0$ deduced from u is surjective; the converse is true if G is flat over S and H has reduced fibers.*[485]

[486] Suppose u flat; then for every $s \in S$, u_s is flat and locally of finite presentation, hence open (EGA IV_2, 2.4.6), so the morphism $u_s^0 : G_s^0 \to H_s^0$ is surjective by 1.3.2. Hence the map $u^0 : G^0 \to H^0$ is surjective.

[485] Compare with (VI_A, 5.6).

[486] We have shortened the original, referring here to 1.3.2. We have also simplified the argument below by removing a reference to the generic flatness theorem (EGA IV_2, 6.9.1), which is not necessary here.

Conversely, suppose the map $u^0 : G^0 \to H^0$ is surjective, G flat over S and H with reduced fibers. Then, for every $s \in S$, the morphism $u^0_s : G^0_s \to H^0_s$ is surjective, hence flat at every point above the generic point η of H^0_s (since $O_{H^0_s,\eta}$ is a field), hence u_s is flat by 1.3. Hence u is flat, by the fiber-wise flatness criterion (EGA IV₃, 11.3.11). ∎

4. Dimension of fibers of groups locally of finite presentation

Proposition 4.1. *Let G be an S-scheme locally of finite type, equipped with an S-section ϵ, and such that for every $s \in S$, one has $\dim G_s = \dim_{\epsilon(s)} G_s$ (which is the case if G is an S-group (1.5)).*

(i) The function $s \mapsto \dim G_s$ is upper semicontinuous on S.

(ii) If, moreover, G is locally of finite presentation over S, this function is locally constructible.

Let $\pi : G \to S$ be the structural morphism. Chevalley's semicontinuity theorem (EGA IV₃, 13.1.3) asserts that the function $x \mapsto \dim_x \pi^{-1}(\pi(x))$ is upper semicontinuous on G. Now, for every $s \in S$, one has

$$\dim G_s = \dim \pi^{-1}(s) = \dim_{\epsilon(s)} \pi^{-1}(\pi(\epsilon(s)));$$

and since the function $s \mapsto \epsilon(s)$ is continuous on S, the composite function $s \mapsto \dim G_s$ is upper semicontinuous on S.

Suppose G locally of finite presentation over S. To show that the function $s \mapsto \dim G_s$ is locally constructible, one sees, reasoning as above, that it suffices to show that the function $x \mapsto \dim_x \pi^{-1}(\pi(x))$ is locally constructible on G, which follows from EGA IV₃, 9.9.1. ∎

Proposition 4.2. *Let $\pi : G \to S$ be an S-scheme locally of finite presentation, equipped with an S-section ϵ and satisfying the following two conditions (which are satisfied if G is an S-group, by 1.5 and (VI_A, 2.4.1)):*

a) For every $s \in S$ and every $x \in G_s$, one has $\dim G_s = \dim_x G_s$ (or, what comes to the same (1.5), for every $s \in S$, all irreducible components of G_s have the same dimension).

b) For every $s \in S$, if one denotes by G^0_s the connected component of G_s containing $\epsilon(s)$, G^0_s is geometrically irreducible.

Let $s \in S$. The following conditions are equivalent:

(i) G is universally open over S at the points of G^0_s.

(i bis) G is universally open over S at every point of a neighborhood of $\epsilon(s)$ in G^0_s.[487]

(ii) The function $t \mapsto \dim G_t$ is constant in a neighborhood of s in S.

[487] One finds in the original: "at the points of $\epsilon(s)$"; it is not sufficient to assume that G is universally open over S at $\epsilon(s)$, as shown by the example given in N.D.E. (19). We have corrected the proof accordingly.

(iii) G^0 is "universally open over S at the points of G^0_s", i.e., given $S' \to S$, $s' \in S'_s$, $g \in G^0_{s'}$ and V an open neighborhood of g in $G' = G_{S'}$, then $\pi(V \cap G'^0)$ is an open neighborhood of s' in S'.[488]

Proof. Clearly (i) ⇒ (i bis). By EGA IV₃, 14.3.3.1 (ii), the set of points of G^0_s where π_G is universally open is closed in G^0_s. Hence, since G^0_s is irreducible, one has (i bis) ⇒ (i).

Let us show that (i) implies (ii). Let T be the set of $t \in S$ such that $\dim G_t = \dim G_s$. By 4.1, T is locally constructible, hence, by EGA IV₁, 1.10.1, to show that T is a neighborhood of s, it suffices to show that every generization s' of s belongs to T.

Let x be the generic point of G^0_s and U an open subset of G containing x. Since π_G is universally open at ξ, by EGA IV₃, 14.3.13, for every generization s' of s, one has $\dim(U \cap G_{s'}) \geq \dim_x(U \cap G_s)$. Taking hypothesis a) into account, this entails $\dim G_{s'} \geq \dim G_s$. Since the function $s \mapsto \dim G_s$ is upper semicontinuous by 4.1, one also has $\dim G_{s'} \leq \dim G_s$, whence $s' \in T$. This proves that (i) ⇒ (ii).

It is clear that (iii) ⇒ (i); let us show that (ii) ⇒ (iii). Since the dimension of fibers is unchanged by extension of the base field, and the formation of G^0 commutes with base change (cf. the proof of 3.3), one may assume $S' = S$ and $s' = s$. Moreover, since every open V of G meeting G^0_s contains the generic point η of G^0_s, one may assume $g = \eta$.

One may further assume S affine. Let U be an affine open neighborhood of $\epsilon(s)$; it is then of finite presentation over S. Replacing S by $\epsilon^{-1}(U)$ and then U by $U \cap \pi^{-1}(S)$, we reduce to the case where $\pi : U \to S$ is of finite presentation and admits $\epsilon : S \to U$ as section. Then, by EGA IV₃, 9.7.12, $G^0 \cap U$ is constructible in U. Then, replacing V by an affine open contained in $V \cap U$, we obtain that $G^0 \cap V$ is constructible in V. Since $\pi : V \to S$ is of finite presentation, $\pi(G^0 \cap V)$ is locally constructible in S, by Chevalley's constructibility theorem (cf. EGA IV₁, 1.8.4).

Hence, by loc. cit., 1.10.1, to show that $\pi(G^0 \cap V)$ is an open neighborhood of s, it suffices to show that for every generization t of s, there exists a generization ξ of η belonging to G^0 (and hence to $G^0 \cap V$). Now the generic point ξ of G^0_t is a generization of η. Indeed, $\epsilon(s)$ belongs to the closure X of ξ in G, so by Chevalley's semicontinuity theorem (cf. EGA IV₃, 13.1.3), one has $\dim_{\epsilon(s)} X_s \geq \dim_\xi X_t$; on the other hand, hypothesis (ii) entails that $\dim_\xi G^0_t = \dim_{\epsilon(s)} G_s$. It follows that one of the irreducible components of X_s containing $\epsilon(s)$ equals G^0_s, whence $\eta \in \xi^-$. This proves that (ii) ⇒ (iii), which proves the proposition. ∎

[489] One can also prove the implication (ii) ⇒ (i) as follows. Since π is locally of finite presentation, there exists an open subset U of G containing $\epsilon(s)$ and an open subset V of S containing s such that $\pi(U) \subset V$ and such that the morphism $\pi' : U \to V$ deduced from π is of finite presentation. Set then $T = \epsilon^{-1}(U)$ and $W = \pi'^{-1}(T) = U \cap \pi^{-1}(T)$. Then the morphism $\pi'' : W \to T$ deduced from π' is of finite presentation

[488] We have added condition (iii), pointed out by O. Gabber; it will be useful later (5.7).

[489] The preceding has been communicated to us by O. Gabber; we have also preserved the proof of the implication (ii) ⇒ (i) given in the original.

and admits as section the morphism $\epsilon'' : T \to W$ deduced from ϵ. Moreover, for every $t \in T$, G^0_t being irreducible, $W \cap G^0_t$ is dense in G^0_t, hence irreducible, hence connected: so it is the connected component of $\pi''^{-1}(t)$ containing $\epsilon''(t)$.

Since $W \cap G^0_t$ is a dense open subset of G^0_t, one has, by 1.5 and EGA IV_2, 5.2.1, $\dim(W \cap G^0_t) = \dim G^0_t = \dim G_t$, so the function $t \mapsto \dim W \cap G^0_t$ is constant in a neighborhood of s in T. Let us show finally that, for every $t \in T$, $W \cap G^0_t$ is geometrically irreducible. Let K be an extension of $\kappa(t)$; then $(W \cap G^0_t) \otimes_{\kappa(t)} K = (W \otimes_{\kappa(t)} K) \cap (G^0_t \otimes_{\kappa(t)} K)$ is a non-empty open subset of $G^0_t \otimes_{\kappa(t)} K$, hence is irreducible since $G^0_t \otimes_{\kappa(t)} K$ is.

We are then in the conditions of application of EGA IV_3, 15.6.6 (ii), which asserts that π'' (hence π) is universally open at the points of $W \cap G^0_s$. But, by EGA IV_3, 14.3.3.1 (ii), the subset of G_s formed by the points where π is universally open is closed in G_s; since it contains $W \cap G^0_s$, it therefore contains its closure G^0_s. ∎

Corollary 4.3. *Let G be a flat S-group locally of finite presentation. Then the function $s \mapsto \dim G_s$ is locally constant on S.*

This follows immediately from 4.2, since every flat morphism locally of finite presentation is universally open (EGA IV_2, 2.4.6). ∎

Corollary 4.4. *Let G be an S-group locally of finite presentation over S at the points of the unit section. Consider the conditions:*

(i) G is smooth over S at the points of the unit section (cf. 3.10).

(ii) For every $s \in S$, G_s is smooth over $\kappa(s)$, and the function $s \mapsto \dim G_s$ is locally constant on S.

(iii) For every $s \in S$, G_s is smooth over $\kappa(s)$, and there exists a neighborhood V of the unit section such that $\pi_V : V \to S$ is universally open.

(iv) For every $s \in S$, G^0_s is smooth over $\kappa(s)$, and G^0 is representable by an open group subscheme of G, universally open over S.

Then one has the following implications: (i) ⇒ (ii) ⇔ (iii) ⇔ (iv).

If one further assumes S reduced, then conditions (i) to (iv) are equivalent, and imply that G^0 is smooth over S.[490]

Proof. Let us show that (i) implies (ii). For every $x \in G$, one has $\dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$, by 1.5. Consequently, by EGA IV_4, 17.10.2, the function

$$x \mapsto \dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$$

is continuous in a neighborhood of the unit section; so the function $s \mapsto \dim G_s$ is continuous on S, hence locally constant on S. Moreover, for every $s \in S$, G_s is smooth over $\kappa(s)$ by 1.3.1.

[490] One cannot do without the hypothesis that S is reduced here: if k is a field, $S = \operatorname{Spec} A$, where $A = k[\delta]$ with $\delta^2 = 0$, and G the closed subgroup $\operatorname{Spec} A[X]/(\delta X)$ of $G_{a,S}$ (which to every A-algebra R associates the subgroup of $r \in R$ such that $\delta r = 0$), then G is an S-group satisfying (ii)–(iv), but is not flat, hence not smooth, over S.

Let us show that (ii) implies (iv). It suffices to show that G^0 is open in G, for then, by 3.4, G^0 will be representable by the group subscheme induced on G^0, and the properties of G^0 cited in the statement will follow from 2.4 and 4.2. Given $s \in S$, construct as in the proof of 3.10, W, T, π'', ϵ'', and W^0. Then, by EGA IV_3, 15.6.7, W^0 is open in W. On the other hand, under hypothesis 4.4 (ii), it follows from 4.2 that π is universally open at every point of W^0, hence (VI_A, 0.1) μ is universally open at every point of $W^0 \times_S W^0$, which shows that $W^0 \cdot W^0$ is open, and one concludes as in the proof of Theorem 3.10.

It is clear that (iv) ⇒ (iii), and (iii) ⇒ (ii) follows from 4.2 applied to V.

Finally, suppose (ii)–(iv) hold and let us show that G^0 is smooth over S if S is reduced.[491] For this, one may assume $G = G^0$. Then G is of finite presentation over S by virtue of 5.5 below. Thus, π_G is of finite presentation, with geometrically integral fibers, of locally constant dimension on S. Then, by EGA IV_3, 15.6.7, the morphism $G \times_S S_{red} \to S_{red}$ deduced from π_G is flat, hence π_G is flat if S is reduced. In this case, $G = G^0$ is smooth over S, by Theorem 3.10. □

5. Separation of groups and homogeneous spaces

Proposition 5.1. *In order that an S-group G be separated, it is necessary and sufficient that the unit section of G be a closed immersion.*

The condition is necessary (EGA I, 5.4.6); it is sufficient by virtue of the following cartesian diagram (cf. (VI_A, 0.3)):

$$\begin{tikzcd}
G \times_S G \arrow[r, "\mu \circ (\mathrm{id}_G \times c)"] \arrow[d, "\Delta_{G/S}"'] & G \\
G \arrow[r, "\pi"'] & S. \arrow[u, "\varepsilon"']
\end{tikzcd}$$

Proposition 5.2. *If S is discrete, every S-group is separated.*

Indeed, S is then equal to $\coprod_{s \in S} \operatorname{Spec} \mathcal{O}_{S,s}$, and by EGA I, 5.5.5, it suffices to show that for every $s \in S$, $G \times_S \operatorname{Spec} \mathcal{O}_{S,s}$ is separated, which follows from (VI_A, 0.3), since $\mathcal{O}_{S,s}$ is a local ring of dimension 0.[492]

Theorem 5.3. *Let S be a scheme, G an S-group scheme locally of finite presentation over S and universally open over S in a neighborhood of the unit section, X an S-scheme on which G acts in such a way that the morphism:*

$$\Phi : G \times_S X \longrightarrow X \times_S X, \qquad (g, x) \mapsto (gx, x)$$

is surjective. Suppose moreover that, for every $s \in S$:

[491] We have expanded the original in what follows.

[492] We have replaced "Artinian" by "of dimension 0" (and mentioned this generalization in (VI_A, 0.3)).

(i) there exists an open subscheme U of X, separated over S, such that U_s is dense in X_s,

(ii) the fiber X_s is locally of finite type over $\kappa(s)$.[493]

Then X is separated over S.

Corollary 5.4. *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that X has fibers locally of finite type and connected. Then:*

(i) X is separated over S.

(ii) If there exists an open subset V of X, quasi-compact over S and meeting every non-empty fiber X_s,[494] *then X is quasi-compact over S.*

Proof. (i) Indeed, let $s \in S$ be such that $X_s \neq \emptyset$. Since the morphism $\Phi_s : G_s \times_{\kappa(s)} X_s \to X_s \times_{\kappa(s)} X_s$ deduced from Φ by base change is surjective, and X_s is connected, X_s is irreducible by (VI_A, 2.5.4). Hence, if U is an affine open of X such that U_s is non-empty, U_s is dense in X_s, and the theorem applies.

To prove (ii), one may assume S affine. Then V is quasi-compact and, by 3.5, there exists a quasi-compact open U of G containing G^0. Let $s \in S$ be such that $X_s \neq \emptyset$. Then X_s is irreducible (VI_A, 2.6.6) and so, since U_s contains G_s^0, the morphism $U_s \times_{\kappa(s)} V_s \to X_s$, $(g,x) \mapsto gx$, is surjective (VI_A, 2.6.4). Consequently, the morphism $U \times_S V \to X$ is surjective, and so X is quasi-compact (since U and V are); as we assumed S affine, hence separated, it follows that X is quasi-compact over S (cf. EGA I, 6.6.4 (v)). ∎

Remark 5.4.1. [495] *We shall see in the course of the proof that the conclusion of Theorem 5.3 holds if one makes only the hypothesis: (i′) there exists an open subscheme U of X, separated over S, such that U_s is dense in every irreducible component of X_s. (The latter is a consequence of (i) if X_s has locally a finite number of irreducible components, which is the case under hypothesis (ii).) On the other hand, we shall see later that 5.4 is also valid under the hypothesis that each fiber X_s is quasi-separated and connected.*

Theorem 5.3A (Raynaud). *Let G be an S-group locally of finite presentation, universally open over S, and with connected fibers. Then G is separated over S. More generally, any S-scheme X locally of finite presentation over S, equipped with an action of G such that the morphism $\Phi : G \times_S X \to X \times_S X$, $(g,x) \mapsto (gx,x)$ is surjective, is separated over S.*

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[493] We have corrected the original by assuming G universally open over S in a neighborhood of the unit section and by adding hypothesis (ii); see below for examples, due to O. Gabber, showing that statements 5.3 and 5.4 of the original are false without additional hypotheses. On the other hand, we note that Thm. 5.3 is a reworked version of Thm. 5.3A below, which appears in the 1965 edition of SGAD, and is due to M. Raynaud, cf. the Notes $(*)$ in Exp. X, 8.5 and 8.8.

[494] Without this hypothesis, one has the following counterexample, pointed out by O. Gabber: let $G = S$ be a local scheme with closed point s, such that $S* = S - s$ is not quasi-compact, and X the disjoint union of s and $S*$.

[495] We have added this remark, cf. N.D.E. (34).

Corollary 5.5. *Let S be a scheme, G an S-group scheme, locally of finite presentation, with connected fibers, and universally open over S. Then G is separated and of finite presentation over S.*[496]

Indeed, by 3.6 and 5.4, G is quasi-compact and separated over S, and since G is locally of finite presentation over S, it is therefore of finite presentation over S. ∎

5.6. Proof of Theorem 5.3.

Before establishing 5.3, let us prove a few lemmas.

Lemma 5.6.0. [497] *(i) Let $A \subset B$ be integral rings, with B integral over A, and let $q \in \mathrm{Spec}(B)$ be such that A is unibranch at the point $p = q \cap A$. Then the morphism $\pi : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is open at the point q.*

(ii) Let X, Y be two irreducible preschemes, $f : X \to Y$ a dominant morphism, x a point of X such that f is quasi-finite at x and $y = f(x)$ is a unibranch point of Y. Then f is open at the point x. In particular, if $Y = \mathrm{Spec}(O_{Y,y})$ is a local prescheme with closed point y, then $f(U) = Y$ for every neighborhood U of x.

Proof. (i) Let K (resp. L) be the fraction field of A (resp. B), A' the normalization of A, and B' the subring of L generated by A' and B. Then B' is integral over A'. Set $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $Y' = \mathrm{Spec}(A')$, $X' = \mathrm{Spec}(B')$, so that one has a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\pi'} & Y' \\ \big\downarrow & & \big\downarrow \varphi \\ X & \xrightarrow{\pi} & Y \end{array}$$

in which all morphisms are integral and surjective.

Since A is unibranch at p, Y' has a unique point p' above p; consequently, if U is an open neighborhood of p' in Y', then $\phi(Y' - U)$ is a closed subset not containing p, so that the complementary open is contained in $\phi(U)$. This shows that $\phi : Y' \to Y$ is open at p', and so it suffices to show that π' is open. We are thus reduced to the case where $A = A'$ is normal.

Let N be a quasi-Galois extension of K containing L, let $\tilde{X} = \mathrm{Spec}(\tilde{B})$, where $\tilde{B}$ is the integral closure of B in N, and let $\Gamma = \mathrm{Aut}(N/K)$. Since $\tilde{X} \to X$ is surjective, it suffices to show that $\tilde{\pi} : \tilde{X} \to Y$ is open. Let U be an open of $\tilde{X}$ and $U' = \bigcup_{g \in \Gamma} gU$. Since Γ acts transitively on the fibers of $\tilde{X} \to Y$ (cf. [BAC], Chap. V, § 2.3, Prop.

[496] Let us point out here the following result ([Ray70a], VI 2.5): if S is normal, G smooth with connected fibers, and if X is a (fppf) homogeneous G-space (i.e. the morphisms $X \to S$ and $G \times_S X \to X \times_S X$ are covering for the (fppf) topology), locally of finite type over S, then X is locally quasi-projective over S. In particular, G is quasi-projective over S. See also N.D.E. (35) in VI_A.

[497] We have added this lemma, communicated by O. Gabber, which improves EGA IV_3, 14.4.1.2 and corrects the proof of loc. cit., 14.4.1.3 without modifying its hypotheses (compare with the erratum (Err_IV, 38) in EGA IV_4).

6), $\tilde{\pi}(U)$ equals $\tilde{\pi}(U')$, and the latter equals the complementary open of the closed $\tilde{\pi}(\tilde{X} - U')$. This proves (i).

(ii) One may assume Y and X reduced. By Zariski's "Main Theorem" (cf. EGA IV_3, 8.12.9), there exist affine open neighborhoods U of x, and $V = \operatorname{Spec}(A)$ of y, such that $f(U) \subset V$, and a factorization:

$$\begin{tikzcd} U \arrow[r, "j"] \arrow[dr, "f"'] & V' \arrow[d, "u"] \\ & V, \end{tikzcd}$$

where j is an open immersion and u is finite. Replacing V' by the closure of $j(U)$, one may assume V' irreducible, hence $V' = \operatorname{Spec}(B)$, where B is an integral A-algebra, finite over A. Moreover, since f is dominant, so is u, so that the morphism $A \to B$ is injective. Since, by hypothesis, A is unibranch at the point y, it follows from (i) that u is open at the point $j(x)$, and hence $f = u \circ j$ is open at the point x. This proves the first assertion of (ii). The second follows, since if Y is a local prescheme with closed point y, every open containing y equals Y. (In the case where $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$, one may also use, instead of EGA IV_3, 8.12.9, the local form of Zariski's Main Theorem, which one finds, e.g., in [Pes66], or [Ray70b], Ch. IV, Th. 1.) $\square$

Lemma 5.6.1.0. [498] *Let $f : X \to S$ be a morphism locally of finite presentation, $s \in S$, and $x \in X_s$. Let $n = \dim_x(X_s)$ and let q be the structural morphism $\mathbb{A}^n_S \to S$.*

Suppose given a morphism $u : X \to \mathbb{A}^n_S$ quasi-finite such that $f = q \circ u$, and suppose f universally open at the generic point z of an irreducible component z of X_s, containing x and of dimension n. Then u is universally open at the point $x \in X$.

Let us note first that: (†) the hypotheses are preserved by any base change $\pi : S' \to S$ covering s (i.e., such that $\pi^{-1}(s) \neq \emptyset$). Indeed, let $S' \to S$ be such a morphism, let

$$\begin{tikzcd} X' \arrow[r, "u'"] \arrow[dr, "f'"'] & \mathbb{A}^n_{S'} \arrow[d, "q'"] \\ & S' \end{tikzcd}$$

be the diagram obtained by base change, and let s' be a point of S' above s and x' a point of $X'_{s'}$ above x. Since $X'_{s'} = X_s \otimes_{\kappa(s)} \kappa(s')$, by lifting of generizations and invariance of dimension under field extension (EGA IV_2, 2.3.4 (i) and 4.1.4), x' is contained in an irreducible component z' of $X'_{s'}$ whose generic point z' lies above

[498] We have made this lemma explicit, used in the proof of Lemma 5.6.1.

z, and one has $n \leq \dim z' \leq \dim_{x'} X'_{s'} \leq n$, whence $\dim z' = n = \dim_{x'} X'_{s'}$. Since f is universally open at x, f' is universally open at x'.

Set $Y = A^n_S$. By EGA IV₃, 14.3.3.1 (i), to prove that u is universally open at x, it suffices to prove that, for every integer $r \geq 0$ and every point x' of

$X' = X[T_1, \cdots, T_r]$ above x, the morphism $u' : X' \to Y' = A^n_S[T_1, \cdots, T_r]$ is open at the point x'. Now, with $S' = S[T_1, \cdots, T_r]$ and q' the projection $Y' \cong A^n_{S'} \to S'$, one is in the situation obtained by the base change $S' \to S$. So, by what precedes, one is reduced to showing that u is open at the point x.

Set $y = u(x)$. Since u is locally of finite presentation (since f and q are, cf. EGA IV₁, 1.4.3), then by EGA IV₁, 1.10.3, it suffices to show that $u(\operatorname{Spec} O_{X,x}) = \operatorname{Spec}(O_{Y,y})$. For this, one may assume S affine and integral. Let then $\pi : S' \to S$ be its normalization; denote by $u' : X' \to Y'$ the morphism deduced from u by base change. Since the morphism $\pi_Y : Y' \to Y$ is integral and surjective, one has

$\operatorname{Spec}(O_{Y,y}) = \bigcup_{y'} \pi_Y(\operatorname{Spec}(O_{Y', y'}))$,

the union being taken over all points of Y' above y; it therefore suffices to show that, for each such y', and every $x' \in X'_{y'}$ above x, one has $u'(\operatorname{Spec} O_{X',x'}) = \operatorname{Spec} O_{Y',y'}$. As the hypotheses are preserved by the base change $\pi : S' \to S$, one is thus reduced to the case of S', i.e., one may assume S integral and normal.

Now, the hypotheses on f imply, by EGA IV₃, 14.3.13, that there exists an irreducible component Z of X containing z (and hence x), dominating S and such that

$$\dim_z(Z_s) = n = \dim(Z_\eta),$$

where η is the generic point of S. Let ξ be the generic point of Z. Since u, and hence also $u_\eta : Z_\eta \to A^n_\eta$, is quasi-finite, the closure in A^n_η of the point $u_\eta(\xi) = u(\xi)$ is of dimension n, hence $u(\xi)$ is the generic point of A^n_η, which is also the generic point of $Y = A^n_S$. Consequently, denoting by g the restriction of u to Z, the morphism $g : Z \to Y$ is quasi-finite and dominant. Since Y is normal, it follows from Lemma 5.6.0 that g is open, so that u is open at every point of Z, in particular at the point x. Consequently, $u(\operatorname{Spec} O_{X,x}) = \operatorname{Spec} O_{Y,y}$, and this completes the proof of Lemma 5.6.1.0. □

The following lemma advantageously replaces EGA IV₃, 14.5.10,[499] in that it is independent of noetherian hypotheses.

Lemma 5.6.1. *Let S be a scheme, $f : X \to S$ an S-scheme locally of finite presentation, s a point of S, x a closed point of X_s. Suppose that f is universally open at the generic point z of an irreducible component z of X_s, containing x and such that* $\dim_x(X_s) = \dim z$. *Then there exists a commutative diagram:*

$$X \xrightarrow{f} S$$

[499]We have corrected the original, which indicated 19.5.10.

```
▲                 ▲
| h               | w
|        φ        |
| ↘             ↗
S″ ──────────→ S′
           π
```

where S' is an affine scheme, w an étale morphism, π a finite surjective morphism, of finite presentation, and $\phi^{-1}(s)$ is formed of a single point s'', such that $h(s'') = x$ and $\kappa(s'') = \kappa(x)$.[500]

Proof. First, one may assume $S = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$, where B is an A-algebra of finite presentation. Let p and q be the prime ideals of A and B corresponding to s and x, respectively, so that $p = A \cap q$. Set $n = \dim_x X_s$, and let $t_1, \cdots, t_n$ be elements of B whose images in $O_{X_s,x} = B_q/pB_q$ form a system of parameters. Then $O_{X_s,x}/(t_1, \cdots, t_n)$ is of finite dimension over $\kappa(x)$ and hence also over $\kappa(s)$, since x is a closed point of the $\kappa(s)$-algebraic scheme X_s. Consequently, the S-morphism

$$u : X \longrightarrow A^n_S = \operatorname{Spec}(A[T_1, \dots, T_n]),$$

defined by $T_i \mapsto t_i$, is of finite presentation, x is isolated in its fiber $u^{-1}(u(x))$, and one has $u(x) = \tau_0(s)$, where $\tau_0 : S \to A^n_S$ denotes the "zero section" of $A^n_S \to S$, corresponding to the morphism of A-algebras $A[T_1, \cdots, T_n] \to A$ that sends each T_i to 0. Since the set of points of X that are isolated in their fiber above A^n_S is open (EGA IV₃, 13.1.4), one may assume, shrinking X, that u is quasi-finite and that $u^{-1}(u(x)) = x$. By Lemma 5.6.1.0, u is universally open at the point x.

Let $B_0 = B/(t_1, \cdots, t_n)$ and let $X_0 = X \times_{A^n_S} \tau_0(S) = \operatorname{Spec} B_0$ and $u_0 : X_0 \to S$ be the morphism deduced from u by the base change $\tau_0 : S \hookrightarrow A^n_S$. Then u_0 is quasi-finite and of finite presentation, universally open at the point x, and x is the unique point of X_0 above s.

Let A' be the henselization of the local ring $A_p = O_{S,s}$, and let $S' = \operatorname{Spec}(A')$, $B'_0 = B_0 \otimes_A A'$, and $X'_0 = X_0 \times_S S' = \operatorname{Spec}(B'_0)$. Then the closed point s' of S' is the unique point of S' above s, one has $\kappa(s') = \kappa(s)$, and X'_0 has a unique point x' above s'; one has $\kappa(x') = \kappa(x)$ and x' is also the unique point of X'_0 above s and above x. Since A' is henselian, by EGA IV₄, 18.5.11, X'_0 is the disjoint sum of two open and closed parts:

$$(*) \qquad X'_0 = V \sqcup W, \qquad \text{where } V = \operatorname{Spec}(O_{X'_0, x'});$$

and the local ring $O_{X'_0,x'}$ is finite and of finite presentation over A'. The restriction π of u'_0 to V is therefore finite and of finite presentation. Moreover, since $u'_0 : X'_0 \to S'$ is open at x', one has $u'_0(V) = \operatorname{Spec}(O_{S',s}) = S'$, so that π is surjective.

This proves the desired result when $S = S'$. In the general case, A' is a filtered direct limit of subalgebras A_i étale over A, and such that $S_i = \operatorname{Spec}(A_i)$ has a unique point s_i above s (and $\kappa(s_i) = \kappa(s)$). Then $B'_0 = \lim B_i$, where $B_i = B_0 \otimes_A A_i$.

[500] Consequently, h induces a section σ of $X \times_S S'' \to S''$ such that $\sigma(s'')$ lies above x; compare with EGA IV₃, 14.5.10.

Set $X_i = \mathrm{Spec}(B_i) = X_0 \times_S S_i$ and $C = O_{X'_0,x'}$. By $(*)$ above, one has $C \cong B'_0/fB'_0$, for some idempotent $f \in B'_0$, and there exist an index i and $f_i \in B_i$ such that f is the image of f_i in B'_0. Set $C_i = B_i/f_iB_i$ and $V_i = \mathrm{Spec}(C_i)$.

Then $C = C_i \otimes_{A_i} A'$, whence $V = V_i \times_{S_i} S'$, and C_i is an A_i-algebra of finite presentation (since the same is true of B_i). Consequently, the morphisms:

$$\begin{tikzcd} X'_0 \arrow[r, "u'_0"] & S' \\ V \arrow[u, "h'"] \arrow[ur, "\pi"'] \end{tikzcd}$$

come, by the base change $S' \to S_i$, from morphisms of finite presentation:

$$\begin{tikzcd} X_i \arrow[r, "u_i"] & S_i \\ V_i \arrow[u, "h_i"] \arrow[ur, "\pi_i"'] \end{tikzcd}$$

For every $j \geq i$, let $V_j = V_i \times_{S_i} S_j$ and let $\pi_j : V_j \to S_j$ be the morphism (of finite presentation) deduced from π_i by base change. Since $\pi : V \to S'$ is finite and surjective, by EGA IV₂, 8.10.5, there exists an index j such that $\pi_j : V_j \to S_j$ is finite and surjective. Then $w : S_j \to S$ is étale affine, S_j has a unique point s_j above s, and V_j has a unique point x_j above s_j (since x' is the unique point of $V = V_j \times_{S_j} S'$ above s'):

$$\begin{tikzcd} X_j \arrow[r] & X_0 \arrow[r] & X \\ V_j \arrow[u, "u_j \; h_j"] \arrow[r, "\pi_j"'] & S_j \arrow[u, "u_0 \; f"] \arrow[r, "w"'] & S \arrow[u, "u \; f"] \arrow[r, "\tau_0"] & \mathbb{A}^n_S. \end{tikzcd}$$

Hence x_j is the unique point of V_j above s, its image under the morphism $V_j \to X_j \to X$ is x, and one has $\kappa(x_j) = \kappa(x)$. ∎

Lemma 5.6.2.0. [501] *Let k be a field and G a non-empty k-scheme of finite type.*

(i) Let K be an extension of k and W a dense open subset of G_K. Denote by π the projection $G_K \to G$; then

$$U = \{g \in G \mid W \text{ contains every maximal point of } \pi^{-1}(g)\}$$

is a dense open subset of G. (N.B. If g is a closed point of G belonging to U, then $\pi^{-1}(g) \subset W$.)

[501] We have added this lemma, communicated by O. Gabber. It allows us to simplify the proof of 5.6.2, and to prove Theorem 5.3, as well as 5.4, in a more general form, see 5.7 and 5.8 below.

(ii) Suppose moreover G geometrically irreducible. Let $\mu : G \times X \to Y$ be a morphism of k-schemes, x a point of X, and Ω an open of Y such that $\mu_x^{-1}(\Omega_{\kappa(x)}) \neq \emptyset$, where μ_x denotes the morphism $G_{\kappa(x)} \to Y_{\kappa(x)}$, $g \mapsto \mu(g, x)$. Then:

$$U = \{g \in G \mid \mu \text{ sends every maximal point of } g \times x \text{ into } \Omega\}$$

is a dense open subset of G, and for every closed point g of G belonging to U, one has $\mu(g \times x) \subset \Omega$ and hence $\mu(g', x) \in \Omega_{\kappa(x)}$ (resp. $\mu(g, x') \in \Omega_{\kappa(g)}$), for every point $g' \in G_{\kappa(x)}$ above g (resp. $x' \in X_{\kappa(g)}$ above x).

(iii) Suppose $G = H^0$, where H is a k-group scheme locally of finite type, acting on a non-empty k-scheme X in such a way that the morphism $H \times_S X \to X \times_S X$, $(h,x) \mapsto (hx, x)$ is surjective. Let U be an open of X. Then:

(a) $G \cdot U$ is an open of X, equal to the union of the irreducible components of X whose generic point belongs to U.

(b) Suppose U dense in every irreducible component of X. Then, for every finite subset A of X, there exists a closed point $g \in G$ such that $g \cdot A' \subset U_{\kappa(g)}$, where A' is the inverse image of A in $X_{\kappa(g)}$. In particular, the morphism $G \times U \to X$ is surjective.

Proof. (i) Let $\bar{k}$ be a separable closure of k and let L be an extension of k containing a copy of $\bar{k}$ and of K. Denote by π_L (resp. ϕ) the projection $G_L \to G$ (resp. $G_L \to G_K$). Since, for every $g \in G$, ϕ sends the maximal points of $\pi_L^{-1}(g)$ surjectively onto those of $\pi^{-1}(g)$, one sees that $U = \{g \in G \mid W_L \text{ contains every maximal point of } \pi_L^{-1}(g)\}$. So, replacing K by L, we reduce to the case where K contains $\bar{k}$.

Since the projection $p : G_K \to G_{\bar{k}}$ is surjective and open, $V = p(W)$ is a dense open of $G_{\bar{k}}$, and since $p^{-1}(g) = \mathrm{Spec}(\kappa(g) \otimes_{\bar{k}} K)$ is irreducible for every $g \in G_{\bar{k}}$ (cf. EGA IV_2, 4.3.3), then for every $g \in V$, the generic point of $p^{-1}(g)$ belongs to W. On the other hand, let $\Gamma = \mathrm{Gal}(\bar{k}/k)$; since the projection $q : G_{\bar{k}} \to G$ is surjective and Γ acts transitively on the fibers of q, then $U = q(V')$, where V' is the intersection of the Γ-conjugates of V.

Now, let Z be the closed $G_{\bar{k}} - V$, with its reduced closed subscheme structure. Since $G_{\bar{k}}$, and hence Z, is of finite presentation over $\bar{k}$, Z comes by base change from a reduced closed subscheme Z_1 of $G \otimes_k k_1$, for some finite Galois extension k_1 of k, so the Γ-conjugates of Z are finite in number, so that their union is again a rare closed F of $G_{\bar{k}}$, and $V' = G_{\bar{k}} - F$ is a dense open of $G_{\bar{k}}$. Hence, since the projection $q : G_{\bar{k}} \to G$ is surjective and open, $U = q(V')$ is a dense open of G. Moreover, for every closed point g of G, the fiber $\pi^{-1}(g)$ is formed of finitely many closed points of G_K, so if $g \in U$ then $\pi^{-1}(g) \subset W$. This proves (i), and (ii) follows from it.

On the other hand, (iii)(a) has been proved in ($\mathrm{VI_A}$, 2.6.4). Finally, if U is dense in every irreducible component of X, then $X = G \cdot U$, hence for every $x \in X$, $\mu_x^{-1}(U_{\kappa(x)})$ is a non-empty open of $G_{\kappa(x)}$, and then (iii)(b) follows from (ii). □

Corollary 5.6.2. *Let S, G, X be as in the preliminary hypotheses of 5.3, and let U be an open of X, $s \in S$, and A a finite part of X_s. Suppose U_s dense in X_s and X_s*

locally of finite type over $\kappa(s)$.[502]

Then there exists a morphism $f : S'' \to S$, composed of a finite surjective morphism $S'' \to S'$ and an étale morphism $S' \to S$, and a morphism $h : S'' \to G$, such that the inverse image A'' of A in $X \times_S S''$ (i.e. in $X_s \times_{\mathrm{Spec}\,\kappa(s)} S''_s$) is contained in $\ell_h^{-1}(U_{S''})$, where ℓ_h denotes the translation of $X_{S''} = X \times_S S''$ defined by the element $h \in G(S'')$.

Proof. Since X_s is locally of finite type over $\kappa(s)$, the connected components of X_s are open, and irreducible (cf. (VI_A, 2.5.4)), so U_s is dense in every irreducible component of X_s.

Hence, by Lemma 5.6.2.0, there exists a closed point $g \in G_s^0$ such that $g \cdot a' \in U_s \otimes_{\kappa(s)} \kappa(g)$ for every $a' \in X_{\kappa(g)}$ above a point of A.

By Lemma 5.6.1, there exists a morphism $f : S'' \to S$, composed of a finite surjective morphism $S'' \to S'$ and an étale morphism $S' \to S$, and a morphism $h : S'' \to G$, such that $f^{-1}(s)$ is formed of a single point s_0, and such that $h(s_0) = g$ and $\kappa(s_0) = \kappa(g)$. Then, denoting by A'' the inverse image of A in $X_s \times_{\mathrm{Spec}\,\kappa(s)} S''_s = X_s \otimes_{\kappa(s)} \kappa(s_0) = X_s \otimes_{\kappa(s)} \kappa(g)$, the translation ℓ_h of $X_{S''}$ sends A'' into $U_s \otimes_{\kappa(s)} \kappa(s_0)$. ∎

Lemma 5.6.3. *Let X be an S-scheme. The following conditions are equivalent:*

(i) X is separated over S.

(ii) For every S-scheme T, every section $\sigma : T \to X_T$ is a closed immersion.

(iii) For every reduced S-scheme T, two S-morphisms f_1 and $f_2 : T \to X$ that coincide on a dense open U of T are equal.

(iv) For every $s \in S$ and every pair of points x_1, x_2 of X_s, there exists a morphism $\phi : S'' \to S' \to S$ and an open subscheme V of $X_{S''}$, separated over S'', such that:

a) $S' \to S$ is open, $S'' \to S'$ closed surjective, and $\phi^{-1}(s) \neq \emptyset$.

b) The inverse image of x_1, x_2 in $X_{S''}$ is contained in V.

(iv′) For every $s \in S$, every pair of points $x_1, x_2 \in X_s$ is contained in an open V of X, separated over S.[503]

Proof. The implication (ii) ⇒ (i) is clear (take $T = X$ and σ = the diagonal section), as is (i) ⇒ (iv′) ⇒ (iv). On the other hand, one has (i) ⇒ (ii) by EGA I, 5.4.6.

(iii) ⇒ (ii). Let $\sigma : T \to X_T$ be a section of $p : X_T \to T$. By EGA I, 5.3.13, σ is an immersion, i.e. an isomorphism of T onto a locally closed subscheme E of X_T. To show that E is closed, one may assume T and E reduced. Let Ē be the reduced closed subscheme of X_T having the closure of E as underlying space,

[502] We have added the hypothesis on X_s and simplified (and corrected) the proof, taking into account the addition 5.6.2.0. Moreover, the proof shows that the conclusion is valid if one assumes only that U_s is dense in every irreducible component of X_s.

[503] We have simplified the formulation of condition (iv) and added condition (iv′). On the other hand, we have added the proof of the implication (i) ⇒ (iii), used in the proof of (iv) ⇒ (iii). Note moreover that if $T = \mathrm{Spec}\,k[\epsilon, x]/(\epsilon^2, \epsilon x)$ (k a field), $X = T_{red} = \mathrm{Spec}\,k[x]$, then the morphisms $\phi_\lambda : T \to X$ defined by $x \mapsto x + \lambda\epsilon$ ($\lambda \in k$) coincide on the dense open $\mathrm{Spec}\,B_x$ but are not equal.

so that E is a dense open subscheme of $\bar{E}$. Then the immersion $i : \bar{E} \hookrightarrow X_T$ and $\sigma \circ p \circ i$ coincide on E, hence on $\bar{E}$ by hypothesis (iii). Hence every point of $\bar{E}$ belongs to $\sigma(T) = E$, whence $E = \bar{E}$.

(i) $\Rightarrow$ (iii). Suppose X separated over S and let T be a reduced S-scheme, f_1, f_2 two S-morphisms $T \to X$ that coincide on a dense open U of T, and g the morphism $T \to X \times_S X$ with components f_1 and f_2. Since $D = \Delta_{X/S}(X)$ is closed, its inverse image under g is a closed subset of T containing U, hence equal to T, and since T is reduced, g factors through D (cf. EGA I, 5.2.2); consequently $f_1 = p_1 \circ g$ equals $p_2 \circ g = f_2$.

(ii) $\Rightarrow$ (iii). Let T be a reduced S-scheme and f_1, f_2 two S-morphisms $T \to X$ that coincide on a dense open U. Since T is reduced, to see that $f_1 = f_2$, it suffices to see that $f_1 = f_2$ set-theoretically. Indeed, suppose this established, and let $t \in T$, V an affine open of X containing $f_1(t) = f_2(t)$, and W the open neighborhood of t equal to the inverse image of V under the continuous map underlying f_1 and f_2; then the morphisms $f_i|_W : W \to V$ coincide on the dense open $U \cap W$ of W. Since V is separated and W reduced, this entails that $f_1|_W = f_2|_W$, whence $f_1 = f_2$.

Let then $t \in T$ and s its image in S; let us show that the points $x_1 = f_1(t)$ and $x_2 = f_2(t)$ of X_s are equal. Let $\phi : S'' \to S' \to S$ and V an open of $X \times_S S''$ as in (iv); set $T' = T \times_S S'$ and $T'' = T \times_S S''$ and denote by $g : T'' \to T$ and $f_i'' : T'' \to X_{S''}$ ($i = 1, 2$) the morphisms obtained by base change.

Since U is dense in T and $T' \to T$ is open, the inverse image U' of U in T' is dense in T'. Let U'' be the inverse image of U' in T'' and let F be the reduced subscheme of T'' having U'' as underlying space. Since $T'' \to T'$ is surjective and closed, the image of F contains U' and is closed, hence equals T'. Consequently, $F \cap g^{-1}(t)$ contains a point u.

For $i = 1, 2$, denote by h_i the restriction to F of f_i''. Then $h_i(u)$ is a point of $X_{S''}$ above $f_i(t) = x_i$, hence belongs to V, since V contains the inverse image of x_1, x_2 in $X_{S''}$.

Then $W = h_1^{-1}(V) \cap h_2^{-1}(V)$ is an open of F, containing u, and the S''-morphisms $h_i|_W : W \to V$ coincide on the dense open $U'' \cap W$ of W. Since V is separated over S'', one has $h_1|_W = h_2|_W$, whence $h_1(u) = h_2(u)$, and hence $f_1(t) = f_2(t)$. This proves (iv) $\Rightarrow$ (iii). ∎

Theorem 5.3 then follows from 5.6.2 and the implication (iv) $\Rightarrow$ (i) of 5.6.3. ∎

Counterexample 5.6.4. *Not every S-group G is separated. Let S be a scheme having a non-isolated closed point s; let G be the scheme obtained by glueing two copies of S along the open $S - s$; one sees easily that G is not separated over S, and that it is equipped with a natural structure of S-group, all of whose fibers are trivial, except the fiber G_s which is isomorphic to the two-element group $\mathbb{Z}/2\mathbb{Z}$.*[504]

[504]We have reproduced this example in (VI_A, 0.3), N.D.E. (5).

Theorem 5.7. [505] *Let S be a scheme, G an S-group locally of finite presentation over S such that the function $s \mapsto \dim G_s$ is locally constant on S, X an S-scheme on which G acts, and U an open of X, separated over S. Then $G^0 \cdot U$ is an open of X, separated over S.*

Proof. Denote by $\mu : G \times_S X \to X$ the action of G on X; it is the composition of the automorphism $(g,x) \mapsto (g,gx)$ of $G \times_S X$ and the projection onto X. Since $G \times_S U$ is an open of $G \times_S X$, by 4.2 (iii), $V = G^0 \cdot U$ is open in X.

Then, proceeding as in the proof of 5.6.2, one deduces from the implication (iv) ⇒ (i) of 5.6.3 that V is separated over S. ∎

Corollary 5.7.1. *Let S and G be as in 5.7, and let σ, τ be two S-sections of G^0 (i.e. $\sigma, \tau \in G^0(S)$). Then the coincidence subscheme $S(\sigma,\tau) \subset S$ of σ and τ (i.e. the inverse image of the diagonal of $G \times_S G$ under the morphism (σ,τ)) is closed.*

Indeed, for every $s \in S$, let U be an affine open of G containing $\epsilon(s)$; then $V = \epsilon^{-1}(U)$ is an open neighborhood of s in S, and since $G^0 U$ is separated, $S(\sigma,\tau) \cap V$ is closed in V. ∎

Remark 5.7.2. *Gabber points out to us that one can show that if S is henselian local, with closed point s, then the intersection of the opens $G^0 U$, where U runs through the affine open neighborhoods of $\epsilon(s)$, is an open group subscheme of G, separated over S.*

5.8. Complements.

[506] Let us begin by recalling Proposition (VI_A, 2.6.6):

Proposition 5.8.1. *Let k be a field, G a k-group locally of finite type acting on a k-scheme X in such a way that the morphism $G \times X \to X \times X$, $(g,x) \mapsto (gx,x)$ is surjective. Suppose X quasi-separated. Then the connected components of X are irreducible.*

Corollary 5.8.2. *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that every fiber X_s is quasi-separated and connected. Then X is separated and quasi-compact over S.*

Indeed, by the preceding proposition, each fiber X_s is irreducible, and the rest of the proof is identical to that of 5.4. ∎

Example 5.8.3. *Fix an algebraically closed field k. Recall first that every "locally Boolean" topological space X (i.e. having a basis of compact open subsets),*

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[505] On the one hand, we have suppressed the corollary 5.6.5, which was a repetition of 5.5. On the other hand, the original stated as Remark 5.7 the corollary 5.7.1 below, referring for the proof to 4.7, a number which does not exist in Lect. Notes 151 (but which appeared in the 1965 edition of SGAD, whose nos. 4.5 and 4.6 became 5.6.1 and 5.6.2); we have added Theorem 5.7, communicated by O. Gabber, which makes precise the aforementioned statement in SGAD.

[506] We have added this paragraph of complements and counterexamples, all communicated by O. Gabber.

equipped with the sheaf of locally constant functions with values in k, is a k-scheme (cf. [DG70], I § 1, 2.12). One then says that X is a locally Boolean k-scheme.

Note moreover that every topological space E admitting a basis of separated opens (and likewise every scheme X) admits a dense separated open. Indeed, since every increasing union of separated opens is a separated open (for a scheme X this follows from 5.6.3), there exists such an open U which is maximal. But then U is dense, for if there existed a non-empty open V such that $U \cap V = \emptyset$, then V would contain a non-empty separated open W, and $U \cup W$ would still be separated, contradicting the maximality of U.

Now, let C be the Cantor space, viewed as the underlying space of the group $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$ equipped with the product topology. For every point p of C, let $C(p)$ be another copy of C, and let X be the space obtained by glueing each $C(p)$ to C along $C - p$; then X is a non-separated locally Boolean k-scheme, and C is a dense open of X.

Let G be the group of automorphisms of the k-scheme X (i.e., of homeomorphisms of X). Then G acts transitively on X. Indeed, X is the union of C and, for each point $p \in C$, of a second point p', which can be characterized as the unique point x of $X - p$ such that $(C - p) \cup x$ is separated. It follows that every automorphism ϕ of C extends to an automorphism ϕ_X of X such that $\phi_X(p') = \phi_X(p)'$ for every p. On the other hand, the map $\theta_p : X \to X$ that exchanges p and p' and fixes the other points is an automorphism of X. Finally, the group of automorphisms of C acts transitively on C: for example, using the group structure of C, it suffices to consider translations.

Hence the discrete k-group G (hence locally of finite type) acts on the k-scheme X in such a way that the morphism $G \times X \to X \times X$, $(g, x) \mapsto (gx, x)$ is surjective, but X is not separated (although C is a dense separated open).

Example 5.8.4. *We retain the notations of the preceding example. Using the description of C as the set of sequences $(u_n)_{n\in\mathbb{N}}$ of elements of $\mathbb{Z}/2\mathbb{Z}$, one sees that C minus a point p is homeomorphic to a countable disjoint union of copies of C. Indeed, using e.g. the group structure of C, one reduces by translation to the case where p is the element 0, i.e. the zero sequence; then $C - 0$ is the disjoint union of the subspaces $C_i = (u_n)_{n\in\mathbb{N}} | u_i = 1 and u_j = 0 for j < i$, for $i \in \mathbb{N}$, each homeomorphic to C. For every non-empty finite subset F of C, one deduces, by induction on $|F|$, that $C - F$ is homeomorphic to a countable disjoint union of copies of C, hence to C minus a point.*

For each F of cardinality 2, let $C(F)$ be another copy of C, denote by q_F the point 0 of $C(F)$, and choose a homeomorphism $\phi_F : C(F) - q_F \xrightarrow{\sim} C - F$; let then X' be the space obtained by glueing each $C(F)$ to C by means of ϕ_F. Then X' is a non-separated locally Boolean k-scheme. Moreover, it follows from the construction that every locally constant function $f : X' \to k$ is constant. Indeed, if $x, y \in C$ and $F = x, y$, every neighborhood of q_F meets every neighborhood of x or y, so if $f : X \to k$ is locally constant, one has $f(x) = f(q_F) = f(y)$, and if $F' = z, t$ with $z, t \in C$, one *likewise has $f(q_{F'}) = f(z) = f(q_{z,x}) = f(x)$. Consequently, X' is connected.*

Moreover, every point $x \in X'$ has a pointed neighborhood homeomorphic to `(C, 0)`*. More precisely, fix for each F a homeomorphism of pointed topological spaces $h_F : (C, 0) \xrightarrow{\sim} (C(F), q_F)$, and denote by T the group of translations of C. Then, if $x = q_F$ one has the homeomorphism q_F, and if $x \in C$, the translation $t_x \in T$ is a homeomorphism $(C, 0) \xrightarrow{\sim} (C, x)$ (and it is the unique element of T having this property).*

Denote by L the free group generated by the h_F and let $G = T * L$ be the "free product" (= coproduct) of T and L. For every $h \in h_{F_{|F|=2}} \cup T$, let $\sigma(h)$ be the generator of G corresponding to h and let $S(h)$ (resp. $B(h)$) be the source (resp. target) of h. It is convenient to also set $\sigma(h_F^{-1}) = \sigma(h_F)^{-1}$ and $S(h_F^{-1}) = B(h_F)$ (resp. $B(h_F^{-1}) = S(h_F)$), and to denote by E the set formed by T, and the h_F and h_F^{-1}.

On the product space $P = G \times X'$ (where G is endowed with the discrete topology), consider the equivalence relation generated by the relations:

`(gσ(h), x) ~ (g, h(x))     when x ∈ S(h)`

for every $h \in E$, and let Z be the quotient space. Then Z is obtained from the disjoint union $\coprod_{g \in G} g \times X'$ by glueing of opens, and hence, for every open Ω of P, its saturate $\tilde{\Omega}$ is open (cf. [BTop], I § 5.1, Example 2). Explicitly, since every open of P is the union of its intersections with the "slices" $g \times X'$, it suffices to consider an open of the form $1 \times W$, where W is an open of X'. In this case, the saturate is the union of $1 \times W$, and of

`{σ(h_1)} × h_1⁻¹(W ∩ B(h_1)),     {σ(h_1) σ(h_2)} × h_2⁻¹(h_1⁻¹(W ∩ B(h_1)) ∩ B(h_2)),`

etc., for all finite sequences $h_1, \cdots, h_n$ of elements of E, hence is open. Therefore the projection $\pi : P \to Z$ is open.

Note moreover that the word $\sigma(h_1) \cdots \sigma(h_n)$ is a reduced word of G, except if one of the h_i is the neutral element of T or if two consecutive h_i belong to T, or if (h_i, h_{i+1}) equals (h_F, h_F^{-1}) or (h_F^{-1}, h_F). So, if $x \in X'$ and if an element $\beta = (\sigma(h_1) \cdots \sigma(h_n), h_n^{-1} \cdots h_1^{-1}(x))$ belongs to $1 \times X'$, then one may assume that each h_i is a translation t_i, and in this case the equality $\sigma(t_1 \cdots t_n) = 1$ entails that $t_1 \cdots t_n = 1$, and hence $\beta = (1, x)$. Since the equivalence relation is compatible with the action of G (acting on $P = G \times X'$ by left translations on the first factor), one deduces that for every $g \in G$, the restriction of π to $g \times X'$ is injective.

Let then $z \in Z$ be arbitrary and let $(g, x) \in P$ be a representative of z. By what precedes, $U = \pi(g \times X')$ is an open neighborhood of z, and the continuous map $g \times X' \to U$ induced by π is open and bijective, hence a homeomorphism. This shows that Z is locally isomorphic to X' (hence also to C), and is therefore still a locally Boolean k-scheme.

Finally, G acts transitively on Z. Indeed, since every $z \in Z$ is G-conjugate to an element of the form $\pi((1, x))$, it suffices to see that every element $(1, x) \in P$ is equivalent to an element $(\sigma(h), 0)$; now if $x \in C$ one may take for h the translation

t_x*, and if* $x = q_F$ *one may take* $h = h_F$*. Hence* Z *is a locally Boolean* k*-scheme equipped with a transitive action of the discrete group* G*, but* Z *is not separated.*

6. Subfunctors and group subschemes[507]

Definition 6.1. *(i) Let* X *be an* S*-functor (i.e., a functor from* $(Sch/S)^\circ$ *into* (Ens)*),* G *an* S*-functor in groups,* u *and* v *two* S*-morphisms from* X *to* G*. The* transporter of u into v*, denoted* $Transp(u,v)$*, is the sub-*S*-functor of* G *defined as follows:*

$$\begin{aligned} Transp(u, v)(S') &= \{g \in G(S') \mid (\mathrm{int}\, g) \circ u_{S'} = v_{S'}\} \\ &= \{g \in G(S') \mid g_{S''}\, u_{S''}(x)\, g_{S''}^{-1} = v_{S''}(x),\ \forall\, x \in X(S''),\ S'' \to S'\}. \end{aligned}$$

In particular, $Transp(u,u)$ *is a sub-*S*-functor in groups of* G*; it is called the* centralizer of u *and is denoted* $Centr(u)$*.*

(ii) Let G *be an* S*-functor in groups,* X *and* Y *two sub-*S*-functors of* G*. The* transporter of X into Y *(resp. the* strict transporter of X into Y*), denoted* $Transp_G(X,Y)$ *(resp.* $Transpstr_G(X,Y)$*), is the sub-*S*-functor of* G *defined as follows:*

$$\begin{aligned} Transp_G(X, Y)(S') &= \{g \in G(S') \mid (\mathrm{int}\, g)(X_{S'}) \subset Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''}\, X(S'')\, g_{S''}^{-1} \subset Y(S''),\ \forall\, S'' \to S'\} \end{aligned}$$

$$\begin{aligned} Transpstr_G(X, Y)(S') &= \{g \in G(S') \mid (\mathrm{int}\, g)(X_{S'}) = Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''}\, X(S'')\, g_{S''}^{-1} = Y(S''),\ \forall\, S'' \to S'\}. \end{aligned}$$

Note that one has

$$Transpstr_G(X, Y) = Transp_G(X, Y) \cap c(Transp_G(Y, X)),$$

where c *denotes the inversion morphism of* G*.*[508]

(iii) Let G *be an* S*-functor in groups,* H *a sub-*S*-functor of* G*,* i *the canonical* S*-morphism* $H \to G$*; the* centralizer *and* normalizer *of* H *in* G *are the following sub-*S*-functors in groups of* G*:*

$$Centr_G H = Centr(i) = Transp(i, i), \qquad Norm_G H = Transpstr_G(H, H).$$

Finally, the center *of* G *is the* S*-functor in groups* $Centr(id_G) = Centr_G G$*; it will be denoted* $Centr\ G$*.*

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[507] On the same theme, see also the results of XI 6, whose natural place would be in the present Exposé VI_B. There one finds in particular representability criteria for certain subfunctors-in-groups of a given group scheme.[^N.D.E-VI_B-48]

[508] If $u : X \to G$ and $v : Y \to G$ are two arbitrary morphisms of S-functors, let $u(X)$ be the image-functor of u, defined by $u(X)(S') = u(S')(X(S')) \subset G(S')$; this is a subfunctor of G, as is the image-functor $v(Y)$. One may then consider the transporter of the image of u into the image of v, $Transp_G(u(X), v(Y))$. We see thus that, in definition (ii), it is not necessary to restrict to subfunctors, i.e. to the case where u and v are monomorphisms. This restriction sometimes imposed in the original repetitions in the hypotheses, such as: "Let $u, v : X \to G$ be morphisms of S-functors, $w : H \to G$ and $w' : K \to G$ monomorphisms, then $Transp(u,v)$, $Centr_G(w) = Centr_G H$ and $Transp_G(H,K)$ verify...", which can be avoided by considering $Transp_G(u(X), v(Y))$. We have made such modifications in 6.2 and, later, in 10.11.

Remark 6.1.1. [509] It follows from the definitions that the functors $Transp(u,v)$ and $Transp_G(X,Y)$ (and hence also $Transpstr_G(X,Y)$) commute with base change: for every $S' \to S$, if G', X', Y', u', v' are deduced from G, X, Y, u, v by base change, then

$$Transp(u, v)_{S'} = Transp(u', v') \quad \text{and} \quad Transp(X, Y)_{S'} = Transp(X', Y').$$

Proposition 6.2. [510] *Let G be an S-group. For a subfunctor T of the functor G, consider the following property:*

(+f) for every $s \in S$, T_s is representable by a closed subscheme of G_s.

Let $u, w : X \to G$ and $v : Y \to G$ be morphisms of S-schemes. Then:

(i) $Transp(u,w)$ and $Centr(u) = Transp(u,u)$ satisfy condition $(+f)$.

(ii) $Transp_G(u(X), v(Y))$ and $Norm_G v(Y)$ satisfy condition $(+f)$ if, for every $s \in S$, v_s is a closed immersion.

(iii) $Transpstr_G(X,Y)$ satisfies condition $(+f)$ if, for every $s \in S$, u_s and v_s are closed immersions.

This follows from Remark 6.1.1 and from Corollary 6.2.5 below.[511]

Definition 6.2.1. *Let $f : X \to S$ be a morphism of schemes. One says that f is* essentially free, *or that X is* essentially free over S, *if one can find a covering of S by affine open sets S_i, for every i an S_i-scheme S'_i affine and faithfully flat over S_i, and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine open sets X'_{ij}, such that for every (i,j), the ring of X'_{ij} is a projective*[512] *module over the ring of S'_i.*

Proposition 6.2.2. *a) If X is essentially free over S, it is flat over S; the converse is true if S is Artinian.*

b) If S is the spectrum of a field, every S-scheme is essentially free over S.

c) If X is essentially free over S, then $X' = X \times_S S'$ is essentially free over S', for every $S' \to S$. The converse is true if $S' \to S$ is faithfully flat and quasi-compact.

The proof is immediate; for the converse in a) one uses the fact that a flat module over an Artinian local ring is free.[513]

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[509] We have added this remark, used implicitly in Proposition 6.2.

[510] We have modified the statement, as indicated in N.D.E. (49).

[511] The original referred to the results of Exp. VIII, § 6. For the reader's convenience, we have reproduced these results (with the exception of VIII, 6.3 and 6.8) in nos. 6.2.1 to 6.2.5 below. Moreover, this was suggested by A. Grothendieck in a Note at the beginning of VIII.6: "The present number is independent of the theory of diagonalizable groups; its natural place would be in VI_B."

[512] On the one hand, we have replaced the word "free" by "projective", as indicated in VIII, Remark 6.8. On the other hand, the notion of essentially free S-scheme is intimately related to the geometric notion of flat and pure S-scheme, introduced and studied by M. Raynaud and L. Gruson ([RG71]); cf. the addition 6.9 below.

[513] Indeed, let $(A, \mathfrak{m})$ be an Artinian local ring, k its residue field, M an arbitrary A-module, $(x_i)_{i \in I}$ elements of M whose images form a basis of $M/\mathfrak{m}M$ over k. Let F be the free A-module with basis $(e_i)_{i \in I}$, and $\phi : F \to M$ the A-morphism defined by $\phi(e_i) = x_i$. Then $Q = Coker\phi$ satisfies $Q = \mathfrak{m}Q$, hence, since $\mathfrak{m}$ is nilpotent, $Q = 0$. Suppose moreover M flat over A; then $K = Ker\phi$ satisfies $K \otimes_A k = 0$,

The introduction of Definition 6.2.1 is justified by the following theorem.

Theorem 6.2.3. *Let S be a scheme, Z an essentially free S-scheme, Y a closed subscheme of Z. Consider the functor $F = \prod_{Z/S} Y/Z : (Sch)^{\circ}_{/S} \to (Ens)$ defined by the following condition: $F(S') = \emptyset$ when $Y_{S'} \neq Z_{S'}$, and $F(S')$ is reduced to a single element otherwise.*[514]

(i) This functor is representable by a closed subscheme T of S.

(ii) If moreover $Y \to Z$ is of finite presentation, then so is $T \to S$.

Let us first note that the functor under consideration is a sheaf for the (fpqc) topology: since $F(S') = \emptyset$ or $\{\mathrm{pt}\}$ for every S', this reduces to verifying that if (S_i) is an open covering of S (resp. $S' \to S$ a faithfully flat and quasi-compact morphism), and if each $Y_{S_i} \to Z_{S_i}$ (resp. $Y_{S'} \to Z_{S'}$) is an isomorphism, then so is $Y \to Z$; this is clear (resp. follows from SGA 1, VIII 5.4 or EGA IV_2, 2.7.1).

Moreover, by SGA 1, VIII 1.9, faithfully flat and quasi-compact morphisms are of effective descent for the fibered category of closed-immersion arrows. This allows us, with the notations of 6.2.1, to limit ourselves to the case where $S = S'_i$.

Let then (Z_j) be a covering of Z by affine open sets such that $\mathcal{O}(Z_j)$ is a projective module over $A = \mathcal{O}(S)$, and let $Y_j = Y \cap Z_j$ and $F_j : (Sch)^{\circ}_{/S} \to (Ens)$ be the functor defined in terms of (Z_j, Y_j) as F is in terms of (Z, Y). It is a subfunctor of the final functor, and one obviously has $F = \bigcap_j F_j$, which reduces us to proving that each F_j is representable by a closed subscheme T_j of S (for then F will be representable by the closed subscheme T intersection of the T_j). We may therefore also suppose Z affine, $Z = \mathrm{Spec}(B)$, where B is a projective A-module. Let L be a free A-module with basis $(e_\lambda)_{\lambda \in \Lambda}$, of which B is a direct factor as an A-module, and let $\phi_\lambda : L \to A$ be the "coordinate" forms relative to this basis. Let E be a generating set of the ideal J of B defining the subscheme Y of Z, and let I be the ideal in A generated by the coordinates $\phi_\lambda(x)$, for $x \in E$. For every A-algebra C, one sees then that the morphism $B \otimes_A C \to (B/J) \otimes_A C$ is an isomorphism if and only if the image of $x \otimes 1$ in $L \otimes_A C$ is zero for every $x \in E$, which amounts to saying that the kernel of $A \to C$ contains the ideal I. This shows that $T = V(I) = \mathrm{Spec}(A/I)$ satisfies the desired condition, which proves (i).

Moreover, if $Y \to Z$ is of finite presentation, one may take E finite, and then I is a finitely generated ideal of A, i.e. the closed immersion $T \to S$ is of finite presentation.

Examples 6.2.4. *Let us give important examples of functors which reduce to functors $\prod_{Z/S} Y/Z$ of the type considered in 6.2.3 and for which it is useful to have representability criteria in what follows. We denote by S a scheme, and by X, Y, Z, etc., schemes over S.*

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i.e. $K = \mathfrak{m}K$, hence $K = 0$.

[514] cf. Exp. II § 1, where this functor is denoted $\prod_{Z/S} Y$; for every $S' \to S$, $F(S') = \Gamma(Y_{S'}/Z_{S'})$, which here equals $id_{Z_{S'}}$ if $Y_{S'} = Z_{S'}$, and is empty otherwise. On the other hand, we have added assertion (ii), used in the proof of 6.5.3.

a) Suppose given an S-morphism

$$(x) \qquad q : X \to \mathrm{Hom}_S(Y, Z),$$

(X acts on Y, with values in Z), i.e. a morphism

$$(xx) \qquad r : X \times_S Y \to Z.$$

Consider a subscheme Z' of Z, whence a monomorphism

$$\mathrm{Hom}_S(Y, Z') \to \mathrm{Hom}_S(Y, Z)$$

which makes the first functor a subfunctor of the second; let X' be the inverse image of this subfunctor under the morphism q. This is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \to Z_T$ factors through Z'_T. This functor X' can be described as follows: set $P = X \times_S Y$, let P' be the inverse image of Z' under $r : P \to Z$; then one has an obvious isomorphism

$$(xxx) \qquad X' \simeq \textstyle\prod_{P/X} P'/P.$$

One thus obtains: if Y is essentially free over S and Z' closed in Z, the subfunctor X' of X is representable by a closed subscheme of X.[515] *If moreover $Z' \to Z$ is of finite presentation, then so is $X' \to X$.*

b) Suppose given two ways of letting X act on Y with values in Z, i.e. two morphisms

$$q_1, q_2 : X \rightrightarrows \mathrm{Hom}_S(Y, Z),$$

and set $X' = Ker(q_1, q_2)$: this is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms $q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$ are equal. Now the data of q_1, q_2 is equivalent to the data of a morphism

$$q : X \to \mathrm{Hom}_S(Y, Z \times_S Z),$$

or equivalently of a morphism $r : X \times_S Y \to Z \times_S Z$; set then $U = Z \times_S Z$, let U' be the diagonal subscheme of $Z \times_S Z$. Then X' is nothing but the inverse image of the subfunctor $\mathrm{Hom}_S(Y, U')$ of $\mathrm{Hom}_S(Y, U)$ under q, hence can be put in the form (xxx), with $P = X \times_S Y$, and P' = inverse image of the diagonal under r, i.e. kernel of $r_1, r_2 : X \times_S Y \rightrightarrows Z$. One is therefore in the conditions of (a).

One sees consequently that: if Y is essentially free over S and Z separated over S, then the subfunctor X' of X is representable by a closed subscheme of X.[516] *If moreover $Z \to S$ is locally of finite type, then $X' \to X$ is of finite presentation.*

c) Suppose given a morphism

$$q : X \to \mathrm{Hom}_S(Y, Y),$$

i.e. "X acts on Y". Let X' be the "kernel" of this morphism, i.e. the subfunctor X' of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \to Y_T$ is the identity. This functor falls under b), as one sees by introducing a second homomorphism

[515] We have added the sentence that follows.
[516] We have added the sentence that follows.

q′ : X → Hom_S(Y, Y)

"by letting X act trivially on Y". Hence: if Y is essentially free and separated over S, the subfunctor X' of X kernel of q is representable by a closed subscheme of X.[517] *If moreover $Y \to S$ is locally of finite type, then $X' \to X$ is of finite presentation.*

d) Under the conditions of c), consider the subfunctor Y' of Y "of the invariants under X", so that $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $q(y) : X_T \to Y_T$ is "the constant T-morphism with value y". Introducing q' as in c), and the homomorphisms corresponding to q and q':

q, q′ : Y ⇒ Hom_S(X, Y),

one sees that Y' is precisely $Ker(q,q')$, and hence falls again under b) (with the roles of X, Y reversed and $Z = Y$).

Consequently, if X is essentially free over S and Y separated over S, then the subfunctor Y' of Y of the invariants under X is representable by a closed subscheme of Y.[518] *If moreover $Y \to S$ is locally of finite type, then $Y' \to Y$ is of finite presentation.*

e) Constructions of the type made explicit in the preceding examples are particularly frequent in group theory. Thus, when G is an S-group scheme acting on the S-scheme X:

$$q : G \to \mathrm{Aut}_S(X),$$

the kernel of q ("the subgroup of G acting trivially") is a closed subscheme of G provided X is essentially free and separated over S (example c)), and the subobject X^G of invariants is a closed subscheme of X, provided G is essentially free over S and X separated over S (example d)).

Let Y, Z be subschemes of X; consider the subfunctor $Transp_G(Y,Z)$ of G ("transporter of Y into Z"), whose points with values in a T over S are those $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subset Z_T$, i.e. induces a morphism $Y_T \to X_T$ factoring through $Y_T \to Z_T$. Hence: if Y is essentially free over S, and Z closed in X, then $Transp_G(Y,Z)$ is a closed subscheme of G (example a)).

One may also consider the strict transporter of Y into Z,[519] *whose points with values in a T over S are those $g \in G(T)$ such that $g(Y_T) = Z_T$, which is nothing but $Transp_G(Y,Z) \cap \sigma(Transp_G(Z,Y))$, where σ is the inversion morphism of G. Consequently,*

if Y and Z are essentially free over S and closed in X, the strict transporter of Y into Z is a closed subscheme of G.

An important case is the one where $X = G$, with G acting on itself by inner automorphisms. If H is a subscheme of G, the strict transporter of H into H is also called

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[517] We have added the sentence that follows.

[518] We have added the sentence that follows.

[519] denoted $Transpstr_G(Y,Z)$.

the normalizer of H in G*, and denoted* $Norm_G H$*. Hence: if* H *is a closed group subscheme of* G*, essentially free over* S*, then* $Norm_G H$ *is representable by a closed group subscheme of* G*.*

Let finally Z *be a subscheme of* G*; then its centralizer* $Centr_G(Z)$ *in* G *is the subfunctor in groups of* G *defined by the procedure of d), when one considers that "*Z *acts on* G*" by the operations induced by those of* G*; hence if* Z *is essentially free over* S *and* G *is separated over* S*,* $Centr_G(Z)$ *is a closed group subscheme of* G*. In particular, if* G *is essentially free and separated over* S*, then the center* C *of* G*, which is none other than* $Centr_G(G)$*, is a closed group subscheme of* G*.*

When S is the spectrum of a field, 6.2.2 b) shows that in examples a) to e) above, the "essentially free" conditions are automatically satisfied; only the separation conditions remain. Recalling that a group scheme over a field is necessarily separated (VI_A, 0.3), one finds for example:

Corollary 6.2.5. *Let* G *be a group scheme over a field* k *and let* Y, Y' *be two subschemes of* G*. Then:*

(i) The centralizer of Y *in* G *is a closed group subscheme of* G*; in particular, this is the case for the center* $Centr_G(G)$ *of* G*.*

(i′) More generally, if $u, v : X \to G$ *are morphisms of schemes,* $Transp_G(u, v)$ *is representable by a closed subscheme of* G*.*

(ii) If Y *is closed, the transporter* $Transp_G(Y', Y)$ *is a closed subscheme of* G*. If* Y' *is also closed, the same conclusion holds for* $Transpstr_G(Y', Y)$*.*

(iii) For every group subscheme[520] H *of* G*,* $Norm_G(H)$ *is a closed group subscheme of* G*.*

Corollary 6.2.6. [521] *Let* k *be a field,* G *a connected algebraic* k*-group. Then* $Centr_G(G)$ *is representable by a closed group subscheme* Z *of* G*, and* G/Z *is an affine algebraic* k*-group.*

Proof. The first assertion is of course contained in 6.2.5 (i), but we shall see that it also follows from the proof of the second assertion. Indeed, G acts by the adjoint representation on the finite-dimensional k-vector spaces $V_n = \mathcal{O}_{G,e}/\mathfrak{m}_e^{n+1}$ (where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$); let K_n denote the kernel of the morphism $\rho_n : G \to GL(V_n)$. By VI_A, 5.4.1, ρ_n induces a closed immersion $G/K_n \hookrightarrow GL(V_n)$, hence each G/K_n is affine. Since G is noetherian, the intersection K of the K_n equals one of the K_n, so G/K is affine.

On the other hand, letting Z denote the center of G, it is clear that $Z \subset K$. Let us show that $Z = K$. Let $\hat{\mathcal{O}}_{G,e}$ denote the completion of $\mathcal{O}_{G,e}$ for the $\mathfrak{m}_e$-adic topology and Ŝ its spectrum (resp. $S = \operatorname{Spec} \mathcal{O}_{G,e}$). Since $\hat{S} \to S$ is faithfully flat and since the two morphisms

[520] Indeed, over a field k, every group subscheme of G is closed, cf. VI_A, 0.5.2. Moreover, 6.2.5 concludes the insertion of the results drawn from VIII § 6.

[521] We have inserted here this corollary (cf. Exp. XII, 6.1), which will be useful later. Moreover, in 6.3 we return to the original text of VI_B.

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K ×_k S → S,    (g, x) ↦ gxg⁻¹    resp.    (g, x) ↦ x
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coincide after base change $\hat{S} \to S$, they coincide, i.e. K acts trivially on $\mathcal{O}_{G,e}$. Now, by 6.2.4 e), the subobject G^K of invariants of G under K (which is none other than $Centr_G(K)$) is a closed subscheme of G, hence defined by a quasi-coherent ideal $\mathcal{J}$ of $\mathcal{O}_G$. As G^K majorizes $S = \operatorname{Spec} \mathcal{O}_{G,e}$ and $\mathcal{J}$ is of finite type (since G is noetherian), there exists an open neighborhood U of e such that $\mathcal{J}|_U = 0$. Then the subgroup $G^K = Centr_G(K)$ contains U, hence also $U \cdot U$, which equals G since G is irreducible (VI_A 0.5). Hence $Centr_G(K) = G$, whence $K \subset Z$ and therefore $Z = K$. ∎

Remark 6.3. *Let k be an algebraically closed field, G a k-group and H a group subscheme of G; assume G and H are of finite type over k and reduced. Then $Norm_G H$ (resp. $Centr_G H$) is representable by a group subscheme of G, whose associated reduced subscheme is none other than the normalizer (resp. the centralizer) of H in G in the sense of the* Bible.

Proposition 6.4. *Let G be an S-group and $u : X \to G$ a monomorphism of S-schemes. Set $T = Transp_G(X,X)$. The following conditions are equivalent:*

(i) T is a sub-S-functor in groups of G.

(ii) $T = Transpstr_G(X,X) = Norm_G X$.

These conditions are satisfied in each of the two following cases:

a) X is of finite presentation over S.

b) T is representable by a scheme of finite presentation over S.

The equivalence of conditions (i) and (ii) follows from the fact that, whatever the morphism $S' \to S$, whatever $t, t' \in T(S')$, one has $tt' \in T(S')$, and from the fact that $Transpstr_G(X,X) = T \cap c(T)$ (cf. 6.1 (ii)).

Let us place ourselves in case a). Let $t \in T(S)$; then $int(t)$ is a monomorphism of X into X, hence an S-automorphism of X (EGA IV$_4$, 17.9.6), so that t belongs to $Transpstr_G(X,X)$, whence a).

In case b), it is clear that $\mu(T \times_S T) \subset T$, and the assertion follows from the following lemma.

Lemma 6.4.2. [522] *Let G be an S-scheme of finite presentation, equipped with an associative law (in the sense of EGA 0_III 8.2.5). Suppose that for every S-scheme S' and every $g \in G(S')$, right and left translations by g in the set $G(S')$ are injective, and that $G(S) \neq \emptyset$. Then G is an S-group.*

It suffices to show that, whatever the S-scheme S', the set $G(S')$ is a group; now from the hypothesis it follows at once that right and left translations by every element $g \in G(S')$ in $G_{S'}$ are S-monomorphisms of $G_{S'}$ into $G_{S'}$. They are therefore S-automorphisms, since G is of finite presentation over S (EGA IV$_4$, 17.9.6), so that right and left translations by g in the set $G(S')$ are bijective, and one is reduced to the following lemma.

[522] We have kept the numbering of the original: there is no n° 6.4.1.

Lemma 6.4.3. *Let G be a non-empty set equipped with an associative law such that right and left translations are bijective. Then G is a group.*

The proof is left to the reader.

Definition 6.5. *Let G be an S-group, H an S-functor, and $u : H \to G$ a monomorphism.*

(i) The connected centralizer of H in G, *denoted $Centr^0_G H$, is the neutral component of the functor $Centr_G H$ (cf. 3.1 and 6.1 (iii)).*

(i′) For every morphism $u : X \to G$, one defines similarly the functor $Centr^0(u)$ (cf. 6.2 (iv)).

(ii) When for every $s \in S$, u_s is a closed immersion, the connected normalizer of H in G, *denoted $Norm^0_G H$, is the neutral component of the functor $Norm_G H$.*

N.B. By 1.4.2, the hypothesis in (ii) is satisfied when H is an S-group scheme, G and H are locally of finite type over S, and u is a quasi-compact morphism of S-groups.

Proposition 6.5.1. *Let G be an S-group locally of finite presentation and quasi-separated over S, H a smooth S-group with connected fibers, and $u : H \to G$ a monomorphism. Let N be the normalizer of H in G (cf. 6.1). According to 6.5.5 below, N is representable by a closed group subscheme of G, of finite presentation over G. This being so, the following conditions are equivalent:*

(i) The canonical morphism $H \to N$ is an open immersion.

(ii) $N^0 = H$ (cf. 3.10).

(iii) For every $s \in S$, one has $H_s = (N_s)^0$.

Condition (i) entails (ii) by Lemma 3.10.1, since $H^0 = H$. On the other hand, it is clear that (ii) entails (iii), since $H_s = (N^0)_s = (N_s)^0$.

Let us show finally that (iii) entails (i). Since $H_s = (N^0)_s$ for every $s \in S$, then $H_s \to N_s$ is an open immersion. Moreover, H and N are locally of finite presentation over S, and H is flat over S, hence (EGA IV₄, 17.9.5), $H \to N$ is an open immersion. ∎

[523] For the reader's convenience, we have included below the results 6.8 to 6.11 of Exp. XI.

Theorem 6.5.2. *Let X be a smooth scheme over S, with geometrically irreducible fibers.*[524]

(i) For every closed subscheme Y of X, the functor $\prod_{X/S} Y/X$ is representable by a closed subscheme T of S.

[523] We have inserted here nos. 6.5.2 to 6.5.5, drawn from Exp. XI, 6.8 to 6.11. This was moreover suggested by A. Grothendieck in a Note at the beginning of XI 6: "The present number does not use the results of nos. 3, 4, 5 (of XI); its natural place would be in VI_B."

[524] On the one hand, we have corrected the original, by replacing "connected" with "irreducible" (cf. the proof). On the other hand, by [RG71] I, 3.3.4 (iii) and 4.1.1, it suffices to assume that X is flat over S, with geometrically irreducible fibers and without embedded components.

(ii) Moreover, if $Y \to X$ is of finite presentation, then so is $T \to S$.

Since $f : X \to S$ is faithfully flat locally of finite presentation, it is covering for the (fpqc) topology. On the other hand, since $T = \prod_{X/S} Y/X$ is obviously a subsheaf of S for the (fpqc) topology, it follows that the question of representability of T by a closed subscheme of S is local in nature on S for the (fpqc) topology,[525] and the same is true of the question of deciding whether T is of finite presentation over S (cf. EGA IV_2, 2.7.1). Up to making the base change $S' \to S$, with $S' = X$, one is reduced to the case where X admits a section ϵ over S. One may moreover suppose S affine and a fortiori quasi-compact. One then has:

Corollary 6.5.3. *Under the conditions of 6.5.2, suppose that S is quasi-compact, that $X \to S$ admits a section ϵ, and that $Y \to X$ is of finite presentation. Then there exists an integer $n \geqslant 0$ such that one has*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n is the n-th infinitesimal neighborhood of the immersion $\epsilon : S \to X$, and $Y_n = Y \cap X_n$.

When Y is of finite presentation over X, this corollary implies 6.5.2: indeed, X being smooth over S, X_n is finite and locally free over S, hence a fortiori "essentially free" over S (cf. 6.2.1), so $\prod_{X_n/S} Y_n/X_n$ is representable by a closed subscheme T of S, of finite presentation over S, by 6.2.3.

Let us first prove 6.5.3 (and hence 6.5.2) when S is noetherian. Let $T_n = \prod_{X_n/S} Y_n/X_n$; then the T_n form a decreasing sequence of closed subschemes of S, and S being noetherian, this sequence is stationary. Let $R = \bigcap_{n\geqslant 0} \prod_{X_n/S} Y_n/X_n$ be their common value for large n; one has obviously $T \subset R$, and it suffices to establish that $R \subset T$. Up to making the base change $R \to S$, one is reduced to the case where $R = S$, i.e. $Y_n = X_n$ for every n, i.e. $Y \supset X_n$ for every n, and one must then prove that $T = S$, i.e. $Y = X$.

Now $Y \supset X_n$ for every n implies (thanks to the fact that X is locally noetherian) that Y is, in a neighborhood of every point of $\epsilon(S)$, an induced open subscheme of X;[526] hence there exists an induced open U of X, containing $\epsilon(S)$, such that $U \subset Y$. By virtue of EGA IV_3, 11.10.10, the fibers of X/S being integral, U is schematically dense in X, hence (Y being a closed subscheme majorizing U) one has $Y = X$. This proves 6.5.3 hence 6.5.2 when S is noetherian.

The general case proceeds by reduction to the preceding case. For every $s \in S$, there exists an affine open neighborhood U of s and an affine open neighborhood V of $\epsilon(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U, and if S' is an affine open neighborhood of s contained in $\epsilon^{-1}(V) \cap f(V)$, and $X' = V \cap f^{-1}(S')$, then X' and S' are affine opens of X resp. S, and X'/S' admits a section. Because of the local nature of 6.5.2 and 6.5.3 we may suppose $S = S'$. Then X' is an affine open of X containing $\epsilon(S)$. As each fiber X_s is supposed irreducible,

[525] See for example the addition 1.7 in Exp. VIII.

[526] see, for example, the proof of 6.2.6.

X'_s is schematically dense in X_s, hence, by EGA IV₃, 11.10.10, X' is schematically dense in X, and similarly, for every base change $S_1 \to S$, $X' \times_S S_1$ is schematically dense in $X \times_S S_1$.

It follows that $\prod_{X/S} Y/X = \prod_{X'/S} Y'/X'$, where $Y' = Y \cap X'$. This reduces us to the case where $X = X'$, so one may suppose S and X affine. Moreover, if $X = \mathrm{Spec}(B)$ and if J is the ideal of B defining Y, then J is a direct limit of its finitely generated subideals, hence Y is the intersection of closed subschemes Y_i of X which are of finite presentation over X, and consequently $\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X$, which reduces us, to prove 6.5.2, to the case where Y is of finite presentation over X, with S and X affine.

But then X and Y over S come by base change $S \to S_0$ from an analogous situation X_0 and Y_0 over S_0, with S_0 noetherian, which reduces us to the case where S is noetherian, already treated. This completes the proof of 6.5.2 and 6.5.3. ∎

Corollary 6.5.4. *Let X be a smooth S-group scheme of finite presentation with connected fibers, Y a group scheme of finite presentation over S, $i : Y \to X$ a monomorphism of S-group schemes.*

(i) Then $\prod_{X/S} Y/X$ is representable by a closed subscheme of finite presentation of S.

(ii) If moreover S is quasi-compact, one has for large enough n:

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n denotes the n-th infinitesimal neighborhood of the unit section $\epsilon : S \to X$, and $Y_n = X_n \cap Y$.

The proof is essentially that of 6.5.3.[527] On the one hand, i is locally of finite presentation (cf. EGA IV₁, 1.4.3). On the other hand, the unit sections of Y and of X induce bijective immersions $S \to Y_n$ and $S \to X_n$, hence isomorphisms of S_{red} with $(Y_n)_{red}$ and $(X_n)_{red}$. Consequently, i_n is quasi-compact hence of finite type, and one has a commutative diagram:

$$\begin{tikzcd} (Y_n)_{red} \arrow[r, "\tau"] \arrow[dr, "\sigma"'] & Y_n \arrow[d, "i_n"] \\ & X_n \end{tikzcd}$$

where σ, τ are closed immersions and τ is surjective. Since i_n is separated (being a monomorphism), it follows that i_n is proper (cf. EGA II, 5.4.3). Hence i_n is a proper monomorphism of finite presentation, hence a closed immersion (cf. EGA IV₃, 8.11.5). Consequently, by virtue of 6.2.3 already used, $\prod_{X_n/S} Y_n/X_n$ is representable by a closed subscheme T_n of S of finite presentation over S, and it remains

[527] We have detailed the original in what follows, adding references to EGA IV.

therefore to prove the last assertion of 6.5.4 in the case where one supposes moreover S affine.

One reduces immediately again to the case where S is noetherian, and one is reduced to proving that one has $R = T$ (with the notations of the proof of 6.5.3), or, again, that $Y \supset X_n$ for every n implies $Y = X$. Now the hypothesis implies that $i : Y \to X$ is étale at the points of the unit section of Y over S, hence Y is smooth over S at the points of the unit section, whence it follows that the open U of points of Y at which Y is smooth over S is an induced open subgroup of Y (cf. 2.3). Then $\tau : U \to X$ is an étale monomorphism by virtue of 2.5, hence an open immersion; since the fibers of X are connected and every open subgroup of an algebraic group is also closed, it follows that τ is a surjective open immersion, i.e. an isomorphism. Hence $U = X$ and a fortiori $Y = X$, which completes the proof of 6.5.4. ∎

Proceeding as in 6.2.4 e), one concludes from 6.5.4:

Corollary 6.5.5. *Let G be an S-group scheme locally of finite type and quasi-separated over S, H a smooth group scheme of finite presentation over S with connected fibers, $i : H \to G$ a monomorphism of S-groups. Then:*

a) $Centr_G(H)$ and $Norm_G(H)$ are representable by closed subschemes of G, of finite presentation over G.

a′) Similarly, for every monomorphism $j : K \to G$ of finite presentation of S-group schemes, $Transp_G(H,K)$ is representable by a closed subscheme of G, of finite presentation over G.

b) If G is quasi-compact, there exists an integer $n \geqslant 0$ such that (if H_n denotes the n-th infinitesimal neighborhood of the unit section of H) one has:

$$Centr_G(H) = Centr_G(H_n) \qquad Norm_G(H) = Norm_G(H_n)$$
$$Transp_G(H, K) = Transp_G(H_n, K) = Transp_G(H_n, K_n).$$

Proof. [528] Let us first note that the hypothesis on G entails that the monomorphism $H \to G$ is of finite presentation (EGA IV_1, 1.2.4 and 1.4.3), as is the diagonal immersion $\Delta_{G/S} : G \to G \times_S G$ (loc. cit. 1.4.3.1). The case of $Norm_G(H)$ therefore reduces to that of the transporter, by taking $H = K$. Taking 6.2.4 e) into account, we shall apply 6.5.4 to the group scheme $X = H_G = H \times_S G$ over the base scheme G, and to the following group subscheme Y.

In the case of $Transp_G(H,K)$, we take for Y the inverse image of K_G under the morphism of G-groups $H \times_S G \to G \times_S G$, $(h,g) \mapsto (ghg^{-1}, g)$. In the case of

$Centr_G(H)$, we take for Y the inverse image of the diagonal subgroup $\Delta_{G/S}(G)_G$ of $(G \times_S G)_G$ under the morphism of G-groups:

$$H \times_S G \to G \times_S G \times_S G, \qquad (h, g) \mapsto (h, ghg^{-1}, g).$$

∎

[528] We have detailed what follows. Moreover, this concludes the insertion of XI, 6.8 to 6.11, i.e. we return in 6.6 below to the original text of VI_B.

Definition 6.6. *Let G be an S-functor in groups, H a sub-S-functor in groups; one says that H is* invariant *(resp.* central*, resp.* characteristic*) in G if $Norm_G H = G$ (resp. if $Centr_G H = G$, resp. if, whatever the S-scheme T and the automorphism $a \in \mathrm{Aut}_{T-gr.}(G_T)$, one has $a(H_T) \subset H_T$), in other words, if, whatever the S-scheme T, the subgroup $H(T)$ of $G(T)$ is invariant in $G(T)$ (resp. central in $G(T)$, resp. invariant under every automorphism of G_T).*

N.B. If H is central (resp. characteristic), then it is invariant.

Remark 6.7. *Let G and H be two S-groups and $u : H \to G$ a monomorphism. For H to be invariant (resp. central) in G, it is necessary and sufficient that the morphism*

(μ ∘ c ∘ pr_2, μ ∘ (u × id_G)) : H ×_S G → G

(defined by $(h, g) \mapsto g^{-1}hg$ whatever $g \in G(S')$ and $h \in H(S')$) factor through u (resp. equal $u \circ pr_1$), and for H to be characteristic in G, it is necessary and sufficient that for every S-scheme T and every T-automorphism of groups a of G_T, $a \circ u(T)$ factor through $u(T)$.

Example 6.8. *Let G be an S-functor in groups. Then Centr G is characteristic and central. If, for every $s \in S$, G_s is representable, then G^0 is characteristic. This follows from the definitions and from 3.3.*

6.9.

[529] In [RG71], I 3.3.3, the authors introduce the geometric notion of pure S-scheme, which is local on S for the étale topology; we refer to loc. cit. for the precise definition. Let us simply point out the following:

a) (loc. cit., Th. 3.3.5) If B is a flat A-algebra of finite presentation, then B is a projective A-module if and only if Spec(B) is pure over Spec(A).

b) Consequently, if $X \to S$ is locally of finite presentation, flat and pure, then X is essentially free over S.

c) (loc. cit., 3.3.4 (iii)) If $X \to S$ is locally of finite type, flat, with geometrically irreducible fibers and without embedded components, then X is pure over S.

Since every group scheme locally of finite type over a field is Cohen–Macaulay (cf. VI_A, 1.1.1), hence without embedded components, one obtains in particular:

d) Every S-group scheme G locally of finite presentation, flat, and with connected fibers is pure over S, hence essentially free over S.

One may then take up again all the statements of 6.2.4 e) taking into account results (b) and (d) above.

7. Generated subgroups; commutator group

In this number, k denotes a fixed field.

[529] We have added this subsection.

Proposition 7.1. *Let G be a k-group, $(X_i)_{i\in I}$ a family of geometrically reduced k-schemes;*[530] *for every $i \in I$, let $f_i : X_i \to G$ be a k-morphism.*

(i) There exists a smallest closed group sub-k-scheme of G majorizing each of the f_i, denoted $\Gamma_G((f_i)_{i\in I})$. It is a geometrically reduced k-scheme, hence smooth in the case where G is supposed locally of finite type over k (1.3.1).

(ii) Set $X = \coprod_{i\in I} X_i$, and let $f : X \to G$ be the morphism whose restriction to X_i is f_i, for every $i \in I$. Set $X^1 = X \sqcup X$, let $f^1 : X^1 \to G$ be the morphism whose restrictions to X are respectively f and $c \circ f$. For every $n > 1$, set

$$X^n = X^1 \times_k X^{n-1} \quad \text{and} \quad f^n = \mu \circ (f^1 \times_k f^{n-1}) : X^n \to G.$$

Then $\Gamma_G((f_i)_{i\in I})$ is the reduced subscheme of G whose underlying space is the closure of the union of the $f^n(X^n)$, for $n \geqslant 1$.

(iii) For every k-scheme S, $\Gamma_G((f_i)_{i\in I})_S$ is the smallest closed group subscheme of G_S majorizing each of the $f_{i,S} : X_S \to G_S$.

(iii′) Moreover, $\Gamma_G((f_i)_{i\in I})_S$ is the smallest group subscheme of G_S majorizing each of the $f_{i,S}$.[531]

Let us first note that, to prove (i), (iii) and (iii′), by defining X and f as in (ii), one is reduced to the case where I is reduced to a single element.

Let H be the reduced subscheme of G with underlying set $\bigcup_{n\geqslant 1} f^n(X^n)$. Then the family of morphisms $f^n : X^n \to H$ is schematically dominant (cf. EGA IV₃, 11.10.4), hence every closed subscheme of G which majorizes the f^n also majorizes H. Moreover, by loc. cit., 11.10.7, H is geometrically reduced. Hence to show (i) and (ii), it suffices to show that H is a group subscheme of G, i.e. that the restriction of c to H and the restriction of μ to $H \times_k H$ factor through the injection $H \to G$.

Since H is geometrically reduced, $H \times_k H$ is reduced, and it suffices to verify that $c(H) \subset H$ and that $\mu(H \times_k H) \subset H$ (set-theoretically). But by EGA IV₃, 11.10.6, the union of the $f^n_{(H)}(X^n \times_k H)$ is schematically dense in $H \times_k H$. Similarly, for every $n \geqslant 1$, the union of the $f^m_{(X^n)}(X^n \times_k X^m)$, for $m \geqslant 1$, is schematically dense in $X^n \times_k H$. Hence it suffices to show that $\mu(f^n_{(H)}(f^m_{(X^n)}(X^n \times_k X^m))) \subset H$ and that $c(f^n(X^n)) \subset H$. Now

$$\mu(f^n_{(H)}(f^m_{(X^n)}(X^n \times_k X^m))) = \mu((f^n \times f^m)(X^n \times_k X^m)) = f^{n+m}(X^{n+m}) \subset H;$$

and, since $c(f^1(X^1)) \subset f^1(X^1)$, one has, whatever n, $c(f^n(X^n)) \subset f^n(X^n) \subset H$. This proves (i) and (ii).

Let us now show (iii). Let G' be a closed group subscheme of G_S majorizing f_S; we must show that G' majorizes H_S, or, what amounts to the same, that $H_S = H_S \times_{G_S} G'$. Set $H'_S = H_S \times_{G_S} G' = G' \cap H_S$. Since G' and H_S both majorize the f^n_S, the same is true of H'_S. Now (EGA IV₃, 11.10.6), since the family of $f^n : X^n \to H$ is schematically

[530] We have replaced, here and in the sequel, the little-used terminology "separable" by the usual terminology "geometrically reduced", cf. EGA IV₂, 4.6.2.

[531] The original stated (iii′) under the additional hypothesis that G be locally of finite type over k, but this can be omitted, by $\mathrm{VI_A}$, 0.5.2.

dominant, the same is true of the family of $f_S^n : X_S^n \to H_S$, so that H'_S, which majorizes each of the f_S^n, is equal to H_S by EGA IV₃, 11.10.1 c). This proves (iii).

Let us show finally that H_S is the smallest group subscheme (not necessarily closed) of G_S majorizing f_S.

Let G' be a group subscheme of G_S majorizing f_S. It must likewise be shown that, if one sets $H'_S = H_S \times_{G_S} G'$, then $H'_S = H_S$. It suffices for this to show that H'_S is closed in H_S and to apply (iii). It suffices therefore to show that H_S and H'_S have the same underlying set, a fortiori it suffices to show that, for every $s \in S$, H_s equals

$H'_s := H'_S \times_S \kappa(s) = H_s \times_{G_s} G'_s.$

Now, by VI_A, 0.5.2, the group sub-$\kappa(s)$-scheme G'_s is closed in G_s. Hence H'_s is closed in H_s, and then the preceding reasoning, applied to H'_s, to H_s and to the f_s^n, shows that $H'_s = H_s$. □

Corollary 7.1.1. [532] *Let G be a k-group locally of finite type and let A, B be two sub-k-groups of G, smooth and of finite type, and i_A (resp. i_B) the inclusion of A (resp. B) into G. Assume that B normalizes A. Then $A \cdot B = \Gamma_G(i_A, i_B)$.*

Indeed, let $H = A \rtimes B$ be the semidirect product of A and B (cf. I, 2.3.5); it is a smooth k-group of finite type. Then the group morphism $u : H \to G$, $(a, b) \mapsto ab$, is quasi-compact, hence, by 1.2, $u(H) = A \cdot B$ is a closed reduced subscheme of G, which is a group in the category $(Sch/k)_{red}$. Now, by EGA IV₃, 11.10.7, $A \cdot B = u(H)$ is geometrically reduced (since H is), so it is a closed subgroup of G. Since obviously $A \cdot B \subset \Gamma_G(i_A, i_B)$, the corollary follows. □

Definitions and remarks 7.2. [533] *(i) Given a k-group G, a family $(X_i)_{i\in I}$ of geometrically reduced k-schemes, and for each $i \in I$ a k-morphism $f_i : X_i \to G$, one calls* closed group subscheme of G generated by the family $(f_i)_{i\in I}$, *and we shall denote in this number $\Gamma_G((f_i)_{i\in I})$, the smallest closed group subscheme of G majorizing each of the f_i. If X is a subscheme of G, geometrically reduced over k, and if f is the immersion $X \hookrightarrow G$, one writes $\Gamma_G(X)$ instead of $\Gamma_G(f)$.*

(i′) With the notations of 7.1 (ii), we shall sometimes set $\Gamma'_G(f) = \bigcup_{n\geqslant 1} f^n(X^n)$. Note that $\Gamma'_G(f)$ is a subset of G stable for the group law (in the sense of 3.0).

(ii) It is clear that if X_1 and X_2 are two geometrically reduced k-schemes and $f_1 : X_1 \to G$ and $f_2 : X_2 \to G$ two k-morphisms such that the sets $f_1(X_1)$ and $f_2(X_2)$ are equal, then $\Gamma_G(f_1) = \Gamma_G(f_2)$.

(iii) Let E be a subset of G such that the reduced subscheme E of G is geometrically reduced. One calls closed group subscheme of G generated by E, *denoted $\Gamma_G(E)$, the group subscheme $\Gamma_G(i)$, where i is the injection of the reduced subscheme E of G into G.*

[532] We have added this corollary, to point out this particular case of 7.1.

[533] We have placed in (i′) the point (viii) of the original, and we have highlighted points (vi) and (vii) in the form of Corollary 7.2.1 and Definition 7.2.2 below.

(iv) Since every group subscheme of G is closed, by VI_A, 0.5.2,[534] *one will speak of "generated group subscheme" instead of "generated closed group subscheme".*

(v) Let X be a geometrically reduced k-scheme and $f : X \to G$ a k-morphism. Suppose that $f(X)$ contains the unit element e of G. Set $X'^1 = X \times_k X$ and $f'^1 = \mu \circ (f \times_k (c \circ f))$, and for $n > 1$,

$$X'^n = X'^1 \times_k X'^{n-1} \quad \text{and} \quad f'^n = \mu \circ (f'^1 \times_k f'^{n-1}).$$

Then $\Gamma_G(f)$ is the reduced subscheme of G whose underlying space is the closure of the union of the $f'^n(X'^n)$, for $n \geqslant 1$.

Indeed, recalling the notation of 7.1 (ii): $X^1 = X \sqcup X$ and $X^n = X^1 \times_k X^{n-1}$, for $n \geqslant 2$, one has the following inclusions, where the first follows from the hypothesis $e \in f(X)$:

$$f^n(X^n) \subset f'^n(X'^n) \subset f^{2n}(X^{2n}), \quad \text{for every } n \geqslant 1.$$

This shows moreover that, for there to exist an integer n such that $f^n(X^n) = \Gamma_G(f)$, it is necessary and sufficient that there exist an integer m such that $f'^m(X'^m) = \Gamma_G(f)$.

From Remark 7.2 (v) one deduces the

Corollary 7.2.1. *Let X be a geometrically reduced and geometrically connected k-scheme, and $f : X \to G$ a k-morphism such that $f(X)$ contains the neutral element of G. Then the k-group $\Gamma_G(f)$ is connected, hence irreducible.*

Indeed, each of the X'^n is then connected, hence the union of the $f'^n(X'^n)$ (which all contain the neutral element) is connected, and the same is true of its closure $\Gamma_G(f)$. Hence $\Gamma_G(f)$ is irreducible, by VI_A, 2.4 (when G, hence also $\Gamma_G(f)$, is locally of finite type over k) and 2.6.5 (iii) (in the general case). ∎

Definition 7.2.2. *Let G be a k-group.*

a) Let A and B be two geometrically reduced group sub-k-schemes of G. The commutator group subscheme of A and B in G*, denoted $(A, B)_G$ or simply (A, B), is the closed group subscheme of G generated by the morphism $\nu : A \times_k B \to G$ defined by: $(a, b) \mapsto aba^{-1}b^{-1}$, for every k-scheme S and $a \in A(S)$, $b \in B(S)$.*

b) Suppose G geometrically reduced over k. The derived group of G*, denoted $D(G)$,*[535] *is the group (G, G).*

N.B. For G to be commutative, it is necessary and sufficient that $D(G)$ be the unit k-group.

Reminders 7.3.0. [536] *Recall that if $\phi : Y \to Z$ is a morphism of S-schemes, the image presheaf $\phi(Y)$ is the S-functor which to every S' over S associates the subset $\phi(Y(S'))$ of $Z(S')$. Note that if T is a subscheme of Z and if the inclusion of*

[534] We have suppressed here the hypothesis that G be locally of finite type over k.
[535] We have changed the notation $\mathrm{Der}(G)$ of the original, in order to avoid any risk of confusion with a space of derivations.
[536] We have added these reminders, and modified accordingly, and detailed, the statement of 7.3.

presheaves $\phi(Y) \hookrightarrow Z$ factors through T, i.e. if $\phi \circ h \in T(S')$ for every S-morphism $h : S' \to Y$, then set-theoretically $\phi(Y) \subset T$ (take $h = id_Y$), and the converse is true if Y is reduced, since in this case ϕ factors through T.

Recall also that, by 6.2, if H is a closed group sub-k-scheme of G, then $Centr_G H$ and $Norm_G H$ are representable by closed group sub-k-schemes of G.

Corollary 7.3. *Let G be a k-group, X a geometrically reduced k-scheme, $f : X \to G$ a k-morphism.*

(i) Let S be a k-scheme and u an endomorphism of the S-group G_S.

(a) If one has $u(f_S(X_S)) \subset \Gamma_G(f)_S$ (set-theoretically), then the morphism $u : \Gamma_G(f)_S \to G_S$ factors through $\Gamma_G(f)_S$.

(b) If u is an automorphism of the S-group G_S and if one has $u(f_S(X_S)) = f_S(X_S)$ (set-theoretically), then u induces an automorphism of $\Gamma_G(f)_S$. In particular, if an element $g \in G(S)$ satisfies $int(g)(f_S(X_S)) = f_S(X_S)$ (set-theoretically), then $g \in Norm_{G_S}(\Gamma_G(f)_S)(S)$.

(c) If $u \circ f_S = f_S$, then the restriction of u to the subgroup $\Gamma_G(f)_S$ of G_S is the identity. In particular, if an element $g \in G(S)$ satisfies $int(g)(f_S) = f_S$, then $g \in Centr_{G_S}(\Gamma_G(f)_S)(S)$.

(ii) Let H be a group subscheme of G; then H centralizes (resp. normalizes) $\Gamma_G(f)$ if and only if, for every $S \to \operatorname{Spec} k$ and $h \in H(S)$, one has $int(h) \circ f_S = f_S$ (resp. $int(h)(f_S(X_S)) \subset \Gamma_G(f)_S$), i.e. if for every $S' \to S$ and $x \in X(S')$, the elements $h_{S'}$ and $f(x)$ of $G(S')$ commute (resp. $h_{S'} f(x) h_{S'}^{-1} \in \Gamma_G(f)(S')$).

(iii) In particular, let Y be a second geometrically reduced k-scheme and $\phi : Y \to G$ a k-morphism. Suppose that, whatever the k-scheme S', for every $x \in X(S')$ and $y \in Y(S')$, the elements $\phi(y)$ and $f(x)$ of $G(S')$ commute (resp. $int(\phi(y))(f(x)) \in \Gamma_G(f)(S')$).

Then $\Gamma_G(\phi)$ is a sub-k-group of $Centr_G \Gamma_G(f)$, resp. $Norm_G \Gamma_G(f)$.

(iv) If g is a k-point of G, then the k-group $\Gamma_G(g)$ is commutative.

(v) Let A and B be two group subschemes of G geometrically reduced over k. If A and B are invariant (resp. characteristic), so is (A, B).

Proof. (i) Let us prove (a). Set $\Gamma_S = \Gamma_G(f)_S$. Then $u^{-1}(\Gamma_S)$ is a closed group sub-S-scheme of G_S, hence, by 7.1 (iii), it suffices to show that f_S factors through $u^{-1}(\Gamma_S)$. Now, since $u(f_S(X_S)) \subset \Gamma_S$ and since X_S is reduced, $u \circ f_S$ factors through Γ_S, hence f_S factors through $u^{-1}(\Gamma_S)$. This proves (a).

Then the first assertion of (b) is a consequence of (a), applied to u and u^{-1} (and in fact it suffices to assume that $u(f_S(X_S)) \subset \Gamma_S$ and $u^{-1}(f_S(X_S)) \subset \Gamma_S$), and the second assertion is the particular case where $u = int(g)$.

Let us prove (c). Consider the morphism of S-groups $G_S \to G_S \times_S G_S$, $g \mapsto (u(g), g)$ and let H be the inverse image of the diagonal; it is a group subscheme of G_S. Since G is separated over k (VI_A 0.3), G_S is separated over S, hence H is closed in

G_S. As, by hypothesis, H majorizes f_S, it contains $\Gamma_G(f)_S$, by 7.1 (iii). This proves the first assertion of (c), and the second is the particular case where $u = int(g)$.

Let us prove (ii). Set $\Gamma = \Gamma_G(f)$ and let i denote the inclusion $\Gamma \hookrightarrow G$. Then H is contained in $N = Norm_G(\Gamma)$ if and only if, for every k-scheme S and $h \in H(S)$, one has $int(h)(\Gamma_S) \subset \Gamma_S$ (this condition applied to h and h^{-1} entailing that $int(h)(\Gamma_S) = \Gamma_S$); and by (i)(a) this is the case if and only if $int(h)(f_S(X_S)) \subset \Gamma_S$.

Similarly, H is contained in $C = Centr_G(\Gamma)$ if and only if, for every k-scheme S and $h \in H(S)$, one has $int(h) \circ i_S = i_S$, and by (i)(c) this is the case if and only if $int(h)(f_S) = f_S$. This proves (ii).

Let us prove (iii). Taking (ii) into account, the hypothesis entails that ϕ factors through C (resp. N); since the latter is a closed subgroup of G, by 6.2, it therefore contains $\Gamma_G(f)$, by 7.1 (i).

Assertion (iv) follows from (iii), when one takes for f and ϕ the closed immersion $\operatorname{Spec} k \hookrightarrow G$ defined by g: in this case, for every k-scheme S, $X(S)$ and $Y(S)$ have only one point, which is sent by f (resp. ϕ) to g.

Let us finally show (v). Let ν be the morphism $G \times G \to G$, $(g,h) \mapsto ghg^{-1}h^{-1}$ and let ν' be its restriction to $A \times B$; by definition (7.2.2), $(A,B) = \Gamma_G(\nu')$. Let S be a k-scheme, and u an inner automorphism (resp. an automorphism of S-group) of G_S. One has $u \circ \nu_S = \nu_S \circ (u \times u)$. On the other hand, the hypothesis entails that u induces an automorphism u_1 of A_S (resp. u_2 of B_S), hence

u(ν′_S(A_S ×_S B_S)) = ν′_S(u_1(A_S) ×_S u_2(B_S)) = ν′_S(A_S ×_S B_S).

Hence, by (i)(b), u induces an automorphism of $(A,B)_S$. This proves (v). ▯

Proposition 7.4. *Let G be a k-group locally of finite type, X a k-scheme of finite type, geometrically reduced and geometrically connected, and $f : X \to G$ a k-morphism such that $f(X)$ contains the neutral element e of G. Then, with the notations of 7.1 (ii), there exists an integer N such that one has (set-theoretically):*

$$f^N(X^N) = \Gamma_G(f).$$

By 7.1 (iii) and EGA IV₂, 2.6.1, we may suppose k algebraically closed. By Corollary 7.2.1, we may suppose $G = G^0$; finally, it suffices to show that there exists an integer N such that one has $f'^N(X'^N) = \Gamma_G(f)$, with the notations of 7.2 (v).

First case. Suppose X irreducible. Then the $f'^n(X'^n)$ form an increasing sequence of irreducible closed sets in the space G, which is noetherian, since $G = G^0$ is

of finite type over k (VI_A 2.4). Hence this sequence is stationary, and there exists an integer m such that $f'^m(X'^m) = \Gamma_G(f)$.

Moreover, since X and G are of finite type over k, the morphisms f'^n are of finite type over k. Consequently, $f'^m(X'^m)$ is constructible in G (EGA IV₁, 1.8.5), hence contains an open U of its closure $\Gamma_G(f)$ (EGA 0_IV, 9.2.3). Then, by VI_A 0.5, one has:

$$\Gamma_G(f) \subset U \cdot U \subset f'^{2m}(X'^{2m}) \subset \Gamma_G(f),$$

whence $f'^{2m}(X'^{2m}) = \Gamma_G(f)$.

Second case. Suppose X has exactly two irreducible components A_1 and A_2. Then, since X is connected and k algebraically closed, there exists $a \in (A_1 \cap A_2)(k)$. Hence the four irreducible parts $A_i \times_k A_j$ $(i, j = 1, 2)$ cover $X \times_k X$, and the image of each of them under the morphism $f'^1 = \mu \circ (f \times_k (c \circ f))$ contains e. If f'^1_{ij} denotes the restriction of f'^1 to the reduced subscheme $A_i \times_k A_j$, set

$$Y = (A_1 \times_k A_1) \times_k (A_1 \times_k A_2) \times_k (A_2 \times_k A_2) \times_k (A_2 \times_k A_1) \quad \text{and}$$
$$g = \mu \circ (\mu \circ (f'^1_{11} \times_k f'^1_{12}) \times_k \mu \circ (f'^1_{22} \times_k f'^1_{21})).$$

Then Y is irreducible, reduced and of finite type, hence we just saw that there exists an integer m such that $g'^m(Y'^m) = \Gamma_G(g)$. Now, for every $n \geqslant 1$, one has $f'^n(X'^n) \subset g'^n(Y^n) \subset f'^{4n}(X'^{4n})$, whence $\Gamma_G(f) = \Gamma_G(g)$ and $f'^{4m}(X'^{4m}) = \Gamma_G(f)$.

General case. Let us argue by induction on the number of irreducible components of X (this number is finite since X, being of finite type over k, is noetherian). Suppose the proposition proved in the case where X has at most $r - 1$ irreducible components, and suppose it has r, namely $A_1, \cdots, A_r$. Then e belongs to the image of one of the A_i; suppose for example that $e \in f(A_1)$. Set then $Y = \Gamma_G(f_r)$, where f_r denotes the restriction of f to the reduced subscheme $X_r = A_1 \cup \cdots \cup A_{r-1}$ (we suppose the numbering of the A_i chosen so that this scheme is connected, which is always possible). Then Y is a subgroup of G, closed, reduced and irreducible, by Corollary 7.2.1.

Set $Z = f(A_r)$, and let $T = Y \cup Z$, equipped with the closed reduced subscheme structure, and i the injection of T into G. It is clear (7.2 (ii)) that $\Gamma_G(i) = \Gamma_G(f)$ and that T is connected (because since X is connected, $A_1 \cup \cdots \cup A_{r-1}$ and A_r have in common a point a, and Y and Z have in common the point $f(a)$). Moreover, $e \in T$, and T has at most two irreducible components, since Y and Z are irreducible. By the induction hypothesis, there exists an integer m such that one has $f_r'^m(X_r'^m) = \Gamma_G(f_r) = Y$. Since $X = X_r \cup A_r$ and since $Z = f(A_r)$ is contained in $f'^m(X'^m)$ (since $e \in f(A_1)$), one therefore has:

$$f(X) \subset Y \cup Z \subset f'^m(X'^m).$$

On the other hand, since T has at most two irreducible components, one has already seen that there exists an integer p such that $i'^p(T'^p) = \Gamma_G(i) = \Gamma_G(f)$. Now, $T \subset f'^m(X'^m)$, hence $f'^{mp}(X'^{mp}) = \Gamma_G(f)$, and one shows, as in the first case, that this last equality entails $f'^{2mp}(X'^{2mp}) = \Gamma_G(f)$. ∎

Lemma 7.5. *Let S be a scheme, G an S-group scheme, X an S-prescheme, $f : X \to G$ an S-morphism. Suppose that X and G are locally of finite presentation over S, and that for every $s \in S$ and every maximal point g of G_s, there exists a point x of X such that $f(x) = g$ and f is flat at x. Then the morphism $\mu \circ (f \times_S f) : X \times_S X \to G$ is covering for the (fppf) topology.*

[537] By EGA IV₃, 11.3.1, the set V of points of X at which f is flat is open and $f|_V$ is an open morphism, hence $U = f(V)$ is an open of G; moreover, by the hypothesis, $U \cap G_s$ is dense in G_s, for every $s \in S$. Denote by ϕ the restriction to $V \times_S V$ of $\mu \circ (f \times_S f)$.

It suffices to show that ϕ is covering for the (fppf) topology, and for this it suffices to show that ϕ is faithfully flat and of finite presentation (cf. IV, 6.3.1 (iv)). Now ϕ is equal to the composite $V \times_S V \longrightarrow U \times_S U \longrightarrow G$, where the first morphism is faithfully flat and locally of finite presentation, since $f|_V$ is. It therefore suffices to prove that the same is true of the restriction of μ to $U \times_S U$.

Now, $G \to S$ being locally of finite presentation and flat, the same is true of $\mu : G \times_S G \to G$ (which is isomorphic to the morphism deduced from $G \to S$ by the base change $G \to S$, cf. VI_A 0.1), hence also of the induced morphism $U \times_S U \to G$. To prove that the latter is surjective, it suffices to look fiber by fiber, where it follows from VI_A 0.5, since $U \cap G_s$ is a dense open of G_s, for every $s \in S$. □

Remark 7.6.0. [538] *Let S be a scheme, H an S-group, and $f : X \to H$ a morphism of S-schemes. The sub-presheaf in groups of H generated by the image of f, denoted $\langle Imf \rangle$, is the sub-S-functor in groups of H which to every S-scheme S' associates the subgroup of $H(S')$ generated by $f(X(S'))$. Since each element of this subgroup can be written as a finite product $f(x_1)^{\epsilon_1} \cdots f(x_n)^{\epsilon_n}$, with $n \in \mathbb{N}*$, $x_i \in X(S')$ and $\epsilon_i = \pm 1$, one sees therefore that if one sets $X^1 = X \sqcup X$ and defines the morphisms $f^n : X^n \to H$ as in 7.1, then $\langle Imf \rangle$ is nothing but the image presheaf of the morphism*

$$f^\infty : X^\infty \to H,$$

where $X^\infty = \coprod_{n \geqslant 1} X^n$ and $f^\infty : X^\infty \to H$ is the S-morphism whose restriction to each X^n is f^n.

Proposition 7.6. [539] *Let S be a scheme, X a flat S-scheme locally of finite presentation over S, H an S-group, locally of finite presentation over S and with reduced fibers, $f : X \to H$ a morphism of S-schemes, and $f^\infty : X^\infty \to H$ the S-morphism introduced above. The following conditions are equivalent:*

(i) H represents the (fppf) S-sheaf associated with the presheaf $\langle Imf \rangle$.

(ii) f^∞ is covering for the (fppf) topology.

(iii) f^∞ is surjective, i.e. $H = \bigcup_{n \geqslant 1} f^n(X^n)$.

If moreover H is quasi-compact, these conditions are also equivalent to the following:

[537] We have detailed the original in what follows.
[538] We have added this definition, which in the original was contained in the statement of Proposition 7.6.
[539] The original stated this result under the hypotheses of the particular case, but the more general form was used implicitly in the proof of 10.12; we have rewritten the statement accordingly.

(iv) There exists an integer n such that $f^n : X^n \to H$ is covering for the (fppf) topology (and a fortiori surjective).

This applies in particular in the case where X is a k-scheme locally of finite type and geometrically reduced, G a k-group locally of finite type, $\phi : X \to G$ a morphism of k-schemes, $H = \Gamma_G(\phi)$, and $f : X \to H$ the morphism induced by ϕ.

Proof. The sheaf considered in (i) is the image sheaf of X^∞ by f^∞, hence, by IV 4.4.3, to say that H represents this sheaf is equivalent to saying that f^∞ is covering, and this implies that f^∞ is surjective. Conversely, suppose f^∞ surjective and let us show that it is then covering for the (fppf) topology.

Let $s \in S$, η a maximal point of the fiber H_s, and $x \in X_s^\infty$ such that $f^\infty(x) = \eta$ (such an x exists, since f^∞ is surjective). Since the fiber H_s is reduced, $\mathcal{O}_{H_s,\eta}$ is a field, hence f_s^∞ is flat at the point x. Since X^∞ and H are locally of finite presentation over S, and X^∞ flat over S, it follows from the fibrewise flatness criterion (EGA IV_3, 11.3.10) that f^∞ is flat at the point x. Hence, by Lemma 7.5, the morphism

$$\mu \circ (f^\infty \times f^\infty) : X^\infty \times_S X^\infty \to H$$

is covering for the (fppf) topology. Now, since $X^n \times_S X^m$ is canonically isomorphic to X^{n+m}, and that, under this isomorphism, $\mu \circ (f^n \times_S f^m)$ corresponds to f^{n+m}, it is clear that $\mu \circ (f^\infty \times_S f^\infty)$ factors through f^∞, so that f^∞ is covering for the (fppf) topology. This proves that (iii) ⇒ (ii), whence the equivalence of conditions (i), (ii) and (iii).

Note moreover that, since the morphisms $X \to S$ and $H \to S$ are locally of finite presentation, the same is true of $f : X \to H$ (cf. EGA IV_1, 1.4.3 (v)), and as $\mu : H \times_S H$ is also of finite presentation (cf. VI_A, 0.1), it follows that each $f^n : X^n \to H$ is so. Hence, by EGA IV_1, 1.9.5 (viii), the $f^n(X^n)$ are ind-constructible parts of H.

Suppose moreover H quasi-compact (then S is also, since $H \to S$ is surjective). Then, by EGA IV_1, 1.9.9, one concludes that there exists p such that $f^p(X^p) = H$. As before, one then deduces, from the fact that the fibers of H are reduced and from Lemma 7.5, that the morphism $\mu \circ (f^p \times_S f^p) : X^p \times_S X^p \to H$ is covering for the (fppf) topology; since this morphism equals $f^{2p} : X^{2p} \to H$, this concludes the proof of 7.6. $\square$

Remark 7.6.1. *Obviously, the equivalent conditions of 7.6 imply that the sheaf F considered is representable. The converse is false in general:*[540] *for example, if k is of characteristic 0, let $G = G_{a,k}$ and let $f : \operatorname{Spec} k \to G$ be the morphism given by the point 1 of $G(k)$; then F is represented by the constant k-group $\mathbb{Z}_k$, while $\Gamma_G(f) = G$, hence the monomorphism $\mathbb{Z}_k \hookrightarrow G$ is not surjective.*

Let us place ourselves, for simplicity, under the hypotheses of the particular case of 7.6, and suppose F representable. Then, by EGA IV_3, 8.14.2, F is locally

of finite presentation over k, hence the question is whether the dominant monomorphism $F \to \Gamma_G(f)$ is an isomorphism, or equivalently, a closed immersion. This will

[540] We have detailed the original in what follows.

be the case, by virtue of 1.4.2, if F is quasi-compact, and, by VI_A, 0.5.1, this will be verified if F is connected, hence, in particular (7.2.1), if X is connected and if $f(X)$ contains the unit element of G.

Lemma 7.7. *Let k be an algebraically closed field, G a k-group locally of finite type, X a geometrically reduced and locally of finite type k-scheme, $f : X \to G$ a k-morphism and H a group subscheme of G such that $H \subset f(X)$. Set*

$$\Gamma' = \bigcup_{n \geqslant 1} f^n(X^n), \qquad \Gamma'_0 = \Gamma' \cap G(k), \qquad H_0 = H(k)$$

and assume H_0 is of finite index in Γ'_0. Then there exists an integer m such that $f^m(X^m) = \Gamma_G(f)$ (cf. 7.6), and $\Gamma_G(f)$ is the union of finitely many translates of H.

For every $n \geqslant 1$, $f^n(X^n)$ is an ind-constructible part of G (EGA IV_1, 1.9.5 (viii)), the same is therefore true of Γ', so that, since G is a Jacobson scheme, Γ'_0 is dense in Γ'. By hypothesis, there exists a finite sequence $a_1, \cdots, a_r$ of points of Γ'_0 such that $\Gamma'_0 = a_1 H_0 \cup \cdots \cup a_r H_0$, whence

$$\Gamma_G(f) = \overline{\Gamma'} = \overline{\Gamma'_0} = \overline{a_1 H_0 \cup \dots \cup a_r H_0} = \overline{a_1 H_0} \cup \dots \cup \overline{a_r H_0} = a_1 H \cup \dots \cup a_r H,$$

the last equality resulting from the fact that translation by a_i is a homeomorphism of G onto G. One has therefore $\Gamma_G(f) = a_1 H \cup \cdots \cup a_r H$. On the other hand, it is clear that there exists an integer p such that each of the a_i ($1 \leqslant i \leqslant r$) belongs to $f^p(X^p)$. Finally, since $H \subset f(X)$, one has, for every i: $a_i H \subset f^{p+1}(X^{p+1})$, so that $f^{p+1}(X^{p+1}) = \Gamma_G(f)$. □

Proposition 7.8. *Let G be a k-group locally of finite type, A and B two geometrically reduced group sub-k-schemes (hence smooth at the generic points of their irreducible components, hence smooth by 1.3) of G. Suppose one of the following conditions a) or b) is satisfied:*

a) A and B are invariant and of finite type over k.

b) A is connected and B is of finite type over k.

Then (A, B) is of finite type over k, and represents the sheaf associated for the (fppf) topology (or (fpqc)) with the presheaf in groups of commutators of A and B in G. Moreover,[541] the k-groups (A, B^0) and (A^0, B) are connected, and one has

$$(A, B)^0 = (A, B^0) \cdot (A^0, B).$$

Proof. By 7.6, to show that (A, B) is the desired associated sheaf, it suffices to show that there exists an integer n such that $\nu^n((A \times_k B)^n) = (A, B)$ (notations of 7.2.2). To show this, as well as to show the two other assertions, we may suppose

k algebraically closed. Indeed, let $\bar{k}$ be an algebraic closure of k. By 7.1 (iii) and VI_A 2.4, one has, with obvious notations:

$$(A, B)_{\bar{k}} = (A_{\bar{k}}, B_{\bar{k}}), \qquad ((A, B)^0)_{\bar{k}} = (A_{\bar{k}}, B_{\bar{k}})^0, \qquad (A, B^0)_{\bar{k}} = (A_{\bar{k}}, B^0_{\bar{k}}),$$

[541] We have made the following precise and, in the proof, detailed the reduction to the case where k is algebraically closed.

Consequently, if one shows that $(A_{\bar{k}}, B_{\bar{k}})$ is of finite type over $\bar{k}$ (resp. that $(A_{\bar{k}}, B^0_{\bar{k}})$ and $(A^0_{\bar{k}}, B_{\bar{k}})$ are connected, and that the morphism $(A_{\bar{k}}, B^0_{\bar{k}}) \times_{\bar{k}} (A_{\bar{k}}, B^0_{\bar{k}}) \to (A_{\bar{k}}, B_{\bar{k}})^0$ is surjective), then the analogous assertions will be true over k, by EGA IV$_2$, 2.7.1 and 2.6.1.

Let then $B_1, \cdots, B_p$ be the connected components of B other than the neutral component B^0 (these are finite in number since B is supposed of finite type over k, hence noetherian), and in case (a), let similarly $A_1, \cdots, A_q$ be those of A. (In case (b), one will consider only $A^0 = A$). Let ν_{ij} be the restriction of ν to $A_i \times_k B_j$. Then each of the A_i and B_j is irreducible (VI_A 2.4.1), so the same is true of $A^0 \times_k B_j$ and $A_i \times_k B^0$. Since the neutral element of G belongs to A^0 and B^0, it belongs to $\nu_{0j}(A^0 \times_k B_j)$ and to $\nu_{i0}(A_i \times_k B^0)$. Then each of the $\Gamma_G(\nu_{0j})$ and $\Gamma_G(\nu_{i0})$ is connected (7.2.1). Similarly, if u_{0j} (resp. u_{i0}) denotes the injection of $\Gamma_G(\nu_{0j})$ (resp. $\Gamma_G(\nu_{i0})$) into G, then

$$(A^0, B) = \Gamma_G((u_{0j})_{j=0}^p) \quad \text{and} \quad (A, B^0) = \Gamma_G((u_{i0})_{i=0}^q)$$

are connected. Moreover, one easily deduces from 7.4 and the preceding constructions that there exists an index r such that (A^0, B) and (A, B^0) are included in $\nu^r((A \times_k B)^r)$. In case b), one has $(A, B) = (A^0, B)$, and we are done.

Let us now place ourselves in case (a).[542] We already know that (A^0, B) and (A, B^0) are smooth and connected sub-k-groups of G, hence of finite type (cf. VI_A, 2.4). On the other hand, since A^0 is a characteristic subgroup of A (cf. VI_A, 2.6.5), it is an invariant subgroup of G and hence, by 7.3 (v), (A^0, B) is an invariant subgroup of G, and similarly for (A, B^0). Hence, by 7.1.1, the subgroup H of G generated by (A, B^0) and (A^0, B) is none other than $(A, B^0) \cdot (A^0, B)$. In particular, one has therefore $H \subset \nu^{2r}((A \times_k B)^{2r})$.

Given a part X of G stable for the group law (cf. 3.0), we shall denote by X_0 the group of k-points of G belonging to X. Set $\Gamma' = \bigcup_{q \geqslant 1} \nu^q((A \times_k B)^q)$. Then, by Proposition 7.9 below, one has:

$$(A^0, B)_0 = (A^0_0, B_0), \quad (A, B^0)_0 = (A_0, B^0_0) \quad \text{and} \quad \Gamma'_0 = (A_0, B_0),$$

so that $H_0 = (A^0_0, B_0) \cdot (A_0, B^0_0)$ is of finite index in Γ'_0 (Bible, Exp. 3, Appendix) since A^0 and B^0 are invariant, and A^0_0 (resp. B^0_0) is of finite index in A_0 (resp. B_0). We are then in the conditions of Lemma 7.7: since $H \subset \nu^{2r}((A \times_k B)^{2r})$, there exists an integer m such that $\nu^{2rm}((A \times_k B)^{2rm}) = (A, B)$, and there exists a finite sequence $a_1, \cdots, a_n$ of k-points of G such that: $(A, B) = a_1 H \cup \cdots \cup a_n H$. Then, since H is of finite type over k, each of the $a_i H$ is quasi-compact, hence their union (A, B) is quasi-compact, hence of finite type over k. Since H is irreducible, the same is true of each of the $a_i H$, and since $e \in H$, it is clear that $H = (A, B)^0$. ∎

Proposition 7.9. *Let k be an algebraically closed field, G a k-group locally of finite type.*

(i) Let A and B be two ind-constructible parts of G. Denote by A_0 the set of rational points of G belonging to A. Then $(A \cdot B)_0 = A_0 \cdot B_0$, the second product being taken in the group $G(k)$.

[542] We have detailed the original in what follows, taking into account the addition 7.1.1.

(ii) Let X be a geometrically reduced and locally of finite type k-scheme, and $f : X \to G$ a k-morphism. Set $\Gamma'_G(f) = \bigcup_{n\geqslant 1} f^n(X^n)$. Then $\Gamma'_G(f)_0$ is the subgroup of $G(k)$ generated by $f(X)_0$.

(iii) In particular, let A and B be two smooth group subschemes of G; set $\Gamma' = \bigcup_{n\geqslant 1} \nu^n(A \times_k B)^n$ (notations of 7.2.2). Then Γ'_0 is the group of commutators of $A(k)$ and $B(k)$ in $G(k)$.

Let us prove (i). It is clear that $A_0 \cdot B_0 \subset (A \cdot B)_0$. Conversely, let $z \in (A \cdot B)_0$. Then $\mu^{-1}(z)$ is a closed subset of $G \times_k G$, and $A \times_k B$ (cf. 3.0) is an ind-constructible part of $G \times_k G$, so that $\mu^{-1}(z) \cap (A \times_k B)$ is a non-empty ind-constructible part of $G \times_k G$; by EGA IV$_3$, 10.4.8, it therefore contains a rational point of $G \times_k G$, whose projections x and y are rational points of G such that $x \in A$, $y \in B$ and $x \cdot y = z$, so that $(A \cdot B)_0 = A_0 \cdot B_0$.

To prove (ii), note that, f^n being locally of finite type, $f^n(X^n)$ is an ind-constructible part of G (EGA IV$_4$, 1.9.5 (viii)). Assertion (i) then allows one to show by induction that, if one sets $A = f(X)_0$, one has: $f^n(X^n)_0 = (A \cup A^{-1})^n$, and consequently,

$$\Gamma'_G(f)_0 = \bigcup_{n \geqslant 1} f^n(X^n)_0 = \bigcup_{n \geqslant 1} (A \cup A^{-1})^n,$$

which is the subgroup of $G(k)$ generated by $A = f(X)_0$.

Finally, (iii) follows from (ii) and the definitions. $\square$

Corollary 7.10. *Under the conditions of 7.8, if k is algebraically closed, then $(A, B)(k)$ is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.*

Indeed, it suffices to apply 7.9 (iii), 7.8 and 7.6.

8. Solvable or nilpotent group schemes

8.1.

Let $\mathcal{C}$ be a category equipped with a topology T (cf. IV § 4). For every presheaf P on $\mathcal{C}$, we denote by $P^\flat$ the associated sheaf.

Let G be a presheaf in groups on $\mathcal{C}$, A and B two sub-presheaves in groups of G, and let $Comm(A,B)$ be the presheaf in groups of commutators of A and B in G; i.e., for every $S \in Ob\mathcal{C}$, $Comm(A,B)(S)$ is the subgroup of $G(S)$ generated by the commutators $aba^{-1}b^{-1}$, with $a \in A(S)$ and $b \in B(S)$. We write

$$Comm_T(A, B) = Comm(A, B)^\flat.$$

In the proof of 8.2, we shall need the following results.[543]

Lemma 8.1.1. *Let $A \subset G$ be sheaves in groups, with A invariant in G.*

(i) $Comm_T(G, A)$ is the smallest invariant sub-sheaf in groups C of G such that the sheaf $(A/C)^\flat$, associated with the quotient presheaf A/C, is central in $(G/C)^\flat$.

[543] These results are mentioned without proof in the original; we have highlighted them in the form of Lemmas 8.1.1 and 8.1.2, and detailed the proofs.

(ii) In particular, $Comm_T(G, G)$ is the smallest invariant sub-sheaf in groups C of G such that the quotient sheaf $(G/C)^\flat$ is commutative.

Obviously, (ii) is the particular case $A = G$ of (i), so it suffices to show (i). Let C be a sub-sheaf in groups of A, invariant in G, and such that the quotient sheaf $(A/C)^\flat$ is central in $(G/C)^\flat$. By Lemma IV 4.4.8.1, the presheaves A/C and G/C are separated, hence, by IV 4.3.11, all the morphisms in the diagram below are monomorphisms:

$$\begin{tikzcd} A/C \arrow[r] \arrow[d] & G/C \arrow[d] \\ (A/C)^\flat \arrow[r] & (G/C)^\flat \end{tikzcd}$$

Since $(A/C)^\flat$ is central in $(G/C)^\flat$, then A/C is central in G/C, whence $Comm(G, A) \subset C$, and hence C contains $Comm_T(G, A)$, by IV 4.3.12.

Conversely, $Comm(G, A)$ is a sub-presheaf in groups of A, invariant in G, and separated (cf. IV 4.3.1, N.D.E. (24)), hence, by IV 4.4.8.2 (i) and IV 4.3.11, $C = Comm_T(G, A)$ is a sub-sheaf in groups of A, invariant in G and containing $Comm(G, A)$. Consequently, the presheaf A/C is central in G/C and hence, by IV 4.4.8.2 (ii), $(A/C)^\flat$ is central in $(G/C)^\flat$. This proves Lemma 8.1.1. ∎

Lemma 8.1.2. *Let G be a sheaf in groups, A, B two sub-presheaves in groups of G.*

(i) The morphism $\tau : Comm(A, B) \to Comm(A^\flat, B^\flat)$ is a covering monomorphism.

(ii) Consequently, one has an isomorphism

$$Comm_T(A, B) \cong Comm_T(A^\flat, B^\flat).$$

Proof. (i) As A (resp. B) is a sub-presheaf of G, then $A^\flat$ (resp. $B^\flat$) is a sub-presheaf of G containing A (resp. B), and it follows that τ is a monomorphism.

Let us show that τ is covering. Let $S \in Ob\mathcal{C}$ and $g \in Comm(A^\flat, B^\flat)(S)$. Then, there exists an integer $n \geqslant 1$ and, for $i = 1, \cdots, n$, elements $a_i' \in A^\flat(S)$, $b_i' \in B^\flat(S)$, and $\epsilon_i \in \pm 1$, such that

$$g = (a_1', b_1')^{\epsilon_1} \cdots (a_n', b_n')^{\epsilon_n},$$

where (a, b) denotes the commutator $aba^{-1}b^{-1}$, and there exists a refinement R of S such that $a_i' \in A(R)$ and $b_i' \in B(R)$ for every i. Then g is the composite morphism

$$R \xrightarrow{(a_1', \ldots, b_n')} (A \times B)^n \xrightarrow{\Phi_{\epsilon_1, \ldots, \epsilon_n}} Comm(A, B),$$

where $\Phi = \Phi_{\epsilon_1, \cdots, \epsilon_n}$ is the morphism defined set-theoretically by:

$$\Phi(a_1, b_1, \ldots, a_n, b_n) = (a_1, b_1)^{\epsilon_1} \cdots (a_n, b_n)^{\epsilon_n},$$

for every $T \in Ob\mathcal{C}$ and $a_i \in A(T)$, $b_i \in B(T)$. This shows that $g \in Comm(A, B)(R)$ and it follows, as in the proof of IV 4.3.11 (i), that $Comm(A, B) \to Comm(A^\flat, B^\flat)$ is covering.

As $Comm(A^{\flat}, B^{\flat}) \to Comm(A^{\flat}, B^{\flat})^{\flat}$ is also a covering monomorphism (IV 4.3.11 (iv)), the same is true of $Comm(A, B) \to Comm(A^{\flat}, B^{\flat})^{\flat}$, hence, by IV 4.3.12, one obtains an isomorphism:

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Comm(A, B)^♭ ⥲ Comm(A^♭, B^♭)^♭.
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This proves Lemma 8.1.2. ∎

Proposition 8.2. *Let $\mathcal{C}$ be a category, T a topology on $\mathcal{C}$, G a sheaf in groups on $\mathcal{C}$, n an integer $\geqslant 0$. The following conditions are equivalent:*

(i) If one sets $K_0 = G$, and for $p \geqslant 1$, $K_p = Comm(K_{p-1}, K_{p-1})$ (resp. $K_p = Comm(G, K_{p-1})$), then K_n is the unit presheaf in groups.

(ii) If one sets $K'_0 = G$, and for $p \geqslant 1$, $K'_p = Comm_T(K'_{p-1}, K'_{p-1})$ (resp. $K'_p = Comm_T(G, K'_{p-1})$), then K'_n is the unit sheaf in groups.

(iii) There exists a sequence $G = H_0 \supset H_1 \supset \cdots \supset H_n$ of invariant sub-sheaves of G, such that, whatever i, the quotient sheaf H_i/H_{i+1} is commutative (resp. central in G/H_{i+1}), and H_n is the unit sheaf.

It is clear that $K_n \subset K'_n$; consequently (ii) entails (i). Let us show that (i) entails (ii). One has $K'_1 = Comm_T(G, G) = K_1^{\flat}$, and one deduces by induction from Lemma 8.1.2 that $K'_p = K_p^{\flat}$ for every p. Consequently, if K_n is the unit presheaf, then $K'_n = K_n^{\flat}$ is the unit sheaf.

Finally, conditions (ii) and (iii) are equivalent by Lemma 8.1.1. ∎

Definition 8.2.1. *When these conditions are satisfied, the sheaf G is said to be* solvable of class n *(resp.* nilpotent of class n*). When there exists an integer n such that these conditions are satisfied, one says that G is* solvable *(resp.* nilpotent*).*

Note that, by condition (i), this does not depend on the topology T.

Proposition 8.3. *Let k be a field, S a non-empty k-scheme, Ω an algebraically closed extension of k, G a smooth k-group of finite type. The following conditions are equivalent:*

(i) G is solvable of class n (resp. nilpotent of class n).

(ii) $G \times_k S$ is solvable of class n (resp. nilpotent of class n).

(iii) The group $G(\Omega)$ is solvable of class n (resp. nilpotent of class n).

(iv) If one sets $K_0 = G$ and considers, for $p \geqslant 1$, the k-groups $K_p = (K_{p-1}, K_{p-1})$ (resp. $K_p = (G, K_{p-1})$) (cf. 7.2.2), then K_n is the unit k-group.

The equivalence of conditions (i) and (ii) follows from Proposition 8.2, given that the formation of the presheaf in groups of commutators commutes with base change (IV 4.1.3).

The equivalence of (i) and (iv) follows from the fact that, by 7.8, the k-group $K_p = (K_{p-1}, K_{p-1})$ (resp. $K_p = (G, K_{p-1})$) represents the sheaf $Comm_T(K_{p-1}, K_{p-1})$ (resp. $Comm_T(G, K_{p-1})$), where T is the (fppf) (or (fpqc)) topology.

To show that conditions (iii) and (iv) are equivalent, one may suppose $k = \Omega$, and then the equivalence of conditions (iii) and (iv) follows from 7.10. ∎

Proposition 8.4. *Let G be an S-group of finite presentation, such that for every $s \in S$, G_s is smooth over $\kappa(s)$. Let T be the set of $s \in S$ such that G_s is solvable (resp. nilpotent).*

(i) Then T is locally constructible in S.

(ii) If one moreover assumes G flat and separated over S (i.e. when G is smooth, quasi-compact and separated over S), then T is closed in S.

Proof. It is clear that one may suppose S affine with ring A. There exists then, by 10.1 and 10.10 b),[544] a noetherian subring A' of A and an A'-group of finite type G' such that $G' \otimes_{A'} A$ is isomorphic to G. By EGA IV₃, 11.2.6 and 8.10.5,[545] if G is flat and separated over S, one may suppose G' flat and separated over $S' = \operatorname{Spec} A'$.[546] As G is of finite presentation over S, then (EGA IV₃, 9.7.7) the set of $s \in S$ such that G_s is geometrically reduced (or, equivalently, smooth over $\kappa(s)$) is locally constructible. Hence, by EGA IV₃, 9.3.3, one may suppose that, for every $s' \in S'$, $G'_{s'}$ is smooth over $\kappa(s')$. On the other hand, if s' denotes the image of s in S', one has: $G'_{s'} \otimes_{\kappa(s')} \kappa(s) \simeq G_s$. Hence, by 8.3, for G_s to be solvable (resp. nilpotent), it is necessary and sufficient that the same be true of $G'_{s'}$. We are therefore reduced to the case where S is a noetherian affine scheme.

Let us show then that T is constructible. By applying the criterion (EGA 0_III, 9.2.3), one sees, reasoning as before, that one is reduced to showing that, in the case where S is noetherian and integral, either T or $S - T$ contains a non-empty open of S.

Suppose then S integral and noetherian, with generic point η. Set, whatever $s \in S$, $K_s^0 = G_s$, and $K_s^p = (K_s^{p-1}, K_s^{p-1})$ (resp. $K_s^p = (G, K_s^{p-1})$). Let us first show that the sequence of closed subschemes K_η^p is stationary. It follows from 7.3 (v) that each of the K_η^p is invariant, hence the sequence of K_η^p is decreasing; this sequence is then stationary since G_η is noetherian; there exists therefore an integer n such that, for every $p \geqslant n$, one has: $K_\eta^p = K_\eta^n$.

On the other hand, by 10.12.1 and 10.13, there exists a non-empty open S' of S and

an S'-group of finite presentation D such that for every $s \in S'$, one has $D_s = K_s^n$ and $(D_s, D_s) = D_s$ (resp. $(G_s, D_s) = D_s$). We may suppose $S' = S$. Then, whatever $s \in S$, and whatever $p \geqslant n$, one has $D_s = K_s^p$, so that G_s is solvable (resp. nilpotent) if and only if D_s is isomorphic to the unit $\kappa(s)$-group.

But by EGA IV₃, 9.6.1 (xi), the set of $s \in S$ such that the structural morphism $D_s \to \operatorname{Spec} \kappa(s)$ is an isomorphism is constructible,[547] hence is either rare or contains a

[544] We shall make use in the course of this proof of results established in number 10, which do not depend, any more than number 9, on the present n° 8.
[545] We have corrected 10.8.5 to 8.10.5.
[546] We have detailed what follows.
[547] Taking into account the fact that S is supposed affine, hence quasi-compact and quasi-separated (cf. EGA 0_III, 9.1.12).

non-empty open of S. We have therefore obtained that T is locally constructible.

Let us show that if, moreover, G is flat and separated over S, then T is closed. Since T is locally constructible, for T to be closed, it is necessary and sufficient that T be stable under specialization (cf. EGA IV_1, 1.10.1).

Let then $s \in S$ and s' a specialization of s in S. Since one has reduced to the case where S is noetherian, then, by EGA II, 7.1.9, there exists a discrete valuation ring A and a morphism $\mathrm{Spec}(A) \to S$ such that s (resp. s') is the image of the generic point α (resp. of the closed point a) of $\mathrm{Spec}(A)$. It suffices then to show that if one sets $G' = G \otimes_S A$, and if G'_α is solvable (resp. nilpotent), then so is G'_a. Note that, since G is flat and separated over S, G'_α is flat and separated over A, so we are reduced to the case where S is the spectrum of a discrete valuation ring A.

Then, since G_α is supposed solvable (resp. nilpotent), there exists an integer q such that K^q_α (with the notations introduced above) is isomorphic to the unit $\kappa(\alpha)$-group. For every n, let $\bar{K}^n_\alpha$ denote the schematic closure (in the sense of EGA IV_2, 2.8.5) of K^n_α in G. Let us show, by induction on p, that $(\bar{K}^p_\alpha)_a \supset K^p_a$. Note first that, since G is flat over A, then G_α is equal to G (EGA IV_2, 2.8.5), so $(\bar{K}^0_\alpha)_a = K^0_a$.

Let $p \geqslant 0$. Suppose we have established that $K^p_a \subset (\bar{K}^p_\alpha)_a$, and denote by $\nu_a, \nu_\alpha, \nu, \bar{\nu}_a$ the following morphisms, defined as in 7.2.2:

$$\begin{array}{llll}
 & \text{solvable case} & & \text{nilpotent case} \\
\nu_a : & K^p_a \times_{\kappa(a)} K^p_a \to G_a, & \text{resp.} & G_a \times_{\kappa(a)} K^p_a \to G_a, \\
\nu_\alpha : & K^p_\alpha \times_{\kappa(\alpha)} K^p_\alpha \to G_\alpha & \text{resp.} & G_\alpha \times_{\kappa(\alpha)} K^p_\alpha \to G_\alpha, \\
\nu : & \bar{K}^p_\alpha \times_A \bar{K}^p_\alpha \to G, & \text{resp.} & G \times_A \bar{K}^p_\alpha \to G, \\
\bar{\nu}_a : & (\bar{K}^p_\alpha)_a \times_{\kappa(a)} (\bar{K}^p_\alpha)_a \to G_a, & \text{resp.} & G_a \times_{\kappa(a)} (\bar{K}^p_\alpha)_a \to G_a.
\end{array}$$

Since ν_α factors through K^{p+1}_α, then ν factors through $\bar{K}^{p+1}_\alpha$, which is obviously a group subscheme of G, hence $\bar{K}^{p+1}_\alpha$ contains $\Gamma_G(\nu)$. By 7.1 (iii), one has $\Gamma_G(\nu)_a = \Gamma_{G_a}(\bar{\nu}_a)$; and, by the induction hypothesis,

$$K^p_a \times_{\kappa(a)} K^p_a \subset (\bar{K}^p_\alpha)_a \times_{\kappa(a)} (\bar{K}^p_\alpha)_a \quad \text{resp.} \quad G_a \times_{\kappa(a)} K^p_a \subset G_a \times_{\kappa(a)} (\bar{K}^p$$

so that $K^{p+1}_a = \Gamma_{G_a}(\nu_a) \subset \Gamma_{G_a}(\bar{\nu}_a) = \Gamma_G(\nu)_a \subset (\bar{K}^{p+1}_\alpha)_a$.

But since K^q_α is isomorphic to the unit $\kappa(\alpha)$-group, and the unit A-group is flat over A and is isomorphic to a closed subscheme of G (since G is separated over A, cf. 5.1), it follows from EGA IV_2, 2.8.5 that the schematic closure $\bar{K}^q_\alpha$ is isomorphic to the unit A-group. As we just saw that $K^q_a \subset (\bar{K}^q_\alpha)_a$, this entails that K^q_a is isomorphic to the unit $\kappa(a)$-group, so that G_a is solvable (resp. nilpotent). ∎

9. Quotient sheaves

The present number is limited essentially to a reminder, in the particular case of homogeneous spaces of groups, of well-known general facts about passage to the quotient by flat equivalence relations (cf. Exp. IV).

Definition 9.1. *Given a monomorphism $u : G' \to G$ of S-groups, one denotes by G/G' (resp. $G'\backslash G$), and calls* right (resp. left) quotient sheaf of G by G', *the sheaf* *(for the (fpqc) topology) quotient of G by the equivalence relation defined by the monomorphism:*

$$G \times_S G' \xrightarrow{\delta \circ (\mathrm{id}_G \times u)} G \times_S G \qquad (\text{resp. } G' \times_S G \xrightarrow{\gamma \circ (u \times \mathrm{id}_G)} G \times_S G),$$

where δ (resp. γ) denotes the automorphism of $G \times_S G$ defined by $(g,h) \mapsto (g,gh)$ (resp. $(h,g) \mapsto (hg,g)$) for $g,h \in G(T)$.

Proposition 9.2. *Let $u : G' \to G$ be a monomorphism of S-groups. Suppose that G/G' is representable by an S-scheme G''. Then:*

(i) The canonical morphism $p : G \to G''$ is covering for the (fpqc) topology.

(ii) If one sets $\epsilon'' = p \circ \epsilon$ (this morphism is called the unit section of G''), the following diagrams are cartesian:

$$\begin{tikzcd} G \times_S G' \arrow[r, "\mu \circ (\mathrm{id}_G \times u)"] \arrow[d, "pr_1"'] & G \arrow[d, "p"] \\ G \arrow[r, "p"] & G'' \end{tikzcd} \qquad \begin{tikzcd} G' \arrow[r, "u"] \arrow[d, "\pi'"'] & G \arrow[d, "p"] \\ S \arrow[r, "\epsilon''"] & G''. \end{tikzcd}$$

In particular, u is an immersion.

(iii) There exists on G'' a unique structure of S-scheme with left operator group G, such that p is a morphism of S-schemes with operator group G.

(iv) If one moreover supposes G' invariant in G, there exists on G'' a unique structure of S-group such that p is a morphism of S-groups.

(v) Let S_0 be an S-scheme; set $G_0 = G \times_S S_0$, and $G'_0 = G' \times_S S_0$; then G_0/G'_0 is representable by $G''_0 = G'' \times_S S_0$.[548]

(vi) Let P be a property for an S-morphism. Suppose P stable under base change; then if $p : G \to G''$ satisfies P, the same is true of the structural morphism $\pi' : G' \to S$.

(vii) Let P be a property for an S-morphism. Suppose P is local in nature for the (fpqc) topology (cf. 2.0 and 2.1.2). Then, for the morphism $p : G \to G''$ to satisfy P, it is necessary and sufficient that the same be true of $\pi' : G' \to S'$.

(viii) Let P be a property for an S-morphism; suppose P is local in nature for the (fpqc) topology, and stable under composition; then, if the structural morphisms $G'' \to S$ and $G' \to S$ satisfy P, the same is true of the structural morphism $G \to S$.

(ix) Suppose G reduced; then G'' is reduced.

(x) For G'' to be separated over S, it is necessary and sufficient that u (or, equivalently, ϵ'') be a closed immersion.

[548] i.e., the quotient is universal, cf. Exp. IV § 3.

(xi) For G' to be flat over S, it is necessary and sufficient that p be a flat morphism (or, equivalently, faithfully flat).

In this case, for G'' to be flat over S, it is necessary and sufficient that G be flat over S.

(xii) For G' to be flat and locally of finite presentation over S, it is necessary and sufficient that $p : G \to G''$ be faithfully flat and locally of finite presentation.

In this case, for G'' to be locally of finite presentation (resp. locally of finite type, of finite type, smooth, étale, unramified, locally quasi-finite, quasi-finite) over S, it suffices that the same be true of G over S (and the condition is also necessary in the first two cases, cf. (viii)).

(xiii) Suppose G' flat and of finite presentation over S.

a) Then p is of finite presentation and faithfully flat;

b) moreover, for G to be of finite presentation over S, it is necessary and sufficient that the same be true of G''.

Recall that the equivalence relation under consideration is universally effective (IV 4.4.9). Then assertions (i), (iii), (iv), (v) and the first assertion of (ii) follow from IV 4.4.3, 5.2.2, 5.2.4, 3.4.5 and 3.3.2 (iii). The second assertion of (ii) follows from the first, as the following cartesian diagram shows, since $(G \times_S G') \times_G S$ is isomorphic to G':

$$\begin{tikzcd} G' \arrow[r, "((\varepsilon \circ \pi'), \mathrm{id}_{G'})"] \arrow[d, "\pi'"] & G \times_S G' \arrow[r, "\mu \circ (\mathrm{id}_G \times u)"] \arrow[d, "pr_1"] & G \arrow[d, "p"] \\ S \arrow[r, "\varepsilon"] & G \arrow[r, "p"] & G''. \end{tikzcd}$$

Finally, it is clear that ϵ'' is an S-section of G'', hence an immersion (EGA I, 5.3.13); by the preceding cartesian diagram, the same is true of u, which completes the proof of (ii). Moreover, (vi) is an immediate consequence of the second cartesian diagram of (ii).

Let us show (vii). By (i), p is covering for the (fpqc) topology; hence, by (ii), to show that p satisfies P, it suffices to show that the first projection $pr_1 : G \times_S G' \to G$ satisfies P, which follows from the fact that P is stable under base change, since pr_1 comes from π' by base change.

It is clear that (viii) follows from (vii), since $\pi = \pi'' \circ p$, where $\pi'' : G'' \to S$ denotes the structural morphism.

Let us show (ix). By (i), p is an epimorphism; since G is reduced, p factors through the immersion $G''_{red} \to G''$, which is therefore also an epimorphism, hence an isomorphism (IV 4.4.4).

Let us show (x). If G'' is separated over S, then ϵ'' is a closed immersion, by EGA I, 5.4.6. On the other hand, one saw in (ii) that ϵ'' is a closed immersion if and only if u is. Finally, if u is a closed immersion, then so is $\delta \circ (id_G \times u) : G \times_S G' \to G \times_S G$; hence, by Lemma 9.2.1 below, G'' is separated over S.

Assertion (xi) follows from (vii) and from EGA IV_2, 2.2.13.

Assertion (xii) follows from (vii), from EGA IV_4, 17.7.5 and 17.7.7, and from the fact that, p being universally open, whatever $s \in S$, if the underlying space of G_s is discrete, the same is true of the underlying space of G''_s.

Finally, assertion (xiii) follows from (vii), (viii), and from EGA IV_4, 17.7.5. □

Lemma 9.2.1. *Let X be an S-scheme and R an equivalence relation defined on X by the monomorphism $v : R \to X \times_S X$. Suppose R effective. Then:*

(i) v is an immersion, and is a closed immersion if $Y = X/R$ is separated.

(ii) Suppose moreover that Y represents the (fpqc) quotient sheaf of X by R[549] *and that v is a closed immersion. Then $Y = X/R$ is separated over S.*

Recall (IV Def. 3.3.2) that the hypothesis "R effective" means that there exists a morphism of S-schemes $p : X \to Y$ such that the natural morphism $R \to X \times_Y X$ is an isomorphism. From this one deduces (EGA I, 5.3.5) the following cartesian diagram:

$$\begin{tikzcd} R \arrow[r, "v"] \arrow[d, "\Delta_{Y/S}"'] & X \times_S X \arrow[d, "p \times p"] \\ Y \arrow[r] & Y \times_S Y. \end{tikzcd}$$

Then, since $\Delta_{Y/S}$ is an immersion (EGA I, 5.3.9), the same is true of v. Similarly, if Y is separated over S, $\Delta_{Y/S}$ is a closed immersion, hence so is v.

Conversely, suppose that v is a closed immersion and that Y represents the (fpqc) quotient sheaf of X by R. Then p is covering for the (fpqc) topology (IV 4.4.3), and hence $p \times p$ is too (by base change, $p \times id_X$ and $id_Y \times p$ are covering, hence so is their composite $p \times p$). Hence, by (fpqc) descent (cf. EGA IV_2, 2.7.1), $\Delta_{Y/S}$ is a closed immersion, i.e. Y is separated over S. □

Remark 9.2.2. *Under the general hypotheses of 9.2, if one assumes G' flat and locally of finite presentation over S, then p is covering for the (fppf) topology,*[550] *by 9.2 (vii), hence assertions (vii) and (viii) of 9.2 can be extended to properties P local in nature for the (fppf) topology.*

Remark 9.3. *a) The question of whether a quotient G/G' is representable or not is often delicate; in this seminar we demonstrate the representability of certain*

[549] As pointed out by O. Gabber, this is used in the proof and must be inserted in the hypotheses.
[550] We have corrected (fpqc) to (fppf).

particular quotients.

In general, in order to assert that the quotient G/G' is representable, it is not sufficient to suppose G and G' of finite presentation over S and G' flat over S. Indeed, suppose moreover G smooth with connected fibers. In this case, if G/G' is a scheme, it is separated, by Corollary 5.4, and hence $G' \hookrightarrow G$ is a closed immersion, by 9.2 (x); consequently, if G' is not closed in G, then G/G' is not representable.

To obtain such a counterexample, one may take for S the spectrum of a discrete valuation ring, and set $G = G_{m,S}$. Consider moreover an integer $n > 1$, invertible on S; then $\mu_n = Ker(G \xrightarrow{n} G)$ is a closed subgroup of G étale over S[551] *(cf. VII_A). Let G' be the open subgroup of μ_n obtained by removing from μ_n the closed part of the closed fiber of μ_n complementary to the origin. Then G' is not closed in G, hence G/G' is not representable. (One can also fabricate such examples where G' is smooth with connected fibers.)*

b) It is not excluded that G/G' be representable on the other hand, when G and G' are of finite presentation over S, and G' is flat over S and closed in G.[552][553] *Under these hypotheses, it is known that G/G' is representable in the following particular cases:*

1° — S is the spectrum of an Artinian ring (cf. VI_A 3.2 and 3.3.2).

2° — G' is proper over S and G quasi-projective over S (cf. V 7.1).

3° — S is locally noetherian of dimension 1 (cf. [An73], Th. 4.C).

10. Passage to the inverse limit in group schemes and in schemes with operator group

10.0.

Let us recall the essential result of EGA IV₃, § 8.8. Suppose given the following situation: `S_0` a quasi-compact and quasi-separated scheme, I a filtered increasing preordered set, $(A_i)_{i\in I}$ an inductive system of quasi-coherent commutative $\mathcal{O}_{S_0}$-algebras, $A = \varinjlim A_i$, $S_i = \operatorname{Spec} A_i$ for $i \in I$, and $S = \operatorname{Spec} A$;[554] then the category of S-schemes of finite presentation is determined up to equivalence by the data of the categories of S_i-schemes of finite presentation, of the functors between these categories $\rho_{ji} : X_i \mapsto X_i \times_{S_i} S_j$ for $i \leqslant j$, and the transitivity isomorphisms $\rho_{kj} \circ \rho_{ji} \xrightarrow{\sim} \rho_{ki}$.

Let us be precise. Given $j \in I$, and an S_j-scheme of finite presentation X_j, we shall set, for every $i \in I$ such that $i \geqslant j$, $X_i = X_j \times_{S_j} S_i$, and $X = X_j \times_{S_j} S$. Then (EGA IV₃, 8.8.2):

[551] see VII_A, 8.4 or VIII, 2.1.

[552] This is too optimistic, as M. Raynaud shows in his thesis (loc. cit. X 14).

[553] The remark (∗) refers to the counterexample X.14 in [Ray70a]. The base there is regular local of dimension 2.

[554] Note that S, being affine over `S_0`, is therefore quasi-compact and quasi-separated, cf. N.D.E. (92) below.

(i) Given $j \in I$, and two S_j-schemes of finite presentation X_j and Y_j, the canonical map $\varinjlim_{i \geqslant j} \mathrm{Hom}_{S_i}(X_i, Y_i) \to \mathrm{Hom}_S(X, Y)$ is bijective.

(ii) For every S-scheme of finite presentation X, there exists an index $j \in I$, an S_j-scheme of finite presentation X_j and an S-isomorphism $X \xrightarrow{\sim} X_j \times_{S_j} S$.

One concludes (EGA IV_3, 8.8.3) that, whenever one has a diagram $\mathcal{D}$ involving a finite number of objects and arrows of the category of S-schemes of finite presentation, one can find an index $i \in I$ and a diagram $\mathcal{D}_i$ in the category of S_i-schemes of finite presentation, such that the diagram $\mathcal{D}$ comes up to isomorphism from the diagram $\mathcal{D}_i$ by base change $S \to S_i$. One can even find i and $\mathcal{D}_i$ such that every cartesian square of $\mathcal{D}$ comes from a cartesian square of $\mathcal{D}_i$.

10.1.

Moreover, a great number of common properties for a morphism, stable under base change, possess the following property:

> Let $u : X \to Y$ be an S-morphism between S-schemes of finite presentation; it comes by base change from an S_j-morphism $u_j : X_j \to Y_j$ between S_j-schemes of finite presentation, by 10.0; then, for u to have property P, it is necessary and sufficient that there exist $i \geqslant j$ such that $u_i = u_j \times_{S_j} S_i$ has property P.

This is so in the case where P is one of the following properties for a morphism: being separated, surjective, radicial, affine, quasi-affine, finite, quasi-finite, proper, projective, quasi-projective, an isomorphism, a monomorphism, an immersion, an open immersion, a closed immersion (EGA IV_3, 8.10.5), flat (EGA IV_3, 11.2.6), smooth, unramified or étale (EGA IV_4, 17.7.8).[555]

Note that this is also the case where P is the property of being covering for the (fppf) topology; indeed, given two S-schemes of finite presentation X and Y, and an S-morphism $u : X \to Y$, it follows from IV, 6.3.1 (i)[556] that, for u to be covering for the (fppf) topology, it is necessary and sufficient that there exist an S-scheme Z and an S-morphism $v : Z \to Y$ faithfully flat and of finite presentation which factors through u.

The aim of this section 10 is to give variants of this kind of results for the category of S-groups of finite presentation, that of S-schemes with operator group, and for certain properties for monomorphisms of groups (being invariant, central with representable quotient sheaf, etc.).

The two preliminary results of this type are the following. (In nos. 10.2 to 10.9 below, we keep the notations introduced in 10.0.)

Lemma 10.2. *Let G_j and H_j be two S_j-groups of finite presentation; set, for every $i \geqslant j$, $G_i = G_j \times_{S_j} S_i$, $G = G_j \times_{S_j} S$, and define similarly H_i and H. Then the canonical map below is bijective:*

[555] We have added the word "flat", and corrected 17.7.6 to 17.7.8.

[556] and from the fact that Y, being of finite presentation over S, is quasi-compact.

$$\varinjlim_{i \geqslant j} \mathrm{Hom}_{S_i\text{-gr.}}(G_i, H_i) \to \mathrm{Hom}_{S\text{-gr.}}(G, H).$$

Lemma 10.3. *Let G be an S-group of finite presentation; then there exist $j \in I$, an S_j-group of finite presentation G_j, and an isomorphism of S-groups $G \cong G_j \times_{S_j} S$.*

Assertions 10.2 and 10.3 are easy consequences of 10.0 and 10.1, taking into account the interpretation[557] of the structure of S-group given in EGA 0_III, 8.2.5 and 8.2.6.

Lemma 10.4. *Let $u : G \to H$ be a morphism of S-groups between S-groups of finite presentation. By 10.3 and 10.2, u comes by base change from a morphism $u_j : G_j \to H_j$ between S_j-groups of finite presentation. Then, for u to be a central monomorphism (resp. an invariant monomorphism), it is necessary and sufficient that there exist $i \geqslant j$ such that $u_i = u_j \times_{S_j} S_i$ has the same property.*

This is an immediate consequence of 10.0 and 10.1, taking into account the characterization given in 6.7 of central or invariant monomorphisms of groups.

Corollary 10.5. *Let G_j be an S_j-group of finite presentation. For $G_j \times_{S_j} S$ to be commutative, it is necessary and sufficient that there exist $i \geqslant j$ such that $G_j \times_{S_j} S_i$ is so.*

Indeed, it amounts to the same to say that an S-group is commutative, or that, considered as a group subscheme of itself, it is central.

Proposition 10.6. *Let G_j be an S_j-group of finite presentation, G'_j a group subscheme of G_j flat and of finite presentation over S_j. For $(G_j \times_{S_j} S)/(G'_j \times_{S_j} S)$ to be representable for the (fpqc) topology, it is necessary and sufficient that there exist $i \geqslant j$ such that $(G_j \times_{S_j} S_i)/(G'_j \times_{S_j} S_i)$ is so.*

This is a consequence of the following more general lemma.

Lemma 10.7. *Let X_j be an S_j-scheme of finite presentation, and R_j an equivalence relation on X_j flat and of finite presentation.*[558] *For the quotient sheaf $(X_j \times_{S_j} S)/(R_j \times_{S_j} S)$ for the topology T = (fppf) or (fpqc) to be representable, it is necessary and sufficient that there exist $i \geqslant j$, such that the quotient sheaf $(X_j \times_{S_j} S_i)/(R_j \times_{S_j} S_i)$ for the topology T is so.*

Taking into account the statements of EGA IV$_2$, 8.8.2, 8.8.3, 8.10.5 and 11.2.6 recalled in 10.0, this lemma is a consequence of the following result.

Lemma 10.8. *Let T be the (fppf) or (fpqc) topology; let X be an S-scheme of finite presentation (resp. locally of finite presentation), R an equivalence relation on X defined by a monomorphism $v : R \to X \times_S X$ such that $pr_1 \circ v$ is flat and of finite presentation (resp. flat and locally of finite presentation). Then the following conditions are equivalent:*

(i) The quotient sheaf X/R for the topology T is representable.

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[557] in terms of commutative diagrams of S-morphisms.

[558] i.e., such that the composite $R_j \hookrightarrow X_j \times_{S_j} X_j \xrightarrow{pr_1} X_j$ is flat and of finite presentation.

(ii) There exist an S-scheme of finite presentation (resp. locally of finite presentation) Y and a faithfully flat morphism $p : X \to Y$ such that the diagram

$$\text{(D)} \qquad \begin{tikzcd} R \arrow[r, "pr_1 \circ v"] \arrow[d, "pr_2 \circ v"'] & X \arrow[d, "p"] \\ X \arrow[r, "p"] & Y \end{tikzcd}$$

is cartesian.

Let us note first that, by IV, 3.3.2 and 4.4.3, for the sheaf X/R for the topology T to be representable by Y, it is necessary and sufficient that the diagram (D) be cartesian and that p be covering for the topology T.

Let us show that (i) entails (ii). Hypothesis (i) implies that the diagram (D) is cartesian, hence that $pr_1 \circ v$ is deduced from p by base change by p, and that p is covering for the (fpqc) topology. Hence, by (fpqc) descent (EGA IV_2, 2.7.1), since $pr_1 \circ v$ is faithfully flat and (locally) of finite presentation, so is p. Then, by EGA IV_4, 17.7.5, since X is (locally) of finite presentation over S, so is Y.

Let us show that (ii) entails (i). It suffices to show that p is covering for the (fppf) topology; now p is faithfully flat by hypothesis, and is locally of finite presentation since X and Y are locally of finite presentation over S (EGA IV_1, 1.4.3 (v)). ∎

Lemma 10.9. *Let G_j be an S_j-group of finite presentation, and $G = G_j \times_{S_j} S$. For G^0 to be representable, it is necessary and sufficient that there exist $i \geqslant j$ such that $(G_i)^0 = (G_j \times_{S_j} S_i)^0$ is so.*

The condition is sufficient, since the functor $G \mapsto G^0$ commutes with base change, by 3.3.

Conversely, suppose G^0 representable. Then, by 3.9, G^0 is open in G and quasi-compact over S, hence of finite presentation over S, since G is. Then, by 10.3 and 10.1, there exist $i \geqslant j$ and an open group subscheme H_i of G_i such that $H_i \times_{S_i} S = G^0$. The structural morphism $G^0 \to S$ is connected, i.e. has geometrically connected fibers (VI_A 2.1.1), hence, by EGA IV_3, 9.3.3 and 9.7.7, up to increasing i, one may suppose that the structural morphism $H_i \to S$ is connected. Then, by 3.10.1, the underlying space of H_i is none other than $(G_i)^0$, and hence H_i represents $(G_i)^0$. ∎

10.10.

Let us recall two very useful particular cases of the situation stated in 10.0 (cf. EGA IV_3, 8.1.2 a) and c)):

a) Given a point x of a scheme X, one sets $S_0 = \operatorname{Spec} \mathbb{Z}$ and considers the filtered decreasing projective system $(S_i)_{i \in I}$ of affine open neighborhoods of x; then $S = \operatorname{Spec} \mathcal{O}_{X,x}$. In particular, if x is the generic point of an integral scheme X, one finds $S = \operatorname{Spec} \kappa(x)$.

b) One sets $S_0 = \operatorname{Spec}\mathbb{Z}$, and considers the family $(A_i)_{i\in I}$ preordered by inclusion of the finitely generated sub-$\mathbb{Z}$-algebras of the ring of an affine scheme S. Given that the A_i are noetherian rings, this allows in many cases to pass from the noetherian case to the general case.

We are now going to give two results concerning the particular case considered in a).[559]

Proposition 10.11. [560] *Let S be an integral scheme with generic point η, G (resp. Y, Z) an S-group (resp. S-schemes) of finite presentation, $u, v, i : Y \to G$ and $j : Z \to G$ morphisms of S-schemes. Suppose that $i_\eta : Y_\eta \to G_\eta$ and $j_\eta : Z_\eta \to G_\eta$ are closed immersions.*

Then, there exists a non-empty open S' of S such that the morphisms $i' : Y' \to G'$ and $j' : Z' \to G'$ obtained by base change are closed immersions, and such that the functors:

`Transp(u′, v′),    Transp_{G′}(i′(Y′), j′(Z′))    and    Transpstr_{G′}(i′(Y′), j′(Z′))`

resp.

`Centr(u′)    and    Norm_{G′} i′(Y′)`

are representable by closed sub-S'-schemes (resp. sub-S'-groups) of G', of finite presentation over S'.

We shall apply the results of 10.1, first in the situation of 10.10 a), then in that of 10.10 b). Since G_η, Y_η, Z_η are flat over the field $\kappa(\eta)$, G_η is separated over $\kappa(\eta)$ (VI_A 0.3), and i_η, j_η are closed immersions, then, by 10.1, there exists an affine open $S' = \operatorname{Spec}(A')$ of S, a noetherian subring A of A', A-schemes G_A, Y_A, Z_A, flat and of finite presentation over A, and morphisms $u_A, v_A, i_A : Y_A \to G_A$ and $j_A : Z_A \to G_A$, such that `G_A` is an A-group, separated over A, i_A and j_A are closed immersions, and $G \times_S S' = G_A \otimes A'$, etc. As the functors considered for S' are deduced by base change from the analogous functors for $\operatorname{Spec}(A)$, it suffices to establish the result for the latter.

By EGA IV₂, 6.9.2, up to replacing A by a localization A_f (and hence S' by the affine open S'_f), one may suppose that G_A, Y_A, Z_A are essentially free over A (in the sense of 6.2.1).[561] It follows then from 6.2.4 b) and e) that, under the hypotheses of the statement, the functors considered are representable by closed sub-A-schemes of `G_A` (hence of finite presentation over A, since A is noetherian and `G_A` of finite presentation

[559] Note that the proof uses also case b).

[560] We have simplified the statement, and treated separately, in Corollary 10.11.1, the case of subgroups.

[561] Indeed, `G_A` being flat and of finite presentation over A, it is covered by affine opens $G_1, \cdots, G_n$ such that each $\mathcal{O}(G_i)$ is a flat and finitely presented A-algebra; then, by EGA IV₂, 6.9.2, there exists $f_i \in A$ such that $\mathcal{O}(G_i)_{f_i}$ is a free module over A_{f_i}; one may then replace $\operatorname{Spec}(A)$ by the affine open $D(f)$, where $f = f_1 \cdots f_n$, and one does the same for `Y_A` and `Z_A`.

over A), and these are sub-A-groups of G_A in the case of $Centr(u_A)$ and of $Norm_{G_A} i_A(Y_A)$. □

Corollary 10.11.1. *Let S be an integral scheme with generic point η, G, H, K S-groups of finite presentation, $i : H \to G$ and $j : K \to G$ two quasi-compact monomorphisms of S-groups. Then, there exists a non-empty open S' of S such that the morphisms $i' : H' \to G'$ and $j' : K' \to G'$ obtained by base change are closed immersions, and such that the functors:*

$$\mathrm{Transp}_{G'}(H', K') \quad \text{and} \quad \mathrm{Transpstr}_{G'}(H', K') \quad (\text{resp. } \mathrm{Centr}_{G'} H' \quad \text{and} \quad \mathrm{Norm}_{G'} H')$$

are representable by closed sub-S'-schemes (resp. sub-S'-groups) of G', of finite presentation over S'.

This follows from the preceding proposition since, by 1.4.2, the hypotheses entail that i_η and j_η are closed immersions.

Proposition 10.12. *Let S be an integral scheme, G an S-group of finite presentation, A and B two group subschemes of G, of finite presentation over S and with smooth generic fiber. Suppose moreover satisfied one of the following conditions:*

a) A has connected generic fiber,

b) A and B are invariant in G.

Then, there exists a non-empty open S' of S and a closed group subscheme D' of $G' = G|_{S'}$, of finite presentation over S', with smooth fibers, which represents the (fppf) sheaf associated with the presheaf in groups of commutators of $A' = A|_{S'}$ and $B' = B|_{S'}$ in G', and D' has connected fibers in case (a), and is invariant in G' in case (b).

In particular, for every $s \in S'$, one has $D'_s = (A_s, B_s)$ with the notations of 7.2.2.

Let η be the generic point of S; set $H_\eta = (A_\eta, B_\eta)$. Since A_η and B_η are smooth, then, by 7.8 in case (a), and 7.3 (v) in case (b), H_η is connected (resp. invariant in G_η).

We are in the situation of 10.0 corresponding to 10.10 (a); hence, by 10.3 and 10.1, there exists a non-empty open S' of S and a sub-S'-group scheme D' of finite presentation and closed in G', such that $D'_\eta = D' \otimes_{S'} \kappa(\eta)$ equals H_η. Moreover, by EGA IV₃, 9.7.7 and 9.3.3, one may suppose that D' has geometrically reduced fibers. In case (a), one may suppose, by EGA IV₃, 9.7.7 and 9.3.3 again, that D' has connected fibers, hence geometrically connected (cf. VI_A, 2.1.1). In case (b), one may suppose, by 10.4, that D' is invariant in G'.

Moreover, we have seen, in the course of the proof of 7.8, that there exists an integer n such that $\nu_\eta^n((A_\eta \times_{\kappa(\eta)} B_\eta)^n) = D'_\eta$, where ν_η and ν_η^n are defined as in 7.2.2 (a) and 7.1 (ii). We may define by the same formulas the morphisms

$$\nu' : A' \times_{S'} B' \to G' \quad \text{and} \quad \nu'^n : (A' \times_{S'} B')^n \to G',$$

and one has $\nu'^n \otimes_{S'} \kappa(\eta) = \nu_\eta^n$.

Consequently, by 10.1, one may choose S' such that the morphism ν'^n is flat and factors through D', and such that the morphism $\nu'^n : (A' \times_{S'} B')^n \to D'$ thus obtained is surjective. Then, by 7.5, the morphism[562]

$$\nu'^{2n} = \mu \circ (\nu'^n \times_{S'} \nu'^n) : (A' \times_{S'} B')^{2n} \to D',$$

is covering for the (fppf) topology. Hence, by 7.6, D' represents the (fppf) sheaf associated with the presheaf of commutators of A' and B' in G'.

Moreover, ν'^n induces, for every $s \in S'$, a surjective morphism $\nu_s^n : (A_s \times_{\kappa(s)} B_s)^n \to D'_s$.[563] Then D'_s is a closed subgroup of G_s containing $\nu_s(A_s \times_{\kappa(s)} B_s)$, hence also (A_s, B_s). As ν_s^n is surjective, D'_s equals (A_s, B_s) and represents, by 7.6, the (fppf) sheaf of commutators of A_s and B_s in G_s. ∎

Corollary 10.12.1. [564] *Let S be an integral scheme, with generic point η, and G an S-group of finite presentation with smooth fibers. Set $K_\eta^0 = G_\eta$ and $K_\eta^p = (K_\eta^{p-1}, K_\eta^{p-1})$ (resp. $K_\eta^p = (G, K_\eta^{p-1})$) for every $p \in \mathbb{N}*$. Fix $n \in \mathbb{N}*$. Then there exists a non-empty open S' of S and a group subscheme D invariant in $G|_{S'}$, of finite presentation and with smooth fibers, such that $D_s = K_s^n$ for every $s \in S'$.*

This follows from 10.12, by induction on n.

Corollary 10.13. *Let S be an integral scheme with generic point η, G an S-group, H an invariant sub-S-group scheme in G; suppose G and H of finite presentation over S and with smooth generic fiber.*[565]

If one has $(H_\eta, H_\eta) = H_\eta$ (resp. $(G_\eta, H_\eta) = H_\eta$), then there exists a non-empty open S' of S such that for every $s \in S'$, one has $(H_s, H_s) = H_s$ (resp. $(G_s, H_s) = H_s$).

Indeed, by the proof of 10.12, there exists a non-empty open S' of S and a sub-S'-group scheme D of $G|_{S'}$, of finite presentation and with smooth fibers, such that $D_s = (H_s, H_s)$ (resp. $D_s = (G_s, H_s)$) for every $s \in S'$. On the other hand, since $D_\eta = H_\eta$ and since D and H are of finite presentation over S', then, by EGA IV₃, 8.8.2.5, there exists a non-empty open S'' of S' such that $D|_{S''} = H|_{S''}$. For every $s \in S''$, one has therefore $H_s = D_s = (H_s, H_s)$ (resp. $H_s = D_s = (G_s, H_s)$). ∎

10.14.

Statements 10.2 and 10.3 concerning the category of S-groups of finite presentation extend to the category of pairs formed by an S-group of finite presentation and an S-scheme of finite presentation with operator group G. To be precise, in the situation recalled at the start of 10.0:

(i) Let $j \in I$ and G_j and G'_j two S_j-groups of finite presentation, H_j (resp. H'_j) an S_j-scheme of finite presentation with operator group G_j (resp. G'_j). Set, for $i \in I$, $i \geqslant j$, $G_i = G_j \times_{S_j} S_i$ and $G = G_j \times_{S_j} S$, and define similarly G'_i, G', H_i, H, H'_i

[562] We have corrected $n+1$ to $2n$ below.
[563] We have detailed the original in what follows.
[564] We have added this corollary, used in the proof of 8.4.
[565] The original assumed G, H of finite presentation and H with smooth fibers; we have modified the hypothesis in order to be able to apply 10.12. We have also detailed the proof.

and H'. Denote by $Dihom_{S-gr.}((G,H),(G',H'))$ the set of di-morphisms of S-groups and S-schemes with operator group of the pair (G,H) into the pair (G',H'). Then the canonical map

$$\varinjlim_{i \geqslant j} Dihom_{S_i\text{-}gr.}((G_i, H_i), (G'_i, H'_i)) \to Dihom_{S\text{-}gr.}((G, H), (G', H'))$$

is bijective.

(ii) Let G be an S-group of finite presentation and H an S-scheme of finite presentation with operator group G; there then exists an index $j \in I$, an S_j-group of finite presentation G_j, an S_j-scheme of finite presentation H_j with operator group G_j and a di-isomorphism of S-groups and S-schemes with operator groups from $(G_j \times_{S_j} S, H_j \times_{S_j} S)$ onto (G,X).

Definition 10.15. [566] *Let T be a topology on (Sch/S), less fine than the canonical topology. Given an S-group scheme G and an S-scheme X with operator group G, one says that X is a* formally homogeneous space under G *(relative to the topology T) if the morphism $\Phi : G \times_S X \to X \times_S X$, defined by $(g,x) \mapsto (gx,x)$ for every $S' \to S$ and $g \in G(S')$, $x \in X(S')$, is an epimorphism in the category of sheaves for the topology T, which amounts to saying that Φ is covering for the topology T (cf. IV 4.4.3).*

One says that X is a homogeneous space *if it is formally homogeneous and if moreover the morphism $X \to S$ is also covering for the topology T.*

In particular, one says that X is a formally principal homogeneous space under G *if Φ is an isomorphism, and that X is a* principal homogeneous bundle *(or G-*torsor*) if Φ is an isomorphism and if moreover the morphism $X \to S$ is covering for the topology T (cf. IV 5.1.5 and 5.1.6 (ii)).*

Proposition 10.16. *We place ourselves in the situation considered at the beginning of 10.0. Let $j \in I$, G_j an S_j-group and X_j an S_j-scheme with operator group G_j. Suppose G_j and X_j of finite presentation over S_j.*

For $X = X_j \times_{S_j} S$ to be a homogeneous space (resp. a principal homogeneous bundle) under $G = G_j \times_{S_j} S$ for the (fppf) topology, it is necessary and sufficient that there exist an index $i \geqslant j$ such that $X_i = X_j \times_{S_j} S_i$ is a homogeneous space (resp. a principal homogeneous bundle) under $G_i = G_j \times_{S_j} S_i$.

Taking into account 10.14 and EGA IV_3, 8.8.2, 8.8.3 and 8.10.5, the statement follows from the property concerning covering morphisms for the (fppf) topology recalled in 10.1.[567]

[566](Added by editors; see source.)
[567](End of section 10.)

11. Affine group schemes

11.0. Reminders.

[568] Let $q : X \to S$ be a quasi-compact and quasi-separated morphism of schemes (cf. EGA IV_1, 1.1 & 1.2), and let F be a quasi-coherent $\mathcal{O}_X$-module. Recall that $q_*(F)$ is a quasi-coherent $\mathcal{O}_S$-module (EGA I, 9.2.1). Moreover, by EGA III, 1.4.15 (completed by EGA IV_1, 1.7.21), one has point (c) below, and the proof of loc. cit. also gives points (a) and (b):

(a) If F is a filtered inductive limit of quasi-coherent submodules F_α, then $q_*(F) = \lim_\alpha q_*(F_\alpha)$.

(b) If E is a flat $\mathcal{O}_S$-module, the canonical morphism $E \otimes_{\mathcal{O}_S} q_*(\mathcal{O}_X) \to q_* q^*(E)$ is an isomorphism.

(c) Let $p : S' \to S$ be a flat morphism, $q' : X' \to S'$ the morphism deduced from q by base change, and F' the inverse image of F on X'. Then the canonical morphism $p^* q_*(F) \to q'_*(F')$ is an isomorphism.

Indeed, let $U = \mathrm{Spec}(A)$ be an arbitrary affine open of S. By hypothesis, $q^{-1}(U)$ is the union of affine opens $V_i = \mathrm{Spec}(B_i)$, for $i = 1, \cdots, n$, and each intersection $V_i \cap V_j$ is the union of finitely many affine opens $W_{ijk} = \mathrm{Spec}(C_{ijk})$. Then $\Gamma(U, q_*(F)) = \Gamma(q^{-1}(U), F)$ is the kernel of the morphism

$$\bigoplus_{i=1}^n \Gamma(V_i, F) \to \bigoplus_{i,j,k} \Gamma(W_{ijk}, F).$$

Point (a) follows, since each of the terms above commutes with filtered inductive limits (since the V_i and W_{ijk} are affine, hence quasi-compact). Let us prove (b): $E = \Gamma(U, E)$ is a flat A-module, and $\Gamma(U, q_* q^*(E))$ is the kernel $K(E)$ of the morphism

$$\bigoplus_{i=1}^n B_i \otimes_A E \to \bigoplus_{i,j,k} C_{ijk} \otimes_A E$$

and since E is flat over A, this kernel is identified with $K(A) \otimes_A E = \mathcal{O}_X(q^{-1}(U)) \otimes_A E$. Finally, if U' is an arbitrary affine open of S' above U, then $A' = \mathcal{O}(U')$ is a flat A-algebra, and one obtains as above that $F(q^{-1}(U)) \otimes_A A' \xrightarrow{\sim} F'(q'^{-1}(U))$.

Notation. Let S be a scheme, X an S-scheme, $f : X \to S$ the structural morphism; we set $\mathcal{A}(X) = f_*(\mathcal{O}_X)$.

Lemma 11.1. *Let X and Y be two S-schemes quasi-compact and quasi-separated over S, $f : X \to S$ and $g : Y \to S$ the structural morphisms. Then the canonical homomorphism*

$$\varphi : \mathcal{A}(X) \otimes_{\mathcal{O}_S} \mathcal{A}(Y) \to \mathcal{A}(X \times_S Y)$$

is an isomorphism in each of the following cases:[569]

a) f and g are affine,

[568] We have added this reminders paragraph.

[569] Note that if k is a field and if X is an infinite sum of copies of $S = \mathrm{Spec}\, k$ (so that X is not quasi-compact), then $\mathcal{A}(X) = k^X$ and the canonical morphism $k^X \otimes k^X \to k^{X \times X}$ is not surjective.

b) g (or f) is flat and affine,

c) g is flat and $f_*(\mathcal{O}_X)$ is a flat $\mathcal{O}_S$-module.

We shall assume, in case (b), that it is g which is flat and affine. Set then $S' = \operatorname{Spec}\mathcal{A}(X)$, $Y' = Y \times_S S'$, $g' = g \times_S S'$, and denote by v the morphism $S' \to S$:

$$\begin{tikzcd} Y' \arrow[r] \arrow[d, "g'"'] & Y \arrow[d, "g"] \\ \operatorname{Spec}\mathcal{A}(X) = S' \arrow[r, "v"] & S \end{tikzcd}$$

In cases (a) and (b), g is affine and so, by EGA II, 1.5.2, one has:

(1) $$g'_*(\mathcal{O}_{Y'}) = v^* g_*(\mathcal{O}_Y) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}.$$

One has the same equality in case (c), by 11.0 (c), since S' is flat over S and g is quasi-compact and quasi-separated.

On the other hand (EGA II 1.2.7), $f : X \to S$ factors through v by means of a morphism $p : X \to S'$, and one has $p_*(\mathcal{O}_X) = \mathcal{O}_{S'}$ and $X \times_S Y = X \times_{S'} Y'$. Since f is quasi-separated, so is p (EGA IV₁, 1.2.2), and since f is quasi-compact and v is quasi-separated, p is also quasi-compact (EGA IV₁, 1.2.4). Consider then the cartesian square:

$$\begin{tikzcd} X \times_S Y \arrow[r, "p'"] \arrow[d, "p"'] & Y' \arrow[d, "g'"] \\ X \arrow[r, "p"'] & S' \end{tikzcd}$$

In cases (b) and (c), Y is flat over S, hence Y' is flat over S'; applying 11.0 (c) again, one obtains:

(2) $$p'_*(\mathcal{O}_{X \times_S Y}) = g'^* p_*(\mathcal{O}_X) = g'^*(\mathcal{O}_{S'}) = \mathcal{O}_{Y'},$$

and one has the same equality in case (a), since in this case p and p' are isomorphisms.

Finally, v being affine, one has, by EGA II, 1.4.7, $v_*(F \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = F \otimes_{\mathcal{O}_S} \mathcal{A}(X)$ for every quasi-coherent $\mathcal{O}_S$-module F. Combined with (2) and (1), this gives:

$$\mathcal{A}(X \times_S Y) = v_* g'_* p'_*(\mathcal{O}_{X \times_S Y}) = v_* g'_*(\mathcal{O}_{Y'}) = v_*(\mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S}$$

$\square$

Corollary 11.2. *The functor $X \mapsto \operatorname{Spec}\mathcal{A}(X)$, from the full subcategory of (Sch/S) formed of S-schemes X flat, quasi-compact and quasi-separated over S, and such that $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$-module, into that of S-schemes flat and affine over S, commutes with finite products, hence transforms S-groups into S-groups.*

Definition 11.3. *Given an S-group G flat, quasi-compact and quasi-separated over S, such that $\mathcal{A}(G)$ is flat over 0_S,*[570] *we shall denote by G_{af}, and call the* affine envelope *of G, the S-group $G_{af} = \operatorname{Spec} \mathcal{A}(G)$.*

Proposition 11.3.1. [571] *The canonical morphism $\tau_G : G \to G_{af}$ is a morphism of S-groups. Moreover, it satisfies the following universal property:*

(i) For every morphism of S-schemes $\phi : G \to H$, where H is affine over S, there exists a unique morphism of S-schemes $\phi' : G_{af} \to H$ such that $\phi = \phi' \circ \tau_G$.

(ii) If moreover H is an S-group and if ϕ is a morphism of S-groups, then so is ϕ'.

11.4.

[572] Let E and F be two quasi-coherent 0_S-modules. Consider the S-functor $\operatorname{Hom}_{O_S}(W(E), W(F))$ (cf. I, 3.1.4), i.e. for every S-scheme $f : X \to S$,

Hom_{O_S}(W(E), W(F))(X) = Hom_{O_X}(W(E)_X, W(F)_X).

Moreover, by I, 4.6.2, one has $W(E)_X = W(f * (E))$ (and similarly for F) and

Hom_{O_X}(W(f*(E)), W(f*(F))) = Hom_{O_X}(f*(E), f*(F)).

One thus obtains (using the adjunction formula for the last equality):

(†) Hom_{O_S}(W(E), W(F))(X) = Hom_{O_X}(f*(E), f*(F)) = Hom_{O_S}(E, f_* f*(F)).

Proposition 11.5. *Let X be an S-scheme quasi-compact and quasi-separated over S, $f : X \to S$ the structural morphism, E and E' two quasi-coherent 0_S-modules. Suppose one of the two following conditions holds:*

a) f is affine,

b) E is flat over 0_S.

Then the canonical morphism $E \bigotimes_{O_S} \mathcal{A}(X) \to f_* f * (E)$ is an isomorphism, and one therefore has

Hom_{O_S}(W(E ′), W(E))(X) = Hom_{O_S}(E ′, E ⊗_{O_S} □(X)).

Indeed, the second assertion follows from 11.4 and the first; this latter follows from EGA II, 1.4.7 in case (a), and from 11.0 (b) in case (b).[573] □

---

[570] Note that if S is a regular locally noetherian scheme of dimension ≤ 2, and X is a flat, quasi-compact and quasi-separated S-scheme, then $\mathcal{A}(X)$ is a flat 0_S-module, cf. [Ray70a], VII 3.2.

[571] We have added this proposition; see also the additional paragraph 12 below for a study of the morphism $G \to G_{af}$ and its kernel.

[572] In 11.4–11.6, we have considered $\operatorname{Hom}_{O_S}(W(E), W(F))$ instead of $\operatorname{Hom}_{O_S}(V(F), V(E))$ (cf. I, 4.6.3) and simplified the original by taking into account the addition 11.0 (b).

[573] We have simplified the original, which used the isomorphism $\operatorname{Hom}_{O_S}(W(E), W(E')) \simeq \operatorname{Hom}_{O_S}(V(E'), V(E))$ then the inclusion of the right-hand side into Hom_S(V(E), V(E ′)) = Hom_{O_S}(E ′, f_* f*(Sym(E))) and applied EGA III, 4.1.15 to $V(E) = \operatorname{Spec}(Sym(E))$ to deduce 11.0 (b).

11.6.

[574] Let G be an S-group and E a quasi-coherent O_S-module. To give E a G-O_S-module structure (i.e. an O_S-linear action of G on $W(E)$, cf. I 4.7.1) is equivalent to giving a morphism of S-functors in monoids $\rho : G \to \mathrm{End}_{O_S}(W(E))$ (indeed, such a ρ necessarily sends G into $\mathrm{Aut}_{O_S}(W(E))$).

Now, by 11.4, giving a morphism of S-functors $\rho : G \to \mathrm{End}_{O_S}(W(E))$ is equivalent to giving an element θ of $\mathrm{Hom}_{O_G}(f*(E), f*(E))$, which corresponds by adjunction to an element δ of $\mathrm{Hom}_{O_S}(E, f_*f*(E))$, where one has denoted by f the projection $G \to S$.

Let $m : G \times_S G \to G$ be the multiplication, δ_G the morphism $O_G \to m_*(O_{G\times_S G})$, and ϕ the projection $G \times_S G \to S$ (which equals $f \circ m$). It is convenient to denote by $\boxtimes$ the "external" tensor product; one thus obtains a morphism

$$id_E \boxtimes \delta_G : \quad f^*(E) = E \boxtimes_{O_S} O_G \to E \boxtimes_{O_S} m_*(O_{G\times G})$$

and by abuse of notation, we shall denote again by $id_E \boxtimes \delta_G$ the composite of the preceding morphism with the canonical morphism $E \boxtimes_{O_S} m_*(O_{G\times G}) \to m_*m*f*(E) = m_*\phi*(E)$.

On the other hand, designate by $h : G \to S$ a second copy of $f : G \to S$ and consider the following commutative diagram, where p, q denote the two projections:

$$\begin{tikzcd} G \times_S G \arrow[r, "q"] \arrow[dr, "p"'] \arrow[dr, phantom, "\varphi", near start, shift left=3ex] & G \arrow[d, "h"] \\ & G \arrow[r, "f"'] & S \end{tikzcd}$$

Denoting again by δ the morphism $E \to h_*h*(E)$, one obtains the morphism

$$\delta \boxtimes id_{O_G} : \quad f^*(E) = E \boxtimes_{O_S} O_G \to h_*h^*(E) \boxtimes_{O_S} O_G = f^* h_* h^*(E)$$

and by abuse we shall denote again by $\delta \boxtimes id_{O_G}$ the composite of this morphism with the canonical morphism $f*h_*h*(E) \to p_*\phi*(E)$.

Then the condition that ρ be compatible with the multiplication is equivalent to saying that, for every open U of S, the diagram below is commutative:

$$\begin{tikzcd} \Gamma(U, E) \arrow[r, "\delta"] \arrow[d, "\delta"'] & \Gamma(U \times_S G, E \boxtimes_{O_S} O_G) \arrow[d, "id_E \boxtimes \delta_G"] \\ \Gamma(U \times_S G, E \boxtimes_{O_S} O_G) \arrow[r, "\delta \boxtimes id_{O_G}"] & \Gamma(U \times_S G \times_S G, E \boxtimes_{O_S} O_G \boxtimes_{O_S} O_G). \end{tikzcd}$$

(1)

[574] We have expanded 11.6, and highlighted the results obtained in the form of Proposition 11.6.1.

Moreover, the unit section $\epsilon : S \to G$ induces a morphism u from 0_G to $\epsilon_*\epsilon*(O_G) = \epsilon_*(O_S)$, and the condition that ρ preserves the unit elements is equivalent to the commutativity of the diagram:

$$\begin{tikzcd}
\Gamma(U, E) \arrow[rr, "\delta"] \arrow[dr, "\simeq"'] & & \Gamma(U \times_S G, E \boxtimes_{O_S} O_G) \arrow[dl, "id_E \boxtimes u"] \\
& \Gamma(U \times_S S, E \boxtimes_{O_S} O_S). &
\end{tikzcd}$$

(2)

One thus sees that giving E a G-0_S-module structure is equivalent to giving a morphism of 0_S-modules $\delta : E \to f_*f*(E)$ satisfying conditions (1) and (2) above, and in this case the morphism $\theta : f*(E) \to f*(E)$, deduced from δ by adjunction, is an isomorphism (since it corresponds to the isomorphism $G \times_S W(E) \xrightarrow{\sim} G \times_S W(E)$ defined set-theoretically by $(g,x) \mapsto (g,gx)$; see also I, 6.5.4).

Suppose now that G is flat, quasi-compact and quasi-separated over S, and that $\mathcal{A}(G)$ is a flat 0_S-module; then, by 11.1 (c), the canonical morphism $\mathcal{A}(G) \otimes_{O_S} \mathcal{A}(G) \to \mathcal{A}(G \times_S G)$ is an isomorphism, and the morphism $\delta_G : \mathcal{A}(G) \to \mathcal{A}(G) \otimes_{O_S} \mathcal{A}(G)$ will be denoted by Δ.

If moreover $E \otimes_{O_S} \mathcal{A}(G) = f_*f*(E)$ (which is the case, by 11.5, if $G \to S$ is affine, or if E is flat over 0_S), one obtains that conditions (1) and (2) are equivalent to the conditions below, which express that $\delta : E \to E \otimes_{O_S} \mathcal{A}(G)$ makes E a right $\mathcal{A}(G)$-comodule (cf. I 4.7.2):

(CM 1) Setting $A = \mathcal{A}(G)$, the diagram below is commutative:

$$\begin{tikzcd}
E \arrow[r, "\delta"] \arrow[d, "\delta"'] & E \otimes A \arrow[d, "id_E \otimes \Delta"] \\
E \otimes A \arrow[r, "\delta \otimes id_A"'] & E \otimes A \otimes A.
\end{tikzcd}$$

(CM 2) Denoting by $\eta : A \to O_S$ the morphism $\mathcal{A}(\epsilon)$, the diagram below is commutative:

$$\begin{tikzcd}
E \arrow[rr, "\delta"] \arrow[dr, "\simeq"'] & & E \otimes A \arrow[dl, "id_E \otimes \eta"] \\
& E \otimes O_S. &
\end{tikzcd}$$

Remark 11.6.A. Recall that one denotes by $V(E)$ the vector fibration on S representing the functor $V(E)$, i.e. for every $S' \to S$, $V(E)(S') = \mathrm{Hom}_{O_{S'}}(E_{S'}, O_{S'})$. As one has, by I, 4.6.2, an anti-isomorphism of S-functors in monoids $\mathrm{End}_{\mathcal{O}_S}(W(E)) \simeq \mathrm{End}_{\mathcal{O}_S}(V(E))$, one sees that if E is a left G-$\mathcal{O}_S$-module, one has a right action $\mu : V(E) \times_S G \to V(E)$ of G on $V(E)$, defined set-theoretically by $(\phi g)(x) = \phi(gx)$, for every $g \in G(S')$, $x \in \Gamma(S', E_{S'})$ and $\phi \in \mathrm{Hom}_{\mathcal{O}_{S'}}(E_{S'}, \mathcal{O}_{S'})$. One thus obtains commutative diagrams:

$$\begin{tikzcd}
V(E) \times_S G \times_S G \arrow[r, "\mu \times \mathrm{id}_G"] \arrow[d, "\mathrm{id} \times m"'] & V(E) \times_S G \arrow[d, "\mu"] \\
V(E) \times_S G \arrow[r, "\mu"] & V(E)
\end{tikzcd}
\qquad
\begin{tikzcd}
V(E) \arrow[d, "\simeq"'] & V(E) \times_S G \arrow[l, "\mu"'] \\
 & V(E) \times_S S. \arrow[u, "\mathrm{id} \times \varepsilon"']
\end{tikzcd}$$

When G is flat, quasi-compact and quasi-separated over S, when $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module, and when one of the conditions of 11.5 is satisfied, one recovers similarly the conditions (CM 1) and (CM 2).

Consequently, we have obtained:

Proposition 11.6.1. *Let G be an S-group flat, quasi-compact and quasi-separated over S, such that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module, and let E be a quasi-coherent $\mathcal{O}_S$-module.*

(i) It amounts to the same to give an $\mathcal{A}(G)$-comodule structure $\delta : E \to E \otimes_{\mathcal{O}_S} \mathcal{A}(G)$ or a G_{af}-$\mathcal{O}_S$-module structure on E (i.e. an $\mathcal{O}_S$-linear action of G_{af} on E). By composition with the morphism of S-groups $G \to G_{af}$, this defines a G-$\mathcal{O}_S$-module structure on E.

(ii) If moreover E is flat, every $\mathcal{O}_S$-linear action of G on E factors through G_{af} and corresponds to a unique $\mathcal{A}(G)$-comodule structure on E.

Lemma 11.7. *Let G be an S-group flat, quasi-compact and quasi-separated over S, such that $A = \mathcal{A}(G)$ is a flat $\mathcal{O}_S$-module. Let E be a quasi-coherent $\mathcal{O}_S$-module, $\delta : E \to E \otimes A$ an A-comodule structure, and $\rho : G \to \mathrm{Aut}_{\mathcal{O}_S}(W(E))$ the action of G on E associated with it.*

Let E_0 be a quasi-coherent $\mathcal{O}_S$-submodule of E such that the restriction δ_0 of μ to E_0 factors through $E_0 \to E_0 \otimes A$, i.e. such that one has a commutative diagram:

$$\begin{tikzcd}
E_0 \arrow[r] \arrow[d, "\delta_0"'] & E \arrow[d, "\delta"] \\
E_0 \otimes A \arrow[r] & E \otimes A.
\end{tikzcd}$$

(N.B. The morphism $E_0 \otimes A \to E \otimes A$ is injective, since A is flat over $\mathcal{O}_S$.)

Then δ_0 makes E_0 an $\mathcal{A}(G)$-comodule, hence defines an action ρ_0 of G on E_0 (which one will call the induced action on E_0 by ρ, and one will say that E_0 is stable *under ρ).*

This follows immediately from the definitions and from 11.6. One will remark, however, that in general the canonical map $W(E_0) \to W(E)$ is not a monomorphism. ∎

Remark 11.7.bis. [575] Let G be a flat S-group and E a quasi-coherent G-O_S-module. Denote by f the morphism $G \to S$ and by δ the morphism $E \to f_*f*(E)$ defined in 11.6. Let E_0 be a quasi-coherent O_S-submodule of E; since f is flat, $f_*f*(E_0)$ is an O_S-submodule of $f_*f*(E)$, and similarly for $\phi = f \times f$. Consequently, if the restriction δ_0 of δ to E_0 factors through $f_*f*(E_0)$, then it makes E_0 a G-O_S-module. In this case, one says that E_0 is a G-stable submodule of E.

Definition 11.8.0. [576] *Let S be a scheme. An O_S-coalgebra is an O_S-module C endowed with two morphisms of O_S-modules $\Delta : C \to C \otimes C$ and $\epsilon : C \to O_S$, satisfying the following two axioms (cf. I 4.2):*

(CO 1) Δ is co-associative: the following diagram is commutative

$$\begin{tikzcd} & C \otimes C \arrow[dr, "\mathrm{id} \otimes \Delta"] & \\ C \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & & C \otimes C \otimes C \\ & C \otimes C \arrow[ur, "\Delta \otimes \mathrm{id}"'] & \end{tikzcd}$$

(CO 2): ϵ is a counit, i.e. the two following composites are the identity

$$C \xrightarrow{\Delta} C \otimes C \xrightarrow{\mathrm{id} \otimes \varepsilon} C \otimes O_S \xrightarrow{\sim} C,$$
$$C \xrightarrow{\Delta} C \otimes C \xrightarrow{\varepsilon \otimes \mathrm{id}} O_S \otimes C \xrightarrow{\sim} C.$$

A (right) C-comodule is an O_S-module E endowed with a morphism of O_S-modules $\mu : E \to E \bigotimes C$ satisfying the axioms (CM 1) and (CM 2) of 11.6.

One says that C (resp. E) is a quasi-coherent coalgebra *(resp. a* quasi-coherent comodule*) if it is a quasi-coherent O_S-module.*

Let A be a commutative ring and C an A-coalgebra; then $C^\vee = \mathrm{Hom}_A(C, A)$ is an A-algebra. We shall denote by ev the natural evaluation map $C \bigotimes_A C^\vee \to A$.

Lemma 11.8. *Let C be an A-coalgebra, V a C-comodule, M an A-submodule of V. Suppose that C is a projective A-module.*[577] *Let $c(M)$ be the image of the morphism of A-modules*

[575] We have added this remark, which generalizes 11.7 and will be useful in 11.10.bis.
[576] Since the statements 11.8 and 11.9 bear solely on the notion of comodule over a coalgebra, we have introduced Definition 11.8.0 and reformulated 11.8 and 11.9 accordingly.
[577] In the original, it is assumed that C is a free A-module; the generalization to the case where C is a projective A-module, pointed out by J.-P. Serre, was mentioned in Remark 11.10.1. We have included this generalization here and in 11.9, and expanded the proofs accordingly.

$$\theta : \quad M \otimes_A C^\vee \xrightarrow{\mu \otimes id} V \otimes_A C \otimes_A C^\vee \xrightarrow{id \otimes ev} V.$$

Then $c(M)$ is the smallest subcomodule of V containing M, and is a finitely generated A-module if M is. One will say that $c(M)$ is the subcomodule generated by M.

Moreover, for every morphism of rings $A \to A'$, if one denotes by M' the image of $M \bigotimes_A A'$ in $V' = V \bigotimes_A A'$, then $c(M')$ is the image of $c(M) \bigotimes_A A'$ in V', hence: "the formation of $c(M)$ commutes with base change".

First, $M \subset c(M)$ by (CM 2), and if N is a subcomodule of V containing M, one has $\mu(M) \subset N \bigotimes C$ and therefore $c(M) \subset N$.

By hypothesis, C is a direct factor of a free A-module L, with basis $(e_i)_{i \in I}$. Denote by ϕ_i the restriction to C of the linear form e_i^*, defined by $e_i^*(e_j) = \delta_{ij}$. Let $x \in M$. One can write:

$$(1) \qquad \mu(x) = \sum_{i \in J} x_i \otimes e_i,$$

where $x_i \in V$ and J is a finite subset of I. Then $x_i = \theta(x \bigotimes \phi_i)$ belongs to $c(Ax)$, and one therefore has $c(Ax) = \sum_{i \in J} Ax_i$. Since C is a direct factor of L, say $L = C \bigoplus R$, whence $V \bigotimes L = (V \bigotimes C) \bigoplus (V \bigotimes R)$, one obtains that

$$(c(Ax) \otimes L) \cap (V \otimes C) = c(Ax) \otimes C.$$

Consequently, $\mu(x)$ can also be written in the form

$$(2) \qquad \mu(x) = \sum_{j \in J} x_j \otimes b_j,$$

with $b_j \in C$. One can write $\Delta(b_j) = \sum_{i \in I} b_{ij} \otimes e_i$, with $b_{ij} \in C$. Then, applying $\mu \bigotimes id$ to (1) (resp. $id \otimes \Delta$ to (2)) and using the axiom (CM 1), one obtains, for every $i \in J$:

$$\mu(x_i) = \sum_{j \in J} x_j \otimes b_{ij} \in c(Ax) \otimes C.$$

This shows that $c(M)$ is a subcomodule of V, and is therefore the smallest subcomodule of V containing M.

It is clear that $c(M)$ is a finitely generated A-module if M is: if $M = Ax_1 + \cdots + Ax_n$ and $\mu(x_k) = \sum_i x_{ik} \bigotimes e_i$, then $c(M)$ is generated by the x_{ik}, for $k = 1, \cdots, n$ and i running through a finite subset of I.

Finally, let $A \to A'$ be a morphism of rings and let M' be the image of $M \bigotimes A'$ in $V' = V \bigotimes A'$. Then $c(M')$ (resp. the image of $c(M) \bigotimes A'$ in V') is the image of the morphism θ' below (resp. of the composite $\theta' \circ \tau$):

$$M \otimes A' \otimes C^\vee \xrightarrow{\tau} M \otimes \mathrm{Hom}_A(C, A') \xrightarrow{\theta'} V'.$$

Now, these two morphisms have the same image. Indeed, let $\psi \in \mathrm{Hom}_A(C, A')$ and $x \in M$. Set $\mu(x) = \sum_{i \in J} x_i \bigotimes e_i$. Then

$$\theta'(x \otimes \psi) = \sum_{i \in J} \psi(e_i) x_i$$

is the image by $\theta' \circ \tau$ of the element $\sum_{i\in J} x \otimes \psi(e_i) \otimes \phi_i$ of $M \otimes A' \otimes C^\vee$. This proves the lemma. ∎

Moreover, one has the following proposition:

Proposition 11.8.bis. [578] *Let A be a noetherian ring, C an A-coalgebra flat over A, V a C-comodule, and M a finitely generated A-submodule of V. Then there exists a subcomodule W of V, finitely generated over A, containing M.*

Indeed, since M is finitely generated, so is `Δ_V(M)`, hence there exists a finitely generated A-submodule M' of V such that `Δ_V(M) ⊂ M′ ⊗_A C`. Let π be the projection $V \to V/M'$ and `Δ̄_V = (π ⊗ id_C) Δ_V`, and let

`W = {x ∈ V | Δ̄_V(x) ∈ M′ ⊗_A C} = Ker Δ̄_V;`

this is an A-submodule of V containing M and contained in M' (since `x = (id_V ⊗ ε) Δ_V(x)`), hence finitely generated over A. Moreover, `(Δ̄_V ⊗ id_C) Δ_V = (π ⊗ Δ_C) Δ_V` vanishes on W, i.e. `Δ_V(W)` is contained in the kernel K of `(Δ̄_V ⊗ id_C)`. But since C is flat over A, one has $K = W \otimes C$, hence W is a subcomodule of V. ∎

Lemma 11.8.1. [579] *Let C be an A-coalgebra, V a C-comodule, M an A-submodule of V, and $f : A \to A'$ a faithfully flat morphism of rings. Suppose that $C' = C \otimes A'$ is a projective A'-module.*

(i) Then there exists a smallest subcomodule $t(M)$ of V containing M, and $t(M)$ is a finitely generated A-module if M is. Moreover, "the formation of $t(M)$ commutes with base change".

(ii) More precisely, C is a projective A-module, and one has $t(M) = c(M)$.

Proof. (ii) By [RG71] (see Proposition 11.8.2 below), C is a projective A-module. One can therefore apply Lemma 11.8: $c(M)$ is the smallest subcomodule of V containing M, it is a finitely generated A-module if M is, and its formation commutes with base change.

To avoid an anachronism ([RG71] being subsequent to SGA 3), let us sketch a direct proof of point (i). Since $A \to A'$ is flat, $M' = M \otimes A'$ is an A'-submodule of $V' = V \otimes A'$, and, since C' is a projective A'-module, $c(M')$ is the smallest subcomodule of V' containing M'. Denote by $V' \otimes A'$ and $A' \otimes V'$ the two $A' \otimes A'$-comodule structures on $V'' = V' \otimes_{A'} (A' \otimes A')$ obtained by the two base changes $A' \rightrightarrows A' \otimes A'$, $a' \mapsto a' \otimes 1$ and $a' \mapsto 1 \otimes a'$. The A'-comodule V' is equipped with an isomorphism of $A' \otimes A'$-comodules $\phi : V' \otimes A' \xrightarrow{\sim} A' \otimes V'$, $(x \otimes a') \otimes b' \mapsto b' \otimes (x \otimes a')$, which is a descent datum, i.e. which satisfies $\phi_{31} = \phi_{32} \circ \phi_{21}$.

Since $M' = M \otimes A'$, ϕ sends $M' \otimes A'$ onto $A' \otimes M'$, and therefore $c(M' \otimes A')$ onto $c(A' \otimes M')$. Since the formation of $c(M')$ commutes with base change, one has $c(M' \otimes A') = c(M') \otimes A'$ and $c(A' \otimes M') = A' \otimes c(M')$. One therefore has

`φ(c(M′) ⊗ A′) = A′ ⊗ c(M′)`

[578] We have added this proposition, taken from [Se68], § 1.5, Prop. 2.
[579] We have added this lemma, which is used in the proof of 11.9.

and it follows that ϕ equips $c(M')$ with a descent datum. By (fpqc) descent, there exists a unique subcomodule $t(M)$ of V such that $c(M') = t(M) \otimes A'$, and $t(M)$ contains M since $t(M) \otimes A'$ contains M'. Moreover, if N is a subcomodule of V containing M, then N contains $t(M)$, since $N \otimes A'$ contains $c(M') = t(M) \otimes A'$. Hence $t(M)$ is the smallest subcomodule of V containing M.

Finally, let $A \to B$ be a morphism of rings. Let $B' = B \otimes A'$ and let M_B (resp. $M'_{B'}$) be the image of $M \otimes B$ in $V_B = V \otimes B$ (resp. of $M' \otimes_{A'} B'$ in $V \otimes B'$); then $M_B \otimes_B B' = M'_{B'}$. On the one hand, the preceding construction, applied to C_B and to the morphism $B \to B'$, gives:

$$c(M_B \otimes_B B') = t(M_B) \otimes_B B' = t(M_B) \otimes A'.$$

On the other hand, since the formation of $c(M')$ commutes with base change, $c(M'_{B'})$ is the image in $V' \otimes_{A'} B' = V \otimes B \otimes A'$ of

$$c(M') \otimes_{A'} B' = t(M) \otimes B \otimes A'.$$

It follows that $t(M_B)$ is the image in V_B of $t(M) \otimes B$. $\square$

Proposition 11.8.2 (Gruson–Raynaud). *Let $f : A \to A'$ be a faithfully flat morphism; then f "descends projectivity", i.e. if M is an A-module and if $M \otimes_A A'$ is a projective A'-module, then M is a projective A-module.*

Indeed, by [RG71] II 2.5.1, f "descends the Mittag-Leffler condition", and therefore, by loc. cit. II 3.1.3, f descends projectivity.[580] $\square$

Proposition 11.9. [581] *Let C be an $\mathcal{O}_S$-coalgebra, E a C-comodule, F an $\mathcal{O}_S$-submodule of E, all quasi-coherent. Suppose given a covering of S by affine opens $U_\alpha = \operatorname{Spec} A_\alpha$, and for each α, a faithfully flat morphism of rings $A_\alpha \to A'_\alpha$ such that $\Gamma(U_\alpha, C) \otimes_{A_\alpha} A'_\alpha$ is a projective A'_α-module.*[582]

Then there exists a smallest quasi-coherent subcomodule $t(F)$ of E containing F, and $t(F)$ is a finitely generated $\mathcal{O}_S$-module if F is. Moreover, for every base change $S' \to S$, if one denotes by F' the image of $F \otimes \mathcal{O}_{S'}$ in $E' = E \otimes \mathcal{O}_{S'}$, then $t(F')$ is the image of $t(F) \otimes \mathcal{O}_{S'}$ in E', i.e. "the formation of $t(F)$ commutes with base change".

Proof. For each α, the $\mathcal{O}_{U_\alpha}$-module $\mathcal{T}_\alpha$ associated with the A_α-module $T_\alpha = t(\Gamma(U_\alpha, F))$ is, by 11.8.1 (i), the smallest quasi-coherent subcomodule of $E|_{U_\alpha}$ containing $F|_{U_\alpha}$, and is a finitely generated $\mathcal{O}_{U_\alpha}$-module if $F|_{U_\alpha}$ is.

For all α, β, set $U_{\alpha\beta} = U_\alpha \cap U_\beta$. Since the construction of $t(M)$ commutes with base change, one has, for all α, β, canonical isomorphisms of $\mathcal{O}_{U_{\alpha\beta}}$-modules

$$\varphi_{\alpha\beta} : \mathcal{T}_\beta \otimes_{A_\beta} \mathcal{O}_{U_{\alpha\beta}} \xrightarrow{\sim} \mathcal{T}_\alpha \otimes_{A_\alpha} \mathcal{O}_{U_{\alpha\beta}}$$

---

[580] Let us point out in passing that assertion II 2.5.2 of loc. cit., more general than II 2.5.1, is corrected in the article [Gr73] (this does not affect the case of faithfully flat morphisms).

[581] As pointed out in N.D.E. (112), we have rewritten the statement for an $\mathcal{O}_S$-coalgebra C (rather than for an S-group G satisfying the hypotheses indicated in Corollary 11.10). Moreover, we have spelled out the proof (the original indicated: "(…) the proposition is a consequence of Lemma 11.8.").

[582] This is the case, for example, when $C = \mathcal{A}(G)$, where G is a reductive S-group, as we shall see in Exp. XXII 5.7.8.

which satisfy the cocycle condition $\phi'_{\alpha\gamma} = \phi'_{\alpha\beta} \circ \phi'_{\beta\gamma}$, where $\phi'_{\alpha\gamma}$ (resp.) denotes the restriction of $\phi_{\alpha\gamma}$ (resp.) to $U_\alpha \cap U_\beta \cap U_\gamma$.

Consequently, the $\mathcal{T}_\alpha$ glue together into a quasi-coherent subcomodule $t(F)$ of E containing F. We leave to the reader the task of verifying that $t(F)$ is the smallest quasi-coherent subcomodule of E containing F, and that its formation commutes with base change.

Definition 11.9.1. [583] *Let S be a scheme and P a quasi-coherent 0_S-module. The following conditions are equivalent:*

(i) For every affine open U of S, $\Gamma(U,P)$ is a projective $O_S(U)$-module.

(ii) There exists a covering (U_α) of S by affine opens such that each $\Gamma(U_\alpha,P)$ is a projective $O_S(U_\alpha)$-module.

(iii) There exists a covering (U_α) of S by affine opens, and faithfully flat morphisms of rings $A_\alpha = O_S(U_\alpha) \to A'_\alpha$, such that, for each α, $\Gamma(U_\alpha,P) \otimes_{A_\alpha} A'_\alpha$ is a projective A'_α-module.

Indeed, it is clear that (i) ⇒ (ii) ⇒ (iii). Conversely, if (iii) is satisfied, 11.8.2 implies that each $\Gamma(U_\alpha,P)$ is a projective $O_S(U_\alpha)$-module, whence (ii). Finally, suppose (ii) satisfied and let $V = \operatorname{Spec} A$ be an arbitrary affine open; it is covered by finitely many affine opens $V_1,\cdots,V_n$, where each $V_i = \operatorname{Spec} A_i$ is contained in at least one $V \cap U_\alpha$, so that $\Gamma(V_i,P)$ is a projective A_i-module. Let $A' = A_1 \times \cdots \times A_n$; then $A \to A'$ is faithfully flat and $\Gamma(V,P) \otimes_A A'$ is a projective A'-module. Hence, by 11.8.2, $\Gamma(V,P)$ is a projective A-module.

When these equivalent conditions are satisfied, one says that P is a *locally projective* 0_S-module.

Corollary 11.10. *Let S be a quasi-compact and quasi-separated scheme, G an S-group, and ρ a linear action of G on a quasi-coherent 0_S-module E. Suppose that:*

(i) G satisfies one of the following conditions:

a) G is affine and flat over S,

b) G is flat, quasi-compact and quasi-separated over S, and E is flat;

(ii) $\mathcal{A}(G)$ is a locally projective 0_S-module.

Then E is an inductive limit of a filtered increasing family of quasi-coherent finitely generated 0_S-submodules of E, stable under G.

By hypothesis (i) and 11.6.1, E is equipped with an $\mathcal{A}(G)$-comodule structure. On the other hand, since S is quasi-compact and quasi-separated, E is the inductive limit of its finitely generated quasi-coherent submodules (EGA I, 9.4.9 and EGA

[583] We have added this definition, taken from [RG71], bottom of p. 82. Thus, in Proposition 11.9, the hypothesis is that the coalgebra C is a locally projective 0_S-module, and we have used this terminology in Corollary 11.10.

IV₁, 1.7.7). Consequently, the corollary follows from Proposition 11.9, applied to the coalgebra $\mathcal{A}(G)$. ∎

Moreover, one has the following proposition:

Proposition 11.10.bis. [584] *Let S be a noetherian scheme, G an S-group flat, quasi-compact and quasi-separated over S, E a quasi-coherent G-$\mathcal{O}_S$-module, M a coherent $\mathcal{O}_S$-submodule of E. Then M is contained in a coherent $\mathcal{O}_S$-submodule stable under G.*

Proof. Denote by f the morphism $G \to S$ and by τ the adjunction morphism $E \to f_*f^*(E)$. By 11.6, the G-$\mathcal{O}_S$-module structure on E is given by an automorphism θ of the $\mathcal{O}_G$-module $f^*(E)$, such that the morphism $\delta = \theta \circ \tau$ satisfies conditions (1) and (2) of 11.6. (The situation considered in [Th87] is more general, in that the author considers a G-scheme X and a G-equivariant $\mathcal{O}_X$-module E (cf. Exp. I, Section 6); here $X = S$ equipped with the trivial action of G.)

Since S is noetherian, E is the filtered inductive limit of its coherent submodules F_α (cf. EGA I, 9.4.9). Then $f^*(E)$ is the filtered inductive limit of the $f^*(F_\alpha)$, which are submodules of $f^*(E)$ since f is flat. Since, moreover, f is quasi-compact and quasi-separated, by 11.0, the $\mathcal{O}_S$-module $f_*f^*(E)$ is quasi-coherent and is the filtered inductive limit of the quasi-coherent submodules $f_*f^*(F_\alpha)$. Consequently, E is the filtered inductive limit of the quasi-coherent $\mathcal{O}_S$-submodules E_α, where E_α denotes the inverse image by δ of $f_*f^*(F_\alpha)$, i.e. the kernel of the composite morphism

$$\delta_\alpha : \quad E \xrightarrow{\delta} f_*f^*(E) \longrightarrow f_*f^*(E/F_\alpha).$$

Since M is coherent and S noetherian, every increasing sequence of submodules of M is stationary, so M is contained in some E_α. Let us show that each E_α is coherent and G-stable.

Let u be the morphism $f^*(E) \to \epsilon_*(E)$ corresponding by adjunction to the identity morphism from E to $f_*\epsilon_*(E) = E$; then the composite morphism

$$E \xrightarrow{\tau} f_*f^*(E) \xrightarrow{f_*(u)} f_*\epsilon_*(E) = E$$

is the identity (cf. 11.6 (2)). Since one has a commutative diagram:

$$\begin{tikzcd} E_\alpha \arrow[r, "\delta"] \arrow[d, "i_\alpha"'] & f_*f^*(F_\alpha) \arrow[r, "f_*(u)"] & F_\alpha \arrow[d, "j_\alpha"] \\ E \arrow[r, "\tau"] & f_*f^*(E) \arrow[r, "f_*(u)"] & E, \end{tikzcd}$$

[584] We have added this proposition, which is a particular case of [Th87], 1.4–1.5. The author makes reference there to an argument of Deligne (cf. [Kn71], III Th. 1.1); one may also note the similarity with the argument of Serre ([Se68], Prop. 2) recalled in 11.8.bis.

where i_α (resp. j_α) denotes the inclusion of E_α (resp. F_α) into E, one deduces that i_α factors through j_α, i.e. E_α is a submodule of F_α, hence is coherent.

Let us finally show that E_α is G-stable (cf. 11.7.bis). Designate by $h : G \to S$ a second copy of $f : G \to S$ and consider the following commutative diagram:

$$\begin{tikzcd} G \times_S G \arrow[r, "q"] \arrow[dr, "p"'] & G \arrow[d, "\varphi", "h"'] \\ & G \arrow[r, "f"'] & S. \end{tikzcd}$$

Then the exact sequence

$$0 \longrightarrow E_\alpha \longrightarrow E \xrightarrow{\delta_\alpha} h_* h^*(E/F_\alpha)$$

gives, since f is flat, the exact sequence

$$(\dagger) \quad 0 \longrightarrow f_* f^*(E_\alpha) \longrightarrow f_* f^*(E) \xrightarrow{f_* f^*(\delta_\alpha)} f_* f^* h_* h^*(E/F_\alpha).$$

Moreover, since h is quasi-compact and quasi-separated, and f flat, the canonical morphism

$$f^* h_* h^*(E/F_\alpha) \to p_* q^* h^*(E/F_\alpha) = p_* \varphi^*(E/F_\alpha)$$

is an isomorphism, so that the right-hand term in (†) is $\phi_*\phi * (E/F_\alpha)$. Resuming the notations of 11.6 and denoting by π_α the projection $E \to E/F_\alpha$, one obtains the commutative diagram below, whose bottom line is exact:

$$\begin{tikzcd} E_\alpha \arrow[rr, "\delta"] \arrow[d, "\delta"'] & & f_* f^*(E) \arrow[d, "f_*(f^*(\pi) \boxtimes \delta_G)"] \\ f_* f^*(E_\alpha) \arrow[r] & f_* f^*(E) \arrow[r, "f_* f^*(\delta_\alpha)"] & \varphi_* \varphi^*(E/F_\alpha) \end{tikzcd}$$

and since $f_*(f*(\pi) \boxtimes \delta_G) \circ \delta$ vanishes on E_α, it follows that δ sends E_α into $f_*f*(E_\alpha)$, i.e. that E_α is G-stable. The proposition is proved. ∎

Remark 11.10.1. [585] In 11.8, it does not suffice to assume that C be a flat A-module, even if A is a principal ring. Indeed, one has the following counterexamples, which were pointed out (independently) to us by O. Gabber and J.-P. Serre.

(a) Let $(A, \mathfrak{m})$ be a discrete valuation ring, K its field of fractions, and G the "extension by zero" A-group of the K-group $(\mathbb{Z}/2\mathbb{Z})_K$. Then the constant group $(\mathbb{Z}/2\mathbb{Z})_A$, and hence also its subgroup G, acts on the free A-module V with basis v_1, v_2 by exchanging v_1 and v_2. Then Av_1 is not a sub-G-module of V,

[585] The first part of the original Remark 11.10.1 has been incorporated into 11.8 and 11.9 (by replacing "free" by "projective"); the counterexamples below correct the second part.

but it is the intersection of the sub-G-modules $Av_1 + \mathfrak{m}^n v_2$, for $n \in \mathbb{N}^*$. Hence there does not exist a smallest sub-G-module of V containing v_1.

(b) Let A be an integral ring, distinct from its field of fractions K, and let G be the flat affine A-group corresponding to the Hopf algebra

```
𝒜(G) = {P ∈ K[T] | P(0) ∈ A},
```

the comultiplication, resp. the counit and the antipode, being defined by `Δ(T) = T ⊗ 1 + 1 ⊗ T`, resp. $\epsilon(T) = 0$ and $\tau(T) = -T$. (N.B. One thus has $G \bigotimes_A K = G_{a,K}$.)

Let V be the free A-module $Av_1 \bigoplus Av_2$ and u the endomorphism of V defined by $u(v_1) = v_2$, $u(v_2) = 0$, so that $u^2 = 0$. Then V is equipped with an $\mathcal{A}(G)$-comodule structure, defined by

```
μ(m) = 1 ⊗ m + T ⊗ u(m).
```

The sub-G-modules of V containing v_1 are exactly the sub-A-modules of the form $Av_1 \bigoplus Iv_2$, for I a non-zero ideal of A; their intersection is Av_1, which is not a sub-G-module. Hence there does not exist a smallest sub-G-module of V containing v_1. (Note moreover that $C = A \bigoplus K \cdot T$ is a sub-coalgebra of $\mathcal{A}(G)$, flat over A, and that the coaction $\mu : V \to V \bigotimes \mathcal{A}(G)$ factors through $V \bigotimes C$, so one also obtains a counterexample for the "very simple" coalgebra C.)

Finally, note that the two preceding examples are particular cases of the following construction. Let A be an integral ring, distinct from its field of fractions K, let B be an A-Hopf algebra, free over A. Denote by $\epsilon : B \to A$ the augmentation of B and $I = Ker(\epsilon)$ the augmentation ideal. Since $B = A \cdot 1 \bigoplus I$, one easily sees that $B' = b \in B \bigotimes_A K | \epsilon(b) \in A$ is a sub-Hopf algebra of `B_K`. If V is a B-comodule, free with basis $(v_1, \cdots, v_n)$ as A-module, and if $\mu_V(v_1) \neq v_1 \bigotimes 1$, then Av_1 is not a subcomodule of V but it is the intersection of the subcomodules $Av_1 + I \cdot V$, for I running through the non-zero ideals of A. Hence there does not exist a smallest subcomodule of V containing v_1.

Proposition 11.11. *Let G be an algebraic group over the field k. The following conditions are equivalent:*

(i) G is affine.

(ii) G is quasi-affine.

(iii) G acts faithfully on a quasi-affine k-scheme X.

(iv) G acts linearly and faithfully on a k-vector space (not necessarily of finite dimension).

(v) G is isomorphic to a closed subgroup of a group $GL(n)_k$.

Proof. One has (i) ⇒ (ii) trivially, and (ii) ⇒ (iii), since G acts faithfully on itself by translations.

Suppose that G acts faithfully on the right on a quasi-affine k-scheme X.[586] Since X is quasi-affine, it is separated and quasi-compact.[587] Similarly, G is separated (VI_A 0.3) and quasi-compact (since of finite type over k). Hence, by 11.1 (c), one has canonical isomorphisms:

`O(X × G) = O(X) ⊗ O(G)` and `O(X × G × G) = O(X) ⊗ O(G) ⊗ O(G).`

One deduces that the morphism $\mu : O(X) \to O(X) \bigotimes O(G)$ induced by the morphism $X \times G \to X$ equips $O(X)$ with a right $O(G)$-comodule structure, i.e. G acts linearly on the left on the k-algebra $O(X)$. Consequently, G also acts on the right on the affine envelope $X_{af} = \operatorname{Spec} O(X)$ of X, and the canonical morphism $X \to X_{af}$ is G-equivariant.

Moreover, X being quasi-affine, $X \to X_{af}$ is an open immersion (EGA II, 5.1.2), hence a fortiori G acts faithfully on X_{af}. One thus obtains that the linear (left) action of G on the k-algebra $O(X) = O(X_{af})$ is faithful. This proves the implication (iii) ⇒ (iv).

Suppose now that G acts faithfully on a k-vector space V. Then, by virtue of 11.10, V is the inductive limit of finite-dimensional vector subspaces V_i, stable under the action of G. If K_i is the kernel of the induced action of G on V_i, i.e. of the morphism $G \to \operatorname{Aut}(V_i)$, then K_i is a closed subscheme of G, and the hypothesis that G acts faithfully is expressed by the fact that the intersection of the K_i is the unit subgroup of G. Since G is noetherian, it follows that one of the K_i is already reduced to the unit group, hence that $G \to \operatorname{Aut}(V_i)$ is a monomorphism. It is therefore a closed immersion by virtue of 1.4.2, which proves that (iv) ⇒ (v). Since (v) ⇒ (i) trivially, this proves 11.11. □

Remark 11.11.1. One can generalize 11.11 as follows. Let S be a regular locally noetherian scheme of dimension ≤ 1, and G a group scheme flat, quasi-compact and quasi-separated over S. (In this case, $\mathcal{A}(G)$ is a torsion-free `O_S`-module, hence flat.)

(a) One then has the equivalence of the following conditions:[588]

(b) G is affine over S.

(ii) G is quasi-affine over S.

(iii) G acts faithfully on a quasi-affine and flat S-scheme.

(iv) G acts linearly and faithfully on a flat quasi-coherent `O_S`-module.

(b) If moreover G is of finite type over S and S noetherian, these conditions imply that G is isomorphic to a closed subgroup of an $\operatorname{Aut}(E)$, where E is a finitely generated locally free `O_S`-module.[589]

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[586] We have expanded the original in what follows; on the other hand, we have chosen to have G act on the right on X in order to obtain a left linear action of G on $O(X)$.

[587] Recall that a k-scheme X is said to be quasi-affine if it is isomorphic to a quasi-compact open of an affine k-scheme (EGA II, 5.1.1).

[588] The equivalence of these conditions is proved in the additional section 12.

[589] This is proved, with various generalizations, in the additional section 13.

Lemma 11.12. *Let k be a field, G an affine k-group. Set $A = \mathcal{A}(G)$. Given $x \in A$, there exists a finitely generated k-subalgebra B of A such that $x \in B$, that $\Delta(B) \subset B \otimes_k B$, and $u(B) \subset B$, where u denotes the involution of A corresponding to the inversion morphism of G.*[590]

One can suppose $x \neq 0$; then $\Delta(x) \neq 0$ since $(\varepsilon \otimes \mathrm{id})\,\Delta(x) = x$, where ϵ denotes the augmentation (counit) of A. Write

$$(1) \qquad \Delta(x) = \sum_{j=1}^n e_j \otimes a_j \qquad \text{with } n \text{ minimal},$$

in which case the e_j (resp. a_j) are linearly independent. Complete $(e_1, \cdots, e_n)$ into a basis $(e_i)_{i \in I}$ of A and set, for $j \in J = 1, \cdots, n$,

$$\Delta(e_j) = \sum_{i \in I} e_i \otimes b_{ij}.$$

Applying $\Delta \otimes \mathrm{id}$ and $\mathrm{id} \otimes \Delta$ to (1), one obtains from the axiom (CO 1) of 11.8.0 (see also (HA 1) in I 4.2) the equalities:

$$\sum_{j \in J} e_j \otimes \Delta(a_j) = \sum_{\ell \in J} \Delta(e_\ell) \otimes a_\ell = \sum_{i \in I} e_i \otimes \Big(\sum_{\ell \in J} b_{i\ell} \otimes a_\ell\Big).$$

Since the e_i are linearly independent, it follows that

$$(2) \qquad \forall\, j \in J, \qquad \Delta(a_j) = \sum_{\ell \in J} b_{j\ell} \otimes a_\ell.$$

Let then B be the finitely generated k-subalgebra of A generated by the b_{ij} and the $u(b_{ij})$, for $i, j \in J$. It is clear that $u(B) = B$.

Applying $\Delta \otimes \mathrm{id}$ and $\mathrm{id} \otimes \Delta$ to (2), one obtains further from (CO 1) the equalities:

$$\sum_{\ell \in J} \Delta(b_{j\ell}) \otimes a_\ell = \sum_{i \in J} b_{ji} \otimes \Delta(a_i) = \sum_{i, \ell \in J} b_{ji} \otimes b_{i\ell} \otimes a_\ell,$$

and since the a_ℓ are linearly independent, one deduces that

$$(3) \qquad \forall\, j, \ell \in J, \qquad \Delta(b_{j\ell}) = \sum_{i \in J} b_{ji} \otimes b_{i\ell}.$$

Since $\Delta \circ u = (u \times u) \circ v \circ \Delta$, where $v(a \otimes b) = b \otimes a$, one therefore also has

$$(4) \qquad \forall\, j, \ell \in J, \qquad \Delta(u(b_{j\ell})) = \sum_{i \in J} u(b_{i\ell}) \otimes u(b_{ji}).$$

Since Δ is an algebra homomorphism, one deduces from (3) and (4) that $\Delta(B) \subset B \otimes_k B$. Finally, the axiom (CO 2) of 11.8.0 (see also (HA 2) in I, 4.2) shows that $a_j = \sum_{i \in I} \epsilon(a_j) b_{ij}$ and that $x = \sum_{j \in J} \epsilon(e_j) a_j$, so that $x \in B$. $\square$

Proposition 11.13. *Let k be a field and G an affine k-group with algebra A. Then G is a projective limit of an increasing filtered system of finitely generated affine k-groups, whose transition morphisms are faithfully flat.*

If B and B' are two finitely generated subalgebras of A stable under Δ and u, then so is the subalgebra generated by B and B'. Hence, by Lemma 11.12, A is the inductive limit of a filtered increasing family $(B_i)_{i \in I}$ of finitely generated subalgebras stable under Δ and u. Then each B_i, equipped with the restriction of u and

[590] On the one hand, this is generalized in Section 13 to the case where G is affine and flat over a regular base of dimension ≤ 2. On the other hand, in what follows we have spelled out and corrected the original.

the morphism $B_i \to B_i \otimes_k B_i$ deduced from Δ, is a Hopf algebra, hence by I 4.2 it is the algebra of an affine k-group G_i, of finite type over k. Finally, since $A = \lim B_i$, one has $G = \lim G_i$ (cf. EGA IV₃ 8.2.3). The transition morphisms are faithfully flat by the following lemma:

Lemma 11.14. *Let k be a field, $u : G \to H$ a morphism between finitely generated affine k-groups, and $u^{\sharp} : B \to A$[591] the corresponding morphism of k-algebras. For u to be faithfully flat, it is necessary and sufficient that $u^{\sharp}$ be injective.*

The condition is obviously necessary (cf. EGA 0_I 6.6.1). Let us show it is sufficient. Set $N = Keru$. Then, by VI_A, 3.3.2 and 5.4.1, G/N is a finitely generated k-group and u factors as $G - -p - - > G/N - -v - - > H$, where p is faithfully flat and v is a closed immersion. Hence, since H is an affine scheme, G/N is an affine scheme and the morphism $v^{\sharp} : B \to O(G/N)$ is surjective (cf. EGA I 4.2.3). Now, since $u^{\sharp}$ is assumed injective, and since $u^{\sharp} = p^{\sharp} \circ v^{\sharp}$, then $v^{\sharp}$ is also injective: it is therefore an isomorphism, as is v, and since p is faithfully flat, so is u. ∎

Definition 11.15. [592] *Let k be a field, G a quasi-compact k-group and V a k-vector space equipped with a k-linear action of G, hence with an $\mathcal{A}(G)$-comodule structure $\delta : V \to V \otimes \mathcal{A}(G)$, by 11.6.1. Let $v \in V$ be non-zero. The following conditions are equivalent:*

(i) There exists $\lambda \in \mathcal{A}(G)$ (necessarily unique) such that $\delta(v) = v \otimes \lambda$.

(ii) For every k-algebra R and every $g \in G(R)$, one has $g \cdot v \in Rv$ (i.e. there exists $f \in R$, necessarily unique, such that one has in $V \otimes R$ the equality $g \cdot v = v \otimes f$).

Indeed, it is clear that (i) ⇒ (ii). Conversely, if (ii) is satisfied and one applies it to $R = \mathcal{A}(G)$ and $g = id_{\mathcal{A}(G)}$, one obtains that there exists a unique $\lambda \in \mathcal{A}(G)$ such that $\delta(v) = v \otimes \lambda$.

If v satisfies these conditions, one says that v is a *semi-invariant vector under G*, and that λ is the *weight* of v; one will also say that "v is a semi-invariant of weight λ".

Denote by Δ the comultiplication of $\mathcal{A}(G)$; then the equality

$$v \otimes \lambda \otimes \lambda = (\delta \otimes id)(\delta(v)) = (id \otimes \Delta)(\delta(v)) = v \otimes \Delta(\lambda)$$

implies that $\Delta(\lambda) = \lambda \otimes \lambda$. Consequently, λ defines a morphism of Hopf algebras

$$\mathcal{A}(G_{m,k}) = k[T, T^{-1}] \to \mathcal{A}(G), \qquad T \mapsto \lambda,$$

and hence a morphism of k-groups $\lambda : G \to G_{m,k}$, i.e. λ is a *character* of G, called the *character associated with the semi-invariant vector v*.

Lemma 11.16.0. [593] *Let k be a field, H an affine k-group, V an H-module of dimension n and U a vector subspace of V of dimension d. Consider the line $D =$*

[591] We have denoted by $u^{\sharp}$ (instead of u°) the morphism $B \to A$ corresponding to $u : \mathrm{Spec}(A) \to \mathrm{Spec}(B)$.
[592] We have added the hypothesis that G be quasi-compact and spelled out the equivalence of conditions (i) and (ii).
[593] We have added this lemma, taken from [DG70], § II.2, 3.5.

$\bigwedge^d U \subset \bigwedge^d V$. *For U to be stable under H, it is necessary and sufficient that D be so.*

Necessity being clear, let us prove sufficiency. One may suppose $d < n$. Let $(e_1, \cdots, e_d)$ be a basis of U, complete it into a basis $(e_1, \cdots, e_n)$ of V. For every k-algebra R, $V_R = V \otimes R$ is a free R-module and one has

$$U_R = \{v \in V_R \mid v \wedge (e_1 \wedge \cdots \wedge e_d) = 0\}$$

(since for $i > d$ the $e_i \wedge e_1 \wedge \cdots \wedge e_d$ are linearly independent in $\bigwedge^{d+1} V_R$). Since $H(R)$ acts on $\bigwedge^\bullet V_R$ by

$$h(x_1 \wedge \cdots \wedge x_s) = h(x_1) \wedge \cdots \wedge h(x_s),$$

it follows that if $H(R)$ stabilizes $D_R = Re_1 \wedge \cdots \wedge e_d$, it also stabilizes U_R. □

Theorem 11.16 (Chevalley). *Let k be a field, G an algebraic affine k-group, H a closed group subscheme of G.*[594] *Then there exist a finite-dimensional G-module V and a line D in V such that $H = Norm_G(D)$, i.e. such that for every k-algebra R,*

$$H(R) = \{g \in G(R) \mid g(D_R) = D_R\}.$$

In other words, there exist finitely many elements $a_1, \cdots, a_n \in \mathcal{A}(G)$, which are semi-invariant, all of the same weight λ, for the "right" action of H (i.e. $a_i(gh) = \lambda(h)a_i(g)$, for every k-algebra R and $g \in G(R)$, $h \in H(R)$), such that H be the largest closed group subscheme of G under which the a_i are semi-invariant.

Denote by Δ (resp. ϵ) the comultiplication (resp. the augmentation) of $A = \mathcal{A}(G)$. Then $H = \operatorname{Spec}(A/I)$, for some ideal I of A, contained in $Ker\epsilon$ and such that $\Delta(I) \subset I \otimes A + A \otimes I$. Let $B = A/I$ and π the projection $A \to B$. Consider the left action of H on A given by $(h\phi)(g) = \phi(gh)$; the corresponding B-comodule structure is given by:

$$\bar{\Delta} : A \xrightarrow{\Delta} A \otimes A \xrightarrow{id_A \otimes \pi} A \otimes B.$$

Then I is a sub-H-module of A, since $\bar{\Delta}(I) \subset I \otimes B$.

On the other hand, A is a finitely generated k-algebra, hence noetherian, hence I admits a finite system of generators $(x_1, \cdots, x_r)$. By 11.8, the x_i are contained in a sub-G-module V of finite dimension over k. Then $W = V \cap I$ is an H-module of finite dimension, whose dimension we denote by d. Since V contains all the x_i, W generates the ideal I.

Set $E = \bigwedge^d V$, let $(w_1, \cdots, w_d)$ be a basis of W, and let $B = (e_0, \cdots, e_n)$ be a basis of E containing the vector $e_0 = w_1 \wedge \cdots \wedge w_d$. The action of G on V canonically determines an action of G on E, hence an A-comodule structure $\rho : E \to E \otimes A$. For $j = 0, \cdots, n$, set $\rho(e_j) = \sum_{i=1}^n e_i \otimes b_{ij}$; one has seen in the proof of 11.12 that

$$(*) \qquad \Delta(b_{ij}) = \sum_{\ell=0}^n b_{i\ell} \otimes b_{\ell j}.$$

[594] On the one hand, we recall that every group subscheme of G is closed (1.4.2). On the other hand, we have stated the result in the usual form: "H is the stabilizer of a line in a representation of G", while keeping the original formulation in terms of a sequence $a_1, \cdots, a_n$ of semi-invariants in $\mathcal{A}(G)$.

Set $a_i = b_{i0}$, i.e. if $(e_0^*, \cdots, e_n^*)$ is the dual basis of B, the a_i are the "matrix coefficients" $g \mapsto e_i^*(ge_0)$. On the other hand, the action of H on E corresponds to:

$$\bar{\rho} : E \xrightarrow{\rho} E \otimes A \xrightarrow{\mathrm{id}_E \otimes \pi} E \otimes B.$$

Since W is stable under H, e_0 is semi-invariant under H, hence $\bar{\rho}(e_0) = e_0 \otimes \pi(a_0)$ and $a_i = b_{i0}$ belongs to I for $i = 1, \cdots, n$. Substituting this into $(*)$, one obtains $\Delta(a_i) = a_i \otimes \pi(a_0)$ for $i = 0, \cdots, n$, i.e. the a_i are semi-invariant under H with weight $\pi(a_0)$. (Moreover, possibly by replacing E by the sub-G-module generated by e_0 (cf. 11.8), one may assume that the a_i are linearly independent.)

Conversely, let $H' = \operatorname{Spec} B'$, where $B' = A/I'$, be a closed group subscheme of G under which each of the a_i is semi-invariant, with weight $\lambda_i \in B'$ (this is the case, in particular, if e_0 is invariant under H' with weight λ'). Let us show that $H' = H$. Denote by π' the projection $A \to B'$; the hypothesis implies that

$$a_i \otimes \lambda_i = (\mathrm{id}_A \otimes \pi')\,\Delta(b_{i0}) = \sum_{\ell=0}^n b_{i\ell} \otimes \pi'(a_\ell),$$

whence $\lambda_i = \pi'(a_0)$ and $a_\ell \in I'$ for $\ell = 1, \cdots, n$, and hence e_0 is semi-invariant under H'. By Lemma 11.16.0, this implies that W is stable under H'. Since the ideal I is generated by W, it is also stable under H', and therefore

$$\Delta(I) \subset I \otimes A + A \otimes I'.$$

Since $I \subset Ker\epsilon$ and $(\varepsilon \otimes \mathrm{id}_A) \circ \Delta = \mathrm{id}_A$, it follows that $I \subset I'$, whence $H' \subset H$. $\square$

Lemma 11.17.0. [595] *Let k be an algebraically closed field, G a reduced algebraic affine k-group, V a finite-dimensional G-module over k, and $v \in V$. Let E be the vector subspace of V generated by the vectors gv, for $g \in G(k)$. Then E is the smallest sub-G-module of V containing v, and hence the morphism $G \times E \to V$, $(g, x) \mapsto gx$, factors through E.*

Proof. By 11.8, we know that there exists a smallest sub-G-module U of V containing v: if $\mu : V \to V \otimes \mathcal{A}(G)$ denotes the comodule structure and if one writes $\mu(v) = \sum_{i=1}^n v_i \otimes a_i$ with the a_i linearly independent, one has $U = Vect(v_1, \cdots, v_n)$. It is clear that U contains E, and that the morphism $G \times U \to V$ factors through U.

Conversely, the inverse image of E by the morphism $\mu_v : g \mapsto gv$ is a closed subset of G which contains the rational points; now these are dense in G, since G is of finite type over k (cf. EGA IV$_3$, 10.4.8), hence $\mu_v^{-1}(E) = G$ and so, since G is reduced, μ_v factors through E, whence $E = U$. $\square$

Theorem 11.17 (Chevalley). *Let k be a field, G an affine k-group (not necessarily of finite type), and N a closed group subscheme of G invariant in G; then the (fpqc) sheaf quotient G/N is representable by an affine k-group.*[596]

[595] We have added this lemma, taken from the proof of thm. 5.6 of [DG70], § III.3 (by abuse of notation, we designate by the same letter E a k-vector space and the k-scheme in modules $W(E) = \operatorname{Spec} S(E^*)$).

[596] For another proof of this theorem, not using the results of VI_A, see [Ta72], Th. 5.2 (see also Remark 11.18.5).

Suppose first G of finite type. By VI_A 3.2 and 5.2, the (fpqc) sheaf quotient G/N is representable by a k-group Q; it therefore remains to show that Q is affine. The proof is done in several stages; suppose first k algebraically closed.[597]

(a) Suppose moreover G reduced and connected, and N reduced. By 11.16, there exist a G-module V, of finite dimension over k, and a line $D = ke_0$ such that $Norm_G(D) = N$; in particular N acts on D via a character $\chi : N \to G_{m,k}$.

Fix $h \in N(k)$. For every $g \in G(k)$, one has $hge_0 = g(g^{-1}hg)e_0 = \chi(g^{-1}hg)ge_0$, hence $\chi(g^{-1}hg)$ is an eigenvalue of h. Hence the continuous map $\phi : G(k) \to k$, $g \mapsto \chi(g^{-1}hg)$, takes only finitely many values, and since $G(k)$ is irreducible (since dense in G), one therefore has $\phi(g) = \phi(e) = \chi(h)$ for every $g \in G(k)$, and so

$$\chi(g^{-1} h g) = \chi(h), \qquad \forall g \in G(k),\ h \in N(k).$$

Let E be the vector subspace of V generated by the vectors ge_0, for $g \in G(k)$; by Lemma 11.17.0, this is the sub-G-module of V generated by e_0.

By what precedes, the two morphisms $N \times E \to E$, $(h,x) \mapsto hx$ and $(h,x) \mapsto \chi(h)x$, coincide on the set of rational points $(N \times E)(k) = N(k) \times E(k)$, which is dense in $N \times E$. Since $N \times E$ is reduced (and E separated), these two morphisms are therefore equal, so N acts on E by homotheties. Consequently, N is contained in the kernel K of the morphism $\rho : G \to GL(\mathrm{End}_k(E))$, defined by $\rho(g)(u) = gug^{-1}$, for every $g \in G(R)$ and $u \in \mathrm{End}_R(E \otimes R)$ (R a k-algebra). On the other hand, if $g \in K(R)$ then $g(Re_0) = Re_0$, whence $g \in N(R)$. This shows that $N = K$. Then, by VI_A 5.4.1, the morphism $G/N \to GL(\mathrm{End}_k(E))$ is a closed immersion, and hence G/N is affine.

(b) Suppose now G and N reduced, G not necessarily connected. Set $N' = N \cap G^0$; then G^0/N' is affine by (a). On the other hand, NG^0 is an invariant subgroup of G and G/NG^0, being a quotient of the finite constant group G/G^0 (cf. VI_A, 5.5.1), is likewise a finite constant group. Hence G/N is the direct sum of the fibers of the morphism $G/N \to G/NG^0$, all isomorphic to NG^0/N, hence to G^0/N', by VI_A, 5.3.3. Hence G/N is affine.

(c) Suppose G reduced, and N arbitrary. The morphism $G \times N \to N$, $(g,h) \mapsto ghg^{-1}$, induces a morphism $(G \times N)_{red} \to N_{red}$; now, since G is reduced and k algebraically closed, one has $(G \times N)_{red} = G \times N_{red}$, hence N_{red} is an invariant subgroup of G. (N.B. this fails when G is not reduced, cf. VI_A, 0.2.)

Hence, by (a), $G' = G/N_{red}$ is affine. On the other hand, by VI_A 5.6.1, $N' = N/N_{red}$ is a finite k-group, hence by Theorem 4.1 of Exp. V, the quotient $G/N = G'/N'$ is affine.

(d) For arbitrary G and N, the equivalence relation deduced from $G \times N \rightrightarrows G$ by the base change $G_{red} \to G$ is:

$$G_{red} \times N' \rightrightarrows G_{red}, \qquad \text{where} \qquad N' = N \cap G_{red}.$$

[597] In what follows, we have spelled out the original (and corrected the erroneous assertion (G/N)_red = G_red / N_red), relying on [DG70], § III.3, 5.6.

Since the underlying spaces are the same (and since the quotient is the quotient ringed space), the morphism $G_{red}/N' \to G/N$ is a homeomorphism. Since G_{red}/N' is reduced (since $p : G_{red} \to G_{red}/N'$ is faithfully flat), it follows that $(G/N)_{red}$ is identified with G_{red}/N', which is affine by (c). Since G/N is of finite type over k (cf. VI_A, 3.3.2), this implies, by EGA I, 5.1.10, that G/N is affine.

Finally, for arbitrary k, let $\bar{k}$ be an algebraic closure of k. Then, by 9.2 (v), $(G \otimes_k \bar{k})/(N \otimes_k \bar{k})$ is isomorphic to $(G/N) \otimes_k \bar{k}$, hence since the former is affine, so is the latter, and so G/N is also affine, by (fpqc) descent (cf. EGA IV_2, 2.7.1). This proves 11.17 when G is of finite type. To extend this to the general case, we shall need the following lemma.[598]

Lemma 11.17.1. *Let $(C_i \to A_i)_{i \in I}$ be a filtered inductive system of ring morphisms, all faithfully flat. Then $A = \lim A_i$ is faithfully flat over $C = \lim C_i$.*

Proof. By [BAC] § I.3, Prop. 9, for a morphism of rings $B \to B'$ to be faithfully flat, it is necessary and sufficient that it be injective and that B'/B be a flat B-module. Since each $C_i \to A_i$ is faithfully flat, one therefore has exact sequences

$$0 \longrightarrow C_i \longrightarrow A_i \longrightarrow A_i / C_i \longrightarrow 0$$

and A_i/C_i is a flat C_i-module, hence $(A_i/C_i) \otimes_{C_i} C$ is a flat C-module. Since inductive limits are exact and commute with the tensor product, one obtains an exact sequence

$$0 \longrightarrow C \longrightarrow A \longrightarrow A/C \longrightarrow 0$$

as well as an isomorphism

$$\lim ((A_i / C_i) \otimes_{C_i} C) = (A/C) \otimes_C C = A/C,$$

from which one deduces that A/C is a flat C-module. Hence A is faithfully flat over C. □

Let us now return to the proof of 11.17 in the general case, i.e. when G is not assumed to be of finite type. Set $\mathcal{A}(G) = A$ and $\mathcal{A}(N) = B = A/J$. By 11.13, A is the inductive limit of a filtered increasing family $(A_i)_{i \in I}$ of finitely generated sub-Hopf algebras, hence G is the projective limit of the algebraic affine k-groups $G_i = \operatorname{Spec}(A_i)$. Denote by Δ (resp. τ) the comultiplication (resp. the antipode) of A, and $\Delta_2 = (\Delta \otimes \mathrm{id}_A) \circ \Delta$.

For every i, $B_i = A_i/(J \cap A_i)$ is a quotient Hopf algebra of A_i, hence $N_i = \operatorname{Spec}(B_i)$ is a closed subgroup of G_i. Moreover, since N is invariant in G, the morphism $G \times N \to G$ defined by $(g, n) \mapsto gng^{-1}$ factors through N, and this is equivalent to saying that the couple (A, J) satisfies the following property:

$$(m_{13} \circ (\Delta \otimes \tau) \circ \Delta_2)(J) \subset A \otimes_k J$$

where m_{13} denotes the map $a_1 \otimes a_2 \otimes a_3 \mapsto a_1 a_3 \otimes a_2$. It follows that $(A_i, A_i \cap J)$ satisfies the analogous property, hence that N_i is invariant in G_i. On the other hand, one has $\lim B_i = B$ and hence $\lim N_i = N$.

[598] We have added this lemma, taken from [DG70], § III.3, 7.1.

By what we have seen previously, each (fpqc) sheaf quotient G_i/N_i is representable by an affine k-group $Q_i = \mathrm{Spec}(C_i)$. Set $C = \lim C_i$. One thus has a filtered projective system of affine k-groups Q_i; its projective limit $Q = \lim Q_i$ is the k-group $\mathrm{Spec}(C)$ (cf. EGA IV$_3$, 8.2.3). One then has an exact sequence of k-groups:

$$1 \longrightarrow N \longrightarrow G \longrightarrow Q.$$

Let us show that Q represents the (fpqc) sheaf quotient of G by N; for this, it suffices to verify that the morphism $G \to Q$ is covering for the (fpqc) topology (cf. IV, 3.3.2.1 and 5.1.7.1). Now, each of the morphisms $G_i \to Q_i$ is faithfully flat (cf. 9.2 (xi)), in other words A_i is faithfully flat over C_i; since $A = \lim A_i$ and $C = \lim C_i$, it follows from Lemma 11.17.1 that A is faithfully flat over C, so that $G \to Q$ is a faithfully flat morphism. Since this morphism is affine, it is quasi-compact, hence covering for the (fpqc) topology. This completes the proof of Theorem 11.17. $\square$

11.18. Complements.

[599] Moreover, one deduces from 11.17 (and from its proof) the following results, taken from [DG70], III § 3.7. Let k be a field. We begin with the following lemma (cf. [An73], 2.3.3.2), which will be useful later (cf. 12.10).

Lemma 11.18.1. *Let $u : G \to G'$ be a morphism of k-groups, $N = Ker(u)$. Suppose G' affine and G of finite type.*

(i) The morphism $\tau : G/N \to G'$ is a closed immersion. In particular, G/N is affine.

(ii) If moreover the morphism $u^{\sharp} : O(G') \to O(G)$ is injective, then τ is an isomorphism. (And hence G' is of finite type and u is faithfully flat.)

Indeed, we know (11.13) that G' is the projective limit of a filtered system of algebraic affine k-groups G'_i. Denote by u_i the composite morphism $G \to G' \to G'_i$ and by N_i its kernel. Then the N_i form a filtered decreasing system of closed subgroups of G, whose intersection is N. Since G is noetherian, there exists an index i such that $N = N_i$. Since G and G_i are of finite type, by VI_A, 3.2 and 5.4.1, the quotient G/N is a finitely generated k-group and u_i is the composite of the projection $p : G \to G/N$, which is faithfully flat, and a closed immersion $\tau_i : G/N \hookrightarrow G_i$.

Consider then the following commutative diagram:

$$\begin{tikzcd}
G \arrow[r, "u"] \arrow[d, "p"'] & G' \arrow[dr, "q_i"] & \\
G/N \arrow[ur, "\tau"] \arrow[rr, "\tau_i"'] & & G_i.
\end{tikzcd}$$

Since $q_i \circ \tau = \tau_i$ is a closed immersion and q_i is separated (G' being separated, cf. VI_A, 0.3), then τ is a closed immersion (cf. EGA I, 5.4.4). It follows that G/N is

[599] We have added this subsection.

affine, and that the morphism $\tau^{\sharp} : O(G') \to O(G/N)$ is surjective, whence (i).

If moreover $u^{\sharp} : O(G') \to O(G)$ is injective, so is $\tau^{\sharp}$, hence $\tau^{\sharp}$ is an isomorphism, hence so is τ (since G' and G/N are affine). This proves (ii). □

Theorem 11.18.2. *Let G and G' be two affine k-groups, with algebras A and A', and let $u : G \to G'$ be a morphism of k-groups, $N = Ker(u)$, and $\phi : A' \to A$ the morphism induced by u.*

(i) If ϕ is injective, then u is faithfully flat and identifies G' with G/N.

(ii) One has $G/N = \mathrm{Spec}(B)$, where $B = \phi(A')$, and hence u is the composite of the faithfully flat morphism $G \to G/N$, corresponding to the inclusion $B \hookrightarrow A$, and of the closed immersion $G/N \hookrightarrow G'$, which corresponds to the surjection $A' \to B$. Moreover, N is defined in G by the ideal AB_+, where B_+ denotes the augmentation ideal of B.

(iii) In particular, if u is a monomorphism, it is a closed immersion.

Proof. (i) Suppose ϕ injective and identify A' with a sub-Hopf algebra of A. By 11.13, A is the filtered union of finitely generated sub-Hopf algebras $A_i = O(G_i)$; denote $G'_i = \mathrm{Spec}(A'_i)$, where $A'_i = A' \cap A_i$, and N_i the kernel of the morphism $G_i \to G'_i$ induced by the inclusion $A'_i \hookrightarrow A_i$. By the preceding lemma, one has $G_i/N_i \cong G'_i$ and one therefore obtains, for every i, an exact sequence

$$1 \longrightarrow N_i \longrightarrow G_i \xrightarrow{p_i} G'_i \longrightarrow 1$$

where p_i is faithfully flat. Since $G = \lim_i G_i$ and $G' = \lim_i G'_i$, one therefore obtains an exact sequence

$$1 \longrightarrow N \longrightarrow G \xrightarrow{p} G/N$$

where one has set $N = \lim_i N_i$. Moreover, by Lemma 11.17.1, p is faithfully flat (and affine), hence G' represents the (fpqc) sheaf quotient of G by N. This proves (i).

In the general case, $B = \phi(A')$ is a sub-Hopf algebra of A; denote by H the k-group $\mathrm{Spec}(B)$ and N' the kernel of the morphism $G \to H$ induced by the inclusion $B \hookrightarrow A$. By (i), H is identified with G/N', and u is therefore the composite of the projection $G \to G/N'$ and the closed immersion $G/N' \hookrightarrow G'$ induced by the surjection $A' \to B$. It follows that $N' = N$. Moreover, by 9.2 (ii), one has a cartesian square:

$$\begin{tikzcd} N \arrow[r] \arrow[d] & G \arrow[d, "p"] \\ \mathrm{Spec}(k) \arrow[r, "\varepsilon"] & G' \end{tikzcd}$$

where ϵ is the unit section of G', which corresponds to the augmentation morphism $B \to k$. It follows that N is defined in G by the ideal AB_+. This proves (ii), and (iii) follows. □

Remark 11.18.3. Let G be an affine k-group and N a normal k-subgroup. Since the morphism $p : G \to G/N$ is faithfully flat and quasi-compact, by IV 3.3.3.2, $O(G/N)$ is the subalgebra of $O(G)$ formed of functions ϕ which are right N-invariant, i.e. which satisfy $\phi(gh) = \phi(g)$, for every k-scheme S and $g \in G(S)$, $h \in N(S)$. Denoting by J the ideal of $A = O(G)$ which defines N, this is equivalent to saying that $\Delta(\varphi) - \varphi \otimes 1 \in O(G) \otimes J$, where Δ is the comultiplication of A.

The preceding theorem can then be reformulated in terms of Hopf algebras as follows.

Corollary 11.18.4. *Let k be a field, A a commutative k-Hopf algebra, $G = \mathrm{Spec}(A)$.*

(i) If B is a sub-Hopf algebra of A, then A is faithfully flat over B.

(ii) The map $N \mapsto O(G/N)$ is a bijection between the set of normal subgroups of G and that of sub-Hopf algebras of A; the inverse map is given by $B \mapsto \mathrm{Spec}(A/AB_+)$. Moreover, if J is the ideal of A defining N, one has

$$O(G/N) = \{x \in A \mid \Delta(x) - x \otimes 1 \in A \otimes J\}.$$

Remarks 11.18.5. (a) A consequence of the preceding theorem is that the category of commutative affine k-groups is abelian. For this, as well as for other results on affine k-groups, we refer to [DG70], § III.3, 7.4 to 7.8.

(b) Let us finally point out that M. Takeuchi has given another proof of results 11.17 to 11.18.4, cf. [Ta72], § 5; he moreover strengthened 11.18.4 (i) above by showing that A is even a projective B-module, cf. [Ta79], Th. 5 (see also [MW94], Th. 3.6).

12. Complements on G_{af} and the "anti-affine" groups

[600] We begin with the following lemma, which extends 11.18.1 to the case where G is not assumed to be of finite type.[601]

Lemma 12.1. *Let k be a field, $u : G \to H$ a monomorphism of k-groups, with H affine. Suppose u quasi-compact. Then u is a closed immersion.*

Proof. By (fpqc) descent, one may suppose k algebraically closed. By VI_A, 6.4, the closed image I of u is a closed group subscheme of H, hence still affine. Hence, replacing H by I, one may suppose u schematically dominant. Since k is algebraically closed, $H' = H_{red}$ is a group subscheme of H; set $G' = G \times_H H'$; then the morphism $u' : G' \to H'$ deduced from u by base change is a quasi-compact monomorphism, and is dominant (the underlying continuous map being the same for u and u'). Hence, by VI_A, 6.2, u' is faithfully flat; it is therefore a quasi-compact faithfully flat monomorphism, hence an isomorphism (cf. IV 1.14).

Hence $u : G \to H$ is a homeomorphism, so it is affine by 2.9.1. Hence G is affine, and so u is a closed immersion by 11.18.2 (iii). ∎

[600] We have added this section.
[601] This lemma was communicated to us by M. Raynaud; it will be used in the proof of Proposition 12.9.

Theorem 12.2. *Let G be an algebraic k-group. Denote by ρ the canonical morphism $G \to G_{af}$ and N its kernel.*

(i) The canonical morphism $G/N \to G_{af}$ is an isomorphism, and hence G_{af} is an algebraic affine group, and ρ is faithfully flat.

(ii) One has a canonical isomorphism $(G/N)_{af} = G_{af}$.

(iii) N is a characteristic subgroup of G.

(iv) $O(N) = k$.

(v) N is smooth, connected, and commutative.

Proof. Point (i) is a particular case of 11.18.1, and point (ii) follows from the universal properties of G_{af} and $(G/N)_{af}$.

Let us prove (iii). For an arbitrary k-scheme $\pi : S \to \operatorname{Spec} k$, consider the cartesian square:

$$\begin{tikzcd} G_S \arrow[r] \arrow[d, "q"'] & G \arrow[d, "p"] \\ S \arrow[r, "\pi"] & \operatorname{Spec} k. \end{tikzcd}$$

Since p is quasi-compact and separated and π flat, by EGA III 1.4.15 and EGA IV_1 1.7.21, one has $q_*(O_{G_S}) = \pi * (O(G)) = \pi * (O(G_{af}))$, and hence, by EGA II, 1.5.2, one has $(G_S)_{af} = (G_{af})_S$. Hence N_S, being the kernel of the canonical morphism $G_S \to (G_S)_{af}$, is invariant under every automorphism of G_S, i.e. N is a characteristic subgroup of G.

To prove (iv), set $N' = Ker(N \to N_{af})$; by (ii), this is an invariant subgroup of G. Since N is algebraic (being a closed subgroup of G), by (i), $N/N' \cong N_{af}$; moreover, by VI_A, 3.2 and 5.3.2, one has an isomorphism of k-groups

$$(G/N') / N_{af} \cong (G/N')/(N/N') \cong G/N.$$

Since N_{af} is affine, the projection $G/N' \to G/N$ is also affine, by 9.2 (vii), and since $G/N = G_{af}$ is affine, so is G/N'. Hence, by the universal property of G_{af}, the projection $p' : G \to G/N'$ factors through $G_{af} = G/N$, whence $N \subset N'$ and hence $N = N'$. Hence N_{af} is the trivial group, whence $O(N) = k$.

Finally, assertion (v) follows from the following lemma. $\square$

Lemma 12.3. *Let k be a field and N an algebraic k-group such that $O(N) = k$. Then N is smooth, connected, and commutative.*

Indeed, one may suppose k algebraically closed. Then $H = N^0_{red}$ is a k-subgroup of N, and the quotient k-scheme $X = N/H$ is finite (hence affine) over k, by VI_A, 5.5.1 and 5.6.1. Moreover, since $p : N \to X$ is faithfully flat, one has $O(X) \subset O(N) = k$. It follows that $N = N^0_{red}$, hence N is smooth (VI_A 1.3.1) and connected.

Let then Z be the center of N. By 6.2.6, N/Z is affine, and one obtains as above that $O(N/Z) = k$, whence $N = Z$. This proves 12.3 and completes the proof of 12.2. □

Let us also state, without proof, the following theorem. (Recall that an *abelian variety* over a field k is a proper, smooth, and connected k-group scheme.)

Theorem 12.4 (Chevalley). *Let k be a perfect field and G an algebraic, smooth and connected k-group. Then there exists an affine, smooth and connected k-subgroup L, invariant in G, such that the quotient G/L is an abelian variety. Moreover, L is unique and its formation commutes with extension of the base field.*

Remarks 12.5. (1) This theorem was announced in 1953 by C. Chevalley, who published his proof in 1960 ([Ch60]). Meanwhile, other proofs were obtained, independently, by I. Barsotti and M. Rosenlicht ([Ba55, Ro56]); see [Se99] for historical comments.

(2) A modern version (i.e. in the language of schemes) of Chevalley's proof has been given by B. Conrad ([Co02]). (Note that in loc. cit., "algebraic group" means smooth and connected k-group scheme.)

(3) On the other hand, a modern version of Rosenlicht's proof has been given by Ngô B.-C. in a course at Orsay in 2005-2006.

(4) If one drops the hypothesis that k is perfect, there still exists a smallest invariant connected affine subgroup L (not necessarily smooth) such that G/L is an abelian variety ([BLR], § 9.2, Thm. 1).

(5) One may also drop the hypothesis that G is smooth over k: indeed, by VII_A, 8.3, there exists an integer $n \geq 1$ such that the quotient $G' = G/(Fr^n G)$ is smooth; then G' contains a subgroup L' as in (4) above, and the inverse image of L' in G still has the same properties. Hence, for every connected algebraic group G over a field k, there exists an exact sequence

$$1 \longrightarrow H \longrightarrow G \longrightarrow A \longrightarrow 1$$

where H is an affine k-group and A a k-abelian variety. Moreover, by [Per76], Cor. 4.2.9, one has such an exact sequence for every connected k-group G (not necessarily algebraic).

(6) Let k be an algebraically closed field and G the semi-direct product of an elliptic curve E by the constant k-group $\pm 1_k$, for the action defined by $(-1) \cdot x = -x$; in this case, if L is a closed invariant subgroup of G such that G/L is connected, then $L = G$.

Remark 12.6. One will say, following [Br09], that a k-group N is *anti-affine* if $O(N) = k$. By 12.3 and 12.4, if k is perfect, every anti-affine algebraic k-group is an extension of an abelian variety by a smooth, connected, commutative affine algebraic k-group. For the precise structure of anti-affine algebraic groups over a perfect field, and various consequences, see the recent articles of M. Brion and C. & F. Sancho de Salas ([Br09, SS09]).

To conclude this section, we shall prove two results due to M. Raynaud, the first being Remark 11.11.1, the second Proposition 2.1 of Exp. XVII, Appendix III. We shall need the following lemma,[602] which improves (for a complete discrete valuation ring R) the flatness criteria given in [BAC], § III.5.

Lemma 12.7. *Let R be a discrete valuation ring, K its field of fractions, π a uniformizer. Let A be a flat R-algebra and M an A-module flat over R. Suppose that:*

(i) $M/\pi M$ is a flat module over $\bar{A} = A/\pi A$,

(ii) $M \otimes_R K$ is a flat module over $A \otimes_R K$.

Then M is a flat A-module.

Proof. By the flatness criterion in the nilpotent case (cf. [BAC], III § 5.2, Th. 1), $M/\pi^n M$ is a flat module over $A/\pi^n A$, for every $n \in \mathbb{N}^*$. It then follows from [RG71], II Lemma 1.4.2.1, that M is a flat A-module. For the reader's convenience, let us briefly indicate the proof. Set $s = \pi^n$ and $P = M/sM$. Since P is flat over A/sA and the latter is of projective dimension 1 over A, it follows from the spectral sequence of composite functors that P is of Tor-dimension ≤ 1 over A. Now, since M is R-flat hence without π-torsion, M_K/M is the inductive limit of the A-modules $\pi^{-n}M/M \cong M/\pi^n M$, and hence M_K/M is also of Tor-dimension ≤ 1. Since one has the exact sequence $0 \to M \to M_K \to M_K/M \to 0$ and by hypothesis `M_K` is flat over `A_K` hence over A, it follows that M is flat.

For completeness, let us also indicate the following simpler proof, pointed out by O. Gabber. Let I be a finitely generated ideal of A; one must show that the morphism $u : M \otimes_A I \to M$ is injective. By hypothesis (ii), $u \otimes_R K$ is injective, hence $Ker(u)$ is an R-module of π-torsion. It therefore suffices to show that the π-torsion part of $M \otimes_A I$ is zero; now this is a quotient of $Tor_1^A(M, I/\pi I)$, as one sees by tensoring with M the exact sequence:

$$0 \longrightarrow I \xrightarrow{\pi} I \longrightarrow I/\pi I \longrightarrow 0.$$

On the other hand, M being without π-torsion (since flat over R), one obtains that $Tor_i^A(M, \bar{A}) = 0$ for every $i \geq 1$. Consequently, if $(P_\bullet)$ is a projective resolution of the A-module M, then $(P_\bullet \otimes_A \bar{A})$ is a projective resolution of the $\bar{A}$-module $\bar{M} = M/\pi M$, and hence for every $\bar{A}$-module N, one has $Tor_i^A(M, N) = Tor_i^{\bar{A}}(\bar{M}, N)$, and this is zero for $i \geq 1$ since $\bar{M}$ is flat over $\bar{A}$. One thus has $Tor_1^A(M, I/\pi I) = 0$, which proves the lemma. ∎

Remark 12.8. Let S be a regular locally noetherian scheme of dimension 1, and X a flat, quasi-separated and quasi-compact S-scheme. Then $\mathcal{A}(X)$ is a flat `O_S`-module. Indeed, one may suppose S local; denote by s its closed point, i the inclusion $X_s \hookrightarrow X$, and π a uniformizer of $R = O(S)$; since X is flat over S, one has an exact sequence of sheaves

[602] This is a version improved by O. Gabber of a statement communicated by M. Raynaud.

$$0 \longrightarrow O_X \xrightarrow{\pi} O_X \longrightarrow i_*(O_{X_s}) \to 0$$

and so, by taking global sections, one obtains that $\mathcal{A}(X)$ is an R-module without π-torsion, hence flat.[603] One obtains moreover that the morphism from $\mathcal{A}((X_{af})_s) = \mathcal{A}(X)/\pi\mathcal{A}(X)$ to $\mathcal{A}(X_s)$, induced by the morphism $X \to X_{af}$, is injective.

One can now prove the following proposition (cf. Remark 11.11.1).

Proposition 12.9. *Let S be a regular locally noetherian scheme of dimension ≤ 1, G an S-group scheme flat, quasi-separated and quasi-compact over S. Then the following conditions are equivalent:*

(i) G is affine over S.

(ii) G is quasi-affine over S.

(iii) G acts faithfully on a quasi-affine and flat S-scheme X.

(iv) G acts linearly and faithfully on a quasi-coherent module E flat over S.

(v) The morphism $\rho : G \to G_{af}$ is a monomorphism.

Proof. The implication (i) ⇒ (ii) is evident, as is (ii) ⇒ (iii) (take $X = G$).

Suppose (iii) holds. Since $\mathcal{A}(G)$ and $\mathcal{A}(X)$ are flat O_S-modules, one obtains, proceeding as in 11.11, that G acts (on the right) faithfully on X_{af} and hence acts (on the left) linearly and faithfully on the flat quasi-coherent O_S-module $\mathcal{A}(X)$.

Moreover, if (iv) holds, by 11.6.1 (ii), the monomorphism $G \to \mathrm{Aut}_{O_S}(E)$ factors through G_{af}, hence $G \to G_{af}$ is a monomorphism.

Finally, suppose (v) holds and let us show that $\rho : G \to G_{af}$ is an isomorphism. Replacing S by one of its connected components, one may suppose S irreducible, with generic point η. Since the formation of G_{af} commutes with flat base changes, one has $(G_{af})_\eta = (G_\eta)_{af}$, and therefore the morphism $G_\eta \to (G_\eta)_{af}$ is a monomorphism, hence a closed immersion by 12.1, hence an isomorphism since $O((G_\eta)_{af}) = O(G_\eta)$. If $S = \mathrm{Spec}(\kappa(\eta))$ we are done; one may therefore suppose $\dim S = 1$.

Let then s be a closed point of S; let us show that $G_s \to (G_{af})_s$ is an isomorphism and that ρ is flat at every point of G_s. For this, one may suppose S local, with closed point s. The morphism $\rho_s : G_s \to (G_{af})_s$ obtained by base change is a monomorphism, hence a closed immersion by 12.1, hence the morphism $O((G_{af})_s) \to O(G_s)$, induced by ρ_s, is surjective. Now, by the preceding remark, it is also injective, hence it is an isomorphism. (In particular, ρ is therefore surjective.)

It then follows from Lemma 12.7 that $\rho : G \to G_{af}$ is faithfully flat. Since G is quasi-compact over S and G_{af} separated over S, ρ is also quasi-compact (cf. EGA I, 6.6.4). Consequently, ρ is a faithfully flat quasi-compact monomorphism, hence an isomorphism. This proves the proposition. ∎

[603] This is true, more generally, if S is regular locally noetherian of dimension ≤ 2, cf. [Ray70a], VII 3.2.

Finally, let us prove Prop. 2.1 of Exp. XVII, Appendix III; taking into account [Per76], Cor. 4.2.5, we have substituted in the hypotheses "quasi-compact and quasi-separated" for "of finite type" (if one assumes G of finite type, one can use 12.1 instead of loc. cit.).

Proposition 12.10. *Let S be a regular locally noetherian scheme of dimension ≤ 1, G an S-group scheme flat, quasi-compact and quasi-separated.*

(i) The canonical morphism $\rho : G \to G_{af}$ is faithfully flat and quasi-compact. Consequently, G_{af} represents the (fpqc) sheaf quotient of G by $N = Ker(\rho)$.

(ii) If moreover G is of finite type over S, then ρ is of finite presentation and G_{af} represents the (fppf) sheaf quotient of G by N and is of finite type over S.

(iii) Suppose moreover G_η affine for every maximal point η of S. Then N is an étale S-group, and is the unit group if G is separated over S.

Proof. First, since G_{af} is affine, hence separated over S, ρ is quasi-compact (cf. EGA I, 6.6.4) and the kernel $N = Ker(\rho)$ is a closed subgroup of G. Moreover, replacing S by one of its connected components, one may suppose S irreducible, with generic point η.

Let us remark that, to prove (i) and (ii), it suffices to show that ρ is faithfully flat, because then, by Exp. IV, 5.1.7.1, G_{af} represents the (fpqc) sheaf quotient of G by N, and if moreover G is of finite type over S, hence of finite presentation (S being locally noetherian), then, by 9.2 (xiii), ρ is of finite presentation (as is $G_{af} \to S$) and hence ρ is covering for the (fppf) topology.

One may therefore suppose $S = \mathrm{Spec}(R)$, where R is a discrete valuation ring (if $\dim S = 1$) or else the field $\kappa(\eta)$ (if $\dim S = 0$). Denote by s the closed point of S. Since the formation of G_{af} commutes with flat base changes, the canonical morphism $G_\eta \to (G_\eta)_{af}$ is identified with the morphism $\rho_\eta : G_\eta \to (G_{af})_\eta$, and since $O((G_{af})_\eta) = O((G_\eta)_{af}) = O(G_\eta)$, then ρ_η is faithfully flat by [Per76], 4.2.5 (see also the addition VI_A, 6.6). If $\dim S = 1$, one similarly obtains that ρ_s is faithfully flat, since the morphism $O((G_{af})_s) \to O(G_s)$ is injective, by Remark 12.8. Hence, by Lemma 12.7, ρ is faithfully flat. This proves (i) and (ii). In particular, N is flat over S.

Suppose now G_η affine. Since ρ_η coincides with the canonical morphism $G_\eta \to (G_\eta)_{af}$, its kernel N_η is the unit group. Let us show that N is étale over S. Since N is flat over S, it remains to see that N_s is étale over $\kappa(s)$, for every point $s \neq \eta$ of S. There is nothing to prove if $S = \mathrm{Spec}(\kappa(\eta))$, so one may suppose $S = \mathrm{Spec}(R)$, where R is a discrete valuation ring. Let s be the closed point of S, K the field of fractions of R, and π a uniformizer. Let $x \in N_s$ and U_x an affine open neighborhood of x in N; since N is flat over S, $U_x \cap N_\eta$ is non-empty, hence equal to $\epsilon(\eta)$, where ϵ denotes the unit section. Hence $A = O(U_x)$ is a flat R-algebra, such that $A_K = K$ and $\pi^{-1} \notin A$ (since π belongs to the maximal ideal of $O_{U,x}$). It follows that $A = R$, and hence the projection $U_x \to S$ is an isomorphism. This proves that N is étale over S; if moreover G is separated over S, then the inverse isomorphism $S \to U_x$ equals the unit section (since they coincide on the dense open subset η of $S = \mathrm{Spec}(R)$), hence N is the unit group. The proposition is proved. ▯

One obtains in particular the following corollary, two other proofs of which are found in [An73], Prop. 2.3.1 and [PY06], Prop. 3.1.

Corollary 12.10.1. *Let R be a discrete valuation ring, K its field of fractions, G an R-group scheme separated, flat and of finite type over R. If G_K is affine, then G is affine.*

Remarks 12.10.2. (a) On the one hand, O. Gabber has pointed out to us examples where G is a flat group of finite type over a discrete valuation ring, whose generic fiber is an abelian variety, and where the kernel N of $G \to G_{af}$ is not smooth.

(b) On the other hand, let us point out that M. Raynaud has given an example, for S the affine plane of dimension 2 over a field k, of a smooth and quasi-affine S-group scheme, with affine and connected fibers, which is not affine over S, cf. [Ray70a], § VII.3, p. 116.

13. Flat affine groups over a regular base of dimension ≤ 2

[604] Let us begin by remarking that the well-known argument which shows that every affine algebraic group over a field k is linear, as well as Lemma 11.12, extend to the case of a group scheme G, affine, flat, and of finite type over a base scheme S which is noetherian, regular, of dimension ≤ 2. For S of dimension ≤ 1, this proves point (b) of Remark 11.11.1. The extension to the case $\dim S = 2$ rests on the following lemma, which was communicated to us by O. Gabber.

Lemma 13.1. *Let S be a normal noetherian scheme, $\mathcal{A}$ a flat quasi-coherent O_S-module, $\mathcal{F}$ a quasi-coherent O_S-submodule of finite type of $\mathcal{A}$, $\mathcal{F}**$ its bidual, and U the flatness locus of $\mathcal{F}$, i.e. the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $O_{S,s}$-module.*

(i) U is an open subset of S and $\mathcal{F}** = j_*j*(\mathcal{F})$, where j denotes the inclusion $U \hookrightarrow X$.

(ii) The canonical morphism $j_*(\mathcal{E}) \otimes_{O_S} \mathcal{A} \to j_*(\mathcal{E} \otimes_{O_U} j*(\mathcal{A}))$ is an isomorphism, for every quasi-coherent O_U-module $\mathcal{E}$.

(iii) In particular, $\mathcal{V} = j_*j*(\mathcal{F})$ is a submodule of $\mathcal{A} = j_*j*(\mathcal{A})$, and the canonical morphism $\mathcal{V} \otimes_{O_S} \mathcal{A} \to j_*j*(\mathcal{F} \otimes_{O_S} \mathcal{A})$ is an isomorphism.

Proof. Replacing S by one of its connected components, we may suppose S integral. By EGA IV₂, 2.1.12, the flatness locus of $\mathcal{F}$, i.e. the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $O_{S,s}$-module, is an open subset U of S; denote by j the inclusion $U \hookrightarrow S$. Since $\mathcal{A}_s$ is flat, hence torsion-free, so is $\mathcal{F}_s$,

so U contains every point of codimension ≤ 1. Consequently, by [BAC], VII, § 4.2, cor. of th. 1, one has $\mathcal{F}** = j_*j*(\mathcal{F})$, and one therefore obtains a monomorphism $\mathcal{F}** \to j_*j*(\mathcal{A})$.

[604] This section has been added.

The proof of (ii) is analogous to that of EGA III, 1.4.15, recalled in 11.0. On the other hand, since S is normal, the morphism $O_S \to j_*j*(O_S)$ is an isomorphism (cf. EGA IV_2, 5.8.6 and 5.10.5). By (ii) applied to $\mathcal{E} = O_U$, one therefore has $\mathcal{A} = j_*j*(\mathcal{A})$. Finally, since $j*(\mathcal{F} \otimes_{O_S} \mathcal{A}) = j*(\mathcal{F}) \otimes_{O_U} j*(\mathcal{A})$, the final assertion of (iii) follows from (ii) applied to $\mathcal{E} = j*(\mathcal{F})$. The lemma is proved. $\square$

Furthermore, recall that a finitely generated R-module M is said to be *reflexive* if the canonical morphism from M to its bidual $M**$ is an isomorphism. When R is a noetherian regular ring of dimension ≤ 2, this entails that M is projective. Indeed, for every finitely generated R-module N, consider a resolution $L_1 \to L_0 \to N \to 0$, where L_0 and L_1 are finitely generated free R-modules; then one has an exact sequence

$$0 \to N^* \to L_0^* \to L_1^* \to Q \to 0,$$

where Q denotes the cokernel of $L_0* \to L_1*$, and since R is of homological dimension ≤ 2 (cf. [BAC], X § 4.2, cor. 1 of th. 1), it follows that $N*$ is projective.

Proposition 13.2. *Let S be a noetherian regular scheme of dimension ≤ 2, G an affine flat S-group, $\mathcal{A}(G)$ its affine algebra.*

(i) If G is of finite type over S, it is isomorphic to a closed subgroup of $H = \mathrm{Aut}_{O_S}(\mathcal{V})$, for some O_S-module $\mathcal{V}$ locally free of finite rank. If moreover S is affine, one may take $\mathcal{V} = O_S^{\oplus d}$ for some d, whence $H = GL_{d,S}$.

(ii) $\mathcal{A}(G)$ is a filtered inductive limit of flat O_S-sub-Hopf-algebras of finite type.

Proof. Let $\mathcal{B}$ be a finitely generated O_S-subalgebra of $\mathcal{A}(G)$. Since every coherent module on an open of S extends to a coherent module on S (cf. EGA I, 9.4.5), there exists a coherent O_S-submodule $\mathcal{M}$ of $\mathcal{B}$ which generates $\mathcal{B}$ as an O_S-algebra (loc. cit., 9.6.5). By 11.10.bis, $\mathcal{M}$ is contained in a coherent G-stable O_S-submodule $\mathcal{F}$. (N.B. Since G is here affine over S, the proof of loc. cit. is written more simply: one may there replace $f_*f*(\mathcal{E})$ by $\mathcal{E} \otimes_{O_S} \mathcal{A}(G)$, etc.)

Let j be the inclusion $U \hookrightarrow S$, where U denotes the flatness locus of $\mathcal{F}$. By Lemma 13.1 and the remarks following it, $\mathcal{V} = j_*j*(\mathcal{F})$ is a locally free O_S-submodule of $\mathcal{A}(G)$, and since the canonical morphism

$$\mathcal{V} \otimes \mathcal{A}(G) \to j_* j^*(\mathcal{V} \otimes \mathcal{A}(G))$$

is an isomorphism, $\mathcal{V}$ is a G-submodule of $\mathcal{A}(G)$. The action of G on $\mathcal{V}$ then induces a morphism of affine S-groups $\rho_{\mathcal{V}} : G \to H = \mathrm{Aut}_{O_S}(\mathcal{V})$ and hence a morphism of O_S-Hopf algebras $\phi_{\mathcal{V}} : \mathcal{A}(H) \to \mathcal{A}(G)$. Denote by $\mathcal{A}_{\mathcal{V}}$ the image of $\phi_{\mathcal{V}}$; this is the affine algebra of a closed subgroup $G_{\mathcal{V}}$ of H, which is the closed image of $\rho_{\mathcal{V}}$. Let us show that $\mathcal{A}_{\mathcal{V}}$ contains $\mathcal{B}$.

The question being local on S, one may suppose $S = \mathrm{Spec}(R)$ and $V = \Gamma(S, \mathcal{V})$ is a free R-module with basis $v_1, \cdots, v_n$; in this case $H \simeq GL_{n,R}$ and $\mathcal{A}(H)$ is generated as an R-algebra by the "matrix coefficients" c_{ij} and the element d^{-1}, where

d denotes the determinant. Let Δ (resp. ϵ) be the comultiplication (resp. the augmentation) of A. For $j = 1, \cdots, n$, write $\Delta(v_j) = \sum_{i=1}^{n} v_i \otimes a_{ij}$; then $a_{ij} = \phi(c_{ij})$

belongs to $Im(\phi)$. On the other hand, since V is an R-submodule of A, one can use the identity $(\epsilon \otimes id_A) \circ \Delta = id_A$, which entails that $v_j = \sum_i \epsilon(v_i) a_{ij}$ belongs to $Im(\phi)$. Since V contains a system of generators of $B = \Gamma(S, \mathcal{B})$, it follows that $B \subset Im(\phi)$.

If G is of finite type over S, one may take $\mathcal{B} = \mathcal{A}(G)$ and ϕ is then surjective, so the morphism of S-groups $G \to H = \mathrm{Aut}_{O_S}(\mathcal{V})$ is a closed immersion.

If moreover S is affine, there exists a locally free 0_S-module $\mathcal{V}'$ of finite rank such that $\mathcal{V} \oplus \mathcal{V}' = O_S^d$ as 0_S-modules. Regarding $\mathcal{V}'$ as a trivial G-module, one may replace $\mathcal{V}$ by O_S^d, and one thus obtains that G is a closed subgroup of $GL_{d,S}$.

Finally, let us return to the case of an arbitrary flat affine S-group G. By EGA I, 9.4.9, $\mathcal{A}(G)$ is the union of its coherent 0_S-submodules $\mathcal{M}$, hence also of the 0_S-sub-Hopf-algebras $\mathcal{A}_{\mathcal{V}}$ as above, whence (ii). $\square$

Example 13.3. *Let R be a discrete valuation ring, with uniformizer π and field of fractions K. Consider the filtered projective system of R-groups:*

$$\cdots \to G_{a,R} \xrightarrow{\times\pi} G_{a,R} \xrightarrow{\times\pi} G_{a,R}$$

(corresponding to the inductive system $R[X_0] \to R[X_1] \to R[X_2] \to \cdots$, where the transition morphisms are given by $X_n = \pi X_{n+1}$). Its projective limit G is a flat affine R-group scheme, not of finite type, whose special fiber is trivial and whose generic fiber is $G_{a,K}$; the affine R-algebra $\mathcal{A}(G)$ is the subring of K[X] formed of polynomials whose constant coefficient belongs to R. (N.B. We have already encountered this example in Remark 11.10.1.) Note that G represents the functor which to every R-algebra B associates the set of sequences $(x_n)_{n\in\mathbb{N}}$ of elements of B such that $x_n = \pi x_{n+1}$ for every n. (In particular, each x_n is indefinitely π-divisible.)

Now let S be a noetherian scheme such that every coherent 0_S-module is the quotient of a locally free 0_S-module of finite type; this is the case, for example, if S is a separated noetherian regular scheme, cf. SGA 6, Exp. II, 2.1.1 and 2.2.7. (One can show that every noetherian regular scheme of dimension ≤ 1 also has this property; on the other hand, it fails when S is the affine plane over a field k with the origin doubled, cf. loc. cit., 2.2.7.2.)

Definition 13.4. *Let G be a flat affine S-group. Following R. W. Thomason ([Th87], 2.1), we say that the pair (G,S) possesses the* equivariant resolution property*, or satisfies (RE), if for every coherent G-0_S-module $\mathcal{F}$, there exists a locally free G-0_S-module $\mathcal{E}$ of finite rank and a G-equivariant epimorphism $\mathcal{E} \to \mathcal{F}$.*

In loc. cit., Th. 3.1, Thomason proves the result below, under the hypothesis that G is essentially free over S (cf. the remark further on). Gabber pointed out to us the simpler proof below, which does not use this hypothesis.

Proposition 13.5. *Let S be a noetherian scheme and G an S-group affine, flat, and of finite type, with affine algebra $\mathcal{A}(G)$. Suppose that (G,S) satisfies (RE). Then:*

(i) G is isomorphic to a closed subgroup of $H = \mathrm{Aut}_{O_S}(\mathcal{V})$, for some 0_S-module $\mathcal{V}$ locally free of finite rank.

(ii) If moreover S is affine, one may take $\mathcal{V} = O_S^{\oplus d}$ for some d, whence $H = GL_{d,S}$.

The proof is analogous to that of 13.2. As in loc. cit., there exists a coherent G-stable 0_S-submodule $\mathcal{F}$ which generates $\mathcal{A} = \mathcal{A}(G)$ as an 0_S-algebra. Replacing S by one of its connected components, we may suppose S connected. By hypothesis (RE), there exists a locally free G-0_S-module $\mathcal{E}$ of rank n, and an epimorphism of $\mathcal{A}$-comodules $\pi : \mathcal{E} \to \mathcal{F}$. Set $H = \mathrm{Aut}_{O_S}(\mathcal{E})$; this is an S-group scheme, locally isomorphic to $GL_{n,S}$. The action of G on $\mathcal{E}$ induces a morphism of affine S-groups $\rho : G \to H$, corresponding to a morphism of 0_S-Hopf algebras $\phi : \mathcal{A}(H) \to \mathcal{A}(G)$. Let us show that ρ is a closed immersion.

The question being local on S, one may suppose $S = \mathrm{Spec}(R)$ and $V = \Gamma(S, \mathcal{E})$ is a free R-module with basis $v_1, \cdots, v_n$; in this case $H \simeq GL_{n,R}$ and $B = \Gamma(S, \mathcal{A}(H))$ is generated as an R-algebra by the "matrix coefficients" c_{ij} and the element d^{-1}, where d denotes the determinant. Let Δ (resp. ϵ) be the comultiplication (resp. the augmentation) of $A = \Gamma(S, \mathcal{A}(G))$, let $\mu : V \to V \bigotimes_R A$ be the A-comodule structure on V, and let $F = \Gamma(S, \mathcal{F})$. For $j = 1, \cdots, n$, write

$$\mu(v_j) = \sum_{i=1}^n v_i \otimes a_{ij}$$

then $\phi(c_{ij}) = a_{ij}$. On the other hand, since $\pi : V \to F$ is a morphism of A-comodules, one has

$$\sum_{i=1}^n \pi(v_i) \otimes a_{ij} = (\pi \otimes_R \mathrm{id}_A)(\mu(v_j)) = \Delta(\pi(v_j))$$

and therefore

$$\pi(v_j) = (\varepsilon \otimes_R \mathrm{id}_A)\, \Delta(\pi(v_j)) = \sum_{i=1}^n (\varepsilon\, \pi(v_i))\, a_{ij} = \varphi\Big(\sum_{i=1}^n (\varepsilon\, \pi(v_i))\, c_{ij}\Big).$$

Hence $\phi(B)$ contains $\pi(V) = F$, which generates A as an R-algebra, and hence ϕ is surjective. This shows that ρ is a closed immersion, whence assertion (i).

If moreover S is affine, there exists a locally free 0_S-module $\mathcal{V}'$ of finite rank such that $\mathcal{V} \bigoplus \mathcal{V}' = O_S^d$ as 0_S-modules. Regarding $\mathcal{V}'$ as a trivial G-module, one may replace $\mathcal{V}$ by O_S^d, and one thus obtains that G is a closed subgroup of $GL_{d,S}$. The proposition is proved. □

Remarks 13.6. (a) For the sake of completeness, let us briefly sketch Thomason's argument ([Th87], Th. 3.1), keeping the preceding notation. The G-equivariant epimorphism $\pi : \mathcal{E} \to \mathcal{F}$ induces a closed immersion $\tau : G \to V(\mathcal{E})$ such that $\tau(g'g) = \tau(g') \cdot g$ (N.B.: G acts on the right on $V(\mathcal{E})$), and one has an isomorphism $H \simeq \mathrm{Aut}_{O_S}(V(\mathcal{E}))^{op}$, which is compatible with the actions of G on the left on $\mathcal{E}$ and on the right on $V(\mathcal{E})$. Now let N be the strict transporter $Transp_H^{str}(\tau(G), \tau(G))$; when G is essentially free over S, it follows from 6.2.4 e) that N is a closed subgroup scheme of H, hence affine over S. Moreover, ρ factors through a

morphism of S-groups $\rho' : G \to N$. On the other hand, for every $S' \to S$ and $h \in N(S')$, set $\pi(h) = \tau(1) \cdot h$ (where 1 is the unit element of $G(S')$); this defines a morphism of S-schemes $\pi : N \to \tau(G)$, which is a retraction of ρ' (when one identifies G with $\tau(G)$). Since N is separated over S, it follows that ρ' is a closed immersion.

(b) It seems that the proof of [Th87], Th. 3.1 requires the hypothesis that G be essentially free over S, which does not appear in loc. cit. (the author invoking in its place the fact that H is essentially free). This hypothesis is however satisfied when G is reductive (cf. Exp. XXII 5.7.8), and so is satisfied in all the cases considered in loc. cit., Cor. 3.2. In particular, Thomason proves in loc. cit., 2.5, that if S is separated noetherian regular of dimension ≤ 2, and if G is affine, of finite presentation, and such that $\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$-module, then (G, S) satisfies (RE); by 13.5, this gives 13.2 under a slightly more restrictive hypothesis.

To conclude, let us point out that the proof of [Th87], 2.5, may be slightly simplified, as follows. (For brevity, we place ourselves in the situation where $S = \mathrm{Spec}(R)$ is affine.)

Proposition 13.7. *Let R be a noetherian regular ring of dimension ≤ 2, C an R-coalgebra, projective as an R-module, and F a C-comodule, of finite type over R. Then F is the quotient of a C-comodule V, projective of finite type over R.*

Proof. Replacing $\mathrm{Spec}(R)$ by one of its connected components, one may suppose R integral, with field of fractions K. Denote by Δ (resp. ϵ) the comultiplication (resp. the augmentation) of C and by $\rho : F \to F \otimes C$ the comodule structure on F. Let $\pi : W \to F$ be a surjective morphism, where W is a free R-module of finite rank. We endow $W \otimes C$ with the comodule structure defined by $id_W \otimes \Delta$, and similarly for $F \otimes C$. Then $\rho : F \to F \otimes C$ is a morphism of C-comodules, which admits $id_F \otimes \epsilon$ as a section.

Let W' be the C-comodule defined by the cartesian square below:

$$\begin{tikzcd}
W' \arrow[r] \arrow[d, "\pi \otimes id_C"'] & F \arrow[d, "\rho"] \\
W \otimes C \arrow[r] & F \otimes C
\end{tikzcd}$$

i.e. W' is identified with the kernel of the morphism $W \otimes C \to (F \otimes C)/\rho(F)$, and the projection $\pi' : W' \to F$, given by $x \mapsto (\pi \otimes \epsilon)(x)$, is surjective. Since F is a finitely generated R-module, there exists a subcomodule V' of W', finitely generated over R, such that $\pi'(V') = F$. Since $W \otimes C$ is R-torsion-free, so is V'; hence, replacing F by V', one may assume at the outset that F is torsion-free.

Applying the preceding construction to this new F, one obtains V' as above. Consider then the subcomodule V, kernel of the morphism

$$W \otimes C \to E = (W \otimes C \otimes K) / (V' \otimes K).$$

Then V contains V', and $Q = (W \otimes C)/V$ is an R-submodule of the K-vector space E; set $Q' = E/Q$. Since $W \otimes C$ and E are flat R-modules, one obtains that,

for every R-module N,

$$\mathrm{Tor}_1^R(V, N) \simeq \mathrm{Tor}_2^R(Q, N) \simeq \mathrm{Tor}_3^R(Q', N)$$

and since R is regular of dimension ≤ 2, the right-hand term is zero. This shows that V is a flat R-module. Let us finally show that V is a finitely generated R-module.

Set $M = W \bigotimes C$; it follows from the definition that V/V' is isomorphic to the R-torsion submodule $(M/V')_{tors}$ of M/V'.

Since M is a projective R-module, there exists a projective R-module P such that $M \bigoplus P$ is a free R-module L. Then $(M/V')_{tors} \simeq (L/V')_{tors}$. On the other hand, since V' is of finite type, there exists a direct factor $L' \simeq R^n$ of L such that $V' \subset L'$, and one therefore also has $(L/V')_{tors} \simeq (L'/V')_{tors}$, and the latter is of finite type since (L'/V') is. Consequently, V is a flat R-module of finite type, hence projective of finite type (R being noetherian). Proposition 13.7 is proved. ∎

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Exposé VII_A. Infinitesimal study: differential operators and restricted p-Lie algebras

by P. Gabriel

In Exposé II we restricted ourselves to the study of first-order differential invariants and did not address certain phenomena specific to characteristic $p > 0$ or to characteristic 0. Our object in part A of this Exposé is to fill in this gap.

Moreover, the infinitesimal study of arbitrary order of a group scheme is related to that of the associated formal group; the object of the second part of this Exposé is to present the first definitions and properties concerning formal groups.

A) Differential operators and restricted p-Lie algebras[606]

1. Differential operators

In this section, as well as in sections 2 and 3, S denotes a fixed scheme and the products considered are cartesian products in the category of S-schemes.[607] If X is an S-scheme, we write $p_{X/S}$, p_X or simply p for the structural morphism of X into S.

1.1.

Let $u : Y \to X$ be a morphism of S-schemes and endow the direct image $u_*(O_Y)$ of the structure sheaf of Y with the O_X-module structure induced by u. The sheaf $H = \mathrm{Hom}_{p_X^{-1}(O_S)}(O_X, u_*(O_Y))$ of $p_X^{-1}(O_S)$-module homomorphisms from O_X into $u_*(O_Y)$ is therefore naturally equipped with a structure of O_X-bimodule: if U is an open of X, f and d sections of O_X and H on U, then fd and df are respectively the morphisms $g \mapsto fd(g)$ and $g \mapsto d(fg)$ from O_X into $u_*(O_Y)$. We shall henceforth write (ad f)(d) in place of $fd - df$.

Definition 1.1.1. *An S-deviation of order $\leqslant n$ is by definition a pair $D = (u, d)$ consisting of a morphism of S-schemes $u : Y \to X$ and a morphism of $p_X^{-1}(O_S)$-modules $d : O_X \to u_*(O_Y)$ such that, for every open U of X and every sequence of $n+1$ sections $f_0, \cdots, f_n \in O_X(U)$, one has in $\mathrm{Hom}_{p_U^{-1}(O_S)}(O_U, u_*(O_Y)|_U)$ the equality:*

(∗_n) (ad f₀)(ad f₁) ⋯ (ad f_n)(d) = 0.

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In this case, we shall also say that d is an S-deviation of u of order $\leqslant n$. In particular, an S-deviation of u of order $\leqslant 0$ is a morphism of O_X-modules from O_X into $u_*(O_Y)$, i.e., an element of $\Gamma(Y, O_Y)$.

---

[606] Part A of the present Exposé had not been treated seriously in the oral seminars.

[607] In particular, if X and Y are two S-schemes, $X \times_S Y$ is denoted simply $X \times Y$. Moreover, let us point out that for the content of sections 1 and 2, one may also consult [DG70], § II.4, nos 5–6; see also [Ja03], § I.7.

[608] One sees easily that this is equivalent to saying that, for every $x \in X$ and $f_0, \cdots, f_n, g \in O_{X,x}$, one has $(adf_0)(adf_1)\cdots(adf_n)(d_x)(g) = 0$. On the other hand, recall that the adjunction isomorphism

θ : Hom_{p_X^{-1}(O_S)}(O_X, u_*(O_Y)) ≅ Hom_{p_Y^{-1}(O_S)}(u^{-1}(O_X), O_Y)

associates with every morphism of $p_X^{-1}(O_S)$-modules $d : O_X \to u_*(O_Y)$ the morphism $d' = \epsilon \circ u^{-1}(d)$, where ϵ is the canonical morphism $u^{-1}u_*(O_Y) \to O_Y$. Conversely, for every $p_Y^{-1}(O_S)$-morphism $d' : u^{-1}(O_X) \to O_Y$, one has $\theta^{-1}(d') = u_*(d') \circ \eta$, where η is the canonical morphism $O_X \to u_*u^{-1}(O_X)$. It follows that d satisfies $(*_n)$ if and only if d' satisfies:

(∗'_n) (ad f₀) ⋯ (ad f_n)(d')(g) = 0

for every open V of Y and $f_0, \cdots, f_n, g \in u^{-1}(O_X)(V)$.

Definition 1.1.2. *A morphism of $p_X^{-1}(O_S)$-modules $d : O_X \to u_*(O_Y)$ is an S-deviation of u if, for every point y of Y, there exist an open neighborhood U of $u(y)$ in X and an open neighborhood V of y in Y satisfying the following conditions:*

a) $u(V) \subset U$;

b) if $v : V \to U$ is the morphism induced by u, there is an integer n such that the morphism $O_U \to v_*(O_V)$ induced by d is an S-deviation of v of order $\leqslant n$.[609]

If d is an S-deviation of u, we also say that the pair $D = (u, d)$ is an S-deviation and it will happen that we write $Y \xrightarrow{D} X$ or $Y \xrightarrow{d} X$ (over u).

When d is the algebra homomorphism $u^{\sharp} : O_X \to u_*(O_Y)$ corresponding to the morphism $u : Y \to X$, we shall also write u in place of D.

Remarks 1.1.3.[610] Let $Dév(u)$ (resp. $Dév_{\leqslant n}(u)$) be the set of S-deviations of u (resp. S-deviations of u of order $\leqslant n$). It is equipped with a natural structure of $O_Y(Y)$-module: if $\lambda \in O_Y(Y)$, λd is the deviation sending f to $\lambda d(f)$, for every section f of O_X on an open U.

For every open V of Y, set $\mathcal{D}év(u)(V) = Dév(u|_V)$, i.e., $\mathcal{D}év(u)(V)$ is the set of

$$d_V \in \mathrm{Hom}_{p_X^{-1}(O_S)}(O_X, (u|_V)_*(O_V)) \simeq \mathrm{Hom}_{p_Y^{-1}(O_S)}((u|_V)^{-1} O_X, O_V) \simeq \mathrm{Hom}_{p_Y^{-1}(O_S)}(u^{-1} O_X, O_Y)(V)$$

such that, for every open U of X, the map $d_V(U) : O_X(U) \to O_Y(u^{-1}(U) \cap V)$ satisfies $(*_n)$. This defines a presheaf of O_Y-modules on Y, and one sees easily that it is a sheaf (more precisely, a subsheaf of $\mathrm{Hom}_{p_Y^{-1}(O_S)}(u^{-1}O_X, O_Y)$).

1.2.

Consider now two S-deviations $D = (u, d)$ and $E = (v, e)$:

$$Z \xrightarrow{v,e} Y \xrightarrow{u,d} X.$$

When U ranges over the opens of X, the composed maps

$$\Gamma(U, O_X) \xrightarrow{d(U)} \Gamma(u^{-1} U, O_Y) \xrightarrow{e(u^{-1} U)} \Gamma(v^{-1} u^{-1} U, O_Z)$$

define an S-deviation of uv which we shall denote de; when d is of order $\leqslant m$ and e of order $\leqslant n$, de is of order $\leqslant m + n$. We shall also write

$$D \circ E = (uv, de) \tag{†}$$

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and we shall say that $D \circ E$ or DE is the *composed S-deviation*. When $d = u^{\sharp}$ (i.e., $D = u$ with the convention of 1.1), one also says that DE is the *image of E by u.*

[609] If X and u are quasi-compact, every S-deviation of u is therefore of order $\leqslant n$, for some integer n.
[610] These remarks have been added, as they will be useful in 1.3, 1.4 and 2.1.
[611] One will note that with this notation, de denotes the composition “d followed by e”.

The map $(D,E) \mapsto D \circ E$ we have just defined will henceforth allow us to speak of the *category of S-deviations*, whose objects are the S-schemes and whose morphisms are the S-deviations.[612]

Definition 1.2.0.[613] *Let $w : Z \to X$ be an S-morphism. An S-derivation of w, or S-derivation of 0_X into $w_*(O_Z)$, is a morphism of $p^{-1}(O_S)$-modules $d : O_X \to w_*(O_Z)$ such that, for every open U of X and $f, g \in O_X(U)$,*

d(fg) = w^♮(f) d(g) + w^♮(g) d(f).

Then d is a deviation of w of order $\leqslant 1$ which vanishes on the unit section of 0_X. We denote by $\mathrm{Dér}_S(w)$ the set of S-derivations of w; it is an $O(Z)$-module.

With the notations of 1.2, take Y equal to $I_Z = \operatorname{Spec} O_Z[t]$, where $t^2 = 0$, and v equal to the zero section $\tau : Z \to I_Z$, defined by the morphism of 0_Z-algebras $O_Z[t] \to O_Z$ sending t to 0, and take e equal to the morphism of 0_Z-modules $\sigma : O_Z[t] \to O_Z$ defined by $\sigma(1) = 0$ and $\sigma(t) = 1$,[614] which it is convenient to denote ∂_t.

If $u : I_Z \to X$ is a morphism satisfying $w = u \circ \tau$, then $\sigma \circ u^{♮}$ is an S-derivation of 0_X into $w_*(O_Z)$. Conversely, to every S-derivation d we associate the morphism $u : I_Z \to X$ such that $u = w$ on the underlying spaces, and

u^♮(f) = w^♮(f) + d(f) t,

for every section f of 0_X on an open U. One thus obtains:

Lemma 1.2.1. *Let $E = (\tau, \partial_t)$ be the deviation of $\tau : Z \to I_Z$ defined above. For every S-morphism $w : Z \to X$, the map $u \mapsto u \circ E$ is a bijection between the S-morphisms $u : I_Z \to X$ such that $u \circ \tau = w$, and the S-derivations of w.*

1.2.2.

Let d be an S-deviation of $u : Y \to X$. On the one hand, d is obviously an S'-deviation of u for every morphism $s : S \to S'$.

On the other hand, let $t : T \to S$ be a morphism with target S, and let $u_T : Y_T \to X_T$ be the morphism deduced from u by base change, and $t_Y : Y_T \to Y$ and $t_X : X_T \to X$ the canonical projections. Then there exists one and only one T-deviation of u_T, which we shall denote d_T or $d \times T$, satisfying the equality $t_X d_T = d t_Y$, in the sense of (†) above, i.e., for every open U of X, one has a commutative diagram:[615]

[612] Often, one considers only the S-deviations of the morphism id_X, which form the algebra of S-differential operators of X, cf. 1.4 below. However, the more general framework of S-deviations provides a convenient "functorial" language for proving statements such as: "if G is an S-group, the algebra of S-differential operators on G invariant under left translation is isomorphic to the algebra of S-deviations of the unit section $\epsilon : S \to G$, cf. 2.1 and 2.4 below."

[613] This paragraph has been expanded, with the number 1.2.0 (resp. 1.2.1) being assigned to this definition (resp. to the lemma which follows).

[614] The following has been added, i.e. the notation ∂_t has been introduced.

[615] Explicitly, if V is an affine open of S and U (resp. U') an affine open of X (resp. T) above V, so that $O_{X\times T}(U \times U') = O_X(U) \otimes_{O_S(V)} O_T(U')$, then $d_T(U \times U')$ is the composition:

0_X(U) ⊗_{0_S(V)} 0_T(U') ––d(U) ⊗ id→ 0_Y(u^{-1} U) ⊗_{0_S(V)} 0_T(U') ——→ 0_{Y × T}(u^{-1} U × U').

The author left to the reader the verification that d_T is well-defined, and the editors do the same.

$$
\begin{tikzcd}
\mathcal{O}(U) \arrow[r, "t_X^{\natural}"] \arrow[d, phantom, "d(U)"] & \mathcal{O}(U \times T) \arrow[d, phantom, "d_T(U \times T)"] \\
\mathcal{O}(u^{-1} U) \arrow[r, "t_Y^{\natural}"] & \mathcal{O}(u^{-1} U \times T).
\end{tikzcd}
$$

If one sets $D = (u, d)$, one will also write $D_T = (u_T, d_T)$ and we shall say that d_T and D_T are deduced from d and D by base change.

1.2.3.

For example, let $u : Y \to X$ and $v : Z \to T$ be two S-morphisms, d and e S-deviations of u and v. One has a commutative diagram

$$
\begin{tikzcd}
X \times T & Y \times T \arrow[l, "u_T"'] \\
X \times Z \arrow[u, "v_X"] \arrow[ur, phantom, "u\times v"] & Y \times Z \arrow[l, "u_Z"'] \arrow[u, "v_Y"']
\end{tikzcd}
$$

and we shall denote by $d \times e$ (the *product* of d and e) the S-deviation of $u \times v$ equal to $d_T e_Y = e_X d_Z$ (with the convention (†) above), i.e., for every open U of $X \times T$, if we denote

by W the open $v_Y^{-1} u_T^{-1} U = u_Z^{-1} v_X^{-1} U$, one has a commutative diagram:

$$
\begin{tikzcd}
\mathcal{O}(U) \arrow[r, "d_T(U)"] \arrow[d, dash, "e_X(U)"'] \arrow[dr, phantom, "(d \times e)(U)"] & \mathcal{O}(u_T^{-1} U) \\
\mathcal{O}(v_X^{-1} U) \arrow[r, "d_Z(v_X^{-1} U)"] & \mathcal{O}(W). \arrow[u, "e_Y(u_T^{-1} U)"']
\end{tikzcd}
$$

If one sets $D = (u, d)$ and $E = (v, e)$, we shall also write $D \times E = (u \times v, d \times e)$.

1.3.

[616] Let $u : Y \to X$ be a morphism of S-schemes. Recall that the adjunction isomorphism

$$\mathrm{Hom}_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y)) \simeq \mathrm{Hom}_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}(\mathcal{O}_X), \mathcal{O}_Y)$$

[616] This paragraph has been expanded with respect to the original; see also N.D.E. (2) in 1.1.1.

associates with every morphism of $p^{-1}(\mathcal{O}_S)$-modules $d : \mathcal{O}_X \to u_*(\mathcal{O}_Y)$ the morphism $d' = \epsilon \circ u^{-1}(d)$, where ϵ is the canonical morphism $u^{-1}u_*(\mathcal{O}_Y) \to \mathcal{O}_Y$.

Let us write $\mathcal{J}_u$ (resp. $\mathcal{I}_u$) for the kernel of the algebra homomorphism $u^\natural : \mathcal{O}_X \to u_*(\mathcal{O}_Y)$ (resp. $u^{\natural\prime} : u^{-1}(\mathcal{O}_X) \to \mathcal{O}_Y$), and let $d : \mathcal{O}_X \to u_*(\mathcal{O}_Y)$ be a morphism of $p^{-1}(\mathcal{O}_S)$-modules. If U is an open of X and $f_0, \cdots, f_n, g \in \mathcal{O}_X(U)$, one sees easily by induction on n that the condition $(*_n)$ is equivalent to the following equality (cf. EGA IV₄, 16.8.8.2):

$$(**_n) \qquad 0 = \sum_{I \subset [\![0,n]\!]} (-1)^{|I|} u^\natural(f_{[\![0,n]\!] - I})\, d(f_I g),$$

where f_I denotes the product of the f_i, for $i \in I$. It follows that if d satisfies $(*_n)$, then d vanishes on the ideal $\mathcal{J}_u^{n+1}$.

Suppose now Y equal to S; then $u : S \to X$ is a section of $p : X \to S$, hence is an immersion (cf. EGA I, 5.3.13). Then, on the one hand, $\epsilon : u^{-1}u_*\mathcal{O}_S \to \mathcal{O}_S$ is an isomorphism, so that $u^{-1}(\mathcal{J}_u) = \mathcal{I}_u$. On the other hand, one has an isomorphism:

$$(\star) u^{-1}(\mathcal{O}_X) \simeq \mathcal{O}_S \bigoplus \mathcal{I}_u.$$

Suppose that d vanishes on $\mathcal{J}_u^{n+1}$. Then $d' = \epsilon \circ u^{-1}(d)$ vanishes on $\mathcal{I}_u^{n+1}$ and hence d' satisfies the analogues $(**'_n)$ and $(*'_n)$ of $(**_n)$ and $(*_n)$, when $f_0, \cdots, f_n \in \mathcal{I}_u(u^{-1}(U))$. Moreover, since $(ada)(\phi) = 0$ for every $a \in \mathcal{O}_S(u^{-1}(U))$ and every morphism of $\mathcal{O}_{u^{-1}(U)}$-modules $\phi : u^{-1}(\mathcal{O}_U) \to \mathcal{O}_{u^{-1}(U)}$, one deduces from $(\star)$ that d' satisfies the analogue $(*'_n)$ of $(*_n)$. It follows that d satisfies $(*_n)$. Consequently, one has obtained:

Lemma. *If $u : S \to X$ is a section of $p : X \to S$, then d is an S-deviation of u of order $\leqslant n$ if and only if d' vanishes on $\mathcal{I}_u^{n+1}$.*

This interpretation may be generalized as follows. Let $u : Y \to X$ be an arbitrary S-morphism and Γ_u the graph of u, i.e., the morphism $Y \to Y \times X$ with components id_Y and u. For every S-deviation d of u of order $\leqslant n$, one obtains by composition

$$Y \xrightarrow{\text{diag.}} Y \times Y \xrightarrow{d_Y} Y \times X \quad (\text{over } u_Y)$$

a Y-deviation of Γ_u of order $\leqslant n$ which we shall denote Γ_d (the graph of d).

Conversely, to every Y-deviation e of Γ_u one associates the composed S-deviation $e_X = pr_2 \circ e$:

$$Y \xrightarrow{e} Y \times X \xrightarrow{pr_2} X \quad (\text{over } \Gamma_u).$$

One sees at once that $(\Gamma_d)_X = d$, and the equality $\Gamma_{e_X} = e$ follows from the fact that e is $\mathcal{O}_Y$- linear.[617] One thus obtains an isomorphism of $\mathcal{O}_Y(Y)$-modules:

$$\{ S\text{-deviations of } u \text{ of order } \leqslant n \} \xrightarrow{\sim} \{ Y\text{-deviations of } \Gamma_u \text{ of order } \leqslant n \}$$
$$d \mapsto \Gamma_d.$$

[617] If λ, f are local sections of $\mathcal{O}_Y$ and $\mathcal{O}_X$, one has $(\Gamma_{e_X})(\lambda \otimes f) = \lambda \cdot e(1 \otimes g)$, and this equals $e(\lambda \otimes g)$ since e is $\mathcal{O}_Y$-linear.

Moreover, one sees easily that d is an S-derivation of u if and only if Γ_d is a Y-derivation of Γ_u.

Let us call I_{Γ_u} the kernel of the algebra homomorphism $(\Gamma_u)^{-1}(O_{Y\times X}) \to O_Y$ corresponding to Γ_u. Taking into account the preceding lemma, one has obtained:

Proposition. *Let $u : Y \to X$ be an S-morphism and $\Gamma_u : Y \to Y \times X$ its graph. The S-deviations of u of order $\leqslant n$ are identified with the Y-deviations of Γ_u of order $\leqslant n$, which are in bijection with*

$$\mathrm{Hom}_{O_Y}((\Gamma_u)^{-1}(O_{Y \times X}) / I_{\Gamma_u}^{n+1}, O_Y).$$

1.3.1.

[618] Let us return to the case where $u : S \to X$ is a section of $p : X \to S$. Then the homomorphism $\phi : u^{-1}(O_X) \to O_S$ admits a section, which we shall denote simply $g \mapsto g \cdot 1$, so that, with the notations of 1.3, one has an isomorphism of O_S-modules:

$$(\star) u^{-1}(O_X) \simeq O_S \oplus I_u,$$

and for every section f of $u^{-1}(O_X)$, $f - \phi(f) \cdot 1$ is a section of I_u.

Let d be an S-deviation of u of order $\leqslant 1$, and d' the O_S-morphism $u^{-1}(O_X) \to O_S$ corresponding to d. If a, b are sections of $u^{-1}(O_X)$, one has:

$$0 = d'((a - \varphi(a) \cdot 1)(b - \varphi(b) \cdot 1)) = d'(ab) - \varphi(a)\, d'(b) - \varphi(b)\, d'(a) + \varphi(ab)\, d'(1).$$

Consequently, one sees that d is an S-derivation of u (cf. 1.2.1 and N.D.E. (2)) if and only if $d'(1) = 0$. One thus obtains:

Lemma. *The S-derivations of u are exactly the S-deviations of u of order 1 which vanish on the unit section of O_X; they correspond to the $O_S(S)$-module*

$$\mathrm{Hom}_{O_S}(I_u/I_u^2, O_S),$$

and one has an isomorphism of $O_S(S)$-modules $D\acute{e}v_{\leqslant 1}(u) \simeq O_S(S) \oplus D\acute{e}r_S(u)$.

Returning to the general case, one deduces, with the notations of 1.3,

Corollary. *Let $u : Y \to X$ be an S-morphism and $\Gamma_u : Y \to Y \times X$ its graph. One has a canonical isomorphism of $O_Y(Y)$-modules*

$$\mathrm{D\acute{e}r}_S(u) \simeq \mathrm{D\acute{e}r}_Y(\Gamma_u) \simeq \mathrm{Hom}_{O_Y}(I_{\Gamma_u} / I_{\Gamma_u}^2, O_Y).$$

[618] This paragraph has been added.

Definition 1.4.

Let X be an S-scheme. We call S-differential operator (resp. S-differential operator of order $\leqslant n$) on X any S-deviation (resp. any S-deviation of order $\leqslant n$) of the identity morphism of X.

According to 1.1, an S-differential operator of order $\leqslant n$ is therefore an endomorphism of $p^{-1}(O_S)$-module of O_X which satisfies the equalities $(*_n)$ of 1.1. We shall denote by $Dif^n_{X/S}$ the $\Gamma(O_S)$-module [619] formed by the S-differential operators of order $\leqslant n$, and by $Dif_{X/S}$ that formed by all the S-differential operators.

As we saw in 1.2, one can compose S-deviations of id_X, which equips $Dif_{X/S}$ with a structure of $\Gamma(O_S)$-algebra; we shall say that this is the *algebra of differential operators of X/S*.

Similarly, for every open V of X, set $\mathcal{D}if_{X/S}(V) = Dif_{V/S} = D\acute{e}v(id_V)$; according to 1.1.3, this defines a sheaf of O_X-modules, called the *sheaf of S-differential operators on X*.[620]

1.4.1.

As we saw in 1.3, one can interpret the differential operators of X/S by means of the graph of the identity morphism of X, i.e., of the diagonal morphism $\Delta = \Delta_{X/S}$ of X into $X \times X$. Let us translate into the present context the statements of 1.3.

Endow $O_{X\times X}$ with the $pr_1^{-1}(O_X)$-algebra structure defined by pr_1, so that $\Delta^{-1}(O_{X \times X})$ is equipped with a structure of algebra over $O_X = \Delta^{-1} pr_1^{-1}(O_X)$. Let $I_{X/S}$ be the kernel of the homomorphism

$$\Delta^{-1}(O_{X \times X}) \xrightarrow{m} O_X$$

adjoint to the homomorphism $O_{X \times X} \to \Delta_*(O_X)$, and let $P^m_{X/S}$ be the O_X-algebra

$$\Delta^{-1}(O_{X \times X}) / I_{X/S}^{m+1}.$$

If V is an affine open of S and U an affine open of X above V, and if one sets

$k = \Gamma(V, O_S)$ and $A = \Gamma(U, O_X)$, one has therefore:

$$\Gamma(U, P^m_{X/S}) = (A \otimes_k A) / I^{m+1},$$

where I is the ideal generated by the elements $a \otimes 1 - 1 \otimes a$, for $a \in A$. This being so, one has according to 1.3 an isomorphism of $O_X(X)$-modules:

$$j_X : Dif^m_{X/S} \xrightarrow{\sim} Hom_{O_X}(P^m_{X/S}, O_X)$$

which one can define as follows: if d belongs to $Dif^m_{X/S}$ and if c is a section of $P^m_{X/S}$ on U of the form $a \otimes b + I^{m+1}$, one has $j_X(d)(c) = a \cdot d(b)$.[621]

[619] In this Exposé, the ring $\Gamma(S, O_S) = O_S(S)$ is denoted $\Gamma(O_S)$.

[620] We have modified the original here, which mentioned the sheaf $U \mapsto Dif_{X_U/U}$, where U ranges over the opens of S; this is the direct image of $\mathcal{D}if_{X/S}$ by the morphism $p_X : X \to S$.

[621] Via this isomorphism, the X-derivations of $\Delta_{X/S}$ correspond, according to 1.3.1, to the S-derivations of id_X, i.e., to the $p^{-1}(O_S)$-derivations of O_X.

1.4.2.

Let d be a differential operator and u a section of X over S. We call *value of d at u* the composed S-deviation

$$S \xrightarrow{u} X \xrightarrow{d} X \text{ (over } id_X).$$

According to 1.3 and 1.4.1, if d is a differential operator of order $\leqslant m$, then du (resp. d) is canonically associated with a morphism of O_S-modules $d' : u^{-1}(O_X)/I_u^{m+1} \to O_S$ (resp. a morphism of O_X-modules $d'' : P^m_{X/S} \to O_X$).

It is clear that one can construct d' from d'' as follows: the square

$$\begin{tikzcd} X \simeq S \times_X X \arrow[r, "u \times X"] \arrow[d, "p"] & X \times X \arrow[d, "pr_1"] \\ S \arrow[r, "u"] & X \end{tikzcd}$$

is cartesian, which allows us to identify X with $S \times_X (X \times X)$, u with $S \times_X \Delta$, hence $u^*(P^m_{X/S})$ with $u^{-1}(O_X)/I_u^{m+1}$. One thus identifies $u^*(d'')$ with a morphism $u^{-1}(O_X)/I_u^{m+1} \to O_S$, which is none other than d'.

1.5.

Set as usual $I_S = \operatorname{Spec} O_S[T]/(T^2)$. Let $\tau : S \to I_S$ be the zero section and σ the canonical deviation of τ which we defined in 1.2.0, i.e. the homomorphism of O_S-modules which vanishes on the unit section of $O_S[T]/(T^2)$ and which sends the class t of T modulo T^2 to the unit section of O_S.

Let X be an S-scheme. To every I_S-automorphism u of $I_S \times X$ inducing the identity on X there is associated by composition a differential operator D_u of X:

$$X \simeq S \times X \xrightarrow{\sigma \times X} I_S \times X \xrightarrow{u} I_S \times X \xrightarrow{pr_2} X.$$

According to II, 3.14, the map $u \mapsto D_u$ is an isomorphism of the $\Gamma(O_S)$-Lie algebra

$$\operatorname{Lie}(\operatorname{Aut} X) := \operatorname{Lie}(\operatorname{Aut} X)(S)$$

onto the $\Gamma(O_S)$-Lie algebra of $p^{-1}(O_S)$-derivations of O_X. The inverse isomorphism associates with every derivation D the automorphism of $I_S \times X$ corresponding to the automorphism $a + bt \mapsto a + (Da + b)t$ of $O_X[T]/(T^2)$.

2. Invariant differential operators on group schemes

2.1.

Let G be an S-group scheme; we denote by ϵ or $\epsilon_G : S \to G$ the unit section of G.

Definition. *Let $U(G)$ be the $\Gamma(O_S)$-module of S-deviations of ϵ_G (or S-deviations of the origin) (cf. 1.1).*

If d and e are two elements of $U(G)$, $d \times e$ is an S-deviation of $\epsilon \times \epsilon : S \simeq S \times S \to G \times G$. The image of $d \times e$ by the multiplication morphism $m : G \times G \to G$ (cf. 1.2) will be called the product of d and e and will be denoted $d \cdot e$.

The $\Gamma(O_S)$-module $U(G)$ is thus equipped with a structure of associative $\Gamma(O_S)$-algebra having ϵ_G as unit element (1.1). We shall say that $U(G)$ is the infinitesimal algebra *of G.*[622]

When T ranges over the schemes above S, the infinitesimal algebra $U(G_T)$ of the T-group $G \times T$ obviously varies contravariantly in T, so that we may speak of the *infinitesimal algebra functor*.

When T ranges over the opens of S, one therefore obtains a presheaf $T \mapsto U(G_T)$ of 0_S-algebras; moreover, according to 1.1.3, this is a sheaf. We shall denote it $\mathcal{U}(G)$ and we shall call it the *sheaf of infinitesimal algebras of G*.

The algebra $U(G)$ is also a covariant functor in G. Indeed, if $u : G \to H$ is a homomorphism of S-groups and d an S-deviation of ϵ_G, the image of d by u is an element $U(u)(d) = ud$ of $U(H)$. The map $U(u) : U(G) \to U(H)$ thus defined is obviously a homomorphism of $\Gamma(O_S)$-algebras. One defines similarly a homomorphism $\mathcal{U}(u)$ from $\mathcal{U}(G)$ to $\mathcal{U}(H)$.

2.2.

Let d be an element of $U(G)$, i.e., an S-deviation of the origin of G. Consider the S-deviation $d \times G$ of $\epsilon \times G : G \simeq S \times G \to G \times G$ obtained from d by base change (1.2.2); the image of $d \times G$ by the multiplication morphism $m : G \times G \to G$ is an S-deviation of $m \circ (\epsilon \times id_G) = id_G$, i.e., an element of $Dif_{G/S}$, which we shall denote d_G.

The map $d \mapsto d_G$ is obviously $\Gamma(O_S)$-linear and the “commutative” diagram below shows that one has $(e \cdot d)_G = d_G \cdot e_G$:[623]

```
                                   m × G
  G × G × G  ───────────────────────────────→  G × G
       △                                        △
       │ G × d × G                              │ G × m
       │ ε × G × G                              │

 G × G                          G × G                      m
```

[622] One now says “the algebra of distributions” (at the origin) of G, cf. [DG70], § II.4, 6.1 and [Ja03], I 7.7.

[623] We have corrected the original, replacing in the diagram $d \times G \times G$ by $G \times d \times G$, so that the composition on the left side of the triangle is $(e \times d) \times G$, and so that the map $d \mapsto d_G$ is an anti-isomorphism of $U(G)$ onto the right-invariant differential operators (cf. 2.3, 2.4 below); on the other hand, by defining ${}_G d$ as the image under m of $G \times d$, one would obtain similarly an isomorphism of $U(G)$ onto the left-invariant differential operators (cf. [DG70], § II.4, Th. 6.5). We have corrected 2.4 and 2.5 accordingly.

```
△  △                                △  △
|  | e × G   m                      |  | d × G   m
|  | ε × G                          |  | ε × G
|                                   |
|          e_G                      |          d_G
G  ──────────────→  G  ──────────────→  G      (over id_G everywhere).
```

The commutativity of the two bottom triangles follows from the definition of d_G and e_G; on the other hand, the composed S-deviation of $e \times G$ and $G \times d \times G$ is $(e \times d) \times G$ (cf. 1.2.2), its image by $m \times G$ is $(e \cdot d) \times G$, and the image of the latter by m is therefore equal to $(e \cdot d)_G$.

One thus obtains an anti-homomorphism $U(G) \to Dif_{G/S}$ of $\Gamma(O_S)$-algebras, called *right translation*.[624]

If $\mathcal{D}if_{G/S}$ denotes the sheaf of S-differential operators on G (cf. 1.4) and p the structural morphism $G \to S$, one defines similarly a "right translation": $\mathcal{U}(G) \to p_*(\mathcal{D}if_{G/S})$.

2.3.

We shall now characterize the differential operators of G over S of the form d_G. Let $g : S \to G$ be a section of the structural morphism of G and g_G the right translation of G by g, i.e., the composed morphism:

```
g_G : G ≃ G × S ──G × g→ G × G ──m→ G.
```

For every differential operator D of G over S, the composition $g_G^{-1} D g_G$ (cf. 1.2) is again an S-deviation of id_G, i.e., an element of $Dif_{X/S}$; we shall denote:

```
D^g = g_G^{-1} D g_G.
```

We shall say that D is *right-invariant* if, for every base change $t : T \to S$ and every section $g : T \to G \times T$, one has $(D_T)^g = D_T$.

Lemma. *For every differential operator D of G over S, the following assertions are equivalent (where m is the multiplication morphism of G):*

(i) D is right-invariant.

(ii) The two following deviations of m are equal: $Dm = m(D \times G)$.

(ii) ⇒ (i): since the condition (ii) is stable under base change, it suffices to show that (ii) entails the equality $D^g = D$ for every section $g : S \to G$. Let h be the morphism $G \times g : G \simeq G \times S \to G \times G$, so that $m \circ h$ is the right translation g_G. The equality $D^g = D$ is equivalent to the equality $g_G \circ D = D \circ g_G$, and this follows from the commutative diagram:

[624] It would be preferable to call this a *left action*. Indeed, let for example d be an S-derivation of the origin; according to 1.2.1, d is the composition of the S-derivation $(\tau, \partial_t) : S \to I_S$ and a morphism $x : I_S \to G$ such that $x \circ \tau = \epsilon$ (i.e. $x \in Lie(G/S)(S)$), and then d_G is the derivation of 0_G which sends a local section ϕ to the section $g \mapsto \partial_t \phi(xg)$. Moreover, with this terminology, one could say: "the left action commutes with right translations".

$$
\begin{tikzcd}
G & G \times G \arrow[l, "m"'] & G \arrow[l, "h"'] \\
G \arrow[u, "{D, id_G}"] & G \times G \arrow[l, "m"'] \arrow[u, "{D \times G, id_{G \times G}}"] & G. \arrow[l, "h"'] \arrow[u, "{D, id_G}"]
\end{tikzcd}
$$

(i) $\Rightarrow$ (ii): take indeed for $t : T \to S$ the structural morphism $p : G \to S$, for section $g : T \to G \times T$ the diagonal morphism $\Delta : G \to G \times G$. The right translation

$\Delta_{G \times G} : G \times G \to G \times G$

is then the morphism from $G \times G$ into $G \times G$ with components m and pr_2. The equality $(D_G)^\Delta = D_G$ is then equivalent to the commutativity of the first square of the following diagram:

$$
\begin{tikzcd}
G \times G \arrow[r, "\Delta_{G \times G}"] & G \times G \arrow[r, "pr_1"] & G \\
G \times G \arrow[r, "\Delta_{G \times G}"] \arrow[u, "{D_G, id_{G \times G}}"] & G \times G \arrow[r, "pr_1"] \arrow[u, "{D_G, id_{G \times G}}"] & G. \arrow[u, "{D, id_G}"]
\end{tikzcd}
$$

The equality (ii) thus follows from the fact that $m = pr_1 \circ \Delta_{G \times G}$.

Consider for example an element d of the infinitesimal algebra $U(G)$. The squares of the diagram

$$
\begin{tikzcd}
G \times G & S \times G \times G \arrow[l] \arrow[r, shift left, "d \times G \times G"] \arrow[r, shift right, "\varepsilon \times G \times G"'] & G \times G \times G \arrow[r, "m \times G"] & G \times G \\
G \arrow[u, "m"] & S \times G \arrow[l] \arrow[r, shift left, "d \times G"] \arrow[r, shift right, "\varepsilon \times G"'] \arrow[u, "S \times m"] & G \times G \arrow[r, "m"] \arrow[u, "G \times m"] & G \arrow[u, "m"]
\end{tikzcd}
$$

are then commutative. Since one has

$m \circ (d \times G) = d_G$ and $(m \times G) \circ (d \times G \times G) = d_G \times G$,

one also has $d_G \circ m = m \circ (d_G \times G)$. Therefore: *for every S-deviation d of the origin, d_G is a right-invariant differential operator.*

2.4. Theorem.

(i) The map $d \mapsto d_G$ is an anti-isomorphism[625] *of the infinitesimal algebra $U(G)$ onto the subalgebra $Dif^G_{G/S}$ of $Dif_{G/S}$ formed by the right-invariant differential operators.*

(ii) Similarly, the map $d \mapsto {}_G d$ is an isomorphism of $U(G)$ onto the subalgebra of $Dif_{G/S}$ formed by the left-invariant differential operators.

Let in fact D be an arbitrary differential operator of G over S and let us denote by D_0 its value at the origin, i.e., the composed deviation $S \xrightarrow{\epsilon} G \xrightarrow{D} G$ (over id_G). The right-invariant differential operator $(D_0)_G$ is then obtained by composition:

$$G \simeq S \times G \xrightarrow{\epsilon \times G} G \times G \xrightarrow{D \times G} G \times G \xrightarrow{m} G \text{ (over } id_{G \times G}).$$

If D is right-invariant, one has $Dm = m(D \times G)$, whence

$$D = D\, m(\epsilon \times G) = m(D \times G)(\epsilon \times G) = (D_0)_G.$$

In particular, the map $d \mapsto d_G$ is surjective.

Conversely, let d be an S-deviation of the origin. One then has a commutative square

$$\begin{tikzcd} G \times G & G \arrow[l, "d \times G"'] \\ G \times S \simeq G \arrow[u, "G \times \epsilon"] & S \arrow[l, "d"'] \arrow[u, "\epsilon"'] \end{tikzcd}$$

whence it follows that $d = m(G \times \epsilon)d = m(d \times G)\epsilon = (d_G)_0$. *A fortiori*, the map $d \mapsto d_G$ is injective. This proves the theorem.

When S varies, Theorem 2.4 obviously implies that the right translation $\mathcal{U}(G) \to p_*(\mathcal{D}if_{G/S})$ is an anti-isomorphism of O_S-algebras of $\mathcal{U}(G)$ onto the sheaf of O_S-algebras $p_*(\mathcal{D}if_{G/S})^G$, which to every open U of S associates $Dif^{G_U}_{G_U/U}$.

2.4.1. Remark.

Consider the commutative diagram

$$\begin{tikzcd} G & G \times G \arrow[l, "\eta"'] \\ S \arrow[u, "p" ', "\epsilon"] & G, \arrow[l, "p"'] \arrow[u, "pr_1", shift left] \arrow[u, "\Delta"', shift right] \end{tikzcd}$$

where η denotes the morphism "$(x, y) \mapsto yx^{-1}$".[626] The latter induces morphisms

[625] We have corrected "isomorphism" to "anti-isomorphism", and added assertion (ii), cf. N.D.E. (17).
[626] i.e., G acts on itself on the left by right translations.

$$\eta' : \eta^{-1}(O_G) \to O_{G \times G} \quad \text{and} \quad \Delta^{-1}(\eta') : p^{-1} \epsilon^{-1}(O_G) \to \Delta^{-1}(O_{G \times G}).$$

For every integer $n \geqslant 1$, set $p^n_{G/S} = \epsilon^{-1}(O_G)/I_\epsilon^{n+1}$ (cf. 1.3 and 1.4 for the notations).[627] Since the square formed by the morphisms ϵ, η, Δ and p is cartesian, $\Delta^{-1}(\eta')$ induces an isomorphism of O_G-modules:

$$p^*(p^n_{G/S}) \xrightarrow{\sim} P^n_{G/S}.$$

The differential operators of G over S of order $\leqslant n$ therefore correspond bijectively to the morphisms of O_G-modules $p^*(p^n_{G/S}) \longrightarrow O_G$, i.e., to the morphisms of O_S-modules

$$p^n_{G/S} \longrightarrow p_*(O_G).$$

In this bijection, the right-invariant differential operators correspond to the composed arrows

$$p^n_{G/S} \longrightarrow O_S \xrightarrow{\text{can.}} p_*(O_G).$$

One thus recovers the isomorphism of Theorem 2.4.

2.5.

[628] Let $Lie(G)$ be the Lie algebra of G;[629] we shall define a morphism of $\Gamma(O_S)$-Lie algebras $\alpha : Lie(G) \to U(G)$.

Let $s : S \to I_S$ be the zero section of $I_S \to S$ and σ the deviation of s defined in 1.2.0. Recall (cf. II, 4.1) that $Lie(G)$ is the set of morphisms $x : I_S \to G$ such that $x \circ s = \epsilon_G$. Then the composition

$$S \xrightarrow{\sigma} I_S \xrightarrow{x} G \quad (\text{over } s)$$

is an S-deviation of ϵ_G, i.e., an element of $U(G)$; with the notations of 1.2 (†), it is denoted σx. Moreover, according to 1.2.1, the map $\alpha : x \mapsto \sigma x$ is an isomorphism of $O_S(S)$-modules from $Lie(G)$ onto the submodule $Dér(\epsilon_G)$ of $U(G)$ formed by the S-derivations of ϵ_G. We shall see that α is a morphism of Lie algebras.[630] Let

$$\rho' : U(G) \longrightarrow Dif_{G/S}$$

be the algebra morphism which to an S-deviation d of ϵ_G associates the left-invariant differential operator ${}_G d \in Dif_{G/S}$, cf. 2.2, N.D.E. (17).

[627] In what follows, we have corrected the original, which referred to the square formed by the morphisms p, p, η, and pr_1, instead of ϵ, η, Δ and p.

[628] In this paragraph, we have modified the order, beginning by defining the map $\alpha : Lie(G) \to U(G)$, and we have corrected the original as indicated in N.D.E. (17).

[629] In this Exposé, if G (resp. X) is an S-group scheme (resp. an S-scheme), the "Lie algebra" $Lie(G)$ (resp. $Lie(\operatorname{Aut} X)$) denotes, with the notations of Exposé II, $Lie(G/S)(S)$ (resp. $Lie(\operatorname{Aut}_S(X)/S)(S)$); it is a $\Gamma(O_S)$-Lie algebra, according to II, 4.11 and 3.14.

[630] See also II, 4.11.

Let $\rho : G \to \mathrm{Aut}_S(G)$ be the homomorphism of group functors which to an S-morphism $g : T \to G$ associates the right translation of G_T by g, i.e. the morphism:

$$G_T \simeq T \times_T G_T \xrightarrow{G_T \times g} G_T \times_T G_T \xrightarrow{m_T} G_T.$$

Recall also (cf. 1.5 and II, 3.14) that $\mathrm{Lie}(\mathrm{Aut}\, G) = \mathrm{Lie}(\mathrm{Aut}_S(G)/S)(S)$ is identified with the infinitesimal automorphisms of G, i.e., with the automorphisms of $I_S \times G$ inducing the identity on G.

Since ρ is a monomorphism, the same holds for the morphism $\mathrm{Lie}(\rho) : \mathrm{Lie}(G/S) \to \mathrm{Lie}(\mathrm{Aut}_S(G)/S)$ (see, for example, Exp. II, N.D.E. (50)), hence $\mathrm{Lie}(\rho) : \mathrm{Lie}(G) \to \mathrm{Lie}(\mathrm{Aut}\, G)$ is injective.

On the other hand, according to 1.5, the map β which to every infinitesimal automorphism u of G associates the differential operator D_u of G:

$$G \simeq S \times G \xrightarrow{\sigma \times G} I_S \times G \xrightarrow{u} I_S \times G \xrightarrow{pr_2} G$$

is an isomorphism of $Lie(\mathrm{Aut}\, G)$ onto the Lie subalgebra of $Dif_{G/S}$ formed by the $p^{-1}(O_S)$-derivations of O_G.

For every $x \in Lie(G)$, one has the commutative square below which determines the image of x by $Lie(\rho)$:

$$\begin{tikzcd}
I_S \times G \arrow[r, "\mathrm{Lie}(\rho)(x)"] \arrow[d, "x \times G"'] & I_S \times G \arrow[d, "pr_2"] \\
G \times G \arrow[r, "m"] & G.
\end{tikzcd}$$

Taking this diagram into account, the image of $Lie(\rho)(x)$ by β is the composed deviation

$$G \simeq S \times G \xrightarrow{\sigma \times G} I_S \times G \xrightarrow{G \times x} G \times G \xrightarrow{m} G \text{ (over } s \times G)$$

which, according to 2.2 N.D.E. (17), is none other than ${}_G(\sigma x) = \rho'(\alpha(x))$. One thus obtains a commutative diagram:

$$\begin{tikzcd}
\mathrm{Lie}(G) \arrow[r, "\mathrm{Lie}(\rho)"] \arrow[d, "\alpha"'] & \mathrm{Lie}(\mathrm{Aut}\, G) \arrow[d, "\beta"] \\
U(G) \arrow[r, "\rho'"] & Dif_{G/S}
\end{tikzcd}$$

where $Lie(\rho)$, β and ρ' are morphisms of Lie algebras. Since ρ' is injective, it follows that α is also a morphism of Lie algebras. Consequently, one has obtained:

Proposition. *α is an isomorphism of $O_S(S)$-Lie algebras, from $Lie(G)$ into the Lie algebra of S-derivations of ϵ_G, itself isomorphic via $Lie(\rho)$ to the Lie algebra of*

S-derivations of G invariant under left translation.[631]

3. Coalgebras and Cartier duality

3.1.

Let S be a scheme (or, more generally, a ringed space). An *$\mathcal{O}_S$-coalgebra*[632] is a pair $(\mathcal{U}, \Delta_{\mathcal{U}})$ consisting of an $\mathcal{O}_S$-module $\mathcal{U}$ and a morphism of $\mathcal{O}_S$-modules $\Delta_{\mathcal{U}} : \mathcal{U} \to \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U}$ (called the *diagonal morphism* or *comultiplication*) such that:

(i) $\sigma \circ \Delta_{\mathcal{U}} = \Delta_{\mathcal{U}}$, where $\sigma(a \otimes b) = b \otimes a$.

(ii) The square

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}}} & \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \\ \Delta_{\mathcal{U}} \downarrow & & \downarrow \mathrm{id}_{\mathcal{U}} \otimes \Delta_{\mathcal{U}} \\ \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} & \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \end{array}$$

is commutative.

(iii) There exists a morphism of $\mathcal{O}_S$-modules $\epsilon_{\mathcal{U}} : \mathcal{U} \to \mathcal{O}_S$, called the *augmentation*, such that the composed morphisms

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes \epsilon_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{O}_S \simeq \mathcal{U}$$
$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \xrightarrow{\epsilon_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{U} \simeq \mathcal{U}$$

are the identity morphism of $\mathcal{U}$.

If $\epsilon_{\mathcal{U}}$ and $\epsilon'_{\mathcal{U}}$ are two augmentations, one has $\epsilon_{\mathcal{U}} = (\epsilon_{\mathcal{U}} \otimes \epsilon'_{\mathcal{U}}) \circ \Delta_{\mathcal{U}} = \epsilon'_{\mathcal{U}}$; the augmentation is therefore uniquely determined by (iii).

If $(\mathcal{U}, \Delta_{\mathcal{U}})$ and $(\mathcal{V}, \Delta_{\mathcal{V}})$ are two $\mathcal{O}_S$-coalgebras, a *morphism* from the first into the second is a morphism of $\mathcal{O}_S$-modules $f : \mathcal{U} \to \mathcal{V}$ such that the diagrams

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{f} & \mathcal{V} \\ \Delta_{\mathcal{U}} \downarrow & & \downarrow \Delta_{\mathcal{V}} \end{array} \qquad \text{and} \qquad \begin{array}{ccc} \mathcal{U} & \xrightarrow{f} & \mathcal{V} \\ \epsilon_{\mathcal{U}} \downarrow & & \downarrow \epsilon_{\mathcal{V}} \end{array}$$

[631] There are examples of Lie algebras $\mathfrak{g}$ over a ring A, such that the map $\mathfrak{g} \to U(\mathfrak{g})$ is not injective, cf. [BLie], § I.2, Ex. 9. The above result shows (since α factors through $Lie(G) \to U(Lie(G)) \to U(G)$) that this cannot happen for "algebraic" Lie algebras, i.e., of the form $Lie(G)$, where G is an A-group scheme.

[632] One also says "cogebra", cf. [BAlg], III § 11.1. On the other hand, one will note that in this Exposé (as well as in VII_B), we place ourselves in the category of cocommutative coalgebras (i.e., those satisfying condition (i)), which is crucial for defining the product and the notion of group coalgebra (cf. 3.1.0 and 3.2).

▼ ▼

□ ⊗ □ —f ⊗ f→ □ ⊗ □ O_S

are commutative. Morphisms of coalgebras compose like morphisms of O_S-modules, so that we shall be able to speak of the *category of O_S-coalgebras*.

3.1.0.

[633] This category has finite products: the final object is the O_S-module O_S, the comultiplication being the identity; the product of two coalgebras (□, Δ_□) and (□, Δ_□) is the tensor product $\mathcal{U} \otimes_{O_S} \mathcal{V}$, the comultiplication being the composed morphism

□ ⊗ □ ——Δ_□ ⊗ Δ_□→ □ ⊗ □ ⊗ □ ⊗ □ ——id_□ ⊗ σ ⊗ id_□→ □ ⊗ □ ⊗ □ ⊗ □

where $\sigma(a \otimes b) = b \otimes a$; the canonical projections of $\mathcal{U} \otimes \mathcal{V}$ onto the factors $\mathcal{U}$ and $\mathcal{V}$ are the morphisms $id_{\mathcal{U}} \otimes \epsilon_{\mathcal{V}}$ and $\epsilon_{\mathcal{U}} \otimes id_{\mathcal{V}}$,[634] and the "diagonal morphism" $\mathcal{U} \to \mathcal{U} \otimes \mathcal{U}$ (corresponding to the pair of morphisms $(id_{\mathcal{U}}, id_{\mathcal{U}})$) is none other than the comultiplication Δ_□.

3.1.1.

Let $\mathcal{A}$ be a commutative O_S-algebra, locally free and of finite type as an O_S-module. If we set

$$\mathcal{A}^* = \mathrm{Hom}_{O_S-Mod.}(\mathcal{A}, O_S),$$

the canonical morphism ϕ from $\mathcal{A}^* \otimes_{O_S} \mathcal{A}^*$ into $(\mathcal{A} \otimes_{O_S} \mathcal{A})^*$ is invertible. If $m : \mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$ is the morphism defining the multiplication of $\mathcal{A}$, one obtains by composition a diagonal morphism

Δ_{□^*} : □^* ——m^*→ (□ ⊗ □)^* ——φ^{-1}→ □^* ⊗ □^*.

This diagonal morphism obviously makes $\mathcal{A}^*$ an O_S-coalgebra whose augmentation is the

transpose morphism of the morphism O_S → □ defined by the unit section of □. Moreover,

it is clear that:

The functor $\mathcal{A} \mapsto \mathcal{A}^*$ is an anti-equivalence of the category of O_S-algebras which are locally free and of finite type as O_S-modules, onto the category of O_S-coalgebras locally free and of finite type as O_S-modules.

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[633]We have added the numbering 3.1.0, for later references.

[634]The following has been added. Let us also recall that, to show that $\mathcal{U} \otimes \mathcal{V}$ is indeed the product of $\mathcal{U}$ and $\mathcal{V}$ in the category of cocommutative O_S-cogebras, one verifies that if one has an arbitrary O_S-cogebra $\mathcal{E}$ and morphisms of cogebras $f : \mathcal{E} \to \mathcal{U}$ and $g : \mathcal{E} \to \mathcal{V}$, then every morphism of cogebras $\phi : \mathcal{E} \to \mathcal{U} \otimes \mathcal{V}$ such that $pr_{\mathcal{U}} \circ \phi = f$ and $pr_{\mathcal{V}} \circ \phi = g$ is necessarily equal to (f ⊗ g) ∘ Δ_□, and this is a morphism of cogebras if and only if it equals (g ⊗ f) ∘ Δ_□.

3.1.2.

To every $\mathcal{O}_S$-coalgebra $\mathcal{U}$ is canonically associated an S-functor

$$\operatorname{Spec}^* \mathcal{U} : (\mathrm{Sch}/S)^\circ \to (\mathrm{Ens}).$$

Note indeed that, for every S-scheme $q : T \to S$, $q^*(\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U})$ is identified with $q^*(\mathcal{U}) \otimes_{\mathcal{O}_T} q^*(\mathcal{U})$, so that $q^*(\Delta_{\mathcal{U}})$ makes $\mathcal{U}_T = q^*(\mathcal{U})$ into an $\mathcal{O}_T$-coalgebra; we can therefore set by definition and with an obvious abuse of notation:[635]

$$(\operatorname{Spec}^* \mathcal{U})(T) = \{ x \in \Gamma(T, \mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1 \text{ and } \Delta_{\mathcal{U}_T}(x) = x \otimes x \}.$$

The sections x of $\mathcal{U}_T$ obviously correspond to the morphisms of $\mathcal{O}_T$-modules $\xi : \mathcal{O}_T \to \mathcal{U}_T$; the conditions $\epsilon(x) = 1$ and $\Delta(x) = x \otimes x$ simply express that ξ is a morphism of coalgebras. One therefore also has:

$$(\operatorname{Spec}^* \mathcal{U})(T) = \operatorname{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{U}_T).$$

In particular, one has the following proposition:[636]

Proposition 3.1.2.1. *Let $\mathcal{A}$ be a commutative $\mathcal{O}_S$-algebra which is locally free of finite type as an $\mathcal{O}_S$-module. Then the S-functor* $\operatorname{Spec}^* \mathcal{A}^*$ *is represented by* $\operatorname{Spec} \mathcal{A}$.

Indeed, for every S-scheme T, one has canonical isomorphisms:

$$(\operatorname{Spec}^* \mathcal{A}^*)(T) = \operatorname{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{A}_T^*) \simeq \operatorname{Hom}_{\mathcal{O}_T\text{-alg.}}(\mathcal{A}_T, \mathcal{O}_T) \simeq (\operatorname{Spec} \mathcal{A})(T).$$

3.2.

An $\mathcal{O}_S$-*coalgebra in groups* (i.e., a group in the category of $\mathcal{O}_S$-coalgebras) consists of the data of an $\mathcal{O}_S$-coalgebra $(\mathcal{U}, \Delta_{\mathcal{U}})$ and three morphisms of $\mathcal{O}_S$-coalgebras $m_{\mathcal{U}} : \mathcal{U} \otimes \mathcal{U} \to \mathcal{U}$, $\eta_{\mathcal{U}} : \mathcal{O}_S \to \mathcal{U}$ and $c_{\mathcal{U}} : \mathcal{U} \to \mathcal{U}$ satisfying the conditions (ii), *(iii)* and (vi) below; on the other hand, the fact that $m_{\mathcal{U}}$ is a morphism of cogebras translates into the commutativity of diagrams (iv) and (v)

below:

$$\text{(iv)}\qquad \begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \\ {\scriptstyle \Delta_{\mathcal{U}} \otimes \Delta_{\mathcal{U}}}\downarrow & & \quad \Delta_{\mathcal{U}} \\ \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} & & \\ \mathrm{id}_{\mathcal{U}} \otimes \sigma \otimes \mathrm{id}_{\mathcal{U}} & & \end{array}$$

---

[635] For every $x \otimes y \in \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \Gamma(T, \mathcal{U}_T)$, its image in $\Gamma(T, \mathcal{U}_T \otimes_{\mathcal{O}_T} \mathcal{U}_T)$ is again denoted $x \otimes y$.

[636] We have added the numbering 3.1.2.1, for later references. Note moreover that the S-functor $\operatorname{Spec}^* \mathcal{U}$ is a sheaf for the Zariski topology (and even for the (fpqc) topology if $\mathcal{U}$ is a quasi-coherent $\mathcal{O}_S$-module).

$$\mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}} \otimes m_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U}$$

(v)
$$\begin{tikzcd}
\mathcal{U} \otimes \mathcal{U} \arrow[rr, "m_{\mathcal{U}}"] \arrow[dr, "\epsilon_{\mathcal{U}} \otimes \epsilon_{\mathcal{U}}"'] & & \mathcal{U} \arrow[dl, "\epsilon_{\mathcal{U}}"] \\
& O_S &
\end{tikzcd}$$

(ii)* The square

$$\begin{tikzcd}
\mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} \arrow[r, "id_{\mathcal{U}} \otimes m_{\mathcal{U}}"] \arrow[d, "m_{\mathcal{U}} \otimes id_{\mathcal{U}}"'] & \mathcal{U} \otimes \mathcal{U} \arrow[d, "m_{\mathcal{U}}"] \\
\mathcal{U} \otimes \mathcal{U} \arrow[r, "m_{\mathcal{U}}"] & \mathcal{U}
\end{tikzcd}$$

is commutative.

(iii)* The two compositions below equal the identity morphism of $\mathcal{U}$:

$$\mathcal{U} \simeq \mathcal{U} \otimes O_S \xrightarrow{id_{\mathcal{U}} \otimes \eta_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}$$
$$\mathcal{U} \simeq O_S \otimes \mathcal{U} \xrightarrow{\eta_{\mathcal{U}} \otimes id_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}.$$

(vi) The composed morphism below is equal to $\eta_{\mathcal{U}} \circ \epsilon_{\mathcal{U}}$:

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{c_{\mathcal{U}} \otimes id_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}.$$

3.2.1.

The morphisms $\eta_{\mathcal{U}}$ and $c_{\mathcal{U}}$ are uniquely determined by $m_{\mathcal{U}}$. On the other hand, conditions (ii)* and (iii)* simply express that $m_{\mathcal{U}}$ makes $\mathcal{U}$ an O_S-algebra having as unit section the image by $\eta_{\mathcal{U}}$ of the unit section of O_S. Condition (iv) also expresses that the diagonal morphism $\Delta_{\mathcal{U}}$ is compatible with multiplication; and indeed, $\Delta_{\mathcal{U}} : \mathcal{U} \to \mathcal{U} \otimes \mathcal{U}$ must be a homomorphism of group coalgebras, which also entails the commutativity of the triangle

(v)*
$$\begin{tikzcd}
& O_S \arrow[dl, "\eta_{\mathcal{U}}"'] \arrow[dr, "\eta_{\mathcal{U}} \otimes \eta_{\mathcal{U}}"] & \\
\mathcal{U} \arrow[rr, "\Delta_{\mathcal{U}}"'] & & \mathcal{U} \otimes \mathcal{U}.
\end{tikzcd}$$

On the other hand, as in every category, the *antipode* $c_{\mathcal{U}}$ is an isomorphism of $\mathcal{U}$ onto the “opposite” group object;[637] in particular, $c_{\mathcal{U}}$ induces an algebra isomorphism of $\mathcal{U}$ onto the opposite algebra $\mathcal{U}^\circ$.

3.2.2.

Since the functor $\mathcal{U} \mapsto \mathrm{Spec}^* \mathcal{U}$ commutes with finite products, it transforms a group coalgebra into a group S-functor; and indeed, for every S-scheme T, the elements $x \in \Gamma(T, \mathcal{U}_T)$ belonging to $(\mathrm{Spec}^* \mathcal{U})(T)$ form a group under the multiplication of the algebra $\Gamma(T, \mathcal{U}_T)$; the inverse of x is none other than $c_{\mathcal{U}}(x)$. According to 3.1.2.1, one has:

Scholium 3.2.2.1.[638] *Let $\mathcal{U}$ be an 0_S-coalgebra in groups, finite and locally free as an 0_S-module. Then the group S-functor* $\mathrm{Spec}^* \mathcal{U}$ *is represented by the S-group, finite and locally free,* $\mathrm{Spec}\, \mathcal{U}^*$.

Remark 3.2.2.2. *Let $\mathcal{L}$ be an 0_S-Lie algebra and $\mathcal{U}(\mathcal{L})$ the enveloping algebra of $\mathcal{L}$, i.e., the sheaf on S associated with the presheaf which to every open V assigns the enveloping algebra $U(\Gamma(V, \mathcal{L}))$ of the Lie algebra $\Gamma(V, \mathcal{L})$.*

Every homomorphism from $\mathcal{L}$ into the Lie algebra underlying an associative 0_S-algebra factors in one and only one way through the canonical morphism from $\mathcal{L}$ into $\mathcal{U}(\mathcal{L})$; moreover, this universal property entails, besides the functoriality of $\mathcal{U}(\mathcal{L})$ in $\mathcal{L}$, that the enveloping algebra of a product of Lie algebras is identified with the tensor product of the enveloping algebras.

In particular, the diagonal morphism $\delta : \mathcal{L} \to \mathcal{L} \times \mathcal{L}$ induces an algebra homomorphism Δ : □(□) → □(□ × □) ≃ □(□) ⊗ □(□). The zero morphism $\mathcal{L} \to 0$ induces a homomorphism $\epsilon : \mathcal{U}(\mathcal{L}) \to \mathcal{U}(0) \simeq O_S$. The isomorphism $x \mapsto -x$ of $\mathcal{L}$ onto the opposite Lie algebra $\mathcal{L}^\circ$ induces an anti-isomorphism c of the algebra $\mathcal{U}(\mathcal{L})$. One then verifies easily that the multiplication m of the algebra $\mathcal{U}(\mathcal{L})$ makes (□(□), Δ) an 0_S-coalgebra in groups with ϵ as augmentation and c as antipode.[639]

3.2.3.

[640] Let $\mathcal{U}$ be an 0_S-coalgebra in groups. We shall see that the group S-functor $G = \mathrm{Spec}^* \mathcal{U}$ is *very good*, in the sense of II, 4.6 and 4.10.

Let M be a free 0_S-module of rank r, and let $T \to S$ be an S-scheme. Since $I_T(M) = \mathrm{Spec}(O_T \bigoplus M_T)$, so that $\pi : I_T(M) \to T$ is affine, one has

π_*(□_{I_T(M)}) = □_T ⊗_{O_T} π_*(O_{I_T(M)}) = □_T ⊗_{O_T} (O_T ⊕ M_T),

and so

(1) Γ(I_T(M), □_{I_T(M)}) ≃ Γ(T, □_T) ⊗_{O(T)} (O(T) ⊕ Γ(T, M_T)).

[637] i.e. endowed with the multiplication $m'_{\mathcal{U}} = m_{\mathcal{U}} \circ \sigma$.
[638] We have added this scholium, implicit in the original.
[639] The group S-functor $\mathrm{Spec}^* \mathcal{U}(\mathcal{L})$ is not representable in general, but one will see later (5.5) that if S is a scheme of characteristic p, if $\mathcal{L}$ is finite locally free over 0_S and if $\mathcal{U}_p(\mathcal{L})$ is its restricted enveloping algebra (cf. 5.3), then $\mathrm{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by a finite and locally free S-group.
[640] We have expanded this paragraph.

Let $(d_1, \cdots, d_r)$ be a basis of M. Then an element $u_0 + \sum_i u_i d_i$ of $\Gamma(I_T(M), \mathcal{U}_{I_T(M)})$ belongs to $G(I_T(M))$ if and only if one has:

$$1 = \varepsilon(u_0 + \sum_i u_i d_i) = \varepsilon(u_0) + \sum_i \varepsilon(u_i) d_i,$$

and

$$(u_0 + \sum_i u_i d_i) \otimes (u_0 + \sum_i u_i d_i) = \Delta(u_0 + \sum_i u_i d_i) = \Delta(u_0) + \sum_i \Delta(u_i) d_i,$$

that is to say:

$$(2) \quad \begin{cases} \varepsilon(u_0) = 1, & \Delta u_0 = u_0 \otimes u_0, \quad (\text{i.e. } u_0 \in G(T)) \\ \varepsilon(u_i) = 0, & \Delta(u_i) = u_i \otimes u_0 + u_0 \otimes u_i, \quad \text{for } i = 1, \dots, r. \end{cases}$$

Moreover, the morphism $G(I_T(M)) \to G(T)$ corresponding to the zero section of $I_T(M) \to T$ is given by $u_0 + \sum_i u_i d_i \mapsto u_0$. From this, combined with (1) and (2), one deduces that, if N is a second free $\mathcal{O}_S$-module of finite rank, the diagram of sets

$$\begin{tikzcd} G(I_T(M \oplus N)) \arrow[r] \arrow[d] & G(I_T(N)) \arrow[d] \\ G(I_T(M)) \arrow[r] & G(T) \end{tikzcd}$$

is cartesian, i.e. G satisfies condition (E) of II, 3.5.

Let us denote by $Prim\Gamma(T, \mathcal{U}_T)$ the sub-$O(T)$-module of $\Gamma(T, \mathcal{U}_T)$ formed by the *primitive elements*, i.e., the elements u which satisfy (with the abuse of notation signaled in 3.1.2):

$$\Delta u = u \otimes 1 + 1 \otimes u, \quad \varepsilon(u) = 0.$$

[641]

Since $(\mathrm{Lie}\ G)(T)$ is the set of elements $u_0 + ud \in G(I_T)$ above the unit element $u_0 = 1$ of $G(T)$, one obtains an isomorphism of $O(T)$-modules, functorial in T:[642]

$$(\mathrm{Lie}\ G)(T) \simeq \mathrm{Prim}\ \Gamma(T, \mathcal{U}_T).$$

On the other hand, one deduces from (1) that

$$\mathrm{Prim}\ \Gamma(I_T(M), \mathcal{U}_{I_T(M)}) \simeq \mathrm{Prim}\ \Gamma(T, \mathcal{U}_T) \otimes_{O(T)} O(I_T(M)),$$

and it follows that the natural morphism of $O(I_T(M))$-modules:

$$(\mathrm{Lie}\ G)(T) \otimes_{O(T)} O(I_T(M)) \to (\mathrm{Lie}\ G)(I_T(M))$$

is an isomorphism, i.e. $\mathrm{Lie}\ G$ is a *good $\mathcal{O}_S$-module* (cf. II, Déf. 4.4).

So G is a good group S-functor (cf. II, Déf. 4.6), and according to II, 4.7.2, $\mathrm{Lie}\ G$ is equipped with an $\mathcal{O}_S$-bilinear "Lie bracket" satisfying the Jacobi identity. It remains to show that G is very good, i.e., that the "bracket" on $(\mathrm{Lie}\ G)(T)$ satisfies $[u, u] = 0$ for every $u \in (LieG)(T)$ (cf. II, 4.10).

[641] Note that the second condition is a consequence of the first, since the first entails that $u = (\mathrm{id} \otimes \varepsilon)\,\Delta(u) = u + \varepsilon(u)$, whence $\epsilon(u) = 0$.

[642] The structure of $\mathcal{O}_S$-module on $\mathrm{Lie}\ G$ is defined in II, Prop. 3.6.

Let u, v be two elements of $(\mathrm{Lie}\ G)(T)$, i.e., two primitive elements of $\Gamma(T, \mathcal{U}_T)$. Set $I = \operatorname{Spec} O_S[d]/(d^2)$ and $I' = \operatorname{Spec} O_S[d']/(d'^2)$. Since the composition law of $G(I \times I')$ is induced by the multiplication of the algebra $\mathcal{U}_{I\times I'}$, one has in $G(I \times I')$ the equality:

$$\begin{aligned}(1 + ud)(1 + vd')(1 + ud)^{-1}(1 + vd')^{-1} &= (1 + ud)(1 + vd')(1 - ud)(1 - vd')\\ &= 1 + (uv - vu)\, dd'\end{aligned}$$

According to the description of the bracket $[u, v]$ given before Prop. 4.8 of Exp. II, one obtains that

$$[u, v] = uv - vu,$$

where the right-hand term is the commutator of u and v in the algebra $\Gamma(T, \mathcal{U}_T)$, whence $[u, u] = 0$. One has thus obtained the following proposition:[643]

Proposition. *Let $\mathcal{U}$ be an O_S-coalgebra in groups. The group S-functor $G = \operatorname{Spec}^* \mathcal{U}$ is very good, and one has an isomorphism $\mathrm{Lie}G \simeq \mathrm{Prim}\mathcal{W}(\mathcal{U})$ of O_S-Lie algebras, where $\mathrm{Prim}\mathcal{W}(\mathcal{U})$ denotes the functor which to every $T \to S$ associates the $O(T)$-Lie algebra formed by the primitive elements of $\mathcal{W}(\mathcal{U})(T) = \Gamma(T, \mathcal{U}_T)$.*

3.3.

Suppose finally that $\mathcal{U}$ is a commutative group coalgebra, i.e., the triangle

$$(\mathrm{i})^* \qquad \begin{tikzcd} \mathcal{U} \otimes \mathcal{U} \arrow[rr, "\sigma"] \arrow[dr, "m_{\mathcal{U}}"'] & & \mathcal{U} \otimes \mathcal{U} \arrow[dl, "m_{\mathcal{U}}"] \\ & \mathcal{U} & \end{tikzcd}$$

is commutative, or in other words that $m_{\mathcal{U}}$ makes $\mathcal{U}$ a commutative O_S-algebra. Conditions (i), (ii), (iii), (iv), (v), (vi), (i), *(ii)*, (iii)* and (v)* then also signify that $\mathcal{U}$ is a cogroup in the category of commutative O_S-algebras. Therefore, if moreover $\mathcal{U}$ is a quasi-coherent O_S-module, then the affine S-scheme $\operatorname{Spec} \mathcal{U}$ is a commutative S-group scheme.

In this case, since the diagonal morphism Δ' of $O_S[T, T^{-1}]$ sends T to $T \otimes T$, the morphisms of S-groups from $\operatorname{Spec} \mathcal{U}$ into $\mathbb{G}_{m,S}$ (I 4.3.2) correspond bijectively to the morphisms of unital O_S-algebras

$$\varphi : O_S[T, T^{-1}] \to \mathcal{U}$$

such that $(\varphi \otimes \varphi) \circ \Delta' = \Delta_{\mathcal{U}} \circ \varphi$ (in this case, $\epsilon_{\mathcal{U}} \circ \phi$ is the neutral element of $\mathbb{G}_{m,S}(S)$, i.e., the augmentation ϵ'). Such a morphism ϕ is determined by the image $\phi(T)$, which must be an invertible element x of $\mathcal{U}$ satisfying $\Delta_{\mathcal{U}}\, x = x \otimes x$ and $\epsilon_{\mathcal{U}}(x) = \epsilon'(T) = 1$. One therefore has:

[643] We have added this proposition, which summarizes the preceding discussion.

$$\mathrm{Hom}_{S\text{-gr.}}(\mathrm{Spec}\,\mathcal{U}, \mathbb{G}_{m,S}) \simeq (\mathrm{Spec}^*\,\mathcal{U})(S)$$

and since this formula remains valid after any base change, this gives:

$$\mathrm{Spec}^*\,\mathcal{U} \simeq \mathrm{Hom}_{S\text{-gr.}}(\mathrm{Spec}\,\mathcal{U}, \mathbb{G}_{m,S}).$$

One has therefore obtained the

Proposition 3.3.0. *If $\mathcal{U}$ is an $\mathcal{O}_S$-coalgebra in commutative groups, quasi-coherent as an $\mathcal{O}_S$-module, then the affine S-scheme $G = \mathrm{Spec}\,\mathcal{U}$ is a commutative S-group scheme, and one has an isomorphism of group S-functors*

$$\mathrm{Spec}^*\,\mathcal{U} \simeq \mathrm{Hom}_{S\text{-gr.}}(G, \mathbb{G}_{m,S}).$$

If one supposes moreover that $\mathcal{U}$ is a locally free $\mathcal{O}_S$-module of finite type, then, according to 3.1.2.1, the group S-functor $\mathrm{Spec}^*\,\mathcal{U}$ is represented by $\mathrm{Spec}\,\mathcal{U}^*$. One thus obtains the

Proposition 3.3.1 (Cartier duality). *The functor*

$$\mathcal{A}(G) \mapsto \mathcal{A}(G)^* = \mathrm{Hom}_{\mathcal{O}_S\text{-Mod.}}(\mathcal{A}(G), \mathcal{O}_S)$$

induces a duality $(*)$ of the category of commutative, finite and locally free S-group schemes; it associates with G the S-group $\mathrm{Hom}_{S-gr.}(G, \mathbb{G}_{m,S})$.

4. "Frobeniuseries"

Let p be a fixed prime number and $(Sch/\mathbb{F}_p)$ the category of schemes of characteristic p, i.e., of schemes above the prime field $\mathbb{F}_p$. Following the general conventions of this Seminar, we identify $(Sch/\mathbb{F}_p)$ with a subcategory of $(Sch/\mathbb{F}_p)^\wedge$ by means of the functor h of I 1.1. We likewise take advantage of the isomorphism from $\mathrm{Hom}(h_X, F)$ onto $F(X)$ defined in I 1.1 to identify these two sets whenever X is an $\mathbb{F}_p$-scheme and F an object of $(Sch/\mathbb{F}_p)^\wedge$.

Notations 4.0.[644] *If T is an $\mathbb{F}_p$-scheme, a T-functor is a morphism $q : F \to T$ of $(Sch/\mathbb{F}_p)^\wedge$ having T as target; for every T-scheme $r : X \to T$, the set of T-morphisms $X \to F$, i.e., of $\mathbb{F}_p$-morphisms $s : X \to F$ such that $q \circ s = r$, will then be denoted $q(r)$, $q(X/T)$, $F(r)$ or $F(X/T)$ (or even $F(X)$ when no confusion will be possible with $\mathrm{Hom}(h_X, F)$).*

4.1.

For every scheme S of characteristic p, we denote by $fr(S)$, or simply fr, the endomorphism of S which induces the identity on the underlying topological space of S and which associates x^p with a section x of $\mathcal{O}_S$ on an open U.

Then the map $fr : S \mapsto fr(S)$ is an endomorphism of the identity functor of $(Sch/\mathbb{F}_p)$,[645] which implies the following results. Let E be an $\mathbb{F}_p$-functor, i.e., an

[644] We have added the numbering 4.0, for later references.

[645] i.e. for every morphism of $\mathbb{F}_p$-schemes $f : Y \to X$, the diagram below is commutative:

f

object of $(Sch/\mathbb{F}_p)^\wedge$; the map which to every $\mathbb{F}_p$-scheme S associates the endomorphism $E(fr(S))$ of $E(S)$ is a functorial endomorphism of E which we shall denote $fr(E)$ or fr; this notation is compatible with the identification of $(Sch/\mathbb{F}_p)$ with a subcategory of $(Sch/\mathbb{F}_p)^\wedge$. Moreover, the map $E \mapsto fr(E)$ is an endomorphism of the identity functor of $(Sch/\mathbb{F}_p)^\wedge$ (which we shall again denote fr).[646]

For every $\mathbb{F}_p$-scheme S and every S-functor $q : X \to S$, we denote by $X^{(p/S)}$ or $X^{(p)}$ the inverse image of X by the base change $fr(S)$:

$$\begin{tikzcd}
X^{(p/S)} \arrow[r, "pr_X"] \arrow[d] & X \arrow[d, "q"] \\
S \arrow[r, "fr(S)"] & S.
\end{tikzcd}$$

The commutative square

$$\begin{tikzcd}
X \arrow[r, "fr(X)"] \arrow[d, "q"'] & X \arrow[d, "q"] \\
S \arrow[r, "fr(S)"] & S
\end{tikzcd}$$

induces an S-morphism, denoted $Fr(X/S)$ (or simply Fr), from X into $X^{(p/S)}$ such that $fr(X) = pr_X \circ Fr(X/S)$:

$$\begin{tikzcd}
X \arrow[dr, "Fr(X/S)"] \arrow[dd, "q"'] & & \\
& X^{(p/S)} \arrow[r, "pr_X"', "fr(X)"] \arrow[d, "q^{(p/S)}"] & X \arrow[d, "q"] \\
S & S \arrow[r, "fr(S)"] & S.
\end{tikzcd}$$

We shall say that $Fr(X/S)$ is the *Frobenius morphism of X relative to S*; it is clear that the map $Fr : X \mapsto Fr(X/S)$ is a functorial homomorphism.

$$\begin{tikzcd}
Y \arrow[r] \arrow[d, "fr(Y)"'] & X \arrow[d, "fr(X)"] \\
Y \arrow[r, "f"] & X.
\end{tikzcd}$$

[646]One says that $fr(X)$ is the "absolute" Frobenius morphism of X, to distinguish it from the "relative" Frobenius morphism $Fr(X/S)$ introduced below.

[647] Let $r : T \to S$ be an S-scheme. For every $\phi \in X(r) = \mathrm{Hom}_S(T, X)$ (cf. 4.0), one has a commutative diagram:

$$\begin{tikzcd} X \arrow[r, "Fr(X/S)"] \arrow[dr, dash, "\varphi", "r"'] & X^{(p/S)} \arrow[r, "pr_X"] \arrow[d, "q^{(p/S)}"] & X \arrow[d, "q"] \\ T \arrow[r] & S \arrow[r, "fr(S)"] & S. \end{tikzcd}$$

According to the definition of $X^{(p/S)}$ as fibered product, pr_X induces a bijection:

$$X^{(p/S)}(r) = \mathrm{Hom}_S(T, X^{(p/S)}) \xrightarrow{\sim} \mathrm{Hom}_S(T, X) = X(fr(S) \circ r).$$

On the other hand, $r \circ fr(T) = fr(S) \circ r$, since $\mathtt{fr}$ is an endomorphism of the identity functor. It follows that the map $Fr(X/S)(r) : X(r) \to X^{(p/S)}(r)$ can be characterized by the commutativity of the following square:

$$\begin{tikzcd} X(r) \arrow[r, "Fr(X/S)(r)"] \arrow[d, "X(fr(T))"'] & X^{(p/S)}(r) \arrow[d, "\sim"] \\ X(r \circ fr(T)) \arrow[r, equal] & X(fr(S) \circ r). \end{tikzcd} \qquad (\dagger)$$

For example, if X is the subscheme of S defined by a quasi-coherent ideal $\mathcal{I}$, then $X^{(p)}$ is the subscheme of S defined by the ideal $\mathcal{I}^p$ generated by the p-th powers of the sections of $\mathcal{I}$; moreover, $Fr(X/S)$ is then the canonical immersion of $\mathrm{Spec}(\mathcal{O}_X/\mathcal{I})$ into $\mathrm{Spec}(\mathcal{O}/\mathcal{I}^p)$.

4.1.1.

[648] Let $t : T \to S$ be a base change and $X_T = X \times_{q,t} T$. Consider the inverse image of X_T by $fr(T)$:

$$\begin{tikzcd} (X_T)^{(p/T)} \arrow[r] & X_T \arrow[r] & X \arrow[d, "q"] \\ T \arrow[r, "fr(T)"] & T \arrow[r, "t"] & S. \end{tikzcd}$$

Since $t \circ fr(T) = fr(S) \circ t$, then $(X_T)^{(p/T)}$ is identified with the inverse image of $X^{(p/S)}$ by t; in other words, one has a canonical isomorphism:

$$X_T^{(p/T)} \xrightarrow{\sim} (X^{(p/S)})_T.$$

[647] We have expanded the original in what follows.
[648] We have expanded the original in what follows.

It is clear that, in this identification, $Fr(X_T/T)$ is identified with the inverse image $Fr(X/S)_T$ of $Fr(X/S)$.

4.1.1.1. In particular, if S is the spectrum of the prime field $\mathbb{F}_p$, $X^{(p/S)}$ is equal to X and $Fr(X/S)$ to $fr(X)$. Consequently, $X_T^{(p/T)}$ is identified with X_T and $Fr(X_T/T)$ with $fr(X)_T$.

For example, if E is a set and E_T the constant T-scheme of type E, one has $E_T^{(p/T)} \simeq E_T$ and $Fr(E_T/T) \simeq id_{E_T}$.

4.1.2.

The functor $X \mapsto X^{(p/S)}$ obviously commutes with products; it therefore transforms an S-group G into an S-group $G^{(p/S)}$; moreover, since Fr is a functorial homomorphism, then

$$Fr(G/S) : G \longrightarrow G^{(p/S)}$$

is a homomorphism of S-groups. We shall denote Fr G its kernel.

If $r : T \to S$ is a scheme above S, it follows from the diagram (†) of 4.1 that the value of Fr G at r is the kernel of the homomorphism

G(fr(T)) : G(r) ▯ G(r ∘ fr(T)).

Now, when T is the scheme I_R of dual numbers over an S-scheme R, $fr(I_R)$ factors as follows:

$I_R \xrightarrow{can.} R \xrightarrow{fr(R)} R \xrightarrow{s} I_R$,

where s is the zero section. It follows that $(FrG)(I_R)$ contains the kernel $Lie(G/S)(R)$ of the morphism $G(s) : G(I_R) \to G(R)$, and that one therefore has: $Lie(G/S) = Lie(FrG/S)$.

4.1.3.

More generally, for every S-functor X, we define the S-functor $X^{(p^n)}$ by recursion on n by means of the formulas:

$X^{(p)} = X^{(p/S)}$ and $X^{(p^n)} = (X^{(p^{n-1})})^{(p)}$.

Similarly, $Fr^n(X/S)$ or Fr^n denote the composed functorial homomorphism

$X \xrightarrow{Fr(X/S)} X^{(p)} \xrightarrow{Fr(X^{(p)}/S)} X^{(p^2)} \longrightarrow \cdots \longrightarrow X^{(p^{n-1})} \xrightarrow{Fr(X^{(p^{n-1})}/S)} X^{(p$

One will note that, according to 4.1.1, $Fr(X^{(p)}/S)$ coincides with $Fr(X/S)^{(p)}$, i.e., the following diagram is commutative:

$X^{(p)} \longrightarrow X$

$Fr(X^{(p)}/S) \qquad Fr(X/S)$

$$X^{(p^2)} \longrightarrow X^{(p)}.$$

If G is a group S-functor, $G^{(p^n)}$ is also one and $Fr^n(G/S)$ is a homomorphism of group S-functors.

Definition. *We shall denote by $Fr^n G$ the kernel of $Fr^n(G/S)$ and we shall say that G is of height $\leqslant n$ if $Fr^n(G/S)$ is zero, i.e., if $Fr^n G = G$.*

Lemma. *The group subfunctor $Fr^n G$ of G is characteristic, i.e., for every S-scheme T, every endomorphism ϕ of the group T-functor G_T induces an endomorphism of $(Fr^n G)_T$.*

Indeed, since the construction of $G^{(p^n)}$ and of $Fr^n(G/S)$ commutes with base changes according to 4.1.1, one may suppose $T = S$; in this case, the assertion follows from the fact that $Fr^n(G/S)$ is a functorial homomorphism.

4.1.4. Examples.

a) Consider first an "abstract" abelian group M and the diagonalizable group $G = D_S(M)$ of type M (I 4.4): for every S-scheme T, $G(T)$ is therefore the abelian group $\mathrm{Hom}_{(Ab)}(M, \Gamma(T, O_T)^\times)$. Since G is the inverse image of the diagonalizable group $D(M)$ over $\mathbb{F}_p$, $G^{(p)}$ is identified with G and $Fr(G/S)(T)$ is identified with the endomorphism $x \mapsto x^p$ of $G(T)$ (4.1.1). In particular, when M is equal to $\mathbb{Z}$, one has $D_S(M) = \mathbb{G}_{m,S}$, so that:

   $Fr\mathbb{G}_{m,S}$ is the S-group $\mu_{p,S}$ which to every S-scheme T associates the group of p-th roots of unity in $\Gamma(T, O_T)^*$.

b) Consider now a scheme S of characteristic p and a sheaf of modules $\mathcal{E}$ on S. According to I 4.6.2, one has a canonical isomorphism

$$\mathcal{W}(\mathcal{E})^{(p)} \simeq \mathcal{W}(\mathcal{E}^{(p)}),$$

where $\mathcal{E}^{(p)}$ is the inverse image of $\mathcal{E}$ by $fr(S)$. For every S-scheme $\pi : T \to S$, the map $Fr(\mathcal{W}(\mathcal{E}))(\pi)$ is determined, according to 4.1 (†), by the commutative triangle

$$\begin{tikzcd}
\Gamma(T, \pi^* fr(S)^* \mathcal{E}) \arrow[rr, "\text{can.}", "\sim"'] & & \Gamma(T, fr(T)^* \pi^* \mathcal{E}) \\
& \Gamma(T, \pi^* \mathcal{E}), \arrow[ul, "Fr(\mathcal{W}(\mathcal{E})/S)(\pi)"] \arrow[ur, "f'"'] &
\end{tikzcd}$$

where f' is the map induced by $fr(T)$.

In particular, if $\mathcal{E}$ is equal to O_S, $\mathcal{W}(\mathcal{E})$ is identified with the additive group $\mathbb{G}_{a,S}$. In this case, one has $\mathcal{E}^{(p)} = \mathcal{E} = O_S$ and the Frobenius morphism $Fr(\mathbb{G}_{a,S}/S)$ sends $x \in \Gamma(T, O_T)$ to x^p. So:

$Fr\mathbb{G}_{a,S}$ is the S-group $\alpha_{p,S}$ which to every S-scheme T associates the group: $x \in \Gamma(T, O_T) | x^p = 0$.

c) One would see similarly that, for every quasi-coherent 0_S-algebra $\mathcal{A}$, $(\operatorname{Spec} \mathcal{A})^{(p)}$ is identified with the spectrum $\operatorname{Spec} \mathcal{A}^{(p)}$ of the inverse image of $\mathcal{A}$ by $fr(S)$. If π denotes the endomorphism $x \mapsto x^p$ of the sheaf of rings 0_S, one has therefore

□^{(p)} = □ ⊗_π 0_S   [^N.D.E-VII_A-44]

and $Fr((\operatorname{Spec} \mathcal{A})/S)$ is induced by the morphism of 0_S-algebras $\mathcal{A} \otimes_\pi O_S \to \mathcal{A}$ defined by $a \otimes_\pi x \mapsto a^p x$.

For every quasi-coherent 0_S-module $\mathcal{E}$, finally, one has canonical isomorphisms

□(□)^{(p)} ≃ □(□^{(p)})   and   □(□)^{(p)} ≃ □(□^{(p)}),

where $\mathcal{S}(\mathcal{E})$ denotes the symmetric algebra of the 0_S-module $\mathcal{E}$.

d) Let $\mathcal{U}$ be an 0_S-coalgebra (3.1) and T a scheme of characteristic p. If $\mathcal{U}^{(p/S)}$ or $\mathcal{U}^{(p)}$ denote the inverse image of the coalgebra $\mathcal{U}$ by $fr(S)$, one has as in b) a canonical isomorphism:

(Spec^* □)^{(p)} ≃ Spec^* □^{(p)}.

If $\mathcal{U}$ is a coalgebra in groups, the value of $Fr(\operatorname{Spec}^* \mathcal{U})$, i.e., of the kernel of the Frobenius morphism $\operatorname{Spec}^* \mathcal{U} \to (\operatorname{Spec}^* \mathcal{U})^{(p)}$, for an S-scheme T is therefore the set of elements γ of

(Spec^* □)(T) = { x ∈ Γ(T, □_T) | ε_{□_T}(x) = 1, Δ_{□_T} x = x ⊗ x }

such that the image in $\Gamma(T, \mathcal{U}_T \otimes_{fr(T)} O_T)$ of the element $\gamma \otimes_{fr(T)} 1$ of $\Gamma(T, \mathcal{U}_T) \otimes_{fr(T)} O(T)$ is equal to 1.

4.2.

[649] We shall now occupy ourselves with a construction close to the preceding one: let S be a scheme of characteristic p, X an S-scheme and X^{ps} the product in the category (Sch/S) of p copies of X.

We then denote by $U_p(X)$ the open subscheme of X^{ps} which is the union of the products U^{ps}, when U ranges over the affine opens of X. A point x of X^{ps} therefore belongs to $U_p(X)$ if and only if the projections $pr_i x$ of x onto the factors of X^{ps} belong to a common affine open of X. For example, if every finite part of X is contained in an affine open, one has $U_p(X) = X^{ps}$.

The symmetric group $\mathfrak{S}_p$ of order p operates on X^{ps} by permutation of factors and leaves stable the open $U_p(X)$. We shall call the *p-fold symmetric product of* X and we shall denote $\Sigma^p X$ the quotient of X^{ps} by $\mathfrak{S}_p$ in the category of ringed spaces. Let $q(X)$, or simply q, be the canonical projection $X^{ps} \to \Sigma^p X$.

[649] For the content of nos 4.2 and 4.3, one may also consult [DG70], § IV.3, nos 4–6.

Then q maps $U_p(X)$ onto an open $V_p(X)$ of the symmetric product, which one may describe as follows (cf. V 4.1). The structure sheaf of $\Sigma^p X$ induces on $V_p(X)$ a scheme structure; the morphism $q'(X) : U_p(X) \to V_p(X)$ induced by $q(X)$ is affine and even integral; when U ranges over the affine opens of X which project into an affine open V of S varying, the $\Sigma^p U$ form an affine covering of $V_p(X)$; if R denotes the affine algebra of V and A that of U, $\Sigma^p U$ has as affine algebra the subalgebra $\Sigma^p A$ of $\bigotimes_R^p A$ formed by the symmetric tensors.

Consider now the diagonal morphism δ of X into $U_p(X)$. If $V = \operatorname{Spec} R$ is an affine open of S and $U = \operatorname{Spec} A$ an affine open of X above V, the restriction of δ to U is defined by the algebra morphism

$$\eta : \bigotimes\nolimits^p_R A \to A, \qquad a_1 \otimes \cdots \otimes a_p \mapsto a_1 a_2 \cdots a_p.$$

One therefore has, if N is the symmetrization operator:

$$\eta(N(a_1 \otimes \cdots \otimes a_p)) = \eta\Big(\sum_{\sigma \in \mathfrak{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}\Big) = p!\, a_1 \cdots a_p = 0.$$

In other words, η vanishes on the subspace $N(\bigotimes_R^p A)$ of $\Sigma^p A$ formed by the symmetrized tensors. Moreover, if f is a symmetric tensor, one has obviously $N(fa) = fN(a)$, which shows that $N(\bigotimes_R^p A)$ is an ideal of $\Sigma^p A$. We shall henceforth denote

$$U^{[p/S]} = \operatorname{Spec}\big(\Sigma^p A / N(\bigotimes\nolimits^p_R A)\big);$$

it is a closed subscheme of $\Sigma^p(U) = V_p(U)$. The union of the $U^{[p/S]}$, when U ranges over the affine opens of X which project into a varying affine open V of S, is a closed subscheme of $V_p(X)$, denoted $X^{[p/S]}$.

Moreover, if $i(X)$ denotes the inclusion of $X^{[p/S]}$ into $V_p(X)$, we have just seen that $q'(X) \circ \delta$ factors through $X^{[p/S]}$, whence a morphism $F^{[p]}(X/S) : X \to X^{[p/S]}$:[650]

$$\begin{tikzcd}
X^p_S \arrow[d, "q(X)"'] & \supset & U_p(X) \arrow[d, "q'(X)"] & & X \arrow[ll, "\delta(X)"'] \arrow[d, "F^{[p]}(X/S)"] \\
\Sigma^p(X) & \supset & V_p(X) & & X^{[p/S]}. \arrow[ll, "i(X)"']
\end{tikzcd}$$

It is clear that $X^{[p/S]}$ is functorial in X and that the map $F^{[p]} : X \mapsto F^{[p]}(X/S)$ is a functorial homomorphism.

4.2.1.

The schemes $X^{[p/S]}$ and $X^{(p/S)}$ are obviously related: let V be an affine open of S with affine ring R and U an affine open of X above V; let A be the affine algebra of U. If π denotes the endomorphism $x \mapsto x^p$ of R, then $U^{(p/S)}$ has $A \otimes_\pi R$ as affine algebra. One verifies moreover that the map

$$a \otimes_\pi \lambda \mapsto \big(\lambda a \otimes \cdots \otimes a \mod N(\bigotimes\nolimits^p_R A)\big)$$

[650] In the original, this morphism (resp. the relative Frobenius morphism) was denoted F' (resp. F).

defines a morphism of R-algebras from $A \otimes_\pi R$ into $\Sigma^p A/N(\otimes^p_R A)$, and the latter induces a morphism $\phi(U) : U^{[p/S]} \to U^{(p/S)}$ such that $\phi(U) \circ F^{[p]}(U/S) = Fr(U/S)$.

"Gluing the pieces", one then obtains a commutative triangle

$$\begin{tikzcd} & X \arrow[dl, "F^{[p]}(X/S)"'] \arrow[dr, "Fr(X/S)"] & \\ X^{[p/S]} \arrow[rr, "\phi(X)"] & & X^{(p/S)}. \end{tikzcd}$$

For example, if X is the subscheme of S defined by a quasi-coherent ideal $\mathcal{I}$, $F^{[p]}(X/S)$ is identified with the identity morphism of X, so that $\phi(X)$ is the canonical immersion of $\mathrm{Spec}(O_S/\mathcal{I})$ into $\mathrm{Spec}(O_S/\mathcal{I}^p)$. One thus sees that $\phi(X)$ is not an isomorphism in general.

However, when M is a free R-module, it is clear that the map

$$M \otimes_\pi R \longrightarrow \Sigma^p M / N(\otimes^p_R M), \qquad m \otimes_\pi \lambda \mapsto (\lambda\, m \otimes \cdots \otimes m \mod N(\otimes^p_R M))$$

is bijective; this map therefore remains bijective when M is R-flat, because every flat module is a filtered direct limit of free modules (Lazard[651] [652]). It follows that

$\phi(X) : X^{[p/S]} \to X^{(p/S)}$ is an isomorphism if X is a flat S-scheme.

4.3.

Consider finally an S-group scheme G in abelian groups. Then the composed morphism $\mu(G) : U_p(G) \hookrightarrow G^{ps} \to G$, which is defined by the multiplication, factors through $V_p(G)$, i.e., there exists a morphism $\nu(G) : V_p(G) \to G$ such that $\nu(G) \circ q'(G) = \mu(G)$, so that one has the following commutative diagram:

$$\begin{tikzcd} G & U_p(G) \arrow[l, "\mu(G)"'] \arrow[d, "q'(G)"] & G \arrow[l, "\delta(G)"'] \arrow[d, "F^{[p]}(G/S)"] \arrow[dd, bend left=60, "Fr(G/S)"] \\ & V_p(G) \arrow[ul, "\nu(G)"'] & G^{[p/S]} \arrow[l, "i(G)"] \arrow[d, "\phi(G)"] \\ & & G^{(p/S)}. \arrow[uull, bend left=30, "Ver(G/S)"] \end{tikzcd}$$

[651] D. Lazard, C. R. Acad. Sc. Paris 258, 1964, p. 6313–6316.

[652] See also: D. Lazard, Bull. Soc. Math. France 97 (1969), 81–128, or: [BAlg], X § 1.6, Th. 1.

When G is S-flat, $\phi(G)$ is an isomorphism and one can define a morphism (called the *Verschiebung*)

$$Ver(G/S) : G^{(p/S)} \longrightarrow G$$

by means of the formula $Ver(G/S) = \nu(G) \circ i(G) \circ \phi(G)^{-1}$. When G ranges over S-flat group schemes in abelian groups, the map $Ver : G \mapsto Ver(G/S)$ is obviously a functorial homomorphism; consequently, $Ver(G/S)$ is a group homomorphism. For every S-scheme T finally, the composed map

$$G(T) \xrightarrow{\delta(G)(T)} U_p(G)(T) \xrightarrow{\mu(G)(T)} G(T)$$

sends $x \in G(T)$ to $p \cdot x$. We may write $p \cdot id_G$ instead of $\mu(G) \circ \delta(G)$, thus obtaining the classical formula:

$$(*) \qquad Ver(G/S) \circ Fr(G/S) = p \cdot id_G.$$

Examples 4.3.1.

(a) When G is a constant S-scheme in abelian groups, we know that $Fr(G/S)$ is identified with the identity morphism of G (cf. 4.1.1.1). One therefore has $Ver(G/S) = pid_G$.

(b) When G is the diagonalizable S-group of type M, $Fr(G/S)$ is equal to pid_G according to 4.1.4 (a); one then sees easily that $Ver(G/S)$ is the identity morphism of G.

(c) When $\mathcal{E}$ is a flat $\mathcal{O}_S$-module and G is the S-group $\mathcal{V}(\mathcal{E})$, the morphism $Ver(G/S)$ is zero, as is $p{\cdot}id_G$. One will see in Exposé VII_B that a commutative algebraic group G over a field k is "unipotent" if and only if the composed homomorphism

$$G^{(p^n)} \xrightarrow{Ver(G^{(p^{n-1})}/S)} G^{(p^{n-1})} \longrightarrow \cdots \longrightarrow G^{(p)} \xrightarrow{Ver(G/S)} G$$

is zero for some n (one has set $G^{(p^n)} = (G^{(p^{n-1})})^{(p)}$, cf. 4.1.3).[653]

4.3.2.

Since the map $Ver : G \mapsto Ver(G/S)$ is a functorial homomorphism when G ranges over S-flat group schemes in commutative groups, the square

$$\begin{tikzcd}
G^{(p)} \arrow[r, "Ver(G/S)"] \arrow[d, "Fr(G/S)^{(p)}"'] & G \arrow[d, "Fr(G/S)"] \\
G^{(p^2)} \arrow[r, "Ver(G^{(p)}/S)"] & G^{(p)}
\end{tikzcd}$$

[653] Since this does not appear explicitly in VII_B, one refers to [DG70], § IV.3, Prop. 4.11.

is commutative, where $Fr(G/S)^{(p)}$ denotes the inverse image of $Fr(G/S)$ by the base change $fr(S)$. According to 4.1.1, one has $Fr(G/S)^{(p)} = Fr(G^{(p)}/S)$, so, according to 4.3 $(*)$ applied to $G^{(p)}$, one obtains:

$$(**)\qquad Fr(G/S) \circ Ver(G/S) = Ver(G^{(p)}/S) \circ Fr(G^{(p)}/S) = p \cdot \mathrm{id}_{G^{(p)}}.$$

4.3.3.

Suppose finally that G is a commutative, finite and locally free S-group; let $\mathcal{A}$ be the $\mathcal{O}_S$-affine algebra of G and π the endomorphism of the sheaf of rings $\mathcal{O}_S$ which sends a section x of $\mathcal{O}_S$ to x^p.[654] We denote by $\Sigma^p\mathcal{A}$ the subalgebra of $\bigotimes^p_{\mathcal{O}_S}\mathcal{A}$ formed by the sections invariant under the action of the symmetric group, by $i(\mathcal{A})$ the inclusion of $\Sigma^p\mathcal{A}$ into the tensor product. Let $\Delta_p(\mathcal{A}) : \mathcal{A} \to \bigotimes^p_{\mathcal{O}_S}\mathcal{A}$ be the morphism obtained by iterating the diagonal morphism of the coalgebra $\mathcal{A}$ (it corresponds to the morphism of multiplication $U_p(G) = G^{ps} \to G$); according to the beginning of paragraph 4.3, $\Delta_p(\mathcal{A})$ factors through $\Sigma^p\mathcal{A}$, i.e., it induces a morphism

$$a(\mathcal{A}) : \mathcal{A} \to \Sigma^p\mathcal{A}$$

such that $i(\mathcal{A}) \circ a(\mathcal{A}) = \Delta_p(\mathcal{A})$.

On the other hand, let $S^p(\mathcal{A})$ be the degree-p component of the symmetric algebra of $\mathcal{A}$ and $q(\mathcal{A}) : \bigotimes^p_{\mathcal{O}_S}\mathcal{A} \to S^p(\mathcal{A})$ the canonical projection. The multiplication $m_p(\mathcal{A}) : \bigotimes^p_{\mathcal{O}_S}\mathcal{A} \to \mathcal{A}$ factors through $S^p(\mathcal{A})$, i.e., it induces a map

$$b(\mathcal{A}) : S^p(\mathcal{A}) \longrightarrow \mathcal{A}$$

such that $b(\mathcal{A}) \circ q(\mathcal{A}) = m_p(\mathcal{A})$.

Since $\Sigma^p\mathcal{A}$ is the affine algebra of $V_p(\mathcal{A})$, then, according to the beginning of 4.3 again, the composed morphism $i(G) \circ \phi(G)^{-1}$ induces an algebra homomorphism

$$r(\mathcal{A}) : \Sigma^p\mathcal{A} \to \mathcal{A} \otimes_\pi \mathcal{O}_S;$$

this homomorphism vanishes on the sections of the form

$$\sum_{\sigma \in \mathfrak{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}$$

and sends $a \otimes \cdots \otimes a$ to $a \otimes_\pi 1$. Similarly, $j(\mathcal{A})$ is the morphism of $\mathcal{O}_S$-modules $a \otimes_\pi 1 \mapsto q(a \otimes \cdots \otimes a)$. One thus obtains the commutative diagram:

$$(*)\qquad \begin{tikzcd}
\mathcal{A} \arrow[r, "\Delta_p(\mathcal{A})"] \arrow[dr, "a(\mathcal{A})"'] & \bigotimes^p_{\mathcal{O}_S}\mathcal{A} \arrow[r, "m_p(\mathcal{A})"] \arrow[d, shift left, "q(\mathcal{A})"] & \mathcal{A} \\
 & \Sigma^p\mathcal{A} \arrow[u, shift left, "i(\mathcal{A})"] \quad S^p(\mathcal{A}) \arrow[ur, "b(\mathcal{A})"'] &
\end{tikzcd}$$

[654] We have modified the order, by first introducing the objects appearing in the diagram that follows.

```
\ r(□)                              j(□) /
 \                                      /
    □ ⊗_π O_S.
```

The composition $r(\mathcal{A}) \circ a(\mathcal{A})$ is associated with the Verschiebung morphism $Ver(G/S)$, while $b(\mathcal{A}) \circ j(\mathcal{A})$ is associated with the Frobenius morphism $Fr(G/S)$.

The commutative diagram $(\mathcal{A})$ above is self-dual; let D indeed be the functor which to every O_S-module $\mathcal{M}$ associates the dual O_S-module $\mathrm{Hom}_{O_S}(\mathcal{M}, O_S)$; it is clear that the image of the diagram $(\mathcal{A})$ by the functor D is none other than the diagram $(D\mathcal{A})$, the morphisms $Dr(\mathcal{A})$, $Da(\mathcal{A})$, $Dj(\mathcal{A})$ and $Db(\mathcal{A})$ being identified respectively with $j(D\mathcal{A})$, $b(D\mathcal{A})$, $r(D\mathcal{A})$ and $a(D\mathcal{A})$. According to 3.3.1, one therefore sees that:

> In the category of commutative, finite and locally free S-groups, Cartier duality interchanges the Frobenius morphism and the Verschiebung.[655]

5. Restricted p-Lie algebras

Let us first recall some results from the *Séminaire Sophus Lie*.[656]

5.1.

Let p be a prime number, R a commutative ring of characteristic p and A an associative R-algebra, but not necessarily commutative. If a and b are two elements of A, we set $[a, b] = ab - ba$ and $a \cdot b = L_a(b) = R_b(a)$. One then has:

$$(\mathrm{ad}\, x^p)(y) = [x^p, y] = (L_x^p - R_x^p)(y) = (L_x - R_x)^p(y) = (\mathrm{ad}\, x)^p(y)$$

whence Jacobson's first formula:

$$\text{(i)} \qquad \mathrm{ad}(x^p) = (\mathrm{ad}\, x)^p.$$

If $a_1, \cdots, a_p$ are p arbitrary elements of A, then, denoting by N the symmetrization operator (cf. 4.2), one has the equalities:

$$(*) \qquad N(a_1 \otimes \cdots \otimes a_p) = \sum_\sigma a_{\sigma(1)} \cdots a_{\sigma(p)} = \sum_\tau [a_{\tau(1)} [a_{\tau(2)} [\cdots [a_{\tau(p-1)}, a_p$$

where σ ranges over the permutations of p letters and τ over those of $(p-1)$ letters. Indeed, the last term equals

$$\sum_\tau \sum_{r=0}^{p-1} \sum_{i_1 < \cdots < i_r} (-1)^s a_{\tau(i_1)} a_{\tau(i_2)} \cdots a_{\tau(i_r)} a_p a_{\tau(j_s)} \cdots$$

where τ ranges over the permutations of $p-1$ letters, $i_1, \cdots, i_r$ the strictly increasing sequences of integers of the interval $[1, p-1]$ and where $j_1, \cdots, j_s$ denotes the strictly increasing sequence whose values are the integers of $[1, p-1]$ different from $i_1, \cdots, i_r$. For a fixed value of r, the sum of the terms $(-1)^s a_{\tau(i_1)} \cdots a_{\tau(i_r)} a_p a_{\tau(j_s)} \cdots a_{\tau(j_1)}$ obviously equals

---

[655] See also [DG70], § IV.3, 4.9.

[656] cf. P. Cartier, Exemples d'hyperalgèbres, Sém. Sophus Lie 1955/56, Exp. 3 (accessible on the Numdam site: http://www.numdam.org).

`(−1)^s (p−1 choose s) ∑_ρ a_{ρ(1)} ··· a_{ρ(r)} a_p a_{ρ(r+1)} ··· a_{ρ(p−1)}`

where ρ ranges over the permutations of $p-1$ letters. Now $(-1)^s(p-1 chooses) = 1$ in $\mathbb{F}_p$, since in $\mathbb{F}_p[x]$ (x an indeterminate) one has: $(x-1)^p = x^p - 1 = (x-1)(x^{p-1} + \cdots + 1)$ and therefore $(x-1)^{p-1} = x^{p-1} + \cdots + 1$. This proves $(*)$.

On the other hand, if x_0 and x_1 are two elements of A, one has

`(x_0 + x_1)^p = x_0^p + x_1^p + ∑ x_{z(1)} x_{z(2)} ··· x_{z(p)},`

where z ranges over the non-constant maps from `[1, p]` into `{0, 1}`. One deduces

`(x_0 + x_1)^p = x_0^p + x_1^p + ∑_{0 < r < p} 1/(r!(p−r)!) N(x_0, …, x_0, x_1, …, x_1)`

(with r factors x_0 and $p-r$ factors x_1).[657] Now, according to $(*)$, one has:

`N(x_0, …, x_0, x_1, …, x_1) = r! (p − 1 − r)! ∑_t [x_{t(1)} x_{t(2)} ··· x_{t(p−1)}, x_1] ···`

(with r factors x_0 and $p-r$ factors x_1), where t ranges over the maps $[1, p-1] \to 0, 1$ taking the value `0` exactly r times. From this one deduces Jacobson's second formula:

`(ii)    (x_0 + x_1)^p = x_0^p + x_1^p − ∑_{0 < r < p} ∑_t (1/r) [x_{t(1)} x_{t(2)} ··· x_{t(p−1)}, x_`

where t ranges over the maps $[1, p-1] \to 0, 1$ taking the value `0` exactly r times.

5.2.

Let now $\mathfrak{g}$ be an R-Lie algebra. One says that a map $x \mapsto x^{(p)}$ from $\mathfrak{g}$ into $\mathfrak{g}$ makes $\mathfrak{g}$ a *restricted p-Lie algebra* over R if the following conditions are satisfied:

(0) $(\lambda x)^{(p)} = \lambda^p \cdot x^{(p)}$, for $\lambda \in R$, $x \in \mathfrak{g}$

(i) $adx^{(p)} = (adx)^p$, for $x \in \mathfrak{g}$

(ii) $(x_0 + x_1)^{(p)} = x_0^{(p)} + x_1^{(p)} - \sum_{0<r<p} \sum_t (1/r)[x_{t(1)} x_{t(2)} \cdots x_{t(p-1)}, x_1] \cdots$

where t ranges over the maps $[1, p-1] \to 0, 1$ taking the value `0` exactly r times ($x_0, x_1 \in \mathfrak{g}$). The map $x \mapsto x^{(p)}$ will then be called the "*symbolic p-th power*".

For example, if A is an associative R-algebra, we saw in 5.1 that one obtains a p-Lie algebra, which we shall denote A_{Lie}, by taking the R-module underlying A and setting, for $x, y \in A$,

`[x, y] = xy − yx    and    x^{(p)} = x^p.`

We shall say that A_{Lie} is the p-Lie algebra underlying A.

In what follows we shall consider mostly sub-p-Lie algebras of p-algebras of the form A_{Lie}; here is an example: let S be a scheme of characteristic $p > 0$ and X an S-scheme. Recall that a derivation of X over S is an endomorphism D of the sheaf of abelian groups `O_X` such that

`D(λ · s) = λ · D(s)    and    D(st) = (Ds) t + s (Dt)`

[657] We have inserted the explanation that follows, taken from [DG70], § II.7, 3.2.

when λ and `s, t` range over the sections of `O_S` and of `O_X` on opens such that the formulas make sense. Leibniz's formula

```
D^n(st) = ∑_{i=0}^n (n choose i) (D^i s)(D^{n−i} t)
```

shows that D^p is again a derivation of X over S, taking into account the equality $(p choose i) \equiv 0 (mod p)$ for $i \neq 0, p$. It follows that:

> The algebra $Dér_{X/S}$ of derivations of X over S is a sub-p-Lie algebra of the $\Gamma(S, O_S)$-algebra of differential operators of X over S.

5.2.1. If $\mathfrak{g}$ and $\mathfrak{h}$ are two p-Lie algebras, a *homomorphism* $h : \mathfrak{g} \to \mathfrak{h}$ is an R-linear map from $\mathfrak{g}$ into $\mathfrak{h}$ such that $h([x, y]) = [h(x), h(y)]$ and $h(x^{(p)}) = h(x)^{(p)}$ if $x, y \in \mathfrak{g}$. The composition of two homomorphisms is again a homomorphism, so that we may speak of the *category of p-Lie algebras* over R.

If (X, R) is a ringed space, we shall say that an R-module $\mathfrak{g}$ is equipped with a structure of p-Lie algebra over R if, for every open U, $\Gamma(U, \mathfrak{g})$ is equipped with a structure of p-Lie algebra over $\Gamma(U, R)$ and if the restrictions are homomorphisms.

5.3.

We are now interested in the left adjoint functor to the functor $A \mapsto A_{Lie}$ of 5.2. Let $\mathfrak{g}$ be a p-Lie algebra over the ring R of characteristic p, $U(\mathfrak{g})$ the enveloping algebra of the Lie algebra underlying $\mathfrak{g}$ (cf. [BLie], I § 2.1) and $i_{\mathfrak{g}}$ (or simply i) the canonical map $\mathfrak{g} \to U(\mathfrak{g})$.

Let A be a unital associative R-algebra. One knows that, for every Lie algebra homomorphism $\phi : \mathfrak{g} \to A_{Lie}$ there exists a unique homomorphism of unital R-algebras $\psi : U(\mathfrak{g}) \to A$ such that $\psi \circ i = \phi$. Moreover, ϕ is a p-Lie algebra homomorphism if and only if ψ vanishes on the elements $i(x)^p - i(x^{(p)})$, when x ranges over $\mathfrak{g}$.

Definition. *One denotes by $U_p^R(\mathfrak{g})$ or simply $U_p(\mathfrak{g})$ the quotient of $U(\mathfrak{g})$ by the two-sided ideal generated by the elements $i(x)^p - i(x^{(p)})$, and $j_{\mathfrak{g}}$ (or simply j) the map $\mathfrak{g} \to U_p(\mathfrak{g})$ composed of $i : \mathfrak{g} \to U(\mathfrak{g})$ and the canonical map $U(\mathfrak{g}) \to U_p(\mathfrak{g})$. One says that $U_p(\mathfrak{g})$ is the* restricted enveloping algebra *of* $\mathfrak{g}$.

According to what precedes, one has the

Proposition. *For every unital associative R-algebra and every p-Lie algebra morphism $\phi : \mathfrak{g} \to A_{Lie}$, there exists a unique homomorphism of unital algebras $\psi : U_p(\mathfrak{g}) \to A$ such that $\psi \circ j = \phi$. In other words, the functor $\mathfrak{g} \mapsto U_p(\mathfrak{g})$ is left adjoint to the forgetful functor $A \mapsto A_{Lie}$.*

5.3.1. With the notations of 5.3, set now $\beta(x) = i(x)^p - i(x^{(p)})$. For every element y of $\mathfrak{g}$, one has, according to 5.1 (i) and 5.2 (i):

```
β(x) i(y) = i(y) β(x) + [β(x), i(y)]
          = i(y) β(x) + (ad i(x))^p i(y) − i((ad x)^p y)
          = i(y) β(x),
```

so $\beta(x)$ belongs to the center of $U(\mathfrak{g})$; in particular, the left ideal generated by the elements $\beta(x)$ is already two-sided.

On the other hand, it is clear that $\beta(\lambda x) = \lambda^p \beta(x)$, for $\lambda \in R$, and it follows from 5.1 (ii) and 5.2 (ii) that, for $x, y \in \mathfrak{g}$,

$$\beta(x + y) = \beta(x) + \beta(y).$$

In particular, if (x_α) is a family of generators of the R-module $\mathfrak{g}$, the left ideal generated by the elements $\beta(x)$ is already generated by the $\beta(x_\alpha)$.

5.3.2. Proposition. [658] *Let $\mathfrak{g}$ be an R-Lie algebra whose underlying R-module is free with basis (x_α). Then the structures of p-Lie algebra on $\mathfrak{g}$ correspond bijectively to the families (y_α) of $\mathfrak{g}$ such that $ady_\alpha = (adx_\alpha)^p$.*

Indeed, if $\mathfrak{g}$ is equipped with a structure of p-Lie algebra $x \mapsto x^{(p)}$, then according to 5.2 (i) and (0), (ii), the $y_\alpha = x_\alpha^{(p)}$ satisfy $ady_\alpha = (adx_\alpha)^p$, and determine the p-Lie algebra structure.

Let us prove the converse. Since $\mathfrak{g}$ is a free R-module, the canonical map $i : \mathfrak{g} \to U(\mathfrak{g})$ is injective, according to the Poincaré–Birkhoff–Witt theorem (cf. [BLie], I § 2.7), so one can identify $\mathfrak{g}$ with an R-submodule of $U(\mathfrak{g})$. Suppose that (y_α) is a family of elements of $\mathfrak{g}$ such that $ady_\alpha = (adx_\alpha)^p$. Let π be the map $r \mapsto r^p$ from R into R, and let $\mathfrak{g} \otimes_\pi R$ be the R-Lie algebra obtained by extension of scalars $\pi : R \to R$.[659]

There then exists an R-linear map γ from $\mathfrak{g} \otimes_\pi R$ into $U(\mathfrak{g})$ which sends $x_\alpha \otimes_\pi 1$ to $x_\alpha^p - y_\alpha$; moreover, since one has, for every $x \in \mathfrak{g}$,

$$(\mathrm{ad}\, x_\alpha^p)(x) = (\mathrm{ad}\, x_\alpha)^p(x) = (\mathrm{ad}\, y_\alpha)(x),$$

γ maps $\mathfrak{g} \otimes_\pi R$ into the center of $U(\mathfrak{g})$. Set, for every $x \in \mathfrak{g}$:

$$x^{(p)} = x^p - \gamma(x \otimes_\pi 1).$$

Then, for every α, one has $x_\alpha^{(p)} = y_\alpha$. If $x = \sum \lambda_\alpha x_\alpha$, one deduces from 5.1 (ii) (by induction on the number of indices α such that $\lambda_\alpha \neq 0$), that

$$x^p - \sum_\alpha \lambda_\alpha^p x_\alpha^p \in \mathfrak{g};$$

denoting this element by z, one then has $x^{(p)} = \sum \lambda_\alpha^p y_\alpha + z$ and therefore $x^{(p)} \in \mathfrak{g}$.

It is clear that the map $x \mapsto x^{(p)}$ satisfies $(\lambda x)^{(p)} = \lambda^p x^{(p)}$. Moreover, since $\gamma(x \otimes_\pi 1)$ is central, then $adx^{(p)} = adx^p$ and therefore, according to Jacobson's first formula (5.1 (i)), one has

$$\mathrm{ad}\, x^{(p)} = (\mathrm{ad}\, x)^p.$$

Finally, according to Jacobson's second formula (5.1 (ii)), the map $x \mapsto x^{(p)}$ satisfies condition (ii) of 5.2. It therefore makes $\mathfrak{g}$ a p-Lie algebra. This proves the proposition.

[658] In this paragraph, we have modified the order, first stating the result, then detailing the proof.
[659] i.e., $xr \otimes_\pi 1 = x \otimes_\pi r^p$ and $r \cdot (x \otimes_\pi 1) = x \otimes_\pi r$, for $x \in \mathfrak{g}$, $r \in R$.

5.3.3. Proposition. *Let $\mathfrak{g}$ be a p-Lie algebra over R whose underlying module is free with basis (x_α). Then the map $j : \mathfrak{g} \to U_p(\mathfrak{g})$ is injective and, if one sets $z_\alpha = j(x_\alpha)$, then $U_p(\mathfrak{g})$ has as basis the monomials*

$$\prod_\alpha z_\alpha^{n_\alpha} \quad \text{where} \quad 0 \leqslant n_\alpha < p,$$

(the n_α are assumed to be zero except for a finite number of them; one assumes the basis to be totally ordered and the products to be performed in increasing order).

Indeed, identify $\mathfrak{g}$ with a submodule of the enveloping algebra $U(\mathfrak{g})$ by means of the canonical map i. For every family $n = (n_\alpha)$ of natural integers, zero except for a finite number of them, set

$$|n| = \sum_\alpha n_\alpha \quad \text{and} \quad x^n = \prod_\alpha x_\alpha^{n_\alpha}.$$

Writing $n_\alpha = m_\alpha + p\ell_\alpha$, with $0 \leqslant m_\alpha < p$, set also

$$T_n = \prod_\alpha x_\alpha^{m_\alpha} \beta(x_\alpha)^{\ell_\alpha}$$

where $\beta(x) = x^p - x^{(p)}$ is the map $\mathfrak{g} \to U(\mathfrak{g})$ defined in 5.3.1.

For every $r \in \mathbb{N}$, denote by U_r the sub-R-module of $U(\mathfrak{g})$ generated by the x^n such that $|n| \leqslant r$. Since the graded ring $\bigoplus_r U_r/U_{r-1}$ is commutative (cf. [BLie], I § 2.6), one sees that, for every n:

$$T_n - \prod_\alpha x_\alpha^{n_\alpha} \in U_{|n|-1}.$$

For every $s \in \mathbb{N}$, the x^n such that $|n| = s$ form, according to the Poincaré–Birkhoff–Witt theorem (loc. cit., § 2.7), a basis of U_s/U_{s-1}, and therefore the same holds for the T_n such that $|n| = s$.

Therefore, when $s = |n|$ varies, the T_n form a basis of $U(\mathfrak{g})$. Now the kernel J of the canonical map $U(\mathfrak{g}) \to U_p(\mathfrak{g})$ is the left ideal of $U(\mathfrak{g})$ generated by the central elements $\beta(x_\alpha)$ (5.3.1). Consequently, the T_n such that $\ell = (\ell_\alpha) \neq 0$ form a basis of J, and the T_n such that $n_\alpha < p$ for every α, form a basis of $U_p(\mathfrak{g}) = U(\mathfrak{g})/J$.

5.3.3 bis. Let $\mathfrak{g}$ be a p-Lie algebra over R and $f : R \to R'$ an extension of the base ring. I claim that there exists on the R'-module $R' \otimes_R \mathfrak{g}$ a p-Lie algebra structure and only one such that

$$(*) \qquad [\lambda \otimes x, \mu \otimes y] = \lambda\mu \otimes [x, y] \quad \text{and} \quad (\lambda \otimes x)^{(p)} = \lambda^p \otimes x^{(p)}.$$

It will follow, in particular, that the functor $\mathfrak{g} \mapsto R' \otimes_R \mathfrak{g}$ is left adjoint to the functor "restriction of scalars from R' to R".

The uniqueness of the p-Lie algebra structure defined by $(*)$ being clear, let us prove existence. When $\mathfrak{g}$ is free with basis (x_α) there exists, according to 5.3.2, one and only one p-Lie algebra structure on the Lie algebra $R' \otimes_R \mathfrak{g}$ such that

$$(1 \otimes x_\alpha)^{(p)} = 1 \otimes x_\alpha^{(p)};$$

this structure is the one we seek.

When $\mathfrak{g}$ is an arbitrary p-Lie algebra, there exists a p-Lie algebra L_0 free (as an R-module) and a surjective homomorphism $q_0 : L_0 \to \mathfrak{g}$; it suffices for example to take for L_0 the p-Lie algebra $R \bigotimes_{\mathbb{F}_p} \mathfrak{g}$, where $\mathbb{F}_p$ denotes the prime field of characteristic p, and for q_0 the homomorphism $\lambda \otimes x \mapsto \lambda x$ ($\mathfrak{g}$ is free over $\mathbb{F}_p$!). The kernel of q_0 is then a p-ideal of L_0, i.e., an ideal of the Lie algebra L_0 which is stable under the endomorphism $x \mapsto x^{(p)}$; there is therefore also a p-Lie algebra L_1 free (as an R-module) and a homomorphism $q_1 : L_1 \to L_0$ whose image is $Ker q_0$, whence the exact sequence:

$$L_1 \xrightarrow{q_1} L_0 \xrightarrow{q_0} \mathfrak{g} \longrightarrow 0.$$

One deduces from this an exact sequence of R'-Lie algebras

$$R' \otimes_R L_1 \xrightarrow{R' \otimes_R q_1} R' \otimes_R L_0 \xrightarrow{R' \otimes_R q_0} R' \otimes_R \mathfrak{g} \longrightarrow 0.$$

Since $R' \bigotimes_R q_1$ is manifestly a homomorphism of p-Lie algebras, the kernel of $R' \bigotimes_R q_0$ is a p-ideal, so that the symbolic p-th power operation of $R' \bigotimes_R L_0$ induces by passage to the quotient a map from $R' \bigotimes_R \mathfrak{g}$ into $R' \bigotimes_R \mathfrak{g}$ (use formula (ii) of 5.2); this last one equips $R' \bigotimes_R \mathfrak{g}$ with the p-Lie algebra structure sought.

5.3.4.

The canonical map $j_{\mathfrak{g}} : \mathfrak{g} \to U_p(\mathfrak{g})$ induces, for every extension $R \to R'$ of the base ring, a homomorphism

$$R' \otimes_R j_{\mathfrak{g}} : R' \otimes_R \mathfrak{g} \to R' \otimes_R U_p(\mathfrak{g}),$$

whence a homomorphism

$$h : U_p(R' \otimes_R \mathfrak{g}) \to R' \otimes_R U_p(\mathfrak{g})$$

such that $h \circ j_{R' \otimes \mathfrak{g}} = R' \bigotimes_R j_{\mathfrak{g}}$. It obviously follows from the universal properties of $R' \bigotimes_R \mathfrak{g}$ and the restricted enveloping algebra that h is an isomorphism, which will allow us to identify $U_p(R' \bigotimes_R \mathfrak{g})$ with $R' \bigotimes_R U_p(\mathfrak{g})$.

In particular, if r is an element of R and if R' is the localized ring R_r, one sees that $\mathfrak{g}_r = R_r \bigotimes_R \mathfrak{g}$ is equipped canonically with a structure of p-Lie algebra over R_r, so that the sheaf $\tilde{\mathfrak{g}}$ on $\operatorname{Spec} R$ is a quasi-coherent p-Lie algebra on $\operatorname{Spec} R$. Moreover, the restricted enveloping algebra $U_p^{R_r}(\mathfrak{g}_r)$ is identified with $U_p^R(\mathfrak{g})_r$, so that the sheaf associated with the presheaf $V \mapsto U_p(\Gamma(V, \tilde{\mathfrak{g}}))$ is quasi-coherent.

Definition. *More generally, if S is a scheme of characteristic p and $\mathcal{G}$ a quasi-coherent p-Lie algebra on $\mathcal{O}_S$, the sheaf associated with the presheaf $V \mapsto U_p(\Gamma(V, \mathcal{G}))$ is quasi-coherent; it will be denoted $\mathcal{U}_p(\mathcal{G})$ and called the* restricted enveloping algebra *of $\mathcal{G}$. If V is affine, $U_p(\Gamma(V, \mathcal{G}))$ is identified with $\Gamma(V, \mathcal{U}_p(\mathcal{G}))$.*

5.4.

The universal character of $U_p(\mathfrak{g})$ entails that $U_p(\mathfrak{g})$ is functorial in $\mathfrak{g}$: every homomorphism of p-Lie algebras $\phi : \mathfrak{g} \to \mathfrak{h}$ induces a homomorphism of unital algebras $U_p(\phi)$ and only one such that $j_{\mathfrak{h}} \circ \phi = U_p(\phi) \circ j_{\mathfrak{g}}$. Here are some examples:

a) If $\mathfrak{h} = 0$, $U_p(\mathfrak{h})$ is identified with the base ring and $U_p(\phi)$ is an algebra homomorphism $\epsilon_{\mathfrak{g}} : U_p(\mathfrak{g}) \to R$ called the *augmentation*.

b) Now take for $\mathfrak{h}$ the algebra $\mathfrak{g}^\circ$ opposite to $\mathfrak{g}$, i.e., $\mathfrak{g}^\circ$ has the same underlying module as $\mathfrak{g}$, the same symbolic p-th power, the bracket of two elements in $\mathfrak{g}^\circ$ being the opposite of the bracket in $\mathfrak{g}$. It is clear that we can identify $U_p(\mathfrak{g}^\circ)$ with the algebra opposite to $U_p(\mathfrak{g})$. Moreover, the isomorphism $x \mapsto -x$ of $\mathfrak{g}$ onto $\mathfrak{g}^\circ$ induces an isomorphism $c_{\mathfrak{g}}$ of $U_p(\mathfrak{g})$ onto $U_p(\mathfrak{g}^\circ) \simeq U_p(\mathfrak{g})^\circ$. One says that $c_{\mathfrak{g}}$ is the *antipode* of $U_p(\mathfrak{g})$.

c) Let finally $\mathfrak{f}$ and $\mathfrak{g}$ be two p-Lie algebras and $\mathfrak{h}$ the p-Lie algebra product $\mathfrak{f} \times \mathfrak{g}$ which has as underlying R-module the direct product $\mathfrak{f} \times \mathfrak{g}$, the bracket and the symbolic p-th power being defined by the formulas

$$[(x, y), (x', y')] = ([x, x'], [y, y']) \quad \text{and} \quad (x, y)^{(p)} = (x^{(p)}, y^{(p)}).$$

If $h_1 : \mathfrak{f} \to \mathfrak{k}$ and $h_2 : \mathfrak{g} \to \mathfrak{k}$ are two p-Lie algebra homomorphisms such that $[h_1(x), h_2(y)] = 0$ for every x of $\mathfrak{f}$ and every y of $\mathfrak{g}$, the map $h_1 + h_2 : (x, y) \to h_1(x) + h_2(y)$ is a p-Lie algebra homomorphism; conversely, every homomorphism from $\mathfrak{f} \times \mathfrak{g}$ into $\mathfrak{k}$ is of this type, which allows us to characterize $\mathfrak{f} \times \mathfrak{g}$ as the solution of a universal problem. For example, the maps

$$h_1 : x \mapsto i_{\mathfrak{f}}(x) \otimes 1 \quad \text{and} \quad h_2 : y \mapsto 1 \otimes i_{\mathfrak{g}}(y)$$

induce a homomorphism $h_1 + h_2$ from $\mathfrak{f} \times \mathfrak{g}$ into the p-Lie algebra underlying $U_p(\mathfrak{f}) \bigotimes U_p(\mathfrak{g})$. It follows from the universal characters of $\mathfrak{f} \times \mathfrak{g}$ and of the restricted enveloping algebras that $h_1 + h_2$ extends to an isomorphism:

$$\varphi : U_p(\mathfrak{f} \times \mathfrak{g}) \xrightarrow{\sim} U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g}).$$

Definition. *If $\mathfrak{f} = \mathfrak{g}$, the diagonal map $\delta : x \mapsto (x, x)$ of $\mathfrak{g}$ into $\mathfrak{g} \times \mathfrak{g}$ induces a homomorphism of $U_p(\mathfrak{g})$ into $U_p(\mathfrak{g} \times \mathfrak{g})$. We shall denote by $\Delta_{\mathfrak{g}}$ the composition of this homomorphism with the isomorphism $\phi : U_p(\mathfrak{g} \times \mathfrak{g}) \xrightarrow{\sim} U_p(\mathfrak{g}) \bigotimes U_p(\mathfrak{g})$.*[660]

One then sees easily that $\Delta_{\mathfrak{g}}$ and the multiplication of the algebra $U_p(\mathfrak{g})$ make $U_p(\mathfrak{g})$ an R-coalgebra in groups (cf. 3.2) which has $\epsilon_{\mathfrak{g}}$ as augmentation and $c_{\mathfrak{g}}$ as antipode.

5.5.

[661] Let now S be a scheme of characteristic p. First, if $\mathcal{U}$ is an O_S-coalgebra in groups and G the group S-functor $\mathrm{Spec}^* \mathcal{U}$, we saw (3.2.3) that, for every $T \to S$, $(\mathrm{Lie}\, G)(T)$ is the Lie subalgebra of $\Gamma(T, \mathcal{U}_T)$ formed by the primitive elements. Now, if x is such an element, one has $\Delta(x^p) = x^p \otimes 1 + 1 \otimes x^p$ (since $(pchoosei) = 0modp$ for $0 < i < p$), i.e. x^p is again a primitive element. It follows, according to

[660] i.e. $\Delta_{\mathfrak{g}}(x) = x \otimes 1 + 1 \otimes x$ for every $x \in \mathfrak{g}$; in particular, the comultiplication $\Delta_{\mathfrak{g}}$ is indeed cocommutative …

[661] We have transformed § 5.4.1 of the original into this § 5.5: on the one hand, Proposition 5.5.1 combines the results of Section 5 and Proposition 3.2.3 and contains Lemma 7.3 of the original; on the other hand, the proof of 5.5.3 (ii) takes up, in expanded form, that of the implication (i) ⇒ (ii) in Theorem 7.4 below.

5.1 and 5.2, that the map $x \mapsto x^p$ equips $(\mathrm{Lie}\ G)(T)$ with a structure of $O(T)$-p-Lie algebra.

Let now $\mathcal{L}$ be an $\mathcal{O}_S$-p-Lie algebra, quasi-coherent on $\mathcal{O}_S$. When V ranges over the opens of S, the structures of group coalgebras previously defined on the sets $U_p(\Gamma(V,\mathcal{L}))$ induce on the associated sheaf, i.e., on the restricted enveloping algebra $\mathcal{U}_p(\mathcal{L})$, a structure of $\mathcal{O}_S$-coalgebra in groups. Moreover, for every S-scheme T, one has an isomorphism $\mathcal{U}_p(\mathcal{L}_T) \xrightarrow{\sim} \mathcal{U}_p(\mathcal{L})_T$.

Denote by $Prim\mathcal{U}_p(\mathcal{L})$ the subpresheaf of $\mathcal{U}_p(\mathcal{L})$ associating with every open V the set of primitive elements of $\mathcal{U}_p(\mathcal{L})(V)$; one sees easily that this is a sheaf. When V ranges over the opens of S, the composed maps

$$\Gamma(V, \mathcal{L}) \xrightarrow{j} \mathrm{Prim}\, U_p(\Gamma(V, \mathcal{L})) \longrightarrow \mathrm{Prim}\, \mathcal{U}_p(\mathcal{L})(V)$$

define a morphism $\mathcal{L} \to Prim\mathcal{U}_p(\mathcal{L})$, which we shall again denote j or $j_{\mathcal{L}}$, and this defines further a morphism of $\mathcal{O}_S$-p-Lie algebras $\mathcal{W}(\mathcal{L}) \to Prim\mathcal{W}(\mathcal{U}_p(\mathcal{L}))$ (cf. 3.2.3).

Proposition 5.5.1. *Let $\mathcal{L}$ be an $\mathcal{O}_S$-p-Lie algebra, locally free as an $\mathcal{O}_S$-module. Then $j_{\mathcal{L}}$ induces an isomorphism of $\mathcal{O}_S$-p-Lie algebras:*

$$\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} Prim\mathcal{W}(\mathcal{U}_p(\mathcal{L})).$$

Proof. Let T be an S-scheme; taking into account the identification $\mathcal{U}_p(\mathcal{L}_T) = \mathcal{U}_p(\mathcal{L})_T$, the task is to show that the map $\Gamma(T,\mathcal{L}_T) \to Prim\Gamma(T,\mathcal{U}_p(\mathcal{L}_T))$ is bijective. Replacing S by T, one is reduced to the case where $T = S$, and it then suffices to show that the morphism of sheaves $j_{\mathcal{L}} : \mathcal{L} \to Prim\mathcal{U}_p(\mathcal{L})$ is an isomorphism. This question being local on S, we may suppose that S is affine with ring R and that $\mathcal{L}$ is the sheaf associated with an R-p-Lie algebra L with basis (x_α). As in 5.3.3, denote by z_α the image of x_α in $U = U_p(L)$ and, for every family $n = (n_\alpha)$ of integers between 0 and $p-1$, zero except for a finite number of them, denote $z^{(n)}$ the product

$$\prod_\alpha z_\alpha^{n_\alpha}/n_\alpha!$$

(one supposes the basis (x_α) totally ordered and the products performed in increasing order).

Since $\Delta(z_\alpha) = z_\alpha \otimes 1 + 1 \otimes z_\alpha$, one sees easily that

$$\Delta(z^{(n)}) = \sum_r z^{(n-r)} \otimes z^{(r)}$$

the sum being taken over the (finite!) set of r such that $0 \leqslant r_\alpha \leqslant n_\alpha$ for every α. Since the $z^{(n)}$ (resp. the $z^{(n)} \bigotimes z^{(m)}$) form a basis of U (resp. of $U \bigotimes U$), one deduces that an element u of U satisfies $\Delta(u) = u \otimes 1 + 1 \otimes u$ if and only if u is a linear combination of the z_α. This proves 5.5.1.

Remark 5.5.2. *Recall (cf. 3.2.2 and 3.2.3), that the group S-functor $G = \mathrm{Spec}^* \mathcal{U}_p(\mathcal{L})$ is very good and that $Lie(G) = Prim\mathcal{W}(\mathcal{U}_p(\mathcal{L}))$. The preceding proposition therefore signifies that $j_{\mathcal{L}}$ induces an isomorphism $\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} Lie(G)$.*

If one supposes moreover that $\mathcal{L}$ is a locally free 0_S-module of finite rank, then $\mathcal{U}_p(\mathcal{L})$ is finite locally free over 0_S, according to 5.3.3, so $\mathrm{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by the S-group $\mathfrak{G}_p(\mathcal{L}) = \mathrm{Spec}\, \mathcal{U}_p(\mathcal{L})^*$ (cf. 3.2.2.1), and one obtains the following more precise proposition:

Proposition 5.5.3. *Let $\mathcal{L}$ be an 0_S-p-Lie algebra, locally free of finite rank as an 0_S-module, let $\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*$ and let $G = \mathfrak{G}_p(\mathcal{L})$ be the affine S-group $\mathrm{Spec}\, \mathcal{A}$.*

(i) $j_{\mathcal{L}}$ induces an isomorphism $\mathcal{W}(\mathcal{L}) \xrightarrow{\sim} Lie(G)$ of 0_S-p-Lie algebras.

(ii) Let $\mathcal{I}$ be the augmentation ideal of $\mathcal{A}$ and $\omega_G = \mathcal{I}/\mathcal{I}^2$ (cf. II, 4.11.4). Then ω_G is identified with $\mathcal{L}^* = \mathrm{Hom}_{O_S}(\mathcal{L}, O_S)$, hence is a locally free 0_S-module of finite rank (and one has $\omega^*_{G/S} = \mathcal{L}$).

Proof. (i) following from 5.5.2, let us prove (ii). Denote by $\eta_{\mathcal{U}}$ and $\epsilon_{\mathcal{U}}$ the unit section and the augmentation of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, by $\eta_{\mathcal{A}}$ and $\epsilon_{\mathcal{A}}$ those of $\mathcal{A}$, and $\mathcal{J} = Ker\epsilon_{\mathcal{U}}$. Then one has:

$$(1)\mathcal{U} = \eta_{\mathcal{U}}(O_S) \oplus \mathcal{J}.$$

Let δ be the morphism defined by the diagram below, where τ and π denote the inclusion and the projection deduced from the decomposition (1):

$$\begin{tikzcd} \mathcal{J} \arrow[r, "\delta"] \arrow[d, "\tau"'] & \mathcal{J} \otimes \mathcal{J} \\ \mathcal{U} \arrow[r, "\Delta"] & \mathcal{U} \otimes \mathcal{U} \arrow[u, "\pi"'] \end{tikzcd}$$

then one has an exact sequence:

$$(*) \qquad 0 \longrightarrow \mathcal{L}^{*} \xrightarrow{j_{\mathcal{L}}} \mathcal{J} \xrightarrow{\delta} \mathcal{J} \otimes \mathcal{J}.$$

Moreover, according to 5.3.3, the 0_S-module $\mathcal{J}/\mathcal{L}$ is locally free and, according to 5.5.1, the sequence $(*)$ remains exact after every base change. So, according to [BAC], II § 3, prop. 6, δ induces an isomorphism of $\mathcal{J}/\mathcal{L}$ onto a submodule Q locally direct factor of $\mathcal{J} \otimes \mathcal{J}$. It follows that $(*)$ gives by duality the exact sequence:

$$(**) \qquad 0 \longleftarrow \mathcal{L}^* \xleftarrow{{}^t j_{\mathcal{L}}} \mathcal{I} \xleftarrow{{}^t\delta} \mathcal{I} \otimes \mathcal{I}.$$

Now the decomposition (1) corresponds by duality to the decomposition:

$$(2)\mathcal{A} = \eta_{\mathcal{A}}(O_S) \oplus \mathcal{I}$$

and the transpose of Δ is the multiplication $m_{\mathcal{A}} : \mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$. Since $\mathcal{I}$ is an ideal of $\mathcal{A}$, $m_{\mathcal{A}}$ sends $\mathcal{I} \otimes \mathcal{I}$ into $\mathcal{I}$; more precisely, taking into account decomposition (2), one has a commutative square

$$\begin{tikzcd} \mathcal{I} \arrow[d, "{}^t\tau"'] & \mathcal{I} \otimes \mathcal{I} \arrow[l, "m'"'] \arrow[d, "{}^t\pi"] \\ \mathcal{A} & \mathcal{A} \otimes \mathcal{A} \arrow[l, "m_{\mathcal{A}}"] \end{tikzcd}$$

which shows that the restriction m' of $m_{\mathcal{A}}$ to $\mathcal{I} \otimes \mathcal{I}$ is the transpose of δ. The exact sequence $(**)$ then gives $\mathcal{I}/\mathcal{I}^2 \simeq \mathcal{L}^*$, and the proposition follows.

6. p-Lie algebra of an S-group scheme

Let S be a scheme of characteristic $p > 0$. In paragraph 5.5 we associated with every quasi-coherent $\mathcal{O}_S$-p-Lie algebra $\mathcal{L}$ a group S-functor $\mathfrak{G}_p(\mathcal{L}) = \mathrm{Spec}^* \mathcal{U}_p(\mathcal{L})$. We shall now see that, for every S-group scheme G, the $\mathcal{O}_S$-Lie algebra $Lie(G/S)$ defined in II 4.11 is naturally equipped with a structure of $\mathcal{O}_S$-p-Lie algebra.

6.1.

Let us first identify $Lie(G/S)(S)$ and $Lie(\mathrm{Aut}\, G/S)(S)$ respectively with Lie subalgebras of $U(G)$ and $Dif_{G/S}$ by means of the injections α and β of 2.5; $Lie(\mathrm{Aut}\, G/S)(S)$ is therefore identified with the $\Gamma(O_S)$-Lie algebra of S-derivations of $\mathcal{O}_G$. According to 5.2, this latter is a sub-p-Lie algebra of $Dif_{G/S}$.

On the other hand, the image of $L = Lie(G/S)(S)$ by the injective algebra morphism $\ell : U(G) \to Dif_{G/S}$, $d \mapsto {}_G d$, is formed by the left-invariant derivations (cf. 2.2, N.D.E. (17), 2.4 and 2.5). If x belongs to L, $\ell(x)^p$ is none other than $\ell(x^p)$, according to loc. cit. Since $\ell(x)^p$ is again a derivation, one sees that x^p belongs to $Lie(G/S)(S)$. Therefore:[662]

$Lie(G/S)(S)$ is a sub-p-Lie algebra of the infinitesimal algebra $U(G)$.

6.1.1. Let $\phi : G \to H$ be a homomorphism of S-group schemes. It is clear that the homomorphisms $Lie(\phi/S)(S)$ and $U(\phi)$ are compatible with the identifications of $Lie(G/S)(S)$ and $Lie(H/S)(S)$ with sub-p-Lie algebras of $U(G)$ and $U(H)$. Since $U(\phi)$ is an algebra homomorphism, one sees therefore that $Lie(\phi/S)(S)$ is a homomorphism of p-Lie algebras.

Similarly, if $s : T \to S$ is a base change, the map from $Lie(G/S)(S)$ into $Lie(G/S)(T)$, which is induced by s, is a homomorphism of p-Lie algebras. One can translate

[662] One can also show directly (without the intermediary of $Dif_{G/S}$) that the Lie algebra of derivations of G at the origin (isomorphic to $Lie(G/S)(S)$ according to 2.5) is stable under raising to the p-th power in $U(G)$: this is done in 6.2 below.

this by saying that the functor $Lie(G/S)$ is equipped with a structure of O_S-p-Lie algebra. In particular, when T ranges over the opens of S, one sees that

the sheaf Lie(G/S) is equipped with a structure of O_S-p-Lie algebra.

6.2.

Following an idea of Demazure, we shall now generalize what precedes to certain group S-functors not necessarily representable. For this, we shall first give another definition of the symbolic p-th power in the Lie algebra of an S-group scheme G.

Let D be a derivation of G at the origin;[663] according to 1.2.1, D is the composition of the S-derivation $\delta = (\tau, \partial_t)$ of the zero section $\tau : S \to I_S$, and a morphism $x : I_S \to G$ such that $x \circ \tau = \epsilon$ (i.e. $x \in Lie(G/S)(S)$). According to the definition we gave in 2.1, D^p is the following composed deviation:

S ≃ S × S × ··· × S ──δ × ··· × δ→ I_S × ··· × I_S ──x × ··· × x→ G × ··· × G ──m^{(p)}→ G

(p copies), where $m^{(p)}$ is the morphism induced by the multiplication $m : G \times G \to G$. Since $I_S \times \cdots \times I_S$ is affine over S and has as affine algebra $B = O_S[d_1, \cdots, d_p]/(d_1^2, \cdots, d_p^2)$, the deviation $\delta \times \cdots \times \delta$ is defined by the morphism of O_S-modules

$$\phi : B \longrightarrow O_S$$

which sends to 1 the monomial $d_1 d_2 \cdots d_p$, and to 0 the other monomials $d_{i_1} \cdots d_{i_r}$, for $0 \leqslant r < p$. On the other hand, if pr_i denotes the projection of I_S^p onto the i-th factor and if x_i is the image in $G(I_S^p)$ of x by $G(pr_i)$, then the composed morphism $m^{(p)} \circ (x \times \cdots \times x)$ is none other than the product $x_1 x_2 \cdots x_p$. Consequently, D^p is also the following composed deviation:

S ──δ × ··· × δ→ I_S × ··· × I_S ──x_1 x_2 ··· x_p→ G.

This description allows us to re-prove that D^p is a derivation of G at the origin. Indeed, since G is a very good group (II 4.11), the images $G(pr_1)(x)$ and $G(pr_2)(x)$ of x in $G(I_S \times I_S)$ commute with each other. It follows that the elements x_i of $G(I_S^p)$ commute pairwise and therefore, for every permutation γ of the factors of I_S^p, one has $(x_1 \cdots x_p) \circ \gamma = x_1 \cdots x_p$; it follows that $x_1 \cdots x_p$ factors through the canonical projection of I_S^p into the symmetric product $\Sigma^p I_S$ (cf. 4.2).

The symmetric product $\Sigma^p I_S$ has as affine algebra the subalgebra A of B which has as basis over O_S the elementary symmetric functions $1 = \sigma_0, \sigma_1, \cdots, \sigma_p$ of $d_1, \cdots, d_p$. Denote by κ the inclusion $A \hookrightarrow B$ and π the morphism of O_S-algebras $A \to O_S[t]/(t^2)$ which annihilates $\sigma_1, \cdots, \sigma_{p-1}$ and sends σ_p to t; then one has $\phi \circ \kappa = \partial_t \circ \pi$ (recall that $\partial_t : O_S[t]/(t^2) \to O_S$ is the morphism of O_S-modules which annihilates 1 and sends t to 1). Consequently, denoting by i the closed immersion $I_S \hookrightarrow \Sigma^p I_S$ defined by π, one has a commutative diagram:

[663] We have expanded the original in what follows.

$$\begin{tikzcd}
S \arrow[r, "\delta \times \cdots \times \delta"] \arrow[d, "\delta"'] \arrow[rr, bend left, "D^p"] & I_S^p \arrow[r, "x_1 \cdots x_p"] \arrow[d, "\mathrm{can.}"] & G \\
I_S \arrow[r, "i"] & \Sigma^p I_S \arrow[r, "y"] & G
\end{tikzcd}$$

which shows that D^p is of the form $y \circ \delta$, so is indeed a derivation of G at the origin.

6.3.

Let $\mathfrak{S}_p$ be the symmetric group of order p and $I_S^p \times \mathfrak{S}_p$ the direct sum of a family of copies of I_S^p indexed by $\mathfrak{S}_p$. We denote by $\pi : I_S^p \times \mathfrak{S}_p \to I_S^p$ the canonical projection and

$$\mu : I_S^p \times \mathfrak{S}_p \to I_S^p$$

the morphism defining the action of $\mathfrak{S}_p$ on I_S^p (i.e., if τ is an element of $\mathfrak{S}_p$, the restriction of μ to $I_S^p \times \tau$ has $pr_{\tau(j)}$ as j-th component). This being so, we lay down the following definition:

Definition. *A functor $X : (Sch/S)^\circ \to (Ens)$ satisfies condition* (F) *if:*

a) X transforms finite direct sums into direct products,

b) for every S-scheme T, the following sequence is exact:

$$X(T \times \Sigma^p I_S) \to X(T \times I_S^p) \rightrightarrows X(T \times I_S^p \times \mathfrak{S}_p),$$

(the two parallel arrows being $X(id_T \times \pi)$ and $X(id_T \times \mu)$).

Every S-scheme satisfies (F); if $\mathcal{F}$ is an $\mathcal{O}_S$-module, $\mathcal{W}(\mathcal{F})$ satisfies (F); every projective limit of functors satisfying (F) also satisfies (F); if Y satisfies (F) and if X is an arbitrary S-functor, $\mathrm{Hom}_S(X, Y)$ satisfies (F).

Let X be a very good group (II 4.10) satisfying condition (F). Denoting by $x : I_S \to X$ a morphism which extends the unit section of X and resuming the notations of 6.2, one sees as above that $x_1 \cdots x_p : I_S^p \to X$ factors through $\Sigma^p I_S$:

$$\begin{tikzcd}
I_S^p \arrow[rr, "x_1 \cdots x_p"] \arrow[dr, "\mathrm{can.}"'] & & X \\
& \Sigma^p I_S \arrow[ur, "\Sigma^p(x)"'] &
\end{tikzcd}$$

and defines by composition a morphism

$$x^{(p)} : I_S \xrightarrow{i} \Sigma^p I_S \xrightarrow{\Sigma^p(x)} X$$

which we shall call the *symbolic p-th power* of x.

The endomorphism $x \mapsto x^{(p)}$ of $Lie(G/S)(S)$ is obviously compatible with base changes and is functorial in G. It would be interesting to know for which G this endomorphism makes $Lie(G/S)(S)$ a p-Lie algebra.

6.4.

The last definition of the symbolic p-th power, which we have just given, is particularly well-suited to computation. Here are some examples:

6.4.1. Let M be an "abstract" abelian group and $D_S(M)$ the diagonalizable S-group of type M (I 4.4.2). For every S-scheme T, one has therefore

$$D_S(M)(T) = \mathrm{Hom}_{(Ab)}(M, O(T)^\times).$$

Let x be an element of $Lie(D_S(M)/S)(S)$, i.e., a homomorphism of abelian groups

$$M \xrightarrow{x} \Gamma(S, O_S + dO_S)^\times$$

of the form $m \mapsto 1 + d\xi(m)$, where $\xi \in \mathrm{Hom}_{(Ab)}(M, O(S))$. With the notations of 6.2 and 6.3, the product $x_1 \cdots x_p$ associates with an element m of M the expression

$$(1 + d_1\xi(m)) \cdots (1 + d_p\xi(m))$$

i.e. $1 + \sigma_1\xi(m) + \sigma_2\xi(m)^2 + \cdots + \sigma_p\xi(m)^p$.

This expression indeed belongs to $O(\Sigma^p I_S)$. Projecting this into $O(S)[d]/(d^2)$ by annihilating $\sigma_1, \cdots, \sigma_{p-1}$ and sending σ_p to d, one sees that $x^{(p)}$ is the following homomorphism from M into $\Gamma(S, O_S + dO_S)^\times$:

$$m \mapsto 1 + d\xi(m)^p.$$

In summary, if one identifies $Lie(D_S(M)/S)(S)$ with $\mathrm{Hom}_{(Ab)}(M, O(S))$ as in II 5.1, the symbolic p-th power associates with ξ the homomorphism $\xi^{(p)} : m \mapsto \xi(m)^p$.

6.4.2. Let $\mathcal{F}$ be an O_S-module and G the group S-functor in abelian groups $\mathcal{W}(\mathcal{F})$ (cf. I, 4.6). Let y be an element of $\mathcal{W}(\mathcal{F})(S) = \Gamma(S, \mathcal{F})$ and y' its image in $\mathcal{W}(\mathcal{F})(I_S)$ by $\mathcal{W}(\mathcal{F})(I_S \to S)$.

One knows (cf. II, 4.4.2 and 4.5.1) that the map $y \mapsto dy'$ is an isomorphism of $O(S)$-modules from $\mathcal{W}(\mathcal{F})(S)$ onto $Lie(\mathcal{W}(\mathcal{F})/S)(S)$. If one sets $x = dy'$, the quantity x_i of 6.2 is none other than $d_i y''$, where y'' denotes the canonical image of y'[664] in $\mathcal{W}(\mathcal{F})(I_S^p)$. Consequently, the product $x_1 \cdots x_p$ is equal here to

$$x_1 + \cdots + x_p = (d_1 + \cdots + d_p)\, y'' = \sigma_1 y''$$

and belongs to $\mathcal{W}(\mathcal{F})(\Sigma^p I_S)$. Since the homomorphism $O(\Sigma^p I_S) \to O(I_S)$, which defines the morphism i of 6.1, annihilates σ_1, one sees that $x^{(p)}$ is zero. Therefore:

For every O_S-module $\mathcal{F}$, the operation $x \mapsto x^{(p)}$ in $Lie\mathcal{W}(\mathcal{F})$ is zero.

[664] We have corrected x to y'.

6.4.3. Let X be an S-scheme, G the group S-functor $\mathrm{Aut}_S\, X$ and D an S-derivation of the structure sheaf $\mathcal{O}_X$. According to 1.5, D can be identified with an I_S-automorphism x of X_{I_S}, inducing the identity on X, which one can describe as follows. If f is a section of $\mathcal{O}_S[d]/(d^2)$ of the form $a + bd$, set $D_{I_S} f = Da + (Db)d$; in other words, D_{I_S} is deduced from D by the base change $I_S \to S$; then the automorphism in question of X_{I_S} is associated with the endomorphism $f \mapsto f + (D_{I_S} f)d = a + (D(a) + b)d$ of $\mathcal{O}_S[d]/(d^2)$.

Similarly, let $D_{I_S^p}$ be the differential operator of $X_{I_S^p}$ deduced from D by the base change $I_S^p \to S$. With the notations of 6.2, the automorphism x_i of $X_{I_S^p}$ is then associated with the endomorphism $f \mapsto f + (D_{I_S^p} f)d_i$ of $\mathcal{O}_S[d_1, \cdots, d_p]/(d_1^2, \cdots, d_p^2)$. The product $x_1 \cdots x_p$ is therefore associated with the endomorphism

$$(1 + d_1 D_{I_S^p})(1 + d_2 D_{I_S^p}) \cdots (1 + d_p D_{I_S^p})$$

i.e., $1 + \sigma_1 D_{I_S^p} + \sigma_2 D_{I_S^p}^2 + \cdots + \sigma_p D_{I_S^p}^p$.

The coefficient of σ_p is $D_{I_S^p}^p$, which means that the Lie algebra isomorphism

$$\mathrm{D\acute{e}r}_S(\mathcal{O}_X) \xrightarrow{\sim} \mathrm{Lie}(\mathrm{Aut}_S\, X), \qquad D \mapsto x$$

(cf. 1.5), is also an isomorphism of p-Lie algebras.

6.4.4. Using the same method, one sees that, for every $\mathcal{O}_S$-module $\mathcal{F}$, the Lie algebra isomorphism

$$\mathrm{Lie}(\mathrm{Aut}_{\mathcal{O}_S\text{-mod.}}\, \mathbb{V}(\mathcal{F}) / S)(S) \xrightarrow{\sim} (\mathrm{End}_{\mathcal{O}_S\text{-mod.}}\, \mathbb{V}(\mathcal{F}))(S).$$

(cf. II 4.8) is also an isomorphism of p-Lie algebras.

6.4.5. [665] Let $\mathcal{U}$ be an $\mathcal{O}_S$-coalgebra in groups and G the group S-functor $\mathrm{Spec}^*\, \mathcal{U}$; suppose that G is representable. In this case, for every $T \to S$, one has defined in 5.5 and 6.1.1 two structures of p-Lie algebra on $L(T) = \mathit{Lie}(G)(T)$. Since one has a commutative diagram

$$\begin{tikzcd}
L(T) \arrow[rr, "\tau"] \arrow[d, "\alpha"'] \arrow[dr, "i"] & & \Gamma(T, \mathcal{O}_T) \\
U(G_T) & U(L(T)) \arrow[l, "\varphi"] \arrow[ur, "\psi"'] &
\end{tikzcd}$$

where $U(L(T))$ is the enveloping algebra of $L(T)$ and ϕ, ψ the algebra morphisms induced by α, τ, one sees that the two p-Lie algebra structures coincide: if one

[665] We have added the numbering 6.4.5, and expanded the original.

identifies $x \in L(T)$ with its image in $U(G_T)$ (resp. $\Gamma(T, \mathcal{U}_T)$), then $x^{(p)}$ is the image of the element x^p of $U(L(T))$ by ϕ (resp. ψ).

7. Radicial groups of height 1

[666] Let S be a scheme of characteristic $p > 0$. We shall say that an $\mathcal{O}_S$-algebra $\mathcal{A}$ (resp. an $\mathcal{O}_S$-p-Lie algebra $\mathcal{L}$) is *finite locally free* if the $\mathcal{O}_S$-module underlying $\mathcal{A}$ (resp. $\mathcal{L}$) is locally free and of finite type. If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, we know (cf. 5.5.2) that the group S-functor $\operatorname{Spec}^* \mathcal{U}_p(\mathcal{L})$ is represented by an S-group scheme $\mathfrak{G}_p(\mathcal{L})$, finite and locally free. We shall see that this S-group scheme is the solution of a universal problem (7.2) and we shall characterize the S-group schemes of the form $\mathfrak{G}_p(\mathcal{L})$ (7.4).

Definition 7.0.

[667] *Let $H = \operatorname{Spec} \mathcal{A}$ be a finite locally free S-group scheme. We say that H is* infinitesimal *if the unit section $\epsilon_H : S \to H$ is a homeomorphism, which is equivalent to saying that the augmentation ideal of $\mathcal{A}$ is locally nilpotent.*

7.1.

[668] Let $\mathcal{L}$ be a finite locally free $\mathcal{O}_S$-p-Lie algebra and let $\mathfrak{G}_p(\mathcal{L})$ be the affine S-group $\operatorname{Spec} \mathcal{U}_p(\mathcal{L})$. According to 5.5, one knows that $\mathcal{L}$ is identified with $Lie\mathfrak{G}_p(\mathcal{L})$.

Consider now a very good group S-functor G satisfying condition (F) of 6.3 and let $\phi : \mathfrak{G}_p(\mathcal{L}) \to G$ be a morphism of group S-functors. According to 6.3, the morphism of $\mathcal{O}_S$-Lie algebras $\operatorname{Lie} \varphi : \operatorname{Lie} \mathfrak{G}_p(\mathcal{L}) \to \operatorname{Lie} G$ is compatible with the symbolic p-th power. If we denote by $\operatorname{Hom}_p(\mathcal{L}, Lie G)$ the set of $\mathcal{O}_S$-Lie algebra morphisms which are compatible with the symbolic p-th power, one therefore has a map

$$\operatorname{Lie} : \operatorname{Hom}_{S\text{-Gr.}}(\mathfrak{G}_p(\mathcal{L}), G) \to \operatorname{Hom}_p(\mathcal{L}, \operatorname{Lie} G), \qquad \varphi \mapsto \operatorname{Lie} \varphi.$$

7.2. Theorem.

If $\mathcal{L}$ is a finite locally free $\mathcal{O}_S$-p-Lie algebra, the map

$$\operatorname{Hom}_{S\text{-gr.}}(\mathfrak{G}_p(\mathcal{L}), G) \to \operatorname{Hom}_p(\mathcal{L}, \operatorname{Lie} G)$$

is bijective in each of the following cases:

(i) G is an S-group scheme;

(ii) G is of the form $\operatorname{Aut}_S X$, where X is an S-scheme;

(iii) G is of one of the forms $\mathcal{W}(\mathcal{F})$ or $\operatorname{Aut}_{O_S-mod} \mathcal{W}(\mathcal{F})$, where $\mathcal{F}$ denotes a quasi-coherent $\mathcal{O}_S$-module.

The proof of the theorem rests on the following lemma:

[666] For the results of this section, one may also consult [DG70], § II.7, nos 3-4.
[667] We have added this definition (cf. [DG70], § II.4, 7.1), which will be used in 7.2.1.
[668] We have simplified the original, taking into account the additions made in 5.5.

Lemma. *If $\mathcal{L}$ is a finite locally free 0_S-p-Lie algebra, the S-group $G = \mathfrak{G}_p(\mathcal{L})$ is annihilated by the Frobenius morphism $Fr : G \to G^{(p)}$. In particular, G is infinitesimal.*

[669] Let in fact $\mathcal{U}$ be the restricted enveloping algebra of $\mathcal{L}$, $\mathcal{A} = \mathcal{U}^*$ the affine algebra of G, and $\mathcal{I} = Ker\epsilon_{\mathcal{A}}$ the augmentation ideal of $\mathcal{A}$. One has

$$(1)\mathcal{A} = \mathcal{I} \bigoplus \eta_{\mathcal{A}}(O_S),$$

where $\eta_{\mathcal{A}}$ denotes the unit section of $\mathcal{A}$, and since $\epsilon_{\mathcal{A}}$ (resp. $\eta_{\mathcal{A}}$) is the transpose of $\eta_{\mathcal{U}}$ (resp. $\epsilon_{\mathcal{U}}$), this decomposition corresponds by duality to the decomposition

$$(2)\mathcal{U} = \mathcal{J} \bigoplus \eta_{\mathcal{U}}(O_S),$$

where $\mathcal{J}$ is the augmentation ideal of $\mathcal{U}$; one therefore has $\mathcal{J} = \mathcal{I}^*$.

Let π denote the endomorphism $x \mapsto x^p$ of 0_S. We must show that the morphism $Fr : G \to G^{(p)}$ factors through the unit section of $G^{(p)}$, which is equivalent to saying (cf. 4.1.4 (c)) that the morphism $\Phi : a \bigotimes_\pi x \mapsto a^p x$ from $\mathcal{I} \bigotimes_\pi O_S$ into $\mathcal{A}$ is zero. Since $\mathcal{A}$ is finite locally free over 0_S, it suffices to see that the transposed morphism ${}^t\Phi$ is zero.

Now Φ is none other than the following composition

□ ⊗_π 0_S ──τ→ □ ⊗_π 0_S ──j(□)→ S^p □ ──b(□)→ □,

where τ is deduced from the inclusion $\mathcal{I} \hookrightarrow \mathcal{A}$, and $b(\mathcal{A})$ and $j(\mathcal{A})$ are defined as in 4.3.3 (i.e. $b(\mathcal{A})$ is induced by the multiplication of $\mathcal{A}$ and $j(\mathcal{A})$ sends $a \bigotimes_\pi 1$ to the image of $a \bigotimes \cdots \bigotimes a$ in $S^p\mathcal{A}$). Since the 0_S-dual module of $S^p\mathcal{A}$ is none other than the submodule $\Sigma^p\mathcal{U}$ of $\bigotimes^p \mathcal{U}$ formed by the sections invariant under the action of the symmetric group of order p, one sees that ${}^t\Phi$ is the following composed morphism:

□ ──a(□)→ Σ^p □ ──r(□)→ □ ⊗_π 0_S ──q→ □ ⊗_π 0_S,

where q is deduced from the projection $\mathcal{U} \to \mathcal{J}$ of kernel $\eta_{\mathcal{U}}(O_S)$, $a(\mathcal{U})$ is induced by the comultiplication of $\mathcal{U}$ and $r(\mathcal{U})$ vanishes on the symmetrized tensors and sends a section $x \bigotimes \cdots \bigotimes x$ to $x \bigotimes_\pi 1$ (confer 4.3.3).

It is clear that ${}^t\Phi \circ \eta_{\mathcal{U}} = 0$ and therefore, according to (2), it remains to see that ${}^t\Phi$ annihilates the augmentation ideal $\mathcal{J}$. Since ${}^t\Phi$ is an algebra morphism and since the ideal $\mathcal{J}$ is generated by $\mathcal{L}$ (identified with its image in $\mathcal{U}$), it suffices to see that ${}^t\Phi(x) = 0$ for every section x of $\mathcal{L}$. Now $-a(\mathcal{U})(x) = (p-1)!a(\mathcal{U})(x)$ is the symmetrization of $x \bigotimes 1 \bigotimes \cdots \bigotimes 1$, so its image by $r(\mathcal{U})$ is zero. This proves the first assertion of the lemma.

The second follows. Indeed, since every local section of $\mathcal{I}$ has p-th power zero and since $\mathcal{I}$ is an 0_S-module of finite type, $\mathcal{I}$ is locally nilpotent (explicitly, if V is an

[669] We have expanded the original in what follows. For another proof, see [DG70], § II.7, 3.9.

affine open of S such that $I = \Gamma(V, \mathcal{I})$ is generated by r elements, then $I^{r(p-1)+1} = 0$), whence $G_{red} = S_{red}$ and therefore the unit section $\epsilon_G : S \to G$ is a homeomorphism.

7.2.1. [670] We shall first prove assertion (ii) of Theorem 7.2. Let $\pi : X \to S$ be an S-scheme. Consider first an arbitrary infinitesimal S-group H. The morphisms ϕ from H into $\operatorname{Aut} X$ correspond bijectively to left actions $\mu : H \times X \to X$ of H on X. For such an action, if ϵ is the unit section of H, the composed morphism

$$X \simeq S \times X \xrightarrow{\varepsilon \times X} H \times X \xrightarrow{\mu} X$$

must be the identity. Since $(H \times X)_{red}$ is identified with X_{red}, one sees that μ must induce the identity on the associated reduced schemes. In particular, μ induces an action of H on each open of X, and one therefore obtains, for every open U of X, affine over S, a morphism of unital associative algebras:

$$\mathcal{A}(U) \longrightarrow \mathcal{A}(H) \otimes \mathcal{A}(U)$$

making $\mathcal{A}(U)$ an $\mathcal{A}(H)$-comodule on the left, in such a way that the restriction maps $\mathcal{A}(U) \to \mathcal{A}(U')$, for $U' \subset U$, are comodule morphisms. Conversely, every datum of this type comes from a unique left action $\mu : H \times X \to X$. On the other hand, one has the following lemma:

Lemma. *Let $X = \operatorname{Spec} \mathcal{C}$ be an affine S-scheme, $H = \operatorname{Spec} \mathcal{A}$ an infinitesimal S-group,*

and $\mathcal{U} = \mathcal{A}^* = \operatorname{Hom}_{O_S}(\mathcal{A}, O_S)$. The left actions of H on X correspond bijectively to the representations of the algebra $\mathcal{U}$ in the O_S-module $\mathcal{C}$ such that one has:

(a) $u(1_{\mathcal{C}}) = \varepsilon(u) \cdot 1_{\mathcal{C}}$

(b) $u(xy) = \sum_i v_i(x)\, w_i(y) \quad \text{if} \quad \Delta u = \sum_i v_i \otimes w_i.$

(In the formulas above, u denotes an arbitrary section of $\mathcal{U}$ on an affine open V of S, x and y sections of $\mathcal{C}$ on V; one denotes by $1_{\mathcal{C}}$ the unit section of $\mathcal{C}$, by ϵ and Δ the augmentation and the diagonal morphism of $\mathcal{U}$.)

Indeed, a left action μ of H on X is defined by a morphism of unital associative algebras:

$$\lambda : \mathcal{C} \to \mathcal{A} \otimes \mathcal{C}$$

making $\mathcal{C}$ an $\mathcal{A}$-comodule on the left. We shall denote by α the composed morphism

$$\mathcal{U} \otimes_{O_S} \mathcal{C} \xrightarrow{\mathcal{U} \otimes \lambda} \mathcal{U} \otimes_{O_S} \mathcal{A} \otimes_{O_S} \mathcal{C} \xrightarrow{\gamma \otimes \mathcal{C}} O_S \otimes_{O_S} \mathcal{C} \simeq \mathcal{C}$$

where γ is the "contraction" of $\mathcal{A}^* \otimes_{O_S} \mathcal{A}$ into O_S. Since $\mathcal{A}$ is finite locally free over O_S, one knows that the map $\lambda \mapsto (\gamma \otimes \mathcal{C})(\mathcal{U} \otimes \lambda)$ is a bijection from $\operatorname{Hom}_{O_S}(\mathcal{C}, \mathcal{A} \otimes \mathcal{C})$ onto $\operatorname{Hom}_{O_S}(\mathcal{U} \otimes \mathcal{C}, \mathcal{C})$. Moreover, one sees easily that the condition that λ define a structure of $\mathcal{A}$-comodule on the left (resp. be a morphism of unital associative

[670] We have expanded the original in what follows.

algebras) is equivalent, by duality, to the condition that α be a representation of $\mathcal{U}$ in $\mathcal{C}$ (resp. that α satisfy conditions (a) and (b)). This proves the lemma.

Moreover, it is clear that, for every representation of $\mathcal{U}$ in the `O_S`-module $\mathcal{C}$, the sections u of $\mathcal{U}$ which satisfy conditions (a) and (b) of the lemma form a subalgebra of $\mathcal{U}$.

In the particular case $H = \mathfrak{G}_p(\mathcal{L})$ of interest to us, these conditions will therefore be satisfied for all sections u of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, if they are true for the sections u of $\mathcal{L}$ (by identifying $\mathcal{L}$ with its image in $\mathcal{U}_p(\mathcal{L})$). Now, if u is a section of $\mathcal{L}$, conditions (a) and (b) simply mean that $u(1_{\mathcal{C}}) = 0$ and that $u(xy) = u(x)y + xu(y)$, i.e. that $\alpha(u)$ is an `O_S`-derivation of $\mathcal{C} = \mathcal{A}(X)$. Assertion (ii)

of 7.2 follows. Indeed, every morphism φ from `□_p = □_p(□)` into `Aut X` defines a

homomorphism of p-Lie algebras $Lie\phi$ from $\mathcal{L} = Lie\mathfrak{G}_p$ into $\pi_*(D\acute{e}r_{O_S} O_X)$, and conversely every datum of this type comes, according to what precedes, from a unique action $\mu : G \times X \to X$.

7.2.2. Let us now show how assertion (i) of Theorem 7.2 follows from (ii). Let G be an S-group scheme. If T is an S-scheme and x an element of $G(T)$, we denote by ℓ_x^T (resp. r_x^T) the left translation (resp. right translation) of `G_T` defined by x. The maps $\ell^T : x \mapsto \ell_x^T$ therefore determine a homomorphism ℓ from G into $\operatorname{Aut} G$. On the other hand, let f be a T-automorphism of `G_T`; one then defines ${}^x f$ as being equal to $(r_x^T)^{-1} f r_x^T$, i.e. for every $T' \to T$ and $g \in G(T')$, $({}^x f)(g) = f(gx)x^{-1}$. In this way G acts on the left on the group S-functor $\operatorname{Aut} G$, hence also on the functors $T \mapsto \operatorname{Hom}_{T-Gr.}(\mathfrak{G}_p(\mathcal{L}_T), \operatorname{Aut} G_T)$ and `T ↦ Hom_p(□_T, Lie(Aut G_T / T))`. On the other hand, the morphism $\ell^T : G_T \to \operatorname{Aut} G_T$ identifies `G_T` with the group of automorphisms of the T-scheme `G_T` commuting with right translations, and the derived morphism $Lie(\ell^T)$ identifies $LieG_T$ with the p-Lie algebra of `O_T`-derivations of O_{G_T} commuting with right translations (cf. II, 4.11.1); they therefore induce commutative squares

```
                                   Lie
Hom_{T-Gr.}(□_p(□_T), G_T)  ────────────→  Hom_p(□_T, Lie(G_T / T))

      □^T                                        Lie □^T

                                   Lie
Hom_{T-Gr.}(□_p(□_T), Aut G_T)  ──────→  Hom_p(□_T, Lie(Aut G_T / T)).
```

The images of the two vertical arrows are the subfunctors formed by the invariants under the action of the S-group G. Since the bottom horizontal arrow is invertible according to 7.2.1 and is compatible with the action of G, the top horizontal arrow is also invertible. This proves 7.2 (i).

7.2.3. Consider now the case where $G = \mathrm{Aut}_{O_S-mod.}\mathcal{W}(\mathcal{F})$.[671] Set $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, $\mathcal{A} = \mathcal{U}^*$ and $H = \mathfrak{G}_p(\mathcal{L}) = \mathrm{Spec}\,\mathcal{A}$. Since H is affine over S then, according to VI_B 11.6.1, a morphism of S-groups from H into $\mathrm{Aut}_{O_S-mod.}\mathcal{W}(\mathcal{F})$ is the same thing as a structure of $\mathcal{A}$-comodule on the right

$\mu : \mathcal{F} \to \mathcal{F} \otimes \mathcal{A}$.

Moreover, since $\mathcal{A}$ is finite locally free over O_S, this is equivalent to the datum of a representation

$$\alpha : \mathcal{U} \longrightarrow \mathrm{End}_{O_S}(\mathcal{F})$$

of $\mathcal{U}$ in $\mathcal{F}$. Finally, according to the universal property of $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$, to give such a morphism α is equivalent to giving its restriction ρ to $\mathcal{L}$ (identified with its image in $\mathcal{U}$), which is a p-Lie algebra morphism from $\mathcal{L}$ into $\mathrm{End}_{O_S}(\mathcal{F})$.

[672] Finally, consider the case where $G = \mathcal{W}(\mathcal{F})$, keeping the preceding notations. First, to give a morphism of S-functors $\phi : H \to \mathcal{W}(\mathcal{F})$ is equivalent to giving an element θ of $\Gamma(H, \mathcal{F} \otimes O_H)$, and since H is finite locally free over S, one has:

$\Gamma(H, \mathcal{F} \otimes O_H) = \Gamma(S, \mathcal{F} \otimes \mathcal{A}) = \mathrm{Hom}_{O_S}(\mathcal{U}, \mathcal{F})$.

The condition that ϕ be a group morphism then translates into the fact that θ, considered as morphism of O_S-modules $\mathcal{U} \to \mathcal{F}$, vanishes on $1_{\mathcal{U}}$ and on $\mathcal{J}^2/\mathcal{J}$, where $\mathcal{J}$ is the augmentation ideal of $\mathcal{U}$, whence

(1) $$\mathrm{Hom}_{S\text{-}gr.}(H, \mathcal{W}(\mathcal{F})) = \mathrm{Hom}_{O_S}(\mathcal{J}/\mathcal{J}^2, \mathcal{F}).$$

On the other hand, consider the quasi-coherent sheaf $[\mathcal{L}, \mathcal{L}]$, image of the morphism $\mathcal{L} \otimes \mathcal{L} \to \mathcal{L}$, $x \otimes y \mapsto [x, y]$; for every affine open V of S, one has $[\mathcal{L}, \mathcal{L}](V) = [\mathcal{L}(V), \mathcal{L}(V)]$. Then one has an exact sequence

(†) $$0 \to [\mathcal{L}, \mathcal{L}] \to \mathcal{L} \xrightarrow{\pi} \mathcal{J}/\mathcal{J}^2 \to 0,$$

where π is the composition of the inclusion $\mathcal{L} \hookrightarrow \mathcal{J}$ and the projection $\mathcal{J} \to \mathcal{J}/\mathcal{J}^2$.

Indeed, the question being local on S, we may suppose that S is affine with ring R and that $L = \mathcal{L}(S)$ is free with basis $(x_1, \cdots, x_r)$. Identifying L with its image in $U = U_p(L)$, let K be the sub-R-module of U direct sum of [L, L] and the submodule with basis the monomials $x_1^{n_1} \cdots x_r^{n_r}$ such that $n_1 + \cdots + n_r \geqslant 2$; one then verifies that K is a two-sided ideal of U. Since K is contained in J^2 (where J is the augmentation ideal of U) and contains all the products $x_i x_j$ (which generate J^2), one deduces that $K = J^2$, whence $J^2 \cap L = [L, L]$ and one has the exact sequence (†).

On the other hand, one knows from 6.4.2 that $Lie\mathcal{W}(\mathcal{F})$ is none other than $\mathcal{W}(\mathcal{F})$, the Lie bracket and the symbolic p-th power being zero. From this and from what precedes one deduces that

(2) $$\mathrm{Hom}_p(\mathcal{L}, \mathcal{F}) = \mathrm{Hom}_{O_S}(\mathcal{L}/[\mathcal{L}, \mathcal{L}], \mathcal{F}) = \mathrm{Hom}_{O_S}(\mathcal{J}/\mathcal{J}^2, \mathcal{F})$$

[671] In what follows, we have expanded (and simplified) the original, taking into account VI_B, 11.6.1.
[672] We have added the following. (The original indicated "the case of $\mathcal{W}(\mathcal{F})$ is analogous").

and this, combined with (1), completes the proof of Theorem 7.2.

7.3. Lemma.

If $\mathcal{L}$ is a finite locally free 0_S-p-Lie algebra, the morphism $j_{\mathcal{L}} : \mathcal{L} \to Lie\mathfrak{G}_p(\mathcal{L})$ of 5.5 is invertible.

[673] For the proof, see 5.5.1.

7.4.

To end this section, we shall give a characterization of the S-group schemes of the form $\mathfrak{G}_p(\mathcal{L})$, where $\mathcal{L}$ is a finite locally free 0_S-p-Lie algebra.

Let G be an S-group scheme, ϵ_G the unit section and $\mathcal{I}'$ the kernel of the morphism $\epsilon_G^{-1}(O_G) \to O_S$ corresponding to ϵ_G. The image of $Lie(G/S)(S)$ in $U(G)$ is identified, according to 2.5 and 1.3.1, with the morphisms of 0_S-modules from $\epsilon_G^{-1}(O_G)$ into 0_S which vanish on the unit section of $\epsilon_G^{-1}(O_G)$ and on $\mathcal{I}'^2$. One thus recovers the canonical isomorphism of $Lie(G/S)(S)$ onto $\mathrm{Hom}_{O_S}(\mathcal{I}'/\mathcal{I}'^2, O_S)$ of II, 3.3 and 4.11.4.[674] We shall set $\omega_{G/S} = \mathcal{I}'/\mathcal{I}'^2$ as in loc. cit., so that the sheaf $Lie(G/S)$ is identified with $\omega_{G/S}^* = \mathrm{Hom}_{O_S}(\omega_{G/S}, O_S)$.[675] Moreover, if $G = \mathfrak{G}_p(\mathcal{L})$, where $\mathcal{L}$ is a finite locally free 0_S-p-Lie algebra, one saw in 5.5.3 that $\omega_{G/S} = \mathcal{L}^*$.

Theorem. *If G is a group scheme over a scheme S of characteristic $p > 0$, the following assertions are equivalent:*

(i) There exists a finite locally free 0_S-p-Lie algebra $\mathcal{L}$ such that $G \simeq \mathfrak{G}_p(\mathcal{L})$.

(i') The 0_S-p-Lie algebra $Lie(G)$ is finite locally free and $G \simeq \mathfrak{G}_p(Lie(G))$.

(ii) G is affine over S, $\omega_{G/S}$ is a locally free 0_S-module of finite type and the affine algebra of G is locally isomorphic to the quotient of the symmetric algebra $S_{O_S}(\omega_{G/S})$ by the ideal generated by the p-th powers of the sections of $\omega_{G/S}$.

(iii) G is locally of finite presentation over S, of height $\leqslant 1$, and $\omega_{G/S}$ is locally free.

(iii') G is locally of finite type over S, of height $\leqslant 1$, and $\omega_{G/S}$ is locally free.

(iv) G is locally of finite presentation and flat over S, of height $\leqslant 1$.[676]

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[673] In order not to modify the numbering, the statement 7.3 has been preserved, although it has been included, with its proof, in 5.5.1.

[674] If G is affine over S and if $\mathcal{I}$ denotes the augmentation ideal of $\mathcal{A}(G)$, then $\mathcal{I}'/\mathcal{I}'^2$ is identified with $\epsilon_G^*(\mathcal{I}/\mathcal{I}^2)$, cf. loc. cit.

[675] We have added the following sentence.

[676] We have added, on the one hand, assertion (i'), implicit in the original, and on the other hand, assertions (iii') and (iv), pointed out by O. Gabber; assertion (iv) takes up a footnote of the original, which indicated: "The condition on $\omega_{G/S}$ is in fact useless, as one sees easily by reducing to the case where S is local with residue field k, and applying the theorem to the case of the group G_k". As pointed out by Gabber, this is inaccurate without a flatness hypothesis: if A is an Artinian local ring of characteristic $p > 0$ and J a proper non-zero ideal of A, let H be the subgroup $\mathrm{Spec}\, A[x]/(x^p, Jx)$ of $\alpha_{p,A}$ (i.e. for every A-algebra R, $H(R) = x \in R | x^p = 0 and Jx = 0$), then H is not flat over A so is not of the form $\mathfrak{G}_p(\mathcal{L})$, where $\mathcal{L}$ is a free p-Lie algebra of finite rank over A.

7.4.1. The equivalence (i) ⇔ (i') follows from 5.5.3 (i), the implications (ii) ⇒ (iii) ⇒ (iii') are clear, and one has (i) ⇒ (iv) since $\mathfrak{G}_p(\mathcal{L})$ is finite locally free and of height $\leqslant 1$, according to 5.5.2 and Lemma 7.2. Let us show that (i) entails (ii). Denote by $\mathcal{J}$ the augmentation ideal of $\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*$. One has already seen in 5.5.3 (ii) that $\omega_{G/S} = \mathcal{J}/\mathcal{J}^2$ is identified with $\mathcal{L}^*$, hence is finite locally free.

Now suppose S affine. There is then a section $\sigma : \omega_{G/S} \to \mathcal{J}$ of the projection $\mathcal{J} \to \mathcal{J}/\mathcal{J}^2$; it induces an algebra morphism $\sigma' : S_{O_S}(\omega_{G/S}) \to \mathcal{A}$ and, according to Lemma 7.2, σ' factors into a morphism

$$\varphi : S_{O_S}(\omega_{G/S}) / K \ \square\ \square,$$

where K denotes the ideal generated by the p-th powers of sections of $\omega_{G/S}$. If one filters $\mathcal{A}$ (resp. $S_{O_S}(\omega_{G/S})/K$) by the powers of $\mathcal{J}$ (resp. of the ideal generated by $\omega_{G/S}$), it is clear that ϕ induces an epimorphism of the associated graded modules. So ϕ is an epimorphism of locally free O_S-modules of the same rank (cf. 5.3.3); so ϕ is an isomorphism. This proves that (i) ⇒ (ii).

7.4.2. Suppose now G of height $\leqslant 1$ and locally of finite presentation over S.[677] Since the Frobenius morphism $Fr : G \to G^{(p)}$ is integral and factors through the unit section of $G^{(p)}$, then G is integral (hence affine) over S. Let then $\mathcal{A} = \mathcal{A}(G)$; since G is supposed locally of finite presentation over S, it follows that G is finite and of finite presentation over S, hence $\mathcal{A}(G)$ is an O_S-module of finite presentation (cf. EGA IV_1, 1.4.7). Let $\mathcal{J}$ be the augmentation ideal of $\mathcal{A}$; since $\mathcal{A} = \eta_{\mathcal{A}}(O_S) \oplus \mathcal{J}$ (where $\eta_{\mathcal{A}}$ is the unit section of $\mathcal{A}$), $\mathcal{J}$ is an O_S-module of finite presentation, and so is $\omega_G = \mathcal{J}/\mathcal{J}^2$. When one supposes G of height $\leqslant 1$ and locally of finite type over S, one obtains similarly that $\mathcal{A}(G)$, $\mathcal{J}$ and $\omega_G = \mathcal{J}/\mathcal{J}^2$ are O_S-modules of finite type.

So, under hypothesis (iii'), one obtains that $\omega_{G/S}$ is finite locally free over O_S, as is $\mathcal{L} = Lie(G/S) = \omega^*_{G/S}$. Let then $\mathcal{B} = \mathcal{U}_p(\mathcal{L})^*$ and $H = \mathfrak{G}_p(\mathcal{L}) = \operatorname{Spec}\mathcal{B}$. According to Theorem 7.2, the identity map of $\mathcal{L}$ corresponds to a group morphism from $H = \mathfrak{G}_p(\mathcal{L})$ to G, hence to a morphism of O_S-algebras $\theta : \mathcal{A} \to \mathcal{B}$. The task is to show that θ, which induces by definition an isomorphism of $\omega_{G/S}$ onto $\omega_{H/S}$, is an isomorphism.

For this, one may restrict to the case where S is affine. There is then a section τ of the projection $\mathcal{J} \to \omega_{G/S}$; it induces an algebra morphism $\tau' : S_{O_S}(\omega_{G/S}) \to \mathcal{A}$ and since every local section of $\mathcal{J}$ has p-th power zero (since $Fr : G \to G^{(p)}$ factors through the unit section of $G^{(p)}$), τ' induces a morphism of O_S-algebras ψ which fits in the commutative diagram below:

```
                                ψ
S_{O_S}(ω_{G/S}) / K  ──────→  □
            \                      |
              \                    | θ
             φ \                   |
```

677 We have expanded (and simplified) the original in what follows.

where K is the ideal generated by the p-th powers of sections of $\omega_{G/S}$. On the one hand, one shows as in 7.4.1 that ψ is an epimorphism of 0_S-modules. On the other hand, we saw in 7.4.1 that $\phi = \theta \circ \psi$ is an isomorphism. The same therefore holds for θ. This proves that (iii') ⇒ (i).

7.4.3. [678] Let us finally show that (iv) entails (iii). It suffices to show that $\omega_{G/S}$ is locally free, so one may suppose S affine with ring R. As remarked at the beginning of 7.4.2, hypothesis (iv) then entails that $G = \operatorname{Spec} A$, for an R-algebra A which is an R-module of finite presentation, as is $\omega_{G/A} = I/I^2$ (where I is the augmentation ideal of A). Since one supposes moreover that G is flat over S, then A is a finite locally free R-module (cf. [BAC] II, § 5.2, Th. 1 and cor. 2) and, according to loc. cit., to show that $\omega_{G/A}$ is locally free of finite rank, it suffices to show that $(\omega_{G/A})_{\mathfrak{m}}$ is flat for every maximal ideal $\mathfrak{m}$ of R. So one may suppose R local and A free of rank $n+1$, hence I free of rank n. Let $\mathfrak{m}$ be the maximal ideal of R and $k = R/\mathfrak{m}$.

Denote by I_k the augmentation ideal of A_k and r the dimension of $\omega_{G_k/k} = I_k/I_k^2$. Let $(e_1, \cdots, e_n)$ be a basis of I_k such that $(e_{r+1}, \cdots, e_n)$ is a basis of I_k^2, and let $x_1, \cdots, x_n$ be elements of I lifting the e_i. According to Nakayama's lemma, $(x_1, \cdots, x_n)$ is a basis of I over R. Let N be the sub-R-module of I with basis $(x_1, \cdots, x_r)$ and let B be the quotient of the symmetric algebra of N by the ideal generated by the elements x^p, for $x \in N$. Since every element of I has p-th power zero, one obtains a morphism of R-algebras

$$\psi : B \longrightarrow A.$$

According to 7.4.2, $\psi \otimes k$ is an isomorphism. It follows that $Coker\psi = 0$ and that, denoting $K = Ker\psi$, the morphism $\tau : K \otimes k \to B \otimes k$ is zero. But since ψ is surjective and A is flat over R, then τ is also injective, whence $K \otimes k = 0$. On the other hand, since A is an R-module of finite presentation, K is an R-module of finite type (cf. [BAC] I, § 2.8, Lemma 9), whence $K = 0$ by Nakayama. So ψ is an isomorphism of R-algebras, and since $\psi^{-1}(I)$ contains the augmentation ideal J of B, it follows that $\psi^{-1}(I) = J$, and therefore ψ^{-1} induces an isomorphism of R-modules from I/I^2 onto $J/J^2 = N$. This proves that $\omega_{G/S}$ is finite locally free, whence the implication (iv) ⇒ (iii). This completes the proof of Theorem 7.4.

Remark 7.5.

[679] It obviously follows from Theorems 7.2 and 7.4 that the functors $G \mapsto Lie(G)$ and $\mathcal{L} \mapsto \mathfrak{G}_p(\mathcal{L})$ induce equivalences, quasi-inverses of each other, between the category of S-groups locally of finite presentation and flat, of height $\leqslant 1$, and the

[678] We have added this paragraph to prove that (iv) ⇒ (iii), cf. N.D.E. (71).
[679] We have added this remark.

full subcategory of that of $\mathcal{O}_S$-p-Lie algebras formed by the finite locally free $\mathcal{O}_S$-p-Lie algebras.

8. The case of a base field

8.1.

Let us now summarize the results obtained in the case where S is the spectrum of a field k of characteristic $p > 0$. Let us then say that a k-group scheme is *algebraic* if the underlying scheme is of finite type over k. In this case, according to Theorem 7.2, one obtains:[680]

Theorem 8.1.1. *The functor $\mathfrak{G}_p$, which to every p-Lie algebra $\mathcal{L}$ of finite dimension over k associates the k-group $\mathfrak{G}_p(\mathcal{L})$, is left adjoint to the functor which to every algebraic k-group G associates $Lie(G)$.*

Combining this with Theorem 7.4, one obtains:

Theorem 8.1.2. *The functors $\mathfrak{G}_p : \mathcal{L} \mapsto \mathfrak{G}_p(\mathcal{L})$ and $G \mapsto Lie(G)$ induce equivalences, quasi-inverses of each other, between the category of p-Lie algebras of finite dimension over k, and that of algebraic k-groups of height $\leqslant 1$.*

Then, since $\mathfrak{G}_p$ is a left adjoint functor, it commutes with inductive limits,[681] hence in particular with the formation of cokernels. On the other hand, if one has two morphisms $\phi : G \to H$ and $\phi' : G' \to H$ between algebraic k-groups of height $\leqslant 1$, then the fibered product $G \times_H G'$ is again an algebraic k-group of height $\leqslant 1$ (since the morphism $Fr : G \to G^{(p)}$ commutes with fibered products). So the inclusion of the category of algebraic k-groups of height $\leqslant 1$ into that of all algebraic k-groups commutes with fibered products, hence in particular with the formation of kernels. From this one deduces the:

Corollary 8.1.3. *The functor $\mathfrak{G}_p$ is exact, in the following sense. If $\pi : \mathcal{L}_1 \to \mathcal{L}_2$ is a surjective morphism between p-Lie algebras of finite dimension over k and if i is the inclusion of $\mathcal{L}_0 = Ker\pi$ in $\mathcal{L}_1$, one has an exact sequence of algebraic k-groups:*

$$1 \longrightarrow \mathfrak{G}_p(\mathcal{L}_0) \xrightarrow{\mathfrak{G}_p(i)} \mathfrak{G}_p(\mathcal{L}_1) \xrightarrow{\mathfrak{G}_p(\pi)} \mathfrak{G}_p(\mathcal{L}_2) \longrightarrow 1.$$

[682]

Indeed, according to what precedes, $\mathfrak{G}_p(i)$ induces an isomorphism of $\mathfrak{G}_p(\mathcal{L}_0)$ onto $Ker(\mathfrak{G}_p(\pi))$ (this kernel being the same in the category of all algebraic k-groups H or in that of H of height $\leqslant 1$), and $\mathfrak{G}_p(\pi) : \mathfrak{G}_p(\mathcal{L}_1) \to \mathfrak{G}_p(\mathcal{L}_2)$ identifies $\mathfrak{G}_p(\mathcal{L}_2)$ with the quotient of $\mathfrak{G}_p(\mathcal{L}_1)$ by $\mathfrak{G}_p(\mathcal{L}_0)$ in the category of algebraic k-groups.

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[680] We have added the numbering 8.1.1 to 8.1.3, to highlight the results stated there.
[681] We have expanded what follows, as well as the proof of the corollary below.
[682] Moreover, according to VI_A, 3.2, $\mathfrak{G}_p(\mathcal{L}_2)$ represents the (fppf) sheaf quotient of $\mathfrak{G}_p(\mathcal{L}_1)$ by $\mathfrak{G}_p(\mathcal{L}_0)$.

Remark 8.1.4. [683] Let $\phi : G \to H$ be a morphism of k-groups and $K = Ker(\phi)$. Suppose ϕ covering for the (fpqc) topology (this will be the case, in particular, if ϕ is covering for a less fine topology, for example the (fppf) topology). Then, on the one hand, ϕ is a K-torsor above H (cf. IV 5.1.7.1). On the other hand, (cf. IV 6.3.1) there exists a covering of H by affine opens S_i, and for each i an affine faithfully flat morphism $T_i \to S_i$ factoring through ϕ. Then $G \times_H T_i$ is T_i-isomorphic to $K \times T_i$, hence faithfully flat over T_i, and therefore, by (fpqc) descent, $G \times_H S_i \to S_i$ is faithfully flat, so that ϕ is faithfully flat.

Conversely, if ϕ is faithfully flat and quasi-compact (resp. and locally of finite presentation), it is covering for the (fpqc) topology (resp. (fppf)), cf. IV 6.3.1. Recall finally that a morphism of sheaves is covering if and only if it is an epimorphism, cf. IV 4.4.3. One therefore obtains, in particular, that a quasi-compact morphism of k-groups is faithfully flat if and only if it is an epimorphism of (fpqc) sheaves.

8.2. Proposition.

Consider an exact sequence[684] *of algebraic groups over a field k of characteristic $p > 0$*

$$1 \to G' \xrightarrow{\tau} G \xrightarrow{\pi} G'' \to 1$$

and the following assertions:

(i) The morphism π is smooth.

(ii) G' is smooth.

(iii) For every integer $n > 0$, the following sequence, induced by τ and π, is exact:

$$1 \to Fr^n G' \to Fr^n G \to Fr^n G'' \to 1.$$

(iv) The morphism $Fr\pi : FrG \to FrG''$ is an epimorphism of (fppf) sheaves.

(v) The morphism $Lie(\pi) : Lie(G) \to Lie(G'')$ is surjective.

Then one has the implications (i) ⇔ (ii) ⇒ (iii) ⇒ (iv) ⇔ (v) and all the assertions are equivalent when G is smooth over k.

Indeed, (i) is equivalent to (ii) according to VI_B 9.2 (vii), and it is clear that (iii) implies (iv). On the other hand, the equivalence of (iv) and (v) follows from 8.1.3.

The implication (ii) ⇒ (iii) follows from the diagram:

$$\begin{tikzcd}
1 \arrow[r] & G' \arrow[r, "\tau"] \arrow[d, "Fr^n(G'/k)"] & G \arrow[r, "\pi"] \arrow[d, "Fr^n(G/k)"] & G'' \arrow[r] \arrow[d, "Fr^n(G''/k)"] & 1 \\
1 \arrow[r] & G'^{(p^n)} \arrow[r, "\tau^{(p^n)}"] & G^{(p^n)} \arrow[r, "\pi^{(p^n)}"] & G''^{(p^n)} \arrow[r] & 1
\end{tikzcd}$$

[683] We have added this remark, pointed out by O. Gabber, which will be useful in 8.3.1.

[684] i.e. π is faithfully flat and i is an isomorphism of G' onto $Ker\pi$, so that G'' represents the (fppf) sheaf quotient of G by G', cf. VI_A, 3.2 and 5.2.

whose two rows are exact: since $Fr^n(G'/k)$ is an epimorphism of (fppf) sheaves according to Corollary 8.3.1 below, π induces an epimorphism of $Fr^n G$ onto $Fr^n G''$ (generalize the snake lemma to sheaves of groups not necessarily commutative).

Similarly, when G is smooth over k, $Fr(G/k)$ is an epimorphism, so if moreover $Fr\pi$ is an epimorphism, the same snake lemma applied to the diagram above for $n = 1$ shows that $Fr(G'/k)$ is an epimorphism, so G' is smooth over k, according to 8.3.1 below.

8.3. Proposition.

If G is a group locally of finite type[685] *over a field k of characteristic $p > 0$, there exists an integer n_0 such that $G/(Fr^n G)$ is smooth over k for $n \geqslant n_0$.*

Since the construction of $G/(Fr^n G)$ commutes with extension of the base field (4.1.1 and VI_A, 3.3.2), we may suppose k perfect. In this case, G_{red} is a k-group locally of finite type (cf. VI_A 0.2) and one has the following commutative and exact diagram, where one has denoted by H the k-scheme $G_{red}\ G$:

$$\begin{tikzcd} 1 \arrow[r] & G_{red} \arrow[r] \arrow[d, "Fr^{n}(G_{red}/k)"] & G \arrow[r] \arrow[d, "Fr^{n}(G/k)"] & H \arrow[d, "Fr^{n}(H/k)"] \\ 1 \arrow[r] & G^{(p^{n})}_{red} \arrow[r] & G^{(p^{n})} \arrow[r] & H^{(p^{n})}. \end{tikzcd}$$

Now H is the spectrum of a finite local k-algebra with residue field k (cf. VI_A, 5.6.1). Consequently, there exists an integer n_0 such that, for every $n \geqslant n_0$, $Fr^n(H/k)$ factors through the "unit" section of $H^{(p^n)}$. It follows that, for $n \geqslant n_0$, $Fr^n(G/k)$ factors through $G_{red}^{(p^n)}$ and therefore, according to VI_A, 5.4.1, one has a commutative diagram

$$\begin{tikzcd} G \arrow[rr, "Fr^n(G/k)"] \arrow[dr, "\pi"'] & & G_{red}^{(p^n)} \\ & G/(Fr^n G) \arrow[ur, "i"'] & \end{tikzcd}$$

where i is a closed immersion (and π is the canonical projection). Since moreover i induces a homeomorphism of the underlying topological spaces, it is therefore an isomorphism. Since k is perfect, $G_{red}^{(p^n)}$ is smooth over k (VI_A, 1.3.1), and therefore $G/(Fr^n G)$ is smooth over k, for every $n \geqslant n_0$.

[685] We have replaced "algebraic" by "locally of finite type".

8.3.1. Corollary. *Let G be a group locally of finite type over a field k of characteristic $p > 0$ and let n be an integer $\geqslant 1$.*[686] *The following conditions are equivalent:*

(i) G is smooth over k.

(ii) $Fr^n(G/k) : G \to G^{(p^n)}$ is an epimorphism of (fppf) sheaves.

(iii) $Fr^n(G/k) : G \to G^{(p^n)}$ is faithfully flat.

First, since G is locally of finite type over k, $Fr^n(G/k)$ is of finite presentation, so the equivalence of (ii) and (iii) follows from 8.1.4. Suppose G smooth over k, hence G reduced. Then, since $Fr^n(G/k)$ is surjective, it is faithfully flat (cf. VI_A, 6.2 or VI_B, 1.3).

Conversely, suppose $Fr^n(G/k)$ faithfully flat. Since $Fr^n(G^{(p^n)}/k)$ is deduced from $Fr^n(G/k)$ by base change (cf. 4.1.3), it is therefore also faithfully flat, as is the composition:

Fr^{2n}(G/k) : G G^{(p^n)} G^{(p^{2n})}.

One thus obtains that, for every $m \in \mathbb{N}$, $Fr^{mn}(G/k) : G \to G^{(p^{mn})}$ is faithfully flat, hence induces an isomorphism $G/(Fr^{mn}G) \simeq G^{(p^{mn})}$ (cf. VI_A, 5.4.1). Now, according to Proposition 8.3, $G^{(p^{mn})}$ is smooth over k for m large, so G is also, by (fpqc) descent (cf. EGA IV_4, 17.7.1).

8.4.

In the two statements which end this Exposé, we return to the case of a field k of arbitrary characteristic.

When k is of characteristic 0 (resp. $p > 0$), let n be an integer $\geqslant 1$ (resp. an integer $\geqslant 1$ and coprime to p); in both cases, we say simply that n is *coprime to the characteristic of k*. Moreover, if G is a group scheme over k, we denote by $n_G : G \to G$ the morphism of k-schemes which sends an element x of $G(T)$ to $x^n \in G(T)$, when T is a k-scheme.

Proposition. *Let G be an algebraic group over a field k and n an integer coprime to the characteristic of k. Then $n_G : G \to G$ is an étale morphism.*

[687] According to VI_B 1.3, it suffices to show that n_G is étale at the origin. Let A be the local ring of G at the origin and I the maximal ideal of A. According to II 3.9.4, the map `Lie(n_G) : Lie(G) → Lie(G)`, which is induced by n_G, is the homothety of ratio n. It is therefore an isomorphism, as is the endomorphism induced by n_G on $I/I^2 = Lie(G)^*$. If k is of characteristic 0, G is smooth over k (VI_B 1.6.1, see also VII_B 3.3.1), so the canonical morphism $S(I/I^2) \to gr_I(A)$ is an isomorphism, where $gr_I(A)$ denotes the graded module associated with the I-adic filtration. It follows that n_G induces an automorphism of $gr_I(A)$, hence also of the completion Â of A, hence n_G is étale at the origin (cf. EGA IV_4, 17.6.3).

[686] We have made explicit the equivalence between (ii) and (iii) and we have expanded the proof.
[687] We have changed in the statement "étale at the origin" to "étale", and we have added the following sentence.

If the characteristic is $p > 0$ and if G is of height $\leqslant 1$, then A is isomorphic to the quotient of the symmetric algebra of $\omega_{G/k} = I/I^2$ by the ideal generated by the p-th powers of the elements of $\omega_{G/k}$ (cf. 7.4); one may then apply the “same” reasoning as in characteristic 0, and one obtains that n_G induces an automorphism of A.

If G is of height $\leqslant r$ and if we suppose our assertion proved for groups of height $\leqslant r-1$, denote by B, A and A' the affine algebras of `Fr G`, G and `G' = Fr G \ G`, and n_B, n_A and $n_{A'}$ the endomorphisms of B, A and A' which are induced by n_{FrG}, n_G and $n_{G'}$.[688] Let $I' = I \cap A'$ be the maximal ideal of A', since one has a cartesian square

$$\begin{tikzcd} \mathrm{Fr}\,G \arrow[r] \arrow[d] & G \arrow[d] \\ e \arrow[r] & G' \end{tikzcd}$$

one has $B = A/I'A$. Observe that $n_{A'}$ (resp. n_B) is none other than the endomorphism induced by n_A on A' (resp. on B). According to VI_A 3.2, A is a faithfully flat A'-module, and since A' is an Artinian local ring (G' being an algebraic k-group of height $\leqslant r-1$), it follows that A is a free A'-module. Since the restriction of n_A to A' is $n_{A'}$, which is an isomorphism according to the inductive hypothesis, it follows from Nakayama’s lemma that n_A will be an automorphism if the endomorphism it induces on $A/I'A$ is one.

Now this endomorphism is none other than `n_B`, which is an automorphism since `B` is of

height $\leqslant 1$. So n_A is an automorphism.

Finally, when G is an arbitrary algebraic group over a field of characteristic $p > 0$, what precedes shows that n_G induces automorphisms of the k-schemes $Fr^r G$; these schemes are affine over k and have as algebras the quotients of the local algebra A by the ideal I^{p^r} generated by the p^r-th powers of the elements of I. Since n_G defines automorphisms of the algebras A/I^{p^r}, one sees by passage to the projective limit that n_G induces an automorphism of $\hat{A}$, hence n_G is étale at the origin (EGA IV_4, 17.6.3).

8.5. Proposition.

Let G be a finite algebraic group, of rank n over the field k. Then $n_G : G \to G$ is the zero morphism of G.

Let us point out at once the following corollary, obtained by combining 8.4 and 8.5:[689]

[688] We have expanded the original in what follows.

[689] We have added this corollary, indicated implicitly in the original by: “(confer 8.4)”. For another proof of the corollary, not using 8.5, see for example [TO70], Lemma 5.

Corollary 8.5.1. *Let G be a finite algebraic group, of rank n over the field k. If n is coprime to the characteristic of k, then G is étale over k.*

Let us now prove 8.5. Let H be a normal subgroup of G of rank m over k. Denote by $\lambda : H \times G \to G$ the morphism induced by the multiplication of G. Then, with the notations of VI_A 3.2, one has a cartesian square:

$$\begin{array}{ccc} H \times G & \xrightarrow{\lambda} & G \\ pr_2 \downarrow & & \downarrow \pi \\ G & \longrightarrow & H \backslash G. \end{array}$$

Since $\pi : G \to H\ G$ is faithfully flat, quasi-compact (VI_A 3.2), and since pr_2 is locally free of rank m, it follows from EGA IV₂, 2.5.2, that $G \to H\ G$ is locally free of rank m. Denoting $r = rg_k(H\ G)$, one therefore has $n = rg_k(G) = rm$.

On the other hand, one has an exact sequence of "abstract" groups

$1 \to H(T) \to G(T) \to (H \backslash G)(T)$

for every k-scheme T; it is therefore clear that n_G is zero if m_H and $r_{H\ G}$ are. If one takes for H the neutral component G_0 of G, then $G_0\ G$ is étale (cf. VI_A 5.5.1), so that one may suppose G étale over k or else infinitesimal (cf. 7.0).

If G is étale, one reduces, by extension of the base field, to the case where k is algebraically closed. In this case, G is a constant group (cf. I 4.1), and the statement is classical.

If G is infinitesimal and non-zero, k is necessarily of characteristic $p > 0$ (cf. VI_B 1.6.1 or VII_B 3.3.1); the subgroups $Fr^n G$ then form a composition series of G, whose quotients are of height $\leqslant 1$.

This reduces us to the case where G is of height $\leqslant 1$. Let then A (resp. L) be the affine algebra (resp. the Lie algebra) of G and $U = U_p(L)$. According to 7.4, one has $G = \mathfrak{G}_p(L)$ whence $A = U^*$; so if $\dim_k L = r$, the rank of G over k is p^r (cf. 5.3.3). We shall therefore study the morphism $p_G : G \to G$ defined by raising to the p-th power; it induces an endomorphism p_A of A and, by duality, an endomorphism p_U of U.

Let I be the augmentation ideal of A; we shall show that $p_A(I) \subset I^p$. Supposing this established, one will therefore have $p_A^r(I) \subset I^{p^r}$. On the other hand, one knows that $I^{r(p-1)+1} = 0$ (since I is generated by r elements of p-th power zero). Since $p^r > r(p-1)$, it follows that $p_A^r(I) = 0$, so p_G^r is the zero morphism. It therefore remains to show the assertion:

$$(*) p_A(I) \subset I^p.$$

For every integer $s \geqslant 1$, we shall denote $m_A^{s-1} : A^{\otimes s} \to A$ (resp. $\Delta_U^{s-1} : U \to U^{\otimes s}$) the map induced by the multiplication m_A of A (resp. the comultiplication

Δ_U of U). Then p_A is equal to the following composition:

$$A \xrightarrow{\Delta_A^{p-1}} A^{\otimes p} \xrightarrow{m_A^{p-1}} A,$$

and since the transpose of m_A (resp. Δ_A) is Δ_U (resp. m_U), the endomorphism $p_U = {}^t p_A$ of U is the following composition:

$$U \xrightarrow{\Delta_U^{p-1}} U^{\otimes p} \xrightarrow{m_U^{p-1}} U.$$

[690] Let J be the augmentation ideal of U; one has $U = k1_U \bigoplus J$ and one will denote by π the projection $U \to J$ of kernel $k1_U$. For every integer $s \geqslant 1$, denote $(I^s)^\perp$ the orthogonal of I^s in $A^* = U$, i.e., $(I^s)^\perp$ is the set of $u \in U$ such that the composition below is zero:

$$I^{\otimes s} \xrightarrow{m_A^{s-1}} I \xrightarrow{u} k.$$

Since the transpose of m_A is Δ_U, one sees that $(I^s)^\perp$ is the vector subspace P_{s-1} formed by the $u \in U$ such that $\Delta_U^{s-1}(u)$ vanishes on $I^{\bigotimes s}$, i.e., denoting by $\bar{\Delta}_U^{s-1}$ the composition of Δ_U^{s-1} and the projection $\pi^{\bigotimes s} : U^{\bigotimes s} \to J^{\bigotimes s}$, one obtains that

$$(I^s)^\perp = P_{s-1} = \operatorname{Ker} \bar{\Delta}_U^{s-1}$$

(see also VII_B, 1.3.6). So, to prove assertion $(*)$, one must show that the transpose map $p_U = {}^t p_A$ sends P_{p-1} into $I^\perp = k1_U$. Since $p_U(1_U) = 1_U$, it suffices to show the assertion below, where P^+_{p-1} denotes $J \cap P_{p-1}$:

$$(**) p_U(P^+_{p-1}) = 0.$$

On the other hand, one shows easily, by induction on s, that P^+_{s-1} is the vector subspace of U generated by the products $x_1 \cdots x_t$, with $1 \leqslant t \leqslant s-1$ and $x_i \in L$ (cf. VII_B 4.3). Now, if $x_1, x_2, \cdots, x_t$ are elements of L, one has:

$$p_U(x_1 x_2 \cdots x_t) = m_U^{p-1}\Big(\prod_{j=1}^t \sum_{i=1}^p 1 \otimes \cdots \otimes x_j \otimes \cdots \otimes 1 \Big) \quad (x_j \text{ in position } i$$

It is clear that the expression $\prod_j \sum_i 1 \bigotimes \cdots \bigotimes x_j \bigotimes \cdots \bigotimes 1$ is a sum of p^t terms x_h indexed by the maps h from $1, \cdots, t$ into $1, \cdots, p$.[691] Such a map h defines an ordered partition p_h of $1, \cdots, t$ into at most p parts. Indeed, denote $i_1 < \cdots < i_r$ the elements of the image of h and, for $s = 1, \cdots, r$, set $I_s = h^{-1}(i_s)$ and $x_{I_s} = \prod_{j \in I_s} x_j$, the product being taken in increasing order. Then h corresponds to the p-tensor

$$1 \bigotimes \cdots \bigotimes x_{I_1} \bigotimes \cdots \bigotimes x_{I_r} \bigotimes \cdots \bigotimes 1$$

(where each x_{I_s} is in position i_s), and its image by m_U^p is the product:

$$x_{I_1} \bigotimes \cdots \bigotimes x_{I_r}$$

[690] We have expanded the original in the paragraph that follows.
[691] We have expanded the original in what follows, replacing the notion of preorder by the equivalent notion of ordered partition.

which depends only on the ordered partition $p = (I_1, \cdots, I_r)$, and which one will denote x_p. For p fixed, x_p is obtained for all choices of $i_1 < \cdots < i_r$ in $1, \cdots, p$, and one therefore obtains the equality

```
p_U(x_1 x_2 ··· x_t) = ∑_p (p choose n(p)) x_p,
```

where p ranges over the set of ordered partitions of $1, \cdots, t$ into at most p parts, and where $n(p)$ denotes the number of parts of p. (One has $1 \leqslant n(p) \leqslant \min(t, p)$.)

When $t < p$, all the terms `(p choose n(p))` are therefore zero, so that $p_U(x_1 \cdots x_t) = 0$. So p_U vanishes on P^+_{p-1}, which proves assertion (∗∗), and hence (∗), and completes the proof of 8.5.

Corollary 8.5.2. [692] *Let S be a reduced scheme and G a finite locally free S-group of rank n. Then $n_G : G \to G$ is the zero morphism of G.*

Indeed, let S' be the sum of the Spec $O_{S,\eta}$, for η ranging over the maximal points of S. Since S is reduced, the morphism $S' \to S$ is schematically dominant, and the same holds for the morphism $f : G_{S'} \to G$, since G is finite locally free over S (cf. EGA IV_3, 11.10.5). Since $G \to S$ is affine, hence separated, the locus of coincidence of n_G and the zero morphism is a closed subscheme of G, and it majorizes f according to 8.5, hence equals G, i.e. n_G is the zero morphism.

Remark 8.5.3. Let us also point out that, according to a theorem of P. Deligne (see [TO70], p. 4), if G is a commutative finite locally free S-group of rank n over an arbitrary base S, then $n_G = 0$.

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Exposé VII_B. Infinitesimal study: formal groups

by P. Gabriel

[692] We have added this corollary, signaled in Exp. VIII, Remark 7.3.1.

B) Formal groups

[693]

The study of formal groups is usually of extreme simplicity. If this does not appear clearly in the pages that follow, the responsibility lies with an arithmetician who claims to know formal groups over "something other than fields".[694] We have therefore unrolled, for formal groups "locally free over inverse limits of artinian rings", the generalities one ordinarily states for formal groups defined over a field. For a more detailed study of the latter, we refer the reader to the 1964/65 algebraic geometry seminar of Heidelberg–Strasbourg.[695]

0. Reminders on pseudocompact rings and modules

This paragraph contains a few technical preliminaries; we recall and complete in it some results from [CA] (*Des catégories abéliennes*, Bull. Soc. Math. France 90, 1962).

0.1.

A *left pseudocompact ring* is a topological ring with unit element, separated and complete, which possesses a basis of neighborhoods of 0 consisting of left ideals ℓ of finite colength (i.e. $length_A(A/\ell) < +\infty$). We shall assume here that A is commutative, so that there is no need to distinguish "between left and right".

In particular, the quotients A/ℓ are artinian rings and A is identified with the topological inverse limit of these rings, each endowed with the discrete topology.

A complete noetherian local ring $(A, \mathfrak{m})$ is obviously pseudocompact.[696]

0.1.1. Every closed ideal I of A is the intersection of the open ideals containing it.[697] Every closed maximal ideal is therefore open. Moreover, if ℓ is an open ideal of A, the maximal ideals of A/ℓ are in bijective correspondence with the maximal ideals $\mathfrak{m}$ of A containing ℓ; these are therefore both open and closed. Consequently, every closed maximal ideal is an open (and hence closed) maximal ideal; the converse being evident. We denote by $\Upsilon(A)$ the set of these ideals.

[693] N.D.E.: Version of 13/10/2024.

[694] N.D.E.: The interest of formal groups over a complete noetherian local ring appears, for example, in the work of Lubin and Tate (cf. [LT65]). The study of formal groups over an arbitrary base, and questions of lifting and deformation, in particular for Barsotti–Tate groups ("p-divisible groups"), has given rise to an abundant literature; cf. for example [LT66, Ta67, Gr74, Me72, La75, Fo77, Il85, Br00]. In particular, the results of the present Exposé are largely taken up again in Chapter I of [Fo77].

[695] N.D.E.: The editors have found only such a seminar dated 1965/66, entitled "Linear algebraic groups", in which the notion of formal group does not appear; see instead [De72].

[696] N.D.E.: (when endowed with the $\mathfrak{m}$-adic topology).

[697] N.D.E.: Indeed, if $x \notin I$, there exists an open ideal ℓ such that $(x + \ell) \cap I = \emptyset$, and then $I + \ell$ is an open ideal not containing x. On the other hand, in what follows, we have made explicit the fact that every "closed maximal" ideal is maximal and open.

If ℓ is an open ideal of A and if $\mathfrak{m} \in \Upsilon(A)$, the localization $(A/\ell)_{\mathfrak{m}}$ is therefore a local ring if $\mathfrak{m}$ contains ℓ and zero otherwise. Since the ring A/ℓ is artinian, it is a direct product of finitely many local rings, which one can write

$$A/\ell \simeq \prod_{\mathfrak{m} \in \Upsilon(A)} (A/\ell)_{\mathfrak{m}}.$$

From this one deduces "canonical" isomorphisms

$$A \simeq \lim_{\ell} (A/\ell) \simeq \lim_{\ell} \prod_{\mathfrak{m}} (A/\ell)_{\mathfrak{m}} \simeq \prod_{\mathfrak{m}} \lim_{\ell} (A/\ell)_{\mathfrak{m}} \simeq \prod_{\mathfrak{m}} A_{\mathfrak{m}},$$

where one has set

$$A_{\mathfrak{m}} = \lim_{\ell} (A/\ell)_{\mathfrak{m}}.$$

This local component $A_{\mathfrak{m}}$ is a filtered inverse limit of artinian local rings, endowed with the discrete topology; it is therefore a local ring which is pseudocompact for the inverse-limit topology.[698]

0.1.2. Let $r(A)$ be the intersection of the open maximal ideals of A, that is, the cartesian product of the ideals $\mathfrak{m}A_{\mathfrak{m}}$ when one identifies A with $\prod_{\mathfrak{m}} A_{\mathfrak{m}}$. For every open ideal ℓ of A, the image of $r(A)$ in A/ℓ is contained in the radical of A/ℓ. Some power of this image is therefore zero, so that $r(A)^n$ is contained in ℓ when n is large enough. The sequence of $r(A)^n$ therefore tends to 0.

The same holds for the sequence of x^n, when x belongs to $r(A)$. In other words, every element of $r(A)$ is topologically nilpotent and the converse is clear. It follows that the sequence with general term $1 + x + \cdots + x^n$ is convergent and converges to $1/(1-x)$ when $x \in r(A)$. This shows that $r(A)$ is the Jacobson radical of A, i.e., the intersection of all maximal ideals of A (cf. Bourbaki, *Algèbre*, Chap. 8, § 6, th. 1).[699]

Remarks.[700] a) If $\mathfrak{p}$ is an open prime ideal of A, then since $A/\mathfrak{p}$ is artinian, $\mathfrak{p}$ is a maximal ideal. Consequently, $\Upsilon(A)$ equals the set of open prime ideals of A.

b) Each $\mathfrak{m}A_{\mathfrak{m}}$ is an ideal of definition of $A_{\mathfrak{m}}$, i.e. an open ideal I such that the sequence of I^n tends to 0 (cf. EGA 0_I, 7.1.2). Consequently, $\mathrm{Spec}(A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}})$, endowed with the topological ring $A_{\mathfrak{m}}$, is an affine formal scheme in the sense of (EGA I, 10.1.2).

[698]N.D.E.: We note that $A_{\mathfrak{m}}$ is the localization of A at the maximal ideal $\mathfrak{m}$. Indeed, the unit element $e_{\mathfrak{m}}$ of $A_{\mathfrak{m}}$ is an idempotent of A not belonging to $\mathfrak{m}$, and since $A = A_{\mathfrak{m}} \times B$, where $B = A(1 - e_{\mathfrak{m}})$, then $A_{\mathfrak{m}}$ is identified with the localization $A_{e_{\mathfrak{m}}}$ and thus also with the localization $S^{-1}A$, where $S = A - \mathfrak{m}$. On the other hand, since $e_{\mathfrak{m}}(1 - e_{\mathfrak{m}}) = 0$, then $\mathfrak{m}$ contains $1 - e_{\mathfrak{m}}$ and hence also B, and so $\mathfrak{m}$ is identified with $\mathfrak{m}A_{\mathfrak{m}} \times B$.

[699]N.D.E.: Indeed, let $x \in r(A)$; if $\mathfrak{m}$ is a maximal ideal not containing x, there exists $y \in A$ such that $1 - xy \in \mathfrak{m}$, and since $xy \in r(A)$, $1 - xy$ is invertible, whence a contradiction. We note the following consequence: if $\Upsilon(A)$ is a finite set $\mathfrak{m}_1, \cdots, \mathfrak{m}_r$, the $\mathfrak{m}_i$ are all the maximal ideals of A.

[700]N.D.E.: We have added these remarks, in order to be able to compare the definition of the formal spectrum $Spf(A)$ given in 1.1 with those of (EGA I, 10.1.2) and (EGA I, 10.4.2).

c) The topological ring A is *admissible* in the sense of (EGA 0_I, 7.1.2) if and only if $r(A)$ is open (hence an ideal of definition), and this is the case if and only if $\Upsilon(A)$ is finite. In this case, the affine formal scheme $Spf(A) = \mathrm{Spec}(A/r(A))$ (cf. EGA I, 10.1.2) has $\Upsilon(A)$, endowed with the discrete topology, as underlying space, and its structure sheaf has as ring of sections on a subset E of $\Upsilon(A)$ the product $\prod_{\mathfrak{m}\in E} A_{\mathfrak{m}}$.

d) Let A be an arbitrary pseudocompact ring. In 1.1 below, the space $\Upsilon(A)$ is endowed with the discrete topology and with the sheaf of rings whose ring of sections on any subset E is $\prod_{\mathfrak{m}\in E} A_{\mathfrak{m}}$. By b), every point then admits an open neighborhood which is an affine formal scheme, so this defines a formal scheme (EGA I, 10.4.2), which we shall denote $Spf(A)$. (For this formal scheme to be affine, it must be quasi-compact, hence $\Upsilon(A)$ must be finite; in this case, $Spf(A)$ coincides with the definition of (EGA I, 10.1.2)).

0.1.3. If A and B are two pseudocompact rings, a *homomorphism* from A to B is, by definition, a continuous map compatible with addition, multiplication, and the unit elements. Such a homomorphism sends a topologically nilpotent element of A to a topologically nilpotent element of B; it therefore maps the radical $r(A)$ of A into the radical $r(B)$ of B.

0.2.

Let A be a (commutative) pseudocompact ring. A *pseudocompact A-module M* is a topological A-module, separated and complete, which possesses a basis of neighborhoods of 0 consisting of submodules M' such that M/M' is of finite length over A.

If M and N are two pseudocompact A-modules, a *morphism* from M to N is by definition a continuous A-linear map. We shall denote by $\mathrm{Hom}_c(M,N)$ the group of these morphisms.

Proposition 0.2.B.[701] *(i) The pseudocompact A-modules form an abelian category, which we shall denote $PC(A)$. (In particular, for every morphism $f : M \to N$, $Im(f)$ is a complete submodule, hence closed in N).*

(ii) The pseudocompact A-modules of finite length (whose topology is therefore discrete) form a system of cogenerators of $PC(A)$.

(iii) Infinite products and filtered inverse limits are exact, i.e., $PC(A)$ satisfies axiom $(AB5*)$.[702]

For the convenience of the reader, let us briefly indicate the steps of the proof. First, one has the following lemma ([CA] § IV.3, Lemma 1; for the proof, see [BEns], III, § 7.4, th. 1 and example 2):

[701] N.D.E.: We have brought out the results of this paragraph in the proposition that follows, and have indicated below the steps of the proof; cf. [CA], § IV.3 or [DG70], § V.2.

[702] N.D.E.: cf. [Gr57], I § 1.5 and Prop. 1.8; one may also consult, for example, [Po73], § 2.8 or [We95], Append. A.4.

Lemma 0.2.C. *Let B be a ring, I a filtered ordered set, (M_i) and (N_i) two inverse systems of left B-modules indexed by I. Let (s_i) be a morphism of inverse systems $(M_i) \to (N_i)$, such that s_i is surjective with artinian kernel for every i. Then the map*

$$\lim s_i : \lim M_i \to \lim N_i$$

is surjective.

Corollary 0.2.D *([CA] § IV.3, Prop. 10 & 11). Let M be a pseudocompact A-module.*

(i) Let K be a closed submodule of M. Then M/K, endowed with the quotient topology, is a pseudocompact A-module.

(ii) Let (M_i) be a filtered decreasing family of closed submodules of M.

(a) The canonical map $M \to \lim M/M_i$ is surjective and has kernel $\bigcap_i M_i$.

(b) For every closed submodule N of M, one has $N + \bigcap_i M_i = \bigcap_i (N + M_i)$.

Proof. Let (L_j) be the filtered decreasing family of open submodules of M. We endow M/K with the quotient topology, i.e. a basis of neighborhoods of 0 is formed by the open submodules $(K + L_j)/K$. Since K is closed, it equals the intersection of the $K + L_j$, so the map

$$\tau : M/K \to \lim_j M/(K + L_j)$$

is injective. It is also open, the right-hand side being the inverse limit of the discrete modules $M/(K + L_j)$. Moreover, for each j, the map $t_j : M/L_j \to M/(K + L_j)$ is surjective with artinian kernel, so by the preceding lemma, the map t in the commutative diagram below is surjective:

$$\begin{tikzcd} M \arrow[r, "p"] \arrow[d, "\tau"'] & \lim_j M/L_j \arrow[d, "t"] \\ M/K \arrow[r] & \lim_j M/(K + L_j). \end{tikzcd}$$

Since p is an isomorphism because M is complete, it follows that τ is surjective, hence is an isomorphism. This proves (i).

Let us prove (ii)(a). By what precedes, one has for every i an isomorphism $M/M_i \xrightarrow{\sim} \lim_j M/(M_i + L_j)$, and so the two horizontal arrows in the commutative diagram below are isomorphisms:

$$\begin{tikzcd} M \arrow[r, "p"] \arrow[d, "g"'] & \lim_j M/L_j \arrow[d, "s"] \\ \lim_i M/M_i \arrow[r] & \lim_{i,j} M/(M_i + L_j). \end{tikzcd}$$

Moreover, for each j, the family of submodules $(M_i + L_j)/L_j$ admits a smallest element, since M/L_j is artinian, so the morphism $s_j : M/L_j \to \lim_i M/(L_j + M_i)$ is surjective; therefore, by the preceding lemma, s is surjective. It follows that g is surjective. Finally, the kernel of g is the inverse limit of the M_i, i.e. their intersection. This proves part (a).

Let us deduce part (b) from it. Since N is a closed submodule (hence separated and complete), it is a pseudocompact module for the topology induced by that of M. Therefore, by (a), the morphisms f and g in the commutative exact diagram below are surjective:

$$\begin{tikzcd} 0 \arrow[r] & N \arrow[r] \arrow[d, "f"] & M \arrow[r] \arrow[d, "g"] & M/N \arrow[r] \arrow[d, "h"] & 0 \\ 0 \arrow[r] & \varprojlim_i N/(N \cap M_i) \arrow[r] & \varprojlim_i M/M_i \arrow[r] & \varprojlim_i M/(N + M_i). & \end{tikzcd}$$

Then, by the "snake lemma", the sequence $0 \to \mathrm{Ker}\ f \to \mathrm{Ker}\ g \to \mathrm{Ker}\ h \to 0$ is exact, and the equality $N + \bigcap_i M_i = \bigcap_i (N + M_i)$ follows.

We can now prove Proposition 0.2.B. Let $f : M \to N$ be a morphism of pseudocompact A-modules. Then $K = Ker(f)$ is a closed submodule of M, hence separated and complete, so K is a pseudocompact module for the topology induced by that of M. By 0.2.D (i), M/K endowed with the quotient topology is pseudocompact.

Let us show that the continuous bijective morphism $M/K \to Im(f)$ is bicontinuous. Identifying M/K with $Im(f)$, it suffices to show that the quotient topology Q is finer than the topology T induced by that of N. Let (L_j) (resp. (N_i)) be the filtered decreasing family of open submodules of M (resp. N) and set $N'_i = N_i \cap Im(f)$. Let $P = (K + L_j)/K$ be a submodule of M/K open for Q. Since $M/(K + L_j)$ is artinian, the family $N'_i + P$ has a smallest element $N'_{i_0} + P$. Since the N'_i are open, hence closed, for T and hence also for Q, it follows from 0.2.D (ii) (b) that

$$N'_{i_0} + P = \bigcap_i (N'_i + P) = P + \bigcap_i N'_i = P,$$

whence $N'_{i_0} \subset P$. This shows that P is open for T, and $M/K \to Im(f)$ is therefore an isomorphism of pseudocompact modules.

In particular, $Im(f)$ is complete for T, hence closed in N. Then, by 0.2.D (i) again, $Coker(f)$ endowed with the quotient topology is pseudocompact. This proves that $PC(A)$ is an abelian category.

On the other hand, arbitrary inverse limits exist in $PC(A)$: if (M_i) is an inverse system of pseudocompact modules, the inverse limit of the M_i has as underlying module the inverse limit of the underlying modules, with the inverse-limit topology. Moreover, if one has a family of exact sequences in $PC(A)$:

$$0 \longrightarrow K_i \longrightarrow M_i \longrightarrow Q_i \longrightarrow 0,$$

then the sequence $0 \to \prod_i K_i \to \prod_i M_i \to \prod_i Q_i \to 0$ is exact. Point (iii) of 0.2.B follows, since in any abelian category where arbitrary products exist, conditions (a) and (b) of 0.2.D are equivalent and equivalent to the exactness of filtered inverse limits (cf. [Mi65], III 1.2–1.9 or [Po73], Chap. 2, Th. 8.6).

Finally, every pseudocompact module M is a submodule of the product $\prod_L M/L$, where L ranges over the open submodules of M, so the objects of finite length form a system of cogenerators of $PC(A)$. (Moreover, every object of length n is isomorphic to a quotient A^n/L, where L is an open submodule of A^n of colength n; these quotients therefore form a *set* of cogenerators.) This completes the proof of 0.2.B.

[703] Let (Ab) be the category of abelian groups and $LF(A)$ the full subcategory of $PC(A)$ formed by the objects of finite length. For every object M of $PC(A)$, denote by h_c^M the functor:

$$LF(A) \to (Ab), \qquad N \mapsto \mathrm{Hom}_c(M, N).$$

By [CA], § II.4, th. 1, Lemma 4 and Cor. 1, one has the following results.[704]

Proposition 0.2.E. *The functor $M \mapsto h_c^M$ is an anti-equivalence of $PC(A)$ onto the category $Lex(LF(A),(Ab))$ of left-exact functors $LF(A) \to (Ab)$.*

Corollary 0.2.F. *(i) An object P of $PC(A)$ is projective if and only if the functor h_c^P is exact (i.e., if and only if the functor $\mathrm{Hom}_c(P,-)$ is exact on $LF(A)$).*

(ii) Let (M_i) be a filtered inverse system[705] of objects of $PC(A)$. For every object N of $LF(A)$, one has a functorial isomorphism in N:

$$\mathrm{Hom}_c(\lim M_i, N) \simeq \mathrm{colim}\, \mathrm{Hom}_c(M_i, N).$$

(iii) Every filtered inverse limit and every product[706] of projective objects of $PC(A)$ is a projective object of $PC(A)$.

Finally, one deduces from 0.2.F the

Corollary 0.2.G. *Let $(M_i)_{i\in I}$ be a family of objects of $PC(A)$. Then $\prod_{i\in I} M_i$ is projective if and only if each M_i is.*

Indeed, for every $N \in Ob LF(A)$, one has $\mathrm{Hom}_c(\prod_i M_i, N) \simeq \bigoplus_i \mathrm{Hom}_c(M_i, N)$.

0.2.1. Each local component $A_{\mathfrak{m}}$ of A is a direct factor of A, hence a projective object of $PC(A)$ (A is manifestly projective). Moreover, $A_{\mathfrak{m}}$ has $S_{\mathfrak{m}} = A_{\mathfrak{m}}/\mathfrak{m}A_{\mathfrak{m}}$ as its unique simple quotient, hence is indecomposable. On the other hand, every simple

[703] N.D.E.: We have inserted into this paragraph Proposition 0.2.E and Corollary 0.2.F, which will be useful in 0.2.2. (In the original, these results appeared in 0.3.)

[704] N.D.E.: Indeed, $PC(A)$ has a set of artinian cogenerators, filtered inverse limits in it are exact, and $LF(A)$ is the subcategory of artinian objects. The dual category $PC(A)^\circ$ therefore has a set of noetherian generators, and filtered direct limits in it are exact. By the proof of [CA] § II.4, th. 1, the functor $M \mapsto \mathrm{Hom}_c(M,-)$ is an anti-equivalence of $PC(A)$ with $Lex(LF(A),(Ab))$. Likewise, Lemma 4 and Cor. 1 of *loc. cit.* state results "dual" to those of Corollary 0.2.F. For a "direct" proof, see [DG70], § V.2, th. 3.1, Lemma 3.5, Cor. 3.3 & 3.4.

[705] N.D.E.: As every infinite product is a filtered inverse limit of finite products, every additive functor that commutes with filtered inverse limits also commutes with infinite products.

[706] N.D.E.: As every infinite product is a filtered inverse limit of finite products, every additive functor that commutes with filtered inverse limits also commutes with infinite products.

object of $PC(A)$ is isomorphic to a unique $S_{\mathfrak{m}}$. By [CA], IV § 3, Cor. 1 of th. 3,[707] one therefore has:

Proposition. *(i) Every projective object of $PC(A)$ is a direct product of indecomposable projective objects, uniquely determined (up to isomorphism).*

(ii) Every indecomposable projective object is isomorphic to a unique $A_{\mathfrak{m}}$ ($\mathfrak{m} \in \Upsilon(A)$).

Definition. *A pseudocompact A-module M is said to be* topologically free *if it is isomorphic to the product of a family (A_i) of copies of A.*

In this case, a family (m_i) of elements of M is called a pseudobasis *of M if the A-linear maps from A_i into M sending the unit element of A_i to m_i extend to an isomorphism of $\prod_i A_i$ onto M.*

0.2.2. [708] If M is a pseudocompact A-module, we shall denote by $M^\dagger$ the A-module (without topology) $\mathrm{Hom}_c(M, A)$.

Conversely, if N is an A-module, we denote by $N^* = \mathrm{Hom}_A(N, A)$ its dual, endowed with the topology of pointwise convergence, i.e., a basis of neighborhoods of 0 in N^* is formed by the following submodules, where $x \in N$ and ℓ is an open ideal of A:

$$V(x, \ell) = \{f \in N^* \mid f(x) \in \ell\}.$$

This makes N^* a pseudocompact A-module. Indeed, one sees first that if $N = A$, then $N^* = A$, endowed with its topology of pseudocompact ring, and if N is a free module $A^{(I)}$, then N^* is the product A^I, endowed with the product topology. On the other hand, for every morphism $\phi : N_1 \to N_2$, the transposed morphism ${}^t\phi : N_2^* \to N_1^*$ is continuous, since the inverse image under ${}^t\phi$ of a submodule $V(x_1, \ell)$ of N_1^* is nothing but the submodule $V(\phi(x_1), \ell)$ of N_2^*. Then, for arbitrary N, taking a presentation

$$A^{(J)} \xrightarrow{\phi} A^{(I)} \xrightarrow{\pi} N \to 0,$$

one sees that N^* is the kernel of the continuous morphism ${}^t\phi : A^I \to A^J$, so N^* is pseudocompact.

When A is artinian (in which case one can take $\ell = 0$ above), one deduces from 0.2.F:

Proposition. *When A is artinian, the functors*

$$P \mapsto P^\dagger \qquad \text{and} \qquad Q \mapsto Q^*,$$

[707] N.D.E.: This refers to the "dual" statements established in *loc. cit.*, § 2, Th. 2 and § 1, Prop. 2; for a "direct" proof, see [DG70], V, § 2, Th. 4.5 and Example 4.6 b).

[708] N.D.E.: We have detailed the results of this paragraph, which play an important role in what follows (cf. 1.2.3, 1.3.5, 2.2.1, etc.).

where P (resp. Q) is a projective object of $PC(A)$ (resp. a projective A-module), establish an anti-equivalence between the category of projective pseudocompact A-modules and that of projective A-modules.[709]

In particular, when A is a field k, $P \mapsto P^\dagger$ is an anti-equivalence between the category of all pseudocompact k-modules (one also speaks of *linearly compact k-vector spaces*) and that of k-vector spaces.[710]

0.3.

[711] Let L and M be two pseudocompact A-modules. The functor

$$LF(A) \to (Ab), \qquad N \mapsto Bil_c(L \times M, N),$$

where $Bil_c(L\times M, N)$ denotes the abelian group of continuous A-bilinear maps from $L \times M$ into N, is left exact.

By 0.2.E, there therefore exists a pseudocompact A-module $L\hat{\otimes}_A M$, unique up to unique isomorphism, which represents this functor, i.e. such that one has a functorial isomorphism, for every object N of $LF(A)$:

$$\mathrm{Hom}_c(L \hat{\otimes}_A M, N) \simeq Bil_c(L \times M, N).$$

Moreover, $L\hat{\otimes}_A M$ is identified with the inverse limit P of the (discrete) A-modules $(L/L') \otimes_A (M/M')$, where L' and M' range respectively over the open submodules of L and M.

Indeed, let $\phi : L \times M \to N$ be a continuous bilinear map of $L \times M$ into an A-module (discrete) of finite length N. By Lemma 0.3.1 below, there exist open submodules L' and M' of L and M such that $\phi(L' \times M) = \phi(L \times M') = 0$. This means that the map $\bar{\phi} : L \otimes_A M \to N$, which is induced by ϕ, is of the form $\phi' \circ p$, where p is the canonical projection of $L \otimes_A M$ onto $(L/L') \otimes_A (M/M')$. If one denotes by $\hat{\phi}$ the composite:

$$P \to (L/L') \otimes_A (M/M') \xrightarrow{\phi'} N,$$

one sees that the map $\phi \mapsto \hat{\phi}$ is a functorial bijection of $Bil_c(L \times M, N)$ onto $\mathrm{Hom}_c(P, N)$, whence $P \simeq L\hat{\otimes}_A M$.

The pseudocompact module $L\hat{\otimes}_A M$ is therefore the separated completion of $L \otimes_A M$ for the linear topology defined by the kernels of the canonical projections of $L \otimes_A M$ onto $(L/L') \otimes_A (M/M')$, and it will be called the *completed tensor product* of L and M.

If x and y belong to L and M, the image of $x \otimes_A y$ in $L\hat{\otimes}_A M$ will be denoted $x\hat{\otimes}_A y$.

[709] N.D.E.: Recall on the other hand that, over an artinian ring, a module is projective if and only if it is flat; see for example (VI_B, N.D.E. (54)) or 0.3.7 below.

[710] N.D.E.: One will note that the direct sum in $PC(k)$ of a family $(V_i)_{i\in I}$ of linearly compact k-vector spaces is $(\prod_{i \in I} V_i^\dagger)^*$.

[711] N.D.E.: We have modified this paragraph, taking into account the additions made in 0.2.

0.3.1. **Lemma 0.3.1.** *Let L, M and N be pseudocompact A-modules, N of finite length. If $\phi : L \times M \to N$ is a continuous A-bilinear map, there exist open submodules L' and M' of L and M such that $\phi(L' \times M) = \phi(L \times M') = 0$.*

Indeed, $\phi^{-1}(0)$ is an open neighborhood of `(0, 0)`, hence contains an open of the form $L_1 \times M_1$, where `L_1` and `M_1` are open submodules of L and M. Since L/L_1 is of finite length, there exist elements $x_1, \cdots, x_r$ of L such that $L_1 + Ax_1 + \cdots + Ax_r = L$. If $M' \subset M_1$ is "small enough", one also has $\phi(x_i, M') = 0$ for every i, because the map $y \mapsto \phi(x_i, y)$ is continuous; from this one deduces $\phi(L, M') = 0$; likewise, $\phi(L', M) = 0$ if L' is small enough.

Corollary 0.3.1.1.[712] *Let M be a pseudocompact A-module.*

(i) For every open submodule M', there exists an open ideal ℓ of A such that $\ell M \subset M'$.

(ii) Consequently, $M \simeq \lim_\ell M/\ell M$, where ℓ ranges over the filtered inverse system of open ideals of A and each $M/\ell M$ is endowed with the quotient topology (which makes it a pseudocompact module, cf. 0.2.D).

Indeed, consider the map $\phi : A \times M \to M/M'$, $(a, m) \mapsto am + M'$; by 0.3.1 there exists an open ideal ℓ of A such that $\ell M \subset M'$, and since M' is also closed, it contains also ℓM. Since the intersection of the open submodules of M is zero, one therefore has $\bigcap_\ell \ell M = (0)$. On the other hand, by 0.2.D, the map $\phi : M \to \lim_\ell M/\ell M$ is surjective; by (the proof of) 0.2.B, ϕ therefore induces an isomorphism $M/Ker(\phi) \xrightarrow{\sim} \lim_\ell M/\ell M$, but we have just seen that $Ker(\phi) = \bigcap_\ell \ell M$ is zero.

Remark 0.3.1.2.[713] *The completed tensor product satisfies the usual associativity condition: if L, M, P are pseudocompact A-modules, one has a canonical isomorphism*

`(L ⊗ˆM) ⊗ˆP ≃ L ⊗ˆ(M ⊗ˆP);`

indeed, these two objects represent the functor that associates to every object N of $LF(A)$ the abelian group of continuous A-trilinear maps from $L \times M \times P$ into N.

0.3.2. Let $L' \xrightarrow{f} L \xrightarrow{g} L'' \to 0$ be an exact sequence and M an object of $PC(A)$. It is clear that for every object N of $LF(A)$, the induced sequences:

`0 → Bil_c(L'' × M, N) → Bil_c(L × M, N) → Bil_c(L' × M, N)`

`0 → Hom_c(L'' ⊗_ˆA M, N) → Hom_c(L ⊗_ˆA M, N) → Hom_c(L' ⊗_ˆA M, N)`

are exact. By 0.2.E, this is equivalent to saying that the sequence

(∗) `L' ⊗_ˆA M —f ⊗ˆM→ L ⊗_ˆA M —g ⊗ˆM→ L'' ⊗_ˆA M → 0`

is exact. Hence:

[712] N.D.E.: We have added this corollary.
[713] N.D.E.: We have added this remark.

Corollary. *For every pseudocompact A-module M, the functor $-\hat{\otimes}_A M$ is right exact.*

In particular, take for L the ring A, for f the inclusion of a closed ideal $\mathfrak{a}$ in A, and for g the canonical projection of A onto $A/\mathfrak{a}$. One can then identify $A\hat{\otimes}_A M$ with M by means of the map $x\hat{\otimes}_A m \mapsto xm$. Since the image of $\mathfrak{a}\hat{\otimes}_A M$ is closed in M (cf. 0.2.B) and the image of $\mathfrak{a} \otimes_A M$ is everywhere dense in $\mathfrak{a}\hat{\otimes}_A M$, the image of $f\hat{\otimes}_A M$ is nothing but the closure $\bar{\mathfrak{a}}M$ of $\mathfrak{a}M$ in M. The exact sequence $(*)$ therefore yields the isomorphism:

$$(A/\mathfrak{a}) \hat{\otimes}_A M \simeq M / \bar{\mathfrak{a}}M.$$

0.3.3. Lemma 0.3.3 (Nakayama's Lemma). *Let A be a pseudocompact ring, M a pseudocompact A-module, and $\mathfrak{a}$ an ideal of A contained in the radical $r(A)$. The equality $\bar{\mathfrak{a}}M = M$ then implies $M = 0$.*

Indeed, suppose $\bar{\mathfrak{a}}M = M$.[714] Let M' be an open submodule of M and $M'' = M/M'$. Since M'' is discrete, $\mathfrak{a}M''$ is closed in M'', hence equal to $\bar{\mathfrak{a}}M''$. By 0.3.2, the canonical map of $M/\bar{\mathfrak{a}}M$ to $M''/\bar{\mathfrak{a}}M''$ is surjective, so one has $\mathfrak{a}M'' = \bar{\mathfrak{a}}M'' = M''$. Since M'' is a finitely generated A-module and $\mathfrak{a} \subset r(A)$, this implies $M'' = 0$ by the usual Nakayama's Lemma. Consequently, every open submodule M' of M equals M, and so M is zero.[715]

0.3.4. From Nakayama's Lemma one draws the usual consequences:

Corollary. *Let $\mathfrak{a}$ be a closed ideal contained in $r(A)$ and $f : M \to N$ a morphism of pseudocompact A-modules.*

(i) f is surjective if the induced map $f' : M/\mathfrak{a}M \to N/\mathfrak{a}N$ is.[716]

(ii) If N is projective, f is invertible if f' is.

Indeed, (i) follows from Lemma 0.3.3 applied to $\operatorname{Coker} f$. For (ii), suppose f' invertible. Then by (i), f is surjective, hence has a section; one then applies 0.3.3 to $\operatorname{Ker} f$.

When A is local with maximal ideal $\mathfrak{m}$, one can also deduce from 0.3.3 the following exchange theorem:

Theorem. *Let A be a local pseudocompact ring, $\mathfrak{m}$ its maximal ideal, M a topologically free A-module with pseudobasis $(m_i)_{i\in I}$ (0.2.1), and N a (closed) direct factor of M. There exists a pseudobasis of M formed of elements of N and of certain m_i.*

Indeed, this is clear when A is a field (one then uses the duality of 0.2.2 and applies the usual exchange theorem); consequently, $N/\mathfrak{m}N$[717] has as a supplement a topologically free module over $A/\mathfrak{m}$ with pseudobasis $(\bar{m}_i)_{i\in J}$, where $\bar{m}_i$ is the image of

[714] N.D.E.: Up to replacing $\mathfrak{a}$ by its closure, one can assume $\mathfrak{a}$ closed.
[715] N.D.E.: since equal to the inverse limit of the $M/M' = 0$.
[716] N.D.E.: Consequently, every pseudocompact A-module is a quotient of a topologically free A-module (one first reduces to the case where A is local), cf. the proof of 0.3.7.
[717] N.D.E.: We have corrected $N/\mathfrak{m}N$ to $N/\bar{\mathfrak{m}}N$, and likewise for $M/\mathfrak{m}M$ below.

m_i in $M/\mathfrak{m}M$ and J is a subset of I. If P denotes the direct product $\prod_{i\in J} Am_i$, the canonical map of $N \oplus P$ to M is "bijective modulo $\mathfrak{m}$"; it is therefore bijective by what precedes (for another proof see [CA], § IV.2, Prop. 8).

0.3.5. Consider now three pseudocompact A-modules L, M and N, where N is of finite length. Endowing the A-module $\mathrm{Hom}_c(M,N)$ with the discrete topology, every element ψ of $\mathrm{Hom}_c(L, \mathrm{Hom}_c(M,N))$ defines a continuous bilinear map $\psi' : (\ell, m) \mapsto \psi(\ell)(m)$ from $L \times M$ to N. One thus obtains a natural isomorphism

$$\mathrm{Hom}_c(L, \mathrm{Hom}_c(M, N)) \xrightarrow{\sim} \mathrm{Hom}_c(L \hat{\otimes}_A M, N), \tag{1}$$

hence another characterization of $\mathrm{Hom}_c(L\hat{\otimes}_A M, N)$, which we shall use when M is the inverse limit of a filtered inverse system of pseudocompact A-modules M_i. Then, by (1) and 0.2.F (ii), one has natural isomorphisms:

$$\begin{aligned}\mathrm{Hom}_c(L \hat{\otimes}_A \lim M_i, N) &\simeq \mathrm{Hom}_c(L, \mathrm{Hom}_c(\lim M_i, N))\\ &\simeq \mathrm{Hom}_c(L, \operatorname{colim} \mathrm{Hom}_c(M_i, N)).\end{aligned} \tag{2}$$

Moreover, since the module *colim* $\mathrm{Hom}_c(M_i, N)$ is discrete, every continuous map with source L factors through a finite-length quotient of L. Consequently, the natural map below is an isomorphism:

$$\operatorname{colim} \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i, N)) \xrightarrow{\sim} \mathrm{Hom}_c(L, \operatorname{colim} \mathrm{Hom}_c(M_i, N)). \tag{3}$$

Finally, by (1) and 0.2.F (ii) again, one has natural isomorphisms:

$$\begin{aligned}\operatorname{colim} \mathrm{Hom}_c(L, \mathrm{Hom}_c(M_i, N)) &\simeq \operatorname{colim} \mathrm{Hom}_c(L \hat{\otimes}_A M_i, N)\\ &\simeq \mathrm{Hom}_c(\lim (L \hat{\otimes}_A M_i), N).\end{aligned} \tag{4}$$

Combining isomorphisms (2), (3), (4), one obtains:

Proposition. *Let (M_i) be a filtered inverse system of objects of $PC(A)$, and let L (resp. N) be an object of $PC(A)$ (resp. $LF(A)$). One has a functorial isomorphism in N:*

$$\mathrm{Hom}_c(L \hat{\otimes}_A \lim M_i, N) \simeq \mathrm{Hom}_c(\lim (L \hat{\otimes}_A M_i), N),$$

and hence an isomorphism:

$$L \hat{\otimes}_A \lim M_i \simeq \lim (L \hat{\otimes}_A M_i).$$

The completed tensor product therefore commutes with filtered inverse limits.[718]

0.3.6. In particular,[719] the completed tensor product commutes with infinite products. For example, since the ring A is the product of its local components $A_\mathfrak{m}$ (0.1.1), every pseudocompact A-module M ($\simeq A\hat{\otimes}_A M$) is identified with the product of the modules $M_\mathfrak{m} = A_\mathfrak{m}\hat{\otimes}_A M$ (the local components of M).

[718] N.D.E.: One thus recovers 0.3.1.1: $L = L\hat{\otimes}_A A = L\hat{\otimes}_A \lim_\ell (A/\ell) \simeq \lim_\ell L/\ell L$, where ℓ ranges over the open ideals of A.

[719] N.D.E.: cf. N.D.E. (12).

Likewise, let M and N be two pseudocompact A-modules. Recall (cf. 0.2.2) that $M^\dagger$ denotes $\mathrm{Hom}_c(M, A)$. Consider the map

$$\varphi : M^\dagger \otimes_A N^\dagger \to (M \hat{\otimes}_A N)^\dagger$$

which associates to an element $f \otimes g$ of $M^\dagger \otimes_A N^\dagger$ the map $m \hat{\otimes} n \mapsto f(m)g(n)$ from $M \hat{\otimes}_A N$ to A. This map ϕ is bijective when M is isomorphic to A.

Corollary. *When A is artinian, ϕ is an isomorphism whenever M is topologically free (or more generally projective).*

Indeed, for N fixed, the functor $M \mapsto (M \hat{\otimes}_A N)^\dagger$ (resp. $M \mapsto M^\dagger \otimes_A N^\dagger$) transforms every direct product into a direct sum, by what precedes and 0.2.F.

Remark 0.3.6.A.[720] *Using 0.2.F in a similar way, one also obtains the following result: Let A be an artinian ring, M, Q objects of $PC(A)$, and N an object of $LF(A)$. Suppose Q projective; then one has natural isomorphisms:*

$$\mathrm{Hom}_c(M, Q) \cong \mathrm{Hom}_A(Q^\dagger, M^\dagger) \qquad \text{and} \qquad Q^\dagger \otimes_A N \cong \mathrm{Hom}_c(Q, N).$$

0.3.7. For every $\mathfrak{m} \in \Upsilon(A)$, the functor $M \mapsto A_{\mathfrak{m}} \hat{\otimes}_A M$ is evidently exact. Since every projective pseudocompact A-module P is a product of modules of the form $A_{\mathfrak{m}}$, it follows that the functor $M \mapsto P \hat{\otimes}_A M$ is exact when P is projective. The converse is true:

Proposition. *Let A be a pseudocompact ring, P a pseudocompact A-module. The following conditions are equivalent:*

(i) P is a projective object of $PC(A)$.

(ii) Each local component $P_{\mathfrak{m}}$ is a topologically free $A_{\mathfrak{m}}$-module.

(iii) The functor $M \mapsto P \hat{\otimes}_A M$ is exact.

Indeed, the equivalence of (i) and (ii) follows from 0.2.F (iii) and 0.2.1, and we have just seen that (ii) ⇒ (iii). Suppose the functor $M \mapsto P \hat{\otimes}_A M$ is exact. Since $P \hat{\otimes}_A M$ is the product of its local components:

$$(P \hat{\otimes}_A M)_{\mathfrak{m}} \simeq P_{\mathfrak{m}} \hat{\otimes}_{A_{\mathfrak{m}}} M_{\mathfrak{m}},$$

one is reduced to the case where the ring A is local. We then prove that P is topologically free.

Let $\mathfrak{m}$ be the maximal ideal of A; then $P/\bar{\mathfrak{m}}P$ is a linearly compact vector space over $A/\mathfrak{m}$, hence a product of copies of $A/\mathfrak{m}$ (cf. 0.2.2). There is therefore a family $(A_i)_{i \in I}$ of copies of A and an isomorphism $\phi : \prod_{i \in I}(A_i/\mathfrak{m}) \xrightarrow{\sim} P/\bar{\mathfrak{m}}P$. Since $\prod_{i \in I} A_i$ is projective, there is a commutative square

$$\begin{array}{ccc} \prod A_i & \xrightarrow{\psi} & P \\ \Big\downarrow{\scriptstyle p} & & \Big\downarrow{\scriptstyle q} \end{array}$$

[720] N.D.E.: We have added this remark, which will be useful in 2.3.1.

$$\begin{array}{ccc}\downarrow & & \downarrow \\ \prod (A_i/\mathfrak{m}) & \xrightarrow{\phi} & P/\bar{\mathfrak{m}}P,\end{array}$$

where p and q denote the canonical projections. Applying Nakayama's Lemma to $Coker\psi$ and noting that $(A/\mathfrak{m})\hat{\otimes}_A\psi$ is nothing but ϕ, one sees that ψ is surjective.[721]

Setting then $B = \prod_{i\in I} A_i$ and $N = Ker\psi$, one has the following commutative and exact diagram:

$$\begin{tikzcd}
\mathfrak{m} \hat{\otimes}_A N \arrow[r] \arrow[d] & \mathfrak{m} \hat{\otimes}_A B \arrow[r] \arrow[d] & \mathfrak{m} \hat{\otimes}_A P \arrow[r] \arrow[d] & 0 \\
A \hat{\otimes}_A N \arrow[r] \arrow[d] & A \hat{\otimes}_A B \arrow[r, "\psi"] \arrow[d] & A \hat{\otimes}_A P \arrow[r] \arrow[d] & 0 \\
(A/\mathfrak{m}) \hat{\otimes}_A N \arrow[r] & (A/\mathfrak{m}) \hat{\otimes}_A B \arrow[r, "\phi"] & (A/\mathfrak{m}) \hat{\otimes}_A P \arrow[r] & 0.
\end{tikzcd}$$

The "snake lemma" applied to the first two rows then shows that, in the bottom row, the morphism $(A/\mathfrak{m})\hat{\otimes}_A N \to (A/\mathfrak{m})\hat{\otimes}_A B$ is a monomorphism. But then, since ϕ is an isomorphism, $(A/\mathfrak{m})\hat{\otimes}_A N$ is zero; whence $N = 0$ (0.3.3) and ψ is an isomorphism.[722]

0.3.8. Corollary 0.3.8. *Let A be a complete noetherian local ring and P a pseudocompact A-module. Then P is topologically free if and only if P is flat over A.*

Indeed, the canonical map of $M \otimes_A P$ into $M\hat{\otimes}_A P$ is bijective when M equals A, hence also when M is noetherian (take a finite presentation of M and use the right exactness of the tensor product and of the completed tensor product).

Now P is flat if and only if the functor $M \mapsto M \otimes_A P$ is exact when M ranges over the noetherian modules. Likewise, we saw in the proof of Proposition 0.3.7 that P is topologically free if the sequence

$$0 \to \mathfrak{m} \hat{\otimes}_A P \to A \hat{\otimes}_A P \to (A/\mathfrak{m}) \hat{\otimes}_A P \to 0$$

is exact. So P is topologically free if and only if the functor $M \mapsto M\hat{\otimes}_A P$ is exact when M ranges over the noetherian modules. The corollary therefore follows from the equality $M \otimes_A P = M\hat{\otimes}_A P$ established above.

0.4.

Let k be a pseudocompact ring; a *topological k-algebra* is a (commutative) topological ring A, equipped with a morphism of topological rings $k \to A$. One says that A is a *profinite k-algebra* if the underlying topological k-module is pseudocompact.

In this case, let ℓ be an open k-submodule of A. The composite map

[721] N.D.E.: This shows the result announced in N.D.E. (22): every pseudocompact A-module M is a quotient of a topologically free A-module. (Since products are exact in $PC(A)$, one reduces to the case where A is local, treated above.)

[722] N.D.E.: A fully analogous proof, using "nilpotent Nakayama's Lemma", shows that if A is artinian and P is a flat A-module, then each local component of P is a free A-module (cf. [BAC], II § 3.2, cor. 2 of prop. 5).

$$\varphi : A \times A \xrightarrow{\text{mult}} A \xrightarrow{\text{can}} A/\ell$$

is continuous, hence by Lemma 0.3.1, there exists an open k-submodule $\mathfrak{n}$ of A such that $\phi(A \times \mathfrak{n}) = 0$. This means that ℓ contains the open ideal $A\mathfrak{n}$ and implies that A is a pseudocompact ring.

Likewise, let M be a topological A-module whose underlying k-module is pseudocompact. If M' is an open k-submodule of M, Lemma 0.3.1 applied to the map

$$A \times M \xrightarrow{\text{mult}} M \xrightarrow{\text{can}} M/M'$$

shows that M' contains an open A-submodule, so that M is also a pseudocompact A-module.[723] Conversely:

Lemma. *Let A be a profinite k-algebra and M a pseudocompact A-module. Then the k-module $M|_k$ obtained by restriction of scalars is pseudocompact.*

Indeed, every pseudocompact A-module of finite length is isomorphic to a quotient A^n/L (where L is an open submodule of A^n), hence is a pseudocompact k-module. Since $M|_k$ is an inverse limit of such modules, it is a pseudocompact k-module.

0.4.1. If A and B are two profinite k-algebras, a *morphism* from A to B is, by definition, a continuous homomorphism of k-algebras. We shall denote by Alp/k the category of profinite k-algebras.

It evidently possesses inverse limits: the algebra underlying an inverse limit is the inverse limit of the underlying algebras; the topology is that of the inverse limit.

It also possesses finite direct limits[724]: for example, if $f : A \to B$ and $g : A \to C$ are two morphisms of profinite k-algebras, the amalgamated sum of B and C over A has $B\hat{\otimes}_A C$ as underlying topological A-module (by 0.4, f and g endow B and C with pseudocompact A-module structures); the multiplication of $B\hat{\otimes}_A C$ is obviously such that $(b\hat{\otimes}c)(b'\hat{\otimes}c') = (bb')\hat{\otimes}(cc')$ if $b, b' \in B$ and $c, c' \in C$.

0.4.2. Definition 0.4.2. *A profinite k-algebra C is said to be of* finite length *if the underlying k-module is of finite length (hence discrete); we denote by Alf/k the full subcategory of Alp/k formed by k-algebras of finite length.*[725]

For every profinite k-algebra A, we denote by h_A the functor:

$$Alf/k \to (\text{Sets}), \qquad C \mapsto \operatorname{Hom}_{Alp/k}(A, C).$$

It is clear that h_A is a left-exact functor[726]. Moreover, the canonical projections $A \to A/\ell$ (where ℓ ranges over the open ideals of A) induce, for every object C of Alf/k, a canonical isomorphism

$$\operatorname{colim}_{\ell} \operatorname{Hom}_{Alf/k}(A/\ell, C) \xrightarrow{\sim} \operatorname{Hom}_{Alp/k}(A, C),$$

[723] N.D.E.: We have added the following lemma, cf. N.D.E. (36) in Proposition 0.5.

[724] N.D.E.: We will see in 0.4.2 that it possesses arbitrary direct limits.

[725] N.D.E.: Caution: k is an object of Alf/k only if k is artinian, cf. 1.2.2 below.

[726] N.D.E.: i.e., which commutes with finite inverse limits.

functorial in C. This means that h_A is the direct limit of the representable functors $h_{A/\ell}$, i.e.,

$$(*) h_A \simeq colim_\ell h_{A/\ell}.$$

If B is another profinite k-algebra, the general properties of the bifunctor Hom and the isomorphism $\mathrm{Hom}(h_C, h_B) = h_B(C)$ for C of finite length give isomorphisms:

$$\begin{aligned}\mathrm{Hom}_{Alp/k}(B, A) &\simeq \lim \mathrm{Hom}_{Alp/k}(B, A/\ell)\\ &\simeq \lim \mathrm{Hom}(h_{A/\ell}, h_B)\\ &\simeq \mathrm{Hom}(colim_\ell\, h_{A/\ell}, h_B);\end{aligned}$$

combined with $(*)$, this shows that the contravariant functor $A \mapsto h_A$ is fully faithful. In fact:

Proposition. *The functor $A \mapsto h_A$ is an anti-equivalence of Alp/k onto the category of left-exact functors from Alf/k to (Sets).*

Indeed, by what precedes, it suffices to show that every left-exact functor $F : Alf/k \to (Sets)$ is isomorphic to a functor of the type h_A; for this, one can construct A as follows (cf. TDTE II, § 3).

Since F is left-exact, for every k-algebra of finite length C and every element ξ of $F(C)$, there is a smallest subalgebra C' of C such that ξ belongs to the image of $F(C')$ in $F(C)$. If one has $C' = C$, one says that the pair (C, ξ) is *minimal*.

The minimal pairs form a category if one takes for morphisms from (C, ξ) to (D, η) the homomorphisms ϕ from C to D such that $(F\phi)(\xi) = \eta$; it is clear that such a ϕ is a surjection and that the category of minimal pairs is "left filtered". Moreover, one can restrict to pairs (C, ξ) such that C belongs to a set containing k-algebras of finite length of each isomorphism type[727]. Hence, the functor $(C, \xi) \mapsto C$, with source category that of minimal pairs and target category that of profinite k-algebras, possesses an inverse limit; one takes for A this inverse limit.

Corollary. *The category Alp/k possesses infinite direct limits.*

Indeed, the category of left-exact functors from Alf/k to (Sets) possesses inverse limits, which are defined "argument by argument", i.e., if (F_i) is an inverse system of such functors, one has, for every object C of Alf/k:

$$(\lim F_i)(C) = \lim F_i(C).$$

[728]

[727] N.D.E.: Indeed, every discrete k-module of length n is isomorphic to k^n/L, where L is an open submodule of k^n of colength n. One can then consider the set of (isomorphism classes of) profinite k-algebra structures on each such module.

[728] N.D.E.: If $(A_i)_{i \in I}$ is a direct system in Alp/k, its direct limit in Alp/k is the separated completion of the k-algebra B, direct limit of the underlying k-algebras, for the topology defined by the ideals I such that $I \cap A_i$ is an open ideal of A_i for every i, and $length_k(B/I) < \infty$. Note that if, for example, K/k is an algebraic extension of fields, of infinite degree, and if (k_i) denotes the filtered direct system of subextensions of finite degree, then the direct limit of the system (k_i) in Alp/k is the zero ring!

0.5.

[729] Let $\phi : k \to \ell$ be a homomorphism of pseudocompact rings (cf. 0.1.3). One can generalize the construction of 0.3 as follows.

Definition 0.5.A. *For every object M of $PC(k)$ (resp. N of $PC(\ell)$), we shall denote by $M\hat{\otimes}_k N$ the separated completion of $M \otimes_k N$ for the linear topology defined by the kernels of the projections:*

$$M \otimes_k N \to (M/M') \otimes_k (N/N'),$$

where M' (resp. N') is an open submodule of M (resp. of N). Then $M\hat{\otimes}_k N$ is a pseudocompact ℓ-module. If $m \in M$ and $x \in N$, we shall denote by $m\hat{\otimes}_k x$ the canonical image of $m \otimes_k x$ in $M\hat{\otimes}_k N$.

This applies in particular when $N = \ell$, in which case we shall say that $M\hat{\otimes}_k \ell$ is the pseudocompact ℓ-module deduced from M by the base change $k \to \ell$.

Remarks 0.5.B. *(i) When one considers such a base change, ℓ will not in general be a profinite k-algebra: a typical example is the case where k is a field and ℓ is an arbitrary extension K of k.*

(ii) However, if the k-module underlying N is pseudocompact (for example if ℓ is a profinite k-algebra) then, by 0.4, every open k-submodule of N contains an open ℓ-submodule of N; consequently, $M\hat{\otimes}_k N$ coincides in this case with the completed tensor product (cf. 0.3) of the pseudocompact k-modules M and N, and the notation therefore does not present any ambiguity.

The k-module $N|_k$ obtained by restriction of scalars is in any case a topological k-module, i.e. the map $k \times N \to N$, $(t, n) \mapsto \phi(t)n$ is continuous. We denote by $\mathrm{Hom}_c(M, N|_k)$ the abelian group of continuous k-module homomorphisms of M into $N|_k$.

Proposition 0.5.C. *For every $M \in ObPC(k)$ and $N \in ObPC(\ell)$, one has a functorial isomorphism*

$$\mathrm{Hom}_{PC(\ell)}(M \hat{\otimes}_k \ell, N) \simeq \mathrm{Hom}_c(M, N|_k).$$

[730]

Indeed, let ϕ be a continuous homomorphism $M \to N|_k$; then the map $\phi' : M \times \ell \to N$, $(m, \lambda) \mapsto \lambda\phi(m)$ is continuous and "bilinear" (i.e., k-linear in the first factor and ℓ-linear in the second). If N' is an open ℓ-submodule of N, one shows as in Lemma 0.3.1 that there exist an open submodule M' of M and an open ideal ℓ' of ℓ such that $\phi'(M' \times \ell)$ and $\phi'(M \times \ell')$ are contained in N'. It follows that ϕ induces a continuous homomorphism of ℓ-modules $\Phi : M\hat{\otimes}_k \ell \to N$, such that $\Phi(m\hat{\otimes}\lambda) = \lambda\phi(m)$, for every $m \in M$ and $\lambda \in \ell$.

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[729] N.D.E.: We have detailed this paragraph.

[730] N.D.E.: Consequently, if ℓ is a profinite k-algebra, the functor $M \mapsto M\hat{\otimes}_k \ell$ is left adjoint to the restriction-of-scalars functor.

Conversely, to every morphism $f : M\hat{\bigotimes}_k\ell \to N$ one associates the morphism $f' : m \mapsto f(m\hat{\bigotimes}_k 1)$ from M to $N|_k$.

One then obtains, as in 0.3.2, 0.3.5, and 0.3.1.2, the:

Corollary 0.5.D. *The functor $PC(k) \to PC(\ell)$, $M \mapsto M\hat{\bigotimes}_k\ell$ is right exact and commutes with filtered inverse limits, i.e., if (M_i) is a filtered inverse system of objects of $PC(k)$, one has a canonical isomorphism:*

$$(\varprojlim M_i) \hat{\otimes}_k \ell \simeq \varprojlim (M_i \hat{\otimes}_k \ell).$$

Moreover, if M, N are pseudocompact k-modules, one has a canonical isomorphism:

$$(M \hat{\otimes}_k N) \hat{\otimes}_k \ell \simeq (M \hat{\otimes}_k \ell) \hat{\otimes}_\ell (N \hat{\otimes}_k \ell).$$

Definition 0.5.E. *Finally, if A is a profinite k-algebra, there is on $A\hat{\bigotimes}_k\ell$ one and only one structure of profinite ℓ-algebra such that, if $a, b \in A$ and $\lambda, \mu \in \ell$,*

$$(a \hat{\otimes}_k \lambda)(b \hat{\otimes}_k \mu) = (ab) \hat{\otimes}_k (\lambda\mu).$$

One says that $A\hat{\bigotimes}_k\ell$ is the profinite ℓ-algebra deduced from A by the extension of scalars (or "base change") $k \to \ell$.

1. Formal varieties over a pseudocompact ring

1.1.

One can associate to every pseudocompact ring A a formal scheme (EGA I, 10.4.2) by proceeding as follows: the underlying topological space is the set $\Upsilon(A)$ of open (hence maximal) prime ideals of A, endowed with the discrete topology; the structure sheaf has the cartesian product $\prod_{\mathfrak{m}\in E} A_{\mathfrak{m}}$ as space of sections on a subset E of $\Upsilon(A)$. The formal scheme thus obtained is denoted $Spf(A)$ (the *formal spectrum* of A).[731]

If A and B are two pseudocompact rings, a morphism from $Spf(B)$ to $Spf(A)$ consists of the datum of a map f from $\Upsilon(B)$ to $\Upsilon(A)$ and of a family of continuous homomorphisms $f_y^{\natural} : A_{f(y)} \to B_y$, for $y \in \Upsilon(B)$. Such a morphism induces a continuous homomorphism $f^{\natural}$ from $A = \prod_{x\in\Upsilon(A)} A_x$ to $B = \prod_{y\in\Upsilon(B)} B_y$. The converse is true:

Proposition. *The contravariant functor $A \mapsto Spf(A)$ is fully faithful.*

Indeed, if $\phi : A \to B$ is a continuous algebra homomorphism, the inverse image $\phi^{-1}(\mathfrak{n})$ of an open maximal ideal of B is an open prime ideal of A, hence maximal in A. One thus obtains a map $\mathfrak{n} \mapsto \phi^{-1}(\mathfrak{n})$ from $\Upsilon(B)$ to $\Upsilon(A)$, and ϕ induces a continuous homomorphism $A_{\phi^{-1}(\mathfrak{n})} \to B_{\mathfrak{n}}$. So ϕ induces a morphism of formal schemes $Spf(\phi) : Spf(B) \to Spf(A)$. One verifies easily that $Spf(\phi)^{\natural} = \phi$, and that $Spf(f^{\natural}) = f$ for every morphism $f : Spf(B) \to Spf(A)$, whence the proposition.

[731] N.D.E.: cf. Remarks 0.1.2.

Although we shall here speak only of formal schemes of the form $Spf(A)$, we shall use the language of formal schemes rather than that of pseudocompact rings, in order to base our assertions on a geometric intuition.

1.2.

Let k be a pseudocompact ring.

Definition 1.2.A.[732] *We shall call a* formal variety over k *any formal scheme* X *over* $Spf(k)$ *which is isomorphic to a formal* k*-scheme* $Spf(A)$ *for some profinite* k*-algebra* A*. The algebra* A *is then isomorphic to the* affine algebra *of* X*, that is, to the algebra of sections of the structure sheaf* O_X *of* X*.*

We denote by Vaf/k *the full subcategory of the category of formal schemes over* $Spf(k)$ *whose objects are the formal* k*-varieties.*[733]

By 1.1, the functor $A \mapsto Spf(A)$ is an anti-equivalence of Alp/k (0.4.1) onto Vaf/k. So, by the corollary of 0.4.2:

Proposition 1.2.B. *The category* Vaf/k *possesses inverse and direct limits.*[734]

For example, let $X \in ObVaf/k$ and $f : Y \to X$ and $g : Z \to X$ be two formal k-varieties over X and let A, B, C be the affine algebras of X, Y, Z; by 0.4.1, the affine algebra of the fiber product $Y \times_X Z$ is identified with $B\hat{\otimes}_A C$,[735] so that the inclusion of Vaf/k in the category of all formal k-schemes commutes with finite inverse limits (cf. EGA I, 10.7).

The direct limits of formal k-varieties correspond to inverse limits of their affine algebras.

Example 1.2.C (Cokernels). Let, for example, $d, e : X \rightrightarrows Y$ be a double arrow of Vaf/k; the affine algebra of $Coker(d,e)$ is isomorphic to the kernel of the homomorphisms induced on the affine algebras of X and Y, but one can also give the following construction of $Coker(d,e)$: the topological space underlying $Coker(d,e)$ is the cokernel of the underlying spaces;[736] if p is the canonical projection of the set underlying Y onto the cokernel and if z belongs to the cokernel, the local algebra of $Coker(d,e)$ at z is the kernel of the double arrow

$$d^\natural, e^\natural : \prod_{p(y) = z} O_{Y, y} \rightrightarrows \prod_{q(x) = z} O_{X, x},$$

where one has set $q = p \circ d = p \circ e$ and where $d^\natural$ and $e^\natural$ are induced by the homomorphisms $d_x^\natural$ and $e_x^\natural$ (notations of 1.1).

[732] N.D.E.: We have added the numbering 1.2.A to 1.2.E, for later references.

[733] N.D.E.: We note that, by the definition of morphisms (1.1), every $X \in ObVaf/k$ is the direct sum in Vaf/k of the pointwise formal varieties $Spf(O_{X,x})$, for $x \in X$.

[734] N.D.E.: In particular, cokernels exist in Vaf/k, see below. We note moreover that a filtered inverse limit in Vaf/k, all of whose transition morphisms are surjective, can be empty, cf. N.D.E. (34).

[735] N.D.E.: We note that $Y \times_X Z$ is the direct sum, for $x \in X$, of the formal varieties $B_x \hat{\otimes}_{O_{X,x}} C_x$, where B_x is the product of the $O_{Y,y}$ for $y \in Y$ such that $f(y) = x$, and C_x is defined analogously. This will be used in 2.5.1.

[736] N.D.E.: i.e., the quotient set of Y by the identifications $d(x) = e(x)$, for every $x \in X$, endowed with the quotient topology, which is here the discrete topology.

Definition 1.2.D. *If $\phi : k \to \ell$ is a homomorphism of pseudocompact rings and X is a formal k-variety with affine algebra A, the formal scheme $X \times_{Spf(k)} Spf(\ell)$, obtained by base change, is a formal ℓ-variety, which we shall also denote $X\hat{\otimes}_k \ell$ and which has as affine algebra the completed tensor product $A\hat{\otimes}_k \ell$ (cf. 0.5 and EGA I, § 10).*

Remark 1.2.E. *Since every formal variety over k decomposes into formal varieties over the local components of k, we shall assume in some proofs that k is a local pseudocompact ring.*

We now give some examples while fixing our terminology.

1.2.1. A *k-functor* will be, by definition, a covariant functor from Alf/k to `(Sets)`. By 1.1 and 0.4.2, one can identify $Vaf/k \simeq (Alp/k)^{\circ}$ with a full subcategory of the category of k-functors, by associating to every formal k-variety X the functor:

```
Alf/k  (Sets),     C ↦ X(C) = Hom_{Vaf/k}(Spf(C), X).
```

We shall encounter later k-functors F that associate to every object C of Alf/k a module $F(C)$ over C and to every morphism $\phi : C \to D$ of Alf/k a k-linear map $F(\phi) : F(C) \to F(D)$ such that, if $x \in F(C)$ and $\lambda \in C$:

```
F(φ)(λ x) = φ(λ) F(φ)(x).
```

By Exposé I, 3.1, such an F is equipped with a structure of O_k-module, if one denotes by O_k the k-functor in rings that associates to every object C of Alf/k the ring underlying C.

Definitions. *(i) An O_k-module F will be said to be* admissible *if every morphism $\phi : C \to D$ of Alf/k induces a bijection of $D \otimes_C F(C)$ onto $F(D)$.*[737]

(ii) One says that F is flat *if it is admissible and if, for every object C of Alf/k, $F(C)$ is a flat C-module.*[738]

For example, if M is a k-module (not necessarily pseudocompact), we shall denote (as in Exposé I, 4.6.1) by $W(M)$ the O_k-module $C \mapsto C \otimes_k M$; then $W(M)$ is flat when M is flat over k.

Moreover, the functor $M \mapsto W(M)$ has as right adjoint the functor that associates to every O_k-module F the k-module $\lim_{\ell} F(k/\ell)$, where ℓ ranges over the open ideals of k.

1.2.2. In what follows, an O_k-module will always be denoted by a boldface letter such as $**F**$; when k is artinian, we shall then write simply F instead of $**F**(k)$. In this case, it goes without saying that the functor $**F** \mapsto F$ induces an equivalence of the category of flat O_k-modules onto that of flat k-modules! The terminology we have adopted has therefore only the goal of allowing us to reason “as if k were always artinian”.

[737]N.D.E.: We have introduced this terminology, cf. N.D.E. (47) in 1.2.3.
[738]N.D.E.: and hence projective, since C is artinian.

In accordance with Exposé I § 3.1, we shall use analogous conventions for other algebraic structures: thus, an O_k-*algebra* (resp. an O_k-coalgebra, resp. an O_k-Lie algebra, resp. an O_k-p-Lie algebra) will consist of the datum of an O_k-module $**M**$ and, for every object C of Alf/k, of a structure of C-algebra (resp. C-coalgebra, resp. C-Lie algebra, resp. C-p-Lie algebra) on $**M**(C)$; one assumes moreover that, for every morphism $\phi : C \to D$ of Alf/k, the map from $D \otimes_C **M**(C)$ to $**M**(D)$ induced by $**M**(\phi)$ is a homomorphism of D-algebras (resp. D-coalgebras, etc.).

Note finally that, if $**F**$ and $**G**$ are two O_k-modules, $**F** \otimes_k **G**$ will denote the O_k-module $C \mapsto **F**(C) \otimes_C **G**(C)$.

1.2.3. [739] We begin with the following lemma.

Lemma 1.2.3.A. *Let $k \to K$ be a morphism of pseudocompact rings, B a pseudocompact K-module, and M a (topology-free) projective k-module. One has a canonical isomorphism of pseudocompact K-modules*

(2) $\theta : \mathrm{Hom}_k(M, k) \hat{\otimes}_k B \xrightarrow{\sim} \mathrm{Hom}_k(M, B).$

Here, $M^* = \mathrm{Hom}_k(M,k)$ is endowed with the topology defined in 0.2.2, which makes it a pseudocompact k-module, and the topology of $\mathrm{Hom}_k(M,B)$ is defined analogously: a basis of neighborhoods of 0 is formed by the following K-submodules, where $x \in M$ and B' is an open K-submodule of B:

$\mathfrak{U}(x, B') = \{f \in \mathrm{Hom}_k(M, B) \mid f(x) \in B'\}.$

Finally, $M^* \hat{\otimes}_k B$ is the pseudocompact K-module deduced from M^* by base change, cf. 0.5.A.

This being so, θ is evidently an isomorphism when $M = k$; moreover, both sides of (2), considered as functors in M, transform direct sums into products (in particular, commute with finite direct sums). One thus obtains that (2) is an isomorphism when M is a free k-module, and then when M is projective.

Definition 1.2.3.B. *Let N be a pseudocompact k-module. We denote by $**V_k^f**(N)$ or $**N**^\dagger$[740] the O_k-module that associates to every $C \in Ob Alf/k$ the C-module $(C \hat{\otimes}_k N)^\dagger$ dual of $C \hat{\otimes}_k N$ (0.2.2), i.e. the C-module*

$\mathrm{Hom}_c(C \hat{\otimes}_k N, C) \simeq \mathrm{Hom}_c(N, C|_k),$

where $C|_k$ denotes the k-module C obtained by restriction of scalars. This O_k-module $**V_k^f**(N)$ will be called the dual *of N.*

If $k_\mathfrak{m}$ is a local component of k, then for every object C of Alf/k,

$\mathrm{Hom}_c(k_\mathfrak{m}, C|_k) = C_\mathfrak{m} = C \otimes_k k_\mathfrak{m}$

is a projective C-module, and moreover one has $D \otimes_C C_\mathfrak{m} = D_\mathfrak{m}$ for every morphism $C \to D$ of Alf/k; so $**V_k^f**(k_\mathfrak{m})$ is a flat O_k-module (cf. 1.2.1). Since every projective

[739] N.D.E.: We have added the lemma that follows and have introduced the numbering 1.2.3.A to 1.2.3.F, for later references.
[740] N.D.E.: In fact, we shall not use this second notation, see N.D.E. (47).

pseudocompact k-module is a product of modules $k_{\mathfrak{m}}$ (cf. Prop. 0.2.1), one deduces from Corollary 0.2.F the:

Corollary 1.2.3.C. *$**V_k^f**(N)$ is a flat O_k-module when N is a projective pseudocompact k-module.*

Definition 1.2.3.D. *Conversely, if $**M**$ is an admissible O_k-module (cf. 1.2.1),*[741] *we call* dual *of $**M**$ and denote by $\Gamma^*(**M**)$ the pseudocompact k-module defined as follows. As ℓ ranges over the open ideals of k, we endow each k/ℓ-module*

$$**M**(k/\ell)^* = \mathrm{Hom}_{k/\ell}(**M**(k/\ell), k/\ell)$$

with the topology described in 0.2.2, which makes it a pseudocompact k-module. Since $**M**(k/\ell) \simeq **M**(k/\ell') \bigotimes (k/\ell)$ for $\ell \supset \ell'$, one has transition morphisms:

$$\mathrm{Hom}_{k/\ell'}(**M**(k/\ell'), k/\ell') \to \mathrm{Hom}_{k/\ell'}(**M**(k/\ell'), k/\ell) = \mathrm{Hom}_{k/\ell}(**M**(k/\ell), k/\ell),$$

and by definition $\Gamma^*(**M**)$ is the inverse limit of this filtered inverse system of pseudocompact k-modules.

From now on, suppose moreover that $**M**$ is a flat O_k-module (cf. 1.2.1); then each $**M**(k/\ell')$ is a projective (k/ℓ')-module, so the transition morphisms above are surjective, and hence so are the projections $\Gamma^*(**M**) \to **M**(k/\ell)^*$, since in $PC(k)$ filtered inverse limits are exact.

If $\tau : k \to K$ is a morphism of pseudocompact rings, we shall denote by $**M** \bigotimes_k K$ or simply $**M_K**$ the O_K-functor defined as follows. If C is a K-algebra of finite length, then the kernel of $k \to C$ is an open ideal ℓ, and one sets $**M_K**(C) = **M**(k/\ell) \bigotimes_k C$; one then has $**M_K**(C) = **M**(k/\ell') \bigotimes_k C$ for every open ideal ℓ' contained in ℓ. One then defines $\Gamma_K^*(**M_K**)$ as the inverse limit, for I ranging over the open ideals of K, of the pseudocompact K-modules:

$$\mathrm{Hom}_{K/I}(**M_K**(K/I), K/I) = \mathrm{Hom}_{k/\ell}(**M**(k/\ell), K/I),$$

where in the right-hand term ℓ is any open ideal of k such that $\tau(\ell) \subset I$. Moreover, by 1.2.3.A, the right-hand side is identified with $**M**(k/\ell)^* \hat{\bigotimes}_k (K/I)$. Since the projections $\Gamma^*(**M**) \to **M**(k/\ell)^*$ are surjective, one sees that the inverse limit of $**M**(k/\ell)^* \hat{\bigotimes}_k (K/I)$ is nothing but the pseudocompact K-module $\Gamma^*(**M**) \hat{\bigotimes}_k K$ (cf. 0.5.A). One has thus obtained that, for every flat O_k-module $**M**$, the formation of $\Gamma^*(**M**)$ commutes with extension of the base, i.e. one has

$$(\star) \qquad \Gamma^*_K(**M** \otimes_k K) \simeq \Gamma^*(**M**) \hat{\otimes}_k K.$$

---

[741] N.D.E.: The editors do not understand the following definition if $**M**$ is not assumed admissible, whence the addition of this hypothesis. On the other hand, we have written $\Gamma^*(**M**)$ instead of $**M**^*$, in order to avoid confusions between M^*, the dual module of the functor $**M**$, and $**N**^\dagger$, the dual functor of the module N, cf. N.D.E. (46). Finally, we have detailed the construction of $\Gamma^*(**M**)$.

Proposition 1.2.3.E. *(i) The functors $N \mapsto **V_k^f**(N)$ and $**M** \mapsto \Gamma^*(**M**)$ define an anti-equivalence between the category of projective pseudocompact k-modules and that of flat O_k-modules.*[742]

(ii) Moreover, if $k \to K$ is a morphism of pseudocompact rings, then the previous anti-equivalence "commutes with base change" in the following sense: if $N \simeq \Gamma^*(**M**)$, then $N \hat{\bigotimes}_k K \simeq \Gamma_K^*(**M** \bigotimes_k K)$.

Proof. On the one hand, one has a natural transformation $\phi_N : N \to \Gamma^*(**V_k^f**(N))$. Since the functor $\Gamma^* \circ **V_k^f**$ commutes with products, by 0.3.5 and 0.2.F, it suffices to verify that ϕ_N is an isomorphism when N is a local component $k_{\mathfrak{m}}$ of k. In this case, for every open ideal ℓ of k contained in $\mathfrak{m}$, the natural morphism

$$(k/\ell)_{\mathfrak{m}} \to \mathrm{Hom}_{k/\ell}(\mathrm{Hom}_c(k_{\mathfrak{m}} \hat{\otimes} k/\ell, k/\ell), k/\ell)$$

is an isomorphism, whence the result.

On the other hand, let $**M**$ be a flat O_k-module. Let us show that $\Gamma^*(**M**) = \lim **M**(k/\ell)^*$ is a projective object of $PC(k)$. Let $N \to N'$ be a surjective morphism between objects of $LF(k)$. By 0.2.F (i) and (ii), it suffices to show that the natural map

$$\mathrm{colim}\, \mathrm{Hom}_c(**M**(k/\ell)^*, N) \to \mathrm{colim}\, \mathrm{Hom}_c(**M**(k/\ell)^*, N')$$

is surjective. But this is clear, because N and N' are k/ℓ_0-modules for some open ideal ℓ_0; so if $\ell \subset \ell_0$, every morphism $**M**(k/\ell)^* \to N'$ lifts to a morphism $**M**(k/\ell)^* \to N$, since $**M**(k/\ell)^*$ is a projective object of $PC(k/\ell)$.

One has a morphism $\psi : **M** \to **V_k^f**(\Gamma^*(**M**))$ of functors from Alf/k to (Sets). Let B be an object of Alf/k; let us show that

$$(1) \qquad \psi(B) : **M**(B) \to **V_k^f**(\Gamma^*(**M**))(B) = \mathrm{Hom}_c(\lim_\ell **M**(k/\ell)^* \hat{\otimes}_k B, B)$$

is a bijection (in the equality above, we have used the fact that $\hat{\bigotimes}$ commutes with filtered inverse limits). Let ℓ_0 be an open ideal of k contained in the kernel of $k \to B$. By Lemma 1.2.3.A, for every $\ell \subset \ell_0$, one has a canonical isomorphism of pseudocompact B-modules:

$$**M**(k/\ell)^* \hat{\otimes}_k B \to \mathrm{Hom}_{k/\ell}(**M**(k/\ell), B)$$

and, since $**M**(B) = **M**(k/\ell) \bigotimes_{k/\ell} B$, the right-hand side equals $\mathrm{Hom}_B(**M**(B), B)$. So the inverse system in (1) is constant for $\ell \subset \ell_0$, and (1) reduces to the canonical morphism

$$**M**(B) \to \mathrm{Hom}_c(\mathrm{Hom}_B(**M**(B), B), B),$$

which is an isomorphism by 0.2.2, since B is artinian and $**M**(B)$ a projective B-module. This proves point (i) of the proposition, and point (ii) follows from the isomorphism (□) established above.

[742] N.D.E.: We have added point (ii) below, as well as the proof that follows. The original stated: "If $**M**$ is a flat O_k-module, it is clear that $**M**$ is nothing other than the dual of $**M**^*$, so the functors $N \mapsto **N**^*$ and $**M** \mapsto **M**^*$ define an anti-equivalence..."

Remark 1.2.3.F. *Let us return to the statement of the proposition, and suppose moreover that N is a topologically free pseudocompact k-module. In this case, one can choose "coherently" bases for the C-modules $**V_k^f**(N)(C)$.*

Indeed, let (n_i) be a pseudobasis of N (0.2.1) and n_i^C the canonical image of n_i in $C\hat{\otimes}_k N$, for $C \in Alf/k$. If one defines the element δ_i^C of $(C \hat{\otimes}_k N)^\dagger$ by the equalities $\delta_i^C(n_i^C) = 1$ and $\delta_i^C(n_j^C) = 0$ for $i \neq j$, the family (δ_i^C) is a basis of $**V_k^f**(N)(C)$; moreover, for every morphism $\phi : C \to D$ of Alf/k, $**V_k^f**(N)(\phi)$ sends δ_i^C to δ_i^D.

1.2.4. [743] For every pseudocompact k-module E, let $\hat{S}_k(E)$ be the *completed symmetric algebra* of E, defined as follows. Let $T_k(E)$ be the direct sum of the pseudocompact k-modules:

$$\hat{\otimes}^n_k E = E \hat{\otimes}_k \cdots \hat{\otimes}_k E \quad (n \geqslant 0;\ \text{if } n = 0,\ \hat{\otimes}^0_k E = k);$$

one makes $T_k(E)$ into a graded k-algebra by defining the multiplication in the usual way; let $S_k(E)$ be the quotient of $T_k(E)$ by the homogeneous ideal whose n-th component is the closed k-submodule of $\hat{\otimes}^n_k E$ generated by the elements:

$$x_1 \hat{\otimes} \cdots \hat{\otimes} x_i \hat{\otimes} x_{i+1} \hat{\otimes} \cdots \hat{\otimes} x_n - x_1 \hat{\otimes} \cdots \hat{\otimes} x_{i+1} \hat{\otimes} x_i \hat{\otimes} \cdots \hat{\otimes} x_n.$$

If one denotes by $S_k^n(E)$ this quotient, $S_k(E)$ is obviously a graded k-algebra with n-th component $S_k^n(E)$.

One endows $S_k(E)$ with the linear topology defined by the set Υ of ideals ℓ such that $S_k(E)/\ell$ is a k-module of finite length and that $S_k^n(E) \cap \ell$ is an open submodule of $S_k^n(E)$ for every n. Then the profinite algebra $\hat{S}_k(E)$ is the separated completion of $S_k(E)$ for this topology.[744]

We denote by $**V_k^f**(E)$ the formal k-variety $Spf(\hat{S}_k(E))$. It represents the k-functor $**V_k^f**(E)$, i.e., for every object C of Alf/k, one has a canonical isomorphism:

$$**V_k^f**(E)(C) = \mathrm{Hom}_c(E, C|_k) \simeq \mathrm{Hom}_{Alp/k}(\hat{S}_k(E), C) = \mathrm{Hom}_{Vaf/k}(Spf(C), **V^f_k**(E)).$$

In all the sequel, we identify $**V_k^f**(E)$ with the k-functor $**V_k^f**(E)$.

1.2.5. By 1.2.4, the zero morphism from E to k is associated with a morphism of profinite algebras $\pi : \hat{S}_k(E) \to k$; this morphism π induces the zero map on $S_k^n(E)$ for $n \geqslant 1$ and defines a section of the structure morphism $**V_k^f**(E) \to Spf(k)$.

[743] N.D.E.: In this paragraph, we have modified the order, by first defining $\hat{S}_k(E)$ and then stating that $**V_k^f**(E)$ represents $**V_k^f**(E)$.

[744] N.D.E.: For example, let k be a field, E a pseudocompact k-vector space, $V = \mathrm{Hom}_c(E, k)$; one has $E \simeq V^*$ (cf. 0.2.2). For every finite-dimensional subspace W of V, let $F(W)$ be the set of closed points of the k-scheme $V(W) = \mathrm{Spec}\, S_k(W^*)$. Denote by $F(V)$ the direct limit of the $F(W)$. Then $\hat{S}_k(E)$ is the product, for $x \in F(V)$, of the pseudocompact k-algebras $\hat{S}_k(E)_x = \lim_W \hat{O}_{W,x}$, where W ranges over the finite-dimensional subspaces of V such that $x \in F(W)$, and $\hat{O}_{W,x}$ denotes the separated completion of the local ring $O_{V(W),x}$ for the topology defined by the ideals of finite codimension (which here coincides with the $\mathfrak{m}$-adic topology). If k is algebraically closed and $(v_i)_{i\in I}$ a basis of V, so that E possesses a pseudobasis $(e_i)_{i\in I}$, then each local component $\hat{S}_k(E)_x$ is isomorphic to the ring of formal series $k[[e_i, i \in I]]$, endowed with the topology defined by the ideals of finite codimension.

We shall denote by $**V_k^{f,0}**(E)$ the formal variety which has as points the images of the points of $Spf(k)$ under the section $Spf(\pi)$ and which has the same local algebras as $**V_k^f**(E)$ at these points.[745]

Then, the affine algebra of $**V_k^{f,0}**(E)$ is the separated completion of $S_k(E)$ for the topology defined by the ideals $\ell \in \Upsilon$ (cf. 1.2.4) that contain $S_k^n(E)$ for n large enough. One deduces that it is the infinite product:

```
k[[E]] = k × E × S^2_k(E) × S^3_k(E) × ⋯
```

On the other hand, let C be an object of Alf/k, u an element of $**V_k^f**(C) = \mathrm{Hom}_c(E,C)$, and $\phi : \hat{S}_k(E) \to C$ the corresponding morphism of profinite k-algebras. Then $Ker(\phi)$ is an open ideal (i.e. ϕ belongs to $**V_k^{f,0}**(C)$) if and only if $Ker(\phi)$ contains $S_k^n(E)$ for n large enough, i.e., if and only if $u(E)$ is contained in the radical $r(C)$ of C. Therefore: for every object C of Alf/k, one has canonical isomorphisms:

```
**V^{f,0}_k**(E)(C) ≃ Hom_{Alp/k}(k[[E]], C) ≃ Hom_c(E, r(C)).
```

1.2.6. A formal k-variety V is said to be *of finite length* if its affine algebra is. Likewise, if S is a scheme, an S-scheme X is said to be of *finite length* if X is finite over S and if the direct image of `O_X` on S is an `O_S`-module of finite length.[746] So, to give an S-scheme of finite length X is "the same thing" as to give a finite set $s_1, \cdots, s_n$ of closed points of S, and at each of these points, an O_{S,s_i}-algebra of finite length.

One sees therefore that the S-schemes of finite length identify with the formal varieties of finite length over the formal scheme Ŝ that follows. The topological space underlying Ŝ is the set of closed points of S endowed with the discrete topology; if s is such a closed point, the structure sheaf $O_{\hat{S}}$ has as stalk $O_{\hat{S},s}$ at s the separated completion $\hat{O}_{S,s}$ of $O_{S,s}$ for the linear topology defined by the ideals of finite colength; one therefore has $\hat{S} = Spf\mathcal{A}(\hat{S})$, where $\mathcal{A}(\hat{S})$ is the product of the $\hat{O}_{S,s}$, for s ranging over the closed points of S, endowed with the product topology.

Definition. *If X is an S-scheme, we denote by $\hat{X}/\hat{S}$ the formal variety over $k = \mathcal{A}(\hat{S})$ defined as follows. The underlying topological space is formed by the points $x \in X$ such that $[\kappa(x) : \kappa(s)] < \infty$, where s is the image of x in S; the local ring $O_{\hat{X}/\hat{S},x}$ is the separated completion of $O_{X,x}$ for the linear topology defined by the ideals I of $O_{X,x}$ such that $O_{X,x}/I$ is of finite length as an $O_{S,s}$-module (N.B. since $[\kappa(x) : \kappa(s)] < \infty$, this is equivalent to saying that $O_{X,x}/I$ is of finite length as an $O_{X,x}$-module).*

Let $Vaf\ell f/\hat{S}$ be the category of formal varieties of finite length over Ŝ (identified with that of S-schemes of finite length). By 1.1 and 1.2.1, the category $Vaf/\hat{S}$ of formal varieties over Ŝ is equivalent to that of left-exact contravariant functors from $Vaf\ell f/\hat{S}$ to `(Sets)`. In particular, for every S-scheme X, the functor $T \mapsto \mathrm{Hom}_{(Sch/S)}(T,X)$, from $Vaf\ell f/\hat{S}$ to `(Sets)`, is such a left-exact functor, hence corresponds to a formal variety over Ŝ. The latter is nothing but $\hat{X}/\hat{S}$:[747]

[745]N.D.E.: We have detailed the rest of this paragraph.
[746]N.D.E.: We have added the following sentence.
[747]N.D.E.: We have amplified the following proposition by inserting in it the fact that the functor $X \mapsto$

Proposition. *For every S-scheme X, the functors*

$$\mathrm{Hom}_{Vaf/\hat{S}}(-, \hat{X}/\hat{S}) \quad \text{and} \quad \mathrm{Hom}_{(Sch/S)}(-, X)$$

have the same restriction to $Vaf\ell f/\hat{S}$. One thus obtains a functor $X \mapsto \hat{X}/\hat{S}$ from (Sch/S) to $Vaf/\hat{S}$ which commutes with finite inverse limits.

Indeed, one sees easily that the formal variety $\hat{X}/\hat{S}$ has the required property, and that the correspondence $X \mapsto \hat{X}/\hat{S}$ is functorial. Let us prove the second assertion.

Set $\ast\ast S \ast\ast = (Sch/S)$ and $\ast\ast V \ast\ast = Vaf/\hat{S}$, and write $\hat{X}$ instead of $\hat{X}/\hat{S}$. We know (1.2.B) that $\ast\ast V \ast\ast$ possesses arbitrary inverse limits. Let $(X_i)_{i\in I}$ be an inverse system in $\ast\ast S \ast\ast$ and suppose that $X = \lim X_i$ exists in $\ast\ast S \ast\ast$ (which is the case if I is finite). Since the functor that associates to every object Y of $\ast\ast S \ast\ast$ (resp. V of $\ast\ast V \ast\ast$) the functor $h^Y = \mathrm{Hom}_{\ast\ast S \ast\ast}(-, Y)$ (resp. $h^V = \mathrm{Hom}_{\ast\ast V \ast\ast}(-, V)$) commutes with inverse limits, one has, for every S-scheme T of finite length, functorial isomorphisms:

$$\mathrm{Hom}_{**S**}(T, X) \simeq \lim \mathrm{Hom}_{**S**}(T, X_i) \simeq \lim \mathrm{Hom}_{**V**}(T, \hat{X}_i) \simeq \mathrm{Hom}_{**V**}(T, \lim \hat{X}_i).$$

Consequently, the functor $X \mapsto \hat{X}/\hat{S}$ commutes with inverse limits when they exist in (Sch/S); in particular, it commutes with finite inverse limits.

1.3.

Proposition 1.3. *Let $f : X \to Y$ be a morphism of formal varieties over k, A and B the affine algebras of X and Y, $g : B \to A$ the morphism induced by f. Then f is a monomorphism of Vaf/k if and only if g is a surjection.*

[748] By 1.1, $A \mapsto Spf(A)$ is an anti-equivalence of Alp/k onto Vaf/k, so f is a monomorphism if and only if g is an epimorphism, and this is the case if g is surjective.

Conversely, suppose that g is an epimorphism and let us show that it is surjective. For every $\mathfrak{n} \in \Upsilon(B)$, set $A_{\mathfrak{n}} = A\hat{\otimes}_B B_{\mathfrak{n}}$; by 0.4, A is a pseudocompact B-module, hence is the product of the $A_{\mathfrak{n}}$ (cf. 0.3.6). Then g is the product of the morphisms $g_{\mathfrak{n}} : B_{\mathfrak{n}} \to A_{\mathfrak{n}}$ deduced from g by base change. These are still epimorphisms, which reduces us to proving the result when B is local with maximal ideal $\mathfrak{n}$. Set $K = B/\mathfrak{n}$.

By Nakayama's Lemma 0.3.3, it suffices to show that the morphism $g\hat{\otimes}_B K$ is surjective; it is deduced from g by base change, so is an epimorphism of Alp/K. One can therefore assume that $B = K$ is a field. Now f is a monomorphism if and only if the diagonal morphism $X \to X \times_Y X$ is an isomorphism, that is, if the homomorphism $x\hat{\otimes}_K x' \mapsto xx'$ is an isomorphism of $A\hat{\otimes}_K A$ onto A. Since K is a field, this implies $A = K$.

$\hat{X}/\hat{S}$ commutes with finite inverse limits (in the original, this appeared in the proof of 1.3.4 — the proof given here is more direct than the original). Moreover, this result will be useful in Section 2 and in 3.3.1.

[748] N.D.E.: We have detailed the beginning of the proof.

Remark.[749] *It follows from the proposition that every monomorphism $f : X \to Y$ of formal varieties is an isomorphism of X onto a (necessarily closed!) formal subvariety of Y.*

1.3.1. The preceding proposition implies in particular that every monomorphism $f : X \to Y$ of Vaf/k is effective (cf. IV 1.3).[750] It is not the same for epimorphisms, as one easily sees by slightly modifying the counterexample of Exposé V, § 2.c);[751] this is why we shall consider a sympathetic class of effective epimorphisms.

Lemma.[752] *Let $f : X \to Y$ be a morphism of formal k-varieties and let A, B be the affine algebras of X, Y and $f^{\natural} : B \to A$ the morphism induced by f. The following conditions are equivalent:*

(i) For every $x \in X$, the homomorphism $f_x^{\natural} : O_{Y,f(x)} \to O_{X,x}$ makes $O_{X,x}$ a topologically free pseudocompact $O_{Y,f(x)}$-module.

(ii) For every $y \in Y$, the local component $A_y = \prod_{f(x)=y} O_{X,x}$ is a topologically free pseudocompact B_y-module.

(iii) $f^{\natural} : B \to A$ makes A a projective pseudocompact B-module.

(iv) The functor $PC(B) \to PC(A)$, $M \mapsto M\hat{\otimes}_B A$, is exact.

If these conditions are satisfied, one says that f is topologically flat.

The implications (i) ⇒ (ii) ⇔ (iii) ⇔ (iv) follow from 0.2.F (iii) and 0.3.7. Conversely, assume (ii) holds and let $x \in X$ and $y = f(x)$. Since $O_{X,x}$ is a direct factor of A_y, it is a projective pseudocompact B_y-module, hence topologically free by 0.2.1 (since B_y is local).

On the other hand, a morphism $f : X \to Y$ of formal k-varieties is said to be *surjective* if it induces a surjection of the underlying sets.

Proposition. *Let $f : X \to Y$ be a surjective and topologically flat morphism of formal k-varieties. Then f is an effective epimorphism (cf. IV 1.3).*

Indeed, let A, B be the affine algebras of X, Y and $g : B \to A$ the morphism induced by f. We must show that Y is identified with the cokernel of $X \times_Y X \rightrightarrows X$, i.e., that for every $\mathfrak{n} \in \Upsilon(B)$, $B_{\mathfrak{n}}$ is identified with the subring of $A_{\mathfrak{n}} = A\hat{\otimes}_B B_{\mathfrak{n}}$ formed by the a such that $a\hat{\otimes}1 = 1\hat{\otimes}a$.

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[749] N.D.E.: We have added this remark.

[750] N.D.E.: i.e., in the opposite category $(Vaf/k)^{\circ} = Alp/k$, the morphism $g : B \to A$ corresponding to f is an effective epimorphism. This is the case, because g is surjective, hence induces (cf. the proof of 0.2.B) an isomorphism of profinite k-algebras $B/I \xrightarrow{\sim} A$, where $I = Ker g$. Consequently, every morphism $\phi : B \to C$ of Alp/k that is zero on I descends to a morphism $\bar{\phi} : A \to C$ such that $\phi = \bar{\phi} \circ g$.

[751] N.D.E.: i.e., let k be a field, $X = Spf(k[[T]])$ and $Y = Spf(B)$, where B is the k-subalgebra of $A = k[[T]]$ generated topologically by T^3 and T^4 (i.e., B is formed by the formal series $\sum a_n T^n$ such that $a_n = 0$ for $n = 1, 2, 5$). Then $X \to Y$ is an epimorphism that is not effective; indeed, the cokernel of $X \times_Y X \rightrightarrows X$ is $Spf(B')$, where B' is the subalgebra of A formed by the a such that $a\hat{\otimes}1 = 1\hat{\otimes}a$, and B' contains T^5.

[752] N.D.E.: We have added this lemma, which explains the terminology "topologically flat".

We can therefore assume B local, with maximal ideal $\mathfrak{n}$. Our hypothesis then means that g makes A a topologically free and nonzero B-module. By Nakayama's Lemma 0.3.3, $A/\mathfrak{n}A$ is not zero, so the morphism $g' : B/\mathfrak{n} \to A/\mathfrak{n}A$ deduced from g is injective. By Lemma 1.3.2 below, B is a direct factor of A as a B-module, say $A = B \oplus A'$; it follows that B is identified with the part of A formed by the a such that $a\hat{\otimes}_B 1 = 1\hat{\otimes}_B a$.

1.3.2. Lemma 1.3.2. *Let B be a local pseudocompact ring, $\mathfrak{n}$ its maximal ideal, M and N two projective pseudocompact B-modules, and g a morphism $M \to N$. If $(B/\mathfrak{n})\hat{\otimes}_B g$ is injective, g is an isomorphism of M onto a direct factor of N.*

Indeed, suppose $g' = (B/\mathfrak{n})\hat{\otimes}_B g$ is injective. Since $B/\mathfrak{n}$ is a field, g' then has a retraction r'. Let p and q be the canonical projections of M and N onto $M/\mathfrak{n}M$ and $N/\mathfrak{n}N$; since N is projective, there exists a morphism $r : N \to M$ such that $p \circ r = r' \circ q$; consequently, r' is deduced from r by passage to the quotient. Then, since $r' \circ g'$ is an isomorphism, so is $r \circ g$, by 0.3.4 (since M is projective). Let s be the inverse isomorphism of $r \circ g$; then $s \circ r$ is a retraction of g.

1.3.3. Proposition 1.3.3. *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of formal k-varieties.*

(i) If f and g are topologically flat, so is $g \circ f$.

(ii) If f and $g \circ f$ are topologically flat and if f is surjective, g is topologically flat.

(iii) If f is topologically flat, so is $f \times_Y Y'$ for every base change $Y' \to Y$.

Assertions (i) and (iii) are clear. To prove (ii), let A, B, C be the affine algebras of X, Y, Z, and $f' : B \to A$ and $g' : C \to B$ the morphisms induced by f and g. Since $g \circ f$ is topologically flat, $f' \circ g'$ makes A a projective pseudocompact C-module; likewise, f' makes A a projective pseudocompact B-module that is also faithful. As P ranges over the pseudocompact C-modules and N over the pseudocompact B-modules, the functors $P \mapsto P\hat{\otimes}_C A$ and $N \mapsto N\hat{\otimes}_B A$ are therefore exact; since the second is moreover faithful, the functor $P \mapsto P\hat{\otimes}_C B$ is exact; by 0.3.7, B is therefore a projective pseudocompact C-module.

1.3.4. Proposition 1.3.4. *Let S be a scheme, Y a locally noetherian S-scheme, and $f : X \to Y$ an S-morphism locally of finite type and faithfully flat, so that f is an effective epimorphism, i.e., the sequence below is exact (cf. IV 6.3.1 (iv) and IV 1.3):*

```
(∗)        X ×_Y X ⇒ X —f→ Y.
```

Then the morphism of formal $\hat{S}$-varieties $\hat{f} : \hat{X}/\hat{S} \to \hat{Y}/\hat{S}$ (cf. 1.2.6) is surjective and topologically flat, and the sequence below, deduced from (∗), is exact:

```
(∗̂)        X̂ ×_{Ŷ} X̂/Ŝ ⇒ X̂/Ŝ —f̂→ Ŷ/Ŝ.
```

Indeed, let y be a point of Y with projection s on S and such that $\kappa(y)$ is a finite extension of the residue field $\kappa(s)$ of s. Since f is surjective and locally of finite type, $f^{-1}(y)$ is non-empty and locally of finite type over $\kappa(y)$; the closed points of $f^{-1}(y)$ are then the points of $\hat{X}/\hat{S}$ projecting onto y. This shows that $\hat{f}$ is surjective.

[753] Let x be a closed point of $f^{-1}(y)$. Since Y is locally noetherian and f locally of finite type, the local ring $O_{Y,y}$ (resp. $O_{X,x}$) is noetherian, so the powers of the maximal ideal are of finite colength, so that the local ring of $\hat{Y}/\hat{S}$ at y (resp. of $\hat{X}/\hat{S}$ at x) is the completion $\hat{O}_{Y,y}$ of $O_{Y,y}$ (resp. $\hat{O}_{X,x}$ of $O_{X,x}$) for the $\mathfrak{m}$-adic topology. Then, since f is flat, $\hat{O}_{X,x}$ is flat over $\hat{O}_{Y,y}$, by (SGA 1, IV 5.8). Hence, by 0.3.8, $\hat{O}_{X,x}$ is a topologically free $\hat{O}_{Y,y}$-module. This shows that $\hat{f}$ is topologically flat.

So, by Proposition 1.3.1, $\hat{f}$ is an effective epimorphism, i.e., the sequence below (where one has written $\hat{X}$ instead of $\hat{X}/\hat{S}$) is exact:

$$\hat{X} \times_{\hat{Y}} \hat{X} \rightrightarrows \hat{X} \xrightarrow{\hat{f}} \hat{Y}.$$

Moreover, by 1.2.6, one has a natural isomorphism (which in particular commutes with the projections on $\hat{X}$):

$$\hat{X} \times_Y X \simeq \hat{X} \times_{\hat{Y}} \hat{X}.$$

Consequently, the sequence $(\hat{*})$ is exact.

1.3.5. Let k be a pseudocompact ring. A formal variety X over k is said to be *topologically flat* if its affine algebra A is a projective pseudocompact k-module, i.e., if the structure morphism $X \to Spf(k)$ is topologically flat.

[754] Let us first note that 0.2.2 and 0.3.6 imply the following result (analogous to VII_A, 3.1.1).

Lemma 1.3.5.A. *Suppose k artinian. The functors $A \mapsto A^\dagger = \mathrm{Hom}_c(A, k)$ and $C \mapsto C^* = \mathrm{Hom}_k(C, k)$ define an anti-equivalence between the category of topologically flat profinite k-algebras, and that of flat k-coalgebras.*

Indeed, if A is a topologically flat profinite k-algebra, then by 0.3.6, one has an isomorphism of k-modules:

$$A^\dagger \otimes_k A^\dagger \cong (A \hat{\otimes} A)^\dagger,$$

so that the multiplication $A \hat{\otimes} A \to A$ induces by duality a k-coalgebra structure on $A^\dagger$. The rest then follows from Proposition 0.2.2.

Let us return to the case of an arbitrary pseudocompact ring k.

Definition 1.3.5.B. *To every formal k-variety X whose affine ring A is a projective pseudocompact k-module, one associates a flat O_k-coalgebra $**H**(X)$, defined as follows.*

[753]N.D.E.: We have detailed what follows; then we have taken advantage of the addition made in 1.2.6.
[754]N.D.E.: We have added the following lemma, used implicitly in the original; on the other hand, we have introduced the numbering 1.3.5.A to 1.3.5.D, for later references.

Denote by $**H**(X)$ the O_k-module $**V_k^f**(A)$, "*dual of A*"; it is a flat O_k-module, since the pseudocompact k-module underlying A is projective (cf. 1.2.3.C). Moreover, by 0.3.6, one has:

$$**V_k^f**(A \hat{\otimes} A) \simeq **V_k^f**(A) \otimes **V_k^f**(A),$$

and so the multiplication of A induces by transposition a diagonal morphism:

$$**H**(X) = **V_k^f**(A) \to **V_k^f**(A \hat{\otimes} A) = **H**(X) \otimes **H**(X),$$

which makes $**H**(X)$ a flat O_k-coalgebra. We shall say that $**H**(X)$ is the *coalgebra of X over O_k*.

Definition 1.3.5.C. *Conversely, to every O_k-coalgebra $**C**$ one can associate a k-functor (cf. 1.2.1) $Spf^*(**C**)$, defined as follows. For every object B of Alf/k, one sets (with the notations of VII_A 3.1):*

$$(1)\quad \begin{aligned} Spf^*(**C**)(B) &= \mathrm{Hom}_{B\text{-coalg.}}(B, **C**(B)) \\ &= \{x \in **C**(B) \mid \varepsilon_{**C**(B)}(x) = 1 \text{ and } \Delta_{**C**(B)}(x) = x \otimes x\}. \end{aligned}$$

[755] Assume moreover that the O_k-module underlying $**C**$ is admissible (cf. 1.2.1), and set

$$(2)\quad A = \Gamma^*(**C**) = \lim_{\ell} **C**(k/\ell)^*.$$

Then the algebra structures on each $**C**(k/\ell)^*$ endow A with a structure of profinite k-algebra. For every object B of Alf/k, one has:

$$(3)\quad \mathrm{Hom}_{Vaf/k}(Spf(B), Spf(A)) = \mathrm{Hom}_{Alp/k}(A, B) = \mathrm{Hom}_{Alp/B}(A \hat{\otimes} B, B).$$

Assume finally that $**C**$ is a flat O_k-module. Then, by 1.2.3.E, $A = \Gamma^*(**C**)$ is a projective pseudocompact k-module. Moreover, we saw in the proof of *loc. cit.* that, if ℓ_0 is an open ideal of k contained in the kernel of $k \to B$, one has isomorphisms

$$(4)\quad A \hat{\otimes} B = \lim_{\ell} **C**(k/\ell)^* \hat{\otimes}_k B \simeq \mathrm{Hom}_{k/\ell_0}(**C**(k/\ell_0), B) \simeq \mathrm{Hom}_B(**C**(B), B),$$

and we shall denote by $**C**(B)^*$ the right-hand term. Finally, by Lemma 1.3.5.A applied to the artinian ring B, one has a natural isomorphism

$$(5)\quad \mathrm{Hom}_{B\text{-coalg.}}(B, **C**(B)) \cong \mathrm{Hom}_{Alp/B}(**C**(B)^*, B).$$

Then, combining (1), (5), (4), (3) and (2), one obtains, when $**C**$ is a flat O_k-coalgebra, an isomorphism of functors:

$$(\star)\quad Spf^*(**C**) \simeq Spf(A) = Spf(\Gamma^*(**C**)).$$

Consequently, if one denotes by $\mathcal{A}(X)$ the affine algebra of a formal k-variety X, one obtains, taking 1.2.3.E into account:

Proposition 1.3.5.D. *(i) The functors $X \mapsto **H**(X) = **V_k^f**(\mathcal{A}(X))$ and $**C** \mapsto Spf^*(**C**) = Spf(\Gamma^*(**C**))$ define an equivalence between the category of topologically flat formal k-varieties and that of flat O_k-coalgebras.*

[755] N.D.E.: We have detailed what follows.

(ii) Moreover, this equivalence "commutes with base change": if $k \to K$ is a morphism of pseudocompact rings, then $X\hat{\bigotimes}_k K$ corresponds to $**H**(X) \bigotimes_k K$.

1.3.6. In the rest of this Exposé, we shall several times define a topologically flat formal k-variety X by exhibiting the coalgebra $**H**(X)$. We shall then need to interpret by means of $**H**(X)$ certain geometric properties of X. We give here an example of this situation: suppose given a section σ of the structure morphism $X \to Spf(k)$ and ask ourselves under what condition σ induces an isomorphism on the underlying topological spaces.[756]

To begin, suppose k artinian. Let (H, Δ, ϵ) be a flat k-coalgebra, $H^+ = Ker(\epsilon)$ and $A = H^*$ the profinite k-algebra dual to H. Suppose given a morphism of k-coalgebras $k \to H$, i.e., an element ϕ of H such that $\epsilon(\phi) = 1$ and $\Delta(\phi) = \phi \otimes \phi$. On the one hand, ϕ defines a continuous algebra morphism $\Phi : A \to k$, and hence a section $\sigma : Spf(k) \to Spf(A)$ of the projection $Spf(A) \to Spf(k)$.

On the other hand, one defines k-submodules of H by setting $H_0 = k\phi$ and, for $n \geqslant 1$,

$$H_n = \{x \in H \mid \Delta(x) - x \otimes \phi \in H_{n-1} \otimes H^+\};$$

this is also valid for $n = 0$ if one sets $H_{-1} = (0)$. One sees, by induction on n, that $H_{n-1} \subset H_n$. We say that $H_0 \subset H_1 \subset \cdots$ is the *filtration of H defined by ϕ*.

Remark. *Since $\Delta(H_n) \subset H_n \bigotimes H_0 \bigoplus H_{n-1} \bigotimes H^+$, one has $\Delta(H_n) \subset H_n \bigotimes H$. Since Δ is cocommutative (i.e. $\sigma \circ \Delta = \Delta$, where $\sigma(a \bigotimes b) = b \bigotimes a$), one also has $\Delta(H_n) \subset H \bigotimes H_n$. When H/H_n is flat over k, it follows that H_n is a sub-coalgebra of H (see also 1.3.6.A (iii) below). But in general, $\Delta : H_n \to H_n \bigotimes H$ does not factor through $H_n \bigotimes H_n$.*[757]

Lemma 1.3.6.A. *Let k be an artinian ring, H a flat k-coalgebra, $A = H^*$ the dual profinite k-algebra, ϕ an element of H such that $\epsilon(\phi) = 1$ and $\Delta(\phi) = \phi \bigotimes \phi$. Let $\Phi : A \to k$ be the continuous algebra morphism, $\sigma : Spf(k) \to Spf(A)$ the section of $Spf(A) \to Spf(k)$, and (H_n) the filtration of H defined by ϕ. Set $I = Ker\Phi$.*

(i) For every $n \geqslant 1$, H_{n-1} is the orthogonal in H of the closure $\bar{I}^n$ of I^n.

(ii) Consequently, σ induces a bijection of the underlying sets if and only if $H = \bigcup_n H_n$.[758]

(iii) If moreover each H/H_n is flat over k, then for every $n \geqslant 0$,

$$(*) \qquad \Delta(H_n) \subset \sum_{i=0}^n H_i \otimes H_{n-i};$$

in particular, each H_n is then a sub-coalgebra of H.

[756]N.D.E.: In Lemma 1.3.6.A that follows, we have detailed the proof of points (i) and (ii), and have added point (iii).

[757]N.D.E.: For example, let k_0 be a field, $k = k_0[T]/(T^n)$, where $n \geqslant 4$, $H = k\phi \bigoplus kx$, with $\epsilon(x) = 0$ and $\Delta(x) = x \bigotimes \phi + \phi \bigotimes x + tx \bigotimes x$, where t is the image of T in k. Then $H_i = k\phi \bigoplus t^{n-i}x$ for $i = 0, \cdots, n$, but for $2 \leqslant i \leqslant n-2$, $\Delta(t^{n-i}x)$ does not belong to the image of $H_i \bigotimes H_i$ in $H \bigotimes H$.

[758]N.D.E.: In this case, one says that the coalgebra H is *connected*, cf. the addition 2.9.

Proof. Note that, for every $x \in H$, the map $A \to k$, $f \mapsto f(x)$ is continuous, so if I^n is annihilated by x, so is its closure $\bar{I}^n$. We set then, for every $n \geqslant 1$,

(I^n)^⊥ = {x ∈ H | f(x) = 0, for every f ∈ I^n}.

Suppose that $\sigma : \mathfrak{m} \mapsto \Phi^{-1}(\mathfrak{m})$ is a bijection of $Spf(k)$ onto $Spf(A)$. Since I is contained in the intersection of the $\Phi^{-1}(\mathfrak{m})$, it follows from 0.1.2 that the sequence of ideals $(\bar{I}^n)$ tends to 0. Let then $x \in H$; since $J(x) = f \in A | f(x) = 0$ is an open and closed k-submodule of A, it contains $\bar{I}^n$ for n large enough; in other words, $x \in (I^n)^{\perp}$ for n large enough.

Conversely, suppose that $H = \bigcup_n (I^n)^{\perp}$. Let $\mathfrak{p}$ be an open prime ideal of A; by the definition of the topology of A (0.2.2), $\mathfrak{p}$ contains an open k-submodule of the form

V(x_1, …, x_s) = {f ∈ A | f(x_1) = ⋯ = f(x_s) = 0}.

By hypothesis, there exists an integer n such that $x_1, \cdots, x_s \in (I^n)^{\perp}$, and so $\bar{I}^n \subset \mathfrak{p}$. Moreover, since k is artinian, $Spf(k)$ is a finite set $\mathfrak{m}_1, \cdots, \mathfrak{m}_r$ and there exists an integer $t \geqslant 1$ such that $(\prod_i \mathfrak{m}_i)^t = 0$, whence

$$\prod_i \Phi^{-1}(\mathfrak{m}_i)^t \subset I.$$

So $\mathfrak{p}$ contains the product of the $\Phi^{-1}(\mathfrak{m}_i)^{tn}$; since $\mathfrak{p}$ is prime, it follows that $\mathfrak{p}$ contains, hence equals, one of the $\Phi^{-1}(\mathfrak{m}_i)$. One has thus shown that:

σ is a bijection H = _n (I^n)^⊥.

[759]

On the other hand, one has $H = k\phi \bigoplus H^+$; denote by π the projection $H \to H^+$ of kernel $k\phi$. For every $n \geqslant 0$, let Δ^n be the "iterated" comultiplication $H \to H^{\otimes(n+1)}$, $\bar{\Delta}^n$ the composite of Δ^n with the projection $\pi^{\otimes(n+1)} : H^{\otimes(n+1)} \to (H^+)^{\otimes(n+1)}$, and

H'_n = Ker(Δ^n) = {x ∈ H | Δ^n(x) ∈ ∑_{i=0}^n H^{⊗(n-i)} ⊗ H_0 ⊗ H^{⊗i}}.

(One sets $\bar{\Delta}^0 = id_H$, whence $H'_0 = H_0$.) One sees easily, by induction on n, that

$$(*) H_n \subset H'_n \subset (I^{(n+1)})^{\perp}.$$

Up to this point, one has not used the hypothesis that H is flat over k. Suppose now H flat, hence projective over k, so that A^† = H by 0.2.2, and let us show that $H_n = (I^{(n+1)})^{\perp}$. This is clear for $n = 0$. Assume it verified for $n < r$. Then H_{r-1} is the kernel of the morphism H → (I^r)^†, and so, since H^+ is flat, the morphism

(H/H_{r-1}) ⊗ H^+ (I^r)^† ⊗ H^+

[759]N.D.E.: This also holds without assuming H cocommutative (k remaining a commutative artinian ring): in this case, a basis of neighborhoods of 0 in $A = H^*$ is given by the two-sided ideals J such that A/J is of finite length over k, and the preceding proof shows that $H = \bigcup_n (I^n)^{\perp}$ if and only if the $\Phi^{-1}(\mathfrak{m}_i)$ are the only open prime ideals of A.

is injective. On the other hand, the hypothesis implies that I is a projective pseudo-compact k-module (since a direct factor of $A = H^*$), whence, by 0.3.6,

$$(I^r \hat{\otimes} I)^\dagger \simeq (I^r)^\dagger \otimes I^\dagger = (I^r)^\dagger \otimes H^+.$$

Then the exact sequence

$$I^r \hat{\otimes} I \to A \to A/I^{r+1} \to 0$$

gives by duality the exact sequence:

$$(1) \qquad (I^r)^\dagger \otimes H^+ \xleftarrow{\delta} H \longleftarrow (A/I^{r+1})^\dagger \longleftarrow 0,$$

where δ is obtained by composing Δ_H with the projection:

$$(2) \qquad H \otimes H \to H \otimes H^+ \to (H/H_{r-1}) \otimes H^+ \hookrightarrow (I^r)^\dagger \otimes H^+.$$

Now, for every $u \in H$, the projection of $\Delta(u)$ onto $H \otimes H^+$ is $\Delta(u) - u \otimes \phi$. Then (1) and (2) show that if u belongs to $(A/I^{r+1})^\dagger = (I^{r+1})^\perp$, then $\Delta(u) - u \otimes \phi$ belongs to the kernel of the map $H \otimes H^+ \to (H/H_{r-1}) \otimes H^+$, that is, to $H_{r-1} \otimes H^+$, so $u \in H_r$. This completes the proof of points (i) and (ii), and also shows that $H_n = Ker(\bar{\Delta}^n)$.

Let us prove (iii). For every $i \geqslant 0$, set $H_i^+ = H_i \cap H^+$. Let $n \geqslant 1$. For every $x \in H_n^+$, $\bar{x} = x - \epsilon(x)\phi$ belongs to H_n^+ and one has:

$$\Delta(x) = \varepsilon(x)\, \varphi \otimes \varphi + x \otimes \varphi + \varphi \otimes x + \bar{\Delta}(x).$$

So it suffices to show that:

$$\bar{\Delta}(H^+_n) \subset \sum_{i=1}^{n-1} H^+_i \otimes H^+_{n-i}.$$

For every $i = 0, \cdots, n-1$, $\bar{\Delta}^n$ factors as:

$$\begin{tikzcd}
H^+ \arrow[r, "\bar{\Delta}"] & H^+ \otimes H^+ \arrow[r, "\bar{\Delta}^i \otimes \bar{\Delta}^{n-i-1}"] \arrow[d] & (H^+)^{\otimes(i+1)} \otimes (H^+)^{\otimes(n-i)} \\
& H^+/H^+_i \otimes H^+/H^+_{n-i-1} \arrow[r, "f"] & H^+ \otimes (H^+)^{\otimes(n-i)} / H^+_i \arrow[u, "g"']
\end{tikzcd}$$

Moreover, since H^+/H_i^+ and $(H^+)^{\otimes(n-i)}$ are flat, the maps f and g above are injective. It follows that $\bar{\Delta}(H_n^+)$ is contained in $H_i^+ \otimes H^+ + H^+ \otimes H_{n-i-1}^+$, for every $i = 0, \cdots, n-1$. Point (iii) then follows from the lemma below, applied to $M = H^+$ and $E_i = H_{i-1}^+$.

Lemma 1.3.6.B. *Let k be a ring, $0 = E_0 \subset E_1 \subset \cdots \subset E_n \subset M$ k-modules. Suppose M/E_i flat for every i. Then one has the equality:*

$$\bigcap_{i=0}^{n} (E_i \otimes M + M \otimes E_i) = \sum_{i=1}^{n} E_i \otimes E_{n-i+1}.$$

Denote by K (resp. S) the left-hand (resp. right-hand) term. One easily sees that $S \subset K$; let us show the converse. For $i = 0, \cdots, n$, set $K_i = K \cap (E_i \otimes M)$. For every $i = 1, \cdots, n$, since M/E_{n-i+1} and E_i/E_{i-1} are flat, the map τ_i below is injective, and the composite map:

$$(E_i/E_{i-1}) \otimes M \to (E_i/E_{i-1}) \otimes (M/E_{n-i+1}) \xrightarrow{\tau_i} (M/E_{i-1}) \otimes (M/E_{n-i+1})$$

has kernel $(E_i/E_{i-1}) \otimes E_{n-i+1}$. Since the image of K_i in $(E_i \otimes M)/(E_{i-1} \otimes M)$ is contained in, and contains, this kernel, one deduces that

$$K_i = K_{i-1} + E_i \otimes E_{n-i+1},$$

whence the lemma.

[760] To finish this paragraph, let us return to an arbitrary pseudocompact ring k. Let $(**H**, \Delta, \epsilon)$ be a flat O_k-coalgebra, $**H**^+ = Ker(\epsilon)$, $A = \Gamma^*(**H**)$ the dual profinite k-algebra, $X = Spf(A)$, so that $**H** = **H**(X)$ (cf. 1.3.5). Suppose given a morphism of O_k-coalgebras $\phi : O_k \to **H**$; it defines a continuous morphism of k-algebras $A \to k$, and hence a section σ of the structure morphism $X \to Spf(k)$.

For every object B of Alf/k, denote $**H**_0(B) = \phi(B) = B\phi_B$, where ϕ_B is the element $\phi(1_B)$ of $**H**(B)$, and one defines sub-O_k-modules $**H**_n$ of $**H**$, by setting, for $n \geqslant 1$,

$$**H**_n(B) = \{u \in **H**(B) \mid \Delta(u) - u \otimes \phi_B \in **H**_{n-1}(B) \otimes **H**^+(B)\}.$$

One thus obtains a filtration $**H**_0 \subset **H**_1 \subset \cdots$ of $**H**(X)$. By what precedes, one has:

Proposition. *In order for σ to induce an isomorphism on the underlying topological spaces, it is necessary and sufficient that $**H**(X)$ be the union of the $**H**_n$.*

1.4. Theorem.

Theorem 1.4. *Let k be a pseudocompact ring and $d_0, d_1 : X_1 \rightrightarrows X$ an equivalence couple in $Vaf_/k$ (cf. Exp. V, § 2.b) such that d_1 is topologically flat.*

(i) The canonical projection of X onto X/X_1 ($= Coker(d_0, d_1)$) is surjective and topologically flat, and the morphism $X_1 \to X \times_{X/X_1} X$ with components d_0 and d_1 is an isomorphism.

(ii) If X is topologically flat over k, then so is X/X_1.

Let us first note that (ii) follows from (i), by 1.3.3 (ii). The proof of (i) occupies paragraphs 1.4.1, 1.4.2 and 1.4.3.

1.4.1. Let us first show that one may reduce to the case where X has a single point. Since we are dealing with an equivalence couple, one sees as in Exp. V, § 3.e), that one defines an equivalence relation on the underlying set of X by declaring two points x, y to be equivalent if there exists a point z of X_1 such that $d_0(z) = y$ and $d_1(z) = x$. One may evidently suppose without loss of generality that X contains a single equivalence class for this relation, in other words that X/X_1 has a single point (see the construction of X/X_1 given in 1.2).

[760] The following has been added; the original was limited to the case where k is artinian.

In this case, let x be an arbitrary point of X and U the formal variety which has x as its only point and the same local ring as X at x. One sees then as in Exp. V, § 6, that the equivalence relation induced by (d_0, d_1) on U again satisfies the hypotheses of the theorem and that it suffices to give the proof for the latter equivalence relation (U is a "quasi-section").

Let us briefly recall the principle of the proof given in Exp. V, § 6. Set $V = d_0^{-1}(U) = U \times_{i,d_0} X_1$, where i is the inclusion of U in X; let u and v be the morphisms with source V induced respectively by d_0 and d_1:

$$X \xleftarrow{\ v\ } V \xrightarrow{\ u\ } U.$$

It is clear that u and v are topologically flat and that u is surjective; since X contains a single equivalence class, v is surjective. If (v_0, v_1) is the inverse image of the equivalence couple (d_0, d_1) under v (cf. V, 3.a)), it follows from V, 3.c) and 3.d) that X/X_1 and the quotient of U by the equivalence relation induced by (d_0, d_1) are both identified with $Coker(v_0, v_1)$. One sees then, as in the proof of V, 6.1, that if the conclusion of Theorem 1.4 is verified for U, it is also verified for X.

1.4.2. We are thus reduced to the case where X has a single point.[761] Consider then the following commutative diagram (cf. V, § 1, (0,1,2)):

$$\begin{tikzcd}
X_2 \arrow[r, shift left, "d'_1"] \arrow[r, shift right, "d'_0"'] \arrow[d, "d'_2"'] & X_1 \arrow[r, "d_0"] \arrow[d, "d_1"] & X \arrow[d] \\
X_1 \arrow[r, shift left, "d_1"] \arrow[r, shift right, "d_0"'] & X \arrow[r] & X/X_1
\end{tikzcd}$$

where X_2 is the fiber product $X_1 \times_{d_1, d_0} X_1$, and where d'_0, d'_1 and d'_2 are respectively the morphisms "$(x, y, z) \mapsto (x, y)$", "$(x, y, z) \mapsto (x, z)$", and "$(x, y, z) \mapsto (y, z)$".[762]

If B, A, A_1 and A_2 denote respectively the affine algebras of X/X_1, X, X_1 and X_2, the preceding diagram induces a commutative diagram:

$$\begin{tikzcd}
A_2 & A_1 \arrow[l, shift right, "j_1"'] \arrow[l, shift left, "j_0"] & A \arrow[l, "i_0"']
\end{tikzcd}$$

j_2 $\qquad i_1 \qquad i$

$i_1 \qquad i$

[761] One may therefore suppose k local.
[762] cf. Exp. V, § 2.b).

$$A_1 \xleftarrow[i_0]{} A \longleftarrow B$$

in which the two rows are exact and the squares determined by (i_0, j_0) and (i_1, j_1) are cocartesian. Since the morphism $X_1 \to X \times X$ with components d_0 and d_1 is a monomorphism by hypothesis, the morphism $A \hat{\otimes}_k A \to A_1$ with components i_0 and i_1 is surjective, by Proposition 1.3.

This means that i_1 makes A_1 into a pseudocompact A-module (assumed topologically free), generated by $i_0(A)$. Since A is local, Lemma 1.4.3 below yields the existence of a topologically free k-module V and a morphism of pseudocompact k-modules $f : V \to A$ such that the morphism

$$\alpha_1 : A \hat{\otimes}_k V \longrightarrow A_1, \qquad a \otimes v \mapsto i_1(a) \cdot i_0(f(v))$$

is invertible. Let $\alpha : B \hat{\otimes}_k V \to A$ and $\alpha_2 : A_1 \hat{\otimes}_k V \to A_2$ be the morphisms:

$$b \otimes v \mapsto i(b) \cdot f(v) \quad \text{and} \quad a_1 \otimes v \mapsto j_2(a_1) \cdot j_0\, i_0(f(v)).$$

In the following commutative diagram, the two rows are therefore exact and the two left-hand squares are cocartesian. Since α_1 is invertible, the same holds for α_2, hence for α.

$$\begin{tikzcd}
A_2 & A_1 \arrow[l, shift right, "j_1"'] \arrow[l, shift left, "j_0"] & A \arrow[l, "i_0"'] \\
A_1 \hat{\otimes}_k V \arrow[u, "\alpha_2"] & A \hat{\otimes}_k V \arrow[l, shift right, "i_1 \otimes V"'] \arrow[l, shift left, "i_0 \otimes V"] \arrow[u, "\alpha_1"] & B \hat{\otimes}_k V. \arrow[l, "i \otimes V"'] \arrow[u, "\alpha"]
\end{tikzcd}$$

This shows on the one hand that A is topologically free over B, with pseudobasis $f(V)$ (cf. 0.2.1), and that one obtains a pseudobasis of A_1 over A (where A_1 is considered as an A-module via i_1) by taking the image under i_0 of $f(V)$; this entails that the morphism $A \hat{\otimes}_B A \to A_1$ with components i_1 and i_0 is invertible:

$$A \hat{\otimes}_B A \simeq A \hat{\otimes}_B B \hat{\otimes}_k V \simeq A \hat{\otimes}_k V \simeq A_1.$$

This proves Theorem 1.4, modulo Lemma 1.4.3 which follows.

1.4.3. Lemma. **Lemma 1.4.3.** *Let k be a pseudocompact ring, A a local profinite k-algebra, A_1 a topologically free A-module and $i_0 : M \to A_1$ a morphism of pseudocompact k-modules. Suppose that the map*

$$A \hat{\otimes}_k M \longrightarrow A_1, \qquad a \otimes m \mapsto a \cdot i_0(m)$$

is surjective. Then there exist a topologically free k-module V and a morphism of pseudocompact k-modules $f : V \to M$ such that the map

$$A \hat{\otimes}_k V \longrightarrow A_1, \qquad a \otimes v \mapsto a \cdot i_0(f(v))$$

is bijective.

Since every pseudocompact k-module is the quotient of a topologically free k-module (cf. N.D.E. (27)), one may suppose without loss of generality that M is topologically free; so let us take for M the direct product of a family $(M_i)_{i\in I}$ of copies of k. In this case, $A \hat{\otimes}_k M$ is none other than the product $\prod_{i \in I} A \hat{\otimes}_k M_i$. Since the map $a \otimes m \mapsto a \cdot i_0(m)$ is surjective and A_1 is projective, the kernel of this map is a direct factor of $A \hat{\otimes}_k M$; since A is local, it follows from the exchange theorem (0.3.4) that this kernel has as supplement a partial product $\prod_{i \in J} A \hat{\otimes}_k M_i$, where J denotes some subset of I. One may therefore take $V = \prod_{i\in J} M_i$.

1.5.

Let k be a pseudocompact ring.

Definition 1.5. *We shall say that a family of morphisms $f_i : X_i \to X$ of $Vaf_{/}k$ is a topologically flat surjective family if the morphism $\coprod_i X_i \to X$ induced by the f_i is surjective and topologically flat; this means that each f_i is topologically flat and that every point of X belongs to the image of at least one of the X_i.*

It follows from 1.3.3 that the topologically flat surjective families define a pretopology on $Vaf_{/}k$ (IV 4.2.5); the corresponding topology will be called the *flat topology* on $Vaf_{/}k$.

By IV, 4.3.5, a functor $F : (Vaf_{/}k)^\circ \to (Ens)$ is a sheaf for the flat topology if and only if F transforms every direct sum into a direct product and the sequence

$$F(Y) \xrightarrow{F(f)} F(X) \underset{F(pr_2)}{\overset{F(pr_1)}{\rightrightarrows}} F(X \times_Y X)$$

is exact for every topologically flat surjective morphism $f : X \to Y$.

By IV, 4.5, Proposition 1.3.1 implies that the flat topology is less fine than the canonical topology, i.e., for every object X of $Vaf_{/}k$, the functor $h_X : T \mapsto \mathrm{Hom}_{Vaf_{/}k}(T, X)$ is a sheaf for the flat topology. (In what follows, one will identify, as usual (cf. Exp. I), X with h_X.)

By IV, 4.6.5, one may reformulate Theorem 1.4 as follows.

Theorem 1.5. *Let k be a pseudocompact ring, $d_0, d_1 : X_1 \rightrightarrows X$ an equivalence couple in $Vaf_{/}k$, and X/X_1 the formal quotient variety (i.e., $Coker(d_0, d_1)$, cf. 1.2). If d_1 is topologically flat, then X/X_1 represents the quotient sheaf for the flat topology.*

1.6.

To complete these generalities on formal varieties, it remains to define briefly the étale formal varieties over k.[763]

[763]The original continued thus: "A formal variety X over k is said to be étale if the diagonal morphism $\Delta_X : X \to X \times X$ is a local isomorphism, that is, if Δ_X induces an isomorphism of $\mathcal{O}_{X\times X, \Delta_X(x)}$ onto

Recollections 1.6.A. *(i)* Let us first recall (cf. EGA 0_{IV}, 19.10.2) that a topological k-algebra A is said to be *formally étale* over k (for the topologies given on k and A, resp. for the discrete topologies) if, for every discrete topological k-algebra C (not necessarily artinian), and every nilpotent ideal J of C, every continuous morphism of k-algebras $A \to C/J$ lifts in a unique way to a continuous morphism $A \to C$ (A being endowed with the given topology, resp. with the discrete topology). One sees at once that this property is preserved by base change, i.e., for every morphism $k \to k'$ of pseudocompact rings, $A \otimes_k k'$ is then formally étale over k'. On the other hand, one sees easily that it suffices to verify the lifting condition for every ideal J of square zero, cf. EGA IV_4, 17.1.2 (ii). One says that A is *étale* over k if it is formally étale over k for the discrete topologies, and if moreover A is a k-algebra of finite presentation (cf. EGA IV_4, 17.3.2 (ii)). In what follows, k being a pseudocompact ring and A a profinite k-algebra, one will use "formally étale" in the sense of the given topologies (unless otherwise stated).

(ii) Recall also that if $F \in k[T]$ is a monic polynomial of degree $d \geqslant 1$, separable (i.e., such that the ideal generated by F and its derivative polynomial F' is $k[T]$), then the k-algebra $B = k[T]/(F)$ (which is free of rank d over k, and endowed with the product topology) is formally étale over k. Indeed, let C be a discrete k-algebra (so that the kernel of $k \to C$ is an open ideal l of k), J an ideal of C of square zero, and $\phi : B \to C/J$ a continuous morphism of k-algebras. Note that, B being a k-module free of finite rank, lB is an open ideal of B, hence every lifting $\Phi : B \to C$ of ϕ is automatically continuous. Let t be the image of T in B and u_0 an arbitrary lifting of $\phi(t)$ in C; then $F(u_0) \in J$ (since $\phi(t)$ is a root of F); on the other hand there exist $G, H \in k[T]$ such that $GF + HF' = 1$, hence $H(u_0)F'(u_0) = 1 - G(u_0)F(u_0)$, and the right-hand term is invertible, since $F(u_0)$ is of square zero, hence $F'(u_0)$ is invertible. Let us seek $h \in J$ such that $x = u_0 + h$ be a root of F; this amounts to $0 = F(u) = F(u_0) + F'(u_0)h$, and as $F'(u_0)$ is invertible, this has the unique solution $h = -F'(u_0)^{-1}F(u_0) \in J$. Of course, the same proof (without making continuity hypotheses on the morphisms $k \to C$, ϕ and Φ) shows that B is also an étale k-algebra.

(iii) Recall finally that if A is a finite product $A_1 \times \cdots \times A_n$, then A is formally étale over k if and only if the A_i are.[764] Indeed, it suffices to see this for $n = 2$, in which case let $e = 1_{A_1}$ be the idempotent such that $A_1 = Ae$ and $A_2 = A(1-e)$, and suppose given a continuous morphism $A \to C/J$, where J is an ideal of square zero. Since the polynomial $F = X^2 - X$ is separable (one has $F' = 2X - 1$ and $(F')^2 - 4F = 1$), the idempotent $\phi(e)$ of C/J lifts in a unique way to an idempotent f of C, whence $C = Cf \bigoplus C(1-f)$, and then to give a lifting of Φ amounts to giving two morphisms $\Phi_1 : A_1 \to Cf$ and $\Phi_2 : A_2 \to C(1-f)$, lifting the restrictions of ϕ to A_1 and A_2.

$O_{X,x}$ for every point x of X. One sees easily with the aid of SGA I that this formulation is equivalent to the following two: the formal variety X is topologically flat, and, for every point $x \in X$, of projection $s \in Spf(k)$, $O_{X,x} \otimes_k \kappa(s)$ is a finite separable extension of the residue field $\kappa(s)$ of s; or also, if A denotes the affine algebra of X, the local components (0.1) of $A \otimes_k (k/l)$ are finite étale algebras over k/l, for every open ideal l of k." In what follows, we have rectified the omission of the flatness hypothesis in the first condition above, and detailed the equivalence of the said conditions.

[764] Let us point out in passing that the proof given in EGA 0_{IV}, 19.3.5 (v) is erroneous.

The same argument shows that if e is an idempotent of k such that $A = Ae$, then A is formally étale over k if and only if it is so over the localization k_e (which is identified with ke).

Let now X be a k-formal variety and A its affine algebra. If x is a point of X (i.e., a maximal open ideal m of A), with image s in $Spf(k)$, one will denote by k_m or k_s the local component of k corresponding to s.

Definition 1.6.B. *The following conditions are equivalent:*

(a) A is formally étale over k.

(b) For every $m \in \Upsilon(A)$, A_m is formally étale over k (or over k_m).

(c) For every open ideal l of k, $A \otimes_k (k/l) = A/Al$ is formally étale over k/l.

(d) For every $m \in \Upsilon(A)$ and every open ideal l of k_m, $A_m/A_m l$ is formally étale over k_m/l.

We shall say that X is *étale over k* if it satisfies these conditions, and one denotes by $Vaf^{é}t_{/}k$ the full subcategory of $Vaf_{/}k$ formed by the formal varieties étale over k.[765]

Note that if $\phi : A \to C/J$ is a continuous morphism of k-algebras, where C is a discrete k-algebra and J an ideal of square zero, then $I = Ker(\phi)$ is an open ideal of A, hence A/I is artinian, hence I is contained in only a finite number of maximal open ideals $m_1, \cdots, m_r$, hence contains the product of the components A_m for $m \neq m_i$, which equals $A(1-e)$ where e denotes the idempotent of A such that $Ae = \prod_{i=1}^{r} A_{m_i}$. Thus $\phi(e) = 1_{C/J}$ and it amounts to the same to give a continuous lifting of ϕ or of the morphism from $Ae \simeq A/A(1-e)$ to C/J, induced by ϕ.

On the other hand, one knows (cf. N.D.E. (24)) that $A \simeq \lim_l A/Al$. Taking these remarks and the preceding recollections into account, one easily obtains the equivalence of the indicated conditions.

Definitions 1.6.C. *Set $\kappa(k) = \prod_{s \in Spf(k)} \kappa(s)$, endowed with the product topology, i.e., the formal variety $Spf(\kappa(k))$ is the direct sum of the* Spec $\kappa(s)$*, for $s \in Spf(k)$. On the other hand, one denotes by $S_{\kappa(k)}$ the scheme direct sum of the* Spec $\kappa(s)$*, for $s \in Spf(k)$.*

For every formal variety X over k, one denotes by $X_\kappa = X \otimes_k \kappa(k)$ the formal variety over $\kappa(k)$ obtained by base change, i.e., X_κ has the same points as X and for every $x \in X$, of projection s on $Spf(k)$, one has $\mathcal{O}_{X_\kappa, x} = \mathcal{O}_{X, x} \otimes_k \kappa(s)$. This base change functor $Vaf_{/}k \to Vaf_{/}\kappa(k)$ sends $Vaf^{é}t_{/}k$ into $Vaf^{é}t_{/}\kappa(k)$ (cf. 1.6.A (i)).

One then has (cf. SGA 1, I 6.2):

Lemma 1.6.D. *For every $Y \in Ob Vaf^{é}t_{/}k$ and $X \in Ob Vaf_{/}k$, the canonical map*

$$\mathrm{Hom}_{Vaf_{/}k}(X, Y) \longrightarrow \mathrm{Hom}_{Vaf_{/}\kappa(k)}(X_\kappa, Y_\kappa)$$

[765] Note that $Vaf^{é}t_{/}k$ is stable under finite inverse limits.

is bijective. In particular, the functor $Vaf^{\acute{e}}t_{/}k \to Vaf^{\acute{e}}t_{/}\kappa(k)$ is fully faithful (and one will see below that it is an equivalence).

Indeed, let A (resp. B) be the affine algebra of X (resp. Y) and r the radical of k, and suppose given a morphism $B \hat{\otimes}_k \kappa(k) \to A \hat{\otimes}_k \kappa(k)$ or, what amounts to the same, a morphism $\phi : B \to A/rA$.

For every open ideal l of k, there exists $n \in \mathbb{N}*$ such that $r^n \subset l$, whence $(rA)^n \subset lA \subset lA$, and since the multiplication map $m^{n-1} : A^n \to A$ is continuous, one also has $(rA)^n \subset lA$, i.e., rA/lA is a nilpotent ideal of A/lA. Consequently, ϕ lifts in a unique way to a morphism $\phi_l : B \to A/lA$. By uniqueness, these morphisms form a projective system, hence give a continuous morphism $\Phi : B \to \lim_l A/lA = A$. Moreover, Φ is unique since if Φ' is a second lifting of ϕ, then Φ' and Φ coincide modulo lA for every l, hence are equal.

Proposition 1.6.E. *(a) Let X be a formal variety over k and A its affine algebra. The following conditions are equivalent:*

(i) X is étale over k.

(ii) X is topologically flat over k and the diagonal morphism $\Delta : X \to X \times X$ is a local isomorphism, i.e., Δ_X induces an isomorphism $O_{X\times X, \Delta(x)} \cong O_{X, x}$ for every point x of X.

(iii) For every $x \in X$, of projection s on $Spf(k)$, $O_{X,x}$ is isomorphic to $k_s[T]/(F)$, where $F \in k_s[T]$ is a monic separable polynomial (cf. 1.6.A (ii)).

(iv) X is topologically flat over k, and, for every point $x \in X$, of projection s on $Spf(k)$, $O_{X, x} \hat{\otimes}_k \kappa(s)$ is a finite separable extension of $\kappa(s)$.

(v) For every open ideal l of k, each local component of $A \hat{\otimes}_k (k/l)$ is a finite étale algebra over the artinian ring k/l.

(b) $Vaf^{\acute{e}}t_{/}\kappa(k)$ is identified with the category of étale schemes over $S_{\kappa(k)}$ (cf. 1.6.C), and the functor $X \mapsto X_\kappa$ induces an equivalence of categories $Vaf^{\acute{e}}t_{/}k \xrightarrow{\sim} Vaf^{\acute{e}}t_{/}\kappa(k)$ (cf. SGA 1, I 6.1).

Proof. (a) Let I denote the kernel of the multiplication morphism $m : A \hat{\otimes}_k A \to A$. Suppose X étale over k, i.e., A formally étale over k. Then, by EGA 0_{IV}, 20.7.4, the Hausdorff completion of I/I^2, for the quotient topology of that of I, is zero, i.e., one has $I = I^2$. Now, for every $x \in X$, I is contained in the maximal ideal $m_{\Delta(x)}$ of $A \hat{\otimes}_k A$, hence the localization $I_{m_{\Delta(x)}}$ is contained in the maximal ideal of $O_{X\times X, \Delta(x)}$, and so, by Nakayama's lemma 0.3, one has $I_{m_{\Delta(x)}} = 0$, and hence $\Delta : X \to X \times X$ is a local isomorphism.

Suppose now that Δ is a local isomorphism and that k is a field κ, and let us show that each $O_{X,x}$ is a finite-dimensional étale κ-algebra. Replacing X by $Spf(O_{X,x})$, one may suppose that $A = O_{X,x}$ is local. One proceeds then as in the proof of EGA IV_4, 17.4.1, (b) $\Rightarrow$ (d''). Let K be a finite normal extension of κ containing the residue field $\kappa(x)$, and let $B = A \hat{\otimes}_\kappa K = A \otimes_\kappa K$ (since $[K : \kappa] < \infty$, then $-\hat{\otimes}_\kappa K$ and $-\otimes_\kappa K$ coincide) and $X_K = Spf(B) = X \hat{\otimes}_\kappa K$. Then $\Delta_K : X_K \to X_K$

$\otimes_K X_K$ is again a local isomorphism, so for every $y \in X_K$, the multiplication $B_y \otimes_K B_y \to B_y$ induces an isomorphism $(B_y \otimes_K B_y)_{m_{\Delta_K(y)}} \cong B_y$. Now, since the residue field of B_y is K (cf. for example VI_A, 1.1.1, N.D.E. (11)), $C = B_y \otimes_K B_y$ is already a local ring (indeed, $n = m_y \otimes_K B_y + B_y \otimes_K m_y$ consists of topologically nilpotent elements, hence is contained in the radical of C, and $C/n = K \otimes_K K = K$ is a field), hence one obtains that the multiplication morphism $B_y \otimes_K B_y \to B_y$ is an isomorphism. Taking a pseudobasis of B_y over K containing the unit element 1, one deduces that $B_y = K$. Since moreover B is finite over A, X_K is a finite set, hence $B = A \otimes_\kappa K$ is the product of a finite number of copies of K, and this entails that A is a finite étale κ-algebra.

One thus obtains that, if κ is a field, every profinite κ-algebra A étale over κ is the product of finite separable extensions K_i of κ, endowed with the product topology, hence the formal variety $Spf(A)$ is the direct sum of the $Spf(K_i) = \operatorname{Spec}(K_i)$, and one deduces that $Vaf^{\acute{e}}t_{/}\kappa$ is identified with the category of étale κ-schemes.

What precedes shows the implication (ii) $\Rightarrow$ (iv) (since $O_{X,x} \otimes_k \kappa(s)$ is formally étale over $\kappa(s)$), and entails point (b) of 1.6.E. Indeed, let k be again an arbitrary pseudocompact ring. By what precedes, $Vaf^{\acute{e}}t_{/}\kappa(k)$ is identified with the category of étale schemes over $S_{\kappa(k)}$. Let us show that $X \mapsto X_\kappa$ induces an equivalence $Vaf^{\acute{e}}t_{/}k \xrightarrow{\sim} Vaf^{\acute{e}}t_{/}\kappa(k)$. Taking 1.6.D into account, it suffices to show that for every $s \in Spf(k)$ and every finite separable extension K of $\kappa(s)$, there exists an étale k_s-algebra A such that $A \otimes_k \kappa(s) \simeq K$. Let ξ be a primitive element of the extension $K/\kappa(s)$, n its degree, and $F \in k_s[T]$ a monic polynomial of degree n whose image $\bar{F}$ in $\kappa(s)[T]$ is the minimal polynomial of ξ. Since $\bar{F}'$ is invertible in $\kappa(s)[T]/(\bar{F})$, it follows from Nakayama's lemma that F' is invertible in $k_s[T]/(F)$, hence F is a separable polynomial and hence, by 1.6.A (ii), $A = k_s[T]/(F)$ is an étale k_s-algebra such that $A \otimes_k \kappa(s) \simeq K$. One thus obtains that every local profinite étale k_s-algebra is free of finite rank over k_s (hence a fortiori topologically free over k), and hence every profinite étale k-algebra is topologically free over k.

On the other hand, condition (v) implies condition (d) of 1.6.B, hence implies (i). One has thus obtained that (i), (iii) and (v) are equivalent, and imply (ii), which implies (iv). Finally, let A be a profinite k-algebra satisfying (iv); let us show that A is formally étale over k. For this, one may suppose A and k local; denote by κ the residue field of k. By hypothesis, $K = A \otimes_k \kappa$ is a finite separable extension of κ, say of degree n. By what we have seen above (and taking Lemma 1.6.D into account), there exists then a k-algebra B free of rank n, formally étale over k, and a continuous morphism $\phi : B \to A$ such that $\varphi \otimes_k \kappa$ is an isomorphism. Since A is topologically flat over k, this entails that $\operatorname{Coker}(\varphi) \otimes_k \kappa = 0$ and also $\operatorname{Ker}(\varphi) \otimes_k \kappa = 0$; by Nakayama's lemma 0.3, one therefore has $Coker(\phi) = 0 = Ker(\phi)$, hence ϕ is an isomorphism (cf. the proof of 0.2.B). This completes the proof of 1.6.E.

Let X be a formal variety over k. For every $x \in X$, of projection s on $Spf(k)$, the residue field $\kappa(x)$ is a finite extension of $\kappa(s)$ and one denotes by $\kappa_e(x)$ the separable closure of $\kappa(s)$ in $\kappa(x)$.

Proposition 1.6.F. *(i) The inclusion of $Vaf^{\acute{e}}t_{/}k$ in $Vaf_{/}k$ has a left adjoint $X \mapsto X_e$:*

the variety X_e has the same points as X, and for every $x \in X$, of projection s on $Spf(k)$, let ξ be a primitive element of $\kappa_e(x)$, n its degree, u an arbitrary lifting of ξ in $O_{X,x}$, and $F \in k[T]$ monic of degree n annihilating u; one sets $O_{X_e,x} = k[T]/(F)$. Then for every $Y \in Ob\,Vaf^{ét}_{/k}$, the canonical maps below are bijective:

$$\begin{tikzcd} \mathrm{Hom}_{Vaf_{/k}}(X, Y) \arrow[r, "\sim"] & \mathrm{Hom}_{Vaf_{/\kappa(k)}}(X_\kappa, Y_\kappa) \\ \mathrm{Hom}_{Vaf_{/k}}(X_e, Y) \arrow[r, "\sim"] & \mathrm{Hom}_{Vaf_{/\kappa(k)}}((X_e)_\kappa, Y_\kappa), \arrow[u, "\sim"'] \end{tikzcd}$$

the vertical arrow being induced by the inclusions $\kappa_e(x) \hookrightarrow \kappa(x)$ for every $x \in X$. This defines, in particular, a morphism $p : X \to X_e$, and every morphism of k-formal varieties $\phi : X \to Y$, with Y étale over k, factors uniquely through p.

(ii) The functor $X \mapsto X_e$ commutes with finite products.

Indeed, (i) follows from 1.6.E (b) and 1.6.D. Let us prove (ii). Taking the equivalence of categories 1.6.E (b) into account, one may suppose that k is a field. In this case, one sees easily that if X is a semi-local k-formal variety, i.e., one whose affine algebra A is semi-local, then the affine algebra of X_e is the largest subalgebra of A which is étale over k; let us denote it by A_e. One thus reduces to seeing that if K, L are two finite-degree extensions of k, then the inclusion $K_e \otimes L_e \subset (K \otimes L)_e$ is an equality. Let p be the characteristic exponent of k (i.e., $p = 1$ if $char(k) = 0$ and $p = char(k)$ otherwise), then for every $x \in K \otimes L$, there exists $n \in \mathbb{N}*$ such that $x^{p^n} \in K_e \otimes L_e$, hence every subalgebra B of $K \otimes L$ is radicial over $K_e \otimes L_e$, and it follows that $K_e \otimes L_e = (K \otimes L)_e$.

Note, keeping the preceding notations, that $O_{X_e,x}$ is not necessarily a subalgebra of $O_{X,x}$, but this is the case when X is topologically flat over k, by the following proposition.

1.6.1. Let Y be an étale k-formal variety and $f : X \to Y$ a morphism of $Vaf_{/}k$. Then one has the cartesian square below, where Γ_f is the graph morphism $X \to X \times Y$, with components id_X and f,

$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] \arrow[d, "f"'] & X \times Y \arrow[d, "f \boxtimes id_Y"] \\ Y \arrow[r, "\Delta_Y"] & Y \times Y. \end{tikzcd}$$

It follows that Γ_f is a local isomorphism, hence that $f = pr_Y \circ \Gamma_f$ is topologically flat if pr_Y is, for example if X is topologically flat over k.

Conversely, since $Y \to k$ is topologically flat, $X \to k$ will be as well if f is so (cf. 1.3.3). Taking in particular f to be the canonical morphism $p : X \to X_e$ of 1.6, one obtains:

Proposition 1.6.1. *Let X be a formal variety over k. The morphism $X \to X_e$ is topologically flat if and only if X is topologically flat over k.*

Remark 1.6.2.[766] When k is a perfect field, the functor $X \mapsto X_e$ is also right adjoint to the inclusion $Vaf^{\acute{e}t}_{/k} \hookrightarrow Vaf_{/k}$. Indeed, in this case one has $O_{X_e,x} = \kappa(x)$ for every $x \in X$, and the canonical projections $O_{X,x} \to \kappa(x)$ define a morphism $s : X_e \to X$, which is a section of $p : X \to X_e$. For every morphism $f : X \to Y$ the diagrams below are commutative:

$$\begin{tikzcd} O_{Y,f(x)} \arrow[r] \arrow[d] & O_{X,x} \arrow[d] \\ \kappa(f(x)) \arrow[r] & \kappa(x) \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "f"] & Y \\ X_e \arrow[u, "s_X"] \arrow[r, "f_e"] & Y_e \arrow[u, "s_Y"'] \end{tikzcd}$$

hence s is functorial in X, and it follows that $X \mapsto X_e$ is right adjoint to the inclusion $Vaf^{\acute{e}t}_{/k} \hookrightarrow Vaf_{/k}$. Hence, being a right adjoint, $X \mapsto X_e$ commutes with inverse limits when they exist in $Vaf^{\acute{e}t}_{/k}$,[767] hence in particular with finite products. (This can also be verified directly: for every k-formal variety X, of affine algebra A, X_e has as affine algebra the quotient of A by its radical $r(A)$, and since k is perfect, the quotient of $O_{X,x} \hat{\otimes} O_{Y,y}$ by its radical is the algebra $\kappa(x) \hat{\otimes}_k \kappa(y)$, since the latter is semi-simple.)

2. Generalities on formal groups

2.1.

Let k be a pseudocompact ring and G a k-formal group, that is, a group of the category $Vaf_{/k}$ of formal varieties over k. Let A be the affine algebra of G. The composition law of G evidently defines a diagonal morphism, that is, a homomorphism of profinite k-algebras $\Delta_A : A \to A \hat{\otimes}_k A$; this homomorphism satisfies the following conditions:

(i) the diagram

$$\begin{tikzcd} A \arrow[r, "\Delta_A"] \arrow[d, "\Delta_A"'] & A \hat{\otimes}_k A \arrow[d, "\Delta_A \hat{\otimes} \mathrm{id}_A"] \\ A \hat{\otimes}_k A \arrow[r, "\mathrm{id}_A \hat{\otimes} \Delta_A"] & A \hat{\otimes}_k A \hat{\otimes}_k A \end{tikzcd}$$

is commutative.

[766] We have inserted this remark here, which in the original appeared in 2.5.2.

[767] This is the case for finite inverse limits.

(ii) there exists an augmentation (necessarily unique), that is, a homomorphism of profinite k-algebras $\epsilon_A : A \to k$ such that the composite maps

$$A \xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{\epsilon_A \hat{\otimes} id_A} k \hat{\otimes}_k A \simeq A$$

and $$A \xrightarrow[\Delta_A]{} A \hat{\otimes}_k A \xrightarrow[id_A \hat{\otimes} \epsilon_A]{} A \hat{\otimes}_k k \simeq A$$

are the identity maps of A.

(iii) there exists an antipodism (necessarily unique), that is, a homomorphism of profinite k-algebras $c_A : A \to A$ such that the composite map

$$A \xrightarrow{\Delta_A} A \hat{\otimes}_k A \xrightarrow{c_A \hat{\otimes} id_A} A \hat{\otimes}_k A \xrightarrow{m_A} A$$

is equal to $\eta_A \circ \epsilon_A$, where m_A denotes the linear map sending $a \hat{\otimes} b$ to ab and η_A the map $\lambda \mapsto \lambda \cdot 1_A$ from k into A.

Conversely, the data of $(\Delta_A, \epsilon_A, c_A)$ satisfying (i)–(iii) endows G with a structure of k-formal group.[768] Explicitly, for every profinite k-algebra B, the set $\mathrm{Hom}_c(A,B)$ of continuous morphisms of k-modules $\phi : A \to B$ is endowed with a group structure, functorial in B, defined by

$$\varphi \cdot \varphi' = m_B \circ (\varphi \hat{\otimes} \varphi') \circ \Delta_A,$$

the neutral element being $\eta_B \circ \epsilon_A$ (where m_B is the multiplication of B and η_B the map $\lambda \mapsto \lambda \cdot 1_B$ from k into B), and $\phi \circ c_A$ being the inverse of ϕ; and the set $\mathrm{Hom}_{Alp/k}(A,B)$ of continuous morphisms of k-algebras $A \to B$ is a subgroup of it (since the algebra B is commutative).

Definition 2.1. *A morphism of k-formal groups $\theta : K \to G$ is, by definition, a morphism of k-formal varieties which respects the group structures.* If B (resp. A) is the affine algebra of K (resp. G) and if $f : A \to B$ is the morphism corresponding to θ, this amounts to saying that f is compatible with the comultiplications, i.e.,

$$(f \hat{\otimes} f) \circ \Delta_A = \Delta_B \circ f$$

(the conditions $\epsilon_B \circ f = \epsilon_A$ and $c_B \circ f = f \circ c_A$ being then automatically satisfied). One will denote by $Grf_{/}k$ the category of k-formal groups.

Notations. *In what follows, we shall call augmentation ideal of A the ideal $I_A = Ker(\epsilon_A)$, and we shall denote by $\omega_{G/k}$ the pseudocompact k-module I_A/I_A^2, that is, the quotient of I_A by the closed ideal generated by the products xy, for $x, y \in I_A$.*

[768] One will also say that A is a *cogroup* in the category of profinite k-algebras. On the other hand, we have detailed what follows; in particular, we have made explicit what a morphism of formal groups $K \to G$ is, cf. Proposition 2.3.1.

2.2.

Let H be a group of the category of coalgebras over O_k, i.e., for every object C of $Alf_{/}k$, $H(C)$ is endowed with a structure of C-coalgebra in groups (cf. VII_A 3.2; following Manin, we shall say *bialgebra*[769] instead of *coalgebra in groups*); moreover, if $\phi : C \to D$ is a morphism of $Alf_{/}k$, the map `D □_C H(C) → H(D)` is a homomorphism of D-bialgebras.

Definition 2.2. *We shall summarize the above properties by saying that H is a* bialgebra over O_k.

It is clear that the functor $H \mapsto Spf*(H)$ of 1.3.5 commutes with finite products. It therefore transforms a bialgebra over O_k into a k-group functor, that is, a (covariant) functor from $Alf_{/}k$ to the category of groups.

And indeed, for every k-algebra C of finite length, the elements of

```
Spf*(H)(C) = Spf*(H(C)) = {x ∈ H(C) | ε(x) = 1 and Δ(x) = x ⊗ x}
```

form a group for the multiplication of the algebra $H(C)$ (cf. VII_A 3.2.2). Note moreover that the condition `Δ(x) = x ⊗ x` entails the equality $\epsilon(x) = \epsilon(x)^2$, hence also $\epsilon(x) = 1$ if C is local and $x \neq 0$.[770]

2.2.1. A bialgebra H over O_k is said to be *flat* if the underlying module is flat (cf. 1.2.1).[771] If H is flat then, by 1.3.5, $A = \Gamma*(H)$ is a topologically flat profinite k-algebra, and $Spf*(H)$ is isomorphic, as a functor from $(Alf_{/}k)^{\circ} = Vaf_{/}k$ to `(Ens)`, to the functor

```
Spf(A) : C ↦ Hom_{Vaf_/k}(Spf(C), Spf(A)).
```

The group structure of $Spf*(H)$ thus endows $\mathcal{G}(H) = Spf(A)$ with a structure of formal group, which is described explicitly as follows.

For every object C of $Alf_{/}k$, since the C-module underlying $H(C)$ is projective, one deduces from Lemma 1.2.3.A, by induction on n, natural isomorphisms:

```
H(C)*^□(n+1) ≃ Hom_C(H(C), (H(C)*)^□ n)
             ≃ Hom_C(H(C), (H(C)^□ n)*) ≃ (H(C)^□(n+1))*.
```

One deduces from this (for $n = 1, 2$) that the C-algebra structure of $H(C)$ endows $H(C)*$ with a diagonal map satisfying conditions 2.1 (i)–(iii), all of this functorially in C.

Consequently, $A = \Gamma*(H) = \lim H(k/l)*$ is endowed with a structure of cogroup in Alp/k, which defines on $\mathcal{G}(H)$ the announced formal group structure.

[769] One also says *bigèbre* (cf. [BAlg], III § 11.4); recall (cf. VII_A, 3.1, N.D.E. (26)) that all the "bialgebras" considered here are assumed cocommutative and equipped with an antipode, i.e., they are in fact cocommutative Hopf algebras.

[770] Since `Δ(x) = x ⊗ x` one has $x = \epsilon(x)x$, hence $\epsilon(x) = 0$ can occur only if $x = 0$.

[771] We have detailed the original in what follows.

Conversely, let G be a topologically flat k-formal group, of affine algebra A, and denote by $H(G)$ the O_k-coalgebra $V_{kf}(A)$ (cf. 1.2.3). The diagonal morphism $\Delta_A : A \to A \hat{\otimes}_k A$ then induces, for every k-algebra C of finite length, a C-linear map

$$V_{kf}(A)(C) \otimes_C V_{kf}(A)(C) \longrightarrow V_{kf}(A)(C)$$

which makes the coalgebra $V_{kf}(A)(C)$ into a C-bialgebra. One says that $H(G)$ is the *covariant bialgebra* of the formal group G.[772] Therefore, by Proposition 1.3.5.D:

Proposition 2.2.1. *(i) The functors $G \mapsto H(G)$ and $H \mapsto \mathcal{G}(H)$ define an equivalence between the category of topologically flat k-formal groups and that of flat O_k-bialgebras.*[773]

(ii) This equivalence "commutes with base change": if $k \to K$ is a morphism of pseudocompact rings, then $H(G \hat{\otimes}_k K) = H(G) \hat{\otimes}_k K$ and $\mathcal{G}(H \hat{\otimes}_k K) = \mathcal{G}(H) \hat{\otimes}_k K$.

When k is an artinian ring and G a topologically flat k-formal group, the functor $H(G)$ is evidently determined by its value $H(G) = H(G)(k)$ at k. One will also say that $H(G)$ is the (covariant) *bialgebra* of G.[774] Consequently, denoting by H_k the category of k-Hopf algebras and $H^{cocom}_{flat/k}$ the full subcategory formed by k-Hopf algebras flat over k and cocommutative, one obtains:

Corollary 2.2.1. *Let k be an artinian ring.*

(i) The functors $G \mapsto H(G)$ and $H \mapsto \mathcal{G}(H) = Spf * H(G)$ define an equivalence between the category of topologically flat k-formal groups and $H^{cocom}_{flat/k}$.

(ii) This equivalence "commutes with base change": if $k \to K$ is a morphism of artinian rings, then $H(G \hat{\otimes}_k K) = H(G) \hat{\otimes}_k K$ and $\mathcal{G}(H \hat{\otimes}_k K) = \mathcal{G}(H) \hat{\otimes}_k K$.

On the other hand, let us denote $H^{com}_{flat/k}$ the full subcategory of H_k formed by the k-Hopf algebras flat over k and commutative, and recall that the functor $K \mapsto O(K)$ is an anti-equivalence of the category of affine k-group schemes onto $H^{com}_{flat/k}$.

2.2.2. Let us suppose for simplicity that k is artinian and let G be a topologically flat k-formal group.[775] Then G is commutative if and only if its affine algebra $\mathcal{A}(G)$ has a cocommutative comultiplication, which is equivalent to saying that the bialgebra $H(G)$ has a commutative multiplication. In this case, $H(G)$ is a commutative and cocommutative Hopf algebra, flat over k; if one sets $D'(G) = \operatorname{Spec} H(G)$, then $D'(G)$ is a commutative k-group scheme, affine and flat over k. Conversely, if T is such a k-group scheme, its affine algebra $O(T)$ is a commutative group in the category of cocommutative k-coalgebras flat over k and hence, by 1.3.5.C, one obtains a topologically flat k-formal group $D(T)$ by setting:

[772] We have added the adjective "covariant" to make it visible that the functor $G \mapsto H(G)$ is covariant; this terminology is used in [Di73], I § 2.14.

[773] Recall that in this Exposé, "bialgebra" means "cocommutative Hopf algebra".

[774] We have introduced here the notation $H^{cocom}_{flat/k}$, which will be useful below.

[775] We have detailed the original in what follows, in order to introduce the notations $D'(G)$ and $D(K)$, cf. [Ca62], § 14.

$$D(T) = Spf^* O(T) = Spf(□), \quad \text{where } □ = O(T)^*.$$

As, by 1.3.5.D, one has canonical isomorphisms $G = Spf * H(G)$ and $H(D(T)) = O(T)$, one obtains canonical isomorphisms:

$$D(D'(G)) = Spf^* O(D'(G)) = Spf^* H(G) = G,$$
$$D'(D(T)) = \mathrm{Spec}\ H(D(T)) = \mathrm{Spec}\ O(T) = T.$$

Moreover, denoting by k-Gr. the category of k-group schemes, one has, by Corollary 2.2.1, functorial isomorphisms:

$$\mathrm{Hom}_{Grf/k}(G, D(T)) \simeq \mathrm{Hom}_{H_k}(H(G), O(T)) \simeq \mathrm{Hom}_{k\text{-Gr.}}(T, D'(G)),$$

One thus obtains:

Proposition (Cartier duality) 2.2.2. *Let k be an artinian ring.*

(i) The functors $G \mapsto D'(G) = \mathrm{Spec}\, H(G)$ and $T \mapsto D(T) = Spf * O(T)$ induce an anti-equivalence between the category $\mathcal{FC}_k$ of commutative topologically flat k-formal groups and the category $\mathcal{AC}_k$ of commutative k-group schemes, affine and flat, i.e., G and $D'(G)$ (resp. T and $D(T)$) are related by the equalities:[776]

$$H(G) = O(D'(G)) \quad \text{and} \quad O(T) = H(D(T)).$$

(ii) This anti-equivalence "commutes with base change": if $k \to K$ is a morphism of artinian rings, then $D'(G □_k K) = D'(G) □_k K$ and $D(T □_k K) = D(T) □_k K$.

(iii) In particular, if k is a field, one obtains an anti-equivalence between the category of commutative k-formal groups and that of commutative affine k-group schemes, which commutes with extension of the base field.

2.3.

Let us now consider an arbitrary k-formal group[777] G, of affine algebra A. Let us still denote by $H(G)$ the O_k-module $V_{kf}(A)$ dual to A and let φ_G denote the functorial homomorphism

$$\phi_G : H(G) □_k H(G) \longrightarrow H(G \times G)$$

induced by the natural map (0.3.6), for every object C of $Alf_{/}k$:

$$(A □_k C)^\dagger \otimes_C (A □_k C)^\dagger \longrightarrow ((A □_k C) □_C (A □_k C))^\dagger.$$

If $m : G \times G \to G$ is the multiplication of G, the composite map

$$H(G) □_k H(G) \xrightarrow{\phi_G} H(G \times G) \xrightarrow{H(m)} H(G)$$

[776] If one denotes by $\mathcal{FC}_k^f$ (resp. $\mathcal{AC}_k^f$) the full subcategory of $\mathcal{FC}_k$ (resp. $\mathcal{AC}_k$) formed by the objects G (resp. T) such that $H(G)$ (resp. $O(T)$) is a finite k-module (and hence finite locally free), then $\mathcal{FC}_k^f$ and $\mathcal{AC}_k^f$ both have the same objects as the category $\mathcal{C}$ of commutative and cocommutative Hopf k-algebras, finite and flat over k, the correspondence $G \mapsto H(G)$ (resp. $T \mapsto O(T)$) being covariant (resp. contravariant), and one thus recovers the "Cartier duality" of the category $\mathcal{AC}_k$, already seen in VII_A, 3.3.1.

[777] That is, not necessarily topologically flat over k.

makes $H(G)$ into an algebra over O_k; for every $C \in Alf_{/}k$, the unit element of $H(G)(C) = (A \hat{\otimes}_k C)^\dagger$ is the augmentation of $A \hat{\otimes}_k C$ (cf. 2.1).[778] If G is not topologically flat over k, φ_G is not necessarily an isomorphism, and hence the morphism $\delta_G : H(G) \to H(G \times G)$ induced by the diagonal morphism "$x \mapsto (x, x)$" from G into $G \times G$ does not necessarily factor through $H(G) \hat{\otimes}_k H(G)$, i.e., $H(G)$ is not necessarily an O_k-bialgebra.

For this reason we shall simply say, in the general case, that $H(G)$ is the "*covariant algebra*" of the formal group G.

Of course, when G is topologically flat over k, φ_G is an isomorphism, and one recovers the structure of O_k-bialgebra on $H(G)$ defined in 2.2.1.

2.3.1. Proposition. Proposition 2.3.1. *Let K and G be two k-formal groups, of affine algebras B and A. Assume K topologically flat over k. Then there exists a canonical bijection from* $\mathrm{Hom}_{Grf/k}(K, G)$ *onto the set of homomorphisms of unital O_k-algebras $h : H(K) \to H(G)$ such that the diagram*

$$(*) \qquad \begin{tikzcd}
H(K) \hat{\otimes}_k H(K) \arrow[r, "h \hat{\otimes} h"] & H(G) \hat{\otimes}_k H(G) \arrow[dr, "\varphi_G"] & \\
 & & H(G \times G) \\
H(K) \arrow[uu, "\Delta_{H(K)}"] \arrow[r, "h"] & H(G) \arrow[ur, "\delta_G"'] &
\end{tikzcd}$$

is commutative.

Since K is topologically flat, $H(K)$ is endowed with a structure of bialgebra (cf. 2.2) and $\Delta_{H(K)}$ is its diagonal morphism; in other words, with the notations of 2.3, one has $\Delta_{H(K)} = \varphi_K^{-1} \circ \delta_K$. When G is also topologically flat over k, our proposition follows from the equivalence of categories established in 2.2.1.

In the general case, one may suppose k artinian and argue on the algebras $H(K) = B^\dagger$ and $H(G) = A^\dagger$. Let $\mathrm{Hom}_c(A, B)$ be the set of continuous k-linear maps from A into B and $\mathrm{Hom}_k(B^\dagger, A^\dagger)$ the set of k-linear maps from $B^\dagger$ into $A^\dagger$.

By 0.3.6.A, one knows that if M, P are pseudocompact k-modules and P is projective, the canonical map

$$\mathrm{Hom}_c(M, P) \longrightarrow \mathrm{Hom}_k(P^\dagger, M^\dagger), \qquad f \mapsto {}^t f$$

(where ${}^t f$ denotes the transpose of f) is bijective. (One will apply this to $M = A \hat{\otimes} A$ and $P = B$, or $M = A$ and $P = B \hat{\otimes} B$.)

[778] We have modified the order of the sentences in what follows.

Let $f \in \mathrm{Hom}_c(A, B)$. Consider the diagrams below, where the squares (0) are commutative, and where the two unnamed vertical arrows are ${}^t(f \widehat{\otimes} f)$.

$$\begin{tikzcd}
A \widehat{\otimes} A \arrow[r, "m_A"] \arrow[d, "f \widehat{\otimes} f"'] \arrow[dr, phantom, "(1)"] & A \arrow[r, "\Delta_A"] \arrow[d, "f"] \arrow[dr, phantom, "(2)"] & A \widehat{\otimes} A \arrow[d, "f \widehat{\otimes} f"] \\
B \widehat{\otimes} B \arrow[r, "m_B"'] & B \arrow[r, "\Delta_B"'] & B \widehat{\otimes} B
\end{tikzcd}$$

$$\begin{tikzcd}
A^\dagger \otimes A^\dagger \arrow[r] & (A \widehat{\otimes} A)^\dagger \arrow[dr, phantom, "(1')"] & A^\dagger \arrow[l, "{}^t m_A = \delta_G"'] \arrow[r, dash, "{}^t\Delta_A = \phi_G"] \arrow[dr, phantom, "(2')"] & (A \widehat{\otimes} A)^\dagger & A^\dagger \otimes A^\dagger \arrow[l] \\
B^\dagger \otimes B^\dagger \arrow[r] \arrow[u, "{}^t f \otimes {}^t f"] \arrow[ur, phantom, "(0)"] & (B \widehat{\otimes} B)^\dagger \arrow[u] & B^\dagger \arrow[l, "{}^t m_B = \delta_K"] \arrow[r, dash, "{}^t\Delta_B = \phi_K"'] \arrow[u, "{}^t f"] & (B \widehat{\otimes} B)^\dagger \arrow[u] & B^\dagger \otimes B^\dagger \arrow[l] \arrow[u, "{}^t f \otimes {}^t f"']
\end{tikzcd}$$

If $f : A \to B$ corresponds to a morphism of formal groups $K \to G$, then squares (1) and (2) are commutative, and $\epsilon_B \circ f = \epsilon_A$; consequently, squares (1') and (2') are commutative and ${}^t f$ sends the unit of $B^\dagger = H(K)$ to that of $A^\dagger = H(G)$, i.e., ${}^t f$ is a morphism of unital k-algebras $H(K) \to H(G)$ such that the diagram $(*)$ of the proposition is commutative.

Conversely, if ${}^t f$ satisfies these conditions, then $\epsilon_B \circ f = \epsilon_A$ and the squares (1') and (2') are commutative. Since, for $M = A \widehat{\otimes} A$ and $P = B$ (resp. $M = A$ and $P = B \widehat{\otimes} B$), the map $g \mapsto {}^t g$ is injective, one deduces that squares (1) and (2) are commutative, hence f is compatible with the multiplications and the diagonal morphisms of A and B. It remains to see that $f(1_A) = 1_B$. But it follows from what precedes that $\epsilon_B f(1) = 1$, $\Delta_B f(1) = f(1) \widehat{\otimes} f(1)$ and $f(1) \cdot f(1) = f(1)$. The first two conditions entail, by 2.1 (iii), that $f(1)$ admits $c_B f(1)$ as inverse in B; consequently $f(1) \cdot f(1) = f(1)$ entails $f(1) = 1$. So $f : A \to B$ is a morphism of Alp/k, compatible with the comultiplications of A and B.

2.3.2. Let us suppose now, for simplicity, that the ring k is artinian. When G is topologically flat over k, the algebra $H(G) = H(G)(k)$ may be characterized by a universal property (due to Cartier). Recall (cf. 1.2.1) that if U is a k-module, one denotes by $W(U)$ the functor which to every k-algebra C of finite length associates the C-module $U \bigotimes_k C$.[779] If U is a k-algebra (associative, with unit element), so is $U \bigotimes_k C$; we shall denote by $W(U)^\times$ the k-group functor which to every $C \in Ob Alf_{/}k$ associates the multiplicative group of the invertible elements of the algebra $U \bigotimes_k C$:

$$W(U)^\times(C) = (U \otimes_k C)^\times.$$

[779] We have added the preceding sentence, and in what follows we have denoted $W(U)^\times$ instead of U_m. Recall on the other hand (cf. 1.2.1) that one calls "k-functor" a covariant functor $Alf_{/}k \to (Ens)$.

Moreover, let us identify G with the k-group functor $C \mapsto \mathrm{Hom}_{\mathrm{Vaf}_{/k}}(\mathrm{Spf}(C), G)$ and denote by $\mathrm{Hom}_{k-Gr.}(G, W(U)^\times)$ the set of homomorphisms of k-group functors from G into $W(U)^\times$. One has the

Proposition 2.3.2. *Let k be an artinian ring. For every formal group G topologically flat over k and for every k-algebra U, there is a canonical isomorphism*

$$\mathrm{Hom}_{k\text{-Gr.}}(G, W(U)^\times) \cong \mathrm{Hom}_{k\text{-Alg.}}(H(G), U).$$

Let us denote by A the affine algebra of G; by hypothesis it is a projective object of $PC(k)$, and $H(G) = A^\dagger$. For every object P of $PC(k)$, let us denote by $U \widehat{\otimes}_k P$ the projective limit of the k-modules $U \otimes_k (P/N)$, where N ranges over the open submodules of P. One has linear maps

$$U \otimes_k (P/N) \longrightarrow \mathrm{Hom}_k((P/N)^*, U)$$

sending $u \otimes x$ to the k-linear map $f \mapsto f(x)u$, and forming a filtered projective system. One obtains therefore, by passage to the projective limit, a morphism

$$(1) \qquad U \widehat{\otimes}_k P \xrightarrow{\ \psi_P\ } \mathrm{Hom}_k(P^\dagger, U)\ .$$

When $P = k$, ψ_k is evidently an isomorphism; moreover, the two sides of (1), considered as functors in P, commute with infinite products (every product being a filtered projective limit of finite products). One obtains therefore that (1) is an isomorphism when P is a product of copies of k, then when P is a projective object of $PC(k)$ (the two sides of (1) commuting with finite direct sums).

Let us now denote by $\mathrm{Hom}_F(G, W(U))$ the set of morphisms of k-functors from G into $W(U)$. Since $G = \mathrm{Spf}(A) = \lim \mathrm{Spf}(A/l)$, where l ranges over the open ideals of A, one has canonical isomorphisms

$$\mathrm{Hom}_F(G, W(U)) = \lim \mathrm{Hom}_F(\mathrm{Spf}(A/l), W(U)) = \lim U \otimes_k (A/l) = U \widehat{\otimes}_k A.$$

By what precedes, one obtains therefore a canonical isomorphism:

$$(2) \qquad \mathrm{Hom}_F(G, W(U)) = U \widehat{\otimes}_k A \xrightarrow{\ \psi_A\ } \mathrm{Hom}_k(H(G), U).$$

For every k-algebra C of finite length, the multiplication makes $U \otimes_k C$ into a monoid with unit, and every morphism of monoids with unit $G(C) \to U \otimes_k C$ is necessarily a morphism of groups $G(C) \to (U \otimes_k C)^\times$. Consequently, one obtains that $\mathrm{Hom}_{k-Gr.}(G, W(U)^\times)$ is the part of $\mathrm{Hom}_F(G, W(U))$ formed by the morphisms of k-functors into monoids with unit element.

It remains to see that these morphisms correspond to the k-linear maps from $H(G)$ into U which preserve multiplication and the unit elements.[780] To simplify the writing, $H(G) = A^\dagger$ will be denoted H and one will write $\widehat{\otimes}$ instead of $\widehat{\otimes}_k$. Let Δ_A, m_A and ϵ_A (resp. Δ_H, m_H and ϵ_H) denote the comultiplication, multiplication and augmentation of A (resp. H). Let $\theta \in \mathrm{Hom}_F(G, W(U))$, denote by γ its image in

[780] The original indicated: "(this) follows from the functorial character of ψ_A". We have detailed this in what follows.

$U \widehat{\otimes} A$ and $\phi : H \to U$ the associated k-linear map. Then θ sends the unit section $s \in G(k)$ onto an element u of U, and since s corresponds to the augmentation $\epsilon : A \to k$, which is the unit element 1_H of H, one sees that $\theta(s) = 1_U$ if and only if $\phi(1_H) = 1_U$.

Moreover, the morphism $\theta \circ m_G : G \times G \to W(U)$ corresponds to the element $(id_U \widehat{\otimes} \Delta_A)(\gamma)$, and this corresponds, by duality, to the map $\phi \circ m_H : H \otimes H \to U$.

On the other hand, the morphism $\theta \circ pr_1 : G \times G \to W(U)$ corresponds to the element $\gamma \widehat{\otimes} 1_A$ of $U \widehat{\otimes} A \widehat{\otimes} A$, which corresponds by duality to $\phi \circ (id_H \otimes \epsilon) : H \otimes H \to U$. Similarly, $\theta \circ pr_2$ corresponds to the element $\tau(\gamma \widehat{\otimes} 1_A)$ of $U \widehat{\otimes} A \widehat{\otimes} A$ (where $\tau(u \widehat{\otimes} a \widehat{\otimes} b) = u \widehat{\otimes} b \widehat{\otimes} a$), which corresponds by duality to $\phi \circ (\epsilon \otimes id_H) : H \otimes H \to U$. Finally, the multiplication map $\mu = m_{U \widehat{\otimes} A \widehat{\otimes} A}$ below:

$$(U \widehat{\otimes} A \widehat{\otimes} A) \times (U \widehat{\otimes} A \widehat{\otimes} A), \quad (u \widehat{\otimes} a_1 \widehat{\otimes} a_2, u' \widehat{\otimes} a_3 \widehat{\otimes} a_4) \mapsto uu' \widehat{\otimes} a_1a_3 \widehat{\otimes} a_2a_4$$

can be seen as the composite of the endomorphism σ_{23} of $(U \otimes U) \widehat{\otimes} A^{\widehat{\otimes} 4}$ which "exchanges the factors a_2 and a_3", and the map

$$(U \otimes U) \widehat{\otimes} A^{\widehat{\otimes} 2} \widehat{\otimes} A^{\widehat{\otimes} 2} \xrightarrow{m_U \widehat{\otimes} m_A \widehat{\otimes} m_A} U \widehat{\otimes} A \widehat{\otimes} A.$$

One deduces that the map $\mu \circ (\phi \circ pr_1, \phi \circ pr_2) : G \times G \to W(U)$ corresponds to the composite map β below:

$$\begin{array}{ccc}
H \otimes H & \xrightarrow{\quad\beta\quad} & U \\
\uparrow & & \uparrow \\
id_H \circ \sigma_{23} \circ id_H & & m_U \\
H \otimes H \otimes H \otimes H & & U \otimes U \\
(id_H \otimes \varepsilon_H) \otimes (\varepsilon_H \otimes id_H) & & \varphi \otimes \varphi \\
H \otimes H & \longrightarrow & .
\end{array}$$

Finally, θ is compatible with the laws of G and of $W(U)\times$ if and only if $\theta \circ m_G$ equals $\mu \circ (\phi \circ pr_1, \phi \circ pr_2)$ which is equivalent, by what precedes, to $\phi \circ m_H = \beta$. Now it is clear that $(\phi \circ m_H)(x \otimes y) = \phi(xy)$, and one sees easily that $\beta(x \otimes y) = \phi(x)\phi(y)$.

2.4.

Let us now return to an arbitrary pseudocompact ring k in order to apply to formal groups the results of 1.4–1.5 on passage to the quotient by a topologically flat equivalence relation.[781]

Let $u : H \to G$ be a monomorphism of k-formal groups, $\mu : G \times G \to G$ the "multiplication" morphism of G, and λ the composite morphism

[781] We have detailed the original in what follows.

$$\lambda : \quad G \times H \xrightarrow{\mathrm{id}_G \times u} G \times G \xrightarrow{\mu} G.$$

Since u is a monomorphism, the couple

$$G \times H \underset{\lambda}{\overset{pr_1}{\rightrightarrows}} G$$

is an equivalence couple in $\mathcal{V}af_{/k}$ (cf. V, 2.b)). Recall (cf. 1.2.C) that the cokernel G/H of this couple is defined as follows.

Let $\mathcal{O}(G)$ and $\mathcal{O}(H)$ be the affine algebras of G and H, $\Delta : \mathcal{O}(G) \to \mathcal{O}(G) \widehat{\otimes}_k \mathcal{O}(G)$ the diagonal morphism of $\mathcal{O}(G)$, and I the kernel of the morphism $u^\natural : \mathcal{O}(G) \to \mathcal{O}(H)$. (One knows, by Proposition 1.3, that $u^\natural$ induces an isomorphism $\mathcal{O}(G)/I \xrightarrow{\sim} \mathcal{O}(H)$.) Then the affine algebra $\mathcal{O}(G/H)$ of G/H is the kernel of the couple of morphisms:

$$\mathcal{O}(G) \underset{(\mathrm{id} \widehat{\otimes} u^\natural)\Delta}{\overset{\tau_1}{\rightrightarrows}} \mathcal{O}(G) \widehat{\otimes}_k \mathcal{O}(H),$$

where $\tau_1(x) = x \widehat{\otimes} 1$, i.e.,

$$\mathcal{O}(G/H) = \{x \in \mathcal{O}(G) \mid \Delta(x) - x \widehat{\otimes} 1 \in \mathcal{O}(G) \widehat{\otimes} I\}.$$

If, moreover, H is topologically flat over k, then pr_1 is topologically flat and one deduces from Theorem 1.4 the following theorem.

Theorem 2.4. *Let $u : H \to G$ be a monomorphism of k-formal groups. Assume H topologically flat over k. Then the projection $p : G \to G/H$ is surjective and topologically flat, one has an isomorphism*

$$(*) \qquad G \times H \xrightarrow{\sim} G \times_{G/H} G$$

and G/H represents the quotient-sheaf for the flat topology.

Consequently, G/H is endowed with a canonical structure of object with group of operators G, such that $p : G \to G/H$ is a morphism of objects with operators. If moreover u identifies H with an invariant subgroup of G, then G/H is endowed with a canonical structure of k-formal group, such that $p : G \to G/H$ is a morphism of k-formal groups, and H is the kernel of p.

Indeed, the first assertion follows from 1.4; the other two from IV, Corollaries 5.2.2 and 5.2.4.

Corollary 2.4.[782] *Let G be a k-formal group, H a formal subgroup of G, A (resp. A/J, B) the affine algebra of G (resp. H, G/H), I_A the augmentation ideal of A, and*

[782] We have made this corollary explicit, since it is used, for example, in 5.2.1/5.2.3.

$I_B = B \cap I_A$. Assume H topologically flat over k. Then J equals AI_B, the closed ideal generated by I_B.

Indeed, the projection $B \to B/I_B$ corresponds to the "unit section" $e : Spf(k) \to G/H$ of G/H. By $(*)$, H is identified with the fiber product $Spf(k) \times_{G/H} G$, and hence its affine algebra A/J is identified with $(B/I_B) \hat{\otimes}_B A \simeq A/AI_B$.

2.4.A. Let G, Q be topologically flat k-formal groups; assume that there exist homomorphisms $\sigma : Q \to G$ and $\pi : G \to Q$ such that $\pi \circ \sigma = id_Q$. In particular, σ is a monomorphism, so Q is a formal subgroup of G (cf. Remark 1.3). Let $N = Ker(\pi)$ and σ' the inclusion $N \hookrightarrow G$. Then G is the semi-direct product of N by Q (cf. I, 2.3.8), i.e., for every $B \in ObAlf_/k$, identifying $N(B)$ and $Q(B)$ with their images in $G(B)$ via σ' and σ, $N(B)$ is an invariant subgroup of $G(B)$ and the map

(1) $\quad \mu : N(B) \times Q(B) \longrightarrow G(B), \quad (x, q) \mapsto xq$

is bijective. Then the morphism of k-formal varieties

(2) $\quad \theta : G(B) \longrightarrow N(B), \quad g \mapsto g \cdot \sigma\pi(g^{-1})$

is a retraction of σ', the inverse of μ is the map

$$(3) g \mapsto (\theta(g), \pi(g))$$

and $\theta \circ \sigma' : N \to G/Q$ is an isomorphism of k-formal varieties. In particular, N is topologically flat over k, by 1.4 (ii). Denote by α (resp. β) the map from $Q \times N$ (resp. $G \times N = N \times Q \times N$) to N defined setwise by $\alpha(q, y) = qyq^{-1}$ (resp. $\beta(x, q, y) = x\alpha(q, y)$). Then one has the following commutative diagram:

$$\begin{tikzcd}
G \times G \arrow[r, "id \times \theta"] \arrow[d, "m_G"] & G \times N \arrow[r, "\theta \times id"] \arrow[dr, "\beta"] & N \times N \arrow[d, "m_N"] \\
G \arrow[rr, "\theta"] & & N.
\end{tikzcd}$$

This may be expressed as follows in terms of the affine algebras A, A_0 and A' of G, Q and N (cf. 5.1.3 below). Let $\rho' : A \to A'$, $\rho : A \to A_0$ and $\tau : A_0 \to A$ be the homomorphisms of k-bialgebras corresponding to σ', σ and π, and let $I = Ker(\rho)$. Then, by the preceding Corollary, A' is identified with $A/A\tau(J_0)$, where J_0 denotes the augmentation ideal of A_0.

On the other hand, let B be the affine algebra of G/Q; it is the kernel of the couple of morphisms

$$A \xrightarrow{\tau_1} A \hat{\otimes}_k A_0,$$

$$\xrightarrow{(\mathrm{id} \otimes \rho)\Delta_A}$$

i.e., $B = \{x \in A \mid \Delta_A(x) - x \otimes 1 \in A \otimes I\}$. Denote by γ the continuous morphism of k-algebras $\theta^{*} : A' \to A$; it is a section of ρ' and an isomorphism of profinite k-algebras from A' onto B. On the other hand, $\tau = \pi^{*}$ identifies A_0 with a sub-bialgebra of A, which is none other than the affine algebra of the quotient $N \backslash G$. One deduces from (1) and (3) that one has an isomorphism of profinite k-algebras

$$(*) \qquad A' \hat{\otimes}_k A_0 \xrightarrow{\sim} A, \qquad a' \otimes a_0 \mapsto \gamma(a')\,\tau(a_0),$$

whose inverse is the map $a \mapsto (\rho' \otimes \rho)\Delta_A(a)$.

Finally, let us identify A' with its image in A via γ, so that the projection $A \to A'$ is then $\gamma\rho'$. Denoting by $\Delta_{A'}$ the comultiplication of A', one deduces from (4) that $\Delta_A(A') \subset A \hat{\otimes} A'$ and that the following diagram is commutative:

$$\begin{array}{ccc} A' & \xrightarrow{\Delta_A} & A \hat{\otimes} A' \\ & & \Big\downarrow{\scriptstyle \gamma\rho' \otimes \mathrm{id}} \\ A' & \xrightarrow{\Delta_{A'}} & A' \hat{\otimes} A' \end{array}$$

(one also has therefore $\gamma\rho' \circ c_A = c_{A'}$, where c_A (resp. $c_{A'}$) is the antipode of A (resp. A')). On the other hand, denoting by m_A the multiplication of A, one deduces from (2) that, for every $a \in A$,

$$\gamma\rho'(a) = (m_A \circ (\mathrm{id} \otimes \tau\rho\, c_A) \circ \Delta_A)(a).$$

2.4.B. Let us suppose, for simplicity, k artinian. Then what precedes is expressed more simply in terms of the covariant bialgebras of G, Q, N. Indeed, since $G = N \times Q$ as k-formal varieties, then $H(G) = H(N) \otimes_k H(Q)$ as k-coalgebras. Moreover, since the multiplication of G is given by

$$(x, q) \cdot (x', q') = (x\,\alpha(q, x'), qq'), \qquad \text{where } \alpha(q, x') = qx'q^{-1},$$

the multiplication of $H(G)$ is given as follows: for every $x \in H(N), q \in H(Q)$,

$$(x \otimes q) \cdot (x' \otimes q') = x\,\varphi(q, x') \otimes qq',$$

where $\varphi : H(Q) \otimes_k H(N)$ is the morphism of k-coalgebras induced by α. Since α is the composite morphism below (where δ_G (resp. m_G) is the diagonal morphism (resp. multiplication) of G, c_Q the inversion morphism of Q, and $v(q \otimes q' \otimes x) = q \otimes x \otimes q'$):

$$Q \times N \xrightarrow{v \circ (\delta_G \times \mathrm{id})} Q \times N \times Q \xrightarrow{\mathrm{id} \times \mathrm{id} \times c_Q} Q \times N \times Q \xrightarrow{m_G} G,$$

one obtains, denoting still by c_Q the antipode of $H(Q)$, that

$$(\star) \qquad \varphi(q \otimes x') = \sum_i q_i\, x'\, c_Q(q'_i) \qquad \text{if } \Delta_{H(Q)}(q) = \sum_i q_i \otimes q'_i.$$

In particular, if M is an abstract group and if $H(Q)$ is the group algebra kM (i.e., $Q = Spf*kM$ is the constant k-formal group M_k), then for every $\gamma \in M$ and $x' \in H(N)$ one has $\phi(\gamma \otimes x') = \gamma x' \gamma^{-1}$, and this defines an action of M on $H(N)$ by Hopf algebra automorphisms.

2.4.1. Proposition. **Proposition 2.4.1.** *Let $f : G \to K$ be a morphism of k-formal groups. If $H = Ker(f)$ is topologically flat over k, the homomorphism $f' : G/H \to K$ induced by f is a monomorphism.*

This is a consequence of the results of Exposé IV;[783] we nonetheless give a direct proof. Let T be a formal variety of finite length over k and t an element of $(G/H)(T)$ such that $f' \circ t$ is the unit element of $K(T)$. We must show that t is the unit element of $(G/H)(T)$. Denote by p the projection $G \to G/H$ and by X the fiber product $T \times_{G/H} G$.

By 2.4, p is surjective and topologically flat, hence so is the morphism $pr_1 : X \to T$, hence pr_1 is an epimorphism by Proposition 1.3.1, hence it suffices to show that $t \circ pr_1$ is the unit element of $(G/H)(X)$. Denote by pr_2 the projection $X \to G$; one has $t \circ pr_1 = p \circ pr_2$, whence the equality $1 = f' \circ t \circ pr_1 = f' \circ p \circ pr_2 = f \circ pr_2$. Then the exact sequence

$$1 \longrightarrow H \longrightarrow G \xrightarrow{f} K$$

shows that pr_2 factors through H, hence $p \circ pr_2$ is the zero morphism. Since $p \circ pr_2 = t \circ pr_1$, this proves the proposition.

One deduces from the proposition the following corollary. Let $O(G), O(K)$ and $O(G/H)$ denote the affine algebras of G, K and G/H; one has seen (2.4) that p induces an injection of $O(G/H)$ into $O(G)$. Moreover, by Proposition 1.3, since $f' : G/H \to K$ is a monomorphism, the morphism $O(K) \to O(G/H)$ is surjective, whence:

Corollary 2.4.1. *Let $f : G \to K$ be a morphism of k-formal groups and $H = Ker(f)$. If H is topologically flat over k, then $O(G/H)$ is the image of $O(K)$ in $O(G)$.*

2.4.2. Let us keep the preceding notations and assume H and G topologically flat over k. Then G is topologically flat over k and over G/H, hence, by 1.3.3, G/H is topologically flat over k. Consequently, by 2.4, the canonical projection q from K onto $Coker(f')$ is topologically flat and f' is an isomorphism from G/H onto $Ker(q)$. It is clear on the other hand that one has $Coker(f) = Coker(f')$. Hence, under the hypothesis that G and $H = Ker(f)$ are topologically flat over k, one has obtained an isomorphism between $Ker(q)$, the image of f, and $G/Ker(f)$, the coimage of f. This entails the theorem below.

Theorem 2.4.2. *Let k be a field. The commutative k-formal groups form an abelian category.*

[783] cf. IV, 5.2.6.

Corollary 2.4.2. *Let k be a field. The commutative affine k-group schemes form an abelian category.*[784]

This follows from the theorem and the equivalence of categories 2.2.2.

2.5.

A k-formal group is said to be *étale* if the underlying formal variety is étale (cf. 1.6); these formal groups have a very simple structure. Indeed, suppose k local; let κ be the residue field of k, κ_s a separable closure of κ, and Γ the topological Galois group of κ_s over κ. Let us call a Γ-*set* the datum of a set E and a continuous operation of Γ on E (i.e., the isotropy group of every element $x \in E$ is an open subgroup of Γ).

For every k-formal variety X, one sets:

$$X(\kappa_s) = \varinjlim_{\ell} X(\ell),$$

where ℓ ranges over the finite extensions of κ contained in κ_s.[785] Then Γ operates continuously on each $X(\ell)$, hence also on $X(\kappa_s)$. Moreover, let $X_\kappa = X \hat{\otimes}_k \kappa$ (cf. 1.6.C); for every ℓ one has $X(\ell) = X_\kappa(\ell)$, whence $X(\kappa_s) = X_\kappa(\kappa_s)$.

Suppose now X étale over k; then X_κ is the κ-formal variety direct sum of the $\operatorname{Spec} \kappa(x)$, for $x \in X$, and if one denotes by X'_κ the κ-scheme direct sum of the $\operatorname{Spec} \kappa(x)$, one sees that $X_\kappa(\kappa_s)$ is none other than $X'_\kappa(\kappa_s) = \operatorname{Hom}_{(Sch/\kappa)}(\operatorname{Spec} \kappa_s, X'_\kappa)$.

Let us denote $\mathcal{C} = (Sch^{ét}_{/\kappa})$ the full subcategory of $(Sch_{/\kappa})$ formed by the étale κ-schemes. One knows that the functor $X' \mapsto X'(\kappa_s)$ is an equivalence of $\mathcal{C}$ onto the category $\mathcal{C}'$ of Γ-sets (cf. SGA 1, V §§ 7–8 or [DG70], § I.4 6.4); it therefore induces an equivalence between the category of $\mathcal{C}$-groups and that of $\mathcal{C}'$-groups; now one sees at once that a $\mathcal{C}'$-group is the same thing as an abstract group G endowed with a continuous operation of Γ by group automorphisms (one will then say that G is a Γ-*group*).

Taking into account the equivalences $Vaf^{ét}_{/k} \simeq Vaf^{ét}_{/\kappa} \simeq (Sch^{ét}_{/\kappa})$ of 1.6.E, one therefore obtains:

Proposition 2.5. *Let k be a local pseudocompact ring, κ its residue field, κ_s a separable closure of κ, and $\Gamma = \mathrm{Gal}(\kappa_s/\kappa)$.*

(i) The functor $X \mapsto X(\kappa_s)$ is an equivalence of the category of étale k-formal varieties onto that of Γ-sets.

(ii) It induces an equivalence of the category of étale k-formal groups onto that of Γ-groups.

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[784] See also the remarks following VI_A, 5.4.3.

[785] Note that if $[\kappa_s : \kappa] = \infty$, κ_s is not a profinite κ-algebra, so the preceding notation is an abuse of notation. On the other hand, we have detailed the original in what follows.

Remark 2.5.A.[786] Let k be a field, k_s a separable closure of k, G an étale k-formal group and M the abstract group $G(k_s)$. Let us denote by X a set of representatives of the orbits of $\Gamma = \mathrm{Gal}(k_s/k)$ in M, and for every $x \in X$ let Γ_x be its stabilizer, which is a subgroup of Γ of finite index, and $L_x = k_s^{\Gamma_x}$, which is an extension of k of degree $[\Gamma : \Gamma_x]$ (see, for example, [BAlg], V § 10.10). Then, by the equivalence of categories above, the affine algebra $\mathcal{A}(G)$ of G is the product of the L_x, endowed with the product topology, and hence the L_x are precisely the simple quotients $\mathcal{A}(G)/m$, where m is a maximal open ideal of $\mathcal{A}(G)$. Since these correspond by duality to the subcoalgebras of $H(G)$, one obtains that $H(G)$ is pointed (i.e., $\dim_k(C) = 1$ for every simple subcoalgebra C) if and only if $L_x = k$ for every x, and in this case $\mathcal{A}(G)$ is the topological algebra k^M, hence $H(G)$ is the group algebra kM, and one has

$$M = \{x \in H(G) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\} = G(k).$$

2.5.1. Let us now suppose k an arbitrary pseudocompact ring. Since the functor $X \mapsto X_e$ of 1.6.F commutes with finite products, it transforms every formal group G into an étale formal group G_e, and since the morphism $p : X \to X_e$ of loc. cit. is functorial in X, then $p : G \to G_e$ is in this case a morphism of formal groups.

Consider the kernel $Ker(p)$;[787] it is the fiber product of the diagram:

$$\begin{tikzcd} & G \arrow[d, "p"] \\ Spf(k) \arrow[r, "\varepsilon"] & G_e \end{tikzcd}$$

where ϵ is the unit section of G_e. Since p induces a bijection on the underlying sets, one deduces (cf. 1.2, N.D.E. (41)) that $Ker(p)$ has as underlying set the image of $Spf(k)$ under ϵ and that for every point s of $Spf(k)$, the local algebra of $Ker(p)$ at the point $\epsilon(s)$ is $O_{G,\varepsilon(s)} \otimes_{O_{G_e,\varepsilon(s)}} k_s$. Moreover, at each point $\epsilon(s)$, the residue field of $O_{G,\epsilon(s)}$ is $\kappa(s)$, and hence $O_{G_e,\epsilon(s)} = k_s$, whence $O_{Ker(p),\epsilon(s)} = O_{G,\epsilon(s)}$. For these reasons we shall say that $Ker(p)$ is the *infinitesimal neighborhood of the origin* of G and we shall write $Ker(p) = G^0$, thereby obtaining an exact sequence:

$$1 \longrightarrow G^0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e.$$

In what follows, we shall say that G is *infinitesimal* if $G = G^0$.[788] This is equivalent to saying that for every $s \in Spf(k)$, or also for every continuous morphism $k \to \kappa$, where κ is a field endowed with the discrete topology, the unit element is the unique element of $G(\kappa(s))$ resp. of $G(\kappa)$.

[786] We have added this remark.
[787] We have detailed the original in what follows.
[788] We have added the following sentence.

Suppose furthermore that G is topologically flat over k.[789] Then, by 1.6.1, the morphism $p : G \to G_e$ is topologically flat, and since it is bijective, it is therefore an effective epimorphism by Proposition 1.3.1. Consequently, G_e is identified with the quotient G/G^0. One has therefore obtained:

Corollary 2.5.1. *Let G be a formal group topologically flat over k. Then G_e is identified with the quotient G/G^0; i.e., one has an exact sequence of formal groups:*

$$1 \longrightarrow G^0 \xrightarrow{\text{incl.}} G \xrightarrow{p} G_e \longrightarrow 1.$$

2.5.2. Suppose k is a perfect field. In this case, one has seen (cf. 1.6.2) that the morphism $p : X \mapsto X_e$ admits a section $s : X_e \to X$ which depends functorially on X; this section is therefore a morphism of formal groups when X is a formal group. One obtains thus:

Proposition 2.5.2. *When k is a perfect field, every k-formal group G is canonically isomorphic to the semi-direct product of an infinitesimal group G^0 and an étale group G_e operating on G^0.*[790]

If, moreover, G is commutative, then G is the product of G^0 and of G_e. By 2.2.2, this canonical decomposition of commutative k-formal groups corresponds to an analogous decomposition of affine commutative k-group schemes; see paragraph 2.5.3 below.

Remark 2.5.2.A.[791] The exact sequence $1 \to G^0 \to G \to G_e \to 1$ is not necessarily split when k is not perfect: let us give the following example, taken from [DG70], § III.6, 8.6. Let k be a non-perfect field of characteristic $p > 0$. Let $\lambda \in k - k^p$, let L_λ be the abelian p-Lie algebra with basis (x, y), the *symbolic p-th power* (cf. VII_A, 5.2) being given by $x^{(p)} = x$ and $y^{(p)} = \lambda x$, let $U_p(L_\lambda) = k[x, y]/(x^p - x, y^p - \lambda x)$ be the restricted enveloping algebra of L_λ (cf. VII_A, 5.3), and let G_λ be the commutative k-formal group with affine algebra $U_p(L_\lambda)$. Then G_λ is a non-split extension of the étale constant k-group $(\mathbb{Z}/p\mathbb{Z})_k$ by the infinitesimal k-group $\alpha_{p,k} = Spf\, k[x]/(x^p - x)$, i.e., one has a non-split exact sequence of commutative k-formal groups:[792]

$$0 \longrightarrow \alpha_{p,k} \longrightarrow G_\lambda \longrightarrow (\mathbb{Z}/p\mathbb{Z})_k \longrightarrow 0.$$

It corresponds by duality to a non-split exact sequence of commutative k-algebraic groups:

$$0 \longrightarrow \mu_{p,k} \longrightarrow D'(G_\lambda) \longrightarrow \alpha_{p,k} \longrightarrow 0,$$

where $\mu_{p,k} = \operatorname{Spec} k[t]/(t^p - 1)$ (and one obtains thus all extensions of $\alpha_{p,k}$ by $\mu_{p,k}$, cf. [DG70], § III.6, 8.6).

[789] We have detailed what follows.

[790] The same argument shows also that, for an arbitrary field k, if all the residue fields of G equal k, then G_e is the constant k-group $(M)_k$, where $M = G(k) = Spf * H(G)$, and $H(G)$ is the semi-direct product of $H(G^0)$ by the group algebra kM, cf. the addition 2.9 further below.

[791] We have added this remark.

[792] $\alpha_{p,k}$, $(\mathbb{Z}/p\mathbb{Z})_k$ and G_λ are also k-algebraic groups, finite and flat over k, cf. N.D.E. (84) in 2.2.2.

2.5.3. Let k be a field.

Definition 2.5.3.A. *One says that a commutative k-group scheme is* unipotent *if it is isomorphic to* $\operatorname{Spec} H(G)$*, where G is an infinitesimal commutative k-formal group.*[793]

On the other hand, following Exp. IX, Definition 1.1, one says that a k-group scheme H is *of multiplicative type* if there exists a scheme S, faithfully flat and quasi-compact over $\operatorname{Spec} k$, such that H_S is a diagonalizable S-group, i.e., is isomorphic to $D_S(M)$ for some "abstract" abelian group M.[794] By (fpqc) descent, this implies that H is affine and commutative. On the other hand, since one may replace S by the residue field of one of its points, one sees that H is of multiplicative type if and only if there exists an extension K of k such that H_K is a diagonalizable K-group.

Proposition 2.5.3.B. *Let T be an affine commutative k-group scheme and let k_s be a separable closure of k.*

(i) For T to be of multiplicative type, it is necessary and sufficient that its dual $D(T)$ be an étale commutative k-formal group.

(ii) Consequently, T is of multiplicative type if and only if $T \otimes_k k_s$ is diagonalizable.

Proof. Let A denote the affine algebra of $D(T)$ and A_0 its local component at the neutral element; then $D(T)$ is étale over k if and only if $A_0 = k$. If K is an extension of k, then the algebra $A_0 \otimes_k K$ is local (since it is a projective limit of local artinian rings), so it coincides with the local component $(A \otimes_k K)_0$; moreover, since the formation of $D(T)$ commutes with base change (cf. 2.2.2), one also has $A \otimes_k K \simeq D(T_K)$.

Suppose T of multiplicative type; then there exists an extension K of k such that $\mathscr{O}(T) \otimes_k K$ is isomorphic, as a K-Hopf algebra, to the group algebra KM, for some abelian group M. Then $A \otimes_k K$ is isomorphic to the algebra K^M, endowed with the product topology, hence one has $K = (A \otimes_k K)_0 = A_0 \otimes_k K$ and this entails that $A_0 = k$, so $D(T)$ is étale.

Conversely, suppose $D(T)$ étale. Then $D(T) \otimes_k k_s = D(T_{k_s})$ is a k_s-constant group M, so its covariant bigèbre is the group algebra $k_s M$ (cf. Remark 2.5.A), so $\mathscr{O}(T) \otimes_k k_s = k_s M$, which proves that T is of multiplicative type, and split by the extension $k \to k_s$. The proposition follows.

By 2.2.2, 2.5.1, and 2.5, one obtains:

Corollary 2.5.3.C. *Let k be a field and G an affine commutative k-group scheme.*

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[793] This coincides with the "usual" notion of unipotent k-group scheme, cf. the addition 2.8 at the end of Section 2.

[794] The original indicated: "let us say that a commutative k-group scheme is of multiplicative type if it is isomorphic to $\operatorname{Spec} H(G)$, where G is an étale commutative k-formal group." We have given here the "usual" definition, drawn from Exp. IX, 1.1, and we have shown its equivalence with the preceding condition; see also [DG70], § IV.1, Th. 2.2.

(i) G contains a subgroup of multiplicative type G_m such that G/G_m is unipotent.

(ii) When k is perfect, there exists furthermore a canonical retraction of G onto G_m, so that G is the product of a unipotent group and a group of multiplicative type.

(iii)[795] *Let k_s be a separable closure of k and $\Gamma = \mathrm{Gal}(k_s/k)$. The category of k-group schemes of multiplicative type is anti-equivalent to the category of continuous Γ-modules.*

2.6.

We shall now study infinitesimal formal groups, to which the following paragraphs are devoted. In this study, Lie algebras play a primordial role.

Suppose first that the base ring k is artinian and let G be a formal group over k. One can give three different interpretations of the Lie algebra of G, all of which we shall use:

a) Let D be the algebra $k[d]/(d^2)$ of dual numbers over k and δ the homomorphism from D to k which annihilates d. For every formal group G over k, $Lie(G)$ is the kernel of $G(\delta)$, so that by definition one has an exact sequence of groups

$$1 \longrightarrow \mathrm{Lie}(G) \longrightarrow G(D) \xrightarrow{G(\delta)} G(k) \longrightarrow 1.$$

b) Let A be the affine algebra of G and $I_A = Ker\epsilon_A$ its augmentation ideal. The group $G(D)$ has as elements the morphisms of profinite k-algebras $f : A \to D$. The condition $G(\delta)(f) = 1$ is equivalent to $\delta \circ f = \epsilon_A$. Since $x - \epsilon_A(x) \cdot 1_A \in I_A$ for every $x \in A$, this is equivalent to $f(I_A) \subset k \cdot d$, hence $Ker(f)$ contains I_A^2 and therefore also $\bar{I}_A^2$ (the closure), so f induces a continuous linear map $f' : I_A/I_A^2 = \omega_{G/k} \to k$ such that, for every $x \in A$, one has the equality

$$f(x) = \epsilon_A(x) \cdot 1_D + f'(\bar{x}) \cdot d,$$

where $\bar{x}$ designates the image of $x - \epsilon_A(x) \cdot 1_A$ in I_A/I_A^2. One sees then that the map $f \mapsto f'$ defines a bijection of $Lie(G)$ onto the topological dual $\omega_{G/k}^\dagger$ of $\omega_{G/k}$ (cf. 0.2.2).

This bijection respects the group structures. Indeed, let f and g be two elements of $Lie(G)$; their product $f \cdot g$ is the composite map $h \circ \Delta_A$, where $h : A \otimes A \to D$ is such that $h(b \otimes b') = f(b)\, g(b')$. Now if $a \in I_A$ one has, by 2.1 (ii), $\Delta_A(a) - a \otimes 1 - 1 \otimes a \in I_A \otimes I_A$, whence $(f \cdot g)(a) = f(a) + g(a)$ (cf. also II 3.10).

In what follows, we identify $Lie(G)$ with $\omega_{G/k}^\dagger$ by means of the bijection $f \mapsto f'$ described above. The group $Lie(G)$ is thus endowed with a structure of k-module.

c) Let $A^\dagger$ and $D^\dagger$ be the k-modules dual to A and D, $\{1_D^\dagger, d^\dagger\}$ the dual basis of the basis $1_D, d$ of D over k (one has $1_D^\dagger = \delta$). Since D is free of finite rank over k, the canonical map

[795] We have added point (iii), a consequence of Proposition 2.5.

$$\mathrm{Hom}_c(A, D) \longrightarrow \mathrm{Hom}_k(D^\dagger, A^\dagger), \qquad f \mapsto {}^t f$$

is bijective. On the other hand, f is determined by the values ${}^t f(1_D*)$ and ${}^t f(d*) = x$.

The condition $G(\delta)(f) = 1$ is equivalent to the equality ${}^t f(1_D*) = \epsilon_A$. One sees easily on the other hand that f is compatible with multiplication if and only if one has (cf. 2.3):

$$(*)\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x).$$

Finally, it is clear that a continuous linear map $f : A \to D$ which is compatible with multiplication and such that $\delta \circ f = \epsilon_A$ sends the unit element of A to that of D.[796] The map $f \mapsto x$ thus permits us to identify $Lie(G)$ with the set of "primitive elements" of $H(G)$ (i.e., the $x \in H(G)$ satisfying relation $(*)$). If x and y are two such elements, one has

$$\begin{aligned}\delta_G(xy) &= \delta_G(x)\,\delta_G(y) = \varphi_G(x \otimes 1 + 1 \otimes x)(y \otimes 1 + 1 \otimes y)\\ &= \varphi_G(xy \otimes 1 + x \otimes y + y \otimes x + 1 \otimes xy),\end{aligned}$$

whence $\delta_G(xy - yx) = \varphi_G((xy - yx) \otimes 1 + 1 \otimes (xy - yx))$.

This shows that the k-module $Lie(G)$ is identified with a Lie subalgebra of $H(G)$: we shall say that $Lie(G)$ is the *Lie algebra* of G.[797]

2.6.1. When k is an arbitrary pseudocompact ring and G a formal group over k, we call *O_k-Lie algebra of G* the functor $Lie(G)$ which associates to every object C of $Alf_{/}k$ the C-Lie algebra of the C-formal group $G' = G \hat{\otimes}_k C$:[798] set $A' = A \hat{\otimes}_k C$, since I_A is a direct factor of A, then $I_{A'}$ equals $I_A \hat{\otimes}_k C = I_A \hat{\otimes}_A A'$, and since $\omega_{G/k} = I_A \hat{\otimes}_A k$ and similarly $\omega_{G'/C} = I_{A'} \hat{\otimes}_{A'} C$, one obtains that $\omega_{G'/C} = \omega_{G/k} \hat{\otimes}_A C$ and hence

$$Lie(G)(C) = \mathrm{Hom}_c(\omega_{G/k} \hat{\otimes}_A C, C) \qquad \text{i.e.} \qquad Lie(G) = V_kf(\omega_{G/k})$$

(with the notations of 1.2.3.B). Therefore, by Proposition 1.2.3.E, $Lie(G)$ is flat over O_k if and only if $\omega_{G/k}$ is a projective pseudocompact k-module.

2.6.2. Conversely, every Lie algebra L over O_k defines a k-group functor. Indeed, let us denote by $U(L)$ the functor which associates to every object C of $Alf_{/}k$ the enveloping algebra $U(L(C))$ of the C-Lie algebra $L(C)$. By VII_A, 3.2.2, $U(L)$ is a

[796] Indeed, $\delta \circ f = \epsilon_A$ entails $f(1) = 1 + \lambda d$, with $\lambda \in k$, and then $f(1) = f(1)^2 = 1 + 2\lambda d$ gives $\lambda = 0$.

[797] Comparing with VII_A 2.5, one sees that if K is a k-group scheme of finite type and if G is the formal completion of K at the origin (i.e., $\mathcal{A}(G)$ is the completion of the local ring $O_{K,e}$ for the m-adic topology, where m is the kernel of the augmentation $\epsilon : O_{K,e} \to k$), then $H(G)$ is identified with the algebra of distributions $U(K)$, and $Lie(G)$ with $Lie(K)$. (The condition that K be of finite type over k is used to ensure that m/m^2 is a k-module of finite length, hence discrete, so that its topological dual coincides with its ordinary dual.) In particular, when G is a finite k-formal group (i.e., such that $\mathcal{A}(G)$ is a finite k-module), in which case G may also be considered as the k-group scheme Spec $\mathcal{A}(G)$, the two definitions of $Lie(G)$ coincide.

[798] We have detailed the original in what follows.

bialgebra over O_k and therefore determines, by 2.2, a k-group functor $Spf * U(L)$ which we shall henceforth denote $\mathcal{G}(L)$. Thus, $\mathcal{G}(L)(C)$ is the group of elements $z \in U(L(C))$ of augmentation 1 and such that $\Delta_{U(L(C))}(z) = z \otimes z$.

Moreover, when L is flat over O_k, one has the following proposition.

Proposition 2.6.2.[799] *Let L be a flat O_k-Lie algebra.*

(i) $\mathcal{G}(L)$ is a formal group topologically flat over k, having $U(L)$ as O_k-bialgebra.

(ii) $\mathcal{G}(L)$ is infinitesimal.

(iii) For every object C of $Alf_/k$, $Lie(\mathcal{G}(L))(C)$ is identified with the set

$$\mathrm{Prim}\, U(L(C)) = \{x \in U(L(C)) \mid \varepsilon(x) = 0 \text{ and } \Delta(x) = x \otimes 1 + 1 \otimes x\}$$

of primitive elements of $U(L(C))$. In particular, one has a natural morphism of O_k-Lie algebras $\tau_L : L \to Lie(\mathcal{G}(L))$.

Indeed, the hypothesis that L be flat over O_k means that for every morphism $C \to D$ of $Alf_/k$, one has $L(D) = L(C) \bigotimes_C D$, and that for each local component C' of C, $L(C')$ is a free C'-module. The first condition entails that $U(L(D)) = U(L(C)) \bigotimes D$ (by the universal property of the tensor product and that of the functor $L \mapsto U(L)$), and the second condition entails, by the Poincaré–Birkhoff–Witt theorem (cf. Bourbaki, *Groupes et algèbres de Lie*, I 2.7), that $U(L(C'))$ is a free C'-module. Therefore the bialgebra $U(L)$ is flat over O_k.

To prove (ii) and (iii), one may suppose k artinian. Set then $L = L(k)$, $U = U(L)$, $U_0 = k \cdot 1_U$ and let U^+ be the two-sided ideal of U generated by the image of L. Set in addition, for every $n \geqslant 1$,

$$U_n = \{u \in U \mid \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+\}.$$

By 1.3.6, it suffices to show that U is the union of the U_n. Now, if one identifies L with its canonical image in U, L is evidently contained in U_1. If $x_1, \cdots, x_n$ are elements of L, one has $\Delta_U(x_1 \cdots x_n) = (x_1 \otimes 1 + 1 \otimes x_1) \cdots (x_n \otimes 1 + 1 \otimes x_n)$, which shows, by induction on n, that the product $x_1 \cdots x_n$ belongs to U_n, hence that $U = \cup_n U_n$. This proves (ii).

On the other hand, let $D = k[d]/(d^2)$ be the algebra of dual numbers over k. By hypothesis one has $L(D) \simeq L \bigotimes D$, whence $U(L(D)) \simeq U \bigotimes D$, by the universal properties of the tensor product and of the enveloping algebra. It follows that $Lie(\mathcal{G}(L))(k)$ is identified with the set of elements $z = 1 + xd$ of $U \bigoplus Ud$ (where $x \in U$) such that $\epsilon(z) = 1$ and $\Delta(z) = z \otimes z$, which is equivalent to $\epsilon(x) = 0$ and $\Delta(x) = x \otimes 1 + 1 \otimes x$, i.e., to $x \in PrimU$. In particular, the map $\tau_L : x \mapsto 1 + dx$ is a morphism of O_k-Lie algebras, from L to $Lie(\mathcal{G}(L))$.

2.6.3. If L is a Lie algebra flat over O_k, the formal group $\mathcal{G}(L)$ may be characterized by a universal property.[800] Indeed, every morphism ϕ from $\mathcal{G}(L)$ into a formal

[799] We have highlighted points (i) and (ii), and added point (iii), which will be useful in 2.6.3 and 3.3.2.
[800] We have modified what follows, taking advantage of the addition made in 2.6.2.

group G induces a morphism $\mathrm{Lie}(\varphi) : \mathrm{Lie}(\mathcal{G}(L)) \to \mathrm{Lie}(G)$; composing it with the morphism $\tau_L : L \to Lie(\mathcal{G}(L))$ (cf. 2.6.2), one obtains a morphism $\phi' : L \to Lie(G)$, and one has:

Proposition 2.6.3. *If G is a k-formal group and L a flat O_k-Lie algebra, the map $\phi \mapsto \phi'$ defined above is a bijection*

$$\mathrm{Hom}_{Grf/k}(\mathcal{G}(L), G) \simeq \mathrm{Hom}_{Lie}(L, Lie(G))$$

where the right-hand term designates the set of morphisms of O_k-Lie algebras from L to $Lie(G)$.

One reduces at once to the case where k is artinian. Set $L = L(k)$. By 2.3.1, $\mathrm{Hom}_{Grf/k}(\mathcal{G}(L), G)$ is in bijection with the set of unital algebra morphisms $\phi : U(L) \to H(G)$ such that the following diagram is commutative:

$$\begin{tikzcd}
U(L) \arrow[r, "h"] \arrow[d, "\Delta_{U(L)}"'] & H(G) \arrow[r, "\delta_G"] & H(G \times G) \\
U(L) \otimes U(L) \arrow[r, "h \otimes h"] & H(G) \otimes H(G) \arrow[ur, "\phi_G"'] &
\end{tikzcd}$$

Now h is determined by its restriction to L, which is a morphism of Lie algebras from L to the Lie algebra underlying $H(G)$, and the commutativity of the diagram means that h sends L into the part of $H(G)$ formed by the x such that $\delta_G(x) = \varphi_G(x \otimes 1 + 1 \otimes x)$, which is none other than $Lie(G)$, cf. 2.6 c).

2.7.

We end these generalities with a statement which goes back to S. Lie and which will serve us in paragraph 5.1. A *formal monoid* over k is by definition a couple (M, m) consisting of a formal variety M and a morphism $m : M \times M \to M$ such that $m(C)$ makes $M(C)$ into a monoid for every object C of $Alf_{/}k$.[801] In particular, the "unit section", which associates to every object C the unit element of $M(C)$, defines a section ϵ_M of the canonical projection $M \to Spf(k)$. We shall say that the formal monoid M is *infinitesimal* if ϵ_M induces a bijection of the underlying sets.

Proposition 2.7. *Every topologically flat infinitesimal k-formal monoid M is a k-formal group.*[802]

[801] Here and in what follows, we have written "monoid" instead of "monoid with unit element" (recall that a monoid is by definition a set endowed with an associative composition law and possessing a unit element).

[802] By the equivalence of categories 1.3.5.D, a monoid in the category of topologically flat k-formal varieties "is the same thing" as a monoid in the category of cocommutative flat O_k-coalgebras, i.e.,

We must show that $M(C)$ is a group for every object C of $Alf_{/}k$. One reduces immediately to the case where k is artinian. Let then $U = H(M)$ be the coalgebra of M (1.3.5); the multiplication $m : M \times M \to M$ induces a morphism of coalgebras $m_U : U \bigotimes U \to U$, which makes U into an associative algebra over k; this algebra has as unit element the image of the unit element of k by the map from k into U induced by the unit section ϵ_M of M. Similarly, the projection $M \to Spf(k)$ induces a homomorphism ϵ_U from U to k; we shall denote U^+ the kernel of ϵ_U.

We must show that there exists an antipodism, that is, a morphism of coalgebras $c_U : U \to U$ such that, for every $u \in U$,

$$(*) \qquad (m_U \circ (c_U \otimes id_U) \circ \Delta_U)(x) = \varepsilon_U(u) \cdot 1_U.$$

Let (U_n) be the filtration of U defined in 1.3.6, set $U_n^+ = U^+ \cap U_n$. Since M is infinitesimal, U^+ is the union of the submodules U_n^+.[803] One sets $c_0(1) = 1$ and $c_1(x) = -x$ if $x \in U_1^+$, i.e., if x is a primitive element. Suppose $c_{n-1} : U_{n-1} \to U$ constructed so as to satisfy $(*)$ for every $x \in U_{n-1}$, and let $x \in U_n^+$. By the proof of Lemma 1.3.6.A, one has $\Delta_U(x) - x \otimes 1 \in U_{n-1} \otimes U^+$ (this is where the hypothesis that U be flat over k intervenes), so one may write $\Delta_U(x) = x \otimes 1 + \Sigma_i y_i \otimes z_i$, with $y_i \in U_{n-1}$; one then sets $c_n(x) = -\Sigma_i c_{n-1}(y_i) z_i$. One obtains thus a k-linear map $c : U \to U$, which is the left inverse of id_U for the monoid law on $\mathrm{End}_k(U)$, defined by $f \cdot g = m_U \circ (f \otimes g) \circ \Delta_U$ (the unit element being the map $\eta : u \mapsto \epsilon(u) \cdot 1_U$). It follows that c is uniquely determined, and is also the right inverse of id_U, i.e., one also has $m_U \circ (c_U \otimes id_U) \circ \Delta_U = \eta$ (without supposing U cocommutative).

2.8. Unipotent group schemes over a field.

Let k be a field. "Recall" that a k-group scheme G is said to be *unipotent* if it satisfies the following two conditions (cf. [DG70], § IV.2, Prop. 2.5):

(a) G is affine.

(b) Every simple $O(G)$-comodule is trivial, i.e., if $\rho : V \to V \bigotimes_k O(G)$ is an $O(G)$-comodule structure on a k-vector space $V \neq 0$, and if there exists no nonzero subspace $W \neq V$ such that $\rho(W) \subset W \bigotimes_k O(G)$, then V is one-dimensional and $\rho(v) = v \bigotimes 1$ for every $v \in V$.

By loc. cit., when G is of finite type over k, this is equivalent to the definition given in Exp. XVII, § 1, namely that G possesses a finite composition series whose successive quotients are isomorphic to k-subgroups of $\mathbb{G}_{a,k}$.

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a cocommutative O_k-bialgebra (in the usual sense, i.e., not necessarily equipped with an antipode). Moreover, by 1.3.6, the hypothesis that M be infinitesimal is equivalent to saying that the corresponding bialgebra is connected. So, if k is an artinian ring, the proposition is equivalent to saying that: every cocommutative connected k-bialgebra, flat over k, is a k-Hopf algebra, i.e., possesses an antipode (and the cocommutativity hypothesis is in fact superfluous, cf. the proof).

[803] The original continued thus: "one shows then easily, by induction on n, that there exists one and only one linear map $c_n : U_n \to U_n$ such that the composite map $m_U \circ (c_n \otimes id_U) \circ \Delta_U : U^+_n \to U$ be zero"; we have detailed the proof, which rests on that of Lemma 1.3.6.A.

Now, for every affine k-group scheme G, the comultiplication of $O(G)$ endows the pseudocompact k-module $A = O(G)*$ with a structure of profinite k-algebra, not necessarily commutative, the unit element 1_A being the augmentation $\epsilon : O(G) \to k$. On the other hand, let $I = f \in A | f(1_{O(G)}) = 0$; this is a two-sided ideal of A, and one has $A = k \cdot 1_A \bigoplus I$, cf. 1.3.6.

Let V be a subspace of A of finite codimension, consider the continuous k-bilinear map $\varphi : A \times A \to A/V$, $(a, b) \mapsto ab + V$; by Lemma 0.3.1, there exist two subspaces L_1, L_2 of finite codimension in A such that V contains AL_2 and $L_1 A$; then $L = L_1 \cap L_2$ is a subspace of A of finite codimension, and V contains the two-sided ideal ALA, which is of finite codimension. This shows that the two-sided ideals of finite codimension form a basis of neighborhoods of 0. One deduces that an $O(G)$-comodule “is the same thing” as a continuous A-module, i.e., an A-module V such that the map $A \times V \to V$ is continuous, V being endowed with the discrete topology. Such a module is evidently the union of submodules V_i of finite dimension over k, each of which is a module over a k-algebra quotient A_i of A, of finite dimension over k. It follows that if M is a simple continuous module, it is of finite dimension over k, and is a faithful simple module over the finite-dimensional k-algebra $A/Ann(M)$; the latter is therefore a finite-dimensional simple k-algebra, i.e., $Ann(M)$ is a maximal open ideal of A. Conversely, let P be an open prime ideal[804] of A; then A/P is a finite-dimensional k-algebra in which the ideal (0) is prime, hence it is a finite-dimensional simple k-algebra, so there exists, up to isomorphism, a unique simple continuous A-module whose annihilator is P. It follows that the map $M \mapsto Ann(M)$ defines a bijection between the isomorphism classes of simple continuous A-modules and the open prime ideals of A. In particular, the A-module A/I, which is one-dimensional over k, is called the “trivial module”; it corresponds to the one-dimensional $O(G)$-comodule V which is trivial, i.e., such that $\rho(v) = v \bigotimes 1_{O(G)}$ for every $v \in V$. One thus obtains the following proposition:

Proposition 2.8.1. *Let k be a field, G an affine k-group scheme and $A = O(G)*$.*

(i) Then G is unipotent if and only if I is the unique open prime ideal of A.[805]

(ii) In particular, if G is commutative, so that $O(G) = H(D(G))$, where $D(G) = Spf(A)$ designates the Cartier dual of G, then G is unipotent if and only if $D(G)$ is infinitesimal.

2.9. Pointed cocommutative Hopf algebras over a field.

Let k be a field, C a k-coalgebra, A the algebra $C*$ endowed with the structure of profinite k-algebra (not necessarily commutative) described in 2.8; by 0.2.2, one has `C = A† = Hom_c(A, k)`. One deduces that the map $D \mapsto D \perp = f \in A | f(D) = 0$ is a bijection from the set of subcoalgebras of C onto that of closed two-sided ideals

[804] Recall that a two-sided ideal P of a ring A is called *prime* if in A/P the product of two nonzero two-sided ideals is nonzero.

[805] This is also equivalent to saying that $k \cdot 1_{O(G)}$ is the unique simple k-subcoalgebra of $O(G)$; see for example [Ab80], 3.1.4. Let us point out in passing a misprint in loc. cit., p. 130, line 4: $M \simeq C * /annM$ is to be replaced by $C * /annM \simeq \mathrm{End}_k(M)$.

(in the sequel, one will simply say "ideals") of A; the inverse bijection being given by $I \mapsto I^{\perp} = \{x \in C = A^{\dagger} \mid x(I) = 0\}$. Since every maximal closed ideal is a maximal open ideal (cf. 0.2.1), every subcoalgebra therefore contains a simple subcoalgebra, necessarily of finite dimension.

Recall that a subcoalgebra D of C is called *irreducible* if it contains only one simple subcoalgebra S_0, which is equivalent to saying that $m_0 = S_0{}^{\perp}$ is the unique maximal open ideal containing $D^{\perp}$, i.e., $D^{\perp} + m = A$ for every maximal open ideal $m \neq m_0$. This is in particular the case if $D = S_0$. Then the sum Σ_0 of all the irreducible subcoalgebras C_i containing S_0 is evidently a subcoalgebra, and it is irreducible because, for every $m \neq m_0$, one has, by 0.2.D:

$$m + \cap_i C_i^{\perp} = \cap_i (m + C_i^{\perp}) = A.$$

One says that Σ_0 is the *irreducible component of C corresponding to C_0*.

Moreover, one says that C is *pointed* if every simple k-subcoalgebra of C is one-dimensional; this is equivalent to saying that for every maximal open ideal m of A, one has $A/m = k$. Recall also that C is said to be *connected* if it is irreducible and pointed. (Let us note in passing that if C is a bialgebra, it is connected if and only if it is irreducible, since $k \cdot 1_C$ is a simple subcoalgebra.)

Suppose henceforth that C is cocommutative. Then A is commutative and is therefore the product of its local components A_m, for $m \in \Upsilon(A)$ (cf. 0.1.1); denote by S_m the simple subcoalgebra $m^{\perp} \simeq (A/m)^*$ and by Σ_m its irreducible component. One may describe Σ_m as follows. Denote by J_m the kernel of the projection $A \to A_m$; it is contained in m and is the smallest closed ideal I of m such that $I + m' = A$ for every $m' \neq m$. Indeed, if I has this property, then I contains $A_{m'}$ for every $m' \neq m$, so contains J_m. Since $A = J_m \bigoplus A_m$, it follows that $\Sigma_m = J_m{}^{\perp}$ is identified with $A_m^{\dagger}$. One can now prove the:

Proposition 2.9.1. *Let k be a field.*

(i) Let G be a k-formal group such that all the residue fields of G equal k. Then G_e is the constant k-group M_k, where $M = G(k) = \{x \in H(G) \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}$, and $H(G)$ is the semi-direct product of $H(G^0)$ by kM (cf. 2.4.B).

(ii) Equivalently: let H be a cocommutative pointed k-Hopf algebra. Then H is the semi-direct product of the irreducible component H_0 of the unit element 1_H by kM, where $M = \{x \in H \mid \varepsilon(x) = 1 \text{ and } \Delta(x) = x \otimes x\}$.

Let us prove (i). Since all the residue fields of G equal k, the projection $\pi : G \to G_e$ admits the section $s : G_e \to G$ defined by $O_{G,g} \to \kappa(g) = k$, for every $g \in G$; moreover, for every $g, h \in G$, $O_{G,g} \otimes_k O_{G,h}$ is local with residue field k, and one therefore obtains that $s \times s$ is a section of the projection $\pi \times \pi : G \times G \to (G \times G)_e = G_e \times G_e$. Since π is a group morphism, it follows that $\pi \circ m_G \circ (s \times s) = m_{G_e} = \pi \circ s \circ m_{G_e}$, and since π is an epimorphism this entails that s is a group morphism. One obtains therefore that $G = G^0 \rtimes G_e$, and hence $H(G)$ is the semi-direct product of $H(G^0)$ by $H(G_e)$. Moreover, since all the residue fields of G equal k, then $H(G_e)$ is the group algebra kM, where $M = G_e(k)$ (cf. 2.5.A). Finally, since G^0 is infinitesimal, the morphism $G(k) \to G_e(k)$ is injective; it is therefore bijective (since it admits a section), so $M = G(k)$. Point (i) follows.

To prove (ii), it remains only to see that H_0 equals $H(G^0)$. Now the unit element 1_H of H is none other than the augmentation $\epsilon_A : A \to k$, which is the unit section of $G(k)$, so the local component of $\mathcal{A}(G)$ corresponding to H_0 is none other than $\mathcal{A}(G^0)$ and therefore, by what was seen above, one has $H_0 = □(G^0)^\dagger = H(G^0)$. This proves the proposition.

Remarks 2.9.2. *(a) The proposition above, contained implicitly in 2.5.2, has been obtained independently by B. Kostant (cf. [Sw69], Preface). Combined with Cartier's Theorem 3.3 below (cf. [Ca62], § 12, Th. 3), also obtained by Kostant (cf. [Sw69], loc. cit.), this result is often called the "Cartier–Gabriel–Kostant theorem".*

(b) In the form (ii), 2.9.1 has been extended by R. G. Heyneman and M. E. Sweedler to the case where one assumes that H is pointed and a direct sum of its irreducible components (but not necessarily cocommutative), cf. [HS69], Th. 3.5.8.

3. Phenomena specific to characteristic 0

Throughout Section 3, we assume that the pseudocompact ring k contains the field of rational numbers $\mathbb{Q}$.

3.1. Lemma.

Lemma 3.1. *Let C be a commutative unital $\mathbb{Q}$-algebra, L a Lie algebra over C whose underlying C-module is free. Then the canonical map $L \to U(L)$ is an isomorphism of L onto the set of primitive elements of $U(L)$.*

Indeed, let us identify L with its canonical image in $U(L)$; let I be a totally ordered set and $(x_i)_{i\in I}$ a basis of L indexed by I; let us denote by $\mathbb{N}^{(I)}$ the set of families $n = (n_i)_{i\in I}$ of natural integers such that n_i is zero except perhaps for a finite number of indices $i_1 < i_2 < \cdots < i_s$ (these indices depend on n); set finally

$$x^n = x_{i_1}^{n_{i_1}} x_{i_2}^{n_{i_2}} \cdots x_{i_s}^{n_{i_s}} \quad \text{and} \quad n! = (n_{i_1}!)(n_{i_2}!)$$

One knows then that the x^n form a basis of $U(L)$ (Poincaré–Birkhoff–Witt theorem) and one sees easily that one has

$$(*) \qquad \Delta_{U(L)}\left(\frac{x^n}{n!}\right) = \sum \frac{x^m}{m!} \otimes \frac{x^{n-m}}{(n-m)!},$$

the sum being extended over all elements m of $\mathbb{N}^{(I)}$ such that $0 \leqslant m \leqslant n$ (i.e., such that $0 \leqslant m_i \leqslant n_i$ for every i). It evidently follows that one has $\Delta_{U(L)}(u) = u \otimes 1 + 1 \otimes u$ if and only if u is a linear combination of the x_i.

3.2.

Suppose now C artinian, of radical r. For every C-algebra U (associative, unital), the ideal rU therefore consists of nilpotent elements; if x belongs to rU, we shall set

$$\exp_U x = 1 + x + \frac{x^2}{2!} + \cdots$$

[806]

One thus obtains a bijection of rU onto $1+rU$; the inverse bijection sends an element $1+y$ of $1+rU$ to

$$\log_U(1+y) = y - \frac{y^2}{2} + \frac{y^3}{3} - \cdots$$

Moreover, it is clear that the map $\exp_U$ is functorial in U.[807]

The ring C still being artinian, suppose U endowed with a structure of bialgebra over C (cf. 2.2). For every primitive element x of rU (cf. VII_A 3.2.3), one has then

$$\begin{aligned}\Delta_U(\exp_U x) &= \exp_{U\otimes U}(\Delta_U(x))\\ &= \exp_{U\otimes U}(x\otimes 1 + 1\otimes x)\\ &= \exp_{U\otimes U}(x\otimes 1)\cdot \exp_{U\otimes U}(1\otimes x)\\ &= ((\exp_U x)\otimes 1)\cdot(1\otimes(\exp_U x))\\ &= (\exp_U x)\otimes(\exp_U x).\end{aligned}$$

One thus sees that the bijection $\exp_U$ transforms a primitive element of rU into an element z of $1+rU$ such that $\Delta_U(z) = z\otimes z$. Conversely, if z satisfies these conditions then, setting $x = \log_U(z)$, the preceding computation shows that $\exp_{U\otimes U}(x\otimes 1 + 1\otimes x)$ equals $z\otimes z = \Delta(\exp_U x) = \exp_{U\otimes U}(\Delta_U(x))$, whence $x\otimes 1 + 1\otimes x = \Delta_U(x)$.[808] Let us moreover note that if $z\in 1+rU$ satisfies $\Delta_U(z) = z\otimes z$, then $\epsilon_U(z)^2 = \epsilon_U(z)$, and since $\epsilon_U(z)$ is invertible (since z is, r being nilpotent), it follows that $\epsilon_U(z) = 1$.

Consider in particular a Lie algebra L flat over C, take for U the enveloping algebra $U(L)$ of L over C, and identify L with its canonical image in U. By Lemma 3.1, L is therefore the set of primitive elements of U (indeed L is a product of free modules over the local components of C). Consider then the C-formal group $\mathcal{G}(L) = Spf * U(L)$, which has $U = U(L)$ as covariant bialgebra (cf. 2.6.2). Let m be a maximal ideal of C and $\bar{C} = C/m$. Since $\mathcal{G}(L)$ is infinitesimal (loc. cit.), the unit element of $\bar{U} = U\bigotimes_C \bar{C}$ is the only element z of $\bar{U}$ such that[809] $\epsilon_{\bar{U}}(z) = 1$ and $\Delta_{\bar{U}}(z) = z\otimes z$. It follows that the elements z of U such that $\epsilon_U(z) = 1$ and $\Delta_U(z) = z\otimes z$ necessarily belong to $1+rU$.

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[806] If $x, x' \in rU$ commute, one has therefore $\exp_U(x+x') = (\exp_U x)(\exp_U x')$.

[807] That is, for every morphism $\phi: U\to V$ of C-algebras, one has $\phi(\exp_U(x)) = \exp_V \phi(x)$.

[808] We have detailed what precedes and added the following sentence.

[809] We have added the condition $\epsilon_U(z) = 1$, omitted in the original.

Finally, since $L \cap rU$ is identified with $rL = L \otimes_C r$, one sees finally that: $\exp_U$ defines a bijection of $L \otimes_C r$ onto the group $\mathcal{G}(L)(C)$. We summarize:

Proposition 3.2. *Let k be a pseudocompact ring containing $\mathbb{Q}$ and L a flat O_k-Lie algebra.*

(i) For every object C of $\mathit{Alf}_{/k}$, denote by $r(C)$ its radical; then the map

$$\exp_{U(L(C))} : \quad L(C) \otimes_C r(C) \longrightarrow \mathcal{G}(L)(C)$$

is bijective and functorial in C and L.

(ii) The natural morphism $L \to \mathrm{Lie}(\mathcal{G}(L))$ (cf. 2.6.2) is an isomorphism of O_k-Lie algebras.[810]

3.2.1. The bijection $\exp_{U(L(C))}$ permits one to define by transport of structure a group law on the set $L(C) \otimes_C r(C)$ (which one identifies with a part of $U(L(C))$ as in 3.2). If x and y are two elements of $L(C) \otimes_C r(C)$, this law is such that

$$x \cdot y = \log((\exp x)(\exp y)) = \log\Big(1 + \sum_{p+q>0} \frac{x^p y^q}{p!\, q!}\Big)$$

$$= \sum_{m \geq 1} \sum_{p_i + q_i > 0} \frac{(-1)^{m-1}}{m} \frac{x^{p_1} y^{q_1}}{p_1!\, q_1!} \cdots \frac{x^{p_m} y^{q_m}}{p_m!\, q_m!} = \sum_{\ell \geq 1} P_\ell(x,y)$$

where $P_\ell(x,y)$ designates the sum of the monomials of total degree ℓ in x and y. One has for example:

$$P_1(x,y) = x + y$$

$$P_2(x,y) = \underbrace{\frac{x^2}{2} + xy + \frac{y^2}{2}}_{\text{(terms with } m=1)} - \underbrace{\frac{1}{2}(x^2 + xy + yx + y^2)}_{\text{(terms with } m=2)} = \frac{1}{2}(xy - yx) = \frac{1}{2}[x,y]$$

and $P_3(x,y)$ is the sum of the three terms below:

$$\underbrace{\frac{x^3}{6} + \frac{x^2y}{2} + \frac{xy^2}{2} + \frac{y^3}{6}}_{(m=1)} - \underbrace{\frac{1}{2}(x^3 + x^2y + yx^2 + xyx + yxy + y^2x + xy^2 + y^3)}_{(m=2)}$$

$$+ \frac{1}{3}(x^3 + x^2y + yx^2 + xyx + yxy + y^2x + xy^2 + y^3)$$

[810] We have added point (ii), an immediate consequence of 3.1.

3

(m=3)

whence

$$P_3(x, y) = \frac{1}{12}(x^2y + yx^2 - 2\,xyx - 2\,yxy + y^2x + xy^2) = \frac{1}{12}([y, x], x] + [y, [y, x]]).$$

One can show more generally that one has the Campbell–Hausdorff formula:[811]

$$P_\ell(x,y) = \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m\cdot \ell} \sum_{p_1,\dots,p_m,\, q_1,\dots,q_{m-1}} \prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}\,(\operatorname{ad} y)^{q_i}}{p_i!\; q_i!} \cdot \frac{(\operatorname{ad} x)^{p_m}}{p_m!}$$

$$+ \sum_{m=1}^{\ell} \frac{(-1)^{m-1}}{m\cdot \ell} \sum_{p_1,\dots,p_{m-1},\, q_1,\dots,q_{m-1}} \prod_{i=1}^{m-1} \frac{(\operatorname{ad} x)^{p_i}\,(\operatorname{ad} y)^{q_i}}{p_i!\; q_i!}\,(x)$$

where the $p_j, q_i \in \mathbb{N}$ verify $p_i + q_i \geqslant 1$ for $i = 1, \cdots, m-1$ and $p_m + \Sigma_{i=1}^{m-1}(p_i + q_i) = \ell - 1$ (i.e., in the sums above, each non-zero "Lie monomial" is of total degree ℓ); for a proof, see N. Jacobson, *Lie Algebras* (Interscience, 1962), § V.5, or [BLie], II § 6.4, Th. 2.

3.3.

If G is a k-formal group of affine algebra A, recall that one denotes by I_A the augmentation ideal of A and by $\omega_{G/k}$ the pseudocompact k-module $I_A / I_A^2 \simeq I_A \otimes_A k$.

Theorem 3.3 (Cartier). *Let k be a pseudocompact ring*[812] *containing $\mathbb{Q}$ and G a k-formal group. The following assertions are equivalent:*

(i) There exists a flat O_k-Lie algebra L such that G is isomorphic to $\mathcal{G}(L)$ (and in this case $L = Lie(G)$ by 3.2).

(ii) There exists a projective pseudocompact k-module ω such that the formal variety underlying G is isomorphic to the formal variety $V_k^{f,0}(\omega)$ (cf. 1.2.5) of affine algebra $k[[\omega]]$ (and in this case $\omega \simeq \omega_{G/k}$).

(iii) G is infinitesimal and $\omega_{G/k}$ is a projective pseudocompact k-module.

(iv) G is infinitesimal and topologically flat over k.

(i) ⇒ (ii): Let $\omega = \Gamma * (L)$ be the pseudocompact k-module dual to L (cf. 1.2.3.D). For every object C of $Alf_{/}k$, we must exhibit an isomorphism from $V_k^{f,0}(\omega)(C)$ onto

[811] We have corrected the formula given, which was erroneous, and added the reference [BLie].

[812] We have removed the hypothesis that k be local (the proof reduces to this case).

$\mathcal{G}(L)(C)$ which is functorial in C. By 1.2.5, $V_k^{f,0}(\omega)(C)$ is identified with the set $\mathrm{Hom}_c(\omega, r(C))$ of continuous k-linear maps from ω into the radical of C. This set is naturally isomorphic to the set $\mathrm{Hom}_c(\omega \hat{\otimes}_k C, r(C))$ of continuous C-linear maps from $\omega \hat{\otimes}_k C$ into $r(C)$; finally, since $\omega \hat{\otimes}_k C$ is a projective pseudocompact C-module, the canonical map

$$(\omega \hat{\otimes}_k C)^\dagger \otimes_C r(C) \longrightarrow \mathrm{Hom}_c(\omega \hat{\otimes}_k C, r(C))$$

is bijective (cf. 0.3.6.A). Since, by 1.2.3.E, $L(C)$ is identified with $V_{kf}(\Gamma^*(L))(C) = (\omega \hat{\otimes}_k C)^\dagger$, one obtains that $V_k^{f,0}(\omega)(C)$ is canonically isomorphic to $L(C) \otimes_C r(C)$, which is canonically isomorphic to $\mathcal{G}(L)(C)$ by Proposition 3.2. This proves implication (i) ⇒ (ii).

(ii) ⇒ (iii): Let ω be a projective object of $PC(k)$ and h an isomorphism from $k[[\omega]]$ onto the affine algebra A of G. Composing h with the augmentation $\epsilon_A : A \to k$, one obtains a homomorphism $\epsilon_A \circ h : k[[\omega]] \to k$, which is determined by its restriction λ to ω; the latter sends ω into the radical r of k. Therefore, the map $x \mapsto x - \lambda(x)$, from ω into the radical of $k[[\omega]]$, extends by the universal property of $k[[\omega]]$ (cf. 1.2.5) to an endomorphism ℓ_λ of $k[[\omega]]$. The equalities $\ell_\lambda \circ \ell_{-\lambda} = \ell_{-\lambda} \circ \ell_\lambda = id$ show that ℓ_λ is an automorphism of $k[[\omega]]$. Consequently, $h \circ \ell_\lambda$ is, like h, an isomorphism from $k[[\omega]]$ onto A, and moreover $\epsilon_A \circ h \circ \ell_\lambda$ sends ω to 0. Replacing h by $h \circ \ell_\lambda$ if necessary, one may therefore suppose that $\epsilon_A \circ h$ vanishes on the closed ideal I of $k[[\omega]]$ generated by ω. In this case, h induces an isomorphism from I/I^2 onto I_A/I_A^2; since $I/I^2 \simeq \omega$, it follows that $\omega_{G/k} = I_A/I_A^2$ is isomorphic to ω, hence projective. It is clear on the other hand that $V_k^{f,0}(\omega)$ is infinitesimal, as is G.

(iii) ⇒ (i): Suppose that G is infinitesimal and that $\omega_{G/k}$ is projective. Let L be the O_k-Lie algebra of G; the underlying O_k-module is $V_{kf}(\omega_{G/k})$, by 2.6.1. Consequently, L is flat over O_k, and $\Gamma * (L) \simeq \omega_{G/k}$, by Proposition 1.2.3.E. Hence, by the proof of (i) ⇒ (ii), the affine algebra of the k-formal group $\mathcal{G}(L)$ is identified with $k[[\omega_{G/k}]]$. On the other hand, by 2.6.3, the identity morphism of L is canonically associated with a morphism of formal groups $\mathcal{G}(L) \to G$, hence with a continuous morphism of k-algebras

$$\varphi : A \longrightarrow k[[\omega_{G/k}]].$$

Let I be the closed ideal of $k[[\omega_{G/k}]]$ generated by $\omega_{G/k}$; let us filter $k[[\omega_{G/k}]]$ (resp. A) by the closures of the ideals I^n (resp. I_A^n). We have to show that ϕ, which by definition induces the identity on $\omega_{G/k} = I_A/I_A^2 = I/I^2$, is an isomorphism.

Since $\omega_{G/k}$ is a projective object of $PC(k)$, there exists a section τ of the canonical projection $I_A \to \omega_{G/k}$. By the universal property of $k[[\omega_{G/k}]]$ (cf. 1.2.5), τ defines a continuous morphism of algebras

$$\psi : k[[\omega_{G/k}]] \longrightarrow A$$

and $\phi \circ \psi$ induces the identity map on $\omega_{G/k}$, hence also on the associated graded of $k[[\omega_{G/k}]]$. It follows that $\phi \circ \psi$ is an isomorphism, by [CA], § V.7, Lemma 1.[813]

[813] cf. 5.1.5 further below; see also [BAC], III, § 2.8, Th. 1 and corollaries. On the other hand, we have

Moreover, ψ induces an isomorphism from I/I^2 onto I_A/I_A^2, hence a surjection of the associated gradeds of $k[[\omega_{G/k}]]$ and A. On the other hand, since G is radicial, I_A is contained in the radical of A, so that the intersection of the I_A^n is zero. Therefore, by loc. cit., ψ is surjective. Then, since $\phi \circ \psi$ is an isomorphism and ψ a surjection, ψ and ϕ are isomorphisms. This proves (iii) ⇒ (i).

Let us note finally that it is clear that (i) or (ii) entail (iv), so that it remains to prove the implication (iv) ⇒ (ii). For this, one may suppose k local, with residue field k_0. Set then $G_0 = G \otimes_k k_0$, $\omega = \omega_{G/k}$, $\omega_0 = \omega \otimes_k k_0$, etc.[814] Since k_0 is a field, the pseudocompact k_0-module ω_0 has a pseudobasis $(y_\lambda)_{\lambda\in\Lambda}$; denoting ω' the topologically free k-module product of copies k_λ of k, for $\lambda \in \Lambda$, and lifting each y_λ to an element x_λ of ω, one obtains a continuous k-linear map $f : \omega' \to \omega$ such that $f_0 = f \otimes_k k_0$ is invertible.[815] Since ω' is a projective pseudocompact k-module, f lifts to a continuous k-linear map $g : \omega' \to I_A$ such that $\pi \circ g = f$, where π is the projection $I_A \to \omega$. By the universal property of $k[[\omega']]$ (cf. 1.2.5), g induces a morphism of topological algebras $\phi : k[[\omega']] \to A$.

Now $\omega' \otimes_k k_0$ is identified with $\omega \otimes_k k_0 = \omega_{G_0/k_0}$ and hence $k[[\omega']] \otimes_k k_0$ is identified with $k_0[[\omega_{G_0/k_0}]]$. Since hypotheses (iii) are satisfied for k_0 and G_0, the proof of (iii) ⇒ (i) shows that $\varphi_0 = \varphi \otimes_k k_0$ is invertible. Since, on the other hand, $k[[\omega']]$ and A are projective pseudocompact k-modules, ϕ is invertible by 0.3.4. (In particular, denoting I the augmentation ideal of $k[[\omega']]$, ϕ induces an isomorphism from $\omega' = I/I^2$ onto $I_A/I_A^2 = \omega$.)

3.3.1. Corollary 3.3.1. *Let S be a locally noetherian scheme over $\mathbb{Q}$ and G an S-group scheme that is flat and locally of finite type.*[816] *If $\omega_{G/S}$ is a locally free O_S-module, G is smooth over S at every point of the unit section.*

Indeed, let x be a point of the unit section, s its image in S, O_x and O_s the local algebras of x and s.[817] Since, by hypothesis, O_s and O_x are noetherian local rings, the $\mathfrak{m}$-adic topology on each of these rings coincides with the "pseudocompact" topology defined by the ideals of finite codimension. Denote then by $\hat{O}_x$ and $\hat{O}_s$ the completions for this topology. By (EGA IV₄, 17.5.3), G is smooth over S at the point x if and only if $\hat{O}_x$ is formally smooth over $\hat{O}_s$, these two algebras being endowed with the $\mathfrak{m}$-adic topology; it therefore suffices to show this latter property. Now $\hat{O}_x$ and $\hat{O}_s$ are the local rings of x and s in the formal varieties $\hat{G}/\hat{S}$ and $\hat{S}$ defined in 1.2.6. Set $k = \hat{O}_s$ and $H = Spf(\hat{O}_x)$; then H is a k-formal variety that is infinitesimal and, since the formation of $\hat{G}/\hat{S}$ commutes with products (loc. cit.), H is an infinitesimal k-formal group. Denote by I the augmentation ideal of O_x. By hypothesis, $\omega_{G/S,x} = I/I^2$ is a locally free O_s-module of finite rank n. Since O_s is noetherian, it follows that $\omega_{H/k}$, which is the completion of $\omega_{G/S,x}$, is identified with $\omega_{G/S} \otimes_{O_S} \hat{O}_s$, hence is a topologically free k-module of rank n. Hence, by the

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detailed the original in what follows.

[814] We have detailed the original in what follows.

[815] We do not know a priori that ω is a projective pseudocompact k-module, but this will follow from what follows: compare with the proof of (iv) ⇒ (iii) in VII_A, 7.4.

[816] N.D.E.: We have added the flatness hypothesis, omitted in the original.

[817] N.D.E.: We have detailed the original in what follows.

implication (iv) ⇒ (ii) of 3.3, $\hat{O}_x$ is isomorphic to $k[[\omega_{H/k}]]$, i.e., to a formal power series algebra $k[[t_1,\cdots,t_n]]$. Finally, this last is formally smooth over k, by (EGA 0_IV, 19.3.3). This proves the corollary.

We thus recover a result obtained otherwise for group schemes locally of finite presentation over an arbitrary scheme S, cf. VI_B, 1.6.

3.3.2. Corollary 3.3.2. *Let k be a field of characteristic 0. The functor $L \mapsto **G**(L)$ is an equivalence from the category of k-Lie algebras onto that of infinitesimal k-formal groups.*[818]

Indeed, when G runs through the infinitesimal k-formal groups, the functor $G \mapsto **G**(LieG)$ is isomorphic, by Theorem 3.3, to the identity functor of the category of infinitesimal k-groups. Likewise, by 3.2 (ii), the functor $L \mapsto Lie(**G**(L)) = Prim(U(L))$ is isomorphic to the identity functor of the category of k-Lie algebras.

3.3.3. Suppose still that k is a field of characteristic 0. Let $\bar{k}$ be an algebraic closure of k and Γ the topological Galois group of $\bar{k}$ over k.

For every k-formal group G, we denote by $**\mathrm{Aut}**_k\,(G)$ the k-group functor that associates to every commutative k-algebra C of finite dimension the group of automorphisms of the C-formal group $G\hat{\otimes}_k C$.[819] Since G is topologically flat over k (since k is a field), i.e., its affine algebra $A = **A**(G)$ is topologically flat over k, or, equivalently, its covariant bialgebra $H = **H**(G)$ is flat over k, this k-functor is a k-formal group. Indeed, since $**\mathrm{Aut}**_k\,(G)$ is identified with the fibered product of the following diagram (cf. Exp. I, 1.7.3):

```
                                   **End**_k(G) × **End**_k(G)
                                                ↓
Spf(k)  →  **End**_k(G) × **End**_k(G)
```

where the vertical (resp. horizontal) arrow is given by $(\phi,\psi) \mapsto (\phi\circ\psi, \psi\circ\phi)$ (resp. is the unit section), it suffices to check that the k-functor $**\mathrm{End}**_k\,(G)$ is represented by an element of Vaf/k, that is (cf. 1.1 and 0.4.2), that it is left exact, i.e., commutes with fibered products of k-algebras. Now to give an element

[818]N.D.E.: Since, by 2.7, 2.2.1 and 1.3.6, an infinitesimal k-formal group G is "the same thing" as a connected cocommutative k-bialgebra H (cf. 2.9), this statement is equivalent to the theorem below, obtained independently by Kostant (cf. 2.9.2). This theorem had been obtained earlier by J. W. Milnor and J. C. Moore ([MM65]), under the additional hypothesis that H is generated as an algebra by its primitive elements (although published in 1965, this text had circulated before 1960, cf. the review [Ja65]), so that it is often called the "Cartier–Kostant–Milnor–Moore theorem".

Theorem (Cartier–Kostant–Milnor–Moore). *Let k be a field of characteristic 0. The functors $L \mapsto U(L)$ and $H \mapsto Prim(H)$ define an equivalence between the category of k-Lie algebras and that of connected cocommutative k-bialgebras.*

On the other hand, if k is an artinian ring containing $\mathbb{Q}$, then 3.2 (ii) and 3.3 (combined with 2.7, 2.2.1 and 1.3.6) similarly show that the functor $L \mapsto **G**(L)$ (resp. $L \mapsto U(L)$) is an equivalence of the category of flat k-Lie algebras onto that of infinitesimal k-formal groups topologically flat (resp. onto that of connected cocommutative k-bialgebras that are flat).

[819]N.D.E.: The original stated: "This k-functor is manifestly left exact (H is topologically flat over k!)". We have detailed this in what follows.

of $**\,\mathrm{End}**_k\,(G)(C)$ is equivalent to giving, say, a morphism of k-algebras $\phi : H \to H \otimes_k C$ that also respects the coalgebra structure, i.e., such that the well-known diagrams are commutative; since H is flat over k, the functor $H \otimes_k -$ commutes with fibered products of k-algebras, and one deduces that the k-functor $**\,\mathrm{End}**_k\,(G)$ is left exact, so we can treat it as a k-formal group.

If L is a k-Lie algebra and G the formal group $**G**(L)$, Theorem 3.3 shows that $**\mathrm{Aut}**_k(G)$ is isomorphic to the k-group functor $**\mathrm{Aut}**_k(L)$ which associates to a finite-dimensional k-algebra C the group of automorphisms of the C-Lie algebra $C \otimes_k L$.

If G is an arbitrary k-formal group, we saw in 2.5.2 that G decomposes canonically into a semi-direct product of an étale formal group G_e and an infinitesimal formal group G^0. This semi-direct product is determined by the data of $L = Lie(G) = Lie(G^0)$, of the Γ-group $M = G_e(\bar{k}) = G(\bar{k})$, and of a morphism of k-formal groups

$$\Phi : G_e \to **\mathrm{Aut}**_k(G^0) \simeq **\mathrm{Aut}**_k(L).$$

Such a morphism sends G_e into the "étale part" $**\,\mathrm{Aut}**_k\,(L)_e$ of $**\,\mathrm{Aut}**_k\,(L)$, cf. 2.5.1. It is therefore determined by the data of a morphism of Γ-groups:

$$\varphi = \Phi(\bar{k}) : \quad M \to (**\mathrm{Aut}**_k L)(\bar{k}) = **\mathrm{Aut}**_{\bar{k}}(L \otimes_k \bar{k}).$$

If we let Γ act on $L \otimes_k \bar{k}$ by the formula $\gamma(\ell \otimes t) = \ell \otimes \gamma(t)$, then ϕ makes M act on $L \otimes_k \bar{k}$ by automorphisms of $\bar{k}$-Lie algebra in such a way that one has $\phi(\gamma(m)) = \gamma \circ \phi(m) \circ \gamma^{-1}$, i.e.:

$$\gamma(m)(\ell \otimes t) = \gamma(m(\ell \otimes \gamma^{-1}(t)))$$

for every $m \in M$, $\gamma \in \Gamma$ and $\ell \otimes t \in L \otimes_k \bar{k}$. We express this last condition by saying that M acts on $L \otimes_k \bar{k}$ *compatibly with* Γ.

One can summarize the situation by means of a "highbrow" statement: let us call Γ-*Lie algebra over* k the data of a triple (L, M, ϕ) formed by a k-Lie algebra L, a Γ-group M, and an action ϕ of M on $L \otimes_k \bar{k}$ that is "compatible with the action of Γ on M and on $L \otimes_k \bar{k}$".

If (L, M, ϕ) and (L', M', ϕ') are two such Γ-Lie algebras, a morphism of the first into the second is a pair (f, θ) formed by a morphism $f : L \to L'$ of k-Lie algebras and a morphism $\theta : M \to M'$ of Γ-groups such that

$$(f \otimes 1)(m \cdot x) = \theta(m) \cdot (f \otimes 1)(x)$$

for every $x \in L \otimes_k \bar{k}$ and $m \in M$. One then obtains:

Theorem. *Let k be a field of characteristic 0. Then the functor that associates to G the triple $(Lie(G), G(\bar{k}), \Phi(\bar{k}))$ is an equivalence of the category of k-formal groups onto that of Γ-Lie algebras.*[820]

[820] N.D.E.: In particular, when $\bar{k} = k$, one thus recovers the "Cartier–Gabriel–Kostant theorem" mentioned in 2.9.2.

4. Phenomena specific to characteristic $p > 0$

Throughout Section 4, we denote by p a prime number, by k a pseudocompact ring of characteristic p, and by π the continuous endomorphism $x \mapsto x^p$ of k.

4.1.

Let X be a k-formal variety with affine algebra A; one denotes by $X^{(p/k)}$ or $X^{(p)}$ the k-formal variety $X\hat{\bigotimes}_\pi k$ obtained by the base change $\pi : k \to k$ (cf. 1.2.D); it has as affine algebra the completed tensor product $A\hat{\bigotimes}_\pi k$. Then, the continuous morphism $A\hat{\bigotimes}_\pi k \to A$ which sends $a\hat{\bigotimes}_\pi \lambda$ onto $a^p\lambda$ induces a morphism of k-formal varieties

$$Fr(X/k) : X \longrightarrow X^{(p/k)}.$$

In what follows, we shall say that $Fr(X/k)$ is the *Frobenius morphism of X relative to k* and we shall often write `Fr` instead of $Fr(X/k)$.

4.1.1. [821] Consider now a scheme S over the prime field $\mathbb{F}_p$ and an S-scheme $\eta : X \to S$; let $fr(S)$ be the "absolute" Frobenius morphism of S (it induces the identity on the underlying topological space and sends every section f of the structure sheaf onto f^p; cf. VII_A 4.1) and let $X^{(p)}$ be the S-scheme $X \times_{fr(S)} S$ (VII_A, loc. cit.), i.e., the structure morphism $X^{(p)} \to S$ is $fr(S) \circ \eta$.

Let Ŝ be the formal scheme whose underlying topological space is the set of closed points of S, endowed with the discrete topology, the local ring $\hat{O}_{S,s}$ at such a closed point s being the separated completion $\hat{O}_{S,s}$ of $O_{S,s}$ for the linear topology defined by the ideals of finite colength (cf. 1.2.6); its affine algebra $k = **A**(\hat{S})$ is therefore the product of the $\hat{O}_{S,s}$, as s runs through the closed points of S. Recall (loc. cit.) that one denotes by $\hat{X}/\hat{S}$ the k-formal variety formed by the points $x \in X$ such that $[\kappa(x) : \kappa(s)] < \infty$, where s is the image of x in S, the local ring $O_{\hat{X}/\hat{S},x}$ being the separated completion $\hat{O}_{X,x}$ of $O_{X,x}$ for the linear topology defined by the ideals I such that $O_{X,x}/I$ is of finite length as $O_{X,x}$-module (and therefore also as $O_{S,s}$-module). One can therefore form, by base change, the k-formal variety $(\hat{X}/\hat{S})^{(p)} = (\hat{X}/\hat{S})\hat{\bigotimes}_\pi k$.

One can also consider the k-formal variety $\hat{X}^{(p)}/\hat{S}$: the underlying set is formed by the $x \in X^{(p)} = X$ such that, denoting by s the image of x in S, the morphism

κ(s) —fr→ κ(s) —η^♯→ κ(x)

makes $\kappa(x)$ an extension of finite degree of $\kappa(s)$; in this case, the same holds for $\eta^{♯} : \kappa(s) \to \kappa(x)$, i.e., x is a point of $\hat{X}/\hat{S}$, and one then has

O_{X̂^{(p)}/Ŝ, x} = Ô_{X, x} ⊗̂_π Ô_{S, s} = Ô_{X^{(p)}, x}

[821] N.D.E.: We have corrected the original, which gave the inclusion $(\hat{X}/\hat{S})^{(p)} \subset \hat{X}^{(p)}/\hat{S}$ instead of the opposite inclusion. Let us point out moreover that this paragraph is not used in the sequel.

(the second equality since $X \mapsto \hat{X}/\hat{S}$ commutes with products, cf. 1.2.6). One therefore sees that: $\hat{X}^{(p)}/\hat{S}$ is identified with a formal subvariety of $(\hat{X}/\hat{S})^{(p)}$. Moreover, equality holds if and only if for every closed point s of S, $\kappa(s)$ is of finite degree over $\kappa(s)^p$, which is the case for example if S is a scheme locally of finite type over a field κ such that $[\kappa : \kappa^p] < +\infty$.

4.1.2. Let G be a k-formal group. Since the functor $X \mapsto X^{(p)} = X\hat{\bigotimes}_\pi k$ commutes with finite products, it transforms a k-formal group into a k-formal group. Moreover, since $Fr(X/k)$ is functorial in X, the morphism

$$Fr : G \longrightarrow G^{(p)}$$

is a homomorphism of k-formal groups. If one sets $G^{(p^{n+1})} = (G^{(p^n)})^{(p)}$, the same holds for the composite morphism

$$Fr^n = Fr^n(G/k) : \quad G \xrightarrow{Fr} G^{(p)} \xrightarrow{Fr} G^{(p^2)} \xrightarrow{Fr} \cdots \xrightarrow{Fr} G^{(p^n)}.$$

Denote by A the affine algebra of G and by I_A its augmentation ideal.

Definitions. *(a) The kernel of Fr^n will be denoted $Fr_n G$. It follows directly from the definitions that $Fr_n G$ is infinitesimal and has as affine algebra the quotient $A/I_A^{\{p^n\}}$, where $I_A^{\{p^n\}}$ denotes the closed ideal of A generated by the p^n-th powers of the elements of I_A.*

(b) One says that G is of height $\leqslant n$ if $I_A^{\{p^n\}} = 0$, that is to say if one has $Fr_n G = G$.

4.2.

It is clear that the Lie algebra $Lie(G)$ of a k-formal group G is a p-Lie subalgebra of the algebra $**H**(G)$ (cf. 2.3). Indeed, one reduces immediately to the case where k is artinian; in this case, $Lie(G)$ is the part of $**H**(G)$ formed by the elements x such that $\phi_G(x \otimes 1 + 1 \otimes x) = \Delta_G(x)$ with the notations of 2.3 and 2.6 (c). One then has

$$\begin{aligned} \varphi_G(x^p \otimes 1 + 1 \otimes x^p) &= \varphi_G((x \otimes 1 + 1 \otimes x))^p \\ &= (\varphi_G(x \otimes 1 + 1 \otimes x))^p = \Delta_G(x)^p = \Delta_G(x^p). \end{aligned}$$

4.2.1. Conversely, every p-Lie algebra L over O_k defines a k-group functor. Denote indeed by $U_p(L)$ the functor that associates to every object C of Alf/k the restricted enveloping algebra $U_p(L(C))$ of the p-Lie algebra $L(C)$ over C (VII_A 5.3). By VII_A 5.4, $U_p(L)$ is an O_k-bialgebra and therefore determines, by 2.2, a k-group functor $Spf^*(U_p(L))$ that we shall henceforth denote $**G**_p(L)$.

Thus, $**G**_p(L)(C)$ is the group of elements $z \in U_p(L(C))$ of augmentation 1 and such that $\Delta_{U_p(L(C))}(z) = z \otimes z$.

4.2.2. Suppose L is a p-Lie algebra that is flat over O_k. Then, taking account of VII_A 5.3.3, one shows as in point (i) of Proposition 2.6.2 that $U_p(L)$ is flat over O_k; hence $**G**_p(L)$ is a topologically flat k-formal group which has $U_p(L)$ as covariant O_k-bialgebra.

[822] By the proof of 2.6.2 (iii), for every k-algebra C of finite length, $Lie(**G**_p(L))(C)$ is the set of primitive elements of $U_p(L)(C) = U_p(L(C))$ (see also VII_A 3.2.3); moreover, by VII_A 5.5.3, the canonical morphism $L \to U_p(L)$ induces an isomorphism of p-Lie algebras

$$\tau_{p,L} : L \xrightarrow{\sim} Lie(**G**_p(L))$$

(compare with 3.1 in characteristic 0).

The formal group $**G**_p(L)$ can be characterized by a universal property. Indeed, every morphism h of $**G**_p(L)$ into a formal group G induces a morphism $Lie(h) : Lie(**G**_p(L)) \to Lie(G)$. Composing this with the isomorphism $\tau_{p,L}$, one obtains a morphism $h' : L \to Lie(G)$.

Proposition. *If k is a pseudocompact ring of characteristic $p > 0$, G a k-formal group, and L a p-Lie algebra that is flat over O_k, the map $h \mapsto h'$ defined above is a bijection*

$$\mathrm{Hom}_{Grf/k}(**G**_p(L), G) \xrightarrow{\sim} \mathrm{Hom}_{p\text{-}Lie}(L, Lie(G))$$

where the right-hand term denotes the set of morphisms of p-Lie algebras from L into $Lie(G)$.

[823] One reduces indeed immediately to the case where k is artinian. Set $L = L(k)$. By 2.3.1, $\mathrm{Hom}_{Grf/k}(**G**_p(L), G)$ is in bijection with the set of morphisms of unitary algebras $h : U_p(L) \to **H**(G)$ such that the following diagram is commutative:

$$\begin{tikzcd}
U_p(L) \arrow[r, "h"] \arrow[d, "\Delta_{U_p(L)}"'] & **H**(G) \arrow[r, "\delta_G"] & **H**(G \times G) \\
U_p(L) \otimes U_p(L) \arrow[r, "h \otimes h"] & **H**(G) \otimes **H**(G) \arrow[ru, "\phi_G"'] &
\end{tikzcd}$$

Now h is determined by its restriction to L, which is a morphism of p-Lie algebras from L into the p-Lie algebra underlying $**H**(G)$, and the commutativity of the diagram means that h sends L into the part of $**H**(G)$ formed by the x such that $\delta_G(x) = \phi_G(x \otimes 1 + 1 \otimes x)$, which is none other than $Lie(G)$, cf. 2.6 (c).

4.3.

We now propose to study in greater detail the k-formal group $**G**_p(L)$ when L is a p-Lie algebra that is flat over O_k.

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[822] N.D.E.: We have detailed what follows.

[823] N.D.E.: The proof is identical to that of 2.6.3.

For this, we first consider a ring C of characteristic p and a p-Lie algebra L over C whose underlying module is free with basis $(x_i)_{i\in I}$. Denote moreover by P the set of families $n = (n_i)_{i\in I}$ of natural integers such that $0 \leqslant n_i < p$ and that the n_i are zero except possibly a finite number of them. If we endow I with a total order and if we call $i_1, i_2, \cdots, i_r$ ($i_1 < i_2 < \cdots < i_r$) the indices i such that $n_i \neq 0$, we can therefore set $|n| = n_{i_1} + \cdots + n_{i_r}$ and

$$x^n = x_{i_1}^{n_{i_1}} \cdot x_{i_2}^{n_{i_2}} \cdots x_{i_r}^{n_{i_r}}, \qquad n! = (n_{i_1}!) \cdots (n_{i_r}!).$$

It is known that the monomials $x^n/n!$ form a basis of $U_p(L)$ (VII_A 5.3.3) and it is clear that one has

$$(*) \qquad \Delta(x^n/n!) = \sum_{m \leqslant n} (x^m/m!) \otimes (x^{n-m}/(n-m)!)$$

the sum being extended to all $m \in P$ such that $m \leqslant n$ (i.e., $m_i \leqslant n_i$ for all $i \in I$).

Formula $(*)$ allows an easy determination of the natural filtration of $U_p(L)$ (cf. 1.3.6). Set $U = U_p(L)$, let U^+ be the two-sided ideal generated by L, and let $U_0 = C \cdot 1_U$. As in 1.3.6, one defines a filtration of U (by C-subcoalgebras) by setting, for $n \geqslant 1$:

$$U_n = \{u \in U \mid \Delta_U(u) - u \otimes 1 \in U_{n-1} \otimes U^+\}.$$

Formula $(*)$ then entails that U_n is the free C-module having as basis the monomials x^m such that $|m| \leqslant n$.

4.3.1. With the notations of 4.3, let us determine the elements ξ of $U = U_p(L)$ such that $\epsilon_U(\xi) = 1$ and $\Delta_U(\xi) = \xi \otimes \xi$. Every element ξ of U is written $\xi = \sum_{n\in P} \xi_n (x^n/n!)$, with $\xi_n \in C$. The condition $\epsilon_U(\xi) = 1$ entails the equality $\xi_0 = 1$, where 0 denotes the element of P all of whose components are zero. The condition $\Delta_{U_p(L)}(\xi) = \xi \otimes \xi$ entails:

$$\sum_{m \leqslant n} \xi_n (x^m/m!) \otimes (x^{n-m}/(n-m)!) = \sum_{q,r} \xi_q \xi_r (x^q/q!) \otimes (x^r/r!)$$

that is to say,

$$\xi_{q+r} = \xi_q \xi_r \quad \text{if } q_i + r_i < p, \quad \text{and} \quad \xi_q \xi_r = 0 \quad \text{otherwise}.$$

If one denotes by ξ_i the value of ξ_n when $n_i = 1$ and $n_j = 0$ for $j \neq i$, these conditions mean that one has

$$\xi_n = \prod_i \xi_i^{n_i} \quad \text{if } n \in P, \quad \text{and} \quad \xi_i^p = 0 \quad \forall\, i \in I.$$

One deduces from this:

Proposition. *Let k be a local pseudocompact ring[824] of characteristic $p > 0$, L a p-Lie algebra that is flat over O_k, C an object of Alf/k, and $\sqrt[p]{0_C}$ the ideal of C formed by the elements x such that $x^p = 0$. There exists a bijection "functorial in C":*

$$L(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \mathbf{G}_p(L)(C).$$

[824]N.D.E.: We have added the hypothesis that k be local, so that every topologically flat pseudocompact k-module be topologically free; cf. the proof.

By Remark 1.2.3.F, one can indeed choose a basis $({}^C x_i)_{i\in I}$ of $L(C)$ over C in such a way that, for every morphism $\phi : C \to D$ of Alf/k, $L(\phi)$ sends ${}^C x_i$ onto ${}^D x_i$. By what precedes, the map

$$\sum_i ({}^C x_i) \otimes \xi_i \mapsto \sum_{n \in P} \Big(\prod_i \xi_i^{n_i}\Big) \cdot \big(({}^C x)^n / n!\big)$$

is a functorial bijection from $L(C) \otimes_C {}^p\sqrt{0_C}$ onto $\ast\ast G\ast\ast_p(L)(C)$.

4.3.2. "There is no Campbell–Hausdorff formula in characteristic $p > 0$". Let us explain ourselves: the functorial isomorphism of 4.3.1 depends on the choice of the bases $({}^C x_i)_{i\in I}$. Unlike what happens in 3.2 (case of characteristic 0), there is no, in general, bijection from $L(C) \otimes_C {}^p\sqrt{0_C}$ onto $\ast\ast G\ast\ast_p(L)(C)$ that is "functorial both in C and in L". The argument that follows shows for example that such an isomorphism does not exist when k is a field containing the (p^2-1)-th roots of unity.

Choose L indeed in such a way that, for every $C \in Alf/k$, $L(C)$ is the abelian p-Lie algebra over C generated by an element x subject to the relation $x^{(p^2)} = 0$. One has therefore

$$L(C) = Cx \oplus Cx^{(p)}, \qquad U_p(L(C)) \cong k[x]/(x^{p^2}).$$

We shall show that the only functorial morphism

$$\chi_C : L(C) \otimes_C {}^p\sqrt{0_C} \to U_p(L(C))$$

that is compatible with the endomorphisms of L and that sends $L(C) \otimes_C {}^p\sqrt{0_C}$ into $\ast\ast G\ast\ast_p(L)(C)$ is the constant morphism of value 1.

Indeed, let ψ_C be the bijection from $L(C) \otimes_C {}^p\sqrt{0_C}$ onto $\ast\ast G\ast\ast_p(L)(C) = Prim(U_p(L(C)))$ given by 4.3.1, i.e.,

$$x \otimes a + x^{(p)} \otimes b \mapsto \sum_{0 \leq i,\, j < p} a^i b^j x^{i+pj}.$$

Composing χ_C with ψ_C^{-1}, one obtains a functorial morphism (in C):

$$\begin{aligned} \theta_C :\ & L(C) \otimes_C {}^p\sqrt{0_C} \to L(C) \otimes_C {}^p\sqrt{0_C} \\ & x \otimes a + x^{(p)} \otimes b \mapsto x \otimes P(a,b) + x^{(p)} \otimes Q(a,b). \end{aligned}$$

Functoriality in C implies that $P(a,b)$ and $Q(a,b)$ are the values at (a,b) of two polynomials $P, Q \in k[X,Y]$ that one can assume of degree $< p$ in X as well as in Y.[825] Let α be an element of k and ℓ_α the p-Lie algebra endomorphism of L that sends x onto αx (and therefore $x^{(p)}$ onto $\alpha^p x^{(p)}$). Then $U_p(\ell_\alpha)$ is the algebra endomorphism that sends x onto αx (and therefore each x^i onto $\alpha^i x^i$, for $i < p^2$), and one sees easily that the square below

$$\begin{array}{ccc} L(C) \otimes_C {}^p\sqrt{0_C} & \xrightarrow{\psi_C} & U_p(L(C)) \\ \Big| & & \Big| \\ \ell_\alpha(C) \otimes_C \mathrm{id} & & U_p(\ell_\alpha)(C) \end{array}$$

[825] N.D.E.: We have detailed the original in what follows.

$$\begin{array}{ccc} \downarrow & & \downarrow \\ L(C) \otimes_C \sqrt[p]{0_C} & \xrightarrow{\psi_C} & U_p(L(C)) \end{array}$$

is commutative. The hypothesis $\chi_C \circ (\ell_\alpha(C) \otimes id) = U_p(\ell_\alpha)(C) \circ \chi_C$ then entails the equalities:

$$P(\alpha a, \alpha^p b) = \alpha P(a, b) \quad \text{and} \quad Q(\alpha a, \alpha^p b) = \alpha^p Q(a, b).$$

Writing $P = \sum_{i,j<p} \lambda_{ij} X^i Y^j$ and $Q = \sum_{i,j<p} \mu_{ij} X^i Y^j$, and taking for C the algebra $k[X,Y]/(X^p, Y^p)$, one deduces that if $\lambda_{ij} \neq 0$ (resp. $\mu_{ij} \neq 0$) then $\alpha^{i-1+pj} = 1$

(resp. $\alpha^{i+p(j-1)} = 1$). Hence, taking for α a primitive root of unity of order $p^2 - 1$, one deduces that $P = \lambda X$ and $Q = \mu Y$, with $\lambda, \mu \in k$. Consequently, one has:

$$\chi_C(x \otimes a + x^{(p)} \otimes b) = \sum_{0 \leqslant i,\, j < p} (\lambda a)^i (\mu b)^j x^{i + pj}.$$

[826] Finally, consider the endomorphism f of L that sends x onto $x + x^{(p)}$; taking $C = k[a]/(a^3)$ and comparing the coefficients of x^p and x^{p+1} in $(\chi_C \circ f(C))(x \otimes a)$ and in $(U_p(f)(C) \circ \chi_C)(x \otimes a)$, one obtains that $\lambda = \mu$ and $\lambda^2 a^2 = 0$, whence $\lambda = \mu = 0$.

4.4.

Theorem 4.4. *Let k be a pseudocompact ring of characteristic $p > 0$ and G a k-formal group. The following assertions are equivalent:*

(i) There exists a p-Lie algebra L that is flat over O_k such that G is isomorphic to $\ast\ast G \ast\ast_p(L)$ (and in this case $L = Lie(G)$ by 4.2.2).

(ii) There exists a projective pseudocompact k-module ω such that the affine algebra of G is isomorphic to the quotient of $k[[\omega]]$ by the closed ideal generated by the x^p, for $x \in \omega$ (and in this case $\omega \simeq \omega_{G/k}$).

(iii) G is of height $\leqslant 1$ and $\omega_{G/k}$ is a projective pseudocompact k-module.

(iv) G is of height $\leqslant 1$ and is topologically flat over k.

(i) $\Rightarrow$ (ii): Let $\omega = \Gamma_*(L)$ (cf. 1.2.3.D) and A the quotient of $k[[\omega]]$ by the closed ideal generated by the x^p, for $x \in \omega$. Every continuous morphism $h : A \to C$, where C is an object of Alf/k, is determined by its restriction h' to ω; this restriction h' sends ω into $\sqrt[p]{0_C}$. One thus obtains a canonical bijection from $\mathrm{Hom}_{Alp/k}(A, C)$ onto the set $\mathrm{Hom}_c(\omega, \sqrt[p]{0_C})$ of continuous k-linear maps from ω into $\sqrt[p]{0_C}$. This last set is canonically isomorphic to $L(C) \otimes_C \sqrt[p]{0_C}$ (see the proof of 3.3). The implication (i) $\Rightarrow$ (ii) thus follows from the functorial bijection $L(C) \otimes_C \sqrt[p]{0_C} \xrightarrow{\sim} \ast\ast G \ast\ast_p(L)(C)$ established in 4.3.1.

For the other implications, consult the proofs of Theorems 3.3 and VII_A 7.4.1, which are analogous.

Remark 4.4.A.[827] *Let G be an infinitesimal k-formal group, with affine algebra A, such that $\omega_{G/k} = I_A/I_A^2$ is a projective pseudocompact k-module. Then there exists*

[826] N.D.E.: We have detailed the original in what follows.
[827] N.D.E.: We have added this remark, used in 4.4.2.

a section $\tau : \omega_{G/k} \to I_A$ of the projection $I_A \to \omega_{G/k}$, and τ induces a continuous morphism of algebras $\psi : k[[\omega_{G/k}]] \to A$ that is surjective, cf. the proof of the implication (iii) ⇒ (i) in 3.3.

4.4.1. Corollary 4.4.1. *If k is a field of characteristic $p > 0$, the functor $L \mapsto **G**_p(L)$ is an equivalence of the category of p-Lie algebras over k onto that of k-formal groups of height $\leqslant 1$.*

Indeed, when G runs through the formal groups of height $\leqslant 1$, the functor $G \mapsto **G**_p(LieG)$ is isomorphic to the identity functor by Theorem 4.4; likewise, we saw in 4.2.2 that the functor $L \mapsto Lie**G**_p(L)$ is isomorphic to the identity functor (see also VII_A, 5.5.3).[828]

4.4.2. Take still k to be a field of characteristic p. Let G be an infinitesimal k-formal group, with affine algebra A. Since G is infinitesimal, every open ideal of A contains $I_A^{\{p^n\}}$ for n large enough, hence G is the inverse limit of the affine algebras $A/I_A^{\{p^n\}}$ of the groups $Fr_n G$ (cf. 4.1.2). Every infinitesimal k-formal group is therefore a direct limit of k-formal groups of finite height.

Suppose G is of height $\leqslant n$ and write $H = G/FrG$.[829] By 2.4 and 2.4.1, $Fr : G \to G^{(p)}$ factorizes through an epimorphism $\pi : G \to H$ followed by a monomorphism $i : H \to G^{(p)}$. One has then the following commutative diagram:

$$\begin{tikzcd}
G \arrow[r, "\pi"] \arrow[d, "Fr"] & H \arrow[r, "Fr^{n-1}"] & H^{(p^{n-1})} \arrow[d, "i^{(p^{n-1})}"] \\
G^{(p)} \arrow[rr, "Fr^{n-1}"] & & G^{(p^n)}
\end{tikzcd}$$

and since the functor $X \mapsto X^{(p)}$ commutes with fibered products, $i^{(p^{n-1})}$ is still a monomorphism. Since $Fr^n(G/k)$ is zero and π is an epimorphism, then $Fr^{n-1}(G^{(p)}/k) \circ i = i^{(p^{n-1})} \circ Fr^{n-1}(H/k)$ is zero, and hence, $i^{(p^{n-1})}$ being a monomorphism, $Fr^{n-1}(H/k)$ is zero, so H is of height $\leqslant n-1$. One therefore sees that: *every k-formal group of finite height possesses a composition series whose quotients are of height $\leqslant 1$, hence can be described by p-Lie algebras over k.*

Finally, the affine algebra A of G is a quotient of $k[[\omega_{G/k}]]$, cf. 4.4.A; hence if $\omega_{G/k}$ is of finite dimension over k, then all the algebras $A/I_A^{\{p^n\}}$ are k-vector spaces of finite dimension. One therefore sees that: *every infinitesimal formal group G over a field of characteristic $p > 0$, such that $\omega_{G/k}$ is of finite dimension over k, is a direct limit of finite formal groups (i.e., of finite length, cf. 1.2.6).*

[828]N.D.E.: If k is an artinian ring of characteristic $p > 0$, the same proof gives an equivalence between the category of p-Lie algebras that are flat over k and that of k-formal groups of height $\leqslant 1$, topologically flat over k.

[829]N.D.E.: We have detailed the original in what follows.

5. Homogeneous spaces of infinitesimal formal groups over a field

5.0.

[830] Assume, for simplicity, that k is a field. Let G be a k-formal group with affine algebra $A = * * A * *(G)$. Let A^{+} be the augmentation ideal of A; for every $a \in A$ we shall denote by $\bar{a} = a - \epsilon_A(a)1_A$ its projection onto A^{+}. Let F be a formal subgroup

of G, defined by the closed ideal J of A, and let π (resp. $\bar{\pi}$) be the projection $A \to A/J$ (resp. the composition of the projections $A \to A^{+} \to A^{+}/J$). Note that, for every $a \in A$, the projection of $\Delta_A(a)$ onto $A \hat{\bigotimes} A^{+}$ is $\Delta_A(a) - a \hat{\bigotimes} 1$.

By Theorem 2.4, one can form the quotient k-formal variety $X = G/F$, its affine algebra B is

```
                               τ₁
B = Ker( A ⇒ A ⊗̂k (A/J) )
            (id_A ⊗ˆπ) Δ_A

  = { a ∈ A | (id_A ⊗ˆπ) Δ_A(a) = a ⊗ˆπ(1) }

  = Ker( (id_A ⊗ˆπ̄) Δ_A ).
```

It is also the subalgebra of A formed by the elements ϕ such that, for every $C \in ObAlf/k$ and $g \in G(C)$, $h \in F(C)$, one has $\phi(gh) = \phi(g)$. For every $g, g' \in G(C)$ and $h \in F(C)$, one has $\phi(gg'h) = \phi(gg')$, hence $\Delta(\phi)$ belongs to the kernel of $id_A \hat{\bigotimes}_k (id_A \hat{\bigotimes} \pi)\Delta_A$, which equals $A \hat{\bigotimes}_k B$ since A is topologically flat over k. One has therefore $\Delta_A(B) \subset A \hat{\bigotimes}_k B$, i.e., the closed subalgebra B is also a left coideal.

On the other hand, B determines F since, by Corollary 2.4.1, one has $J = AB^{+}$, i.e., J is the closed ideal generated by $B^{+} = B \cap A^{+}$. One thus obtains an injective map Q from the set $* * F * *$ of formal subgroups of G into the set $* * B * *$ of closed subalgebras B of A that are left coideals. The question then arises of determining the image of this map, and Proposition 5.1 below shows that Q is bijective when G is infinitesimal. In fact, this is true for every k-formal group G.

Indeed, recall (cf. 2.2.1) that the functor $G \mapsto **H**(G)$ is an equivalence between the category of k-formal groups and that of cocommutative k-Hopf algebras; if F is a formal subgroup of G, defined by the closed ideal J of A, then the Hopf subalgebra $**H**(F)$ of $H = **H**(G)$ is the orthogonal of J for the duality between $A = H^{*}$ and `H = A^†` (cf. 0.2.2). On the other hand, if B is a closed subalgebra of A that is also a left coideal, then its orthogonal $I = B^{\perp}$ is a coideal of H (i.e., $\Delta_H(I) \subset I \bigotimes H + H \bigotimes I$ and $\epsilon_H(I) = 0$) and a left ideal. Denote by $* * H * *$ (resp. $* * I * *$) the set of Hopf subalgebras (resp. left ideals that are coideals) of H. For every $I \in * * I * *$, one will denote by π_I (resp. $\bar{\pi}_I$) the projection $H \to H/I$ (resp. the composition of the projections $H \to H^{+} \to H^{+}/I$), where H^{+} is the augmentation ideal of H.

[830] N.D.E.: We have added subsection 5.0, in order to express in the language of cocommutative Hopf algebras the proposition 5.1 that follows, and to cite the results obtained since in this direction.

Let K be a Hopf subalgebra of H and $K^+ = K \cap H^+$. If F is the formal subgroup corresponding to K, then $J = K^\perp$ and A^+/J is identified with the dual of K^+, and since $B = Q(F)$ is the kernel of the map

$$A \xrightarrow{\Delta_A} A \widehat{\otimes}_k A \xrightarrow{id \widehat{\otimes} \bar{\pi}} A \widehat{\otimes}_k (A^+/J)$$

one obtains that Q corresponds by duality to the map Φ which to K associates the image of

$$H \xleftarrow{m_H} H \otimes_k H \xleftarrow{id \widehat{\otimes} can.} H \otimes_k K^+$$

i.e., the left ideal HK^+, which is also a coideal. One sees similarly that the map which to $B \in **B**$ associates AB^+ corresponds by duality to the map Ψ which to every $I \in **I**$

associates the kernel of $(id \bigotimes_k \bar{\pi}_I) \circ \Delta_H$, i.e., one has

$$\Psi(I) = \{ x \in H \mid (id \otimes_k \pi_I)\, \Delta_H(x) = x \otimes \pi_I(1) \}.$$

One then has the following theorem:

Theorem 5.0.1. *Let k be a field and H a cocommutative Hopf algebra. Then the maps Φ and Ψ above are inverse bijections between the set of Hopf subalgebras of H and that of left ideals that are coideals.*

This theorem was first proved by K. Newman, cf. [Ne75], Th. 4.1 (where the word "cocommutative" was forgotten). Its proof uses "the Cartier–Gabriel–Kostant theorem" (cf. 2.9) to reduce to the case where H is connected, then the existence in that case of a "Sweedler basis" (cf. [Sw67], Th. 3), a result dual to the Dieudonné–Cartier theorem 5.2.2 below. Another proof, shorter, was given by H. J. Schneider [Sch90], Th. 4.15. A generalization was then obtained by A. Masuoka when one assumes only that the coradical H_0 of H (i.e., the sum of the simple subcoalgebras) is commutative [Ma91], Th. 1.3 (3).

Let us point out finally that for a commutative k-Hopf algebra, corresponding therefore to an affine k-group scheme G, one cannot expect an analogue of 5.0.1 without additional hypotheses, since for a k-subgroup F of G, the quotient G/F is not necessarily affine. But M. Takeuchi established in [Tak72], Th. 4.3 (resp. [Tak79], Th. 3), an analogous bijection between the set of k-subgroups F of G which are invariant (resp. such that G/F is affine), and that of subalgebras B of $O(G)$ such that $\Delta(B) \subset A \bigotimes B$ and which are stable under the antipode (resp. and such that $B \to A$ is faithfully flat).

5.1.

Let k be a pseudocompact ring.[831] Let G be a topologically flat infinitesimal k-formal group, A its affine algebra, B a closed subalgebra of A, $X = Spf(B)$, and $r : G \to X$ the epimorphism induced by the inclusion of B into A. One proposes

[831] N.D.E.: In the original, it is supposed in 5.1 that k is a field. In fact, this hypothesis can be replaced by flatness hypotheses; we have modified accordingly nos. 5.1 to 5.1.5.

to see under what condition r makes X the right quotient of G by a subgroup H (cf. 2.4).[832]

Proposition. *Let G be a topologically flat infinitesimal k-formal group, A its affine algebra, I_A the augmentation ideal of A, B a closed subalgebra of A, and $J_B = AI_B$, where $I_B = B \cap I_A$. Assume that A is topologically flat over k, as well as J_B^n/J_B^{n+1} for every $n \geqslant 0$. Then the following two assertions are equivalent:*

(i) For every $x \in B$, $\Delta_A(x) - x\hat{\otimes}1$ belongs to $A\hat{\otimes}_k J_B$.

(ii) The sequence below (where $\tau_1(a) = a\hat{\otimes}1$ and π is the projection $A \to A/J_B$) is exact:

$$(*)\qquad B \longrightarrow A \underset{(\pi\hat{\otimes}\mathrm{id}_A)\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A\hat{\otimes}_k (A/J_B)$$

that is, B is the set of all $x \in A$ such that $\Delta_A(x) - x\hat{\otimes}1$ belongs to $A\hat{\otimes}_k J_B$.

In this case, $H = Spf(A/J_B)$ is a formal subgroup of G, and the sequence below (where λ is the restriction to $G \times H$ of the multiplication of G) is exact:

$$(**)\qquad G \times H \underset{\lambda}{\overset{pr_1}{\rightrightarrows}} G \longrightarrow Spf(B)$$

that is, $Spf(B)$ is isomorphic to G/H.

Set $\bar{A} = A/J_B$ and $H = Spf(\bar{A})$; then H is a formal subvariety of G. Since $J_B \subset I_A$, the augmentation ϵ_A induces a continuous morphism of k-algebras $\bar{\epsilon} : \bar{A} \to k$.

If (i) is satisfied, then $\Delta_A(I_B) \subset I_B\hat{\otimes}1 + A\hat{\otimes}J_B$, and hence Δ_A induces by passage to the quotient a diagonal morphism $\bar{\Delta}$. Then $\bar{\Delta}$ and $\bar{\epsilon}$ endow H with a structure of formal submonoid of G. Since G is infinitesimal, so is H; hence, by Proposition 2.7, H is a formal subgroup of G. It then follows from the definition of G/H (cf. 2.4) that (ii) entails the last assertion of the proposition.

On the other hand, it is clear that (ii) implies (i). The proof of the converse occupies paragraphs 5.1.1 to 5.1.5.

5.1.1. Let us first consider the following category $**C**$: an object of $**C**$ is a pair (A, J) formed by a profinite k-algebra A and a closed ideal J of A; a morphism $\psi : (A, J) \to (A', J')$ of $**C**$ is a continuous homomorphism of k-algebras $A \to A'$ that sends J into J'. If one associates to (A, J) the pair $(Spf(A/J), Spf(A))$, one obtains evidently an anti-equivalence of $**C**$ onto the category of pairs (Z, Y) formed by a k-formal variety Y and a formal subvariety Z, a morphism $\phi : (Z, Y) \to (Z', Y')$ being a morphism of k-formal varieties $Y \to Y'$ that sends Z into Z'.

[832] N.D.E.: We have replaced "left" by "right" and have modified the statement of Proposition 5.1, in order to bring out more clearly, on the one hand, the equivalent conditions (i), (ii), and, on the other hand, the conclusion $Spf(B) \simeq G/H$.

A *cogroup structure* on an object (A, J) of $**C**$ consists of the data of a structure of formal group on $Spf(A)$ such that the following conditions are realized (notations of 2.1):

(1) $\Delta_A(J) \subset J\hat{\otimes}_k A + A\hat{\otimes}_k J$;

(2) $\epsilon_A(J) = 0$;

(3) $c_A(J) \subset J$.

These conditions also mean that $H = Spf(A/J)$ is a formal subgroup of $G = Spf(A)$.[833]

Suppose moreover that A is local, i.e., that $Spf(A)$ is an infinitesimal formal group. Then, if $J \neq A$, conditions (2) and (3) are consequences of (1). Indeed, if J is a closed ideal distinct from A, it is contained in the augmentation ideal I_A, hence (2) is verified, and $M = Spf(A/J)$ is a formal submonoid of G. Since G is infinitesimal, it follows from 2.7 that M is a formal subgroup of G, i.e., condition (3) is verified.

5.1.2. Denote by $Alpg/k$ the category of *graded profinite k-algebras*: an object of this category consists of the data of a sequence $A_0, A_1, \cdots, A_n, \cdots$ of pseudocompact k-modules and of a profinite algebra structure on the product $\prod_{n\geqslant 0} A_n$ such that one has $A_n \cdot A_m \subset A_{m+n}$ (A_n being identified with a part of $\prod_{i\geqslant 0} A_i$ by means of the canonical injection); a morphism $\psi : (A_n) \to (B_n)$ is a sequence of continuous linear maps $\psi_n : A_n \to B_n$ such that one has $\psi_{m+n}(a\cdot a') = \psi_m(a)\cdot\psi_n(a')$ if $a \in A_m$ and $a' \in A_n$.

Definitions. *It is clear that two graded profinite k-algebras (A_n) and (B_n) have a coproduct*[834] *in $Alpg/k$, which has as n-th component the product $\prod_{i=0}^n A_i\hat{\otimes}_k B_{n-i}$ of the pseudocompact k-modules $A_i\hat{\otimes}_k B_{n-i}$. This coproduct will be denoted $(A_n)\hat{\otimes}_k(B_n)$.*

Then, a cogroup structure on an object (A_n) of $Alpg/k$ is the data of continuous k-linear maps $\Delta_n : A_n \to \prod_{i=0}^n A_i\hat{\otimes}_k A_{n-i}$ and $\epsilon : A_0 \to k$, which induce on $\prod_{n\geqslant 0} A_n$ (setting $\epsilon(A_i) = 0$ for $i \geqslant 1$) a cogroup structure in Alp/k.

Finally, for every object (A, J) of $**C**$, one denotes by $Gr_J(A)$ the graded profinite algebra associated with the filtration of A by the closures J^n of the powers of J; one has therefore $Gr_J(A)_n = J^n/J^{n+1}$ and the multiplication of $Gr_J(A)$ is induced by that of A.

Lemma.[835] *Let U, V be two pseudocompact k-modules, with U topologically flat, and let $U = U_0 \supset U_1 \supset \cdots$ and $V = V_0 \supset V_1 \supset \cdots$ be two decreasing sequences of closed k-submodules. Filter the completed tensor product $W = U\hat{\otimes}_k V$ by means of the closed submodules*

[833]N.D.E.: Note that if (A', J') is a second cogroup of $**C**$, corresponding to a pair $H' \subset G'$ of k-formal groups, then to give a morphism of cogroups $(A', J') \to (A, J)$ amounts to giving a morphism of k-formal groups $G \to G'$ that sends H into H'.

[834]N.D.E.: We have replaced "direct sum" by "coproduct".

[835]N.D.E.: In the original, the lemma is stated when k is a field, the proof in this case being left to the reader.

$$W_n = U_n \hat{\otimes}_k V_0 + U_{n-1} \hat{\otimes}_k V_1 + \cdots + U_0 \hat{\otimes}_k V_n.$$

Suppose that each U_i/U_{i+1} is topologically flat over k (so that U/U_n and hence U_n are also so, for every n). Then, for every n, one has an isomorphism

$$W_n / W_{n+1} \simeq \bigoplus_{i+j=n} (U_i/U_{i+1}) \hat{\otimes}_k (V_j/V_{j+1}).$$

Proof. Set $W_{i,j} = U_i \hat{\bigotimes} V_j$ and $\bar{W}_{i,j} = (U_i/U_{i+1}) \hat{\bigotimes}_k (V_j/V_{j+1})$, for every $i, j \geqslant 0$. Let us show by induction on n that the natural map

$$\pi_n : W_n \to \bigoplus_{i+j=n} \bar{W}_{i,j}$$

is surjective and that the inclusion $W_{n+1} \subset Ker(\pi_n)$ is an equality. For $n = 0$, the projection

$$\pi_0 : U_0 \hat{\otimes}_k V_0 \to (U_0/U_1) \hat{\otimes}_k (V_0/V_1)$$

is surjective and, since U_0, U_0/U_1 and hence U_1 are topologically flat over k, one sees that $Ker(\pi_0) = U_0 \hat{\bigotimes}_k V_1 + U_1 \hat{\bigotimes}_k V_0$ and that, moreover, $U_0 \hat{\bigotimes}_k V_1 \cap U_1 \hat{\bigotimes}_k V_0 = U_1 \hat{\bigotimes}_k V_1$.

Suppose then $n > 0$ and the result established for $n - 1$. Set $M_0 = U_0 \hat{\bigotimes}_k V_n$ and $S_0 = \sum_{i=1}^{n} U_i \hat{\bigotimes}_k V_{n-i}$. One has $S_0 \subset U_1 \hat{\bigotimes}_k V_0$ and hence, by what precedes applied to $V_0 \supset V_n$ instead of $V_0 \supset V_1$, one has

$$M_0 \cap S_0 \subset U_0 \hat{\otimes}_k V_n \cap U_1 \hat{\otimes}_k V = U_1 \hat{\otimes}_k V_n$$

from which one deduces that $M_0 \cap S_0 = U_1 \hat{\bigotimes}_k V_n$. Since $W_n = M_0 + S_0$, one obtains a commutative diagram with exact rows, where one has set $U'_i = U_{i+1}$ and $W'_{i,n-1-i} = W_{i+1,n-1-i}$ for $i = 0, \cdots, n-1$:

$$\begin{tikzcd}
0 \arrow[r] & S_0 \arrow[r] \arrow[d, "\pi'_{n-1}"] & W_n \arrow[r] \arrow[d, "\pi_n"] & (U_0/U_1) \hat{\otimes}_k V_n \arrow[r] \arrow[d, "p"] & 0 \\
0 \arrow[r] & \bigoplus_{i=0}^{n-1} \bar{W}'_{i,n-1-i} \arrow[r] & \bigoplus_{i=0}^{n} \bar{W}_{i,n-1} \arrow[r] & \bar{W}_{0,n} \arrow[r] & 0.
\end{tikzcd}$$

Then p is surjective, with kernel $(U_0/U_1) \hat{\bigotimes}_k V_{n+1}$. Moreover, by the induction hypothesis applied to the sequence (U'_i), π'_{n-1} is surjective, with kernel equal to $W'_n = \sum_{i=1}^{n} W_{i,n+1-i}$. It follows that π_n is surjective, and that the inclusion $W_{n+1} \subset Ker(\pi_n)$ is an equality. This proves the lemma.

Let us return to an object (A, J) of $**C**$ and note that, by 0.2.G, the hypothesis that each J^n/J^{n+1} is topologically flat over k is equivalent to saying that $Gr_J(A)$ is topologically flat over k.

Corollary. *Let $**P**$ be the full subcategory of $**C**$ formed by the objects (A, J) such that A and $Gr_J(A)$ are topologically flat over k. Then the functor $**P** \to Alpg/k$, $(A, J) \mapsto Gr_J(A)$, commutes with finite coproducts, hence transforms a cogroup of $**P**$ into a cogroup of $Alpg/k$.*

In particular, if k is a field then, for every cogroup (A, J) of $**C**$, $Gr_J(A)$ is a cogroup of $Alpg/k$.[836]

[836] N.D.E.: To a pair $H \subset G$ of k-formal groups, one therefore associates the "formal completion $\hat{G}_H$ of G

5.1.3. Identify every profinite k-algebra Γ with the graded profinite k-algebra $(\Gamma_n)_{n\geqslant 0}$ such that $\Gamma_0 = \Gamma$ and $\Gamma_n = 0$ if $n > 0$. In particular, if $(A_n)_{n\geqslant 0}$ is a graded profinite k-algebra, we shall consider A_0 indifferently as a profinite k-algebra or as a graded profinite k-algebra. We shall then denote by $\rho : (A_n) \to A_0$ the morphism of $Alpg/k$ such that $\rho_0 = id_{A_0}$ and $\rho_n = 0$ if $n > 0$. Similarly, $\tau : A_0 \to (A_n)$ will denote the section of ρ such that $\tau_0 = id_{A_0}$ and $\tau_n = 0$ if $n > 0$.

Every cogroup structure on $(A_n)_{n\in\mathbb{N}}$ induces a cogroup structure on A_0 such that ρ and τ are homomorphisms of cogroups. In this case, denote by I_0 the augmentation ideal of A_0 and set $A'_n = A_n/I_0A_n$ for every $n \geqslant 0$ (so that $A'_0 = k$).

Then, $(A'_n)_{n\in\mathbb{N}}$ is a cogroup in $Alpg/k$ (note that, since $A_0 = I_0 \bigoplus k\cdot 1$, then $I_0 \hat{\bigotimes}_k A_n \simeq I_0A_n$ is a direct factor of A_n, for every n). Since τ is a section of ρ, then, by 2.4.A, the cogroup $(A_n)_{n\in\mathbb{N}}$ is the "semi-direct coproduct" of A_0 and of the cogroup $(A'_n)_{n\in\mathbb{N}}$. More precisely, $(A'_n)_{n\in\mathbb{N}}$ is isomorphic, as an object of $Alpg/k$, to the kernel of the pair:

$$(A_n) \underset{(id\,\hat{\otimes}\,\rho)\,\Delta}{\overset{\tau_1}{\rightrightarrows}} (A_n)\,\hat{\otimes}_k\, A_0$$

(where $\Delta : (A_n) \to (A_n)\hat{\bigotimes}(A_n)$ is the comultiplication of (A_n) and $\tau_1(x) = x\hat{\bigotimes}1$), and, identifying A'_n with its image in A_n, the map

$$(A'_n\,\hat{\otimes}_k\, A_0) \xrightarrow{\sim} (A_n), \qquad a'_n\,\hat{\otimes}\, a_0 \mapsto a'_n a_0$$

is an isomorphism in $Alpg/k$. (N.B. This is not an isomorphism of cogroups, but $\Delta(A') \subset A\hat{\bigotimes}A'$ and $(\gamma'_\rho\hat{\bigotimes}id)\circ\Delta|_{A'} = \Delta'$, where Δ' is the comultiplication of A' and γ'_ρ the projection $A' \to A'$, cf. 2.4.A.)

5.1.4. Let (A, J) be an object of $**C**$ and $(A_n) = Gr_J(A)$ the object of $Alpg/k$ associated, i.e., $A_n = J^n/J^{n+1}$ for every $n \geqslant 0$. It is clear that the algebra $**A** = \prod_{n\geqslant 0} A_n$ is generated by A_0 and A_1, that is to say that, for $n \geqslant 1$, the map

$$(1)\qquad A_1\,\hat{\otimes}_{A_0}\, A_1\,\hat{\otimes}_{A_0} \cdots \hat{\otimes}_{A_0}\, A_1 \to A_n$$

defined by multiplication is surjective.

Suppose moreover that (A, J) is a cogroup of $**C**$ and that A and the quotients J^n/J^{n+1} are flat over k. Then, by Corollary 5.1.2, $Gr_J(A)$ is a cogroup of $Alpg/k$. Therefore, by 5.1.3, if one sets

$$(2)\qquad A'_n = \{\, x \in J^n/J^{n+1} \mid \Delta(x) - x\,\hat{\otimes}\,1 \in \sum_{i=1}^n (J^{n-i}/J^{n-i+1})\,\hat{\otimes}\,(J^i/J^{i+1}) \,\},$$

then the map $(A'_n\hat{\bigotimes}A_0) \to (A_n)$, $a'_n\hat{\bigotimes}a_0 \mapsto a'_n a_0$, is an isomorphism in $Alpg/k$.[837]

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along H", which is a k-formal group; moreover, one shall see in 5.1.3–5.1.4 that the inclusion $\sigma : H \hookrightarrow \hat{G}_H$ admits a retraction $\pi : \hat{G}_H \to H$ and that the k-formal group $N = Ker(\pi)$ is identified, as a formal variety, with the completion of the homogeneous space G/H along the unit section. This will be useful in 5.2.2.

[837] N.D.E.: Let G, H and $\hat{G}_H$ be the k-formal groups corresponding to A, $A_0 = A/J$ and $Gr_J(A)$; then $\tau : A_0 \hookrightarrow Gr_J(A)$ corresponds to a retraction $\pi : \hat{G}_H \to H$ of the inclusion $H \hookrightarrow \hat{G}_H$, and what precedes means that $\hat{G}_H$ is the semi-direct product of $N = Ker(\pi)$ by H.

Denoting by I_0 the augmentation ideal of A_0, one deduces from (1) and the commutative diagram below, where $A_1'^{\hat{\otimes} n}$ denotes $A'_1 \hat{\otimes}_k \cdots \hat{\otimes}_k A'_1$ (n factors):

$$\begin{array}{ccccc}
A_1 \hat{\otimes}_{A_0} \cdots \hat{\otimes}_{A_0} A_1 & \cong & A_1'^{\hat{\otimes} n} \hat{\otimes}_k A_0 & \cong & (A_1'^{\hat{\otimes} n} \hat{\otimes}_k I_0) \oplus A_1'^{\hat{\otimes} n} \\
\Big\downarrow{\scriptstyle m} & & \Big\downarrow{\scriptstyle m' \hat{\otimes} \mathrm{id}} & & \Big\downarrow{\scriptstyle m' \hat{\otimes} \mathrm{id} \oplus m'} \\
A_n & \cong & A'_n \hat{\otimes}_k A_0 & \cong & (A'_n \hat{\otimes}_k I_0) \oplus A'_n
\end{array}$$

that the map

$$(3) \qquad m' : A'_1 \hat{\otimes}_k \cdots \hat{\otimes}_k A'_1 \to A'_n$$

induced by multiplication is surjective; in other words, the profinite k-algebra $\mathbf{A}' = \prod_{n \geqslant 0} A'_n$ is generated by its terms of degree 1.

[838] Let us now return to the hypothesis (i) of Proposition 5.1: let G be a topologically flat infinitesimal k-formal group, A its affine algebra, I_A the augmentation ideal of A, B a closed subalgebra of A, and $J = AI_B$, where $I_B = B \cap I_A$. Denote by H the formal subgroup $Spf(A/J)$ and π the projection $A \to A/J$. Suppose that A/J^n is topologically flat over k, for every $n \geqslant 1$, and that B is contained in the kernel $\tilde{B}$ of the pair:

$$A \underset{(\mathrm{id}_A \hat{\otimes}_k \pi)\,\Delta_A}{\overset{\tau_1}{\rightrightarrows}} A \hat{\otimes}_k (A/J).$$

Let $(A_n) = Gr_J(A)$, let $I_0 = I_A/J$ be the augmentation ideal of $A_0 = A/J$, and define $(A'_n)_{n \in \mathbb{N}}$ as in (2) above. Denote by $(B_n)_{n \in \mathbb{N}}$ (resp. $(\tilde{B}_n)_{n \in \mathbb{N}}$) the object of $Alpg/k$ associated to the filtration of B (resp. $\tilde{B}$) induced by that of A, i.e., defined by the ideals $B \cap J^n$ (resp. $\tilde{B} \cap J^n$). Then, it is clear that $B_n \subset \tilde{B}_n \subset A'_n$ for every n, and that

$$\mathbf{B} = \prod_{n \geqslant 0} B_n \subset \tilde{\mathbf{B}} = \prod_{n \geqslant 0} \tilde{B}_n$$

are subalgebras of $\mathbf{A}' = \prod_{n \geqslant 0} A'_n$.

On the other hand, J (resp. J^2) is the image in A of $I_B \hat{\otimes}_k A$ (resp. of $I_B^2 \hat{\otimes}_k A$). Consequently, the map

$$(I_B/I_B^2) \hat{\otimes}_k A \to J/J^2 = A_1 \simeq A'_1 \hat{\otimes}_k A_0$$

is surjective, and it factors through $(I_B/I_B^2) \hat{\otimes}_k A_0$. Since $A_0 = k \cdot 1 \bigoplus I_0$, one deduces from the commutative and exact diagram:

$$\begin{array}{ccc}
(I_B/I_B^2) \hat{\otimes}_k A_0 & \cong & ((I_B/I_B^2) \hat{\otimes}_k I_0) \oplus (I_B/I_B^2) \\
\downarrow & & \downarrow \\
J/J^2 & \cong & (A'_1 \hat{\otimes}_k I_0) \oplus A'_1 \\
\downarrow & & \downarrow \\
0 & & 0
\end{array}$$

[838] N.D.E.: We have detailed the original in what follows, and we have put at the end the "supplement" $I_B^n = B \cap J^n$ (which is not necessary to establish Proposition 5.1).

that A'_1 is the image of I_B/I_B^2, so that one has $B_1 = A'_1$. Since $* * A * *'$ is generated by A'_1, it follows that, for every $n \geqslant 1$, one has

$$A'_n \subset B_n \subset \tilde{B}_n \subset A'_n,$$

whence $B_n = \tilde{B}_n = A'_n$.[839]

Finally, since the formal group G is infinitesimal, one has $\mathfrak{r}(A) = I_A$ and hence the ideals I_A^n tend to 0 (cf. 0.1.2); a fortiori, the ideals J^n tend to 0, and hence the induced filtrations on $\tilde{B}$ and B are separated. Moreover, since B is a closed subalgebra of A, it is complete for the topology defined by the ideals $B \cap J^n$. Consequently, it follows from [CA], § V.7, Lemma 1 (see also 5.1.5 below) that $B = \tilde{B}$. This completes the proof of Proposition 5.1.

One has moreover the following supplement. For every n, $J^n = AI_B^n$ is the image in A of $A \hat{\otimes}_B I_B^n$ and also of $A \hat{\otimes}_B (B \cap J^n)$. Now, by hypothesis, the affine algebra A/J of the formal subgroup H is topologically flat over k. Hence, by Theorem 2.4, the morphism $G = Spf(A) \to G/H = Spf(B)$ is surjective and topologically flat; one has therefore

$$A \hat{\otimes}_B I_B^n = J^n = A \hat{\otimes}_B (B \cap J^n),$$

and this entails that $I_B^n = B \cap J^n$ for every n. This also follows from the fact that the maps

$$I_B^n / I_B^{n+1} \to A'_i = (B \cap J^n)/(B \cap J^{n+1})$$

are surjective, and from 5.1.5 (ii) below, applied to $B'_n = I_B^n$ and $B_n = B \cap J^n$.

5.1.5. Lemma 5.1.5.[840] *(i) Let M and N be two abelian groups filtered by decreasing sequences of subgroups $(M_n)_{n\in\mathbb{Z}}$ and $(N_n)_{n\in\mathbb{Z}}$. Suppose that the union of the M_n (resp. N_n) equals M (resp. N), that the intersection of the M_n (resp. N_n) is zero, and that M is complete for the topology defined by the M_n. Let $f : M \to N$ be a morphism of filtered groups.*

a) If f induces a surjection of the associated graded modules, then f is a surjection and N is complete for the topology defined by the N_n.

b) If f induces an injection of the associated graded modules, then f is an injection.

(ii) Let B be an abelian group, $B = B'_0 \supset B'_1 \supset \cdots$ and $B = B_0 \supset B_1 \supset \cdots$ two separated filtrations of B by subgroups such that $B'_n \subset B_n$ for every n. Suppose B is complete for the topology defined by the filtration (B'_n).

If the map $B'_i/B'_{i+1} \to B_i/B_{i+1}$ is surjective for every i, then $B'_n = B_n$ for every n.

[839] N.D.E.: With the notations of N.D.E. [^N.D.E-VII_B-C-31], this entails that the formal completion of G/H along the unit section (which has $\prod_n B_n$ as affine algebra) is isomorphic, as a formal variety, to the k-formal group N.

[840] N.D.E.: The original stated only point (ii); for the reader's convenience, we have stated in (i) Lemma 1 of [CA], § V.7.

Indeed, (i) is Lemma 1 of [CA], § V.7 (see also [BAC], III, § 2.8), and (ii) follows from it by taking $M = B'_n \supset B'_{n+1} \supset \cdots$ and $N = B_n \supset B_{n+1} \supset \cdots$.

5.2.

In all the rest of Section 5, k denotes a perfect field of characteristic $p > 0$.

We set $* * \bar{N} * * = \mathbb{N} \cup \infty$. If B is a profinite k-algebra and $r \in * * \bar{N} * *$, we denote by $((x^{p^r}))_{x\in\mathfrak{r}(B)}$ the closed ideal of B generated by the elements x^{p^r}, where x runs through the radical $\mathfrak{r}(B)$ of B. If $r = \infty$, we use the same notation, with the convention that $((x^{p^\infty}))_{x\in\mathfrak{r}(B)}$ is the zero ideal. In both cases, B_r denotes the quotient $B/((x^{p^r}))_{x\in\mathfrak{r}(B)}$.

We say that B is *of height* $\leqslant r$ if $((x^{p^r}))_{x\in\mathfrak{r}(B)}$ is the zero ideal; if this holds and r is finite, we say that B is *of finite height*.

Consider in particular the case where B is of the form $k[[\omega]]$, ω being a pseudo-compact k-vector space (cf. 1.2.5).[841] We then say that B is a *formal power series algebra* and that B_r is a *truncated formal power series algebra* ($r \in * * \bar{N} * *$; we therefore agree to say that $B = B_\infty$ is also "truncated"). If $B = k[[\omega]]$, we also write $((x^{p^r}))_{x\in\omega}$ instead of $((x^{p^r}))_{x\in\mathfrak{r}(B)}$.

Notations. *Let ω be a pseudocompact k-vector space filtered by an increasing sequence of closed vector subspaces*

$$0 = \omega_0 \subset \omega_1 \subset \omega_2 \subset \omega_3 \subset \cdots$$

(a) The closed ideal of $k[[\omega]]$ generated by the elements x^{p^r}, where r runs through $\mathbb{N}$ and x runs through ω_r, will be denoted $((x^{p^r}))_{r\in\omega_r}$.

(b) On the other hand, we shall denote by ${}^r\omega$ the filtered pseudocompact vector space such that

$${}^r\omega_i = \omega_i \quad \text{if } i < r \quad \text{and} \quad {}^r\omega_i = \omega \quad \text{if } i \geqslant r.$$

Theorem (Dieudonné–Cartier). *Let $H \to G$ be a monomorphism of infinitesimal formal groups over a perfect field k of characteristic $p > 0$. Let B be the affine algebra of the homogeneous space G/H and suppose one of the following three conditions verified:*[842]

(i) B is of finite height (this is the case in particular if G is of finite height).

(ii) B is a complete noetherian local ring.

(iii) B is a reduced ring.

Then B is isomorphic to the completed tensor product of a finite family of truncated formal power series algebras.

[841] N.D.E.: In the original, the author uses "linearly compact vector space", which is equivalent to "pseudocompact vector space" (cf. [BAC], § III.2, Exercises 15 a), 19 a), and 20 d)). We have preferred to keep the terminology "pseudocompact" used so far.

[842] N.D.E.: On the one hand, we have replaced $H\backslash G$ by G/H, and likewise in the proof; on the other hand, we have added condition (iii).

The proof of this theorem occupies paragraphs 5.2.1 to 5.2.5.

5.2.1. Let A be the affine algebra of G, I_A its augmentation ideal, and $I = B \cap I_A$. By 2.4, one has $H = Spf(A/AI)$ and $B = x \in A|\Delta(x) - 1\hat{\otimes}x \in A\hat{\otimes}AI$. Set $\omega = I/I^2$. One denotes by I_r the closed sub-ideal of I formed by the x such that $x^{p^r} = 0$, and by ω_r the canonical image of I_r in ω. We shall prove:

Proposition. *If there exists a continuous section $\sigma : \omega \to I$ of the projection $I \to I/I^2$, such that $\sigma(\omega_r) \subset I_r$ for every r, then B is isomorphic to $k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$.*

Such a section indeed extends into a continuous morphism $k[[\omega]] \to B$, which factors through $B' = k[[\omega]]/((x^{p^r}))_{x \in \omega_r}$. We prove from 5.2.2 to 5.2.5 that the morphism

$$\varphi : B' = k[[\omega]]/((x^{p^r}))_{x \in \omega_r} \to B$$

thus obtained is an isomorphism.

5.2.1.A. [843] For each $r \in \mathbb{N}$, set $\bar{I}_r = I^2 + I_r$, so that $\bar{I}_r/I^2 \simeq \omega_r$; one then has a commutative diagram with exact rows:

$$0 \longrightarrow \bar{I}_{r-1} \longrightarrow \bar{I}_r \longrightarrow \bar{I}_r/\bar{I}_{r-1} \longrightarrow 0 \wr \downarrow 0 \longrightarrow \omega_{r-1} \longrightarrow \omega_r \longrightarrow \omega_r/\omega_{r-1} \longrightarrow 0$$

and since k is a field, the rows are split: one can complete a pseudobasis B_{r-1} of $\bar{I}_{r-1}$ into a pseudobasis $B_{r-1} \cup B'_r$ of $\bar{I}_r$, and then the closed subspace S_r with pseudobasis B'_r is a supplement of $\bar{I}_{r-1}$ in $\bar{I}_r$, and the projection $\pi : I \to \omega$ induces an isomorphism of S_r onto a supplement ω'_r of ω_{r-1} in ω_r. Denote by I_∞ the closed ideal $\bigcup_r \bar{I}_r$; it admits similarly a supplement S_∞ in I, and π induces an isomorphism of I_∞/I^2 (resp. of S_∞) onto the closure ω_∞ of the union of the ω_r (resp. onto a supplement ω'_∞ of ω_∞ in ω). Denote by η the isomorphism $S_\infty \xrightarrow{\sim} \omega'_\infty$. One then obtains continuous linear maps:

$$\begin{array}{ccc} I^2 \times S_\infty \times \bigoplus^c_r S_r & \xrightarrow{\varphi} & I \\ \downarrow{\scriptstyle \eta \times \theta} & & \\ \omega = \omega'_\infty \times \omega_\infty & & \end{array}$$

where $\bigoplus^c_r S_r$ is the direct sum of the S_r in $PC(k)$, i.e., $(\prod_r S_r^\dagger)^*$ (cf. N.D.E. (16) of 0.2.2) and where $\theta : \bigoplus^c_r S_r \to \omega_\infty$ is induced by the maps $S_r \xrightarrow{\sim} \omega'_r \hookrightarrow \omega_\infty$. One therefore sees that a sufficient condition (but not necessary, see below) to obtain a section $\sigma : \omega \to I$ as desired is that θ be an isomorphism. By duality (cf. 0.2.2), this amounts to saying that the linear map $\omega_\infty^\dagger \to \prod_r S_r^\dagger$ is bijective.

[843] N.D.E.: We have added paragraphs 5.2.1.A and 5.2.1.B.

5.2.1.B. Denote, as before, $\omega_\infty = \bigcup_{n\in\mathbb{N}} \omega_n$. A second case in which a section $\sigma : \omega \to I$ as desired exists is the case where ω_∞ possesses a pseudobasis $\ast\ast B \ast\ast_\infty$ that is a union of pseudobases of the ω_n/ω_{n-1}, for $n \in \mathbb{N}^*$ (one can then complete it by a pseudobasis $\ast\ast B \ast\ast'_\infty$ of ω/ω_∞ to obtain a pseudobasis of ω compatible with the filtration). Setting $V = \omega_\infty^\dagger$ and denoting by V_n the orthogonal in V of ω_n, this amounts to saying that, in the category of "ordinary" k-vector spaces, the decreasing separated filtration

$$V = V_0 \supset V_1 \supset V_2 \supset \cdots$$

is split, i.e., that V is the direct sum, for $n \in \mathbb{N}$, of subspaces F_n such that $F_n \simeq V_n/V_{n+1}$. This is not necessarily the case: for example, if V is the space $S = k^{\mathbb{N}}$ of sequences of elements of k and S_n the subspace of sequences (u_i) such that $u_i = 0$ for $i < n$, so that $\dim S_n/S_{n+1} = 1$, then S is not isomorphic to the direct sum of the S_n/S_{n+1} since S is not of countable dimension (on the other hand, S is here the product of the S_n/S_{n+1}, cf. 5.2.1.A). It is however the case if V is of countable dimension.[844] Indeed, let $(e_n)_{n\in\mathbb{N}}$ be a basis of V; we shall construct by induction on n an increasing function $g : \mathbb{N} \to \mathbb{N}$ and subspaces F_i, for $i = 0, \cdots, g(n)$, such that $F_i \simeq V_i/V_{i+1}$ and that $F_{\leqslant g(n)} = \bigoplus_{i=0}^{g(n)} F_i$ is a supplement of $V_{g(n)+1}$ containing $e_0, \cdots, e_n$; one will then have $V = \bigoplus_{i\geqslant 0} F_i$. Let $n+1 \in \mathbb{N}$; we may assume the assertion established for n (the assertion being vacuous for $n = -1$). If $e_{n+1} \in F_{\leqslant g(n)}$, set $g(n+1) = g(n)$; otherwise write $e_{n+1} = f + x$ with $f \in F_{\leqslant g(n)}$ and $x \in V_{g(n)+1}$ nonzero. Let then j be the smallest integer such that $x \in V_j - V_{j+1}$; for $i = g(n)+1, \cdots, j$, choose a supplement F_i of V_{i+1} in V_i, so that $e_{n+1} \in F_j$; one then sets $g(n+1) = j$.

5.2.1.C. [845] In particular, the two preceding conditions (5.2.1.A and B) are verified when the filtration of ω is stationary, i.e., when there exists an integer n_0 such that $\omega_n = \omega_{n_0}$ for $n_0 \leqslant n < +\infty$. In this case, one obtains an isomorphism of $k[[\omega]]/((x^{p^r}))_{x\in\omega_r}$ onto the completed tensor product:

$$\left(k[[\omega_1]]/((x^p))_{x\in\omega_1}\right) \hat{\otimes} \left(k[[\omega'_2]]/((x^{p^2}))_{x\in\omega'_2}\right) \hat{\otimes} \cdots \hat{\otimes} \left(k[[\omega'_{n_0}]]/((x^{}\right.$$

where ω'_n (resp. ω'_∞) is a supplement of ω_{n-1} in ω_n (resp. of $\omega_\infty = \omega_{n_0}$ in ω). The filtration of ω is obviously stationary in case (i), i.e., if $\omega_r = \omega$ for r large enough, and in case (ii), i.e., if ω is of finite dimension, and also in case (iii), i.e., if $I_r = 0$ for every r (and in this case B will be isomorphic to the formal power series algebra

$k[[\omega]]$). The remarks above therefore imply our theorem, modulo points 5.2.2–5.2.5 below.

5.2.2. Suppose first that B is of height $\leqslant 1$, that is to say that $x^p = 0$ if $x \in I$. By 5.1.4, the graded ring $Gr_I(B)$ associated with B for the filtration $I \supset I^2 \supset I^3 \supset \cdots$ is endowed with a cogroup structure in the category $Alpg/k$, i.e., the profinite algebra $\ast\ast C \ast\ast = \prod_{n\geqslant 0} Gr_I(B)_n$ is the affine algebra of a k-formal group N. It is clear that

[844] N.D.E.: This paragraph is the fruit of discussions with J.-M. Fontaine and E. Bouscaren; in particular Bouscaren indicated to us the proof that follows.

[845] N.D.E.: We return here to the original, which we have shortened taking account of the preceding additions.

one has $\omega_{N/k} = I/I^2$ and that N is infinitesimal of height $\leqslant 1$. By 4.4, the identity map of $\omega_{N/k}$ therefore induces an isomorphism of $B' = k[[\omega_{N/k}]]/((x^p))_{x\in\omega_{N/k}}$ onto $**C**$. This implies in particular that the map ϕ of 5.2.1 induces an isomorphism of the associated graded rings of B' and B when one filters B' and B by the powers of the augmentation ideal. Hence ϕ is an isomorphism, by [CA], § V.7, Lemma 1 (see also 5.1.5).

5.2.3. Suppose now B of finite height $\leqslant r$. Let π be the isomorphism $x \mapsto x^p$ of k onto k. The linear map of $B\hat{\otimes}_\pi k$ into B which sends $b\hat{\otimes}_\pi x$ onto $b^p x = (bx^{1/p})^p$ has a closed image which is none other than the closed subalgebra $B^p = b^p|b \in B$ of B. Set $J = B^p \cap I = B^p \cap I_A$.

[846] Denote by G_1 the kernel of the Frobenius morphism $Fr : G \to G^{(p)}$ and by HG_1 the formal subgroup of G inverse image of the formal subgroup $H^{(p)}$ of $G^{(p)}$. Then HG_1 is defined by the closed ideal generated by the p-th powers of elements of AI, which equals AJ. On the other hand, since the formation of G/H commutes with base change (since G/H represents the sheaf-quotient for the flat topology, cf. 2.4), then $(G/H)^{(p)} = G^{(p)}/H^{(p)}$, and one therefore has commutative diagrams:

```
G ——Fr——→ G^{(p)}                    A ←——a ⊗ˆ1 ↦ a^p———— A ⊗ˆ_π k
↓                  ↓                    ↑                          ↑
G/H ——Fr——→ G^{(p)}/H^{(p)}       B ←——b ⊗ˆ1 ↦ b^p———— B ⊗ˆ_π k
```

from which one deduces that B^p is the affine algebra of the quotient G/HG_1.[847] Denote provisionally by $**C**$ the affine algebra of the quotient HG_1/H. Since the formation of G/H commutes with base change, one has a cartesian square:

$$HG_1 \longrightarrow G \downarrow\downarrow HG_1/H \longrightarrow G/H$$

whence an isomorphism $A\hat{\otimes}_B **C** \simeq A/AJ = A\hat{\otimes}_B(B/BJ)$, and since A is topologically free over B (by 2.4, since A and B are local), it follows that the natural morphism $B/BJ \to **C**$ is an isomorphism, hence B/BJ is the affine algebra of HG_1/H,[848] and of course $B/BJ = B_1$ is of height $\leqslant 1$ since $J = ((x^p))_{x\in\mathrm{r}(B)}$.

Let $B' = k[[\omega]]/((x^{p^r}))_{x\in\omega_r}$, $\phi : B' \to B$ the morphism introduced in 5.2.1, B'^p the subalgebra $x^p|x \in B'$, and J' the augmentation ideal of B'^p. Then, one has a commutative diagram:

```
B′ ——φ——→ B
↓                     ↓
B′_1 = B′/B′J′  B_1 = B/BJ
```

and, by 5.2.2, ϕ_1 is an isomorphism.

[846] N.D.E.: We have added what follows.
[847] N.D.E.: As indicated in the original, this also follows from Proposition 5.1, but we have preferred to indicate the argument above, which does not use the implication (i) ⇒ (ii) of loc. cit.
[848] N.D.E.: As indicated in the original, this also follows from Proposition 5.1, but we have preferred to indicate the argument above, which does not use the implication (i) ⇒ (ii) of loc. cit.

On the other hand, by 2.4, A is topologically flat over $B = **A**(G/H)$ and over $B^p = **A**(G/HG_1)$; hence, by 1.3.3, B is topologically flat over B^p. Moreover, by 5.2.4 below, the morphism $B'^p \to B^p$ induced by ϕ is an isomorphism. One can then apply 0.3.4 to the pseudocompact ring $B'^p = B^p$ and to the pseudocompact B^p-modules $M = B'$, $N = B$: by what precedes, $\phi_1 = \phi\hat{\otimes}_{B^p} k$ is an isomorphism, and it follows that ϕ is an isomorphism. This proves 5.2 when B is of finite height, modulo point 5.2.4 below.

5.2.4. For every pseudocompact vector space V over k, we denote by ${}^\pi V$ the space $V\hat{\otimes}_\pi k$ deduced from V by the extension $x \mapsto x^p$ of the field of scalars.[849] One then has a commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & {}^\pi I^2 \arrow[r, "\alpha"] \arrow[d, "u"] & {}^\pi I \arrow[r, "\beta"] \arrow[d, "v"] & {}^\pi \omega \arrow[r] \arrow[d, "w"] & 0 \\ 0 \arrow[r] & J^2 \arrow[r, "\gamma"] & J \arrow[r, "\delta"] & \bar{\omega} \arrow[r] & 0 \end{tikzcd}$$

where one has set $\bar{\omega} = J/J^2$ and where the maps u, v, w are induced by the linear map $x\hat{\otimes}a \mapsto x^p a$ from ${}^\pi B$ into B^p. Since u and v are surjections, w is a surjection and has as kernel the image ${}^\pi\omega_1$ of ${}^\pi I_1 = Ker(v)$.

Then, setting $J_n = x \in J | x^{p^n} = 0$ and $\bar{\omega}_n = \delta(I_n)$, one has $J_n = v({}^\pi I_{n+1})$ and $\bar{\omega}_n = w({}^\pi\omega_{n+1})$, for every $n \geqslant 0$. The section ${}^\pi\sigma : {}^\pi\omega \to {}^\pi I$, which is induced by the section σ of 5.2.1, therefore defines by passage to the quotient a section $\tau : \bar{\omega} \to J$ that is compatible with the filtrations of J and $\bar{\omega}$. Since B^p is of height $\leqslant r-1$, this section induces, by induction hypothesis, an isomorphism

$$\psi : B'' = k[[\omega]] / ((x^{p^n}))_{x \in \bar{\omega}_n} \to B^p.$$

Now B'' is identified with B'^p and ψ with the morphism $B'^p \to B^p$ induced by ϕ, hence our

theorem is proved when B is of finite height.

Remark 5.2.4.A.[850] *Suppose B of height $\leqslant r+1$ (with $r \in \mathbb{N}^*$). Then $I_{r+1} = I$ and one has an isomorphism*

$$(1) \qquad B \simeq (k[[S_1]]/((x_1^p))_{x_1 \in S_1}) \hat{\otimes}_k \cdots \hat{\otimes}_k (k[[S_r]]/((x_r^{p^r}))_{x_r \in S_r}) \hat{\otimes}_k (k[[S_{\ldots}$$

where each S_n is a supplement of $I^2 + I_{n-1}$ in $I^2 + I_n$. Then $\omega = I/I^2$ is identified with $\prod_{i=1}^{r+1} S_i$, and one sees easily that, for $n = 1, \cdots, r+1$, the image ω_n of I_n in ω is identified with $\prod_{i=1}^{n} S_i$.

This has the following consequence. Let $B_r = B/\check{J}_r$, where $\check{J}_r = ((x^{p^r}))_{x \in \mathrm{r}(B)}$, and let $\mathfrak{m} = I/\check{J}_r$ be the augmentation ideal of B_r; since $\check{J}_r \subset I^2$, then $\omega(r) = \mathfrak{m}/\mathfrak{m}^2$ is identified with ω. For $n = 1, \cdots, r$, denote by $\omega(r)_n$ the image in $\omega(r)$ of $\mathfrak{m}^n$; it is

[849] N.D.E.: Since k is perfect, one can identify ${}^\pi V$ with the abelian group V on which k acts by $\lambda \cdot v = \lambda^{1/p} v$.
[850] N.D.E.: We have added this remark, used in 5.2.5.

also the image in ω of $x \in I | x^{p^n} \in \check{J}_r$, hence $\omega(r)_n$ contains ω_n. On the other hand, it follows from isomorphism (1) that one has $\check{J}_r = ((x_{r+1}^{p^r}))_{x_{r+1} \in S_{r+1}}$, whence

$$(2) \qquad B_r \simeq (k[[S_1]]/((x_1^p))_{x_1 \in S_1}) \hat{\otimes}_k \cdots \hat{\otimes}_k (k[[S_r]]/((x_r^{p^r}))_{x_r \in S_r}) \hat{\otimes}_k (k[[S$$

and therefore, by what precedes, $\omega(r)_n$ is identified with $\prod_{i=1}^n S_i$ for $n = 1, \cdots, r-1$. One thus obtains that the inclusion $\omega_n \subset \omega(r)_n$ is an equality, for $n = 1, \cdots, r-1$.

5.2.5. It remains to consider the case where B is of infinite height, and where the projection $I \to \omega$ possesses a section σ compatible with the filtrations of ω and I. Consider the morphism

$$\varphi : B' = k[[\omega]] / ((x^{p^n}))_{x \in \omega_n} \to B$$

induced by σ; it suffices to show that for every $r \in \mathbb{N}$, the map $\phi_r : B'_r \to B_r$ induced by ϕ is invertible.[851]

For every $r \in \mathbb{N}^*$, denote by G_r the kernel of the iterated Frobenius morphism $G \to G^{(p^r)}$ and by HG_r the formal subgroup of G inverse image of the formal subgroup $H^{(p^r)}$ of $G^{(p^r)}$, so that HG_r is defined by the closed ideal generated by the p^r-th powers of elements of AI, which equals $\check{A}J_r$, where $\check{J}_r = x^{p^r} | x \in I$. One obtains then, exactly as in 5.2.3, that $B_r = B/B\check{J}_r$ is the affine algebra of HG_r/H (and is of course of height $\leqslant r$).

Denote by $\mathfrak{m}(r) = I/B\check{J}_r$ the augmentation ideal of B_r; the canonical projection of B onto B_r obviously induces an isomorphism of $\omega = I/I^2$ onto $\omega(r) = \mathfrak{m}(r)/\mathfrak{m}(r)^2$, which allows us to identify these two spaces. Let $\omega(r)_n$ be the image in $\omega(r)$ of the closed ideal $\mathfrak{m}(r)_n = y \in \mathfrak{m}(r) | y^{p^n} = 0$; it is also the image in ω of the closed ideal $I(r)_n = x \in I | x^{p^n} \in B\check{J}_r$.[852] It is clear that $\omega_n(r) = \omega$ if $n \geqslant r$; let us show that $\omega_n(r) = \omega_n$ if $n < r$. For every r, n, the sequence below is exact:

$$0 \to I(r)_n \cap I^2 \to I(r)_n \to \omega(r)_n \to 0.$$

Moreover, for n fixed, one has $\bigcap_r I(r)_n = I_n$, since $\bigcap_r B\check{J}_r = 0$. Since in $PC(k)$ filtered inverse limits are exact (cf. 0.2), it follows that, for every n, one has

$$(*) \omega_n = \bigcap_r \omega(r)_n.$$

On the other hand, by Remark 5.2.4.A, one has $\omega(r)_n = \omega(r+1)_n$ if $n < r$. Combined with $(*)$, this entails that $\omega(r)_n = \omega_n$ if $n < r$.

Consequently, the vector space ω filtered by the subspaces $(\omega(r)_n)_{n \geqslant 0}$ is none other than ${}^r\omega$ (Notations 5.2). A fortiori, the map $\sigma(r)$ composed of $\sigma : \omega \to I$ and of the projection $I \to \mathfrak{m}(r)$ is compatible with the filtrations $(\omega(r)_n)$ and $(\mathfrak{m}(r)_n)$ of ω and $\mathfrak{m}(r)$. Since $k[[{}^r\omega]]/((x^{p^n}))_{x \in {}^r\omega_n}$ is none other than B'_r and $\phi_r : B'_r \to B_r$ is

[851] N.D.E.: Indeed, B' (resp. B) is the inverse limit of the B'_r (resp. B_r). On the other hand, we have modified the original in what follows, taking account of the addition made in 5.2.3.
[852] N.D.E.: We have detailed the original in what follows.

the morphism induced by $\sigma(r)$, the result already established for algebras of finite height shows that ϕ_r is an isomorphism.

5.2.6. Definition 5.2.6.[853] *Let $(A_\lambda)_{\lambda\in\Lambda}$ be a family of profinite k-algebras, each endowed with an augmentation $\epsilon_\lambda : A_\lambda \to k$ (this is the case in particular if each A_λ is local with residue field k). One defines then the infinite completed tensor product*

$$**A** = \hat{\otimes}^c_{\lambda \in \Lambda} A_\lambda$$

as the inverse limit in Alp/k of the $**A**_F = \hat{\otimes}^c_{\lambda\in F} A_\lambda$, for F running through the finite subsets of Λ, the transition morphisms $**A**_{F'} \to **A**_F$, for $F' = F \cup \lambda$, being $id \hat{\otimes} \epsilon_\lambda$.

In particular, if $\Lambda = \mathbb{N}^*$ and if one denotes by X_n the k-formal variety $Spf(A_n)$, then $Spf(**A**)$ represents the functor that to every $C \in Alf/k$ associates the set of "finite" sequences of elements of $\prod_{n\geqslant 1} X_n(C)$, i.e., sequences

$$(x_1, x_2, \dots) \in \prod_{n \geqslant 1} X_n(C)$$

such that $x_n = \epsilon_n$ for n large enough, where ϵ_n denotes, by abuse of notation, the composite of $\epsilon_n : A_n \to k$ and of the structure morphism $k \to C$. (If moreover each A_n is a quotient of an algebra $k[[\omega'_n]]$, one can denote by 0 the unique morphism $A_n \to C$ that vanishes on ω'_n, and one therefore obtains the set of sequences such that "$x_n = 0$" for n large enough.)

5.3.

Remarks. *(a) Let us call **stationary** every profinite k-algebra that is the completed tensor product of a family of truncated formal power series algebras.*[854]

If G is an infinitesimal k-formal group and B the affine algebra of a homogeneous space of G, it follows from Theorem 5.2 that the algebra $B/((x^{p^r}))_{x\in\mathrm{r}(B)}$ is stationary for every integer $r \in \mathbb{N}$. This implies in particular that B is an inverse limit of stationary algebras.[855]

[853] N.D.E.: We have added this number, in order to define the infinite tensor products used in 5.3 (a).

[854] N.D.E.: The author no doubt had in mind a tensor product $**A** = \hat{\otimes}^c_{n\in\mathbb{N}^*} k[[\omega'_n]]/((x^{p^n}))_{x\in\omega'_n}$, where the ω'_n are arbitrary pseudocompact k-vector spaces. In this case, one sees without difficulty that $\omega = \omega_{**A**/k}$ is identified with the product of the ω'_n, and the filtration $(\omega_n)_{n\in\mathbb{N}^*}$ is given by $\omega_n = \prod_{i=1}^n \omega'_i$, i.e., one is in case 5.2.1.B. For this reason, it would be preferable to name these algebras **stable** (rather than "stationary"), cf. [Di73], II § 2.9, p. 75. If for example $\dim \omega'_n = 1$ for every $n \in \mathbb{N}^*$, then $**A**$ represents the functor that to every $C \in Alf/k$ associates the set of sequences $(x_n)_{n\in\mathbb{N}^*}$ of elements of C such that $x_n^{p^n} = 0$, and $x_n = 0$ for n large enough. Let us note finally that this case (i.e., the case where $\omega = \prod_{n\in\mathbb{N}^*} \omega'_n$) corresponds to the case studied, in the dual situation of connected cocommutative Hopf algebras, by M. E. Sweedler, cf. [Sw67], Th. 3.

[855] N.D.E.: But such an inverse limit is not necessarily a stable profinite k-algebra (in the sense of the previous N.D.E.). For example, let S be the "ordinary" k-vector space of sequences $(u_1, u_2, \cdots)$ of elements of k and let $\omega = S^*$; then ω is the direct sum in $PC(k)$ of copies k_n of k, for $n \in \mathbb{N}^*$, i.e., one is in case 5.2.1.A. If one denotes by x_n the element of ω defined by $x_n(u) = u_n$, for every sequence $u = (u_i)_{i\in\mathbb{N}^*}$, then the k-algebra $**A** = k[[\omega]]/((x_n^{p^n}))_{n\in\mathbb{N}^*}$ is such that $\omega_{**A**/k} = \omega$ and $\omega_n = \prod_{i=1}^n kx_i$, but is not stable: $Spf(**A**)$ represents the functor that to every $C \in Alf/k$ associates

(b) I do not know if, with the notations of 5.2.1, one can choose A and B in such a way that there exists no section $\sigma : \omega \to I$ compatible with the filtrations.[856] *One will however note that one can have for ω any pseudocompact vector space filtered by an increasing sequence of closed subspaces. Indeed, if $\omega_1 \subset \omega_2 \subset \cdots \subset \omega$ is such a filtered space, one can define in the algebra $B = A = k[[\omega]]/((x^{p^r}))_{x\in\omega_r}$ a diagonal morphism $\Delta_A : A \to A\hat{\otimes}_k A$ satisfying conditions (i), (ii), (iii) of 2.1; it suffices to set $\Delta_A(y) = y\hat{\otimes}1 + 1\hat{\otimes}y$ when y is the image in A of an element of ω.*

5.4.

Corollary 5.4. *Let G be an algebraic group over a perfect field k of characteristic $p > 0$, H an algebraic subgroup of G, e the image of the neutral element of G in G/H, and A the local algebra of G/H at e. Then $\hat{A}$ is isomorphic to an algebra of the form*

k[[X_1, …, X_r, … X_s]] / (X_1^{p^{n_1}}, …, X_r^{p^{n_r}}).

Indeed, consider the infinitesimal formal groups $\hat{G} = Spf(\hat{O}_{G,e})$ and $\hat{H} = Spf(\hat{O}_{H,e})$; by 1.3.4, the completion Â of $A = O_{G/H,e}$ is isomorphic to $* * A * *(\hat{G}/\hat{H})$, and the corollary therefore follows from Theorem 5.2 (ii).[857]

5.5. Complements.

[858] Recall the following definitions. On the one hand, one says that a noetherian local ring A is a *complete intersection* if the completion Â is a quotient of a complete regular noetherian local ring B by an ideal I generated by a regular sequence of elements of B (cf. EGA IV₄, 19.3.1). On the other hand, let $\tau : Y \hookrightarrow X$ be a closed immersion of schemes. If $y \in Y$, one says that τ is a *regular immersion at the point* y if the kernel of $O_{X,y} \to O_{Y,y}$ is generated by a regular sequence; if moreover X is locally noetherian and if τ is a regular immersion at every point, one says that τ is a regular immersion (cf. loc. cit., Prop. 16.9.10 and Déf. 16.9.2).

Corollary 5.5.1. *If G is an algebraic group over a field k, the local ring $O_{G,e}$ is a complete intersection.*

Indeed, by EGA IV₄, 19.3.4, one can assume k algebraically closed. If $car(k) = 0$, we already know that G is smooth (cf. 3.3.1 or VI_B, 1.6.1) and hence $O_{G,e}$ is a k-algebra of formal power series, by EGA IV₄, 17.5.3 (d″). If $car(k) = p > 0$, it follows from 5.4, applied to $H = e$, that $O_{G,e}$ is a complete intersection.

Remarks 5.5.2. *Let k be a field, G a smooth k-algebraic group, and H a closed subgroup of G.*

a) We saw in Exp. III, 4.15, that the immersion $H \hookrightarrow G$ is regular; this can also be deduced from 5.4, as follows. As in loc. cit., one can assume k algebraically closed,

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the set of "infinite" sequences $(x_n)_{n\in\mathbb{N}^*}$ of elements of C such that $x_n^{p^n} = 0$ for every $n \in \mathbb{N}^*$.

[856] N.D.E.: The editors do not know either, outside the cases considered in 5.2.1.A and B.

[857] N.D.E.: See also [DG70], § III.3, Th. 6.1.

[858] N.D.E.: We have added this subsection, in order to give some consequences of 5.4, mentioned in Exposés III and VI_A.

and it suffices to show that the kernel I of $O_{G,e} \to O_{H,e}$ is generated by a regular sequence. Set $A = O_{G,e}$, $\hat{G} = Spf(\hat{A})$ and $\hat{H} = Spf(\hat{O}_{H,e})$. Since A is noetherian, one has an exact sequence

$$0 \to I \otimes_A \hat{A} \to \hat{A} \xrightarrow{\pi} A(\hat{H}) \to 0$$

and Â is faithfully flat over A. Hence, by EGA IV_4, 16.9.10 (ii) and 19.1.5 (ii), it suffices to show that the kernel $\hat{I} = I \bigotimes_A \hat{A}$ of π is generated by a regular sequence of elements of Â.

Now, since G is smooth, Â is reduced; by 5.4, the subalgebra $B = **A**(\hat{G}/\hat{H})$ is therefore isomorphic to a formal power series algebra $k[[x_1, \cdots, x_n]]$, and hence the unit section of $\hat{G}/\hat{H}$ is defined in B by the regular sequence $(x_1, \cdots, x_n)$. Since Â is noetherian, the ideal J of Â generated by $x_1, \cdots, x_n$ is closed, hence equal to Î, by Corollary 1.4. Moreover, since Â is topologically flat, hence flat over B (cf. 0.3.8), then $(x_1, \cdots, x_n)$ is a regular sequence in Â, by EGA IV_4, 19.1.5 (ii).

b) One can also deduce from 5.2 (ii) the following more precise result. Suppose k perfect. By 5.2 (ii) applied to the algebra $C = **A**(H)$, there exists a basis $(y_1, \cdots, y_{r+s})$ of ω_H and integers $1 \leqslant n_1 \leqslant \cdots \leqslant n_r$ such that $**A**(H)$ is isomorphic to the quotient of $k[[y_1, \cdots, y_{r+s}]]$ by the ideal generated by the $y_i^{p^{n_i}}$ for $i = 1, \cdots, r$. Lift the y_i to elements x_i of ω_G and complete $(x_1, \cdots, x_{r+s})$ into a basis $(x_1, \cdots, x_n)$ of ω_G. Since $**A**(G)$ is reduced, the morphism $k[[x_1, \cdots, x_n]] \to **A**(G)$ is an isomorphism, by 5.2 (iii). One thus obtains: *there exists a "system of coordinates" $(x_1, \cdots, x_n)$ of G (i.e., an isomorphism $**A**(G) \simeq k[[x_1, \cdots, x_n]]$) such that H is defined by the equations $x_i^{p^{n_i}} = 0$ for $i = 1, \cdots, r$ and $x_i = 0$ for $i > r + s$* ("Dieudonné's theorem", compare with [Di55], § 19, Th. 6 and [Di73], II § 3.2, Prop. 3 and what precedes it).

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Exposé VIII. Diagonalizable groups

by A. Grothendieck

> Version 1.1 of 8 November 2009: additions in 1.2, 1.4, 1.7, 3.1, 3.4, 4.5.1, 6.4, 6.8 — 1.5.1 and section 7 to be revised.[860]

1. Biduality

Let C be a category, which we identify, as is usual, with a full subcategory of $\hat{C} = \mathrm{Hom}(C^{\circ}, (Ens))$ (cf. Exp. I). Let I be a commutative group functor, i.e. an object of $\hat{C}$ endowed with a commutative group structure (cf. I, 2.1).[861] For every $X \in Ob(\hat{C})$, the object $\mathrm{Hom}(X, I)$ carries a commutative group structure induced by that of I. For every group G in $\hat{C}$, let

$$D(G) = \mathrm{Hom}_{gr.}(G, I)$$

be the subobject of $\mathrm{Hom}(G, I)$ defined, for every $S \in Ob(C)$, by

$$(*) D(G)(S) = \mathrm{Hom}_{S\text{-}gr.}(G_S, I_S),$$

[860] Version notice of the 2009 re-edition.

[861] N.D.E.: The original has been expanded in what follows.

where $G_S = G \times S$ and $I_S = I \times S$ are considered as S-groups, i.e. groups in $\hat{C}/S$. Then $D(G)$ is a $\hat{C}$-subgroup of $\mathrm{Hom}(G, I)$. In this way one obtains a contravariant functor D from the category of $\hat{C}$-groups to the category of commutative $\hat{C}$-groups.

The right-hand side of $(*)$ can also be interpreted as the subset of $\mathrm{Hom}(G \times S, I)$ consisting of morphisms $G \times S \to I$ which are "multiplicative with respect to the first argument G". Moreover, the preceding formulas remain valid more generally when S is an arbitrary object of $\hat{C}$, not necessarily coming from C.

If we now take for S a group in $\hat{C}$, which we shall denote G', then in the left-hand side $\mathrm{Hom}(G', D(G))$ of $(*)$ we can single out the subset $\mathrm{Hom}_{gr}.(G', D(G))$ consisting of morphisms that respect the group structures of G' and $D(G)$.

It then corresponds to the subset of $\mathrm{Hom}(G \times G', I)$ consisting of morphisms that are multiplicative with respect to the first and with respect to the second argument — which one may call *bilinear morphisms from $G \times G'$ to I*, or *pairings of G and G' with values in I*. One thus finds

$$(**) \quad \mathrm{Hom}_{gr.}(G', D(G)) \simeq \mathrm{Hom}_{bil.}(G \times G', I),$$

which is an isomorphism functorial in the pair (G, G'). Since the right-hand side is symmetric in G and G', one deduces a functorial bijection

$$(***) \quad \mathrm{Hom}_{gr.}(G', D(G)) \simeq \mathrm{Hom}_{gr.}(G, D(G')).$$

In other words, "it amounts to the same thing" to give a group homomorphism $G' \to D(G)$ or a group homomorphism $G \to D(G')$, both reducing in effect to the datum of a pairing $G \times G' \to I$.

Applying this to the case $G' = D(G)$ and to the identity homomorphism $G' \to D(G)$, one finds a canonical homomorphism

$$(****) G \to D(D(G)).$$

Definition 1.0.[862] *We say that G is* reflexive *(relative to I) if the preceding homomorphism is an isomorphism. We note that this implies that G is commutative.*

One thus sees that:

Proposition 1.0.1. *The functor D induces an anti-equivalence of the category of reflexive $\hat{C}$-groups with itself.*

In particular, if G, H are two reflexive groups, D induces an isomorphism

$$\mathrm{Hom}_{gr.}(G, H) \simeq \mathrm{Hom}_{gr.}(D(H), D(G))$$

(it even suffices that H be reflexive, as one sees from formula $(***)$).

Definition 1.0.2. *As usual, we shall then say that a C-group G is* reflexive *if it is reflexive as a $\hat{C}$-group (without worrying whether $D(G)$ is representable or not).*

[862] N.D.E.: We have added the numbering 1.0, 1.0.1, ... in order to highlight the definitions and statements that occur there.

One thus obtains, by D, an anti-equivalence of the category of reflexive C-groups G such that $D(G)$ is representable, with itself.

Remark 1.0.3. *To conclude these generalities, let us point out that the formation of duals $D(G)$ commutes with base extension, which therefore transforms reflexive groups into reflexive groups.*

We shall be interested henceforth in the case where $C = (Sch)/S$, the category of preschemes over S, and $I = G_{m,S}$, the "multiplicative group over S" (cf. Exp. I). For every ordinary group M, we consider the S-group M_S. One sees at once that for every prescheme in groups J over S, there is a canonical isomorphism (functorial in M and J, and compatible with base extension):

Hom_{S-gr.}(M_S, J) = Hom_gr.(M, J(S)).

Applying this to $J = I = G_{m,S}$ and to a variable S' over S, one finds a functorial isomorphism:

$$(1.0.4) D(M_S)(S') \xrightarrow{\sim} \mathrm{Hom}_{gr.}(M, G_m(S')).$$

One thus recovers the functor already considered in I, 4.4, also denoted $D_S(M)$, which is representable for M commutative, since

D_S(M) = D(M_S) = Spec O_S(M),

where $O_S(M)$ denotes the algebra of the group M with coefficients in O_S. (Let us note moreover, in the general case, that $D(M)$ does not change if one replaces M by its abelianization, so that no information is lost by assuming M commutative.)

Definition 1.1. *A prescheme in groups G over S is said to be* diagonalizable *if it is isomorphic to a scheme of the form $D_S(M) = D(M_S) = \mathrm{Hom}_{S-gr.}(M_S, G_m)$ for some suitable commutative group M.*

We say that G is locally diagonalizable *if every point of S admits an open neighborhood U such that $G|U$ is diagonalizable.*

Theorem 1.2. *Let Γ be a constant commutative group scheme over S, i.e. isomorphic to a group scheme of the form M_S, where M is an ordinary commutative group. Then Γ is reflexive, i.e. the canonical homomorphism*

$$\Gamma \to D(D(\Gamma))$$

is an isomorphism.[863] *The diagonalizable group $D(M_S)$ is therefore also reflexive.*

Taking the definitions into account, this follows from the following statement (which one will apply to a prescheme S' over S):

Corollary 1.3. *Let $G = D(M_S)$. Then every homomorphism of S-groups*

[863] N.D.E.: We have added the sentence that follows.

$$u : G \to G_{m,S}$$

is defined by a uniquely determined section of M_S *over S, i.e. by a uniquely determined locally constant map from S to M.*

Proof. Since by definition

$$G_{m,S} = GL(1)_S = \mathrm{Aut}_{O_S-mod.}(O_S),$$

one sees that giving a group homomorphism $G \to G_{m,S}$ is equivalent to giving on 0_S a structure of G-0_S-module compatible with the natural 0_S-module structure on 0_S (cf. I, 4.7). By I, 4.7.3, this also amounts to giving an M-grading on 0_S, i.e. a decomposition of 0_S as a direct sum of modules L_m ($m \in M$).

Now it is well known that a direct factor of a locally free module of finite type is locally free of finite type; therefore each L_m is, in a neighborhood of each point of S, either zero or free of rank 1, and in the latter case identical to 0_S in that neighborhood. Let S_m be the open subset of S consisting of the points where this second alternative occurs. Expressing that 0_S is the direct sum of the L_m, one sees that the union of the S_m is S, and that the S_m are pairwise disjoint. Hence giving a group homomorphism $G \to G_{m,S}$ is equivalent to giving a decomposition of S as a union of pairwise disjoint open subsets S_m ($m \in M$), i.e. to giving a locally constant map from S to M. This establishes 1.3, hence 1.2.

Corollary 1.4. *A diagonalizable group is reflexive; the same therefore holds for a locally diagonalizable group. If M, N are two ordinary commutative groups, the natural homomorphism*

Hom_{S-gr.}(M_S, N_S) → Hom_{S-gr.}(D(N_S), D(M_S))

is bijective.

[864] The preceding isomorphism being compatible with base extension, one deduces an isomorphism of S-functors in groups:

(1)   Hom_{S-gr.}(M_S, N_S) ≅ Hom_{S-gr.}(D(N_S), D(M_S)).

For every S-prescheme T, one has

Hom_{S-gr.}(M_S, N_S)(T) = Hom_gr.(M, Γ(N_T/T)),

and, by I 1.8, $\Gamma(N_T/T)$ is the abelian group of locally constant maps $T \to N$. On the other hand, let $\mathrm{Hom}_{gr}.(M,N)_S$ be the constant S-group associated with the ordinary abelian group $\mathrm{Hom}_{gr}.(M,N)$. One has an evident homomorphism of S-functors in commutative groups:

(2)   Hom_gr.(M, N)_S -θ→ Hom_{S-gr.}(M_S, N_S),

[864] N.D.E.: We have expanded what follows.

which is always a monomorphism. Moreover, it is an isomorphism if M is of finite type.[865]

From the foregoing one deduces point (a) of the following corollary; point (b) follows from it by the descent results "recalled" in 1.7.[866]

Corollary 1.5. *a) Let M, N be two ordinary commutative groups, with M of finite type. Then one has an isomorphism*

$$\mathrm{Hom}_{gr.}(M, N)_S \xrightarrow{\sim} \mathrm{Hom}_{S\text{-}gr.}(D(N_S), D(M_S));$$

consequently $\mathrm{Hom}_{S-gr.}(D(N_S), D(M_S))$ *is representable.*

b) More generally, if G, H are locally diagonalizable, with H of finite type, then $\mathrm{Hom}_{S-gr.}(G, H)$ *is representable.*

[867] From 1.5 one concludes:

Corollary 1.6. *Under the conditions of 1.5, if S is connected, one has*

$$\mathrm{Hom}_{S\text{-}gr.}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Hom}_{gr.}(M, N)$$

and

$$\mathrm{Isom}_{S\text{-}gr.}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Isom}_{gr.}(M, N).$$

1.7. Descent of representability

[868] In this paragraph we "recall" some descent results that will be used frequently in what follows.

Scholium 1.7.1. *Let S be a prescheme and X, Y, T S-preschemes. If (T_i) is an open covering of T, and if we set $T_{ij} = T_i \cap T_j = T_i \times_T T_j$, then, since giving a morphism of T-preschemes $X_T \to Y_T$ is local on T, one has an exact sequence of sets:*

$$(1) \qquad \mathrm{Hom}_T(X_T, Y_T) \to \prod_i \mathrm{Hom}_{T_i}(X_{T_i}, Y_{T_i}) \rightrightarrows \prod_{i,j} \mathrm{Hom}_{T_{ij}}(X_{T_{ij}}, Y_{T_{ij}})$$

i.e. $\mathrm{Hom}_S(X, Y)$ *is a local S-functor, that is, a sheaf on $(Sch)/S$ endowed with the Zariski topology.*

[865] N.D.E.: Indeed, if one denotes by $F(M)$ (resp. $G(M)$) the left-hand (resp. right-hand) side of (2), and if $M = M_1 \oplus M_2$, one has a canonical isomorphism $F(M) = F(M_1) \oplus F(M_2)$, and likewise for G. This reduces us to verifying that θ is an isomorphism when $M = \mathbb{Z}/r\mathbb{Z}$, for an integer $r \geqslant 0$. In this case, $F(M) = ({}_rN)_S$, where ${}_rN$ is the kernel of $r \cdot id_N$, and, for every $T \to S$, the homomorphism

$$F(M)(T) = \Gamma({}_rN_T/T) \to G(M)(T) = {}_r\Gamma(N_T/T)$$

is bijective, whence the desired result.

[866] N.D.E.: The original stated after 1.5: "One concludes more generally that if G, H are locally diagonalizable, with H of finite type, then $\mathrm{Hom}_{S-gr.}(G, H)$ is representable." We have included this assertion in the statement of 1.5, and have made its proof explicit in 1.7.

[867] N.D.E.: One should add a statement 1.5.1 treating the functor $Isom_{S-gr.}(G, H)$, considered in X 5.10 and 5.11...

[868] N.D.E.: We have added this paragraph, to make explicit the "local on S" character of the representability of certain sheaves on S, used many times in the sequel (and implicitly in the original).

More generally, by IV 4.5.13, $\mathrm{Hom}_S(X,Y)$ *is a sheaf on* $(Sch)/S$ *for every topology coarser than the canonical topology — for example, for the (fpqc) topology.*

If G, H *are* S*-preschemes in groups, one deduces that the subfunctor* $\mathrm{Hom}_{S-gr.}(X,Y)$ *is a sheaf for the (fpqc) topology (hence a fortiori a local functor).*

Lemma 1.7.2.[869] *Let* F *be a local* S*-functor.*

(i) Suppose there exists an open covering (S_i) *of* S *such that the restriction* $F_i = F \times_S S_i$ *of* F *to each* S_i *is representable by an* S_i*-prescheme* X_i*. Then* F *is representable by an* S*-prescheme* X*.*

(ii) Suppose F *is an (fpqc) sheaf and that there exists a faithfully flat quasi-compact morphism* $S' \to S$ *such that the restriction* $F' = F \times_S S'$ *of* F *is representable by an* S'*-prescheme* X'*. Then* X' *is endowed with a descent datum (cf. IV 2.1) relative to* $S' \to S$*.*

If moreover this descent datum is effective (which is the case if X' *is affine over* S'*), then* F *is representable by an* S*-prescheme* X*.*

Proof. (i) It follows from the hypothesis that $X_i \times_S S_j$ and $X_j \times_S S_i$ both represent the restriction of F to $S_{ij} = S_i \times_S S_j$, hence, by the Yoneda lemma, there is a unique isomorphism of S_{ij}-preschemes

$$c_{ji} : X_i \times_S S_j \xrightarrow{\sim} X_j \times_S S_i;$$

one then has isomorphisms of preschemes over $S_{ijk} = S_i \times_S S_j \times_S S_k$:

$$X_i \times_S S_j \times_S S_k \xrightarrow{c_{ji} \times \mathrm{id}_{S_k}} X_j \times_S S_i \times_S S_k \cong X_j \times_S S_k \times_S S_i \xrightarrow{c_{kj} \times \mathrm{id}_{S_i}} X_k \times_S S_j \times_S S_i$$

$$X_i \times_S S_k \times_S S_j \xrightarrow{c_{ki} \times \mathrm{id}_{S_j}} X_k \times_S S_i \times_S S_j$$

and since all these objects represent the restriction of F to S_{ijk}, this diagram is commutative, i.e. the c_{ji} satisfy the usual cocycle relation $c_{kj} \circ c_{ji} = c_{ki}$.

It follows that the X_i glue together into an S-prescheme X such that $X \times_S S_i = X_i$ for every i. For every Y over S_i, one therefore has

$$(*) \qquad F(Y) = F_i(Y) = \mathrm{Hom}_{S_i}(Y, X \times_S S_i) = \mathrm{Hom}_S(Y, X) = h_X(Y).$$

Then, for $Y \to S$ arbitrary, the $Y_i = Y \times_S S_i$ form an open covering of Y; set $Y_{ij} = Y_i \times_Y Y_j = Y \times_S S_{ij}$. Since F (resp. h_X) is a local functor by hypothesis (resp. because the Zariski topology is coarser than the canonical topology), $F(Y)$ and $h_X(Y)$ both identify, in view of $(*)$, with the kernel of the double arrow:

$$\prod_i F(Y_i) \rightrightarrows \prod_{i,j} F(Y_{ij})$$

$$\prod_i h_X(Y_i) \rightrightarrows \prod_{i,j} h_X(Y_{ij}).$$

This proves (i).

[869] N.D.E.: See also Remark XI 3.4.

(ii) It follows from the hypothesis that $F''_1 = F' \times_{S'} S''_1$ (where $S''_1 = S'' = S' \times_S S'$ is considered as S'-prescheme via the first projection) is represented by $X''_1 = X' \times_{S'} S''_1$; similarly, $F''_2 = F' \times_{S'} S''_2$ is represented by $X''_2 = X' \times_{S'} S''_2$. Now $F''_1 = F \times_S S'' = F''_2$, hence there exists a (unique) S''-isomorphism $c : X''_1 \xrightarrow{\sim} X''_2$. Then, denoting by q_i (resp. p_{ji}) the projection of $S''' = S' \times_S S' \times_S S'$ onto the i-th factor (resp. onto the factors i and j), $X'''_i = X' \times_{S'} S'''_i$ (where $S'''_i = S'''$ considered as S'-prescheme via q_i), and $p^*_{ji}(c) : X'''_i \xrightarrow{\sim} X'''_j$ the isomorphism of S'''-preschemes deduced from c by base change, one obtains a diagram of isomorphisms of S'''-preschemes:

$$\begin{tikzcd} X'''_1 \arrow[r, "p_{21}^*(c)"] \arrow[dr, "p_{31}^*(c)"'] & X'''_2 \arrow[d, "p_{32}^*(c)"] \\ & X'''_3 \end{tikzcd}$$

and since all these objects represent the restriction of F to S''', this diagram is commutative, i.e. the usual cocycle relation $p^*_{32}(c) \circ p^*_{21}(c) = p^*_{31}(c)$ holds, i.e. c is a descent datum on X' relative to $S' \to S$ (cf. IV 2.1).

Suppose moreover that this descent datum is effective, i.e. that there exists an S-prescheme X such that $X' \simeq X \times_S S'$ (by SGA 1, VIII 2.1, this is the case if X' is affine over S'[870]). Then, for every $Y \to S'$, one has

$$(**) \qquad F(Y) = F'(Y) = \mathrm{Hom}_{S'}(Y, X \times_S S') = \mathrm{Hom}_S(Y, X) = h_X(Y).$$

Then, for $Y \to S$ arbitrary, set $Y' = Y \times_S S'$ and $Y'' = Y' \times_Y Y' \simeq Y \times_S S''$. Then $Y' \to Y$, like $S' \to S$, is faithfully flat and quasi-compact, hence an M-effective epimorphism (where M = family of faithfully flat quasi-compact morphisms), i.e. the equivalence relation

$$Y' \times_Y Y' \rightrightarrows Y'$$

is M-effective and has quotient Y. Since F (resp. h_X) is an (fpqc) sheaf by hypothesis (resp. because the (fpqc) topology is coarser than the canonical topology), $F(Y)$ and $h_X(Y)$ both identify, in view of $(**)$, with the kernel of the double arrow

$$F(Y') \rightrightarrows F(Y' \times_Y Y')$$

$$h_X(Y') \rightrightarrows h_X(Y' \times_Y Y').$$

This proves (ii).

Corollary 1.7.3. *Let F be an (fpqc) sheaf on $(Sch)/S$. Suppose there exists an open covering (S_i) of S and, for each i, a faithfully flat quasi-compact morphism $S'_i \to S_i$ such that the restriction $F'_i = F \times_S S'_i$ is representable by an S'_i-prescheme X'_i affine over S'_i. Then F is representable by an S-prescheme X affine over S (such that $X \times_S S'_i = X'_i$ for every i).*

[870] N.D.E.: For another effectivity criterion, see later X 5.4–5.6.

If moreover each $X'_i \to S'_i$ is a closed immersion (resp. a finite étale morphism), the same holds for $X \to S$.

The first assertion follows from 1.7.2. For the second, it suffices to verify that each morphism $X \times_S S_i \to S_i$ is a closed immersion (resp. finite and étale), which follows from EGA IV_2, 2.7.1 (resp. and IV_4, 17.7.3).

Remark 1.7.4. *Assertion 1.5 (b) follows, as announced, from 1.7.1 and 1.7.2 (i).*

2. Schematic properties of diagonalizable groups

They are summarized in the following.

Proposition 2.1. *Let S be a non-empty prescheme, M an ordinary commutative group, $G = D(M_S)$ the diagonalizable S-group defined by M. Then:*

a) G is faithfully flat over S, and affine over S (a fortiori quasi-compact over S).

b) M of finite type $\Leftrightarrow$ G of finite type over S $\Leftrightarrow$ G of finite presentation over S.

c) M finite $\Leftrightarrow$ G finite over S $\Leftrightarrow$ G of finite type over S and annihilated by an integer $n > 0$. Then $deg(G/S) = Card(M)$.

c′) M a torsion group $\Leftrightarrow$ G integral over S.

d) $M = 0 \Leftrightarrow G =$ unit S-group.

e) M of finite type, and the order of its torsion subgroup is prime to the residue characteristics of S $\Leftrightarrow$ G is smooth over S.

The verification of (a) to (d) is trivial, and is left to the reader. Let us prove (e). If G is smooth over S, it is locally of finite presentation over S, hence of finite presentation over S since it is affine over S, hence M is of finite type. So we may already assume M of finite type, hence G of finite presentation over S. Then[871] G is smooth over S if and only if its geometric fibers are, which reduces us to the case where S is the spectrum of an algebraically closed field k. Writing $M = T \times L$, with T the torsion subgroup and L free, $L \simeq \mathbb{Z}^r$, one has $D(M) = D(T) \times D(L)$, where $D(L) \simeq G_m^r$ is smooth over k. So $G = D(M)$ is smooth over k if and only if $D(T)$ is, which means, since $D(T)$ is finite over k of degree equal to the order n of T, that $D(T)(k)$ has n elements. Now T is isomorphic to a sum of groups $\mathbb{Z}/n_i\mathbb{Z}$, n being the product of the n_i, so $D(T)$ is the product of the $D(\mathbb{Z}/n_i\mathbb{Z}) = \mu_{n_i}$ (group scheme of n_i-th roots of unity), hence

$$card(D(T)(k)) = \prod_i card\, \mu_{n_i}(k),$$

where $card \mu_{n_i}(k) =$ (number of n_i-th roots of unity in k) $\leqslant n_i$, equality being attained if and only if n_i is prime to the characteristic p of k. Hence one has $card(D(T)(k)) = n$ (where $n = \prod_i n_i$) if and only if all the n_i are prime to p, i.e. if and only if n is prime to p. QED.

[871] N.D.E.: (since G is flat over S, by (a)).

3. Exactness properties of the functor D_S

Theorem 3.1. *Let S be a prescheme, and*

$0 \to M' \xrightarrow{u} M \xrightarrow{v} M'' \to 0$

an exact sequence of ordinary commutative groups. Consider the sequence of transposed homomorphisms:

$0 \to D_S(M'') \xrightarrow{v^t} D_S(M) \xrightarrow{u^t} D_S(M') \to 0.$

(i) v^t induces an isomorphism of $D_S(M'')$ with the kernel of u^t, and u^t is faithfully flat and quasi-compact.

(ii)[872] *$D_S(M')$ represents the (fpqc) quotient sheaf $D_S(M)/D_S(M'')$.*

Let M denote the family of faithfully flat quasi-compact morphisms. First, (ii) follows from (i) (cf. IV, 4.6.5.1). Indeed, the equivalence relation in $D_S(M)$ defined by u^t is the same as that defined by the subgroup $Ker(u^t) = D_S(M'')$; since $u^t \in M$, this equivalence relation is M-effective (cf. IV, 3.3.2.1), and therefore $D_S(M')$ represents the quotient sheaf for the (fpqc) topology (cf. IV, 4.6.5).

The first assertion of (i) is a trivial consequence of the definition of the functors $D_S(-)$; more generally, for any exact sequence

$$M' \to M \to M'' \to 0$$

(without zero on the left), one will have a transposed exact sequence:

$$0 \to D_S(M'') \to D_S(M) \to D_S(M').$$

(This is valid more generally in the context of the beginning of §1.) On the other hand, since $D_S(M)$ and $D_S(M')$ are affine over S, u^t is necessarily an affine morphism, a fortiori quasi-compact (whatever the homomorphism $u : M' \to M$). The second assertion of (i) will therefore follow from point (a) of the following:

Corollary 3.2. *Let S be a non-empty prescheme, $u : M' \to M$ a homomorphism of ordinary commutative groups, $u^t : G \to G'$ the transposed homomorphism. Then:*

a) For u to be a monomorphism, it is necessary and sufficient that u^t be faithfully flat.

b) For u to be an epimorphism, it is necessary and sufficient that u^t be a monomorphism (and then u^t is even a closed immersion).

To prove (a), one notes that if u is a monomorphism, then $O_S(M)$ is a module over $O_S(M')$ admitting a non-empty basis (namely, the system of sections defined by any system of representatives of M modulo M'), a fortiori it is faithfully flat. Conversely,

[872] N.D.E.: We have added what follows, and have detailed the proof accordingly.

if this is the case, then $u^t : O_S(M') \to O_S(M)$ is injective, which (for $S \neq \emptyset$) implies that $u : M' \to M$ is injective.

To prove (b), one notes that if u is an epimorphism, then $O_S(M') \to O_S(M)$ is surjective, hence u^t is a closed immersion and a fortiori a monomorphism. Conversely, if this is the case, then $Ker u^t$ = unit group; now setting $M'' = Coker u$, we have seen that $Ker u^t \simeq D_S(M'')$, hence by 2.1 (d) one has $M'' = 0$, hence u is an epimorphism.

One concludes from 3.1 in the usual way:

Corollary 3.3. *Let $M' - u \to M - v \to M''$ be an exact sequence of ordinary commutative groups; consider the transposed sequence*

$$G'' \xrightarrow{v^t} G \xrightarrow{u^t} G'.$$

Then v^t induces a faithfully flat quasi-compact morphism from G'' to $Ker u^t$, and the latter is a diagonalizable group isomorphic to $D_S(v(M)) = D_S(Coker u)$.

Corollary 3.4. *Let S be a prescheme, $u : G \to H$ a homomorphism of S-preschemes in locally diagonalizable groups, with H of finite type over S. Set $G' = Ker u$. Then:*

a) G' is locally diagonalizable; it is of finite type over S if G is.

b) The quotient G/G' "exists"; more precisely, the equivalence relation defined by G' in G is M-effective (where M = set of faithfully flat quasi-compact morphisms, cf. IV, 3.4). Moreover, G/G' is locally diagonalizable, of finite type over S.

c) The homomorphism $u : G \to H$ factors uniquely as

$$G \xrightarrow{v} G/G' \xrightarrow{w} H,$$

where v is the canonical homomorphism (so v is faithfully flat and quasi-compact). Moreover w is a closed immersion, and a fortiori a monomorphism.

Finally, the quotient $H' = H/Im w = Coker w = Coker u$ exists; more precisely, the equivalence relation defined by G/G' in H is M-effective, and H' is of finite type over S.

The first assertion of (c) is a consequence of (b), by definition of the quotient G/G' (cf. IV, 3.2.3).[873] Let us show that the (fpqc) quotient sheaf $\tilde{G}/\tilde{G}'$ is representable. This may be verified locally on S, as may all the other assertions; we may therefore suppose G and H diagonalizable, of the form $D_S(M)$ and $D_S(N)$.

Since H is of finite type over S, N is of finite type by 2.1 (b), hence by 1.5 u is defined by a homomorphism $u_0 : N \to M$. Then, in virtue of 3.1 and 3.2, G' is isomorphic to $D_S(Coker u_0)$ and $\tilde{G}/\tilde{G}'$ is representable by $D_S(Im u_0)$; moreover, considering the exact sequence

$$0 \to Ker\, u_0 \to N \xrightarrow{w_0} Im\, u_0 \to 0,$$

[873] N.D.E.: The original has been expanded in what follows.

one obtains that w is a closed immersion, and that the quotient $H' = H/Imw$ is $D_S(Keru_0)$; the latter is of finite type over S since N, and hence $Keru_0$, is of finite type.

Remarks. The existence-of-quotients result 3.4 will be substantially generalized in §5.

On the other hand, one will note that in the present § and the preceding one, the hypothesis $S \neq \emptyset$ was used only to ensure the validity of certain converses, enabling one to deduce from certain hypotheses on diagonalizable S-groups properties of the corresponding ordinary groups. The "direct"-sense results are valid without restriction on S, and the proofs given here apply in the general case.

Corollary 3.5. *Let G be a diagonalizable group prescheme over S, and let n be an integer $\neq 0$. Then the subgroup ${}_{n}G$ of G, kernel of the homomorphism $n \cdot id_G : G \to G$, is integral over S, and finite over S if G is of finite type over S.*

Indeed, if $G = D_S(M)$, then ${}_{n}G = D_S(M/nM)$ by 3.1, and one concludes by 2.1 (b), (c), (c′).

4. Torsors under a diagonalizable group

Let S be a prescheme, and $G = D_S(M)$ a diagonalizable group over S. We propose to determine the G-torsors (or principal homogeneous G-bundles) on S, in the sense of the "faithfully flat quasi-compact topology" (cf. Exp. IV, 5.1). Recall that a prescheme P over S with operator group G is called a *torsor* or *principal homogeneous* if every point of S admits an open neighborhood U and a faithfully flat quasi-compact morphism $S' \to U$ such that $P' = P \times_S S'$ is, as a bundle with operators, isomorphic to $G' = G \times_S S'$ (acting on itself by right translations). Since G is affine over S, it follows from SGA 1, VIII 5.6 that P is necessarily affine over S. Note also that since G is itself faithfully flat and quasi-compact over S, P is principal homogeneous under G if and only if it is "formally principal homogeneous", and if moreover it is faithfully flat and quasi-compact over S (cf. IV, 5.1.6).

Recall on the other hand (Exp. I, 4.7.3) that giving an S-prescheme P affine over S with operator group $G = D_S(M)$ amounts to giving a quasi-coherent commutative M-graded algebra on S, i.e. a quasi-coherent algebra $\mathcal{A}$ on S endowed with a direct sum decomposition (as a module):

$$\mathcal{A} = \bigoplus_{m \in M} \mathcal{A}_m,$$

with

$$\mathcal{A}_m \cdot \mathcal{A}_{m'} \subseteq \mathcal{A}_{m+m'} \quad \text{for } m, m' \in M.$$

This said, the answer to the problem posed above is given by:

Proposition 4.1. *For the prescheme P with operator group $G = D_S(M)$, defined by the M-graded algebra $\mathcal{A}$, to be a principal homogeneous bundle under G, it is necessary and sufficient that $\mathcal{A}$ satisfy the following conditions:*

a) For every $m \in M$, $\mathcal{A}_m$ is an invertible module on S.

b) For $m, m' \in M$, the homomorphism

$$\mathcal{A}_m \otimes_{O_S} \mathcal{A}_{m'} \to \mathcal{A}_{m+m'}$$

induced by the multiplication in $\mathcal{A}$, is an isomorphism.

The necessity of the conditions is immediate by descent, since they are satisfied in the case where P is the trivial principal homogeneous bundle, i.e. $\mathcal{A} = O_S(M)$. For the sufficiency, one notes that (a) already implies that P is faithfully flat over S, it is in any case quasi-compact over S (being affine over S), so it remains to verify that it is formally principal homogeneous under G, i.e. that the well-known homomorphism

$$P \times_S G \to P \times_S P$$

is an isomorphism. Now on affine algebras, this homomorphism is given explicitly as the homomorphism

$$\mathcal{A} \otimes \mathcal{A} \to \mathcal{A}(M) = \mathcal{A} \otimes O_S(M)$$

which in bidegree (m, n) (where $m, n \in M$) is given by

$$x_m \otimes y_n \mapsto x_m y_n \otimes e_n.$$

From the standpoint of degrees, this homomorphism is compatible with the homomorphism $M \times M \to M \times M$ given by

$$(m, n) \mapsto (m + n, n),$$

which is an isomorphism. This shows that (b) expresses precisely (independently of (a)) that P is formally principal homogeneous, and establishes 4.1.

Note also that one obtains, by faithfully flat descent:

Corollary 4.2. *The conditions of 4.1 imply that the homomorphism*

$$O_S \to \mathcal{A}_0$$

is an isomorphism.

If for example $M = \mathbb{Z}$, then under the conditions of 4.1 one sees that $\mathcal{A}$ is essentially known when one knows $\mathcal{A}_1 = \mathcal{L}$, namely

$$\mathcal{A} \simeq \bigoplus_{n \in \mathbb{Z}} \mathcal{L}^{\otimes n}$$

(isomorphism of graded algebras). One thereby recovers the well-known result:

Corollary 4.3. *There is an equivalence between the category of principal homogeneous bundles P on S with group $G_{m,S}$, and the category of invertible modules $\mathcal{L}$ on S (taking as morphisms, for the definition of each category, the isomorphisms of the structures in play). One obtains two quasi-inverse functors by associating* *with each P the degree-1 component of its $\mathbb{Z}$-graded affine algebra, and with each $\mathcal{L}$ the spectrum of the $\mathbb{Z}$-graded algebra $\bigoplus_{n\in\mathbb{Z}} \mathcal{L}^{\otimes n}$.*

In particular:

Corollary 4.4. *The group of classes of principal homogeneous bundles on S with group $G_{m,S}$ is isomorphic to the group* $\mathrm{Pic}(S)$ *of classes of invertible modules on S, i.e. to $H^1(S, O_S^\times)$.*

Taking into account that $G_{m,S}$ is the scheme of automorphisms of the module O_S, one sees that 4.4 is equivalent to the following statement, which is one of the variants of Hilbert's "Theorem 90":

Corollary 4.5. *Every principal homogeneous bundle on S with group G_m is locally trivial (in the sense of the Zariski topology).*

Remark 4.5.1. *One will note that the preceding statement is no longer true in general for a group such as μ_n, or for a "twisted form" of G_m; for example, the unique twisted form of G_m over the field $\mathbb{R}$ of reals gives a group of 1-cohomology equal to $\mathbb{Z}/2\mathbb{Z}$.*

[874] Indeed, let S^1 be the kernel of the norm morphism $N : \prod_{\mathbb{C}/\mathbb{R}} G_{m,\mathbb{C}} \to G_{m,\mathbb{R}}$; this is a $\mathbb{C}/\mathbb{R}$-twisted form of $G_{m,\mathbb{R}}$. The equation $N(z) = -1$ in $\prod_{\mathbb{C}/\mathbb{R}}(G_{m,\mathbb{C}})$ defines an S^1-torsor X over $\mathrm{Spec}(\mathbb{R})$, locally trivial for the étale topology, but not trivial since $X(\mathbb{R}) = \emptyset$. Let us show that $H^1_{\acute{e}t}(\mathbb{R}, S^1) \cong \mathbb{Z}/2\mathbb{Z}$. One has an exact sequence of smooth commutative $\mathbb{R}$-group schemes:

$$1 \to S^1 \to \prod_{\mathbb{C}/\mathbb{R}} G_{m,\mathbb{C}} \to G_{m,\mathbb{R}} \to 1$$

which gives rise to a long exact sequence of étale cohomology (or of Galois cohomology):

$$0 \to S^1(\mathbb{R}) \to \mathbb{C}^\times \xrightarrow{N} \mathbb{R}^\times \to H^1_{\acute{e}t}(\mathbb{R}, S^1) \to H^1_{\acute{e}t}(\mathbb{R}, \prod_{\mathbb{C}/\mathbb{R}} G_{m,\mathbb{C}}) \to \cdots$$

Now (see for example XXIV, 8.4), $H^1_{\acute{e}t}(\mathbb{R}, \prod_{\mathbb{C}/\mathbb{R}} G_{m,\mathbb{C}}) \simeq H^1_{\acute{e}t}(\mathbb{C}, G_{m,\mathbb{C}})$, and the latter is zero by 4.5 (or, here, because $\mathbb{C}$ is algebraically closed). One thus obtains an isomorphism $H^1_{\acute{e}t}(\mathbb{R}, S^1) \simeq \mathbb{R}^\times/N(\mathbb{C}^\times) \simeq \pm 1$.

We shall need in the following § the following result:

Proposition 4.6. *Under the conditions of 4.1, conditions (a) and (b) are equivalent to the following conditions:*

a′) $O_S \to \mathcal{A}_0$ is an isomorphism.

b′) For every m in M (it suffices: in a system of generators of M), one has

$$\mathcal{A}_m \cdot \mathcal{A}_{-m} = \mathcal{A}_0.$$

The necessity being evident, taking 4.2 into account,[875] we shall reduce to proving:

[874] N.D.E.: We have added what follows.
[875] N.D.E.: We have corrected 4.1 to 4.2.

Corollary 4.7. *Let $A = \bigoplus_{n\in\mathbb{Z}} A_n$ be a $\mathbb{Z}$-graded ring such that*

$$A_1 \cdot A_{-1} = A_0.$$

Then the A_n are invertible A_0-modules, and for $n, n' \in \mathbb{Z}$, the homomorphism

$$A_n \otimes_{A_0} A_{n'} \to A_{n+n'}$$

induced by the multiplication in A, is an isomorphism.

By hypothesis, there exist $f_i \in A_1$, $g_i \in A_{-1}$, such that

$$(*) \sum_i f_i g_i = 1.$$

As the conclusion to be established is local on $\mathrm{Spec}(A_0)$,[876] and as, by $(*)$, $\mathrm{Spec}(A_0)$ is covered by the affine opens $D(f_i g_i)$, one is reduced to the case where there exists an element $f \in A_1$ invertible in A. Then for every $n \in \mathbb{Z}$, f^n is an element of A_n invertible in A, hence defines an isomorphism $h \mapsto f^n h$ from A_0 onto A_n. Moreover, this shows that one obtains an isomorphism $A_0[t, t^{-1}] \to A$ of graded A_0-algebras by sending t to f, which completes the proof of 4.7.

Then, under the conditions of 4.6, 4.7 already implies that the $\mathcal{A}_m$ $(m \in M)$ are invertible. To prove condition 4.1 (b), one may therefore suppose that $\mathcal{A}_m$ and $\mathcal{A}_{m'}$ admit bases f_m and $f_{m'}$, having inverses $f_m^{-1} \in \Gamma(\mathcal{A}_{-m})$ and $f_{m'}^{-1} \in \Gamma(\mathcal{A}_{-m'})$. Then the product by $f_m^{-1} f_{m'}^{-1} \in \Gamma(\mathcal{A}_{-m-m'})$ defines a homomorphism $\mathcal{A}_{m+m'} \to \mathcal{A}_0 \simeq \mathcal{O}_S$, sending the image of $f_m \otimes f_{m'}$ to the section 1 of $\mathcal{O}_S$. In the diagram

$$\begin{tikzcd}
\mathcal{A}_m \otimes \mathcal{A}_{m'} \arrow[r, "u"] \arrow[rr, bend left, "w"] & \mathcal{A}_{m+m'} \arrow[r, "v"] & \mathcal{A}_0 \simeq \mathcal{O}_S
\end{tikzcd}$$

w and v are therefore epimorphisms of invertible sheaves, hence isomorphisms, hence u is an isomorphism. QED.

5. Quotient of an affine scheme by a diagonalizable group acting freely

[877] We denote by M the set of faithfully flat quasi-compact morphisms, and we recall that torsors are considered in the sense of the (fpqc) topology.

Theorem 5.1. *Let S be a prescheme, M an ordinary commutative group, $G = D_S(M)$ the diagonalizable group over S defined by it, P an S-prescheme affine over S on which G acts freely on the right.*

[876] N.D.E.: We have expanded the passage that follows.
[877] N.D.E.: We have added the sentence that follows.

Then the equivalence relation defined by G in P is M-effective (cf. IV, 3.4), i.e. the quotient $X = P/G$ exists and P is a torsor over X with group $G_X = D_X(M)$. Moreover, P/G is affine over S; more precisely, if P is defined by the M-graded algebra $\mathcal{A}$, then P/G is isomorphic to $\operatorname{Spec}(\mathcal{A}_0)$*, where $\mathcal{A}_0 = \mathcal{A}^G$ is the degree-0 component of $\mathcal{A}$.*

Proof. Set $X = \operatorname{Spec}(\mathcal{A}_0)$. Then one has a natural morphism $P \to X$, deduced from $\mathcal{A}_0 \to \mathcal{A}$, which is invariant under the action of G. In this way, P becomes an X-prescheme, affine over X, with operator group $G_X = D_X(M)$, and the hypothesis that G acts freely on P/S implies that G_X acts freely on P/X. Everything reduces to showing that P is a principal homogeneous bundle under G_X, using the fact that $\mathcal{B}_0 = O_X$, where $\mathcal{B}$ is the M-graded algebra on X defining P/X. We may then suppose $X = S$ and S affine, so P is affine, given by an M-graded ring A whose homogeneous part of degree m will be denoted A_m, so that $S = \operatorname{Spec}(A_0)$. Taking 4.6 into account, it remains to verify that one has:

$$(*) \qquad A_m \cdot A_{-m} = A_0 \quad \text{for every } m \in M.$$

One observes moreover by a direct computation that $(*)$ is equivalent to saying that $P \times_S G \to P \times_S P$ is a closed immersion, and not only a monomorphism (under the hypothesis that G acts freely), i.e. that the homomorphism on affine rings

$$\theta : A \otimes_{A_0} A \to A(M)$$

is surjective[878]. This gives 5.1 when one supposes that the equivalence relation defined by G in P is closed. We shall, however, show that this hypothesis is already a consequence of the fact that G acts freely (which is moreover implicitly contained in Theorem 5.1, since $G \times_S P$ must be isomorphic to $P \times_X P$, which is closed in $P \times_S P$ since X is affine over S hence separated over S).

Let $R = P \times_S G$. The hypothesis that G acts freely, i.e. that $R \to P \times_S P$ is a monomorphism, can be written as saying that the diagonal morphism

$$R \to R \times_{(P \times_S P)} R = R_0$$

is an isomorphism. One has $R = \operatorname{Spec}(A(M))$ and

$$R_0 = \operatorname{Spec}(A(M \times M)/K)$$

[879] where K is the ideal generated by the elements of the form

$$x_m(e_{m,0} - e_{0,m}), \quad \text{with } m \in M,\ x_m \in A_m;$$

let $\phi : A(M \times M) \to A(M)$ be the surjective ring homomorphism defined by

$$x\, e_{m,n} \mapsto x\, e_{m+n} \quad (m, n \in M,\ x \in A)$$

[878] N.D.E.: Indeed, for every $b \in A$, $a_m \in A_m$, one has $\theta(b \otimes a_m) = ba_m \otimes e_m$, hence the surjectivity of θ is equivalent to $(*)$.

[879] N.D.E.: We have denoted by K the ideal denoted J in the original, in order to distinguish it from the ideals J_p of A_0 appearing in $(***)$. We have also made explicit later that the relations $(**)$ are equivalent to the equality $K = K'$ (see what follows).

(where the e_m, resp. $e_{m,n} = e_m \otimes e_n$, are the elements of the canonical basis of $A(M)$, resp. $A(M \times M)$). Then the diagonal morphism $R \to R_0$ corresponds to the homomorphism

$$\bar{\varphi} : A(M \times M)/K \to A(M)$$

obtained by passing to the quotient by K. Now the kernel of ϕ is the ideal K' generated by the

$$d_m = e_{m,0} - e_{0,m}.$$

One has $K \subseteq K'$, and the hypothesis that G acts freely on P, i.e. that $\bar{\phi}$ is an isomorphism, is equivalent to the equality $K' = K$, which is expressed by the relations

$$(**) \qquad d_m \in K = \sum_p A(M \times M)\, A_p\, d_p, \qquad \text{for every } m \in M.$$

Using the natural tri-grading of $A(M \times M)$, and the fact that the first degree of d_m is zero, this means that one can write d_m as a sum of elements of the form

$$f\, e_{r,s}\, (e_{p,0} - e_{0,p}) \qquad \text{with } f \in A_{-p} \cdot A_p,$$

and using the fact that the total degree of d_m is m, one can restrict to terms such that

$$r + s + p = m.$$

Hence one must have, for every $m \in M$, an expression:

$$(***) \qquad \begin{cases} d_m = e_{m,0} - e_{0,m} = \sum_{r,s} \lambda_{r,s}(e_{m-s,s} - e_{r,m-r}) \\ \text{with } \lambda_{r,s} \in J_p = A_p \cdot A_{-p} \subseteq A_0,\ p = m - (r + s). \end{cases}$$

One must conclude the relation $(*)$, i.e. the relations

$$(****) \qquad J_n = A_0 \qquad \text{for every } n \in M.$$

Now for this, it suffices to establish the same relation modulo every maximal ideal of A_0. As the hypotheses made are invariant under such a reduction, one may already suppose that A_0 is a field.

Lemma 5.2. *Under the preceding conditions (with A_0 a field), if $M \neq 0$, there exists a $p \in M - 0$ such that $J_p = A_0$.*

Indeed, if this were not so, the sum in the right-hand side of $(***)$ would be zero for every $m \in M$, which is absurd.

Corollary 5.3. *Under the preceding conditions, but without supposing any longer that A_0 is a field, there exist finitely many elements $p_i \in M - 0$ such that $\sum_i J_{p_i} = A_0$.*

Indeed, one applies the result 5.2 to the situations deduced from A/A_0 by reduction modulo the maximal ideals of A_0.[880]

[880] N.D.E.: It follows from 5.2 that $\sum_{p \neq 0} J_p = A_0$, hence 1 is written as a finite sum $\sum_i x_i$, with $x_i \in J_{p_i}$.

Corollary 5.4. *Suppose again that A_0 is a field. Then for every subgroup N of M such that $N \neq M$, there exists a $p \in M - N$ such that $J_p = A_0$.*

Indeed, let $M' = M/N$, and consider the M'-graded ring A', whose underlying ring is A, and whose grading is given by

$$A'_{m'} = \bigoplus_{m \in h^{-1}(m')} A_m,$$

where $h : M \to M' = M/N$ is the canonical homomorphism. Geometrically, this construction amounts to considering the subgroup $G' = D_S(M')$ of G, and the structure on P of scheme with operator group G' induced by the action of G. It is then evident that G' acts freely on P', i.e. the pair (M', A') satisfies the hypotheses of 5.3. One thus obtains

$$1 = \sum_i f_i g_i \quad \text{with } f_i \in A'_{m'_i},\ g_i \in A'_{-m'_i} \text{ and } m'_i \in M' - \{0\},$$

whence at once the conclusion 5.4 by taking the components of the right-hand side along A_0, and using that A_0 is a field.

Let us now note that

$$J_p \cdot J_q \subseteq J_{p+q} \quad \text{and } J_p = J_{-p},$$

so if N denotes the set of $m \in M$ such that $J_p = A_0$, one sees that N is a subgroup of M. Using 5.4 one sees that it is equal to M. This completes the proof of Theorem 5.1.

As we have pointed out in the course of the proof, Theorem 5.1 implies:

Corollary 5.5. *Under the conditions of 5.1, the graph morphism*

$$P \times_S G \to P \times_S P$$

is a closed immersion.

One concludes at once:

Corollary 5.6. *Let σ be a section of P over S. Then the morphism $g \mapsto \sigma \cdot g$ from G to P defined by σ is a closed immersion.*

Corollary 5.7. *Let G, H be two S-groups, with G diagonalizable, H affine over S, and let $u : G \to H$ be a homomorphism of S-groups that is a monomorphism. Then u is a closed immersion, $H/G = X$ exists and H is a principal homogeneous bundle over X with group G_X; finally, X is affine over S.*

Corollary 5.8. *Under the conditions of 5.1, if P is of finite type (resp. of finite presentation) over S, the same holds for $X = P/G$.*

Indeed, it follows from the hypothesis that the fibers of G_X are of finite type, hence G_X is of finite presentation over X by 2.1 (b), hence P being a torsor under G_X is of finite presentation over X[881]. Since it is also faithfully flat over X, our conclusion then follows from Exp. V, Prop. 9.1.

[881] N.D.E.: cf. EGA II, 2.7.1 (vi).

6. Essentially free morphisms, and representability of certain functors of the form $\prod_{Y/S} Z/Y$[882]

Definition 6.1. *Let $f : X \to S$ be a morphism of preschemes. We say that f is* essentially free*, or also that X is* essentially free over S*, if one can find a covering of S by affine opens S_i, for each i an S_i-prescheme S'_i affine and faithfully flat over S_i, and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine opens X'_{ij}, such that for every (i,j), the ring of X'_{ij} is a free module over the ring of S'_i.*[883]

Proposition 6.2. *a) If X is essentially free over S, it is flat over S, the converse being true if S is Artinian.*

b) If S is the spectrum of a field, every S-prescheme is essentially free over S.

c) If X is essentially free over S, and if $S' \to S$ is a base change morphism, then $X' = X \times_S S'$ is essentially free over S'. The converse is true if $S' \to S$ is faithfully flat and quasi-compact.

The proof is immediate, using for the converse in (a) the fact that a flat module over an Artinian local ring is free.[884]

Proposition 6.3. *Let H be an S-prescheme in groups that is diagonalizable (more generally, that becomes diagonalizable by suitable faithfully flat quasi-compact extension of every affine open of S, i.e. H is "of multiplicative type", cf. IX 1.1). Then H is essentially free over S.*

Indeed, if H is diagonalizable, it is affine over S and defined by an algebra which is a free 0_S-module.

The introduction of Definition 6.1 is justified by the following:

Theorem 6.4. *Let S be a prescheme, Z an essentially free S-prescheme, Y a closed subprescheme of Z. Consider the following functor*[885]

$$F = \prod_{Z/S} Y/Z : (Sch)^{\circ}/S \to (Ens),$$
$$F(S') = \Gamma(Y_{S'}/Z_{S'}) = \begin{cases} \varnothing & \text{if } Z_{S'} \neq Y_{S'}; \\ \{id_{Z_{S'}}\} & \text{if } Z_{S'} = Y_{S'}. \end{cases}$$

This functor is representable by a closed subprescheme of S.[886]

[882]The present § is independent of the theory of diagonalizable groups; its natural place would be in VI_B.

[883]N.D.E.: One can replace "free" by "projective", cf. 6.8 below. On the other hand, this notion is to be compared with that of S-prescheme flat and pure, introduced and developed in [RG71]; see in particular *loc. cit.*, Part I, 3.3.12 and Part II, 3.1.4.1.

[884]N.D.E.: Indeed, let (A, m) be an Artinian local ring, k its residue field, M an arbitrary A-module, $(x_i)_{i \in I}$ elements of M whose images form a basis of M/mM over k. Let F be the free A-module with basis $(e_i)_{i \in I}$, and $\phi : F \to M$ the A-morphism defined by $\phi(e_i) = x_i$. Then $Q = Coker\phi$ satisfies $Q = mQ$, whence, since m is nilpotent, $Q = 0$. Suppose moreover M flat over A; then $K = Ker\phi$ satisfies $K \otimes_A k = 0$, i.e. $K = mK$, whence $K = 0$.

[885]N.D.E.: cf. II.1, where this functor is denoted $\prod_{Z/S} Y$.

[886]N.D.E.: We have corrected "Z" to "S".

[887] Let us first note that F is a sheaf for the (fpqc) topology: since $F(S') = \emptyset$ or $\{\mathrm{pt}\}$ for every S', this reduces to verifying that if (S_i) is an open covering of S (resp. $S' \to S$ a faithfully flat quasi-compact morphism), and if each $Y_{S_i} \to Z_{S_i}$ (resp. if $Y_{S'} \to Z_{S'}$) is an isomorphism, the same holds for $Y \to Z$; now this is clear (resp. follows from SGA 1, VIII 5.4 or EGA IV_2, 2.7.1).

Moreover, by SGA 1, VIII 2.1 and 5.5, faithfully flat quasi-compact morphisms are of effective descent for the fibered category of closed immersion arrows. This allows us to restrict ourselves, with the notation of 6.1, to the case where $S = S'_i$.

Let (Z_j) be a covering of Z by affine opens such that $O(Z_j)$ is a free module over $A = O(S)$, and let $Y_j = Y \cap Z_j$ and $F_j : (Sch)^\circ/S \to (Ens)$ be the functor defined in terms of (Z_j, Y_j) as F in terms of (Z, Y). It is a subfunctor of the final functor, and one evidently has $F = \bigcap_j F_j$, which reduces us to

proving that each F_j is representable by a closed subscheme T_j of S (for then F will be representable by the closed subscheme T intersection of the T_j). One may therefore suppose Z also affine, $Z = \mathrm{Spec}(B)$, where B is a free A-module. Let J be a subset of B defining the subscheme Y of Z, and let K be the ideal in A generated by the $u_i(J) \subseteq A$, where the $u_i : B \to A$ are the coordinate forms with respect to the chosen basis. One observes at once that $T = V(K) = \mathrm{Spec}(A/K)$ satisfies the desired condition, which completes the proof.

Examples 6.5. Let us give some important examples of functors that reduce to functors $\prod_{Z/S} Y/Z$ of the type envisaged in 6.4 and for which it is useful in the sequel to have criteria of representability. We denote by S a prescheme, by X, Y, Z, etc., preschemes over S.

a) Suppose given an S-morphism

```
(*)     q : X → Hom_S(Y, Z),
```

("X acts on Y with values in Z"), i.e. a morphism

```
(**)    r : X ×_S Y → Z.
```

Consider a subprescheme Z' of Z, whence a monomorphism

```
Hom_S(Y, Z′) → Hom_S(Y, Z)
```

which makes the first functor a subfunctor of the second; let X' be the inverse image of this subfunctor by `(*)`, that is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \to Z_T$ factors through Z'_T. This functor X' can be described as follows: set $P = X \times_S Y$, let P' be the inverse image of Z' by $r : P \to Z$; then one has an evident isomorphism

$$(***) X' \simeq \prod_{P/X} P'/P.$$

[887]N.D.E.: We have expanded the original in what follows; see also 1.7.3.

One thus obtains: *if Y is essentially free over S and Z' closed in Z, the subfunctor X' of X is representable by a closed subprescheme of X.*

b) Suppose given two ways of making X act on Y with values in Z, i.e. two morphisms

$$q_1, q_2 : X \rightrightarrows \mathrm{Hom}_S(Y, Z),$$

and set $X' = Ker(q_1, q_2)$: this is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms $q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$ are equal. Now giving q_1, q_2 is equivalent to giving a morphism

$$q : X \to \mathrm{Hom}_S(Y, Z \times_S Z),$$

or, again, a morphism $r : X \times_S Y \to Z \times_S Z$; setting $U = Z \times_S Z$, let U' be the diagonal subprescheme of $Z \times_S Z$; then X' is none other than the inverse image of the subfunctor $\mathrm{Hom}_S(Y, U') \to \mathrm{Hom}_S(Y, U)$ by q, hence can be put in the form (***), with $P = X \times_S Y$ and P' = inverse image of the diagonal by r, i.e. kernel of

$$X \times_S Y \rightrightarrows Z.$$

One is thus in the conditions of (a). One sees consequently that: *if Y is essentially free over S and Z separated over S, the subfunctor X' of X is representable by a closed subprescheme of X.*

c) Suppose given a morphism

$$q : X \to \mathrm{Hom}_S(Y, Y),$$

i.e. "X acts on Y". Let X' be the "kernel" of this morphism, i.e. the subfunctor X' of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \to Y_T$ is the identity. This functor is amenable to (b), as one sees by introducing a second homomorphism

$$q' : X \to \mathrm{Hom}_S(Y, Y)$$

"by making X act trivially on Y". Hence: *if Y is essentially free over S and separated over S, the kernel subfunctor of q is representable by a closed subprescheme of X.*

d) Under the conditions of (c), consider the subfunctor Y' of Y "of invariants under X", so $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $q(y) : X_T \to Y_T$ is "the T-constant morphism with value y". Introducing q' as in (c), and the homomorphisms corresponding to q and q':

$$q, q' : Y \rightrightarrows \mathrm{Hom}_S(X, Y),$$

one sees that Y' is precisely $Ker(q, q')$, and is therefore amenable again to (b) (with the roles of X, Y reversed and $Z = Y$).

Consequently, *if X is essentially free over S, Y separated over S, then the subfunctor Y' of Y of invariants under X is representable by a closed subprescheme of Y.*

e) Constructions of the type explained in the preceding examples are especially frequent in group theory. Thus, when G is an S-prescheme in groups acting on the S-prescheme X:

$$q : G \to \mathrm{Aut}_S(X),$$

the kernel of q ("the subgroup of G acting trivially") is a closed subscheme of G provided that X is essentially free and separated over S (example (c)), and the subobject X^G of invariants is a closed subprescheme of X, provided that G is essentially free over S and X separated over S[888] (example (d)).

Let Y, Z be subpreschemes of X; consider the subfunctor $Transp_G(Y,Z)$ of G ("transporter of Y into Z") whose points with values in a T over S are the $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subseteq Z_T$ i.e. induces a morphism $Y_T \to X_T$ factoring through $Y_T \to Z_T$. Hence: *if Y is essentially free over S, and Z closed in X, then $Transp_G(Y,Z)$ is a closed subprescheme of G* (example (a)).

One may also consider the strict transporter of Y into Z,[889] whose points with values in a T over S are the $g \in G(T)$ such that $g(Y_T) = Z_T$, which is nothing but $Transp_G(Y,Z) \cap \sigma(Transp_G(Z,Y))$, where σ is the symmetry of G. Consequently, if Y and Z are essentially free over S and closed in X, the strict transporter of Y into Z is a closed subprescheme of G.

An important case is that where $X = G$, with G acting on itself by inner automorphisms. If H is a subprescheme of G, the strict transporter of H into H is also called the *normalizer of H in G*, and denoted $Norm_G H$. Hence: *if H is a closed subprescheme of G in groups, essentially free over S, then $Norm_G H$ is representable by a closed subprescheme of G in groups.*

Let finally Z be a subprescheme of G; then its *centralizer $Centr_G(Z)$* in G is the subfunctor of G in groups defined by the procedure of (d), considering "Z acts on G" via the action induced by that of G; hence *if Z is essentially free over S and G separated over S, $Centr_G(Z)$ is a closed subprescheme of G in groups*.

In particular, if G is essentially free and separated over S, then *the center C of G, which is none other than $Centr_G(G)$, is a closed subprescheme of G in groups*.

When S is the spectrum of a field, 6.3 (b) shows that in examples (a) to (e) above, the conditions "essentially free" are automatically satisfied; only separation conditions remain. Recalling that a prescheme in groups over a field is necessarily separated, one finds for example:

Corollary 6.7.[890] *Let G be a prescheme in groups over a field k. Then:*

– *For every subprescheme Z of G, the centralizer of Z in G is a closed subprescheme of G in groups; this is in particular the case for the center $Centr_G(G)$ of G.*

– *More generally, if $u, v : X \to G$ are morphisms of preschemes, $Transp_G(u,v)$ is representable by a closed subprescheme of G.*

[888] N.D.E.: We have corrected "S" to "S".
[889] N.D.E.: denoted $Transp_G^{str}(Y,Z)$.
[890] N.D.E.: We have kept the numbering of the original: there is no 6.6.

– *For two subpreschemes Y, Z of G, with Z closed, $Transp_G(Y,Z)$ is a closed subprescheme of G. If Y is also closed, the same conclusion holds for $Transp_G^{str}(Y,Z)$.*

– *For every subprescheme in groups*[891] *H of G, the normalizer $Norm_G(H)$ is a closed subscheme of G in groups.*

Remark 6.8.[892] *Let A be a commutative ring, M an A-module, $M^\vee = \mathrm{Hom}_A(M,A)$; endow the A-module $\mathrm{End}_A(M)$ with the topology of pointwise convergence (discrete), i.e. a basis of neighborhoods of 0 is formed by the following A-submodules, where $n \in \mathbb{N}$ and $m_1, \cdots, m_n \in M$:*

$$K(m_1, \dots, m_n) = \{ u \in \mathrm{End}_A(M) \mid u(m_i) = 0 \ \text{ for } i = 1, \dots, n \}.$$

We say that M is a quasi-free A-module *if the image of the canonical morphism $\Theta : M \otimes_A M^\vee \to \mathrm{End}_A(M)$ contains id_M in its closure, i.e. if the following condition holds:*

$$(*) \quad \begin{cases} \text{for all } m_1, \dots, m_n \in M, \text{ there exist } x_1, \dots, x_r \in M \text{ and } f_1, \dots, f_r \in M^\vee \\ \text{such that } m_i = \Theta(\sum_{s=1}^r x_s \otimes f_s)(m_i) = \sum_{s=1}^r f_s(m_i)\, x_s \text{ for } i = 1, \dots, n. \end{cases}$$

(In this case, $Im\Theta$ is dense in $\mathrm{End}_A(M)$, since for every $u \in \mathrm{End}_A(M)$ one has

$$u(m_i) = \sum_{s=1}^r f_s(m_i)\, u(x_s) = \Theta(\sum_{s=1}^r u(x_s) \otimes f_s)(m_i).)$$

Let us note first that this property is stable under base change. Indeed, let $\phi : A \to A'$ be a morphism of rings, $M' = M \otimes_A A'$, and $m'_1, \cdots, m'_n \in M'$. Then $m'_i = \sum_j m_{ij} \otimes b_{ij}$ ($m_{ij} \in M$, $b_{ij} \in A'$); by hypothesis, there exist $x_1, \cdots, x_r \in M$ and $f_1, \cdots, f_r \in \mathrm{Hom}_A(M,A)$ such that $m_{ij} = \sum_s x_s f_s(m_{ij})$ for all i, j. Denote by $\phi \circ f_s$ the image of f_s in $\mathrm{Hom}_A(M,A') = \mathrm{Hom}_{A'}(M',A')$; then for every $i = 1, \cdots, n$ one has:

$$(\sum_s x_s \otimes \varphi \circ f_s)(m'_i) = \sum_{s,j} x_s \otimes \varphi(f_s(m_{ij}))\, b_{ij} = \sum_j (\sum_s x_s f_s(m_{ij})) \otimes b_{ij} =$$

which proves that M' is quasi-free over A'.

Let us also note that every projective A-module P is quasi-free (there exist A-morphisms $A^{(I)} - \pi \to P - \tau \to A^{(I)}$ such that $\pi \circ \tau = id_P$; denote by (e_i) the canonical basis of $A^{(I)}$ and f_i the linear form $e_i^* \circ \tau$ on P; if $m_1, \cdots, m_n \in P$, there exists a finite subset J of I such that $m_k = \sum_{i \in J} f_i(m_k)\pi(e_i)$ for $k = 1, \cdots, n$).

Then, Theorem 6.4 remains valid if in the statement of Definition 6.1 the word "free" is replaced by "projective" or, more generally, by "quasi-free". Indeed, proceeding as in the proof of 6.4, one is reduced to proving:

Lemma 6.8.1. *Let M be a quasi-free A-module, N a submodule, F the covariant functor $(A-algebras) \to (Ens)$ such that $F(B) = pt$ if $M_B = (M/N)_B$, and $F(B) = \emptyset$ otherwise. Then there exists an ideal K of A such that $F(B) = pt$ if and only if the morphism $A \to B$ factors through A/K, i.e. one has a functorial isomorphism in B:*

$$F(B) \simeq \mathrm{Hom}_{A-alg.}(A/K, B).$$

[891] N.D.E.: Indeed, over a field k, every subprescheme in groups of G is closed, cf. VI_A, 0.6.1.

[892] N.D.E.: We have expanded the original in what follows; in particular we have added Lemma 6.8.1.

Proof. Let (n_j) be a system of generators of N, and let F_j be the subfunctor of the final functor e ($e(B) = pt$ for every B) corresponding to the submodule An_j. One has $F(B) = pt$ if and only if the image of each n_j in `M_B` is zero, hence F is the intersection of the functors F_j. This reduces us to the case where N is generated by one element n.

Let $\phi : A \to B$ be an A-algebra; if the image $n \otimes 1$ of n in `M_B` is zero, then for every $f \in M^\vee = \mathrm{Hom}_A(M, A)$ one has $0 = f(n) \otimes 1 = \phi(f(n))$. On the other hand, since M is quasi-free, there exist $x_1, \cdots, x_r \in M$ and $f_1, \cdots, f_r \in M^\vee$ such that $n = \sum_s f_s(n)x_s$, whence $n \otimes 1 = \sum_s x_s \otimes \phi(f_s(n))$. It follows that $n \otimes 1 = 0$ if and only if ϕ factors through A/K, where K is the ideal generated by the $f_s(n)$ for $s = 1, \cdots, r$. This proves the lemma.

7. Appendix. On monomorphisms of preschemes in groups

The result proved in the present § is unnecessary for the sequel of the seminar, except for X 8.8 and XV and XVI. It rests in an essential way on the existence theorem for quotient groups of Exposé VI_A.[893]

Corollary 5.7 leads one to ask under what conditions one can assert that a monomorphism $u : G \to H$ of S-groups is an immersion, or even a closed immersion.

We have seen in VI_B 1.4.2 that this is the case if S is the spectrum of a field, provided that G is of finite type over k and H is locally of finite type over k. One easily concludes that the same result remains valid if one only supposes S Artinian.[894]

On the other hand, it is easy to give examples of bijective monomorphisms that are not immersions, with S being for example the affine line over a field, or the spectrum of a discrete valuation ring. One will take for example $H = (\mathbb{Z}/2\mathbb{Z})_S$, $G = G_1 \times_S G_2$, where `G_1` is the open subgroup of H, the complement of the closed point x distinct from `0` of the fiber $H_s \simeq \mathbb{Z}/2\mathbb{Z}$ (where s denotes a fixed closed point of S), and `G_2` is the closed subscheme of H that is the sum of the subscheme reduced to the unit section, and the closed reduced subscheme defined by the closed point x. One easily verifies that `G_2` is indeed stable under the multiplication of H, hence is a group scheme. The immersions $G_i \to H$ $(i = 1, 2)$ then define a homomorphism of S-groups $G = G_1 \times G_2 \to H$, which is obviously a bijective monomorphism (and moreover a local immersion), but is not an immersion. (One observes that G and H are reduced, G having three disjoint irreducible components, while H has only two irreducible components.) One will note that $G_1 \to H$ also gives an example of an open subgroup G of H which is not closed (contrary to what occurs for algebraic groups over a field). The theory of the degeneration of elliptic curves provides further examples of this latter phenomenon, with moreover H smooth over S of relative dimension 1, and G with connected fibers.

It is possible[895] on the other hand that, as soon as one assumes G flat over S, and

[893]N.D.E.: cf. Exp. VI_A, Theorems 3.2 and 3.3.2.
[894]N.D.E.: take into account the additions made in VI_B...
[895]In fact, M. Raynaud has constructed a counter-example, with G smooth with connected fibers, cf. XVI

(say) G and H of finite presentation over S, a monomorphism $u : G \to H$ of S-groups is automatically an immersion. We shall prove a result of this nature, under supplementary hypotheses.

Let us note first that one may assume S affine, and (thanks to the finite presentation hypothesis on G and H, which allows one to reduce to the case of the spectrum of a ring of finite type over $\mathbb{Z}$[896]) S Noetherian. Then G and H are Noetherian. To say that $u : G \to H$ is a closed immersion (resp. an immersion) then amounts to saying that u is a monomorphism (which is true by hypothesis) and that u is proper (resp. and that u is

proper at every point of $u(G)$)[897]. The valuative criterion of properness assures us that it suffices to verify that for every base change $S' \to S$, with S' the spectrum of a discrete valuation ring, complete if one wishes, the morphism $u' : G' \to H'$ has the same properness property. (The case of local properness was forgotten in EGA II 7.3, and will appear as an erratum in EGA IV[898].) This therefore reduces us to the case where S itself is the spectrum of a complete discrete valuation ring — subject to the supplementary hypotheses on G, H, u that we are led to formulate being stable under base change.

Let then s (resp. s_0) be the closed point (resp. the generic point) of S. Then the homomorphisms induced on the fibers

u_s : G_s → H_s   and   u_{s_0} : G_{s_0} → H_{s_0}

are closed immersions, since these are monomorphisms of group schemes of finite type over fields (VI_B 1.4.2). We may therefore identify G_{s_0} with a closed subscheme of H_{s_0}. Now we have the following result:

Lemma 7.1. *Let S be the spectrum of a discrete valuation ring, s_0 its generic point, H an S-prescheme, L_0 a closed subprescheme of the generic fiber H_{s_0}, so that L_0 is also a subprescheme of H.*

Then the schematic closure $\bar{L}_0$ in H (i.e. the smallest closed subprescheme of H majorizing L_0, cf. EGA I 9.5) exists and is also the unique closed subprescheme of H, flat over S, whose generic fiber is L_0. Moreover, the formation of $\bar{L}$ is functorial with respect to a variable couple (H, L_0), and commutes with the formation of cartesian products over S.

In particular, if H is an S-group and L_0 is a subgroup of H_{s_0}, then $\bar{L}$ is a subgroup of H.

The proof is immediate and left to the reader[899]. Applying this to the situation H, G_{s_0}, one sees that the monomorphism $u : G \to H$ factors as $G \to L \to H$, where L is a subgroup of H that is a closed subprescheme, flat over S, and where $G \to L$

1.1 (c). If one does not assume G has connected fibers, one may take $S = \mathrm{Spec}(\mathbb{Z}_2)$, $G = (\mathbb{Z}/2\mathbb{Z})_S$ deprived of the non-neutral closed point, $H = (\mu_2)_S$.

[896]N.D.E.: cf. EGA IV₃, § 8, and Exp. VI_B, § 10.

[897]N.D.E.: cf. EGA IV₃, 8.11.5 and 15.7.1.

[898]Cf. EGA IV₃, 15.7.

[899]Cf. EGA IV₂, 2.8.

induces an isomorphism on the generic fibers. Then $u : G \to H$ is an immersion (resp. a closed immersion) if and only if $u' : G \to L$ is. This therefore reduces us to the case where H is flat over S and $u_{s_0} : G_{s_0} \to H_{s_0}$ is an isomorphism (subject to the supplementary hypotheses we may have to formulate on G, H, u being respected when H is replaced by a closed subprescheme in groups). As then H is the schematic closure of H_{s_0}, if u is an immersion, then $u(G)$ will be a subprescheme[900] of H majorizing H_{s_0}, hence its schematic closure will also be H, and consequently $u(G)$ will be an open subprescheme of H. Hence, we shall in fact have to prove that u is an open immersion (resp. an isomorphism, if we want to establish that u is a closed immersion). Since G and H are flat

over S, it amounts to the same thing to say that the morphisms induced on the fibers are open immersions, resp. isomorphisms (cf. SGA 1, I 5.7), and since this is already the case for u_{s_0}, one is reduced to proving that u_s is an open immersion, resp. an isomorphism.

Let us note first that, since G and H are flat over S, the dimension of their fibers remains constant (VI_B 4.3). Since G and H have the same generic fiber, it follows that this dimension is the same for G and H, hence $u_s : G_s \to H_s$ is a monomorphism of algebraic groups of the same dimension. One concludes easily that G_s is open in H_s, and in fact is set-theoretically a union of connected components of H_s (one is reduced to the case where the base field is algebraically closed, and G and H reduced, hence smooth over k, where it is immediate…). Hence $H_s - G_s$ is closed in H_s, hence in H, hence its complement `H_0` in H is open, and is obviously an open stable under the group law of H. Hence, up to replacing H by `H_0`, one may, in order to prove that u is an immersion, reduce (with the usual proviso) to the case where, in addition to the preceding hypotheses, one assumes u bijective, i.e. $u_s : G_s \to H_s$ bijective. One is therefore in any case reduced to proving that u or again u_s is an isomorphism, possibly under the hypothesis of bijectivity.

Suppose therefore first that u is bijective. If H_s is reduced, one can evidently conclude that u_s is an isomorphism, since G_s identifies with a closed subscheme of H_s having the same underlying set. In particular, if $k = \kappa(s)$ is of characteristic zero, every algebraic group over k is reduced by Cartier (cf. VI_B, 1.6.1, or VII_B, 3.3.1, or EGA IV_4, 16.12.2 and 17.12.5), and one has thus obtained:

Proposition 7.2. *Let $u : G \to H$ be a homomorphism of preschemes in groups of finite presentation over S. Suppose that u is a monomorphism, G flat over S, and the residue fields of S of characteristic zero. Then u is an immersion.*

When $\kappa(s)$ is of characteristic $p > 0$, we shall restrict to the case where G is commutative. Then (with the reductions made) H is also commutative, since it is the schematic closure of H_{s_0} which is isomorphic to G_{s_0}, hence commutative. For every integer $n \geqslant 0$, we set $S_n = \mathrm{Spec}(V/m^{n+1})$ (where V is the valuation ring defining S and m its maximal ideal), $G_n = G \times_S S_n$, $H_n = H \times_S S_n$. For every integer $m > 0$, we also introduce the subgroups ${}_mG$ and ${}_mH$ of G and H, kernels of the m-th power.

[900] N.D.E.: We have suppressed the word “closed”.

One defines similarly ${}_m(G_n) = ({}_mG)_n$, which one denotes simply ${}_mG_n$, and likewise for H.

By virtue of VI_A 3.2, one can form the quotients $Q_n = H_n/G_n$. Then Q_n is a commutative group scheme over S_n, flat over S_n, and $H_n \to Q_n$ is a faithfully flat morphism with kernel G_n. Since the formation of quotients commutes with base extension[901], one has

$$Q_n \simeq Q_m \times_{S_m} S_n \quad \text{for } m \geqslant n,$$

in particular the fiber $Q_n \times_{S_n} S_0$ is none other than $Q_0 = H_0/G_0$. Since G_0 has the same

underlying set as H_0, then Q_0 is set-theoretically reduced to a single point: it is a purely infinitesimal group. Consequently, each Q_n is finite and flat over S_n. Hence Q_n is defined by an algebra C_n over $V_n = V/m^{n+1}$ which is a free module of finite type over this ring, and for $m \geqslant n$ one has $C_n = C_m \otimes_{V_m} V_n$, an isomorphism also respecting the diagonal map. One thus obtains a free module of finite type $C = \varprojlim C_n$ over $V = \varprojlim V_n$, and the diagonal maps of the C_n define a diagonal map of C, so that $Q = \mathrm{Spec}(C)$ becomes a group scheme finite and flat over S, such that

$$Q \times_S S_n = Q_n$$

for every n.

Lemma 7.3.[902] *Let K be a field, Q a finite group scheme over K, of degree n. Then the n-th power morphism in Q is zero.*

Cf. VII_A 8.5.

Remark 7.3.1. *Statement 7.3 retains a sense for a group scheme Q/S finite and locally free over S, S being an arbitrary base prescheme. It would be interesting to find a proof in this general case.*

One will note that 7.3 (i.e. VII_A, 8.5) proves in any case that the envisaged statement is true if S is reduced, as one sees by applying 7.3 to the fibers of Q at the maximal points (i.e. generic points of the irreducible components) of S.[903]

In particular, under the conditions of the preceding proof, where S is the spectrum of a discrete valuation ring and Q is commutative, one finds that $n \cdot id_Q = 0$. Moreover, here n is a power p^{ν_0} of the residue characteristic, and one finds:

Corollary 7.4. *With Q and the Q_n as above, and their common degree being p^{ν_0}, one will have $p^{\nu_0} \cdot id_Q = 0$ and consequently $p^{\nu_0} \cdot id_{Q_n} = 0$ for every n.*

[901] N.D.E.: Make this explicit in Exp. V and VI_A...

[902] N.D.E.: We have suppressed here the hypothesis that Q be commutative, and have modified 7.3.1 accordingly. Note that if Q is not commutative, the n-th power morphism is not in general a group homomorphism.

[903] N.D.E.: Add this in VII_A — On the other hand, the statement is true for every S if Q is commutative, cf. Deligne's theorem in [TO70], p. 4.

Corollary 7.5. *Suppose moreover that G_0 is smooth over k, and the homomorphism $p \cdot id_{G_0}$ flat (which amounts to saying, in virtue of the structure of algebraic groups over an algebraically closed field $\bar{k}$, that $G_{\bar{k}}$ contains no subgroup isomorphic to the additive group). Then for every $\nu \geqslant \nu_0$ and $n > 0$, the group ${}_{p^\nu}H_n$ is flat over S_n.*

Indeed, it follows from $p^{\nu_0} \cdot id_{Q_n} = 0$ that $p^\nu \cdot id_{H_n}$ factors through G_n, so that one has a commutative diagram:

```
              p^ν · id_{H_n}
H_n  ──────────────────────────────→  H_n
  \                                    ↑
   \ v_n                              u_n
u_n \                                  |
     ↘   p^ν · id_{G_n}                |
     G_n  ───────────────────────────→ G_n.
```

I claim that v_n is flat. Indeed, since H_n and G_n are flat over S_n, one is reduced to verifying that v_0 is flat (SGA 1, IV 5.9), so one may assume that $n = 0$, whence $S_n = \mathrm{Spec}(k)$. Since $p \cdot id_{G_0}$, and hence $p^\nu \cdot id_{G_0}$, is flat, its image is an open induced subgroup G_0' of G_0, and since $u_0 : G_0 \to H_0$ is surjective, it follows that v_0 takes its values in G_0', hence may be considered as a homomorphism into G_0'. Since its composition with u_0 is an epimorphism, it is itself an epimorphism, hence a flat homomorphism into G_0', hence a flat homomorphism into G_0. Hence v_n is flat, hence $Ker v_n = {}_{p^\nu}H$ is flat over S. QED.

Remark. We have not explicitly used the fact that G_0 is smooth over k; but it is easy to see that this is a consequence of the fact that $p \cdot id_G$ is flat — this is why we made this condition explicit in the hypothesis of Corollary 7.5.

Lemma 7.6. *Let $u : G \to H$ be a surjective monomorphism of commutative algebraic groups over a field k of characteristic $p > 0$; consider the (purely infinitesimal) group $Q = H/G$. Then there exists an integer ν_1 such that for $\nu \geqslant \nu_1$, the sequence*

$$0 \to {}_{p^\nu}G \to {}_{p^\nu}H \to Q \to 0$$

is exact.

It suffices to ensure exactness at Q, and for this to ensure that the homomorphism

$$(*) O_{Q,e} \to O_{{}_{p^\nu}H,e}$$

of local rings at the neutral elements is injective (N.B. recall that Q is set-theoretically reduced to the element e). Now one has a natural homomorphism

$$(**) O_{{}_{p^\nu}H,e} \to O_{H,e}/m^{p^\nu},$$

where m is the augmentation ideal (i.e. the maximal ideal) of $O_{H,e}$, as one sees by noting that $p^{\nu} \cdot id_H$ vanishes on the kernel of "the iterated Frobenius homomorphism" F^{ν}; the composite of the homomorphisms (*) and (**) is also equal to the natural composite

$$(***) O_{Q,e} \to O_{H,e} \to O_{H,e}/m^{p^{\nu}}.$$

Now $O_{Q,e} \to O_{H,e}$ is injective, since $H \to Q$ is an epimorphism hence is flat; on the other hand $O_{Q,e}$ is Artinian, and finally the intersection in $O_{H,e}$ of the $m^{p^{\nu}}$ is reduced to 0, hence so is the intersection of their traces on $O_{Q,e}$. Consequently one of these traces is reduced to 0, which proves that (***) is injective, and a fortiori (*) is injective.

Lemma 7.7. *Under the conditions of 7.5 there exists a ν_1 such that, for $\nu \geqslant \nu_1$ and every n, the sequence of S_n-groups*

$$0 \to {}_{p^{\nu}}G_n \to {}_{p^{\nu}}H_n \xrightarrow{w_n} Q_n \to 0$$

is exact (more precisely, w_n is faithfully flat and its kernel is ${}_{p^{\nu}}G_n$).

One takes ν_1 as in 7.6 applied to $u_0 : G_0 \to H_0$, and $\nu_1 \geqslant \nu_0$ (where p^{ν_0} = rank Q_0). It only remains to verify that w_n is faithfully flat. Now by virtue of 7.5, ${}_{p^{\nu}}H$ is flat over S_n, and since Q_n is too, one is reduced to verifying that $w_0 : {}_{p^{\nu}}H_0 \to Q_0$ is faithfully flat, i.e. is an epimorphism, which is true by the choice of ν_1.

Corollary 7.8. *Suppose moreover H separated over S, more generally that ${}_{p^{\nu}}H$ is separated over S for every ν, and that ${}_{p^{\nu}}G$ is finite over S for every ν. Then the group schemes ${}_{p^{\nu}}G$ and ${}_{p^{\nu}}H$ are finite and flat over S for $\nu \geqslant \nu_1$, and one has an exact sequence*

$$0 \to {}_{p^{\nu}}G \to {}_{p^{\nu}}H \xrightarrow{w} Q \to 0.$$

Since $u : G \to H$ is surjective, so is the induced morphism ${}_{p^{\nu}}G \to {}_{p^{\nu}}H$, and since the first member is finite over S and the second separated over S, it follows that the second member is finite over S. To then verify that ${}_{p^{\nu}}G$ and ${}_{p^{\nu}}H$ are also flat over S, it suffices to verify that this is the case for ${}_{p^{\nu}}G_n$ and ${}_{p^{\nu}}H_n$ for every n, which is contained in 7.5. Finally, the exact sequence of 7.8 comes from the exact sequences 7.7 for variable n.

Taking the generic fibers in the exact sequence 7.8 and recalling that $u_{s_0} : G_{s_0} \to H_{s_0}$ is an isomorphism, one finds Q_{s_0} = unit group, whence (since Q is flat over S) Q = unit group, whence Q_s = unit group, hence $u_s : G_s \to H_s$ is an isomorphism. Hence:

Proposition 7.9. *Let $u : G \to H$ be a homomorphism of preschemes in groups of finite presentation over the prescheme S. Suppose:*

a) u is a monomorphism.

b) G is flat over S.

c) For every $s \in S$ such that $\kappa(s)$ is of residue characteristic $p > 0$, one wants the following conditions to be satisfied for the homomorphism $u' : G' \to H'$ of preschemes in groups over $S' = \mathrm{Spec}(O_{S,s})$ deduced from $u : G \to H$ by the base change $S' \to S$:

– G' is commutative,

– the special fiber $G'_0 = G_s$ is smooth over $\kappa(s)$,

– for every integer $\nu > 0$, ${}_{p^\nu}G'$ is finite over S' and ${}_{p^\nu}H'$ is separated over S'.

Under these conditions, u is an immersion.

It suffices to remark that, in (c), the condition that ${}'_{pG}{}_0$ be finite over S' implies that ${}'_{pG}{}_0$ is finite over $\kappa(s)$, which already implies that $G'_0 \otimes_{\kappa(s)} \overline{\kappa(s)}$ contains no subgroup isomorphic to the additive group, so that one is under the hypotheses of 7.5.

Remark 7.10. *Examples of M. Raynaud (XVI 1.1 (a) and (b)) show that in (c) one cannot drop either the hypothesis that the ${}_{p^\nu}G'$ are finite over S' or the hypothesis that the ${}_{p^\nu}H'$ are separated over S'.*

We now want conditions ensuring that u is a closed immersion. We therefore preserve the hypotheses preceding 7.9 but no longer assume u surjective (only u an isomorphism on the generic fibers). We already know that $u : G \to H$ is an open immersion, hence also $u_0 : G_0 \to H_0$, whose image therefore contains the connected component of the identity element $(H_0)^0$. Note that, since `G_0` is smooth over k and "without additive component" over the algebraic closure of k, the same holds for `H_0`. Now we have:

Lemma 7.11. *Let H be a commutative algebraic group over a field k, such that $H \otimes_k \bar{k}$ contains no subgroup isomorphic to G_a. Let $n = \text{degree } H/H^0$; then the homomorphism*

$$ {}_{nH} \to H/H^0 $$

is surjective.

One may indeed assume k algebraically closed, and then this follows from the well-known fact that $H^0(k)$ is a divisible group.

Suppose now (returning to our situation $u : G \to H$) that the ${}_{nG}$ ($n > 0$) are proper over S, and the ${}_{nH}$ are separated over S. Note on the other hand that the ${}_{nH}$ are flat over S. Indeed, it suffices to see this at the points above s; one is then reduced to proving that $n \cdot id_H$ is flat at the points above s, and for this one is reduced to verifying that $n \cdot id_{H_0}$ is flat, which is equivalent (as we have already noted) to the fact that $(H_0)^0$ is smooth[904] over k and has no G_a-component over the algebraic closure of k. As $(H_0)^0 = (G_0)^0$, this follows from the analogous hypothesis made on

[904] N.D.E.: We have updated the terminology, replacing "simple" by "smooth" (see, for example, footnote (1) of A. Grothendieck in SGA 1, Exp. II).

G_0. On the other hand, the $_nH$ are separated over S since H is, hence the morphism $_nG \to {_nH}$ is proper, hence its image is closed. As this image contains the generic fiber of $_nH$ (since $u_{s_0} : G_{s_0} \to H_{s_0}$ is an isomorphism) and $_nH$ is flat over S, hence identical to the closure of its generic fiber, it follows that $_nG \to {_nH}$ is surjective, hence _nG_0 → _nH_0 is surjective. Recalling that $G_0 \supset (H_0)^0$ and applying 7.11, one finds that $G_0 \to H_0$ is surjective, hence u is surjective. One thus obtains:

Proposition 7.12. *With the notation of 7.9, suppose conditions (a) and (b) hold, as well as condition (c′) (stronger than (c)), obtained by requiring that for every integer $n > 0$ (not only of the form p^ν), $'_{nG}$ be finite over S' and $'_{nH}$ separated over S'. Under these conditions, u is a closed immersion.*

Remark 7.13. *a) One easily verifies that the separation hypothesis made on the $_nH$ implies in fact that H is separated over S. This is moreover formally contained in 7.12 by taking for G the unit S-group. Note also that when S is locally Noetherian, one may in 7.9 restrict to assuming H locally of finite type over S (in*

place of: of finite type over S), the proof given applying as it stands; for 7.12 one will moreover assume that the fibers of H are of finite type.

b) Using 7.12, it is not difficult to prove that if $u : G \to H$ is a monomorphism of S-preschemes in groups of finite presentation, with G diagonalizable (or more generally, "of multiplicative type") and H separated over S, then u is a closed immersion — which constitutes a satisfactory generalization of the first conclusion stated in 5.7. When G is smooth over S, it suffices to apply 7.12, and the general case reduces easily to that one. When one no longer assumes H separated over S, one can still show that u is an immersion; in the case where G is a torus, this fact moreover follows also from what follows.

c) When in 7.9 one assumes G has connected fibers, one can in condition 7.9 (c) drop the hypothesis that the $_{p^\nu}H'$ be separated over S'. Indeed, with the reductions made in the proof, one may assume H flat over S and u bijective, hence H with connected fibers, hence H separated over S by virtue of Raynaud's theorem (VI_B 5.5).[905]

Bibliography

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[905] N.D.E.: specify in VI_B § 5 this attribution, which appeared in SGAD 1965, indicating the modifications between SGAD and Lect. Notes 151.
[906] N.D.E.: additional references cited in this Exposé.

Footnotes

Exposé IX. Groups of multiplicative type: homomorphisms into a group scheme

by A. Grothendieck

> Version 1.0 of 8 November 2009: additions in proof of 3.6 bis, 4.4–7, 5.0–6, 6.1, 7.1, 8.2. 8.1 and 4.5 to be revised.[907]

1. Definitions

Definition 1.1. *Let S be a prescheme and G an S-prescheme in groups. One says that G is a* group of multiplicative type *if G is locally diagonalizable in the sense of the faithfully flat quasi-compact topology (cf. IV 6.3), i.e. if for every $s \in S$, there exist an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that $G' = G \times_S S'$ is a diagonalizable S'-group (I 4.4 and VIII 1.1).*

— *One says that G is* of quasi-isotrivial multiplicative type *if it is even locally diagonalizable in the sense of the étale topology (IV 6.3), i.e. if in the preceding definition one can even take $S' \to U$ étale surjective, or again (which amounts to the same, as one sees by taking the disjoint-sum scheme of the various S' attached to the various $s \in S$) if there exists $S' \to S$ étale and surjective such that $G' = G \times_S S'$ is a locally diagonalizable S'-group.*

— *If one can even choose $S' \to S$ étale surjective and finite, one says that G is* of isotrivial multiplicative type.

— *Finally, one says that G is a* group of locally trivial *(resp.* locally isotrivial*) multiplicative type if every $s \in S$ admits an open neighborhood U such that $G \times_S U$ is a diagonalizable U-group (resp. a group of isotrivial multiplicative type, i.e. there exists an étale surjective finite morphism $S' \to U$ such that $G \times_S S'$ is a diagonalizable S'-group).*

Comments 1.2. One will note that the five preceding notions all derive from the notion of diagonalizable group by the process of localization, in the sense of five different "topologies" associated with (Sch).

We agree, generally, that when the word "locally" is not made explicit by specifying the topology envisaged, the Zariski topology is meant, in accordance with received terminology. In the terminology introduced here, "of locally trivial multiplicative type" is equivalent to "locally diagonalizable" of VIII 1.1, and likewise "trivial" translates as "diagonalizable". Among the five notions introduced, one has the following diagram of implications, which results from the corresponding relations between the topologies in play:

[907] Version 1.0 of 8 November 2009: additions in proof of 3.6 bis, 4.4–7, 5.0–6, 6.1, 7.1, 8.2. 8.1 and 4.5 to be revised.

```
         multiplicative type (general)
                      ⇑
               quasi-isotrivial
                      ⇑
locally isotrivial ⇐ isotrivial
       ⇑                  ⇑
locally trivial ⇐    trivial
```

From the practical point of view, let us point out at once that all the groups of multiplicative type we shall encounter will be quasi-isotrivial; thus, we shall see in the following Exposé (X 4.5) that when G is of finite type over S, then G is automatically quasi-isotrivial, but we shall give examples where it is not locally isotrivial. We shall likewise see there that G may be locally trivial without being isotrivial nor, a fortiori, trivial (which easily implies that the inclusions of the preceding diagram are strict).

On the other hand, we shall also see there (X 5.16) that when S is locally noetherian and normal (more generally geometrically unibranch), every group of multiplicative type of finite type over S is necessarily isotrivial, and moreover trivial as soon as it is locally trivial (or even merely trivial on a dense open). This explains that most groups of multiplicative type one will encounter in practice will doubtless be isotrivial, all the more so since we shall see later that the maximal tori of semisimple group schemes are automatically isotrivial.

Definition 1.3. *Let S, G be as in 1.1. One says that G is a* torus *if it is locally isomorphic, in the sense of the faithfully flat quasi-compact topology, to a group of the form G_m^r (where r is an integer $\geqslant 0$).*

With the notation of 1.1, this means therefore that one can choose S' such that G' be isomorphic to a group of the form $(G_{m,S'})^r$. One will note that the integer r depends on $s \in S$: it is the dimension of the fiber $G_s = G \otimes_S \kappa(s)$. It is a locally constant function of s, as one verifies at once. There is occasion to generalize this remark:

Definition 1.4. *Let G be a diagonalizable group scheme over a field k, so that G is isomorphic to a group $D_k(M)$, where M is an ordinary commutative group, defined up to isomorphism by this condition — more precisely $M \simeq \mathrm{Hom}_{k-gr.}(G, G_{m,k})$ (VIII 1.3). The isomorphism class of M is called the* type *of the diagonalizable group G; it is evidently invariant under extension of the base field.*

If now G is a group scheme of multiplicative type over an arbitrary prescheme S, then for every $s \in S$, there exists an extension k of $\kappa(s)$ such that $G \times_S \mathrm{Spec}(k)$ is a diagonalizable k-group;[908] *its type is then independent of the chosen extension by the preceding remark, and will be called the* type of G at s, *or* type of G_s. *In particular, if G itself is diagonalizable, hence isomorphic to a group of the form $D_S(M)$, then for every $s \in S$, the type of G at s equals the class of the group M.*

[908] *N.D.E.* Indeed, there exists by hypothesis an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that $G_{S'}$ is diagonalizable; then for every $s' \in S'$ above s, $\kappa(s')$ is an extension of $\kappa(s)$ and $G_{s'} = G \times_S \mathrm{Spec}(\kappa(s'))$ is diagonalizable.

In general, if G is a group of multiplicative type over S and M an ordinary commutative group, one will say that G is of type M *if it is of type M at every point $s \in S$, in other words if G is locally isomorphic to $D_S(M)$ in the sense of the faithfully flat quasi-compact topology.*

Remark 1.4.1.[909] *a) One sees immediately that for a given group of multiplicative type G over S, the function $s \mapsto typeof G_s$ is locally constant on S: indeed, with the notation of 1.1, if G' is of type M, then G is of type M on the neighborhood U of s. It follows that one has a canonical partition of S into a disjoint sum of preschemes S_i, such that for every i, G_{S_i} is of type M_i, where the M_i are pairwise non-isomorphic commutative groups.*

b) In particular, if S is connected, the type of the fibers of G is constant, i.e. there exists a commutative group M such that G is of type M. Finally, if G is a torus, the type of G at s is characterized by the integer $\dim G_s$ (indeed, G_s is of type $\mathbb{Z}^n$, where $n = \dim G_s$).

Remark 1.5. *a) It is trivial on definitions 1.1 and 1.3 that these are "compatible with base extension". Thus, if G is a prescheme in groups over S, and $S' \to S$ is a base-change morphism, then if G is of multiplicative type (resp. of isotrivial multiplicative type, etc.), the same holds for G'.*

b) When moreover $S' \to S$ is faithfully flat and quasi-compact, then if G' is of multiplicative type, resp. a torus, the same holds for G. If moreover $S' \to S$ is étale (resp. finite étale), and G' quasi-isotrivial (resp. isotrivial), the same holds for G.

c) Finally, returning to an arbitrary base-change morphism $f : S' \to S$, if $s' \in S'$ and $s = f(s')$, then the type of G' at s' is equal to that of G at s, since $G'_{s'} = G_s \bigotimes_{\kappa(s)} \kappa(s')$.

2. Extension of certain properties of diagonalizable groups to groups of multiplicative type

The extensions in question are essentially trivial consequences of the results of the preceding Exposé, given Definitions 1.1 and the "local" nature (in the sense of the faithfully flat quasi-compact topology) of the results concerned.

Definition 2.0.[910] *We denote by M the set of faithfully flat quasi-compact morphisms.*

Proposition 2.1. *Let G be a group of multiplicative type over a prescheme S. One has the following:*

a) G is faithfully flat over S, and affine over S (a fortiori quasi-compact over S).

b) G of finite type over $S \Leftrightarrow G$ of finite presentation over $S \Leftrightarrow$ for every $s \in S$, the type of G at s is given by a commutative group of finite type.

[909] *N.D.E.* The number 1.4.1 has been added to the remarks that follow, for later references.
[910] *N.D.E.* This definition has been added, since it appears in propositions 2.3 and 2.7.

c) G finite over S ⇔ for every $s \in S$, the type of G at s is given by a finite commutative group ⇔ (if S quasi-compact) G is of finite type over S and is annihilated by an integer $n > 0$.

c′) G integral over S ⇔ for every $s \in S$, the type of G at s is given by a torsion commutative group.

d) G is the unit group ⇔ for every $s \in S$, the type of G at s is given by the zero commutative group.

e) G is smooth over S ⇔ for every $s \in S$, the type of G at s is given by a commutative group of finite type whose torsion subgroup is of order prime to the characteristic of $\kappa(s)$.

These statements follow from VIII 2.1, taking into account that the properties in question descend along faithfully flat quasi-compact morphisms (cf. SGA 1, VIII or EGA IV_2, § 2).

Using VIII 3.5, one obtains likewise:

Proposition 2.2. *Let G be a group of multiplicative type and of finite type over S; then for every integer $n \neq 0$, the kernel ${}_nG$ of $n \cdot id_G$ is a group of multiplicative type, finite over S.*

Proposition 2.3. *Let G be a group of multiplicative type over the prescheme S, operating freely on the right on the S-prescheme X affine over S. Then:*

a) The equivalence relation defined by G in X is M-effective (IV 3.4), and $Y = X/G$ is affine over S.

b) If moreover X is of finite presentation (resp. of finite type) over S, the same holds for Y.

The first assertion follows from VIII 5.1, which treats the case where G is diagonalizable, and from IV 3.5.2, which allows one to reduce to that case, given that faithfully flat quasi-compact morphisms $S' \to S$ are morphisms of effective descent for the fibered category of affine schemes over others, i.e. for every Y' affine over S' endowed with a descent datum relative to $S' \to S$, this descent datum is effective, i.e. Y' comes from a Y affine over S (cf. SGA 1, VIII 2.1).

For the second assertion one is likewise reduced to the diagonalizable case VIII 5.8, since the finiteness conditions envisaged descend along faithfully flat quasi-compact morphisms (SGA 1, VIII 3.3 and 3.6[911]). Proceeding as in VIII, Corollaries 5.5 to 5.7, one deduces from 2.3:

Corollary 2.4. *Under the conditions of 2.3, the graph morphism*

$$X \times_S G \to X \times_S X$$

is a closed immersion. For every section σ of X over S, the corresponding morphism $G \to X$, $g \mapsto \sigma \cdot g$ is a closed immersion.

[911] *N.D.E.* Cf. also V, 9.1 or EGA IV_2, 2.7.1.

In particular:

Corollary 2.5. *Let $u : G \to H$ be a monomorphism of S-preschemes in groups, with G of multiplicative type and H affine over S. Then u is a closed immersion, $H/G = Y$ exists and is affine over S, and finally H is a principal homogeneous bundle over Y with group G_Y.*

Remark 2.6. *Let $u : G \to H$ be a monomorphism of S-preschemes in groups, with G of multiplicative type and of finite type over S, H of finite presentation over S and separated over S. Then one can show that u is a closed immersion, using VIII 7.12 (see remark VIII 7.13 b)).*

Proposition 2.7. *Let $u : G \to H$ be a homomorphism of S-groups of multiplicative type, with H of finite type over S. Then:*

(i) $G_0 = Ker u$ is an S-group of multiplicative type, the equivalence relation defined by G_0 in G is M-effective, hence (IV 3.4) u factors as

$$G \to G/G_0 = I \to H,$$

where $I \to H$ is a monomorphism[912] of S-groups; I is of multiplicative type, the equivalence relation in H defined by I is M-effective, consequently $H_0 = H/I$ exists, and moreover H_0 is of multiplicative type.

(ii) H_0 and I are of finite type, and the same holds for G_0 if G is.

Proof. Proceeding as for 2.3, one is reduced to the case where G and H are diagonalizable, and then 2.7 reduces to VIII 3.4.

Let us note the following corollaries:

Corollary 2.8. *a) Let S be a prescheme. Then the category of S-groups of multiplicative type and of finite type is an abelian category.*

b) Let $u : G \to H$ be a homomorphism in this category; for it to be a monomorphism (resp. an epimorphism, resp. an isomorphism) in this category, it is necessary and sufficient that u be a monomorphism of preschemes (resp. that u be faithfully flat, resp. that u be an isomorphism of preschemes).

It suffices to note that the product of two S-groups of multiplicative type is again of multiplicative type (which is immediate), and evidently of finite type over S if both factors are. The rest of 2.8 follows immediately from 2.7; the detail is left to the reader.

Corollary 2.9. *Let $u : G \to H$ be a homomorphism of S-groups of multiplicative type and of finite type. Let U be the set of points $s \in S$ such that $u_s : G_s \to H_s$ is a monomorphism (resp. faithfully flat, resp. an isomorphism). Then U is both open and closed, and the induced homomorphism $u_U : G_U \to H_U$ is a monomorphism (resp. faithfully flat, resp. an isomorphism).*

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[912] *N.D.E.* and even a closed immersion.

Let P (resp. Q) be the kernel (resp. cokernel) of u. By 2.7, Q exists and P and Q are of multiplicative type; moreover the formation of P and of Q commutes with every base change $S' \to S$, in particular with the formation of fibers. On the other hand, u is a monomorphism (resp. faithfully flat, resp. an isomorphism) if and only if $P = 0$ (resp. $Q = 0$, resp. $P = Q = 0$). Thanks to these remarks, one is therefore reduced to verifying the following: if R is an S-group of multiplicative type, then the set U of $s \in S$ such that $R_s = 0$ is open and closed, and $R_U = 0$. But this is contained in Remark 1.4.1 a).

Corollary 2.10. *Let G be an S-group of multiplicative type and of finite type, and H, H' two subgroups of multiplicative type.*

a) $H'' = H \cap H'$ is a subgroup of multiplicative type of G.

b) Let U be the set of $s \in S$ such that $H_s \subseteq H'_s$ (resp. $H_s = H'_s$). Then U is open and closed, and $H_U \subseteq H'_U$ (resp. $H_U = H'_U$).

Of course the intersection sign in $H \cap H'$ denotes the intersection in the functorial sense, i.e. $H \times_G H'$, which is evidently an S-subgroup of G; likewise the inclusion sign (resp. equality sign) denotes the order relation (resp. the equality) between subfunctors of H (and not inclusion (resp. equality) of underlying sets).

Applying first 2.7 to the inclusion morphism $H \to G$, one finds that H is of finite type; likewise H' is of finite type. It then follows from 2.8 that $H \cap H'$ is of multiplicative type and of finite type (one will note that the canonical functor from the category envisaged in 2.8 to the category $(Sch)/S$ commutes with finite projective limits).

The formation of $H \cap H'$ commutes with base extension, in particular with the formation of fibers. On the other hand, $H \subseteq H'$ (resp. $H = H'$) is equivalent to $H'' = H$ (resp. $H'' = H$ and $H'' = H'$). Consider then the canonical homomorphisms $H'' \to H$ and $H'' \to H'$; then U is the set of $s \in S$ such that the induced homomorphism $H''_s \to H_s$ is an isomorphism (resp. such that $H''_s \to H_s$ and $H''_s \to H'_s$ are isomorphisms). By 2.9, it follows that U is open and closed, and that $H''_U \to H_U$ (resp. $H''_U \to H_U$ and $H''_U \to H'_U$) are isomorphisms. Hence the desired conclusion.

Proposition 2.11. *Let S be a prescheme, G an S-group of multiplicative type and of finite type over S, H a subgroup of multiplicative type of G, and $K = G/H$ (which is a group of multiplicative type, quotient of G).*

(i) Suppose G is trivial, i.e. diagonalizable. Then there exists a partition of S into open-and-closed parts S_i, such that for every i, H_{S_i} and K_{S_i} are diagonalizable. In particular, if S is connected, H and K are diagonalizable.

(ii) Same statement as in (i), replacing "diagonalizable" by "isotrivial", provided that S is connected, or locally connected, or quasi-compact.

(iii) Suppose G is locally trivial (resp. locally isotrivial, resp. quasi-isotrivial); then the same holds for K and H.

Proof. (i) By hypothesis $G = D_S(M)$, where M is a commutative group of finite type. For every quotient group M_i of M, let $H_i = D_S(M_i)$ be the corresponding diagonalizable subgroup of G (VIII 3.1). Let S_i be the set of $s \in S$ such that $H_s = (H_i)_s$; by 2.10, S_i is open and closed, and one has $H_{S_i} = (H_i)_{S_i}$, hence H_{S_i} is diagonalizable, hence so is K_{S_i} (VIII 3.1). Evidently the S_i are pairwise disjoint; we claim that they cover S. This follows from the fact that for every s, H_s is diagonalizable, as a subgroup of the diagonalizable group G_s (cf. 8.1 below[913]); hence H_s is of the form $D_{\kappa(s)}(M_i)$, by VIII, 1.5 and 3.2 b). Restricting to the family of non-empty S_i, the conclusion (i) appears.

(ii) By hypothesis, there exists $S' \to S$ étale finite surjective such that $G_{S'}$ is diagonalizable. Hence every point of S' has an open-and-closed neighborhood U' such that $H_{U'}$ and $K_{U'}$ are diagonalizable. Then the image U of U' in S is open and closed, and $S' \to S$ still induces an étale finite surjective morphism $U' \to U$; hence one sees that every point s of S has an open-and-closed neighborhood U such that H_U and K_U are isotrivial. The conclusion (ii) follows at once.

(iii) Follows at once from (i) and (ii) and the definitions. Note moreover that the "quasi-isotrivial" case will also follow from the more general fact that "finite type ⇒ quasi-isotrivial", announced in 1.2 (cf. X 4.5).

3. Infinitesimal properties: lifting and conjugation theorems

The fundamental infinitesimal properties of groups of multiplicative type follow from the following theorem:

Theorem 3.1. *Let S be an affine scheme, H an S-group of multiplicative type, F a quasi-coherent sheaf on S on which H operates (I 4.7). Then one has*

$$H^i(H, F) = 0 \quad \text{for} \quad i > 0,$$

where H^i is the Hochschild cohomology studied in Exp. I, § 5.

Indeed, by loc. cit., 5.3, the Hochschild cohomology is described, in terms of the affine rings A of S and B of G, and the module M over A defining F, as the cohomology of a complex of A-modules $C^\bullet(H, M)$, the formation of which commutes with every base change $A \to A'$. Consequently, for a base change $A \to A'$ with A' flat over A, one has

$$H^i(H', F') = H^i(H, F) \otimes_A A',$$

and consequently, if one supposes even A' faithfully flat over A, in order to prove that $H^i(H, F)$ is zero, it suffices to prove that the same holds for $H^i(H', F')$. This remark reduces us to verifying 3.1 in the case where G is diagonalizable; in that case this was proved in I 5.3.3.

Using the results of Exposé III, we shall deduce from 3.1 various consequences of geometric nature:

[913] *N.D.E.* The erroneous reference to IV 4.7.5 has been corrected. Note that N° 8 of the present Exposé is independent of N°s 3 to 7.

Theorem 3.2. *Let S be an affine scheme, S_0 an affine subscheme defined by an ideal J, H a group of multiplicative type over S, G an arbitrary prescheme in groups over S, $u, v : H \to G$ two homomorphisms of groups, $u_0, v_0 : H_0 \to G_0$ the homomorphisms deduced from u, v by the base change $S_0 \to S$. If $J^2 = 0$ and $u_0 = v_0$, then there exists a $g \in G(S)$ such that $v = int(g) \circ u$ and $g_0 = e$.*

This follows from III 2.1 (ii), and from 3.1 (vanishing of H^1).

Corollary 3.3. *Under the preliminary conditions of 3.2, suppose moreover G smooth over S, but assume only J nilpotent (not necessarily of square zero). If there exists $g_0 \in G_0(S_0)$ such that $v_0 = int(g_0) \circ u_0$, then g_0 lifts to a $g \in G(S)$ such that $v = int(g) \circ u$.*

By induction on the integer n such that $J^n = 0$, one is reduced to the case where J is of square zero. Moreover, G being smooth over S, one can lift g_0 to an element g' of $G(S)$. Replacing v by $v' = int(g') \circ u$, one is then reduced to the situation of 3.2.

Corollary 3.4. *Under the preliminary conditions of 3.2, with J nilpotent, suppose moreover u is central (for example G commutative). Then $u_0 = v_0$ implies $u = v$.*

Indeed, the reduction to the case where J has square zero is again immediate, and then it suffices to apply 3.2. In particular:

Corollary 3.5. *Let S, S_0, H, G be as in 3.2, with J nilpotent. Let $u : H \to G$ be a homomorphism of S-groups such that $u_0 : H_0 \to G_0$ is the unit homomorphism. Then u is the unit homomorphism.*

Theorem 3.6. *Let S be an affine scheme, S_0 a subscheme defined by a nilpotent ideal J, H a group of multiplicative type over S, G a prescheme in groups smooth over S, $u_0 : H_0 \to G_0$ a homomorphism of the S_0-groups deduced by the base change $S_0 \to S$.*

Then there exists a homomorphism $u : H \to G$ of S-groups which lifts u_0 (and by 3.3 any two such liftings u, u' are conjugate by an element of $G(S)$ reducing along the unit element of $G(S_0)$).

This follows from III 2.1 and 2.3, and from 3.1 (vanishing of H^2 for the existence of the lift of u_0, vanishing of H^1 for the uniqueness up to conjugation).

One can prove in the same way the following variants of 3.2 to 3.6 (which in fact entail the preceding statements, by applying them to the graph subgroups of the homomorphisms envisaged in 3.2 to 3.6):

Theorem 3.2 bis. *Let S be an affine scheme, S_0 a subscheme defined by an ideal J, G an S-prescheme in groups, H, H' subpreschemes in groups, of multiplicative type, G_0, H_0, H'_0 the groups over S_0 deduced by the base change $S_0 \to S$. If $J^2 = 0$ and $H_0 = H'_0$, then there exists $g \in G(S)$ such that $H' = int(g)(H)$ and $g_0 = e$.*

Corollary 3.3 bis. *Under the preliminary conditions of 3.2 bis, suppose moreover G smooth over S, and J nilpotent (not necessarily of square zero). Let $g_0 \in G_0(S_0)$ be such that $H'_0 = int(g_0)(H_0)$; then g_0 lifts to $g \in G(S)$ such that $H' = int(g)(H)$.*

Corollary 3.4 bis. *Under the preliminary conditions of 3.2 bis, suppose J nilpotent, H central (for example G commutative). Then $H_0 = H_0'$ implies $H = H'$.*

Theorem 3.6 bis. *Let S be an affine scheme, `S_0` a subscheme defined by a nilpotent ideal J, G an S-prescheme in groups smooth over S, `H_0` a subprescheme in groups of $G_0 = G \times_S S_0$, of multiplicative type. Then:*

(a) There exists a subprescheme in groups H of G, flat over S, such that $H \times_S S_0 = H_0$.

(b) Such an H is necessarily of multiplicative type.

(c) Finally, by 3.2 bis, any two such liftings H, H' of `H_0` are conjugate by an element $g \in G(S)$ reducing along the unit element of $G(S_0)$.

Let us show how one may prove 3.2 bis (of which 3.3 bis and 3.4 bis are an immediate consequence) and assertion (a) of 3.6 bis. The latter follows from 3.1 (vanishing of H^2) and from III 4.37 (ii). Similarly 3.2 bis follows from 3.1 (vanishing of H^1) and from III 4.37 (i), at least when G and H are smooth over S.

But using the results of the following Exposé, one can very simply deduce 3.2 bis and 3.6 bis, in the general form stated here, from 3.2 resp. 3.6. For 3.2 bis, it suffices indeed to note that by X 2.1, H and H' are isomorphic, which reduces us to 3.2.

For 3.6 bis, one notes that `H_0` is necessarily of finite type over `S_0`:[914] indeed, as a subprescheme of `G_0`, which is smooth over `S_0`, `H_0` is locally of finite type over `S_0`; but, by 2.1 a), `H_0` is affine over `S_0`, hence is of finite type over `S_0`. Then, by X 4.5, `H_0` is quasi-isotrivial, hence comes, by X 2.1, from a group of multiplicative type H over S.[915] Then, by 3.6, there exists a homomorphism of S-groups $u : H \to G$ which lifts the immersion $H_0 \hookrightarrow G_0$; since H and `H_0` (resp. G and `G_0`) have the same underlying topological space, and since, for every $h \in H$, the morphism $O_{G,u(h)} \to O_{H,h}$ is surjective (since it is so after reduction modulo J, which is nilpotent), u is also an immersion (cf. EGA I, 4.2.2).

Finally, for every lifting H of `H_0` to a flat subgroup of G, H is necessarily of multiplicative type by X 2.3,[916] which also proves assertion (b) of 3.6 bis. (The reader will verify that the results 3.2 bis to 3.6 bis are not used in Exposé X, hence that there is no vicious circle!)

Proposition 3.8.[917] *Let S be a prescheme, `S_0` a closed subprescheme defined by a locally nilpotent ideal J, G an S-group of multiplicative type, X an S-prescheme locally of finite presentation. Suppose that G operates on X, in such a way that $G_0 = G \times_S S_0$ operates trivially on $X_0 = X \times_S S_0$. Under these conditions, G operates trivially on X.*

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[914] *N.D.E.* The original indicated: "by 2.1 b), since its fibers are so". The editors did not understand this argument, and substituted the one that follows.
[915] *N.D.E.* The following sentence has been spelled out.
[916] *N.D.E.* Note that X 2.3 depends in an essential way on Theorem 3.6 of the present Exposé.
[917] *N.D.E.* The numbering of the original has been preserved: there is no N° 3.7.

The proof is that of III 2.4 b); one can also reduce to loc. cit. by noting that 3.8 reduces at once to the case where G is diagonalizable, which is the case envisaged in loc. cit.

Corollary 3.9. *Let S, S_0 be as above, and $u : G \to H$ a homomorphism of S-groups, with G of multiplicative type and H locally of finite presentation over S. Suppose that the homomorphism $u_0 : G_0 \to H_0$ deduced by base change $S_0 \to S$ is central. Then u is central.*

Indeed, it suffices to apply 3.8 by taking $X = H$ and making G operate on H by $(g, h) \mapsto int(u(g)) \cdot h = u(g)hu(g)^{-1}$.

4. The density theorem

This theorem (cf. 4.7 below)[918], together with the theorem[919] of N° 7, will be the essential tool in the present Exposé and the two following, for passing from the infinitesimal properties of groups of multiplicative type, which have just been developed, to the "finite" properties.

Definition 4.1. *Let X be a prescheme. A family $(Z_i)_{i\in I}$ of subpreschemes of X is said to be* schematically dense *if for every open U of X, and every closed subprescheme* `U_0` *of U which majorizes the $Z_i \cap U$, one has $U_0 = U$.*

One says that a subprescheme Z of X is schematically dense in X *if such is the case of the family reduced to Z.*

One sees immediately (cf. EGA IV₃, 11.10.1) that the definition is equivalent to saying that for every open U of X, every section f of `O_U` which is zero on the $Z_i \cap U$ is zero, which also means that the intersection of the kernels of the canonical homomorphisms

$$u_i : O_X \longrightarrow (v_i)_*(O_{Z_i})$$

is zero, where $v_i : Z_i \to X$ is the canonical immersion.[920] When X lies over a prescheme S, this is again equivalent to saying that for every open U of X and every pair (u, v) of morphisms from U into an S-prescheme Y separated over S that coincide on the $Z_i \cap U$, one has $u = v$. (Indeed, the relation $u = v$ is equivalent to $U_0 = U$, where `U_0` is the inverse image of the diagonal of $Y \times_S Y$ by the S-morphism $X \to Y \times_S Y$ defined by (u, v); this diagonal is a closed subprescheme of $Y \times_S Y$, hence `U_0` is a closed subprescheme of U, majorizing the $Z_i \cap U$ by the hypothesis on (u, v); hence if the family (Z_i) is schematically dense, one will have $U_0 = U$, hence $u = v$; the converse implication is seen by simply taking $Y = \operatorname{Spec} O_S[T]$.)

[918] All the results from 4.1 to 4.6 are contained in EGA IV₃, 11.10, to which we refer the reader for an exposition in form of the notion of schematic density.
[919] *N.D.E.* The "algebraicity" theorem 7.1.
[920] *N.D.E.* All the results on schematic density stated in EGA IV₃, § 11.10 follow from the results on "separating families of homomorphisms" proved in loc. cit., § 11.9, to which we shall refer in certain *N.D.E.* that follow.

With the terminology introduced in EGA I, end of N° 9.5, to say that the subprescheme Z of X is schematically dense also means that X is identical to the adherence subprescheme of Z in X.

Lemma 4.2. *Let X be a flat S-prescheme, $(Z_i)_{i\in I}$ a family of subpreschemes of X flat over S. Let `S_0` be a subprescheme of S, defined by a nilpotent ideal J; suppose the modules J^n/J^{n+1} locally free over `S_0`.*

Let X_0, Z'_i be deduced from X, Z_i by the base change $S_0 \to S$. Then, if the family $(Z'_i)_{i\in I}$ is schematically dense in `X_0`, the family $(Z_i)_{i\in I}$ is so in X.

Suppose $J^{n+1} = 0$ (where $n \geqslant 0$); let us argue by induction on n, the assertion being trivial for $n = 0$. Defining S_m, X_m, Z^i_m by reduction modulo J^{m+1} as is customary, the induction hypothesis already implies that $(Z^i_{n-1})_{i\in I}$ is schematically dense in X_{n-1}. Replacing X by an induced open if necessary, we are reduced to proving that every section f of `O_X` which vanishes on the Z_i is zero.

Now the section f_{n-1} of $O_{X_{n-1}} = O_X/J^n O_X$ defined by f vanishes on the Z^i_{n-1}, hence is zero, hence f is a section of $J^n O_X$. Since X is flat over S, one has

```
J^n O_X □ E ⊗_{O_{S_0}} O_{X_0},            where E = J^n = J^n/J^{n+1}.
```

Likewise, since each Z_i is flat over S, the restriction f_i of f to Z_i may be regarded as a section of:

```
J^n O_{Z_i} □ E ⊗_{O_{S_0}} O_{Z'_i}.
```

By hypothesis, the f_i are zero. Now E is locally free by hypothesis, hence so is $F = E \otimes_{O_{S_0}} O_{X_0}$.

Hence `f` is a section of the locally free module `F` over `X_0`, such that for every `i` its

restriction to Z'_i is zero. Since $(Z'_i)_{i\in I}$ is schematically dense in `X_0`, it follows at once that f is zero. QED.

Corollary 4.3. *Let X be a locally noetherian S-prescheme flat over S, $(Z_i)_{i\in I}$ a family of subpreschemes of X flat over S. Suppose that for every $s \in S$, the family $(Z_{i,s})_{i\in I}$ of fibers at s is schematically dense in X_s. Then the family $(Z_i)_{i\in I}$ is schematically dense in X.*

Replacing X by an induced open if necessary, one is reduced to proving that every section f of `O_X` whose restrictions to the Z_i are zero is itself zero.[921] For this, it suffices to prove that for every $x \in X$, the image of f in $O_{X,x}$ is zero. Denote by m_x the maximal ideal of $O_{X,x}$, and by m_s that of $O_{S,s}$, where s is the image of x in S. Since $O_{X,x}$ is noetherian, one has $\bigcap_{n\in\mathbb{N}} m_x^n = 0$, hence a fortiori $\bigcap_{n\in\mathbb{N}} O_{X,x} m_s^n = 0$.

So it suffices to show that, for every integer n, the section induced by f on $X \times_S \mathrm{Spec}(O_{S,s}/m_s^{n+1})$ is zero. By hypothesis, the family of fibers $Z_{i,s} = Z_i \otimes_{O_{S,s}} \kappa(s)$ is schematically dense in the fiber $X_s = X \otimes_{O_{S,s}} \kappa(s)$. This reduces us to the case where S is the spectrum of a local ring whose maximal ideal J is nilpotent, $S_0 = \mathrm{Spec}\,\kappa(s)$, and the hypotheses of 4.2 are satisfied; hence the conclusion.

[921] *N.D.E.* The original has been spelled out in what follows.

Lemma 4.4. *Let S be a locally noetherian prescheme, X a locally noetherian S-prescheme flat over S, $(Z_i)_{i\in I}$ a family of subpreschemes of X flat over S. Suppose that for every $s \in S$, the family $(Z_{i,s})_{i\in I}$ of fibers at s is schematically dense in X_s.*

Then, for every base change $S' \to S$ (S' not necessarily locally noetherian), the family $(Z'_i)_{i\in I}$ is schematically dense in X' (i.e. the family $(Z_i)_{i\in I}$ is universally schematically dense in X relative to S, cf. EGA IV₃, Def. 11.10.8).

Proof.[922] Let f' be a section of $O_{X'}$ over an open U' of X', vanishing on the Z'_i; we must show that f' is zero. Let $s' \in S'$; it suffices to prove that f' is zero at every point of U' above s'. Let s be the image of s' in S; replacing S, S' by the spectra of the local rings at s, s' if necessary, one may suppose S, S' local with s, s' as closed points. One may moreover suppose X affine, hence

```
S = Spec(A),    S′ = Spec(A′),    X = Spec(B),    X′ = Spec(B′),    B′ = B ⊗_A A′
```

where A, A' are local, $A \to A'$ is a local homomorphism, and `A`, `B` are noetherian. One may also suppose U' affine, of the form $X'_{g'}$, with $g' \in B'$.

For every sub-A-algebra A'' of A', consider $S'' = \mathrm{Spec}(A'')$ and $X'', Z^i_{n''}$ deduced from X, Z_i by the base change $S'' \to S$, giving the diagram:

$$Z_i \leftarrow Z^i_{n''} \leftarrow Z'_i \downarrow\downarrow\downarrow X \leftarrow X'' \leftarrow X' \downarrow\downarrow\downarrow S \leftarrow S'' \leftarrow S'.$$

We restrict in what follows to those A'' which are localizations of finite-type sub-A-algebras A''_1 of A' at $m' \cap A''_1$ (where m' is the maximal ideal of A'), so that the homomorphisms $A \to A'' \to A'$ are local. Note that these A'' form an increasing filtered family of subalgebras whose union is A', hence their direct limit is also A'. One has therefore likewise $B' = \lim_{A''} B \otimes_A A''$. Hence there exist an A'' and an element $g'' \in B'' = B \otimes_A A''$ whose image in B' is g'.

[923] By Lemma 4.5 below, the property that the family of fibers $(Z_{i,s})_{i\in I}$ be schematically dense in X_s is preserved by every base change $S'' \to S$; consequently, since each S'' is locally noetherian, if one replaces S by an appropriate S'', and X, Z_i by X'', Z''_i, the hypotheses of 4.4 will be preserved.

Hence, replacing S by S'' (and X, Z_i by X'', Z''_i) if necessary, one may suppose that g' comes from $g \in B$. Replacing X by the open $X_g = U$ if necessary, one may therefore suppose $U' = X'$. One sees likewise that one may suppose that f' comes from a section f of `O_X` over X.

Let $Y = V(f)$ be the subscheme of X defined by f, and $Y_i = Z_i \times_X Y$ its intersection with Z_i, which is a closed subscheme of Z_i, equal to $V(f_i)$, where f_i is the section of O_{Z_i} induced by f. Denoting by Y', Y'_i the S'-schemes deduced from the preceding by base change, one will again have

[922] *N.D.E.* For another proof, using a reduction to the case where S' is locally noetherian (and 4.3 and 4.5 as here), see EGA IV₃, 11.9.16 and 11.9.12 (N.B. in the last line of the proof of 11.9.16, replace 11.9.5 by 11.9.12).
[923] *N.D.E.* The following sentence has been added, and the original spelled out in what follows.

$$Y' = V(f'), \qquad Y'_i = Z'_i \times_{X'} Y' = V(f'_i),$$

and one has the analogous relations for Y'', Y''_i. The hypothesis that f' vanishes on the Z'_i is expressed by the relations $Y'_i = Z'_i$ for every i.[924]

Let m be the maximal ideal of A. For every integer $n \geqslant 0$, introduce the subscheme $S_n = \mathrm{Spec}(A/m^{n+1})$ of S, and the schemes X_n, Y_n, Z^i_n, Y^i_n deduced from X, Y, Z_i, Y_i by the base change $S_n \to S$. In general, for every S-prescheme P, we shall set $P_n = P \times_S S_n$.

For every n, and $i \in I$, consider the functor

$$F^i_n = \prod_{Z^i_n/S_n} Y^i_n/Z^i_n : (Sch)^\circ/S_n \to (Ens)$$

defined by (cf. VIII, 6.4): for every P over S_n,

$$F^i_n(P) = \Gamma((Y^i_n)_P / (Z^i_n)_P) = \begin{cases} \varnothing & \text{if } (Y^i_n)_P \neq (Z^i_n)_P; \\ \{\mathrm{id}\} & \text{if } (Y^i_n)_P = (Z^i_n)_P. \end{cases}$$

Since S_n is local artinian, and Z^i_n flat hence essentially free over S_n, then, by VIII, 6.4, each F^i_n is representable by a closed subprescheme T^i_n of S_n.

[925] The completion Â of the local ring A is noetherian since A is, and it is the projective limit of the A_n. Denote by K^i_n the ideal of $A_n = A/m^{n+1}$ defining T^i_n and K^i the projective limit of the K^i_n; it is an ideal of Â.

For i fixed, $m \geqslant n$ and every S_n-prescheme P, one has

$$(Y^i_n)_P = Y_i \times_S S_n \times_{S_n} P = Y_i \times_S P = (Y^i_m)_P,$$

and likewise $(Z^i_n)_P = (Z^i_m)_P$; it follows that F^i_n is the restriction $F^i_m \times_{S_m} S_n$ of F^i_m to $(Sch)/S_n$, whence $T^i_n = T^i_m \times_{S_m} S_n$. One therefore has a commutative diagram with exact rows:

$$\begin{tikzcd} 0 \arrow[r] & K^i_m \arrow[r] \arrow[d] & A_m \arrow[r] \arrow[d] & \mathcal{O}(T^i_m) \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & K^i_n \arrow[r] & A_n \arrow[r] & \mathcal{O}(T^i_n) \arrow[r] & 0 \end{tikzcd}$$

where all the vertical arrows are surjective. This has the following consequences: on the one hand, the projective system $(K^i_n)_{n\in\mathbb{N}}$ satisfies the Mittag-Leffler condition, hence the projective limit of the $\mathcal{O}(T^i_n)$ identifies with the topological ring quotient $\hat{A}/K^i$ (cf. EGA 0_III, § 13.2). On the other hand, the map $K^i \to K^i_n$ is surjective (cf. [BEns], III, § 7.4, Prop. 5), whence it follows that $K^i_n \simeq (K^i + m^{n+1}\hat{A})/m^{n+1}\hat{A}$ and $\mathcal{O}(T^i_n) \simeq (\hat{A}/K^i) \otimes_A (A/m^{n+1})$.

In other words, $(T^i_n)_n$ is an inductive system of affine artinian schemes, and the inductive limit $T^i = \lim_n T^i_n$ is a closed formal subscheme of the formal scheme $\hat{S} = Spf(\hat{A})$ (cf. EGA I, § 10), whose reduction modulo m^{n+1} is T^i_n.

[924] *N.D.E.* The original added: "at least at every point x' of X' above s'", but this restriction seems unnecessary.

[925] *N.D.E.* The original has been spelled out and corrected, to show that the projective limit of the rings $\mathcal{O}(T^i_n)$ defines a closed formal subscheme of the formal scheme $Spf(\hat{A})$.

Let T be the closed formal subscheme of $\hat{S}$ intersection of the T^i, that is, defined by $K = \sum_i K^i$. Since $\hat{A}$ is noetherian, there exists a finite part J of I such that one has $K = \sum_{i\in J} K^i$ (i.e. $T = \bigcap_{i\in J} T^i$). Note then that for every n, one has $K_n = \sum_{i\in J} K_n^i$ where K_n denotes the image of K in A_n.

Recall that f_i denotes the image of $f \in B = O(X)$ in $O(Z_i)$, and f_i' its image in $O(Z_i') = O(Z_i) \otimes_A A'$; the hypothesis $Y_i' = Z_i'$ is equivalent to the vanishing of f_i'. Since $O(Z_i) \otimes_A A'$ is the inductive limit of the subalgebras $O(Z_i) \otimes_A A''$ (where A'' satisfies the conditions made explicit above), there exists therefore an A'' such that $Y_i'' = Z_i''$. A priori, A'' depends on i, but one can find an A'' that works for all $i \in J$, since J is finite. Set $S'' = \operatorname{Spec}(A'')$ and $S''_{(n)} = S'' \times_S S_n$.[926]

Since $Y_i'' \times_{S''} S''_{(n)} = (Y_n^i)_{S''_{(n)}}$ equals $Z_i'' \times_{S''} S''_{(n)} = (Z_n^i)_{S''_{(n)}}$ for every $i \in J$, it follows from the definition of the T_n^i that $S''_{(n)} \to S_n$ factors through T_n^i for every $i \in J$, hence also through $T_n = \bigcap_{i\in J} T_n^i$, hence also through T_n^i for every $i \in I$. Denoting by f'' the image of f in $B'' = B \otimes_A A''$, and $f''_{(n)}$ its image in $B''_{(n)} = B''/m^{n+1}B''$, this means that for every n, $f''_{(n)}$ vanishes on the $Z_i \times_S S''_{(n)}$.

Since the morphisms $A \to A'' \to A'$ are local, the image s'' of s' in S'' is the closed point of S'', corresponding to the maximal ideal m'' of B'', and the image of s'' in S is s. Fix n and denote now by S_n'' the n-th infinitesimal neighborhood of s'' in S'', that is, $\operatorname{Spec}(A''/m''^{n+1})$. Set $B_n'' = B''/m''^{n+1}B''$ and $Z''_{n,i} = Z_i \times_S S_n'' = \operatorname{Spec}(B_n'')$. Then the image f_n'' of $f''_{(n)}$ in B_n'' vanishes on $Z''_{n,i}$, for every i. Now, by Lemma 4.5 below, the family of fibers

```
Z_i''' = Z″_{n,i} ×_{S″_n} κ(s″) = Z_{i,s} ×_{κ(s)} κ(s″)
```

is schematically dense in the fiber $X_0''' = X_n'' \times_{S_n''} \kappa(s'') = X_s \times_{\kappa(s)} \kappa(s'')$. Hence $S_n'', X_n'', (Z''_{n,i})_{i\in I}$, and $S_0''' = \operatorname{Spec} \kappa(s'')$ satisfy the hypotheses of Lemma 4.2; it follows therefore that $f_n'' = 0$, i.e. $f'' \in m''^{n+1}B''$, for every n.

Since B is noetherian and $B'' = B \otimes_A A''$ is a localization of a finite-type B-algebra, B'' is noetherian, hence its local rings are separated for their usual topology. It follows that f'' is zero at the points x'' of X'' such that $m''B''$ is contained in the maximal ideal of $O_{X'',x''}$, i.e. at the points of X'' above s''. A fortiori, f' is zero at the points x' of X' above s'. QED.

Lemma 4.5.0.[927] *Let X be an S-prescheme, $(Z_i)_{i\in I}$ a family of subpreschemes of X, and $S' \to S$ a faithfully flat morphism. If the family $(Z_i')_{i\in I}$ is schematically dense in X', then $(Z_i)_{i\in I}$ is schematically dense in X.*

This follows from EGA IV₃, 11.10.5 (i) and 11.9.10 (i). It remains to give the proof of:

Lemma 4.5. *Let X be a locally noetherian prescheme*[928] *over a field k, $(Z_i)_{i\in I}$ a family of subpreschemes of X, k' an extension of k, X' and Z_i' the preschemes*

[926] *N.D.E.* The original has been spelled out and corrected, distinguishing between $S''_{(n)}$ and the subscheme $S_n'' = S_n''(s'')$ introduced below.
[927] *N.D.E.* This lemma has been inserted here, since it is used in the proof of 4.5 and 4.7.
[928] *N.D.E.* This hypothesis is in fact superfluous, cf. EGA IV₃, 11.10.6 and 11.9.13.

deduced from X and Z_i by the base change k'/k. For $(Z_i)_{i\in I}$ to be schematically dense in X, it is necessary and sufficient that $(Z'_i)_{i\in I}$ be so in X'.

The "if" follows from 4.5.0; let us prove the "only if".[929] First, one may suppose $X = \mathrm{Spec}(B)$ affine. For every i, let $(Z_{ij})_{j\in J_i}$ be a covering of Z_i by affine opens; replacing $(Z_i)_{i\in I}$ by the family $(Z_{ij})_{(i,j)\in J}$, where $J = \coprod_{i\in I} J_i$, one may also suppose the Z_i affine.

Let $g \in B' = B \otimes_k k'$ and $t' \in B'_g$ be a section of $O_{X'}$ on the affine open $U' = X'_g$, vanishing on the $Z'_i \cap U'$, i.e. whose image in each $(O(Z_i) \otimes_k k')_g$ is zero. There exists a finite-type sub-k-algebra A of k' such that $g \in B_A = B \otimes_k A$ and $t' \in (B_A)_g$. The map $O(Z_i) \otimes_k A \to O(Z_i) \otimes_k k'$ being injective, so is the map

(O(Z_i) ⊗_k A)_g (O(Z_i) ⊗_k k′)_g;

given the hypothesis, this implies that the image of t' in each $(O(Z_i) \otimes_k A)_g$ is zero. This reduces us to showing that the family $(Z_{iA})_{i\in I}$ is schematically dense in X_A,[930] which follows from EGA IV₃, 11.9.10 b).

Let us note in passing the following result, which will serve in a later Exposé:[931]

Corollary 4.6. *Let X be a flat S-prescheme, U an open subset of X. Suppose that for every $s \in S$, U_s is schematically dense in X_s, and that X is locally noetherian, or locally of finite presentation over S. Then U is schematically dense in X.*

Suppose for simplicity U retrocompact in X (cf. EGA IV₃, 11.10.10 and 11.9.17 for the general case). The case X locally noetherian is included only for memory; it is contained in 4.3.

In the second case envisaged, one may evidently suppose S and X affine; then X and U are of finite presentation over $S = \mathrm{Spec}(A)$. The ring A is the inductive limit of its finite-type sub-$\mathbb{Z}$-algebras A_i. The "patented procedure" already used (cf. EGA IV₃, 8.8.2 and 8.10.5 (iii)) shows that there exist an i, an affine scheme X_i over $S_i = \mathrm{Spec}(A_i)$ and an open U_i in X_i, from which X, U are deduced by base change $S \to S_i$. Let E_i be the part of S_i consisting of $s \in S_i$ such that $(U_i)_s$ is schematically dense in $(X_i)_s$.

[932] By EGA IV₂, 5.9.9 and 5.10.2, E_i is the set of $s \in S_i$ such that $(U_i)_s$ contains the set $Ass O(X_i)_s$ of points "associated" with the structure sheaf of $(X_i)_s$, and by EGA IV₃, 11.9.17.1 this condition is of constructible nature, i.e. E_i is a constructible part of S_i.

[933] By 4.5, the inverse image of E_i by $S_j \to S_i$ (resp. by $S \to S_i$) is E_j (resp. the set E of $s \in S$ such that U_s is schematically dense in X_s). Moreover, by hypothesis,

[929] *N.D.E.* The original has been spelled out in what follows.
[930] *N.D.E.* The original continued with: "Using 4.3 for the $X_{A'}$, where A' is a quotient of A by a power of a maximal ideal, and Hilbert's Nullstellensatz, one is reduced to the case where k' is a finite extension of k". One could spell out this reduction, and pursue this approach, since the case of a finite extension k'/k is somewhat simpler than the more general case treated in EGA IV₃, 11.9.10 b).
[931] *N.D.E.* Specify this ...
[932] *N.D.E.* The original has been spelled out in what follows.
[933] *N.D.E.* The original has been spelled out in what follows.

$E = S$, which is also the inverse image by each $S \to S_i$ of S_i. By EGA IV₃, 8.3.11, this implies that there exists $j \geqslant i$ such that $E_j = S_j$, i.e. such that for every $s \in S_j$, $(U_j)_s$ is schematically dense in $(X_j)_s$.

Then, by 4.4 applied to (S_j, X_j, U_j) and to the base change $S \to S_j$, it follows that U is schematically dense in X. One will note moreover that we use 4.4 here only in the case of a family with finite index set (in fact, reduced to one element), in which case the proof of 4.4 simplifies considerably, as the reader will check.

Theorem 4.7. *Let S be a prescheme, G a group of multiplicative type and of finite type over S. For every integer $n > 0$, let ${}_{nG}$ be the kernel of $n \cdot id_G$. Then the family of subpreschemes $({}_{nG})_{n>0}$ of G is schematically dense (Def. 4.1).*

We distinguish two cases.

a) Case S locally noetherian. Then by 4.3 one is reduced to the case where S is the spectrum of a field k. By 4.5.0, one may suppose k algebraically closed and G diagonalizable, i.e. of the form $D_k(M)$, where M is a commutative group of finite type. Then M is of the form $\Gamma \times \mathbb{Z}^r$, with Γ finite, hence G is of the form $G_1 \times G_2$, with $G_1 = D(\Gamma)$ and $G_2 = G_m^r$; and for n large multiplicatively (namely n a multiple of the order of Γ) one will have _nG = G_1 × _nG_2 since one will have _nG_1 = G_1.

Applying again 4.3 to the projection $G \to G_1$, one is reduced to the case of G_2, i.e. to the case where $G = G_m^r$. Since G is then reduced, it amounts to the same to say that $({}_{nG})_{n>0}$ is schematically dense in G, or that the union of the ${}_{nG}$ is dense in G for the ordinary topology. Since _nG = (_nG_m)^r, one is reduced to the case of $G = G_m$, hence G irreducible of dimension 1. Then this follows from the fact that the union of the ${}_{nG}$ (equal to the set of roots of unity in k) is infinite.

b) General case. For every point s of S, there exist an open neighborhood U of s and a faithfully flat quasi-compact morphism $S' \to U$ such that $G' = G_{S'}$ is diagonalizable, i.e. of the form $D_{S'}(M)$. By 4.5.0 again, one may reduce to the case of G', so one may suppose G diagonalizable; hence it comes from the absolute group $H = D_{\mathbb{Z}}(M)$ over $\mathrm{Spec}(\mathbb{Z})$ by base change $S \to \mathrm{Spec}(\mathbb{Z})$. By the proof of a), for every $s \in \mathrm{Spec}(\mathbb{Z})$, the family (_nH_s)_{n > 0} is schematically dense in H; it now suffices to apply 4.4.[934]

Corollary 4.8. *a) Let $u, v : H \to G$ be two homomorphisms of S-preschemes in groups, with H of multiplicative type and*[935] *of finite type, and suppose that for every integer $n > 0$, the restrictions of u and v to ${}_{nH}$ are identical. Then $u = v$.*

b) Let H_1, H_2 be two subgroups of multiplicative type and of finite type of a prescheme in groups G, and suppose that for every integer $n > 0$, one has _nH_1 = _nH_2; then $H_1 = H_2$.

[934] *N.D.E.* One will need in X 4.3 this result for S non locally noetherian (namely, $S = \mathrm{Spec}(\hat{A} \otimes_A \hat{A})$, where $\hat{A}$ is the completion of the noetherian local ring A).
[935] *N.D.E.* “Diagonalizable” has been replaced by “of multiplicative type”.

The first assertion follows from the second, by consideration of the subgroups H_1 and H_2 of $H \times G$, graphs of u and v. To prove b), let $H = H_1 \cap H_2$; this is a subprescheme in groups of H_i ($i = 1, 2$); one must show that it is identical to H_i. Now the hypothesis means that it majorizes the _nH_i. One is therefore reduced to proving the:

Corollary 4.9. *Let G be an S-group of multiplicative type and of finite type, and H a subprescheme in groups which majorizes the ${}_{n}G$, $n > 0$. Then $H = G$.*

By 4.7, one is reduced to proving that the subprescheme H is closed, or again that it equals G set-theoretically. This reduces us to the case where S is the spectrum of a field; but then every subprescheme in groups of G is closed (VI_B 1.4.2), whence the conclusion.

Remark 4.10. *Under the conditions of 4.7, let m be an integer > 0 having the following properties: for every $s \in S$, m is not a power of the characteristic of $\kappa(s)$, and if G_s is of type M, the prime divisors of the torsion of M divide m. (N.B. This second condition is always satisfied if G is a torus.) Then the proof given shows that in the statement of 4.7 and the Corollaries 4.8 and 4.9, one may restrict attention to subgroups of the form ${}_{(m^r)}G$, with $r > 0$.*

Moreover, one sees at once that when G is smooth over S, then for every point $s \in S$, there exist an open neighborhood U of s and an integer $m > 0$, prime to all the residual characteristics of U, satisfying the preceding conditions for G_U. (Take for example for m the order of the torsion of the type of G at s, cf. 1.4.1 a) and 2.1 e).) Under these conditions, one finds ${}_{(m^r)}G$ that are finite and étale over S, and whose family is schematically dense, provided one restricts to over U. This remark allows, in certain cases (notably those involving the theorems 3.2 and 3.2 bis, cf. XI 6, but not those involving the theorems 3.6 and 3.6 bis), to dispense with the theorem 3.1, which involves Hochschild cohomology.

5. Central homomorphisms of groups of multiplicative type

Lemma 5.0.[936] *Let (A, m) be a noetherian local ring, $S = \mathrm{Spec}(A)$, and H a finite scheme over S, so $H = \mathrm{Spec}(B)$, where B is finite over A. Let K be a subscheme of H, such that by reduction modulo m^{n+1} one has $K_n = H_n$ for every n. Then $K = H$.*

Let s be the closed point of S. Note first that K is a closed subscheme of H. Indeed, it is a priori a closed subscheme of an open U of H. But K, hence U, contains the fiber $K_s = H_s$; since the morphism $H \to S$ is finite, hence closed, it follows that the complement of U is empty, i.e. $U = H$. Hence K is defined by an ideal I of B. The hypothesis entails that I is contained in $m^n B$ for every n; since B is a finite A-module, it is separated for the m-adic topology, whence $I = 0$.

Theorem 5.1. *Let $u, v : H \to G$ be two homomorphisms of S-preschemes in groups, with H of multiplicative type and of finite type, and S locally noetherian or G of*

[936] *N.D.E.* This lemma has been made explicit, since it is used several times in the sequel (implicitly in the original).

finite presentation over S. Let $s \in S$ be such that $u_s = v_s$, and suppose u_s central. Then there exists an open neighborhood U of s such that $u_U = v_U$.

We distinguish the two cases:

a) S locally noetherian.[937] Let $K = Ker(u, v)$ be the inverse image of the diagonal of $G \times_S G$ by the morphism (u, v); this is a subprescheme in groups of H. We wish to find U such that $K_U = H_U$. Note that, since S is locally noetherian and H of finite type over S, H is locally noetherian, hence the immersion $K \hookrightarrow H$ is of finite type (cf. EGA I, 6.3.5). Hence K is of finite type over S, hence of finite presentation over S, since S is locally noetherian. Consequently, by EGA IV3, 8.8.2.4, to show that there exists an open neighborhood U of s such that $K_U = H_U$, it suffices to show that $K_{S_0} = H_{S_0}$, where S_0 is the spectrum of $A = O_{S,s}$. One may therefore suppose S local with closed point s. Replacing the noetherian local ring A by its completion Â if necessary, which introduces a base change $\hat{S} \to S$ faithfully flat and quasi-compact, one may even, if one wishes, suppose[938] A complete.

By 3.4 one has $u_n = v_n$ for every n, where as usual the index n indicates reduction modulo m^{n+1} (m being the maximal ideal of A). For every integer $m > 0$, denoting by ${}_{m}u, {}_{m}v$ the homomorphisms ${}_{m}H \to G$ induced by u, v, one therefore also has $({}_{m}u)_n = ({}_{m}v)_n$. This being true for every n, and ${}_{m}H$ being finite over S by 2.2, it follows that ${}_{m}u = {}_{m}v$ by 5.0. This being true for every m, one has therefore $u = v$ by 4.7.

b) G of finite presentation.[939] Since H is also of finite presentation over S, then, by EGA IV3, 8.8.2, we may again suppose S local with closed point s and prove that then $u = v$. If $f : S' \to S$ is a faithfully flat quasi-compact morphism, and if one denotes by f_H the morphism $H' \to H$ and $u', v' : H' \to G'$ the morphisms deduced from u, v, then the equality $u' = v'$ entails $u' \circ f_H = v' \circ f_H$, hence $u = v$, since f_H is an epimorphism. Hence, by making a faithfully flat quasi-compact extension of the base, one may suppose moreover H diagonalizable, hence of the form $D_S(M)$, with M a commutative group of finite type.

Introduce, as in the proof of 4.6, the increasing filtered family of finite-type sub-$\mathbb{Z}$-algebras A_i of $A = O_{S,s}$, and $S_i = \operatorname{Spec}(A_i)$.[940] Note that $H = D_S(M)$ comes, for every i, from the diagonalizable group $H_i = D_{S_i}(M)$. Since G is of finite presentation over S, then, by EGA IV3, 8.8.2 (see also VI_B, 10.2 and 10.3), there exist an index i, a prescheme in groups G_i of finite presentation over S_i, and morphisms of S_i-groups $u_i, v_i : H_i \to G_i$ from which u, v come by base change. Let s_i be the image of s in S_i and let $\rho_i : H_i \times_{S_i} G_i \to G_i$ be the morphism of S_i-preschemes defined by $\rho_i(h, g) = u_i(h) g u_i(h)^{-1}$. Then, since u_s is central, $\rho_s = \rho_i \times_{\kappa(s_i)} \kappa(s)$ equals the second projection; so the same holds for ρ_i (since $\kappa(s_i) \to \kappa(s)$ is faithfully flat and

[937] *N.D.E.* The original has been spelled out in what follows.

[938] *N.D.E.* Because $K = H$ if $K_{\hat{S}} = H_{\hat{S}}$, cf. SGA 1, VIII 5.4; but the sequel does not use the hypothesis "A complete".

[939] *N.D.E.* The original has been spelled out in what follows.

[940] *N.D.E.* The original has been spelled out in what follows.

quasi-compact), i.e. $(u_i)_{s_i}$ is central. Similarly, since $u_s = v_s$ one has $(u_i)_{s_i} = (v_i)_{s_i}$. One can then apply a) to the situation over S_i, whence the announced conclusion.

Corollary 5.2. *Let $u : H \to G$ be a homomorphism of S-preschemes in groups, with H of multiplicative type and of finite type, and S locally noetherian or G of finite presentation over S. Let $s \in S$ and suppose that u_s is the unit homomorphism. Then there exists an open neighborhood U of s such that u_U is the unit homomorphism.*

Corollary 5.3. *Let u, H, G be as in 5.2, suppose moreover G separated over S. Let U be the set of $s \in S$ such that u_s is the unit homomorphism. Then U is an open-and-closed part of S, and $u_U : H_U \to G_U$ is the unit homomorphism.*

It remains only to prove that U is closed. Now let $K = Ker(u)$; since G is separated over S, K is a closed subprescheme of H, and U is the set of $s \in S$ such that $K_s = H_s$. One then sees easily that U is closed, for example as an application of VIII 6.4 (H being essentially free over S by VIII 6.3), or by noting that one may suppose S reduced and U dense in S, hence schematically dense in S, which implies H_U schematically dense in H since H is flat over S,[941] and since u and the unit homomorphism coincide on H_U, they coincide on H.

Corollary 5.4. *Let S be a prescheme, H and G two S-groups, with H of multiplicative type and of finite type and G separated and of finite presentation over S, $\pi : S' \to S$ a*[942] *universal effective epimorphism (for example, a faithfully flat quasi-compact morphism, or a morphism admitting a section, cf. IV 1.12) with geometrically connected fibers.*[943]

Let $u' : H' \to G'$ be a central homomorphism of the S'-groups deduced from H, G by the base change $S' \to S$. Then there exists a unique homomorphism $u : H \to G$ of S-groups such that $u \times_S S' = u'$. When S'/S admits a section g, then u is the morphism deduced from u' by the base change $g : S \to S'$.

Since π is a universal effective epimorphism, so is $H' \to H$, whence the uniqueness of u, cf. the beginning of the proof of 5.1 b). If π admits a section g, then $u' = \pi^*(u)$ entails $u = g^*\pi^*(u) = g^*(u')$.

For the existence of u, one is reduced, by IV 2.3, to showing that the two homomorphisms $u''_1, u''_2 : H'' \to G''$ of S''-groups deduced from u' by the two base changes $pr_1, pr_2 : S'' = S' \times_S S' \to S'$, are identical. Now they coincide on the diagonal of S''; more precisely the inverse images of u''_1 and u''_2 by the diagonal morphism $S' \to S''$ are identical (since both equal u').[944] Since u''_1 and u''_2 are central, one may apply 5.3 to the morphism $u''_1(u''_2)^{-1}$. There exists therefore an open-and-closed part U of S'', containing the diagonal of S'', such that u''_1 and u''_2 coincide above U.

[941]*N.D.E.* And of finite presentation over S, cf. EGA IV_3, 11.10.5 (ii) and 11.9.10 (ii).
[942]*N.D.E.* "Morphism covering for the faithfully flat quasi-compact topology" has been replaced by "universal effective epimorphism", so as to be able to apply this to the morphism $K \to S$ of 5.5; the beginning of the proof has been modified accordingly.
[943]*N.D.E.* This is the case, for example, if $S = \operatorname{Spec} k$ and $S' = \operatorname{Spec} k'$, where k is a field and k' a radicial extension of k, cf. X, Prop. 1.4.
[944]*N.D.E.* The following sentence has been added.

Now the fibers of S'/S being geometrically connected, the same holds for those of S''/S, which are therefore a fortiori connected; whence it follows that U (containing the diagonals of the said fibers) contains the said fibers, hence is equal to S'', which completes the proof.

Corollary 5.5. *Let S be a prescheme, K an S-prescheme in groups locally of finite type*[945] *and with connected fibers, H a subgroup of multiplicative type and of finite type, invariant in K. Then H is a central subgroup of K.*

[946] Note first that $\pi : K \to S$ has geometrically connected fibers, since for a group scheme locally of finite type over a field, connected implies geometrically connected (cf. VI_A 2.4). One can then apply 5.4 by taking $G = H$ and $S' = K$, to the homomorphism of K-groups $u' : H_K \to H_K$ defined set-theoretically by $(h, k) \mapsto (khk^{-1}, k)$, which is central since H is commutative. The inverse image of u' by the unit section $\epsilon : S \to K$ is the identity homomorphism of K, hence by 5.4 the same holds for $H_K \to H_K$; hence H is central in K.

Let us state the variants of the preceding results for central subgroups of multiplicative type. One obtains, by proceeding as for the preceding results (and using 3.2 bis):

Theorem 5.1 bis. *Let G be an S-prescheme in groups, H_1 and H_2 two sub-preschemes in groups of multiplicative type and of finite type, with $(H_1)_s$ central. Suppose S locally noetherian or G of finite presentation over S. Then for every $s \in S$ such that $(H_1)_s = (H_2)_s$, there exists an open neighborhood U of s such that $H_1|U = H_2|U$.*

Corollary 5.3 bis. *Under the preceding conditions, suppose moreover G separated over S. Let U be the set of $s \in S$ such that $(H_1)_s = (H_2)_s$. Then U is an open-and-closed part of S, and $H_1|U = H_2|U$.*

Corollary 5.4 bis. *Let S be a prescheme, G a prescheme in groups separated and of finite presentation over S, $S' \to S$ a covering morphism for the faithfully flat quasi-compact topology, with geometrically connected fibers, H' a subgroup of multiplicative type and of finite type of $G' = G \times_S S'$.*

Then there exists a unique subgroup H of G such that $H' = H \times_S S'$, and H is of multiplicative type and of finite type over S, by 1.1 and 2.1 b). If $S' \to S$ admits a section g, then H is the inverse image of the subgroup H' of G' by $g : S \to S'$.

Theorem 5.6. *Let S be a prescheme, $u : H \to G$ a homomorphism of S-preschemes in groups, with H of multiplicative type and of finite type, G of finite presentation over S.*

a) Suppose G has connected fibers, and let $s \in S$ be such that $u_s : H_s \to G_s$ is a central homomorphism. Then there exists an open neighborhood U of s such that the homomorphism $H|U \to G|U$ induced by u is central.

[945] *N.D.E.* The hypothesis "locally of finite type" has been added to agree with the reference VI_A, 2.4. In fact, one can show that over a field, every connected group scheme is geometrically connected.
[946] *N.D.E.* The original has been spelled out in what follows.

b) Suppose that for every $s \in S$, $u_s : H_s \to G_s$ is central. Then u is central.

[947] Let us prove (a). Proceeding exactly as in the proof of 5.1 (b), one may suppose S local with closed point s, then $H = D_S(M)$ diagonalizable, then that there exist a finite-type sub-$\mathbb{Z}$-algebra A_i of $A = O_{S,s}$ and a morphism $u_i : D_{S_i}(M) \to G_i$ such that u_{s_i} is central (where s_i is the image of s in S_i) and from which u comes by base change. This reduces us to the case where S is local noetherian with closed point s, and one must then prove that u is central.

Let K be the subprescheme $Ker(v, w)$, where $v, w : H \times_S G \rightrightarrows G$ are defined by

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v(h, g) = g,      w(h, g) = int(u(h)) · g = u(h) g u(h)^{-1}.
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Then K is a sub-G-group of the G-group $H_G = H \times_S G$; we wish to show that it is equal to H_G itself. By 4.9, one is reduced to proving that it majorizes the $_m(H_G) = (_mH) \times_S G$ for every integer $m > 0$, which reduces us to the case where $H = {}_mH$, hence H finite over S.

Let e be the unit element of the fiber G_s; then $S^0 = \mathrm{Spec}(O_{G,e})$ is a local noetherian scheme (G being of finite presentation over S noetherian); set $S^0_n = S^0 \times_S S_n$, where $S_n = \mathrm{Spec}(A/m^{n+1})$. Then $K_{S^0} = K \times_G S^0$ is a subprescheme of $H_{S^0} = H \times_S S^0$, and, by 3.9, one has $K_{S^0_n} = H_{S^0_n}$ for every n.[948] Since H_{S^0} is finite over S^0, one concludes from 5.0 (applied to the noetherian local ring B) that $K_{S^0} = H_{S^0}$.

On the other hand, since $H \times_S G$ is noetherian (being of finite presentation over S noetherian), the immersion $K \hookrightarrow H \times_S G$ is of finite type (cf. EGA I, 6.3.5), so that K is of finite type, hence of finite presentation over G. Then the equality $K_{S^0} = H_G \times_G S^0$ entails, by EGA IV₃, 8.8.2.4, that there exists an open neighborhood W of e in G such that $K \times_G W = H_G \times_G W = H \times_S W$. Hence K majorizes the open neighborhood $V = H \times_S W$ of the unit section of G_H over H.

For every $t \in H$, the fiber G_t (being a $\kappa(t)$-algebraic group) is Cohen–Macaulay (VI_A, 1.1.1), hence without embedded components; as it is moreover connected, hence irreducible (VI_A, 2.4), it has its generic point as unique associated point. Hence, by EGA IV₂, 3.1.8, the open V_t is schematically dense in G_t. By 4.3, V is therefore schematically dense in G_H. Moreover, since K is a sub-H-group of G_H, it induces on each fiber G_h a subgroup K_h, and since the latter majorizes an open neighborhood of the unit element and G_h is connected, it follows that $K_h = G_h$, hence $K = G_H$ set-theoretically. Thus K is a closed subprescheme of G_H which majorizes the schematically dense subprescheme V, hence $K = G_H$, which proves a).

Finally, b) is a direct consequence of 3.9, given that to verify that a subprescheme K of $P = G \times_S H$ is identical to P, it suffices to verify that for every base change $S' \to S$, where S' is an artinian quotient of a local ring of S, one has $K' = P'$.

[947] *N.D.E.* The original has been spelled out in the proof of a).

[948] *N.D.E.* The original has been spelled out and corrected, by replacing "unit section of $(G_H)_n$ over H_n" by $H_{S^0_n}$.

6. Monomorphisms of groups of multiplicative type, and canonical factorization of a homomorphism of such a group

Lemma 6.1. *Let S be a quasi-compact prescheme, G an S-prescheme in commutative groups, of finite presentation and quasi-finite over S. Then there exists an integer $n > 0$ such that $n \cdot id_G = 0$, i.e. $G = {}_nG$.*

If S is the spectrum of a field k, then G is finite over k, and by VII_A 8.5, it suffices to take $n = deg(G/k)$. Suppose now $S = \mathrm{Spec}(A)$ local artinian; let k be the residue field of A, $G_0 = G \otimes_A k$, and $n_0 = deg(G_0/k)$. Distinguish two cases.

a) k is of characteristic zero. Then `G_0` is separable over k, hence G is unramified over S. Then the unit section of G is an open immersion, hence ${}_nG = Ker(n \cdot id_G)$ is an open subscheme of G; therefore in order that it be equal to G, it suffices that it be so set-theoretically; one may then take $n = n_0$.

b) k is of characteristic $p > 0$. Denote by m the maximal ideal of A and $S_r = \mathrm{Spec}(A/m^{r+1})$ for every r. Let m be an integer such that $m^{m+1} = 0$; we claim that one may take $n = p^m n_0$. Indeed, proceeding by induction on m, and setting $n' = p^{m-1} n_0$, one may suppose that $n' \cdot id_G$ induces the zero endomorphism in $G_{m-1} = G \times_S S_{m-1}$.

[949] Denote by E the set of endomorphisms u of G having this property and K the kernel of the morphism of functors in groups $G \to \prod_{S_{m-1}/S} G$ (cf. III 0.1.2). Then E identifies with $\mathrm{Hom}_{S-gr.}(G, K)$; in particular the abelian group law on E is induced by that of K. Now, by III 0.9, K is the S-functor in groups which to every $g : T \to S$ associates the k-vector space

$$\mathrm{Hom}_{O_{T_0}}(g_0^*(\Omega^1_{G_0/S_0}), m^m O_T);$$

one has therefore $pu = 0$ for every $u \in E$. Hence one has $pn' \cdot id_G = 0$, i.e. $p^m n_0 \cdot id_G = 0$.

Suppose now S noetherian (one reduces to this in 6.1 by the customary reduction to the noetherian case[950]). It then suffices to combine the foregoing with the

Lemma 6.2. *Let X be a quasi-finite prescheme over S noetherian, and consider an increasing filtered family of subpreschemes $(X_i)_{i \in I}$ having the following property:*

`for every s ∈ S and n □ 0, setting S_{s,n} = Spec(O_{S,s}/m^{n+1}), there exists an i ∈ I such that X_i ×_S S_{s,n} = X ×_S S_{s,n}.`

Under these conditions, there exists an $i \in I$ such that $X_i = X$.

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[949] *N.D.E.* The original has been spelled out in what follows, by making explicit the results of Exp. III that are used.

[950] *N.D.E.* Since S is quasi-compact, it is covered by a finite number of affine opens, hence one is reduced to the case where $S = \mathrm{Spec}(A)$. Then, since G is of finite presentation over S, one can apply EGA IV_3, 8.8.2.

Since S is noetherian, there exists a maximal open U such that one has $X|U = X_i|U$ for i large; we shall show that $U = S$. In other words, we shall show that if $U \neq S$, one can find a U_1 strictly larger than U, and an i such that $X|U_1 = X_i|U_1$. Localizing at a maximal point s of $S - U$, one is reduced to the case where S is local with closed point s, and $U = S - s$. (Indeed, if one writes $S^0 = \mathrm{Spec}(O_{S,s})$ and if there exists i such that $X \times_S S^0 = X_i \times_S S^0$ then,[951] there exists an open neighborhood V of s such that $X|V = X_i|V$; hence taking i large enough so that $X_i|U = X|U$, one will have $X|W = X_i|W$, where $W = U \cup V$.)

Then, for i large, since $X|U = X_i|U$, one sees that X_i is a closed subprescheme of X defined by an ideal $I^{(i)}$ of support contained in $X_s = X_0 = X \times_S S_0$ (where $S_n = \mathrm{Spec}(A/m^{n+1})$, $S = \mathrm{Spec}(A)$). Since X_0 is quasi-finite over S_0, X_0 is a finite closed part of X noetherian, hence $I^{(i)}$ is a module of finite length. It follows that there exists an integer $n \geqslant 0$ such that $I^{(i)} \cap m^{n+1} O_X = 0$. On the other hand, by the hypothesis in 6.2, one may suppose (by enlarging i if necessary) that the image of $I^{(i)}$ in $O_{X_n} = O_X/m^{n+1}O_X$ is zero. This implies $I^{(i)} = 0$, hence $X_i = X$. QED.

Lemma 6.3. *Let S be a prescheme, K a prescheme in groups over S, locally of finite presentation, $s \in S$ such that K_s is quasi-finite (resp. unramified) over $\kappa(s)$ at the unit element. Then there exists an open neighborhood U of s such that $K|U$ is locally quasi-finite (resp. unramified) over U.*[952]

Let V be the set of points x of K such that, denoting by t the image of x in S, the fiber K_t is quasi-finite (resp. unramified) over $\kappa(t)$ at x, i.e. such that x is isolated in K_t (resp. and its local ring in K_t is a separable extension of $\kappa(t)$). One knows that V is open since K is locally of finite presentation over S,[953] hence if ϵ denotes the unit section of K, $\epsilon^{-1}(V)$ is open. By hypothesis it contains s, hence is an open neighborhood U of s. The latter does the trick; in other words, $t \in U$ implies that K_t is locally quasi-finite (resp. unramified) over $\kappa(t)$: indeed, since K_t is a group locally of finite type over $\kappa(t)$, this follows from the fact that it is quasi-finite (resp. unramified) over $\kappa(t)$ at the point $\epsilon(t)$, cf. VI_B 1.3.

Combining 6.1 and 6.3, one finds the

Theorem 6.4. *Let S be a prescheme, H an S-group of multiplicative type and of finite type, K a closed subprescheme in groups, of finite presentation over S. Let $s \in S$ be such that K_s is finite.*

Then there exists an open neighborhood U of s such that $K|U$ is contained in ${}_{nH}|U$ for some n, and a fortiori (${}_{nH}$ being finite over S) such that $K|U$ is finite over U.

Using Nakayama's lemma, one deduces:

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[951] *N.D.E.* Above, $X - s$ has been corrected to $S - s$. Recall on the other hand that a quasi-finite morphism is supposed to be of finite type, cf. EGA II, 6.2.3 (and EGA III₂, Err_III 20 for the definition of "locally quasi-finite"). Hence here (S being noetherian), X is of finite presentation over S, each immersion $X_i \hookrightarrow X$ is of finite type (by EGA I, 6.3.5), hence X_i is of finite presentation over S, and one may apply EGA IV₃, 8.8.2.

[952] Cf. VI_B 2.5 (ii) for a more general statement.

[953] *N.D.E.* Cf. EGA IV₃, 13.1.4, and EGA IV₄, 17.4.1. Moreover, t (instead of s) has been used to denote the image of x.

Corollary 6.5. *With the preceding notation, suppose that K_s is the unit group. Then there exists an open neighborhood U of s such that $K|U$ is the unit group.*

Corollary 6.6. *Let $u : H \to G$ be a homomorphism of S-preschemes in groups, with H of multiplicative type and of finite type, and G separated over S. Suppose moreover S locally noetherian, or G locally of finite presentation*[954] *over S.*

Let $s \in S$ be such that $u_s : H_s \to G_s$ is a monomorphism; then there exists an open neighborhood U of s such that u induces a monomorphism $H|U \to G|U$.

Indeed, let $K = Ker(u)$; the hypothesis on u_s means that K_s is the unit group, the conclusion that $K|U$ is the unit group, for a suitable U. Now G being separated over S, K is a closed subgroup of H; and in the case where one does not suppose S locally noetherian but rather G locally of finite presentation over S, K is locally of finite presentation over S,[955] hence of finite presentation over S since it is separated over S (H being so) and quasi-compact over S (being closed in H which is quasi-compact over S). One may therefore apply 6.5, whence 6.6.

Remark 6.6.1.[956] *Under the hypotheses of 6.6, one will note that when moreover G is affine over S (resp. of finite presentation over S), $H|U \to G|U$ is even a closed immersion, as was pointed out in 2.5 (resp. 2.6).*

Corollary 6.7. *Let $u : H \to G$ be a homomorphism of S-preschemes in groups, with H of multiplicative type and of finite type; suppose S locally noetherian or G locally of finite presentation over S.*

If all the homomorphisms induced on the fibers $u_s : H_s \to G_s$ are monomorphisms, then u is a monomorphism.

The reasoning is the same as in 6.6; the hypothesis that G be separated over S is here unnecessary to ensure that K is closed in H, since the hypothesis that the u_s be monomorphisms implies that K reduces set-theoretically to the unit section of G.

Theorem 6.8. *Let $u : H \to G$ be a homomorphism of S-preschemes in groups, with H of multiplicative type and of finite type, and G separated over S. Suppose moreover S locally noetherian or G of finite presentation over S.*

Then $K = Ker(u)$ is a subgroup of multiplicative type and of finite type of H, and u factors as

$$H \xrightarrow{u'} H/K \xrightarrow{u''} G,$$

where H/K is of multiplicative type and of finite type, u' is the canonical homomorphism and is faithfully flat (and affine), and u'' is a monomorphism.

[954] *N.D.E.* It in fact suffices to suppose G locally of finite type over S; by EGA IV_4, 1.4.3 (v), this entails ($H \to S$ being of finite presentation) that $H \to G$ is locally of finite presentation, hence so is $K \to S$, which is deduced from it by base change.

[955] *N.D.E.* It in fact suffices to suppose G locally of finite type over S; by EGA IV_4, 1.4.3 (v), this entails ($H \to S$ being of finite presentation) that $H \to G$ is locally of finite presentation, hence so is $K \to S$, which is deduced from it by base change.

[956] *N.D.E.* The numbering 6.6.1 has been added, for later references.

(N.B. As remarked in 6.6.1, u'' is a closed immersion if G is affine over S or of finite presentation over S.) It suffices to prove that K is of multiplicative type, the rest of the proposition then following from 2.7 and IV 5.2.6.

Suppose first G of finite presentation over S. This hypothesis being stable under base change, one is reduced, to prove that K is of multiplicative type, to the case where H is diagonalizable, i.e. $H = D_S(M)$, where M is a commutative group of finite type. Let $s \in S$; then K_s is a closed subgroup of $H_s = D_{\kappa(s)}(M)$, hence

by 8.1[^N.D.E-IX-45] is of the form $D_{\kappa(s)}(N)$, where N is a quotient group of M. Set

$K' = D_S(N)$; then K' is a subgroup of multiplicative type of H. Let $v' : K' \to G$ be induced by u.[957] Then v'_s is the unit homomorphism by construction, hence by 5.2 there exists an open neighborhood U of s such that $v'|U : K'|U \to G|U$ is the unit homomorphism. Hence, replacing S by U if necessary, one may suppose that v' is the unit homomorphism, hence that u factors as

$H \xrightarrow{u'} H/K' \xrightarrow{u''} G.$

Now, since u''_s is deduced from u_s by factoring through $H_s \to H_s/Ker(u_s)$, then u''_s is a monomorphism (IV 5.2.6), hence by 6.6 there exists an open neighborhood U of s such that $u''|U : (H/K')|U \to G|U$ is a monomorphism. Hence, restricting S if necessary, one sees that u'' is a monomorphism, hence $Ker(u) = Ker(u') = K'$, which proves that $\mathrm{Ker}\ u$ is of multiplicative type.

The same proof is valid if, instead of supposing G of finite presentation over S, one supposes S locally noetherian — at least in the case where H is diagonalizable. In the case where one does not make this hypothesis on H, one must show that one can find a covering base change $S' \to S$, with S' locally noetherian, that trivializes H. This will indeed be seen in the following Exposé (X 4.6).

7. Algebraicity of formal homomorphisms into an affine group

Theorem 7.1. *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, $S = \mathrm{Spec}(A)$, $S_n = \mathrm{Spec}(A/I^{n+1})$, H, G S-preschemes in groups, with H of isotrivial multiplicative type, and G affine.*

Then the canonical map

$$(*) \qquad \theta : \mathrm{Hom}_{S\text{-gr.}}(H, G) \longrightarrow \varprojlim_n \mathrm{Hom}_{S_n\text{-gr.}}(H_n, G_n)$$

is bijective (where H_n, G_n are the S_n-groups deduced from H, G by the base change $S_n \to S$).

Suppose first H diagonalizable, hence of the form

$$H = \mathrm{Spec}\, A^{(M)},$$

[957] *N.D.E.* "Induced by G" has been corrected to "induced by u".

where $B = A^{(M)}$ is the algebra of the commutative group M with coefficients in A. One also has $G = \mathrm{Spec}(C)$, where C is an A-algebra equipped with a diagonal map (satisfying the well-known axioms). Then the homomorphisms of S-groups $H \to G$ correspond bijectively to the homomorphisms of A-algebras $\varphi : C \to B$ compatible with the diagonal maps, i.e. such that, for every $f \in C$,

$$\Delta_H(\varphi(f)) = (\varphi \otimes \varphi)(\Delta_G(f))$$

where Δ_H and Δ_G are the diagonal maps. One has an analogous description for the homomorphisms of S_n-groups $H_n \to G_n$, defined by certain homomorphisms of A_n-algebras $\varphi_n : C_n \to B_n$ (where one sets $A_n = A/I^{n+1}$, $B_n = B \otimes_A A_n$, $C_n = C \otimes_A A_n$). Set

$$\hat{B} = \lim_n B_n \quad \text{and} \quad \hat{C} = \lim_n C_n;$$

then $\hat{B} = \lim_n A_n^{(M)}$ identifies with a submodule of the product A^M (namely, that formed by the families $(a_m)_{m \in M}$ of elements of A which tend to 0 (for the I-adic topology) along the filter of complements of finite parts of M). This already implies that the canonical homomorphism $B \to \hat{B}$ is injective,[958] since the projective system $\theta(\varphi) = (\varphi_n)_{n \geqslant 0}$ defines a homomorphism $\hat{\varphi} : \hat{C} \to \hat{B}$ such that the diagram below is commutative:

$$\begin{array}{ccc} C & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ \hat{C} & \xrightarrow{\hat{\varphi}} & \hat{B}. \end{array}$$

Let us prove that θ is surjective: take a projective system $(\varphi_n)_{n \geqslant 0}$ and let us prove that it comes by reduction from a φ of the first member. A priori, (φ_n) defines a homomorphism on the completed algebras

$$\hat{\varphi} : \hat{C} \longrightarrow \hat{B},$$

and all that remains is to see that its composite $\Phi : C \to \hat{C} \xrightarrow{\hat{\varphi}} \hat{B}$ with the canonical homomorphism $C \to \hat{C}$ sends C into B. Indeed, if this is the case, one finds a homomorphism of A-algebras $\varphi : C \to B$, reducing along the φ_n, from which one concludes at once that it is compatible with the diagonal maps (since the φ_n are, and $B \otimes_A B \to \hat{B} \hat{\otimes}_A B$ is injective, as one sees, as above, by replacing M by $M \times M$).

Note that the diagonal homomorphism Δ_H of H defines, on passing to the completions, a homomorphism

$$\hat{\Delta}_H : \hat{B} \to \hat{B} \hat{\otimes}_A B,$$

and one has a commutative diagram (deduced by passage to the projective limit from the analogous diagrams defined by the φ_n):

[958] *N.D.E.* What follows has been spelled out.

$$\begin{tikzcd}
C \arrow[r, "\Phi"] \arrow[d, "\Delta_G"'] & \hat{B} \arrow[d, "\hat{\Delta}_H"] \\
C \otimes_A C \arrow[r, "\Psi"] & \hat{B} \hat{\otimes}_A B,
\end{tikzcd}$$

where Ψ is the composite

$$C \otimes_A C \xrightarrow{\Phi \otimes \Phi} \hat{B} \otimes_A \hat{B} \to \hat{B} \hat{\otimes}_A B$$

(the last arrow being the obvious canonical homomorphism). It follows that for every $f \in C$, $\Phi(f)$ is an element of $\hat{B}$ whose image under $\hat{\Delta}_H$ is a "decomposable" element of $\hat{B}\hat{\bigotimes}_A B$, i.e. is in the image of $\hat{B} \bigotimes_A \hat{B}$.

Denote by $(e_m)_{m\in M}$ the canonical basis of $A^{(M)}$ and $(e_{m,m'})$ that of $A^{(M\times M)} = A^{(M)} \bigotimes_A A^{(M)}$. Since $\Delta_H(e_m) = e_m \bigotimes e_m = e_{m,m}$ for every m, it suffices now to apply the

Lemma 7.2. *Let A be a noetherian ring, M a set, $(a_{m,m'})$ a family of elements of A indexed by $M \times M$, such that*

(i) $a_{m,m'} = 0$ if $m \neq m'$ (i.e. the support of the family is contained in the diagonal of $M \times M$),

(ii) One has

$$a_{m,m'} = \sum_i b^i_m c^i_{m'}$$

where the b^i, c^i are finitely many elements of A^M (i.e. $(a_{m,m'})$ belongs to the image of the canonical homomorphism $A^M \bigotimes_A A^M \to A^{M\times M}$).

Under these conditions, the support of the family $(a_{m,m'})$ is finite.

By (i), the family $(a_{m,m'})$ is determined by knowledge of the $a_m = a_{m,m}$. Set, for every $x = (x_n)_{n\in M} \in A^{(M)}$:

$$(u \cdot x)_m = \sum_{m'} a_{m,m'} x_{m'},$$

i.e. interpret $(a_{m,m'})$ as the matrix of a homomorphism $u : A^{(M)} \to A^M$. Then by (i) one has simply

$$(u \cdot x)_m = a_m x_m.$$

On the other hand, by (ii) one has

$$u \cdot x \in \sum_i A \cdot b^i,$$

hence $u(A^{(M)})$ remains in a finitely generated A-module. Consequently, denoting by $(e_m)_{m\in M}$ the canonical basis of $A^{(M)} \subset A^M$, the $a_m e_m$ remain in a finitely generated A-module. Since A is noetherian, the module they generate is itself of finite type, which implies (since the e_m are linearly independent) that all but finitely many of the a_m are zero. This proves 7.2 and consequently 7.1 in the case where H is diagonalizable.

Let us now prove the general case of 7.1, where one supposes only H isotrivial, i.e. there exists a finite étale surjective morphism $S' \to S$ such that $H' = H \times_S S'$ is diagonalizable. We shall use only the fact that $S' \to S$ is finite and covering (for the faithfully flat quasi-compact topology, or simply for the canonical topology of (Sch)) — thus the “étale” hypothesis could be replaced by “flat”.

Let $S'' = S' \times_S S'$; introduce likewise S'_n and $S''_n = S'' \times_S S_n = S'_n \times_{S_n} S'_n$, and $H', G', H'', G'', H'_n, G'_n, H''_n, G''_n$ deduced from H and G by the base changes one guesses. Note that H' and H'' are now diagonalizable. Note also that S' hence S'' is affine, and that if $S' = \mathrm{Spec}(A')$, $S'' = \mathrm{Spec}(A'')$, then A' and A'' are separated and complete for the topology defined by IA' resp. by IA'' (since A' and A'' are finite over A). Since $S' \to S$ and $S'_n \to S_n$ are covering, one obtains a commutative diagram of maps of sets whose two rows are exact:

$$\begin{tikzcd}
\mathrm{Hom}_{S\text{-gr.}}(H, G) \arrow[r] \arrow[d, "u"] & \mathrm{Hom}_{S'\text{-gr.}}(H', G') \arrow[r, shift left] \arrow[r, shift right] \arrow[d, "u'"] & \mathrm{Hom}_{S''\text{-gr.}}(H'', G'') \arrow[d, "u''"] \\
\varprojlim_n \mathrm{Hom}_{S_n\text{-gr.}}(H_n, G_n) \arrow[r] & \varprojlim_n \mathrm{Hom}_{S'_n\text{-gr.}}(H'_n, G'_n) \arrow[r, shift left] \arrow[r, shift right] & \varprojlim_n \mathrm{Hom}_{S''_n\text{-gr.}}(H''_n, G''_n).
\end{tikzcd}$$

(N.B. The second row is exact as a projective limit of exact diagrams, relative to the various $S'_n \to S_n$.) By what has already been proved (the diagonalizable case), u' and u'' are bijective. The same therefore holds for u, which completes the proof of 7.1.

Corollary 7.3. *Under the conditions of 7.1, suppose moreover G smooth over S, and let $u_0 : H_0 \to G_0$ be a homomorphism of S_0-groups. Then there exists a homomorphism of S-groups $u : H \to G$ which lifts u_0. Any two such liftings u, u' are conjugate by an element g of $G(S)$ reducing along the unit element of $G_0(S_0)$.*

Follows from the conjunction of 3.6 and 7.1. To construct a u, one constructs step by step u_n, which is possible by 3.6, and by 7.1 the system (u_n) comes from a u. Given two liftings u and u', to construct g such that $u' = int(g)u$, $g_0 = 1$, one constructs step by step g_n, such that $g_0 = 1$, g_n is deduced from g_{n+1} by reduction, $u'_n = int(g_n)u_n$; this is possible thanks to 3.6. Since A is separated and complete, the g_n come from a $g \in G(S)$; and to prove that $u' = int(g)u$, it suffices to use the injectivity in assertion 7.1.

Remark 7.4. *One will compare 7.1 with EGA III 5.4.1, which implies that the statement 7.1 is valid if, instead of supposing H of multiplicative type and G affine, one supposes H proper over S, and G separated and locally of finite type over S. Having a statement like 7.1 without a properness hypothesis is rather exceptional, and must here be interpreted as one of the aspects of the great “rigidity” of the structure of a group of multiplicative type. The analogous statement with $G = H = G_a$ (additive group) is false in general, as one sees by taking A of characteristic $p > 0$, and defining the u_n by reduction mod I^{n+1} from an additive formal series*

$$u(T) = \sum_{n \geqslant 0} a_n T^{p^n},$$

where the a_n are elements of A that tend to 0 for the I-adic topology, but not for the discrete topology. Statement 7.1 also becomes false if one suppresses the hypothesis G affine, even for $H = G_m$ (multiplicative group); one sees an example of this (with A a complete discrete valuation ring) by starting from an elliptic curve over the field of fractions K of A, which reduces (in the Néron–Kodaira reduction theory, *say) along the group G_m over the residue field k:*[959] *one will thus have a commutative smooth group scheme G over S, whose special fiber is $G_{m,k}$ (which, thanks to 3.6, allows one to define a projective system of isomorphisms $u_n : H_n \xrightarrow{\sim} G_n$, where $H = G_{m,A}$), but whose generic fiber is an abelian variety, so that there exists no homomorphism of S-groups $H \to G$ other than 0.*

8. Subgroups, quotient groups, and extensions of groups of multiplicative type over a field[960]

Scholium 8.0.[961] *Let k be a field, H an affine k-group scheme. Denote by $k[H]$ its affine algebra and $X(H) = \mathrm{Hom}_{k-gr.}(H, G_{m,k})$ the group of characters of H. By the lemma of independence of characters, $X(H)$ is a free part of $k[H]$. It follows that H is diagonalizable if and only if $X(H)$ generates $k[H]$ as a k-vector space.*

Proposition 8.1. *Let G be a diagonalizable group scheme (resp. of multiplicative type) over a field k.*[962] *Then for every subgroup scheme H of G, H and G/H are diagonalizable (resp. of multiplicative type).*

Definition 1.1 reduces us at once to the non-resp. case. An easy passage to the limit, using VI_B 11.13 and VIII 3.1, reduces us to the case where G is of finite type over k.[963] By VI_B 11.16,[964] one can find a finite family of non-zero elements

$$f_i = \sum a_{im} m, \quad a_{im} \in k \tag{8.1.1}$$

of the affine ring $k^{(M)}$ of $G = D_k(M)$, such that the points of H (with values in an arbitrary k-algebra k') are the points $g \in G(k')$ such that one has

$$\tau_g f_i = \lambda_i(g) f_i, \quad \text{with } \lambda_i(g) \in k', \tag{8.1.2}$$

where τ_g denotes the translation by g. Now one has

$$\tau_g f_i = \sum a_{im} \chi_m(g) m, \tag{8.1.3}$$

where $\chi_m : G \to G_{m,k}$ is the character associated with m, so that (8.1.2) is equivalent to the relation

$$\chi_m(g) = \chi_{m'}(g) \quad \text{if } m, m' \in Z_i, \tag{8.1.4}$$

Z_i denoting the set of $m \in M$ such that $a_{im} \neq 0$. This relation may also be written

$$\chi_{m'-m}(g) = 1 \quad \text{if } m, m' \in Z_i.$$

[959] *N.D.E.* One could spell out such an example ...
[960] Added in July 1969. This regretful section is independent of N°s 3 to 7.
[961] *N.D.E.* This scholium has been added.
[962] *N.D.E.* "Over a field k" has been added, being implicit in the original; on the other hand, "algebraic group" has been replaced by "group scheme", since G is not supposed to be of finite type.
[963] *N.D.E.* Spell out this "passage to the limit": this uses 8.0 and also the equality $G/H = \lim_i G_i/H_i$, cf. VI_B, proof of 11.17. To see that $H = \lim_i (H \cap G_i)$, does one use the fact that H is closed in G? Using this, one can give a direct proof ...
[964] *N.D.E.* It would doubtless be necessary to rewrite VI_B 11.16 in the usual, more pleasant form. In particular, a single f_i suffices below ...

Denoting by N the subgroup of M generated by all the $m' - m$ (i varying, and $m, m' \in Z_i$), one deduces from the definition of $D_k(M)$ ($\simeq G$) that H identifies with $D_k(M/N)$. It then follows from VIII 3.1 that G/H identifies with $D_k(N)$. QED.

Proposition 8.2.[965] *Let k be a field, H, K k-group schemes of multiplicative type and of finite type, and G a k-group scheme such that one has an exact sequence*

$$1 \longrightarrow H \longrightarrow G \longrightarrow K \longrightarrow 1$$

(which entails that G is of finite type over k).

(i) If one supposes G commutative or K connected, then G is of multiplicative type.

(ii) If K and H are diagonalizable, with K a torus, then G is diagonalizable.

For the proof of (i), the reader is referred to XVII 7.1.1, of which 8.2 (i) is a special case; the case of a field is treated in part (i) of the proof of XVII 7.1.1 (not using the results of the following Exposés).

(ii) Suppose now K and H diagonalizable:

$K \simeq D_k(M)$ and $H \simeq D_k(N)$.

[966] Suppose G of multiplicative type (which is the case by (i) if K is a torus). Then, by X 1.4, G is isotrivial over k, i.e. there exists a finite separable extension k'/k such that $G' = G \times_k k'$ is diagonalizable, so $G' = D_{k'}(E)$ for some commutative group E, and one has an exact sequence

$$0 \longrightarrow D_{k'}(N) \longrightarrow D_{k'}(E) \longrightarrow D_{k'}(M) \longrightarrow 0.(1)$$

Hence, by VIII, 3.1 and 3.2, M is a subgroup of E and one has an exact sequence

$$0 \longrightarrow M \longrightarrow E \longrightarrow N \longrightarrow 0.(2)$$

For a given extension E_0 of N by M, consider the diagonalizable k-group $G_0 = D_k(E_0)$; then the k-functor in groups A of automorphisms of the extension

$$1 \longrightarrow H \longrightarrow G_0 \longrightarrow K \longrightarrow 1,(3)$$

i.e. the sub-functor in groups of $\mathrm{Aut}_{k-gr.}(G_0)$ whose points on a k-prescheme T are the $\phi \in \mathrm{Aut}_{T-gr.}(G_T)$ inducing the identity on H_T and on K_T, identifies with $\mathrm{Hom}_{k-gr.}(K, H)$, which is, by VIII 1.5, the constant k-group of value $L = \mathrm{Hom}_{gr.}(N, M)$.

One sees therefore that the classification of extensions G of K by H which, over a separable closure k_s of k, become isomorphic to the extension (3), is the same as

[965] *N.D.E.* The statement, as well as its proof, has been spelled out.
[966] *N.D.E.* The statement, as well as its proof, has been spelled out.

that of the k-torsors for the étale topology under the constant group L_k. Denoting $\Gamma = \mathrm{Gal}(k_s/k)$, these are classified by the Galois cohomology group (L being a trivial Γ-module):

$$H^1_{ét}(k, L_k) = H^1(\Gamma, L) = \mathrm{Hom}_{gr.\ top.}(\Gamma, L).$$

If moreover K is a torus, M is torsion-free, hence so is L, whence $\mathrm{Hom}_{gr.top.}(\Gamma, L) = 0$; it follows that every extension of K by H is already diagonalizable.

Remark 8.3. *As already pointed out in the proof of 8.2, the first assertion is stated and proved over an arbitrary base in XVII 7.1.1. On the other hand, the second assertion generalizes, with essentially the same proof, to the case of an integral normal base scheme (or more generally, geometrically unibranch), using X 5.13 below.*

Finally, one can also consider 8.1 as a corollary of the (markedly less trivial) result 6.8, also valid over an arbitrary base scheme.[967]

Bibliography

[968]

[BEns] N. Bourbaki, *Théorie des ensembles*, Chap. I–IV, Hermann, 1970.

Exposé X. Characterization and classification of groups of multiplicative type

by A. Grothendieck

> Version xy of 6 November 2009: Addenda placed in Section 9, reviewed through 8.8.[969]

1. Classification of isotrivial groups. Case of a base field

Recall (IX 1.1) that the group of multiplicative type H over the prescheme S is said to be *isotrivial* if there exists an étale, finite, surjective morphism $S' \to S$ such that $H' = H \times_S S'$ is diagonalizable. When S is connected, S' decomposes as a finite sum of connected components S'_i, and one can therefore (possibly by replacing S' by one of the S'_i) assume S' connected. Finally, one knows that S' can be dominated by an S'_1 finite, étale and connected, which is Galois, i.e. a principal homogeneous bundle over S with group Γ_S, where Γ is an ordinary finite group (cf. SGA 1, V N°s 7 & 3 when S is locally noetherian, and EGA IV_4, 18.2.9 in the general case).[970] We assume S' chosen in this way, and we propose to determine

[967] *N.D.E.* Note however that the proof of 6.8 uses 8.1!

[968] *N.D.E.* Additional references cited in this Exposé.

[969] Version xy of 6 November 2009: Addenda placed in Section 9, reviewed through 8.8.

[970] *N.D.E.* More precisely, this follows from the "axiomatic conditions of a Galois theory", cf. SGA 1, V N° 4 g); when S is locally noetherian, the verification of the axioms is done in *loc. cit.*, N°s 7 & 3, in

the groups of multiplicative type H over S which are "trivialized" by S', i.e. such that $H' = H \times_S S'$ is diagonalizable.[971] By descent theory (cf. SGA 1, VIII 5.4), the category of these H is equivalent to the category of diagonalizable groups H' over S', equipped with operations of Γ on H' compatible with the operations of Γ on S'. (N.B. As the groups under consideration are affine over the base, the question of effectivity of a descent datum is answered in the affirmative, cf. SGA 1, VIII 2.1.) Now S' being connected, the contravariant functor

$$M \mapsto D_{S'}(M)$$

is an anti-equivalence of the category of ordinary commutative groups with the category of diagonalizable groups over S' (cf. VIII 1.6), a quasi-inverse functor being $H \mapsto \mathrm{Hom}_{S'-gr.}(H, G_{m,S'})$.[972]

Proposition 1.1. *Let S be a connected prescheme, S' a connected principal cover of S with group Γ (finite). Then the category of groups of multiplicative type over S split[973] by S' is anti-equivalent to the category of Γ-modules, i.e. of ordinary commutative groups M equipped with a homomorphism $\Gamma \to \mathrm{Aut}_{gr.}(M)$.*

One concludes from this in a standard manner:

Corollary 1.2. *Let S be a connected prescheme, $\xi : \mathrm{Spec}(\Omega) \to S$ a "geometric point" of S, i.e. a homomorphism into S of the spectrum of an algebraically closed field Ω; consider the fundamental group of S at ξ (cf. SGA 1 V, N° 7):*

$$\pi = \pi_1(S, \xi).$$

Then the category of isotrivial groups of multiplicative type H over S is anti-equivalent to the category of "Galois modules" under π, i.e. of π-modules M such that the stabilizer in π of every point of M is an open subgroup.

In this correspondence, to the isotrivial group of multiplicative type H is associated the group $M = \mathrm{Hom}_{\Omega-gr.}(H_\xi, G_{m,\Omega})$, where H_ξ is the fiber of H at ξ; this group is naturally equipped with operations of $\pi_1(S, \xi)$.

Remark 1.3. We shall see below (cf. 5.16) that if S is normal, or more generally geometrically unibranch, then every group of multiplicative type and of finite type over S is necessarily isotrivial, so that the classification principle 1.2 is applicable to groups of multiplicative type and of finite type over S, which correspond to Galois π-modules that are of finite type over $\mathbb{Z}$. For the moment, let us confine ourselves to the following result:

particular 3.7, which rests on SGA 1, I 10.9. This last result is proved, without noetherian hypotheses, in EGA IV_4, 18.2.9.

[971] *N.D.E.* In the sequel, one will say that an S-group of multiplicative type is "split" if it is "trivial" in the sense of IX 1.2, i.e., if it is a diagonalizable S-group.

[972] *N.D.E.* One has corrected S to S'.

[973] *N.D.E.* One has replaced, here and in the sequel, "splittés" by "déployés", "splitte" by "déploie", etc. (In English: "split".)

Proposition 1.4. *Let k be a field, H a group of multiplicative type and of finite type over k; then H is isotrivial, i.e. there exists a finite separable extension k' of k which splits H.*

Consequently, by 1.2, if π is the topological Galois group of an algebraic closure Ω of k, the category of groups of multiplicative type and of finite type H over k is anti-equivalent to the category of Galois modules M under π that are of finite type as $\mathbb{Z}$-modules.

It follows first from the fact that H is of finite type over k and from the "principle of the finite extension" (cf. EGA IV₃, 9.1.4) that there exists a finite extension k' of k which splits H. Let us recall the principle of the proof: by hypothesis there exist a diagonalizable group of finite type G over k, a faithfully flat S' over $S = \mathrm{Spec}(k)$, and an isomorphism of S'-groups $H_{S'} \simeq G_{S'}$. Possibly replacing S' by the residue field of a point of S', we may suppose that S' is the spectrum of an extension K of k. The latter is the inductive limit of its finitely generated subalgebras A_i, from which it readily follows (cf. EGA IV₃, 8.8.2.4) that u comes from an A_i-isomorphism $u_i : H_{A_i} \simeq G_{A_i}$ for i large enough. By the Nullstellensatz, there exists a quotient ring k' of A_i which is a finite extension of k. The latter therefore splits H.

Then k' is a radicial extension of a separable extension k'_s of k. By IX 5.4, the isomorphism $u' : H_{k'} \simeq G_{k'}$ comes from an isomorphism $H_{k'_s} \simeq G_{k'_s}$, which proves that k'_s splits H and establishes 1.4.

Remark 1.5. Statement 1.2 yields in particular a characterization of isotrivial tori over S of relative dimension n: setting $\pi = \pi_1(S, \xi)$, they correspond to classes (up to "equivalence") of representations $\pi \to GL(n, \mathbb{Z})$ with kernel an open subgroup of π.

1.6. Even when S is an algebraic curve, there can exist over S tori (of relative dimension 2) that are not locally isotrivial (and *a fortiori* not isotrivial); there can also exist locally trivial tori that are not isotrivial. (Note however that such phenomena can present themselves only if S is not normal, as already signalled in 1.3.) Let for example S be an irreducible algebraic curve (over an algebraically closed field to fix ideas) having an ordinary double point a, let S' be its normalization, and b and c the two points of S' above a. One then constructs a principal homogeneous bundle P over S, with structural group $\mathbb{Z}$, connected, by attaching together an infinity of copies of S' along the diagram

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......          b           b           b           b   ......
                       c           c           c
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(N.B. This is a principal bundle in the sense of the étale topology.) Now one has a homomorphism

$$\varphi : \mathbb{Z} \to GL(2, \mathbb{Z}), \qquad \varphi(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix},$$

which permits the construction of a torus T over S, of relative dimension 2, from the trivial torus G_m^2 over P and from the descent datum on the latter deduced from ϕ.

(N.B. one will note that the projection $P \to S$ is covering for the étale topology and *a fortiori* for the canonical topology of (Sch), and that the descent datum under consideration is necessarily effective, since $G^2_{m,P}$ is affine over P.) It is not difficult to prove that T is not isotrivial in a neighborhood of a[974] (it is however trivial on $S - a$).

One finds a variant of this construction by taking for S a curve having two irreducible components S_1 and S_2 intersecting in two points a' and a'', which permits the construction of a principal homogeneous bundle P over S with group $\mathbb{Z}_S$, connected and locally trivial, hence an associated torus T which is locally trivial, but not isotrivial.

2. Infinitesimal variations of structure

[975] Let us begin by recalling the following result (cf. SGA 1, I 8.3 in the case S locally noetherian, EGA IV$_4$, 18.1.2 in general):

Recall 2.0. *Let S be a prescheme, S_0 a sub-prescheme having the same underlying set. Then the functor*

$$X \mapsto X_0 = X \times_S S_0$$

is an equivalence between the category of étale preschemes over S and the analogous category over S_0.

Proposition 2.1. *Let S be a prescheme, S_0 a sub-prescheme having the same underlying set (i.e. defined by a nilideal I). Then the functor*

$$H \mapsto H_0 = H \times_S S_0$$

from the category of preschemes in groups of multiplicative type over S to the analogous category over S_0, is fully faithful.

Moreover, it induces an equivalence between the category of quasi-isotrivial groups of multiplicative type over S and the analogous category over S_0.

Let us first prove the full faithfulness, i.e. that if H, G over S are of multiplicative type, then

$$\mathrm{Hom}_{S\text{-gr.}}(H, G) \to \mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0)$$

is bijective. The question being local on S, we may suppose S affine; there then exists a faithfully flat quasi-compact morphism $S' \to S$ which splits H and G. Let $S'' = S' \times_S S'$; denote by H', G' resp. H'', G'' the groups deduced from H, G by the base change $S' \to S$ resp. $S'' \to S$; define S'_0 and S''_0 similarly, the latter being also isomorphic to $S'_0 \times_{S_0} S'_0$. One then finds a commutative diagram with exact rows:

$$\begin{array}{ccccc}
\mathrm{Hom}_{S\text{-gr.}}(H, G) & \to & \mathrm{Hom}_{S'\text{-gr.}}(H', G') & \rightrightarrows & \mathrm{Hom}_{S''\text{-gr.}}(H'', G'') \\
\downarrow u & & \downarrow u' & & \downarrow u'' \\
\mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0) & \to & \mathrm{Hom}_{S'_0\text{-gr.}}(H'_0, G'_0) & \rightrightarrows & \mathrm{Hom}_{S''_0\text{-gr.}}(H''_0, G''_0),
\end{array}$$

[974] *N.D.E.* See 7.3 below.

[975] *N.D.E.* One has added this "recall", for later references.

so to prove that u is bijective, it suffices to prove that u' and u'' are, which reduces us to the case where H and G are diagonalizable, hence of the form $D_S(M)$ and $D_S(N)$, where M and N are ordinary commutative groups. One will therefore have likewise $H_0 = D_{S_0}(M)$, $G_0 = D_{S_0}(N)$. One then has a commutative diagram[976]

$$\begin{tikzcd} \mathrm{Hom}_{S\text{-gr.}}(N_S, M_S) \arrow[r] \arrow[d] & \mathrm{Hom}_{S\text{-gr.}}(D_S(M), D_S(N)) \arrow[d] \\ \mathrm{Hom}_{S_0\text{-gr.}}(N_{S_0}, M_{S_0}) \arrow[r] & \mathrm{Hom}_{S_0\text{-gr.}}(D_{S_0}(M), D_{S_0}(N)), \end{tikzcd}$$

where the horizontal arrows are isomorphisms by VIII 1.4, so we are reduced to proving that the homomorphism

$$(\times) \qquad \mathrm{Hom}_{S\text{-gr.}}(N_S, M_S) \to \mathrm{Hom}_{S_0\text{-gr.}}(N_{S_0}, M_{S_0})$$

is bijective, i.e. to proving that the functor $M_S \mapsto M_{S_0}$, from preschemes in constant commutative groups over S to preschemes in constant commutative groups over S_0, is fully faithful. Now $(\times)$ identifies also with the natural map

$$\mathrm{Hom}_{\mathrm{gr.}}(N, \Gamma(M_S)) \to \mathrm{Hom}_{\mathrm{gr.}}(N, \Gamma(M_{S_0}))$$

deduced from $\Gamma(M_S) \to \Gamma(M_{S_0})$; this latter map is obviously bijective (since $\Gamma(M_S)$ = set of locally constant maps from S to M, depends only on the topological space underlying S), whence the desired conclusion.

To prove the second assertion of 2.1, it remains to see that every group H_0 of multiplicative type over S_0 which is quasi-isotrivial comes from a quasi-isotrivial group of multiplicative type H over S. To see this, let $S'_0 \to S_0$ be an étale surjective morphism which splits H_0.

One knows (cf. 2.0) that there exists an étale morphism $S' \to S$ and an S_0-isomorphism $S' \times_S S_0 \simeq S'_0$, so that one may suppose that S'_0 comes from S' by reduction. Since H'_0 is diagonalizable, one sees at once that it is isomorphic to the group deduced by base change $S'_0 \to S'$ from a diagonalizable group H' over S' (N.B. if $H'_0 = D_{S'_0}(M)$, one takes $H' = D_{S'}(M)$). Set as usual $S'' = S' \times_S S'$, $S''' = S' \times_S S' \times_S S'$, and define S''_0, S'''_0 similarly, deduced from the preceding by the base change $S_0 \to S$ and isomorphic also to the fibered square resp. cube of S'_0 over S_0. Using the full faithfulness already proved, in the cases (S'', S''_0) and (S''', S'''_0), one sees that the natural descent datum on H'_0 relative to $S'_0 \to S_0$ (cf. IV 2.1) comes from a well-determined descent datum on H' relative to $S' \to S$. This descent datum is effective since H' is affine over S' (SGA 1, VIII 2.1), so there exists an S-group H such that $H \times_S S' = H' = D_{S'}(M)$, and H is therefore of quasi-isotrivial multiplicative type.

One then verifies easily, using now the full-faithfulness result for (S', S'_0), that the given isomorphism between H'_0 and $H' \times_{S'} S'_0$ comes from an isomorphism between H_0 and $H \times_S S_0$. (For a more formal exposition of results of this kind, see Giraud's article in preparation[977] on descent theory.)

Corollary 2.2. *Let H be an S-group of multiplicative type, and $H_0 = H \times_S S_0$. For*

[976] *N.D.E.* One has corrected the original by exchanging M and N in the right-hand terms.

[977] Cf. J. Giraud, *Méthode de la descente*, Bull. Soc. Math. France, Mémoire 2 (1964).

H to be quasi-isotrivial (resp. locally isotrivial, resp. isotrivial, resp. locally trivial, resp. trivial), it is necessary and sufficient that H_0 be so.

The "only if" is trivial; the "if" has already been seen in the quasi-isotrivial case, since thanks to the full faithfulness, it suffices to know that every quasi-isotrivial group over S_0 lifts to a quasi-isotrivial group over S. The same argument works for "trivial". For the case "isotrivial", one takes up the reasoning establishing the second assertion of 2.1, but taking $S_0' \to S_0$ étale surjective and finite. The cases "locally isotrivial" and "locally trivial" follow at once from the cases "isotrivial" and "trivial".

One can generalize 2.2 somewhat when I is nilpotent, without supposing *a priori* H of multiplicative type:[978]

Corollary 2.3. *Suppose the ideal I defining S_0 locally nilpotent. Let H be a prescheme in groups over S, flat over S, and $H_0 = H \times_S S_0$. For H to be of quasi-isotrivial multiplicative type, it is necessary and sufficient that H_0 be so.*

Indeed, suppose H_0 is of quasi-isotrivial multiplicative type; let us prove that H is so. The question being local for the étale topology, and the category of étale preschemes over S being equivalent to the category of étale preschemes over S_0 by the functor $S' \mapsto S' \times_S S_0$ (cf. 2.0), one is at once reduced to the case where H_0 is diagonalizable, hence isomorphic to a group $D_{S_0}(M)$. Let $G = D_S(M)$; one then has an isomorphism $u_0 : H_0 \xrightarrow{\sim} G_0$; I claim it comes from a unique homomorphism $u : H \to G$, which will therefore be an isomorphism since u_0 is one (H and G being flat over S, and I locally nilpotent[979]); this will establish 2.3. Now one has (cf. VIII 1 (xxx))

$$\mathrm{Hom}_{S\text{-}gr.}(H, G) \simeq \mathrm{Hom}_{S\text{-}gr.}(M_S, \mathrm{Hom}_{S\text{-}gr.}(H, G_{m,S})),$$

and the second member identifies also with

$$\mathrm{Hom}_{gr.}(M, \mathrm{Hom}_{S-gr.}(H, G_{m,S})),$$

so the homomorphism

$$(\times) \qquad \mathrm{Hom}_{S\text{-}gr.}(H, G) \to \mathrm{Hom}_{S_0\text{-}gr.}(H_0, G_0)$$

is isomorphic to the homomorphism

$$\mathrm{Hom}_{gr.}(M, \mathrm{Hom}_{S\text{-}gr.}(H, G_{m,S})) \to \mathrm{Hom}_{gr.}(M, \mathrm{Hom}_{S_0\text{-}gr.}(H_0, G_{m,S_0}))$$

deduced from the restriction homomorphism

$$(\times\times) \qquad \mathrm{Hom}_{S\text{-}gr.}(H, G_{m,S}) \to \mathrm{Hom}_{S_0\text{-}gr.}(H_0, G_{m,S_0});$$

[978] *N.D.E.* One recalls that 2.3 is used to prove Theorem IX 3.6 bis.

[979] *N.D.E.* Since u_0 is an isomorphism, it suffices to see that for every $h \in H$, the morphism $\phi : O_{G,u(h)} \to O_{H,h}$ is bijective. It is so by reduction modulo $I = I_s$ (where s is the image of h in S), so its cokernel C verifies $C = IC$, whence $C = 0$ since I is nilpotent. Then, since $O_{H,h}$ is flat over $O_{S,s}$, the kernel K of ϕ also verifies $K = IK$, whence $K = 0$.

so to prove that (×) is bijective, it suffices to prove that (××) is. The question is local on S; we may therefore suppose S affine. Now $G_{m,S}$ being commutative and smooth over S, the situation is governed by IX 3.6, which completes the proof.

Corollary 2.4. *Let A be an artinian local ring with residue field k, $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(k)$.*

(i)[980] *Let H be an S-prescheme in groups, flat and locally of finite type, such that $H_0 = H \times_S S_0$ is of multiplicative type. Then H is of multiplicative type, of finite type and isotrivial. In particular, every S-prescheme in groups H of multiplicative type and of finite type is isotrivial.*

(ii) The functor $H \mapsto H_0$ is an equivalence between the category of groups of multiplicative type of finite type over A and the analogous category over k.

Indeed, let H be as in (i). Then H_0 is of multiplicative type and locally of finite type, hence of finite type, hence isotrivial by 1.4. Therefore, by 2.3 and 2.2, H is of multiplicative type (hence of finite type) and isotrivial. Assertion (ii) then follows from 2.1.

Corollary 2.5. *Let $S' \to S$ be a faithfully flat and radicial morphism.*

(i) The functor $H \mapsto H' = H \times_S S'$, from the category of groups of multiplicative type over S to the analogous category over S', is fully faithful.

Moreover, it induces an equivalence between the subcategories formed by the quasi-isotrivial groups of multiplicative type.

(ii) If H is of multiplicative type, for it to be quasi-isotrivial (resp. locally isotrivial, resp. isotrivial, resp. locally trivial, resp. trivial) it is necessary and sufficient that H' be so.

Indeed, let $S'' = S' \times_S S'$ and $S''' = S' \times_S S' \times_S S'$; then the hypothesis that $S' \to S$ is radicial implies that the diagonal immersions $S' \to S''$ and $S' \to S'''$ are surjective, so the base change by either of these immersions is governed by 2.1 and 2.2. Taking into account that $S' \to S$ is a morphism of effective descent for the fibered category of groups of multiplicative type over preschemes (since it is faithfully flat and quasi-compact[981]), our assertion follows formally from 2.1 and 2.2 (cf. for a formal argument the already-cited Giraud article).

3. Finite variations of structure: complete base ring

Lemma 3.1. *Let A be a noetherian ring, equipped with an ideal I such that A is separated and complete for the I-adic topology, $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A/I)$, G and H two S-group schemes, with G of multiplicative type and isotrivial, H affine over S, flat over S at the points of $H_0 = H \times_S S_0$, H_0 of quasi-isotrivial multiplicative type. Then the natural map*

[980] *N.D.E.* One has rewritten the statement keeping only the non-trivial implication, to bring it into focus.

[981] *N.D.E.* Spell out this point ....

$$\operatorname{Hom}_{S\text{-gr.}}(G, H) \to \operatorname{Hom}_{S_0\text{-gr.}}(G_0, H_0)$$

is bijective.

For every integer $n \geqslant 0$, let $S_n = \operatorname{Spec}(A/I^{n+1})$, and let G_n, H_n be the groups deduced from G, H by the base change $S_n \to S$. Since G/S is of isotrivial multiplicative type and H affine over S, then, by IX 7.1, the natural homomorphism

$$\operatorname{Hom}_{S\text{-gr.}}(G, H) \to \varprojlim_n \operatorname{Hom}_{S_n\text{-gr.}}(G_n, H_n)$$

is bijective. On the other hand, by 2.3, the H_n are of quasi-isotrivial multiplicative type, and by 2.1, the transition homomorphisms in the projective system $(\operatorname{Hom}_{S_n-gr.}(G_n, H_n))_n$ are isomorphisms, whence 3.1 at once.

Theorem 3.2. *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(A/I)$. Then:*

(i) The functor

$$H \mapsto H_0 = H \times_S S_0$$

is an equivalence between the category of isotrivial groups of multiplicative type over S and the analogous category over S_0.

(ii) Let H be an S-group of multiplicative type and of finite type; for H to be isotrivial, it is necessary and sufficient that H_0 be so.

Proof. For (i) one can either take up the proof of 2.1, or use 1.2, in either case using the fact that the functor

$$S' \mapsto S'_0 = S' \times_S S_0$$

from the category of finite étale schemes over S to the category of finite étale schemes over S_0 is an equivalence (SGA 1, I 8.4), which one can also state (reducing to the case S connected, i.e. S_0 connected), by choosing a geometric point ξ of S_0, by saying that the canonical homomorphism

$$\pi_1(S_0, \xi) \to \pi_1(S, \xi)$$

is an isomorphism.

Let us prove (ii), i.e. that if H_0 is isotrivial, then H is so. By (i), there exists an isotrivial group of multiplicative type G over S and an S_0-isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

[982] Since H is of finite type, so are H_0 and G_0; therefore, by IX 2.1 b), the type of G at each point of S is an abelian group of finite type, and so G is of finite type over S. On the other hand, by 3.1, u_0 comes from a homomorphism of S-groups

[982] *N.D.E.* One has spelled out the original in what follows.

$$u : G \longrightarrow H.$$

Finally, since G, H are of multiplicative type and of finite type over S, and since u_0 is an isomorphism, then, by IX 2.9, u is an isomorphism (taking into account that every neighborhood of S_0 in S equals S).

Corollary 3.3. *Let A be a complete noetherian local ring with residue field k.*

(i) Every group of multiplicative type and of finite type over A is isotrivial.

(ii) The functor $H \mapsto H \otimes_A k = H_0$ is an equivalence between the categories of groups of multiplicative type and of finite type over A and over k.

[983] First, (i) follows from 3.2 (ii) and 1.4; then (ii) follows from 3.2 (i), taking into account the fact that H is of finite type if and only if H_0 is (cf. the proof of 3.2 (ii)).

Remark 3.3.1. One will note that 3.3 yields, by 1.4, a complete classification of groups of multiplicative type and of finite type over A in terms of the topological Galois group of an algebraic closure Ω of k.

Remarks 3.4. Under the hypotheses of 3.2 (i.e. A noetherian, separated and complete for the I-adic topology, but without further hypothesis on A/I), it will follow from N° 5 that the functor $H \mapsto H_0$, from the category of groups of multiplicative type and of finite type over S to the analogous category over S_0, is again fully faithful (without hypotheses of isotriviality).[984]

However, it is not in general essentially surjective; in fact, there can exist an S_0-group H_0 of multiplicative type and of finite type, locally trivial if one wishes (but not isotrivial), which does not come by reduction from a group of multiplicative type H over S.

To see this, let us take up either of the examples 1.6 of a non-isotrivial group of multiplicative type over a non-normal curve. One may obviously take this curve affine, say S_0, and assume that it is embedded in the affine space of dimension 2, hence defined by an ideal J in $k[X, Y]$. We shall take for A the completion of this ring for the J-preadic topology, so that A is a regular ring of dimension 2, *a fortiori* normal. We shall see in 5.16 that it follows that every group of multiplicative type and of finite type over $S = \mathrm{Spec}(A)$ is isotrivial; therefore H_0, which is of finite type and not isotrivial, does not come from a group of multiplicative type H over S (since H would necessarily be of finite type, hence isotrivial).

Lemma 3.5.[985] *Let S be a prescheme, $u : G \to H$ a homomorphism of S-preschemes in groups, locally of finite presentation and flat over S, U the set of $s \in S$ such that the induced homomorphism on fibers $u_s : G_s \to H_s$ is flat (resp. smooth, resp. unramified, resp. étale, resp. quasi-finite).*

[983] *N.D.E.* One has spelled out the original in what follows.
[984] *N.D.E.* Specify this reference, possibly adding a corollary in N° 5 ….
[985] Cf. also VI_B 2.5 for more systematic developments of this nature.

Then U is open, and the restriction $u|U : G|U \to H|U$ is a flat (resp. smooth, resp. unramified, resp. étale, resp. quasi-finite) morphism.

Indeed, let V be the set of points at which u is flat (resp. ...). One knows that V is open (cf. SGA 1, I to IV in the locally noetherian case, EGA IV in general[986]), and that for $x \in G$ above $s \in S$, one has $x \in V$ if and only if $u_s : G_s \to H_s$ is flat (resp. ...) at x (same reference; in the flat, smooth, étale cases, one uses here the flatness of G and H over S). Since u_s is a homomorphism of locally algebraic groups, this also means that u_s is flat (resp. ...) everywhere (Exp. VI_B, 1.3), i.e. $s \in U$. Therefore U is the inverse image of the open set V by the unit section of G, hence open, and $V = G|U$, so $G|U \to H|U$ is flat (resp. ...), which completes the proof. (N.B. In the "unramified" or "quasi-finite" case, the flatness hypothesis on G and H is unnecessary.)

Lemma 3.6. *Let H be a commutative algebraic group over a field k, admitting an open subgroup G of multiplicative type. Then the family of subschemes ${}_nH$ $(n > 0)$ of H is schematically dense; in particular, if one has ${}_nH = {}_nG$ for every $n > 0$, then $H = G$.*

Here ${}_nH$ denotes the kernel of $n \cdot id_H$.[987] Let $\bar{k}$ be an algebraic closure of k; it suffices to show that the family $({}_nH_{\bar{k}})_{n>0}$ is schematically dense in $H_{\bar{k}}$, for then the family $({}_nH)_{n>0}$ will be so in H (cf. IX 4.5). Thus, one may suppose k algebraically closed, hence G of the form $D_k(M)$, M a finitely generated ordinary commutative group.

Let $M_0 = M/Torsion(M)$; then $T = D_k(M_0)$ is the largest torus contained in G, and H/T is finite, so $H(k)/T(k)$ is annihilated by an integer $\nu > 0$. One can find a finite number of elements $g_i \in H(k)$ such that

```
H = ⨆_i g_i · G,
```

and one will have $g_i^\nu \in T(k)$. Since k is algebraically closed, $\nu \cdot id_T$ is surjective on $T(k) \simeq k^{*d}$, so up to replacing the g_i by $g_i t_i^{-1}$, where $t_i \in T(k)$ is such that $t_i^\nu = g_i^\nu$, one may suppose that $g_i^\nu = 1$. If then n is a multiple of ν, one will have

```
_n H ⊇ g_i · _n G,
```

and since (for $n > 0$ variable) the family of ${}_nG$ is schematically dense in G by IX 4.7, conclusion 3.6 follows.

Theorem 3.7. *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(A/I)$, and let H be an affine S-group scheme of finite type, flat over S at the points of `H_0`, such that $H_0 = H \times_S S_0$ is of multiplicative type and isotrivial.*

Then there exists an open and closed subgroup G of H, of isotrivial multiplicative type (and of finite type), such that $G_0 = H_0$.

[986] *N.D.E.* Specify these references: Since G, H are locally of finite presentation, u is locally of finite presentation (IV_1, 1.4.3 (v)); next one applies IV_3, 11.3.10 and 13.1.4 for flat and quasi-finite, IV_4, 17.4.1, 17.5.1 and 17.6.1 for unramified, smooth and étale; compare with VI_B 2.5.

[987] *N.D.E.* One has added the following sentence.

By 3.2 (i), there exists an isotrivial group of multiplicative type G over S and an isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

By 3.1, u_0 comes from a unique homomorphism of S-groups

$$u : G \longrightarrow H.$$

Using IX 6.6, one sees that u is a monomorphism (since if U is the set of $s \in S$ such that $u_s : G_s \to H_s$ is a monomorphism, then U is an open neighborhood of S_0 hence identical to S, and $G|U \to H|U$ is a monomorphism). By IX 2.5, u is even a closed immersion.

[988] Therefore G is of finite type, hence of finite presentation over S. Then, by lemma 3.5 in the "étale" case, one sees that there exists an open set U neighborhood of S_0, hence identical to S, such that $G|U \to H|U$ is étale, so u is étale, hence an open immersion (since it is an étale monomorphism[989]), which completes the proof.

Corollary 3.8. *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A/I)$, and let H be an S-prescheme in groups of finite type, affine and flat over S.*

For H to be of multiplicative type and isotrivial, it is necessary and sufficient that H_0 be so, and that H satisfy one of the following conditions a) and b) (which are therefore equivalent given the preceding conditions):

a) The fibers of H are of multiplicative type, and of constant type on each connected component of S.

b) H is commutative and the ${}_nH$ ($n > 0$) are finite over S.

These last conditions are also implied by the following:

c) The fibers of H are connected.

Of course, if H is of multiplicative type (and isotrivial), conditions a) and b) are verified by IX 1.4.1 a) and 2.1 c), so only the "if" requires proof. We shall use the subgroup G indicated in 3.7. When condition c) is satisfied, one obviously has $G = H$ and we are done.

In case b), one notes that the immersion $u : G \to H$ induces an open immersion

$${}_n u : {}_n G \to {}_n H$$

which induces an isomorphism $({}_nG)_0 \xrightarrow{\sim} ({}_nH)_0$; since ${}_nH$ is finite over S, this implies at once that ${}_nu$ is an isomorphism (the complement of its image is finite over S and

[988] *N.D.E.* One has added the following sentence.
[989] *N.D.E.* Cf. EGA IV₄, 17.9.1.

reduces to $\emptyset$). By 3.6 it follows that the morphisms induced on fibers $u_s : G_s \to H_s$ are isomorphisms, so u is surjective, hence an isomorphism.

Finally, in case a), one may assume S connected,[990] and it follows that for every $s \in S$, $u_s : G_s \to H_s$ is a monomorphism of algebraic groups of multiplicative type and of the same type over $\kappa(s)$.[991] I claim that such a homomorphism is necessarily an isomorphism (which will again complete the proof[992]). Indeed, one may suppose, possibly extending the base field, that the two groups over $\kappa(s)$ are diagonalizable, and then this follows from VIII 3.2 b) and from the fact that a surjective homomorphism of isomorphic finitely generated $\mathbb{Z}$-modules $M \to N$ is necessarily bijective.

Corollary 3.9. *Let A be a noetherian ring equipped with an ideal I such that A is separated and complete for the I-adic topology, and let H be an S-prescheme in groups of finite type, affine and flat over S, with connected fibers.*

For H to be an isotrivial torus, it is necessary and sufficient that H_0 be so.

4. Case of an arbitrary base. Quasi-isotriviality theorem

Let A be a local ring. Recall that one says, after Nagata, that A is *henselian* if every algebra B finite over A is a product of local algebras B_i finite over A.

Recall 4.0.[993] *Let A be a henselian local ring, k its residue field, $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(k)$, and ξ a geometric point of S_0. Then, the functor*

$$X \mapsto X_0 = X \times_S S_0$$

is an equivalence between the category of étale covers of S and the analogous category over S_0 (cf. EGA IV₄, § 18.5). Consequently (cf. SGA 1, V), one has $\pi_1(S_0, \xi) = \pi_1(S, \xi)$.

Suppose moreover A noetherian and denote by A' its completion, $S' = \mathrm{Spec}(A')$. Then A' is a complete noetherian local ring, hence henselian (loc. cit., 18.5.14), and the functor

$$X \mapsto X' = X \times_S S',$$

from the category of étale covers of S to the analogous category over S', fits into a commutative diagram

$$\begin{tikzcd}
\mathrm{Rev.ét.}(S) \arrow[rr] \arrow[dr, "\simeq"'] & & \mathrm{Rev.ét.}(S') \arrow[dl, "\simeq"] \\
& \mathrm{Rev.ét.}(S_0) &
\end{tikzcd}$$

[990] *N.D.E.* The original indicated: "i.e. S_0 connected". Note that, since A is separated for the I-adic topology, S_0 meets every connected component of S (and since A is complete, the connected components of S_0 and S are in bijection).
[991] *N.D.E.* since G_s and H_s are of the same type for every $s \in S_0$, hence for every s.
[992] *N.D.E.* because u will once again be a surjective open immersion.
[993] *N.D.E.* One has spelled out this recall, and added remark 4.0.1.

so is also an equivalence of categories, whence $\pi_1(S_0, \xi) = \pi_1(S', \xi)$.

Remark 4.0.1. Since S is connected (A being local), it follows from 1.2 that the category of isotrivial groups of multiplicative type over S is equivalent to the analogous category over S_0 (and also over S' if moreover A is noetherian).

Lemma 4.1. *Let A be a henselian local ring with residue field k, $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(k)$.*

(i) The functor

$$H \mapsto H_0 = H \times_S S_0$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S_0.

(ii) If moreover A is noetherian, denoting by A' its completion and $S' = \mathrm{Spec}(A')$, the functor

$$H \mapsto H' = H \times_S S'$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S'.

As in 4.0, the second assertion is a consequence of the first; let us prove the latter. One already knows that the functor under consideration is essentially surjective, since every group of multiplicative type H_0 over $S_0 = \mathrm{Spec}(k)$, finite hence of finite type over k, is isotrivial by 1.4, hence comes from an isotrivial group of multiplicative type over S by 4.0.1.

It remains to prove full faithfulness, i.e. that for two finite groups G, H of multiplicative type over S, the map below is bijective:

$$\mathrm{Hom}_{S\text{-gr.}}(G, H) \to \mathrm{Hom}_{S_0\text{-gr.}}(G_0, H_0),$$

i.e.,[994] denoting by F the S-functor $\mathrm{Hom}_{S-gr.}(G, H)$, that the natural map

$$\mathrm{Hom}_S(S, F) \to \mathrm{Hom}_{S_0}(S_0, F_0)$$

induced by the base change $S_0 \to S$ is bijective. For this, given recall 4.0, it suffices to prove:

Lemma 4.2. *Let G, H be two finite groups of multiplicative type over a prescheme S. Then $F = \mathrm{Hom}_{S-gr.}(G, H)$ is representable by a finite étale scheme over S.*

[995] Let $f : S' \to S$ be a faithfully flat quasi-compact morphism such that G' and H' are diagonalizable. It suffices to show that $F_{S'}$ is representable by a preschem X' étale and finite (hence affine) over S', since the descent datum on X' relative to f (cf. VIII 1.7.2) will then be effective (by SGA 1 VIII, 2.1), whence the existence of an S-prescheme X such that $X \times_S S' = X'$, which represents F, and is étale and finite over S (cf. SGA 1, V 5.7 and EGA IV_4, 17.7.3 (ii)).

[994] *N.D.E.* One has added what follows.
[995] *N.D.E.* One has spelled out the original in what follows.

One may therefore suppose that $G = D_S(M)$ and $H = D_S(N)$, where M and N are finite commutative groups (cf. VIII 2.1 c)). Then, $K = \mathrm{Hom}_{gr.}(N, M)$ is a finite abelian group and, by VIII 1.5, one has an isomorphism

$$\mathrm{Hom}_{S-gr.}(G, H) \simeq K_S,$$

which completes the proof of 4.2 and hence of 4.1.

Proposition 4.3.0.[996] *Let S be a henselian local scheme, s its closed point, $g : X \to S$ a morphism locally of finite type, x an isolated point of the fiber X_s.*

(i) Then $\mathcal{O}_{X,x}$ is finite over $\mathcal{O}_{S,s}$. (In particular, if the residue extension $\kappa(x)/\kappa(s)$ is trivial, then $\mathcal{O}_{S,s} \to \mathcal{O}_{X,x}$ is surjective, by Nakayama's lemma.)

(ii) If moreover g is separated, then $X' = \mathrm{Spec}(\mathcal{O}_{X,x})$ is an open and closed subscheme of X, i.e. one has a decomposition as a disjoint sum $X = X' \sqcup X''$.

Proof. By the local form of Zariski's main theorem given in [Pes66], or [Ray70, Ch. IV, Th. 1], x has an affine open neighborhood $U = \mathrm{Spec}(B)$ of finite type and quasi-finite over $A = \mathcal{O}_{S,s}$, and there exists an open immersion $U \hookrightarrow Y = \mathrm{Spec}(C)$, where C is a finite A-algebra. (N.B. to reach this conclusion, one may also use Chevalley's semi-continuity theorem (EGA IV_3, 13.1.4), then the form of Zariski's main theorem given in loc. cit., 8.12.8.)

Since A is henselian, Y is the disjoint sum of local schemes $Y_1, \cdots, Y_n$, each finite over S, and the points of Y above s are the closed points $y_1, \cdots, y_n$. Therefore $x = y_i$ for some i, and $\mathcal{O}_{X,x} = \mathcal{O}_{U,x} = C_i$ is finite over A. Moreover, $X' = C_i$ is an open subscheme of U hence of X.

Suppose moreover g separated. Then, since the morphism $X' \to S$ is finite (C_i being finite over A), so is the immersion $X' \to X$ (cf. EGA II, 6.1.5 (v)), so X' is also closed in X.

Lemma 4.3. *Let A be a noetherian henselian local ring, A' its completion, $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$, s the closed point of S, G and H two S-preschemes in groups, with G of multiplicative type and of finite type over S, H locally of finite type and separated over S, H_s of multiplicative type, and H flat over S at the points of H_s.*

Let G', H' be deduced from G, H by the base change $S' \to S$. Then the natural map below is bijective:

Hom_{S-gr.}(G, H) □ Hom_{S´-gr.}(G´, H´).

Since S' is faithfully flat and quasi-compact over S, one knows by SGA 1, VIII 5.2 that this map is a bijection of the first member onto the part of the second formed by the $u' : G' \to H'$ such that the two inverse images u''_1, $u''_2 : G'' \to H''$ of u' on $S'' = S' \times_S S'$ (by the projections pr_1, pr_2 of S'' onto S') are equal.

[996] *N.D.E.* One has added this recall, used in the proof of 4.3 (the reference to EGA IV_4 18.5.11 not being fully satisfactory).

Therefore everything reduces to proving that for every homomorphism of S'-groups $u' : G' \to H'$, one has $u''_1 = u''_2$. By the density theorem IX 4.8 it suffices to prove that u''_1 and u''_2 coincide on ${}_nG''$ for every integer $n > 0$. (N.B. one needs here in an essential way the density theorem in a case where the base prescheme, here S'', is not locally noetherian.)

This reduces us, replacing G by ${}_nG$, to the case where there exists an integer $n > 0$ such that $n \cdot id_G = 0$, hence where G is finite over S. Let likewise ${}_nH$ be the kernel of the n-th power morphism ϕ_n in H. (N.B. we have not assumed H commutative, so ϕ_n (resp. ${}_nH$) is not necessarily a homomorphism of groups (resp. a subgroup of H).) Since ${}_nH$ is defined as the fibered product of $\phi_n : H \to H$ and the unit section $\epsilon : S \to H$, its formation commutes with every base change $T \to S$, i.e. one has $({}_nH)_T = {}_n(H_T)$.

[997] We denote by an index m on the right the reductions modulo m^{m+1}, where m is the maximal ideal of A. Then H_m is flat over S_m (since H is so over S at the points of H_0); therefore, by 2.4, since H_0 is of multiplicative type and of finite type, so is H_m. Therefore each $({}_nH)_m = {}_n(H_m)$ is a subgroup of H_m, of multiplicative type, finite and flat over S_m.

In particular, $({}_nH)_0 = {}_n(H_0)$ is finite over S_0; since S is henselian local and ${}_nH$ is (like H) locally of finite type and separated over S, then, by 4.3.0, the local rings of ${}_nH$ at the points of $({}_nH)_0$ are finite over S and one has a decomposition as a disjoint sum of open sets

(∗) _n H = _n H⁺ ⨿ _n H⁻,

where $Z = {}_nH^+$ is finite over S, and ${}_nH^-$ lies above $S - s$.

Note that, for every finite S-scheme Y (such as G), every S-morphism $Y \to {}_nH$ factors through Z, and that the formation of the decomposition (∗) commutes with the base change $S' \to S$ (where $S' = \mathrm{Spec}(A')$, A' the completion of A).

Then $Z' = Z \times_S S'$ is a finite scheme over S', as is $P' = Z' \times_{S'} Z'$. Denote by ν the restriction to P' of the multiplication of H' and by σ the automorphism of P' which exchanges the two factors. Since P' is finite over S' and H' separated and locally of finite type over S', then, by EGA II 5.4.3 and IV₁ 1.1.3, $Y = \nu(P')$ is a closed subscheme of H', universally closed and quasi-compact, hence of finite type, hence proper over S'. Moreover, $Y \to S'$ has finite fibers (since $P' \to S'$ does). Therefore, since S' is noetherian, the morphism $Y \to S'$ is finite (cf. EGA III, 4.4.2). Since $Z' \subseteq Y$ and $Z'_m = Y'_m$ for every m, then $Z' = Y$, by lemma IX 5.0, so Z' is a subgroup of H'. Likewise, the kernel $K = Ker(\nu, \nu \circ \sigma)$ is a closed subscheme of P', such that $K_m = P'_m$ for every m (since $Z'_m = {}_nH_m$ is commutative), so $K = P'$, i.e. Z' is a commutative subgroup of H'.

On the other hand, since each reduction $Z'_m = {}_nH_m$ is flat over S_m, then, by the "local criterion of flatness" (cf. EGA 0_III, 10.2.2 or [BAC], III § 5, Example 1 and Th. 1), Z' is flat over S'. Since moreover $Z'_0 = {}_nH_0$ is a finite group of multiplicative

[997] *N.D.E.* One has spelled out the original in what follows, on the basis of indications by M. Raynaud.

type over S_0, hence isotrivial by 1.4, then Z' is of multiplicative type (and isotrivial) over S', by 3.8 b). Since $S' \to S$ is faithfully flat and quasi-compact, one deduces that the multiplication $Z \times_S Z \to H$ factors through Z and makes Z a subgroup of H, finite over S and of multiplicative type (since Z' is).

Finally, by the remarks made above on decomposition $(*)$, Z' is the "finite part" of ${}_nH'$, and so the morphism $u' : G' \to {}_nH'$ takes its values in Z'. Since Z is of multiplicative type and finite over S, as is $G = {}_nG$, then, by 4.1, u' comes from a $u : G \to Z$, and therefore $u''_1 = u''_2$. This completes the proof of 4.3.

Lemma 4.4.0.[998] *Let A be a ring, $S = \mathrm{Spec}(A)$, H an S-group scheme of multiplicative type and of finite type, quasi-isotrivial (resp. isotrivial), (A_i) a filtered family of subrings of A of which A is the inductive limit, $S_i = \mathrm{Spec}(A_i)$.*

Then there exist an index i and an S_i-group scheme H_i of multiplicative type and of finite type, quasi-isotrivial (resp. isotrivial), such that $H = H_i \times_{S_i} S$.

Theorem 4.4. *Let S be a prescheme, H an S-prescheme in groups affine and of finite presentation over S, and $s \in S$. Assume:*

α) H is flat over S at the points of H_s.

β) H_s is of multiplicative type.

Then there exist an étale morphism $S' \to S$, a point s' of S' above s such that the extension $\kappa(s')/\kappa(s)$ is trivial, and an open and closed subgroup G' of $H' = H \times_S S'$, of multiplicative type and of finite type, isotrivial, such that $G'_{s'} = H'_{s'}$.

[999] **a)** Let us provisionally denote $T = \mathrm{Spec}(\mathcal{O}_{S,s})$ and show that, by "descent of conclusions", one can reduce to the case where $S = T$. Indeed, suppose we have found an étale morphism $g : T' \to T$, a point $s' \in T'$ and a subgroup G' of $H' = H_{T'}$ satisfying the conditions of the statement; replacing T' by an affine open neighborhood of s', one may suppose T' of finite presentation over T.

Since G' is isotrivial, there exists an étale finite surjective morphism $f : \tilde{T} \to T'$ such that $\tilde{G} = G' \times_{T'} \tilde{T}$ is isomorphic to $D_{\tilde{T}}(M)$, where M is a finitely generated abelian group. Since T', H, and $\tilde{T}$, G' are of finite presentation over $T = \mathrm{Spec}(\mathcal{O}_{S,s})$, and $\mathcal{O}_{S,s}$ is the inductive limit of the subalgebras $A_i = \mathcal{O}_S(S_i)$, where S_i runs through the affine open neighborhoods of s in S, then, by EGA IV₃, 8.8.2 (and Exp. VI_B, 10.2 and 10.3), there exist an index i, S_i-preschemes (resp. an S_i-prescheme in groups) of finite presentation S'_i and $\tilde{T}_i$ (resp. G'_i), and morphisms $g_i : S'_i \to S_i$ and $f_i : \tilde{T}_i \to S'_i$ (resp. a morphism of S_i-preschemes in groups $u_i : G'_i \to H \times_S S'_i$), from which f and g (resp. $u : G' \to H'$) come by the base change $T' \to S_i$. Moreover, taking i large enough, g_i will be étale, f_i étale finite surjective, and u_i an open and closed immersion (cf. EGA IV, 8.10.5 and 17.7.8).

Then, $\tilde{G}$ comes from the groups $\tilde{G}_i = G_i \times_{S_i} \tilde{T}_i$ and $D_{\tilde{T}_i}(M)$; therefore, by EGA IV₃, 8.8.2 (i) (and VI_B, 10.2), there exists an index $j \geqslant i$ such that $\tilde{G}_j \simeq D_{\tilde{T}_j}(M)$, so

[998] *N.D.E.* One has added this lemma, used several times in the sequel.

[999] *N.D.E.* One has spelled out the reductions that follow (the original indicated: "One reduces at once to the case where S is the spectrum of a noetherian local ring A, and where s is the closed point of S.").

G_j is of isotrivial multiplicative type. Denote by s'_j the image of s' in S'_j. Then the étale morphism $S'_j \to S_j \hookrightarrow S$, the point s'_j and the open and closed subgroup G'_j of $H \times_S S'_j$, verify the conditions of the statement. This shows that one may suppose $S = \mathrm{Spec}(A)$ local, with closed point s.

b) Then, A is the inductive limit of local rings A_i which are localizations of $\mathbb{Z}$-algebras of finite type; denote $S_i = \mathrm{Spec}(A_i)$. Let us show that the hypotheses α) and β) "descend" to some A_i. Since $H \to S$ is of finite presentation, the set of $x \in H$ such that H is flat over S at x is an open set W, which contains H_s by hypothesis, hence contains a quasi-compact open set V containing H_s (since H_s being affine, one can cover it by a finite number of affine open sets contained in W). Then the open immersion $\tau : V \hookrightarrow H$ is of finite presentation, so $V \to S$ is too.

Therefore, by EGA IV₃, 8.8.2 (and Exp. VI_B, 10.2 and 10.3), there exist an index i, an S_i-prescheme V_i (resp. an S_i-prescheme in groups H_i) of finite presentation over S_i, and an S_i-morphism $\tau_i : V_i \to H_i$ from which V, H and τ come by base change $S \to S_i$; moreover, taking i large enough, τ_i will be an open immersion and V_i will be flat over S_i, by EGA IV₃, 8.10.5 and 11.2.6. Denote by s_i the image of s in S_i; since the open immersion $(V_i)_{s_i} \hookrightarrow (H_i)_{s_i}$ gives, by the base change $\kappa(s_i) \to \kappa(s)$, the equality $V_s = H_s$, one already has $(V_i)_{s_i} = (H_i)_{s_i}$, i.e. $(H_i)_{s_i} \subseteq V_i$, so H_i is flat over S_i at the points of $(H_i)_{s_i}$. Finally, $(H_i)_{s_i}$ is of multiplicative type, since $H_s = (H_i)_{s_i} \otimes_{\kappa(s_i)} \kappa(s)$ is. Therefore the triple (S_i, H_i, s_i) verifies the hypotheses of 4.4, and if the desired assertion is verified for this triple, it will also be verified, by base change, for (S, H, s). This reduces us to the case where A is local and noetherian. Let us now distinguish two cases.

1°) *A is local noetherian and henselian.* Let Â be its completion, $\hat{S} = \mathrm{Spec}(\hat{A})$, and $\hat{H} = H \times_S \hat{S}$. Applying theorem 3.7, one finds an Ŝ-group Ĝ of multiplicative type, isotrivial and of finite type, and a homomorphism of Ŝ-groups

$$\hat{u} : \hat{G} \longrightarrow \hat{H}$$

which is an open immersion and a closed immersion, such that û induces an isomorphism $\hat{u}_0 : \hat{G}_0 \xrightarrow{\sim} \hat{H}_0$.

By remark 4.0.1, the base change functor by $\hat{S} \to S$ induces an equivalence between the category of isotrivial groups of multiplicative type over S, and over Ŝ; in particular Ĝ "comes from" an S-group of multiplicative type G, isotrivial and of finite type. By 4.3, û comes from a homomorphism

$$u : G \longrightarrow H;$$

moreover u is an open and closed immersion and induces an isomorphism $u_0 : G_0 \to H_0$, since this is so after the faithfully flat quasi-compact base change $\hat{S} \to S$. This therefore proves 4.4 in this case (taking of course, in the conclusion of 4.4, $S' = S$ and $s' = s$).

2°) *A is local noetherian.* The reduction to case 1°) is immediate, by applying 1°) to the "henselized" ring A^h of A. More precisely, one sees easily (using SGA 1, I § 5[1000]) that the local rings $\mathcal{O}_{S',s'}$ of the S-preschemes étale S' equipped with a point s' above s such that $\kappa(s')/\kappa(s)$ is trivial, form a filtered inductive system, whose inductive limit is a noetherian henselian local ring A^h (called the "henselized" ring of A); for details of this construction (due to Nagata in the normal case), cf. SGA 4, VIII § 4[1001]. The sorites of EGA IV₃ § 8 then allow, as in part a) of the proof, to deduce from a known result on the inductive limit A^h of the $\mathcal{O}_{S',s'}$, an analogous result on one of the (S', s'), which proves 4.4.

Corollary 4.5. *Let S be a prescheme, H an S-prescheme in groups of multiplicative type and of finite type. Then H is quasi-isotrivial, i.e. is split by an étale surjective morphism $S' \to S$.*

Indeed, let $s \in S$. By 4.4, there exist an étale morphism $S' \to S$, an $s' \in S'$ above s such that $\kappa(s) = \kappa(s')$, and a subgroup G' of H', of isotrivial multiplicative type and of finite type, such that $G'_{s'} = H'_{s'}$. Since G' and H' are of multiplicative type and of finite type, then, by IX 2.9, there exists an open neighborhood U' of s' such that $G'|U' = H'|U'$.

[1002] Suppose moreover S local henselian, with closed point s; then, by EGA IV₄, 18.5.11 b), there exists a section σ of $S' \to S$ such that $\sigma(s) = s'$. (N.B. one can see directly that $B = \mathcal{O}_{S',s'}$ equals $A = \mathcal{O}_{S,s}$ as follows: by 4.3.0 (i), one has $B \simeq A/I$, and since B is a finitely presented and flat A-algebra, I is a finitely generated ideal (cf. EGA IV₁, 1.4.7), and $I = Im$ (where m is the maximal ideal of A), whence $I = 0$.) Therefore H is already isotrivial. One thus obtains:

Corollary 4.6. *Let A be a henselian local ring, k its residue field, and π the topological Galois group of an algebraic closure of k.*

(i) Every group of multiplicative type and of finite type over $S = \mathrm{Spec}(A)$ is isotrivial.

(ii) The category of these groups over S is equivalent (via the functor $H \mapsto H \otimes_A k$) to the analogous category over $S_0 = \mathrm{Spec}(k)$; it is therefore anti-equivalent, by 1.4, to the category of Galois modules under π which are of finite type as $\mathbb{Z}$-modules.

Corollary 4.7. *Under the conditions of 4.4 suppose moreover that one of the following conditions is verified:*

a) For every generization t of s, H_t is of multiplicative type and of the same type as H_s.

b) H is commutative and the ${}_nH$ $(n > 0)$ are finite over S in a neighborhood of s.

c) For every generization t of s, the fiber H_t is connected.

Then there exists an open neighborhood U of s such that $H|U$ is of multiplicative type.

[1000] Or EGA IV₄, § 18.6.
[1001] Or EGA IV₄, § 18.6.
[1002] *N.D.E.* One has spelled out the original in what precedes and what follows.

Taking into account that an étale morphism is open, one is reduced to proving that there exists (with the notations of the conclusion of 4.4) an open neighborhood U' of s' such that $G'|U' = H'|U'$. Set $S'' = \mathrm{Spec}(\mathcal{O}_{S',s'})$; since G' and H' are of finite presentation over S', it suffices to show, by EGA IV$_3$, 8.8.2, that $G'' = H''$. We may therefore suppose $S = S''$, and then hypotheses (a), (b), (c) become those of the lemma below, whose proof is the same as that of 3.8:

Lemma 4.7.1. *Let S be a local scheme, s its closed point, H an S-group scheme of finite type, G an open and closed subgroup of H, of multiplicative type, such that $G_s = H_s$. Suppose moreover one of the following conditions verified:*

a) The fibers of H are of multiplicative type and of the same type as H_s.

b) H is commutative and the ${}_nH$ ($n > 0$) are finite over S.

c) The fibers of H are connected.

Then $G = H$.

Corollary 4.8. *Let S be a prescheme, H an S-prescheme in groups affine, flat and of finite presentation over S, with fibers of multiplicative type.*

For H to be of multiplicative type, it is necessary and sufficient that it satisfy one of the two following conditions (equivalent given the preceding conditions):

a) The type of H_s (cf. IX 1.4) is a locally constant function of $s \in S$.

b) H is commutative, and the ${}_nH$ ($n > 0$) are finite over S.

Moreover, these conditions a), b) are implied by the following:

c) The fibers of H are connected.

In particular, one finds:

Corollary 4.9. *Let S be a prescheme, H an S-prescheme in groups flat and of finite presentation over S. Suppose moreover H affine over S and with connected fibers.*[1003] *If $s \in S$ is such that H_s is a torus, there exists an open neighborhood U of s such that $H|U$ is a torus.*

In particular, if all the fibers of H are tori, H is a torus.

5. Scheme of homomorphisms from one group of multiplicative type to another. Twisted constant groups and groups of multiplicative type

Definition 5.1. *a) Let S be a prescheme, R a prescheme in groups over S. One says that R is a* twisted constant group *over S if it is locally isomorphic, in the sense of the faithfully flat quasi-compact topology, to a constant group scheme, i.e. of the form M_S with M an ordinary group.*

[1003] *N.D.E.* This result is generalized in 8.1: it suffices in fact to suppose that the fibers of H are affine and connected.

b) One says that the twisted constant group R over S is quasi-isotrivial, resp. isotrivial, resp. locally isotrivial, resp. locally trivial, resp. trivial, *if in the definition above one can replace the faithfully flat quasi-compact topology by the étale topology, resp. the global finite étale topology, resp. the finite étale topology, resp. the Zariski topology, resp. the coarsest (or "chaotic") topology, cf. IX 1.1 and 1.2, and IV 6.6.*

To say that R is quasi-isotrivial (resp. isotrivial) thus means that there exists an étale surjective (resp. and finite) morphism $S' \to S$ such that $R' = R \times_S S'$ is a constant group over S; to say that it is trivial means that R is a constant group.

c) One defines as in VIII 1.4 the type *of a twisted constant group R over S at an $s \in S$; it is a class of ordinary groups up to isomorphism, which for variable s is a locally constant function of s, hence constant if S is connected. One will also say that R is "of type M" if all the fibers of R are of type M.*

Beware that R is quasi-compact over S only if it is finite over S, i.e. if its type at every $s \in S$ is a finite group.[1004]

*d) The case most interesting for us is that where R is commutative. We shall then say that R is "*finitely generated*" if its type at each point $s \in S$ is given by a finitely generated $\mathbb{Z}$-module, a notion which should not be confused with the schematic notion "R of finite type over S" (cf. above).*

Remark 5.2. We shall also have to consider S-preschemes X which are locally isomorphic (for the faithfully flat quasi-compact topology) to constant preschemes, independently of any group structure. We shall then say that X is a *twisted constant bundle* over S, and we extend to these preschemes the terminology introduced in 5.1. Of course one will note that when X is endowed with an S-group structure, the meaning of the expressions "twisted constant", "isotrivial" etc. changes, depending on whether one takes into account or not the group structure over S. The same holds if one considers on X any other species of algebraic structure (for example that of Galois principal bundle that will be considered in the following number).

Proposition 5.3. *Let R be a commutative twisted constant group over S.*

(i) The functor $H = D_S(R)$ (cf. VIII 1) is representable and is a group of multiplicative type over S.

(ii) For every $s \in S$, the type of R at s equals that of H at s.

(iii) For R to be quasi-isotrivial (resp. isotrivial, resp. trivial, resp. locally isotrivial, resp. locally trivial), it is necessary and sufficient that H be so.

Remark 5.3.1.[1005] One may worry about seeing in 5.3 the term "type" used in two different senses depending on whether we speak of R or H; fortunately, when an S-prescheme in groups G is at the same time a twisted constant group and of

[1004] *N.D.E.* See lemma 5.12 below.
[1005] *N.D.E.* One has moved this remark here, appearing in the original after the proof.

multiplicative type, its type in either sense is the same, thanks to the fact that a finite ordinary commutative group is isomorphic to its dual!

Proof. Since the families covering for the faithfully flat quasi-compact topology are families of effective descent for the fibered category of affine preschemes in groups over variable base preschemes (SGA 1, VIII 2.1), one sees that H is representable (and is affine over S), since it is so "locally"[1006] (because it is so when R is constant, and then H is a diagonalizable group).

The fact that H is of multiplicative type is then evident by definition, as is the fact that the type of R and H at $s \in S$ is the same. Finally, since $H_{S'} = D_{S'}(R_{S'})$, the last assertion is reduced to the "trivial" case, i.e. to verifying that R is trivial if and only if H is, which follows at once from the biduality theorem VIII 1.2.

To finish making precise the correspondence between twisted constant groups and groups of multiplicative type, one must start from a group of multiplicative type H, and study $R = D_S(H)$. If the latter is representable, it is obviously a twisted constant group, and one will have $H \simeq D_S(R)$. In other words:

Scholie 5.4.0. *The functor $R \mapsto D_S(R)$ is an anti-equivalence*[1007] *between the category of twisted constant groups over S and that of groups of multiplicative type H over S such that $D_S(H)$ is representable.*

I do not know whether this condition on H is always satisfied; we shall see however that it is satisfied when H is quasi-isotrivial, in particular when H is of finite type.

Lemma 5.4. *Let $S' \to S$ be a faithfully flat morphism locally of finite presentation, X' an S'-prescheme separated, locally of finite presentation and locally quasi-finite over S'.*

Then every descent datum on X' relative to $S' \to S$ is effective, i.e. there exists an S-prescheme X, and an S'-isomorphism $X \times_S S' \simeq X'$ compatible with the descent datum.

We have already noted that the hypothesis on $S' \to S$ implies that it is a morphism covering for the faithfully flat quasi-compact topology (IV 6.3.1 (iv)), *a fortiori* a morphism of effective descent.

When $X' \to S'$ is quasi-compact, hence of finite presentation and quasi-finite, then it is a quasi-affine morphism (cf. SGA 1, VIII 6.2[1008]), and effectivity follows in this case from SGA 1, VIII 7.9. In the general case, one reduces at once to the case where S and S' are affine. One covers X' by affine open sets U_i'; let V_i' be the saturation of U_i' for the equivalence relation in X' defined by the descent datum, i.e. $V_i' = q_2(q_1^{-1}(U_i'))$, where q_1, q_2 are the two projections of $X_1'' = X' \times_{S'} (S'', p_1)$ onto X' ($q_1 = pr_1$, and q_2 is deduced from the first projection of $X_2'' = X' \times_{S'} (S'', p_2)$ thanks to the given descent isomorphism $X_1'' \simeq X_2''$). Since $S' \to S$ is faithfully flat quasi-compact locally of finite presentation, the same holds for $p_1 : S'' = S' \times_S S' \to$

[1006] *N.D.E.* cf. VIII § 1.7.
[1007] *N.D.E.* One has corrected "equivalence" to "anti-equivalence".
[1008] *N.D.E.* (when S' is locally noetherian, and EGA IV₃, 8.11.2 in general).

S', hence also for q_1 and q_2, which are consequently open morphisms (SGA 1 IV 6.6[1009]). Consequently, V'_i is an open and quasi-compact part of X'. By what we have already seen, the descent data induced on the V'_i are effective, whence it follows that the descent datum on X' is so (SGA 1, VIII 7.2).

Corollary 5.5. *A faithfully flat morphism of finite presentation $S' \to S$ is a morphism of effective descent for the fibered category of twisted constant groups (over variable base preschemes).*

Indeed, this amounts to asserting the effectivity of a descent datum under the conditions of 5.4, when X' is a constant S'-prescheme.

Theorem 5.6. *Let S be a prescheme, G and H two S-preschemes in groups of quasi-isotrivial multiplicative type, with G[1010] of finite type.*

Then $\mathrm{Hom}_{S-gr.}(H,G)$ *is representable by, and is a quasi-isotrivial twisted constant group over S;*[1011] *for every $s \in S$, if the type at s of G (resp. H) is M (resp. N), that of* $\mathrm{Hom}_{S-gr.}(H,G)$ *is* $\mathrm{Hom}_{gr.}(M,N)$.

One proceeds as in 4.2, using the fact that the assertion is established (VIII 1.5) when G and H are trivial. The necessary effectivity criterion is furnished by 5.5 (in the case of an étale surjective morphism $S' \to S$).

In particular, taking $G = G_{m,S}$, one finds:

Corollary 5.7. *(i) Let H be a quasi-isotrivial S-group of multiplicative type; then the S-group $D_S(H)$ is representable and is a quasi-isotrivial twisted constant group over S.*

(ii) The functors $R \mapsto D_S(R)$ and $H \mapsto D_S(H)$ are anti-equivalences, quasi-inverse to one another, between the category of quasi-isotrivial twisted constant S-groups and that of quasi-isotrivial S-groups of multiplicative type.

(iii) These functors induce anti-equivalences between the subcategories formed by the isotrivial, resp. locally isotrivial, resp. locally trivial, resp. trivial groups.

The last assertion is included only for the record, being already contained in 5.3.

Moreover, since every S-group of multiplicative type and of finite type is quasi-isotrivial by 4.5, one deduces from 5.6:

Corollary 5.8. *Let S be a prescheme, G and H two S-preschemes in groups of multiplicative type and of finite type. Then* $\mathrm{Hom}_{S-gr.}(H,G)$ *is representable and is a finitely generated quasi-isotrivial twisted constant S-group.*

Let us note also that in 5.3, R is finitely generated if and only if $H = D_S(R)$ is of finite type (IX 2.1 b)). By 4.5, H is then quasi-isotrivial, hence R is quasi-isotrivial. One thus finds:

[1009] *N.D.E.* (when S' is locally noetherian, and EGA IV₂, 2.4.6 in general).

[1010] *N.D.E.* One has corrected H to G.

[1011] *N.D.E.* One has added what follows.

Corollary 5.9. *The functors of 5.7 induce anti-equivalences quasi-inverse to one another between the category of S-groups H of multiplicative type of finite type, and that of finitely generated twisted constant S-groups R; moreover, every such group R is quasi-isotrivial.*

Corollary 5.10. *Let H, G be two S-preschemes in groups of multiplicative type and of finite type.*

(i) Then $Isom_{S-gr.}(H,G)$ is representable by an open and closed subprescheme of $\mathrm{Hom}_{S-gr.}(H,G)$*, and it is a twisted constant S-prescheme.*

(ii) In particular, $\mathrm{Aut}_{S-gr.}(H)$ *is representable and is a twisted constant S-group (in general non-commutative).*

This follows from 5.8 and from VIII 1.6.[1012]

Recall 5.11.0.[1013] Recall that if X is a locally noetherian prescheme, its connected components (which are always closed) are open. Indeed, let C be a connected component of X, $x \in C$ and U a noetherian open neighborhood of x; then U has only a finite number of connected components, hence the connected component U_x of x in U is open in U hence in X; since $U_x \subseteq C$, this shows that C is open in X.

Proposition 5.11. *Let S be a prescheme, R a commutative twisted constant group over S, $H = D_S(R)$ the group of multiplicative type that it defines. Consider the following conditions:*

(i) H is isotrivial (i.e. R is isotrivial).

(ii) R is the union of subpreschemes both open and closed R_i, which are quasi-compact over S (and then necessarily finite over S).

(iii) The connected components of R are finite over S.

a) Then (i) ⇒ (ii) ⇒ (iii).[1014]

b) One has (i) ⇔ (ii) ⇔ (iii) if S is locally noetherian.

c) Finally, (i) ⇔ (ii) if R is finitely generated (i.e. if H is of finite type over S), at least if S is quasi-compact or if its connected components are open.

Decomposing first S into a prescheme sum of preschemes S_i on each of which H is of constant type (cf. IX 1.4.1), we are reduced to the case where H hence R is of constant type M. We shall need:

Lemma 5.12. *Let S be a prescheme, R a twisted constant prescheme over S. Then every closed subprescheme Z of R which is quasi-compact over S is finite over S.*

[1012] *N.D.E.* Correct this reference by treating the case of the functor $Isom_{S-gr.}(H,G)$ in an addition 1.5.1 in VIII ....

[1013] *N.D.E.* One has added this recall, used several times in the sequel. (See also EGA I, 6.1.9; note however that the proof of *loc. cit.* seems unnecessarily complicated.)

[1014] *N.D.E.* Compare with the examples of 1.6 ....

Indeed, one reduces to the case where R is constant, hence of the form I_S, where I is a set, hence the filtered increasing union of the J_S, where J runs through the finite subsets of I. One may moreover suppose S affine; then Z is quasi-compact, hence contained in one of the J_S. Since J_S is finite over S, the same holds for Z.

Lemma 5.12 already establishes the parenthesized assertion in 5.11 (ii).[1015] The implication (ii) ⇒ (iii) is then clear, since the connected components of R are closed; by 5.11.0 they are open and closed if S is locally noetherian (R being étale, hence locally of finite type over S), whence (iii) ⇒ (ii) in this case.

Let us prove that (i) ⇒ (ii). For this, let $S' \to S$ be an étale finite surjective morphism splitting H hence R (cf. 5.3), so that R' is isomorphic to $M_{S'} = \coprod_{m\in M} R'_m$, where the R'_m are disjoint open sets of R', S'-isomorphic to S'. Let $g : R' \to R$ be the projection, which is a finite étale surjective morphism, hence an open and closed morphism; then the $R_m = g(R'_m)$ are open and closed parts of R, and obviously quasi-compact over S since the R'_m are so.

Finally, suppose H of finite type over S, and let us prove (ii) ⇒ (i). The case where the connected components of S are open reduces at once to the case where S is connected, so we may suppose S quasi-compact or connected. Since M is finitely generated, one can write $M = \mathbb{Z}^r \times N$, where r is an integer $\geqslant 0$ and N a finite abelian group. Let $G = D_S(M)$; consider the preschemes

$$P = \mathrm{Isom}_{S\text{-gr.}}(H, G) \subset \mathrm{Hom}_{S\text{-gr.}}(H, G) = Q$$

(cf. 5.8 and 5.10). One has isomorphisms

$$Q \simeq \mathrm{Hom}_{S\text{-gr.}}(M_S, R) \simeq \mathrm{Hom}_{S\text{-gr.}}(\mathbb{Z}^r_S, R) \times \mathrm{Hom}_{S\text{-gr.}}(N_S, R) \simeq R^r \times E,$$

where $E = \mathrm{Hom}_{S-gr.}(N_S, R)$ is finite over S[1016]. It follows that Q is the union of subpreschemes both open and closed Q_i finite over S. Therefore P is the union of the subpreschemes both open and closed $P_i = P \cap Q_i$, finite over S. Since they are étale over S, their images in S are parts S_i both open and closed, and they cover S. If S is connected or quasi-compact, there therefore exists a finite set of indices i such that the S_i cover S; let S' be the union of the corresponding P_i. Then $S' \to S$ is finite étale surjective, and setting $P' = P \times_S S' = \mathit{Isom}_{S'-gr.}(H', G')$, one sees that P' has a section over S', hence there exists an isomorphism of S'-groups

$$H' = H \times_S S' \simeq G' = G \times_S S' = D_{S'}(M),$$

which proves that S' splits H. This completes the proof of 5.11.[1017]

Remark 5.11 bis. Let us note moreover that one can, when H is of finite type over S and of constant type, give the following isotriviality criterion (in which it is no longer necessary to make any restriction on S): $H = D_S(R)$ is isotrivial if and only if R is the union of a sequence of parts both open and closed finite over S.[1018]

[1015] *N.D.E.* One has spelled out the original in what follows.

[1016] *N.D.E.* because it is a twisted constant group of type $\mathrm{End}_{gr.}(N)$ (cf. 5.6 and 5.8).

[1017] *N.D.E.* modulo the verification that (ii) ⇒ (i) when S is locally noetherian and R is not finitely generated ....

[1018] *N.D.E.* Spell out this point: M being a finitely generated $\mathbb{Z}$-module, it is countable, and the proof of

Lemma 5.13. *Let S be a locally noetherian and connected prescheme, P a quasi-isotrivial twisted constant S-prescheme, Z a part both open and closed of P, such that there exists an $s \in S$ with Z_s finite. Then Z is finite over S.*

Let us first not suppose S connected: let U be the set of $s \in S$ such that Z_s is finite; we shall prove that U is both open and closed, and that $Z|U$ is finite over U. This is a statement essentially equivalent to 5.13 but has the advantage of being "of local nature" on S for the étale topology (say), which reduces us to the case where P is constant, i.e. of the form I_S, where I is a set. (N.B. the locally noetherian hypothesis is not lost by an étale base change; this is where we use the quasi-isotriviality of P over S.)

One may moreover suppose S connected, since its connected components are open (S being locally noetherian). But then one necessarily has $Z = J_S$, where J is a part of I, and one therefore has $U = \emptyset$ or $U = S$, depending on whether J is infinite or finite, which gives the desired conclusion.

Recalls 5.14.0.[1019] Let S be a locally noetherian prescheme, and let $\tilde{S}$ be the normalization of S_{red} (cf. EGA II, 6.3.8). Recall that S is said to be *geometrically unibranch* (cf. EGA 0_IV, § 23.2 and IV_2, § 6.15) if the canonical morphism $\tilde{S} \to S$ is radicial (and hence a universal homeomorphism); in particular, the connected components of S are irreducible.

Suppose then S connected, hence irreducible, let η be its generic point and let $f : P \to S$ be a flat and locally quasi-finite morphism. Let P_i be the irreducible components of P, and ξ_i the generic point of P_i. Since P is flat over S, each ξ_i lies above η (cf. EGA IV_2, 2.3.4), and so $(P_i)_\eta = P_i \cap P_\eta$ is the closure of ξ_i in P_η. Since the fiber P_η is discrete, one therefore has $(P_i)_\eta = \xi_i$. This applies in particular when f is étale; in this case, P is also locally noetherian and geometrically unibranch (cf. EGA IV_4, 17.5.7), so its irreducible components are its connected components and are open (and closed).

Corollary 5.14. *Let S be a locally noetherian and geometrically unibranch prescheme, P a quasi-isotrivial twisted constant S-prescheme. Then the connected components of P are finite over S.*

One may obviously suppose S connected hence irreducible, and let η be its generic point. By what precedes, each connected component P_i of P is open and closed, and meets the fiber P_η at a single point. Therefore 5.13 applies and shows that each P_i is finite over S.

Theorem 5.16.[1020] *Let S be a locally noetherian and geometrically unibranch*

5.11 (i) ⇒ (ii) shows that R is the union of a countable family of open and closed parts, finite over S. One would need to see the converse ....

[1019] *N.D.E.* The original treated in 5.14 the case where S is locally noetherian and normal, and signalled in Remark 5.15 that the reasoning applies, more generally, if one supposes only S geometrically unibranch instead of normal. One has modified the statement of 5.14 (and also 5.16) accordingly, and one has added these "recalls", drawn from EGA IV_4, 18.10.6 and 18.10.7, which show that the proof of 5.14 applies verbatim to the geometrically unibranch case.

[1020] *N.D.E.* One has removed remark 5.15, rendered obsolete by the addition of 5.14.0 (cf. the preceding *prescheme. Then every S-group of multiplicative type and of finite type is isotrivial.*

One may indeed suppose S connected, hence H of constant type M. It suffices to apply 5.14 to $P = Isom_{S-gr.}(H, G)$, where $G = D_S(M)$, then to argue as in the proof of 5.11 (ii) ⇒ (i). One may also apply 5.14 to $P = R = D_S(H)$ (cf. 5.9), then use 5.11.

6. Infinite Galois principal covers and the enlarged fundamental group

(The results of the present N° and of the following will no longer be used in the sequel of this Seminar.)

Let S be a prescheme; we propose to determine the principal homogeneous bundles P over S with structural group of the form G_S, the constant S-group defined by an ordinary group G (not necessarily finite), which we shall also call *Galois principal bundles over S with group G*. We take "principal bundle" in the sense of the faithfully flat quasi-compact topology (cf. Exp. IV, Def. 5.1.5), but we shall note that for such a P, the structural morphism $P \to S$ is necessarily étale and surjective, hence covering for the étale topology; consequently P is also locally trivial for the étale topology (cf. IV, Prop. 5.1.6).[1021]

We shall assume that S is a sum of connected preschemes, i.e. that its connected components are open, which reduces us at once to the case where S is connected. We shall then choose a "geometric point" ξ of S, i.e. an S-scheme ξ which is the spectrum of an algebraically closed field $\Omega = \kappa(\xi)$. Then for every Galois principal bundle P over S with group G, P_ξ is a Galois principal bundle over the algebraically closed field $\kappa(\xi)$, hence is trivial. We shall therefore make the initial problem precise by proposing to determine the category of Galois principal bundles over S *pointed* above ξ, i.e. endowed with an S-homomorphism $\xi \to P$, i.e. with a trivialization of P_ξ. For fixed G, the set of classes of such bundles, up to an isomorphism respecting the base point, will be denoted by $\bar{\pi}^1(S, \xi; G)$. Then the set $\bar{\pi}^1(S; G)$ of isomorphism classes of Galois principal bundles over S with group G (without a specified base point) is isomorphic to the set of orbits of G in $\bar{\pi}^1(S, \xi; G)$ (taking into account the natural operations of G on this set, corresponding to translation by G of the marked point in a pointed Galois principal bundle P):

π¹(S; G) = π¹(S, ξ; G)/G.

For every morphism $S' \to S$ which is a morphism of universal effective descent for the fibered category of twisted constant preschemes over a variable base (for example $S' \to S$ faithfully flat and locally of finite presentation, cf. 5.4; for other examples cf. SGA 1 IX), we propose to determine the subsets of the preceding sets, denoted $\bar{\pi}^1(S'/S, \xi; G)$ and $\bar{\pi}^1(S'/S; G)$, formed by the Galois principal bundles over S which become trivial on S' (or, as one says, are "split" by S'). One will in fact

N.D.E.), and in 5.16 one has replaced "normal" by "geometrically unibranch".

[1021] *N.D.E.* Note that P is supposed to be a prescheme — in the *a priori* more general case of a sheaf (fpqc) P which is a G_S-torsor, is P necessarily representable?

determine the category of Galois principal bundles P over S which are split by S'. Of course, one will then have

$$\bar{\pi}^1(S, \xi; G) = \varinjlim_{S'} \bar{\pi}^1(S'/S, \xi; G),$$

where in the second member, S' runs over a cofinal system in the set of S'/S covering for the étale topology (for example, when S is quasi-compact, the set of S' over S which are quasi-compact and with étale and surjective structural morphism). Likewise, the category of Galois principal bundles over S will be the inductive limit of the subcategories defined by the S' (formed of the bundles which are split over S').

Thanks to the hypothesis made on S'/S, the category of Galois principal bundles over S split by S' is equivalent to the category of trivial Galois principal bundles over S' (hence of the form $G_{S'}$, where G acts by right translations), endowed with a descent datum relative to $S' \to S$. The datum of a base point on a Galois principal bundle P over S split by S' amounts, in terms of the corresponding trivial bundle P' over S' and its descent datum, to the datum of a trivialization of $P' \times_{S'} S'_\xi$ compatible with the induced descent datum, relative to $S'_\xi \to \xi$ (N.B. we have set $S'_\xi = S' \times_S \xi$), i.e. a section σ of P'_ξ over S'_ξ compatible with the descent datum. There is then advantage, for an arbitrary S-prescheme S' (for which one no longer supposes that $S' \to S$ is a morphism of universal effective descent for the fibered category of twisted constant bundles...), in defining $\bar{\pi}^1(S'/S; G)$ and $\bar{\pi}^1(S'/S, \xi; G)$ as the set of classes, up to isomorphism, of the structures with descent data just specified. One then obtains, for these functors in G, a very simple simplicial description, in terms of the fibered square and cube S'' and S''' of S' over S, which we shall sketch below (cf. 6.3).

The important conclusion to retain will be the following:

Proposition 6.1. *Suppose that the connected components of S' and S'' are open, and, for example, that the quotient set of $\pi_0(S')$ by the equivalence relation induced by the two projections $\pi_0(S'') \to \pi_0(S')$ is a single point.*[1022]

(i) The functor $G \mapsto \bar{\pi}^1(S'/S, \xi; G)$, from the category of groups to the category of sets, is representable by a group, denoted $\pi_1(S'/S, \xi)$ and called the fundamental group of S at ξ relative to $S' \to S$. *One thus has a functorial bijection:*

$$\bar{\pi}^1(S'/S, \xi; G) \simeq \mathrm{Hom}_{gr.}(\pi_1(S'/S, \xi), G).$$

(ii) This group has a set of generators in bijection with $\pi_0(S'')$, and is described in terms of these generators by relations in bijection with the elements of $\pi_0(S''')$.[1023]

[1022] *N.D.E.* One has added the hypothesis that the connected components of S' be open, as well as the second hypothesis. This latter means that the simplicial set $K_\bullet = \pi_0(S_\bullet)$ defined in 6.3 is connected; in fact a weaker hypothesis suffices, namely that the cone $\tilde{K}_\bullet$ of a certain morphism of simplicial sets $k_\bullet \to K_\bullet$ be connected (cf. *loc. cit.*).

[1023] *N.D.E.* One has suppressed the superfluous hypothesis that the connected components of S''' be open. On the other hand, the description given later (cf. ...) gives as the natural set of generators the set $\pi_0(S'') \coprod \pi_0(S'_\xi)$; one then reduces to $\pi_0(S'')$ by means of the relations between these generators arising from the 2-cells. See for example [Kan58], § 19 for a finer description.

In particular, $\pi_1(S'/S,\xi)$ is finitely generated (resp. of finite presentation) if $\pi_0(S'')$ (resp. as well as $\pi_0(S''')$) is finite.

(iii) The category of Galois principal bundles over S split by S', with base point above ξ, is equivalent to the category of ordinary groups G, endowed with a homomorphism of $\pi_1(S'/S,\xi)$ into G.

The proof is given below, cf. ….

6.2. When S is a connected locally noetherian prescheme, which implies that every étale prescheme S' over S is locally noetherian hence has its connected components open, one concludes from what precedes[1024] that the functor $G \mapsto \bar{\pi}^1(S,\xi;G)$, from the category of ordinary groups to the category of sets, is strictly pro-representable (cf. *Séminaire Bourbaki*, February 1960, N° 195, §§ A.2 and A.3), i.e. there exists a projective system

$$\Pi = \Pi_1(S;\xi) = (\pi_i)_{i \in I}$$

of ordinary groups over a filtered increasing index set I, which is "strict" (i.e. with surjective transition morphisms $\pi_j \to \pi_i$), and an isomorphism of functors in G

$$\pi^{\mathrm{I}}(S, \xi; G) \simeq \lim_i \mathrm{Hom}_{gr.}(\pi_i, G).$$

The second member is also simply denoted $\mathrm{Hom}_{pro-gr.}(\Pi, G)$ (cf. loc. cit.).

In the case where the projective limit $\pi = \lim \pi_i$ is "sufficiently large", more precisely when the canonical homomorphisms $\Pi \to \Pi_i$ are surjective, it is appropriate to endow Π with the projective limit topology of the discrete topologies of the Π_i, and the preceding isomorphism is also written:

$$\pi^{\mathrm{I}}(S, \xi; G) \simeq \mathrm{Hom}_{gr.\ top.}(\pi, G),$$

where the second member denotes the set of homomorphisms of topological groups, it being understood that G is endowed with the discrete topology.

The hypothesis just formulated on the projective system Π is verified, as is well known, when the π_i are finite groups (cf. [BEns], III § 7.4, Th. 1). This last condition obviously also means that every Galois principal bundle over S is isotrivial, i.e. is split by an étale surjective finite morphism. This is the case when S is geometrically unibranch (for example normal), as follows at once from 5.14.[1025] In the case where the π_i are finite, the group π coincides also with the fundamental group $\pi_1(S,\xi)$ introduced in SGA 1, V.

Also, in the favorable case ($\pi \to \pi_i$ surjective) one could call π the *enlarged fundamental group of S at ξ*. Outside of this favorable case, π itself is hardly of interest, and the role of the usual fundamental group is played by the projective system Π itself, which one will call the *enlarged fundamental pro-group of S at ξ*. (Any terminological suggestion better than "enlarged" is welcome![1026]). One will note that

[1024] *N.D.E.* One could spell out this deduction; I is in bijection with a cofinal set of morphisms $S' \to S$ covering for the étale topology ….

[1025] *N.D.E.* One has corrected 5.13 to 5.14.

[1026] *N.D.E.* Some authors speak of the "true" fundamental group.

knowledge of this pro-group is more precise than that of the usual fundamental group $\pi_1(S, \xi)$ of SGA 1 V; more precisely, the latter is the projective limit of the projective system formed by the finite quotients of the π_i.

6.3. Let us indicate rapidly the "computation" of $\pi_1(S'/S, \xi)$. Let S_i be the $(i+1)$-th fibered power of S' over S (i.e. $S_0 = S'$, $S_1 = S''$, etc.). One has between the S_i obvious simplicial operations, which make $(S_i)_{i\in\mathbb{N}}$ a simplicial object of $(Sch)/S$.

Transforming this simplicial object by the functor "set of connected components"

$$\pi_0 : (Sch)/S \longrightarrow (Ens),$$

one finds a simplicial set $K_\bullet = (K_i)_{i\in\mathbb{N}}$, with $K_i = \pi_0(S_i)$.

Likewise, the $S_{i,\xi}$ (= $(i+1)$-th fibered power of S'_ξ over ξ) form a simplicial object of $(Sch)/\xi$ hence of $(Sch)/S$, moreover endowed with a natural homomorphism of simplicial objects into $(S_i)_{i\in\mathbb{N}}$, whence a simplicial set $k_\bullet$ (with $k_i = \pi_0(S_{i,\xi})$) and a canonical homomorphism

$$k_\bullet \longrightarrow K_\bullet.$$

We can form a new simplicial set by taking the cone of this morphism (cf. 9.5.1):

$$\tilde{K}_\bullet = Cone(k_\bullet \longrightarrow K_\bullet).$$

[1027] In this way, one obtains a "pointed simplicial set" $\tilde{K}_\bullet$ (i.e. a simplicial set endowed with a homomorphism $\tilde{\xi} : \tilde{e}_\bullet \to \tilde{K}_\bullet$, where $\tilde{e}_\bullet$ is the final simplicial set). We can construct its well-known combinatorial invariants $\pi_0(\tilde{K}_\bullet, \tilde{\xi})$ and $\pi_1(\tilde{K}_\bullet, \tilde{\xi})$, whose construction involves only the components of degree $\leqslant 1$ resp. of degree $\leqslant 2$. These invariants are defined without restriction on S or S'. One then verifies without difficulty, when the connected components of S_0 and S_1 are open and $\tilde{K}_\bullet$ is connected,[1028] that $\pi_1(\tilde{K}_\bullet, \tilde{\xi})$ represents the functor $G \mapsto \bar{\pi}^1(S'/S, \xi; G)$, i.e. one has:

π_1(S′/S; ξ) ≃ π_1(K̃_•, ξ̃).

Let us also signal that when the morphism $S' \to S$ is "universally submersive" (cf. SGA 1, IV 2.1), and the connected components of S' are open, then[1029] the simplicial set $K_\bullet$, hence also $\tilde{K}_\bullet$, is connected.

[1027] *N.D.E.* One has corrected the original, which considers $\tilde{K}_\bullet = K_\bullet/k_\bullet$ whereas on the one hand $k_\bullet$ is already contractible (cf. 9.7.1) and on the other hand the morphism $k_\bullet \to K_\bullet$ is not in general injective. It is an epimorphism if S'/S is étale finite (cf. 9.7.3).

[1028] *N.D.E.* One has added the hypothesis that $\tilde{K}_\bullet$ be connected; for the proof, see the addendum below (Section 9).

[1029] *N.D.E.* One has modified the sequel, taking into account the correction made above, cf. N.D.E. (53). (The original was: "$\pi_0(\tilde{K}_\bullet, \tilde{\xi})$ is also canonically isomorphic to the set $\pi_0(S, \xi)$ of connected components of S, pointed by the connected component of ξ in S".)

Examples 6.4. It remains to give examples of enlarged fundamental groups. Let us take up the examples of 1.6, i.e. let k be an algebraically closed field, C_1 a complete rational curve over k, having exactly one singular point, this point being an ordinary double point, and C_2 a curve which is the union of two irreducible components, isomorphic to P^1_k and meeting at exactly two points, which are ordinary double points of C_2. In either case, the enlarged fundamental group of the curve is isomorphic to $\mathbb{Z}$.[1030]

In general, there would be reason to take up (simplifying and rectifying them) the results of SGA 1 IX 5, in the framework of the enlarged fundamental group. The examples of loc. cit., 5.5 would give as many examples of enlarged fundamental pro-groups which are not profinite. Thus, if instead of an ordinary double point, one took in the first example a double point with n distinct branches,[1031] one would find as enlarged fundamental group the free (discrete!) group on $n-1$ generators.

7. Classification of twisted constant preschemes and of groups of multiplicative type of finite type in terms of the enlarged fundamental group

7.0. Let S be a prescheme, which we still assume locally noetherian, to ensure that S and certain preschemes over S that we shall consider (notably those étale over S, more generally those locally of finite type over S) are locally connected.

Proposition 7.0.1. *Every twisted constant prescheme X over S which is locally trivial for the (fppf) topology (i.e. which is split by a faithfully flat morphism locally of finite presentation $S' \to S$) is quasi-isotrivial (i.e. one can even choose $S' \to S$ étale surjective).*

Indeed, one may suppose S connected, hence X of type I, where I is a fixed set. Therefore $X' = X \times_S S'$ is isomorphic to $I_{S'}$, hence $I_{S'}$ is endowed with a descent datum relative to $S' \to S$, i.e. one has an isomorphism $I_{S''} \xrightarrow{\sim} I_{S''}$ satisfying the usual transitivity condition. Now, $S'' = S' \times_S S'$ is locally noetherian hence locally connected, whence it follows that the automorphisms of $I_{S''}$ correspond to the sections of $G_{S''}$, where $G = \mathrm{Aut}(I)$ is the group of permutations of I.

In this way, one obtains a descent datum on $G_{S'}$ (considered as a trivial Galois principal bundle) relative to $S' \to S$. By 5.4 this descent datum is effective, whence a Galois principal bundle P over S, with group G. By construction, it represents the functor $\mathit{Isom}_S(I_S, X)$ in the category of preschemes over S which are locally

[1030] *N.D.E.* One could spell this out: first, taking into account 5.14, the enlarged fundamental group of the projective line P^1_k is zero, i.e. P^1_k is "truly" simply connected. Next, one would have to extend to the case of the enlarged fundamental group and the category of principal bundles (not necessarily finite), Corollary 5.4 of SGA 1 IX and the discussion that follows. (Let Γ_1 and Γ_2 be two copies of P^1_k, a_i, b_i two distinct points of Γ_i, C''_2 the disjoint union of Γ_1 and Γ_2, C'_2 the curve obtained by identifying a_1 with a_2; then C_2 is obtained from C'_2 by additionally identifying b_1 with b_2. The discussion following *loc. cit.*, extended to the enlarged case, then shows that the enlarged fundamental group of C'_2 (resp. C_2) is zero (resp. $\mathbb{Z}$).

[1031] *N.D.E.* This is an n-fold point with n distinct tangents, for example the curve $X^n - Y^n = X^{n+1}$.

noetherian. Consequently, the étale surjective base change $P \to S$ splits X, so X is indeed quasi-isotrivial.

Remark 7.0.2. Beware that even if S is the spectrum of a field, it is not true in general that every twisted constant bundle over S is quasi-isotrivial. It suffices for example to take for X the scheme sum of a sequence of schemes of the form $\mathrm{Spec}(k_i)$, where the k_i are separable extensions of k of strictly increasing degrees.

The proof given above shows at the same time that the classification of twisted constant bundles X over S, quasi-isotrivial and of type I, is equivalent to that of Galois principal bundles over S with group $G = \mathrm{Aut}(I)$. It is even an equivalence of categories.

It can be put in a more convenient form as in SGA 1 V. For this, suppose S connected and equipped with a geometric point ξ. Consequently the enlarged fundamental pro-group $\Pi = \Pi_1(S, \xi)$ is defined. Moreover, for every quasi-isotrivial twisted constant bundle X over S, let $I = X(\xi)$ be its set-theoretic fiber at ξ. Therefore X is of type I, and consequently associated as we have just said with a Galois principal bundle $P = Isom_S(I_S, X)$ over S, with group $G = \mathrm{Aut}(I)$.

By the definition of Π, one therefore obtains a canonical homomorphism of Π into G, i.e. of one of the Π_i into G. Since G is the group of permutations of $I = X(\xi)$, this means that Π "acts continuously on $I = X(\xi)$", it being understood that the π_i (i large) act on I, in a way compatible with the transition morphisms.

We leave to the reader the verification that every S-morphism $X \to Y$ between quasi-isotrivial twisted constant bundles over S induces a map $X(\xi) \to Y(\xi)$ compatible with the operations of Π, and that the functor thus obtained is an equivalence of categories:

Proposition 7.0.3. *Let S be a locally noetherian connected prescheme, ξ a geometric point of S, $\Pi = \Pi_1(S, \xi)$ the enlarged fundamental pro-group of S at ξ. Then the functor*

$$X \mapsto X(\xi)$$

is an equivalence between the category of quasi-isotrivial twisted constant bundles over S and the category of sets on which Π acts continuously.

This functor is compatible with the operations of finite sums and finite inverse limits. It follows for example that the twisted constant groups (or rings etc.) quasi-isotrivial over S correspond to the ordinary groups (resp. rings etc.) on which the pro-group Π acts continuously. In particular:

Corollary 7.0.4. *The category of commutative twisted constant groups quasi-isotrivial over S is equivalent to the category of "Π-modules", i.e. of commutative groups M on which Π acts continuously.*

Using now 5.7 one concludes the:

Theorem 7.1. *Let S be a locally noetherian connected prescheme, ξ a geometric point of S, $\Pi = \Pi_1(S,\xi)$ the enlarged fundamental pro-group of S at ξ. Then the functor*

$$G \mapsto \mathrm{Hom}_{\kappa(\xi)-gr.}(G_\xi, G_{m,\xi})$$

induces an anti-equivalence of the category of quasi-isotrivial groups of multiplicative type over S with the category of Π-modules.

Using 4.5 one concludes:

Corollary 7.2. *The preceding functor induces an anti-equivalence of the category of groups of multiplicative type and of finite type over S with the category of Π-modules which are of finite type over $\mathbb{Z}$.*

Example 7.3. Take for example for S a complete rational curve over an algebraically closed field, having exactly one multiple point with $n+1$ distinct branches. By 6.4, the enlarged fundamental group $\Pi(S,\xi)$ is a free group on n generators. Therefore, by 7.2, the classification of tori of relative dimension m over S is equivalent to the classification of systems of n endomorphisms $A_1, \cdots, A_n$ of the $\mathbb{Z}$-module $\mathbb{Z}^m$, up to automorphism of $\mathbb{Z}^m$. Except for $n \leqslant 1$ or $m \leqslant 1$, an explicit classification of such systems seems hopeless. One can at least define a multitude of non-trivial invariants for such a system, such as the characteristic polynomials of the A_i.[1032]

Remark 7.4. If one makes no hypothesis on S, it remains true that for a given ordinary commutative group M of finite type over $\mathbb{Z}$, the category of groups of multiplicative type of type M over S is anti-equivalent to the category of Galois principal bundles over S with group $G = \mathrm{Aut}_{gr.}(M)$. This follows easily from 5.9 and 5.10.

Remark 7.5. The theory of the fundamental pro-group that we have sketched in the present two numbers will be written more advantageously in the framework of general sites. In this form, it applies equally well, for example, to ordinary topological spaces, and gives a satisfactory theory at least for a locally connected topological space (not necessarily locally simply connected). In this case too it seems that one cannot be content with defining a fundamental group, and that a pro-group is needed. Finally, let us note that, once one has available the language of topologies and descent (which is really at the bottom of these questions), the exposition sketched here is also technically simpler than that of SGA 1 V, and should therefore in principle replace it.

8. Appendix: Elimination of certain affineness hypotheses

Our aim is to prove the following generalization of 4.9.

[1032] *N.D.E.* In particular, this shows that the tori of relative dimension 2 considered in 1.6 are not isotrivial.

Theorem 8.1. *Let S be a prescheme, H an S-prescheme in groups flat and of finite presentation, with connected and affine fibers.*[1033] *Let $s \in S$; suppose that H_s is a torus. Then there exists an open neighborhood U of s such that $H|U$ is a torus.*

One concludes immediately:

Corollary 8.2. *Let H be an S-prescheme in groups, flat and of finite presentation over S. For H to be a torus, it is necessary and sufficient that its fibers be tori.*

Remark 8.3. Even when S is the spectrum of a discrete valuation ring, one cannot abandon in 8.1 the hypothesis that the fibers of H (here the generic fiber) be affine, since there are examples of smooth groups over S whose generic fiber is an elliptic curve, and whose special fiber is G_m.

Proof of 8.1. One may obviously suppose $S = \mathrm{Spec}(A)$ affine, which reduces us by the standard procedure (cf. EGA IV_2, 8.8.2) to the case where S is moreover noetherian. We begin by proving 8.2 in this case.

By 4.9 we are reduced to proving that H is affine over S. We may therefore suppose[1034] A noetherian local, and since the completion $\hat{A}$ is faithfully flat over A, we are reduced by descent to the case where A is a complete noetherian local ring. Let S' be the normalization of S_{red}; one knows by Nagata that it is finite over S (EGA 0_IV, 23.1.5); moreover $S' \to S$ is surjective, so $H' = H \times_S S' \to H$ is finite and surjective, so to prove that H is affine, it suffices to show that H' is so (EGA II, 6.7.1).

[1035] Replacing S by a connected component of S', this reduces us to the case where A is a noetherian (semi-local) normal and integral ring. Moreover, possibly replacing A by its normalization in a finite separable extension of its field of fractions, one may suppose that the generic fiber H_η of H is diagonalizable, i.e. that one has an isomorphism

$$u_\eta : H_\eta \xrightarrow{\sim} T_\eta,$$

where $T = G^r_{m,S}$. Now we have:

Lemma 8.4. *Let S be a locally noetherian normal and irreducible prescheme, with generic point η, H a prescheme in groups over S smooth and with connected fibers, T a prescheme in groups of multiplicative type and of finite type over S, $u_\eta : H_\eta \to T_\eta$ a homomorphism of algebraic groups over $\kappa(\eta)$.*

Then u_η extends to a homomorphism of groups $u : H \to T$.

Possibly replacing S by its normalization in a finite separable extension of its function field, one may suppose T diagonalizable (which is moreover the case in the application we have in view). Then T is a closed subgroup of a group of the form $G^r_{m,S}$, which reduces us to the case where $T = G_{m,S}$.

[1033] *N.D.E.* Note that the hypotheses entail that H is separated over S, by VI_B, Th. 5.3 or Cor. 5.5.
[1034] *N.D.E.* by EGA IV_2, 8.8.2 again.
[1035] *N.D.E.* One has spelled out the original in what follows.

It all amounts to proving that u_η, considered as a rational map of H into $T = G_{m,S}$, is everywhere defined (since the morphism $u : H \to T$ extending it is then necessarily a homomorphism of groups). One may consider u_η as an invertible section f of the structural sheaf of H_η, and one must show that it extends to an invertible section of the structural sheaf of H. Now H, being smooth over the normal S, is normal (SGA 1, II 3.1), so it suffices to find a closed part Z of H of codimension $\geqslant 2$ such that f extends to an invertible section of the structural sheaf of $H-Z$. This reduces us at once to the case where S is the spectrum of a discrete valuation ring A (by localizing at points of codimension 1 of S).

Let t be a uniformizer of A, t' the section of `O_H` that it defines, so that the special fiber `H_0` is equal to $V(t')$. By hypothesis `H_0` is smooth over the residue field k, and connected. Then f is a rational function on H which has neither zeros nor poles in $H - H_0$; since $H_0 = V(t')$ is an irreducible divisor, there exists an integer $n \in \mathbb{Z}$ such that $t'^n f = t^n f$ has neither zeros nor poles, i.e. is an invertible section of `O_H`. It therefore defines a morphism $v : H \to G_{m,S}$, and since $v_\eta = t'^n u_\eta$ and u_η is a homomorphism of groups, v transforms the unit section of H into a section of $G_{m,S}$ whose value at the generic point of S is t^n; since it is a question of a section of $G_{m,S}$, t^n must be a unit, i.e. $n = 0$, so v extends u_η, which completes the proof of 8.4.

Applying this lemma to the present case, one finds a homomorphism of groups

```
u : H  T = G^r_{m,S}
```

which induces on the generic fibers an isomorphism. Let us prove that u is an isomorphism.

Lemma 8.5. *For every integer $n > 0$ prime to the residual characteristic of A, ${}_nH = Ker(n \cdot id_H)$ is finite over S.*

If n is prime to the residual characteristic of A, it is prime to all the residual characteristics of points of S. Therefore $n \cdot id_H$ induces on every fiber of H an étale morphism; consequently $n \cdot id_H$ is étale (SGA 1, I 5.9), so its kernel ${}_nH$ is étale over S. On the other hand, ${}_nH$ is separated over S since H is.[1036] [1037] Moreover, all its fibers have the same rank n^r, since the fibers of H are tori, all of the same dimension r (H being smooth over S). One concludes that ${}_nH$ is finite over S (SGA 1, I 10.9 or EGA IV_4, 18.2.9).

Therefore, by the preceding lemma, $u({}_nH)$ is a closed part of T. Since on the generic fiber T_η, this part is identical to ${}_n(T_\eta)$, it follows that it contains the closure of this part, namely ${}_nT$. Now for every $s \in S$, the ${}_n(T_s)$ are dense in T_s; since $u_s(H_s)$ is a closed part (VI_B 1.4.2) containing them, one sees that $u_s(H_s) = T_s$, so u is surjective. Since a surjective homomorphism of tori of the same dimension over a field is flat,[1038] it follows that u induces on each fiber a flat morphism, so u is

[1036] By Raynaud's theorem VI_B 5.3.

[1037] *N.D.E.* Spell out this point, in connection with the modifications in VI_B § 5.

[1038] *N.D.E.* This follows, for example, from VIII 3.2 a); more generally, if $f : G \to H$ is a surjective morphism of algebraic groups over a field k and if H is reduced, then f is flat (cf. VI_B 1.3).

flat (SGA 1 I 5.9). Consequently, $K = Ker(u)$ is flat over S,[1039] hence equal to the closure of its generic fiber K_η. Now K_η is the unit group by construction, and since K is separated over S (since H is), its unit section is closed, whence it follows that K is the unit group. Therefore u is a monomorphism; since we have seen that it is faithfully flat, it is therefore an isomorphism (cf. SGA 1, I 5.1 or EGA IV$_4$, 17.9.1). This proves that H is a torus, hence completes the proof of 8.2.[1040]

Remark 8.5.1. Instead of invoking 8.5 one can also invoke Zariski's "Main Theorem", which directly implies that u is an open immersion, hence an isomorphism.[1041]

To prove 8.1, we are reduced, thanks to the quasi-isotriviality theorem, to the case where S is local, s its closed point,[1042] and to proving that, with the hypotheses made elsewhere, H is then a torus. By 8.2 already proved, we are reduced to proving that the fibers of H are tori. One may suppose S the spectrum of a complete noetherian local ring A. By 3.3 there exists for every $n > 0$ a group of multiplicative type T_n finite over S, and an isomorphism $(T_n)_s \simeq {}_n(H_s)$, where s is the closed point of S. Proceeding as in 3.1 and using the fact that T_n is finite over S,[1043] one sees that the preceding isomorphism comes from a homomorphism $u_n : T_n \to H$, moreover uniquely determined. (Pass to the limit over the artinian quotients of A.)

Moreover, by uniqueness properties, $u_n : T_n \to H$ is deduced from $u_m : T_m \to H$ by restriction to ${}_n(T_m) \simeq T_n$ when m is a multiple of n. It follows from IX 6.6 that the u_n are monomorphisms, so the T_n are subgroups of H, and for m a multiple of n, one has $T_n = {}_n(T_m)$.

Therefore for every $t \in S$, the $(T_n)_t$ are subgroups of H_t, of type $(\mathbb{Z}/n\mathbb{Z})^r$ (where $r = \dim H_s = \dim H_t$), such that for m a multiple of n, one has $(T_n)_t = {}_n(T_m)_t$. The fact that H_t is a torus now follows from:

Lemma 8.6. *Let H be a smooth affine algebraic group over an algebraically closed field k, $(T_n)_{n>0}$ a family of subgroups of multiplicative type of H, such that for every integer $n > 0$, T_n is of type $(\mathbb{Z}/n\mathbb{Z})^r$, and for every multiple m of n, one has $T_n = {}_n(T_m)$.*

Under these conditions, H contains a torus of dimension $\geqslant r$ containing the T_n, so if H is connected of dimension $\leqslant r$, H is a torus of dimension r.

This is an exercise on affine algebraic groups, which we shall treat by reference to *Bible*. We confine ourselves to considering the T_n for n prime to the characteristic. Let K be the closure of the union of the T_n in H, endowed with the induced reduced structure; then standard arguments show that K is a commutative algebraic subgroup of H. By *Bible*, § 4.5, Th. 4, K is therefore isomorphic to a product $K_u \times K_s$, with K_u "unipotent" and K_s diagonalizable. Every diagonalizable subgroup of K is

[1039] *N.D.E.* One has spelled out the original in what follows.
[1040] *N.D.E.* At least in the case envisaged so far, namely S locally noetherian.
[1041] *N.D.E.* The editors did not understand this remark, not understanding why u would *a priori* have finite fibers and be surjective ....
[1042] *N.D.E.* Spell out this reduction ....
[1043] *N.D.E.* Spell out this point ....

contained in K_s, so the T_n are subgroups of K_s, hence $K = K_s$. Write $K = D(M)$, with M an ordinary commutative group of finite type over $\mathbb{Z}$; then $T_n \subseteq K$ means that M admits a quotient group isomorphic to $(\mathbb{Z}/n\mathbb{Z})^r$. Since this is true for every integer n prime to the characteristic of k (it would suffice for powers of a fixed prime number), it follows that M is of rank $\geqslant r$, so K contains a torus of dimension r, say T. When H is connected of dimension r, it follows that $T = H$, which completes the proof of 8.6. Thus 8.1 is proved.

Remarks 8.7. Using 8.1, it should not be difficult to give analogous generalizations of 4.7 and 4.8. A more interesting study would be that of the situation in 8.1 where one abandons the hypothesis that the fibers of H be affine. One can show that there then exists an open neighborhood U of s such that $H|U$ is commutative and that for every $t \in U$, the geometric fiber H_t is an extension of an abelian variety by a torus.[1044] Of course, in questions of this kind, one may restrict oneself to the case where S is the spectrum of a discrete valuation ring, s its closed point, t its generic point.

One can generalize this result as follows. For every algebraic group G connected smooth over an algebraically closed field k, a well-known theorem of Chevalley tells us that G is (in a unique way) an extension of an abelian variety A by a smooth connected affine group V. Let us denote by *abelian rank* (resp. *reductive rank*, resp. *nilpotent rank*, resp. *semisimple dimension*) of G, and denote by $\rho_{ab}(G)$ (resp. $\rho_r(G)$, resp. $\rho_n(G)$, resp. $d_s(G)$), the dimension of A, resp. the dimension of the maximal tori of G, resp. the dimension of the Cartan subgroups[1045] of G, resp. the dimension of the quotient of G (or also of V) by its radical (cf. *Bible* for all these notions). Let us also introduce the *unipotent rank* $\rho_u(G) = \rho_u(V) = \rho_n(G) - \rho_r(G) - \rho_{ab}(G)$. When G is not over an algebraically closed field, we still denote by the same names and the same notations the corresponding invariants for $G_{\bar{k}}$, where $\bar{k}$ is the algebraic closure of k.

This being so, let G be a smooth group scheme over the spectrum S of a discrete valuation ring; let ρ_{ab} etc. (resp. ρ'_{ab} etc.) be the invariants associated with the special fiber (resp. with the generic fiber); then one has the inequalities:

$$\begin{cases} \rho_{ab} \leqslant \rho'_{ab} \\ \rho_r + \rho_{ab} \leqslant \rho'_r + \rho'_{ab} \\ d_s \leqslant d'_s \end{cases} \qquad \begin{cases} \rho_n \geqslant \rho'_n \\ \\ \rho_u \geqslant \rho'_u. \end{cases}$$

It amounts to the same to say that if G is smooth of finite type over an arbitrary base S, the functions $s \mapsto \rho_{ab}(s)$, $\rho_{ab}(s) + \rho_r(s)$, $d_s(s)$ are lower semi-continuous, and the functions $s \mapsto \rho_n(s)$, $\rho_u(s)$ are upper semi-continuous.[1046]

The same results probably remain valid without supposing G smooth over S, but simply flat of finite presentation over S, by agreeing to denote, for an algebraic group G over an algebraically closed field k, by $\rho_{ab}(G)$ etc. the corresponding invariants of G^0_{red}.

[1044] *N.D.E.* Give a reference here?
[1045] *N.D.E.* Recall the definition of “Cartan subgroup” ….
[1046] *N.D.E.* Give a reference for these results?

In this Seminar, we present some results of this kind for G affine over S, or more generally with affine fibers:

in this case, we shall verify the semi-continuity properties for ρ_r, ρ_n hence for

$\rho_u = \rho_n - \rho_r$, and the continuity of ρ_s in a neighborhood of a point s whose fiber is a reductive group.[1047]

One can generalize 8.2 when one supposes already G commutative, as follows:

Theorem 8.8.[1048] *Let G be an S-prescheme in commutative groups which is flat and of finite presentation over S, with affine connected fibers. Let $s \in S$, and suppose that*

a) if $\bar{k}$ denotes an algebraic closure of $\kappa(s)$, $(G_{\bar{k}})_{red}$ is a torus.

b) There exists a generization t of s such that G_t is smooth over $\kappa(t)$.

Under these conditions, there exists an open neighborhood U of s such that $G|U$ is a torus.

(Note moreover that if one supposes only that for every generization t of s, G_t is affine resp. connected, one easily derives that the same conclusion is valid for t in an open neighborhood of s).

Proof of 8.8. It suffices to prove that G_s is smooth over $\kappa(s)$. Indeed, since G is flat of finite presentation over S, it then follows that G is smooth over S above a neighborhood of s (cf. 3.5), but then one is under the conditions of 8.1.

To prove that G_s is smooth over $\kappa(s)$, the usual procedure reduces us to the case where S is affine noetherian. Choosing a homomorphism $S' \to S$ from a spectrum of a discrete valuation ring S' into S, sending the closed point to s and the generic point η to t, we are reduced to the case where S is itself the spectrum of a discrete valuation ring, which may be assumed moreover complete with algebraically closed residue field, and where s and $t = \eta$ are respectively the closed point and the generic point of S.

Therefore G is flat, separated, of finite type over S, the generic fiber G_η is smooth and connected, and the special fiber G_0 is such that $T_0 = (G_0)_{red}$ is a torus. Let m be an integer > 0 prime to the residual characteristic at s, hence also to that at η; one knows then that $m \cdot id_G$ is a fiber-by-fiber étale morphism (VII_A 8.4), hence an étale morphism since G is flat over S (SGA 1 I 5.9), so its kernel ${}_mG$ is étale over S, and since G is separated[1049] [1050] of finite type over S, so is ${}_mG$. Since $({}_mG)_0 = {}_m(T_0)$, its degree is m^r, where r = dim T_0 = dim G_0 = dim G. It follows that the rank of $({}_mG)_\eta = {}_m(G_\eta)$ is $\geqslant m^r$, which proves already (using 8.6) that G_η

[1047] *N.D.E.* Give a reference for these results?

[1048] *N.D.E.* G. Prasad and J.-K. Yu have generalized (subject to some additional hypotheses) this result without supposing G commutative and replacing in hypothesis a) and the conclusion "torus" by "reductive group", cf. [PY06], Th. 6.2.

[1049] By Raynaud's theorem VI_B 5.3.

[1050] *N.D.E.* cf. N.D.E. (62) in 8.5.

is a torus of dimension r, since the two fibers of ${}_mG$ have the same rank, hence as in 8.5 that ${}_mG$ is finite over S.[1051]

Note that since S is complete with algebraically closed residue field, the finite étale cover ${}_mG$ decomposes completely, so through every point of ${}_mG_0$ passes a section of G over S; in particular, the set of points of G_0 through which passes a section of G over S is everywhere dense. Now we have this:

Lemma 8.9. *Let S be a locally noetherian regular prescheme of dimension 1, G an S-prescheme in groups flat and locally of finite type, such that G_η is smooth over $\kappa(\eta)$ for every maximal point η of S (which implies that G is reduced). Suppose that the normalized scheme X of G is finite over G (this is the case, by Nagata, if S is the spectrum of a complete discrete valuation ring, cf. EGA IV$_2$, 7.7.4), and let G' be the open set of X formed by the points at which X is smooth over S. With these notations:*

a) If the projection $G' \to S$ is surjective, then there exists on G' a unique structure of S-prescheme in groups such that the canonical morphism $G' \to G$ is a homomorphism of groups.

b) Suppose that for every closed point s of S, the set of points of G_s^0 through which passes an étale quasi-section is dense in G_s^0 for the Zariski topology. Then X is regular, one is under the conditions of a), and the map $\Gamma(G'/S) \to \Gamma(G/S)$ is bijective.

In the case of interest to us, this lemma applies and gives us a homomorphism of groups $u : G' \to G$, where G' is smooth over S, and u_η is an isomorphism $G'_\eta \xrightarrow{\sim} G_\eta$. Possibly replacing G' by an open subgroup, one may suppose that G'_0 is connected, and since $u_0 : G'_0 \to G_0$ induces a surjective morphism $G'_0 \to T_0$, where T_0 is a torus of the same dimension as G'_0, one easily concludes that G'_0 is a torus (for example using 8.6, or by referring to *Bible*, § 7.3, Th. 3 a)). Consequently, by 8.2 G' is a torus, but then by IX 6.8, `Ker u` is a subgroup of multiplicative type of G', and since its generic fiber is reduced to the unit group, it is the unit group, hence u is a monomorphism. Using now VIII 7.9 it follows that u is an immersion. Being surjective, and G being reduced, it follows that u is an isomorphism, which completes the proof of 8.8.

It remains to prove 8.9. To prove a), note that the uniqueness of the group law on G' making $u : G' \to G$ a homomorphism of groups is clear, since one knows the group law of G' on the generic fiber (supposing S irreducible, which is permissible). For existence, one reduces easily to the case where S is local, the spectrum of a discrete valuation ring A, and thanks to uniqueness, and to the fact that the operation of integral closure commutes with an étale base extension, one can make étale base extensions on S, which reduces us to the case where A is "strictly local" i.e. henselian with separably closed residue field. The same reduction applies for b), but under the hypothesis made in b), one can now replace "étale quasi-section" by "section".

[1051] *N.D.E.* revise the preceding sentence ....

Note that G' being smooth over S and X normal, $G' \times_S X$ is normal (since smooth over X which is normal), so the composite morphism $G' \times_S X \to G \times_S G \to G$ factors as

```
p : G′ ×_S X  X.
```

Let us prove that this last morphism induces on the open set $G' \times_S G'$ of $G' \times_S X$ a morphism

```
G′ ×_S G′  G′.
```

One must therefore show that $G'_0 \times_k G'_0$ is mapped into the open set G'_0 of `X_0`; it suffices to see that for every point g'_0 of G'_0 with values in k, the morphism

$$(\times) h'_0 \mapsto p_0(g'_0, h'_0)$$

from G'_0 into `X_0` takes its values in G'_0. Now since G' is smooth over S and S is henselian, every g'_0 as above is induced by a section g' of G' over S, and one sees at once that the morphism $(\times)$ above is then induced by the morphism $h' \mapsto p(g', h')$ of G' into X, itself deduced by transport of structure from the automorphism $h \mapsto g \cdot h$ of G, left translation by the section g of G image of g'. So $h' \mapsto p(g', h')$ is itself an automorphism of X, hence maps G' into G', which proves our assertion.

It remains to prove that the composition law thus obtained on G' is a group law. Associativity follows at once from the associativity of the generic fiber (isomorphic to that of G). On the other hand, the symmetry automorphism of the S-prescheme G induces an automorphism of X, which therefore leaves G' stable and induces an automorphism σ of G'. One then verifies that the latter has the properties of an inverse for the composition law on G', since this again reduces to verifying the commutativity of certain diagrams involving fibered powers of G' over S, and the latter being smooth over S, it suffices to verify the commutativity on the generic fiber, which is clear. This proves part a) of 8.9.

Let us prove b). Let Z' be the set of $x \in X$ such that $O_{X,x}$ is non-regular; it is a closed part by a theorem of Nagata (EGA IV_2, 6.12.6), obviously contained in `X_0`; let Z be its image in G, which is therefore a closed part of `G_0`. Then Z is a rare part of `G_0`, i.e. contains no maximal point y of `G_0`. Indeed, since `G_0` is defined in G by an equation $t = 0$ (where t is a uniformizer of the discrete valuation ring A), $O_{G,y}$ is of dimension 1 by the Hauptidealsatz, so for every x of X above y, $O_{X,x}$ is of dimension 1, hence a discrete valuation ring since X is normal hence regular in codimension 1. On the other hand, it is evident that for every section g of G over S, Z' is stable under the automorphism of X defined by transport of structure from the left translation by g in G, so Z is stable under the left translation in `G_0` defined by g_0. Now by hypothesis the set of such g_0 is dense in G_0^0. Since Z is stable under translation by these g_0, and is a rare closed set, it follows at once that $Z = \emptyset$, whence $Z' = \emptyset$, so X is regular. Consequently, X is smooth over S at every point through which passes a section. Now every section of G over S lifts in a unique way to a section of X over S hence of G' over S. This is so in particular for the unit section, which proves that the image of G' in S is S, i.e. that one is under the conditions of (a). This completes the proof of 8.9 and hence of 8.8.

9. Addenda

[1052]

9.1. Simplicial sets, topoi, groupoids, and topological spaces

Notations 9.1.1.[1053] *Let E be a simplicial set. One can associate with it the following objects:*

(2) a topos $T = E^{\sim}$, obtained from the topoi naturally associated with the sets E_i, by the procedure described in [Del74], 6.3.1 (see also [Ill72], VI.5.2 and SGA 4 VI.7);

(3) a groupoid $G = \Pi(E)$, whose objects are the elements of `E_0` (the "vertices"), and whose arrows are defined in [GZ67], II.7;

(4) a topological space $X = |E|$ (a cellular complex), called the "geometric realization" (or "topological") (cf. loc. cit., III.1).

Let us note that a sheaf on T is nothing other than a simplicial set over E.

9.2. Locally constant sheaves; descent data

Definitions 9.2.1. *One calls a* locally constant sheaf *on:*

(1) a simplicial set E, every morphism of simplicial sets $E' \to E$ such that for every $i \in \mathbb{N}$ and every $e \in E_i$, the face operators d induce isomorphisms $E'_e \xrightarrow{\sim} E'_{d(e)}$ between the fibers (cf. [AM69], § 10);

(2) a topos T, every object F of T such that there exists an epimorphism $U \to 1$ and an isomorphism $F \times U \simeq f^*I \times U$, where I is a set and $f : T \to Ens$ is the final morphism (cf. SGA 4, IX.2);

(3) a groupoid G, every presheaf on G, that is, every contravariant functor from G into the category of sets (cf. [GZ67], append. I.1.2);

(4) a topological space X, every sheaf of sets on X, locally constant in the usual sense.

Finally:

(5) one calls a descent datum *on a simplicial set E the datum of a sheaf F on $E_0^{\sim}$ (that is, a set-valued function on `E_0`) and of an isomorphism $\alpha : p_1^*F \xrightarrow{\sim} p_2^*F$, where $p_1, p_2 : E_1^{\sim} \to E_0^{\sim}$ are the morphisms (deduced from the) faces, satisfying the usual cocycle relation (cf. Exp. IV, 2.1 (1) and infra).*

[1052] *N.D.E.* This additional section was written by Fabrice Orgogozo (following indications by Ofer Gabber).

[1053] PP: I have replaced "FGA, *Technique de descente* I, 1.6" by "Exp. IV, 2.1".

The morphisms between these five types of objects, as well as the associated inverse-image functors, are defined in an obvious manner.

9.3. Some equivalences of categories

Let E be a simplicial set and let T, G and X be the associated topos, groupoid and topological space.

Proposition 9.3.1. *The categories of locally constant sheaves on E, T, G, X as well as the category of descent data on E are equivalent.*

Sketch of proof. Let us denote by (1) through (5) the categories of objects defined in the preceding paragraph.

– (1) ⇔ (5). This is a particular case of [AM69], 10.6 (see also [GZ67], append. I.2.3, [Fri82], ?.5.6, or [Ill72], VI.8.1.6). An equivalence of categories is given by the functor associating with the object $(E' \to E)$ the pair (E'_0, α) where E'_0 is considered as a sheaf on `E_0` and where α is the unique isomorphism $p_1^* E'_0 \xrightarrow{\sim} p_2^* E'_0$ whose fiber at each $y \in E_1$ — with images denoted x_1 and x_2 by the two projections — is given by the isomorphisms $(E'_0)_{x_1} \leftarrow (E'_1)_y \to (E'_0)_{x_2}$.

– (5) ⇔ (3). Evident: one of the two relations defining the morphisms in the groupoid G associated with E is a cocycle relation.

– (1) ⇔ (4). Cf. [GZ67], append. I.3.2.1.

– (1) ⇔ (2). One must show that an object over E is a locally constant sheaf in the simplicial sense if and only if it is so as a sheaf on $T = E^{\sim}$. This follows from the fact that the locally constant sheaves $F_\bullet$ on T are nothing other than the cartesian sheaves, that is, those for which the arrows $([n] \to [m])^* F_m \to F_n$ are isomorphisms. This last point is a particular case of a general fact on simplicial topoi together with the fact that every sheaf on $E_0^{\sim}$ is locally constant. (See also [Ill72], VI.8.1.6.) QED.

For every group H, the equivalences of categories above induce equivalences between the categories of H-torsors, these being locally constant sheaves equipped with an action of H, with fibers isomorphic to H acting on itself by translation.

9.4. Fundamental groups and groupoids

It follows from the preceding equivalences of categories that for every group H and every simplicial set E, the sets of isomorphism classes of H-torsors $H^1(E, H)$, $H^1(E^{\sim}, H)$, $\pi_0(\mathrm{Hom}(\Pi(E), BH))$ and $H^1(|E|, H)$ are naturally in bijection. Recall that one denotes by `BH` the punctual groupoid associated with H and by π_0 the functor associating with a category the set of isomorphism classes of its objects.

Likewise, if e is a point of E, the preceding equivalences induce bijections between the sets of isomorphism classes of H-torsors trivialized over e, denoted respectively $H^1(E rel e, H)$, $H^1(E^{\sim} rel e^{\sim}, H)$, $\mathrm{Hom}(\pi_1(\Pi(E), e), H)$ and $H^1(|E| rel |e|, H)$. Recall that one denotes by $\pi_1(\Pi(E), e)$ the group $Isom_{\Pi(E)}(e, e)$. For variable

H, these functors are represented, in the connected case, by a group which one denotes by $\pi_1(E,e)$. The group $\pi_1(E,e)$ is isomorphic to $\pi_1(\Pi(E),e)$ and $\pi_1(|E|,|e|)$, hence also to the fundamental group of a simplicial set as defined by Kan (cf. e.g. [May67], 16.1 or [Ill72], I.2.1.1). (Recall that the set $H^1(E,H)$ is in turn isomorphic to the set $H^1(\pi_1(E,e),H)$ of morphisms to H modulo conjugation, also denoted $\mathrm{Hom}\,ext(\pi_1(E,e),H)$.)

9.5. Cones

Definitions 9.5.1. *(1) Let $f : E' \to E$ be a morphism of simplicial sets. Recall (cf. for example [Del74], 6.3.1) that one denotes by $C(f)$ the pointed simplicial set whose underlying set in degree $n \geqslant 0$ is*

$$E_n \sqcup \left(\coprod_{i<n} E'_i\right) \sqcup \star,$$

where $\star$ is a singleton. We leave to the reader the task of defining the simplicial maps, the definition of the faces in rank less than or equal to two being recalled below. (See also [GZ67], VI.2 for a pointed variant.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on E equipped with a trivialization of the inverse image on E'. The simplicial set $C(f)$ is naturally pointed by $\star \to C(f)$.

(2) Let $f : T' \to T$ be a morphism of topoi. Denote by $C(f)$ the topos whose objects are quintuples $(F, F', A, \alpha : f^*F \to F', \beta : A \to \Gamma(T', F'))$, where F (resp. F') is an object of T (resp. T'), A is a set, and α (resp. β) is a morphism in T' (resp. Ens). (See also [Del80], 4.3.4 and [Ill72], III.4 for a variant of this construction.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on T equipped with a trivialization of the inverse image on T'. The topos $C(f)$ is naturally pointed by the fiber functor sending the quintuple $(F, F', A, \alpha : f^*F \to F', \beta : A \to \Gamma(T', F'))$ to the set A.

(3) Let $f : G' \to G$ be a morphism of groupoids. Denote by $C(f)$ the colimit of the diagram

$$G \leftarrow G' \to B_1$$

where B_1 is the punctual category (one object, one arrow). The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on G equipped with a trivialization of the inverse image on G'.

(4) Let $f : X' \to X$ be a morphism of topological spaces. Denote by $C(f)$ the colimit of the diagram

$$\star \leftarrow X' \times \{0\} \to X' \times [0,1] \leftarrow X' \times \{1\} \to X.$$

The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on X equipped with a trivialization of the inverse image on X'. The topological space $C(f)$ is naturally pointed by $\star \to C(f)$.

9.5.2. Descent data on the cone of a simplicial map Let $f : E' \to E$ be a morphism of simplicial sets and $C(f)$ its cone (cf. supra, (1)). Let us use the letter p (resp. q, r) to denote the face maps of E' (resp. E, $C(f)$). Before stating the proposition below, let us explicit the faces r in degree less than or equal to two in terms of p and q. By convention, p_{ij} (resp. p_i) is the face map $E'_2 = E'_{\{1,2,3\}} \to E'_1 = E'_{\{1,2\}}$ (resp. $E'_1 = E'_{\{1,2\}} \to E'_0 = E'_{\{1\}}$) corresponding to the increasing map $\{1,2\} \to \{1,2,3\}$ (resp. $\{1\} \to \{1,2\}$) of image `\{i, j\}` (resp. `\{i\}`)[1054]. The same convention is adopted for the faces of E and $C(f)$.

The morphism

$$r_1 : C(f)_1 = E_1 \coprod E'_0 \coprod \star \to C(f)_0 = E_0 \coprod \star$$

(resp. r_2) is:

– $q_1 : E_1 \to E_0$ (resp. $q_2 : E_1 \to E_0$) on `E_1`;

– $E'_0 \to \star$ (resp. $f_0 : E'_0 \to E_0$) on E'_0;

– $\star \to \star$ on $\star$.

Likewise the morphism

$$r_{21} : C(f)_2 = E_2 \coprod E'_1 \coprod E'_0 \coprod \star \to C(f)_1 = E_1 \coprod E'_0 \coprod \star$$

(resp. r_{32}, r_{31}) is:

– $q_{21} : E_2 \to E_1$ (resp. q_{32}, q_{31}) on `E_2`;

– $p_1 : E'_1 \to E'_0$ (resp. $f_1 : E'_1 \to E_1$, $p_2 : E'_1 \to E'_0$) on E'_1;

– $E'_0 \to \star$ (resp. $Id : E'_0 \to E'_0$, $Id : E'_0 \to E'_0$) on E'_0.

Let H be a descent datum on $C(f)$, that is, a sheaf `H_0` on $C(f)_0 = E_0 \coprod \star$ equipped with an isomorphism $\gamma : r_1^* H_0 \xrightarrow{\sim} r_2^* H_0$ between its inverse images on $C(f)_1 = E_1 \coprod E'_0 \coprod \star$ satisfying the cocycle relation $r_{31}^*\gamma = r_{32}^*\gamma \circ r_{21}^*\gamma$.

The restriction of the isomorphism γ to `E_1` (resp. E'_0) is a descent datum $\beta : q_1^* G_0 \xrightarrow{\sim} q_2^* G_0$ where `G_0` is the restriction of `H_0` to `E_0` (resp. a trivialization $t : H_0(\star) \xrightarrow{\sim} f^* G_0 =: F_0$ where $H_0(\star)$ is the constant sheaf with stalk $H_0(\star)$). The restriction of the cocycle relation satisfied by γ to `E_2` (resp. E'_1) is the cocycle relation for β (resp. $p_2^*(t) = f_1^*(\beta) \circ p_1^*(t)$). This latter relation means that the trivialization t is compatible with the descent datum $f_1^*(\beta)$ induced by β on `F_0`. From this it follows:

Proposition 9.5.3. *Let $f : E' \to E$ be a morphism of simplicial sets and $C(f)$ its cone, pointed by $\star$. The category of descent data on $C(f)$ trivialized over $\star$ is equivalent to the category of descent data on E equipped with a trivialization of the descent datum induced on E'.*

[1054] This convention departs from the current simplicial norm but is closer to the notations used by Grothendieck in his theory of descent.

9.6. Representability of the functor π̄¹(S′/S, ξ; −)

The notations are those of page 90 and from now on we suppose the connected components of $S_0 = S'$ and $S_1 = S' \times_S S'$ open. Under this hypothesis, one has the following explicit combinatorial description, which follows immediately from the recalls above.

Proposition 9.6.1. *For every group G, the set $\bar{\pi}^1(S'/S, \xi; G)$ of isomorphism classes of descent data equipped with a trivialization above ξ is in natural bijection with the relative non-abelian cohomology set*

```
H^1(K rel k, G) := H^1(K̃ rel ⋆, G).
```

It follows that this functor is representable by a group, denoted $\pi_1(K, k)$ (the "*relative fundamental group*"), as soon as the simplicial set $\tilde{K} = C(k \to K)$ is connected. One verifies immediately that this is so if and only if the map $\pi_0(k) \to \pi_0(K)$ is surjective. It suffices in particular that the simplicial set K be connected. As indicated in the text, this is the case if the scheme S is connected and the morphism $S' \to S$ is universally submersive.

9.7. Contractibility of k, counter-example to the injectivity of k_• → K_•

The two following propositions make the N.D.E. (53) more precise.

Proposition 9.7.1. *Let κ be an algebraically closed field, and X a connected κ-scheme. For every integer $i \geqslant 0$, denote by X_i the $(i+1)$-th fibered power of X over κ. For every integer $i \geqslant 0$, the canonical map*

$$\pi_0(X_i) \to \pi_0(X)^{i+1}$$

is a bijection. In particular, the simplicial set $\pi_0(X_\bullet)$ is contractible.

It suffices to prove the following lemma, which follows by passage to the limit from the Künneth formula. [Spell out?]

Lemma 9.7.2. *Let κ be an algebraically closed field and X, Y two connected κ-schemes. Then the fibered product $X \times_\kappa Y$ is connected.*

Proposition 9.7.3. *Let X be a connected scheme, $X' \to X$ a finite étale morphism, and x a geometric point of X localized at a point x. The canonical morphism $\pi_0(X'_x) \to \pi_0(X')$ is surjective.*

Indeed, every connected component of X' has image an open-closed of X hence meets the fiber X'_x. The morphism $\pi_0(X'_x) \to \pi_0(X')$ is therefore surjective, as is the morphism $\pi_0(X'_x) \to \pi_0(X'_x)$.

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Exposé XI. Representability criteria. Applications to multiplicative-type subgroups of affine group schemes

by A. Grothendieck

0. Introduction

As we have already seen examples in Exp. X, N°s 4, 5, the representability of certain functors, such as certain functors of type $\mathrm{Hom}_S(X,Y)$ and various variants, plays an important role in many questions concerning preschemes in groups.

Among the results particularly useful in this theory, let us mention (in addition to the questions of representability of quotients, studied in Exposés V and VI_B and

[1055] *N.D.E.* additional references cited in this Exposé; the references [AM69] and following concern the Addenda.

in Exp. VIII 5) the question of the representability of functors of the form $\prod_{X/S} Y/X$ (where Y is a subobject of X), studied in Exp. VIII 6 in a very elementary case, of which we shall give variants in N° 6 of the present Exposé; these results give us the representability of various centralizers, normalizers, transporters.

Less elementary representability criteria, using results that will appear in EGA VI, are indicated in 6.12 and in Exp. XV, XVI, where we shall give a criterion of representability of quotients G/H in cases not covered by the preceding Exposés (a criterion that was not developed in the oral Exposés).

Our principal object in the present Exposé is the proof of theorems 4.1 and 4.2, which furnish a typical example of a non-projective construction technique (close to the one that will be developed in EGA VI). It has indeed appeared, since the oral Exposé and the writing of the present text, that the affine hypotheses made in 4.1 and 4.2 can to a large extent be eliminated (cf. XV), and that, on the other hand, one can, for the essential part of the theory developed in the following Exposé, do without 4.1 and 4.2. It might finally be interesting to prove the analogue of these results for a general reductive (for instance semisimple) group prescheme instead of a group of multiplicative type, in which case 4.1 and 4.2 will doubtless be the key result for the proof.

1. Reminders on smooth, étale, and unramified morphisms

The reader is referred to EGA IV §§ 17 & 18, and pending its publication, to SGA 1, I, II, III (where it is however appropriate to replace certain noetherian hypotheses, troublesome in applications, by hypotheses of finite presentation).

Definition 1.1. *Let S be a prescheme, F a functor $(Sch)^\circ/S \to (Ens)$. One says that F is* formally smooth *(resp.* formally unramified [1056]*, resp.* formally étale*) if for every S-prescheme S', affine (in the absolute sense), and every subscheme S'' of S' defined by a nilpotent ideal J, the map*

$$F(S') \longrightarrow F(S'')$$

is surjective (resp. injective, resp. bijective). A prescheme X over S is said to be formally smooth over S *(resp.* formally unramified over S*, resp.* formally étale over S*) if the corresponding functor is formally smooth (resp. formally unramified, resp. formally étale); one says that X is* smooth over S *(resp.* unramified over S*, resp.* étale over S*) if it satisfies the preceding condition and if moreover X is locally of finite presentation over S.*

The interest of these definitions for X resides in the fact that, on the one hand, they are expressed in a remarkably simple way in terms of the functor represented by X (and in practice, X is often given as the object over S representing some explicit

[1056] One will now rather say "net" instead of "unramified".

functor), and that, on the other hand, they are also expressed by remarkable properties concerning the local structure of X, which we shall recall in the following statements (for the proof, we refer to loc. cit.).

Proposition 1.2. *Let X be a prescheme locally of finite presentation over S. Then:*

(i) For X to be smooth over S, it is necessary and sufficient that X be flat over S, and that its geometric fibers $X \otimes_S \operatorname{Spec}(\kappa(s))$ be regular schemes. More generally, for X to be smooth over S in a neighborhood of the point $x \in X$ (one then says that X is smooth over S at x*), it is necessary and sufficient that X be flat over S at x and X_s be smooth over $\kappa(s)$ at x, i.e. $X_s \otimes_{\kappa(s)} \bar{\kappa}(s)$ be regular at the points (or simply, a point) above x.*

(ii) Suppose S, and hence X, locally noetherian; let $x \in X$ and let $s \in S$ be its image in S. Then the smooth nature of X over S at x is detected on the local homomorphism $A = O_{S,s} \to B = O_{X,x}$ of noetherian local rings (or even on the local homomorphism $\hat{A} \to \hat{B}$ of their completions), by the following characteristic property: B is flat over A, and $B \otimes_A k$ is geometrically regular over k (the residue field of A), i.e. for every finite extension k' of k, $(B \otimes_A k) \otimes_k k' = B \otimes_A k'$ is a regular semi-local ring. When the residual extension $k(B)/k(A)$ is trivial, these conditions are equivalent also to the following: $\hat{B}$ is isomorphic as $\hat{A}$-algebra to an algebra of formal power series $\hat{A}[[t_1, \cdots, t_n]]$.

Thus, from the "formal" point of view, the structure of X over S is that of the typical affine space $S[t_1, \cdots, t_n]$ over S.

Proposition 1.3. *Let X be a prescheme locally of finite presentation over S. Then:*

(i) The following conditions are equivalent:

a) X is unramified over S.

b) The diagonal morphism $X \to X \times_S X$ is an open immersion.

c) $\Omega^1_{X/S} = 0$.

d) The geometric fibers of X/S are discrete and reduced, i.e. isomorphic to sums of copies of the base field.

e) For every $s \in S$, the fiber $X \otimes_S \operatorname{Spec} \kappa(s) = X_s$ is unramified over $\kappa(s)$, or equivalently is isomorphic to a sum of spectra of finite separable extensions of $\kappa(s)$.

(ii) One has analogous conditions of pointwise nature for X to be unramified over S at a given point x (i.e. in a neighborhood of the said point); for example, it is necessary and sufficient that $O_{X_s,x}$ be a finite separable extension of $\kappa(s)$, which is expressed also in terms of the local homomorphism $A = O_{S,s} \to B = O_{X,x}$ by the condition that $B \otimes_A k$ be a finite separable extension of k (residue field of A), i.e. $r(A)B = r(B)$ and $k(B)$ is a finite separable extension of $k(A)$. When A, and hence also B, is noetherian, and $k(A) \xrightarrow{\sim} k(B)$, this means also that $\hat{A} \to \hat{B}$ is surjective.

Thus, from the “formal” point of view, to say that X is unramified over S means that X is essentially a subprescheme of S. Observe also that conditions d) and e) are expressed solely in terms of the fibers of X/S.

Proposition 1.4. *Let X be a prescheme locally of finite presentation over S. For X to be étale over S, it is necessary and sufficient that it be smooth over S and unramified over S (trivial by definition), which permits applying criteria 1.2 and 1.3. One finds in particular:*

*(i) For X to be étale over S, it is necessary and sufficient that it be flat over S and unramified over S.

Analogous local criterion for X to be étale over S at a given point x.*

(ii) Suppose moreover S, and hence X, locally noetherian. Then the fact that X is étale over S at a point x is detected on the local homomorphism $A = \mathcal{O}_{S,s} \to B = \mathcal{O}_{X,x}$ (and even on the local homomorphism $\hat{A} \to \hat{B}$) by the following characteristic property: B is flat over A, and $B \otimes_A k$ is a finite separable extension of k (residue field [1057] of A). When $k(A) \xrightarrow{\sim} k(B)$, this condition means simply that $\hat{A} \to \hat{B}$ is an isomorphism.

Thus, from the “formal” point of view, to say that X is étale over S means simply that X is locally isomorphic to S.

Remark 1.5. When X is locally of finite presentation over S and one is given a point $x \in X$, then the fact that X is smooth (resp. unramified, resp. étale) over S at x, i.e. in a neighborhood of x, is detected on the functor $X : (Sch)^\circ/S \to (Ens)$ by the following property: for every S' over S, S' the spectrum of a local ring, every subscheme S'' of S' defined by a nilpotent ideal, and every S-morphism $u'' : S'' \to X$ sending the closed point s'' of S'' to x, there exists at least one (resp. at most one, resp. exactly one) S-morphism $u : S' \to X$ which extends it. This statement shows in particular that in definition 1.1 one may restrict to S' which are local schemes (provided that the functor in question is representable by a prescheme locally of finite presentation over S). On the other hand, when S is locally noetherian, one may even, in the preceding pointwise criterion, restrict to S' which are local artinian schemes, and one may thus impose the same restriction on S' in definition 1.1 (provided that S is locally noetherian and the functor in question is represented by a prescheme locally of finite type over S).

Remark 1.6. Of course, given a morphism $f : X \to Y$ of preschemes, one will say that f is smooth (resp. ...) if f makes X into a smooth (resp. ...) Y-prescheme. When X and Y are S-preschemes and f an S-morphism of finite presentation, then these properties of f are expressed immediately in terms of the morphism of functors $F, G : (Sch)^\circ/S \to (Ens)$ defined by f: f is smooth (resp. unramified, resp. étale) if for every S-prescheme S', S' affine (in the absolute sense), and every subscheme S'' of S' defined by a nilpotent ideal J, the map

F(S′) □ F(S″) ×_{G(S″)} G(S′)

[1057] *N.D.E.* “Residual extension” has been replaced by “residue field”.

deduced from the commutative square

$$F(S') \longrightarrow G(S') \downdownarrows F(S'') \longrightarrow G(S'')$$

is surjective (resp. injective, resp. bijective), i.e. for every commutative square of morphisms of functors over S (where S', S'' are as above and $i : S'' \to S'$ is the canonical immersion):

$$\begin{tikzcd} S'' \arrow[r, "i"] \arrow[d, "u''"'] & S' \arrow[d, "v"] \\ F \arrow[r, "f"] & G \end{tikzcd} \qquad \text{(Q)}$$

there exists at least one (resp. at most one, resp. exactly one) morphism $u : S' \to F$ making the two corresponding triangles commutative: $u'' = ui$ and $v = fu$.

This property for a homomorphism of functors $f : F \to G$ over S keeps its meaning even when the functors in question are not representable; one will say (if it is satisfied) that the homomorphism of functors $f : F \to G$ is *formally smooth* (resp. *formally unramified*, resp. *formally étale*). One will note that this depends only on the homomorphism of functors $(Sch)^\circ \to (Ens)$ defined by F, G (cf. I 1.4.1), and not on the structural morphisms $F \to S$ and $G \to S$. An equivalent way of expressing the preceding definition, somewhat more manageable in applications, is the following: the morphism of functors $F \to G$ over S is said to be formally smooth (resp. formally unramified, resp. formally étale) when, for every S' over S and every morphism $S' \to G$, the functor $F' = F \times_G S'$ over S' is formally smooth (resp. formally unramified, resp. formally étale).

Remark 1.7. Under the conditions of 1.6, when one is given a "point of F" with values in a field k, i.e. an element x of $F(\mathrm{Spec}(k))$ or, what amounts to the same, a morphism $\mathrm{Spec}(k) \to F$, one says likewise that f is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) *at* x, if the condition of the preceding definition is satisfied whenever S' is a local scheme and the morphism $S'' \to F$ in the diagram (Q) above is "compatible with the marked points" in the following sense: if k' denotes the residue field of S' and of S'' at their closed point, the diagram

$$\begin{tikzcd} \mathrm{Spec}(k') \arrow[d, "j"'] & \mathrm{Spec}(k) \arrow[d, "x"] \\ S'' \arrow[r] & F \end{tikzcd}$$

(where $j : \mathrm{Spec}(k') \to S''$ denotes the canonical immersion) can be completed in a commutative diagram

```
      Spec(k″)
     /        \
    /          \
```

```
        ↓                   ↓
    Spec(k′)            Spec(k)
        |                   |
        ↓                   ↓
       S″ ───────────────→ F              ,
```

where k'' is the spectrum of a field. When F, G are representable by S-preschemes X, Y and the S-morphism $f : X \to Y$ is locally of finite presentation, this condition means precisely (by virtue of 1.5) that $f : X \to Y$ is smooth at the point of X image of $\mathrm{Spec}(k)$ by $x : \mathrm{Spec}(k) \to X$.

Remark 1.8. When the condition of the preceding definition is satisfied while restricting to local artinian S', one will say that $F \to G$ is *infinitesimally smooth* (resp. *infinitesimally unramified*, resp. *infinitesimally étale*) *at* x, and one says that $F \to G$ is infinitesimally smooth (resp. …) if it is so at every point x, in other terms if the condition envisaged in 1.6 is satisfied whenever S' is local artinian. This variant of the preceding notions is technically useful, since it is often easier to verify, being a weaker notion, while being frequently sufficient (for example if $F \to G$ is a morphism locally of finite presentation, with G representable by a prescheme locally noetherian…) to entail the strong condition.

Smooth morphisms of preschemes behave in a remarkably simple way with respect to differential calculus. We confine ourselves here to recalling the following property:

Proposition 1.9. *Let $f : X \to S$ be a smooth morphism of preschemes. Then $\Omega^1_{X/S}$ is a locally free module of finite type over X, and its rank at a point $x \in X$ equals the dimension of the fiber X_s (where $s = f(x)$) in a neighborhood of the point x.*

This dimension is called the *relative dimension of X over S at x*. One will note that it is zero (when f is smooth at x) if and only if f is étale at x. This dimension is computed in practice still in terms of the functor F represented by X, in the following way. Let ξ be a point of F with values in a field k, "localized at x", i.e. a morphism $\mathrm{Spec}(k) \to F = X$ whose image is x. Consider the algebra $D(k) = k[t]/(t^2)$ of dual numbers over k, considered as an S-prescheme, and consider the map

$$F(D(k)) \longrightarrow F(k)$$

deduced from the augmentation $D(k) \to k$, and let finally $F(D(k), \xi)$ be the inverse image of $\xi \in F(k)$ under this map. Then this set is naturally endowed with a structure of vector space over k (in fact, it is the vector space dual of $\Omega^1_{X/S} \otimes_{O_{X,x}} k$), whose dimension is the relative dimension of X over S at x.

To make explicit the vector law on $F(D(k), \xi)$, it is convenient to introduce more generally, as in Exp. II, for every vector space V over k, the algebra $D_k(V) = k + V$ (with V an ideal of square zero), and to consider $F(D_k(V), \xi)$, the inverse image of ξ by $F(D_k(V)) \to F(k)$, as a covariant functor in V, with values in $(\mathtt{Ens})$. It then suffices that this functor commute with products of two factors (which means that

F transforms certain amalgamated sums of a very special type into fibered products, compare Exp. II — a condition always satisfied when F is representable), to conclude that the $F(D_k(V), \xi)$ and in particular $F(D(k), \xi)$ are endowed with vector structures over k. One can thus define the relative dimension of F over X at the "point" ξ, under conditions appreciably broader than the representability of F.

In the present Exposé, the fact that certain functors which we shall make explicit are representable by preschemes smooth over S will serve us mainly through the intermediary of the following result, which will be for us the technical intermediary to pass from constructions on the completion of the local ring A of a point s of a noetherian scheme S to a local ring A' over A étale over A, which will yield in particular a means of passing from s to neighboring points:

Proposition 1.10. *Let $f : X \to S$ be a smooth morphism of preschemes, s a point of S, and x a point of X above s such that $\kappa(x)$ be a finite separable extension of $\kappa(s)$. Then there exists a subscheme S' of S, étale over S, passing through x. Hence one can find an étale morphism $S' \to S$, a point s' of S' above s of residue extension equal to $\kappa(x)/\kappa(s)$, and an S-morphism $S' \to X$ sending s' to x.*

To construct S', one simply takes a system $g_1, \cdots, g_n$ of sections of 0_X on a neighborhood of x, which induce at x a regular system of parameters of the local ring of the fiber X_s at x; the subscheme S' defined by the g_i is then étale over S at x, and provided one shrinks S', it will then be étale over S.

We shall use 1.10 when $\kappa(x) = \kappa(s)$, i.e. x is rational over $\kappa(s)$, i.e. x can be considered as a section of X_s over Spec($\kappa(s)$). Then 1.10 is in the nature of a theorem of extension of sections (after étale extension of the base). It takes a particularly simple form in the following special case:

Corollary 1.11. *Under the conditions of 1.10, suppose that S is the spectrum of a henselian local ring, and that $\kappa(x) = \kappa(s)$. Then there exists a section of X over S passing through x (uniquely determined if X is in fact étale over S at x).*

Indeed, S being henselian, it follows under the conditions of 1.10 that S' contains an open subscheme which is finite over S and whose fiber at s is reduced to x. As it is étale over S, it follows that it is isomorphic to S, whence the conclusion. — One will note that when S is the spectrum of a complete local ring, 1.10 or 1.11 is more or less the equivalent of the classical "Hensel's lemma", and one sometimes refers to it by this name.

2. Examples of formally smooth functors drawn from the theory of multiplicative-type groups

We are going to interpret, in the language introduced in the preceding N°, the results stated in IX 3 concerning infinitesimal extensions of a homomorphism of a group of multiplicative type (consequences of the vanishing of the Hochschild cohomology of such a group, established in Exposé I).

Proposition 2.1. *Let S be a prescheme, H a group of multiplicative type over S, G an S-prescheme in groups smooth over S; consider the functor over S*

$$M_H = \mathrm{Hom}_{S-gr.}(H, G)$$

(whose value at S' over S is the set of homomorphisms of S'-groups from $H_{S'}$ to $G_{S'}$). This functor is formally smooth over S.

Cf. IX 3.6. More generally:

Corollary 2.2. *Let S, G be as above, and consider a homomorphism $u : H_1 \to H_2$ of group schemes of multiplicative type over S, whence with the preceding notation a morphism of functors over S:*

$$M_{H_2} \longrightarrow M_{H_1} \qquad (M_{H_i} = \mathrm{Hom}_{S\text{-gr.}}(H_i, G)),$$

given by $w \mapsto w \circ u$. This homomorphism is formally smooth.

Indeed, by virtue of definitions 1.6, this is equivalent to the following statement: when S is affine, S' a subscheme defined by a nilpotent ideal, let

$$v : H_1 \longrightarrow G$$

be a homomorphism of S-groups, and

$$w' : (H_2)_{S'} \longrightarrow G_{S'}$$

a homomorphism of S'-groups such that $w' u_{S'} = v_{S'}$; there then exists a homomorphism of S-groups

$$w : H_2 \longrightarrow G$$

extending w', and such that $wu = v$. To see this, one begins by extending w' to a homomorphism of S-groups $w_0 : H_2 \to G$, which is possible by 2.1; consider then $v' = w_0 u : H_1 \to G$. It is such that $v'_{S'} = v_{S'}$ by hypothesis on w', hence by virtue of VIII 3.6 there exists an element g of $G(S)$, whose image in $G(S')$ is the unit element, and such that $v = int(g)v'$, whence $v = (int(g)w_0)u$, and it will therefore suffice to take $w = int(g)w_0$.

Corollary 2.3. *With the notation of 2.1, consider $M_H = M$ as a functor with group of operators G (G operating by $(v, g) \mapsto int(g) \circ v$). Then the corresponding morphism*

$$R : G \times_S M \longrightarrow M \times_S M$$

defined by $(g, v) \mapsto (int(g) \circ v, v)$ is a formally smooth morphism.

By means of a base change $S' \to S$, this is equivalent to the following statement:

Corollary 2.4. *With the notation of 2.1, let $v_1, v_2 : H \to G$ be two morphisms of S-groups, and let $Transp(v_1, v_2)$ be the subfunctor of G formed of the g such that $int(g) \circ v_1 = v_2$. Then this functor is formally smooth over S. In particular (if $v_1 = v_2 = v$) the functor $Centr(v)$, subgroup of G formed of the h such that $int(h)v = v$, is formally smooth over S.*

(N.B. The pair (v_1, v_2) can be considered as a section over S of the second member in the morphism R of 2.3, and $Transp(v_1, v_2)$ as the inverse image functor of the said section under R.) Statement 2.4 itself is equivalent to the following: when S is affine and S' is a subscheme of S defined by a nilpotent ideal, for every $g_0 \in G(S')$ such that $int(g_0)(v_1)_{S'} = (v_2)_{S'}$, g_0 extends to a $g \in G(S)$ such that $int(g)v_1 = v_2$. To prove this, one begins by extending g_0 to a section g' of G over S, which is possible since G is smooth over S; one sets $v_2' = int(g')v_1$, one notes that v_2 and v_2' have the same restriction over S', hence by the result already invoked IX 3.6 there exists a $g'' \in G(S)$, inducing the unit section over S', and such that $v_2 = int(g'')v_2'$, whence `v_2 = int(g″) int(g′) v_1 = int(g″ g′) v_1`, so that it suffices to take $g = g''g'$.

Proposition 2.1 bis. *Let S be a prescheme, G an S-prescheme in groups smooth over S, and consider the functor $M : (Sch)^\circ/S \to (Ens)$:*

`M(S′) = set of multiplicative-type subgroups of G_{S′}.`

Then M is formally smooth over S.

Cf. IX 3.6 bis.

Corollary 2.2 bis. *Let n be an integer, and consider the morphism of functors*

$$\phi_n : M \longrightarrow M$$

defined by

`φ_n(H) = ₙH = Ker(n · id_H).`

Then ϕ_n is a formally smooth morphism. If for every integer $p > 0$, M_p denotes the subfunctor of M such that $M_p(S')$ is the set of multiplicative-type subgroups H of $G_{S'}$ such that ${}_pH = H$, then the morphism induced by ϕ_n:

$$M_{np} \longrightarrow M_n$$

is formally smooth.

The second assertion is trivially contained in the first and is included only for the convenience of a later reference. The proof of the first is analogous to that of 2.2, invoking this time IX 3.6 bis.

Corollary 2.3 bis. *With the notation of 2.1 bis, consider M as a functor with group of operators G (G operating by $(g, H) \mapsto int(g)(H)$). Then the corresponding morphism*

$R : G \times_S M \to M \times_S M$

defined by $(g, v) \mapsto (int(g)v, v)$ *is formally smooth.*

This is equivalent to the following statement:

Corollary 2.4 bis. *With the notation of 2.1 bis, let H_1, H_2 be two multiplicative-type subgroups of G, and let $Transp_G(H_1, H_2)$ be the subfunctor of G formed of the g such that $int(g)H_1 = H_2$. Then this functor is formally smooth over S. In particular, if $H_1 = H_2 = H$, the subfunctor $Norm_G(H)$ of G, normalizer of H, is formally smooth over S.*

The proof is analogous to that of 2.4, invoking again IX 3.6 bis.

Proposition 2.5. *Let S be a prescheme, G an S-prescheme in groups smooth over S, K a sub-S-prescheme in groups, smooth over S or of multiplicative type, H an S-prescheme in groups of multiplicative type, $u : H \to G$ a homomorphism of S-groups, and denote by $Transp_G(u, K)$ the subfunctor of G whose value, at an S' over S, is formed of the $g \in G(S')$ such that $int(g)u_{S'} : H_{S'} \to G_{S'}$ factors through $K_{S'}$. Then this functor is formally smooth over S.*

The proof is analogous to that of 2.2, invoking IX 3.6 and X 2.1 (the latter in the case K of multiplicative type).

3. Auxiliary representability results

Proposition 3.1. *Let $F \to S$ be a functor over the prescheme S. The following conditions are equivalent:*

(i) F is representable, and $F \to S$ is an open immersion (one also says simply that $F \to S$ is an open immersion).

(ii) F is a sheaf for the faithfully flat quasi-compact topology, "commutes with inductive limits of rings", $F \to S$ is a monomorphism, and finally the following condition is satisfied: for every local prescheme S' over S, of residue field k, and every S-morphism $\mathrm{Spec}(k) = S'' \to F$*, there exists an S-morphism $S' \to S$ which extends it.*

(iii) (When S is locally noetherian.) F is a sheaf for the faithfully flat quasi-compact topology, "commutes with inductive limits of rings", "commutes with adic projective limits of local artinian rings", $F \to S$ is a monomorphism, and is infinitesimally étale (cf. 1.8).

Let us first make precise two points of terminology.

Remark 3.2. One says that a functor [1058] F over S "*commutes with inductive (understood: filtered) limits of rings*" if for every filtered projective system $(S'_i)_{i \in I}$ of S-affine schemes above an affine open of S, with rings A'_i, the natural homomorphism

[1058] *N.D.E.* contravariant.

$$(*) \quad \varinjlim F(S'_i) \to F(S') \quad (\text{where } S' = \operatorname{Spec} A',\ A' = \varinjlim A'_i)$$

is bijective. One will note that S' is none other than the projective limit of (S'_i) in the category of preschemes (and even of all ringed spaces), so that the condition envisaged is in the nature of a right-exactness condition (commutation with certain inductive limits in $(Sch)^\circ/S$), just as the condition of being a sheaf for some topology. One will pay attention to the fact that the condition envisaged is essentially relative, i.e. involves the morphism $F \to S$ and not only the functor $F : (Sch)^\circ \to (Ens)$; more precisely, in $(*)$, $F(S'_i)$ and $F(S')$ denote $\mathrm{Hom}_S(S'_i, F)$, $\mathrm{Hom}_S(S', F)$. Thus, when F is representable, the condition envisaged means that F is locally of finite presentation over S. (And we have used several times, in the last two Exposés, the fact that a functor represented by an S-prescheme locally of finite presentation commutes with inductive limits of rings.)

Remark 3.3. One says that a functor [1059] F over S *commutes with adic projective limits of local artinian rings* if for every S' over S which is the spectrum of a complete noetherian local ring A', setting $S'_n = \operatorname{Spec}(A'/rad(A')^{n+1})$, the natural map

$$(**) F(S') \longrightarrow \varprojlim F(S'_n)$$

is bijective. One will note that this condition, which is in the nature of a left-exactness condition, is satisfied whenever F is representable. One sees easily that, contrary to the condition of commutation with inductive limits of rings, it is intrinsic to F as an element of $Ob(\hat{S}ch)$, i.e. it does not involve the morphism $F \to S$.

Remark 3.4. Let F be a functor over S which is a sheaf for the Zariski topology, or as one also says, which is "of local nature". (It suffices for this that F be a sheaf for a finer topology, such as the faithfully flat quasi-compact topology.) Let (S_i) be a covering of S by opens; then one easily verifies (by a method of gluing pieces) that F is representable if and only if the $F_i = F \times_S S_i$ are, which allows for example reduction to the case where S is affine. Suppose that the functor F of local nature commutes with inductive limits of rings. Then, for F to be representable, it is necessary and sufficient that its restriction to the category of preschemes locally of finite presentation over S be representable. The "necessary" was pointed out in 3.2; the "sufficient" amounts to this: if X is a prescheme locally of finite presentation over S and $X \to F$ a morphism such that, for every S' locally of finite presentation over S, the induced morphism

$$\mathrm{Hom}_S(S', X) \to \mathrm{Hom}_S(S', F)$$

is bijective, then $X \to F$ is an isomorphism. Now this results easily from the fact that X and F are two functors of local nature which commute with inductive limits of rings.

Let us now prove 3.1. The implications (i) ⇒ (ii) and (i) ⇒ (iii) are evident; let us prove the inverse implications.

[1059] *N.D.E.* contravariant.

One has (ii) ⇒ (i). Let U indeed be the set of $s \in S$ such that the canonical monomorphism $\mathrm{Spec}(\kappa(s)) \to S$ factors through F. By virtue of the last condition (ii), for every $s \in U$, the canonical monomorphism $\mathrm{Spec}(O_{S,s}) \to U$ factors through F. Noting that $O_{S,s}$ is the inductive limit of the rings of affine neighborhoods of s, it follows from the fact that F commutes with inductive limits of rings that for every $s \in S$, there exists an open neighborhood U_s such that the canonical immersion $U_s \to S$ factors through F. This implies $U_s \subset U$, so U is open. As $F \to S$ is a monomorphism, and F is of local nature, the S-morphisms $U_s \to F$ glue on the intersections $U_s \cap U_{s'}$ ($s, s' \in U$), hence come from an S-morphism $U \to F$. It remains to prove that this is an isomorphism, hence that every S-morphism $S' \to F$ factors uniquely through U (where S' is an S-prescheme). As $F \to S$ and $U \to S$ are monomorphisms, it amounts to the same to say that the structural morphism $S' \to S$ factors through U, which reduces us to the case where S' is the spectrum of a field, hence reduced to a single point s'. Let s be the point of S below s'; I say that the S-morphism $S' \to F$ factors through $\mathrm{Spec}(\kappa(s)) = S_0 \to F$ (which implies $s \in U$ and will prove what we want).

It amounts to the same, since $S' \to S_0$ is covering for fpqc and F is a sheaf for this topology, that the two composites

$$S'' = S' \times_{S_0} S' \rightrightarrows S' \to F$$

are the same, which follows from the fact that $F \to S$ is a monomorphism.

One has (iii) ⇒ (ii) (when S is locally noetherian). It suffices to prove the last condition of (ii), and moreover (by virtue of the preceding proof) it suffices to do so when S' is of the form $\mathrm{Spec}(O_{S,s})$, with $s \in S$. Let $A = O_{S,s}$, $A_n = A/m^{n+1}$, $S_n = \mathrm{Spec}(A_n)$. Then it follows from the hypothesis that $F \to S$ is infinitesimally smooth, that the given morphism $S_0 \to F$ extends to morphisms $S_n \to F$. As $F \to S$ is a monomorphism, one thus obtains an element of $\varprojlim F(S_n)$, and since F commutes with adic projective limits of local artinian rings, the $S_n \to F$ come from a morphism $\mathrm{Spec}(\hat{A}) = \hat{S}' \to F$. As $F \to S$ is a monomorphism, F a sheaf for fpqc, and $\hat{S}' \to S'$ covering for the said topology, this morphism $\hat{S}' \to F$ factors through $S' \to F$, which completes the proof.

Proposition 3.5. *Let S be a locally noetherian prescheme, $F \to S$ a functor over S, $(X_i, u_i)_{i \in I}$ a family of S-morphisms $u_i : X_i \to F$, where the X_i are preschemes locally of finite type over S. Suppose the following conditions are satisfied:*

a) F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, commutes with adic projective limits of local artinian rings.

b) The $u_i : X_i \to F$ are monomorphisms, and are infinitesimally étale (cf. 1.8).

c) The family of the u_i is "set-theoretically surjective".

Under these conditions, F is representable by a prescheme locally of finite type over S (and the u_i are open immersions, which make the family of the X_i into an open covering of F).

Remark 3.6. Proceeding as at the end of Remark 1.7, one defines, for every functor $F : (Sch)^\circ \to (Ens)$, an "underlying set" $set(F)$ as a quotient set of the set of points of F with values in fields (for the equivalence relation made precise in 1.7). When F is representable by X, one recovers the underlying set of X. Evidently $set(F)$ depends functorially on F, so that if $G \to F$ is a morphism of functors, one will say that this morphism is *set-theoretically surjective* if the induced map $set(G) \to set(F)$ is surjective. This means therefore also that every point of F with values in a field k "comes from" a point of G with values in a suitable extension of k. This definition extends immediately to the case of a family of morphisms $G_i \to F$, which makes precise the meaning of c).

Let us prove 3.5. For this, let us introduce, for $(i, j) \in I \times I$,

$X_{i,j} = X_i \times_F X_j$,

and consider the projections

$v_{i,j} : X_{i,j} \to X_i$ and $w_{i,j} : X_{i,j} \to X_j$.

I say that these last are representable by open immersions. To see it, one applies criterion 3.1 (iii): $X_{i,j}$ satisfies the three exactness conditions (being a sheaf, commuting with $\varinjlim$ of rings and with adic $\varprojlim$ of local artinian rings), since F, X_i, X_j satisfy them, and these conditions are stable under finite projective limits, in particular under fibered products; since $X_i \to F$ is a monomorphism, so is $v_{i,j} : X_{i,j} \to X_i$ deduced from it by base change $X_j \to F$, and symmetrically $v_{j,i}$ is a monomorphism; finally the "infinitesimally étale" condition is also preserved by base change. This proves that one is under the conditions of 3.1 (iii).

We may now use the X_i, $X_{i,j}$, $v_{i,j}$ and $w_{i,j}$ to construct in the usual way an S-prescheme X such that the X_i are identified with opens of X, the $X_{i,j}$ with the intersections $X_i \cap X_j$, and the $v_{i,j}$, $w_{i,j}$ with the canonical immersions. Note that X is also the quotient of

$$Y = \coprod_i X_i$$

by the equivalence relation $R = \coprod_{i,j} X_{i,j}$ (the two projections $v, w : R \rightrightarrows Y$ being defined by the $v_{i,j}$ resp. the $w_{i,j}$). More precisely, F being a sheaf for fpqc, the $u_i : X_i \to F$ come from a $u : Y \to F$, and R is none other than the equivalence relation defined by u, $R = Y \times_F Y$; finally the quotient $X = Y/R$ is also a quotient in the category of fpqc-sheaves (and even, in the category of sheaves for the Zariski topology): it suffices to use the definitions of "quotient" and "sheaf" to convince oneself. Consequently, u factors uniquely through a morphism

$$\bar{u} : X \longrightarrow F,$$

and this morphism is a monomorphism. It remains to show that it is an isomorphism. As F is of local nature, one may suppose S affine, and as moreover F commutes with inductive limits of rings, it suffices to verify that for every T affine of finite type over S, every morphism $T \to F$ factors through X (cf. 3.4). For this, consider $G = X \times_F T \to T$. It suffices to prove that this is an isomorphism. Now, T

being noetherian, one sees as above that this is an open immersion (N.B. $X \to F$ is infinitesimally étale, as follows at once from the fact that the induced morphisms $u_i : X_i \to F$ are). Now by hypothesis $X \to F$ is set-theoretically surjective, and one sees at once that this is a condition stable under base change, hence $G \to T$ is set-theoretically surjective, hence an isomorphism since it is an open immersion. QED.

Proposition 3.7. *Let S be a locally noetherian prescheme, I a filtered increasing index set, $(T_i)_{i \in I}$ a projective system of S-preschemes locally of finite type, $T = \varprojlim T_i$ the projective limit functor, F a functor over S, $u : F \to T$ an S-morphism. Suppose the following conditions are satisfied:*

a) F is a sheaf for the faithfully flat quasi-compact topology, commutes with inductive limits of rings, and with adic projective limits of local artinian rings.

b) The morphism $u : F \to T$ is a monomorphism.

b′) The morphism $u : F \to T$ is infinitesimally étale.

c) For every point ξ of F with values in the spectrum of a field k, denoting by $\xi_i \in T_i(\mathrm{Spec}(k))$ its image and by t_i the corresponding element of T_i, there exists an $i \in I$ such that for $j \geqslant i$ the transition morphism $p_{i,j} : T_j \to T_i$ is étale at t_j.

d) For every prescheme X locally of finite type over S, and every S-morphism $X \to F$, the set of $x \in X$ at which this morphism is infinitesimally étale is open.

Under these conditions, F is representable by a prescheme locally of finite type over S.

Let us note immediately that in the case which will occupy us in the following N°, one will verify conditions c) and d) through the following corollary:

Corollary 3.8. *Granting a), b), b′), conditions c) and d) are implied by the following:*

c′) The T_i are smooth over S, and the transition morphisms $p_{i,j} : T_j \to T_i$ are smooth.

d′) For every point ξ of F with values in the spectrum of a field k, let $t_i(\xi)$ be the element of T_i defined by ξ, $d_i(\xi)$ the relative dimension of T_i over S at $t_i(\xi)$, and $d(\xi) = \sup d_i(\xi)$. Then:

1°) for every ξ as above, one has $d(\xi) < +\infty$, and

2°) for every prescheme X locally of finite type over S and every S-monomorphism $v : X \to F$, the function $x \mapsto d(\xi_x)$ on X is locally constant (where for $x \in X$ one denotes by ξ_x the point of F with values in $\kappa(x)$ induced by v).

Let us prove 3.7. Let us place ourselves under the conditions of c); let t be the element of $set(F)$ defined by ξ (cf. 3.6) and set

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O_t = lim_→ O_{T_i, t_i}.
```

Then using the condition stated in c), one sees easily that O_t is a noetherian local ring (EGA 0_IV 10.3.1.3). Its residue field $\kappa(t)$ is the inductive limit of the residue fields $\kappa(t_i)$, and k is an extension of it. If η denotes the spectrum of $\kappa(t)$, ξ that of k, one has a commutative diagram

$$\begin{array}{ccc} \xi & \longrightarrow & F \\ \downarrow & & \downarrow \\ \eta & \longrightarrow & T \end{array} ,$$

whose definition is evident and left to the reader. As $\xi \to \eta$ is covering for the fpqc topology, F is a sheaf for it by virtue of a), and $F \to T$ is a monomorphism by virtue of b), one sees at once that the morphism $\eta \to T$ factors (uniquely) through a morphism $\eta \to F$. Let us note now the

Lemma 3.9. *Under the conditions of 3.7, let $T_0 = \mathrm{Spec}(A)$ be a noetherian local scheme, $T_0'' = \mathrm{Spec}(k(A))$, $i : T_0'' \to T_0$ the canonical immersion, and suppose given a commutative square of morphisms*

$$\begin{array}{ccc} T_0'' & \longrightarrow & F \\ {\scriptstyle i}\downarrow & & \downarrow \\ T_0 & \longrightarrow & T \end{array} .$$

Then there exists a unique T-morphism $T_0 \to F$.

The proof is that of 3.1 (iii) ⇒ (ii) (where the fact that the S of the cited statement be representable was not used), using that F is a sheaf for the fpqc topology, commutes with adic projective limits of local artinian rings (in this case the $A/rad(A)^{n+1}$), and that $F \to T$ is a monomorphism and is infinitesimally étale.

We shall apply lemma 3.9 to the case where $T_0 = \mathrm{Spec}(O_t)$, hence $T_0'' = \mathrm{Spec}(\kappa(t))$, noting that we have just constructed $T_0'' \to F$, and that the canonical homomorphisms $O_{T_i,t_i} \to O_t$ define a projective system of morphisms $\mathrm{Spec}(O_t) \to T_i$, whence a canonical morphism $\mathrm{Spec}(O_t) \to T$. The commutativity of the corresponding square (×) is trivial (since by definition of T, it suffices to verify it with T replaced by T_i), whence a unique T-morphism

$$v' : T_0 = \mathrm{Spec}(O_t) \to F.$$

As F commutes with inductive limits of rings, this morphism factors through

$$v'_i : T'_i = \mathrm{Spec}(O_{T_i, t_i}) \to F$$

for i large. For such an i, one has

$$T_0 \simeq T'_i \qquad \text{i.e.} \qquad O_{T_i, t_i} \simeq O_t.$$

Indeed, as $T_0 \to T_i'$ is faithfully flat quasi-compact, hence an effective epimorphism, it suffices to see that it is a monomorphism. Now if one has two morphisms $R \rightrightarrows T_0$ (with R representable) having the same composite with $T_0 \to T_i'$, they have the same composite with $T_0 \to T$ (which factors through $T_0 \to T_i'$), hence the same composite with $T_0 \to T_j$ for every j, hence with $T_0 \to T_j'$ for every j, hence are equal because T_0 is the projective limit of the T_j' in the category (Sch).

Consider the diagram

$$\begin{tikzcd} T_0 \arrow[r, "v'"] \arrow[d] & F \\ T'_i \arrow[r] \arrow[ur, "v'_i"] & T_i \end{tikzcd},$$

where the unspecified arrows are the obvious ones. The square is commutative and so is the upper triangle, hence as $T_0 \to T'_i$ is an epimorphism, it follows that the lower triangle is commutative. Now O_{T_i,t_i} is the inductive limit of the affine rings of affine open neighborhoods of t_i in T_i, hence as F commutes with inductive limits of rings, v'_i comes from a morphism

$$v_t : U_t \longrightarrow F$$

from an open neighborhood U_t of t_i in T_i. One sees at once that, restricting this neighborhood if necessary, v_t is necessarily a T_i-morphism (T_i being locally of finite type over S, hence also commuting with inductive limits of rings). Then the composite of v_t with $F \to T_i$ is a monomorphism, hence v_t is a monomorphism. I say that it is infinitesimally étale at t_i. Since $F \to T$ is infinitesimally étale, it amounts to the same to say that the composite $T_i \to F \to T$ is infinitesimally étale at t_i, or again that the induced morphism $T'_i \to T$ is infinitesimally étale, or finally (since $T_0 \xrightarrow{\sim} T'_i$) that $T_0 \to T$ is infinitesimally étale, which is immediate, this morphism being the projective limit of the infinitesimally étale morphisms $T'_i \to T_i$.

Let us now apply condition d), which had not yet been used; it follows that v_t is infinitesimally étale in a neighborhood of t, hence, restricting U_t to a smaller open if necessary, one may suppose that v_t is a monomorphism infinitesimally étale.

For $t \in set(F)$ variable, the family of morphisms v_t is amenable to 3.5, which implies the conclusion of 3.7.

Let us prove now 3.8, supposing satisfied conditions c′) and d′) of 3.8. Then with the notation of d′), one will have $d_i(\xi)$ = constant for i large, hence the relative dimension of T_j over T_i at t_j, equal to $d_j(\xi) - d_i(\xi)$, is zero, hence $T_j \to T_i$ is étale at t_j, which proves condition c). Consequently the preceding proof applies to give, for each $t \in set(F)$, an index i, an open neighborhood U_t of t_i, and a morphism $U_t \to F$ which is a monomorphism, infinitesimally étale at t_i, and all reduces to proving that this morphism is infinitesimally étale in a neighborhood of t_i. Now with the notation of d′) 2°), where one takes $X = U_t$, one may suppose, replacing U_t by the connected component of t_i in U_t if necessary, that for every $x \in U_t$, one has

$$d(\xi_x) = d(\xi_{t_i}) \text{ for every } x \in U_t.$$

Now $d(\xi_{t_i})$ = relative dimension of T_i over S at t_i, and as the relative dimension of T_i over S remains constant on U_t, one will have also, for every $x \in U_t$:

$$(*) \quad d(\xi_x) = \text{relative dimension of } U_t \text{ over } S \text{ at } x.$$

On the other hand, with the notation of the proof of 3.7, one sees at once that for every local noetherian prescheme $R_0 = \mathrm{Spec}(A)$, setting $R_0'' = \mathrm{Spec}\, k(A)$, every morphism $R_0 \to F$ such that the induced morphism $R_0'' \to F$ factors through F (i.e. sends the closed point of R_0 to $t \in set(F)$) factors (uniquely in an obvious way) through T_0. (The proof is that of 3.1 or 3.9: one uses that $T_0 \to F$ is a monomorphism infinitesimally étale at t, and that T_0 is a sheaf for fpqc commuting with adic $\varprojlim$ ...). Applying this result to the morphism $\mathrm{Spec}(O_{U_t,x}) = R_0 \to F$ induced by v_t and to the point $t_x \in set(F)$ image of the closed point of R_0, i.e. image of x by v_t, one finds a factorization

$\mathrm{Spec}(O_{U_t, x}) \to \mathrm{Spec}(O_{t_x}) \to F$

and as the second arrow is infinitesimally étale at t_x, to prove that the composite is so at x, it suffices to prove that the first is so at t_x. Now thanks to formula (×) above, it is an S-homomorphism of localizations, at two points x, t_x, of smooth S-preschemes U_t, U_{t_x} of the same relative dimension d over S at t, t_x. This morphism is induced by an S-morphism

$$w : U \longrightarrow V$$

where U is an open neighborhood of x in U_t, and $V = U_{t_x}$ (EGA I 6.5.1). It all reduces to proving that this morphism is étale at x. Moreover, U and V are equipped with monomorphisms $U \to F$, $V \to F$, and one sees at once that, restricting U further if necessary, w is an F-morphism, hence w is a monomorphism. It now suffices to prove the

Lemma 3.10. *Let U, V be two smooth S-preschemes, of the same relative dimension d over S, and $w : U \to V$ an S-morphism which is a monomorphism; then w is an open immersion (and a fortiori is étale).*

By virtue of SGA 1, I 5.7, one is reduced to the case where S is the spectrum of a field, which one may suppose algebraically closed. By virtue of SGA 1, I 5.1, it suffices to prove that w is étale, and it suffices to prove this at the closed points of U. Let x be such a point, $y = w(x)$; then, taking a regular system of parameters $f_1, \cdots, f_d$ of $O_{V,y}$, one sees that $A = O_{U,x}/\sum f_i O_{U,x}$ is the trivial extension of $k = O_{V,y}/\sum f_i O_{V,y}$ (for w being a monomorphism, so is the structural morphism $\mathrm{Spec}(A) \to \mathrm{Spec}(k)$ deduced from it by base change). As $O_{U,x}$ is a regular local ring of dimension d, it follows that the f_i form a regular system of parameters of this ring, which immediately implies that w is étale at x and completes the proof of 3.10, hence of 3.8.

Corollary 3.11. *Under the conditions of 3.8, for every quasi-compact open U of F separated over S, there exists an $i \in I$ such that for every $j \geqslant i$ the morphism $u_j|U : U \to T_j$ is an open immersion. In particular, if the T_i are quasi-affine over S, then every open U of F quasi-compact over S, i.e. of finite type over S, is quasi-affine over S.*

The proof of 3.7 shows that for every $t \in F$, there exists an $i \in I$ such that u_i :

$F \to T_i$ is a local isomorphism at t, and then u_j is a local isomorphism at x for every $j \geqslant i$. By reason of quasi-compactness, one may choose i independent of $x \in U$. It remains to prove that for i large, $u_i|U : U \to T_i$ is a monomorphism. Now as $U \to T$ is a monomorphism, one sees that the intersection of the equivalence relations $U \times_{T_i} U \subset U \times_S U$ is reduced to the diagonal, and as $U \times_S U$ is a noetherian prescheme and the $U \times_{T_i} U$ closed subpreschemes, it follows that one of these $U \times_{T_i} U$ is already reduced to the diagonal, i.e. $u_i|U$ is a monomorphism. This proves the first assertion in 3.11, and the second is an immediate consequence of it.

Proposition 3.12. *Let S be a prescheme, G an affine S-prescheme in groups.*

a) Let F be the functor $(Sch)^{\circ}/S \to (Ens)$ such that, for every T over S,

```
F(T) = set of multiplicative-type subgroups of G_T which are finite over T.
```

Suppose S locally noetherian or G of finite presentation over S. Then the functor F is representable and is affine over S. If G is of finite presentation over S, then F is locally of finite presentation over S.

b) Let H be an S-prescheme in groups of multiplicative type, and finite over S. Then $\mathrm{Hom}_{S-gr.}(H, G)$ *is representable. It is affine over S, and if G is of finite type (resp. of finite presentation) over S, the same holds for* $\mathrm{Hom}_{S-gr.}(H, G)$.

Remark 3.13. Except for the precision that $\mathrm{Hom}_{S-gr.}$ is affine, and in the case where G is of finite presentation over S (which will suffice for us), 3.12 is an immediate consequence of the theory of Hilbert schemes (A. Grothendieck, *Techniques de construction et théorèmes d'existence en Géométrie Algébrique: IV Les Schémas de Hilbert*, Séminaire Bourbaki, May 1961, N° 221). It even suffices that G be quasi-projective over S; in case a), one may also represent the larger functor

```
F′(T) = { set of subpreschemes in groups of G_T,
          flat, proper and of finite presentation over T }
```

(the canonical monomorphism $F \to F'$ is an "open immersion", as follows from criterion 3.1 and from X 4.7 b), so that the representability of F' entails that F is representable by an open of F'); in case b), one may restrict to supposing that H is projective and of finite presentation over S. In both cases, one obtains a functor locally of finite presentation over S. In the present Exposé, 3.12 is only a technical lemma for proving a key result in the following N°, so we shall sketch an easy direct proof of 3.12, not using Hilbert schemes.

Let us first prove b). It will suffice to prove that $\mathrm{Hom}_S(H, G)$ is representable (independently of any group structure on H, G), and has the supplementary properties stated for $\mathrm{Hom}_{S-gr.}(H, G)$, since using also the same result for $\mathrm{Hom}_S(H \times_S H, G)$, one makes explicit the subfunctor $\mathrm{Hom}_{S-gr.}(H, G)$ of $\mathrm{Hom}_S(H, G)$ by a finite projective limit (in fact, with the help of fibered products) involving G, $\mathrm{Hom}_S(H, G)$ and $\mathrm{Hom}_S(H \times_S H, G)$, which we leave to the reader to make explicit. On the other hand, one will have

$$H = \mathrm{Spec}(B),$$

where B is a sheaf of algebras over S which is locally free as a sheaf of modules (this is the only hypothesis on H that we shall need to retain). As the representability question envisaged is local on S, we may suppose S affine with ring A. On the other hand, one will have $G = \mathrm{Spec}(C)$, where C is an A-algebra. When $G = S[t] = G_0$, t an indeterminate, the functor `Hom` is none other than

$$T \mapsto \Gamma(T, B_T),$$

which is representable (B being locally free) by the vector bundle $V(B^\vee)$, where $B^\vee$ is the sheaf of modules dual to B. When $G = S[(t_i)]$, with (t_i) a (not necessarily finite) family of indeterminates, one will have therefore $G = G_0^I$ (product over S of a family of copies of `G_0`), which is representable by the affine scheme

`Hom_S(H, G)^I = V(B^v)^I = V(B^{v(I)}).`

In the general case, G will be isomorphic to a closed subscheme of a scheme of the form $S[(t_i)]$, i.e. C will be a quotient of an A-algebra of the form $A[(t_i)]$. Let (F_j) be a system of generators of the ideal by which one divides. Suppose B free of rank n, which is permissible by covering S by smaller affine opens. Choose a basis $(e_k)_{1 \leqslant k \leqslant n}$ of B; then writing the n components in this basis of $F_j((x_i))$, for $x_i = \sum x_{ik} e_k$ (x_{ik} indeterminate coefficients, taken in an unspecified algebra A' over A), one finds, for each F_j, n polynomials $F_{j,k}$ in the $(x_{i,k})_{i \in I, 1 \leqslant k \leqslant n}$, with coefficients in A. One verifies at once that $\mathrm{Hom}_S(H, G)$ is represented by the spectrum of the quotient of the polynomial ring $A[(x_{i,k})]$ by the ideal generated by the $F_{j,k}$. This proves at once b).

Let us prove a). For every ordinary finite commutative group M, let `F_M` be the subfunctor of F obtained by restricting to subgroups of `G_T` which are of multiplicative type and of type M. One sees easily, by a gluing argument like the one which served in 3.5 (which we should have stated by unscrewing a bit more!), that it suffices to verify that the `F_M` are representable; then F will be representable by the prescheme sum of the `F_M`, where M runs over the set of classes of finite commutative groups up to isomorphism. (F is in fact the sum of the `F_M` in the category of sheaves...).

From now on we suppose M fixed, and we shall write F instead of `F_M`. Let $H = D_S(M)$; consider the subfunctor $F' = Imm_{S-gr.}(H, G)$ of $\mathrm{Hom}_{S-gr.}(H, G)$ whose value at T is the set of homomorphisms of T-groups $H_T \to G_T$ which are closed immersions. We know already that $\mathrm{Hom}_{S-gr.}(H, G)$ is representable by an affine S-prescheme by virtue of b), and using IX 6.8, one sees at once that F' is representable by an open and closed subprescheme of the latter, hence it is also affine over S. Consider finally the canonical morphism $F' \to F$, which associates to each monomorphism $H_T \to G_T$ the image group. One verifies at once, by virtue of the definitions, that in this way F' becomes a principal homogeneous bundle (in the category of fpqc sheaves) over F, with group $\Gamma_F = \Gamma_S \times_S F$, where $\Gamma = \mathrm{Aut}_{gr.}(M)$, whence it follows "by descent" that this morphism is representable (i.e. for every morphism $T \to F$, with T representable, $T \times_F F'$ over T is representable — in fact, representable by a principal galois bundle under Γ). Hence by virtue of IV 4.6.6, F is representable if and only if the quotient F'/F'', where $F'' = F' \times_F F'$, exists in (Sch) and is universally effective for faithfully flat quasi-compact morphisms, or what amounts to the same, if and only if the quotient F'/Γ exists and is universally effective for the said morphisms. Now as F' is affine over S, one has seen in V 4.1 that the condition in question is indeed satisfied. This proves the representability of F in a).

As for the complement, relative to the case where one supposes G of finite presentation over S, it follows at once from the preceding proof, taking into account that by virtue of b), F' is then locally of finite presentation over S, so that one may apply Exposé V[1060].

4. The scheme of multiplicative-type subgroups of a smooth affine group

The principal result of the present Exposé is constituted by

Theorem 4.1. *Let S be a prescheme, G an S-prescheme in groups smooth and affine over S, F the functor $(Sch)^{\circ}/S \to (Ens)$ defined by*

F(T) = set of multiplicative-type subgroups of G_T.

Then the functor F is representable, and is smooth and separated over S.

Let us signal at once the following variant:

Corollary 4.2. *Let G, H be two S-preschemes in groups, with G smooth and affine over S, H of multiplicative type and of finite type. Then* $\mathrm{Hom}_{S-gr.}(H,G)$ *is representable, and is smooth and separated over S.*

Indeed, when H is smooth over S, so is $H \times_S G$, and one may apply 4.1 to this latter. By the consideration of graph subgroups, $\mathrm{Hom}_{S-gr.}(H,G)$ becomes a subfunctor F' of the functor F of "multiplicative-type subgroups of $H \times_S G$", namely $F'(T)$ = set of multiplicative-type subgroups K of $(H \times_S G)_T = H_T \times_T G_T$ such that the homomorphism induced by the first projection

$$K \longrightarrow H_T$$

is an isomorphism. By virtue of IX 2.9, the canonical morphism $F' \to F$ is an open immersion, hence F being representable, so is F', which will be representable by an open of F; and F being separated and smooth over S, so will F' be. In the case where H is finite over S, it suffices to apply 3.12 b) for the representability of $\mathrm{Hom}_{S-gr.}(H,G)$. When H is a product $H_1 \times_S H_2$, with H_1 smooth over S and H_2 finite over S, then $\mathrm{Hom}_{T-gr.}(H_T,G_T)$ is identified with the subset of $\mathrm{Hom}_{T-gr.}((H_1)_T,G_T) \times \mathrm{Hom}_{T-gr.}((H_2)_T,G_T)$ formed of pairs (u_1,u_2) such that $u_1 \cdot u_2 = u_2 \cdot u_1$, whence it follows that $\mathrm{Hom}_{S-gr.}(H,G)$ is representable by a closed

[1060] *N.D.E.* Reference to verify/make precise.

subprescheme of $\mathrm{Hom}_{S-gr.}(H_1, G) \times_S \mathrm{Hom}_{S-gr.}(H_2, G) = X$, as one sees by applying VIII 6.5 b), where one takes $Y = H$, $Z = G$, q_1 and q_2 being defined respectively by $(u_1, u_2) \mapsto u_1 \cdot u_2$ and $(u_1, u_2) \mapsto u_2 \cdot u_1$.

In the general case, the question being local on S for the Zariski topology, one may suppose that S is affine, and H of constant type over S. Then H being quasi-isotrivial (X 4.5), one can find an étale surjective morphism $S' \to S$, S' affine, that splits H, i.e. such that $H' = H_{S'}$ is diagonalizable. Then the preceding result applies, since a diagonalizable group is the product of a diagonalizable torus by a diagonalizable finite group. It remains to see that the descent datum obtained on the S'-prescheme $\mathrm{Hom}_{S'-gr.}(H', G') = X'$ is effective. This is seen by the reasoning of X 5.4, noting that in loc. cit., the hypothesis that $X' \to S'$ was separated and locally quasi-finite served only to ensure that every open part of X' quasi-compact over S' was quasi-affine over S'. Now this property is still verified in the present case, as follows easily from the fact that this is so for the functor F of 4.1 (which will be seen in the course of the proof of 4.1). This (or any other variant of this little unscrewing) establishes the representability of $\mathrm{Hom}_{S-gr.}(H, G)$, and at the same time the fact that it is separated over S. It is locally of finite presentation over S, as one sees for example (as was pointed out in 3.2) thanks to the fact that this functor "commutes with inductive limits of rings". Finally, this functor being formally smooth over S (by virtue of 2.1), it is smooth over S.

Remark 4.3. We have here deduced 4.2 from 4.1, which is really immediate only when H is also smooth over S. For the deduction to be made without contortions for the general case, the representability result 4.1 would have to be established without supposing G smooth over S, but only affine of finite presentation over S. (Of course, then F will no longer be smooth in general over S!) There is little doubt that 4.1 remains true under these more general hypotheses, but the proof appears to have to be more delicate (failing the possibility of invoking 3.8)[1061][1062]. Let us point out however that when G is a closed subgroup of an affine and smooth group `G_0` over S, then the functor F representing the multiplicative-type subgroups of G is representable by a closed subprescheme of the prescheme representing the analogous functor `F_0` for `G_0` (amenable to 4.1), as one sees easily by applying VIII 6.4. This raises also the question: is an affine S-group scheme G, which is affine and of finite presentation over S, isomorphic to a subgroup scheme of some $GL(n)_S$, n suitable? This is true when S is the spectrum of a field, cf. VI_B 11.11, but unfortunately false in general, even for tori, cf. 4.6. Finally, note that one could also prove 4.2 directly by exactly the same method as 4.1.

Let us now prove 4.1. Since the functor F is evidently of local nature, one may suppose S affine, so that $S = \mathrm{Spec}(A)$, A a ring. Considering A as the inductive limit of its subrings of finite type over $\mathbb{Z}$, and noting that G comes from a smooth

[1061] It is in fact proved for G flat and quasi-affine over S with connected fibers, provided one restricts to the central subtori of G (XV 8.8).

[1062] *N.D.E.* In the case H smooth (not necessarily affine) over a normal locally noetherian base S, M. Raynaud has shown that the largest representable open of the functor of subtori of H is a disjoint sum of smooth and affine opens over S. This is theorem IX.9.26 in *Faisceaux amples sur les schémas en groupes et sur les espaces homogènes*, Lecture Notes Maths. 119 (1970).

and affine group over such a subring (EGA IV 8), one is reduced to the case where S is noetherian. For every integer $n > 0$, let T_n be the functor defined as F, but restricting to subgroups H of multiplicative type of G_T such that $n \cdot id_H = 0$, i.e. such that ${}_nH = H$. Order the set I of integers > 0 by the divisibility relation. When m is a multiple of n, define

$$p_{n,m} : T_m \longrightarrow T_n$$

by $p_{n,m}(H) = {}_nH = \mathrm{Ker}(n \cdot id_H)$. In this way the T_n form a projective system of

functors over S. By virtue of 3.12 a), the functors T_n are representable, affine and of finite type over S. Define likewise morphisms

$$u_n : F \longrightarrow T_n,$$

by the relation $u_n(H) = {}_nH$. In this way, one obtains a morphism

$u : F \to T = \varprojlim T_n$,

where the $\varprojlim$ is taken in the category of functors over S. But let us point out at once that, the T_n being representable and affine over S, so is T (it will be the spectrum of the inductive limit of the quasi-coherent algebras on S which define the T_n). Of course, T is not in general of finite type over S.

We are going to apply 3.7 and are reduced to verifying conditions a) to d) of 3.7, which will imply that F is representable by a prescheme locally of finite type over S. It then follows from 2.1 bis that F is even smooth over S, and as F is a subfunctor of T which is affine over S, it follows that F is separated over S (being separated over T, which is separated over S). Let us prove at once the complement invoked above, namely that every open U of F quasi-compact over S is quasi-affine over S: this follows from 3.11 and from the fact that the T_n are affine over S.

Let us verify therefore the conditions of 3.7.

a) F is a sheaf for the faithfully flat quasi-compact topology, by descent theory SGA 1, VIII, which applies here since multiplicative-type groups over T are affine over S. It commutes with inductive limits of rings by the general carpet [1063] EGA IV 8. Let us show that it commutes with adic projective limits of local artinian rings. When dealing with, instead of the functor F, the analogous functor envisaged in 4.2, this property is none other than that of IX 7.1 in the special case where A is a complete noetherian local ring, equipped with an ideal of definition for its usual topology (N.B. This is exactly where the hypothesis G affine intervenes in an essential way). In the present case, we are reduced to proving the

[1063] the French *tapis* is the standard SGA idiom for an underlying body of foundational results that one rolls out; "the general machinery of EGA IV §8" would be an equivalent rendering.

Lemma 4.4. *Let A be a complete noetherian local ring, with maximal ideal m, G a group scheme affine over $S = \mathrm{Spec}(A)$. For every integer $n \geqslant 0$, let $S_n = \mathrm{Spec}(A/m^{n+1})$, $G_n = G \times_S S_n$. Let for every n, H_n be a subgroup of multiplicative type and of finite type of G_n, such that for $m \geqslant n$, H_n is deduced from H_m by reduction. Under these conditions, there exists a unique multiplicative-type subgroup H in G which reduces along the H_n.*

By virtue of X 3.2 there exists a group of multiplicative type H over S, necessarily of finite type and isotrivial, determined up to unique isomorphism, equipped with an isomorphism $H \times_S S_0 \simeq H_0$ (using the fact that H_0 is isotrivial, being of finite type over a field, cf. X 1.4). By virtue of X 2.1, for every n, the isomorphism $H \times_S S_0 \simeq H_0$ lifts to a unique isomorphism $H \times_S S_n \simeq H_n$. Having said this, by virtue of IX 7.1 already cited, the homomorphisms $H_n \to G_n$ come from a unique homomorphism of S-groups $u : H \to G$. By virtue of IX 6.6, this last is a monomorphism since $u_0 : H_0 \to G_0$ is one. This completes the proof of 4.4.

b) The morphism $u : F \to T$ is a monomorphism. This follows from the density theorem IX 4.7, in the form of corollary 4.8 b). One will pay attention to the fact that it is essential, for the application we make of it here, to dispose of this result over a not necessarily noetherian base. (N.B. As the functor T does not commute with inductive limits of rings, it is not possible to reduce to that a priori.)

b′) The morphism $u : F \to T$ is infinitesimally étale; in other terms one has the

Lemma 4.5. *Let A be an artinian local ring of residue field k, $S = \mathrm{Spec}(A)$, I an ideal $\subset rad(A)$, $S_0 = \mathrm{Spec}(A/I)$, G an S-prescheme in groups smooth over S, $G_0 = G \times_S S_0$; let us give ourselves a multiplicative-type subgroup H_0 of G_0 and for every integer $n > 0$, a multiplicative-type subgroup $H(n)$ of G such that*

1°) for every multiple m of n, $H(n) = {}_nH(m)$, and

2°) $H(n)_0 = {}_nH_0$.

Under these conditions, there exists a multiplicative-type subgroup H of G and one only such that ${}_nH = H(n)$ for every n.

Uniqueness is already contained in b). For existence, an immediate recurrence reduces us to the case where $Jm = 0$, m being the maximal ideal of A. Let $k = A/m$, $S_0 = \mathrm{Spec}(k)$, $G_0 = G \times_S S_0$, g_0 the Lie algebra of G_0, h_0 that of H_0. One has a canonical isomorphism of groups:

g_0 ⊗_k J ≃ Ker(G(S) □ G(S_0)),

cf. Exp. III. By virtue of IX 3.6 bis and 3.7, there exists a multiplicative-type subgroup H of G reducing along H_0, and such an H is determined modulo inner automorphism by an element of $N = Ker(G(S) \to G(S_0))$. Thus the set P of liftings H of H_0 is a principal homogeneous set under the group N/M, where M is the group of $g \in N$ such that $int(g) \cdot H = H$.

Now one sees easily (Exp. III) that this subgroup is none other than the vector subspace of $g_0 \otimes_k J$ formed of the invariants under `H_0`, when `H_0` operates on $g_0 \otimes_k J$ by the representation induced by the adjoint representation of `G_0`. Likewise, the set $P(n)$ of liftings of $H(n)_0 = {}_nH_0$ is a principal homogeneous set under $N/M(n)$, where $M(n)$ is the vector subspace of $g_0 \otimes_k J = N$ formed of the invariants under ${}_nH_0$. Using the density theorem Exp. IX 4.7, one sees easily that for n large (in the sense of the order relation put on the set of integers $n > 0$, namely the divisibility relation) one has $M = M(n)$. Consequently, the natural map $H \mapsto {}_nH$ from P to $P(n)$, which is compatible with the operations of N and hence with the homomorphism $N/M \to N/M(n)$ on the groups of operators, is bijective for n large. The conclusion of 4.5 results from this at once.

To verify c) and d) of 3.7, we use 3.8, which reduces us to verifying c′) and d′) below.

c′) The T_n are smooth over S, and the transition morphisms $p_{n,m} : T_m \to T_n$ are smooth.

This is none other than 2.2 bis.

d′) With the notation of 3.8, a point ξ of F with values in a field k over S is none other than a multiplicative-type subgroup `H_0` of $G_0 = G_k$. Taking up again the reasoning of b′) above, one sees that the integer $d(\xi)$ envisaged in 3.8 is none other than the dimension of $g_0/g_0^{H_0}$, where g_0 is the Lie algebra of `G_0` and $g_0^{H_0}$ is the vector subspace of invariants under `H_0`. It is therefore a finite integer, i.e. condition d′) 1°) of 3.8 is verified. With the notation of d′) 2°), the datum of a morphism $v : X \to F$ amounts to the datum of a multiplicative-type subgroup H of `G_X`. For $x \in X$, the integer $d(\xi_x)$ is then none other than the dimension of $(g \otimes \kappa(x))^{H_x}$, where g is the sheaf of Lie algebras of `G_X` (which is locally free of finite type over X since `G_X` is smooth over X, and $g \otimes \kappa(x)$ is none other than the Lie algebra of the fiber G_x of `G_X` at x), and H_x is the fiber of H at x. Now g being a locally free module on S on which the group of multiplicative type H operates, one sees at once that the subfunctor g^H of invariants under H is given by a locally direct factor subsheaf, hence locally free, of g. (By descent, one is reduced to the case where H is diagonalizable, and where one applies Exp. I 4.7.3, noting that the subsheaf of invariants corresponds to the component of degree zero.) Consequently $d(\xi_x)$ = rank at x of g^H, hence it is a function locally constant in x. This completes the proof of condition d′).

We have thus verified the conditions of 3.7, which completes the proof of 4.1.

Remark 4.6. When $G = GL(n)_S$, one can give a noticeably simpler and more explicit direct proof of 4.1, by using I 4.7.3. The proof shows moreover that in this case, the modular scheme is a scheme that is a sum of a family of affine schemes over S. Proceeding as was said in 4.3, one deduces the same result whenever G is a closed subgroup of a group of the form $GL(n)_S$. One should however beware of thinking that the preschemes which represent the functors in 4.1 or 4.2 are always sums of a family of affine schemes over S. Let for example H be a multiplicative-type group of finite type over a locally noetherian prescheme S; then by virtue of X 5.11, H is isotrivial if and only if $\mathrm{Hom}_{S-gr.}(H, G_m)$ is a sum of a family of affine preschemes over S. Now one has pointed out (X 1.6) that there can exist tori H (of relative dimension 2) which are not isotrivial; for such a torus, $\mathrm{Hom}_{S-gr.}(H, G_m)$ is therefore not a sum of S-preschemes affine over S, and one sees likewise that the twisted constant group "dual" $\mathrm{Hom}_{S-gr.}(G_m, H)$ is not such a sum either (for if two twisted constant commutative groups of finite presentation R, R' are dual to each other, $R' = \mathrm{Hom}_{S-gr.}(R, \mathbb{Z}_S)$, one sees easily that one is isotrivial if and only if the other is). This last point shows also that such an H is not isomorphic to a subgroup of a group of the form $GL(n)_S$; more precisely, one can show that a multiplicative-type group of finite type over S locally noetherian connected is isomorphic to a subgroup of a group of the form $\mathrm{Aut}_{Mod}(E)$ (with E a locally free module of finite type over S) if and only if it is isotrivial. Finally, taking $G = H \times G_m$ in the two preceding examples, one finds an example where the prescheme representing the functor F of 4.1 is not a sum of a family of S-preschemes affine over S (with G a torus of relative dimension 3 if one wishes).

5. First corollaries of the representability theorem

Let H, G be as in 4.2. Set

$$M = \mathrm{Hom}_{S-gr.}(H, G),$$

which is an S-prescheme smooth, separated over S, by virtue of 4.2. Note that G operates on the functor $M = \mathrm{Hom}_{S-gr.}(H, G)$ by

$$(g, u) \mapsto \mathrm{int}(g) \circ u,$$

whence a canonical morphism

$$(\times) \qquad G \times_S M \to M \times_S M,$$

whose components are the preceding morphism $G \times_S M \to M$, and the second projection $G \times_S M \to M$.

Corollary 5.1. *The preceding morphism* $(\times)$ *is smooth.*

This follows from 4.2 and 2.3. This statement is equivalent to the following:

Corollary 5.2. *Let* $u_1, u_2 : H \rightrightarrows G$ *be two homomorphisms of* S*-groups. Then the subfunctor* $Transp(u_1, u_2)$ *of* G *(cf. 2.4) is representable by a closed subprescheme of* G*, smooth over* S*.*

It remains only to prove that $Transp(u_1, u_2) \to G$ is indeed a closed immersion, which follows from the fact that it is the kernel of a pair of morphisms $G \rightrightarrows M$ (namely $g \mapsto int(g)u_1$ and the "constant" morphism $g \mapsto u_2$), and from the fact that M is separated over S. In particular:

Corollary 5.3. *Let* $u : H \to G$ *be a morphism of* S*-groups. Then* $Centr_G(u) = Transp(u, u)$ *is representable by a closed subprescheme in groups of* G*, smooth over* S*. Moreover,* $G/Centr_G(u)$ *is representable by an open subprescheme of* M*.*

It remains to prove this last point. Now the morphism

$$g \mapsto int(g) \circ u$$

from G to M is smooth of finite type by virtue of 5.2; it is therefore an open morphism (EGA IV 6.6), and if U denotes its image, with the structure induced by M, the induced morphism $G \to U$ is smooth, surjective, of finite type, hence covering for the faithfully flat and quasi-compact topology. Moreover, it is evident that the preceding morphism $G \to M$ makes G into a formally principal homogeneous sheaf under $Centr_G(u)_M$, which implies that the sheaf $G/Centr_G(u)$ is indeed representable by U.

Corollary 5.4. *Let* u_1, u_2 : H ⇉ G *be two homomorphisms of* S*-groups. The following conditions are equivalent:*

(i) u_1 *and* u_2 *are conjugate locally for the étale topology.*

(i bis) u_1*,* u_2 *are conjugate locally in the sense of the faithfully flat quasi-compact topology.*

(ii) For every $s \in S$*, denoting by* $\bar{s}$ *the spectrum of an algebraic closure of* $\kappa(s)$*, the morphisms* u_{1, s¯}, u_{2, s¯} : H_{s¯} ⇉ G_{s¯} *are conjugate by an element of* $G(\kappa(\bar{s}))$*.*

(ii bis) The structural morphism $Transp(u_1, u_2) \to S$ *is surjective.*

(iii) $Transp(u_1, u_2)$ *is a torsor under the action of the* S*-prescheme in groups smooth of finite type* $Centr_G(u_1)$*.*

(i) ⇒ (i bis) and (ii) ⇒ (ii bis) are trivial (the second thanks to the Nullstellensatz); on the other hand (i bis) ⇒

(ii) by the "principle of finite extension" (EGA IV 9). On the other hand (ii bis) ⇒ (iii) thanks to the fact that $Transp(u_1, u_2)$ is smooth over S hence flat over S, and of finite type hence quasi-compact over S; it is therefore faithfully flat quasi-compact over S if and only if its structural morphism is surjective. As, on the other hand, it is formally principal homogeneous under $Centr_G(u_1)$, which is faithfully flat and quasi-compact over S, one sees that this last condition is equivalent also to saying that $Transp(u_1, u_2)$ is a torsor under $Centr_G(u_1)$ (understood: in the sense of the faithfully flat and quasi-compact topology). Finally (iii) ⇒ (i) thanks to the fact that $Transp(u_1, u_2)$ is smooth over S and to "Hensel's lemma" in the form 1.10.

Remark 5.5. For $u_1 = u : H \to G$ fixed, the functor $(Sch)^\circ/S \to (Ens)$ which associates to every T over S the set of homomorphisms of T-groups $u_2 : H_T \to G_T$ which are conjugate to $u_T : H_T \to G_T$ locally for the étale topology, is precisely representable by the open of M, isomorphic to $G/Centr_G(u)$, envisaged in 5.3.

Let us sketch the variants of the preceding results, obtained by application of 4.1 instead of 4.2. Let therefore G be a prescheme in groups smooth and affine over S, and let us now denote by M the S-prescheme smooth, separated over S, which represents the functor envisaged in 4.1. One has again operations of G on M:

$$(g, H) \mapsto \mathrm{int}(g)(H),$$

whence as above a morphism

$$(\times\ \mathrm{bis}) \qquad G \times_S M \to M \times_S M.$$

Corollary 5.1 bis. *The preceding morphism is smooth.*

This follows from 4.1 and 2.3 bis. One concludes again:

Corollary 5.2 bis. *Let H_1, H_2 be two multiplicative-type subgroups of G. Then the subfunctor $Transp_G(H_1, H_2)$ of G is representable by a closed subprescheme of G, smooth over S.*

In particular:

Corollary 5.3 bis. *Let H be a multiplicative-type subgroup of G. Then the subfunctor in groups $Norm_G(H)$ of G is representable by a closed subprescheme of G, smooth over S. Moreover, the quotient $G/Norm_G(H)$ is representable by an open subprescheme of M.*

Corollary 5.4 bis. *Let H_1, H_2 be two multiplicative-type subgroups of G. The following conditions are equivalent:*

(i) H_1 and H_2 are conjugate locally for the étale topology.

(i bis) H_1 and H_2 are conjugate locally for the faithfully flat quasi-compact topology.

(ii) For every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, the subgroups $H_{1,\bar{s}}, H_{2,\bar{s}}$ of $G_{\bar{s}}$ are conjugate by an element of $G(\kappa(\bar{s}))$.

(ii bis) The structural morphism $Transp(H_1, H_2) \to S$ is surjective.

(iii) $Transp(H_1, H_2)$ is a principal homogeneous bundle under the action of the S-prescheme in groups smooth of finite type $Norm_G(H_1)$.

Remark 5.5 bis. Remark 5.5 transposes likewise to the present case.

Remark 5.6. Note that to establish the result 5.2, and consequently also the first assertion in 5.3, as well as 5.4, the reference to 4.2 can be replaced by a reference to VIII, 6.4, whose proof is much easier. This shows also that the hypothesis G affine over S is unnecessary there. Moreover, in N° 6, we shall show how a variant of this method permits extending these results to the case of certain groups H more general than groups of multiplicative type. These same observations extend to the variants 5.2 bis etc. On the other hand, the result 5.8 that follows uses in an essential way all the hypotheses made (notably G affine and smooth over S, H of multiplicative type), and the lecturer knows of no other proof of it than via the representability theorems 4.1 or 4.2[1064].

5.7. Since the morphism $(\times)$ resp. $(\times bis)$ is smooth hence open, its image is open. Let R be this image, equipped with the structure induced by $M \times_S M$; one easily

[1064] The situation has changed since the writing of this text, cf. XV and XIX N° 6.

verifies that R is an equivalence relation in M, which is none other moreover than the one made explicit in 5.4 resp. 5.4 bis. It would be interesting to know whether the quotient sheaf M/R (which is formally étale over S) is representable (it is then representable by a prescheme étale over S); it is so for example when S is the spectrum of a field. One will note moreover that R may not be closed in $M \times_S M$ (even when G is quasi-finite over S...), which means (when M/R is representable) that M/R is then not separated over S.

Theorem 5.8. *Let S be a prescheme, G an S-prescheme smooth and affine over S, $s \in S$. Then:*

a) For every multiplicative-type subgroup H^0 of G_s, there exists an étale morphism $S' \to S$, a point s' of S' above s such that the residue extension $\kappa(s')/\kappa(s)$ be trivial, and a multiplicative-type subgroup H' of $G' = G \times_S S'$, such that $H'_{s'} = H^0 \otimes_{\kappa(s)} \kappa(s')$.

b) For every group homomorphism $u^0 : H_s \to G_s$, where H is an S-prescheme in groups of multiplicative type and of finite type, there exists an étale morphism $S' \to S$, a point s' of S' above s such that the residue extension $\kappa(s')/\kappa(s)$ be trivial, and a group homomorphism $u' : H' \to G'$, such that $u'_{s'} : H'_{s'} \to G'_{s'}$ be equal to $u^0 \otimes_{\kappa(s)} \kappa(s')$.

This results from 4.1 resp. 4.2, and from Hensel's lemma in the form 1.10.

Proposition 5.9. *Let S be a prescheme, G an S-prescheme smooth and affine over S, H a multiplicative-type subgroup of G. Consider $Centr_G(H) \hookrightarrow Norm_G(H)$, which by 5.3 and 5.3 bis are closed subpreschemes in groups of G, smooth over S. Then the first group is an open and closed subprescheme of the second, and the quotient sheaf*

$$W_G(H) = Norm_G(H)/Centr_G(H)$$

is representable by an open subprescheme in groups of $\mathrm{Aut}_{S-gr.}(H)$; it is therefore an S-prescheme in groups quasi-finite, étale and separated over S.

Consider indeed the evident homomorphism

$$\theta : Norm_G(H) \longrightarrow \mathrm{Aut}_{S-gr.}(H),$$

whose kernel is by definition $Centr_G(H)$. As $\mathrm{Aut}_{S-gr.}(H)$ is representable by an S-prescheme in groups étale and separated over S (X 5.10), its unit section is an open and closed immersion, hence its inverse image under θ is an open and closed subgroup of $Norm_G(H)$. I say moreover that θ is a smooth morphism: this indeed results formally from the definitions, and from the fact that $Norm_G(H)$ is smooth over S and $\mathrm{Aut}_{S-gr.}(H)$ is étale over S. One concludes as in 5.3 that the image of θ is an open U of $\mathrm{Aut}_{S-gr.}(H)$ and that, equipped with the induced structure, U represents the quotient sheaf $Norm_G(H)/Centr_G(H)$. This latter is therefore étale and separated over S since $\mathrm{Aut}_{S-gr.}(H)$ is, and it is quasi-finite over S, being quasi-compact over S as image of $Norm_G(H)$ which is. This completes the proof of 5.9.

Corollary 5.10. *For every $s \in S$, let*

$$w(s) = \mathrm{rank}\ \mathrm{Norm}_{G_s}(H_s)/\mathrm{Centr}_{G_s}(H_s)$$

(which is also the index of $Centr_{G(\bar{k})}(H(\bar{k}))$ in $Norm_{G(\bar{k})}(H(\bar{k}))$, where $\bar{k}$ is an algebraic closure of $\kappa(s)$). Then the function $s \mapsto w(s)$ is lower semicontinuous. For it to be constant in a neighborhood of the point s, it is necessary and sufficient that $W_G(H)$ be finite over S in a neighborhood of s.

Indeed, for every S-prescheme W which is étale, of finite type and separated over S, it is true that the function $s \mapsto w(s) = rank[W_s : \kappa(s)]$ is lower semicontinuous, and that it is constant in a neighborhood of the point s if and only if W is finite over S in a neighborhood of s (a fact pointed out in SGA 1, I 10.9, and whose proof, which offers no difficulty, will be found in EGA IV[1065]).

Remark 5.11.[1066] Let G be a prescheme in groups affine and smooth over S, H a multiplicative-type subgroup; then 5.3 and 5.3 bis imply that the quotients

$$G/\mathrm{Centr}_G(H) \text{ and } G/\mathrm{Norm}_G(H)$$

are preschemes smooth over S and quasi-affine over S, since in both cases, the modular prescheme M in which the quotient is embedded is such that every open U of M quasi-compact over S is quasi-affine over S (3.11). We shall see moreover in the following Exposé [1067] that for every U as above, the schematic closure $\bar{U}$ of U in M is even affine over S, which shows that the quotients envisaged are affine over S provided the open U in the corresponding modular scheme (of homomorphisms of group from H to G, resp. of multiplicative-type subgroups of G) is closed in M. This is for example the case if H is a "maximal torus" in the second case envisaged, or when S is the spectrum of a field, cf. XII.5.4[1068]. I do not know whether the quotients envisaged are affine over S in general.

6. On a rigidity property for homomorphisms of certain group schemes, and the representability of certain transporters[1069]

Proposition 6.1. *Let S be a locally noetherian prescheme, H a commutative S-prescheme in groups, of finite type over S, E a set of integers > 0, stable under multiplication, and suppose the following conditions satisfied:*

a) For every $n \in E$, the subgroup ${}_nH = Ker(n \cdot id_H)$ is finite and flat over S.

b) For every $s \in S$, the family of the ${}_nH_s$ ($n \in E$) is schematically dense in H.

[1065] EGA IV 15.5.1 and 18.10.7.

[1066] *N.D.E.* For a generalization to the non-affine case, see M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Lecture Notes Math. 119 (1970), IX.2.8.

[1067] *N.D.E.* Cf. remark XII.5.7. Let us mention here the following generalization. Without an affinity hypothesis on G, if H is a torus and S is locally noetherian, U is affine and smooth by a theorem of Raynaud (loc. cit., IX.2.6 and IX.2.8). Indeed, $G/Norm_G(H)$ is an open of M, and the largest representable open of M is a disjoint sum of opens smooth and affine over S.

[1068] *N.D.E.* Under the hypothesis that G is of locally constant reductive rank.

[1069] The present N° does not use the results of N°s 3, 4, 5; its natural place would be in VI_B.

Let Y be a closed subprescheme of H. Then the subfunctor $\prod_{H/S} Y/H$ of S (compare VIII, 6) is representable by a closed subprescheme T of S, and if S is noetherian, there exists an $n \in E$ such that

$$(\times) \qquad \prod_{H/S} Y/H = \prod_{{}_nH/S} Y \cap {}_nH/S.$$

Indeed, by virtue of VIII 6.4, as by condition a) ${}_nH$ is finite and flat and a fortiori "essentially free" over S, it follows that the second member of $(\times)$ is representable by a closed subprescheme T_n of T. Of course, for $n \in E$, with E ordered by divisibility, the T_n form a decreasing family of closed subpreschemes of S, hence if S is noetherian (which we may suppose) it is stationary for n large. Let T be the value of T_n for n large; I say that T indeed represents the first member of $(\times)$, which will complete the proof. One is reduced to proving that if S' is a prescheme over S such that $(Y \cap {}_nH)_{S'} = ({}_nH)_{S'}$ for every $n \in E$, i.e. such that $Y_{S'} \supset {}_nH_{S'}$ for every $n \in E$, then $Y_{S'} = H_{S'}$. Now this is indeed the case, for by virtue of IX 4.4 the family of the ${}_nH_{S'}$ is schematically dense in $H_{S'}$, taking conditions a) and b) into account.

Theorem 6.2. *Let S be a prescheme, H, E as in 6.1 satisfying conditions a), b), $u : H \to G$ a homomorphism of S-groups from H to an S-prescheme in groups locally of finite type over S, finally K a closed subprescheme in groups of H. Consider the subfunctor $Transp_G(u,K)$ of G (cf. 2.5). Then this last is representable by a closed subprescheme of G, and if G is noetherian (for example S noetherian and G of finite type over S), there exists an integer $n \in E$ such that $Transp_G(u,K) = Transp_G(u|_{{}_nH},K)$. If finally G is smooth over S, and K smooth over S or of multiplicative type, and if H, E satisfy the following condition stronger than a):*

a′) For every $n \in E$, the subgroup ${}_nH$ of H is of multiplicative type,

then $Transp_G(u,K)$ is smooth over S.

Consider indeed the G-group $H_G = H \times_S G$, which evidently satisfies the conditions of 6.1 (with S replaced by G, and H by H_G), and the closed subprescheme Y of H_G, inverse image of $K \subset G$ under

$$H_G = H \times_S G \to G$$

defined by $(h, g) \mapsto \mathrm{int}(g) \cdot u(h)$. Then $Transp_G(u, K)$ is none other than

$\prod_{H_G/G} Y/H_G$ (compare VIII, examples 6.5 e)). Hence the first assertions result from 6.1, and moreover, one sees that for every quasi-compact open U of G, there exists $n \in E$ such that $Transp_G(u,K)$ and $Transp_G(u|_{{}_nH},K)$ have the same trace on U. To verify the last assertion of 6.2, one can therefore replace H by an ${}_nH$, and then it suffices to apply 2.5, which applies since ${}_nH$ is supposed of multiplicative type over S.

Remark 6.3. The preceding proof uses only the very elementary result VIII 6.4, and moreover (for the last part) 2.5, that is, when K is smooth over S, the infinitesimal result IX 3.6, hence the vanishing of the cohomology of multiplicative-type groups. One will note that in the most important cases (cf. 6.7) one can suppose even the $n \in E$ prime to the residual characteristics of S, i.e. invertible in $\mathcal{O}_S$, hence the ${}_nH$ finite étale over S, and then the cohomological result invoked is practically trivial, so that 6.2 is then independent of the theory of multiplicative-type groups.

6.4. One sees as usual that 6.1 extends to the case where one has an S-prescheme S' over S (not necessarily locally noetherian), and a closed subprescheme Y' of $H' = H_{S'}$, provided that $Y' \to H'$ is of finite presentation i.e. the ideal defining Y' is of finite type: then $\prod_{H'/S'} Y'/H'$ is representable by a closed subprescheme T' of S', such that $T' \to S'$ is of finite presentation, and if S' is quasi-compact, there exists $n \in E$ such that the relation analogous to (×) is valid. It follows also that the first statement in 6.2 is valid without supposing G locally of finite type over S, provided that the immersion $K \to G$ is of finite presentation.

6.5. As announced in 5.6, theorem 6.2 permits extending to the preschemes in groups satisfying a′) and b) above, certain results established by another method and under more restrictive conditions for groups of multiplicative type. This is the case for results 5.2, for the beginning of 5.3, for 5.4, and for the bis variants of the preceding results 5.9 and 5.10. This is also the case for the results of IX, N°s 5 and 6, with the exclusion of IX 6.8 (already false for a homomorphism of abelian schemes $u : H \to G$, over the spectrum S of a discrete valuation ring with residue characteristic $p > 0$: it can happen that `Ker u` have as generic fiber the unit group, and as special fiber a radicial group not reduced to the unit group).

6.6. We have just given an example of a rigidity property for groups of multiplicative type which is not shared by abelian schemes. Another example, extremely important, is in the fact that the existence theorem of infinitesimal extensions of homomorphisms IX 3.6 is no longer valid when H is an abelian scheme. Thus, an abelian scheme $\neq 0$ over a field admits non-trivial infinitesimal variations, contrary to what occurs for a group of multiplicative type — which is the infinitesimal aspect of the fact that there exists a "theory of moduli" (moreover far from being completed) for abelian varieties, while the theory of moduli for groups of multiplicative type is empty. Another "global" aspect of this infinitesimal difference is that if H is an abelian scheme over S locally noetherian, and G is an S-prescheme in groups commutative locally of finite type over S, then one can show that $\mathrm{Hom}_{S-gr.}(H, G)$ is representable by a prescheme locally of finite type over S, but contrary to what happens for H of multiplicative type, this prescheme is not étale over S, but only unramified over S. Thus, if S is for example the spectrum of a complete discrete valuation ring, H and G abelian schemes over S, there can exist homomorphisms $H_0 \to G_0$ on the special fibers which do not come "by specialization" from a homomorphism on the generic fibers.

6.7. Theorem 6.2 applies whenever H is an abelian scheme over S, or more generally an extension of such a scheme by a torus. Indeed, the question being local on S, one may suppose that there exists a prime number ℓ prime to the residue characteristics of S, and one sees that it then suffices to take for E the set of powers of ℓ to satisfy conditions a′) and b).

A reasoning analogous to that of 6.1 gives us the

Theorem 6.8. *Let X be a prescheme smooth over S, with geometric fibers* *connected non-empty. Then for every closed subprescheme Y of X, the functor $\prod_{X/S} Y/X$ is representable by a closed subprescheme T of S. If Y is of finite presentation over X, then T is of finite presentation over S.*

As $f : X \to S$ is faithfully flat locally of finite presentation, it is covering for the fpqc topology. As on the other hand $T = \prod_{X/S} Y/X$ is evidently a subsheaf of S (for the fpqc topology), it follows that the question of representability of T by a closed subprescheme of S is of local nature on S for the fpqc topology, and the same holds for the question of deciding whether T is of finite presentation over S. Doing then the base change $S' \to S$, with $S' = X$, one is reduced to the case where X admits a section e over S. One may moreover suppose S affine and a fortiori quasi-compact. One then has:

Corollary 6.9. *Under the conditions of 6.8, suppose that S be quasi-compact and that X admit a section e over S. Let, for every integer $n \geqslant 0$, X_n be the subprescheme of X, infinitesimal neighborhood of order n of the section e. Suppose Y of finite presentation over X. Then there exists an integer $n \geqslant 0$ such that one has $\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n$ (where $Y_n = Y \cap X_n$).*

This corollary indeed implies 6.8 when Y is of finite presentation over X, thanks to VIII 6.4: for, X being smooth over S, X_n is finite and locally free over S and a fortiori is "essentially free" over S in the sense of VIII 6.1, hence $\prod_{X_n/S} Y_n/X_n$ is representable by a closed subprescheme of S. Moreover, the proof of loc. cit., or the reduction to the noetherian case, immediately shows us that the said closed subprescheme of S is of finite presentation over S.

Let us prove first 6.9, hence 6.8, when S is noetherian. Let $T_n = \prod_{X_n/S} Y_n/X_n$. Then the T_n form a decreasing sequence of closed subpreschemes of S, and S being noetherian, this sequence is stationary. Let $R = \bigcap_{n\geqslant 0} \prod_{X_n/S} Y_n/X_n$ their common value for n large; one has evidently $T \subset R$, and it suffices to establish that one has $R \subset T$. Doing the base change $R \to S$, one is reduced to the case where $R = S$, i.e. $Y_n = X_n$ for every n i.e. $Y \supset X_n$ for every n, and to prove then $T = S$ i.e. $Y = X$. Now $Y \supset X_n$ for every n implies (thanks to the fact that X is locally noetherian) that Y is, in a neighborhood of each point of $e(S)$, an induced open subprescheme of X, hence there exists an induced open U of X, containing $e(S)$, such that $U \subset Y$. By virtue of IX 4.3, the fibers of X/S being integral, U is schematically dense in X, hence (Y being a closed subprescheme majorizing U) one has $Y = X$. This proves 6.9, hence 6.8, in this case.

The general case proceeds by reduction to the preceding case. For every $s \in S$, there exists an affine open neighborhood U of s and an affine open neighborhood V of $e(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U, and if S_0 is an affine open neighborhood of s contained in $e^{-1}(V) \cap f(V)$, and $X_0 = V \cap f^{-1}(S_0)$, then X_0 and S_0 are affine opens of X resp. S, and X_0/S_0 admits a section. Because of the local nature of 6.8 and 6.9, one may suppose $S = S_0$. I say that one then has $\prod_{X/S} Y/X = \prod_{X_0/S_0} Y_0/X_0$, where $Y_0 = Y \cap X_0$; indeed, by virtue of IX 4.6, X_0 is schematically dense in X (at least when X is quasi-separated over S so that X_0 is retrocompact in X; but in fact one can show without difficulty that

IX 4.6 remains valid without the retrocompactness hypothesis), and likewise for every base change $S_1 \to S$, $X_0 \times_S S_1$ is schematically dense in $X \times_S S_1$, whence at once the announced equality. This reduces us to the case where $X = X_0$, so one may suppose S and X affine. Moreover, if $X = \mathrm{Spec}(B)$ and if J is the ideal of B which defines Y, J is the inductive limit of its sub-ideals of finite type, hence Y is the intersection of closed subpreschemes Y_i of X which are of finite presentation over S, and consequently $\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X$, which reduces us, to prove 6.8, to the case where Y is of finite presentation over X. It then suffices to prove 6.9 with S and X affine. But then X and Y over S come by base change $S \to S_0$ from an analogous situation `X_0` and `Y_0` over `S_0`, with `S_0` noetherian, which reduces us to the case where S is noetherian, which has already been treated. This completes the proof of 6.8 and 6.9.

Corollary 6.10. *Let X be an S-prescheme in groups smooth of finite presentation, with connected fibers, Y a prescheme in groups of finite presentation over S, $i : Y \to X$ a monomorphism of S-preschemes in groups, making Y therefore a subgroup of X. Then $\prod_{X/S} Y/X$ is representable by a closed subprescheme of finite presentation of S. If S is quasi-compact, denoting for every integer $n \geqslant 0$ by X_n the infinitesimal neighborhood of order n of the unit section of X, and setting $Y_n = X_n \cap Y$, one has for n large enough: $\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n$.*

The proof is essentially that of 6.9. Note that the unit sections of X and of Y induce bijective immersions $S \to X_n$ and $S \to Y_n$, hence induce isomorphisms of S_{red} with $(X_n)_{red}$ and $(Y_n)_{red}$, which implies that the injection $Y_n \to X_n$ is proper, hence, being a monomorphism of finite presentation, is a closed immersion. Consequently, by virtue of VIII 6.4 already used, $\prod_{X_n/S} Y_n/X_n$ is representable by a closed subprescheme of S of finite presentation over S, and it remains therefore to prove the last assertion of 6.10 in the case where one supposes moreover S affine. One reduces immediately again to the case where S is noetherian, and one is reduced to proving that one then has $R = T$ (with the notation of the proof of 6.9), or again that $Y \supset X_n$ for every n implies $Y = X$. Now the hypothesis implies that $i : Y \to X$ is étale at the points of the unit section of Y over S, hence Y is smooth over S at the points of the unit section, whence it follows that the open `Y_0` of points of Y at which Y is smooth over S is an induced open subgroup of Y. Then $Y_0 \to X$ is a monomorphism étale by virtue of X 3.5, hence an open immersion; now the fibers of X being connected and every open subgroup of an algebraic group being also closed, it follows that this is a surjective open immersion i.e. an isomorphism. Hence $Y_0 = X$ and a fortiori $Y = X$, which completes the proof of 6.10.

Proceeding as in VIII 6.5, one concludes from 6.10:

Corollary 6.11. *Let G be an S-prescheme in groups locally of finite type and quasi-separated over S, H a prescheme in groups smooth of finite presentation over S with connected fibers, $i : H \to G$ a monomorphism of S-groups, making H therefore a subgroup of G. Then:*

a) $Centr_G(H)$ and $Norm_G(H)$ are representable by closed subpreschemes of G of finite presentation over G, and likewise, for every monomorphism $j : K \to G$ of *finite presentation of S-preschemes in groups, $Transp_G(H, K)$ is representable by a closed subprescheme of G of finite presentation over G.*

b) If G is quasi-compact, in the various cases envisaged in a), there exists an integer $n \geqslant 0$ such that (if H_n denotes the infinitesimal neighborhood of order n of the unit section of H) one has

```
Centr_G(H) = Centr_G(H_n)
Norm_G(H) = Norm_G(H_n)
Transp_G(H, K) = Transp_G(H_n, K) = Transp_G(H_n, K_n).
```

One applies 6.10 to the prescheme in groups $X = H_G = H \times_S G$ above the base prescheme G, and to the subprescheme in groups Y inverse image of the diagonal subgroup of $(G \times_S G)_G$ over G by a suitable homomorphism of G-groups of X into $(G \times_S G)_G$ (in the case of `Centr`), resp. the inverse image of `K_G` by a suitable homomorphism of G-groups of X into `G_G` (in the case of `Transp`). The case of `Norm` reduces to the transporter by taking $K = H$, the hypothesis on G ensuring that $H \to G$ is of finite presentation (hence $Y \to X$ is of finite presentation); in the case of `Centr`, the hypothesis made on G ensures that the diagonal group of $G \times_S G$ is of finite presentation over $G \times_S G$, whence again the fact that Y is of finite presentation over X.

Remark 6.12. One can prove (using rather delicate results of EGA VI[1070]) that if X is a prescheme flat of finite presentation over S, which is proper over S or with non-empty connected geometric fibers, then for every closed subprescheme Y of X of finite presentation over S, $\prod_{X/S} Y/X$ is representable by a closed subprescheme of S of finite presentation over S. Likewise, if X is an S-prescheme in groups flat of finite presentation with connected fibers, and $i : Y \to X$ a monomorphism of S-groups, with Y an S-prescheme in groups of finite presentation, then $\prod_{X/S} Y/X$ is representable by a closed subprescheme of S of finite presentation over S. In particular, 6.11 a) remains valid by replacing the hypothesis "H smooth over S" by "H flat over S".

Exposé XII. Maximal tori, the Weyl group, Cartan subgroups, the reductive center of smooth and affine group schemes

by A. Grothendieck

From the present Exposé onwards, in contrast to the preceding Exposés, we make use of the known results on the structure of smooth algebraic groups over an algebraically closed field k, and above all of Borel's theory of affine algebraic groups, expounded in the Séminaire Chevalley 56/57: "Classification des groupes de Lie

[1070]Cf. also J.P. Murre, *Representation of unramified functors. Applications*, Sém. Bourbaki N° 294 (May 1965), th. 3 (p. 13).

algébriques".[1071] Following the usage in the theory of algebraic groups, we refer to this Séminaire by the sigil *BIBLE*. For the next Exposés, we shall need in particular the results of *BIBLE* 4, 5, 6, 7 (the number after *BIBLE* refers to the Exposé number). It seems, moreover, that the theory of schemes brings no notable simplification to Borel's theory as exposed in *BIBLE*. That is why it has not seemed useful to reproduce it in the present Séminaire, our aim here being to deduce, from results known over algebraically closed fields, analogous results valid over an arbitrary base prescheme. (Things will be different for Chevalley's structure theory of semisimple groups, which, it appears, is best treated *ab ovo* over an arbitrary base prescheme.)

In the oral Exposés (which we have followed in Nos. 1 to 4) we restricted ourselves to affine group preschemes over S, relying in an essential way on the representability theorems of Exp. XI N° 4. In the present notes (cf. Nos. 6 to 8) we show rapidly how the affine hypotheses can be eliminated by a simpler method that does not use the most delicate results of Exposé XI. For other generalizations of the results contained in the present Exposé, see also Exposés XV and XVI. (It is evident that the content of Exposés XI, XII, XV, XVI ought to be completely rewritten.)

1. Maximal tori

1.0. Let first G be an algebraic group over an algebraically closed field k. By a *maximal torus of G* one means an algebraic subgroup T of G which is a torus (meaning here, since k is algebraically closed, that it is isomorphic to a group of the form G_m^r), and which is maximal for this property. Note that, k being perfect, G_{red} is a subgroup of G, smooth over k, so it is essentially an algebraic group in the sense of *BIBLE*. Moreover, every reduced subgroup of G (such as a torus) is automatically a subgroup of G_{red}, so the maximal tori of G coincide with the maximal tori of G_{red}. When G is affine, hence G_{red} is affine, a fundamental theorem of Borel tells us that two maximal tori of G are conjugate by an element of $G(k) = G_{red}(k)$ (*BIBLE* 6, th. 4 c)); in particular they have the same dimension. We shall call the common dimension of the maximal tori of G the *reductive rank of G*. Note moreover that the restriction that G be affine is unnecessary, as follows from a known theorem of Chevalley, asserting that every connected smooth algebraic group over k is an extension of an abelian variety by an affine group; cf. N° 6. In Nos. 1 to 4 we shall mostly restrict ourselves to affine group preschemes over the base.

Let G be a smooth algebraic group over k, and T a maximal torus of G; the centralizer $C(T)$ of T in G will be called the *Cartan subgroup of G associated with T*. For us this is a group subpreschema defined thanks to (VIII, 6.7), but one will note that, G being smooth over k, the same holds for the Cartan subgroup C by (XI, 5.3); hence in this case C is the unique group subpreschema of G smooth over k (i.e. an algebraic subgroup of G in the sense of *BIBLE*) such that $C(k)$ is the subgroup of $G(k)$ centralizing $T(k)$, i.e. it is essentially the centralizer in the sense of *BIBLE*. By

[1071] N.D.E.: This Séminaire has now been published, in a revised form by P. Cartier, as volume 3 of the *Œuvres de Chevalley* (Springer, 2005).

the conjugation theorem already cited, the Cartan subgroups associated with the various maximal tori are conjugate to each other, hence have the same dimension. We shall call their common dimension the *nilpotent rank of G*; it equals that of G_{red}. Let $\rho_r(G)$ and $\rho_n(G)$ denote the reductive and nilpotent ranks of G; then one has

$$\rho_r(G) \leqslant \rho_n(G),$$

and the difference

$$\rho_u(G) = \rho_n(G) - \rho_r(G) = \dim C/T$$

might be called, when G is affine, the *unipotent rank of G*. When G is smooth, affine, and connected, then C is a nilpotent connected algebraic group (*BIBLE* 6, th. 6 a) and c)), hence (*BIBLE* 6, th. 2) isomorphic to the product $C_s \times C_u$, where $C_s = T$ is the maximal torus we started with (which is *a fortiori* a maximal torus in C), and where C_u is a smooth unipotent subgroup, i.e. a successive extension of groups isomorphic to G_a (*BIBLE* 6 th. 1 cor. 1 and 7, th. 4). One then has also, in this case:

$$\rho_u(G) = \dim C_u.$$

Remark 1.1. *Besides the three notions of rank we have just made precise for an affine algebraic group, there are two others that are useful, namely the* semisimple rank $\rho_s(G)$, *which is by definition the reductive rank of the quotient G/R, where R is the radical of G, and finally the* infinitesimal rank $\rho_i(G)$, *which is by definition the nilpotent rank of the Lie algebra of G (which will be defined and studied in the following Exposé). We shall not use them in the present Exposé. Let us only signal the inequalities:*

$$\rho_s \leqslant \rho_r \leqslant \rho_n \leqslant \rho_i.$$

Lemma 1.2. *Let G be an algebraic group over the algebraically closed field k, T an algebraic subgroup of G, k' an algebraically closed extension of k, G' and T' the groups obtained from G, T by extension of the base. For T to be a maximal torus of G, it is necessary and sufficient that T' be a maximal torus of G'.*

This is an immediate consequence of the "principle of finite extension" (EGA IV, 9.1.1).

Definition 1.3. *Let S be a prescheme, G an S-prescheme in groups of finite type, T a group subpreschema of G. One says that T is a* maximal torus of G *if*

a) T is a torus (IX, 1.3) and

b) for every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, $T_{\bar{s}}$ is a maximal torus in $G_{\bar{s}}$.

Remarks 1.4. *It follows from 1.2 that when S is the spectrum of an algebraically closed field one recovers the usual definition, and that property 1.3 is stable under* *any base change. Note that, by (X, 8.8), one may in Definition 1.3 replace condition a) by:*

a′) T is of finite presentation and flat over S.

One should beware that a maximal torus in the sense of Definition 1.3 is indeed maximal among the subtori of G (as follows at once from (IX, 2.9)), but that the converse cannot be valid—the base S being, say, connected—unless G effectively admits a maximal torus in the sense of 1.3, which is not the case in general (even if $S = \mathrm{Spec}(\mathbb{Z})$ and G is "semisimple"; see also 1.6). We shall see however in XIV that the converse holds when S is artinian, or when S is a local scheme and G is "reductive": in this case, every torus of G is contained in a maximal torus.

Definition 1.5. *Let G be an algebraic group over a field k. The* reductive rank *(resp.* nilpotent rank*, resp.* unipotent rank*, etc.) *of* G is the reductive rank (resp. ...) of $G_{\bar{k}}$, where $\bar{k}$ is an algebraic closure of k.*

One will note that, in view of 1.2 and the commutation of the formation of $Centr_G(T)$ with base extension, the notions of rank introduced in 1.5 are invariant under extension of the base field; on the other hand, for k algebraically closed, they coincide with those introduced at the beginning of the present number.

Remark 1.6. *It is not difficult to construct an affine smooth group scheme G over the spectrum S of a discrete valuation ring, whose generic fiber is isomorphic to G_m, and whose special fiber is isomorphic to G_a. Such a G contains no torus except the trivial torus T (reduced to the unit subgroup), which is evidently not a maximal torus. More precisely, in the special fiber $G_0 = G_{a,k}$, T_0 is indeed a maximal torus, but in the generic fiber $G_1 = G_{m,K}$, T_1 is no longer a maximal torus (k is the residue field, K the fraction field of the valuation ring under consideration). One sees also from this example that the reductive rank of G_s ($s \in S$) is not a continuous function of s. One has however the following results:*

Theorem 1.7. *Let S be a prescheme, G an affine smooth S-prescheme in groups over S. For every $s \in S$, consider $\rho_r(s) = \rho_r(G_s)$ and $\rho_n(s) = \rho_n(G_s)$, the reductive and nilpotent ranks of G_s (1.5). With these notations, one has the following:*

a) The function ρ_r on S is lower semicontinuous, the function ρ_n on S is upper semicontinuous, hence the function $\rho_u = \rho_n - \rho_r$ is upper semicontinuous.

b) The following conditions (stable under arbitrary base change) are equivalent:

(i) The function ρ_r on S is locally constant.

(ii) There exists, locally for the étale topology, a maximal torus in G.

(ii bis) There exists, locally for the faithfully flat quasi-compact topology, a maximal torus in G.

c) Let T_1, T_2 be two maximal tori in G (which implies that one is under the conditions of b)). Then T_1, T_2 are conjugate locally for the étale topology, i.e. there exists an étale surjective morphism $S' \to S$ such that the subgroups $(T_1)_{S'}$ and $(T_2)_{S'}$ of $G_{S'}$ are conjugate by a section of $G_{S'}$ over S'.

d) Under the conditions of b), i.e. when ρ_r is locally constant, the same holds for ρ_n (hence also for $\rho_u = \rho_n - \rho_r$).

Proof. a) Note that for any morphism $S' \to S$, setting $G' = G \times_S S'$, the functions ρ'_r etc. on S' defined in terms of G' as ρ_r etc. in terms of G are obtained simply by composing the latter with $S' \to S$. When $S' \to S$ is faithfully flat quasi-compact, it follows that ρ' is continuous, resp. upper semicontinuous, resp. lower semicontinuous, if and only if ρ is, the Zariski topology of S being in effect a quotient of that of S' (SGA 1, VIII 4.3). Consequently, the assertions of a) are local for the faithfully flat quasi-compact topology. Let then $s \in S$; we wish to show that the set U of $t \in S$ such that $\rho_r(t) \geqslant \rho_r(s)$ (resp. $\rho_n(t) \leqslant \rho_n(s)$) is a neighborhood of s. By the principle of finite extension, there exists a finite extension k of $\kappa(s)$ such that G_k admits a maximal torus. There then exist an open neighborhood U of s, and a finite surjective locally free morphism $S' \to U$ such that the fiber S'_s is $\kappa(s)$-isomorphic to $\mathrm{Spec}(k)$ (cf. EGA III, 10.3.2, where the noetherian hypothesis is manifestly unnecessary). Since $S' \to S$ is an open morphism (SGA 1, IV 6.6), we are reduced to the case where $S' = S$, i.e. to the case where there exists a maximal torus T_s in G_s. Furthermore, by (XI, 5.8 a)), possibly replacing S by an S' étale over S equipped with a point s' above s, we may suppose that T_s is the fiber at s of a subtorus T of G. Then for every $t \in S$, $\rho_r(t) = \rho_r(G_t) \geqslant \dim T_t = \dim T_s = \rho_r(G_s) = \rho_r(s)$, which proves that ρ_r is lower semicontinuous. On the other hand, by (XI, 5.3), the functor

$$C = \mathit{Centr}_G(T)$$

is representable by $C = \mathit{Centr}_G(T)$,[1072] a closed group subprescheme of G smooth over S. So there exists a neighborhood U of s such that $t \in U$ implies $\dim C_t = \dim C_s = \rho_n(s)$. The upper semicontinuity of ρ_n is then a consequence of the relation

$\rho_n(t) \leqslant \dim C_t$ for every $t \in S$,

which is itself contained in the following purely geometric lemma:

Lemma 1.8. *Let k be a field, G an affine smooth algebraic group over k, T a torus in G, C its centralizer; then one has*

$$\rho_n(G) \leqslant \dim C.$$

Indeed, one may suppose k algebraically closed, so that T is contained in a maximal torus T_0. Let C_0 be the centralizer of the latter; then $C_0 \subset C$, hence $\dim C_0 \leqslant \dim C$. QED.

b) If ρ_r is locally constant, then for every torus T in G, and every $s \in S$, if T_s is a maximal torus in G_s, then there exists an open neighborhood U of s such that $T|U$ is a maximal torus in $G|U$. Using now the reasoning of a), one sees that (i)

[1072] N.D.E.: modification made to introduce the prescheme $C = \mathit{Centr}_G(T)$ representing the functor $\mathit{Centr}_G(T)$.

$\Rightarrow$ (ii bis). On the other hand (ii bis) $\Rightarrow$ (i), for when G admits a maximal torus T, then it is evident that $\rho_r(s) = \dim T_s$ is a locally constant function of s; now we signaled at the beginning of the proof of a) that the question of continuity of ρ_r was local for the faithfully flat topology. Hence (i) $\Leftrightarrow$ (ii bis), and obviously (ii) $\Rightarrow$ (ii bis); it remains to prove the converse implication (i) $\Rightarrow$ (ii). For this, introduce the functor F of (XI, 4.1),[1073] which is therefore a smooth separated prescheme over S, and consider the subfunctor $\mathcal{T}$ of F, whose value for an S' over S is the set of maximal tori in $G_{S'}$. Using the previous remark that, granted (i), a torus in $G_{S'}$ which is maximal in the fiber of some point $s \in S'$ is maximal above an open neighborhood of s, one sees that $\mathcal{T}$ is representable by an open subpreschema of F, and is therefore smooth and separated over S. Since the structural morphism $\mathcal{T} \to S$ is evidently surjective, it admits a section locally for the étale topology by (XI, 1.10), which proves (i) $\Rightarrow$ (ii).

c) This is an immediate consequence of (XI, 5.4 bis), taking into account Borel's conjugation theorem recalled at the beginning of the number.

d) By the remark at the beginning of the proof in a), and taking into account b), we may suppose that there exists a maximal torus T in G. If C is its centralizer, then C is representable and smooth over S by (XI, 5.3), so the function $s \mapsto \rho_n(s) = \dim C_s$ is indeed locally constant.

The proof of 1.7 is complete. We shall refer to the conditions envisaged in 1.7 b) by saying that in this case G is *of locally constant reductive rank*. Let us note:

Corollary 1.9. *Let G be as in 1.7, and let $s \in S$ be such that $\rho_u(s) = 0$, i.e. $\rho_r(s) = \rho_n(s)$ (i.e. the Cartan subgroups of $G_{\bar{k}}$ are tori, where $\bar{k}$ is an algebraic closure of $\kappa(s)$). Then there exists an open neighborhood U of s over which ρ_r and ρ_n are constant; in particular, for every $t \in U$, the unipotent rank $\rho_u(t)$ of G_t is zero.*

Indeed, this follows immediately from 1.7 a) and from the inequality $\rho_r(t) \leqslant \rho_n(t)$ for every $t \in S$.

Note also that we have proved, at the same time as b), the

Corollary 1.10. *Let G be as in 1.7, and suppose G of locally constant reductive rank. Consider the functor*

$$\mathcal{T} : (Sch)^{\circ} \longrightarrow (Ens),$$

such that for every S' over S, one has

□(S′) = the set of maximal tori of G_{S′}.

Then $\mathcal{T}$ is representable by a smooth, separated prescheme of finite type over S.

It remains to verify that $\mathcal{T}$ is of finite type over S. The question being local for the faithfully flat quasi-compact topology, we may suppose that G admits a maximal torus T. By (XI, 5.3 bis) and the bis version of 5.5, $Norm_G(T)$ and $G/Norm_G(T)$ are

[1073] N.D.E.: This is the functor of multiplicative-type subgroups of G.

representable by preschemes $Norm_G(T)$ and $G/Norm_G(T)$, and $\mathcal{T}$ is isomorphic to $G/Norm_G(T)$.[1074] The morphism $G \to \mathcal{T}$ defined by $g \mapsto int(g)(T)$ being surjective, and G quasi-compact over S, the same holds for $\mathcal{T}$, which completes the proof.

Remarks 1.11. *a) The prescheme $\mathcal{T}$ of 1.10 will rightly be called the* prescheme of maximal tori of G. *We shall see in N° 5 that it is in fact affine over S. One can verify this directly when G is isomorphic to a closed group subpreschema of a prescheme of the form $GL(n)$, using (XI, 4.6) and remarking that, by the reasoning of the proof of 1.7 b), $\mathcal{T}$ identifies with a subpreschema both open and closed of the prescheme F which represents the multiplicative-type subgroups of G.*

b) It is possible to give a proof of 1.7—and hence of 1.9—not using the delicate results of XI, but only the easy results (XI, 3.12) and (XI, 6.2), working only with multiplicative-type subgroups finite over the base (morally, the groups ${}_nT$ where T is a maximal torus); compare N° 7. The same remark applies to the proof of 1.10.

We end this number by giving examples in which there exists a unique maximal torus.

Proposition 1.12. *Let G be an S-prescheme in groups of multiplicative type and of finite type. Then G admits a unique maximal torus, and every torus in G is contained in this maximal torus.*

Uniqueness obviously follows from the latter assertion, which characterizes the maximal torus as the largest subtorus of G. From uniqueness it follows that the existence question is local for the faithfully flat quasi-compact topology, which allows us to suppose G diagonalizable, i.e. of the form $D_S(M)$, with M a finitely generated commutative group. Let `M_0` be the quotient of M by its torsion subgroup; I claim that the torus $T = D_S(M_0)$ in G is a maximal torus and a largest subtorus. Indeed, a subtorus T' of G is locally diagonalizable for the fpqc topology, hence to prove $T' \subset T$ we may suppose T' diagonalizable, hence of the form $D_S(N)$, where N is a free quotient of M (VIII 1.4 and 3.2 b)), hence N is a quotient of `M_0`, hence $T' \subset T$. Since the construction of T as $D_S(M_0)$ is compatible with any base extension, this shows at the same time that T is a maximal torus of G, and completes the proof. In the case where G is smooth over S, 1.12 may be generalized:

Proposition 1.13. *Let G be an S-prescheme in groups of finite presentation over S. Suppose that G admits, locally for the faithfully flat quasi-compact topology, a central maximal torus. Then G admits (globally) a unique maximal torus,*[1075] *and it is the largest subtorus of G.*

Here again, uniqueness is trivial from the last assertion, and renders existence a local question for fpqc, which allows us to suppose that G admits a central maximal torus T. We show that every torus R of G is contained in T.

This follows from the

[1074] N.D.E.: modification made to introduce the preschemata $Norm_G(T)$ and $G/Norm_G(T)$.
[1075] N.D.E.: which is central!

Lemma 1.14. *Let G be an S-prescheme in groups of finite presentation over S, T a maximal torus of G, R a subtorus of G, and suppose that T and R commute. Then $R \subset T$.*

Indeed, since R and T commute, the morphism $R \times_S T \to G$ defined by $(r,t) \mapsto r \cdot t$ is a group homomorphism, hence by (IX, 6.8) it admits an image subgroup T' in G, which is a multiplicative-type group quotient of $R \times_S T$, hence a torus, evidently containing T. Since T is a maximal torus, one has $T' = T$ (1.4), hence $R \subset T$, which proves 1.14 and hence 1.13. In particular, using 1.7 b):

Corollary 1.15. *Let G be a commutative, smooth, and affine S-prescheme in groups over S, of locally constant reductive rank. Then G admits a unique maximal torus, and this latter contains every subtorus of G.*

Corollary 1.16. *Let G be a smooth and affine S-prescheme in groups over S. Suppose that for every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure $\bar{k}$ of $\kappa(s)$, the geometric fiber $G_{\bar{s}}$ is a connected nilpotent algebraic group (in the sense of BIBLE, i.e. the group $G(\bar{k})$ of its $\bar{k}$-valued points is nilpotent). Suppose moreover that the reductive rank of G is locally constant. Then G admits a unique maximal torus T; moreover, T is central and is the largest subtorus of G.*

By 1.7 b), G admits a maximal torus locally for fpqc, and by 1.13 we are reduced to proving that a maximal torus of G is central. By (IX, 5.6 b)), we are reduced to the case where S is the spectrum of a field, which one may suppose algebraically closed. Then $T(k)$ lies in the center of $G(k)$ by *BIBLE* 6 th. 2, which implies that T lies in the center of G, since $\mathit{Centr}_G(T)$ is a closed subscheme of G (VIII 6.6) which contains the points of $G(k)$, hence is identical to G by the Nullstellensatz (since G is reduced).

Finally, for later reference, let us signal the following trivial proposition (of which we have already made implicit use):

Proposition 1.17. *Let $G \supset H \supset T$ be preschemes in groups of finite type over S. Equivalent conditions:*

(i) T is a maximal torus of G.

(ii) T is a maximal torus of H, and for every $s \in S$, one has equality of the reductive ranks $\rho_r(G_s) = \rho_r(H_s)$.

2. The Weyl group

Let first k be an algebraically closed field, and let G be a smooth affine algebraic group over k. If T is a maximal torus, C its centralizer and N its normalizer, then by (XI, 5.9) these are smooth closed subgroups of G, and C is an open subgroup of N, so that $W = N/C$ is a finite étale group over k, hence determined by the group $W(k)$ of its k-valued points, as $W = W(k)_k$ (constant group defined by the ordinary finite group $W(k)$). The finite group $W(k)$ will be called the *geometric Weyl group* (or simply the *Weyl group* if no confusion is to be feared) of G relative to T. By Borel's conjugation theorem, the Weyl groups relative to the various maximal tori are isomorphic to each other; this is why one sometimes speaks of "the" Weyl group of G, without specifying a maximal torus. Since the formation of C, N, and N/C commutes with any base extension, one sees that if k' is an algebraically closed extension of k, the geometric Weyl group of $G_{k'}$ relative to $T_{k'}$ is canonically isomorphic to that of G relative to T; consequently, "the" geometric Weyl group of G (which is properly speaking an isomorphism class of ordinary finite groups) coincides with that of G'.

This allows one, when G is a smooth affine algebraic group over an arbitrary field k, to speak of the *geometric Weyl group of* G as the isomorphism class of the geometric Weyl group of $G_{k'}$, where k' is any algebraically closed extension of k. When G admits a maximal torus T, then one may evidently form as above $C = Centr_G(T)$, $N = Norm_G(T)$, $W = N/C$ which is a finite étale group over k, called the *Weyl group of* G *relative to* T; the geometric Weyl group is then nothing other than the class of the group of points of W with values in any algebraically closed extension k' of k. Here, knowledge of the geometric Weyl group $W(k')$ evidently no longer suffices in general to reconstruct the algebraic group W: one must also know the operations on $W(k')$ of the Galois group of the separable algebraic closure of k in k'.

When finally G is an S-prescheme in groups over an arbitrary base S, G smooth and affine over S, and if T is a maximal torus of G, then (XI, 5.9) still allows us to form the group

```
W(T) = N(T)/C(T),
```

which is an étale, separated, quasi-finite S-prescheme in groups over S. Its geometric fibers (relative to the algebraic closures of the residue fields $\kappa(s)$, $s \in S$) are the geometric Weyl groups of the fibers G_s. Consequently, (XI, 5.10) gives information on the variation of these groups with $s \in S$. We can make this information more precise and generalize it as follows:

Theorem 2.1. *Let S be a prescheme, G a smooth and affine S-prescheme in groups over S. For every $s \in S$, let $w(s)$ be the geometric Weyl group of G_s, which is therefore an isomorphism class of finite groups. In the set E of such classes, introduce the following preorder relation: $w \leqslant w'$ if and only if w and w' are represented by finite groups W and W' such that W is isomorphic to a quotient of a subgroup of W'. With this convention:*

a) The function $s \mapsto w(s)$ from S to E is lower semicontinuous.

b) Suppose that the reductive rank of G is locally constant. Then the following conditions are equivalent:

(i) The function $s \mapsto w(s)$ is locally constant.

(i bis) The function $s \mapsto cardw(s)$ is locally constant.

(ii) There exists, locally for the étale topology (or merely for the fpqc topology), a maximal torus T, such that $W(T)$ is finite over its base.

(ii bis) For every S' over S, and every maximal torus T of $G_{S'}$, the associated Weyl group $W(T)$ is finite over S'.

Proof. a) Proceeding as in 1.7 a), one is reduced, in order to prove that for every $s \in S$ there exists an open neighborhood U of s such that $t \in U$ implies $w(t) \geqslant w(s)$, to the case where there exists a torus R in G such that R_s is a maximal torus in G_s. Let

$$W(R) = Norm_G(R)/Centr_G(R)$$

as in (XI, 5.9); this is an étale, separated, quasi-finite group prescheme over S. For every $t \in S$, let $w'(t) \in E$ be its geometric fiber at t. Since R_s is a maximal torus in G_s, and the formation of `Norm`, `Centr`, $Norm/Centr$ is compatible with any base extension, in particular with passage to the fibers, one sees that one has

$$w(s) = w'(s);$$

I claim moreover that, for t near s, one has the inequalities

`w(t) □ w′(t)   and   w′(t) □ w′(s),`

which will suffice to establish a). These two inequalities are contained in the two following lemmas:

Lemma 2.2. *Let S be a prescheme, W an étale, separated, quasi-finite S-prescheme in groups over S. For every $s \in S$, let $f(s)$ be the class of the geometric fiber of W at s, which is an element of the ordered set E of isomorphism classes of finite groups, introduced in 2.1. Then the function $f : S \to E$ is lower semicontinuous. For it to be constant in a neighborhood of $s \in S$, it is necessary and sufficient that the same hold for the function $s \mapsto card f(s)$, and for this it is necessary and sufficient that W be finite over S above an open neighborhood U of s.*

This result is a refinement, in terms of groups, of the one invoked in the proof of (XI, 5.10); we content ourselves with a sketch of the proof (which is of the most standard type). As usual, one is reduced to the case S affine noetherian. One sees at once that the function f is constructible (EGA 0_III, 9.3.1 and 9.3.2, and the sundries of EGA IV, 9), and by (EGA 0_III, 9.3.4) one is reduced, for the semicontinuity, to proving that if t is a generalization of s, one has $f(t) \geqslant f(s)$. This reduces us, thanks to (EGA II, 7.1.7), to the case where S is the spectrum of a discrete valuation ring, which one may suppose complete with algebraically closed residue field. But then, s denoting the closed point of S, since G is étale and separated over S, it contains a subscheme G' both open and closed, finite over S, such that $G'_s = G_s$ (EGA II, 6.2.6), and one sees at once that G' is here a subgroup of G. Moreover G', being étale finite over $S = \mathrm{Spec}(V)$, with V complete with algebraically closed residue field, is a constant group, hence of the form `A_S`, where $A = G'(\kappa(s)) = G(\kappa(s))$ has class $f(s)$. If B is the geometric fiber of G at the generic point t of S, one has therefore a canonical monomorphism $A \to B$, which proves $f(s) \leqslant f(t)$. (N.B. this proof in fact proves the semicontinuity for an order relation on E finer than the one indicated in 2.1.) The fact that f is continuous at s if and only if $s \mapsto card f(s)$ is, follows from the fact that for $w, w' \in E$, the relations $w \leqslant w'$ and $card w = card w'$ imply $w = w'$. The fact that this condition is equivalent to the finiteness of W on a neighborhood of s is then independent of the group structure on W, and has been signaled after (XI, 5.10); its proof can moreover easily be carried out by the preceding arguments, using the valuative criterion of properness (EGA II, 7.3.8).

Lemma 2.3. *Let G be a smooth affine algebraic group over an algebraically closed field k, $R \subset T$ two subtori, $W(R)$ and $W(T)$ the two associated finite groups as in (XI, 5.9), quotients of the normalizer by the centralizer. Then $W(R)$ is isomorphic to a quotient of a subgroup of $W(T)$.*

Indeed, consider the diagram

$$\begin{tikzcd}
T \arrow[r, hook] & C(T) \arrow[r, hook] \arrow[d] & C(R) \arrow[d] \\
& N(T) \cap N(R) \arrow[r, hook] \arrow[d] & N(R) \\
& N(T) &
\end{tikzcd}$$

then $(N(T) \cap N(R))/C(T)$ is a subgroup of $W(T) = N(T)/C(T)$, and one has an obvious homomorphism:

$$(N(T) \cap N(R))/C(T) \to W(R) = N(R)/C(R),$$

and everything reduces to proving that this last is surjective, i.e. that for every k-valued point g of $N(R)$, there exists a k-valued point c of $C(R)$ such that cg normalizes T, i.e. such that

$$\mathrm{int}(c)(\mathrm{int}(g)T) = T.$$

Now for this it suffices to note that $\mathrm{int}(g)T$ is a torus of $N(R)$ hence of $C(R)$ (which is an open subgroup thereof). Then T and $\mathrm{int}(g)T$ are maximal tori of $C(R)$, since they are maximal in G, and one concludes by Borel's conjugation theorem.

This proves 2.3 and thereby 2.1 a).

b) We have already signaled that (i) and (i bis) are trivially equivalent; they imply (ii bis) by the converse to 2.2 or to (XI, 5.10) at choice; (ii bis) ⇒ (ii) thanks to 1.7 b); finally (ii) ⇒ (i bis), for one sees as in 1.7 a) that the condition (i) is local for fpqc, which allows us to suppose that G admits a maximal torus T such that $W(T)$ is finite over S, and one concludes again by (XI, 5.10). This completes the proof of 2.1.

3. Cartan subgroups

Definition 3.1. *Let G be a smooth S-prescheme in groups of finite type over the prescheme S. By a* Cartan subgroup *of G one means a group subpreschema C of G, smooth over S, such that for every $s \in S$, denoting by $\bar{s}$ the spectrum of an algebraic closure of $\kappa(s)$, $C_{\bar{s}}$ is a Cartan subgroup (cf. N° 1) of $G_{\bar{s}}$.*

It is immediate that if C is a Cartan subgroup of G, then for every S' over S, $C_{S'}$ is a Cartan subgroup of $G_{S'}$. If S is the spectrum of an algebraically closed field, one recovers the notion recalled in N° 1. Finally, one verifies at once that the property for a group subpreschema C of G to be a Cartan subgroup is local in nature for the faithfully flat quasi-compact topology.

Theorem 3.2. *Let G over S be as in 3.1, and suppose that G is affine over S and of locally constant reductive rank. Then the map*

$$T \mapsto Centr_G(T)$$

induces a bijection from the set of maximal tori of G onto the set of Cartan subgroups[1076] *of G. If C corresponds to T, then T is the unique maximal torus*

of C.

If T is a maximal torus of G, the functor $Centr_G(T)$ is representable, by (XI, 5.3) or (XI, 6.2), by a closed smooth subpreschema of G, $C = Centr_G(T)$, and it follows from the definitions that C is a Cartan subgroup of G. Moreover, T is obviously a maximal torus of C, and being central, it is the unique maximal torus of C (1.13). Hence the map $T \mapsto Centr_G(T)$ from the set of maximal tori into the set of Cartan subgroups is injective; it remains to prove that it is surjective. Let then C be a Cartan subgroup of G; we prove that it is of the form $Centr_G(T)$, for T a maximal torus of G. For this it suffices to find a maximal torus T of C, for then T will be a maximal torus of G (since for every $s \in S$, G_s and C_s have the same reductive rank). By (IX, 5.6 b)) T lies in the center of C, hence $C \subset C' = Centr_G(T)$, and then C is a smooth subgroup of the smooth group C' over S, coinciding with C' fiber by fiber, whence $C = C'$. Now since G is of locally constant reductive rank, the same holds for C, hence C admits a maximal torus locally for the étale topology by 1.7 b), and since this torus is central by the preceding reasoning, it follows by 1.13 that C does admit a maximal torus, which completes the proof.

Corollary 3.3. *Let G be a smooth and affine S-prescheme in groups over S of locally constant reductive rank, and let $\mathcal{C} : (Sch)^\circ/S \to (Ens)$ be the functor defined by*

$$\mathcal{C}(S') = \text{the set of Cartan subgroups of } G_{S'}.$$

Then the functor $\mathcal{C}$ is isomorphic to the functor $\mathcal{T}$ of 1.10, hence is representable by the same smooth, separated prescheme of finite type over S.

Corollary 3.4. *Under the conditions of 3.2, if $C = Centr_G(T)$, one has*

$$Norm_G(C) = Norm_G(T).$$

Remark 3.5. *When G is not of locally constant reductive rank, it is nevertheless possible that G admit Cartan subgroups (for example if G has connected nilpotent fibers, G is a Cartan subgroup of itself, but is not necessarily of locally constant*

[1076] The proof given here actually only proves 3.2 for the closed Cartan subgroups of G. However, 7.1 a) establishes 3.2 in the form stated, and implies that the Cartan subgroups of G are closed.

reductive rank, cf. 1.6). In XV, we develop the theory of Cartan subgroups without supposing G affine over S or of locally constant reductive rank, using the theory of regular elements of the next Exposé. See also Nos. 6 and 7 for the elimination of the affine hypothesis.

4. The reductive center

Definition 4.1. *Let S be a prescheme, G an S-group of finite presentation over S, with affine fibers, Z a group subpreschema of G. One says that Z is a* reductive center of G *if*

(i) Z is central, and of multiplicative type, and

(ii) for every base change $S' \to S$ and every central homomorphism $u : H \to G_{S'}$, where H is a group of multiplicative type and of finite type over S', u factors through $Z_{S'}$.

(For a variant of this notion when the fibers of G are not supposed affine, cf. 8.6.)

Note that such a Z is necessarily of finite type over S (since its fibers are), hence is uniquely determined as the largest central multiplicative-type subgroup of G. It is easy to give examples where G (smooth and affine over S) admits a largest central multiplicative-type subgroup Z, but where Z is not a reductive center (i.e. G does not admit a reductive center), cf. for example 1.6; it follows however easily from (IX, 6.8) that a subgroup Z of G is a reductive center of G if and only if it is a largest central multiplicative-type subgroup, and retains this property under any base change; see also 4.3 below.

It is evident from 4.1 that if Z is the reductive center of G, then for every base change $S' \to S$, $Z_{S'}$ is the reductive center of $G_{S'}$. From this, and from the uniqueness of the reductive center, follows, thanks to the theory of faithfully flat quasi-compact descent (SGA 1, VIII):

Proposition 4.2. *Let G be an S-prescheme in groups of finite presentation over S and with affine fibers. If G admits a reductive center locally for the faithfully flat quasi-compact topology, then it admits a reductive center. For Z to be a reductive center of G, it is necessary and sufficient that it be so locally for the fpqc topology.*

Let us also note:

Proposition 4.3. *Let G be an S-prescheme in groups of finite presentation and with affine fibers, Z a group subpreschema. For Z to be a reductive center of G, it is necessary and sufficient that it be of multiplicative type, and that for every $s \in S$, Z_s be a reductive center of G_s.*

Indeed, it follows first from (IX, 5.6 b)) that Z is then central. Since the conditions envisaged are stable under base change, it remains to prove that any central homomorphism $u : H \to G$, with H of multiplicative type and of finite type, factors through Z. Now, Z being central, u and the canonical immersion $v : Z \to G$ define a group homomorphism `w : H ×_S Z  G.`

By (IX, 6.8) this last admits an image group K, which is a multiplicative-type subgroup of G, and everything reduces to proving that $K = Z$. Now this is true fiber by fiber by the hypothesis on Z, and it now suffices to apply (IX, 5.1 bis), which completes the proof of 4.3.

One will note that in criterion 4.3, the hypothesis that for every $s \in S$, Z_s is the reductive center of G_s is in fact purely geometric, i.e. it suffices to verify it on the algebraic closure of $\kappa(s)$, as follows from the second assertion of 4.2.

Theorem 4.4. *Let G be an affine algebraic group over a field k. Then G*

admits a reductive center.

Since the center of G is representable by a closed subgroup of G (VIII, 6.7), one is immediately reduced to the case where G is commutative. Furthermore, by 4.2 one may suppose k algebraically closed. We shall see in XVII that in this case G is written as a product $Z \times U$, where Z is of multiplicative type and U is "unipotent", from which it will follow at once that Z is indeed a reductive center of G. We shall give here a proof independent of the structure theorem for commutative algebraic groups in the general form just indicated.

Note that it follows from (IX, 6.8) that the set of multiplicative-type subgroups H of G is filtered upward. We show that it admits a largest element.

When G is smooth over k, one applies the structure theorem *BIBLE* 4 th. 4,

$$G = G_s \times G_u,$$

with G_s of multiplicative type and G_u "unipotent", which means here that G_u admits a composition series whose factors are subgroups (smooth if one insists) of G_a. (Indeed, G_u/G_u^0 is a unipotent group by *BIBLE* 4 cor. to th. 3, hence a p-group where p is the characteristic exponent, thanks to *BIBLE* 4 prop. 4; on the other hand, G_u^0 admits a composition series with connected smooth quotients of dimension 1 by *BIBLE* 6 th. 1 cor. 1, and these are isomorphic to G_a by *BIBLE* 7 th. 4.) Now one sees easily (cf. the lemma below) that every homomorphism from a multiplicative-type group to G_a, and hence also to G_u, is trivial, which proves indeed that every multiplicative-type subgroup of G is contained in G_s.

In the general case, consider the subgroup

$$G_0 = G_{red}$$

of G, which is smooth over k, hence by the foregoing admits a largest multiplicative-type subgroup `Z_0`. The multiplicative-type subgroups of G containing `Z_0` correspond to multiplicative-type subgroups of $G^0 = G/Z_0$. Now one has an exact sequence

$$0 \longrightarrow G_0/Z_0 \longrightarrow G/Z_0 \longrightarrow G/G_0 \longrightarrow 0,$$

where by the foregoing, G_0/Z_0 has no multiplicative-type subgroup except the unit group. Since a subgroup of a multiplicative-type group is of multiplicative type (IX, 8), it follows that for every multiplicative-type subgroup H of G/Z_0, one has $H \cap (G_0/Z_0) = 0$, hence $H \to G/G_0$ is injective. Since G/G_0 is a radicial algebraic group, hence finite over k, this implies that H is itself radicial and of rank bounded by that of G/G_0. This implies that the family of multiplicative-type subgroups of G admits a largest element, say Z.

I claim that G/Z has no multiplicative-type subgroup other than the unit subgroup. This follows from the fact (proved in 7.1.1) that a commutative extension of two multiplicative-type algebraic groups is of multiplicative type: if then Z' is the inverse image in G of a multiplicative-type subgroup of G/Z, then Z' is of multiplicative type, hence $Z' = Z$ by the maximal character of Z, hence $Z'/Z = 0$.

It now easily follows from the "principle of finite extension" that for any extension K of k, $(G/Z)_K = G_K/Z_K$ likewise has no multiplicative-type subgroup except the unit group.

We can now prove that Z is a reductive center of G. Indeed, let $u : H \to G_S$ be a homomorphism of S-groups, where S is a prescheme over k and H a multiplicative-type group of finite type over S; we prove that u factors through `Z_S`. This amounts to saying that the composite homomorphism $H \to G_S \to (G/Z)_S = G_S/Z_S$ is zero. Now setting $U = G/Z$, I claim that every homomorphism $u : H \to U_S$ is zero. Indeed, by (IX, 5.2) one is reduced to the case where S is the spectrum of a field, and by (IX, 6.8) this then follows from the fact that `U_K` has no multiplicative-type subgroup other than `0`. This completes the proof of 4.4. It only remains to give the proof of the

Lemma 4.4.1. *Let H be a multiplicative-type S-prescheme in groups; then every homomorphism of S-groups $u : H \to G_a$ is trivial.*

Indeed, consider the module $O_S^2 = E$ as an extension of $O_S = E'$ by $O_S = E''$; then G_a identifies with the prescheme of automorphisms of this extension, hence a homomorphism $u : H \to G_a$ identifies with an H-module structure on E respecting the extension structure, i.e. such that E' is stable under H and the operations induced by H on E' and E'' are trivial. By (I, 4.7.3) it follows that H operates trivially on E, hence u is trivial. QED.

Remark 4.4.2. *If G is a nonzero abelian variety over k, then G does not admit a reductive center in the sense of 4.1 with the restriction "G affine" omitted, because for n coprime to the characteristic, ${}_nG$ is étale over k of order prime to the characteristic, hence of multiplicative type; now the family of ${}_nG$ is schematically dense in G, so that if there were a reductive center, it would be identical to G, which is absurd, since G is not of multiplicative type. This is the reason why it is appropriate in 4.1 to impose the restriction that G have affine fibers.*

Lemma 4.5. *Let S be a prescheme, G a smooth S-prescheme in groups over S, affine over S, with connected fibers, T a maximal torus of G, $u : H \to G$ a central homomorphism, with H a multiplicative-type S-prescheme in groups of finite type. Then u factors through T.*

Indeed, let C be the centralizer of T, which is a closed group subpreschema of G smooth over S (XI, 5.3), hence affine over S. Since T is in the center of C, it is invariant, and one may consider the quotient group $C/T = U$, which is representable (VIII, 5.1). Since u is central, it factors through C, and everything reduces to proving that the composite homomorphism $H \to C \to U = C/T$ is trivial. By (IX, 5.2) one is reduced to the case where S is the spectrum of a field, which one may suppose algebraically closed. But then by *BIBLE* 6, th. 2, U is a connected algebraic group (smooth over k) which is "unipotent", meaning, as we have already observed, that it admits a composition series with quotients isomorphic to G_a. Hence every homomorphism from a multiplicative-type group of finite type H to U is trivial. This proves 4.5.

Corollary 4.6. *Let G be as in 4.5. If G admits a reductive center, the latter is contained in every maximal torus of G.*

Theorem 4.7. *Let S be a prescheme, G an S-prescheme in groups, smooth over S, affine over S, with connected fibers.*

a) For every $s \in S$, let $z(s)$ be the type of the reductive center of G_s (which is defined by 4.4). Order the set E of isomorphism classes of finitely generated $\mathbb{Z}$-modules, by declaring that the class of M is greater than that of M' if M' is

isomorphic to a quotient of M. Then the map $S \to E$, $s \mapsto z(s)$, is lower semicontinuous.

b) For G to admit a reductive center Z, it is necessary and sufficient that the preceding function $z : S \to E$ be locally constant. If this is so, G/Z is representable (cf. VIII, 5.1), and G/Z admits the unit subgroup as reductive center.

c) Suppose that the reductive rank of G is locally constant (cf. 1.7 b)). Then G admits a reductive center Z. If G/Z is representable (for example, G affine over S), then moreover the maximal tori T of G (resp. the Cartan subgroups C of G) and T' (resp. C') of $G' = G/Z$ are in bijective correspondence, to T (resp. C) corresponding $T' = T/Z$ (resp. $C' = C/Z$), and to T' (resp. C') corresponding $T = \varphi^{-1}(T')$ (resp. $C = \varphi^{-1}(C')$), where $\varphi : G \to G'$ is the canonical homomorphism.

d) Let T be a maximal torus of G, let g denote the Lie algebra of G, and consider the homomorphism

$$\theta : T \longrightarrow GL(g),$$

induced by the adjoint representation of G (II, 4). Then the kernel of θ is a reductive center of G.

Proof. a) and b). The proof of a) and of the first assertion of b) is essentially identical to that of 1.7 a) and b), and we refrain from reproducing the reasoning here. We signal only that one must in a) appeal to (IX, 5.6) (using the fact that G has connected fibers). Let us prove the second assertion of b), i.e. that if Z is a reductive center of G, then $G' = G/Z$ admits the unit subgroup as reductive center. Note immediately that by (VIII, 5.1) the quotient group G/Z is indeed representable; it is affine over S, of finite presentation over S (VIII, 5.8) and even smooth over S (for it suffices to see this on the fibers, G' being flat of finite presentation over S; now on the fibers we signaled in (VI_B, 9.2.(xii)) that a quotient of a smooth algebraic group over a field k is smooth); finally, G having connected fibers, the same holds for G'. Thus G' satisfies the same starting hypotheses as G. To see that the unit subgroup of G' is a reductive center, we may restrict if we wish to the case where S is the spectrum of a field (4.3). Let Z' be a central multiplicative-type subgroup of G'; everything reduces to proving that Z' is reduced to the unit group. Let Z_1 be the inverse image of Z' in G, and consider the operations of Z_1 on G induced by inner automorphisms of G. Since Z is central, Z operates trivially, so Z_1 operates by way of operations of Z' on G. Moreover Z being in the center of G, Z_1 and hence Z' operate trivially on Z; in addition Z' being in the center of G', the operations of Z' on the quotient $G/Z = G'$ are also trivial. Since Z is central, it follows at once that the operations of Z' on G come from a group homomorphism

$$u : Z' \to \mathrm{Hom}_{S\text{-}gr}(G', Z),$$

by

$$r(z') \cdot g = g \cdot u(z')(\phi(g)),$$

where $r : Z' \to \mathrm{Aut}_{S-gr}(G)$ is the envisaged representation of Z' by automorphisms of G, and $\varphi : G \to G'$ the canonical homomorphism. Now the datum of a group homomorphism u as above amounts to the datum of a group homomorphism

$$v : G' \to \mathrm{Hom}_{S\text{-}gr}(Z', Z),$$

on the other hand by (X, 5.8) the right-hand side is representable and is an étale group over S, hence its unit section is an open immersion, hence $\mathrm{Ker}\ v$ is an open subgroup of G'. Since G' has connected fibers, it is equal to G', hence v is zero, hence u is zero, hence Z' operates trivially on G, hence the same holds for Z_1, which is therefore central in G. Thus Z_1 is a commutative extension of a multiplicative- type group Z' by a multiplicative-type group Z, hence (since we are over a field) is a multiplicative-type group (7.1.1). Being central in G, it is contained in the reductive center Z, i.e. $Z_1 = Z$, whence $Z' =$ unit group, which completes the proof of b).

d) Let $Z = Ker\theta$, which is a multiplicative-type subgroup of T (e.g. by (IX, 6.8)). By 4.5, every central homomorphism $u : H \to G$, with H of multiplicative type and of finite type, factors through T, hence through Z. Since the formation of Z is compatible with any base change, it remains to prove that Z is central, i.e. that the centralizer C of Z equals G. Now by (XI, 5.3), C is a closed smooth subgroup of G; on the other hand, since Z operates trivially on g, one sees that

$Lie(C) = g$. One easily concludes that $C = G$: indeed, the immersion $C \to G$ is étale, since it is so fiber by fiber (being an unramified homomorphism of smooth algebraic groups of the same dimension, VI_B, 1.3), and one may apply (X, 3.5). Thus $C \to G$ is an étale closed immersion, hence an open immersion (SGA 1, I 5.1), and since it is also a closed immersion and G has connected fibers, it is an isomorphism.

e) By 1.7 b), G admits a maximal torus locally for the fpqc topology, hence by 4.2 and by d) which we have just proved, G admits a reductive center Z. We saw in d) that every maximal torus T of G contains Z. One sees at once that $T' = T/Z$ is a torus; to prove that it is a maximal torus in G', one is reduced by Definition 1.3 to the case where S is the spectrum of an algebraically closed field, and then the assertion is contained in *BIBLE* 7 th. 3 a). Conversely, let T' be a maximal torus of G', and let $T = \varphi^{-1}(T')$; we prove that T is a maximal torus of G. The question being local for the fpqc topology, we may suppose that G already admits a maximal torus, say `T_0`, so that by the foregoing, $T'_0 = T_0/Z_0$ is a maximal torus of G'. By the conjugation theorem 1.7 c), T' and T'_0 are locally conjugate for the fpqc topology, hence (since $\varphi : G \to G'$ is covering for this topology, so every section of G' lifts locally to a section of G) T and `T_0` are also locally conjugate. Since `T_0` is a maximal torus, the same therefore holds for T. One proceeds analogously for the Cartan subgroups. This completes the proof.

Let us give a useful translation of d), in the case where T is diagonalizable, hence of the form $D_S(M)$, where M is a finitely generated free $\mathbb{Z}$-module. Then (I, 4.7.3) the T-module g decomposes as a direct sum of T-submodules g_m ($m \in M$):

`g = ⨁_{m ∈ M} g_m,`

which are necessarily locally free. Suppose that for every $m \in M$, the rank of g_m is constant (which is the case in particular if S is connected). Then the set R of $m \in M$ such that $g_m \neq 0$ ("roots") is finite. With this stated:

Corollary 4.8. *Under the preceding conditions and with the preceding notations, the reductive center of G is the intersection of the kernels of the root characters $m \in R$ on T. One thus has an isomorphism*

$$Z \simeq D_S(N),$$

where N is the quotient of M by the subgroup generated by R.

Corollary 4.9. *Let G be an algebraic group over an algebraically closed field k. Suppose G smooth over k, connected affine, with reductive center reduced to the unit group, and that the Lie algebra g of G is nilpotent. Then G is "unipotent", i.e. admits a composition series with quotients isomorphic to G_a.*

By *BIBLE* 6, th. 4, cor. 3 it suffices to prove that a maximal torus T of G is reduced to the unit group, or equivalently that the Lie algebra t of T is reduced to `0`. Now it follows from the fact that the T-module g decomposes according to the characters

α of T (I, 4.7.3) that for every $t \in t$, the operation $ad_g(t)$ on g is semisimple. If then g is nilpotent, $ad_g(t)$ is zero. Now by 4.7 d), the reductive center of G being reduced to the identity element, the homomorphism $T \to GL(g)$ is a monomorphism, hence induces an injective application on the Lie algebras, which means (II, 4.5) that for $t \in t$, the relation $ad_g(t) = 0$ implies $t = 0$. This proves that $t = 0$ and completes the proof.

Proposition 4.10. *Let G be an affine, smooth, connected algebraic group over an algebraically closed field k. Then the reductive center of G is the intersection of the maximal tori of G.*

Of course, this is the intersection in the schematic sense (or equivalently, in the sense of subfunctors of G), i.e. the largest closed subpreschema of G majorized by the maximal tori of G. It follows from the noetherian character of G that this is also the intersection of a suitable finite set of maximal tori of G.

Let Z be the intersection in question; Z is a closed subgroup of a torus, hence of multiplicative type; on the other hand by 4.5 it contains the reductive center of G. To prove that it is equal, it remains to prove that it is central. Since G is connected, it suffices by (IX, 5.5) to prove that Z is invariant. Now by construction Z is invariant under the $int(g)$, with $g \in G(k)$, hence the normalizer N of Z (cf. VIII, 6.7) is a closed subgroup of G containing the rational points of G. Since G is reduced, one has $N = G$, which completes the proof.

Proposition 4.11. *Let S be a prescheme, G a smooth S-prescheme in groups over S, affine over S, with connected fibers, and of zero unipotent rank, which implies (1.9) that*

G is of locally constant reductive rank, hence (1.10) that the "prescheme of maximal tori of G" is defined, say $\mathcal{T}$, and is smooth, separated, of finite type over S. Let G operate on $\mathcal{T}$ via inner automorphisms, whence a homomorphism of group functors

$$u : G \longrightarrow \mathrm{Aut}_S(\mathcal{T}).$$

Under these conditions, the following three subfunctors of G are identical:

(i) The reductive center Z of G.

(ii) The center Z' of G.

(iii) The kernel $Z'' = Ker u$ of the preceding homomorphism.

In particular, the center of G is representable by a multiplicative-type subgroup of G.

Proof. One has trivially $Z \subset Z' \subset Z''$; it remains to prove that $Z'' \subset Z$, which amounts to proving (the hypotheses being stable under base change) that every section g of G over S which operates trivially on $\mathcal{T}$ is a section of Z. Introducing $G' = G/Z$ and using 4.7 b) and c), which imply in particular that the prescheme $\mathcal{T}'$ of maximal tori of G' is canonically isomorphic to $\mathcal{T}$, one is reduced to the case where $G = G'$, i.e. to the case where the reductive center Z of G is zero. (N.B. note that by 4.7 c), the unipotent rank of G' equals that of G, hence is zero since that of G is zero.) It thus remains to prove in this case that g is the unit section of G. The usual reduction procedure brings us to the case where S is noetherian, and even to the case where S is local artinian (since to verify that g is the unit section it suffices to verify when S is locally noetherian that this is so after any base change $S' \to S$ with S' local artinian). Now in this case the kernel Z'' of u is representable (VIII, 6.2 a) and 6.5 c)) (Note that Z'' is representable by (XI, 6.8)). To prove that Z'' is reduced to the unit subgroup, it suffices by Nakayama to prove that the same holds for its fiber Z''_0. This thus reduces us to the case where S is the spectrum of a field k, which one may evidently suppose algebraically closed. Now Z'' is contained in the stabilizer of every point of $\mathcal{T}(k)$, i.e. in the normalizer $N(T)$ of every maximal torus T of G. Since the reductive rank of G is zero, it follows by (XI, 5.9) that T is an open subgroup of $N(T)$, hence the Lie algebra of $N(T)$ is identical to that of T. Hence the Lie algebra of Z'' is contained in that of T. On the other hand, it follows from 4.10 that the intersection of the Lie algebras of the maximal tori T of G is none other than the Lie algebra of the reductive center Z, hence here zero, since we have assumed $Z = 0$. Consequently, the Lie algebra of Z'' is zero, i.e. Z'' is étale over k. Moreover Z'' is evidently invariant in G, and since G is connected it follows easily that Z'' is in the center of G. It is therefore in $T = Centr_G(T)$ for every maximal torus T, hence in the intersection of the maximal tori, i.e. in $Z = 0$ by 4.10, which completes the proof.

5. Application to the scheme of multiplicative-type subgroups[1077]

Theorem 5.1. *Let S be a prescheme, G a smooth and affine S-prescheme in groups over S, M the "prescheme of multiplicative-type subgroups of G", representing the functor explicit in (XI, 4.1). For every integer $n > 0$, let T_n be the subfunctor of M such that $T_n(S')$ = the set of multiplicative-type group subpreschemata H of $G_{S'}$ such that $n \cdot id_H = 0$, so that T_n is representable, and is affine over S (XI, 3.12 a)). Let $u_n : M \to T_n$ be the morphism defined by $u_n(H) = {}_nH$ (where ${}_nH = Ker(n \cdot id_H)$). With these notations, one has the following:*

a) Every subpreschema U of M of finite type over S is contained in a closed subpreschema of finite type over S, and every closed subpreschema X of M of finite type over S is affine over S.

b) Suppose S quasi-compact, and let X be a closed subpreschema of M of finite type over S. Then there exists an integer $n > 0$ such that for every multiple m of n, the induced morphism

$$u_m|X : X \longrightarrow T_m$$

[1077] For a generalization of the results of the present number to the case where G is not supposed affine over S, cf. M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes*, Chap. IX.

is a closed immersion.

Proof. a) To prove the first assertion of a), one takes for X the schematic closure of U (EGA I, 9.5.1 and 9.5.3), which is defined since the immersion $i : U \to M$ is quasi-compact (because U is of finite type over S and M is separated over S (4.1)). It therefore remains to prove that such an X is affine over S, which will prove at the same time the second assertion of a). In the foregoing form, one sees that the question is local on S, which one may therefore suppose affine. Then U, being of finite type over S, is contained in a quasi-compact open of M, and this brings us to the case where U itself is an open, of finite type over S, i.e. quasi-compact. (N.B. A closed subscheme of a scheme affine over S is affine over S.)

It suffices to prove that such a U is majorized by a closed subpreschema of M which is affine. In this form, the usual reduction procedure brings us at once to the case where S is noetherian. One proceeds similarly for b), which reduces to the case where S is affine noetherian.

Let us take up again the inverse limit T of the T_n used in the proof of (XI, 4.1), which is an affine prescheme (but not of finite type) over S, and whose local rings are noetherian, as was seen at the beginning of the proof of (IX, 3.7). (N.B. For $t \in T$, $t = (t_n)$, the transition morphisms $T_m \to T_n$ being smooth and the dimension of the T_m being bounded, there exists an n_0 such that the $T_n \to T_{n_0}$ are étale at t_n for every n multiple of n_0.) The proof of *loc. cit.*, or (XI, 3.11), show that the canonical morphism

$$u : M \longrightarrow T$$

is an immersion, and induces isomorphisms on the local rings (but one will note that u is not in general an open immersion nor a quasi-compact morphism). Let U be a quasi-compact open of M; we shall prove that its closure in T is contained in M, which proves (if X denotes the schematic closure of U in M) that $u|X : X \to T$ is a closed immersion, hence that X is affine, and that will prove a). Since U is noetherian, it has only finitely many irreducible components, and every element $t \in T$ in the closure of U is a specialization of an element x of U. Since the local ring of t in T is noetherian, the same holds for the local ring A of t in $\bar{x}$ endowed with the induced reduced structure.

The canonical morphism of the noetherian integral local scheme $S' = \mathrm{Spec}(A)$ into T then sends the closed point of S' to t, the generic point to $x \in M$, and one must show under these conditions that $t \in M$. Possibly replacing A by the quotient of a suitable complete local A-algebra flat over A (EGA 0_III, 10.3.1) by a minimal prime ideal, we may suppose that A is complete with algebraically closed residue field, (and we could even reduce to the case where it is moreover a discrete valuation ring thanks to EGA II, 7.1.7). One is thus reduced (making the change of notation: S' denoted by S) to the

Lemma 5.2. *Let S be the spectrum of a complete noetherian integral local ring with algebraically closed residue field, G a smooth and affine S-prescheme in groups* *over S, $(H(n))_{n>0}$ a system of multiplicative-type subgroups of G, indexed by the integers $n > 0$, η the generic point of S. Suppose:*

a) If m is a multiple of n, one has $H(n) = {}_nH(m)$.

b) There exists a multiplicative-type subgroup H_η of G_η such that one has $H(n)_\eta = {}_n(H_\eta)$ for every $n > 0$.

Under these conditions, there exists a multiplicative-type subgroup H of G such that for every $n > 0$, one has $H(n) = {}_nH$.

For each n, let $C_n = Centr_G(H(n))$, which is a closed group subpreschema of G, smooth over S (XI, 5.3). The set of integers $n > 0$ being ordered by divisibility, the C_n form a decreasing family of closed subpreschemata of G, and since G is noetherian, this family is stationary. Let C be the common value of the C_n for n large. Then C satisfies the same conditions as G; moreover the $H(n)$ are in fact subgroups of C. Hence, replacing G by C, we may suppose that the $H(n)$ are central subgroups of G.

Let then s be the closed point of S, and Z_s the reductive center of G_s (defined thanks to 4.4). By the definition (4.1) Z_s contains the $H(n)_s$. Since A is complete, Z_s comes from a multiplicative-type group subpreschema Z of G (XI, 5.8). I claim that Z contains the $H(n)$. Indeed, since $H(n)$ is central, it commutes with Z in G, hence one deduces a group homomorphism $Z \times H(n) \to G$, which by (IX, 6.8) admits an image group which is a multiplicative-type subgroup K of G containing Z, and everything reduces to proving that $K = Z$, which follows from $K_s = Z_s$ and (IX, 5.1 bis).

One is thus reduced to proving the analogue of 5.2, but with G replaced by a group Z of multiplicative type and of finite type over S (not necessarily smooth over S). Since A is complete with algebraically closed residue field, Z is diagonalizable (X, 3.3 and 1.4), hence of the form $D_S(M)$, with M a finitely generated commutative group. Consequently every multiplicative-type subgroup H of Z is diagonalizable (IX, 2.11 (i)), hence of the form $D_S(N)$, where N is a quotient group of M (VIII, 3.2). Thus the $H(n)$ correspond to quotient groups $M(n)$ of M, and the existence of a multiplicative-type subgroup H of Z such that $H(n) = {}_nH$ for every n amounts to that of a quotient group N of M such that $M(n) = N/nN$ for every n. Now this follows at once from the fact that there exists a subgroup H_η of G_η such that $H(n)_\eta = {}_n(H_\eta)$ for every n. This completes the proof of 5.2 and hence of a).

b) We already know that $u|X : X \to T$ is a closed immersion. It easily follows that for m large, the composite $u_m|X : X \to T_m$ is also a closed immersion. Since we will not need this fact in the sequel, we omit the details of the proof.

Corollary 5.3. *With the notations of 5.1, let U be a part of M both open and closed, of finite type over S. Then U is affine over S for the structure induced by M, and if S is quasi-compact, there exists an integer $n > 0$ such that for every multiple m of n, the induced morphism $u_m|U : U \to T_m$ is an open and closed immersion (i.e. an isomorphism onto an open and closed part of T_m, equipped with the induced structure).*

The first assertion follows from 5.1 a), the second from 5.1 b), taking into account that $u_m : M \to T_m$ is smooth (XI, 2.2 bis) and that a smooth immersion, i.e. étale, is an open immersion.

Corollary 5.4. *Let S be a prescheme, G a smooth and affine S-prescheme in groups over S, of locally constant reductive rank (1.7 b)), $\mathcal{T}$ the "prescheme of maximal tori of G" (1.10). Then $\mathcal{T}$ is smooth and affine over S. If T is a maximal torus of G, $N(T)$ its normalizer, then $G/N(T)$ (XI, 5.3 bis) is affine over S. The same holds for $G/C(T)$ (where $C(T)$ is the centralizer of T) provided that $W(T) = N(T)/C(T)$ is finite over S (cf. 2.1 b)).*

The second assertion is contained in the first, since by the conjugation theorem, $G/N(T)$ is isomorphic to $\mathcal{T}$ (XI, 5.5 bis). For the first assertion, one notes that by construction, $\mathcal{T}$ is isomorphic to an open and closed subpreschema of M (for one may suppose the reductive rank of G constant and equal to r, and then $\mathcal{T}$ is the subpreschema of M corresponding to the subtori of relative dimension r, i.e. the largest subpreschema of M over which the "universal multiplicative-type subgroup" $H \subset G_M$ is a torus of relative dimension r). One may therefore apply 5.3. Finally, for the last assertion, one notes that $G/C(T)$ is finite over $G/N(T) \simeq (G/C(T))/W(T)$, hence is affine since $G/N(T)$ is.

Corollary 5.5. *Let G be a smooth and affine algebraic group over a field k. Then the scheme M of multiplicative-type subgroups of G is a direct sum of affine schemes over S. For every multiplicative-type subgroup H of G, if C and N denote respectively its centralizer and its normalizer, the quotients G/C and G/N are affine.*

Using (XI, 5.1 bis), one sees that the saturation under G operating on M of any finite closed part of M is open: indeed, one is reduced to the case where k is algebraically closed, hence to the case of the trajectory of a k-rational point x, but then by *loc. cit.* the morphism $g \mapsto g \cdot x$ from G to M is smooth hence open, hence its image is open. Let U be the union of the classes of the closed points of M for the equivalence relation defined by the operations of G. Then U is open and contains every closed point of M, hence by the Nullstellensatz is identical to M. Thus M is a disjoint union of opens, which are therefore necessarily closed, hence M is the sum prescheme of preschemata M_i, each of which is a G-trajectory of a closed point, hence is quasi-compact, hence of finite type. By 5.3 the M_i are therefore affine. If H is a multiplicative-type subgroup of G, it corresponds to a k-rational point x of M, and G/N identifies with the trajectory of x under G (XI, 5.5 bis). It is therefore affine by the foregoing. Since C is an open subgroup of N (XI, 5.9), G/C is finite over G/N (since the latter is isomorphic to $(G/C)/(N/C)$), hence affine since G/C is.

This proof shows at the same time:

Corollary 5.6. *Under the conditions of 5.5 for G and H, the subscheme U of M "of multiplicative-type subgroups of G that are locally conjugate to H" is an open and closed subpreschema of G.*

In picturesque language, every multiplicative-type subgroup H' of G that is a limit of subgroups of G conjugate to H, is itself conjugate to H.

Remarks 5.7. *Let S be a prescheme, G a smooth and affine S-prescheme over S, H an S-prescheme of multiplicative type and of finite type, and set $M = \mathrm{Hom}_{S-gr}(H, G)$, which by (XI, 4.2) is representable and is smooth over S and separated over S. One can then prove for M a result entirely analogous to 5.1, either by reducing to 5.1 by an argument analogous to that of (XI, 4.2), or by proceeding directly by an argument modeled on that of 5.1. From this one deduces corresponding variants for 5.3, 5.5, 5.6, which the reader will formulate.*

6. Maximal tori and Cartan subgroups of not-necessarily-affine algebraic groups (algebraically closed base field)

Lemma 6.1. *Let G be a connected algebraic group over a field k, Z an algebraic subgroup of finite index in its center; then G/Z is affine.*

One may suppose that Z is the center of G, since a scheme finite over an affine scheme is affine. Consider the vector spaces

```
P_n = P_n(G) = O_{G,e}/m_{G,e}^{n+1}   (n an integer □ 0),
```

where $m_{G,e}$ is the maximal ideal; then G operates on the $P_n(G)$ by the adjoint representation, and if Z_n is the kernel of the corresponding homomorphism

$$ad_n : G \longrightarrow GL(P_n),$$

one verifies easily (using the fact that G is connected) that Z is the intersection of the Z_n, hence (G being noetherian) equal to one of the Z_n. But ad_n defines, on passage to the quotient, a monomorphism $G/Z_n \to GL(P_n)$, which is therefore a closed immersion, hence G/Z_n is affine, and consequently G/Z is.

Lemma 6.2. *Let G be a smooth algebraic group over the field k, Z a central algebraic subgroup, $G' = G/Z$, $u : G \to G'$ the canonical homomorphism, T a multiplicative-type subgroup in G, $T' = u(T)$ the image group, C the centralizer of T in G, C' that of T' in G'. Then one has*

$$C'^0 \subset u(C).$$

One may suppose k algebraically closed. Let

$$C_1 = (u^{-1}(C')^0)_{red},$$

it suffices to prove that $C_1 \subset C$, since $u(C_1)$ is connected of finite index in C' hence equal set-theoretically to C'^0, hence equal to C'^0 since C' and consequently C'^0 are smooth.

Consider the morphism

$$(g,t) \mapsto gtg^{-1}t^{-1}$$

from $G \times G$ to G; it induces a morphism

$$\varphi : C_1 \times T \to Z_1, \quad \text{where } Z_1 = Z_{red},$$

since the left-hand side being reduced, it suffices to see that for $g \in C_1(k)$, $t \in T(k)$, one has $gtg^{-1}t^{-1} \in Z(k)$, which comes from the fact that g centralizes T modulo Z. One sees easily (by computing on k-rational points) that $\varphi(g,t)$ is additive in g and additive in t, hence is "bilinear"; I claim that (with Z_1 smooth and C_1 connected) this homomorphism is identically zero, which will prove indeed that $C_1 \subset C$. Using the density theorem for T, we are reduced to the case where T is finite, i.e. where there exists an integer $n > 0$ such that $n \cdot id_T = 0$. Note that φ is defined by a group homomorphism

$$C_1 \longrightarrow \mathrm{Hom}_{S-gr}(T, Z_1),$$

now Z_1 being commutative and smooth, the right-hand side is representable by an étale algebraic group over k, and C_1 being connected, every group homomorphism from C_1 to the latter is zero. QED.

Corollary 6.3. *Under the preceding conditions, suppose C' connected; then one has*

$$C' = u(C), \quad C = u^{-1}(C').$$

Indeed, then $C' = C'^0$; on the other hand C obviously contains Z, hence is equal to $u^{-1}(u(C))$.

Lemma 6.4. *Let G be an algebraic group over the field k, Z a central algebraic subgroup such that $G/Z = G'$ is a torus. Then G is commutative, and if k is algebraically closed, there exists a torus T in G such that $u(T) = G'$, where u is the canonical homomorphism $G \to G/Z = G'$.*

One may suppose k algebraically closed. Consider again the morphism $G \times G \to G$ defined by $(g,h) \mapsto ghg^{-1}h^{-1}$; then (since Z is central) this morphism factors through a morphism $G' \times G' \to G$, and since G' is commutative, this latter takes its values in Z, and even in $Z_1 = Z_{red}$, since $G' \times G'$ is reduced. One sees as above that the morphism $\varphi : G' \times G' \to Z_{red}$ thus obtained is bilinear, hence zero, which proves that G is commutative. To find a torus T of G such that $u(T) = G'$, one may (replacing G by G^0_{red} if necessary) suppose G smooth and connected, and moreover (replacing Z by Z^0_{red} if necessary) that Z is smooth and connected. By a well-known theorem of Chevalley,[1078] Z is an extension of an abelian variety by an affine group, which reduces us at once to proving our assertion in each of the two following cases: 1°) Z is affine, 2°) Z is an abelian variety. In case 1°), G is affine and the

[1078] N.D.E.: references to be given here.

result is well known (*BIBLE* 7 th. 3 a)). Suppose then that Z is an abelian variety. Since every homomorphism from the additive group G_a to the torus $G' = R$ or to the abelian variety Z is trivial, it follows that the same holds for every homomorphism from G_a to G, hence G does not contain a subgroup isomorphic to G_a. By Chevalley's structure theorem already invoked, one has an exact sequence

$$(*) 0 \longrightarrow T \longrightarrow G \longrightarrow A \longrightarrow 0,$$

where A is an abelian variety, and T an affine smooth connected group. Since the latter is commutative and contains no additive subgroup, it follows that T is a torus (and it is evidently the unique maximal torus of G). Everything reduces to proving that every epimorphism

$$u : G \longrightarrow R,$$

where R is a torus, satisfies $u(T) = R$. Set

$\mathrm{Hom}_{gr}(T, G_m) = M, \quad \mathrm{Hom}_{gr}(R, G_m) = P,$

(these are finitely generated free $\mathbb{Z}$-modules which recover T, R up to isomorphism by $T = D_k(M)$, $R = D_k(P)$), and let

$$B = Ext^1_{k-gr}(A, G_m),$$

(B is also the set of k-rational points of the abelian variety dual to A). One evidently has

$(**) \quad \mathrm{Ext}^1(A, T) = \mathrm{Hom}_{gr}(M, B), \quad \mathrm{Ext}^1(A, R) = \mathrm{Hom}_{gr}(P, B),$

in particular the extension G of A by T is given by a homomorphism

$$\theta : M \longrightarrow B.$$

On the other hand the exact sequence $(*)$ gives the exact sequence

$0 \to \mathrm{Hom}(A, R) \to \mathrm{Hom}(G, R) \to \mathrm{Hom}(T, R) \to \mathrm{Ext}^1(A, R),$

moreover $\mathrm{Hom}(A, R) = 0$, and $\mathrm{Hom}(T, R) \to Ext^1(A, R)$ identifies with the homomorphism

$\mathrm{Hom}(P, M) \to \mathrm{Hom}(P, B)$

deduced from $\theta : M \to B$. Setting

$$N = Ker(\theta),$$

one therefore finds a canonical bijection

Hom(G, R) ≃ Hom(P, N) ≃ Hom(S, R),   where S = D_k(N),

which can be described geometrically at once as follows:

Lemma 6.5. *Let G be an extension of an abelian variety A by a torus $T = D_k(M)$, defined by a homomorphism $\theta : M \to B = Ext^1(A, G_m)$ (base field k algebraically closed). Let $N = Ker\theta$, $S = D_k(N)$ the corresponding torus, isomorphic to T/U where $U = D_k(M/N)$; consider the extension $H = G/U$ of A by S. This extension splits,*[1079] *hence one has a unique projection of H onto S, whence a unique homomorphism*

$$v : G \longrightarrow S$$

extending the canonical homomorphism $T \to S$. With this stated, for every torus R and every homomorphism $u : G \to R$, there exists a unique homomorphism $u' : S \to R$ such that $u = u'v$. In particular, one has $Im(u) = Im(u') = u(T)$.

This shows in particular that if u is an epimorphism, the same holds for its restriction to T, which completes the proof of 6.4.

Theorem 6.6. *Let G be a smooth and connected algebraic group over an algebraically closed field k.*

a) The maximal tori of G are conjugate.

b) Let T be a torus in G; then its centralizer is smooth and connected.

c) The map $T \mapsto Centr_G(T)$ establishes a bijective correspondence between the set of maximal tori T of G and the set of Cartan subgroups (N° 1) of G. For an algebraic subgroup C of G to be a Cartan subgroup, it is necessary and sufficient that it be smooth, nilpotent, and set-theoretically equal to its connected normalizer (and then it is even equal to its connected normalizer); one then has $C = Centr_G(T)$, where T is the unique maximal torus of C, and $Norm_G(C) = Norm_G(T)$.

d) Let $v : G \to H$ be an epimorphism of smooth connected algebraic groups; then the maximal tori (resp. the Cartan subgroups) of H are the images of the maximal tori (resp. of the Cartan subgroups) of G. If T is a maximal torus of G and C its centralizer, then $v(C)$ is the centralizer of $v(T)$.

e) Under the conditions of d), suppose that $\mathrm{Ker}\ v$ is a central subgroup of G; then $T \mapsto v(T)$ (resp. $C \mapsto v(C)$) establishes a bijective correspondence between the set of maximal tori (resp. the Cartan subgroups) of G and the set of maximal tori (resp. Cartan subgroups) of H. The Cartan subgroups of G contain the center of G and a fortiori $\mathrm{Ker}\ v$, and are the groups of the form $v^{-1}(C')$, where C' is a Cartan subgroup of H.

Let us also state at once the following immediate consequence of c):

[1079] N.D.E.: i.e. splits!

Corollary 6.7. *For G to be nilpotent (i.e. the group $G(k)$ nilpotent) it is necessary and sufficient that the tori in G be central, or equivalently that G have only a single maximal torus (and then this latter is the largest subtorus of G).*

Proof of 6.6. Let Z be a central algebraic subgroup of G, let $G' = G/Z$, and $u : G \to G'$ the canonical homomorphism. Then G' is a smooth and connected group. If T' is a subtorus of G', it follows from 6.4 that $u^{-1}(T')$ is commutative and that T' is the image of a subtorus of $u^{-1}(T')$, hence of a subtorus of G. Since $u^{-1}(T')$ is commutative, it obviously admits a largest subtorus T (since the sum of two subtori gives a third containing both), and one therefore has $u(T) = T'$. From this it follows immediately that for every maximal torus T of G, its image $T' = u(T)$ is a maximal torus of G', and that $T \mapsto u(T)$ is a bijective correspondence between maximal tori of G and maximal tori of G'.

We now make

$$Z = Centr(G)^0_{red},$$

then G' is affine by 6.1. Since the maximal tori of G' are then conjugate, the same holds for those of G, which proves a). Moreover, for G to be nilpotent, resp. have only a single maximal torus, it is necessary and sufficient that G' satisfy the same condition; now G' being affine, the two conditions in question on G' are equivalent (*BIBLE* 6 th. 4 cor. 2), hence the same holds for the conditions in question on G. Moreover, if G has only a single maximal torus T, this latter is invariant hence central, and since every torus in G is contained in a maximal torus, it is central. Conversely, if every torus is central, the same holds for the maximal tori, and by the conjugation theorem a) there is only a single maximal torus. This proves 6.7.

Let T be any torus of G, $T' = u(T)$; then $C' = Centr_{G'}(T')$ is connected (*BIBLE* 6 th. 6 a)), hence by 6.3 the centralizer C of T is equal to $u^{-1}(C')$, hence connected (since Z is connected), which proves b). If T is maximal, hence T' maximal, then we know that C' is nilpotent, hence C (which is a central extension of C') is nilpotent. Moreover, T is a maximal torus of C, hence by 6.7 it is the unique maximal torus of C; consequently the map $T \mapsto Centr_G(T)$ from the set of maximal tori of G into the set of Cartan subgroups is bijective. Moreover one has

```
Centr_G(T) = C ⊂ Norm_G(C) ⊂ Norm_G(T)
```

and since we know that the centralizer of T is of finite index in its normalizer (cf. XI, 5.9, whose reasoning is valid without affine hypothesis, using only the representability of the two functors in question, as was signaled in (XI, 6.5)), and that C is smooth and connected, we conclude

$$C = Norm_G(C)^0.$$

Moreover, by the bijective correspondence between maximal tori and Cartan subgroups, we see that $Norm_G(C)$ and $Norm_G(T)$ have the same k-valued points, and since the latter is smooth, one has

$$Norm_G(T) = Norm_G(C).$$

To complete the proof of c), it remains to prove that if C is a smooth nilpotent connected subgroup of G of finite index in its normalizer, then C is a Cartan subgroup. Now since Z is central, the normalizer of C contains Z, and since Z is smooth and connected, we conclude $Z \subset C$, whence $C = u^{-1}(C')$, where $C' = u(C)$. One then has

$$Norm_G(C) = u^{-1}(Norm_{G'}(C')),$$

which proves that C' is nilpotent connected of finite index in its normalizer, hence by *BIBLE* 7 th. 1 it is a Cartan subgroup of G', whence at once that C is a Cartan subgroup of G.

Let us prove e): we are under the conditions of the start of the proof (setting $Ker u = Z$, $G' = H$); we have already seen that $T \mapsto u(T)$ is a bijective correspondence between maximal tori of G and maximal tori of G'. Taking into account the bijective correspondence between maximal tori and Cartan subgroups just proved, we deduce a bijective correspondence between Cartan subgroups C of G and Cartan subgroups C' of G', by making correspond to $C = Centr_G(T)$ the group $C' = Centr_{G'}(T')$, where $T' = u(T)$. Since C' is connected by b), it follows by 6.3 that $C' = u(C)$, and $C = u^{-1}(C')$, which proves e).

It remains to prove d). Let T be a maximal torus of G; we prove that $v(T)$ is a maximal torus of H (which, taking into account the conjugation theorem a), will imply that the maximal tori of H are all of the form $v(T)$ as above). Let then R be a torus in H containing $v(T)$; we prove $R = v(T)$. Replacing H by R and G by $v^{-1}(R)^0_{red}$, we may suppose $R = H$, i.e. that H is a torus, and we are reduced to proving that then $v(T) = H$. Let again $Z = Centr(G)^0_{red}$, $G' = G/Z$, and $H' = H/v(Z)$:

$$\begin{tikzcd} e \arrow[r] & Z \arrow[r] \arrow[d, "v''"] & G \arrow[r, "u"] \arrow[d, "v"] & G' \arrow[r] \arrow[d, "v'"] & e \\ e \arrow[r] & v(Z) \arrow[r] & H \arrow[r, "u'"] & H' \arrow[r] & e. \end{tikzcd}$$

We already know that $u(T)$ is a maximal torus of G', and since G' is affine, so $v' : G' \to H'$ is an epimorphism of affine smooth connected groups, $v'(u(T))$ is a maximal torus of H' (*BIBLE* 7 th. 3 a)) hence equal to H', i.e. $u'(v(T)) = H'$; hence to prove $v(T) = H$ it suffices to show that $v(T) \supset v(Z)$. Now $v(Z)$ is a smooth connected subgroup of H, hence a torus, and $v'' : Z \to v(Z)$ is an epimorphism, hence by 6.4 one has $v(Z) = v''(S)$, where S is a torus of Z. Now S, being central in G, is evidently contained in the maximal torus T, whence $v(Z) \subset v(T)$. This proves assertion d) in the case of maximal tori.

Taking into account the bijective correspondence between maximal tori and Cartan subgroups, it remains to prove that if T is a maximal torus of G and C its centralizer, then $v(C)$ is the centralizer of $v(T)$. For this, take up again the diagram (D) above (where of course H is no longer supposed to be a torus), let $T' = u(T)$, $C' = u(C)$; we have already seen in e) that C' is the centralizer of the maximal torus T', hence (G' being affine, hence H' being affine) $v'(C')$ is the centralizer of the maximal torus $v'(T')$ of H' (*BIBLE* 7 th. 3 a)), i.e. $u'(v(C))$ is the centralizer of $u'(v(T))$; since C contains Z, $v(C)$ contains $v(Z)$; $v(C)$ is therefore the inverse image by u' of $u'(v(C))$, i.e. of the centralizer of $u'(v(T))$, hence is the centralizer of $v(T)$ as follows from e) applied to $u' : H \to H'$ and to the maximal torus $v(T)$ of H. This completes the proof of 6.6.

7. Application to not-necessarily-affine smooth group preschemes

Theorem 7.1. *Let S be a prescheme, G an S-prescheme in groups smooth, separated and of finite type over S. Suppose that G admits, locally for the faithfully flat quasi-compact topology, a maximal torus. Then:*

a) The map

$T \mapsto \mathrm{Centr}_G(T)$

induces a bijection from the set of maximal tori of G onto the set of Cartan

subgroups of G. If C corresponds to T, then T is the unique maximal torus of C.

b) Let T, T' be two maximal tori of G, C, C' the corresponding Cartan subgroups; then one has

$\mathrm{Transp}_G(T, T') = \mathrm{Transp}_G(C, C') = \mathrm{Transp}_G(T, C')$,

the first two terms are also identical to the strict transporters; finally the functor in question is representable by a closed subpreschema of G smooth over S. The tori T, T' and the Cartan subgroups C, C' are conjugate locally for the étale topology.

c) There exists, locally for the étale topology, a maximal torus of G and a Cartan subgroup of G.

d) Suppose that every finite part of a fiber G_s of G is contained in an affine open of G (for example G quasi-projective over S, or S artinian); then the functor $\mathcal{T} : (Sch)^{\circ}/S \to (Ens)$ defined in 1.10 (functor of maximal tori of G), isomorphic to the functor $\mathcal{C} : (Sch)^{\circ}/S \to (Ens)$ of Cartan subgroups of G, is representable by a smooth, separated prescheme of finite type over S, which is quasi-projective over S when G is, and is affine over S when G is affine over S, or when S is artinian.

e) Let $u : G \to G'$ be a morphism of S-preschemata in groups, where G' is smooth, separated of finite type over S, and suppose that for every $s \in S$, one has $u_s(G_s) = G'_s$, i.e. u_s is faithfully flat (Exp. VI[1080]). Then for every maximal torus T of G, $u(T)$ is a maximal torus of G'; if G' has connected fibers, then for every Cartan subgroup C of G, $u(C)$ is a Cartan subgroup of G', and if C is the centralizer of T, $u(C)$ is the

[1080] N.D.E.: reference not located in VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), proposition II.5.1.(c).

centralizer of $u(T)$. In either case, the induced morphism $T \to u(T)$, resp. $C \to u(C)$, is faithfully flat.

f) Under the conditions of e), suppose moreover that $\mathrm{Ker}\ u$ is a central

subgroup of G. Then $T \mapsto u(T)$ is a bijective map from the set of maximal tori of G onto the set of maximal tori of G', and if G' has connected fibers, $C \mapsto u(C)$ is likewise a bijective map from the set of Cartan subgroups of G onto the set of Cartan subgroups of G'; if $C' = u(C)$, one has $C = u^{-1}(C')$.

Remarks 7.2. *We shall see in XV that the conclusion of d) remains valid without the restrictive condition on the finite parts of the G_s. We shall also prove there the conclusions concerning only the Cartan subgroups contained in b), c), d), e), when one supposes only that G admits, locally for the faithfully flat quasi-compact topology, a Cartan subgroup (but not necessarily a maximal torus). For this we shall moreover need to use 7.1 in the case where S is artinian. Moreover, the proof of c) and d) (in the case S not artinian) is considerably simplified by using the method of XV.*

Proof of 7.1. a) Proceeding as in 3.2, one sees that for every maximal torus T of G, $C = Centr_G(T)$ is indeed a Cartan subgroup of G, and T is determined in terms of C as the unique maximal torus of C, so that it remains only to show that every Cartan subgroup C of G is defined by a maximal torus of G, or equivalently that C admits a central maximal torus. The question being still local for the fpqc topology, we may suppose that G admits a maximal torus T. Then by (XI, 6.2), $Transp_G(T, C)$ is representable by a closed subpreschema S' of G smooth over S; moreover $S' \to S$ is surjective, as follows from 6.6 c). Hence, replacing S by S', we may suppose that there exists a section g of G such that $ad(g) \cdot T \subset C$; but then $ad(g) \cdot T$ is a maximal torus of C, which is central fiber by fiber by 6.6 c), hence central by (IX, 5.6 b)). QED.

b) Let T, T' be two maximal tori of G, and C, C' the corresponding Cartan subgroups of G. Then the following conditions are equivalent:

(1) $T \subset T'$ (2) $T = T'$ (3) $T \subset C'$ (4) $C \subset C'$ (5) $C = C'$.

This follows trivially from a). Using the same result after arbitrary base change, one deduces the identity stated in b) between various transporters and strict transporters. Moreover, we already noted in a) that $Transp_G(T, C')$ is representable by a closed subpreschema of G, smooth over S, and that its structural morphism is surjective. Consequently, by Hensel there exists locally for the étale topology a section of this prescheme over S, hence T and T' on the one hand, C and C' on the other hand, are conjugate locally for the étale topology, which proves b).

c) Suppose first that S is artinian, local. When G admits a maximal torus T, then it follows from the conjugation theorem proved in b) that the functor $\mathcal{T}$ of maximal tori of G is representable by the homogeneous space G/N, where N is the normalizer of T in G, which is smooth by b). Moreover, as we noted in (VI_A, 3.2.1), since $\mathcal{T}$ is a homogeneous space under G, every finite part of $\mathcal{T}$ is contained in an affine open. In the general case, there exists a finite extension k' of the residue field k of S such that $G_{k'}$ has a maximal torus; then k' comes from k by a finite flat base change $S' \to S$, and the maximal torus of $G_{k'}$ lifts to a maximal torus of $G_{S'}$ (XI, 2.1 bis), so the functor $\mathcal{T}_{S'}$ is representable by a prescheme over S', smooth separated and of finite type over S', every finite part of which is contained in an affine open. Hence the natural descent datum on $\mathcal{T}_{S'}$ is effective, hence $\mathcal{T}$ is representable, and by descent one sees that $\mathcal{T}$ is smooth over S, separated and of finite type over S. From smoothness it follows, thanks to Hensel, that $\mathcal{T}$ admits a section locally for the étale topology. This proves c) and d) in the case S artinian (N.B. we shall prove below that in this case, $\mathcal{T}$ is in fact affine over S).

Suppose now S arbitrary. To prove c) and d), which are assertions local on S for the Zariski topology, we may suppose that there exists a prime ℓ prime to the residue characteristics of S, that S is affine, and that the reductive rank of the fibers of G (which is evidently locally constant, thanks to the hypothesis of local existence for fpqc of a maximal torus) is constant, say r. Let $S' \to S$ be a faithfully flat and quasi-compact morphism, S' affine, such that $G_{S'}$ admits a maximal torus `T_0`. Let `C_0` be its centralizer; I claim that there exists a suitable power n of ℓ such that one has also $C = Centr_G({}_nT)$. Indeed, to see this, one is reduced at once to the case S' noetherian, where this was seen in (XI, 6.2). Fix n thus, set

```
M = (ℤ/nℤ)^r,    P = Hom_{S-gr}(M_S, G),
```

then P is evidently representable as a closed subpreschema of finite presentation of $({}_nG)^r$, where

```
_n G = Hom_{S-gr}((ℤ/nℤ)_S, G),
```

which is also the kernel of the n-th power morphism on G, hence representable by a closed subpreschema of finite presentation of G. Moreover P is smooth over S by (XI, 2.1). Let $s \in S$; by what was seen above, there exists a finite separable extension k'' of $\kappa(s)$ such that there exists a maximal torus T'' in $G_{k''}$; moreover, since ${}_nT''$ is étale over k'' (n being prime to the characteristic of k'') we may suppose (replacing k'' by a finite separable extension if necessary) that ${}_nT''$ is isomorphic to $M_{k''}$. Let $S'' \to S$ be an étale morphism such that there exists $s'' \in S''$ above $s \in S$, giving rise to the residual extension $\kappa(s'') \simeq k''$. One thus has a section of $P_{S''} \otimes \kappa(s'')$ over $\mathrm{Spec}(\kappa(s''))$, hence by Hensel, replacing (S'', s'') by (S''', s''') étale over it if necessary, we may suppose that there exists a section of $P_{S''}$ over S'', i.e. an element of $P(S'')$, extending the given section. In other words, one has a homomorphism $M_{S''} \to G_{S''}$ which induces an isomorphism $M_{\kappa(s'')} \simeq {}_nT''$. By (IX, 6.4) (which here reduces to a simple application of Nakayama's lemma) this homomorphism is a closed immersion above an open neighborhood of s', which may be supposed equal to S''. Let H'' be its image, and consider its centralizer C'' in $G_{S''}$, which is a closed smooth subpreschema of G, by (XI, 6.2). Let us also note:

Lemma 7.3. *Under the preceding conditions for ℓ, n, r, for every prescheme S'' over S and every maximal torus T'' of $G_{S''}$, one has*

$$Centr_{G_{S''}}(T'') = Centr_{G_{S''}}({}_nT'').$$

Indeed, by faithfully flat descent from $S' \times_S S''$, one is reduced to the case where $G_{S''}$ admits a maximal torus T_1'' for which the preceding relation is true. But since T'' is locally conjugate to T_1'' for the étale topology by b), it follows that the same relation is true for T''.

Applying the foregoing result for $\mathrm{Spec}(\kappa(s''))$ instead of S'', one sees that the fiber $C''_{s''}$ is a Cartan subgroup of G. Note now that G being smooth over S, the union of the neutral components of the fibers G_s is an open of G (by a general result of EGA IV on smooth morphisms[1081]), evidently stable under the group law of G; this is therefore a subgroup of G for the prescheme structure induced by G. Moreover, G^0 satisfies the preliminary hypotheses of G, and there is an evident bijective correspondence between the maximal tori of G and those of G^0. Hence to prove c) and d), we may suppose G has connected fibers, which we shall do. Then the Cartan subgroups of G have connected fibers (6.6 a)). This being so, I claim that C''^0 is a Cartan subgroup of $G_{S''}$ above an open neighborhood of s'' in S'' (which will complete the proof of c)). This follows indeed from the

Lemma 7.4. *Under the conditions of 7.1, suppose G has connected fibers; let C be a closed group subpreschema of G, smooth over S, and s an element of S such that C_s is a Cartan subgroup of G_s; then C^0 is a Cartan subgroup of G above an open neighborhood of s.*

One easily reduces to the case where S is local and s is its closed point, and to proving that then C is a Cartan subgroup of G; then by flat descent to the case where G admits a maximal torus, say T. Then by (XI, 6.2), $Transp_G(T, C)$ is representable by a closed subpreschema of G smooth over S. Moreover, by the hypothesis on C_s, the fiber of the transporter at s is non-empty (taking into account the conjugation theorem 6.6 a)). This reduces us by faithfully flat descent to the case where this transporter admits a section over S, hence to the case where C contains a maximal torus T of G. But then T is central in C^0 by (IX, 5.6 a)), hence $C^0 \subset Centr_G(T)$, and since this is an inclusion of smooth group schemes over S having the same relative dimension (namely, the dimension of their common fiber at s) and with connected fibers, it is an equality, which completes the proof.

d) We keep the preceding notations and hypotheses for ℓ, n, r, and the connectedness of the fibers of G. Let $Q : (Sch)^\circ/S \to (Ens)$ be the functor defined by

Q(S′) = the set of multiplicative-type subgroups of G_{S′} of type equal to (ℤ/nℤ)^r = M (IX, 1.4).

Then

T □ _n T

is a morphism

[1081] N.D.E.: to specify this reference.

$$\varphi : \mathcal{T} \longrightarrow Q,$$

which is a monomorphism by 7.3. I claim that Q is representable by a separated prescheme of finite presentation over S. Indeed, as we signaled in the proof of (XI, 3.12 a)), one has an isomorphism

$$Q \simeq P_0/\Gamma$$

where P_0 is the open and closed subpreschema of the prescheme $P = \mathrm{Hom}_{S-gr}(M_S, G)$ introduced in c) corresponding to the monomorphisms $M_{S'} \to G_{S'}$ (cf. IX, 6.8), and where $\Gamma = (\mathrm{Aut}_{gr}(M))_S$. The hypothesis that every finite part of a fiber G_s is contained in an affine open of G, being stable under passage to a closed subpreschema and under cartesian products, is evidently inherited by $({}_nG)^r$ hence by P, hence by P_0, so that Q is representable by a prescheme of finite presentation over S (cf. V).[1082] One sees in the same way that if G is quasi-projective (resp. affine) over S, the same holds for Q. In any case, Q is separated over S. Now one has the

Lemma 7.5. *The homomorphism $\varphi : \mathcal{T} \to Q$ above is representable by an open immersion.*

In other words, one must prove that if H is a multiplicative-type subgroup of G, of type M, then there exists an open part U of S such that for every S' over S, $H_{S'}$ is of the form ${}_nT_{S'}$, for a suitable maximal torus $T_{S'}$ of $G_{S'}$, if and only if $S' \to S$ factors through U. One may evidently suppose that the nilpotent rank of the fibers of G is constant (since by b), it is locally constant), say r'. Let $C = Centr_G(H)$, which is a closed group subpreschema of G smooth over S (XI, 6.2). Then replacing S by the open and closed part of the points at which C is of relative dimension r', we may suppose C of relative dimension r' everywhere. One then sees at once that H is of the form ${}_nT$, for a maximal torus T of G, if and only if C^0 admits a central maximal torus T of relative dimension r everywhere and $H = {}_nT$, which gives another expression of the subfunctor U of S we wish to represent (in replacing in the preceding criterion S by an S' over S). Moreover by flat descent, we may suppose that G admits a maximal torus T_1. Let R be the subfunctor $Transp_G(T_1, Centr_{C^0}(C^0))$ of $Transp_G(T_1, C)$; the latter is representable by a smooth prescheme over S by (XI, 6.2), and the former is representable by an induced open subpreschema, as follows at once from (IX, 5.6 a)); in particular it is smooth over S. Consequently its structural morphism into S is open, hence its image is open, and replacing S by the said image (equipped with the induced structure) we may suppose the structural morphism surjective. Then by 1.13 applied to C^0, C^0 admits a central maximal torus T since it admits one locally for fpqc, which will evidently be of relative dimension equal to that of T_1, i.e. r. Thus, the condition to be expressed for H is the equality $H = {}_nT$, which by (IX, 2.10) amounts again to taking a suitable open (and closed) part of S.

[1082] N.D.E.: to specify this reference.

Lemma 7.5 therefore implies that $\mathcal{T}$ is representable by a separated prescheme locally of finite presentation over S, and even of finite presentation over S, as one sees by taking up again the proof of 7.5 to assure oneself that φ is in fact a quasi-compact open immersion, or by reducing by faithfully flat quasi-compact descent to the case where G admits a maximal torus, and where $\mathcal{T}$ is therefore isomorphic to $G/Norm_G(T)$. This last expression, or at choice (XI, 2.1), show moreover that $\mathcal{T}$ is smooth over S. Finally, if G is quasi-projective over S, the same holds for Q hence also for $\mathcal{T}$. If G is affine over S, the assertion that $\mathcal{T}$ is then affine over S is recorded for memory, being established in 5.4 (N.B. I do not know whether without the affine hypothesis on G/S, it is possible to choose n in such a way that in 7.5 the open immersion is also a closed immersion). For the assertion that $\mathcal{T}$ is affine over S if S is artinian, one is reduced to the case where S is the spectrum of a field (EGA I, 6.1.7), which one may suppose algebraically closed. Then thanks to f), which will be proved below, it suffices to prove the same assertion for $G/Centr(G)$; now this last is affine by 6.1, so that one is under the preceding conditions. This completes the proof of d).

e) By (IX, 6.8), we know that there exists a subtorus T' of G' such that u induces a faithfully flat morphism $T \to T'$ (which characterizes T' as the subsheaf $u(T)$ of G'). Let C be the centralizer of T, C' that of T'; we prove that the morphism $C \to C'$ is flat, and faithfully flat if G' has connected fibers. Since C, C' are flat of finite presentation over S, one is reduced to the case of a base field (SGA 1, I 5.9), which one may evidently suppose algebraically closed. Moreover one may suppose G, G' connected (replacing them by G^0 and G'^0 if necessary, which does not change C^0 and C'^0), and it then suffices to apply 6.6 d), taking into account a).

f) Taking into account a) and e), one may restrict to proving the assertion concerning the Cartan subgroups. Now since a Cartan subgroup of G is the centralizer of a maximal torus, it contains the center of G and a fortiori `Ker u`, hence is of the form $u^{-1}(C')$, where $C' = u(C)$ is the Cartan subgroup of G' envisaged in e). Hence the map $C \mapsto u(C)$ is injective; to show that it is bijective, it suffices to see that for every Cartan subgroup C' of G', $u^{-1}(C')$ is a Cartan subgroup of G. The question being local for fpqc, we may suppose that G admits a Cartan subgroup `C_1`, hence $u(C_1) = C_1'$ is a Cartan subgroup of G', hence locally conjugate to C' for fpqc by b), and since $u^{-1}(C_1') = C_1$ is a Cartan subgroup of G, it follows that $u^{-1}(C')$ is a Cartan subgroup of G. QED.

One can also, to prove that $u^{-1}(C')$ is a Cartan subgroup, note that it is flat over S since u is (SGA 1, I 5.9), which reduces us by definition to the case of a base field, and one may apply 6.6 e).

Corollary 7.6. *Under the conditions of 7.1 e), G' admits, locally for the étale topology, a maximal torus, hence satisfies the preliminary conditions for G. If moreover `Ker u` is central, then the functors $\mathcal{T}_G$, $\mathcal{C}_G$ of maximal tori of G and Cartan subgroups of G are isomorphic to the analogous functors $\mathcal{T}_{G'}$, $\mathcal{C}_{G'}$ for G' (hence, in the case where they are representable, they are represented by isomorphic S-preschemata).*

Remark 7.7. *a) Contrary to what happens in the case where G is affine over S (it suffices, in fact, that G have affine fibers, as one will see in XVI), it is not true that the fact that G has locally constant reductive rank implies that G admits, locally for fpqc, a maximal torus. For example, let S be the spectrum of a discrete valuation ring,* `G_1` *and* `G_2` *smooth separated group schemes of finite type over S such that the generic fiber of* `G_1` *is an elliptic curve, the special fiber a group G_m, and the generic fiber of* `G_2` *a group G_m, the special fiber a group G_a, and take $G = G_1 \times_S G_2$. Then the two fibers of G have reductive rank 1, but one sees at once that G does not admit a maximal torus locally for fpqc. It is on the other hand very plausible that the following condition (for a smooth separated group of finite type over a prescheme S) is sufficient for the existence of a maximal torus locally for the étale topology: the reductive rank and the abelian rank of the fibers of G are locally constant functions.*

b) In the proof of 7.1 (notably a)) we have invoked (XI, 6.2) in cases where S is not supposed locally noetherian. However, in the cases of application of (XI, 6.2) envisaged, the reduction to the case S noetherian affine is immediate.

Here is a variant of 7.1 b):

Proposition 7.8. *Let G be an S-prescheme in groups smooth of finite presentation with connected fibers, H a smooth S-prescheme in groups, $i : H \to G$ a monomorphism of S-groups (making H an S-subgroup of G). Then for every Cartan subgroup C*[1083] *of G, $Transp_G(C, H)$ is representable by a closed subpreschema of G smooth over S.*

The fact that the transporter is representable by a closed subpreschema of G of finite presentation is contained in (XI, 6.11), taking into account that C has connected fibers since G does (6.6 b)). To show that the transporter is smooth over S, one is reduced by the standard procedure to the case where S is affine noetherian, then to the case where S is artinian local, and by descent to the case where the residue field of S is algebraically closed. But then C admits a maximal torus T, which is a maximal torus of G. We may suppose that the reductive rank and the nilpotent rank of the fiber `H_0` are equal to those of `G_0` (otherwise the transporter would be empty), but then one sees at once (using the connectedness of C and the fact that the centralizer in H of a maximal torus of H is smooth) that one has

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Transp_G(C, H) = Transp_G(T, H)
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and since we know that the second member is smooth (XI, 2.5), the same holds for the first. The preceding reasoning shows more generally part b) of the

Proposition 7.9. *Let G and $i : H \to G$ be as in 7.8, suppose moreover that for every $s \in S$, H_s is connected and has the same reductive rank and the same nilpotent rank as G_s (i.e. H_s contains a Cartan subgroup of G_s). Then one has the following:*

a) $Norm_G(H)$ is representable by a closed subpreschema $Norm_G(H)$ of G of finite

[1083] The proof shows that it suffices to suppose that C is a smooth subgroup of G each of whose geometric fibers is the connected centralizer of a subgroup of a maximal torus of `G_S`.

presentation over S, and the canonical monomorphism $H \to Norm_G(H)$ is an open immersion; consequently i is an immersion, and one has

$$H = Norm_G(H)^0.$$

b) For every Cartan subgroup C of G, $Transp_G(C,H)$ is a closed subpreschema of G, smooth over S, with surjective structural morphism. If C is the centralizer of a maximal torus T of G, one has moreover

```
Transp_G(T, H) = Transp_G(C, H).
```

c) Let C be a group subpreschema of H. For it to be a Cartan subgroup of H, it is necessary and sufficient that it be a Cartan subgroup of G.

d) Suppose that G admits, locally for the étale topology, or for the fpqc topology,

a Cartan subgroup (resp. a maximal torus); then the same holds for H.

Proof. a) The representability of $Norm_G(H)$ by a closed subpreschema $Norm_G(H)$ of G of finite presentation over S is contained in (XI, 6.11). Since H is smooth hence flat locally of finite presentation over S, to verify that $H \to Norm_G(H)$ is an open immersion, one is reduced to verifying it on the fibers (VI_B, 2.6), which reduces us to the case where S is the spectrum of an algebraically closed field k. One is then reduced (Exp. VI[1084]) to verifying that the corresponding homomorphism on the Lie algebras is an isomorphism, or equivalently that $(g/h)^H = 0$, where g and h are the Lie algebras of G and of H, and the exponent H denotes the invariants under H (cf. II, 5.2.3 (i)). Now H contains by hypothesis a Cartan subgroup C of G, the centralizer of the maximal torus T of G, and it suffices therefore to prove that one has

$$(g/h)^T = 0,$$

which follows, taking into account complete reducibility of the representations of T (I, 4.7.3), from the analogous relation $(g/c)^T$, where $c = Lie(C)$. As for this latter, equivalent to

$$g^T = c,$$

it means that the centralizer C and the normalizer N of T have the same Lie algebra, which follows from the fact that C is an open subgroup of N (XI, 5.9). This completes the proof of a).

b) As we have signaled, the proof was given in 7.8.

[1084]N.D.E.: reference not located in VI_B; see M. Demazure and P. Gabriel, *Groupes algébriques*, I, Masson (1970), corollary II.5.6.

c) Taking into account the fact that H is a subpreschema of G, the assertion reduces trivially to the case where S is the spectrum of an algebraically closed field, in which case it follows at once from the hypothesis made on H.

d) One uses b), c), and the "Hensel lemma" (XI, 1.10).

Corollary 7.10. *Let G and H be as in 7.9, and suppose that for every algebraically closed field k over S, every element of $G_k(k)$ which normalizes H_k is in $H_k(k)$. Then H is a closed subpreschema of G, and is its own normalizer.*

This follows trivially from 7.9 a). We shall apply 7.10 in particular to the Borel subgroups (more generally, to the parabolic subgroups) of G.

Corollary 7.11. *Let G, H be as in 7.9. Then H contains every group subpreschema Z of G which is central in G, flat and of finite presentation over S.*

One may as usual reduce to the case S affine noetherian, then to the case S artinian, which implies that one is under the conditions of 7.1. By 7.9 c) one may suppose that H contains a Cartan subgroup C of G, and one is reduced to proving that $C \supset Z$. Now since S is artinian, $G' = G/Z$ is representable by a prescheme in groups of finite type over S, the canonical morphism $u : G \to G'$ being faithfully flat and its kernel being Z (VI_A, 3.2). Obviously G' is smooth over S, and one may apply 7.1 f), which implies that C is of the form $u^{-1}(C')$, hence contains Z.

Corollary 7.12. *Let G, G' be two S-preschemata in groups smooth of finite presentation, $u : G \to G'$ a faithfully flat group homomorphism (i.e. for every $s \in S$, u_s is faithfully flat); suppose* Ker u *central and G with connected fibers. Then the map*

$$H' \mapsto H = u^{-1}(H')$$

establishes a bijective correspondence between group subpreschemata H' of G', smooth of finite presentation over S, having the same reductive rank and the same

nilpotent rank as G' at every $s \in S$, and the set of group subpreschemata H of G, smooth of finite presentation over S, having the same reductive rank and the same nilpotent rank as G at every $s \in S$. For H to have connected fibers, it is necessary and sufficient that H' do.

Let H be a group subpreschema of G having the properties just specified. Then by 7.10, H contains $Z = Ker u$, hence by the theory of faithfully flat descent, is of the form $u^{-1}(H')$, where H' is a well-determined group subpreschema of G', and one verifies at once, taking into account 6.6 e), that this latter has the properties stated above. Moreover, if H has connected fibers, the same evidently holds for $H' = H/Z$. It therefore remains to prove that if one starts from a subgroup H' of G' having the stated properties, then $u^{-1}(H') = H$ has the same properties in G; and that if H' has connected fibers, the same holds for H. Taking into account 6.6 e), one is reduced to proving that H is smooth over S (resp. and has connected fibers). Now H is already flat over S as the inverse image of H' which is, by the flat morphism u, hence one is reduced to verifying that the geometric fibers of H are smooth (resp. and connected) which reduces us to the case where S is the spectrum of an algebraically closed field. Then H' contains a Cartan subgroup C'

of G', hence H contains the inverse image C of C', which is a Cartan subgroup of G by 6.6 e), hence C is smooth and connected. Consequently 7.11 follows from the following lemma:

Lemma 7.13. *Let G, G' be two preschemata in groups flat of finite presentation over S, $u : G \to G'$ a group homomorphism which is faithfully flat, C' a group subpreschema of G' of finite presentation over S, such that $C = u^{-1}(C')$ is smooth over S (resp. has connected fibers). Then for every group subpreschema H' of G' of finite presentation over S, containing C', and such that H' is smooth over S (resp. has connected fibers), its inverse image $H = u^{-1}(H')$ is smooth over S (resp. has connected fibers).*

As we have remarked above, this statement reduces at once to the case where S is the spectrum of a field. Note then that H is a principal bundle of base H/C, group C (Exp. VI_B, 9); on the other hand $H/C \simeq H'/C'$ (cf. Exp. IV), and H' being smooth (resp. connected) the same holds for H'/C' hence for H/C. Since the same holds for C by hypothesis, it follows at once that the same holds for the bundle H. QED.

8. Semisimple elements; union and intersection of maximal tori in not-necessarily-affine group schemes

Throughout this number, G denotes a smooth S-prescheme in groups of finite presentation over S, with connected fibers.

Suppose first that S is the spectrum of an algebraically closed field. When G is affine, one has defined in *BIBLE* 4 N° 4 the notion of semisimple element of $G(k)$; one verifies at once that this notion is invariant under algebraically closed extension k'/k of the base field. Moreover, one has seen in *BIBLE* 6 th. 5 (c) that $g \in G(k)$ is semisimple if and only if it is contained in a maximal torus of G. When G is no longer supposed affine, G is written canonically (thanks to Chevalley) as an extension of an abelian variety by a smooth and connected affine algebraic group:

$$(*) \qquad e \to V \to G \to A \to e.$$

We shall say that an element g of $G(k)$ is *semisimple* if it is a semisimple element of $V(k)$. Since the maximal tori of V are obviously identical to the maximal tori of G, it amounts to the same to say that g belongs to a maximal torus of G. This is evidently still a notion invariant under algebraically closed extension k'/k of the base field k.

Suppose now that S is the spectrum of an arbitrary field k, and let $g \in G$. Then (choosing an algebraically closed extension K of $\kappa(g)$) one sees that g is the image of a geometric point g' of G with values in an algebraically closed extension K of k, and we shall say that g is *semisimple* if g' is semisimple, which is independent of the particular choice of g', thanks to what was said above. If k' is an extension of k, then the set of semisimple elements of $G_{k'}$ is the inverse image of the set G^{ss} of semisimple elements of G.

Suppose finally S arbitrary; then a point $g \in G$ is said to be *semisimple* if it is semisimple in its fiber G_s. If $S' \to S$ is any morphism, then $G^{ss}_{S'}$ is the inverse image of G^{ss}. Suppose that the functor $\mathcal{T}$ defined in 1.10 (functor of maximal tori) is representable by a prescheme of finite presentation over S (which is the case for example if G admits locally for fpqc a maximal torus, by 7.1 d), at least if G is quasi-projective over S). Consider the canonical maximal torus T of $G_{\mathcal{T}}$ ("universal maximal torus of G"), and the morphism

$$u : T \longrightarrow G$$

induced by the projection $G_{\mathcal{T}} \to G$. Then it follows at once from the definition that G^{ss} is none other than the image of the preceding morphism. We shall conclude:

Proposition 8.1. *The set G^{ss} of semisimple elements of G is locally constructible (hence constructible if S, hence G, is quasi-compact and quasi-separated).*

One reduces as usual to the case where S is affine noetherian; moreover one may suppose (by the usual noetherian criterion of constructibility (EGA 0_III, 9.2.3)) that S is integral, and restrict to proving that there exists a nonempty open U of S such that $G^{ss}|U$ is constructible. Taking U small enough, and replacing it if necessary by a finite covering, one may suppose that G is separated over S and contains a maximal torus T. But then by 7.1 d) the functor $\mathcal{T}$ is representable by a prescheme of finite presentation over S, and the same therefore holds for T, whose image in G is consequently constructible. QED.

Suppose again that k is a field, and consider the subgroup H of G generated by the preceding morphism (cf. VI_B, 1.2). It is a smooth group subscheme of G, connected since T is, whose formation is evidently compatible with any base field extension (cf. VI), that of T being so. When k is algebraically closed, one sees at once that H is also the algebraic subgroup of G generated by the maximal tori of G, or equivalently by the tori of G, and it is also the smallest algebraic subgroup of G that contains the semisimple elements of $G(k)$. (In fact, these characterizations of H remain valid as soon as k is an infinite field, thanks to the fact, proved in XIV, that the set of points of T rational over k is dense in T). Moreover, H is invariant in G, for to see this, one may restrict to the case where k is algebraically closed, and then (H and G being smooth over k) it suffices to verify that H is stable under the inner automorphisms $int(g)$, $g \in G(k)$, which is evident. (One could even show that H is a characteristic subgroup of G, i.e. stable under $\mathrm{Aut}_{k-gr}(G)$.) It then follows at once from 6.6 d) that the reductive rank of G/H is zero; more precisely, if K is an invariant algebraic subgroup of G, it follows at once from the fact that for k algebraically closed, the maximal tori of G/K are the direct images of the maximal tori of G, that the reductive rank of G/K is zero if and only if K contains all the maximal tori of G (k being supposed algebraically closed), or equivalently if and only if K contains H (k arbitrary): hence H is the smallest invariant algebraic subgroup of G such that G/H is of zero reductive rank. Another evident characterization of H is the following: it is the smallest algebraic subgroup of G having the same reductive rank as G. Let us note finally that H is affine: indeed, to see this one may again suppose k algebraically closed, and taking up the exact sequence

($*$) at the beginning of the N°, one notes that the maximal tori of G are contained in V, hence the same holds for H, hence V being affine, H is. Let us summarize the results obtained:

Proposition 8.2. *Let G be a smooth and connected algebraic group over the field k, and let $\bar{k}$ be an algebraic closure of k. There exists an algebraic subgroup H of G such that $H_{\bar{k}}$ is the algebraic subgroup of $G_{\bar{k}}$ generated by the maximal tori (or equivalently the tori) of $G_{\bar{k}}$. The group H is also characterized as the smallest algebraic subgroup of G having the same reductive rank as G, or the smallest invariant algebraic subgroup of G such that the reductive rank of G/H is zero. It is a smooth connected invariant and affine subgroup of G, whose formation commutes with any extension of the base field.*

In order to use the characterization of H in terms of G/H, it is appropriate to make explicit the

Corollary 8.3. *Let G be a smooth and connected algebraic group over an algebraically closed field k. For G to be of zero reductive rank (i.e. with the notations of 8.2, for one to have $H = (e)$) it is necessary and sufficient that G be an extension of an abelian variety by a smooth connected unipotent algebraic group (i.e. successive extension of groups isomorphic to the additive group G_a).*

Indeed, thanks to Chevalley's exact sequence ($*$), one is reduced to proving that the reductive rank of the smooth connected affine group V is zero if and only if V is unipotent, which is contained in *BIBLE* 6.4 th. 4 cor. 3. Hence with the notations of 8.2, H is the smallest invariant algebraic subgroup of G such that $(G/H)_{\bar{k}}$ is an extension of an abelian variety by a smooth connected affine unipotent algebraic group. One concludes also:

Corollary 8.4. *With the notations of 8.2, for one to have $G = H$ (i.e. $G_{\bar{k}}$ generated by its maximal tori) it is necessary and sufficient that G be affine and that every homomorphism from $G_{\bar{k}}$ to the additive group be trivial.*

Remarks 8.5. *a) Let V be the largest smooth connected affine algebraic subgroup of $G_{\bar{k}}$ (so that $G_{\bar{k}}$ is an extension of an abelian variety by V). It is well known (Rosenlicht[1085]), if k is not perfect, that V is in general not "defined over k", i.e. that there does not in general exist an algebraic subgroup V of G such that $V_{\bar{k}} = V$. However, when V is generated by its maximal tori, i.e. when V admits no quotient group isomorphic to the additive group $G_{a,\bar{k}}$, there exists such a V, namely the group H of 8.2. One thus sees that in this question of rationality, as in many others (cf. for example XIV, N° 6), all the troubles come from the unipotent groups, i.e. from the additive group, while it is the presence of (enough) multiplicative-type groups that on the contrary ensures that things go well.*

b) One may also introduce, with the notations of 8.2, the inverse image H' in G of the commutator subgroup of G/H; then H' is the smallest invariant algebraic subgroup of G such that $(G/H')_{\bar{k}}$ is a commutative extension of an abelian variety by a smooth connected affine unipotent algebraic group. H' is again a smooth connected affine

[1085]N.D.E.: indicate references here.

subgroup of G. Let H'' be the inverse image in G of $p^n(G/H')$ for n large, p being the characteristic (assumed > 0); then one sees easily that H''/H' is an abelian variety, and H'' is the smallest invariant algebraic subgroup of G such that G/H'' is a commutative smooth connected affine unipotent algebraic group. This being so, one sees easily that for $V = H_{\bar{k}}$ (notations of a)), i.e. for every homomorphism from V to the additive group to be trivial, it is necessary and sufficient that $H'' = G$, or equivalently that every homomorphism from $G_{\bar{k}}$ to the additive group be trivial.

To finish, we shall generalize to smooth groups with connected fibers the notion of reductive center developed in N° 4, inspired by 4.10. Let $\mathcal{Z}$ be the subfunctor of G defined by

$\mathcal{Z}(S')$ = *Set of sections g' of $G_{S'}$ over S' such that for every S'' over*

S' and every maximal torus T'' of $G_{S''}$, the inverse image g'' of g' by $S'' \to S'$ is a section of T'' over S''.

Introducing the functors $\mathcal{T}(S')$ = set of maximal tori of $G_{S'}$, and $T(S')$ = set of pairs (T', g'), where T' is a maximal torus of $G_{S'}$ and g' a section of T' over S', one sees that T is a subfunctor of $\mathcal{T} \times_S G = \mathcal{T}_G$, and with these notations, one may also write

$$\mathcal{Z} = \prod_{\mathcal{T}_G/G} T/\mathcal{T}_G.$$

Using (XI, 6.8) and 7.1 d) which ensures the representability of $\mathcal{T}$ by a smooth S-prescheme under certain conditions, one could conclude the representability of $\mathcal{Z}$ under certain conditions, which we shall however obtain by a more direct route below.

Definition 8.6. *Let G be a smooth S-prescheme in groups of finite presentation with connected fibers. One says that G admits a* reductive center *if the preceding functor $\mathcal{Z}$ (which is evidently a subgroup of G) is representable by a multiplicative-type group. One then says that $\mathcal{Z}$ is the* reductive center of G.

One will note that if Z is a reductive center of G, then for every base change $S' \to S$, $Z_{S'}$ is a reductive center of $G_{S'}$; on the other hand, the existence of a reductive center is evidently a local question for the fpqc topology. As for the terminology "reductive center", note that Z is in any case central, since evidently $\mathcal{Z}$ is invariant under

$$\mathrm{Aut}_S(G)$$

and *a fortiori* it is an invariant subgroup of G, and one applies (IX, 5.5).

Lemma 4.5 must be replaced here by:

Lemma 8.7. *Let $u : H \to G$ be a central homomorphism, where H is of multiplicative* *type and of finite type over S; suppose that for every algebraically closed field k over S, $u_k(H_k)$ is contained in the largest affine smooth connected subgroup of G_k. Then u factors through every maximal torus T of G (hence u factors in fact through the subfunctor $\mathcal{Z}$ of G defined above).*

Using an easy variant of (IX, 5.1 bis) (where the sign = would be replaced by an inclusion sign), one is reduced to the case where S is the spectrum of a field k, which one may suppose algebraically closed. (Reduce to the case S affine noetherian, then artinian, then use (IX, 3.6).) Since T is contained in the largest affine smooth connected subgroup V of G, one is then reduced to the case where $G = V$, i.e. where G is affine, where the result was proved in 4.5.

Proposition 8.8. *Let G be a smooth S-prescheme in groups of finite presentation with connected fibers.*

a) If S is the spectrum of a field k, then G admits a reductive center Z. When G is an extension of an abelian variety by a smooth connected affine algebraic group V (for example if k is algebraically closed), then Z is also the reductive center of V, and is the largest central multiplicative-type subgroup of V.

b) Let Z be a multiplicative-type group subpreschema of G. Then Z is a reductive center of G if and only if for every $s \in S$, Z_s is a reductive center of G_s. Then Z is the largest multiplicative-type subgroup K of G such that for every $s \in S$, K_s is contained in the reductive center of G_s; more generally, for every homomorphism $u : H \to G$, with H of multiplicative type and of finite type over S, such that for every algebraically closed field k over S, $u_k : H_k \to G_k$ factors through the largest smooth connected affine subgroup of G_k, u factors through Z (and in particular, u is central).

c) If G admits, locally for the fpqc topology, a maximal torus, then G admits a reductive center.

d) Let T be a maximal torus of G. Then $T \cap Centr(G) = Ker(T \to GL(g))$ and this is a reductive center of G.

e) Let Z be a reductive center of G, and suppose $G' = G/Z$ representable (for example S artinian); then G' admits the unit subgroup as reductive center, and $T' \mapsto T = u^{-1}(T')$ establishes a bijective correspondence between maximal tori of G' and maximal tori of G.

Proof. a) Suppose that S is the spectrum of a field k. To prove the existence of a reductive center, one may suppose k algebraically closed, and one is reduced consequently to the case where G is an extension of an abelian variety by a smooth connected affine algebraic group V. Since for every S' over k, the maximal tori of $G_{S'}$ are those of $V_{S'}$ (by (IX, 5.2) and the fact that over an algebraically closed field, a homomorphism from a torus to an abelian variety is trivial), it follows that the functor $\mathcal{Z}$ defined above in terms of G is the same as the one defined in terms of V. One is thus reduced to the case G affine. Since $\mathcal{T}$ is representable and necessarily "essentially free" over k (VIII, 6.1), it follows that $\mathcal{T}_G$ is essentially free over G, and since T is a closed subscheme of $\mathcal{T}_G = G_{\mathcal{T}}$ (by e.g. (VIII, 5.7)) it follows by (VIII, 6.4) that $\mathcal{Z} = \prod_{\mathcal{T}_G/G} T/\mathcal{T}_G$ is representable by a closed subscheme of G. It is therefore a group subscheme of G; I claim that it is of multiplicative type: indeed one may suppose k algebraically closed; then G admits a maximal torus T, and by definition one will have $Z \subset T$, hence Z is of multiplicative type as an algebraic subgroup of a multiplicative-type group (IX, 8). This proves that Z is a reductive center of G. The fact that it is the largest central multiplicative-type subgroup of V is contained in 8.7.

b) The "only if" being trivial, let us prove that if Z is a multiplicative-type subgroup of G such that for every $s \in S$, Z_s is the reductive center of G_s, then Z is a reductive center of G. One must first prove that Z is contained in every maximal torus of G (which will thus remain true after every base change): this is an immediate consequence of 8.7. Next, one must prove that if g is a section of G over S such that for every S' over S and every maximal torus T' of $G_{S'}$, $g_{S'}$ is a section of T', then g is a section of Z. Note that one could reduce as usual (taking into account that $\prod_{\mathcal{T}_G/G} T/\mathcal{T}_G = \mathcal{Z}_0$ is an fpqc sheaf that commutes with inductive limits of rings) to the case S affine, then S noetherian, and finally S artinian. Then $\mathcal{T}$ is representable by a smooth prescheme over S by 7.1 d), so proceeding as in a), one sees that $\prod_{\mathcal{T}_G/G} T/\mathcal{T}_G$ is representable by a closed subscheme Z_0 of G. We have already seen that $Z \subset Z_0$; on the other hand by hypothesis on Z one has Z_k = Z_0_k (where k is the residue field); now Z being flat over S, it follows $Z = Z_0$, which proves that Z is a reductive center of G. The other assertions of b) are contained in 8.7.

c) Reduces immediately to d).

d) Of course, g denotes the Lie algebra of G, and $T \to GL(g)$ the homomorphism induced by the adjoint representation of G. One has trivially

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T ∩ Centr_G ⊂ Ker(T → GL(g)),
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the proof of the reverse inclusion is the same as in 4.7 d), we do not repeat it here. Let Z be the group in question; as $Ker(T \to GL(g))$ it is of multiplicative type (cf. for example (IX, 6.8)). To prove that it is a reductive center of G, one is reduced by b) to the case where S is the spectrum of a field. Let then Z_0 be the reductive center (which exists by a)); one has evidently $Z_0 \subset T \cap Centr(G) = Z$; on the other hand since Z is a central multiplicative-type subgroup contained in the smooth connected affine subgroup T, it follows from a) that $Z_0 \supset Z$, hence $Z_0 = Z$. QED.

e) Under the conditions of 7.1 f) set $Z = Keru$ and suppose that for every algebraically closed field k over S, Z_k is contained in a maximal torus of G_k. Then one verifies easily that the map $T' \mapsto u^{-1}(T')$ induces a bijective correspondence between the set of maximal tori of G' and the set of maximal tori of G. Applying this to the situation 8.8 e), the desired conclusion follows at once.

Remarks 8.9. *a) The proof given of 8.8 is independent of the results of N° 4, and in particular that of 8.8 a) does not use 4.4 (whose proof is a little burdensome).*

b) One sees easily that the subgroup $\mathcal{Z}$ of G envisaged in 8.6 is always central *(whether it is representable or not), and it may be tempting to call it the reductive center of G in all cases.*

c) One may also generalize 4.9; one finds the following statement: Let G be a smooth and connected algebraic group over an algebraically closed field. For G to be an extension of an abelian variety by a smooth connected unipotent algebraic group (i.e. for the reductive rank of G to be zero, cf. 8.3) it is necessary and sufficient that the reductive center of G be reduced to the unit group and that the Lie algebra of G be nilpotent.

9. Complement: action of a group scheme and fixed points

The aim of this section, added in January 2008, is to study fixed points under the action of a group scheme.

9.1. Representability of the functor of fixed points

Let S be a scheme, G an S-group scheme acting on an S-scheme X. One defines the subfunctor X^G of X of fixed points of X under G: for every S-scheme T, $X^G(T)$ is the subset of $X(T)$ consisting of fixed points under G (VIII, 6.e).

Recall the notion of "S-pure scheme" introduced in ([G-R], § 3.3, [R]). Let G be a flat S-group scheme of finite presentation. Since the fibers of G have no embedded components, the notion of purity for G takes the following form. The scheme G is *S-pure* if for every strictly henselian local S-scheme T with closed point t, every generic point $\tilde{x}$ of a fiber of $G \times_S T$ specializes to a point of G_t, i.e. the closure of $\tilde{x}$ in $G \times_S T$ meets G_t.

Examples. (1) If G is quasi-finite and separated over S, G is S-pure if and only if G is finite over S.

(2) If $S = \mathrm{Spec}(R)$ and $G = \mathrm{Spec}(A)$, G is S-pure if and only if A is a projective R-module ([G-R], 3.3.5). In particular G is S-pure if G is diagonalizable.

(3) A multiplicative-type group scheme is S-pure since the notion of purity is local for the étale topology on S.

(4) The scheme G is S-pure if G is S-proper or if the fibers of G are irreducible or if S is semi-local artinian.

(5) In particular, an S-reductive group scheme G is S-pure.

Proposition 9.2. *Let G be an S-group scheme of finite presentation, S-flat and S-pure, acting on an S-scheme X of finite presentation. Then the functor X^G of fixed points of X under G is representable by a closed subscheme of X, of finite presentation over S.*

Indeed, let a in $X(S)$ and let H be the group subscheme of G fixing a. Then a is fixed under G if and only if $H = G$. Now the subfunctor C of S of coincidences (precisely, if T is an S-scheme, $C(T) = \emptyset$ if $H \times_S T \to G \times_S T$ is bijective, and $C(T) = \emptyset$

otherwise) of H with G is representable by a closed subscheme of S, defined by a finitely generated sheaf of ideals ([G-R], 4.1.1). One applies this result with $S = X$ taking for a the universal point of X (i.e. the identity in $X(X) = \mathrm{Hom}_S(X, X)$).

9.3. Infinitesimal obstruction

Under the hypothesis that G is a flat S-group scheme of finite presentation acting (on the left) on a smooth S-prescheme X, we shall now study the formal smoothness of the functor X^G.

One supposes here that S is equipped with a closed subscheme S_0 defined by a sheaf of quasi-coherent ideals $\mathcal{J}$ of square zero. One writes $j : S_0 \to S$, $G_0 = G \times_S S_0$, $X_0 = X \times_S S_0$.

Let ϵ_0 be a point of $X^G(S_0)$ that lifts to a point of $X(S)$. We shall study the obstruction to lifting ϵ_0 to a point of $X^G(S)$. Since X is smooth, one knows that the liftings of ϵ_0 to $X(S)$ form a principal homogeneous space, trivial for the abelian group

$$\mathrm{Hom}_{O_{S_0}}(\epsilon_0^*(\Omega^1_{X_0/S_0}), \mathcal{J})$$

(III, 0.2, 0.3). One sets

$$L_0 = \mathrm{Hom}_{O_{S_0}}(\epsilon_0^*(\Omega^1_{X_0/S_0}), \mathcal{J}),$$

this is an O_{S_0}-module. Moreover, given that ϵ_0 is fixed under G, L_0 is naturally a G_0-O_{S_0}-module. One writes $\rho_0 : G_0 \to \mathrm{Aut}_{S_0}(L_0)$ (resp. $\rho : G \to \mathrm{Aut}_S(j_*L_0)$) for the associated representation.

Lemma 9.4. *There exists a certain class $c(\epsilon_0) \in H^1(G, j_*L_0) \simeq H^1(G_0, L_0)$, defined canonically by ϵ_0, such that ϵ_0 lifts to $X^G(S)$ if and only if $c(\epsilon_0) = 0$.*

The adjunction $H^1(G, j_*L_0) \simeq H^1(G_0, L_0)$ is that of lemma (III, 1.1.2). We work with the small fppf site on S. For every flat scheme T of finite presentation over S, one writes $G_T = G \times_S T$, $X_T = X \times_S T$ for the corresponding objects over T. Consider the sheaf $\mathcal{A}$ on the small fppf site of S such that, for every T flat and of finite presentation over S, one has:

$\square(T) = \{\text{set of liftings of } \varepsilon_0 \times_S T \text{ in } X(T)\}.$

Given that the formation of L_0 commutes with flat base changes, the foregoing considerations indicate that $\mathcal{A}(T)$ is a trivial principal homogeneous space under the abelian group $H^0(T, j_*L_0)$. Again from the fact that ϵ_0 is fixed, for every T flat and of finite presentation over S, and every g in $G(T)$, g acts by affine automorphisms on $\mathcal{A}(T)$, compatibly with the action of G on j_*L_0. That is to say:

$g(a_T + v) = g(a_T) + \rho(g)(v), \quad v \in H^1(T, j_*L_0).$

Since G is flat of finite type one may apply these considerations taking $T = G$ and for g the universal point id_G of G. One thus obtains an action of the fppf sheaf G on $\mathcal{A}$.

Let a be a lifting of ϵ_0 in $X(S)$. One writes $g^\sharp$ for the universal point of G, and one defines $v^\sharp \in H^0(G, j_*L_0)$ by

ρ(g^♯)(v^♯) = g^♯ . a_G − a_G.

For every S-prescheme Y, one sets $z(Y) : G(Y) \to H^0(Y, j_*L_0)$, $g \mapsto v^\sharp(g)$. This defines the 1-cocycle z in $Z^1(G, L)$ (Exposé I, § 5).

Its class $c(\epsilon_0)$ in $H^1(G, j_*L_0)$ does not depend on the choice of a. In particular, if ϵ_0 lifts to $X^G(S)$, one has $c(\epsilon_0) = 0$. Conversely, if $c(\epsilon_0) = 0$, then there exists $w \in H^0(S, j_*L_0)$ such that $z(Y)(g) = g.w_Y - w_Y$ for every S-prescheme Y and every $g \in G(Y)$. Applying this to $Y = G$ and to $g^\sharp$, one concludes that $a - w \in X^G(S)$. We have thus established that ϵ_0 lifts in $X^G(S)$ if and only if $c(\epsilon_0) = 0$.

Remark 9.5. *In the case where G is affine and flat of finite type over S affine, one can refine this obstruction into a class $\tilde{c}(\epsilon_0) \in H^1_{G_0}(X_0, L_0)$ where $H^1_{G_0}(X_0, L_0)$ is the G_0-equivariant cohomology group defined by Wevers ([W], app. C). In this case, it is not necessary to suppose that ϵ_0 lifts to $X(S)$.*

9.6. Smoothness of fixed points

Theorem 9.7. *Let G be an S-group scheme flat, of finite type over a noetherian base S, acting on a smooth S-scheme X. Suppose that G is S-pure, and for every geometric point s above S (with algebraically closed residue field), one has*

$$H^1(G_s, (\Omega^1_{X_s})*) = 0.$$

Then the functor X^G of fixed points of X under G is representable by a closed subscheme of X smooth over S.

Let us prove Theorem 9.7. We know that X^G is representable by a closed subscheme of X of finite presentation over S. We may suppose $S = \mathrm{Spec}(A)$ local. Following the smoothness criterion (SGA 1, III, 3.1.iii bis), it suffices to verify the formal smoothness of X^G for $S' = \mathrm{Spec}(A')$ and the closed subscheme $S''_0 = \mathrm{Spec}(A'/I') = \mathrm{Spec}(A''_0)$, where A' is an artinian local ring and I' an ideal of square zero of A'. By dévissage, one reduces to the case where I' is annihilated by the maximal ideal of A'. In particular, if k denotes the residue field of A', I' is a k-vector space.

One is therefore given an element $\epsilon''_0 \in X^G(S''_0)$ which one wishes to lift to $X^G(S')$. One then writes $X_0 = X \times_S S'$, $G' = G \times_S S'$, $X''_0 = X \times_S S''_0$, $G''_0 = G \times_S S''_0$. Given that S' is affine and that X_0 is smooth over S', $\epsilon''_0 \in X^G(S''_0)$ lifts to $X(S')$. By Lemma 9.4, the class

$$c(\epsilon_0) \in H^1(G''_0, L''_0)$$

is the obstruction to lifting ϵ''_0 in $X^G(S')$, where $L''_0 = \mathrm{Hom}_{O_{S''_0}}(\epsilon''_0{}^*(\Omega^1_{X''_0/S''_0}), I')$. Given that L''_0 is a k-vector space, one has a canonical isomorphism $H^1(G''_0, L''_0) \simeq H^1(G''_0 \times_{A''_0} k, L''_0)$. It suffices to show that $H^1(G''_0 \times_{A''_0} k, L''_0) = 0$. Writing $\bar{k}$ for an algebraic closure of k, one has an isomorphism (IX, 3.1 in the affine case, lemma 9.11 in general)

$$H^1(G''_0 \times_{A''_0} k, L''_0) \otimes_k \bar{k} \simeq H^1(G''_0 \times_{A''_0} \bar{k}, L''_0 \otimes_k \bar{k}).$$

One then remarks that the representation $L''_0 \otimes_k \bar{k}$ is a direct sum of $(\Omega^1_{X_0 \times_k \bar{k}})^*$. By hypothesis one has $H^1(G''_0 \times_k A''_0, (\Omega^1_{X_0 \times_k \bar{k}})^*) = 0$, which implies $H^1(G''_0 \times_{A''_0} k, L''_0) = 0$.

Corollary 9.8. *Let G be a flat S-group scheme of finite presentation acting on a smooth S-scheme X of finite presentation. Suppose that G admits a composition series whose factors are of one of the following types:*

(1) S-abelian scheme (i.e. G is smooth over S and its fibers are abelian varieties),

(2) S-group scheme of multiplicative type,

(3) S-group scheme finite, étale, of degree invertible on S,

(4) S-reductive group if S is a $\mathbb{Q}$-scheme.

Then the functor X^G of fixed points of X under G is representable by a closed subscheme of X, of finite presentation, smooth over S.

Indeed, if $1 \to G' \to G \to G'' \to 1$ is an exact sequence of S-groups satisfying the hypotheses of the theorem, the S-group G'' acts on the functor $X^{G'}$, and the functor X^G is nothing other than that of fixed points of $X^{G'}$ under G''. Thus one is reduced to the case of an S-group of one of the four types above. By passage to the limit, one may suppose S noetherian. We shall verify the cohomological criterion stated above for a fiber G_s above a geometric point s of S and a finite-dimensional linear representation V of G_s.

In the case of an abelian scheme, every map from G_s^n to V is constant, and consequently the $H^i(G_s, V)$ are zero for $i > 0$.

In the case of a multiplicative-type group scheme, G_s is a diagonalizable group. Theorem (I, 5.3.3) shows that the $H^i(G_s, V)$ are zero for $i > 0$.

In the two following cases, the argument is the same since it rests on the semisimplicity of the linear representations of G_s. Indeed in the case of an S-group scheme finite, étale, of degree invertible on S, G_s is a constant finite group of degree invertible in the residue field $\kappa(s)$; it is well known that every linear representation of G_s is semisimple.

For the reductive case, the residue field $\kappa(s)$ is supposed (algebraically closed) of characteristic zero. One knows then that every linear representation of G_s is semisimple (see [T-Y], theorem 27.3.3).

Remark 9.9. *This applies in particular to the case of an action of G/S on a smooth S-group scheme H. If G is a multiplicative-type group acting by inner transformations* *on H smooth and affine over S, one recovers then the corollary (XI, 5.3) stating the smoothness of the centralizer of G in H.*

Remark 9.10. *In the case where $S = \mathrm{Spec}(k)$ and G is a linearly reductive algebraic group defined over the algebraically closed field k, this result is due independently to Fogarty ([F], theorem 5.4) and Iversen ([I], proposition 1.3).*

The proof of Theorem 9.7 uses the following lemma, well known in the case of an affine group scheme (IX, 3.1).

Lemma 9.11. *Let G be a group scheme over an affine scheme $S = \mathrm{Spec}(A)$, and $f : S' = \mathrm{Spec}(A') \to S$ a flat morphism. Set $G' = G \times_S S'$. Then for every quasi-coherent G-0_S-module M, one has canonical isomorphisms*

$$H^i(G, M) \otimes_A A' \simeq H^i(G', M \otimes_A A') \quad (i \geqslant 0).$$

Given that the chains of degree $n \geqslant 1$ for Hochschild cohomology are determined by their value at the point $(id, \cdots, id) \in G(G) \times \cdots \times G(G)$, one has an isomorphism of A-modules $C^n(G, M) \simeq \Gamma(G^n, M \bigotimes_A O_{G^n})$. The cohomology group $H^n(G, M)$ is the n-th cohomology group of the complex

$$L \to \Gamma(G, M \otimes_A O_G) \to \Gamma(G^2, M \otimes_A O_{G^2}) \to \Gamma(G^3, M \otimes_A O_{G^3}) \to \cdots$$

Since A' is flat over A, by (EGA IV_1, 1.7.21) one has

$$\Gamma(G^n, M \otimes_A O_{G^n}) \otimes_A A' \simeq \Gamma(G'^n, (M \otimes_A A') \otimes_{A'} O_{G'^n}).$$

This implies the required assertion by taking cohomology.

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Footnotes

Exposé XIII. Regular elements of algebraic groups and Lie algebras

by A. Grothendieck

1. An auxiliary lemma on varieties with operators

Let S be a prescheme, G an S-prescheme in groups acting on the left on an S-prescheme V, W a closed S-subprescheme of V, N its stabilizer in G — the subgroup of G whose points, with values in an S' over S, are the $g \in G(S')$ such that $g \cdot W_{S'} = W_{S'}$. We endow $(Sch)/S$ with the faithfully flat quasi-compact topology, and identify G, V, W with the corresponding sheaves (cf. IV). We shall therefore argue in the category of sheaves on $(Sch)/S$, and in this section the term “locally” refers to the topology we have just specified on $(Sch)/S$. Note that N is a sheaf; consider the quotient sheaf G/N. One sees at once that it is isomorphic to the following functor: to every S' over S, one associates the set of subsheaves W' of $V_{S'}$ that are locally conjugate to $W_{S'}$ by the group G. Let X be the subsheaf of $G/N \times_S V$ whose value, for every S' over S, is the set of pairs (W', v), where W' is as above and v is a section of W' over S' (hence a section of $V_{S'}$ over S'). Let Z be the inverse image of X in $G \times_S V$, so that we have the cartesian diagram

$$(*)\qquad \begin{tikzcd} Z \arrow[r] \arrow[d] & X \arrow[d, "i"] \\ G \times_S V \arrow[r] & G/N \times_S V, \end{tikzcd}$$

where i is the canonical immersion, and the second horizontal arrow comes from the canonical morphism $G \to G/N$.

Since this morphism sends the point $g \in G(S')$ to the subsheaf $g \cdot W_{S'}$ of $V_{S'}$, one

sees that $Z(S')$ is the set of pairs $(g, v) \in G(S') \times V(S')$ such that $v \in g \cdot W_{S'}(S')$. Consequently, Z is isomorphic to the sheaf $G \times_S W$, by means of the isomorphism

$$G \times_S W \xrightarrow{\sim} Z$$

defined by $(g, w) \mapsto (g, g \cdot w)$. Thus the preceding cartesian diagram gives the cartesian diagram

$$(**)\qquad \begin{tikzcd} G \times_S W \arrow[r, "q"] \arrow[d, "\lambda"'] & X \arrow[d, "i"] \\ G \times_S V \arrow[r] & G/N \times_S V \end{tikzcd}$$

where $\lambda(g,w) = (g, g\cdot w)$, hence $q(g,w) = (\bar{g}, g\cdot w)$, where ḡ denotes the image of g under the canonical map $G(S') \to (G/N)(S')$. One sees finally from diagram $(*)$ that $Z \to X$ makes Z into a principal bundle with base X and group N acting on the right by $(g,v)\cdot n = (gn, v)$, so that in $(**)$, $q : G\times_S W \to X$ makes $G\times_S W$ into a principal bundle with base X and group N acting on the right by

$$(g, w)\cdot n = (gn, n^{-1}\cdot w).$$

We summarize the principal morphisms above in the following diagram:

$$\begin{tikzcd}
 & & G\times_S W \arrow[dl, "q"'] \arrow[dr, "\varphi"] & \\
G/N & X \arrow[l, "p"'] \arrow[rr, "\psi"] \arrow[d, "i"] & & V \\
 & G/N\times_S V \arrow[ul, "pr_1"] \arrow[urr, "pr_2"'] & &
\end{tikzcd} \tag{D}$$

where $\psi = pr_2 \circ i$ and $\varphi = \psi\circ q$, i.e. $\varphi(g,w) = g\cdot w$. If v is a section of V over S, the subsheaf X_v of X, inverse image of this section by ψ, is given by $X_v(S')$ = set of subsheaves W' of $V_{S'}$ which are locally conjugate to $W_{S'}$ by G and which contain the section $v_{S'}$ of $V_{S'}$, while the subsheaf of $G\times_S W$ inverse image of v by φ is isomorphic to the subsheaf M_v of G given by $M_v(S')$ = set of $g\in G(S')$ such that $v_{S'}\in g\cdot W(S')$, i.e. such that $g^{-1}v_{S'}\in W(S')$. If v is a section of W and not merely of V, then M_v evidently contains N.

In these explicit descriptions we have not used the fact that G, V, W are representable (nor that the site over which we work is defined in terms of preschemes!). But suppose now that N is representable and faithfully flat and quasi-compact over S, and that G/N is representable. When S is the spectrum of a field, and G is of finite type over k, one knows that this hypothesis is necessarily satisfied (VIII 6 and VI_B.11.18). One then sees from the cartesian diagram $(**)$, using the theory of faithfully flat quasi-compact descent and the fact that $Z\to G\times_S V$ is a closed immersion, that X is representable (it is obtained by descent of the closed subprescheme Z of $G\times_S V$ by the faithfully flat quasi-compact morphism $G\times_S V\to G/N\times_S V$). Hence diagram (D) is a diagram of morphisms of preschemes over S.

We henceforth assume that S is the spectrum of a field k, and that G, V, W are of finite type over k. Let $\mathfrak{n}$ be the Lie algebra of N, so that $\dim N \leqslant rank\mathfrak{n}$, with equality if and only if N is smooth over k (Exp VI [1087]). Let $a\in W(k)$, and consider the subscheme M_a of G defined above, containing N, and isomorphic to $\varphi^{-1}(a)$; we shall denote by $\mathfrak{m}_a$ its Zariski tangent space at the identity element e of N, so that

[1087] Not having identified this reference, we refer to Theorem II.5.2.1 of the book: M. Demazure & P. Gabriel, *Groupes algébriques* I, Masson (1970).

one has

$$(1) \qquad \mathfrak{n} \subset \mathfrak{m}_a, \qquad \dim N \leq \operatorname{rank} \mathfrak{n} \leq \operatorname{rank} \mathfrak{m}_a.$$

Lemma 1.1. *With the preceding notation:*

a) Consider the following conditions:

- *(i)* $\mathfrak{n} = \mathfrak{m}_a$ *and* N *is smooth over* k.
- *(i bis)* $\dim N = rank_k \mathfrak{m}_a$.
- *(ii) The morphism* $\psi : X \to V$ *is unramified at* $(\bar{e}, a)$.
- *(iii)* M_a *and* N *coincide in a neighborhood of* e.

Then one has the implications (i) ⇔ *(i bis)* ⇒ *(ii)* ⇔ *(iii).*

b) Suppose $\varphi : G \times_S W \to V$ *is smooth at* (e, a). *Then* M_a *is smooth over* k *at* e, *and* ψ *is smooth at* $(\bar{e}, a)$.

Proof. a) The equivalence of (i) and (i bis) follows at once from the relations (1) and from the fact already noted that N is smooth over k if and only if $\dim N = rank_k \mathfrak{n}$. On the other hand, consider the inclusion morphism $N \to M_a$; it is well known [1088] that if N is smooth over k at e and the tangent map at e is surjective, then $N \to M_a$ is smooth at e, hence (being an immersion) is an isomorphism at e, which shows that (i) implies (iii). To prove the equivalence of (ii) and (iii), consider as above $X_a = \psi^{-1}(a)$ and use the isomorphism $M_a \simeq \varphi^{-1}(a) \simeq q^{-1}(X_a)$ to obtain a morphism $p_a : M_a \to X_a$ which makes M_a a principal homogeneous bundle with group N_{X_a}. Consider the following diagram

$$\begin{tikzcd} M_a \arrow[d, "p_a"'] & N \arrow[l, "j_a"'] \arrow[d] \\ X_a & S = \operatorname{Spec}(k), \arrow[l, "j_a'"'] \end{tikzcd}$$

where $\operatorname{Spec}(k) \to X_a$ is defined by the point $(\bar{e}, a)$ of X, and $j_a : N \to M_a$ is the canonical immersion. To say that ψ is unramified at $(\bar{e}, a)$ means that j'_a is an open immersion, or equivalently that it induces an isomorphism $S \xrightarrow{\sim} \operatorname{Spec}(\mathcal{O}_{X_a, \bar{a}})$. Since p_a is flat, it is equivalent to say that the morphism deduced from the preceding by the base change $\operatorname{Spec}(\mathcal{O}_{M_a, e}) \to \operatorname{Spec}(\mathcal{O}_{X_a, \bar{a}})$ is an isomorphism; now this deduced morphism is none other than the morphism $\operatorname{Spec}(\mathcal{O}_{N,e}) \to \operatorname{Spec}(\mathcal{O}_{M_a, e})$, which proves the equivalence of (ii) and (iii).

One will note moreover that the proof shows that conditions (ii), (iii) imply the following condition, apparently stronger than (iii):

- (iii bis) N *is an open and closed subscheme of* M_a.

b) The first assertion follows from the fact that M_a is isomorphic to $\varphi^{-1}(a)$, the second from the fact that q is flat and $q(e, a) = (\bar{e}, a)$.

[1088] See, for example, EGA IV₄, Th. 17.11.1 d).

2. Density theorem and theory of the regular points of G

We shall apply the constructions and notation of the preceding section to the case where G is a connected smooth algebraic group over k, where $V = G$ on which G acts by inner automorphisms, and where W is a connected smooth algebraic subgroup H of G. We shall denote by $\mathfrak{g}$ the Lie algebra of G, by $\mathfrak{h}$ that of H, by N the normalizer of H in G, and by $\mathfrak{n}$ the Lie algebra of N. If $a \in G(k)$, we shall denote, as in section 1, by M_a the symmetric of its transporter into H, so that if $a \in H(k)$, one has $N \subset M_a$; in this case, we denote by $\mathfrak{m}_a$ the Zariski tangent space of M_a at the identity element e of G. Note that

$$\mathfrak{h} \subset \mathfrak{n} \subset \mathfrak{m}_a \quad \text{for } a \in H(k).$$

We shall need the following:

Lemma 2.0. *In order that $H = N^0$, it is necessary and sufficient that $(\mathfrak{g}/\mathfrak{h})^H = 0$ (where the first member denotes the subspace of invariants under the action of H deduced from the adjoint representation). When this condition is satisfied, N is smooth and $\dim X = \dim G$. In any case, $\dim X \leqslant \dim G$, and this inequality is an equality if and only if H is of finite index in N.*

Indeed, one has seen (II 5.2.3 (i)) that $\mathfrak{n}$ equals the inverse image of $(\mathfrak{g}/\mathfrak{h})^H$ under the morphism $\mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$, hence $(\mathfrak{g}/\mathfrak{h})^H = 0$ is equivalent to $\mathfrak{h} = \mathfrak{n}$, which is also equivalent (since H is a smooth connected algebraic subgroup of the algebraic group N) to $H = N^0$ (cf. VI.2). This evidently implies that N is smooth. On the other hand, one has

$$\dim X = (\dim G - \dim N) + \dim H = \dim G - (\dim N - \dim H),$$

hence

$$\dim X \leqslant \dim G,$$

with equality if and only if $\dim H = \dim N$, i.e. if and only if H is of finite index in N. This is the case in particular if $H = N^0$, which completes the proof of 2.0.

Theorem 2.1. *Let G be a smooth connected algebraic group over the algebraically closed field k, H a connected smooth algebraic subgroup, N its normalizer, $\mathfrak{g}$, $\mathfrak{h}$, $\mathfrak{n}$ the Lie algebras, $X = G \times^N H$ the scheme (fibered over G/N with typical fiber H) introduced in section 1, $\psi : X \to G$ the canonical morphism (whose image is also the image of $\varphi : G \times H \to G$ defined by $\varphi(g,h) = int(g)h = ghg^{-1}$). The following conditions are all equivalent:*

- *(i) H contains a Cartan subgroup (XII 1) C of G.*
- *(i bis) H has the same reductive rank and the same nilpotent rank (XII 1) as G.*
- *(ii) H contains a maximal torus T of G, and $(\mathfrak{g}/\mathfrak{h})^T = 0$.*
- *(iii) The set of conjugates of H containing a given maximal torus is finite and non-empty, and H is of finite index in its normalizer.*

- *(iv) There exists $a \in H(k)$ contained only in finitely many conjugates of H (or merely such that $\psi^{-1}(a)$ has an isolated point), and H is of finite index in its normalizer.*
- *(iv bis) The morphism $\psi : X \to G$ is generically quasi-finite (i.e. there exists a dense open of X on which ψ is quasi-finite), and H is of finite index in its normalizer.*
- *(v) There exists a dense open U in G such that for every $x \in U(k)$, the set of conjugates of H containing x is finite and non-empty, i.e. $\psi : X \to G$ is dominant and generically quasi-finite.*
- *(vi) There exists a dense open U of G such that every $x \in U(k)$ is contained in a conjugate of H, i.e. $\psi : X \to G$ is dominant.*
- *(vii) There exists $a \in H(k)$ such that the subspace of $\mathfrak{g}/\mathfrak{h}$ of fixed points of $ad_{\mathfrak{g}/\mathfrak{h}}(a)$ is zero.*

Furthermore, these conditions imply that H is its own connected normalizer, i.e. N is smooth and $\dim H = \dim N$, *and that $\psi : X \to G$ is generically étale.*

Proof. By 2.0, one has $\dim X \leqslant \dim G$, with equality if and only if $\dim H = \dim N$, i.e. H of finite index in N. From the inequality $\dim X \leqslant \dim G$ it follows that ψ is dominant if and only if it is dominant and generically quasi-finite, or again if and only if ψ is generically quasi-finite and $\dim X = \dim G$. As this latter equality means also, by 2.0, that H is of finite index in N, we have proved the equivalence of (vi), (v), (iv bis). The equivalence of (iv) and (iv bis) is immediate.

The equivalence of (i) and (i bis) is immediate from the definitions, and is left to the reader. On the other hand, if H contains a Cartan subgroup C of G, then it contains the maximal torus T of C, which is a maximal torus of G. Since C is the centralizer of T, its Lie algebra $\mathfrak{c}$ is given by

$$\mathfrak{c} = \mathfrak{g}^T$$

(II, 5.2.3 (ii)). Hence as $H \supset C$, so that $\mathfrak{h} \supset \mathfrak{c}$, it follows that $\mathfrak{c} \subset \mathfrak{g}^T$, which is equivalent (by I, 4.7.3) to the relation

(✱) $(\mathfrak{g}/\mathfrak{h})^T = 0.$

Conversely, suppose that H contains the maximal torus T and that the preceding relation holds, i.e. that $\mathfrak{h} \supset \mathfrak{c}$; I claim that H contains the centralizer C of T (which will establish (i) $\Leftrightarrow$ (ii)). This follows from the:

Lemma 2.1.1. *Let G be a smooth algebraic group over the field k, T a subgroup of multiplicative type of G, C its connected centralizer (equal to the centralizer of T if G is connected and T is a torus, (XII 6.6 b))), H a smooth subgroup of G containing T. In order that H contain C, it is necessary and sufficient that its Lie algebra $\mathfrak{h}$ contain that $\mathfrak{c}$ of C.*

Indeed, one knows (XI 2.4) that $Centr_G(T)$ is smooth over k, hence C is smooth over k; similarly $Centr_H(T)$ is smooth over k, and $Centr_H(T) = Centr_G(T) \cap H$ has $\mathfrak{h} \cap \mathfrak{c}$ as Lie algebra; hence the hypothesis implies that the smooth subgroup $Centr_H(T)$ of the smooth group $Centr_G(T)$ has the same Lie algebra, so it contains the connected component C of the latter, and thus H contains C. *QED*.

Let us prove the equivalence of (i) and (iii), which amounts to proving that if H contains the maximal torus T of G, then the condition $H \supset C$ (which is also equivalent to $(*)$ above, as we have just seen) is equivalent to the fact that H is of finite index in its normalizer and that the set of conjugates of H containing T is finite. If H contains C, hence if one has $(*)$, then a fortiori

$$(**) \qquad (\mathfrak{g}/\mathfrak{h})^H = 0,$$

and we know that $\mathfrak{n}$ is the inverse image of the first member of the preceding relation under the canonical homomorphism $\mathfrak{g} \to \mathfrak{g}/\mathfrak{h}$ (II 5.2.3 (i)), so that the preceding relation means, by 2.0, that $H = N^0$, and a fortiori H is of finite index in its normalizer. Now consider the diagram of subgroups

$$\begin{tikzcd}
T \arrow[r] & N(T) \cap H \arrow[r] \arrow[d] & H \arrow[d] \\
& N(T) \cap N(H) \arrow[r] \arrow[d] & N(H) \\
& N(T). &
\end{tikzcd}$$

Using the theorem of conjugacy of maximal tori in H (XII 6.6 a)), one sees that every conjugate of H containing T is a conjugate of H by an element of $N(T)(k)$, so the set of conjugates of H containing T is in bijective correspondence with the set of points of $N(T)/N(T) \cap N(H)$ with values in k; but as $H \supset C$, one has $N(T) \cap H \supset C$, hence the preceding set is a quotient of $(N(T)/C)(k)$, which is a finite set, hence is finite. This proves (i) ⇒ (iii). Conversely, suppose (iii), i.e. $N(T)/N(T) \cap N(H)$ finite and $N(H)/H$ finite. Using again the conjugacy theorem in H, one sees again that the homomorphism

$$N(T) \cap N(H)/N(T) \cap H \longrightarrow N(H)/H$$

induced by the preceding diagram is bijective on points with values in k (in fact, it is an isomorphism), hence as the latter is finite, so is the former, so $N(T) \cap H$ is of finite index in $N(T)$, hence contains $C = N(T)^0$, hence $H \supset C$. Thus (i), (i bis), (ii), (iii) are equivalent conditions.

Let us prove that (ii) ⇒ (vii). One sees at once that conditions (ii) and (vii) are each invariant under an extension $k \to k'$ of the base field, with k' algebraically closed, which allows us to assume that k has infinite transcendence degree over its prime subfield. Then it is well known (and easily verified) that there exists an element a of $T(k)$ such that the subgroup of $T(k)$ it generates is dense in T for the Zariski topology. One easily concludes that $(\mathfrak{g}/\mathfrak{h})^T = (\mathfrak{g}/\mathfrak{h})^{ad(a)}$, and as the first member is zero by hypothesis, one concludes (vii).

Let us prove (vii) ⇒ (vi). This implication is contained in the following result, which sharpens 2.1:

Corollary 2.2. *Let G be a smooth algebraic group over a field k, H a smooth algebraic subgroup, N its normalizer in G, $\varphi : G \times H \to G$ the morphism defined by $\varphi(g,h) = ad(g)h = ghg^{-1}$, $\psi : X = G \times^N H \to G$ the morphism deduced from φ by passage to the quotient (cf. section 1), $a \in H(k)$. The following conditions are equivalent:*

- *(i) φ is smooth at (e,a).*
- *(ii) ψ is étale at $(\bar{e},a)$, and N is smooth over k.*
- *(iii) $(\mathfrak{g}/\mathfrak{h})^{ad(a)} = 0$ (where $\mathfrak{g}$, $\mathfrak{h}$ are the Lie algebras of G, H).*

These conditions imply $H^0 = N^0$.

We know that the smoothness of φ (which is a morphism between smooth k-preschemes) at a k-rational point is equivalent to the surjectivity of the tangent map at that point. Now an immediate calculation shows that this tangent map is written (using the usual identifications of the tangent spaces at points of G and H with the Lie algebras of G and H)

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dϕ(ξ, η) = (id − ad(a)) · ξ + η,
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regarded as a map from $\mathfrak{g}\times\mathfrak{h}$ to $\mathfrak{g}$. Surjectivity therefore amounts to the surjectivity of $(id - ad(a))$ on $\mathfrak{g}/\mathfrak{h}$, i.e. to (iii). Now (iii) implies a fortiori $(\mathfrak{g}/\mathfrak{h})^H = 0$, i.e. (cf. 2.0, where the connectedness hypothesis at the beginning of the section is unnecessary) $H^0 = N^0$. From this we deduce that N is smooth, and $\dim H = \dim N$, whence $\dim X = \dim G$. Now since $q : G \times H \to X$ is flat and $\psi \circ q = \varphi$ is smooth at (e,a), it follows that ψ is smooth at $q(e,a) = (\bar{e},a)$, hence étale at this point by reasons of dimension. So we have proved (i) ⇔ (iii) ⇒ (ii); on the other hand (ii) ⇒ (i), because the smoothness of N implies that of q. *QED*.

Let us finally prove (vi) ⇒ (i), which, together with the implications already established, will prove the theorem. Suppose first that G is affine. Let U be a non-empty open of G such that $x \in U(k)$ implies that x is contained in a conjugate of H. Let C be a Cartan subgroup of G. Using the implication (i) ⇒ (iv) for C in place of H (this is Borel's "density theorem"), it follows that one can find a conjugate of C which meets U, so one can assume $U \cap C \neq \emptyset$, i.e. that there exists a non-empty open V in C such that for every $x \in V(k)$, x is contained in a conjugate of H. Write C as a product

$$C = T \cdot C_u$$

where T is the maximal torus of C (which is a maximal torus of G) and C_u is the unipotent part of C, T being in the center of C (*Bible* 6 th. 2). We can again assume that k has infinite transcendence degree over its prime subfield, which allows us to find an element t of $T(k)$ that is in the projection of V onto T (which is a non-empty open of T), i.e. $t \cdot C_u \cap V \neq \emptyset$, and such that t "generates" T. Since every algebraic subgroup of G containing a product $t \cdot u$ ($t \in T(k)$, $u \in C_u(k)$) contains the two factors (*Bible* 4 th. 3), it follows, with the preceding choice of t, and taking

$t \cdot u \in V(k)$, that there exists a conjugate of H containing t, hence T. So we may already assume that one has

$$T \subset H.$$

If W is the open of C_u inverse image of V by $u \mapsto t \cdot u$, one sees therefore that for every element x of $(T \cdot W)(k)$, there exists a conjugate of H containing T and x. As we have already noted, such a conjugate is of the form $int(g) \cdot H$, where $g \in N(T)(k)$. Consider then the morphism

$$f : N(T) \times H \to G$$

defined by $f(g,h) = int(g) \cdot h = ghg^{-1}$; then the image of f contains $T \cdot W$, so as $N(T)$ is a finite union of translates $C \cdot g_i$ (where $g_i \in N(T)(k)$) since C is of finite index in $N(T)$, it follows that there exists a dense open V_0 of $C = T \cdot C_u$ which is contained in the image of $(C \cdot g_i) \times H$ by f. Replacing H by $int(g_i) \cdot H$ if necessary, we may assume $g_i = e$, i.e. $f(C \times H) \supset V_0$. So for every $u \in V_0(k)$, there exist $v \in C(k)$ and $h \in H(k)$ such that

$$v^{-1} h v = u, \text{ whence } vuv^{-1} \in H(k),$$

hence, setting

$$C_0 = C \cap H = \mathrm{Centr}_H(T),$$

$vuv^{-1} \in C_0$, whence $int(v) \cdot C_0 \supset u$. This proves that the union of the conjugates of C_0 in C (by elements of $C(k)$) is dense, which implies (as we have seen for the pair (G,H) instead of (C,C_0)) that C_0 is of finite index in its normalizer in C. By *Bible* 7, lemma of section 1, it follows that $C_0 = C$, hence $C = H$, which proves (vi) ⇒ (i) when G is affine.

In the general case, we proceed by induction on $n = \dim G$, the assertion being trivial if $n = 0$. Let Z be the center of G, and distinguish two cases:

1°) $\dim Z \cap H > 0$: then setting $G' = G/(Z \cap H)$, one has $\dim G' < n$; on the other hand the hypothesis (vi) on H implies the same condition for the image H' of H in G', so H' contains a Cartan subgroup C' of G', hence H contains the inverse image C of C', which is a Cartan subgroup by XII 6.6 e).

2°) $\dim Z \cap H = 0$: then the canonical morphism $H \to G/Z$ is a finite morphism, and as G/Z is affine by virtue of XII 6.1, it follows that H is affine, hence every homomorphism from H into an abelian variety is zero (and even every morphism of preschemes from H to an abelian variety is zero): this follows from the fact that an affine smooth connected algebraic group over an algebraically closed field is a rational variety, or simply that it is the union of its Borel subgroups (*Bible* 6 th. 5 b)), and it follows very easily from the structure theorems *Bible* 6.2 and 6.3 that an affine smooth connected solvable group is a rational variety. Now use Chevalley's structure theorem for G, according to which G is an extension of an abelian variety A by a smooth affine group. Then the image of H in A is zero, H being affine; on the other hand it is identical to A, because the union of its conjugates in A must be dense. So $A = 0$, hence G is affine, and we are reduced to the case already treated. This completes the proof of 2.1.

Corollary 2.3. *Assume that the equivalent conditions of 2.1 hold.*

a) Let $k(X)$ (resp. $k(G)$) be the field of rational functions of X (resp. G); then $k(X)$ is a finite separable extension of $k(G)$. Denote its degree by d.

b) Let T be a maximal torus of G contained in H (which exists by form 2.1 (ii)), and let C be the corresponding Cartan subgroup of G. Then $C \subset H$. On the other hand, $Norm_G(T)$ is a smooth subgroup of G and `Norm_G(T) ∩ Norm_G(H) = Norm_{Norm_G(H)}(T)` is a smooth subgroup of it, of finite index equal to d (defined in a)). The number of conjugates of H containing a given maximal torus or a given Cartan subgroup is equal to d.

c) Let U be the largest open of G such that $\psi : X \to G$ induces a morphism $\psi^{-1}(U) \to U$ that is finite and étale. Then U is a dense open, and for $g \in G(k)$, one has $g \in U(k)$ if and only if there exist exactly d conjugates of H containing g, or again if and only if there exist at least d distinct conjugates H_i of H containing g such that for every i, $(\mathfrak{g}/\mathfrak{h}_i)^{ad(g)} = 0$ (where $\mathfrak{h}_i = Lie(H_i)$).

Assertion a) comes from the fact that ψ is generically étale (which was stated at the end of 2.1); this also implies that the open U introduced in c) is non-empty, i.e. dense, and the two characterizations stated for the elements of $U(k)$ (taking into account that ψ is separated, X integral and G integral normal, SGA 1 I 10.11, and that $(\mathfrak{g}/\mathfrak{h}_i)^{ad(g)} = 0$ means that ψ is étale at the point x_i of $\psi^{-1}(g)$ corresponding to H_i). If H contains the maximal torus T of G, then the centralizers of T in H and G have the same dimension, and are smooth and connected (XII 6.6 b)), hence are equal, which proves that $C \subset H$. Moreover, one knows that the normalizer of T in a smooth group containing it is smooth (XI, 2.4 bis), so $Norm_G(T)$ and $Norm_{Norm_G(H)}(T)$ are smooth (N.B. we have noted that $N = Norm_G(H)$ is smooth at the end of the statement of 2.1); also $Norm_N(T)$ contains C, which is of finite index in $N(T)$, so it is of finite index in $N(T)$. Using the theorem of conjugacy for maximal tori in H, one sees that the index in question equals the number of conjugates of H containing T, or what amounts to the same, containing C. Now since the union of the conjugates of C in G is dense (by 2.1 (i) ⇒ (vi) applied to C instead of H), and the open U defined in c) is evidently stable under inner automorphisms, one sees that $C \cap U \neq \emptyset$. Proceeding as in the proof of the implication (vi) ⇒ (ii) of 2.1, one concludes that (up to a harmless change of base field) there exists $g \in (C \cap U)(k)$ such that every conjugate of H containing g contains T, and hence also C. So the conjugates of H containing C are those containing g, and as $g \in U(k)$, their number equals d, which completes the proof of b). One will note that we have in fact shown that the set of conjugates of H containing T is a homogeneous set under the group of rational points of

$$W_G(T) = Norm_G(T)/Centr_G(T),$$

which proves in particular that

d □ order of the Weyl group of G.

Corollary 2.4. *With the notation of 2.1, the following conditions are equivalent:*

- *(i) $\psi : X \to G$ is a birational morphism.*
- *(ii) There is exactly one conjugate of H containing a given Cartan subgroup of G.*
- *(iii) H contains a Cartan subgroup C of G, and $Norm_G(H) \supset Norm_G(C)$.*
- *(iv) There exists a non-empty open V of G such that $g \in V(k)$ implies that g is contained in exactly one conjugate of H.*

This is clear from 2.1 and 2.3.

Corollary 2.5. *Assume that the conditions of 2.4 hold, and let $g \in G(k)$. The following conditions are equivalent:*

- *(i) $g \in U(k)$, where U is defined in 2.3 c), i.e. g is contained in exactly one conjugate of H.*
- *(ii) The set of conjugates of H containing g is finite and non-empty.*
- *(iii) The scheme $\psi^{-1}(g)$ "of conjugates of H containing g" has an isolated point.*
- *(iv) There exists a conjugate H' of H containing g, and one has $(\mathfrak{g}/\mathfrak{h}')^{ad(g)} = 0$, where $\mathfrak{h}' = Lie(H')$.*

Finally, U is also the largest open of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$.

The equivalence of (i), (ii), (iii) and the last assertion follow from the "Main Theorem"[1089] applied to the birational morphism $\psi : X \to G$, given that G is normal. The equivalence of these conditions with (iv) follows at once from the last assertion of 2.1 characterizing the set of points of X at which ψ is étale.

Theorem 2.6. *Let G be a smooth connected algebraic group over an algebraically closed field k, C a Cartan subgroup, associated to a maximal torus T, $N = Norm_G(C) = Norm_G(T)$ (cf. XII 8.4), let $X = G \times^N C$, where N acts on the left factor G by right translations, and on the right factor C by inner automorphisms, and let $\psi : X \to G$ be the canonical morphism.*

a) The morphism ψ is birational.

b) Let U be the largest open of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$ (cf. 2.5). Let

ρ = ρ_ν(G) = dim C

be the nilpotent rank of G. Then for every $g \in G(k)$, the multiplicity of the eigenvalue 1 in $ad(g)$ acting on $\mathfrak{g}$ is at least equal to ρ, and for it to equal ρ, it is necessary and sufficient that $g \in U(k)$.

[1089] of Zariski!

Proof. Since condition 2.1 (i) is satisfied, one can apply 2.4 (iii) ⇒ (i), which establishes a). In *Bible* 7 (in the case where G is affine) the points of $U(k)$ are called the *regular points* of $G(k)$, and we shall follow this terminology, calling U the *open of regular points* of G. (N.B. The proof given in the *Bible* that the set in question is itself open is incorrect, but we have obtained it in the present Exposé under more general conditions.)

Let us prove b); for this, introduce for every $g \in G(k)$ the characteristic polynomial

$$P(ad(g), t) = t^n + c_1(g)\, t^{n-1} + \cdots + c_n(g);$$

one sees at once (replacing k by any algebra over k) that the $c_i(g)$ come from well-determined sections

$$c_i \in \Gamma(G, \mathcal{O}_G).$$

When $g \in G(k)$ is an element contained in a Cartan subgroup (for example a regular element), which we may assume to be C, then by 2.5 (iv) one sees that $(\mathfrak{g}/\mathfrak{c})^{ad(g)} = 0$ if and only if g is regular (where $\mathfrak{c}$ denotes the Lie algebra of C); on the other hand, since C is nilpotent, one sees at once that $ad_{\mathfrak{c}}(g)$ has only the eigenvalue 1, which proves that the multiplicity of the eigenvalue 1 in $ad_{\mathfrak{g}}(g)$ is $\geqslant \rho$, and equal exactly to $\dim C = \rho$ if and only if g is regular. In particular, the polynomial above is divisible by $(t-1)^{\rho}$. Since the relation of divisibility by $(t-1)^{\rho}$ is expressed by linear relations (with integer coefficients) among the coefficients of the polynomial, and these relations are satisfied for $g \in U(k)$, U being a dense open, it follows (G being reduced) that they hold for all g; in fact one has a relation

$$(\dagger) \qquad t^n + c_1\, t^{n-1} + \cdots + c_n = (t-1)^{\rho}\,(t^{n-\rho} + b_1\, t^{n-\rho-1} + \cdots + b_{n-\rho})$$

in the ring of polynomials over $\Gamma(G, \mathcal{O}_G)$; in particular for every $g \in G(K)$, $ad(g)$ has the eigenvalue 1 with multiplicity at least ρ. Moreover, we have seen that equality holds if g is regular; let us prove the converse. To this end, suppose first G affine, and write g as a product

$$g = g_s g_u$$

of its semisimple part and its unipotent part (*Bible* 4 section 4); then

$$ad(g) = ad(g_s) ad(g_u)$$

is the analogous decomposition of $ad(g)$ (loc. cit. cor. to th. 3), so $ad(g)$ and $ad(g_s)$ have the same eigenvalues (counted with multiplicity); in particular the eigenvalue 1 appears with the same multiplicity in $ad(g)$ and in $ad(g_s)$.

Moreover, by *Bible* 7 th. 2 cor. 1, g is regular if and only if g_s is. Hence to prove b), we may assume $g = g_s$, i.e. g semisimple, so contained in a maximal torus by *Bible* 6 th. 5 c), and a fortiori in a Cartan subgroup — the case already treated. This proves b) in the case G affine. In the general case, let $Z = Centr(G)_{red}$; then by XII 6.6 e) the Cartan subgroups of G are the inverse images of those of $G' = G/Z$, hence g is regular in G if and only if its image g' in G' is regular in G'. On the other hand, since Z is smooth, the Lie algebra $\mathfrak{g}'$ of G' is none other than $\mathfrak{g}/\mathfrak{z}$, where $\mathfrak{z} = Lie(Z)$, and $ad(g')$ is none other than $ad(g)_{\mathfrak{g}/\mathfrak{z}}$; hence the multiplicity of the eigenvalue 1 in $ad(g)$ equals $d = \dim Z$ plus the multiplicity of the eigenvalue 1 in $ad(g')$, whence at once the first is equal to the nilpotent rank of G if and only if the second is equal to the nilpotent rank of G'. Thus we are reduced to the case of G'; now G' being affine by XII 6.1, this case has already been treated. This completes the proof of 2.6.

Corollary 2.7. *With the notation of the preceding proof,*[1090] *let*

$b = 1 + b_1 + \cdots + b_{n-\rho} \in \Gamma(G, \mathcal{O}_G)$.

Then the open of regular points of G is given by

$$U = G_b$$

(set of points of G at which b is invertible); in particular U is an affine open if G is affine.

Corollary 2.8. *Let H be a smooth connected algebraic subgroup of G containing a Cartan subgroup of G.*

a) Let C be an algebraic subgroup of H. In order that C be a Cartan subgroup of H, it is necessary and sufficient that it be a Cartan subgroup of G.

b) Let $g \in G(k)$, and let d be the integer introduced in 2.3. In order that g be a regular point of G, it is necessary and sufficient that there exist exactly d conjugates H_i of H containing g, and that for each i, g be a regular element of H_i; or again, that there be at most d conjugates of H containing g, and that g be regular in one of them. If so, and if C is the unique Cartan subgroup of G containing g, then the conjugates of H containing g are the conjugates of H containing C.

c) Let $g \in H(k)$; in order that g be regular in G, it is necessary and sufficient that it be regular in H, and that one have $(\mathfrak{g}/\mathfrak{h})^{ad(g)} = 0$.

Let us prove a). Under either hypothesis on C, the unique maximal torus T of C is a maximal torus of H and of G (H having the same reductive rank as G); hence as $Centr_H(T) \subset Centr_G(T)$ are smooth connected groups of the same dimension, they are equal, so it amounts to the same to say that C is equal to one or the other of these two groups, which proves a).

Let us prove b). Suppose first g regular in G, let C be the unique Cartan subgroup of G containing g. Then by 2.3 b) there exist exactly d conjugates H_i of H containing C. Since $(\mathfrak{g}/\mathfrak{c})^{ad(g)} = 0$, i.e. $ad(g)$ has no eigenvalue +1 on $\mathfrak{g}/\mathfrak{c}$, one has a fortiori $(\mathfrak{g}/\mathfrak{h}_i)^{ad(g)} = 0$, hence by 2.3 c) there are exactly d conjugates of H containing g,

[1090] cf. (†) above.

namely the H_i. For such an H_i, a Cartan subgroup of H_i containing g is a Cartan subgroup of G containing g by a), hence equals C, which proves that g is regular in H_i. Conversely, suppose that there are at most d conjugates H_i of H containing g, and that g is regular in one of them, which we may assume to be H. Let us prove that g is regular in G. Since g is regular in H, it is contained in a unique Cartan subgroup C of H; by a) this is a Cartan subgroup of G. Let C' be a Cartan subgroup of G containing g; let us prove $C' = C$ (which will prove that g is regular in G). Indeed, by 2.3 b) there exist exactly d conjugates of H containing C', and as these latter contain g, they are necessarily the H_i, hence the H_i and in particular H contain C'. Hence C, C' are two Cartan subgroups of H (by a)) that contain the same regular element g of H, so they are equal. *QED*.

Let us prove c). Denoting by $\nu(u)$ the nullity[1091] of $id - u$, for an endomorphism of a finite-dimensional vector space, one has

$$\nu(ad(g)_{\mathfrak{g}}) = \nu(ad(g)_{\mathfrak{h}}) + \nu(ad(g)_{\mathfrak{g}/\mathfrak{h}}),$$

and the two terms on the right-hand side are respectively $\geqslant$ nilpotent rank of H (equal to the nilpotent rank ρ of G by a)) and $\geqslant 0$; hence one has $\nu(ad(g)_{\mathfrak{g}}) = \rho$ if and only if $\nu(ad(g)_{\mathfrak{h}}) = \rho$ and $\nu(ad(g)_{\mathfrak{g}/\mathfrak{h}}) = 0$, i.e. g is regular in G if and only if g is regular in H and $ad(g)_{\mathfrak{g}/\mathfrak{h}}$ has no non-zero invariants. *QED*.

Remarks 2.9. *In the statement of 2.1, one cannot weaken condition (iii) by assuming only that H contains a maximal torus and is of finite index in its normalizer, even if one further requires that this normalizer be smooth, i.e. that $H = N^0$, and even when G is affine solvable. An example is furnished by the group G of matrices of the form*

$$g = \begin{pmatrix} t & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix},$$

and the subgroup H of matrices of the preceding form with $b = c = 0$ (N.B. the Cartan subgroup of G is here the matrices g with $a = c = 0$).

Remarks 2.10. *Let G be a smooth algebraic group over k, H a smooth algebraic subgroup, but we no longer assume H and G connected. Suppose that H^0 contains a Cartan subgroup of G^0. Then $(\mathfrak{g}/\mathfrak{h})^H \subset (\mathfrak{g}/\mathfrak{h})^{H^0} = 0$, hence $H^0 = N^0$ (N is the normalizer of H), in particular N is smooth. However, one easily constructs examples, with G connected, where H has a connected component H_i such that for no $h \in H_i(k)$ one has $(\mathfrak{g}/\mathfrak{h})^{ad(h)} = 0$, i.e. the morphism $(g,h) \mapsto ad(g)\cdot h$ from $G \times H_i$ to G is étale (nor even quasi-finite) at no point (take for example for H the normalizer of the maximal torus in $SL(2)_k$). Similarly, even if the image H'' of H in the finite group $G'' = G/G^0$ equals G'', it is not necessarily true that the union of conjugates of H in G is dense (take for G the semidirect product of $\mathbb{Z}/2\mathbb{Z}$ with $G^0 = SL(2)_k$ on which it acts by "symmetry", and for H the semidirect product $\mathbb{Z}/2\mathbb{Z} \cdot T$, where T*

[1091] i.e. the dimension of its nil-space.

is a maximal torus of G^0). On the other hand, if one does not assume a priori that H^0 contains a Cartan subgroup of G^0, but that the union of conjugates of H in G is dense, then H^0 necessarily contains a Cartan subgroup of G^0: to verify this, one may evidently suppose G connected, and it suffices to redo the proof of 2.1 (vi) ⇒ (i), which is valid without assuming H connected.

3. The case of an arbitrary base prescheme

Suppose first that we are over a base field k, not necessarily algebraically closed. Since conditions 2.1 (i bis), (iv bis), (v), (vi) are invariant under extension of the base field, one sees by passing to the algebraic closure $\bar{k}$ of k that they are equivalent to each other, and equivalent to the fact that $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$. When this condition is satisfied, then (with the notation of 2.3) it will still be true that $k(X)$ is a finite separable extension of $k(G)$, of degree d independent of any extension of the base. If U is the largest open of G such that ψ induces a morphism $\psi^{-1}(U) \to U$ that is finite and étale, then the formation of U commutes with extension of the base field. If ψ is birational, then U is also the largest open of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$, and if then $g \in U(k)$, there exists a subgroup H' of G and only one, conjugate to H over the algebraic closure $\bar{k}$ of k, such that $H' \ni g$.

A point $g \in G(k)$ is said to be *regular* if it is regular as an element of $G(\bar{k}) = G_{\bar{k}}(\bar{k})$. More generally, the construction of 2.7 gives us an open of G, whose formation commutes with any extension of the base field, called the *open of regular points* of G, which is also characterized by the fact that for every algebraically closed extension K of k and every point $g \in G(K)$, g is a regular point of G_K if and only if $g \in U(K)$. If $g \in U(k)$,[1092] one sees that g is contained in one and only one Cartan subgroup of G, as we shall show under more general conditions below.

Let G be a smooth, separated, finite-type prescheme in groups with connected fibers over the prescheme S; consider the functor $\mathcal{C} : (Sch)^\circ/S \to (Ens)$ defined by

$\mathcal{C}(S') =$ set of Cartan subgroups (XII 3.1) of $G_{S'}$.

Suppose this functor is representable by a smooth prescheme over S; we give in XV a necessary and sufficient condition for this to be so, but we already know that this hypothesis is satisfied if G is affine over S with locally constant reductive rank (XII 3.3), or more generally if G admits locally for the fpqc topology a maximal torus (XII 7.1 a)), for example if S is the spectrum of a field. Let X be the Cartan subgroup of the $\mathcal{C}$-prescheme in groups $G_{\mathcal{C}}$, the "universal Cartan subgroup" of G. As a prescheme over S, X therefore represents the functor

$X(S') =$ set of pairs (C, g), C a Cartan subgroup of $G_{S'}$ and g a section of C over S'.

Consider the canonical projection morphism $(C, g) \mapsto g$

$$\psi : X \to G.$$

[1092] correction of $G(k)$ to $U(k)$.

One then has the:

Theorem 3.1. *Under the preceding conditions on G, and with the preceding notation, let U be the set of $g \in G$ such that g is a regular element of its fiber G_s.*

Then U is open, and it is also the largest open U of G such that ψ induces an isomorphism $\psi^{-1}(U) \xrightarrow{\sim} U$.

Let us first prove that U is open. From the hypothesis of representability of $\mathcal{C}$ as a smooth prescheme over S, since its structural morphism is evidently surjective, one concludes at once that G admits locally for the étale topology a Cartan subgroup, and that the nilpotent rank of the fibers of G is locally constant. The same holds for the dimension of the fibers of G, and up to localizing on S, one may assume that both are constant, say ρ and n. Consider then the Killing polynomial

$$P_G(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t], \quad \text{where } A = \Gamma(G, \mathcal{O}_G).$$

The restriction of this polynomial to the fibers G_s of G, and in particular to the fibers at the maximal points of S, is divisible by $(t-1)^\rho$, which is expressed by the fact that certain linear combinations with integer coefficients of the c_i vanish on the fibers G_s. When S is reduced (which we may assume in order to establish that U is open), it follows that they are themselves zero, hence the Killing polynomial itself is divisible by $(t-1)^\rho$, say

$$P_G(t) = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho}).$$

Let b be the sum of the coefficients $b_0 = 1, b_1, \cdots, b_{n-\rho}$ of the second factor; then by 2.7 applied to the fibers of G, one sees that

$$U = G_b,$$

which indeed proves that U is open.

To prove that $\psi^{-1}(U) \to U$ is an isomorphism, we are reduced by SGA 1, I 5.7 to verifying it fiber by fiber, which reduces to the case of a base field, which we may assume algebraically closed. Then there exists a Cartan subgroup C of G, and if N is its normalizer, $\mathcal{C}$ is identified by the conjugacy theorem XII 7.1 a) and b) with G/N, and the morphism $\psi : X \to G$ considered here is none other than that defined in section 2. One concludes then by 2.6 b). The same reasoning also shows that U is the largest open of G such that ψ induces an isomorphism $\psi^{-1}(U) \to U$.

Corollary 3.2. *Under the conditions of 3.1, let g be a regular section of G, i.e. such that for every $s \in S$, $g(s)$ is a regular point of G_s. Then there exists one and only one Cartan subgroup C of G such that g is a section of C.*

Indeed, the hypothesis on g means that g is a section of U, and the conclusion means that there exists a unique section of X lifting it, which is just another way of expressing that $\psi^{-1}(U) \to U$ is an isomorphism.

Note now that the open $\psi^{-1}(U)$ of the Cartan subgroup X of $G_{\mathcal{C}}$ is none other than the open of X consisting of the points of X that are regular in $G_{\mathcal{C}}$ (by which we mean: regular in their fiber). One thus obtains a natural "fibration" of the dense open U of regular points of G over the prescheme $\mathcal{C}$, the fibers being dense opens of Cartan subgroups of the fibers of $G_{\mathcal{C}}$ (namely the opens of regular points in G). One finds for example the following result (which will be considerably refined in the following Exposé):

Corollary 3.3. *Let G be a smooth connected algebraic group over the field k, $\mathcal{T}$ the scheme of maximal tori of G ($\simeq$ the scheme of Cartan subgroups of G). Then the field of functions $k(G)$ of G is isomorphic to the field of functions of a smooth connected affine nilpotent algebraic group C over the field of functions $k(\mathcal{T})$ of $\mathcal{T}$, namely C = "the generic Cartan subgroup of G". If G is affine of zero unipotent rank, i.e. if the Cartan subgroups of G_K are tori, then $k(G)$ is a unirational extension of $k(\mathcal{T})$.*

Of course, by *generic Cartan subgroup* of G, we mean (by abuse of language) the Cartan subgroup of $G_{k(\mathcal{T})}$ generic fiber of X over $\mathcal{T}$. It remains only to prove the last assertion of 3.3, which is contained in the following well-known result (due to Chevalley):

Lemma 3.4. *Let k be a field, T a torus over k, $k(T)$ the field of rational functions on T; then $k(T)$ is a unirational extension of k, i.e. is contained in a pure transcendental extension of k.*

Indeed, let k' be a finite separable extension of k that splits[1093] T (X 1.4); then $T \otimes_k k'$ is a rational variety, i.e. admits a dense open isomorphic to a dense open of affine space $\mathbb{A}^n_{k'}$; hence T′ = ∏_{Spec(k′)/Spec(k)} T_{k′}/Spec(k′) is a rational variety (because it admits a dense open isomorphic to a dense open of $\prod_{\mathrm{Spec}(k')/\mathrm{Spec}(k)} \mathbb{A}^n_{k'}$, which is isomorphic to affine space of dimension mn over k, where $m = [k' : k]$). Consider the norm homomorphism from T' to T (defined whenever T is a commutative group scheme over k); the composite $T \to T' \to T$ is the m-th power in T, hence dominant, hence $T' \to T$ is dominant, which proves that T is unirational.

Let us return to the conditions of 3.1, but assume further that G admits, locally for the fpqc topology, a maximal torus (XII 7.1). Let Y be the maximal torus of the Cartan subgroup X of $G_{\mathcal{C}}$, so that the morphism $\psi : X \to G$ induces a morphism $Y \to G$ whose image is set-theoretically the semisimple elements of the fibers of G (XII 8). Finally, it follows from 3.1 that the restriction of ψ to the open Y_{reg} of regular points of Y induces a closed immersion

$$Y_{reg} \to U = G_{reg}.$$

Making explicit the meaning of $Z = Y_{reg}$ as a functor on S, one finds:

Corollary 3.5. *Let G be a smooth, separated, finite-type S-prescheme in groups with connected fibers over the prescheme S, admitting locally for the fpqc topology a maximal torus. Let $Z : (Sch)^{\circ}/S \to (Ens)$ be the functor defined by*

[1093] splits (?), or: "over which T is split".

$Z(S')$ = set of regular sections of $G_{S'}$ over S' that are contained in a maximal torus of $G_{S'}$.

Then Z is representable by a closed sub-prescheme, smooth over S, of the open $U = G_{reg}$ of G introduced in 3.1.

To finish, let us note the following result, which sharpens the density theorem 2.1 (i) ⇒ (vi):

Corollary 3.6. *Under the conditions of 3.5, let C be a Cartan subgroup of G, and consider the morphism*

$$\phi : Z \times C \to G$$

defined by $\varphi(g,h) = ad(g)h = ghg^{-1}$. Then φ is dominant.

It evidently suffices to prove this fiber by fiber, which reduces us to the case where S is the spectrum of an algebraically closed field. Let T be the maximal torus of C, t_0 an element of $T(k)$ regular in G, c_0 an element of $C(k)$ regular in G; consider $\varphi^{-1}(\varphi(t_0,c_0))$, whose k-rational points are the pairs (t,c) with $t \in Z(k)$, $c \in C(k)$, such that

$$ad(t)\, c = ad(t_0)\, c_0 \quad \text{i.e.} \quad c = ad(t^{-1} t_0)\, c_0,$$

which are therefore in bijective correspondence with the $t \in Z(k)$ such that $ad(t^{-1}t_0)c_0 \in C$, or equivalently (since c_0 is regular) such that $t^{-1}t_0 \in N$ (normalizer of C), i.e. $t \in N$. One obtains an open and closed part of this fiber by restricting to the $t \in Z(k)$ such that $t \in C(k)$. So we have found a connected component of $\varphi^{-1}\varphi(t_0,c_0)$ isomorphic to T (N.B. that the preceding set-theoretic reasoning indeed gives an isomorphism of schemes is seen by replacing points with values in k by points with values in an arbitrary k-prescheme); hence the generic fiber of φ is of dimension $\leqslant \dim T$, so $Im(\varphi)$ is of dimension [illegible] $\dim Z \times C - \dim T = \dim Z + \dim C - \dim T$; now one has $\dim Z = \dim Y = \dim$ [illegible] $+ \dim T = \dim G - \dim C + \dim T$, whence finally $\dim Im(\varphi) \geqslant \dim G$, so φ is dominant. *QED*.

Remarks 3.7. *One will note that the reasoning shows moreover that the connected component at (t_0,c_0) of the fiber $\varphi^{-1}(\varphi(t_0,c_0))$ is isomorphic to T, in particular is smooth over k, and has the same dimension as the generic fiber, which implies that φ is in fact smooth at (t_0,c_0) (which one ought to be able to verify also by calculating the tangent map). It follows that under the conditions of 3.6 the induced morphism $Z \times_S C_{reg} \to G_{reg}$ (where one has set $C_{reg} = C \cap G_{reg}$) is a smooth morphism. One sees similarly that the analogous morphism $Z \times T_{reg} \to Z$ (where T is a maximal torus of G) is smooth; more generally, for every smooth connected invariant algebraic subgroup H of C containing a regular element c_0 of $G(k)$, the image of $Z \times H \to G$ is dense in that of $G \times H \to G$.*

4. Lie algebras over a field: rank, regular elements, Cartan subalgebras

In what follows of this Exposé, we take up the theory developed by Chevalley in his book *Théorie des Groupes de Lie III* (Act. Sc. Ind. 1226, Paris 1955), the technique of schemes allowing us to eliminate the hypothesis of characteristic zero. We begin by recalling in the present section certain well-known notions and results.

Let $\mathfrak{g}$ be a Lie algebra over a ring k. For every $a \in \mathfrak{g}$, one denotes by $ad(a)$ the endomorphism

$$\mathrm{ad}(a) \cdot x = [a, x]$$

of $\mathfrak{g}$, which is a derivation of the Lie algebra $\mathfrak{g}$. Now for any derivation D of the Lie algebra $\mathfrak{g}$, the nil-space of D, i.e. the union of the kernels of the iterates of D, is a Lie subalgebra of $\mathfrak{g}$, as one sees from the Leibniz formula

$$D^m([x, y]) = \sum_{0 \leq p \leq m} \binom{m}{p} [D^p x, D^{m-p} y].$$

We shall set

$$\mathrm{Nil}(a, \mathfrak{g}) = \text{nil-space of } \mathrm{ad}(a) = \bigcup_{m \geq 0} \mathrm{Ker}\, \mathrm{ad}(a)^m;$$

when no confusion need be feared, we shall denote it simply $Nil(a)$, and we shall call it the *nil-space of a (in $\mathfrak{g}$)*.

Proposition 4.1. *For every $a \in \mathfrak{g}$, its nil-space $Nil(a, \mathfrak{g})$ is a Lie subalgebra of $\mathfrak{g}$, equal to its own normalizer.*

It remains to prove that it is its own normalizer, i.e. that every element of $\mathfrak{g}/Nil$ annihilated by the adjoint representation of Nil on $\mathfrak{g}/Nil$ is zero, which is trivial (since every element in this quotient annihilated by $Ad(a)$ is zero).

In what follows of this section, we suppose that k is a field, and that $\mathfrak{g}$ is of finite dimension over k. We shall denote by $W(\mathfrak{g})$ the scheme over k defined by $\mathfrak{g}$, whose points in the k-algebra A are the elements of $\mathfrak{g} \bigotimes_k A$. If $a \in \mathfrak{g}$, the characteristic polynomial of $ad(a)$ is also called the *characteristic polynomial* or *Killing polynomial* of a in $\mathfrak{g}$, namely

$$P_{\mathfrak{g}}(a, t) = t^n + c_1(a)\, t^{n-1} + \cdots + c_n(a),$$

where $n = rank_k \mathfrak{g}$, the $c_i(a) \in k$. Taking this polynomial also for $a \in \mathfrak{g} \bigotimes_k A$,[1094] where A is any k-algebra, one sees that the $c_i(a)$ come from well-determined sections c_i of the structural sheaf of $W(\mathfrak{g})$, i.e. from elements of the symmetric algebra $A = Sym_k(\mathfrak{g}^\vee)$, where $\mathfrak{g}^\vee$ is the dual of the k-module $\mathfrak{g}$. (When k is an infinite field, the c_i are determined by knowing the corresponding polynomial functions $\mathfrak{g} \to k$, but this is no longer the case if k is a finite field.) Let r be the largest integer such that the Killing polynomial

$$P_{\mathfrak{g}}(t) = t^n + c_1\, t^{n-1} + \cdots + c_n \in A[t]$$

[1094] $W \otimes_k A$ has been corrected to $\mathfrak{g} \otimes_k A$.

is divisible by t^r, i.e. one has:

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_{n-r} t^r, \qquad c_{n-r} \neq 0.$$

The integer r is called the *nilpotent rank* of the Lie algebra $\mathfrak{g}$. It is invariant under extension of the base field.

Proposition 4.2. *Let r be the nilpotent rank of $\mathfrak{g}$, and $a \in \mathfrak{g}$. Then one has*

$$\mathrm{rank}_k \, Nil(a, \mathfrak{g}) \geq r,$$

with equality if and only if

$$c_{n-r}(a) \neq 0.$$

In this case, $Nil(a, \mathfrak{g})$ is a nilpotent Lie algebra (and we shall see in 5.7 b) the converse, when $\mathfrak{g}$ is the Lie algebra of an algebraic group G smooth over k).

The first assertion is trivial, since by definition $rank_k Nil(a, \mathfrak{g})$ = multiplicity of the zero root in $P_{\mathfrak{g}}(a, t)$. Let us prove that if $c_{n-r}(a) \neq 0$, then $Nil(a)$ is nilpotent, which also means that for every $x \in Nil(a)$, $ad_{Nil(a)}(x)$ is a nilpotent endomorphism. One may assume k algebraically closed; then as $ad(a)_{\mathfrak{g}/Nil(a)}$ is injective, there exists a non-empty open U of $W(Nil(a))$ such that for every $x \in U(k)$, $ad(x)_{\mathfrak{g}/Nil(a)}$ is injective, hence $Nil(x) \subset Nil(a)$; one may further suppose U contained in the open of points where c_{n-r} does not vanish (since this open is non-empty by $c_{n-r}(a) \neq 0$), and then $Nil(x)$ having the same dimension as $Nil(a)$, one will have $Nil(x) = Nil(a)$. Consequently, for every $x \in U(k)$, $ad(x)_{Nil(a)}$ is nilpotent, and by the principle of extension of algebraic identities, this remains true for every $x \in Nil(a)$, hence $Nil(a)$ is nilpotent.

One says that the element a of $\mathfrak{g}$ is *regular* if $c_{n-r}(a) \neq 0$, i.e. if $rank_k Nil(a, \mathfrak{g}) = r$. When k is infinite, this also means that $rank_k Nil(a, \mathfrak{g})$ is the smallest possible (for a varying in $\mathfrak{g}$). In any case, the notion of regular element of $\mathfrak{g}$ is invariant under extension of the base field, and the set of points of $W(\mathfrak{g})$ that are regular (i.e. that come from regular points of $W(\mathfrak{g})$ with values in a suitable extension of k) is open, since identical to $W(\mathfrak{g})_{c_{n-r}}$ (set of points where c_{n-r} is invertible).

Corollary 4.3. *Let a be a regular element of $\mathfrak{g}$, and $\mathfrak{h}$ a Lie subalgebra of $\mathfrak{g}$ containing a. Then $\mathfrak{h}$ is nilpotent if and only if $\mathfrak{h} \subset Nil(a, \mathfrak{g})$; in particular, $Nil(a, \mathfrak{g})$ is a maximal nilpotent subalgebra of $\mathfrak{g}$.*

Since $Nil(a)$ is nilpotent, the relation $\mathfrak{h} \subset Nil(a)$ indeed implies that $\mathfrak{h}$ is nilpotent; and conversely, if $\mathfrak{h}$ is nilpotent, it is contained in the nil-space of its element a, i.e. $\mathfrak{h} \subset Nil(a)$.

Proposition 4.4. *Suppose k infinite. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. Consider the following conditions:*

- *(i) $\mathfrak{d}$ is maximal nilpotent and contains a regular element of $\mathfrak{g}$.*
- *(i bis) $\mathfrak{d}$ is of the form $Nil(a, \mathfrak{g})$, where a is a regular element of $\mathfrak{g}$.*
- *(ii) $\mathfrak{d}$ is nilpotent and of the form $Nil(a, \mathfrak{g})$, where $a \in \mathfrak{g}$.*

- *(ii bis)* $\mathfrak{d}$ *is nilpotent, and there exists* $a \in \mathfrak{d}$ *such that* $ad(a)_{\mathfrak{g}/\mathfrak{d}}$ *is injective.*
- *(iii)* $\mathfrak{d}$ *is nilpotent and identical to its own normalizer.*

One has the implications:

`(i) ⇔ (i bis) ⇒ (ii) ⇔ (ii bis) ⇔ (iii)`

(and we shall see in 5.7 a) that if $\mathfrak{g}$ *is the Lie algebra of a smooth algebraic group, then all the preceding conditions are equivalent).*

The equivalence of (i) and (i bis) is trivial by 4.3, and these conditions trivially imply (ii). The equivalence of (ii) and (ii bis) is also trivial, as is (ii bis) ⇒ (iii) (cf. 4.1). It remains to prove the implication (iii) ⇒ (ii bis), the only one moreover that uses the fact that k is infinite, and which follows at once from:

Lemma 4.5. *Let* $\mathfrak{d}$ *be a nilpotent Lie algebra over an infinite field* k*, acting on a finite-dimensional vector space* V*. Suppose that for every* $x \in \mathfrak{d}$*, the endomorphism* $u(x)$ *is non-injective. Then there exists a non-zero element* v *of* V *annihilated by* $\mathfrak{d}$*.*

One may suppose k algebraically closed and $\mathfrak{d}$ of finite dimension. One knows then that V is a direct sum of finitely many non-zero stable subspaces V_i ($1 \leqslant i \leqslant n$), such that for every i, and every $x \in \mathfrak{d}$, $u(x)|V_i$ has a single eigenvalue $\lambda_i(x)$ (cf. Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Exercise 22). Let $c_i(x)$ be the constant term of the characteristic polynomial of $u(x)|V_i$, so that $\lambda_i(x) = 0$ if and only if $c_i(x) = 0$. Then c_i is a polynomial function on $\mathfrak{d}$, and the hypothesis means that $\mathfrak{d}$ is the union of the sets of zeros of the c_i. So one of the c_i is zero, which reduces us (replacing V by V_i) to the case where V is such that the $u(x)$ ($x \in \mathfrak{d}$) are nilpotent. But then Engel's theorem (Bourbaki loc. cit. th. 1) implies that there exists a non-zero v in V annihilated by $\mathfrak{d}$. *QED*.

One sees easily that (k being always an infinite field) conditions (i) (i bis) of 4.4 are invariant under any extension of the base field. If they are satisfied, one will say that $\mathfrak{d}$ is a *Cartan subalgebra* of $\mathfrak{g}$; in the general case (k not necessarily infinite) one will say that $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{g}$ if it becomes a Cartan subalgebra under one (and hence any) extension of the base field $k \to k'$, with k' infinite. This thus implies that $\mathfrak{d}$ is nilpotent and equal to its own normalizer.

Proposition 4.6. *a) Let* a *be an element of* $\mathfrak{g}$*. If* a *is regular, it is contained in one and only one Cartan subalgebra of* $\mathfrak{g}$ *(and we shall see in 6.1 d) the converse when* k *is algebraically closed and* $\mathfrak{g}$ *is the Lie algebra of a smooth algebraic group).*

b) Let $\mathfrak{d}$ *be a Cartan subalgebra of* $\mathfrak{g}$*,* a *an element of* $\mathfrak{d}$*; then* a *is regular in* $\mathfrak{g}$ *if and only if* $ad(a)_{\mathfrak{g}/\mathfrak{d}}$ *is injective.*

Indeed, for a) one notes that if a is regular, then $Nil(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ (because this is true over an infinite extension k' of k), and it follows then at once from 4.3 that every Cartan subalgebra of $\mathfrak{g}$ containing a is identical to the preceding one. For b) one notes that the nullity[1095] of $ad(a)_{\mathfrak{g}}$ is equal to the sum

[1095] i.e. the dimension of the nil-space.

of the nullities of $ad(a)_{\mathfrak{d}}$ and of $ad(a)_{\mathfrak{g}/\mathfrak{d}}$, and as the first equals r, the sum is equal to r if and only if $ad(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective. *QED*.

Corollary 4.7. *Let a be a regular element of $\mathfrak{g}$, $\mathfrak{d}$ a Cartan subalgebra of $\mathfrak{g}$ containing a, A an algebra over k, $\mathfrak{g}_A$ and $\mathfrak{d}_A \subset \mathfrak{g}_A$ the A-Lie algebras deduced from $\mathfrak{g}$, $\mathfrak{d}$ by base change, a_A the image of a in $\mathfrak{g}_A$. Let u be an automorphism of $\mathfrak{g}_A$. In order that $u(\mathfrak{d}_A) = \mathfrak{d}_A$, it is necessary and sufficient that one have $u(a_A) \in \mathfrak{d}_A$.*

The condition is trivially necessary; let us prove that it is also sufficient. If it is satisfied, then $\mathfrak{d}' = u(\mathfrak{d}_A)$ is a Lie subalgebra containing a_A, and every element b of which is such that $ad(b)_{\mathfrak{d}'}$ is nilpotent (because $\mathfrak{d}'$ is isomorphic to $\mathfrak{d}_A$, which has this property, as follows at once from the definition of "nilpotent" in Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, def. 1). Taking $b = a_A$, one sees that the nil-space $Nil(b, \mathfrak{g}_A)$ contains $\mathfrak{d}'$; on the other hand it is equal to $\mathfrak{d}_A$, and as $\mathfrak{d}'$ is locally a direct factor in the module $\mathfrak{g}_A$ ($\mathfrak{d}$ being so), hence in $\mathfrak{d}_A$, and as it is a projective module of the same rank r as the latter, one concludes that it equals it. *QED*.

Proposition 4.8. *Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$.*

a) The following conditions are equivalent if k is infinite:

- *(i) $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.*
- *(ii) $\mathfrak{h}$ contains a regular element a of $\mathfrak{g}$, and an element b such that $ad(b)_{\mathfrak{g}/\mathfrak{h}}$ is injective.*
- *(iii) $\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$, and contains a regular element of $\mathfrak{g}$.*

These conditions are invariant under extension of the base field k.

b) Suppose these conditions are satisfied over a suitable infinite extension k' of k. Let $a \in \mathfrak{h}$; then a is regular in $\mathfrak{g}$ if and only if it is regular in $\mathfrak{h}$ and $ad(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, i.e. if and only if $Nil(a, \mathfrak{h}) = Nil(a, \mathfrak{g})$ and this is a Cartan subalgebra of $\mathfrak{h}$.

c) Under the conditions of b), let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$; in order that it be a Cartan subalgebra of $\mathfrak{h}$, it suffices that it be a Cartan subalgebra of $\mathfrak{g}$ (and we shall see in 5.8 that the condition is also necessary if $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G, and $\mathfrak{h}$ the Lie algebra of a smooth algebraic subgroup H of G).

One sees at once that conditions (ii) and (iii) of a) are invariant under extension of the base field k (assumed infinite), and that in statements b) and c), one may assume k infinite, which we shall do. If $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = Nil(a, \mathfrak{g})$, then a is a regular element of $\mathfrak{g}$ such that $ad(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, hence (i) $\Rightarrow$ (ii). Conversely, if (ii) is satisfied, then for an element a "sufficiently general" of $\mathfrak{h}$, a satisfies simultaneously the two conditions of (ii), hence $Nil(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ and is contained in $\mathfrak{h}$, so one has (i). So (i) and (ii) are equivalent. Suppose them satisfied, let a be a variable element of $\mathfrak{h}$; then

`(∗)   rank_k Nil(a, □) = rank_k Nil(a, □) + rank_k Nil(ad(a)_{□/□}),`

on the other hand, the two terms of the right-hand side are respectively $\geqslant r_0 =$

nilpotent rank of $\mathfrak{h}$, and $\geqslant 0$, the equalities being moreover attained[1096] for an element "sufficiently general" of $\mathfrak{h}$. Moreover, one has also $rank_k Nil(a, \mathfrak{g}) \geqslant r =$ nilpotent rank of $\mathfrak{g}$, with equality attained for an element "sufficiently general" of $\mathfrak{h}$, and attained if and only if a is regular in $\mathfrak{g}$. One concludes from this that one has $r = r_0$, and that a is regular if and only if the two terms of the right-hand side of $(*)$ equal respectively r_0 and 0, i.e. if and only if a is regular in $\mathfrak{h}$ and $ad(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective, which proves b), and c) follows trivially by taking an element a in $\mathfrak{d}$ regular in $\mathfrak{g}$, so that $Nil(a, \mathfrak{g}) = \mathfrak{d}$. Moreover, the preceding result shows that (i) $\Rightarrow$ (iii); finally (iii) $\Rightarrow$ (i), because under (iii), an element sufficiently general a of $\mathfrak{h}$ is regular in $\mathfrak{h}$ and in $\mathfrak{g}$, hence $Nil(a, \mathfrak{h}) \subset Nil(a, \mathfrak{g})$ are respectively Cartan subalgebras of $\mathfrak{h}$ and of $\mathfrak{g}$, and as they have the same rank over k, they are identical, which proves (i). This completes the proof of 4.8.

5. The case of the Lie algebra of a smooth algebraic group: density theorem

Let G be a smooth algebraic group over the field k, and $\mathfrak{g}$ its Lie algebra. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. Let G act on $W(\mathfrak{g})$ by the adjoint representation, and consider the subscheme $W(\mathfrak{h})$. The construction of section 1 leads us to introduce

$$N = Norm_G(\mathfrak{h}) = Norm_G(W(\mathfrak{h})),$$

which is an algebraic subgroup of G (not necessarily smooth),

$$\mathfrak{n} = Lie(N),$$

and the scheme

$$X = G \times^N W(\mathfrak{h})$$

quotient of $\tilde{X} = G \times W(\mathfrak{h})$ by N acting on the right by $(g, x) \cdot n = (gn, Ad(n^{-1})x)$. We consider the canonical morphisms

$$\begin{tikzcd} G \times W(\mathfrak{h}) \arrow[d, "q"'] \arrow[dr, "\phi"] & \\ X \arrow[r, "\psi"'] & W(\mathfrak{g}). \end{tikzcd}$$

Theorem 5.1. *With the preceding notation, suppose k infinite. Consider the following conditions:*

- *(i) $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.*
- *(ii) There exists $a \in \mathfrak{h}$ such that $ad(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective.*
- *(iii) $\varphi : G \times W(\mathfrak{h}) \to W(\mathfrak{g})$ is generically smooth.*

[1096] "equality" has been replaced by "the equalities".

- *(iv) The preceding morphism φ (or also $\psi : X \to W(\mathfrak{g})$) is dominant, and $\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$.*
- *(v) $\psi : X \to W(\mathfrak{g})$ is generically smooth and $\mathfrak{h} = \mathfrak{n}$.*
- *(vi) $\psi : X \to W(\mathfrak{g})$ is dominant, and $\mathfrak{h} = \mathfrak{n}$.*
- *(vii) $\psi : X \to W(\mathfrak{g})$ is dominant.*
- *(viii) $\psi : X \to W(\mathfrak{g})$ is dominant, and N is smooth.*
- *(ix) $\psi : X \to W(\mathfrak{g})$ is dominant, and $\mathfrak{h} = \mathfrak{n}$, and $\mathfrak{h}$ is the Lie algebra of a smooth algebraic subgroup H of G.*
- *(x) $\psi : X \to W(\mathfrak{g})$ is generically quasi-finite, and $\mathfrak{n} = \mathfrak{h}$.*
- *(xi) $\psi : X \to W(\mathfrak{g})$ is generically étale, and $\mathfrak{n} = \mathfrak{h}$.*
- *(xii) There exists a smooth algebraic subgroup H of G with Lie algebra $\mathfrak{h}$, and $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.*
- *(xiii) There exists $a \in \mathfrak{h}$ such that N and the transporter M_a of a to $\mathfrak{h}$ (cf. section 1) coincide in a neighborhood of e, and □ = □.*

One then has the following diagram of implications:

```
 (xi) ⇔ (xii) ⇔ (xiii)  ⇒  (viii) ⇔ (ix) ⇔ (x)
                                                ↓
  (i) ⇔ (ii) ⇔ (iii) ⇔ (iv)  ⇒ (v)  ⇒ (vi) ⇒ (vii).
```

When k is of characteristic zero, all the conditions considered are equivalent. Finally, one has

(xi) ⇔ [(i) and (viii)] ⇔ [(v) and (viii)].

Let us first note the trivial implications:

(v) ⇒ (vi) ⇒ (vii),   (ix) ⇒ (vi),   (xi) ⇔ [(x) and (v)].

Let us prove the equivalence of conditions (i) to (iv) and that they imply (v). The implication (i) ⇒ (ii) is trivial. On the other hand (iii) means, when k is algebraically closed, that there exists a k-rational point of $G \times W(\mathfrak{h})$ at which the tangent map to φ is surjective, and one sees at once that this point may be taken of the form (e, a), where $a \in \mathfrak{h}$ (up to transforming it by an operation of $G(k)$). One concludes that if k is infinite (not necessarily algebraically closed) this condition (evidently sufficient) of generic smoothness is still necessary. Now the tangent map is easily calculated: identifying the tangent space to $W(\mathfrak{h})$ at a with $\mathfrak{h}$, it is the map

$(\xi, x) \mapsto [\xi, a] + x$

from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. To say that it is surjective also means that $ad(a)_{\mathfrak{g}/\mathfrak{h}}$ is surjective, or what amounts to the same, injective. This proves the equivalence of conditions (ii) and (iii). Moreover, (ii) evidently implies $\mathfrak{h} = \mathfrak{n}$, and (iii) implies that ψ is generically smooth, since if φ is smooth at a point u, it follows (q being flat) that ψ is smooth at $q(u)$. So (ii), (iii) imply (v). Let us prove that they imply (i). For this, note that since ψ is dominant, and the set of regular points of $\mathfrak{g}$ is open dense, it follows that $\mathfrak{h}$ contains regular elements of $\mathfrak{g}$, hence a "sufficiently general" element b of $\mathfrak{h}$ is regular in $\mathfrak{g}$ and satisfies $ad(b)_{\mathfrak{g}/\mathfrak{h}}$ injective, so $Nil(b, \mathfrak{g}) \subset \mathfrak{h}$, hence $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = Nil(b, \mathfrak{g})$. So (i), (ii), (iii) are equivalent; finally (i) ⇔ (iv), since we have already noted that if $\mathfrak{h}$ contains a Cartan subalgebra, it has the same rank as $\mathfrak{g}$ (4.6), so (i) ⇒ (iv); conversely, if (iv) is satisfied, then $\mathfrak{h}$ contains a regular element of $\mathfrak{g}$, and since it has the same rank as $\mathfrak{g}$, it contains a Cartan subalgebra by 4.6.

Let us prove the equivalence of conditions (viii) to (x). Let us first remark the following facts:

Lemma 5.2. *a) If N is smooth, then*

`dim X □ dim G,`

with equality if and only if $\mathfrak{h} = \mathfrak{n}$.

b) If $\mathfrak{h} = \mathfrak{n}$*, then*

`dim X □ dim G,`

with equality if and only if N is smooth.

These assertions follow at once from the formula

`dim X = dim G − dim N + dim_k □,`

and from the fact that $\dim N = \dim_k \mathfrak{n}$ is equivalent to N smooth.

This established, (viii) ⇒ (x), since (viii) implies $\dim X \geqslant \dim G$, hence by 5.2 a) the equality of these dimensions and $\mathfrak{h} = \mathfrak{n}$, hence (x); and one sees similarly (x) ⇒ (viii) by applying 5.2 b). On the other hand, (viii) implies (ix), since it implies $\mathfrak{h} = \mathfrak{n}$, hence $\mathfrak{h}$ is the Lie algebra of the smooth algebraic subgroup N of G; and conversely (ix) ⇒ (viii), since H normalizes its Lie algebra $\mathfrak{h}$, so it is contained in N, and as H is smooth and has the same Lie algebra as N, it follows that N is smooth.

Let us finally prove the equivalence of conditions (xi), (xii), (xiii) and the fact that they entail (iii) (which will complete the establishment of our diagram of implications). One has (xi) ⇔ (xiii), since if $\mathfrak{n} = \mathfrak{h}$, then by 5.2 b) one has $\dim X \geqslant \dim G$, hence (xi) is then equivalent (given that $W(\mathfrak{g})$ is normal) to the fact that ψ is generically unramified, which is equivalent also to (xiii) by (1.1 (ii) ⇔ (iii)), proceeding as above for the proof of (ii) ⇔ (iii). Since (xi) ⇒ (x) ⇒ (viii) by what we have seen, one sees that (xi) implies that N is smooth, i.e. $q : G \times W(\mathfrak{h}) \to X$ is smooth, hence the composite $\varphi = \psi \circ q$ is generically smooth, i.e. one has (iii). Since (iii) ⇒ (i), it also follows that (xi) ⇒ (xii).[1097] Finally (xii) ⇒ (xi), since one evidently has (xii) ⇒ (i), so as one has seen (i) ⇒ (iii) ⇒ (v), one has (xii) ⇒ (v); it follows that one has also (xii) ⇒ (ix), and as one has seen (ix) ⇒ (x), it follows that (xii) ⇒ ((v) and (x)), hence (xii) ⇒ (xi) since generically étale = generically smooth + generically quasi-finite.

Finally, when k is of characteristic 0, then (vii) ⇒ (viii), since by a theorem of Cartier, N is automatically smooth (VI_B 1.6.1), and [(viii) and (x)] ⇒ (xi), since in characteristic zero, for a morphism of integral preschemes, generically étale = dominant and generically quasi-finite. This shows that in this case, all the conditions (i) to (xiii) are equivalent.

[1097] because $H = N$ works.

Corollary 5.3. *Under the equivalent conditions (viii) to (x), there exists a unique smooth connected algebraic subgroup H of G whose Lie algebra is $\mathfrak{h}$, and one has*

$$Norm_G(H) = Norm_G(\mathfrak{h}) = N, \quad H = N^0.$$

Indeed, $H = N^0$ will satisfy the required conditions; on the other hand if H satisfies them, then (since H normalizes its Lie algebra $\mathfrak{h}$) one has $H \subset N$, hence as this is an inclusion of smooth groups having the same Lie algebra, with H connected, one will have $H = N^0$. For the identity $Norm_G(H) = Norm_G(\mathfrak{h})$, one may assume k algebraically closed; then from what one has just seen, it follows immediately that the points of the two groups with values in k are the same; on the other hand the inclusions $H \subset Norm_G(H) \subset N$ show that $Norm_G(H)$ and N have the same Lie algebra, hence they are identical.

Corollary 5.4. *Under the equivalent conditions (i) to (iv), let $a \in \mathfrak{h}$. Then the following conditions are equivalent, and are realized if a is regular in $\mathfrak{g}$:*

- *(i) φ is smooth at (e, a).*
- *(ii) $M_a = Transp_G(a, \mathfrak{h})$ is smooth at e, and $\dim_e(M_a) = rank_k \mathfrak{h}$.*
- *(iii) $ad(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective (or again, bijective).*

When one is under the equivalent conditions (xi) to (xiii), let H be the algebraic subgroup of G considered in 5.3. Then the preceding conditions are also equivalent to the following conditions:

- *(iv) ψ is étale at $(\bar{e}, a)$.*
- *(v) Denoting by M_a^0 the connected component of e in the transporter M_a of a to $\mathfrak{h}$, endowed with the structure induced by M_a, one has*

$$H = M_a^0.$$

Evidently (i) ⇒ (ii) since M_a is isomorphic to the fiber $\varphi^{-1}(a)$, the point e corresponding to (e, a); and one has (ii) ⇒ (i), since (ii) implies that φ is "equidimensional" at (e, a) (i.e. the dimension of the fiber passing through this point equals that of the generic fiber), which implies ($G \times W(\mathfrak{h})$ and $W(\mathfrak{g})$ being regular) that it is flat at (e, a), hence smooth since its fiber is at this point. The equivalence of (i) and (iii) has been seen in the proof of 5.1 as resulting from the simple calculation of the tangent map. Moreover, one has seen in 4.8 b) that "a regular in $\mathfrak{g}$" ⇒ (iii). Under conditions (xi) to (xiii), since $q : G \times W(\mathfrak{h}) \to X$ is smooth (N being smooth), it follows that (i) is equivalent to ψ smooth at $(\bar{e}, a)$, and as ψ is generically étale, this is equivalent to (iv). Finally, as was noted at the end of the proof of 1.1, (iv) implies that N is the prescheme induced on M_a by an open and closed part of M_a, whence (v); finally (v) ⇒ (ii) trivially (or again (v) ⇒ (iv) by 1.1, since ψ is dominant and $W(\mathfrak{g})$ normal, "unramified" is here equivalent to "étale"). This completes the proof of 5.4.

Corollary 5.5. *Let G be a smooth algebraic group over a field k, H a smooth algebraic subgroup such that its Lie algebra $\mathfrak{h}$ contains (at least over a suitable extension of k) a Cartan subalgebra of the Lie algebra $\mathfrak{g}$ of G. Let K be a connected* *algebraic subgroup of G (not necessarily smooth), with Lie algebra $\mathfrak{k}$; suppose that $\mathfrak{k}$ contains a regular element a of $\mathfrak{g}$ (at least over a suitable extension of k). Then H contains K if and only if $\mathfrak{h}$ contains $\mathfrak{k}$.*

Indeed, by 5.4 (iii) ⇒ (v) one has $H = M_a^0$ (N.B. of course, this relation being invariant under extension of the base field, it is valid without the hypothesis that the latter be infinite!); on the other hand $\mathfrak{k} \subset \mathfrak{h}$ evidently implies $K \subset M_a$, hence as K is connected, $K \subset M_a^0$, whence $K \subset H$. *QED*.

Corollary 5.6. *Let G, H be as in 5.5, and let K be an algebraic subgroup of G; suppose H connected and K smooth. Then K contains H if and only if $\mathfrak{k}$ contains $\mathfrak{h}$.*

Indeed, if $\mathfrak{k} \supset \mathfrak{h}$, then K satisfies the hypothesis considered in 5.5 for H; on the other hand H evidently satisfies the condition considered for K in 5.5. The conclusion then follows from 5.5.

Corollary 5.7. *Let $\mathfrak{g}$ be the Lie algebra of a smooth algebraic group G over a field k. Then:*

a) Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. In order that $\mathfrak{d}$ be a Cartan subalgebra, it is necessary and sufficient that $\mathfrak{d}$ be nilpotent and equal to its own normalizer.

b) Let a be an element of $\mathfrak{g}$; in order that a be regular, it is necessary and sufficient that $Nil(a, \mathfrak{g})$ be nilpotent.

Up to making an extension of the base field, one may assume k infinite. Given 4.4, one is reduced for a) to proving that if $\mathfrak{d}$ is nilpotent and contains an element a such that $ad(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective, then $\mathfrak{d}$ is a Cartan subalgebra. Now by 5.1 (ii) ⇒ (i), $\mathfrak{d}$ contains a Cartan subalgebra $\mathfrak{d}' = Nil(a, \mathfrak{g})$, and by 4.3 one concludes from the fact that $\mathfrak{d}$ is nilpotent that $\mathfrak{d} = \mathfrak{d}'$. To prove b), one notes that $Nil(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ by a), hence a is regular.

Corollary 5.8. *Let G be a smooth algebraic group over a field k, H a smooth algebraic subgroup, $\mathfrak{g}$, $\mathfrak{h}$ the Lie algebras; suppose that after suitable extension of the base field, $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$; then it is a Cartan subalgebra of $\mathfrak{h}$ if and only if it is a Cartan subalgebra of $\mathfrak{g}$.*

Given 4.8 c), it remains to show that if $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$, it is a Cartan subalgebra of $\mathfrak{g}$; for this one is reduced to showing that $\mathfrak{d}$ contains an element a regular in $\mathfrak{g}$, assuming (which is permissible) k algebraically closed. But since there is a dense open in $\mathfrak{h}$ consisting of regular points of $\mathfrak{g}$, our assertion follows from 5.1 (i) ⇒ (vii) applied to $(\mathfrak{h}, \mathfrak{d})$.

6. Cartan subalgebras and subgroups of type (C), relative to a smooth algebraic group

For simplicity, we restrict ourselves in the following theorem to the case of an algebraically closed base field (the case of an arbitrary base prescheme being treated in the next Exposé):

Theorem 6.1. *Let G be a smooth algebraic group over an algebraically closed field k, $\mathfrak{g}$ its Lie algebra. Then:*

a) The Cartan subalgebras of $\mathfrak{g}$ are conjugate.

b) Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then its normalizer N in G is smooth, and $D = N^0$ is the only smooth connected subgroup of G whose Lie algebra is $\mathfrak{d}$. One has

$$\mathrm{Norm}_G(\mathfrak{d}) = \mathrm{Norm}_G(D) = N, \quad \text{hence } D = \mathrm{Norm}_G(D)^0.$$

c) With $\mathfrak{d}$ as in b), set as in section 5: $X = G \times^N W(\mathfrak{d})$, and consider the canonical morphism

$$\psi : X \to W(\mathfrak{g}),$$

(whose fiber at $a \in \mathfrak{g}$ has as points with values in k the set of Cartan subalgebras of $\mathfrak{g}$ containing a). Then ψ is a birational morphism.

d) With the notation of c), let U be the largest open of $W(\mathfrak{g})$ such that ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Then for $a \in \mathfrak{g}$, the following conditions are equivalent:

- *(i) $a \in U(k)$.*
- *(ii) a is contained in one and only one Cartan subalgebra of $\mathfrak{g}$.*
- *(iii) The set of Cartan subalgebras of $\mathfrak{g}$ containing a is finite and non-empty.*
- *(iv) The fiber $\psi^{-1}(a)$ has an isolated point.*
- *(v) (If $a \in \mathfrak{d}$) The morphism ψ is étale (or merely: quasi-finite) at the point $(\bar{e}, a)$.*
- *(vi) a is a regular element of $\mathfrak{g}$.*
- *(vii) With the notation of 5.4 (iii), one has $N = M_a$.*

Proof. Let us apply 5.1 when $\mathfrak{h}$ is a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$; let us show that the strongest conditions (xi) to (xiii) are then satisfied. This is evident, either in form (xiii) given 4.7 (which implies that for a regular element a of $\mathfrak{g}$ contained in $\mathfrak{d}$, one has $N = M_a$), or in form (xi) = (x) + (i), since condition (i) is trivial and condition (x) follows from the fact that a regular point of $\mathfrak{g}$ is contained in a single Cartan subalgebra of $\mathfrak{g}$ (4.6 a)), and a fortiori in a single conjugate of $\mathfrak{d}$. Then b) follows from 5.3, and c) follows from the fact that ψ is generically étale and that a sufficiently general point (more precisely, a regular point) of $\mathfrak{g}$ is contained in a single Cartan subalgebra of $\mathfrak{g}$. Under these conditions, the equivalence of conditions (i) to (v) on a is an immediate consequence of Zariski's Main Theorem for the separated birational morphism ψ, given that $W(\mathfrak{g})$ is normal and X is integral. The equivalence of (v) and (vi) is a particular case of 5.4 (iii) $\Leftrightarrow$ (iv) (reducing to the case where $a \in \mathfrak{d}$ by transforming a by a suitable element $g \in G(k)$), given 4.6

b). Moreover, by 5.4, (v) and (vi) are also equivalent to $N \supset M_a^0$, and by 4.7 already invoked, this even implies $N = M_a$. This proves d).

Of course, b), c), and the equivalence of (i), (iv), (v), (vi), (vii) remain valid over an arbitrary base field. Let us now prove a) using the fact that k is algebraically closed. By 5.1 (i) ⇒ (vii), $\psi : X \to W(\mathfrak{g})$ is dominant, hence there exists a dense open V of $W(\mathfrak{g})$ such that every $a \in V(k)$ is the image of an element of $X(k)$, i.e. is contained in a conjugate of $\mathfrak{d}$. Applying this result to a second Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}$, one sees that one may take V such that every element of $V(k)$ is a conjugate of an element of $\mathfrak{d}$ and of an element of $\mathfrak{d}'$. Taking a regular element in $V(k)$, it follows that there is a conjugate $\mathfrak{d}''$ of $\mathfrak{d}'$ which contains an element of $\mathfrak{d}$ regular in $\mathfrak{g}$, hence which equals $\mathfrak{d}$ by 4.6 a). This completes the proof of 6.1.

Definition 6.2. *Let G be a smooth algebraic group over a field k. One calls a* sub-group of type (C) *of G any smooth connected algebraic subgroup whose Lie algebra is a Cartan subalgebra of $\mathfrak{g}$. One calls the* infinitesimal rank *of G the nilpotent rank of its Lie algebra $\mathfrak{g}$, equal to the dimension of any subgroup of type (C) of G.*

From 6.1 b) one concludes at once:

Corollary 6.3. *Under the conditions of 6.2, the map $D \mapsto \mathfrak{d} = LieD$ establishes a bijective correspondence between subgroups of type (C) of G, and Cartan subalgebras $\mathfrak{d}$ of $\mathfrak{g}$. If D is a subgroup of type (C) of G, D is its own connected normalizer:*

$$D = Norm_G(D)^0.$$

Combining 6.1 (a) and 6.3, one finds:

Corollary 6.4. *If k is algebraically closed, the subgroups of type (C) of G are conjugate to each other.*

Corollary 6.5. *Let G be a smooth algebraic group over algebraically closed k, H a smooth algebraic subgroup of G, $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras. In order that $\mathfrak{h}$ contain a Cartan subalgebra of $\mathfrak{g}$, it is necessary and sufficient that H contain a subgroup D of type (C) of G.*

This is evidently sufficient, and is also necessary, since in order that one have $H \supset D$, it is necessary and sufficient that $\mathfrak{h} \supset \mathfrak{d}$, by 5.5.

Remarks 6.6. *a) One will note that the connected subgroups H of G described in 6.5 correspond bijectively to the Lie subalgebras $\mathfrak{h}$ of $\mathfrak{g}$ satisfying the strongest condition (xii) of 5.1 (and by 5.5, inclusion relations between such subgroups can already be recognized on their Lie algebras).*

b) Suppose still that k is algebraically closed, and let D be a subgroup of type (C) of G; then it is easy to show that D contains a Cartan subgroup C of G: indeed, let $V = \mathfrak{g}/\mathfrak{d}$; then for a "general" element a of $\mathfrak{d}$, $ad(a)$ acting on V is injective (it suffices that a be regular in $\mathfrak{g}$); from this one easily concludes that for $g \in D(k)$ "general", $Ad(g)$ acting on V has no non-zero invariants, which allows one to apply 2.1 (vii) ⇒ (i). In fact, we shall see in the following Exposé a more precise result:

every Cartan subgroup C of G is contained in one and only one subgroup of type (C) of G.

*c) One should not confuse in general Cartan subgroups and subgroups of type (C): the subgroups of type (C) of G are Cartan subgroups if and only if they are nilpotent (Cartan subgroups being indeed maximal nilpotent subgroups), and it may happen that $\mathfrak{g}$ is nilpotent without G being so (example: $SL(2)_k$ for k of characteristic 2); then the Cartan subalgebras of $\mathfrak{g}$ are identical to $\mathfrak{g}$, i.e. G is a subgroup of type (C) of itself if G is connected, but it is not a Cartan subgroup of G! On the other hand, we shall see in XIV that if G is a semisimple adjoint group, then its subgroups of type (C) are its Cartan subgroups. Similarly, in characteristic 0, without restriction on the smooth algebraic group G over k, the same conclusion is valid: this follows at once from the fact that in characteristic 0, a connected algebraic group is nilpotent if (and only if) its Lie algebra is. This follows at once from the fact that in characteristic zero, the center Z of the connected algebraic group G has as Lie algebra the center of $\mathfrak{g}$ (for one obtains a priori the space of invariants of $\mathfrak{g}$ under the adjoint representation of G; now G being connected and k of characteristic zero, these are also the "invariants" in the infinitesimal sense of the adjoint representation of $\mathfrak{g}$, i.e. the center of $\mathfrak{g}$), and that by Cartier's theorem (cf. VI_B 1.6.1) Z is smooth.

Exposé XIV. Regular elements (continuation). Applications to algebraic groups

by A. Grothendieck, with an Appendix by J.-P. Serre

1. Construction of Cartan subgroups and of maximal tori for a smooth algebraic group

Theorem 1.1. *Let G be a smooth algebraic group over a field k. Then G admits a maximal torus T, hence a Cartan subgroup $C = Centr_G(T)$.*

By virtue of (XII 3.2), it amounts to the same to find a maximal torus T of G or a Cartan subgroup C of G. Moreover, since the maximal tori of G are those of G^0, we may assume G connected. We distinguish two cases:

1°) *The field k is finite.* Let T be the scheme of maximal tori of G (Exp XII 1.10), which is a smooth scheme over k. Note that G operates on T via inner automorphisms, and by virtue of the conjugation theorem (XII 6.6 a)), two points of $T_{\bar{k}}$ rational over $\bar{k}$ are congruent under $G_{\bar{k}}(\bar{k})$. Taking into account that T is smooth over k, hence $T_{\bar{k}}$ smooth over $\bar{k}$, it follows that $T_{\bar{k}}$ is isomorphic to $G_{\bar{k}}/N$, where N is the stabilizer of an element of $T_{\bar{k}}(\bar{k})$, i.e. the normalizer of a maximal torus T of $G_{\bar{k}}$. Consequently, T is a "homogeneous space" under the action of the group G. A well-known theorem of Lang (Amer. J. Math. 78, 1956, pp. 555–563) tells us that every homogeneous space under a smooth connected algebraic group over a finite field k admits a rational point. In particular, T admits a rational point, i.e. G admits a maximal torus T. *QED*.

2°) *The field k is infinite.* We shall use the

Lemma 1.2. *Let G be a smooth algebraic group over a field k. Then G admits a subgroup of type (C) (XIII 6.2).*

By virtue of (XIII 6.3), this amounts to saying that $\mathfrak{g}$ contains a Cartan subalgebra $\mathfrak{d}$. This is trivial if k is infinite, for then $\mathfrak{g}$ contains a regular element a, and one takes $\mathfrak{d} = Nil(a, \mathfrak{g})$. The case of finite k is handled exactly as in the proof of 1°) above, but requires the prior construction of the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ and the fact that this scheme is smooth over k, which will be carried out below (2.16).

To establish 1.1 in case 2°), where we have placed ourselves, it suffices in any event to know 1.2 for k infinite. Let us also record for the record:

Lemma 1.3. *Let G be a smooth, connected, affine algebraic group whose reductive center (XII 4.1 and 4.4) is reduced to the unit group. Then G is nilpotent if (and only if) its Lie algebra $\mathfrak{g}$ is nilpotent.*

This is contained in (XII 4.9).

We can now give a procedure for constructing Cartan subgroups of G (also valid when k is finite, granting 1.2 in that case). Suppose first that G is affine. We proceed by induction on $n = \dim G$, the assertion being trivial if $n = 0$. So suppose $n > 0$ and the assertion proved for dimensions $n' < n$. Let Z be the reductive center of G, and let

$u : G \to G' = G/Z$

be the canonical homomorphism. By virtue of (XII 4.7 c)), one has a bijective correspondence $C' \mapsto C = u^{-1}(C')$ between Cartan subgroups of G' and Cartan subgroups of G. So, replacing G by G' if necessary, we may suppose that the reductive center Z of G is reduced to the unit element (since this is the case for that of G' by XII 4.7 b)).

By virtue of 1.3, G admits a subgroup D of type (C). We know that over the algebraic closure $\bar{k}$ of k, $D_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$ (XIII 6.6 b)), hence every Cartan subgroup of D is a Cartan subgroup of G (XIII 2.8 a)). So we are reduced to finding a Cartan subgroup of D. If $\dim D = \dim G$, i.e. $D = G$, then the Lie algebra of G is a Cartan subalgebra of itself, hence is nilpotent, hence by 1.3 G is nilpotent, hence is a Cartan subgroup of itself. If $\dim D < \dim G$, then by the induction hypothesis there exists a Cartan subgroup of D, which is therefore a Cartan subgroup of G, which completes the proof in the case where G is affine. In the general case, let Z be the center of G; then $G/Z = G'$ is affine (XII 6.1) and for every Cartan subgroup C' of G', its inverse image C in G is a Cartan subgroup of G (XII 6.6 e)). One is reduced to finding a Cartan subgroup of the smooth affine algebraic group G', a case already treated.

Corollary 1.4. *Let G be a smooth group scheme of finite type over an artinian scheme S. Then G admits a maximal torus T, hence a Cartan subgroup $C = Centr_G(T)$. Every torus S in G is contained in a maximal torus.*

One may indeed suppose S local of residue field k; then by virtue of 1.1, $G_0 = G \times_S \operatorname{Spec}(k)$ admits a maximal torus T_0, and by virtue of (IX 3.6 bis) and (X 2.3), T_0 comes from a torus T of G, which is evidently a maximal torus. The last statement follows from this, applying the preceding result to the centralizer of S, which is indeed smooth over k (XI 2.4).

Remarks 1.5. a) We shall give below (3.20, 3.21 and XV) variants of 1.4 in the case where S is not assumed artinian.

b) I do not know whether every algebraic group G (not necessarily smooth) over a field k admits a maximal torus. The question only arises in characteristic $p > 0$, and using 1.1 for a smooth quotient group of the form $G' = G/I$, where I is a suitable radicial subgroup of G (for example, the kernel of a sufficiently high power of the Frobenius homomorphism), one is reduced, by taking the inverse image in G of a maximal torus of G', to the case where $(G_{\bar{k}})_{red}$ is a torus ($\bar{k}$ always denoting the algebraic closure of k). It is easy to see that the answer is affirmative when G is commutative (or more generally nilpotent): then G admits a unique maximal torus, which one may construct for example by descent from the maximal torus of $G_{\bar{k}}$ [1098].

c) In the case where G is affine, and k is perfect or G solvable, 1.1 is known and due to Rosenlicht; his proof is very different from the one given here.

d) When k is infinite, 1.1 is a consequence of the much more precise result that the scheme T of maximal tori of G is a rational variety, proved below (6.1). The method is essentially a combination of the proof of 1.1 and of the explicit description of the structure of the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$. To reach the desired result, we must first generalize to the case of an arbitrary base prescheme certain results from (XIII 4 to 6) (this is the goal of the next two sections), and refine the previous construction proving 1.1, by using the fact that every Cartan subgroup of G is contained in one and only one subgroup of type (C) of G (Nos 4, 5).

2. Lie algebras over an arbitrary prescheme: regular sections and Cartan subalgebras

Lemma 2.1. *Let A be a ring, $\mathfrak{d}$ a Lie algebra over A, and for every $s \in \operatorname{Spec}(A)$, let $\mathfrak{d}(s) = \mathfrak{d} \otimes_A k(s)$ be the corresponding Lie algebra over the residue field $k(s)$. Suppose the A-module $\mathfrak{d}$ is of finite presentation. The following conditions are equivalent:*

- *(i) For every $s \in \operatorname{Spec}(A)$, $\mathfrak{d}(s)$ is nilpotent.*
- *(ii) For every $x \in \mathfrak{d}$, $ad(x)$ is nilpotent.*
- *(iii) There exists an integer $N \geqslant 0$ such that for every sequence $x_1, \cdots, x_N$ of elements of $\mathfrak{d}$, one has*

[1098] M. Raynaud gave a negative answer to the question raised here, cf. (XVII Example 5.9.c)).

$$ad(x_1)ad(x_2)\cdots ad(x_N) = 0.$$

When A is a field, the equivalence of (i) and (iii) is the definition of "nilpotent", that of (ii) and (iii) is a well-known consequence of Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chap. I, § 4, N° 2). In the general case, one has trivially (iii) ⇒ (ii), and (ii) ⇒ (i) thanks to the preceding result and to the fact that (ii) is stable by passage to quotients and by localization. It remains to prove (i) ⇒ (iii). When A is local artinian with maximal ideal $\mathfrak{m}$, let $n > 0$ be an integer such that $\mathfrak{m}^n = 0$, let N be an integer such that condition (iii) is satisfied for $\mathfrak{d}(s) = \mathfrak{d} \otimes_A A/\mathfrak{m}$, and take $N' = nN$; one sees at once that this integer satisfies (iii). When A is noetherian, there exist finitely many elements $s_i \in \mathrm{Spec}(A)$ and ideals of definition $\mathfrak{q}_i \subset A_{s_i}$ such that the natural map

$$\mathfrak{d} \to \prod_i \mathfrak{d} \otimes_A A_i, \quad \text{where } A_i = A_{s_i}/\mathfrak{q}_i,$$

is injective; then by what precedes there exists for every i an integer N_i satisfying (iii) for the Lie algebra $\mathfrak{d}\otimes_A A_i$, and taking N to be the largest of the N_i, one satisfies (iii) for $\mathfrak{d}$. Finally, the general case reduces to the noetherian case by the usual procedure explained in (EGA IV 8).

Definition 2.2. *Let S be a prescheme, $\mathfrak{d}$ a quasi-coherent Lie algebra over S that is a module of finite presentation. One says that $\mathfrak{d}$ is locally nilpotent if for every $s \in S$, the Lie algebra $\mathfrak{d}(s)$ over $k(s)$ is nilpotent. One says that $\mathfrak{d}$ is strictly locally nilpotent if it is locally free, and if on every open U of S on which it is of constant rank r, its Killing polynomial reduces to $P_{\mathfrak{d}}(t) = t^r$.*

Of course, if $\mathfrak{d}$ is a locally free Lie algebra (as a module) over S, one defines its Killing polynomial as a polynomial

$$P_{\mathfrak{d}} \in A[t], \quad \text{where } A = \Gamma(\mathrm{Sym}_{\mathcal{O}_S}(\mathfrak{d})) \simeq \Gamma(W(\mathfrak{d}))$$

is the ring of sections of the structure sheaf of the vector bundle $W(\mathfrak{d})$ defined by $\mathfrak{d}$.

It is evident that the two notions just introduced in 2.2 are stable by base change, and of local nature for the fpqc topology. Let us note:

Proposition 2.3. *If $\mathfrak{d}$ is strictly locally nilpotent, it is locally nilpotent. The converse is true when $\mathfrak{d}$ is locally free and S is reduced.*

The proof is immediate.

Definition 2.4. *Let S be a prescheme, $\mathfrak{g}$ a Lie algebra over S that is a locally free module of finite type. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$; one says that it is a Cartan subalgebra if it satisfies the following conditions:*

- *(i) $\mathfrak{d}$ is a locally direct factor submodule (hence also locally free of finite type).*
- *(ii) For every $s \in S$, $\mathfrak{d}(s)$ is a Cartan subalgebra of $\mathfrak{g}(s)$.*

Definition 2.5. *Let S, $\mathfrak{g}$ be as in 2.4. A section a of $\mathfrak{g}$ is called quasi-regular if for every $s \in S$, $a(s) \in \mathfrak{g}(s)$ is a regular element of the Lie algebra $\mathfrak{g}(s)$ over $k(s)$.*

One says that a is a regular section if it is quasi-regular and contained in a Cartan subalgebra of $\mathfrak{g}$.

Notions 2.4 and 2.5 are again stable by base change, and of local nature for the fpqc topology. Only the last assertion, and only for the notion "regular section", requires a proof, and follows from the fact that the Cartan subalgebra containing a given regular section is uniquely determined. More precisely:

Proposition 2.6. *Let S, $\mathfrak{g}$ be as in 2.4 and let a be a quasi-regular section of $\mathfrak{g}$. Then there exists at most one Cartan subalgebra of $\mathfrak{g}$ containing a. For one to exist, i.e. for a to be a regular section, it is necessary and sufficient that a satisfy the following condition: $\mathfrak{d} = Nil(a, \mathfrak{g})$ is a locally direct factor submodule of $\mathfrak{g}$, and $ad(a)$ induces an automorphism of $\mathfrak{g}/\mathfrak{d}$. In this case, $\mathfrak{d}$ is the unique Cartan subalgebra of $\mathfrak{g}$ containing a.*

Suppose indeed that a is contained in the Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$. Then $ad(a)_{\mathfrak{g}/\mathfrak{d}}$ is bijective on each fiber, hence (since $\mathfrak{g}/\mathfrak{d}$ is locally free of finite type) it is an automorphism of $\mathfrak{g}/\mathfrak{d}$; on the other hand by virtue of 2.1 $ad(a)_{\mathfrak{d}}$ is locally nilpotent, hence $\mathfrak{d} = Nil(a, \mathfrak{g})$, which proves the uniqueness of $\mathfrak{d}$, and the necessity of the stated regularity criterion. For sufficiency, note that the hypothesis made on a implies that the formation of $Nil(a, \mathfrak{g})$ commutes with base extension and in particular with passage to fibers, which shows in particular that the fibers $\mathfrak{d}(s)$ of $\mathfrak{d} = Nil(a, \mathfrak{g})$ are Cartan subalgebras of the $\mathfrak{g}(s)$; moreover $\mathfrak{d}$ is a Lie subalgebra of $\mathfrak{g}$ by virtue of (XIII 4.1), hence it is a Cartan subalgebra.

Corollary 2.7. *Under the conditions of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, a a section of $\mathfrak{d}$ that is regular in $\mathfrak{g}$, u an automorphism of $\mathfrak{g}$. In order for u to preserve $\mathfrak{d}$, it is necessary and sufficient that $u(a)$ be a section of $\mathfrak{d}$.*

Indeed, by transport of structure $u(a)$ is then a regular section of $\mathfrak{g}$, contained in the two Cartan subalgebras $\mathfrak{d}$ and $u(\mathfrak{d})$, which are therefore identical.

Corollary 2.8. *Under the conditions of 2.4, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then for every $s \in S$ such that $k(s)$ is infinite, there exists an open neighborhood V of s and a regular section a of $\mathfrak{g}$ over V such that $\mathfrak{d} = Nil(a, \mathfrak{g}|V)$ (i.e. such that a is a section of $\mathfrak{d}|V$).*

Indeed, the fact that $k(s)$ is infinite ensures the existence of a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}(s)$; one can extend it to a section a of $\mathfrak{d}$ on an open neighborhood U of s, and since $ad(a)_{\mathfrak{g}/\mathfrak{d}}$ induces an automorphism of $\mathfrak{g}(s)/\mathfrak{d}(s)$, it follows that, on restricting V, $ad(a)_{\mathfrak{g}/\mathfrak{d}}$ is an automorphism, which implies that a is a quasi-regular section of $\mathfrak{g}|V$ satisfying the desired condition.

Let $\mathfrak{g}$ be as in 2.4; then examination of its Killing polynomial implies at once that the function

$$s \mapsto \text{nilpotent rank of } \mathfrak{g}(s)$$

on S is upper semi-continuous. We are above all interested in the case when this function is in fact continuous, i.e. locally constant. Here are some variants of this property:

Proposition 2.9. *Let S, $\mathfrak{g}$ be as in 2.4. Consider the following conditions:*

- *(C_0) The nilpotent rank of the $\mathfrak{g}(s)$ ($s \in S$) is a locally constant function of s.*
- *(C_1) There locally exists, for the fpqc topology, a Cartan subalgebra of $\mathfrak{g}$.*
- *(C_1') Like (C_1), with "fpqc topology" replaced by "étale topology".*
- *(C_2) Condition (C_0) is satisfied, and for every S' over S, every quasi-regular section of $\mathfrak{g}_{S'} = \mathfrak{g} \otimes_S S'$ is regular.*
- *(C_3) Every $s \in S$ has an open neighborhood V on which the Killing polynomial of $\mathfrak{g}$ is of the form*

$$t^r(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r})$$

where for every $s \in V$, $c_{n-r}(s) \in Sym(\mathfrak{g}(s)^\vee)$ is non-zero.

With these notations, one has the implications

$$(C_3) \Rightarrow (C_2) \Rightarrow (C_1') \Rightarrow (C_1) \Rightarrow (C_0),$$

and when S is reduced, the five conditions considered are equivalent.

Let us also note that the conditions considered are manifestly stable by base change, and of local nature for the fpqc topology.

The implications (C_1') ⇒ (C_1) ⇒ (C_0) are trivial, the implication (C_0) ⇒ (C_3) when S is reduced is immediate. Note moreover:

Corollary 2.10. *Suppose condition (C_0) satisfied. Let U be the set of elements of $W(\mathfrak{g})$ that are regular in their fiber. Then U is open; in particular, for every section a of $\mathfrak{g}$ over S, the set V of $s \in S$ such that $a(s) \in \mathfrak{g}(s)$ is regular, is open.*

Indeed, the first assertion follows from the second (applied to $\mathfrak{g}_{S'}$ for every base change S' over S). For the second, since one may suppose here S reduced, hence (C_3) verified, it suffices to examine the Killing polynomial of a in $\mathfrak{g}$.

The implication (C_2) ⇒ (C_1') now follows easily from b) in the more precise statement that follows:

Corollary 2.11. *Suppose condition (C_2) satisfied. Then:*

a) For every $s \in S$ and every Cartan subalgebra $\mathfrak{d}_0$ of $\mathfrak{g}(s)$ such that $\mathfrak{d}_0$ contains a regular element of $\mathfrak{g}(s)$ (condition automatically satisfied if $k(s)$ is infinite), there exists an open neighborhood V of s and a subalgebra $\mathfrak{d}$ of $\mathfrak{g}|V$ whose fiber at s is $\mathfrak{d}_0$. If S_1 is a subprescheme of S containing s, and if one has already extended $\mathfrak{d}_0$ to a Cartan subalgebra $\mathfrak{d}_1$ of $\mathfrak{g} \otimes_S S_1$, then one can find an open neighborhood V of s in S and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|V$ such that $\mathfrak{d} \otimes_V (S_1 \cap V)$ is equal to $\mathfrak{d}_1|(S_1 \cap V)$.

b) For every $s \in S$ such that $\mathfrak{g}(s)$ contains a regular element (condition automatically satisfied if $k(s)$ is infinite), there exists an open neighborhood V of s and a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}|V$.

Statement b) follows from a) by taking $\mathfrak{d}_0 = Nil(a_0, \mathfrak{g}(s))$, a_0 being a regular element of $\mathfrak{g}(s)$. To prove a), let us say the second formulation, one considers a regular element a_0 of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$, one extends it to a neighborhood of s in S_1 as a section of $\mathfrak{d}_1$, and one extends this last to a section of $\mathfrak{g}$ in a neighborhood of s. By virtue of 2.10, this section is quasi-regular in an open neighborhood V of s, hence regular by virtue of (C_2), hence by virtue of 2.6 $\mathfrak{d} = Nil(a, \mathfrak{g}|V)$ satisfies the required conditions.

Remark 2.12. I do not know if statement 2.11 a) and b) is valid without the hypothesis of existence of regular points (when $k(s)$ is finite).

It remains to prove the implication (C_3) ⇒ (C_2). Let us also note the following equivalent form of (C_3):

- (C_3') One has (C_0), i.e. the nilpotent rank of the $\mathfrak{g}(s)$ ($s \in S$) is locally constant, and on every open V of S where this rank has value r, the Killing polynomial of $\mathfrak{g}|V$ is divisible by t^r.

It is necessary to show that this condition implies that every quasi-regular section a of $\mathfrak{g}$ is regular. Taking 2.6 into account, this is contained in the following lemma (applied to the endomorphism $ad(a)$ of $\mathfrak{g}$), (iv) ⇒ (iii):

Lemma 2.13. *Let A be a ring, M a projective A-module of finite type, u an endomorphism of M. The following conditions are equivalent:*

- *(i) M is the direct sum of two stable submodules M', M'' such that $u|M'$ is nilpotent and $u|M''$ is an automorphism of M''.*
- *(ii) There exists an integer $n > 0$ such that $Im u^n + Ker u^n = M$.*
- *(iii) The nil-space $Nil(u) = \bigcup_{n>0} Ker u^n$ is a direct factor in M, and $M = Nil(u) + u(M)$.*

These conditions are implied by the following (and are equivalent to it when A is reduced):

- *(iv) Locally on* Spec(A) *(for the Zariski topology) the characteristic polynomial $P_u(t)$ of u can be put in the form*

$$t^r(t^{n-r} + c_1 t^{n-r-1} + \cdots + c_{n-r}),$$

where c_{n-r} is invertible.

The equivalence of (i) (ii) (iii) is immediate and is recorded for the record. The fact that (i) implies (iv) when A is reduced follows from the fact that in this case a nilpotent endomorphism of a projective module of rank r has characteristic polynomial t^r, while in any case the characteristic polynomial of an automorphism of a projective module of finite type has as constant term the determinant of u up to sign (locally on Spec(A)), hence an invertible element of A. Finally, to prove (iv) ⇒ (i), one notes that M is a module over the polynomial ring $A[t]$, by letting t act as u, and the well-known identity

$$P(u) = 0$$

shows that M is annihilated by $P\,A[t]$, hence can be considered as a module over $A[t]/PA[t]$. Now writing $P = t^r Q$, where the constant term of Q is invertible, one sees at once that

$$P\,A[t] = t^r A[t] \cap Q\,A[t],$$

hence $A[t]/PA[t]$ decomposes into the product of the rings $A[t]/t^r A[t]$ and $A[t]/QA[t]$, whence a corresponding decomposition of M as a sum of two $A[t]$-modules, i.e. as a sum of two A-submodules M' and M'' stable under u; this is the decomposition envisaged in (i).

This completes the proof of 2.13, hence of 2.9.

Corollary 2.14. *When condition (C_3) of 2.9 is satisfied, the Cartan subalgebras of* $\mathfrak{g}$ *are strictly nilpotent (2.2).*

The proof is immediate.

Remark 2.15. a) Let us point out that one can prove a converse of 2.14: (C_3) is equivalent to the fact that for every S' over S, every quasi-regular section of $\mathfrak{g}_{S'}$ is regular and every Cartan subalgebra of $\mathfrak{g}_{S'}$ is strictly nilpotent, or again, every quasi-regular section of $\mathfrak{g}_{S'}$ is contained in a strictly nilpotent Cartan subalgebra of $\mathfrak{g}_{S'}$.

b) Contrary to the other conditions (C_0) to (C_2), condition (C_3) is of "infinitesimal nature". More precisely, when S is locally noetherian, $\mathfrak{g}$ satisfies condition (C_3) if and only if for every base change $S' \to S$ with S' local artinian (if one wishes, S' the spectrum of an artinian quotient of a local ring of S), $\mathfrak{g}_{S'}$ satisfies the same condition. Similarly, when (C_0) is satisfied, the condition for a section of $\mathfrak{g}$ to be regular is of infinitesimal nature.

c) When $\mathfrak{g}$ is the Lie algebra of a smooth prescheme in groups of finite presentation over S, then we shall see that conditions (C_0), (C_1), (C_1'), (C_2) on $\mathfrak{g}$ are equivalent (5.2 a)); I do not know what is the case in general (except that, even for S local artinian, (C_0) does not imply (C_1)). However, even in the case where $\mathfrak{g}$ comes from a G, and S being local artinian, it is not true in general that (C_2) implies (C_3), since $\mathfrak{g}$ can be nilpotent without being strictly nilpotent. One obtains an example of this fact starting from a smooth affine group scheme G over the spectrum S of a discrete valuation ring such that the Lie algebra of the generic fiber is non-nilpotent (for example the generic fiber is an adjoint semisimple group), and that of the special fiber is nilpotent (for example, the special fiber being a vector group): then for n large enough, the Lie algebra of $G_n = G \times_S S_n$ is not strictly nilpotent, however it is nilpotent.

Let us still place ourselves under the conditions of 2.4, and let

$$\mathcal{D} : (Sch)^\circ/S \longrightarrow (Ens)$$

be the functor defined by

$\mathcal{D}(S')$ = set of Cartan subalgebras of $\mathfrak{g}_{S'}$.

Let us also introduce the functor

$X(S') =$ set of pairs $(\mathfrak{d}, a)$, where $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{g}_{S'}$ and a a section of $\mathfrak{d}$.

One thus has two projections $(\mathfrak{d}, a) \mapsto \mathfrak{d}$ and $(\mathfrak{d}, a) \mapsto a$:

$p : X \to \mathcal{D}$ and $\psi : X \to W(\mathfrak{g})$.

When (C_0) is satisfied, we also consider the open U of regular points of $W(\mathfrak{g})$ (cf. 2.10) and condition (C_2) is then expressed by the fact that the morphism

$$\psi^{-1}(U) \longrightarrow U$$

induced by ψ is an isomorphism (a priori, it is a monomorphism thanks to 2.6). Note that it is trivial that the morphism $p : X \to \mathcal{D}$ is representable by a projection of vector bundles (i.e. for every S-morphism $S' \to \mathcal{D}$, corresponding to a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}_{S'}$, $X \times_{\mathcal{D}} S'$ is representable by a vector bundle over S', namely $W(\mathfrak{d})$); so if $\mathcal{D}$ is representable, the same is true of X. Now one has:

Theorem 2.16. *Let S be a prescheme, $\mathfrak{g}$ a Lie algebra over S that is a locally free $\mathcal{O}_S$-module of finite type; suppose condition (C_0) of 2.9 is satisfied.*

a) The functor $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ defined above is representable by a quasi-projective prescheme of finite presentation over S. The same is true of the functor X defined above.

b) When condition (C_2) of 2.9 is satisfied, $\mathcal{D}$ and X are smooth over S, and the morphism $\psi^{-1}(U) \to U$ induced by ψ is an isomorphism.

c) Still assuming condition (C_2) satisfied, let $s \in S$, $\mathfrak{d}_0$ a Cartan subalgebra of $\mathfrak{g}(s)$, corresponding to a point d of $\mathcal{D}$ rational over $k(s)$. Suppose that $\mathfrak{d}_0$ contains a regular point of $\mathfrak{g}(s)$ (condition automatically satisfied if $k(s)$ is infinite). Let r be the infinitesimal rank of $\mathfrak{g}(s)$, n its rank over $k(s)$; then there exists an open neighborhood V of d in $\mathcal{D}$ that is S-isomorphic to an open V' of $S[t_1, \cdots, t_{n-r}]$.

Proof. One may suppose $\mathfrak{g}$ of constant rank n and of constant nilpotent rank r. The assertions made about X in a) and b) follow at once from the assertions made about $\mathcal{D}$ and from the fact that X is a vector bundle over $\mathcal{D}$ defined by a locally free module, and are stated only for the record.

a) The functor $\mathcal{D}$ is a subfunctor of the functor $Grass_{n-r}(\mathfrak{g})$ whose value at S' is the set of locally free quotient modules of rank $n-r$ of $\mathfrak{g}_{S'}$, and it is well known that this latter functor is representable by a projective prescheme smooth over S (cf. for example Séminaire Cartan 1960/61, Exp. 12, Nos 2 and 3, whose constructions transpose as they stand to the case of preschemes) [1099] [1100]. So one is reduced to a relative problem, namely the following: given a locally free quotient module of rank $n-r$ of $\mathfrak{g}$, or equivalently, a locally free submodule $\mathfrak{d}$ of rank r that is locally a direct factor, represent the following functor: $F(S') = \emptyset$ if $\mathfrak{d}$ is not a Cartan subalgebra of $\mathfrak{g}_{S'}$, $F(S') = \emptyset$

[1099] Cf. also EGA I, 2nd edition (to appear in North Holland Publishing Co.).

[1100] See § I.1.3 of M. Demazure and P. Gabriel, *Groupes algébriques*, Masson (1970).

otherwise. In fact, we shall see that F is representable by a subprescheme of finite presentation of S (which will show that $\mathcal{D} \to Grass$ is representable by an immersion of finite presentation, and will finish proving a)). One begins by expressing the condition that $\mathfrak{d}_{S'}$ is a Lie subalgebra of $\mathfrak{g}_{S'}$; one sees at once that this is expressed by the fact that $S' \to S$ factors through a certain closed subprescheme S_1 of S, of finite presentation over S (whose local equations on S can be written immediately using a basis of $\mathfrak{g}$ adapted to the submodule $\mathfrak{d}$). One may therefore suppose that one has already $S = S_1$. One must then express that $\mathfrak{d}_{S'}$ contains locally for fpqc a quasi-regular section of $\mathfrak{g}_{S'}$, and for this one considers $V = W(\mathfrak{d}) \cap U$, where U is the open of regular points of $W(\mathfrak{g})$ (2.10); then the structure morphism $V \to S$ being smooth and quasi-compact, its image S_1 is an open part of S and the immersion morphism $S_1 \to S$ is quasi-compact i.e. of finite presentation. The condition envisaged on S' is then expressed by saying that $S' \to S$ factors through S_1. So one is reduced to the case where $S = S_1$, and using the theory of descent, to the case where $\mathfrak{d}$ admits a section a that is a quasi-regular section of $\mathfrak{g}$. It remains finally to express that the section $a_{S'}$ of $\mathfrak{g}_{S'}$ deduced from a satisfies $ad(a_{S'})_{\mathfrak{g}_{S'}/\mathfrak{d}_{S'}}$ bijective, which amounts again to saying that $S' \to S$ factors through a certain open subprescheme of finite presentation of S, namely `S_D`, where D is the determinant of $ad(a)_{\mathfrak{g}/\mathfrak{d}}$. But then one sees at once that $\mathfrak{d}|S_D$ is a Cartan subalgebra of $\mathfrak{g}|S_D$, hence `S_D` represents the functor F, which proves a).

b) Is immediate thanks to 2.11 a) and (XI 1.5). Of course b) is also a consequence of the more precise statement c).

c) Let a_0 be a regular point of $\mathfrak{g}(s)$ contained in $\mathfrak{d}_0$; extend it to a section a of $\mathfrak{g}$ on an open neighborhood V of s; one may evidently suppose $V = S$. On the other hand let M_0 be a complement of the vector space $\mathfrak{d}_0$ in $\mathfrak{g}(s)$; then in an open neighborhood V of S there exists a submodule M of $\mathfrak{g}$, a direct factor of $\mathfrak{g}|V$, such that $M(s) = M_0$, and one may again suppose $V = S$. Let now $\mathcal{V}$ be the subfunctor of $\mathcal{D}$ such that $\mathcal{V}(S')$ is the set of Cartan subalgebras $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ satisfying the two following conditions:

- 1°) $\mathfrak{d}'$ is a complement of $M_{S'}$, and
- 2°) the unique section of $(a_{S'} + M_{S'}) \cap \mathfrak{d}'$ is a regular section of $\mathfrak{g}_{S'}$.

Condition 1°) corresponds to an open V_1 of $\mathcal{D}$ (induced by the open of $Grass_{n-r}(\mathfrak{g})$ defined by the same condition 1°)); the conjunction of 1°) and 2°) corresponds to an open of V_1 by virtue of 2.10 and (C₂). Hence $\mathcal{V}$ is represented by an open subprescheme V of $\mathcal{D}$, evidently containing d.

On the other hand let $\mathcal{V}'$ be the subfunctor of $W(M)$ defined by

$\mathcal{V}'(S') = setofsectionsu'ofM_{S'}$ such that:

- (i) $a_{S'} + u'$ is a regular section of $\mathfrak{g}_{S'}$, and
- (ii) the unique Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ containing $a_{S'} + u'$ is a complement of $M_{S'}$.

Then condition 1°) corresponds to an open subprescheme V_1' of $W(M)$, namely the inverse image of the open U of regular points of $W(\mathfrak{g})$ (cf. 2.10) by the translation morphism $m \mapsto a + m$. The conjunction of (i) and (ii) corresponds to an open V' of V_1', namely the inverse image of V by the obvious morphism $V_1' \to \mathcal{D}$ (associating to u' the unique Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}_{S'}$ containing $a_{S'} + u'$). The restriction of this last morphism to V' is a morphism

$$V' \longrightarrow V,$$

which is evidently an isomorphism. This proves c).

Corollary 2.17. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field k. Then the scheme $\mathcal{D}$ of Cartan subalgebras of $\mathfrak{g}$ (2.16 a)) is quasi-projective, smooth and irreducible. When $\mathfrak{g}$ contains a regular element (for example when k is infinite), $\mathcal{D}$ is a rational variety, i.e. its function field is a pure extension of k.*

The fact that $\mathcal{D}$ is irreducible follows from the fact that one has a surjective morphism $\psi^{-1}(U) \to \mathcal{D}$, and $\psi^{-1}(U)$ is irreducible, being isomorphic to the open U of $W(\mathfrak{g})$. The assertion on the function field is an immediate consequence of c).

Remark 2.18. I do not know if this conclusion remains valid if k is finite, without supposing that $\mathfrak{g}$ contains a regular point; compare 2.12. One can prove that this is the case when $\mathfrak{g}$ is the Lie algebra of an algebraic group G smooth over k, at least when $G/radical$ is an "adjoint" semisimple group, by using a result of Chevalley pointed out below (cf. Appendix). It is plausible that this result remains valid without restriction on G; it would suffice for this that the cited result of Chevalley be proved for every (not necessarily adjoint) semisimple algebraic group.

3. Subgroups of type (C) of group preschemes over an arbitrary prescheme

Theorem 3.1. *Let S be a prescheme, G a smooth S-prescheme in groups, $\mathfrak{g}$ its Lie algebra (which is a locally free module of finite type over S), $\mathfrak{h}$ a Lie subalgebra of $\mathfrak{g}$ that is (as a module) a locally direct factor in $\mathfrak{g}$, and such that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Let a be a quasi-regular section of $\mathfrak{g}$ (2.5). Then*

$$M_a = Transp_G(a, \mathfrak{h})$$

(subfunctor of G whose S'-valued points are the $g \in G(S')$ such that $ad(g) \cdot a_{S'} \in \Gamma(S', \mathfrak{h}_{S'})$) is representable by a closed subprescheme [1101] of G smooth over S, whose structure morphism to S is surjective.

Consider the canonical morphism

$\phi : G \times_S W(\mathfrak{g}) \to W(\mathfrak{g})$

[1101] One has written $Transp_G(a, \mathfrak{h})$.

given by $(g, x) \mapsto ad(g) \cdot x$; then M_a is S-isomorphic to $\varphi^{-1}(a)$, the inverse image of a (considered as a section of $W(\mathfrak{g})$ over S) by φ. It will suffice for the smoothness of M_a to show that φ is smooth at points of $G \times_S W(\mathfrak{h})$ lying over $Im(a)$; more generally φ is smooth at every point lying over a point of $W(\mathfrak{g})$ that is regular in its fiber $W(\mathfrak{g}(s))$ over S. To see this, since the source and target of φ are smooth, hence flat locally of finite presentation over S, one is reduced to making the verification fiber by fiber, which reduces us to the case where S is the spectrum of an algebraically closed field, G thus being a locally algebraic group over k, $\mathfrak{h}$ a Lie subalgebra of its Lie algebra $\mathfrak{g}$, containing a Cartan subalgebra of $\mathfrak{g}$, and a a regular point of $\mathfrak{g}$. One may evidently suppose (taking into account that φ is a G-morphism) that the point of $G \times W(\mathfrak{h})$ envisaged is of the form (e, a). One may evidently suppose G connected, hence of finite type over k, but then our assertion is none other than (XIII 5.4). Moreover, the fact that M_a is a closed subprescheme of G (of finite presentation over S) is trivial, since M_a is the inverse image of $W(\mathfrak{h})$ by the morphism $g \mapsto ad(g) \cdot a$ of G into $W(\mathfrak{g})$. The surjectivity of the structure morphism $M_a \to S$ also reduces to the case of an algebraically closed base field, but then $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ by hypothesis, which is therefore conjugate to the Cartan subalgebra $\mathfrak{d} = Nil(a, \mathfrak{g})$ by the conjugation theorem (XIII 6.1 a)), hence $\mathfrak{h}$ contains a conjugate of a. This completes the proof of 3.1.

Corollary 3.2. *Let G, $\mathfrak{g}$ be as in 3.1, with G of finite type over S; let $\mathfrak{k}$ and $\mathfrak{h}$ be two Lie subalgebras of $\mathfrak{g}$, locally direct factors (as modules); suppose that one is under one of the two following hypotheses:*

a) For every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent and contains a regular element of $\mathfrak{g}_{\bar{s}}$; the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$.

b) $\mathfrak{k}$ is a Cartan subalgebra of $\mathfrak{g}$.

Then $Transp_G(\mathfrak{k}, \mathfrak{h})$ is a closed subprescheme of G smooth over S; moreover, in case a), its structure morphism to S is surjective.

The fact that the transporter is representable by a closed subprescheme of finite presentation of G is immediate, and left to the reader. Let us first prove smoothness in case a). Suppose first that there exists a section a of $\mathfrak{k}$ that is quasi-regular in $\mathfrak{g}$. Then it suffices to apply 3.1 and the

Lemma 3.3. *Under the conditions of 3.2 a), if a is a section of $\mathfrak{k}$ quasi-regular in $\mathfrak{g}$, then one has*

`Transp_G(□, □) = Transp_G(a, □).`

Indeed, taking the definitions into account, this amounts to showing that if a is moreover a section of $\mathfrak{h}$, then one has $\mathfrak{k} \subset \mathfrak{h}$. Now since by hypothesis $\mathfrak{k}$ is locally nilpotent, it follows from 2.1 that $\mathfrak{k} \subset Nil(a, \mathfrak{g})$; on the other hand $Nil(a, \mathfrak{g}) \subset \mathfrak{h}$ because $ad(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective (being so fiber by fiber by virtue of (XIII 4.8 b))). Whence the conclusion. In the general case, one reduces to the case where S is affine noetherian by the standard procedure, then to the case where S is local artinian (smoothness being a property of infinitesimal nature), and by flat descent to the case where its residue field is infinite, hence the fiber K_0 admits an element that is regular in $\mathfrak{g}_0$. One lifts this element to an element of $K = \Gamma(\mathfrak{k})$, which reduces us to the preceding case. Thus, in case a) we have proved the smoothness of the transporter; as for the fact that its structure morphism is surjective, it reduces to the case where S is the spectrum of an algebraically closed field, hence where K contains a regular point of $\mathfrak{g}$, and one applies 3.3 and 3.1.

To prove b), one is reduced by the definition of smoothness (XI 1.1) to proving that if S is affine, S_0 a subscheme defined by a quasi-coherent nilpotent ideal J, g_0 an element of $G(S_0)$ that transports $\mathfrak{k}_0$ into $\mathfrak{g}_0$, then g_0 lifts to an element g of $G(S)$ that transports $\mathfrak{k}$ into $\mathfrak{g}$. Now the hypothesis on g_0 implies that one is under the conditions of a), already treated. This completes the proof.

Of course, when in 3.2 b) $\mathfrak{h}$ satisfies the stronger hypothesis of 3.1, then (and only then) the structure morphism $Transp_G(\mathfrak{d}, \mathfrak{h}) \to S$ is surjective. Using Hensel's lemma (XI 1.10), one concludes from 3.1 and 3.2:

Corollary 3.4. *Under the conditions of 3.1 for G and $\mathfrak{h}$, and supposing G of finite type over S:*

a) For every quasi-regular section a of $\mathfrak{g}$, there locally exists for the étale topology a conjugate of a that is a section of $\mathfrak{h}$.

b) For every Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$, $\mathfrak{d}$ is locally for the étale topology conjugate to a subalgebra of $\mathfrak{h}$.

In particular, when $\mathfrak{h}$ is itself a Cartan subalgebra of $\mathfrak{g}$, one finds:

Corollary 3.5. *Let G be a smooth S-prescheme in groups of finite type, $\mathfrak{g}$ its Lie algebra, $\mathfrak{d}$ and $\mathfrak{d}'$ two Cartan subalgebras of $\mathfrak{g}$. Then $Transp_G(\mathfrak{d}, \mathfrak{d}')$ is identical to the strict transporter of $\mathfrak{d}$ into $\mathfrak{d}'$, and is a closed subprescheme of G smooth over S, with surjective structure morphism. Locally for the étale topology, $\mathfrak{d}$ and $\mathfrak{d}'$ are conjugate.*

The fact that the transporter is identical here to the strict transporter follows trivially from the fact that $\mathfrak{d}$ and $\mathfrak{d}'$ are locally direct factors in $\mathfrak{g}$ and have the same rank at every point. Hence 3.5 is a particular case of 3.4. In particular, if $\mathfrak{d} = \mathfrak{d}'$:

Corollary 3.6. *Let G be as in 3.5, and let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Then $Norm_G(\mathfrak{d})$ is a closed subprescheme in groups of G smooth over S, whose Lie algebra is identical to $\mathfrak{d}$.*

Indeed, this last assertion amounts to saying that $\mathfrak{d}$ is its own normalizer in $\mathfrak{d}$, which follows at once from the fact that this is true fiber by fiber.

Corollary 3.7. *Let G, $\mathfrak{g}$ be as in 3.5. Then conditions (C_2), (C_1'), (C_1) of 2.9 are equivalent; in other words, if $\mathfrak{g}$ admits locally for the fpqc topology a Cartan subalgebra, then every quasi-regular section of $\mathfrak{g}$ is regular.*

Indeed, let a be a quasi-regular section; let us prove that it is regular. The question being local for the fpqc topology, one may suppose that $\mathfrak{g}$ admits a Cartan subalgebra $\mathfrak{d}$. By virtue of 3.4 a), a is then locally for the étale topology conjugate to a section of $\mathfrak{d}$, which reduces us to the case where a is a section of $\mathfrak{d}$, where the conclusion is trivial from the definition.

Definition 3.8. *Let G be a smooth prescheme in groups over a prescheme S. One calls a subgroup of type (C) of G a subprescheme in groups D of G, smooth over S, with connected fibers, such that $\mathfrak{d} = Lie(D)$ is a Cartan subalgebra (2.4) of $\mathfrak{g} = Lie(G)$, i.e. such that for every $s \in S$, D_s is a subgroup of type (C) of the algebraic group G_s (XIII 6.2).*

Theorem 3.9. *Let G be a smooth S-prescheme in groups of finite presentation, $\mathfrak{g}$ its Lie algebra. Then:*

a) The map

$$D \mapsto \mathfrak{d} = Lie(D)$$

establishes a bijective correspondence between subgroups of type (C) of G and Cartan subalgebras of $\mathfrak{g}$.

b) If D and $\mathfrak{d}$ correspond, one has

$$Norm_G(D) = Norm_G(\mathfrak{d}),$$

it is a closed subprescheme of G smooth over S, and one has

$$D = Norm_G(D)^0 = Norm_G(\mathfrak{d})^0.$$

c) Two subgroups of type (C) D and D' of G are conjugate locally for the étale topology.

Proof. Let D be a subgroup of type (C) of G, and $\mathfrak{d}$ its Lie algebra; then $D \subset Norm_G(\mathfrak{d})$, and by virtue of Definition 3.8 and 3.6 this is an inclusion of preschemes in groups smooth over S, inducing an isomorphism on the Lie algebras. Since D has connected fibers, one therefore has $D = Norm_G(\mathfrak{d})^0$. Hence the map envisaged in a) is injective; let us prove that it is surjective. So let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$; then by virtue of 3.6 $Norm_G(\mathfrak{d}) = N$ is a closed subprescheme in groups of G smooth over S, admitting $\mathfrak{d}$ as Lie algebra. Since G is of finite presentation over S, the same is true of N, hence (as was pointed out in XII after 7.3) the union of the connected components of the fibers of N is the underlying set of an open subprescheme in groups N° of N, which is evidently a subgroup of type (C) of G having Lie algebra $\mathfrak{d}$. This proves a); the first assertion of b) follows from it at once; and the formula $D = Norm_G(\mathfrak{d})^0$ has already been proved. Finally, c) follows from a) and 3.5.

Corollary 3.10. *Suppose that $\mathfrak{g}$ admits locally for the fpqc topology a Cartan subalgebra (or equivalently, by virtue of 3.9 a), that G admits locally for the fpqc topology a subgroup of type (C)). Consider the functor $\mathcal{D} : (Sch)^\circ/S \to (Ens)$ defined by*

$\mathcal{D}(S')$ = set of subgroups of type (C) of $G_{S'}$. Then this functor is representable by a quasi-projective and smooth prescheme over S, with connected geometric fibers. When S is the spectrum of a field k, hence G a smooth algebraic group over k, and $\mathfrak{g}$ admits a regular point (condition automatically satisfied if k is infinite), then $\mathcal{D}$ is a rational variety over k.

Indeed, by virtue of 3.9 a), the functor $\mathcal{D}$ is canonically isomorphic to the functor envisaged in 2.16; on the other hand by virtue of 3.7 condition (C_2) is satisfied. Hence 3.10 follows from 2.16 and 2.17.

Corollary 3.11. *Let D be a subgroup of type (C) of G, and let N be its normalizer in G. Then the quotient sheaf G/N is canonically isomorphic to the functor $\mathcal{D}$ of 3.10, hence representable by a quasi-projective and smooth prescheme over S, with connected geometric fibers.*

Proposition 3.12. *Let G be a smooth S-prescheme in groups of finite presentation over S, H, K two subpreschemes in groups, smooth of finite presentation over S, with K having connected fibers; $\mathfrak{g}$, $\mathfrak{k}$, $\mathfrak{h}$ the corresponding Lie algebras. Suppose that one of the two following conditions is realized for the geometric fibers of these latter:*

a) For every $s \in S$, $\mathfrak{k}_s$ contains a regular element of $\mathfrak{g}_s$, and $\mathfrak{h}_s$ contains a Cartan subalgebra of $\mathfrak{g}_s$.

b) For every $s \in S$, $\mathfrak{k}_s$ contains a Cartan subalgebra of $\mathfrak{g}_s$.

Under these conditions, in order to have $H \supset K$, it is necessary and sufficient that one have $\mathfrak{h} \supset \mathfrak{k}$.

Of course, we need only prove that if $\mathfrak{h} \supset \mathfrak{k}$, then $H \supset K$. In case b), the inclusion $\mathfrak{h} \supset \mathfrak{k}$ shows that one is in fact under the conditions of a), so it suffices to prove a). Proceeding as in 3.2 a) by reduction to the case S local artinian, one is reduced to the case where there exists a section a of $\mathfrak{k}$ that is quasi-regular in $\mathfrak{g}$. In this case, proceeding as in (XIII 5.5), one is reduced to the following statement:

Corollary 3.13. *Let G be a smooth S-prescheme in groups of finite presentation over S, H a subprescheme in groups of G smooth of finite presentation over S, $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras, a a section of $\mathfrak{h}$ that is quasi-regular in $\mathfrak{g}$. Suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_s$ contains a Cartan subalgebra of $\mathfrak{g}_s$. Let*

$$M_a = Transp_G(a, \mathfrak{h}),$$

which is a closed subprescheme of G smooth over S (cf. 3.1), so that M_a^0 (union of the connected components of the identity element in the fibers of M_a) is an open part of M_a, which we shall endow with the structure induced by M_a. One then has $H^0 = M_a^0$.

Evidently one has $H \subset M_a$, whence $H^0 \subset M_a^0$. Since this is an inclusion of preschemes smooth over S, to prove that it is an equality, one is reduced to verifying it on the fibers, which reduces us to the case where S is the spectrum of an algebraically closed field, a case treated in (XIII 5.4).

Corollary 3.14. *Let G be a smooth S-prescheme in groups of finite presentation, D a subgroup of type (C) of G, H a subprescheme in groups of G smooth over S and of finite presentation over S, $\mathfrak{d}$ and $\mathfrak{h}$ their Lie algebras. Then one has*

`Transp_G(D, H) = Transp_G(□, □),`

and this functor is representable by a closed subprescheme of G smooth over S.

Indeed, the identity between the two transporters follows from 3.12 b), which allows one to apply 3.2.

Corollary 3.15. *Let G, H be as in 3.14 and suppose that for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a Cartan subalgebra of $\mathfrak{g}_{\bar{s}}$. Suppose moreover that $\mathfrak{g}$ admits locally for the fpqc topology a Cartan subalgebra. Then locally for the étale topology, H contains a subgroup of type (C) of G.*

By virtue of 3.7 and 3.9 a), G admits locally for the étale topology a subgroup of type (C), so one may suppose that G admits such a subgroup, say D. Then the hypothesis on $\mathfrak{h}$ means also that the structure morphism of the transporter considered in 3.14 is surjective (taking into account the conjugation theorem XIII 6.1 a)). One concludes by Hensel's lemma (XI 1.10).

Corollary 3.16. *Let G, H, K be as in 3.12 a); suppose moreover that for every $s \in S$, the geometric fiber $\mathfrak{k}_{\bar{s}}$ is nilpotent (i.e. $\mathfrak{k}$ is locally nilpotent). Then one has*

`Transp_G(K, H) = Transp_G(□, □),`

and this functor is representable by a closed subprescheme of G smooth over S, whose structure morphism to S is surjective. H contains locally for the étale topology a subgroup conjugate to K.

The identity of the two transporters is again contained in 3.12 a), the assertion on its structure is then none other than 3.2 a), and the last assertion of 3.16 is then a consequence of Hensel's lemma.

Remarks 3.17. a) In 3.12 and 3.16, one may replace the hypothesis that K is smooth over S by the following weaker hypothesis: the sheaf $e_K * (\Omega^1_{K/S}) = \omega^1_K$ of relative 1-differentials along the unit section is locally free. In this way, 3.12 contains (XIII 5.5).

b) Let G, $\mathfrak{g}$, $\mathfrak{h}$ be as in 3.1, with G of finite presentation over S. Then $N = Norm_G(\mathfrak{h})$ is not necessarily smooth over S along the unit section, or equivalently, there does not necessarily exist a subprescheme in groups H of S smooth over S whose Lie algebra is $\mathfrak{h}$, even if S is the spectrum of a field. When such an H exists, so that one has (taking H with connected fibers) $N = Norm_G(H)$, I do not know whether N is smooth over S. In this question, one may evidently reduce to the case where S is local artinian.

To conclude this section, let us examine the case where G is "semisimple" over S:

Theorem 3.18. *Let S be a prescheme, G a smooth S-prescheme in groups over S, whose geometric fibers are "adjoint" semisimple groups, i.e. semisimple with reductive center (XII 4.1 and 4.4) reduced to the unit group. Then the subgroups of type (C) of G are identical with its maximal tori, hence also with its Cartan subgroups (XII 3.1).*

Taking definitions into account, one is reduced to the case where S is the spectrum of an algebraically closed field, and to proving then that for a maximal torus T of G, the Lie algebra $\mathfrak{t}$ of T is a Cartan subalgebra of $\mathfrak{g}$, that is to say (taking into account the inequality $\texttt{nilpotent rank of } \mathfrak{g} \le \texttt{nilpotent rank of G = dim T = rank } \mathfrak{g} \texttt{ = r}$) that there exists $x \in \mathfrak{t}$ with $\dim Nil(x, \mathfrak{g}) = r$. Since $\mathfrak{t}$ is abelian and *a fortiori* nilpotent, it amounts to the same to say that there exists $a \in \mathfrak{t}$ such that $ad(a)_{\mathfrak{g}/\mathfrak{t}}$ is injective (XIII 5.7 a)). Now consider the characters α of T that intervene in the representation of T induced by the adjoint representation of G. The structure theory of the semisimple group G (*Bible*, 13 th. 1 a) and th. 3, cor. 2), more precisely the "big cell" of G, semidirect product of T and subgroups P_α isomorphic to the additive group G_a, preserved by T and corresponding to the "root" characters of G for the torus T, shows that the eigenspace of $\mathfrak{g}$ relative to the trivial character is none other than $\mathfrak{t}$, and the other eigenspaces are of dimension 1, the characters α associated being none other than the roots of G for T. By virtue of the computation of the reductive center of G as the intersection of the kernels of the characters of T that intervene in the adjoint representation of G (XII 4.8), one sees that the fact that G is adjoint is interpreted as saying that the roots generate the lattice $M = \mathrm{Hom}(T, G_m)$. Now a well-known lemma of the theory of roots tells us that every root is part of a system of simple roots, hence of a basis of the group generated by the roots, and consequently of a basis of the dual M of T [1102]. We conclude:

Corollary 3.19. *If G is an adjoint semisimple algebraic group over an algebraically closed field k, T a maximal torus of G, then for every root α of G with respect to T, $\alpha : T \to G_m$, the corresponding homomorphism $\alpha' : \mathfrak{t} \to k$ is non-zero.*

This result is essentially equivalent to Theorem 3.18, for given $t \in \mathfrak{t}$, $ad(t)$ is semisimple and its eigenvalues in $\mathfrak{g}/\mathfrak{t}$ are none other than the $\alpha'(t)$, hence $ad(t)_{\mathfrak{g}/\mathfrak{t}}$ is injective if and only if the $\alpha'(t)$ are $\neq 0$, and there exists $t \in \mathfrak{t}$ having this property if and only if all the α' are $\neq 0$.

Corollary 3.20. *Let S be a prescheme, G an S-prescheme in groups, smooth, of finite presentation over S, whose geometric fibers are connected reductive algebraic groups (i.e. extensions of a semisimple group by a torus). Then for every $s \in S$ there exists an open neighborhood U of s such that $G|U$ admits a maximal torus.* [1103]

We shall see indeed in XVI that the hypothesis just made on G implies that G is affine over S and of locally constant reductive rank; hence (XII 4.7 c)) G admits a

[1102] One has replaced $\hat{T}$ by T.

[1103] This result, as well as 2.11 on which it rests, generalizes immediately to the case where s is replaced by a finite part of S contained in an affine open.

reductive center Z, $G' = G/Z$ is a smooth and affine group over S whose reductive center is reduced to the unit group, and finally the maximal tori of G and of G' are in bijective correspondence. Moreover, one sees at once that the geometric fibers of G' are connected semisimple groups, and moreover adjoint by definition (their reductive center being trivial). So it suffices to limit oneself to the case where G is semisimple and adjoint, and by virtue of 3.18 one is reduced to finding an open neighborhood U of s and a subgroup of type (C) of $G|U$, or equivalently (3.9 a)) a Cartan subalgebra of $\mathfrak{g}|U$. Now this is possible if $k(s)$ is infinite, since by virtue of 3.7 $\mathfrak{g}$ satisfies condition (C_2) of 2.9, hence one may apply 2.11 b). In fact, statement 3.20 remains valid without supposing $k(s)$ infinite. Indeed, by the preceding argument, it suffices to know that for every adjoint semisimple group G over a finite field k, the Lie algebra $\mathfrak{g}$ of G contains a regular element. Now this statement has been proved by Chevalley (using the properties of the Coxeter element of the Weyl group…), cf. the Appendix below by J.-P. Serre.

Remarks 3.21. a) Statement 3.20 remains valid, with essentially the same proof, replacing G by a closed subprescheme in groups H smooth over S, having everywhere the same rank as G (for example a "parabolic subgroup" of G), provided that $k(s)$ is infinite. I do not know if here again the hypothesis that $k(s)$ is infinite is superfluous. One can show that this is the case at least if H is parabolic, thanks to the construction of the radical U of H and of the quotient H/U, which reduces us to the semisimple case. Unfortunately, the method of regular elements seems here powerless, since one easily constructs examples (for example with the projective group and its "standard" Borel subgroup) where the Lie algebra $\mathfrak{h}$ of H does not contain any regular element.

b) The proof of 3.20 in fact shows a more precise result (by invoking 3.9 b)) in the case where $k(s)$ is infinite, namely that every maximal torus T_0 of G_s comes from a maximal torus T on an open neighborhood of s. I do not know if this statement remains valid when $k(s)$ is no longer supposed infinite; the difficulty obviously coming from the fact that the Lie algebra $\mathfrak{t}_0$ of T_0 does not in general contain a regular element of the Lie algebra $\mathfrak{g}_0$ of the fiber G_s. An affirmative answer to this problem would imply the following statement (which is proved only in the case of an infinite residue field or when A is separated and complete): Let A be a local ring, with residue field k, M an "Azumaya algebra" over A, i.e. an algebra such that M is a free module of finite type over A, and $M_0 = M \otimes_A k$ a central simple algebra over k, D_0 a commutative subalgebra of M_0 separable over k, such that $[M_0 : k] = ([D_0 : k])^2$; then there exists a commutative subalgebra D of M, which is a direct factor module in M and such that $D \otimes_A k = D_0$ (?). (Note that the datum of M is equivalent to the datum of a principal homogeneous bundle under the projective group $PGL(n)_A$, whence an "inner" twisted form G of $PGL(n)$, and the maximal tori of G correspond bijectively to the commutative subalgebras D of M étale over A of rank n over A.)

c) Applying 3.20 to the centralizer of a subtorus Q of G (G a reductive S-prescheme in groups), one deduces that every such Q is contained, locally for the Zariski topology, in a maximal torus of G.

4. A digression on Borel subgroups

Definition 4.1. *Let G be a smooth algebraic group over an algebraically closed field. One calls a Borel subgroup of G a smooth, solvable, connected algebraic subgroup that is maximal for these properties.*

When G is affine, one thus recovers the terminology of (*Bible* 6 def. 1). Let us note at once that if Z is a connected and smooth subgroup of G contained in the center (or more generally, solvable and invariant), then for every Borel subgroup B of G, the image BZ of $B \times Z$ by the morphism $(b, z) \mapsto bz$ of $B \times Z$ into G is a smooth, solvable, connected subgroup of G containing B, hence identical to B, hence B contains Z, hence B is the inverse image of an algebraic subgroup B' of $G' = G/Z$, and it is immediate that B' is a Borel subgroup of G'. Taking $Z = Centr(G^0)^\circ_{red}$, $G' = G/Z$ is affine (XII 6.1), hence the Borel subgroups of G' are conjugate and for such a B', G'/B' is a projective variety (*Bible* 6 th. 4). Consequently:

Proposition 4.2. *Let G be as in 4.1. Then the Borel subgroups of G are conjugate. If B is a Borel subgroup, then G/B is a projective variety. The maximal tori of B (resp. the Cartan subgroups of B, G being connected) are maximal tori of G (resp. Cartan subgroups of G).*

It remains to prove the last assertion, and one may evidently suppose $G = G^0$. For the Cartan subgroups, it follows from the analogous assertion in G' (*Bible* 6 th. 4 cor. 4) and from (XII 6.6 e)). For the maximal tori, it follows from the preceding, since by (XII 6.6 c)) the maximal tori of a smooth algebraic group are the maximal tori of its Cartan subgroups.

Corollary 4.3. *Suppose G connected. Then every element of G is contained in a Borel subgroup of G.*

One is reduced to the same statement in G', which is well known (*Bible* 6 th. 5 d)).

Corollary 4.4. *Let B be a Borel subgroup of G, C a Cartan subgroup of B, N its normalizer in G; then $N \cap B = C$.*

Indeed, $N \cap B = Norm_B(C)$, so one is reduced to showing that when G is connected and solvable, a Cartan subgroup C is its own connected normalizer. Now with the preceding notations, C is the inverse image of a Cartan subgroup C' of G', so one is reduced to the case where G is affine. Since one knows that the normalizer of a Cartan subgroup is smooth (C being its own connected normalizer, cf. for example XII 6.6 c)), it suffices to see that C and N have the same k-valued points, which is none other than (*Bible* th. 6 d)).

Definition 4.5. *Let G be a smooth prescheme in groups of finite presentation over a prescheme S. One calls a Borel subgroup of G any subprescheme in groups B of G, smooth of finite presentation, such that for every $s \in S$, the geometric fiber $B_{\bar{s}}$ is a Borel subgroup of $G_{\bar{s}}$.*

This is therefore, as one verifies at once, a notion stable by base change and of local nature for the fpqc topology (for if k' is an algebraically closed extension of an algebraically closed field k, then an algebraic subgroup B of the smooth algebraic group G over k is a Borel subgroup of G if and only if $B_{k'}$ is one of $G_{k'}$). It follows from this definition that if G is a smooth algebraic group over an arbitrary field k, B a Borel subgroup of G, then G/B is a projective variety, every maximal torus T of B is a maximal torus of G, its normalizer in B is identical to its centralizer C, and is a Cartan subgroup of G when G is connected.

Remarks 4.6. Unfortunately, it is no longer true in general (even if G is affine over S and S is the spectrum of the algebra of dual numbers over an algebraically closed field k) that two Borel subgroups of G are conjugate locally for the fpqc topology. As a consequence of this regrettable fact, let us point out that if G is a smooth, affine, connected algebraic group over a non-perfect field k, it is not possible in general to define in a natural way a homogeneous space $\mathcal{D}$ under G, playing the role of a flag variety, i.e. of the variety of Borel subgroups of G (which, over the algebraic closure $\bar{k}$ of k, would therefore be isomorphic to $G_{\bar{k}}/B$, where B is a Borel subgroup of $G_{\bar{k}}$). Indeed, when the quotient of $G_{\bar{k}}$ by its radical R is an adjoint semisimple group, then the kernel of $G_{\bar{k}} \to \mathrm{Aut}_{\bar{k}}(\mathcal{D})$ is the radical of $G_{\bar{k}}$, hence if $\mathcal{D}$ came from a homogeneous space D under G, the radical R would come from a subgroup R of G. Now one easily constructs examples where $G_{\bar{k}}/R$ is adjoint but R is not "defined over k". It is easy to see that under these conditions, the functor $\mathfrak{B} : (Sch)^{\circ}/S \to (Ens)$ such that $\mathfrak{B}(S')$ = set of Borel subgroups of $G_{S'}$ is not representable by a smooth S-prescheme. From the infinitesimal point of view (III § 3), the non-validity of the conjugation theorem is expressed by the fact that if B is a Borel subgroup of the smooth algebraic group G, the cohomology group $H^1(B, \mathfrak{g}/\mathfrak{b})$ [1104] can be different from zero.

We shall see on the other hand in a later exposé that when G is semisimple, or more generally reductive, such unpleasant phenomena do not occur. It is these phenomena no doubt, as well as the absence of good existence theorems, that explain why Borel subgroups play only a relatively effaced role in the study of general group schemes from the scheme-theoretic point of view, while they will dominate the theory of semisimple group schemes in the later exposés.

Proposition 4.8. [1105] *Let G be a smooth S-prescheme in groups of finite presentation with connected fibers, B a Borel subgroup of G; then B is identical to its own normalizer, and is a closed subprescheme of G.*

Indeed, by virtue of (XII 7.10), one is reduced to proving that over an algebraically closed field k, every element of $G(k)$ that normalizes B is in $B(k)$, which for G affine is a fundamental result of Chevalley (*Bible* 9 th. 1); the general case reduces to it at once by the reduction already used in 4.2.

Remark 4.8.1. One can generalize Definition 4.5 by also introducing the notion of parabolic subgroup of G: one calls thus a subprescheme in groups P of G, smooth

[1104] Where $\mathfrak{b}$ is the Lie algebra of B.
[1105] There is no number 4.7.

and of finite presentation over S, such that for every $s \in S$, the geometric fiber P_s is a parabolic subgroup of G_s, i.e. contains a Borel subgroup of G_s. Proposition 4.8 extends (with the same reduction proof to the "set-theoretic" statement, which is known) to the case of a parabolic subgroup P of G. Let us note the following consequence of this result (cf. XVI). If P is a parabolic subgroup of G, then G/P is representable by a quasi-projective prescheme of finite presentation over S (N.B. one assumes G to have connected fibers). Moreover G/P is evidently smooth over S, and moreover with connected and proper geometric fibers, whence one can conclude easily, using (EGA III 5.5.1), that $D = G/P$ is in fact proper, hence projective, over S. Moreover, if its relative dimension is n, it is known that the invertible sheaf $\Omega^n_{D/S}$ is such that its inverse induces on the geometric fibers of D/S ample sheaves, hence (EGA III 4.7.1) $(\Omega^n_{D/S})^{-1}$ is ample on D relative to S.

One sees easily, by reduction to the affine case and to the case of an algebraically closed base field, that if $u : G \to G'$ is an epimorphism of smooth algebraic groups, then for every Borel subgroup B of G, $u(B) = B'$ is a Borel subgroup of G'. We are interested in the case where one obtains in this way a bijective correspondence between Borel subgroups of G and of G':

Proposition 4.9. *Let G, G' be two smooth S-preschemes in groups of finite presentation with connected fibers, $u : G \to G'$ a faithfully flat (i.e. surjective) homomorphism of groups* [1106]. *Suppose one is in one of the following two cases (where one has set $N = Ker u$):*

a) N is central in G.

b) S is the spectrum of a field k, and if $\bar{k}$ denotes an algebraic closure of it, $N_{\bar{k}} = Ker u_{\bar{k}}$ is contained in the radical of $G_{\bar{k}}$, i.e. in the largest smooth connected solvable invariant subgroup of $G_{\bar{k}}$.

Then the map $B' \mapsto u^{-1}(B')$ induces a bijection of the set of Borel subgroups of G' with the analogous set for G.

Case b) follows at once from the correspondence between algebraic subgroups of G' and algebraic subgroups of G containing N, and from the fact that when k is algebraically closed, the Borel subgroups of G contain the radical of G (which is immediate by the reasoning preceding 4.2).

Let us prove case a). By virtue of (XII 7.12), the map $H' \mapsto H = u^{-1}(H')$ establishes a bijective correspondence between subpreschemes in groups H' of G' that are smooth of finite presentation over S, with connected fibers, and that have at every $s \in S$ the same reductive rank and the same nilpotent rank as G', and the subpreschemes in groups H of G having the analogous properties. Now Borel subgroups (of G', or of G) have the properties in question. It remains to prove that if H', H correspond, then H' is a Borel subgroup of G' if and only if H is one of G. By definition, this question reduces to the case where S is the spectrum of an algebraically closed field. Now since N is central in G hence in H, it follows that H is solvable if and only if H' is. Finally, taking the correspondence between algebraic

[1106] Add a reference here?

subgroups of G' and algebraic subgroups of G containing N into account, one sees at once that H' has the maximal character of Definition 4.1 if and only if H does, which completes the proof.

Corollary 4.10. *With the notations of 4.7, if B' and B are Borel subgroups of G' and G which correspond, one has*

$$\mathfrak{b} = Lie(u)^{-1}(\mathfrak{b}')$$

where $\mathfrak{g}$, $\mathfrak{g}'$, $\mathfrak{b}$, $\mathfrak{b}'$ are the Lie algebras of G, G', B, B', and where $Lie(u) : \mathfrak{g} \to \mathfrak{g}'$ is the homomorphism deduced from u.

This statement follows trivially from the definitions and from the relation $u^{-1}(B') = B$.

We can now prove the principal result of the present section:

Theorem 4.11. *Let G be a smooth algebraic group over an algebraically closed field k, $\mathfrak{g}$ its Lie algebra. Then $\mathfrak{g}$ is equal to the union of the Lie algebras $\mathfrak{b}$ of the Borel subgroups B of G.*

One may evidently suppose G connected. Let R be the radical of G and let $G' = G/R$. Then 4.9 b) and 4.10 reduce us to proving Theorem 4.11 for G' instead of G, i.e. one may suppose G semisimple. Let then Z be the center of G, identical to the reductive center, and let $G' = G/Z$. The same reasoning (now using 4.9 a)) reduces us to proving the theorem for G', i.e. one may suppose G semisimple adjoint. Let B be a Borel of G, T a maximal torus of B hence of G, and let $\mathfrak{b}$ and $\mathfrak{t}$ be the Lie algebras. By virtue of 3.18, T is a subgroup of type (C) of G, i.e. $\mathfrak{t}$ is a Cartan subalgebra of $\mathfrak{g}$, hence the union of the conjugates of $\mathfrak{t}$ is dense in $\mathfrak{g}$ (XIII 5.1 (i) ⇒ (vii)). *A fortiori* the union of the conjugates of $\mathfrak{b}$ is dense in $\mathfrak{g}$. Now let X be the closed subscheme of $G/B \times W(\mathfrak{g})$ whose k-valued points are the (g', x) such that $x \in Ad(g) \cdot \mathfrak{b}$ (cf. XIII 1). Then the morphism $\psi : X \to W(\mathfrak{g})$ induced by the second projection is proper since G/B is proper over k; on the other hand we have just seen that it is dominant, hence it is surjective, which proves 4.11.

The only result of the present section that we shall use in the rest of this exposé is the following corollary:

Corollary 4.12. *Let k be an infinite field, G a smooth algebraic group over k, T a maximal torus of G, $\mathfrak{g}$ and $\mathfrak{t}$ the Lie algebras, $u : G \to GL(V)$ a linear representation (V a finite-dimensional vector space over k), whence a representation of Lie algebras $u' : \mathfrak{g} \to \mathfrak{gl}(V)$, making V into a $\mathfrak{g}$-module. Then the minimum of the nullity of $u'(x)$ ($x \in \mathfrak{g}$) is attained for an element $x \in \mathfrak{t}$.*

One is reduced at once to the case where k is algebraically closed. One may evidently suppose G connected, and replacing G by $G/(Keru)_{red}$ if necessary, one may suppose G affine. Using 4.11 and the conjugation theorem of maximal tori of G, one is reduced to the case where G is moreover solvable. Then G is a semidirect product $T \cdot V$, where V is the "unipotent part" of G, which is a smooth connected unipotent group (*Bible* 6 th. 3). Hence $\mathfrak{g}$ is the direct sum (as a vector space) of the Lie subalgebras $\mathfrak{t}$ and $\mathfrak{v} = Lie(V)$. By virtue of the Lie-Kolchin theorem (*Bible* 6 th. 1), $\mathfrak{v}$ admits a composition series by stable subspaces $\mathfrak{v}_i$ such that $\mathfrak{v}_i/\mathfrak{v}_{i+1} = w_i$ is of dimension 1. Then for each i, one has an induced representation $u_i : G \to GL(w_i) = G_m$ and the corresponding homomorphism of Lie algebras $u_i' : \mathfrak{g} \to k$, so that for every $x \in \mathfrak{g}$, the nullity of $u'(x)$ is equal to the number of i such that $u_i'(x) = 0$. Since V is unipotent, the u_i are trivial on V, hence the u_i' are trivial on $\mathfrak{v}$, which proves that if $x = t + v$ ($t \in \mathfrak{t}$, $v \in \mathfrak{v}$), then $u_i'(x) = u_i'(t)$ for all i, hence the nullity of $u'(x)$ is equal to that of $u'(t)$. Assertion 4.12 follows at once.

5. Relations between Cartan subgroups and Cartan subalgebras

Applying 4.12 to the adjoint representation of G, one finds:

Theorem 5.1. *Let G be a smooth algebraic group over an infinite field, T a maximal torus of G, $\mathfrak{g} \supset \mathfrak{t}$ the Lie algebras; then $\mathfrak{t}$ contains a regular element of $\mathfrak{g}$.*

Corollary 5.2. *Let G be a smooth S-prescheme in groups of finite presentation, $\mathfrak{g}$ its Lie algebra.*

a) Conditions (C_0) to (C_2) of 2.9 on $\mathfrak{g}$ are equivalent; in particular, if the infinitesimal rank of the fibers of $\mathfrak{g}$ at the points of S is locally constant, then $\mathfrak{g}$ admits locally for the étale topology a Cartan subalgebra, hence (by 3.9 a)) G admits locally for the étale topology a subgroup of type (C).

b) Let H be a subprescheme in groups of G smooth of finite presentation over S, with connected fibers having the same reductive rank as G at each $s \in S$ (for example, H is a maximal torus or a Cartan subgroup of G); let D be a subgroup of type (C) of G; then one has $H \subset D$ if and only if one has $\mathfrak{h} \subset \mathfrak{d}$.

c) Suppose condition (C_0) is satisfied, i.e. the infinitesimal rank of G is locally constant. Let H be a subprescheme in groups of G smooth of finite presentation over S, with connected nilpotent fibers having the same reductive rank as G at each point (for example, H is a maximal torus or a Cartan subgroup of G). Then H is contained in one and only one subgroup D of type (C) of G.

d) Suppose that G admits locally for fpqc a Cartan subgroup; then the same is true of every subgroup D of type (C) of G.

Proof. a) Suppose condition (C_0) is satisfied, and let us prove that the same is true of (C_2), i.e. that every quasi-regular section a of $\mathfrak{g}$ is regular. One reduces as usual to the case S affine noetherian, then, the question being “infinitesimal” (2.15 b)), to the case where S is local artinian (in which case (C_0) is moreover trivially satisfied). One may suppose moreover that the residue field k of S is infinite. Note that by virtue of 3.7 it suffices to establish that $\mathfrak{g}$ admits a Cartan subalgebra. Let T be a maximal torus of G; by virtue of 5.1 there exists a quasi-regular element t contained in the Lie algebra $\mathfrak{t}$ of T; let us prove that it is regular, which will complete the proof. Consider the linear representation of T in $\mathfrak{g}$ induced by the adjoint representation of G; there exists therefore a finite set $(u_i)_{i\in I}$ of characters of T, such that $\mathfrak{g}$ decomposes as a direct sum of submodules $\mathfrak{g}_i$ stable under T, T operating on $\mathfrak{g}_i$ by u_i (cf. I § 4.7.3). Let $u'_i : \mathfrak{t} \to A$ be the homomorphism of Lie algebras deduced from $u_i : T \to G_m$ (N.B. A denotes the ring of S). Consider the homomorphisms u_{i0} and u'_{i0} deduced from the preceding by passage to the fibers, i.e. by the base change $A \to k$. Let I' be the set of $i \in I$ such that $u'_{i0} \neq 0$, and let $I'' = I - I'$. The fact that t is regular is expressed by the condition $u'_{i0}(t_0) \neq 0$ for all $i \in I'$, hence $u'_i(t)$ invertible for $i \in I'$. The Cartan subalgebra of $\mathfrak{g}_0$ defined by t, i.e. the kernel subspace of the semisimple endomorphism $ad(t_0)$ of $\mathfrak{g}_0$, is $\sum_{i\in I''}(\mathfrak{g}_i)_0$. Consider

$$\mathfrak{d} = \sum_{i \in I''} \mathfrak{g}_i;$$

then $ad(t)$ is nilpotent in $\mathfrak{d}$; on the other hand it is an automorphism of $\mathfrak{g}/\mathfrak{d} \simeq \sum_{i\in I'} \mathfrak{g}_i$. By virtue of 2.6, t is therefore regular.

b) Follows from 3.12 a), H satisfying the hypothesis that every geometric fiber of $\mathfrak{h}$ contains a regular element of that of $\mathfrak{g}$, thanks to 5.1.

c) By virtue of b), one is reduced to proving that $\mathfrak{h}$ is contained in one and only one Cartan subalgebra of $\mathfrak{d}$. By virtue of a) we know moreover that $\mathfrak{g}$ satisfies (C_2). One is thus reduced to the:

Lemma 5.3. *Let $\mathfrak{g}$ be a Lie algebra over S that is a locally free module of finite type and satisfies condition (C_2) of 2.9. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ satisfying the following conditions: it is a locally direct factor module, it is locally nilpotent, and for every $s \in S$, the geometric fiber $\mathfrak{h}_{\bar{s}}$ contains a regular element of $\mathfrak{g}_{\bar{s}}$. Then $\mathfrak{h}$ is contained in one and only one Cartan subalgebra of $\mathfrak{g}$.*

(N.B. in the case of interest to us, $\mathfrak{h}$ satisfies the stated conditions: it is locally nilpotent because H has nilpotent fibers, and the condition on regular elements follows from 5.1.) Since 5.3 is local for the fpqc topology, it suffices to prove that at a point $s \in S$ such that $\kappa(s)$ is infinite, there exists an open neighborhood U of s such that existence and uniqueness hold for every base change $S' \to S$ factoring through U. Take a regular element of $\mathfrak{g} \otimes \kappa(s)$ contained in $\mathfrak{h} \otimes \kappa(s)$, extend it to a section a of $\mathfrak{h}$ on an open neighborhood U of s; thanks to (C_2) one may suppose this section is regular (2.10). One may suppose $U = S$. A Cartan subalgebra of $\mathfrak{g}$ containing $\mathfrak{h}$ contains a, hence is identical to $\mathfrak{d} = Nil(a, \mathfrak{g})$ (2.6), whence uniqueness. Moreover since $\mathfrak{h}$ is locally nilpotent, one indeed has $\mathfrak{h} \subset \mathfrak{d}$, which proves existence.

d) Is a particular case of (XII 7.9 d)).

Corollary 5.4. *Let G be a smooth and connected algebraic group over an algebraically closed field k, H a connected algebraic subgroup such that $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$, i.e. such that H contains a subgroup of type (C) of G. Then the number of conjugates of H containing a regular element of $G(k)$ is equal to the number of conjugates of $\mathfrak{h}$ containing a regular element of $\mathfrak{g}$.*

Indeed, let g be a regular element of $G(k)$, C the unique Cartan subgroup of G containing g (XIII 2); then the conjugates of H containing g are those containing

C (XIII 2.8 b)). Similarly, let x be a regular element of $\mathfrak{g}$; then if a conjugate $\mathfrak{h}'$ of $\mathfrak{h}$ contains x, then $ad(x)_{\mathfrak{g}/\mathfrak{h}'}$ is injective (XIII 5.4), hence $\mathfrak{h}'$ contains $Nil(x, \mathfrak{g}) = \mathfrak{d}$, hence the number of conjugates of $\mathfrak{h}$ containing x is equal to the number of conjugates containing the Cartan subalgebra $\mathfrak{d}$. Moreover $\mathfrak{d}$ is the Lie algebra of a subgroup D of type (C) of G. One may evidently suppose $C \subset D$, and assertion 5.4 will follow from this: in order for H to contain C, it is necessary and sufficient that $\mathfrak{h}$ contain $\mathfrak{d}$. Indeed, by virtue of (XIII 5.5) the relation $\mathfrak{h} \supset \mathfrak{d}$ implies $H \supset D$ and *a fortiori* $H \supset C$. Conversely, by 5.1 $\mathfrak{t}$ contains a regular element x of $\mathfrak{g}$; then $H \supset C$ implies $\mathfrak{h} \ni x$, hence as already pointed out, this implies $\mathfrak{h} \supset \mathfrak{d}$. This completes the proof.

Let G be as in 5.2 and suppose that the infinitesimal rank of the fibers of G remains locally constant (condition (C_0)). Then thanks to 5.2 c), one finds a homomorphism of functors on $(Sch)^{\circ}/S$:

$$\mathcal{C} \longrightarrow \mathcal{D}$$

where

$\mathcal{C}(S') =$ set of Cartan subgroups of $G_{S'}$
$\mathcal{D}(S') =$ set of subgroups of type (C) of $G_{S'}$.

By virtue of 3.10 and 5.2 a), $\mathcal{D}$ is representable by a smooth and quasi-projective prescheme over S. Consider the subgroup D of type (C) of $G_{\mathcal{D}}$ playing the universal role with respect to G/S; one may then consider the functor $\mathcal{C}_D : (Sch)^{\circ}/\mathcal{D} \to (Ens)$ defined in terms of the $\mathcal{D}$-group D as $\mathcal{C}$ in terms of the S-group G. One then has the following result:

Proposition 5.5. *Under the preceding conditions, considering $\mathcal{C}$ as a functor over the prescheme $\mathcal{D}$, then $\mathcal{C}$ is $\mathcal{D}$-isomorphic to the functor $\mathcal{C}_D$ "of Cartan subgroups of D".*

This follows at once from the:

Corollary 5.6. *Let G be a smooth S-prescheme in groups of finite presentation, D a subgroup of type (C) of G. Then there is a bijective correspondence between Cartan subgroups of G contained in D, and Cartan subgroups of D (more precisely, for a subgroup H of G, H is a Cartan subgroup of G if and only if it is a Cartan subgroup of D).*

Indeed, this is a particular case of (XII 7.9 c)), taking into account that over an algebraically closed field, a subgroup of type (C) of G contains a Cartan subgroup of G.

For the following section, the main result obtained here is 5.2 c) for H a Cartan subgroup of G, which permits stating 5.5 and thus furnishes a useful "dévissage" of $\mathcal{C}$.

6. Applications to the structure of algebraic groups

Theorem 6.1. *Let G be a smooth algebraic group over a field k. Consider the scheme $\mathcal{T}$ of maximal tori of G, isomorphic to the scheme $\mathcal{C}$ of Cartan subgroups of G, which is a homogeneous space under G^0, and a smooth affine connected algebraic scheme (XII 7.1 d)). Then $\mathcal{C}$ is a rational variety, i.e. the field of rational functions of $\mathcal{C}$ is a pure extension of k.*

We shall first give the proof in the case where k is infinite. One may evidently suppose G connected, since $\mathcal{T}$ hence $\mathcal{C}$ does not change on replacing G by G^0. Moreover, by virtue of (XII 7.6), $\mathcal{C}$ does not change on dividing G by a central subgroup. This allows us, first dividing by the center of G, to suppose G affine (XII 6.1), then, dividing by its reductive center (XII 4.1 and 4.4), to suppose that the reductive center of G is trivial (XII 4.7 b)). Moreover we proceed by induction on $n = \dim G$, supposing the theorem proved for dimensions $n' < n$. If G is nilpotent, then $\mathcal{C}$ is reduced to a single point rational over k, and 6.1 is trivial. In the contrary case, the Lie algebra of G is non-nilpotent (1.3), hence the Cartan subalgebras of $\mathfrak{g}$ are of dimension $n' < n$, hence the subgroups of type (C) of G are of dimension $n' < n$. Consider then the morphism

$$\mathcal{C} \longrightarrow \mathcal{D}$$

envisaged in 5.5. We know by 3.10 (k being an infinite field, hence $\mathfrak{g}$ containing a regular element) that $\mathcal{D}$ is a rational variety, i.e. the field K of rational functions on $\mathcal{D}$ is a pure extension of k. Consider the fiber of $\mathcal{C}$ over the generic point x of $\mathcal{D}$; by virtue of 5.5 this is the scheme of Cartan subgroups of a certain smooth and connected algebraic group D_x over $K = \kappa(x)$ (namely $D_x =$ "the generic subgroup of type (C) of G"). The field L of rational functions on $\mathcal{C}$ is therefore isomorphic to the field of rational functions on $\mathcal{C}_{D_x}$, which by the induction hypothesis (since $\dim D_x = n' < n$) is a pure extension of K. So by transitivity L is a pure extension of k.

When k is finite, a different proof is needed. One may still suppose G is affine and connected. Note that k is perfect; it follows at once that the radical R of $G_{\bar{k}}$ is "defined over k", i.e. comes from a subgroup R of G. Suppose first $R \neq G$, i.e. G non-solvable, and let

$u : G \to G' = G/R$

be the canonical morphism. Consider the corresponding morphism $C \mapsto u(C)$

$$v : \mathcal{C}_G \longrightarrow \mathcal{C}_{G'}$$

(whose definition is immediate by virtue of (XII 7.1 e))). Let x be the generic point of $\mathcal{C}_{G'}$; then the fiber $v^{-1}(x)$ is identified with the scheme of Cartan subgroups of G_K (where $K = \kappa(x)$) whose image in G'_K is a certain Cartan subgroup C'_x (namely, "the generic Cartan subgroup of G'"). It is therefore also the scheme of Cartan subgroups of $H = u_K^{-1}(C')$ (XII 7.9 c)), and since K is here an infinite extension of k, it follows from the already proved part of 7.1 that the field L of rational functions of $\mathcal{C}_G$, equal to that of $\mathcal{C}_H$, is a pure transcendental extension of K. To prove that it is a pure transcendental extension of k, it therefore suffices to prove that this is the case of K, i.e. one is reduced to the case where G is semisimple. One may moreover suppose that G is adjoint (dividing G by its reductive center if necessary). But then by virtue of 3.18 one has $\mathcal{C}_G \simeq \mathcal{D}_G$, and by virtue of 3.10 it suffices to prove that $\mathfrak{g}$ admits a regular point, which (as was pointed out in 3.20) is an unpublished result of Chevalley [1107].

It only remains to treat the case where k is finite, G connected affine solvable. One in fact has a more general result:

Corollary 6.2. *Let G be a smooth solvable algebraic group over a field k; then the variety $\mathcal{C}$ of Cartan subgroups of G is isomorphic to an affine space* Spec $k[t_1, \cdots, t_N]$.

One may still suppose G connected and affine. Let G_∞ be the smallest of the groups appearing in the descending central series of G (by G_i such that $G_{i+1} = [G, G_i]$); it is thus the smallest invariant algebraic subgroup of G such that G/G_∞ is nilpotent. Let C be a Cartan subgroup of G (it exists by virtue of 1.1); then the image of C in G/G_∞ is a Cartan subgroup of it, hence is equal to G/G_∞, consequently the morphism $g \mapsto Ad(g) \cdot C$ of G_∞ into $\mathcal{C}$ is an epimorphism, and identifies $\mathcal{C}$ with the homogeneous space $G_\infty/N \cap G_\infty$, where N is the normalizer of C in G (moreover equal to C as we recalled in 4.4, but this is of little importance here). Note that $U = G_\infty$ is evidently a smooth connected "unipotent" algebraic group (since over the algebraic closure of k, it is contained in the unipotent part of G, by virtue of the known structure of smooth affine solvable groups, *Bible* 6 th. 3). When k is perfect (the only case needed to establish 6.1), it follows very easily that U is even k-unipotent, i.e. admits a composition series by algebraic subgroups U_i such that U_i/U_{i+1} is isomorphic to G_a. In fact, Rosenlicht has proved that this result remains valid for a group of the form G_∞ as above, without restriction on k (M. Rosenlicht, *Questions of rationality for solvable algebraic groups over non perfect fields*, Annali di Matematica 1963, pp. 97–120, theorem 4 cor. 2), a result clearly more delicate which we shall admit here. It now suffices to apply the following lemma, doubtless well known to specialists:

Lemma 6.3. *Let U be a smooth connected algebraic group over a field k, $X = U/V$ a homogeneous space under U having a rational point over k. Suppose U is k-unipotent. Then as a k-scheme, X is isomorphic to an affine space* Spec $k[t_1, \cdots, t_N]$.

Indeed, let $(U_i)_{0 \leqslant i \leqslant n}$ be a composition series of U by smooth connected subgroups, with $U_n = (e)$, $U_0 = U$, $U_i/U_{i+1} \simeq G_a$, the U_i invariant in U. Then the $K_i = U_i V$ are algebraic subgroups of U (not necessarily smooth nor connected if V is not, but no matter), and K_{i+1} is invariant in K_i. One has a canonical morphism $U_i/U_{i+1} \to K_i/K_{i+1}$ that is an epimorphism, which proves that K_i/K_{i+1} is either reduced to the unit group, or isomorphic to G_a/H_i, where H_i is a finite subgroup of G_a, which by virtue of Rosenlicht (loc. cit., th. 2) implies that $K_i/K_{i+1} \simeq G_a$ (a result moreover

[1107] Cf. the Appendix, by J.-P. Serre.

immediate if k is perfect). Set now $X_i = U/K_i$; let us prove by induction on i that X_i is isomorphic to an affine space. Indeed, if this is so for X_i, let us prove that the same is true of X_{i+1}. If $K_i/K_{i+1} = e$ one has $X_i = X_{i+1}$ and this is trivial. Otherwise, X_{i+1} is a principal bundle with base X_i and structure group $G_a \simeq K_i/K_{i+1}$. So, X_i being affine, X_{i+1} is a trivial bundle, hence is isomorphic to $X_i \times G_a$, which again proves that X_{i+1} is isomorphic to an affine space. This proves 6.2 and consequently completes the proof of 6.1.

Corollary 6.4. *Let G be a smooth algebraic group over an infinite field k. Then the set of k-rational points of $\mathcal{C}$ (notations of 6.1) is dense for the Zariski topology. The union of the Cartan subgroups of G is dense in G.*

The first assertion is valid for any unirational variety over an infinite field, and is moreover the most important "arithmetic" consequence of the unirationality results for us here. The second assertion follows from the first and from the density theorem (XIII 2.1).

Corollary 6.5. *Let G be a smooth connected algebraic group over k. Then the variety Z of regular semisimple points of G (XIII 3.5) is a unirational variety. In particular, if k is infinite, the set of k-rational points of Z is dense in Z.*

Indeed, Z is an open of a torus over $\mathcal{C}$, hence its function field L is the function field of a torus defined over the function field K of $\mathcal{C}$ (namely the "generic maximal torus" of G); it is therefore a unirational extension of K (XIII 3.4), and since K is a pure extension of k by virtue of 6.1, L is a unirational extension of k.

Corollary 6.6. *Let G be a smooth connected algebraic group over k, and let H be the subgroup of G generated by the subvariety Z of regular semisimple points, i.e. (XII 8.2) the smallest invariant algebraic subgroup of G such that G/H has reductive rank zero (i.e. is, over the algebraic closure of k, an extension of an abelian variety by a smooth connected unipotent group). Then H is a unirational variety. In particular, if $G = H$, i.e. (XII 8.4) if G is affine and if over the algebraic closure $\bar{k}$, there exists no non-trivial homomorphism of $G_{\bar{k}}$ to G_a, then G is a unirational variety, hence if k is infinite, the set of its k-rational points is dense.*

This follows at once from 6.5, for it is immediate that if one has smooth connected k-preschemes Z_i that are unirational varieties and morphisms $u_i : Z_i \to G$, then the algebraic subgroup of G generated by the u_i (VI_B 7.1) is a unirational variety.

As an interesting particular case of 6.5 or 6.6 (whichever one chooses), let us note:

Corollary 6.7. *Let G be a smooth connected affine algebraic group of unipotent rank zero; then G is a unirational variety.*

One can refine 6.4 as follows:

Corollary 6.8. *Let G be a smooth connected algebraic group over the infinite field k, and let C be a Cartan subgroup of G. Then the union of the conjugates of C by regular semisimple elements of $G(k)$ is dense in G.*

This follows at once from 6.4 and from (XIII 3.6), which says that the morphism

$\varphi : Z \times C \to G$ defined by $\varphi(t,c) = ad(t)c$ is dominant. This result also implies (without supposing k infinite):

Corollary 6.9. *Let G be a smooth connected algebraic group over k, H a smooth connected algebraic subgroup of G such that H has the same reductive rank and the same nilpotent rank as G (i.e. over the algebraic closure $\bar{k}$ of k, $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$). If H is a unirational variety, the same is true of G. If $H(k)$ is dense in H, $G(k)$ is dense in G.*

Indeed, the morphism $\varphi : Z \times H \to G$ defined by $\varphi(t,h) = ad(t)h$ is dominant. Now by virtue of 6.5, Z is a unirational variety, and by hypothesis the same is true of H, hence of $Z \times H$, whence the first result. The second is proved analogously.

We now recover the following well-known result (due to Chevalley, in characteristic 0, to Rosenlicht in characteristic $p > 0$):

Corollary 6.10. *Let G be a smooth connected affine algebraic group over a perfect field k. Then G is a unirational variety, hence if k is moreover infinite, $G(k)$ is dense in G.*

Indeed, by virtue of 1.1, G admits a Cartan subgroup C. By virtue of 6.9 it suffices to prove that this latter is a unirational variety. Now k being perfect, one sees at once by Galois descent from the case k algebraically closed (*Bible* 6 th. 2) that one has $C = T \times C_u$, where T is the maximal torus of C and C_u a smooth connected unipotent group. One already knows that T is a unirational variety (XIII 3.4); it remains to see that the same is true of C_u. Now k being perfect, C_u is even k-unipotent, and one may apply 6.3.

Remarks 6.11. a) One knows (Rosenlicht) examples of twisted forms of G_a over a non-perfect field that have only finitely many rational points, hence *a fortiori* are not unirational varieties. On the other hand, Chevalley has given an example of a torus over a field of characteristic zero that is not a rational variety. By contrast, it follows from Chevalley's theory of semisimple groups that over an algebraically closed field, every smooth connected affine algebraic group is a rational variety. Let us note moreover that the question of unirationality only arises in any event for affine algebraic groups, a unirational algebraic group being necessarily affine by Chevalley's structure theorem.

b) With the notations of 6.6, it is tempting to try to give a unirationality condition on G in terms of the group G/H (which is unipotent if G is supposed affine). It is evidently necessary that this latter be unirational; is this condition also sufficient? Note that an example of Rosenlicht (loc. cit.) shows that a smooth connected unipotent algebraic group U may be a rational variety without being k-unipotent.

c) It would be interesting to study, over a finite field k, questions of the "density" type, such as the following (raised by Rosenlicht): Let G be a smooth and connected algebraic group over k; is G generated by its Cartan subgroups?

[1108] [1109]

In this question, one can reduce to the case G affine, by dividing by the center. The answer would be affirmative in the semisimple case, if one could refine the existence result for regular points of Chevalley pointed out in 3.20, so as to obtain a regular element of $\mathfrak{g}$ that does not belong to $\mathfrak{h} = Lie(H)$, where H is a smooth algebraic subgroup of G and $\neq G$ given in advance (G an adjoint semisimple group over the finite field k).

Appendix. Existence of regular elements over finite fields

by J.-P. Serre

In all that follows, k denotes a finite field, and $\bar{k}$ its algebraic closure; the Galois group of $\bar{k}/k$ is denoted $\mathcal{G}$; one recalls that, if $q = Card(k)$, $\mathcal{G}$ is topologically generated by the "Frobenius" element $F : x \mapsto x^q$.

We propose to prove the following theorem [1110]:

Theorem. *Let G be an adjoint semisimple group defined over k, and let $\mathfrak{g}$ be its Lie algebra. The k-Lie algebra $\mathfrak{g}(k)$ contains a regular element.*

Remarks. (1) It is well to recall that "adjoint" means that the center of G is trivial (as a subgroup scheme of G). In view of the Chevalley seminar, this also means that, if T is a maximal torus of G, the group $X(T)$ of characters of T (defined over $\bar{k}$) is generated by the set R of roots. It follows in particular that the rank of the Lie algebra $\mathfrak{g}$ is equal to the dimension of T (i.e. to the rank of G).

[1108] This question has since been resolved affirmatively by Steinberg.

[1109] Not having identified this result in the *Collected Papers* of R. Steinberg, we give a proof based on the Bruhat decomposition. Let G/k be a semisimple group defined over the finite field k. The point is to show that G is generated by its k-Cartan subgroups. This question is stable by central isogeny and by restriction of scalars; one is therefore reduced to the case where G is geometrically simple (in the same way as in Lemma 1 of the appendix below). Denote by $G^{\sharp}$ the subgroup of G generated by its k-subtori. The group G is quasi-split (Lang) and therefore admits a Killing couple (T, B). By definition, one has $T \subset G^{\sharp}$; in particular T normalizes $G^{\sharp}$. The big cell $R_u(B^-)TR_u(B)$ of G indicates that it suffices to verify that $R_u(B) \subset G^{\sharp}$. Let S be the maximal split k-torus of T. One considers the relative root system $\Phi(G,S)$ and a basis Δ_k. Given $\alpha \in \Phi(G,S)$, denote by U_α the unipotent subgroup associated to α (cf. A. Borel, *Linear Algebraic Groups*, second edition (1991), Springer, Prop. 21.9). Since the k-group $R_u(B)$ is generated by the k-unipotent groups U_α ($\alpha \in \Delta_k$), one is reduced to verifying that $U_\alpha \subset G^{\sharp}$. A glance at the classification shows that there exists a semisimple group G_α of quasi-split type A_1, ${}^2A_1$, ${}^3A_1$ or ${}^2A_2$ such that $U_\alpha \subset G_\alpha \subset G$. It is therefore permissible to suppose that $G = PGL_2$ or $G = SU_3(K)$, where K denotes the unique quadratic extension field of k. The group $PGL_2^{\sharp}$ containing the standard split torus T, the possibilities up to conjugation under $G(k)$ are the following: $PGL_2^{\sharp} = T$, $PGL_2^{\sharp} = B$, or $PGL_2^{\sharp} = PGL_2$. The case $PGL_2^{\sharp} = T$ is excluded since $PGL_2^{\sharp}$ contains the k-torus $R_{K/k}(G_m)/G_m$. The preceding discussion indicates that if $PGL_2^{\sharp} = B$, then $PGL_2^{\sharp} = PGL_2$. Hence $PGL_2^{\sharp} = PGL_2$. If $G = SU_3(K)$, $\Delta_k = \alpha$ and the possibilities for $G^{\sharp}$ (up to conjugation) are the following: $G^{\sharp} = T$, $G^{\sharp} = B$, $G^{\sharp} = U_{2\alpha} \rtimes T$, $G^{\sharp} = \langle U_{2\alpha}, U_{-2\alpha}, T\rangle = SL_2 \cdot T$, or $G^{\sharp} = G$. The case $G^{\sharp} = B$ is excluded as for PGL_2. The case $SL_2 \cdot T$ is excluded because $G^{\sharp}$ contains the k-torus $R^1_{K/k}(R^1_{K_3/K}(G_m))$, K_3/K denoting the extension of degree 3 of K, and the isogeny class of this torus is irreducible. Moreover, this torus also excludes the cases $G^{\sharp} = T$ and $G^{\sharp} = U_{2\alpha} \rtimes T$. One concludes that $G^{\sharp} = G$.

[1110] This theorem is due to Chevalley. The editor wishes to express his gratitude to American Express which, by misplacing a trunk of Chevalley manuscripts, obliged him to reconstruct the proof.

(2) The editor does not know whether the hypothesis that G is adjoint is indispensable.

Lemma 1. *It suffices to prove the theorem when G is geometrically simple.*

(One says that G is geometrically simple if $G \otimes \bar{k}$ does not contain any normal subgroup scheme smooth over $\bar{k}$, apart from G and e; an equivalent condition: the associated root system R is irreducible.)

One may first suppose G indecomposable over k. The group $G \otimes \bar{k}$ is then a product of geometrically simple components S that are permuted transitively by the Galois group $\mathcal{G}$. If H is the subgroup of $\mathcal{G}$ fixing one of these components S, this component is defined over the field K corresponding to H (i.e. comes by extension of scalars from a subscheme of $G \otimes K$), and a standard argument shows that $G = \prod_{K/k}(S)$ (i.e. $R_{K/k}(S)$, for readers used to the notation of Weil). Likewise, the Lie algebra $\mathfrak{g}$ of G is identified with $\prod_{K/k} \mathfrak{s}$, where $\mathfrak{s}$ is the Lie algebra of S. If the theorem is true for S, there exists $x \in \mathfrak{s}(K) = \mathfrak{g}(k)$ that is regular in $\mathfrak{s}$; one then verifies easily that it is regular in $\mathfrak{g}$. *QED*.

From now on, one supposes G geometrically simple and one chooses a maximal torus T of G (one knows this is possible). One denotes by X the group of characters of $T \otimes \bar{k}$, R its root system, W its Weyl group, and E the group of automorphisms of X preserving R. The group W is a normal subgroup of E.

If T' is another torus of G, one denotes by X', R', W', E' the group of characters, root system, ..., corresponding. If one chooses $y \in G(\bar{k})$ such that $y \cdot (T \otimes \bar{k}) \cdot y^{-1} = T' \otimes \bar{k}$, one can identify X', R', W', E' with X, R, W, E thanks to $int(y)$. Changing y modifies this identification by an automorphism of X corresponding to an element of W.

The canonical generator $x \mapsto x^q$ of $\mathcal{G}$ operates on X' while preserving R'; it therefore defines an element $f_{T'}$ of E'. In particular one sets $f = f_T$. When one identifies E' with E as just stated, the element $f_{T'}$ is transformed into an element f' of E, defined up to replacement by $wf'w^{-1}$ with $w \in W$. We are going to compare this element to f:

Lemma 2. *One has $f' \equiv f mod W$; conversely, every $f' \in E$ satisfying this condition can be obtained from a maximal torus T' of G.*

Set $\bar{T} = T \otimes \bar{k}$, $\bar{T}' = T' \otimes \bar{k}$, and let $y \in G(\bar{k})$ be such that $y\bar{T}y^{-1} = \bar{T}'$; since $y^q\bar{T}y^{-q} = \bar{T}'$, one concludes that $n = y^{-1}y^q$ belongs to the normalizer $N(T)$ of T. The effect of f' on the points of $\bar{T}(\bar{k})$ is then the following:

$$f'(t) = y^{-1}(yty^{-1})^q y = y^{-1} y^q t^q y^{-q} y = n t^q n^{-1}.$$

If $w \in W$ is the element defined by n, this shows that f' and $f \circ w$ have the same effect on the points of $\bar{T}$, hence also on its characters, and one has $f' = f \circ w$, whence $f' \equiv f mod W$. Conversely, if $w \in W$ is given, one represents it by an element $n \in N(T)(\bar{k})$; thanks to a classical theorem of Lang, one can write n in the form $n = y^{-1}y^q$, with $y \in G(\bar{k})$; the torus $\bar{T}' = y\bar{T}y^{-1}$ is then defined over k, and the preceding calculation shows that the corresponding f' is equal to $f \circ w$.

Lemma 3. *Let X, R, W, E be as above (R being irreducible), and let $\varphi \in E/W$. Then there exists an element $f \in E$ representing φ, and a family $\theta_1, \cdots, \theta_n$ of roots enjoying the two following properties:*

(1) $(\theta_1, \cdots, \theta_n)$ is a basis of X.

(2) R is the union of the orbits of the θ_i under the powers of f (i.e. every $a \in R$ can be written $a = f^m\theta_i$, with suitable m and i).

The proof will be given a little further on.

Here now is a lemma of linear algebra:

Lemma 4. *Let V be a vector space over k, and let $(\theta_1, \cdots, \theta_n)$ be a basis of the dual of $V \otimes \bar{k}$. There exists $x \in V$ such that $\theta_i(x) \neq 0$ for all i.*

Let $V*$ be the dual of V, and let W_i be the subspace of $V * \otimes \bar{k}$ generated by the conjugates of θ_i; this subspace is defined over k, i.e. of the form $W_i \otimes \bar{k}$. The obvious application $W_1 \otimes \cdots \otimes W_n \to \wedge^n V*$ is not zero (otherwise its extension to $\bar{k}$ would be also, which is absurd since $\theta_i \in W_i$ and $\theta_1 \wedge \cdots \wedge \theta_n \neq 0$). There exists therefore a basis of $V*$ formed of elements $u_i \in W_i$. Let $x \in V$ be such that $u_i(x) \neq 0$ for all i (for example $u_i(x) = 1$). The element x answers the question, for if one had $\theta_i(x) = 0$, the conjugates of θ_i would also vanish at x, and the same would be true of u_i, which is not the case.

End of the proof of the theorem.

By combining Lemmas 2 and 3, one can choose a torus T whose element f satisfies the properties of Lemma 3. If Y is the dual of X, the Lie algebra $\mathfrak{t}(\bar{k})$ of T [1111] is canonically identified with $Y \otimes \bar{k}$, and this operation is compatible with the action of the Galois group (the latter operating on $Y \otimes \bar{k}$ thanks to its action on Y and on $\bar{k}$). Let $V = \mathfrak{t}(k)$ be the Lie algebra of T over k. An element $x \in V$ is regular if and only if it is not annihilated by any root $\alpha \in R$, or rather by any of the linear forms $\alpha \in V * \otimes \bar{k}$ defined canonically by the $\alpha \in R$. By Lemma 4, one can find such an x not annihilated by any of the roots θ_i; but every root is conjugate to a θ_i (this is what condition (2) of Lemma 3 expresses); it follows that x is not annihilated by any root, and it is indeed a regular element.

Proof of Lemma 3.

It relies on properties of Coxeter transformations. Let us briefly recall what this involves:

Let $a_1, \cdots, a_n$ be a simple root system of R, and, for every i, let r_i be the symmetry corresponding to a_i. Set:

$$c = r_1 \cdots r_n.$$

[1111] One has changed $\bar{T}$ to T.

One has $c \in W$; of course the element c depends on the choice of the simple system $(a_1, \cdots, a_n)$ as well as on the order of the a_i; however one shows that its conjugacy class does not depend on any of these choices. One calls c the *Coxeter element* of the system considered. One shows (we shall admit it) that c does not have 1 as an eigenvalue.

Lemma 5. *Set* $\theta_i = r_n \cdots r_{i+1}(a_i)$*. Then:*

(a) The θ_i *form a basis of the group* X *generated by the roots.*

(b) One has $\theta_i > 0$ *and* $c(\theta_i) < 0$ *for all* i [1112].

(c) Every root $a \in R$ *such that* $a > 0$ *and* $c(a) < 0$ *is equal to one of the* θ_i*.*

(d) R *is the union of the orbits of the* θ_i *under the powers of* c*.*

It is clear that θ_i is of the form $a_i + \sum_{j>i} m_{ij} a_j$, with $m_{ij} \in \mathbb{Z}$, which proves (a). Assertions (b) and (c) are consequences of the following remark: the symmetry r_i preserves the sign of every root distinct from $\pm a_i$, and changes the sign of $\pm a_i$.

Finally, for (d) one remarks that an orbit of c cannot be entirely formed of positive (resp. negative) roots, since, taking the sum of these roots one would find a non-zero element of X invariant under c, and we have admitted that c does not have 1 as an eigenvalue. There is therefore necessarily in every orbit an element $a > 0$ such that $c(a) < 0$, and one applies (c).

Remark. We have sketched the preceding proof only to facilitate the reader's task; one could have limited oneself to referring to the canonical texts on Coxeter (cf. for example Koszul, Séminaire Bourbaki, 1959/1960, exposé 191). Said texts contain other results: the orbits of c all have the same number h of elements, and each contains only one θ_i. In particular $h = Card(R)/n$.

Let us now return to the proof of Lemma 3. Distinguish three cases:

(1) *The element* $\varphi \in E/W$ *given is equal to the identity element.*

One must then take $f \in W$; one chooses $f = c$; it works by Lemma 5.

(2) *The system* R *is of type* A_n*, with* n *even* $\geqslant 2$*, and* φ *is the unique non-trivial element of* E/W*.*

One knows that $-1 \notin W$; one then takes $f = -c$. A simple calculation shows that c is of order $h = n+1$; hence its order is odd. If $a \in R$ is any root, one has $a = c^m \theta_i$ for a pair (m, i), cf. Lemma 5; by adding h to m if necessary, one may suppose m even, and one sees that one has then $a = (-c)^m \theta_i = f^m \theta_i$. The orbits of the θ_i therefore indeed fill R.

(3) *The element* $\varphi \in E/W$ *is non-trivial, and* R *is of one of the following types:*

- A_n, n odd ≥ 3
- D_4
- D_n, $n \geq 5$

[1112] For the order relation defined by $(a_1, \cdots, a_n)$.

- E_6.

(A glance at the classification shows that these are indeed all the cases (with A_n, n even) where E/W is non-trivial.)

Let S be a simple root system, which we shall not number for the moment. One knows that E is the semidirect product of W and the group Ψ of permutations of S that leave invariant the Cartan matrix (or the Dynkin diagram, it is the same thing). The group E/W is thereby identified with Ψ, and in particular φ corresponds to an element $\psi \in \Psi$. One observes by inspection of the Dynkin diagrams (cf. figures above) that every orbit of ψ in S is formed of roots that are pairwise orthogonal (i.e. not linked in the diagram). [Note that this would not be the case for A_n (n even), which obliged us to treat this case separately.] If σ is such an orbit, the symmetries r_a, $a \in \sigma$, commute with each other; their product will be denoted ρ_σ. It is clear that ρ_σ commutes with ψ.

This being so, let us choose on S a total order such that every orbit is a segment for this order relation; this amounts to numbering the elements of S: $S = a_1, \cdots, a_n$. Let $\theta_1, \cdots, \theta_n$ be the roots defined above, and $c = r_1 \cdots r_n$ the corresponding Coxeter element. The element c is the product of the ρ_σ, the σ being arranged in a certain order; it follows that it commutes with ψ. One then sets $f = c\psi$. One further remarks that ψ permutes the θ_i among themselves. Indeed, one has $\psi(\theta_i) > 0$ since $\theta_i > 0$ and $c(\psi(\theta_i)) = \psi(c(\theta_{i'})) < 0$, hence (Lemma 5, (c)), $\psi(\theta_i)$ is equal to some θ_j. It is now immediate that $f = c\psi$ answers the question. Indeed, if $a \in R$, one has $a = c^m\theta_i$ for a pair (m, i), whence $a = f^m(\psi^{-m}\theta_i) = f^m\theta_j$ for some j. *QED*.

Remark. One can prove that, except in case (2), every orbit of f has exactly $h = Card(R)/n$ elements and contains one and only one θ_i. In case (2), some of the θ_i are superfluous.

Exposé XV. Complements on sub-tori of a group scheme. Application to smooth groups

by M. Raynaud [1113]

0. Introduction

This Exposé complements and partially recasts Exposés XI and XII; the contents of Exposés XIII and XIV are not indispensable. Continuing the effort undertaken in XII, we shall work with S-preschemes in groups that are not necessarily affine and not necessarily separated over S.

Sections 1, 2, 3, 4 are devoted to the study of sub-tori of a group prescheme. We obtain theorems of infinitesimal lifting (§ 2) and global lifting (§ 4), in which an

[1113] This Exposé and the two following ones (Exp. XVI and XVII) do not correspond to oral lectures of the seminar. They develop, with some additional material, the substance of (succinct) manuscript notes of A. Grothendieck, written in 1964 on the occasion of the present seminar.

essential role is played by points of finite order (§ 1).

Sections 5, 6 and 7 are independent of the preceding ones. The consideration of infinitesimal neighborhoods leads to the representability of the functor of smooth subgroups equal to their connected normalizer (§ 5). In §§ 6 and 7, we turn more specifically to Cartan subgroups.

Finally, in § 8 we give a necessary and sufficient condition for the functor of sub-tori of a smooth group, or that of maximal tori, to be representable.

1. Lifting of finite subgroups

1.1. Finite, smooth and central subgroups of multiplicative type

Proposition 1.1. *Let S be an affine scheme, S_0 a closed subscheme of S defined by an ideal of square zero, G an S-prescheme in groups, H_0 a subgroup scheme of $G_0 = G \times_S S_0$ which is smooth over S_0, of finite type, and of multiplicative type. Then, in order that there exist a subgroup scheme of G, of multiplicative type, which lifts H_0, it is necessary and sufficient that there exist a subscheme H' of G, flat over S, which lifts H_0.*

The necessity of the condition is clear; let us prove sufficiency. The group H_0 of multiplicative type is quasi-isotrivial (X 4.5); by Exp. X 2.1, there exist an S-group H of multiplicative type and an S_0-isomorphism of groups:

$u_0 : H \times_S S_0 \xrightarrow{\sim} H_0.$

Since H' is flat over S and $H' \times_S S_0$ is of finite presentation over S_0 (Exp. IX 2.1 b)), H' is of finite presentation over S; moreover, its fibers are smooth, so H' is smooth over S (EGA IV 17.5.1). Since S is affine, u_0 therefore lifts to an S-morphism of preschemes:

$$u : H \longrightarrow H'.$$

It then follows from Exp. III 2.1 and Exp. IX 3.1 that the composed morphism $v_0 : H \times_{S_0} S \xrightarrow{\sim} H_0 \to G_0$

also lifts to an S-morphism of groups:

$$v : H \longrightarrow G.$$

Since v_0 is an immersion, so is v. The image of H by v is therefore a subgroup scheme of G, of multiplicative type, which lifts H_0.

Proposition 1.2. *Let S be a prescheme, S_0 a subprescheme of S defined by a locally nilpotent sheaf of ideals, G an S-prescheme in groups, flat and of finite presentation over S, and H_0 a subgroup scheme of $G_0 = G \times_S S_0$ which is smooth, finite over S_0, of multiplicative type and central. Then there exists a unique subgroup scheme H*

of G, of multiplicative type, which lifts H_0. Moreover H is central. (See XVII App. III, 1).

Proposition 1.2 bis. *Let S, G, S_0, G_0 be as above, H an S-group scheme of multiplicative type, smooth and finite over S, and $u_0 : H_0 = H \times_S S_0 \to G_0$ a central homomorphism. Then u_0 lifts uniquely to a homomorphism $u : H \to G$. Moreover u is central.*

The existence of the lifting u in 1.2 bis is easily deduced from 1.2 by considering the graph of u_0. The lifting u is unique and central by Exp. IX 3.4 and Exp. IX 5.1.

Proof of 1.2. The uniqueness of H and the fact that H is central follow from Exp. IX 5.6 b) and Exp. IX 3.4 bis. Given uniqueness, in order to prove the existence

of H we may assume S affine and S_0 defined by an ideal of square zero, and it suffices (1.1) to find a subscheme of G, flat over S, which lifts H_0.

Since H_0 is smooth and finite over S_0, we may assume — possibly after restricting S — that there exists an integer $n > 0$, invertible on S, such that $H_0 = {}_nH_0$. Consider the n-th power morphism in G:

$$u : G \to G, \qquad x \mapsto x^n.$$

We still denote by ${}_nG$ the "kernel of u", that is, the inverse image under u of the unit section of G (N.B. u is not in general a group morphism). Granting for a moment the following lemma:

Lemma 1.3. *Let k be a field, G a group scheme locally of finite type over k, n an integer prime to the characteristic of k, u the n-th power morphism in G. Then u is étale at every point x of G belonging to the center of G.*

Since G is flat and of finite presentation over S, it follows from the preceding lemma and from EGA IV 17.8.2 that if x is a point of G projecting to s in S and belonging to the center of G_s, the morphism u is étale at x. If moreover x is a point of ${}_nG$, then ${}_nG$ is étale over S at x. By hypothesis, the group H_0 is central and contained in ${}_nG_0$, so it is in fact contained in the largest open subset V of ${}_nG$ which is étale over S. Since H_0 and $V \times_S S_0 = V_0$ are étale over S_0, H_0 is an open subset of V_0 (EGA IV 17.8.7 and 17.9.1).

But then the open subprescheme of V having the same underlying space as H_0 is a subscheme of G, flat over S, which lifts H_0.

It remains to prove Lemma 1.3. For this, note that the largest open subset of G on which u is étale is invariant under base field extension (EGA IV 17.8.2); this allows us to reduce to the case where x is rational over k. Let t_x denote translation by x, which is a k-automorphism of the scheme G. Since x is in the center of G, we have the commutative diagram:

$$\begin{array}{ccc} G & \xrightarrow{u} & G \\ \Big\downarrow{\scriptstyle t_x} & & \Big\downarrow{\scriptstyle t_{x^n}} \end{array}$$

$$\downarrow \qquad\qquad \downarrow$$
$$G \xrightarrow{\ u\ } G.$$

It therefore suffices to show that u is étale at the origin, but this was seen in VII_A 8.4.

1.2. Global lifting of finite groups

Lemma 1.4. *Let A be a local ring, separated and complete for the topology defined by its maximal ideal $\mathfrak{m}$, let $S = \operatorname{Spec}(A)$, $S_n = \operatorname{Spec}(A/\mathfrak{m}^n)$. Then for every prescheme X (resp. every S-prescheme X), the canonical map*

$$(*) \qquad \operatorname{Hom}(S, X) \to \varprojlim_n \operatorname{Hom}(S_n, X)$$

(resp.

$$(**) \qquad \Gamma(X/S) \to \varprojlim_n \Gamma(X_n/S_n), \qquad \text{where } X_n = X \times_S S_n)$$

is bijective.

Statement (**) is an easy consequence of (). *Let us prove* ().

Let $u_n : S_n \to X$ ($n \in \mathbb{N}$) be a coherent system of morphisms. The image y of the closed point of S_n is therefore independent of n, and u_n factors through $\operatorname{Spec}(\mathcal{O}_y)$. The morphisms u_n define, by passage to the projective limit, a ring morphism

$$\tilde{u} : \mathcal{O}_y \to \varprojlim_n (A/\mathfrak{m}^n) = A.$$

This shows that (*) is surjective; it is injective as soon as A is separated for the $\mathfrak{m}$-adic topology.

Corollary 1.5. *Let A be a complete noetherian local ring, $\mathfrak{m}$ its maximal ideal, $S = \operatorname{Spec}(A)$, $S_n = \operatorname{Spec}(A/\mathfrak{m}^n)$, X a finite scheme over S and Y an S-prescheme. Then the canonical map*

$$\operatorname{Hom}_S(X, Y) \to \varprojlim_n \operatorname{Hom}_{S_n}(X_n, Y_n)$$

(where $X_n = X \times_S S_n$ and similarly $Y_n = \cdots$) is bijective.

Indeed, it follows from EGA II 6.2.5 that X is a finite sum of local S-schemes finite over S. This reduces us to the case where X itself is the spectrum of a complete noetherian local ring. But $\operatorname{Hom}_S(X, Y) = \Gamma(Z/X)$ where Z is the X-prescheme $Y \times_S X$, and we apply the preceding proposition.

Proposition 1.6. *Let A, S, S_n be as above, and let G and M be two S-preschemes in groups, with M finite over S. Then:*

a) The canonical map

$$\operatorname{Hom}_{S\text{-gr}}(M, G) \to \varprojlim_n \operatorname{Hom}_{S_n\text{-gr}}(M_n, G_n)$$

is bijective.

b) If M is of multiplicative type and G is smooth over S, the canonical map

$$\varphi : \mathrm{Hom}_{S\text{-gr}}(M, G) \to \mathrm{Hom}_{S_0\text{-gr}}(M_0, G_0)$$

is surjective. Moreover, if $\phi(u) = \phi(u') = u_0$, then u and u' are conjugate by an element of $G(S)$ reducing to the unit element of $G(S_0)$.

c) If M is of multiplicative type and smooth over S, if G is flat of finite type over S, and if $u_0 : M_0 \to G_0$ is a central homomorphism, then u_0 lifts uniquely to a homomorphism

$$u : M \longrightarrow G.$$

Moreover u is central if G has connected fibers.

Proof. a) Follows from 1.5, from the fact that $M \times_S M$ is finite over S, and from the characterization of group morphisms as those rendering commutative the well-known diagram:

$$\begin{tikzcd} M \times_S M \arrow[r, "u \times u"] \arrow[d] & G \times_S G \arrow[d] \\ M \arrow[r, "u"] & G. \end{tikzcd}$$

b) By Exp. IX 3.6, one can construct a coherent system of homomorphisms $u_n : M_n \to G_n$ lifting a homomorphism $u_0 : M_0 \to G_0$. Hence the first assertion of b), in view of a).

If now u and u' are two liftings of u_0, then u_n and u'_n are conjugate

by an element g_n of $G(S_n)$ lifting the unit element of $G(S_0)$ (Exp. IX 3.6); *loc. cit.* also implies that one may choose the g_n in a coherent way, and hence (1.4) coming from a section g of $G(S)$. The morphisms u and $int(g) \cdot u'$ then coincide modulo $\mathfrak{m}^n$ for every n, so they coincide (1.5).

c) The existence and uniqueness of u follow from a) and 1.2 bis. If G has connected fibers, u is central by Exp. IX 5.6 a).

2. Infinitesimal lifting of sub-tori

2.1. Statement of the theorem

We shall give a theorem of infinitesimal lifting of sub-tori of a group prescheme which does not appeal to smoothness hypotheses (in contrast to Exp. IX 3.6 bis) and which moreover answers a very natural question[1114]: does it suffice to be able to lift "enough" points of finite order of a sub-torus T_0 in order to be assured of being able to lift T_0 (infinitesimally, of course)?

[1114] The idea of approximating a torus by its finite subgroups appears in Grothendieck's proof of the connectedness of the centralizers of tori; see § 4.6 of *Bible*.

Theorem 2.1. *Let S be a noetherian affine scheme, S_0 a closed subscheme of S defined by an ideal J of square zero, G an S-prescheme in groups of finite type, $G_0 = G \times_S S_0$, T_0 a sub-torus of G_0, q an integer > 0 invertible on S. Suppose that for every integer n equal to a power of q, there exists a subscheme M_n of G, flat over S, such that $M_n \times_S S_0 = {}_nT_0$. Then there exists a sub-torus T of G such that $T \times_S S_0 = T_0$.*

Theorem 2.1 will be useful to us through the following two corollaries:

Corollary 2.2. *Let S be a locally noetherian prescheme, S_0 a closed subprescheme of S defined by a locally nilpotent sheaf of ideals, G an S-prescheme in groups of finite type, T_0 a sub-torus of $G_0 = G \times_S S_0$, q an integer > 0 invertible on S; finally, with the integer n ranging over powers of q, let (M_n) be a coherent system*

of S-subgroup schemes of G, of multiplicative type, which lifts the ${}_nT_0$ (N.B. The system of subgroups of multiplicative type (M_n) is said to be coherent if $M_m = {}_m(M_n)$ whenever the integer m divides n.) Then there exists one and only one sub-torus T of G such that $T \times_S S_0 = T_0$ and ${}_nT = M_n$ for every n.

Corollary 2.3. *Let G be an S-prescheme in groups, flat and of finite presentation over S, S_0 a closed subprescheme of S defined by a sheaf of ideals of finite type and locally nilpotent, T_0 a central torus of $G_0 = G \times_S S_0$. Then there exists one and only one sub-torus T of G lifting T_0. Moreover T is central.*

Remark 2.4. We leave to the reader the task of formulating the analogue of statements 2.1, 2.2, 2.3 in which, instead of lifting a sub-torus of G_0, one is given a torus T over S and one proposes to lift a morphism

$$u_0 : T_0 \longrightarrow G_0$$

(one reduces to the preceding cases by considering the graph of u_0).

Let us show how Corollaries 2.2 and 2.3 are deduced from 2.1.

Proof of Corollary 2.2. The uniqueness of T follows from Exp. IX 4.8 b) and Exp. IX 4.10. To prove the existence of T, we may therefore reduce to the case where S is affine, hence noetherian, and where S_0 is defined by an ideal of square zero.

Lemma 2.5. *Let G be an S-prescheme in groups, of finite presentation, and H a subgroup scheme of G, of multiplicative type, smooth over S. Then $Centr_G(H)$ is representable by a subgroup prescheme of G, of finite presentation.*

The lemma follows from Exp. VIII 6.5 e), without smoothness hypothesis on H, when G is separated over S. In the present case, one notes that the assertion to be proved is local for the fpqc topology, which allows us to assume H diagonalizable, hence of the form $D_S(M)$. We may also assume S affine, then S noetherian thanks to EGA IV 8. Since H is smooth over S and of finite type, the order q of the torsion subgroup of M is invertible on S (Exp. VIII 2.1 e)). It is then immediate (cf. Exp. IX 4.10) that the subgroups ${}_nH$, where n ranges over powers of q, are schematically dense in H (Exp. IX 4.1). But ${}_nH$ is a completely decomposed covering of

S (i.e. is isomorphic to a finite direct sum of copies of S), so $Centr_G({}_nH) = Z_n$ is representable as the intersection of the centralizers in G of the sections of ${}_nH$ over S. It then suffices to apply the lemma:

Lemma 2.5 bis. *Let S be a noetherian prescheme, G an S-prescheme in groups of finite type, H a subgroup of G of multiplicative type, (H_i) an increasing filtered family of subgroups of G of multiplicative type, and suppose that $Z_i = Centr_G(H_i)$ is representable by a subgroup prescheme of G. Then the family of Z_i is stationary.*

If moreover the H_i are schematically dense in H, one has $Z_i = Centr_G(H)$ for i large enough.

To see that the family of Z_i is stationary, it suffices to show that the family of underlying sets $ens(Z_i)$ is stationary. Indeed, the stationary value will then be a closed subset of an open subset U of G; and, possibly replacing G by U, we are reduced to studying a decreasing filtered family of closed subpreschemes of a noetherian prescheme. An easy constructibility argument reduces us to the case where S is integral. We must then show that the family of $ens(Z_i)$ is stationary above some non-empty open subset of S. Now the generic fiber of G is separated (Exp. VI_A 0.3), so, possibly restricting S, we may assume G separated over S (EGA IV 8). But then Z_i is closed in G (Exp. VIII 6.5 e)).

To establish the last assertion of the lemma, denote by Z the stationary value of the family Z_i. It is clear that $Centr_G(H)$ is a subfunctor of Z; let us show that Z centralizes H. Let E be the subprescheme of $H \times_S Z$ which is the kernel of the pair of morphisms:

$$H \times_S Z \rightrightarrows G$$
$$(h, c) \mapsto c$$
$$(h, c) \mapsto hch^{-1}.$$

The prescheme E majorizes $H_i \times_S Z$ for every i. On the other hand, the H_i are flat over S, so (EGA IV 11.10.9) for every point s of S, the $(H_i)_s$ are schematically dense in H_s and the $H_i \times_S Z$ are schematically dense in $H \times_S Z$. Since G_s is separated, E_s is closed in $(H \times_s Z)_s$ and therefore equal to it. But then E is closed in

$H \times_S Z$, so equals $H \times_S Z$. This says that Z centralizes H.

Let us return to the proof of 2.2. By 2.5, $Z_n = Centr_G(M_n)$ is representable, and by 2.5 bis the decreasing family of subpreschemes Z_n is stationary; let Z be its stationary value. The group Z majorizes T_0 and the M_n. Possibly replacing G by Z, we may therefore assume the M_n central.

We are then in the conditions of application of Theorem 2.1, and there exists a subtorus T of G lifting T_0. The groups ${}_nT$ and M_n are then two liftings of ${}_nT_0$, hence are conjugate (Exp. IX 3.2 bis) and consequently coincide, M_n being central. The torus T answers the question.

Proof of Corollary 2.3. The uniqueness of T follows from Exp. IX 5.1 bis, and the fact that T is central follows from IX 5.6 b). This remark allows us, by the usual procedure, to reduce to the case where S is affine (so S_0 is defined by a nilpotent ideal of finite type), then to the case where S is noetherian. Possibly restricting S, we may assume that there exists an integer q invertible on S. Corollary 2.3 is then a consequence of 2.2 and of 1.2.

Remark 2.6. One easily shows that Corollary 2.3 remains true if one replaces the torus T_0 by a smooth central subgroup of multiplicative type of G_0.

2.2. Proof of 2.1

a) Reduction to the case $T_0 = G_{m,S_0}$.

Thanks to 1.1, we may assume that M_n is a subgroup of multiplicative type. Using Exp. IX 3.2 bis, we may assume that the family of the M_n is coherent (2.2). The centralizer Z_n of M_n in G is representable (2.5), and the filtered family of Z_n is stationary (2.5 bis). Possibly replacing G by Z_n for n large enough, we may therefore assume T_0 and the M_n central. The uniqueness of the lifting of T_0 is then assured (Exp. IX 5.1 bis).

Proceeding as in the proof of 1.1, we may assume that there exist an S-torus T and an S_0-isomorphism

$u_0 : T \times_S S_0 \xrightarrow{\sim} T_0,$

and it is equivalent to lift u_0 or to lift T_0. In view of uniqueness, it suffices to prove the existence of a lifting of u_0 after performing a faithfully flat affine extension $S' \to S$ of finite type (fpqc descent), which allows us to assume $T = G^r_{m,S}$ (Exp. X 4.5). If the restriction of u_0 to each factor G_m lifts to an S-morphism — necessarily central — one immediately deduces a lifting of u_0. In short, we may assume $T_0 = G_{m,S_0}$.

b) Definition of the obstruction to the existence of a lifting of T_0.

To prove 2.1, it suffices by 1.1 to find a subscheme of G, flat over S, which lifts T_0. We shall see that one can define the obstruction to the existence of such a lifting as an element of a certain $Ext^1(\cdot,\cdot)$ of sheaves of modules.

Let U be an open subset of G such that T_0 is closed in U, and let us still denote by U (resp. U_0) the open subscheme of G (resp. G_0) having U as underlying space. The sheaf $\mathcal{O}_{T_0}$, viewed as a sheaf on U, is therefore a quotient of $\mathcal{O}_{U_0}$. Let h be the canonical epimorphism:

$$h : \mathcal{O}_{U_0} \longrightarrow \mathcal{O}_{T_0}.$$

Lemma 2.7. *The canonical map*

$\tilde{h} = \mathrm{id}_J \otimes h : J \otimes_{S_0} \mathcal{O}_{U_0} \to J \otimes_{S_0} \mathcal{O}_{T_0}$

factors (evidently uniquely) as $i \circ j_U$, where j_U is the canonical epimorphism

$J \otimes_{S_0} \mathcal{O}_{U_0} \to J\mathcal{O}_U \simeq J\mathcal{O}_{U_0}$.

We must show that $\tilde{h}(K) = 0$, where K is the kernel of j_U. Now for every integer n equal to a power of q, we have an epimorphism

$$h_n : \mathcal{O}_{T_0} \longrightarrow \mathcal{O}_{{}_nT_0}$$

and since ${}_nT_0$ lifts to a scheme M_n flat over S, the canonical morphism

$$j_n : J \otimes_{S_0} \mathcal{O}_{{}_nT_0} \to J\mathcal{O}_{M_n}$$

is an isomorphism.

The commutative diagram below:

$$\begin{tikzcd}
K \arrow[r] & J \otimes_{S_0} \mathcal{O}_{U_0} \arrow[r, "j_U"] \arrow[d, "\tilde{h}"] & J\mathcal{O}_U \subset \mathcal{O}_U \\
& J \otimes_{S_0} \mathcal{O}_{T_0} \arrow[d, "\tilde{h}_n"] \arrow[ur, "i"] & \\
& J \otimes_{S_0} \mathcal{O}_{{}_nT_0} \arrow[r, "j_n", "\cong"'] & J\mathcal{O}_{M_n} \subset \mathcal{O}_{M_n}
\end{tikzcd}$$

shows that $\tilde{h}(K)$ is contained in $Ker\tilde{h}_n$ for every n, hence is contained in $\bigcap_n Ker\tilde{h}_n$, and it suffices to show that this last intersection is zero. Now the sheaf $\mathcal{O}_{T_0}$ is equal to the sheaf $\mathcal{O}_{S_0}[\mathbb{Z}]$, the algebra of the group $\mathbb{Z}$ with coefficients in $\mathcal{O}_{S_0}$, while $\mathcal{O}_{{}_nT_0}$ is the quotient algebra $\mathcal{O}_{S_0}[\mathbb{Z}/n\mathbb{Z}]$.

Let $a = \Sigma_{m\in\mathbb{Z}} a_m \otimes m$ be an element of $J \otimes_{S_0} \mathcal{O}_{T_0}$. The a_m are then sections of J, almost all zero. Take n large enough that the indices m for which a_m is non-zero have distinct images in $\mathbb{Z}/n\mathbb{Z}$. Then if $a \in Ker\tilde{h}_n$, one necessarily has $a = 0$. This proves that $\bigcap_n Ker\tilde{h}_n = 0$, and proves 2.7.

Let then K_0 be the kernel of h; consider the following diagram:

$$\begin{tikzcd}
& & 0 \arrow[d] & & \\
& J\mathcal{O}_U \arrow[d, "\cong"'] & K_0 \arrow[d] & & \\
0 \arrow[r] & J\mathcal{O}_{U_0} \arrow[r] \arrow[d, "i"] & \mathcal{O}_U \arrow[r, "h"'] & \mathcal{O}_{U_0} \arrow[r] \arrow[d] & 0 \\
& J \otimes_{S_0} \mathcal{O}_{T_0} & & \mathcal{O}_{T_0} \arrow[d] & \\
& & & 0. &
\end{tikzcd}$$

The sheaf $\mathcal{O}_U$ defines an element Φ of the group $Ext^1_{\mathcal{O}_U}(\mathcal{O}_{U_0}, J\mathcal{O}_{U_0})$. Let Ψ be the element of

$Ext^1_{\mathcal{O}_U}(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$ deduced from Φ by bifunctoriality of $Ext^1(\cdot,\cdot)$ through the morphisms $K_0 \to \mathcal{O}_{U_0}$ and $i : J\mathcal{O}_{U_0} \to J \otimes_{S_0} \mathcal{O}_{T_0}$. It follows from Exp. III 4.1 and from the infinitesimal flatness criterion (cf. Exp. III 4.3) that there exists a subscheme of U, flat over S, which lifts T_0, if and only if Ψ is zero.

But note that T_0 is affine, so it suffices (Exp. III 4.5 and 4.6) to show that there locally on U exists a subscheme flat over S lifting T_0. In short, it suffices to show that the image of Ψ in the sheaf

$$\mathcal{E} = \mathcal{E}xt^1_{\mathcal{O}_U}(K_0, J \otimes_{S_0} \mathcal{O}_{T_0})$$

is zero. We shall still denote by Ψ this element of $\Gamma(U, \mathcal{E})$.

c) Reduction to the case S local artinian with algebraically closed residue field.

Since K_0 and $J \otimes_{S_0} \mathcal{O}_{T_0}$ are coherent sheaves, so is the sheaf $\mathcal{E}$; moreover $\mathcal{E}$ has its support in T_0, since this is the case for $J \otimes_{S_0} \mathcal{O}_{T_0}$. To show that the section Ψ of $\mathcal{E}$ is zero, it suffices to see that for every point u of T_0, the image of

Ψ in the fiber of $\mathcal{E}$ at the point u is zero. But the formation of the $Ext^i(\cdot,\cdot)$ of coherent sheaves commutes with flat extensions of the base[1115], so we are reduced to proving the existence of a lifting of $T_0 \cap \operatorname{Spec} \mathcal{O}_u$ flat over S. Let s be the projection of u on S; we may then replace S by $\operatorname{Spec} \mathcal{O}_s$ and G by $G \times_S \operatorname{Spec} \mathcal{O}_s$.

Possibly again making a faithfully flat extension, we may assume that $\mathcal{O}_s$ has an algebraically closed residue field (EGA 0_III, 10.3.1).

Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_s$ and $S_n = \operatorname{Spec} \mathcal{O}_s/\mathfrak{m}^n$. Suppose we have shown that for every $n > 0$ the obstruction to lifting $T_0 \times_S S_n = (T_0)_n$ to a subscheme of $G_n = G \times_S S_n$, flat over S_n, is zero, and let T_n be the unique flat lifting of $(T_0)_n$ over S_n which is a sub-torus of G_n. It is clear that if $n > n'$, one has

$$(T_n) \times_{S_n} S_{n'} = T_{n'}.$$

If then u is a point of T_0 projecting onto s, it follows from the lemma below, applied to the coherent system of liftings $(T_n \cap \operatorname{Spec} \mathcal{O}_u)$ of $(T_0)_n \cap \operatorname{Spec} \mathcal{O}_u$, that there indeed exists a lifting of $T_0 \cap \operatorname{Spec} \mathcal{O}_u$ flat over $\mathcal{O}_s$. We are therefore reduced to proving that Ψ is zero when $S = S_n$ is the spectrum of an artinian local ring with algebraically closed residue field, and to proving:

Lemma 2.8. *Let $A \to B$ be a local homomorphism of noetherian local rings, $\mathfrak{m}$ the maximal ideal of A, J an ideal of square zero of A, M a B-module of finite type, $A_0 = A/J$, $B_0 = B/JB$, $M_0 = M/JM$, N_0 a quotient B_0-module of M_0 which is flat over A_0. For every integer $n > 0$, let A_n, $A_{0,n}$, etc., be the objects obtained by base extension*

[1115] cf. EGA 0_III, 12.3.5.

$A \to A/\mathfrak{m}^n = A_n$, and let J_n be the image of J in A_n. For every integer $n > 0$, let N_n be a quotient B_n-module of M_n, flat over A_n, lifting $N_{0,n}$, and suppose that for $n \geqslant n'$, N_n' is obtained from N_n by base extension $A_n \to A_n'$. Then there exists a B-module N, quotient of M, flat over A, lifting N_0.

Proof of 2.8. Let P_0 be the kernel of the epimorphism $M_0 \to N_0$. For every $n > 0$, we have the following commutative diagram in which the horizontal rows are exact:

$$\begin{tikzcd}
 & & & & 0 \arrow[d] & \\
 & & & & P_{0,n} \arrow[d] & \\
 & 0 \arrow[r] & J_n M_n \arrow[r] & M_n \arrow[r] \arrow[d] & M_{0,n} \arrow[r] \arrow[d] & 0 \\
J_n \otimes_{A_{0,n}} M_{0,n} \arrow[urr] \arrow[drr] & & & & & \\
 & 0 \arrow[r] & J_n \otimes_{A_{0,n}} N_{0,n} \arrow[r] & N_n \arrow[r] \arrow[d] & N_{0,n} \arrow[r] \arrow[d] & 0 \\
 & & & 0 & 0 &
\end{tikzcd} \qquad (*)_n$$

Moreover, by hypothesis, the diagram $(*)_{n+1}$ reduces modulo $\mathfrak{m}^n$ to $(*)_n$.

The Artin–Rees lemma (Bourbaki, *Algèbre commutative*, Chap. 3 § 3 cor. 1) shows that the filtration defined on JM (resp. $J \otimes_{A_0} M_0$ and $J \otimes_{A_0} N_0$) by the kernels of the morphisms

$$JM \to J_n M_n, \quad (\text{resp. } J \otimes_{A_0} M_0 \to J_n \otimes_{A_{0,n}} M_{0,n} \text{ and } J \otimes_{A_0} N_0 \to J_n \otimes_{A_{0,n}} N_{0,n})$$

is $\mathfrak{m}B$-good, so that, passing to the projective limit on the diagrams $(*)_n$ and denoting by $\hat{Q}$ the separated completion of a B-module Q for the $\mathfrak{m}B$-adic topology, one obtains the following commutative diagram, where the two horizontal rows are still exact:

$$\begin{tikzcd}
 & & & & 0 \arrow[d] & \\
 & & & & \hat{P}_0 \arrow[d] & \\
 & 0 \arrow[r] & \hat{J}M \arrow[r] & \hat{M} \arrow[r] \arrow[d] & \hat{M}_0 \arrow[r] \arrow[d] & 0 \\
\hat{J} \otimes_{A_0} M_0 \arrow[urr] \arrow[drr] & & & & & \\
 & 0 \arrow[r] & \hat{J} \otimes_{A_0} N_0 \arrow[r] & \varprojlim_n N_n \arrow[r] & \hat{N}_0 \arrow[r] & 0
\end{tikzcd}$$

▼ ▼

0 0

(̂)

Now $(B, \mathfrak{m}B)$ is a Zariski ring and $J \otimes_{A_0} N_0$ is of finite type over B, so it is separated for the $\mathfrak{m}B$-adic topology. The diagram (*̂) then shows that the morphism

$J \otimes_{A_0} M_0$ □ $J \otimes_{A_0} N_0$

deduced from the epimorphism $M_0 \to N_0$ factors through JM:

$$\begin{tikzcd} J \otimes_{A_0} M_0 \arrow[rr, "\text{can.}"] \arrow[dr] & & JM \arrow[dl] \\ & J \otimes_{A_0} N_0. & \end{tikzcd}$$

Under these conditions, it follows from Exp. III 4.1 and Exp. III 4.3 that there exists a B-module quotient N of M, flat over A, lifting N_0, if and only if a certain element g of $E = Ext^1_B(P_0, J \otimes_{A_0} N_0)$ is zero. More precisely, the exact sequence

$$0 \longrightarrow JM \longrightarrow M \longrightarrow M_0 \longrightarrow 0$$

defines an element f of F, where F is the B-module $Ext^1_B(M_0, JM)$, and g is the image of f

under the natural morphism $F \to E$ arising by bifunctoriality from the morphisms

P_0 □ M_0 and JM □ $J \otimes_{A_0} N_0$.

It follows from the diagram (*̂) and from Exp. III 4.1 that the image of g in $\hat{E}$, canonically identified with $Ext^1_{\hat{B}}(\hat{P}_0, \hat{J} \otimes_{A_0} N_0)$, is zero. But E is of finite type over B,

so is separated for the $\mathfrak{m}B$-adic topology, and consequently g is indeed equal to 0, which completes the proof of 2.8.

d) Study of $\mathcal{E}$. We therefore suppose that S is the spectrum of a local artinian ring A with algebraically closed residue field k. Let A_0 be the ring of S_0, $\mathfrak{m}_0$ the maximal ideal of A_0. Since S_0 is artinian, T_0 is closed in G (Exp. VI_B 1.4.2); we may therefore take the open subset U equal to G, so that

□ = $Ext^1_{□_G}(K_0, J \otimes_{S_0}$ □$_{T_0})$.

Let B_0 be the ring of the affine S_0-scheme $T_0 = G_{m,S_0}$, and B_0' the ring of the special fiber $T_0 \times_{S_0} \mathrm{Spec}(k) = G_{m,k}$ of T_0. The sheaf $\mathcal{E}$ is a coherent $\mathcal{O}_{T_0}$-module, so is defined by a B_0-module of finite type which we shall denote E.

Consider the graded module associated to E and to the $\mathfrak{m}_0 B_0$-adic filtration:

E_r = □$_0^r$ E / □$_0^{r+1}$ E.

Each E_r is therefore a B_0'-module of finite type, and $E_r = 0$ for r large enough, since S_0 is artinian.

Let then g be a section of G above S which on S_0 induces a section g_0 of T_0. Translation (on the left, to fix ideas) by the element g defines an "automorphism of the situation" from the viewpoint of the obstruction problem under consideration. In particular, to g corresponds an S_0-automorphism of the sheaf $\mathcal{E}$ leaving the obstruction Ψ fixed. More precisely, g defines a semi-linear automorphism of the B_0-module E (relative to the A_0-automorphism of B_0 defined by translation by g_0 in the group T_0). By reduction modulo $\mathfrak{m}_0^{r+1}$, g then defines a semi-linear automorphism of E_r (relative to the k-automorphism of B'_0 defined by translation by $g_0 \times_{S_0} \operatorname{Spec}(k)$ in $T_0 \times_{S_0} \operatorname{Spec}(k)$).

Lemma 2.9. *For every integer $r \geqslant 0$, E_r is a locally free B'_0-module.*

Let x be a point of T_0, $\kappa(x)$ its residue field, $(E_r)_x$ "the fiber" of E_r at x, equal to $E_r \otimes_{B'_0} \kappa(x)$, $\ell(x)$ the rank of $(E_r)_x$ over $\kappa(x)$, ℓ the maximum value of $\ell(x)$ as x ranges over the points of T_0. Let L be the largest closed subscheme of $\operatorname{Spec} B'_0 = G_{m,k}$ above which E_r is locally free of rank ℓ (TDTE IV Lemma 3.6). Let β be a point of $L(k)$ (there is one, L being of finite type over k algebraically closed) and let α be a point of $G_{m,k}(k)$ of order equal to a power n of q. The point α is therefore rational over k, and since by hypothesis ${}_nT_0$ lifts to a subscheme M_n étale over S, there exists a section a of G above S which lifts α and which, above S_0, is a section of T_0. The remarks made above then show that the fibers of E_r are k-isomorphic at the points β and $\alpha\beta$ of $G_{m,k}(k)$. But the points of order a power of q are dense in $G_{m,k}$, and similarly their translates by β. Since L is closed in $G_{m,k}$ and $G_{m,k}$ is reduced, L equals $G_{m,k}$ and E_r is locally free over B'_0 of rank ℓ.

e) End of the proof of Theorem 2.1.

Let $K_0(n)$ be the sheaf of ideals of $\mathcal{O}_{G_0}$ defining the closed subscheme ${}_nT_0$. The sheaf K_0[1116] is therefore a subsheaf of $K_0(n)$. Set, for simplicity,

$$R = J \otimes_{S_0} \mathcal{O}_{T_0} \quad \text{and} \quad R(n) = J \otimes_{S_0} \mathcal{O}_{{}_nT_0}.$$

The sheaf $R(n)$ is therefore a quotient of R, and one has the diagram of morphisms:

$$\begin{tikzcd}
 & & K_0 \arrow[d] & \\
 & & K_0(n) \arrow[d] & \\
0 \arrow[r] & J\mathcal{O}_G \arrow[r] \arrow[d] & \mathcal{O}_G \arrow[r] & \mathcal{O}_{G_0} \arrow[r] & 0 \\
 & R \arrow[d] & & & \\
 & R(n) & & &
\end{tikzcd}$$

[1116]introduced after Lemma 2.7.

Using then the bifunctoriality of $Ext^1_{\mathcal{O}_G}(\cdot,\cdot)$, one obtains the following commutative diagram:

$$\begin{tikzcd}
Ext^1_{\mathcal{O}_G}(\mathcal{O}_{G_0}, J\mathcal{O}_G) \arrow[dr] & & \\
& Ext^1_{\mathcal{O}_G}(K_0(n), R) \arrow[r] \arrow[d] & Ext^1_{\mathcal{O}_G}(K_0, R) = \mathcal{E} \arrow[d] \\
& Ext^1_{\mathcal{O}_G}(K_0(n), R(n)) \arrow[r] & Ext^1_{\mathcal{O}_G}(K_0, R(n)).
\end{tikzcd}$$

Let us again denote by Φ the element of $Ext^1_{\mathcal{O}_G}(\mathcal{O}_{G_0}, J\mathcal{O}_G)$ defined by the exact sequence

$$0 \longrightarrow J\mathcal{O}_G \longrightarrow \mathcal{O}_G \longrightarrow \mathcal{O}_{G_0} \longrightarrow 0,$$

so that Ψ is the image of Φ in $\mathcal{E}$. Since ${}_nT_0$ lifts to a subscheme M_n of G, flat over S, the image of Φ in the sheaf $Ext^1_{\mathcal{O}_G}(K_0(n), R(n))$ is zero (Exp. III 4.1); *a fortiori*, the image of Φ in $Ext^1_{\mathcal{O}_G}(K_0, R(n))$, which is also the image of Ψ, is zero.

Lemma 2.10. *The canonical morphism*

$$Ext^1_{\mathcal{O}_G}(K_0, R) \otimes_{B_0} \mathcal{O}_{{}_nT_0} \to Ext^1_{\mathcal{O}_G}(K_0, R(n))$$

is injective for every integer n.

Indeed, the affine scheme $T_0 = G_{m,S_0}$ has ring

$$B_0 = A_0[T, T^{-1}].$$

The subscheme ${}_nT_0$ is defined by the vanishing of the following section of B_0:

$$h(n) = T^n - 1,$$

which is regular (EGA 0_IV 15.2.2) and remains regular after any base change $S_0' \to S_0$. We therefore have an exact sequence of sheaves:

$$0 \to \mathcal{O}_{T_0} \xrightarrow{h(n)} \mathcal{O}_{T_0} \to \mathcal{O}_{{}_nT_0} \to 0.$$

Since ${}_nT_0$ is flat over S, one obtains, by tensoring with J over $\mathcal{O}_{S_0}$, the exact sequence

$$0 \to R \xrightarrow{h(n)} R \to R(n) \to 0,$$

then the exact sequence of Ext:

$$\cdots \to Ext^1_{\mathcal{O}_G}(K_0, R) \xrightarrow{h(n)} Ext^1_{\mathcal{O}_G}(K_0, R) \to Ext^1_{\mathcal{O}_G}(K_0, R(n)),$$

which completes the proof of the lemma.

The foregoing shows that for every integer n equal to a power of q, the image of Ψ in $\mathcal{E}/h(n)\mathcal{E}$ is zero. To show that Ψ is zero, it suffices to see that if $\Psi \in \mathfrak{m}_0^r\mathcal{E}$, then

$\Psi \in \mathfrak{m}_0^{r+1}\mathcal{E}$. Let $\bar{\Psi}$ be the image of Ψ in E_r. There exists an element $\Psi(n)$ of $\mathcal{E}$ such that one has

$$\Psi = \Psi(n) \cdot h(n) \quad (n \text{ equal to a power of } q).$$

We noted that the image $\bar{h}(n)$ of $h(n)$ in B_0' is again B_0'-regular. Since E_s is locally free over B_0' for every s (2.9), multiplication by $\bar{h}(n)$ in E_r is injective. One deduces immediately that $\Psi(n)$ is in $\mathfrak{m}_0^r\mathcal{E}$. Let $\bar{\Psi}(n)$ be its image in E_r, so that one has the relation

$$\bar{\Psi} = \bar{h}(n)\ \bar{\Psi}(n) \quad (n \text{ equal to a power of } q).$$

This shows that the set F of points of $G_{m,k}$ at which $\bar{\Psi}$ takes the value 0 contains the dense set of points of order a power of q. Moreover F is a closed set (since E_r is locally free over B_0') and $G_{m,k}$ is reduced, so $\bar{\Psi}$ is zero.

This completes the proof of Theorem 2.1.

3. Characterization of a sub-torus by its underlying set

3.1. Statement of the theorem

Notation. If X is a prescheme, $ens(X)$ denotes the underlying set of X. If X and S' are two S-preschemes, $X_{S'} = X \times_S S'$, $u : X_{S'} \to X$ the canonical morphism, E a subset of $ens(X)$, one denotes by $E_{S'}$ (or $E \times_S S'$) the subset of $ens(X_{S'})$ equal to $u^{-1}(E)$.

Theorem 3.1. *Let S be a locally noetherian prescheme, G an S-prescheme in groups of finite type, q an integer > 0 invertible on S, E a subset of $ens(G)$. Consider the following assertions concerning E:*

- *(i) The set E is the underlying set of a sub-torus T of G.*
- *(ii) a) For every point s of S, there exists a sub-torus T_s of G_s such that $E \cap ens(G_s) = E_s$ is the underlying set of T_s.*
- - b) As the integer n ranges over powers of q, there exists a coherent family (cf. 2.2) M_n of subgroup schemes of G, of multiplicative type, such that for every point s of S one has*

$$(M_n)_s = {}_nT_s.$$

- *(iii) a) As in (ii) a).*
- - b) The set E is locally closed in $ens(G)$, and the dimension of the fibers of E over S is locally constant.*
- *(iv) a) As in (ii) a).*
- - b) For every S-scheme S' which is the spectrum of a complete discrete valuation ring with algebraically closed residue field, $E_{S'}$ is the underlying set of a sub-torus of $G_{S'}$.*

Then one has the following implications:

- *A) (i) ⇔ (ii) ⇒ (iii) ⇒ (iv).*

- *B) If G is separated over S, one has (iii) $\Leftrightarrow$ (iv), and moreover E is closed.*
- *C) Conditions (i), (ii), (iii) (and also (iv) if G is separated over S) are equivalent in the following two cases:*
- - 1st case: a) The prescheme S is reduced, or G is flat over S, and*
- - b) For every point s of S, T_s is a central torus of G_s.*

- - 2nd case: S is normal.*

Moreover, in the two cases above, the torus T with underlying set E is unique.

Remarks 3.2. a) When S is reduced, it is unnecessary in (ii) to assume that the family M_n is coherent, this condition being entailed by the other hypotheses. Indeed, if the integer m divides n, the subgroups ${}_mM_n$ and M_m are étale over S, hence reduced, and have the same underlying space, so they coincide.

b) If S is not assumed normal, it is no longer true in general that (iii) $\Rightarrow$ (i), even when S is reduced, geometrically unibranched and G is a smooth group scheme over S. Indeed, consider the Borel subgroup of $SL_{2,S}$ formed by matrices of the form

$$(tu)(0t^{-1}),$$

where S is the affine curve over a field k with ring

$$k[x,y]/(y^2 - x^3).$$

Consider then the set E obtained as follows: above the "cusp of S" ($x = y = 0$) we take the diagonal torus ($u = 0$). Above the complementary open subset ($x \neq 0$) we take the torus deduced from the diagonal torus by conjugation by the element

$$(1y/x)(01).$$

The set E so obtained satisfies (iii) a); on the other hand it is closed in G, and the reduced subscheme having E as underlying set has equations

$$xu + y(t - t^{-1}) = 0u^2 - x(t - t^{-1})^2 = 0.$$

This is therefore not a sub-torus of G, since the fiber above the cusp is not reduced.

Plan of the proof of 3.1. In a first part we shall establish the following "easy" implications:

$$\begin{array}{ccccc} & & (ii) & & \\ & \nearrow & & \searrow & \\ (i) & & & & (iv) \\ & \searrow & & \nearrow & \\ & & (iii) & & \end{array}$$

and [(iv) ⇒ (iii) and E closed if G is separated over S].

The proof of the more delicate implications will be carried out in three stages:

- I) Reduction of the implication (iii) ⇒ (i) (under the hypotheses of C), 1st case) to the case where S is normal.
- II) (iii) ⇒ (ii) when S is normal.
- III) (ii) ⇒ (i).

3.2. Proof of the "easy" assertions contained in 3.1

(i) ⇒ (ii) and (iii) is trivial.

(ii) ⇒ (iv). Possibly replacing S by S', we may assume that S is the spectrum of a discrete valuation ring. Let t be the generic point of S and s the closed point.

Since E is locally closed, there exists a subprescheme of G which is reduced and whose underlying space is E; we shall denote it E. The generic fiber E_t of E is therefore equal to the sub-torus T_t of rank r of G_t (by (iii) a)). Let E' be the schematic closure of E_t in E. The prescheme

E' is irreducible, and its closed fiber E'_s is non-empty (it contains the unit section of G_s), so E'_s is of dimension r (EGA IV 14.3.10). But then E'_s is a closed subscheme of E_s, and the latter is of dimension r (by (iii) b)) and irreducible (since it has the same underlying space as T_s), so E'_s has the same underlying space as E_s, and consequently $E' = E$, which proves that E is flat over S.

Let now E'' be the schematic closure of E_t in G. Then E'' is a subprescheme in groups of G, flat over S (Exp. VIII 7.1), which majorizes E', hence E. The closed

fiber E''_s is an algebraic subgroup of G_s of dimension r (*loc. cit.*). Since T_s is a closed irreducible subset of G_s of dimension r, E_s has the same underlying set as the connected component $(E''_s)^0$ of E''_s. Let $(E'')^0$ be the "connected component" of E'', i.e. the open subgroup of E'' complementary to the union of the irreducible components of E''_s not containing the origin. One then has

$$E = ens[(E'')^0].$$

Since E and E'' are reduced, one has even $E = (E'')^0$. Finally E is a subgroup prescheme of G, flat and of finite type over S, with connected fibers, hence separated (Exp. VI_B 5.2), whose generic fiber is a torus T_t, and whose reduced closed fiber is a torus T_s; but then E is a torus (Exp. X 8.8).

(ii) ⇒ (iv). One may again assume that S is the spectrum of a discrete valuation ring, and we keep the notation introduced above. The schematic closure in G of T_t is a subgroup prescheme T of G, flat over S, which majorizes M_n for every integer n equal to a power of q. Consequently the closed fiber of T is a closed subscheme of G_s majorizing $(M_n)_s$ for every n, hence majorizing T_s. For dimension reasons, the "connected component" T^0 of T has E as underlying set, and one concludes as above that T^0 is a sub-torus of G.

(iii) ⇒ [(iii) and E closed], if G is separated over S. Let us show that E is closed.[1117] First let us prove the lemma:

Lemma 3.3. *If the conditions stated in 3.1 (iv) are satisfied, E is a locally constructible part of $ens(G)$.*

By the usual method, we are reduced to studying the case where S is noetherian, integral, with generic point η. Possibly restricting S, we may assume (Exp. VI_B 10.10) that there exists a subgroup scheme H of G, flat over S, with connected fibers, whose generic fiber H_η is equal to T_η. To prove 3.3 it then suffices to show that $ens(H) = E$. Now if s is a point of S, there exists, by EGA II 7.1.9, an S-scheme S', the spectrum of a discrete valuation ring, that one may assume complete with algebraically closed residue field, whose generic point t' projects onto η and whose closed point s' projects onto s. By (iv) b), there exists a sub-torus $T_{S'}$ of $G_{S'}$ having $E_{S'}$ as underlying space. The two subprescheme in groups $T_{S'}$ and $H_{S'} = H \times_S S'$ of $G_{S'}$ are flat over S', have connected fibers, and the same generic fiber $T_{t'}$, so they coincide with the connected component of the schematic closure

of $T_{t'}$ in $G_{S'}$. Consequently $ens(H_s) = E_s$, so $ens(H) = E$, which proves the lemma.

This being so, knowing that E is locally constructible, in order to see that E is closed it suffices to show that every specialization in G of a point of E is a point of E. By the usual technique we are reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. But then the sub-torus of G of underlying space E ((iv) b)) is closed in G since G is separated (Exp. VIII 7.12).

3.3. Continuation of the proof of 3.1

I) Reduction of (iii) ⇒ (i) (C, 1st case) to the case where S is normal.

a) Reduction to the case S affine reduced. We therefore assume that for every point s of S, E_s is the underlying space of a central sub-torus of G_s. The uniqueness of T then follows from Exp. IX 5.1 bis, and moreover T will necessarily be central (*loc. cit.*). In view of uniqueness, to prove the existence of T we may assume S noetherian, affine of ring A. If S is not reduced, by hypothesis G is flat over S. By 2.3 it then suffices to solve the problem for S_{red} and $G \times_S S_{red}$. We may therefore assume in addition that S is reduced.

[1117] The fact that (iv) ⇒ (iii) will appear in the course of the proof.

b) Reduction to the case where the ring A is of finite type over $\mathbb{Z}$. Let us first prove two lemmas:

Lemma 3.4. *Let k be a field, G a k-algebraic group, E a subset of G, k' an extension of k, $T_{k'}$ a sub-torus of $G_{k'}$ having $E_{k'}$ as underlying space. Then, if $T_{k'}$ is central or if k is perfect, E is the underlying set of a sub-torus of G_k.*

Indeed, by fpqc descent it suffices to show that the two inverse images of $T_{k'}$ in $G_{k''}$, where $k'' = k' \otimes_k k'$, coincide. Now they have the same underlying space, namely the inverse image of E. If $T_{k'}$ is central, the lemma is a consequence of Exp. IX 5.1 bis. If k is perfect, k'' is reduced and the two inverse images of $T_{k'}$, being smooth over k'', are reduced, hence coincide.

Remark 3.5. It follows from the preceding lemma that in the statement of 3.1 (iv), property (iv) a) is a consequence of (iv) b) in the two following cases:

1. One assumes that the residue fields of the points of S are perfect.
2. For every S' as in (iv) b), one assumes that the torus with underlying space $E_{S'}$ is central in $G_{S'}$.

Lemma 3.6. *Let S be a prescheme projective limit of affine schemes S_i (cf. EGA IV 8), H an S-group scheme of multiplicative type and of finite type. Then there exist an index i, an S_i-group scheme H_i of multiplicative type and of finite type, and an S-isomorphism*

$$H_i \times_{S_i} S \xrightarrow{\sim} H.$$

If moreover H is isotrivial, one may assume H_i isotrivial.

Since H is of finite type over S, H is in fact of finite presentation over S (Exp. IX 2.1 b)); there therefore exist an index ℓ and an S_ℓ-group scheme H_ℓ such that $H_\ell \times_{S_\ell} S$ is isomorphic to H (Exp. VI_B 10). Setting $H_i = H_\ell \times_{S_\ell} S_i$, one therefore has $H \cong H_i \times_{S_i} S$ for every $i \geqslant \ell$.

Since H is of finite type over S, H is quasi-isotrivial (Exp. X 4.5), hence trivialized by an étale surjective morphism $S' \to S$. Using the quasi-compactness of S, one easily sees that there exist a covering of S by a finite number of affine open subsets S_α, and for every α an étale, surjective and finitely presented morphism $S'_\alpha \to S_\alpha$ trivializing $H|S_\alpha$. This covering (S_α) of S then comes from a covering $(S_{i,\alpha})$ of S_i for i large enough (EGA IV 8). Possibly replacing S_i by $S_{i,\alpha}$ and S by S_α, we may therefore assume that H is trivialized by an étale surjective morphism $S' \to S$ of finite presentation. For i large enough, there then exist a prescheme S'_i, an étale surjective morphism $S'_i \to S_i$ of finite presentation, and an S-isomorphism $S'_i \times_{S_i} S \to S'$ (EGA IV 17.16). Set then for j large enough:

$$S'_j = S'_i \times_{S_i} S_j, \qquad H'_j = H_j \times_{S_j} S'_j, \qquad H' = H \times_S S'.$$

Given the choice of S', there exist a finitely generated abelian group M and an S'-isomorphism of group schemes $D_{S'}(M) \xrightarrow{\sim} H'$. Since the S'_i are quasi-compact and $S' = \lim S'_i$, it follows from Exp. VI_B 10 that there exist an index j and an S'_j-isomorphism of group schemes

$$D_{S'_j}(M) \xrightarrow{\sim} H'_j.$$

But this says that H_j is a quasi-isotrivial group of multiplicative type.

When H is isotrivial, one proceeds analogously, using a trivialization of H by a finite étale morphism $S' \to S$.

This being so, we can carry out the announced reduction b). The ring A of S is a filtered inductive limit of its subrings A_i of finite type over $\mathbb{Z}$. Let $S_i = \operatorname{Spec} A_i$, $u_j : S \to S_j$ and $u_{jk} : S_k \to S_j$ ($k \geqslant j$) the transition morphisms. Since S is noetherian and G is of finite presentation over S, there exist an index i and an S_i-prescheme in groups G_i, of finite type over S_i, such that G is S-isomorphic to $G_i \times_{S_i} S$. Similarly, since E is locally closed in G, we may assume that there exists a locally closed subset E_i of G_i such that $E = E_i \times_{S_i} S$ (EGA IV 8.3.11). For every $j > i$, let $G_j = G_i \times_{S_i} S_j$ and $E_j = E_i \times_{S_i} S_j$, and let Q_j be the set of points s of S_j such that $(E_j)_s$ is the underlying set of a central sub-torus of $(G_j)_s$.

It follows from 3.4 that $Q_k = u_{jk}^{-1}(Q_j)$ for $k \geqslant j$, and by hypothesis $ens(S) = u_j^{-1}(Q_j)$ for $j \geqslant i$. Moreover, I claim that Q_j is ind-constructible (EGA IV 1.9.4). Indeed, since S_j is noetherian, it suffices (EGA IV 1.9.10) to see that if S is a noetherian integral scheme with generic point η, and if E_η is the underlying set of a central sub-torus T_η of G_η, there exists a neighborhood U of η such that for every point s of U, E_s has the same property. Now, possibly restricting S, we may assume that the

center Z of G is representable and that there exists a subgroup scheme T of G whose generic fiber is T_η (VI_B § 10). But then $ens(T)$ and E are two locally closed subsets of $ens(G)$ which coincide on the generic fiber; one may therefore find a neighborhood U of η such that, above U, $ens(T) = E$ (EGA IV 8.3.11). For the same reasons, one may assume that above U, T is central, since T is a torus (3.6).

Knowing now that Q_j is ind-constructible, it follows from EGA IV 8.3.4 that $Q_j = ens(S_j)$ for j large enough. We may then replace S, G, E by S_j, G_j, E_j, which reduces us to the case where S is an affine reduced scheme of finite type over $\mathbb{Z}$.

c) Reduction to the case where S is the spectrum of a complete reduced noetherian local ring. Owing to the uniqueness of the torus of underlying set E, the usual technique (EGA IV 8) and Lemma 3.6 allow us to replace S by the spectrum of the local ring A of a point of S. Let $\hat{S}$ be the spectrum of the completion $\hat{A}$ of A for the topology defined by the maximal ideal. Since A is the localization of a finitely generated algebra over $\mathbb{Z}$, $\hat{S}$ is still

reduced (EGA IV 7.6.5). I claim that it suffices to solve the problem after the base change $\hat{S} \to S$. Indeed if $\hat{T}$ is the sub-torus of $G_{\hat{S}}$ with underlying space $E_{\hat{S}}$, its two inverse images in $G_{\hat{S}} \times_S \hat{S}$ are two central sub-tori with the same underlying space, so they coincide (Exp. IX 5.1 bis), and by fpqc descent, $\hat{T}$ comes from a torus T of G which answers the question (cf. 3.4).

d) A descent lemma. Let us recall the following properties of finite morphisms which were noted in TDTE I: Let S and S' be two preschemes and $u : S' \to S$ a finite morphism. Then:

1. The morphism u is an epimorphism if and only if the canonical morphism of sheaves $\mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'})$ is injective.
2. The morphism u is an effective epimorphism (Exp. IV 1.3) if and only if the canonical diagram $0 \longrightarrow \mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'}) \rightrightarrows u_*(\mathcal{O}_{S'}) \otimes_{\mathcal{O}_S} u_*(\mathcal{O}_{S'})$ is exact.
3. If moreover S is noetherian and if u is an epimorphism, u is the composite of a finite sequence of effective finite epimorphisms.

We are then in a position to prove the following lemma, whose proof uses a technique of non-flat descent:

Lemma 3.7. *Let S be a locally noetherian prescheme, G an S-prescheme in groups of finite type, S' a prescheme and $u : S' \to S$ a finite morphism. For every S-prescheme T, let $M(T)$ denote the set of subgroups of multiplicative type of G_T*

(resp. the set of central subgroups of multiplicative type of G_T), so that M is in a natural way a contravariant functor defined on Sch/S. Then, if u is an effective epimorphism (resp. an epimorphism), the diagram of sets

$$(*) \qquad M(S) \to M(S') \rightrightarrows M(S' \times_S S')$$

is exact.

Proof. i) **Reduction to the case where u is an effective epimorphism.** We are then interested in the functor of central subgroups of multiplicative type of G. The injectivity of $M(S) \to M(S')$ is a local question on S, and, this injectivity being granted, the exactness of (*) becomes a local problem on S. We may therefore assume S affine noetherian.

Let us study the case where $u : S' \to S$ is the composite of two finite epimorphisms:

$$S' \xrightarrow{v} S'' \xrightarrow{w} S.$$

I claim that if the two diagrams

$$(*)' \qquad M(S'') \to M(S') \rightrightarrows M(S' \times_{S''} S')$$
$$(*)'' \qquad M(S) \to M(S'') \rightrightarrows M(S'' \times_S S'')$$

are exact, then so is (*).

Indeed the injectivity of $M(S) \to M(S')$ is clear. If now T' is an element of $\mathit{Ker}M(S') \rightrightarrows M(S' \times_S S')$, *a fortiori* T' belongs to $\mathit{Ker}M(S') \rightrightarrows M(S' \times_{S''} S')$, so by exactness of (*)′ comes from a unique element T'' of $M(S'')$. It suffices

to show that T'' belongs to $\mathit{Ker}M(S'') \rightrightarrows M(S'' \times_S S'')$, since, by exactness of (*)″, T'' will descend to S. Let T''_1 and T''_2 be the two inverse images of T'' in $M(S'' \times_S S'')$. Since these are two central subgroups of multiplicative type of $G_{S'' \times_S S''}$, to show that $T''_1 = T''_2$ it suffices to see that they have the same fibers (Exp. IX 5.1 bis). Consider the commutative diagram

$$\begin{array}{ccc} S' & \longleftarrow & S' \times_S S' \\ | & & | \end{array}$$

v   v × v
▼   ▼
S″ ←——— S″ ×_S S″.

The morphism v is a finite epimorphism, hence is dominant (property (a) above) and closed, hence is surjective, and the same is true of $v \times v$. Let x'' be a point of $S'' \times_S S''$ and x' a point of $S' \times_S S'$ above x''. It follows from commutativity of the diagram above that the two inverse images of $(T_1'')_{x''}$ and $(T_2'')_{x''}$ in $G_{x'}$ coincide, so $(T_1'')_{x''} = (T_2'')_{x''}$ (EGA IV 2.2.15).

What precedes, and an immediate induction on the number of factors of a factorization of $u : S' \to S$ into effective epimorphisms (property (c) recalled above), show that to prove the exactness of (*) in the case of central subgroups of multiplicative type, we may restrict to the case where u is an effective epimorphism. Finally, using once again Exp. IX 5.1 bis, one sees that it suffices to prove 3.7 when M is the functor of subgroups of multiplicative type of G and u an effective epimorphism.

ii) **Injectivity of** $M(S) \to M(S')$**.** Since $u : S' \to S$ is an epimorphism, the canonical

morphism $\mathcal{O}_S \longrightarrow u_*(\mathcal{O}_{S'})$ is injective; moreover an S-group of multiplicative type is flat over S; the injectivity of $M(S) \to M(S')$ is therefore a consequence of the more general following lemma:

Lemma 3.8. *Let $f : X \to S$ and $g : S' \to S$ be two morphisms of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-module, $X' = X \times_S S'$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, $Q(\mathcal{F})$ the set of quotient $\mathcal{O}_X$-modules $\mathcal{G}$ of $\mathcal{F}$ which are quasi-coherent and flat over S, $Q(\mathcal{F}')$ the analogue relative to $\mathcal{F}'$, X' and S'. Suppose g is quasi-compact and $\mathcal{O}_S \to g_*(\mathcal{O}_{S'})$ injective; then the canonical map*

Q(□) □ Q(□′)
□ ↦ □ ⊗_{□_S} □_{S′}

is injective.

Indeed, one may assume S affine, then X affine. The morphism g being quasi-compact, S' is then a union of a finite number of affine open subsets S_i'. Possibly replacing S' by $\coprod S_i'$ (an operation which preserves the injectivity of $\mathcal{O}_S \to g_*(\mathcal{O}_{S'})$), one may assume S' affine. One is then reduced to proving the following lemma, whose proof is immediate:

Lemma 3.9. *Let $A \to A'$ be an injective homomorphism of rings, M an A-module, $N = M/P$ a quotient A-module flat over A, $M' = M \otimes_A A'$, $N' = N \otimes_A A' = M'/P'$. Then P is the inverse image of P' under the canonical homomorphism $M \to M'$ (so N and P are known when one knows N' and P').*

iii) **Exactness of 3.7 (*) at** $M(S')$**.** Let T' be an element of $Ker M(S') \rightrightarrows M(S' \times_S S')$. Suppose we have proved iii) when S is the spectrum of a noetherian local ring. For every point s of S, there then exists a subgroup of multiplicative type T_s of $G \times_S \operatorname{Spec}(\mathcal{O}_{S,s})$ which comes by descent from $T' \times_{S'} u^{-1} \operatorname{Spec}(\mathcal{O}_{S,s})$. By (Exp. VI_B § 10 and 3.6), there exist a neighborhood U of s in S and a subgroup scheme T of G above U which is of multiplicative type and which extends T_s. Let U' be the inverse image of U in S'. One then knows two subschemes of $G'|U'$: $T'|U'$ and $T \times_U U'$, which coincide on $u^{-1}(\mathrm{Spec}(\mathcal{O}_{S,s}))$. If one regards $u^{-1}(\mathrm{Spec}(\mathcal{O}_{S,s}))$ as the projective limit of the schemes $u^{-1}(V)$ where V ranges over the open neighborhoods of s in U, it follows from EGA IV 8 that there exists an open neighborhood V of s in U such that $T \times_U U'$ and T' coincide above $u^{-1}(V)$. So with the hypotheses made, T' descends locally on S; but, owing to the uniqueness proved in ii), T' then descends globally on S. In short, it suffices to prove

iv) when S is the spectrum of a noetherian local ring.

Let then Ŝ denote the spectrum of the completion of the ring of S, and let $S'' = S' \times_S S'$, $\hat{S}' = \hat{S} \times_S S'$, $\hat{S}'' = \hat{S} \times_S S'' = \hat{S}' \times_{\hat{S}} \hat{S}'$. I claim it suffices to show that the diagram

$$(*) M(\hat{S}) \longrightarrow M(\hat{S}') \rightrightarrows M(\hat{S}'')$$

is exact at $M(\hat{S}')$. This follows from the commutative diagram below, in which the second row is exact at $M(\hat{S}')$ by hypothesis, the first two columns are exact (fpqc descent), and the map f is injective as follows from ii) applied to the finite epimorphism

$$\hat{S}' \times_{\hat{S}} \hat{S}' \to \hat{S} \times_S \hat{S}$$

deduced from $S' \to S$ by the flat base change $\hat{S} \times_S \hat{S} \to S$:

$$\begin{tikzcd}
M(S) \arrow[r] \arrow[d] & M(S') \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & M(S'') \arrow[d] \\
M(\hat{S}) \arrow[r] \arrow[d, shift left] \arrow[d, shift right] & M(\hat{S}') \arrow[r, shift left] \arrow[r, shift right] \arrow[d, shift left] \arrow[d, shift right] & M(\hat{S}'') \\
M(\hat{S} \times_S \hat{S}) \arrow[r, "f"] & M(\hat{S}' \times_{S'} \hat{S}') &
\end{tikzcd}$$

(the diagram-chase is left to the reader). It follows from the characterization (b) of effective finite epimorphisms that the morphism $\hat{S}' \to \hat{S}$, deduced from $S' \to S$ by the flat base change $\hat{S} \to S$, is again an effective finite epimorphism. We are therefore reduced to the case where furthermore S is the spectrum of a complete noetherian local ring.

Let S_0 denote the reduced subscheme of S whose underlying space is the closed point of S. Let T' be an element of $Ker M(S') \rightrightarrows M(S'')$, T'_0 its image in $M(S'_0)$. Since $S''_0 = S'_0 \times_{S_0} S'_0$ is faithfully flat over S_0, there exists an S-subgroup of multiplicative type T'_{S_0} of G_{S_0} whose inverse image in $G_{S''_0}$ is $T'_{S'_0}$ (fpqc descent). But S is local complete noetherian, so there exist an S-group of multiplicative type T and an S_0-morphism $u_0 : T \times_S S_0 \to T'_{S_0}$ (Exp. X 3.3). The inverse image u'_0 of u_0 above S'_0 extends uniquely to an S'-isomorphism $u' : T_{S'} \to T'$, still by Exp. X 3.3 (note that S', being finite over S local complete, is the sum of a finite number of complete local schemes). The two inverse images of u' above S'' are two morphisms of $T_{S''}$

into T'' which coincide on $S_0 \times_S S''$, so they coincide (*loc. cit.*). Since T is flat over S and $S' \to S$ is a finite effective epimorphism, it follows from TDTE I page 8 that the diagram

```
Hom_S(T, G) → Hom_{S′}(T_{S′}, G_{S′}) ⇒ Hom_{S″}(T_{S″}, G_{S″})
```

is exact. So u' comes from a morphism of preschemes $u : T \to G$. Similarly $T \times_S T$ is flat over S, and consequently the map

```
Hom_S(T ×_S T, G) → Hom_{S′}(T_{S′} ×_{S′} T_{S′}, G_{S′})
```

is injective. One deduces immediately that u is a morphism of groups, since this is so of u'. Moreover u is a monomorphism. Indeed, note first that `Ker u` is flat over S, since to establish this fact one may assume S artinian local (EGA 0_III 10.2.2), hence G separated (Exp. VI_A 0.3); but then `Ker u` is of multiplicative type (Exp. IX 6.8), hence is flat over S. Since $Ker(u) \times_S S' = Ker(u') = 0$, one has $Ker(u) = 0$ (3.8). But u being a monomorphism is an immersion (Exp. VIII, remarks 7.13), and the image group $u(T)$ is indeed an element of $M(S)$ whose image in $M(S')$ is T'. This completes the proof of 3.7.

e) End of the proof of I).

We are reduced by reduction c) to the case where S is the spectrum of a complete reduced noetherian local ring A. Let S' be the spectrum of the normalization A' of A, which is finite over A by Nagata (EGA 0_IV 23.1.5); $S' \to S$ is an epimorphism since A embeds injectively into A'. Suppose then that there exists a torus T' of $G_{S'}$ having $E_{S'}$ as underlying space. The two inverse images of T' in $G_{S' \times_S S'}$ are two central sub-tori with the same underlying space, so they coincide (Exp. IX 5 bis); hence, by 3.7, T' comes from a central sub-torus T of `G_S` which evidently has E as underlying space. It therefore suffices to prove the existence of T', which reduces us to the case where S is normal and completes the proof of I).

II) Proof of (iii) ⇒ (ii) when S is normal.

We may restrict to the case where S is integral; let t be its generic point. For every integer n equal to a power of q, let ${}_nG$ be the subprescheme of G "kernel" of the n-th power morphism in G. Since E is locally closed in G (by (iii) b)), the intersection of $ens({}_nG)$ with E is locally closed in $ens(G)$; let us then denote by $E(n)$ the reduced subprescheme of G having $ens({}_nG) \cap E$ as underlying space.

Let us show that the structural morphism $E(n) \to S$ is separated and universally open. For these two properties we have a valuative criterion (EGA II 7.2.3 and

EGA IV 14.5.8). Let then S' be an S-scheme which is the spectrum of a complete discrete valuation ring with algebraically closed residue field. We have shown that 3.1 (iii) ⇒ 3.1 (iv), so there exists a sub-torus T' of $G_{S'}$ having $E_{S'}$ as underlying space. Now ${}_nT'$ is finite and étale over S', hence separated and universally open over S', and the same is true of $E(n) \times_S S'$, which has the same underlying space as ${}_nT'$.

Moreover, the fibers of $E(n)$ have the same number of geometric points, namely

r^n if r is the rank of the torus T_t. Finally, the generic fiber $E(n)_t$, being reduced, is equal to ${}_nT_t$, hence is étale over $\mathrm{Spec}(t)$. Since S is normal, it then follows from SGA I 10.11 that $E(n)$ is an étale covering of S.

If s is a point of S, $E(n)_s$ is étale over $\mathrm{Spec}\,\kappa(s)$, hence reduced, and consequently coincides with the group of multiplicative type ${}_nT_s$. Let us show that $E(n)$ is a subgroup prescheme of G. Indeed, let m be the morphism

$$E(n) \times_S E(n) \to G$$

induced by the multiplication in G. The underlying map $ens(m)$ factors through $E(n)$, so m factors through the prescheme $E(n)$, since its source $E(n) \times_S E(n)$ is étale over S, hence reduced. It then follows from Exp. X 4.8 a) that $E(n)$ is a subgroup of multiplicative type. As already noted (3.2 a)), the family of subgroups $E(n)$ is necessarily coherent. We have therefore proved that (iii) ⇒ (ii) when S is normal.

III) Proof of (ii) ⇒ (i).

In fact we shall show that there exists a unique sub-torus T of G with underlying space equal to E and such that ${}_nT = M_n$ for every n equal to a power of q. The uniqueness of T follows simply from Exp. IX 4.8 b). To establish the existence of T, in view of uniqueness, we may successively assume:

a) S is noetherian.

b) The M_n are central subgroups. Indeed, it suffices to replace G by $Z_n = Centr_G(M_n)$ for n large enough (2.5 and 2.5 bis).

c) S is the spectrum of a local ring. Suppose indeed the problem solved after every base change $\mathrm{Spec}(\mathcal{O}_{S,s}) \to S$ where s ranges over the points of S. Let T_s be the sub-torus of $G \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$ thus obtained. For every s there then exist an open neighborhood U of s and a sub-torus T of $G|U$ extending T_s (Exp. VI_B § 10 and 3.6). We have shown that 3.1

(ii) ⇒ 3.1 (iv); since S is noetherian, E is therefore constructible (3.3). Consequently, possibly restricting U, we may assume that $ens(T) = E \times_S U$ (EGA IV 9.5.2). But then, for every integer n equal to a power of q, ${}_nT$ and $M_n \times_S U$ are two subgroups of multiplicative type of $G|U$ with the same fibers, hence which coincide, M_n being central (Exp. IX 5.3 bis). In short, with the hypotheses made, there exists a solution locally on S, hence by uniqueness, there exists a global solution.

d) S is the spectrum of a complete noetherian local ring, and if s is the closed point, T_s is trivial. This follows from EGA 0_III 10.3.1 and from fpqc descent.

e) S is reduced. One applies 2.2.

f) S is normal. One applies Nagata's finiteness theorem (EGA 0_IV 23.1.5) and Lemma 3.7 in the case of central sub-tori.

These reductions being made, since T_s is trivial, $(M_n)_s$ is trivial, so M_n is trivial (Exp. X 3.3). If t is a point of S, the subgroups ${}_nT_t$ of T_t are therefore trivial for every n equal to a power of q, and it follows easily from Exp. X 1.4 that T_t itself is trivial. It then suffices to prove the lemma:

Lemma 3.10. *Under the hypotheses of 3.1 (ii), suppose in addition that S is noetherian and normal and that for every point s of S, T_s is a trivial torus. Then there exists a unique sub-torus T of G with underlying space equal to E, such that ${}_nT = M_n$ for every n equal to a power of q. Moreover T is trivial.*

The uniqueness of T follows from the fact that T, being smooth over S, is reduced. To prove the existence, we may assume S irreducible with generic point η. Let r be the rank of T_η, $T^0 = G^r_{m,S}$, u_η an isomorphism of T^0_η onto T_η, ${}_nu_\eta$ the restriction of u_η to ${}_nT^0_\eta$. Since ${}_nT^0$ and M_n are trivial, there exists a unique extension ${}_nu$ of ${}_nu_\eta$ to an S-isomorphism of ${}_nT^0$ onto M_n. I claim that for every point s of S there exists a group isomorphism, necessarily unique:

$$u_s : T^0_s \xrightarrow{\sim} T_s$$

extending ${}_nu_s$ for every n equal to a power of q. Indeed, let S_1 be an S-scheme,

spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point t_1 projects onto η and whose closed point s_1 projects onto s (EGA II 7.1.9). It follows from the proof of 3.1 (ii) ⇒ 3.1 (iv) that there exists a sub-torus T_1 of G_{S_1} such that $(M_n)_{S_1} = {}_nT_1$ for every n equal to a power of q. Since S_1 is normal, T_1 is isotrivial (Exp. X 5.16), and it follows from the classification of isotrivial tori (Exp. X 1.2) and from SGA 1 V 8.2 that T_1, having trivial generic fiber, is trivial. One may therefore extend the isomorphism $u_\eta \times_\eta t_1$ to an S_1-isomorphism of group schemes

$$u_1 : T^0 \times_S S_1 \xrightarrow{\sim} T_1.$$

The restriction of u_1 to ${}_nT^0 \times_S S_1$ on the one hand, and ${}_nu \times_S S_1$ on the other, coincide on the generic fiber by construction, hence coincide. The restriction u_{1,s_1} of u_1 to the closed fiber thus realizes the desired extension after the residue field extension $\kappa(s) \to \kappa(s_1)$. It then follows from Exp. IX 4.8 a) and from fpqc descent that u_{1,s_1} descends to $\kappa(s)$ to a group isomorphism $u_s : T^0_s \xrightarrow{\sim} T_s$ answering the question.

Moreover, T^0 is smooth over S which is normal, hence is normal. It then follows from an easy extension criterion for rational maps (EGA IV4 20.4.6) that for the existence of an S-morphism $u : T^0 \to G$ whose restriction to T^0_s, for every point s of S, is the composite morphism

$$T^0_s \xrightarrow{u_s} T_s \hookrightarrow G_s,$$

it is necessary and sufficient that this be the case after every base change $S_1 \to S$ where S_1 is the spectrum

of a complete discrete valuation ring with algebraically closed residue field.

Now in the present case, a reasoning analogous to the one just made shows that the extension condition is indeed satisfied. Let u denote the S-morphism $T^0 \to G$ thus obtained.

Let us show that u is indeed a morphism of groups. Let m_{T^0} (resp. m_G) be the morphism defining the multiplication in T^0 (resp. G). We must verify that $m_G \circ (u \times_S u) = u \circ m_{T^0}$. Now the subprescheme of coincidences of these two morphisms is a subprescheme of $T^0 \times_S T^0$, which majorizes the fibers (since u_s is a group morphism), so has the same underlying space as $T^0 \times_S T^0$, hence is equal to it, since $T^0 \times_S T^0$ is smooth over S, hence reduced.

Finally, note that u is a monomorphism (since this is so on the fibers), hence is an immersion (Exp. VIII 7.9). The image of T^0 by u is then a sub-torus of G having E as underlying set.

This completes the proof of Theorem 3.1.

4. Characterization of a sub-torus T by the subgroups ${}_nT$

4.1. Statement of the main theorem

Theorem 4.1. *Let S be a locally noetherian connected prescheme, G an S-prescheme in groups of finite type over S, q an integer $>$ 1 invertible on S, r a positive integer. For every integer n equal to a power of q, let $M(n)$ be a subgroup scheme of G, of multiplicative type and of type $(\mathbb{Z}/n\mathbb{Z})^r$. Suppose:*

- *a) The family of subgroups $M(n)$ is coherent, that is, if the integer m divides n, one has*

$$ {}_mM(n) = M(m). $$

- *b) There exists a point s of S and a sub-torus T_s of G_s such that*

$M(n)_s = {}_nT_s$ for every n.

- *c) For every point t of S, there exists a closed affine subscheme F_t of G_t majorizing $M(n)_t$ for every n.*

Then there exists one and only one sub-torus T of G such that ${}_nT = M(n)$ for every n equal to a power of q.

One has an analogous theorem concerning the lifting of morphisms:

Theorem 4.1 bis. *Let S, G and q be as above, T an S-torus, and for every integer n equal to a power of q, let $u(n)$ be an S-group morphism ${}_nT \to G$. Suppose:*

- *a) The family of morphisms $u(n)$ is coherent, i.e., if m divides n, one has*

$$ u(m) = u(n)|_{{}_mT}. $$

- *b) There exists a point s of S and a group morphism $u_s : T_s \to G_s$ such that $u_s|_{{}_nT_s} = u(n)_s$ for every n equal to a power of q.*
- *c) For every point t of S, there exists a closed affine subscheme F_t of G_t majorizing $u(n)_t({}_nT_t)$ for every n.*

Then there exists a unique group morphism $u : T \to G$ such that for every n equal to a power of q, the restriction of u to ${}_nT$ is equal to $u(n)$.

Remark 4.2. Using the lower semicontinuity of the abelian rank of a flat group prescheme of finite type over the spectrum of a discrete valuation ring (cf. Exp. X 8.7), one can, in the statements of 4.1 and 4.1 bis, weaken condition c) by simply requiring that the required closed affine subscheme F_t exists for every maximal point t of S.

Let us show how 4.1 bis follows from 4.1. Let $G' = G \times_S T$. For every integer n equal to a power of q, consider the group morphism

$$v(n) : {}_nT \longrightarrow G'$$

whose projections to G and T are respectively $u(n)$ and the canonical immersion ${}_nT \to T$. The morphism $v(n)$ is therefore an immersion; let $M(n)$ be the image subgroup. It is clear that the family of subgroups $M(n)$ is coherent in the sense of 4.1, that the group $M(n)_s$ is equal to

${}_nT'_s$, where T'_s is the sub-torus of G'_s which is the graph of u_s, and that for every point t of S, the closed affine subscheme $F_t \times_t T_t$ of G'_t majorizes the subgroups $M(n)_t$ for every n. By 4.1, there therefore exists a sub-torus T' of G' such that ${}_nT' = M(n)$ for every n equal to a power of q. Let f be the restriction to T' of the projection of G' to T, and $f(n)$ the restriction of f to $M(n)$. One has

$$f(n) \circ v(n) = id_{{}_nT}.$$

The fiber at s of T' is the torus already denoted T'_s, equal to the graph of u_s (this follows from Exp. IX 4.8 b)), so f_s is an isomorphism. But `Ker f` and `Coker f` are groups of multiplicative type (Exp. IX 2.7) of constant type, S being connected, hence reduced to the unit group, and f is an isomorphism. Let v be the inverse isomorphism of f. One has

$$v|_{{}_nT} = v(n).$$

Consequently, the composite of v and the projection of G' onto G is a morphism $u : T \to G$ answering the question. The foregoing proves the existence of the morphism u; as for uniqueness, it follows in any case from Exp. IX 4.8 a).

4.2. Application

We propose to generalize Theorem 7.1 of Exp. IX.

Let A be a complete noetherian local ring, I its maximal ideal,

$S = \operatorname{Spec} A$, $S_m = \operatorname{Spec}(A/I^m)$. For every prescheme X, set $X_m = X \times_S S_m$.

Let then G be an S-prescheme in groups of finite type, T an S-torus, q an integer invertible on S, and $u_m : T_m \to G_m$ ($m \in \mathbb{N}$) a coherent family of group morphisms. With n ranging over the powers of q, denote by $u_m(n)$ the restriction of u_m to ${}_nT_m$, and by $u(n)$ the unique group morphism

$$u(n) : {}_nT \longrightarrow G$$

extending the morphisms $u_m(n)$ for every m (1.6 a)). We shall say that the family (u_m), $m \in \mathbb{N}$, is *admissible* if for every point t of S there exists a closed affine subprescheme F_t of G_t majorizing $u(n)_t({}_nT_t)$ for every n equal to a power of q (this property is independent of q, as we shall see).

Proposition 4.3. *With the notation above, the canonical map*

$$\mathrm{Hom}_{S\text{-gr}}(T, G) \to \varprojlim_{m} \mathrm{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

induces an isomorphism of the source onto the subset of the target consisting of "admissible" coherent families.

Indeed, it suffices to apply 4.1 bis, taking for s the closed point of S.

Corollary 4.4. *With the notation above, suppose in addition that G has affine fibers; then the canonical map*

$$\mathrm{Hom}_{S\text{-gr}}(T, G) \to \varprojlim_{m} \mathrm{Hom}_{S_m\text{-gr}}(T_m, G_m)$$

is an isomorphism.

Indeed, if G has affine fibers, every coherent family is "admissible".

Remark 4.5. When G is separated, one may in 4.3 replace the ring A by an I-adic noetherian ring; one may indeed use EGA III 5.4.1 instead of 1.6 a).

4.3. Proof of 4.1

Lemma 4.6. *Let k be a field, G a k-algebraic group, q an integer prime to the residue characteristic of k, r an integer > 0, $M(n)$ (n ranging over the powers of q) a coherent family of subgroups of G, of multiplicative type and of type $(\mathbb{Z}/n\mathbb{Z})^r$. Then there exists a smallest closed subscheme H of G majorizing the $M(n)$ for every n. Moreover, H is a smooth, connected and commutative algebraic subgroup of G "whose formation is compatible with base field extension".*

Let M be the subset of $ens(G)$ which is the union of the underlying sets of the subgroups $M(n)$ for every n, and H the reduced closed subscheme of G having the closure of M as underlying space. Since $M(n)$ is étale over k hence reduced, $M(n)$ is contained in H, and consequently H is the smallest closed subscheme of G majorizing $M(n)$ for every n. Now let $\bar{k}$ be an algebraically closed extension of k. By

construction the subschemes $M(n)$ are schematically dense in H (Exp. IX 4.1); the $M(n)_{\bar{k}}$ are therefore schematically dense in $H_{\bar{k}}$ (Exp. IX 4.5); moreover $M(n)_{\bar{k}}$ is reduced, and it follows that $H_{\bar{k}}$ is necessarily equal to the closed reduced subscheme of $G_{\bar{k}}$ having as underlying space the closure of $M_{\bar{k}}$. This proves that H is geometrically reduced (EGA IV 4.6.1). The family $M(n)$ being coherent, M is stable under the group law; moreover $M \times_k M$ is dense in $ens(H \times_k H)$ and $H \times_k H$ is reduced (EGA IV 4.6.5); one immediately deduces that H is an algebraic subgroup of G. Moreover H is smooth over S, since it is geometrically reduced (Exp. VI_A 1.3.1), and H is commutative since the $M(n)$ are commutative. It remains to see that H is connected. Let H^0 be the connected component of H, m the number of geometric points of H/H^0, q^s the exponent of q in the decomposition of m into prime factors. For every integer n equal to a power of q, ${}_{q^s}M(n)$ is then contained in H^0. But the family $M(n)$ is coherent and $M(n)$ is of type $(\mathbb{Z}/n\mathbb{Z})^r$, so ${}_{q^s}M(nq^s) = M(n)$. Hence H^0 majorizes $M(n)$ for every n and consequently equals H.

This being so, we shall make condition c) of 4.1 more precise. Indeed, by what precedes, we may consider the smallest closed subscheme H_t of G_t majorizing $M(n)_t$ for every n. The formation of H_t commuting with base field extension (4.6), $H_t \otimes_t \bar{t}$ is contained in the affine closed subset F_t, hence is affine. In short, H_t is affine. On the other hand, we know (4.6) that H_t is a smooth, connected, commutative algebraic group. It then follows from the structure of smooth, commutative, connected affine algebraic groups over an algebraically closed field (*Bible* 4 Th. 4),

that H_t is a direct product of a torus T_t and a unipotent group U_t. But then $M(n)_t$ is necessarily contained in T_t, so $H_t = T_t$, and consequently H_t is a torus.

We can now begin the proof of 4.1.

a) **Uniqueness of the solution:** it suffices to apply Exp. IX 4.8 b).

b) **Case where S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field k.** Denote by K the field of fractions of A, s the closed point of S, t the generic point, J the maximal ideal of A, $S_m = \mathrm{Spec}(A/J^m)$, $X_m = X \times_S S_m$.

Let us distinguish two cases:

1st case: The point s of 4.1 b) is the generic point t of S. Let then T' be the schematic closure in G of the torus T_t. The closed fiber T'_s is therefore an algebraic subgroup of G_s, of dimension r, majorizing $M(n)_s$ for every n, hence T'_s majorizes H_s. But H_s is a torus containing $M(n)_s$, so H_s has rank at least r. Consequently H_s has the same underlying space as the connected component of T'_s. The "connected component" $(T')^0$ of T' is then a subgroup prescheme of G, flat, separated (VI_B 5.2), whose generic fiber T_t is a torus and whose reduced closed fiber H_s is a torus. But then $(T')^0$ is a torus (Exp. X 8.8) which we denote by T. The groups ${}_nT$ and M_n

are smooth over S, hence reduced, and since they have the same underlying space, they coincide. So the torus T is the solution of the problem.

2nd case: The point s of 4.1 b) is the closed point s of S. Possibly replacing G by

the schematic closure in G of the smallest algebraic subgroup H_t majorizing the family $M(n)_t$ (4.6), we may assume G_t affine.

For every integer $m \geqslant 0$, it follows from 2.2 that there exists a unique sub-torus T_m of G_m lifting T_s and such that for every integer n equal to a power of q, one has ${}_nT_m = M(n)_m$. Moreover, let $T^0 = G^r_{m,S}$. Since k is algebraically closed, T_s is trivial, and there exists a k-isomorphism $u_s : T^0_s \xrightarrow{\sim} T_s$. The morphism u_s lifts uniquely to an S_m-isomorphism $u_m : T^0_m \to T_m$ (Exp. IX 3.3). The family of morphisms u_m, $m \in \mathbb{N}$, defines in the limit a morphism $\hat{u}$ of formal completions $\hat{T}^0$ and $\hat{G}$ of T^0 and G along their closed fibers:

$$\begin{tikzcd}
\hat{T}^0 \arrow[r, "\hat{u}"] \arrow[d, "i"] & \hat{G} \arrow[d, "j"] \\
T^0 & G,
\end{tikzcd}$$

where i and j denote the canonical morphisms.

We shall show that the morphism $\hat{u}$ is algebraizable. For this we shall reduce to the case where the group G is affine.

Lemma 4.7. *Let S be the spectrum of a discrete valuation ring A, X and Y two S-preschemes, quasi-compact, quasi-separated and flat over S. Then the canonical map*

$$\Gamma(X, \mathcal{O}_X) \otimes_A \Gamma(Y, \mathcal{O}_Y) \to \Gamma(X \times_S Y, \mathcal{O}_{X \times_S Y})$$

is an isomorphism.

Let $f : X \to S$ (resp. $g : Y \to S$) be the structural morphism. Since X (resp. Y)

is quasi-compact and quasi-separated, it follows from EGA I 9.2.2 and from EGA IV 1.7.4 that $f_*(\mathcal{O}_X)$ (resp. $g_*(\mathcal{O}_Y)$) is a quasi-coherent $\mathcal{O}_S$-algebra, which therefore corresponds to an affine S-scheme $\tilde{X}$ (resp. $\tilde{Y}$). By hypothesis Y is flat over S; it then follows from EGA III 1.4.15, in view of EGA IV 1.7.21, that the canonical map (deduced from the canonical morphism $X \to \tilde{X}$)

$$\Gamma(\tilde{X} \times_S Y, \mathcal{O}_{\tilde{X} \times_S Y}) \to \Gamma(X \times_S Y, \mathcal{O}_{X \times_S Y})$$

is an isomorphism. But X being flat over S, $\tilde{X}$ is flat over S (since flat over A is equivalent to torsion-free). Applying once again EGA III 1.4.15 with the roles of X and Y exchanged, one obtains an isomorphism

$$\Gamma(\tilde{X}, \mathcal{O}_{\tilde{X}}) \otimes_A \Gamma(\tilde{Y}, \mathcal{O}_{\tilde{Y}}) \simeq \Gamma(\tilde{X} \times_S \tilde{Y}, \mathcal{O}_{\tilde{X} \times_S \tilde{Y}}) \to \Gamma(\tilde{X} \times_S Y, \mathcal{O}_{\tilde{X} \times_S Y}),$$

whence the lemma.

In the present case, the S-group G is flat over S and of finite type, hence quasi-compact and quasi-separated. One may therefore apply the lemma to $G \times_S G$:

$$\Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G) \xrightarrow{\sim} \Gamma(G \times_S G, \mathcal{O}_{G \times_S G}).$$

To the morphism $m_G : G \times_S G \to G$ defining the multiplication in G there corresponds therefore a morphism

$$\Gamma(G, \mathcal{O}_G) \to \Gamma(G \times_S G, \mathcal{O}_{G \times_S G}) \xrightarrow{\sim} \Gamma(G, \mathcal{O}_G) \otimes_A \Gamma(G, \mathcal{O}_G),$$

hence an S-morphism

$$m_{\tilde{G}} : \tilde{G} \times_S \tilde{G} \to \tilde{G},$$

where $\tilde{G}$ denotes the affine S-scheme having $\Gamma(G, \mathcal{O}_G)$ as A-algebra. It is formal, from there, to verify that $m_{\tilde{G}}$ endows $\tilde{G}$ with a structure of S-group scheme such that the canonical morphism $v : G \to \tilde{G}$ is an S-group morphism.

Remark 4.8. $\tilde{G}$ plays the role of a largest "affine quotient" of G; moreover one can show that $\tilde{G}$ is indeed a quotient of G for fpqc, hence is of finite type over S (XVII App. III, 2).

In the case at hand, the generic fiber G_t of G is affine; it then follows from EGA I 9.3.3 that v_t is an isomorphism. Since $\tilde{G}$ is affine, the coherent family of morphisms

$$w_m = v_m u_m : T^0_m \to \tilde{G}_m \qquad (m \in \mathbb{N})$$

comes from a unique S-group morphism (Exp. IX 7.1)

$$w : T^0 \longrightarrow \tilde{G}.$$

Let T_t be the sub-torus of G_t equal to $v_t^{-1} w_t(T_t^0)$. The torus T_t is therefore of rank at most r (as the image of a torus of rank r). Let us show that T_t majorizes $M(n)_t$ for every n. Indeed, let $u(n)_m$ be the S_m-isomorphism $({}_nT^0)_m \xrightarrow{\sim} M(n)_m$ obtained by restriction of u_m to $({}_nT^0)_m$. The coherent family of morphisms $u(n)_m$ comes from a unique S-isomorphism $u(n) : {}_nT^0 \xrightarrow{\sim} M(n)$ (since $M(n)$ is finite over S). For every integer $m \geqslant 0$, one then has the equalities

$$w_m|_{({}_nT^0)_m} = (v_m u_m)|_{({}_nT^0)_m} = (v \circ u(n))_m.$$

Consequently, $w|_{{}_nT^0} = v \circ u(n)$ (1.6 a)). In particular, one has $w_t|_{({}_nT^0)_t} = v_t \circ u(n)_t$, so $v_t^{-1} w_t({}_nT^0)_t = u(n)_t({}_nT^0)_t = M(n)_t$.

This indeed proves that T_t majorizes $M(n)_t$, and entails that T_t is of rank r. One concludes

as in the first case already studied, by considering the schematic closure in G of T_t, namely T''. Since T_t majorizes $M(n)_t$, T''_s majorizes $M(n)_s$, hence majorizes T_s (density theorem). On the other hand T'' is flat over S and T_t is of dimension r, so T''_s is of dimension r (Exp. VI_B 4). In short, T_s has the same underlying space as the connected component of T''_s, and one concludes as in the first case that the connected component of T'' is a sub-torus of G answering the question.

c) **End of the proof of 4.1.**

To prove the existence of the sub-torus T, we may assume S reduced (2.2). In view of 3.1 (ii) ⇒ (i), it then suffices to prove that the set U of points x of S such that there exists a sub-torus T_x of G_x with ${}_nT_x = M(n)_x$ for every n equal to a power of q, is equal to $ens(S)$. The torus T_x in question is necessarily unique, and by fpqc descent, it suffices to prove its existence after extension of the residue field of x (Exp. IX 4.8 b)).

This being so, since S is locally noetherian and connected and since U is non-empty (it contains the point s of 4.1 b)), one is reduced, by an immediate argument, to proving that if x and x' are two points of S, with x a specialization of x', and if one of the two points belongs to U, then both points are in U. By the usual technique (EGA II 7.1.9) one reduces to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, a case which has been treated in b).

This completes the proof of Theorem 4.1.

5. Representability of the functor of smooth subgroups identical to their connected normalizer

Proposition 5.1. *Let S be a prescheme, G an S-prescheme in groups of finite presentation, H a subgroup prescheme of G, smooth with connected fibers. Then:*

- *a) The normalizer N of H in G is representable by a closed subprescheme of G of finite presentation over S.*
- *b) The following conditions are equivalent:*
    - *i) The canonical immersion $H \to N$ is an open immersion.*
    - *ii) The group N is smooth along the unit section, and its connected component, which is then representable (Exp. VI_B 3.10), is equal to H.*
    - *iii) For every point s of S, one has $H_s = (N_s)^0$.*

Proof. The group prescheme H is locally of finite presentation over S (H is smooth over S) and has connected fibers, hence is of finite presentation over S (Exp. VI_B 5.3.3). Assertion a) is then a consequence of Exp. XI 6.11. The equivalence of the conditions appearing in b) is included here for the record and was proved in VI_B 6.5.1.

This being so, we can state the main theorem of this section:

Theorem 5.2. *Let S be a prescheme, G an S-prescheme in groups of finite presentation over S, $\mathcal{L}_G$ (or simply $\mathcal{L}$ if there is no ambiguity) the S-functor such that for every S-prescheme S' one has:*

> *$\mathcal{L}_G(S')$ = set of subgroup preschemes of $G_{S'}$, smooth over S', with connected fibers, which are identical to their connected normalizer.*

Then the functor $\mathcal{L}$ is representable by an S-prescheme, a union of an increasing family $(U_i)_{i\in\mathbb{N}}$ of open subpreschemes, quasi-projective and of finite presentation over S, hence a fortiori separated over S.

Initial reductions.

For every integer $r \geqslant 0$, let $\mathcal{L}^r$ be the subfunctor of $\mathcal{L}$ such that for every S-prescheme S' one has:

> $\mathcal{L}^r(S')$ = *set of subgroup preschemes of $G_{S'}$, smooth, with connected fibers, identical to their connected normalizer and of relative dimension r.*

Since the dimension of the fibers of a smooth group is a locally constant function, the canonical monomorphism $\mathcal{L}^r \to \mathcal{L}$ is representable by an immersion both open and closed. It therefore suffices to show that for every r, $\mathcal{L}^r$ is representable by an S-prescheme having the properties stated above, since $\mathcal{L}$ will then be representable by the S-prescheme sum $\coprod_{r\in\mathbb{N}} \mathcal{L}^r$.

For every integer $n \geqslant 0$, every S-prescheme S' and every subgroup prescheme H of $G_{S'}$, we denote by $H^{(n)}$ the n-th normal invariant of the unit section $S' \to H$ of H (EGA IV 16.1.2), so that $H^{(n)}$ is a sheaf of $\mathcal{O}_{S'}$-modules of finite type corresponding to the n-th infinitesimal neighborhood of the unit section of H. If H is smooth over S' of relative dimension r, $H^{(n)}$ is a locally free $\mathcal{O}_S$-module whose rank $\phi(n,r)$ depends only on n and r. Moreover, since H is a subprescheme of $G_{S'}$, one has a canonical epimorphism, compatible with base extension:

$$G^{(n)} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \simeq G_{S'}^{(n)} \to H^{(n)}.$$

Introduce then the projective S-scheme

$$P_{\phi(n, r)} = \mathrm{Grass}_{\phi(n, r)}(G^{(n)})$$

(EGA I 2nd ed. 9.7; cf. also Séminaire Cartan 1960/61, Exp. N° 14 by A. Grothendieck). It then follows from the preceding remarks that the map

$$H \mapsto H^{(n)}$$

defines a canonical morphism

$$u_{n, r} : \mathcal{L}^r \to P_{\phi(n, r)}.$$

The group G acts in a natural way on $G^{(n)}$, hence on $P_{\phi(n,r)}$, by means of the representation

$$\mathrm{int} : G \to \mathrm{Aut}_{S\text{-gr}}(G), \qquad g \mapsto \mathrm{int}(g).$$

Moreover, if S' is a quasi-compact S-prescheme and H an element of $\mathcal{L}^r(S')$, one knows

(Exp. XI 6.11) that for n large enough,

$$N = Norm_{G_{S'}}(H) = Norm_{G_{S'}}(H^{(n)}).$$

For each integer $n \geqslant 0$, introduce the subfunctor $\mathcal{L}^r_n$ of $\mathcal{L}^r$ such that for every S-prescheme S' one has:

> *$\mathcal{L}^r_n(S')$ = set of subgroups H of $G_{S'}$ belonging to $\mathcal{L}^r(S')$ such that $Norm_{G_{S'}}(H) = Norm_{G_{S'}}(H^{(n)})$.*

Representability of $\mathcal{L}^r_n$.

Since the integer r is fixed until the end of the proof of 5.2, we shall not repeat it in the notation. Thus we shall write $\mathcal{L}$, $\mathcal{L}_n$, P_n, u_n instead of $\mathcal{L}^r$, $\mathcal{L}^r_n$, $P_{\phi(n,r)}$, $u_{n,r}$.

Let v_n be the restriction of u_n to the subfunctor $\mathcal{L}_n$ of $\mathcal{L}$. It follows from the definition of $\mathcal{L}_n$ and from 5.1 b) ii) that v_n is a monomorphism. In fact one has the following lemma:

Lemma 5.3. *The morphism v_n is an immersion of finite presentation. A fortiori, $\mathcal{L}_n$ is representable by an S-prescheme, quasi-projective and of finite presentation over S.*

Possibly replacing S by P_n, we are reduced by the usual technique to proving the following assertion: Let $Q \in P_n(S)$ and consider the subfunctor F of the functor h_S represented by the final object S of Sch/S, such that for every S-prescheme S' one has:

- *$F(S') = h_S(S')$ (set with one element) if there exists $H \in \mathcal{L}_n(S')$ such that $H^{(n)} = Q_{S'}$,*
- *$F(S') = \emptyset$ otherwise.*

Then the canonical monomorphism $F \to S$ is an immersion of finite presentation.

Let us first transform the definition of the functor F. For this, note that the normalizer of Q in G is representable by a subgroup prescheme N of finite presentation over S (namely the inverse image of the point Q of $P_n(S)$ under the morphism $G \to P_n$, $g \mapsto g \cdot Q$). I claim that the functor F coincides with the following subfunctor of h_S:

> *$F(S') = h_S(S')$ if:*
>
> - *(a) $N_{S'}$ is smooth along the unit section and of relative dimension r,*
> - *(b) $(N_{S'})^{(n)}$ (which is then canonically an element of $P_n(S')$) is equal to $Q_{S'}$.*
>
> *$F(S') = \emptyset$ otherwise.*

Indeed, denote temporarily by `F_1` the functor F in the first formulation, and `F_2` the functor F in the second formulation. Then:

i) $F_1(S') = h_S(S') \Rightarrow F_2(S') = h_S(S')$.

Indeed, let $H \in \mathcal{L}_n(S')$ be the subgroup of $G_{S'}$ such that $H^{(n)} = Q_{S'}$. Then

```
Norm_{G_{S′}}(H) = Norm_{G_{S′}}(H^{(n)}) = Norm_{G_{S′}}(Q_{S′}) = N_{S′}.
```

So by 5.1 b) ii), $N_{S'}$ is smooth along the unit section and its connected component is H. Consequently $N_{S'}$ is of relative dimension r; and since H is open in $N_{S'}$ (by 5.1 b) i)), one has $(N_{S'})^{(n)} = H^{(n)} = Q_{S'}$. In short, $F_2(S') = h_S(S')$.

ii) $F_2(S') = h_S(S') \Rightarrow F_1(S') = h_S(S')$.

By hypothesis, $N_{S'}$ is smooth along the unit section, of relative dimension r; its connected component is then representable (Exp. VI_B 3.10) by a subgroup prescheme H, smooth over S', with connected fibers of dimension r. Since H is invariant in $N_{S'}$ and open in $N_{S'}$, one has the following inclusions:

```
N_{S′} ⊂ Norm_{G_{S′}}(H) ⊂ Norm_{G_{S′}}(H^{(n)}) = Norm_{G_{S′}}((N_{S′})^{(n)}) = Norm_{G_{S′}}(Q_
```

These inclusions are therefore equalities. The first inclusion then shows that H is equal to its connected normalizer, and the second shows that H is an element of $\mathcal{L}_n(S')$. This says that $F_1(S') = h_S(S')$.

Implications i) and ii) entail $F_1 = F_2$. We keep the second definition of the functor F, and we shall first "represent condition a)" by an immersion of finite presentation. For this it suffices to apply the:

Lemma 5.4. *Let S be a prescheme, X an S-prescheme locally of finite presentation over S, $\sigma : S \to X$ a section of X, r an integer $\geqslant 0$, and $L : (Sch/S)^\circ \to (Ens)$ the subfunctor of S defined as follows:*

- *$L(S') = h_S(S')$ if $X_{S'}$ is smooth along the section $\sigma_{S'}$ and of relative dimension r at the points of $\sigma_{S'}(S')$.*
- *$L(S') = \emptyset$ otherwise.*

Then:

- *a) The monomorphism $L \to S$ is an immersion of finite presentation.*
- *b) Let $\mathcal{J}$ be the conormal sheaf relative to the immersion $S \to X$ (EGA IV 16.1.2); assume that for every point s of S, $\mathcal{J} \otimes_{\mathcal{O}_{S,s}} \kappa(s)$ is of rank at most r. Then the immersion $L \to S$ is a closed immersion.*

Proof. The functor L is of local nature on S, which reduces us to the case where S is affine. Possibly replacing X by a neighborhood of $\sigma(S)$, we may assume X of finite presentation over S, then (EGA IV 8.9) S noetherian (noting, in case b), that the formation of the conormal sheaf commutes with base extension (EGA IV 16.6.4) and that the rank of the fibers of $\mathcal{J}$ is a constructible function on S). This being so, for every S-prescheme S', denote $\mathcal{J}_0 = \mathcal{J}_{S'}$ the conormal sheaf relative to the section $\sigma_{S'}$, $S_\bullet(\mathcal{J}_0)$ (canonically isomorphic to $S_\bullet(\mathcal{J})_{S'}$) the symmetric algebra of $\mathcal{J}_0$ over $\mathcal{O}_{S'}$, $Gr_\bullet(\sigma_{S'})$ the sheaf of graded $\mathcal{O}_{S'}$-algebras associated to $\sigma_{S'}$ (EGA IV 16), and for every integer $n \geqslant 0$ let $\sigma_{n,S'} : S_n(\mathcal{J}_0) \to Gr_n(\sigma_{S'})$ be the canonical epimorphism.

It follows from EGA IV 17.12.3 and from EGA 0_IV 19.5.4 that, for $X_{S'}$ to be smooth along the section $\sigma_{S'}$ and of relative dimension r at the points of $\sigma_{S'}(S')$,

it is necessary and sufficient that:

- i) $\mathcal{J}_0$ be a locally free $\mathcal{O}_{S'}$-sheaf of rank r.
- ii) For every integer $n \geqslant 1$, $\sigma_{n,S'}$ be an isomorphism.

Now it follows from TDTE IV, Lemma 3.6, that "the functor that makes $\mathcal{J}$ locally free of rank r" is representable by a subprescheme S_1, closed in S in case b). Replacing S by S_1, we are reduced to the case where $\mathcal{J}$ is locally free of rank r. Let us then proceed by induction on the integer n. Suppose that we have represented by a closed subprescheme S_{n-1} of S the "functor that makes the morphisms $\sigma_{q,S'}$ injective for every integer $q \leqslant n-1$", and let us show that the "subfunctor[1118] that makes $\sigma_{n,S'}$ injective" is representable by a closed subprescheme S_n of S_{n-1}. Replacing S by S_{n-1}, we may assume that $\sigma_{q,S}$ is bijective for $q \leqslant n-1$. But then the $(n-1)$-st normal invariant $X^{(n-1)}$ relative to the section σ_s admits a composition series $X^{(0)}, \cdots, X^{(n-1)}$ whose successive quotients $Gr_0(\sigma_s), \cdots, Gr_{n-1}(\sigma_s)$ are locally free over $\mathcal{O}_{S'}$, hence flat; so $X^{(n-1)}$ is flat over $\mathcal{O}_S$. Since the formation of $X^{(i)}$ commutes with base extension (EGA IV 16), one has, for every S-prescheme S':

`Gr_n(σ_{S′}) = Ker(X_{S′}^{(n)} → X_{S′}^{(n-1)}) = (Gr_n(σ))_{S′}    and    σ_{n, S′} = (σ_{n, S})_{S`

So the functor that interests us is "the one that makes the morphism $\sigma_{n,S}$ injective".

This functor is of local nature on S, which allows us to assume S affine and $S_n(\mathcal{J})$ free over $\mathcal{O}_S$. But then it is clear that the functor in question is representable by the closed subscheme of S defined by the ideal generated by the coordinates of $Ker\sigma_{n,S}$ with respect to a basis of $S_n(\mathcal{J})$.

Since S is noetherian, the decreasing sequence S_n of closed subpreschemes of S_1 is stationary, and the stationary value represents the functor L, which completes the proof of 5.4.

Let us return to the question of representability of $\mathcal{L}_n$. Replacing S by a suitable subprescheme S_1, we may therefore assume N smooth along the unit section and of relative dimension r. The functor $\mathcal{L}_n$ is then the "functor of coincidences" of two sections of P_n above S, h and g, corresponding to the sheaves Q and $N^{(n)}$ (condition b) appearing in the definition of the functor F_2 above). It is therefore representable by the closed subprescheme of S, inverse image of the diagonal of $P_n \times_S P_n$ by the morphism of finite presentation $h \times_S g$. This completes the proof of 5.3.

Study of the transition morphisms $\mathcal{L}_n \to \mathcal{L}_m$.

If a subgroup H of $G_{S'}$ belongs to $\mathcal{L}_n(S')$, it belongs *a fortiori* to $\mathcal{L}_m(S')$ for every $m \geqslant n$, whence natural monomorphisms

`u^m_n : □_n □ □_m    for m □ n.`

Lemma 5.5. *The morphism u_n^m is an open immersion.*

Possibly changing S, we are reduced to the following problem: Let $H \in \mathcal{L}_m(S)$,

[1118] We have replaced "functor" by "subfunctor". Should one rather write "which makes $\sigma_{q,S'}$ injective for $q \leqslant n$"?

$$N = Norm_G(H) = Norm_G(H^{(m)}), \qquad N' = Norm_G(H^{(n)}),$$

and let $D : (Sch/S)^{\circ} \to (Ens)$ be the functor of coincidences of N and N', defined by $D(S') = h_S(S')$ if $N_{S'} = N'_{S'}$, and $D(S') = \emptyset$ otherwise. We must show that $D \to S$

is an open immersion. Now I claim that D is also the subfunctor of S which "makes the immersion $H \to N'$ open". Indeed, if $N_{S'} = N'_{S'}$, then $H_{S'} \to N'_{S'}$ is indeed an open immersion since this is so for $H_{S'} \to N_{S'}$ (Prop. 5.1). Conversely, if $H_{S'} \to N'_{S'}$ is an open immersion, H having connected fibers, $H_{S'}$ is the connected component of $N'_{S'}$ (Exp. VI_B 3.10) and consequently is invariant in $N'_{S'}$, so $N'_{S'} \subset N_{S'}$. Since in any case N' majorizes N, one has $N_{S'} = N'_{S'}$. The group preschemes H and N' are of finite presentation over S and H is flat over S; the fact that $D \to S$ is an open immersion then follows from Exp. VI_B 2.6.

End of the proof of 5.2.

The functors $\mathcal{L}_n$ being representable and the transition morphisms u_n^m being compatible with one another and representable by open immersions, there exists an S-prescheme X, union of an increasing sequence of open subsets $(X_i)_{i\in\mathbb{N}}$, such that X_i represents the functor $\mathcal{L}_i$, and such that if one identifies $\mathcal{L}_i$ with X_i, the inclusion $X_i \to X_j$ $(j \geqslant i)$ is identified with u_i^j. To conclude that X represents the functor $\mathcal{L}$, it then suffices to remark that in the category of sheaves on Sch/S equipped with the Zariski topology (Exp. IV 6.1) one has

$$X = \varinjlim X_i \quad \text{and} \quad \mathcal{L} = \varinjlim \mathcal{L}_i.$$

Remark 5.6. With the preceding notation, suppose in addition that S has all residue characteristics zero. Then, for every integer $r \geqslant 0$, the functor $\mathcal{L}^r$ is equal to $\mathcal{L}_1^r$, hence is representable by an S-prescheme of finite presentation and quasi-projective over S. Indeed, it suffices to show that if $H \in \mathcal{L}^r(S')$, the canonical immersion

$$H \longrightarrow N = Norm_{G_{S'}}(H^{(1)})$$

is an open immersion (since this entails $N = Norm_{G_{S'}}(H)$, hence $H \in \mathcal{L}_1^r(S')$). Since H is flat over S, and H and N are of finite presentation over S, it suffices (Exp. VI_B 2.6) to show that for every point s of S, $H_s \to N_s$ is an open immersion. Now it follows easily[1119] from Cartier's theorem (Exp. VI_B 1.6) that if G is an algebraic group over a field of characteristic 0 and H is a connected algebraic subgroup, one has

$$Norm_G(H) = Norm_G(\mathrm{Lie}\, H) = Norm_G(H^{(1)}).$$

On the other hand, if S has non-zero residue characteristics, the subfunctors $\mathcal{L}_n^r$ of $\mathcal{L}^r$ may form a strictly increasing sequence (even when S is quasi-compact) and in this case $\mathcal{L}^r$ is not representable by an S-prescheme quasi-compact over S. Take for example the algebraic group G, defined over a field k of characteristic $p > 0$,

[1119] One may apply, for example, Proposition II.6.1 of the book by M. Demazure and P. Gabriel, *Groupes algébriques* I, Masson & North Holland (1970).

equal to the semidirect product of the torus $T = G_m \times G_m$ by the unipotent group $U = G_a \times G_a$, the action of T on U being defined by $(t, t', u, u') \to (tu, t'u')$. For every integer $n > 0$, consider the smooth connected subgroup U_n of U of equation $u' = u^{p^n}$, and the sub-torus T_n of T of equation $t' = t^{p^n}$. It is immediate to verify that T_n acts on U_n and that the subgroup G_n of G, equal to $T_n \cdot U_n$, is smooth, connected and equal to its normalizer in G. Now all the groups G_n, for $n \geqslant m$, are distinct but have the same infinitesimal neighborhood of order p^m.

Remark 5.7. There exists on $\mathcal{L}$ a canonical invertible sheaf L,

whose restriction to every open subset U of $\mathcal{L}$ quasi-compact over S is S-ample. Indeed, consider the subgroup prescheme H of $G_{\mathcal{L}}$ smooth over $\mathcal{L}$, with connected fibers, equal to its connected normalizer and universal for these properties. I claim that one may take for L the sheaf $(det(LieH))^{-1}$ (recall that if $\mathcal{F}$ is a sheaf of $\mathcal{O}_S$-modules on a prescheme S that is locally free of finite rank, $det(\mathcal{F})$ denotes the invertible $\mathcal{O}_S$-module whose restriction to the open-closed subprescheme S_r of S $(r \geqslant 0)$ where $\mathcal{F}$ is of rank r is equal to $\bigwedge^r(\mathcal{F})$). We keep the notation from the proof of 5.2. To prove the assertion made on L, we may restrict to the functor $\mathcal{L}_n^r$ and prove that $L|_{\mathcal{L}_n^r}$ is S-ample. Consider the canonical immersion $v_n : \mathcal{L}_n^r \to P_n$, and let Q be the locally free sheaf on P_n universal for the Grassmannian P_n. By construction, one has $v_n^*(Q) = H^{(n)}$ (where now H denotes the subgroup prescheme of $G_{\mathcal{L}_n^r}$ universal for the functor $\mathcal{L}_n^r$). Now $det(Q)$ is the canonical ample sheaf on P_n (EGA I 2nd ed. 9.7), so $detH^{(n)}$ is ample relative to $\mathcal{L}_n^r$ (EGA II 4.6.13 i) bis). Let $\mathcal{J}$ still denote the conormal sheaf, equal to $H^{(1)}$, and $S^q(\mathcal{J})$ the homogeneous part of degree q of the symmetric algebra of $\mathcal{J}$ over $\mathcal{O}_{\mathcal{L}_n^r}$. Since H is smooth over $\mathcal{L}_n^r$, it is immediate that one has a canonical isomorphism:

$$\det H^{(n)} \simeq \prod_{1 \leqslant q \leqslant n} \det S^q(\mathcal{J}).$$

On the other hand, one proves that for every locally free sheaf $\mathcal{J}$ of rank r and for every integer $q > 0$, there exists a canonical isomorphism:

$$\det S^q(\mathcal{J}) \simeq (\det \mathcal{J})^{\otimes s},$$

where $s > 0$ is an integer depending only on r and q. Finally, one obtains $detH^{(n)} \simeq (det\mathcal{J})^{\otimes s}$ for a suitable integer $s > 0$, hence (EGA II 4.5.6), $\det \mathcal{J} = (\det \mathrm{Lie}\, H)^{-1}$ is indeed S-ample.

Remark 5.8. Let S be a prescheme, G and H two S-preschemes in groups of finite presentation over S, $i : H \to G$ an S-homomorphism of groups which is a monomorphism. If H is smooth over S with connected fibers, one knows (Exp. XI 6.11) that $N = Norm_G(H)$ is representable by a closed subgroup prescheme of G, of finite presentation over S. Suppose in addition that N is smooth along the unit section and has the same relative dimension over S as H. The connected component N^0 of N is then representable by an open subgroup prescheme of N, smooth over S (Exp. VI_B 3.10). The monomorphism i evidently factors through N^0. In fact one has $H = N^0$. Indeed, for every point s of S, one has $H_s = (N^0)_s$, these two algebraic groups being connected, smooth, of the same dimension. Since H is flat over S, one deduces that $H \to N^0$ is an isomorphism (EGA IV 17.9.5). Finally H is a subgroup prescheme of G. We have therefore shown that the functor $\mathcal{L}$ introduced in this section is identical to the functor of subgroups H of G, smooth over S, with connected fibers and equal to their connected normalizer.

6. Functor of Cartan subgroups and functor of parabolic subgroups

When G is a smooth, connected algebraic group defined over an algebraically closed field k, one has defined the sub-tori of G, the maximal sub-tori, the Cartan subgroups (Exp. XII 1), the Borel subgroups (Exp. XIV 4.1), the parabolic subgroups (Exp. XIV 4.8 bis). We extend these notions to the case of a group prescheme over an arbitrary base, as follows:

Definition 6.1. *Let S be a prescheme, G an S-prescheme in groups of finite presentation over S, H an S-prescheme in groups, $i : H \to G$ an S-monomorphism making H a subgroup of G. We say that H is a sub-torus of G (resp. a maximal sub-torus of G, a Cartan subgroup, a Borel subgroup, a parabolic subgroup) if:*

- *i) H is smooth over S.*
- *ii) For every geometric point s above S, H_s is a sub-torus of $(G_s)^0_{red}$ (resp. a maximal sub-torus, a Cartan subgroup, a Borel subgroup, a parabolic subgroup).*

Remarks 6.1 bis.

a) If the S-group H is a sub-torus of G (resp. ...), its fibers are connected, and consequently H is of finite presentation over S (Exp. VI_B 5.3.3).

b) If H is a sub-torus of G, then H is a torus in the sense of Exp. IX, as follows immediately from Exp. X 8.1. Moreover the monomorphism $i : H \to G$ is an immersion (cf. 8.3 below).

c) If G is smooth over S with connected fibers and if H is a Cartan subgroup of G (resp. a Borel subgroup, a parabolic subgroup), the monomorphism $i : H \to G$ is an immersion, so that our definitions coincide with those introduced in Exp. XII and Exp. XIV. Indeed, H is then identical to its connected normalizer (by XII 6.6 c), XIV 4.8 and 4.8 bis), and it suffices to apply 5.8.

Definition 6.1 ter. *Let S be a prescheme, G an S-prescheme in groups locally of finite type, s a point of S. The **nilpotent rank** of G at the point s, denoted $\rho_n(s)$, is the dimension of the Cartan subgroups of $(G_s)^0_{red}$. One similarly defines the reductive rank $\rho_r(s)$, the unipotent rank $\rho_u(s)$, the abelian rank $\rho_{ab}(s)$ (cf. Exp. X 8.7).*

If now G is a smooth connected algebraic group defined over an algebraically closed field k, recall that the **radical** of G, denoted $rad(G)$, is the largest invariant, smooth, connected, solvable algebraic subgroup of G; $G/rad(G)$ is then semisimple (use Exp. XII 6.1 to reduce to the affine case). If G is moreover affine, one defines the **unipotent radical** $rad_u(G)$ of G as the largest invariant, smooth, connected, unipotent algebraic subgroup of G; $G/rad_u(G)$ is then reductive.

Proposition 6.2. *Let S be a prescheme, G and H two S-preschemes in groups of finite presentation over S, $i : H \to G$ an S-monomorphism making H a subgroup of G, and let $P(s)$ be one of the following properties concerning the point s of S:*

- *i) $(G_s)^0_{red}$ is an abelian variety (resp. is affine, is a torus, is unipotent).*
- *ii) $(H_s)^0_{red}$ is a maximal torus of G_s.*
- *iii) $(H_s)^0_{red}$ is the centralizer in $(G_s)^0_{red}$ of a torus of G_s (resp. is a Cartan subgroup of G_s).*
- *iv) $(H_s)^0_{red}$ is a Borel subgroup (resp. a parabolic subgroup) of G_s.*
- *v) $(H_s)^0_{red}$ is the radical of G_s (resp. $(G_s)^0_{red}$ is semisimple).*
- *vi) G_s is affine and $(H_s)^0_{red}$ is the unipotent radical of G_s (resp. $(G_s)^0_{red}$ is reductive).*

Then the set E of points s of S for which $P(s)$ is true is locally constructible (EGA 0_III 9.1.11).

Remark 6.2.1. This proposition complements Exp. VI_B § 10. Moreover, one can further specify the structure of E by using semi-continuity theorems (cf. Exp. X 8.7); we shall see an example a little later.

Proof of 6.2. Note that if S is the spectrum of a field, E is invariant under extension of that field. A standard reduction (EGA IV 9) then allows us to reduce to the case where S is noetherian, integral, with generic point η. One must show that E or `ens(S) \ E` contains a neighborhood of η (EGA IV 9.2.1). One may assume S affine of ring A and field of fractions K. If L is a finite extension of K, it is immediate that there exists an A-subalgebra B of L, finite over A, having L as field of fractions. The canonical morphism $S' \to S$, where $S' = \operatorname{Spec} B$, is dominant, of finite presentation, so the image of a non-empty open subset of S' contains a non-empty open subset of S (EGA IV 1.8.4). From the viewpoint that interests us, we may therefore replace S by S', hence replace K by a finite extension L. Thus we may choose L so that $(G_L)_{red}$ and $(H_L)_{red}$ are smooth over L (EGA IV 4.6.6). Possibly restricting S', we may assume that G_{red} and H_{red} are group preschemes smooth over S (Exp. VI_B § 10 and EGA IV 17). In view of the properties to be proved, we may replace G and H by their reduced connected components (Exp. VI_B 10.9), hence assume G and H smooth over S with connected fibers. Finally we may assume H is a closed subgroup prescheme of G (Exp. VI_B 10.4).

Proof of i). Possibly after a finite extension of K, we may assume that G_η admits a "Chevalley decomposition", i.e. is an extension of an abelian variety D_η by a smooth connected linear algebraic group F_η (Séminaire Bourbaki 1956/57 N° 145). By Exp. VI_B 10.16, there exists an open neighborhood U of η

such that this generic extension comes from an extension $1 \to F \to G|_U \to D \to 1$. One may further assume F and D smooth over U with connected fibers, F affine over U and D proper over U (EGA IV 8, 9 and 17). For every point s of U, (D_s, F_s) is then the "Chevalley decomposition" of G_s. Consequently, G_s is an abelian variety (resp. is affine) if and only if F_s (resp. D_s) is the unit group, which is a constructible property (EGA IV 9.2.6.1).

To establish the last two assertions of i), we may, by the preceding, assume G affine over S. Let q be a prime number invertible on S and ${}_qG$ the "kernel" of the q-th power morphism in G. It follows easily from the structure of affine algebraic groups that G_s is a torus (resp. is unipotent) if and only if a) ${}_qG_s$ is quasi-finite, which is a constructible property (EGA IV 9.3.2), and b) ${}_qG_s$ has r^q geometric points, where r denotes the relative dimension of G over S (resp. ${}_qG_s$ has a single point). Now the function $s \mapsto (\text{number of geometric points of } {}_qG_s)$ is constructible (EGA IV 9.7.9). This completes the proof of (i).

Proof of iii). a) **Case of a centralizer of a torus.** Suppose $H_\eta = Centr_{G_\eta}(T_\eta)$, where T_η is a torus of G_η, and let us show that H_s is the centralizer of a sub-torus of G_s for s in a neighborhood of η. By i), possibly restricting S,

we may assume that T_η comes from a sub-torus T of G. But then $Z = Centr_G(T)$ is representable (Exp. XI 6.11) by a subgroup prescheme of G. Since H and Z coincide generically, they coincide over a neighborhood of η. This proves to us that the set E of points s of S such that H_s is the centralizer in G_s of a sub-torus of G_s is ind-constructible (EGA IV 1), and this result will suffice for us to establish, in Lemma 6.6 below, that E is an open part of S; *a fortiori*, E will indeed be a locally constructible part of S.

iii) b) **Case of a Cartan subgroup.** Suppose H_η is a Cartan subgroup of G_η, and let us show that H_s is a Cartan subgroup of G_s for every point s of a neighborhood of η. The group H_η is the centralizer in G_η of a torus of G_η and is nilpotent (Exp. XII 6.6). By a) and Exp. VI_B 8.4, H_s has the same properties for every point s of a neighborhood U of η. For every point s of U, the group H_s thus has the same reductive rank as G_s, and its unique maximal torus is central (Exp. XII 6.7); it is therefore the centralizer of a maximal torus of G_s, i.e. a Cartan group of G_s.

Suppose now that H is not a Cartan subgroup of G, and let us show that H_s is not a Cartan subgroup of G_s for s in a neighborhood U of η. In view of the assertion provisionally admitted in (a) above, we may restrict to the case where H is the centralizer in G of a sub-torus T. But then H_η contains a Cartan subgroup C_η of H_η. We have just seen that, possibly restricting S, C_η extends to a Cartan subgroup C of G, which one may assume contained in H. By hypothesis H_η strictly majorizes C_η, so H_s strictly majorizes C_s for s in a neighborhood U of η (EGA IV 9.5.2); *a fortiori*, H_s is not

a Cartan subgroup of G_s for s in U.

Proof of ii). Suppose H_η is a maximal torus of G_η and let C_η be its centralizer in G_η. By i) and iii), H is a torus over a neighborhood U of η and $C = Centr_{G|U}(H|U)$ is a Cartan subgroup of $G|U$. To prove that H is a maximal torus of G over a neighborhood of η, we may then replace G by C, then by the linear component F of a Chevalley decomposition of C (cf. i)). Let q be an integer invertible on S, ${}_qF$ the kernel of the q-th power morphism in F. Since F_s is affine, nilpotent, smooth and connected, F_s is the direct product of its maximal torus T_s by a unipotent group (*Bible* 6-04), so ${}_qF_s = {}_qT_s$. Since H_η is a maximal torus, ${}_qH_\eta = {}_qF_\eta$, and consequently ${}_{q}H = {}_{q}F$ over a neighborhood V of η. For every point s of V, ${}_{q}H_{s} = {}_{q}T_{s}$, so $H_s = T_s$ is a maximal torus.

Suppose now that H_η is not a maximal torus of G_η. By i), we may restrict to the case where H_η is a torus, then assume that it is contained in a strictly larger torus T_η. The latter extends to a torus T strictly majorizing H over a neighborhood U of η. *A fortiori*, H_s is not a maximal torus for $s \in U$.

Proof of iv). Possibly restricting S, we may assume that the center Z of G is representable (Exp. VI_B 10.11) and flat over S, as well as the quotient G/Z (*loc. cit.*). The property "H_s majorizes Z_s" is constructible (EGA IV 9.5.2) and every parabolic subgroup of G_s contains Z_s (Exp. XIV 4.9 a)); this allows us to replace G by G/Z, hence assume G affine over S (Exp. XII 6.1 and i)). We may further

assume that G/H is representable, but then H_s is a parabolic subgroup of G_s if and only if $(G/H)_s$ is proper (*Bible* 6 Th. 4 b)), which is an ind-constructible property (EGA IV 9.3.5). So E is ind-constructible, and this will suffice for us to prove that E is open (Lemma 6.6), hence locally constructible.

Let us now examine the case of Borel subgroups. If H_η is a Borel subgroup of G_η, i.e. a solvable parabolic subgroup of G_η, what precedes and Exp. VI_B 8.4 entail that these properties remain true at every point s of a neighborhood of η. If now H_η is not a Borel subgroup of G_η, to prove that the same holds at points s of a neighborhood of η, we may restrict (in view of what precedes) to the case where H_η is a parabolic subgroup, then assume that H_η contains a Borel subgroup B_η. We have just shown that the latter extends to a Borel subgroup B of G over a neighborhood U of η. Since H_η strictly majorizes B_η, H_s strictly majorizes B_s at every point of an open subset V, and H_s is not a Borel subgroup of G_s for $s \in V$.

Proof of v). Suppose H_η is the radical of G_η. The group H_η is then invariant in G_η, solvable (smooth and connected); the same is therefore true for H_s for s belonging to a neighborhood U of η (Exp. VI_B 8 and 10), so for $s \in U$, H_s is contained in the radical of G_s. Replacing G by G/H (Exp. VI_B 10), we must

prove that if G_η is semisimple, then G_s is semisimple at every point of a neighborhood V of η. Using i) and ii), one may assume that G is affine over S and possesses a maximal torus T. Let W be the Weyl group of T (Exp. XII 2), which is quasi-finite and étale over S, hence finite and étale over an open subset V. It then follows from the elementary properties of roots (Exp. XIX 1.12) that G is semisimple over V.

Suppose now that H_η is not the radical of G_η. Possibly replacing K by a finite extension L, we may assume that G_η admits a radical R_η. By what precedes, R_η extends to a subgroup prescheme R of $G|_U$ such that for every $s \in U$, R_s is the radical of G_s. By hypothesis, $R_\eta \neq H_\eta$. So $R_s \neq H_s$ for $s \in V$. It remains to prove that if G_η is not semisimple, neither is G_s at neighboring points, but this is a particular case of what precedes (take H = unit group).

Proof of vi). The proof is entirely analogous to that of v), in view of i), and is left to the care of the reader.

Corollary 6.3. *Let S_0 be a quasi-compact prescheme, $(S_i)_{i \in L}$ a projective system of S_0-preschemes, affine over S_0, $S = \lim S_i$ (EGA IV 8.2), G_0 a group prescheme of finite presentation over S_0, $G_i = G_0 \times_{S_0} S_i$, $G = G_0 \times_{S_0} S$, H a subgroup of G. Then, if H is a sub-torus of G (resp. a maximal sub-torus, a Cartan subgroup, a Borel subgroup, a parabolic subgroup), there exists an index $i \in L$ and a subgroup $\mathtt{H}_i$ of $\mathtt{G}_i$ such that $H = H_i \times_{S_i} S$ and $\mathtt{H}_i$ is a sub-torus of $\mathtt{G}_i$ (resp. ...).*

Indeed, H is smooth with connected fibers, hence of finite presentation over S (Exp. VI_B 5.3.3). By Exp. VI_B § 10, there exist $i \in L$ and a subgroup $\mathtt{H}_i$ of $\mathtt{G}_i$, smooth over S, such that $H = H_i \times_{S_i} S$. Corollary 6.3 then follows from Definition 6.1, from 6.2, and from EGA IV 9.3.3.

Corollary 6.3 bis. *Let S be a prescheme, G an S-prescheme in groups of finite presentation over S. Then the functions ρ_n, ρ_r, ρ_u, ρ_{ab} (cf. 6.1 ter) are locally constructible functions on S.*

It suffices to show (EGA IV 9.) that if S is a noetherian integral scheme with generic point η, the functions in question are constant on a neighborhood of η. Possibly replacing S by a scheme S' finite over S and dominating S, we may assume that G_η admits a Cartan subgroup C_η with a Chevalley decomposition $1 \to L_\eta \to C_\eta \to A_\eta \to 1$. The argument made in 6.2 i) proves that this decomposition extends to a Chevalley decomposition over a neighborhood of η:

$$1 \longrightarrow L \longrightarrow C \longrightarrow A \longrightarrow 1.$$

Moreover, one may assume that C is a Cartan subgroup of G (6.3) and that the maximal torus T_η of L_η extends to a maximal torus T of L (6.3). The corollary follows immediately from this and from the definitions.

6.4.0. Let then S be a prescheme, G an S-prescheme in groups of finite presentation over S, $\mathcal{L}$ the functor of subgroups of G, smooth, with connected fibers and equal to their connected normalizer (cf. § 5); $\mathcal{L}$ is representable (5.2 and 5.8). The remainder of this

section is devoted to the study of certain subfunctors of $\mathcal{L}$. More precisely, we introduce the subfunctors of $\mathcal{L}$, denoted $\mathcal{L}_C$ (resp. $\mathcal{CT}$, resp. $\mathcal{P}$), defined as follows: for every S-prescheme S', $\mathcal{L}_C(S')$ (resp. $\mathcal{CT}(S')$, resp. $\mathcal{P}(S')$) is the set of subgroups H of $G_{S'}$, smooth over S', with connected fibers, equal to their connected normalizer, and such that for every point s' of S', H'_s contains a Cartan subgroup (6.1) of G'_s (resp. is the centralizer in $(G'_s)^0_{red}$ of a sub-torus of G'_s, resp. is a parabolic subgroup (6.1) of G'_s).

Theorem 6.4. *Let S be a prescheme, G an S-prescheme in groups of finite presentation over S. Then the S-functors $\mathcal{L}_C$, $\mathcal{CT}$, $\mathcal{P}$ above (6.4.0) are representable by S-preschemes of finite presentation over S, quasi-projective over S.*

Remark 6.5. If G has smooth fibers, for example if G is smooth over S, or if the residue characteristics of S are zero (Exp. VI_B 1.6.1), every subgroup H of G, smooth over S with connected fibers, such that for every point s of S, H_s contains a Cartan subgroup of $(G_s)^0$, is necessarily equal to its connected normalizer (and consequently is an element of $\mathcal{L}_C(S)$). Indeed, using 5.1 iii), one may assume that S is the spectrum of a field, in which case the property was noted at the end of the statement of Exp. XIII 2.1.

Note that one has natural monomorphisms

$$\begin{tikzcd}
\mathcal{CT} \arrow[dr] & & \\
& \mathcal{L}_C \arrow[r] & \mathcal{L}. \\
\mathcal{P} \arrow[ur] & &
\end{tikzcd}$$

Let us show that these monomorphisms are open immersions of finite presentation (which will already prove, in view of 5.2, that $\mathcal{L}_C$, $\mathcal{CT}$, $\mathcal{P}$ are representable by S-preschemes, union of an increasing sequence of open subpreschemes, quasi-projective and of finite presentation over S). Now this will follow, by the usual technique, from the following lemma:

Lemma 6.6. *Let S be a prescheme, G an S-prescheme in groups of finite presentation, H a subgroup of G, smooth with connected fibers. Then the set of points s of S such that H_s contains a Cartan subgroup of G_s (resp. is equal to the centralizer in $(G_s)^0_{red}$ of a torus of G_s) is an open subset of S. If moreover H is equal to its connected normalizer[1120], the set of points s of S such that H_s is a parabolic subgroup of G_s is also an open subset of S.*

The assertion to be proved is local on S, which allows us to assume S affine, then, by EGA IV 8.1 and 6.3, S noetherian. Denote by E the set of points of S having the property in question. It then follows from the assertions effectively proved in 6.2 that E is ind-constructible. But S is noetherian, so to prove that E is open, it suffices to show that E is stable under generizations (EGA IV 1.10.1). Using EGA II 7.1.9, we are finally led to prove that if S is the spectrum of a discrete valuation ring, and if the closed point s belongs to E, then so does the generic point t.

We shall need the following lemma:

Lemma 6.7. *Let A be a complete noetherian local ring, $S = \operatorname{Spec} A$, s the closed point of S, H an S-prescheme in groups, smooth with connected fibers, T_s a sub-torus of H_s. Then:*

- *i) There exists a closed subgroup prescheme C of H, smooth, with connected fibers, such that $C_s = Centr_{H_s}(T_s)$.*
- *ii) For every point t of S, C_t is the centralizer in H_t of a sub-torus T_t of H_t.*

Proof of 6.7. Let T^0 be an S-torus such that there exists an isomorphism $u_0 : T^0_s \xrightarrow{\sim} T_s$ (Exp. X 4.6). Let $\mathfrak{m}$ be the maximal ideal of A, $A_n = A/\mathfrak{m}^n$, $S_n = \operatorname{Spec} A_n$, $H_n = H \times_S S_n$, etc. Since H is smooth over S, for every integer $n > 0$ there exists an S_n-group morphism $u_n : T^0_n \to H_n$ lifting u_0 (Exp. IX 3.6), and one may assume by induction on n that u_n lifts u_{n-1}. Moreover, let q be a prime number invertible on S; for every integer ℓ equal to a power of q, denote by ${}_\ell u_n$ the restriction of u_n to

[1120]hypothesis in fact superfluous, cf. XVII App. III 3.

the subgroup ${}_\ell T_n^0$. For fixed ℓ and variable n, the morphisms ${}_\ell u_n$ form a projective system, hence come from a unique S-group morphism ${}_\ell u : {}_\ell T^0 \to H$ (1.6 a)). Since H is separated (Exp. VI_B 5.2) and u_0 is a monomorphism, ${}_\ell u$ is a monomorphism (Exp. IX 6.8), and even a closed immersion since it is finite, ${}_\ell T^0$ being finite over S. Denote by $M(\ell)$ the image group. It is clear that the family of subgroups of multiplicative type $M(\ell)$ is coherent in the sense of 4.1. Let $C_\ell = Centr_H(M(\ell))$, which is representable by a subgroup prescheme (2.5), closed (H is separated),

smooth over S (Exp. XI 2.4). The C_ℓ form a filtered decreasing family of closed subschemes, hence stationary, H being noetherian. The stationary value is a smooth closed subgroup C such that $C_s = Centr_{H_s}(T_s)$ by the density theorem (Exp. IX 4.7). It remains to show that for every point t of S, C_t is the centralizer in H_t of a subtorus T_t (which will entail that C_t is connected). But this will follow from the more precise lemma below, applied to the family $M(\ell)_t$ of subgroups of H_t:

Lemma 6.8. *Let G be a connected algebraic group defined over a field k, r an integer > 0, q an integer prime to the characteristic*[1121] *of k, $M(\ell)$ (ℓ ranging over the powers of q) a coherent family of subgroups of G, of multiplicative type and of type $(\mathbb{Z}/\ell\mathbb{Z})^r$ (cf. 4.6), M the algebraic subgroup of G generated by the $M(\ell)$ (*loc. cit.*), T the unique maximal torus of M (cf. 3.4). Then*

`Centr_G(T) = Centr_G(M) = Centr_G(M(□))    for □ large enough.`

The last equality is clear. To prove the first, introduce the center Z of G, $G' = G/Z$, M' (resp. T') the image of M (resp. T) in G', K the inverse image of T' in G (i.e. the algebraic subgroup of G generated by T and Z). It evidently suffices to prove that $K \supset M$, hence that $T' = M'$. Now, M is smooth and connected (4.6) and G' is affine (Exp. XII 6.1), so M' is a direct product of its maximal torus

T' (Exp. XII 6.6 d)) and a unipotent group (*Bible* 4 Th. 4) (one may assume k algebraically closed). The image of $M(\ell)$ in M' is therefore necessarily contained in T'. So the inverse image of T' in M majorizes $M(\ell)$ for every ℓ, hence equals M. Consequently $M' = T'$. This proves 6.8 and hence 6.7.

This being so, let us prove 6.6. We have reduced to the case where S is the spectrum of a discrete valuation ring A, which one may further assume complete with algebraically closed residue field. Possibly replacing A by its normalization in a finite extension of its field of fractions, one may assume that $(G_t)_{red}$ is smooth[1122]. It is clear that to prove 6.6 one may replace G by the connected component of the schematic closure in G of $(G_t)^0_{red}$, hence assume G flat over S with connected fibers, and G_t smooth.

a) Suppose H_s is the centralizer in $(G_s)_{red}$ of a torus T_s, and let us show that H_t is then the centralizer in G_t of a sub-torus of G_t. By Lemma 6.7, there exists a subgroup scheme C of H, smooth over S, whose closed fiber is H_s and such that $C_t = Centr_{H_t}(T_t)$, where T_t is a sub-torus of H_t. Since H is smooth over S with connected fibers, one concludes for dimension reasons

[1121] we have removed the word "residue".

[1122] details or references to be given here...

that $C = H$. Keeping the notation of 6.7, one has $H = C = C_\ell$ for ℓ large. Consider similarly $C'_\ell = Centr_G(M(\ell))$ (2.5), and let C' be the stationary value of C'_ℓ for ℓ large (2.5 bis). The group scheme C' majorizes H and is such that $C'_t = Centr_{G_t}(T_t)$ (6.8) and $C'_s = Centr_{G_s}(T_s)$. The hypothesis made on H_s implies $\dim H_s = \dim C'_s$. Moreover, $\dim H_s = \dim H_t$ (since H is smooth over S) and $\dim C'_t \leqslant \dim C'_s$ (Exp. VI_B 4.1),

so $\dim H_t = \dim C'_t$. But G_t being smooth and connected, $C'_t = Centr_{G_t}(T_t)$ is smooth and connected, so finally $H_t = C'_t = Centr_{G_t}(T_t)$.

b) Suppose H_s contains a Cartan subgroup C_s of G_s, i.e. the centralizer in $(G_s)_{red}$ of a maximal torus T_s of G_s. By 6.7, there exists a subgroup scheme C of H, smooth over S, with connected fibers, lifting C_s. It evidently suffices to prove that C_t contains a Cartan subgroup of G_t. Now by a) applied with $H = C$, C_t is the centralizer in G_t of a sub-torus of G_t, hence contains a Cartan subgroup of G_t.

c) Suppose H_s is a parabolic subgroup of G_s. Let $N = Norm_G(H)$. By hypothesis, H is equal to its connected normalizer, so N_s is smooth, and consequently equals $Norm_{(G_s)_{red}}(H_s) = H_s$ (Exp. XII 8 bis). But then N is flat over S. We shall see in Exp. XVI that, under these conditions, G/N is representable. Since $H_s = N_s$ is a parabolic subgroup of G, $(G/N)_s$ is proper. Since (G/N) has connected fibers and is flat over S, it follows from EGA III 5.5.1 that (G/N) is proper over S. So $(G/N)_t = G_t/N_t$ is proper over t, and the same is true of G_t/H_t, since N_t/H_t is finite. It then suffices to apply the following lemma:

Lemma 6.9. *Let k be a field, G a k-algebraic group, H a smooth connected algebraic subgroup of G,

$N = Norm_G H$; then if $\dim H = \dim N$ and if G/H is proper, H is a parabolic subgroup of G.*

Indeed, one may assume k algebraically closed and G smooth and connected. The center Z of G is contained in N, and the hypothesis $\dim H = \dim N$ entails that $Z^0 = (Z)^0_{red}$ is contained in H, so $G' = G/Z^0$ is affine (Exp. XII 6.1). Replacing G by G' and H by its image in G', one is reduced to the case where G is affine (Exp. XIV 4.9), and Lemma 6.9 then follows from *Bible* 6 Th. 4. We have therefore proved Lemma 6.6.

To complete the proof of 6.4, we must prove that the S-preschemes representing $\mathcal{L}_C$ (resp. $\mathcal{CT}$, resp. $\mathcal{P}$) are of finite presentation over S. This assertion is local on S, which allows us to assume S affine, then, G being of finite presentation, S noetherian (EGA IV 8.9). We have just seen that the natural inclusions $\mathcal{CT} \to \mathcal{L}$ (resp. $\mathcal{P} \to \mathcal{L}$) are immersions, so the same is true of the inclusions $\mathcal{CT} \to \mathcal{L}_C$ (resp. $\mathcal{P} \to \mathcal{L}_C$), and consequently it suffices to prove that $\mathcal{L}_C$ is representable by an S-prescheme of finite presentation.

Let us resume the notation introduced in 5.2. For every integer $n \geqslant 0$, let $\mathcal{L}_n$ be the subfunctor of $\mathcal{L}$ such that

$$\mathcal{L}_n(S') = \{H \in \mathcal{L}(S') \text{ such that } Norm_{G_{S'}}(H) = Norm_{G_{S'}}(H^{(n)})\}.$$

The S-functor $\mathcal{L}_n$ is therefore representable by an open subprescheme of $\mathcal{L}$, sum of the $\mathcal{L}_n^r$. Each $\mathcal{L}_n^r$ is of finite presentation over S (5.3) and is empty for $r > \sup_{s\in S} \dim G_s$ (which is a finite number, S being quasi-compact), so $\mathcal{L}_n$ is

of finite presentation over S. It suffices to prove that $\mathcal{L}_C$ is contained in $\mathcal{L}_n$ for n large enough.

For every point s of S, let $d(s)$ be the smallest integer n (finite or infinite) such that $\mathcal{L}_{C,G_s} \subset \mathcal{L}_{n,G_s}$. It suffices to show that the function d is bounded on S, since if M is an upper bound, $\mathcal{L}_C$ will be set-theoretically contained in $\mathcal{L}_M$, so $\mathcal{L}_C$ will be contained in $\mathcal{L}_M$, since the latter is an open subset of $\mathcal{L}$. An immediate constructibility argument reduces us to proving that if S is noetherian integral with generic point t, then d is bounded on a neighborhood of t.

a) **Reduction to the case where G is smooth over S.** Proceeding as in 6.2, one sees that, possibly changing S, one may assume that $(G)_{red}$ is a group prescheme smooth over S, which we shall denote G'. Set $X = \mathcal{L}_{C,G}$, $X' = \mathcal{L}_{C,G'}$ and let H (resp. H') be the subgroup prescheme of G_X (resp. $G'_{X'}$) universal for the functor $\mathcal{L}_{C,G}$ (resp. $\mathcal{L}_{C,G'}$). Since H is smooth over X, $H \times_X X_{red}$ is reduced, hence contained in $G'_{X_{red}}$, and is an element of $\mathcal{L}_{C,G'}(X_{red})$, whence a canonical morphism $p : X_{red} \to X'$. It is clear that p is a monomorphism; let us show that p is even an immersion. Let N' be the normalizer of H' in $G_{X'}$. The set of points s of X' such that the immersion $H'_s \to N'_s$ is an open immersion is an open subset U, and $H'|_U \to N|_U$ is an open immersion (Exp. VI$_B$ 2.5 and EGA IV 17.9.5). It follows from 5.1 iii) that U is the largest open subset of X' above which H' is equal to its connected normalizer in $G_{X'}$, so $H'|_U \in \mathcal{L}_{C,G}(U)$. One immediately deduces that p is an isomorphism of X_{red} onto U_{red}. If one shows that $\mathcal{L}_{C,G}$ is of finite type when G is smooth, X' will be of finite type over S,

so X_{red} will be of finite type (S is noetherian) and consequently will be contained in $\mathcal{L}_G^M$ for M large enough, and the same will hold of X.

b) **Case where S is the spectrum of an algebraically closed field k of characteristic $p > 0$ and G is a smooth algebraic group over k.** Instead of using the infinitesimal neighborhoods $H^{(n)}$ of a subgroup prescheme H of G, we shall use the radicial subgroups $F_n(H)$, kernels of the iterates of the Frobenius morphism in H (Exp. VII$_A$ 4), which is legitimate here, since $F_n(H)$ is contained in the infinitesimal neighborhood of order p^n of the unit section of H. If T is a sub-torus of G, $F_n(T) = {}_{p^n}(T)$, and it is immediate by duality that $Centr_G({}_{p^n}(T))$ is equal to $Centr_G(T)$ for n large enough. One then has the following more precise proposition:

Proposition 6.10. *Let k be a field of characteristic $p > 0$, G a smooth k-algebraic group, T a maximal torus of $G_{\bar{k}}$, m the smallest integer such that*

$$Centr_G({}_{p^m}(T))^0 = Centr_G(T)^0.$$

Then, for every k-prescheme S and every $H \in \mathcal{L}_C(S)$, one has

$$Norm_{G_S}(H) = Norm_{G_S}(F_m(H)).$$

A fortiori, $\mathcal{L}_C$ is contained in $\mathcal{L}^{p^m}$.

Since H is smooth over S, of finite presentation over S and has a constant nilpotent rank (namely that of G), we shall see in the following section (7.3) that the functor $\mathcal{C}_H$

of Cartan subgroups of H is representable by an S-prescheme, smooth over S (the reader will verify that the proof given of this property does not use the fact that $\mathcal{L}_{C,H}$ is of finite type over S). It then follows from Exp. XIII 3.1 that one may consider the open subset U of regular points of H.

Let S' be an S-prescheme, g an element of $G(S')$ normalizing $F_m(H)_{S'}$. To prove that g normalizes $H_{S'}$, it suffices to prove that $int(g)U_{S'}$ is contained in $H_{S'}$; indeed, $H_{S'} \cap int(g)H_{S'}$ will then contain an open subgroup of int(g)H, hence will be equal to

int(g)H, since the latter has connected fibers. Possibly replacing S' by a suitable S'', then S'' by S, we are reduced to proving that if $g \in G(S)$ normalizes $F_m(H)$ and if $u \in U(S)$, then $int(g)u \in H(S)$.

Now let C be the unique Cartan subgroup of H_S "containing" u (Exp. XIII 3.2). It suffices to show that $C' = int(g)C \subset H$. Since $H \in \mathcal{L}_{C,G}(S)$, C is also a Cartan subgroup of G; but the latter admits maximal tori, so (Exp. XII 7.1 (a)) C is the centralizer in G_S^0 of its unique maximal torus T. It follows from the definition of m (and from the fact that any two maximal tori of G are locally conjugate for fpqc (Exp. XII 7.1)) that

$$C = (Centr_G(T))^0 = Centr_G({}_{p^m}(T))^0,$$

whence by conjugation by g:

$$C' = Centr_G(int(g)({}_{p^m}(T)))^0.$$

But $int(g)({}_{p^m}(T))$ is a subgroup of multiplicative type of $int(g)(F_m(H)) = F_m(H)$ (g normalizes $F_m(H)$), hence is contained in H. It then follows from Exp. XIII 2.1 (which is proved when the base is a field, but extends immediately to the case of an arbitrary base) that for C' to be contained in H it suffices that

Lie C′ ⊂ Lie H.

Now Lie C′ = int(g)(Lie C) ⊂ int(g)(Lie H).

On the other hand, if $m \geqslant 1$, which it is permissible to assume, one has (using Exp. VII_A 4.1.2):

$$\mathrm{Lie}\,H = \mathrm{Lie}(F_m(H)) = \mathrm{Lie}(\mathrm{int}(g)(F_m(H))) = \mathrm{int}(g)(\mathrm{Lie}\,H).$$

So $LieC' \subset LieH$, which completes the proof of 6.10.

We shall need another definition of the integer m introduced in 6.10.

Lemma 6.11. *Let G be a smooth algebraic group defined over an algebraically closed field k of characteristic $p > 0$, T a maximal torus of G, $\mathfrak{g} = LieG$, and R the family of non-zero characters of T appearing in the representation of T in $\mathfrak{g}$ induced by the adjoint representation of G. For every element $r \in R$, denote by e_r the largest integer n such that p^n divides r in the group of characters of T. Then $m = \sup_{r \in R}(e_r + 1)$ if $R \neq \emptyset$, and $m = 0$ otherwise.*

Indeed, $Centr_G(T)$ is smooth and contained in $Centr_G({}_{p^m}(T))$, so

$$\begin{aligned}(\mathrm{Centr}_G\,T)^0 = (\mathrm{Centr}_G({}_{p^m}(T)))^0 &\Leftrightarrow \mathrm{Lie}(\mathrm{Centr}_G\,T) = \mathrm{Lie}(\mathrm{Centr}_G(F_m(T)))\\ &\Leftrightarrow \mathfrak{g}^T = \mathfrak{g}^{{}_{p^m}(T)} \qquad (\text{Exp. II } 5.2.3).\end{aligned}$$

Now with the usual notation,

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \coprod_{r \in R} \mathfrak{g}_r.$$

So $\mathfrak{g}^T = \mathfrak{g}_0$ and $\mathfrak{g}^{p^m(T)} = \mathfrak{g}_0 + \coprod_{r \in R_0} \mathfrak{g}_r$, where R_0 is the subset of R consisting of characters of T whose restriction to ${}_{p^m}(T)$ is zero. But a non-zero character r of T has trivial restriction to ${}_{p^m}(T)$ if and only if $m \leqslant e_r$, whence the lemma.

c) **Return to the proof of 6.4.** We have reduced (by point a) and the section preceding it) to the case where S is a noetherian integral scheme and G is smooth over S. We must show that the function d is bounded on a neighborhood of the generic point t of S. Possibly changing S, we may assume that G admits a trivial maximal torus (6.2) $T = D_S(M)$. Let then $\mathfrak{g} = \coprod_{\lambda \in M} \mathfrak{g}_\lambda$ be the decomposition of the Lie algebra of G according to the characters of T, and let R be the finite set of non-zero characters of M such that $\mathfrak{g}_\lambda \neq 0$. Let us distinguish two cases:

1st case: the point t, and hence all points of S, have residue characteristic $p > 0$. It is clear, from what precedes, that the function d is then bounded by p^m, where m is defined as in 6.11.

2nd case: the point t has residue characteristic zero. For every $\lambda \in R$, let n_λ be the largest integer dividing λ in the group M, and set $n = \prod n_\lambda$, $\lambda \in R$. For every prime number q dividing n, denote by S_q the closed subset of S consisting of points of S whose residue characteristic equals q, and let U be the non-empty open subset (it contains t) complementary in S to the union of the S_q. If now s is a point of U, either s

has residue characteristic zero, in which case $d(s) = 1$ (5.6), or s has residue characteristic $p > 0$ not dividing n, so the integer m relative to the group G_s, defined in 6.11, is at most one. Moreover, it follows from Exp. VII[1123] that

$$\mathrm{Norm}_{G_{S'}}(F_1(H)) = \mathrm{Norm}_{G_{S'}}(\mathrm{Lie}\,H) = \mathrm{Norm}_{G_{S'}}(H^{(1)}).$$

[1123] argument to be made explicit...

Finally, it follows from 6.10 that if $H \in \mathcal{L}_{C,G_S}(S')$, one has $Norm_{G_{S'}}(H) = Norm_{G_{S'}}(H^{(1)})$, and consequently $d(s) \leqslant 1$, so d is bounded by 1 on U.

This completes the proof of 6.4.

Corollary 6.12. *Let S be a prescheme and G an S-prescheme in groups of finite presentation. Suppose that the nilpotent rank (resp. the dimension of the Borel subgroups) of the fibers of G is a locally constant function on S. Then the functor $\mathcal{C}$ of Cartan subgroups of G (resp. the functor $\mathcal{B}$ of Borel subgroups of G) is representable by an S-prescheme of finite presentation over S.*

Indeed, possibly restricting S, we may assume that the nilpotent rank ν of the fibers is constant. But then it is clear that $\mathcal{C}$ is represented by the open-closed subprescheme of the prescheme X representing $\mathcal{L}_C$ (6.4) above which the universal subgroup of G_X relative to the functor $\mathcal{L}_C$ is of relative dimension ν. The proof is analogous for the functor $\mathcal{B}$, in view of the representability of the functor $\mathcal{P}$.

7. Cartan subgroups of a smooth group

Proposition 7.1. *Let S be a prescheme, G an S-prescheme in groups of finite presentation, smooth over S, with connected fibers.*

- *i) Let $\mathcal{CT}$ be the S-functor $(Sch/S)^{\circ} \to Ens$ such that for every S-prescheme S': > $\mathcal{CT}(S')$ = set of subgroup preschemes H of $G_{S'}$, smooth over S', such that for every point s' of S', > H'_s is the centralizer in G'_s of a sub-torus of G'_s.*
- *Then $\mathcal{CT}$ is representable by an S-prescheme of finite presentation over S, smooth and quasi-projective over S.*
- *ii) If S is artinian and $H \in \mathcal{CT}(S)$, then H is the centralizer in G of a sub-torus of G. If S is the spectrum of a Henselian local ring and $H \in \mathcal{CT}(S)$, then H is the centralizer in G of a subgroup of multiplicative type of G, étale over S.*
- *iii) If L is an S-prescheme in groups, smooth and of finite presentation over S, $i : L \to G$ an S-group monomorphism, and H an element of $\mathcal{CT}(S)$, then $Transp_G(H, L)$ and $Transp_G^{str}(H, L)$ (cf. Exp. VIII 6.5 e)) are representable by closed subpreschemes of G, smooth over S.*
- *iv) If $H \in \mathcal{CT}(S)$, H is closed in G, $N = Norm_G(H)$ is representable by a closed subgroup prescheme of G, smooth over S;*

N/H is representable by an S-prescheme in groups, separated over S, étale and of finite type over S; G/N is representable by an S-prescheme smooth and quasi-projective over S.

- *v) Let G' be an S-prescheme in groups of finite presentation over S and $u : G \to G'$ a faithfully flat S-group morphism, so that G' satisfies the same hypotheses as G (Exp. VI_B 9). Then if $H \in \mathcal{CT}_G(S)$, the image of H by u is representable by a subgroup prescheme H' of G' which is an element of $\mathcal{CT}_{G'}(S)$. Moreover, $H \to H'$ is faithfully flat, and if H is the centralizer in G of a torus T, H' is the centralizer in G' of the torus $T' = u(T)$.*
- *vi) Under the conditions of v), consider the S-morphism*

$\tilde{u} : \mathcal{CT}_G \to \mathcal{CT}_{G'}, \quad H \mapsto H' = u(H).$

- *Then $\tilde{u}$ is a quasi-compact faithfully flat morphism; if moreover $\mathrm{Ker}\ u$ is central, $\tilde{u}$ is an isomorphism, the inverse isomorphism being $H' \mapsto u^{-1}(H')$.*

Proof of ii). For the first assertion, we may assume S local artinian with closed point s. Let $H \in \mathcal{CT}(S)$ and let T_s be the maximal central torus of H_s, which is already defined over $\kappa(s)$ (cf. 3.4). Since $H_s \in \mathcal{CT}(s)$, one has $H_s = Centr_{G_s}T_s$. The group H is smooth, so T_s lifts uniquely to a sub-torus T of H, central

in H (Exp. IX 3.6 bis and Exp. IX 5.6). But then $H' = Centr_G T$ majorizes H and has the same fiber as H; since H is flat over S, one has $H = H'$ (Exp. VI_B 2.5).

Suppose now that S is the spectrum of a Henselian local ring, which one may assume noetherian by the usual reductions. Denote by s the closed point of S, T_s the maximal central torus of H_s, q an integer invertible on S, ℓ a power of q, T an S-torus having a closed fiber isomorphic to T_s (Exp. X 4.6). Moreover, let ${}_\ell H$ be the "kernel" of the ℓ-th power morphism in H, and let U_ℓ be the largest open subset of ${}_\ell H$ étale over S. It then follows from 1.3 and from the fact that H is flat over S that U_ℓ majorizes ${}_\ell T_s$. Since S is Henselian, there exists a unique S-morphism

$$u_\ell : {}_\ell T \longrightarrow U_\ell$$

which, on the closed fiber, induces the canonical immersion ${}_\ell T_s \to (U_\ell)_s$. By uniqueness, one easily sees that u_ℓ is an S-group morphism, central (Exp. IX 5.6 a)). Proceeding then as in 6.6 and 6.7, one shows that u_ℓ is an immersion, and that if M_ℓ is the image group $u_\ell({}_\ell T)$, then $Centr_G(M_\ell)$ is equal to H for ℓ large enough (this is where the noetherian hypothesis on S is used).

Proof of i). The group G is smooth over S with connected fibers, so if $H \in \mathcal{CT}(S)$, H has connected fibers (Exp. XII 6.6 b)) and is equal to its connected normalizer (6.5), so that the functor $\mathcal{CT}$ defined in 7.1 i) coincides with the functor also denoted $\mathcal{CT}$ defined in 6.4.0. So, by Theorem 6.4, $\mathcal{CT}$ is representable by an S-prescheme of finite presentation and quasi-projective over S. It remains to show that this prescheme is smooth over S. One first reduces by EGA IV 8

to the case where S is affine noetherian. Thanks to Exp. XI 1.5, it then suffices to prove that if S is the spectrum of a local artinian ring, S_0 a subscheme defined by a nilpotent ideal, H_0 an element of $\mathcal{CT}(S_0)$, then H_0 lifts to a subgroup prescheme H of G, smooth over S. Now by ii), $H_0 = Centr_{G_0}T_0$, where T_0 is a sub-torus of G_0. Since G is smooth, T_0 lifts to a sub-torus T of G (Exp. IX 3.6 bis), and it suffices to take $H = Centr_G(T)$, which is indeed smooth over S (Exp. XI 2.4 and Lemma 2.5).

Proof of iii). Since H is smooth with connected fibers, by Exp. XI 6.11, $Transp_G(H, L)$ is representable by a closed subprescheme of G of finite presentation over S. To show that this transporter is smooth, one reduces, as above, to proving that if S is local artinian, S_0 a closed subprescheme of S, $g_0 \in G(S_0)$ such that $int(g_0)H_0 \subset L_0$, then g_0 lifts to $g \in G(S)$ such that $int(g)H \subset L$. The group G being smooth over S, there exists a section g_1 of G lifting g_0; let $H' = int(g_1)H$. So this is an element of $\mathcal{CT}(S)$ such that $H'_0 \subset L_0$. By ii), H' is the centralizer in G of a torus T' of G. Since L is smooth, the torus T'_0 of L_0 lifts to a torus T''_0 of L (Exp. XI 3.6 bis). The group $Centr_L T''_0$ is contained in $Centr_G T''_0$, has the same fiber as the latter (namely H'_0) and is smooth, so equals $Centr_G T''_0 = H'_0$. The sub-tori T' and T''_0 of G are two liftings of T'_0, so are conjugate by an element h of $G(S)$ reducing to the unit section of G_0 (Exp. IX 3.3 bis); the same is therefore true of their centralizers H' and H''_0 in G. The section $g = hg_1$ lifts g_0 and one has $int(g)H \subset L$.

If now $g \in Transp_G(H, L)(S)$, for $int(g)H = L$ it is necessary and sufficient that for every $s \in S$, $\dim H_s = \dim L_s$. It follows that if U denotes the open-closed subprescheme of S above which the fibers of H have the same dimension as those of L, the strict transporter of H in L, $Transp_G^{str}(H, L)$, is representable by the S-prescheme

$$U \times_S Transp_G(H, L).$$

Proof of iv). To see that if $H \in \mathcal{CT}(S)$, H is closed in G, one may assume S affine noetherian, then S spectrum of a complete local ring (EGA IV 8); but then H is the centralizer in G of a subgroup of multiplicative type (by ii)), hence is closed since G is separated over S (Exp. VI_B 5.2).

By iii), $N = Norm_G(H) = Transp^{str}_G(H, H)$ is representable by a subgroup prescheme smooth over S and closed in G. Consider the S-morphism

$$G \to \mathcal{CT}, \qquad g \mapsto int(g)H.$$

It follows from iii) that this morphism is smooth, so its image is an open subset U of $\mathcal{CT}$. One then proves as in Exp. XI 5.3 that G/N is representable by U, hence in particular is quasi-projective.

Let us now study the quotient N/H. Thanks to EGA IV 8, to prove that N/H is representable, one may assume S affine noetherian, then S the spectrum of a local ring A, of finite (Krull[1124]) dimension.

We shall proceed by increasing induction on the dimension of S. If $\dim S = 0$, the property follows from Exp. VI_A § 4. Note now that if N/H is representable, it is separated over S (since H is closed in N by iv)), of finite type and étale over S (since N is smooth over S, of finite type and H is open in N), so N/H is necessarily quasi-affine over S (SGA 1 VIII.6.2). By effective descent of quasi-affine schemes (*loc. cit.* 7.9), one may replace A by its completion, hence assume S spectrum of a complete noetherian local ring. Let s be its closed point and $U = S \setminus s$. By induction hypothesis, $(N|_U)/(H|_U)$ is representable by a U-group K. Let then T_s be the maximal central torus of H_s, q an integer invertible on S, ℓ a power of q, M_ℓ the unique central subgroup of multiplicative type of H lifting ${}_\ell T_s$ (cf. ii)). Choose ℓ large enough that $Centr_G(M_\ell) = H$, and let $N' = Norm_G(M_\ell)$. Since $Centr_G(M_\ell) = H$, one has $N' \subset Norm_G(H)$. Moreover, one immediately verifies that T_s (so also $(M_\ell)_s$) is a characteristic subgroup of H_s (i.e. stable under $\mathrm{Aut}_{gr}(H_s)$), so N_s normalizes $(M_\ell)_s$ and consequently $N_s = N'_s$. The proof of Exp. XI 5.9 then

[1124] (Krull's)

shows that the quotient $N'/H = Norm_G(M_\ell)/Centr_G(M_\ell)$ is representable by an S-group K'. Since N' is smooth over S and $N'_s = N_s$, N' is an open subgroup of N (Exp. VI_B 2.5) containing H, so the image of $N'|_U$ in K is an open subgroup, isomorphic to $K'|_U$. Let L be the S-prescheme obtained by gluing K and K' along the previous isomorphism, and let p be the S-morphism $N \to L$ obtained by gluing the canonical projections $N|_U \to K$ and $N' \to K'$.

It is clear that (L, p) represents the quotient N/H.

Proof of v). Suppose first that S is the spectrum of a field k. The image H' of H is then a smooth subgroup of G'. We must show that $H' \in \mathcal{CT}_{G'}(S)$, which will follow from the following more precise lemma:

Lemma 7.2. *Let $u : G \to G'$ be an epimorphism of smooth connected k-algebraic groups, T a torus of G, T' its image in G'. Then*

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u(Centr_G T) = Centr_{G′} T′.
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Denote $H = Centr_G T$, $H' = Centr_{G'} T'$, $H'' = u(H)$. One has $H'' \subset H'$. To prove that $H' = H''$, one may assume the base field algebraically closed, and it suffices to prove that every Cartan subgroup of H' is contained in H''. Indeed, H'' will then contain the open subset of regular points of H', so H'' will be an open subgroup of H' and consequently will be equal to H' since the latter has connected fibers. So let C' be a Cartan subgroup of H'; C' is also a Cartan subgroup of G', since H' is the centralizer of a torus T', hence has the same reductive rank and the same nilpotent rank as G'. Set $K = (u^{-1}(C'))^0_{red}$. Since T' is in the center of H', C' contains T', so K contains T. Let then C be a Cartan subgroup of K containing T. The torus T is contained in the unique maximal torus of C, which is central in C (Exp. XII 6.6 c)), so C is contained in $H = Centr_G T$. Using now the fact that any two Cartan subgroups of G are conjugate and that the image of a Cartan subgroup of G is a Cartan subgroup of G' (Exp. XII 6.6), one

deduces that C is also a Cartan subgroup of G; its image is therefore a Cartan subgroup of G'; as it is contained in C', one has $u(C) = C'$, so C' is indeed contained in H''.

We have therefore established v) when S is the spectrum of a field k. Let us now study the general case. Since G, H and G' are of finite presentation over S, to prove that $u(H)$ is representable and is an element of $\mathcal{CT}_{G'}(S)$, one reduces by the usual technique to the case where S is affine noetherian, then to the case where S is the spectrum of a local ring. By fpqc descent of subpreschemes of G', one may even assume that S is the spectrum of a complete noetherian local ring A.

Let us resume the notation of ii), namely: let T_s be the maximal central torus of H_s (s is the closed point of S), M_ℓ a subgroup of multiplicative type of H lifting ${}_\ell T_s$ such that $H = Centr_G M_\ell$. Let T'_s be the image of T_s in G'_s. Since G' is separated over S (Exp. VI_B 5.2), the image of M_ℓ by u is a subgroup of multiplicative type M'_ℓ of G' (Exp. IX 6.8). Set then $H' = Centr_{G'} M'_\ell$, which is a smooth subgroup prescheme of G'. For every integer ℓ' equal to a power of q, there exists ℓ such that $(M'_\ell)_s$ majorizes ${}_{\ell'}T'_s$, so we may assume ℓ chosen large enough that $H'_s = Centr_{G'_s} T'_s = u_s(H_s)$, where the last equality follows from 7.2. The restriction of u to H, namely v, evidently factors through H'. Let us prove that $v : H \to H'$ is a flat morphism. Since H and H' are flat over S and $H_s \to H'_s$ is flat, v is flat on a neighborhood of H_s (EGA IV 11.3.10 and 11.3.1). The morphism v is therefore flat on an open subgroup of H (Exp. VI_B 2.2), so v is flat, H having connected fibers.

The set-theoretic image of H is therefore an open subset of H' (necessarily equal to $(H')^0$) which, equipped with its induced structure, represents the image sheaf $u(H)$ (for the fpqc topology). The fact that $u(H)$ is an element of $\mathcal{CT}_{G'}(S)$ then follows from 7.2.

Suppose now that H is the centralizer in G of a torus T, and let T' be the image of T by u, which is a sub-torus of G' (Exp. IX 6.8). The image of H by u is contained in $Centr_{G'}(T')$, coincides fiber by fiber with the latter (7.2), and is smooth over S, so $u(H) = Centr_{G'}(T')$.

Proof of vi). To show that ũ is a faithfully flat S-morphism, knowing that $\mathcal{CT}_G$ and $\mathcal{CT}_{G'}$ are smooth over S (by i)), it suffices to verify it on the geometric fibers. We are therefore reduced to the case where S is the spectrum of an algebraically closed field k. Let $H' \in \mathcal{CT}_{G'}(k)$, T' its maximal central torus, T a sub-torus of G whose image is T', $H = Centr_G(T)$, so that $H' = u(H)$ (7.2), $N = Norm_G(H)$, $N' = Norm_{G'}(H')$. We have shown in iv) that G/N (resp. G'/N') canonically identifies with open neighborhoods of H in $\mathcal{CT}_G$

(resp. of H' in $\mathcal{CT}_{G'}$). Under these identifications, the restriction of ũ to G/N coincides with the natural morphism

$$w : G/N \longrightarrow G'/N'$$

deduced from u by passage to the quotient. Now w is an epimorphism of homogeneous spaces under G, hence is faithfully flat. This proves that ũ is a flat morphism such that $\tilde{u}(\mathcal{CT}_G)$, which is therefore an open subset of $\mathcal{CT}_{G'}$, contains every point of $\mathcal{CT}_{G'}(k)$. Since $\mathcal{CT}_{G'}$ is of finite type over k, one deduces that ũ is surjective, hence faithfully flat.

The complementary assertions contained in vi) in the case where `Ker u` is central follow from Exp. XII 7.12.

The following theorem generalizes Theorem 7.1 of Exp. XII:

Theorem 7.3. *Let S be a prescheme, G an S-prescheme in groups of finite presentation over S, smooth, with connected fibers, and consider the S-functor $\mathcal{C} : (Sch/S)^\circ \to Ens$ such that:*

> *$\mathcal{C}(S')$ = set of Cartan subgroups of $G_{S'}$.*

- *i) The following conditions are equivalent:*
  - *a) The functor $\mathcal{C}$ is representable.*
  - *b) The functor $\mathcal{C}$ is representable by an S-prescheme, smooth, quasi-projective, of finite presentation over S, with affine fibers.*

- *c) The group G admits locally for the étale topology a Cartan subgroup.*
- *d) The group G admits locally for the faithfully flat topology a Cartan subgroup.*
- *e) The nilpotent rank of the fibers of G is a locally constant function on S.*

- *ii) If the preceding conditions hold, any two Cartan subgroups of G are locally conjugate for the étale topology.*

The set of regular points of the fibers of G (Exp. XIII 2.7) is an open subset G_{reg}, of finite presentation over S, and every section of G_{reg} over S is contained in a unique Cartan subgroup of G.

- *iii) Let G' be an S-prescheme in groups of finite presentation over S and $u : G \to G'$ a faithfully flat S-group morphism, so that G' is smooth over S with connected fibers. Then if C is a Cartan subgroup of G, $u(C) = C'$ is representable by a Cartan subgroup of G', and $C \to C'$ is faithfully flat.*
- *iv) Under the conditions of i) and iii) the morphism*

$$\tilde{u} : \mathcal{C}_G \to \mathcal{C}_{G'}, \qquad C \mapsto u(C) = C'$$

is faithfully flat. If moreover $Ker(u)$ is central, $\tilde{u}$ is an isomorphism.

- *v) For any complementary information concerning the transporters, or the relations with maximal tori, one may consult 7.1 and Exp. XII 7.1.*

Proof. i) We shall show that b) ⇒ c) ⇒ d) ⇒ e) ⇒ b) ⇒ a) ⇒ d).

b) ⇒ c). Let s be a point of S. Since C_s is smooth over $\kappa(s)$, there exist points of C_s whose residue field is a finite separable extension of $\kappa(s)$. Applying Exp. XI 1.10, one sees that there exist an open neighborhood U of s and an étale surjective morphism

$U' \to U$ such that $G_{U'}$ admits a Cartan subgroup.

c) ⇒ d) is clear.

d) ⇒ e). Let $s \in S$. By hypothesis, there exists an S-prescheme S', flat over S, whose image contains s, such that $G_{S'}$ admits a Cartan subgroup. Let s' be a point of S' above s. The nilpotent rank of the fibers of $G_{S'}$ is therefore constant on Spec $\mathcal{O}_{S',s'}$, and consequently the nilpotent rank of the fibers of G is constant on Spec $\mathcal{O}_{S,s}$ which is the image of Spec $\mathcal{O}_{S',s'}$ (EGA IV 2.3.4 ii)). Let r be its value. It follows from 6.3 bis that the set E_r of points x of S such that the nilpotent rank of G_x equals r is an ind-constructible part of S, hence contains a neighborhood of s (EGA IV 1.10.1).

e) ⇒ b). The assertion is local on S, so one may assume that the nilpotent rank of the fibers of G is constant and equal to r. For every S-prescheme S' there is then identity between the Cartan subgroups of $G_{S'}$ and the subgroup preschemes of $G_{S'}$ smooth over S' of relative dimension r, whose geometric fibers are centralizers of a torus. Since $\mathcal{CT}$ is representable by 7.1, $\mathcal{C}$ is representable by the open-closed subprescheme of $\mathcal{CT}$ representing the subfunctor of $\mathcal{CT}$ consisting of groups of relative dimension r. The other assertions in b) are contained in 7.1 i), except for the fact that the fibers of $\mathcal{C}$ are affine, which itself follows from Exp. XII 7.1 d).

It is clear that b) ⇒ a). Let us show that a) ⇒ d). It follows from 6.2 iii) that the functor $\mathcal{C}$

commutes with filtered inductive limits of rings, so if $\mathcal{C}$ is representable, it is necessarily representable by an S-prescheme locally of finite presentation (EGA IV 8.14.2). To prove that $\mathcal{C}$ is smooth over S, one is reduced to showing that if S is affine and if S_0 is the closed subscheme defined by a nilpotent ideal J, then every Cartan subgroup C_0 of $G_0 = G \times_S S_0$ lifts to a Cartan subgroup C of G. But the existence of C_0 entails that condition e) is satisfied for S_0, hence for S which has the same underlying space, and one concludes from the fact that d) ⇒ b). Since $\mathcal{C}$ is smooth over S and $\mathcal{C} \to S$ is surjective, one sees that a) ⇒ d).

ii) Let C and C' be two Cartan subgroups of G. Then $Transp_G(C, C')$ is representable by a prescheme smooth over S (7.1 iii)) with non-empty fibers (cf. Exp. XII 6.6 a) and c)). The fact that C and C' are locally conjugate for the étale topology is then a consequence of Hensel's lemma (Exp. XI 1.10).

The other assertions of ii) are consequences of XIII 3.1 and XIII 3.2, in view of i).

iii) Let C be a Cartan subgroup of G. One knows (7.1 v)) that $u(C)$ is representable by a smooth subgroup C' of G'. Since the fibers of C' are Cartan subgroups of the fibers of G' (Exp. XII 6.6 d)), C' is a Cartan subgroup of G'.

iv) To prove that the morphism $\tilde{u}$ is faithfully flat, one proceeds as in 7.1 vi).

If now `Ker u` is central and C' is a Cartan subgroup of G',

$u^{-1}(C') = C$ is smooth, with connected fibers (7.1 vi)) and its fibers are Cartan subgroups (Exp. XII 6.6 f)), so C is a Cartan subgroup of G, which completes the proof of iv), in view of 7.1 vi).

8. Representability criterion for the functor of sub-tori of a smooth group

8.0. In this section, if S is a prescheme and G an S-prescheme in groups, $\mathcal{T}_G$ (or simply $\mathcal{T}$ if there is no ambiguity) denotes the S-functor $(Sch/S)^\circ \to Ens$ such that, for every S-prescheme S', one has

$\mathcal{T}(S')$ = *set of sub-tori of* $G_{S'}$.

One similarly defines $\mathcal{T}_C$ as the functor of central sub-tori of G.

Proposition 8.1. *Let S be a locally noetherian prescheme and G an S-prescheme in groups of finite type. Then the following conditions are equivalent:*

- *i) The functor $\mathcal{T}_G$ "commutes with adic limits of local artinian rings", i.e. for every S-prescheme of the form* Spec$(A) = S'$*, where A is a complete noether-*

ian local ring for the topology defined by its maximal ideal $\mathfrak{m}$, the canonical map

$$\mathcal{T}(S') \to \varprojlim_{n} \mathcal{T}(S'_n) \qquad (\text{where } S'_n = \mathrm{Spec}(A/\mathfrak{m}^n))$$

is bijective.

- *ii) As in i) but restricted to the case where A is a complete discrete valuation ring with algebraically closed residue field.*
- *iii) As in ii), but one restricts to the subfunctor $\mathcal{T}^{(1)}$ of $\mathcal{T}$ relative to sub-tori of G of relative dimension 1.*

- *i bis) For every S-prescheme S' as in i) and every S'-torus T, the canonical map*

$$\mathrm{Hom}_{S'\text{-gr}}(T, G_{S'}) \to \varprojlim_{n} \mathrm{Hom}_{S'_n\text{-gr}}(T_{S'_n}, G_{S'_n})$$

is bijective.

- *ii bis) As i bis), but one restricts to the case where A is a complete discrete valuation ring with algebraically closed residue field.*
- *iii bis) As ii bis), but one restricts to the case where T is the multiplicative group G_m.*

Remark 8.2. One has an analogous proposition restricting to central sub-tori of G and to central homomorphisms of a torus into G.

Proof. We shall use the following lemma:

Lemma 8.3. *Let S be a prescheme, G an S-prescheme in groups, T an S-torus, $u : T \to G$ an S-group morphism. Assume in addition that G is of finite presentation over S, or else that S is locally noetherian and G locally of finite type. Then:*

- *a) $\mathrm{Ker}\ u$ is a subgroup of multiplicative type of T.*
- *b) The quotient $T' = T/Ker u$ is a torus.*
- *c) The canonical monomorphism $T' \to G$ deduced from u by passage to the quotient is an immersion.*

This lemma is a consequence of Exp. IX 6.8 when G is separated over S. In the general case, one reduces as usual to the case where S is noetherian. Let us first show that $K = Ker u$ is flat. We may assume that S is the spectrum of a local artinian ring (EGA 0_III 10.2.6), in which case G is separated (Exp. VI_B 5.2), so K is of multiplicative type (Exp. IX 6.8) and *a fortiori* flat over S. Let us now prove that $\mathrm{Ker}\ u$ is closed in T, which reduces us to the case where S is the spectrum of a discrete valuation ring (EGA II 7.2.1). Since T is flat with connected fibers, u factors through the connected component of the schematic closure in G of the generic fiber of G. We may therefore assume G flat with connected fibers, but then G is separated (Exp. VI_B 5.2), so K is closed. This being so, it follows from Exp. X 4.8 b) that K is a subgroup of multiplicative type of T. The quotient $T' = T/K$ is then representable and T' is a group of multiplicative type (Exp. IX 2.7) whose fibers are tori, so it is a torus. The fact that the monomorphism $T' \to G$ is an immersion then follows from Exp. VIII 7.9.

Proof of 8.1. i) ⇒ i bis). Set $T_n = T_{S'_n}$, $G_n = G_{S'_n}$, and let $(u_n)_{n\in\mathbb{N}}$ be an element of `lim_{←n} Hom_{S′_n-gr}(T_n, G_n)`. For every integer n, $u_n(T_n)$ is therefore a sub-torus T'_n of G_n (Lemma 8.3). By hypothesis, there exists a unique sub-torus T' of G lifting T'_n for every n. Since T' has affine fibers, one concludes thanks to 4.4.

i bis) ⇒ i). Let $(T_n)_{n\in\mathbb{N}}$ be an element of `lim_{←n} ▯(S_n)`. By Exp. X 4.6,

there exist an S'-torus T^0 and an S'_0-isomorphism $u_0 : T^0_0 \xrightarrow{\sim} T_0$. Since T_n is smooth over S'_n, u_0 lifts to an S'_n-morphism $u_n : T^0_n \to T_n$ (Exp. IX 3.6), and one may assume the family $(u_n)_{n\in\mathbb{N}}$ coherent, hence coming from a morphism $u : T^0 \to G$. The image of T^0 by u is then a sub-torus of G (Lemma 8.3) lifting T_n for every n.

The implications i) ⇒ ii) ⇒ iii) on the one hand, and i bis) ⇒ ii bis) ⇒ iii bis) on the other, are evident. The implication iii) ⇒ iii bis) is proved as i) ⇒ i bis). It therefore suffices to prove: iii bis) ⇒ ii bis) ⇒ i bis).

ii bis) ⇒ i bis). With the terminology introduced in 4.3, assertion i bis) is true if and only if every element $(u_n)_{n\in\mathbb{N}}$ of `lim_{←n} Hom_{S_n-gr}(T_n, G_n)` is "admissible". For every point t of S' distinct from the closed point s of S', there exists an S-scheme S'', spectrum of a complete discrete valuation ring with algebraically closed residue field, whose generic point projects to t and whose closed point projects to s (EGA II 7.1.9). One immediately deduces a valuative criterion for a family of morphisms to be admissible. This says that ii bis) ⇒ i bis).

iii bis) ⇒ ii bis). Let T be an S-torus, where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. The torus T is then trivial (Exp. X 4.6), i.e. isomorphic to $(G_{m,S})^r$ for a suitable integer r. Let $(u_n)_{n\in\mathbb{N}}$ be an element of `lim_{←n} Hom_{S_n-gr}((G_m)ʳ_{S_n}, G_n)`. By hypothesis, the restrictions of u_n to each

factor of $(G_m)^r_S$ come from a group morphism $G_{m,S} \to G$. Whence a product morphism $(G_m)^r_S \to G^r$ which, composed with the morphism $G^r \to G$ defined by the composition law in G, gives a morphism

$$v : (G_m)^r_S \longrightarrow G.$$

Given the existence of the group morphism u_n, it is clear that $u_n = v_n$. It remains to see that v is a group morphism, which translates into the fact that two obvious morphisms $f, g : X = (G_m)^r_S \times_S (G_m)^r_S \to G$ coincide. Let Z be the subscheme of coincidences of f and g. Since $(G_m)^r_S$ has connected fibers and $v((G_m)^r_S)$ contains the unit section, one sees as in 8.3 that v factors through the connected component of G, which allows us to assume G separated (Exp. VI_B 5.2), so Z is closed. On the other hand, since v_n is a group morphism, one has $f_n = g_n$ for every n, so Z contains a neighborhood of the closed fiber of X (EGA I 10.9.4), hence is schematically dense in X, X having smooth and irreducible fibers (Exp. IX 4.6), and consequently $Z = X$, so $f = g$.

Definition 8.4. *Let S be a locally noetherian prescheme, G an S-prescheme in groups of finite type, and S the set of S-schemes S', spectrum of a complete discrete* *valuation ring with algebraically closed residue field, of closed point s, of generic point t,*

such that $(G_t)^0_{red}$ is smooth and admits a Chevalley decomposition, i.e. is an extension of an abelian variety by a smooth, connected linear algebraic subgroup L_t (this decomposition is then unique). If $S' \in \mathcal{S}$, denote by G' (resp. L) the schematic closure in $G_{S'}$ of G_t (resp. L_t).

Under these conditions, we shall say that

> *"the abelian part of G does not degenerate into a toric part", or more briefly that G satisfies property AT,*

if for every $S' \in \mathcal{S}$, L_s has the same reductive rank as G'_s. (Intuitively, suppose that the quotient $A = G'/L$ is representable, in which case A is a flat group prescheme such that $(A_t)^0_{red}$ is an abelian variety. The condition "AT" then means that A_s has reductive rank zero, hence $(A_s)^0_{red}$ is an extension of an abelian variety by a unipotent group.)

Similarly, assuming in addition G has connected fibers, we shall say that

> *G satisfies property ATC*

if for every $S' \in \mathcal{S}$, the schematic closure Z of the center Z_t of G_t satisfies AT.

These technical definitions are justified by the following proposition:

Proposition 8.5. *Let S be a locally noetherian prescheme, G an S-prescheme in groups of finite type. Then:*

- *a) For the functor $\mathcal{T}$ of sub-tori of G to "commute with adic limits of local artinian rings" (cf. 8.1), it is necessary and sufficient that G satisfy property AT (8.4).*
- *b) If G has connected fibers, for the functor $\mathcal{T}_C$ of central sub-tori of G to commute with adic limits of local artinian rings,*

it is necessary and sufficient that G satisfy property ATC (8.4).

We shall need the following technical lemma:

Lemma 8.6. *Let S be the spectrum of a discrete valuation ring, s the closed point of S, G an S-prescheme in groups, flat and of finite type, T_s a sub-torus of G_s. Then there exists an S-scheme S', spectrum of a discrete valuation ring faithfully flat over S, with closed point s', and a subgroup scheme C of $G_{S'}$, flat over S', commutative, with connected fibers, such that C'_s majorizes $T_s \times_s s'$.*

Possibly replacing S by S', spectrum of a discrete valuation ring faithfully flat over S, we may assume that T_s is equal to $(G_m)^r_s$ and that the transcendence degree of $\kappa(s)$ over the prime field is $\geqslant r$ (EGA 0_III 10.3.1 and EGA II 7.1.9). There then exists an element x of $T_s(s)$ such that every algebraic subgroup of G_s "containing" x majorizes T_s (cf. Exp. XIII proof of 2.1 (ii) ⇒ (vii)). Since G is flat over S, a quasi-section passes through x (EGA IV 14.5.8), and consequently, possibly replacing S by the spectrum of a discrete valuation ring faithfully flat over S, one may assume that there exists a section $\tilde{x}$ of G above S lifting x. Let C_t be the commutative algebraic subgroup of G_t generated by $\tilde{x}_t$ (Exp. VI_B 7), and let C be the schematic closure of C_t in G. It is clear that C_s contains x, hence majorizes T_s, and consequently, the "connected component" of C will be a flat and commutative group scheme answering the question.

Proof of 8.5 a). Suppose the functor $\mathcal{T}$ commutes with adic limits of artinian rings and let us show that G satisfies property AT. Let then $S' \in \mathcal{S}$, and let T_s be a maximal torus of G'_s. We must prove that T_s is contained in L_s. The formation of L and G' evidently commutes with faithfully flat extensions $S'' \to S'$ of discrete valuation rings. Possibly changing S', we may therefore, by Lemma 8.6, assume that there exists a subgroup scheme C of $G'_{S'}$, flat and commutative, such that C_s majorizes T_s. But then T_s is a central sub-torus of C_s, and consequently lifts infinitesimally to a central sub-torus (Cor. 2.3). Given the hypothesis made on G, T_s lifts to a sub-torus T of G. Evidently T_t is contained in the linear component L_t of G_t, so T is contained in L.

Suppose now that G satisfies property AT, and let us show that condition 8.1 iii bis) is verified. Let then S be the spectrum of a complete discrete valuation ring A with algebraically closed residue field, $\mathfrak{m}$ the maximal ideal of A, $S_n = \mathrm{Spec}(A/\mathfrak{m}^n)$, u_n, $n \in \mathbb{N}$, a coherent system of group morphisms $u_n : (G_{m,S})_n \to G_n = G \times_S S_n$. Let q be a prime number invertible on S. The integer ℓ ranging over the powers of q, there exists a unique S-group morphism

$${}_\ell u : {}_\ell G_{m,S} \to G$$

lifting $u_n|_{\ell(G_{m,S})_n}$ for every n (Prop. 1.6 a)). Consequently, if there exists an S-group morphism $u : G_{m,S} \to T$ lifting u_n for every n, its restriction to ${}_\ell G_{m,S}$ is uniquely determined. By the density theorem (Exp. IX 4.8 (a)), this proves

the uniqueness of u and the fact that to prove the existence of u, we may allow a faithfully flat extension of the base. Now $(G_t)^0_{red}$ admits a Chevalley decomposition, and this is already defined over a finite extension L of the field of fractions K of A. Possibly replacing S by the normalization of S in L, we may therefore assume $S \in \mathcal{S}$. The morphism ${}_\ell u_t$ factors through $(G_t)_{red}$, so ${}_\ell u$ factors through the schematic closure G'' of $(G_t)_{red}$ in G'. Still by the density theorem, one deduces that u_n factors through G''_n for every n. Since G has property AT, every sub-torus of G'_s, and *a fortiori* of G''_s, is contained in L_s, so u_s factors through L_s. I claim that for every n, u_n factors through L_n. Indeed, since L_t is invariant in $(G'')_t$, L is invariant in G''; on the other hand, L is flat over S, so for every integer n, the quotient $H_n = G''_n/L_n$ is representable (Exp. VI_A § 4). The image of $(G_{m,S})_n$ in H_n is a sub-torus of H_n (Exp. IX 6.8) whose closed fiber is zero, so this image is zero and consequently u_n factors through L_n. But since L_t is affine, one deduces that the family u_n is admissible (Prop. 4.3), which completes the proof.

The proof of 8.5 b) is entirely analogous to that of 8.5 a), in view of 8.2 and Exp. IX 5.6 a).

Proposition 8.7. *Let S be a locally noetherian prescheme, G an S-prescheme in* *groups of finite type. Then:*

- *i) If G has connected fibers and satisfies AT, it satisfies ATC.*
- *ii) If G satisfies AT, every subgroup prescheme of G satisfies AT.*
- *iii) If G is flat over S and the abelian rank of the fibers of G is a locally constant function on S, then G satisfies AT.*

i) follows immediately from 8.5 and Exp. IX 5.6 a).

ii) Let H be a subgroup prescheme of G. To prove that H satisfies property AT, we may assume that S is the spectrum of a complete discrete valuation ring and we must show that every coherent family of S_n-group morphisms

$u_n : (G_{m,S})_n \to H_n$

is admissible (8.5 and 8.1 iii bis)). Now by hypothesis, G satisfies property AT, so there exists an S-group morphism

$$u : (G_m)_S \longrightarrow G$$

lifting u_n for every n. Proceeding as in the proof of 8.5, one sees that $u|_{{}_\ell G_{m,S}}$ (where ℓ ranges over the powers of a prime q invertible on S) factors through H. By density, one deduces that on the generic fiber, u_t factors through H_t. Since $(G_m)_S$ is reduced, it indeed follows that u factors through H.

iii) Let $S' \in \mathcal{S}$ (notation of 8.4). The group L is flat over S, its generic fiber L_t is affine; under these conditions, one can show that L_s is necessarily affine (XVII App. III, 2). Since G and L are flat, one has $G = G'$ and the dimension of the fibers of G and L is constant on S (Exp. VI_B 4), so that one has the inequalities:

abelian rank $G_s \leq \dim G_s - \dim L_s = \dim G_t - \dim L_t =$ abelian rank G_t.

The hypothesis entails that these inequalities are equalities. It follows that $(G_s/L_s)^0_{red}$ is an abelian variety, hence has reductive rank zero, and consequently L_s has the same reductive rank as G_s.

We are now in a position to prove the main theorems of this section:

Theorem 8.8. *Let G be a group prescheme of finite type over a locally noetherian prescheme S. Suppose G flat over S with connected fibers. Then:*

- *a) For the functor $\mathcal{T}_C$ of central sub-tori of G to be representable, it is necessary and sufficient that G have property ATC (8.4). Moreover, in this case, $\mathcal{T}_C$ is representable by an S-prescheme étale and separated over S.*
- *b) Under the conditions of a), for every S-torus T, the functor $\mathrm{Hom}^{cent}_{S-gr}(T, G)$ of central homomorphisms of T into G is representable by an S-prescheme étale and separated over S.*

Theorem 8.9. *Let S be a locally noetherian prescheme and G an S-prescheme in groups of finite type, smooth over S.*

- *a) For the functor $\mathcal{T}$ of sub-tori of G to be representable, it is necessary and sufficient*

that G have property AT (8.4). Moreover, in this case, $\mathcal{T}$ is representable by an S-prescheme smooth and separated over S; more precisely, the structural morphism $\mathcal{T} \to S$ admits a canonical factorization

$$\mathcal{T} \xrightarrow{u} Y \xrightarrow{v} S,$$

where v is a smooth and quasi-projective morphism (hence of finite type) and u is an étale separated morphism.

- *b) Under the conditions of a), for every S-torus T, the functor* $\mathrm{Hom}_{S-gr}(T,G)$ *of homomorphisms of T into G is representable by an S-prescheme smooth and separated over S.*

Proof of 8.8 a). If the functor $\mathcal{T}_C$ is representable, it commutes with adic limits of artinian rings, and consequently (8.2 and 8.5 b)) G has property ATC. To establish the converse, we shall use the following result, which will be proved in EGA VI, and is also found in Murre's exposé: Sém. Bourbaki, May 1965, N° 294, Theorem 1, corollary 2.

Lemma 8.10. *Let S be a locally noetherian prescheme and F a contravariant functor defined on Sch/S with values in the category of sets. For F to be representable by an étale separated S-prescheme, it is (necessary and) sufficient that F satisfy the following five properties:*

- *i) F is a sheaf for the fpqc topology (Exp. IV).*
- *ii) F commutes with inductive limits of rings (Exp. XI 3.2).*
- *iii) F commutes with adic limits of local artinian rings (8.1 i)).*
- *iv) F satisfies the "valuative criterion of separation", i.e. for every S-scheme S' which is the spectrum of a discrete valuation ring with generic point t, the canonical map $F(S') \to F(t)$ is injective.*
- *v) F is infinitesimally étale (Exp. XI 1.8).*

Let us show that the functor $\mathcal{T}_C$ of 8.8 satisfies the five conditions of 8.10.

i) The functor $\mathcal{T}$ (resp. $\mathcal{T}_C$) is a sheaf for the fpqc topology as soon as G is of finite presentation over S. Indeed, every monomorphism of a torus into G is then an immersion (Exp. VIII 7.9), and the property follows from fpqc descent of subpreschemes.

ii) Corollary 6.3 proves that the functor $\mathcal{T}$ commutes with inductive limits of rings if G is of finite presentation over S; one immediately deduces that the same is true of $\mathcal{T}_C$.

iii) By 8.5, condition iii) is precisely equivalent to property ATC.

iv) follows simply from the fact that if S is the spectrum of a discrete valuation ring, two sub-tori of G having the same generic fiber coincide and more precisely coincide with the connected component of the schematic closure in G of their generic fiber.

v) follows from 2.3 since G is flat over S.

Proof of 8.8 b). Proceeding as in Exp. XI 4.2, one sees that it suffices to prove that the product group $T \times_S G$ again satisfies property ATC, which is immediate from the definition.

Proof of 8.9 a). Possibly replacing G by its connected component (Exp. VI_B 3.10), we may assume G has connected fibers. If T is a sub-torus of G, its centralizer $Centr_G(T)$ is then representable (Exp. XI 6.11) by a subgroup prescheme of G, smooth over S (Exp. XI 2.4), with connected fibers (Exp. XII 6.6 b)), hence is an element of $\mathcal{CT}(S)$, where $\mathcal{CT}$ is the functor defined in 7.1 i). It is clear that the map

$$T \mapsto Centr_G(T)$$

defines an S-morphism

$$u : \mathcal{T} \longrightarrow \mathcal{CT}.$$

Since $\mathcal{CT}$ is representable by an S-prescheme smooth and quasi-projective over S (7.1 i)), it suffices to prove that the morphism u is representable by a separated and étale morphism.

After a suitable base change $S' \to S$ (with $S' = \mathcal{CT}$, hence S' locally noetherian),

we are reduced to the following problem: let S be a locally noetherian prescheme, G an S-prescheme in groups, smooth and of finite type over S, with connected fibers, H a subgroup of G, smooth over S with connected fibers. Consider the subfunctor X of $\mathcal{T}_{C,H}$ such that, for every S-prescheme S', $X(S')$ is the set of central sub-tori T of $H_{S'}$ such that $Centr_{G_{S'}}(T) = H_{S'}$. We shall show that X is representable by an S-prescheme étale and separated over S.

Indeed, by hypothesis, G satisfies property AT, hence so does H (8.7 ii)); and since H has connected fibers, H also satisfies property ATC (8.7 i)). On the other hand H is smooth over S, hence flat. By 8.8 a), $\mathcal{T}_{C,H}$ is representable by an S-prescheme étale and separated over S. It then suffices to show that the canonical monomorphism $X \to \mathcal{T}_{C,H}$ is representable by an open immersion.

Set $S' = \mathcal{T}_{C,H}$ and let K be the centralizer in $G_{S'}$ of the "universal" central torus of $H_{S'}$. The group K is a smooth group scheme over S' with connected fibers, majorizing $H_{S'}$. By definition, $X = \prod_{K/S'} H_{S'}/K$, which is indeed representable by the open-closed subprescheme of S' above which $H_{S'}$ and K have the same relative dimension.

One proves 8.9 b) analogously to 8.8 b).

Corollary 8.11. *Let S be a prescheme, G an S-prescheme in groups smooth over S and of finite presentation. Then, if the abelian rank of the fibers of G is a locally constant function on S (in particular if the fibers of G are affine), the functor*

of sub-tori of G is representable by an S-prescheme, smooth and separated over S, and the same is true of the functor $\mathrm{Hom}_{S-gr}(T, G)$ for every S-torus T.

Indeed, the assertion is local on S, so we may assume S affine and the abelian rank of the fibers of G constant. One may then find (Exp. VI_B § 10) a noetherian affine scheme S_0 and an S_0-prescheme in groups G_0, smooth and of finite type over S_0, such that G is S-isomorphic to $G_0 \times_{S_0} S$. Moreover, the abelian rank of the fibers of an S-prescheme in groups of finite presentation over S is an ind-constructible function (6.3 bis). By a standard argument (EGA IV 8), one concludes that in the present case, one may assume the abelian rank of the fibers of G_0 constant on S_0. But then G_0 has property AT (8.7 iii)), and consequently (8.10) the functor of sub-tori of G_0 is representable by an S_0-prescheme smooth and separated over S_0, whence the announced property for G. As for the functor $\mathrm{Hom}_{S-gr}(T, G)$, one proceeds analogously.

Generalization of 8.9.

The functor $\mathcal{T}$ of sub-tori of a smooth group G not being necessarily representable, we shall state sufficient conditions for a subfunctor of $\mathcal{T}$ to be representable.

Proposition 8.12. *Let S be a locally noetherian prescheme, G an S-prescheme in groups, smooth over S, with connected fibers, and F an S-subfunctor of the functor $\mathcal{T}$ of*

sub-tori of G, satisfying the following properties:

- *i) F is a sheaf for the fpqc topology (Exp. IV).*
- *ii) F commutes with inductive limits of rings (Exp. XI 3.2).*
- *iii) F commutes with adic limits of local artinian rings (Exp. XI 3.3).*
- *iv) F is infinitesimally smooth over S (Exp. XI 1.8).*
- *v) F is stable under inner automorphisms of G, i.e. for every S-prescheme S', one has:*

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T ∈ F(S′) and g ∈ G(S′) ⇒ int(g)T ∈ F(S′).
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Then F is representable by an S-prescheme, smooth and separated over S.

Proposition 8.12 bis. *Let S and G be as above, T an S-torus, and F a subfunctor of $\mathrm{Hom}_{S-gr}(T, G)$, satisfying the following properties:*

- *i), ii), iii), iv) as above.*
- *v) F is stable under inner automorphisms of G, i.e. for every S-prescheme S', one has:*

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u ∈ F(S′) and g ∈ G(S′) ⇒ int(g) u ∈ F(S′).
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Then F is representable by an S-prescheme smooth and separated over S.

Proof of 8.12. (The proof of 8.12 bis is entirely analogous and is left to the care of the reader.)

Let $u : F \to \mathcal{CT}$ (7.1 i)) be the S-morphism which to every element T of $F(S')$ associates the element $Centr_{G_{S'}}(T)$ of $\mathcal{CT}(S')$. Since $\mathcal{CT}$ is representable by an S-prescheme smooth and quasi-projective (7.1 i)), we are reduced to proving the representability of u. After base change $\mathcal{CT} \to S$, we are reduced to the following problem: given S and G as above, H a smooth subgroup of G with connected fibers, we must represent the functor F_H such that $F_H(S')$ = set of elements $T \in F(S')$ such that

$$H_{S'} = Centr_{G_{S'}}(T).$$

We shall show that F_H is étale and separated over S. To do this, it suffices to verify the five conditions of 8.10.

Conditions i), ii) and iii) of 8.10 follow easily from 8.12 i), ii) and iii). One has already remarked that $\mathcal{T}_{C,H}$ is a separated and infinitesimally étale functor, so F_H, which is a subfunctor of $\mathcal{T}_{C,H}$, is separated and infinitesimally unramified (Exp. XI 1.8). It therefore suffices to show that F_H is infinitesimally smooth (*loc. cit.*). Let S be the spectrum of a local artinian ring, S_0 a subscheme of S defined by a nilpotent ideal, T_0 an element of $F_H(S_0)$; let us prove that T_0 lifts to an element T of $F_H(S)$. By hypothesis (8.12 iv)), T_0 lifts to an element T' of $F(S)$. On the other hand, H being smooth, T_0 lifts to a sub-torus T'' of H_S (Exp. IX 3.6 bis), which is conjugate to T' by an element $g \in G(S)$ (*loc. cit.*), so $T'' \in F(S)$ (8.12 v)). Since $Centr_{G_S}(T'')$ is smooth over S and coincides with H_{S_0} above S_0, T'' lies in the

center of H_S (Exp. IX 5.6 a)) and its centralizer in G is equal to H_S; in short, $T'' \in F_H(S)$, and one takes $T = T''$.

Corollary 8.13. *Let S be a prescheme, G an S-prescheme in groups smooth and of finite presentation over S, with connected fibers, and let $\mathcal{T}_D$ be the subfunctor of $\mathcal{T}$ whose set of points with values in an S-prescheme S' is the set of sub-tori T of $G_{S'}$ such that for every point s' of S', T'_s is contained in the derived subgroup (Exp. VI_B 7) of G'_s. Then $\mathcal{T}_D$ is representable by an S-prescheme smooth and separated over S.*

Corollary 8.13 bis. *Let S and G be as above, T an S-torus, and let $\mathrm{Hom}^{der}_{S-gr}(T,G)$ be the subfunctor of $\mathrm{Hom}_{S-gr}(T,G)$ whose set of points with values in S' is the set of S'-morphisms $u : T_{S'} \to G_{S'}$ such that for every point s' of S', u'_s factors through the derived group of G'_s. Then this functor is representable by an S-prescheme smooth and separated over S.*

Corollary 8.13 and Corollary 8.13 bis are proved analogously; let us prove 8.13 bis for example. By the usual procedure, we reduce to the case where S is noetherian. Let us verify that the five conditions of 8.12 bis are satisfied:

Conditions i) and iv) follow immediately from the corresponding properties of the functor $\mathrm{Hom}_{S-gr}(T,G)$. Condition v) is satisfied since the derived group of an algebraic group is invariant (Exp. VI_B § 7). To establish ii), we are reduced by a standard reduction (EGA IV 8) to proving that if S is a noetherian integral scheme with generic point η and $u : T \to G$ is an S-group morphism which on the generic fiber

factors through the derived group of G_η, then there exists a neighborhood U of η such that, for every point s of U, u_s factors through the derived group of G_s. But this follows immediately from Exp. VI_B 10.12. To establish iii), let us resume the notation of 4.3. To show that an element $(u_m)_{m\in\mathbb{N}}$ of $\lim_{\leftarrow m} \mathrm{Hom}^{der}_{S_m-gr}(T_m, G_m)$ is "admissible", in the sense of 4.3, and comes from an element of $\mathrm{Hom}^{der}_{S-gr}(T, G)$, one reduces immediately to the case where S is the spectrum of a complete discrete valuation ring (cf. 8.1). Let t be the generic point and s the closed point of S, D the schematic closure in G of the derived group D_t of G_t.

a) D_s contains the derived group of G_s. Indeed, since D_t is invariant in G_t and G is flat over S, D is invariant in G. Moreover the morphism

$$G \times_S G \to G, \quad (x, y) \mapsto x y x^{-1} y^{-1}$$

factors through D_t on the generic fiber, hence factors through D. Consequently the algebraic group G_s/D_s is commutative, whence assertion a).

b) If $u_m \in \mathrm{Hom}^{der}_{S_m-gr}(T_m, G_m)$, then u_m factors through D_m. Indeed, by hypothesis, u_s factors through the derived group of G_s, hence *a fortiori* factors through D_s by a). Since D_m is flat over S_m and invariant in G_m, the quotient group $H_m = G_m/D_m$ is representable (Exp. VI_A § 4). Since the image of T_m in H_m is a torus (Exp. IX 6.8) whose closed fiber is zero, the image of T_m is zero; this says that u_m factors through D_m.

c) The family $(u_m)_{m\in\mathbb{N}}$ is "admissible" and lifts to a morphism $T \to G$ belonging to $\mathrm{Hom}^{der}_{S-gr}(T, G)$.

By what precedes and 4.1 bis, it suffices to prove that D_t is an affine algebraic group, which follows from the following lemma:

Lemma 8.14. *Let G be a smooth connected algebraic group defined over a field k. Then the derived group D of G is affine.*

Since the formation of D commutes with base field extension (Exp. VI_B 7), one may assume k algebraically closed. But then G is an extension of an abelian variety A by a linear group L. Since A is commutative, D is necessarily contained in L, hence is affine.

Maximal tori.

Theorem 8.15. *Let S be a locally noetherian prescheme and G an S-prescheme in groups, smooth and of finite type over S. Then the following conditions are equivalent:*

- *i) The S-functor $\mathcal{T}_M$, whose set of points with values in an S-prescheme S' is the set of maximal tori of $G_{S'}$ (Exp. XII 1.3), is representable.*

- *ii) The preceding functor $\mathcal{T}_M$ is representable by an S-prescheme smooth and quasi-projective over S with affine fibers.*
- *iii) The group G admits locally for the étale topology a maximal torus.*
- *iv) The group G admits locally for the faithfully flat topology a maximal torus.*
- *v) The group G has property AT (8.4), and the reductive rank of its fibers is a locally constant function on S.*

Proof. ii) ⇒ i) is clear.

i) ⇒ iii). Indeed, since G is of finite presentation over S, it follows from 6.3 that $\mathcal{T}_M$ commutes with inductive limits of rings, hence is locally of finite presentation if it is representable (EGA IV 8.14). Moreover $\mathcal{T}_M$ is formally smooth (Exp. XI 2.1 bis). So if it is representable, it is representable by a prescheme smooth over S, and iii) then follows from Hensel's lemma (Exp. XI 1.10).

ii) ⇒ iv) is clear.

iii) ⇒ v). Let s be a point of S. By hypothesis, there exists an S-prescheme S', flat over S, whose image contains s, such that $G_{S'}$ admits a maximal torus T'. Let s' be a point of S' above s. The reductive rank of the fibers of $G_{S'}$ is therefore constant on $\operatorname{Spec} \mathcal{O}_{S',s'}$, and consequently the reductive rank of the fibers of G is constant on the image of $\operatorname{Spec} \mathcal{O}_{S',s'}$, which is $\operatorname{Spec} \mathcal{O}_{S,s}$ (EGA IV 2.3.4 ii)). Now the reductive rank of the fibers of G is a locally constructible function on S (6.3 bis), so this rank is constant on a neighborhood of s (EGA IV 1.10.1).

To see that G has property AT, consider an S-scheme S_1, spectrum of a discrete valuation ring, with closed point s_1 projecting to the point s above. The prescheme $S_1' = S_1 \times_S S'$ is faithfully flat over S_1, and $G_{S_1'}$ admits a maximal torus. Let A (resp. A') be the ring of S_1 (resp. S_1'). Regarding A' as an inductive limit of its finitely generated A-subalgebras, it follows from 6.3 that there exists an S_1-scheme S_1'' such that $G_{S_1''}$ admits a maximal torus and the structural morphism $S_1'' \to S_1$ is of finite type and surjective. Using now EGA II 7.1.9, we may assume that S_1' is the spectrum of a discrete valuation ring. But then it is clear that $G_{S_1'}$, hence also G_{S_1}, has property AT. Since this is true for every S-prescheme S_1 spectrum of a discrete valuation ring, G has property AT.

v) ⇒ i). Indeed, by 8.9, the functor $\mathcal{T}$ of sub-tori of G is representable, and it is clear that $\mathcal{T}_M$ is representable by the open-closed subprescheme of $\mathcal{T}$ representing the subfunctor of tori of rank r, where r denotes the reductive rank of G (which one may assume constant).

iii) ⇒ ii). Indeed, if condition iii) is realized, we may use the results of Exp. XII 7.1. The functor $\mathcal{T}_M$ is therefore canonically isomorphic to the functor of Cartan subgroups of G, and it suffices to apply 7.3 i).

Remark 8.16. One can show that the prescheme $\mathcal{T}_M$ of maximal tori of G is affine over S[1125], which generalizes Exp. XII 5.4.

Corollary 8.17. *Let S be a prescheme, G an S-prescheme in groups,*

smooth and of finite presentation over S. Suppose that the abelian rank and the reductive rank of the fibers of G are locally constant functions on S; then G satisfies the (equivalent) properties i) to iv) of 8.15.

We may assume that the abelian rank and the reductive rank of the fibers of G are constant. Proceeding as in 8.11, and in view of 6.3 bis, we reduce to the case where S is noetherian. But then G has property AT (8.7), and one concludes by 8.15 v).

Corollary 8.18. *Let S be a prescheme, G an S-prescheme in groups, smooth over S. Then the following conditions are equivalent:*

- *a) The unipotent rank ρ_u and the abelian rank ρ_{ab} (6.1 ter) of the fibers of G are locally constant functions on S.*
- *b) The unipotent rank ρ_u is locally constant and G admits locally for the fpqc topology maximal tori.*
- *c) The reductive rank ρ_r (6.1 ter) and the abelian rank ρ_{ab} of the fibers of G are locally constant functions on S.*

Remarks 8.19. Under the hypotheses of 8.18, a more refined argument, using the lower semicontinuity of the abelian rank (announced in Exp. X 8.7), shows that if two of the three ranks ρ_u, ρ_r, ρ_{ab} are locally constant, then so is the third.

Proof of 8.18. Possibly replacing G by its connected component, we may assume G of finite presentation over S (Exp. VI_B 5.3.3).

a) ⇒ c). Let s be a point of S. Since ρ_{ab} is locally constant, it follows from 8.11 that, possibly after an étale extension covering S, we may assume that there exists a sub-torus T of G whose fiber T_s is a maximal torus of G_s. Let $C = Centr_G(T)$, which is a subgroup prescheme of G, smooth over S with connected fibers. For every point t of S, C_t evidently contains a Cartan subgroup C'_t of G_t. Possibly restricting S, we may assume ρ_u, ρ_{ab}, and the relative dimension of C over S are constant. One then has the inequalities

$$\begin{aligned} \dim C_s = \dim C_t \geq \dim C'_t \geq \rho_u(t) + \rho_{ab}(t) + \dim T_t \\ = \rho_u(s) + \rho_{ab}(s) + \dim T_s = \rho_\nu(s) = \dim C_s. \end{aligned}$$

One deduces that $C_t = C'_t$, hence that T is a maximal torus of G, and *a fortiori* $\rho_r(t) = \rho_r(s)$.

c) ⇒ b) by 8.17.

d) ⇒ a). Indeed, since G admits locally for the fpqc topology maximal tori, it admits locally for the fpqc topology Cartan subgroups, and consequently (Exp. XII 7.3) the nilpotent rank $\rho_\nu = \rho_u + \rho_r + \rho_{ab}$ is locally constant. Since ρ_r and ρ_{ab} are locally constant, ρ_u is locally constant.

[1125] cf. M. Raynaud, *Faisceaux amples sur les schémas en groupes et les espaces homogènes* (thesis, to appear)(N.D.E.: see Lecture Notes Math. 119 (1970), Springer), in particular IX 2.9.

Exposé XVI. Groups of zero unipotent rank

by M. Raynaud[1126]

1. An immersion criterion

1.1. Examples of monomorphisms of group preschemes that are not immersions.

We are going to construct a prescheme S, two S-group preschemes G and H, and an S-group monomorphism $u : G \to H$ that is not an immersion. The groups G and H will be of finite presentation over S, flat over S, and G will even be smooth over S (cf. Exp. VIII, paragraph and footnote $(*)$ preceding 7.1; see also XVII App. III, 4).

a) Take first for S the spectrum of a discrete valuation ring A, of unequal characteristic and of residue characteristic equal to 2. Let t be the generic point of S and s the closed point. Take for G the open subgroup of the constant group $G_0 = (\mathbb{Z}/2\mathbb{Z})_S$ obtained by removing the point of the closed fiber $(\mathbb{Z}/2\mathbb{Z})_s$ distinct from the identity element, and take for H the group of multiplicative type $(\mu_2)_S$ of second roots of unity (Exp. I 4.4.4). In view of the choice of S, H_t is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_t$ while H_s is a radicial group. One has an evident morphism $G_0 \to H$ defined by the section (-1) of H, whence a morphism $u : G \to H$ which is a monomorphism but is not an immersion (otherwise it would necessarily be an isomorphism (cf. Exp. VIII 7)). Here, Exp. VIII 7.9 does not apply, since ${}_2G = G$ is not finite over S.

b) Denote by K the S-group prescheme, étale and non-separated, obtained from the unit group by "doubling" the unique point of the closed fiber. Take for H the product group $(\mu_2)_S \times_S K$. Let a (resp. b) be the unique section of μ_2 (resp. K) over S distinct from the unit section, and let $h = (a, b)$ be the corresponding section of H. The datum of h defines an S-group morphism

$u : G = (\mathbb{Z}/2\mathbb{Z})_S \to H.$

It is clear that u is a monomorphism and is not an immersion. Here, Exp. VIII 7.9 does not apply, since ${}_2H = H$ is not separated over S.

c) More interesting is the following example, in which G is a smooth group with connected fibers (and even $G = (\mathbb{G}_a)_S$) (cf. Exp. VIII 7.10).

Take for S the spectrum of a discrete valuation ring A of equal characteristic 2, and let π be a uniformizer of A and $\mathfrak{m}$ the maximal ideal of A.

Consider the additive group $(\mathbb{G}_a)_S$, the product group $(\mathbb{G}_a \times \mathbb{G}_a)_S$ of ring $A[X, Y]$, and let G_1 be the closed subgroup of equation

$$(1) X^2 + X - \pi Y = 0.$$

[1126] Cf. footnote on page 1 of Exp. XV.

The group G_1 is smooth over S, its generic fiber is isomorphic to $(\mathbb{G}_a)_t$ and its closed fiber is isomorphic to the product of $(\mathbb{G}_a)_s$ by $(\mathbb{Z}/2\mathbb{Z})_s$. One may take for the factor $(\mathbb{Z}/2\mathbb{Z})_s$ the subgroup N_s of $(G_1)_s$ of equation

$$(2) Y = X^2 + X = 0.$$

Let N' and N'' be two group subschemes of G_1, isomorphic to $(\mathbb{Z}/2\mathbb{Z})_S$, whose closed fibers coincide with N_s and whose generic fibers are distinct. The groups N' and N'' are therefore defined by the data of two sections of G_1 over S with coordinates (a', b') (resp. (a'', b'')) such that

$$(3) \quad \begin{aligned} &a'^2 + a' - \pi b' = a''^2 + a'' - \pi b'' = 0,\\ &a' \text{ and } a'' \equiv 1 \pmod{[illegible]},\ a' \neq a'',\\ &b' \text{ and } b'' \equiv 0 \pmod{[illegible]}. \end{aligned}$$

(One may take, for example, $(1, 0)$ and $(1 + \pi^2, \pi + \pi^3)$.)

The quotient groups $G' = G_1/N'$ and $G'' = G_1/N''$ are representable (Exp. V 4.1). Let u be the canonical morphism

$$u : G_1 \to G' \times_S G'',$$

and let H be the group subscheme of $G' \times_S G''$ equal to the schematic closure in $G' \times_S G''$ of $u_t((G_1)_t)$, so that u factors through H. The kernel of u is $K = N' \cap N''$, whose generic fiber is the unit group and whose closed fiber is equal to N_s; in particular, K is not flat over S. On the generic fiber, u_t is therefore an isomorphism of $(G_1)_t$ onto H_t. On the other hand, I claim that H_s is not smooth. Indeed, if H_s were smooth, then since u_s is a morphism with finite kernel and $(G_1)_s$ and H_s have the same dimension (namely 1), u_s would be a flat morphism; since G_1 and H are flat, u would be a flat morphism and consequently $\mathrm{Ker}\ u$ would be flat over S, which is not the case. It is clear that the restriction of u to the connected component G of G_1 is a monomorphism (the fibers of $K \cap G$ are unit groups, so $K \cap G$ is the unit group (Exp. VI_B 2.9)), but it is not an immersion (otherwise it would be an open immersion and H_s would be smooth). Here, Exp. VIII 7.9 does not apply, since ${}_2G = G$ is not finite over S.

For lovers of equations, let us say that in the example above one may take for H the closed subgroup of $(\mathbb{G}_a \times_S \mathbb{G}_a)_S$ with ring $A[V, W]/(F)$, where

$$F = (a''b' - b''a')(a''^2 V - a'^2 W) - (a''V - a'W)^2$$

(where a', a'', b', b'' satisfy (3)). For G, one takes the group $(\mathbb{G}_a)_S$ with ring $A[T]$, and for morphism $u : G \to H$ the S-morphism defined by the maps

$$V \mapsto a'(a'T + \pi T^2), \qquad W \mapsto a''(a''T + \pi T^2).$$

Remark 1.2. *The preceding construction is inspired by the method of Koizumi–Shimura.*[1127] *It does not apply when S is the spectrum of a ring of unequal charac-*

[1127]N.D.E.: S. Koizumi & G. Shimura, Specialization of abelian varieties, *Sci. Papers Coll. Gen. Ed. Univ. Tokyo* **9** (1959), 187–211.

teristic, the points of finite order of G being then "too rigid".

1.3. Statement of the immersion criterion.

Theorem 1.3. *Let S be a locally noetherian prescheme, G an S-group prescheme, smooth over S, of finite type, possessing locally for the fpqc topology Cartan subgroups (Exp. XV 6.1 and 7.3 (i)), H an S-group prescheme locally of finite type, $u : G \to H$ an S-group monomorphism. Then:*

a) If G has connected fibers, in order that u be an immersion, it is necessary and sufficient that for every S-scheme S' which is the spectrum of a complete discrete valuation ring with algebraically closed residue field, and for every Cartan subgroup C of $G_{S'}$, the restriction of $u_{S'}$ to C be an immersion.

b) In order that u be an immersion (resp. a closed immersion), it is necessary and sufficient that for every S' as above and every Cartan subgroup C of the connected component $(G_{S'})^0$ of $G_{S'}$, the restriction of $u_{S'}$ to the normalizer (Exp. XI 6.11) N of C in $G_{S'}$ be an immersion (resp. a closed immersion).

Before proving 1.3, let us state some applications.

Corollary 1.4. *Let S be a prescheme, G an S-group prescheme smooth over S, of finite presentation over S, of zero unipotent rank (Exp. XV 6.1 ter) and possessing locally for the fpqc topology maximal tori, H an S-group prescheme, $u : G \to H$ an S-group monomorphism. Suppose further that either H is of finite presentation over S, or S is locally noetherian and H locally of finite type. Then:*

a) If G has connected fibers, u is an immersion.

b) If H is separated over S and if for every S' over S and every maximal torus T of $G_{S'}$ the Weyl group

$$W = Norm_{G_{S'}}(T)/Centr^0_{G_{S'}}(T)$$

is representable (a condition always realized if G has connected fibers (Exp. XV 7.1 (iv))) and is finite over S, then u is a closed immersion.

Corollary 1.5. *Let S be a prescheme, G an S-group prescheme smooth over S, of finite presentation over S, with connected fibers, H an S-group prescheme, $u : G \to H$ an S-group monomorphism. Suppose either H of finite presentation over S, or S locally noetherian and H locally of finite type. Then:*

a) If G is reductive, i.e. is affine over S with reductive fibers (cf. Exp. XIX), u is an immersion, and a closed immersion if H is separated.

b) If G has affine, solvable, connected fibers, of zero unipotent rank, u is an immersion, and a closed immersion if H is separated.

Proof of 1.4 from 1.3.

Reduction to the noetherian case. If S is locally noetherian, the reduction is immediate, the properties to prove being local on S. In the second case, G and H are of finite presentation over S. Replacing S by a smaller open if necessary, we may assume S affine. By Exp. VI_B § 10, there exist a noetherian scheme S_0, an S_0-group prescheme G_0 smooth over S_0 (with connected fibers in case a)), of finite type over S_0, an S_0-group prescheme H_0 of finite type (separated in case b)), and an S_0-group morphism

$$u_0 : G_0 \longrightarrow H_0,$$

such that G, H, u are obtained from G_0, H_0, u_0 by a base extension $S \to S_0$. The fact that u is a monomorphism translates to $Ker u$ = unit group; we may therefore assume that u_0 is a monomorphism. Since G has zero unipotent rank ρ_u and possesses, locally for the fpqc topology, maximal tori, the abelian rank ρ_{ab} of the fibers of G is locally constant (Exp. XV 8.18). But ρ_u and ρ_{ab} are locally constructible functions (Exp. XV 6.3 bis). A standard argument (cf. EGA IV 8.3.4) shows that one may choose S_0 and G_0 so that the unipotent rank (resp. the abelian rank) of the fibers of G_0 is zero (resp. locally constant). But then G_0 possesses locally maximal tori for the fpqc topology (Exp. XV 8.18), and the functor $\mathcal{TM}_{G_0}$ of maximal tori of G_0 is representable by an S_0-scheme X_0, of finite type over S_0 (Exp. XV 8.15). Let T_0 be the "universal" maximal torus of $(G_0)_{X_0}$, X the S-prescheme $X_0 \times_{S_0} S$, $T = T_0 \times_{S_0} S$ the universal maximal torus for G. By hypothesis, in case b), the Weyl group relative to T is representable and finite over X. These two properties are compatible with projective limits of preschemes (Exp. VI_B 10.1 iii) and EGA IV 8.10.5). One may therefore choose S_0 so that the Weyl group of T_0 is finite over X_0. Under these conditions it is clear that to prove 1.4 we may replace S, G, u, H by S_0, G_0, u_0, H_0, hence assume S noetherian.

Let us use the valuative criterion provided by 1.3. We are reduced to the case where S is the spectrum of a discrete valuation ring and where G_0 possesses a Cartan subgroup C.

Proof of 1.4 a). We must show that the restriction of u to C is an immersion (1.3 a)), which reduces us to the case where $G = C$ has nilpotent connected fibers. Standard reductions (cf. Exp. VIII proof of 7.1) reduce us to the case where H is flat over S, u is an isomorphism on the generic fiber, and u is an isomorphism of underlying spaces on the closed fiber. The group H is then with connected fibers, hence separated over S (Exp. VI_B 5.2). Granting for a moment the lemma:

Lemma 1.6. *Let S be a prescheme, G an S-group prescheme smooth over S, with connected, nilpotent fibers, of zero unipotent rank. Then:*

i) G is commutative.

ii) For every integer $n > 0$, ${}_nG$ is a group prescheme, flat, quasi-finite over S, finite over S if and only if the abelian rank, or the reductive rank, of G is locally constant on S.

iii) For every integer $q > 0$ invertible on S, the family of subgroups ${}_{q^n}G$ is universally schematically dense in G relative to S (EGA IV 11.10.8).

Lemma 1.6 applies to the group G, and since G possesses locally maximal tori, the reductive rank of the fibers of G is locally constant; hence (1.6 ii)), for every integer $n > 0$, ${}_nG$ is finite over S. The fact that u is an immersion then follows from Exp. VIII 7.9.

Proof of 1.6. Let us first examine the case where S is the spectrum of a field:

Lemma 1.7. *Let k be a field of characteristic p, G an algebraic k-group, smooth, connected, nilpotent, of zero unipotent rank. Then G is a commutative group, extension of an abelian variety A by a torus T. For every integer $n > 0$, ${}_nG$ is a finite group, étale if $(n,p) = 1$, defined by a k-algebra of rank $n^{\rho_r + 2\rho_{ab}}$. If $(q,p) = 1$, the family of subgroups ${}_{q^n}G (n \in \mathbb{N})$ is schematically dense in G.*

Proof of 1.7. Let Z be the center of G. The group G/Z is affine (Exp. XII 6.1), of zero unipotent rank, smooth and connected, hence is a torus (*Bible* 4 th. 4); but then G is commutative (Exp. XII 6.4). If T is the unique maximal torus of G (Exp. XV 3.4), it follows at once from Chevalley's theorem (Sém. Bourbaki 1956/57, No. 145) that G is an extension of an abelian variety A by T. For every integer $n > 0$, raising to the nth power is an epimorphism in T; one deduces an exact sequence

$$0 \to {}_nT \to {}_nG \to {}_nA \to 0.$$

A classical theorem of Weil (A. Weil: *Variétés abéliennes et courbes algébriques*, § IX th. 33 cor. 1) tells us that ${}_nA$ is a finite group defined by a k-algebra of rank $n^{2\rho_{ab}}$. Since ${}_nT$ is a finite group of rank n^{ρ_r}, one deduces the announced structure of ${}_nG$.

Now let H be the smallest closed subscheme of G majorizing ${}_{q^n}G$ for every n. It follows from the foregoing and from Exp. XV 4.6 that H is a smooth, connected subgroup, hence of the same type as G. Raising to the nth power in H is an epimorphism (since ${}_qH$ is finite), so that one has the exact sequence

$$0 \to {}_qH \to {}_qG \to {}_q(G/H) \to 0.$$

It follows that ${}_q(G/H) = 0$, hence $G/H = 0$. That is to say, the subgroups ${}_{q^n}G$ are schematically dense in G.

Continuation of the proof of 1.6 i). To show that G is commutative, we reduce by the usual procedure to the case where S is noetherian, then to the case where S is the spectrum of a local ring with closed point s. The center of G is representable by a closed group subscheme Z of G (Exp. XI 6.11). To show that $Z = G$, it suffices to show that $Z = G$ after reduction by every power of the maximal ideal of the ring of S (for Z will then be an open subgroup of G, hence equal to G since G has connected fibers). This reduces us to the case where S is local artinian.

Let q be an integer invertible on S. For every integer n, ${}_{q^n}G_s$ is a subgroup of multiplicative type of G_s, étale and central (1.7). Since G is smooth, ${}_{q^n}(G_s)$ lifts to an S-group subscheme étale and central M_n of G (Exp. XV 1.2). The family of flat subgroups M_n, $n \in \mathbb{N}$, is then schematically dense in G (1.7 and EGA IV 11.10.9). Since Z is closed in G and majorizes all the M_n, we have $Z = G$.

1.6 ii). To see that ${}_nG$ is flat and quasi-finite over S, it suffices to show that raising to the nth power in G is a flat and quasi-finite morphism. Since G is flat over S, we are reduced to the case where S is the spectrum of a field (EGA IV 11.3.10), and we noted in the proof of 1.7 that raising to the nth power was an epimorphism. Moreover ${}_nG$ is separated over S, since G is separated over S (Exp. VI_B 5.2). For ${}_nG$ to be finite over S, it is necessary and sufficient that the fibers of ${}_nG$ be the spectra of finite algebras, of locally constant rank (one sees this immediately by reducing to the case where S is the spectrum of a henselian local ring). But, by 1.7, this condition amounts to saying that the abelian rank, or the reductive rank, of the fibers of G is locally constant.

1.6 iii). To see that the family of subgroups ${}_{q^n}G (n \in \mathbb{N})$ is universally schematically dense in G, we again reduce to the case S noetherian. Taking 1.7 and 1.6 b) into account, it suffices to apply EGA IV 11.10.9.

Proof of 1.4 b). We have reduced to the case where S is the spectrum of a discrete valuation ring and where G_0 possesses a Cartan subgroup C. Let $N = Norm_G C = Norm_G T$, where T is the unique maximal torus of C (Exp. XII 7.1 a) and b)). Since H is separated over S, it follows from 1.6 ii) and from Exp. VIII 7.12 that the restriction of u to C is a closed immersion. On the other hand, to prove that u is a closed immersion, it suffices to show that this is the case for $u|N$ (1.3 b)). Since by hypothesis $W = N/C$ is representable by a finite group scheme, this will follow from the following lemma, applied to the exact sequence

$$0 \longrightarrow C \longrightarrow N \longrightarrow W \longrightarrow 1$$

(note that a proper immersion is a closed immersion).

Lemma 1.8. *Let S be a prescheme, G an S-group prescheme of finite presentation over S, extension of a group prescheme G'', proper and of finite presentation over S, by a group prescheme G', of finite presentation and flat over S:*

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1.$$

Let on the other hand H be an S-group prescheme of finite presentation over S (or locally of finite presentation if S is locally noetherian) and $u : G \to H$ an S-group morphism. Then if the restriction of u to G' is proper, u is proper.

Proof of 1.8. We reduce as usual to the case where S is noetherian (Exp. VI_B § 10 and Exp. XV 6.2) and we may then apply the valuative criterion of properness (EGA II 7.3.8). We therefore suppose that S is the spectrum of a discrete valuation ring A, with closed point s and generic point t. Let $x \in G(t)$ and $h \in H(S)$ be such that $u_t(x) = h(t)$. We must show that x comes from a unique element x of $G(S)$. It even suffices to prove the existence and uniqueness of x after a faithfully flat extension of the discrete valuation ring A. Now let y be the projection of x in $G''(t)$. Since G'' is proper over S, y comes from a unique element y of $G''(S)$. Let X be the inverse image of y in G. The prescheme X is faithfully flat over S (since G' is faithfully flat over S as is the morphism $G \to G''$ (Exp. VI_B § 9)). Replacing if necessary the discrete valuation ring by a faithfully flat one, we may assume that X has a section ℓ over S (EGA IV 14.5.8). Replacing x by $x\ell_t^{-1}$, we may assume that $x \in G'(t)$. But then the existence and uniqueness of x follow from the fact that $u|G'$ is proper.

Proof of 1.5.

a) We shall see in Exp. XIX that if G is reductive, G possesses locally for the faithfully flat topology maximal tori, has a finite Weyl group, and has zero unipotent rank. Assertion a) follows, in view of 1.4.

b) If G has affine fibers of zero unipotent rank, G possesses maximal tori locally for the fpqc topology (Exp. XV 8.18). It then suffices to apply 1.4, taking into account the fact that G, having connected, smooth, affine, solvable fibers, has unit Weyl group (*Bible* 6 th. 1 cor. 3).

Proof of 1.3 a).

Since the morphism u is already a monomorphism, in order to see that u is an immersion, it suffices to show that u is proper at the points of $u(G)$ (EGA IV 15.7.1); for that we may use the valuative criterion of local properness (EGA IV 15.7.5). We are thus reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, with closed point s and generic point t. Since G is flat over S, standard reductions (cf. Exp. VIII proof of 7.1) allow us to reduce to the case where H is flat over S, u_t is an isomorphism, and u_s is an isomorphism of underlying spaces. To say that u is an immersion is then equivalent to saying that u is an isomorphism.

By hypothesis, the group G possesses locally Cartan subgroups for the fpqc topology, so we may speak of the open set U of regular points of G (Exp. XV 7.3 i) and ii)).

Lemma 1.9. *With the preceding hypotheses, in order that u be an immersion, it suffices that $u|U$ be an immersion.*

Indeed, to say that $u|U$ is an immersion means that there exist an open V of H and a closed F of V such that $u|U$ factors through F and induces an isomorphism of U onto F. Since H_t, and hence also V_t, is irreducible, one necessarily has $V_t = F_t$. But then F_t majorizes the schematic closure of V_t in V, which is equal to V since H is flat over S. In short, $V = F$. It follows that $u|U$ is an open immersion, hence u is an open immersion (VI_B 2.6).

It remains to show that $u|U$ is an immersion, and for that we apply the valuative criterion of local properness. Replacing S by the spectrum of a faithfully flat discrete valuation ring, we must show that if h is a section of H over S whose image $h(S)$ is contained in $u(U)$, then h is the image of a section g of U over S. It suffices to show that h is contained in the image of a Cartan subgroup C of G. Indeed, by hypothesis

$u|C$ is an immersion, so h is the image of a section g of C, which is necessarily a section of U.

Let $a \in U(s)$ (resp. $b \in U(t)$) be such that $u_s(a) = h(s)$ (resp. $u_t(b) = h(t)$). Since u_t is an isomorphism, $h(t)$ is a regular point of H_t, so is contained in a unique Cartan subgroup D_t of H_t (Exp. XIII 3.2). Let D be the schematic closure of D_t in H.

Lemma 1.10. *i) D_s is a nilpotent algebraic group (Exp. VI_B § 8).*

ii) $\dim D_s = \nu$ *= nilpotent rank of G = nilpotent rank of $(H_s)_{red}$.*

iii) $h(s) \in D_s(\kappa(s))$.

i) Since H is separated (Exp. VI_B 5.2), so is D. Moreover, D is flat over S, and its generic fiber is nilpotent. The fact that D_s is nilpotent then follows from Exp. VI_B 8.4.

ii) Since $(H_s)_{red} \simeq G_s$, these two groups have the same nilpotent rank. Since G possesses, locally for the fpqc topology, Cartan subgroups, G_s and G_t have the same nilpotent rank ν. Finally, since D is flat over S, one has $\dim D_s = \dim D_t = \nu$ (Exp. VI_B § 4).

iii) Since h_t factors through D_t, h evidently factors through D, whence iii).

This being said, the following lemma proves that $(D_s)_{red}$ is a Cartan subgroup of $(H_s)_{red} \cong G_s$:

Lemma 1.11. *Let G be a smooth and connected algebraic group defined over an algebraically closed field k, D a smooth nilpotent algebraic subgroup containing a regular element a of $G(k)$. Then D^0 is contained in a Cartan subgroup of G. If moreover* $\dim D$ *is equal to the nilpotent rank of G, then D is a Cartan subgroup of G (and hence is connected).*

Let Z be the center of G, $G' = G/Z$ (which is affine, Exp. XII 6.1), a', D' the images of a, D in G'. It follows immediately from the correspondence between Cartan subgroups of G and Cartan subgroups of G' (Exp. XII 6.6 e)) that a' is a regular element of G' and that it suffices to prove the lemma for the pair D', G'; this allows us to assume G affine.

Let s then be the semisimple component of a, which belongs to $D(k)$ (*Bible* 4 th. 3) and is regular (*Bible* 7 th. 2 cor. 1). By *Bible* 6 th. 2, s centralizes D^0, hence D^0 is contained in the connected centralizer of s, which is a Cartan subgroup of G (*Bible* 7 th. 2). If now $\dim D$ = nilpotent rank of G, D^0 is therefore a Cartan subgroup of G, equal to $Centr_G(T)$, where T is the unique maximal torus of D^0 (Exp. XII 6.6). But D is nilpotent, hence centralizes T (*Bible* 6 th. 2), hence $D = D^0$.

Let us now work with the group G. The group $C_t = u_t^{-1}(D_t)$ is the unique Cartan subgroup of G_t containing b. Let C be the schematic closure of C_t in G.

By functoriality of the schematic closure, $u|C$ factors through D, so $u_s(C_s) \subset D_s$. Since u_s is an isomorphism of G_s onto $(H_s)_{red}$, since `dim C_s = dim C_t = ν = dim`

D_s (1.10), and since D_s is connected (1.11), u_s furnishes an isomorphism of $(C_s)_{red}$ onto $(D_s)_{red}$, hence $(C_s)_{red}$ is a Cartan subgroup of G_s. The following lemma shows that C is in fact a Cartan subgroup of G.

Lemma 1.12. *Let S be the spectrum of a discrete valuation ring, G an S-group prescheme smooth, of finite type, C a group sub-prescheme of G, flat over S, such that the generic fiber C_t is a Cartan subgroup of G_t and the geometric reduced closed fiber $(C_s)_{red}$ is a Cartan subgroup of G_s. Then C is a Cartan subgroup of G.*

We must show that C is smooth over S, and it suffices to establish this point after a faithfully flat extension of the base. The hypotheses on C imply that the nilpotent rank of the fibers of G is constant, and we may therefore speak of the open set U of regular points of G (Exp. XV 7.3). Since $(C_s)_{red}$ is a Cartan subgroup of G_s, C_s contains a regular point a_s of G_s. But C is flat over S, so (EGA IV 14.5.8), replacing the base by a faithfully flat extension if necessary, we may assume that a_s is an element of $G(s)$ lifting to a section a of $C(S)$. Since U is open and contains $a(s)$, U contains a. Let C' be the unique Cartan subgroup of G containing a (Exp. XV 7.3). One necessarily has $C_t = C'_t$, and consequently C' and C coincide with the connected component of the schematic closure of C_t in G (note that C' and C are flat over S).

End of the proof of 1.3 a). By hypothesis, the restriction of u to the Cartan subgroup C of G is an immersion. It is clear that $h(S)$ is contained in $u(C) = D$, hence h is the image of a section g of $C(S)$. QED.

Proof of 1.3 b).

We shall again use the valuative criterion of local properness (EGA IV 15.5) in the case of an immersion (resp. the valuative criterion of properness (EGA II 7.3.6) in the case of a closed immersion). This reduces us to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. Let s be the closed point and t the generic point of S. Replacing H by the schematic closure in H of $u_t(G_t)$, we may assume that H is flat over S and that u_t is an isomorphism.

Let G^0 be the connected component of G. By 1.2 a), $u|G^0$ is an immersion; denote by H^0 the group sub-prescheme of H image of G^0. One has therefore $H^0_s = (H_s)^0_{red}$.

To verify the valuative criterion, we must show that every section h of H over S such that $h(S)$ is contained in $u(G)$ in the case of an immersion (resp. every section h of H over S in the case of a closed immersion) is the image of a section g of G over S.

Let C be a Cartan subgroup of G, $D = u(C)$ the Cartan subgroup of H^0 image of C. The S-group $D' = int(h)D$ has connected fibers, hence its underlying space is contained in that of H^0; moreover, being smooth over S, it is reduced, and consequently D' is a smooth group sub-prescheme of H^0. The fibers of D' are Cartan subgroups of the fibers of H^0, so D' is a Cartan subgroup of H^0, and $C' = u^{-1}(D')$ is a Cartan subgroup of G. But $Transp^{str}_G(C, C')$ is smooth and surjective over S (Exp. XII 7.1 b)). Since S is henselian with algebraically closed residue field, there exists $g \in G(S)$ such that $int(g)C = C'$. Replacing h by $u(g)^{-1}h$, we may assume that $h(t)$ normalizes D_t (and that $h(s)$ is the image of an element of the normalizer of C_s in G_s in the case of an immersion). Let $N = Norm_G(C)$. By hypothesis $u|N$ is an immersion (resp. a closed immersion), so h is the image of a section of N over S, which completes the proof.

2. A representability theorem for quotients

Let us "recall" the following result:

Theorem 2.1. *Let S be a prescheme, X and Y two S-preschemes, $f : X \to Y$ an S-morphism. Suppose we are in one of the following two cases:*

a) The morphism f is locally of finite presentation.

b) The prescheme S is locally noetherian and X is locally of finite type over S.

Then the following conditions are equivalent:

i) There exist an S-prescheme X' and a factorization of f:

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S-morphism, locally of finite presentation, and f'' is a monomorphism.

ii) The (first) projection

$$p_1 : X \times_Y X \to X$$

is a flat morphism.

Moreover, if the preceding conditions are satisfied, (X', f') is a quotient of X by the equivalence relation defined by f (for the fpqc topology), so that the factorization $f = f'' \circ f'$ of i) is unique up to isomorphism.

The proof of this delicate theorem will be found in EGA V; one may also consult the lecture by J.-P. Murre, *Séminaire Bourbaki*, May 1965, No. 294 th. 2 cor. 2, where the case Y locally noetherian, X of finite type over Y is treated. We shall see that one may reduce to this case.

Let us first make some remarks:

a) i) $\Rightarrow$ ii) is trivial. Indeed, the first projection

$$p'_1 : X \times_{X'} X \to X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism, since f'' is a monomorphism, and p'_1 is flat, since f' is flat, so p_1 is flat.

b) The assertions of 2.1 are local on Y (and so are local on S); they are also local on X, as follows easily from the fact that a flat morphism locally of finite presentation is open (EGA IV 11.3.1).

c) Under the hypotheses of 2.1 a), in view of the foregoing, we are reduced to the case where X and Y are affine and f of finite presentation. Replacing S by Y, we may assume X and Y of finite presentation over S. We then reduce to the case S noetherian thanks to EGA IV 11.2.6.

d) Under the hypotheses of 2.1 b), we may assume S, X, Y affine, S noetherian and X of finite type over S. Consider Y as a filtered projective limit of affine schemes Y_i of finite type over S. The schemes $X \times_{Y_i} X$ form a decreasing filtered family of closed subschemes of $X \times_S X$, whose projective limit is $X \times_Y X$. Since $X \times_S X$ is noetherian, one has $X \times_{Y_i} X = X \times_Y X$ for i large enough, so that

$$f_i : X \xrightarrow{f} Y \to Y_i$$

satisfies the hypotheses of 2.1 ii) if f does. Since the equivalence relation defined by f on X coincides with that defined by f_i, it is clear that it suffices to prove ii) ⇒ i) for f_i, which reduces us to the case where Y is of finite type over S.

Application to group preschemes

Theorem 2.2. *Let S be a prescheme, G an S-group prescheme locally of finite presentation over S, acting on an S-prescheme X. Let $\xi \in X(S)$ be a section of X over S such that the stabilizer H of ξ in G is an S-group sub-prescheme of G, flat over S. Then, if X is locally of finite type over S, or if S is locally noetherian, the homogeneous space quotient G/H is representable by an S-prescheme, locally of finite presentation over S, and the S-morphism*

$$f : G \to X, \quad g \mapsto g \cdot \xi$$

factors as

$$\begin{tikzcd} & G \arrow[dl, "p"'] \arrow[dr, "f"] & \\ G/H \arrow[rr, "i"] & & X, \end{tikzcd}$$

where p is the canonical projection, which is a faithfully flat morphism locally of finite presentation, and i is a monomorphism. (For definiteness, we have assumed that G acts on the left on X.)

Proof. The morphism f makes G into an X-prescheme. By definition of the stabilizer of ξ, the morphism

$$G \times_S H \to G \times_X G, \quad (g, h) \mapsto (g, gh)$$

is an isomorphism. Since H is flat over S, $G \times_S H$ is flat over G, so the first projection $p_1 : G \times_S G \to G$ is a flat morphism. Furthermore, if X is locally of finite type over S, f is locally of finite presentation (EGA IV 1.4.3 v)); otherwise S is assumed locally noetherian. It then suffices to apply 2.1 to the morphism f.

It remains to see that G/H is locally of finite presentation over S, but this follows immediately from Exp. V 9.1.

Corollary 2.3. *Let S be a prescheme, G and H two S-group preschemes, $u : G \to H$ an S-homomorphism. Suppose G locally of finite presentation over S and that either H is locally of finite type over S, or S is locally noetherian. Then, if $K = Ker(u)$ is flat over S, the quotient group G/K is representable by an S-group prescheme locally of finite presentation over S, and u factors as*

$$\begin{tikzcd} G \arrow[rr, "u"] \arrow[dr, "p"'] & & H \\ & G/K \arrow[ur, "i"'] & \end{tikzcd}$$

where p is the canonical projection and i a monomorphism.

Proof. One applies 2.2 with $X = H$ and ξ the unit section of H.

Corollary 2.4. *Let S be a prescheme, G an S-group prescheme of finite presentation over S, H an S-group prescheme smooth over S with connected fibers (hence of finite presentation over S, by VI_B 5.5), $i : H \to G$ a monomorphism of S-groups, so that H is a subgroup of G.*

Suppose that $N = Norm_G(H)$ (which is representable by a closed group subprescheme of G, of finite presentation over S (Exp. XI 6.11)) is flat over S. Then G/N is representable by an S-prescheme of finite presentation over S and quasi-projective over S.

All the assertions to be proved, except the last, are local on S. To establish them, we may therefore assume S quasi-compact and the relative dimension of H over S constant and equal to r. Let us proceed as in XV § 5. For every integer $n > 0$, let $G_{(n)}$ (resp. $H_{(n)}$) be the nth normal invariant of the unit section of G (resp. of H) (EGA IV 16). The sheaf of $\mathcal{O}_S$-modules $H_{(n)}$ is a quotient of $G_{(n)}$, and H being smooth over S of dimension r, $H_{(n)}$ canonically defines an element ξ_n of $X_n = Grass_{\varphi(n,r)} G_{(n)}$ (EGA I 2nd ed. § 9) for a suitable integer $\varphi(n,r)$. On the other hand, G acts naturally on $G_{(n)}$ (and hence also on X_n) via the representation

$$G \to \underline{Aut}_{S\text{-gr}}(G), \qquad g \mapsto \operatorname{int}(g).$$

Since S is quasi-compact, there exists an integer m such that for $n \geqslant m$ one has

$$N = Norm_G(H) = Norm_G H_{(n)} \qquad \text{(Exp. XI 6.11 b)).}$$

That is to say, N is the stabilizer of ξ_n for n large. The representability of G/N therefore follows from 2.2. The fact that G/N is of finite presentation over S is a consequence of Exp. V 9.1. It remains to see that G/N is quasi-projective over S, and for that we shall exhibit a canonical invertible sheaf on G/N, S-ample. Consider the functor $F : (Sch/S)^{\circ} \to Ens$ such that for every S-prescheme T one has

$$F(T) = \text{set of } T\text{-subgroups } H \text{ of } G_T, \text{ representable, smooth over } T, \text{ with connected fibers.}$$

Such a group H is of finite presentation over S (Exp. VI_B 5.5) and $H \to G_T$ is a quasi-affine morphism (EGA IV 8.11.2). By effective descent of quasi-affine morphisms (SGA 1, VIII 7.9), one deduces that F is a sheaf for the fpqc topology. Since G/N is the sheaf associated to a subfunctor of F, one sees that there is a canonical monomorphism $G/N \to F$. There exists therefore a subgroup H_0 of $G \times_S (G/N)$, representable, smooth over G/N, with connected fibers, “universal” for the functor G/N. I claim that the invertible sheaf $L = (det(LieH_0))^{-1}$ is S-ample. In this form, the assertion becomes local on S, and the proof is analogous to the one given in (Exp. XV 5.8).

The following corollary was announced in Exp. XIV 4.8 bis.

Corollary 2.5. *Let S be a prescheme, G an S-group prescheme smooth and of finite presentation over S, with connected fibers, P a parabolic subgroup of G (Exp. XIV 4.8 bis). Then G/P is representable by an S-prescheme smooth and projective over S.*

It only remains to show (*loc. cit.*) that G/P is representable by an S-prescheme quasi-projective, of finite presentation, which follows from 2.4, given that $P = Norm_G(P)$ (*loc. cit.*) is therefore flat over S.

3. Groups with flat center

Proposition 3.1. *Let S be a prescheme, G an S-group prescheme smooth and of finite presentation over S, with connected fibers. Suppose that the center Z of G (which is representable by Exp. XI 6.11) is flat over S. For every integer $n > 0$, let $G_{(n)}$ denote the locally free $\mathcal{O}_S$-module equal to the nth normal invariant of G along the unit section, and let ρ_n be the natural “adjoint representation” of G in $GL_S(G_{(n)})$. Then:*

a) The quotient $G' = G/Z$ is representable by an S-group prescheme, smooth, of finite presentation over S, quasi-affine, with affine connected fibers.

b) The representation ρ_n factors through G':

$$\begin{tikzcd} G \arrow[rr, "\rho_n"] \arrow[dr] & & GL_S(G_{(n)}) \\ & G/Z \arrow[ur, "i_n"'] & \end{tikzcd}$$

If S is quasi-compact, i_n is a monomorphism for n large enough.

c) The functor $\mathcal{T}_{G'}$ of subtori of G' is representable by an S-prescheme, smooth over S, which is a sum of a family of preschemes affine over S if S is quasi-compact.

Proof. The assertions of a) are local on S, which allows us to assume S quasi-compact. By Exp. XI 6.11 b), Z is equal to $Ker\rho_n$ for n large enough, so this kernel is flat over S. The fact that G/Z is representable and that i_n is a monomorphism therefore follows from 2.3. Since G is smooth and of finite presentation over S, the same is true of G' (Exp. VI_B § 9). The group $GL_S(G_{(n)})$ is affine over S, and a monomorphism of finite presentation is quasi-affine (EGA IV 8.11.2), hence G' is quasi-affine over S and has affine fibers. Since G' is smooth over S, the functor $\mathcal{T}_{G'}$ is formally smooth (Exp. XI 2.1 bis). Taking into account Exp. XI 4.6 and 4.3, the assertions of 3.1 c) will therefore follow from the next lemma:

Lemma 3.2. *Let S be a prescheme, G and H two S-group preschemes of finite presentation over S, $u : G \to H$ an S-group monomorphism, $\mathcal{T}_G$ (resp. $\mathcal{T}_H$) the functors of subtori of G (resp. of H) (cf. Exp. XV § 8). Then the map $T \mapsto u(T)$ defines a monomorphism (XV 8.3 c))*

$$\tilde{u} : \mathcal{T}_G \longrightarrow \mathcal{T}_H$$

which is representable by a closed immersion of finite presentation.

By the usual procedure, we are reduced to the following problem: let T be a subtorus of H, $T' = G \times_T H$ its inverse image in G, $u_T : T' \to T$ the S-group monomorphism deduced from u. One must show that the S-functor $\prod_{T/S}(T'/T)$ is representable by a closed sub-prescheme of S, of finite presentation. But T, being a torus, is smooth over S with connected fibers, and it suffices to apply Exp. XI 6.10.

Theorem 3.3. *Let S be a prescheme, G an S-group prescheme smooth and of finite presentation over S, with connected fibers, Z the center of G. Suppose that Z is flat over S and that the unipotent rank (Exp. XV 6.1 ter) of G is equal to that of Z. Then:*

a) The group $G' = G/Z$ is representable, and if S is quasi-compact, the canonical morphism $i_n : G' \to GL_S(G_{(n)})$ (cf. 3.1 b)) is an immersion for n large.

b) The group G' is quasi-affine over S, with affine fibers, the center of G' is the unit group, and G' has zero unipotent rank.

c) The group G' possesses locally for the étale topology maximal tori, and these are also Cartan subgroups of G'. The functor $\mathcal{TM}_{G'}$ of maximal tori of G' (Exp. XV § 8) is representable by an S-prescheme smooth and affine over S.

d) The group G possesses locally for the étale topology Cartan subgroups, and the functor $\mathcal{C}_G$ of Cartan subgroups of G is representable by an S-prescheme, smooth and affine over S.

Proof. By 3.1, the group G' is representable by an S-prescheme, smooth and quasi-affine over S, with affine fibers. In view of the correspondence between Cartan subgroups of the fibers of G and of the fibers of G' (Exp. XII 6.6 e)), the hypothesis on the unipotent rank of Z implies that G' has zero unipotent rank. Using now Exp.

XV 8.18, one sees that G' possesses locally for the étale topology maximal tori. The fact that i_n is an immersion for n large then follows from 3.1 b) and 1.4 a). This completes the proof of a).

Let us now show c). Since G' has zero unipotent rank, it is clear that every maximal torus of G' is also a Cartan subgroup of G'. The functor of maximal tori of G' is representable by a sub-prescheme both open and closed of the functor of subtori of G'; moreover it is of finite type over S (Exp. XV 8.15); it follows from 3.1 c) that this functor is representable by an S-prescheme, smooth and affine over S.

Since G' possesses, locally for the étale topology, Cartan subgroups, the same is true of G, and the functor of Cartan subgroups of G is canonically isomorphic to that of G' (Exp. XV 7.3 iv)), so c) ⇒ d).

It remains to show b), and more precisely, it remains to prove that the center Z' of G' is the unit group. Since Z' is representable (Exp. XI 6.11) and of finite presentation over S, it suffices to show that the fibers of Z' are reduced to the unit group (Exp. VI_B 2.9), which reduces us to the case where S is the spectrum of an algebraically closed field. The center Z' is evidently contained in every Cartan subgroup of G', hence in every maximal torus of G' by c), and is therefore of multiplicative type. Moreover, we shall see in Exp. XVII that if G is a connected algebraic group with center Z, the center Z' of G/Z is unipotent. In the present case, Z' being both of multiplicative type and unipotent, is reduced to the unit group (cf. Exp. XVII).

Examples of groups whose center is flat

Proposition 3.4. *Let S be a prescheme, G an S-group prescheme of finite presentation over S, smooth, with connected fibers, Z the center of G. Then Z is flat over S in the following two cases:*

a) The unipotent rank ρ_u (Exp. XV 6.1 ter) of the fibers of G is zero.

b) i) S is reduced.

ii) The dimension of the fibers of Z is a locally constant function on S.

iii) The unipotent rank of G is equal to the unipotent rank of Z.

Proof of 3.4 a). We shall prove at the same time:

Proposition 3.5. *Under the hypotheses of 3.4 a), suppose moreover that G possesses a maximal torus T and let $C = Centr_G(T)$ be the Cartan subgroup of G associated to T (Exp. XII 7.1). Then:*

(i) $Z \cap T$ is a subgroup of multiplicative type of G.

(ii) C is commutative and equal to $T \cdot Z$.

(iii) If the quotients $Z/Z \cap T$ and C/T are representable, they are representable by abelian preschemes (i.e. S-group preschemes, smooth over S, whose fibers are

abelian varieties) and the canonical monomorphism $Z/Z \cap T \to C/T$ is an isomorphism.

Using the general properties of passage to the limit proved in Exp. VI_B § 10 and Exp. XV 6.2, 6.3, and 6.3 bis, we reduce as usual to the case where S is noetherian (note that the assertions of 3.4 and 3.5 are local on S). We noted in 3.1 that the hypotheses on G imply that Z is representable. To show that Z is flat over S, we reduce by EGA 0_III 10.2.6 to the case where S is local artinian. But then, G being smooth, G possesses locally for the étale topology maximal tori (Exp. XV 8.17). Replacing S by a finite flat extension if necessary (which is allowable for proving that Z is flat), we may assume that G possesses a maximal torus T.

Let us show, in the same way, that to prove 3.5 we may reduce to the case S artinian.

i) Since Z is closed in G (Exp. XI 6.11), $Z \cap T$ is a closed group subscheme of T. It follows from Exp. X 4.8

b) that $Z \cap T$ is of multiplicative type if and only if it is flat over S. As before, it suffices to establish that $Z \cap T$ is flat when S is artinian.

ii) Since G has zero unipotent rank, every Cartan subgroup C of G satisfies the hypotheses of Lemma 1.6, hence is commutative. The fact that $C = T \cdot Z$ will follow from iii).

iii) If C/T is representable, C/T is smooth over S (Exp. VI_B 9) and its fibers are abelian varieties (1.7), so C/T is an abelian prescheme. To show that the canonical monomorphism

$i : Z/Z \cap T \to C/T$

is an isomorphism, it suffices to verify this when S is local artinian. Indeed, one will then deduce in succession that $Z/Z \cap T$ is flat over S (EGA 0_III 10.2.6), then that i is flat (EGA IV 11.3.10), then that i is an open immersion (EGA IV 17.9.1), then that i is an isomorphism (since C/T has connected fibers).

We henceforth assume that S is local artinian with closed point s and that G possesses a maximal torus T. Let $C = Centr_G(T)$. The group C is a Cartan subgroup of G (Exp. XII 7.1) and majorizes Z.

The algebraic group $M_s = Z_s \cap T_s$ is a subgroup of multiplicative type of G_s which is central. Since T is smooth over S, M_s lifts to a group subscheme M of T, of multiplicative type (Exp. IX 3.6 bis) and contained in the center of G (Exp. IX 3.9), hence contained in $Z \cap T$. Since S is artinian and M and T are flat over S, the quotient groups $Z \cap T/M$, $Z' = Z/M$, and $A = C/T$ are representable (Exp. VI_A §§ 4 and 5). By construction, $(Z \cap T/M)_s$ is the unit group, so $Z \cap T = M$ (Exp. VI_B 2.9); a fortiori it is flat over S, which proves 3.5 i).

By passage to the quotient, one has a canonical monomorphism $i : Z' \to A$, which is therefore a closed immersion (Exp. VI_B 1.4.2). We have already noted that A is an abelian scheme. It remains to show that i is an isomorphism. This will prove 3.5 iii) and will imply that Z' is flat over S, hence that Z is flat over S (as an extension of Z' by the flat group M (Exp. VI_B § 9)). Since Z_s has the same abelian rank as G_s (Exp. XII 6.1), i_s is an epimorphism, hence an isomorphism. Let q be an integer > 0, invertible on S. By the density theorem of 1.6 iii), to see that $i(Z') = A$, it suffices to show that for every integer n equal to a power of q, $i(Z')$ majorizes ${}_nA$. Now let M_s^0 be the connected component of M_s. It is immediate by duality that raising to the q-th power in M_s^0 is an epimorphism. One deduces immediately that if m_0 is the exponent of q in the factorization into prime factors of $card(M_s/M_s^0)$, the image of ${}_{nm_0}Z_s$ in $A_s \cong Z_s/M_s$ majorizes ${}_nA_s$. There exists therefore a subgroup of multiplicative type $M_s(n)$ of Z_s whose image in A_s is ${}_nA_s$. As before, one sees that $M_s(n)$ lifts to a subgroup of G, central and of multiplicative type, $M(n)$. The image of $M(n)$ in A is a subgroup of multiplicative type (Exp. IX 6.8), hence necessarily equal to ${}_nA$, since it is so on the reduced fiber (Exp. IX 5.1 bis). This completes the proof of 3.4 a) and of 3.5.

Proof of 3.4 b). The assertion to be proved is local on S; we may therefore assume S affine with ring A. Considering A as an inductive limit of its sub-$\mathbb{Z}$-algebras of finite type, we reduce as above to the case where S is noetherian reduced.

To show that Z is flat, we have at our disposal a valuative criterion of flatness (EGA IV 11.8.1), which allows us to reduce to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, with generic point t and closed point s. Let Z' be the schematic closure in Z of Z_t. We must show that $Z' = Z$, and it suffices even to show that $Z_s = Z'_s$ (Exp. VI_B 2.6). By (Exp. VI_B § 4), one has the inequalities

$$\dim Z_t = \dim Z'_t = \dim Z'_s \leq \dim Z_s.$$

But by hypothesis $\dim Z_t = \dim Z_s$, hence $\dim Z'_s = \dim Z_s$ and consequently Z'_s majorizes $(Z_s)^0_{red}$. It follows that $G'_s = G_s/Z'_s$ is affine (Exp. XII 6.1). In view of the correspondence between Cartan subgroups of G and of G' (Exp. XII 6.6 e)), hypothesis 3.4 iii) implies that the unipotent rank of G'_s is zero, and consequently its Cartan subgroups are also its maximal tori. The image Z''_s of Z_s in G'_s is a central algebraic subgroup of G'_s, hence contained in every Cartan subgroup of G'_s, that is, in every maximal torus; Z''_s is therefore a subgroup of multiplicative type of G'_s. Under these conditions, we shall show in (Exp. XVII § 7) that there exists a finite subgroup of multiplicative type M_s of Z_s whose image in Z_s/Z'_s is Z''_s (we used this fact in the proof of 3.4 a) when Z''_s is étale). Since G is smooth over S and S is the spectrum of a complete local ring, M_s lifts to an S-subgroup of multiplicative type M of G (Exp. IX 3.6 bis and Exp. XV 1.6 b)) which is central (Exp. IX 5.6 a)). So M_t is contained in $Z'_t = Z_t$, and since M is flat, M is contained in Z'. The group M_s is therefore contained in Z'_s, but this implies that Z''_s is the unit group, i.e. $Z_s = Z'_s$. This completes the proof of 3.4 b).

Example of a smooth group prescheme with connected fibers whose center is not flat

Let S be the spectrum of a discrete valuation ring A, π a uniformizer of A, t the generic point of S, s the closed point. Let G be the smooth and affine S-group with ring $B = A[T, T^{-1}, U]/F$ with $F = 1 - T + \pi U$, the composition law being defined by

$((t, u), (t', u')) \mapsto (tt', \pi uu' + u + u')$.

The generic fiber G_t is therefore isomorphic to the multiplicative group $(\mathbb{G}_m)_t$, while the closed fiber G_s is isomorphic to the additive group $(\mathbb{G}_a)_s$. The function T is invertible in $\Gamma(G, \mathcal{O}_G)$ and defines an S-group morphism $\varphi : G \to (\mathbb{G}_m)_S$ which is an isomorphism on the generic fiber and the zero morphism on the closed fiber. The datum of φ allows one to construct the semidirect product group $H = G \cdot (\mathbb{G}_a)_S$ (note that $(\mathbb{G}_m)_S$ acts on $(\mathbb{G}_a)_S$). The center Z of H is the unit group on the generic fiber and is equal to H_s on the closed fiber, hence is not flat over S.

4. Groups with affine fibers, of zero unipotent rank

Theorem 4.1. *Let S be a prescheme, G an S-group prescheme smooth over S, of finite presentation, with affine, connected fibers, of zero unipotent rank (Exp. XV 6.1 ter). Then:*

a) The center Z of G is a group of multiplicative type (and is therefore a reductive center of G (Exp. XII 4.1)).

b) G satisfies conditions a) to d) of 3.3.

c) G possesses locally for the étale topology maximal tori, and these are also Cartan subgroups of G. The functor $\mathcal{TM}_G$ of maximal tori of G is representable by an S-prescheme smooth and affine over S.

d) G is quasi-affine over S. Moreover, if T is a maximal torus of G, $N = Norm_G T$ its normalizer in G, $W = N/T$ the Weyl group relative to T, the following conditions are equivalent:

(i) G is affine over S.

(ii) $G' = G/Z$ is affine over S.

(iii) N is affine over S.

(iv) W is affine over S.

These conditions are always realized if S is locally noetherian of dimension $\leqslant 1$.

Proof. Since G has affine fibers (and so abelian rank zero) and zero unipotent rank, G possesses locally for the étale topology maximal tori (Exp. XV 8.18), and every maximal torus of G is evidently a Cartan subgroup of G, which proves the first part of c). To see that Z is of multiplicative type, we may assume, by what precedes, that G possesses a maximal torus T. Since T is also a Cartan subgroup, T majorizes Z (since $T = Centr_G(T)$ by Exp. XII 7.1), and $Z = Z \cap T$ is of multiplicative type by 3.5 i). This proves a). Assertion b) is clear, given a). On the other hand the functor $\mathcal{TM}_G$ is isomorphic to the functor of Cartan subgroups of G (Exp. XII 7.1), hence is smooth and affine over S (3.3 c)), which completes the proof of c). It remains to prove d).

Proof of d). Since G' is quasi-affine (3.5 b)) and Z is of multiplicative type, hence affine over S, G is quasi-affine (Exp. VI_B § 9).

i) ⇔ ii). If G is affine, G' is affine by Exp. IX 2.3. If G' is affine, G is affine as an extension of an affine group by an affine group (Exp. VI_B § 9).

ii) ⇔ iii). If G is affine, N is affine since closed in G (Exp. XI 6.11 a)). Moreover G/N is isomorphic to the functor $\mathcal{TM}_G$ (conjugacy of maximal tori, cf. Exp. XII 7.1 b)), so is affine over S by c). Hence if N is affine over S, G is affine, as an "extension" of an affine homogeneous space by an affine group (Exp. VI_B § 9).

iii) ⇔ iv). If N is affine, $W = N/T$ is affine (Exp. IX 2.3) and conversely by Exp. VI_B § 9.

Moreover, N/T is representable by an S-group prescheme, étale, separated over S, of finite type (Exp. XV 7.1 iv)), hence quasi-finite over S. The last assertion of d) will therefore follow from the following lemma, applied with $X = W$ and $Y = S$:

Lemma 4.2. *Let S be a locally noetherian prescheme of dimension 1, X and Y two S-preschemes locally of finite type over S, $u : X \to Y$ an S-morphism quasi-finite and separated. Suppose that for every point s of S the morphism $u_s : X_s \to Y_s$ deduced from u by the base change* $\operatorname{Spec}\kappa(s) \to S$ *is finite. Then u is an affine morphism.*

The assertion to be proved is local on Y, which allows us to assume Y (and hence also X) of finite type over S. By EGA IV 8, one sees that it suffices to prove 4.2 when S is the spectrum of a local ring A. By fpqc descent we may assume A complete, then A reduced (EGA II 1.6.4), then A normal (Nagata's theorem (EGA 0_IV 22) and Chevalley's theorem (EGA II 6.7.1)). If A is a field, u is finite by hypothesis, hence affine. Otherwise A is a discrete valuation ring; let s be the closed point of S, t the generic point, π a uniformizer of A. Let y be a point of Y_s. Applying EGA IV 8 once more, we may replace Y by the spectrum Y' of $\mathcal{O}_{Y,y}$ and X by $X' = X \times_Y Y'$; we may even assume $\mathcal{O}_{Y,y}$ complete. Since u is quasi-finite and separated, by EGA II 6.2.6 X' is a sum of two schemes X_1 and X_2 with X_1 finite over Y' (hence affine) and X_2 such that $u(X_2)$ does not contain the closed point y of Y'. I claim that under these conditions X_2 does not meet the closed fiber X'_s. Indeed, by hypothesis $X'_s \to Y'_s$ is finite, hence the restriction of this morphism to the closed subset $(X_2)_s$ is finite and its image in $(Y')_s$ is a closed subset. Since this image does not contain the closed point of the local scheme $(Y')_s$, $(X_2)_s$ is empty. Since u_t is finite, the restriction to $(X_2)_t = X_2$ of the morphism $X'_t \to Y'_t$ is finite. Moreover, one has $Y'_t = Y'_\pi$, so the open immersion $Y'_t \to Y'$ is affine. In short, the composed morphism $X_2 \to Y'_t \to Y'$ is affine, and it follows that the morphism $X' = X_1 \sqcup X_2 \to Y'$ is affine.

Corollary 4.3. *Let S be a prescheme, G an S-group prescheme smooth and of finite presentation over S, with affine, solvable, connected fibers, of zero unipotent rank. Then G is affine over S. If moreover the center of G is the unit group and if S is quasi-compact, the canonical morphism $i_n : G \to GL_S(G_{(n)})$ (cf. 3.1 b)) is a closed immersion for n large enough.*

To prove that G is affine over S, we may assume that G possesses a maximal torus T

(4.1 c)). Since G has solvable fibers, the Weyl group relative to T is the unit group (*Bible* 6 th. 1 cor. 3) and condition 4.1 d) iv) is satisfied. If the center of G is the unit group, i_n is a monomorphism for n large enough (3.1), hence is a closed immersion (1.5 b)).

5. Application to reductive and semisimple groups

Definition 5.1. *An S-group prescheme G is said to be reductive (resp. semisimple) if G is smooth and affine over S, with reductive (resp. semisimple) fibers.*

5.1.1. Reductive groups will be systematically studied from Exp. XIX onwards. In this section, we shall need the following properties, which will be proved in Exp. XIX (without using the developments of the present Exposé).

a) *Let S be a prescheme, G an S-group prescheme smooth and affine over S, with connected fibers, s a point of S such that G_s is reductive (resp. semisimple). Then there exists a neighborhood U of s such that $G|U$ is reductive (resp. semisimple).*

b) *If G is reductive, G possesses locally for the étale topology maximal tori, and if T is a maximal torus of G, the Weyl group $W = Norm_G(T)/T$ is finite over S.*

c) *A reductive group has zero unipotent rank.*

This being admitted, we propose to improve assertion a) above.

Theorem 5.2. *Let S be a prescheme, G an S-group prescheme smooth and of finite presentation over S, with connected fibers, s a point of S. Then:*

(i) If the fibers of G are affine and G_s is reductive, there exists an open neighborhood U of s such that $G|U$ is reductive.

(ii) If G_s is semisimple, there exists an open neighborhood U of s such that $G|U$ is semisimple.

Using Exp. XV 6.2 i) and Exp. VI_B § 10, one reduces to the case where S is noetherian.

In case (ii), consider the center Z of G, which is representable (Exp. XI 6.11 a)). Since G_s is semisimple, it is well known that Z_s is finite. Consequently (Exp. VI_B § 4), there exists a neighborhood U of s such that Z is quasi-finite over U. For every point t of U, G_t is then affine (Exp. XII 6.1). Replacing S by a smaller open if necessary, we may therefore assume that the fibers of G are affine, both in case (ii) and in case (i).

To prove 5.2 it suffices, by a), to prove that G is affine over S. Since G has affine fibers, we know that the functor of subtori of G is representable by a smooth S-prescheme (Exp. XV 8.11 and 8.9). Replacing S by an étale extension covering s (which is allowable), we may therefore assume that there exists a subtorus T of G such that T_s is a maximal torus of G_s. But then $C = Centr_G(T)$ is a smooth subgroup of G, with connected fibers, majorizing T, such that $C_s = T_s$ (since C_s is a Cartan subgroup of G_s and G_s has zero unipotent rank by c)). It follows that $C = T$, hence T is a Cartan subgroup and a maximal torus of G; a fortiori, the unipotent rank of G is zero and one may apply 4.1.

By 4.1 d), it suffices to show that the Weyl group W relative to T is affine over a neighborhood U of s. In fact we shall see that W is even finite over a neighborhood of s. Since W is quasi-finite over S (Exp. XV 7.1 iv)), to say that W is finite over a neighborhood U of s is equivalent to saying that the morphism $W \to S$ is proper at s (EGA IV 15.7 and EGA III 4.4.2); to establish this we have at our disposal the valuative criterion of local properness (EGA IV 15.7), which reduces us to the case where S is the spectrum of a discrete valuation ring, with closed point s. But then, by the last assertion of 4.1 d), G is affine over S, and we may apply property a) recalled above to conclude that G is reductive (resp. semisimple). Using now property b), we conclude that W is indeed finite over S, which completes the proof of 5.2.

6. Applications: extension of certain rigidity properties of tori to groups of zero unipotent rank

Proposition 6.1. *Let S be a prescheme, G an S-group prescheme smooth over S, of finite presentation, with connected affine fibers, and whose center is of multiplicative type (for example G of zero unipotent rank, cf. 4.1 a)). Then every closed normal subgroup K of G, of finite presentation over S, quasi-finite over S, is finite over S.*

Indeed, for every geometric point s above S, $(K_s)_{red}$ is a finite étale subgroup of G_s, normal hence central (since G_s is connected). Consequently $K_0 = Z \cap K$ (where Z denotes the center of G) has the same underlying space as K, and it suffices to show that K_0 is finite over S. Now K_0 is a closed, quasi-finite subgroup of the multiplicative-type group Z, hence is finite, as one sees immediately (locally on S, K_0 will be majorized by ${}_nZ$ for a suitable integer n, and ${}_nZ$ is finite over S).

Corollary 6.2. *Let S and G be as above, K an S-group sub-prescheme of G, of finite presentation over S, normal and closed in G, s a point of S. If K_s is finite (resp. is the unit group), there exists an open neighborhood U of s such that $K|U$ is finite over S (resp. is the unit group).*

In view of the upper semi-continuity of the dimension of the fibers of K (Exp. VI_B § 4), we may already assume, replacing S by a smaller open if necessary, that K is quasi-finite over S, hence finite (6.1). If moreover K_s is the unit group, one deduces easily by Nakayama's lemma that K is the unit group over $\operatorname{Spec} \mathcal{O}_s$, hence over a neighborhood of s.

Proposition 6.3. *Let S be a prescheme, G an S-group prescheme smooth and of finite presentation over S, with affine, connected fibers, of zero unipotent rank, H an S-group prescheme, s a point of S, $u : G \to H$ an S-group morphism such that $Ker u_s$ is central. Suppose moreover that H is of finite presentation over S or that S*

is locally noetherian and G is locally of finite type over S. Then there exists an open neighborhood U of s such that if $K = Keru$, $K|U$ is central, of multiplicative type. Moreover, $G' = (G|U)/(K|U)$ is representable and the morphism $u' : G' \to H|U$ deduced from u by passage to the quotient is an immersion.

Proof. Let Z be the center of G, and $K_0 = K \cap Z$.

a) K_0 is a subgroup of multiplicative type over a neighborhood U of s. Indeed, replacing S by an étale extension over a neighborhood U of s if necessary (which is legitimate), we may assume that G possesses a maximal torus T (4.1 c)). But then $T \cap K$ is a group of multiplicative type (Exp. XV 8.3) whose fiber at s is central by hypothesis, so $T \cap K$ is central over a neighborhood of s (Exp. IX 5.6 a)) and consequently coincides with K_0.

b) Let us show that $K_0 = K$ over a neighborhood U of s. By a) we may already assume that K_0 is of multiplicative type, hence flat over S. Since K₀_s = K_s, the natural immersion $K_0 \to K$ is then open over a neighborhood U of s (Exp. VI_B 2.6). If $t \in U$, the image of K_t in the group $G'_t = G_t/Z_t$ is then an étale finite group, normal in G'_t (since K_t is normal in G_t), hence central, hence reduced to the unit group (3.3 b)). This says that K_t is contained in Z_t, hence is equal to K₀_t, whence $K = K_0$ over U.

c) The representability of G' then follows from 2.3; the fact that u' is an immersion is contained in 1.3 a), taking 4.1 c) applied to G' into account.

Proposition 6.4. *Let S be a prescheme, G an S-group prescheme smooth, of finite presentation over S, with connected fibers, s a point of S, such that G_s is generated by its subtori (Exp. XII 8.2) (for example G_s affine of zero unipotent rank), H an S-group prescheme, u and $v : G \to H$ two S-homomorphisms such that $u_s = v_s$ and such that u_s is central. Suppose moreover that H is of finite presentation over S, or that S is locally noetherian. Then there exists a neighborhood U of s such that $u|U = v|U$.*

We reduce as usual to the case where S is noetherian (to study the condition "G_s is generated by its subtori", one uses Exp. VI_B 7.4). Since G has connected fibers, to show that $u = v$ over a neighborhood U of s, it suffices

to show that u = v after reduction by every power of the maximal ideal of the local ring

$\mathcal{O}_{S,s}$, which reduces us to the case where S is local artinian.

But then, the functor of maximal subtori of G is representable by an S-scheme X, smooth of finite type over S (Exp. XV 8.17). Let T be the maximal subtorus of G_X, universal for the functor X. The hypothesis made on G_s means that the algebraic subgroup of G_s generated (Exp. VI_B § 7) by the $\kappa(s)$-morphism $f : T_s \to G_s$ (composite of the immersion $T_s \to G_{X_s}$ and the canonical projection $G_{X_s} \to G_s$) is equal to G_s itself. But X_s is geometrically connected (since if T is a maximal torus of G_s, N its normalizer in G_s, X_s is isomorphic to G_s/N), and the image of T_s under f contains the unit section of G_s; it then follows from Exp. VI_B 7.4 that there exist an integer $N > 0$ and an S-morphism $f^N : T^N \to G$ (where $T^N = T \times_S \cdots \times_S T$ (N factors) and f^N depends only on f and on the multiplication law of G) such that $(f^N)_s$ is surjective.

Consider the base change $X \to S$ and the restrictions of u_X and v_X to the subtorus T of G_X. By hypothesis, $u_{X_s} = v_{X_s}$ and $u_{X_s} : T_{X_s} \to H_{X_s}$ is a central homomorphism. It then follows from Exp. IX 5.1 that $u_X|T = v_X|T$. Making explicit the definition of f^N and using the fact that u and v are homomorphisms, one deduces immediately that $uf^N = vf^N$. But the $\kappa(s)$-morphism f_s^N is surjective and G_s is smooth, hence reduced; consequently f_s^N is generically flat. Since T is smooth over S, hence flat, there exists a non-empty open set V of T such that $f^N|V$ is flat (EGA IV 11.3.10). The image of V is an open W of G (EGA IV 11.3.1) and $f^N : V \to W$ is faithfully flat, of finite presentation, hence covering for the fpqc topology. The equality $uf^N = vf^N$ then implies $u|W = v|W$. Since W is schematically dense in G (EGA IV 11.10.10) and H is separated over S (Exp. VI_A 0.3), one deduces immediately that $u = v$.

Exposé XVII. Unipotent algebraic groups. Extensions between unipotent groups and groups of multiplicative type

by M. Raynaud[1128]

0. Some notations

In the present chapter, we shall mostly consider algebraic groups defined over a field k. The number $p \geqslant 0$ will always denote the characteristic of k, F_p the prime field with p elements if $p > 0$, $\bar{k}$ an algebraically closed extension of k, q a prime number distinct from p.

For every S-prescheme, $(G_a)_S$ (resp. $(G_m)_S$) denotes the additive group (resp. the multiplicative group) over S (cf. Exp. I, 4.3). For every integer $n > 0$, $(\mu_n)_S$ (resp. $(\mathbb{Z}/n\mathbb{Z})_S$) denotes the group of n-th roots of unity (Exp. I, 4.4.4) (resp. the constant group over S associated with the abstract group $\mathbb{Z}/n\mathbb{Z}$ (Exp. I, 4.1)). The group $(\mathbb{Z}/n\mathbb{Z})_S$ is finite and étale over S; the group $(\mu_n)_S$ is flat and finite over S, and is étale over S if and only if n is invertible on S (Exp. VIII, 2.1).

If S is a prescheme of characteristic $p > 0$, for every integer $n > 0$ and every S-prescheme in groups G, we denote by $F_n(G)$ the radicial sub-S-prescheme in groups of G equal to the kernel of the n-th iterate of the Frobenius morphism relative to G (Exp. VII_A). In particular, if $G = (G_a)_S$ we set $F_n(G) = (\alpha_{p^n})_S$, which is a radicial S-group, flat and finite over S, representing the following functor: for every S-prescheme S', $(\alpha_{p^n})_S(S')$ is the set of $x' \in \Gamma(S', O_{S'})$ such that $x'^{p^n} = 0$.

The group $(\mu_{p^n})_S$, already defined, is canonically isomorphic to $F_n(G_m)_S$.

[1128] cf. note on page 1 of Exposé XV.

When there is no ambiguity about the base scheme S, we shall simply write G_a, G_m, α_{p^n}, etc. instead of $(G_a)_S$, $(G_m)_S$, $(\alpha_{p^n})_S$, etc.

If G is an S-prescheme in commutative groups, for every integer $n > 0$, ${}_nG$ is the sub-prescheme in groups of G equal to the kernel of raising to the n-th power in G.

For the convenience of the reader, we have gathered in an appendix some properties of algebraic groups proved in Exp. VI and VII, as well as elementary properties of Hochschild cohomology, which will be useful in this chapter.

1. Definition of unipotent algebraic groups

Definition 1.1. *An algebraic group G defined over an algebraically closed field k is said to be* unipotent *if G admits a composition series whose successive quotients are isomorphic to algebraic subgroups of $(G_a)_k$.*

Proposition 1.2. *Let k be an algebraically closed field, K an algebraically closed extension of k, and G an algebraic group defined over k. Then, in order that G be unipotent, it is necessary and sufficient that G_K be unipotent.*

The necessity of the condition is clear, since a composition series gives a composition series by extension of the base.

The proof of sufficiency is standard: K is the direct limit of its finitely generated k-subalgebras. By Exp. VI_B § 10, one can find a finitely generated k-subalgebra A of K, a composition series G_i of G_S ($S = \operatorname{Spec} A$), and immersions $u_i : H_i = G_i/G_{i+1} \to (G_a)_S$. To prove that G is unipotent, it then suffices to make a k-base extension $A \to k$, which is possible because $\mathrm{Hom}_{k-alg}(A, k)$ is non-empty, since A is a non-zero k-algebra of finite type over the algebraically closed field k.

Definition 1.3. *Let G be an algebraic group defined over a field k. We shall say that G is* unipotent *if there exists an algebraically closed extension $\bar{k}$ of k such that $G_{\bar{k}}$ is unipotent (Definition 1.1).*

By 1.2, the property is independent of the chosen algebraically closed extension $\bar{k}$.

Definition 1.4. *Let G be an algebraic group defined over a field k and H an algebraic group defined over an extension k' of k. We shall say that H is a* form *of G over k' if the algebraic groups $G_{\bar{k}'}$ and $H_{\bar{k}'}$ become isomorphic over $\bar{k}'$ (as above, one sees that the property does not depend on the choice of the algebraically closed extension $\bar{k}'$ of k'). We shall also say that H is a* twisted *form of G.*

We are now in a position to describe the algebraic subgroups of G_a.

Proposition 1.5. *Let k be a field of characteristic $p \geqslant 0$. Then an algebraic subgroup H of $(G_a)_k$ is of one of the following types:*

(i) $H = 0$.

(ii) $H = G_a$.

(iii) (If $p > 0$) H is an extension of a twisted constant group $(\mathbb{Z}/p\mathbb{Z})^r$ by a radicial group α_{p^n} (r and n integers $\geqslant 0$, $r+n > 0$). If moreover k is perfect, this extension is necessarily trivial.

Proof. If H is of dimension 1, it is clear that $H = G_a$. Otherwise H is of dimension 0 and consequently is an extension[1129] of an étale group `H''` by its identity component H', which is a radicial group. To describe $H''_{\bar{k}}$, it suffices to know the abstract group $H''(\bar{k})$, isomorphic to $H(\bar{k})$.

Now this latter is a finite subgroup of $G_a(\bar{k})$, hence is zero if $p = 0$, and of the form $(\mathbb{Z}/p\mathbb{Z})^r$ otherwise, since it is annihilated by p. The group H' is closed in G_a and is defined by a single equation (the ring `k[T]` of G_a is principal), which over $\bar{k}$ admits `0` as its only root, so this equation is of the form $T^n = 0$. Compatibility with the group law entails that $(T+T')^n$ belongs to the ideal generated by T^n and T'^n in the ring `k[T, T']`, hence: if $p = 0$, one has $n = 1$ and H' is the trivial group; if $p > 0$, one has $n = p^m$ and $H' = \alpha_{p^m}$.

The last assertion of 1.5 follows, more generally, from the lemma:

Lemma 1.6. *If k is a perfect field, every extension H of an étale algebraic group* `H''` *by a radicial group H' is trivial. Moreover, there is a unique lift of* `H''` *in H, namely H_{red}.*

Indeed, since k is perfect, the reduced k-scheme H_{red} is an algebraic subgroup of H (Exp. VI_A, 0.2) which is geometrically reduced, hence smooth over k (Exp. VI_A, 1.3.1), hence étale, since H is of dimension 0. To see that the canonical projection $H_{red} \to H''$ is an isomorphism, it suffices to verify it after base extension $k \to \bar{k}$, in which case it suffices to show that one has an isomorphism on the $\bar{k}$-valued points, which is clear. The last assertion follows from the fact that every lift of `H''` in H, being étale over k, is reduced, hence is necessarily contained in H_{red}.

Note that α_{p^n} is a multiple extension of groups isomorphic to α_p. From 1.5 we then deduce the corollary:

Corollary 1.7. *In order that an algebraic group G defined over an algebraically closed field k be unipotent, it is necessary and sufficient that it possess a composition series whose successive quotients are isomorphic to G_a if $p = 0$, and to one of the groups G_a, $\mathbb{Z}/p\mathbb{Z}$, α_p if $p > 0$. (We shall call these the* elementary unipotent groups*.)*

2. First properties of unipotent groups

Proposition 2.1. *A unipotent algebraic group defined over a field k is affine over k.*

By fpqc descent of affine morphisms, it suffices to prove 2.1 when k is algebraically closed. In this case, G is by definition a multiple extension of affine algebraic groups, hence is affine, by Exp. VI_B, 9.2 (viii) applied to affine morphisms.[1130]

[1129]N.D.E.: reference to VII_A?
[1130]N.D.E.: verify this reference.

Proposition 2.2.

i) The property of being unipotent for an algebraic group is invariant under base field extension.

ii) Every algebraic subgroup of a unipotent group is unipotent.

iii) Every algebraic quotient group of a unipotent group is unipotent.

iv) Every extension of a unipotent algebraic group by a unipotent algebraic group is likewise unipotent.

Proof. i) follows immediately from 1.3 and 1.2. To establish the other properties, we may suppose the field k algebraically closed. Then iv) is evident from Definition 1.3.

Let then G be a unipotent algebraic group, G' an algebraic subgroup of G, `G''` an algebraic quotient group of G, and G_i ($i = 1, \cdots, n$) a composition series of G such that $H_i = G_i/G_{i+1}$ is an elementary unipotent group (1.7).

To prove ii), consider the composition series of G' induced by that of G: $G'_i = G_i \cap G'$. The group G'_i/G'_{i+1} is identified with an algebraic subgroup of H_i, hence is isomorphic to a subgroup of G_a, and consequently G' is unipotent.

To prove iii), consider the composition series of `G''` image of that of G: G''_i = image of G_i in `G''`. The group G''_i/G''_{i+1} is then a quotient of H_i, and it suffices to prove the lemma:

Lemma 2.3. *If H is an elementary unipotent group (1.7) defined over a field k, every algebraic quotient group* `H''` *of H is zero or is isomorphic to H over k.*

Proof. If $H = G_a$ in characteristic 0, or if $H = \alpha_p$ or $\mathbb{Z}/p\mathbb{Z}$ ($p > 0$), it suffices to note that it follows from 1.5 that H has no algebraic subgroups other than 0 and H (observe that a non-zero algebraic subgroup of α_p is defined by a k-algebra of k-rank at least p (1.5 iii)), hence is equal to α_p).

Now let $H = G_a$ ($p > 0$), so that one has an exact sequence:

$$0 \longrightarrow N \longrightarrow G_a \longrightarrow H'' \longrightarrow 0.$$

If $N = G_a$, then $H'' = 0$. Otherwise, proceeding by induction on the length of a composition series of N, one may assume that $N = \alpha_p$, or that N is a form of $(\mathbb{Z}/p\mathbb{Z})^r$ (1.5).

a) If $N \simeq \alpha_p$, the proof of 1.5 iii) shows that N is necessarily the kernel of the Frobenius morphism F in G_a, and one concludes using the exact sequence:

`0 → α_p → G_a —F→ G_a → 0.`

b) If $N \simeq \mathbb{Z}/p\mathbb{Z}$, it is immediate that there exists $a \in k*$ such that N is the closed subscheme of $G_a = \operatorname{Spec} k[X]$ defined by the equation $X^p - aX = 0$. One then concludes using the exact sequence:

```
0 → N → G_a —P→ G_a → 0,
```

where P is the Artin–Schreier morphism $x \mapsto x^p - ax$.

c) If N is a form of $(\mathbb{Z}/p\mathbb{Z})^r$, there exists a finite Galois extension k' of k that trivializes N. By b) and an evident induction on r, G_a/N is a form of G_a trivialized by k'. It then suffices to apply the following lemma:

Lemma 2.3 bis. *Let k be a field, G a k-algebraic group which is a form of G_a, trivialized by a finite separable extension k' of k. Then G is isomorphic to G_a.*

Indeed, the group of k'-automorphisms of the algebraic group $(G_a)_{k'}$ is the group of non-zero homotheties $(k')^\times$ (*Bible*, Exp. 9, Lemma 1), and the Galois cohomology group $H^1(k'/k, G_m)$ is zero (one may suppose that k' is a Galois extension of k), by Hilbert's Theorem 90. Lemma 2.3 bis then follows from the classification of k-forms of an algebraic group (cf. J.-P. Serre, *Cohomologie galoisienne*, Chap. III, 1.3).

This completes the proof of 2.2.

Proposition 2.4. *Let k be a field, M a k-group of multiplicative type (Exp. VIII) and of finite type, U a unipotent k-algebraic group. Then:*

i) $\mathrm{Hom}_{k-gr}(M, U) = e$ *(a fortiori* $\mathrm{Hom}_{k-gr}(M, U) = e$*).*

ii) $\mathrm{Hom}_{k-gr}(U, M) = e$.

To prove i) we must establish that for every prescheme S over k, $\mathrm{Hom}_{S-gr}(M_S, U_S) = e$. But this follows from the following lemma:

Lemma 2.5. *Let S be a prescheme, M an S-group of multiplicative type and of finite type over S, U an S-prescheme in groups of finite presentation over S with unipotent fibers. Then* $\mathrm{Hom}_{S-gr}(M, U) = e$.

Indeed, under the hypotheses made, in order that an S-morphism of groups $u : M \to U$ be the unit homomorphism, it suffices that the restriction of u to the fibers of M over the points of S be the zero morphism (Exp. IX, 5.2). We are thus reduced to the case where S is the spectrum of a field, which we may further assume to be algebraically closed. In view of Definition 1.1, we may restrict to $U = G_a$, in which case the property has already been proved (Exp. XII, 4.4.1).

Let us now prove 2.4 ii). Let $u : U \to M$ be a k-morphism of groups. The image $u(U)$ is representable by an algebraic subgroup `U''` of M (Exp. VI_B, 5.4).[1131] The group `U''` is unipotent as a quotient of a unipotent group (2.2 iii)) and is of multiplicative type as a subgroup of a group of multiplicative type

(cf. *Bible*, Exp. 4, Th. 2 cor. 1, or Exp. IX, 6.8), so `U''` is the trivial group by 2.4 i).

Remark 2.6. *Keeping the notations of 2.4, it is no longer true in general that the functor* $\mathrm{Hom}_{k-gr}(U, M)$ *is equal to e. Thus, take a prescheme S such that $\Gamma(S, O_S)$ contains a non-zero element ϵ such that $\epsilon^2 = 0$ (for example the spectrum of the*

[1131] N.D.E.: verify this reference.

algebra of dual numbers of a ring A). For every S' over S, the map $u \mapsto 1 + \epsilon_{S'} u$ defines a homomorphism, functorial in S', from the additive group $\Gamma(S', O_{S'})$ to the multiplicative group $\Gamma(S', O_{S'}^{\times})$, hence defines an S-morphism of groups $(G_a)_S \to (G_m)_S$, and since $\epsilon \neq 0$, this morphism is not zero.

Let us recall (Exp. VII_A § 3) that when G is an S-prescheme in commutative groups, finite and flat over S, one has Cartier duality at one's disposal, and G is reflexive in the sense of Exp. VIII § 1. More precisely, the functor $G \mapsto \mathrm{Hom}_{S-gr}(G, G_m)$ is representable

by an S-prescheme in commutative groups $D(G)$, finite and flat over S, and the canonical morphism $G \to D(D(G))$ is an isomorphism. In particular, one obtains:

(i) $D(\mu_n) \simeq \mathbb{Z}/n\mathbb{Z}$ and consequently $D(\mathbb{Z}/n\mathbb{Z}) \simeq \mu_n$.

(ii) If S is of characteristic $p > 0$, $D(\alpha_p) \simeq \alpha_p$.

3. Unipotent groups acting on a vector space

Let us recall (Exp. I, 4.6.1) that if S is a prescheme and M a sheaf of 0_S-modules, one denotes by $W(M)$ the S-functor $(Sch/S)^{\circ} \to (Ens)$ defined by the condition $W(M)(S') = \Gamma(S', M \otimes O_{S'})$ for every S-prescheme S'.

Moreover, let us recall that if an S-group acts on an S-functor V, one defines the S-functor V^G of invariants of V under G as the subfunctor of V whose set of points with values in some S' over S is the set of $x \in V(S')$ such that $x_{S''}$ is fixed under $G(S'')$ for every S'' over S'.

This being so, one has the following lemma:

Lemma 3.1. *Let S be a prescheme, G an S-prescheme in groups, affine over S, defined by the quasi-coherent 0_S-algebra A. Suppose that G acts on a quasi-coherent sheaf of 0_S-modules M, and let $\mu : M \to A \otimes_{O_S} M$ be the comorphism defining the action of G on M (Exp. I, 4.7.2), and $\nu : M \to A \otimes_{O_S} M$ the morphism $x \mapsto \mu(x) - 1 \otimes x$. Then:*

i) $M^G(S) = \Gamma Ker(\nu)$.

ii) If S is the spectrum of a field k, M^G is of the form $W(N)$, where N is a vector subspace of M.

iii) If S is the spectrum of a field k, every element x of M is contained in a finite-dimensional k-vector subspace of M, stable under the action of G.

Proof. iii) is included for the record and has already been proved in Exp. VI_B, 11.2.

i) It is clear that $M^G(S)$ contains $\Gamma Ker(\nu)$. To establish the converse, one may suppose S affine with ring B. Let $m \in M^G(S)$. Then for every B-algebra B' and every u in $\mathrm{Hom}_{B-alg}(A, B')$ (u corresponds to an element of $G(\mathrm{Spec}\, B')$), one has:

$$(u \otimes 1_M)\, \nu(m) = 0 \quad \text{in} \quad B' \otimes_B M.$$

Taking in particular $B' = A$ and u the identity of A, one finds $\nu(m) = 0$.

ii) Let N be the kernel of ν, equal to $M^G(k)$ by i). Every k-prescheme S is flat over k, so one has:

$$M^G(S) = \Gamma \operatorname{Ker}(\nu \otimes_k S) = \Gamma(N \otimes_k S).$$

So M^G is isomorphic to the functor $W(N)$.

Proposition 3.2. *Let G be a unipotent algebraic group defined over a field k, acting on a k-vector space V. Then if $V \neq 0$, one has $V^G \neq 0$.*

In view of 3.1 ii) and iii), one may suppose k algebraically closed and V of finite dimension over k.

Let $0 \to G' \to G \to G'' \to 0$ be an exact sequence of algebraic groups and suppose that G acts on a sheaf V (for the fpqc topology). Of course $V^G \subset V^{G'}$,

and the quotient presheaf G/G' acts naturally on $V^{G'}$. But $V^{G'}$ is a sheaf (as kernel of the well-known pair of morphisms $V \rightrightarrows \mathrm{Hom}(G', V)$); consequently, the sheaf associated with G/G', that is, G'', acts on $V^{G'}$, and it is immediate to verify that $(V^{G'})^{G''} = V^G = (V^{G'})^{G/G'}$.

This remark allows us to reduce, in order to prove 3.2, to the case where G is an elementary unipotent group (1.7).

a) $G = G_a$, $p = 0$. It follows from (*Bible* 4 prop. 4) that a morphism from G_a into the linear group $GL(V)$ is given by an exponential map:

$$T \mapsto \sum_{q=0}^{\infty} T^q n^q / q!$$

where n is a nilpotent endomorphism of V. But then $V \neq 0 \Rightarrow Kern \neq 0$, and it is clear that every vector of V annihilated by n is left fixed by G.

Suppose now $p > 0$.

b) $G = \alpha_p$. The group α_p being a radicial group of height 1 (Exp. VII_A § 7), giving a representation of α_p in V amounts to giving a representation of the p-Lie algebra α_p in $gl(V)$ (App. II 2.2), that is, here, to giving an element X of $\mathrm{End}(V)$ such that $X^p = 0$ (App. II 2.1). But then $V \neq 0 \Rightarrow W = Ker(X) \neq 0$, and still by (App. II 2.2), one has $W = V^{\alpha_p}$.

c) $G = \mathbb{Z}/p\mathbb{Z}$. A representation of G in V is equivalent to giving an element x of $\mathrm{Aut}(V)$ such that $x^p = 1$, i.e. $(1-x)^p = 0$, so x is of the form $1 + n$ with n nilpotent, and $W = Kern$ is left fixed by x.

d) $G = G_a$. Let G_i, $i \in I$, be the increasing filtered family of étale algebraic subgroups of G_a, hence isomorphic to $(\mathbb{Z}/p\mathbb{Z})^{r_i}$ (prop. 1.5), and let $V_i = V^{G_i}$. Since V is of finite, non-zero dimension, and V_i is non-zero by c), the decreasing filtered family of the V_i is stationary, and $W = \bigcap_{i \in I} V_i \neq 0$. Now one has the lemma:

Lemma 3.3. *The family of étale subgroups of $(G_a)_S$ (S an S-prescheme of characteristic $p > 0$) is schematically dense in G (Exp. IX, 4.1).*

By Exp. IX, 4.4, it suffices to prove the lemma when S is the spectrum of the prime field F_p. In this case, it suffices to consider the family of étale subgroups G_n ($n \geqslant 1$) with equation $X^{p^n} - X = 0$, which is schematically dense in $(G_a)_{F_p}$ since it contains every closed point.

This being so, let us return to the proof of 3.2 d). If $w \in W$, its stabilizer in G is an algebraic subgroup of G_a that majorizes G_i for every $i \in I$, hence is equal to G_a (3.3), and consequently $W = V^G$.

One deduces immediately from 3.2 the

Corollary 3.4. *Let k be a field, G a unipotent k-algebraic group acting on a finite-dimensional k-vector space V. Then V possesses a sequence of vector subspaces V_i, defined over k, stable under G,*

```
0 = V_0 ⊂ V_1 ⊂ … ⊂ V_n = V,
```

such that G acts trivially on V_{i+1}/V_i. One may further suppose V_{i+1}/V_i of dimension 1.

We shall now summarize and complement the properties of unipotent groups already proved, in the following theorem:

Theorem 3.5. *Let G be an algebraic group defined over a field k. The following properties are equivalent:*

i) G is unipotent.

ii) G possesses a composition series, defined over k, whose successive quotients are isomorphic to G_a if $p = 0$ (resp. to α_p, G_a, or twisted $(\mathbb{Z}/p\mathbb{Z})^r$ (1.4) if $p > 0$).

iii) As in ii), but one further assumes the composition series to be central.

iv) G possesses a characteristic composition series (Exp. VI_B) defined over k, whose successive quotients are isomorphic to $(G_a)^r$ if $p = 0$ (resp. to $(\alpha_p)^r$, twisted $(G_a)^s$, or twisted $(\mathbb{Z}/p\mathbb{Z})^t$, taken in this order, if $p > 0$).

v) G is isomorphic to an algebraic subgroup of the group $Trigstr(n)_k$ of strictly upper-triangular matrices of the linear group $GL(n)_k$, for some suitable integer $n \geqslant 0$.

vi) G is affine and, for every linear representation of G in a non-zero finite-dimensional k-vector space V, one has $V^G \neq 0$.

Proof.

- i) ⇒ vi) by 2.1 and 3.2.
- ii) ⇒ v). Since the algebraic group G is affine, G is an algebraic subgroup of a suitable linear group $GL(V)$ (Exp. VI_B, 11.3). Applying 3.4 to the representation of G in V defined by this embedding, one finds v).

iii) ⇒ iii). One knows that the algebraic group $Trigstr(n)$ possesses a central composition series with successive quotients isomorphic to G_a. The composition series induced on G gives property iii), taking 1.5 into account.

iv) ⇒ ii) ⇒ i) and iv) ⇒ i) is clear.

We shall prove i) ⇒ iv) shortly, but first let us note some consequences of what has been proved.

Definition 3.6. *We shall say that a p-Lie algebra g ($p > 0$) (cf. Exp. VII_A § 5) is* unipotent *if the map $x \mapsto x^{(p)}$ is nilpotent, i.e. if for every $x \in g$, there exists an integer $n > 0$ such that $x^{(p^n)} = 0$.*

Corollary 3.7. *A unipotent algebraic group G is nilpotent (Exp. VI_B § 8); its Lie algebra g is nilpotent (Bourbaki,* Groupes et algèbres de Lie*, Chap. 1 § 4) and is isomorphic to a Lie algebra of nilpotent endomorphisms of a finite-dimensional vector space.*

In characteristic $p > 0$, g is a unipotent p-Lie algebra (3.6).

Since i) ⇒ v), it suffices to prove 3.7 when $G = Trigstr(n)$. We have already used the fact that $Trigstr(n)$ is a nilpotent algebraic group. Moreover, the Lie algebra h of $Trigstr(n)$ consists of the upper-triangular endomorphisms of V having zeros on the principal diagonal. They are therefore nilpotent and consequently h is nilpotent (Bourbaki, *loc. cit.* Chap. 1 § 4 cor. 3). If $p > 0$, since the p-th power in the p-Lie algebra $gl(V) = \mathrm{End}(V)$ coincides with the p-th power of the endomorphisms of V (Exp. VII_A, 6.4.4), one sees that h is unipotent.

Corollary 3.8. *Let k be an algebraically closed field and G a smooth and affine algebraic group over k. Then the following properties are equivalent:*

i) G is unipotent.

*ii) $G(k)$ consists of unipotent elements (*Bible* 4 prop. 4 cor. 1), that is, G is unipotent in the sense of the *Bible*.*

i) ⇒ ii) because G is isomorphic to an algebraic subgroup of a group $Trigstr(n)$ by 3.2 i) ⇒ v).

ii) ⇒ i). Let G be an algebraic group, unipotent in the sense of the *Bible*, and let G^0 denote its identity component. The maximal tori of G^0, being made up of unipotent elements, are trivial, so G^0 is equal to its Cartan subgroups. Consequently, G^0 is solvable (*Bible* 6 Th. 6), hence triangularizable (*Bible* 6, Th. 1). In brief, G^0 is an algebraic subgroup of a group $Trigstr(n)$, and is therefore unipotent in the sense of the present Exposé.

The group $(G/G^0)(k)$ is a finite group made up of unipotent elements; it is therefore zero if $p = 0$ and equal to a finite p-group if $p > 0$ (*Bible* 4 prop. 4). But then G/G^0 is unipotent in the sense of the present Exposé as a multiple extension of groups isomorphic to $\mathbb{Z}/p\mathbb{Z}$. This proves that G is unipotent.

End of the proof of 3.5. Let us prove that i) ⇒ iv).

a) $p > 0$. Consider the increasing sequence of algebraic subgroups of G:

$\{e\} \subset F(G) \subset F_2(G) \subset \cdots \subset F_n(G) \subset G^0 \subset G.$

One thus obtains a characteristic composition series of G (App. II 1), and for n large enough, $G/F_n(G)$ is smooth (App. II 3.1), so that the successive quotients are, in order:

(1) radicial groups of height 1,

(2) a smooth and connected group,

(3) an étale group.

To prove i) ⇒ iv) it therefore suffices to prove the:

Lemma 3.9. *Let G be a unipotent algebraic group defined over a field k of characteristic $p > 0$. Then G possesses a characteristic composition series, defined over k, whose successive quotients are isomorphic to:*

i) $(\alpha_p)^r$ if G is radicial.

ii) Twisted $(G_a)^r$ if G is smooth and connected.

iii) Twisted $(\mathbb{Z}/p\mathbb{Z})^r$ if G is étale.

Proof. i) The group G is radicial. Filtering G by the $F_n(G)$, one reduces to the case where G is radicial of height 1. Since G is nilpotent (3.5 i) ⇒ iii)), and since the center of an algebraic group is representable (Exp. VIII, 6.5 e)), one may consider the ascending central series of G, evidently characteristic in G, which reduces us to the case where G is moreover commutative.

Let $g = Lie(G)$. The p-th power morphism π is therefore additive on g (Exp. VII_A); we shall reduce to the case where it is zero. For every prescheme S over k, set $g_S = g \otimes_k O_S$, and let h_S be the subsheaf of abelian groups of g_S image of g_S under π_S. Finally, let $\bar{h}_S$ be the subsheaf of O_S-modules of g_S generated by h_S. It is clear that $\bar{h}_S = \bar{h}_k \otimes_k O_S$ and that $\bar{h}_S$[1132] is a characteristic sub-p-Lie algebra of g_S (i.e. is stable under the S-functor $\mathrm{Aut}_{p-Lie}(g)$). It then follows from App. II 2.2 that $\bar{h}_k$ is the Lie algebra of an algebraic subgroup H of G that is characteristic in G.

Moreover, taking 3.7 and Lemma 3.9 bis below into account, if $G \neq e$, H is distinct from G, because $\bar{h}_k$ is distinct from g. On the other hand, if $G/H = G''$, one has $LieG'' = g/\bar{h}_k$ (App. II 2.2), and consequently, the p-th power is zero in Lie G''. Proceeding by induction on $\dim LieG$, we are thus reduced to the case where Lie G is a p-Lie algebra in which the p-th power is zero. But then Lie G'' is isomorphic to $Lie(\alpha_p)^r$ for some suitable integer $r \geqslant 0$ (App. II 2.1), and consequently (App. II 2.2), G'' is isomorphic to $(\alpha_p)^r$. It remains to prove the:

Lemma 3.9 bis. *Let k be a field of characteristic $p > 0$, g a commutative, unipotent (3.6), finite-dimensional p-Lie algebra over k, and h the sub-p-Lie algebra of g generated by the image of the p-th power in g. Then if $g \neq 0$, one has $h \neq g$.*

[1132] N.D.E.: h has been changed to $\bar{h}$.

Indeed, since g is commutative, h is simply the k-vector subspace of g generated by the $X^{(p)}$ ($X \in g$). If $g \neq 0$ and is unipotent, there exists $X \in g$, $X \neq 0$, such that $X^{(p)} = 0$.

Let $X_1, \cdots, X_n$ be a basis of a supplement in g of the line `kX`. The Lie algebra h is then the k-vector subspace of g generated by $X_1^{(p)}, \cdots, X_n^{(p)}$, hence has dimension at most $n = \dim g - 1$.

Proof of 3.9 ii). G is smooth and connected. In this case, the descending central series of G is representable by characteristic smooth connected algebraic subgroups G_i (Exp. VI_B, 8.3 and 7.4), and $G_i = 0$ for i large enough, since G is nilpotent (3.5 i) ⇒ iii)). It suffices to prove 3.9 for the groups G_i/G_{i+1}, which reduces us to the case where moreover G is commutative. For every integer $n > 0$, let G_n be the algebraic subgroup of G which is the image of G under the p^n-th power morphism. The group G_n is therefore smooth, connected and characteristic, and it follows from Definition 1.1 of unipotent groups that $G_n = 0$ for n large enough. Replacing G by G_n/G_{n+1}, one may further suppose that G is annihilated by raising to the p-th power. But then, by (J.-P. Serre, *Groupes algébriques et corps de classes*, Chap. VII, prop. 11), G is a form of $(G_a)^r$ for some suitable integer r.

Proof of 3.9 iii). G is étale. Proceeding as in ii), one reduces to the case where G is commutative, then to the case where G is annihilated by p, but then $G_{\bar{k}}$ is isomorphic to $(\mathbb{Z}/p\mathbb{Z})^r$.

Proof of 3.5 i) ⇒ iv) in the case b) $p = 0$. The group G is then smooth and connected, and proceeding as in 3.9 ii), one reduces to the case where G is moreover commutative. One then has the following more precise result:

Lemma 3.9 ter. *Let k be a field of characteristic 0, G a commutative unipotent k-algebraic group, $g = Lie G$. Then there exists a canonical isomorphism:*

$$\exp : W(g) \xrightarrow{\sim} G.$$

The morphism `exp` *is the unique homomorphism $W(g) \to G$ that induces the identity on Lie algebras.*

Since G is unipotent, G is realized as an algebraic subgroup of $Trigstr(n)$ for some suitable integer n (3.5 i) ⇒ v)). The choice of such an embedding allows one to identify g with a Lie subalgebra of $gl(n)$ consisting of nilpotent endomorphisms. Whence a k-morphism:

```
exp : W(g) → GL(n),    T ↦ Σ_{i ≥ 0} T^i / i!.
```

Since G is commutative, the morphism `exp` is a homomorphism. Let G' be the algebraic group image of $W(g)$ under the morphism `exp`. If one canonically identifies `Lie W(g)` with g, the linear tangent map of `exp` is simply the injection $g \to gl(n)$. Consequently $Lie(G \cap G') = g \cap g = g$. Since $G \cap G'$ is smooth (Exp. VI_B, 1.6.1) and G is connected (being a multiple extension of groups G_a (3.5 i) ⇒ ii)), one necessarily has $G = G'$. The kernel $Ker(\exp)$ is a unipotent étale group, hence is the trivial group, and consequently exp is an isomorphism of $W(g)$ onto G.

If $h : W(g) \to G$ is another homomorphism such that $Lie(h)$ is the identity map of g, the morphism $h - \exp$ is a homomorphism (G being commutative) whose linear tangent map is therefore zero. Since k is of characteristic 0 and $W(g)$ is connected, it again follows from Cartier's theorem that one necessarily has $h = \exp$.

Proposition 3.10. *Let G be an algebraic group defined over an algebraically closed field k. Then the following properties are equivalent:*

i) Every morphism of a group of multiplicative type M into G is the zero morphism.

i bis) G has no non-zero subgroups of multiplicative type.

ii) a) if $p = 0$: $G(k)$ contains no points of finite order other than e;

b) if $p \neq 0$: for every prime number $q \neq p$, $x \in G(k)$ and $x^q = e$ imply $x = e$,

for every $X \in g = Lie\ G$ such that $X^{(p)} = X$, one has $X = 0$.

ii bis) a) if $p = 0$: as in ii) a);

b) if $p \neq 0$: every finite subgroup of $G(k)$ is a p-group,

for every $X \in g = Lie\ G$ such that $X^{(p)} = X$, one has $X = 0$.

Proof.

i) $\Leftrightarrow$ i bis), since the image of a group of multiplicative type is of multiplicative type (Exp. IX, 2.7).

ii) $\Leftrightarrow$ ii bis). Because if an ordinary finite group H has order not a power of p, there exists a prime number $q \neq p$, and an element x of H distinct from e, such that $x^q = e$ (Sylow's theorem, cf. J.-P. Serre, *Corps locaux*, Chap. IX § 2).

iii) $\Rightarrow$ ii) follows from the following lemma:

Lemma 3.11. *Let G be an algebraic group defined over a field k.*

a) If k contains the n-th roots of unity, with n an integer prime to p, one has:

$$\mathrm{Hom}_{k\text{-gr}}(\mu_n, G) \simeq \mathrm{Hom}_{k\text{-gr}}(\mathbb{Z}/n\mathbb{Z}, G) \simeq {}_n G(k)$$

(points of order n of $G(k)$).

b) If $p > 0$, $\mathrm{Hom}_{k\text{-gr}}(\mu_p, G) \simeq \{X \in g = Lie\ G,$ such that $X^{(p)} = X\}$.

Indeed, to prove a) we note that μ_n is then isomorphic over k to $\mathbb{Z}/n\mathbb{Z}$, and b) is a consequence of App. II 2.1.

ii) $\Rightarrow$ i bis). By Lemma 3.11, ii) is equivalent to the fact that for every prime number r, G does not contain any subgroups μ_r, which entails i bis) by reason of the:

Lemma 3.12. *Let G be a diagonalizable S-group, of finite type over S and distinct from the trivial group. Then there exists a prime number r and a subgroup of G isomorphic to $(\mu_r)_S$.*

Let $G = D_S(M)$, where M is a finitely generated abelian group, hence extension of a free group `M''` by a finite group M'. If $M'' \neq 0$, it is clear that M admits quotients isomorphic to $\mathbb{Z}/r\mathbb{Z}$ for every integer r. If $M'' = 0$, M' admits a quotient isomorphic to $\mathbb{Z}/r\mathbb{Z}$ for every prime number r dividing the order of M.

One then deduces the lemma by duality.

We have seen (Prop. 2.4) that a unipotent group satisfies the equivalent conditions of 3.10. The aim of the next section is to prove the converse.

4. A characterization of unipotent groups

As announced, we shall show that an algebraic group G defined over an algebraically closed field k, which does not contain any non-zero subgroup of multiplicative type, is unipotent. In fact, it suffices that it does not contain certain particular "elementary" subgroups of multiplicative type, which depend on the hypotheses made on G. Before stating the general theorem, let us study in detail some special cases.

4.1. Smooth, connected, affine algebraic groups

Proposition 4.1.1. *Let k be a field, G a smooth, connected, affine k-algebraic group, $g = LieG$. Then the following properties are equivalent:*

i) G is unipotent.

ii) G possesses a central composition series whose successive quotients are forms of G_a.

iii) G possesses a central, characteristic composition series whose successive quotients are forms of $(G_a)^r$.

iv) There exists an integer $n > 1$ such that $G_{\bar{k}}$ does not contain any subgroup isomorphic to μ_n.

v) Every maximal torus of G is the trivial group.

Suppose moreover that G is an algebraic subgroup of a linear group $GL(n)$. Then the preceding conditions are also equivalent to:

vi) $g \subset gl(n)$ consists of nilpotent endomorphisms.

vii) g is nilpotent and its center contains no non-zero semisimple endomorphism.

Proof. ii) ⇒ i) is clear, and i) ⇒ iii) was seen in 3.9. The implication iii) ⇒ ii) will follow from the following lemma:

Lemma 4.1.2. *Let k be a field, G a k-algebraic group which is a form of $(G_a)^r$. Then:*

a) G is realized as an algebraic subgroup of the group $(G_a)^n$ for some suitable integer n.

b) G possesses a composition series whose successive quotients are forms of G_a.

Indeed, by hypothesis, there exists an extension k' of k such that $G_{k'}$ is isomorphic to $(G_{a,k'})^r$. By the principle of finite extension (EGA IV 9.1.1), one may suppose that k' is a finite extension of k. But then, for a), it suffices to consider the canonical closed immersion (EGA V[1133]):

$$G \hookrightarrow \prod_{k'/k} (G_{k'})/k' \simeq (G_{a,k})^n \quad (\text{with } n = r\deg(k'/k)).$$

To prove b), in view of a), one may suppose that G is a closed subgroup of $G' = (G_a)^n$. If $G \neq 0$, there exists a hyperplane h of $g' = Lie G'$ not containing g. Let H be the vector subgroup $W(h)$ of $W(g') = G'$. Since H is defined by an equation in G', $H \cap G$ is defined by an equation in G, and one has the inequalities:

$$\begin{aligned}\dim G - 1 &\leq \dim(G \cap H)\\ &\leq \dim \mathrm{Lie}(G \cap H) = \dim_k(g \cap h) = \dim_k(g) - 1 = \dim G - 1.\end{aligned}$$

Hence $\dim(G \cap H) = \dim \mathrm{Lie}(G \cap H)$, and consequently $G \cap H$ is smooth. The group $G_1 = (G \cap H)^0$

is an algebraic subgroup of G, smooth and connected, such that G/G_1 is smooth, connected, of dimension 1, hence a form of G_a (4.1 i) ⇒ iii)). One finishes by induction on the dimension of G.

Before continuing the proof of 4.1, let us note that the equivalence i) ⇔ ii) and 2.3 bis entail the following corollary:

Corollary 4.1.3. *If k is a perfect field, a smooth and connected k-algebraic group is unipotent if and only if it possesses a composition series whose successive quotients are isomorphic to G_a.*

Continuation of the proof of 4.1.

i) ⇒ iv) by 2.4 i).

ii) ⇒ v). By Exp. XIV, 4.1, G possesses a maximal torus T defined over k. Now if $r = \dim T$, $({}_nT)_{\bar{k}}$ is isomorphic to $(\mu_n)^r$. Hence $r = 0$.

iii) ⇒ i) as remarked in the proof of 3.8.

iv) ⇒ vi). By 3.4, G is in fact contained in an algebraic subgroup of $GL(n)$ isomorphic to $Trigstr(n)$, hence g consists of nilpotent endomorphisms.

v) ⇒ vii). Indeed, g is nilpotent by Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chap. I § 4 cor. 3).

vi) ⇒ v). Let T be a maximal sub-torus of G (Exp. XIV, 1.1), t its Lie algebra. The embedding of G in $GL(n)$ defines a representation of T which is necessarily

[1133] N.D.E.: give another reference here...

semisimple (this is seen on an algebraic closure $\bar{k}$ of k, and one applies Exp. I, 4.7.3).

Hence if $X \in t$, X is a semisimple endomorphism in $gl(n)$. One sees immediately that this entails that the map

```
ad X : Y ↦ [X, Y]
```

is a semisimple endomorphism of $gl(n)$, hence of g. Since, moreover, this endomorphism is nilpotent (g being nilpotent), `ad X` is zero, so X is central. But then t is central and consists of semisimple endomorphisms, hence is zero by hypothesis; *a fortiori*, T is the trivial group.

Remark 4.1.4.

a) We shall give below (4.3.1) an infinitesimal characterization of unipotent groups in characteristic $p > 0$, which is independent of an embedding in $GL(n)$.

b) When k is perfect, conditions ii) and iii) of 4.1.1 simplify by reason of the following lemma:

Lemma 4.1.5. *If k is a perfect field, every k-algebraic group G which is a form of $(G_a)^r$ is isomorphic to $(G_a)^r$.*

The lemma follows from 3.9 ter if the characteristic p of k is zero, and from 2.3 bis if $r = 1$. In the general case ($p > 0$), realize G as an algebraic subgroup of $(G_a)^n$ for some suitable integer n (4.1.2), and argue by induction on the integer $n - r$. If $r = n$, one indeed has $G = (G_a)^r$. Otherwise the quotient group G_a^n/G is a non-zero smooth connected unipotent group which, taking 4.1.1 i) ⇒ ii) and 2.3 bis into account,

possesses a composition series with quotients isomorphic to G_a. One deduces that there exists an algebraic subgroup `G_1` of $(G_a)^n$, smooth and connected, containing G, such that $G_a^n/G_1 = G_a$. By induction it suffices to show that `G_1` is isomorphic to $(G_a)^{n-1}$. Now it is immediate to verify that a homomorphism from $\mathrm{Spec}\, k[X_1, \cdots, X_n] = G_a^n$ to $G_a = \mathrm{Spec}\, k[T]$ is defined by an additive polynomial of the form:

$$\Sigma_{i,j} a_{i,j} X_i^{p^j}.$$

Since `G_1` is smooth, the linear part of this polynomial is non-zero. Possibly making a linear change of the coordinates X_i, we may assume that `G_1` is an algebraic subgroup of $(G_a)^n$ defined by the equation:

```
(*)   P(X) = −X_1 + Σ_{j=1}^q a_j X_1^{p^j} + Q(X_2, …, X_n) = 0,
                                              with   Q(X_2, …, X_n) = Σ_{i>1, j>0} b_{i,j} X_i^{p^j}.
```

Proceed then by induction on the degree of P. If $degP = 1$, it is clear that $G_1 \xrightarrow{\sim} G_a^{n-1}$. Otherwise, since k is perfect, one has $Q(X) = Q_1(X)^p$ and we may define an endomorphism v of G_a^n by the formulas:

$$X_i \mapsto X_i \quad \text{for } i > 1, \qquad X_1 \mapsto \sum_{j=1}^q a_j^{1/p} X_1^{p^{j-1}} + Q_1(X).$$

It is clear that v induces an isomorphism on G_1 and that $v(G_1)$ has equation in G_a^n:

$$(*)_1 \quad P_1(X) = -X_1 + \sum_{j=1}^q a_j^{1/p} X_1^{p^j} + Q_1(X_2, \dots, X_n) = 0.$$

Let us then distinguish two cases:

1st case. $deg(Q) = degP > p^q$. Then $degP_1 < degP$ and one wins by the induction hypothesis.

2nd case. $degP = p^q$ ($a_q \neq 0$). One then has $degP_1 = degP = p^q$ and $degQ_1 < p^q$. (One cannot have $Q = 0$, otherwise G_1 would not be connected.) If the polynomial Q_1 has no linear part, one can iterate the preceding transformation. Continuing the process, one finally obtains an equation of the form:

$$(*)_s \quad P_s(X) = -X_1 + \sum_{j=1}^q a_j^{1/p^s} X_1^{p^j} + Q_s(X),$$

where $Q_s(X) = Q(X_2, \cdots, X_n)^{1/p^s}$ is an additive polynomial having a non-zero linear part, and moreover $degQ_s < p^q$. Suppose, for example, that the coefficient of X_2 in Q_s is non-zero, and let $-L$ be the linear part of $P_s(X)$. Possibly making a linear change of coordinates, the equation of G_1 becomes:

$$P'(X) = -L + \sum_{j=1}^q a_j^{1/p^s} X_1^{p^j} + Q'(L, X_3, \dots, X_n),$$

where Q' is an additive polynomial without linear part, and $deg(Q') < p^q$. But then we are reduced to the first case.[1134]

4.2. Radicial groups

Proposition 4.2.1. *Let G be a radicial algebraic group defined over a field k of characteristic $p > 0$. Then the following conditions are equivalent:*

i) G is unipotent.

ii) G possesses a central composition series with successive quotients isomorphic to α_p.

iii) G possesses a central and characteristic composition series with successive quotients isomorphic to $(\alpha_p)^r$.

iv) $G_{\bar{k}}$ contains no subgroup isomorphic to μ_p.

v) $g = LieG$ is a unipotent p-Lie algebra (3.6).

Proof. iii) ⇒ ii) ⇒ i) is clear, i) ⇒ iii) is 3.9 i), and i) ⇒ iv) by 2.4 i).

We shall need the following lemma on abelian p-Lie algebras:

Lemma 4.2.2. *Let g be an abelian, finite-dimensional p-Lie algebra over a perfect field k. Then g can be written uniquely as a direct sum of a sub-p-Lie algebra r on which the p-th power is bijective, and a sub-p-Lie algebra u, unipotent (3.6). (The algebra r will be called the* reductive part *of g and u the* unipotent part*.)* The

[1134] N.D.E.: replacing X_1 by L.

formation of r and u is compatible with extension of the field k. If moreover k is algebraically closed, r admits a basis e_i such that $e_i^{(p)} = e_i$.*

The proof of this lemma is easy and left to the reader (cf. Bourbaki, *Groupes et algèbres de Lie*, Chap. I § 1 exercise 23). Let us simply say that u is the kernel of a suitable iterate of the map $X \mapsto X^{(p)}$, and that r is the image of the same iterate.

This being so, let us prove iv) ⇒ v). Possibly making an extension of the base field, one may suppose k algebraically closed.

Let then X be an element of g and h the sub-p-Lie algebra generated by X in g. The algebra h is evidently commutative, and its reductive part (4.2.2) is zero, otherwise by *loc. cit.*, h would contain an element $Y \neq 0$ such that $Y^{(p)} = Y$, and consequently (App. II 2.1 and 2.2) G would contain a subgroup isomorphic to μ_p contrary to hypothesis. Hence h, and consequently g, is a unipotent p-Lie algebra.

v) ⇒ i). This is the least trivial implication of 4.2.1.

w) Case where G is of height 1 (Exp. VII_A, 4.1.3). Since G is radicial, it is affine, hence isomorphic to an algebraic subgroup of a linear group $GL(V)$ (Exp. VI_B, § 11). This embedding identifies g with a sub-p-Lie algebra of End(V), the p-th power of X in g coinciding with the p-th power of the endomorphism X (Exp. VII_A, 6.4.4). Since g is unipotent by hypothesis, g is therefore an algebra of nilpotent endomorphisms of V, and by Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chap. I § 4 th. 1) is a Lie subalgebra of the Lie algebra h of the group of strictly upper-triangular matrices $Trigstr(n)$ with respect to a suitable basis of V. Since G is of height 1, one then deduces from App. II 2.2 that G itself is an algebraic subgroup of $Trigstr(n)$, hence is unipotent (3.5 v) ⇒ i)).

x) General case. Proceed by induction on the height h of G[1135]. The case $h = 1$ has just been treated. Suppose $h > 1$, and set $G' = FG$ and $G'' = G/G'$. The group G' is of height 1 and $LieG' = LieG$ is unipotent, hence G' is unipotent by a).

To show that G is unipotent, it therefore suffices to prove that G'' is unipotent (2.2). But G'' is of height $h - 1$, hence, by induction hypothesis, it suffices to show that $\mathrm{Lie}\, G''$ is unipotent. Since iv) ⇒ v), it suffices to show that $G''_{\bar{k}}$ does not contain any group isomorphic to μ_p. Let then a subgroup of $G''_{\bar{k}}$ isomorphic to μ_p, H its inverse image in $G_{\bar{k}}$. The group G' being unipotent, we shall prove in § 5 that the extension:

$$e \to G'_{\bar{k}} \to H \to \mu_p \to e$$

is necessarily trivial (the proof given of this fact is independent of the results of the present section). Briefly, the group μ_p lifts in H, but being of height 1 it is necessarily contained in $G'_{\bar{k}} = FG_{\bar{k}}$, whence a contradiction, G' being unipotent.

[1135] i.e. the smallest integer such that $F^h = id_G$, cf. App. II 1.

4.3. Connected affine groups in characteristic $p > 0$

Proposition 4.3.1. *Let G be a connected affine algebraic group over a field k of characteristic $p > 0$. Then the following conditions are equivalent:*

i) G is unipotent.

ii) G possesses a composition series with successive quotients isomorphic to α_p and G_a (taken in this order).

iii) G admits a characteristic composition series with successive quotients isomorphic to $(\alpha_p)^r$ and $(G_a)^s$ (taken in this order).

iv) $G_{\bar{k}}$ contains no subgroup isomorphic to μ_p.

v) $g = LieG$ is unipotent (3.6).

vi) g is nilpotent, and the reductive part of the center of g (4.2.2) is trivial.

vi bis) g is nilpotent, and every subgroup of multiplicative type of the identity component of the center of G is zero.

vi ter) G is nilpotent, and every subgroup of multiplicative type of the identity component of the center of G is zero.

Proof. It is clear that iii) ⇒ ii) ⇒ i). To establish i) ⇒ iii), we shall need the following lemma:

Lemma 4.3.2. *Let k be a field of characteristic $p > 0$, $n \in \mathbb{N}$, k' a radicial extension of k such that $(k')^{p^n}$ is contained in k; for every k-prescheme X (resp. every k'-prescheme X'), denote by $X^{(p^n)}$ (resp. X'^{ϕ}) the k-prescheme deduced from X (resp. X') by the base change:*

$F^n : k \to k$, $x \mapsto x^{p^n}$, (resp. $\varphi : k' \to k$, $x' \mapsto x'^{p^n}$).

Then, for every k-prescheme X, there exists a functorial isomorphism:

$$(X_{k'})^{\phi} \xrightarrow{\sim} X^{(p^n)}.$$

Consequently, if X and Y are two k-preschemes such that there exists a k'-isomorphism $u' : X_{k'} \xrightarrow{\sim} Y_{k'}$, then there exists a k-isomorphism $v : X^{(p^n)} \xrightarrow{\sim} Y^{(p^n)}$. If moreover X and Y are equipped with structures of k-preschemes in groups and if u' is a k'-homomorphism, then v is a k-homomorphism.

The lemma follows simply from the transitivity of base changes and from the fact that the composite morphism $k \to k' \xrightarrow{\phi} k$ is equal to F^n.

Continuation of the proof of 4.3.1.

i) ⇒ iii). Proceed by induction on $\dim G$. If $\dim G = 0$, since G is connected, it is radicial, and one applies 3.9 i). If $\dim G > 0$, there exists an integer $m \geqslant 0$ such that the quotient $G/F_m G$ is a smooth group (App. II 3.1), evidently connected and non-zero. Applying 4.1.1 i) ⇒ iii) to the latter, one sees that there exists an algebraic subgroup G' of G which is characteristic and connected, and such that the quotient $G'' = G/G'$ is a form of G_a^r $(r > 0)$. By 4.1.5, if K is a perfect closure of k, one has $G''_K \xrightarrow{\sim} (G_{a,K})^r$. Since `G''` is of finite type over k, there exists a finite radicial extension k' of k such that $G''_{k'} \xrightarrow{\sim} (G_{a,k'})^r$ (Exp. VI_B § 10). Let $n > 0$ be such that $(k')^{p^n} \subset k$. Keeping the notations of 4.3.2, one deduces that there exists a k-isomorphism of algebraic groups:

`(G_{a, k})^r = (G_{a, k'})^{rφ} □ (G'')^{(p^n)}.`

Consider then the Frobenius homomorphism relative to `G''` (Exp. VII_A § 4)

$$F^n : G'' \longrightarrow (G'')^{(p^n)}.$$

Since `G''` (and hence also $(G'')^{(p^n)}$) is smooth over k, and F^n is radicial, F^n is an epimorphism for the fpqc topology, so that $(G'')^{(p^n)}$ is identified with $G''/F_n(G'')$. Finally we have shown that $G''/F_n(G'')$ is isomorphic, as an algebraic group, to $(G_a)^r$.

The inverse image G'_n of $F_n(G'')$ in G is a subgroup of G, connected, characteristic, of dimension strictly less than that of G, to which we may apply the induction hypothesis.

i) ⇒ iv) by 2.4 i).

ii) ⇒ i). Consider G as an extension of a smooth connected group `G''` by a radicial group G' (App. II 3.1). The group G' is unipotent (4.2.1 iv) ⇔ i)). It suffices to see that `G''` is unipotent, and for that it suffices to show that $G''_{\bar{k}}$ does not contain a subgroup isomorphic to μ_p (4.1.1 i) ⇔ iv)). Now if $G''_{\bar{k}}$ contained a subgroup μ_p, the latter would lift in $G_{\bar{k}}$, by the result (5.1) — already used — proved in § 5, whence a contradiction with iv).

iii) ⇒ v) by 3.7.

iv) ⇒ vi). Indeed, since $(adX)^{p^r} = ad(X^{(p^r)})$ (VII_A, 5.2), `ad X` is nilpotent if g is unipotent, hence g is nilpotent by Engel's theorem (Bourbaki, *Groupes et algèbres de Lie*, Chap. I § 4). On the other hand, if g is unipotent, evidently so is its center, whose reductive part is then trivial (4.2.2).

v) ⇒ iv). Indeed, if $G_{\bar{k}}$ contains a subgroup isomorphic to μ_p, there exists a non-zero element X of its Lie algebra such that $X^{(p)} = X$ (App. II 2.1), hence $X^{(p^r)} = X$ for every $r > 0$. Since `ad X` is nilpotent (because g is nilpotent and $(adX)^{p^r} = adX^{(p^r)}$), necessarily $adX = 0$, so X belongs to the reductive part of the center of $g_{\bar{k}}$, whence a contradiction with vi).

vi) ⇒ vi ter) follows from 2.4 i) and from 3.5 i) ⇒ iii).

vi ter) ⇒ vi bis). Indeed, if G is nilpotent, so is its subgroup `F G`. It also follows from App. II 2.2 that `F G` is nilpotent if and only if $LieFG = LieG$ (Exp. VII_A) is nilpotent.

vi bis) ⇒ vi). Let Z be the identity component of the center of G, and let r be the reductive component of the center of g. We must show that $r = 0$. Now it is immediate that r is a characteristic sub-p-Lie algebra of $g = LieFG$ (i.e. stable under the functor $\mathrm{Aut}_{p-Lie}(g)$); hence r is the Lie algebra of a characteristic radicial subgroup R of `F G` (App. II 2.2). On the other hand, it follows from the last assertion of 4.2.2 and from App. II 2.1 that R is a form of $(\mu_p)^r$. The group R being characteristic in `F G`, which is itself a characteristic subgroup of G (App. II 1), R is *a fortiori* invariant in G, hence is central, G being connected (Exp. IX, 5.5). Hence by hypothesis vi bis), R is zero, and so therefore is r.

4.4. Étale groups

The following proposition is an easy consequence of Sylow's theorems and of the structure of finite q-groups (cf. J.-P. Serre, *Corps locaux*, Chap. IX § 1).

Proposition 4.4.1. *Let G be a finite étale algebraic group defined over an algebraically closed field k. Then in order that G be unipotent, it is necessary and sufficient that for every prime number q distinct from the characteristic p of k, G does not contain a subgroup isomorphic to μ_q.*

4.5. Abelian varieties

Let G be an abelian variety defined over an algebraically closed field k. Then the following conditions are equivalent:

i) G is unipotent.

ii) $G = 0$.

iii) There exists an integer n, prime to the characteristic p of k, such that G does not contain a subgroup isomorphic to μ_n.

Indeed, if G is an abelian variety of dimension d, one knows (cf. S. Lang, *Abelian varieties*, Chap. IV § 3 th. 6) that the group ${}_nG(k)$ (n an integer prime to p) is isomorphic to $(\mathbb{Z}/n\mathbb{Z})^{2d}$, hence is isomorphic to $(\mu_n)^{2d}$. Whence iii) ⇒ ii), and ii) ⇒ i) ⇒ iii) are obvious.

4.6. General case

If G and H are two algebraic groups defined over an algebraically closed field k, we shall denote by $P(G,H)$ the property: "there is no algebraic subgroup of G isomorphic to H". One then obtains the following characterizations of unipotent groups:

Theorem 4.6.1. *Let G be an algebraic group defined over an algebraically closed field k of characteristic p. Then:*

i) If G is smooth, affine, and connected:

```
G is unipotent ⟺ ∃ n > 1 such that P(G, μ_n) is true ⟺ P(G, G_m) is true.
```

ii) If G is smooth and connected:

G is unipotent $\iff \exists$ n prime to p such that P(G, μ_n) is true.

iii) If G is smooth:

G is unipotent $\iff$ for every prime number n ≠ p, P(G, μ_n) is true.

iv) G affine connected and $p > 0$: G unipotent $\iff P(G, \mu_p)$ is true.

v) G connected and $p > 0$:

G unipotent $\iff \exists$ n prime to p such that P(G, μ_n) is true and P(G, μ_p) is true.

vi) G an arbitrary algebraic group:

G is unipotent $\iff$ for every prime number n, P(G, μ_n) is true.

Proof. i) follows from 4.1.1, and iv) from 4.3.1. We shall prove vi); ii), iii), v) are proved analogously and are left to the reader.

Let then G be an algebraic group. If G is unipotent, $P(G, \mu_n)$ is true for every $n > 1$ (2.4 i)). Conversely, suppose $P(G, \mu_n)$ is true for every prime number n, and let us show that G is unipotent. Let G^0 be the identity component of G. If G^0 is smooth, it follows from a classical theorem of Chevalley[1136] that G^0 is an extension of an abelian variety A by an affine, connected, smooth group L. If G is not smooth, which presupposes $p > 0$ (Exp. VI_B, 1.6.1), there exists an integer $n > 0$ such that $G'' = G^0/F_n(G)$ is smooth (App. II 3.1). Then G'' is an extension of an abelian variety A by a smooth connected linear group L''. Denote by L the inverse image of L'' in G^0, which is still affine and connected, since $F_n(G)$ is radicial. In all cases, G therefore possesses a composition series:

$$0 \subset L \subset G^0 \subset G$$

such that L is affine and connected, $G^0/L = A$ is an abelian variety, and G/G^0 an étale group.

If $P(G, \mu_n)$ is true, *a fortiori* $P(L, \mu_n)$ is true, hence L is unipotent (4.1.1 and 4.3.1). If A is non-zero, there exists a prime number n and a subgroup of A isomorphic to μ_n (4.5); by 5.1 below, this subgroup lifts in G, which contradicts the hypothesis $P(G, \mu_n)$; hence $A = 0$. Finally if G/G^0 is not unipotent, there exists an integer q and a subgroup of G/G^0 isomorphic to μ_q (4.4.1). One deduces as above that $P(G, \mu_q)$ is not true; hence G/G^0 is unipotent, and consequently so is G.

5. Extension of a group of multiplicative type by a unipotent group

5.1. Statement of the theorem

Definition 5.1.0. *Let k be a field, G a k-algebraic group. Following the terminology introduced by Rosenlicht (Questions of rationality for solvable algebraic*

[1136] *Sém. Bourbaki* n° 145, 1956/57.

groups over non perfect fields, Annali di Math. 61 (1963)), we shall say that G is "k-solvable" if it satisfies the following equivalent conditions:

i) G possesses a composition series with successive quotients isomorphic to G_a.

ii) G possesses a characteristic composition series (G_i) such that the successive quotients G_i/G_{i+1} are commutative and possess a composition series with successive quotients isomorphic to G_a.

The fact that i) ⇒ ii) is proved in *loc. cit.* In fact, one may take as composition series (G_i) the composition series introduced in the proof of 3.9 ii).

Theorem 5.1.1. *Let k be a field, U a unipotent k-algebraic group, H a k-group of multiplicative type, E a k-algebraic group extension of H by U, so that one has the exact sequence:*

$$1 \longrightarrow U \longrightarrow E \longrightarrow H \longrightarrow 1.$$

Then:

i) The extension E is trivial in each of the following cases:

a) k is algebraically closed.

b) k is perfect and one of the groups U or H is connected.

c) U is k-solvable.

d) U is smooth and H is connected.

ii) If H' and H'' are two lifts of H in E, then H' and H'' are conjugate by an element of $U(k)$ in each of the following cases:

a) k is algebraically closed and H is smooth.

b) k is algebraically closed and U is smooth.

(We shall indicate in the course of the proof other cases where the conclusion of ii) is true without assuming k algebraically closed.)

If U is an algebraic group (resp. a commutative algebraic group) defined over a field k, we denote by $H^1(k,U)$ (resp. $H^i(k,U)$, $i \geqslant 0$) the pointed set (resp. the i-th group) of Galois cohomology of k with values in U (cf. J.-P. Serre, *Cohomologie Galoisienne*, Lecture Notes Maths. n° 5, Springer).

If S is a prescheme, H an S-prescheme in commutative groups, G an S-prescheme in groups acting on H, we denote by $H^i(G,H)$[1137] the i-th Hochschild cohomology group of G with values in H (App. I 1).

To prove 5.1.1, we shall proceed in several steps.

[1137]N.D.E.: H_0^i has been replaced by H^i, to be in accord with the notations of App. I and of Exp. I.

5.2. Proof of 5.1.1 i) and ii) in the case U smooth and H étale

Lemma 5.2.1. *With the notations of 5.1.1, if H is étale, the canonical morphism $E \to H$ possesses a section $s : H \to E$, defined over k, in the following cases:*

a) k is algebraically closed.

b) k is perfect and U is smooth and connected.

c) U is "k-solvable".

a) follows simply from the fact that $E(k) \to H(k)$ is surjective. One has b) ⇒ c) by 4.1.2 b) and 2.3 bis. It therefore suffices to treat case c). Let x be a point of H; x is both an open and closed part of H, and the induced subscheme is isomorphic to $\operatorname{Spec} K$, where K is a finite separable extension of k. Let X be the inverse image in E of the scheme x. The K-scheme X is a principal homogeneous K-space under the group U_K. But U_K possesses a composition series with successive quotients isomorphic to $(G_a)_K$, hence $H^1(K, U_K) = 0$ (J.-P. Serre, *Cohomologie Galoisienne*, Chap. III, prop. 6), and consequently X is trivial, hence has a K-rational point. One thus obtains a k-section of $E \to H$ over x, for every point x of H, whence the existence of a section $H \to E$.

Lemma 5.2.3.[1138] *With the notations of 5.1.1, suppose H étale. Then:*

i) The extension E is trivial in each of the following cases:

a) U is commutative and $E \to H$ possesses a section.

b) k is algebraically closed.

c) k is perfect and U is connected.

d) U is "k-solvable".

Moreover, in each of the above cases, two lifts H' and H'' of H in E are conjugate by an element of $U(k)$.

ii) Let R be the k-functor $(Sch/k)^\circ \to (Ens)$ such that, for every k-prescheme S, $R(S)$ is the set of lifts of H_S in E_S. Then if U is commutative, R is representable by a non-empty affine k-scheme. The group U acts by inner automorphisms on R, and this action makes R into a homogeneous space under U (for the fpqc topology (cf. Exp. IV)).

Proof of i). We shall reduce cases b), c), and d) to case a).

Case b) Since U possesses a characteristic composition series with successive commutative quotients (3.5 i) ⇔ iv)), one immediately reduces to the case where U is commutative. Moreover $E \to H$ possesses a section by 5.2.1 a), and we are reduced to case a).

Case d) One proceeds similarly using 5.1.0 ii) and 5.2.1 c).

[1138] N.D.E.: there is no number 5.2.2.

Case c) Let E_{red} be the reduced subgroup associated with E. Since H is smooth, E_{red} is an extension of H by $U_0 = U \cap E_{red}$, and one may replace E by E_{red}. But U, and therefore also `U_0`, is connected, and it is clear that `U_0` is the identity component of E_{red}, hence is smooth. But then `U_0` is smooth and connected, k is perfect, hence `U_0` is k-solvable (4.1.4 b)) and we are reduced to case d).

The preceding reductions show that in cases b), c), and d) one may suppose that U possesses a characteristic composition series (U_i) such that U_i/U_{i+1} is commutative and such that the maps $U_i(k) \to (U_i/U_{i+1})(k)$ are surjective (in cases c) and d), this last point comes from $H^1(k, G_a) = 0$). An immediate dévissage then shows that it suffices to prove the conjugacy of two lifts H' and `H''` in E when U is commutative. In short, it suffices to prove i) a). In this case, the triviality of the extension E is ensured if $H^2(H, U) = 0$ (App. I 3.1), and the conjugacy of H' and `H''` if $H^1(H, U) = 0$. Now we have the following lemma:

Lemma 5.2.4. *Let S be a prescheme, U an S-prescheme in commutative groups, H an S-prescheme in groups, étale, finite, of rank n, acting on U. Then the groups $H^i(H, U)$ ($i > 0$) are annihilated by n in the following two cases:*

a) H is a constant S-group (Exp. I 4.1).

b) S is the spectrum of a field.

Proof of a). The group H is by hypothesis the constant group associated with an ordinary group H of order n. It is clear[1139] that $H^i(H, U)$ is isomorphic to the i-th cohomology group $H^i(H, U(S))$ of the group H, with values in the ordinary group $U(S)$, and it is classical that these groups are annihilated by n (J.-P. Serre, *Corps locaux*, Chap. VIII, prop. 4 cor. 1).

Proof of b). Let $x \in H^i(H, U)$ ($i > 0$), which one represents by an i-cocycle $f : H^{(i)} \to$
U

(where $H^{(i)}$ denotes the product, over k, of i copies of H). If K is a finite Galois extension of k that decomposes H, it follows from a) that nf_K is a coboundary. More precisely, an easy computation shows that if one defines the K-morphism $F_K : H_K^{(i-1)} \to U_K$ by the formula:

```
F_K(h_1, …, h_{i−1}) = Σ_{h ∈ H(K)} f_K(h_1, …, h_{i−1}, h),
```

one has, up to sign:

```
d(F_K) = n f_K   (d coboundary operator).
```

Now an immediate Galois argument shows that `F_K` comes from a k-morphism $F :$ $H^{(i-1)} \to U$, and consequently one has $d(F) = nf$.

Corollary 5.2.4 bis. *With the notations of 5.2.4, suppose moreover that U is flat and of finite presentation over S, with unipotent fibers, and that n is prime to the*

[1139] N.D.E.: This is Proposition III.6.4.2 of the book by M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

residue characteristics of S. Then, in cases a) and b) above, one has $H^i(H,U) = 0$ for $i > 0$.

It suffices to show that raising to the n-th power in U is an isomorphism, since this will entail that multiplication by n in $H^i(H,U)$ is both an isomorphism and the zero morphism, hence $H^i(H,U) = 0$. Now, under the hypotheses on U, it suffices to verify that raising to the n-th power is an isomorphism on the fibers of U (EGA IV 17.9.5), which reduces us to the case where S is the spectrum of a field k of characteristic p. Since $(n,p) = 1$, raising to the n-th power in U is an étale morphism (Exp. VII), and is a monomorphism (2.4 i)), hence an open immersion (EGA IV 17.9.1).

This already proves that the restriction to the identity component U^0 is an isomorphism; since U/U^0 is a finite p-group, we are done.

This finishes the proof of 5.2.3 i) a), since H, being an étale group of multiplicative type, is of order prime to p.

Proof of 5.2.3 ii). It is clear that R is a sheaf for the fpqc topology. Moreover, taking the descent of affine schemes into account, the assertions of 5.2.3 ii) are local for the fpqc topology. We may therefore suppose k algebraically closed. The group H is then decomposed and the extension E is trivial (i b)); let `H_0` be a lift of H in E. For every k-prescheme S, and every lift `H''` of `H_S` in `E_S`, $H_{0,S}$ and `H''` are conjugate by an element of $U(S)$, since $H^1(H_S, U_S) = 0$ (5.2.4 bis). Let then U^{H_0} be the sheaf of invariants of U under `H_0`, which is representable by an algebraic subgroup of U (Exp. VIII 6.5 d)). It follows from the preceding remarks that the k-morphism:

```
U  R,   u ↦ int(u) H_0   (u ∈ U(S))
```

defines a k-isomorphism $U/U^{H_0} \xrightarrow{\sim} R$. This proves the representability of R and the fact that R is affine (2.1).

Remark 5.2.5. *One can show that the assertions of 5.2.3 ii) remain true when U is not commutative, but we shall not need this to prove 5.1.1.*

5.3. Study of the case H smooth

Proposition 5.3.1. *The assertions contained in 5.2.3 i) remain true when one replaces the hypothesis "H étale" by "H smooth".*

Proceeding as in the proof of 5.2.3 i), one reduces to the case where moreover U is commutative.

Let `N_0` be the set of integers `> 0` prime to p, ordered by divisibility. For every $n \in N_0$, ${}_nH$ is an étale group and the family of ${}_nH$ ($n \in N_0$) is schematically dense in H, since H is smooth (Exp. IX, 4.10). Denote by E_n the inverse image of ${}_nH$ in E, so that E_n is an extension of ${}_nH$ by U, and finally let R_n be the k-functor of lifts of ${}_nH$ in E_n (cf. 5.2.3 ii)). If n divides m, it is clear that one has a natural k-morphism $R_m \to R_n$, so the R_n form an inverse system of k-functors. Since R_n is representable by a non-empty affine k-scheme (5.2.3 ii)), and since a filtered direct limit of non-zero rings is non-zero, the functor $R = \lim R_n$ is representable by a non-empty affine k-scheme (EGA IV 8 and 1.9.1). There therefore exists an extension K of k and a point $u \in R(K)$. The image u_n of u in $R_n(K)$ corresponds to a lift H'_n of $({}_nH)_K$ in $(E_n)_K$. By construction, $H'_n = {}_n(H'_m)$ if n divides m. Set $U_n = (U_K)^{H'_n}$. The choice of H'_n allows one to identify $(R_n)_K$ with U_K/U_n. But the family of H'_n is filtered increasing, hence the family of U_n is filtered decreasing and consequently is stationary for n large enough (U_K is noetherian). It follows that the family of $(R_n)_K$ is stationary, and consequently so is the family of R_n.

In brief, one has $R_n = R$ for n large enough.

Under the hypotheses made, it follows from 5.2.3 i) that $R_n(k)$ is non-empty. One can therefore find a coherent system of lifts H'_n of ${}_nH$ for $n \in N_0$. Denote by H' the smallest closed subscheme of E majorizing H'_n for every n (Exp. VI_B § 7). The argument made in Exp. XV 4.6 shows that H' is a smooth commutative algebraic group whose formation commutes with base field extension. To show that H' is a lift of H in E, we may therefore suppose k algebraically closed. By *Bible* 4 th. 4, H' is then the direct product of a (smooth) group of multiplicative type M and a unipotent group V. The groups H'_n are then necessarily contained in M (2.4) and in view of the definition of H' this entails $H' = M$. So H' is of multiplicative type and consequently $H' \cap U = 0$. The morphism $H' \to H$ is therefore a monomorphism; moreover, it follows from the density theorem (Exp. IX, 4.10) that it is an epimorphism, hence an isomorphism.

Let now H' and H'' be two lifts of H in E. For every $n \in N_0$, denote by $T_n = Transp({}_nH', {}_nH'')$ the transporter of ${}_nH'$ into ${}_nH''$, which is representable by a closed subscheme of U (Exp. VIII 6.5 e)). The T_n form a decreasing filtered family of closed subschemes of U, non-empty by 5.2.3 i a). Let T be the stationary value. Under the hypotheses of 5.2.3 i), $T_n(k)$ is non-empty. There therefore exists an element u of $U(k)$ such that ${}_nH'' = int(u)_nH'$ for every $n \in N_0$. But then $H'' = int(u)H'$ (Exp. IX, 4.8 b).

5.4. Study of the case U radicial

Proposition 5.4.1. *If k is a perfect field of characteristic $p > 0$, and if U is radicial, the extension E of 5.1.1 is trivial.*

Using a characteristic composition series of U, we may limit ourselves to the case where U is equal to $(\alpha_p)^r$ (3.9).

It follows from App. II 2.2 and 2.1 that one has isomorphisms of k-functors:

Aut_{k-gr}(α_p)^r Aut_{p-Lie}(Lie(α_p)^r) GL(Lie(α_p)^r).

Consider then a k-vector space V of rank r, the k-scheme of vector groups $W(V)$ (Exp. I 4.6), and identify $(\alpha_p)^r$ with F V. The preceding remarks then show that the action of H on F V, defined by the extension E, comes from a linear representation v of H in V. Consider then the exact sequence:

(*) 0 → F V → V —F→ V^{(p)} → 0,

where F is the Frobenius morphism. Then `(*)` is an exact sequence of H-groups, provided one makes H act on the factor $V^{(p)}$ via the linear representation $H \xrightarrow{v} GL(V) \xrightarrow{F} GL(V^{(p)})$.

Since the field k is perfect, the morphism $F : V \to V^{(p)}$ induces a surjective map $V(k) \to V^{(p)}(k)$. It then follows from App. I 2.1 that the exact sequence `(*)` defines an exact sequence:

```
(**)    H¹(H, V^{(p)}) □ Ext_{alg}(H, F V) □ Ext_{alg}(H, V).
```

Let us show that $Ext_{alg}(H, V) = 0$. Let then `E_0` be an algebraic group extension of H by V:

```
1 → V → E_0 —h→ H → 1.
```

The scheme `E_0` is a torsor with base H and group G_a^r, hence defines an element of $H^1(H, O_H^r)$ (in the sense of the cohomology of coherent sheaves). Since H is affine, one has $H^1(H, O_H^r) = 0$ (EGA III § 1). That is to say, $E_0 \to H$ possesses a section. Consequently, the group $Ext_{alg}(H, V)$ is isomorphic to $H^2(H, V)$ (App. I 3.1). Now $H^i(H, V) = H^i(H, W(V)) = 0$ for $i > 0$ (Exp. IX, 3.1). One then concludes, by the exact sequence `(**)`, that $Ext_{alg}(H, FV) = 0$, hence that E is a trivial extension.

5.5. Proof of 5.1.1 i)

If k is of characteristic 0, U is k-solvable (4.1.3) and H is smooth; the fact that E is a trivial extension therefore follows from 5.3.1 and 5.2.3 d). One proves similarly that two lifts of H in E are conjugate by an element of $U(k)$.

Henceforth, we therefore suppose that k is a field of characteristic $p > 0$.

Proof of i) b): *Case k perfect, U connected.*

We shall reduce to the case where U is radicial. For this, note that since k is perfect, H_{red} is smooth; let `E_0` be its inverse image in E. It then follows from 5.3.1 and

5.2.3 i) c) that the extension:

$$1 \longrightarrow U \longrightarrow E_0 \longrightarrow H_{red} \longrightarrow 1$$

is trivial. Let `H_1` be a lift of H_{red} in E. By App. II 3.1, there exists an integer $n > 0$ such that $E^{(n)} = E/F_n(E)$ is smooth; let `E''` be the algebraic subgroup of E generated by `H_1` and $F_n(E)$ (i.e. the inverse image in E of the image of `H_1` in $E^{(n)}$). Denote by `H''` the image of `E''` in H. Then I claim that $H'' = H$. Indeed, denote by R the image of $F_n(E)$ in H, so that `H''` is generated by R and H_{red}. The group H/R is a quotient of $E^{(n)}$ hence is smooth; consequently the canonical morphism

$$H_{red} \longrightarrow H/R$$

is an epimorphism, so H is generated by R and H_{red}, hence equals H''. One thus obtains an exact sequence:

(†) 1 □ U'' = U ∩ E'' □ E'' □ H □ 1.

But E'' has the same underlying space as H_1, so U'' is radicial. Moreover, it is clear that it suffices to prove that the extension (†) is trivial, which follows from 5.4.1.

Proof of i) b): *Case k perfect, H connected.*

The group U is an extension of an étale group by its identity component U^0. The case U connected having just been treated, it suffices to examine the case U étale. One then has the more precise lemma:

Lemma 5.5.1. *With the notations of 5.1.1, suppose moreover U étale. Then:*

i) If H is connected, there exists a unique lift of H in E, namely the identity component E^0 of E.

ii) If k is algebraically closed, E is trivial, and two lifts of H in E are conjugate by an element of $U(k)$.

i) The formation of the identity component commuting with base field extension, we may limit ourselves to the case k algebraically closed. If H is a torus, E is trivial (5.3.1 and 5.2.3 i b)), and it is clear that E^0 is the unique lift of H. This already proves that, in the general case, $E^0 \cap U$ is radicial; since moreover it is étale (U being étale), it is zero. The morphism $E^0 \to H$ is therefore a monomorphism, flat (since E^0 is open in E) and surjective (H is connected), hence an isomorphism. If now H' is another lift of H, H' is connected, hence contained in E^0, and consequently equal to E^0.

ii) Let H_0 be the identity component of H. By i), E^0 is the unique lift of H_0 in E. Set $E_0 = E/E^0$, $H_0' = H/H_0$, so that one has the extension:

$$1 \longrightarrow U \longrightarrow E_0 \longrightarrow H_0' \longrightarrow 1.$$

H_0' being étale, this extension is trivial (5.2.3 i b)). If H_1' is a lift of H_0' in E_0, H_1 its inverse image in E, it is clear that H_1 lifts H in E. If H_2 is a second lift of H in E, it contains E^0; by 5.2.3 i b), the image of H_2 in E_0 is conjugate to H_1' by an element u of $U(k)$, whence immediately $H_2 = int(u)H_1$.

Remark 5.5.2. *Under the hypotheses of 5.5.1 i), it is easy to see that E^0 centralizes U.*

Proof of 5.1.1 i) a). Using the composition series

$$1 \longrightarrow U^0 \longrightarrow U \longrightarrow U/U^0 \longrightarrow 1,$$

i) a) follows from the conjunction of i) b) and 5.5.1 ii).

Before proving 5.1.1 c) and d), we shall first establish 5.1.1 ii).

Proof of 5.5.1 ii) a). For lack of a satisfactory general statement, we shall describe a certain number of cases where, when H is smooth, two lifts of H in E are conjugate:

Proposition 5.6.1. *With the notations of 5.1.1, suppose moreover H smooth, and let H' and H'' be two lifts of H in E. Then H' and H'' are conjugate by an element of $U(k)$ in each of the following cases:*

a) k is algebraically closed.

b) U is commutative.

c) k is perfect and U is connected.

d) U is k-solvable.

e) The centralizer of H' in U is k-solvable.

f) The group of multiplicative type H is trivialized by a finite Galois extension K of k of degree prime to p.

Proof. a), b), c), d) follow from 5.3.1.

e) Let Z be the centralizer of H' in U, T the transporter of H' into H''. By a), T is non-empty, so T is a principal homogeneous space under Z, and the hypothesis on Z entails that it is trivial.

f) Proceeding as in 5.3.1, one sees that it suffices to consider the case H étale. Suppose first H diagonalizable, defined by the ordinary group M, of order m prime to p. The data of the two lifts H' and H'' defines a 1-cocycle of M with values in $U(\bar{k})$, that is, a map h from M to $U(\bar{k})$ such that $h(mn) = h(m)\cdot{}^m h(n)$ for every pair m, n of elements of M. The groups H' and H'' are conjugate by an element of $U(k)$ if and only if there exists $a \in U(\bar{k})$ such that

$$h(m) = a^{-1} ({}^m a).$$

Now the abstract group $U(\bar{k})$ possesses a composition series with successive quotients commutative and annihilated by a power of p (one may suppose $p > 0$ in view of 5.6.1 c)). One deduces immediately in this case that h is a coboundary.

Let us now examine the general case. Still denote by Z the centralizer of H' in U, and by T the transporter of H' into H'', which is a torsor under Z. By hypothesis, there exists a finite Galois extension K of k, with Galois group G, of order prime to p, that trivializes H. By the preceding study, H'_K and H''_K are conjugate by an element of $U(K)$,

hence T_K is trivial. Consequently T is defined by an element of $H^1(G, Z)$. For the same reasons as above, the hypothesis on G entails that $H^1(G, Z) = e$, hence T is trivial.

5.7. Proof of 5.1.1 ii) b)

Lemma 5.7.1. *Let S be a prescheme, G an S-prescheme in groups, separated and smooth over S, H an S-group of multiplicative type acting on G. Then the S-functor $Z = G^H$ of invariants of G under H is representable by a sub-S-prescheme in groups of G, closed and smooth over S.*

The fact that G^H is representable by a closed subprescheme in groups of G follows from Exp. VIII 6.5 d). Consider then the semidirect product $K = G \times_S H$. The centralizer of H in K is then equal to $Z \times_S H$. To prove that Z is smooth, we must verify that if S is affine, S' is a closed subscheme of S defined by a square-zero ideal, and u_0 is an element of $Z(S')$, then there exists an element u of $Z(S)$ lifting u_0. Now since G is smooth over S, there exists an element u of $G(S)$ lifting u_0. Let i be the canonical immersion of H into K and consider the two S-morphisms of groups

`H ⇒ K,   i   and   int(u) i = j.`

Since u_0 belongs to $Z(S')$, one has $i_{S'} = j_{S'}$. By Exp. IX, 3.2, there exists $v \in K(S)$ lifting the unit section of $K(S')$, and such that

`i = int(v) j = int(vu) i.`

So $vu = (u', v')$ belongs to $(Z \times_S H)(S)$. So u' belongs to $Z(S)$ and lifts u_0.

Lemma 5.7.2. *Let k be a field, H a k-algebraic group, and let $P(H)$ be the following property:*

> *"for every smooth unipotent k-group U, and for every extension E of H by U, two lifts of H in E are conjugate by an element of $U(k)$".*

Then if H is an algebraic group extension of a group of multiplicative type `H''` *by a group of multiplicative type H', and if $P(H')$ and $P(H'')$ are true, $P(H)$ is true.*

Indeed, let U be a smooth unipotent k-group, E an extension of H by U, `H_1` and `H_2` two lifts of H in E, H_1' and H_2' the corresponding lifts of H'. Since $P(H')$ is true, there exists $u \in U(k)$ such that $H_2' = int(u)H_1'$. Possibly replacing `H_1` by $int(u)H_1$, we may suppose $H_1' = H_2'$, which we allow ourselves to denote simply `H_0`. Let $E_0 = Centr_E H_0$, which is equal to

`U^{H_0} · H_1 = U^{H_0} · H_2.`

By 5.6.1, $U^{H_0} = U_0$ is smooth. Consider then the extension

$$1 \longrightarrow U_0 \longrightarrow E_0 \longrightarrow H \longrightarrow 1.$$

By construction `H_0` is central in `E_0`, hence invariant. By passage to the quotient, one obtains the exact sequence:

$$1 \longrightarrow U_0 \longrightarrow E_0' \longrightarrow H'' \longrightarrow 1.$$

Since U_0 is smooth, and since $P(H'')$ is true, the two images of H_1 and H_2

in E'_0 are conjugate by an element u of $U_0(k)$, but then $H_1 = int(u)H_2$.

To prove 5.1.1 ii) b), note then that, since k is algebraically closed, H possesses a composition series whose successive quotients are smooth or isomorphic to μ_p when $p > 0$. By repeated use of 5.7.2, we are reduced to the case where H is smooth or equal to μ_p. In the first case, it suffices to apply 5.1.1 ii) a). There remains the case $H = \mu_p$. Since U is smooth, U possesses a characteristic composition series with successive quotients étale or isomorphic to $(G_a)^r$ (3.9). If U is étale, one applies 5.5.1. There finally remains the case $H = \mu_p$, $U = G_a^r$.

We must show that $H^1(\mu_p, U) = 0$. The method used in 5.4.1 no longer applies here, since μ_p does not in general act linearly on $(G_a)^r$. Let us fix the notations: H' denotes a lift of H in E, $e = LieE$, $u = LieU$, $h = LieH$, $h' = LieH'$. Let X be a non-zero element of h such that $X^{(p)} = X$ (App. II 3.1), and let X' be its lift in h'.

Since μ_p is a radicial group of height 1, there is a bijective correspondence between the set of lifts of H in E and the set A of $Y \in e$ such that $Y^{(p)} = Y$ and which project to X (App. II 2.2). Likewise if $Y \in A$ corresponds to the lift H'' of H in E, and if $u \in U(k)$, then $int(u)H' = H''$ if and only if $Ad(u)X = Y$. Let then B be the subset of $Y \in e$ of the form Ad(u) X, where $u \in U(k)$. One has evidently $B \subset A$, and everything reduces to showing that $A = B$ if k is

algebraically closed.

a) *Study of A.* Since u is commutative and normalized by h, Jacobson's formula (Exp. VII_A 5.2) gives simply, for $u \in u$:

(X' + u)^{(p)} = X'^{(p)} + u^{(p)} + (ad X')^{p−1}(u) = X' + (ad X')^{p−1}(u),

so that $X' + u \in A \Leftrightarrow u = (adX')^{p-1}(u)$.

Now let $u = \bigoplus_{n \in \mathbb{Z}/p\mathbb{Z}} u_n$ be the canonical decomposition of u under the action of H', which one may also write:

u = u_0 ⊕ ⊕_{n ∈ (ℤ/pℤ)^×} u_n.

If $u \in u_0$, one has $adX'(u) = 0$. If $u \in u_n$ ($n \in (\mathbb{Z}/p\mathbb{Z})^\times$), one has $(adX')^{p-1}(u) = u$. Finally, $Y = X' + u$ is an element of A if and only if $u \in \bigoplus_{n \in (\mathbb{Z}/p\mathbb{Z})^\times} u_n$. Note that A is the set of k-rational points of an irreducible subscheme of $W(e)$, of dimension equal to $rgu - rgu_0 = rgu - rgCentr_u(X')$.

b) *Study of B.* We shall need the following lemma:

Lemma 5.7.3. *(Rosenlicht). Let U be a unipotent algebraic group over a field k acting on a quasi-affine k-scheme X. Then the orbit of every point $x \in X(k)$ is closed in X (by* orbit *of x we mean the subset of $ens(X)$ image of G under the morphism $g \mapsto g \cdot x$).*

By functoriality, G acts on the affine envelope of X (that is, $\operatorname{Spec}\Gamma(X, O_X)$), which allows us to suppose X affine. One may further suppose k algebraically closed, X reduced, and U smooth (note that U_{red} acts on X_{red} if k is perfect). Let Y be the schematic image of G (EGA I 9.5.1) under the morphism $g \mapsto g \cdot x$, which is a closed and reduced subscheme of X on which G acts. It follows easily from EGA IV 1.8.6 that the orbit of x is an open part Z of Y, dense in Y. We must show that $Z = ens(Y)$. Let F be the closed reduced subscheme of Y having `Y \ Z` as underlying space. One has therefore $F = V(J)$, where J is a non-zero ideal of $\Gamma(Y, O_Y)$. Since G is smooth, G acts on F, hence on J, and consequently (3.2) $J^G \neq 0$. If a is a non-zero element of J^G, a is necessarily constant on the orbit Z, hence is constant on Y, Z being dense in Y. But then the ideal J contains k, and $F = \emptyset$.

This being so, let us apply the preceding lemma to the group U acting on the affine space $W(e)$ via the adjoint representation. One obtains that the orbit of X' is the underlying set of a closed sub-prescheme of $W(e)$. Moreover, the stabilizer Z of X' is the centralizer of X' in U, and one has a closed immersion:

$$U/Z \longrightarrow W(e).$$

Let us note that the orbit of X' is the underlying space of a closed subscheme of $W(e)$ of dimension equal to $\dim U - \dim Z$.

c) *End of the proof of 5.1.1 ii) b).* When k is algebraically closed, the canonical map $U(k) \to (U/Z)(k)$ is surjective,

so that by point b) above, B is the set of k-rational points of a closed subscheme of $W(e)$ of dimension $\dim U - \dim Z$. Taking point a) into account, to prove that $A = B$, it then suffices to show that one has:

`rg u - rg Centr_u(X')  dim U - dim Centr_U(X').`

Now since U is smooth, one has $\dim U = rgu$. On the other hand, one has (Exp. II 5.3.3):

`dim Centr_U(X')  rg Centr_u(X'),`

whence the result (note that one in fact has `dim Centr_U(X') = rg Centr_u(X')`, which gives another proof that Z is smooth (5.7.1)).

5.8. End of the proof of 5.1.1 i)

It remains to prove i) c) and i) d).

Proof of 5.1.1 i) d) (U smooth, H connected). We shall need the following lemma:

Lemma 5.8.1. *With the notations of 5.1.1, suppose U smooth and H radicial, and let* `H_1` *be an algebraic subgroup of H that possesses a lift H_1' in E. Then H possesses a lift H' in E majorizing H_1'.*

By induction on the height of H/H_1, one may suppose H/H_1 of height 1. Let $C' = Centr_E(H_1')$. I claim that the canonical morphism $C' \to H$ is an epimorphism. To establish this point, we may suppose k algebraically closed; but then, the extension E is trivial (5.1.1 i a)); let `H''` be a lift of H in E, H_1'' the inverse image of `H_1` in `H''`.

The groups H_1' and H_1'' are two lifts of `H_1` in E, hence are conjugate by an element of $U(k)$, since k is algebraically closed and U smooth (5.1.1 ii b)). It is clear then that to prove the assertion on C', it suffices to prove it for $C'' = Centr_E(H_1'')$. But in this case, `C''` majorizes `H''`, and the property is clear. Moreover, it follows from 5.7.1 that $C \cap U$ is smooth. It is clear then that we may replace E by C, hence suppose H_1' central. But then, possibly passing to the quotient by H_1', we may suppose $H_1 = 0$ and H of height 1.

Since U is smooth, one has the exact sequence of p-Lie algebras (App. II 3.2):

```
(*)    0 → Lie U → Lie E → Lie H → 0.
```

Taking App. II 2.2 into account, saying that E is trivial is equivalent to saying that `(*)` is a trivial extension of p-Lie algebras. Suppose $H \neq 0$ (hence $LieH \neq 0$) and suppose we have found a non-zero sub-p-Lie algebra h_1 of $h = LieH$ that lifts to a sub-p-Lie algebra h_1' of `Lie E`. By *loc. cit.*, there exists a subgroup `H_1` of H such that $LieH_1 = h_1$, and a lift H_1' of `H_1` in E such that $LieH_1' = h_1'$. Applying again the reduction described above, one is reduced to the same problem, where one has replaced H by H/H_1. Since H is of height 1, `Lie(H/H_1) = Lie H / Lie H_1` (*loc. cit.*), so $rgLie(H/H_1) \leqslant rgLieH - 1$. In brief, proceeding by induction on the rank of `Lie H`, one sees that it suffices, when $h \neq 0$,

to find a non-zero Lie subalgebra of `Lie H` that lifts in `Lie E`. Now we have the following lemma:

Lemma 5.8.2. *Let k be a field of characteristic $p > 0$, and ϕ a surjective morphism of finite-rank p-k-Lie algebras $g \to h$. Then:*

i) If $h_{\bar{k}}$ is reductive (4.2.2) and $\neq 0$, there exists a reductive Lie subalgebra h_1 of g whose image in h is non-zero.

ii) If k is perfect and if u is a unipotent element of h (i.e. there exists n such that $u^{(p^n)} = 0$), then u lifts to a unipotent element of g.

Take X a non-zero element of h in case i) and u in case ii), and let X' be a lift of X in g. The sub-p-Lie algebra of g generated by X' is an abelian p-Lie algebra (Exp. VII) h_0.

Case i). It is clear from the description given in 4.2.2 that the reductive part (*loc. cit.*) of $h_{0,\bar{k}}$ is already defined over k; denote it r_0. I claim that the image of r_0 in h is non-zero. To establish this point we may suppose k algebraically closed, so that $h_0 = r_0 \bigoplus u_0$ (u_0 the unipotent part of h_0). If the image of r_0 in h were zero, the image of h_0 in h would be unipotent, hence would be zero since h is reductive (cf. 2.4 ii)); now by construction it contains X.

Case ii). One proceeds similarly, exchanging the roles of r_0 and u_0.

End of the proof of 5.8.1. Supposing $H \neq 0$, by 5.8.2 there exists a non-zero reductive Lie subalgebra h_1' of `Lie E`. Since `Lie U` is unipotent (4.3 i)), one necessarily has $(LieU) \cap h_1' = 0$, so h_1' is a lift of a sub-p-Lie algebra of `Lie H`.

End of the proof of 5.1.1 i d). By 5.8.1, there exists a family of algebraic subgroups H'_n ($n \in \mathbb{N}$) of E, such that H'_{n+1} majorizes H'_n and H'_n lifts $F_n H$. The decreasing sequence of subgroups $Centr_E(H'_n)$ is stationary; let C be the stationary value. The center Z of C majorizes H'_n for every n, hence the image of Z in H majorizes $F_n(H)$ for every n, and is consequently an open subgroup of H (Exp. VII_A § 4), hence is equal to H since H is connected. To prove that E is trivial, we may therefore replace E by Z, hence suppose E commutative. We shall then see in 7.2.1 that E contains a maximal subgroup of multiplicative type M, whose formation commutes with base field extension. Since $E_{\bar{k}}$ is a trivial extension (5.1.1 i) a)) and U is unipotent, it is clear that M is the unique lift of H in E.

Proof of 5.1.1 i) c) (U k-solvable). Since $(H/H)^0(k)$ is of order prime to p, it is immediate by duality that there exists an integer n such that ${}_nH$ is an étale subgroup and the canonical morphism ${}_nH \to H/H^0$ is an epimorphism, so that $H/{}_nH$ is connected. By 5.2.3 d), there exists a lift H' of ${}_nH$ in E. One shows, as at the beginning of the proof of 5.8.1, that $C = Centr_E(H')$

is a subgroup of E such that $C \to H$ is an epimorphism and such that $C \cap U$ is smooth. Replacing E by C and passing to the quotient by H', one is reduced to the case where U is smooth and H connected, that is, to case i) d).

This finishes the proof of 5.1.1.

5.9. Counterexamples

Let us first indicate a procedure for obtaining non-trivial extensions of a k-group of multiplicative type H by a unipotent k-group U. Suppose given an action of H on U, and let E be the semidirect product $E = U \cdot H$. Suppose moreover that one is given an element of $H^1(k, U)$ represented by a 1-cocycle a. The group U acts by inner automorphisms on E. The datum of a therefore defines a k-form of E denoted E_a. Suppose moreover that U is commutative; then U acts trivially on U and on the quotient $E/U = H$, so that E_a is still an extension of H by U. Suppose, for simplicity, that H is an étale diagonalizable group, so that the Galois group G of $\bar{k}/k$ acts trivially on $H(\bar{k})$; the action of G on the points of $E_a(\bar{k})$ is then given by the formula:

$$ {}^g(u, h) = ({}^g u + a(g) - {}^h a(g), h) \quad (g \in G,\ u \in U(\bar{k}),\ h \in H(\bar{k})). $$

If h is a point of $H(k)$, X its inverse image in E_a, X is therefore a torsor under U defined by the class of the 1-cocycle of G with values in U, $g \mapsto a(g) - {}^h a(g)$. It follows that if there exists a point h of $H(k)$ such that the preceding 1-cocycle is non-trivial,

the extension E_a is non-trivial. We shall apply this construction in two particular cases:

a) *Non-trivial extension of an étale diagonalizable group H by $\mathbb{Z}/p\mathbb{Z}$.*

Take for H the group $(\mathbb{Z}/p\mathbb{Z})^\times$ acting by multiplication on $U = \mathbb{Z}/p\mathbb{Z}$. Let k be a field of characteristic p, K an extension of k with Galois group G isomorphic to

$\mathbb{Z}/p\mathbb{Z}$, a a non-zero element of $H^1(G,U) = \mathrm{Hom}(G,U)$. The group E_a then answers the question.

b) *Example of a non-trivial extension of an étale diagonalizable group H by a smooth connected unipotent group U.*

Take for k a non-perfect field such that there exists a k-form U of G_a with $H^1(k,U) \neq 0$. For example (cf. J.-P. Serre, *Cohomologie Galoisienne*), one may take for k the field of fractions of a discrete valuation ring of equal characteristic $p > 0$, and for U the algebraic subgroup of $G_a \times_k G_a$ with equation $X^p + X + tY^p = 0$, where t denotes a uniformizer of k. Indeed, supposing for simplicity that k contains the $(p-1)$-th roots of unity, one has an exact sequence of algebraic groups:

$$0 \to \mathbb{Z}/p\mathbb{Z} \to U \to G_a \to 0, \quad (x, y) \mapsto y,$$

hence an exact sequence of cohomology:

$$G_a(k) \xrightarrow{d} H^1(k, \mathbb{Z}/p\mathbb{Z}) \to H^1(k, U) \to 0,$$

where d sends x to the principal homogeneous space under $\mathbb{Z}/p\mathbb{Z}$ with equation

$$X^p + X + t x^p = 0.$$

Moreover, one knows that $H^1(k,\mathbb{Z}/p\mathbb{Z})$ is isomorphic to $k/P(k)$ (where $P(x) = x^p + x$), hence $H^1(k,U)$ is isomorphic to $k/(P(k) + tk^p)$. Suppose moreover $p \neq 2$; it is then clear that t^2 is an element of k not belonging to $P(k)+tk^p$, hence $H^1(k,U) \neq 0$.

On the other hand μ_{p-1} acts on U by the formula:

$$(h, x, y) \mapsto (hx, hy).$$

Denote by G the Galois group of the extension K defined by the equation

$$X^p + X + t^2 = 0,$$

and let $a \in H^1(G,U)$ be the non-zero element described above. One verifies immediately that E_a is then a non-trivial extension of μ_{p-1} by U.

c) *Non-trivial extension of G_m by α_p.*

By 5.1.1 i) b), such an extension can only exist over a non-perfect field k. Let then k be a non-perfect field, G the semidirect product of $U = G_a$ by $H = G_m$, with G_m acting on G_a by homotheties. Since U is invariant in G, so is FU. Let $G'' = G/FU$. The group G'' is isomorphic to $G_a \cdot G_m$, where G_m acts on G_a by the formula:

$$(h, u) \mapsto h^p u.$$

The functor T_G (resp. $T_{G''}$) of subtori of G (resp. G'') (cf. Exp. XV) is isomorphic to G_a, and the morphism $T_G \to T_{G''}$ deduced from the morphism $G \to G''$ is identified with the morphism $u \mapsto u^p$. It follows that if T'' is a subtorus of G'' corresponding to a point x of $k \simeq G_a(k)$ such that $x^{1/p}$ is not in k, the inverse image E of T'' in G will be an extension of a torus T'' by $FU = \alpha_p$, will not possess maximal tori defined over k, hence will not be trivial. One finds for E the subgroup of $G = \operatorname{Spec} k[U, T, T^{-1}]$ with equation $U^p = x - xT^p$.

Remark 5.9.1. *This last example shows that a non-smooth algebraic group defined over a non-perfect field does not necessarily possess maximal tori, and thus answers the question raised in Exp. XIV, 1.5 b).*

d) Let us now give an example of a trivial extension E of a group of multiplicative type H by a unipotent group U, and of two lifts H' and H'' of H that are not conjugate by an element of $U(k)$.

Take for E the semidirect product of $U = G_a$ by $\mu_p = \operatorname{Spec} k[T]/(T^p - 1)$, the action of μ_p on $G_a = \operatorname{Spec} k[U]$ being defined by the comorphism:

$U \mapsto (U + U^p)\, T - U^p.$

The centralizer of μ_p in U is then the étale group Z of equation $U + U^p = 0$. It follows that if k is not algebraically closed, the canonical map $U(k) \to (U/Z)(k)$ is not in general surjective,

hence, taking 5.1.1 ii) b) into account, two lifts of μ_p in E are not necessarily conjugate by an element of $U(k)$.

Here is another example, with k algebraically closed of characteristic $p > 0$. Let G be the radicial group semidirect product of μ_p by α_p, where μ_p acts on α_p by "homotheties". One then has an exact sequence of algebraic groups, with μ_p as operator group:

$0 \to \alpha_p \to G_a \to G_a \to 0, \quad x \mapsto x^p,$

where μ_p acts by homotheties on the first term G_a, and trivially on the second. The exact cohomology sequence (App. I, prop. 11) furnishes here the exact sequence:

$0 \to G_a(k) \to H^1(\mu_p, \alpha_p) \to H^1(\mu_p, G_a).$

Since the last term is zero (I 5.3.3), one sees that $H^1(\mu_p, \alpha_p)$ is non-zero, hence two lifts of μ_p in G are not necessarily conjugate.

6. Extension of a unipotent group by a group of multiplicative type

6.1. Statement of the theorem

Theorem 6.1.1. *Let k be a field, U a unipotent k-algebraic group, H a k-group of multiplicative type, E a k-algebraic group extension of U by H, so that one has the exact sequence*

$$1 \longrightarrow H \longrightarrow E \longrightarrow U \longrightarrow 1.$$

Then the extension E is trivial and there exists a unique lift of U in E in each of the following cases:

A) The group U is smooth and one of the following conditions is satisfied:

i) U is connected and the canonical morphism $E \to U$ possesses a section.

ii) U possesses a composition series with successive quotients isomorphic to G_a.

iii) H is étale.

iv) k is perfect.

B) $U = \alpha_p$ and k is perfect.

C) E is commutative and k is perfect.

6.2. Proof of 6.1.1 A)

Let us first establish three lemmas.

Lemma 6.2.1. *Let S be a prescheme, E an S-prescheme in groups, extension of an S-prescheme in groups U with connected fibers, by an S-group of multiplicative type and of finite type H (i.e. U is the quotient of E by H for the fpqc topology). Then E is a central extension.*

Indeed, since H is commutative, the group U acts by inner automorphisms on H, via an S-morphism of groups

$$u : U \longrightarrow \mathrm{Aut}_{S-gr}(H).$$

The functor $\mathrm{Aut}_{S-gr}(H)$ is representable by an étale S-scheme (Exp. X 5.10), and consequently the unit section is both an open and closed immersion. Since U has connected fibers, one deduces that u is the unit morphism.

Lemma 6.2.2. *With the notations of 6.1.1, if E is trivial, there exists a unique lift of U in E.*

Let U' and U'' be two lifts of U in E. To show that $U' = U''$, we may suppose k algebraically closed, and it suffices to show that $H^1(U, H) = 0$. If U is connected, U centralizes H (6.2.1), hence

$H^1(U, H) = \mathrm{Hom}_{k\text{-}gr}(U, H) = 0$

by 2.4 ii). In the general case, denote by U'_1 the unique lift in E of the connected component U_0 of U, and let $N = Norm_E(U'_1)$. If $g \in E(k)$, $int(g)U'_1$ is a lift of U_0, hence is equal to U'_1, and consequently $N(k) \supset E(k)$. Moreover, N majorizes H (6.2.1) and U'_1, whence immediately the fact that $N = E$. Passing to the quotient by U'_1,

one is reduced to the case where U is étale. In this case, k being algebraically closed, $H^1(U, H)$ is identified with the ordinary cohomology group

$H^1(U(k), H(k))$[1140], and consequently is zero, since $U(k)$ is a finite p-group and $H(k)$ is uniquely p-divisible.

Corollary 6.2.3. *To prove that E is trivial, it suffices to show that E becomes trivial after a finite separable extension of the base field; in particular, one may suppose H diagonalizable.*

Lemma 6.2.4. *For every central extension E of U by a diagonalizable k-group H to be trivial, it suffices that every central extension of U by G_m be trivial.*

Indeed, by induction on r, one notes first that the hypothesis entails that E is trivial if $H = G_m^r$. In the general case, H embeds in G_m^r for a suitable integer r (this is immediate by duality); let $H'' = G_m^r/H$. One obtains the exact sequence (App. I 2.1):

$$Z^1(U, H'') \to Ext_{alg}(U, H) \to Ext_{alg}(U, G_m^r) = 0$$

(where U acts trivially on H, H'', G_m^r). But $Z^1(U, H'') = \mathrm{Hom}_{k-gr}(U, H'') = 0$ (2.4 ii)), hence $Ext_{alg}(U, H) = 0$.

Proof of 6.1.1 A) i). Since $E \to U$ possesses a section, E defines an element f of $H^2(U, H)$ (App. I 3.1). We must show that f is zero, and for this, it suffices to show that a 2-cocycle $f : U \times_k U \to H$ is a constant morphism, which will follow from the following lemma:

Lemma 6.2.5. *Let U be a smooth connected unipotent k-algebraic group, H a k-group of multiplicative type; then every k-morphism (of preschemes)*

$$f : U \longrightarrow H$$

is constant.

To prove this lemma, we may suppose k algebraically closed. We proceed by increasing induction on $\dim U$. If $\dim U > 0$, U possesses a composition series (cf. 3.9):

$$1 \to U' \to U \xrightarrow{\pi} U'' \to 1$$

with $U' \simeq G_a$ and $\dim U'' < \dim U$. It suffices to show that f factors through U''. Since the graph of the equivalence relation defined by π is smooth over k, hence reduced, it suffices to show that if $x, y \in U(k)$ have the same image z in $U''(k)$, then $f(x) = f(y)$. Now $\pi^{-1}(z)$ is isomorphic to G_a, hence the restriction of f to $\pi^{-1}(z)$ factors through a reduced irreducible component of H, hence through a k-scheme isomorphic to $(G_m)^r$. It then suffices to note that every morphism from G_a into $(G_m)^r$ is constant, since every invertible regular function on G_a is constant.

Proof of 6.1.1 A) ii). Thanks to 6.2.1, 6.2.3, and 6.2.4, we may suppose that $H = G_m$. By hypothesis, U possesses a composition series $U \supset U_1 \supset \cdots$ such that U_i/U_{i+1} is

[1140]N.D.E.: see, for example, Proposition III.6.4.2 of the book by M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson & North-Holland (1970).

isomorphic to G_a. Let E_1 be the inverse image of U_1 in E. By induction on $\dim U$, one may suppose the extension E_1 is trivial; let U_1' be the unique

lift of U_1. Proceeding as in the proof of 6.2.4, one shows that U_1' is invariant in E. After passing to the quotient by U_1', one is reduced to the case where $U = G_a$. The S-scheme E (where $S = U$) is then a torsor under the S-group $G_m \times_k S$, hence possesses a section, since $\mathrm{Pic}(S) = 0$ (Exp. VIII 4.3). The extension E is then trivial by A) i).

Proof of 6.1.1 A) iii) (U smooth, H étale). Suppose first U connected. The group $U_{\bar{k}}$ then possesses a composition series with successive quotients isomorphic to G_a, hence $E_{\bar{k}}$ is trivial by A) ii). Since H is étale, it is clear that the unique lift of $U_{\bar{k}}$ in $E_{\bar{k}}$ is the connected component of $E_{\bar{k}}$, hence this lift is already defined over k. In the general case, possibly passing to the quotient by the connected component of E, one is reduced to the case where E is étale, then to the case where E is completely decomposed (6.2.3). Since $U(k)$ is a p-group and $H(k)$ is of order prime to p, one may take as a lift of $U(k)$ the Sylow p-subgroup of $E(k)$.

Proof of 6.1.1 A) iv) (U smooth, k perfect). If U is connected, U possesses a composition series with successive quotients isomorphic to G_a (4.1.2 b)), and one applies A) ii). In the general case, what precedes allows us to reduce to the case where U is étale, then to the case where U is completely decomposed and H diagonalizable (6.2.3). Using now a characteristic composition series of H ($H \supset H^0 \supset F_n(H) \supset 0$), one reduces to the case where H is of one of the three following types:

a) H is étale.

b) $H = G_m^r$.

c) H is radicial.

In case a), one applies A) iii); in case c), one applies 1.6. Finally in case b), one notes that by Hilbert's Theorem 90, $E \to U$ possesses a section, so that it suffices to show that $H^2(U, G_m^r) = H^2(U(k), G_m^r(k)) = 0$. Now $U(k)$ is a finite p-group, while $G_m^r(k)$ is uniquely p-divisible (since k is perfect).

6.3. Proof of 6.1.1 B) and C)

Thanks to 6.2.1, 6.2.3, 6.2.4, one sees that it suffices to prove B) when $H = G_m$. One therefore has an exact sequence:

$$1 \longrightarrow G_m \longrightarrow E \longrightarrow \alpha_p \longrightarrow 1.$$

Since G_m is smooth, one deduces an exact sequence of p-Lie algebras (App. II 3.2):

```
(*)    0  Lie G_m  Lie E  Lie α_p  0.
```

The group α_p being of height 1, the extension E is trivial if and only if (App. II 2.2) the exact sequence (*) of p-Lie algebras is split. One knows that $Lie \alpha_p$ is generated by an element $X \neq 0$ such that $X^{(p)} = 0$ (App. II 2.1); it therefore suffices to show that X lifts to an element Z of `Lie E` such that $Z^{(p)} = 0$. Now by 5.8.2 ii), there exists a unipotent element Z of `Lie E` lifting X. Since the unipotent part of `Lie E` is clearly at most of dimension 1, one necessarily has $Z^{(p)} = 0$.

Proof of 6.1.1 C). If U_1' is a lift in E of an algebraic subgroup `U_1` of U, U_1' is invariant in E, since E is supposed commutative, and we may pass to the quotient by U_1'. Using a composition series of U (3.5 ii)),

the preceding remark allows us to reduce to the case where U is smooth or equal to α_p. But then E is trivial by A) iv) and B).

6.4. Examples of extensions of a unipotent group U by a group of multiplicative type H that are non-trivial

In view of 6.1.1 A) iv), the problem only arises in characteristic $p > 0$.

a) $H = G_m$, U is a non-trivial form of G_a (cf. App. III, § 5).

Let k be a non-perfect field of characteristic 2, u an element of k such that $u^{1/2}$ does not belong to k. Consider the affine group E with ring $k[X, Y, (X^2 + uY^2)^{-1}]$, where multiplication is given by the comorphism:

```
(X, Y) ↦ (XX' + uYY', XY' + YX').
```

The group E is smooth, connected, commutative, of dimension 2; the subscheme $Y = 0$ defines a subgroup $H \simeq G_m$. The kernel K of squaring in E has equation:

$$X^2 + uY^2 = 1,$$

hence is of dimension 1. The group K contains the unipotent radical of E (defined over $\bar{k}$) but also the contribution of H which is isomorphic to μ_2. Since K is reduced over k, the unipotent radical of E is not defined over k,

and $U = E/H$ does not lift in E. (One verifies immediately that U is the form of G_a having ring $k[V, W]/(V + uV^2 + W^2)$, the morphism $E \to U$ corresponding to the comorphism: $V \mapsto Y^2/(X^2 + Y^2u)$, $W \mapsto XY/(X^2 + Y^2u)$).

b) $H = G_m$, $U = \mathbb{Z}/2\mathbb{Z}$, k non-perfect of characteristic 2.

Choosing k and u as in a), consider the subgroup E of `GL_2` generated by the element X such that:

$$X = (01) = (u^{1/2}0)(0u^{-1/2}).(u0)(0u^{1/2})(u^{1/2}0)$$

The group E is an extension of $\mathbb{Z}/2\mathbb{Z}$ by G_m, but this extension is not trivial because the unipotent part of X is not defined over k.

c) $H = \mu_p$, $U = \alpha_p$, k non-perfect.

Let e be the commutative p-Lie algebra generated by two elements X and Y such that $X^{(p)} = X$ and $Y^{(p)} = aX$. By App. II 2.2, e is the p-Lie algebra of an algebraic group E extension of α_p by μ_p, but this extension is trivial if and only if there exists $b \in k$ such that $b^p = a$ (since one then has $(bX + Y)^{(p)} = 0$).

d) $H = \mu_2$, $U = \alpha_2 \times \alpha_2$, E non-commutative, k a field of characteristic 2.

Consider the special linear group $SL_{2,k}$ and let $E = F(SL_{2,k})$. The group E is a radicial group of height 1, whose Lie algebra is generated by three elements

X, Y, Z satisfying the following relations:

```
[X, Y] = Z,    [X, Z] = [Y, Z] = 0,
X^{(p)} = Y^{(p)} = 0,    Z^{(p)} = Z.
```

Consequently, E is a central extension of $U \simeq \alpha_2 \times \alpha_2$ by μ_2. Each factor α_2 of U lifts uniquely in E, but U itself does not lift in E, because $[X, Y] \neq 0$.

7. Nilpotent affine algebraic groups

7.1. Extensions of groups of multiplicative type

Proposition 7.1.1. *Let S be a prescheme, H and K two S-preschemes in groups of multiplicative type and of finite type, E a prescheme in groups extension of K by H (i.e. K is the quotient of E by H for the fpqc topology). Then E is of multiplicative type in the following two cases:*

a) E is commutative.

b) The fibers of K are connected.

Proof. i) *Case where S is the spectrum of a field k.*

The assertion to prove is local for the fpqc topology, which allows us to suppose k algebraically closed, hence H and K diagonalizable. Note that K acts trivially on H by inner automorphisms: this is clear in case a) and follows from 6.2.1 in case b). By induction on the length of a suitable composition series of K, one is reduced to the case where K is of one of three types: a) $K = G_m$, b) $K = \mu_p$, c) $K = \mu_q$ with $(q, p) = 1$, and in this last case, E is commutative. Using now an embedding of H into G_m^r, one deduces that E embeds into an extension of K by G_m^r. One may therefore suppose $H = G_m^r$.

a) If $K = G_m$, E is a smooth, connected, affine (Exp. VI_B 9.2) algebraic group of unipotent rank zero, hence is a torus.

b) $K = \mu_p$. In this case E is a trivial extension.

Indeed, since H is smooth, the canonical morphism $Lie E \to Lie \mu_p$ is surjective (App. II 3.2), and it suffices to apply 5.8.2 i), taking App. II 2.2 into account.

c) $K = \mu_q$, with $(q, p) = 1$ and E commutative. Here again the extension E is trivial. Indeed, let $x \in E(k)$ be a lift of a generator $\bar{x}$ of $\mu_q(k)$. The element

x^q is an element of $H(k)$, hence is of the form y^q, $y \in H(k)$ (note that $G_m^r(k)$ is q-divisible). Since E is commutative, $y^{-1}x$ is a lift of $\bar{x}$ of order q.

ii) *General case.* The groups H and K are flat, affine, and of finite presentation over S (Exp. IX 2.1), and consequently so is E (Exp. VI_B 9.2). Using then the general technique of VI_B § 10, we reduce to the case where S is noetherian. To show that E is of multiplicative type, it suffices then to prove that E is commutative and that ${}_nE$ is finite over S for every n (Exp. X 4.8 b).

a) E is commutative. One must verify that the morphism:

$$E \times_S E \longrightarrow E, \qquad (x, y) \mapsto [x, y] = xyx^{-1} y^{-1}$$

factors through the unit section of E, and it suffices to verify this when S is the spectrum of a local Artinian ring. But then E is of multiplicative type by i) and Exp. X 2.3.

b) ${}_nE$ is finite over S. Indeed, one has the exact sequence:

$$0 \to {}_nH \to {}_nE \to {}_nK \xrightarrow{u} H_n$$

(where H_n is the cokernel of raising to the n-th power in H). One knows that H_n is of multiplicative type (Exp. IX 2.7), hence separated, and that ${}_nH$ and ${}_nK$ are finite over S; $\operatorname{Ker} u$ is a closed subgroup of ${}_nK$, hence is finite over S, and ${}_nE$ is finite over S, as an extension of a finite group by a finite group (Exp. VI_B 9.2).

7.2. Structure of commutative affine algebraic groups

Theorem 7.2.1. *Let k be a field, G a commutative affine k-algebraic group. Then:*

a) G contains a largest subgroup of multiplicative type M. The group M is characteristic in G and G/M is unipotent, and its formation commutes with extension of the field k.

b) If k is perfect, G is the direct product of M and a unipotent algebraic subgroup U, and this in a unique manner.

Proof. i) *k algebraically closed.*

When G is smooth, 7.2.1 b) is well-known (*Bible* § 4 Th. 4). If G is radicial of height 1, to the decomposition of $\operatorname{Lie} G$ described in 4.2.2 corresponds, taking 4.3.1 v) and App. II 2.2 into account, a decomposition of G of the type 7.2.1 b). In the general case, G admits a composition series whose successive quotients are smooth or radicial of height 1 (App. II 3.1). To prove 7.2.1 b), it then suffices to note that if one has an exact sequence of commutative algebraic groups:

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0,$$

where G' (resp. G'') is a product of a group of multiplicative type by a unipotent group, $G' = M' \cdot U'$ (resp. $G'' = M'' \cdot U''$), then so is G. Indeed, consider the exact sequence:

$$0 \longrightarrow G'/U' \longrightarrow G/U' \longrightarrow G'' \longrightarrow 0.$$

By 7.1.1 a), the inverse image in G/U' of `M''` is a subgroup of multiplicative type `M_1`. The group `M_1` lifts to a subgroup M of G (5.1.1 i) a)). Likewise, using this time 6.1.1 C), one proves that there exists a unipotent subgroup U of G, extension of `U''` by U'. It is clear that $G = M \cdot U$.

ii) *General k.* By i), $G_{\bar{k}}$ is a direct product of a group of multiplicative type $M_{\bar{k}}$ by a unipotent group $U_{\bar{k}}$. Set $S = \bar{k} \times_k \bar{k}$ and let `M_1` and `M_2` be the two inverse images of $M_{\bar{k}}$ under the two projections $S \rightrightarrows \bar{k}$. The group $G_S/M_2 = (G/M_{\bar{k}})_S$ has unipotent fibers, so the image of `M_1` in G_S/M_2 is zero (2.4 i)) and `M_2` majorizes `M_1`. Likewise `M_1` majorizes `M_2`, and finally $M_1 = M_2$. By fpqc descent, it follows that $M_{\bar{k}}$ comes from an algebraic subgroup M of G. It is clear that M is of multiplicative type, that G/M is unipotent, and that the formation of M is compatible with every extension of the field k. For every k-prescheme S, every subgroup of multiplicative type H of `G_S` is contained in `M_S`. Indeed, by 2.5, its image in the group with unipotent fibers $(G/M)_S = G_S/M_S$ is zero. Taking in particular

for H the transform of `M_S` under an automorphism of `G_S`, one deduces that M is characteristic in G.

Finally if k is perfect, G/M lifts in G to a unipotent group, and this in a unique manner by 6.1.1 C).

Remark 7.2.2.

i) If k is not perfect, the unipotent component of $G_{\bar{k}}$ is not necessarily defined over k, as shown by the example 6.4 a).

ii) Unlike what happens for the multiplicative-type component M, the unipotent component U is not in general characteristic in G (whatever the characteristic of k). Of course, the uniqueness of the decomposition 7.2.1 b) entails that U is invariant under every k-automorphism of G. But if U and M are such that there exists a k-prescheme S and a non-zero S-homomorphism $h : U_S \to M_S$ (cf. 2.6), one deduces an S-automorphism of `G_S`, $(u, m) \mapsto (u, h(u) + m)$, which does not leave `U_S` invariant.

iii) If G is finite over k, G/M corresponds by Cartier duality (2.6) to the connected component of the dual $D(G)$ of G.

7.3. Structure of nilpotent affine algebraic groups

Theorem 7.3.1. *Let k be a field, G a nilpotent (Exp. VI_B § 8), affine, connected k-algebraic group. Then G possesses a largest subgroup of multiplicative type M. The group M is central and characteristic, and G/M is a unipotent algebraic group.*

Let Z be the center of G, M the largest subgroup of multiplicative type of Z (7.2.1). Since Z is characteristic in G and M characteristic in Z, M is characteristic in G. It suffices to show that G/M is unipotent. By induction on the length of the ascending central series of G, this will follow, more generally, from the following lemma:

Lemma 7.3.2. *(Rosenlicht). Let G be a connected k-algebraic group, Z its center. Then the center Z' of $G' = G/Z$ is unipotent.*

Proof. We may suppose k algebraically closed. It then suffices to show that Z' does not contain a subgroup isomorphic to μ_ℓ for every prime number ℓ (4.6.1 vi)). Let then μ_ℓ be a subgroup of Z', N its inverse image in G. Since μ_ℓ is central in G', N is invariant in G.

i) *Case where* $(\ell, p) = 1$. One may find an element x of $N(k)$ and an integer n having the following properties:

a) x lifts a generator $\bar{x}$ of μ_ℓ.

b) $x^{\ell^n} \in Z^0(k)$;

(it suffices to choose a lift of $\bar{x}$ whose image in $N/Z^0(k)$ belongs to the ℓ-Sylow subgroup).

Raising to the ℓ-th power in the commutative group Z^0 is an étale morphism, hence $Z^0(k)$ is ℓ-divisible. Consequently, possibly multiplying x by an element of $Z^0(k)$, one may suppose $x^{\ell^n} = 0$. The group N is then generated by two commuting commutative groups (Z and the group generated by x), hence is commutative. The group ${}_{\ell^n}N$ is a group of multiplicative type, characteristic in N, hence invariant in G, and consequently central, G being connected (Exp. IX, 5.5). Hence ${}_{\ell^n}N$ is contained in Z, which contradicts the fact that its image in G' contains μ_ℓ.

ii) $\ell = p$. There then exists an integer n such that the image of $F_n N$ in G' contains μ_p, so that one has the exact sequence:

$$(*) \qquad 1 \to K \to F_n N \to \mu_p \to 1.$$

The group K is contained in Z, hence is commutative; it then follows from 7.2.1 and from 7.1.1 b) and 5.5.1 that there exists a subgroup of multiplicative type contained in $F_n N$ whose image in the quotient by K is μ_p. One deduces, as in i), that $F_n N$ is commutative. The multiplicative-type component of $F_n N$ (7.2.1) is characteristic in $F_n N$, hence invariant in G, hence central (Exp. IX, 5.5); since its image in G' contains μ_p, one obtains a contradiction.

A. Appendix I. Hochschild cohomology and extensions of algebraic groups

A.1. Definition of cohomology groups

Let k be a field, G a k-algebraic group, $(Ab)_G$ the abelian category of commutative k-algebraic groups on which G acts. If $A \in Ob(Ab)_G$, the functor $h_A : (Sch/k)^\circ \to (Ens)$ canonically defined by A is a G-$\mathbb{Z}$-module in the sense of I 3.2. We may therefore consider the standard complex $C^\bullet(G, A)$ of algebraic cochains of G with values in A (Exp. I 5.1), as well as the group of i-cocycles $Z^i(G, A)$, of i-coboundaries

$B^i(G,A)$, and the i-th cohomology group $H^i(G,A)$. As usual, $H^0(G,A)$ is identified with the group $A^G(k)$, where A^G is the k-functor of invariants of A under G. The group $H^1(G,A)$ classifies the principal homogeneous spaces under A, trivial, on which G acts.

The functor $A \mapsto H^\bullet(G,A)$ is not in general a cohomological functor from the category $(Ab)_G$ to Ab; however, one has the following proposition:

Proposition A.1.1. *Let $0 \to A' \xrightarrow{u} A \xrightarrow{v} A'' \to 0$ be an exact sequence in $(Ab)_G$. Then:*

a) If v possesses a section (that is, if there exists a k-morphism of preschemes $s : A'' \to A$ such that $vs = 1_{A''}$), one has the usual exact cohomology sequence:

$$\dots \to H^i(G, A) \to H^i(G, A'') \xrightarrow{d} H^{i+1}(G, A') \to \dots$$

b) If $A(k) \to A''(k)$ is surjective, one has the exact sequence:

$$0 \to A'^G(k) \to A^G(k) \to A''^G(k) \xrightarrow{d} H^1(G, A') \to H^1(G, A) \to H^1(G, A''). \tag{1}$$

Proof. a) One notes that the existence of a section entails the exactness of the sequence of complexes:

$$0 \to C^\bullet(G, A') \to C^\bullet(G, A) \to C^\bullet(G, A'') \to 0.$$

b) If $x'' \in A''^G(k)$, its inverse image in A is a principal homogeneous space under A', trivial (since $A(k) \to A''(k)$ is supposed surjective), on which G acts, hence defines an element $d(x'') \in H^1(G,A')$. The exactness of the sequence (1) is then immediate.

A.2. The group $Ext_{alg}(G,A)$

Let A and G be two k-algebraic groups, E and E' two k-algebraic group extensions of G by A. These two extensions are said to be *isomorphic* if there exists a k-morphism of groups $u : E \to E'$ making the diagram

$$\begin{tikzcd} & E \arrow[dr] \arrow[dd, "u"] & \\ A \arrow[ur] \arrow[dr] & & G \\ & E' \arrow[ur] & \end{tikzcd}$$

commutative. The group E acts on A by inner automorphisms, and if A is commutative, this action factors through G, so G acts on A. Conversely, if $A \in Ob(Ab)_G$, we denote by $Ext_{alg}(G,A)$ the set of classes of algebraic extensions E of G by A for which the action of G on A defined by E, and that coming from the structure

of object of $(Ab)_G$, coincide.

$Ext_{alg}(G,A)$ is in a natural way a bifunctor, covariant in A and contravariant in G. More precisely:

a) If $f : A \to B$ is a morphism in $(Ab)_G$, and if E represents an element of $Ext_{alg}(G, A)$, one defines $f_*(E) \in Ext_{alg}(G, B)$ as the class of the extension of G by B equal to the quotient of the semidirect product $B \cdot E$ (E acting on B through G) by the algebraic subgroup image of A under the morphism (f, i) (this quotient is representable by Exp. VI_A § 5), so that one has a commutative diagram:

$$\begin{tikzcd} 1 \arrow[r] & A \arrow[r, "i"] \arrow[d, "f"] & E \arrow[r] \arrow[d] & G \arrow[r] \arrow[d, "1_G"] & 1 \\ 1 \arrow[r] & B \arrow[r] & f_*(E) \arrow[r] & G \arrow[r] & 1. \end{tikzcd}$$

b) If $g : H \to G$ is a k-morphism of k-algebraic groups, and if E is an extension of G by A, the fiber product $E \times_G H$ is naturally an extension of H by A, denoted $g^*(E)$. One therefore has a commutative diagram:

$$\begin{tikzcd} 1 \arrow[r] & A \arrow[r] \arrow[d, "1_A"] & g^*(E) \arrow[r] \arrow[d] & H \arrow[r] \arrow[d, "g"] & 1 \\ 1 \arrow[r] & A \arrow[r] & E \arrow[r] & G \arrow[r] & 1. \end{tikzcd}$$

Adapting the proofs given in J.-P. Serre, *Groupes algébriques et corps de classes*, Chap. VII, one endows $Ext_{alg}(G, A)$ with a natural abelian group structure, functorial in A and G.

Proposition A.2.1. *Let $0 \to A' \to A \to A'' \to 0$ be an exact sequence in $(Ab)_G$. Then:*

a) One has a canonical exact sequence of abelian groups:

$$Z^1(G, A) \to Z^1(G, A'') \xrightarrow{d} Ext_{alg}(G, A') \to Ext_{alg}(G, A) \to Ext_{alg}(G, A'').$$

b) If $A(k) \to A''(k)$ is surjective, one deduces from a) the exact sequence:

$$(2) \quad H^1(G, A) \to H^1(G, A'') \xrightarrow{d} Ext_{alg}(G, A') \to Ext_{alg}(G, A) \to Ext_{alg}(G, A'').$$

The exact sequence of a) generalizes the usual exact sequence of $\mathrm{Hom}(\cdot, \cdot)$ valid in the framework of commutative extensions (*loc. cit.*) and is proved in the same way. Let us simply recall the definition of the coboundary $d : Z^1(G, A'') \to Ext_{alg}(G, A')$. For this, consider the extension

$$1 \to A' \to A \cdot G \to A'' \cdot G \to 1,$$

deduced in the obvious way from the exact sequence $0 \to A' \to A \to A'' \to 0$. If $u \in Z^1(G, A'')$, u defines in the usual way a section homomorphism $u : G \to A'' \cdot G$. One then has $d(u) = u^*(A \cdot G)$.

A.3. Comparison of $H^2(G, A)$ and $Ext_{alg}(G, A)$

It is well known, in the case of abstract groups, that there exists a functorial isomorphism between the abelian groups $H^2(G, A)$ and $Ext(G, A)$. Likewise in the present case, if A is an element of $(Ab)_G$, to every 2-cocycle $u \in Z^2(G, A)$ one can associate a structure of algebraic group on the prescheme $A \times_k G$ which makes it an element of $Ext_{alg}(G, A)$. Moreover this extension is trivial if and only if $u \in B^2(G, A)$ (cf. Exp. III 1.2.2). Let us recall that the composition law on $A \times G$ is defined by the formula:

$$(a, g)(a', g') = (a + {}^g a' + u(g, g'), gg').$$

It is clear that the extensions of G by A thus obtained are not arbitrary, since they possess a section. But conversely, if $E \in Ext_{alg}(G, A)$ possesses a section s, E is isomorphic to the extension of G by A associated with the 2-cocycle u such that:

$$u(g, g') = s(g)\, s(g')\, s(gg')^{-1}.$$

One finally obtains the following proposition:

Proposition A.3.1. *There exists a functorial isomorphism between the bifunctors with values in abelian groups:*

$$(G, A) \mapsto H^2(G, A) \quad \text{and} \quad (G, A) \mapsto Ext_s(G, A),$$

where $Ext_s(G, A)$ denotes the subgroup of $Ext_{alg}(G, A)$ formed by the classes of extensions of G by A that possess a section.

B. Appendix II. Reminders and complements on radicial groups

Let p be a prime number > 1 and let S be an F_p-prescheme.

B.1. The Frobenius morphism

For every S-prescheme X and every integer $n > 0$, denote by $P_n : X \to X$ the F_p-endomorphism corresponding to raising to the p^n-th power in O_X, and denote by $X^{(n)}$ the S-prescheme inverse image of X under the morphism $P_n : S \to S$. There then exists a unique S-morphism $F_n : X \to X^{(n)}$ making the diagram

$$\begin{tikzcd}
X \arrow[rr] \arrow[dr, "F_n"'] & & X \\
& X^{(n)} \arrow[ur] \arrow[d, "P_n"] & \\
& S \arrow[r, "P_n"'] & S
\end{tikzcd}$$

commutative.

It is clear that F_n is identified with the "n-th iterate" of $F_1 = F$, called the Frobenius morphism of X/S.

If G is an S-prescheme in groups, $G^{(n)}$ is an S-prescheme in groups and $F_n : G \to G^{(n)}$ is an S-morphism of groups. Its kernel $F_n(G)$ is a characteristic sub-prescheme in groups of G (i.e. stable under the functor $\mathrm{Aut}_{S-gr}(G)$), and radicial over S. If G is an S-prescheme in groups that is radicial, one says that G is

of height $\leqslant h$ if $F_h(G) = G$.

B.2. Groups and p-Lie algebras

If G is an S-prescheme in groups, $Lie(G)$ (Exp. II) is naturally a restricted p-Lie algebra (Exp. VII_A §§ 5 and 6). In particular one has the following result (cf. Exp. VII_A):

Proposition B.2.1. *i) $Lie(\alpha_p)_S = Lie(G_a)_S$*[1141] *is a sheaf of $\mathcal{O}_S$-modules, free over $\mathcal{O}_S$ of rank 1, generated by an element X such that $X^{(p)} = 0$.*

ii) $Lie(\mu_p)_S = Lie(G_m)_S$ is a sheaf of $\mathcal{O}_S$-modules, free over $\mathcal{O}_S$ of rank 1, possessing a canonical basis X such that $X^{(p)} = X$.

Let us now recall the fundamental result proved in Exp. VII_A § 7:

Theorem B.2.2. *Suppose S affine with ring A. Then the functor*

$$G \mapsto \mathrm{Lie}\, G$$

establishes an equivalence of categories between, on the one hand, the category of S-preschemes in groups G of finite presentation and flat over S, of height 1, whose Lie algebra is locally free over $\mathcal{O}_S$,

and, on the other hand, the category of restricted p-A-Lie algebras locally free of finite rank.

Moreover, if G is as above and if H is an S-prescheme in groups of finite presentation, the canonical morphism:

$$\mathrm{Hom}_{S\text{-gr}}(G, H) \xrightarrow{\sim} \mathrm{Hom}_{p\text{-}A\text{-Lie}}(\mathrm{Lie}\, G(S), \mathrm{Lie}\, H(S))$$

is an isomorphism.

B.3. Radicial groups and smooth groups

We now suppose that S is the spectrum of a field k of characteristic p.

Let us recall that in Exp. VI_A § 5, it was shown that if G is a k-algebraic group, H an algebraic subgroup of G, then the sheaf G/H (sheaf for the fpqc topology) is representable. Recall then (VII_A 8.3):

Proposition B.3.1. *Let G be a k-algebraic group. Then there exists an integer m such that for every $n \geqslant m$, the algebraic group $G/F_n(G)$ is smooth over k.*

Proposition B.3.2. *Consider an exact sequence of k-algebraic groups:*

$$1 \to G' \to G \xrightarrow{u} G'' \to 1$$

and the following assertions:

i) The morphism u is smooth.

ii) G' is smooth over k.

iii) For every integer $n > 0$, one has the exact sequence:

[1141] N.D.E.: the definition of α_p, already given at the start of this Exposé, has been deleted here.

$$1 \longrightarrow F_n(G') \longrightarrow F_n(G) \longrightarrow F_n(G'') \longrightarrow 1.$$

iv) The morphism $FG \to FG''$ is an epimorphism.

v) The morphism $(LieG)(k) \to (LieG'')(k)$ is surjective.

Then one has the following implications:

i) ⇔ ii) ⇒ iii) ⇒ iv) ⇔ v).

Moreover, if G is smooth, the five assertions are equivalent.

i) ⇔ ii) by Exp. VI_B 9.2 vii).

ii) ⇒ iii). If G' is smooth, $F_n : G' \to G'^{(n)}$ is an epimorphism and iii) follows from the snake lemma diagram.

iii) ⇒ iv) is clear.

iv) ⇔ v) by Theorem B.2.2.

v) ⇒ ii) when G is smooth. Indeed `G''` is then smooth, and v) entails that one has `dim G' = dim_k(Lie G')(k)`, hence G' is smooth.

C. Appendix III. Remarks and complements concerning Exposés XV, XVI, XVII

C.1.

It may be that propositions 1.2 and 1.2 bis of XV remain true if one removes the hypothesis that H' is smooth over S'. This is in particular the case if G is finite, flat, and commutative.

C.2. Complement to XV 4.8

The following proposition, as well as Theorems C.3.1 and C.4.1 below, will appear in a paper in preparation by M. Raynaud on group schemes over a discrete valuation ring.[1142]

Proposition C.2.1. *Let S be the spectrum of a discrete valuation ring, t its generic point, G an S-prescheme in groups of finite type and flat, $\tilde{G} = \operatorname{Spec}\Gamma(G, O_G)$, $u : G \to \tilde{G}$ the canonical morphism. Then:*

(1) $\tilde{G}$ is naturally an S-group scheme and u is a homomorphism.

(2) $Ker(u)$ is flat over S and $(\tilde{G}, u)$ is an fpqc quotient of G by $Ker(u)$, so that $\tilde{G}$ is the largest affine quotient of G.

(3) If G_t is affine, $Ker(u)$ is an étale group over S, equal to the trivial group if and only if G is separated over S. In particular, an S-group scheme G, flat, of finite type, separated, with affine generic fiber, is affine.

[1142] N.D.E.: comments to be added here, including references to VI_B...

C.3.

In the statement of XV 6.6, the hypothesis that H is equal to its connected normalizer is unnecessary for the set of points s of S such that H_s is a parabolic subgroup of G_s to be an open set. Indeed, taking up the proof given in paragraph c) preceding Lemma 6.9, and denoting by N the schematic closure of N_t in G. Then N is a closed subscheme in groups of G, flat over S, majorizing H and contained in N. It then follows from the theorem below

that G/N is representable. Since G_s/N_s is proper and connected, one finishes as in *loc. cit.*

Theorem C.3.1.[1143] *Let S be the spectrum of a henselian discrete valuation ring, G an S-prescheme in groups locally of finite type, H a closed sub-prescheme in groups of G, flat over S. Then G/H is representable.*

C.4. Complement to (XVI 1.1)

The following theorem refines (VIII 7.9).

Theorem C.4.1. *Let S be the spectrum of a discrete valuation ring of residue characteristic $p > 0$, and let G be a commutative, smooth S-group scheme of finite type and separated over S. Then the following conditions are equivalent:*

(i) ${}_pG$ is finite over S.

(ii) For every S-prescheme S' and for every S'-prescheme in groups H', of finite presentation over S', separated over S', every S'-monomorphism $u : G_{S'} \to H'$ is an immersion.

C.5.

The example (XVII 6.4 a)) provides an example of a smooth group over a field k, whose unipotent radical is not defined over k. The following proposition gives a general method for obtaining such groups:

Proposition C.5.1. *Let k be a field, K a finite radicial extension of k of degree > 1, H a connected, smooth K-algebraic group of dimension r, $G = \prod_{K/k} H/K$, which is a k-algebraic group, smooth, connected, of dimension $r[K : k]$. Let $u : G_K \to H$ be the canonical homomorphism, and $R = Ker(u)$. Then:*

i) The morphism u is an epimorphism and R is a smooth, unipotent, connected algebraic group.

ii) If U is a smooth algebraic subgroup of G such that U_K majorizes R, then $U = G$.

Corollary. *Keep the notations of the preceding proposition.*

a) If H is not unipotent, the unipotent radical of $G_{\bar{k}}$ is not defined over k.

[1143] N.D.E.: See Theorem 4C in: S. Anantharaman, *Schémas en groupes, espaces homogènes et espaces algébriques sur une base de dimension 1*, Bull. Soc. Math. France, Mém. 33 (1976), 5-79.

b) If H is a non-zero abelian variety, G is not an extension of an abelian variety by a smooth linear group.

c) If H is not solvable, the solvable radical of $G_{\bar{k}}$ is not defined over k, and G does not possess a Borel subgroup defined over k.

Let us first show how the corollary follows from the proposition.

a) Let U be a unipotent radical of G. Then U_K is the unipotent radical of G_K, hence majorizes R, since R is smooth unipotent connected by i), hence $U = G$ by ii). Now G_K admits H as a quotient, and H is not unipotent by hypothesis, hence G is not unipotent, whence a contradiction.

b) If G is an extension of an abelian variety by a linear group L, smooth over k, necessarily L_K majorizes R, hence $L = G$. Now G cannot be a linear group since G_K possesses a quotient that is a non-zero abelian variety.

c) Let S (resp. B) be a solvable radical of G (resp. a Borel subgroup of G); then S_K (resp. B_K) majorizes R, hence $S = G$ (resp. $B = G$). But then G is solvable, which contradicts the fact that G_K possesses a quotient that is not solvable.

Proof of Proposition C.5.1.

i) Let us begin by describing the morphism u. The data of a K-scheme

$$f : S \longrightarrow \mathrm{Spec}(K)$$

allows one to construct the diagram:

```
Spec(K) ←—h———— T ——jT——→ X
   ↓ j           ↓ s
   ↓             ↓
Spec(k) ←——g———— S          (where g = j ∘ f)
```

where $T = \mathrm{Spec}(K) \times_{\mathrm{Spec}(k)} S$, h and jT are the two projections, and s is the section of T over S such that $h \circ s = f$. The map

$$u(S) : G_K(S) \longrightarrow H(S)$$

is simply the composite map:

$$G_K(S) \xrightarrow{\sim} H(T) \to H(S),$$

where the last arrow is defined by the section s.

Take in particular for S the spectrum of an algebraic closure $\bar{k}$ of k and for f the unique k-morphism $K \to \bar{k}$, so that T is a local Artinian scheme. To prove i) it suffices to do so after extension $K \to \bar{k}$ of the base field. Now it is clear that $G_S =$

$\prod_{T/S} H_T/T$ represents the Greenberg functor of H_T relative to S (M. J. Greenberg, *Schemata over local rings*, Ann. of Maths. 73, 1961, p. 624-648). The description made above then shows that, by means of this last identification, u_S is the canonical transition morphism:

$$\mathrm{Green}(H_T) \xrightarrow{\sim} H_S = H_T \times_T S.$$

Assertion i) then follows from the fact that H is smooth over K and from (M. J. Greenberg, *Schemata over local rings II*, Ann. of Maths. 78, 1963, p. 256-266).

ii) To establish ii) we may suppose k separably closed. Let U be a smooth algebraic subgroup of G such that $U_K \supset R$, and let us show that $U = G$.

Since G is connected, we may suppose U connected. Let V be the smooth connected algebraic subgroup $u(U_K)$ of H, and $V' = \prod_{K/k} V/K$, which is a smooth connected algebraic subgroup of G. The group V'_K is an algebraic subgroup of G_K, and the canonical morphism $V'_K \to V$ is simply the restriction of u to V'_K. By hypothesis, U_K majorizes R, *a fortiori* U_K majorizes $Ker(V'_K \to V)$; on the other hand, by construction, the image of U_K in H is equal to V. One deduces that U_K majorizes V'_K, hence U majorizes V'. On the other hand, the canonical isomorphism

$$G(k) \hookrightarrow G(K) \xrightarrow{u(K)} H(K)$$

evidently sends $U(k)$ into $V(K) = V'(k)$; that is to say, $U(k)$ is contained in $V'(k)$. Since U is smooth and k is separably closed, this entails $U \subset V'$, whence $U = V'$. One then has the equalities:

$$([K : k] - 1) \dim V = \dim Ker(V'_K \to V) = \dim R = ([K : k] - 1) \dim H.$$

Since $K \neq k$, one concludes that $\dim V = \dim H$, whence $V = H$ and finally $U = V' = G$.

Footnotes

Exposé XVIII. Weil's theorem on the construction of a group from a rational law

by Michael Artin

0. Introduction

This Exposé is devoted to the well-known theorem of Weil [1] which gives the construction of a group variety from a birational law. It seems that the generalization of this result to the case where the base is the spectrum of a discrete valuation ring was already known to several people; one may consult, for example, [2]. Here we shall prove the theorem for a flat group of finite presentation over an arbitrary base prescheme S. Since the hypotheses must be made with somewhat more care, the statement is not in the form given by Weil; but when S is the spectrum of a field, one sees that the form given here is essentially equivalent to Weil's. We use the following suggestion of Grothendieck: given a “group germ” X over a prescheme S, one is to construct the group as the quotient of $X \times_S X$ by a suitable equivalence relation.

1. “Recollections” on rational maps

Let X/S be a relative prescheme and $U \subseteq X$ an open subset. We say that U is *schematically dense in X relatively to S*, or that $U \hookrightarrow X$ is *relatively schematically dense*, if for every base change $S' \to S$ the open subset $U \times_S S'$ is schematically dense in $X \times_S S'$. For the definition and properties of this notion we refer to Exp. IX, § 4.[1144]

Proposition 1.1. *(i) A finite intersection, as well as a union of a non-empty family of opens that are schematically dense relatively to S, is schematically dense relatively to S.*

(ii) If $U \subseteq X$ is schematically dense relatively to S and if $S \to T$ is a morphism, then $U_T \subseteq X_T$ is schematically dense relatively to T.

(iii) Let $U \subseteq V \subseteq X$ be open immersions. For U to be schematically dense in X relatively to S, it is necessary and sufficient that it be so in V and that V be so in X.

(iv) If $U \subseteq X$ and $V \subseteq Y$ are relatively schematically dense, then $U \times_S V$ is schematically dense in $X \times_S Y$ relatively to S.

In Exp. IX one finds the following criterion:

Proposition 1.2. *Let X/S be locally of finite presentation and flat, and let U be an open subset of X. If for each $s \in S$ the fiber U_s is schematically dense in X_s, then U is schematically dense in X relatively to S.*

In particular:

Corollary 1.3. *If X/S is locally of finite presentation and flat, and if each fiber X_s is “without embedded component”, then U is schematically dense in X relatively to S if and only if for each $s \in S$, U_s is dense in X_s in the topological sense.*

One easily deduces from the definition the following:

Proposition 1.4. *Let $U \subseteq X$ be schematically dense relatively to S, and let $f, g : X \to Y$ be two morphisms of functors over S, where Y satisfies one of the following conditions:*

(i) Y is a prescheme over S and each fiber Y_s is separated.

(ii) Y is a presheaf[1145] on S such that the diagonal morphism $Y \to Y \times_S Y$ is representable by a closed immersion.[1146]

[1144] cf. also EGA IV_3, 11.9 and 11.10 (notably 11.10.8), where one says “universally schematically dense relatively to S”.

[1145] N.D.E.: to be made precise for which topology: a priori (fpqc).

[1146] N.D.E.: We recall here the definition of a closed immersion. A morphism of S-functors $F \to G$ is *relatively representable* if for every S-morphism $T \to G$, where T is an S-prescheme, the T-functor

Then if $f = g$ on U, one has $f = g$.

Proof. In case (i), the standard argument following IX 4.1 shows that f and g are equal on each fiber, so that the subprescheme $Ker(f, g)$ of X is set-theoretically equal to X, hence closed; and since it majorizes U, it is equal to X, i.e. $f = g$. Case (ii) is proved as in *loc. cit.*[1147]

Definition 1.5.[1148] *Let X be a prescheme over S and Y a presheaf over S. By a* rational map $f : X \dashrightarrow Y$ over S *we mean an equivalence class of morphisms $f : U \to Y$ over S, where U is an open of X that is schematically dense relatively to S, and where one sets $(f : U \to Y) \sim (f' : U' \to Y)$ if and only if there exists an open $U'' \subseteq U' \cap U$, schematically dense relatively to S, such that $f = f'$ on* U''*.*

This definition is made in such a way that one may define in the obvious way the rational map $f \times_S S'$ for any base change $S' \to S$.

If U is an open of X, we say that the rational map f is *defined on U* if there exists a morphism representing f whose set of definition contains U.

Definition 1.5.1.[1149] *If Y satisfies one of the conditions (i), (ii) of 1.4, it is clear that there exists a largest open U of X on which f is defined, and this U is schematically dense relatively to S. It is called the* domain of definition of f over S, *and is denoted $Dom(f)$.*

This notion does not commute with base change in general, but one has:

Proposition 1.6. *Let X be an S-prescheme and Y an S-functor satisfying one of the hypotheses (i), (ii) of 1.4. Let $f : X \dashrightarrow Y$ be a rational map over S, let $S' \to S$ be a flat morphism locally of finite presentation, and set $f' = f \times_S S'$. Then*

Dom(f') = Dom(f) ×_S S'.

Proof. Set $U = Dom(f)$. It is clear that $V' = Dom(f')$ contains $U \times_S S'$. Let V be the image of V', which is an open of X because $X' \to X$ is open. We must show that $V = U$, that is, we must find a morphism $V \to Y$ that represents f. Set $S'' = S' \times_S S'$, and

X'' = X ×_S S'',   U'' = U ×_S S'',   V'' = V' ×_V V'.

Then U'' is schematically dense in X'' relatively to S'', hence U'' is schematically dense in V'' relatively to S, since $U'' \subseteq V'' \subseteq X''$. The restriction of $f' : V' \to Y'$ to U' is deduced from $f : U \to Y$ by base change. The two morphisms $V'' \to Y$ deduced from g by base change are equal on U'', hence are equal. Now since $V' \to V$ is flat and locally of finite presentation, it is fppf-covering (Exp. IV 6.3), and one finds the morphism $V \to Y$ by descent.

$F_T = F \times_S T$ is representable by a T-prescheme. A morphism of S-functors $F \to G$ is an *open* (resp. *closed*) *immersion* if it is relatively representable and if for every S-morphism $T \to G$, where T is an S-prescheme, the T-morphism $F_T \to T$ is an open (resp. closed) immersion.

[1147] N.D.E.: To be made precise...

[1148] cf. EGA IV_4, 20.5, where one says "pseudo-morphism of X into Y relatively to S", in order not to conflict with EGA I, 7.12.

[1149] N.D.E.: The number 1.5.1 has been added.

We shall make frequent use of the following triviality:

Proposition 1.7. *Let X/S be faithfully flat, locally of finite presentation, and let $U \subseteq X$ be a relatively schematically dense open. Then there exists a base change $S' \to S$ which is fppf-covering, and a section $x \in X(S')$ which is contained in $U(S')$.*

Indeed, U/S is fppf-covering. One takes $S' = U$ and the S-graph of the inclusion of U in X as the section.

2. Local determination of a group morphism

Let G and H be groups and let $U \subseteq G$ be a subset such that $U \cdot U = G$. Then if f and g are homomorphisms from G to H such that $f = g$ on U, one has $f = g$. Similarly:

Proposition 2.1. *Let S be a site (cf. Exp. IV[1150]), G a sheaf of groups on S, and $U \subseteq G$ a subsheaf of sets such that the morphism $U \times U \to G$ induced by multiplication is an epimorphism. Then if f and g are homomorphisms from G to a sheaf of groups H that are equal on U, one has $f = g$.*

Corollary 2.2. *Let G be an S-group prescheme locally of finite presentation and flat, let $U \subseteq G$ be an open that is relatively schematically dense, and let H be a sheaf of groups on S for the fppf topology. Then every homomorphism $f : G \to H$ is determined by its restriction to U.*

Indeed, since G is flat over S, the composition law on G is a flat morphism (VI_B.9.2.xi), and it follows that $U \times_S U \to G$ is faithfully flat and locally of finite presentation, hence fppf-covering, hence an epimorphism.

Proposition 2.3. *Let G be a group prescheme locally of finite presentation and flat over S, $U \subseteq G$ a relatively schematically dense open, and H a sheaf of groups for the fppf topology. Write m_G for the multiplication of G and m_H for that of H.*

Let $f : U \to H$ be an S-morphism and assume that there exists a relatively schematically dense open V of $m_G^{-1}(U) \cap U \times_S U$ such that the diagram

$$\begin{tikzcd} V \arrow[r, "(f \times f)|V"] \arrow[d, "m_G"'] & H \times_S H \arrow[d, "m_H"] \\ U \arrow[r, "f"] & H \end{tikzcd}$$

is commutative. Then there exists a morphism of S-groups $\bar{f} : G \to H$ (necessarily unique) extending f, in each of the following situations:

(i) H is representable.

(ii) The diagonal morphism $H \to H \times_S H$ is representable by a closed immersion.[1151]

[1150]N.D.E.: i.e. a category equipped with a topology, cf. IV, § 4.2.
[1151]N.D.E.: See the note at 1.4.

(iii) For each section $a \in U(S)$, the open $pr_2((a \times_S U) \cap V)$ is relatively schematically dense in U[1152]*, and this statement remains true after every base change $S' \to S$.*

Proof. Note first that V is relatively schematically dense in $G \times_S G$. Indeed, V is relatively schematically dense in

$$m_G^{-1}(U) \cap (U \times_S G) \cap (G \times_S U),$$

and the three factors are deduced from $U \subseteq G$ by an obvious base change $G \times_S G \to G$, which implies the assertion by 1.1.

To construct a morphism $\bar{f} : G \to H$, it suffices, since $U \times_S U \to G$ is fppf-covering, to find a morphism from $U \times_S U$ to H such that the two morphisms induced on $(U \times_S U) \times_G (U \times_S U)$ are the same.

We take $m_H \circ (f \times f) : U \times_S U \to H$. We must verify that whenever we have sections $a, b, c, d \in U(S')$, $S' \to S$ arbitrary, such that $ab = cd$, we also have $f(a)f(b) = f(c)f(d)$. By hypothesis, this is true if (a, b) and (c, d) are contained in $V(S')$, since in that case $f(a)f(b) = f(ab)$ and $f(c)f(d) = f(cd)$. Therefore it is true whenever $(a, b, c, d) \in (V \times_G V)(S')$.

I claim that $V \times_G V$ is an open of $(U \times_S U) \times_G (U \times_S U)$ schematically dense relatively to S. This will complete the proof in cases (i) and (ii) by 1.4, the facts that the morphism $\bar{f}$ so constructed extends f and that $\bar{f}$ is a homomorphism being evident, again by 1.4.

Indeed, we write

$$V \times_G V = (V \times_G (G \times_S G)) \cap ((G \times_S G) \times_G V).$$

By symmetry and 1.1, it suffices to verify that $V \times_G (G \times_S G)$ is relatively schematically dense in $(G \times_S G) \times_G (G \times_S G)$. But this last prescheme is S-isomorphic to $G \times_S G \times_S G$, the morphism being given by $(a, b, c, d) \mapsto (a, b, c)$. Thus what must be shown is that $V \times_S G$ is schematically dense in $G \times_S G \times_S G$ relatively to S, which is a consequence of the fact that V is relatively schematically dense in $G \times_S G$.

It remains to treat case (iii). To prove $f(a)f(b) = f(c)f(d)$, it is permitted to make an fppf-covering base change. Suppose we have a section $x \in G(S')$, $S' \to S$ fppf-covering, such that (b, x), (a, bx), (d, x), (c, dx) are all in V. Then one will have

$$f(a)f(b)f(x) = f(a)f(bx) = f(abx) = f(cdx) = f(c)f(dx) = f(c)f(d)f(x),$$

whence the desired equality.

[1153] To find such an x, set, for every $z \in U(S')$,

$$V_z = pr_2((z \times_{S'} U_{S'}) \cap V_{S'}).$$

Then the hypotheses on x say that $x \in W$, where

$$W = V_b(S') \cap V_d(S') \cap b^{-1} V_a(S') \cap d^{-1} V_c(S').$$

[1152] Or in G, which amounts to the same by virtue of 1.1 (iv).

[1153] N.D.E.: The beginning of the following sentence has been added.

By (iii), W is relatively schematically dense in G. Hence the existence of a section x after an fppf-covering extension follows from Proposition 1.7.

Let us verify that the morphism so constructed is multiplicative. Let $a, b \in G(S')$, $S' \to S$ arbitrary, and write $S_0 = S$. Choose an fppf-covering base change $S' \to S$ and a section $x \in G(S')$ such that $x \in U(S')$ and $ax^{-1} \in U(S')$, that is, $x \in (a^{-1}U)^{-1}(S')$. Choose moreover another fppf-covering extension $S'' \to S'$ and a section $y \in G(S'')$ such that

$$y,\ by^{-1},\ aby^{-1} \quad \text{belong to } U(S''),$$

and

$$by^{-1} \in V_x, \quad xby^{-1} \in V_{ax^{-1}}.$$

Then by the definition of $\bar{f}$, one has on S''

$$\bar{f}(a) = f(ax^{-1})f(x), \quad \bar{f}(b) = f(by^{-1})f(y), \quad \bar{f}(ab) = f(aby^{-1})f(y).$$

Moreover,

$$\bar{f}(a)\bar{f}(b) = f(ax^{-1})f(x)f(by^{-1})f(y) = f(ax^{-1})f(xby^{-1})f(y) = f(aby^{-1})f(y) = \bar{f}(ab),$$

whence multiplicativity.

The fact that $\bar{f}$ extends f is now easy. Let $a \in U(S')$, write $S_0 = S$, and choose $S' \to S$ fppf-covering and sections $x, y \in G(S')$ such that (x, y), (ax, y), (a, xy) are in $V(S')$. Then

$$\bar{f}(xy) = f(x)f(y) = f(xy), \quad \bar{f}(axy) = f(ax)f(y) = f(axy).$$

Hence

$$\bar{f}(a) = \bar{f}(axy)\bar{f}((xy)^{-1}) = \bar{f}(axy)\bar{f}(xy)^{-1} = f(axy)f(xy)^{-1} = f(a).$$

Remark 2.3.1. *In many cases hypothesis (iii) is true, for it will even be true if one replaces U by a smaller open U^0, still relatively schematically dense in G. For example one has:*

Proposition 2.4. *The situation being as in 2.3, suppose that each geometric fiber of G/S is irreducible. Let $U^0 = pr_1 V$ and*

$$V^0 = V \cap m_G^{-1}(U^0) \cap (U^0 \times_S U^0).$$

Then $U^0 \subseteq U$ is an open relatively schematically dense of G and the objects U^0, $f|U^0$ and V^0 satisfy hypothesis (iii).

Proof. U^0 is open because $G \times_S G \to G$ is flat and locally of finite presentation. All the other verifications are trivial save hypothesis (iii).

Let $a \in U(S')$, write $S_0 = S$. To verify that $pr_2((a \times_S U^0) \cap V^0)$ is relatively schematically dense in U^0, it suffices to do so fiber by fiber by Corollary 1.3, that is, it suffices to treat the case where S is the spectrum of a field, and in that case it suffices to show that U^0 is non-empty, because G is irreducible and "without embedded components" (cf. VI_A, 1.1.1). Now

$$pr_2((a \times_S U^0) \cap V^0) = pr_2((a \times_S U^0) \cap V) \cap pr_2((a \times_S U^0) \cap m_G^{-1}(U^0))$$

and the second term of the right-hand side is dense in G. Hence it suffices to show that $pr_2((a \times_S U^0) \cap V)$ is dense in G, that is, non-empty, which is clear because $a \in U^0 = pr_1 V$.

3. Construction of a group from a rational law

3.0.

We are given a prescheme X/S and a rational map $X \times_S X \dashrightarrow X$ over S, and we seek a group G/S and a birational map relatively to S[1154]

$$X \dashrightarrow G$$

that commutes with the composition laws. We treat only the case where X/S satisfies the following hypothesis:

(♦) X/S is faithfully flat, of finite presentation, and with separated fibers "without embedded components".

(Note that the latter two hypotheses are properties that hold for a group prescheme.[1155])

We shall often suppress the symbol S in fiber products.

Let X/S be a prescheme with the properties (♦) above, and let W be a subprescheme of finite presentation of $X \times X \times X$ having the following property:

(∗) The three morphisms $W \to X \times X$ given by the projections of X^3 onto X^2 are open immersions, schematically dense relatively to S.

Notations. We shall use the following terminology. Given sections $a, b, c \in X(S')$, $S' \to S$ arbitrary, such that $(a, b, c) \in W(S')$, we write:

$$c = ab, \quad b = a^{-1}c, \quad a = cb^{-1}.$$

Definition 3.0.1.[1156] *We say, given a section $(a, b) \in X^2(S')$, that* ab is defined *if and only if there exists a section $c \in X(S')$ such that $(a, b, c) \in W(S')$, i.e. if and only if (a, b) lies in $pr_{12}W(S')$. Similarly, to say that $a^{-1}b$ or ab^{-1} is defined has the analogous meaning, and one extends this terminology to products of several factors as well.*

[1154] N.D.E.: To make explicit the notion of birational map (relatively to S), perhaps in an addition 1.5.2?
[1155] N.D.E.: cf. VI_A, to be made precise...
[1156] N.D.E.: We have introduced the numbering 3.0.1 and 3.0.2.

Remark 3.0.2. *Let us note immediately the following fact: by (i), W defines a rational map $X^2 \dashrightarrow X$ over S (the one given by $(a,b) \mapsto ab$). It may well happen that this rational map has a domain of definition larger than $pr_{12}W$. Nevertheless, we say that* `ab` *is defined only if $(a,b) \in pr_{12}W(S')$.*

Definition 3.1. *A* group germ *is a prescheme X/S having the properties (♦) above together with a subprescheme W of finite presentation of $X \times X \times X$ having the following properties:*

(i) W satisfies the property $(*)$ above.

(ii) For each section $a \in X(S)$, the sets

```
pr_i((a × X × X) ∩ W),    i = 2, 3,
pr_i((X × a × X) ∩ W),    i = 3, 1,
pr_i((X × X × a) ∩ W),    i = 1, 2,
```

are schematically dense in X relatively to S, and this statement remains true after every base change $S' \to S$. (Hypothesis (i) implies that these sets are opens of X.) Intuitively, this says that "`ax` *is defined for x sufficiently general", etc.*

(iii) The law is associative, i.e. if $(a,b,c) \in X^3(S')$, $S' \to S$ arbitrary, is such that `(ab)c` *and $a(bc)$ are defined, one has $(ab)c = a(bc)$.*

Remarks. *(a) In (i), the condition that W be schematically dense in X^2 relatively to S may be replaced by the condition that for each $s \in S$ the fiber W_s be dense in X_s^2 in the topological sense, thanks to the hypotheses on X and to 1.2.*

(b) Condition (iii) is equivalent to the following: let V_1 (resp. V_2) be the open of X^3 where the rational map $(a,b,c) \mapsto a(bc)$ (resp. $(a,b,c) \mapsto (ab)c$) is defined[1157]*. Then there exists an open $V \subseteq V_1 \cap V_2$ which is schematically dense in X^3 relatively to S on which the two preceding maps coincide. This is a consequence of 1.4 because, of course, V_1 and V_2 are schematically dense in X^3 relatively to S.*

(c) Hypothesis (ii) will serve below to ensure that X will be a subobject of the group prescheme that it defines. In many cases one may deduce (ii) from (i), provided that one replaces X by an open X^0 relatively schematically dense, and W by a relatively schematically dense open of $W \cap X^{03}$. In fact one has:

Proposition 3.2. *Suppose that each geometric fiber of X/S is irreducible, and let W be a subprescheme of X^3 satisfying condition (i) of 3.1. Then there exists an open X^0 of X relatively schematically dense and an open W^0 of $W \cap (X^0 \times X^0)$ relatively schematically dense such that the pair (X^0, W^0) satisfies conditions (i) and (ii). If (iii) is satisfied for (X, W), it is satisfied for (X^0, W^0).*

Proof. Set $X^0 = \bigcap_{i=1}^3 pr_i W$. Each $pr_i W$ is open in X because $W \to X^2$ is an open immersion and the projections $X^2 \to X$ are flat and of finite presentation. Moreover, $pr_i W$ is relatively schematically dense in X because W is so in X^2 and the projections

[1157] In the sense explained in 3.0.1; one could also replace these opens by the domains of definition (cf. 1.5) of the rational maps in question.

$X^2 \to X$ are surjective (it suffices to check density in the topological sense). Take $W^0 = W \cap X^{03}$.

To verify that (i) holds, note that

$$W^0 = (W \cap (X^0 \times X^0 \times X)) \cap (W \cap (X \times X^0 \times X^0)) \cap (W \cap (X^0 \times X \times X^0)).$$

Now $W \cap (X^0 \times X^0 \times X) \simeq pr_{12} W \cap (X^0 \times X^0)$, hence is relatively schematically dense in W. Similarly, the other terms of the right-hand side are relatively schematically dense in W, and consequently W^0 is relatively schematically dense in W. Hence $pr_{ij} W^0$ is relatively schematically dense in $pr_{ij} W$, hence in $X \times X$, hence in $X^0 \times X^0$.

To verify condition (ii), let $a \in X^0(S')$ and write $S_0 = S$. We must show that, for example, $pr_2((a \times X^0 \times X^0) \cap W^0)$ is schematically dense in X^0 relatively to S. By 1.3, it suffices to verify this fiber by fiber, that is, it suffices to treat the case where S is the spectrum of a field, and in that case it suffices to verify that the open is non-empty, because the fibers of X/S are irreducible and "without embedded components". Since $pr_2((a \times X \times X) \cap W)$ is non-empty (as a is a section of X^0), this open is dense in X. One has

$$pr_2((a \times X \times X) \cap W) = pr_2((a \times X) \cap pr_{12} W).$$

Hence,

$$pr_2((a \times X) \cap pr_{12} W) \cap X^0 = pr_2((a \times X^0) \cap pr_{12} W) = pr_2((a \times X^0 \times X) \cap W)$$

is dense in X, hence in X^0.

Similarly, $pr_3((a \times X \times X^0) \cap W)$ is dense in X, hence in $pr_3((a \times X \times X) \cap W)$, i.e. $(a \times X \times X^0) \cap W$ is dense in $(a \times X \times X) \cap W$, i.e. $pr_2((a \times X \times X^0) \cap W)$ is dense in $pr_2((a \times X \times X) \cap W)$, hence dense in X, hence dense in X^0.

Now since

$$\begin{aligned} pr_2((a \times X^0 \times X^0) \cap W^0) &= pr_2((a \times X^0 \times X^0) \cap W) \\ &= pr_2((a \times X^0 \times X) \cap W) \cap pr_2((a \times X \times X^0) \cap W), \end{aligned}$$

it is indeed dense in X^0, as was to be shown. The other assertions of (ii) follow by symmetry, and the fact that condition (iii) is preserved is trivial.[1158] Proposition 3.2 is proved.

3.2.1.

[1159]

Let us now fix a group germ (X, W) over S. We must make some preliminary remarks on the situation, which we have gathered together below. We shall use these rules often without explicit mention in the sequel.

Let $a \in X(S')$, and write $S_0 = S$. Then one obtains a (bi)rational map ϕ over S from X to itself by associating to a section x the section ax if it is defined. By 3.1

[1158] N.D.E.: The following sentence has been added.
[1159] N.D.E.: We have added the numbering 3.2.1, as well as 3.2.2.

(ii), the domain of definition of ϕ contains the relatively schematically dense open $pr_2((a \times X \times X) \cap W)$, and ϕ defines an isomorphism of this open onto the open (where $a^{-1}x$ is defined) $pr_3((a \times X \times X) \cap W)$. This remark is generalized in the obvious way in Rule 1.

Rule 1. *Let $P = P(x, t_1, \cdots, t_n)$ be a "product" of the symbols $x, t_1, \cdots, t_n$ obtained recursively as follows: $P_0 = x$; P_{i+1} is one of the following expressions:*

$$P_i t, \quad t P_i, \quad P_i^{-1} t, \quad t P_i^{-1}, \quad P_i t^{-1}, \quad t^{-1} P_i,$$

where t is one of the t_j; $P_r = P$. Let $a_1, \cdots, a_n \in X(S')$. Then there exists a relatively schematically dense open U of $X_{S'}$ such that the product $P(x, a_1, \cdots, a_n)$ is defined (in the sense of Remark 3.0.2) for a section $x \in X(S'')$ if and only if $x \in U(S'')$, and the map $x \mapsto P(x, (a))$ gives an isomorphism of U onto another relatively schematically dense open, denoted $P(U, (a))$, of $X_{S'}$.

Rule 2. *Let $a, b \in X(S')$. Then:*

If ab is defined, so is $a^{-1}(ab)$, and $a^{-1}(ab) = b$.

If $a^{-1}b$ is defined, so is $a(a^{-1}b)$, and $a(a^{-1}b) = b$.

If ba^{-1} is defined, so is $(ba^{-1})a$, and $(ba^{-1})a = b$.

Rule 3. *Let $a, b, b' \in X(S')$. If $ab = ab'$, if $ba = b'a$, if $a^{-1}b = a^{-1}b'$ or if $ba^{-1} = b'a^{-1}$, then $b = b'$. Here it is implicit that the equality relation implies that both sides are defined.*

Rule 4. *Let $a, b, c \in X(S')$. Then whenever both sides are defined, one has*

$$a((ba)^{-1} c) = b^{-1} c.$$

Similarly:

$$(c(ab)^{-1}) a = cb^{-1}, \quad a^{-1}(ab^{-1}) c = b^{-1} c, \quad (c(b^{-1} a)^{-1}) a = cb^{-1}.$$

Rule 5. *All the following associativity laws are true, whenever both sides are defined:*

$$(a^{-1}b)c = a^{-1}(bc), \quad (ab^{-1})c = a(b^{-1}c), \quad (ab)c^{-1} = a(bc^{-1}),$$
$$(ab)^{-1}c = b^{-1}(a^{-1}c), \quad (a^{-1}b)c^{-1} = a^{-1}(bc^{-1}), \quad (ab^{-1})c^{-1} = a(cb)^{-1}.$$

3.2.2. Verification of the rules.

(1) is by an obvious induction on the length r of P, the case $r = 1$ being a direct consequence of 3.1 (ii).

(2) Trivial from the definition.

(3) Indeed, by Rule 2, for example in the first case, one has

$$b = a^{-1}(ab) = a^{-1}(ab') = b'.$$

(4) Let us verify, for example, the first relation. On the right-hand side left multiplication by b is defined and gives c, by Rule 2. Suppose it is also defined on the left-hand side. Then one will have

```
b(a((ba)⁻¹ c)) = (ba)((ba)⁻¹ c) = c.
```

Indeed `(ba)` is defined by hypothesis because it figures in the expression. Hence the middle member is defined and equal to c by Rule 2, and equal to the left-hand member by associativity (cf. 3.1 (iii)). Hence Rule 3 implies that the desired equality is true if this multiplication by b is defined.

Now fix a and b. Then Rule 1 implies that $b(a((ba)^{-1}c))$ is well defined for c "in" an open U of X relatively schematically dense; hence on this relatively schematically dense open the two rational maps $c \mapsto b^{-1}c$ and $c \mapsto a((ba)^{-1}c)$ are equal. By 1.4, they are equal on every common domain of definition, whence the desired result.

(5) The same kind of argument as the preceding. For example one verifies $(ab^{-1})c^{-1} = a(cb)^{-1}$ as follows. If right multiplication by c is defined on the right-hand side, one has equality by Rule 4. Since it suffices to verify such a formula on a relatively schematically dense open, one reduces to the case where this multiplication is well defined.

3.2.3.

Consider now the relation R on $X \times X$ obtained by setting, for $a, b, a', b' \in X(S')$, $(a, b) \sim (a', b') mod R(S')$ if and only if there exists $S' \to S$ fppf-covering and a section $x \in X(S')$ such that `(xa)b` and `(xa')b'` are defined and equal. Then R is an equivalence relation.

Indeed, this relation is evidently symmetric. By Rule 1, the product `(xa)b` is defined if x is "in" a suitable relatively schematically dense open. Hence 1.7 asserts that there exists $S' \to S$ fppf-covering and an $x \in X(S')$ such that `(xa)b` is defined. The relation is therefore reflexive, and transitivity is a consequence of the following lemma:

Lemma 3.3. *Let* `x, y, a, b, a', b'` *be sections of $X(S')$ such that* `(xa)b`, `(xa')b'`, `(ya)b`, `(ya')b'` *are defined. If $(xa)b = (xa')b'$ then $(ya)b = (ya')b'$.*

Indeed, the lemma says that one may test $(a, b) \sim (a', b')$ with any $x \in X(S')$, $S' \to S$ fppf-covering, such that both products are defined. Given $a, b, a', b', a'', b'' \in X(S)$, one may, by Rule 1, 1.1 and 1.7, find an fppf-covering extension $S' \to S$ and a section $x \in X(S')$ such that the three products in question are defined, whence transitivity.

Proof of the lemma. Let us write formally to begin with:

```
(za)b = ((zx⁻¹)x) a b = ((zx⁻¹)(xa)) b = (zx⁻¹)((xa)b),
(za')b' = ((zx⁻¹)x) a' b' = ((zx⁻¹)(xa')) b' = (zx⁻¹)((xa')b').
```

One verifies that these equalities are indeed true if the members are defined, by the appropriate rules[1160]. It follows that $(za)b = (za')b'$ if all these expressions are defined. Moreover, by Rule 1 and the hypotheses already made, these expressions are well defined if $z \in X(S'')$ is in $V(S'')$, where V is a certain open of X relatively schematically dense (we have taken $S_0 = S$). Hence the two rational maps from X to itself given by $z \mapsto (za)b$ and $z \mapsto (za')b'$ are equal, whence $(ya)b = (ya')b'$.

Lemma 3.4. *Consider the rational map $\phi : X^3 \dashrightarrow X$ over S defined by $(a,b,c) \mapsto c^{-1}(ab)$. Let U be the domain of definition of ϕ and consider the graph Γ of the morphism $f : U \to X$ induced by ϕ, which is a subscheme of X^4. Then a section $(a,b,c,d) \in X^4(S)$ is in $\Gamma(S)$ if and only if $(a,b) \sim (c,d)$.*

Proof. Note first that the rational map ϕ is the same as the one given by the formula $(a,b,c) \mapsto (xc)^{-1}((xa)b)$ for an arbitrary section $x \in X(S)$. This is the same as saying that one has $c^{-1}(ab) = (xc)^{-1}((xa)b)$ whenever both sides are defined. We leave the verification to the reader.

We show then that the map ϕ is defined at a section $(a,b,c) \in X^3(S)$ if and only if there exists d with $(a,b) \sim (c,d)$, and that one then has $d = \phi(a,b,c)$. Indeed, suppose that $(a,b) \sim (c,d)$. To verify that ϕ is defined, it is permitted to make an fppf-covering base change (Proposition 1.6), and one may therefore assume that there exists a section $x \in X(S')$ such that `(xa)b` and `(xc)d` are defined and equal. It follows (Rule 2) that $(xc)^{-1}((xa)b)$ is defined and equal to d.

Conversely, suppose ϕ is defined at the section (a,b,c) and let $d = \phi(a,b,c)$. Choose $S' \to S$ fppf-covering and a section $x \in X(S')$ such that `(xa)b` and `(xc)d` are defined. We want to show that they are equal. For this, it suffices to show that the two rational maps over S from X to itself, given by $b \mapsto (xa)b$ and $b \mapsto (xc)\phi(a,b,c)$, are the same, which follows from the remark of the first paragraph.

Proposition 3.5. *The equivalence relation $R \rightrightarrows X^2$ is representable, and it is a flat relation of finite presentation, that is, the projections of R onto X^2 are flat morphisms of finite presentation.*

Proof. We may suppose S affine. Since (X, W) is of finite presentation over S, we may descend the whole situation to an S of finite type over $\operatorname{Spec}\mathbb{Z}$, hence noetherian. We may therefore suppose S noetherian. Then it is trivial that the graph Γ of (3.4) is of finite presentation over X^2. The projection $pr_{12}|\Gamma$ is flat because $\Gamma \xrightarrow{\sim} pr_{123}\Gamma$, which is an open of X^3, and the projection $pr_{12} : X^3 \to X^2$ is flat because X is flat over S.

I claim that Γ represents R. Note that there is something to prove, because the domain of definition of a rational map does not in general commute with base change. Let $S'' \to S$. What is clear, by (3.4) applied to `S''`, is that $R(S'') \supset \Gamma(S'')$, because $\phi \times_S S''$ is certainly defined on $U \times_S S''$. Let then $(a,b,c,d) \in R(S'')$. We must show that $(a,b,c,d) \in \Gamma(S'')$. The verification of this is done locally for, say, the étale topology. We may therefore suppose `S''` strictly local, i.e. the spectrum of a henselian ring with separably closed residue field. Moreover, by applying (1.6)

[1160]N.D.E.: to be made precise...

and the usual passage-to-the-limit standard arguments, we reduce to the case S strictly local and $S'' \to S$ local. Suppose we have a section $x \in X(S)$ such that over S'' the products (xa)b and (xc)d are defined. This will imply that $(xc)^{-1}((xa)b)$ is defined, and equal to d. Now there exists an open V of X^3 relatively sch. dense such that $(xc)^{-1}((xa)b)$ is defined if and only if $(a,b,c) \in V(S'')$, and one has $V \subseteq U$. Hence $(a,b,c,d) \in \Gamma(S'')$ if such an x exists. By (1.6) it is permitted to make a base change $S' \to S$ fppf-covering to find such an x. Since S is strictly local, one can[1161] find an $S' \to S$ faithfully flat, local, and finite and a section $x \in X(S')$ which "passes through" an arbitrary closed point of the closed fiber of X/S. Now to say that (xa)b and (xc)d are defined means that over S'', x lies in a certain relatively sch. dense open, which is verified on the closed fiber of $X_{S''}$. Hence it works.

Let now G be the quotient of X^2 by R as a sheaf for the fppf topology. We shall define a composition law on G as follows. Let $(g,g') \in G(S')$ be represented by a section ((a, b), (c, d)) of $X^2 \times X^2(S'')$, $S'' \to S'$ fppf-covering. Suppose moreover that X admits a section x over S'' such that $a(b(cx))$ and $x^{-1}d$ are defined, which is permitted by Rule 1 and (1.7), and we call gg' the class in $G(S')$ represented by the section $(a(b(cx)), x^{-1}d)$ of $X^2(S'')$.

Let us verify that gg' does not depend on the choice of the section x and the representative ((a, b), (c, d)). Indeed, let $(a',b') \sim (a,b)$, $(c',d') \sim (c,d)$, and x' be such that a'(b'(c'x')) and $x'^{-1}d'$ are defined. We may suppose that all are sections over S''. We must show that

(a(b(cx)), x⁻¹d) ~ (a'(b'(c'x')), x'⁻¹d'),

that is, that for a suitable section $z \in X(S''')$, $S''' \to S''$ suitably fppf-covering,

$$(z(a(b(cx))))(x^{-1}d) = (z(a'(b'(c'x'))))(x'^{-1}d').$$

Whenever all the products are defined, one has

$$(z(a(b(cx))))(x^{-1}d) = ((za)(b(cx)))(x^{-1}d) == (((za)b)(cx))(x^{-1}d) = ((((za)b)c)x)(x^{-1}d) == (((za)b)c)(x(x^{-1}d))$$

and the same identities are true with the primes. Now by Rule 1 and (1.7) there exists such a z. One must therefore show that

(((za)b)c)d = (((za')b')c')d'.

But $(za)b = (za')b'$ because $(a,b) \sim (a',b')$ (3.3) and one has the desired equality because $(c,d) \sim (c',d')$.

Consider the natural morphism $i : X \to G$ defined as follows. For $a \in X(S')$, one chooses $S'' \to S'$ fppf-covering and a section $b \in X(S'')$ such that ab^{-1} is defined, and one sets

[1161] combining Exp. VI_A, 1.1.1 and EGA IV₄, 17.16.2.

i(a) = class in G of (ab⁻¹, b).

One easily verifies that this class, which is a priori in $G(S'')$, does not depend on the choice of b and so gives a well-determined element of $G(S')$.

The reader will give himself the pleasure of verifying the following:

Proposition 3.6. *The morphism i commutes with the composition laws of X and of G, i.e. if $a, b \in X(S')$ are sections such that ab is defined, one has $i(a)i(b) = i(ab)$.*

The goal of this section is the following theorem:

Theorem 3.7.[1162] *Let (X, W) be a group germ over S, with X/S faithfully flat, of finite presentation, and with separated fibers without embedded component. Then with the notations above one has*

(i) G is a sheaf of groups.

(ii) $i : X \to G$ is representable by an open immersion.

(iii) G is representable locally on S for the fppf topology.

(iv) If G/S is representable, then it is a flat group of finite presentation, and $i : X \to G$ is schematically dense relatively to S.

Note that G is evidently characterized by properties (i), …, (iv); one may therefore forget the explicit construction of G.

The proof proceeds in several stages:

Lemma 3.8. *(i) Let c be a section of $X(S)$; then the morphism from the prescheme $c \times X$ (which is S-isomorphic to X) into G given by $c \times X \hookrightarrow X \times X \to G$ is a monomorphism.*

(ii) Let c_i, $i \in I$, be sections of $X(S)$, let $Z = \sqcup_i c_i \times X$, and call $R^0 \rightrightarrows Z$ the equivalence relation induced on Z by the obvious morphism $Z \to X^2$. Then R^0 is a "gluing" of the $c_i \times X \simeq X$.

Proof.

(i) For two sections (c, a) and (c, a') of $(c \times X)(S')$ to have the same image in G, one must have $(c, a) \sim (c, a')$, that is, $(xc)a \sim (xc)a'$ for a suitable $x \in X(S'')$, $S'' \to S'$ fppf-covering, whence $a = a'$ by Rule 3.

(ii) Let c_i, c_j be two sections and consider the birational map ψ_{ji} of X to itself over S given by the formula $x \mapsto c_j^{-1}(c_i x)$. This is the same map as the one given by $x \mapsto (yc_j)^{-1}((yc_i)x)$, for $y \in X(S)$, as one easily sees. Moreover, one verifies that ϕ_{ji} is defined at $b \in X(S')$ if and only if there exists b' such that $(c_i, b) \sim (c_j, b')$, and then $b' = \phi_{ji}(b)$. Let U_{ji} be the domain of definition over S of ϕ_{ji}. It remains to show that this domain of definition is universal, i.e. that

[1162] N.D.E.: If X is smooth, separated over S, faithfully flat of finite presentation over S, then G/S is representable by an S-group scheme smooth and of finite type over S. This is Theorem 6.6.1 of the book *Néron models* of Bosch–Lütkebohmert–Raynaud, Springer (1990).

if $b \in X(S'')$, $S'' \to S$ arbitrary, and if ϕ_{ji} is defined at b, then $b \in U_{ji}(S'')$. It comes to the same to show that if $b, b' \in X(S'')$ are such that $(c_i, b) \sim (c_j, b')$, then $b \in U_{ji}(S'')$. We leave the verification of this fact, which is analogous to that of (3.5), to the reader.

Lemma 3.9. *Suppose that c_i, $i \in I$, are sections of $X(S)$ such that $\sqcup_i c_i \times X \to G$ is surjective as a morphism of sheaves. Then G is representable and flat, of finite presentation over S, and the structural morphism $X^2 \to G$ is flat and of finite presentation.*

Proof. The fact that G is representable is an immediate consequence of (3.8), and it follows that $\sqcup_i c_i \times X \to G$ is an open covering. To show that $X^2 \to G$ is flat, it suffices to do so locally, hence to show that the rational map $X^2 \dashrightarrow c_i \times X$ induces a flat morphism on its domain of definition. Now this rational map is given by $(a, b) \mapsto (c_i, ((xc_i)^{-1}(xa))b)$, x an arbitrary section, and if it is defined at $(a, b) \in X^2(S')$, one can find $S'' \to S'$ fppf-covering and a section $x \in X(S'')$ such that $((xc_i)^{-1}(xa))b$ is defined. One sees easily that this is therefore a flat morphism.

It similarly follows that it is locally of finite presentation, hence fppf-covering. Now by construction, the relation R is effective. Hence by (3.5), the morphism $X^2 \to G$ becomes of finite presentation after the base change $G \leftarrow X^2$, which is fppf-covering; hence $X^2 \to G$ is of finite presentation. Let us show that $G \to S$ is flat and of finite presentation. It is flat and locally of finite presentation since G is covered by the $c_i \times X \simeq X$. Now X^2/S is quasi-compact, and $X^2 \to G$ is surjective. This shows that G/S is quasi-compact. To show that $G \to S$ is quasi-separated, note that one has the following cartesian diagram

$$\begin{tikzcd}
R \arrow[r, "\alpha"] \arrow[d, "\gamma"'] & X^2 \times_S X^2 \arrow[d, "\beta"] \\
G \arrow[r, "\Delta"] & G \times_S G.
\end{tikzcd}$$

One has γ surjective and α, β quasi-compact, hence $\beta\alpha$ is quasi-compact, hence Δ is quasi-compact.[1163]

Lemma 3.10. *Let c_i, $i \in I$, be sections of $X(S)$. For $\sqcup_i c_i \times X \to G$ to be surjective as a morphism of fppf-sheaves, it suffices that the following condition be satisfied:*

For each $S' \to S$ and $(a, b) \in X^2(S')$, there exist an open covering S'_ν, $\nu \in N$, of S' and a function $N \to I$ ($\nu \mapsto i(\nu)$) such that $(c_{i(\nu)}^{-1} a)b$ is defined on S'_ν.

Proof. Let $S'' \to S$ be arbitrary, and $g \in G(S'')$. Choose $S' \to S''$ fppf-covering and a section $(a, b) \in X^2(S')$ representing g. Take the open covering S'_ν of S' that exists by the hypothesis of the lemma. Then on each S'_ν one has $(a, b) \sim (c_{i(\nu)}, (c_{i(\nu)}^{-1} a)b)$,

[1163] N.D.E.: Another way to conclude here is by faithfully flat descent (EGA IV_2, 2.7.1), since β is fppf-covering.

hence g is represented by a section of $[c_{i(\nu)} \times X](S')$ on S'_ν, which proves surjectivity, because the family of morphisms $S'_\nu \to S''$ is fppf-covering.

Lemma 3.11. *Let Y/S be a prescheme of finite presentation, and let a_i, $i \in I$, be sections of $Y(S)$. Let s_0, s_1 be points of S such that s_0 is a specialization of s_1, and Y_j the fiber of Y/S at the point s_j. Let C_j be the closure in Y_j of the set of points $a_i(s) \cap Y_j$. Then one has $\dim C_1 \geqslant \dim C_0$.*

Proof. It suffices to make the verification after a base change $S' \to S$ with chosen points s'_0 and s'_1 such that $s'_j \mapsto s_j$ and s'_0 is a specialization of s'_1. One is

therefore (EGA II 7.1.4) reduced to the case where S is the spectrum of a valuation ring A, s_0 the closed point of S, and s_1 the generic point of S. Now let V be the closure of C_1 in Y. It is clear that $C_0 \subseteq V$, and so the lemma is a consequence of the "well-known" fact that an irreducible closed subprescheme V of a prescheme Y/S of finite presentation satisfies $\dim V \times_S s_1 \geqslant \dim V \times_S s_0$ if S is the spectrum of a valuation ring and if $V \times_S s_1 \neq \emptyset$ (EGA IV 13.1.6).

Lemma 3.12. *Suppose that S is the spectrum of a local ring with closed point s_0, and let c_i, $i \in I$, be sections such that the closure C_0 of the set $c_i(s) \cap X_0$ in the closed fiber X_0 is of dimension equal to $\dim X_0 = n$. Then the condition of Lemma 3.9 is satisfied.*

Proof. Note first that the fibers of X/S all have the same dimension n, which results from EGA IV 12.1.1 (i) and from the fact that X has a rational composition law. Lemma (3.11) therefore implies that for each morphism $S_1 \to S$ with S_1 the spectrum of a field, the dimension of the closure of the set $c_i \times_S S_1$ in X_{S_1} is equal to n. Let us verify the condition of (3.10). Let $(a, b) \in X^2(S')$. For $(c_i^{-1}a)b$ to be defined, it is necessary and sufficient that c_i be contained in a certain open $U \subseteq X_{S'}$ which is sch. dense relatively to S' (Rule 1). We must show that this is true for a suitable i, locally on S'. It therefore suffices to treat the case where S' is the spectrum of a local ring, and then the fact that $c_i \in U(S')$ is verified on the closed fiber. We are thus reduced to the case $S' = \operatorname{Spec} k$, k a field. Now with the notations above, take $S' = S_1$.

One has $\dim C_1 = \dim X_{S_1}$, and U is relatively sch. dense in X_{S_1}. Hence $U \cap C_1 \neq \emptyset$, whence $U \cap c_p \times_S S_1 \neq \emptyset$, and we are done.

The proof of the theorem is now easy. Note first the following consequence of the finiteness of Lemma (3.9): if A_i is an inductive system of rings over S, if $\tilde{A} = \lim A_i$, and if the hypotheses of (3.9) are satisfied for $S = \operatorname{Spec} \tilde{A}$, then one may descend the object that represents the quotient G of $R \rightrightarrows X^2$ to one of $S_i = \operatorname{Spec} A_i$ with the finiteness and flatness properties stated in (3.9). This is the usual passage to the limit (EGA IV 8 and 11). It follows that for the proof of (iii) and (iv) of (3.7), one may restrict to the case $S = \operatorname{Spec} A$, with A a strictly local ring. Let then x_0 be a closed point of the closed fiber X_0 of X/S. There exists[1164] an extension A' of A, local, free and finite, and a section of $X' = X \times_S S'$ passing through the unique point x'_0 of X' above x_0. Note that $X'_0 \to X_0$ is radicial since S is strictly local, hence

[1164] cf. note at the bottom of page 15, Exp. VII.

the residue field of A separably closed. It follows that there exists an inductive system A_i of local rings, flat and finite over S, such that, setting $\tilde{A} = \lim A_i$ and $\tilde{X} = X \times_S \operatorname{Spec} \tilde{A}$, $\tilde{X}$ has a set of sections which induces a dense set on the closed fiber $\tilde{X}_0$. For each closed point x_0 of X_0 one takes an extension $A(x_0)$ such that the corresponding base change of X admits a section "passing through x_0", and one takes as inductive system the system of finite tensor products of the $A(x_0)$. Note that Ã is local, being a limit of local rings. Hence by (3.12) and (3.10) one has the quotient G

over $\operatorname{Spec} \tilde{A}$, hence over one of $\operatorname{Spec} A_i$ by the remarks above, hence locally for the fppf topology.

In fact it follows from the constructions that locally for the fppf topology one can find a finite set of sections c_i $(i = 1, \cdots, r)$ such that G is covered by the $c_i \times X$. (ii) and (iv) follow easily from this fact, and we leave the verification of (i) to the reader.

Corollary 3.13. *The sheaf of groups G determined by a group germ (X, W) over S is representable in each of the following situations:*

(i) S is artinian.

(ii) For each local scheme $S' \to S$ at a closed point s of S, $X_{S'}$ has a set of sections that induces on the closed fiber a set whose closure is of dimension $\dim X_s$.

(iii) S is strictly local, and X/S is smooth.

(iv) There exists $S' \to S$ fppf-covering such that $G_{S'}$ is representable and affine over S.

Indeed, (ii) is a consequence of (3.10) and (3.12), (iii) follows directly from (ii) and "Hensel's lemma", (iv) from the descent of affine schemes, and (i) from the descent of group schemes, which is possible here because one knows that every finite subset of a group over a field is contained in an affine open (Exp. VI).

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Footnotes

Exposé XIX. Reductive groups: generalities

by M. Demazure

[1165] The remainder of this Séminaire is devoted to the study of reductive groups. Its principal aim is the generalization of Chevalley's classical results (*Bible* and

[1165] *N.D.E.*: Version of 13/10/2024.

Tôhoku)[1166] to arbitrary base schemes, the two central results being the uniqueness theorem (Exp. XXIII, Th. 4.1 and Cor. 5.1 to 5.10) and the existence theorem (Exp. XXV, Th. 1.1) for "pinned" reductive group schemes corresponding to prescribed "root data". The proofs employed are inspired by Chevalley's, the technique of schemes permitting one to lend them increased efficiency.

The results of the first volume of the *Bible* (Exp. 1 to 13) will be used systematically. By contrast, we shall prove directly over an arbitrary scheme the results of the second volume (in particular the isomorphism theorem); a knowledge of the proofs over an algebraically closed field is therefore not absolutely indispensable.

In the proof of these two fundamental results, we shall use only the most elementary results from the theory of groups of multiplicative type, contained essentially in Exp. VIII and IX; we shall, on the other hand, make essential use of the results of Exposé XVIII.[1167] The reader specifically interested in the existence and uniqueness theorems may, on a first reading, skip Exposés X to XVII.

1. Reminders on groups over an algebraically closed field

1.1. In this number, k will always denote an algebraically closed field. As announced above, the only results from *Bible* used in what follows are found in volume 1 (Exposés 1 to 13). All the results of *Bible* (*loc. cit.*) concerning semisimple groups are valid more generally for reductive groups (definition below) and their proof is identical, with the following innocuous modifications:

- Exp. 9, § 4, Definition 3: see 1.6.1 below.
- Exp. 12, § 4, Theorem 2, e): suppress "finite".
- Exp. 13, § 3, Theorem 2: replace "rank" by "semisimple rank".
- Exp. 13, § 4, Corollary 2 to Theorem 3: replace "rank" by "reductive rank".

1.2. Let G be a smooth affine connected k-group. The *radical* of G (*Bible*, § 9.4, Prop. 2)[1168] is the reduced subgroup associated with the neutral component of the intersection of the Borel subgroups of G; it is also the largest smooth connected solvable invariant subgroup of G; we shall denote it $rad(G)$.

The *unipotent radical* of G is the unipotent part of the radical of G; it is also the largest smooth connected unipotent invariant subgroup of G; we shall denote it $rad_u(G)$.

1.3. Let G be a smooth affine connected k-group, and Q a torus of G. Then the centralizer $Centr_G(Q)$ of Q in G is a closed subgroup of G (Exp. VIII, 6.7), smooth (Exp. XI, 2.4), and connected (cf. *Bible*, § 6.6, Th. 6 a) or [Ch05], § 6.7, Th. 7 a)).

[1166] *N.D.E.*: See the bibliographical references at the end of this Exposé. In particular, a re-edition of the *Séminaire Chevalley 1956–58*, cited [Bible], revised by P. Cartier, was published in 2005, cf. [Ch05].

[1167] *N.D.E.*: More precisely, Proposition XVIII 2.3 (extension of a "generic homomorphism" between groups) is used in XXII 4.1.11 and then in Exp. XXIII (proof of the uniqueness theorem), and also in Exp. XXIV; Theorem XVIII 3.7 (construction of a group from a group germ) is used only in Exp. XXV.

[1168] *N.D.E.*: The reference 9-06 (= Exp. 9, p. 6) has been replaced by § 9.4 (= Exp. 9, § 4), which applies equally well to [Bible] and to [Ch05]. When the numbering of [Ch05] differs from [Bible], which is the case in Exp. 6, both references will be indicated explicitly.

One has the fundamental relation:

$$rad_u(Centr_G(Q)) = rad_u(G) \cap Centr_G(Q).$$

First,[1169] $U = rad_u(G) \cap Centr_G(Q)$ is an invariant unipotent subgroup of $Centr_G(Q)$; let us show that it is smooth and connected. If we make Q operate on $rad_u(G)$ by inner automorphisms, U is none other than the scheme of invariants of this operation. But this scheme is smooth and connected, by Lemma 1.4 below.

Consequently, U is a closed subgroup of $rad_u(Centr_G(Q))$. On the other hand, by *Bible*, Exp. 12, § 3, Cor. to Th. 1, one has $rad_u(Centr_G(Q))(k) \subset rad_u(G)(k)$. The equality (1.3.1) follows.

Let us signal a particular case of the foregoing result: if G is a smooth affine connected k-group and if T is a maximal torus of G, then

$$Centr_G(T) = T \cdot (Centr_G(T) \cap rad_u(G)).$$

Indeed (cf. *Bible*, § 6.6, Th. 6 c) or [Ch05], § 6.7, Th. 7 c)), $Centr_G(T)$ is a solvable group, hence the semi-direct product of its maximal torus T and its unipotent radical.

By the density theorem (cf. *Bible*, § 6.5, Th. 5 a) or [Ch05], § 6.6, Th. 6 a)), the union of the $Centr_G(T)$, as T runs through the set of maximal tori of G, contains a dense open subset of G; it therefore follows from (1.3.2):

Corollary 1.3.3. *If G is a smooth affine connected k-group, then the union of the $T \cdot rad_u(G)$, where T runs through the set of maximal tori of G, contains a dense open subset of G.*

Notations 1.4.0. [1170] *We shall systematically use the following notation: if S is a scheme and s a point of S, we denote by $\bar{s}$ the spectrum of an algebraic closure $\kappa\bar{s}$ of $\kappa(s)$.*

Lemma 1.4. [1171] *Let S be a scheme, Q an S-torus operating on an S-group scheme H, separated and smooth over S.*

(i) The functor of invariants H^Q is representable by a closed subscheme of H, smooth over S.

(ii) If moreover H is affine over S, and has connected fibers, then H^Q does too.

Proof. First, by VIII 6.5 (d),[1172] since H is separated over S and Q essentially free over S, H^Q is representable by a closed subscheme of H. In particular, if H is affine over S, then so is H^Q.

[1169] *N.D.E.*: The following has been detailed.

[1170] *N.D.E.*: These notations, which figured in 2.3 of the original, have been inserted here; on the other hand, the statement and proof of 1.4 have been detailed.

[1171] *N.D.E.*: These notations, which figured in 2.3 of the original, have been inserted here; on the other hand, the statement and proof of 1.4 have been detailed.

[1172] *N.D.E.*: See also the additions made in VI_B, 6.2.4 (d) and 6.5.5 (a).

Consider now the semi-direct product $G = H \cdot Q$; it is smooth and separated over S. Then, the centralizer $Centr_G(Q)$ is representable by a closed subscheme of G, of finite presentation over G, by Exp. XI 6.11 (a),[1173] and it is smooth over S by Exp. XI 2.4.

Since $Centr_G(Q) = H^Q \cdot Q$, it follows that H^Q is smooth over S. (Indeed, using the section $\sigma : H^Q \to Centr_G(Q)$ deduced from the unit section ϵ_Q of Q, one sees at once that H^Q is formally smooth over S; and since moreover ϵ_Q is of finite presentation, H^Q is locally of finite presentation over S.) This proves (i).

Finally, suppose H affine over S and with connected fibers. Then each geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa\bar{s}$-group, hence, by the first assertion of 1.3, so is

$$Centr_{G_{\bar{s}}}(Q_{\bar{s}}) = (Centr_G(Q))_{\bar{s}} = (H^Q)_{\bar{s}} \cdot Q_{\bar{s}},$$

and this entails that $(H^Q)_{\bar{s}}$ is connected.

1.5. We recall that the *reductive rank* of the smooth affine k-group G is the common dimension of the maximal tori of G. We shall denote it $rgred(G/k)$ or $rgred(G)$. In order that $rgred(G/k) = 0$, it is necessary and sufficient that G be unipotent (i.e. that $G = rad_u(G)$), by *Bible*, § 6.4, Cor. 1 to Th. 4 or [Ch05], § 6.5, Cor. 1 to Th. 5.

If H is an invariant subgroup of the smooth affine k-group G, then the quotient G/H is affine and smooth (Exp. VI_B, 11.17 and VI_A, 3.3.2 (iii)). Moreover (*Bible*, § 7.3, Th. 3, a) and c)), one has

$$rgred(G) = rgred(G/H) + rgred(H).$$

Definition 1.6.1. [1174] *One says that the k-group G is* reductive *if it is smooth, affine, and connected, and if $rad(G)$ is a torus, i.e. if $rad_u(G) = e$.*

Lemma 1.6.2. *Let G be a reductive k-group.*

(i) If Q is a torus of G, then $Centr_G(Q)$ is reductive.

(ii) In particular, if T is a maximal torus of G, then $Centr_G(T) = T$.

(iii) The center of G is contained in every maximal torus, hence is diagonalizable.

(iv) The radical of G is the (unique) maximal torus of $Centr(G)$.

Proof. Indeed, (i) follows from (1.3.1), (ii) from (1.3.2), and (iii) follows from (ii). Finally, the maximal torus of $Centr(G)$ (i.e. the neutral component $Centr(G)^0$) is a smooth connected solvable subgroup, invariant in G, hence contained in $rad(G)$; conversely, $rad(G)$ is an invariant torus in G, hence central (*Bible*, § 4.3, Cor. to Prop. 2), whence (iv).

[1173] *N.D.E.*: See also the additions made in VI_B, 6.2.4 (d) and 6.5.5 (a).

[1174] *N.D.E.*: 1.6 has been transformed into 1.6.1 to 1.6.3. Note that in this Séminaire every reductive (resp. semisimple, cf. 1.8) k-group is, by definition, connected.

Remark 1.6.3. *If G is reductive and if* $\dim(G) \neq rgred(G)$*, then* $\dim(G) - rgred(G) \geqslant 2$*. Indeed, this difference is always even (cf. 1.10 below).*

1.7. Let G be a smooth affine connected k-group and H a smooth connected invariant subgroup. Then

$$rad(H) = (rad(G) \cap H)^0_{red} \quad \text{and} \quad rad_u(H) = (rad_u(G) \cap H)^0_{red},$$

as one sees at once. In particular, if G is reductive, so is H.

Let $f : G \to G'$ be a surjective morphism of smooth affine connected k-groups. Then

$$f(rad_u(G)) = rad_u(G').$$

In particular, if G is reductive, so is G'.

Let us prove (1.7.2). First, f sends $rad_u(G)$ into $rad_u(G')$. Introducing $H = (f^{-1}(rad_u(G')))^0_{red}$, which contains $rad_u(G)$, one has $rad_u(H) = rad_u(G)$ and one is reduced to the case where $G = H$, i.e. where G' is unipotent. Since the union of the $T \cdot rad_u(G)$ (as T runs through the set of maximal tori of G) is dense in G, the union of the $f(T)f(rad_u(G))$ is dense in G'; but $f(T)$ consists of semisimple elements, so $f(T) = e$, G' being unipotent; this shows that $f(rad_u(G))$ is dense in G'. Therefore, by *Bible*, § 5.4, Lemma 4 or [Ch05], § 6.1, Lemma 1,[1175] $f(rad_u(G))$ is an open subgroup of G'; the latter being connected, it follows that $f(rad_u(G)) = G'$. (N.B.: one can prove under the same hypotheses that $f(rad(G)) = rad(G')$.)

1.8. One says that the k-group G is *semisimple* if it is smooth, affine, and connected and if $rad(G) = e$, i.e. if G is reductive and $Centr(G)$ is finite. If G is an arbitrary smooth affine connected k-group, then $G/rad(G)$ is semisimple (*Bible*, § 9.4, Prop. 2), and $G/rad_u(G)$ is reductive. One calls the *semisimple rank of* G and denotes by $rgss(G/k)$ or $rgss(G)$ the reductive rank of $G/rad(G)$.

If G is reductive, one therefore has

$$rgred(G) = rgss(G) + \dim(rad(G)).$$

If G is a smooth affine connected k-group and if Q is a central subtorus of G, then G/Q is semisimple if and only if G is reductive and $Q = rad(G)$. Indeed, if G/Q is semisimple, $Q \supset rad(G)$, so $rad(G)$ is a torus, so G is reductive, so $rad(G)$ is the maximal torus of $Centr(G)$, so $rad(G) = Q$. If G is reductive and if Q is a central subgroup then (G/Q is reductive and) $rgss(G) = rgss(G/Q)$.

1.9. If K is an algebraically closed extension of k and if G is a smooth affine connected k-group, then G is reductive (resp. semisimple) if and only if G_K is, and one has

$$rgred(G/k) = rgred(G_K/K), rgss(G/k) = rgss(G_K/K).$$

[1175] *N.D.E.*: The following has been detailed.

1.10. Let G be a smooth connected k-group and let T be a torus of G.[1176] Denote by g the k-Lie algebra of G, i.e. $g = Lie(G) = Lie(G/k)(k)$; likewise denote $t = Lie(T)$. Then g decomposes under the action of T (via $T \to G$ and the adjoint operation of G) as

$$g = g_0 \oplus \bigoplus_{\alpha \in R} g_\alpha,$$

where the $\alpha \in R$ are non-trivial characters of T and the g_α are $\neq 0$. The characters $\alpha \in R$ are called the *roots* of G with respect to T. By Exp. II, 5.2.3 (i), one has

$$g_0 = Lie(Centr_G(T)).$$

In particular,[1177] since $Centr_G(T)$ is connected (1.3 in the case where G is affine, and Exp. XII 6.6 b) in the general case), T is its own centralizer if and only if $g_0 = Lie(T)$.

This condition entails that T is maximal and that $Centr(G) \subset T$, hence by Exp. XII 8.8 d) one has:

$$Centr(G) = Ker(T \to GL(g)) = \bigcap_{\alpha \in R} Ker(\alpha);$$

moreover $Centr(G)$ is then affine, hence $G \to G/Centr(G)$ is affine; since the latter group is affine (Exp. XII 6.1),[1178] so is G.

When G is reductive and T maximal, the roots in the preceding sense coincide with the roots in the sense of *Bible*, § 12.2, Def. 1; the latter are indeed roots in the sense of this number (*Bible*, § 13.2, Th. 1, c)) and there are $\dim(G) - \dim(T)$ of them (*Bible*, § 13.4, Cor. 2 to Th. 3). Moreover, if α is a root, so is $-\alpha$ (*Bible*, § 12.2, Cor. to Prop. 1). (As usual, one writes the group structure of $\mathrm{Hom}_{k-gr.}(T, G_{m,k})$ indifferently additively or multiplicatively.) It follows that, for G reductive,

$$\dim(G) - rgred(G) = Card(R)$$

is always even.

The semisimple rank of the reductive group G is the rank of the subset R of the Q-vector space $\mathrm{Hom}_{k-gr.}(T, G_{m,k}) \otimes Q$ (*Bible*, § 13.3, Th. 2).

Lemma 1.11. *Let k be an algebraically closed field, G a smooth affine connected k-group, T a torus of G, $W(T) = Norm_G(T)/Centr_G(T)$ the Weyl group of G with respect to T. The following conditions are equivalent:*

(i) G is reductive, T is maximal, $rgss(G) = 1$.

(ii) G is reductive, T is maximal, $G \neq T$; there exists a subtorus Q of T, of codimension 1 in T, central in G.

(iii) G is not solvable and $\dim(G) - \dim(T) \leqslant 2$.

[1176] *N.D.E.*: The following sentence has been added.
[1177] *N.D.E.*: The following has been detailed.
[1178] *N.D.E.*: See also the addition VI_B, 6.2.6.

(iv) $W(T) \neq e$ *and* $\dim(G) - \dim(T) \leqslant 2$*.*

(v) $W(T) = (Z/2Z)_k$ *and* $\dim(G) - \dim(T) = 2$*.*

Moreover, under these conditions, there are exactly two roots of G with respect to T; they are opposite. Under the conditions of (ii), $Q = rad(G)$.

Proof. One has obviously (v) ⇒ (iv). One has (iv) ⇒ (iii) by *Bible*, § 6.1, Cor. 3 to Th. 1 or [Ch05], § 6.2, Cor. 3 to Th. 2. Let us prove (iii) ⇒ (ii): let U be the unipotent radical of G; one knows that G/U is reductive and is not a torus (otherwise G would be solvable); one therefore has, by (1.10.4),

$$\dim(G/U) - \mathrm{rgred}(G/U) = \mathrm{Card}(R) \geqslant 2;$$

but

$$\mathrm{rgred}(G) = \mathrm{rgred}(G/U) + \mathrm{rgred}(U) = \mathrm{rgred}(G/U),$$

whence

$$\begin{aligned}\dim(G) - \dim(U) - \mathrm{rgred}(G) &= \dim(G/U) - \mathrm{rgred}(G/U)\\ &\geqslant 2 \geqslant \dim(G) - \dim(T) \geqslant \dim(G) - \mathrm{rgred}(G).\end{aligned}$$

This gives $\dim(U) = 0$, hence G is reductive, $rgred(G) = \dim(T)$, hence T is maximal, $\dim(G) - \dim(T) = 2$. By *Bible*, § 10.4, Prop. 8, there exists a singular torus Q of codimension 1 in T; then $Centr_G(Q)$ is reductive (1.6.2 (i)), non-solvable (definition of a singular torus), hence $\dim(Centr_G(Q)) - \mathrm{rgred}(G) \geqslant 2$, which proves that $G = Centr_G(Q)$ and Q is central in G.

Let us prove (ii) ⇒ (i). One knows that G/Q is reductive (1.7) and that $rgred(G/Q) = 1$, hence $rgss(G/Q) = 0$ or 1. The first case is impossible, for it would entail $rgss(G) = 0$, hence $G = T$;

one therefore has $rgred(G/Q) = rgss(G/Q) = 1$, which proves that G/Q is semisimple; hence Q is the radical of G and $rgss(G) = 1$.

Let us finally prove (i) ⇒ (v). If Q is the radical of G, one has $\dim(T) - \dim(Q) = 1$ and Q is central in G, hence $G = Centr_G(Q)$, which proves that Q is a singular torus; by *Bible*, § 11.3, Th. 2, one has $W_G(T) = (Z/2Z)_k$; by *Bible*, § 12.1, Lemma 1, one has $\dim(G) - \dim(T) = 2$. There are therefore at most two roots of G with respect to T; but there are at least two, opposite to each other (1.10).

Proposition 1.12. *Let k be an algebraically closed field, G a smooth connected k-group, T a torus of G, R the set of roots of G with respect to T, and*

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \quad \text{with } \mathfrak{g}_\alpha \neq 0,$$

the decomposition of $\mathfrak{g}$ under $Ad(T)$. For each $\alpha \in R$, let T_α be the maximal torus of $Ker(\alpha)$[1179] *and $Z_\alpha = Centr_G(T_\alpha)$. The following conditions are equivalent:*

(i) G is affine, reductive; T is maximal.

(ii) $\mathfrak{g}_0 = \mathfrak{t}$, each Z_α ($\alpha \in R$) is reductive.

[1179] *N.D.E.*: i.e., the connected component of $Ker(\alpha)$.

(iii) $g_0 = t$, each g_α ($\alpha \in R$) is of dimension 1; and if $\alpha, \beta \in R$ and $q \in Q$ are such that $\beta = q\alpha$, then $q = \pm 1$; for each $\alpha \in R$, there exists $w_\alpha \in G(k)$ which normalizes T, centralizes T_α, but does not centralize T.

Moreover, under these conditions, each Z_α is of semisimple rank 1 and one has $Lie(Z_\alpha) = t \bigoplus g_\alpha \bigoplus g_{-\alpha}$.

Proof. If G is affine and reductive and if T is maximal, each Z_α is affine and reductive (1.6.2 (i)), with maximal torus T; moreover, T is its own centralizer (1.6.2 (ii)), hence $g_0 = t$, which proves (i) ⇒ (ii).

On the other hand $g_0 = t$ entails that T is maximal and G affine (cf. 1.10), hence each Z_α is affine, smooth, and connected, by 1.3. In any case, one has, by Exp. II, 5.2.2,

$$\mathrm{Lie}(Z_\alpha) = g^{T_\alpha} = g_0 \oplus \bigoplus_{\beta \in R \cap Q\alpha} g_\beta.$$

One therefore has

$$\mathrm{Lie}(Z_\alpha) \supset t \oplus g_\alpha \oplus g_{-\alpha},$$

which entails $Z_\alpha \neq T$. Since T_α is a subtorus of codimension 1 in T, central in Z_α, one obtains by 1.11, applied to Z_α, the equivalence (ii) ⇔ (iii) and the supplements.

It remains to prove (ii) ⇒ (i); one already knows that (ii) entails that T is maximal and G affine. Let U be its unipotent radical; it is invariant in G, its Lie algebra is invariant under $Ad(T)$. One therefore has

$$\mathrm{Lie}(U) = (\mathrm{Lie}(U) \cap g_0) \oplus \bigoplus_{\alpha \in R} (\mathrm{Lie}(U) \cap g_\alpha).$$

By (1.3.1), one has

$$U \cap T = U \cap \mathrm{Centr}_G(T) = \mathrm{rad}_u(\mathrm{Centr}_G(T)) = \mathrm{rad}_u(T) = \{e\},$$
$$U \cap Z_\alpha = U \cap \mathrm{Centr}_G(T_\alpha) = \mathrm{rad}_u(\mathrm{Centr}_G(T_\alpha)) = \mathrm{rad}_u(Z_\alpha) = \{e\}.$$

One therefore has[1180]

$$\mathrm{Lie}(U) \cap g_0 = \mathrm{Lie}(U) \cap t = \mathrm{Lie}(U \cap T) = 0,$$
$$\mathrm{Lie}(U) \cap g_\alpha \subset \mathrm{Lie}(U) \cap \mathrm{Lie}(Z_\alpha) = \mathrm{Lie}(U \cap Z_\alpha) = 0;$$

whence $Lie(U) = 0$ and $U = e$, i.e. G is reductive.

Corollary 1.13. *Let k be an algebraically closed field, G a smooth connected k-group, T a torus of G, R the set of roots of G with respect to T, and*

$$g = g_0 \oplus \bigoplus_{\alpha \in R} g_\alpha$$

as above. For each $\alpha \in R$, let T_α be the maximal torus of $Ker(\alpha)$ and $Z_\alpha = Centr_G(T_\alpha)$. The following conditions are equivalent:

(i) G is affine, semisimple; T is maximal.

(ii) $g_0 = t$, each Z_α is reductive, and $\bigcap_{\alpha \in R} Ker(\alpha)$ is finite.

[1180] *N.D.E.*: Note that if U, L are two subgroup schemes of G, one has $\mathrm{Lie}(U) \cap \mathrm{Lie}(L) = \mathrm{Lie}(U \cap L)$.

[1181] This follows from the equalities $rad(G) = Centr(G)^0$ and $Centr(G) = \bigcap_{\alpha \in R} Ker(\alpha)$, established in 1.6.2 (iv) and (1.10.3).

2. Reductive group schemes. Definitions and first properties

Scholie 2.1. *If G is a group scheme over S, the following properties are equivalent:*

(i) G is affine and smooth over S, with connected fibers.

(ii) G is affine and flat over S, of finite presentation over S, with geometrically integral fibers.

These properties are stable under base change and local for (fpqc).

[1182] Indeed, suppose (i) is satisfied. Since G is affine and smooth over S, it is of finite presentation over S; and since its fibers are smooth and connected, they are geometrically integral, by VI_A, 2.4.

Conversely, if (ii) is satisfied, the fibers of G are connected and geometrically reduced, hence smooth (VI_A, 1.3.1); then G is smooth over S, by EGA IV_4, 17.5.2.

Of course, these properties are stable under base change: cf. EGA II, 1.5.1 for "affine", IV_1, 1.6.2 (iii) for "of finite presentation", IV_2, 2.1.4 for "flat", and IV_2, 4.6.5 (i) for "with geometrically integral fibers".

On the other hand, these properties are clearly local for the Zariski topology, so it suffices to verify that if $S' \to S$ is a faithfully flat quasi-compact morphism and if $G_{S'} \to S'$ has the indicated properties, then so does $G \to S$. This follows from EGA IV_2, 2.5.1 for "flat", 2.7.1 (vi) and (xiii) for "of finite presentation" and "affine", and 4.6.5 (i) for "with geometrically integral fibers" (and also EGA IV_4, 17.7.3 (ii) for "smooth").

2.2. Let G be an S-group scheme satisfying the preceding conditions, and Q a torus (cf. IX, Def. 1.3) of G.[1183] Then, by XI, 6.11 a) and XI, 2.4, $Centr_G(Q)$ is representable by a closed subgroup scheme of G (hence affine over S), of finite presentation and smooth over S; moreover, since each geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa\bar{s}$-group, then, by the first assertion of 1.3, so is

$$Centr_{G_{\bar{s}}}(Q_{\bar{s}}) = (Centr_G(Q))_{\bar{s}}.$$

Lemma 2.3. *Let S be a scheme, G an S-group scheme smooth and affine over S, with connected fibers, T a torus of G. The set of $s \in S$ such that G_s is a reductive s-group, of semisimple rank 1 and with maximal torus T_s, is an open subset U of S.*

Proof. Since G and T are flat over S, the function

[1181] *N.D.E.*: The following sentence has been added.
[1182] *N.D.E.*: The following has been added.
[1183] *N.D.E.*: The following has been detailed.

$$s \mapsto \dim(G_s/s) - \dim(T_s/s)$$

is locally constant; let U_1 be the open subset of points of S where it equals 2.

[1184] By 6.3, the Weyl group

$$W_G(T) = Norm_G(T)/Centr_G(T)$$

is representable by an S-group scheme étale and separated over S, and the function

s ↦ number of points of W_G(T)_s

is lower semicontinuous. Let U_2 be the set of points of S where this function is > 1; it is open. By 1.11, the set of s such that G_s is reductive, of semisimple rank 1, with maximal torus T_s, is $U = U_1 \cap U_2$; moreover, for every $s \in U$, $W_G(T)_s$ has exactly two points.

Consequently (cf. SGA 1, I 10.9 and EGA IV₃, 15.5.1 and IV₄, 18.12.4), $W_G(T)_U$ is étale and finite over U.

Remark 2.4. *The group $W_G(T)_U$ is isomorphic to $(Z/2Z)_U$.*

Indeed, by what precedes, it is étale and finite over U; since the functor of automorphisms of $(Z/2Z)_U$ is the trivial group, it suffices to verify the assertion locally; but the assertion is immediate if the algebra A defining $W_G(T)_U$ is a free O_U-module.[1185]

Notations and reminders 2.5.0. [1186] *Let S be a scheme, G an S-group scheme, $\epsilon : S \to G$ the unit section of G. One has seen in II, § 4.11 that the functor $Lie(G/S)$ is representable by the vector fibration $V(\omega^1_{G/S})$ (where $\omega^1_{G/S} = \epsilon * (\Omega^1_{G/S})$), and one denotes*

$$g = Lie(G/S) = (\omega^1_{G/S})^V$$

the sheaf of sections of this vector fibration. Suppose moreover that G is smooth over S; then $\omega^1_{G/S}$ and hence $Lie(G/S)$ are locally free O_S-modules of finite type, and one has (cf. I 4.6.5.1):

Lie(G/S) = W(Lie(G/S)),

[1184] *N.D.E.*: The following has been detailed. Note, on the other hand, that section 6 is independent of the rest of this Exposé.

[1185] *N.D.E.*: The hypothesis entails that A is locally free of rank 2, and since the augmentation ideal I is a direct factor of A, then, replacing U by a sufficiently small affine open subset Spec(R), one may assume that $I = \Gamma(U, I)$ is a free R-module of rank 1. If x is a generator of I, one then has $x^2 = ax$ for a certain $a \in R$, and the hypothesis that $A = \Gamma(U, A)$ be étale entails that a is invertible in R, and one then sees easily that A is the affine algebra of the constant R-group $Z/2Z$ (compare with the first lines of [TO70]).

[1186] *N.D.E.*: This paragraph of notations and reminders has been added.

i.e. for every S-scheme S',

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Lie(G/S)(S′) = Γ(S′, Lie(G/S) ⊗ O_{S′}).
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By II 4.1.1.1, the adjoint action of G endows $Lie(G/S) = W(Lie(G/S))$ with a G-O_S-module structure, hence $Lie(G/S)$ is a G-O_S-module (cf. I 4.7.1). If moreover G is affine over S, this amounts to saying, by I 4.7.2, that $Lie(G/S)$ is an $A(G)$-comodule.

If T is a torus over S (IX Def. 1.3), one says that T is split *("trivial" in the original) if it is isomorphic to $G^r_{m,S}$ for some integer $r \geqslant 0$, and one says that T is* trivialized *if one has fixed such an isomorphism or, more generally, an isomorphism $T \simeq D_S(M)$, where M is a free abelian group of rank r.*

Let us finally recall (XII Def. 1.3) that a torus T of G is called maximal *if, for every $s \in S$, $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.*

Theorem 2.5. *Let S be a scheme, G an S-group scheme affine and of finite presentation over S, with connected fibers, and s_0 a point of S. Suppose G flat over S at $\epsilon(s_0)$ and the geometric fiber $G_{\bar{s}_0}$ (reduced and) reductive (resp. semisimple). Then there exists an open subset U of S containing s_0 and an étale surjective morphism $S' \to U$ such that:*

(i) G_U is smooth over S, with geometrically reductive (resp. semisimple) fibers, of constant reductive rank and semisimple rank.

(ii) $G_{S'}$ possesses a split[1187] *maximal torus T, and the Weyl group (cf. 6.3)*

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W_{G_{S′}}(T) = Norm_{G_{S′}}(T) / Centr_{G_{S′}}(T) = Norm_{G_{S′}}(T) / T
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is finite over S'.

Proof.[1188] Denote by $\pi : G \to S$ the structure morphism and $\epsilon : S \to G$ the unit section. Since G is flat over S at $\epsilon(s_0)$ and $G_{\bar{s}_0}$ reduced (hence smooth over $\bar{s}_0$, cf. VI_A 1.3.1), G is smooth over S at the point $\epsilon(s_0)$ (EGA IV$_4$, 17.5.1), i.e. there exists an open neighborhood V of $\epsilon(s_0)$ such that $\pi|V$ is smooth. Then, $S' = \epsilon^{-1}(V)$ is an open subset of S, and $G_{S'}$ is smooth over S' at the points of $\epsilon(S')$. Since G has connected fibers, $G_{S'}$ is smooth over S', by VI_B, 3.10. So, replacing S by S', one may assume G smooth over S.

By Exp. XI, Th. 4.1, the functor of subgroups of multiplicative type of G is representable by an S-scheme $\mathcal{M}$, smooth and separated over S. Denote by r_0 the rank of $G_{\bar{s}_0}$ and consider the open subscheme $\mathcal{M}_{r_0}$ of $\mathcal{M}$, which represents the functor of subtori of rank r_0 of G (IX 1.4). Smoothness entails that $\mathcal{M}_{r_0}$ admits a rational point over a finite separable extension of $\kappa(s_0)$ (cf. EGA IV$_4$, 17.15.10 (iii)). Thus there exists $S' \to S$ étale equipped with a point s_0' mapping to s_0 such that $G_{s_0'}$ admits a torus of rank r_0. Therefore, replacing S by S', one may assume that G_{s_0} admits a torus of rank r_0.

[1187] *N.D.E.*: "Split" has been added.
[1188] *N.D.E.*: The proof has been detailed.

By "Hensel's lemma" (cf. XI, 1.10), the smoothness of $\mathcal{M}_{r_0}$ permits one to lift this torus to an S'-torus T of G, where $S' \to S$ is étale equipped with a point s_0' mapping to s_0. By Exp. X, 4.5 (see also 6.1[1189]), there exists an étale morphism $f : S'' \to S'$ splitting T and such that $f^{-1}(s_0') \neq \emptyset$. Since an étale morphism is open and the assertions of (i) are local for the étale topology, one may therefore assume that G possesses a split torus T,[1190] maximal at s_0. Write therefore $T = D_S(M)$ and let

$$\mathfrak{g} = \bigoplus_{m \in M} \mathfrak{g}_m$$

be the decomposition of $\mathfrak{g} = Lie(G/S)$ under $Ad(T)$ (Exp. I, 4.7.3).

Put $\mathfrak{t} = Lie(T)$ and, for every $m \in M$, denote $\mathfrak{g}_m(s_0) = \mathfrak{g}_m \otimes_{O_S} \kappa(s_0)$.[1191] Let R be the set of $m \in M$, $m \neq 0$, such that $\mathfrak{g}_m(s_0) \neq 0$.[1192] Since $G_{\bar{s}_0}$ is reductive, one has $\mathfrak{g}_0(s_0) = \mathfrak{t}(s_0)$, hence

$$\mathfrak{g}(s_0) = \mathfrak{t}(s_0) \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^{\alpha}(s_0).$$

Since the modules in question are locally free, one may, by shrinking S if necessary, assume the $\mathfrak{g}_\alpha$ free and

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha.$$

One recalls, cf. Exp. XII, 1.12, that a group of multiplicative type possesses a unique maximal torus (this is moreover essentially trivial by descent, the diagonalizable case being evident). Let then T_α be the maximal torus of the kernel of α and $Z_\alpha = Centr_G(T_\alpha)$.[1193] By 2.2, Z_α is affine and smooth over S, with connected fibers, and by 1.12, its fiber $(Z_\alpha)_{\bar{s}_0}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}_0}$. By 2.3, there therefore exists an open subset U_α of S, containing s_0, such that $Z_\alpha | U_\alpha$ has reductive fibers.

Set $U = \bigcap_{\alpha \in R} U_\alpha$. By 1.12, for every $s \in U$, $G_{\bar{s}}$ is reductive, with maximal torus $T_{\bar{s}}$ and the set of roots of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ is identified with R, so that

$$\mathrm{rgred}(G_{\bar{s}}) = \dim(T) = \mathrm{rg}(M), \qquad \mathrm{rgss}(G_{\bar{s}}) = \mathrm{rg}(R) \quad (\text{cf. } 1.10).$$

One has therefore proved (i) and the first assertion of (ii); it remains to prove that the Weyl group $W_{G_U}(T_U)$ is finite over U, i.e. "that it has the same number of points in each geometric fiber" (cf. SGA 1, I 10.9 and EGA IV_3, 15.5.1 and IV_4, 8.12.4).

For this, it suffices to remark that the geometric fiber of this group at $s \in U$ is determined by the situation $R \subset M$, as the constant group associated with the "abstract Weyl group of this root system", and in particular is independent of the point s, cf. *Bible*, § 11.3, Th. 2 (see also Exp. XXII, no. 3).

Corollary 2.6. *Let G be an S-group affine and smooth over S, with connected fibers. The set of points $s \in S$ such that $G_{\bar{s}}$ is reductive (resp. semisimple) is an open subset U of S, and the functions*

[1189] *N.D.E.*: which is independent of the rest of this Exposé.
[1190] *N.D.E.*: In all this volume, "trivial torus" has been replaced by "split torus".
[1191] *N.D.E.*: The preceding sentence has been added.
[1192] *N.D.E.*: Note that, $\mathfrak{g}$ being a finite locally free O_S-module, R is a finite set.
[1193] *N.D.E.*: The following sentence has been detailed.

$s \mapsto \mathrm{rgred}(G_{\bar{s}}/\bar{s}), \qquad s \mapsto \mathrm{rgss}(G_{\bar{s}}/\bar{s})$

are locally constant on U.

Definition 2.7. *An S-group (= S-group scheme) G is called* reductive *(resp.* semisimple*) if it is affine and smooth over S, with geometrically connected fibers, and reductive (resp. semisimple).*

The property of being reductive (resp. semisimple) for an S-group G is stable under base change and local for the (fpqc) topology.

2.8. Let G be a reductive S-group. For every torus (resp. maximal torus) Q of G, $Z(Q) = Centr_G(Q)$ is reductive (resp. equals Q). This follows at once from 2.2 and 1.6. Applying 2.5 to $Centr_G(Q)$, one deduces from it that Q is contained (locally for the étale topology) in a maximal torus.

Remark 2.9. *Using as usual the technique of EGA IV₃, § 8, the finite-presentation hypotheses, and Theorem 2.5, one sees that if G is a reductive group over S, there exists an open covering of S, say U_i, such that each G_{U_i} comes by base change from a reductive group over a noetherian ring (in fact, a finitely generated algebra over $\mathbb{Z}$). Likewise, if G possesses a split maximal torus T, one may further assume that T_{U_i} comes from a split maximal torus of the preceding group, ...*

3. Roots and root systems of reductive group schemes

3.1. Let S be a scheme, T an S-torus operating linearly on a locally free $\mathcal{O}_S$-module of finite type F (cf. I, § 4.7). For every character α of T (i.e. $\alpha \in \mathrm{Hom}_{S-gr.}(T, G_{m,S})$), one defines a subfunctor of $W(F)$ by

$$W(F)^\alpha(S') = \{ x \in W(F)(S') \mid t\cdot x = \alpha(t)\, x \text{ for all } t \in T(S''),\ S'' \to S' \}.$$

Lemma. [1194] *Then $W(F)^\alpha = W(F^\alpha)$, where F^α is a submodule of F, locally a direct summand in F, hence also locally free.*

Indeed, the assertion is local for the (fpqc) topology, and one may assume $T = D_S(M)$, where M is a (free) abelian group of finite type. Then α is identified with a locally constant function from S to M (Exp. VIII 1.3), and shrinking S if necessary, one may assume that this function is constant. One is then reduced to Exp. I, 4.7.3.

Definition 3.2. *Let S be a scheme, G an S-group scheme smooth and with connected fibers,*[1195] *T a torus of G. Denote $\mathfrak{g} = Lie(G/S)$ and let T operate on $\mathfrak{g}$ through the adjoint representation of G.*

One says that the character α of T is a root *of G with respect to T if the following equivalent conditions are satisfied:*

(i) For every $s \in S$, $\alpha_{\bar{s}}$ is a root of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$ (1.10).

[1194] *N.D.E.*: This statement has been made into a lemma, in order to put it in evidence.

[1195] *N.D.E.*: The original added the hypothesis that G be of finite presentation over S, which does not seem to be used in what follows. In any case, G being smooth over S and with connected fibers, it is quasi-compact and separated over S ($\mathrm{VI_B}$, 5.5), hence of finite presentation over S.

(ii) α is non-trivial on each fiber and $g^\alpha(s) \neq 0$ for every $s \in S$.

Lemma 3.3. *Let S be a scheme, T an S-torus, α a character of T. The following conditions are equivalent:*

(i) α is non-trivial on each fiber, that is to say: for every $s \in S$, $\alpha_{\bar{s}}$ is distinct from the unit character of $T_{\bar{s}}$.

(ii) For every $S' \to S$, $S' \neq \emptyset$, $\alpha_{S'}$ is distinct from the unit character of $T_{S'}$.

(iii) The morphism α is faithfully flat.

[1196] It is clear that (ii) ⇒ (i), and one sees easily that (iii) ⇒ (i). One has (i) ⇒ (ii), for if $s' \in S'$ lies over s and if $\alpha_{s'}$ is the unit character, then so is α_s. Finally, to prove (i) ⇒ (iii), one reduces to the case where T is diagonalizable and concludes by Exp. VIII 3.2 a).

3.4. [1197] Let G be a reductive S-group scheme, T a maximal torus of G. Let α be a root of G with respect to T. Then, by 2.5.0 and 1.12, g^α is a locally free $\mathcal{O}_S$-module of rank one. Moreover, by 1.10, $-\alpha$ is also a root of G with respect to T. In particular, if G is of semisimple rank 1, one has by 1.11 and 1.12:

Lemma 3.5. *Let S be a scheme, G a reductive S-group of semisimple rank 1, T a maximal torus of G. If α is a root of G with respect to T, then $-\alpha$ is also one, and one has*

$$g = t \oplus g^\alpha \oplus g^{-\alpha},$$

where g^α and $g^{-\alpha}$ are locally free of rank 1.

Definition 3.6. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, R a set of roots of G with respect to T. One says that R is a* root system *of G with respect to T if the following equivalent conditions are satisfied:*

(i) For every $s \in S$, $\alpha \mapsto \alpha_{\bar{s}}$ is a bijection of R onto the set of roots of $G_{\bar{s}}$ with respect to $T_{\bar{s}}$.

(ii) The elements of R are distinct on each fiber (i.e. if $\alpha, \alpha' \in R$, $\alpha \neq \alpha'$, then $\alpha - \alpha'$ ($= \alpha\alpha'^{-1}$) is non-trivial on each fiber), and for every $s \in S$, one has

$$\dim(G_s/\kappa(s)) - \dim(T_s/\kappa(s)) = \mathrm{Card}(R).$$

(iii) One has $g = t \bigoplus \coprod_{\alpha \in R} g^\alpha$.

The equivalence of these various conditions is trivial.

Lemma 3.7. *Let S be a scheme, G a reductive S-group, T a maximal torus of G, R a root system of G with respect to T. Every root of G with respect to T is locally on S equal to an element of R.*

[1196] *N.D.E.*: The original has been simplified; it invoked Exp. IX, 5.2. Let us note, however, that *loc. cit.*, 5.3 shows that if S is connected, then the conditions of the lemma are verified as soon as (i) is verified at one point s of S.

[1197] *N.D.E.*: The following has been modified, in order to recall the hypotheses of 3.2 and to add that T is a maximal torus.

Proof. Visible on condition (iii).

Put $\mathcal{M} = \mathrm{Hom}_{S-gr.}(T, G_{m,S})$; it is a twisted constant S-group scheme (Exp. X 5.6). If G admits a root system R with respect to T, then the inclusion $R \hookrightarrow \mathcal{M}(S)$ defines a morphism $R_S \to \mathcal{M}$, where R_S is the constant S-scheme defined by R; thanks to condition (ii), one sees easily that this morphism is an open and closed immersion whose image is none other than $\bigcup_{\alpha \in R} \alpha(S)$ (each $\alpha \in R$ being considered as a section $S \to \mathcal{M}$).

Let $\mathcal{R}$ be the functor of roots of G with respect to T: by definition, $\mathcal{R}(S')$ is the set of roots of $G_{S'}$ with respect to $T_{S'}$ for every $S' \to S$; if $S' = \emptyset$, one sets $\mathcal{R}(\emptyset) = e$, and if $S' \neq \emptyset$ then the inclusion $R \hookrightarrow \mathcal{M}(S')$ identifies R with a root system of $G_{S'}$ with respect to $T_{S'}$, and therefore, by 3.7, one has

$$\mathcal{R}(S') = \mathrm{Hom}_{\text{loc. const.}}(S', R),$$

which shows that $\mathcal{R}$ is representable by R_S.

If one now no longer supposes necessarily that G possesses a root system relative to T, $\mathcal{R}$ is in any case a subsheaf of $\mathcal{M}$ for (fpqc). Locally for this topology, G possesses a root system with respect to T (take for example T split and use the proof of Th. 2.5). By Exp. IV 4.6.8 and the theory of descent of open (resp. closed) subschemes,[1198] one obtains:

Proposition 3.8. *Let S be a scheme, G an S-group, T a maximal torus of G. The functor $\mathcal{R}$ of roots of G with respect to T is representable by a twisted constant finite S-scheme (Exp. X 5.1) which is an open and closed subscheme of* $\mathrm{Hom}_{S-gr.}(T, G_{m,S})$.

For $R \subset \mathrm{Hom}_{S-gr.}(T, G_{m,S})$ to be a root system of G with respect to T, it is necessary and sufficient that the morphism $R_S \to \mathrm{Hom}_{S-gr.}(T, G_{m,S})$ defined by the preceding inclusion induce an isomorphism $R_S \xrightarrow{\sim} \mathcal{R}$.

3.9. Let S be a scheme, G a reductive S-group, T a maximal torus of G, α a root of G with respect to T (i.e. a section of $\mathcal{R}$). Consider the kernel $Ker(\alpha)$ of α, its unique maximal torus T_α, and the centralizer of the latter, $Z_\alpha = Centr_G(T_\alpha)$. It is an S-group closed in G, reductive (2.8) of semisimple rank 1 (1.12). Moreover,

$$\mathrm{Lie}(Z_\alpha/S) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

so $\alpha, -\alpha$ is a root system of Z_α with respect to T.

3.10. Let S be a scheme, G a reductive S-group, T a maximal torus of G, α a root of G with respect to T. If $q \in Q$ and if $q\alpha$ is a root of G with respect to T, then $q = 1$ or $q = -1$. This follows at once from 1.12.

4. Roots and vector group schemes

4.1. Let S be a scheme, F a locally free $\mathcal{O}_S$-module of finite type. The S-scheme $W(F)$ is smooth over S. Its Lie algebra is canonically isomorphic to F. Indeed, one has a canonical isomorphism

[1198] *N.D.E.*: see SGA 1, VIII 4.4 or EGA IV_2, 2.7.1.

$$W(F) \xrightarrow{\sim} Lie(W(F)/S) = W(Lie(W(F)/S))$$

(Exp. II, 4.4.1 and 4.4.2). We shall always identify F and $Lie(W(F)/S)$.

Lemma 4.2. *Let S be a scheme, V a vector fibration over S, smooth over S. Then there exists a unique isomorphism of 0_S-modules*

$$\exp : W(Lie(V/S)) \xrightarrow{\sim} V$$

inducing the identity on the Lie algebras. [1199]

Proof. Indeed, $V = V(F)$ for some quasi-coherent 0_S-module F. Since V is smooth over S, then $F \simeq \omega^1_{V/S}$ is locally free of finite type, and therefore

$$Lie(V/S) = V(\omega^1_{V/S}) \simeq W(Lie(V/S)).$$

Moreover, by Exp. II *loc. cit.*, one has a canonical isomorphism

$$V \xrightarrow{\sim} Lie(V/S) \simeq W(Lie(V/S))$$

and one has at once the uniqueness of exp, since W is fully faithful.

4.3. Notations. If V is a vector bundle over S, one will denote by $V\times$ the open subset of V obtained by removing the zero section. Write the group law of V in multiplicative notation. The action of 0_S on V defining the module structure will then be written exponentially

$$0_S \times_S V \to V, \qquad (x, v) \mapsto v^x.$$

One has $(vv')^x = v^x v'^x$, $v^{x+x'} = v^x v^{x'}$, $v^{xx'} = (v^x)^{x'}$. In particular, if one restricts the operator law to $G_{m,S}$, then $V\times$ is stable and is therefore endowed with a

[1199] *N.D.E.*: We have kept the original proof; one can also detail it as follows. Let F be a quasi-coherent 0_S-module, $V = V(F)$. Denote by π the projection $V \to S$ and ϵ the zero section $S \to V$. Then $\Omega^1_{V/S} = \pi * F$, whence

(1) $\omega^1_{V/S} = \epsilon^* \Omega^1_{V/S} \simeq \epsilon^* \pi^* F \simeq F,$

and therefore $Lie(V/S) = (\omega^1_{V/S})^V \simeq F^V$. If one supposes V smooth over S then, by (1), F is locally free of finite type, and therefore

(2) $V = V(F) \simeq W(F^\vee) \simeq W(Lie(V/S)).$

Now, if U is an S-scheme equipped with a section $\tau : S \to U$, and if one is given an isomorphism $\phi : V \xrightarrow{\sim} U$ of "pointed" S-schemes, i.e. such that $\phi \circ \epsilon = \tau$ (for example if U is an S-group), then ϕ induces an isomorphism $d\phi : F^V \xrightarrow{\sim} Lie(U/S)$, whence an isomorphism $\psi : W(F^V) \xrightarrow{\sim} W(Lie(U/S))$ making the appropriate diagram commute, which permits one to identify U with $W(Lie(U/S))$. Since the functor W is fully faithful, there exists a unique isomorphism $\exp : W(Lie(U/S)) \xrightarrow{\sim} U$ such that $Lie(\psi^{-1} \circ \exp) = id$. In fact, one can see directly that $Lie(\psi) = id$, so that $\exp = \psi$.

structure of object with group of operators $G_{m,S}$. We shall again write this law as $(z, v) \mapsto v^z$.

Definition 4.4.1. [1200] *Let L be an invertible module on S and $W(L)$ the associated vector bundle. Then $W(L)\times$ is a principal homogeneous bundle (locally trivial) under $G_{m,S}$. One denotes $\Gamma(S, L)\times = W(L) \times (S)$.*

Scholie 4.4.2. *There is a bijective correspondence between the isomorphisms of 0_S-modules $O_S \simeq L$, the isomorphisms of 0_S-modules $O_S \simeq W(L)$,*[1201] *and the sections $S \to W(L)\times$.*

This correspondence is effected by $f \mapsto W(f) \mapsto W(f)(1)$. It is compatible with extension of the base. One may therefore consider $W(L)\times$ as the "scheme of trivializations of $W(L)$".

4.5. Let S be a scheme, T a torus over S, P an S-group with group of operators $G_{m,S}$ (for example a vector bundle), α a character of T. One denotes $T \cdot_\alpha P$ the semi-direct product of P by T, where T operates on P by the composite morphism

```
T --α--> G_{m,S} ----> Aut_{S-gr.}(P).
```

Definition 4.6. *Let S be a scheme, G an S-group scheme, T a subgroup of G, α a character of T, L an 0_S-module. Let*

```
p : W(L) → G          [^N.D.E-XIX-35]
```

be a morphism of S-group functors. One says that p is normalized by T with multiplier *α if it satisfies the following equivalent conditions:*

(i) p is a morphism of objects with group of operators T, if one makes T operate on $W(L)$ by α and on G by inner automorphisms. In other words, for every $S' \to S$ and all $t \in T(S')$ and $x \in W(L)(S') = \Gamma(S', L \otimes O_{S'})$, one has

```
int(t) p(x) = p(α(t) x).
```

(ii) The morphism $T \cdot_\alpha W(L) \to G$ defined by the product in G (i.e. by $(t, x) \mapsto t \cdot p(x)$) is a morphism of groups.

Lemma 4.7. *Under the conditions of 4.6, suppose moreover that G is smooth and has connected fibers,*[1202] *T is a maximal torus of G, L is invertible. If p is a monomorphism and α non-zero on each fiber, then α is a root of G with respect to T.*

Proof. Indeed $Lie(p) : L \to g$ is a monomorphism which factors through g^α.

Proposition 4.8. *Under the conditions of 4.7, suppose that G is reductive, and that p is a monomorphism. Then α is a root of G with respect to T and $Lie(p)$ induces an isomorphism*

[1200] *N.D.E.*: For later references, 4.4 has been transformed into 4.4.1 and 4.4.2.

[1201] *N.D.E.*: Here, one identifies the vector bundle $W(L)$ with the functor in 0_S-modules that it represents.

[1202] *N.D.E.*: cf. N.D.E. (28).

$$Lie(p) : L \xrightarrow{\sim} g^{\alpha}.$$

Proof. Indeed, by virtue of 4.7 and the fact that g^{α} is invertible, it suffices to prove that α is non-zero on each fiber. Let then $s \in S$ be such that $\alpha_s = 0$ (= 1 in multiplicative notation). If X is a non-zero section of $L \otimes_S \kappa(s)$, then $p(X)$ is a unipotent element $\neq e$ of $G(\bar{s})$ which centralizes $T_{\bar{s}}$, which is impossible, since the latter is its own centralizer.

Corollary 4.9. *Under the conditions of 4.8, there exists a monomorphism of groups with operators T (i.e. normalized by T with multiplier α)*

$$W(g^{\alpha}) \to G$$

which induces on the Lie algebras the canonical morphism $g^{\alpha} \to g$.

We shall see shortly that 4.9 is verified in fact whenever G is a reductive group and α a root of G with respect to T, and that such a morphism is unique.

Reminder 4.10. *Let k be an algebraically closed field, G a reductive k-group, T a maximal torus of G, α a root of G with respect to T. There exists a monomorphism*

$$p : G_{a,k} \to G$$

normalized by T with multiplier α.

See *Bible*, § 13.1, Th. 1.

4.11. Let us conclude this section with a technical result that will be useful to us. Let S be a scheme, and L an invertible `O_S`-module. Let q be an integer `> 0` such that $x \mapsto x^q$ is an endomorphism of the S-group $G_{a,S}$. (If $S \neq \emptyset$, one has $q = 1$, or $q = p^n$, p being a prime number that is zero on S; this follows at once from the elementary fact: the gcd of the binomial coefficients `(q choose i)`, for $i \neq 0, q$, is p if $q = p^n$, p prime, and 1 in the contrary case.) The morphism defined by the q-th power

$$L \to L^{\otimes q}$$

is a morphism of sheaves of abelian groups. It defines by base change a morphism of S-group schemes:

$$W(L) \to W(L^{\otimes q}).$$

In particular, if L' is another invertible module and if one has a morphism of `O_S`-modules

$$h : L^{\otimes q} \to L',$$

one deduces from it a morphism of S-group schemes:

$$W(L) \to W(L'), \qquad x \mapsto h(x^q).$$

These notations laid down, one has:

Proposition 4.12. *Let S be a scheme, T (resp. T') an S-torus, L (resp. L') an invertible $\mathcal{O}_S$-module, α (resp. α') a character of T (resp. T').*[1203] *Let $f : T \to T'$ be a morphism of groups and $g : W(L) \to W(L')$ a morphism of S-schemes (not necessarily a morphism of S-groups) satisfying the following condition:*

$$(\star) \qquad g(\alpha(t)\, x) = \alpha'(f(t))\, g(x)$$

for all $x \in W(L)(S')$, $t \in T(S')$, $S' \to S$. Let $s_0 \in S$ be such that $\alpha_{s_0} \neq 0$.

a) Suppose that g sends the zero section to the zero section and that for every integer $n > 0$, one has $(\alpha' \circ f)_{s_0} \neq n\alpha_{s_0}$. Then $g = 0$ in a neighborhood of s_0.

b) Suppose that g is a morphism of groups such that $g_{s_0} \neq 0$. Then there exist an open subset U of S containing s_0 and an integer $q > 0$ such that $x \mapsto x^q$ is an endomorphism of $G_{a,U}$ and $(\alpha' \circ f)_U = q\alpha_U$.

c) Suppose that $(\alpha' \circ f)_{s_0} = q\alpha_{s_0}$, where q is an integer > 0 such that $x \mapsto x^q$ is an endomorphism of $G_{a,S}$. Then there exist an open subset U of S containing s_0 and a unique morphism of $\mathcal{O}_S$-modules

$$h : L^{\otimes q}|U \to L'|U$$

such that g_U is the composite morphism

$$W(L)_U \xrightarrow{x \mapsto x^q} W(L^{\otimes q})_U \xrightarrow{W(h)} (L')_U.$$

Proof. Let us prove (a). Since the conclusion is local on S, one may assume that

$W(L) = W(L') = G_{a,S}$ and therefore that g is expressed as a polynomial

$$g(X) = \sum_{n \geqslant 0} a_n X^n, \qquad a_n \in \Gamma(S, \mathcal{O}_S).$$

The condition $(\star)$ linking f and g is written as an identity in $\Gamma(S', \mathcal{O}_{S'})[X]$:

$$\sum_{n \geqslant 0} a_n\, \alpha'(f(t))\, X^n = \sum_{n \geqslant 0} a_n\, \alpha(t)^n X^n,$$

that is, for every $n \geqslant 0$, every $S' \to S$ and every $t \in T(S')$,

$$a_n(\alpha'(f(t)) - \alpha(t)^n) = 0.$$

[1203] *N.D.E.*: "consider the semi-direct product $T \cdot_\alpha W(L)$, resp. $T' \cdot_{\alpha'} W(L')$" has been removed. On the other hand, the hypothesis in (b) that g be a morphism of groups, combined with $(\star)$, amounts to saying that the morphism $(t, x) \mapsto (f(t), g(x))$ is a morphism of groups from $T \cdot_\alpha W(L)$ to $T' \cdot_{\alpha'} W(L')$.

For each $n \geqslant 0$, let S_n be the set of $s \in S$ such that $(\alpha' \circ f)_s = n\alpha_s$. One knows (Exp. IX 5.3) that the S_n are open and closed, and that $(\alpha' \circ f)_{S_n} = n\alpha_{S_n}$. Moreover, since $\alpha_{s_0} \neq 0$, one may, shrinking S if necessary, assume that α is non-zero on each fiber (same reference), which entails that the S_n are disjoint. Shrinking S if necessary, one may therefore assume that one is in one of the two following cases: there exists an n such that $S = S_n$, or all the S_n are empty.

Let $m \geqslant 0$ be such that $S_m = \emptyset$; I claim that then $a_m = 0$; indeed $\alpha' \circ f$ and $m\alpha$ are distinct on each fiber of S, and one has:

Lemma 4.13. *Let S be a scheme, T an S-torus, α and α' two characters of T distinct on each fiber; there exists a family $S_i \to S$ covering for (fpqc), and for each i a $t_i \in T(S_i)$, such that $\alpha(t_i) - \alpha'(t_i) = 1$.*

Proof. The lemma is trivial, by reduction to the diagonalizable case, then to the case $T = G_{m,S}$.

Let us resume the proof of Proposition 4.12; we have just proved (a). In cases (b) and (c), there exists an n such that $S = S_n$ (with $n = q$ in (c)). By the preceding result, one has therefore $a_m = 0$ for $m \neq n$, which proves that g is written

```
g(X) = a_n X^n,     a_n ∈ Γ(S, O_S).
```

This proves (c) at once. In case (b), one knows that $a_n(s_0) \neq 0$, one may therefore assume a_n invertible on S, which entails that $x \mapsto x^n$ is an endomorphism of $G_{a,S}$ (by virtue of the hypothesis of (b)) and completes the proof.

5. An instructive example

5.1. Let k be an algebraically closed field of characteristic `0`. Put $A = k[t]$, the polynomial ring in one variable over k, and $S = \mathrm{Spec}(A)$. Consider the following Lie algebra g over `O_S`: as a module, it is free of dimension `3`, with basis `{X, Y, H}`; the multiplication table is

```
[X, Y] = 2t H,    [H, X] = X,   [H, Y] = −Y.
```

For $s \in S$, $s \neq s_0$ (the point defined by $t = 0$), the fiber $g(s) = g \otimes_A \kappa(s)$ is isomorphic to the Lie algebra of the group $PGL_{2,\kappa(s)}$. For $s = s_0$, it is a solvable Lie algebra.

5.2. Let G_1 be the group scheme of automorphisms of g: for every $S' \to S$, $G_1(S')$ is the group of automorphisms of the $O_{S'}$-Lie algebra $g \otimes_{O_S} O_{S'}$. It is a closed subscheme of the group $GL(g)$ of automorphisms of the `O_S`-module g. Let $S' \to S$ and $u \in M_3(\Gamma(S', O_{S'}))$ considered as an endomorphism of the `O_S`-module $g \otimes_{O_S} O_{S'}$:

```
u(X) = a X + b Y + e H,
u(Y) = b′ X + a′ Y + e′ H,
u(H) = c X + c′ Y + d H.
```

One sees at once that u is a section of G_1 if and only if $det(u)$ is invertible and one has the relations:[1204]

```
(1)  a(d − 1) = e c,            (1′)  a′(d − 1) = e′ c′,
(2)  b(d + 1) = e c′,           (2′)  b′(d + 1) = e′ c,
(3)  e = 2t(b c − a c′),        (3′)  e′ = 2t(b′ c′ − a′ c),
(4)  2t c = e b′ − a e′,        (4′)  2t c′ = b e′ − e a′,
                      (5)  2t(a a′ − b b′) = 2t d.
```

Lemma 5.3. *The relations (1), (1′), (2), (2′) imply*

```
det(u) = a a′ (2 − d) + b b′ (d + 2),
a a′ − b b′ = d · det(u).
```

Proof. Indeed, the first assertion is obtained at once by inserting the relations (1), (1′), (2), (2′) into the expansion of $det(u)$:

```
det(u) = a a′ d + b e′ c + b′ c′ e − a′ e c − a e′ c′ − b b′ d
       = a a′ d + b b′ (d + 1) + b b′ (d + 1) − a a′ (d − 1) − b b′ d − a a′ (d − 1)
       = a a′ (d − d + 1 − d + 1) + b b′ (d + 1 + d + 1 − d)
       = a a′ (2 − d) + b b′ (d + 2).
```

Multiplying this relation by d, one obtains

```
d · det(u) = a a′ (2d − d²) + b b′ (d² + 2d).
```

But the relation (1) × (1′) = (2) × (2′) gives at once

```
a a′ (d − 1)² = b b′ (d + 1)².
```

Combining the two preceding relations, one finds at once the second sought formula.

5.4. Consider then $G = G_1 \cap SL(g)$. It is the closed subgroup of G_1 defined by the equation $det(u) = 1$. It is therefore an affine group over S.

Proposition 5.5. *The group G is smooth over S.*

To prove the proposition, we shall need the following lemmas.

Lemma 5.6. *Let U be the open subset of $\mathrm{End}_A(g) \simeq W(M_3(O_S))$ defined by the condition "the product a d is invertible", i.e. the open subset $\mathrm{End}_A(g)_f$, where f is the function defined by $f(u) = ad$. Then $U \cap G$ is the closed subscheme of U defined by the six equations:*

```
(1),  (2),  (2′),  (3),  (3′)   and   (D) : a a′ − b b′ = d.
```

Proof. It is first clear that these six relations are verified by every "point" of G (Lemma 5.3), in particular by every "point" of $U \cap G$. Conversely, it must be shown that if $u \in U(S')$ (for every $S' \to S$), and if u verifies the six conditions of the statement, then $det(u) = 1$ and u also verifies (1′), (4), (4′) and (5).

[1204] *N.D.E.*: The equality $[u(X), u(Y)] = 2tu(H)$ (resp. $[u(H), u(X)] = u(X)$, resp. $[u(H), u(Y)] = -u(Y)$) gives the relations (4), (4′), and (5) (resp. (1)–(3), resp. (1′)–(3′)).

One has first (D) ⇒ (5). By (2) and (2′), one has

$$b\, b'\, (d + 1) = b\, c\, e' = b'\, c'\, e.$$

But by (3) and (3′), writing $2t(bc - ac')(b'c' - a'c)$ in two ways:

$$(b\, c - a\, c')\, e' = (b'\, c' - a'\, c)\, e.$$

Combining with the preceding relation, this gives $ac'e' = a'ce$. But by (1), $a'ce = a'a(d-1)$, which proves $a(a'(d-1)-e'c') = 0$ and entails (1′), since a is supposed invertible.

Thus, (1), (2), (2′) and (1′) are verified, hence by Lemma 5.3 and (D) one has $d(det(u) - 1) = 0$. Since d is supposed invertible, this entails $det(u) = 1$.

Let us prove (4) and (4′). Let us do it for example for (4′), the other calculation being deduced from it by symmetry.

By (3), (3′) and (D), one has at once

$$a'\, e + b\, e' = -2t(a\, a' - b\, b')\, c' = -2t\, d\, c'.$$

Combining (1′) and (2), one has also[1205]

$$a'\, e + b\, e' = -d(b\, e' - e\, a'),$$

which completes the proof of (4′), d being supposed invertible.

Lemma 5.7. *G is smooth over S along the unit section.*

Proof. By 5.6 and SGA 1, II 4.10, it suffices to prove that the differentials of the functions

$$\begin{array}{lll} a(d - 1) - e\, c, & b(d + 1) - e\, c', & b'(d + 1) - e'\, c, \\ e - 2t(b\, c - a\, c'), & e' - 2t(b'\, c' - a'\, c), & a\, a' - b\, b' - d, \end{array}$$

at the points of the unit section of G are linearly independent. But denoting by a capital letter the differential of the corresponding lower-case, these are[1206]

$$D, \quad 2B, \quad 2B', \quad E + 2t\, C', \quad E' + 2t\, C, \quad A + A' - D,$$

which are indeed linearly independent modulo every $(t - \lambda)$, $\lambda \in k$.[1207]

Lemma 5.8. *For $s \in S$, $s \neq s_0$, the fiber G_s is connected and semisimple.*

Proof. [1208] Indeed, since $s \neq s_0$, $g(s)$ is isomorphic to the Lie algebra of $PGL_{2,\kappa(s)}$ and, on the other hand, one has $G_s = (G_1)_s$; but it is known that the group of automorphisms of the Lie algebra of PGL_2 over a field of characteristic 0 is PGL_2 itself, which is connected and semisimple.

Lemma 5.9. *The fiber G_{s_0} is solvable and has two connected components which are of the following form:*

[1205] *N.D.E.*: The sign has been corrected.
[1206] *N.D.E.*: $A + A' + B$ has been corrected to $A + A' - D$.
[1207] *N.D.E.*: "$(t - a)$, $a \in A$" has been corrected to "$(t - \lambda)$, $\lambda \in k$".
[1208] *N.D.E.*: The following sentence has been slightly modified.

$$G^0_{s_0} = \begin{pmatrix} a & 0 & 0 \\ 0 & a^{-1} & 0 \\ c & c' & 1 \end{pmatrix} \quad \text{and} \quad G^-_{s_0} = \begin{pmatrix} 0 & b & 0 \\ b^{-1} & 0 & 0 \\ c & c' & -1 \end{pmatrix}.$$

Proof. Indeed, one has $e = e' = 0$, since $t = 0$ at s_0. One then solves immediately the equations (1), (1′), … (5) and (D).

Lemma 5.10. *The matrix*

$$w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

is a section of G over S, such that $w(s_0) \in G^-_{s_0}$.

Let us now prove 5.5.[1209] Denote by G^0 the union of the neutral components of the fibers of G (that is to say the complement of $G^-_{s_0}$); since G is smooth over S along the unit section (5.7), G^0 is an open subgroup of G, smooth over S, by VI_B, 3.10. Since, by translation, G is evidently smooth at the points of $w(S)$, G is smooth over S.

5.11. Consider the morphism $G_{m,S} \to G^0$ defined by

$$z \mapsto \begin{pmatrix} z & 0 & 0 \\ 0 & 1/z & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is a monomorphism that defines a torus T of G^0. I claim that one has

$$T = Centr_G(T) = Centr_{G^0}(T).$$

It suffices indeed to verify the first equality. Since these are smooth subgroups of G over S, it suffices to verify that they have the same geometric points. For the fibers at points $s \neq s_0$, this follows from the fact that $PGL_{2,\kappa(s)}$ is reductive and that T_s is a maximal torus of it for reasons of dimensions (cf. 1.11). On the fiber at s_0, the computation is done immediately. It follows in particular that T is a maximal torus of G and of G^0.

5.12. The section w of G defined in 5.9 normalizes T. It follows at once (cf. 2.4) that the Weyl group of G is isomorphic to $(Z/2Z)_S$, and in particular finite over S.

$$W_G(T) = Norm_G(T) / T = (Z/2Z)_S.$$

On the other hand, $W_{G^0}(T)$ is not finite over S: it "lacks a point" above s_0.

5.13. The open immersion $G^0 \to G$ is not a closed immersion (since G^0 is dense in G); it is however an affine morphism (and therefore G^0 is affine over S). Indeed, since $G^0_{s_0}$ is closed in G_{s_0}, which is closed in G, the complement U of $G^0_{s_0}$ in G is open; G^0 and U form an open covering of G and it suffices to verify that the immersions

[1209] *N.D.E.*: The following sentence has been modified, by adding the reference to VI_B, 3.10.

$G^0 \to G^0$ and $G^0 \cap U \to U$ are affine; for the first this is trivial, for the second, one notes that $U \cap G^0$ is defined in U by the equation $b \neq 0$.

We have therefore constructed an affine smooth S-group, with connected fibers, G^0, possessing a maximal torus T which is its own centralizer and whose Weyl group $W_{G^0}(T)$ is not finite (compare with Theorem 2.5).

6. Local existence of maximal tori. The Weyl group

In the course of the proof of 2.5, we used a result from Exp. XI on the local existence for the étale topology of maximal tori; the proof of Exp. XI uses a fine representability result (XI 4.1). In the particular case which occupies us, one may give another proof, based on the ideas of Exp. XII no. 7, and of a much more elementary nature.

Proposition 6.1. *Let S be a scheme, G a smooth, affine S-group scheme with connected fibers over S, s_0 a point of S such that the maximal tori of the geometric fiber $G_{\bar{s}_0}$ are their own centralizer. There exists an étale morphism $S' \to S$ covering s_0, and a split maximal torus T of $G_{S'}$.*

Proof. First, one may assume S affine.[1210] Since G is of finite presentation over S, one may assume S noetherian, then local, then henselian with separably closed residue field (cf. EGA IV, 8.12, § 8.8, and § 18.8). Set therefore $S = \mathrm{Spec}(A)$, A henselian with separably closed residue field $k = \kappa(s_0)$. Choose a maximal torus T_0 of $G_0 (= G_k)$ (one exists, for example because the scheme of maximal tori of G_0 is smooth over k, Exp. XII, 7.1 c)); since k is separably closed, T_0 is split (cf. X 1.4) and is therefore given by a monomorphism of groups

$$f_0 : G^r_{m,k} \to G_0.$$

Let m be an integer $>$ 1 prime to the characteristic of k. By Exp. VIII 6.7, for every $h > 0$, $Centr_{G_0}({}_{m^h}T_0)$ is representable by a closed subscheme of G_0. Since the ${}_{m^h}T_0$ are schematically dense in T_0 (cf. Exp. IX 4.10) and G_0 is noetherian, there exists an h such that

$$Centr_{G_0}({}_{m^h}T_0) = Centr_{G_0}(T_0) = T_0.$$

Put $n = m^h$; since n is invertible on S, ${}_nG_{m,S}$ is isomorphic to $(Z/nZ)_S$; f_0 therefore defines a monomorphism of groups

$$u_0 : (Z/nZ)^r_k \to G_0$$

such that $Centr_G(u_0) = T_0$. Now the S-functor

[1210] *N.D.E.*: The preceding sentence as well as the reference to EGA IV in what follows have been added.

$$\mathcal{P} = \mathrm{Hom}_{S-gr.}((Z/nZ)^r_S, G)$$

is representable by an S-scheme of finite type (as a closed subscheme of r copies of ${}_nG = \mathrm{Hom}_{S-gr.}((Z/nZ)_S, G) = Ker(G - -n - - > G)$). But $\mathcal{P}$ is smooth over S (Exp. IX 3.6), hence $u_0 \in \mathcal{P}(k)$ lifts to a section $u \in \mathcal{P}(S)$ (Hensel's lemma, Exp. XI 1.11):

$$u : (Z/nZ)^r_S \to G.$$

Consider $H = Centr_G(u)$; it is a closed subgroup scheme of G, by Exp. VIII 6.5 e), and one has $H_0 = T_0$ by hypothesis.[1211] Moreover, H is smooth over S: indeed, let $S' = \mathrm{Spec}(A)$ be an affine scheme over S, $u' : ({}_nG_{m,S'})^r \to G_{S'}$ the morphism deduced from u by base change, $S'_J = \mathrm{Spec}(A/J)$, where J is an ideal of square zero, and let $x \in H(S'_J)$; since G is smooth, x lifts to an element g of $G(S')$; then $v = int(g)(u')$ verifies $v_J = u'_J$ and therefore, by IX 3.2, there exists an element g' of $G(S')$ such that $g'_J = e$ and $int(g')(v) = u'$; then $h = g'g$ belongs to $H(S')$ and verifies $h_J = x$.

Let then H^0 be the neutral component of H; it is a subgroup scheme of G, smooth and with connected fibers, whose special fiber is a torus. By Exp. X 8.1, it is a torus, necessarily split (Exp. X 4.6). Put $H^0 = T$ and let $C = Centr_G(T)$, which is a closed subgroup of G (Exp. VIII 6.5 e)), smooth (Exp. XI 2.4). Consider C^0 (in fact $C^0 = C$, but we do not need to know it); then $C^0 \supset T$ and these are two smooth groups with connected fibers. They coincide at s_0, hence in a neighborhood. Shrinking S if necessary, one may therefore assume $C^0 = T$, hence *a fortiori* T maximal.

Remark 6.2. *The proof shows in particular that the reductive rank of $G_{\bar{s}}$ is constant in a neighborhood of $s = s_0$.*

Proposition 6.3. *Let S be a scheme, G an S-group scheme smooth and of finite presentation over S, Q a subtorus of G.*

(i) $Centr_G(Q)$ and $Norm_G(Q)$ are representable by closed subgroup schemes, smooth (and therefore of finite presentation) over S.

(ii) $Centr_G(Q)$ is an open and closed subscheme of $Norm_G(Q)$. The quotient $W_G(Q) = Norm_G(Q)/Centr_G(Q)$ is representable by an open subgroup scheme of $\mathrm{Aut}_{S-gr.}(Q)$; it is therefore an S-group scheme quasi-finite, étale and separated over S.

(iii) For every $s \in S$, put

$$w(s) = Norm_{G(\bar{s})}(Q(\bar{s}))/Centr_{G(\bar{s})}(Q(\bar{s})).$$

[1211] *N.D.E.*: The reference VIII 6.5 e) has been added in what precedes, and the following sentence has been detailed.

Then $s \mapsto w(s)$ is lower semicontinuous, and is constant in a neighborhood of s if and only if $W_G(Q)$ is finite over S in a neighborhood of s.

Proof. By Exp. XI 6.11, $Centr_G(Q)$ and $Norm_G(Q)$ are representable by closed subschemes of finite presentation of G. These are smooth by Exp. XI 2.4 and 2.4 bis, which proves (i). Assertions (ii) and (iii) are then proved as in Exp. XI 5.9 and 5.10, whose proof uses in fact only (i) and not the fine theorems Exp. XI 4.1 and 4.2.

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Footnotes

Exposé XX. Reductive groups of semisimple rank 1

by M. Demazure

1. Elementary systems. The groups U_α and $U_{-\alpha}$

Recollection 1.1. *Let $S = \mathrm{Spec}(k)$, where k is an algebraically closed field, and let G be a reductive S-group of semisimple rank 1, T a maximal torus (necessarily split) of G. One then has*

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha},$$

where α and $-\alpha$ are the roots of G with respect to T. Moreover, there exist two group monomorphisms

$$p_\alpha : G_{a,S} \to G \quad \text{and} \quad p_{-\alpha} : G_{a,S} \to G$$

such that

$$t\, p_\alpha(x)\, t^{-1} = p_\alpha(\alpha(t)\, x) \quad \text{and} \quad t\, p_{-\alpha}(x)\, t^{-1} = p_{-\alpha}(\alpha(t)^{-1} x),$$

for every $S' \to S$ and all $t \in T(S')$, $x \in G_a(S')$, and such that the morphism

$$G_{a,S} \times_S T \times_S G_{a,S} \to G,$$

defined by $(y, t, x) \mapsto p_{-\alpha}(y) t p_\alpha(x)$*, is radicial and dominant (*Bible*, § 13.4, cor. 2 to th. 3).*

Since the tangent map at the identity is bijective, this morphism is also separable, hence birational; by Zariski's "Main Theorem" (EGA III_1, 4.4.9), it is therefore an open immersion.

Lemma 1.2. *Let* S *be a scheme,* G *an* S*-group scheme,* T *a torus of* G*,* Q *a subtorus of* T*,* α *a character of* T *inducing on* Q *a non-trivial character on each fiber. Let* $p_\alpha : G_{a,S} \to G$ *(resp.* $p_{-\alpha} : G_{a,S} \to G$*) be a group morphism normalized by* T *with multiplier* α *(resp.* $-\alpha$*). Suppose that the morphism*

$$u : G_{a, S} \times_S T \times_S G_{a, S} \to G,$$

defined set-theoretically by $u(y, t, x) = p_{-\alpha}(y) t p_\alpha(x)$*, is an open immersion. Let finally* q *be an integer* $\geqslant 0$ *and*

$$p : G_{a, S} \to G$$

a group morphism such that for every $S' \to S$ *and all* $t \in Q(S')$*,* $x \in G_a(S')$ *one has*

$$int(t)(p(x)) = p(\alpha(t)^q x).$$

Then there exists a unique $v \in G_a(S)$ *such that* $p(x) = p_\alpha(v x^q)$*.*

Indeed, let Ω be the image of u and $U = p^{-1}(\Omega)$. This is an open subset of $G_{a,S}$ containing the zero section. For every section t of Q, the automorphism of $G_{a,S}$ defined by multiplication by $\alpha(t)$ leaves U globally fixed. We have $U = G_{a,S}$; indeed, it suffices to check this when S is the spectrum of an algebraically closed field k; then $\alpha : Q(k) \to k*$ is surjective, which proves at once that $U(k) \supset k*$, hence $U = G_{a,k}$. There therefore exist three morphisms

$$a : G_{a, S} \to G_{a, S}, \quad b : G_{a, S} \to T, \quad c : G_{a, S} \to G_{a, S},$$

such that

$$p(x) = p_{-\alpha}(a(x))\, b(x)\, p_\alpha(c(x)).$$

The condition on p translates to

$$a(\alpha(t) x) = \alpha(t)^{-q} a(x),$$
$$b(\alpha(t) x) = b(x),$$
$$c(\alpha(t) x) = \alpha(t)^q c(x).$$

For the same reason as before, one therefore has for every $S' \to S$ and every $z \in G_m(S')$,

$$a(z x) = z^{-q} a(x), \quad b(z x) = b(x), \quad c(z x) = z^q c(x),$$

hence

$$z^q a(z) = a(1), \quad b(z) = b(1), \quad c(z) = z^q c(1).$$

Since $G_{m,S}$ is schematically dense in $G_{a,S}$, one has at once for every $x \in G_a(S')$, $S' \to S$:

```
x^q a(x) = a(1) = a(0) = 0,    hence a = 0,
c(x) = x^q c(1) = ν x^q,       for some ν ∈ G_a(S),
b(x) = b(1) = b(0) = e,        hence b = e,
```

which completes the proof.

Definition 1.3. *Let S be a scheme. By an S-*elementary system* one means a triple (G, T, α) where*

(i) G is a reductive S-group of semisimple rank 1 (Exp. XIX 2.7),

(ii) T is a maximal torus of G,

(iii) α is a root of G with respect to T (Exp. XIX 3.2).

One thus has a direct sum decomposition (Exp. XIX 3.5)[1212]

```
g = t ⊕ g_α ⊕ g_{-α},
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g_α and $g_{-\alpha}$ being locally free of rank one.

1.4. If (G, T, α) is an S-elementary system, then $(G_{S'}, T_{S'}, \alpha_{S'})$ is an S'-elementary system for every $S' \to S$. If (G, T, α) is an S-elementary system, then $(G, T, -\alpha)$ is also one.

If S is a scheme, G a reductive S-group, T a maximal torus of G, α a root of G with respect to T, then (Exp. XIX 3.9), (Z_α, T, α) is an S-elementary system.

Let (G, T, α) be an S-elementary system. The invertible module g_α is canonically endowed with a T-module structure. One therefore has also a T-module structure on the vector bundle $W(g_\alpha)$. On the other hand, the inner automorphisms of T define on G a structure of group with group of operators T.

Theorem 1.5. *Let (G, T, α) be an S-elementary system.*

(i) There exists a unique morphism of groups with group of operators T

$$\exp : W(g_\alpha) \to G$$

which induces on the Lie algebras the canonical morphism $g_\alpha \to g$.[1213]

In other words, exp is the unique morphism satisfying the following conditions: for every $S' \to S$ and every $t \in T(S')$, $X, X' \in W(g_\alpha)(S')$, one has

```
exp(X + X′) = exp(X) exp(X′),
int(t)(exp(X)) = exp(α(t) X),
Lie(exp)(X) = X.
```

[1212] N.D.E.: of 0_S-modules.

[1213] N.D.E.: We shall see later (Cor. 5.9) that exp is an isomorphism of $W(g_\alpha)$ onto a closed subgroup of G.

(ii) If one defines analogously (in the S-elementary system $(G, T, -\alpha)$)

$$\exp : W(g_{-\alpha}) \to G,$$

then the morphism

$$W(g_{-\alpha}) \times_S T \times_S W(g_\alpha) \to G$$

defined set-theoretically by $(Y, t, X) \mapsto \exp(Y) \cdot t \cdot \exp(X)$ *is an open immersion.*

Suppose we have proved the existence of the desired exp morphisms, and let us prove the other assertions of the theorem. We first prove (ii). Since both sides are of finite presentation and flat over S, it suffices to do so when S is the spectrum of an algebraically closed field (SGA 1, I 5.7 and VIII 5.5). Let then $S = \operatorname{Spec} k$. Let $X \in \Gamma(S, g_\alpha)^\times$, $Y \in \Gamma(S, g_{-\alpha})^\times$. It suffices to prove that the morphism

$$G_{a,k} \times_k T \times_k G_{a,k} \to G, \quad (y, t, x) \mapsto \exp(yY)\, t \exp(xX)$$

is an open immersion. Now by 1.1 and 1.2, there exist $a, b \in k$ such that

$$\exp(yY) = p_{-\alpha}(a y) \quad \text{and} \quad \exp(xX) = p_\alpha(b x).$$

Since $\exp : W(g_{-\alpha}) \to G$ induces a monomorphism on Lie algebras, one has $a \neq 0$; likewise $b \neq 0$, and one is reduced to 1.1.

The uniqueness of the morphism exp may be proved locally on S; one then reduces to the case where g_α and $g_{-\alpha}$ are free, and one has only to apply 1.2 (with $Q = T$ and $q = 1$).

It remains then to prove the existence of the desired morphism exp. Let us first remark that, by virtue of the theory of faithfully flat descent and the uniqueness assertion just proved, it suffices to demonstrate this existence locally on S for the (fpqc) topology. By the usual arguments using finite presentation, one reduces to the case where S is noetherian, then to the case where it is noetherian local. By virtue of the preceding remark, one can therefore content oneself with proving the existence of the desired morphism exp when $S = \operatorname{Spec}(A)$, with A a complete noetherian local ring with algebraically closed residue field k. Let then $p_0 : G_{a,k} \to G_k$ be a monomorphism of k-groups normalized by T_k with multiplier $\alpha_0 = \alpha \otimes_A k$ (one exists by 1.1). One knows (1.1 and 1.2) that the corresponding morphism $T_k \cdot_{\alpha_0} G_{a,k} \to G_k$ is an immersion, hence in particular a monomorphism. Let us provisionally admit the following two lemmas:

Lemma 1.6. *Let S be a scheme, G an S-group of finite presentation, T an S-torus, α a character non-trivial on each fiber of T, s_0 a point of S. Let*

$$f : T \cdot_\alpha G_{a,S} \to G$$

be an S-group morphism such that f_{s_0} is a monomorphism and the restriction of f to T is a monomorphism. There exists an open neighborhood U of s_0 such that $f|U$ is a monomorphism.

Lemma 1.7. *Let A be a complete noetherian local ring with algebraically closed residue field k, (G, T, α) an A-elementary system, $p_0 : G_{a,k} \to G_k$ a morphism of k-groups normalized by T_k with multiplier $\alpha \otimes_A k$. There exists a group morphism $p : G_{a,A} \to G$ normalized by T with multiplier α.*

Let p be the morphism whose existence is asserted by 1.7. Let $f : T \cdot_\alpha G_{a,S} \to G$ be the corresponding morphism. It satisfies the hypotheses of 1.6, hence is a monomorphism; in particular p is a monomorphism. One then concludes by Exp. XIX 4.9.

Proof of 1.6. Denote by $\epsilon : S \to T \cdot_\alpha G_{a,S}$ the unit section. Since f is unramified at $\epsilon(s_0)$, it is so at $\epsilon(s)$ for all s in an open neighborhood U of s_0; $f|U$ is therefore unramified (Exp. X 3.5)[1214], hence its kernel $Ker(f)_U$ is unramified over U. To prove that $f|U$ is a monomorphism, it suffices therefore[1215] to prove that $Ker(f)_U$ is radicial over U, which is a set-theoretic question. One is thus reduced to proving:

Lemma 1.8. *Let k be an algebraically closed field; let N be an invariant subgroup of $T \cdot_\alpha G_{a,k}$ (α a non-trivial character of the torus T), étale over k and such that $N \cap T = e$. Then $N = e$.*

One has $int(t')(t, x) = (t, \alpha(t')x)$. If (t, x) is a point of N, with $x \neq 0$, then `(t, z x)` is also a point of N for $z \in k*$, and (t, x) is not isolated; hence N is not quasi-finite. One therefore has set-theoretically $N \subset T$, and we are done.

Proof of 1.7. Let m be the radical of A, and $S_n = \mathrm{Spec}(A/m^{n+1})$, $n \geqslant 0$. We first show, by induction on n, that p_0 can be extended for each n to a morphism of S_n-groups

`p_n : G_{a, S_n} → G_{S_n}`

normalized by T_{S_n} with multiplier α_{S_n}, the p_n further satisfying the condition $p_{n+1} \times_{S_{n+1}} S_n = p_n$.

Let $H = T \cdot_\alpha G_{a,S}$. The morphism $H_{S_n} \to G_{S_n}$ defined by p_n is denoted f_n. Let us admit the following lemma:

Lemma 1.9. *If (G, T, α) is a k-elementary system, k algebraically closed, and if $p : G_{a,k} \to G$ is a monomorphism normalized by T with multiplier α, one has*

`H²(T ·_α G_{a, k}, g) = 0.`

(One lets $T \cdot_\alpha G_{a,k}$ act on g through the morphism $T \cdot_\alpha G_{a,k} \to G$ defined by p, and the adjoint representation of G).

Then, by virtue of Exp. III 2.8, f_n extends to a morphism of S_{n+1}-groups

$$f'_{n+1} : H_{S_{n+1}} \to G_{S_{n+1}}.$$

[1214] N.D.E.: see also VI_B, 2.5.
[1215] N.D.E.: by EGA IV₄, 17.9.1.

Now f'_{n+1} and the canonical immersion of $T_{S_{n+1}}$ into $G_{S_{n+1}}$ have the same restriction to T_{S_n}. By Exp. III 2.5, there exists an element $g \in G(S_{n+1})$ such that $g \times_{S_{n+1}} S_n = e$ and such that $f_{n+1} = int(g) \circ f'_{n+1}$ restricts to T_{n+1} along the canonical immersion of T_{n+1}. Let p_{n+1} be the restriction of f_{n+1} to $G_{a,S_{n+1}}$. It is a morphism normalized by $T_{S_{n+1}}$ with multiplier $\alpha_{S_{n+1}}$, which extends p_n.

One has thus constructed a coherent system (f_n) and one must now algebraize it. Now one has:

Lemma 1.10. *Let A be a complete noetherian local ring, $\mathfrak{m}$ its maximal ideal, $S = \mathrm{Spec}(A)$, $S_n = \mathrm{Spec}(A/\mathfrak{m}^{n+1})$, T an S-torus, α a non-zero character of T, X an affine S-scheme on which T acts. Let T act on $G_{a,S}$ via α. Let q be an integer $\geqslant 0$, and let $(f_n)_{n\geqslant 0}$ be a coherent system of morphisms*

$$f_n : G_{a,S_n}^q \to X_{S_n}$$

of objects with operators T_{S_n}. There exists a unique morphism of objects with operators T

$$f : G_{a,S}^q \to X$$

which induces the f_n (compare with Exp. IX 7.1).

Corollary 1.11. *If X is a group with group of operators T and if the f_n are group morphisms, then f is one too.*

It suffices to apply the uniqueness assertion of the lemma to the two morphisms $G_{a,S}^{2q} \to X$ deduced from f in the usual way.

Proof of 1.10. Suppose T split, which is moreover the case in the application of 1.10 to the proof of 1.5. One knows (Exp. I 4.7.3.1) that $X \mapsto A(X)$ realizes an equivalence between the category of affine S-schemes equipped with a T-operation and the opposite category of S-graded algebras of type $M = \mathrm{Hom}_{S-gr.}(T, G_{m,S})$.

One therefore has gradings

$$B = A(X) = \bigoplus_{m \in M} B_m \quad \text{and} \quad C = A(G_{a,S}^q) = \bigoplus_{m \in M} C_m.$$

One sees at once that each C_m is free of finite type over A. (Indeed, one has $C_m = 0$ if m is not a multiple of α, and if $m = d\alpha$, C_m is isomorphic to the A-module of homogeneous polynomials of degree d, in q variables.) Set

$$\hat{B}_m = \varprojlim_n B_m \otimes_A (A/\mathfrak{m}^{n+1}),$$
$$\hat{C}_m = \varprojlim_n C_m \otimes_A (A/\mathfrak{m}^{n+1}),$$
$$\hat{B} = \bigoplus_{m \in M} \hat{B}_m, \quad \hat{C} = \bigoplus_{m \in M} \hat{C}_m.$$

One then has canonical morphisms of M-graded algebras

$$g_B : B \to \hat{B} \quad \text{and} \quad g_C : C \to \hat{C}.$$

It follows from the remark made above that g_C is an isomorphism. To give a coherent system (f_n) as in the statement is equivalent to giving a morphism of graded A-algebras

$$\hat{F} : \hat{B} \to \hat{C}.$$

To find a morphism f as in the statement is equivalent to finding a morphism of graded A-algebras $F : B \to C$ rendering commutative the diagram

$$\begin{array}{ccc} B & \xrightarrow{F} & C \\ {\scriptstyle g_B}\downarrow & & \downarrow{\scriptstyle g_C} \\ \hat{B} & \xrightarrow{\hat{F}} & \hat{C}. \end{array}$$

Since g_C is an isomorphism, the existence and uniqueness of F are immediate. This proves 1.10.

To complete the proof of 1.5, it remains only to prove 1.9.

1.12. Proof of 1.9. One has $g = t \oplus g_\alpha \oplus g_{-\alpha}$. As explained in 1.9, consider g as a $(T \cdot_\alpha G_{a,k})$-module. It is clear that $t \oplus g_\alpha$ is a submodule of g, the quotient being isomorphic to $g_{-\alpha}$ as a k-vector space and even as a T-module. It is clear that $G_{a,k}$ acts trivially on this quotient, which is of dimension 1 (for every group morphism from $G_{a,k}$ to $G_{m,k}$ is trivial). Similarly g_α is a submodule of $t \oplus g_\alpha$, the quotient being isomorphic to t as a T-module, $G_{a,k}$ acting trivially on it. To summarize:

Lemma 1.13. *Under the conditions of 1.9, g admits a composition series as $(T \cdot_\alpha G_{a,k})$-module whose successive quotients are*

$$g_{-\alpha}, t, g_\alpha,$$

viewed as $(T \cdot_\alpha G_{a,k})$-modules via the projection $T \cdot_\alpha G_{a,k} \to T$.

One is therefore reduced to computing the cohomology of $T \cdot_\alpha G_{a,k}$ acting via the projection $T \cdot_\alpha G_{a,k} \to T$ and the character β of T (here $\beta = 0$, α, or $-\alpha$) on $W(k)$.[1216] Let $k[x_1, \cdots, x_n]$ denote the algebra of polynomials over k in n variables and $k_q[x_1, \cdots, x_n]$ the subspace of homogeneous polynomials of degree q.

Lemma 1.14. *With the preceding notations, one has $H^n(T \cdot_\alpha G_{a,k}, k) = H^n(C^*_{\alpha,\beta})$, where the complex $C^*_{\alpha,\beta}$ is defined by*

$$C^n_{\alpha,\beta} = \begin{cases} k_q[x_1, \dots, x_n] & \text{if } \beta = q\alpha, \text{ with } q \in \mathbb{N}^*; \\ 0 & \text{otherwise,} \end{cases}$$

and

$$\begin{aligned} \delta f(x_1, x_2, \dots, x_{n+1}) = {} & f(x_2, \dots, x_{n+1}) \\ & + \sum_{i=1}^n (-1)^i f(x_1, \dots, x_i x_{i+1}, \dots, x_{n+1}) \\ & + (-1)^{n+1} f(x_1, \dots, x_n). \end{aligned}$$

[1216] N.D.E.: We have added the sentence that follows.

Indeed, the functor $M \mapsto H^0(T, M)$ is exact on the category of T-modules (and the $H^q(T, -)$ vanish), by Exp. I 5.3.2. It follows, as in the usual case of group cohomology, that $H^n(T \cdot_\alpha G_{a,k}, k)$ can be computed as the n-th cohomology group of the complex of cochains of $G_{a,k}$ in k, invariant under T, i.e. satisfying

$$f(\alpha(t) x_1, \dots, \alpha(t) x_n) = \beta(t) f(x_1, \dots, x_n).$$

This indeed gives the announced complex.

To prove 1.9, it therefore suffices to prove that $H^2(C^*_{\alpha,\beta}) = 0$ for $\beta = 0, \alpha, -\alpha$, which is done at once.

Remark 1.15. *One can explicitly compute the groups $H^n(C^*_{\alpha,\beta})$ for $\beta = q\alpha$ (see M. Lazard,* Lois de groupes et analyseurs*, Annales E.N.S., 1955). In particular, one finds $H^n(C^*_{\alpha,q\alpha}) = 0$ for $n > q$.*

Notations 1.16. *The image of the canonical immersion*

$$W(\mathfrak{g}_{-\alpha}) \times_S T \times_S W(\mathfrak{g}_\alpha) \to G$$

will be denoted Ω. It is an open subset of G containing the unit section. The image of

$$W(\mathfrak{g}_{-\alpha}), \quad \text{resp. } W(\mathfrak{g}_\alpha), \quad \text{resp. } W(\mathfrak{g}_{-\alpha}) \times_S T, \quad \text{resp. } T \times_S W(\mathfrak{g}_\alpha)$$

will be denoted[1217]

$$U_{-\alpha}, \quad \text{resp. } U_\alpha, \quad \text{resp. } U_{-\alpha} \cdot T, \quad \text{resp. } T \cdot U_\alpha.$$

Then U_α (resp. $U_{-\alpha}$) is a subgroup of G canonically endowed with a vector bundle structure, and one has

$$\mathrm{int}(t)(x) = x^{\alpha(t)} \quad (\text{resp. } x^{-\alpha(t)}),$$

for every $S' \to S$, $t \in T(S')$, $x \in U_\alpha(S')$ (resp. $x \in U_{-\alpha}(S')$).

One has canonical isomorphisms

$$T \cdot U_\alpha \cong T \cdot_\alpha U_\alpha \quad \text{and} \quad T \cdot U_{-\alpha} \cong T \cdot_{-\alpha} U_{-\alpha}.$$

The open set Ω is stable under $int(T)$: one has

$$\mathrm{int}(t')(y \cdot t \cdot x) = y^{-\alpha(t')} \cdot t \cdot x^{\alpha(t')}.$$

Corollary 1.17. *One has $Lie(U_\alpha/S) = \mathfrak{g}_\alpha$ and $Lie(U_{-\alpha}/S) = \mathfrak{g}_{-\alpha}$. The isomorphisms*

$$W(\mathfrak{g}_\alpha) \xrightarrow{\exp} U_\alpha \quad \text{and} \quad W(\mathfrak{g}_{-\alpha}) \xrightarrow{\exp} U_{-\alpha}$$

are those of Exp. XIX 4.2.

Corollary 1.18. *The open set Ω is relatively schematically dense in G (cf. XVIII, § 1).*

[1217] N.D.E.: We have replaced $P_{-\alpha}$ and P_α by $U_{-\alpha}$ and U_α.

Clear by Exp. XVIII, 1.3.[1218]

Corollary 1.19. *The center of G is $Centr(G) = Ker(\alpha)$. It is therefore a closed subgroup of G, of multiplicative type and of finite type.*

The second assertion follows from the first by Exp. IX 2.7. Let us therefore prove the first. The inner automorphism defined by a section of $Ker(\alpha)$ acts trivially on Ω (last formula of 1.16), hence on G by 1.18. Conversely, if $g \in G(S)$ centralizes G, then it centralizes T and U_α, hence is a section of T (Exp. XIX 2.8) annihilating α; since this also holds after any base change, one indeed has $Centr(G) = Ker(\alpha)$.

Corollary 1.20. *For there to exist a monomorphism $p_\alpha : G_{a,S} \to G$ normalized by T with multiplier α, it is necessary and sufficient that the O_S-module g_α be free. More precisely, one has a bijection given by*

$$X_\alpha \mapsto (x \mapsto \exp(x X_\alpha)) \quad \text{and} \quad p_\alpha \mapsto \mathrm{Lie}(p_\alpha)$$

between $\Gamma(S, g_\alpha)^\times$ and the set of monomorphisms p_α as above (which is also the set of isomorphisms of vector group schemes $G_{a,S} \xrightarrow{\sim} U_\alpha$).[1219]

Corollary 1.21. *The subgroups U_α and $U_{-\alpha}$ of G commute on no fiber.*

Indeed, if $(U_\alpha)_s$ and $(U_{-\alpha})_s$ commute, then Ω_s is a subgroup of G_s, hence $\Omega_s = G_s$[1220] and G_s is solvable, which contradicts the hypothesis that G_s is reductive of semisimple rank 1.

2. Structure of elementary systems

Theorem 2.1. *Let S be a scheme, (G, T, α) an S-elementary system. There exists a morphism of O_S-modules*

$$g_\alpha \otimes_{O_S} g_{-\alpha} \to O_S, \quad (X, Y) \mapsto \langle X, Y\rangle,$$

and a morphism of S-groups

$$\alpha^* : G_{m,S} \to T$$

such that for every $S' \to S$ and all $X \in \Gamma(S', g_\alpha \otimes O_{S'})$, $Y \in \Gamma(S', g_{-\alpha} \otimes O_{S'})$ one has the equivalence:

$$\exp(X) \cdot \exp(Y) \in \Omega(S') \iff 1 + \langle X, Y\rangle \in G_m(S'),$$

and under these conditions one has the formula:

$$\text{(F)} \quad \exp(X) \cdot \exp(Y) = \exp\left(\frac{Y}{1 + \langle X, Y\rangle}\right) \cdot \alpha^*(1 + \langle X, Y\rangle) \cdot \exp\left(\frac{X}{1 + \langle X, Y\rangle}\right).$$

Moreover, the morphisms $(X, Y) \mapsto \langle X, Y\rangle$ and $\alpha*$ are uniquely determined, the former is an isomorphism, hence puts the modules g_α and $g_{-\alpha}$ in duality, and one has $\alpha \circ \alpha* = 2$ (squaring in $G_{m,S}$).

[1218] N.D.E.: see also EGA IV_3, 11.10.10.

[1219] N.D.E.: Indeed, on the one hand, $Lie(G_{a,S}) = O_S$ and $Lie(p_\alpha)$ is an element of $\mathrm{Hom}_{O_S}(O_S, g_\alpha) = \Gamma(S, g_\alpha)$.

[1220] N.D.E.: by 1.18.

In view of the uniqueness assertions of the theorem, it suffices to do the proof locally on S. One can therefore assume g_α and $g_{-\alpha}$ free on S. Take then $X \in \Gamma(S, g_\alpha)^\times$, $Y \in \Gamma(S, g_{-\alpha})^\times$ and set $p_\alpha(x) = \exp(xX)$, $p_{-\alpha}(y) = \exp(yY)$, for $x, y \in G_a(S')$, $S' \to S$. By 1.5 and 1.21, it suffices to prove:

Proposition 2.2. *Let S be a scheme, G an S-group, T a torus of G, α a character of T non-trivial on each fiber, $p_\alpha : G_{a,S} \to G$ (resp. $p_{-\alpha} : G_{a,S} \to G$) a group monomorphism normalized by T with multiplier α (resp. $-\alpha$). Suppose that:*

(i) The morphism $G_{a,S} \times_S T \times_S G_{a,S} \to G$ defined by $(y, t, x) \mapsto p_{-\alpha}(y) t p_\alpha(x)$ is an open immersion. Denote its image by Ω.

(ii) For every $s \in S$, $(p_\alpha)_s(G_{a,\kappa(s)})$ and $(p_{-\alpha})_s(G_{a,\kappa(s)})$ do not commute.

Then there exist $a \in G_a(S)$ and $\alpha* \in \mathrm{Hom}_{S-gr.}(G_{m,S}, T)$, uniquely determined, having the following properties: for every $S' \to S$ and all $x, y \in G_a(S')$, one has

$$p_\alpha(x)\, p_{-\alpha}(y) \in \Omega(S') \iff 1 + a x y \in G_m(S'),$$

and, under this condition, one has the formula

$$p_\alpha(x)\, p_{-\alpha}(y) = p_{-\alpha}\left(\frac{y}{1 + a x y} \right) \cdot \alpha^*(1 + a x y) \cdot p_\alpha\left(\frac{x}{1 + a x y} \right).$$

Moreover, a is invertible (i.e. $a \in G_m(S)$) and $\alpha \circ \alpha* = 2$.

Proof:

A) Consider the morphism

$$G_{a,S}^2 \to G$$

defined by $(x, y) \mapsto p_\alpha(x) p_{-\alpha}(y)$. Let U be the inverse image of Ω under this morphism. It is an open subset of $G_{a,S}^2$ containing $0 \times_S G_{a,S}$ and $G_{a,S} \times_S 0$. There therefore exist uniquely determined S-scheme morphisms

$$A : U \to G_{a,S}, \quad C : U \to G_{a,S}, \quad B : U \to T$$

satisfying the set-theoretic relation:

$$p_\alpha(u)\, p_{-\alpha}(v) = p_{-\alpha}(A(u, v))\, B(u, v)\, p_\alpha(C(u, v)).$$

One has immediately the relations

$$A(0, v) = v, \quad A(u, 0) = 0, \quad C(u, 0) = u, \quad C(0, v) = 0,$$
$$B(u, 0) = B(0, v) = e.$$

Let S' be a separated S-scheme and let $t \in T(S')$ be a point of T. Since $\Omega_{S'}$ is stable under $int(t)$, then, by the last formula of 1.16, $U_{S'}$ is stable under the automorphism $(x, y) \mapsto (\alpha(t)x, \alpha(t)^{-1}y)$ of $G_{a,S'}^2$, and one has the relations:

$$A(\alpha(t)\, u, \alpha(t)^{-1}\, v) = \alpha(t)^{-1}\, A(u, v),$$
$$C(\alpha(t)\, u, \alpha(t)^{-1}\, v) = \alpha(t)\, C(u, v),$$
$$B(\alpha(t)\, u, \alpha(t)^{-1}\, v) = B(u, v).$$

Since α is faithfully flat, one deduces that for every $S' \to S$ and every $z \in G_m(S')$, $U_{S'}$ is stable under the transformation $(x, y) \mapsto (zx, z^{-1}y)$ and one has

$$A(zu, z^{-1}v) = z^{-1} A(u, v),$$
$$C(zu, z^{-1}v) = z\, C(u, v),$$
$$B(zu, z^{-1}v) = B(u, v).$$

Suppose first that v is invertible; setting $z = v$, one deduces that if (u, v) is a section of U, then $(uv, 1)$ is also one, and one has

$$A(uv, 1) = v^{-1} A(u, v), \qquad B(uv, 1) = B(u, v).$$

Let then V be the open set of $G^2_{a,S}$ defined by[1221]

$$(u, v) \in V(S') \iff (u, v),\ (uv, 1) \text{ and } (1, uv) \text{ belong to } U(S').$$

Since U is an open set of $G^2_{a,S}$ containing $0 \times_S G_{a,S}$ and $G_{a,S} \times_S 0$, then V is a neighborhood of the zero section of $G^2_{a,S}$ and we have just seen that the morphisms

$$(u, v) \mapsto A(u, v) \text{ and } (u, v) \mapsto v\, A(uv, 1)$$
$$\text{resp. } (u, v) \mapsto B(u, v) \text{ and } (u, v) \mapsto B(uv, 1)$$

coincide on $V \cap (G_{a,S} \times_S G_{m,S})$. Since $G_{a,S} \times_S G_{m,S}$ is schematically dense in $G^2_{a,S}$, these morphisms therefore coincide on V.

One knows that $A(0, 1) = 1$; it follows that there exists an open set W_1 of $G_{a,S}$ containing the zero section such that for every section x of W_1, $A(x, 1)$ is invertible; setting $A(x, 1)^{-1} = F(x)$, one obtains that if $(u, v) \in V(S')$[1222] and $uv \in W_1(S')$, $S' \to S$, then $A(u, v) = vA(uv, 1) = vF(uv)^{-1}$. Arguing similarly for C, one obtains that there exists an open set W_2 of $G_{a,S}$ containing the zero section, and an element E[1223] of $O(W_2)^\times$, such that $C(u, v) = uC(1, uv) = uE(uv)^{-1}$, if $(u, v) \in V(S')$ and $uv \in W_2(S')$. Consequently, setting $W = W_1 \cap W_2$, one obtains:

There exists an open set W of $G_{a,S}$ containing the zero section, and S-morphisms

$$F : W \to G_{m,S}, \quad F(0) = 1,$$
$$H : W \to T, \quad H(0) = e,$$
$$E : W \to G_{m,S}, \quad E(0) = 1,$$

such that if $(u, v) \in V(S')$ and $uv \in W(S')$, $S' \to S$, one has

$$(+) \qquad p_\alpha(u)\, p_{-\alpha}(v) = p_{-\alpha}(v F(uv)^{-1})\, H(uv)\, p_\alpha(u E(uv)^{-1}).$$

B) Let us now use the associativity of G to write

$$p_\alpha(u)\, p_{-\alpha}(v)\, p_{-\alpha}(w) = p_\alpha(u)\, p_{-\alpha}(v + w).$$

There exists an open set L of $G^3_{a,S}$, containing the unit section, such that $(u, v, w) \in L(S')$ is equivalent to

[1221] N.D.E.: We have added the condition "$(1, uv) \in U(S')$".
[1222] N.D.E.: here and in what follows, we have replaced $U(S')$ by $V(S')$.
[1223] N.D.E.: We have denoted by E the element denoted G in the original, since G already denotes the S-group under consideration.

$$(u, v) \in V(S'), \quad (u E(u v)^{-1}, w) \in V(S'), \quad (u, v + w) \in V(S'),$$
$$u v \in W(S'), \quad u w E(u v)^{-1} \in W(S'), \quad u(v + w) \in W(S').$$

Using then the formula (+), one writes at once for $(u, v, w) \in L(S')$ the relations:

$$\begin{aligned}
&(1) \quad E(u v + u w) = E(u w E(u v)^{-1}) E(u v),\\
&(2) \quad H(u v + u w) = H(u w E(u v)^{-1}) H(u v),\\
&(3) \quad (v + w) F(u v + u w)^{-1} = \alpha(H(u v)^{-1}) w F(u w E(u v)^{-1})^{-1} + v F(u v)^{-1}.
\end{aligned}$$

It is immediate from the definition of L that $(1, 0, 0) \in L(S)$. Consider therefore

$$L \cap_T (1 \times_S G_{a, S} \times_S G_{a, S}) = 1 \times_S M;$$

M is an open set of $G^2_{a,S}$, containing the section $(0, 0)$, and for $(v, w) \in M(S')$, one has $v, wE(v)^{-1}, v + w \in W(S')$ and

$$\begin{aligned}
&(1') \quad E(v + w) = E(w E(v)^{-1}) E(v),\\
&(2') \quad H(v + w) = H(w E(v)^{-1}) H(v),\\
&(3') \quad (v + w) F(v + w)^{-1} = \alpha(H(v))^{-1} w F(w E(v)^{-1})^{-1} + v F(v)^{-1}.
\end{aligned}$$

Consider finally the morphism from M to $G^2_{a,S}$ defined set-theoretically by $(v, w) \mapsto (v, wE(v)^{-1})$.[1224] It preserves the section $(0, 0)$ and induces an isomorphism of M onto an open set N of $G^2_{a,S}$ containing the zero section (the inverse isomorphism being given by $(x, y) \mapsto (x, yE(x))$).[1225] One has thus proved the following assertion:

There exists an open set N of $G^2_{a,S}$, containing the zero section, such that if $(x, y) \in N(S')$, then x, y and $x + yE(x)$[1226] belong to $W(S')$ and:

$$\begin{aligned}
&(1'') \quad E(x + y E(x)) = E(x) E(y),\\
&(2'') \quad H(x + y E(x)) = H(x) H(y),\\
&(3'') \quad (x + y E(x)) F(x + y E(x))^{-1} = x F(x)^{-1} + \alpha(H(x))^{-1} y E(x) F(y)^{-1}.
\end{aligned}$$

C) Arguing similarly with left associativity, one demonstrates the following assertion:[1227]

There exists an open set N' of $G^2_{a,S}$, containing the zero section, such that if $(x, y) \in N'(S')$, then x, y and $x + yF(x)$[1228] belong to $W(S')$, and

$$\begin{aligned}
&(4'') \quad F(x + y F(x)) = F(x) F(y),\\
&(5'') \quad H(x + y F(x)) = H(x) H(y),\\
&(6'') \quad (x + y F(x)) E(x + y F(x))^{-1} = x E(x)^{-1} + \alpha(H(x))^{-1} y F(x) E(y)^{-1}.
\end{aligned}$$

We are therefore led to solve the "functional equation" (1″).

Lemma 2.3. *Let S be a scheme, W an open set of $G_{a,S}$ containing the unit section, $F : W \to G_{m,S}$ an S-morphism. Suppose that $F(0) = 1$ and that there exists an open*

[1224] N.D.E.: We have corrected what follows.

[1225] N.D.E.: that is, we have made the "change of variables" $x = v$, $y = wE(v)^{-1}$, i.e. $v = x$, $w = yE(x)$.

[1226] N.D.E.: We have corrected $y\ E(x)$ to $x + yE(x)$.

[1227] N.D.E.: that is, one writes the equalities resulting from $p_\alpha(t)p_\alpha(u)p_{-\alpha}(v) = p_\alpha(t + u)p_{-\alpha}(v)$ and sets $v = 1$ and $x = u$, $t = yF(u)$ (i.e. $y = tF(u)^{-1}$).

[1228] N.D.E.: We have corrected $y\ F(x)$ to $x + yF(x)$.

set N of $G^2_{a,S}$ containing the zero section such that for $(x,y) \in N(S')$, x, y, and $x + yF(x)$[1229] *belong to $W(S')$ and that one has:*

(†) $F(x + y\,F(x)) = F(x)\,F(y)$.

(i) If S is the spectrum of a field k, there exists $a \in k$ such that $F(x) = 1 + ax$.

(ii) If $a = F'(0) \in \Gamma(S, O_S)$ is invertible, then $F(x) = 1 + ax$.

By the hypotheses, we can differentiate the given equation at $x = 0$ (resp. at $y = 0$) and we find that

(∗) $F'(y)\,(1 + y\,F'(0)) = F'(0)\,F(y) \quad \text{for } (0, y) \in N(S')$,

resp.

$F'(x)\,F(x) = F(x)\,F'(0) \quad \text{for } (x, 0) \in N(S')$.

Since F takes its values in G_m, the second relation gives us

(∗′) $F'(x) = F'(0) \quad \text{for } (x, 0) \in N(S')$;

whence, by the first,

$F'(0)(1 + y\,F'(0)) = F'(0)\,F(y) \quad \text{for } (y, 0), (0, y) \in N(S')$.

If $a = F'(0)$ is invertible, this gives us

$F(y) = 1 + a\,y$,

for y a section of an open set of W containing the unit section, hence schematically dense in W, which proves (ii). This also proves (i) when $F'(0) \neq 0$.

If $F'(0) = 0$, then, by (∗′), $F'(x) = 0$ when x is "near 0", hence $F' = 0$ by schematic density. If k is of characteristic 0, F is a rational fraction with zero derivative, hence constant and equal to $F(0) = 1$.

If k is of characteristic p, and if F is not constant,[1230] there exists an integer $n > 0$ and a rational fraction $F_1 \in k(X)$ such that $F_1'(X) \neq 0$ and

$$F(X) = F_1(X^{p^n}) = F_1(X)^{p^n}.$$

Substituting in the functional equation, one finds

(†_1) $F_1(x + y\,F_1(x)^{p^n}) = F_1(x)\,F_1(y)$.

Differentiating at $x = 0$, one finds

$$(*_1) F_1'(y) = F_1'(0) F_1(y),$$

and differentiating (†_1) at $y = 0$, one obtains

[1229] N.D.E.: We have corrected $y\,F(x)$ to $x + yF(x)$.

[1230] N.D.E.: We have corrected what follows.

$$F'_1(x)\, F_1(x)^{p^n} = F_1(x)\, F'_1(0). \tag{$*'_1$}$$

Since, by hypothesis, $F'_1(X)$ is an invertible element of $k(X)$, one deduces from these two equalities that

$$F_1(X)^{p^n} = 1,$$

hence F_1 is a constant, contradicting the initial hypothesis. This shows that F is constant, and equal to $1 = F(0)$.

D) Suppose S is the spectrum of a field. If $F'(0) = 0$, then $F = 1$. Formula (5″) then gives us $H(x + y) = H(x)H(y)$, which shows that H extends to a group morphism $G_{a,S} \to T$ (Exp. XVIII 2.3), which is necessarily constant of value e. On the other hand, by Lemma 2.3, one will also have $E(x) = 1 + bx$ for some $b \in k$. But then (6″) gives, for $(x, y) \in N(S')$,

$$(x + y)\, E(x + y)^{-1} = x\, E(x)^{-1} + y\, E(y)^{-1},$$

hence,[1231] by Exp. XVIII 2.3 again, $x \mapsto xE(x)^{-1}$ extends to a morphism of k-groups $G_{a,k} \to G_{a,k}$, hence $x/(1 + bx) = cx$ for some $c \in k$, whence $b = 0$ (and $c = 1$).

This shows that F, H, E are constant of value $(1, e, 1)$ in a neighborhood of the unit section, hence everywhere, which by (+) shows that U_α and $U_{-\alpha}$ commute, contrary to hypothesis (ii).

If S is now arbitrary, one has therefore proved that $F'(0)$ is non-zero on no fiber, hence is invertible. The same evidently holds for $E'(0)$, which by Lemma 2.3 shows that there exist $a, b \in G_m(S)$ such that

$$F(x) = 1 + a x, \qquad E(x) = 1 + b x, \qquad \text{for } x \in W(S'). \tag{$\blacklozenge_1$}$$

E) The rest is now easy. Substituting the preceding results into (3″), one finds

$$y\, \alpha(H(x))\, (1 + a y) = y\, (1 + a x + a y(1 + b x))\, (1 + a x).$$

This formula is valid for every section (x, y) of N. But since $G_{a,S} \times_S G_{m,S}$ is schematically dense in $G^2_{a,S}$, one deduces

$$(1 + a y)\, \alpha(H(x)) = (1 + a x + a y(1 + b x))\, (1 + a x).$$

Setting $y = 0$, this gives $\alpha(H(x)) = (1 + ax)^2$. Substituting this into the preceding equality, one finds[1232]

$$a^2\, x\, y = a\, b\, x\, y.$$

Since $G_{m,S}$ is schematically dense in $G_{a,S}$, one deduces $a^2 = ab$, whence, since a is invertible,

[1231] N.D.E.: We have added the sentence that follows. One can also see by a direct calculation that the preceding equality entails $0 = xyb(2 + (x + y)b)$, hence $0 = b(2 + (x + y)b)$, and finally $b = 0$.

[1232] N.D.E.: In the three preceding equalities, we have corrected the original by replacing the factor $(1 + bx)$ on the right by $1 + ax$. Substituting the third equality into the second and taking into account that $1 + ax$ is invertible, one obtains the equality $a^2xy = abxy$.

$(\blacklozenge_2)\quad a = b \quad \text{and} \quad \alpha(H(x)) = (1 + a x)^2.$

Since a is invertible, $x \mapsto 1 + ax$ is an automorphism of $G_{a,S}$; one can therefore find an open set W' of $G_{a,S}$ containing the section 1 and a morphism

$$P : W' \to T$$

such that $P(1 + ax) = H(x)$.[1233]

Substituting in the relation (2′), one finds at once for $(x, y) \in N(S')$,

$$P(1 + a x + a y) = P((1 + a x + a y) / (1 + a x))\, P(1 + a x),$$

which proves that there exists an open neighborhood of 1 in $G_{m,S}$ such that one has for x and y in this neighborhood $P(x)P(y) = P(xy)$. By Exp. XVIII 2.3, there exists a group morphism

$(\blacklozenge_3)\quad \alpha^* : G_{m,S} \to T$

extending P. Since $\alpha(H(x)) = (1 + ax)^2$ near the section 0, one has $\alpha(\alpha*(z)) = z^2$ near the section 1, hence

$(\blacklozenge_4)\quad \alpha \circ \alpha^* = 2.$

F)[1234] Assembling the results (+) and ($\blacklozenge_1$ — $\blacklozenge_4$), one sees that there exist $a \in G_m(S)$ and $\alpha* \in \mathrm{Hom}_{S-gr.}(G_{m,S}, T)$ such that $\alpha \circ \alpha* = 2$ and that, if $(u, v) \in V(S')$ and $uv \in W(S')$, then $1 + auv$ is invertible and

$$p_\alpha(u)\, p_{-\alpha}(v) = p_{-\alpha}\big(v / (1 + a u v) \big) \cdot \alpha^*(1 + a u v) \cdot p_\alpha\big(u / (1 + a u v) \big).$$

Consider the open set V' of $G^2_{a,S}$ defined by "$1 + auv$ invertible", i.e. $V' = (G^2_{a,S})_f$ where $f(u, v) = 1 + auv$. The two sides of the preceding formula define morphisms from V' to G which coincide in a neighborhood of the section 0, hence coincide on V'. The preceding formula is therefore valid for every section (u, v) of V'. In particular, it follows that $V' \subset U$, where U is the open set introduced at the beginning of A).

Let us prove that $U = V'$. Returning to the notations of A), one has a morphism

$$A : U \to G_{a,S}$$

which, on V', is defined by $A(u, v) = v(1 + auv)^{-1}$. To show that $U = V'$, which is a set-theoretic question, one is reduced to the case where S is the spectrum of a field k, hence to the obvious assertion: the domain of definition of the rational map $G^2_{a,k} \to G_{a,k}$ defined by the rational fraction $Y/(1 + aXY)$ is the open set defined by the function $1 + aXY$.

G) One has thus proved the existence of a and $\alpha*$, as well as the two additional properties announced. It remains to prove uniqueness. Let then a' and $\alpha*'$ also satisfy the required conditions. If $u, v \in G_a(S')^2$, one has at once:

[1233] N.D.E.: that is, we have made the change of variables $x' = 1 + ax$, i.e. $x = (x' - 1)/a$.

[1234] N.D.E.: We have slightly modified what follows, since an open set V was already introduced in A).

$$1 + a u v \text{ invertible} \Rightarrow 1 + a' u v \text{ invertible and } v / (1 + a u v) = v / (1 + a' u v);$$

one therefore has for every section u of $G_a(S')$

$$1 + a u \text{ invertible} \Rightarrow 1 + a u = 1 + a' u,$$

which proves at once $a = a'$.

With the same notations, one then has

$$1 + a u \text{ invertible} \Rightarrow \alpha^*(1 + a u) = \alpha^{*\prime}(1 + a u),$$

hence also $\alpha* = \alpha*'$.

Corollary 2.4. *Let* $\exp(Y)t\exp(X)$ *and* $\exp(Y')t'\exp(X')$ *be two elements of* $\Omega(S')$*. Then their product is in* $\Omega(S')$ *if and only if* $u = 1 + \langle X, Y'\rangle$ *is invertible, and one has then*

$$\text{(F}'\text{)} \quad \exp(Y)\, t \exp(X) \cdot \exp(Y')\, t' \exp(X') = \exp(Y + u^{-1} \alpha(t)^{-1} Y') \cdot t\, t'\, \alpha^*(u) \cdot \exp(u^{-1} \alpha(t')^{-1} X + X').$$

Remark 2.5. *One can also write formula (F) of Theorem 2.1 without invoking the morphisms exp. Indeed, transporting through these morphisms the duality* $g_\alpha \otimes g_{-\alpha} \to O_S$*, one obtains a canonical pairing of vector bundles:*

$$U_\alpha \times_S U_{-\alpha} \to G_{a, S},$$

which we shall still denote $(x, y) \mapsto \langle x, y\rangle$*. One therefore has*

$$\langle \exp X, \exp Y\rangle = \langle X, Y\rangle.$$

If $x \in U_\alpha(S')$*,* $y \in U_{-\alpha}(S')$ *and if* $1 + \langle x, y\rangle \in G_m(S')$*, one has*

$$\text{(F)} \quad x \cdot y = y^{(1 + \langle x, y\rangle)^{-1}} \cdot \alpha^*(1 + \langle x, y\rangle) \cdot x^{(1 + \langle x, y\rangle)^{-1}}.$$

Corollary 2.6. *The pairing*

$$W(g_\alpha) \times_S W(g_{-\alpha}) \to G_{a, S}$$

defines a pairing of principal $G_{m,S}$*-bundles*

$$W(g_\alpha)^\times \times_S W(g_{-\alpha})^\times \to G_{m, S}.$$

This pairing will be denoted $(X, Y) \mapsto \langle X, Y\rangle$*, or more simply* $(X, Y) \mapsto XY$*.*

For every section $X \in \Gamma(S, g_\alpha)^\times$, there therefore exists a unique section X^{-1} of $\Gamma(S, g_{-\alpha})^\times$ such that $XX^{-1} = 1$. One has $(zX)^{-1} = z^{-1}X^{-1}$. The morphism

$$s : W(g_\alpha)^\times \to W(g_{-\alpha})^\times$$

thus defined is therefore an isomorphism of schemes, compatible with the isomorphism $s : z \mapsto z^{-1}$ on the operator groups.

Definition 2.6.1. *One says that* X *and* $s(X) = X^{-1}$ *are* paired.

Apply Corollary 2.4 to $Y = 0 = X'$ and $Y' = aX^{-1}$, $a \in O_S(S)$. Then $u = 1 + a$ and $u^{-1}Y' = u^{-1}(u-1)X^{-1} = (1-u^{-1})X^{-1}$, whence:

Corollary 2.7. *Let $X \in \Gamma(S, g_\alpha)^\times$ and $u \in \Gamma(S, O_S)^\times$. One has*

$$\alpha^*(u) = \exp((u^{-1} - 1) X^{-1}) \exp(X) \exp((u - 1) X^{-1}) \exp(-u^{-1} X).$$

Definition 2.8. *The morphism $\alpha*$ is called the* coroot *associated with the root α.*

Remark 2.9. *If (G, T, α) is an S-elementary system, $(G, T, -\alpha)$ is also one. By Theorem 2.1 one therefore has a duality between $g_{-\alpha}$ and g_α, and a coroot $(-\alpha)*$. Taking the inverse of formula (F), one proves at once*

$$\langle X, Y\rangle = \langle Y, X\rangle, \qquad (-\alpha)^* = -\alpha^*.$$

Let us now pass to the Lie algebra of G. The root α and the coroot $\alpha*$ define the linear forms

$$O_S \xrightarrow{\alpha^*} t \xrightarrow{\alpha} O_S.$$

One will denote $H_\alpha = \alpha * (1)$. One calls α the *infinitesimal root* associated with α, and H_α the corresponding *infinitesimal coroot*.

Lemma 2.10. *Let $S' \to S$ and $X, X' \in W(g_\alpha)(S')$, $H \in W(t)(S')$, $Y, Y' \in W(g_{-\alpha})(S')$, $t \in T(S')$. One has*

(1) $\quad \mathrm{Ad}(t)\, H = H, \quad \mathrm{Ad}(t)\, X = \alpha(t)\, X, \quad \mathrm{Ad}(t)\, Y = \alpha(t)^{-1}\, Y.$

(2) $\quad \begin{cases} \mathrm{Ad}(\exp(X))\, H = H - \alpha(H)\, X, \quad \mathrm{Ad}(\exp(X))\, X' = X', \\ \mathrm{Ad}(\exp(X))\, Y = Y + \langle X, Y\rangle H_\alpha - \langle X, Y\rangle X. \end{cases}$

(2′) $\quad \begin{cases} \mathrm{Ad}(\exp(Y))\, H = H + \alpha(H)\, Y, \quad \mathrm{Ad}(\exp(Y))\, Y' = Y', \\ \mathrm{Ad}(\exp(Y))\, X = X + \langle X, Y\rangle H_{-\alpha} - \langle X, Y\rangle Y. \end{cases}$

(3) $\quad [H, X] = \alpha(H)\, X, \quad [H, Y] = -\alpha(H)\, Y, \quad [X, Y] = \langle X, Y\rangle H_\alpha.$

$$(4) H_{-\alpha} = -H_\alpha.$$

$$(5) \alpha(H_\alpha) = 2.$$

The proof of these various formulas is either trivial or an immediate consequence of formula (F) of 2.1.

Corollary 2.11. *Suppose H_α is non-zero on every fiber (which is in particular the case if 2 is invertible on S, by (5)). Then $X_\alpha \in \Gamma(S, g_\alpha)^\times$ and $X_{-\alpha} \in \Gamma(S, g_{-\alpha})^\times$ are paired if and only if $[X_\alpha, X_{-\alpha}] = H_\alpha$.*

2.12. Let (G, T, α) be an S-elementary system. We know (1.19) that the center of G is $Centr(G) = Ker(\alpha)$, a group of multiplicative type and of finite type. If Q is a subgroup of multiplicative type of $Centr(G)$, the quotient G/Q is affine over S (Exp. IX 2.5), smooth over S (Exp. VI_B 9.2) with connected reductive fibers of semisimple rank 1 (Exp. XIX 1.8).

Set $G' = G/Q$; this is a reductive S-group of semisimple rank 1; $T' = T/Q$ is a maximal torus of it. The open set $U_{-\alpha} \cdot T \cdot U_\alpha$ of G is stable under Q, and one sees at once that the quotient is isomorphic to $U_{-\alpha} \times_S (T/Q) \times_S U_\alpha$. If one denotes by α' the character of T' induced by α, it follows that the morphism derived from the canonical morphism $G \to G'$ induces isomorphisms

$$\mathfrak{g}_\alpha \xrightarrow{\sim} \mathfrak{g}'_{\alpha'} \quad \text{and} \quad \mathfrak{g}_{-\alpha} \xrightarrow{\sim} \mathfrak{g}'_{-\alpha'}.$$

In particular, α' is a root of G' with respect to T'. Hence, denoting by α/Q the character $T/Q \to G_{m,S}$ induced by α, one has:

Lemma 2.13. *If Q is a subgroup of multiplicative type of $Ker(\alpha)$, then*

$$(G/Q, T/Q, \alpha/Q)$$

is an elementary system.

Lemma 2.14. *Under the preceding conditions, the following diagrams are commutative:*

$$\begin{tikzcd}
W(\mathfrak{g}_\alpha) \arrow[r, "\exp"] \arrow[d, "\mathrm{can}"] & G \arrow[d, "\mathrm{can}"] & W(\mathfrak{g}_{-\alpha}) \arrow[l, "\exp"'] \arrow[d, "\mathrm{can}"] \\
W(\mathfrak{g}'_{\alpha'}) \arrow[r, "\exp"] & G' & W(\mathfrak{g}'_{-\alpha'}) \arrow[l, "\exp"']
\end{tikzcd}$$

$$\begin{tikzcd}
\mathfrak{g}_\alpha \otimes \mathfrak{g}_{-\alpha} \arrow[r, "\sim"] \arrow[d, "\mathrm{can}"] & \mathcal{O}_S \arrow[d, "\mathrm{id}"] \\
\mathfrak{g}'_{\alpha'} \otimes \mathfrak{g}'_{-\alpha'} \arrow[r, "\sim"] & \mathcal{O}_S
\end{tikzcd}$$

$$\begin{tikzcd}
& T \arrow[dr, "\alpha"] & \\
G_{m,S} \arrow[ur, "\alpha^*"] \arrow[d, "\mathrm{can}"] & & G_{m,S} \\
G_{m,S} \arrow[dr, "\alpha'^*"'] & & G_{m,S} \arrow[u] \\
& T' \arrow[ur, "\alpha'"'] &
\end{tikzcd}$$

3. The Weyl group

Notations 3.0.[1235] If (G, T, α) is an S-elementary system, one will denote

$$N = \mathrm{Norm}_G(T), \qquad W = \mathrm{Norm}_G(T)/T,$$

(cf. Exp. XIX 6.3); N is a closed subgroup of G, smooth over S. One will denote by $N^\times = N - T$ the open subscheme of N induced on the complement of T.[1236] Let R

[1235] N.D.E.: We have added the numbering 3.0, for later references.
[1236] N.D.E.: We have replaced Q by the more suggestive notation $N^\times$.

be the (unique) maximal torus of $Ker(\alpha)$, and T' the image of $\alpha * : G_{m,S} \to T$, which is a subtorus of dimension 1 of T.

The morphism

$$T' \times_S R \to T$$

induced by the product in T is surjective (hence faithfully flat); indeed, one is reduced to checking this on the geometric fibers, and this follows at once from the formula $\alpha \circ \alpha * = 2$.

Theorem 3.1. *With the preceding notations:*

(i) W is isomorphic to the constant group $(\mathbb{Z}/2\mathbb{Z})_S$.

(ii) $N^\times$ is a principal homogeneous bundle locally trivial under T, on the left by the law $(t,q) \mapsto tq$ (resp. on the right by the law $(q,t) \mapsto qt$).

(iii) One has the formula

$$\mathrm{int}(w)\, t = t \cdot \alpha^*(\alpha(t)^{-1})$$

for $w \in N^\times(S')$, $t \in T(S')$, $S' \to S$. In the decomposition $T_{S'} = T'_{S'} \cdot R_{S'}$, $int(w)$ induces the identity on $R_{S'}$ and the symmetry on $T'_{S'}$. One has the relations

$$\alpha \circ \mathrm{int}(w) = \alpha^{-1}, \quad \mathrm{int}(w) \circ \alpha^* = (\alpha^*)^{-1}.$$

(iv) For $X \in W(g_\alpha)^\times(S')$, $S' \to S$, set

$$w_\alpha(X) = \exp(X)\exp(-X^{-1})\exp(X).$$

Then $w_\alpha(X) \in N^\times(S')$ and the morphism $w_\alpha : W(g_\alpha)^\times \to N^\times$ thus defined satisfies

$$w_\alpha(zX) = \alpha^*(z)\, w_\alpha(X) = w_\alpha(X)\, \alpha^*(z)^{-1},$$

for $z \in G_m(S')$, $X \in W(g_\alpha)^\times(S')$, $S' \to S$.

(v)[1237] *For $X, Y \in W(g_\alpha)^\times(S')$, $S' \to S$, one has, with the notations of 2.6,*

$$w_\alpha(X)\, w_\alpha(Y) = \alpha^\vee(-X Y^{-1}).$$

In particular,

$$w_\alpha(X)^2 = \alpha^*(-1) \in {}_2T(S) \cap \mathrm{Centr}(G)(S),$$
$$w_\alpha(X)^{-1} = w_\alpha(-X) = \alpha^*(-1)\, w_\alpha(X).$$

(vi) If one defines analogously, for $Y \in W(g_{-\alpha})^\times(S')$,

$$w_{-\alpha}(Y) = \exp(Y)\exp(-Y^{-1})\exp(Y),$$

one has (in addition to formulas analogous to the preceding)

$$w_{-\alpha}(X^{-1}) = w_\alpha(X)^{-1} = w_\alpha(-X),$$
$$w_\alpha(X)\, w_{-\alpha}(Y) = \alpha^*(XY).$$

[1237] N.D.E.: We have corrected the first formula of the original.

Proof. (i) has already been seen in Exp. XIX 2.4; it follows at once that $N^\times$ is indeed a principal homogeneous bundle under T for the laws defined in (ii); the fact that it is locally trivial (for the Zariski topology) follows notably from (iv).

Let us prove (iii); if $w \in N^\times(S)$, it is clear that $\alpha \circ int(w)$ is a root of G with respect to T, hence is locally equal to α or $-\alpha$; since on each fiber it is $-\alpha$ (*Bible*, 12-05, proof of the corollary to prop. 1), one has $\alpha \circ int(w) = -\alpha$. By transport of structure, one deduces

$$-\alpha^* = int(w)^{-1} \circ \alpha^* = int(w) \circ \alpha^*,$$

since $int(w)^2 = int(w^2)$ and w^2 is a section of T. Therefore $int(w)$ induces the symmetry on T'; since R is central, $int(w)$ induces the identity on R. The formula of (iii) defines a morphism $T \to T$ which satisfies the same properties, hence coincides with $int(w)$.

Let us prove (iv). One has successively

$$\begin{aligned} w_\alpha(X)\, t\, w_\alpha(X)^{-1} &= \exp(X) \exp(-X^{-1}) \exp(X)\, t \exp(-X) \exp(X^{-1}) \exp(-X) \\ &= \exp(X) \exp(-X^{-1}) \exp(X - \alpha(t) X) \exp(\alpha(t)^{-1} X^{-1}) \exp(-\alpha(t) X)\, t. \end{aligned}$$

By application of formula (F), one has

$$\exp(-X^{-1}) \exp((1 - \alpha(t)) X) = \exp((\alpha(t)^{-1} - 1) X)\, \alpha^*(\alpha(t)^{-1}) \exp(-\alpha(t)^{-1} X^{-1}).$$

Substituting in the preceding relation, one finds

$$\begin{aligned} int(w_\alpha(X))\, t &= \exp(\alpha(t)^{-1} X)\, \alpha^*(\alpha(t)^{-1}) \exp(-\alpha(t) X)\, t \\ &= \exp(a X)\, \alpha^*(\alpha(t)^{-1})\, t, \end{aligned}$$

where

$$a = \alpha(t)^{-1} - (\alpha \circ \alpha^*)(\alpha(t)^{-1})\, \alpha(t),$$

but $\alpha \circ \alpha* = 2$, which gives at once $a = 0$ and $w_\alpha(X) \in N^\times(S')$.

Let us prove the second assertion of (iv). One has[1238]

$$\begin{aligned} \alpha^*(z)\, w_\alpha(X) &= \exp(z^2 X) \exp(-z^{-2} X^{-1}) \exp(z^2 X)\, \alpha^*(z) \\ &= \exp(z X) \exp((z^2 - z) X) \exp(-z^{-2} X^{-1}) \exp(z^2 X)\, \alpha^*(z) \\ &= \exp(z X) \exp(-z^{-1} X^{-1})\, \alpha^*(z)^{-1} \exp((z^3 - z^2) X) \exp(z^2 X)\, \alpha^*(z) \\ &= \exp(z X) \exp(-z^{-1} X^{-1}) \exp(z X) = w_\alpha(z X). \end{aligned}$$

Let us prove (v). By virtue of the preceding result, the first formula of (v) follows at once from the second; let us prove the second:

[1238] N.D.E.: The first equality follows from 1.5 (i) which, combined with the equality $\alpha \circ \alpha* = 2$, gives the formulas

$$(\dagger) \quad \alpha^*(z) \exp(X)\, \alpha^*(z)^{-1} = \exp(z^2 X), \qquad \alpha^*(z) \exp(X^{-1})\, \alpha^*(z)^{-1} = \exp(z^{-2} X),$$

the third equality follows from formula (F), and the fourth from (†), again. Finally, an analogous calculation shows that $w_\alpha(X)\alpha * (z^{-1}) = w_\alpha(zX)$.

$$
\begin{aligned}
w_\alpha(X)^2 &= \exp(X)\exp(-X^{-1})\exp(2X)\exp(-X^{-1})\exp(X)\\
&= \exp(X)\exp(-X^{-1})\exp(X^{-1})\,\alpha^*(-1)\exp(-2X)\exp(X)\\
&= \exp(X)\,\alpha^*(-1)\exp(-X) = \alpha^*(-1),
\end{aligned}
$$

since $\alpha(\alpha*(-1)) = (-1)^2 = 1$, which proves that $\alpha*(-1) \in Centr(G)(S)$.

Let us prove (vi). The first assertion is a particular case of the second; let us prove the second. Both sides of this formula define morphisms $W(g_\alpha)^\times \times_S W(g_{-\alpha})^\times \to G$. To prove that they coincide, it suffices to do so on a non-empty open set on each fiber (Exp. XVIII 1.4); it therefore suffices to verify the relation when $1 + XY$ is invertible. One then has successively:

$$
\begin{aligned}
w_\alpha(X)\,w_{-\alpha}(Y) &= \exp(X)\exp(-X^{-1})\exp(X)\exp(Y)\exp(-Y^{-1})\exp(Y)\\
&= \exp(X)\exp(-X^{-1})\exp(Y/(1+XY))\,\alpha^*(1+XY)\exp(X/(1+XY))\exp(-Y^{-1})\exp(Y\\
&= \exp(X)\exp(-X^{-1}/(1+XY))\,\alpha^*(1+XY)\exp(-Y^{-1}/(1+XY))\exp(Y)\\
&= \exp(-X^{-2}Y^{-1})\,\alpha^*(XY/(1+XY))\exp(X+Y^{-1})\,\alpha^*(1+XY)\exp(-Y^{-1}/(1+XY))\ e\\
&= \exp(-X^{-2}Y^{-1})\,\alpha^*(XY)\exp((Y^{-1}+X)/(1+XY)^2)\exp(-Y^{-1}/(1+XY))\exp(Y)\\
&= \alpha^*(XY)\exp(-Y)\exp(Y) = \alpha^*(XY).
\end{aligned}
$$

Corollary 3.2. *Let $n \in \mathbb{Z}$, $n \neq 0$. For every $w \in G(S)$, the following conditions are equivalent:*

(i) $w \in N^\times(S)$,

(ii) one has $int(w) \circ n\alpha* = -n\alpha*$ (recall that $(n\alpha*)(z) = \alpha*(z)^n$).

One has (i) ⇒ (ii) (assertion (iii) of Theorem 3.1); conversely, one may suppose that $N^\times$ admits a section, and one is reduced to proving:

Lemma 3.3. *One has $Centr_G(n\alpha*) = T$ for $n \neq 0$.*

Indeed, the image T' of $n\alpha*$ is a subtorus of G. It follows (Exp. XIX 2.8) that $Centr_G(n\alpha*)$ is a reductive subgroup of G containing T. Since on each fiber one has $Centr_{G_s}(n\alpha*_s) \neq G_s$, then $Centr_{G_s}(n\alpha*_s) = T_s$ (Exp. XIX 1.6.3)[1239], hence $Centr_G(n\alpha*) = T$, since these are smooth subgroups of G.

Remark 3.4. *The construction of w_α and the fact that $w_\alpha(X)$ normalizes T rely only on formula (F). In particular, if G is an S-group satisfying the conditions of 2.2, $Norm_G(T)$ differs from T on each fiber. It follows that if G is an affine S-group with connected fibers satisfying the conditions of 2.2, it is reductive of semisimple rank 1. Indeed, it is smooth in a neighborhood of the unit section, hence smooth, and one can apply the criterion of Exp. XIX 1.11.*

3.5. Before stating the following theorem, let us make a few remarks. We identify as usual $g_{-\alpha}$ with $(g_\alpha)^{\otimes -1}$. Similarly, we shall identify $\mathrm{Hom}_{O_S}(g_{-\alpha}, g_\alpha)$ with $(g_\alpha)^{\otimes 2}$ and hence

$$Isom_{O_S\text{-}mod.}(W(g_{-\alpha}), W(g_\alpha)) \cong W((g_\alpha)^{\otimes 2})^\times.$$

[1239]N.D.E.: The hypothesis $Centr_{G_s}(n\alpha*_s) \neq G_s$ entails that $\dim Centr_{G_s}(n\alpha*_s) - \dim T_s < 2$, but this difference is even, by *loc. cit.*

If $w \in N^\times(S)$, then $Ad(w)$ permutes g_α and $g_{-\alpha}$ (3.1, (iii)), hence defines an isomorphism:

$$a_\alpha(w) : g_{-\alpha} \xrightarrow{\sim} g_\alpha,$$

which we shall therefore identify with a section $a_\alpha(w) \in \Gamma(S, (g_\alpha)^{\otimes 2})^\times$. This construction is compatible with base change and therefore defines a morphism

$$a_\alpha : N^\times \to W((g_\alpha)^{\otimes 2})^\times,$$

such that $a_\alpha(w)Y = Ad(w)Y$ for all $w \in N^\times(S')$, $Y \in \Gamma(S', g_{-\alpha})^\times$, $S' \to S$.

Theorem 3.6. *(i) One has*

$$\mathrm{int}(w)\exp(Y) = \exp(a_\alpha(w)\,Y)$$

for every $S' \to S$ and all $w \in N^\times(S')$, $Y \in W(g_{-\alpha})(S')$.

(ii) One has

$$a_\alpha(t\,w) = \alpha(t)\,a_\alpha(w), \qquad a_\alpha(w\,t) = \alpha(t)^{-1}\,a_\alpha(w).$$

(iii) If one defines analogously $a_{-\alpha} : N^\times \to W((g_{-\alpha})^{\otimes 2})^\times$, one has

$$a_{-\alpha}(w) = a_\alpha(w)^{-1}.$$

[1240]

(iv) For every $X \in W(g_\alpha)^\times(S')$, $S' \to S$, one has

$$a_\alpha(w_\alpha(X)) = -X^2.$$

Assertion (i) is trivial, by the characterization of the morphisms exp given in 1.5. Assertion (ii) is immediate, as is (iii). Let us prove (iv): let $X \in \Gamma(S', g_\alpha)^\times$, $Z \in \Gamma(S', g_\alpha)$; by definition[1241]

$$a_\alpha(w_\alpha(X))^{-1}(Z) = Ad(w_\alpha(X))(Z) = Ad(\exp(X))\,Ad(\exp(-X^{-1}))\,Ad(\exp(X))(Z).$$

Applying formulas (2′) and (2) of Lemma 2.10, as well as the equalities $H_{-\alpha} = -H_\alpha$, $\alpha(H_\alpha) = 2$ (*loc. cit.* (4) and (5)) and $\langle X, X^{-1}\rangle = 1$ (2.6), one obtains that the right-hand side equals successively:

$$\begin{aligned} Ad(\exp(X))\,Ad(\exp(-X^{-1}))(Z) &= Ad(\exp(X))(Z + \langle X^{-1}, Z\rangle\,(H_\alpha - X^{-1})) \\ &= Z + \langle X^{-1}, Z\rangle\,(H_\alpha - 2X - X^{-1} - H_\alpha + X) \\ &= Z - \langle X^{-1}, Z\rangle\,X - \langle X^{-1}, Z\rangle\,X^{-1}. \end{aligned}$$

[1240] N.D.E.: that is, $a_{-\alpha}(w)$ and $a_\alpha(w)$ are paired, cf. 2.6.1.
[1241] N.D.E.: We have detailed what follows.

But $Z = \langle X^{-1}, Z\rangle X$ and $\langle X^{-1}, Z\rangle X^{-1} = X^{-2}Z$, hence this shows that $a_\alpha(w_\alpha(X))^{-1} = -X^{-2}$, whence $a_\alpha(w_\alpha(X)) = -X^2$.

Corollary 3.7. *One has in particular*

$$\mathrm{int}(w_\alpha(X))\, \exp(X) = \exp(-X^{-1}),$$

whence (by the definition of $w_\alpha(X)$):

$$w_\alpha(X) \exp(X)\, w_\alpha(X)^{-1} = \exp(-X)\, w_\alpha(X) \exp(-X),$$

or, by an immediate calculation,

$$(w_\alpha(X) \exp(X))^3 = e.$$

Corollary 3.8. *Let $X \in \Gamma(S, g_\alpha)^\times$ and $n \in \mathbb{Z}$, $n \neq 0$. Then $w_\alpha(X)$ is the unique section $w \in G(S)$ satisfying*

(i) $int(w) \circ n\alpha* = -n\alpha*$.

(ii) $(w \exp(X))^3 = e$.

One knows that $w_\alpha(X)$ does satisfy these conditions. Conversely, let $w \in G(S)$ satisfy (i) and (ii). By 3.2 and 3.1 (ii), one knows that there exists $t \in T(S)$ such that $w = w_\alpha(X)t$. Set $u = \exp(X)$. One then has

$$w\, u\, w^{-1} = w_\alpha(X)\, t \exp(X)\, t^{-1}\, w_\alpha(X)^{-1} = \exp(-\alpha(t)\, X^{-1}),$$

and on the other hand

$$\begin{aligned} u^{-1}\, w\, u^{-1} &= \exp(-X)\, w_\alpha(X)\, t \exp(-X) \\ &= \exp(-X)\, w_\alpha(X) \exp(-X) \exp(X - \alpha(t)\, X)\, t \\ &= \exp(-X^{-1}) \exp(X - \alpha(t)\, X)\, t = \exp(-X^{-1})\, t \exp(H). \end{aligned}$$

Now $(wu)^3 = e \Leftrightarrow wuw^{-1} = u^{-1}wu^{-1}$; comparing the two decompositions of this element on $U_{-\alpha} \cdot T \cdot U_\alpha$, one extracts $t = e$.

Remark 3.9. *One can summarize a number of results of this number by the following diagram of principal homogeneous (left) bundles*

$$\begin{array}{ccccc} W(g_\alpha)^\times & \xrightarrow{w_\alpha} & N^\times & \xrightarrow{a_\alpha} & W((g_\alpha)^{\otimes 2})^\times \\ \downarrow & & \downarrow & & \downarrow \\ G_{m,S} & \xrightarrow{\alpha*} & T & \xrightarrow{\alpha} & G_{m,S}. \end{array}$$

Note that a_α is faithfully flat (α being so) and that w_α is a monomorphism if and only if $\alpha*$ is a monomorphism. We leave the reader the task of writing the corresponding diagrams for the right principal bundle structures, as well as the analogous diagrams for the root $-\alpha$, and of studying the relations between these various diagrams.

Lemma 3.10. *Let S be a scheme, q an integer > 0 such that $x \mapsto x^q$ defines an endomorphism of $G_{a,S}$, (G, T, α) and (G', T', α') two S-elementary systems, $f : G \to G'$ an S-group morphism. Let*

$$h : (\mathfrak{g}_\alpha)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'_{\alpha'}$$

be an isomorphism of $\mathcal{O}_S$-modules and

$$h^\vee : (\mathfrak{g}_{-\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'_{-\alpha'}$$

the contragredient isomorphism. For every $S' \to S$ and every $X \in W(\mathfrak{g}_\alpha)(S')$, suppose:

$$f(\exp(X)) = \exp(h(X^q)).$$

Then the following conditions are equivalent:

(i) $f(\alpha * (z)) = \alpha' * (z)^q$.

(ii) $f(w_\alpha(Z)) = w_{\alpha'}(h(Z^q))$.

(iii) $f(\exp(Y)) = \exp(h^\vee(Y^q))$.

(Each condition is to be read: for every $S' \to S$ and every $z \in G_m(S')$, $Z \in W(\mathfrak{g}_\alpha)^\times(S')$, $Y \in W(\mathfrak{g}_{-\alpha})(S')$, one has ...).

Indeed, (i) ⇒ (ii) by 3.8, (ii) ⇒ (iii) by 3.7, (iii) ⇒ (i) by 2.7.

Proposition 3.11. *Let S be a scheme, $q \in \mathbb{Z}$, $q > 0$, such that $x \mapsto x^q$ defines an endomorphism of $G_{a,S}$, (G, T, α) and (G', T', α') two S-elementary systems, $f : G \to G'$ an S-group morphism. The following conditions on f are equivalent:*

(i) The restriction of f to T factors through a morphism $f_T : T \to T'$ making the diagram

$$\begin{tikzcd}
G_{m,S} \arrow[r, "\alpha^*"] \arrow[d, "q"] & T \arrow[r, "\alpha"] \arrow[d, "f_T"] & G_{m,S} \arrow[d, "q"] \\
G_{m,S} \arrow[r, "\alpha'^*"] & T' \arrow[r, "\alpha'"] & G_{m,S}
\end{tikzcd}$$

commutative.

(ii) There exists an (unique) isomorphism of $\mathcal{O}_S$-modules

$$h : (\mathfrak{g}_\alpha)^{\otimes q} \to \mathfrak{g}'_{\alpha'}$$

such that $f(\exp(X)) = \exp(h(X^q))$, $f(\exp(Y)) = \exp(h^\vee(Y^q))$ for all $X \in W(\mathfrak{g}_\alpha)(S')$, $Y \in W(\mathfrak{g}_{-\alpha})(S')$, $S' \to S$ (it follows that f also satisfies the equivalent conditions of 3.10).

One has (ii) ⇒ (i). Indeed, by 3.10, condition (ii) entails $f \circ \alpha * = q\alpha' *$, hence, by 3.3, $f|T$ factors through T'. It remains to prove $\alpha'(f(t)) = \alpha(t)^q$, which follows at once from the fact that f induces a morphism of groups $T \cdot U_\alpha \to T' \cdot U_{\alpha'}$.

Let us prove (i) ⇒ (ii). Let $X \in \Gamma(S, g_\alpha)$, $Y \in \Gamma(S, g_{-\alpha})$. Set $p^+(x) = f(\exp(xX))$ and $p^-(y) = f(\exp(yY))$; these are group morphisms

$$p^+, p^- : G_{a, S} \to G.$$

Now one has

$$\begin{aligned} int(\alpha'^*(z))^q (p^+(x)) &= int(f_T(\alpha^*(z))) (f(\exp(x X))) \\ &= f(int(\alpha^*(z))(\exp(x X))) \\ &= f(\exp(z^2 x X)) = p^+(z^2 x). \end{aligned}$$

Applying Lemma 1.2 (with $Q = \alpha' * (G_{m,S})$), one deduces that there exists a section $X' \in \Gamma(S, g'_{\alpha'})$ such that

$$f(\exp(x X)) = p^+(x) = \exp(x^q X').$$

Similarly, there exists a section $Y' \in \Gamma(S, g'_{-\alpha'})$ such that

$$f(\exp(y Y)) = \exp(y^q Y').$$

Writing now that f is a group morphism, hence that it respects formula (F), one obtains at once

$$X^q Y^q = (X Y)^q = X' Y'.$$

One concludes easily that $X^q \mapsto X'$ and $Y^q \mapsto Y'$ define isomorphisms h and $h^\vee$ as announced.

Proposition 3.12. *Let (G, T, α) be an S-elementary system, $w \in N^\times(S)$; set*

$$\Omega_0 = \Omega \cap int(w^{-1})(\Omega).$$

Let d be the function on Ω defined by

$$d(\exp(Y) \cdot t \cdot \exp(X)) = \alpha(t)^{-1} + X Y.$$

Then $\Omega_0 = \Omega_d$, and one has for $\exp(Y) \cdot t \cdot \exp(X) \in \Omega_0(S')$ the following formula (set $z = d(\exp(Y) \cdot t \cdot \exp(X))$):

$$(\star) \quad int(w)(\exp(Y) \cdot t \cdot \exp(X)) = \exp(z^{-1} a_\alpha(w)^{-1} X) \cdot t\, \alpha^*(z) \cdot \exp(z^{-1} a_\alpha(w) Y).$$

Moreover, one has $d \circ int(w) = d^{-1}$.

Indeed, one has at once[1242]

$$\begin{aligned} int(w)(\exp(Y) \cdot t \cdot \exp(X)) &= \exp(a_\alpha(w) Y) \cdot t\, \alpha^*(\alpha(t)^{-1}) \cdot \exp(a_\alpha(w)^{-1} X) \\ &= \exp(a_\alpha(w) Y) \cdot \exp(\alpha(t)\, a_\alpha(w)^{-1} X) \cdot t\, \alpha^*(\alpha(t)^{-1}). \end{aligned}$$

By 2.1, this is a section of Ω if and only if $1 + \alpha(t)XY$ is invertible, which proves indeed the equality $\Omega_0 = \Omega_d$; applying then formula (F) of *loc. cit.*, one deduces by an immediate calculation the announced formula (□). Finally, it follows from (□) that one has

[1242] N.D.E.: We have corrected the original by swapping $a_\alpha(w)$ and $a_\alpha(w)^{-1}$, and we have detailed the proof of the equality $d \circ int(w) = d^{-1}$.

$$(d \circ \mathrm{int}(w))(\exp(Y) \cdot t \cdot \exp(X)) = \alpha(t\, \alpha^*(z))^{-1} + z^{-2} X Y = z^{-2} (\alpha(t)^{-1} + X Y) = z^{-1},$$

whence the last assertion.

N.B. One notes that the function d is independent of the choice of w.

4. The isomorphism theorem

Theorem 4.1. *Let S be a scheme, $q \in \mathbb{Z}$, $q > 0$, such that $x \mapsto x^q$ is an endomorphism of $G_{a,S}$, (G, T, α) and (G', T', α') two S-elementary systems. Let*

$$h : (\mathfrak{g}_\alpha)^{\otimes q} \to \mathfrak{g}'_{\alpha'} \quad \text{and} \quad h^\vee : (\mathfrak{g}_{-\alpha})^{\otimes q} \to \mathfrak{g}'_{-\alpha'}$$

be two isomorphisms contragredient to each other. Let $f_T : T \to T'$ be an S-group morphism making the diagram

$$\begin{tikzcd}
G_{m,S} \arrow[r, "q"] \arrow[d, "\alpha^*"] & G_{m,S} \arrow[d, "\alpha'^*"] \\
T \arrow[r, "f_T"] \arrow[d, "\alpha"] & T' \arrow[d, "\alpha'"] \\
G_{m,S} \arrow[r, "q"] & G_{m,S}
\end{tikzcd}$$

commutative.

There exists a unique morphism of S-groups $f : G \to G'$ which extends f_T and satisfies

$$f(\exp(X)) = \exp(h(X^q))$$

for every $X \in W(\mathfrak{g}_\alpha)(S')$, $S' \to S$. Moreover, this morphism also satisfies

$$f(\exp(Y)) = \exp(h^\vee(Y^q)) \quad \text{and} \quad f(w_\alpha(Z)) = w_\alpha(h(Z^q)),$$

for every $S' \to S$ and all $Y \in \Gamma(S', \mathfrak{g}_{-\alpha})$, $Z \in \Gamma(S', \mathfrak{g}_\alpha)^\times$.

If $f : G \to G'$ extends f_T, then $f \circ \alpha* = (\alpha'*)^q$. If moreover f satisfies the second condition, then it satisfies the two others as well by 3.10. It follows that f is determined on Ω by the relation

$$f(\exp(Y)\, t\, \exp(X)) = \exp(h^\vee(Y^q))\, f_T(t)\, \exp(h(X^q)).$$

Since Ω is schematically dense in G, this already proves the uniqueness of f. To prove its existence, it suffices, by virtue of Exp. XVIII 2.3, to prove that the preceding formula defines a "generically multiplicative" morphism from Ω to G'. Now, by 2.4, this amounts to verifying that $\alpha' \circ f = \alpha^q$, which follows from the fact that f extends f_T.

Scholie 4.2. *One can also interpret 4.1 as follows: consider the category E of S-elementary systems and the category D of tuples*

$$(G_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} G_{m,S}, L),$$

where T is a torus, α and $\alpha*$ are group morphisms such that $\alpha \circ \alpha* = 2$, and L is an invertible $\mathcal{O}_S$-module (the reader will specify the morphisms of the two categories under consideration). One defines a functor $E \to D$ by

$$(G, T, \alpha) \mapsto (G_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} G_{m,S}, g_\alpha).$$

The preceding theorem says that this functor is fully faithful. It is in fact an equivalence of categories, as one will see in the next number. One already has:

Corollary 4.3. *If $q = 1$ and if f_T is an isomorphism, then f is an isomorphism.*

Corollary 4.4. *If $q = 1$ and if f_T is faithfully flat with kernel Q (cf. Exp. IX 2.7), then f is faithfully flat (quasi-compact) with kernel Q, hence identifies G' with G/Q.*

Indeed, if f_T is faithfully flat with kernel Q, then

$$Q = \mathrm{Ker}(f_T) \subset \mathrm{Ker}(f_T \circ \alpha') = \mathrm{Ker}(\alpha).$$

Introducing the S-elementary system $(G/Q, T/Q, \alpha/Q)$ of 2.13, one is reduced by 2.14 to proving that f/Q induces an isomorphism of G/Q onto G', which follows at once from 4.3.

5. Examples of elementary systems, applications

5.1. Let S be a scheme, L an invertible $\mathcal{O}_S$-module. Consider the group GL over S defined by

$$\mathrm{GL}(S') = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; a, d \in G_a(S'),\ b \in W(L)(S'),\ c \in W(L^{-1})(S'),\ ad - bc \in G_m(S') \right\}$$

equipped with the usual matrix multiplication law. It is locally isomorphic to $GL_{2,S}$. It is therefore an S-group scheme, affine and smooth over S, with connected fibers.

Remark. Let L' and L'' be two invertible sheaves on S, such that $L = L' \otimes L''^{-1}$.[1243] Then one has an isomorphism of S-groups:

$$\mathrm{GL} \xrightarrow{\sim} \mathrm{GL}(L' \oplus L'')$$

defined as follows: if x (resp. y) is a section of L' (resp. L'') on an open set V of S, one has

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

5.2. One will denote by SL the closed subgroup of GL defined by the relation $ad - bc = 1$. It is also an S-group scheme, affine and smooth over S, with connected fibers (isomorphic to $SL(L' \oplus L'')$ by the preceding isomorphism).

[1243] N.D.E.: We have corrected $L'' \otimes L'^{-1}$ to $L' \otimes L''^{-1}$ and we have detailed the sentence that follows.

Likewise, consider the morphism $G_{m,S} \to GL$ defined by $z \mapsto (z0/0z)$. It is a central monomorphism; by passage to the quotient, one deduces a group PL, smooth and affine over S, with connected fibers (cf. Exp. VIII 5.7).

One can see that, by passage to the quotient from the isomorphism of the preceding remark, PL is identified with the group of automorphisms of the projective bundle $P(L' \oplus L'')$ (cf. EGA, II 4.2.7). One will denote by i and p the canonical morphisms

$$SL \xrightarrow{i} GL \xrightarrow{p} PL;$$

i is a closed immersion, p is faithfully flat and affine.

5.3. Consider the group morphisms

$$t_G : G^2_{m,S} \to GL, \quad t_G(z, z') = \begin{pmatrix} z & 0 \\ 0 & z' \end{pmatrix},$$
$$t_S : G_{m,S} \to SL, \quad t_S(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix},$$
$$t_P : G_{m,S} \to PL, \quad t_P(z) = p(t_G(z, 1)).$$

These are group monomorphisms, which define in each group a (split) torus of relative codimension 2. For every $s \in S$, let

$$X \in \Gamma(s, L \otimes s)^{\times};$$

then the section $(0X/-X^{-1}0)$ of GL_s normalizes $t_G(G^2_{m,S})$ and does not centralize it; one concludes from Exp. XIX 1.6 that GL is reductive, of semisimple rank 1, with maximal torus $t_G(G^2_{m,S})$.

One argues similarly for SL and PL, and one sees that SL (resp. PL) is reductive, of semisimple rank 1, with maximal torus $t_S(G_{m,S})$ (resp. $t_P(G_{m,S})$).

5.4. Reasoning as usual, one determines at once the Lie algebra of these various groups and the adjoint action of the chosen maximal torus. Let us do it for GL; this is immediate by Exp. II 4.8: $Lie(GL/S)$ is the Lie algebra of the matrices below:

$$Lie(GL/S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; a \text{ and } d \text{ sections of } O_S,\ b \text{ section of } L,\ c \text{ section of } L^{-1} \right\}$$

with the usual bracket; one has

$$Ad(t_G(z, z')) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & z z'^{-1} b \\ z' z^{-1} c & d \end{pmatrix}.$$

Denote $Lie(GL/S) = g$. Let $\alpha_G : t_G(G^2_{m,S}) \to G_{m,S}$ be the character defined by

$$\alpha_G(t_G(z, z')) = z z'^{-1}.$$

One sees at once from the preceding relation that α_G is a root of GL with respect to $t_G(G^2_{m,S})$ and that the morphism

$$u : L \to g \quad (\text{resp. } u^- : L^{-1} \to g)$$

defined by $u(X) = (0X/00)$ (resp. $u^-(X) = (00/X0)$) is an isomorphism of L onto g_{α_G} (resp. of L^{-1} onto $g_{-\alpha_G}$).

One has thus proved that $(G, t_G(G^2_{m,S}), \alpha_G)$ is an S-elementary system.

Setting likewise

$$\alpha_S(t_S(z)) = z^2, \qquad \alpha_P(t_P(z)) = z,$$

one proves that $(SL, t_S(G_{m,S}), \alpha_S)$ and $(PL, t_P(G_{m,S}), \alpha_P)$ are elementary systems, and one defines isomorphisms of L (resp. L^{-1}) with the corresponding direct summands of the Lie algebras of SL and PL.

5.5. Set $\exp(0X/00) = (1X/01)$. One has thus defined a morphism

$$W(g_{\alpha_G}) \to GL$$

which induces on the Lie algebras the canonical morphism, hence is the unique morphism of this type (1.5). Similarly, one sets $\exp(00/Y0) = (10/Y1)$. Carrying out the explicit calculation of formula (F), one finds

$$\left\langle \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} \right\rangle = XY, \qquad \alpha^*_G(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = t_G(z, z^{-1}).$$

[1244]

The open set $N^\times = N^\times_G$ (defined before 3.1) is:

$$N^\times_G(S') = \left\{ \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \;\middle|\; P \in W(g_\alpha)^\times(S'),\ Q \in W(g_{-\alpha})^\times(S') \right\},$$

the morphism w_{α_G} (cf. 3.1 (iv)) is given, for every $X \in W(g_\alpha)^\times(S')$, by

$$w_{\alpha_G}(X) = (0X); (-X^{-1}0)$$

the morphism a_{α_G} (cf. 3.5) is given by:

$$\text{if } w = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \in N^\times_G(S'), \text{ then } a_{\alpha_G}(w) = PQ^{-1} \in W((g_\alpha)^{\otimes 2})^\times(S'),$$

that is, for every $Y \in W(g_{-\alpha})^\times(S')$, one has $a_{\alpha_G}(w)(Y) = PQ^{-1}Y \in W(g_\alpha)^\times(S')$.

5.6. We leave the reader the task of carrying out the same computations in SL and PL. One finds the same duality formula and the coroots

$$\alpha^*_S(z) = t_S(z), \qquad \alpha^*_P(z) = t_P(z^2).$$

Denote by p_T the morphism induced by $p : GL \to PL$ on $t_S(G_{m,S})$, i.e.

$$p_T(t_S(z)) = t_P(z^2).$$

One therefore has the commutative diagram:[1245]

[1244] N.D.E.: We have detailed what follows.

[1245] N.D.E.: where t_S and t_P are isomorphisms.

```
                G_{m, S}
             id /   \ 2
               /     \
        α*_S /         \ α*_P
             /           \
t_S(G_{m, S})  ——p_T——→  t_P(G_{m, S})
     ↑                          ↑
    t_S                        t_P
     |                          |
G_{m, S}                   G_{m, S}
             α_S \   / α_P
                  \ /
               G_{m, S}.
              /   2  \
             id
```

One recognizes in the central part the commutative diagram of 4.1[1246] relative to the canonical morphism $p \circ i : SL \to PL$, which induces a morphism of the preceding S-elementary systems.

5.7. Let now (G, T, α) be any S-elementary system. Consider the commutative diagram:

```
              G_{m, S}
           id ↙    ↘ 2
         ↙       α*       ↘
       ↙                      ↘
    ↙                             ↘
G_{m, S}  ——α*——→  T  ——α——→  G_{m, S}
    ↘                             ↗
      ↘                       ↗
         ↘       α        ↗
          2 ↘   ↗ id
              G_{m, S}
```

Combining the two preceding diagrams, one obtains a commutative diagram:

```
                    G_{m, S}
               α*_S ↙  α*  ↘ α*_P
                     ↙            ↘
                  ↙                  ↘
          ↙    α* ∘ t_S⁻¹       t_P ∘ α     ↘
t_S(G_{m, S})  ————→  T  ————→  t_P(G_{m, S})
          ↘                                  ↗
              ↘                          ↗
                  ↘     α          ↗
```

[1246]N.D.E.: with $q = 1$.

$$\alpha_S \searrow \qquad \nearrow \alpha_P$$
$$G_{m,S}.$$

Using 4.1, one therefore has:

Proposition 5.8. *Let S be a scheme, (G,T,α) an S-elementary system. Set $L = g_\alpha$ (and hence $L^{-1} = g_{-\alpha}$).*

(i) There exists a unique group morphism $f : SL \to G$ satisfying the following equivalent conditions:

(a) $f\begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = \alpha^*(z), \quad f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X);$

(b) $f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), \quad f\begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} = \exp(Y);$

(c) $f\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), \quad f\begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix} = w_\alpha(X).$

(ii) There exists a unique group morphism $g : G \to PL$ satisfying

$$g(t) = \begin{pmatrix} \alpha(t) & 0 \\ 0 & 1 \end{pmatrix}, \quad g(\exp(X)) = p\begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

Moreover, one has

$$g(\exp(Y)) = p\begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}, \quad g(w_\alpha(X)) = p\begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

The morphism g is faithfully flat quasi-compact with kernel $Ker(\alpha) = Centr(G)$, and $g \circ f$ is the canonical morphism $SL \to PL$.

Note that conditions (b) of (i) give an explicit description of the duality between g_α and $g_{-\alpha}$.

Corollary 5.9. *Let (G,T,α) be an S-elementary system. The subgroups $T \cdot U_\alpha$, $T \cdot U_{-\alpha}$, U_α and $U_{-\alpha}$ are closed.*

Since U_α is a closed subgroup scheme of $T \cdot U_\alpha$, it suffices to make the verification for the latter. By Noether's theorem (Exp. IV 5.3.1 and 6.4.1), it suffices to prove that $(T \cdot U_\alpha)/Ker(\alpha)$ is a closed subgroup of $G/Ker(\alpha)$. By virtue of 5.8, one is therefore reduced to proving that the subgroup of PL (or of GL, which amounts to the same by a new application of Noether's theorem) defined by $c = 0$ is closed, which is trivial.

Consequently, the morphisms $\exp$ of Theorem 1.5 (i) are closed immersions.

N.B. The corollary also follows from the fact that $T \cdot U_\alpha$ and $T \cdot U_{-\alpha}$ are "Borel subgroups" of G (cf. Exp. XII 7.10).

5.10. Let L be an invertible $\mathcal{O}_S$-module and

$$G_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} G_{m,S}$$

a diagram of groups[1247] such that $\alpha \circ \alpha* = 2$. Let R be the maximal torus of $Ker(\alpha)$ and $K = \alpha *^{-1}(R)$. Then K is a subgroup of multiplicative type of $G_{m,S}$; by virtue of $\alpha \circ \alpha* = 2$, it is even a subgroup of $\mu_{2,S}$. In particular the morphism

$$K \to SL, \quad z \mapsto \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}$$

is central. One has therefore a central group monomorphism:

$$K \to R \times SL, \quad z \mapsto \left(\alpha^*(z), \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}\right).$$

Consider the group $G = (R \times SL)/K$ obtained by passage to the quotient. It is an affine and smooth group over S, with connected fibers. It is immediate that the sequence

$$1 \to K \to R \times t_S(G_{m,S}) \xrightarrow{u} T \to 1$$

where $u(x, t_S(z)) = x\alpha*(z)$ is exact. The image of $R \times t_S(G_{m,S})$ in G is therefore a torus T' isomorphic to T. One then shows without difficulty that if α' is the character of T' deduced from α by the preceding isomorphism, (G, T', α') is an S-elementary system, that $g'_{\alpha'}$ is isomorphic to L, and that $\alpha'*$ is obtained from $\alpha*$ by the isomorphism $T \xrightarrow{\sim} T'$. One has therefore constructed an S-elementary system (G, T', α') such that the corresponding object

$$(G_{m,S} \xrightarrow{\alpha'*} T' \xrightarrow{\alpha'} G_{m,S}, \ g'_{\alpha'})$$

of the category D defined in 4.2 is isomorphic to

$$(G_{m,S} \xrightarrow{\alpha*} T \xrightarrow{\alpha} G_{m,S}, \ L).$$

One has therefore proved the

Theorem 5.11. *In the notations of 4.2, the functor*

$$(G, T, \alpha) \mapsto (G_{m,S} \xrightarrow{\alpha*} T \xrightarrow{\alpha} G_{m,S}, \ g_\alpha)$$

is an equivalence of categories between E and D.

6. Generators and relations for an elementary system

6.1. Let S be a scheme, (G, T, α) an S-elementary system. Let $X \in W(g_\alpha)^\times(S)$ and $u = \exp(X)$; one has seen in 3.8 that the element $w = w_\alpha(X)$ satisfies in particular the relation

$$(wu)^3 = e.$$

[1248]

[1247] N.D.E.: T being a torus.
[1248] N.D.E.: We have added the sentence that follows.

One denotes by s_α the automorphism of T induced by $int(w)$; according to Theorem 3.1 (iii), for every $S' \to S$ and $t \in T(S')$, one has

$$s_\alpha(t) = int(w)(t) = t \cdot \alpha^*(\alpha(t)^{-1}).$$

Theorem 6.2. *Let H be an S-sheaf of groups for (fppf). Let*

$$f_T : T \to H, \qquad f_\alpha : U_\alpha \to H$$

be group morphisms and $h \in H(S)$ a section of H. For there to exist a (necessarily unique) group morphism

$$f : G \to H$$

extending f_T and f_α and satisfying $f(w) = h$, it is necessary and sufficient that the following conditions be satisfied:

(i) For every $S' \to S$, every $t \in T(S')$ and every $x \in U_\alpha(S')$, one has

$$(1) \qquad f_T(t)\, f_\alpha(x)\, f_T(t)^{-1} = f_\alpha(t\, x\, t^{-1}) = f_\alpha(x^{\alpha(t)}).$$

(in other words, f_T and f_α extend to a group morphism from the semidirect product $T \cdot U_\alpha$ into H).

(ii) For every $S' \to S$ and every $t \in T(S')$, one has

$$(2) \qquad h\, f_T(t)\, h^{-1} = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^*(\alpha(t)^{-1})).$$

(iii) One has the two relations in $H(S)$:

$$(3) h^2 = f_T(\alpha * (-1)), (4) (h f_\alpha(u))^3 = e.$$

Proof. Denote additively U_α and $U_{-\alpha}$ and multiplicatively their vector structure. If f satisfies the conditions of the statement, one necessarily has for every $y \in U_{-\alpha}(S')$,

$$f(y) = f(w^{-1}\, w\, y\, w^{-1}\, w) = h\, f_\alpha(w^{-1}\, y\, w)\, h^{-1}.$$

Let then $f_{-\alpha} : U_{-\alpha} \to H$ be the morphism defined by

$$(*_1) \qquad f_{-\alpha}(y) = h\, f_\alpha(w^{-1}\, y\, w)\, h^{-1}.$$

It is a group morphism. On the other hand, f is determined on the big cell Ω by

$$f(y\, t\, x) = f_{-\alpha}(y)\, f_T(t)\, f_\alpha(x).$$

This shows the uniqueness of f; since the conditions of the statement are manifestly necessary, let us show that they are sufficient.

One has by (4)

$$h\, f_\alpha(u)\, h^{-1}\, h^2 = f_\alpha(-u)\, h^{-1}\, f_\alpha(-u).$$

Now, by (3) and (1), $h^2 = f_T(\alpha * (-1))$ commutes with $f_\alpha(-u)$, which gives

$$h\, f_\alpha(u)\, h^{-1} = f_\alpha(-u)\, h\, f_\alpha(-u).$$

But, by definition, $hf_\alpha(u)h^{-1} = f_{-\alpha}(wuw^{-1})$; by 3.7, since $u = \exp(X)$ and $w = w_\alpha(X)$, one has

$$(*_2) \qquad w\,u\,w^{-1} = -\tilde{u},$$

where ũ denotes the element paired with u. One obtains therefore:

$$(*_3) \qquad f_{-\alpha}(-\tilde{u}) = f_\alpha(-u)\,h\,f_\alpha(-u).$$

Let now t be a section of T over a variable $S' \to S$. Apply $int(f_T(t))$ to the preceding formula. One obtains on the left-hand side[1249]

$$\begin{aligned} f_T(t)\,f_{-\alpha}(-\tilde{u})\,f_T(t)^{-1} &= f_T(t)\,h\,f_\alpha(u)\,h^{-1}\,f_T(t)^{-1} \\ &= h\,(h^{-1}\,f_T(t)\,h)\,f_\alpha(u)\,(h^{-1}\,f_T(t)^{-1}\,h)\,h^{-1} \\ &= h\,f_T(s_\alpha(t))\,f_\alpha(u)\,f_T(s_\alpha(t))^{-1}\,h^{-1} = h\,f_\alpha(\alpha(s_\alpha(t))\,u)\,h^{-1} \end{aligned}$$

by (2) and (1); then since $s_\alpha(t) = t \cdot \alpha*(\alpha(t)^{-1})$ and $\alpha \circ \alpha* = 2$, this equals

$$h\,f_\alpha(\alpha(t)^{-1}\,u)\,h^{-1}.$$

Finally, by $(*_1)$ and $(*_2)$ one has

$$h\,f_\alpha(\alpha(t)^{-1}\,u)\,h^{-1} = f_{-\alpha}(\alpha(t)^{-1}\,w\,u\,w^{-1}) = f_{-\alpha}(-\alpha(t)^{-1}\,\tilde{u}).$$

The right-hand side of $(*_3)$ gives

$$f_\alpha(-\alpha(t)\,u) \cdot f_T(t)\,h\,f_T(t)^{-1}\,h^{-1} \cdot h \cdot f_\alpha(-\alpha(t)\,u)$$

and since $hf_T(t)^{-1}h^{-1} = f_T(s_\alpha(t^{-1})) = f_T(t \cdot \alpha*(\alpha(t)))$, this equals

$$f_\alpha(-\alpha(t)\,u) \cdot f_T(\alpha*(\alpha(t))) \cdot h \cdot f_\alpha(-\alpha(t)\,u).$$

Comparing the two expressions obtained, one obtains

$$f_{-\alpha}(-\alpha(t)^{-1}\,\tilde{u}) = f_\alpha(-\alpha(t)\,u) \cdot f_T(\alpha*(\alpha(t))) \cdot h \cdot f_\alpha(-\alpha(t)\,u).$$

Since $\alpha : T \to G_{m,S}$ is faithfully flat and H is a separated presheaf, one can replace $-\alpha(t)^{-1}$ by an arbitrary section of $G_{m,S}$, and one obtains the

Lemma 6.2.1. *For every $z \in G_m(S')$, $S' \to S$, one has*

$$f_{-\alpha}(z\,\tilde{u}) = f_\alpha(z^{-1}\,u) \cdot f_T(\alpha*(-z^{-1})) \cdot h \cdot f_\alpha(z^{-1}\,u).$$

Let now $x, y \in G_a(S')$, $S' \to S$; suppose y and $(1 + xy)$ invertible. Applying the lemma first to $z = y$, one obtains[1250]

$$f_\alpha(x\,u)\,f_{-\alpha}(y\,\tilde{u}) = f_\alpha((x + y^{-1})\,u) \cdot f_T(\alpha*(-y^{-1})) \cdot h \cdot f_\alpha(y^{-1}\,u).$$

Now $x + y^{-1} = y^{-1}(1 + xy)$. Applying the lemma to $z = y/(1 + xy)$, one finds

$$f_\alpha((x + y^{-1})\,u) = f_{-\alpha}(\,y/(1 + x\,y) \cdot \tilde{u}\,)\,f_\alpha(-(x + y^{-1})\,u) \cdot h \cdot f_T(\alpha*(-y\,/\,(1 + x\,y))).$$

Substituting in the preceding equality, one obtains

$$f_\alpha(x\,u)\,f_{-\alpha}(y\,\tilde{u}) = f_{-\alpha}(\,y\,/\,(1 + x\,y) \cdot \tilde{u}\,)\,f_\alpha(-(x + y^{-1})\,u) \cdot h^{-1} \cdot f_T(\alpha*(1 + x\,y)^{-1}) \cdot h \cdot$$

[1249]N.D.E.: We have corrected what follows.
[1250]N.D.E.: We have detailed the calculations that follow.

Since $h^{-1}f_T(t)h = f_T(s_\alpha(t))$ by (2) (note that $s_\alpha^2 = id$) and since $s_\alpha \circ \alpha* = -\alpha*$ (cf. 6.2.1), this equals

$$f_{-\alpha}(y / (1 + x y) \cdot \tilde{u})\ f_\alpha(-(x + y^{-1})\ u) \cdot f_T(\alpha^*(1 + x y)) \cdot f_\alpha(y^{-1}\ u).$$

Finally, since for all $x' \in U_\alpha(S')$ and $z \in G_m(S')$ one has

$$f_\alpha(x')\ f_T(\alpha^*(z)) = f_T(\alpha^*(z))\ f_\alpha(z^{-2}\ x'),$$

one obtains

$$\begin{aligned} f_\alpha(x\ u)\ f_{-\alpha}(y\ \tilde{u}) &= f_{-\alpha}(y / (1 + x y) \cdot \tilde{u}) \cdot f_T(\alpha^*(1 + x y)) \cdot f_\alpha(-y^{-1}\ (1 + x y)^{-1} / (1 + x \\ &= f_{-\alpha}(y / (1 + x y) \cdot \tilde{u}) \cdot f_T(\alpha^*(1 + x y)) \cdot f_\alpha(x / (1 + x y) \cdot u). \end{aligned}$$

One has thus proved:

Lemma 6.2.2. *Let $S' \to S$. If $a \in U_\alpha(S')$, $b \in U^\times_{-\alpha}(S')$, and $1 + ab \in G_m(S')$, one has*

$$f_\alpha(a)\ f_{-\alpha}(b) = f_{-\alpha}(b / (1 + a b))\ f_T(\alpha^*(1 + a b))\ f_\alpha(a / (1 + a b)).$$

By schematic density, this formula remains valid when $b \in U_{-\alpha}(S')$, $1 + ab$ being always invertible. Consider then the morphism

$$f : \Omega \to H$$

defined by $f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x)$.

It follows at once from 6.2.2, from condition 6.2 (i), and from formula (F′) of 2.4 that if $g, g' \in \Omega(S')$ and $gg' \in \Omega(S')$, one has $f(gg') = f(g)f(g')$. By Exp. XVIII 2.3 (iii) and 2.4[1251], there therefore exists a group morphism $G \to H$ extending f. Denote it also by f; it answers the question; it suffices to prove, indeed, that $f(w_\alpha) = h$. Now $w_\alpha = u \cdot (-\tilde{u}) \cdot u$, whence, by $(*\_3)$:[1252]

$$f(w_\alpha) = f_\alpha(u)\ f_{-\alpha}(-\tilde{u})\ f_\alpha(u) = h.$$

Remark 6.3. *We shall complete these results in Exp. XXIII 3.5.*

Exposé XXI. Root data

by M. Demazure

[1253] This Exposé collects, in the absence of a suitable reference,[1254] some known results on root data (= “abstract” root systems), most of which will be used in what follows.

Notations. — We denote by Q_+ the set of positive (or zero) rational numbers; one has $\mathbb{Z} \cap Q_+ = \mathbb{N}$. Let V be a $\mathbb{Q}$-vector space; if A (resp. B) is a subset of $\mathbb{Q}$ (resp. V), we denote by $A \cdot B$ the image of $A \otimes B$ under the morphism $\mathbb{Q} \otimes V \to V$, in other

[1251]N.D.E.: Note that each geometric fiber of G is connected, for example by 1.1.
[1252]N.D.E.: We have simplified the original by invoking $(*\_3)$.
[1253]N.D.E.: Version of 13/10/2024.
[1254]N.D.E.: For the results on root systems (§§ 1–5), one may consult [BLie], Chap. VI.

words the set of linear combinations of elements of B with coefficients in A. We write $-B = -1 \cdot B$. We denote by $E - F$ the set of elements of E that do not belong to F.

1. Generalities

1.1. Definitions, first properties

Definition 1.1.1. *Let M and $M*$ be two free $\mathbb{Z}$-modules of finite type in duality. Write $V = M \otimes \mathbb{Q}$, $V* = M * \otimes \mathbb{Q}$; these are two $\mathbb{Q}$-vector spaces in duality. We identify M (resp. $M*$) with a subset of V (resp. $V*$). The canonical bilinear form on $M * \times M$ (resp. $V * \times V$) is denoted $(,)$.*

Let R be a finite subset of M. Suppose we are given a map $\alpha \mapsto \alpha*$ of R into $M*$; the set of $\alpha*$, for $\alpha \in R$, is denoted $R*$. To each $\alpha \in R$ we associate the endomorphism s_α (resp. $s*_\alpha$) of M and V (resp. $M*$ and $V*$) given by the formulas:

(1) $s_\alpha(x) = x - (\alpha^*, x)\alpha$, i.e. $s_\alpha = \mathrm{id} - \alpha^* \otimes \alpha$;
(1*) $s^*_\alpha(u) = u - (u, \alpha)\alpha^*$, i.e. $s^*_\alpha = \mathrm{id} - \alpha \otimes \alpha^*$.

One says that the pair $(R, R*)$ (more precisely the pair $(R, R \to M*)$) is a root datum *in $(M, M*)$, or that $(M, M*, R, R*)$ is a* root datum, *if the following axioms are satisfied:*

(DR I) For each $\alpha \in R$, one has $(\alpha^*, \alpha) = 2$.
(DR II) For each $\alpha \in R$, one has $s_\alpha(R) \subset R$, $s^*_\alpha(R^*) \subset R^*$.

One says that R is the root system *of the root datum $\mathcal{R} = (M, M*, R, R*)$. The elements of R (resp. $R*$) are called the* roots *(resp.* coroots*) of the root datum.*

Remark 1.1.2. *Axiom (DR I) is equivalent to any of the following properties:*

(2) $s_\alpha s_\alpha = \mathrm{id}$, (2*) $s^*_\alpha s^*_\alpha = \mathrm{id}$,
(3) $s_\alpha(\alpha) = -\alpha$, (3*) $s^*_\alpha(\alpha^*) = -\alpha^*$.

Remark 1.1.3. *Axioms (DR I) and (DR II) entail*

$$R = -R, \quad R^* = -R^*, \quad 0 \notin R, \quad 0 \notin R^*.$$

Lemma 1.1.4. *The map $R \to R*$ is a bijection. More generally, if $\alpha, \beta \in R$ and $(\alpha*, x) = (\beta*, x)$ for all $x \in R$, then $\alpha = \beta$.*

Proof. Indeed, one then has $s_\beta(\alpha) = \alpha - 2\beta$, $s_\alpha(\beta) = \beta - 2\alpha$. One deduces at once

$$s_\beta s_\alpha(\alpha) = 2\beta - \alpha = \alpha + 2(\beta - \alpha), \qquad s_\beta s_\alpha(\beta - \alpha) = s_\beta(\beta - \alpha) = \beta - \alpha,$$

whence $(s_\beta s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha) \in R$ by (DR II). Since R is finite, one has $\beta - \alpha = 0$.

Corollary 1.1.5. *The inverse map $R* \to R$ defines a root datum*

$$\mathcal{R}^* = (M^*, M, R^*, R)$$

called the dual *of $\mathcal{R}$.*[1255]

[1255] N.D.E.: Note that $(\alpha*)* = \alpha$.

Definition 1.1.6. *We denote by $\Gamma_0(R)$ the subgroup of M generated by R. We denote by $V(R)$ the vector subspace of V generated by R, that is to say $\Gamma_0(R) \otimes \mathbb{Q}$. Applying these definitions to $\mathcal{R}*$, one constructs similarly $\Gamma_0(R*)$ and $V(R*)$.*

One calls reductive rank *of $\mathcal{R}$ the number*

$rgred(\mathcal{R}) = rank(M) = dim(V) = dim(V*) = rank(M*) = rgred(\mathcal{R}*)$.

One calls semisimple rank *of $\mathcal{R}$ the number*

$rgss(\mathcal{R}) = rank(R) = rank(\Gamma_0(R)) = dim(V(R))$.

One thus has $rgss(\mathcal{R}) \leqslant rgred(\mathcal{R})$.

We shall see below that $rgss(\mathcal{R}) = rgss(\mathcal{R}*)$, that is to say that $V(R)$ and $V(R*)$ have the same dimension.

Definition 1.1.7. *One says that $\mathcal{R}$ is* semisimple *(resp.* trivial*) if $rgss(\mathcal{R}) = rgred(\mathcal{R})$ (resp. $rgss(\mathcal{R}) = 0$). For $\mathcal{R}$ to be trivial it is therefore necessary and sufficient that R be empty. The trivial root datum of reductive rank zero is denoted $0 = (0, 0, \emptyset, \emptyset)$.*

Definition 1.1.8. *We denote by $W(\mathcal{R})$ the group of transformations of M generated by the s_α, $\alpha \in R$. We call it the* Weyl group *of $\mathcal{R}$. We write*

$$W*(\mathcal{R}) = W(\mathcal{R}*).$$

Then $W(\mathcal{R})$ operates on R, $\Gamma_0(R)$, $V(R)$, M and V. If $w \in W(\mathcal{R})$ and $x \in M$ (resp. $x \in V$), one has $wx - x \in \Gamma_0(R)$ (resp. $wx - x \in V(R)$); this is immediate from formula (1). Likewise for $W*(\mathcal{R})$.

Lemma 1.1.9. *For all $\alpha \in R$, $x \in V$, $u \in V*$, one has*

(4) $(s*_\alpha(u), s_\alpha(x)) = (u, x)$.

Proof. Indeed, taking the scalar product of (1) and (1*), one finds that the left-hand side equals $(u, x) + (u, \alpha)(\alpha*, x)((\alpha*, \alpha) - 2) = (u, x)$.

Remark 1.1.10. *If one assumes $0 \notin R$ and $0 \notin R*$, then 1.1.9 is equivalent to (DR I).*

Corollary 1.1.11. *Denote by $h \mapsto h^\vee$ the isomorphism of $GL(M)$ onto $GL(M*)$ that associates to h its contragredient. Then formula (4) is also written*

$$(5)(s_\alpha)^\vee = s*_\alpha.$$

Corollary 1.1.12. *The preceding isomorphism induces an isomorphism of $W(\mathcal{R})$ onto $W*(\mathcal{R})$.*

Scholie 1.1.13. *By virtue of the preceding result, we shall identify W and $W*$, and we shall regard W as a group of transformations of R, $R*$, M, $M*$, $\Gamma_0(R)$, $\Gamma_0(R*)$, V, $V*$, $V(R)$, $V(R*)$. We shall write s_α for $s*_\alpha$.*

1.2. The map p

Lemma 1.2.1. *Let $p : M \to M*$ (resp. $V \to V*$) be the linear map defined by*

$$p(x) = \sum_{u \in R*} (u, x)\, u. \tag{6}$$

Write $\ell(x) = (p(x), x)$. One has the following properties:

$$\ell(x) \geqslant 0, \quad \ell(\alpha) > 0 \quad \text{for } \alpha \in R. \tag{7}$$

$$\ell(w x) = \ell(x), \quad \text{for } w \in W. \tag{8}$$

$$(p(x), y) = (p(y), x), \quad \text{for } x, y \in V. \tag{9}$$

$$\ell(\alpha)\, \alpha* = 2\, p(\alpha), \quad \text{for } \alpha \in R. \tag{10}$$

Proof. The first three relations are immediate.[1256] Let us prove the last. By (1), for $u \in V*$:

$$\begin{aligned} (u, \alpha)^2\, \alpha* &= (u, \alpha)\, u - (u, \alpha)\, s_\alpha(u) \\ &= (u, \alpha)\, u + (u, s_\alpha(\alpha))\, s_\alpha(u) \\ &= (u, \alpha)\, u + (s_\alpha(u), \alpha)\, s_\alpha(u) \quad (\text{since } s_\alpha^2 = \mathrm{id}). \end{aligned}$$

Since s_α is a permutation of $R*$ (by (DR II)), it remains only to sum over $u \in R*$ to conclude.

Scholie 1.2.2. *Relation (10) says that $\alpha \mapsto \ell(\alpha)\alpha*$ is the restriction to R of a linear map from M into $M*$. In particular, one has*

$$(-\alpha)* = -\alpha * .$$

Corollary 1.2.3. *The map p induces an isomorphism of $V(R)$ onto $V(R)*$.*

Proof. Indeed, p sends $V(R)$ into $V(R*)$. We thus have

$$\dim(V(R)) \geqslant \dim(V(R*)).$$

Applying this inequality to the dual root datum, we deduce

$$\dim(V(R)) = \dim(V(R*)),$$

so p, being surjective, is also bijective.

Corollary 1.2.4. *One has* $\dim(V(R)) = \dim(V(R*))$*, hence*

$$rgss(\mathcal{R}) = rgss(\mathcal{R}*).$$

[1256] N.D.E.: (8) follows from (4), (2) and (DR II).

Corollary 1.2.5. *The bilinear form (u, x) is nondegenerate on $V(R*) \times V(R)$, and so puts these $\mathbb{Q}$-vector spaces in duality.*

Proof. Indeed, if $(u, x) = 0$ for all $u \in R*$, then $p(x) = 0$.

Corollary 1.2.6. *The symmetric bilinear form $(p(x), y)$ is positive nondegenerate on $V(R)$.*

Corollary 1.2.7. *W operates faithfully on R (and hence on the other sets of 1.1.13).*

Proof. Indeed, let $u \in V*$. Let $w \in W$; suppose that $w(\alpha) = \alpha$ for all $\alpha \in R$, and let us prove that $w(u) = u$. One has

$(w(u) - u, \alpha) = (w(u), \alpha) - (u, \alpha) = (u, w^{-1}(\alpha)) - (u, \alpha) = 0.$

But $w(u) - u \in V(R*)$. If it is orthogonal to all the roots, it is zero by 1.2.5.

Corollary 1.2.8. *The group W is finite.*

Proposition 1.2.9. *The operations of W respect the correspondence between roots and coroots. In other words, for $\alpha \in R$ and $w \in W$, one has*

$$w(\alpha*) = w(\alpha)*.$$

Proof. It suffices to verify this for $w = s_\beta$, $\beta \in R$, that is to say to verify the formula

$$s_\beta(\alpha*) = s_\beta(\alpha)*.$$

Now $\ell(s_\beta(\alpha)) \cdot s_\beta(\alpha)*/2$ equals:

$p(s_\beta(\alpha)) = \Sigma_{u \in R*} (u, s_\beta(\alpha))\, u = \Sigma_{u \in R*} (s_\beta(u), \alpha)\, u = \Sigma_{u \in R*} (u, \alpha)\, s_\beta(u) = s_\beta(p(\alpha)$

since $\ell(s_\beta(\alpha)) = \ell(\alpha)$, one obtains $s_\beta(\alpha)* = s_\beta(2p(\alpha)/\ell(\alpha)) = s_\beta(\alpha*)$.

Corollary 1.2.10. *If $\alpha \in R$ and $w \in W$, one has*

$w\, s_\alpha\, w^{-1} = s_{w(\alpha)}.$

Proof. Indeed, $ws_\alpha w^{-1}(x) = x - (\alpha*, w^{-1}(x))w(\alpha) = x - (w(\alpha*), x)w(\alpha)$, and this last term equals, by 1.2.9:

$x - (w(\alpha)*, x)\, w(\alpha) = s_{w(\alpha)}(x).$

Corollary 1.2.11. *Let $R' \subset R$. For $(M, M*, R', R'*)$ to be a root datum, it is necessary and sufficient that $\alpha, \beta \in R'$ imply $s_\alpha(\beta) \in R'$.*

2. Relations between two roots

2.1. Proportional roots

Proposition 2.1.1. *Let α and β be two roots. The following conditions are equivalent:*

(i) There exists $k \in \mathbb{Q}$ such that $\alpha = k\beta$.

(ii) $s_\alpha = s_\beta$.

Moreover, under these conditions one has $\alpha* = k^{-1}\beta*$, and k is equal to one of the numbers 1, -1, 2, -2, 1/2, $-1/2$.

Proof. Suppose first (i). One has first

$$\alpha^* = \square(\alpha)^{-1} \cdot 2\, p(\alpha) = k^{-2}\, \square(\beta)^{-1} \cdot 2\, k\, p(\beta) = k^{-1}\, \beta^*.$$

This entails at once $s_\alpha = s_\beta$. Conversely, if $s_\alpha = s_\beta$, then

$$\alpha = s_\alpha(-\alpha) = s_\beta(-\alpha) = (\beta^*, \alpha)\, \beta - \alpha,$$

whence $2\alpha = (\beta*, \alpha)\beta$, with $(\beta*, \alpha) \in \mathbb{Z}$, so (ii) entails (i). Finally, if $\alpha = k\beta$, then $\alpha* = k^{-1}\beta*$, whence

$$(\alpha^*, \beta) = 2\, k^{-1}, \qquad (\beta^*, \alpha) = 2\, k,$$

so 2k and $2k^{-1}$ are integers and we are done.

Application 2.1.2. *The root data $\mathcal{R}$ with $rgss(\mathcal{R}) = 1$ are of one of the following two types:*

(i) Type A_1: *there exists $\alpha \in M$ such that the roots are α and $-\alpha$. The coroots are then $\alpha*$ and $-\alpha*$.*

(ii) Type A_1': *there exists $\alpha \in M$ such that the roots are α, $-\alpha$, 2α, -2α. The coroots are then $\alpha*$, $-\alpha*$, $\alpha*/2$, $-\alpha*/2$.*

Definition 2.1.3. *One says that $\alpha \in R$ is* indivisible *if $\alpha/2 \notin R$. One says that $\mathcal{R}$ is* reduced *if every root is indivisible.*

For $\mathcal{R}$ to be reduced it is necessary and sufficient that $\mathcal{R}*$ be so. If α is indivisible and if $w \in W$, then $w(\alpha)$ is indivisible.

Definition 2.1.4. *Let $\alpha \in R$. If α is indivisible, set $ind(\alpha) = \alpha$. Otherwise, set $ind(\alpha) = \alpha/2$.*

Corollary 2.1.5. *If $\alpha \in R$ is indivisible and if $k\alpha \in R$, where $k \in \mathbb{Q}$, then $k \in \mathbb{Z}$.*

Proposition 2.1.6. *Let $\mathcal{R}$ be a root datum. Then*

$$\mathrm{ind}(\square) = (M, M^*, \mathrm{ind}(R), \mathrm{ind}(R)^*)$$

is a reduced root datum, and one has

$$W(ind(\mathcal{R})) = W(\mathcal{R}).$$

Proof. Indeed, $ind(\mathcal{R})$ is a root datum by 1.2.11, since the Weyl group permutes indivisible roots. The second assertion follows from 2.1.1.

Remark 2.1.7. *If $\mathcal{R}$ is not reduced, one has $ind(R)* \neq ind(R*)$ and so $ind(\mathcal{R}*) \neq ind(\mathcal{R})*$.*

2.2. Orthogonal roots

Lemma 2.2.1. *Let α and β be two roots. One has*

$$\ell(\alpha)\,(\alpha^*, \beta) = \ell(\beta)\,(\beta^*, \alpha). \tag{11}$$

Proof. This follows at once from 1.2.1, formulas (9) and (10).

Corollary 2.2.2. *Let $\alpha, \beta \in R$. The following conditions are equivalent:*

(i) $(\alpha*, \beta) = 0$, *(i bis)* $(\beta*, \alpha) = 0$,

(ii) $(p(\alpha), \beta) = 0$,

(iii) $s_\alpha(\beta) = \beta$, *(iii bis)* $s_\beta(\alpha) = \alpha$,

(iv) $s_\alpha(\beta*) = \beta*$, *(iv bis)* $s_\beta(\alpha*) = \alpha*$,

(v) $s_\alpha \neq s_\beta$ *and* s_α *and* s_β *commute.*

Proof. All the equivalences are immediate, except those involving (v). Let us show that (i) (and (i bis)) entail (v). One has

$$s_\alpha s_\beta(x) = x - (\beta^*, x)\,\beta - (\alpha^*, x)\,\alpha + (\beta^*, x)\,(\alpha^*, \beta)\,\alpha.$$

If $(\alpha*, \beta) = 0$, then $(\beta*, \alpha) = 0$ and $s_\alpha s_\beta(x) = s_\beta s_\alpha(x)$.

Suppose conversely (v). One has

$$s_\alpha = s_\beta\, s_\alpha\, s_\beta = s_{s_\beta(\alpha)} \quad \text{(by 1.2.10).}$$

By 2.1, there exists $k \in \mathbb{Q}$ such that $\alpha = k s_\beta(\alpha) = k\alpha - k(\beta*, \alpha)\beta$. Since $s_\alpha \neq s_\beta$, β and α are not proportional by 2.1.1, hence $(\beta*, \alpha) = 0$.

Definition 2.2.3. *Two roots satisfying the equivalent conditions of 2.2.2 are called* orthogonal.

Remark 2.2.4. *The roots α and β are orthogonal if and only if the coroots $\alpha*$ and $\beta*$ are orthogonal.*

Lemma 2.2.5. *If α and β are two orthogonal roots, then $\alpha + \beta \in R$ if and only if $\alpha - \beta \in R$.*

Proof. Indeed, $s_\beta(\alpha + \beta) = s_\beta(\alpha) + s_\beta(\beta) = \alpha - \beta$.

Lemma 2.2.6. *Let α and β be two non-orthogonal roots. If one defines $\ell(\gamma*)$ for a coroot $\gamma*$ as ℓ of the root $\gamma*$ of $\mathcal{R}*$, one has the relation*

$$\ell(\alpha)\,\ell(\alpha^*) = \ell(\beta)\,\ell(\beta^*). \tag{12}$$

Proof. Indeed, multiplying formula (11) by the corresponding formula for $\mathcal{R}*$, and using the equality $(\gamma*)* = \gamma$ for all $\gamma \in R$, one finds:

$$(\beta^*, \alpha)\,(\alpha^*, \beta)\,\ell(\alpha)\,\ell(\alpha^*) = (\beta^*, \alpha)\,(\alpha^*, \beta)\,\ell(\beta)\,\ell(\beta^*).$$

2.3. General case

Proposition 2.3.1. *If α and β are any two roots, one has*

$$(13) 0 \leqslant (\alpha*, \beta)(\beta*, \alpha) \leqslant 4.$$

If α and β are neither proportional nor orthogonal, one has

$$1 \leqslant (\alpha*, \beta)(\beta*, \alpha) \leqslant 3.$$

Proof. Indeed, one has $4(p(\alpha), \beta)^2 = \ell(\alpha)\ell(\beta)(\alpha*, \beta)(\beta*, \alpha)$. On the other hand, by 1.2.6 the symmetric bilinear form $(p(x), y)$ is positive nondegenerate on $V(R)$, whence $(p(x), y)^2 \leqslant \ell(x)\ell(y)$.[1257]

Corollary 2.3.2. *Let α and β be two non-orthogonal roots. If $\ell(\alpha) = \ell(\beta)$, there exists $w \in W$ such that $w(\alpha) = \beta$.*

Proof. Indeed, if α and β are proportional, then since $\ell(\alpha) = \ell(\beta)$, one has $\alpha = \beta$ or $\alpha = -\beta$; in this case one takes $w = 1$ or $w = s_\alpha$. If α and β are neither proportional nor orthogonal, then by formula (11) and 2.3.1:

$(\beta*, \alpha) = (\alpha*, \beta) = \pm 1.$

If $(\beta*, \alpha) = (\alpha*, \beta) = 1$, one takes $w = s_\beta s_\alpha s_\beta$. If $(\beta*, \alpha) = (\alpha*, \beta) = -1$, one takes $w = s_\alpha s_\beta$.

Corollary 2.3.3. *If α and β are two roots, if $\alpha \neq \beta$ and $(\beta*, \alpha) > 0$ (resp. if $\alpha \neq -\beta$ and $(\beta*, \alpha) < 0$), then $\alpha - \beta$ (resp. $\alpha + \beta$) is a root.*

Proof. The second case is deduced from the first by changing β into $-\beta$. If α and β are proportional and $(\beta*, \alpha) > 0$, then $\beta = \alpha$, $2\beta = \alpha$, or $\beta = 2\alpha$. The first case is excluded. In the others, one has respectively $\alpha - \beta = \beta \in R$ or $\alpha - \beta = -\alpha \in R$. If α and β are not proportional, $(\alpha*, \beta)$ and $(\beta*, \alpha)$ are two strictly positive integers whose product is at most 3. One of them is therefore equal to 1. If $(\beta*, \alpha) = 1$, one has $\alpha - \beta = s_\beta(\alpha) \in R$; if $(\alpha*, \beta) = 1$, one has $\alpha - \beta = -s_\alpha(\beta) \in R$.

Lemma 2.3.4. *Let α and β be two non-proportional roots. If $\beta - \alpha$ is not a root, then $\beta + k\alpha \in R$ for $k = -(\alpha*, \beta)$, but not for $k = -(\alpha*, \beta) + 1$.*

Proof. Indeed, $\beta - (\alpha*, \beta)\alpha = s_\alpha(\beta) \in R$, but $\beta + (-(\alpha*, \beta) + 1)\alpha = s_\alpha(\beta - \alpha) \notin R$.

Proposition 2.3.5. *Let α and β be two non-proportional roots. The set of integers k such that $\beta + k\alpha \in R$ is an interval $[p, q]$ ($p \leqslant 0$, $q \geqslant 0$), and one has $p + q = -(\alpha*, \beta)$.*

Proof. For the first assertion it suffices, for example, to prove that if $\beta + k\alpha \in R$, k integer > 0, then $\beta + (k-1)\alpha \in R$. If $k = 1$, this is trivial. If $k \geqslant 2$, one has

$(\alpha*, \beta + k\alpha) = (\alpha*, \beta) + 2k \geqslant -3 + 4 > 0,$

[1257] N.D.E.: with equality if and only if x and y are proportional.

and one concludes by 2.3.3. Let then $[p, q]$ be the interval in question. Applying 2.3.4 to $\beta + p\alpha$, one finds

$$q - p = -(\alpha^*, \beta + p\,\alpha) = -(\alpha^*, \beta) - 2p.$$

[1258]

Remark 2.3.6. *The previous formula contains the qualitative statements 2.2.5 and 2.3.3.*

Complements 2.3.7.[1259] *By 1.2.1 (9) and 1.2.6, the bilinear form on $V(R)$ defined by*

$$\langle x, y\rangle = (p(x), y)$$

is symmetric and positive definite. By 1.2.1 (10), for any $\alpha \in R$ and $y \in V(R)$,

$$\langle \alpha, y\rangle = (\ell(\alpha)/2)\ (\alpha^*, y),$$

where $\ell(\alpha) = \langle \alpha, \alpha\rangle$. Consequently, one deduces from 2.3.3 the following corollary.

Corollary. *Let $\alpha \neq \beta$ in R. If $\alpha - \beta \notin R$, then $\langle \alpha, \beta\rangle \leqslant 0$.*

3. Simple roots, positive roots

3.1. Systems of simple roots

Lemma 3.1.1. *Let α and α_i, $i = 1, \cdots, n$, be roots. Assume α is distinct from the α_i. If one has a relation*

$$\alpha = \Sigma_i\, q_i\, \alpha_i, \qquad q_i \in \mathbb{Q}_+,$$

there exists an index i such that $q_i \neq 0$, $(\alpha*, \alpha_i) > 0$, and $\alpha - \alpha_i \in R$.

Proof. Indeed, one writes $2 = (\alpha*, \alpha) = \Sigma_i q_i (\alpha*, \alpha_i)$, which proves the first two assertions. The third then follows from 2.3.3.

Proposition 3.1.2. *Let α and α_i, $i = 1, \cdots, n$, be roots. If*

$$\alpha = \Sigma_i\, m_i\, \alpha_i, \qquad m_i \in \mathbb{N}_+,$$

there exists a sequence $\beta_1, \cdots, \beta_m$ of roots taken from the α_i (not necessarily pairwise distinct) such that if one denotes

$$\gamma_p = \Sigma_{i=1}^{p} \beta_i, \qquad p = 1, \ldots, m,$$

one has $\gamma_p \in R$ and $\gamma_m = \alpha$.

Proof. We argue by induction on the integer $\Sigma m_i = m'$. If α equals one of the α_i, say α_{i_0} (which is automatic if $m' = 1$), one takes $m = 1$, $\beta_1 = \alpha_{i_0}$. Otherwise, one applies the previous lemma and there exists an index i such that $m_i \neq 0$ and $\alpha - \alpha_i$ is a root. One then has $(m_i - 1) \in \mathbb{N}$ and

[1258] N.D.E.: We have corrected $+2p$ to $-2p$.

[1259] N.D.E.: We have added these complements, useful for the proof of 3.1.5.

$$\alpha - \alpha_i = (m_i - 1)\alpha_i + \sum_{j \neq i} m_j \alpha_j.$$

It remains only to apply the induction hypothesis to $\alpha - \alpha_i$.

Corollary 3.1.3. *Let $R' \subset R$. The following conditions are equivalent:*

(i) If $\alpha, \beta \in R'$ and $\alpha + \beta \in R$, then $\alpha + \beta \in R'$.

(ii) $(\mathbb{N} \cdot R') \cap R \subset R'$.

Proof. Indeed, one has clearly (ii) ⇒ (i). The converse follows at once from the proposition.

Definition 3.1.4. *A set of roots satisfying the conditions of 3.1.3 is called* closed.

Proposition 3.1.5. *Let $\Delta \subset R$ be a set of roots. The following conditions are equivalent:*

(i) The elements of Δ are indivisible, linearly independent, and

$$R \subset (\mathbb{Q}_+ \cdot \Delta) \cup (-\mathbb{Q}_+ \cdot \Delta).$$

(ii) The elements of Δ are linearly independent and

$$R \subset (\mathbb{N} \cdot \Delta) \cup (-\mathbb{N} \cdot \Delta).$$

(iii) Every root is written in a unique way as a linear combination of the elements of Δ, with integer coefficients all of the same sign.

[1260] One has obviously (ii) ⇒ (iii). Denote by $\alpha_1, \cdots, \alpha_n$ the (distinct) elements of Δ.

Proof of (i) ⇒ (ii). Let $\alpha \in R$. One has thus a unique expression

$$\varepsilon\alpha = \sum_i q_i \alpha_i \qquad q_i \in \mathbb{Q}_+,\ \varepsilon = \pm 1.$$

Let us show that the q_i are integers. This is certainly true if all but one of them are zero (cf. 2.1.5). Otherwise, α is distinct from the α_i and, applying 3.1.1, one finds i_0 such that $\alpha' = \epsilon\alpha - \alpha_{i_0} \in R$ and $q_{i_0} \neq 0$. This gives

$$\alpha' = (q_{i_0} - 1)\alpha_{i_0} + \sum_{i \neq i_0} q_i \alpha_i.$$

Since at least one of the q_i, $i \neq i_0$, is nonzero, (i) entails $q_{i_0} - 1 \geqslant 0$. One repeats the operation for α' and after finitely many steps one has shown that the q_i are integers.

Proof of (iii) ⇒ (i). Let us show that $\alpha_1, \alpha_2, \cdots, \alpha_n$ are linearly independent over $\mathbb{Q}$. In the contrary case, one would have an equality

$$x = \sum_{i \in I} a_i \alpha_i = \sum_{j \in J} b_j \alpha_j,$$

where I, J are two disjoint subsets of $1, \cdots, n$, at least one of them, say I, being nonempty, and $a_i, b_j \in \mathbb{N}*$. By Corollary 2.3.7, one has $\langle \alpha_i, \alpha_j \rangle \leqslant 0$ if $i \in I$ and $j \in J$, whence $\langle x, x \rangle \leqslant 0$ and so $x = 0$. Let $i_0 \in I$; then the equalities

[1260]N.D.E.: In what follows, we have modified the order and detailed the proof of the implication (iii) ⇒ (i).

$$\alpha_{i_0} = \alpha_{i_0} + \sum_{i \in I} a_i \alpha_i = \alpha_{i_0} + \sum_{j \in J} b_j \alpha_j$$

entail (by (iii)) $a_i = 0 = b_j$ for all i, j, a contradiction. This shows that the elements of Δ are linearly independent over $\mathbb{Q}$; let us show they are also indivisible.

So let $\beta \in \Delta$ be divisible. One has $\beta/2 \in R$, whence

$$\beta/2 = \varepsilon \sum_{\alpha \in \Delta} m_\alpha \alpha, \qquad m_\alpha \in \mathbb{N}, \ \varepsilon = \pm 1,$$

so also $\beta = 2\epsilon\Sigma_\alpha m_\alpha \alpha$, whence by uniqueness $m_\alpha = 0$ if $\alpha \neq \beta$ and $2\epsilon m_\beta = 1$, a contradiction.

Definition 3.1.6. *A set Δ of roots satisfying the conditions of 3.1.5 is called a* system of simple roots, *or a* base of $\mathcal{R}$.

If $w \in W$ and if Δ is a system of simple roots, then $w(\Delta)$ is a system of simple roots.

Remark 3.1.7. *This definition involves only R and not $\mathcal{R}$, and in fact involves only $ind(R)$.*

Remark 3.1.8. *If Δ is a system of simple roots, Δ is a basis of the free abelian group $\Gamma_0(R)$. One has thus $Card(\Delta) = rgss(\mathcal{R})$.*

Remark 3.1.9. *The conditions of 3.1.5 are then evidently equivalent to:*

(i′) The elements of Δ are indivisible, of number $\leqslant rgss(\mathcal{R})$, and $R \subset (Q_+ \cdot \Delta) \cup (-Q_+ \cdot \Delta)$.

(ii′) $Card(\Delta) \leqslant rgss(\mathcal{R})$ and $R \subset (\mathbb{N} \cdot \Delta) \cup (-\mathbb{N} \cdot \Delta)$.

Corollary 3.1.10. *If Δ is a system of simple roots, then $ind(\Delta*)$ is a system of simple coroots (i.e. a system of simple roots of $\mathcal{R}*$).*

Proof. Indeed, if $\beta = \Sigma_{\alpha\in\Delta} a_\alpha \alpha$ ($a_\alpha \in \mathbb{N}$), then $p(\beta) = \Sigma_{\alpha\in\Delta} a_\alpha p(\alpha)$, whence, by 1.2.1 (10):

$$\beta* = \sum_{\alpha \in \Delta} a_\alpha \left(\ell(\alpha)/\ell(\beta)\right) \alpha*,$$

which shows that $ind(\Delta*)$ satisfies (i′).

[1261] By 2.1.1, if $\alpha*$ is not indivisible, then $\alpha* = 2u*$, where $u* \in ind(\Delta*)$ (and $u = 2\alpha$). One deduces:

Corollary 3.1.11. *If Δ is a system of simple roots and if $\beta = \Sigma_{\alpha\in\Delta} a_\alpha \alpha$ ($a_\alpha \in \mathbb{Z}$) is the expression of β along Δ, then $2a_\alpha \ell(\alpha)$ is divisible by $\ell(\beta)$ (and even by $2\ell(\beta)$ if $\alpha*$ is indivisible).*

Before continuing to state the properties of systems of simple roots, let us show that such systems exist.

[1261] N.D.E.: We have added the following sentence, and in 3.1.11 we have replaced $4a_\alpha\ell(\alpha)$ by $2a_\alpha\ell(\alpha)$.

3.2. Systems of positive roots

Definition 3.2.1. *A subset $R_+ \subset R$ is called a* system of positive roots *of R (or of $\mathcal{R}$, cf. Remark 3.2.2) if it satisfies the following conditions:*

(P 1) $R_+ \cap -R_+ = \emptyset$.

(P 2) $R_+ \cup -R_+ = R$.

(P 3) $(Q_+ \cdot R_+) \cap R \subset R_+$.

In particular, such a set is closed. We shall see later (3.3.8) that in fact a closed subset satisfying (P 1) and (P 2) also satisfies (P 3), and hence is a system of positive roots. If $w \in W$ and if R_+ is a system of positive roots, then $w(R_+)$ is a system of positive roots.

Remark 3.2.2. *This definition involves only R. One will also say that R_+ is a system of positive roots of $\mathcal{R}$.*

Remark 3.2.3. *From (P 1) and (P 2), one obtains at once*

$$Card(R_+) = Card(R)/2.$$

It follows that if R_+ and R'_+ are two systems of positive roots with $R_+ \subset R'_+$, then $R_+ = R'_+$.

Remark 3.2.4. *If R_+ is a system of positive roots, $R*_+$ is a system of positive coroots (i.e. a system of positive roots of $\mathcal{R}*$).*

This follows at once from 1.1.4 and 1.2.2.

Definition 3.2.5. *Let Δ be a system of simple roots. Set*

$P(\Delta) = (Q_+ \cdot \Delta) \cap R = (\mathbb{N}_+ \cdot \Delta) \cap R.$

Proposition 3.2.6. *If Δ is a system of simple roots, $P(\Delta)$ is a system of positive roots. If Δ is a system of simple roots and R_+ is a system of positive roots, one has the equivalence:*

$\Delta \subset R_+ \iff R_+ = P(\Delta).$

Proof. The first assertion is immediate. If $\Delta \subset R_+$, then $P(\Delta) \subset R_+$ by (P 3), so $P(\Delta) = R_+$ by 3.2.3. The rest is trivial.

Remark 3.2.7. *Systems of positive roots exist: let $\geqslant$ be a structure of totally ordered vector space on $V(R)$. The set of roots $\geqslant 0$ for this order relation is a system of positive roots.*

Theorem 3.2.8. *Let R_+ be a system of positive roots. There exists a unique system of simple roots $\mathcal{S}(R_+)$ such that $\mathcal{S}(R_+) \subset R_+$, i.e. such that $R_+ = P(\mathcal{S}(R_+))$.*

Proof. Uniqueness follows at once from:

Lemma 3.2.9. *Let Δ be a system of simple roots. Then for $\alpha \in P(\Delta)$ to belong to Δ, it is necessary and sufficient that α not be a sum of two elements of $P(\Delta)$.*

Proof. This lemma follows at once from the definitions and from 3.1.2.

Let us now prove the existence of $\mathcal{S}(R_+)$. Consider the set of subsets T of R_+ such that $T = ind(T)$ and $(Q_+ \cdot T) \supset R_+$. This set is nonempty, since it contains $ind(R_+)$. Let Δ be a minimal element of this set for the inclusion relation. We show that Δ is a system of simple roots, that is to say, by 3.1.5 (i), that Δ is a free subset of $M \otimes \mathbb{Q}$.

Lemma 3.2.10. *If $\alpha, \beta \in \Delta$ and $q\alpha + q'\beta \in R$, where $q, q' \in \mathbb{Q}$, then $qq' \geqslant 0$.*

Proof. Indeed, if $qq' < 0$, one can write (possibly exchanging α and β)

$$a\,\alpha - b\,\beta \in R_+, \quad a, b \in Q^*_+,$$

so by hypothesis there exists a relation

$$a\,\alpha - b\,\beta = \Sum_{\gamma \in \Delta} c(\gamma)\,\gamma, \quad c(\gamma) \in Q_+.$$

If $a \leqslant c(\alpha)$, it is written

$$-\beta = ((c(\alpha) - a)/b)\,\alpha + \Sum_{\gamma \neq \alpha} (c(\gamma)/b)\,\gamma.$$

Then this element belongs to $(Q_+ \cdot \Delta) \cap R$, which is contained in R_+ by (P 3). But then $\beta \in R_+ \cap -R_+$, which contradicts (P 1).

If, on the contrary, $a > c(\alpha)$, one writes

$$(a - c(\alpha))\,\alpha = b\,\beta + \Sum_{\gamma \neq \alpha} c(\gamma)\,\gamma,$$

which proves $\Delta \subset Q_+ \cdot (\Delta - \alpha)$, contrary to the minimal character of Δ.

Recall (cf. 2.3.7) that the bilinear form on $V(R)$ defined by $\langle x, y\rangle = (p(x), y)$ is a Euclidean inner product; moreover, for $\alpha \in R$ one has $\langle \alpha, y\rangle = \ell(\alpha)(\alpha*, y)/2$, so that $\langle \alpha, y\rangle$ and $(\alpha*, y)$ have the same sign.

Lemma 3.2.11. *If $\alpha, \beta \in \Delta$, then $(\beta*, \alpha) \leqslant 0$ and so $\langle \beta, \alpha\rangle \leqslant 0$.*

Proof. Indeed, $s_\beta(\alpha) = \alpha - (\beta*, \alpha)\beta \in R$, whence $(\beta*, \alpha) \leqslant 0$ by 3.2.10.

Let us now prove that Δ is free. In the contrary case, 3.2.11 entails, as in the proof of 3.1.5, the existence of a non-trivial relation

$$\Sum_{\alpha \in \Delta} m(\alpha)\,\alpha = 0, \quad m(\alpha) \in \mathbb{N},$$

whence $-\alpha_0 = (m(\alpha_0) - 1)\alpha_0 + \Sigma_{\alpha \neq \alpha_0} m(\alpha)\alpha$, if $m(\alpha_0) \geqslant 1$. Then, by (P 3), α_0 belongs to $R_+ \cap -R_+$, contradicting (P 1).

This shows that Δ is a base of $\mathcal{R}$ and completes the proof of Theorem 3.2.8.

Corollary 3.2.12. *Let R_+ be a system of positive roots, Δ a base of $\mathcal{R}$, and $w \in W$. One has:*

$$\Delta \subset R_+ \quad \square \quad R_+ = P(\Delta) \quad \square \quad \Delta = \square(R_+)$$

$$P(ind(\Delta^*)) = P(\Delta)^*, \quad \square(R^*_+) = ind(\square(R_+)^*);$$

$$\mathcal{S}(w(R_+)) = w(\mathcal{S}(R_+)), \quad P(w(\Delta)) = w(P(\Delta)).$$

Definition 3.2.13. *If one has chosen a system of simple roots Δ, the elements of $P(\Delta)$ will be called* positive. *If one has chosen a system of positive roots R_+, the elements of $\mathcal{S}(R_+)$ will be called* simple.

Corollary 3.2.14. *Let R_+ be a system of positive roots. Let $\alpha \in R_+$. The following conditions are equivalent:*

(i) α is simple (i.e. $\alpha \in \mathcal{S}(R_+)$).

(ii) α is not a sum of two elements of R_+.

(iii) $R_+ - \alpha$ is closed (cf. 3.1.4).

Proof. The equivalence of (i) and (ii) follows at once from 3.2.9. The equivalence of (ii) and (iii) is immediate.

Definition 3.2.15. *Let Δ be a system of simple roots. The sum of the coefficients of the decomposition of a root α along Δ is called the* order of α *relative to Δ and is denoted $ord_\Delta(\alpha)$.*

One has the equivalences:

$$\alpha \in P(\Delta) \iff ord_\Delta(\alpha) > 0, \quad \alpha \in \Delta \iff ord_\Delta(\alpha) = 1.$$

Lemma 3.2.16. *Let Δ be a system of simple roots and $\alpha \in P(\Delta)$. There exists a sequence $\alpha_i \in \Delta$, $i = 1, \cdots, m$, such that*

$$\alpha_1 + \alpha_2 + \cdots + \alpha_p \in P(\Delta) \quad \text{for } p = 1, \ldots, m,$$
$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = \alpha.$$

Moreover, for every such sequence α_i, one has $m = ord_\Delta(\alpha)$.

Proof. Trivial by 3.1.2.

3.3. Characterization and conjugacy of systems of positive roots

Lemma 3.3.1. *If $\alpha \in P(\Delta)$, $\beta \in \Delta$, and if α and β are not proportional, then $s_\beta(\alpha) \in P(\Delta)$.*

Proof. Indeed, $s_\beta(\alpha) = \alpha - (\beta*, \alpha)\beta$. Since at least one simple root other than β appears in the decomposition of α with a nonzero (hence strictly positive) coefficient, it also appears in the decomposition of $s_\beta(\alpha)$ with the same coefficient, so $s_\beta(\alpha)$ is also positive.

Corollary 3.3.2. *If $\beta \in \Delta$, the symmetry s_β interchanges the elements of $P(\Delta)$ not proportional to β.*

Lemma 3.3.3. *If $\alpha \in P(\Delta) - \Delta$, α indivisible, there exists $\beta \in \Delta$ such that $s_\beta(\alpha) \in P(\Delta)$ and $ord_\Delta(s_\beta(\alpha)) < ord_\Delta(\alpha)$.*

Proof. Indeed, by 3.1.1 there exists $\beta \in \Delta$ such that $(\beta*, \alpha) > 0$. Since $\alpha \notin \Delta$ and is indivisible, α cannot be proportional to β. Hence $s_\beta(\alpha) \in P(\Delta)$, by 3.3.1, and one has $ord_\Delta(s_\beta(\alpha)) = ord_\Delta(\alpha) - (\beta^*, \alpha) < ord_\Delta(\alpha)$.

Corollary 3.3.4. *If $\alpha \in P(\Delta)$ is indivisible, there exists a sequence $\beta_p \in \Delta$, $p = 1, \cdots, q$, such that*

$$\alpha_p = s_{\beta_p} \cdots s_{\beta_1}(\alpha) \in P(\Delta) \quad \text{for } p = 1, \dots, q,$$

and $\alpha_q \in \Delta$.

Proof. This follows from 3.3.3 by induction on $ord_\Delta(\alpha)$.

Proposition 3.3.5. *The Weyl group is generated by the s_α, for $\alpha \in \Delta$. Every indivisible root is conjugate to a simple root by an element of the Weyl group.*

Proof. The second assertion follows at once from 3.3.4. The first follows from it by 1.2.10 and 2.1.1.

Proposition 3.3.6. *Let R_+ be a system of positive roots. Let $P' \subset R$ satisfy (P 2) and be closed. Then there exists $w \in W$ such that $w(R_+) \subset P'$.*

Let us state the corollaries at once.

Corollary 3.3.7. *The Weyl group operates transitively on the set of systems of positive roots (resp. of systems of simple roots).*

Corollary 3.3.8. *For a subset of R to be a system of positive roots, it is necessary and sufficient that it satisfy (P 1) and (P 2) and be closed.*

Corollary 3.3.9. *If one endows $\Gamma_0(R)$ with a structure of ordered group such that every root is > 0 or < 0, the set of positive roots for this order structure is a system of positive roots.*

Proof of 3.3.6. One can find $w \in W$ such that $Card(w(R_+) \cap P')$ is maximal, hence by replacing R_+ by $w(R_+)$ one may assume that

$$(*) \qquad Card(R_+ \cap P') \geq Card(R_+ \cap s_\alpha(P'))$$

for all $\alpha \in \mathcal{S}(R_+) = \Delta$. Let us show that $R_+ \subset P'$. Otherwise, P' being closed, there exists $\alpha \in \Delta$, $\alpha \notin P'$. But P' satisfying (P 2), one has then $-\alpha \in P'$ (so $-2\alpha \in P'$ if 2α is a root). For every $\beta \in R_+ \cap P'$, one has $\beta \neq \alpha$; if 2α is not a root, then $s_\alpha(\beta) \in R_+$ (by 3.3.1), so

$$s_\alpha(R_+ \cap P') \subset R_+ \cap s_\alpha(P');$$

but $R_+ \cap s_\alpha(P')$ also contains $\alpha = s_\alpha(-\alpha)$, contradicting inequality $(*)$. If 2α is a root, one argues similarly.

To study sets of roots satisfying (P 2) and closed, one is therefore reduced to the case where they contain a set of positive roots.

Proposition 3.3.10. *Let R_+ be a system of positive roots and P' a closed subset of R containing R_+. If one denotes $\Delta = \mathcal{S}(R_+)$ and $\Delta' = \Delta \cap -P'$, then P' is the union* *of R_+ and the set of roots that are linear combinations with negative coefficients of the elements of Δ'.*

Proof. We prove the assertion by induction on the order of a root $\gamma \in P' - R_+$. If $ord_\Delta(\gamma) = -1$, then $-\gamma \in \Delta$ and $\gamma \in -\Delta'$. If $ord_\Delta(\gamma) < -1$, there exists, by 3.1.2, $\beta \in \Delta$ such that $-\gamma - \beta \in R$; then

$$0 < ord_\Delta(-\gamma - \beta) = ord_\Delta(-\gamma) - 1 < ord_\Delta(-\gamma)$$

and the first inequality shows that $-\gamma-\beta \in R_+$. Hence, since $R_+ \cap -R_+ = \emptyset$ and $\gamma+\beta$ is a sum of two roots of P', it is an element of $P'-R_+$ with $ord_\Delta(\gamma+\beta) > ord_\Delta(\gamma)$. So, by the induction hypothesis, $\gamma+\beta$ is a linear combination with negative coefficients of the elements of Δ'. Since $\gamma = (\gamma+\beta) - \beta$, it suffices to verify that $\beta \in -P'$. Now $\gamma \in P'$ and $-(\gamma+\beta) \in R_+ \subset P'$, so $-\beta = \gamma - (\gamma+\beta)$ belongs to P'.

Definition 3.3.10.1.[1262] *One says that a subset R' of R is* symmetric *if $-R' = R'$.*

Scholie 3.3.11. *Let P' be a set of roots satisfying (P 2) and closed. There exists a system of simple roots Δ and a subset Δ' of Δ such that*

$$P' = R \cap (\mathbb{N} \cdot \Delta \cup -\mathbb{N} \cdot \Delta').$$

If one denotes $R_{\Delta'} = (\mathbb{Z} \cdot \Delta') \cap R$, which is a closed and symmetric subset of R, then P' is the disjoint union of $R_{\Delta'}$ and of the closed part of $P(\Delta)$ formed by the $\alpha \in P(\Delta) - \mathbb{N}\cdot\Delta'$, i.e. the positive roots which in the decomposition along Δ "contain at least one root of $\Delta - \Delta'$".

3.4. Closed and symmetric sets of roots

Proposition 3.4.1. *Let $\mathcal{R} = (M, M*, R, R*)$ be a root datum and R' a closed and symmetric subset of R. Then:*

(i) $\mathcal{R}' = (M, M*, R', R'*)$ is a root datum;

(ii) for every system of positive roots R_+ of R, $R'_+ = R_+ \cap R'$ is a system of positive roots of R';

(iii) the Weyl group $W(\mathcal{R}')$ of $\mathcal{R}'$ is the subgroup of $W(\mathcal{R})$ generated by the s_α, for $\alpha \in R'$.

Proof. The first assertion is trivial by 1.2.11, the second follows from 3.3.8, the third is evident.

Corollary 3.4.2. *Let $w \in W(\mathcal{R}')$. The order of w is the smallest integer $n > 0$ such that $w^n(\alpha') = \alpha'$ for every $\alpha' \in R'$.*

Proof. It suffices to apply 1.2.7 to the root datum $\mathcal{R}'$.

Corollary 3.4.3. *Let α and β be two non-proportional roots. Let n be the smallest integer > 0 such that $(s_\alpha s_\beta)^n(\alpha) = \alpha$ and $(s_\alpha s_\beta)^n(\beta) = \beta$. Then the subgroup $W_{\alpha,\beta}$ of W generated by s_α and s_β is defined by the relations:*

[1262] N.D.E.: We have inserted this definition here.

$$s_\alpha^2 = 1, \quad s_\beta^2 = 1, \quad (s_\alpha s_\beta)^n = 1.$$

Proof. Taking 3.4.2 into account, it suffices to verify:

Lemma 3.4.4. *Let G be the group generated by two elements x and y subject to the relations $x^2 = y^2 = 1$. Every normal subgroup of G containing neither x nor y is generated (as a normal subgroup) by an element of the form $(xy)^n$, where $n > 0$.*

[1263] *Proof.* Indeed, every element of G is written $(xy)^n$, or $(yx)^n = (xy)^{-n}$, or:

$$\text{(a)} \quad x\,(yx)^{2n} \quad \text{or} \quad y\,(xy)^{2n+1}$$

$$\text{(b)} \quad y\,(xy)^{2n} \quad \text{or} \quad x\,(yx)^{2n+1},$$

where $n \in \mathbb{N}$. Now the elements of type (a) (resp. (b)) are conjugate to x (resp. to y).

Remark 3.4.5. *One computes the integer n immediately: if one sets*

$$(\alpha^*, \beta) = p, \quad (\beta^*, \alpha) = q,$$

one has[1264]

$$(s_\alpha s_\beta)(\alpha) = (pq - 1)\,\alpha - q\,\beta, \quad (s_\alpha s_\beta)(\beta) = p\,\alpha - \beta.$$

The integer n is therefore the order of the matrix

$$\begin{pmatrix} pq - 1 & p \\ -q & -1 \end{pmatrix}.$$

If $pq = 0$, then $p = q = 0$, by 2.2.2, whence $n = 2$. Otherwise, by 2.3.1, pq equals 1, 2, or 3, and one finds, respectively, $n = 3$, 4, or 6.

N.B. By writing that the order of the previous matrix is finite, one recovers inequality (13) of 2.3.1.

Definition 3.4.6. *Let Δ be a system of simple roots and $\Delta' \subset \Delta$. We write*

$$R_{\Delta'} = R \cap (\mathbb{Q} \cdot \Delta') = R \cap (\mathbb{Z} \cdot \Delta').$$

Lemma 3.4.7. *$R_{\Delta'}$ is closed and symmetric, Δ' is a system of simple roots of the root datum $(M, M*, R_{\Delta'}, R*_{\Delta'})$, whose Weyl group is the subgroup $W_{\Delta'}$ of W generated by the s_α, for $\alpha \in \Delta'$. One has $\Delta \cap R_{\Delta'} = \Delta'$.*

Proof. Trivial.

Proposition 3.4.8. *Let $R' \subset R$. The following conditions are equivalent:*

(i) There exists a vector subspace V' of V (or of $V(R)$) such that $R' = R \cap V'$.

(ii) There exists a system of simple roots Δ of R and a subset Δ' of Δ such that $R' = R_{\Delta'}$.

[1263] N.D.E.: We have corrected the original in what follows.
[1264] N.D.E.: We have corrected the original in what follows.

More precisely, under these conditions, every system of simple roots Δ' of $(M, M, R', R'*)$ is contained in a system of simple roots Δ of R, and one has $R' = R_{\Delta'}$.*

Proof. One has obviously (ii) ⇒ (i). Suppose (i) verified: then R' is closed and symmetric, so $(M, M*, R', R'*)$ is a root datum. Let Δ' be a system of simple roots of this datum and $R'_+ = P(\Delta')$. If $V' = V$, then Δ' is a system of simple roots of $\mathcal{R}$ and we are done. Otherwise, there exists $x \in V*$ such that

$$(x, V') = \{0\}, \quad (x, \alpha) \neq 0 \quad \text{for all } \alpha \in R - R'.$$

Put $R_+(x) = \alpha \in R | (x, \alpha) > 0$ and $R_+ = R_+(x) \cup R'_+$. For every $\alpha \in R$, one has the equivalences

$$(x, \alpha) > 0 \iff \alpha \in R_+(x),$$
$$(x, \alpha) < 0 \iff \alpha \in -R_+(x),$$
$$(x, \alpha) = 0 \iff \alpha \in R'.$$

It follows at once from 3.3.8 that R_+ is a system of positive roots of R. Put $\Delta = \mathcal{S}(R_+)$. It evidently suffices to prove $\Delta' \subset \Delta$. Otherwise let $\alpha \in \Delta' - \Delta$. Then, by 3.2.14, there exist $\beta, \gamma \in R_+$ such that $\alpha = \beta + \gamma$. If $\beta, \gamma \in R_+(x)$, one has $\alpha \in R_+(x)$, which is absurd since $(x, \Delta') = 0$. If β or γ, say β, belongs to R'_+, then, since R' is symmetric and closed, $\gamma = \alpha - \beta$ belongs to $R_+ \cap R' = R'_+$; but then α is not simple in R'_+.

Lemma 3.4.9. *Under the preceding conditions, let $\alpha \in P(\Delta) - R'$. For every $w \in W_{\Delta'}$, one has $w(\alpha) \in P(\Delta) - R'$.*

Proof. It indeed suffices to verify this for $w = s_\beta$, $\beta \in \Delta'$, in which case it follows from 3.3.1 and from the fact that $s_\beta(R') = R'$.

Lemma 3.4.10. *Let $w \in W$. Under the conditions of 3.4.8, the following conditions are equivalent:*

(i) $w \in W_{\Delta'}$.

(ii) For every $m \in M$, $w(m) - m \in V'$.

(iii) For every $\alpha \in R$, $w(\alpha) - \alpha \in V'$.

Proof. One has obviously (i) ⇒ (ii) ⇒ (iii). Let us prove (iii) ⇒ (i). Let then $w \in W$ be such that $w(\alpha) - \alpha \in V'$ for all $\alpha \in R$. Write $w = s_{\alpha_n} \cdots s_{\alpha_1}$, with $\alpha_i \in \Delta$, and let us prove by induction on n that each $\alpha_i \in \Delta'$. One may assume that $w' = s_{\alpha_{n-1}} \cdots s_{\alpha_1} \in W_{\Delta'}$. One has then, for all $\alpha \in R$,

$$w(\alpha) - \alpha = w'(\alpha) - \alpha - (\alpha^*_n, w'(\alpha))\, \alpha_n.$$

Taking $\alpha = w'^{-1}(\alpha_n)$, one finds $2\alpha_n \in V'$, whence $\alpha_n \in \Delta'$, so $w = s_{\alpha_n} w'$ belongs to $W_{\Delta'}$.

3.5. Miscellaneous remarks

Proposition 3.5.1. *Let R_+ be a system of positive roots. Denote*

$$\rho_{R_+} = (1/2) \sum_{\alpha \in \mathrm{ind}(R_+)} \alpha.$$

Then $(\beta*, \rho_{R_+}) = 1$ *for every* $\beta \in \mathcal{S}(R_+)$.

Proof. Indeed, one can write

$$2\rho_{R_+} = \beta + \Sigma_{\alpha \in ind(R_+),\ \alpha \neq \beta}\ \alpha,$$

so $s_\beta(2\rho_{R_+}) = 2\rho_{R_+} - 2\beta$, by 3.3.2.

Corollary 3.5.2. *Put* $\rho*_{R_+} = (1/2)\Sigma_{\alpha* \in ind(R*_+)}\alpha*$. *Then:*

(i) $(\rho*_{R_+}, \alpha) > 0$ *for every* $\alpha \in R_+$ *(i.e.* $\rho*_{R_+} \in \mathcal{C}(R_+)$*, cf. 3.6.8).*

(ii) For every $\alpha \in R$*, one has* $ord_{\mathcal{S}(R_+)}(\alpha) = (\rho*_{R_+}, \alpha)$.[1265]

Remark 3.5.3. *If* $w \in W$*, one has* $\rho_{w(R_+)} = w(\rho_{R_+})$ *and* $\rho*_{w(R_+)} = w(\rho*_{R_+})$.

Proposition 3.5.4. *Let* α *and* γ *be two non-proportional roots, with* α *indivisible. There exists a system of simple roots containing* α *and a root* β *such that* $\gamma = a\alpha + b\beta$*, with* $a, b \in \mathbb{N}$.

Proof. Indeed, let us construct a basis of the vector space $V(R)$ containing $\alpha_1 = \alpha$, $\alpha_2 = \gamma$. Consider the lexicographic order with respect to this basis. Denoting by R_+ the set of roots > 0, it is clear that R_+ is a system of positive roots and[1266] that the smallest element of R_+ not proportional to α is simple. This element is of the form

$$\beta = p\,\alpha + q\,\gamma, \qquad 0 < q \leqslant 1.$$

One thus has $\gamma = q^{-1}\beta - q^{-1}p\alpha$, and since $q > 0$, one has $q^{-1} \in \mathbb{N}*$ and $-q^{-1}p \in \mathbb{N}$.

Let us finally make two remarks about the group $\Gamma_0(R)$.

Proposition 3.5.5. *Let* G *be an abelian group and* $f : R \to G$ *a map satisfying the following two conditions:*

(i) If $\alpha \in R$*,* $f(-\alpha) = -f(\alpha)$.

(ii) If $\alpha, \beta, \alpha + \beta \in R$*,* $f(\alpha + \beta) = f(\alpha) + f(\beta)$.

Then there exists a unique group homomorphism $\bar{f} : \Gamma_0(R) \to G$ *such that* $\bar{f}(\alpha) = f(\alpha)$ *for* $\alpha \in R$.

Proof. Indeed, if Δ is a system of simple roots of R, and if $\beta \in R$ is written $\Sigma_{\alpha \in \Delta} a(\alpha)\alpha$, it follows at once from 3.2.16 that $f(\beta) = \Sigma_{\alpha \in \Delta} a(\alpha) f(\alpha)$. Now Δ is a basis of $\Gamma_0(R)$.

Proposition 3.5.6. *Let* Δ *be a system of simple roots. There exists on* $\Gamma_0(R)$ *a structure of totally ordered group such that the roots* > 0 *are the elements of* $P(\Delta)$ *and* $\alpha \mapsto ord_\Delta(\alpha)$ *is an increasing function.*

Proof. Indeed, let $\alpha_1, \alpha_2, \cdots, \alpha_n$ ($n = rgss(\mathcal{R})$) be the elements of Δ. For $x \in \Gamma_0(R)$, one has a decomposition

[1265] N.D.E.: Indeed, by 3.1.5, it suffices to verify the formula for $\alpha \in \Delta$. Now, by 3.5.1 applied to $\mathcal{R}*$, one has in this case $(\rho*_{R_+}, \alpha) = 1 = ord_{\mathcal{S}(R_+)}(\alpha)$.

[1266] N.D.E.: We have modified the original, to take into account the case where 2α could be a root.

$x = \Sigma_i m_i(x) \alpha_i$.

It suffices to take the lexicographic order with respect to the functions Σm_i, m_n, m_{n-1}, ..., m_2.

Remark 3.5.7. *The first roots are in order:*

$$\alpha_1, \alpha_2, \cdots, \alpha_n;$$

one then has (if these are roots) $2\alpha_1, \alpha_1 + \alpha_2, \alpha_1 + \alpha_3, \cdots$.

3.6. Weyl chambers

Lemma 3.6.1. *Let V be a finite-dimensional $\mathbb{R}$-vector space.*[1267] *Let f_i be independent linear forms. Set*

$C = \{x \in V \mid f_i(x) > 0\}$.

Then C is a maximal convex subset of $X = V - \bigcup_i f_i^{-1}(0)$.

Proof. Trivial.

Definition 3.6.2. *A subset C of V describable by the procedure of 3.6.1 will be called (here) a* chamber *of V.*

Definition 3.6.3. *One says that the hyperplane H of V is a* wall *of C if $H \cap (\bar{C} - C)$ contains a nonempty open subset of H.*

Remark 3.6.4. *For a convex subset, the closure is described without appeal to the topology of V: it is the set of endpoints of all open segments contained in the given subset.*

Lemma 3.6.5. *Under the conditions of 3.6.1, one has*

$\bar{C} = \{x \in V \mid f_i(x) \geq 0\}$.

The walls of C are the hyperplanes $f_i^{-1}(0)$.

Proof. The first assertion is clear. The second then follows from the fact that $\bar{C} - C \subset \bigcup_i f_i^{-1}(0)$ and that the $f_i^{-1}(0)$ are obviously walls of C.

Proposition 3.6.6. *Let C be a chamber of V. If H_i, $i = 1, 2, \cdots, n$, are the distinct walls of C, then for every system of linear forms u_i such that $H_i = u_i^{-1}(0)$, there exist $\epsilon_i \in -1, +1$ such that C is defined by*

$C = \{x \in V \mid \varepsilon_i u_i(x) > 0\}$.

For every wall H of C, one has $H \cap C = \emptyset$ and $H \cap \bar{C} = C_H$, where C_H is a chamber in H. The walls of C_{H_i} are the $H_j \cap H_i$, for $j \neq i$.

Proof. This follows trivially from the lemma.

[1267] N.D.E.: We have replaced $\mathbb{Q}$ by $\mathbb{R}$.

Definition 3.6.7. *The* C_{H_i} *are the* faces *of* C*.*

Let now $\mathcal{R} = (M, M*, R, R*)$ be a root datum. We put $V*_{\mathbb{R}} = M * \bigotimes \mathbb{R}$.

Definition 3.6.8.[1268] *For every* $\alpha \in R$*, set*

$$H_\alpha = \{x \in V*_{\mathbb{R}} \mid (x, \alpha) = 0\}.$$

We denote $X = V*_{\mathbb{R}} - \bigcup_{\alpha \in R} H_\alpha$*. For every* $x \in X$*, set*

$$R_+(x) = \{\alpha \in R \mid (x, \alpha) > 0\}.$$

For every system of positive roots R_+*, denote*

$$\mathcal{C}(R_+) = \{x \in V*_{\mathbb{R}} \mid (x, \alpha) > 0 \text{ for all } \alpha \in R_+\}.$$

Proposition 3.6.9. *(i) For every* $x \in X$*,* $R_+(x)$ *is a system of positive roots. For every system of positive roots* R_+*,* $\mathcal{C}(R_+)$ *is a chamber in* $V*_{\mathbb{R}}$*. The* $\mathcal{C}(R_+)$ *are the maximal convex subsets of* X*.*

(ii) Let Δ *be a system of simple roots. One has*

$$\mathcal{C}(P(\Delta)) = \{x \in V*_{\mathbb{R}} \mid (x, \alpha) > 0 \quad \text{for all } \alpha \in \Delta\}.$$

The walls of $\mathcal{C}(P(\Delta))$ *are the hyperplanes* $H_\alpha = \alpha^{-1}(0)$*, for* $\alpha \in \Delta$*; its faces are the*

$$C_\alpha = \{x \in V*_{\mathbb{R}} \mid (x, \alpha) = 0, (x, \beta) > 0 \text{ for } \beta \in \Delta, \beta \neq \alpha\}.$$

(iii) One has the equivalence

$$R_+(x) = R_+ \iff x \in \mathcal{C}(R_+).$$

Proof. It is first clear that $R_+(x)$ is a system of positive roots. Since $x \in \mathcal{C}(R_+(x))$, the union of the $\mathcal{C}(R_+(x))$ is X. Property (iii) is immediate; it follows that the $\mathcal{C}(R_+)$ form a partition of X. The first assertion of (ii) is evident. It follows at once that $\mathcal{C}(R_+)$ is a chamber in V, which proves the rest of (i). Putting $C = \mathcal{C}(P(\Delta))$, it remains only to remark that

$$\bar{C} \cap (V*_{\mathbb{R}} - \bigcup_{\alpha \in R} \alpha^{-1}(0)) = \bar{C} \cap (V*_{\mathbb{R}} - \bigcup_{\alpha \in \Delta} \alpha^{-1}(0))$$

to complete the proof of (ii) by 3.6.1.

Definition 3.6.10. *The* $\mathcal{C}(R_+)$ *are called the* Weyl chambers *of the root datum. On the other hand, for every Weyl chamber* C*, one sets* $R_+(C) = R_+(x)$ *for any* $x \in C$*.*

Corollary 3.6.11. *The maps* $R_+ \mapsto \mathcal{C}(R_+)$ *and* $C \mapsto R_+(C)$ *realize a bijective correspondence between systems of positive roots and Weyl chambers.*

This correspondence is invariant under the Weyl group:

Lemma 3.6.12. *If* $w \in W(\mathcal{R})$*, one has* $\mathcal{C}(w(R_+)) = w(\mathcal{C}(R_+))$*.*

Corollary 3.6.13. *The correspondences* $\Delta \leftrightarrow R_+ \leftrightarrow C$ *are isomorphisms of homogeneous spaces under* $W(\mathcal{R})$*.*

We shall see later that these homogeneous spaces are principal (5.5).

[1268] N.D.E.: We have added the definition of the hyperplanes H_α.

Remark 3.6.14. *If C is a Weyl chamber, then $-C$ is also one, called the* opposite *of C. There exists therefore a $w_0 \in W$ (and in fact a unique one, cf. 5.5) such that $w_0(C) = -C$; one calls it the* symmetry *of the root datum with respect to the Weyl chamber C (or with respect to $R_+(C)$ or $\mathcal{S}(R_+(C))$...).*

4. Reduced root data of semisimple rank 2

4.0.[1269] Let $\mathcal{R}$ be a root datum of semisimple rank 2. Let α, β be a system of simple roots. Assume $\ell(\alpha) \leqslant \ell(\beta)$. One then has by 2.3.1 and 3.2.1 four possibilities:

Type	$\ell(\beta)/\ell(\alpha)$	$\ell(\beta*)/\ell(\alpha*)$	$(\beta*, \alpha)$	$(\alpha*, \beta)$
$A_1 \times A_1$	$-$	$-$	0	0
A_2	1	1	-1	-1
B_2	2	1/2	-1	-2
G_2	3	1/3	-1	-3

It follows from 3.4.5 that the order of $s_\alpha s_\beta$ is respectively 2, 3, 4, 6.

Let us study each of these systems separately and give the list of indivisible roots.

Type $A_1 \times A_1$. The indivisible roots are α, β, $-\alpha$, $-\beta$. The corresponding coroots are $\alpha*$, $\beta*$, $-\alpha*$, $-\beta*$.

Type A_2. The indivisible positive roots are as follows:

root γ	α	β	$\alpha + \beta$
$\ell(\gamma)/\ell(\alpha)$	1	1	1
coroot $\gamma*$	$\alpha*$	$\beta*$	$\alpha * + \beta*$
$\ell(\gamma*)/\ell(\beta*)$	1	1	1

The half-sum of the indivisible positive roots is $\rho = \alpha + \beta$.

Type B_2. The indivisible positive roots are as follows:

root γ	α	β	$\alpha + \beta$	$2\alpha + \beta$
$\ell(\gamma)/\ell(\alpha)$	1	2	1	2
coroot $\gamma*$	$\alpha*$	$\beta*$	$2\alpha * + \beta*$	$\alpha * + \beta*$
$\ell(\gamma*)/\ell(\beta*)$	2	1	2	1

The half-sum of the indivisible positive roots is $\rho = (4\alpha + 3\beta)/2$.

Type G_2. The indivisible positive roots are as follows:

[1269] N.D.E.: We have added the numbering 4.0 for later references.

root γ	α	β	$\alpha+\beta$	$2\alpha+\beta$	$3\alpha+\beta$	$3\alpha+2\beta$
$\ell(\gamma)/\ell(\alpha)$	1	3	1	1	3	3
coroot $\gamma*$	$\alpha*$	$\beta*$	$\alpha*+3\beta*$	$2\alpha*+3\beta*$	$\alpha*+\beta*$	$\alpha*+2\beta*$
$\ell(\gamma*)/\ell(\beta*)$	3	1	3	3	1	1

The half-sum of the indivisible positive roots is $\rho = 5\alpha + 3\beta$.

Proposition 4.1.[1270] *Let n be the order of $s_\alpha s_\beta$. Set $u_0 = 0$ and, for $p \in \mathbb{N}$,*

$$u_{2p+1} = u_{2p} + (s_\alpha s_\beta)^p(\alpha);$$
$$u_{2p+2} = u_{2p+1} + (s_\alpha s_\beta)^p s_\alpha(\beta).$$

Then:[1271]

(i) $u_{k+2n} = u_k$, for every $k \in \mathbb{N}$.

(ii) $u_0 = 0$, $u_1 = \alpha$, $u_{2n-1} = \beta$, $u_{2n} = 0$.

(iii) If $1 < k < 2n-1$, one has $u_k = a_k\alpha + b_k\beta$, with $a_k, b_k \in \mathbb{N}*$.

Proof. Assertion (i) follows from $(s_\alpha s_\beta)^n = 1$ and $u_{2n} = 0$.

Let us prove (ii) and (iii). Direct computation gives, in the four cases, the following sequences of values:[1272]

$(A_1 \times A_1)$ $\quad 0,\ \alpha,\ \beta + \alpha,\ \beta,\ 0.$

(A_2) $\quad 0,\ \alpha,\ \beta + 2\alpha,\ 2\beta + 2\alpha,\ 2\beta + \alpha,\ \beta,\ 0.$

(B_2) $\quad 0,\ \alpha,\ \beta + 3\alpha,\ 2\beta + 4\alpha,\ 3\beta + 4\alpha,\ 3\beta + 3\alpha,\ 2\beta + \alpha,\ \beta,\ 0.$

(G_2) $\quad 0,\ \alpha,\ \beta + 4\alpha,\ 2\beta + 6\alpha,\ 4\beta + 9\alpha,\ 5\beta + 10\alpha,\ 6\beta + 10\alpha,\ 6\beta + 9\alpha,\ 5\beta + 6\alpha,\ 4\beta + 4\alpha,\ 2\beta + \alpha,\ \beta,\ 0.$

Lemma 4.2. *Set $w_{2p} = (s_\beta s_\alpha)^p$, $w_{2p+1} = s_\alpha(s_\beta s_\alpha)^p$, so that $w_0, \cdots, w_{2n-1}$ are the distinct elements of W. Let $u_0, \cdots, u_{2n-1}$ be the elements of V defined in 4.1. For every $x \in V*$, one has*

$$x - w_k(x) = n_k\,\alpha^* + m_k\,\beta^*,$$

with $n_k, m_k \in \mathbb{Q}$ and $n_k + m_k = (x, u_k)$.[1273]

[1270] The following numbers 4.1 to 4.4 are used in the proof of 5.1. There exist nowadays simpler proofs of 5.1 (see [BLie], § V.3, Th. 1). *N.D.E.: we have specified the reference and placed here this Note, which in the original appeared in 4.2.*

[1271] N.D.E.: Let ρ be the half-sum of the positive roots (cf. 3.5.1); if one sets, as in 4.2 below, $w_0 = 1$, $w_1 = s_\alpha$, $w_2 = s_\beta s_\alpha$, $w_3 = s_\alpha s_\beta s_\alpha$, etc., then u_k is nothing other than $\rho - w_{k-1}(\rho)$, which proves (i) since $w_{2n} = id$.

[1272] N.D.E.: We have corrected the original, to be in agreement with the convention $\ell(\alpha) \leqslant \ell(\beta)$ of 4.0.

[1273] N.D.E.: Taking into account N.D.E. (17), this follows immediately from 3.5.1 and 1.1.9: one has $n_k + m_k = (x - w_k(x), \rho) = (x, \rho - w_{k-1}(\rho)) = (x, u_k)$.

Proof. The proof is by induction on k. If $k = 0$, the formula is trivially verified. Let us carry out, for example, the passage from w_{2p} to w_{2p+1}. One has $w_{2p+1} = s_\alpha w_{2p}$, whence

$$x - w_{2p+1}(x) = x - w_{2p}(x) + w_{2p}(x) - s_\alpha w_{2p}(x) = n_{2p} \alpha^* + m_{2p} \beta^* + (w_{2p}(x), \alpha)$$

So

$$\begin{aligned} n_{2p+1} + m_{2p+1} &= n_{2p} + m_{2p} + ((s_\beta s_\alpha)^p(x), \alpha) \\ &= (x, u_{2p}) + (x, (s_\alpha s_\beta)^p(\alpha)) = (x, u_{2p+1}). \end{aligned}$$

Corollary 4.3.[1274] *Let $x \in V*$. For every $w \in W$, set*

$$x - w(x) = a_w \alpha^* + b_w \beta^*.$$

If $(x, \alpha) \geqslant 0$ and $(x, \beta) \geqslant 0$, then $a_w + b_w \geqslant 0$. If moreover $(x, \alpha) > 0$ (resp. $(x, \beta) > 0$), then $a_w + b_w > 0$ for $w \neq 1, s_\beta$ (resp. for $w \neq 1, s_\alpha$).

Proof. This follows at once from 4.1 and 4.2.

Corollary 4.4. *Let $\mathcal{R}$ be any root datum and Δ a system of simple roots. Let γ be a positive root and α, β two simple roots; let $W_{\alpha,\beta}$ be the subgroup of W generated by s_α and s_β. If*

$$\mathrm{ord}_\Delta(s_\alpha(\gamma)) < \mathrm{ord}_\Delta(\gamma), \qquad \mathrm{ord}_\Delta(s_\beta(\gamma)) \geqslant \mathrm{ord}_\Delta(\gamma),$$

then, for every $w \in W_{\alpha,\beta}$, one has $ord_\Delta(w(\gamma)) \leqslant ord_\Delta(\gamma)$; moreover $ord_\Delta(w(\gamma)) < ord_\Delta(\gamma)$ if $w \neq 1$, $w \neq s_\beta$.

Proof. Indeed, consider the dual root datum $\mathcal{R}*$, then the datum $(M*, M, R'*, R')$, where R' is the set of roots which are rational linear combinations of α and β. Applying 4.3 to this datum, one finds the announced corollary.[1275]

5. The Weyl group: generators and relations

Let $\mathcal{R}$ be a root datum. Since the Weyl group is the same for this datum and for the corresponding reduced datum, one may assume $\mathcal{R}$ reduced to study the Weyl group.

Let $\Delta = \alpha_1, \cdots, \alpha_n$ be a system of simple roots ($n = rgss(\mathcal{R})$). Let n_{ij} be the order of the element $s_{\alpha_i} s_{\alpha_j}$ of W. In particular, one has $n_{ii} = 1$, and we saw in 3.4.2 and 3.4.3 that the subgroup W_{ij} of W generated by s_{α_i} and s_{α_j} was defined by the relations:

$$s_{\alpha_i}^2 = s_{\alpha_j}^2 = (s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1.$$

Theorem 5.1. *The group W is the group generated by the elements s_{α_i}, $i = 1, 2, \cdots, n$, subject to the relations $(s_{\alpha_i} s_{\alpha_j})^{n_{ij}} = 1$.*

[1274] N.D.E.: We have corrected the statement.

[1275] N.D.E.: Indeed, the hypotheses are equivalent to saying that $(\gamma, \alpha*) > 0$ and $(\gamma, \beta*) \geqslant 0$; then $\gamma - w(\gamma)$ belongs to $\mathbb{N}\alpha + \mathbb{N}\beta$ for every $w \in W_{\alpha,\beta}$, and is $\neq 0$ if $w \notin 1, s_\beta$.

Proof. We have already seen that the theorem is true when $n = 2$; we shall use this remark in the course of the proof. Introduce the group $\tilde{W}$ generated by elements T_i, $i = 1, 2, \cdots, n$, subject to the relations $(T_iT_j)^{n_{ij}} = 1$. One has in particular $n_{ii} = 1$, whence $T_i^2 = 1$. Let $p : \tilde{W} \to W$ be the group morphism that sends T_i to s_{α_i}. One knows that p is surjective; we shall show it is injective.

Lemma 5.2. *One can define in a unique manner for each $\alpha \in P(\Delta)$ an element $T_\alpha \in \tilde{W}$ such that one has the following properties:*

(i) $p(T_\alpha) = s_\alpha$,

(ii) $T_{\alpha_i} = T_i$,

(iii) if β and α are two positive roots such that $s_{\alpha_i}(\alpha) = \beta$, then $T_iT_\alpha T_i = T_\beta$.

Proof. Remark first that it follows from 1.2.10 and 3.3.6 that (i) is a consequence of (ii) and (iii), and that (ii) and (iii) completely determine the T_α. We shall make the construction by induction on $ord_\Delta(\alpha)$. If $ord_\Delta(\alpha) = 1$, then $\alpha \in \Delta$ and one sets $T_\alpha = T_i$ if $\alpha = \alpha_i$. Consider the hypothesis:

(H_p) *there exist T_α, for $\alpha \in P(\Delta)$, $ord_\Delta(\alpha) \leqslant p$, satisfying (ii) and condition (iii) whenever $ord_\Delta(\alpha) \leqslant p$, $ord_\Delta(\beta) \leqslant p$.*

This is verified for $p = 1$: indeed, if α and $s_{\alpha_i}(\alpha) = \beta$ are simple, α_i and α are orthogonal, so if one denotes $\alpha_j = \alpha = \beta$, one has $n_{ij} = 2$, whence

```
T_i T_j T_i = T_j.
```

Suppose $p > 1$ and (H_{p-1}) verified.

A) Construction of the T_α for $ord_\Delta(\alpha) \leqslant p$. It evidently suffices to do this for $ord_\Delta(\alpha) = p$. There then exists $\alpha_i \in \Delta$ such that $s_{\alpha_i}(\alpha) \in P(\Delta)$ and $ord_\Delta(s_{\alpha_i}(\alpha)) < p$ (3.3.3). One then sets

```
(⋆)    T_α = T_i T_{s_{α_i}(α)} T_i.
```

Let us verify that T_α depends only on α. Let $\alpha_j \in \Delta$ be such that $s_{\alpha_j}(\alpha) \in P(\Delta)$ and $ord_\Delta(s_{\alpha_j}(\alpha)) < p$. Let us prove that

```
(+)    T_i T_{s_{α_i}(α)} T_i = T_j T_{s_{α_j}(α)} T_j.
```

Let us distinguish two cases.

(1) Suppose α is a linear combination of α_i and α_j. Then the same is true of $s_{\alpha_i}(\alpha)$ and $s_{\alpha_j}(\alpha)$, and by (H_{p-1}), $T_{s_{\alpha_i}(\alpha)}$ and $T_{s_{\alpha_j}(\alpha)}$ are written as words in T_i and T_j. Since the projection of (+) in W is verified, and since the theorem is true for $n = 2$, so p is injective on the subgroup of $\tilde{W}$ generated by T_i and T_j, (+) is indeed verified.

(2) Suppose α is not a linear combination of α_i and α_j. Then if $w \in W_{\alpha_i,\alpha_j}$, the $w(\alpha)$ will all be positive (cf. 3.4.9). The relation to be verified is also written

```
(++)    (T_i T_j)^{n_{ij} − 1} T_{s_{α_i}(α)} (T_j T_i)^{n_{ij} − 1} = T_{s_{α_j}(α)}.
```

Now it follows from 4.4 that the $w(\alpha)$ are all of order $< p$ for $w \in W_{\alpha_i,\alpha_j}$, $w \neq 1$. One can therefore apply hypothesis (H_{p-1}) $2(n_{ij}-1)$ times, and we are done.

B) Verification of (H_p).[1276] One must verify that if $\alpha_j \in \Delta$ and if $\beta = s_{\alpha_j}(\alpha)$ satisfies $ord_\Delta(\beta) \leqslant p$, then $T_j T_\alpha T_j = T_\beta$. If $ord_\Delta(\beta) < p$, this follows from what precedes (since $T_j^2 = 1$), so one may suppose $ord_\Delta(\beta) = p = ord_\Delta(\alpha)$. In this case, α and α_j are orthogonal, so $\beta = \alpha$, and it is a matter of seeing that one has

$$(\dagger) \qquad T_j T_\alpha T_j = T_\alpha.$$

By $(\star)$ above, one has $T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i$, so it remains only to verify the following equality:

$$(+{+}+) \qquad T_j T_i T_{s_{\alpha_i}(\alpha)} T_i T_j = T_\alpha = T_i T_{s_{\alpha_i}(\alpha)} T_i.$$

Write $m = n_{ij}$ and $s_i = s_{\alpha_i}$, $s_j = s_{\alpha_j}$. One has $T_j T_i = (T_i T_j)^{m-1}$ and, by 4.4, one has $ord_\Delta(w(\alpha)) < p$ for every $w \in W_{ij}$ distinct from $1 = (s_i s_j)^m$ and from $s_j = s_i(s_i s_j)^{m-1}$. One deduces, by the induction hypothesis, that

$$T_j (T_i T_j)^{m-2} T_{s_i(\alpha)} (T_j T_i)^{m-2} T_j = T_{s_j (s_i s_j)^{m-2} s_i(\alpha)} = T_{s_i s_j(\alpha)}$$

(the last equality since $s_j(\alpha) = \alpha$), whence finally

$$(T_i T_j)^{m-1} T_{s_i(\alpha)} (T_j T_i)^{m-1} = T_i T_{s_i(\alpha)} T_i$$

which proves $(+{+}+)$.

Lemma 5.3. *Let $h \in \tilde{W}$. Write it*

$$h = T_{\alpha_1} \cdots T_{\alpha_m}$$

with the $\alpha_i \in \Delta$, not necessarily distinct, in such a way that m is minimal. Then

$$p(h)(\alpha_m) \in -P(\Delta).$$

Proof. Indeed, since $p(T_{\alpha_m})(\alpha_m) = s_{\alpha_m}(\alpha_m) = -\alpha_m$, if $p(h)(\alpha_m)$ were positive, there would exist an index k, $1 \leqslant k \leqslant m-1$, such that

$$u = s_{\alpha_{k+1}} \cdots s_{\alpha_m}(\alpha_m) = -s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) \in -P(\Delta),$$

and $s_{\alpha_k}(u) \in P(\Delta)$. But then one has necessarily $u = -\alpha_k$ (3.3.1), whence

$$s_{\alpha_{k+1}} \cdots s_{\alpha_{m-1}}(\alpha_m) = \alpha_k,$$

which entails by (iii)

$$T_{\alpha_k} T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}} T_{\alpha_m} = T_{\alpha_{k+1}} \cdots T_{\alpha_{m-1}},$$

[1276] N.D.E.: We have detailed the original in what follows.

and this contradicts the minimal character of m.

Let now $h \in \tilde{W}$ be such that $p(h)(P(\Delta)) \subset P(\Delta)$. By Lemma 5.3, one has $p(h) = 1$, which proves Theorem 5.1 and moreover the

Corollary 5.4. *If R_+ is a system of positive roots and if $w \in W$ is such that $w(R_+) = R_+$, then $w = 1$.*

Corollary 5.5. *The Weyl group operates freely and transitively on the set of systems of positive roots (resp. of systems of simple roots, resp. of Weyl chambers).*

Let us now choose a system of simple roots Δ. Put $R^+ = P(\Delta)$.[1277]

For every pair of simple roots $(\alpha, \beta) \in \Delta \times \Delta$, denote by $R_{\alpha,\beta}$ the set of roots that are linear combinations of α and β. Write $R^+_{\alpha,\beta} = R^+ \cap R_{\alpha,\beta}$ and let $W_{\alpha,\beta}$ be the Weyl group of $R_{\alpha,\beta}$, that is to say, the subgroup of W generated by s_α and s_β.

Theorem 5.6 (Tits). *Let α and β be two simple roots and let $w \in W$ be such that $w(\alpha) = \beta$. There exists a sequence of simple roots $\alpha_0, \cdots, \alpha_m$ and a sequence $w_0, \cdots, w_{m-1}$ of elements of W satisfying the following conditions:*

(i) $\alpha_0 = \alpha$, $\alpha_m = \beta$.

(ii) $w = w_{m-1} w_{m-2} \cdots w_0$.

(iii) $w_i(\alpha_i) = \alpha_{i+1}$, for $0 \leqslant i \leqslant m-1$.

(iv) For every i, $0 \leqslant i \leqslant m-1$, such that $\alpha_i \neq \alpha_{i+1}$, one has $w_i \in W_{\alpha_i, \alpha_{i+1}}$.

(v) For every i, $0 \leqslant i \leqslant m-1$, such that $\alpha_i = \alpha_{i+1}$, there exists a simple root β_i such that $w_i \in W_{\alpha_i, \beta_i}$.

Proof. Set[1278]

$$M(w) = \mathrm{Card}(R^+ \cap w^{-1}(-R^+)) = \mathrm{Card}(\{\alpha \in R^+ \mid w(\alpha) \in -R^+\}).$$

If $M(w) = 0$, then $w(R^+) = R^+$, so $w = 1$ by 5.4 and the theorem is trivial ($m = 0$; assertions (iii) to (v) are empty). We argue by induction on $M(w)$. If $M(w) > 0$, there exists $\beta_0 \in \Delta$ such that $w(\beta_0) \in -R^+$. Set $\alpha_0 = \alpha$. Consider the set

$$A = w^{-1}(R^+) \cap R_{\alpha_0, \beta_0}.$$

This is a system of positive roots of R_{α_0,β_0}. There thus exists $w_0 \in W_{\alpha_0,\beta_0}$ such that

$$w_0^{-1}(R^+_{\alpha_0,\beta_0}) = A.$$

Put $w' = w w_0^{-1}$. By 3.4.9, one has at once

$$R^+ - R^+_{\alpha_0, \beta_0} = w_0(R^+ - R^+_{\alpha_0, \beta_0}),$$

whence

[1277] N.D.E.: We have written R^+ here instead of R_+, in order to be able to write $R^+_{\alpha,\beta} = R_{\alpha,\beta} \cap R^+$.
[1278] N.D.E.: In what follows, we have detailed the original, and we have corrected equality (1).

$$(1) \qquad (R^+ - R^+_{\alpha_0,\beta_0}) \cap w^{-1}(-R^+) = w_0((R^+ - R^+_{\alpha_0,\beta_0}) \cap w'^{-1}(-R^+)).$$

On the other hand,

$$(2) \qquad \beta_0 \in R^+_{\alpha_0,\beta_0} \cap w^{-1}(-R^+),$$

and, since $w_0(R_{\alpha_0,\beta_0}) = R_{\alpha_0,\beta_0}$, one has $w_0(-A) = R_{\alpha_0,\beta_0} \cap w'^{-1}(-R^+)$, whence

$$(2') \qquad R^+_{\alpha_0,\beta_0} \cap w'^{-1}(-R^+) = R^+_{\alpha_0,\beta_0} \cap w_0(-A) = R^+_{\alpha_0,\beta_0} \cap -R^+_{\alpha_0,\beta_0} = \varnothing.$$

It follows from (1), (2), (2′) that $M(w') < M(w)$.

Set $\alpha_1 = w_0(\alpha_0)$; let us show that $\alpha_1 \in \Delta$, that is to say $\alpha_0 \in w_0^{-1}(\Delta)$. One knows that $w(\alpha_0) \in \Delta$, so $\alpha_0 \in w^{-1}(\Delta)$, hence also $\alpha_0 \in w^{-1}(\Delta) \cap R_{\alpha_0,\beta_0}$; so α_0 is a simple root of $A = w^{-1}(R^+) \cap R_{\alpha_0,\beta_0} = w_0^{-1}(R^+_{\alpha_0,\beta_0})$, hence belongs to

$$w_0^{-1}(\Delta \cap R^+_{\alpha_0,\beta_0}) = w_0^{-1}(\{\alpha_0, \beta_0\})$$

(see 3.4.8). So $\alpha_1 = w_0(\alpha_0)$ equals α_0 or β_0. If $\alpha_1 \neq \alpha_0$, one has $\alpha_1 = \beta_0$ and $w_0 \in W_{\alpha_0,\alpha_1}$.

Finally, one has $\beta = w'(\alpha_1)$, with $M(w') < M(w)$, and one concludes by induction.

6. Morphisms of root data

6.1. Definition

Let $\mathcal{R} = (M, M*, R, R*)$ and $\mathcal{R}' = (M', M'*, R', R'*)$ be two root data. Let $f : M' \to M$ be a linear map and ${}^t f : M* \to M'*$ the transposed map.

Definition 6.1.1. *One says that f is a* morphism *from $\mathcal{R}'$ to $\mathcal{R}$ and writes $f : \mathcal{R}' \to \mathcal{R}$, if f induces a bijection of R' onto R and ${}^t f$ a bijection of $R*$ onto $R'*$.*

Then ${}^t f$ is a morphism of the dual root data:

$${}^t f : \mathcal{R}* \longrightarrow \mathcal{R}'* .$$

One sees easily that if f is a morphism from $\mathcal{R}'$ to $\mathcal{R}$, and if one writes $\alpha = f(\alpha')$ for $\alpha' \in R'$, one has $\alpha'* = {}^t f(\alpha*)$. Indeed, one sees immediately that if one writes p and p' for the maps of 1.2.1 respective to $\mathcal{R}$ and $\mathcal{R}'$, one has $p' = {}^t f \circ p \circ f$, and the assertion sought follows at once. We leave to the reader the task of proving the statements that follow, which are almost all trivial.

Proposition 6.1.2. *Let $f : \mathcal{R}' \to \mathcal{R}$ be a morphism of root data. If $\alpha' \in R'$ and $\alpha = f(\alpha')$, then $\alpha'* = {}^t f(\alpha*)$. Moreover, f induces isomorphisms:*

$$R' \xrightarrow{\sim} R, \qquad \Gamma_0(R') \xrightarrow{\sim} \Gamma_0(R), \qquad V(R') \xrightarrow{\sim} V(R),$$

and ${}^t f$ induces isomorphisms:

$$R* \xrightarrow{\sim} R'*, \qquad \Gamma_0(R*) \xrightarrow{\sim} \Gamma_0(R'*), \qquad V(R*) \xrightarrow{\sim} V(R'*),$$

the last being the transpose of the corresponding morphism induced by f. The map $s_{\alpha'} \mapsto s_{f(\alpha')}$ extends to an isomorphism $W(\mathcal{R}') \xrightarrow{\sim} W(\mathcal{R})$ compatible with the operations of these two groups on the sets of 1.1.13.

Proposition 6.1.3. *The maps*

$$\Delta' \mapsto f(\Delta'), \quad R'_+ \mapsto f(R'_+), \quad C' \mapsto ({}^t f \otimes \mathbb{R})^{-1}(C')$$

define bijective correspondences between systems of simple roots, systems of positive roots, and Weyl chambers for $\mathcal{R}'$ and $\mathcal{R}$. These correspondences are compatible with the action of the Weyl groups and with the correspondences

$$\mathcal{S}(R_+) \leftrightarrow R_+ \leftrightarrow \mathcal{C}(R_+).$$

Lemma 6.1.4. *Morphisms compose. For the morphism $f : \mathcal{R}' \to \mathcal{R}$ to be an isomorphism, it is necessary and sufficient that $f : M' \to M$ be bijective.*

6.2. Isogenies

Definition 6.2.1. *A morphism $f : \mathcal{R}' \to \mathcal{R}$ of root data is called an* isogeny *if $f : M' \to M$ is injective with finite cokernel.*

If f is an isogeny, then ${}^t f$ is an isogeny.

Definition 6.2.2. *Let $f : \mathcal{R}' \to \mathcal{R}$ be an isogeny. Set $K(f) = Coker(M' \to^f M)$.*

Lemma 6.2.3. *One has a natural pairing*

$$K(f) \times K({}^t f) \to \mathbb{Q}/\mathbb{Z},$$

which puts these two finite groups in duality.

Proof. This is classical.

Lemma 6.2.4. *If $f : \mathcal{R}' \to \mathcal{R}$ is a morphism, then $rgss(\mathcal{R}') = rgss(\mathcal{R})$. If moreover f is an isogeny, one has also $rgred(\mathcal{R}') = rgred(\mathcal{R})$.*

Proof. Trivial.

Lemma 6.2.5. *Every morphism of semisimple root data is an isogeny.*

Proof. This follows at once from the fact that f must induce an isomorphism of $V' = V(R')$ onto $V = V(R)$.

If $\mathcal{R}'$ and $\mathcal{R}$ are semisimple, every isogeny $f : \mathcal{R}' \to \mathcal{R}$ defines a commutative diagram:

$$\begin{tikzcd}
\Gamma_0(R') \arrow[r] \arrow[d, "\subset"] & \Gamma_0(R) \arrow[d, "\subset"] \\
M' \arrow[r, "f"'] & M.
\end{tikzcd}$$

If $M = \Gamma_0(R)$, then f is necessarily an isomorphism.

Definition 6.2.6. *A root datum is called* adjoint *(resp.* simply connected*) if $M = \Gamma_0(R)$, resp. $M* = \Gamma_0(R*)$.*

An adjoint or simply connected root datum is therefore semisimple. On the other hand, $\mathcal{R}$ is adjoint (resp. simply connected) if and only if $\mathcal{R}*$ is simply connected (resp. adjoint). By virtue of the preceding result, one has:

Proposition 6.2.7. *Let $\mathcal{R}$ be a semisimple root datum. The following conditions are equivalent:*

(i) $\mathcal{R}$ is adjoint (resp. simply connected).

(ii) Every isogeny $\mathcal{R}' \to \mathcal{R}$ (resp. $\mathcal{R} \to \mathcal{R}'$) is an isomorphism.

Proposition 6.2.8. *Let $\mathcal{R}$ be an adjoint (resp. simply connected) root datum. Every indivisible root (resp. coroot) is an indivisible element of M (resp. $M*$).*

Proof. Indeed, every indivisible root is part of a basis of $\Gamma_0(R)$, by 3.3.5.

6.3. Radical and coradical

Let $\mathcal{R}$ be a root datum. Set

$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for all } \alpha^* \in R^*\};$
$N^* = M^*/(V(R^*) \cap M^*).$

Lemma 6.3.1. *Consider the canonical morphisms:*

N □ M,   M* □ N*.

They are transposes of each other and $N*$ is identified with the dual of N.

Proof. This is immediate, taking 1.2.5 into account.

Definition 6.3.2. *One calls* coradical *of $\mathcal{R}$ and denotes by $corad(\mathcal{R})$ the trivial root datum*

$$corad(\mathcal{R}) = (N, N*, \emptyset, \emptyset).$$

If one sets $\mathcal{R}_0 = (M, M*, \emptyset, \emptyset)$ (this is a trivial root datum), one has therefore a morphism

$$corad(\mathcal{R}) \longrightarrow \mathcal{R}_0.$$

Definition 6.3.3. *One calls* radical *of $\mathcal{R}$ and denotes by $rad(\mathcal{R})$ the trivial root datum:*

$$rad(\mathcal{R}) = corad(\mathcal{R}*)*.$$

[1279]

One thus has a diagram

$$\begin{tikzcd} corad(\mathcal{R}) \arrow[rr] \arrow[dr] & & rad(\mathcal{R}) \arrow[dl] \\ & \mathcal{R}_0 & \end{tikzcd}$$

whose transpose is the corresponding diagram for $\mathcal{R}*$.

Lemma 6.3.4. *The canonical morphism $u : corad(\mathcal{R}) \to rad(\mathcal{R})$ is an isogeny.*

Definition 6.3.5. *Set $N(\mathcal{R}) = K(u) = M/((V(R) \cap M) + \bigcap_{\alpha \in R} Ker(\alpha*))$. One then has a canonical pairing*

$$N(\mathcal{R}) \times N(\mathcal{R}*) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

Lemma 6.3.6. *One has $rgred(rad(\mathcal{R})) = rgred(corad(\mathcal{R})) = rgred(\mathcal{R}) - rgss(\mathcal{R})$, and the following conditions are equivalent:*

(i) $\mathcal{R}$ is semisimple,

(ii) $rad(\mathcal{R}) = 0$,

(iii) $corad(\mathcal{R}) = 0$.

6.4. Products of root data

Definition 6.4.1. *Let $\mathcal{R} = (M, M*, R, R*)$ and $\mathcal{R}' = (M', M'*, R', R'*)$ be two root data. One calls* product root datum *of $\mathcal{R}$ and $\mathcal{R}'$ and denotes by $\mathcal{R}'' = \mathcal{R} \times \mathcal{R}'$ the root datum $(M'', M''*, R'', R''*)$ where*

$$M'' = M \times M', \qquad M''* = M* \times M'*,$$
$$R'' = (R \times 0) \cup (0 \times R'), \qquad R''* = (R* \times 0) \cup (0 \times R'*),$$

the map $\alpha \mapsto \alpha*$ being the obvious map.

Proposition 6.4.2. *Under the preceding conditions, one has canonical isomorphisms*

$$\Gamma_0(R'') \simeq \Gamma_0(R) \times \Gamma_0(R'), \qquad V(R'') \simeq V(R) \times V(R'),$$
$$W(\mathcal{R}'') \simeq W(\mathcal{R}) \times W(\mathcal{R}'),$$

etc., and the equalities

$$rgred(\mathcal{R}'') = rgred(\mathcal{R}) + rgred(\mathcal{R}'), \qquad rgss(\mathcal{R}'') = rgss(\mathcal{R}) + rgss(\mathcal{R}').$$

One also has a canonical isomorphism of root data

$$(\mathcal{R} \times \mathcal{R}')* \simeq \mathcal{R}* \times \mathcal{R}'*.$$

[1279] N.D.E.: That is, if one denotes $P = y \in M*|(y,\alpha) = 0 forall \alpha \in R$ and $P* = M/(V(R) \cap M)$, one has $rad(\mathcal{R}) = (P*, P, \emptyset, \emptyset)$.

The preceding definitions extend at once to a product of several factors. One has at once:

Proposition 6.4.3. *Let $\mathcal{R} = \mathcal{R}_1 \times \mathcal{R}_2 \times \cdots \times \mathcal{R}_n$ be a product of root data. The following conditions are equivalent:*

(i) $\mathcal{R}$ is semisimple (resp. simply connected, resp. adjoint, resp. reduced).

(ii) Each $\mathcal{R}_i$ is semisimple (resp. simply connected, resp. adjoint, resp. reduced).

Let us consider the following special case: let $\mathcal{R}_0$ be a trivial root datum and $\mathcal{R}_1$ a semisimple root datum. One then has a commutative diagram

$$\begin{tikzcd} & \mathcal{R}_0 \times \mathcal{R}_1 \arrow[dr] & \\ \mathcal{R}_1 \arrow[ur] \arrow[rr, "\mathrm{id}"] & & \mathcal{R}_1. \end{tikzcd}$$

Lemma 6.4.4. *One has canonical isomorphisms*

$$\begin{tikzcd} \mathrm{corad}(\mathcal{R}_0 \times \mathcal{R}_1) \arrow[rr] \arrow[dr, "\simeq"'] & & \mathrm{rad}(\mathcal{R}_0 \times \mathcal{R}_1) \arrow[dl, "\simeq"] \\ & \mathcal{R}_0. & \end{tikzcd}$$

In particular, $N(\mathcal{R}_0 \times \mathcal{R}_1) = 0$.

We shall see later that if conversely $N(\mathcal{R}) = 0$, then the root datum $\mathcal{R}$ is a product of a semisimple datum by a trivial datum.

6.5. Induced and coinduced root data

Let $\mathcal{R} = (M, M*, R, R*)$ be a root datum. Let $N \subset M$ be a subgroup containing the roots, i.e. such that

$$\Gamma_0(R) \subset N \subset M.$$

The canonical linear map $i_N : N \to M$ gives by transposition a linear map

$${}^t i_N : M* \longrightarrow N*.$$

Set $R_N = R$ and $R*_N = {}^t i_N(R*)$.

Lemma 6.5.1. $\mathcal{R}_N = (N, N*, R_N, R*_N)$ *is a root datum, and i_N a morphism.*

Proof. Let us first show that ${}^t i_N$ induces an isomorphism of $R*$ onto $R*_N$. If $\alpha, \beta \in R$ and ${}^t i_N(\alpha*) = {}^t i_N(\beta*)$, one has $(\alpha*, x) = (\beta*, x)$ for all $x \in N$, in particular for $x \in R$, which gives $\alpha = \beta$ by 1.1.4. The rest follows without difficulty.

Definition 6.5.2. $\mathcal{R}_N$ *is called the* root datum induced by $\mathcal{R}$ on N.

Lemma 6.5.3. *Let $f : \mathcal{R}' \to \mathcal{R}$ be a morphism. Set $N = f(M') \subset M$. Then f factors uniquely through i_N.*

In particular, isogenies $\mathcal{R}' \to \mathcal{R}$, up to isomorphism, correspond bijectively to the subgroups of finite index of $M/\Gamma_0(R)$, which makes 6.2.7 more precise.

Let now $N*$ be a subgroup of $M*$ containing $R*$. One defines the root datum *coinduced* by $\mathcal{R}$ on $N*$ by

$$\mathcal{R}^{N*} = (\mathcal{R}*_{N*})*,$$

and one has a canonical morphism:

$$p_{N*} : \mathcal{R} \longrightarrow \mathcal{R}^{N*}.$$

Lemma 6.5.4. *Let $f : \mathcal{R}' \longrightarrow \mathcal{R}$ be a morphism. There exist subgroups $N \subset M$ and $N'* \subset M'*$ such that f factors as*

$$\begin{tikzcd}
\mathcal{R}' \arrow[r, "f"] \arrow[d, "p_{N'*}"'] & \mathcal{R} \\
\mathcal{R}'^{N'*} \arrow[r, "f_0", "\sim"'] & \mathcal{R}_N, \arrow[u, "i_N"']
\end{tikzcd}$$

where f_0 is an isomorphism.

Proof. Indeed, one takes $N = f(M')$ as in 6.5.3. The morphism $M' \to N$ obtained is surjective, so its transpose is injective. One takes the image of the latter as $N'*$.

Let us now treat certain special cases. If one takes $N = \Gamma_0(R)$, one writes $\mathcal{R}_N = ad(\mathcal{R})$. If one takes $N = V(R) \cap M$, one writes $\mathcal{R}_N = ss(\mathcal{R})$. One thus has a diagram:

$$ad(\mathcal{R}) \longrightarrow ss(\mathcal{R}) \longrightarrow \mathcal{R}.$$

Set $d\acute{e}r(\mathcal{R}) = ss(\mathcal{R}*)*$ and $sc(\mathcal{R}) = ad(\mathcal{R}*)*$; by duality, one obtains a diagram:

$$\mathcal{R} \longrightarrow d\acute{e}r(\mathcal{R}) \longrightarrow sc(\mathcal{R}).$$

Proposition 6.5.5. *(i) In the first row of the diagram*

$$\begin{tikzcd}
ad(\mathcal{R}) \arrow[r] & ss(\mathcal{R}) \arrow[r] \arrow[d] & d\acute{e}r(\mathcal{R}) \arrow[r] & sc(\mathcal{R}) \\
 & \mathcal{R} \arrow[ur] & &
\end{tikzcd}$$

the four data are semisimple and the three morphisms are isogenies.

(ii) $ad(\mathcal{R})$ is an adjoint datum, and $\mathcal{R}$ is adjoint if and only if $ad(\mathcal{R}) \to \mathcal{R}$ is an isomorphism.

(iii) $sc(\mathcal{R})$ is a simply connected datum, and $\mathcal{R}$ is simply connected if and only if $\mathcal{R} \to sc(\mathcal{R})$ is an isomorphism.

(iv) The following conditions are equivalent:

(a) $\mathcal{R}$ is semisimple,

(b) $ss(\mathcal{R}) \to \mathcal{R}$ is an isomorphism,

(c) $\mathcal{R} \to d\acute{e}r(\mathcal{R})$ is an isomorphism.

Let us pause briefly on the morphism $ss(\mathcal{R}) \to d\acute{e}r(\mathcal{R})$. Referring to the construction of $ss(\mathcal{R})$ and $d\acute{e}r(\mathcal{R})$, it is easy to prove the

Lemma 6.5.6. *Let $h : ss(\mathcal{R}) \to d\acute{e}r(\mathcal{R})$ be the canonical isogeny. One has $K(h) \simeq N(\mathcal{R})$.*

6.5.7. Consider other special cases of induced data. Set

$$N = \{x \in M \mid (\alpha^*, x) = 0 \text{ for } \alpha \in R\} \times \Gamma_0(R);$$

one knows that the sum is direct by 1.2.5. It follows that the root datum $\mathcal{R}_N$ is identified with the product $ad(\mathcal{R}) \times corad(\mathcal{R})$.

One can do the same by replacing $\Gamma_0(R)$ by $V(R) \cap M$, then dualize these two constructions. One thus obtains a diagram of root data:

$$\begin{tikzcd}
ad(\mathcal{R}) \arrow[r] \arrow[d] & ss(\mathcal{R}) \arrow[rr] \arrow[d] & & d\acute{e}r(\mathcal{R}) \arrow[r] \arrow[d] & sc(\mathcal{R}) \arrow[d] \\
ad(\mathcal{R}) \times corad(\mathcal{R}) \arrow[r] \arrow[d] & ss(\mathcal{R}) \times corad(\mathcal{R}) \arrow[r] \arrow[d] & \mathcal{R} \arrow[r] & d\acute{e}r(\mathcal{R}) \times rad(\mathcal{R}) \arrow[r] \arrow[d] & sc(\mathcal{R}) \times rad(\mathcal{R}) \arrow[d] \\
ad(\mathcal{R}) \arrow[r] & ss(\mathcal{R}) \arrow[rr] & & d\acute{e}r(\mathcal{R}) \arrow[r] & sc(\mathcal{R});
\end{tikzcd}$$

which is commutative, as one verifies at once. This diagram is self-dual in an obvious sense. The horizontal morphisms are isogenies. The composites of the vertical arrows are the identity.

Lemma 6.5.8. *Let h_1 and h_2 be the canonical isogenies:*

$$ss(\mathcal{R}) \times corad(\mathcal{R}) \xrightarrow{h_1} \mathcal{R} \xrightarrow{h_2} d\acute{e}r(\mathcal{R}) \times rad(\mathcal{R}).$$

One has $K(h_1) \simeq K(h_2) \simeq N(\mathcal{R})$.

Proof. This is trivial from the definitions.

Corollary 6.5.9. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

(i) $N(\mathcal{R}) = 0$, i.e. $corad(\mathcal{R}) \to rad(\mathcal{R})$ is an isomorphism.

(ii) $h : ss(\mathcal{R}) \to d\acute{e}r(\mathcal{R})$ is an isomorphism.

(iii) $h_1 : ss(\mathcal{R}) \times corad(\mathcal{R}) \to \mathcal{R}$ is an isomorphism.

(iv) $h_2 : \mathcal{R} \to d\acute{e}r(\mathcal{R}) \times rad(\mathcal{R})$ is an isomorphism.

(v) $\mathcal{R}$ is the product of a semisimple datum and a trivial datum.

Let us also state a trivial consequence of the preceding remarks:

Corollary 6.5.10. *For every root datum $\mathcal{R}$, there exist isogenies*

$\mathrm{ad}(\mathcal{R}) \times \mathcal{R}_0 \to \mathcal{R} \to \mathrm{sc}(\mathcal{R}) \times \mathcal{R}_0,$

where $\mathcal{R}_0$ is "the" trivial root datum of rank $rgred(\mathcal{R}) - rgss(\mathcal{R})$.

Let us finally note a result that may be useful:

Lemma 6.5.11. *Let $\mathcal{R}$ be a root datum, Δ a system of simple roots, Δ' a subset of Δ; consider the root datum (cf. 3.4.7)*

$\mathcal{R}_{\Delta'} = (M, M^*, R_{\Delta'}, R^*_{\Delta'}).$

(i) If $\mathcal{R}$ is simply connected, then $d\acute{e}r(\mathcal{R}_{\Delta'})$ is simply connected.

(ii) If $\mathcal{R}$ is adjoint, then $ss(\mathcal{R}_{\Delta'})$ is adjoint.

Proof. The two assertions are evidently equivalent by duality. The second reduces to verifying the formula:

$$M \cap V(R_{\Delta'}) = \Gamma_0(R_{\Delta'});$$

now, if $M = \Gamma_0(R)$, the two sides are equal to the subgroup of M generated by Δ'.

6.6. Weights

Definition 6.6.1. *Let $\mathcal{R}$ be a root datum. Set*[1280]

$\Lambda(\mathcal{R}) = \{x \in V(R) \mid (\alpha^*, x) \in \mathbb{Z} \text{ for all } \alpha^* \in R^*\}.$

The elements of $\Lambda(\mathcal{R})$ are called the weights *of $\mathcal{R}$. The weights of $\mathcal{R}*$ are called the* coweights *of $\mathcal{R}$.*

One has $\Gamma_0(R) \subset \Lambda(\mathcal{R})$ and $\Lambda(\mathcal{R})$ is stable under $W(\mathcal{R})$.

Lemma 6.6.2. *The bilinear map $V * \times V \to \mathbb{Q}$ induces a duality*

$$\Gamma_0(R*) \times \Lambda(\mathcal{R}) \longrightarrow \mathbb{Z}.$$

Proof. Trivial.

Corollary 6.6.3. *Let $\Delta* = (\alpha*_1, \cdots, \alpha*_n)$ be a system of simple coroots. Let p_i, $i = 1, 2, \cdots, n$, be the elements of $V(R)$ defined by*

$$(\alpha*_i, p_j) = \delta_{ij},$$

[1280]N.D.E.: We have replaced $\Gamma(\mathcal{R})$ by $\Lambda(\mathcal{R})$, to avoid any risk of confusion with $\Gamma_0(R)$.

(whence $s_{\alpha_i}(p_i) = p_i - \alpha_i$ and $s_{\alpha_i}(p_j) = p_j$ for $i \neq j$).[1281] *Then $\Lambda(\mathcal{R})$ is the free abelian group generated by the p_i.*

The p_i are called the *fundamental weights* corresponding to the system of simple coroots $\Delta*$.

Corollary 6.6.4. *For every $\alpha* \in \Delta*$, one has therefore $(\alpha*, \Sigma_i p_i) = 1$, hence $\Sigma_i p_i = \rho_{R_+}$ (cf. 3.5.1), where $R_+ = P(ind(\Delta))$.*

Corollary 6.6.5. *For every $x \in V(R)$, one has $x = \Sigma_i(\alpha*_i, x)p_i$.*

Remark that $R* \subset \Gamma_0(R*)$ and $R \subset \Lambda(\mathcal{R})$, so $(\Lambda(\mathcal{R}), \Gamma_0(R*), R, R*)$ is a root datum.

Corollary 6.6.6. *The canonical morphism $\Gamma_0(R*) \to M*$ is the transpose of the morphism $x \mapsto \Sigma_i(\alpha*_i, x)p_i$, which defines a morphism of root data, and one has a commutative diagram:*

$$\begin{tikzcd} & sc(\mathcal{R}) \arrow[d] \\ \mathcal{R} \arrow[ur] \arrow[dr] & \\ & (\Lambda(\mathcal{R}), \Gamma_0(R*), R, R*). \end{tikzcd}$$

One thus has an explicit description of $sc(\mathcal{R})$ in terms of the weights of $\mathcal{R}$. Similarly, one finds a commutative diagram:

$$\begin{tikzcd} ad(\mathcal{R}) \arrow[d] \arrow[dr] & \\ & \mathcal{R} \\ (\Gamma_0(R), \Lambda(\mathcal{R}*), R, R*). \arrow[ur] & \end{tikzcd}$$

Corollary 6.6.7. *For $\mathcal{R}$ to be simply connected, it is necessary and sufficient that $M = \Lambda(\mathcal{R})$.*

Remark 6.6.8. *One has $\Lambda(\mathcal{R}) \cap M = V(R) \cap M$. For $\mathcal{R}$ to be semisimple, it is therefore necessary and sufficient that $M \subset \Lambda(\mathcal{R})$.*

From the results of 6.5 there also follows:

Corollary 6.6.9. *For $\mathcal{R}$ to be the product of a simply connected datum by a trivial datum, it is necessary and sufficient that $M \supset \Lambda(\mathcal{R})$.*

Consider now the canonical isogeny

$$f : ad(\mathcal{R}) \longrightarrow sc(\mathcal{R}),$$

and set $Z(\mathcal{R}) = K(f)$. One has $Z(\mathcal{R}) \simeq Z(sc\mathcal{R}) \simeq Z(ad\mathcal{R})$.

Corollary 6.6.10. *One has a canonical isomorphism $Z(\mathcal{R}) = \Lambda(\mathcal{R})/\Gamma_0(R)$. More precisely, one has an exact sequence of $W(\mathcal{R})$-modules:*

[1281] Warning: if the datum is not reduced, the α_i do not form a system of simple roots.

$$0 \longrightarrow \Gamma_0(R) \longrightarrow \Lambda(\mathcal{R}) \longrightarrow Z(\mathcal{R}) \longrightarrow 0.$$

Corollary 6.6.11. *One has a canonical pairing*

$$Z(\mathcal{R}*) \times Z(\mathcal{R}) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

which puts these groups in duality.

Remark 6.6.12. *One has* $Z(\mathcal{R} \times \mathcal{R}') \simeq Z(\mathcal{R}) \times Z(\mathcal{R}')$. *Consider in particular simply connected data* $\mathcal{R}_i$, $i = 1, 2, \cdots, n$, *and a trivial datum* $\mathcal{R}_0$. *Set* $\mathcal{R} = \mathcal{R}_0 \times \mathcal{R}_1 \times \cdots \times \mathcal{R}_n$. *Let* $\mathcal{R} = (M, M*, R, R*)$, $\mathcal{R}_0 = (M_0, M*_0, \emptyset, \emptyset)$. *One has*

$$M/\Gamma_0(R) \simeq M_0 \times Z(\mathcal{R}_1) \times \cdots \times Z(\mathcal{R}_n).$$

6.7. Automorphisms

An automorphism of $\mathcal{R}$ is, by 6.1.4, an automorphism of M, say u, such that $u(R) = R$ and ${}^t u(R*) = R*$. In particular, every element w of $W(\mathcal{R})$ defines an automorphism of $\mathcal{R}$.

Lemma 6.7.1. $W(\mathcal{R})$ *is a normal subgroup of* $\mathrm{Aut}(\mathcal{R})$. *More precisely, if* $u \in \mathrm{Aut}(\mathcal{R})$ *and* $\alpha \in R$, *one has*

$$u\, s_\alpha\, u^{-1} = s_{u(\alpha)}.$$

Proof. The proof is the same as that of 1.2.10.

Proposition 6.7.2. *Let* Δ *be a system of simple roots. Set*

$$E_\Delta(\mathcal{R}) = \{u \in \mathrm{Aut}(\mathcal{R}) \mid u(\Delta) = \Delta\}.$$

Then $\mathrm{Aut}(\mathcal{R})$ *is the semidirect product of* $W(\mathcal{R})$ *by* $E_\Delta(\mathcal{R})$.

Proof. This follows at once from the fact that $W(\mathcal{R})$ operates simply transitively on the systems of simple roots, and from the fact that if Δ is a system of simple roots of $\mathcal{R}$, then $u(\Delta)$ is a system of simple roots for every automorphism u of $\mathcal{R}$.

We shall see later a simpler description of $E_\Delta(\mathcal{R})$ in the case of reduced and irreducible root data.[1282]

Definition 6.7.3. *One denotes by* $\mathrm{Aut}_s(\mathcal{R})$[1283] *the set of* $u \in \mathrm{Aut}(\mathcal{R})$ *such that the following diagram is commutative:*

$$\begin{tikzcd} \mathcal{R} \arrow[rr, "u"] \arrow[dr] & & \mathcal{R} \arrow[dl] \\ & \mathrm{rad}(\mathcal{R}). & \end{tikzcd}$$

[1282] N.D.E.: when $\mathcal{R}$ is simply connected or adjoint, see 7.4.5.
[1283] N.D.E.: The exponent s is meant to suggest "semisimple".

One writes $E_\Delta^s(\mathcal{R}) = E_\Delta(\mathcal{R}) \cap \mathrm{Aut}_s(\mathcal{R})$.

Remark 6.7.4. *If* $u \in \mathrm{Aut}(\mathcal{R})$*, one has therefore* $u \in \mathrm{Aut}_s(\mathcal{R})$ *if and only if* $(u - id)(M) \subset V(R)$*. In particular* $W(\mathcal{R}) \subset \mathrm{Aut}_s(\mathcal{R})$*. It follows at once:*

Proposition 6.7.5. *The group* $\mathrm{Aut}_s(\mathcal{R})$ *is the semidirect product of* $W(\mathcal{R})$ *by* $E_\Delta^s(\mathcal{R})$*, for every system of simple roots* Δ*.*

To every automorphism of $\mathcal{R}$ is associated by functoriality an automorphism of $ad(\mathcal{R})$. One thus has a canonical morphism

$$\mathrm{Aut}(\mathcal{R}) \longrightarrow \mathrm{Aut}(ad(\mathcal{R})).$$

Lemma 6.7.6. *The morphism* $\mathrm{Aut}_s(\mathcal{R}) \to \mathrm{Aut}(ad(\mathcal{R}))$ *is injective.*

Proof. Let u be an automorphism of M such that $(u-id)(M) \subset V(R)$ and ${}^t u(\alpha*) = \alpha*$ for $\alpha* \in R*$. For every $x \in M$, one has

$(\alpha*, u(x) - x) = ({}^t u(\alpha*) - \alpha*, x) = 0,$

so $u(x) - x = 0$, by 1.2.5.

Lemma 6.7.7. *The group* $\mathrm{Aut}_s(\mathcal{R})$ *is finite.*

Proof. Indeed, it suffices for us to prove that $\mathrm{Aut}(\mathcal{R})$ is finite if $\mathcal{R}$ is adjoint. Since M is then generated by R, every automorphism of $\mathcal{R}$ is determined by the permutation of R that it defines.

Remark 6.7.8. *One sees at once that* $\mathrm{Aut}(\mathcal{R})$ *(resp.* $E_\Delta(\mathcal{R})$*) is finite if and only if* $rgred(\mathcal{R}) - rgss(\mathcal{R}) \leqslant 1$*.*

6.8. p-morphisms of reduced root data

In this number, p is an integer > 0 fixed once and for all.

Definition 6.8.1. *Let* $\mathcal{R} = (M, M*, R, R*)$ *and* $\mathcal{R}' = (M', M'*, R', R'*)$ *be two reduced root data. One says that a group morphism*

$$f : M' \longrightarrow M$$

is a p-morphism *from* $\mathcal{R}'$ *to* $\mathcal{R}$*, if the following conditions are satisfied: there exist a bijection*

$$u : R \xrightarrow{\sim} R'$$

and a map $q : R \to p^n, n \in \mathbb{N}$ *such that:*

(i) $f(u(\alpha)) = q(\alpha)\alpha$ *for every* $\alpha \in R$*.*

(ii) ${}^t f(\alpha'*) = q(\alpha)u(\alpha)*$ *for every* $\alpha \in R$.[1284]

Corollary 6.8.2. *A 1-morphism is nothing other than a morphism.*

Corollary 6.8.3. *The transpose of a p-morphism is a p-morphism.*

Lemma 6.8.4. *If* $w \in W(\mathcal{R})$, $\alpha \in R$, *one has* $q(w(\alpha)) = q(\alpha)$. *The map* $s_\alpha \mapsto s_{u(\alpha)}$ *extends to an isomorphism* $u : W(\mathcal{R}) \to W(\mathcal{R}')$ *such that*

$$u(w(\alpha)) = u(w)(u(\alpha)).$$

Proof. It suffices to prove that for $\alpha, \beta \in R$, one has $u(s_\alpha(\beta)) = s_{u(\alpha)}u(\beta)$ and $q(s_\alpha(\beta)) = q(\beta)$. Now one has successively:

```
f(s_{u(α)} u(β)) = f(u(β)) − (u(α)*, u(β)) f(u(α))
                 = q(β) β − q(β) q(α)⁻¹ (α*, β) q(α) α
                 = q(β) (β − (α*, β) α) = q(β) s_α(β).
```

If $\gamma = u^{-1}(s_{u(\alpha)}u(\beta))$, one has therefore $q(\gamma)\gamma = f(u(\gamma)) = q(\beta)s_\alpha(\beta)$. The two roots γ and $s_\alpha(\beta)$ are therefore proportional (over $\mathbb{Q}$), so equal or opposite, but $q(\gamma)$ and $q(\beta)$ are positive. One thus has $q(\gamma) = q(\beta)$ and $\gamma = s_\alpha(\beta)$.

Definition 6.8.5. *The* $q(\alpha)$ *are called the* radical exponents *of* f.

Example 6.8.6. *Let* $\mathcal{R}$ *be a reduced root datum and* $q = p^n$ *(*$n \in \mathbb{N}$*). Then the multiplication by* $q : M \to M$, $x \mapsto qx$ *is a p-morphism whose radical exponents are all equal to* q *(and* $u = id$*); one denotes it*

$$q : \mathcal{R} \longrightarrow \mathcal{R}.$$

Proposition 6.8.7. *In the notation of 6.8.1,* u *realizes an isomorphism of the set of systems of simple roots (resp. of positive roots) of* R *onto the corresponding set for* R'.

Proof. This follows at once from 3.1.5 (resp. 3.2.1).

7. Structure

7.1. Decomposition of a root datum

Proposition 7.1.1. *Let* $\mathcal{R}$ *be a root datum,* Δ *a system of simple roots.*

(i) Let R' *and* R'' *be two closed and symmetric sets of roots forming a partition of* R. *If one denotes* $\Delta' = \Delta \cap R'$, $\Delta'' = \Delta \cap R''$, *then* $R' = R_{\Delta'}$, $R'' = R_{\Delta''}$, *and every root of* Δ' *is orthogonal to every root of* Δ''.

(ii) Let Δ' *and* Δ'' *be two subsets of* Δ *forming a partition of* Δ *and orthogonal. Then* $R' = R_{\Delta'}$ *and* $R'' = R_{\Delta''}$ *form a partition of* R.

[1284] N.D.E.: Note that these two conditions entail $q(\alpha)(u(\alpha)*, u(\beta)) = q(\beta)(\alpha*, \beta)$, for all $\alpha, \beta \in R$.

Proof. Let us first prove (i).

Lemma 7.1.2. *Under the conditions of (i), if α, β, and $\alpha + \beta$ are roots, they all three belong to R' or all three to R''.*

Proof. Suppose for example $\alpha + \beta \in R'$. Then one cannot have $\alpha, \beta \in R''$, since R'' is closed; suppose therefore $\alpha \in R'$. Then $-\alpha \in R'$ and $\beta = (\beta + \alpha) - \alpha \in R'$.

Let us now show that $R' = R_{\Delta'}$ by induction on the order of a positive root $\alpha \in R' \cap P(\Delta)$. If $ord_\Delta(\alpha) = 1$, then $\alpha \in R' \cap \Delta = \Delta'$. If $ord_\Delta(\alpha) > 1$, there exists $\beta \in \Delta$ such that $\alpha - \beta \in R$. By the lemma, one has $\beta \in \Delta'$, $\alpha - \beta \in R'$, so $\alpha - \beta \in R_{\Delta'}$ by induction, and finally $\alpha = (\alpha - \beta) + \beta \in R_{\Delta'}$.

Let us finally show that Δ' and Δ'' are orthogonal. If $\alpha \in \Delta'$ and $\beta \in \Delta''$, then $(\beta*, \alpha) \leqslant 0$. If $(\beta*, \alpha) \neq 0$, then $\beta + \alpha$ is a root, contrary to the lemma.

Let us prove (ii). If Δ' or Δ'' is empty, this is immediate. Otherwise, if $R_{\Delta'}$ and $R_{\Delta''}$ do not form a partition of R, there exists a root α of the form

$$\alpha = \sum_i m'_i \alpha'_i + \sum_j m''_j \alpha''_j, \quad m'_i \in \mathbb{Z}_+, \ m''_j \in \mathbb{Z}_+,$$

where one denotes by α'_i (resp. α''_j) elements of Δ' (resp. Δ''). Applying 3.1.2, one deduces a relation of the form (possibly inverting Δ' and Δ''):

$$\delta = \gamma + \beta, \quad \gamma \in R_{\Delta'}, \quad \beta \in \Delta'', \quad \delta \in R.$$

But since $(\beta*, \gamma) = 0$, $\gamma - \beta$ is also a root by 2.2.5, which is impossible.

Proposition 7.1.3. *Let $\mathcal{R}$ be a root datum. The following conditions are equivalent:*

(i) There exists no non-trivial partition of R into two closed and symmetric subsets.

(ii) For some (resp. every) system of simple roots Δ of R, there exists no partition of Δ into two non-empty orthogonal subsets.

(iii) The natural representation of $W(\mathcal{R})$ in $V(R)$ is irreducible.

(iv) For every pair (α, β) of roots, there exists a sequence of roots $\alpha_0, \alpha_1, \alpha_2, \cdots, \alpha_n$, with $\alpha = \alpha_0$, $\alpha_n = \beta$, such that the roots α_i and α_{i+1} ($i = 0, \cdots, n-1$) are non-orthogonal.

Proof. One has (i) ⇔ (ii) by 7.1.1. One has obviously (iv) ⇒ (ii). Conversely, if (ii) is verified for Δ, condition (iv) is verified for $\alpha, \beta \in \Delta$. Now for every root, there exists a simple root not orthogonal to it (3.1.1 for example). On the other hand (iii) ⇒ (i): in effect, under the conditions of 7.1.1, $V(R')$ is stable under $W(\mathcal{R})$. It remains to prove (i) ⇒ (iii).

So let H be a vector subspace of $V(R)$, stable under $W(\mathcal{R})$. For every $\alpha \in R$, the equation $s_\alpha(H) = H$ gives at once $\alpha \in H$, or $\alpha* \in H^\perp$ (orthogonal of H in $V(R*)$, which is in duality with $V(R)$). If one sets $R' = \alpha \in R | \alpha \in H$ and $R'' = \alpha \in R | \alpha* \in H^\perp$, one has realized a partition of R into two closed and symmetric subsets.

Definition 7.1.4. *A root datum (resp. a root system) satisfying the equivalent conditions of 7.1.3 and of semisimple rank* $\neq 0$ *is called* irreducible.

Corollary 7.1.5. *For every root datum* $\mathcal{R}$*, there exists a unique partition (up to order) of* R *into closed, symmetric, and irreducible subsets.*

Corollary 7.1.6. *Every adjoint (resp. simply connected) root datum is a product of irreducible adjoint (resp. simply connected) root data.*

Proof. It suffices to see this in the adjoint case. The assertion then follows from the fact that under the conditions of 7.1.1, one has

$$\Gamma_0(R) = \Gamma_0(R') \times \Gamma_0(R'').$$

Corollary 7.1.7. *For every root datum (resp. reduced root datum)* $\mathcal{R}$*, there exists an isogeny* $\mathcal{R} \to \mathcal{R}'$*, where* $\mathcal{R}'$ *is a product of a trivial root datum and of irreducible simply connected root data (resp. and reduced).*

7.2. Properties of irreducible root data

Definition 7.2.1. *Let* $\mathcal{R}$ *be an irreducible root datum. For every* $\alpha \in R$*, set*

$$long(\alpha) = \ell(\alpha)/\ell(\alpha_0),$$

where $\alpha_0 \in R$ *is such that* $\ell(\alpha_0)$ *is minimal; one says that* $long(\alpha)$ *is the* length *of* α*.*

Lemma 7.2.2. *Let* $\mathcal{R}$ *be an irreducible root datum. The Weyl group operates transitively on the set of roots of the same length.*

Proof. Indeed, let $\alpha, \beta \in R$. Since the representation of W on $V(R)$ is irreducible, α cannot be orthogonal to all the $w(\beta)$, $w \in W$. There exists therefore $w \in W$ with $w(\beta)$ non-orthogonal to α. Now $\ell(w(\beta)) = \ell(\beta)$, and one concludes by 2.3.2.

Lemma 7.2.3. *If* $\mathcal{R}$ *is irreducible and reduced, then* $long(R)$ *is {1}, {1, 2}, or {1, 3}.*

Proof. By virtue of the remark used above, for every $\alpha, \beta \in R$, there always exists $w \in W$ such that $w(\beta)$ is not orthogonal to β. One thus has $\ell(\alpha)/\ell(\beta) = 1, 2, 3, 1/2,$ or $1/3$ (by 2.3.1). One has therefore $long(\alpha) = 1, 2,$ or 3, but if $long(\alpha) = 2$, $long(\beta) = 3$, then $\ell(\alpha)/\ell(\beta) = 2/3$, which is impossible.

Remark 7.2.4. *Reasoning similarly, one proves the following result: if* $\mathcal{R}$ *is irreducible and not reduced with* $rgss(\mathcal{R}) > 1$*, one has* $long(R) = 1, 2, 4$*. If one sets* $long^{-1}(i) = R_i$*, then* $ind(R) = R_1 \cup R_2$*,* $R_4 = 2R_1$*, and two non-proportional roots of R_1 are orthogonal. Conversely, if* R *is an irreducible and reduced system such that* $long(R) = 1, 2$*, set* $long^{-1}(i) = R_i$ *and suppose that two non-proportional roots of R_1 are orthogonal; then* $R \cup 2R_1$ *is irreducible, not reduced, and* $ind(R \cup 2R_1) = R$*.*

Lemma 7.2.5. *If $\mathcal{R}$ is an irreducible root datum, $\mathcal{R}*$ also is, and the product $long(\alpha) long(\alpha*)$ is constant as α runs over R.*

Proof. This follows at once from 7.1.3 (iv) and 2.2.6.

Definition 7.2.6. *Let $\mathcal{R}$ be any root datum. One calls* length *of $\alpha \in R$ and denotes by $long(\alpha)$ the length of α in its irreducible component.*

Lemma 7.2.7. *There exists a unique group homomorphism $u : \Gamma_0(R) \to \Gamma_0(R*)$ such that $u(\alpha) = long(\alpha)\alpha*$ for $\alpha \in R$.*

Proof. By 3.5.5, it suffices to verify that if $\alpha, \beta, \alpha + \beta \in R$, one has

```
long(α) α* + long(β) β* = long(α + β) (α + β)*.
```

But α, β, and $\alpha + \beta$ are in the same irreducible component of R by 7.1.2, and one is reduced to 1.2.2.

Remark 7.2.8. *Let u be as in 7.2.7. For $\alpha, \beta \in R$, one has $(u(\alpha), \beta) = (u(\beta), \alpha)$. Indeed, this amounts to seeing that*

```
long(α) (α*, β) = long(β) (β*, α),
```

which is obviously verified if α and β are orthogonal. If α and β are not orthogonal, then they are in the same irreducible component of R, and one is reduced to 1.2.1, formula (9).

Remark 7.2.9. *The symmetric bilinear form `(u(x), y)` is positive nondegenerate on $\Gamma_0(R)$.*

Proof. Indeed, let R_i be the irreducible components of R. One has

$$\Gamma_0(R) = \prod_i \Gamma_0(R_i),$$

and the bilinear form `(u(x), y)` is the product of the forms

$$2^{-1}\ell(\alpha_i)^{-1}(p(x), y)$$

on the $\Gamma_0(R_i)$, where $\ell(\alpha_i)$ is the minimum of $\ell(\alpha)$ for $\alpha \in R_i$. Now these various symmetric bilinear forms are positive nondegenerate (1.2.6).

7.3. Cartan matrix

Let $\mathcal{R}$ be a root datum. If Δ is a system of simple roots, one calls *Cartan matrix* of $\mathcal{R}$ relative to Δ the square matrix on the index set Δ defined by

```
a_{α,β} = (α*, β),    for α, β ∈ Δ.
```

Let us first remark that if Δ' is another system of simple roots and w an element of $W(\mathcal{R})$ such that $w(\Delta) = \Delta'$, one has

```
(w(α)*, w(β)) = (α*, β),
```

so the Cartan matrix of $\mathcal{R}$ relative to Δ' is obtained from that relative to Δ by the isomorphism $\Delta \to \Delta'$ on the index set defined by w. It follows that, up to isomorphism on the index set, the Cartan matrix depends only on $\mathcal{R}$.

Proposition 7.3.1. *The Cartan matrix has the following properties:*

(i) $a_{\alpha,\alpha} = 2$, $a_{\alpha,\beta} \leqslant 0$ *for* $\alpha \neq \beta$.

(ii) $a_{\alpha,\beta} = 0$ *entails* $a_{\beta,\alpha} = 0$.

(iii) There exist strictly positive integers m_α *(=* $long(\alpha)$*) such that the matrix*

$$(m_\alpha a_{\alpha,\beta})$$

is symmetric, positive, and nondegenerate.

(iv) The diagonal minors of the matrix $(a_{\alpha,\beta})_{\alpha,\beta\in\Delta}$*, i.e. the determinants*

```
det(a_{α,β})_{α, β ∈ Δ'}     for Δ' ⊂ Δ,
```

are strictly positive.

(v) One has $s_\alpha(\beta) = \beta - a_{\alpha,\beta}\alpha$ *and* $s_\alpha(\beta*) = \beta * - a_{\beta,\alpha}\alpha*$.

Proof. Indeed, (v) is a definition, (i) follows from 3.2.11, (ii) from 2.2.2, (iii) from 7.2.9, (iv) follows at once from (iii) by the relation

```
det(m_α a_{α,β})_{α, β ∈ Δ'} = (∏_{α ∈ Δ'} m_α) det(a_{α,β})_{α, β ∈ Δ'}.
```

Proposition 7.3.2. *Let* $\mathcal{R}$ *and* $\mathcal{R}'$ *be two simply connected (resp. adjoint) reduced root data,* Δ *(resp.* Δ'*) a system of simple roots of* $\mathcal{R}$ *(resp.* $\mathcal{R}'$*), and* $u : \Delta \to \Delta'$ *an isomorphism such that if one denotes by* $(a_{\alpha,\beta})$ *and* $(a'_{\alpha',\beta'})$ *the Cartan matrices of* $\mathcal{R}$ *and* $\mathcal{R}'$ *relative to* Δ *and* Δ'*, one has:*

$$a'_{u(\alpha),u(\beta)} = a_{\alpha,\beta}.$$

Then there exists a unique isomorphism of $\mathcal{R}$ *onto* $\mathcal{R}'$ *that induces* u *on* Δ*.*

Proof. It evidently suffices to make the proof in the adjoint case. Then $M = \Gamma_0(R)$ and $M' = \Gamma_0(R')$ are the free abelian groups generated by Δ and Δ'. There exists therefore a unique isomorphism of groups of M onto M' that induces u on Δ. Denote it also u. Let us show that $u(R) \subset R'$. Every root α of R is written $s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1})$ with $\alpha_i \in \Delta$. One has obviously

$$u(\alpha) = s_{u(\alpha_1)} \cdots s_{u(\alpha_n)}(u(\alpha_{n+1})),$$

by virtue of the hypothesis on u and the relations (v) of 7.3.1.

It remains to prove that ${}^t u(R'*) \subset R*$, which follows from the fact that the elements of $M*$ (resp. $M'*$) are determined by the duality with R or Δ (resp. R' or Δ'), by 1.2.5.

Corollary 7.3.3. *A reduced simply connected or adjoint root datum is determined up to isomorphism by its Cartan matrix.*

Corollary 7.3.4. *Let $\mathcal{R}$ be a reduced simply connected (resp. adjoint) root datum, and Δ a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the set Δ that leave the Cartan matrix invariant.*

Remark 7.3.5. *The question of the existence of a root datum corresponding to a given Cartan matrix satisfying (i), (ii), and (iv) (for example) is not easily resolved directly, without using the classification.*

7.4. Dynkin diagrams

Definition 7.4.1. *One calls a* Dynkin diagram structure *(the word "schéma" having been banished for obvious reasons) on a finite set Δ the data of a set of pairs of distinct elements of Δ, called* linked pairs, *and of a map from Δ to the set {1, 2, 3}. The notion of isomorphism of such structures is obvious.*

Definition 7.4.2. *Let $\mathcal{R}$ be a root datum and Δ a system of simple roots. One calls* Dynkin diagram *of $\mathcal{R}$ relative to Δ the set Δ, two simple roots being linked if and only if they are non-orthogonal, with each root being assigned its length.*

Proposition 7.4.3. *Dynkin diagram and Cartan matrix determine each other biuniquely.*

Proof. Indeed, the equivalence

$$\alpha \text{ and } \beta \text{ are not linked} \iff a_{\alpha,\beta} = 0,$$

and the relation

$$\mathrm{long}(\alpha)\, a_{\alpha,\beta} = \mathrm{long}(\beta)\, a_{\beta,\alpha},$$

(with $\inf long(\alpha) = 1$ in each connected component of the diagram) determine the $a_{\alpha,\beta}$ as a function of the linkages and lengths, and conversely (the details of the verification are left to the reader).

Corollary 7.4.4. *A reduced simply connected or adjoint root datum is determined by its Dynkin diagram.*

Corollary 7.4.5. *Let $\mathcal{R}$ be a reduced simply connected or adjoint root datum and Δ a system of simple roots. The group $E_\Delta(\mathcal{R})$ is identified with the group of automorphisms of the Dynkin diagram of $\mathcal{R}$ relative to Δ, that is to say with the group of permutations of Δ preserving lengths and linkages.*

Remark 7.4.6. *One classifies, with the usual method,*[1285][1286] *the various connected Dynkin diagrams, and one shows that each effectively corresponds to an*

[1285] Cf. Bourbaki, *Groupes et algèbres de Lie*, Chap. VI N° 4.2 or *Séminaire Sophus Lie*.
[1286] N.D.E.: For a slightly different proof, see also [De80].

irreducible reduced simply connected root datum. One finds the well-known types:

$$1 - 1 - 1 - \dots - 1 - 1 - 1 \qquad A_n, \quad n \geq 1.$$

$$2 - 2 - 2 - \dots - 2 - 2 \Rightarrow 1 \qquad B_n, \quad n \geq 2.$$

$$1 - 1 - 1 - \dots - 1 - 1 \Leftarrow 2 \qquad C_n, \quad n \geq 3.$$

$$\begin{array}{ll} & \quad 1 \\ & / \\ 1 - 1 - 1 - \dots - 1 - 1 & \\ & \backslash \\ & \quad 1 \end{array} \qquad D_n, \quad n \geq 4.$$

$$\begin{array}{c} \phantom{1 - 1 - {}}1\phantom{{} - 1 - 1 - 1} \\ \phantom{1 - 1 - {}}|\phantom{{} - 1 - 1 - 1} \\ 1 - 1 - 1 - 1 - 1 \end{array} \qquad E_n, \quad n = 6, 7, 8.$$

$$1 - 1 \Rightarrow 2 - 2 \qquad F_4.$$

$$1 \Rrightarrow 3 \qquad G_2.$$

By 7.4.5, one finds at once the corresponding group $E_\Delta(\mathcal{R})$; one has:

$E_\Delta(\mathcal{R}) = \{e\}$ for $A_1, B_n, C_n, E_7, E_8, F_4, G_2$.
$E_\Delta(\mathcal{R}) = \mathbb{Z}/2\mathbb{Z}$ for A_n $(n \geq 2)$, D_n $(n \geq 5)$, E_6.
$E_\Delta(\mathcal{R}) = S_3$ for D_4.

7.5. Complements on p-morphisms

Let $f : \mathcal{R} \to \mathcal{R}'$ be a p-morphism (cf. 6.8). It is clear from the definitions that the bijection $u : R \xrightarrow{\sim} R'$ associated with f makes systems of simple roots, systems of positive roots, irreducible components (etc.) of R and R' correspond. Suppose therefore, for simplicity, R and R' irreducible.

Lemma 7.5.1. *If R and R' are irreducible, there exists $k \in \mathbb{Q}$ such that for every $\alpha \in R$*

$$k \, long(u(\alpha)) = q(\alpha)^2 \, long(\alpha).$$

Proof. Indeed, one has $long(\alpha)(\alpha*, \beta) = long(\beta)(\beta*, \alpha)$ and, similarly,

$$long(u(\alpha)) \, (u(\alpha)^*, u(\beta)) = long(u(\beta)) \, (u(\beta)^*, u(\alpha)).$$

One deduces at once that for non-orthogonal α and β, one has

$$q(\alpha)^2 \, long(\alpha) / long(u(\alpha)) = q(\beta)^2 \, long(\beta) / long(u(\beta))$$

and one concludes then by 7.1.3 (iv).

Remark 7.5.2. *It follows from 7.2.2 and 6.8.4 that $q(\alpha)$ depends only on $long(\alpha)$. One then sees easily that if $q(\alpha)$ is not constant, then $q(\alpha)long(\alpha)$ is constant, which shows that then $p = 2$ or 3. A glance at the diagrams of the preceding number shows that there are four possible cases (we denote by the same letter a Dynkin diagram and the corresponding simply connected reduced root datum):*

$p = 2$, $\quad B_n \xrightarrow{f_1} C_n$, $\quad C_n \xrightarrow{f_2} B_n$ $\quad$ (with $C_2 = B_2$).
$p = 2$, $\quad F_4 \xrightarrow{g} F_4$.
$p = 3$, $\quad G_2 \xrightarrow{h} G_2$.

The reader will note that $f_1 \circ f_2$, $f_2 \circ f_1$, $g \circ g$, and $h \circ h$ are p-morphisms of the form described in 6.8.6.

7.5.3. One sees at once from the preceding description that if $\mathcal{R}$ and $\mathcal{R}'$ are two reduced root data of semisimple rank $\leqslant 2$ and if one has a p-morphism from $\mathcal{R}'$ to $\mathcal{R}$, then $\mathcal{R}$ and $\mathcal{R}'$ are of the same type. More precisely, one has the following table.

Notations. Let $f : \mathcal{R}' \to \mathcal{R}$ be a p-morphism. We denote by q (resp. q_1) any positive power of p. We use for the rank-2 systems the notations of number 4 (we denote by α, β the simple roots, with $\ell(\alpha) \leqslant \ell(\beta)$).

Type	p	values of f	values of ${}^t f$
Trivial	any	−	−
A_1	any	$f(\alpha') = q\alpha$	${}^t f(\alpha*) = q\alpha'*$
$A_1 \times A_1$	any	$f(\alpha') = q\alpha$	${}^t f(\alpha*) = q\alpha'*$
		$f(\beta') = q_1\beta$	${}^t f(\beta*) = q_1\beta'*$
A_2, B_2, G_2	any	$f(\alpha') = q\alpha$	${}^t f(\alpha*) = q\alpha'*$
		$f(\beta') = q\beta$	${}^t f(\beta*) = q\beta'*$
B_2	$p = 2$	$f(\alpha') = q\beta$	${}^t f(\alpha*) = 2q\beta'*$
		$f(\beta') = 2q\alpha$	${}^t f(\beta*) = q\alpha'*$
G_2	$p = 3$	$f(\alpha') = q\beta$	${}^t f(\alpha*) = 3q\beta'*$
		$f(\beta') = 3q\alpha$	${}^t f(\beta*) = q\alpha'*$

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Footnotes

Exposé XXII. Reductive groups: splittings, subgroups, quotient groups

by M. Demazure

[1287] This Exposé consists of two parts. The first (1 through 5.5) gathers the technical results needed for the proof of the uniqueness and existence theorems. The second (5.6 to the end) will not be used in that proof; the end of section 5 will be used in particular in Exposé XXVI on parabolic subgroups; section 6 establishes, in the scheme-theoretic setting, the classical results on the derived group of a reductive group.

1. Roots and coroots. Split groups and root data

Theorem 1.1. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, α a root of G relative to T.*

(i) There exists a unique morphism of groups with operator group T

$$\exp\alpha : W(g\alpha) \to G$$

inducing on the Lie algebras the canonical morphism $g\alpha \to g$. This morphism is a closed immersion. The corresponding morphism

$T \cdot_\alpha W(g\alpha) \to G$

is also a closed immersion.

If $p\alpha : Ga,S \to G$ is a monomorphism normalized by T with multiplier α, there exists a unique $X\alpha \in \Gamma(S, g\alpha)^\times$[1288] such that $p\alpha(x) = \exp\alpha(xX\alpha)$; one has $Lie(p\alpha)(1) = X\alpha$, and the two preceding formulas set up a bijective correspondence between $\Gamma(S, g\alpha)^\times$ and the set of monomorphisms $Ga,S \to G$ normalized by T with multiplier α.

(ii) There exists a unique duality (denoted $(X, Y) \mapsto XY$)

$g\alpha \otimes_{O_S} g_{-\alpha} \to O_S$,

and a unique morphism of groups

$$\alpha^* : Gm,S \to T,$$

such that formula (F) of Exp. XX 2.1 holds. One has

$\alpha \circ \alpha^* = 2, \qquad (-\alpha)^* = -\alpha^*,$

and α^* is given by the formula of Exp. XX 2.7.

Indeed, a morphism normalized by T with multiplier α factors necessarily through the closed subgroup $Z\alpha = Centr_G(T\alpha)$ of G (cf. Exp. XIX 3.9). Now $(Z\alpha, T, \alpha)$ is an S-elementary system (Exp. XX 1.4), and one is reduced to the results of Exposé XX (1.5, 2.1, and 5.9).

[1287] Version of 13/10/2024.

[1288] The set $\Gamma(S, g\alpha)^\times$ is defined in XIX 4.4.1.

Remark 1.2. *Part (i) of Theorem 1.1 remains valid if one only assumes that α is a character of T, non-trivial on each fiber. Indeed, one then has a decomposition $S = S' \coprod S''$ such that $\alpha|S'$ is a root of $G_{S'}$ relative to $T_{S'}$ and $g\alpha|S'' = 0$. If $S = S'$, one is reduced to 1.1; if $S = S''$ the result is trivial; the general case follows immediately.*

Notations 1.3. *As in Exposé XX, we denote by $U\alpha$ the image of $W(g\alpha)$; it is a closed subgroup of G, equipped canonically with a vector-space structure. We shall call it the* vector group associated with the root α. *We say that $\alpha*$ is the* coroot associated with α. *Sections $X\alpha \in \Gamma(S, g\alpha)$ and $X_{-\alpha} \in \Gamma(S, g_{-\alpha})$ are said to be* paired *if $X\alpha \cdot X_{-\alpha} = 1$. Then $X\alpha \in \Gamma(S, g\alpha)\times$ and likewise for $X_{-\alpha}$. The corresponding morphisms $p\alpha$ and $p_{-\alpha}$ are contragredient to one another in the sense of XX, 1.5,*[1289] *and one has*

$$p\alpha(x)\, p_{-\alpha}(y) = p_{-\alpha}(y / (1 + xy)) \cdot \alpha^*(1 + xy) \cdot p\alpha(x / (1 + xy)).$$

Proposition 1.4. *Under the conditions of 1.1, let $w \in Norm_G(T)(S)$. Then $\beta = \alpha \circ int(w)^{-1} : T \to Gm,S$ is a root of G relative to T, $\beta* = int(w) \circ \alpha*$ is the corresponding coroot, and the following diagram is commutative:*

$$\begin{tikzcd}
W(g\alpha) \arrow[r, "\exp\alpha"] \arrow[d, "Ad(w)"'] & G \arrow[d, "int(w)"] \\
W(g\beta) \arrow[r, "\exp\beta"] & G.
\end{tikzcd}$$

Trivial: transport of structure.

Definitions 1.5. *(a) Under the conditions of 1.1, we denote by $s\alpha$ the automorphism of T defined by*

$$s\alpha(t) = t \cdot \alpha * (\alpha(t))^{-1}.$$

We denote by $(,)$ the canonical pairing:

$$\mathrm{Hom}_{S\text{-gr.}}(Gm,S,\ T) \times \mathrm{Hom}_{S\text{-gr.}}(T,\ Gm,S) \to \mathrm{Hom}_{S\text{-gr.}}(Gm,S,\ Gm,S) = \mathbb{Z}_S.$$

Then $s\alpha$ operates on $\mathrm{Hom}_{S-gr.}(T, Gm, S)$, resp. $\mathrm{Hom}_{S-gr.}(Gm, S, T)$, by the following formulas, where χ (resp. u) denotes an arbitrary section of $\mathrm{Hom}_{S-gr.}(T, Gm, S)$ (resp. of $\mathrm{Hom}_{S-gr.}(Gm, S, T)$):

$$s\alpha(\chi) = \chi - (\alpha^*, \chi)\, \alpha,$$
$$s\alpha(u) = u - (u, \alpha)\, \alpha^*.$$

One has $s\alpha \circ s\alpha = id$ and $s_{-\alpha} = s\alpha$.

(b) If $X \in \Gamma(S, g\alpha)\times$, then the inner automorphism $w\alpha(X)$ of T defined by

$$w\alpha(X) = \exp\alpha(X)\, \exp_{-\alpha}(-X^{-1})\, \exp\alpha(X)$$

[1289] That is, if one denotes by $\langle , \rangle$ the duality between $g\alpha$ and $g_{-\alpha}$ and if one identifies $g\alpha$ with $U\alpha$ via $p\alpha(x) = \exp(xX\alpha)$, and similarly for $-\alpha$, then $\langle p\alpha(x), p_{-\alpha}(y)\rangle = xy$.

(cf. Exp. XX 3.1) coincides with sα (loc. cit.). One then concludes from 1.4:

Corollary 1.6. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, α and β two roots of G relative to T. Then*

$$s_\alpha(\beta) = \beta - (\alpha^*, \beta)\,\alpha$$

is a root of G relative to T, and the corresponding coroot is

$$s_\alpha(\beta)^* = s_\alpha(\beta^*) = \beta^* - (\beta^*, \alpha)\,\alpha^*.$$

Corollary 1.7. *Under the preceding conditions, $\alpha* = \beta*$ implies $\alpha = \beta$.*

Indeed, if $\alpha* = \beta*$, one has (cf. XXI.1.4)

$$s_\beta(\alpha) = \alpha - 2\beta, \qquad s_\alpha(\beta) = \beta - 2\alpha,$$

and from this one deduces immediately

$$(s_\beta\, s_\alpha)^n(\alpha) = \alpha + 2n(\beta - \alpha).$$

If $\beta \neq \alpha$, there exists $s \in S$ such that $\alpha_s \neq \beta_s$. But then the preceding formula shows that there exist infinitely many distinct roots of G_s relative to T_s, which is impossible.

Definitions 1.8.0.[1290] *If $u : Gm,S \to T$ is a morphism of groups, we shall say that u is a* coroot *of G relative to T if there exists a root α of G relative to T such that $\alpha* = u$. Consider the functor $R*$ of coroots of G relative to T defined as follows:*

$$R^*(S') = \text{the set of coroots of } G_{S'} \text{ relative to } T_{S'}.$$

If R is the functor of roots of G relative to T (Exp. XIX 3.8), one has a canonical morphism $R \to R*$. By virtue of 1.7 and Exp. XIX 3.8, one has:

Corollary 1.8. *The canonical morphism $R \to R*$ is an isomorphism. In particular, $R*$ is representable by a finite twisted constant S-scheme, which is an open and closed subscheme of* $\mathrm{Hom}_{S-gr}(Gm, S, T)$.

This leads us to the following definition:

Definition 1.9. *Let S be a scheme, T an S-torus. We call* twisted root datum *in T the data:*

(i) a finite subscheme R of $\mathrm{Hom}_{S-gr.}(T, Gm, S)$,

(ii) a finite subscheme $R*$ of $\mathrm{Hom}_{S-gr.}(Gm, S, T)$,

(iii) an isomorphism $R \xrightarrow{\sim} R*$ denoted $\alpha \mapsto \alpha*$,

satisfying the following conditions:

(DR 1) For every $S' \to S$ and every $\alpha \in R(S')$, one has $\alpha \circ \alpha^* = 2$.

(DR 2) For every $S' \to S$ and every $\alpha, \beta \in R(S')$, one has

$$\alpha - (\beta^*, \alpha)\,\beta \in R(S'), \qquad \alpha^* - (\alpha^*, \beta)\,\beta^* \in R^*(S').$$

[1290] We have added the numbering 1.8.0 to highlight these definitions.

Moreover, if for $\alpha \in R(S')$ ($S' \neq \emptyset$) one has $2\alpha \notin R(S')$, the root datum is said to be reduced.

Proposition 1.10. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, R (resp. $R*$) the scheme of roots (resp. of coroots) of G relative to T. Then $(R, R*)$ is a reduced twisted root datum in T.*

The only thing that remains to be checked is that this twisted root datum is reduced. This was done in Exp. XIX 3.10.

1.11.

Let $T = D_S(M)$ be a trivialized torus. If we denote by $M*$ the abelian group dual to M, we have canonical isomorphisms (cf. Exp. VIII 1.5):

$$\mathrm{Hom}_{S-gr.}(T, Gm, S) \xrightarrow{\sim} M_S, \mathrm{Hom}_{S-gr.}(Gm, S, T) \xrightarrow{\sim} M*_S,$$

hence isomorphisms of groups:

```
Hom_{S-gr.}(T, Gm,S) → Hom_{loc.const.}(S, M),
Hom_{S-gr.}(Gm,S, T) → Hom_{loc.const.}(S, M*).
```

A character of T (resp. a morphism of groups $Gm, S \to T$) will be called *constant* (relative to the given trivialization) if the preceding isomorphism transforms it into a constant map from S to M (resp. $M*$).

1.12.

With the same notations, let $(M, M*, R, R*)$ be a root datum (Exp. XXI). Then $(R_S, R*_S)$ is a twisted root datum in T. Conversely, if $(R, R*)$ is a twisted root datum in a torus T, a *splitting* of this root datum is the data of an ordinary root datum $(M, M*, R, R*)$ together with an isomorphism $T \simeq D_S(M)$ that transforms $(R, R*)$ into $(R_S, R*_S)$.

Definition 1.13. *Let S be a scheme, G an S-reductive group, T a maximal torus of G. We call* splitting *of G relative to T the data*

(i) of an abelian group M and an isomorphism $T \simeq D_S(M)$,

(ii) of a system of roots R of G relative to T (Exp. XIX 3.6),

satisfying the following two conditions:

(D_1) S is non-empty and the roots $\alpha \in R$ (resp. the corresponding coroots) are identified with constant functions from S to M (resp. $M*$).

(D_2) The $g\alpha$ ($\alpha \in R$) are free `0_S`*-modules.*

We say that G is splittable *relative to T if there exists a splitting of G relative to T. By a* splitting *of G we mean the data of a maximal torus T of G and a splitting of G relative to T. We say that G is* splittable *if there exists a splitting of G. By an S*-split group *we mean an* S*-reductive group equipped with a splitting; we denote it by a symbol of the form* (G, T, M, R)*, or simply* G *when there is no risk of confusion.*

Condition (D_1) entails that R (resp. $R*$) is canonically identified with a subset of M (resp. $M*$).

Proposition 1.14. *Let* S *be a scheme (non-empty),* (G, T, M, R) *an* S*-split group. Then*

`R(G, T, M, R) = (M, M*, R, R*)`

is a reduced root datum (Exp. XXI 1.1 and 2.1.3); it is a splitting of the twisted root datum of 1.10.

This is a trivial consequence of 1.10 and Exp. XIX 3.7.

We shall sometimes write, for simplicity, $R(G, T, M, R) = R(G)$. We shall systematically use the notations V, $V(R)$, W, ... of Exp. XXI.

Remark 1.15. *(a) If* S *is connected non-empty (resp. if* $\mathrm{Pic}(S) = 0$*), the condition (*D_1*) (resp. (*D_2*)) is automatically satisfied.*

(b) If (G, T, M, R) *is an* S*-split group, then for every* $S' \to S$*,* $S' \neq \emptyset$*,* $(G_{S'}, T_{S'}, M, R)$ *is an* S'*-split group, and* $R(G, T, M, R) = R(G_{S'}, T_{S'}, M, R)$*.*

1.16.

Let $T = D_S(M)$ be a trivialized torus. The Lie algebra t of T is canonically identified (Exp. II 5.1.1) with

$$t \simeq M * \bigotimes O_S.$$

For every morphism of groups $u : T \to Gm, S$, $Lie(u)$ is a linear form

`Lie(u) : t → O_S = Lie(Gm,S/S).`

In particular, if u is defined by an element $\alpha \in M$, then $Lie(u)$ is the linear form α on $M * \bigotimes O_S$ defined by α:

`α(m ⊗ x) = (m, α) x.`

Symmetrically, for every morphism of groups $h : Gm, S \to T$, $Lie(h)$ is an `O_S`-morphism $O_S = Lie(Gm, S/S) \to t$, canonically defined by the section

`H = Lie(h)(1) ∈ Γ(S, t).`

In particular, if h is defined by an element $m \in M*$, one has

$$H = Lie(h)(1) = m \bigotimes 1.$$

Comparing the two definitions, one finds in particular

`α(H) = (h, α) · 1 ∈ Γ(S, O_S).`

1.17.

These definitions apply in particular to the case where T is the maximal torus of a split group. Each root $\alpha \in R$ defines an *infinitesimal root* $\alpha \in \mathrm{Hom}_{O_S}(t, O_S)$ with

$\alpha(m \otimes x) = (m, \alpha)\, x.$

Each coroot $\alpha* \in R*$ defines an *infinitesimal coroot*

$H\alpha \in \Gamma(S, t), \quad H\alpha = \alpha^* \otimes 1.$

For $\alpha, \beta \in R$, one has the relation

$$\alpha(H\beta) = (\beta*, \alpha) \cdot 1,$$

and in particular

$$\alpha(H\alpha) = 2.$$

In particular, if 2 is invertible on S, then α and $H\alpha$ are non-zero on each fiber.

2. Existence of a splitting. Type of a reductive group

Proposition 2.1. *Let S be a scheme, G an S-reductive group, T a maximal torus of G. Suppose T is split. Then G is locally splittable relative to T: for every $s_0 \in S$, there exists an open neighborhood U of s_0 such that the U-group G_U is splittable relative to T_U.*

Indeed, write $T \simeq D_S(M)$ and

$g = \bigoplus_{m \in M} g^m.$

Let $R = m \in M | m \neq 0, g^m(s_0) \neq 0$. Shrinking S if necessary (replacing it by an open neighborhood of s_0), we may suppose that the $g\alpha$, $\alpha \in R$, are free, and that the g^m, $m \neq 0$, $m \notin R$, are zero. We then have

$g = t \oplus \bigoplus_{\alpha \in R} g\alpha,$

with the $g\alpha$ free of rank 1. It follows that R is a system of roots of G relative to T (Exp. XIX 3.6). The coroots $\alpha*$ corresponding to the $\alpha \in R$ are then identified with locally constant functions on S with values in $M*$. Shrinking S further, we may take them constant, and we are done.

Note that the proof gives immediately:

Proposition 2.2. *Let S be a non-empty connected scheme with* $\mathrm{Pic}(S) = 0$*, for example* $\mathrm{Spec}(\mathbb{Z})$ *or a local scheme (in particular the spectrum of a field). If G is an S-reductive group possessing a split maximal torus T, then G is splittable relative to T.*

We deduce immediately from 2.1 and the fact that a reductive group locally possesses maximal tori for the étale topology (Exp. XIX 2.5):

Corollary 2.3. *Let S be a scheme, G an S-reductive group (resp. and T a maximal torus of G). Then G is locally splittable (resp. locally splittable relative to T) for the étale topology on S.*

Corollary 2.4. *Let k be a field, G a k-reductive group. There exists a finite separable extension K/k such that G_K is splittable.*

Remark 2.5. *Using 2.1 and the remark Exp. XIX 2.9, one immediately proves the following result: let $G = (G, T, M, R)$ be an S-split group; there exists a cover of S by open sets U_i such that each split group G_{U_i} arises by base change from a split group over a noetherian ring (in fact, a finitely generated $\mathbb{Z}$-algebra). We will furthermore prove that every split group over S already arises from a $\mathbb{Z}$-split group (Exp. XXV).*

2.6.

Let k be an algebraically closed field and G a k-reductive group. One knows (e.g. by 2.4) that there exist splittings of G. Let (G, T, M, R) and (G, T', M', R') be two splittings of G; the root data $R(G, T, M, R)$ and $R(G, T', M', R')$ are then isomorphic.

Indeed, one sees first that one can reduce to the case where $T = T'$ (because there exists $g \in G(k)$ such that $T' = int(g)T$, and one verifies easily that if one transports a splitting by an automorphism of G, one obtains a root datum isomorphic to the initial datum); but since $S = \mathrm{Spec}(k)$ is connected, the isomorphism $D_k(M) \xrightarrow{\sim} T \xrightarrow{\sim} D_k(M')$ arises from a unique isomorphism $M \simeq M'$; for the same reason, there is at most one system of roots of G relative to T.

Definition 2.6.1.[1291] *If G is a k-reductive group (k an algebraically closed field), we call* type of G *the isomorphism class of the root datum defined by an arbitrary splitting of G; if G is a torus, of type M in the sense of Exp. IX 1.4, then the type of G as reductive group is given by the trivial root datum $(M, M*, \emptyset, \emptyset)$.*

By 1.15 (b),[1292] the type is invariant under (algebraically closed) extension of the base field.

Definition 2.7. *If G is an S-reductive group and $s \in S$, we call* type of G at s *the type of the s-reductive group G_s.*

For every $S' \to S$ and every $s' \in S'$ projecting to $s \in S$, the type of $G_{S'}$ at s' is equal to the type of G at s.

If G is splittable, and if (G, T, M, R) is a splitting of G, then the type of G at s is the isomorphism class of $R(G, T, M, R)$ by 1.15 (b).[1293] It then follows immediately from 2.3:

[1291] We have added the number 2.6.1 for later references.
[1292] We have corrected the original, which referred to 1.17.
[1293] We have corrected the original, which referred to 1.17.

Proposition 2.8. *Let G be an S-reductive group ($S \neq \emptyset$). The function*

```
s ↦ type of G at s
```

is locally constant on S. In particular, there is a partition of S into non-empty open subschemes such that on each of them G is of constant type. More precisely, let E be the set of types of the fibers of G; for every $t \in E$, let S_t be the set of points $s \in S$ where G is of type t; then $(S_t)_{t\in E}$ is a partition of S and each S_t is open and closed (and non-empty).

3. The Weyl group

3.1.

Let S be a scheme, G an S-reductive group, T a maximal torus of G. Then

$$W_G(T) = Norm_G(T)/T$$

is a finite étale S-group (Exp. XIX 2.5). The morphism $n \mapsto int(n)$ induces, by passage to the quotient, a canonical monomorphism (which is in fact an open immersion):

$$W_G(T) \to \mathrm{Aut}_{S-gr.}(T).$$

3.2.

Suppose now that G is splittable relative to T. Choose a splitting, say (G, T, M, R). We then have a canonical isomorphism (Exp. VIII 1.5)

$$\mathrm{Aut}_{S-gr.}(T) \simeq (\mathrm{Aut}_{gr.}(M))_S.$$

In particular, if W is the Weyl group of the root datum $R(G)$ (Exp. XXI 1.1.8), we have a monomorphism

$$W_S \to \mathrm{Aut}_{S-gr.}(T).$$

3.3.

For each root $\alpha \in R$, the symmetry $s\alpha \in W$ operates on M by

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sα(x) = x - (α*, x) α,
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hence on T (via the preceding morphism) by

$$s\alpha(t) = t \cdot \alpha * (\alpha(t))^{-1}.$$

On the other hand, since $g\alpha$ is assumed free, there exists $X \in \Gamma(S, g\alpha)\times$. Consider $w\alpha(X) \in Norm_G(T)(S)$ (Exp. XX 3.1). One has (loc. cit.)

$$int(w\alpha(X))(t) = s\alpha(t).$$

Since W is generated by the $s\alpha$, $\alpha \in R$, it follows from the preceding remarks that if we regard W and $Norm_G(T)(S)/T(S)$ as groups of automorphisms of T, we have

$$W \subset Norm_G(T)(S)/T(S) \subset W_G(T)(S).$$

By definition of the constant group W_S associated with W (cf. I 1.8), we thus have a commutative diagram

$$\begin{tikzcd} W_S \arrow[rr] \arrow[dr] & & W_G(T) \arrow[dl] \\ & \mathrm{Aut}_{S\text{-gr.}}(T). & \end{tikzcd}$$

Proposition 3.4. *Let S be a scheme, (G, T, M, R) an S-split group, W the Weyl group of the root datum $R(G)$. Then the canonical monomorphism*

$$W_S \to W_G(T) = Norm_G(T)/T$$

is an isomorphism.

These are both étale groups over S; it therefore suffices to check that for every $s \in S$, $W_S(s) \to W_G(T)(s)$ is an isomorphism.[1294] The latter assertion follows, for example, from *Bible*, § 11.3, th. 2.

Remark 3.5. *Using 2.3, the preceding proposition gives a new proof of the fact that the Weyl group of a maximal torus of an S-reductive group G is finite over S (Exp. XIX 2.5 (ii)).*[1295]

3.6.

Under the conditions of 3.1, for every $w \in W_G(T)(S)$, we denote by N_w[1296] the fiber product of the following diagram:

$$\begin{tikzcd} N_w \arrow[r] \arrow[d] & Norm_G(T) \arrow[d, "w"] \\ S \arrow[r] & W_G(T). \end{tikzcd}$$

[1294]Indeed, since W_S and $W_G(T) = Norm_G(T)/T$ are étale over S, the morphism $f : W_S \to W_G(T) = Norm_G(T)/T$ is étale (EGA IV_4, 17.3.4); if furthermore each f_s is an isomorphism, then, by loc. cit., 17.9.1, f is a surjective open immersion, hence an isomorphism.

[1295]Indeed, let T be a maximal torus of G. The fact that $W_G(T)$ is finite over S is local for the (fpqc) topology (EGA IV_2, 2.7.1), so a fortiori for the étale topology. By 2.3, one may therefore suppose that G is split relative to T, in which case the assertion follows from 3.4.

[1296]We have replaced Q_w by N_w, just as in XX 3.0 we had replaced Q by $N\times$.

This is an open and closed subscheme of $Norm_G(T)$, which is a principal homogeneous bundle under T on the left (resp. on the right) by the law $(t,q) \mapsto tq$ (resp. $(q,t) \mapsto qt$). If $n \in N_w(S)$, one has

$$N_{ww'} = n \cdot N_{w'}, \qquad N_{w'w} = N_{w'} \cdot n.$$

3.7.

In particular, if α is a root of G relative to T, $N_{s\alpha}$ is none other than what was denoted $N\times$ in Exp. XX 3.0. If $g\alpha$ is free on S, one then has $N_{s\alpha}(S) \neq \emptyset$.

By 3.4 and condition (D_2) of the splitting, we deduce:

Corollary 3.8. *Under the conditions of 3.4, the morphism*

$$Norm_G(T)(S) \to W_G(T)(S) = Hom_{loc.cons.}(S, W)$$

is surjective. In particular, for every $w \in W$, there exists $n_w \in Norm_G(T)(S)$ such that $int(n_w)|T = w$.

4. Homomorphisms of split groups

4.1. The "big cell"

4.1.1. Let (G,T,M,R) be a split S-reductive group. Choose a system of positive roots (Exp. XXI 3.2.1) $R+$ of the root datum $R(G)$. Set $R- = -R+$.

Choose a total ordering on $R+$ (resp. $R-$) and consider the morphism induced by the product in G

$$u : \prod_{\alpha \in R-} U_\alpha \times_S T \times_S \prod_{\alpha \in R+} U_\alpha \to G.$$

This is an open immersion. Indeed, since both sides are flat and of finite presentation over S, it suffices to verify this on each geometric fiber (SGA 1, I 5.7 and VIII 5.4); one is thus reduced to the case where S is the spectrum of an algebraically closed field; but, by *Bible*, § 13.4, cor. 2 to th. 3, u is radicial and dominant; since the tangent map of u at the origin is an isomorphism (definition of a system of roots), u is birational; but G being normal, one may apply Zariski's "Main Theorem" (EGA III_1, 4.4.9) and u is an open immersion.

Let us show that the image Ω of this open immersion is independent of the ordering chosen on $R+$ (resp. $R-$). Since this is a question of comparing open subsets of G, one is reduced to proving that they have the same geometric points, so one may again assume that S is the spectrum of an algebraically closed field. Then the assertion is none other than *Bible*, § 13, prop. 1 (c) and th. 1 (a).

We have thus proved:

Proposition 4.1.2. *Let (G,T,M,R) be an S-split group. Let $R+$ be a system of positive roots of R. There exists an open subset Ω_{R+} of G such that for every total ordering on $R+$ (resp. $R-$), the morphism induced by the product in G*

$$\prod_{\alpha \in R-} U\alpha \times_S T \times_S \prod_{\alpha \in R+} U\alpha \to G$$

is an open immersion with image Ω_{R+}.

Remark 4.1.3. *One can translate 4.1.2 as follows: choose, for every* $\alpha \in R$, *an isomorphism of vector groups* $p\alpha : Ga,S \xrightarrow{\sim} U\alpha$ *(cf. 1.19); then the morphism (set* $N = Card(R+) = Card(R-)$*)*

$$Ga,S^N \times_S T \times_S Ga,S^N \to G$$

defined set-theoretically by

$$((x\alpha)_{\alpha \in R-}, t, (x\alpha)_{\alpha \in R+}) \mapsto \prod_{\alpha \in R-} p\alpha(x\alpha) \cdot t \cdot \prod_{\alpha \in R+} p\alpha(x\alpha)$$

is an open immersion whose image depends only on $R+$ *(and not on the choice of the* $p\alpha$ *or the orderings on* $R+$ *and* $R-$*).*

Notation 4.1.4. *We write* $\Omega_{R+} = \prod_{\alpha\in R-} U\alpha \cdot T \cdot \prod_{\alpha\in R+} U\alpha$.[1297]

Proposition 4.1.5. *The scheme* Ω_{R+} *is of finite presentation over* S *(hence retro-compact in* G*) and is universally schematically dense in* G *relative to* S *(cf. Exp. XVIII 1).*

The first assertion is trivial. Then,[1298] Ω_{R+} is flat and of finite presentation over S, and contains the unit section, hence meets each fiber of G in a non-empty open subset, hence in a dense one; the second assertion follows from Exp. XVIII 1.3.[1299]

Corollary 4.1.6. *Let* (G, T, M, R) *be a split* S*-reductive group. Then*

$$Centr(G) = \bigcap_{\alpha \in R} Ker(\alpha).$$

1300

Consequently, $Centr(G)$ *is representable by a closed diagonalizable subgroup of* G.

The second assertion follows immediately from the first. To prove the latter, one may invoke Exp. XII 4.8 and 4.11; one may also proceed directly as follows.[1301]

Let $S' \to S$. If $t \in T(S')$ and $\alpha(t) = 1$ for every $\alpha \in R$, then $int(t)$ induces the identity on $T_{S'}$ and on each $(U\alpha)_{S'}$, $\alpha \in R$, hence also on $(\Omega_{R+})_{S'}$, hence on $G_{S'}$ by schematic density, whence $t \in Centr(G)(S')$.

Conversely, since $Centr_{G_{S'}}(T_{S'}) = T_{S'}$ (cf. Exp. XIX 2.8), if $g \in G(S')$ centralizes $T_{S'}$ and the $(U\alpha)_{S'}$, it is a section of $T_{S'}$ that annihilates the $\alpha \in R$.

Corollary 4.1.7. *Let* S *be a scheme,* G *an* S*-reductive group. Then the center of* G *is representable by a closed subgroup of* G*, of multiplicative type; it is also "the intersection of the maximal tori of* G*" in the following sense: for every* $S' \to S$*,*

[1297]And we call it the "big cell" corresponding to $R+$.
[1298]We have expanded the original in what follows.
[1299]See also EGA IV_3, 11.10.10.
[1300]We have added the following sentence.
[1301]We have expanded the original in what follows.

$Centr(G)(S')$ is the set of $g \in G(S')$ whose inverse image in $G(S'')$, for every $S'' \to S'$, is contained in all the $T(S'')$, where T runs through the set of maximal tori of $G_{S''}$.

Taking into account 2.3, the first assertion follows from 4.1.6 by descent.[1302] Let us prove the second assertion. Let H be "the intersection of the maximal tori of G" in the preceding sense. One obviously has $Centr(G) \subset H$.[1303] Then, by descent, it suffices to prove $Centr(G) = H$ in the case where G is split. Since H is contained in the intersection of the maximal tori of G in the usual sense, this follows from the following remark: if (G, T, M, R) is a splitting, $\alpha \in R$ and $X \in \Gamma(S, g\alpha)^{\times}$, then $int(\exp \alpha(X))(T) \cap T = Ker(\alpha)$, as a trivial computation shows. (Cf. also Exp. XII 8.6 and 8.8 for a more general statement.)

Remark 4.1.8. *In what follows, we shall systematically identify, in the split case, T with $D_S(M)$. Then $Centr(G)$ is none other than $D_S(M/\Gamma_0(R))$, where $\Gamma_0(R)$ is the subgroup of M generated by R (cf. Exp. XXI 1.1.6). If $\alpha_1, \alpha_2, \cdots, \alpha_n$ is a system of simple roots of R, one immediately has (cf. Exp. XX 1.19):*

$$Centr(G) = \bigcap Ker(\alpha_i) = \bigcap Centr(Z_{\alpha_i}).$$

Proposition 4.1.9. *Let S be a scheme, (G, T, M, R) an S-split group, Q an S-torus, α_0 a character of Q, L an invertible $\mathcal{O}_S$-module,*

$$f : Q \to T, \quad p : W(L) \to G$$

morphisms of groups satisfying the set-theoretic relation

$$p(\alpha_0(q)\, x) = int(f(q)) \cdot p(x),$$

for all $q \in Q(S')$, $x \in W(L)(S')$, $S' \to S$. Suppose that f separates the elements of R in the following sense: if $\alpha, \alpha' \in R$ and $m, m' \in \mathbb{Z}$, then $m\alpha \circ f = m'\alpha' \circ f$ implies $m\alpha = m'\alpha'$.[1304] Finally, let $s \in S$ be such that $(\alpha_0)_s \neq e$ and $p_s \neq e$.

There then exist an open set U of S containing s, an integer $q > 0$ such that $x \mapsto x^q$ is an endomorphism of $G_{a,U}$, a root $\alpha \in R$, and an isomorphism of $\mathcal{O}_U$-modules

$$h : (L|U)^{\otimes q} \xrightarrow{\sim} g\alpha|U$$

such that

(i) $(\alpha \circ f)|U = (q\alpha_0)|U$,

(ii) $p(X) = \exp \alpha(h(X^q))$ for every $X \in W(L)(S')$, $S' \to U$.

Moreover, once U is chosen, q, α and h are uniquely determined.

[1302] Indeed, the representability of the center by a closed subscheme of G is local for the (fpqc) topology (SGA 1, VIII 5.2 and 5.4), so a fortiori for the étale topology, and the same holds for the property "of multiplicative type".

[1303] We have expanded the original in what follows.

[1304] Note that this is equivalent to the hypothesis: if $\alpha, \alpha' \in R$, $m, m' \in \mathbb{Z}$ and if $(m\alpha \circ f)_{\bar{s}} = (m'\alpha' \circ f)_{\bar{s}}$ for every geometric point $\bar{s}$ of S, then $m\alpha = m'\alpha'$. In particular, this separation hypothesis is stable under base change.

Shrinking S if necessary, we may suppose that α_0 is non-zero on every fiber of S. Choose a system of positive roots $R+$ of R and let $V = p^{-1}(\Omega_{R+})$. This is an open subset of $W(L)$ containing the zero section and stable under multiplication by every $\alpha_0(q)$, $q \in Q(S')$, $S' \to S$. Since α_0 is non-trivial on every fiber, it follows immediately that $V = W(L)$, hence that p factors through Ω_{R+}. Choose an arbitrary ordering on $R+$ and $R-$; all products will be taken in this ordering. We thus have unique morphisms

```
aα : W(L) → Uα,    α ∈ R,
b  : W(L) → T
```

such that

```
p(x) = ∏_{α ∈ R−} aα(x) · b(x) · ∏_{α ∈ R+} aα(x).
```

Writing the covariance condition under Q, one obtains immediately

```
aα(α₀(q) x) = α(f(q)) aα(x),    α ∈ R,
b(α₀(q) x)  = b(x)
```

for all $x \in W(L)(S')$, $q \in Q(S')$, $S' \to S$. The second condition gives at once $b = e$.

Now let $\alpha \in R$ be such that $(a\alpha)_s \neq e$ (we know such an α exists, since p_s is supposed $\neq e$). Applying Exp. XIX 4.12 (a), one deduces that there exists an integer $n > 0$ such that $(\alpha \circ f)_s = (n\alpha_0)_s$. Shrinking S, one can assume $\alpha \circ f = n\alpha_0$ (Exp. IX 5.3). But then, for every $\alpha' \in R$, $\alpha' \neq \alpha$, one has $(\alpha' \circ f)_s \neq m\alpha_0$ for every integer $m > 0$, by virtue of the hypothesis made on f (and the fact that the only roots proportional to α are α and $-\alpha$). Applying again Exp. XIX 4.12 (a), this time to $a\alpha'$, one deduces that $a\alpha'$ is zero in a neighborhood of S; since R is finite, one may, shrinking S again, suppose the $a\alpha'$ zero for $\alpha' \in R$, $\alpha' \neq \alpha$. One then has $p = a\alpha$, and one may apply Exp. XIX 4.12 (b), then (c), which gives the announced result (the uniqueness assertions are obvious).

Remark 4.1.10. *The condition imposed on f in 4.1.9 is satisfied in particular when f is surjective (= faithfully flat).*

Proposition 4.1.11. *Let (G, T, M, R) be an S-split group, $R+$ a system of positive roots of R, Ω_{R+} the corresponding "big cell".*

(i) Let H be a separated S-group functor[1305] *for (fppf). If $f, g : G \rightrightarrows H$ are two morphisms of groups that coincide on Ω_{R+}, then $f = g$.*

(ii) Let H be an S-sheaf of groups for (fppf) and $f : \Omega_{R+} \to H$ an S-morphism satisfying the following condition: for every $S' \to S$ and every $x, y \in \Omega_{R+}(S')$ such that $xy \in \Omega_{R+}(S')$, one has $f(xy) = f(x)f(y)$. There then exists a (unique, by (i)) morphism of groups $\bar{f} : G \to H$ extending f.

Indeed, by 4.1.5, (i) (resp. (ii)) follows immediately from Exp. XVIII 2.2 (resp. 2.3 and 2.4).

[1305] Recall (cf. Exp. IV, 4.3.5) that an S-prefunctor H is separated for a topology T if for every $S' \to S$ and every family of S-morphisms $(S'_i \to S')_{i \in I}$ covering for T, the map $H(S') \to \prod_i H(S'_i)$ is injective.

Remark 4.1.12. *If $\alpha \in R+$, one has*

(†) Ω_{R+} ∩ Zα = U_{-α} · T · Uα.

[1306] Indeed, for every $S' \to S$, if $g = \prod_{\beta \in R-} p\beta(x\beta) \cdot t \cdot \prod_{\beta \in R+} p\beta(x\beta)$ is an element of $\Omega_{R+}(S')$ and if $t' \in T\alpha(S'')$, then

t′ g t′⁻¹ = ∏_{β ∈ R−} pβ(β(t′) xβ) · t · ∏_{β ∈ R+} pβ(β(t′) xβ)

and since α and $-\alpha$ are the only two elements of R that take the value 1 on $T\alpha$, we obtain that g belongs to $Z\alpha = Centr(T\alpha)$ if and only if $x\beta = 0$ for $\beta \neq \pm\alpha$.

By (†), one deduces from XX 2.1 that if $X \in \Gamma(X, g\alpha)$ and $Y \in \Gamma(S, g_{-\alpha})$, one has:

expα(X) exp_{-α}(Y) ∈ Ω_{R+}(S) ⇔ 1 + XY invertible.

4.2. Morphisms of split groups

Definition 4.2.1. *Let S be a scheme (non-empty), (G, T, M, R) and (G', T', M', R') two S-split groups. We say that the morphism of S-groups $f : G \to G'$ is* compatible with the splittings, *or defines a* morphism of split groups, *if the restriction of f to T factors through a morphism $f_T : T \to T'$ of the form $f_T = D_S(h)$, where $h : M' \to M$ is a morphism of groups satisfying the following condition:*

there exists a bijection $d : R \xrightarrow{\sim} R'$[1307] and for each $\alpha \in R$ an integer $q(\alpha) > 0$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of Ga,S and that

h(d(α)) = q(α) α, ᵗh(α*) = q(α) d(α)*.

Remark 4.2.2. *It is immediate that h, d, $q(\alpha)$ for $\alpha \in R$, are uniquely determined by f. One writes $h = R(f)$. The $q(\alpha)$ are the* root exponents *of f (or of h).*

Let p be the prime number (if it exists) that is zero on S; set $p = 1$ if there is no prime number zero on S. Then $R(f)$ is a p-morphism of reduced root data in the sense of Exp. XXI 6.8. One has thus defined a functor R from the category of S-split groups to that of reduced root data (equipped with p-morphisms).

Proposition 4.2.3. *Under the conditions of 4.2.1, one has the following properties:*

(i) For every $\alpha \in R$, there exists a unique isomorphism of O_S-modules

$$f\alpha : (g\alpha)^{\otimes q(\alpha)} \xrightarrow{\sim} g'_{d(\alpha)}$$

such that

$$f(\exp\alpha(X)) = \exp_{d(\alpha)}(f\alpha(X^{q(\alpha)}))$$

for every $X \in W(g\alpha)(S')$, $S' \to S$.

[1306]We have expanded the original in what follows.
[1307]We have replaced the notation $u : R \to R'$ by $d : R \to R'$.

(ii) For every $\alpha \in R$, one has $q(-\alpha) = q(\alpha)$ and $f\alpha$ and $f_{-\alpha}$ are contragredient to one another.

(iii) For every $\alpha \in R$, every $Z \in W(g\alpha) * (S')$, $S' \to S$, one has

$$f(w\alpha(Z)) = w_{d(\alpha)}(Z^{q(\alpha)}).$$

By hypothesis the diagram

$$\begin{tikzcd}
G_{m,S} \arrow[r, "\alpha^*"] \arrow[d, "q(\alpha)"] & T \arrow[r, "\alpha"] \arrow[d, "f_T"] & G_{m,S} \arrow[d, "q(\alpha)"] \\
G_{m,S} \arrow[r, "d(\alpha)^*"] & T' \arrow[r, "d(\alpha)"] & G_{m,S}
\end{tikzcd}$$

is commutative. It follows that f maps $Ker(\alpha)$ into $Ker(d(\alpha))$, hence $T\alpha$ into $T'_{d(\alpha)}$, hence $Z\alpha$ into $Z'_{d(\alpha)}$. There is then nothing more to do than apply Exp. XX 3.10 and 3.11 to the groups $Z\alpha$ and $Z'_{d(\alpha)}$.

Proposition 4.2.4. *The morphism f induces a morphism f_N of $Norm(T)$ into $Norm_{G'}(T')$, hence a morphism f_W of $W_G(T)$ into $W_{G'}(T')$; the latter is an isomorphism. More precisely, if we denote by $d : W(R(G)) = W \to W' = W(R(G'))$ the isomorphism extending $s\alpha \mapsto s_{d(\alpha)}$ (Exp. XXI 6.8.4), we have a commutative diagram of isomorphisms:*

$$\begin{tikzcd}
W_G(T) \arrow[r, "f_W"] & W_{G'}(T') \\
W_S \arrow[u] \arrow[r, "d_S"] & W'_S \arrow[u]
\end{tikzcd}$$

This follows immediately from 3.4, Exp. XXI 6.8.4, and (iii) above.

Remark 4.2.5. *With the notations of 4.2.3, the restriction of f to Ω_{R+} (for a system of positive roots $R+$) is written explicitly: it maps Ω_{R+} to $\Omega'_{d(R+)}$ ($d(R+)$ is a system of positive roots of R' by Exp. XXI 6.8.7) and is given by the set-theoretic formula:*

$$f\left(\prod_{\alpha \in R-} \exp\alpha(X\alpha) \cdot t \cdot \prod_{\alpha \in R+} \exp\alpha(X\alpha)\right) = \prod_{\alpha \in R-} \exp_{d(\alpha)}(f\alpha(X\alpha^{q(\alpha)})) \cdot f_T(t) \cdot \prod_{\alpha \in R+} \exp_{d(\alpha)}(f\alpha(X\alpha^{q(\alpha)})).$$

Proposition 4.2.6. *(i) f is surjective (= faithfully flat in the present case, cf. VI_B 3.11) if and only if f_T is.*

(ii) One has $Ker(f) \subset \Omega_{R+}$.

Let us prove (i): if f is surjective, then $f_T(T) = f(T)$ is a maximal torus of G' (indeed $f(T)$ is a subtorus of a maximal torus T' (Exp. IX 6.8); to verify that $f(T) = T'$, one reduces to the case of an algebraically closed field, where this is *Bible*, § 7.3, th. 3 (a)).

If f_T is surjective, the preceding formula shows that f induces a surjection from $\Omega = \Omega_{R+}$ onto $\Omega' = \Omega'_{d(R+)}$.[1308] Since the fibers of G' are connected, it follows (cf. Exp. VI_A, 0.5) that f is surjective.

Let us prove (ii) and for this admit a result to be proved below (5.7.4): choose for each $w \in W$ an $n_w \in Norm_G(T)(S)$ representing it; then the open sets $n_w\Omega$ ($w \in W$) form a cover of G. It is then enough to prove that $Ker(f) \cap n_w\Omega \neq \emptyset$ implies $w = 1$. If $x \in \Omega(S')$, $S' \to S$ and $f(n_w x) = 1$, then $f(x) = f(n_w)^{-1}$; but $f(x) \in \Omega'(S')$ and $f(n_w)^{-1} \in Norm_{G'}(T')(S')$. By 4.2.4, one is reduced to proving:

Lemma 4.2.7. *Under the conditions of 4.1.2, one has $\Omega \cap Norm_G(T) = T$.*

Let

$$x = \prod_{\alpha \in R-} p_\alpha(x_\alpha) \cdot t \cdot \prod_{\alpha \in R+} p_\alpha(x_\alpha) = v\, t\, u \in \Omega(S').$$

If x normalizes $T_{S'}$, one has for every $t' \in T(S')$,

$$x\, t'\, x^{-1} = t'' \in T(S'),$$

that is, $xt' = t''x$, which is written

$$v\,(t\,t')\,(t'^{-1}\,u\,t') = (t''\,v\,t''^{-1})\,(t''\,t)\,u,$$

which gives $t'^{-1}ut' = u$, hence $u \in Centr_G(T)(S') = T(S')$, that is $u = 1$. Similarly $v = 1$.

Corollary 4.2.8. *One has*

$$Ker(f) = \prod_{\alpha \in R-} K_\alpha \cdot Ker(f_T) \cdot \prod_{\alpha \in R+} K_\alpha,$$

where for each $\alpha \in R$, K_α denotes the finite S-group

$$K_\alpha = Ker(U_\alpha \to U_\alpha^{\otimes q(\alpha)}) \simeq \alpha_{q(\alpha), S}.$$

To apply this corollary, let us set:

Definition 4.2.9. *Let S be a scheme, G and G' two S-reductive groups. A morphism of S-groups $f : G \to G'$ that is faithfully flat and finite (i.e. surjective with finite kernel over S) is called an* isogeny. *If moreover $Ker(f)$ is a central subgroup of G, one says that f is a* central isogeny.

Proposition 4.2.10. *Let $f : G \to G'$ be a morphism of split groups. For f to be an isogeny (resp. a central isogeny) it is necessary and sufficient that f_T be an isogeny, i.e. that $R(f)$ be injective with finite cokernel (resp. and that for every $\alpha \in R$, one has $q(\alpha) = 1$).*

Indeed, by 4.2.8, $Ker(f)$ is finite over S if and only if $Ker(f_T)$ is finite over S, and $Ker(f) \subset T$ if and only if each $q(\alpha) = 1$ ($Ker(f)$ is then central since of multiplicative type and invariant, cf. Exp. IX 5.5).

[1308] We have expanded the reference to Exp. VI in what follows.

Remark 4.2.11. *(a) One thus sees that $f : G \to G'$ is a central isogeny if and only if $R(f) : R(G') \to R(G)$ is an isogeny in the sense of Exp. XXI 6.2; moreover one has in this case (with the notations of loc. cit.):*

```
Ker(f) = D_S(K(R(f))),    K(R(f)) = Coker(R(f)).
```

(b) If G and G' are semisimple, every morphism of split groups $f : G \to G'$ is an isogeny.[1309]

(c) If $f : G \to G'$ is faithfully flat and finite and if G is reductive (resp. semisimple), then G' is also. It is indeed of finite presentation over S (Exp. V 9.1), affine over S (EGA II 6.7.1), smooth over S (Exp. VI 9.2), with connected reductive (resp. semisimple) geometric fibers by Exp. XIX 1.7.

Definition 4.2.1 may seem arbitrary. It is justified by the following proposition (which we state, for simplicity, for semisimple groups).

We say that a morphism $f : G \to G'$ of S-reductive groups is *splittable* if there exist splittings of G and G' with which f is compatible. One then has:

Proposition 4.2.12. *Let S be a scheme, G and G' two semisimple S-groups, $f : G \to G'$ a morphism of groups. Let $s \in S$. The following conditions are equivalent:*

(i) $Ker f_s$ is finite ($\Leftrightarrow e$ is isolated in `Ker f(s)`) and f_s is surjective, i.e. f_s is an isogeny.

(ii) f_s is splittable.

(iii) There exists an étale morphism $S' \to S$ covering s such that $f_{S'} : G_{S'} \to G'_{S'}$ is splittable.

One has obviously (iii) $\Leftrightarrow$ (ii); (ii) $\Rightarrow$ (i) follows from 4.2.11 (b) (this is where the hypothesis that G and G' are semisimple intervenes — the other implications are valid for reductive groups).

Let us now prove (i) $\Rightarrow$ (iii). One may suppose G and G' split in such a way that f induces a morphism $f_T : T \to T'$ (2.3 and Exp. XIX 2.8); shrinking S, one may suppose that $f_T = D_S(h)$, where h is a morphism of groups $M' \to M$. Let $\alpha \in R$, and consider the composite morphism

```
p : W(gα) --expα--> G --f--> G′.
```

Since $Ker(p_s)$ is finite, $p_s \neq e$. On the other hand f_{T_s} is surjective; one may therefore apply 4.1.9, and there exist an open subset V_α of S containing s, a root $\alpha' \in R'$, an integer $q(\alpha)$ such that $x \mapsto x^{q(\alpha)}$ is an endomorphism of Ga, V_α, and an isomorphism of O_{V_α}-modules

$$f\alpha : (g\alpha)^{\otimes q(\alpha)}|V_\alpha \xrightarrow{\sim} g'_{\alpha'}|V_\alpha$$

such that $f(\exp\alpha(X\alpha)) = \exp_{d(\alpha)}(f\alpha(X\alpha^{q(\alpha)}))$ and $\alpha' \circ f_T = h(\alpha') = q(\alpha)\alpha$. One may replace S by the intersection of the V_α, for $\alpha \in R$. Set $\alpha' = d(\alpha)$. It is clear that

[1309] Indeed, since G and G' are semisimple, f_T is surjective with finite kernel, so f is faithfully flat with finite kernel by 4.2.6 (i) and 4.2.8.

$d : R \to R'$ is a bijection, because the kernel of h is finite (f_{T_S} being surjective). It only remains to prove that $f_T \circ \alpha * = q(\alpha)\alpha' *$, which is done by a trivial modification of the argument used in Exp. XX 3.11.

In any case, as one has seen in the course of the demonstration, one has (i) ⇒ (iii). Therefore:

Corollary 4.2.13. *Let S be a scheme, $f : G \to G'$ an isogeny of reductive groups. Then f is locally splittable for the étale topology.*

4.3. Central quotients of reductive groups

Let us first consider a particular case.

Proposition 4.3.1. *Let S be a scheme, (G, T, M, R) an S-split group, N a subgroup of M containing R, $Q = D_S(M/N) \subset Centr(G)$. Then:*

(i) $G' = G/Q$ is an S-reductive group, and $T' = T/Q$ is a maximal torus of it;

(ii) if one identifies T' with $D_S(N)$, then $R \subset N$ is a system of roots of G' relative to T', (G', T', N, R) is a splitting of G', and $R(G')$ is canonically identified with the induced root datum (Exp. XXI 6.5) $R(G)^N$;

(iii) the canonical morphism $G \to G'$ is compatible with the splittings, with root exponents 1, and gives by functoriality the canonical morphism (loc. cit.) $R(G)^N \to R(G)$.

One knows that $G' = G/Q$ is representable by an affine group scheme over S (Exp. VIII 5.7), smooth over S (Exp. VI_B 9.2), with connected reductive geometric fibers (as quotients of reductive groups, cf. Exp. XIX 1.7); G' is therefore an S-reductive group.

It is clear that $T' = T/Q \simeq D_S(N)$ is a maximal torus of it. Note next that, choosing a system of positive roots $R+$ of R, the open set Ω_{R+} of 4.2 is stable under Q and that one has a canonical isomorphism

$\Omega_{R+}/Q \simeq \prod_{\alpha \in R-} U\alpha \times_S (T/Q) \times_S \prod_{\alpha \in R+} U\alpha,$

and that Ω_{R+}/Q is an open subset of G' containing the unit section (cf. Exp. IV, 4.7.2 and 6.4.1).

It follows that, if one denotes by g' the Lie algebra of G' and by $\bar{\alpha}$ the character of T/Q induced by α (or, which amounts to the same, defined by $\alpha \in N$ in the identification $T/Q = D_S(N)$), the canonical morphism $g \to g'$ induces for each $\alpha \in R$ an isomorphism

$$g\alpha \xrightarrow{\sim} g'_{\bar{\alpha}}.$$

One has thus proved that R is a system of roots of G' relative to T', and one finishes the proof without difficulty.

Corollary 4.3.2. *Let S be a scheme, G an S-reductive group, Q a normal subgroup of multiplicative type of G. Then Q is central in G, the quotient G/Q is representable by an S-reductive group, and the canonical morphism $G \to G/Q$ is locally splittable for the étale topology (with root exponents equal to 1).*

The first assertion follows from Exp. IX 5.5; the others are local for the étale topology and one is reduced to 4.3.1.

Definition 4.3.3. *Let G be an S-reductive group. We say that G is* adjoint *(resp.* simply connected*) if for every $s \in S$, the type of G at s is given by an adjoint (resp. simply connected) root datum, i.e. (Exp. XXI 6.2.6) such that M be generated by R (resp. $M*$ generated by $R*$).*

Proposition 4.3.4. *(i) An adjoint (resp. simply connected) reductive group is semisimple.*

(ii) If T is a maximal torus of the adjoint (resp. simply connected) reductive group G and α is a root of G relative to T, then the infinitesimal root α is non-zero on every fiber (resp. $\alpha*$ is a monomorphism and the infinitesimal coroot $H\alpha$ is non-zero on every fiber).

Indeed, (i) is trivial; (ii) is checked on geometric fibers and follows immediately from Exp. XXI 6.2.8.

Proposition 4.3.5. *(i) For the reductive group G to be adjoint, it is necessary and sufficient that $Centr(G) = e_S$.*

(ii) For every reductive group G, the quotient group $G/Centr(G)$ is an adjoint reductive group.

Indeed, one may suppose G split; then (i) is trivial (since $Centr(G) = D_S(M/\Gamma_0(R))$), and (ii) follows from 4.3.1.

Definition 4.3.6. *Let G be an S-reductive group. We call* adjoint group of *G and denote by $ad(G)$ the group $G/Centr(G)$. We call* radical of *G and denote by $rad(G)$ the maximal torus (unique by Exp. XII 1.12) of $Centr(G)$. We call* semisimple group associated with *G the quotient $G/rad(G)$.*

The preceding definitions are compatible with base change. If $s \in S$, $rad(G)_s$ is indeed the radical of G_s in the usual sense (Exp. XIX 1.6).

4.3.7. If (G, T, M, R) is a split group, then $rad(G) = D_S(M/N)$, where $N = M \cap V(R)$, so the semisimple group associated with G (and similarly the adjoint group of G) is equipped with a canonical splitting (4.3.1) and one has a diagram of split groups

$$G \to ss(G) \to ad(G)$$

corresponding to the canonical diagram of root data (Exp. XXI 6.5.5)

$$ad(R(G)) \to ss(R(G)) \to R(G).$$

Remark 4.3.8. *Let (G, T, M, R) be an S-split adjoint (resp. simply connected) group, Δ a system of simple roots of R. Then the family $\alpha_{\alpha\in\Delta}$, resp. $\alpha*_{\alpha\in\Delta}$, induces an isomorphism*

`T □ (Gm,S)^Δ,    resp.    (Gm,S)^Δ □ T.`

Indeed, $M = \Gamma_0(R)$ (resp. ...) and Δ is a basis of the free abelian group $\Gamma_0(R)$ (Exp. XXI 3.1.8).

Remark 4.3.9. *The radical of a reductive group is a characteristic subgroup (i.e. stable under* $\mathrm{Aut}_{S-gr.}(G)$*), by its very definition.*

5. Subgroups of type (R)

We are especially interested in reductive groups, but some of the results we shall establish are valid more generally for a wider class of groups: groups of type (RR).

5.1. Groups of type (RR)

Definition 5.1.1. *Let S be a scheme, G an S-group scheme. We say that G is* of type (RR) *if it satisfies the following conditions:*

(i) G is smooth and of finite presentation over S and has connected fibers.

(ii) G locally possesses maximal tori for the (fpqc) topology.

(iii) For every $s \in S$, every maximal torus T of G_s, and every root of G_s relative to T_s (Exp. XIX 1.10), $Lie(G_s)^\alpha$ is of rank 1 (as a vector space over $\kappa(s)$).

(iv) For every $s \in S$ and every maximal torus T of G_s, let R denote the set of roots of G_s relative to T and $\Gamma_0(R)$ the subgroup of $\mathrm{Hom}_{s-gr.}(T, Gm, s)$ *generated by R; then the content*[1310] *of every root $\alpha \in R$ in the free abelian group $\Gamma_0(R)$ (which is therefore a positive integer) is invertible on S.*

Recall 5.1.1.1.[1311] *Let us recall that if G is a smooth connected algebraic group over an algebraically closed field k, a* Cartan subgroup *of G is the centralizer of a maximal torus of G (XII 1.0), and such a subgroup is smooth and connected: for this, as well as for other characterizations of Cartan subgroups, see Bible, § 7.1, Th. 1 in the affine case and Exp. XII Th. 6.6 in the general case. If S is an arbitrary scheme and G an S-smooth group of finite type, a* Cartan subgroup *of G is an S-smooth subgroup C of G such that, for every $s \in S$, C_s is a Cartan subgroup of G_s (XII Def. 3.1).*

Remark 5.1.2. *(a) By virtue of Exp. XII 7.1 (where the hypothesis that G is separated holds here since G has connected fibers, cf. Exp. VI_B 5.5), (i) and (ii) entail that G locally possesses, for the étale topology, maximal tori (resp. Cartan subgroups), conjugate locally for the étale topology.*

[1310] The content of the root α is the positive generator of the ideal $f(\alpha), f \in \Gamma_0(R)*$ of $\mathbb{Z}$; it is the largest integer $c > 0$ such that $\alpha/c \in \Gamma_0(R)$.

[1311] We have added this recall.

(b) The Cartan subgroups of G have connected fibers (Exp. XII 6.6).

(c) If G is affine over S, (i) and (ii) are respectively equivalent to

(i′) G is smooth over S with connected fibers.

(ii′) The reductive rank of the fibers of G is locally constant (Exp. XII 1.7).

(d) By Exp. XII 8.8 (c) and (d), G has a reductive center Z and, for every $s \in S$, with the notations of (iv), one has[1312] *$Z_s = \bigcap_{\alpha\in R} Ker(\alpha)$, whence*

$$\mathrm{Hom}_{s-gr.}((T/Z)_s, Gm, s) \simeq \Gamma_0(R).$$

(e) Condition (iv) holds in particular in the following two cases:

(1) S is of characteristic 0;

(2) every root is an indivisible element of the group generated by the roots.

(f) Being of type (RR) is stable under base change and is local for the (fpqc) topology.

From remarks (c) and (e), one immediately deduces:

Proposition 5.1.3. *A reductive group is of type (RR).*

Proposition 5.1.4. *Let S be a scheme, G an S-group of type (RR), Q a central subgroup of G of finite presentation over S such that the quotient G/Q is representable (for instance G affine over S and Q of multiplicative type (IX 2.3) or else S artinian (VI_A 3.3.2)); then G/Q is an S-group of type (RR).*

Indeed, G/Q is smooth over S (Exp. VI_B 9.2), of finite presentation over S (Exp. V 9.1) and with connected fibers, so condition (i) holds. On the other hand, condition (ii) follows from Exp. XII 7.6; it remains to verify conditions (iii) and (iv).

Let $G' = G/Q$, $u : G \to G'$ the canonical morphism, $T' = u(T)$ the maximal torus of G' image of T (cf. Exp. XII 7.1 (e)); for each $\alpha \in R$, denote again by α the character of T' defined by α (one has $Q \cap T \subset \bigcap_{\alpha\in R} Ker(\alpha)$ according to 5.1.2 (d)). Let us first prove:

Lemma 5.1.5. *Under the conditions of 5.1.4, let $T = D_S(M)$ be a trivialized maximal torus of G, and suppose that the decomposition of $g = Lie(G)$ under $Ad(T)$ is of the form*

$$g = g^0 \oplus \bigoplus_{\alpha \in R} g\alpha, \qquad R \subset M - \{0\},$$

where for every $s \in S$, $g\alpha(s) \neq 0$ for $\alpha \in R$ (so $g\alpha$ is an invertible O_S-module for every $\alpha \in R$ and R is the set of roots of G_s relative to T_s for every $s \in S$).

Then the Lie algebra g' of G' decomposes under $Ad(T')$ as follows:

[1312] The original stated "under the conditions of (iv)", but the last condition of (iv) does not seem to be used here.

$$g' = g'^0 \oplus \coprod_{\alpha \in R} g'^{\alpha},$$

and $Lie(u)$ induces an isomorphism of $g\alpha$ onto g'^{α}.

Indeed, let $p = Lie(u) : g \to g'$. One immediately has $p(g\alpha) \subset g'^{\alpha}$ for every $\alpha \in R$, and $p(g^0) \subset g'^0$. Since

$$Ker(p) = Lie(Q) \subset Lie(Centr_G(T)) = g^0,$$

p induces a monomorphism from $g\alpha$ into g'^{α}, for every $\alpha \in R$.

To prove the lemma, it suffices to do so when S is the spectrum of an algebraically closed field, and by virtue of the preceding remarks, it then suffices to prove that $rg(g') = rg(g'^0) + Card(R)$. Now set $C = Centr_G(T)$, $C' = Centr_{G'}(T')$; by Exp. XII 7.1 (e), u induces a faithfully flat morphism $C \to C'$ of kernel Q. One thus has

$$\dim C' + \dim Q = \dim C.$$

But G, G', C and C' are smooth, so

$$\begin{aligned} \dim G = rg(g) &= rg(g^0) + Card(R) = \dim C + Card(R) \\ &= \dim Q + \dim C' + Card(R) = \dim Q + rg(g'^0) + Card(R) \end{aligned}$$
$$rg(g') = \dim G' = \dim G - \dim Q$$

which entails

$$rg(g') = rg(g'^0) + Card(R),$$

that is, the desired relation.

Let us return to the proof of 5.1.4; one may suppose that S is the spectrum of an algebraically closed field. Let T be a maximal torus of G; applying 5.1.5, one immediately has (iii) and (iv) for G/Q.

To use the preceding proposition, let us introduce a definition:

Definition 5.1.6. *We say that the S-group scheme G is* of type (RA) *if it is of type (RR) and if it further satisfies the following condition (iv′) (stronger than (iv)):*

(iv′) For every $s \in S$ and every maximal torus T of G_s, every root of G_s relative to T has a content in $\mathrm{Hom}_{s-gr.}(T, Gm, s)$ that is invertible on S.

Remarks 5.1.7. *(a) An S-reductive adjoint group is of type (RA).*

(b) If S is of characteristic 0, every group of type (RR) is of type (RA).

(c) Being of type (RA) is stable under base change and is local for the (fpqc) topology.

Remark (a) above generalizes to:

Proposition 5.1.8. *Let S be a scheme, G an S-group of type (RR), Z its reductive center, suppose the quotient G/Z representable (for instance G affine over S, or S artinian); then G/Z is of type (RA).*

Indeed, this follows immediately from 5.1.4, 5.1.5, and 5.1.2 (d).

Remark 5.1.9. *If G is of type (RR) and if T is a maximal torus of G, one may apply Exp. XIX 6.3. In particular $W_G(T)$ is étale, quasi-finite and separated over S.*

5.2. Subgroups of type (R)

Definition 5.2.1. *Let S be a scheme, G a smooth S-group scheme of finite presentation with connected fibers,*[1313] *H a group subscheme of G. We say that H is* of type (R) *if:*

(i) H is smooth, of finite presentation over S and with connected fibers.[1314]

(ii) For every $s \in S$, H_s contains a Cartan subgroup of G_s.

This notion is stable under base change and local for the (fpqc) topology.

Recall 5.2.2. *(cf. Exp. XII 7.9) Under the preceding conditions:*

(a) $H = Norm_G(H)^0$.

(b) If G locally possesses, for the étale topology, Cartan subgroups (resp. maximal tori), so does H, and the Cartan subgroups (resp. maximal tori) of H are Cartan subgroups (resp. maximal tori) of G.

Examples 5.2.3. *(a) Borel subgroups: a* Borel subgroup *of G is a subgroup H of type (R) whose geometric fibers are Borel subgroups of those of G.*[1315]

(b) Parabolic subgroups: a parabolic subgroup *of G is a subgroup of type (R) whose geometric fibers contain Borel subgroups.*

Other examples are given by the following propositions.

Proposition 5.2.4. *Let G be as in 5.2.1, $K \subset H$ two group subschemes of G, H assumed to be of type (R). Then K is a subgroup of type (R) of H if and only if it is a subgroup of type (R) of G.*

[1316] Indeed, let $s \in S$. Since H is of type (R), every maximal torus of H_s is a maximal torus of G_s, and so likewise for Cartan subgroups.

Proposition 5.2.5. *Let G be as in 5.2.1, T a maximal torus of G, Q a subtorus of T, $Z = Centr_G(Q)$. If H is a subgroup of type (R) of G containing T, then $H \cap Z$ is a subgroup of type (R) of Z.*

[1313] The hypothesis that G (resp. H) be of finite presentation over S is automatically verified because G (resp. H) being smooth over S with connected fibers, is quasi-compact and separated over S (VI_B 5.5), so of finite presentation over S.

[1314] The hypothesis that G (resp. H) be of finite presentation over S is automatically verified because G (resp. H) being smooth over S with connected fibers, is quasi-compact and separated over S (VI_B 5.5), so of finite presentation over S.

[1315] This amounts to saying that H is a smooth subgroup of G, each of whose geometric fibers H_s is a Borel subgroup of G_s (since every Borel subgroup of G_s is connected and contains a Cartan subgroup of G_s).

[1316] We have expanded what follows.

Let us first recall that Z is a closed group subscheme of G, of finite presentation (Exp. XI 6.11), with connected fibers (Exp. XII 6.6), smooth over S (Exp. XI 2.4), so it satisfies the conditions imposed in Definition 5.2.1. Similarly, $H \cap Z$ is of finite presentation, smooth and has connected fibers (since $H \cap Z = Centr_H(Q)$); moreover $H \cap Z \supset Centr_G(T)$.

Proposition 5.2.6. *Let S be a scheme, G an S-group of type (RR) (resp. (RA)), H a subgroup of type (R) of G. Then H is an S-group of type (RR) (resp. (RA)).*

Indeed, (i) is clear, (ii) follows from 5.2.2 (b), (iii) and (iv) (resp. (iv′)) are to be verified when S is the spectrum of an algebraically closed field. Then H contains a maximal torus T of G (and hence also $C = Centr_G(T)$),[1317] and the assertions to be proved follow immediately from:

Lemma 5.2.7. *Let S be a scheme, G an S-group of type (RR), T a maximal torus of G equipped with a trivialization $T \simeq D_S(M)$, and suppose that*

$$g = g^0 \oplus \coprod_{\alpha \in R} g\alpha$$

(the $g\alpha$ being then invertible O_S-modules).

Let H be a subgroup of type (R) containing $C = Centr_G(T)$ (i.e. containing T). Then $h = Lie(H/S)$ is locally on S of the form

$$g^0 + \coprod_{\alpha \in R'} g\alpha = g_{R'};$$

more precisely, let, for each $s \in S$, $R'(s) = \alpha \in R | g\alpha(s) \subset h(s)$. Then $R'(s)$ is a locally constant function of s; if U is an open subset of S on which $R'(s) = R'$, one has

$$h_U = g^0_U \oplus \coprod_{\alpha \in R'} g\alpha_U.$$

Indeed, h is a submodule of g, locally a direct factor, containing g^0 and stable under T.

5.3. Strict transporter of two subgroups of type (R). Applications

Recall 5.3.0.[1318] *Let S be a scheme, G an S-smooth group, $g = Lie(G/S)$ and h a sub-O_S-module of g that is locally a direct factor. The O_S-algebra $A = Sym(\omega^1_{G/S})$ is locally free, so the S-scheme $Lie(G/S) = W(g) = \mathrm{Spec}(A)$ is essentially free in the sense of Exp. VIII, 6.1. Since $W(h)$ is a closed subscheme of $Lie(G/S)$, of finite presentation over $Lie(G/S)$, then $N = Norm_G(h)$ is representable by a closed group subscheme of G, of finite presentation over G, by Exp. VIII, 6.5 (a). (See also the additions 6.2.3 and 6.2.4 (a) in Exp. VI_B.) On the other hand, by Exp. II 5.3.1, one has $Lie(N/S) = Norm_{Lie(G/S)}(h)$.*

Finally, by Exp. VI_B 3.10, if N is smooth over S at the points of the unit section, then the group subfunctor N^0 (defined in VI_B 3.1) is represented by an open group

[1317]By hypothesis, H contains $Centr_G(T')$ for some maximal torus T' of G; then T and T' are conjugate in H, so H also contains $C = Centr_G(T)$.

[1318]We have added this recall, which is used in the proofs of 5.3.1 and 5.3.4.

subscheme of N, smooth over S.

Proposition 5.3.1. *Let S be a scheme, G an S-group of type (RA) (5.1.6), H a subgroup of type (R) of G, $g \supset h$ their Lie algebras.*

Then $Norm_G(h)$ (which is representable by a closed subscheme of G of finite presentation over S according to 5.3.0) is smooth over S at every point of the unit section, and one has

$$Norm_G(h)^0 = H.$$

[1319] Proof. Set $N = Norm_G(h)$ and $n = Lie(N/S)$. One has $H \subset N$ and, by Exp. II 5.3.1, one has for every $s \in S$

$$h(s) \subset n(s) = Norm_{g(s)}(h(s)).$$

Now, by 5.3.2 below, one has $h(s) = Norm_{g(s)}(h(s))$, and since H is smooth over S, one has $\dim_{\kappa(s)} h(s) = \dim H_s$ (cf. [DG70], § II.5, Th. 2.1). One thus obtains

$$\dim_{\kappa(s)} n(s) = \dim_{\kappa(s)} h(s) = \dim H_s \leq \dim N_s$$

whence $N_s^0 = H_s^0 = H_s$ (H having connected fibers). It follows that the group subfunctor N^0 (defined in VI_B, 3.1) is represented by the smooth S-group H. This proves 5.3.1, modulo the following lemma:

Lemma 5.3.2. *Under the conditions of 5.2.7, if G is of type (RA), one has, for every $s \in S$,*

$$Norm_{g(s)}(h(s)) = h(s).$$

Indeed, one is reduced to the case where S is the spectrum of a field, so where $h = g_{R'}$ for some $R' \subset R$. But one already has

$$Transp_g(t, h) = h.$$

Indeed, if $H \in t$ and $X \in g\alpha$, one has $[H, X] = \alpha(H)X$, where $\alpha : t \to O_S$ is the derived morphism of α. Now condition (iv′) says precisely that $\alpha \neq 0$ for every $\alpha \in R$.

Corollary 5.3.3. *Let S be a scheme, G an S-group of type (RA), H and H' two subgroups of type (R) of G, h and h' their Lie algebras. Then*

$$H = H' \Leftrightarrow h = h'.$$

Corollary 5.3.4. *Under the conditions of 5.2.7, with G of type (RA), the maps*

$$H \mapsto Lie(H/S), \quad h \mapsto Norm_G(h)^0$$

[1319] We have expanded the original in what follows.

realize a bijective correspondence between the set of subgroups of type (R) of G containing T and the set of Lie subalgebras of g containing g^0, stable under T, and whose normalizer in G is smooth over S at every point of the unit section.

[1320] Indeed, let h be a Lie subalgebra of g having the above properties. By 5.3.0, $H = Norm_G(h)^0$ is a smooth S-group scheme. Moreover, since $C = Centr_G(T)$ stabilizes each $g\alpha$ and has connected fibers (XII 6.6), one has $C \subset H$. Therefore H is a subgroup of G of type (R). By Exp. II 5.3.1, one has $Lie(H) = Norm_g(h)$. Finally, by the proof of 5.3.2, one has $Norm_g(h) = h$.

Corollary 5.3.5. *Let S be a scheme, G an S-group of type (RR) (5.1.1), T a maximal torus of G, H and H' two subgroups of type (R) of G, both containing T. Then*

H = H′ ⇔ h = h′.

By virtue of the finite presentation hypotheses, one reduces as usual (cf. EGA IV₃, § 8 and Exp. VI_B § 10) to the case where S is noetherian; it suffices then to verify that $h = h'$ implies $H_{S'} = H'_{S'}$ for every S' spectrum of an artinian quotient of a local ring of S;[1321] one is thus reduced to the case where S is artinian, and one may apply 5.1.8. Let $u : G \to G' = G/Z$ be the canonical morphism and $T' = T/Z$ the maximal torus of G' corresponding to T. By Exp. XII 7.12, there exist subgroups of type (R) H_1 and H_1' of G', containing T', such that $H = u^{-1}(H_1)$ and $H' = u^{-1}(H_1')$. It suffices to prove that $H_1 = H_1'$. But by 5.2.7 and 5.1.5, one has

$$Lie(H_1) = Lie(H_1'),$$

and one reduces to 5.3.3.

Remark 5.3.6. *The fact that H and H' contain the same maximal torus is essential for the validity of 5.3.5 when G is not of type (RA). Example: maximal tori of SL_{2},k for k of characteristic 2.*[1322]

Corollary 5.3.7. *Let S be a scheme, G an S-group of type (RR), T a maximal torus of G, H and H' two subgroups of type (R) of G, both containing T. The set U of $s \in S$ such that $H_s = H'_s$ is open and closed in S and $H_U = H'_U$.*

Indeed, this follows immediately from 5.3.5 and 5.2.7.

Corollary 5.3.8. *The "functor of subgroups of type (R) containing T", where T is a given maximal torus in a group G of type (RR), is formally unramified (Exp. XI 1.1).*

[1320] We have added the following proof.

[1321] Indeed, let $g \in G$, s its image in S, m the maximal ideal of $O_{G,g}$, n that of $O_{S,s}$, and I (resp. I') the kernel of the morphism of $O_{G,g}$ into $O_{H,g}$ (resp. $O_{H',g}$) (the latter being the zero ring if $g \notin H$, resp. $g \notin H'$). Since $O_{G,g}$ is noetherian, I and I' are closed for the m-adic topology, so a fortiori for the n-adic topology, so it suffices to show that $I + n^n O_{G,g} = I' + n^n O_{G,g}$ for every $n \in \mathbb{N}$.

[1322] Indeed, if k is algebraically closed, all the maximal tori of $G = SL_2, k$ are conjugate under $G(k)$, and all have for Lie algebra the line $k \cdot id \subset M_2(k)$ (which is invariant under the adjoint action).

Theorem 5.3.9. *Let G be an S-group of type (RR) (5.1.1), H and H' two subgroups of type (R) (5.2.1). Let $Transt_G(H,H')$ be the strict transporter of H into H' defined by*

```
Transt_G(H, H′)(S′) = {g ∈ G(S′) | int(g) H_{S′} = H′_{S′}}.
```

Then $Transt_G(H,H')$ is representable by a closed subscheme of G, which is smooth and of finite presentation over S.

The fact that $Transt_G(H,H')$ is representable by a closed subscheme of G, of finite presentation over S, follows from Exp. XI 6.11 (a). To prove that it is smooth over S, we must show that if S is affine and S_0 is the closed subscheme defined by a nilpotent ideal J, and if $g_0 \in G(S_0)$ and $int(g_0)H_0 = H'_0$, there exists $g \in G(S)$ projecting to g_0 and such that $int(g)H = H'$. Since the question of smoothness is local for the étale topology, we may suppose that H contains a maximal torus T of G.

Then T_0 is a maximal torus of H_0, hence $int(g_0)T_0$ is a maximal torus of H'_0. By Exp. IX 3.6 bis, there exists a torus T' of H' such that $T'_0 = int(g_0)T_0$; by Exp. IX 3.3 bis, there thus exists $g \in G(S)$ projecting to g_0 and such that $int(g)T = T'$. Replacing H by `int(g) H` if necessary, we may therefore suppose that H and H' contain the same maximal torus T and that $H_0 = H'_0$. But then $H = H'$ by 5.3.7. QED.

Corollary 5.3.10. *Let G be an S-group of type (RR), H a subgroup of type (R) of G. Then $Norm_G(H)$ is representable by a closed group subscheme of G, of finite presentation and smooth over S.*

Using now the reasoning that, in Exp. XI, served to deduce 5.4 bis from 5.2 bis, one obtains:

Corollary 5.3.11. *Under the hypotheses of 5.3.9, the following conditions are equivalent:*

(i) H and H' are locally conjugate in G for the étale topology.

(i bis) idem for the (fpqc) topology.

(ii) For every $s \in S$, H_s and H'_s are conjugate by an element of $G(s)$.

(ii bis) The structural morphism $Transt_G(H,H') \to S$ is surjective.

(iii) $Transt_G(H,H')$ is a principal homogeneous bundle under the action of the smooth S-group scheme of finite presentation $Norm_G(H)$.

Let us simply remark that the non-trivial assertion (iii) ⇒ (i) is Hensel's lemma.

Using now Bible, § 6.4, th. 4 (= [Ch05], § 6.5 th. 5) and § 9.3, th. 1, one obtains by 5.3.10 and 5.3.11:

Corollary 5.3.12. *Let G be an S-group of type (RR). The Borel subgroups of G are closed in G, equal to their normalizer, and conjugate locally for the étale topology.*

Definition 5.3.13. *Let S be a scheme, G a smooth S-group scheme of finite presentation with connected fibers.*[1323] *By a* Killing couple *of G we mean a couple $T \subset B$, where T is a maximal torus of G and B a Borel subgroup of G containing T.*

Using now the conjugacy of the maximal tori in B (cf. 5.1.2 (a) and 5.2.6, for example), one has:

Corollary 5.3.14. *Let G be an S-group of type (RR). The Killing couples of G are conjugate locally for the étale topology.*

Corollary 5.3.15. *Let G be an S-group of type (RR). Let T be a maximal torus of G, $W_G(T) = Norm_G(T)/Centr_G(T)$ the corresponding Weyl group (Exp. XIX 6.3). The "functor of Borel subgroups of G containing T" is formally principal homogeneous under $W_G(T)$.*

This follows immediately from 5.3.14 and from the fact that if B is a Borel subgroup of G containing T, one has

$$Norm_G(T) \cap B = Centr_G(T),$$

cf. Exp. XIV 4.4.

Proposition 5.3.16. *Let G be an S-group of type (RR), H a subgroup of type (R), $N = Norm_G(H)$ its normalizer (5.3.10). Let T be a maximal torus of H, $W_H(T)$ and $W_N(T)$ the corresponding Weyl groups (étale, quasi-finite, separated by Exp. XIX 6.3). One has the following exact sequence of sheaves (for the étale topology) (the morphisms are induced by the morphisms $Norm_H(T) \to Norm_N(T) \to N/H$):*

$$1 \to W_H(T) \to W_N(T) \to N/H \to 1.$$

The only non-trivial point is that the last arrow is an epimorphism. So let $n \in N(S')$, $S' \to S$. The two maximal tori T and `int(n) T` of H are conjugate in H locally for the étale topology. There thus exists a covering family $S_i' \to S'$ and for each i an $h_i \in H(S_i')$ such that $int(h_i)T = int(n)T$. Hence $nh_i^{-1} \in Norm_N(T)$, which gives the desired result.

Remark 5.3.17. *One can describe $W_N(T)$ as follows: suppose we are reduced to the situation of 5.2.7, with $h = g_{R'}$. Then $W_N(T)$ equals $Norm_W(R')$, the sheaf of sections of $W = W_G(T)$ that, acting on R, normalize R'. Indeed, by 5.3.5, one has*

`Norm_N(T) = Norm_G(H) ∩ Norm_G(T) = Norm_G(h) ∩ Norm_G(T).`

Corollary 5.3.18. *Let G be an S-group of type (RR), H a subgroup of type (R). Suppose that "the Weyl groups of G are finite", i.e. that for every $S' \to S$ and every maximal torus T of $G_{S'}$, the étale S'-scheme $Norm_{G_{S'}}(T)/Centr_{G_{S'}}(T)$ is finite (cf. Exp. XIX 6.3 (iii)). The following conditions are equivalent:*

(i) H is closed in G.

[1323] Cf. footnote (25).

(ii) $Norm_G(H)/H$ is representable by a finite étale S-scheme.

(iii) "The Weyl groups of H are finite".

Indeed, one may suppose that H possesses a maximal torus T. By 5.3.10, $N = Norm_G(H)$ is closed in G, so $W_N(T)$ is closed in $W_G(T)$ and hence finite over S. One obviously has (i) ⇒ (iii), and (iii) ⇒ (ii) by the exact sequence of 5.3.16. Finally, (ii) ⇒ (i), because if N/H is finite, it is separated, so H is closed in N, hence in G.

Remark 5.3.19. *When G is reductive, the preceding conditions on H seem always satisfied. We prove them below in most cases.*

5.4. Subgroups of type (R) of a split reductive group (generalities)

5.4.1. If H is a subgroup of type (R) of the reductive group G, then H contains locally, for the étale topology, a maximal torus of G (5.2.2). By 2.3, one may, locally for the étale topology, suppose G split relative to this torus. Let then (G, T, M, R) be an S-split group, H a subgroup of type (R) of G containing T. By 5.3.5, such a subgroup is characterized by its Lie algebra, which (5.2.7) is locally on S of the form $g_{R'}$:

$$g_{R'} = t \oplus \coprod_{\alpha \in R'} g_\alpha.$$

Definition 5.4.2. *Let (G, T, M, R) be an S-split group. We shall say that the subset R' of R is* of type (R) *if $g_{R'}$ is the Lie algebra of a subgroup of type (R) of G containing T. This subgroup, uniquely determined by R', is denoted $H_{R'}$.*

Lemma 5.4.3. *Under the preceding conditions, one has the following equivalences:*

$$\begin{aligned}
H \cap Z_\alpha = T &\iff \alpha \notin R', \quad -\alpha \notin R',\\
H \supset U_\alpha &\iff \alpha \in R',\\
H \cap U_\alpha = e &\iff \alpha \notin R',\\
H \supset Z_\alpha &\iff \alpha \in R', \quad -\alpha \in R'.
\end{aligned}$$

Indeed, $H \cap Z\alpha$ is a subgroup of type (R) of $Z\alpha$, by 5.2.5; but a subgroup of type (R) of $Z\alpha$ containing T is locally equal to one of the following subgroups: T, $T \cdot U\alpha$, $T \cdot U_{-\alpha}$, $Z\alpha$, by 5.3.5.

Lemma 5.4.4. *Under the conditions of 5.4.2, let $R+$ be a system of positive roots; choose orderings on $R' \cap R+$ and $R' \cap -R+$. The morphism*

$$\Omega_{R+, R'} = \prod_{\alpha \in R' \cap -R+} U_\alpha \times_S T \times_S \prod_{\alpha \in R' \cap R+} U_\alpha \to G$$

induced by the product in G induces an open immersion

$$\Omega_{R+,R'} \to H_{R'}.$$

Indeed, by 5.4.3, this morphism factors through $H_{R'}$ and thus induces an immersion $\Omega_{R+,R'} \to H_{R'}$. One then argues as in 4.1.1.

Proposition 5.4.5. *Let (G,T,M,R) be an S-split group. Let R' and R'' be two subsets of R of type (R).*

(i) $H_{R'} \cap H_{R''}$ is smooth at every point of the unit section, $R' \cap R''$ is of type (R), and one has

$$(H_{R'} \cap H_{R''})^0 = H_{R' \cap R''}.$$

(ii) One has the equivalence

$$H_{R'} \subset H_{R''} \Leftrightarrow R' \subset R''.$$

Indeed, (ii) follows immediately from (i). To prove (i), it suffices to show that $H_{R'} \cap H_{R''}$ is smooth at every point of the unit section: its neutral component (cf. Exp. VI_B 3.10) will then be a subgroup of type (R) containing T, hence equal to $H_{R' \cap R''}$; but $\Omega_{R+,R'} \cap \Omega_{R+,R''} = \Omega_{R+,R' \cap R''}$ is an open subset of $H_{R'} \cap H_{R''}$ containing the unit section and smooth over S.

Corollary 5.4.6. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, s a point of S. If H and H' are two subgroups of type (R) of G containing T such that $H_s \subset H'_s$, there exists an open subset U of S containing s such that $H_U \subset H'_U$.*

Indeed, one may suppose G split relative to T. The assertion then follows immediately from 5.4.5 (ii).

One is led to ask which subsets R' of R are of type (R). One may suppose the group adjoint; one then has to verify that $g_{R'}$ is a Lie algebra and that its normalizer is smooth at every point of the unit section. The most important case is given by:

Theorem 5.4.7. *Every closed subset R' of R is of type (R). (Recall, cf. Exp. XXI 3.1.4, that $R' \subset R$ is called* closed *if $\alpha, \beta \in R'$, $\alpha + \beta \in R$ entail $\alpha + \beta \in R'$.)*

Remark 5.4.8. *We shall see later (Exp. XXIII 6.6) that if $6 \cdot 1_S \neq 0$[1324] (for example, if S has residue characteristic distinct from 2 and 3), the fact that $g_{R'}$ is a Lie algebra already entails that R' is closed, so R' is of type (R) if and only if it is closed. Theorem 5.4.7 thus gives all subsets of type (R) "independent of the characteristic".*

Let us first prove:

Lemma 5.4.9. *Choose for each $\alpha \in R$ an $X\alpha \in \Gamma(S, g\alpha)\times$. Let $\alpha, \beta \in R$, with $\alpha + \beta \neq 0$ and let q be the largest integer i such that $\alpha + i\beta \in R$. There exist sections $M\alpha, \beta, i \in \Gamma(S, O_S)$, uniquely determined, such that*

$$\mathrm{Ad}(\exp\alpha(xX\alpha))(X\beta) = X\beta + \sum_{i=1}^{q} M\alpha,\beta,i\, x^i X_{\beta+i\alpha},$$

for every $x \in Ga(S')$, $S' \to S$.

Indeed, $x \mapsto Ad(\exp\alpha(xX\alpha))(X\beta)$ defines a morphism $Ga,S \to W(g) \simeq Gm_{a,S}$. There thus exist sections $Y_n \in \Gamma(S, g)$, uniquely determined, such that

$$\mathrm{Ad}(\exp\alpha(xX\alpha))(X\beta) = \sum_{n \geq 0} x^n Y_n.$$

Applying the inner automorphism defined by a section t of T, one finds immediately

[1324] In fact, it suffices (cf. loc. cit.) that 2 and 3 be non-zero on S.

$$\mathrm{Ad}(t)(Y_n) = \beta(t)\,\alpha(t)^n\,Y_n,$$

which entails $Y_n \in \Gamma(S, g^{\beta+n\alpha})$. Since α and β are not proportional, none of the $\beta{+}n\alpha$ is zero; one thus has $Y_n = 0$ for $n > q$, $Y_n = M\alpha,\beta,nX_{\beta+n\alpha}$ for $0 \leqslant n \leqslant q$, where $M\alpha,\beta,n \in Ga(S)$ is uniquely determined. Setting $x = 0$ in the formula obtained, one finds $Y_0 = X\beta$, which completes the proof.

Remark 5.4.10. *Differentiating the preceding formula at $x = 0$, one finds*

$$[X\alpha, X\beta] = \begin{cases} N\alpha,\beta\, X_{\alpha+\beta}, & \text{where } N\alpha,\beta = M\alpha,\beta,1, \text{ if } \alpha+\beta \in R, \\ 0 & \text{if } \alpha+\beta \notin R,\ \alpha+\beta \neq 0. \end{cases}$$

Let us now prove 5.4.7. If R' is a closed subset of R, then $g_{R'}$ is a Lie subalgebra of g, by 5.4.10 and Exp. XX 2.10, formula (3). By 5.4.9 and Exp. XX 2.10, formula (2), $U\alpha$ normalizes $g_{R'}$ for each $\alpha \in R'$. Choose a system of positive roots $R+$ and consider the open set Ω_{R+} of 4.1.2; let $\Omega_{R+,R'}$ be the closed subscheme of Ω_{R+} defined as follows:

$$\Omega_{R+,R'} = \prod_{\alpha \in R' \cap -R+} U\alpha \cdot T \cdot \prod_{\alpha \in R' \cap R+} U\alpha.$$

The canonical immersion $\Omega_{R+,R'} \to G$ factors through $i : \Omega_{R+,R'} \to Norm_G(g_{R'})$. Suppose G adjoint; the tangent map of i at the points of the unit section is bijective by 5.3.2; in particular, the morphism i is étale at every point of the unit section, hence is a local immersion[1325] at every point of the unit section, hence $Norm_G(g_{R'})$ is smooth at every point of the unit section, as was to be shown.

5.5. Borel subgroups of a split reductive group

Proposition 5.5.1. *Let (G, T, M, R) be an S-split group. For every system of positive roots $R+$ of R, H_{R+} (which exists by 5.4.7) is a Borel subgroup of G and, for every ordering on $R+$, the morphism induced by the product in G*

$$T \times_S \prod_{\alpha \in R+} U\alpha \to G$$

is a closed immersion with image H_{R+}. One writes $B_{R+} = H_{R+}$.

By definition of the Borel subgroups, the first assertion may be verified by replacing S by the spectrum of an algebraically closed field. Let B be the Borel subgroup of G containing T and corresponding to the system of positive roots $R+$ (Bible, § 10.4, prop. 9); the Lie algebra of B is g_{R+}; one therefore has $B = H_{R+}$ by 5.3.5.

Let us prove the second assertion: the morphism in the statement induces an open immersion $i : T \times_S \prod_{\alpha \in R+} U\alpha \to H_{R+}$ (5.4.4). Now i is surjective (Bible, § 15.1, cor. 1 to prop. 1).

Corollary 5.5.2. *Choose an arbitrary ordering on $R+$ and for each $\alpha \in R+$ an $X\alpha \in \Gamma(S, g\alpha)\times$. Let $\alpha, \beta \in R+$. For each pair $(i, j) \in \mathbb{N}* \times \mathbb{N}*$ such that $i\alpha + j\beta \in R$, there exists a unique section*

[1325] By EGA IV₄, 17.11.2, i is étale at every point of the unit section (and $Norm_G(g_{R'})$ is smooth at every point of the unit section). Furthermore, let V be the largest open subset of $\Omega_{R+,R'}$ on which i is étale; since i is a monomorphism, i_V is an open immersion (ibid., 17.9.1).

$$C_{i,j,\alpha,\beta} \in \Gamma(S, O_S)$$

such that, for all $x, y \in Ga(S')$, $S' \to S$, one has

$$\exp_\alpha(xX_\alpha)\ \exp_\beta(yX_\beta)\ \exp_\alpha(xX_\alpha)^{-1} = \exp_\beta(yX_\beta) \prod_{i,j \in \mathbb{N}^*,\ i\alpha+j\beta \in R} \exp_{i\alpha+j\beta}(C_{i,j,\alpha,\beta}\, x^i y^j X_{i\alpha+j\beta}).$$

If $\alpha = \beta$, the assertion is trivial. Suppose therefore $\alpha \neq \beta$; then, by virtue of the proposition, there exist unique morphisms

$$F_0 : G^2_{a,S} \to T, \qquad F_\gamma : G^2_{a,S} \to G_{a,S} \qquad (\gamma \in R_+)$$

such that one has

$$\exp(xX_\alpha)\ \exp(yX_\beta)\ \exp(xX_\alpha)^{-1} = F_0(x, y) \prod_{\gamma \in R_+} \exp(F_\gamma(x, y) X_\gamma).$$

Let $t \in T(S')$, $S' \to S$. Let $int(t)$ act on this formula; one immediately has the relations

$$(1) \quad F_0(\alpha(t)\, x, \beta(t)\, y) = F_0(x, y),$$
$$(2) \quad F_\gamma(\alpha(t)\, x, \beta(t)\, y) = \gamma(t)\, F_\gamma(x, y).$$

Since α and β are two linearly independent characters (over $\mathbb{Q}$) of T, one concludes as usual from the first relation that F_0 is constant, so $F_0(x, y) = e$. Write next

$$F_\gamma(x, y) = \sum a_{ij}\, x^i y^j, \qquad \text{with } a_{ij} \in \Gamma(S, O_S).$$

Substituting in relation (2) and identifying the polynomials of the two sides, one finds

$$a_{ij}\, (\alpha(t)^i \beta(t)^j - \gamma(t)) = 0.$$

If $\gamma \neq i\alpha + j\beta$, one knows (Exp. XIX 4.13) that there exists an $S' \to S$ faithfully flat quasi-compact and a $t \in T(S')$ such that $\alpha(t)^i\beta(t)^j - \gamma(t) = 1$. One thus has $a_{ij} = 0$ on S', hence on S. If $\gamma = i\alpha + j\beta$, one sets $a_{ij} = C_{i,j,\alpha,\beta}$. Setting $x = 0$ (resp. $y = 0$), one finds $C_{0,1,\alpha,\beta} = 1$ (resp. $C_{1,0,\alpha,\beta} = 0$).

Remark 5.5.3. *Differentiating at $y = 0$ and comparing with 5.4.9, one finds*

$$C_{i,1,\alpha,\beta} = M\alpha, \beta, i.$$

Corollary 5.5.4. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, $\alpha \neq \beta$ two roots of G relative to T such that $\alpha + \beta$ is non-trivial on every fiber. Order the set of $i\alpha + j\beta$ ($i, j \in \mathbb{N}*$) in an arbitrary way. For all $i, j \in \mathbb{N}*$ such that $i\alpha + j\beta \in R$, there exists a unique morphism of O_S-modules*

$$f_{\alpha,\beta,i,j} : (g_\alpha)^{\otimes i} \otimes (g_\beta)^{\otimes j} \to g^{i\alpha+j\beta}$$

such that for every $S' \to S$ and every $X \in W(g_\alpha)(S')$, $Y \in W(g_\beta)(S')$ one has (the exp on the right being taken in the sense of 1.2[1326]):

[1326] That is, if $g^{i\alpha+j\beta} = 0$ on a connected component of S, the corresponding exponential is 1 on this component.

$$\exp_\alpha(X)\exp_\beta(Y)\exp_\alpha(-X) = \exp_\beta(Y)\prod_{(i,j)} \exp_{i\alpha+j\beta}(f_{\alpha,\beta,i,j}(X^i \otimes Y^j)).$$

The assertion is local for (fpqc). By § 2, one may therefore suppose G split relative to T, and α and β constant in the splitting. Since $\alpha+\beta \neq 0$, there exists a system of positive roots $R+$ containing α and β (Exp. XXI 3.5.4), and one is reduced to 5.5.2.

Corollary 5.5.5. *Let S be a scheme, G an S-reductive group.*

(i) G possesses Borel subgroups locally for the étale topology. If T is a maximal torus of G, then G also possesses Borel subgroups containing T locally for the étale topology.

(ii) If T is a maximal torus of G, the "functor of Borel subgroups of G containing T" is representable by a principal homogeneous bundle under $W_G(T)$.

(iii) If (G,T,M,R) is split, every Borel subgroup B of G containing T is locally on S of the form B_{R+}, where $R+$ is a system of positive roots of R.

(iv) If $T \subset B$ is a Killing couple of G, there exists a covering family $S_i \to S$ for the étale topology, and for each i a splitting $(G_{S_i}, T_{S_i}, M_i, R_i)$ and a system of positive roots R_{i+} of R_i such that $B_{S_i} = B_{R_{i+}}$.

Indeed, (i) follows from 2.3 and 5.5.1, (ii) from (i) and 5.3.15, (iii) from (ii) and 5.5.1, (iv) from (iii) and 2.3.

Lemma 5.5.6. *Choose on the group $\Gamma_0(R)$ generated by the roots a structure of totally ordered group such that the positive roots are the elements of $R+$ (cf. Exp. XXI 3.5.6).*[1327] *Let $\alpha_1 < \cdots < \alpha_N$ be the elements of $R+$. Consider the isomorphism*

$$f : T \times_S U_{\alpha_1} \times_S \cdots \times_S U_{\alpha_N} \to B_{R+}$$

induced by the product in G. Set for $i = 1, \cdots, N$:

$$U_{\geqslant i} = f(U_{\alpha_i} \times_S \cdots \times_S U_{\alpha_N}).$$

(i) Each $U_{\geqslant i}$ is a normal subgroup of B_{R+}.

(ii) For $1 \leqslant i \leqslant N-1$, $U_{\geqslant i}$ is identified with the semi-direct product

$$U_{\geqslant i} = U_{\alpha_i} \cdot U_{\geqslant i+1}.$$

(iii) B_{R+} is identified with the semi-direct product

$$B_{R+} = T \cdot U_{\geqslant 1}.$$

(iv) For $1 \leqslant i \leqslant N-1$, the inner automorphisms of $U_{\geqslant 1}$ act trivially on $U_{\geqslant i}/U_{\geqslant i+1}$ (which is identified with U_{α_i} by (ii)).

Let us first prove by induction on i the following assertion:

$U_{\geqslant i}$ is a normal subgroup of B_{R+}, semi-direct product of U_{α_i} and $U_{\geqslant i+1}$.

[1327] Note that such an ordering is necessarily compatible with ord_Δ, where $\Delta = S(R+)$ (cf. XXI 3.2.15).

The assertion is true for $i = N$; suppose it for $i + 1$ and let us prove it for i. One has (as schemes)

$$U_{\geqslant i} = U_{\alpha_i} \cdot U_{\geqslant i+1};$$

it is clear first that $U_{\geqslant i}$ is stable under the inner automorphisms of B_{R+}. This is clear for $int(t)$, $t \in T(S)$; it suffices to verify it for $int(x)$, $x \in U\alpha(S)$, $\alpha \in R+$. Now $U_{\geqslant i+1}$ is supposed normal, so it suffices to see that $int(x)U_{\alpha_i} \subset U_{\geqslant i}$. By 5.5.2, if $y \in U_{\alpha_i}(S')$, one has $y^{-1}xyx^{-1} \in U_{\geqslant i+1}(S')$, which entails $int(x)(y) \in U_{\geqslant i}(S')$.

Let us now prove that $U_{\geqslant i}$ is a subgroup of B_{R+}. If $x, y \in U_{\geqslant i}(S)$, one may write $x = px'$, $y = qy'$, with $p, q \in U_{\alpha_i}(S)$, and $x', y' \in U_{\geqslant i+1}(S)$. One has

$$xy = p x' q y' = pq(q^{-1} x' q) y' \in U_{\alpha_i}(S') U_{\geqslant i+1}(S');$$

similarly $x^{-1} = p^{-1}(px'^{-1}p^{-1}) \in U_{\alpha_i}(S')U_{\geqslant i+1}(S')$. We have thus proved (i) and (ii), as well as (iv) along the way. As for (iii), it is a trivial consequence of 5.5.1.

Lemma 5.5.7. *With the preceding notations, choose for each $1 \leqslant i \leqslant N$ an $X_i \in \Gamma(S, g^{\alpha_i})\times$ and consider the isomorphism*

$$a : G^N_{a,S} \to U_{\geqslant 1}$$

defined set-theoretically by

$$a(x_1, \dots, x_N) = \exp_{\alpha_1}(x_1 X_1) \cdots \exp_{\alpha N}(x_N X_N).$$

There exists a unique family (Q_i), $i = 1, \cdots, N$, of polynomials

$$Q_i = Q_i(x_1, \dots, x_N, y_1, \dots, y_N)$$

with coefficients in $\Gamma(S, O_S)$ such that one has set-theoretically

$$a(x_1, \dots, x_N)\, a(y_1, \dots, y_N) = a(Q_1(x_1, \dots, y_N), \dots, Q_N(x_1, \dots, y_N)).$$

Furthermore, the Q_i have coefficients in the subring of $\Gamma(S, O_S)$ generated by the $C_{i,j,\alpha,\beta}$ of 5.5.2 ($\alpha, \beta \in R+$, $i, j \in \mathbb{N}*$) and each Q_i is of the form

$$Q_i(x_1, \dots, y_N) = x_i + y_i + Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}).$$

The existence and uniqueness of the Q_i follow immediately from the fact that a is an isomorphism of schemes. Denoting z, z', z'' sections of $U_{\geqslant i+1}$, one has

$$\begin{aligned} &a(x_1, \dots, x_N)\, a(y_1, \dots, y_N) = \\ &\quad a(x_1, \dots, x_{i-1}, 0, \dots, 0) \exp(x_i X_i)\, z \cdot a(y_1, \dots, y_{i-1}, 0, \dots, 0) \exp(y_i X_i)\, z'; \end{aligned}$$

using 5.5.6 (i) and (iv), the right-hand side is written

$$a(x_1, \dots, x_{i-1}, 0, \dots, 0)\, a(y_1, \dots, y_{i-1}, 0, \dots, 0) \exp((x_i + y_i) X_i)\, z'';$$

which gives, reusing 5.5.6,

$$Q_i(x_1, \dots, x_N, y_1, \dots, y_N) = x_i + y_i + Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}),$$

with

$$Q'_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}) = Q_i(x_1, \dots, x_{i-1}, 0, \dots, 0, y_1, \dots, y_{i-1}, 0, \dots, 0);$$

that is, the precise form requested.

Let us prove finally the assertion on the coefficients of the polynomials Q_i. Let A be the subring of $\Gamma(S, O_S)$ generated by the $C_{i,j,\alpha,\beta}$ ($\alpha, \beta \in R+$, $i, j \in \mathbb{N}*$). Let us prove by descending induction on i that if $x_1 = \cdots = x_{i-1} = 0$ and $y_1 = \cdots = y_{i-1} = 0$, that is, if $a(x_1, \cdots, x_N)$ and $a(y_1, \cdots, y_N)$ are sections of $U_{\geqslant i}$, then the polynomials

$$R_j(x_i, \dots, x_N, y_i, \dots, y_N) = Q_j(x_1, \dots, x_N, y_1, \dots, y_N)$$

have coefficients in A. This is trivial for $i = N$ and also for $j < i$ (because $R_j = 0$ for $j < i$). Let $i < N$, suppose the assertion verified for $i + 1$, and let us prove it for i (and $j \geqslant i$). One has

$$a(0, \dots, 0, x_i, \dots, x_N) = \exp(x_i X_i)\, a(0, \dots, 0, x_{i+1}, \dots, x_N) = \exp(x_i X_i)\, Z_i.$$

Similarly write

$$a(0, \dots, y_i, \dots, y_N) = \exp(y_i X_i)\, T_i.$$

One has

$$a(0, \dots, x_i, \dots, x_N)\, a(0, \dots, y_i, \dots, y_N) = \exp((x_i + y_i) X_i)\, int(\exp(-y_i X_i))(Z_i) \cdot T_i.$$

Now

$$int(\exp(-y_i X_i))(Z_i) = int(\exp(-y_i Y_i))(\exp(x_{i+1} X_{i+1}) \cdots \exp(x_N X_N))$$

is a product of $N - i - 1$ sections of $U_{\geqslant i+1}$ whose coefficients in the decomposition $U_{\geqslant i+1} = U_{\alpha_{i+1}} \cdots U_{\alpha N}$ are polynomials in y_i and $x_{i+1}, \dots, x_N$ with coefficients in A (by 5.5.2). Applying the induction hypothesis, one deduces that the coefficients of

$$int(\exp(-y_i X_i))(Z_i) \cdot T_i$$

are also polynomials with coefficients in A, which finishes the proof.

Let us remark that the preceding induction immediately gives a proof of:

Lemma 5.5.8. *With the notations of 5.5.6, let, for each $i = 1, \cdots, N$, a morphism of groups*

$$f_i : U_{\alpha_i} \to H,$$

where H is an S-group functor. For the morphism

$$f : U_{\geqslant 1} \to H$$

defined by

$$f(\exp(x_1 X_1) \cdots \exp(x_N X_N)) = f_1(\exp(x_1 X_1)) \cdots f_N(\exp(x_N X_N))$$

to be a morphism of groups, it is necessary and sufficient that for every pair $i < j$, one has

$$f_j(\exp(x_j X_j))\, f_i(\exp(x_i X_i))\, f_j(\exp(-x_j X_j)) = f(\exp(x_j X_j) \exp(x_i X_i) \exp(-x_j X_j)).$$

5.6. Subgroups of type (R) with solvable fibers

Proposition 5.6.1. *Let (G, T, M, R) be an S-split group, R' a subset of R of type (R) (5.4.2), $H_{R'}$ the corresponding subgroup of G. The following conditions are equivalent:*

(i) $H_{R'}$ has solvable geometric fibers.

(ii) There exists a system of positive roots $R+$ such that $R' \subset R+$, hence $H_{R'} \subset B_{R+}$ (cf. 5.4.5).

(iii) $R' \cap -R' = \emptyset$.

(iv) For every ordering on R', the morphism induced by the product in G

$$T \times_S \prod_{\alpha \in R'} U\alpha \to H_{R'}$$

is an isomorphism.

(v) $H_{R'} \cap Norm_G(T) = T$.

(vi) For every subset R'' of R, of type (R), one has (cf. 5.4.5)

$$H_{R'} \cap Norm_G(H_{R''}) = H_{R' \cap R''}.$$

We shall prove these equivalences according to the logical scheme

$$(iii) \Leftrightarrow (ii) \Rightarrow (vi) \Rightarrow (iv) \Rightarrow (v) \Rightarrow (i) \Leftrightarrow (ii).$$

One obviously has (ii) ⇒ (iii) and (vi) ⇒ (v) (take $R'' = \emptyset$). By 5.4.6, it suffices to verify (i) ⇒ (ii) on geometric fibers; now if S is the spectrum of an algebraically closed field, $H_{R'}$ is contained in a Borel subgroup containing T, so of the form H_{R+} (5.5.5 (iii)).

Similarly (iii) ⇒ (i) is verified on geometric fibers; suppose (iii) is satisfied; if $H_{R'}$ were not solvable, there would exist a subtorus Q of T, of codimension 1 in T, such that $Centr_{H_{R'}}(Q)$ is not solvable (Bible, § 10.4, prop. 8); now $Centr_{H_{R'}}(Q)$ has Lie algebra $g_{R''}$, where R'' is the set of roots of R' vanishing on Q, so $R'' = \emptyset$ or α (by virtue of (iii)); hence $Centr_{H_{R'}}(Q)$, which is a subgroup of type (R) of G, is T or $T \cdot U\alpha$, hence solvable, contrary to the hypothesis.

Similarly (ii) ⇒ (iv) is verified on geometric fibers (since these are flat S-schemes of finite presentation); by Bible, § 13.2, th. 1 d), the morphism in question is bijective; it induces an isomorphism on the tangent spaces at the origin, and one concludes as usual (cf. 4.1.1).

One has (iv) ⇒ (v) by 4.2.7. To prove (v) ⇒ (i), one is again reduced to the case where S is the spectrum of an algebraically closed field, and one concludes by Bible, § 10.3, cor. to prop. 6 and § 9.3, cor. 3 to th. 1.

It thus remains only to prove the assertion (ii) ⇒ (vi). One may reduce to the case where G is adjoint. One then has, by 5.3.3,

$\mathrm{Norm}_G(H_{R''}) = \mathrm{Norm}_G(g_{R''}) \subset \mathrm{Transp}_G(t, g_{R''})$.

[1328] By 5.4.5, it suffices to prove

(x) $H_{R'}(S) \cap \mathrm{Transp}_{G(S)}(t, g_{R''}) \subset H_{R' \cap R''}(S)$.

Let us first prove a lemma.

Lemma 5.6.2. *In the notations of 5.5.7, let*

$u = \exp(x_1 X_1) \cdots \exp(x_N X_N)$

where $x_i \in Ga(S)$. Let m be an integer, $1 \leqslant m \leqslant N$, such that $x_i = 0$ for $i < m$.

(a) If $H \in \Gamma(S,t)$, the component of $Ad(u)\ H$ on g^{α_m} is

$$-\alpha_m(H)x_mX_m.$$

(b) If $Y \in \Gamma(S, g^{-\alpha_m})$, the component of $Ad(u)\ Y$ on t is (with the notations of Exp. XX 2.6)

$x_m \langle X_m, Y\rangle H_{\alpha_m}$.

Denote indeed $u = g^{\alpha_{m+1}} + \cdots + g^{\alpha_N}$. By 5.4.9, one has

$\mathrm{Ad}(\exp(x_i X_i))\, u \subset u, \quad \text{for } i > m.$

By Exp. XX 2.10, one has

$\mathrm{Ad}(\exp(x_i X_i))\, H = H - \alpha_i(H)\, x_i\, X_i$.

This immediately gives, modulo u,

$\mathrm{Ad}(u)\, H = \mathrm{Ad}(\exp(x_m X_m))\, H = H - \alpha_m(H)\, x_m\, X_m$,

which entails the first result.

Likewise denote[1329] $n = g^{\alpha_1} + \cdots + g^{\alpha_N}$ and $u_1 = \exp(x_{m+1}X_{m+1}) \cdots \exp(x_N X_N)$. For $i > m$, one has $\alpha_i > \alpha_m$ so, by 5.4.9, one has, modulo n,

$\mathrm{Ad}(u_1)\, Y \equiv Y, \quad \text{whence} \quad \mathrm{Ad}(u)\, Y \equiv \mathrm{Ad}(\exp(x_m X_m))\, Y.$

Applying Exp. XX 2.10, one therefore obtains, modulo n,

$\mathrm{Ad}(u)\, Y - Y \equiv x_m \langle X_m, Y\rangle H_{\alpha_m}$

[1328] The inclusion follows from $t \subset g_{R''}$.
[1329] We have corrected the original in what follows.

whence the second result.

Let us return to the proof of inclusion (x). Suppose that there exists $h \in H_{R'}(S)$, $h \notin H_{R' \cap R''}(S)$, such that

$$Ad(h)t \subset g_{R''}.$$

One may write

$h = t \exp(x_1 X_1) \cdots \exp(x_N X_N)$.

Since $h \notin H_{R' \cap R''}(S)$, there exists a smallest m such that

$t \exp(x_1 X_1) \cdots \exp(x_{m-1} X_{m-1}) \in H_{R' \cap R''}(S) \quad \text{and} \quad \alpha_m \notin R'', \quad x_m \neq 0$.

Then $h' = \exp(x_m X_m) \cdots \exp(x_N X_N)$ also satisfies the conditions imposed on h above. But by 5.6.2, for every $H \in \Gamma(S,t)$ the component of $Ad(h')H$ on g^{α_m} is $-\alpha_m(H)x_m X_m$. By virtue of the hypothesis on h and on m, one has therefore $\alpha_m(H) = 0$ for every $H \in \Gamma(S,t)$, which is impossible (because G is supposed adjoint and α_m is therefore non-zero on every fiber).

Remark 5.6.2. bis. *Resume the notations of 5.6.2. If $Ad(u)$ is the identity on t and on $g^{-\alpha_m}$, one has $x_m = 0$. Indeed, one has $x_m \alpha_m = 0$ and $x_m H_{\alpha_m} = 0$; if $\alpha_m \notin 2M$, then α_m is non-zero on every fiber; if $\alpha_m \in 2M$, then $\alpha*_m \notin 2M*$ and H_{α_m} is non-zero on every fiber; in each case, this entails $x_m = 0$. It follows that $u = e$ if $Ad(u)$ operates trivially on g.*

Remark 5.6.3. *If H is a subgroup of type (R) of the reductive group G, with solvable geometric fibers, then H is closed in G and $Norm_G(H)/H$ is representable by a separated finite étale S-scheme.*

This follows from 5.3.18 and, at one's choice, 3.5 or Exp. XIX 2.5 (ii).

Corollary 5.6.4. *Let (G,T,M,R) be a split reductive group. If $R' \subset R$ is closed and $R' \cap -R' = \emptyset$, then R' is contained in a system of positive roots.*[1330]

Indeed, R' is of type (R) by 5.4.7, so the result follows from 5.6.1.

Corollary 5.6.5. *Under the conditions of 5.6.1, the product in G induces an isomorphism*

$\prod_{\alpha \in R'} U_\alpha \xrightarrow{\sim} U_{R'}$,

where $U_{R'}$ is a closed group subscheme of G, smooth over S, with connected and unipotent geometric fibers, independent of the choice of ordering on R'. Moreover, $H_{R'}$ is the semi-direct product $T \cdot U_{R'}$ ($U_{R'}$ normal).

Indeed, if $R' \subset R+$, then $H_{R'} \cap U_{\geqslant 1}$ (notations of 5.5.6) is a closed subgroup of G of finite presentation, normal in $H_{R'}$. By 5.6.1 (iv), one has $H_{R'} = T \cdot U_{R'}$, which entails the other assertions.

[1330] This is in fact a statement about root systems, completing Exposé XXI (cf. [BLie], VI § 1.7, Prop. 22) and proved here by an indirect route.

Remark 5.6.6. *In particular, U_{R+} is the group $U_{\geqslant 1}$ of 5.5.6.*

Let us extract some corollaries of the preceding results concerning groups of type $U_{R'}$.

Corollary 5.6.7. *Let (G, T, M, R) be a split reductive group, R' and R'' two subsets of R of type (R), with $R' \cap -R' = \emptyset$.*

(i) One has

$$U_{R'} \cap \mathrm{Norm}_G(H_{R''}) = U_{R' \cap R''}.$$

(ii) Suppose R' closed. If the conditions $\alpha \in R'$, $\beta \in R''$ and $\alpha+\beta \in R$ entail $\alpha+\beta \in R'$, then $H_{R''}$ normalizes $U_{R'}$.

Indeed, (i) follows immediately from 5.6.5 and 5.6.1 (vi). To prove (ii), it suffices, by 5.4.4, to show that T and each $U\beta$, $\beta \in R''$, normalize $U_{R'}$.[1331] For T, this is trivial; for $U\beta$, it follows from 5.5.2 and Exp. XXI 3.1.2.[1332]

Corollary 5.6.8. *Let (G, T, M, R) be an S-split group, $R+$ a system of positive roots, α a simple root of $R+$ (i.e. an element of $R+$ such that $R + -\alpha$ is closed). Denote*

$$U_{\hat{\alpha}} = U_{R+-\alpha}.$$

Then

(i) $U_{\hat{\alpha}}$ is a normal subgroup of B_{R+}.

(ii) U_{R+} is the semi-direct product of $U_{\hat{\alpha}}$ by $U\alpha$.

(iii) $U_{-\alpha}$ normalizes $U_{\hat{\alpha}}$.

(iv) $Z\alpha$ normalizes $U_{\hat{\alpha}}$.

If one defines similarly $U_{-\hat{\alpha}} = U_{R---\alpha}$ (where $R- = -R+$), one has

$$\Omega_{R+} = U_{-\hat{\alpha}} \cdot U_{-\alpha} \cdot T \cdot U\alpha \cdot U_{\hat{\alpha}}.$$

Indeed, (ii) follows from 5.6.5, and (i) from 5.6.7 (ii). Similarly, (iii) follows from 5.5.2 (indeed, if $\beta \in R+$, $\beta \neq \alpha$, no combination $i(-\alpha) + j\beta$, with $i, j > 0$, can be negative because β contains at least one simple root $\neq \alpha$). Then (iv) follows from (i) and (iii), because $U_{-\alpha} \cdot T \cdot U\alpha$ is schematically dense in $Z\alpha$. Finally, the last assertion follows from (ii) and its analogue for U_{R-}.

Let us return to the general situation.

[1331] Indeed, G having connected fibers is separated over S by VI_B 5.5, so by XI 6.11 (see also the addition 6.5.5 in VI_B), $Norm_G(U_{R'})$ is represented by a closed group subscheme N of G. If N contains T and the $U\beta$, for $\beta \in R''$, it then contains the big cell $\Omega_{R+,R''}$; now this is schematically dense in $H_{R''}$ by 5.4.4 and EGA IV_3, 11.10.10 (the fibers of $H_{R''}$ are integral and $\Omega_{R+,R''}$ contains a non-empty open subset of each fiber). It follows that $H_{R''} \subset N$, hence $H_{R''}$ normalizes $U_{R'}$.

[1332] One must see that, under the hypotheses of (ii), if $\alpha \in R'$ and $\beta \in R''$, then all roots of the form $i\alpha + j\beta$ with $i, j \in \mathbb{N}*$ belong to R', and for this we have replaced the reference XXI 2.3.5 by XXI 3.1.2. This may also be seen directly by inspection of the rank 2 root systems.

Proposition 5.6.9. *Let S be a scheme, G an S-reductive group, H a subgroup of type (R) with solvable geometric fibers.*

(i) $D_S(H) = \mathrm{Hom}_{S-gr.}(H, Gm,S)$ is representable by a twisted constant S-group, whose type at $s \in S$ is $\mathbb{Z}^{rg_{red}(G_s)}$. The biduality morphism (Exp. VIII § 1)

$$f : H \to D_S(D_S(H))$$

is smooth and surjective.

(ii) The kernel H_u of f is the largest closed normal group subscheme of H, smooth over S, with connected and unipotent geometric fibers. We call it the unipotent part *of H and write also $H_u = rad_u(H)$.*

Then H_u is also the sheaf of commutators of H: every morphism of groups from H to an S-presheaf of commutative groups separated for (fppf) vanishes on H_u and thus factors through $H/H_u = D_S(D_S(H))$.

(iii) If T is a maximal torus of H, the morphism $T \to H$ induces isomorphisms $D_S(H) \xrightarrow{\sim} D_S(T)$ and $T \xrightarrow{\sim} D_S(D_S(H))$. Furthermore, H is identified with the semi-direct product of H_u by T.

(iv) In the situation of 5.6.1, if $H = H_{R'}$, then $H_u = U_{R'}$.

The assertions of the proposition are local for the étale topology (Exp. X 5.5). One may therefore reduce to the case of 5.6.1. One then knows (5.6.5) that $H_{R'}$ is the semi-direct product of $U_{R'}$ by T. Let us show that $U_{R'}$ is the sheaf of commutators of $H_{R'}$: since $H_{R'}/U_{R'} = T$ is commutative, it suffices to prove that every morphism of groups $\phi : H_{R'} \to V$ as in (ii) vanishes on $U_{R'}$. It suffices to prove that ϕ vanishes on each $U\alpha$, $\alpha \in R'$. Now if $t \in T(S')$, $X \in W(g\alpha)(S')$, one has

$$1 = \varphi(t \exp_\alpha(X)\, t^{-1} \exp_\alpha(-X)) = \varphi(\exp_\alpha((\alpha(t) - 1) X)).$$

Since $\alpha : T \to Gm,S$ is faithfully flat, one deduces immediately that ϕ vanishes on $U\times_\alpha$; but every section of $U\alpha$ is locally the sum of two sections of $U\times_\alpha$. One thus has

$$\mathrm{Hom}_{S\text{-}gr.}(H, V) = \mathrm{Hom}_{S\text{-}gr.}(H/U_{R'}, V)$$

for every V as above. Applying this result to $V = Gm,S$, one deduces immediately (i) and (iii), then (iv) and the second assertion of (ii). It now suffices to prove the first assertion of (ii); the only non-trivial fact is that every closed normal subgroup U of H, smooth over S with connected unipotent geometric fibers, is a subgroup of H_u. Now one first has:

Lemma 5.6.10. *Let G be an S-reductive group, T a maximal torus, U a group subscheme of G, smooth over S, with unipotent geometric fibers, normalized by T. Then $U \cap T = e$.*

Indeed, since $T = Centr_G(T)$, one has $U \cap T = U^T$ (invariants under $int(T)$). Applying Exp. XIX 1.4, one deduces that $U \cap T$ is smooth over S, but it is also radicial over S:

for every $s \in S$, $U(s) \cap T(s)$ consists of elements that are simultaneously unipotent and semisimple. This proves the lemma.

Let us return to the proof of 5.6.9 (ii). If U is a normal subgroup of H as above, then the semi-direct product $T \cdot U$ is a subgroup of type (R) of G, with solvable geometric fibers. One may therefore suppose it of the form $H_{R''}$, with $R'' \subset R'$. It suffices to prove $U = U_{R''}$ and one is therefore reduced to the case where $H = T \cdot U$; but the quotient H/U being commutative, U is a subsheaf of the sheaf of commutators of H, which is H_u. QED.

Let us remark that we have in fact just proved:

Proposition 5.6.11. *Let S be a scheme, G an S-reductive group, T a maximal torus of G. The maps*

$$H \mapsto H_u, \quad U \mapsto T \cdot U$$

are mutually inverse bijections between the set of subgroups H of type (R) of G containing T and having solvable geometric fibers, and the set of subgroups U of G, smooth over S, normalized by T, with connected and unipotent geometric fibers.[1333]

In particular, when (G,T,M,R) is split, the groups $H_{R'}$ and $U_{R'}$ correspond.

Corollary 5.6.12. *Let S be a scheme, (G,T,M,R) an S-split group (resp. and $R+$ a system of positive roots of R defining the Borel subgroup B).*

Every smooth group subscheme of G with connected and unipotent geometric fibers (resp. every smooth group subscheme of B_u) normalized by T is locally on S of the form $U_{R'}$, where R' is a subset of R contained in a system of positive roots (resp. a subset of $R+$) of type (R).

For the "resp." case, it suffices to note that the geometric fibers of the given group are unipotent and connected, by Bible, § 13.2, th. 1 (d).

Proposition 5.6.9 has moreover the following corollary:

Corollary 5.6.13. *Let S be a scheme, G an S-reductive group, H a subgroup of type (R) with solvable geometric fibers, $Tor(H)$ the functor of maximal tori of H:*

$$Tor(H)(S') = \{\text{maximal tori of } H_{S'}\}.$$

Then $Tor(H)$ is representable by an affine and smooth S-scheme, which is a principal homogeneous bundle under H_u for the law $(h,T) \mapsto int(h)T$.[1334]

Indeed, if T and T' are two maximal tori of $H_{S'}$, there exists a unique section $h \in H_u(S')$ such that $int(h)T = T'$. The uniqueness of h follows immediately from the equality

[1333] We have removed the hypothesis that U be closed, which is automatically verified. Indeed, for such a U, one has $U \cap T = e$ by 5.6.10, so the semi-direct product $H = T \cdot U$ is a subgroup of type (R) of G (cf. 5.2.1), with solvable geometric fibers. So, by 5.6.3, H is closed in G, and since U is closed in H, it is closed in G.

[1334] This re-proves and refines XII 1.10 (for G reductive).

$$Norm_G(T) \cap H_u = e$$

(cf. for example 5.6.1); it therefore suffices to prove the existence of h locally for the étale topology. By 5.2.6 and 5.1.2 (a), one may suppose T and T' conjugate by a section of H, whence the desired conclusion since $H = H_u \cdot T$ by 5.6.9 (iii). It follows that $Tor(H)$ is a principal homogeneous sheaf under H_u, which is affine and smooth over S, which immediately entails the statement.[1335]

5.7. Bruhat's theorem

Recall 5.7.1. *Let k be an algebraically closed field, G a k-reductive group, B a Borel subgroup of G, T a maximal torus of B, $N = Norm_G(T)$. Then*

$$G(k) = B(k)\, N(k)\, B(k);$$

this is Bruhat's theorem (Bible, § 13.4, cor. 1 to th. 3); more precisely, with the notations of 3.6, the sets

$$B(k)\, N_w(k)\, B(k) = B_u(k)\, N_w(k)\, B_u(k)$$

form, as w runs through $(N/T)(k)$, a partition of $G(k)$. If B' is another Borel subgroup of G containing T, the sets $B'(k)N_w(k)B(k)$ also form a partition of $G(k)$. Indeed, if $y \in N(k)$ is such that $int(y)B = B'$, one has

$$y\, B(k)\, N_w(k)\, B(k) = B'(k)\, N_{yw}(k)\, B(k).$$

Definition 5.7.2. *Let (G,T,M,R) be an S-split group, $R-$ a system of positive roots of R, $B' = B_{R-}$ the Borel group it defines. For $w \in W$, one writes (cf. 5.6.5):*

$$R^w_- = R- \cap w(R-), \qquad B'u_w = U_{R^w_-} = \prod_{\alpha \in R^w_-} U\alpha.$$

If $n_w \in Norm_G(T)(S)$ is a representative of w (3.8), one may also write

$$B'u_w = B'u \cap \mathrm{int}(n_w)\, B'u.$$

Lemma 5.7.3. *Let (G,T,M,R) be an S-split group, $R+$ a system of positive roots of R, $R- = -R+$, B (resp. B') the Borel subgroup of G defined by $R+$ (resp. $R-$). Let $w \in W$, N_w and $B'u_w$ the corresponding subschemes of G (3.8 and 5.7.2).*

(i) The sheaf $B' \cdot N_w \cdot B$, image of the morphism

$$B' \times_S N_w \times_S B \to G$$

induced by the product in G, is representable by a subscheme of G (in fact a closed subscheme of the open set $n_w\Omega_{R+}$).

(ii) The morphism

$$B'u_w \times_S N_w \times_S B^u \to G,$$

[1335] By SGA 1, VIII 2.1 and EGA IV_4, 17.7.1.

induced by the product in G, is an immersion with image the preceding subscheme.

Let us first show that the morphism of (ii) is an immersion. By definition, $int(n_w)^{-1}$ induces a closed immersion of $B'u_w$ into $B'u$, so the morphism

$$(u, b) \mapsto n_w^{-1} u n_w b$$

induces a closed immersion

$$B'u_w \times_S B \to \Omega_{R+}.$$

This immediately entails that the morphism of (ii) induces a closed immersion of the first member into the open set $n_w\Omega_{R+}$. To prove (i), it suffices to see that

$$B'(S) N_w(S) B(S) = B'u_w(S) N_w(S) B^u(S).$$

Now, if $\alpha \in R$, one has $int(n_w)U\alpha(S) = U_{w(\alpha)}(S)$, so if $w^{-1}(\alpha) \in R+$,

$$\begin{aligned} U\alpha(S) N_w(S) B(S) &= U\alpha(S) n_w T(S) B^u(S) \\ &= n_w U_{w^{-1}(\alpha)}(S) T(S) B^u(S) \\ &= n_w B(S) = N_w(S) B^u(S). \end{aligned}$$

This immediately entails, in view of the definition of R_-^w, the desired assertion.

Theorem 5.7.4. *Let S be a scheme, (G,T,M,R) an S-split group, B the Borel subgroup defined by the system of positive roots $R+$, B' the Borel subgroup defined by $R- = -R+$.*

(i) (Bruhat's theorem) The schemes $B'u_w \cdot N_w \cdot B$ form, as w runs through W, a partition of the underlying set of G.

(ii) For each $w \in W$, let n_w be a representative of w in $Norm_G(T)(S)$ (3.8); then the open sets $n_w\Omega = n_w B'u \cdot T \cdot B^u$ form, as w runs through W, a cover of G.

The two assertions are verified on geometric fibers, where one concludes by 5.7.1 and 5.7.3.

Remark 5.7.5. *(i) entails that if S is the spectrum of a field, $G(S)$ is the disjoint union of the $B'u_w(S) \cdot T(S) \cdot B^u(S)$. The corresponding assertion for an arbitrary S (even local or artinian) is obviously false. Note however that (ii) entails that if S is local, $G(S)$ is the union of the $n_w\Omega(S)$. In fact:*

Corollary 5.7.6. *Let Δ be a system of simple roots of the split group G over the local scheme S.*

(i) Then $G(S)$ is generated by $T(S)$ and the $U\alpha(S)$, $\alpha \in \Delta \cup -\Delta$.

(ii) If G is simply connected (4.3.3), $G(S)$ is already generated by the $U\alpha(S)$, $\alpha \in \Delta \cup -\Delta$.

Indeed, let H be the subgroup of $G(S)$ generated by the $U\alpha(S)$, $\alpha \in \Delta \cup -\Delta$. Let us first remark that H contains a representative of each $s\alpha$ ($\alpha \in \Delta$) in $Norm_G(T)(S)$ (Exp. XX 3.1), hence a representative n_w of each $w \in W$.

Since every $\alpha \in R$ is written $w(\alpha_0)$ with $w \in W$, $\alpha_0 \in \Delta$, one has

$U\alpha(S) = \mathrm{int}(n_w)\, U_{\alpha_0}(S) \subset H.$

The subgroup generated by H and $T(S)$ thus contains $\Omega(S)$ and is therefore the whole of $G(S)$, by the remark made above.

If now G is simply connected, let us prove that $H \supset T(S)$. By Exp. XX 2.7, H contains $\alpha * (Gm(S))$ for every $\alpha \in \Delta$, and it suffices to apply 4.3.8.

Remark 5.7.6.1. *Instead of taking, for each $\alpha \in \Delta$, $U\alpha(S)$ and $U_{-\alpha}(S)$, one may content oneself with taking $U\alpha(S)$ and a representative $w\alpha$ of the symmetry $s\alpha$, or else $U\alpha(S)$ and a section of $U\times_{-\alpha}$, ….*

Corollary 5.7.7. *If G is of semisimple rank 1, choose a $u\alpha \in U \times_\alpha (S)$. Then Ω and $u\alpha\Omega$ form a cover of G.*

Indeed, if $u_{-\alpha}$ is the section of $U_{-\alpha}$ paired with $u\alpha$ (cf. 1.3), one has, by 5.7.4 (ii),

$G = \Omega \cup u_{-\alpha}^{-1}\, u\alpha\, u_{-\alpha}^{-1}\, \Omega,$

whence

$G = u_{-\alpha}\, G = u_{-\alpha}\, \Omega \cup u\alpha\, u_{-\alpha}^{-1}\, \Omega = \Omega \cup u\alpha\, \Omega.$

Corollary 5.7.8. *Let S be a scheme, G an S-reductive group. Then G is essentially free over S (Exp. VIII 6.1).*[1336]

Indeed, the assertion is local for the (fpqc) topology; one may suppose G split. Then G admits a cover by open subsets isomorphic to $G_{a,S}^N \times_S G_{m,S}^n$, which are essentially free.

Lemma 5.7.9. *Under the conditions of 5.7.4, let α be a simple root of $R+$ and $u\alpha \in U \times_\alpha (S)$. For every $v \in U_{-\alpha}(S)$, one has*

$\Omega \cdot v \subset \Omega \cup u\alpha \cdot \Omega.$

We have to compare two open subsets of G; it suffices to do so when S is the spectrum of a field k. One thus has to prove

$\Omega(k)\, v \subset \Omega(k) \cup u\alpha\, \Omega(k).$

Now

$\Omega(k)\, v = B'u(k)\, T(k)\, B^u(k)\, v = U_{-\alpha}^{*}(k)\, U_{-\alpha}(k)\, T(k)\, U\alpha(k)\, U_{\alpha}^{*}(k)\, v$
$\quad \subset U_{-\alpha}^{*}(k)\, Z\alpha(k)\, U_{\alpha}^{*}(k)\, v.$

(One uses the decomposition of 5.6.8.) Applying now 5.6.8 (iii) and using 5.7.7 for the group $Z\alpha$, one obtains

$\Omega(k)\, v \subset U_{-\alpha}^{*}(k)\, Z\alpha(k)\, v\, U_{\alpha}^{*}(k) \subset U_{-\alpha}^{*}(k)\, Z\alpha(k)\, U_{\alpha}^{*}(k)$
$\quad \subset U_{-\alpha}^{*}(k)\, U_{-\alpha}(k)\, T(k)\, U\alpha(k)\, U_{\alpha}^{*}(k) \quad \cup \quad U_{-\alpha}^{*}(k)\, u\alpha\, U_{-\alpha}(k)\, T(k)\, U\alpha(k)\, U_{\alpha}^{*}(k).$

Using again 5.6.8 (iii) (for $R-$ instead of $R+$), one obtains the result.

[1336] See also the additions made in VI_B, 6.2.1 to 6.2.6 and 6.5.2 to 6.5.5.

Proposition 5.7.10. *Under the conditions of 5.7.4, choose for each simple root α a $u\alpha \in U\times_\alpha(S)$. Let U_1 be the submonoid of $B^u(S)$ generated by the $u\alpha$. The open sets $u\Omega$, for $u \in U_1$, form a cover of G.*

Once again, one may suppose that S is the spectrum of a field k; by virtue of 5.7.6, it suffices to prove that $\bigcup_{u\in U_1} u\Omega(k)$ is stable under right multiplication by $T(k)$, $U\alpha(k)$, $U_{-\alpha}(k)$ (for α simple). In the first two cases, this is trivial. In the last, it follows from the lemma.

Remark 5.7.11. *Let us point out a particular case of 5.7.2. If $w = s\alpha$ is the symmetry with respect to the simple root α, then*

R− ∩ sα(R−) = R− − {−α}

(Exp. XXI 3.3.1), and, in the notations of 5.6.8, one therefore has

$$B'u_{s\alpha} = U_{-\hat{\alpha}}.$$

Remark 5.7.12. *In fact, the proof of 5.7.10 immediately gives the following statement: under the conditions of 5.7.10, let Γ be a submonoid of $G(S)$; for the open sets $g\Omega$ ($g \in \Gamma$) to form a cover of G, it is necessary and sufficient that for every $s \in S$ and every simple root α, one has*

(uα)_s B′u(s) ⊂ Γ · B′u(s) · T(s) · B^u(s).

Remark 5.7.13. *By 5.5.5 (iii), reasoning as in 5.7.1, one obtains immediately the following variant of 5.7.4: let (G,T,M,R) be an S-split group, B and B' two Borel subgroups of G containing T; for every $w \in W$, the sheaf $B'\cdot N_w\cdot B$ is representable by a subscheme of G; these subschemes form, for $w \in W$, a partition of the underlying set of G. One may also give the analogue of 5.7.3 (ii): one must set*

B′u_w = B′u ∩ int(n_w) B̃^u

where $\tilde{B}$ is the Borel subgroup "opposite" to B relative to T (cf. 5.9.2).

Proposition 5.7.14. *Let S be a scheme, G an S-reductive group, and*

$$Ad : G \to GL_{O_S}(g)$$

its adjoint representation. Then $Ker(Ad) = Centr(G)$ (in other words, the canonical homomorphism deduced from Ad by passage to the quotient:

Ad : G/Centr(G) = ad(G) → GL_{O_S}(g)

is a monomorphism).[1337]

One may suppose G split. Choose on $\Gamma_0(R)$ a total ordering structure compatible with the group structure and let $R+$ be the set of positive roots. By virtue of 5.7.4 (ii) and 4.1.6, it suffices to prove that if n_w is a representative of the element w of

[1337] It is in fact a closed immersion, by Exp. XVI 1.5 (a).

W, if $u \in U(S)$, $t \in T(S)$, $v \in U_-(S)$, and if $Ad(n_w vtu) = id$, then $w = e$, $v = e$, $u = e$. For each $m \in R \cup 0$, set

$$g_{>m} = \bigoplus_{n > m} g^n, \qquad g_{<m} = \bigoplus_{n < m} g^n.$$

Let $X \in \Gamma(S, g^m)$; write $Ad(tu)X = Ad(v^{-1}n_w^{-1})X$. Now

$$Ad(t)\, Ad(u)\, X - m(t)\, X \in \Gamma(S, g_{>m}),$$
$$Ad(v^{-1}\, n_w^{-1})\, X - Ad(n_w^{-1})\, X \in \Gamma(S, g_{<w^{-1}(m)}).$$

If $w \neq e$, there exists $\alpha \in R$ such that $w^{-1}(\alpha) < \alpha$, and setting $m = \alpha$, one obtains a contradiction because

$$Ad(tu)\, X \in \Gamma(S, g^\alpha + g_{>\alpha}) \cap \Gamma(S, g^{w^{-1}(\alpha)} + g_{<w^{-1}(\alpha)}) = 0.$$

Hence $w = e$, and one may choose $n_w = e$; one then has

$$Ad(v^{-1})\, X - X \in \Gamma(S, g_{<m} \cap (g^m + g_{>m})) = 0,$$

whence $Ad(v)X = X$ for every $X \in \Gamma(S, g^m)$, so $Ad(v) = id$. Similarly $Ad(u) = id$. One concludes by 5.6.2 bis.

5.8. Schemes associated with a reductive group

Theorem 5.8.1. *Let S be a scheme, G an S-reductive group. Let H be the functor of subgroups of type (R) of G: for every $S' \to S$, $H(S')$ is the set of subgroups of type (R) of $G_{S'}$ (cf. 5.2.1). Then H is representable by a quasi-projective S-scheme of finite presentation over S.*

[1338] Let $G' = G/Centr(G)$ be the adjoint group of G (4.3.6) and u the morphism $G \to G'$. By Exp. XII 7.12, the map $H' \mapsto u^{-1}(H')$ establishes a bijection between the set of subgroups of type (R) of G' and of G (and this remains valid after any base change). Thus, replacing G by G', one may suppose that G is adjoint. Consider then the morphism

$$u : H \to Grass(g)$$

which associates with each subgroup of type (R) its Lie algebra (which is a sub-$\mathcal{O}_S$-module of g that is locally a direct factor.[1339]). Then u is a monomorphism by 5.3.3. It suffices to prove that it is representable by an immersion of finite presentation, in other words to prove the following assertion: for every $S_1 \to S$, given a sub-$\mathcal{O}_S$-module locally a direct factor h of g_{S_1}, the $S' \to S_1$ such that $h_{S'}$ is the Lie algebra of a subgroup of type (R) of $G_{S'}$ are exactly those that factor through some subscheme Σ of finite presentation of S_1. Replacing S_1 by S, we reduce to $S_1 = S$, and we may furthermore suppose S affine; then there exists a noetherian affine scheme S_0 such that G (resp. h) arises by base change from an adjoint S_0-reductive group G_0 (resp. a sub-$\mathcal{O}_S$-module locally a direct factor h_0 of $g_0 = Lie(G_0/S_0)$). It suffices to show that there exists a subscheme Σ_0 of S_0 with the required properties (because one

[1338] We have expanded the original in what follows.

[1339] Cf. Exp. II 4.11.8.

will then have $\Sigma = \Sigma_0 \times_{S_0} S$). Replacing S by S_0, one may therefore suppose S affine and noetherian (note that then every subscheme of S is of finite presentation over S). Finally, replacing S by a sufficiently small open set, one may suppose that g is free of rank n and that h is a direct factor, free of rank r.

One must first write that $h_{S'}$ is a Lie subalgebra of $g_{S'}$, i.e. that the morphism induced by the Lie bracket:

```
φ : h ⊗ h --[,]--> g
```

factors through h. If $(e_1, \cdots, e_n)$ is a basis of g such that $(e_1, \cdots, e_r)$ is a basis of h, then ϕ is given by sections a_k^{ij} of `O_S` (where $i, j = 1, \cdots, r$ and $k = 1, \cdots, n$), and the preceding condition is equivalent to saying that $S' \to S$ factors through the closed subscheme of S defined by the equations $a_k^{ij} = 0$ for $k = r+1, \cdots, n$ and $i, j = 1, \cdots, r$. Replace S by this closed subscheme.

Then, by 5.3.0, $N = Norm_G(h)$ is a closed group subscheme of G of finite presentation over G. One must now write (cf. 5.3.1) that $N_{S'}$ is smooth at every point of the unit section, of relative dimension $r = rank(h)$, and that the inclusion of $h_{S'}$ in $Lie(N_{S'}/S')$ is an equality.

[1340] Since N is affine over S (being closed in G), the unit section $\epsilon : S \to N$ is a closed immersion, so $\epsilon(S)$ is defined by a quasi-coherent ideal J of $A(N)$. Note that $n_{S/N} = J/J^2$ is identified with $\epsilon * (\Omega^1_{N/S})$, so its formation commutes with every base change $S' \to S$. By the equivalence (c′) ⇔ (a) in EGA IV₄, 17.12.1 (applied to $f : N \to S$ and $j = \epsilon$), $N_{S'} \to S'$ is smooth, of relative dimension r, at every point of $\epsilon(S')$ if and only if $n_{S'/N'} = n_{S/N} \otimes_{O_S} O_{S'}$ is locally free of rank r and the morphism $\phi_n(S') : Sym^n(n_{S'/N'}) \to J^n_{S'}/J^{n+1}_{S'}$ is an isomorphism for every $n \geqslant 1$. Denote $K_n(S') = Ker\phi_n(S')$. By TDTE I, Lemma 3.6, $n_{S/N} \otimes_{O_S} O_{S'}$ is locally free of rank r if and only if $S' \to S$ factors through some subscheme Z of S. Replacing S by Z, we may therefore suppose that $n_{S/N} = J/J^2$ is locally free of rank r. Then, for every $S' \to S$, one has $(J^2/J^3) \otimes_{O_S} O_{S'} = J^2_{S'}/J^3_{S'}$ and hence $K_2(S') = K_2(S) \otimes_{O_S} O_{S'}$. It follows that $\phi_2(S')$ is an isomorphism if and only if $S' \to S$ factors through the closed subscheme S_2 of S defined by the ideal generated by the image of $Sym^2(n_{S/N}) * \otimes K_2(S)$ in `O_S`. Then, over S_2, J^2/J^3 is isomorphic to $Sym^2(n_{S/N})$, hence locally free, and the same argument shows that $\phi_3(S')$ is an isomorphism if and only if $S' \to S$ factors through some closed subscheme S_3 of S, etc. One obtains thus that $N_{S'} \to S'$ is smooth, of relative dimension r, at every point of $\epsilon(S')$ if and only if $S' \to S$ factors through the closed subscheme Z intersection of the S_n. But then, for every $S' \to Z$, $Lie(N_{S'}/S')$ is locally a direct factor of rank r of $g_{S'}$, and so the inclusion $h_{S'} \subset Lie(N_{S'}/S')$ is an equality. One then sets $H = N^0$.

Replacing S by Z, it remains only to express that $H_{s'}$ is of the same reductive rank as $G_{s'}$ at every point $s' \in S'$, or, equivalently, that H_s is of the same reductive rank

[1340]The original next indicated that, denoting $n_{N/G}$ the inverse image of the conormal sheaf $N_{N/G}$ by the unit section $S \to N$, the condition that $n_{N_{S'}/G_{S'}} \to \omega^1_{G_{S'}/S'}$ be universally injective is equivalent to the fact that $S' \to S$ factors through some open subscheme of S. We were unable to justify this point, in view of the fact that the formation of $N_{N/G}$ does not commute with base change, and we have replaced this argument with the one that follows, indicated by O. Gabber.

as G_s at every point s of the (set-theoretic) image of S' in S. Now this condition defines an open subset of S (Exp. XIX 6.2).

Remark. *In general, the scheme H is not smooth over S. It is however smooth if 6 is invertible on S, or if there exists a prime number p such that $p \cdot 1_S = 0$ (i.e. if S is of characteristic $p > 0$).*

Corollary 5.8.2. *Let S be a scheme, G an S-reductive group, H a subgroup of type (R) of G. (Recall (5.3.10) that $Norm_G(H)$ is representable by a closed group subscheme of G, smooth over S.)*

Then the quotient sheaf $G/Norm_G(H)$ is representable by a quasi-projective S-scheme, smooth and of finite presentation over S (which is in fact an open subset of H).

Indeed, consider the morphism

$$f : G \to H,$$

defined set-theoretically by $f(g) = int(g)H$. By 5.3.9, this morphism is smooth and of finite presentation, hence open. Let $V = f(G)$ equipped with its structure of open subscheme of H. The morphism $f : G \to V$ is covering and of kernel $Norm_G(H)$, which proves that $G/Norm_G(H)$ is representable by V (cf. Exp. IV 4.6.5).

Corollary 5.8.3. *Let S be a scheme, G an S-reductive group. Consider the functors $Tor(G)$, $Bor(G)$, $Kil(G)$ defined by*

Tor(G)(S′) = {maximal tori of G_{S′}},
Bor(G)(S′) = {Borel subgroups of G_{S′}},
Kil(G)(S′) = {Killing couples of G_{S′} (cf. 5.3.13)}.

(i) $Tor(G)$, $Bor(G)$, $Kil(G)$ are representable by smooth S-schemes of finite presentation, with integral geometric fibers, and respectively affine, projective, affine over S.

(ii) The canonical morphism $Kil(G) \to Tor(G)$ (resp. $Kil(G) \to Bor(G)$) is finite étale surjective (resp. affine smooth surjective).

(iii) Let T be a maximal torus of G (resp. B a Borel subgroup of G, resp. $B \supset T$ a Killing couple of G). The morphism

G → Tor(G),   resp. G → Bor(G),   resp. G → Kil(G)

defined by

g ↦ int(g) T,   resp. g ↦ int(g) B,   resp. g ↦ (int(g) B, int(g) T)

induces an isomorphism

G/Norm_G(T) Tor(G),   resp. G/B Bor(G),   resp. G/T Kil(G).

One sees first that (iii) follows from the conjugacy theorem for maximal tori (resp. Borel subgroups, resp. Killing couples) and the fact that

Norm_G(B) = B,   Norm_G(B) ∩ Norm_G(T) = T,

all results previously established (5.1.2, 5.3.12, 5.3.14, 5.6.1).

It follows first that the canonical morphisms

Tor(G) → H,   Bor(G) → H

are representable, locally for the étale topology, by open immersions (5.8.2 and 5.1.2, resp. 5.5.5), hence by descent that $Tor(G)$ and $Bor(G)$ are representable by open subsets of H. Similarly $Kil(G)$ is locally (for the étale topology) representable by a scheme affine over the base (Exp. IX 2.3), hence representable by an affine S-scheme, by descent of affine schemes.[1341]

The assertions of (ii) follow immediately from 5.5.5 (ii) and 5.6.13. It already follows that $Tor(G)$ is affine over S (EGA II 6.7.1). It thus remains only to prove the fact that $Bor(G)$ is projective over S. We already know that it is quasi-projective; it remains to prove that it is proper;[1342] but it has connected fibers, so by EGA IV_3, 15.7.10, one is reduced to proving it on geometric fibers; if S is the spectrum of an algebraically closed field, one has $Bor(G) = G/B$ by (iii) and one concludes by Bible, § 6.4, th. 4 (or [Ch05], § 6.5, th. 5).

Remark 5.8.4. *Under the conditions of 5.8.3, let Q be a central subgroup of multiplicative type of G. The obvious morphisms define isomorphisms*

Tor(G) ≃ Tor(G/Q),   Bor(G) ≃ Bor(G/Q),   Kil(G) ≃ Kil(G/Q).

Corollary 5.8.5. *Let S be a scheme, G an S-reductive group, P a group subscheme of G, smooth and of finite presentation over S. The following conditions are equivalent:*

(i) For each $s \in S$, P_s is a parabolic subgroup of G_s (i.e. the quotient scheme G_s/P_s is proper over s, or equivalently P_s contains a Borel subgroup of G_s, cf. Bible, § 6.4, th. 4 or [Ch05], § 6.5, th. 5).

(ii) The quotient sheaf G/P is representable by an S-scheme that is smooth and projective over S.

Moreover, under these conditions, P is closed in G, with connected fibers, and one has $P = Norm_G(P)$.

One obviously has (ii) ⇒ (i). If (i) holds, then $P(s) = Norm_{G(s)}(P_s)$ and P_s is connected (for the first point, cf. Bible, § 12.3, lemme 4;[1343] the second point follows from this, because $P' = P_s^0$ is a parabolic subgroup of G_s normalized by P_s, whence $P'(s) = P(s)$ and thus $P' = P_s$); it follows that P is of type (R), and that P equals $Norm_G(P)$, hence is closed in G. By 5.8.2, $G/P = G/Norm_G(P)$ is representable by a quasi-projective S-scheme. Its fibers are connected and proper, hence it is projective by the reasoning of 5.8.3.

[1341] Cf. SGA 1, VIII 2.1.

[1342] One may suppose S affine and, since $Bor(G)$ is of finite presentation over S by (i), reduce to the case where S is noetherian; one is then under the hypotheses of EGA IV_3, 15.7.10.

[1343] We have expanded what follows.

Remark 5.8.6. *The statements 5.8.1, 5.8.2, 5.8.5 are valid for an S-group of type (RA), or for an S-group of type (RR) satisfying 5.1.8.*[1344]

Remark 5.8.7. *Through the inner automorphisms of G, one has canonical operations:*

```
G → Aut_S(Tor(G)),    G → Aut_S(Bor(G)),    G → Aut_S(Kil(G)),
```

which, in the situation of 5.8.3 (iii), are identified with the canonical operations

```
G → Aut_S(G/Norm_G(T)),    G → Aut_S(G/B),    G → Aut_S(G/T).
```

One concludes in particular that

```
Ker(G → Aut_S(Tor(G))) = Ker(G → Aut_S(Bor(G))) = Ker(G → Aut_S(Kil(G))) = Centr(G).
```

It is indeed clear that $Centr(G)$ operates trivially on each of the three schemes. Conversely, the kernel of $G \to \mathrm{Aut}_S(Kil(G))$ is "the intersection of the maximal tori of G" in the sense of 4.1.7, hence equals $Centr(G)$ (loc. cit.). For $Bor(G)$, one notes that "the intersection of the Borel subgroups of G" is also "the intersection of its maximal tori" (see the following section). For $Tor(G)$, one uses Exp. XII 4.11.

5.9. Properties peculiar to Borel subgroups

Most of these properties will be generalized in Exp. XXVI to parabolic subgroups.

Definition 5.9.1. *Let S be a scheme, G an S-reductive group, B and B' two Borel subgroups of G. We say that B and B' are in* general position *(or that B' is in general position relative to B) if $B \cap B'$ is a torus (necessarily maximal) of G.*

If T is a maximal torus of G contained in B and B', we say that B and B' are opposite *(relative to T) if $B \cap B' = T$.*

Proposition 5.9.2. *Let S be a scheme, G an S-reductive group, B a Borel subgroup of G, T a maximal torus of B. There exists a unique Borel subgroup B' of G, opposite to B relative to T.*

If (G, T, M, R) is a splitting of G relative to T and if $B = B_{R+}$ (5.5.1), then $B' = B_{-R+}$.

By faithfully flat descent, it suffices to prove the proposition in the split case, when $B = B_{R+}$ (5.5.5 (iv)). Then B_{-R+} is indeed opposite to B (4.1.2); let us show that it is the only Borel subgroup of G containing T that is opposite to B. If B' is a Borel subgroup of G containing T, then B' is locally on S of the form $B_{R'+}$, where $R'+$ is another system of positive roots of R (5.5.5 (iii)). If $R'+ \neq -R+$, there exists $\alpha \in R' + \cap R+$, so that $U\alpha \subset B_{R+} \cap B_{R'+}$.

Proposition 5.9.3. *Let S be a scheme, G an S-reductive group, B a Borel subgroup of G.*

(i) If B' is a Borel subgroup of G, the following conditions are equivalent:

[1344] Indeed, the proof of 5.8.1 only uses 5.3.3 (valid for a group of type (RA)) and XIX 6.2 which, by XII 1.7 (b), is also valid for groups of type (RR).

(a) B' is in general position relative to B (5.9.1).

(b) $B'u \cap B^u = e$.

(b′) $B'u \cap B = e$.

(c) The product in G induces an open immersion $B'u \times_S B \to G$.

(c′) The canonical morphism $B'u \to G/B$ is an open immersion.

(ii) The functor $Opp(B)$:

$S' \mapsto \{\text{Borel subgroups of } G_{S'} \text{ in general position relative to } B_{S'}\}$

is representable by an open subscheme of $Bor(G)$ (5.8.3). The morphism

$$Opp(B) \to Tor(B)$$

defined by $B' \mapsto B \cap B'$ is an isomorphism. In particular (5.6.13) the inner automorphisms of B^u equip $Opp(B)$ with a structure of principal homogeneous bundle under B^u.

Let us first examine (i). One has (a) ⇒ (c): indeed, (c) is local for the étale topology; by 5.5.5 (iv), one reduces to the case where G is split relative to $B \cap B'$ and B of the form B_{R+}; by 5.9.2, one then has $B'u = U_{-R+}$ and one is reduced to 4.1.2.

One trivially has (c′) ⇔ (c) ⇒ (b′) ⇒ (b). It therefore remains to prove (b) ⇒ (a). Let us first prove (ii): the second assertion is a formal consequence of 5.9.2, the third follows immediately from this by 5.6.13; let us prove the first; it is local for the étale topology and one may therefore suppose that B possesses a maximal torus T; let B'_0 be opposite to B relative to T (5.9.2).

By what precedes, the morphism $B^u \to Bor(G)$ induced by the canonical morphism $G \to G/B'_0 \to Bor(G)$ (5.8.3) induces an isomorphism $B^u \xrightarrow{\sim} Opp(B)$. One thus has a commutative diagram

$$\begin{tikzcd} B^u \arrow[r] \arrow[d] & G/B'_0 \arrow[d] \\ Opp(B) \arrow[r] & Bor(G). \end{tikzcd}$$

Now the morphism $B^u \to G/B'_0$ is an open immersion (by (i) (a) ⇒ (c′)), which finishes the proof of (ii). Let us immediately note the corollary:

Corollary 5.9.4. *Let G be an S-reductive group and B and B' two Borel subgroups of G. If $s \in S$ is such that B_s and B'_s are in general position, there exists an open subset V of S containing s such that B_V and B'_V are in general position.*

It only remains to prove (b) ⇒ (a). By virtue of the preceding corollary, it suffices to do so when S is the spectrum of an algebraically closed field k. One may suppose G split relative to a maximal torus T of B. Let B'_0 be the Borel subgroup opposite to B. Since the Borel subgroups of G are conjugate under $G(k)$, there exists $g \in G(k)$

such that $int(g)B'_0 = B'$. By Bruhat's theorem (5.7.4), one may write $g = bnb'$, with $b \in B(k)$, $b' \in B'_0(k)$, $n \in Norm_G(T)(k)$. One thus has

$$B' = int(b)\, int(n)\, B'_0$$

and $B' \cap B = int(b)(int(n)B'_0 \cap B)$. If $n \notin T(k)$, $int(n)B'u_0 \cap B^u \neq e$ (cf. proof of 5.9.2); it follows that (b) entails $B' \cap B = int(b)(B'_0 \cap B) = int(b)T$. QED.

Proposition 5.9.5. *Let S be a scheme, G an S-reductive group, B a Borel subgroup of G, B^u its unipotent part. There exists a sequence of subgroups of B:*

$$U_0 = B^u \supset U_1 \supset \cdots \supset U_n \supset \cdots$$

possessing the following properties:

(i) Each U_i is smooth, with connected fibers, characteristic in B; the inner automorphisms of B^u act trivially on the (sheaf) quotients U_i/U_{i+1}.

(ii) For each $i \geqslant 0$, there exists a locally free $\mathcal{O}_S$-module E_i and an isomorphism of S-sheaves of groups

$$U_i/U_{i+1} \xrightarrow{\sim} W(E_i).$$

(iii) For every $s \in S$, $(U_n)_s = e$ for $n \geqslant \dim(B^u_s)$.

Suppose first that there exists a splitting (G,T,M,R) of G and a system of positive roots $R+$ of R such that $B = B_{R+}$. We denote by Δ the set of simple roots of $R+$; for each $\alpha \in R+$, we write $ord(\alpha)$ for the sum of the coefficients of α in the basis Δ of $\Gamma_0(R)$; it is the *order* of α relative to $R+$. One has $ord(\alpha) \leqslant Card(R+)$. For every $i > 0$, let $R^{(i)}$ be the set of roots of order $> i$; it is a closed set of positive roots, so one may construct (5.6.5)

$$U_i = U_{R^{(i)}}.$$

If $\alpha \in R+$ and $\beta \in R^{(i)}$, then $\alpha + \beta \in R^{(i+1)}$. It follows, by 5.5.2, that each U_i is a normal subgroup of B and that the inner automorphisms of B^u operate trivially on U_i/U_{i+1}. This group is moreover identified with

$$\prod_{ord(\alpha)=i+1} U\alpha$$

and is thus equipped with a vector-space structure.

If B is of the form $B_{R'}$ for another splitting (G,T',M',R') of G, let us show that the groups U'_i constructed as above using the new splitting coincide with the U_i and that the vector-space structures on the successive quotients also coincide. By 5.6.13, there exists $b \in B^u(S)$ such that $T' = int(b)T$; the assertion to be proved is local on S and one may therefore suppose that the isomorphism $T \xrightarrow{\sim} T'$ induced by $int(b)$ arises by duality from an isomorphism of root data

$h : (M', M'^*, R', R'^*) \square (M, M^*, R, R^*).$

It is clear that the roots of $R'+$ are the $\alpha \circ int(b) = h(\alpha)$, $\alpha \in R+$, and that the simple roots of $R'+$ are the $\alpha \circ int(b) = h(\alpha)$, $\alpha \in \Delta$, hence $ord(h(\alpha)) = ord(\alpha)$ for $\alpha \in R+$. On the other hand, it is clear by transport of structure that the vector groups $U'_{h(\alpha)}$ are none other than the $int(b)U\alpha$. One thus has $int(b)U_i = U'_i$; but U_i being invariant, this gives $U_i = U'_i$.

Similarly the isomorphism of vector groups

$$int(b) : U_i/U_{i+1} \xrightarrow{\sim} U'_i/U'_{i+1}$$

is the identity, by virtue of what has already been proved.

Let us now treat the general case. There exists a covering family for the étale topology $S_i \to S$ and for each i a splitting (G_i, T_i, M_i, R_i) and a system of positive roots R_{i+} of R_i such that $B \times_S S_i = B_{R_{i+}}$ (5.5.5, (iii)). For each i, one thus has a family

$B_{S_i} = U_{i,0} \supset U_{i,1} \supset \cdots \supset U_{i,j} \supset \cdots$

and vector-space structures on the $U_{i,j}/U_{i,j+1}$. By descent, it suffices to prove that for every pair (i, i') and every j, one has

$U_{i,j} \times_{S_i} S_{ii'} = U_{i',j} \times_{S_{i'}} S_{ii'}$

(one writes $S_{ii'} = S_i \times_S S_{i'}$) and that the vector-space structures on the quotients

$(U_{i,j}/U_{i,j+1}) \times_{S_i} S_{ii'}$ and $(U_{i',j}/U_{i',j+1}) \times_{S_i} S_{ii'}$

coincide. Now if $S_{ii'} = \emptyset$, this is trivial; if $S_{ii'} \neq \emptyset$, then one is in the situation studied above: $B \times_S S_{ii'}$ is defined by the system of positive roots R_{i+} (resp. $R_{i'+}$) in the splitting $(G_{S_{ii'}}, T_i \times_{S_i} S_{ii'}, M_i, R_i)$ (resp. in the splitting $(G_{S_{ii'}}, T_{i'} \times_{S_{i'}} S_{ii'}, M_{i'}, R_{i'})$).

Corollary 5.9.6. *If S is affine, $H^1(S, B^u) = e$, i.e. every principal bundle under B^u possesses a section.*

Indeed, S decomposes as a direct sum of subschemes on each of which B^u is of constant relative dimension. One may therefore, by 5.9.5 (iii), suppose that there exists an n such that $U_n = e$. Since, by TDTE I, B.1.1,[1345]

$H^1(S, U_i/U_{i+1}) = H^1(S, W(E_i)) = 0,$

one has $H^1(S, B^u) = 0$.

Corollary 5.9.7. *If S is affine, B possesses maximal tori. If T is a maximal torus of B, one has $H^1(S, T) = H^1(S, B)$.*

[1345] This is the corollary on page 18 of TDTE I.

The first assertion follows immediately from 5.9.6 and 5.6.13; the second follows in standard fashion.[1346]

Corollary 5.9.8. *If G is an S-reductive group, the canonical morphism (cf. 5.8.3)*

$$Kil(G) \to Bor(G)$$

possesses sections over every affine open subset.

Corollary 5.9.9. *Under the conditions of 5.9.5, suppose S affine; then there exists a locally free O_S-module E such that B^u is, as a scheme, S-isomorphic to $W(E)$.*

Let us show by induction on i that B^u/U_i is S-isomorphic to $W(E_0 \bigoplus \cdots \bigoplus E_{i-1})$. This is clear for $i = 0$; suppose $i \geqslant 1$. Then B^u/U_i is a principal homogeneous bundle of base $X = B^u/U_{i-1}$ under the group $(U_{i-1}/U_i)_X$. Since B^u/U_{i-1} is affine, by the induction hypothesis, and since $U_{i-1}/U_i = W(E_{i-1})$, this bundle is trivial. One thus has (at least) an isomorphism of S-schemes

$$B^u/U_i \simeq (B^u/U_{i-1}) \times_S W(E_{i-1}) = W(E_0 \oplus \cdots \oplus E_{i-1}).$$

One concludes immediately by condition (iii) of 5.9.5.

Corollary 5.9.10. *Let S be a semi-local scheme, s_i its closed points, B a Borel subgroup of the S-reductive group G. The canonical map*

$$B^u(S) \to \prod_i B^u(\operatorname{Spec} \kappa(s_i))$$

is surjective.

Indeed, if $S = \operatorname{Spec}(A)$, $\kappa(s_i) = A/p_i$ and if E is given by the A-module E, one has

$$B^u(S) = E \otimes A, \qquad B^u(\operatorname{Spec} \kappa(s_i)) = E \otimes_A (A/p_i).$$

The assertion then follows immediately from the fact that $A \to \prod_i A/p_i$ is surjective.

5.10. Subgroups of type (R) with reductive fibers

Proposition 5.10.1. *Let (G, T, M, R) be an S-split group, R' a subset of R of type (R) (5.4.2), $H_{R'}$ the corresponding subgroup of G. The following conditions are equivalent:*

(i) $H_{R'}$ is reductive (i.e. has reductive geometric fibers).

(ii) One has $R' = -R'$, i.e. R' is symmetric.

Moreover, under these conditions, $(H_{R'}, T, M, R')$ is a splitting of $H_{R'}$; for every system of positive roots $R+$ of R, $R'+ = R' \cap R+$ is a system of positive roots of R' and

[1346] Indeed, one has an exact sequence $H^1(S, B^u) \to H^1(S, B) \xrightarrow{\pi} H^1(S, B/B^u) = H^1(S, T)$, see [Se64], I § 5.5, Prop. 38 or [Gi71], III Prop. 3.3.1. Now $H^1(S, T) \to H^1(S, B)$ is a section of π, so π is surjective; on the other hand $H^1(S, B^u) = 0$ by 5.9.6.

$$B_{R+} \cap H_{R'} = H_{R'+}$$

is a Borel subgroup of $H_{R'}$, whose unipotent part is

$$U_{R+} \cap H_{R'} = U_{R'+}.$$

One has obviously (i) ⇒ (ii) (it suffices to check it fiber by fiber and R' is a system of roots of $H_{R'}$ relative to T). To prove (ii) ⇒ (i), one remarks by 5.4.3 that

H_{R′} ∩ Zα = Centr_{H_{R′}}(Tα) = Zα

for every $\alpha \in R'$ and one applies the criterion of Exp. XIX 1.12.

If $R+$ is a system of positive roots of R, then $R'+ = R + \cap R'$ is obviously a closed subset of R' such that $R' + \cup - R'+ = R'$ and $R' + \cap - R'+ = \emptyset$, hence a system of positive roots of R'. The other two assertions follow respectively from 5.6.1 (vi) and 5.6.7 (i).

Corollary 5.10.2. *Let S be a scheme, G an S-reductive group, H an S-reductive subgroup such that for every $s \in S$, G_s and H_s have the same reductive rank. Then H is closed in G, $Norm_G(H)$ is smooth over S, $Norm_G(H)/H$ is representable by a finite étale S-scheme.*

If T is a maximal torus of H and B a Borel subgroup of G containing T, then $B \cap H$ is a Borel subgroup of H, whose unipotent part is $(B \cap H)_u = B^u \cap H$.

The first assertions follow immediately from 5.3.10 and 5.3.18, via the fact that the Weyl groups of G and H are finite (Exp. XIX 2.5). The other assertions are local for the étale topology and reduce to the case studied in 5.10.1.

Proposition 5.10.3. *Let S be a scheme, G an S-reductive group.*

(a) If Q is a torus of G, $Centr_G(Q)$ is a subgroup of type (R) of G with reductive fibers. If $Q \subset Q'$ are two tori of G, then $Centr_G(Q) \supset Centr_G(Q')$.

(b) If H is a subgroup of type (R) of G with reductive fibers, then $rad(H)$ (4.3.6) is a torus of G. If $H \subset H'$ are two subgroups of type (R) of G with reductive fibers, then $rad(H) \supset rad(H')$.

(c) If Q is a torus of G, one has

rad(Centr_G(Q)) ⊃ Q   and   Centr_G(rad(Centr_G(Q))) = Centr_G(Q).

(d) If H is a subgroup of type (R) of G with reductive fibers, one has

Centr_G(rad(H)) ⊃ H   and   rad(Centr_G(rad(H))) = rad(H).

Indeed, (a) follows immediately from Exp. XIX 2.8. To prove (b), it suffices to note that $rad(H') \subset H$, because H contains (locally for (fpqc)) a maximal torus of G, hence of H'. The first assertion of (c) (resp. (d)) is trivial, the second follows by the usual reasoning.

This proposition leads to the following definition:

Definition 5.10.4. *Let S be a scheme, G an S-reductive group, H a reductive subgroup of type (R) of G, and Q a subtorus of G.*[1347]

1) We say that H is a critical subgroup *if it is the centralizer of its radical.*

2) We say that Q is a C-critical torus *if it is the radical of its centralizer.*

It then follows from Proposition 5.10.3:

Corollary 5.10.5. *(i) For every subtorus Q of G, $Centr_G(Q)$ is critical.*

(ii) For every subgroup of type (R) with reductive fibers H of G, $rad(H)$ is a C-critical torus of G.

(iii) The maps

```
Q ↦ Centr_G(Q),    H ↦ rad(H)
```

are mutually inverse bijections between the set of C-critical tori of G and the set of its critical reductive subgroups of type (R).

(iv) If Q is a torus of G, $rad(Centr_G(Q))$ is the smallest C-critical torus of G containing Q.

(v) If H is a reductive subgroup of type (R) of G, $Centr_G(rad(H))$ is the smallest critical reductive subgroup of type (R) of G containing H.

Remark 5.10.5.1.[1348] *(1) A torus T of G is a critical subgroup of G if and only if it is a maximal torus.*

(2) In what follows, "critical torus" means "C-critical torus".

Proposition 5.10.6. *Let (G,T,M,R) be an S-split group, R' a subset of R. The following conditions are equivalent:*

(i) R' is of type (R), $H_{R'}$ is reductive and critical.

(ii) There exists a system of simple roots Δ of R and a subset Δ' of Δ such that R' is the set of elements of R that are linear combinations of elements of Δ'.

(iii) R' is closed, symmetric, and every system of simple roots of R' is the intersection with R' of a system of simple roots of R.

Indeed, by Exp. XXI 3.4.8, (ii) and (iii) are equivalent and are also equivalent to the fact that R' is the intersection of R with a $\mathbb{Q}$-subspace of $M \otimes \mathbb{Q}$. Now this last condition is entailed by (i): if $H_{R'} = Centr_G(Q)$, then R' is the set of elements of R that vanish on Q (Exp. II 5.2.3 (ii)). Finally, this condition entails (i), because

[1347] We have modified the original, introducing the terminology "C-critical torus" instead of "critical torus", in order to avoid confusions in later references (cf. Exp. XXVI, 3.9). We have also expanded the statement of 5.10.5 and added Remark 5.10.5.1.

[1348] We have modified the original, introducing the terminology "C-critical torus" instead of "critical torus", in order to avoid confusions in later references (cf. Exp. XXVI, 3.9). We have also expanded the statement of 5.10.5 and added Remark 5.10.5.1.

$rad(H_{R'})$ is the maximal torus of $\bigcap_{\alpha\in R'} Ker(\alpha)$, hence $Centr_G(rad(H_{R'}))$ is none other than $H_{R''}$ where R'' is the intersection of R with the subspace generated by R'.

5.10.7. Let us summarize some of the preceding results: let (G,T,M,R) be an S-split group, and let Δ be a system of simple roots of R and $R+$ the corresponding system of positive roots; choose a subset Δ' of Δ, denote R' the set of elements of R that are linear combinations of elements of Δ' and set $R'+ = R' \cap R+$. Let $T_{\Delta'}$ be the maximal torus of $\bigcap_{\alpha\in\Delta'} Ker(\alpha)$ and $Z_{\Delta'} = Centr_G(T_{\Delta'})$.

Then $Z_{\Delta'}$ is a reductive subgroup of G, with radical $T_{\Delta'}$; $(Z_{\Delta'},T,M,R')$ is an S-split group; $B_{R+} \cap Z_{\Delta'}$ is the Borel subgroup of $Z_{\Delta'}$ defined by the system of positive roots $R'+$ (or by the system of simple roots Δ') and its unipotent part is $U_{R+} \cap Z_{\Delta'} = U_{R'+}$.

Remark 5.10.8. *Under the conditions of 5.10.4, let Q be a critical torus of G, $L = Centr_G(Q)$ its centralizer. Since $Q = rad(L)$, then Q is a characteristic subgroup of L; it follows immediately that*

$$Norm_G(L) = Norm_G(Q),$$

hence also

```
Norm_G(L)/L = Norm_G(Q)/Centr_G(Q) = W_G(Q).
```

By 5.10.2, one deduces:

Proposition 5.10.9. *Let S be a scheme, G an S-reductive group, Q a critical torus of G. The Weyl group $W_G(Q)$ is (étale) finite over S.*

Remark 5.10.10. *Under the conditions of 5.10.7, one may make explicit*

$$W_G(T_{\Delta'}) = Norm_G(Z_{\Delta'})/Z_{\Delta'}.$$

It is the constant group associated with the quotient W_1/W_2, where W_1 is the subgroup of W formed by the elements that normalize the subgroup of M generated by Δ', and W_2 is the subgroup of W generated by the $s\alpha$, $\alpha \in \Delta'$.

5.11. Subgroups of type (RC)

Definition 5.11.1. *Let S be a scheme, G an S-reductive group. A group subscheme H of G is called* of type (RC) *if it is of type (R), i.e. (5.2.1) satisfies the following two conditions:*

(i) H is smooth over S, with connected fibers;

(ii) for every $s \in S$, H_s contains a maximal torus of G_s;

and if it further satisfies the following condition:

(iii) for every $s \in S$ and every maximal torus T of H_s, the set of roots of H_s relative to T is a closed subset of the set of all roots of G_s relative to T.

Remark 5.11.2. *As we have already noted in 5.4.8, condition (iii) is a consequence of the others when 6 is invertible on S.*[1349]

Lemma 5.11.3. *Let (G, T, M, R) be an S-split group and R' a closed subset of R. Let*

$R_1 = \{\alpha \in R' \mid -\alpha \in R'\}$ and $R_2 = \{\alpha \in R' \mid -\alpha \notin R'\}$.

Then R_1 and R_2 are closed. Consider the groups $H_{R'}$, H_{R_1} and U_{R_2} (5.4.7 and 5.6.5), which are smooth with connected fibers.

(i) The group U_{R_2} is normal in $H_{R'}$ and $H_{R'}$ is the semi-direct product of U_{R_2} by H_{R_1}.

(ii) H_{R_1} is reductive, U_{R_2} has connected and unipotent geometric fibers; every normal subgroup of $H_{R'}$, smooth over S and with connected and unipotent geometric fibers, is contained in U_{R_2}, and every reductive subgroup of $H_{R'}$ containing T is contained in H_{R_1}.

(iii) One has $U_{R_2} \cap Norm_G(H_{R_1}) = e$.

One first has (iii) by 5.6.7 (i). The first assertion of (i) follows from 5.6.7 (ii). Since $U_{R_2} \cap H_{R_1} = e$ by (iii), the semi-direct product $H_{R_1} \cdot U_{R_2}$ is a subgroup of $H_{R'}$; but these are two subgroups of type (R) of G, containing T, and they have the same Lie algebra $g_{R'}$; they thus coincide by 5.3.5, finishing the proof of (i).

Let us now prove (ii); the first two assertions are simply 5.10.1 and 5.6.5. Let U be a group subscheme of $H_{R'}$, smooth and of finite presentation, normal (hence normalized by T), with connected and unipotent geometric fibers; by 5.6.12, one has, locally on S, $U = U_{R''}$, where R'' is a subset of R' such that $R'' \cap -R'' = \emptyset$. If $U \not\subset U_{R_2}$, then $R'' \not\subset R_2$, so there exists $\alpha \in R''$ such that $-\alpha \in R'$. Then $Z\alpha \subset H_{R'}$ (5.4.3), so $Z\alpha$ normalizes U. But U contains $U\alpha$ and $Z\alpha$ possesses a section w such that $int(w)U\alpha = U_{-\alpha}$; this entails $-\alpha \in R''$, contradicting the hypothesis $R'' \cap -R'' = \emptyset$.

Finally, if L is a reductive subgroup of $H_{R'}$ containing T, one has locally on S, $L = H_{R'''}$, with R''' symmetric contained in R', hence contained in R_1.

Proposition 5.11.4. *Let S be a scheme, G an S-reductive group, H a group subscheme of G of type (RC).*

(i) H is closed in G, $Norm_G(H)/H$ is representable by a finite étale S-group scheme.

(ii) H possesses a largest normal group subscheme that is smooth and of finite presentation over S, with connected and unipotent geometric fibers; we call it the unipotent radical *of H and denote it $rad_u(H)$. The quotient sheaf $H/rad_u(H)$ is representable by an S-reductive group.*

(iii) If T is a maximal torus of H, H possesses a reductive subgroup L containing T of type (RC) possessing the two following properties:

[1349] I.e. when every residue characteristic of S is > 3.

(a) Every reductive subgroup of H containing T is contained in L.

(b) H is the semi-direct product $H = L \cdot rad_u(H)$, i.e. the canonical morphism $L \to H/rad_u(H)$ is an isomorphism.

Furthermore, L is the unique reductive subgroup of H containing T and satisfying one or the other of the two preceding conditions. Finally, one has the following equalities:

$$\mathrm{Norm}_H(L) = L, \quad \mathrm{Norm}_H(T) = \mathrm{Norm}_L(T), \quad W_H(T) = W_L(T),$$

in particular $W_H(T)$ is finite over S.

Proof. Let us first note that (i) is local for the étale topology. Therefore, by Corollary 5.3.18, (i) is a consequence of the last assertion of (iii).

The assertions of (ii) are local for the étale topology. One may therefore suppose to be in the situation of 5.11.3, where one concludes immediately by (i) and (ii).

By virtue of the uniqueness assertions contained in it, (iii) is also local for the étale topology and one may again reduce to the situation of 5.11.3, where properties (a) and (b) have been verified. The uniqueness of an L satisfying (a) is trivial; the uniqueness of an L satisfying (b) is obvious, given (a). The equality $Norm_H(L) = L$ is none other than 5.11.3 (iii); if a section of H normalizes T, then it normalizes L, by uniqueness of L, hence is a section of L by what was just demonstrated, which proves the second equality; the third is then trivial.

Proposition 5.11.5. *Let S be a scheme, G an S-reductive group, $\mathrm{H}c$ the functor of subgroups of type (RC) of G, which is a subfunctor of the functor H of 5.8.1.*

(i) $\mathrm{H}c$ is representable by an open subscheme of H, smooth, quasi-projective, and of finite presentation over S.

(ii) There exists a finite étale S-scheme $C\ell_c$ and a morphism

$$c\ell : Hc \to C\ell_c,$$

smooth, quasi-projective, of finite presentation, surjective, with connected geometric fibers, possessing the following property:

For every $S' \to S$ and every $H, H' \in Hc(S')$, $c\ell(H) = c\ell(H')$ if and only if H and H' are conjugate in G locally for the étale topology (or, equivalently by 5.3.11, if for every $s \in S$, H_s and H'_s are conjugate by an element of $G(s)$).

(iii) $C\ell_c$ and $c\ell$ are determined (up to unique isomorphism) by the preceding conditions.

(iv) If (G, T, M, R) is a splitting of G, let E be the set of conjugacy classes modulo W of closed subsets of R; then there exists an isomorphism $C\ell_c \xrightarrow{\sim} E_S$ such that, for every closed subset R' of R, $c\ell(H_{R'})$ corresponds to the canonical image of R' in $E_S(S) = \mathrm{Hom}_{loc.const.}(S, E)$.

It is first clear that Hc is a sheaf for the étale topology and that (ii) entails that $C\ell_c$ is none other than the quotient sheaf of Hc by the equivalence relation defined by conjugation.

This entails first (iii), as well as the fact that it suffices to verify (i) and (ii) locally for the étale topology. One thus reduces to the situation of (iv); let us first construct a morphism

$$f : Hc \to E_S.$$

It suffices to construct an application $Hc(S) \to E_S(S)$ functorial in S; so let H be a subgroup of type (RC) of G; since H locally for the étale topology possesses maximal tori, and since the maximal tori of G are conjugate locally for the étale topology, there exists a covering family $S_i \to S$ and for each i a $g_i \in G(S_i)$ and a closed subset R_i of R such that $int(g_i)(H \times_S S_i) = H_{R_i} \times_S S_i$; each R_i defines a section η_i of E_{S_i}, i.e. an element of $E_S(S_i)$; it now suffices to prove that the family (η_i) arises from a section $\eta = f(H)$ of E_S over S, and that the latter depends only on H.

For this, one is reduced to proving that $H_{R'}$ and $H_{R''}$ are conjugate locally for the étale topology if and only if R' and R'' are conjugate by an element of the Weyl group W, which is trivial.

For every $\eta \in E$, there exists an $H_0 \in Hc(S)$ such that $f(H_0) = \eta$: it suffices to take $H_0 = H_{R'}$ where R' is a closed subset of R whose image in E is η. If $H \in H_0(S')$, $S' \to S$, H is conjugate to H_0 locally for the étale topology if and only if $f(H) = \eta$ (as one sees immediately by the preceding argument), which shows that $f^{-1}(\eta)$ is identified with the quotient $G/Norm_G(H_0)$, which by 5.8.2 is an open subset of H, smooth, quasi-projective of finite presentation over S, with connected and non-empty fibers. Since E_S is the sum of the open subschemes images of the sections corresponding to the $\eta \in E$, Hc is identified with the sum of the $f^{-1}(\eta)$, $\eta \in E$, which proves (i) and (ii). Finally, (iv) holds by construction.

Corollary 5.11.6. *If $u \in C\ell_c(S')$, $S' \to S$, $c\ell^{-1}(u)$ is an S'-scheme that is smooth, quasi-projective, of finite presentation, with non-empty connected fibers; it is an open subset of* Hc *and a "homogeneous" scheme under $G_{S'}$ (by inner automorphisms). In particular, if $H \in c\ell^{-1}(u)(S')$, the morphism $G_{S'} \to (Hc)_{S'}$ defined by $g \mapsto int(g)H$ identifies $G_{S'}/Norm_{G_{S'}}(H)$ with $c\ell^{-1}(u)$.*

Examples 5.11.7. *In particular, one has two canonical sections u_t, u_b of $C\ell_c$ corresponding respectively to maximal tori ($R' = \emptyset$) and to Borel subgroups (R' = system of positive roots). The S-schemes $c\ell^{-1}(u_t)$ and $c\ell^{-1}(u_b)$ are none other than the S-schemes $Tor(G)$ and $Bor(G)$ introduced in 5.8.3. We will see other examples in Exp. XXVI.*

Remark 5.11.8. *One may construct an S-scheme $C\ell$, of finite presentation and unramified, and a morphism $H \to C\ell$ smooth and surjective, with connected geometric fibers, enjoying properties analogous to 5.11.5 (ii) and (iii).*

6. The derived group

6.1. Preliminaries

In this section, we fix a scheme S, an S-split group (G, T, M, R), a system of positive roots $R+$ of R, and write

```
B = B_{R+},     B_- = B_{R−},     U = B^u,     U_- = (B_-)^u,
Ω = Ω_{R+} = U_- · T · U.
```

6.1.1. We denote by T' the subtorus of T "image of the family $\alpha*$, $\alpha \in R$"; in other words, T' is the image of the morphism of groups

$$G_{m,S}^{R} \to T$$

defined by $(z_\alpha)_{\alpha \in R} \mapsto \prod_{\alpha \in R} \alpha * (z_\alpha)$. One sees immediately that if Δ denotes the set of simple roots of $R+$, the morphism

$$G_{m,S}^{\Delta} \to T'$$

defined in the same way is surjective with finite kernel. If we identify T with $D_S(M)$, then T' is identified with $D_S(M/N)$, where

$$N = M \cap V(R*)^{\perp}$$

(we denote by $V(R*)^{\perp}$ the orthogonal of $V(R*)$ in the duality between V and $V*$).

Lemma 6.1.2. *The morphism defined by the product in T*

```
rad(G) ×_S T′ → T
```

is an isogeny (cf. 4.2.9).

Indeed, the canonical morphism $rad(T) \to T/T'$ arises by duality from the morphism of commutative groups

```
M ∩ V(R*)^⊥ → M/(M ∩ V(R)),
```

which one sees immediately to be injective with finite cokernel (cf. Exp. XXI 6.3).

Definition 6.1.3. *One writes $\Omega' = U_{-} \cdot T' \cdot U$; it is a closed subscheme of $\Omega = U_{-} \cdot T \cdot U$.*

Lemma 6.1.4. *Let α be a simple root and $w_\alpha \in Norm_G(T)(S)$ lifting s_α. One has*

```
int(w_α) Ω′ ∩ Ω ⊂ Ω′.
```

It suffices for us to prove that if $g \in \Omega'(S)$ and $int(w_\alpha)g \in \Omega(S)$, then $int(w_\alpha)g \in \Omega'(S)$. By 5.6.8, write

```
g = a · exp_{-α}(Y) · t · expα(X) · b,
```

with $a \in U_{-\hat{\alpha}}(S)$, $Y \in \Gamma(S, g^{-\alpha})$, $t \in T'(S)$, $X \in \Gamma(S, g^{\alpha})$, $b \in U_{\hat{\alpha}}(S)$. One then has

$$\mathrm{int}(w_\alpha)\, g = \mathrm{int}(w_\alpha)\, a \cdot \mathrm{int}(w_\alpha)(\exp_{-\alpha}(Y)\, t \exp\alpha(X)) \cdot \mathrm{int}(w_\alpha)\, b.$$

By virtue of 5.6.8 (iv), one has

$$\mathrm{int}(w_\alpha)\, a \in U_{-\hat{\alpha}}(S), \qquad \mathrm{int}(w_\alpha)\, b \in U_{\hat{\alpha}}(S).$$

It follows that one has the following equivalences (setting $h = \exp_{-\alpha}(Y) t \exp \alpha(X)$):

$$\begin{aligned} \mathrm{int}(w_\alpha)\, g \in \Omega(S) &\Leftrightarrow \mathrm{int}(w_\alpha)\, h \in \Omega(S) \\ \mathrm{int}(w_\alpha)\, g \in \Omega'(S) &\Leftrightarrow \mathrm{int}(w_\alpha)\, h \in \Omega'(S). \end{aligned}$$

One is therefore reduced to the case where $g = h$. Since one has (4.1.12)

$$Z_\alpha \cap \Omega = U_{-\alpha} \cdot T \cdot U_\alpha, \qquad Z_\alpha \cap \Omega' = U_{-\alpha} \cdot T' \cdot U_\alpha,$$

one is reduced to proving the following assertion:

$$\mathrm{int}(w_\alpha)\, h \in (U_{-\alpha} \cdot T \cdot U_\alpha)(S) \Rightarrow \mathrm{int}(w_\alpha)\, h \in (U_{-\alpha} \cdot T' \cdot U_\alpha)(S).$$

Now this latter follows immediately from Exp. XX 3.12, which shows that the component on T of $int(w_\alpha)h$ is of the form $t \cdot \alpha * (z) \in T'(S)$.

Lemma 6.1.5. *For every $w \in Norm_G(T)(S)$, there exists an open subset V_w of G containing the unit section, such that*

$$\mathrm{int}(w)\, \Omega' \cap V_w \subset \Omega'.$$

Choose, for each simple root α, an $n_\alpha \in Norm_G(T)(S)$ lifting s_α. For every point $s \in S$, there exist an open subset V of S containing s, a $t \in T(V)$ and over V a relation

$$w = n_{\alpha_1} \cdots n_{\alpha_p}\, t, \qquad \text{with the } \alpha_i \text{ simple}.$$

One may obviously content oneself with making the demonstration for $V = S$; it is done by induction on p. If $p = 0$, then $w \in T(S)$ and one takes $V_w = G$; suppose therefore $w = n_\alpha \cdot w'$, w' satisfying the conclusion of the lemma; there thus exists an open subset $V_{w'}$ of G, containing the unit section, such that $int(w')\Omega' \cap V_{w'} \subset \Omega'$. One may then write

$$\begin{aligned} \mathrm{int}(w)\, \Omega' \cap (\mathrm{int}(n_\alpha)\, V_{w'} \cap \Omega) &= \mathrm{int}(n_\alpha)(\mathrm{int}(w')\, \Omega' \cap V_{w'}) \cap \Omega \\ &\subset \mathrm{int}(n_\alpha)\, \Omega' \cap \Omega \subset \Omega', \end{aligned}$$

by 6.1.4. One then takes $V_w = int(n_\alpha)V_{w'} \cap \Omega$ and we are done.

Lemma 6.1.6. *There exists an open subset V_0 of G, containing the unit section, such that for every $S' \to S$, one has*

$$U(S')\, U_-(S') \cap V_0(S') \subset \Omega'(S').$$

Let n_0 be an element of $Norm_G(T)(S)$ lifting the symmetry w_0 of the Weyl group,[1350] that is, such that $int(n_0)U = U_-$ (cf. Exp. XXI 3.6.14); then $n_0^2 \in T(S)$. Let us show that the open set $V_0 = V_{n_0}$ of 6.1.5 answers the question. Indeed

[1350] The symmetry w_0 is defined in XXI 3.6.14.

$$U(S') U_-(S') = \mathrm{int}(n_0)(\mathrm{int}(n_0)^{-1} U(S') \cdot \mathrm{int}(n_0)^{-1} U_-(S'))$$
$$= \mathrm{int}(n_0)(U_-(S') \cdot U(S')) \subset \mathrm{int}(n_0)\, \Omega'(S').$$

Whence

$$U(S') U_-(S') \cap V_0(S') \subset \mathrm{int}(n_0)\, \Omega'(S') \cap V_0(S') \subset \Omega'(S').$$

Lemma 6.1.7. *Consider the morphism*

$$f : \Omega = U_- \cdot T \cdot U \to T/T'$$

composite of the second projection and the canonical morphism of T into T/T'. Then f is "generically multiplicative": there exists an open subset V of $\Omega \times_S \Omega$, containing the unit section (and hence relatively schematically dense, Exp. XVIII 1.3) such that for every $S' \to S$ and every $(x, y) \in V(S')$, one has $xy \in \Omega(S')$ and $f(xy) = f(x)f(y)$.

Let x and y be two sections of Ω over S'. Write

$$x = u\, t\, v, \quad y = u'\, t'\, v', \quad \text{with } u, u' \in U_-(S'),\ t, t' \in T(S'),\ v, v' \in U(S').$$

Let V_0 be the open set of 6.1.6 and V the open subset of $\Omega \times_S \Omega$ defined by "$vu' \in V_0(S')$" (it is the inverse image of V_0 by the morphism $\Omega \times_S \Omega$ written set-theoretically $(x, y) \mapsto vu'$). Then V answers the question. Indeed, for $(x, y) \in V(S')$, one has

$$xy = (u\, t\, v)(u'\, t'\, v') = (u\, t)(v\, u')(t'\, v').$$

But $vu' \in \Omega'(S')$, whence

$$xy \in U_-(S') \cdot t \cdot \Omega'(S') \cdot t' \cdot U(S') \subset U_-(S') \cdot tt'\, T'(S') \cdot U(S'),$$

which shows that $xy \in \Omega(S')$ and that

$$f(xy) = f(tt') = f(t)\, f(t') = f(x)\, f(y).$$

Proposition 6.1.8. *There exists a morphism of groups*

$$f : G \to T/T'$$

inducing on T the canonical projection. The kernel $Ker(f)$ of f is a closed group subscheme of G smooth over S and with connected fibers. Every morphism of groups from G to a presheaf of commutative groups on S, separated for (fppf), vanishes on $Ker(f)$.

The first assertion follows immediately from 4.1.11. One has immediately $Ker(f) \cap \Omega = \Omega'$, which proves that $Ker(f)$ is smooth over S at every point of the unit section.[1351] By 5.6.9 (ii), every morphism ϕ from G to a presheaf of commutative groups separated for (fppf) vanishes on U and U_-. By Exp. XX 2.7, ϕ therefore vanishes also on T', hence on Ω'. Taking the notations of 5.7.10, one sees that the monoid U_1 is contained in $Ker(f)(S)$, which shows that

[1351] We have added "at every point of the unit section" as well as the reference to VI_B 3.10 at the end of the proof. Furthermore, in the next sentence we have replaced "prescheme" with "presheaf".

$Ker(f) = \bigcap_{u \in U_1} u\,\Omega'$.

It follows on the one hand that every ϕ as above vanishes on $Ker(f)$, and on the other hand that $Ker(f)$ has connected fibers, hence is smooth over S by Exp. VI_B 3.10.

6.2. Derived group of a reductive group

Theorem 6.2.1. *Let S be a scheme, G an S-reductive group.*

(i) $D_S(G) = \mathrm{Hom}_{S-gr.}(G, Gm, S)$ is representable by a twisted constant S-group, whose type at $s \in S$ is $\mathbb{Z}^{rg_{red}(G_s) - rg_{ss}(G_s)}$.

(ii) Write $corad(G) = D_S(D_S(G))$, which is therefore an S-torus. The biduality morphism (cf. Exp. VIII § 1)

$$f_0 : G \to corad(G)$$

is smooth and surjective.

(iii) The composite morphism

$$rad(G) \to G \to corad(G)$$

is an isogeny (cf. 4.2.9).

(iv) The kernel of f_0, denoted

$$d\acute{e}r(G) = Ker(f_0)$$

is a closed group subscheme of G, semisimple over S, called the derived group *of G. If G is semisimple, one has $d\acute{e}r(G) = G$.*

(v) Every morphism of groups from G to an S-presheaf of commutative groups, separated for (fppf), vanishes on $d\acute{e}r(G)$ and thus factors through f_0.

Proof. All assertions of the theorem are local for the étale topology; one may thus reduce to the case where G is split over S. Consider then the morphism f of 6.1.8. By the last assertion of 6.1.8, one immediately has an isomorphism

$$\mathrm{Hom}_{S-gr.}(G, Gm, S) \xrightarrow{\sim} \mathrm{Hom}_{S-gr.}(T/T', Gm, S),$$

which proves (i), then (ii) and gives a commutative diagram

$$\begin{tikzcd} G \arrow[r, "f_0"] \arrow[dr, "f"'] & corad(G) \arrow[d, "\wr"] \\ & T/T'. \end{tikzcd}$$

One then has (v) by 6.1.8, and (iii) by 6.1.2. One also has $Ker(f) = Ker(f_0)$, which by 6.1.8 entails that $dér(G)$ is smooth over S and has connected fibers; it remains to verify that its fibers are semisimple; now they are reductive by Exp. XIX 1.7, as invariant subgroups of reductive groups. By (iii), $rad(G) \cap dér(G)$ is finite, which indeed entails that the fibers of $dér(G)$ are semisimple.

Remark 6.2.2. *(a) By construction, in the case where G is split, $dér(G)$ is the (fppf) subsheaf of G generated by the $U\alpha$, $\alpha \in R$. (It even suffices to take the $U\alpha$, $\alpha \in \Delta \cup -\Delta$, where Δ is a basis of R.)*

(b)[1352] Let C be the presheaf of commutators of G, i.e. the S-group functor that associates with every $S' \to S$ the group of commutators of $G(S')$ (i.e. the subgroup of $G(S')$ generated by the elements $xyx^{-1}y^{-1}$, for $x, y \in G(S')$), and let $\tilde{C}$ be the associated (fppf) sheaf. Since the quotient $G/dér(G) = T/T'$ is commutative, then $dér(G)$ contains C and hence $\tilde{C}$ (cf. Exp. IV 4.3.12).

On the other hand, the quotient presheaf $G/\tilde{C}$ is separated (Exp. IV 4.4.8.1), and therefore by (v) one has $dér(G) \subset \tilde{C}$, whence $dér(G) = \tilde{C}$, i.e. $dér(G)$ is the (fppf) sheaf of commutators of G.

Let us finally note that C, being a subpresheaf of G, is separated, but is not equal to $dér(G)$ in general: for example, $dér(SL_2) = SL_2$ but $SL_2(F_2) \simeq S_3$ is not equal to its derived group.

(c) When S is the spectrum of an algebraically closed field k, $dér(G)(k)$ is the group of commutators of $G(k)$ (Exp. VI_B 7.10).

6.2.3. Consider now the two exact sequences

$$1 \to rad(G) \to G \to ss(G) \to 1, 1 \to dér(G) \to G \to corad(G) \to 1.$$

Since $rad(G)$ is central in G, the product in G defines a morphism of groups

```
u : rad(G) ×_S dér(G) → G
```

which is covering by virtue of 6.2.1 (iii), hence surjective and flat (Exp. VI_B 9.2 (xi)).[1353] Its kernel is isomorphic to $rad(G) \cap dér(G)$, which is also the kernel of $rad(G) \to corad(G)$, hence is a finite subgroup of multiplicative type of $rad(G)$.

One reasons similarly for the morphism

```
G → corad(G) ×_S ss(G),
```

whose kernel is $dér(G) \cap rad(G)$. One thus has:

Proposition 6.2.4. *Let G be an S-reductive group. The morphisms*

[1352] In what follows, we have expanded the original, and suppressed the assertion that "$dér(G)$ is the separated (fppf) presheaf of commutators of G".
[1353] Indeed, G is the (fppf) quotient of $rad(G) \times_S dér(G)$ by $Ker(u)$, which is a group of multiplicative type, hence flat over S. Therefore, by VI_B 9.2 (xi), the morphism u is flat.

$$\mathrm{rad}(G) \times_S \mathrm{d\acute{e}r}(G) \to G, \quad G \to \mathrm{corad}(G) \times_S \mathrm{ss}(G), \quad \mathrm{rad}(G) \to \mathrm{corad}(G)$$

are central isogenies, and their kernels are isomorphic.

Corollary 6.2.5. *The following conditions are equivalent:*

(i) G is the product of a semisimple group and a torus.

(ii) $rad(G) \times_S d\acute{e}r(G) \xrightarrow{\sim} G$.

(iii) $G \xrightarrow{\sim} corad(G) \times_S ss(G)$.

(iv) $rad(G) \cap d\acute{e}r(G) = e$.

6.2.6. Let us return provisionally to the case of a split group. Let us keep the notations of 6.1. Set $N = M \cap V(R*)^{\perp}$. One thus has $T' = D_S(M/N)$. One has seen that $U_{-} \cdot T' \cdot U$ was an open neighborhood of the unit section of $d\acute{e}r(G)$. One thus has

$$\mathrm{Lie}(\mathrm{d\acute{e}r}(G)/S) = \mathfrak{t}' \oplus \coprod_{\alpha \in R} \mathfrak{g}^{\alpha}.$$

Since the characters induced on T' by the $\alpha \in R$ are non-zero and distinct (cf. Exp. XXI 1.2.5 — one has moreover already used this fact in 6.1.2), R is a system of roots of $d\acute{e}r(G)$ relative to T'. It is then immediate (since $U\alpha \subset d\acute{e}r(G)$) that the exp morphisms of $d\acute{e}r(G)$ "are" those of G and similarly for the coroots.

It follows:

Proposition 6.2.7. *In the preceding notations, $(d\acute{e}r(G), T', M/N, R)$ is a split group with root datum $d\acute{e}r(R(G))$. The canonical morphism $d\acute{e}r(G) \to G$ gives by functoriality the canonical morphism of root data $R(G) \to d\acute{e}r(R(G))$ of Exp. XXI 6.5.*

N. B. The reader may, as an exercise, construct the diagram of split groups corresponding to the three left columns of the diagram of root data of Exp. XXI 6.5.7.

Proposition 6.2.8. *Let S be a scheme, G an S-reductive group, $d\acute{e}r(G)$ its derived group.*

(i) For every maximal torus T of G, $T \cap d\acute{e}r(G)$ is a maximal torus of $d\acute{e}r(G)$. For every maximal torus T' of $d\acute{e}r(G)$, $Centr_G(T') = rad(G) \cdot T'$ is a maximal torus of G. The two preceding constructions are inverse to one another and set up a bijective correspondence between maximal tori of G and of $d\acute{e}r(G)$.

(ii) For every Borel subgroup B of G, $B \cap d\acute{e}r(G)$ is a Borel subgroup B' of $d\acute{e}r(G)$. One has $B'u = B^u$. For every Borel subgroup B' of $d\acute{e}r(G)$, $Norm_G(B') = rad(G) \cdot B'$ is a Borel subgroup of G. The preceding applications are inverse to one another and set up a bijective correspondence between Borel subgroups of G and of $d\acute{e}r(G)$.

By the local conjugacy theorem for maximal tori and the construction of the derived group, the only assertion that remains to prove in (i) is the following: if T is a maximal torus of G, then

$$T = (T \cap \mathrm{d\acute{e}r}(G)) \cdot \mathrm{rad}(G) = \mathrm{Centr}_G(T \cap \mathrm{d\acute{e}r}(G)).$$

The first equality is trivial (one reduces to the split case); the second follows from this immediately, because $rad(G)$ is central in G, so $T = Centr_G(T) = Centr_G(T \cap dér(G))$. One reasons similarly for (ii).

6.3. Subgroups with commutative quotients

6.3.1. Let G be an S-reductive group. If H is a group subsheaf of G, the following conditions are equivalent:

- H contains $dér(G)$.
- H is normal and G/H is commutative.

In this case, the canonical morphism $f_0 : G \to corad(G)$ sends H to a subsheaf $f_0(H)$ of $corad(G)$; one has

$$G/H \simeq corad(G)/f_0(H), \quad H/dér(H) \simeq f_0(H),$$
$$dér(G) = dér(H), \quad H = f_0^{-1}(f_0(H)).$$

Since $dér(G)$ is smooth over S and has connected fibers,[1354] by Exp. IV, 5.3.1 and 6.3.1, and Exp. IV_B 9.2, the map $H \mapsto f_0(H)$ establishes a bijective correspondence between group subschemes (resp. closed group subschemes) of G containing $dér(G)$, smooth over S and with connected fibers, and group subschemes (resp. closed group subschemes) of $corad(G)$, smooth over S and with connected fibers.

Now, if H' is a group subscheme of $corad(G)$, smooth over S with connected fibers, then H' is of finite presentation over S (Exp. VI_B 5.5) and its fibers are tori (since those of $corad(G)$ are), so by Exp. X 8.2, H' is a subtorus of $corad(G)$, hence is closed in $corad(G)$ (Exp. IX 2.6).

Consequently, every subgroup of G, smooth with connected fibers and containing $dér(G)$, is closed in G.[1355]

6.3.2. If H is a closed group subscheme of G, smooth over S, with connected fibers, and normal in G, then H is reductive. If furthermore $H \supset dér(G)$, then $dér(H) = dér(G)$ and $f_0(H)$ is identified with $corad(H)$. One has thus proved:

Proposition 6.3.3. *Let G be an S-reductive group. Every group subscheme H of G, normal in G, with commutative quotient (i.e. containing $dér(G)$), smooth over S, with connected fibers,[1356] is closed and reductive. One has $dér(H) = dér(G)$ and $f_0(H)$ is identified with $corad(H)$; one has*

$$G/H \simeq corad(G)/corad(H), \quad H = (H \cap rad(G)) \cdot dér(G).$$

[1354] We have expanded the original in what follows. In particular, we have added the conclusion (implicit in the original) that every subgroup of G, smooth with connected fibers and containing $dér(G)$, is closed in G.

[1355] We have expanded the original in what follows. In particular, we have added the conclusion (implicit in the original) that every subgroup of G, smooth with connected fibers and containing $dér(G)$, is closed in G.

[1356] We have removed the hypothesis that H be retrocompact in G, which is automatically verified because, by VI_B 5.5, G and H are separated and quasi-compact over S, so $H \hookrightarrow G$ is quasi-compact by EGA IV$_1$, 1.1.2 (v).

Furthermore, $H \mapsto f_0(H)$ defines a bijection between the set of subgroups H of G possessing the preceding properties and the set of subtori of $corad(H)$.

By a further application of Noether's theorem (Exp. IV, 5.3.1 and 6.3.1), one deduces:

Proposition 6.3.4. *Let S be a scheme, G an S-reductive group, T a maximal torus of G. For every subgroup H of G as above, $T \cap H$ is a maximal torus of G and one has*

$$G/H \simeq T/(T \cap H), \qquad H = (T \cap H) \cdot \mathrm{d\acute{e}r}(G).$$

Furthermore, $H \mapsto T \cap H$ is a bijection between the set of subgroups H of G as above and the set of subtori of T containing $T \cap d\acute{e}r(G)$.

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Editor's Notes

Exposé XXIII. Reductive groups: uniqueness of pinned groups

by M. Demazure

[1357]

The aim of this Exposé is the proof of the uniqueness theorem (Theorem 4.1). This was proved by Chevalley in the case of an algebraically closed field; the method of reduction to rank two used here is also due to Chevalley (see *Bible*, Exp. 23 and 24). Along the way, we obtain an explicit description of reductive groups by generators and relations (3.5).

1. Pinnings

Definition 1.1. *Let S be a scheme and (G, T, M, R) an S-split group (XXII, 1.13). One calls* pinning[^N.D.E-XXIII-butterfly] *of this split group the datum of a system*

[1357] N.D.E.: Version of 13/10/2024.

Δ of simple roots of R and, for each $\alpha \in \Delta$, a section $X_\alpha \in \Gamma(S, g_\alpha)^\times$.[^N.D.E-XXIII-butterfly-anchor]

In other words, a pinning of the reductive group G over the nonempty scheme S is the datum of:

(i) a maximal torus T,

(ii) an abelian group M and an isomorphism $T \simeq D_S(M)$,

(iii) a system of roots R of G with respect to T,

(iv) a system of simple roots Δ of R,

(v) an $X_\alpha \in \Gamma(S, g_\alpha)^\times$ for every $\alpha \in \Delta$, that is, of a

$$u_\alpha = \exp_\alpha(X_\alpha) \in U_\alpha^\times(S) \qquad \text{for every } \alpha \in \Delta,$$

satisfying condition (D 1) of Exp. XXII, 1.13 (indeed condition (D 2) of *loc. cit.* is automatically satisfied[1358]).

Every split group admits a pinning; in particular, every reductive group is locally pinnable for the étale topology.

1.2.

If G is a pinned S-group — that is, an S-split group equipped with a pinning — it is canonically endowed with the system of positive roots R^+ defined by Δ, the corresponding Borel subgroup $B = B_{R^+}$, the opposite Borel subgroup $B^- = B_{R^-}$, the unipotent groups $U = B_u$, $U^- = (B^-)_u$, the open set $U^- \cdot T \cdot U$, etc. Likewise, for each $\alpha \in \Delta$, one has a canonical isomorphism of vector groups

$$p_\alpha : G_{a,S} \xrightarrow{\sim} U_\alpha, \qquad x \mapsto \exp_\alpha(x X_\alpha) = u_\alpha^x,$$

normalized by T with multiplier α, and whose datum is equivalent to that of X_α (Exp. XXII, 1.1).

By duality, one deduces an $X_{-\alpha} \in \Gamma(S, g_{-\alpha})^\times$ and an isomorphism

$$p_{-\alpha} : G_{a,S} \xrightarrow{\sim} U_{-\alpha}$$

which is the contragredient of the preceding one (Exp. XXII, 1.3). We shall set (Exp. XX, 3.1)

$$w_\alpha = w_\alpha(X_\alpha) = p_\alpha(1)\, p_{-\alpha}(-1)\, p_\alpha(1) = p_{-\alpha}(-1)\, p_\alpha(1)\, p_{-\alpha}(-1).$$

One then has (*loc. cit.* 3.1, 3.7)

$$w_\alpha^2 = \alpha^*(-1), \qquad \mathrm{int}(w_\alpha)\, t = s_\alpha(t) = t \cdot \alpha^*(\alpha(t)^{-1}),$$

$$\mathrm{int}(w_\alpha)\, p_\alpha(x) = p_{-\alpha}(-x) = p_{-\alpha}(x)^{-1},$$
$$\mathrm{Ad}(w_\alpha)\, X_\alpha = -X_{-\alpha},$$

[1358] N.D.E.: It is implied by condition (v), i.e. the existence of a section $X_\alpha \in \Gamma(S, g_\alpha)^\times$.

$$\operatorname{int}(w_\alpha)\, p_{-\alpha}(x) = p_\alpha(-x) = p_\alpha(x)^{-1},$$
$$\operatorname{Ad}(w_\alpha)\, X_{-\alpha} = -X_\alpha.$$

We shall use the preceding notation systematically in what follows.

Definition 1.3. *Let S be a scheme and $(G, T, M, R, \Delta, (X_\alpha))$ and $(G', T', M', \cdots)$ two pinned S-groups. One says that the morphism of split groups (Exp. XXII, 4.2.1)*

$$f : G \to G'$$

is compatible with the pinnings, *or that it defines a* morphism of pinned groups, *if the bijection $d : R \to R'$ associated with it (cf. loc. cit.) satisfies $d(\Delta) = \Delta'$ and if, for every $\alpha \in \Delta$, one has*

$$f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(X'_{d(\alpha)}), \quad \text{i.e.} \quad f(u_\alpha) = u'_{d(\alpha)}.$$

[1359]

1.4.

If we denote by $q(\alpha)$ the integer of *loc. cit.*, we therefore have

$$f(p_\alpha(x)) = p'_{d(\alpha)}(x^{q(\alpha)}) \quad \text{for} \quad \alpha \in \Delta,$$

and consequently

$$f(p_{-\alpha}(x)) = p'_{-d(\alpha)}(x^{q(\alpha)}), \quad f(w_\alpha) = w'_{d(\alpha)}.$$

Recall (Exp. XXII, 4.2) that, for every $\alpha \in R$ and all $z \in G_m(S')$, $t \in T(S')$,

$$f(\alpha^*(z)) = d(\alpha)^*(z)^{q(\alpha)} = d(\alpha)^*(z^{q(\alpha)}), \quad d(\alpha)(f(t)) = \alpha(t)^{q(\alpha)}.$$

1.5.

We shall call a *pinned root datum* a root datum endowed with a system of simple roots, and a *p-morphism of pinned root data* a p-morphism of root data (Exp. XXI, 6.8) sending simple roots to simple roots.

If G is a pinned S-group, we denote by $R(G)$ the corresponding pinned root datum (this is the root datum of Exp. XXII, 1.14 equipped with Δ). Let p be the integer defined in Exp. XXII, 4.2.2. One then has:

Scholium 1.6. *The correspondence $G \mapsto R(G)$ defines a functor from the category of pinned reductive S-groups to that of pinned root data (with p-morphisms as morphisms).*

[1359]N.D.E.: We have denoted by d the bijection $R \xrightarrow{\sim} R'$ (instead of u), in order to avoid the notation $u'_{u(\alpha)}$.

1.7. The pinned groups Z_{Δ_1}

Let Δ_1 be a subset of the system of simple roots Δ of the pinned group G. Let T_{Δ_1} be the maximal torus of $\bigcap_{\alpha\in\Delta_1} Ker(\alpha)$; set

$$Z_{\Delta_1} = Centr_G(T_{\Delta_1}).$$

Set $R' = \mathbb{Z}\cdot\Delta_1 \cap R$; one knows (Exp. XXII, 5.10.7) that Z_{Δ_1} is a reductive S-group with radical T_{Δ_1}, that (T, M, R') is a splitting of it, and Δ_1 a system of simple roots. It follows that $(Z_{\Delta_1}, T, M, R', \Delta_1, (X_\alpha)_{\alpha\in\Delta_1})$ is a pinned S-group. We shall always equip Z_{Δ_1} with this pinning. In particular, we shall consider the groups

$$Z_\alpha = Z_{\{\alpha\}}, \qquad Z_{\alpha\beta} = Z_{\{\alpha,\beta\}}.$$

We denote by $B_{\Delta_1} = B \cap Z_{\Delta_1}$; one knows (*loc. cit.*) that this is the canonical Borel subgroup[1360] of Z_{Δ_1}, and that its unipotent part is $U_{\Delta_1} = U \cap Z_{\Delta_1}$. In particular,

$$B_\alpha = T\cdot U_\alpha.$$

We shall denote

$$U_{\alpha\beta} = U_{\{\alpha,\beta\}} = U \cap Z_{\alpha\beta} = \prod_{\gamma\in R^+_{\alpha\beta}} U_\gamma,$$

where $R^+_{\alpha\beta}$ denotes the set of positive roots that are linear combinations of α and β.

Now let $f : G \to G'$ be a morphism of pinned S-groups. If $d : R \to R'$ is the corresponding bijection and if Δ_1 is a subset of Δ, then $d(\Delta_1) = \Delta'_1$ is a subset of Δ', and it is clear that f sends T_{Δ_1} into $T'_{\Delta'_1}$, hence Z_{Δ_1} into $Z'_{\Delta'_1}$. The corresponding S-morphism

$$f_{\Delta_1} : Z_{\Delta_1} \to Z'_{d(\Delta'_1)}$$

is compatible with the canonical pinnings; it defines a morphism of pinned root data

$$R(f_{\Delta_1}) : R(Z_{\Delta_1}, T, M, \dots) \to R(Z_{\Delta'_1}, T', M', \dots)$$

and the underlying morphisms $M' \to M$ of $R(f)$ and $R(f_{\Delta_1})$ coincide.

1.8. Study of the group $Norm_G(T)$

For each pair (α,β) of simple roots, denote by $n_{\alpha\beta}$ the order of the element $s_\alpha s_\beta$ of the Weyl group W. In particular, $n_{\alpha\alpha} = 1$. One has therefore $(w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S)$.

Definition 1.8.1. *For every* $(\alpha,\beta)\in\Delta\times\Delta$*, one sets*

$$t_{\alpha\beta} = (w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S).$$

[1360] N.D.E.: i.e., the Borel subgroup of Z_{Δ_1} corresponding to $R'_+ = R' \cap R_+$.

Moreover, one sets (Exp. XX, 3.1)

$$t_\alpha = t_{\alpha\alpha} = w_\alpha^2 = \alpha^*(-1) \in T(S).$$

Proposition 1.8.2. *Let H be an S-functor in groups transforming direct sums of schemes into products (for example a sheaf for the Zariski topology). Let*

$$f_T : T \to H$$

be a morphism of groups and $h_\alpha (\alpha \in \Delta)$ elements of $H(S)$. In order that there exist a morphism of groups (necessarily unique)

$$f_N : Norm_G(T) \to H$$

inducing f_T on T and such that $f(w_\alpha) = h_\alpha$ for $\alpha \in \Delta$, it is necessary and sufficient that the following two conditions be satisfied:

(i) For every $\alpha \in \Delta$,

$$f_T(s_\alpha(t)) = h_\alpha f_T(t) h_\alpha^{-1}$$

for every $t \in T(S')$, $S' \to S$ (i.e. $f_T \circ s_\alpha = int(h_\alpha) \circ f_T$).

(ii) For every $(\alpha, \beta) \in \Delta \times \Delta$,

$$f_T(t_{\alpha\beta}) = (h_\alpha h_\beta)^{n_{\alpha\beta}}.$$

Proof. Indeed, equip (Sch) with the topology $\mathcal{T}$ generated by the pretopology whose covering families are the direct sums; the hypothesis of the statement says that H is a $\mathcal{T}$-sheaf. Let L be the free group with generators $(m_\alpha)_{\alpha\in R}$ and L_1 the invariant subgroup generated by the elements $(m_\alpha m_\beta)^{n_{\alpha\beta}}$, $(\alpha, \beta) \in \Delta \times \Delta$. Let $g : L \to W$ be the morphism defined by $g(m_\alpha) = s_\alpha$; one knows (Exp. XXI, 5.1) that g induces an isomorphism $\bar{g}$ of L/L_1 onto W. Make L act on T through g (or, equivalently, by $m_\alpha \cdot t = s_\alpha(t)$). Let L_S be the constant group defined by L; consider the semidirect product $T \cdot L_S = N$ for the preceding operation. One has a morphism of S-groups

$$h : T \cdot L_S = N \to Norm_G(T)$$

unique such that $h(m_\alpha) = w_\alpha$, $h(t) = t$ for every $t \in T(S')$, $S' \to S$. Let N_1 be the invariant subsheaf of groups of N generated by the

$$t_{\alpha\beta}^{-1} \cdot (m_\alpha m_\beta)^{n_{\alpha\beta}}, \qquad (\alpha, \beta) \in \Delta \times \Delta.$$

One evidently has $N_1 \subset Ker h$; consider the induced morphism

$$h_1 : N/N_1 \to Norm_G(T).$$

Let us prove that h_1 is an isomorphism. Since h induces on T the canonical immersion, which is a monomorphism, the canonical morphism

$$T \to N/N_1$$

is also a monomorphism, hence induces an isomorphism of T onto TN_1/N_1. For the same reason, h_1 induces an isomorphism of TN_1/N_1 onto T; to prove that h_1 is an isomorphism, it therefore suffices to verify that the corresponding morphism

$$h_2 : N/TN_1 \to Norm_G(T)/T$$

is an isomorphism. Now `TN_1` is the invariant $\mathcal{T}$-subsheaf of groups of N generated by T and the $(m_\alpha m_\beta)^{n_{\alpha\beta}}$, that is, the $\mathcal{T}$-subsheaf generated by T and `L_1`, that is, $T \cdot (L_1)_S$. The morphism h_2 is therefore identified with the morphism

$$\bar{g} : L_S/(L_1)_S \to W_S$$

which is an isomorphism by construction.

The proof of 1.8.2 is now easy; the conditions are evidently necessary. Let us prove that they are sufficient. Condition (i) shows that there exists a morphism

$$u : N \to H$$

such that $u(m_\alpha) = h_\alpha$ for $\alpha \in \Delta$, and $u|_T = f_T$. Condition (ii) says that u vanishes on `N_1`, which yields the result at once.

1.9. Faithfulness of the functor R

Proposition 1.9.1. *The functor R of 1.6 is faithful: if*

`f, g : G ⇒ G'`

are two morphisms of pinned groups such that $R(f) = R(g)$, then $f = g$.

Indeed, f and g coincide on T, $U_\alpha(\alpha \in \Delta)$ and $U_{-\alpha}(\alpha \in \Delta)$; it therefore suffices to apply:

Lemma 1.9.2. *Let S be a scheme, G a pinned reductive S-group, H an S-presheaf in groups, separated for (fppf). Let*

`f, g : G ⇒ H`

be two morphisms of S-groups. The following conditions are equivalent:

(i) $f = g$.

(ii) f and g coincide on T, on each $U_\alpha(\alpha \in \Delta)$, on each $U_{-\alpha}(\alpha \in \Delta)$.

(iii) f and g coincide on T, on each $U_\alpha(\alpha \in \Delta)$, and $f(w_\alpha) = g(w_\alpha)$ for each $\alpha \in \Delta$.

Indeed, (i) ⇒ (ii) is trivial, (ii) ⇒ (iii) follows at once from the definition of w_α (1.2). It remains to prove (iii) ⇒ (i). If $\alpha \in R$, there exists a sequence $\alpha_i \subset \Delta$ with $\alpha = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_n}(\alpha_{n+1})$, hence

$$U_\alpha = \operatorname{int}(w_{\alpha_1} \cdots w_{\alpha_n})\, U_{\alpha_{n+1}},$$

which shows that f and g coincide on each U_α. It follows that f and g coincide on Ω, hence coincide (Exp. XXII, 4.1.11).

Remark 1.9.3. *If G is semisimple, one may, in (ii) and (iii), drop the hypothesis that f and g coincide on T. Indeed, G is generated as an (fppf) sheaf by the U_α, $\alpha \in R$ (Exp. XXII, 6.2.2 (a)).*

2. Generators and relations for a pinned group

In this section, we fix a pinned S-group G. If $\alpha, \beta \in \Delta$, we shall use the notation Z_α, $Z_{\alpha\beta}$, $U_{\alpha\beta}$, $R^+_{\alpha\beta}$ of 1.7.

Theorem 2.1. *Let S be a scheme, G a pinned S-group, H an S-sheaf in groups for (fppf). Let*

$$f_N : \operatorname{Norm}_G(T) \to H, \qquad f_\alpha : U_\alpha \to H, \qquad \alpha \in R,$$

be morphisms of groups. In order that there exist a morphism of groups (necessarily unique)

$$f : G \to H$$

extending f_N and the $f_\alpha(\alpha \in R)$, it is necessary and sufficient that the following conditions be satisfied:

(i) For every $\alpha \in \Delta$ and every $\beta \in R$,

$$\operatorname{int}(f_N(w_\alpha)) \circ f_\beta = f_{s_\alpha(\beta)} \circ \operatorname{int}(w_\alpha).$$

(ii) For every $\alpha \in \Delta$, there exists a morphism of groups

$$F_\alpha : Z_\alpha \to H$$

extending f_α, $f_{-\alpha}$ and $f_N|_{Norm_{Z_\alpha}(T)}$.

(iii) For every pair $(\alpha, \beta) \in \Delta \times \Delta$, there exists a morphism of groups $U_{\alpha\beta} \to H$ inducing f_γ on U_γ for every $\gamma \in R^+_{\alpha\beta}$ (i.e. $U_\gamma \subset U_{\alpha\beta}$).

2.1.1. Proof

The conditions of the statement are evidently necessary. Choose on the other hand a structure of totally ordered group on $\Gamma_0(R)$ such that the roots > 0 are the elements of R^+ (Exp. XXI, 3.5.6); every product indexed by a subset of R will be taken relative to this order.

Denote by f_T the restriction of f_N to T, and consider the morphism

$$f : \Omega \to H$$

defined set-theoretically by

$$f\Big(\prod_{\alpha \in R^-} y_\alpha \cdot t \cdot \prod_{\alpha \in R^+} x_\alpha\Big) = \prod f_\alpha(y_\alpha) \cdot f_T(t) \cdot \prod f_\alpha(x_\alpha).$$

Any morphism satisfying the conditions of the statement must extend f; on the other hand any morphism of groups $f : G \to H$ extending f also extends f_N: indeed, extending f_α and $f_{-\alpha}$, it satisfies $f(w_\alpha) = F_\alpha(w_\alpha) = f_N(w_\alpha)$, and it extends f_T by hypothesis. By Exp. XXII, 4.1.11 (ii), one is therefore reduced to proving:

Proposition 2.1.2. *The morphism $f : \Omega \to H$ defined above is "generically multiplicative"; more precisely, for every $S' \to S$ and all $x, y \in \Omega(S')$ such that $xy \in \Omega(S')$, one has $f(xy) = f(x)f(y)$.*

Lemma 2.1.3. *Let $n \in Norm_G(T)(S')$, $S' \to S$, and let $\alpha, \beta \in R$ be such that $int(n)U_\alpha = U_\beta$ (i.e. $n(\alpha) = \beta$); then*

$$\mathrm{int}(f_N(n)) \circ f_\alpha = f_\beta \circ \mathrm{int}(n).$$

Indeed, it suffices to verify the formula for a system of generators of the sheaf $Norm_G(T)$; it is true for each w_α, $\alpha \in \Delta$ (by 2.1 (i)), so it suffices to do it for $n \in T(S')$. This is trivial by 2.1 (ii) if α is simple; otherwise, one takes a $w \in W$ such that $w^{-1}(\alpha) \in \Delta$; writing w as a product of simple reflections,[1361] one is reduced to proving that if the formula is true for α and for every n, it is also true for $w_{\alpha_0}(\alpha)$ and $t \in T(S')$, where $\alpha_0 \in \Delta$. Now, by 2.1 (i),

$$\begin{aligned}\mathrm{int}(f_N(t)) \circ f_{w_{\alpha_0}(\alpha)} &= \mathrm{int}(f_N(t\, w_{\alpha_0})) \circ f_\alpha \circ \mathrm{int}(w_{\alpha_0}^{-1}) \\ &= f_{w_{\alpha_0}(\alpha)} \circ \mathrm{int}(t\, w_{\alpha_0}) \circ \mathrm{int}(w_{\alpha_0}^{-1}).\end{aligned}$$

Lemma 2.1.4. *The restriction of f to U (resp. U^-) is a morphism of groups. In particular, this restriction is independent of the order chosen on the roots.*

It suffices to give the proof for U. By virtue of Exp. XXII, 5.5.8, it suffices to verify that for every pair $\alpha < \beta$ of positive roots, one has for all $x_\alpha \in U_\alpha(S')$, $x_\beta \in U_\beta(S')$, $S' \to S$,

$$f_\beta(x_\beta)\, f_\alpha(x_\alpha)\, f_\beta(x_\beta^{-1}) = f(x_\beta\, x_\alpha\, x_\beta^{-1}).$$

By Exp. XXII, 5.5.2, there exist $x_\gamma \in U_\gamma(S')$ ($\gamma = i\alpha + j\beta \in R$, $i > 0$, $j \geqslant 0$[1362]) such that

[1361] N.D.E.: We have replaced "fundamental symmetries" by "simple reflections".

[1362] N.D.E.: We have corrected $i, j > 0$ to $i > 0$, $j \geqslant 0$ (since x_α appears in the product on the right).

$$x_\beta\, x_\alpha\, x_\beta^{-1} = \prod_\gamma x_\gamma,$$

and we must therefore verify the relation

$$f_\beta(x_\beta)\, f_\alpha(x_\alpha)\, f_\beta(x_\beta^{-1}) = \prod_{\gamma = i\alpha + j\beta,\ i > 0,\ j \geq 0} f_\gamma(x_\gamma).$$

By Exp. XXI, 3.5.4, there exists $w \in W$ such that $w(\alpha) = \alpha_0 \in \Delta$, $w(\beta) \in A_{\alpha_0\beta_0}$ (notation of 1.7), where $\beta_0 \in \Delta$. Lifting w to an $n \in Norm_G(T)(S)$ (by Exp. XXII, 3.8), it suffices to verify the preceding relation after conjugation by $f_N(n)$. By 2.1.3, one is therefore reduced to the case $\alpha, \beta \in R^+_{\alpha_0\beta_0}$, a case in which we conclude by condition 2.1 (iii).

Lemma 2.1.5. *Let $n \in Norm_G(T)(S)$ be such that $int(n)U = U^-$ (i.e. n is the symmetry of the Weyl group*[1363] *(Exp. XXI, 3.6.14)). For every $u \in U(S')$, $S' \to S$ (resp. $u \in U^-(S')$, $S' \to S$), one has*

$$f(n\, u\, n^{-1}) = f_N(n)\, f(u)\, f_N(n^{-1}).$$

Immediate by 2.1.3 and 2.1.4.

Lemma 2.1.6. *Let $u \in B(S')$, $v \in B^-(S')$, $g \in \Omega(S')$, $S' \to S$. Then*

$$f(v\, g\, u) = f(v)\, f(g)\, f(u).$$

Indeed, set $v = v_1 t_1$, $g = v_2 t_2 u_2$, $u = t_3 u_3$, with $v_i \in U^-(S')$, $t_i \in T(S')$, $u_i \in U(S')$. One has

$$f(v)\, f(g)\, f(u) = f(v_1)\, f_T(t_1)\, f(v_2)\, f_T(t_2)\, f(u_2)\, f_T(t_3)\, f(u_3),$$

on the one hand, and

$$\begin{aligned} f(v\, g\, u) &= f(v_1\, t_1\, v_2\, t_1^{-1}\, t_1\, t_2\, t_3\, t_3^{-1}\, u_2\, t_3\, u_3) \\ &= f(v_1 \cdot t_1\, v_2\, t_1^{-1})\, f_T(t_1\, t_2\, t_3)\, f(t_3^{-1}\, u_2\, t_3 \cdot u_3). \end{aligned}$$

Using 2.1.4 to decompose the two extreme terms of this last expression, one obtains

$$f(v\, g\, u) = f(v_1)\, f(t_1\, v_2\, t_1^{-1})\, f_T(t_1\, t_2\, t_3)\, f(t_3^{-1}\, u_2\, t_3)\, f(u_3).$$

One is then reduced to the obvious formulas

$$f(t_1\, v_2\, t_1^{-1}) = f_T(t_1)\, f(v_2)\, f_T(t_1)^{-1}, \qquad f(t_3^{-1}\, u_2\, t_3) = f_T(t_3)^{-1}\, f(u_2)\, f$$

which follow from the definition of f and from 2.1.3.

Lemma 2.1.7. *Let $\alpha \in \Delta$ and $g \in \Omega(S') \cap int(w_\alpha)^{-1}\Omega(S')$, $S' \to S$. Then*

$$f(w_\alpha\, g\, w_\alpha^{-1}) = f_N(w_\alpha)\, f(g)\, f_N(w_\alpha)^{-1}.$$

Indeed, let $g \in \Omega(S')$, $S' \to S$. Write, by Exp. XXII, 5.6.8,

$$g = a\, x_{-\alpha}\, t\, x_\alpha\, b,$$

with $a \in U^-_{-\hat\alpha}(S')$, $x_{-\alpha} \in U_{-\alpha}(S')$, $t \in T(S')$, $x_\alpha \in U_\alpha(S')$, $b \in U_{\hat\alpha}(S')$. By 2.1.3, 2.1.4 and the fact that s_α permutes the positive roots $\neq \alpha$ (cf. Exp. XXI, 3.3.2), one has

[1363] N.D.E.: relative to Δ.

$$int(w_\alpha)\, a \in U_{-\alpha}^{-}(S'), \qquad int(w_\alpha)\, b \in U_{\alpha}(S')$$

and the formula to be proved is true for $g = a$ or $g = b$. By 2.1.6, one is therefore reduced to proving the asserted formula when $g = x_{-\alpha} t x_\alpha \in Z_\alpha(S')$. But then "everything happens in Z_α", and one concludes by condition (ii) of 2.1.

Lemma 2.1.8. *Let $n \in Norm_G(T)(S)$. For every $S' \to S$ and every $g \in \Omega(S')$ such that $int(n)g \in \Omega(S')$, one has*

$$f(n g n^{-1}) = f_N(n)\, f(g)\, f_N(n)^{-1}.$$

This is trivial if $n \in T(S)$ (by 2.1.3). The two sides of the preceding formula define morphisms of $\Omega \cap int(n)^{-1}\Omega$ into H; to verify that they coincide, it suffices to verify that there exists an open V_n of Ω containing the unit section such that $int(n)V_n \subset \Omega$, and that $f \circ int(n)$ and $int(f_N(n)) \circ f$ coincide on V_n. By virtue of the structure of $Norm_G(T)$, it suffices to prove that if the preceding assertion is true for an $n' \in Norm_G(T)(S)$ and if $\alpha \in \Delta$, it is true for $n = n' w_\alpha$. Set

$$V_n = \Omega \cap int(w_\alpha)^{-1} V_{n'}.$$

One has $int(n)V_n \subset int(n')V_{n'} \subset \Omega$. If $g \in V_n(S')$, then

$$int(n)\, g = int(n')\, int(w_\alpha)\, g.$$

Now $int(w_\alpha)g \in V_{n'}(S')$, hence by hypothesis

$$f(int(n')\, int(w_\alpha)\, g) = int(f_N(n'))\, f(int(w_\alpha)\, g);$$

since $int(w_\alpha)g \in \Omega(S')$, one may apply 2.1.7, which gives

$$f(int(w_\alpha)\, g) = int(f_N(w_\alpha))\, f(g),$$

and one concludes at once.

Let us now prove 2.1.2. Let $x, x' \in \Omega(S')$; write as usual

$$x = v t u, \qquad x' = v' t' u',$$

whence

$$x x' = v t (u v') t' u'.$$

By 2.1.6 and the relation $U^-(S')\Omega(S')U(S') = \Omega(S')$, one is reduced to proving that if $uv' \in \Omega(S')$, then $f(uv') = f(u)f(v')$. Let $n \in Norm_G(T)(S')$ be as in 2.1.5 (ii). One has

$$f(u) = f_N(n)^{-1} f(n u n^{-1}) f_N(n), \qquad f(v') = f_N(n)^{-1} f(n v' n^{-1}) f_N(n),$$

by *loc. cit.*, whence

$$f(u) f(v') = f_N(n)^{-1} f(n u n^{-1}) f(n v' n^{-1}) f_N(n).$$

But $nun^{-1} \in U^-(S')$, $nv'n^{-1} \in U(S')$, so that

$$f(n u n^{-1}) f(n v' n^{-1}) = f((n u n^{-1})(n v' n^{-1})) = f(n u v' n^{-1}),$$

which gives

$$f(u)\, f(v') = f_N(n)^{-1} f(n u v' n^{-1}) f_N(n).$$

If $uv' \in \Omega(S')$, one may apply 2.1.8 and one is done.

Remark 2.2. *Instead of giving oneself f_N, one may give oneself a morphism of groups $f_T : T \to H$ and sections $h_\alpha \in H(S)(\alpha \in \Delta)$ satisfying the conditions of 1.8.2. One must then replace condition (ii) of the theorem by:*

(ii') There exists a morphism of groups $F_\alpha : Z_\alpha \to H$ extending

$$f_\alpha, \quad f_{-\alpha}, \quad f_T \quad \text{and satisfying} \quad F_\alpha(w_\alpha) = h_\alpha.$$

We shall now give the preceding theorem a more explicit form. Keep the preceding notation.

Theorem 2.3. *Let S be a scheme, G a pinned S-group. Let H be an S-sheaf in groups for (fppf),*

$$f_T : T \to H, \quad f_\alpha : U_\alpha \to H, \quad \alpha \in \Delta,$$

morphisms of groups, and

$$h_\alpha \in H(S), \quad (\alpha \in \Delta),$$

sections of H. In order that there exist a morphism of groups

$$f : G \to H$$

(necessarily unique) inducing f_T and the $f_\alpha(\alpha \in \Delta)$ and satisfying $f(w_\alpha) = h_\alpha$ for every $\alpha \in \Delta$, it is necessary and sufficient that the following conditions be satisfied:

(i) For every $S' \to S$, every $t \in T(S')$, every $\alpha \in \Delta$ and every $x \in U_\alpha(S')$,

$$f_T(t)\, f_\alpha(x)\, f_T(t)^{-1} = f_\alpha(\mathrm{int}(t)\, x). \qquad (1)$$

(ii) For every $\alpha \in \Delta$, every $S' \to S$, every $t \in T(S')$,

$$h_\alpha\, f_T(t)\, h_\alpha^{-1} = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^*\alpha(t)^{-1}). \qquad (2)$$

(iii) For every $(\alpha, \beta) \in \Delta \times \Delta$,

$$(h_\alpha h_\beta)^{n_{\alpha\beta}} = f_T(t_{\alpha\beta}).(3)$$

(iv) For every $\alpha \in \Delta$, (recall that we write $u_\alpha = \exp_\alpha(X_\alpha)$)

$$(h_\alpha f_\alpha(u_\alpha))^3 = e.(4)$$

(v) For every $(\alpha, \beta) \in \Delta \times \Delta$, $\alpha \neq \beta$, there exists a morphism of groups

$$f_{\alpha\beta} : U_{\alpha\beta} \to H$$

satisfying the two following conditions:

a) One has

$$f_{\alpha\beta}|_{U_\alpha} = f_\alpha, \qquad f_{\alpha\beta}|_{U_\beta} = f_\beta, \tag{5}$$

b) For every $\gamma \in R^+_{\alpha\beta}$, $\gamma \neq \alpha$ (resp. $\gamma \neq \beta$), and every $x \in U_\gamma(S')$, $S' \to S$,

$$\mathrm{int}(h_\alpha)\, f_{\alpha\beta}(x) = f_{\alpha\beta}(\mathrm{int}(w_\alpha)\, x), \tag{6}$$
$$(\text{resp. } \mathrm{int}(h_\beta)\, f_{\alpha\beta}(x) = f_{\alpha\beta}(\mathrm{int}(w_\alpha)\, x)).$$

Proof. The uniqueness is clear by 1.9.2. Let us prove existence.

Lemma 2.3.1. *There exists a morphism of groups*

$$f_N : Norm_G(T) \to H$$

extending f_T and satisfying $f_N(w_\alpha) = h_\alpha$.

This is precisely what is asserted by 1.8.2, taking into account conditions (2) and (3).

Lemma 2.3.2. *There exists a morphism of groups*

$$F_\alpha : Z_\alpha \to H, \qquad \alpha \in \Delta,$$

extending f_T, f_α and satisfying $F_\alpha(w_\alpha) = h_\alpha$, hence extending $f_N|_{Norm_{Z_\alpha}(T)}$.

This is clear from Exp. XX, 6.2 and conditions (1), (3) and (4).

Lemma 2.3.3. *For every $(\alpha,\beta) \in \Delta \times \Delta$, $\alpha \neq \beta$, every $n \in Norm_{Z_{\alpha\beta}}(T)(S)$ such that $n(\alpha) = \alpha$ (resp. $n(\alpha) = \beta$), i.e. $int(n)U_\alpha = U_\alpha$ (resp. $int(n)U_\alpha = U_\beta$), every $S' \to S$ and every $x \in U_\alpha(S')$,*

$$\mathrm{int}(f_N(n))\, f_\alpha(x) = f_\alpha(\mathrm{int}(n)\, x)$$

(resp.

$$\mathrm{int}(f_N(n))\, f_\alpha(x) = f_\beta(\mathrm{int}(n)\, x)).$$

Indeed, there exist a $t \in T(S)$ and a sequence $\alpha_i \subset \alpha, \beta$ such that $n = tw_{\alpha_k} \cdots w_{\alpha_1}$. The condition is satisfied for $n = t$ by condition (1). One may therefore suppose $n = w_{\alpha_k} \cdots w_{\alpha_1}$. We proceed by induction on k, i.e. we suppose the assertion proved for every n which can be written as a product of fewer than $k-1$ simple reflections (and satisfies the hypotheses of the lemma). Consider the roots

$$\gamma_i = s_{\alpha_i} \cdots s_{\alpha_1}(\alpha).$$

If all the γ_i are positive, i.e. belong to $R^+_{\alpha\beta}$, one may apply condition (v) of 2.3; using the notation of 2.3 (v), one concludes at once by induction that

$$\mathrm{int}(f_N(w_{\alpha_i} \cdots w_{\alpha_1}))\, f_\alpha(x) = f_{\alpha\beta}(\mathrm{int}(w_{\alpha_i} \cdots w_{\alpha_1})\, x),$$

which for $i = k$ is the desired property. If not all the γ_i are positive, there exists an index $j < k$ such that

$$\gamma_j \in R^+, \qquad \gamma_{j+1} \notin R^+.$$

Since $\gamma_{j+1} = s_{\alpha_j}(\gamma_j)$, it follows at once that $\gamma_j = \alpha_j$, by Exp. XXI, 3.3.2, hence that γ_j is α or β, and one may decompose n as

$$n = n' \cdot n'', n' = w_{\alpha_k} \cdots w_{\alpha_{j+1}}, n'' = w_{\alpha_j} \cdots w_{\alpha_1},$$

n' and n'' satisfying the hypotheses of the lemma and being a product of fewer than $k-1$ reflections, hence satisfying by the induction hypothesis the conclusion of the lemma.

Lemma 2.3.4. *Let $\alpha \in R$. If $n, n' \in Norm_G(T)(S)$ and $\beta, \beta' \in \Delta$ satisfy $n(\alpha) = \beta$ and $n'(\alpha) = \beta'$, one has*

$$int(f_T(n)^{-1}) f_\beta(n x n^{-1}) = int(f_T(n')^{-1}) f_{\beta'}(n' x n'^{-1}),$$

for every $x \in U_\alpha(S')$, $S' \to S$.

It suffices to verify that if $n(\alpha) = \beta$, $\alpha, \beta \in \Delta$, then $int(f_T(n)) \circ f_\alpha = f_\beta \circ int(n)$. Now, by Tits's lemma (Exp. XXI, 5.6), there exists a sequence of simple roots $\alpha_0 = \alpha, \alpha_1, \cdots, \alpha_m = \beta$, and a sequence of elements $w_i \in W$, $i = 0, 1, \cdots, m-1$, with

$$n = w_{m-1} \cdots w_0,$$
$$w_i(\alpha_i) = \alpha_{i+1}, \qquad i = 0, 1, \ldots, m-1,$$

the following condition being moreover satisfied: for every $i = 0, 1, \cdots, m-1$, there exists a simple root β_i such that

$$w_i \in W_{\alpha_i \beta_i}, \qquad \alpha_{i+1} = \alpha_i \quad \text{or} \quad \beta_i.$$

One is then reduced by induction to the case treated in the preceding lemma.

Lemma 2.3.5. *There exists a family of morphisms of groups $f'_\gamma : U_\gamma \to H$, $\gamma \in R$, satisfying*

(i) For $\alpha \in \Delta$, $f'_\alpha = f_\alpha$ and $f'_{-\alpha} = F_\alpha|_{U_{-\alpha}}$, where F_α is defined in 2.3.2.

(ii) For $\alpha, \beta \in \Delta$ and $\gamma \in R^+_{\alpha\beta}$, $f'_\gamma = f_{\alpha\beta}|_{U_\gamma}$.

(iii) For every $n \in Norm_G(T)(S)$ and $\alpha, \beta \in R$ such that $n(\alpha) = \beta$,

$$int(f_N(n)) f'_\alpha(x) = f'_\beta(int(n) x)$$

for every $x \in U_\alpha(S')$, $S' \to S$.

For every root $\alpha \in R$, there exists an $n \in Norm_G(T)(S)$ such that $n(\alpha) \in \Delta$. Then define $f'_\alpha(x)$ as the expression of 2.3.4. This is independent of the choice of n, and f'_α is indeed a morphism of groups. Property (iii) is satisfied by construction. The first assertion of (i) is clear (take $n = e$), the second too (take $n = w_\alpha$ and apply

2.3.2); if $\gamma \in R^+_{\alpha\beta}$ ($\alpha, \beta \in \Delta$), there exists $n \in Norm_{Z_{\alpha\beta}}(T)(S)$ such that $n(t) = \alpha$ or β; one then applies (iii) and conditions (5) and (6) and has proved (ii).

2.3.6.

Let us now prove the theorem by showing that the conditions of 2.1 are satisfied, for the morphisms f_N and f'_α, $\alpha \in R$.

- 2.1 (i) follows from 2.3.5 (iii),
- 2.1 (ii) follows from 2.3.5 (i) and 2.3.2,
- 2.1 (iii) follows from 2.3.5 (ii) and from the fact that $f_{\alpha\beta}$ is a morphism of groups.

An obvious corollary is:

Corollary 2.4. *Let S be a scheme, G a pinned S-group of semisimple rank $\geqslant 1$, H an (fppf) S-sheaf in groups. For each $(\alpha, \beta) \in \Delta \times \Delta$, let*

$$F_{\alpha\beta} : Z_{\alpha\beta} \to H$$

be a morphism of groups. In order that there exist a morphism of groups

$$f : G \to H$$

inducing the $F_{\alpha\beta}$, it is necessary and sufficient that for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$F_{\alpha\beta}|_{Z_\alpha} = F_{\alpha\alpha}.$$

The condition is evidently necessary. Suppose it satisfied. Set $f_T = F_{\alpha\alpha}|_T$ (which does not depend on α, since $F_{\alpha\alpha}|_T = F_{\alpha\beta}|_T = F_{\beta\beta}|_T$). Set

```
p_α = F_{αα}|_{U_α},      h_α = F_{αα}(w_α),      f_{αβ} = F_{αβ}|_{U_{αβ}}.
```

The conditions of 2.3 are evidently satisfied: (1), (2), (4) "are verified" in Z_α, (3), (5) and (6) in $Z_{\alpha\beta}$. There therefore exists a morphism $f : G \to H$ extending f_T, the $f_\alpha (\alpha \in \Delta)$ and satisfying $f(w_\alpha) = h_\alpha$; it coincides with $F_{\alpha\beta}$ on generators of $Z_{\alpha\beta}$, hence satisfies $f|_{Z_{\alpha\beta}} = F_{\alpha\beta}$.

One also has the following technical corollary.

Corollary 2.5. *Let S be a scheme, G and G' two pinned S-groups of semisimple rank 2, q an integer > 0 such that $x \mapsto x^q$ is an endomorphism of $G_{a,S}$, $h : R(G') \to R(G)$ a q-morphism of pinned root data. Choose for each $\alpha \in R^+$ an $X_\alpha \in \Gamma(S, g_\alpha)^\times$ and an $X'_{d(\alpha)} \in \Gamma(S, g'_{d(\alpha)})^\times$ (extending the canonical choices for $\alpha \in \Delta$). Suppose the following conditions are realized:*

(i) If $\Delta = \alpha, \beta$, then $D_S(h)(t_{\alpha\beta}) = t'_{d(\alpha)d(\beta)}$.

(ii) For every $\alpha \in \Delta$ and every $\beta \in R^+$, $\beta \neq \alpha$ (whence $s_\alpha(\beta) \in R^+$), if $z \in G_m(S)$ is defined by

$$Ad(w_\alpha) X_\beta = z X_{s_\alpha(\beta)},$$

one has also

$$Ad(w'_{d(\alpha)}) X'_{d(\beta)} = z^{q(\beta)} X'_{d(s_\alpha(\beta))}.$$

(iii) There exists a morphism of groups $f : U \to U'$ such that for every $\alpha \in R^+$,

$$f(\exp(x X_\alpha)) = \exp(x^{q(\alpha)} X'_{d(\alpha)})$$

for every $x \in G_a(S')$, 'S' $\to S$.

Then there exists a morphism of pinned groups $g : G \to G'$ such that $R(g) = h$.

Indeed, one defines $f_\alpha : U_\alpha \to G'$ by

$$f_\alpha(\exp(x X_\alpha)) = \exp(x^{q(\alpha)} X'_{d(\alpha)});$$

one sets $f_T = D_S(h)$, $h_\alpha = w'_{d(\alpha)}$. The conditions of 2.3 are satisfied (note that $q(s_\alpha(\beta)) = q(\beta)$ (Exp. XXI, 6.8.4) and that one always has $D_S(h)(t_\alpha) = t'_{d(\alpha)}$), and one concludes at once.

Remark 2.6. *One may make condition (v) of 2.3 more precise as follows. One must first verify:*

(a) For every word in w_α and w_β such that the corresponding word transforms α or β into α or β, the relation of type 2.3.3 corresponding is satisfied. In fact the proof of 2.3.3 shows that it suffices to verify it for the words in w_α and w_β that are minimal (in the sense that any nontrivial initial sub-word does not satisfy the imposed conditions).

If condition (a) is satisfied, one may now define for each $\gamma \in R^+_{\alpha\beta}$ an $f_\gamma : U_\gamma \to H$ as in 2.3.5; one must then write:

(b) The morphism defined by the f_γ

$$U_{\alpha\beta} = \prod_{\gamma \in R^+_{\alpha\beta}} U_\gamma \to H$$

is a morphism of groups. By Exp. XXII, 5.5.8, (b) is entailed by:

(b') The preceding morphism respects the commutation relations between U_γ and U_δ for $\gamma, \delta \in R^+_{\alpha\beta}$, $\gamma > \delta$ (i.e. the relations in $C_{i,j,\gamma,\delta}$ of Exp. XII, 5.5.2).

It is clear that conversely the set of conditions (a) and (b') is equivalent to (v).

One may even reduce the preceding conditions to conditions bearing only on the elements h_α, h_β, $f_\alpha(u_\alpha)$, $f_\beta(u_\beta)$ of H. A condition of type (a) will be written for example, if $s_\alpha s_\beta s_\alpha(\beta) = \alpha$:

$$int(h_\alpha h_\beta h_\alpha) f_\beta(x) = f_\alpha(int(w_\alpha w_\beta w_\alpha) x); \qquad (1)$$

for every $x \in U_\alpha(S')$, $S' \to S$. In particular, for $x = u_\beta$, one has $int(w_\alpha w_\beta w_\alpha)u_\beta = u_\alpha^z$ for a certain section z of $G_m(S)$, and the preceding relation will give

```
int(h_α h_β h_α) f_β(u_β) = f_α(u_α)^z.                (1')
```

Let us show that, conversely, taking into account conditions (i) to (iv) of 2.3, (1’) entails (1). If $t \in T(S')$, $S' \to S$, make $int(t)$ act on (1’); taking into account conditions (i) and (ii), one obtains (1) for $x = int(t)u_\beta = u_\beta^{\beta(t)}$. It suffices to remark now that $\beta : T \to G_{m,S}$ is faithfully flat, hence that condition (1) is certainly true for $x \in U_\alpha(S')^\times$, $S' \to S$. Since it is additive in x, and every section of U_α can be locally written as a sum of two sections of $U_\alpha^\times$, one concludes that (1’) + (i) + (ii) ⇒ (1).

One reasons similarly with conditions of type (b). It is then necessary to use the fact that if γ and γ' are two distinct (hence linearly independent over $\mathbb{Q}$) positive roots, the morphism $T \to G_{m,S}^2$ of components γ and γ' is faithfully flat. We leave the details of this transposition to the reader.

3. Groups of semisimple rank 2

3.1. Generalities

Lemma 3.1.1. *Let S be a scheme, G a pinned S-group, α and β two roots of G, with $\alpha + \beta \neq 0$.*

(i) If $\alpha + \beta \notin R$, then

```
exp(X_α) exp(X_β) = exp(X_β) exp(X_α)
```

for all $X_\alpha \in W(g_\alpha)(S')$, $X_\beta \in W(g_\beta)(S')$, $S' \to S$.

(ii) If $\alpha + \beta$ and $\alpha - \beta$ are not roots, then

```
w_α(z_α) exp(X_β) w_α(z_α)^{-1} = exp(X_β)
```

for all $X_\beta \in W(g_\beta)(S')$, $z_\alpha \in W(g_\alpha)^\times(S')$, $S' \to S$, and

```
w_α(z_α) w_β(z_β) = w_β(z_β) w_α(z_α)
```

for all $z_\alpha \in W(g_\alpha)^\times(S')$ and $z_\beta \in W(g_\beta)^\times(S')$, $S' \to S$.

(iii) Let $X_\alpha \in W(g_\alpha)^\times(S')$, $X_\beta \in W(g_\beta)^\times(S')$, and $w \in Norm_G(T)(S')$ such that $w(\alpha) = \beta$; define $z \in G_m(S')$ by

```
Ad(w) X_α = z X_β.
```

Then

```
int(w) exp(x X_α) = exp(x z X_β),
int(w) exp(y X_α^{-1}) = exp(y z^{-1} X_β^{-1}),
int(w) w_α(X_α) = β^*(z) w_β(X_β).
```

In particular, if $z = \pm 1$, then

$$int(w)w_\alpha(X_\alpha) = w_\beta(X_\beta)^z.$$

(iv) If one sets $t_\alpha = \alpha^*(-1)$, $t_\beta = \beta^*(-1)$, then

$$s_\alpha(t_\beta) = t_\beta t_\alpha^{(\beta^*, \alpha)}, \qquad \beta(t_\alpha) = (-1)^{(\alpha^*, \beta)}.$$

Proof. (i) follows at once from Exp. XXII, 5.5.2; (ii) from Exp. XX, 3.1 and from (i) applied to (β,α), $(\beta,-\alpha)$, $(-\beta,\alpha)$, $(-\beta,-\alpha)$; (iii) is evident from the definitions. For the last assertion of (iii), note that $\beta^*(-1) = w_\beta(X_\beta)^{-2}$. Finally, (iv) is trivial.

Proposition 3.1.2 (Groups of type $A_1 \times A_1$). *Let S be a scheme, G a pinned S-group of type $A_1 \times A_1$, and denote $\Delta = R^+ = \alpha, \beta$.*

(i) One has

$$t_{\alpha\beta} = (w_\alpha w_\beta)^2 = t_\alpha t_\beta = (w_\beta w_\alpha)^2 = t_{\beta\alpha}.$$

(ii) One has

$$Ad(w_\alpha) X_\beta = X_\beta, \qquad Ad(w_\beta) X_\alpha = X_\alpha.$$

(iii) U_α and U_β commute (i.e. U is commutative).

Indeed, by assertion (ii) of the lemma, $w_\alpha w_\beta = w_\beta w_\alpha$, whence $(w_\alpha w_\beta)^2 = w_\alpha^2 w_\beta^2 = t_\alpha t_\beta$, which is (i). By assertion (ii) of the lemma, one has also (ii); finally, (iii) is assertion (i) of the lemma.

3.1.3.

Let us make condition (v) of 2.3 explicit here. Using the method exposed in 2.6, one obtains the two following groups of conditions, setting $v_\alpha = f_\alpha(u_\alpha)$ for $\alpha \in \Delta$:

$$\text{(A)} \quad h_\alpha v_\beta h_\alpha^{-1} = v_\beta, \quad h_\beta v_\alpha h_\beta^{-1} = v_\alpha \qquad\qquad \text{(B)} \quad v_\alpha v_\beta = v_\beta v_\alpha.$$

3.2. Groups of type A_2

Proposition 3.2.1. *Let S be a scheme, G a pinned S-group of type A_2, and denote $\Delta = \alpha, \beta$, $R^+ = \alpha, \beta, \alpha+\beta$.*

(i) One has

$$t_{\alpha\beta} = (w_\alpha w_\beta)^3 = e = (w_\beta w_\alpha)^3 = t_{\beta\alpha}.$$

(ii) Set $Ad(w_\beta)X_\alpha = X_{\alpha+\beta}$. Then

$$Ad(w_\alpha) X_\beta = -X_{\alpha + \beta}, \qquad Ad(w_\alpha) X_{\alpha + \beta} = X_\beta, \qquad Ad(w_\beta) X_{\alpha + \beta} = -X_\alpha.$$

(iii) Set $p_{\alpha+\beta}(x) = \exp(xX_{\alpha+\beta}) = int(w_\beta)p_\alpha(x)$. Then

$$p_\beta(y)\, p_\alpha(x) = p_\alpha(x)\, p_\beta(y)\, p_{\alpha + \beta}(x y).$$

3.2.2.

The proof occupies Nos. 3.2.2 to 3.2.7. First, $(\beta^*,\alpha) = (\alpha^*,\beta) = -1$, whence $\alpha(t_\beta) = \beta(t_\alpha) = -1$.

Set $Ad(w_\beta)X_\alpha = X_{\alpha+\beta}$; one has at once

$$\mathrm{Ad}(w_\beta) X_{\alpha + \beta} = \alpha(t_\beta) X_\alpha = -X_\alpha.$$

Set

$$\mathrm{Ad}(w_\alpha) X_\beta = z X_{\alpha + \beta}, \qquad z \in G_m(S),$$

whence

$$\mathrm{Ad}(w_\alpha) X_{\alpha + \beta} = \beta(t_\alpha) z^{-1} X_\beta = -z^{-1} X_\beta.$$

We know (Exp. XXII, 5.5.2) that there exists a unique section $A \in G_a(S)$ such that

$$p_\beta(y) p_\alpha(x) = p_\alpha(x) p_\beta(y) p_{\alpha + \beta}(A x y). \qquad (+)$$

To prove (ii) and (iii), it therefore remains to show that $A = -z = 1$.

3.2.3.

Make $int(w_\beta)$ act on the preceding formula (+); one obtains:

$$p_{-\beta}(-y) p_{\alpha + \beta}(x) = p_{\alpha + \beta}(x) p_{-\beta}(-y) p_\alpha(-A x y). \qquad (++)$$

3.2.4.

By definition of $p_{\alpha+\beta}$,

$$w_\beta p_\alpha(x) w_\beta^{-1} = p_{\alpha + \beta}(x),$$

which is written

$$p_\beta(1) p_{-\beta}(-1) p_\beta(1) p_\alpha(x) p_\beta(-1) p_{-\beta}(1) p_\beta(-1) = p_{\alpha + \beta}(x).$$

Since p_β and $p_{\alpha+\beta}$ commute, $\alpha + 2\beta$ not being a root, this is also written

$$p_\beta(1) p_\alpha(x) p_\beta(-1) = p_{-\beta}(1) p_{\alpha + \beta}(x) p_{-\beta}(-1).$$

Using now (+) in the left member and (++) in the right, one obtains:

$$p_\alpha(x) p_\beta(1) p_{\alpha + \beta}(A x) p_\beta(-1) = p_{\alpha + \beta}(x) p_{-\beta}(1) p_\alpha(A x) p_{-\beta}(-1).$$

Since $\alpha + 2\beta$ (resp. $\alpha - \beta$) is not a root, the left (resp. right) member is written

$$p_\alpha(x) p_{\alpha + \beta}(A x) \qquad \text{resp. } p_{\alpha + \beta}(x) p_\alpha(A x)$$

and the right-hand term equals $p_\alpha(Ax)p_{\alpha+\beta}(x)$, since $2\alpha+\beta$ is not a root. Therefore

$$p_\alpha(x) p_{\alpha + \beta}(A x) = p_\alpha(A x) p_{\alpha + \beta}(x),$$

which gives (by XXII 4.1.3) $A = 1$.

3.2.5.

Now make $int(w_\alpha)$ act on formula (+); using the fact that $A = 1$, one finds

$$p_{\alpha + \beta}(z y) p_{-\alpha}(-x) = p_{-\alpha}(-x) p_{\alpha + \beta}(z y) p_\beta(-z^{-1} x y). \qquad (+++)$$

3.2.6.

Write now, as a moment ago,

$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{\alpha + \beta}(z y),$$

whence, since p_α and $p_{\alpha+\beta}$ commute,

$$p_\alpha(1) p_\beta(y) p_\alpha(-1) = p_{-\alpha}(1) p_{\alpha + \beta}(z y) p_{-\alpha}(-1).$$

Using now (+) and (+++), this is also written

$$p_\beta(y) p_{\alpha + \beta}(-y) = p_{\alpha + \beta}(z y) p_\beta(-z^{-1} y),$$

and since p_β and $p_{\alpha+\beta}$ commute, this gives $z = -1$.

3.2.7.

We have therefore proved (ii) and (iii). Let us prove (i). One has

$$w_\alpha w_\beta w_\alpha = w_\alpha w_\beta w_\alpha^{-1} w_\alpha^2 = w_{\alpha + \beta}^{-1} t_\alpha,$$

whence

$$\begin{aligned} w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{\alpha + \beta}^{-1} t_\alpha w_\beta = w_\beta w_{\alpha + \beta}^{-1} w_\beta^{-1} \cdot s_\beta(t_\alpha) \cdot t_\beta \\ &= w_\alpha \cdot t_\alpha t_\beta \cdot t_\beta = w_\alpha t_\alpha = w_\alpha^{-1}, \end{aligned}$$

which gives

$$(w_\alpha w_\beta)^3 = (w_\beta w_\alpha)^3 = e,$$

which completes the proof.

Remark 3.2.8. *Condition (v) of 2.3 translates here as (setting $v_\alpha = f_\alpha(u_\alpha)$):*

(A) $\quad int(h_\alpha h_\beta h_\alpha) v_\beta = v\beta \to h_\alpha v_\beta h_\alpha^{-1} = h_\beta v_\alpha h_\beta^{-1}$ $\qquad$ (B)

$\quad int(h_\beta h_\alpha h_\beta) v_\alpha = v\alpha$

`v_β v_α = v_α v_β ·`

`v_β · h_β v_α h_β^{-`

`v_α · h_β v_α h_β^{-`

Setting $v_{\alpha+\beta} = int(h_\beta)v_\alpha$, the three last conditions are also written

$$\text{(B)} \quad \begin{aligned} &v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha + \beta}, \\ &v_\alpha v_{\alpha + \beta} = v_{\alpha + \beta} v_\alpha, \\ &v_\beta v_{\alpha + \beta} = v_{\alpha + \beta} v_\beta. \end{aligned}$$

3.3. Groups of type B_2

Proposition 3.3.1. *Let S be a scheme, G a pinned S-group of type B_2, and denote $\Delta = \alpha, \beta$, $R^+ = \alpha, \beta, \alpha + \beta, 2\alpha + \beta$.*

(i) One has

$$t_{\alpha\beta} = (w_\alpha w_\beta)^4 = t_\alpha = (w_\beta w_\alpha)^4 = t_{\beta\alpha}.$$

(ii) If one sets

$$\mathrm{Ad}(w_\beta) X_\alpha = X_{\alpha + \beta}, \qquad \mathrm{Ad}(w_\alpha) X_\beta = X_{2\alpha + \beta},$$

one has:

$$\mathrm{Ad}(w_\alpha) X_{\alpha + \beta} = -X_{\alpha + \beta},$$
$$\mathrm{Ad}(w_\alpha) X_{2\alpha + \beta} = X_\beta,$$
$$\mathrm{Ad}(w_\beta) X_{\alpha + \beta} = -X_\alpha,$$
$$\mathrm{Ad}(w_\beta) X_{2\alpha + \beta} = X_{2\alpha + \beta}.$$

(iii) Set

$$p_{\alpha + \beta}(x) = \exp(x X_{\alpha + \beta}) = \mathrm{int}(w_\beta) p_\alpha(x),$$
$$p_{2\alpha + \beta}(x) = \exp(x X_{2\alpha + \beta}) = \mathrm{int}(w_\alpha) p_\beta(x).$$

Then:

$$p_\beta(y) p_\alpha(x) = p_\alpha(x) p_\beta(y) p_{\alpha + \beta}(x y) p_{2\alpha + \beta}(x^2 y),$$
$$p_{\alpha + \beta}(y) p_\alpha(x) = p_\alpha(x) p_{\alpha + \beta}(y) p_{2\alpha + \beta}(2 x y).$$

3.3.2.

The proof occupies Nos. 3.3.2 to 3.3.6. One has $(\beta^*, \alpha) = -1$, $(\alpha^*, \beta) = -2$, whence $\alpha(t_\beta) = -1$, $\beta(t_\alpha) = 1$.

Note in passing that $\beta(t_\alpha) = \alpha(t_\alpha) = 1$, which shows that t_α is a section of $Centr(G)$. Set

$$\mathrm{Ad}(w_\beta) X_\alpha = X_{\alpha + \beta}, \qquad \mathrm{Ad}(w_\alpha) X_\beta = X_{2\alpha + \beta}.$$

One has at once

$$\mathrm{Ad}(w_\beta) X_{\alpha + \beta} = \alpha(t_\beta) X_\alpha = -X_\alpha,$$
$$\mathrm{Ad}(w_\alpha) X_{2\alpha + \beta} = \beta(t_\alpha) X_\beta = X_\beta.$$

Since $(2\alpha + \beta) + \beta$ and $(2\alpha + \beta) - \beta$ are not roots,

$$\mathrm{Ad}(w_\beta) X_{2\alpha + \beta} = X_{2\alpha + \beta}.$$

There exists a scalar $k \in G_m(S)$ such that

$$\mathrm{Ad}(w_\alpha) X_{\alpha + \beta} = k X_{\alpha + \beta}.$$

On the other hand, by Exp. XXII, 5.5.2, there exist sections $A, B, C \in G_a(S)$ such that

$$p_\beta(y) p_\alpha(x) = p_\alpha(x) p_\beta(y) p_{\alpha + \beta}(A x y) p_{2\alpha + \beta}(B x^2 y), \tag{1}$$
$$p_{\alpha + \beta}(y) p_\alpha(x) = p_\alpha(x) p_{\alpha + \beta}(y) p_{2\alpha + \beta}(C x y). \tag{2}$$

It is therefore necessary, in (ii) and (iii), to prove that $A = B = 1$, $C = 2$, $k = -1$.

3.3.3.

Make $int(w_\alpha)$ act on formula (2). One finds

$$p_{2\alpha + \beta}(y) p_{-\alpha}(-x) = p_{-\alpha}(-x) p_{2\alpha + \beta}(y) p_{\alpha + \beta}(A k x y) p_\beta(B x^2 y). \tag{3}$$

Transforming (2) likewise, one obtains

$$p_{\alpha+\beta}(ky)\, p_{-\alpha}(-x) = p_{-\alpha}(-x)\, p_{\alpha+\beta}(ky)\, p_\beta(Cxy). \tag{4}$$

Transforming (1) by $int(w_\beta)$,

$$p_{-\beta}(-y)\, p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)\, p_{-\beta}(-y)\, p_\alpha(-Axy)\, p_{2\alpha+\beta}(Bx^2y). \tag{5}$$

3.3.4.

Write

$$w_\beta\, p_\alpha(x)\, w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Since $\alpha + 2\beta$ is not a root, this gives

$$p_\beta(1)\, p_\alpha(x)\, p_\beta(-1) = p_{-\beta}(1)\, p_{\alpha+\beta}(x)\, p_{-\beta}(-1).$$

Using (1) in the left member and (5) in the right, one obtains

$$p_\alpha(x)\, p_\beta(1)\, p_{\alpha+\beta}(Ax)\, p_{2\alpha+\beta}(Bx^2)\, p_\beta(-1) =$$
$$p_{\alpha+\beta}(x)\, p_{-\beta}(1)\, p_\alpha(Ax)\, p_{2\alpha+\beta}(-Bx^2)\, p_{-\beta}(-1).$$

Since p_β commutes with $p_{\alpha+\beta}$ and $p_{2\alpha+\beta}$ on the one hand, and $p_{-\beta}$ commutes with p_α and $p_{2\alpha+\beta}$ on the other hand, this is written

$$p_\alpha(x)\, p_{\alpha+\beta}(Ax)\, p_{2\alpha+\beta}(Bx^2) = p_{\alpha+\beta}(x)\, p_\alpha(Ax)\, p_{2\alpha+\beta}(-Bx^2).$$

Transforming the right member by (2), one obtains

$$p_\alpha(x)\, p_{\alpha+\beta}(Ax)\, p_{2\alpha+\beta}(Bx^2) = p_\alpha(Ax)\, p_{\alpha+\beta}(x)\, p_{2\alpha+\beta}((AC - B)x^2),$$

which gives

$$A = 1, \qquad C = 2B.$$

3.3.5.

Write now

$$w_\alpha\, p_\beta(y)\, w_\alpha^{-1} = p_{2\alpha+\beta}(y).$$

Since $3\alpha + \beta$ is not a root, this gives

$$p_\alpha(1)\, p_\beta(y)\, p_\alpha(-1) = p_{-\alpha}(1)\, p_{2\alpha+\beta}(y)\, p_{-\alpha}(-1).$$

Using (1) in the left member, (3) in the right, one obtains

$$p_\beta(y)\, p_{\alpha+\beta}(-Ay)\, p_{2\alpha+\beta}(By) = p_{2\alpha+\beta}(y)\, p_{\alpha+\beta}(Aky)\, p_\beta(By).$$

Since $p_{\alpha+\beta}$, $p_{2\alpha+\beta}$ and p_β commute, this gives at once

$$B = 1, \qquad -A = Ak,$$

whence finally

$$A = 1, \qquad B = 1, \qquad C = 2, \qquad k = -1.$$

3.3.6.

We have therefore proved (ii) and (iii). Let us prove (i). Taking into account the equality $s_\beta(t_\alpha) = t_\alpha t_\beta^{(\alpha^*,\beta)} = t_\alpha$ (since $(\alpha^*,\beta) = 2$),[1364] one has successively:

$$\begin{aligned}
w_\alpha w_\beta w_\alpha &= w_\alpha w_\beta w_\alpha^{-1} t_\alpha = w_{2\alpha+\beta} t_\alpha,\\
w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{2\alpha+\beta} w_\beta^{-1} \cdot s_\beta(t_\alpha) \cdot t_\beta = w_{2\alpha+\beta} t_\alpha t_\beta,\\
w_\alpha w_\beta w_\alpha w_\beta w_\alpha w_\beta w_\alpha &= w_\alpha w_{2\alpha+\beta} w_\alpha^{-1} \cdot s_\alpha(t_\alpha t_\beta) \cdot t_\alpha = w_\beta \cdot t_\alpha \cdot t_\beta t_\alpha \cdot t\\
&= w_\beta^{-1} t_\alpha,
\end{aligned}$$

whence

$$(w_\beta w_\alpha)^4 = t_\alpha,$$

and

$$(w_\alpha w_\beta)^4 = s_\beta(t_\alpha) = t_\alpha,$$

which completes the proof.

Remark 3.3.7. *Condition (v) of 2.3 translates here as follows, setting $v_{\alpha+\beta} = int(h_\beta)v_\alpha$ and $v_{2\alpha+\beta} = int(h_\alpha)v_\beta$:*

(A)
$$\begin{aligned}
int(h_\alpha h_\beta h_\alpha)\, v_\beta &= v_\beta,\\
int(h_\beta h_\alpha h_\beta)\, v_\alpha &= v_\alpha,
\end{aligned}$$

(B)
$$\begin{aligned}
v_\beta v_\alpha &= v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta},\\
v_{\alpha+\beta} v_\alpha &= v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2,\\
v_{\alpha+\beta} v_\beta &= v_\beta v_{\alpha+\beta},\\
v_{2\alpha+\beta} v_\alpha &= v_\alpha v_{2\alpha+\beta},\\
v_{2\alpha+\beta} v_\beta &= v_\beta v_{2\alpha+\beta},\\
v_{2\alpha+\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{2\alpha+\beta}.
\end{aligned}$$

3.4. Groups of type G_2

Proposition 3.4.1. *Let S be a scheme, G a pinned S-group of type G_2, and denote $\Delta = \alpha,\beta$, $R^+ = \alpha,\beta,\alpha+\beta,2\alpha+\beta,3\alpha+\beta,3\alpha+2\beta$.*

(i) One has

$$t_{\alpha\beta} = (w_\alpha w_\beta)^6 = e = (w_\beta w_\alpha)^6 = t_{\beta\alpha}.$$

(ii) If one sets

$$\begin{aligned}
&Ad(w_\beta) X_\alpha = X_{\alpha+\beta}, \quad Ad(w_\alpha) X_{\alpha+\beta} = X_{2\alpha+\beta},\\
&Ad(w_\alpha) X_\beta = -X_{3\alpha+\beta}, \quad Ad(w_\beta) X_{3\alpha+\beta} = X_{3\alpha+2\beta},
\end{aligned}$$

one has:

$$\begin{aligned}
&Ad(w_\alpha) X_{2\alpha+\beta} = -X_{\alpha+\beta}, \quad Ad(w_\alpha) X_{3\alpha+\beta} = X_\beta,\\
&Ad(w_\alpha) X_{3\alpha+2\beta} = X_{3\alpha+2\beta}, \quad Ad(w_\beta) X_{\alpha+\beta} = -X_\alpha,\\
&Ad(w_\beta) X_{2\alpha+\beta} = X_{2\alpha+\beta}, \quad Ad(w_\beta) X_{3\alpha+2\beta} = -X_{3\alpha+\beta}.
\end{aligned}$$

[1364] In the source: "since $(\alpha^*,\beta) = 2$". The correct expression $(\alpha^*,\beta) = -2$ was used in the previous section; here Demazure exploits the resulting formula $s_\beta(t_\alpha) = t_\alpha t_\beta^{-2} = t_\alpha$ (using $t_\beta^2 = e$, which holds in B_2).

(iii) If one sets

```
p_{α + β}(x) = exp(x X_{α + β}) = int(w_β) p_α(x),
p_{2α + β}(x) = exp(x X_{2α + β}) = int(w_α w_β) p_α(x),
p_{3α + β}(x) = exp(x X_{3α + β}) = int(w_α) p_β(-x),
p_{3α + 2β}(x) = exp(x X_{3α + 2β}) = int(w_β w_α) p_β(-x),
```

one has:

```
p_β(y) p_α(x) = p_α(x) p_β(y) p_{α + β}(x y) p_{2α + β}(x^2 y) p_{3α + β}(x^3 y) p_{3α + 2β}(x^3 y^2)
p_{α + β}(y) p_α(x) = p_α(x) p_{α + β}(y) p_{2α + β}(2 x y) p_{3α + β}(3 x^2 y) p_{3α + 2β}(3 x y^2),
p_{2α + β}(y) p_α(x) = p_α(x) p_{2α + β}(y) p_{3α + β}(3 x y),
p_{3α + β}(y) p_β(x) = p_β(x) p_{3α + β}(y) p_{3α + 2β}(-x y),
p_{2α + β}(y) p_{α + β}(x) = p_{α + β}(x) p_{2α + β}(y) p_{3α + 2β}(3 x y).
```

3.4.2.

The proof occupies Nos. 3.4.2 to 3.4.9. One has $(\beta^*, \alpha) = -1$, $(\alpha^*, \beta) = -3$, whence $\beta(t_\alpha) = \alpha(t_\beta) = -1$. Define $X_{\alpha+\beta}$, $X_{2\alpha+\beta}$, $X_{3\alpha+\beta}$ and $X_{3\alpha+2\beta}$ as in (ii). One has at once

```
Ad(w_β) X_{α + β} = α(t_β) X_α = -X_α,
Ad(w_α) X_{2α + β} = α(t_α) β(t_α) X_{α + β} = -X_{α + β},
Ad(w_α) X_{3α + β} = -β(t_α) X_β = X_β,
Ad(w_β) X_{3α + 2β} = α(t_β)^3 β(t_β) X_{3α + β} = -X_{3α + β}.
```

Finally, since $(3\alpha + 2\beta) \pm \alpha$ and $(2\alpha + \beta) \pm \beta$ are not roots,

```
Ad(w_α) X_{3α + 2β} = X_{3α + 2β},    Ad(w_β) X_{2α + β} = X_{2α + β},
```

which completes the proof of (ii).

3.4.3.

By virtue of Exp. XXII, 5.5.2, there exist scalars[1365]

```
A, B, C, D, E, F, G, H, J ∈ G_a(S),
```

such that

```
p_β(y) p_α(x) = p_α(x) p_β(y) p_{α + β}(A x y) p_{2α + β}(B x^2 y) p_{3α + β}(C x^3 y) p_{3α + 2β}(D x
p_{α + β}(y) p_α(x) = p_α(x) p_{α + β}(y) p_{2α + β}(E x y) p_{3α + β}(F x^2 y) p_{3α + 2β}(G x y^2),
p_{2α + β}(y) p_α(x) = p_α(x) p_{2α + β}(y) p_{3α + β}(H x y),
p_{3α + β}(y) p_β(x) = p_β(x) p_{3α + β}(y) p_{3α + 2β}(J x y).
```

3.4.4.

Make $int(w_\beta)$ act on (1), (3) and (4):

[1365] N.D.E.: We introduce here absolute constants $A, B, \cdots, J$. These constants will be determined in the pages that follow; their values are $A = B = C = D = 1$, $E = 2$, $J = -1$, $F = G = H = 3$, cf. 3.4.8.

```
p_{-β}(-y) p_{α + β}(x) =
    p_{α + β}(x) p_{-β}(-y) p_α(-A x y) p_{2α + β}(B x^2 y) p_{3α + 2β}(C x^3 y) p_{3α + β}(-D x^3 y^
p_{2α + β}(y) p_{α + β}(x) = p_{α + β}(x) p_{2α + β}(y) p_{3α + 2β}(H x y),
p_{3α + 2β}(y) p_{-β}(-x) = p_{-β}(-x) p_{3α + 2β}(y) p_{3α + β}(-J x y).
```

Making $int(w_\alpha)$ act on (1), one finds

```
p_{3α + β}(-y) p_{-α}(-x) =
    p_{-α}(-x) p_{3α + β}(-y) p_{2α + β}(A x y) p_{α + β}(-B x^2 y) p_β(C x^3 y) p_{3α + β}(D x^3 y^2
```

3.4.5.

Write

```
w_β p_α(x) w_β^{-1} = p_{α + β}(x),
```

that is, $\alpha + 2\beta$ not being a root,

```
p_β(1) p_α(x) p_α(-1) = p_{-β}(1) p_{α + β}(x) p_{-β}(-1).                 (9)
```

Transforming the left member of (9) by (1), then (4), one obtains:

```
p_β(1) p_α(x) p_β(-1)
   = p_α(x) p_β(1) p_{α + β}(A x) p_{2α + β}(B x^2) p_{3α + β}(C x^3) p_{3α + 2β}(D x^3) p_β(-1)
   = p_α(x) p_β(1) p_{α + β}(A x) p_{2α + β}(B x^2) p_β(-1) p_{3α + β}(C x^3) p_{3α + 2β}((D - C J) x
   = p_α(x) p_{α + β}(A x) p_{2α + β}(B x^2) p_{3α + β}(C x^3) p_{3α + 2β}((D - C J) x^3).
```

Transforming the right member of (9) by (5), then (7):

```
p_{-β}(1) p_{α + β}(x) p_{-β}(-1)
   = p_{α + β}(x) p_{-β}(1) p_α(A x) p_{2α + β}(-B x^2) p_{3α + 2β}(-C x^3) p_{3α + β}(-D x^3) p_{-β}
   = p_{α + β}(x) p_α(A x) p_{2α + β}(-B x^2) p_{3α + 2β}(-C x^3) p_{3α + β}((C J - D) x^3).
```

Using now (2), this right member becomes

```
p_α(A x) p_{α + β}(x) p_{2α + β}(A E x^2) p_{3α + β}(A^2 F x^3) p_{3α + 2β}(A G x^3) ×
    p_{2α + β}(-B x^2) p_{3α + 2β}(-C x^3) p_{3α + β}((C J - D) x^3)
= p_α(A x) p_{α + β}(x) p_{2α + β}((A E - B) x^2) p_{3α + β}((A^2 F + C J - D) x^3) p_{3α + 2β}((A G
```

So (9) rewrites:

```
p_α(x) p_{α + β}(A x) p_{2α + β}(B x^2) p_{3α + β}(C x^3) p_{3α + 2β}((D - C J) x^3) =
   p_α(A x) p_{α + β}(x) p_{2α + β}((A E - B) x^2) p_{3α + β}((A^2 F + C J - D) x^3) p_{3α + 2β}((A G
```

which gives

```
A = 1,    E = 2 B,    C + D = F + C J,    F = G.
```

3.4.6.

Write now

```
w_α p_β(y) w_α^{-1} = p_{3α + β}(-y),
```

that is, $4\alpha + \beta$ not being a root,

$$p_\alpha(1)\, p_\beta(y)\, p_\alpha(-1) = p_{-\alpha}(1)\, p_{3\alpha + \beta}(-y)\, p_{-\alpha}(-1). \qquad (13)$$

Transform the left member by (1):

$$p_\alpha(1)\, p_\beta(y)\, p_\alpha(-1) = p_\beta(y)\, p_{\alpha + \beta}(-A y)\, p_{2\alpha + \beta}(B y)\, p_{3\alpha + \beta}(-C y)\, p_{3\alpha + 2\beta}(-D y^2).$$

Transform the right member of (13) successively by (8), (6) and (4):

$$\begin{aligned}
&p_{-\alpha}(1)\, p_{3\alpha + \beta}(-y)\, p_{-\alpha}(-1)\\
&\quad = p_{3\alpha + \beta}(-y)\, p_{2\alpha + \beta}(A y)\, p_{\alpha + \beta}(-B y)\, p_\beta(C y)\, p_{3\alpha + 2\beta}(D y^2)\\
&\quad = p_{3\alpha + \beta}(-y)\, p_{\alpha + \beta}(-B y)\, p_{2\alpha + \beta}(A y)\, p_{3\alpha + 2\beta}(-A B H y^2)\, p_\beta(C y)\, p_{3\alpha + 2\beta}(D y^
\end{aligned}$$

$$\begin{aligned}
&\quad = p_\beta(C y)\, p_{3\alpha + \beta}(-y)\, p_{\alpha + \beta}(-B y)\, p_{2\alpha + \beta}(A y)\, p_{3\alpha + 2\beta}((D - C J - A B H) y^2)\\
&\quad = p_\beta(C y)\, p_{\alpha + \beta}(-B y)\, p_{2\alpha + \beta}(A y)\, p_{3\alpha + \beta}(-y)\, p_{3\alpha + 2\beta}((D - C J - A B H) y^2).
\end{aligned}$$

So (13) rewrites:

$$\begin{aligned}
&p_\beta(C y)\, p_{\alpha + \beta}(-B y)\, p_{2\alpha + \beta}(A y)\, p_{3\alpha + \beta}(-y)\, p_{3\alpha + 2\beta}((D - C J - A B H) y^2) =\\
&\quad p_\beta(y)\, p_{\alpha + \beta}(-A y)\, p_{2\alpha + \beta}(B y)\, p_{3\alpha + \beta}(-C y)\, p_{3\alpha + 2\beta}(-D y^2),
\end{aligned}$$

whence

$$C = 1, \quad A = B, \quad D - C J - A B H = -D.$$

Taking into account the results already obtained:

$$A = B = C = 1, \quad E = 2, \quad F = G, \quad D + 1 = F + J, \quad 2 D = H + J.$$

3.4.7.

Write

$$w_\beta\, p_{3\alpha + \beta}(x)\, w_\beta^{-1} = p_{3\alpha + 2\beta}(x),$$

that is,

$$p_\beta(1)\, p_{3\alpha + \beta}(x)\, p_\beta(-1) = p_{-\beta}(1)\, p_{3\alpha + 2\beta}(x)\, p_{-\beta}(-1).$$

Transforming the left member by (4), the right by (7), one obtains:

$$p_{3\alpha + \beta}(x)\, p_{3\alpha + 2\beta}(-J x) = p_{3\alpha + 2\beta}(x)\, p_{3\alpha + \beta}(-J x),$$

so $J = -1$.

3.4.8.

Write finally

$$w_\alpha\, p_{\alpha + \beta}(y)\, w_\alpha^{-1} = p_{2\alpha + \beta}(y),$$

that is,

$$p_\alpha(1)\, p_{\alpha + \beta}(y)\, p_\alpha(-1) = p_{-\alpha}(1)\, p_\alpha(-1)\, p_{2\alpha + \beta}(y)\, p_\alpha(1)\, p_{-\alpha}(-1).$$

Transforming the left member by (2), the right by (3), one obtains:

$$\begin{aligned}
&p_{\alpha + \beta}(y)\, p_{2\alpha + \beta}(-E y)\, p_{3\alpha + \beta}(F y)\, p_{3\alpha + 2\beta}(-G y^2) =\\
&\quad p_{-\alpha}(1)\, p_{2\alpha + \beta}(y)\, p_{3\alpha + \beta}(H y)\, p_{-\alpha}(-1).
\end{aligned}$$

It is immediate to see that, if one makes $p_{-\alpha}(-1)$ commute with $p_{3\alpha+\beta}(Hy)$, then $p_{2\alpha+\beta}(y)$, one does not introduce in the right member any new terms in $p_{3\alpha+\beta}$. The latter therefore writes, denoting by empty parentheses quantities whose exact value does not matter to us:

$$p_{\alpha + \beta}(\ )\ p_{2\alpha + \beta}(\ )\ p_\beta(\ )\ p_{3\alpha + \beta}(H y)\ p_{3\alpha + 2\beta}(\ ).$$

Comparing with the left member, one has at once $F = H$, whence by the previous results $2D = D + 1$, so $D = 1$, and finally $F = G = H = 2D - J = 3$, which completes the determination of the coefficients $A, \cdots, J$ and the proof of (iii).

3.4.9.

Let us prove (i) finally, in the usual manner. One has successively:

$$w_\alpha\, w_\beta\, w_\alpha = w_\alpha\, w_\beta\, w_\alpha^{-1}\, t_\alpha = w_{3\alpha + \beta}^{-1} \cdot t_\alpha,$$
$$w_\beta\, (w_\alpha\, w_\beta)^2 = w_\beta\, w_{3\alpha + \beta}^{-1}\, w_\beta^{-1} \cdot s_\beta(t_\alpha) \cdot t_\beta = w_{3\alpha + 2\beta}^{-1} \cdot t_\alpha\, t_\beta \cdot t_\beta = w$$
$$w_\alpha\, (w_\beta\, w_\alpha)^3 = w_\alpha\, w_{3\alpha + 2\beta}^{-1}\, w_\alpha^{-1} = w_{3\alpha + 2\beta}^{-1},$$
$$w_\beta\, (w_\alpha\, w_\beta)^4 = w_\beta\, w_{3\alpha + 2\beta}^{-1}\, w_\beta^{-1} \cdot t_\beta = w_{3\alpha + \beta} \cdot t_\beta,$$
$$w_\alpha\, (w_\beta\, w_\alpha)^5 = w_\alpha\, w_{3\alpha + \beta}\, w_\alpha^{-1} \cdot s_\alpha(t_\beta) \cdot t_\beta = w_\beta \cdot t_\beta\, t_\alpha \cdot t_\alpha = w_\beta^{-1}.$$

Whence

$$(w_\alpha\, w_\beta)^6 = (w_\beta\, w_\alpha)^6 = e.$$

Remark 3.4.10. *Condition (v) of 2.4 is composed of*

$$\text{(A)}\quad \begin{aligned} &\mathrm{int}(h_\alpha\, h_\beta\, h_\alpha\, h_\beta\, h_\alpha)\, v_\beta = v_\beta,\\ &\mathrm{int}(h_\beta\, h_\alpha\, h_\beta\, h_\alpha\, h_\beta)\, v_\alpha = v_\alpha,\end{aligned}$$

and, setting

$$v_{\alpha + \beta} = \mathrm{int}(h_\beta)\, v_\alpha, \qquad v_{2\alpha + \beta} = \mathrm{int}(h_\alpha\, h_\beta)\, v_\alpha,$$
$$v_{3\alpha + \beta} = \mathrm{int}(h_\alpha)\, v_\beta^{-1}, \qquad v_{3\alpha + 2\beta} = \mathrm{int}(h_\beta\, h_\alpha)\, v_\beta^{-1},$$

the commutation relations:

$$\text{(B)}\quad \begin{aligned}
&v_\beta\, v_\alpha = v_\alpha\, v_\beta\, v_{\alpha + \beta}\, v_{2\alpha + \beta}\, v_{3\alpha + \beta}\, v_{3\alpha + 2\beta},\\
&v_{\alpha + \beta}\, v_\alpha = v_\alpha\, v_{\alpha + \beta}\, v_{2\alpha + \beta}^2\, v_{3\alpha + \beta}^3\, v_{3\alpha + 2\beta}^3,\\
&v_{2\alpha + \beta}\, v_\alpha = v_\alpha\, v_{2\alpha + \beta}\, v_{3\alpha + \beta}^3,\\
&v_{3\alpha + \beta}\, v_\alpha = v_\alpha\, v_{3\alpha + \beta},\\
&v_{3\alpha + 2\beta}\, v_\alpha = v_\alpha\, v_{3\alpha + 2\beta},\\
&v_{\alpha + \beta}\, v_\beta = v_\beta\, v_{\alpha + \beta},\\
&v_{2\alpha + \beta}\, v_\beta = v_\beta\, v_{2\alpha + \beta},\\
&v_{3\alpha + \beta}\, v_\beta = v_\beta\, v_{3\alpha + \beta}\, v_{3\alpha + 2\beta}^{-1},\\
&v_{3\alpha + 2\beta}\, v_\beta = v_\beta\, v_{3\alpha + 2\beta},\\
&v_{2\alpha + \beta}\, v_{\alpha + \beta} = v_{\alpha + \beta}\, v_{2\alpha + \beta}\, v_{3\alpha + 2\beta}^3,\\
&v_{3\alpha + \beta}\, v_{\alpha + \beta} = v_{\alpha + \beta}\, v_{3\alpha + \beta},\\
&v_{3\alpha + 2\beta}\, v_{\alpha + \beta} = v_{\alpha + \beta}\, v_{3\alpha + 2\beta},\\
&v_{3\alpha + \beta}\, v_{2\alpha + \beta} = v_{2\alpha + \beta}\, v_{3\alpha + \beta},
\end{aligned}$$

$$v_{3\alpha+2\beta}\, v_{2\alpha+\beta} = v_{2\alpha+\beta}\, v_{3\alpha+2\beta},$$
$$v_{3\alpha+2\beta}\, v_{3\alpha+\beta} = v_{3\alpha+\beta}\, v_{3\alpha+2\beta}.$$

3.5. Explicit form of the generators-and-relations theorem

Theorem 3.5.1. *Let S be a scheme, G a pinned S-group, T its maximal torus, Δ its system of simple roots, $u_\alpha \in U_\alpha(S)^\times$ and $w_\alpha \in Norm_G(T)(S)$ the elements defined by the pinning ($\alpha \in \Delta$). Let*

$$f_T : T \to H, \qquad f_\alpha : U_\alpha \to H, \qquad \alpha \in \Delta$$

be morphisms of groups, H being an S-sheaf in groups for (fppf); let $h_\alpha \in H(S)$, ($\alpha \in \Delta$) be sections of H, and set $v_\alpha = f_\alpha(u_\alpha)$, $\alpha \in \Delta$. In order that there exist a morphism of groups

$$f : G \to H$$

extending f_T, $f_\alpha(\alpha \in \Delta)$ and satisfying $f(w_\alpha) = h_\alpha$ ($\alpha \in \Delta$) (and then necessarily unique), it is necessary and sufficient that the following conditions be satisfied:

(i) For every $S' \to S$, every $\alpha \in \Delta$, every $t \in T(S')$ and every $x \in U_\alpha(S')$,

$$\mathrm{int}(f_T(t))\, f_\alpha(x) = f_\alpha(\mathrm{int}(t)\, x) = f_\alpha(x^{\alpha(t)}). \tag{1}$$

(ii) For every $\alpha \in \Delta$, every $S' \to S$ and every $t \in T(S')$,

$$\mathrm{int}(h_\alpha)\, f_T(t) = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^*\alpha(t)^{-1}). \tag{2}$$

(iii) For every $\alpha \in \Delta$,[1366]

$$h_\alpha^2 = f_T(\alpha^*(-1)), (3_1) (h_\alpha v_\alpha)^3 = e. (4)$$

(iv) For all $\alpha, \beta \in \Delta$, $\alpha \neq \beta$, such that $(\alpha^*, \beta) = 0$ (resp. $(\alpha^*, \beta) = -1$, resp. $(\alpha^*, \beta) = -2$, resp. $(\alpha^*, \beta) = -3$):

(a) the relation

$$\begin{array}{ll} (h_\alpha h_\beta)^2 = f_T(\alpha^*(-1)\,\beta^*(-1)) & \text{if } (\alpha^*, \beta) = 0; \\ (h_\alpha h_\beta)^3 = e, & \text{if } (\alpha^*, \beta) = -1; \\ (h_\alpha h_\beta)^4 = f_T(\alpha^*(-1)), & \text{if } (\alpha^*, \beta) = -2; \\ (h_\alpha h_\beta)^6 = e, & \text{if } (\alpha^*, \beta) = -3. \end{array} \tag{3_2}$$

(b) The relations (A) and (B) of 3.1.3 (resp. 3.2.8, resp. 3.3.7, resp. 3.4.10).

This follows at once from 2.3, 2.6 and the calculations done in each particular case.

Remark 3.5.2. *One may present the preceding results in a slightly different way: one gives oneself morphisms, for $\alpha \in \Delta$,*

[1366] N.D.E.: The relations (3_1) and (3_2) form the description of the normalizer of the torus, (3_1) being, like (4), in a group of rank 1, while (3_2) is in a group of rank 2.

$$a_\alpha : T \cdot U_\alpha \to H, \qquad b_\alpha : \mathrm{Norm}_{Z_\alpha}(T) \to H,$$

and one sets

$$h_\alpha = b_\alpha(w_\alpha), \qquad v_\alpha = a_\alpha(u_\alpha);$$

then the conditions to be verified are the following:

(1) all the $a_\alpha (\alpha \in \Delta)$ and all the $b_\alpha (\alpha \in \Delta)$ have the same restriction to T;

(2) conditions (4) and (iv) of 3.5.1 above are satisfied.

3.5.3.

Various applications of this theorem will be given in the following Exposé. Let us mention one here: Theorem 3.5.1 gives a description by generators and relations of G in the category of S-sheaves for (fppf); in other words, consider for each $S' \to S$ the group $H(S')$ generated by $T(S')$, $U_\alpha(S')$, $\alpha \in R$, and w_α, $\alpha \in R$, subjected to relations analogous to (1), (2), (3_1), (4), (3_2), (A), (B); then G is none other than the sheaf associated with the presheaf $S' \mapsto H(S')$.

In particular, if S' is the spectrum of an algebraically closed field k, one has $G(S') = H(S')$ (an immediate consequence of the Nullstellensatz in the form: "a sieve over an algebraically closed field, covering for (fppf), is trivial"), so that 3.5.1 yields at once an explicit description by generators and relations of the "abstract" group $G(k)$.[1367]

4. Uniqueness of pinned groups: fundamental theorem

Theorem 4.1. *Let S be a nonempty scheme. The functor R of 1.6 is fully faithful: let G and G' be two pinned S-groups, p*[1368] *an integer > 0 such that $x \mapsto x^p$ is an endomorphism of $G_{a,S}$, and $h : R(G') \to R(G)$ a p-morphism of pinned root data. There exists a unique morphism of pinned groups*

$$f : G \to G'$$

such that $R(f) = h$.

The uniqueness is proved in 1.9. It suffices to prove existence. By hypothesis, one has a bijection $d : R \xrightarrow{\sim} R'$ and a map $q : R \to p^n | n \in \mathbb{N}$ such that

$$h(d(\alpha)) = q(\alpha)\, \alpha \quad \text{and} \quad {}^t h(\alpha^*) = q(\alpha)\, d(\alpha)^*$$

for every $\alpha \in R$. In particular, the semisimple ranks of G and G' coincide.

[1367] N.D.E.: For an arbitrary field k and G simply connected, R. Steinberg gave a presentation of the group $G(k)$ in [St62], Th. 3.2, see also [St67], § 6, Th. 8.

[1368] N.D.E.: To avoid a notational problem later on, we have replaced q by p, so that in what follows, q (resp. q_1) is an arbitrary power of p.

4.1.1.

Suppose $rg_{ss}(G) = rg_{ss}(G') = 0$. Then G and G' are tori: one has $G = T = D_S(M)$, $G' = T' = D_S(M')$ and h is simply a morphism of ordinary groups $h : M' \to M$. One then takes $f = D_S(h)$.

4.1.2.

Suppose $rg_{ss}(G) = rg_{ss}(G') = 1$. Consider then

$$f_T = D_S(h) : T \to T'.$$

By hypothesis, one has a commutative diagram, where $\alpha' = d(\alpha)$:

$$\begin{tikzcd}
G_{m,S} \arrow[r, "\alpha^*"] \arrow[d, "q(\alpha)"] & T \arrow[r, "\alpha"] \arrow[d, "f_T"] & G_{m,S} \arrow[d, "q(\alpha)"] \\
G_{m,S} \arrow[r, "\alpha'^*"] & T' \arrow[r, "\alpha'"] & G_{m,S}.
\end{tikzcd}$$

One then applies Exp. XX, 4.1.

4.1.3.

Suppose $rg_{ss}(G) = rg_{ss}(G') = 2$. Then, by Exp. XXI, 7.5.3, one knows all the possibilities for $h : R(G') \to R(G)$. Let us examine them successively, verifying each time the conditions of 2.5.

Denote $\Delta = \alpha, \beta$, $\Delta' = \alpha', \beta'$ in such a way that $d(\alpha) = \alpha'$, $d(\beta) = \beta'$.

4.1.4. G and G' of type $A_1 \times A_1$

One has

$$h(\alpha') = q\,\alpha, \qquad h(\beta') = q_1\,\beta.$$

Let us show that the conditions of 2.5 are satisfied: (ii) and (iii) follow from 3.1.2 (ii) and (iii); let us prove (i).

One has[1369]

$$D_S(h)\, t_{\alpha\beta} = D_S(h)(t_\alpha t_\beta) = {}^t h(\alpha^*)(-1) \cdot {}^t h(\beta^*)(-1) = \alpha'^*((-1)^q) \cdot \beta'^*((-1)^{q_1}).$$

Now, the hypothesis that $x \mapsto x^p$ is an endomorphism of $G_{a,S}$ (and $S \neq \emptyset$) entails that $p = 1$ or that S is of characteristic p; in all cases $(-1)^q = -1$ (if q is even, $p = 2$ and $1 = -1$). Consequently,

$$D_S(h)\, t_{\alpha\beta} = \alpha'^*(-1) \cdot \beta'^*(-1) = t'_{\alpha'\beta'}.$$

This shows that condition 2.5 (i) is satisfied.

[1369] N.D.E.: We have spelled out what follows.

4.1.5. G and G' of type A_2

One has

$$h(\alpha') = q\,\alpha, \qquad h(\beta') = q\,\beta.$$

Set $X_{\alpha+\beta} = Ad(w_\beta)X_\alpha$ and $X'_{\alpha'+\beta'} = Ad(w_{\beta'})X_{\alpha'}$. Let us verify the conditions of 2.5. For (i), one reasons as above, using 3.2.1 (i); for (ii), it is immediate by 3.2.1 (ii); it remains to verify (iii). We must check that

$$p_\alpha(x)\, p_\beta(y)\, p_{\alpha+\beta}(z) \mapsto p'_{\alpha'}(x^q)\, p'_{\beta'}(y^q)\, p'_{\alpha'+\beta'}(z^q)$$

is a morphism of groups. The only nontrivial commutation relation is that of 3.2.1 (iii), which is written

$$p_\beta(y)\, p_\alpha(x) = p_\alpha(x)\, p_\beta(y)\, p_{\alpha+\beta}(x\,y),$$
$$p'_{\beta'}(y^q)\, p'_{\alpha'}(x^q) = p'_{\alpha'}(x^q)\, p'_{\beta'}(y^q)\, p'_{\alpha'+\beta'}(x^q\,y^q).$$

4.1.6.

One reasons similarly for G and G' of type B_2 (resp. G_2), when the radicial exponents are equal, using 3.3.1 (resp. 3.4.1); it remains, therefore, to treat — in order to complete the case of groups of rank 2 — the two exceptional cases of Exp. XXI, 7.5.3.

4.1.7. G and G' of type B_2, S of characteristic 2, $q(\alpha) = 2q$, $q(\beta) = q$

The positive roots are $\alpha, \beta, \alpha+\beta, 2\alpha+\beta$ and $\alpha', \beta', \alpha'+\beta', \alpha'+2\beta'$ (note that the "short" simple roots are α and β'). One has

$$h(\alpha') = 2\,q\,\alpha, \qquad h(\beta') = q\,\beta, \qquad h(\alpha'+\beta') = q(2\alpha+\beta), \qquad h(\alpha'+2\beta') = 2\,q(\alpha+\beta),$$

which gives

$$d(\alpha+\beta) = \alpha'+2\beta', \qquad q(\alpha+\beta) = 2\,q,$$
$$d(2\alpha+\beta) = \alpha'+\beta', \qquad q(2\alpha+\beta) = q.$$

Set

$$X_{\alpha+\beta} = Ad(w_\beta)\, X_\alpha, \qquad X_{2\alpha+\beta} = Ad(w_\alpha)\, X_\beta,$$
$$X'_{\alpha'+\beta'} = Ad(w'_{\alpha'})\, X'_{\beta'}, \qquad X'_{\alpha'+2\beta'} = Ad(w'_{\beta'})\, X'_{\alpha'}.$$

Let us now verify the conditions of 2.5.

(i) Since S is of characteristic 2, one has $-1 = 1$ on S, hence $t_{\alpha\beta} = t_\alpha = \alpha^*(-1) = e = \beta'^*(-1) = t'_{\beta'} = t'_{\alpha'\beta'}$ (cf. 3.3.1 (i)).

(ii) One has by construction

$$Ad(w_\alpha)\, X_\beta = X_{2\alpha+\beta}, \qquad Ad(w'_{\alpha'})\, X'_{\beta'} = X'_{\alpha'+\beta'} = X'_{d(2\alpha+\beta)};$$
$$Ad(w_\beta)\, X_\alpha = X_{\alpha+\beta}, \qquad Ad(w'_{\beta'})\, X'_{\alpha'} = X'_{\alpha'+2\beta'} = X'_{d(\alpha+\beta)}.$$

By 3.3.1 (ii) and the fact that $-1 = 1$ on S, one has on each side

```
Ad(w_α) X_{α + β} = X_{α + β},    Ad(w'_{α'}) X'_{d(α + β)} = Ad(w'_{α'}) X'_{α' + 2β'} = X'_{α' + 2β
Ad(w_α) X_{2α + β} = X_β,    Ad(w'_{α'}) X'_{d(2α + β)} = Ad(w'_{α'}) X'_{α' + β'} = X'_{β'} = X'_{d(
Ad(w_β) X_{α + β} = X_α,    Ad(w'_{β'}) X'_{d(α + β)} = Ad(w'_{β'}) X'_{α' + 2β'} = X'_{α'} = X'_{d(α
Ad(w_β) X_{2α + β} = X_{2α + β},    Ad(w'_{β'}) X'_{d(2α + β)} = Ad(w'_{β'}) X'_{α' + β'} = X'_{α' +
```

(iii) By 3.3.1 (iii), one sees that the only nontrivial commutation relation in U (resp. U') is

```
p_β(y) p_α(x) = p_α(x) p_β(y) p_{α + β}(x y) p_{2α + β}(x^2 y),
```

resp.

```
p'_{α'}(y') p'_{β'}(x') = p'_{β'}(x') p'_{α'}(y') p'_{α' + β'}(x' y') p'_{α' + 2β'}(x' y'^2).
```

We must verify that the morphism

```
p_α(x) p_β(y) p_{α + β}(z) p_{2α + β}(t) ↦ p'_{α'}(x^{2q}) p'_{β'}(y^q) p'_{α' + 2β'}(z^{2q}) p'_{α' +
```

is a morphism of groups; one sees at once that this amounts to seeing that

```
p'_{β'}(y^q) p'_{α'}(x^{2q}) = p'_{α'}(x^{2q}) p'_{β'}(y^q) p'_{α' + 2β'}((x y)^{2q}) p'_{α' + β'}((x
```

which is none other than the second relation above (setting $y' = x^{2q}$, $x' = y^q$).

4.1.8. G and G' of type `G_2`, S of characteristic 3, $q(\alpha) = 3q$, $q(\beta) = q$

The positive roots are $\alpha, \beta, \alpha+\beta, 2\alpha+\beta, 3\alpha+\beta, 3\alpha+2\beta$ on the one hand, $\alpha', \beta', \alpha'+\beta', \alpha'+2\beta', \alpha'+3\beta', 2\alpha'+3\beta'$ on the other (as in the preceding case, the short simple roots are α and β'). One has

```
h(α') = 3 q α,    h(β') = q β,    h(α' + β') = q(3α + β),
h(α' + 2β') = q(3α + 2β),    h(α' + 3β') = 3 q(α + β),
h(2α' + 3β') = 3 q(2α + β),
```

which gives

```
d(α + β) = α' + 3β',    q(α + β) = 3 q,
d(2α + β) = 2α' + 3β',    q(2α + β) = 3 q,
d(3α + β) = α' + β',    q(3α + β) = q,
d(3α + 2β) = α' + 2β',    q(3α + 2β) = q.
```

Set

```
X_{α + β} = Ad(w_β) X_α,    X_{2α + β} = Ad(w_α) X_{α + β},
X_{3α + β} = -Ad(w_α) X_β,    X_{3α + 2β} = Ad(w_β) X_{3α + β};
X'_{α' + β'} = -Ad(w'_{α'}) X'_{β'},    X'_{α' + 2β'} = Ad(w'_{β'}) X'_{α' + β'},
X'_{α' + 3β'} = Ad(w'_{β'}) X'_{α'},    X'_{2α' + 3β'} = Ad(w'_{α'}) X'_{α' + 3β'}.
```

Let us now verify the conditions of 2.5.

(i) $t_{\alpha\beta} = e$ and $t'_{\alpha'\beta'} = e$ by 3.4.1 (i).

(ii) One has by construction

```
Ad(w_β) X_α = X_{α + β},    Ad(w'_{β'}) X'_{α'} = X'_{α' + 3β'} = X'_{d(α + β)};
Ad(w_α) X_β = -X_{3α + β},    Ad(w'_{α'}) X'_{β'} = -X'_{α' + β'} = -X'_{d(3α + β)};
Ad(w_β) X_{3α + β} = X_{3α + 2β},    Ad(w'_{β'}) X'_{d(3α + β)} = Ad(w'_{β'}) X'_{α' + β'} = X'_{α' +
Ad(w_α) X_{α + β} = X_{2α + β},    Ad(w'_{α'}) X'_{d(α + β)} = Ad(w'_{α'}) X'_{α' + 3β'} = X'_{2α' +
```

By 3.4.1 (ii), one has on each side:

```
Ad(w_α) X_{2α + β} = -X_{α + β},    Ad(w'_{α'}) X'_{d(2α + β)} = Ad(w'_{α'}) X'_{2α' + 3β'} = -X'_{α'
      …                                              …
Ad(w_β) X_{3α + 2β} = -X_{3α + β},    Ad(w'_{β'}) X'_{d(3α + 2β)} = Ad(w'_{β'}) X'_{α' + 2β'} = -X'_{
```

(The dots replace four verifications of the same kind.)

(iii) The only nontrivial commutation relations in U and U' are, by 3.4.1 (iii) (and taking into account $3 = 0$ on S):

```
p_β(y) p_α(x) = p_α(x) p_β(y) p_{α + β}(x y) p_{2α + β}(x^2 y) p_{3α + β}(x^3 y) p_{3α + 2β}(x^3 y^2)
p_{α + β}(y) p_α(x) = p_α(x) p_{α + β}(y) p_{2α + β}(-x y),
p_{3α + β}(y) p_β(x) = p_β(x) p_{3α + β}(y) p_{3α + 2β}(-x y);


p'_{α'}(y') p'_{β'}(x') =
    p'_{β'}(x') p'_{α'}(y') p'_{α' + β'}(-x' y') p'_{α' + 2β'}(-x'^2 y') p'_{α' + 3β'}(-x'^3 y') p'_{
p'_{α' + β'}(y') p'_{β'}(x') = p'_{β'}(x') p'_{α' + β'}(y') p'_{α' + 2β'}(x' y'),
p'_{α' + 3β'}(y') p'_{α'}(x') = p'_{α'}(x') p'_{α' + 3β'}(y') p'_{2α' + 3β'}(x' y').
```

We must verify that the morphism $\phi : U \to U'$ defined by

```
φ(p_α(x) p_β(y) p_{α + β}(t) p_{2α + β}(u) p_{3α + β}(v) p_{3α + 2β}(w))
  = p'_{α'}(x^{3q}) p'_{β'}(y^q) p'_{α' + 3β'}(t^{3q}) p'_{2α' + 3β'}(u^{3q}) p'_{α' + β'}(v^q) p'_{α
```

is a morphism of groups. Now one verifies immediately that the three last commutation relations are also written

```
p'_{β'}(y^q) p'_{α'}(x^{3q}) =
    p'_{α'}(x^{3q}) p'_{β'}(y^q) p'_{α' + 3β'}((x y)^{3q}) p'_{2α' + 3β'}((x^2 y)^{3q}) p'_{α' + β'}(
p'_{α' + 3β'}(y^{3q}) p'_{α'}(x^{3q}) = p'_{α'}(x^{3q}) p'_{α' + 3β'}(y^{3q}) p'_{2α' + 3β'}(-(x y)^{
p'_{α' + β'}(y^q) p'_{β'}(x^q) = p'_{β'}(x^q) p'_{α' + β'}(y^q) p'_{α' + 2β'}(-(x y)^q);
```

which shows that ϕ is indeed a morphism of groups and completes the proof of 4.1.7.

4.1.9. Case where G and G' are of semisimple rank > 2

For each root $\alpha \in \Delta$, denote $\alpha' = d(\alpha) \in \Delta' = d(\Delta)$. For each $(\alpha, \beta) \in \Delta \times \Delta$, consider the pinned groups of semisimple rank $\leqslant 2$, $Z_{\alpha\beta}$ and $Z'_{\alpha'\beta'}$. The morphism of groups $M' \to M$ underlying h defines a p-morphism of root data

```
h_{αβ} : R(Z_{αβ}) → R(Z'_{α' β'}).
```

By virtue of the preceding results, there exists therefore a morphism of pinned groups

```
f_{αβ} : Z_{αβ} → Z'_{α' β'}
```

such that $R(f_{\alpha\beta}) = h_{\alpha\beta}$. Let us prove that the $f_{\alpha\beta}$ satisfy the gluing condition of 2.5; indeed $f_{\alpha\beta}|_{Z_\alpha}$ and $f_{\alpha\alpha}$ are two morphisms of pinned groups

$$Z_\alpha \to Z'_{\alpha'}$$

corresponding to the same morphism of pinned root data, and therefore coincide by the uniqueness result already proved. By 2.5 there exists therefore a morphism of groups

$$f : G \to G'$$

extending the $f_{\alpha\beta}$. This is evidently a morphism of pinned groups such that $R(f) = h$, which completes the proof of Theorem 4.1.

5. Corollaries of the fundamental theorem

The most important is:

Corollary 5.1. *Let S be a nonempty scheme, G and G' two pinned S-groups, h an isomorphism of pinned root data*

$$h : R(G') \xrightarrow{\sim} R(G).$$

There exists a unique isomorphism of pinned S-groups

$$f : G \xrightarrow{\sim} G'$$

such that $R(f) = h$.

Note that 5.1 also follows from 3.5.1 (the relations of 3.5.1 may be written using only the datum of $R(G)$); note also that 5.1 follows from the most elementary part of the proof of 4.1 (one does not need to consider the "exceptional isogenies" of 4.1.7 and 4.1.8).

Corollary 5.2 ("Uniqueness theorem"). *Let S be a scheme, G and G' two splittable S-groups (Exp. XXII, 1.13). If G and G' are of the same type (Exp. XXII, 2.6), they are isomorphic.*

Corollary 5.3. *Let S be a scheme, G and G' two splittable S-groups. The following conditions are equivalent:*

(i) G and G' are isomorphic.

(ii) G and G' are isomorphic locally for the (fpqc) topology.

(iii) There exists an $s \in S$ such that the s-groups G_s and G'_s are of the same type.

Indeed, one evidently has (i) ⇒ (ii) ⇒ (iii). On the other hand, (iii) entails that $R(G) = R(G_s) = R(G'_s) = R(G')$, hence that G and G' satisfy the condition of 5.2.

Corollary 5.4 ("Uniqueness of the Chevalley schemes"). *Let G and G' be two reductive groups over $\mathbb{Z}$ possessing split maximal tori.*[1370] *The following conditions are equivalent:*

(i) G and G' are isomorphic.

(ii) There exists $s \in \operatorname{Spec}(\mathbb{Z})$ such that G_s and G'_s are of the same type.

(iii) $G_{\mathbb{C}}$ and $G'_{\mathbb{C}}$ are of the same type.

Indeed, G and G' are splittable by Exp. XXII, 2.2.

Corollary 5.5 ("Existence of outer automorphisms"). *Let S be a scheme, G a pinned S-group, h an automorphism of the pinned root datum $R(G)$. There exists a unique automorphism u of G respecting its pinning and such that $R(u) = h$.*

Let us make the preceding corollary explicit.

Corollary 5.5.bis. *Let S be a scheme, G a split reductive S-group, R^+ a system of positive roots of G.*

Choose for each simple root α an isomorphism of vector groups $p_\alpha : G_{a,S} \xrightarrow{\sim} U_\alpha$. Let h be an automorphism of M permuting the positive roots and the corresponding coroots: if $\alpha \in R^+$, $h(\alpha) \in R^+$ and $h^\vee(\alpha^*) = h(\alpha)^*$. There exists a unique automorphism u of G inducing $D_S(h)$ on T and such that $u \circ p_\alpha = p_{h(\alpha)}$ for every simple root α.

Corollary 5.6. *Let G and G' be two reductive S-groups. The following conditions are equivalent:*

(i) G and G' are isomorphic locally for the (fpqc) topology.

(i bis) G and G' are isomorphic locally for the étale topology.

(ii) For every $s \in S$, G_s and G'_s are isomorphic.

(iii) The functions $s \mapsto$ type of G_s and $s \mapsto$ type of G'_s are equal.

Indeed (i bis) ⇒ (i) trivially, (i) ⇒ (ii) by the principle of finite extension (EGA IV_3, 9.1.4), (ii) ⇒ (iii) trivially; it remains to prove (iii) ⇒ (i bis). Now one may suppose G and G' splittable (Exp. XXII, 2.3), in which case the assertion follows from 5.3.

Corollary 5.7. *Let S be a scheme, G a reductive S-group, G' an S-group that is affine, smooth, and with connected fibers. Let $s \in S$ be such that G_s and G'_s are isomorphic; there exists then an étale $S' \to S$ covering s such that $G_{S'}$ and $G'_{S'}$ are isomorphic.*

Indeed, by Exp. XIX 2.5 and Exp. XXII 2.1, one may suppose G and G' reductive splittable and one is reduced to 5.3.

[1370] In fact, one may prove that every $\mathbb{Z}$-torus is split.[^N.D.E-XXIII-14]

In the case where S is the spectrum of a field, one deduces from 5.6 and 5.7:

Corollary 5.8. *Let k be a field and G and G' two reductive k-groups. The following conditions are equivalent:*[1371]

(i) G and G' are of the same type.

(ii) $G \otimes_k \bar{k}$ and $G' \otimes_k \bar{k}$ are isomorphic.

(iii) There exists a finite separable extension K of k such that $G \otimes_k K$ and $G' \otimes_k K$ are isomorphic.

Corollary 5.9. *Let S be a nonempty scheme and R a root datum. The following conditions are equivalent:*[1372]

(i) There exists a pinned S-group of type R.

(ii) There exists an S-group of type R.

(iii) There exists locally for (fpqc) a reductive S-group of type R.

It is evidently a matter of proving (iii) ⇒ (i). To simplify the proof, suppose that there exist a faithfully flat quasi-compact morphism $S' \to S$ and a reductive S'-group G' of type R. One may suppose G' splittable; fix a pinning E' of G'; denote $R = R(G', E')$. The two pullbacks of (G', E') to $S'' = S' \times_S S'$ are pinned groups (G''_1, E''_1), (G''_2, E''_2); one has canonical isomorphisms $p_i : R(G''_i, E''_i) \xrightarrow{\sim} R$, whence an isomorphism

$$p = p_2^{-1} \circ p_1 : R(G''_1, E''_1) \to R(G''_2, E''_2).$$

By the uniqueness theorem, there exists a unique isomorphism

$$f : (G''_1, E''_1) \xrightarrow{\sim} (G''_2, E''_2)$$

such that $R(f) = p$. One has therefore on G' a gluing datum; this is a descent datum.

Indeed, one must verify a compatibility condition between the pullbacks of f on S''', but it suffices to make this verification on the transforms of these arrows by the functor R, since the latter is fully faithful. Now p is indeed a descent datum, by construction, which shows that f is one too. Since G' is affine, this descent datum is effective; since the pinning of G' is stable under the descent datum, one verifies easily that there exists a pinned S-group (G, E) which gives (G', E') by base extension and which is therefore of type R.

N.B. Naturally, in the language of fibered categories, the preceding proof simplifies (and becomes intelligible).

[1371] N.D.E.: Another proof of this corollary, not using the reduction to the case of groups of rank 2, was given by M. Takeuchi ([Ta83], Th. 4.6); see also [Ja87], II 1.14.

[1372] N.D.E.: This corollary is rendered superfluous by Exp. XXV, which shows the existence of a split group of type R over Spec($\mathbb{Z}$), hence over any base S. (In fact, after XXV 1.3 one finds a reference to the present corollary to ensure that the reductive $\mathbb{Z}$-group obtained is split, but this already follows from XXII 2.2; cf. N.D.E. (3) of XXV.)

Corollary 5.10. *Let S be a nonempty scheme. Let R be a pinned root datum such that there exists a reductive S-group of type R. Then there exists a pinned S-group of type R, unique up to a unique isomorphism.*

Definition 5.11. *Under the preceding conditions, we shall denote by $\acute{E}p_S(R)$ the unique pinned S-group of type R, by $T_S(R)$ its canonical maximal torus, by $B_S(R)$ its canonical Borel subgroup, etc.*

If one has a morphism $S' \to S$ (S' nonempty), one may identify $\acute{E}p_{S'}(R)$ with $\acute{E}p_S(R) \times_S S'$, etc. In particular, if $\acute{E}p_{\mathrm{Spec}(\mathbb{Z})}(R)$ exists (we shall see that this is always the case), one denotes it $\acute{E}p(R)$, and one has

```
Ép_S(R) = Ép(R) ×_{Spec(ℤ)} S.
```

One says that $\acute{E}p(R)$ is the *Chevalley group scheme of type R*.

5.12.

It therefore comes to the same thing to say that the S-sheaf in groups G is a reductive S-group of type R, or to say that it is locally isomorphic (for the étale or (fpqc) topology) to $\acute{E}p_S(R)$. Likewise, by virtue of the conjugation theorems, it comes to the same thing to say that (G,T) is a reductive S-group of type R equipped with a maximal torus, or to say that it is locally isomorphic to $(\acute{E}p_S(R), T_S(R))$; likewise with Borel subgroups or Killing couples.

6. Chevalley systems

The explicit calculations of No. 3 have important numerical consequences. We first lay down the following definition.

Definition 6.1. *Let S be a scheme, (G,T,M,R) a split S-group. One calls* Chevalley system *of G a family $(X_\alpha)_{\alpha\in R}$ of elements*

$$X_\alpha \in \Gamma(S, g_\alpha)^\times$$

satisfying the following condition:

(SC) for every pair $\alpha, \beta \in R$,

$$Ad(w_\alpha(X_\alpha))X_\beta = \pm X_{s_\alpha(\beta)}.$$

Recall (Exp. XX, 3.1) that

```
w_α(X_α) = exp(X_α) exp(-X_α^{-1}) exp(X_α).
```

Note that (SC) entails in particular $X_{-\alpha} = \pm X_\alpha$ for $\alpha \in R$, by virtue of the relation (Exp. XX, 3.7) $Ad(w_\alpha(X_\alpha))X_\alpha = -X_\alpha^{-1}$.

Proposition 6.2. *Every split group admits a Chevalley system. More precisely, let $(\Delta, (X'_\alpha)_{\alpha\in\Delta})$ be a pinning (1.1) of the split group (G, T, M, R); there exists then a Chevalley system $(X_\alpha)_{\alpha\in R}$ of G such that $X_\alpha = X'_\alpha$ for $\alpha \in \Delta$.*

Let us first show that it suffices to verify condition (SC) for $\alpha \in \Delta$; for every $\alpha \in R$, there exists a sequence $\alpha_i \subset \Delta$ such that $\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1})$, whence, applying condition (SC) for each of the α_i,

$$X_\alpha = \pm \operatorname{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))\, X_{\alpha_{n+1}}.$$

By 3.1.1 (iii),

$$w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n})\, w_{\alpha_{n+1}}(X_{\alpha_{n+1}})\, w_{\alpha_n}(X_{\alpha_n})^{-1} \cdots w_{\alpha_1}(X_{\alpha_1})$$
$$= \alpha^*(\pm 1)\, w_\alpha(X_\alpha).$$

Now, it suffices to note that $w_{\alpha_i}(X_{\alpha_i})^{-1} = \alpha_i^*(-1) w_{\alpha_i}(X_{\alpha_i})$ and that for every pair of roots (β, γ), $\beta(\gamma^*(-1)) = (-1)^{(\gamma^*, \beta)} = \pm 1$, which entails that if (SC) is satisfied for the pairs (α_i, γ) $(\gamma \in R)$, it is so for every pair (α, β), $(\beta \in R)$.

Let us now construct a system $(X_\alpha)_{\alpha\in R}$ in the following manner. For every $\alpha \in R$, choose a sequence $\alpha_i \subset \Delta$ as above, and define X_α by

$$X_\alpha = \operatorname{Ad}(w_{\alpha_i}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}))\, X'_{\alpha_{n+1}}.$$

To verify (SC), it suffices to prove:

Lemma 6.3. *Let $(G, T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$ be a pinned S-group; let $\alpha_i (0 \leqslant i \leqslant n+1)$ be a sequence of simple roots such that*

$$int(s_{\alpha_1} \cdots s_{\alpha_n})(\alpha_{n+1}) = \alpha_0.$$

Then

$$\operatorname{Ad}(w_{\alpha_1} \cdots w_{\alpha_n})\, X_{\alpha_{n+1}} = \pm X_{\alpha_0}.$$

Reasoning as in 2.3.4 with the help of Tits's lemma (Exp. XXI 5.6), one is reduced to verifying lemma 6.3 in the two following cases:

a) G is of semisimple rank at most 2, or b) $w_{\alpha_1} \cdots w_{\alpha_n}$ is a section of T.

In case a), note that 6.3 is a consequence of 6.2 and that 6.2 has been verified in part (i) of 3.1.2 (resp. 3.2.1, resp. 3.3.1, resp. 3.4.1).

It therefore remains to prove 6.3 in case b), or, what amounts to the same thing, that if α_i is a sequence of simple roots such that $s_{\alpha_1} \cdots s_{\alpha_n} = id$, then $t = w_{\alpha_1} \cdots w_{\alpha_n}$ satisfies $\alpha(t) = \pm 1$ for every root $\alpha \in R$. By virtue of the structure of the Weyl group (Exp. XXI, 5.1), the word $s_{\alpha_1} \cdots s_{\alpha_n}$ in the free group generated by the s_α, $\alpha \in \Delta$, is in the invariant subgroup generated by the $(s_\alpha s_\beta)^{n_{\alpha\beta}}$, $(\alpha, \beta) \in \Delta \times \Delta$. One is therefore reduced to the case where $s_{\alpha_1} \cdots s_{\alpha_n}$ is of the form

$$s_{\alpha_1} \cdots s_{\alpha_i} (s_{\alpha_{i+1}} s_{\alpha_{i+2}})^{n_{\alpha_{i+1} \alpha_{i+2}}} s_{\alpha_i} \cdots s_{\alpha_1}.$$

Then

$$t = s_{\alpha_1} \cdots s_{\alpha_i}(t_{\alpha_{i+1} \alpha_{i+2}}),$$

and one is reduced to verifying that for every pair of simple roots (α_1, α_2) and every root $\beta \in R$, $\beta(t_{\alpha_1\alpha_2}) = \pm 1$, which is trivial, in view of the values of $t_{\alpha_1\alpha_2}$ calculated in No. 3 (part (i) of 3.1.2, 3.2.1, 3.3.1, 3.4.1).

Proposition 6.4. *Let (G, T, M, R) be a split S-group, $(X_\alpha)_{\alpha\in R}$ a Chevalley system of G. Let α and β be two non-proportional roots; suppose*

$$\mathrm{long}(\alpha) \leqslant \mathrm{long}(\beta), \quad \text{i.e.} \quad |(\beta^*, \alpha)| \leqslant |(\alpha^*, \beta)|.$$

Let q and $p-1$ be the integers $\geqslant 0$ such that the set of roots of the form $\beta + k\alpha$, $k \in \mathbb{Z}$, is

$$\{\beta - (p-1)\alpha, \dots, \beta, \dots, \beta + q\alpha\}.$$

(cf. Exp. XXI, 2.3.5; by loc. cit., one has therefore $-(\alpha^*, \beta) = q - p + 1$). Then the commutation relation between U_α and U_β is given by the following table (which exhausts the possible cases, since the length of the preceding root chain is $p + q - 1 \leqslant 3$),

where, for each $\gamma \in R$, one writes $p_\gamma(x) = \exp(xX_\gamma)$:

(p, q)	$p_\beta(y)\, p_\alpha(x)\, p_\beta(-y)\, p_\alpha(-x)$
$(-, 0)$	$= e$
$(1, 1)$	$= p_{\alpha+\beta}(\pm xy)$
$(1, 2)$	$= p_{\alpha+\beta}(\pm xy)\, p_{2\alpha+\beta}(\pm x^2 y)$
$(1, 3)$	$= p_{\alpha+\beta}(\pm xy)\, p_{2\alpha+\beta}(\pm x^2 y)\, p_{3\alpha+\beta}(\pm x^3 y)\, p_{3\alpha+2\beta}(\pm x^3 y^2)$
$(2, 1)$	$= p_{\alpha+\beta}(\pm 2xy)$
$(2, 2)$	$= p_{\alpha+\beta}(\pm 2xy)\, p_{2\alpha+\beta}(\pm 3x^2 y)\, p_{\alpha+2\beta}(\pm 3xy^2)$
$(3, 1)$	$= p_{\alpha+\beta}(\pm 3xy)$.

Proof. By virtue of Exp. XXI, 3.5.4, there exists a system of simple roots Δ of R satisfying: $\alpha \in \Delta$, and there exist $\alpha' \in \Delta$ and $a, b \in \mathbb{Q}_+$ such that $\beta = a\alpha + b\alpha'$. Consider the pinning $(\Delta, (X_\alpha)_{\alpha\in\Delta})$ of G. The commutation relation to be verified is a relation between elements of $U_{\alpha\alpha'}$; one is therefore reduced to the explicit calculations of No. 3, and one concludes at once by condition (SC).

Corollary 6.5 (Chevalley's rule). *Let S be a scheme, $(X_\alpha)_{\alpha\in R}$ a Chevalley system of the split S-group G. If $\alpha, \beta, \alpha + \beta \in R$, then*

$$[X_\alpha, X_\beta] = \pm p\, X_{\alpha+\beta},$$

where p is the smallest integer > 0 such that $\beta - p\alpha$ is not a root.

Indeed, since the assertion is symmetric in α and β, one may suppose $long(\alpha) \leqslant long(\beta)$, and one is reduced to 6.4.

Corollary 6.6. *Let S be a scheme such that $6 \cdot 1_S \neq 0$ and (G, T, M, R) a split S-group. If R' is a subset of R such that*

$$\mathfrak{g}_{R'} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R'} \mathfrak{g}_\alpha \qquad (*)$$

is a Lie subalgebra of $\mathfrak{g}$, then R' is a closed subset of R (Exp. XXI, 3.1.4).

[1373] Indeed, let $(X_\gamma)_{\gamma\in R^+}$ be a Chevalley system of $\mathfrak{g}$, and let $\alpha, \beta \in R'$ be such that $\alpha+\beta \in R$. By 6.5 and 6.4,

$$[X_\alpha, X_\beta] = \pm p\, X_{\alpha+\beta}, \quad \text{with } p \in \{1, 2, 3\}$$

and since neither 2 nor 3 are zero on S, this entails, by (*), that $\mathfrak{g}_{R'}$ contains $\mathfrak{g}_{\alpha+\beta}$, hence $\alpha+\beta \in R'$.[1374]

6.7.

It is possible to make explicit the exact value of the various $\pm$ of this section, thanks to the study of the group $Norm_G(T)$, and more precisely of the "extended Weyl group":

$$W^* = N(\mathbb{Z}), \quad \text{where } N = \mathrm{Norm}_{\mathrm{\acute{E}p}_{\mathbb{Z}}(R)}(T_{\mathbb{Z}}(R))$$

(cf. 5.11), which is an extension of $W(R)$ by an abelian group of type $(2, 2, \cdots, 2)$, which is "responsible for the signs".[1375] [1376]

Remark 6.7.1.[1377] *Note that, by point (i) of 3.1.2 and 3.n.1 ($n = 2, 3, 4$), the elements w_α and w_β of $N(\mathbb{Z})$ satisfy the "braid relations":*

$$w_\alpha w_\beta \cdots = w_\beta w_\alpha \cdots \quad (n_{\alpha\beta} \text{ factors in each term}).$$

(See also [Ti66], [BLie], § IX.4, Ex. 12, and [Sp98], 9.3.2.)

Bibliography

[1378]

[BLie] N. Bourbaki, *Groupes et algèbres de Lie*, Chap. IX, Masson, 1982.

[Ja87] J. C. Jantzen, *Representations of algebraic groups*, Academic Press, 1987; 2nd ed., Amer. Math. Soc., 2003.

[Sp98] T. A. Springer, *Linear algebraic groups*, 2nd ed., Birkhäuser, 1998.

[1373] N.D.E.: We have added the following proof. Note that it suffices that 2 and 3 be nonzero on S; for example the result is valid for $S = \mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$.

[1374] N.D.E.: On the other hand, let us point out that if $2 = 0$ on S and R is of type C_n, then the set R' of short roots (which is a root system of type D_n) is not closed in R, but is a part of type (R) of R, symmetric (cf. XXII 5.4.2 and 5.4.10), i.e. it corresponds to a subgroup $H_{R'}$ of type (R) of G with reductive fibers: this is in particular the case for the natural embedding (in characteristic 2) of $SO(2n)$ into $Sp(2n)$. Likewise, if $2 = 0$ on S and R is of type F_4 (resp. if $3 = 0$ on S and R is of type G_2), the set R' of short roots (which is a root system of type D_4 (resp. A_2)) is not closed in R, but corresponds to a subgroup $H_{R'}$ of type (R) of G, with reductive fibers.

[1375] Cf. J. Tits, *Sur les constantes de structure et le théorème d'existence des algèbres de Lie semi-simples*, Publ. Math. I.H.É.S. **31** (1966), 21–58.

[1376] N.D.E.: see also [Ti66].

[1377] N.D.E.: We have added this remark.

[1378] N.D.E.: additional references cited in this Exposé.

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Exposé XXIV. Automorphisms of reductive groups

by M. Demazure

[1379]

The first part of this Exposé (nos 1 to 5) is a direct consequence of the existence, for a reductive group, of "sufficiently many outer automorphisms", a result which is itself a consequence of the weakest form of the isomorphism theorem for pinned groups. The second part (nos 6 and 7) presents two applications of the more precise results of the preceding Exposé; in particular, no 7 uses the theorem of generators and relations in its explicit form. Finally, we have given in an appendix (no 8) some results on "Galois" cohomology used in the text.

Let us specify our cohomological notation: if S is a scheme and G an S-group scheme, we write $H^1(S,G)$ for the first cohomology set of S with coefficients in G, computed for the (fpqc) topology; this is also the set of isomorphism classes of (fpqc) principal homogeneous sheaves under G. We write $H^1_{ét}(S,G)$ for the corresponding set for the étale topology; this is therefore the part of $H^1(S,G)$ formed by classes of homogeneous sheaves under G which are quasi-isotrivial (= locally trivial for the étale topology). We write $Fib(S,G)$ for the part of $H^1(S,G)$ formed by classes of representable sheaves (principal homogeneous bundles). We thus have the inclusions

$$H^1_{ét}(S, G) \subset H^1(S, G),$$
$$Fib(S, G) \subset H^1(S, G).$$

If every principal homogeneous sheaf under G is representable (for example if G is quasi-affine over S, cf. SGA 1, VIII 7.9), then $Fib(S,G) = H^1(S,G)$.

If $S' \to S$ is a covering morphism for the (fpqc) topology, we write $H^1(S'/S,G)$ for the kernel of the canonical map $H^1(S,G) \to H^1(S',G_{S'})$. It is known that $H^1(S'/S,G)$ can be computed simplicially (TDTE I, § A.4), which implies that when $S' \to S$ is covering for the étale topology, $H^1(S'/S,G)$ is also the kernel of $H^1_{ét}(S,G) \to H^1_{ét}(S',G_{S'})$.

[1379] N.D.E.: Version of 13/10/2024.

Finally, following Exp. VIII, 4.5, we call "theorem 90" the following assertion: "every principal homogeneous sheaf under $G_{m,S}$ is representable and locally trivial", an assertion equivalent to "$H^1(S, G_{m,S}) = \mathrm{Pic}(S)$", or again to "$H^1(S, G_{m,S}) = 0$ for S local (or more generally semi-local)".

1. The automorphism scheme of a reductive group

1.0.

It is convenient first to make certain definitions of the preceding Exposé more precise. Let S be a nonempty scheme, G an S-reductive group, $R = (M, M*, R, R*, \Delta)$ a pinned reduced root datum (Exp. XXIII 1.5). We call *pinning of G of type R*, or *R-pinning of G*, the datum:

(i) of an isomorphism of $D_S(M)$ onto a maximal torus T of G (or, what amounts to the same, of a monomorphism $D_S(M) \to G$ whose image is a maximal torus T of G), identifying R with a root system of G relative to T (Exp. XIX, 3.6) and $R*$ with the corresponding set of coroots,

(ii) for each $\alpha \in \Delta$, of an $X_\alpha \in \Gamma(S, g_\alpha)^\times$.

For G to possess an R-pinning, it is necessary and sufficient that it be splittable and of type R (Exp. XXII, 2.7).

If $u : G \to G'$ is an isomorphism of S-reductive groups, to every R-pinning E of G there corresponds by "transport of structure" an R-pinning $u(E)$ of G'. If $v : R' \to R$ is an isomorphism of pinned root data, to every R'-pinning E of G' there corresponds by transport of structure an R-pinning $v(E)$ of G.

Let us call a *pinned group* a triple (G, R, E) where G is an S-reductive group, R is a pinned reduced root datum, and E is a pinning of G of type R. We call an *isomorphism of the pinned group (G, R, E) onto the pinned group (G', R', E')* a pair (u, v) where u is an isomorphism $u : G \to G'$ and v is an isomorphism of pinned root data $v : R' \to R$, such that $u(E) = v(E')$.[1380]

N. B. If S is nonempty, v is uniquely determined by u, and we shall also say by abuse of language that u is an isomorphism of pinned groups. In particular, if (G, R, E) is a pinned group, an automorphism of (G, R, E) is therefore an automorphism u of G such that there exists an automorphism v of R with $u(E) = v(E)$; it is therefore an automorphism of G normalizing T, inducing on T an automorphism of the form $D_S(h)$ where h is an automorphism of M, and permuting among themselves the elements X_α, $\alpha \in \Delta$. (As is easily seen, the preceding conditions moreover characterize the automorphisms of (G, R, E).)

One has an obvious contravariant functor

$\Phi : (G, R, E) \mapsto R, \quad (u, v) \mapsto v$

[1380] N.D.E.: Hence $Lie(u)(X'_{v(\alpha)}) = X_\alpha$, for every $\alpha \in R'$.

and the principal result of the preceding Exposé (Exp. XXIII, 4.1) shows us that this is a fully faithful functor (we shall moreover see in the next Exposé that it is an equivalence of categories). It follows in particular that the group of automorphisms of (G, R, E) is canonically isomorphic to the group of automorphisms of the pinned root datum R (cf. Exp. XXIII, 5.5).

1.1.

Let S be a scheme; endow $(Sch)/S$ with the (fpqc) topology and consider the S-group sheaf $\mathrm{Aut}_{S-gr.}(G)$, where G is an S-group scheme. One has an exact sequence of S-group sheaves

```
1 → Centr(G) → G —int→ Aut_{S-gr.}(G)
```

which defines a monomorphism

```
j : G / Centr(G) → Aut_{S-gr.}(G).
```

The image sheaf of j is the sheaf of *inner automorphisms* of G; for an automorphism u of G to be inner, it is necessary and sufficient that there exist a covering family $S_i \to S$ and for each i a $g_i \in G(S_i)$ such that $int(g_i) = u_{S_i}$. In this case, if v is another automorphism of G, one sees at once that $int(v)u = vuv^{-1}$ is the inner automorphism defined by the family $g_i' = v(g_i)$. It follows that the image of j is normal in $\mathrm{Aut}_{S-gr.}(G)$. The quotient group sheaf, denoted $Autext(G)$, is the sheaf of *outer automorphisms* of G. One thus has an exact sequence

```
1 → G / Centr(G) → Aut_{S-gr.}(G) → Autext(G) → 1.
```

The preceding definitions are all compatible with base change. They are naturally valid in any site.

1.2.

Let S be a scheme and (G, R, E) a pinned S-reductive group. Let E be the (abstract) group of automorphisms of the pinned root datum R, i.e. the group of automorphisms of R normalizing Δ. By Exp. XXIII, 5.5, one has a canonical monomorphism

$$E \to \mathrm{Aut}_{S-gr.}(G)$$

which associates to $h \in E$ the unique automorphism u of the pinned group G such that $\Phi(u) = h$. This monomorphism canonically defines a monomorphism of sheaves

$$(*) a : E_S \to \mathrm{Aut}_{S-gr.}(G).$$

For an automorphism u of G to be a section of the image sheaf of a, it is necessary and sufficient that the following conditions be satisfied:

(i) u normalizes T. One then knows that u permutes the roots of G relative to T. If $\alpha \in R$, then $u(\alpha) : t \mapsto \alpha(u^{-1}(t))$ is locally on S of the form $t \mapsto \beta(t)$, with $\beta \in R$. The second condition is then written as follows:

(ii) If $\alpha \in \Delta$ and if U is an open set of S such that $u(\alpha)_U \in R$, then $u(\alpha)_U \in \Delta$ and

$$Lie(u_U)(X_\alpha)_U = (X_{u(\alpha)})_U.$$

It follows at once from the definitions that the sections of $a(E_S)$ normalize the subgroups of G defined by the pinning: T, B, B^-, U, U^-.

These definitions being set, one has:

Theorem 1.3. *Let S be a scheme, G an S-reductive group. Consider the canonical exact sequence of S-group sheaves*[1381]

1 → ad(G) → Aut_{S-gr.}(G) —p→ Autext(G) → 1.

(i) $\mathrm{Aut}_{S-gr.}(G)$ *is representable by a smooth and separated S-scheme.*

(ii) $Autext(G)$ is representable by a finitely generated twisted constant S-scheme (Exp. X, 5.1).

(iii) If G is splittable, the preceding exact sequence splits. More precisely, for every pinning of G, the morphism (cf. 1.2 (*))

p ∘ a : E_S → Autext(G)

is an isomorphism.

Let us first show how the theorem follows from the following lemma:

Lemma 1.4. *Under the hypotheses of (iii),* $\mathrm{Aut}_{S-gr.}(G)$ *is the semi-direct product $a(E_S) \cdot ad(G)$.*[1382]

The lemma immediately implies the theorem when G is splittable. Since G is locally splittable for the étale topology (Exp. XXII, 2.3), hence also for the (fppf) topology, and since the latter is "of effective descent" for the fibered category of twisted constant morphisms (Exp. X, 5.5), one deduces (ii) in the general case (cf. Exp. IV, 4.6.8). To deduce (i), one notes that $ad(G)$ is affine over S, hence the morphism p is affine when $\mathrm{Aut}_{S-gr.}(G)$ is representable, and one concludes by descent of affine schemes.[1383]

It therefore remains only to prove 1.4. For this, it suffices to prove:

Lemma 1.5. *If (R, E) and (R', E') are two pinnings of the S-reductive group G, there exists a unique inner automorphism u of G over S transforming one pinning into the other (i.e. such that there exists $v : R' \xrightarrow{\sim} R$ with $u(E) = v(E')$, cf. 1.0).*

[1381] N.D.E.: One recalls that $ad(G) = G/Centr(G)$ denotes the adjoint group of G, cf. XXII 4.3.6.
[1382] N.D.E.: We have replaced here and in the sequel the notation $int(G)$ by $ad(G)$.
[1383] N.D.E.: cf. SGA 1, VIII 2.1.

1.5.1. Uniqueness. It suffices to prove that if G is a pinned S-group and if $int(g)$ is an automorphism of pinned group ($g \in G(S)$), then $int(g) = id$. Now one has first $int(g)T = T$, $int(g)B = B$, so $g \in Norm_G(T)(S) \cap Norm_G(B)(S) = T(S)$ (cf. for example Exp. XXII, 5.6.1). It follows that $int(g)$ normalizes each U_α and that

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Lie(int(g)) X_α = Ad(g) X_α = α(g) X_α
```

for every $\alpha \in \Delta$. One thus has $\alpha(g) = 1$ for $\alpha \in \Delta$, so $g \in \bigcap_{\alpha\in\Delta}(Ker\alpha)(S) = Centr(G)(S)$ (Exp. XXII, 4.1.8). QED

1.5.2. Existence. It suffices to prove this locally for the (fpqc) topology. Let

```
(T, M, R, Δ, (X_α)_{α ∈ Δ})    and    (T', M', R', Δ', (X'_{α'})_{α' ∈ Δ'})
```

be the two pinnings. By conjugacy of maximal tori, one may assume $T = T'$. Up to restricting S, one may assume that the isomorphism $D_S(M) \simeq D_S(M')$ comes from an isomorphism $M \simeq M'$ carrying R onto R', and one is reduced to the situation $T = T'$, $M = M'$, $R = R'$. Since the systems of simple roots are conjugate by the Weyl group (Exp. XXI, 3.3.7), one may also assume $\Delta = \Delta'$. There then exists for each $\alpha \in \Delta$ a scalar $z_\alpha \in G_m(S)$ such that $X'_\alpha = z_\alpha X_\alpha$, and it suffices to construct locally for (fpqc) a section t of T such that $\alpha(t) = z_\alpha$ for each $\alpha \in \Delta$. But the morphism $T \to (G_{m,S})^\Delta$ with components $\alpha, \alpha \in \Delta$ is the dual of an injection $\mathbb{Z}^\Delta \to M$, hence faithfully flat, which completes the proof of 1.5.2 and so of 1.3.

Corollary 1.6. *Let S be a scheme, G an S-reductive group. The following conditions are equivalent:*

(i) $\mathrm{Aut}_{S-gr.}(G)$ *is affine (resp. of finite type, resp. of finite presentation, resp. quasi-compact) over S.*

(ii) $Autext(G)$ is finite over S.

(iii) For every $s \in S$, one has $rg_{red}(G_s) - rg_{ss}(G_s) \leqslant 1$.

Indeed, since $ad(G)$ is affine, flat and of finite presentation over S, the morphism $p : \mathrm{Aut}_{S-gr.}(G) \to Autext(G)$ is affine, faithfully flat and of finite presentation.

If $Autext(G)$ is finite over S, it is affine over S, hence so is $\mathrm{Aut}_{S-gr.}(G)$, which proves (ii) ⇒ (i). If $\mathrm{Aut}_{S-gr.}(G)$ is quasi-compact over S, it is of finite presentation over S (being in any case locally of finite presentation and separated over S); by Exp. V, 9.1, $Autext(G)$ is then of finite presentation over S, hence finite, which proves (i) ⇒ (ii). Finally, to prove the equivalence of (ii) and (iii), one may assume G split, and one is reduced to Exp. XXI, 6.7.8.

Corollary 1.7. *Let S be a scheme and G an S-reductive group. Then* $\mathrm{Aut}_{S-gr.}(G)^0 \simeq ad(G)$.

Corollary 1.8. *Let S be a scheme, G an S-reductive group, H an S-group scheme smooth, affine and with connected fibers over S. Then the S-functor*

$$Isom_{S-gr.}(G, H)$$

is representable by a smooth and separated S-scheme (which is affine over S if G is semisimple).

Indeed, let U be the set of points s of S such that H_s is reductive; this is open (Exp. XIX, 2.6); if S' is an S-scheme, $H_{S'}$ is reductive if and only if $S' \to S$ factors through U. It follows that the canonical morphism $Isom_{S-gr.}(G,H) \to S$ factors through U. One may therefore assume $S = U$ and one is reduced to:

Corollary 1.9. *Let S be a scheme, G and G' two S-reductive groups. Then*

$$F = Isom_{S-gr.}(G,G')$$

is representable by a smooth and separated S-scheme (affine if G or G' is semisimple). Moreover, S decomposes as a sum of two open subschemes S_1 and S_2 such that $F_{S_1} = \emptyset$ and such that F_{S_2} is a principal homogeneous bundle on the left (resp. right) under $\mathrm{Aut}_{S_2-gr.}(G'_{S_2})$ *(resp.* $\mathrm{Aut}_{S_2-gr.}(G_{S_2})$*).*

Indeed, let S_2 be the set of points of S where G and G' are of the same type, and let S_1 be its complement.

Since the type of a reductive group is a locally constant function, S_1 and S_2 are open. It is clear that $F_{S_1} = \emptyset$, and one may assume $S = S_2$. By Exp. XXIII, 5.6, F is a principal homogeneous sheaf under $\mathrm{Aut}_{S-gr.}(G)$, locally trivial for the étale topology. It follows that $F_0 = F/ad(G)$ is a principal homogeneous sheaf under $Autext(G)$, locally trivial for the étale topology, hence representable (Exp. X, 5.5). Since $ad(G)$ is affine, F is therefore also representable.

Let us remark that in the course of the proof, we have obtained:

Corollary 1.10. *Let S be a scheme, G and G' two S-reductive groups of the same type at each $s \in S$. Then $ad(G)$ operates freely (on the right) in $Isom_{S-gr.}(G,G')$, the quotient sheaf*

`Isomext(G, G') = Isom_{S-gr.}(G, G') / ad(G)`

is representable by a twisted constant S-scheme, which is a principal homogeneous bundle under $Autext(G)$ (and which is therefore finite over S if G is semisimple). Moreover, the isomorphism

`Isom_{S-gr.}(G, G') ≃ Isom_{S-gr.}(G', G)`

defined by $u \mapsto u^{-1}$ induces an isomorphism

`Isomext(G, G') ≃ Isomext(G', G).`

Remark 1.11. If $Isomext(G,G')(S) \neq \emptyset$, one says that G is an *inner twisted form* of G'; then G' is an inner twisted form of G; one can then reduce the structure group of $Isom_{S-gr.}(G,G')$ to $ad(G)$. More precisely, let $u \in Isomext(G,G')(S)$, considered as a section $u : S \to Isomext(G,G')$. Write

$$Isomint_u(G,G')$$

for the inverse image under the canonical morphism $Isom_{S-gr.}(G,G') \to Isomext(G,G')$ of the closed subscheme of $Isomext(G,G')$ image of u. The natural operation of $ad(G)$ on $Isomint_u(G,G')$ endows this scheme with a structure of principal homogeneous bundle; by extension of the structure group[1384] $ad(G) \to \mathrm{Aut}(G)$, $Isomint_u(G,G')$ recovers $Isom_{S-gr.}(G,G')$.

By Hensel's lemma (Exp. XI, 1.11), 1.8 gives at once:

Corollary 1.12. *Let S be a henselian local scheme, G an S-reductive group, G' a smooth affine S-group with connected fibers, s the closed point of S. If G_s and G'_s are isomorphic $\kappa(s)$-algebraic groups, then G and G' are isomorphic. More precisely, every $\kappa(s)$-isomorphism $G_s \simeq G'_s$ comes from an S-isomorphism $G \simeq G'$.*

Applying now 1.7 (resp. 1.12) to the scheme of dual numbers over a field, one deduces from Exp. III, 2.10 (resp. 3.10) point (i) (resp. (ii)) of the following corollary.

Corollary 1.13. *Let k be a field and G a k-reductive group.*

(i) If G is adjoint, one has $H^1(G, Lie(G/k)) = 0$.[1385]

(ii) One has $H^2(G, Lie(G/k)) = 0$.

Remark 1.14. (i) The assertion concerning H^1 was known (Chevalley); the one concerning H^2 had been proved in most cases of the classification by Chevalley.

(ii) In fact, the conjunction of 1.13 and the uniqueness theorem over an algebraically closed field is essentially equivalent to the uniqueness theorem. A direct proof of 1.13 would therefore give a way to deduce the general uniqueness theorem from Chevalley's uniqueness theorem over a field.

The existence of reductive groups of all types over all schemes (Exp. XXV) shows that the obstructions to lifting a k-reductive group G over Artinian rings with residue field k (which by Exp. III, 3.8 are elements of

$$H^3(G, Lie(G/k) \otimes V) \simeq H^3(G, Lie(G/k)) \otimes V,$$

where V is a certain k-vector space (equipped with the trivial action of G)) are zero. This seems to suggest that $H^3(G, Lie(G/k)) = 0$. Here too, a direct proof of this fact (if it is true) would doubtless give a way to deduce the general existence theorem from the existence theorem over a field (Chevalley's Tôhoku).

Corollary 1.15. *Let k be a field,*[1386] *G a k-reductive group. Consider k as a trivial G-module. Then*

$$H^1(G, k) = H^2(G, k) = 0.$$

[1384] N.D.E.: We have corrected $ad(G) \to Autext(G)$ to $ad(G) \to \mathrm{Aut}(G)$.

[1385] N.D.E.: We have corrected the original, which stated (i) without assuming G adjoint. Following the characterization of the $H^i(G,-)$ as derived functors of $\mathrm{Hom}_G(k,-)$ (Exp. I, 5.3.1; see also [Ja87], I 4.16), one has $H^i(G,V) = Ext^i_G(k,V)$ for every G-module V; now if $char(k) = p > 0$ and $G = SL_{p,k}$, then $Lie(GL_p/k)$ is a non-trivial extension of k by $g = Lie(SL_p/k)$, so $H^1(G,g) \neq 0$. See also the addition 1.15.1 below.

[1386] N.D.E.: We have replaced "scheme" by "field".

[1387] Since $H^i(G_{\bar{k}}, \bar{k}) = H^i(G, k) \otimes \bar{k}$, one may assume k algebraically closed. An element of $H^1(G, k)$ is nothing but a morphism of k-groups $\phi : G \to G_{a,k}$. Then $\phi(G) = G/Ker(\phi)$ is a smooth, connected and reductive subgroup (cf. XIX 1.7) of $G_{a,k}$, hence trivial. So $H^1(G, k) = 0$.

Consider now the k-reductive group $H = G \times_k G_{m,k}$. One has $Lie(H/k) = Lie(G/k) \oplus k$, a decomposition stable under H. For any H-module V, one has

$$H^i(H, V) = H^i(G, H^0(G_{m,k}, V))$$

(this follows from the characterization of $H^i(H, -)$ as derived functors of $H^0(H, -) = H^0(G, H^0(G_{m,k}, -))$, and from the fact that the functor $H^0(G_{m,k}, -)$ is exact, cf. Exp. I, 5.3.1 and 5.3.3). In particular, one has for every i

$$H^i(H, Lie(H/k)) = H^i(G, Lie(G/k)) \oplus H^i(G, k)$$

whence $H^2(G, k) = 0$ by applying 1.13 (ii) to the reductive group H.

Corollary 1.15.1. [1388] *Let k be a field, G a k-reductive group, Z its center and $R = (M, M*, R, R*)$ the type of G. Then one has*

$$H^1(G, Lie(G/k)) \simeq Ext^1_{\mathbb{Z}}(M / \Gamma_0(R), k).$$

In particular, $H^1(G, Lie(G/k)) = 0$ if and only if Z is smooth over k.

Indeed, let g (resp. z, resp. g_{ad}) be the Lie algebra of G (resp. Z, resp. $G_{ad} = G/Z$). It follows from 1.15 (and from its proof) that

$$H^1(G, g) = H^1(G_{ad}, g) \simeq H^1(G_{ad}, g/z).$$

Set $C = Coker(g \to g_{ad})$; one has an exact sequence

$$0 \to g/z \to g_{ad} \to C \to 0.$$

Since $H^1(G_{ad}, g_{ad}) = 0$ (1.13) and $H^0(G_{ad}, g_{ad}) = 0$ (cf. Exp. II, 5.2.3), one obtains

$$H^1(G_{ad}, g/z) \simeq H^0(G_{ad}, C).$$

To compute the right-hand term, one may assume k algebraically closed. Let (G, T, M, R) be a splitting of G; then $Z = D_k(M/\Gamma_0(R))$ and C is identified with

$$Coker(Hom_{\mathbb{Z}}(M, k) \to Hom_{\mathbb{Z}}(\Gamma_0(R), k)) \simeq Ext^1_{\mathbb{Z}}(M / \Gamma_0(R), k),$$

equipped with the trivial action of G_{ad}. The corollary follows, since Z is not smooth over k if and only if $char(k) = p > 0$ and $M/\Gamma_0(R)$ has p-torsion (Exp. IX, 2.1).

Definition 1.16. *Let S be a scheme, G an S-reductive group. We call a* form of G over S *an S-group scheme G' locally isomorphic to G for the (fpqc) topology (it*

[1387] N.D.E.: Because of the correction made in 1.13, we have given another proof in the case of H^1. On the other hand, it follows from a theorem of G. Kempf that $H^i(G, k) = 0$ for every $i > 0$, cf. [Ja87], II 4.5 and 4.11.

[1388] N.D.E.: We have added this corollary, drawn from remarks of O. Gabber, which makes 1.13 (i) more precise.

amounts to the same to say (cf. Exp. XXIII, 5.6) that G' is locally isomorphic to G for the étale topology, or again that G' is an S-reductive group of the same type as G at every point of S).

Corollary 1.17. *Let S be a scheme, G an S-reductive group.*

(i) The functor

$$G' \mapsto Isom_{S-gr.}(G, G')$$

is an equivalence between the category of forms of G over S and the category of principal homogeneous bundles under $\mathrm{Aut}_{S-gr.}(G)$.

(ii) If $S' \to S$ is a covering morphism, forms of G trivialized by S' and bundles trivialized by S' correspond.

(iii) Every principal homogeneous sheaf under $\mathrm{Aut}_{S-gr.}(G)$ is representable and quasi-isotrivial (i.e. locally trivial for the étale topology).

The first assertion is formal in the category of sheaves (for (fpqc) for example). On the other hand, every sheaf locally isomorphic to G (for (fpqc)) is representable (since G is affine over S) and locally isomorphic to G for the étale topology. Finally, for every form G' of G, the S-sheaf $Isom_{S-gr.}(G, G')$ is representable, by 1.8. The corollary follows at once.

Corollary 1.18. *The set of isomorphism classes of forms of the reductive group G over S is isomorphic to*

$$H^1(S, \mathrm{Aut}_{S\text{-}gr.}(G)) = H^1_{\text{ét}}(S, \mathrm{Aut}_{S\text{-}gr.}(G)) = \mathrm{Fib}(S, \mathrm{Aut}_{S\text{-}gr.}(G)).$$

If $S' \to S$ is a covering morphism, the subset formed by forms trivialized by S' is isomorphic to $H^1(S'/S, \mathrm{Aut}_{S-gr.}(G))$.

Corollary 1.19. *Let S be a scheme, R a pinned reduced root datum such that $\acute{E}p_S(R)$[1389] exists (condition automatically satisfied, cf. Exp. XXV). Write*

$$A_S(R) = \mathrm{Aut}_{S\text{-}gr.}(\acute{E}p_S(R)) = \mathrm{ad}(\acute{E}p_S(R)) \cdot E(R)_S.$$

(i) The set of isomorphism classes of S-reductive groups of type R (Exp. XXII, 2.7) is isomorphic (by Exp. XXIII, 5.12) to

$$H^1(S, A_S(R)) = H^1_{\text{ét}}(S, A_S(R)) = \mathrm{Fib}(S, A_S(R)).$$

(ii) If $S' \to S$ is a covering morphism, the subset formed by the classes of groups splittable over S' is isomorphic to $H^1(S'/S, A_S(R))$.

Remark 1.20. With the preceding notation, to every S-reductive group of type R is canonically associated a right principal homogeneous bundle under $A_S(R)$:

$$Isom_{S-gr.}(\acute{E}p_S(R), G) = P.$$

[1389] N.D.E.: cf. XXIII, Definition 5.11.

Note that P is to be interpreted as the "scheme of pinnings of G of type R" (cf. 1.0). Moreover P is also a principal homogeneous bundle (on the left) under $\mathrm{Aut}_{S-gr.}(G)$, a structure that appears at once in the description above.

Proposition 1.21. *Let S be a henselian local scheme, s its closed point. The functor*

$$G \mapsto G_s$$

induces a bijection of the set of isomorphism classes of S-reductive groups onto the set of isomorphism classes of $\kappa(s)$-reductive groups.

In particular, for every S-reductive group G, there exists a finite surjective étale morphism $S' \to S$ such that $G_{S'}$ is splittable.

Using the existence of the $\acute{E}p_S(R)$ (Exp. XXV), one is reduced to proving that if one writes $H = A_S(R)$, the canonical map

$$\mathrm{Fib}(S, H) \to \mathrm{Fib}(\kappa(s), H_s)$$

is bijective (and that every element of $Fib(S, H)$ has the property indicated above). Now, every finite part of H is contained in an affine open set (it is indeed trivial for a constant group, and H is affine above a constant group); one may therefore use the result proved in the appendix (8.1).

2. Automorphisms and subgroups

Let us introduce a notation: if $H = \mathrm{Aut}_{S-gr.}(G)$, and if X is a subfunctor of G, we write

$$\mathrm{Aut}_{S\text{-gr.}}(G, X) = \mathrm{Norm}_H(X), \qquad \mathrm{Aut}_{S\text{-gr.}}(G, \mathrm{id}_X) = \mathrm{Centr}_H(X).$$

If Y is a second subfunctor of G, one defines similarly $\mathrm{Aut}_{S\text{-gr.}}(G, X, Y) = \mathrm{Aut}_{S\text{-gr.}}(G, X) \cap \mathrm{Aut}_{S\text{-gr.}}(G, Y)$, and if G' is a second S-group and X' a subfunctor of G', one writes $Isom_{S-gr.}(G, X; G', X')$ for the subfunctor of $Isom_{S-gr.}(G, G')$ defined by: for every $S' \to S$,

$$\mathrm{Isom}_{S\text{-gr.}}(G, X; G', X')(S') = \{ u \in \mathrm{Isom}_{S\text{-gr.}}(G, G')(S') \mid u(X_{S'}) = X'_{S'} \}$$

and one defines similarly $Isom_{S-gr.}(G, X, Y; G', X', Y')$, etc.[1390]

Proposition 2.1. *Let S be a scheme, G an S-reductive group, T a maximal torus of G (resp. B a Borel subgroup of G, resp. $B \supset T$ a Killing couple of G). Write T_{ad} (resp. B_{ad}) for the maximal torus (resp. Borel subgroup) of $ad(G)$ corresponding to T (resp. B):*

$$B_{ad} \simeq B / \mathrm{Centr}(G) = B / \mathrm{Centr}(B),$$
$$T_{ad} \simeq T / \mathrm{Centr}(G).$$

[1390] N.D.E.: We have made explicit the preceding definitions (the original indicated "One defines similarly $\mathrm{Aut}_{S-gr.}(G, X, Y)$, ..., $Isom_{S-gr.}(G, X; G', X')$, ...").

Then $\mathrm{Aut}_{S-gr.}(G,T)$ *(resp.* $\mathrm{Aut}_{S-gr.}(G,B)$*, resp.* $\mathrm{Aut}_{S-gr.}(G,B,T)$*) is representable by a closed subscheme of* $\mathrm{Aut}_{S-gr.}(G)$*, smooth over* S*, and the exact sequence of Theorem 1.3 induces exact sequences:*

1 → Norm_{ad(G)}(T_ad) → Aut_{S-gr.}(G, T) → Autext(G) → 1;
1 → B_ad → Aut_{S-gr.}(G, B) → Autext(G) → 1;
1 → T_ad → Aut_{S-gr.}(G, B, T) → Autext(G) → 1.

By descent of closed subschemes,[1391] one is at once reduced to the case where G is pinned and where $B \supset T$ is its canonical Killing couple (cf. Exp. XXII, 5.5.5 (iv)). Since the group E of 1.2 normalizes B and T, the result follows at once from the normalization theorems in $ad(G)$ (Exp. XXII, 5.3.12 and 5.6.1).

Using now the conjugation theorems (cf. Exp. XXIII, 5.12), and arguing as in no 1, one deduces:

Corollary 2.2. *Let* S *be a scheme,* G *and* G' *two* S*-reductive groups of the same type at each point. Let* $B \supset T$ *(resp.* $B' \supset T'$*) be a Killing couple of* G *(resp.* G'*).*

(i) The S*-functor* $Isom_{S-gr.}(G,T;G',T')$ *is representable by a closed smooth subscheme of* $Isom_{S-gr.}(G,G')$ *which is principal homogeneous under* $\mathrm{Aut}_{S-gr.}(G,T)$*.*

Moreover, $Norm_{ad(G)}(T_{ad})$ *operates freely on this scheme, and one has a canonical isomorphism*

Isom_{S-gr.}(G, T; G', T') / Norm_{ad(G)}(T_ad) ≃ Isomext(G, G').

(ii) The S*-functor* $Isom_{S-gr.}(G,B;G',B')$ *is representable by a closed smooth subscheme of* $Isom_{S-gr.}(G,G')$ *which is principal homogeneous under* $\mathrm{Aut}_{S-gr.}(G,B)$*.*

Moreover, B_{ad} *operates freely on this scheme and one has a canonical isomorphism*

Isom_{S-gr.}(G, B; G', B') / B_ad ≃ Isomext(G, G').

(iii) The S*-functor* $Isom_{S-gr.}(G,B,T;G',B',T')$ *is representable by a closed smooth subscheme of* $Isom_{S-gr.}(G,G')$*, principal homogeneous under* $\mathrm{Aut}_{S-gr.}(G,B,T)$*.*

Moreover, T_{ad} *operates freely on this scheme and one has a canonical isomorphism*

Isom_{S-gr.}(G, B, T; G', B', T') / T_ad ≃ Isomext(G, G').

Arguing again as in no 1, one deduces:

Corollary 2.3. *Let* S *be a scheme,* G *an* S*-reductive group,* $B \supset T$ *a Killing couple of* G*. The functor*

(G', T') ↦ Isom_{S-gr.}(G, T; G', T'),

resp.

(G', B') ↦ Isom_{S-gr.}(G, B; G', B'),

resp.

[1391] N.D.E.: cf. SGA 1, VIII 1.9.

$$(G', B', T') \mapsto \mathrm{Isom}_{S\text{-}gr.}(G, B, T; G', B', T'),$$

is an equivalence between the category of pairs (G', T') (resp. of pairs (G', B'), resp. of triples (G', B', T')), where G' is a form of G and T' a maximal torus of G' (resp. B' a Borel subgroup of G', resp. $B' \supset T'$ a Killing couple of G'), and the category of principal homogeneous bundles under the S-group H, where $H = \mathrm{Aut}_{S-gr.}(G,T)$ (resp. $H = \mathrm{Aut}_{S-gr.}(G,B)$, resp. $H = \mathrm{Aut}_{S-gr.}(G,B,T)$).

Moreover, every principal homogeneous sheaf under H is representable and quasi-isotrivial, so that one has

$$H^1(S, H) = H^1_{\text{ét}}(S, H) = \mathrm{Fib}(S, H).$$

Remark 2.4. Under the conditions of 2.2, the morphism noted set-theoretically $u \mapsto u(T)$ (resp. $u \mapsto u(B)$, resp. $u \mapsto (u(B), u(T))$) induces an isomorphism

$$\mathrm{Isom}_{S\text{-}gr.}(G, G') / \mathrm{Aut}_{S\text{-}gr.}(G, T) \simeq \mathrm{Tor}(G')$$

$$\text{resp.} \quad \mathrm{Isom}_{S\text{-}gr.}(G, G') / \mathrm{Aut}_{S\text{-}gr.}(G, B) \simeq \mathrm{Bor}(G'),$$

$$\text{resp.} \quad \mathrm{Isom}_{S\text{-}gr.}(G, G') / \mathrm{Aut}_{S\text{-}gr.}(G, B, T) \simeq \mathrm{Kil}(G').$$

The proof is immediate: it suffices to do it locally for (fpqc), so one may assume $G \simeq G'$, and one is reduced to Exp. XXII, 5.8.3 (iii).

Remark 2.5. The preceding results are at once interpreted in terms of restriction of the structure group: if G' is a form of G, corresponding to the principal bundle $Isom_{S-gr.}(G, G')$, to give a restriction of the structure group of this bundle to $\mathrm{Aut}_{S-gr.}(G,T)$ amounts to giving a maximal torus T' of G', the bijections suggested above being that of 2.4 on the one hand, the map $T' \mapsto Isom_{S-gr.}(G,T;G',T')$ on the other. Similarly for Borel subgroups and Killing couples.

Proposition 2.6. *Let S be a scheme, G and G' two S-reductive groups of the same type at each point, T (resp. T') a maximal torus of G (resp. G'). Then T_{ad} operates freely on $Isom_{S-gr.}(G,T;G',T')$, the quotient*

$$P = \mathrm{Isom}_{S\text{-}gr.}(G, T; G', T') / T_{ad}$$

is representable; it is a principal homogeneous bundle under

$$A = \mathrm{Aut}_{S\text{-}gr.}(G, T) / T_{ad},$$

where A is representable by a twisted constant S-scheme, extension of $Autext(G)$ by $W_{ad(G)}(T_{ad}) = Norm_{ad(G)}(T_{ad})/T_{ad}$. Moreover, if one makes A operate on T in the obvious way, the bundle associated to P is none other than T'.[1392]

The first part of the proposition follows at once from the preceding results. To prove the second, one notes that there is an obvious morphism $P \times_S T \to T'$ (defined by $(u,t) \mapsto u(t)$); to show that after passage to the quotient by A it induces an isomorphism, one may once again assume $(G,T) \simeq (G',T')$, in which case it is immediate.

[1392] N.D.E.: Therefore, if P possesses a section, then T and T' are isomorphic; in particular T_P and T'_P are isomorphic. See the addition 4.5.1 where this remark is developed in the case where $G = \acute{E}p_S(R)$.

In an entirely analogous way, one has:

Proposition 2.7. *Let S be a scheme, G and G' two S-reductive groups of the same type at each point, $B \supset T$ (resp. $B' \supset T'$) a Killing couple of G (resp. G'). If one makes $\mathrm{Aut}_{S-gr.}(G,B,T)/T_{ad} \simeq Autext(G)$ operate in the obvious way on T, the bundle associated to $Isomext(G,G')$ is none other than T'.*

Corollary 2.8. *Let G and G' be two S-reductive groups that are inner twisted forms of each other; let $B \supset T$ (resp. $B' \supset T'$) be a Killing couple of G (resp. G'). Then T and T' are isomorphic.*[1393]

Remark 2.9. It is not true in general that B and B' are isomorphic; they are however inner twisted forms of each other (cf. no 5).

One can develop "Isomint" variants[1394] of the preceding results. Let us point out one:

Proposition 2.10. *Let S be a scheme, G and G' two S-reductive groups of the same type at each point, $B \supset T$ (resp. $B' \supset T'$) a Killing couple of G (resp. G'). Let $u \in Isomext(G,G')(S)$; consider*

`Isomint_u(G, B, T; G', B', T') = Isom_{S-gr.}(G, B, T; G', B', T') ∩ Isomint_u(G, G').`

It is a smooth and affine S-scheme which is a principal homogeneous bundle under T_{ad}. In particular, $Isomint_u(G,B,T;G',B',T')(S) \neq \emptyset$ if and only if the corresponding element of $H^1(S,T_{ad})$ is zero.

To conclude this section, let us prove two results which will be useful later:

Proposition 2.11. *Let S be a scheme, G an S-reductive group, T a maximal torus of G. The obvious morphism*

$$T_{ad} \xrightarrow{\sim} \mathrm{Aut}_{S-gr.}(G, id_T)$$

is an isomorphism.

This follows from the preceding statements; moreover, one has given a direct proof in the course of the proof of 1.5.2.

Corollary 2.12. *Under the preceding conditions, there exists an equivalence between the category of pairs `(G', f)`, where G' is a form of G and f is an isomorphism of T onto a maximal torus of G', and the category of principal homogeneous bundles under T_{ad}.*

Corollary 2.13. *If $H^1(S,T_{ad}) = 0$, and if G' is a form of G possessing a maximal torus isomorphic to T, then G' is isomorphic to G.*

Corollary 2.14. *Let S be a scheme such that $\mathrm{Pic}(S) = 0$, and G an S-reductive group of constant type. For G to be splittable, it is necessary and sufficient that G possess a split maximal torus.*

[1393] N.D.E.: by 2.7, since $Isomext(G,G')$ possesses a section (cf. the preceding N.D.E.).
[1394] N.D.E.: Recall that $Isomint_u(G,G')$ is defined in 1.11.

Let G be a reductive group, $rad(G)$ its radical (Exp. XXII, 4.3.9); since $rad(G)$ is central and characteristic in G, one has a canonical morphism

$$q : \mathrm{Autext}(G) \to \mathrm{Aut}_{S\text{-gr.}}(rad(G)).$$

Proposition 2.15. *Let G be an S-reductive group. The following sequence is exact:*

$$1 \to ad(G) \to \mathrm{Aut}_{S\text{-gr.}}(G, id_{rad(G)}) \xrightarrow{p} \mathrm{Autext}(G) \xrightarrow{q} \mathrm{Aut}_{S\text{-gr.}}(rad(G))$$

and $Ker(q) = Im(p)$ is an open and closed subscheme of $Autext(G)$, finite over S.

One may assume G split. The first assertion is immediate; the second follows from Exp. XXI, 6.7.5 and 6.7.7.

Writing $H = \mathrm{Aut}_{S-gr.}(G, id_{rad(G)})$, one deduces:

Corollary 2.16. *There exists an equivalence between the category of pairs (G', f), where G' is a form of G and f is an isomorphism of $rad(G)$ onto the radical of G', and the category of principal bundles under a certain S-group scheme H, where H is such that there exists an exact sequence*

$$1 \to ad(G) \to H \to F \to 1,$$

where the S-group F is étale and finite over S.

3. Dynkin scheme of a reductive group. Quasi-split groups

3.1.

Recall (Exp. XXI, 7.4.1) that a *Dynkin diagram* is a finite set endowed with the structure defined by a set of pairs of distinct elements (*bonds*) and a map to $\{1, 2, 3\}$ (*lengths*). To each pinned reduced root datum R is associated a Dynkin diagram $\Delta(R)$, whose underlying set is the set of simple roots.

3.2.

Let S be a scheme. An *S-Dynkin scheme* is a finite twisted constant S-scheme Δ, equipped with the structure defined by a subscheme L of $\Delta \times_S \Delta$ having empty intersection with the diagonal, and by a morphism $\Delta \to 1, 2, 3_S$. If S' is a connected[1395] S-scheme, $\Delta(S')$ is naturally endowed with a structure of Dynkin diagram.

One defines at once the following notions: isomorphism of two Dynkin schemes, base change of a Dynkin scheme, constant Dynkin scheme associated to a Dynkin diagram.

Every descent datum on a Dynkin scheme for the étale topology is effective.

[1395]N.D.E.: We have added the hypothesis that S' be connected.

3.3.

We propose to associate to each S-reductive group G an S-Dynkin scheme. Suppose first that G is splittable over S; for every pinning E of G, write $\Delta(E)$ for the constant Dynkin scheme associated to the pinned root datum defined by E; if E and E' are two pinnings of G, there exists by 1.5 a unique inner automorphism of G over S transforming E into E'; this automorphism of G defines an isomorphism $a_{EE'} : \Delta(E) \xrightarrow{\sim} \Delta(E')$; the $a_{EE'}$ evidently form a transitive system, so that one may identify the $\Delta(E)$ (i.e. take the inductive limit); the result is a constant Dynkin scheme denoted $Dyn(G)$. If now G is an arbitrary S-reductive group, there exists a covering family for the étale topology $S_i \to S$ such that G_{S_i} is splittable. Arguing as previously, one therefore has a canonical descent datum on the $Dyn(G_{S_i})$, allowing one to construct by descent an S-Dynkin scheme $Dyn(G)$.

3.4.

This construction satisfies the following properties (which moreover essentially characterize it):

(i) To each S-reductive group is associated a Dynkin scheme $Dyn(G)$; to every isomorphism $u : G \xrightarrow{\sim} G'$ is functorially associated an isomorphism $Dyn(u) : Dyn(G) \xrightarrow{\sim} Dyn(G')$.

(ii) If S' is an S-scheme and G an S-reductive group, one has

$$Dyn(G \times_S S') \simeq Dyn(G) \times_S S'.$$

(iii) If E is a pinning of G, defining the pinned root datum with Dynkin diagram Δ, one has

$$Dyn(G) \simeq \Delta_S.$$

(iv) If u is an inner automorphism of G, $Dyn(u)$ is the identity automorphism of $Dyn(G)$.

3.5.

Let S be a scheme, G an S-reductive group. It is clear that the functor $\mathrm{Aut}_{Dyn}(Dyn(G))$ of automorphisms of $Dyn(G)$ for the structure of Dynkin scheme is representable by a finite twisted constant S-scheme. By 3.4 (i) and (ii), one has a canonical morphism

$$\mathrm{Aut}_{S\text{-gr.}}(G) \to \mathrm{Aut}_{Dyn}(Dyn(G)),$$

which, by (iv), factors through a morphism

$$\mathrm{Autext}(G) \to \mathrm{Aut}_{Dyn}(Dyn(G)).$$

More generally, if G and G' are two S-reductive groups, one has a canonical morphism

```
Isomext(G, G') → Isom_{Dyn}(Dyn(G), Dyn(G'));
```

in particular, if G' is an inner twisted form of G (1.11), the Dynkin schemes $Dyn(G)$ and $Dyn(G')$ are isomorphic.

3.6.

If G is semisimple (resp. adjoint or simply connected), the morphism

```
Autext(G) → Aut_{Dyn}(Dyn(G))
```

is a monomorphism (resp. an isomorphism). Indeed, one may assume G pinned and one is reduced to the corresponding result for pinned reduced root data (cf. Exp. XXI, 7.4.5).

One has an analogous result for `Isom`'s; whence it follows in particular that two semisimple adjoint (resp. simply connected) S-groups are inner twisted forms of each other if and only if their Dynkin schemes are isomorphic.

3.7.

One can give a different construction of the Dynkin scheme associated to a reductive group. Let R be a pinned reduced root datum, G an S-reductive group of type R; write $\Delta(R)$ for the Dynkin diagram defined by the root datum R. One has (3.5) a canonical morphism

```
A_S(R) = Aut_{S-gr.}(Ép_S(R)) → Aut_{Dyn}(Δ(R)_S).
```

The S-reductive group G corresponds (1.17) to a bundle $Isom_{S-gr.}(Ép_S(R), G)$, principal homogeneous under $A_S(R)$. The bundle under $\mathrm{Aut}_{Dyn}(\Delta(R)_S)$ associated corresponds to a form on S of $\Delta(R)_S$: this is $Dyn(G)$; in other words, this associated bundle is none other than `Isom_{Dyn}(Δ(R)_S, Dyn(G))`. In this last form, the proof is immediate.

3.8. Dynkin scheme and Killing couples.

Let S be a scheme, G an S-reductive group, $B \supset T$ a Killing couple of G. There exists a canonical morphism

```
i : Dyn(G) → Hom_{S-gr.}(T, G_{m, S})
```

which identifies $Dyn(G)$ with the "scheme of simple roots of B relative to T"; this morphism is defined at once by descent from the pinned case. Note moreover that the datum of T and of i allows one to reconstruct B ("biunivocal correspondence between systems of simple roots and systems of positive roots").

It follows from the preceding description of $D = Dyn(G)$ that there exists a canonical root of `B_D` with respect to `T_D`: this root α_D is the image under $i(D)$ of the identity morphism of D. One thereby deduces a canonical invertible `O_D`-module g_D:

$$g_D = (g \otimes_{O_S} O_D)^{\alpha_D}.$$

In the pinned case, one has

$$D = \coprod_{\alpha \in \Delta} S_\alpha,$$

where each S_α is a copy of S, and g_D is then the O_D-module which induces g_α on S_α, for every $\alpha \in \Delta$.

3.9. Quasi-pinnings. Quasi-pinned groups.

If G is an S-reductive group, we call *quasi-pinning of G* the datum:

(i) of a Killing couple $B \supset T$ of G,

(ii) of a section $X \in \Gamma(Dyn(G), g_D)^\times$.

We say that an S-reductive group is *quasi-splittable* if it possesses a quasi-pinning. We call *quasi-pinned group* a reductive group equipped with a quasi-pinning.

Let $B \supset T$ be a Killing couple of the S-reductive group G; then G is quasi-pinnable relative to this Killing couple if and only if g_D possesses a non-zero section at every point, i.e. if the element of $\mathrm{Pic}(Dyn(G))$ defined by g_D is zero. Suppose in particular that S is semi-local; then $Dyn(G)$ is also semi-local, so $\mathrm{Pic}(Dyn(G)) = 0$. One deduces:

Proposition 3.9.1. *Let S be a semi-local scheme, G an S-reductive group. For G to be quasi-splittable, it is necessary and sufficient that it possess a Borel subgroup.*

[1396] Indeed, S is affine so, by the first assertion of Exp. XXII, 5.9.7, if G possesses a Borel subgroup B, it also possesses a Killing couple $B \supset T$. Then, since $\mathrm{Pic}(D) = 0$, g_D possesses a section X nowhere zero.

Let still A be a semi-local ring and $S = \mathrm{Spec}(A)$; remark now that for every S-reductive group G the morphism $Bor(G) \to S$ is surjective (since G_s possesses Borel subgroups, for every $s \in S$) and smooth and projective (Exp. XXII, 5.8.3), so it possesses sections after a finite surjective étale extension of the base.[1397] One

[1396]N.D.E.: We have detailed the reference to Exp. XXII, 5.9.7.

[1397]N.D.E.: The original referred to EGA IV, § 24, which has not appeared. The point is to use "Bertini's theorem". Let us detail the argument, which was indicated to us by O. Gabber. Let $X \to S$ be a surjective, smooth and projective morphism; replacing S by a connected component, one may assume that X/S has constant relative dimension d at every point (cf. EGA IV_4, 17.10.2). One may also assume that X is a closed subscheme of a projective space $P^n_S = P(E)$, where E is a free A-module of rank $n+1$ (cf. EGA II, 5.3.3). Let $s_1, \dots, s_r$ be the closed points of S; by Bertini's theorem (see for example [Jou83], I 6.10), there exists an open set U of the product $P = P(E^*)^d$, with non-empty fibers, such that for every point $u = (f_1, \dots, f_d)$ of U_{s_i}, the intersection of $X_{\kappa(u)}$ with the d hyperplanes of $P^n_{\kappa(u)}$ defined by u is étale over $\kappa(u)$. Up to shrinking U, one may assume that U is the complement in the affine space A^{nd}_S of the locus of zeros $V(Q)$ of a certain polynomial Q of degree $m > 0$. One then sees easily (by induction on the number of variables) that $V(Q)$ cannot contain all rational points of $A^{nd}_{\kappa(s_i)}$ if $|\kappa(s_i)| > m$. Since the morphism $A \to \prod_i \kappa(s_i)$ is surjective, one can find a monic polynomial $R \in A[X]$ of degree m whose image in $\kappa(s_i)[X]$ is irreducible if $\kappa(s_i)$ is finite, and possesses m distinct roots if $\kappa(s_i)$ is infinite. Put $A' = A[X]/(R)$ and $S' = \mathrm{Spec}(A')$; then $S' \to S$ is étale, finite and surjective, and the residue field at each of its closed points $s'_1, \dots, s'_t$ is of cardinality $\geqslant 2m$. Then U possesses a rational point u_i over each closed point of S', and since $A' \to \prod_i \kappa(s_i)$ is surjective, these lift to a section u of $P_{S'}$. Write Z for the deduces:

Corollary 3.9.2. *Let S be a semi-local scheme, G an S-reductive group. There exists an étale, finite and surjective morphism $S' \to S$ such that $G_{S'}$ is quasi-splittable.*

Remark 3.9.3. Under the preceding conditions, let T be a maximal torus of G (cf. Exp. XIV, 3.20); then one may moreover assume that $G_{S'}$ is quasi-splittable relative to $T_{S'}$: it suffices to apply the preceding argument to the "scheme of Borel subgroups containing T", which is finite and étale over S (Exp. XXII, 5.5.5 (ii)).

3.10.

Let E and E' be two quasi-pinnings of the S-reductive group G. There exists a unique inner automorphism of G transforming E into E'. Indeed, one is at once reduced to the split case, where the assertion has already been proved (1.5; it suffices indeed to note that for an inner automorphism of G it amounts to the same to respect a pinning or the underlying quasi-pinning). One concludes as in no 1 that a

quasi-pinning of the S-reductive group G defines a splitting h of the exact sequence:

$$1 \to \mathrm{ad}(G) \to \mathrm{Aut}_{S\text{-gr.}}(G) \underset{p}{\overset{h}{\rightleftarrows}} \mathrm{Autext}(G) \to 1,$$

the image of h being the subgroup of $\mathrm{Aut}_{S-gr.}(G)$ which leaves invariant the quasi-pinning.

Similarly if G and G' are two quasi-pinned S-groups, one defines in a natural way the subfunctor

$$Isom_{S-gr.q-ép.}(G, G')$$

of $Isom_{S-gr.}(G, G')$; the projection of $Isom_{S-gr.}(G, G')$ onto $Isomext(G, G')$ induces an isomorphism

$$(*) \qquad Isom_{S\text{-gr. q-ép.}}(G, G') \xrightarrow{\sim} Isomext(G, G').$$

Theorem 3.11. *Let S be a scheme, R a pinned reduced root datum such that $Ép_S(R)$ exists (cf. Exp. XXV), E the group of its automorphisms. Consider the three following categories:*

(i) The category Rev *of principal Galois coverings of S with group E (the morphisms are isomorphisms of E-bundles).*

(ii) The category Redext *whose objects are the S-reductive groups of type R (Exp. XXII, 2.7), the morphisms from G to G' being the elements of $Isomext(G, G')(S)$.*

intersection of $X_{S'}$ with the d hyperplanes of $P^n_{S'}$ defined by u, and V for the open set of Z formed by the points at which Z is étale over S'. By EGA IV_3, 11.3.8, V contains the fibers $Z_{s'_i}$ for every i; since $\pi : Z \to S'$ is proper, it follows that the closed set $\pi(Z - V)$ is empty, whence $V = Z$. Then $Z \to S'$ is surjective, étale and proper, hence finite, as is the composite $Z \to S$, and this yields the desired section of $X \to S$.

(iii) The category `Qép` *of quasi-pinned S-reductive groups of type R (the morphisms are the isomorphisms respecting the quasi-pinnings).*

These three categories are equivalent: more precisely, one has a diagram of functors, commutative up to functorial isomorphisms

$$\begin{tikzcd} \mathtt{Rev} \arrow[rr, "\mathtt{q\acute{e}p}"] \arrow[dr, dash, "\mathtt{rev}"'] & & \mathtt{Q\acute{e}p} \arrow[dl, dash, "\mathtt{i}"] \\ & \mathtt{Redext} & \end{tikzcd}$$

We shall describe below these three functors, leaving to the reader the task of verifying the commutativity of the diagram.[1398]

3.11.1. The functor i. It is the obvious functor: $i(G) = G$, $i(f)$ = image of f under the isomorphism of 3.10 (*).

3.11.2. The functor `qép`. Let S' be a principal Galois covering of S with group E. Let $(G, T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$ be a pinning of $G = \acute{E}p_S(R)$ and $(X'_\alpha)_{\alpha\in\Delta}$ the corresponding pinning of $G' = G_{S'} = \acute{E}p_{S'}(R)$. By Exp. XXIII 5.5 bis, there exists a unique morphism of groups

`θ : E → Aut_{S-gr.}(Ép_S(R)) = A(R)(S)`

such that, for every $h \in E$, $\theta(h)$ induces $D_S(h)$ on T and $Lie(\theta(h))(X_\alpha) = X_{h(\alpha)}$ for every $\alpha \in \Delta$. Taking into account the action of E on S', one thereby obtains an action of E on G', compatible with the action of E on S' and respecting the canonical quasi-pinning of G'.[1399] Since $S' \to S$ is covering for the (fpqc) topology, it is an effective descent morphism for the fibered category of affine morphisms, and one writes

$$q\acute{e}p(S') = Q - \acute{E}p_{S'/S}(R)$$

for the quasi-pinned S-group obtained by Galois descent.[1400]

[1398]N.D.E.: We have sketched in 3.11.4 the verification of the fact that the functors i, `rev` and `qép` are fully faithful, and that the composite $rev \circ i \circ q\acute{e}p$ is isomorphic to the identity functor of `Rev`.

[1399]N.D.E.: We have expanded the original in the foregoing, to show that the action of E on G' is obtained by combining the natural action of E on $\acute{E}p_{S'}(R)$ and the given action on S'. The canonical Killing couple of G' is preserved by this action, as is the quasi-pinning given by the element $\tilde{X}'$ of $\Gamma(\Delta_{S'}, Lie(G'/S')_{\Delta_{S'}})$, equal to X'_α on the copy of S' indexed by α (indeed, since E acts on $\Delta_{S'}$ by permutation of the copies of S', one has indeed $h(\tilde{X}') = \tilde{X}'$ for every $h \in E$).

[1400]N.D.E.: see, for example, TDTE I, p. 22, Example 1. One can also describe $q\acute{e}p(S')$ as the twist of $G = \acute{E}p_S(R)$ by the E-torsor S'/S, i.e. the (fpqc) sheaf quotient of $S' \times_S G$ by the right action of E defined by $(s', g)\cdot h = (s'h, h^{-1}(g))$. Since G is affine over S and since E acts on G by group automorphisms, this sheaf is representable by an S-group $G^{\square}$, which is a "twisted" form of G, and $D^{\square} = Dyn(G^{\square})$ is the twist of the constant Dynkin scheme Δ_S by the torsor S'/S. Since E normalizes B and T, one obtains likewise a pair $(B^{\square}, T^{\square})$ which is a Killing couple of $G^{\square}$. On the other hand, let $g = Lie(G/S)$ and $g^{\square} = Lie(G^{\square}/S)$;

3.11.3. The functor rev. Let G be an S-reductive group of type R. One writes

$$rev(G) = Isomext(\acute{E}p_S(R), G);$$

this is a principal homogeneous bundle under E_S (cf. 1.10 and 1.19), that is, an object of Rev.

3.11.4. [1401] It is clear, by the isomorphism 3.10 (*), that the functor i is fully faithful. On the other hand, if G_0, G, G' are S-reductive groups, the natural morphism

$$Isom(G_0, G) \times_S Isom(G, G') \to Isom(G_0, G')$$

induces a morphism

$$(\star) \quad Isomext(G_0, G) \times_S Isomext(G, G') \to Isomext(G_0, G')$$

since, for every $S' \to S$, if $g_0 \in G_0(S')$, $g \in G(S')$, $u \in Isom(G_0, G)(S')$, and $v \in Isom(G, G')(S')$, one has the equality $v \circ int(g) \circ u \circ int(g_0) = v \circ u \circ int(u^{-1}(g)\, g_0)$. Taking $G_0 = \acute{E}p_S(R)$, one obtains a map

$$Isomext(G, G')(S) \to Hom_{Rev}(Isomext(\acute{E}p_S(R), G), Isomext(\acute{E}p_S(R), G'))$$

and to show that this is a bijection, one may assume by (fpqc) descent that G and G' are pinned, in which case it is evident.

Similarly, if S_1, S_2 are two objects of Rev and if one writes $S_i \times^E \acute{E}p_S(R)$ for the S-group $q\acute{e}p(S_i)$ obtained by twisting $\acute{E}p_S(R)$ by the E-torsor S_i (cf. N.D.E. (21)), one has a natural map

$$Hom_{Rev}(S_1, S_2) \to Hom_{Q\acute{e}p}(S_1 \times^E \acute{E}p_S(R), S_2 \times^E \acute{E}p_S(R)),$$

and to show that this is a bijection, one may assume by (fpqc) descent that S_1 and S_2 are trivial E-bundles, in which case it is evident.

Thus the three functors of the diagram are fully faithful. Moreover, for every object S' of Rev, one has a natural morphism

$$S' \to Isomext(\acute{E}p_S(R), S' \times^E \acute{E}p_S(R))$$

and to see that this is an isomorphism, one may assume by (fpqc) descent that S' is a trivial E-bundle, in which case it is evident.

One thus has an isomorphism of functors $Id_{Rev} \to rev \circ i \circ q\acute{e}p$. Since the functor rev is fully faithful one obtains therefore, for every S-reductive group H (not necessarily quasi-split), a bijection

---

for every $U \to S$, the sections of $g^{\square}$ on U are the E-equivariant S-morphisms $U \times_S S' \to W(g)$. Since $D^{\square} \times_S S'$ is E-isomorphic to $\Delta \times S'$, equipped with the action $(\alpha, s') \cdot h = (h^{-1}(\alpha), s'h)$, one obtains that the morphism given by $(\alpha, s') \mapsto X_\alpha$ is a section of $g^{\square}$ on $D^{\square}$ which is a quasi-pinning, i.e. a section nowhere zero of $(g^{\square} \otimes_{O_S} O_{D^{\square}})^{D^{\square}}$ (cf. 3.8).

[1401] N.D.E.: We have added this number.

$$Isomext(Q\text{-}\acute{E}p_{rev(H)/S}(R), H)(S) \simeq Aut_{Rev}(rev(H))$$

functorial in H. In particular, the identity morphism of $rev(H)$ corresponds to a "canonical" element u_0 of $Isomext(Q - \acute{E}p_{rev(H)/S}(R), H)(S)$; moreover, every $u \in Isomext(Q - \acute{E}p_{rev(H)/S}(R), H)(S)$ corresponds to an automorphism ϕ_u of $rev(H)$, and $q\acute{e}p(\phi_u)$ is the unique automorphism of pinned S-group of $Q - \acute{E}p_{rev(H)/S}(R)$ such that $u_0 \circ q\acute{e}p(\phi_u) = u$. We have thus proved theorem 3.11.

Let us develop one of the corollaries of 3.11:

Corollary 3.12. *For every S-reductive group G, there exists a quasi-pinned S-group $G_{q-\acute{e}p.}$ and an "outer isomorphism" $u \in Isomext(G_{q-\acute{e}p.}, G)(S)$. The pair $(G_{q-\acute{e}p.}, u)$ is unique up to a unique isomorphism.*

Indeed, one may assume G of constant type R, and one takes $G_{q-\acute{e}p.} = Q - \acute{E}p_{rev(G)/S}(R)$.

3.12.1. Note that the datum of the pair $(G_{q-\acute{e}p.}, u)$ allows one to define canonically the S-scheme (cf. 1.11)

$$Q = Isomint(G_{q-\acute{e}p.}, G), [^{N}.D.E - XXIV - 23]$$

which is a principal homogeneous bundle under $ad(G_{q-\acute{e}p.})$, and whose datum "is equivalent" to that of the inner twisted form G of $G_{q-\acute{e}p.}$. Moreover, Q is none other than the "scheme of quasi-pinnings of G", a definition which accounts for its structure of principal homogeneous bundle (on the left) under $ad(G)$ (3.10) — compare with 1.20.

Proposition 3.13. *Let S be a scheme, G a semisimple adjoint (resp. simply connected) S-group, $B \supset T$ a Killing couple of G, $Dyn(G)$ the Dynkin S-scheme*

of G. There exists a canonical isomorphism of S-group schemes

$$T \simeq \prod_{Dyn(G)/S} G_{m, Dyn(G)}.$$

(Recall (cf. Exp. II, § 1) that the right-hand side is by definition the S-functor which to $S' \to S$ associates $G_m(Dyn(G) \times_S S')$, or, what amounts to the same, $G_m(Dyn(G_{S'}))$.)

First proof. Let us do it for simplicity in the adjoint case. Consider the composite morphism

$$T \to \prod_{D/S} T_D \to \prod_{D/S} G_{m, D},$$

where the first morphism is the canonical morphism, the second is $\prod_{D/S} \alpha_D$ (we have written $D = Dyn(G)$). To verify that this morphism is an isomorphism, one may assume G split; now, in this case, this is none other than the morphism $T \to (G_{m,S})^{\Delta}$ with components α, for $\alpha \in \Delta$, and the latter is an isomorphism (Exp. XXII, 4.3.8).

Second proof. By 2.8, 3.5 and 3.11, one may assume that $G = Q - \acute{E}p_{S'/S}(R)$, T being the canonical maximal torus. On S', one has by Exp. XXII 4.3.8, an isomorphism

$T_{S'} \xrightarrow{\sim} (G_{m,S'})^{\Delta}$, defined by the simple roots (resp. the simple coroots). The group E operates on the right-hand side by permutation of Δ. Now the bundle associated to S'/S by $E \to \mathrm{Aut}(\Delta)$ is $Dyn(G)$ (3.7), and one concludes at once.

Using Prop. 8.4 of the appendix, one deduces:

Corollary 3.14. *Under the preceding conditions, one has*

$H^1(S, T) \xrightarrow{\sim} H^1(Dyn(G), G_m) \xrightarrow{\sim} \mathrm{Pic}(Dyn(G))$.

In particular, $H^1(S,T) = 0$ when S is semi-local.

Remark 3.15. Let S be a scheme, G (resp. G') an S-reductive group, $B \supset T$ (resp. $B' \supset T'$) a Killing couple of G (resp. G'), $u \in Isomext(G,G')(S)$. Set (cf. 2.10)

$P = Isomint_u(G, B, T; G', B', T')$;

this is a principal homogeneous bundle under T_{ad} (by $(f,t) \mapsto f \circ int(t)$).

Let on the other hand $D = Dyn(G) = Dyn(G')$ (identified via u (3.5)), and let $L = Isom_D(g_D, g'_D)$ be the principal homogeneous bundle under $G_{m,D}$ defined by the invertible O_D-module

$Hom_{O_D}(g_D, g'_D) = g'_D \otimes (g_D)^\vee$.

Each $f \in P(S')$ defines an isomorphism of $g_D \otimes_{O_S} O_{S'}$ onto $g'_D \otimes_{O_S} O_{S'}$, whence a canonical morphism

$$P \to \prod_{D/S} L.$$

This morphism is an isomorphism, compatible with the isomorphism of operator groups

$T_{ad} \xrightarrow{\sim} \prod_{D/S} G_{m,D}$

defined above. Indeed, it suffices to verify it in the case where the two groups are pinned, where it is easy.

It follows in particular that in the isomorphism

$$H^1(S,T_{ad}) \xrightarrow{\sim} \mathrm{Pic}(Dyn(G))$$

of 3.14, the class of the bundle P is transformed into $c\ell(g'^D) - c\ell(g_D)$. The bundle P is therefore trivial if and only if g'^D and g_D are isomorphic.

If (G,B,T) is quasi-splittable, for example if one takes for G the group $G'_{q-ép.}$, with its canonical Killing couple, it follows that the image of the class of P is none other than the obstruction to the quasi-splitting of G' defined in 3.9.

3.16. Symmetries

3.16.1. Let S be a scheme, G an S-reductive group, $B \supset T$ a Killing couple of G. Write $D = Dyn(G)$. Recall (Exp. XXII, 5.9.1) that there exists a unique Borel subgroup B^- of G such that $B \cap B^- = T$. If $X \in \Gamma(D, g_D)^\times$ defines a quasi-pinning of G relative to (B,T) (3.9), then $Y = -X^{-1} \in \Gamma(D, (g_D)^\vee)^\times$ defines a quasi-pinning of G relative to (B^-, T); one says that this is the *opposite quasi-pinning*.

If R is a pinned reduced root datum and if E is an R-pinning of the S-reductive group G, one defines an R-pinning E^- said to be *opposite to* E in the following way: one keeps the same maximal torus T, one takes the opposite of the isomorphism $D_S(M) \xrightarrow{\sim} T$, and one "pins" by $Y_\alpha = -X_\alpha^{-1} \in \Gamma(S, g^{-\alpha})^\times$, for $\alpha \in \Delta$. The quasi-pinning underlying E^- is the quasi-pinning opposite to the quasi-pinning underlying E.

Remark. In the notation of Exp. XIX, 3.1, if one sets

```
w_α(X_α) = exp(X_α) exp(−X_α^{-1}) exp(X_α),
```

one has $w_\alpha(X_\alpha) = w_{-\alpha}(Y_\alpha)$ (loc. cit., 3.1 (vi)).

Proposition 3.16.2. *Let S be a scheme, G an S-reductive group.*

(i) Let T be a maximal torus of G; there exists a unique

```
i_T ∈ (Aut_{S-gr.}(G, T) / T_ad)(S) ⊂ Aut_{S-gr.}(T)
```

such that $i_T(t) = t^{-1}$ for every section t of T.

(ii) Let (B,T) be a Killing couple of G; there exists a unique section

```
w_{B, T} ∈ (Norm_G(T) / T)(S) = W_G(T)(S)
```

such that $int(w_{B,T})(B) = B^-$ (with the obvious abuse of language).

(iii) Let $Q = (B,T,X)$ be a quasi-pinning of G, $Q^- = (B^-,T,Y)$ the opposite quasi-pinning; there exists a unique inner automorphism of G

$$n_Q \in Norm_{ad(G)}(T_{ad})(S) \subset \mathrm{Aut}_{S-gr.}(G)$$

such that $n_Q(Q) = Q^-$, that is, $n_Q(T) = T$, $n_Q(B) = B^-$, $n_Q(X) = Y$.

(iv) Let (R,E) be a pinning of G, (R,E^-) the opposite pinning; there exists a unique automorphism of G

```
u_E ∈ Aut_{S-gr.}(G, T) ⊂ Aut_{S-gr.}(G)
```

such that $u_E(E) = E^-$, i.e. $u_E(t) = t^{-1}$ for every section t of T, and $Ad(u_E)X_\alpha = Y_\alpha$ for every $\alpha \in \Delta$.

Proof. (ii) follows from Exp. XXII, 5.5.5 (ii), then (iii) follows from 3.10, and (iv) follows from Exp. XXIII, 4.1. Finally to prove (i), one may assume G pinned. Existence follows from (iv) for example, uniqueness from the fact that an automorphism of G inducing the identity on T is given by a section of T_{ad} (2.11).

Corollary 3.16.3. *One has*

$$i_T^2 = w_{B,T}^2 = n_Q^2 = u_E^2 = e.$$

Moreover, i_T (resp. u_E) is $\neq e$ if $G \neq e$, and $w_{B,T}$ (resp. n_Q) is $\neq e$ if G is not a torus.

Corollary 3.16.4. *In the situation of (iii) (resp. (iv)), n_Q projects onto $w_{B,T}$ (resp. u_E projects onto i_T) by the canonical morphism*

$$Norm_{ad(G)}(T_{ad}) \to W_{ad(G)}(T_{ad}) \simeq W_G(T),$$

resp.

$$\mathrm{Aut}_{S\text{-}gr.}(G, T) \to \mathrm{Aut}_{S\text{-}gr.}(G, T) / T_{ad}.$$

Corollary 3.16.5. *The preceding definitions are compatible with base change, and are functorial under isomorphism (in an obvious sense).*

Proposition 3.16.6. *(i) One can define uniquely for each reductive group G over a scheme S an element*

$$s_G \in Autext(G)(S)$$

in such a way that this construction is functorial in G under isomorphism, is compatible with base change, and is such that whenever T is a maximal torus of the S-reductive group G, s_G is the image of i_T under the canonical morphism

$$\mathrm{Aut}_{S\text{-}gr.}(G, T) / T_{ad} \to Autext(G).$$

(ii) One has $s_G^2 = e$, and s_G is a central element of $Autext(G)$.

(iii) Under the conditions of 3.16.2 (ii), if one identifies $\mathrm{Aut}_{S-gr.}(G, B, T)/T_{ad}$ with $Autext(G)$ (2.2), one has

$$w_{B,T}\, i_T = i_T\, w_{B,T} = s_G.$$

(iv) Under the conditions of 3.16.2 (iv), if one identifies $\mathrm{Aut}_{S-gr.-ép}(G, R, E)$ with $Autext(G)$ (1.3 (iii)), one has

$$n_E\, u_E = u_E\, n_E = s_G.$$

Proof. (i) is proved without difficulty by descent. On the other hand, since i_T is evidently a central section of square e in $\mathrm{Aut}_{S-gr.}(T)$, (ii) follows immediately; (iii) is a consequence of (iv) by descent. Finally, under the conditions of (iv), it is clear that $n_E u_E = u_E n_E$ and that this automorphism of G respects the pinning; modulo the identification made, it is therefore equal to its image in $Autext(G)$; but n_E is inner and u_E projects onto s_G.

Remark 3.16.7. (i) One determines s_G explicitly in each case of the classification thanks to (iii): for each irreducible pinned root datum it suffices to compose the symmetry through the origin with the symmetry in the Weyl group (i.e. the element w_0 of the Weyl group such that $w_0(\Delta) = -\Delta$). One finds the following results: one has $s_G = e$ except for A_n ($n \geqslant 2$), D_n (n odd) and E_6, in which case s_G is the unique non-trivial "outer automorphism".

(ii) The automorphism u_E is the one used to construct "the compact real forms" in the theory of semisimple Lie algebras.

Remark 3.16.8. We defined in 3.16.1 an involution in the S-scheme $Q = Isomint(G_{q-ép.}, G)$ of quasi-pinnings of G (cf. 3.12.1); by transport of structure from $G_{q-ép.}$ to G, one sees at once that this involution is given by the action of an element of $ad(G_{q-ép.})(S)$: the element n_0 defined (3.16.2 (iii)) by the canonical quasi-pinning of $G_{q-ép.}$.

In the same way, one has defined an involution in the S-scheme

$$P = Isom_{S-gr.}(\acute{E}p_S(R), G)$$

of R-pinnings of G (cf. 1.20). Arguing as before, one sees that this involution is given by the action of the automorphism u_0 of $\acute{E}p_S(R)$ defined (3.16.2 (iv)) by the canonical pinning of $\acute{E}p_S(R)$.

4. Isotriviality of reductive groups and of principal bundles under reductive groups

4.1. Definitions. Isotriviality theorem

Definition 4.1.1. *Let S be a scheme, G an S-group scheme, P a principal homogeneous bundle under G. One says that P is* locally isotrivial *(resp.* semi-locally isotrivial*) if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open set) there exist an open set U of S containing s (resp. F) and a finite surjective étale morphism $S' \to U$ such that $P_{S'}$ is trivial.*

Definition 4.1.2. *Let S be a scheme, G an S-reductive group. One says that G is* locally isotrivial *(resp.* semi-locally isotrivial*) if for every point $s \in S$ (resp. every finite set F of points of S contained in an affine open set) there exist an open set U of S containing s (resp. F) and a finite surjective étale morphism $S' \to U$ such that $G_{S'}$ is splittable.*

Remark 4.1.3. (i) The equivalence of categories 4.1.1 of 1.17 respects by definition local (resp. semi-local) isotriviality.

(ii) Add to the conditions of 4.1.1: G locally of finite presentation over S. Then the principal bundle P (or the reductive group G) is locally isotrivial (resp. semi-locally isotrivial) if and only if for every $S' \to S$, S' local (resp. semi-local), $P_{S'}$ is isotrivial (or $G_{S'}$ isotrivial), that is to say, if there exists $S'' \to S'$ finite surjective étale such that $P_{S''}$ is trivial (or $G_{S''}$ split).

(iii) In the case of tori, definition 4.1.2 coincides with that of Exp. IX, 1.1.

4.1.4. Recall (Exp. XXII, 4.3 and 6.2) that for every reductive group G, we have introduced the groups $rad(G)$, $corad(G)$ and $dér(G)$. The groups $rad(G)$ and $corad(G)$ are tori, which are split when G is split; moreover, there exists an isogeny $rad(G) \to corad(G)$. The S-group $dér(G)$ is semisimple, one has $G/dér(G) = corad(G)$; it follows that for every principal homogeneous bundle P under G, $P/dér(G)$ is a principal homogeneous bundle under $corad(G)$. This said, one has:

Theorem 4.1.5. *Let S be a scheme, G an S-reductive group of constant type.*

(i) The following conditions are equivalent:

(a) G is locally (resp. semi-locally) isotrivial.

(b) The torus $rad(G)$ is.

(b') The torus $corad(G)$ is.

(c) The Galois covering of S associated to G (3.11) is.

If T is a maximal torus of G, these conditions are also equivalent to

(d) The torus T is locally (resp. semi-locally) isotrivial.

(ii) Let P be a principal homogeneous bundle under G; for P to be locally (resp. semi-locally) isotrivial, it is necessary and sufficient that the $corad(G)$-principal bundle $P/dér(G)$ be so.

Corollary 4.1.6. *The conditions of (i) are satisfied when G is semisimple or when S is locally noetherian and normal (or more generally geometrically unibranch). The conditions of (ii) are satisfied when G is locally (resp. semi-locally) isotrivial.*

For (i), the first assertion is trivial from (b), the second follows from (c) and Exp. X, 5.14 and 5.15. For (ii), it suffices to note that by virtue of theorem 90, a principal bundle under a split torus is semi-locally isotrivial.

4.2. Proof: the semisimple case

Let us first prove, for later reference:

Proposition 4.2.1. *Let S be a scheme, G an S-reductive group, T a maximal torus of G (resp. B a Borel subgroup, resp. $B \supset T$ a Killing couple of G), P a principal homogeneous bundle under G, G' the twisted form of G associated to P (via the morphism $int : G \to \mathrm{Aut}_{S-gr.}(G)$). One has canonical isomorphisms*

$$P / \mathrm{Norm}_G(T) \xrightarrow{\sim} \mathrm{Tor}(G'), \quad P / B \xrightarrow{\sim} \mathrm{Bor}(G'), \quad P / T \xrightarrow{\sim} \mathrm{Kil}(G').$$

By construction, G' is the quotient of $P \times_S G$ by a certain action of G ($((p, g'))g = (pg, g^{-1}g'g)$); one therefore has a morphism $P \times_S G \to G'$, that is, a morphism

$$P \to \mathrm{Hom}_S(G, G'),$$

which, as one sees at once, factors through a morphism `f : P → Isom_{S-gr.}(G, G'),`

(to verify this assertion, one may assume $G = P$, in which case one has $G' = G$, $f = int$). The set-theoretic map $p \mapsto f(p)(T)$ defines a morphism

$$P \to Tor(G').$$

To verify that this morphism induces an isomorphism $P/Norm_G(T) \xrightarrow{\sim} Tor(G')$ as announced, one may again assume $P = G$ in which case one is reduced to Exp. XXII, 5.8.3 (iii). (In fact, *loc. cit.* should be replaced by the statement above.) One argues similarly for `Bor` and `Kil`.

Proposition 4.2.2. *Let S be a semi-local scheme, G an S-semisimple group of constant type.*

(i) G is isotrivial.

(ii) Every principal bundle under G is isotrivial.

Let us prove (i). Up to making a finite surjective étale extension of the base, one may, by 3.9.2, assume G quasi-split. But then G is isotrivial by construction (3.11, the group E being finite). To prove (ii), one may, by (i), assume G split; one is then reduced to:

Lemma 4.2.3. *Let S be a semi-local scheme. Every principal bundle under a split reductive group is isotrivial.*

Indeed, with the notation of 4.2.1, where $B \supset T$ denotes the canonical Killing couple of the split G, the S-scheme $Kil(G')$ possesses a section, after finite surjective étale extension of the base, by 3.9.2. One may therefore reduce the structure group of G to T; but T is split, so $H^1(S,T) = 0$ (theorem 90).

Corollary 4.2.4. *Let S be a scheme and*

$$1 \to G \to G' \to G'' \to 1$$

an exact sequence of S-group schemes (for (fpqc)), G being semisimple of constant type. Let P be a principal homogeneous bundle under G', suppose the sheaf[1402] *P/G associated representable (for example `G''` affine over S). For P to be locally isotrivial (resp. semi-locally isotrivial), it is necessary and sufficient that P/G be so (as bundle under `G''`).*

If P is trivial, P/G is too, which shows that the condition is necessary. Conversely, suppose S local (resp. semi-local) and P/G isotrivial, so trivialized by a finite surjective étale extension S' of S. Extending the base to S', one may reduce the structure group of $P_{S'}$ to $G_{S'}$. But S' is still semi-local and $G_{S'}$ semisimple of constant type, so the obtained bundle is isotrivial (4.2.2).

[1402] N.D.E.: We have replaced "bundle" by "sheaf", and then G' by `G''`.

4.3. Proof: general case.

Let us first note that 4.1.5 (ii) follows at once from the application of 4.2.4 to the exact sequence

$$1 \to d\acute{e}r(G) \to G \to corad(G) \to 1.$$

Let us therefore prove (i). If G is split, $rad(G)$ and $corad(G)$ are split, as is $rev(G)$; so (a) implies (b), (b'), and (c).

4.3.1. One has (c) ⇒ (a). Let R be the type of G; one has an exact sequence

$$1 \to ad(\acute{E}p_S(R)) \to A_S(R) \to E_S \to 1.$$

Applying 4.2.4 to the bundle $P = Isom_{S-gr.}(\acute{E}p_S(R), G)$ and to the associated bundle $rev(G) = P/ad(\acute{E}p_S(R))$, one has (c) ⇒ (a).

4.3.2. One has (b) ⇒ (a). It suffices to prove that if $rad(G)$ is split, G is semi-locally isotrivial. Now let $G_0 = \acute{E}p_S(R)$; consider the category of pairs `(G', f)`, where G' is a form of `G_0` and f is an isomorphism of $rad(G_0)$ onto $rad(G)$.

One knows (2.16) that this category is equivalent to the category of principal homogeneous bundles under a certain S-group H extension of a finite twisted constant group by a semisimple group. It suffices to prove that every principal bundle under H is semi-locally isotrivial; this follows at once from 4.2.4.

4.3.3. One has (b') ⇒ (a). One can argue as previously (it will moreover be the same group H that arises). One can also see that (b) and (b') are equivalent: a torus isogenous to a locally split torus is also locally split (cf. Exp. IX, 2.11 (iii)).

4.3.4. One has (d) ⇒ (a). It suffices to prove that a group of constant type possessing a split maximal torus is semi-locally isotrivial; this follows at once from 2.14.

4.3.5. One has (a) ⇒ (d). It suffices to prove that a maximal torus of a split group is semi-locally isotrivial. More precisely:

Lemma 4.3.6. *Let S be a semi-local scheme, G an S-reductive group, `T_0` a split maximal torus of G, $W_0 = Norm_G(T_0)/T_0$ (this is a locally constant S-group, constant if G is of constant type, by 2.14), T a maximal torus of G.*

There exists a morphism $S' \to S$ which is principal homogeneous under `W_0` (hence étale finite and surjective, and even principal Galois if G is of constant type) such that $T_{S'}$ is conjugate to $(T_0)_{S'}$ by an element of $G(S')$ (and so in particular split).

Indeed, one knows that $Transp_G(T_0, T)$ is a principal homogeneous bundle under $Norm_G(T_0)$ (cf. for example Exp. XI, 5.4 bis). Set $S' = Transp_G(T_0, T)/T_0$; this is a principal homogeneous bundle under W_0. Extending the base from S to S', one can reduce the structure group of $Transp_G(T_0, T)$ to T_0. Now S' is semi-local and T_0 split, so $Transp_G(T_0, T)$ possesses a section over S'.[1403] QED

4.4. Use of the existence of maximal tori

Using Grothendieck's existence theorem for maximal tori (Exp. XIV, 3.20), one can considerably sharpen the preceding results. Let us state at once:

Theorem 4.4.1. *Let S be a semi-local scheme, R a pinned reduced root datum, W its Weyl group, E the group of its automorphisms. (Recall that E operates naturally on W and that the semi-direct product $A = W \cdot E$ is identified with the group of automorphisms of R non-pinned, cf. Exp. XXI, 6.7.2).*

(i) Every principal homogeneous bundle under $\acute{E}p_S(R)$ is trivialized by a principal Galois covering $S' \to S$ of group W.

(ii) Let G be an S-reductive group of type R; let $rev(G) = Isomext(\acute{E}p_S(R), G)$ be the associated Galois covering of S with group E. Let W_0 be the form of W_S associated to $rev(G)$. There exists a morphism $S' \to S$, which is a principal homogeneous bundle under W_0, such that $G_{S'}$ is quasi-splittable (i.e. possesses a Borel subgroup).

(iii) Every S-reductive group of type R is split by a principal Galois covering $\tilde{S} \to S$ with group A.

Let us first state:

Proposition 4.4.2. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, P a principal homogeneous bundle under G, G' the twisted form of G associated to P (one then has, cf. 4.2.1, a canonical isomorphism $P/Norm_G(T) \xrightarrow{\sim} Tor(G')$). Let T' be a maximal torus of G', defining a composite morphism*

$$S \to Tor(G') \simeq P / Norm_G(T).$$

Consider the canonical morphisms

$$P \to P/T \to P / Norm_G(T)$$

and take their inverse images by the preceding morphism:

$$\begin{tikzcd} P \arrow[r] & P/T \arrow[r] & P / Norm_G(T) \\ H \arrow[r] \arrow[u] & S' \arrow[r] \arrow[u] & S. \arrow[u] \end{tikzcd}$$

Then S' (resp. H) is a principal homogeneous bundle over S (resp. S') under $W_G(T)$ (resp. $T_{S'}$). Moreover, if one makes $W_G(T)$ operate on T in the obvious way, the bundle associated to S' is isomorphic to T'.

The first two assertions are trivial, the last is proved like the corresponding assertion of 2.6.

[1403] N.D.E.: by "theorem 90".

Corollary 4.4.3. *Let S be a semi-local scheme, G an S-reductive group, T a maximal torus of G. Suppose one of the two following conditions is satisfied:*

(i) T is split.

(ii) T is contained in a Borel subgroup of G, and G is either adjoint, or simply connected.

Let moreover P be a principal homogeneous bundle under G. There exists an S-scheme S', which is a principal homogeneous bundle under $W_G(T)$, such that $P_{S'}$ is trivial.

Indeed, if G' is the form of G associated to P, then G' possesses a maximal torus T' (Exp. XIV, 3.20). Resuming the notation of the preceding proposition, one sees that $H^1(S', T_{S'}) = 0$ (by theorem 90 for (i), by 3.14 for (ii)). The morphism $H \to S'$ possesses a section, so $P_{S'}$ also possesses a section over S'. QED

Let us now prove the theorem. Assertion (i) is a particular case of the preceding corollary (take $G = \acute{E}p_S(R)$, endowed with its canonical split torus). Let us prove (ii). One knows (3.12) that G is an inner twisted form of

$$G_0 = Q - \acute{E}p_{rev(G)/S}(R).$$

If `T_0` is the canonical maximal torus of `G_0`, $W_{G_0}(T_0)$ is the group `W_0` described in the statement. The form G of `G_0` corresponds to a principal homogeneous bundle P under $ad(G_0)$ ($P = Isomint(G_0, G)$). The group $W_{ad(G_0)}(T_0^{ad})$ is canonically isomorphic to `W_0`, and one obtains the wanted result by applying 4.4.3 to the situation $(ad(G_0), T_0^{ad}, P)$, hypothesis (ii) of 4.4.3 being well verified. Let us finally prove (iii). Resume the notation of (ii); one has a diagram

```
      rev(G)
     ↗      ↘
  E_S        W_0
     ↘      ↗
       S'
```

One knows that $G_{S'}$ is isomorphic to $(G_0)_{S'}$ and that $(G_0)_{rev(G)}$ is splittable. If one sets $\tilde{S} = S' \times_S rev(G)$, $G_{\tilde{S}}$ is splittable, and it only remains to verify that $\tilde{S}$ is indeed a principal Galois covering of S with group A, which follows from the following more general lemma (naturally valid in any site):

Lemma 4.4.4. *Let S be a scheme, G and H two S-group schemes, $G \to \mathrm{Aut}_{S-gr.}(H)$ an action of G on H, E a G-principal homogeneous bundle, F an `H_0`-principal homogeneous bundle, where `H_0` is the form of H associated to E. Then $E \times_S F$ is endowed naturally with a structure of principal homogeneous bundle under the semi-direct product $H \cdot G$.*

Write $(e, g) \mapsto eg$ (resp. $(f, u) \mapsto fu$) for the action (on the right) of G on E (resp. of `H_0` on F). Write

$$p : E \times_S H \to H_0$$

for the canonical projection (H_0 is by definition the quotient of $E \times_S H$ by G acting on it by the formula $(e,h)g = (eg, g^{-1}hg)$). Consider the morphism

$$r : E \times_S F \times_S H \times_S G \to E \times_S F$$

defined set-theoretically by

$$r(e, f, h, g) = (eg, f \cdot p(e, h)).$$

The morphism r indeed defines an action of the semi-direct product $H \cdot G$ on $E \times_S F$. Indeed, one has set-theoretically

$$r(r(e, f, h, g), h', g') = (egg', f\, p(e, h)\, p(eg, h'));$$

but

$$p(e, h)\, p(eg, h') = p(e, h)\, p(e, g h' g^{-1}) = p(e, h g h' g^{-1}),$$

whence

$$r(r(e, f, h, g), h', g') = r(e, f, h g h' g^{-1}, g g'),$$

which had to be proved.

To prove now that this law is indeed a law of principal homogeneous bundle, one may assume that E and F are trivial, in which case one sees at once that $E \times_S F$ is also a trivial bundle.

4.5. Isotriviality of maximal tori of semisimple groups[1404]

Let us bring out here the following result, contained implicitly in 2.6 (cf. the N.D.E. (13)) and which was mentioned at the end of Exp. IX 1.2. Recall (XXIII 5.11) that, for every scheme S and every reduced root datum R, one writes $\acute{E}p_S(R)$ for the unique pinned S-group of type R (which exists by Exp. XXV) and $T_S(R)$ its canonical maximal torus.

Proposition 4.5.1. *Let G be an S-semisimple group of constant type R; assume that G possesses a maximal torus T (which is the case if S is semi-local, by XIV 3.20).*

(i) Then $P = Isom_{S-gr.}(\acute{E}p_S(R), T_S(R); G, T)/T_S(R)_{ad}$ is a principal S-bundle under the finite group $\mathrm{Aut}(R)$*, and T_P is a split maximal torus of G_P. Consequently, T is isotrivial.*

(ii) Moreover, for every finite set F of points of P contained in an affine open set, there exists an open set U of P containing F such that G_U is split.

Proof. The exact sequence

$$1 \to T_S(R)_{ad} \to \mathrm{Aut}_{S\text{-gr.}}(\acute{E}p_S(R), T_S(R)) \to \mathrm{Aut}_{S\text{-gr.}}(T_S(R))$$

[1404] N.D.E.: We have added this no 4.5.

induces for every $S' \to S$ a morphism of bundles $P_{S'} \to Isom_{S'-gr.}(T_{S'}(R), T_{S'})$, so if $P_{S'}$ possesses a section, then $T_{S'}(R)$ and $T_{S'}$ are isomorphic. This is in particular the case for $S' = P$.

Note moreover that $\mathrm{Aut}_{S-gr.}(\acute{E}p_S(R), T_S(R))/T_S(R)_{ad}$ is none other than the constant S-group $\mathrm{Aut}(R)_S$, and that P is identified with the $\mathrm{Aut}(R)_S$-torsor $Isom(R, \tilde{R}(G,T))$, where $\tilde{R}(G,T)$ denotes the twisted root datum associated to (G,T) (cf. XXII 1.10).

Then (ii) follows from (i) and from 2.14, taking into account the fact that the g_α (α ranging over the roots of R) are free $\mathcal{O}_U$-modules of rank 1 on a sufficiently small affine open set U containing F.

5. Canonical decomposition of an adjoint or simply connected group

In this section, we shall use the results of no 1 to generalize to the case of schemes a classical decomposition of adjoint (resp. simply connected) groups. So as not to overburden the exposition indefinitely, the proofs are sketched and the details are left to the reader; in fact it is always a matter of absolutely standard proofs in the theory of principal bundles: reduction of the structure group, twisting, etc.

5.1.

Recall (Exp. XXI, 7.4) that a Dynkin diagram is a disjoint union of its connected components, which are Dynkin diagrams. Moreover, every non-empty connected Dynkin diagram corresponding to a root datum is isomorphic to one of the standard diagrams (A_n, B_n, ..., G_2) which were exhibited in Exp. XXI, 7.4.6. In the sequel, we shall be interested only in Dynkin diagrams whose connected components are of one of the preceding types. Let T be the set of these standard diagrams. For every Dynkin diagram D, let $n(t) = n_D(t)$ be the number of connected components of D isomorphic to t, where $t \in T$. The *type* of D is by definition $\sum_{t \in T} n_D(t) \cdot t$.

A Dynkin diagram of type t is said to be *simple of type t*, a Dynkin diagram of type $n \cdot t$ is said to be *isotypic of type t*. Let D_0 be the set of connected components of D and let

$$a : D_0 \to T$$

be the obvious map. The type of D is none other than $\sum_{x \in D_0} a(x)$.

5.2.

Let S be a scheme, D an S-Dynkin scheme (verifying the restrictive condition stated above). The cokernel of the pair of morphisms $L \rightrightarrows D$ ($L =$[1405] scheme of bonds of D) is written D_0. This is the "scheme of connected components" of D (it

[1405]N.D.E.: We have replaced R by L, as in 3.2.

exists trivially when D is constant; the general case is deduced from this by descent); it is a finite twisted constant S-scheme. One has a canonical morphism

$$a : D_0 \to T_S.$$

For $t \in T$, set $a^{-1}(t) = D_{0,t}$; this is a subscheme of D_0, whose inverse image in D, written D_t, is the *isotypic component of type t* of the Dynkin scheme D. Each D_t is a subscheme of D, and one has

$$D = \coprod_{t \in T} D_t.$$

Note that the degree of $D_{0,t}$ at $s \in S$ is $n(s,t)$, if the type of D at s is $\sum_t n(s,t) \cdot t$.

5.3.

In what follows, we shall consider only semisimple adjoint (resp. simply connected) groups. To simplify the language, we shall state the results for adjoint groups; all statements will remain valid if one substitutes everywhere "simply connected" for "adjoint".

Recall that an adjoint reduced root datum is determined up to isomorphism by the type of its Dynkin diagrams. We shall therefore say that an adjoint root datum R (resp. a semisimple adjoint group G) is *of type* $\sum n(t) \cdot t$ if its Dynkin diagrams are (resp. if its type is given by an adjoint root datum of type $\sum n(t) \cdot t$). We shall say that R or G is *simple of type t* (resp. *isotypic of type t*) if its type is t (resp. $n \cdot t$, $n > 0$).

If G is a semisimple adjoint group, we shall use the symbols $Dyn_0(G)$ and $Dyn_{0,t}(G)$ in the sense defined in 5.2.

5.4.

Let t_i, $i = 1, 2, \ldots, n$, be distinct elements of T, and let R_i be a pinned adjoint root datum isotypic of type t_i. Consider the product $R = R_1 \times \cdots \times R_n$ (Exp. XXI, 6.4.1). Let S be a scheme such that the different $\acute{E}p_S(R_i)$ exist (cf. Exp. XXV). One verifies at once that there exists a canonical isomorphism

$$(*) \qquad \acute{E}p_S(R) = \acute{E}p_S(R_1) \times_S \cdots \times_S \acute{E}p_S(R_n).$$

Moreover, if E_i denotes the group of automorphisms of R_i, E the group of automorphisms of R, D_i (resp. D) the Dynkin diagram of R_i (resp. R), one has isomorphisms:

$$E_i \simeq \mathrm{Aut}(D_i), \qquad D = \coprod D_i, \qquad E \simeq \prod E_i \simeq \mathrm{Aut}(D).$$

Combining with (*) and 1.4, one sees that (*) induces an isomorphism

$$A_S(R) \simeq A_S(R_1) \times_S \cdots \times_S A_S(R_n).$$

Proposition 5.5. *Let S be a scheme, G an S-semisimple adjoint group. There exists a unique decomposition*

$$G \simeq \prod_{t \in T} G_t,$$

where G_t is an S-semisimple adjoint group isotypic of type t. Moreover, the preceding decomposition induces isomorphisms

$$\mathrm{Dyn}_t(G) \simeq \mathrm{Dyn}(G_t), \qquad \mathrm{Aut}_{S\text{-gr.}}(G) \simeq \prod_{t \in T} \mathrm{Aut}_{S\text{-gr.}}(G_t).$$

This has indeed been proved above when G is split. In the general case, one may assume G of constant type R. Using the preceding decomposition of $A_S(R)$ and 1.17, one deduces the wanted decomposition of G. The other results are then proved by descent.

Remark 5.6. More generally, if G and H are two semisimple adjoint groups, one has canonical isomorphisms as follows (the diagram is commutative):

$$\begin{tikzcd}
\mathrm{Isom}_{S\text{-gr.}}(G, H) \arrow[r, "\sim"] \arrow[d] & \prod_{t \in T} \mathrm{Isom}_{S\text{-gr.}}(G_t, H_t) \arrow[d] \\
\mathrm{Isomext}(G, H) \arrow[r, "\sim"] \arrow[d, "\wr"'] & \prod_{t \in T} \mathrm{Isomext}(G_t, H_t) \arrow[d, "\wr"] \\
\mathrm{Isom}_{\mathrm{Dyn}}(\mathrm{Dyn}(G), \mathrm{Dyn}(H)) \arrow[r, "\sim"] & \prod_{t \in T} \mathrm{Isom}_{\mathrm{Dyn}}(\mathrm{Dyn}(G_t), \mathrm{Dyn}(H_t)).
\end{tikzcd}$$

Remark 5.7. One can give an intrinsic characterization of G_t, which we state below without proof: it is the largest reductive subgroup of G invariant (and moreover characteristic) and isotypic of type t.

The preceding proposition allows one to reduce the study of semisimple adjoint groups to that of isotypic semisimple adjoint groups. It is this study that we shall now undertake.

5.8.

Let R be a pinned reduced adjoint simple root datum of type t, I a nonempty finite set, R^I the product root datum of copies R_i of R, for $i \in I$. Write E for the group of automorphisms of R, which is identified with the group of automorphisms of the Dynkin diagram D of R. The Dynkin diagram of R^I is identified with D^I, which shows that one has an exact sequence:

$$1 \to \mathrm{Aut}(D)^I \to \mathrm{Aut}(D^I) \to \mathrm{Aut}(I) \to 1,$$

where $\mathrm{Aut}(I)$ denotes the group of permutations of I. It follows that the canonical isomorphism

$$\acute{E}p_S(R)^I \simeq \acute{E}p_S(R^I)$$

induces an exact sequence

$$1 \to A_S(R)^I \to A_S(R^I) \to \mathrm{Aut}(I_S) \to 1,$$

the last group being the S-constant group associated to $\mathrm{Aut}(I)$. Note moreover that I is canonically identified with the set of connected components of D^I.

If G is an S-semisimple group of type R^I, defining (cf. 1.17) a principal homogeneous bundle P under $A_S(R^I)$, the definition of $Dyn(G)$ by descent (3.7), and that of $Dyn_0(G)$ (5.2), shows that the bundle associated to P by the morphism $A_S(R^I) \to \mathrm{Aut}(I_S)$ is none other than $Isom_S(I_S, Dyn_0(G))$, corresponding to the form $Dyn_0(G)$ of I_S. Using again the equivalence of categories 1.17 and the preceding exact sequence, one deduces by a formal argument that there exists a $Dyn_0(G)$-reductive group of type R, say G_0, and an S-isomorphism $G \simeq \prod_{Dyn_0(G)/S} G_0$.

Proposition 5.9. *Let S be a scheme, G an S-semisimple adjoint group isotypic of type t. There exist a $Dyn_0(G)$-semisimple adjoint group G_0 simple of type t, and an S-isomorphism (unique)*

$$G \simeq \prod_{Dyn_0(G)/S} G_0.$$

Moreover, this isomorphism induces an exact sequence

$$1 \to \prod_{Dyn_0(G)/S} \mathrm{Aut}_{S\text{-gr.}}(G_0) \to \mathrm{Aut}_{S\text{-gr.}}(G) \to \mathrm{Aut}_S(Dyn_0(G)) \to 1.$$

One may evidently assume G of constant type $n \cdot t$. The first assertion has already been proved (the uniqueness assertion is evident). The second is then deduced from the split case by descent.

One can combine 5.5 and 5.9:

Proposition 5.10. *Let S be a scheme, G an S-semisimple adjoint group, $D = Dyn(G)$ its Dynkin scheme.*

(i) There exists a canonical decomposition

$$G \simeq \prod_{t \in T} \prod_{D_{0,t}/S} G_{0,t} \simeq \prod_{Dyn_0(G)/S} G_0,$$

where each $G_{0,t}$ is a $Dyn_{0,t}(G)$-simple adjoint group of type t (resp. where G_0 is a $Dyn_0(G)$-semisimple adjoint group whose type at $x \in Dyn_0(G)$ is $a(x) \in T$ (the morphism $a : Dyn_0(G) \to T_S$ was defined in 5.2)).

(ii) The preceding decompositions induce isomorphic exact sequences (written vertically), where $\mathrm{Aut}_{S,a}(D_0)$ denotes the scheme of automorphisms of $Dyn_0(G)$ which commute with the morphism $a : Dyn_0(G) \to T_S$:

$$\begin{tikzcd}
1 \arrow[d] & 1 \arrow[d] \\
\prod_{t \in T} \prod_{D_{0,t}/S} \mathrm{Aut}_{\mathrm{gr.}}(G_{0,t}) \arrow[r, phantom, "\simeq"] \arrow[d] & \prod_{Dyn_0(G)/S} \mathrm{Aut}_{\mathrm{gr.}}(G_0) \arrow[d] \\
\mathrm{Aut}_{S\text{-gr.}}(G) \arrow[r] \arrow[d] & \mathrm{Aut}_{S\text{-gr.}}(G) \arrow[d] \\
\prod_{t \in T} \mathrm{Aut}_S(Dyn_{0,t}(G)) \arrow[r, phantom, "\simeq"] & \mathrm{Aut}_{S,a}(Dyn_0(G))
\end{tikzcd}$$

$$\begin{array}{ccc} \downarrow & \qquad\qquad & \downarrow \\ 1 & & 1 \end{array}$$

Corollary 5.11. *Under the preceding conditions, the three following categories are equivalent:*

(i) the category of principal homogeneous bundles under G.

(ii) the category of principal homogeneous bundles under G_0.

(iii) the product category, for $t \in T$, of the categories of principal homogeneous bundles under the $G_{0,t}$.

This is deduced formally from the preceding decompositions and from 8.4.

Corollary 5.12. *One has canonical isomorphisms*

$$\mathrm{Tor}(G) \simeq \prod_{t \in T} \prod_{D_{0,t}/S} \mathrm{Tor}(G_{0,t}) \simeq \prod_{Dyn_0(G)/S} \mathrm{Tor}(G_0),$$

and similarly replacing Tor by Bor (resp. Kil).

Trivial starting from the split case.

Remark 5.13. The canonical morphism

$$Dyn(G) \to Dyn_0(G)$$

allows one to consider $Dyn(G)$ as a $Dyn_0(G)$-Dynkin scheme; in fact it is the Dynkin scheme $Dyn(G_0)$ of G_0.

Similarly if $T \subset B$ is a Killing couple of G, corresponding canonically to the Killing couple $T_0 \subset B_0$ of G_0, one verifies that the obstructions to the quasi-splitting of G and G_0, which lie (3.9) in $\mathrm{Pic}(Dyn(G)) = \mathrm{Pic}(Dyn(G_0))$ coincide. One deduces:

Corollary 5.14. *The following conditions are equivalent:*

(i) G is quasi-splittable,

(ii) G_0 is quasi-splittable,

(iii) each $G_{0,t}$, $t \in T$, is quasi-splittable.

6. Automorphisms of Borel subgroups of reductive groups

Lemma 6.1. *Let S be a scheme, (G,T,M,R) a split S-group, Δ a system of simple roots of R, and B the corresponding Borel subgroup. Then B_u is generated as (fppf) group sheaf by the U_α, $\alpha \in \Delta$.*

Let H be the group subsheaf of B_u generated by the U_α, $\alpha \in \Delta$. Let us prove $H \supset U_\beta$ ($\beta \in R^+$) by induction on the integer $ord(\beta) = ord_\Delta(\beta) > 0$ (cf. Exp. XXI, 3.2.15). The assertion is verified by hypothesis for $ord(\beta) = 1$. Suppose then $ord(\beta) > 1$ and $U_\gamma \subset H$ as soon as $ord(\gamma) < ord(\beta)$. There exists $\alpha \in \Delta$ such that $\beta - \alpha \in R^+$

(Exp. XXI, 3.1.1). Let p be the largest integer such that $\beta - p\alpha = \beta' \in R^+$. One has $U_\alpha, U_{\beta'} \subset H$, $\beta' - \alpha \notin R$. One is therefore reduced to proving:

Lemma 6.2. *Resume the notation of Exp. XXIII, 6.4. Suppose* $p = 1$*, that is,* $\beta - \alpha$ *not a root. If* H *is a group subsheaf of* B_u *such that* $U_\alpha, U_\beta \subset H$*, then* $U_{i\alpha+j\beta} \subset H$ *whenever* $i\alpha + j\beta \in R$*,* $i > 0$*,* $j > 0$*.*

Let us distinguish four cases according to the value of $q = 0, 1, 2, 3$. In the sequel x and y denote two arbitrary sections of $G_{a,S'}$, $S' \to S$.

If $q = 0$, there is nothing to prove.

If $q = 1$, one has $p_{\alpha+\beta}(\pm x) = p_\beta(-y)p_\alpha(-x)p_\beta(y)p_\alpha(x) \in H(S')$, so $U_{\alpha+\beta} \subset H$.

If $q = 2$, one similarly has

$$p_{\alpha+\beta}(\pm xy)\, p_{2\alpha+\beta}(\pm x^2y) = p_\beta(-y)\, p_\alpha(-x)\, p_\beta(y)\, p_\alpha(x) \in H(S'),$$

whence, up to changing certain signs,

$$F(x, y) = p_{\alpha+\beta}(xy)\, p_{2\alpha+\beta}(x^2y) \in H(S').$$

Since $p_{\alpha+\beta}$ and $p_{2\alpha+\beta}$ commute, one then has $F(x,1)F(1,-x) = p_{2\alpha+\beta}(x^2 - x)$.[1406] If $a \in G_a(S)$, the equation $X^2 - X = a$ defines a free surjective extension of S (this is $\operatorname{Spec} O_S[X]/(X^2 - X - a)$); one therefore has $p_{2\alpha+\beta}(a) \in H(S')$ so $U_{2\alpha+\beta} \subset H$, hence also $U_{\alpha+\beta} \subset H$ (since $F(1,a) \in H(S')$).

If $q = 3$, one similarly has

$$F(x, y) = p_{\alpha+\beta}(xy)\, p_{2\alpha+\beta}(x^2y)\, p_{3\alpha+\beta}(x^3y)\, p_{3\alpha+2\beta}(x^3y^2) \in H(S');$$

and since

$$p_{3\alpha+2\beta}(\pm x) = F(1, 1)^{-1}\, p_\beta(-x)\, F(1, 1)\, p_\beta(x) \in H(S'),$$

one obtains that $U_{3\alpha+2\beta} \subset H$ and so

$$K(x, y) = p_{\alpha+\beta}(xy)\, p_{2\alpha+\beta}(x^2y)\, p_{3\alpha+\beta}(x^3y) \in H(S').$$

Computing then

$$K(x + y, 1)\, K(-x, 1)\, K(1, y)^{-1}$$

modulo $U_{3\alpha+2\beta}$, one finds

$$p_{2\alpha+\beta}(2x^2 + y^2 + 2xy - y)\, p_{3\alpha+\beta}(y^3 + 3xy^2 + 3x^2y - y) \in H(S').$$

If $a \in G_a(S)$, the "equations"

$$x^2 = -xy - y + 1 - a$$
$$y^2 = 3y - 2 + 3a$$

[1406] N.D.E.: We have corrected $F(-1,x)$ to $F(1,-x)$.

define a free surjective extension of S; one then has $y^3 + 3xy^2 + 3x^2y - y = 0$ and the preceding expression gives $p_{2\alpha+\beta}(a) \in H(S')$.[1407] One has thus proved that H contains $U_{2\alpha+\beta}$ and $U_{3\alpha+2\beta}$ and that

$$E(x, y) = p_{\alpha+\beta}(xy)\, p_{3\alpha+\beta}(x^3y) \in H(S').$$

Since $p_{\alpha+\beta}(xy)$ and $p_{3\alpha+\beta}$ commute, one is reduced to the preceding computation, i.e. $E(1,x)E(1,-x) = p_{3\alpha+\beta}(x^3 - x)$, whence $U_{3\alpha+\beta} \subset H$, then $U_{\alpha+\beta} \subset H$.

Remark 6.2.1. The preceding proof shows that one could have replaced the hypothesis "H contains U_α and U_β" by "H contains U_α or U_β, and H is invariant under $int(U_\alpha)$ and $int(U_\beta)$".

Theorem 6.3. *Let S be a scheme, G and G' two S-semisimple groups, B (resp. B') a Borel subgroup of G (resp. G'). Every isomorphism $B \xrightarrow{\sim} B'$ is induced by a unique isomorphism $G \xrightarrow{\sim} G'$.*

The assertion is local for the étale topology and one may assume G split: $G = (G, T, M, R)$, B being defined by the system of positive roots R^+ of R. The given isomorphism $u : B \xrightarrow{\sim} B'$ induces an isomorphism of T onto a maximal torus T' of B', hence of G'. The given isomorphism $T \simeq D_S(M)$ gives an isomorphism $T' \simeq D_S(M)$ in which the elements of R^+ become the constant roots of B' with respect to T', hence the elements of R the constant roots of G' with respect to T'. Since G and G' are semisimple, the coroots are determined by duality (Exp. XXI, 1.2.5), which proves that (T', M, R) is a splitting of G' such that $R(G) = R(G')$.

Applying Exp. XXIII, 5.1 (uniqueness theorem), one deduces that there exists a unique isomorphism $G \xrightarrow{\sim} G'$ coinciding with u on T and the U_α, $\alpha \in \Delta$. By 5.1, the restriction of this isomorphism to B is u. QED

Remark 6.3.1. (i) Using Exp. XXII, 4.1.9 and arguing as in *loc. cit.* 4.2.12, one can in the statement of the theorem replace "isomorphism" by "isogeny" (resp. "central isogeny").

(ii) The theorem is false for reductive groups. Take for example $G = G' = SL_2 \times G_m$ identified with the group of matrices below:

$$\left\{ \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & h \end{pmatrix} : ad - bc = 1,\ h \text{ invertible} \right\};$$

take for $B = B'$ the Borel subgroup defined by $c = 0$ and for u the automorphism of B below: $u(a, b, d, h) = (a, b, d, ah)$.

Corollary 6.4. *The functor $(G, B) \mapsto B$ from the category formed by pairs (G, B), where G is an S-semisimple group and B a Borel subgroup of G, to the category of S-group schemes (morphisms being isomorphisms) is fully faithful.*

[1407] N.D.E.: We have corrected $p_{3\alpha+\beta}(a)$ to $p_{2\alpha+\beta}(a)$, and detailed the end of the argument.

Corollary 6.5. *Let S be a scheme, G an S-semisimple group, B a Borel subgroup of G, B' an S-group locally isomorphic to B for (fpqc). Then B' is locally isomorphic to B for the local finite étale topology*[1408] *and there exists an S-semisimple group G' of which B' is a Borel subgroup; G' is unique up to a unique isomorphism inducing the identity on B'.*

Corollary 6.6. *Let S be a scheme, G an S-semisimple group, B a Borel subgroup of G. Then $\mathrm{Aut}_{S-gr.}(B)$ is representable by an affine and smooth S-scheme, $Autext(B)$ is representable by an étale and finite S-scheme, and the obvious morphisms induce an isomorphism of exact sequences*

$$\begin{tikzcd} 1 \arrow[r] & B_{ad} \arrow[r] \arrow[d, "\simeq"] & \mathrm{Aut}_{S\text{-}gr.}(G, B) \arrow[r] \arrow[d, "\simeq"] & \mathrm{Autext}(G) \arrow[r] \arrow[d, "\simeq"] & 1 \\ 1 \arrow[r] & B_{ad} \arrow[r] & \mathrm{Aut}_{S\text{-}gr.}(B) \arrow[r] & \mathrm{Autext}(B) \arrow[r] & 1. \end{tikzcd}$$

This follows at once from 2.1 and from the preceding results. The reader is left the task of developing the analogues of the results of nos 1, 2, 3, 4 in the framework above.

Remark 6.7. If S is a scheme and B an S-group, B can therefore be a Borel subgroup only of a unique semisimple group, well determined by B. It therefore remains to characterize the S-groups B that are Borel subgroups of semisimple groups.[1409]

7. Representability of the functors $\mathrm{Hom}_{S-gr.}(G, H)$, for G reductive

7.1. The split case

7.1.1. Let S be a scheme, G a pinned S-group, T its maximal torus, Δ the set of simple roots and, for $\alpha \in \Delta$,

$$u_\alpha \in U_\alpha^\times(S), \qquad w_\alpha \in \mathrm{Norm}_G(T)(S),$$

the elements defined by the pinning.

Let on the other hand H be an S-group scheme; we are interested in the functor $\mathrm{Hom}_{S-gr.}(G, H)$, and more precisely in the morphism

$$q : \mathrm{Hom}_{S\text{-}gr.}(G, H) \to \mathrm{Hom}_{S\text{-}gr.}(T, H).$$

Let then

$$f_T : T \to H$$

[1408] N.D.E.: denoted (étf) in Exp. IV 6.3. In other words, for every $s \in S$ there exist an open neighborhood U of s and a finite surjective étale morphism $V \to U$ such that $B' \times_S V \simeq B \times_S V$.

[1409] N.D.E.: One can ask whether every smooth affine S-group, each of whose geometric fibers is a Borel subgroup of a semisimple group, is a Borel subgroup of a semisimple S-group. (We have suppressed the "counter-example" given in the original, for S = scheme of dual numbers over a field k, which was erroneous, as M. Demazure pointed out to us.)

be a morphism of S-groups; consider $q^{-1}(f_T) = F$. This is the functor defined by

F(S') = { f ∈ Hom_{S'-gr.}(G_{S'}, H_{S'}) | f = (f_T)_{S'} on T_{S'} }.

One has a morphism of S-functors

$$i : F \to H^{2n},$$

where $n = Card(\Delta)$, defined set-theoretically by $i(f) = (f(u_\alpha), f(w_\alpha))_{\alpha\in\Delta}$. By Exp. XXIII, 1.9, i is moreover a monomorphism.

Proposition 7.1.2. *If H is separated over S, F is representable and i is a closed immersion.*

The usual technique of relative representability[1410] shows us that it suffices to prove that, given sections

v_α, h_α ∈ H(S),   for α ∈ Δ,

the S-schemes S' such that there exists an S'-homomorphism

$$f : G_{S'} \to H_{S'}$$

extending $(f_T)_{S'}$ and verifying $f(u_\alpha) = v_\alpha$, $f(w_\alpha) = h_\alpha$, are exactly those that factor through a certain closed subscheme of S. One may evidently assume S affine.

To simplify the sequel, let us say that a morphism $Y \to X$[1411] of affine schemes verifies condition (L) if Y is the spectrum of an $O(X)$-algebra which is a free $O(X)$-module. It is clear that if one restricts to the category of affine schemes, a product, or a composite of morphisms verifying (L) verifies (L), and that (L) is stable under base change.

Lemma 7.1.3. *Suppose S affine, and let $\alpha \in \Delta$. Consider the morphism*

a : T ×_S T → G_{a, S}

defined set-theoretically by $a(t,t') = \alpha(t) + \alpha(t')$.

(i) a is faithfully flat and of finite presentation.

(ii) Let R be the fibered square of a. The structure morphism $R \to S$ verifies (L).

It is first clear that the morphism a is surjective, since $\alpha : T \to G_{m,S}$ is. It is trivial that a is of finite presentation. Since α verifies (L), it suffices, to prove (i) and (ii), to show that the morphism

u : G_{m, S}² → G_{a, S}

[1410] N.D.E.: See for example the proof of XXII 5.8.1.

[1411] N.D.E.: We have corrected $X \to Y$ to $Y \to X$.

defined set-theoretically by $u(x,y) = x + y$ verifies (L). In other words, it suffices to verify that for every ring A, the ring $A[X, Y, X^{-1}, Y^{-1}]$ is a free module over its subring $A[X+Y]$.[1412] Now A[X, Y] is a free module with basis {1, X} over the subring $B = A[X+Y, XY]$ (one has $X^2-(X+Y)X+XY = 0$), so $A[X^{\pm 1}, Y^{\pm 1}] = A[X,Y]_{XY}$ is free over $B_{XY} = A[X+Y, (XY)^{\pm 1}]$ with basis {1, X}, and free over $A[X+Y]$ with basis the elements $(XY)^p$ and $X(XY)^p$, for $p \in \mathbb{Z}$.

Lemma 7.1.4. *Let $\alpha \in \Delta$, and let $b : T \times_S T \to H$ be the morphism defined set-theoretically by*

```
b(t, t') = int(f_T(t)) v_α · int(f_T(t')) v_α.
```

Let $f_\alpha : U_\alpha \to H$ be an S-morphism. The following conditions are equivalent:[1413]

(i) f_α is a morphism of groups; one has $f_\alpha(u_\alpha) = v_\alpha$ and

```
f_T(t) f_α(x) f_T(t)^{-1} = f_α(α(t) x)
```

for all $t \in T(S')$, $x \in U_\alpha(S')$, $S' \to S$.

(ii) One has $f_\alpha(u_\alpha) = v_\alpha$ and the relation

```
f_α(a(t, t') u_α) = b(t, t')
```

for all $t, t' \in T(S')$, $S' \to S$.

If f_α verifies (i), one has $f_\alpha(\alpha(t)u_\alpha) = int(f_T(t))v_\alpha$, which entails (ii) at once. Conversely, suppose (ii) verified and let us prove the various conditions of (i); let us first prove the last. Since a is covering for (fpqc), it suffices to prove that if $t, t', t'' \in T(S')$, one has

```
f_T(t) f_α(a(t', t'') u_α) f_T(t)^{-1} = f_α(α(t) a(t', t'') u_α);
```

now this also reads

```
f_T(t) b(t', t'') f_T(t)^{-1} = b(t t', t t''),
```

an evident property from the definition of b. It remains to prove that f_α is a morphism of groups. Now the preceding property gives at once

```
f_α(α(t) u_α) f_α(α(t') u_α) = (f_T(t) f_α(u_α) f_T(t)^{-1}) · (f_T(t') f_α(u_α) f_T(t')^{-1})
                                = b(t, t') = f_α(α(t) u_α + α(t) α(t') u_α).
```

One therefore has $f_\alpha(x + x') = f_\alpha(x)f_\alpha(x')$, whenever x and x' are sections of the open subset $U_\alpha^\times$ of U_α, which is schematically dense; one concludes then by Exp. XVIII, 1.4.

[1412]N.D.E.: We have simplified the proof of the original.

[1413]N.D.E.: In what follows, the group law of U_α is written additively, that is, if one writes $p_\alpha : G_{a,S} \xrightarrow{\sim} U_\alpha$ for the isomorphism such that $p_\alpha(1) = u_\alpha$ and if $x = p_\alpha(z)$, then $\alpha(t)x$ (resp. $a(t,t')u_\alpha$) denotes $p_\alpha(\alpha(t)z)$ (resp. $p_\alpha(a(t,t')) = \alpha(t)u_\alpha + \alpha(t')u_\alpha$).

7.1.5. Let us fix provisionally an $\alpha \in \Delta$. The morphism α is faithfully flat quasi-compact, so $G_{\alpha,S}$ is identified with the quotient of $T \times_S T$ by the equivalence relation R defined by α. Let

$$R \rightrightarrows^{i_1}_{i_2} T \times_S T$$

be this equivalence relation.

For the morphism b to factor through α,[1414] it is necessary and sufficient that $b \circ i_1 = b \circ i_2$; that is, if one writes K for the kernel of the pair of morphisms

$$R \rightrightarrows^{b \circ i_1}_{b \circ i_2} H,$$

it is necessary and sufficient that $K = R$. Now H is assumed separated over S, so K is a closed subscheme of R. Recall, on the other hand, that R is essentially free over S (7.1.3).

7.1.6. By the foregoing, if S' is an S-scheme, for there to exist over S' an f_α verifying the conditions of 7.1.4 (i) (and then necessarily unique), it is necessary and sufficient that $K_{S'} = R_{S'}$, and that the morphism f_α obtained verify $f_\alpha(u_\alpha) = v_\alpha$.

The first condition is equivalent to the fact that $S' \to S$ factors through a certain closed subscheme of S (Exp. VIII, 6.4[1415]); if one replaces S by this closed subscheme, the second condition defines once again a closed subscheme of S (equality of two sections of H, now H is assumed separated over S).

Up to replacing S by this closed subscheme, one may therefore assume that there exists a morphism $f_\alpha : U_\alpha \to H$ verifying the conditions of 7.1.4 (i). Taking the intersection of the subschemes of S obtained for each $\alpha \in \Delta$, one may assume this condition verified for every $\alpha \in \Delta$.

7.1.7. Similarly, consider for each $\alpha \in \Delta$ the two morphisms of S-groups

$$f_T \circ \mathrm{int}(w_\alpha),\ \mathrm{int}(h_\alpha) \circ f_T : T \to H.$$

Since H is separated over S and T essentially free over S, the same reasoning as previously shows that, up to replacing S by a closed subscheme, one may assume that for every $\alpha \in \Delta$ one has

$$f_T \circ \mathrm{int}(w_\alpha) = \mathrm{int}(h_\alpha) \circ f_T.$$

7.1.8. Using now the theorem of generators and relations (Exp. XXIII, 3.5), one sees that there exists a homomorphism of groups $f : G \to H$ verifying the required conditions if and only if a certain finite set of algebraic relations between the sections $h_\alpha, v_\alpha, f_T(\alpha^*(-1))$ $(\alpha \in \Delta)$ of H is satisfied:

$$R_i\big((h_\alpha)_\alpha, (v_\alpha)_\alpha, (f_T(\alpha^*(-1)))_\alpha \big) = e, \qquad i = 1, \dots, n.$$

[1414] N.D.E.: i.e. for condition (ii) of 7.1.4 to be satisfied.

[1415] N.D.E.: see also the addition in Exp. VI_B, 6.2.3.

Since H is separated over S, this defines once again a closed subscheme of S, and one is done.

Corollary 7.1.9. *Let S be a scheme, G a split S-reductive group, T its maximal torus, H an S-group scheme separated over S. Let further P be a property of morphisms such that:*

(i) A closed immersion verifies P.

(ii) The composite of two morphisms verifying P also verifies P.

(iii) P is stable under base change.

(iv) The structure morphism $H \to S$ verifies P.

Then the canonical morphism

$$\mathrm{Hom}_{S\text{-gr.}}(G, H) \to \mathrm{Hom}_{S\text{-gr.}}(T, H)$$

is relatively representable by a separated morphism verifying P.[1416]

Indeed, one may assume G pinned; the structure morphism $H^{2n} \to S$ verifies P and one concludes by 7.1.2.

Corollary 7.1.10. *Let S be a scheme, G a split S-reductive group, and H an S-group scheme smooth and with affine fibers.*[1417] *Then the functor* $\mathrm{Hom}_{S-gr.}(G, H)$ *is representable by an S-scheme locally of finite presentation and separated over S.*

Indeed, since H is smooth over S, one can consider its neutral component H^0, which has affine fibers, is smooth, separated and of finite presentation over S (Exp. VI_B, 3.10 and 5.5). Since G has connected fibers, one may replace H by H^0. Since G and H are then of finite presentation, one may assume S noetherian, and one applies 7.1.9 taking for P the property "of finite type". But, by Exp. XV, 8.9, one knows that $\mathrm{Hom}_{S-gr.}(T, H)$ is representable by a separated S-scheme locally of finite type.

Remark 7.1.11. If H is affine over S, one can replace the reference to Exp. XV by Exp. XI, 4.2.

7.2. General case

Proposition 7.2.1. *Let S be a scheme, G an S-reductive group, T a maximal torus of G. Let on the other hand H be an S-group scheme, such that the structure morphism $H \to S$ verifies the following condition:*

(+) Each point $s \in S$ possesses an open neighborhood U such that the morphism $H_U \to U$ is quasi-projective.

Then the canonical morphism

[1416] N.D.E.: i.e. for every $S' \to \mathrm{Hom}_{S-gr.}(T, H)$, with S' representable, $\mathrm{Hom}_{S-gr.}(G, H) \times_S S'$ is representable and the morphism of S-schemes $\mathrm{Hom}_{S-gr.}(G, H) \times_S S' \to S'$ is separated and verifies P.

[1417] N.D.E.: We have suppressed the hypothesis that H be quasi-separated, which is superfluous.

$$\mathrm{Hom}_{S\text{-gr.}}(G, H) \to \mathrm{Hom}_{S\text{-gr.}}(T, H)$$

is relatively representable by a morphism verifying (+).

When G is splittable relative to T, one applies 7.1.9 taking for P the property (+) above. When G is locally isotrivial, for example semisimple (4.2.2), one notes that the assertion of the theorem is local for the local finite étale topology (indeed, the property of being quasi-projective is local for the global finite étale topology and ensures the effectiveness of descent for this topology, cf. SGA 1, VIII, 7.7). Finally, in the general case, one uses the following lemma:

Lemma 7.2.2. *Let S be a scheme, G an S-reductive group, T a maximal torus of G, G' the derived group of G (Exp. XXII, 6.2), $T' = T \cap G'$ the maximal torus of G' corresponding to T (Exp. XXII, 6.2.8). Then the diagram*

$$G \leftarrow G' \upuparrows T \leftarrow T'$$

is an amalgamated sum in the category of S-group sheaves: that is, for every S-group sheaf H, the following square is cartesian:

$$\begin{array}{ccc} \mathrm{Hom}_{S\text{-gr.}}(G, H) & \to & \mathrm{Hom}_{S\text{-gr.}}(G', H) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{S\text{-gr.}}(T, H) & \to & \mathrm{Hom}_{S\text{-gr.}}(T', H). \end{array}$$

Indeed, if $rad(G)$ is the radical of G, then $rad(G) \subset T$, so

$$rad(G) \cap G' = rad(G) \cap T' = K,$$

and the product in G induces isogenies (Exp. XXII, 6.2)

$$i : G' \times_S rad(G) \to G, \qquad j : T' \times_S rad(G) \to T.$$

Let $f_{G'} : G' \to H$, $f_T : T \to H$, and $f_{T'} : T' \to H$ be morphisms of S-groups such that $f_{G'}|_{T'} = f_T|_{T'} = f_{T'}$. Let us show that there exists a unique morphism of S-groups $f : G \to H$ inducing $f_{G'}$ and f_T. Let $f_{rad} = f_T|_{rad(G)}$[1418] and let

$$f_1 = f_{G'} \cdot f_{rad} : G' \times_S rad(G) \to H.$$

For f to exist (and it will evidently be unique), it is necessary and sufficient that f_1 induce the identity on the kernel of i, that is, that $f_{G'}$ and f_{rad} coincide on K; but the existence of f_T shows by the same argument that $f_{T'}$ and f_{rad} coincide on K. It only remains to note that $f_{G'}$ and $f_{T'}$ evidently coincide on $K \subset T'$.

Arguing now as in 7.1.10, one deduces from 7.2.1:

Corollary 7.2.3. *Let S be a scheme, G an S-reductive group, H an S-group scheme smooth and quasi-projective over S with affine fibers. Then* $\mathrm{Hom}_{S-gr.}(G,H)$ *is representable by an S-scheme locally of finite presentation and separated over S.*

[1418]N.D.E.: We have replaced f_r by f_{rad}.

7.3. Phenomena specific to characteristic 0

If G and H are two smooth S-groups, g and h their Lie algebras, one has a canonical morphism

$$\mathrm{Lie} : \mathrm{Hom}_{S\text{-gr.}}(G, H) \to \mathrm{Hom}_{O_S\text{-Lie alg.}}(g, h),$$

where the right-hand side has an obvious definition.

Proposition 7.3.1. *Let S be a scheme of characteristic 0 (i.e. a $\mathbb{Q}$-scheme), G an S-reductive group, H an S-group scheme smooth and quasi-projective over S with affine fibers.*

(i) $\mathrm{Hom}_{S-gr.}(G,H)$ *is representable by a smooth and separated S-scheme over S.*

(ii) If G is semisimple, this scheme is affine and of finite presentation over S.

(iii) If G is simply connected (Exp. XXII, 4.3.3), the canonical morphism

$$\mathrm{Lie} : \mathrm{Hom}_{S\text{-gr.}}(G, H) \to \mathrm{Hom}_{O_S\text{-Lie alg.}}(g, h)$$

is an isomorphism.

Lemma 7.3.2. *Let k be a field of characteristic 0, G a k-reductive group, V a finite-dimensional k-vector space, $G \to GL(V)$ a linear representation of G in V. One has*

$$H^0(G, V) = H^0(g, V), \qquad H^i(G, V) = 0, \qquad \text{for } i > 0.$$

The first equality is true in general for a smooth connected group;[1419] in the case of a reductive group, one can prove it as follows: one may assume k algebraically closed, hence G splittable, hence G generated by subgroups isomorphic to $G_{m,k}$,[1420] and it suffices to verify the assertion for this group, which is easy.

From this first equality it follows that $H^0(G,V)$ is an exact functor in V when G is semisimple; g is indeed then a semisimple Lie algebra and one applies [BLie], § I.6, exercise 1 (b). The assertion remains true when G is reductive; indeed, if one introduces the radical R of G[1421] and the quotient G/R which is semisimple, one has $H^0(G,V) = H^0(G/R, H^0(R,V))$, and $H^0(G,-)$ is composed of two exact functors. Applying then Exp. I, 5.3.1, one deduces $H^i(G,V) = 0$ for $i > 0$.

Remark 7.3.3. Under the preceding conditions, if G is semisimple, one has $H^1(g,V) = H^2(g,V) = 0$, cf. [BLie], *loc. cit.* (b) and (d).

7.3.4. Let us now prove the proposition. Already, by 7.2.3, $\mathrm{Hom}_{S-gr.}(G,H)$ is representable by an S-scheme locally of finite presentation and separated over S;

[1419] N.D.E.: Indeed, it is clear that $V^G \subset V^g$. On the other hand, $H = Centr_G(V^g)$ is a closed subgroup of G, smooth since $char(k) = 0$; by Exp. II 5.3.1, one has $Lie(H) = Centr_g(V^g) = g$, and since H is smooth and G connected this entails $H = G$, whence $V^g \subset V^G$ (see also [DG70], § II.6, Prop. 2.1 (c)).

[1420] N.D.E.: Indeed, the union of the maximal tori of G is dense in G, cf. *Bible*, § 6.5, Th. 5 (= [Ch05], § 6.6, Th. 6).

[1421] N.D.E.: Note that R is a torus.

to show that it is smooth, it suffices to prove that it is infinitesimally smooth (Exp. XI, 1.8),[1422] which follows from Exp. III, 2.8 (i) by 7.3.2. We have thus proved (i).

Let us show that (ii) follows from (iii). It suffices first to prove that $\mathrm{Hom}_{S-gr.}(G,H)$ is affine over S; it will then be of finite presentation over S, since it is smooth over S by (i); in any case `Hom_{O_S\text{-Lie alg.}}(g, h)` is representable by a closed subscheme of

`Hom_{O_S\text{-mod.}}(g, h) ≃ W(g^v ⊗_{O_S} h)`

which is an affine S-scheme, and the desired conclusion appears when G is simply connected.

In the general case, one may assume G split, so $G \simeq \acute{E}p_S(R)$; introducing the simply connected root datum $sc(R)$ (Exp. XXI, 6.5.5), and using the existence theorem (Exp. XXV, 1.1), one constructs a simply connected S-group $\tilde{G}$ and a central isogeny $\tilde{G} \to G$. The kernel K of this isogeny is a finite diagonalizable S-group (so a twisted constant S-group, S being of characteristic 0) and $\mathrm{Hom}_{S-gr.}(K,H)$ is (trivially) representable by a separated S-scheme (if $K \simeq (\mathbb{Z}/n\mathbb{Z})_S$, then $\mathrm{Hom}_{S-gr.}(K,H)$ is isomorphic to $Ker(H \to^n H)$). One has an exact sequence of "pointed S-schemes":

`1 → Hom_{S-gr.}(G, H) →^u Hom_{S-gr.}(G̃, H) → Hom_{S-gr.}(K, H),`

so u is a closed immersion, hence $\mathrm{Hom}_{S-gr.}(G,H)$ is affine over S.

7.3.5. Let us finally prove (iii). Arguing as in the proof of (i) and using 7.3.3, one can show that the S-scheme `Hom_{O_S\text{-Lie alg.}}(g, h)` is smooth over S. To prove (iii), one may therefore assume $S = \mathrm{Spec}(k)$, where k is an algebraically closed field of characteristic 0; it even suffices to prove that the morphism `Lie` is bijective on k-valued points and that it induces a bijection on tangent spaces at two corresponding points. Let us first show that

`Hom_{k-gr.}(G, H) → Hom_{k\text{-Lie alg.}}(g, h)`

is bijective.

If $u : G \to H$ is a morphism of k-groups, the graph of u is a connected subgroup of $G \times_k H$ which is determined by its Lie algebra which is the graph of $Lie(u)$; the map is therefore injective. Conversely, if $v : g \to h$ is a morphism of Lie algebras, the graph k of v is a Lie subalgebra of $g \times h$, isomorphic to g. In particular, since g is its own derived algebra, the same is true of k and so, by a theorem of Chevalley ([Ch51], § 14, Th. 15), k is the Lie algebra of a connected subgroup K of $G \times_k H$.[1423] Moreover, since $k \simeq g$ is semisimple, G is a semisimple k-group. Since

`dim(K) = dim(k) = dim(g) = dim(G)`

[1422] N.D.E.: i.e. that for every S-scheme S' local artinian, with closed point s', every morphism of $\kappa(s')$-groups $G_{s'} \to H_{s'}$ lifts to a morphism of S'-groups $G_{S'} \to H_{S'}$. By Exp. III 2.8, this follows from the vanishing of $H^2(G_{s'}, V)$, where $V = Lie(H_{s'}/\kappa(s'))$.

[1423] N.D.E.: see also [Bo91], II 7.9. Moreover, we have added the sentence that follows.

and since $K \cap H$ is finite (because its Lie algebra is zero), the canonical morphism $pr_1 : K \to G$ is finite and dominant; since G is connected, pr_1 is surjective; it is therefore an isogeny. Since G is simply connected and K semisimple then, by Exp. XXI 6.2.7, pr_1 is an isomorphism, so K is the graph of a morphism of k-groups $u : G \to H$ such that $Lie(u) = v$.

Finally, let $u : G \to H$ be a morphism of k-groups. The tangent space to $\mathrm{Hom}_{k-gr.}(G,H)$ at u is identified with $Z^1(G,h)$ (cf. Exp. II, 4.2; G operates on h by $Ad_H \circ u$); similarly, one can prove that the tangent space to $\mathrm{Hom}_{k\text{-Lie alg.}}(g, h)$ at $Lie(u)$ is identified with $Z^1(g,h)$. We must therefore prove that the canonical map $Z^1(G,h) \to Z^1(g,h)$ is bijective. Now one has a commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & h^G & \to & h & \to & Z^1(G, h) & \to & H^1(G, h) \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \to & h^g & \to & h & \to & Z^1(g, h) & \to & H^1(g, h) \end{array}$$

and by 7.3.2 and 7.3.3 one has $h^G = h^g$ and $H^1(G,h) = 0 = H^1(g,h)$, whence the desired conclusion.

Remark 7.3.6. It is presumably the case that if S is a locally noetherian scheme, G an S-semisimple simply connected group, H an S-group scheme smooth, $\hat{G}$ and $\hat{H}$ their formal completions along the unit section, every homomorphism $\hat{G} \to \hat{H}$ comes from a unique homomorphism of G into H, which would generalize 7.3.1 (iii). When S is the spectrum of a field k and H is affine and of finite type, this follows from a theorem of Dieudonné ([Di57], § 15, th. 4).[1424]

7.4. An example

By way of example, we shall determine

$$\mathrm{Hom}_{S\text{-gr.}}(SL_{2,S}, SL_{2,S}).$$

7.4.1. Recall (Exp. XX, no 5) that $SL_{2,S}$ is the S-group scheme formed by the matrices $(ab;cd)$ over S satisfying $ad - bc = 1$. A maximal torus T is defined by the monomorphism $\alpha^* : G_{m,S} \to SL_{2,S}$

[1424]N.D.E.: See Exp. VII_B for the definition of the formal groups $\hat{G}$ and $\hat{H}$. Suppose that $S = \mathrm{Spec}(k)$, where k is a field. By *loc. cit.*, 2.2.1, to give a morphism of k-formal groups $v : \hat{G} \to \hat{H}$ is equivalent to giving a morphism of k-Hopf algebras $\phi : Dist(G) \to Dist(H)$, where $Dist(G)$ denotes the algebra of distributions (at the origin) of G, cf. Exp. VII_A, 2.1 or [DG70], § II.4, 6.1 or [Ja87], I 7.7 (it is called the "hyperalgebra" of G in [Di57] and [Ta75]). Theorem 4 of [Di57] (see also [Ta75], 0.3.4 (f) and (g)) generalizes in this context the theorem of Chevalley used in 7.3.5; one thereby obtains that there exists a closed connected k-subgroup K of $G \times H$ such that $\hat{K}$ equals the graph of v; since $Dist(K) \to Dist(G)$ is an isomorphism, $K \to G$ is a finite étale morphism. One then deduces that K is semisimple and then, by Exp. XXI 6.2.7, that $K \to G$ is an isomorphism (since G is simply connected); see also [Ta75], 1.8 and 2.2. More generally, *loc. cit.* studies the k-groups G having the property (SC): every finite étale morphism of k-groups $G' \to G$, with G' connected, is an isomorphism. Note finally that what precedes shows that every $Dist(G)$-module V of finite dimension is, uniquely, a G-module; for an extension to the case of a split reductive k-group G (or even of a Borel subgroup of G) see [Ja87], II 1.20 (and the references therein).

$$\alpha^*(z) = (z0)(0z^{-1}).$$

A root of G with respect to T is defined by $\alpha(\alpha^*(z)) = z^2$, a corresponding monomorphism

```
p : G_{a, S} → SL_{2, S}
```

being defined by

$$p(x) = (1x)(01).$$

Finally, the representative of the Weyl group corresponding to $u = p(1)$ is

$$w = (01)(-10).$$

Recall on the other hand (Exp. XX, 6.2) that $SL_{2,S}$ is generated by T, $p(G_{a,S})$ and w, subject to the relations

```
α^*(z) p(x) α^*(z^{-1}) = p(z² x),
w α^*(z) w^{-1} = α^*(z^{-1}),
w² = α^*(−1),
(w u)³ = 1.
```

7.4.2. Let f be an endomorphism of $SL_{2,S}$. It defines first a homomorphism $f \circ \alpha^*$: $G_{m,S} \to SL_{2,S}$. The kernel $Ker(f \circ \alpha^*)$ is a closed subgroup of $G_{m,S}$, hence is locally on S equal to a $\mu_{n,S}$ ($n \geqslant 1$) or to $G_{m,S}$. Up to restricting S, one may therefore assume that there exists an $n \in \mathbb{N}$ and a monomorphism

```
f' : G_{m, S} → SL_{2, S}
```

such that $f \circ \alpha^*(z) = f'(z^n)$.

By the conjugacy of monomorphisms $G_{m,S} \to SL_{2,S}$, one can, after an étale-covering extension of the base, find a section g of G such that $f \circ \alpha^*(z) = int(g) \circ \alpha^*(z^n)$. Transforming f by $int(g)$, one is therefore reduced to the case where there exists an $n \in \mathbb{N}$ such that $f \circ \alpha^*(z) = \alpha^*(z^n)$.

7.4.3. Consider now the morphism

```
f ∘ p : G_{a, S} → SL_{2, S}.
```

It verifies the condition

```
α^*(z^n) (f ∘ p)(x) α^*(z^n)^{-1} = (f ∘ p)(z² x).
```

If $n = 0$, it follows at once that $f \circ p$ is invariant under the homotheties of $G_{a,S}$, hence constant. If $n \neq 0$, one can apply Exp. XXII, 4.1.9; $x \mapsto x^n$ is an endomorphism of $G_{a,S}$, there exists a $\lambda \in G_{m,S}$ such that

```
f ∘ p(x) = p(λ x^n);
```

this relation is moreover valid for $n = 0$, taking $\lambda = 1$. Up to again extending S, one can find a $z \in G_{m,S}$ such that $z^2 = \lambda$. Replacing f by $int(\alpha^*(z)) \circ f$, one is therefore reduced to the case where one has

```
f ∘ α^*(z) = α^*(z^n),    f ∘ p(x) = p(x^n).
```

7.4.4. Finally, one must have $f(w)\alpha^*(z^n)f(w)^{-1} = \alpha^*(z^n)^{-1}$ and $(f(w)u)^3 = 1$. By Exp. XX, 3.8, this entails $f(w) = w^n$. Since G is generated by T, $p(G_{a,S})$ and w, this entails that for every $(ab; cd) \in G(S')$, $S' \to S$, one has

```
f( a b ) = ( a^n b^n )
  ( c d )   ( c^n d^n ).
```

7.4.5. Summarizing the preceding discussion, one sees that locally on S for the étale topology, one can find for every endomorphism f of $SL_{2,S}$ an (inner) automorphism ϕ of $SL_{2,S}$, and an integer $n \geqslant 0$ such that $f = \phi \circ F_n$, where

```
F_n( a b ) = ( a^n b^n )
   ( c d )   ( c^n d^n ).
```

Note that if $f = \phi \circ F_n$, the integer n is well determined by a fiber f_s of f, for example by $Ker(f_s)$. It follows that n is a locally constant function on S.

7.4.6. One deduces at once that if f is an endomorphism of $SL_{2,S}$, then S decomposes canonically as a sum of open and closed subschemes `S_0`, `S_1`, S_{p^n} (where p^n ranges over the set of positive powers of prime numbers) such that:

(i) f_{S_0} is the zero morphism,

(ii) f_{S_1} is an isomorphism (= an inner automorphism),

(iii) S_{p^n} is of characteristic p and $f_{S_{p^n}}$ decomposes uniquely in the form $\phi \circ F_p^n$, where ϕ is an inner automorphism and F_p the Frobenius endomorphism of SL_{2,F_p}.

7.4.7. In other words, $\mathrm{Hom}_{\mathbb{Z}-gr.}(SL_{2,\mathbb{Z}}, SL_{2,\mathbb{Z}})$ is the sum scheme:

(i) of a scheme isomorphic to $\mathrm{Spec}(\mathbb{Z})$,

(ii) of a scheme isomorphic to $\mathrm{Aut}_{\mathbb{Z}-gr.}(SL_{2,\mathbb{Z}}) \simeq ad(SL_{2,\mathbb{Z}})$,

(iii) for each prime number p and each integer $n > 0$, of a scheme isomorphic to $\mathrm{Aut}_{F_p-gr.}(SL_{2,F_p}) \simeq ad(SL_{2,F_p})$.

7.4.8. It follows in particular that

(i) $\mathrm{Hom}_{F_p-gr.}(SL_{2,F_p}, SL_{2,F_p})$ has an infinite number of connected components, and so is not quasi-compact.

(ii) If S is a scheme of unequal characteristics, $\mathrm{Hom}_{S-gr.}(SL_{2,S}, SL_{2,S})$ is not flat over S.

8. Appendix: Cohomology of a smooth group over a henselian ring. Cohomology and the functor $\prod$

Proposition 8.1. *Let S be a henselian local scheme, s its closed point, G an S-group scheme smooth such that every finite subset of G is contained in an affine open set.*[1425] *Then*

(i) If P is a principal homogeneous bundle under G, there exists an $S' \to S$ finite, surjective étale that trivializes P. One has $Fib(S, G) \simeq H^1_{ét}(S, G)$.

(ii) For every finite surjective étale morphism $S' \to S$, the canonical map

$$H^1(S'/S, G) \to H^1(S' \otimes_S \kappa(s) / \kappa(s), G_s)$$

is bijective.

(iii) The canonical map

$$Fib(S, G) \to Fib(\kappa(s), G_s)$$

is bijective.

8.1.2. If K is a finite separable extension of $\kappa(s)$, there exists an $S' \to S$ finite surjective étale such that $K \simeq S' \otimes_S \kappa(s)$.[1426] If P is a principal homogeneous bundle under G, then P is smooth over S, hence P_s smooth over $\kappa(s)$; there therefore exists a finite separable extension K of $\kappa(s)$ such that P_K possesses a section (cf. EGA IV$_4$, 17.15.10). Representing K as said above, one sees that $P_{S'}$ possesses a section by "Hensel's lemma" (cf. EGA IV$_4$, 18.5.17), which proves the first part of (i).

Conversely, if P is a principal homogeneous sheaf under G for the étale topology, there exists an $S' \to S$ finite surjective étale that trivializes P (indeed every covering family of a henselian local scheme for the étale topology is dominated by a family reduced to a morphism $S' \to S$ of the wanted form).

By virtue of the descent hypothesis made on G, P is representable (SGA 1, VIII, 7.6), which completes the proof of (i), and shows that (ii) entails (iii). It only remains to prove (ii).

8.1.3. The map of (ii) is injective: let P and Q be two principal homogeneous bundles under G trivialized by S'. Consider the S-group sheaf $H = \mathrm{Isom}_{G\text{-bundles}}(P, Q)$; since $H_{S'}$ is isomorphic to $G_{S'}$, H is representable, by the second

[1425] This last condition is in fact unnecessary (cf. A. Grothendieck, *Groupe de Brauer III*, in *Dix Exposés sur la cohomologie des schémas*, North-Holland, 1968, theorem 11.7 and remarks 11.8.3).

[1426] N.D.E.: Note that S' is still local and henselian. Furthermore, we have kept the numbering of the original: there is no no 8.1.1.

hypothesis on G, cf. above. If $H(\kappa(s)) \neq \emptyset$, then $H(S) \neq \emptyset$ by Hensel's lemma, so P and Q are isomorphic.

8.1.4. Let us finally prove that the map of (ii) is surjective, or equivalently that the canonical map

$$Z^1(S'/S, G) \to Z^1(S' \otimes_S \kappa(s) / \kappa(s), G_s)$$

is surjective. Let Z^1 be the S-functor defined by

$$Z^1(T) = Z^1(S' \times_S T / T, G_T);$$

the preceding map is identified with the map

$$Z^1(S) \to Z^1(\kappa(s));$$

by a further application of Hensel's lemma, it suffices to prove that Z^1 is representable by a smooth S-scheme.

8.1.5. Let us prove that Z^1 is representable by an S-scheme locally of finite presentation. Let C_i, $i = 0, 1, \ldots$, be the S-functor defined by

$$C_i(T) = C_i(S' \times_S T / T, G_T),$$

that is

$$C_i(T) = G((S' \times_S T / T)^{i+1}) = G((S'/S)^{i+1} \times_S T),$$

or again

$$C_i = \mathrm{Hom}_S((S'/S)^{i+1}, G).$$

Since Z^1 is obtained from C_1 and C_2 by fibered products, it suffices to prove that C_i, $i = 1, 2$, is representable by an S-scheme locally of finite presentation.

8.1.6. If $T \to S$ is a finite surjective étale morphism that decomposes S', then

$$C_i \times_S T = \mathrm{Hom}_T((S' \times_S T / T)^{i+1}, G_T)$$

is representable by a product of n copies of G_T, where n is the degree of $(S'/S)^{i+1}$. Applying once more the hypothesis on G, one deduces that C_i is indeed representable by an S-scheme locally of finite presentation (SGA 1, VIII, *loc. cit.*).

8.1.7. To prove that Z^1 is smooth, we must now, by definition, prove that if T is an affine S-scheme, T_0 the closed subscheme defined by an ideal J of square zero, the canonical map

$$Z^1(T) \to Z^1(T_0)$$

is surjective. Since G is smooth, the canonical map $G(T) \to G(T_0)$ is surjective, and it suffices to prove that the canonical map

$$H^1(S' \times_S T / T, G_T) \to H^1(S' \times_S T_0 / T_0, G_{T_0})$$

is bijective.

Changing notation slightly and generalizing the hypotheses, it now suffices for us to prove:

Lemma 8.1.8. *Let S and S' be two affine schemes, $S' \to S$ a faithfully flat morphism, J an ideal of square zero on S, S_0 the closed subscheme it defines, G a smooth S-group. Set $S'_0 = S' \times_S S_0$, $G_0 = G \times_S S_0$. The canonical map*

$$H^1(S'/S, G) \to H^1(S'_0/S_0, G_0)$$

is bijective.

Remark 8.1.9. If one assumes G commutative, the same assertion is valid for all H^i, $i > 0$, with an analogous proof.

Proof. Let M_0 be the quasi-coherent O_{S_0}-module $M_0 = Lie(G_0/S_0) \otimes_{O_{S_0}} J$. For each S_0-prescheme T_0, set $M_0(T_0) = H^0(T_0, M_0 \otimes_{O_{S_0}} O_{T_0})$, and let

$$M = \prod_{S_0/S} M_0 \quad \text{and} \quad \bar{G} = \prod_{S_0/S} G_0$$

be the S-group functors defined by $M(T) = M_0(T_0)$ and $\bar{G}(T) = G_0(T_0)$, where $T_0 = T \times_S S_0$. By Exp. III, 0.9 and (0.6.2), there exists for every affine S-scheme T an exact sequence, functorial in T:

$$1 \to M(T) \to G(T) \to \bar{G}(T) \to 1.$$

We have to study the map

$$H^1(S'/S, G) \to H^1(S'_0/S_0, G_0) = H^1(S'/S, \bar{G}).$$

Suppose first G commutative. One then has an exact cohomology sequence

$$H^1(S'/S, M) \to H^1(S'/S, G) \to H^1(S'/S, \bar{G}) \to H^2(S'/S, M);$$

but

$$H^i(S'/S, M) = H^i(S'_0/S_0, M_0) = H^i(S'_0/S_0, M_0),$$

and one knows (TDTE I, B, Lemma 1.1) that $H^i(S'_0/S_0, M_0) = 0$ for $i \neq 0$.

If now G is not commutative, we must use the exact sequence of non-abelian cohomology. If $u \in Z^1(S'/S, \bar{G})$, one knows that the elements of $H^1(S'/S, G)$ having the same image in $H^1(S'/S, \bar{G})$ as the class of u are in the image of the corresponding coboundary map:

$$H^1(S'/S, M_u) \to H^1(S'/S, G),$$

where M_u is the S-functor M "twisted by u". Similarly, if v is an element of $Z^1(S'/S, \bar{G})$, there exists a "coboundary"

$$\Delta(v) \in H^2(S'/S, M_v),$$

where M_v is the S-functor M "twisted by v", such that $\Delta(v) = 0$ if and only if the class of v is in the image of $H^1(S'/S, G)$. It suffices to prove that one has $H^1(S'/S, M_u) = H^2(S'/S, M_v) = 0$.

Now recall (Exp. III 0.8) that the action of G on M defined by the exact sequence is none other than that which is deduced functorially from the adjoint representation of G_0:

$$ad : G_0 \to \mathrm{Aut}_{O_{S_0}}(Lie(G_0/S_0)).$$

The element u (resp. v) therefore acts in $M_{S'\times_S S'}$ via an $S'_0 \times_{S_0} S'_0$-automorphism of $Lie(G_0/S_0)$. Since u (resp. v) is a cocycle, this automorphism is a descent datum; write L_u (resp. L_v) for the quasi-coherent O_{S_0}-module obtained. One verifies at once that for $T \to S$, one has

$$M_u(T) = H^0(T_0, L_u \otimes_{O_{S_0}} J \otimes_{O_{S_0}} O_{T_0})$$

and the same relation replacing u by v. One therefore has

$$H^1(S'/S, M_u) = H^1(S'_0/S_0, L_u \otimes J),$$
$$H^2(S'/S, M_v) = H^2(S'_0/S_0, L_v \otimes J),$$

and both are indeed zero by virtue of the result already used.

Proposition 8.2. *Let C be a category possessing fibered products, equipped with a topology coarser than the canonical topology, $S' \to S$ a morphism of C, G' an S'-group sheaf, G the S-group sheaf $\prod_{S'/S} G'$. Let $H^1_S(S', G') \subset H^1(S', G')$ be the set of classes of principal homogeneous sheaves under G' which are trivialized by a sieve of S' obtained by base change from a suitable covering sieve of S. The canonical map $H^1(S, G) \to H^1(S', G')$ defined by the functor*

$$P \mapsto P \times_S S'$$

induces a bijection

$$H^1(S, G) \xrightarrow{\sim} H^1_S(S', G');$$

the inverse bijection is defined by the functor $P' \mapsto \prod_{S'/S} P'$.

For every object X of C/S, one has by definition a functorial isomorphism in X

$$\mathrm{Hom}_S(X, G) \xrightarrow{\sim} \mathrm{Hom}_{S'}(X \times_S S', G').$$

One therefore has for each S-object T a functorial bijection in T

$$H^1(T/S, G) \xrightarrow{\sim} H^1(T'/S', G').$$

Replacing now the unique morphism $T \to S$ by an arbitrary covering family of S and passing to the inductive limit, one deduces the first part of the statement. The second part is deduced without difficulty.

Lemma 8.3. *Under the conditions of 8.2, the assertion $H^1_S(S', G') = H^1(S', G')$ is local on S: suppose there exists a covering family $S_i \to S$ such that for every i, one has $H^1_{S_i}(S' \times_S S_i, G') = H^1(S' \times_S S_i, G')$. Then $H^1_S(S', G') = H^1(S', G')$.*

Indeed, let P' be a principal homogeneous sheaf under G'. Set

$$P'_i = P' \times_{S'} (S' \times_S S_i);$$

by hypothesis, there exists a covering family $S_{ij} \to S_i$ such that for each j, $P' \times_{S'} (S' \times_S S_{ij})$ possesses a section. But $S_{ij} \to S$ is a covering family of S, and P' is indeed trivialized by the covering family of S' obtained from this one by base change.

Proposition 8.4. *Let $S' \to S$ be a finite étale morphism of schemes. Let G' be an S'-group sheaf, G the S-group sheaf $\prod_{S'/S} G'$. For the étale (resp. local finite étale, resp. (fpqc)) topology, the functors*

$$\begin{array}{ccc} P & \mapsto & P \times_S S' \\ \prod_{S'/S} P' & \leftarrow\!\shortmid & P' \end{array}$$

induce inverse bijections of each other:

$$H^1(S, G) \simeq H^1(S', G').$$

By 8.2, it suffices to show that $H^1_S(S', G') = H^1(S', G')$. By 8.3, it suffices to do this locally for the local finite étale topology; one may therefore assume that S' is a finite direct sum of copies of S, say I_S, where I is a suitable finite set. Then G' is given by a family $(G_i)_{i \in I}$ of sheaves on S and

$$H^1(S', G') \simeq \prod_{i \in I} H^1(S, G_i).$$

On the other hand

$$H^1(S, G) \simeq \prod_{i \in I} H^1(S, G_i),$$

whence, by 8.2, $H^1_S(S', G') = H^1(S', G')$. QED

Remarks 8.5. One can interpret 8.2 and 8.3 by the following exact sequence (f is the given morphism $S' \to S$)

$$1 \to H^1(S, f_*(G')) \to H^1(S', G') \to H^0(S, R^1 f_*(G')).$$

In the commutative case, this exact sequence follows from the Leray spectral sequence; it is still valid in the non-commutative case (cf. Giraud's thesis[1427]).

In this form, one sees that the result is still valid if f is only assumed finite, or simply integral, the topology being the étale topology, because for every G', one has

$$R^1 f_*(G') = \text{final sheaf},$$

[1427] N.D.E.: See § V 3.1.4 of [Gi71].

by SGA 4, VIII, 5.3.[1428]

On the other hand, this result becomes false if one takes a topology such as (fpqc) or (fppf), even if $S = \mathrm{Spec}(k)$, k algebraically closed field of characteristic $p \neq 0$, $S' = \mathrm{Spec}(k[t]/t^2)$, $G' = \mu_p$ or α_p.

Similarly, 8.2 becomes false, even for the étale topology, if one suppresses in it the hypothesis that f is finite, as one sees by taking for f an open immersion; for example if $S = \mathrm{Spec}(V)$, V a complete discrete valuation ring with algebraically closed residue field, S' being the open subset induced at the generic point, and G' the constant group $(\mathbb{Z}/n\mathbb{Z})_{S'}$, with n prime to the residue characteristic of V, one has $H^1(S, G) = 0$, $H^1(S', G') \neq 0$. Moreover, replacing S' by $S \coprod S'$, one deduces an analogous example, with $S' \to S$ étale surjective, hence covering for the topology considered.

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Footnotes

Exposé XXV. The existence theorem

by M. Demazure

[1430] For the sake of completeness, we give in this Exposé a proof of the existence theorem for split groups. Like Chevalley's original proof ("Sur certains schémas

[1428] N.D.E.: This reference also refers to § V 3.1.4 of [Gi71].

[1429] N.D.E.: We have added to these three references, figuring in the original, the following references.

[1430] N.D.E.: Version of 13/10/2024.

de groupes semi-simples", *Séminaire Bourbaki*, May 1961, no. 219), it rests on the existence of complex semisimple algebraic groups of all possible types. The principle of a more satisfactory proof, establishing directly the existence of a simply connected split semisimple $\mathbb{Z}$-group corresponding to a given Cartan matrix, was given by Cartier (unpublished).[1431] Let us point out, however, that the difficulty is not to give an explicit construction of a group scheme, but to verify that the group thus constructed really meets the required conditions, that is, essentially, that its fibers are indeed smooth and reductive.

1. Statement of the theorem

Theorem 1.1. *Let S be a non-empty scheme. The functor*

$$G \mapsto R(G)$$

is an equivalence of the category of pinned S-reductive groups with the category of pinned reduced root data.

By virtue of the uniqueness theorem (Exp. XXIII, 4.1), the preceding statement is equivalent to:

Corollary 1.2 (Existence of split groups). *For every reduced root datum R, there exists a split reductive $\mathbb{Z}$-group G such that $R(G) \simeq R$.*[1432]

In particular:

Corollary 1.3. *Let k be a field. For every splittable reductive k-group G_k, there exists a reductive $\mathbb{Z}$-group G such that $G \otimes_{\mathbb{Z}} k \simeq G_k$.*

Conversely, let us remark first that to prove 1.2 it suffices, by Exp. XXII 4.3.1 and Exp. XXI 6.5.10, to consider the case where the root datum R is simply connected (and even irreducible if one insists, by Exp. XXI, 7.1.6).

On the other hand, under the conditions of 1.3, the group scheme G is of constant type (since $\mathrm{Spec}(\mathbb{Z})$ is connected), hence of type $R(G_k)$; by Exp. XXIII, 5.9, it follows that the validity of 1.3 for a given group G_k entails the existence of a split $\mathbb{Z}$-group of type $R(G_k)$.[1433]

[1431] N.D.E.: Such a principle was also sketched by B. Kostant [Ko66]; a complete proof, using quantum groups, has been given recently by G. Lusztig [Lu09].

[1432] N.D.E.: Let us also point out that the work [BT84] of F. Bruhat and J. Tits contains a variant of Chevalley's construction (*loc. cit.*, 2.2.3–2.2.5 and § 3.2), which yields in particular a smooth affine $\mathbb{Z}$-group G with connected fibers, possessing a split maximal torus, and whose generic fiber is a reductive $\mathbb{Q}$-group of type R; the fact that the geometric fibers of G are reductive follows from the study of the unipotent radical of a special fiber carried out in *loc. cit.*, 4.6.12 and 4.6.15 (valid for more general G, associated with a valued root datum), but it is simpler to deduce it from the description of $\mathfrak{g} = \mathit{Lie}(G)$, from Exp. XIX 1.12 (iii), and from the existence of the elements $w_\alpha(X)$ of Exp. XX 3.1 (iv) (cf. [BT84], 3.2.1).

[1433] N.D.E.: In fact, Exp. XXIII, 5.9 is not necessary because the present Exposé constructs, for every semisimple $\mathbb{Q}$-group $G_{\mathbb{Q}}$, a semisimple $\mathbb{Z}$-group G of the same type as $G_{\mathbb{Q}}$ and equipped with a split maximal torus T; hence, by Exp. XXII 2.2, G is split.

To prove 1.2, and hence 1.1, it therefore suffices to prove 1.3 when k is of characteristic zero (for example $k = \mathbb{C}$) and G_k is simply connected (and in particular semisimple), as well as:

Proposition 1.4. *For every simply connected reduced root datum R, there exists a semisimple algebraic $\mathbb{C}$-group of type R.*

One may prove 1.4 in the following way. One knows first that there exists a complex semisimple Lie algebra g of type R, cf. for example N. Jacobson, *Lie Algebras*, ch. VII, Th. 5.[1434] Then $G = \mathrm{Aut}(g)^0$ is a semisimple algebraic $\mathbb{C}$-group of type $ad(R)$.[1435] By *Bible*, § 23.1, Prop. 1, one deduces from this the existence of a semisimple $\mathbb{C}$-group of type R.

The rest of this Exposé is devoted to the proof of 1.3 for k of characteristic zero and G_k semisimple. It will be carried out in two stages: construction of a "piece of $\mathbb{Z}$-group scheme" (n° 2), and study of the group obtained by application of the "theorem of Weil" (n° 3).

To avoid confusion, we shall not use in n° 2 the abbreviated notation X_k to denote the k-scheme $X \otimes_{\mathbb{Z}} k$, where X is a $\mathbb{Z}$-scheme.

2. Existence theorem: construction of a piece of group

2.1.

Let us choose, once and for all, a splitting of G_k, denoted

(G_k, T_k, M, R)

(cf. Exp. XXII, 1.13), a system of simple roots Δ of R (defining the system of positive roots R^+), a Chevalley system $(X_{\alpha,k})_{\alpha\in R}$ of G_k (Exp. XXIII, 6.1 and 6.2) satisfying the following supplementary condition (cf. XX 2.6): for every $\alpha \in R$, one has

$$X_{\alpha,k}X_{-\alpha,k} = 1.$$

Finally, let us choose on the subgroup of M generated by R a total order relation compatible with the group structure, such that the roots > 0 are the elements of R^+. We then denote the roots

[1434] N.D.E.: Let Δ be a base of R and $\tilde{g}$ the complex Lie algebra generated by generators $(e_\alpha, f_\alpha, h_\alpha)_{\alpha\in\Delta}$ subject to the relations $[h_\alpha, h_\beta] = 0$, $[e_\alpha, f_\beta] = h_\alpha$ if $\beta = \alpha$ and $= 0$ otherwise, $[h_\alpha, e_\beta] = (\alpha*, \beta)e_\beta$ and $[h_\alpha, f_\beta] = -(\alpha*, \beta)f_\beta$. In *loc. cit.*, g is defined as the quotient of $\tilde{g}$ by the intersection of the kernels of the finite-dimensional irreducible representations of $\tilde{g}$; in [Se66], § VI.5, Th. 9 (see also [BLie], VIII, § 4.3, Th. 1) it is shown that g is the quotient of $\tilde{g}$ by the relations $ad(e_\alpha)^{1-(\alpha*,\beta)}(e_\beta) = 0$ and $ad(f_\alpha)^{1-(\alpha*,\beta)}(f_\beta) = 0$. For an explicit description of the structure constants (in particular the choice of signs), see Exp. XXIII 6.5 and 6.7 as well as [Ti66], § 4, Th. 1 and (for types A, D, E) [Sp98], 10.2.5.

[1435] N.D.E.: One may suppose R irreducible, hence g simple. By a general argument, one knows that $Lie(G)$ is the complex Lie algebra of derivations of g (cf. [DG70], § II.4, Prop. 2.3); but these are all inner (cf. [BLie], I § 6.1, Cor. 3 of Prop. 1), hence $Lie(G) = g$; consequently G has no invariant subgroup of dimension > 0, hence G is semisimple. Its root system is then the same as that of g; moreover, the center of G acts simultaneously trivially and faithfully on g, hence is trivial, hence G is adjoint.

$-\alpha_n < -\alpha_{n-1} < \cdots < -\alpha_1 < \alpha_1 < \alpha_2 < \cdots < \alpha_n.$

For $\alpha \in R$, we denote by $U_{\alpha,k}$ the vector group corresponding to the root α, and by

$$p_{\alpha,k} : \mathbb{G}_{a,k} \xrightarrow{\sim} U_{\alpha,k}$$

the isomorphism of vector groups defined by $X_{\alpha,k}$.

2.2.

The splitting of G_k includes in particular an isomorphism of k-groups

$$T_k \simeq D_k(M).$$

Put $T = D(M)$; this is a $\mathbb{Z}$-torus, and the previous isomorphism may be regarded as an isomorphism

$T_k \simeq T \otimes_{\mathbb{Z}} k.$

One has

$$\mathrm{Hom}_{\mathbb{Z}-gr.}(T, \mathbb{G}_m) = M,$$

and one will regard the elements of $R \subset M$ as characters of T. Likewise, one will regard the elements of $R*$ as morphisms of $\mathbb{Z}$-groups $\mathbb{G}_m \to T$.

2.3.

For each $\alpha \in R^+$, let $\mathbb{G}_a(\alpha)$ be a copy of the group $\mathbb{G}_a$; consider the $\mathbb{Z}$-scheme

$$U = \mathbb{G}_a(\alpha_1) \times \cdots \times \mathbb{G}_a(\alpha_n).$$

If U_k denotes the unipotent part of the Borel group B_k of G_k defined by R^+, denote by

$a : U \otimes_{\mathbb{Z}} k \xrightarrow{\sim} U_k$

the isomorphism of k-schemes defined by

$a(x_1, \ldots, x_n) = p_{\alpha_1,k}(x_1) \cdots p_{\alpha_n,k}(x_n).$

2.4.

The group law of U_k translates into relations of the form

$a(x_1, \ldots, x_n) \cdot a(y_1, \ldots, y_n) = a(z_1, \ldots, z_n),$

where each z_h $(h = 1, \cdots, n)$ is expressed as a polynomial

$$z_h = x_h + y_h + Q_h(x_1, \dots, x_{h-1}, y_1, \dots, y_{h-1}),$$

the coefficients of Q_h being integers (Exp. XXII, 5.5.8 and Exp. XXIII, 6.4). Moreover $Q_h(x_1, \cdots, x_{h-1}, 0, \cdots, 0) = 0$.

Let us equip U with the composition law defined by the preceding formulas (which are indeed “defined over $\mathbb{Z}$”). Since this law induces on U_k its group law, it is associative, and (0) is a unit element (indeed, the two preceding assertions are expressed by relations between the polynomials Q_h, and $\mathbb{Z} \to k$ is injective). Let us show that this is a group law: if (x_i) is a section of U (over some S), one computes the inverse (y_i) of (x_i) by the recurrence formulas

$$y_i = -x_i - Q_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1})$$

which are still “defined over $\mathbb{Z}$”.

In summary, we have constructed on U a group law such that the previous isomorphism a is an isomorphism of groups.

For each $\alpha \in R^+$, consider the morphism

$$p_\alpha : \mathbb{G}_a \to U$$

defined by $p_\alpha(x) = (x_i)$ where

$$x_i = x \quad \text{if } \alpha_i = \alpha,$$
$$x_i = 0 \quad \text{if } \alpha_i \neq \alpha.$$

This is a closed immersion and a group homomorphism; we denote its image by U_α. One has $(x_i) = p_{\alpha_1}(x_1) \cdots p_{\alpha_n}(x_n)$, which proves that U is identified with the product

$$U = U_{\alpha_1} \cdot U_{\alpha_2} \cdots U_{\alpha_n}.$$

2.5.

Let $T = D(M)$ act on each U_α through the character α; one verifies at once that this defines an action of T on the group U, and one can construct the semi-direct product $B = T \cdot U$. One has a canonical isomorphism of k-groups

$$B \otimes_{\mathbb{Z}} k \cong B_k.$$

If we now take any order on R^+, the morphism

$$\prod_{\alpha \in R^+} U_\alpha \to U$$

defined by the product in U is still an isomorphism. Indeed, since both sides are flat $\mathbb{Z}$-schemes of finite presentation, it suffices to verify the assertion on geometric fibers; one is then reduced to Lazard’s theory (*Bible*, § 13.1, Prop. 1): one considers

U as a group with operators T, and one uses the fact that the U_α are pairwise non-isomorphic as groups with operators (since the characters $\alpha \in R^+$ of T are pairwise distinct on each fiber).

2.6.

Replacing R^+ by $R^- = -R^+$, one constructs similarly the groups U^-, B^-, U_α ($\alpha \in R^-$) and the isomorphisms

$$p_\alpha : \mathbb{G}_a \xrightarrow{\sim} U_\alpha, \quad \alpha \in R^-.$$

Let us finally introduce the product scheme

$$\Omega = U^- \times T \times U;$$

one has a canonical isomorphism of k-schemes

$$\Omega \otimes_{\mathbb{Z}} k \xrightarrow{\sim} U^-_k \times_k T_k \times_k U_k \simeq \Omega_k,$$

where Ω_k is the "big cell" of G_k (Exp. XXII, 4.1.2).

From now on, we identify $\Omega \otimes_{\mathbb{Z}} k$ with Ω_k by the previous isomorphism; we regard U^-, T, U as subschemes of Ω, through their unit sections. We denote by $e = ((0), e, (0))$ the "unit section" of Ω.

Our goal now is to put a piece-of-group law on Ω.

Lemma 2.7. *Let $\alpha \in \Delta$, and let $w_{\alpha,k}$ be the element of $Norm_{G_k}(T_k)(k)$ defined by $X_{\alpha,k}$ (recall that by definition*

$$w_{\alpha,k} = p_{-\alpha,k}(-1)\, p_{\alpha,k}(1)\, p_{-\alpha,k}(-1)).$$

There exist an open subset V_α of Ω containing the section e, and a morphism

$$h_\alpha : V_\alpha \to \Omega,$$

satisfying the following conditions:

(i) $h_\alpha(e) = e$,

(ii) $(h_\alpha) \otimes_{\mathbb{Z}} k$ coincides with the restriction of $int(w_{\alpha,k})$ to $V_\alpha \otimes_{\mathbb{Z}} k \subset G_k$.

(iii) One has $T \subset V_\alpha$, and h_α sends T into T. For every $\beta \in R$, one has $U_\beta \subset V_\alpha$ and h_α sends U_β into $U_{s_\alpha(\beta)}$.

By virtue of the definition of a Chevalley system (Exp. XXIII, 6.1), there exists for each $\beta \in R$ an integer $e_\beta = \pm 1$ such that

$$int(w_{\alpha,k})\, p_{\beta,k}(x) = p_{s_\alpha(\beta),k}(e_\beta x)$$

for every $x \in \mathbb{G}_a(S)$, $S \to \mathrm{Spec}(k)$.

Let S be any scheme, and write an arbitrary element of $\Omega(S)$ in the form

$$u = \Big(\prod_{\beta\in R^-,\ \beta\neq-\alpha} p_\beta(x_\beta) \cdot p_{-\alpha}(x_{-\alpha}),\ \ t,\ \ p_\alpha(x_\alpha) \cdot \prod_{\beta\in R^+,\ \beta\neq\alpha} p_\beta(x_\beta) \Big),$$

where one has chosen some (arbitrary) order on $R^- - -\alpha$ and $R^+ - \alpha$ (cf. 2.5).

One defines a morphism $d : \Omega \to \mathrm{Spec}(\mathbb{Z})$ by

$$d(u) = \alpha(t) + x_\alpha x_{-\alpha}.$$

Let V_α be the open subset Ω_d (that is, the open subset of Ω defined by "$d(u)$ invertible"); it contains e, T, and each U_β, $\beta \in R$. Let

$$h_\alpha : V_\alpha \to \Omega$$

be the morphism defined by $h_\alpha(u) = (a(u), b(u), c(u))$ where

$$a(u) = \Big(\prod_{\beta\in R^-,\ \beta\neq-\alpha} p_{s_\alpha(\beta)}(e_\beta x_\beta) \Big) \cdot p_{-\alpha}(-x_\alpha d(u)^{-1}),$$

$$b(u) = t \cdot \alpha^*(d(u)),$$

$$c(u) = p_\alpha(-x_{-\alpha} d(u)^{-1}) \cdot \Big(\prod_{\beta\in R^+,\ \beta\neq\alpha} p_{s_\alpha(\beta)}(e_\beta x_\beta) \Big).$$

Since s_α permutes the positive roots (resp. negative roots) distinct from α (resp. $-\alpha$), $c(u)$ (resp. $a(u)$) is a section of U (resp. U^-) and the previous morphism is well-defined. It satisfies (i) and (iii) trivially. As for (ii), this follows at once from the definition of the e_β, $\beta \in R$, and from Exp. XX, 3.12.

Lemma 2.8. *There exist open subsets V and V' of Ω and morphisms*

$$h : V \to \Omega, \quad h' : V' \to \Omega,$$

satisfying the following conditions:

(i) V and V' contain e, and $h(e) = h'(e) = e$.

(ii) The morphism induced by $h' \circ h : h^{-1}(V') \to \Omega$ is the restriction of the identity morphism.

(iii) The k-morphisms $h \otimes_{\mathbb{Z}} k$ and $h' \otimes_{\mathbb{Z}} k$ are the restrictions to $V \otimes_{\mathbb{Z}} k$ and $V' \otimes_{\mathbb{Z}} k$ of automorphisms of the group G_k.

(iv) V and V' contain U, T and U^-; h and h' send U into U^-, U^- into U, and T into T.

Let w^0 be the element of the Weyl group of G_k that transforms R^+ into R^-. Write

$$w^0 = s_{\alpha_n} \cdots s_{\alpha_1}, \quad \alpha_i \in \Delta$$

(no relation to the numbering of 2.1). Put

$$w_0 = w_{\alpha_n,k} \cdots w_{\alpha_1,k} \in \mathrm{Norm}_{G_k}(T_k)(k).$$

Define by recursion on $i \leq n$ an open subset V_i of Ω and a morphism $h_i : V_i \to \Omega$ by $V_0 = \Omega$, $h_0 = id$, and, for $i = 0, \cdots, n-1$,

$$V_{i+1} = h_i^{-1}(V_{\alpha_{i+1}}), \quad h_{i+1} = h_{\alpha_{i+1}} \circ h_i,$$

where the notations V_{α_j} and h_{α_j} are those of 2.7.

Take $V = V_n$ and $h = h_n$. The conditions in (i), (iii) and (iv) bearing on V and h are indeed verified; for (i) and (iii) this follows at once from 2.8, for (iv) from the fact that $h \otimes_{\mathbb{Z}} k$ is the restriction of $int(w_0)$ to $V \otimes_{\mathbb{Z}} k$.

Since $(w^0)^2 = 1$, one also has

$$w^0 = s_{\alpha_1} \cdots s_{\alpha_n} = (s^3_{\alpha_1}) \cdots (s^3_{\alpha_n}).$$

Setting

$$w_0' = (w_{\alpha_1,k})^3 \cdots (w_{\alpha_n,k})^3,$$

and carrying out the same construction as above, one deduces from it an open subset V' and a morphism h' also satisfying (i), (iii), (iv). Moreover, $h' \otimes_{\mathbb{Z}} k$ is the restriction of $int(w_0')$ to $V' \otimes_{\mathbb{Z}} k$. But for each simple root $\alpha \in \Delta$, one has $(w_{\alpha,k})^4 = e$ (cf. Exp. XX, 3.1), hence $w_0' \cdot w_0 = e$, which shows that $h' \circ h$ induces the identity morphism on a non-empty open subset of $\Omega \otimes_{\mathbb{Z}} k$. But since Ω is smooth and of finite presentation over $\mathbb{Z}$, $\Omega \otimes_{\mathbb{Z}} k$ is schematically dense in Ω, which proves (ii).

Proposition 2.9. *There exist an open subset V_1 of $\Omega \times \Omega$, an open subset V_2 of Ω, and morphisms*

$$\pi : V_1 \to \Omega, \quad \sigma : V_2 \to \Omega,$$

having the following properties:

(i) If $x \in \Omega(S)$, then (e, x) and (x, e) are sections of V_1, and

$$\pi(e, x) = \pi(x, e) = x.$$

(ii) V_2 contains e and $\sigma(e) = e$.

(iii) π_k and σ_k are the restrictions of the morphisms $G_k \times_k G_k \to G_k$ and $G_k \to G_k$ defined by the product (resp. the inverse).

Proof. Let (v, t, u) and (v', t', u') be two sections of Ω. Then $h(u)$ is a section of U^-, $h(v')$ is a section of U by 2.8 (iv), and one can therefore regard $(h(u), e, h(v'))$ as a section of Ω. Let V_1 be the open subset of $\Omega \times \Omega$ defined by the condition:

$$((v, t, u), (v', t', u')) \in V_1(S) \Leftrightarrow (h(u), e, h(v')) \in V'(S)$$

(notation of 2.8). If $((v, t, u), (v', t', u'))$ is a section of V_1, then $h'(h(u), e, h(v'))$ is defined; it is a section of Ω which one can decompose:

$$h'(h(u), e, h(v')) = (v'', t'', u'').$$

One then sets

$$\pi((v, t, u), (v', t', u')) = (v \cdot {}^t v'' t^{-1}, t t'' t', t'^{-1} u'' t' \cdot u').$$

The verification of (i) is immediate (by 2.8 (ii)). To verify the condition in (iii) bearing on π, one sees that

$$h'(h(u), e, h(v')) = u\, v' = v''\, t''\, u''$$

when $u \in U(S)$, $v \in U^-(S)$, $S \to k$, by virtue of 2.8 (iii) and (ii). One constructs σ similarly: if (v, t, u) is a section of Ω, $h(u^{-1})$ is a section of U^-,

$h(v^{-1})$ is a section of U, $h(t^{-1})$ is a section of T, so $(h(u^{-1}), h(t^{-1}), h(v^{-1}))$ is a section of Ω and one can define an open subset V_2 of Ω by

$$(v, t, u) \in V_2(S) \Leftrightarrow (h(u^{-1}), h(t^{-1}), h(v^{-1})) \in V'(S)$$

and a morphism $\sigma : V_2 \to \Omega$ by

$$\sigma(v, t, u) = h'(h(u^{-1}), h(t^{-1}), h(v^{-1})).$$

One verifies the conditions on σ as above.

Corollary 2.10. *π is "generically associative" and σ is a "generic inverse": if $x, y, z \in \Omega(S)$ and if the expressions below are defined (which always happens over an open subset of Ω containing the unit section), one has:*

$$\pi(x, \pi(y, z)) = \pi(\pi(x, y), z), \quad \pi(x, \sigma(x)) = e = \pi(\sigma(x), x).$$

Indeed, the two sides of each of these formulas define morphisms between smooth $\mathbb{Z}$-schemes of finite presentation, which coincide on the generic fibers, by 2.10 (iii).

Corollary 2.11. *Let $\alpha \in R$. For all S and all $x, y \in \mathbb{G}_a(S)$ such that $(p_\alpha(x), p_{-\alpha}(y)) \in V_1(S)$ and $1 + xy \in \mathbb{G}_m(S)$ (which defines an open subset of $\mathbb{G}^2_{a,S}$ containing the section $(0, 0)$), one has:*

$$\pi(p_\alpha(x), p_{-\alpha}(y)) = \left(p_{-\alpha}\left(\frac{y}{1 + xy} \right), \alpha^*(1 + xy), p_\alpha\left(\frac{x}{1 + xy} \right) \right).$$

The proof is the same as before, by Exp. XX, 2.1.

3. Existence theorem: end of the proof

To simplify language, let us pose the following definition.

Definition 3.1. *Let S be a scheme and G an S-group scheme. One says that G is* admissible *if there exists an open immersion of S-schemes $i : \Omega_S = \Omega \times S \to G$ satisfying the following conditions:*

(i) The diagram

$$\begin{tikzcd}
(V_1)_S \arrow[r, "i \times_S i"] \arrow[d, "\pi"] & \Omega_S \times_S \Omega_S \arrow[r, "i \times_S i"] & G \times_S G \arrow[d, "\pi_G"] \\
\Omega_S \arrow[rr, "i_S"] & & G,
\end{tikzcd}$$

where π_G denotes the multiplication morphism in G, is commutative.

(ii) There exists a finite set of sections $a_j \in \Omega(S)$ such that the $i(a_j) \cdot i(\Omega_S)$ cover G.

By Weil's "theorem" (Exp. XVIII, 3.13 (iii) and (iv)), one has:

Lemma 3.2. *If for every scheme S étale and of finite type over $\mathbb{Z}$, every admissible S-group is affine, then there exists an admissible affine $\mathbb{Z}$-group.*

Now one has:

Lemma 3.3. *Let S be a scheme and G an admissible S-group. Then G is smooth and of finite presentation over S, with affine connected semisimple fibers.*

Since Ω_S is smooth and of finite presentation over S, with connected fibers, the same holds for G, by condition (ii). To verify 3.3, one may therefore suppose that S is the spectrum of a field K. Let us identify Ω_K with its image in G. It is clear that Ω_K is the product

$$\prod_{\alpha\in R^-} U_{\alpha,K} \cdot T_K \cdot \prod_{\alpha\in R^+} U_{\alpha,K}$$

of the subgroups T_K and $U_{\alpha,K}$ ($\alpha \in R$) of G. The Lie algebra of G is therefore identified with the direct sum

$$Lie(T_K) \oplus \bigoplus_{\alpha\in R} Lie(U_{\alpha,K}).$$

Since the inner automorphism defined by a section of T_K acts on $U_{\alpha,K}$, and therefore on $Lie(U_{\alpha,K})$, through the character

$$\alpha \in R \subset M \simeq \mathrm{Hom}_{K\text{-gr.}}(T_K, \mathbb{G}_{m,K}),$$

the previous decomposition of $Lie(G)$ is exactly the decomposition under the adjoint action of T. The roots of G_K with respect to T_K are therefore the $\alpha \in R$. Apply Exp. XIX, 1.13. Let T_α be the maximal torus of $Ker(\alpha) \subset T_K$, and let $Z_\alpha = Centr_G(T_\alpha)$; it suffices for us to prove that each Z_α is reductive. Now $Z_\alpha \cap \Omega_K$ is none other than

$$\prod_{\beta\in R^-,\ \beta|_{T_\alpha} = e} U_{\beta,K} \cdot T_K \cdot \prod_{\beta\in R^+,\ \beta|_{T_\alpha} = e} U_{\beta,K};$$

but the roots vanishing on T_α are the rational multiples of α, hence α and $-\alpha$; this proves

$$Z_\alpha \cap \Omega_K = U_{-\alpha,K} \cdot T_K \cdot U_{\alpha,K}.$$

To prove that Z_α is reductive, it suffices, by Exp. XX 3.4, to prove that $U_{\alpha,K}$ and $U_{-\alpha,K}$ do not commute, which follows at once from 2.11.

It follows from 3.2 and 3.3 that the proof will be complete if we prove:

Lemma 3.4. *If S is a locally noetherian scheme of dimension ≤ 1, and if G is a smooth S-group of finite type with affine connected semisimple fibers, then G is affine (and therefore semisimple).*

Nota. In Exp. XVI, it was seen that 3.4 is true without hypothesis on S, but the proof is relatively delicate; since here we need only the particular case 3.4, we give a direct proof of it.

Consider the Lie algebra g of G, which is a locally free $\mathcal{O}_S$-module, and the adjoint representation of G

$$Ad : G \to GL_{\mathcal{O}_S}(g).$$

To prove that G is affine over S, it suffices to prove that the morphism Ad is affine. Since G is smooth with connected fibers, it is separated over S (VI_B 5.5), so the morphism Ad is separated. Using a result proved in the appendix (see 4.1), it suffices to prove that the morphism Ad is quasi-finite. One is thus reduced to the case where S is the spectrum of a field; in this case G is affine, hence semisimple, and one is reduced to Exp. XXII 5.7.14.

4. Appendix

We used in the course of the proof the following proposition:

Proposition 4.1. *Let S be a locally noetherian scheme of dimension ≤ 1, let G and H be two S-group schemes of finite type, and $f : G \to H$ a quasi-finite separated group morphism. If G is flat over S,*[1436] *then f is affine.*

We shall give the proof only in the case where G is smooth over S, a hypothesis which is indeed verified in the application we have made of the proposition.

4.2.

By EGA II, 1.6.4, one may suppose S reduced. By the usual techniques of passage to the limit,[1437] one may suppose S local. If $\dim(S) = 0$, the assertion is trivial,[1438] suppose $\dim(S) = 1$. By faithfully flat descent, one may suppose that S is complete with algebraically closed residue field. Replacing S by its normalization $\tilde{S}$ if necessary, one may (EGA II, 6.7.1 and EGA 0_IV 23.1.5) suppose S normal.[1439] One is therefore reduced to the case where S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field.

4.3.

Let η (resp. s) be the generic (resp. closed) point of S. Consider the image $f_\eta(G_\eta)$ of G_η in H_η. It is a closed subgroup scheme of H_η. Let H' be the schematic closure in H of $f_\eta(G_\eta)$. Since $H' \to H$ is affine (it is a closed immersion), one may replace H by H' and so suppose H flat over S and f_η surjective. Since f_η is finite, and G and H are flat over S, one has

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dim(G_s) = dim(G_η) = dim(H_η) = dim(H_s).
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[1436] N.D.E.: We have added the flatness hypothesis, which had been omitted.

[1437] N.D.E.: cf. EGA IV_3, 8.10.5 (viii).

[1438] N.D.E.: Indeed, if $S = \mathrm{Spec}(k)$ (k a field), then f is the composite of the projection $p : G \to G/Ker(f)$ and a closed immersion i, and since $Ker(f)$ is finite over k, p is finite (VI_B 9.2).

[1439] N.D.E.: Indeed, the irreducible components $S_1, \cdots, S_r$ of S are complete integral noetherian local schemes, hence, by a theorem of Nagata (cf. EGA 0_IV, 23.1.5) the normalization $\tilde{S}_i$ is finite over S_i, and therefore $\tilde{S}$ is finite over S. Then, by a theorem of Chevalley (cf. EGA II, 6.7.1), if $f \times_S \tilde{S}$ is an affine morphism, then so is f.

4.4.

Let $H_s^0, \cdots, H_s^n$ be the irreducible components of H_s, where H_s^0 denotes the neutral component, and let $z_0, \cdots, z_n$ be their generic points. Since each local ring $\mathcal{O}_{H,z_i}$ is of dimension ≤ 1, the morphism $G \times_H \mathcal{O}_{H,z_i} \to \mathcal{O}_{H,z_i}$ is affine because quasi-finite and separated (cf. Exp. XVI, Lemma 4.2), so G is affine over H in a neighborhood of z_i. Denoting by V the largest open subset of H such that $G|_V$ is affine over V, it follows

that V contains all the z_i,[1440] so contains at least one closed point $y_i \in H_s^i$ (and one has $\kappa(y_i) = \kappa(s)$ since $\kappa(s)$ is algebraically closed).

On the other hand, V is obviously stable under the translation defined by any element $g \in G(S)$. But one has $\dim(G_s) = \dim(H_s)$ and f_s is finite, so

$$f(s) : G_s^0(s) \to H_s^0(s)$$

is surjective. Since A is complete and G smooth over S, the canonical map $G^0(S) \to G_s^0(s)$ is surjective; since $H_s^0(s)$ acts transitively on each $H_s^i(s)$, it follows that $V \supset H_s(s)$, hence (since $\kappa(s)$ is algebraically closed) $V \supset H_s$.[1441]

Since one obviously has $V \supset H_\eta$, since f_η is finite, one therefore has $V = H$. *QED*.

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[1440] N.D.E.: The following sentence has been added.

[1441] N.D.E.: When G is not assumed smooth over S, one may proceed as follows. Let h_0 be a rational point of H_s^0 and g_0 a rational point of G_s^0 such that $f(g_0) = h_0$. By VI_B 5.6.1, there exists a commutative diagram

$$\begin{tikzcd} S'' \arrow[r, "g"] \arrow[d, "\pi"'] \arrow[dr, "\varphi"] & G \arrow[d] \\ S' \arrow[r, "w"] & S \end{tikzcd}$$

where w is étale and surjective, π finite and surjective, and $\phi^{-1}(s)$ is formed of a single point s'' such that $g(s'') = g_0$. Then the morphism $G_{S''} \to H_{S''}$ is affine above a neighborhood of $h_0 y_i$, and the same holds for $G_{S'} \to H_{S'}$ (EGA II, 6.7.1), and then for $G \to H$ by faithfully flat descent (EGA IV_2, 2.7.1 (xiii)). Hence $h_0 y_i \in V$, and it follows that V contains $H_s(s)$, and therefore H_s.

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Footnotes

Exposé XXVI. Parabolic subgroups of reductive groups

by M. Demazure

[1442]

This Exposé studies the parabolic subgroups of an S-reductive group G. Its essential result is the conjugacy theorem (5.4). The essential tool is the notion of *transversal position* of two parabolic subgroups, a notion studied systematically in §4. Another fact plays an important role: the decomposition of the unipotent radical $rad^u(P)$ of a parabolic subgroup P as successive extensions of vector groups (2.1)[1443].

Various schemes associated with G are studied in §3; §6 deals with the split subtori[1444] of G and their relations with parabolic subgroups.

Finally, in §7, we briefly expound how, over a semi-local base, one formulates the "relative theory" of reductive groups as presented over fields in the article of A. Borel and J. Tits, *Groupes réductifs*, Publications Mathématiques de l'IHÉS, no. 27. In that article, cited [BT65] in what follows, the reader will moreover find, in the case of a base field, other results that have not been touched on here.

1. Recollections. Levi subgroups

Definition 1.1. *Let S be a scheme, G an S-reductive group, P a sub-S-group-scheme of G. One says that P is a* parabolic subgroup *of G if*

(i) P is smooth over S,

[1442] *N.D.E.* Version of 13/10/2024.

[1443] *N.D.E.* Since the Levi subgroups of P form a torsor under $rad^u(P)$ (1.9), this entails, when S is semi-local, that P has a Levi subgroup and hence a maximal torus (2.4).

[1444] *N.D.E.* The terminology "trivial torus" has been replaced by that of "split torus".

(ii) for each $s \in S$, the s-quotient-scheme G_s/P_s is proper (i.e. Bible, §6.4, Th. 4 (= [Ch05], §6.5, Th. 5), P_s contains a Borel subgroup of G_s).

Proposition 1.2 (Exp. XXII, 5.8.5). *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G. Then P is closed in G, with connected fibers, and*

$$P = Norm_G(P).$$

Moreover, the quotient sheaf G/P is representable by an S-scheme that is smooth and projective over S.

Proposition 1.3 (Exp. XXII, 5.3.9 and 5.3.11). *Let S be a scheme, G an S-reductive group, P and P' two parabolic subgroups of G. The following conditions are equivalent:*

(i) P and P' are conjugate in G, locally for the étale (resp. (fpqc)) topology.

(ii) For each $s \in S$, P_s and P'_s are conjugate by an element of $G(s)$.

(iii) The strict transporter $Transt_G(P, P')$ of P into P' (defined by

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Transt_G(P, P')(S') = { g ∈ G(S') | int(g) P_{S'} = P'_{S'} }
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for every $S' \to S$) is a closed subscheme of G, smooth and of finite presentation over S, which is a principal homogeneous bundle on the right under P and on the left under P'.

Proposition 1.4. *Let S be a nonempty scheme, (G, T, M, R) an S-split reductive group, R' a subset of R. The following conditions on R' are equivalent:*

(i) $g_{R'} = t \bigoplus \bigoplus_{\alpha \in R'} g^{\alpha}$ is the Lie algebra of a parabolic subgroup of G containing T (necessarily unique, Exp. XXII, 5.3.5).

(ii) R' is of type (R) (Exp. XXII, 5.4.2) and contains a system of positive roots.

(iii) R' is a closed subset of R and satisfies: if $\alpha \in R - R'$, then $-\alpha \in R'$ (that is, $R = R' \cup (-R')$).

(iv) There exist a system of simple roots Δ, and a subset A of Δ such that R' is the union of the set of positive roots and the set of negative roots that are linear combinations of the elements of A.

(v) R' contains a system of simple roots of R; moreover, if $\Delta \subset R'$ is a system of simple roots of R and one sets

$$A = (-R') \cap \Delta,$$

then R' is the union of the set of positive roots and the set of negative roots that are linear combinations of the elements of A.

One has (i) ⇔ (ii) by Exp. XXII, 5.4.5 (ii) and 5.5.1. One has (iii) ⇒ (ii) by Exp. XXI, 3.3.6 and Exp. XXII, 5.4.7. One trivially has (v) ⇒ (iv) ⇒ (iii). One has (iii) ⇒ (v) by Exp. XXI, 3.3.6 and 3.3.10.

It therefore remains to prove that (i) implies that R' is a closed subset of R. But this last assertion can be verified on any geometric fiber; one may therefore assume that S is the spectrum of an algebraically closed field.

Let P be the parabolic subgroup of G containing T whose Lie algebra is $\mathfrak{g}_{R'}$. Since the Borel subgroups of P are the Borel subgroups of G contained in P, it follows from Bible, §12.3, Th. 1, and from Exp. XXII, 5.4.5 (i), that if one denotes by U the unipotent radical of P, then $T \cdot U$ is the subgroup of G containing T whose Lie algebra is

$$\mathfrak{g}_{R''} = \mathfrak{t} \oplus \coprod_{\alpha \in R''} \mathfrak{g}^{\alpha},$$

where R'' is the intersection of the systems of positive roots of R contained in R'. In particular, it follows that R'' is closed and that $R'' \cap (-R'') = \emptyset$. On the other hand, the group $H = P/U$ is reductive, the canonical image $\bar{T}$ of T is a maximal torus of it ($T \to \bar{T}$ is an isomorphism), and one has an isomorphism of $\bar{T}$-modules, i.e. of M-graded vector spaces,

$$\mathrm{Lie}(H) \simeq \mathfrak{t} \oplus \coprod_{\alpha \in R_s} \mathfrak{g}^{\alpha},$$

where R_s is the complement of R'' in R'. It follows that R_s is naturally identified with the set of roots of H relative to $\bar{T}$, and in particular satisfies $R_s = -R_s$. One immediately deduces that

$$R'' = \{ \alpha \in R' \mid -\alpha \notin R' \}, \qquad R_s = \{ \alpha \in R' \mid -\alpha \in R' \}.$$

Let us now show that R' is closed. Let $\alpha, \beta \in R'$ such that $\alpha + \beta \in R$; let us prove that $\alpha + \beta \in R'$. If $\alpha, \beta \in R''$, then $\alpha + \beta \in R''$ because R'' is closed. If $\alpha \in R_s$, $\beta \in R''$, and if $\alpha + \beta \notin R'$, then $\alpha + \beta \in -R''$, and one has $-\alpha = -(\alpha + \beta) + \beta \in R''$ because R'' is closed, which entails $-\alpha \in R'' \cap R_s$ and contradicts the fact that $R_s \cap R'' = \emptyset$. It therefore remains to study the case where $\alpha, \beta \in R_s$. If $\alpha + \beta \notin R'$, then $\alpha+\beta \in -R''$. But, as $\alpha+\beta \neq 0$, there exists a system of positive roots of the root system R_s containing α and β, hence a Borel subgroup of $H = P/U$ containing the canonical image of U_α and U_β. Its inverse image in P is a Borel subgroup containing U_α, U_β and U, hence U_α, U_β and $U_{-(\alpha+\beta)}$, which is impossible.

Corollary 1.5. *A parabolic subgroup of a reductive group is of type (RC) (Exp. XXII, 5.11.1).*

Proposition 1.6 (Exp. XXII, 5.11.4). *Let S be a scheme, G an S-reductive group, P a subgroup of G of type (RC)*[1445].

(i) P possesses a largest invariant sub-group-scheme that is smooth and of finite presentation over S, with connected unipotent geometric fibers. It is a characteris-

[1445] *N.D.E.* “Parabolic subgroup” has been replaced by “subgroup of type (RC)”, as in Exp. XXII, 5.11.4, because it will be useful later on (4.5.1, 6.17) to be able to apply this statement to $P \cap P'$, when P, P' are two parabolic subgroups such that $P \cap P'$ is of type (RC).

tic subgroup of P, called the unipotent radical *of P, denoted $rad^u(P)$. The quotient sheaf $P/rad^u(P)$ is representable by an S-reductive group.*

(ii) If T is a maximal torus of P, P possesses a reductive subgroup L containing T such that:

(a) Every reductive subgroup of P containing T is contained in L.

(b) P is the semidirect product $L \cdot rad^u(P)$, i.e. the canonical morphism $L \to P/rad^u(P)$ is an isomorphism.

Moreover, L is the unique subgroup (resp. reductive subgroup) of P containing T and satisfying (b) (resp. (a)). Finally, one has

```
Norm_P(L) = L,        Norm_P(T) = Norm_L(T).
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1.7.

A subgroup L of P satisfying condition (b) above is called a *Levi subgroup* of P. It is a maximal reductive subgroup of P; indeed, it is reductive, since isomorphic to $P/rad^u(P)$. Let us show that it is maximal for this property. Let L' be a reductive subgroup of P containing L; to prove that $L' = L$, one may reason locally for the (fpqc) topology, and hence assume that L has a maximal torus T, and one is reduced to 1.6 (ii).

If L and L' are two Levi subgroups of P, then L and L' are conjugate in P, locally for the (fpqc) topology. Indeed, locally for this topology, one may assume that L (resp. L') has a maximal torus T (resp. T'); since T and T' are conjugate in P locally for the (fpqc) topology, one may assume $T = T'$, and one then has $L = L'$, by 1.6 (ii). But since $P = L \cdot rad^u(P)$ and $Norm_P(L) = L$, one immediately deduces:

Corollary 1.8. *Let P be a subgroup of type (RC)*[1446] *of the S-reductive group G. If L and L' are two Levi subgroups of P, there exists a unique $u \in rad^u(P)(S)$ such that $int(u)L = L'$.*

Let us denote by $Lev(P)$ the functor of Levi subgroups of P: for $S' \to S$, $Lev(P)(S')$ is the set of Levi subgroups of $P_{S'}$. From 1.8 one deduces:

Corollary 1.9. *Let P be a subgroup of type (RC)*[1447] *of the S-reductive group G. Then $Lev(P)$ is a principal homogeneous bundle under the S-group $rad^u(P)$, and in particular is representable by an S-scheme that is smooth and affine over S, with integral geometric fibers.*

It follows immediately from 1.6:

Corollary 1.10. *Let P be a subgroup of type (RC)*[1448] *of the S-reductive group G. The functor $Tor(P)$ of maximal tori of P is representable by an S-scheme that is smooth and affine, the "relation $L \supset T$" defines a morphism*

[1446] *N.D.E.* cf. N.D.E. (3).
[1447] *N.D.E.* cf. N.D.E. (3).
[1448] *N.D.E.* cf. N.D.E. (3).

$$Tor(P) \longrightarrow Lev(P),$$

the fiber of which over $L \in Lev(P)(S)$ is identified with $Tor(L)$ (Exp. XXII, 5.8.3).

The first assertion of 1.10 is a consequence of the other two and of Exp. XXII, 5.8.3.

Definition 1.11. *Let S be a nonempty scheme, G an S-reductive group, P a parabolic subgroup of G, $E = (T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$ a pinning of G. One says that E is* adapted *to P, or that E is a* pinning of the pair (G, P), *if $P \supset T$ and if the Lie algebra of P is of the form $t \bigoplus \bigoplus_{\alpha\in R'} g^\alpha$, where R' is a subset of R containing R^+.*

In particular, if $T \subset B$ is the Killing pair defined by the pinning, one has $T \subset B \subset P$.

Under the preceding conditions, one sets $\Delta(P) = \Delta \cap (-R')$; then, by Exp. XXII, 5.4.3, one has:

$$\alpha \in \Delta(P) \iff \alpha \in \Delta \text{ and } U_{-\alpha} \subset P \iff \alpha \in \Delta \text{ and } U_{-\alpha} \cap P \neq e.$$

It follows immediately from 1.4 (v) and Exp. XXII, 5.11.3 and 5.10.6:

Proposition 1.12. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, $(T, M, R, \Delta, (X_\alpha)_{\alpha\in\Delta})$ a pinning of G adapted to P, $\Delta(P)$ the subset of Δ defined above.*

(i) The unipotent radical $rad^u(P)$ of P is none other than

$$U_{R''} = \prod_{\alpha \in R''} U_\alpha,$$

where R'' is the set of positive roots that, in their decomposition along Δ, involve at least one element of $\Delta - \Delta(P)$[1449] with a nonzero coefficient.

(ii) The unique Levi subgroup L of P containing T is none other than

$$Z_{\Delta(P)} = Centr_G(T_{\Delta(P)}),$$

where $T_{\Delta(P)}$ is the maximal torus of $\bigcap_{\alpha\in\Delta(P)} Ker(\alpha)$; moreover, one has $T_{\Delta(P)} = rad(L)$.

Corollary 1.13. *Every Levi subgroup L of the parabolic subgroup P of the reductive group G is a* critical subgroup *of G, i.e. satisfies (Exp. XXII, 5.10.4):*

$$L = Centr_G(rad(L)).$$

This follows immediately from 1.12 and from the following lemma, which is contained in 1.4 and Exp. XXII, 5.4.1:

Lemma 1.14. *Locally for the étale topology, every pair (G, P), where P is a parabolic subgroup of the reductive group G, may be pinned (1.11).*

[1449] *N.D.E.* $\Delta(P)$ has been corrected to $\Delta - \Delta(P)$.

Let us note:

Proposition 1.15. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, E and E' two pinnings of G adapted to P. The unique inner automorphism of G over S that transforms E into E' (Exp. XXIV, 1.5) comes from P, via the morphism*

$$P \to P/\mathrm{Centr}(P) = P/\mathrm{Centr}(G) \to G/\mathrm{Centr}(G).$$

Indeed, it suffices to reason as in Exp. XXIV, 1.5, using:

Lemma 1.16 (Exp. XXII, 5.3.14 and 5.2.6). *The maximal tori (resp. Borel subgroups, resp. Killing pairs) of a parabolic subgroup P of the S-reductive group G are conjugate in P, locally for the étale topology.*

Proposition 1.17. *Let S be a scheme, G an S-reductive group, P and P' two parabolic subgroups of G, B a Borel subgroup contained in P and P'. If P and P' are conjugate in G locally for the étale topology, then $P = P'$.*

Indeed, one may assume that there exists $g \in G(S)$ such that $int(g)P = P'$. Then B and $int(g)^{-1}B$ are two Borel subgroups of P. Possibly after extending S, by 1.16 one may assume that there exists $p \in P(S)$ such that $int(p)int(g^{-1})B = B$. Then

$$p\, g^{-1} \in \mathrm{Norm}_G(B)(S) = B(S),$$

and $g \in B(S) \cdot p \subset P(S)$, hence $P' = int(g)P = P$.

Remark 1.18. *If P and P' are two parabolic subgroups of G containing a common Borel subgroup, then $P \cap P'$ is again a parabolic subgroup of G. Indeed, it is smooth along the unit section (Exp. XXII, 5.4.5), and it contains a Borel subgroup.*

Proposition 1.19. *Let S be a scheme, G an S-reductive group, G' its derived group (Exp. XXII, 6.2.1).*

(i) The maps

$$P \mapsto P' = P \cap G' \quad \text{and} \quad P \mapsto P' \cdot \mathrm{rad}(G) = \mathrm{Norm}_G(P')$$

are mutually inverse bijections between the set of parabolic subgroups of G and the set of parabolic subgroups of G'. One has $rad^u(P) = rad^u(P')$.

(ii) Let P be a parabolic subgroup of G and $P' = P \cap G'$. The maps

$$L \mapsto L' = L \cap G' = L \cap P'$$
$$L' \mapsto L' \cdot \mathrm{rad}(G) = \mathrm{Centr}_G(\mathrm{rad}(L'))$$

are mutually inverse bijections between the set of Levi subgroups of P and the set of Levi subgroups of P'. Moreover, one has $rad(L') = (rad(L) \cap G')^0$.

The proof (by reduction to the split case, for example) is straightforward and is left to the reader, together with that, immediate, of:

Proposition 1.20. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, L a Levi subgroup of P. The maps*

$$Q \mapsto Q \cap L = Q', \qquad Q' \mapsto Q' \cdot rad^u(P)$$

are mutually inverse bijections between the set of parabolic subgroups of G contained in P and the set of parabolic subgroups of L. Moreover, the Levi subgroups of Q' are the Levi subgroups of Q contained in L.

One may complete 1.6 as follows:

Proposition 1.21. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G.*

(i) P possesses a largest invariant subgroup that is smooth and of finite presentation over S, with connected solvable geometric fibers. It is a characteristic subgroup of P, called the radical *of P, and denoted $rad(P)$. The quotient sheaf $P/rad(P)$ is representable by an S-semisimple group.*

(ii) If L is a Levi subgroup of P, $rad(P)$ is the semidirect product of $rad^u(P)$ and $rad(L)$; one has $rad(L) = L \cap rad(P)$, hence $L = Centr_G(L \cap rad(P))$, and $P/rad(P) \simeq L/rad(L)$.

Indeed, assertion (i) being local, one may assume that G has a Levi subgroup L, and one is reduced to proving that $R = rad^u(P) \cdot rad(L)$ has the properties announced in (i), which is immediate. For (ii), it only remains to show that $rad(L) = L \cap rad(P)$, which follows immediately from the fact that $L \cap rad(P)$ is smooth and has connected fibers, L being the centralizer of a torus[1450].

2. Structure of the unipotent radical of a parabolic subgroup

Proposition 2.1. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, $rad^u(P)$ its unipotent radical. There exists a sequence of sub-group-schemes of $rad^u(P)$*

$$rad^u(P) = U_0 \supset U_1 \supset U_2 \supset \cdots \supset U_n \supset \cdots$$

possessing the following properties:

(i) Each U_i is smooth, with connected fibers, characteristic and closed in P. The commutator of a section of U_i and a section of U_j is a section of U_{i+j+1} (over a varying $S' \to S$).

(ii) For each $i \geqslant 0$, there exists a locally free $\mathcal{O}_S$-module E_i and an isomorphism of S-group sheaves

$$U_i/U_{i+1} \xrightarrow{\sim} W(E_i).$$

Moreover, the automorphisms of P (over a varying $S' \to S$) act linearly on $W(E_i)$.

(iii) For every $s \in S$, one has $U_{n,s} = e$ for $n > \dim(rad^u(P)_s)$.

[1450] *N.D.E.* cf. XIX 1.4.

2.1.1.

Suppose first that the pair (G, P) is pinnable. Let $(T, M, R, \Delta, \cdots)$ be a pinning of G adapted to P; let $\Delta(P)$ be the part of Δ defined by P. Let $\alpha_1, \cdots, \alpha_p$ be

the elements of $\Delta(P)$, and $\beta_1, \cdots, \beta_q$ the elements of $\Delta - \Delta(P)$. Every root $\gamma \in R$ is written uniquely as

$$\gamma = a_1 \alpha_1 + \cdots + a_p \alpha_p + b_1 \beta_1 + \cdots + b_q \beta_q.$$

Set

$$a(\gamma) = b_1 + \cdots + b_q.$$ [^N.D.E-XXVI-7]

It follows immediately from the definitions that the following properties hold (cf. 1.12):

(i) $U_\gamma \subset P \Leftrightarrow a(\gamma) \geqslant 0$.

(ii) $U_\gamma \subset rad^u(P) \Leftrightarrow a(\gamma) > 0$.

(iii) $a(n\gamma + m\gamma') = na(\gamma) + ma(\gamma')$ for all $n, m \in \mathbb{Z}$.

For $i > 0$, let R_i be the set of roots $\gamma \in R$ such that $a(\gamma) > i$. Each R_i is a closed set of roots satisfying $R_i \cap (-R_i) = \emptyset$. Consider (Exp. XXII, 5.6.5) the S-group

$$U_i = U_{R_i} = \prod_{\gamma \in R_i} U_\gamma.$$

It is a closed sub-group-scheme of G, smooth over S, with connected fibers.

Let $\alpha, \beta \in R$; consider the commutation relation of Exp. XXII, 5.5.2

$$p_\alpha(x)\, p_\beta(y)\, p_\alpha(-x) = p_\beta(y) \prod_{n, m \in \mathbb{N}^*} p_{n\alpha + m\beta}(C_{n, m, \alpha, \beta}\, x^n y^m),$$

where each p_γ is an isomorphism of vector groups $G_{a,S} \xrightarrow{\sim} U_\gamma$. Let us first remark that if $a(\alpha) > i$ and $a(\beta) > j$, one has

$$a(n\alpha + m\beta) = n\, a(\alpha) + m\, a(\beta) \geqslant n(i + 1) + m(j + 1) > i + j + 1$$

when n and m are > 0. It follows that the commutator of a section of U_α and a section of U_β is a section of U_{i+j+1} (over a varying $S' \to S$), which indeed entails $(U_i, U_j) \subset U_{i+j+1}$. For each $i \geqslant 0$, the quotient U_i/U_{i+1} is therefore commutative; it is naturally identified with

$$U_i / U_{i+1} \simeq \prod_{a(\gamma) = i+1} U_\gamma \simeq W(E_i),$$

where E_i is the direct sum of the g^γ for $a(\gamma) = i + 1$.

Let us return to the above commutation formula, and suppose $a(\alpha) \geqslant 0$, $a(\beta) > i$. If $n, m \in \mathbb{N}^*$,

– either $a(n\alpha + m\beta) > i + 1$,

– or $a(n\alpha + m\beta) = i + 1$, in which case one necessarily has $m = 1$.

This proves first that $int(p_\alpha(x))$ respects U_i (hence also U_{i+1}), and then that, in the expression of $int(p_\alpha(x))p_\beta(y)$, only terms of the form $p_{n\alpha+\beta}(C_{n,1,\alpha,\beta}x^n y)$ occur modulo U_{i+1}, which are therefore linear in y. It follows that the inner automorphisms defined by sections of U_α act linearly on the quotient U_i/U_{i+1} identified with $W(E_i)$. As this is also trivially true for the inner automorphisms defined by sections of T, and as P is generated by T and the U_α, $a(\alpha) \geqslant 0$, one deduces that:

(i) each U_i is invariant in P,

(ii) the inner automorphisms defined by sections of P act linearly on $U_i/U_{i+1} \simeq W(E_i)$.

2.1.2.

Now let $(T', M', R', \Delta', \cdots)$ be a new pinning of G adapted to P.

By 1.15, there exists an inner automorphism of G coming from P that transforms the old pinning into the new one. Possibly after extending S, one may assume that this inner automorphism is of the form $int(p)$, $p \in P(S)$. If one takes up the preceding constructions using the new pinning, it is clear that the groups U_i' and the isomorphisms $U_i'/U_{i+1}' \simeq W(E_i')$ obtained are deduced from U_i and $U_i/U_{i+1} \simeq W(E_i)$ by transport of structure via $int(p)$. It follows from remarks (i) and (ii) above that one therefore has $U_i' = U_i$, and that the two vector structures constructed on $U_i/U_{i+1} = U_i'/U_{i+1}'$ coincide.

This shows that the groups U_i and the vector structures on the quotients U_i/U_{i+1} are independent of the pinning considered (and in particular invariant under every automorphism of P, as one sees easily).

We have therefore proved the proposition when the pair (G, P) is pinnable (part (iii) is trivial, since by Exp. XXI 3.1.2, the set $a(\gamma)|\gamma \in R$ is an interval of $\mathbb{Z}$, hence one cannot have $\dim(U_{i,s}) = \dim(U_{i+1,s})$ unless $U_{i,s} = e$).

2.1.3.

In the general case, there exists a covering family for the (fpqc) topology $S_j \to S$ such that each pair (G_{S_j}, P_{S_j}) is pinnable (1.14). By the preceding, one has descent data on the $rad^u(P_{S_j})_i$, compatible with the vector structures of the quotients, and one concludes by descent of closed subschemes (resp. locally free modules)[1451].

Corollary 2.2. *Let S be an affine scheme, G an S-reductive group, P a parabolic subgroup of G. One has*

$$H^1(S, rad^u(P)) = 0,$$

i.e. every principal homogeneous bundle under $rad^u(P)$ is trivial.

[1451] *N.D.E.* cf. SGA 1, VIII 1.3, 1.9 and 1.10.

Indeed, S decomposes as a sum of subschemes on each of which $rad^u(P)$ is of constant relative dimension. One may

therefore, by (iii), assume that there exists n such that $U_n = e$. Since $H^1(S, U_i/U_{i+1}) = H^1(S, W(E_i)) = 0$ by TDTE I, B, 1.1 (or SGA 1, XI 5.1), one concludes at once.

Corollary 2.3. *Under the preceding conditions, P possesses a Levi subgroup L. If L is a Levi subgroup of P, the canonical map*

$$H^1(S, L) \to H^1(S, P)$$

is bijective (cf. the introduction of Exp. XXIV for the definition of $H^1(S, -)$).

The first assertion follows from 2.2 and 1.9. The canonical map $H^1(S, L) \to H^1(S, P)$ is surjective, because P is the semidirect product $L \cdot rad^u(P)$. To prove that it is injective, it suffices to see that for every principal homogeneous bundle Q under L, one has $H^1(S, rad^u(P)_Q) = 0$, where the subscript Q denotes twisting by the L-bundle Q. This can be proved in two ways: one may take up the proof of 2.2, using the fact that the vector structures on the U_i/U_{i+1} are invariant under L; one may also remark that $rad^u(P)_Q$ is identified with the unipotent radical of the parabolic subgroup P_Q of G_Q, and apply 2.2 to P_Q.

Corollary 2.4. *Let S be a semi-local scheme, G an S-reductive group, P a parabolic subgroup of G. There exists a maximal torus T of G contained in P.*

Indeed, in view of 2.3, P has a Levi subgroup L, and it suffices to prove that L has a maximal torus, which follows from Exp. XIV, 3.20.

Corollary 2.5. *Let S be an affine scheme, G an S-reductive group, P a parabolic subgroup of G. There exists a locally free $\mathcal{O}_S$-module E such that $rad^u(P)$ is isomorphic as an S-scheme to $W(E)$.*

Indeed, let us prove by induction on i that one has an isomorphism of S-schemes

$$rad^u(P) / U_i \simeq W(E_0 \oplus E_1 \oplus \cdots \oplus E_{i-1}).$$

This is clear for $i = 0$. Assume $i > 0$; then $rad^u(P)/U_i$ is a principal homogeneous bundle with base $rad^u(P)/U_{i-1}$, under the group

$$(U_{i-1} / U_i)_{rad^u(P) / U_{i-1}} \simeq W(E_{i-1} \otimes \mathcal{O}_{rad^u(P) / U_{i-1}}).$$

Now the base is affine (e.g. by the induction hypothesis), so this bundle is trivial (TDTE I or SGA 1 XI, *loc. cit.*), and there exists an isomorphism of S-schemes $rad^u(P)/U_i \simeq (rad^u(P)/U_{i-1}) \times_S (U_{i-1}/U_i)$, which completes the proof.

Corollary 2.6. *Let S be a semi-local scheme, s_i the set of its closed points, G an S-reductive group, P a parabolic subgroup of G. The canonical map*

$$rad^u(P)(S) \to \prod_i rad^u(P)(\operatorname{Spec} \kappa(s_i))$$

is surjective.

Indeed, if $S = \mathrm{Spec}(A)$, $\kappa(s_i) = A/p_i$, and if E is given by the projective (hence flat) module E, we must prove that the map

$$E \to \prod_i E \otimes_A A/p_i$$

is surjective. It suffices to do this when $E = A$, in which case it is well known (cf. Bourbaki, *Alg. Comm.* Chap. II, §1, no. 2, Proposition 5).

Corollary 2.7. *Let k be an infinite field, G a k-reductive group, P a parabolic subgroup of G; then $rad^u(P)(k)$ is dense in $rad^u(P)$.*

Corollary 2.8. *Let S be a semi-local scheme, s_i the set of its closed points, G an S-reductive group, P a parabolic subgroup of G, and L_i a Levi subgroup of P_{s_i} for each i. There exists a Levi subgroup L of P inducing L_i for each i.*

Indeed, let L_0 be a Levi subgroup of P (2.3). For each i, let $u_i \in rad^u(P)(\mathrm{Spec}(\kappa(s_i)))$ be such that $int(u_i)L_{0,s_i} = L_i$ (1.8); if $u \in rad^u(P)(S)$ induces u_i for each i (2.6), then $L = int(u)L_0$ answers the question.

Corollary 2.9. *In the situation of 2.1, let moreover H be a sub-group-scheme of G, smooth and of finite presentation over S, with connected fibers, such that $P \cap H$ contains locally for the (fpqc) topology a maximal torus of G. Then for each $i \geqslant 0$, there exists a locally direct factor submodule F_i of E_i such that the isomorphism $U_i/U_{i+1} \xrightarrow{\sim} W(E_i)$ induces an isomorphism of groups*

$$(U_i \cap H) / (U_{i+1} \cap H) \xrightarrow{\sim} W(F_i).$$

Indeed, H is a subgroup of type (R) of G (Exp. XXII, 5.2.1). On the other hand, the assertion to be proved is local for the (fpqc) topology, and one may assume G split relative to a maximal torus of $P \cap H$; one may even reduce to the situation of 2.1.1, with H defined by a subset R' of R. Taking up the notation of *loc. cit.*, one sees by Exp. XXII, 5.6.7 (ii) that $U_i \cap H = \prod_{\alpha \in R_i \cap R'} U_\alpha$, hence that $(U_i \cap H)/(U_{i+1} \cap H)$ is identified with $\prod_{\alpha \in R', a(\alpha) = i+1} U_\alpha$, which yields the result.

Corollary 2.10. *In the situation of 2.9, the conclusions of 2.2, 2.5, 2.6, 2.7 are also valid when $rad^u(P)$ is replaced by $rad^u(P) \cap H$.*

Corollary 2.11.[1452] *Let G be an S-reductive group, P a parabolic subgroup, H a subgroup of type (RC) of P such that $rad^u(H) = rad^u(P) \cap H$. Then statements 2.2 to 2.8 are also valid when P is replaced by H.*

3. Scheme of parabolic subgroups of a reductive group

3.1.

Let E be a finite twisted constant S-scheme (Exp. X, 5.1). Consider the S-functor $Of(E)$, where $Of(E)(S')$ is the set of open and closed subschemes of $E_{S'}$ (or, equivalently, the set of open and closed subsets of $E_{S'}$); then $Of(E)$ is representable by a finite twisted constant S-scheme. Indeed, if $E = A_S$, where A is a finite set, one

[1452] *N.D.E.* Corollary 2.11 has been added; it will be used in 4.5.1 and 6.17.

has immediately $Of(E) \simeq P(A)_S$ (where $P(A)$ denotes the power set of A), and one concludes by descent of open and closed subschemes. One trivially has:

$$Of(E_{S'}) = Of(E)_{S'}, \qquad Of(E \times_S E') = Of(E) \times_S Of(E').$$

3.2.

Let S be a scheme, G an S-reductive group. The functor $Par(G)$ of parabolic subgroups of G is defined by

$$Par(G)(S') = \text{the set of parabolic subgroups of } G_{S'}.$$

In particular, $G \in Par(G)(S)$, $Bor(G) \subset Par(G)$. We propose to define a morphism

$$t : Par(G) \longrightarrow Of(Dyn(G))$$

possessing the following properties:

(i) t is functorial in G (with respect to isomorphisms) and commutes with base change.

(ii) If $(T, M, R, \Delta, \cdots)$ is a pinning of G adapted to the parabolic subgroup P (1.11), the canonical isomorphism $Dyn(G) \simeq \Delta_S$ (Exp. XXIV, 3.4 (iii)) transforms $t(P)$ into $\Delta(P)_S$ (notations of 1.11, 1.12).

Let first P be a parabolic subgroup of G and $(T, M, R, \Delta, \cdots)$ a pinning of G adapted to P. One defines $t(P)$ by (ii); the subscheme $t(P)$ of $Dyn(G)$ thus constructed is independent of the pinning chosen. Indeed, if $(T', M', R', \Delta', \cdots)$ is another pinning of G adapted to P, the unique inner automorphism of G transforming the first pinning into the second comes from P (1.15); the canonical isomorphism $\Delta \xrightarrow{\sim} \Delta'$ therefore transforms $\Delta(P)$ into $\Delta'(P)$, which entails the announced result.

If now one no longer assumes (G, P) to be pinnable, it follows immediately from 1.14 and the definition of $Dyn(G)$ (Exp. XXIV, 3.3) that one may define by descent an open and closed subscheme $t(P)$ of $Dyn(G)$, unique, such that for every $S' \to S$ for which $(G, P)_{S'}$ is pinnable, one has $t(P)_{S'} = t(P_{S'})$.

Theorem 3.3. *Let S be a scheme, G an S-reductive group,*

$$t : Par(G) \longrightarrow Of(Dyn(G))$$

the morphism defined above.

(i) For two parabolic subgroups P and P' of G to be conjugate locally for the (fpqc) topology (cf. 1.3), it is necessary and sufficient that $t(P) = t(P')$.

(ii) $Par(G)$ is representable, and the morphism t is smooth, projective, with integral geometric fibers.

By 3.2 (i), and the fact that inner automorphisms of G act trivially on $Dyn(G)$ (Exp. XXIV, 3.4 (iv)), one indeed has $t(P) = t(P')$ when P and P' are conjugate. Conversely, let P and P' be two parabolic subgroups of G such that $t(P) = t(P')$; let us prove that P and P' are conjugate in G, locally for the (fpqc) topology; one may first

assume that the pairs (G, P) and (G, P') are pinnable (1.14); by conjugacy of pinnings in G (Exp. XXIV, 1.5), one may assume that there exists a pinning (T, M, R, Δ) of G adapted to P and P'. Then $t(P) = t(P')$ implies $\Delta(P) = \Delta(P')$, hence $P = P'$ (cf. 1.4 (v)). We have therefore proved (i). To demonstrate (ii), let us take up the notation of Exp. XXII, 5.11.5[1453].

We have a canonical morphism $Par(G) \to H_c$, and it is clear (e.g. by reduction to the pinned case) that it fits into a cartesian square (where the vertical arrows are monomorphisms)

$$\begin{tikzcd} Par(G) \arrow[r, "t"] \arrow[d] & Of(Dyn(G)) \arrow[d] \\ H_c \arrow[r, "c\ell"] & C\ell_c. \end{tikzcd}$$

Now (*loc. cit.*) H_c is representable and the morphism $c\ell$ is smooth, quasi-projective, of finite presentation, with integral geometric fibers, hence the same holds for t.

It remains to prove that t is proper; but this is now a local assertion for the (fpqc) topology, and one may reduce to the pinned case $G = (G, T, M, R, \Delta, \cdots)$. One then has $Dyn(G) \simeq \Delta_S$, and it suffices to prove that for every subset Δ_1 of Δ, the S-scheme $t^{-1}((\Delta_1)_S)$ is proper over S. Now if P_1 is the parabolic subgroup of G containing T such that $\Delta(P_1) = \Delta_1$, it follows from (i) that the morphism $G \to Par(G)$ defined set-theoretically by $g \mapsto int(g)P_1$ induces an isomorphism of $G/Norm_G(P_1)$ onto $t^{-1}((\Delta_1)_S)$. Now, by 1.2, $G/Norm_G(P_1) = G/P_1$ is projective over S.

Definition 3.4. *$Of(Dyn(G))$ is called the* scheme of types of parabolics *of G; $t(P)$ is called the* type *of P.*

Corollary 3.5. *The S-functor $Par(G)$ is representable by an S-scheme that is smooth and projective over S. The decomposition*

$$Par(G) \longrightarrow Of(Dyn(G)) \longrightarrow S$$

is the Stein factorization (EGA III, 4.3.3) of the structural morphism $Par(G) \to S$.

Corollary 3.6. *For each $t \in Of(Dyn(G))(S)$, the S-scheme*

$$Par_t(G) = t^{-1}(t)$$

[1453] *N.D.E.* One recalls (cf. *loc. cit.*) that $H_c = H_c(G)$ denotes the functor of subgroups of G of type (RC), $C\ell_c = C\ell_c(G)$ the functor of "conjugacy classes" of such subgroups, and that $c\ell : H_c \to C\ell_c$ is the canonical projection.

of parabolic subgroups of G of type t is smooth and projective over S, homogeneous under G. If P is a parabolic subgroup of G, one has a canonical isomorphism $G/P \xrightarrow{\sim} Par_{t(P)}(G)$. One has $Par_{\emptyset}(G) = Bor(G)$, $Par_{Dyn(G)}(G) \xrightarrow{\sim} S$.

Remark 3.7. *The S-scheme $Of(Dyn(G))$ is equipped with a natural order structure (the relation of domination, here set-theoretic inclusion, between subschemes). This order structure is a lattice; in particular, the infimum of two open and closed subschemes of $Dyn(G)_{S'}$ is obviously their intersection. Let us moreover remark that if B is a Borel subgroup of G, one may define the functor X of parabolic subgroups of G containing B. The morphism $X \to Of(Dyn(G))$ induced by t is an isomorphism (for the structure of "ordered scheme"), by virtue of the assertion $P \subset Q \Rightarrow t(P) \subset t(Q)$ and of:*

Lemma 3.8. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G, t' a section of $Of(Dyn(G))$ over S, such that $t(P) \subset t'$. There exists a unique parabolic subgroup P' of G, containing P, and such that $t(P') = t'$.*

Possibly after extending the base, one may assume that P contains a Borel subgroup B of G. The uniqueness of P' then follows from 1.17. To demonstrate existence, one may place oneself in the split case, in which case the assertion is evident, cf. §1.

Remarks 3.8.1. *(i) The assertion analogous to 3.8 obtained by reversing the inclusions is obviously false. It would, for instance, entail that every group of type A_1 has a Borel subgroup, which is not the case, cf. Exp. XX, §5.*

(ii) It follows immediately from the preceding that $t(P) \subset t(Q)$ means that locally for (fpqc) or (ét), P is conjugate to a subgroup of Q (it suffices moreover to verify the assertion on geometric fibers). Moreover, one shall see in §5 that the étale topology may be replaced by the Zariski topology.

3.9.

The preceding discussions may be taken up in the case of critical subgroups. Let us recall (Exp. XXII, 5.10.4 and 5.10.5) that a reductive subgroup H of the reductive group G is *critical* if $H = Centr_G(rad(H))$, that a subtorus Q of G is a *C-critical torus* if $Q = rad(Centr_G(Q))$, and that critical subgroups and C-critical tori[1454] are in bijective correspondence (via $H \mapsto rad(H)$ and $Q \mapsto Centr_G(Q)$).

If (G, T, M, R) is a split S-group, the subgroup of G containing T corresponding to the part R' of R (Exp. XXII, 5.4.2) is critical if and only if R' is "vectorial" (that is, the intersection of R with a vector subspace of $M \otimes \mathbb{Q}$), cf. Exp. XXII, 5.10.6.

If G is an arbitrary S-reductive group, one defines, as in Exp. XXII, 5.11.5, an étale finite S-scheme $C\ell_{crit}$, which in the split case will be the constant scheme associated with the set of vectorial parts of R modulo the action of the Weyl group.

[1454] *N.D.E.* "Critical torus" has been replaced here by "C-critical torus", cf. *loc. cit.*. In the sequel, we shall simply write "critical torus" instead of "C-critical torus".

If $Crit(G)$ denotes the "functor of critical subgroups" of G, one has a canonical morphism $Crit(G) \longrightarrow C\ell_{crit}$, which fits into a cartesian diagram[1455]

$$\begin{tikzcd} Crit(G) \arrow[r, "c\ell"] \arrow[d] & C\ell_{crit} \arrow[d] \\ H_c \arrow[r, "c\ell"] & C\ell_c. \end{tikzcd}$$

Proposition 3.10. *Let S be a scheme, G an S-reductive group, $Crit(G)$ the functor of its critical subgroups, $C\ell_{crit}$ and $c\ell : Crit(G) \to C\ell_{crit}$ the étale finite S-scheme and the morphism defined above.*

(i) For critical subgroups H and H' of G to be conjugate (locally for the (fpqc) topology), it is necessary and sufficient that $c\ell(H) = c\ell(H')$.

(ii) $Crit(G)$ is representable and the morphism $c\ell$ is smooth, affine, with integral geometric fibers.

This is proved as 3.3, with the exception of the assertion "$c\ell$ is affine". It suffices to prove that $Crit(G)$ is affine over S. Now $Crit(G)$ is naturally identified with the S-functor of critical tori of G, and one therefore has a canonical monomorphism

$$Crit(G) \longrightarrow M$$

where M is the scheme of subgroups of multiplicative type of G (Exp. XI, 4.1). To prove that $Crit(G)$ is affine over S, it suffices, by Exp. XII 5.3, to show that this morphism is an open and closed immersion, or equivalently, by making the base change $M \to S$, to prove the following assertion: if Q is a subgroup of multiplicative type of the reductive group G, the $S' \to S$ such that $Q_{S'}$ is a critical torus of $G_{S'}$ are those that factor through a certain open and closed subscheme of S. Now to say that Q is a critical torus is to say:

(1) that Q is a torus,

(2) Q being a torus, that $rad(Centr_G(Q))$, which is also a torus, is of the same relative dimension as Q.

These two conditions are indeed of the type envisaged.

Corollary 3.11. *The S-functor $Crit(G)$ is representable by an S-scheme that is smooth and affine over S.*

Corollary 3.12. *Let H be a critical subgroup of the S-reductive group G. Then $G/Norm_G(H)$ and G/H are representable by S-schemes that are affine and smooth over S.*

The first assertion follows from 3.11, the second from the first and from Exp. XXII, 5.10.2.

[1455] *N.D.E.* cf. 3.3, N.D.E. (10) for the notations H_c and $C\ell_c$.

Corollary 3.13. *Let Q be a subtorus of the S-reductive group G. Then $G/Centr_G(Q)$ is representable by an S-scheme that is smooth and affine over S. The same holds for $G/Norm_G(Q)$ if Q is a critical subtorus of G.*

Indeed, $H = Centr_G(Q)$ is critical (Exp. XXII, 5.10.5), and one has $Norm_G(H) = Norm_G(Q)$ if Q is critical (*loc. cit.* 5.10.8).

3.14.

By the conjugacy of Levi subgroups of parabolic subgroups of G, there exists a unique morphism

$$u : Of(Dyn(G)) \longrightarrow C\ell_{crit}$$

such that for every parabolic subgroup P of G, and every Levi subgroup L of P, one has $c\ell(L) = u(t(P))$, and such that this holds after every base change.

3.15.

Let S be a scheme, G an S-reductive group. Consider the S-functors[1456]:

```
PL(S')  = { pairs P ⊃ L, P parabolic of G_{S'}, L Levi subgroup of P };
PT(S')  = { pairs P ⊃ T, P parabolic of G_{S'}, T maximal torus of P };
CT(S')  = { pairs C ⊃ T, C critical subgroup of G_{S'}, T maximal torus of C };
PLT(S') = { triples P ⊃ L ⊃ T, (P, T) ∈ PT(S'), L Levi subgroup of P }.
```

One has evident morphisms between these functors and the functors $Par(G)$, $Crit(G)$,

$Tor(G)$ already introduced, and one has a commutative diagram in the form of a truncated cube (see the following figure):

```
                         g (ét.)
              CT  ←──────────────  PLT
             /   \                   \
          a/      \b                  \ (aff.)
          /         \                   \
         /            \                   \
        /  (aff.)    (ét.) k               \
       /       f (ét.)                       \
  Crit(G) ←──────────  PL                   PT     (e isom.)
      |                 |                    |
      | cℓ (aff.)       |                    |
      ▼                 ▼                    ▼
   Cℓ_{crit}        Tor(G) ←──── h (ét.) — PT
          \            |                    /
         q \   (ét.)   |    d (aff.)       /
```

[1456] *N.D.E.* LT has been changed to CT.

```
    \          |               /
u \     | r (aff.)           /  c (aff.)
        ▼                   /
u (ét.) S                  /
    \                     /
       \ p (ét.)         /
                        /
              t (proj.)
    Of(Dyn(G)) ────── Par(G)
```

Figure 3.15.1.

Theorem 3.16. *(cf. Figure 3.15.1).*

(i) All the morphisms of the diagram are smooth, surjective, and of finite presentation.

(ii) All the morphisms of the diagram, with the exception of t, are affine; the morphism t is projective.

(iii) All the morphisms of the diagram are either étale finite or with integral geometric fibers: the morphisms f, g, h, k, p, q and u are étale finite, the morphisms a, b, c, d, r and $c\ell$ are with integral geometric fibers, the morphism e is an isomorphism.

(iv) The square (a, b, f, g) is cartesian.

Proof. It is first clear that e is an isomorphism, by 1.6 (ii). On the other hand, (iv) is evident.

The morphism a is smooth, affine, with integral geometric fibers: indeed, by base change $Crit(G) \to S$, it suffices to verify that the morphism $Tor(L_0) \to S$, where L_0 is the universal critical subgroup, has these properties; but L_0 is reductive (by definition), and one is reduced to Exp. XXII, 5.8.3. The morphism b therefore has the same properties, by virtue of (iv).

The morphism d is also smooth, affine, with integral geometric fibers, by 1.9; the same therefore holds for $c = dbe^{-1}$. The morphism r has these same properties (Exp. XXII, 5.8.3), as does the morphism $c\ell$ (3.10).

On the other hand, we have already proved that the morphisms p and q are étale finite surjective (3.1 and 3.9). If we prove that f and k are étale finite surjective, the same properties will hold for g (by (iv)) and for h (since $h = kge^{-1}$); as the properties stated for t were established in 3.3 (ii), it therefore only remains to prove that f (resp. k) is étale finite surjective; let us give the proof for k, that for f being analogous.

It suffices to prove that if T is a maximal torus of G, the functor C of critical subgroups of G containing T is representable by an étale finite S-scheme with non-empty fibers; one may assume G split with respect to T; let then E be the set of vectorial parts of R (root system of the splitting); C is representable by E_S (3.9), which completes the proof.

Corollary 3.17. *All the functors of the preceding diagram are representable by S-schemes that are smooth over S, and they are all affine over S, with the exception of $Par(G)$.*

Remark 3.18. *(i) The fact that the morphism $f : PL \to Crit(G)$ is étale surjective implies that a subgroup of G is critical if and only if it is, locally for the étale topology, a Levi subgroup of a parabolic subgroup of G.*

On the other hand, one should not believe that in general the map $f(S) : PL(S) \to Crit(G)(S)$ is surjective: it may very well happen that a critical subgroup C of G does not come on S from a parabolic subgroup of G; for example, a maximal torus is not always contained in a Borel subgroup (example: a non-split form of SL_2, cf. Exp. XX, §5).

(ii) Similarly, it may happen that the morphism $u : Of(Dyn(G)) \to C\ell_{crit}$ is not an isomorphism: two parabolic subgroups of distinct types may have Levi subgroups of the same type; for example: in a group of type A_2, there are two types of parabolic subgroups whose Levi subgroups are of semisimple rank 1 (corresponding to the two vertices of the diagram), whereas there is only one type of critical subgroup of rank 1.

The analogous example with a group of type A_3 shows that, even over an algebraically closed field, non-isomorphic parabolic subgroups may have Levi subgroups of the same type.[1457]

Let us close this section with an application to the theory of principal bundles.

Lemma 3.20.[1458] *Let S be a scheme, G an S-reductive group, F a principal homogeneous bundle under G, G_F the corresponding twisted form of G. Identify $Dyn(G)$ and $Dyn(G_F)$ (Exp. XXIV, 3.5). Let P be a parabolic subgroup of G. One has a canonical isomorphism*

$$F/P \xrightarrow{\sim} Par_{t(P)}(G_F).$$

In particular, for the structural group of F to reduce to P, it is necessary and sufficient that G_F have a parabolic subgroup of type $t(P)$.

The proof goes exactly as in Exp. XXIV, 4.2.1.

3.21.

If S is a scheme, G an S-reductive group, and if $t \in Of(Dyn(G))(S)$, one denotes by $H^1_t(S,G)$ the subset of $H^1(S,G)$ formed of classes of principal bundles F under G such that the associated group G_F has a parabolic subgroup of type t. If G itself has

[1457] *N.D.E.* i.e. the parabolic subgroups $P = ((****), (0***), (00**), (000*))$ and $P' = ((****), (0***), (0***), (000*))$ of GL_4 are not isomorphic (because $rad^u(P) \neq rad^u(P')$), but their Levi subgroups $L = ((**00), (**00), (00*0), (000*))$ and $L' = ((*000), (0**0), (0**0), (000*))$ are conjugate by the element ((0 0 1 0), (0 1 0 0), (1 0 0 0), (0 0 0 1)).

[1458] *N.D.E.* The numbering of the original has been preserved: there is no §3.19.

a parabolic subgroup P of type t, $H^1_t(S, G)$ is none other than the image of $H^1(S, P)$ in $H^1(S, G)$, an image which therefore does not depend on the P chosen.

4. Relative position of two parabolic subgroups

4.1. A preliminary result

Lemma 4.1.1. *Let k be a field, G a k-reductive group, P and P' two parabolic subgroups of G.*

(i) Then $P \cap P'$ is smooth, of the same reductive rank as G, and contains a maximal torus T of G.

(ii) The set R' of roots of $P \cap P'$ with respect to T is a closed subset of the set R of roots of G with respect to T.[1459]

Suppose first k algebraically closed. Let B (resp. B') be a Borel subgroup

of P (resp. P'). One knows that there exists $g \in G(k)$ such that $int(g)B = B'$. On the other hand, if T_0 is a maximal torus of B and one sets $N = Norm_G(T_0)$, one knows (Bruhat's theorem, Bible, §13.4, Cor. 1 to Th. 3) that $G(k) = B(k)N(k)B(k)$. One therefore sees that there exist $b, b_1 \in B(k)$ and $n \in N(k)$ such that $g = bnb_1$, hence $int(b)int(n)B = B'$. One then has

P ∩ P' ⊃ B ∩ B' = int(b)(B ∩ int(n) B') ⊃ int(b) T_0.

Now suppose k arbitrary. Applying the preceding result, one sees that $P_{\bar{k}} \cap P'_{\bar{k}}$ contains a maximal torus of $G_{\bar{k}}$; by Exp. XXII, 5.4.5, one deduces that $(P \cap P')_{\bar{k}}$ is smooth "along the unit section", hence smooth since one is over a field (Exp. VI_A 1.3.1), hence $P \cap P'$ is smooth. By Exp. VI_A 2.3.1, the neutral component $(P \cap P')^0$ of $P \cap P'$ is therefore an open subgroup of $P \cap P'$, smooth over S. One may then apply Exp. XIV, 1.1 to it, whence (i).

Finally, the set R_P (resp. $R_{P'}$) of roots of P (resp. P') with respect to T is closed and one has $R' = R_P \cap R_{P'}$, whence (ii).

Remark 4.1.2. *One may prove ([BT65], 4.5) that $P \cap P'$ is connected;*[1460] *this will be used in 4.5.1.*

Remark 4.1.3. *The preceding lemma is not true over an arbitrary scheme. Indeed, let G be, for example, a reductive group over an algebraically closed field k, and let B be a Borel subgroup of G. Take $S = X$ as base and consider the Borel subgroups B_1 and B_2 of G_X, where $B_1 = B_X$ and $B_2 = int(g_0)B_1$, g_0 being the canonical (diagonal) section of G_X. For each $g \in X(k)$, the fiber of $B_1 \cap B_2$ at g is none other than $B \cap int(g)B$. If one assumes $B \neq G$, the dimension of this fiber varies with g, hence $B_1 \cap B_2$ cannot be smooth over X.*

[1459] *N.D.E.* Point (ii) has been added; it will be useful in 4.5.1.
[1460] *N.D.E.* In view of the details added in 4.5.1, the original has been modified here (which stated "we shall not use this fact").

4.2. Transversal position

Theorem 4.2.1. *Let S be a scheme, G an S-reductive group, P and Q two parabolic subgroups of G. The following conditions on the pair (P, Q) are equivalent:*

(i) $\mathrm{Lie}(P/S) + \mathrm{Lie}(Q/S) = \mathrm{Lie}(G/S)$.

(ii) The canonical morphism $P \times_S Q \to G$ is smooth.

(ii') The canonical morphism $P \to G/Q$ is smooth.

(iii) The canonical morphism $P \times_S Q \to G$ is open.

(iii') The canonical morphism $P \to G/Q$ is open.

(iv) The canonical morphism $P \times_S Q \to G$ is fiberwise dominant.

(iv') The canonical morphism $P \to G/Q$ is fiberwise dominant.

(v) For every $s \in S$, "$P_s \cap Q_s$ is of minimum dimension", i.e. one has

$$\dim(P_s \cap Q_s) = \dim P_s + \dim Q_s - \dim G_s.$$

(vi) There exists a covering family for the étale topology $S_i \to S$, and for each i a Borel subgroup B_i of P_{S_i} and a Borel subgroup B'_i of Q_{S_i}, such that $B_i \cap B'_i$ is a maximal torus of G_{S_i}.

(vii) There exists a covering family for the étale topology $S_i \to S$, and for each i a splitting $(T_i, \cdots, R_i)$ of G_{S_i} and a system of positive roots R_i^+ of R_i, such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by a subset $R_i^{(1)}$ (resp. $R_i^{(2)}$) of R containing R_i^+ (resp. $-R_i^+$) (see Exp. XXII, 5.4.2 and 5.2.1 for the definitions).

Proof. We shall prove the theorem following the logical diagram

```
(i) ⇔ (vii) ⇔ (vi)
                  \
                   \
(ii') ⇒ (iii') ⇒ (iv') ⇒ (v)
                  /
                 /
(ii)  ⇒ (iii)  ⇒ (iv)
```

One trivially has (ii) ⇒ (iii) and (ii') ⇒ (iii'). If (iii) holds, the set-theoretic image of the morphism $P \times_S Q \to G$ is an open set of G containing the unit section; since the fibers of G are connected, this image is dense on each fiber, which proves (iv). One similarly has (iii') ⇒ (iv').

One has (ii') ⇒ (ii), by the cartesian diagram

```
P ×_S Q ──────→ G
   |            |
 pr_1|          |
```

$$\begin{array}{ccc} \downarrow & & \downarrow \\ P & \longrightarrow & G/Q. \end{array}$$

On the other hand (iv) or (iv') implies (v), by the theory of dimension (cf. EGA IV_2, 5.6.6). One notes that one may indeed assume $S = \mathrm{Spec}(k)$, k an algebraically closed field, and that every non-empty fiber of the morphism (iv), resp. (iv'), at a point of $G(k)$, resp. of $(G/Q)(k)$, is isomorphic to $P \cap Q$ (as an immediate computation shows).

One has (vi) ⇒ (vii), by Exp. XXII, 5.5.1 (iv) and 5.9.2.

One has (vii) ⇒ (i), because to verify that $\mathrm{Lie}(P) + \mathrm{Lie}(Q) = \mathrm{Lie}(G)$, one may reason locally for (fpqc); thus if (vii) is satisfied one may assume G split, $P \supset B_{R^+}$ and $Q \supset B_{-R^+}$ (usual notations), in which case one already has

$$\mathrm{Lie}(B_{R^+}) + \mathrm{Lie}(B_{-R^+}) = \mathrm{Lie}(G).$$

Let us prove that (i) implies (ii').

Let $u : P \to G/Q$ be the canonical morphism; to prove that u is smooth, it suffices to do so on the geometric fibers of u, since P and G/Q are smooth over S, and one may therefore assume that S is the spectrum of an algebraically closed field. As the morphism u is compatible with the obvious action of P (one has $u(pp') = pu(p')$) it suffices, by a translation argument (cf. the proof of VI_B 1.3), to verify that u is smooth at $e \in P(k)$, i.e. (SGA 1, II 4.7) that the tangent map to u at e is surjective; but the latter is naturally

identified with the canonical map $\mathrm{Lie}(P) \to \mathrm{Lie}(G)/\mathrm{Lie}(Q)$, which is surjective if (i) is satisfied.

It therefore only remains to verify the last assertion, namely (v) ⇒ (vi). Suppose first that S is the spectrum of an algebraically closed field. By 4.1.1, there exists a maximal torus T contained in P and Q; let R (resp. R_1, resp. R_2) be the set of roots of G (resp. P, resp. Q) relative to T.

One has:

$$\dim(G) = \dim(T) + \mathrm{Card}(R), \qquad \dim(P) = \dim(T) + \mathrm{Card}(R_1),$$
$$\dim(Q) = \dim(T) + \mathrm{Card}(R_2), \qquad \dim(P \cap Q) = \dim(T) + \mathrm{Card}(R_1 \cap R_2),$$

by Exp. XXII, 5.4.4 and 5.4.5 for example. The condition of (v) is therefore equivalent to

$$\mathrm{Card}(R_1 \cap R_2) = \mathrm{Card}(R_1) + \mathrm{Card}(R_2) - \mathrm{Card}(R),$$

that is $R_1 \cup R_2 = R$. To demonstrate (vi), it suffices, by Exp. XXII, 5.9.2 and 5.4.5, to prove that $R_1 \cap -R_2$ contains a system of positive roots of R. We are therefore reduced to proving:

Lemma 4.2.2. *Let R be a "root system" (e.g. the set of roots of a root datum in the sense of Exposé XXI). Let R_1 and R_2 be two closed subsets of R each containing a system of positive roots. If $R_1 \cup R_2 = R$, then $R_1 \cap -R_2$ contains a system of positive roots.*

Indeed, since $R_1 \cap -R_2 = R_3$ is evidently closed, and by Exp. XXI, 3.3.6, it suffices to show that $R_3 \cup -R_3 = R$. Now one knows that $R_1 \cup -R_1 = R = R_2 \cup -R_2$, and one concludes thanks to the following elementary fact: if `A, A', B, B'` are four subsets of a set E, and if $A \cup A' = B \cup B' = A \cup B = A' \cup B' = E$, then $(A \cap B') \cup (A' \cap B) = E$.

This completes the proof of (v) ⇒ (vi) in the case where the base is the spectrum of an algebraically closed field. Let

us now return to the general case and assume (v) is satisfied. Let $s \in S$; by the preceding, one may find a Borel subgroup $\bar{B}$ (resp. $\bar{B}'$) of P_s (resp. Q_s) such that $\bar{B} \cap \bar{B}'$ is a maximal torus of G_s.

Since the S-scheme `Bor(P) ≃ Bor(P/rad^u(P))` of Borel subgroups of P is smooth, one may, applying "Hensel's lemma" (cf. Exp. XI 1.10) and reasoning locally for the étale topology (i.e. replacing S by an $S' \to S$ that is étale and covering s, and s by a point of its fiber in S'), assume that there exists a Borel subgroup B of P projecting onto $\bar{B}$; one may similarly assume that there exists a Borel subgroup B' of Q projecting onto $\bar{B}'$. Since $B_s \cap B'_s$ is a maximal torus of G_s, there exists an open neighborhood U of s in S such that $B_U \cap B'_U$ is a maximal torus of `G_U` (Exp. XXII, 5.9.4), which proves (vi). QED.

Definition 4.2.3. *A pair (P, Q) satisfying the equivalent conditions (i) to (vii) of Theorem 4.2.1 is said to be in* transversal position. *One also says that P is in transversal position relative to Q, or, by abuse of language, that P and Q are in (mutual) transversal position.*

In view of (vi), this definition coincides in the case of Borel subgroups with that of Exp. XXII, 5.9.1.

Corollary 4.2.4. *Let S be a scheme, G an S-reductive group, P and Q two parabolic subgroups of G.*

(i) For (P, Q) to be in transversal position, it is necessary and sufficient that for each point s of S, the pair (P_s, Q_s) be in transversal position; if $S' \to S$ is a surjective morphism, and if $(P_{S'}, Q_{S'})$ is in transversal position, then (P, Q) is in transversal position.

(ii) There exists an open subscheme U of S having the following property: for a morphism $S' \to S$ to factor through U, it is necessary and sufficient that $(P_{S'}, Q_{S'})$ be in transversal position.

(iii) Consider the subfunctors

```
Gen(G)   ⊂ Par(G) ×_S Par(G),
Gen(/Q)  ⊂ Par(G),
Gen(P/Q) ⊂ G
```

defined as follows: for $S' \to S$, $Gen(G)(S')$ is the set of pairs of parabolic subgroups of $G_{S'}$ in transversal position, $Gen(/Q)(S')$ is the set of parabolic subgroups of $G_{S'}$ in transversal position relative to $Q_{S'}$, $Gen(P/Q)(S')$ is the set of $g \in G(S')$ such that $int(g)P_{S'}$ is in transversal position relative to $Q_{S'}$.

Each of these functors is representable by a universally schematically dense open subscheme over S[1461] *(cf. Exp. XVIII §1) of the corresponding S-scheme $Par(G) \times_S Par(G)$, resp. $Par(G)$, resp. G.*

Assertions (i) follow at once from the description 4.2.1 (i) of the term "transversal position". To demonstrate (ii), one takes $U = S - Supp(Coker u)$ where u is the canonical morphism $\mathrm{Lie}(P) \oplus \mathrm{Lie}(Q) \to \mathrm{Lie}(G)$.

Since one has cartesian diagrams

$$\begin{tikzcd} G \arrow[r, "f"] & Par(G) \\ Gen(P/Q) \arrow[u] \arrow[r] & Gen(/Q) \arrow[u] \end{tikzcd} \qquad \begin{tikzcd} Par(G) \arrow[r, "f'"] & Par(G) \times_S Par(G) \\ Gen(/Q) \arrow[u] \arrow[r] & Gen(G) \arrow[u] \end{tikzcd}$$

(where $f(g) = int(g)P$ and $f'(R) = (R, Q)$), it suffices to verify (iii) in the case of $Gen(G)$.

Let then P_0 be the canonical parabolic subgroup of $G_{Par(G)}$; set

$$X = Par(G) \times_S Par(G), \qquad P = pr_1^*(P_0), \qquad Q = pr_2^*(P_0);$$

applying assertion (ii) to the parabolic subgroups P and Q of G_X, one constructs an open subscheme U of X, which, as one verifies at once, is indeed identified with $Gen(G)$. It remains to verify the density assertion, which can be done on geometric fibers[1462]; one may therefore assume $S = \mathrm{Spec}(k)$, k an algebraically closed field. As $Par(G)$ is smooth, it suffices to verify that $Gen(G)$ meets each irreducible component of $Par(G) \times_S Par(G)$; in other words, by 3.3, it suffices to see that if $t, t' \in Of(Dyn(G))(S)$, there exists a pair (P, P') in transversal position with $t(P) = t$, $t(P') = t'$. Now this is immediate: one chooses a pair (B, B') of Borel subgroups of G such that $B \cap B'$ is a maximal torus (one splits G and applies Exp. XXII, 5.9.2) then one applies 3.8 to construct $P \supset B$ and $P' \supset B'$, with $t(P) = t$, $t(P') = t'$; P and P' are of the desired types and are in transversal position by 4.2.1 (vi).

Corollary 4.2.5. *Let S be a scheme, G an S-reductive group, P and Q two parabolic subgroups of G, the pair (P, Q) being in transversal position.*

(i) Let P' and Q' be two parabolic subgroups of G, of the same type as P and Q respectively. For the pair (P', Q') to be in transversal position, it is necessary and sufficient that it be conjugate to the pair (P, Q), locally for the étale topology. (N. B. One shall see in §5 that the étale topology can be replaced by the Zariski topology.)

(ii) The canonical morphism $P \times_S Q \to G$ induces a smooth and surjective morphism $P \times_S Q \to Gen(Q/P)$, and an isomorphism

$$(P \times_S Q) / (P \cap Q) \simeq Gen(Q/P)$$

[1461] *N.D.E.* "Relatively dense" has been replaced by "universally schematically dense over S", cf. EGA IV_3, Def. 11.10.8.
[1462] *N.D.E.* cf. EGA IV_3, 11.10.10.

(where $P \cap Q = R$ operates on $P \times_S Q$ by $(p,q)r = (pr, r^{-1}q)$).

(iii) The canonical morphism $P \to Par_{t(Q)}(G)$ (defined set-theoretically by $p \mapsto int(p)Q$) induces a smooth and surjective morphism $P \to Gen(/P) \cap Par_{t(Q)}(G)$, and an

isomorphism

$$P / (P \cap Q) \simeq Gen(/P) \cap Par_{t(Q)}(G).$$

(iv) The canonical morphism $G \to Par_{t(P)}(G) \times_S Par_{t(Q)}(G)$ (defined set-theoretically by $g \mapsto (int(g)P, int(g)Q)$) induces a smooth and surjective morphism $G \to Gen(G) \cap (Par_{t(P)}(G) \times_S Par_{t(Q)}(G))$ and an isomorphism

$$G / (P \cap Q) \simeq Gen(G) \cap (Par_{t(P)}(G) \times_S Par_{t(Q)}(G)).$$

Let us demonstrate (i). It is clear that the condition is sufficient; let us prove that it is necessary. Let then (P', Q') be in transversal position. Since P and P' are conjugate locally for the étale topology, we may assume $P = P'$, and it suffices to prove that if Q and Q' are two parabolic subgroups of G, in transversal position relative to P, and of the same type, then they are conjugate, locally for the étale topology, by a section of P. Using 4.2.1 (vi), one may assume that there exist Borel subgroups B, B', B_1, B_1' of P, P, Q, Q' respectively, such that $B \cap B_1 = T$ and $B' \cap B_1' = T'$ are maximal tori of G. Now the Killing pairs (B,T) and (B', T') of P are conjugate locally in P for the étale topology (1.16), and one may assume $B = B'$, $T = T'$, in which case one has $B_1 = B_1'$ by Exp. XXII, 5.9.2, hence $Q = Q'$ by 3.8.

Assertions (ii), (iii) and (iv) are proved in a parallel fashion. Let us demonstrate, for instance, (ii); let $g \in Gen(Q/P)(S)$, i.e. let $g \in G(S)$ be such that $int(g)\,Q$ is in transversal position relative to P. By the preceding proof, Q and $int(g)\,Q$ are conjugate locally for the étale topology, by a section of P. Reasoning

locally for this topology, one may assume that there exists $p \in P(S)$ such that $int(g)Q = int(p)Q$, hence $p^{-1}g \in Norm_G(Q)(S) = Q(S)$; which proves the existence of $q \in Q(S)$ such that $g = pq$. We have therefore proved that the morphism considered in (ii) is covering for the étale topology. Comparing with 4.2.1 (ii), one deduces that it is smooth and surjective. On the other hand, an immediate reasoning shows that the equivalence relation defined on $P \times_S Q$ by the morphism $P \times_S Q \to G$ is the equivalence relation associated with the action of the group $R = P \cap Q$ (operating by $(p,q)r = (pr, r^{-1}q)$), which demonstrates the last assertion of (ii) (since a smooth surjective morphism is an effective epimorphism, for example).

Remark 4.2.6. *If P and Q are in transversal position, one will often denote by $P \cdot Q$ the open subset $Gen(Q/P)$ of G, a notation justified by 4.2.5 (ii).*

Proposition 4.2.7. *Let S be a scheme, G an S-reductive group, P and Q two parabolic subgroups of G, the pair (P,Q) being in transversal position.*

(i) The group $P \cap Q$ is smooth over S (and in fact has connected fibers by 4.1.2); introducing then $(P \cap Q)^0$ (cf. Exp. VI_B 3.10), this is a subgroup of type (RC) of

G (Exp. XXII, 5.11.1), whose unipotent radical (loc. cit. 5.11.4) decomposes as a direct product

$$\mathrm{rad}^u((P \cap Q)^0) = (\mathrm{rad}^u(P) \cap Q) \times_S (P \cap \mathrm{rad}^u(Q)).$$

(ii) If S is affine, $H^1(S, rad^u((P \cap Q)^0)) = 0$. If S is semi-local, $P \cap Q$ contains a maximal torus of G.

Indeed, $P \cap Q$ is smooth by 4.2.1 (ii) and the cartesian diagram

$$\begin{array}{ccc} P \times_S Q & \longrightarrow & G \\ \uparrow & & \uparrow \\ P \cap Q & \longrightarrow & S. \end{array}$$

To verify the announced assertions on $(P \cap Q)^0$, one may reason locally for the étale topology, hence by 4.2.1 (vii) assume to have chosen a splitting (G, T, M, R) of G, such that P and Q contain T and are defined respectively by subsets R_1 and R_2 of R, R_1 containing a system of positive roots R^+, and R_2 containing the opposite system R^-. Let Δ be the set of simple roots of R^+; denote

$$A_1 = \Delta \cap -R_1, \qquad A_2 = \Delta \cap R_2, \qquad A = A_1 \cap A_2.$$

By 1.4 (v) and 1.12, one has

$$R_1 = R^+ \cup (R^- \cap -\mathbb{N}A_1), \qquad R_2 = (R^+ \cap \mathbb{N}A_2) \cup R^-,$$
$$\mathrm{rad}^u(P) = \prod_{\alpha \in R_1,\ \alpha \notin -R_1} U_\alpha, \qquad \mathrm{rad}^u(Q) = \prod_{\alpha \in R_2,\ \alpha \notin -R_2} U_\alpha.$$

By Exp. XXII, 5.6.7, one therefore has

$$\mathrm{rad}^u(P) \cap Q = \prod_{\alpha \in R_1 \cap R_2,\ \alpha \notin -R_1} U_\alpha = \prod_{\alpha \in K_2} U_\alpha,$$
$$\mathrm{rad}^u(Q) \cap P = \prod_{\alpha \in R_1 \cap R_2,\ \alpha \notin -R_2} U_\alpha = \prod_{\alpha \in K_1} U_\alpha,$$

where K_2 is the set of positive roots that are linear combinations of the elements of A_2, but not linear combinations of the elements of A, and K_1 the set of negative roots that are linear combinations of the elements of A_1, but not linear combinations of the elements of A. It is clear that if $\alpha \in K_2$, $\beta \in K_1$, $\alpha + \beta$ is never a root, nor zero, which entails that the two groups above commute.

On the other hand, one knows by Exp. XXII, 5.4.5, that $H = (P \cap Q)^0$ is defined by the set of roots $R_1 \cap R_2$, namely

$$R_1 \cap R_2 = (R^+ \cap \mathbb{N}A_2) \cup (R^- \cap -\mathbb{N}A_1).$$

Since $R_1 \cap R_2$ is closed, H is of type (RC) by definition (Exp. XXII 5.11.1), and by *loc. cit.* 5.11.3 and 5.11.4, one has

$$\mathrm{rad}^u(H) = \prod_{\alpha \in K} U_\alpha,$$

where K is the set of $\alpha \in R_1 \cap R_2$ such that $\alpha \notin -(R_1 \cap R_2)$. As the symmetric part of $R_1 \cap R_2$ is evidently $R \cap \mathbb{Z}A$, one sees at once that $K = K_1 \cup K_2$, which completes the proof of (i).

The first assertion of (ii) then follows from (i) and 2.10; let us demonstrate the second. Since $(P \cap Q)^0/rad^u((P \cap Q)^0)$ is reductive, it has a maximal torus $\bar{T}$ if the base is semi-local. The inverse image of $\bar{T}$ in $(P \cap Q)^0$ is a subgroup N of type (R) of G with solvable fibers, and one has $N^u = rad^u((P \cap Q)^0)$ (Exp. XXII, 5.6.9). The scheme of maximal tori of N is a principal homogeneous bundle under N^u (*loc. cit.* 5.6.13), hence has a section, since $H^1(S, N^u) = 0$.

4.3. Opposite parabolic subgroups

4.3.1.

If G is an S-reductive group, one has defined in Exp. XXIV, 3.16.6, a canonical "outer automorphism" s_G of order $\leqslant 2$[1463] of G, hence a canonical automorphism of order $\leqslant 2$ of $Dyn(G)$, hence also an automorphism of order $\leqslant 2$ of $Of(Dyn(G))$, which we shall also denote s_G or simply s. Two types of parabolic subgroups $t, t' \in Of(Dyn(G))(S)$ will be said to be *opposite* when $t = s_G(t')$.

Theorem 4.3.2. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G.*

(a) If L is a Levi subgroup of P, there exists a unique parabolic subgroup P' of G such that $P \cap P' = L$.

(b) For every parabolic subgroup Q of G, the following conditions are equivalent:

(i) For every $s \in S$, $((P \cap Q)_s)^0$ (which is smooth by 4.1.1) is reductive.

(ii) $P \cap Q$ is a Levi subgroup of P and of Q.

(iii) P and Q are of opposite types, and the pair (P, Q) is in transversal position (cf. 4.2.3).

(iv) P and Q are of opposite types and $rad^u(P) \cap Q = e$.

(v) $rad^u(P) \cap Q = rad^u(Q) \cap P = e$.

(vi) The canonical morphism $rad^u(P) \times_S Q \to G$ is an open immersion.

(vi') The canonical morphism $rad^u(P) \to G/Q$ is an open immersion.

(vii) There exists a covering family for the étale topology $S_i \to S$, and for each i a splitting (T_i, M_i, R_i) of G_{S_i}, and a subset $R_i^{(1)}$ of R_i such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by $R_i^{(1)}$ (resp. by $R_i^{(2)} = -R_i^{(1)}$).

Proof. Let us first demonstrate the second part of the theorem; one sees first that (iii) ⇔ (vii) by 4.2.1 (vii) and the definition of s_G in the split case (Exp. XXII, 3.16.2 (iv)); one trivially has (ii) ⇒ (i); one has (vi') ⇒ (vi) by base change $G \to G/Q$.

Now suppose (vii) holds, and let us prove all the other conditions; as they are local for the étale topology, one may assume $G = (G, T, M, R)$ split, P defined by the part

[1463] *N.D.E.* "Of order 2" has been replaced by "of order $\leqslant 2$" because s_G may be trivial (for example, if G is of type A_1, B_n, C_n, ...).

R' of R and Q by the part $-R'$. If L is the subgroup of type (R) of G containing T defined by $R' \cap -R'$, it is clear by Exp. XXII, 5.11.3 that L is a Levi subgroup common to P and Q. But $P = L \cdot rad^u(P)$, $Q = L \cdot rad^u(Q)$, and by Exp. XXII, 5.6.7, $rad^u(P) \cap Q = Q \cap rad^u(P) = e$; hence $P \cap Q = L$, and we have proved (ii) and (v). Since P and Q are in transversal position, the canonical morphism $P \to G/Q$ induces an open immersion $P/P \cap Q \to G/Q$ (4.2.1); but the canonical morphism $rad^u(P) \to P/P \cap Q = P/L$ is an isomorphism, so we have proved (vi'). In view of what was already seen, all assertions are therefore consequences of (vii).

It now suffices to prove that any one of the assertions (i), (iv), (v), (vi) implies (vii); as we have already proved the equivalence of (ii) and (iii), it suffices to give the proof on geometric fibers, and one may assume that S is the spectrum of an algebraically closed field. By 4.1.1, there exists a maximal torus T of G contained in $P \cap Q$. Let R (resp. `R_1`, resp. `R_2`) be the set of roots of G (resp. P, resp. Q) relative to T.

Let R_1^a be the asymmetric part of `R_1` (i.e. $R_1^a = \alpha \in R_1 | -\alpha \notin R_1$). Introduce R_2^a similarly. We must prove that $R_1 = -R_2$.

– Condition (i) entails that $R_1 \cap R_2$ is symmetric; let $\alpha \in R_1$; if $\alpha \notin R_2$, then $\alpha \in -R_2$, and if $\alpha \in R_2$, then $\alpha \in R_1 \cap R_2 = -(R_1 \cap R_2) \subset -R_2$; one therefore has $R_1 \subset -R_2$, hence by symmetry $R_1 = -R_2$.

– Condition (iv) entails $Card(R_1) = Card(R_2)$ and $R_1^a \cap R_2 = \emptyset$; the second condition is equivalent to $R_2 \subset -R_1$; the first then gives $R_2 = -R_1$.

– Condition (v) entails $R_1^a \cap R_2 = R_2^a \cap R_1 = \emptyset$, hence $R_2 \subset -R_1$ and $R_1 \subset -R_2$, which again gives $R_2 = -R_1$.

– Condition (vi) entails `Lie(rad^u(P)) ⊕ Lie(Q) = Lie(G)`, which entails that R is the disjoint union of `R_2` and

R_1^a, hence that $R_2 = -R_1$.

This completes the proof of the second part of the theorem. Let us prove the first; let us first remark that, by virtue of (vii) ⇒ (ii), we have already proved the existence locally for the étale topology of the sought group P'; it therefore remains to prove its uniqueness, and this can also be done locally for the étale topology. One may therefore assume G split relative to a maximal torus T of L, and P (resp. P') defined by a part `R_1` (resp. R_1') of the root system R.

By hypothesis $R_1 \cap R_1'$ is symmetric; reasoning as above, one derives $R_1' = -R_1$, which proves that P' is determined by P and L and completes the proof.

Definition 4.3.3. *Two parabolic subgroups of G satisfying the equivalent conditions (i) to (vii) of 4.3.2 are said to be* opposite. *If P is a parabolic subgroup of G, and if L is a Levi subgroup of P (resp. and if T is a maximal torus of P), one calls* parabolic subgroup opposite to P relative to L (resp. T) *the unique parabolic subgroup Q of G such that $P \cap Q = L$ (resp. such that $P \cap Q$ is the unique Levi subgroup of P containing T, cf. 1.6, or, equivalently, such that $P \cap Q$ contains T and that P and Q are opposite).*

By 4.3.2 (iii), one derives from 4.2.4 and 4.2.5 parallel results; let us give one sample.

Corollary 4.3.4. *Let S be a scheme, G an S-reductive group, P and Q two parabolic subgroups of G.*

(i) For P and Q to be opposite, it is necessary and sufficient that for every point $s \in S$, P_s and Q_s be opposite. If $S' \to S$ is a surjective morphism, and if $P_{S'}$ and $Q_{S'}$ are opposite, then P and Q are opposite.

(ii) The functor $Opp(G)$, such that for $S' \to S$, $Opp(G)(S')$ is the set of pairs of opposite parabolic subgroups of $G_{S'}$, is representable by an open subscheme of $Par(G)^2$. The functor $Opp(/P)$ such that for $S' \to S$, $Opp(/P)(S')$ is the set of parabolic subgroups of $G_{S'}$ opposite to $P_{S'}$, is representable by a universally schematically dense open subscheme over S[1464] of $Par_{s(t(P))}(G)$.

(iii) Suppose P and Q are opposite; let P' and Q' be two parabolic subgroups of G, P' being of the same type as P. For P' and Q' to be opposite, it is necessary and sufficient that locally for the étale topology, the pair `(P', Q')` be conjugate to the pair (P, Q). (N. B. One shall see in §5 that the étale topology can be replaced by the Zariski topology.)

Corollary 4.3.5. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G.*

(i) The morphism $Opp(/P) \to Lev(P)$ (cf. 1.9) defined set-theoretically by $Q \mapsto P \cap Q$, is an isomorphism; $Opp(/P)$ is a principal homogeneous bundle under $rad^u(P)$ ($rad^u(P)$ operating by inner automorphisms). If S is affine, there exists a parabolic subgroup of G opposite to P.

(ii) Suppose S semi-local; let s_i be the set of its closed points; for each i, let Q_i be a parabolic subgroup of G_{s_i}, opposite to P_{s_i}. There exists a parabolic subgroup Q of G, opposite to P, and such that $Q_{s_i} = Q_i$ for each i.

(iii) The morphism $Opp(G) \to PL$ (cf. 3.15) defined set-theoretically by $(P, Q) \mapsto (P, P \cap Q)$ is an isomorphism.

All this follows from the first part of the theorem and from 1.9, 2.3 and 2.8.

Remark 4.3.6. *Let P and Q be two opposite parabolic subgroups of G, and let $P \cdot Q$ be the open subscheme of G, sheaf-theoretic image of $P \times_S Q$, introduced in 4.2.6. The "product morphism" $G \times_S G \to G$ induces isomorphisms:*

`rad^u(P) ×_S Q  P · Q  P ×_S rad^u(Q).`

This follows indeed from 4.3.2 (or 4.2.5 (ii)) and from the fact that $P \cap Q$ is a Levi subgroup of P and of Q, hence that $P = rad^u(P) \cdot (P \cap Q)$ and $Q = rad^u(Q) \cdot (P \cap Q)$.

One has similarly a commutative diagram

[1464] *N.D.E.* "Relatively dense" has been replaced by "universally schematically dense over S", cf. EGA IV_3, Def. 11.10.8.

$$\begin{tikzcd} \mathrm{rad}^u(P) \arrow[r] \arrow[d] & G/Q \arrow[d] \\ \mathrm{Opp}(/P) \arrow[r] & \mathrm{Par}_{t(Q)}(G), \end{tikzcd}$$

where the vertical arrows are induced by $g \mapsto int(g)Q$.

4.4. Osculating position

Proposition 4.4.1. *Let S be a scheme, G an S-reductive group, P and Q two parabolic subgroups of G. The following conditions are equivalent:*

(i) $P \cap Q$ is a parabolic subgroup of G.

(ii) $P \cap Q$ contains locally for the étale topology a Borel subgroup of G.

(iii) $P \cap Q$ contains locally for the étale topology a maximal torus of G, and, for every $S' \to S$ and every maximal torus T of $G_{S'}$ contained in $P_{S'}$ and $Q_{S'}$, the opposite of $P_{S'}$ relative to T is in transversal position relative to $Q_{S'}$.

(iv) There exists a covering family for the étale topology $S_i \to S$, and for each i a maximal torus T_i of G_{S_i} contained in P_{S_i} and Q_{S_i}, and such that the opposite of P_{S_i} relative to T_i is in transversal position relative to Q_{S_i}.

(v) There exists a covering family for the étale topology $S_i \to S$, and for each i a splitting (T_i, M_i, R_i) of G_{S_i} such that P_{S_i} (resp. Q_{S_i}) is the subgroup of type (R) of G_{S_i} containing T_i and defined by a set of roots $R_i^{(1)}$ (resp. $R_i^{(2)}$), $R_i^{(1)} \cap R_i^{(2)}$ containing a system of positive roots of R.

Moreover, if these conditions are satisfied, one has $t(P \cap Q) = t(P) \cap t(Q)$ (with the notations of 3.2).

One has (v) ⇒ (ii) and (iii) ⇒ (iv) trivially. On the other hand, (ii) ⇒ (i) by 1.18. One has (iv) ⇒ (v): indeed, one may assume G split, P (resp. Q) defined by the set of roots R_1 (resp. R_2); the opposite of P is then defined by $-R_1$, and one is reduced to Lemma 4.2.2. One proves (i) ⇒ (iii) by splitting, in the same way. Finally, the last assertion of the theorem can be proved locally for the étale topology; one may assume that $P \cap Q$ contains a Borel subgroup B of G, and one is reduced to 3.7.

Definition 4.4.2. *Two parabolic subgroups of G satisfying conditions (i) to (v) of 4.4.1 are said to be in* osculating position.

Corollary 4.4.3. *Let P and Q be two parabolic subgroups in osculating position, and let P' and Q' be two parabolic subgroups of G, of the same type as P and Q respectively. For P' and Q' to be in osculating position, it is necessary and sufficient that the pair (P', Q') be conjugate to the pair (P, Q), locally for the étale topology.*

It suffices to prove that if P and P' are in osculating position with respect to Q, they are conjugate, locally for (ét), by a section of Q. Now $P \cap Q$ and $P' \cap Q$ are two parabolic subgroups of the same type contained in Q, hence are conjugate locally for (ét) by a section of Q, by part (ii) of the lemma below. One may therefore assume $P \cap Q = P' \cap Q$; one then has $P = P'$, by part (i) of the same lemma:

Lemma 4.4.4. *Let P, P' and Q be three parabolic subgroups of the S-reductive group G.*

(i) For $P = P'$, it is necessary and sufficient that P and P' be in osculating position and of the same type.

(ii) If $P \subset Q$, $P' \subset Q$, and if $g \in G(S)$ is such that $int(g)\ P$ and P' are in osculating position, then $g \in Q(S)$.

Part (i) follows trivially from the last assertion of 4.4.1. Let us demonstrate (ii): Q and $int(g)\ Q$ contain $P' \cap int(g)P$, hence are in osculating position; they coincide by (i), hence $g \in Norm_G(Q)(S) = Q(S)$.

Let us remark that assertions (iii) and (iv) of the theorem immediately yield:

Corollary 4.4.5. *Let P, P' and Q be three parabolic subgroups of the S-reductive group G, containing the same maximal torus T of G. Suppose P and P' opposite relative to T. For Q to be in osculating position relative to P, it is necessary and sufficient that it be in transversal position relative to P'. Under these conditions $P \cap Q$ is also in transversal position relative to P'.*

Corollary 4.4.6. *Let P and Q be two parabolic subgroups of G containing the same maximal torus T. For P and Q to be in transversal position, it is necessary and sufficient that there exist two parabolic subgroups $P' \subset P$ and $Q' \subset Q$ of G, opposite relative to T; one may even choose $t(P') = t(P) \cap s(t(Q))$.*[1465]

The condition is evidently sufficient (4.2.1 (i) and 4.3.2 (iii)). Let us show that it is necessary; let P^- (resp. Q^-) be the opposite of P (resp. Q) relative to T. By 4.4.5, $P^- \cap Q$ is in transversal position relative to P and Q^-, hence also relative to $P \cap Q^-$ by a further application of 4.4.5; moreover

$$t(P^- \cap Q) = t(P^-) \cap t(Q) = s(t(P)) \cap s(t(Q^-)) = s(t(P) \cap t(Q^-)) = s(t(P \cap Q^-)),$$

hence $P^- \cap Q = P'$ and $P \cap Q^- = Q'$ are opposite (4.3.2 (iii)); but $P' \cap Q' \supset T$, hence they are indeed opposite relative to T.

4.5. Standard position

In this section, we briefly indicate how some of the preceding results generalize.

Proposition 4.5.1. *If P_1 and P_2 are two parabolic subgroups of the S-reductive group G, the following conditions are equivalent:*[1466]

(i) $P_1 \cap P_2$ is smooth.

(ii) $P_1 \cap P_2$ is a subgroup of type (R) (or of type (RC)) of G.

(iii) $P_1 \cap P_2$ contains locally for the (fpqc) topology a maximal torus of G.

[1465] *N.D.E.* Recall that the involution s was defined in 4.3.1.

[1466] *N.D.E.* The proof of the equivalence of these conditions has been added (as well as condition (v), used implicitly in 6.17 of the original); consequently, §4.5.1 has been transformed into Proposition 4.5.1 plus Definition 4.5.1.1.

(iv) $P_1 \cap P_2$ contains locally for the Zariski topology a maximal torus of G.

When S is semi-local, these conditions are moreover equivalent to:

(v) $P_1 \cap P_2$ contains a maximal torus of G.

Proof.[1467] Evidently, (iv) ⇒ (iii) (and (v) ⇒ (iv) when S is semi-local), and (ii) ⇒ (i) according to Exp. XXII, Def. 5.2.1. We shall show (i) ⇒ (ii) ⇒ (iii), then that (iii) entails (i) and (iv) (and also (v) when S is semi-local). Set $K = P_1 \cap P_2$. By 4.1.1 and [BT65], 4.5, each geometric fiber K_s contains a maximal torus T_s of G_s and is connected, and the set of roots of K_s with respect to T_s is a closed subset of the set of roots of G_s with respect to T_s. Hence, if K is smooth over S, then it is a subgroup of type (RC) (hence a fortiori of type (R)), cf. Exp. XXII 5.2.1 and 5.11.1. We therefore have (i) ⇔ (ii).

If K is a subgroup of type (R), it contains locally for the étale topology a maximal torus of G, by Exp. XXII 2.2 and Exp. XIX 6.1. So (ii) ⇒ (iii).

Suppose (iii) holds and let us show that K is smooth. By (fpqc) descent, one may assume that $G = (G, T, M, R)$ is split, where T is a maximal torus contained in $P \cap Q$, and that there exist two closed subsets `R_1` and `R_2` of R such that $P = H_{R_1}$ and $Q = H_{R_2}$.

Since K has connected fibers, it follows from Exp. XXII 5.4.5 that K equals $H_{R_1 \cap R_2}$, hence is smooth over S and of type (RC).

Let R_1^s be the symmetric part of `R_1` and $R_1^a = R_1 - R_1^s$; then $rad^u(P_1) = H_{R_1^a}$. As noted in the proof of [BT65], 4.4, $R' = (R_1^s \cap R_2) \cup R_1^a$ is a closed subset of R such that $R' \cup (-R') = R$; hence, by Exp. XXI 3.3.6, R' contains a system of positive roots of R. Moreover, the symmetric part of R' is $R_1^s \cap R_2^s$, which is contained in $R_1 \cap R_2$.

It follows from the preceding that $P' = K \cdot rad^u(P)$ is a parabolic subgroup of G, and that K is a subgroup of type (RC) of P' such that $rad^u(K) = K \cap rad^u(P')$. Hence, by 2.11, K has a Levi subgroup L. It then follows from Exp. XIV 3.20 and 3.21 that K satisfies assertion (iv), as well as assertion (v) when S is semi-local. This proves 4.5.1.

Definition 4.5.1.1.[1468] *When the preceding conditions are satisfied, one says that P and Q are in* mutual standard position; *this is, for instance, the case if P and Q are in transversal position, or in osculating position, or if the base is the spectrum of a field. It is a notion stable under base extension and local for the (fpqc) topology.*

4.5.2.

Let (P, Q) and `(P', Q')` be two pairs of parabolic subgroups of G, in standard position, and let H be the subfunctor of G defined as follows: $H(S')$ is the set of

[1467] *N.D.E.* The proof of the equivalence of these conditions has been added (as well as condition (v), used implicitly in 6.17 of the original); consequently, §4.5.1 has been transformed into Proposition 4.5.1 plus Definition 4.5.1.1.

[1468] *N.D.E.* see N.D.E. (23). Moreover, for an example of parabolic subgroups `P, Q` that are not in standard position, see 7.11 below.

$g \in G(S')$ such that $int(g)P = P'$ and $int(g)Q = Q'$. It is a closed subscheme of G, smooth over S and formally principal homogeneous under $P \cap Q$. One deduces that the following conditions are equivalent:

(i) (P, Q) and (P', Q') are conjugate locally for the (fpqc) topology,

(ii) (P, Q) and (P', Q') are conjugate locally for the étale topology.

(iii) (P, Q) and (P', Q') are conjugate on each geometric fiber.

One then says that the pairs (P, Q) and (P', Q') have the same *type of mutual position*. This is a notion stable under base change and local for the (fpqc) topology.

4.5.3.

Let $Stand(G)$ be the subfunctor of $Par(G) \times_S Par(G)$ "formed of pairs in mutual standard position". Then $Stand(G)$ is representable, there exists an étale finite S-scheme `TypeStand` ("scheme of types of mutual standard position"), and a morphism, smooth, of finite presentation, with irreducible geometric fibers (and hence in particular faithfully flat)

```
t_2 : Stand(G) □ TypeStand
```

which is a quotient of $Stand(G)$ by the action of G: two sections of $Stand(G)$ (over an arbitrary $S' \to S$) have the same type of mutual position if and only if they have the same image under t_2. One has a commutative diagram

```
Stand(G) ————————t_2————→ TypeStand
   |                          |
   |                          | q
   ▼                          ▼
Par(G) ×_S Par(G) —t × t—→ Of(Dyn(G)) ×_S Of(Dyn(G)),
```

where the morphism q may be described by descent as follows: if (P, Q) is a pair of parabolic subgroups of G in mutual standard position, and if T is a maximal torus of $P \cap Q$, then the morphism $Norm_G(T) \to Stand(G)$ defined set-theoretically by $n \mapsto (P, int(n)Q)$ induces an isomorphism

$$W_P(T) \backslash W_G(T) / W_Q(T) \simeq q^{-1}(t(P), t(Q)).$$

(The left-hand side denotes the sheaf of double cosets ...). These assertions are proved without difficulty (remark in particular that $t_2^{-1}(t_2(P, Q)) \simeq G/(P \cap Q)$).

4.5.4.

Now let P be a fixed parabolic subgroup of G, and let `Par(G; P)` be the functor of parabolic subgroups of G in standard position relative to P. For each $t \in Of(Dyn(G))(S)$, set similarly `Par_t(G; P) = Par(G; P) ∩ Par_t(G)`. One sees at once that the two preceding functors are obtained from $Stand(G)$ by fibered products, and are therefore representable by S-schemes that are smooth and of finite presentation over S, with non-empty fibers. One has a canonical morphism t_P induced by t_2 (i.e. $t_P(Q) = t_2(P, Q)$)

$$t_P : Par_t(G; P) \to q^{-1}(t(P), t)$$

which is smooth and of finite presentation, with irreducible geometric fibers. The canonical morphism $Par_t(G;P) \to Par_t(G)$ is a surjective monomorphism, and may therefore be considered as a *cellular decomposition* of $Par_t(G)$ (indexed by the set of connected components of $q^{-1}(t(P), t)$).

4.5.5.

Suppose now that the type t is of the form $t(Q)$, where Q is a parabolic subgroup of G in standard position relative to P, and that $P \cap Q$ contains a maximal torus T.

Then $Par_t(G) \simeq G/Q$ and $q^{-1}(t(P), t) \simeq W_P(T)\, W_G(T)/W_Q(T)$, which gives a diagram

$$\begin{tikzcd}
Par_{t(Q)}(G; P) \arrow[r, "f"] \arrow[d, "i"'] & W_P(T) \backslash W_G(T) / W_Q(T) \\
Par_{t(Q)}(G) \arrow[r, "\sim"] & G/Q
\end{tikzcd}$$

where i is a surjective monomorphism, and where f is smooth and of finite presentation, with irreducible geometric fibers. Moreover, if Q_1 and Q_2 are two sections of $Par_{t(Q)}(G;P)$ (over an $S' \to S$), i.e. two parabolic

subgroups of $G_{S'}$ conjugate (locally for (fpqc)) to Q, and in standard position relative to $P_{S'}$, then Q_1 and Q_2 are conjugate by a section of P (locally for (fpqc)) if and only if $f(Q_1) = f(Q_2)$. If S is the spectrum of an algebraically closed field k, one thereby finds the relation

$$P(k) \backslash G(k) / Q(k) \simeq W_P(T)(k) \backslash W_G(T)(k) / W_Q(T)(k).$$

More generally, if one assumes that the scheme $W_P(T)\, W_G(T)/W_Q(T)$ is constant and of the form E_S (which happens, for instance, when G is split relative to T and S is connected), the $f^{-1}(e)$, $e \in E$, form a decomposition of $Par_{t(Q)}(G;P)$ into open and closed subschemes, which are homogeneous spaces under P, smooth and of finite presentation over S, with irreducible geometric fibers.

4.5.6.

Let us return to the general situation of 4.5.4. The scheme $q^{-1}(t(P), t)$[1469] always has two distinguished sections, corresponding respectively to the types "transversal position" and "osculating position". The inverse image of the first section is a relatively dense open subscheme of $Par_t(G;P)$, as one has seen above; it is the cell of maximum relative dimension of the decomposition.

[1469] *N.D.E.* $t^{-1}(t(P), t)$ has been corrected to $q^{-1}(t(P), t)$ and, lower down, $Par_t(G)$ to $Par_t(G;P)$ (twice).

The inverse image of the second section is presumably a closed subscheme of $Par_t(G;P)$; it is the cell of minimum relative dimension of the decomposition.

5. Conjugacy theorem

Theorem 5.1. *Let S be a semi-local scheme, G an S-reductive group, P and P' two opposite parabolic subgroups (4.3.3) of G. Then*

`rad^u(P)(S) · P' · P = G,`

i.e. the union of the open subsets $uP' \cdot P$ (4.3.6), for u running through $rad^u(P)(S)$, is all of G.

The proof is carried out in several steps:

5.1.1.

It suffices to give the proof in the case where S is the spectrum of a field k; this follows at once from 2.6.

5.1.2.

Let $L = P \cap P'$. Suppose L has a Borel subgroup `B_L`; let T be a maximal torus of `B_L` (Exp. XXII, 5.9.7); one verifies at once that $B = B_L \cdot rad^u(P)$ is a Borel subgroup of P; let B' be the Borel subgroup of G opposite to B relative to T (i.e. such that $B \cap B' = T$). One has $B' \subset P'$ as one verifies at once by splitting G relative to T. Let us prove that

`(x)    B^u(S) · B' · B ⊂ rad^u(P)(S) · P' · P.`

Since one has $B^u(S) \subset P(S) = rad^u(P)(S) \cdot L(S) \subset rad^u(P)(S) \cdot P'(S)$, it suffices to prove that $B' \cdot B \subset P' \cdot P$, which is evident. It follows from (x) that it suffices to prove 5.1

for the pair `(B, B')`.

5.1.3.

The theorem is true if k is algebraically closed; indeed, the condition of 5.1.2 is satisfied, and one concludes by Exp. XXII, 5.7.10.

5.1.4.

The theorem is true when k is an infinite field. Indeed, $rad^u(P)(k)$ is dense in $rad^u(P)(\bar{k})$ by 2.7, and the theorem is true for $\bar{k}$.

5.1.5.

We are therefore reduced to the case where k is a finite field. Now $Bor(L)$ is a smooth homogeneous space of L; it follows from Lang's theorem (Am. J. of Maths., 78, 1956[1470]) that L has a Borel subgroup B_L. By 5.1.2, one may therefore assume that $P = B$ and $P' = B'$ are Borel subgroups. One writes $T = B \cap B'$.

5.1.6.

Let K be the algebraic closure of k; choose a pinning of the triple (G_K, B_K, T_K), let R^+ (resp. Δ) be the set of positive roots (resp. simple). By Exp. XXII, 5.7.2, it suffices to prove that for every $\alpha \in \Delta$, one has

$$(1) \qquad u_\alpha B'^u(K) \subset B^u(k) \cdot B'^u(K) \cdot B(K).$$

Let α_i be the various roots conjugate to α over k (these are elements of Δ, since B is "defined over k"), and let R' be the set of roots that are linear combinations of the α_i. Set $R'^- = R^- \cap R'$. As "R' is defined

over k", there exists a subtorus Q of T such that Q_K is the maximal torus of the common kernel of the α_i.

Set $Z = Centr_G(Q)$, $B_Z = B \cap Z$, $B'_Z = B' \cap Z$ (cf. Exp. XXII, 5.10.2). Let us show that it suffices to verify the sought assertion in Z, that is

$$(2) \qquad u_\alpha \cdot B'^u_Z(K) \subset B^u_Z(k) \cdot B'^u_Z(K) \cdot B_Z(K).$$

One has $(B'^u_Z)_K = \prod_{\alpha \in R'^-} U_\alpha$; let R'' be the complement of R' in R, and set

$$V = \prod_{\alpha \in R'' \cap R^-} U_\alpha.$$

One immediately has $(B'^u)_K = (B'^u_Z)_K \cdot V$, and $(B_Z)_K$ normalizes V (Exp. XXII, 5.6.7). One therefore derives from (2) successively

$$\begin{aligned} u_\alpha \cdot B'^u(K) = u_\alpha \cdot B'^u_Z(K)\, V(K) &\subset B^u_Z(k)\, B'^u_Z(K)\, B_Z(K)\, V(K) \\ &\subset B^u_Z(k)\, B'^u_Z(K)\, V(K)\, B_Z(K), \end{aligned}$$

which yields (1) at once.

We are therefore reduced to the case where $G = Z$, that is, where the Galois group of K over k acts transitively on the simple roots.

5.1.7.

The assertion to be proved is equivalent to the fact that G/B is the union of the translates by $B^u(k)$ of the open image of B'^u, an assertion that does not change if one replaces G by its adjoint group (or by any group yielding the same adjoint group). One may therefore assume G adjoint.

5.1.8.

Consider then the Dynkin diagram of G_K. The Galois group acts transitively on this Dynkin diagram. But this Galois group has only cyclic quotients, and the Dynkin diagram has no cycles. It follows immediately that this diagram is of type nA_1,

[1470] *N.D.E.* See also [DG70], §III.5, 7.4.

$n \geqslant 0$, or mA_2, $m \geqslant 0$. Using the canonical decomposition of Exp. XXIV, 5.9, one may write

$$G = \prod_{D/K} G_0,$$

where D is a finite K-scheme and `G_0` is either a torus, of type `A_1`, or of type `A_2`.

By Exp. XXIV, 5.12, B comes from a Borel subgroup `B_0` of `G_0`, T from a maximal torus `T_0` of `G_0`; B' comes from the Borel subgroup B'_0 of `G_0` opposite to `B_0` relative to `T_0`. One has

```
B'^u(k) = B^u_0(D),
B'^u · T · B^u = ∏_{D/k} B'^u_0 · T_0 · B^u_0,
```

and it suffices to prove the sought assertion for the triple (G_0, B_0, T_0).

5.1.9.

One may therefore assume that G is of type $\emptyset$, `A_1`, or `A_2`. Since G has a Borel subgroup B, G is quasi-splittable relative to B (Exp. XXIV, 3.9.1), hence splittable if it is of type $\emptyset$ or `A_1`. Since the theorem has already been proved in the split case (Exp. XXII 5.7.10), only the case `A_2` remains to be treated. By Exp. XXIV, 3.11 there exists a morphism $E \to \mathrm{Spec}(k)$, principal Galois bundle under the group $\mathbb{Z}/2\mathbb{Z}$ of automorphisms of the Dynkin diagram of type `A_2`, such that $G = Q - \acute{E}p_{E/\mathrm{Spec}(k)}(A_2)$. If E has a section, G is splittable and the theorem is proved. Otherwise, one necessarily has $E = \mathrm{Spec}(k')$, where k' is a quadratic extension of k. Finally, as one saw in 5.1.7, one may assume G simply connected (i.e. that G is a form of $SL_{3,k}$).

5.1.10.

One is therefore in the following situation: one has a finite field k, a quadratic extension k' of k. The group $SL_{3,k'}$ of 3×3 matrices of determinant 1 is pinned as follows: the maximal torus is the group of diagonal matrices, the Borel subgroup is the group of upper triangular matrices, the "pins" are the elements:

$$u_\alpha = (110)u_\beta = (100)(010)(011)(001)(001).$$

One sees at once that the big cell Ω is defined by

```
( a b c )
( d e f ) ∈ Ω(S) ⇔ a and ae − bd are invertible,
( g h i )
```

and that

```
B^u(k) = { ( 1 x z )                                   }
         { ( 0 1 x̄ ) :  x, z ∈ k',  z + z̄ = x x̄ }
         { ( 0 0 1 )                                   }.
```

We must prove inclusion (1) of 5.1.6, that is to say, show that for all $a,b,c \in K$ (algebraic closure of k), there exist $x,z \in k'$ such that $z+\bar{z}=x\bar{x}$ and

$$(1)(1xz)(110)(100)(01\bar{x})(010)(a10) \subset \Omega(K).(001)(001)(bc1)$$

– If $a \neq -1$, one takes $x=z=0$.

– If $a=-1$, the conditions to be realized are written

```
{ z + z̄ = x x̄,
{ b z − x ≠ 0,
{ (b + c) z + b x − 1 ≠ 0.
```

Let q be the number of elements of k ($q \geqslant 2$). One knows that for every $m \in k$, the equation $z+\bar{z}=m$, with $z \in k'$, has q solutions.

– If $b=0$, take $x=1$; one must solve $z+\bar{z}=1$, $cz \neq 1$, which is always possible by the preceding remark.

– If $b \neq 0$, take $x=0$; one must solve

```
z + z̄ = 0,    z ≠ 0,    (b + c) z ≠ 1.
```

This is always possible if $q \geqslant 3$. If $k=\mathbb{F}_2$, it is possible if $b+c \neq 1$; one may take $z=1$.

– It remains to treat the case $k=\mathbb{F}_2$, $b+c=1$, $b \neq 0$. The system is then written

```
z + z̄ = x x̄,     b z ≠ x,     z + b x ≠ 1.
```

If $b=1$ (resp. $b \notin k'$), let us take $x=1$; then the last two conditions are written $z \neq b^{-1}, 1-b$, and they are consequences of $z+\bar{z}=1$, which has solutions. Finally, if $b \in k'-k$, one may take $x=b$, $z=\bar{b}$. QED.

Corollary 5.2. *Let S be a semi-local scheme, G an S-reductive group, P and P' two opposite parabolic subgroups of G. The canonical map*

$$rad^u(P)(S) \cdot rad^u(P')(S) \longrightarrow (G/P)(S)$$

is surjective (in particular, one has $(G/P)(S)=G(S)/P(S)$). Every parabolic subgroup Q of G, of the same type as P, is of the form `int(u u') P` *with $u \in rad^u(P)(S)$ and $u' \in rad^u(P')(S)$.*

The second assertion is evidently equivalent to the first; let us demonstrate the latter. Let s_i be the closed points of S, let V be the open subscheme of G/P image of P' (and isomorphic to $rad^u(P')$, cf. 4.3.6), and let $x \in (G/P)(S)$. By 5.1, there exists for each i a section $u_i \in rad^u(P)(\kappa(s_i))$ such that $u_i x_{s_i}$ is a section of V_{s_i}. If $u \in rad^u(P)(S)$ lifts the u_i (2.6), `u x` is a section of V, since such an assertion is verified on the closed fibers. But $rad^u(P')(S) \xrightarrow{\sim} V(S)$ is bijective, and one concludes at once.

Corollary 5.3. *Let S be a semi-local scheme, G an S-reductive group, P, P' and Q three parabolic subgroups of G. There exists $g \in G(S)$ such that $int(g)\ Q$ is in transversal position relative to P and P'.*

With the notation of 4.2.4 (ii), one must verify that the universally schematically dense open subscheme over S[1471] $Gen(Q/P) \cap Gen(Q/P')$ of G has a section over S. Let us in fact choose a parabolic subgroup of G opposite to Q (4.3.5 (i)), say Q_1, and set $U = rad^u(Q)$, $U' = rad^u(Q_1)$. We shall show that there exists $g \in U(S)U'(S)$ answering the question; in this form, it follows from 4.2.4 (i) and 2.6 that it suffices to verify the assertion on the fibers at the closed points of S, and one may therefore assume that S is the spectrum of a field k.

If k is algebraically closed, there exists $g \in G(k)$ answering the question; now g is written $u\ u'\ q$ with $u \in U(k)$, $u' \in U'(k)$, $q \in Q(k)$ (5.2), and one has $int(uu')Q = int(g)Q$.

If k is infinite, consider the open subset V of $U \times_k U'$ defined by the cartesian diagram

$$\begin{tikzcd} U \times_k U' \arrow[r] & G \\ V \arrow[u] \arrow[r] & Gen(Q/P) \cap Gen(Q/P'); \arrow[u] \end{tikzcd}$$

since $V(\bar{k}) \neq \emptyset$ by what was just seen, V is dense in $U \times_k U'$, hence has a section by 2.7.

If k is finite, P (resp. P') has a Borel subgroup B (resp. B'), by virtue of

Lang's theorem (cf. 5.1.5), since the schemes $Bor(P) \simeq Bor(P/rad^u(P))$ and $Bor(Q) \simeq Bor(Q/rad^u(Q))$ are smooth. If B_1 is a Borel subgroup opposite to B (4.3.5 (i)), there exist $a \in B^u(k)$ and $a_1 \in B_1^u(k)$ such that $int(aa_1)B = B'$ (5.2); then $B_0 = int(aa_1)B_1 = int(a)B_1$ is opposite to B' and to B; if Q_0 is the unique parabolic subgroup of G containing B_0 and of the same type as Q (3.8), Q_0 is in transversal position relative to P and P' (4.2.1 (vi)). On the other hand, by 5.2, Q_0 is written $int(u\ u')\ Q$ with $u' \in U'(k)$, $u \in U(k)$, which is what was to be demonstrated.

Corollary 5.4. *Let S be a semi-local scheme, G an S-reductive group, P and Q two parabolic subgroups of G. There exists $g \in G(S)$ such that $int(g)\ P$ is in osculating position relative to Q, i.e. (4.4.2) that $int(g)P \cap Q$ is a parabolic subgroup of G.*

Indeed, by 4.3.5 (i), there exists a parabolic subgroup P' of G opposite to P. By 5.3, there exists a parabolic subgroup P_1' of G of the same type as P', in transversal position relative to P and Q. If T is a maximal torus of $P_1' \cap Q$ (4.2.7 (ii)), and if P_1 is the opposite of P_1' relative to T, then P_1 and Q are in osculating position, by 4.4.5. On the other hand, P and P_1 being opposite to P_1', there exists $g \in rad^u(P_1')(S)$ such that $int(g)P = P_1$ (4.3.5 (i)). QED.

[1471] *N.D.E.* "Relatively dense" has been replaced by "universally schematically dense over S", cf. EGA IV₃, Def. 11.10.8.

Let us remark, moreover, that for the same reason there exists $u \in rad^u(P)(S)$ such that $int(u)P' = P'_1$, hence g is written `int(u) u'` with $u' \in rad^u(P')(S)$, which gives $P_1 = int(uu'u^{-1})P = int(uu')P$ and re-proves 5.2 in passing.

Statements 5.3 and 5.4 are the essential results of this section. Let us first state some consequences of 5.4.

Corollary 5.5. *Let S be a semi-local scheme, G an S-reductive group.*

(i) If P and Q are two parabolic subgroups of G and if $t(P) \subset t(Q)$ (cf. 3.3), there exists $g \in G(S)$ such that $int(g)P \subset Q$.

(ii) Let

`P_1 ⊃ P_2 ⊃ · · · · · · · ⊃ P_n    and    P'_1 ⊃ P'_2 ⊃ · · · · · · · ⊃ P'_n`

be two chains of parabolic subgroups of G such that $t(P_i) = t(P'_i)$. There exists $g \in G(S)$ such that $int(g)P_i = P'_i$ for each i.

(iii) Let `P, Q, P', Q'` be four parabolic subgroups of G such that $t(P) = t(P')$ and $t(Q) = t(Q')$. If the pairs `(P, P')` and `(Q, Q')` are in transversal (resp. osculating) position, there exists $g \in G(S)$ such that $int(g)P = P'$ and $int(g)Q = Q'$.

(iv) Let P and P' be two parabolic subgroups of the same type, L (resp. L') a Levi subgroup of P (resp. P'). There exists $g \in G(S)$ such that $int(g)P = P'$ and $int(g)L = L'$.

Proof: (i) follows at once from 5.4; (ii) is proved by induction on n, the case $n = 0$ being trivial; one may therefore assume $P_i = P'_i$ for $i = 1, \cdots, n-1$; by 5.2 there exists $g \in G(S)$ such that $int(g)P_n = P'_n$; but then P_n and $int(g)P_n$ are contained in $P_{n-1} = P'_{n-1}$, hence $g \in P_{n-1}(S)$ (4.4.4 (ii)) and $int(g)P_i = P'_i$ for $i = 1, \cdots, n$.

On the other hand, (iv) follows at once from 5.2 and 1.8. Let us demonstrate (iii) in the case of "transversal position"; the assertion is a consequence of (iv) when the types of P and Q are opposite (4.3.3 (iii)); in the general case, by 4.2.7 (iii) and 4.4.6, one may find parabolic subgroups P_1, P'_1, Q_1, Q'_1 of `P, P', Q, Q'` respectively, such that `P_1` and P'_1 are opposite, as are `Q_1` and Q'_1, and such that $t(P_1) = t(P'_1)$; there therefore exists $g \in G(S)$ such that $int(g)P_1 = P'_1$, $int(g)Q_1 = Q'_1$, and one may assume $P_1 = P'_1$ and $Q_1 = Q'_1$; but then P and P' are in osculating position and of the same type, hence $P = P'$ (4.4.4 (i)); for the same reason $Q = Q'$.

It remains to demonstrate assertion (iii) in the case of "osculating position". By the conjugacy theorem (5.2), one may assume $P = P'$; by the same theorem, one may find $g \in G(S)$ such that $int(g)Q = Q'$; but then $g \in P(S)$ by 4.4.4 (ii) and one has $int(g)P = P = P'$.

Definition 5.6. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G. One says that P is* minimal *if whenever Q is a parabolic subgroup of G contained in P, one has $Q = P$.*

One should note that this is not in general a notion stable under passage to fibers.

Corollary 5.7. *Let S be a semi-local scheme, G an S-reductive group.*

(i) Let $t, t' \in Of(Dyn(G))(S)$. If there exist in G a parabolic subgroup of type t and a parabolic subgroup of type t', there exists a parabolic subgroup of type $t \cap t'$. In particular, there exists a smallest element $t_{\min}$ in the set of $t(P)$, P running through the set of parabolic subgroups of G.

(ii) Every parabolic subgroup of G contains a minimal parabolic subgroup. For a parabolic subgroup of G to be minimal, it is necessary and sufficient that it be of type $t_{\min}$. Two minimal parabolic subgroups of G are conjugate by an element of $G(S)$.

This follows at once from 5.4 and 5.5 (i).

Remark 5.8. *A parabolic subgroup opposite to a minimal parabolic subgroup is also minimal; this entails $s(t_{\min}) = t_{\min}$.*

Corollary 5.9. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G. The canonical morphism*

$$G \longrightarrow G/P = X$$

makes G an X-bundle locally trivial (in the sense of Zariski) with group P_X. If L is a Levi subgroup of P, the canonical morphism (cf. 3.12)

$$G \longrightarrow G/L = Y$$

makes G a Y-bundle locally trivial (in the sense of Zariski) with group L_Y.

It suffices to prove that if one has a morphism $S' \to S$, where S' is local and a morphism $S' \to X$ (resp. $S' \to Y$), it lifts to a morphism $S' \to G$. In other words, one may assume S local, and one must show that the map $G(S) \to X(S)$ (resp. $G(S) \to Y(S)$) is surjective.

The first assertion was demonstrated in 5.2; let us demonstrate the second. Let $y \in Y(S)$; its canonical image in $X(S)$ comes from a $g \in G(S)$; the projection y' of g in $Y(S)$ therefore has the same projection as y in $X(S)$. There exists therefore a unique $u \in rad^u(P)(S)$ such that $y'u = y$, and the projection of g u in $Y(S)$ is indeed y.

Corollary 5.10. *Let S be a semi-local scheme, G an S-reductive group.*

(i) Let P be a parabolic subgroup of G, L a Levi subgroup of P. The canonical maps (cf. 3.21) induce bijections

H^1(S, L) □ H^1(S, P) □ H^1_{t(P)}(S, G).

(ii) Let $t, t' \in Of(Dyn(G))(S)$; one has (cf. 3.21)

H^1_t(S, G) ∩ H^1_{t'}(S, G) = H^1_{t ∩ t'}(S, G).

(iii) If P and Q are two parabolic subgroups of G in osculating position, the following canonical diagram is cartesian and composed of injections:

$$\begin{tikzcd} H^1(S, P) \arrow[r] & H^1(S, G) \\ H^1(S, P \cap Q) \arrow[r] \arrow[u] & H^1(S, Q). \arrow[u] \end{tikzcd}$$

Let us demonstrate (i). The map $H^1(S,L) \to H^1(S,P)$ is bijective by 2.3; the map $H^1(S,P) \to H^1_{t(P)}(S,G)$ is surjective (3.21); let us show that it is injective, i.e. that the canonical map $H^1(S,P) \to H^1(S,G)$ is injective. Let Q be a principal bundle under P, Q_1 the associated principal bundle under G, P' and G' the corresponding twisted forms of P and G. It is clear that G' is an S-reductive group and that P' is a parabolic

subgroup of it. The set of elements of $H^1(S,P)$ that have the same image as the class of Q in $H^1(S,G)$ is naturally identified with the kernel of the canonical map $H^1(S,P') \to H^1(S,G')$, and this, by the exact sequence of cohomology, with the set of orbits of $G'(S)$ in $(G'/P')(S)$ (for these arguments of non-abelian cohomology, see Giraud's thesis[1472]). But $G'(S)$ acts transitively on $(G'/P')(S)$ by 5.2.

Let us demonstrate (ii): let Q be a principal homogeneous bundle under G and G_Q the corresponding twisted form of G. By definition (3.21), we must prove that G_Q has a parabolic subgroup of type $t \cap t'$ if and only if it has parabolic subgroups of type t and t', which is none other than the conjunction of 3.8 and 5.7 (i). Finally, (iii) follows at once from (i) and (ii).

Let us now state a consequence of 5.3.

Corollary 5.11. *Let S be a semi-local scheme, G an S-reductive group, P a parabolic subgroup of G. If $P \neq G$, there exist at least three distinct parabolic subgroups of G of the same type as P; in other words, $P \neq G$ entails $(G(S) : P(S)) \geqslant 3$.*

Indeed, let P' be a parabolic subgroup of G opposite to P (4.3.5 (i)). Since $P \neq G$, one has $rad^u(P') \neq e$ (by 4.3.2 for example). By 2.1, $rad^u(P')(S) \neq e$; let then $u \in rad^u(P')(S)$, $u \neq e$. Then $int(u)P \neq P$, and by 5.3, there exists a P_1, of the same type as P, and opposite to P and $int(u)P$; then P_1, P and $int(u)P$ are three distinct parabolic subgroups of G, of the same type as P.

6. Parabolic subgroups and split tori

Proposition 6.1. *Let S be a scheme, G an S-reductive group, $g = Lie(G/S)$, and Q a split subtorus of G. Write $Q = D_S(M)$ and let*

$$g = \coprod_{\alpha \in M} g^\alpha$$

be the decomposition of g under the action of Q. Let M_1 be a subset of M such that $0 \in M_1$ and that $\alpha, \beta \in M_1 \Rightarrow \alpha + \beta \in M_1$.

(i) There exists a unique smooth subgroup H_{M_1} of G, with connected fibers, containing $Centr_G(Q)$, and whose Lie algebra is $\bigoplus_{\alpha \in M_1} g^\alpha$.

[1472] *N.D.E.* See [Gi71], §III.3.

(ii) One has the following implications:

$$\begin{array}{lcl}
M_1 = \{0\} & \Longrightarrow & H_{M_1} = \mathrm{Centr}_G(Q), \\
M_1 = -M_1 & \Longrightarrow & H_{M_1} \text{ is reductive}, \\
M_1 \cup (-M_1) = M & \Longrightarrow & H_{M_1} \text{ and } H_{-M_1} \text{ are parabolic subgroups of } G, \\
 & & \text{opposite, with common Levi subgroup } H_{M_1 \cap -M_1}.
\end{array}$$

To demonstrate (i) and (ii), which are local for the (fpqc) topology, one may assume that Q is contained in a maximal

torus T of G; one may furthermore split G relative to T. Assertion (i) then follows at once from Exp. XXII, 5.3.5, 5.4.5 and 5.4.7; the assertions of (ii) follow from Exp. XXII, 5.3.5, 5.10.1, 5.11.3 and, in this Exposé, 1.4 and 4.3.2.

Corollary 6.2. *Let S be a scheme, G an S-reductive group, Q a split subtorus of G. There exists a parabolic subgroup of G of which $Centr_G(Q)$ is a Levi subgroup.*

Indeed, writing $Q = D_S(M)$, one chooses a total order structure on the group M, one calls M_1 the set of positive elements of M; the group H_{M_1} answers the question.

Corollary 6.3. *If the S-reductive group G has a non-central split subtorus, it has a proper parabolic subgroup (i.e. $\neq G$).*

By 5.9 and 5.10, one derives from 6.2:

Corollary 6.4. *Let S be a scheme, G an S-reductive group, Q a split subtorus of G. The canonical morphism $G \to G/Centr_G(Q)$ is a locally trivial fibration.*

If S is semi-local, the map $G(S) \to (G/Centr_G(Q))(S)$ is surjective, and the map $H^1(S, Centr_G(Q)) \to H^1(S, G)$ is injective.

6.5.

Suppose S connected. If T is an S-torus and T' and T'' are two split subtori of T, their product $T' \cdot T''$[1473] is also a split subtorus of T. Indeed,

it is identified with the quotient of $T' \times_S T''$ by $T' \cap T''$, a quotient that is split by Exp. IX, 2.11. It follows that T has a largest split subtorus; one denotes it T_{spl}.

Lemma 6.6. *Let S be a connected scheme, T an isotrivial S-torus, T_{spl} its largest split subtorus. The following conditions are equivalent:*

(i) There exists a homomorphism $T \to G_{m,S}$ distinct from e.

(ii) $T_{spl} \neq e$.

Since T is assumed isotrivial, there exist a finite group Γ, a connected principal Galois covering $S' \to S$ of group Γ, and an isomorphism $T_{S'} \simeq D_{S'}(M)$; M is then endowed with a structure of Γ-module, and one has a natural isomorphism $\mathrm{Hom}_S(T, G_{m,S}) = H^0(\Gamma, M)$.

[1473] *N.D.E.* Multiplicative notation has been adopted, i.e. "their sum $T' + T''$" has been replaced by "their product $T' \cdot T''$".

On the other hand, let V be the vector subspace of $M \otimes \mathbb{Q}$ generated by elements of the form $g(m) - m$, $g \in \Gamma$, $m \in M$. One verifies at once that $(T_{spl})_{S'}$ is identified with $D_{S'}(M/M \cap V)$. Assertion (i) is therefore equivalent to $H^0(\Gamma, M) \neq 0$, or equivalently to $H^0(\Gamma, M \otimes \mathbb{Q}) \neq 0$, whereas assertion (ii) is equivalent to $M \neq M \cap V$, or equivalently to $M \otimes \mathbb{Q} \neq V$. Now one has $M \otimes \mathbb{Q} = H^0(\Gamma, M \otimes \mathbb{Q}) \bigoplus V$, as one verifies at once (consider the projector $M \otimes \mathbb{Q} \to H^0(\Gamma, M \otimes \mathbb{Q})$ that sends x to the average of the transforms of x by Γ).

Lemma 6.7. *Let S be a scheme, G an S-reductive group, P a parabolic subgroup of G such that $P \neq G$, L a Levi subgroup of P, Q its radical. There exists a homomorphism $Q \to G_{m,S}$ distinct from e.*

Consider the unipotent radical U of P; it is invariant under $int(P)$, hence under $int(Q)$. Consider the invertible O_S-module $det(Lie(U))$, the "maximum exterior power" of the locally free O_S-module $Lie(U)$. The adjoint representation defines a homomorphism of groups

$f : Q \to Aut(det(Lie(U))) = G_{m, S}$.

If $P \neq G$, then $U \neq e$. Let us choose $s \in S$ such that $U_s \neq e$. Splitting G_s relative to a maximal torus containing Q_s, one sees at once that $f_s \neq e$.

Proposition 6.8. *Let S be a connected semi-local scheme, G an S-reductive group, P a parabolic subgroup of G, L a Levi subgroup of P, Q its radical, Q_{spl} the largest split subtorus of Q (i.e. the largest central split subtorus of L). Then*

$$L = Centr_G(Q_{spl}).$$

Set $L' = Centr_G(Q_{spl})$; this is a reductive subgroup of G containing L; moreover, $P' = P \cap L'$ is a parabolic subgroup of L', of Levi subgroup L (1.20). If $L' \neq L$, then $L' \not\subset P$ (since L is a maximal reductive subgroup of

P, cf. 1.7), hence $P' \neq L'$.

Let G_1 be the derived group of G, and $P_1 = P' \cap G_1$. By 1.19, P_1 is a parabolic subgroup of the semisimple group G_1, $L_1 = L \cap G_1$ is a Levi subgroup of it, and

$Q_1 = rad(L_1) = (rad(L) \cap G_1)^0 = (Q \cap G_1)^0$.

Since L_1 has a maximal torus T_1 (Exp. XIV, 3.20), and the latter is isotrivial (Exp. XXIV, 4.1.5), Q_1, being a subtorus of T_1, is also isotrivial (Exp. IX, 2.11); since $P_1 \neq G_1$, one may apply 6.7 and 6.6, and one therefore has $(Q_1)_{spl} \neq e$, whence $(Q_{spl} \cap G_1)^0 \neq e$, hence $Q_{spl} \not\subset rad(L')$ (since $rad(L') \cap G_1$ is finite), which is contradictory with the definition of L'.

Corollary 6.9. *Let S be a connected semi-local scheme, G an S-reductive group, Q a critical subtorus of G (i.e. such that $rad(Centr_G(Q)) = Q$). For $Centr_G(Q)$ to be a Levi subgroup of a parabolic subgroup of G, it is necessary and sufficient that $Centr_G(Q) = Centr_G(Q_{spl})$, that is, $Lie(G)^Q = Lie(G)^{Q_{spl}}$.*

This follows from 6.2 and 6.8.

Corollary 6.10. *Let S be a connected semi-local scheme, G an S-reductive group, L a subgroup of G. The following conditions are equivalent:*

(i) There exists a parabolic subgroup of G of which L is a Levi subgroup.

(ii) There exists a split subtorus of G of which L is the centralizer.

(iii) There exists a homomorphism $G_{m,S} \to G$ of which L is the centralizer.

Indeed, one has (i) ⇒ (ii) by 6.8, and (iii) ⇒ (i) by 6.1; it remains to prove (ii) ⇒ (iii). Suppose then $L = Centr_G(Q)$, with $Q = D_S(M)$; write $g = Lie(G/S)$ and

$$g = \coprod_{\alpha \in M} g^\alpha,$$

and let R be the set of $\alpha \in M - 0$ such that $g^\alpha \neq 0$.

Since R is finite and does not contain 0, there exists a homomorphism $u : M \to \mathbb{Z}$ such that $u(\alpha) \neq 0$ for each $\alpha \in R$. By duality, u gives a homomorphism $G_{m,S} \to Q$, hence a homomorphism $f : G_{m,S} \to G$. One has $Centr_G(f) \supset Centr_G(Q)$; these are two smooth subgroups of G, with connected fibers; their Lie algebras coincide (since both are equal to g^0); they therefore coincide, by a customary argument.

Corollary 6.11. *Let S be a connected semi-local scheme, G an S-reductive group. The maps*

$$L \mapsto rad(L)_{spl}, \qquad Q \mapsto Centr_G(Q)$$

are mutually inverse bijections, which reverse the natural order structures, between the set of subgroups L of G that are Levi subgroups of parabolic subgroups of G and the set of split subtori Q of G such that $rad(Centr_G(Q))_{spl} = Q$.

Corollary 6.12. *Let S be a connected semi-local scheme, G an S-reductive group. Consider the following assertions:*

(i) There exists a parabolic subgroup of G distinct from G.

(ii) G has a non-central split subtorus.

(ii bis) G has a non-central split subtorus of relative dimension 1.

(iii) There exists a homomorphism of groups $G_{a,S} \to G$ that is a closed immersion.

Then one has (i) ⇔ (ii) ⇔ (ii bis) ⇒ (iii).

The only new assertion is (i) ⇒ (iii). Let then P be a parabolic subgroup of G, distinct from G. Then $U = rad^u(P) \neq e$. Consider the last non-trivial subgroup U_n of the composition series of U (2.1). One has an isomorphism $U_n \simeq W(E_n)$, where E_n is a locally free O_S-module, hence free[1474]. Since $E_n \neq 0$, there exists a locally direct-factor monomorphism $O_S \to E_n$, hence a closed immersion $G_{a,S} = W(O_S) \hookrightarrow W(E_n) \simeq U_n$, which yields (iii) at once.

[1474] *N.D.E.* Since S is assumed semi-local and connected.

Remark 6.12.1. *When S is the spectrum of a field of characteristic 0, it follows from the Jacobson–Morozov theorem that (iii) ⇒ (ii bis). The preceding four conditions are then equivalent ("Godement's criterion", cf. [BT65], 8.5).*[1475]

Definition 6.13. *Let S be a connected semi-local scheme, G an S-reductive group. One says that G is* anisotropic *if G contains no split subtorus not reduced to e.*

Corollary 6.14. *Let S be a connected semi-local scheme. For the S-reductive group G to be anisotropic, it is necessary and sufficient that it have no parabolic subgroup $P \neq G$, and that its radical be anisotropic.*

Using now 6.6, Exp. XXIV, 4.1.5, and Exp. XXII, 6.2, one deduces:

Corollary 6.15. *Let S be a connected semi-local scheme, G an isotrivial S-reductive group (e.g. G semisimple, or S normal (Exp. X 5.16)). For G to be anisotropic, it is necessary and sufficient that G have no parabolic subgroup $P \neq G$, and that* $\mathrm{Hom}_{S-gr.}(G, G_{m,S}) = e$.

Proposition 6.16. *Let S be a connected semi-local scheme, G an S-reductive group. The maximal split subtori of G are the largest central split subtori of the Levi subgroups of the minimal parabolic subgroups of G. Two such tori are conjugate by an element of $G(S)$.*

Let Q be a maximal split subtorus of G.[1476] Then, by 6.2, $L = Centr_G(Q)$ is a Levi subgroup of a parabolic subgroup P of G, and since `Q ⊂ rad(Centr_G(Q))_{spl}`, the maximality of Q entails `Q = rad(Centr_G(Q))_{spl}`. By 6.11, L is a minimal element of the set of Levi subgroups of parabolic

subgroups of G, hence P is a minimal parabolic subgroup of G by 1.20. It then follows from 5.7 and 5.5 (iv) that two subtori such as Q are conjugate by a section of $G(S)$. The conjugacy of the Q and of the pairs (P, L) then entails the first assertion of 6.16.

Corollary 6.17. *Let S be a connected semi-local scheme, P and P' two minimal parabolic subgroups in standard position (4.5.1.1). Then $P \cap P'$ contains a common Levi subgroup of P and P'.*

Indeed, $P \cap P'$ contains a maximal torus T of G, by 4.5.1 (v); let L be the unique Levi subgroup of P containing T. One has

`rad(P) ∩ T = rad(P) ∩ L = rad(L)`

by 1.21, hence $rad(P) \cap T$ contains $rad(L)_{spl}$, which is a maximal split subtorus of G, hence is necessarily equal to T_{spl}. One therefore has $L = Centr_G(T_{spl})$, and by symmetry L is also a Levi subgroup of P'.

Remark 6.18. *It follows from 1.21 that the parabolic subgroup P of G is minimal if and only if $rad(P)$ contains a maximal split subtorus of G; then, by the proof of 6.17, if T is a maximal torus of P, T_{spl} is a maximal split torus of G and of $rad(P)$,*

[1475] This is more generally true when S is the spectrum of a perfect field (Tits).[^N.D.E-XXVI-31]
[1476] *N.D.E.* The following sentence has been spelled out.

and $Centr_G(T_{spl})$ is a Levi subgroup of P. Moreover, every Levi subgroup of P is obtained in this way.

7. Relative root datum

In this section, S will denote a non-empty connected semi-local scheme, G an S-reductive group, Q a maximal split subtorus of G, and L the centralizer of Q in G, i.e. $L = Centr_G(Q)$.

7.1.

Since Q is the largest central split subtorus of L, every section of $G(S)$ that normalizes L also normalizes Q. One therefore has (cf. 7.1.1)

$$Norm_G(L)(S) = Norm_G(Q)(S).$$

On the other hand, one saw in 6.4 that the map $G(S) \to (G/L)(S)$ is surjective. It follows that one has a canonical identification

```
W_G(Q)(S) = (Norm_G(Q) / Centr_G(Q))(S) ≃ Norm_G(L)(S) / L(S).
```

One will denote by M the group $\mathrm{Hom}_{S-gr.}(Q, G_{m,S})$, so that one has a canonical isomorphism $Q \simeq D_S(M)$. One will denote by W the group of automorphisms of M defined by $W_G(Q)(S)$. One therefore has isomorphisms

```
W ≃ W_G(Q)(S) ≃ Norm_G(L)(S) / L(S).
```

7.1.1.

One does not in general have $Norm_G(L) = Norm_G(Q)$. Take, for example, for S the spectrum of a field k, having a quadratic extension k', for G the unitary group $Q - \acute{E}p_{k'/k}(A_2)$ (cf. Exp. XXIV, 3.11.2).

Since the minimal parabolic subgroups of G are its Borel subgroups, their Levi subgroups are maximal tori, and one has $(Norm_G(L)/L)(k) = S_3$.

On the other hand, since G is not split, the maximal split tori of G are of dimension $\leqslant 1$, hence isomorphic to $G_{m,k}$. Since $Norm_G(Q)/L$ acts faithfully on Q, one has $(Norm_G(Q)/Q)(k) = \mathbb{Z}/2\mathbb{Z}$.

7.2.

If P is a parabolic subgroup of G with Levi group L (one exists by 6.2), P is necessarily minimal (cf. 6.18). By the conjugacy of minimal parabolic subgroups of G (5.7), of that of Levi subgroups of a parabolic subgroup (1.8), and of the equalities $P = Norm_G(P)$ and $Norm_G(L) \cap P = L$ (1.6), one obtains: the set of (minimal) parabolic subgroups of G with Levi group L is principal homogeneous under the group W.

7.3.

The Lie algebra of G decomposes under the action of Q as

$$Lie(G) = Lie(L) \oplus \bigoplus_{\alpha \in R} Lie(G)^{\alpha},$$

where R is the set of non-zero characters of Q such that $Lie(G)^{\alpha} \neq 0$ (roots of G relative to Q).

Let us denote by M^* the group $\mathrm{Hom}_{S-gr.}(G_{m,S}, Q)$, which is in duality with M and on which W acts naturally by transport of structure.

Theorem 7.4. *With the notations of 7.3, there exists a unique map $\alpha \mapsto \alpha^*$ from R to M^* that defines on (M, M^*) a root datum (Exp. XXI, 1.1) whose Weyl group is W.*

Moreover, the parabolic subgroups P of G with Levi group L and the systems of positive roots R^+ of R correspond bijectively via the relation

$$Lie(P) = Lie(L) \oplus \bigoplus_{\alpha \in R^+} Lie(G)^{\alpha}.$$

7.4.1.

Suppose first that the existence of the map $\alpha \mapsto \alpha^*$ sought has been proved. By Exp. XXI, 3.4.10, s_α is the unique element of W such that for every $m \in M$, $s_\alpha(m) - m$ is a rational multiple of α, which shows that s_α is determined by α; as one then has[1477] $\alpha^*(m)\alpha = m - s_\alpha(m)$, one sees that α^* is determined by α, which proves the uniqueness of the map $\alpha \mapsto \alpha^*$.

7.4.2.

Let $\alpha \in R$ and let L_α (resp. H_α, resp. $H_{-\alpha}$) be the unique smooth subgroup of G with connected fibers containing L and such that (cf. 6.1 (i))

$$\begin{aligned} Lie(L_\alpha) &= Lie(L) \oplus \bigoplus_{\gamma \in \mathbb{Z}\alpha \cap R} Lie(G)^{\gamma}, \\ Lie(H_\alpha) &= Lie(L) \oplus \bigoplus_{\gamma \in \mathbb{N}\alpha \cap R} Lie(G)^{\gamma}, \\ Lie(H_{-\alpha}) &= Lie(L) \oplus \bigoplus_{\gamma \in -\mathbb{N}\alpha \cap R} Lie(G)^{\gamma}. \end{aligned}$$

L_α is a reductive subgroup of G, H_α and $H_{-\alpha}$ are parabolic subgroups of it with Levi subgroup L, and H_α and $H_{-\alpha}$ are opposite relative to L (cf. 6.1 (ii)).

By 7.2, there therefore exists $s_\alpha \in Norm_{L_\alpha}(Q)(S)/L(S) \subset W$ such that $s_\alpha(H_\alpha) = H_{-\alpha}$. One has $s_\alpha(\alpha) = -\alpha$ (because α (resp. $-\alpha$) is the common divisor of the elements of R occurring in H_α (resp. $H_{-\alpha}$)), and one has $s_\alpha^2 = id$ (since $s_\alpha^2(H_\alpha)$ and H_α are both opposite to $H_{-\alpha}$ relative to L). One has therefore constructed an $s_\alpha \in W$ satisfying the following properties:

(x) $s_\alpha(\alpha) = -\alpha, \quad s_\alpha^2 = id;$

(xx) s_α can be represented by an element of $L_\alpha(S)$.

Let us moreover remark that s_α is constructed canonically from α, and in particular that

[1477] *N.D.E.* $\alpha^*(m)$ has been corrected to $\alpha^*(m)\alpha$.

(xxx) for every w ∈ W, one has w s_α w^{-1} = s_{w(α)}.

7.4.3.

We now propose to prove the assertion:

(xxxx) for every m ∈ M, s_α(m) − m ∈ ℤα.

Since S is connected, this assertion is local for the (fpqc) topology. One may therefore assume that $L_\alpha = G_1$ is splittable relative to a maximal torus T_1 of L. Let then (G_1, T_1, M_1, R_1) be such a splitting. The monomorphism $Q \to T_1$ identifies M with a quotient of M_1; let $p : M_1 \to M$ be the canonical map.

The image of R_1 under p consists, possibly with the inclusion of zero, of the roots of G_1 with respect to Q

(hence of the elements of R that are integer multiples of α); one therefore has $p(R_1) \subset \mathbb{Z}\alpha$. By (xx), there exists an element of $Norm_{G_1}(L)(S)$ that induces s_α on Q. By Exp. XXII, 5.10.10, there therefore exists a section $w \in (W_1)_S(S)$ that induces s_α on Q (one denotes by W_1 the Weyl group of the root datum $(M_1, R_1, \cdots)$). Possibly after restricting S, one may therefore assume that there exists $w \in W_1$ inducing s_α on Q, hence satisfying $p(w(m_1)) = s_\alpha(p(m_1))$ for every $m_1 \in M_1$. But, by definition of W_1, w is a product of reflections with respect to elements of R_1, hence $w(m_1) - m_1$ is a linear combination with integer coefficients of elements of R_1. It follows that $s_\alpha(p(m_1)) - p(m_1)$ is a linear combination with integer coefficients of the elements of $p(R_1) \subset \mathbb{Z}\alpha$, hence an integer multiple of α, as was to be proved.

7.4.4.

One may therefore define an element $\alpha^* \in M^*$ by[1478]

α^*(m) α = m − s_α(m).

By (x), one has $(\alpha^*, \alpha) = 2$; on the other hand, it follows from (xxx) that for every pair $(\alpha, \alpha') \in R \times R$, one has $s_\alpha(\alpha') \in R$ and $s_\alpha(\alpha'^*) = s_\alpha(\alpha')^*$, which proves (cf. Exp. XXI, 1.1) that the constructed map $\alpha \mapsto \alpha^*$ indeed defines a root datum on (M, M^*).

7.4.5.

Let W' be the Weyl group of this root datum (the group of transformations of M generated by the s_α); one has $W' \subset W$.

Let on the other hand $\geqslant$ be a total order relation on the free abelian group M; set $R^+ = \alpha \in R | \alpha \geqslant 0$. One knows that R^+ is a system of positive roots of R. Let $w \in W$, represented by an $n \in Norm_G(L)(S) = Norm_G(Q)(S)$. Set $P = H_{R^+}$

(notation of 6.1); by *loc. cit.*, P is a parabolic subgroup of G with Levi subgroup L. One obviously has $int(n)P = H_{w(R^+)}$. It then follows from 7.3 that $w(R^+) = R^+$ entails $w = e$. As the group W' acts transitively on the systems of positive roots

[1478] *N.D.E.* $\alpha^*(m)$ has been corrected to $\alpha^*(m)\alpha$.

of R (Exp. XXI, 3.3.7) and the stabilizer in W of R^+ is the identity, one concludes immediately that $W = W'$. One also concludes that $W = W'$ acts in a simply transitive way both on the set of systems of positive roots of R and on the set of parabolic subgroups of G with Levi group L, which entails the last assertion of 7.4. QED.

7.5.

If P and P_1 are two minimal parabolic subgroups of G, with Levi subgroups L and L_1, and if one denotes by Q and Q_1 the maximal central split tori of L and L_1, then the pairs (P, Q) and (P_1, Q_1) are conjugate: there exists $g \in G(S)$ such that $int(g)P = P_1$ and $int(g)Q = Q_1$.

Indeed, P and P_1 are conjugate (5.7) and one may therefore assume $P = P_1$; then L and L_1 are conjugate by a section of $P(S)$ (1.8). Moreover, if g and g' are two sections of G conjugating the pairs (P, Q) and (P_1, Q_1), then $g'^{-1}g$ normalizes P and Q, hence P and L; but

$$\mathrm{Norm}_G(P) \cap \mathrm{Norm}_G(L) = P \cap \mathrm{Norm}_G(L) = L = \mathrm{Centr}_G(Q).$$

The isomorphism $Q \xrightarrow{\sim} Q_1$ induced by $int(g)$ is therefore independent of g.

Let R and R_1 be the root data defined thanks to 7.4 in $\mathrm{Hom}_{S-gr.}(Q, G_{m,S})$ and $\mathrm{Hom}_{S-gr.}(Q_1, G_{m,S})$, and let R^+ and R_1^+ be the systems of positive roots corresponding to P and P_1. The canonical isomorphism $Q \xrightarrow{\sim} Q_1$ defined above transforms (R, R^+) into (R_1, R_1^+). One immediately deduces that one may define the *pinned relative root datum*[1479] of G over S, by identifying the various (R, R^+) by means of the transitive system of isomorphisms described above.

From now on, we shall denote $(M, M^*, R, R^*, R^+) = R(G/S)$ this pinned root datum; for each pair $P \supset Q$ as above, one therefore has a canonical isomorphism $M \xrightarrow{\sim} \mathrm{Hom}_{S-gr.}(Q, G_{m,S})$ transforming R (resp. R^+) into the set of roots of G (resp. of P) relative to Q, and $W(R)$ into $W_G(Q)(S)$.

7.6.

Let still Q be a maximal split torus of G, P a (minimal) parabolic subgroup of G with Levi group $Centr_G(Q)$, (M, M^*, R, R^*, R^+) the corresponding pinned root datum (7.4), and Δ the set of simple roots of R^+. For every $A \subset \Delta$, let $R_A \subset R$ be the set

$$R_A = R^+ \cup (\mathbb{Z}A \cap R^-)$$

consisting of positive roots and negative roots that are linear combinations of elements of A. It is a closed set (Exp. XXI, 3.1.4) of roots, and every closed set containing R^+ is uniquely of this form (Exp. XXI, 3.3.10). By 6.1,

[1479] *N.D.E.* Recall (Exp. XXIII 1.5) that a pinned root datum is a root datum endowed with the choice of a system of positive roots (or of simple roots).

there exists a unique subgroup P_A of G, smooth and with connected fibers, containing $Centr_G(Q)$ and such that

$$Lie(P_A) = Lie(G)^0 \oplus \coprod_{\alpha \in R_A} Lie(G)^\alpha.$$

It then follows at once from 6.1, from the conjugacy of minimal parabolics, and from the fact that the set of roots of a parabolic subgroup of G containing Q is closed (which is deduced at once from 1.4 by splitting), that:

Proposition 7.7. *(i) The map $A \mapsto P_A$ is a bijection of the set of subsets of Δ onto the set of parabolic subgroups of G containing P. This bijection preserves the natural order relations of inclusion.*

(ii) Every parabolic subgroup of G is conjugate by a section of $G(S)$ to a unique P_A.

7.8.

Let $P \supset Q$ be as above. Consider the relative root datum (7.5) of G over S and the canonical isomorphism

$$f : (M, M^*, R, R^*, R^+) \xrightarrow{\sim} (M, M^*, R, R^*, R^+).$$

The set Δ of simple roots of R^+ is transformed into the set Δ of simple roots of R^+, hence every subset A of Δ into a subset $f(A) \subset \Delta$.

Definition 7.9.0.[1480] *Let H be an arbitrary parabolic subgroup of G. By 7.7 (ii), it is conjugate to a unique P_A. Let us denote $t_r(H) = f(A) \subset \Delta$. One verifies at once using the conjugacy theorems that $t_r(H)$ is independent of the choice of the pair $P \supset Q$. One says that it is the* relative type *of H.*

Proposition 7.9. *(i) The map $H \mapsto t_r(H)$ induces a bijection between the set of conjugacy classes (under $G(S)$) of parabolic subgroups of G, and the set of subsets of Δ.*

(ii) Let H be a parabolic subgroup of G, P a minimal parabolic subgroup contained in H, Q the maximal central split torus of a Levi subgroup of P, Δ the set of simple roots of P relative to Q, and $f : \Delta \xrightarrow{\sim} \Delta$ the canonical isomorphism. Then, for every $\alpha \in \Delta$, one has the equivalence:

$$f(\alpha) \in t_r(H) \Leftrightarrow Lie(H)^{-\alpha} \neq 0$$

and one has $H = P_A$, where $A = f^{-1}(t_r(H))$.

(iii) If H and H' are two parabolic subgroups of G containing P, then (see 3.8.1 (ii) and 5.5 (i) for other equivalent conditions):

$$t_r(H) \subset t_r(H') \Leftrightarrow t(H) \subset t(H')$$

[1480] *N.D.E.* The numbering 7.9.0 has been added to highlight the definition of "relative type".

7.10.

One can study the relative positions of two minimal parabolic subgroups; the results are as follows (one refers to

4.5.2 for the notation $t_2(P, P_1)$):

(1) If P, P_1, P', P_1' are four minimal parabolic subgroups of G, then $t_2(P, P_1) = t_2(P', P_1')$ (i.e. (P, P_1) and (P', P_1') are conjugate locally for (fpqc)) if and only if there exists $g \in G(S)$ such that $int(g)P = P'$ and $int(g)P_1 = P_1'$.

(2) Let us fix in particular a minimal parabolic subgroup P of G of Levi subgroup L, and let T be a maximal torus of L. Consider the scheme $Par_{t_{\min}}(G; P)$ of minimal parabolic subgroups of G in standard position relative to P. One has a morphism (cf. 4.5.5)

```
f : Par_{t_{min}}(G; P)  W_P(T) \ W_G(T) / W_P(T)
```

whose fibers are “the orbits of P in $Par_{t_{\min}}(G; P)$”. By (1), f therefore induces a monomorphism

```
P(S) \ Par_{t_{min}}(G; P)(S) ↪ (W_P(T) \ W_G(T) / W_P(T))(S).
```

The image of this morphism is identified with W; this is the Bruhat theorem: each orbit of $P(S)$ in $Par_{t_{\min}}(G; P)(S)$ contains one and only one parabolic subgroup of G with Levi group L (that is, of the form `int(n) P`, where $n \in Norm_G(L)$).

(3) In other words, let $E(S)$ be the set of $g \in G(S)$ such that `int(g) P` and P are in mutual standard position. Then $E(S)$ has a partition into double cosets modulo $P(S)$ indexed by W:

```
E(S) = P(S) · W · P(S)
```

(obvious notation). Setting $U = rad^u(P)(S)$, one may also write

```
E(S) = U(S) · Norm_G(L)(S) · U(S),
```

thereby exhibiting a partition of $E(S)$ into double cosets modulo $U(S)$, indexed by $Norm_G(L)(S)$.

(4) If S is the spectrum of a field, then $E(S) = G(S)$, and one recovers [BT65], 5.15.

Counterexamples 7.11. *Let $S = \mathrm{Spec}(\mathbb{Z}/4\mathbb{Z})$, $G = SL_{2,S}$. Let B be the standard Borel subgroup formed of matrices `(a b; c d)` with $c = 0$. Let $g = (1; -21) \in$ G(S)[scil.: g = (1, 1; −2, 1)]; setB′ = int(g) B. ThenB(S) = B′(S), andB ∩ B′does not contain a maximal torus[^N.D.E-XXVI-37]. This shows on the one hand that two distinct minimal parabolic subgroups may have the same group of sections, and on the other hand that there exists no general criterion allowing one to recognize whether two minimal parabolic subgroupsPandP′are in standard position, using only the groupsP(S)andP′(S). In particular, the* *part $E(S)$ of $G(S)$ does not seem to be definable using only the situation $\{G(S), P(S), Norm\_G(L)(S)\}$ (in the preceding case, this part is defined by $c \neq 2'$*[1481]*).*

7.12.

One now proposes to study the variation of $R(G/S)$ with S. Let then S' be an S-scheme, also semi-local connected and non-empty. Let Q be a maximal split torus of G; then $Q_{S'}$ is a split torus of $G_{S'}$; let Q' be a maximal split torus of $G_{S'}$ containing $Q_{S'}$. Set

$$M = \mathrm{Hom}_{S\text{-gr.}}(Q, G_{m, S}) \simeq \mathrm{Hom}_{S'\text{-gr.}}(Q_{S'}, G_{m, S'}),$$
$$M' = \mathrm{Hom}_{S'\text{-gr.}}(Q', G_{m, S'}).$$

The monomorphism $Q_{S'} \to Q'$ induces an epimorphism $u : M' \to M$. Denote

$$L = \mathrm{Centr}_G(Q), \qquad L' = \mathrm{Centr}_{G_{S'}}(Q'),$$

one has $L' \subset L_{S'}$.

If H is a subgroup of G containing L, then $H_{S'}$ contains L', and one has

$$\mathrm{Lie}(H) = \mathrm{Lie}(L) \oplus \bigoplus_{\alpha \in R_H} \mathrm{Lie}(H)^\alpha,$$
$$\mathrm{Lie}(H_{S'}) = \mathrm{Lie}(L') \oplus \bigoplus_{\alpha' \in R'_{H_{S'}}} \mathrm{Lie}(H_{S'})^{\alpha'},$$

where R_H (resp. $R'_{H_{S'}}$) denotes the set of roots of H (resp. $H_{S'}$) relative to Q (resp. Q'). One immediately derives that

$$R_H \subset u(R'_{H_{S'}}) \subset R_H \cup 0.$$

Taking $H = G$, one sees first that $R \subset u(R') \subset R \cup 0$; taking then for H a minimal parabolic subgroup P of Levi subgroup L, one sees that $R'_{H_{S'}}$ contains a system of positive roots of R', hence (7.4) that there exists a minimal parabolic subgroup P' of $G_{S'}$ of Levi subgroup L' contained in $P_{S'}$. One has therefore constructed a diagram

$$Q_{S'} \subset L_{S'} \subset P_{S'} \cap \cup \cup Q' \subset L' \subset P'.$$

If R^+ (resp. Δ) is the system of positive roots (resp. simple) of R defined by P,

and if one defines similarly R'^+ and Δ', one easily verifies that

$$R^+ \subset u(R'^+) \subset R^+ \cup \{0\}, \qquad \Delta \subset u(\Delta') \subset \Delta \cup \{0\}.$$

Let now $w \in W \simeq Norm_G(Q)(S)/Centr_G(Q)(S)$, and $n \in Norm_G(Q)(S)$ a representative of w. One has $int(n)Q = Q$ hence $int(n)L = L$, hence $int(n)L_{S'} = L_{S'}$. Then Q' and $int(n)\, Q'$ are two maximal split tori of $L_{S'}$, hence are conjugate by a section $x \in L(S')$, and one has $int(nx)Q' = Q'$, hence $nx \in Norm_{G_{S'}}(Q')(S')$. Let w' be the image of $n' = nx$ in $W' \simeq Norm_{G_{S'}}(Q')(S')/Centr_{G_{S'}}(Q')(S')$. It is clear that the

[1481] *N.D.E.* More generally, for every S-scheme S', $E(S')$ is defined by the condition: "c is zero or invertible".

operation of w' on M' is compatible with the projection $u : M' \to M$ and that the induced operation on M coincides with that defined by w.

Using now the definition of relative root data and the conjugacy theorems, one proves without difficulty:

Theorem 7.13. *Let S and S' be two non-empty connected semi-local schemes, $S' \to S$ a morphism of S-schemes, G an S-reductive group,*

R(G/S) = (M, M^*, R, R^*, R^+),   R(G_{S'}/S') = (M', M'^*, R', R'^*, R'^+)

the pinned relative root data. There exists a canonical homomorphism

$$u : M' \longrightarrow M$$

satisfying the following conditions:

(i) u is surjective.

(ii) For every $w \in W$, there exists an element w' of W' compatible with u and which induces w on M.

(iii) For every subset X of M, denote $X^\wedge = X \cap (M - 0)$. Then

u(R'^+)^∧ = R^+,   u(Δ')^∧ = Δ.

(iv) For every parabolic subgroup H, consider $t_r(H) \subset \Delta$ and $t_r(H_{S'}) \subset \Delta'$. Then

t_r(H_{S'}) = (u^{-1}(t_r(H) ∪ {0})) ∩ Δ' = { α' ∈ Δ' | u(α') ∈ t_r(H) or u(α') = 0 }.

Remark 7.14. *If G is splittable over S, its maximal split tori are maximal tori, and the relative notions introduced here then coincide with the absolute notions already introduced. The preceding theorem therefore gives a description of the relative root datum $R(G/S)$ and of the relative type t_r, via the absolute root datum and the absolute type of the group $G_{S'}$, S' being chosen in such a way that $G_{S'}$ is splittable (cf. Exp. XXIV, 4.4.1). We refer to [BT65], 6.12 et seq. for this description.*

7.15.

Let S be a Henselian local scheme, s_0 its closed point, S_0 the spectrum of the residue field of s_0, identified with a closed subscheme of S; for every object X above S, let us denote by X_0 the object above S_0 deduced from X by base change. Let finally G be an S-reductive group. For every parabolic subgroup $\bar{P}$ of G, $\bar{P}_0$ is a parabolic subgroup of G_0; conversely, for every parabolic subgroup P of G_0, there exists a parabolic subgroup $\bar{P}$ of G such that $\bar{P}_0 = P$ (this follows from Hensel's lemma and from the fact that $Par(G)$ is a smooth S-scheme); in particular (cf. 5.7), a parabolic subgroup $\bar{P}$ of G is minimal if and only if $\bar{P}_0$ is minimal. Such a subgroup $\bar{P}$ of G being chosen, an analogous reasoning shows that the maximal split subtori of $\bar{P}_0$ are of the form T_0, where T is a maximal split subtorus of $\bar{P}$. It follows without difficulty that the relative root data of G over S and of G_0 over S_0 are canonically isomorphic, so that the theory of parabolic subgroups of G reduces to that of parabolic subgroups of `G_0`.

Let us remark, moreover, that every `S_0`-reductive group is of the form `G_0` (Exp. XXIV, Prop. 1.21), which conversely allows one to reduce the study of parabolic subgroups of an `S_0`-reductive group to the corresponding study over S.

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Footnotes

SGA 3: Group Schemes

Séminaire de Géométrie Algébrique du Bois-Marie, 1962–1964.

M. Demazure and A. Grothendieck, with collaborators J.-E. Bertin, P. Gabriel, M. Raynaud, M. Artin, J.-P. Serre. 2011 SMF *Documents Mathématiques* re-edition by Demazure (Tomes I, II, III, with N.D.E. footnotes).

Abstract

This Séminaire develops the theory of group schemes over an arbitrary base prescheme, in two complementary parts. Part I (Exposés I–VII_B, Tome I) lays foundations: schematic and categorical syntax, faithfully flat quasi-compact descent, fibered categories and sheaves, the construction of quotient schemes, generalities on algebraic groups and group schemes, and the infinitesimal study of differential operators and formal groups, including the correspondence between height-1 radicial group schemes and restricted p-Lie algebras. Part II (Exposés VIII–XXVI, Tomes II and III) develops the structure theory: groups of multiplicative type (VIII–XII); regular elements in algebraic groups and Lie algebras (XIII–XIV); smoothness and maximal tori (XV); groups of zero unipotent rank, unipotent algebraic groups,

[1482] *N.D.E.* The references that follow have been added to this reference, which appears in the original.

and extensions (XVI–XVII); Weil's theorem on the construction of a group from a rational law (XVIII); and, in Tome III, the structure theory of reductive group schemes — generalities (XIX), the rank-1 case (XX), root data (XXI), splitting and subgroup structure (XXII), uniqueness of pinned groups (XXIII), automorphisms (XXIV), the existence theorem (XXV), and parabolic subgroups (XXVI). The Séminaire generalizes the Borel–Chevalley structure theory of affine algebraic groups to arbitrary base preschemes, with Chevalley's "Tôhoku" groups over $\mathbb{Z}$ as a guiding example.

Reading order

Files are numbered so alphanumeric order matches reading order.

- Exposé XVII — Unipotent algebraic groups; extensions (Raynaud)
- Exposé XVIII — Weil's theorem on the construction of a group from a rational law (Artin)

Tome III — Reductive groups (Exposés XIX–XXVI)

- Tome III errata
- Exposé XIX — Reductive groups: generalities (Demazure)
- Exposé XX — Reductive groups of semisimple rank 1 (Demazure)
- Exposé XXI — Root data (Demazure)
- Exposé XXII — Reductive groups: splittings, subgroups, quotients (Demazure)
- Exposé XXIII — Reductive groups: uniqueness of pinned groups (Demazure)
- Exposé XXIV — Automorphisms of reductive groups (Demazure)
- Exposé XXV — The existence theorem (Demazure)
- Exposé XXVI — Parabolic subgroups of reductive groups (Demazure)
- Tome III index

Reference convention

Following the source's §7 convention (see the Introduction), SGA 3 is cited as `(Exp. N, M.K)` where N is the Exposé Roman numeral and `M.K` is the decimal numbering inside that Exposé — for example `(XII, 3.2)` for statement 3.2 of Exposé XII. A bare `5.7.11` refers to the proposition of that name in the same Exposé. The split Exposés are cited as `VI_A`, `VI_B`, VII_A, VII_B (matching the source's typeset subscript notation).

Cross-volume citations follow the source's abbreviations: `EGA X, x.y.z` for the *Éléments de Géométrie Algébrique*, `SGA n, X y.z` for other SGA volumes, *Bible* for the *Séminaire Chevalley* 1956/58, and TDTE for Grothendieck's *Techniques de descente et théorèmes d'existence* (Bourbaki, 1959–62). See the glossary for the full list.

Editorial conventions

- **Terminology.** SGA 3 retains the historical SGA/EGA distinction between *prescheme* (`préschéma`) and *scheme* (`schéma`, in the older sense of separated prescheme), matching SGA 1 and SGA 2 (Exposés I–XIII). The translation honors this throughout all 26 Exposés. As Grothendieck notes in the Introduction (§ 6), SGA 3 also deliberately replaces the usual hypothesis "of finite type" by "of finite presentation"; the translation preserves this distinction.

- **Editor footnotes (N.D.E.).** The 2011 SMF *Documents Mathématiques* re-edition (M. Demazure, ed.) added footnotes recording corrections, updates, modern references, and the current status of questions raised in the original 1962–1964 seminar. In this translation, original footnotes from the Séminaire authors (Demazure, Grothendieck, Bertin, Gabriel, Raynaud, Artin, Serre) use slugs like [$^{I}-3-1$], while editor footnotes use [$^{N}.D.E-I-3$] so the two are visibly distinct.

- **Page marks.** HTML comments <!-- original page N --> mark the start of page N in the 2011 SMF re-edition, whose pagination is the reference for the errata sheets.
- **Mathematics.** Mathematics is written with Unicode and wrapped in backticks where formatter mangling is a risk (most identifier-rich expressions). Displayed equations use fenced ```text blocks, optionally pinned with <!-- label: eq:III.<Exp>.X.Y -->.
- **Split Exposés (VI and VII).** Exposés VI and VII are split into A and B parts in the source. Filenames keep the split (06A-, 06B-, 07A-, 07B-); cross-references use VI_A, VI_B, VII_A, VII_B.

Provenance

This is a translation, not a critical edition. The authoritative text remains the French 2011 SMF *Documents Mathématiques* re-edition, derived from the SLN 151, 152, 153 of 1970 as re-typeset by M. Demazure. For any claim that matters mathematically, consult the source: this English version exists to make the volume readable, not to replace it.

Glossary and Translation Ledger — SGA 3

This file records the translation choices made for the SGA 3 Markdown translation. It is the authoritative reference for the parallel editors of individual Exposes; the glossary should be consulted before drafting and must obey it where it applies.

Core terminology

French	English	Note
préschéma	prescheme	Historical SGA/EGA term. Used throughout SGA 3, matching SGA 1 and SGA 2 I–XIII.
schéma	scheme	Older sense: separated prescheme. The 2011 re-edition retains this.
S-préschéma	S-prescheme	A prescheme over S. Sometimes typeset S-préschéma; preserve hyphen.
S-schéma en groupes	S-group scheme	The relative form. Title-level for the whole Séminaire.
faisceau	sheaf	Standard.
faisceau (quasi-)cohérent	(quasi-)coherent sheaf	Standard.
espace annelé	ringed space	Standard.

French	English	Note
application	map	“Function” only in elementary set-theoretic contexts.
morphisme	morphism	Standard.
morphisme étale / lisse / plat / propre	étale / smooth / flat / proper morphism	Standard; matches SGA 1 glossary.
de présentation finie	of finite presentation	SGA 3 deliberately replaces *de type fini* by *de présentation finie* (cf. Intro §6); preserve.
de type fini	of finite type	Use only when the source uses *de type fini*.
revêtement étale	étale covering	Per SGA 1 glossary.
immersion / immersion fermée / ouverte	immersion / closed immersion / open immersion	Standard.
ouvert / fermé / localement fermé	open / closed / locally closed	Standard.
voisinage	neighborhood	American spelling.
recouvrement	covering	“Cover” only in casual prose.
corps	field	False friend: not “body”.
anneau	ring	Commutative by SGA convention unless stated.
anneau local (noethérien)	(noetherian) local ring	Standard.
anneau local régulier	regular local ring	Standard.
corps résiduel	residue field	Denoted $\kappa(x)$ or $\kappa(A)$ (per SGA 1).
catégorie	category	Standard.
foncteur / foncteur dérivé	functor / derived functor	Standard.
foncteur exact / exact à gauche / à droite	exact / left exact / right exact functor	Standard.
sous-foncteur	subfunctor	Standard.
foncteur représentable	representable functor	Central to SGA 3.
représentation (d’un groupe par un foncteur)	representation	Be careful: SGA 3 uses *représenter* both for representable functors and for representations of groups; pick by context.
catégorie abélienne	abelian category	Standard.

French	English	Note
limite projective / inductive	inverse / direct limit	Modern English.
fibre	fiber	American spelling.
fibre géométrique	geometric fiber	Standard. (Used heavily in Tome III for reductive groups.)
spécialisation	specialization	American spelling.
canonique	canonical	Do *not* translate as “natural” unless the source specifically appeals to naturality.
fonctoriel	functorial	Standard.
naturel	natural	Reserve for genuine naturality.
à isomorphisme près	up to isomorphism	Standard.
nécessaire et suffisant	necessary and sufficient	Sometimes “if and only if” in theorem statements.
si et seulement si	if and only if	“Iff” only if local idiom permits.

SGA 3 topics: foundations (Tome I)

French	English	Note
schéma en groupes	group scheme	Title-level term.
S-groupe	S-group	Shorthand for *S-schéma en groupes*. Preserve.
schéma en monoïdes	monoid scheme	Standard.
schéma en anneaux	ring scheme	Standard.
structure algébrique	algebraic structure	Per Exposé I title.
cohomologie des groupes	group cohomology	Per Exposé I title.
univers (Univers)	universe	Set-theoretic universe (Grothendieck universe).
catégorie `(Ens)`	category of sets `(Ens)`	Keep the source notation `(Ens)`. Gloss only at first use.
catégorie `(Sch)`	category of schemes `(Sch)`	Keep notation. The source writes $(Sch)/S$ for S-schemes.
catégorie `(Ann)`	category of rings `(Ann)`	Keep notation.

French	English	Note
catégorie `(Gr)`	category of groups `(Gr)`	Keep notation.
Ĉ	Ĉ	Notation for $\mathrm{Hom}(C^\circ, (Ens))$. OCR may produce `Cb`; restore the hat.
C°	C° (opposite category)	Keep ˚ Unicode; OCR may give o or `0`.
foncteur contravariant	contravariant functor	Standard.
topologie fpqc	fpqc topology	"Faithfully flat quasi-compact." Acronym kept; spell out on first use per Exposé.
topologie fppf	fppf topology	"Faithfully flat of finite presentation." Same convention.
topologie étale	étale topology	Standard.
topologie de Zariski	Zariski topology	Standard.
descente fidèlement plate	faithfully flat descent	Matches SGA 1 glossary.
catégorie fibrée	fibered category	American spelling.
faisceau (au sens d'une topologie)	sheaf (in the sense of a topology)	Standard.
faisceau associé	associated sheaf	Standard.
sous-foncteur représentable	representable subfunctor	Standard.
objet en groupes	group object	When the source uses *objet en groupes* rather than *schéma en groupes*.
algèbre de Hopf	Hopf algebra	Standard.
algèbre affine	affine algebra	Standard.
spectre	spectrum	Standard.
$\mathrm{Spec}(A)$	$\mathrm{Spec}(A)$	Keep as-is; the source sometimes writes $\mathrm{Spec}\,A$ without parens.
extension infinitésimale	infinitesimal extension	Per Exposé III title.
déformation infinitésimale	infinitesimal deformation	Standard.
fibré tangent	tangent bundle	Per Exposé II title. Context may favor "tangent sheaf"; flag first occurrence.
fibré normal	normal bundle	Standard.

French	English	Note
fibré principal	principal bundle	Standard.
algèbre de Lie	Lie algebra	Standard. $Lie(G/S)$ notation preserved.
p-algèbre de Lie	p-Lie algebra	Modern: *restricted Lie algebra*; the source's term is *p-algèbre de Lie*. Keep it; flag with a one-time gloss.
schéma en groupes radiciel de hauteur 1	radicial group scheme of height 1	Per Intro §1. *Radiciel* = purely inseparable.
radiciel	radicial	Loanword (per EGA); not "radical".
opérateur différentiel	differential operator	Per Exposé VII_A title.
groupe formel	formal group	Per Exposé VII_B title.
S-foncteur	S-functor	A functor from $(Sch)^{\circ}/S$ to $(\mathtt{Ens})$. Preserve.
passage au quotient	passage to the quotient	Standard.
schéma quotient	quotient scheme	Per Exposé V title.
relation d'équivalence	equivalence relation	Standard.
groupoïde	groupoid	Standard.
sous-groupe distingué / normal	normal subgroup	Standard (English does not distinguish).
centralisateur / normalisateur	centralizer / normalizer	American spelling.
transporteur	transporter	Standard.
stabilisateur	stabilizer	American spelling.

SGA 3 topics: multiplicative-type and Lie-algebra material (Tome II)

French	English	Note
groupe diagonalisable	diagonalizable group	American spelling. Per Exposé VIII title.
groupe de type multiplicatif	group of multiplicative type	Per Exposé IX–X titles.
caractère / cocaractère	character / cocharacter	Standard.

French	English	Note
$D(M)$	$D(M)$	The Cartier dual of the abstract group M. Notation preserved.
groupe constant	constant group	Standard.
groupe tordu	twisted group	Standard.
forme tordue	twisted form	Standard.
critère de représentabilité	representability criterion	Per Exposé XI title.
tore	torus	Standard. Plural: *tori*.
tore maximal	maximal torus	Standard.
sous-tore	subtorus	Standard.
variété des tores maximaux	variety of maximal tori	Standard.
groupe de Weyl	Weyl group	Standard.
sous-groupe de Cartan	Cartan subgroup	Standard.
centre / centre réductif	center / reductive center	American spelling. $Z(G)$, $rad(G)$, $rad_u(G)$ notations preserved.
élément régulier	regular element	Per Exposés XIII–XIV titles.
élément semi-simple / unipotent	semisimple / unipotent element	"Semisimple" one word, no hyphen.
polynôme caractéristique	characteristic polynomial	Standard.
sous-groupe lisse	smooth subgroup	Standard.
groupe (algébrique) lisse	smooth (algebraic) group	Standard.
groupe de rang unipotent nul	group of zero unipotent rank	Per Exposé XVI title.
groupe algébrique unipotent	unipotent algebraic group	Per Exposé XVII title.
extension (de groupes algébriques)	extension (of algebraic groups)	Standard.
loi rationnelle	rational law	Per Exposé XVIII title (Weil's theorem on rational laws).
théorème de Weil	Weil's theorem	Standard.

SGA 3 topics: reductive groups (Tome III)

French	English	Note
groupe réductif	reductive group	Per Tome III title and Exposés XIX–XXVI.

French	English	Note
groupe semi-simple	semisimple group	One word, no hyphen.
groupe résoluble	solvable group	American spelling.
groupe à rang semi-simple 1	group of semisimple rank 1	Per Exposé XX title.
déploiement / déployé	splitting / split	Standard.
groupe déployable	splittable group	Standard.
sous-groupe parabolique	parabolic subgroup	Per Exposé XXVI title.
sous-groupe de Borel	Borel subgroup	Standard.
sous-groupe radiciel	root subgroup	The U_α of a reductive group.
donnée radicielle	root datum	Per Exposé XXI title. Plural: *root data*.
système de racines	root system	Standard.
racine	root	Standard. (Not "radical" — that is *radical*.)
coracine	coroot	Standard.
poids	weight	Standard.
groupe épinglé	pinned group	Per Exposé XXIII title. *Épinglage* = *pinning*; "framed group" only in passing.
épinglage	pinning	Standard (in this translation).
unicité des groupes épinglés	uniqueness of pinned groups	Per Exposé XXIII title.
automorphisme (d'un groupe réductif)	automorphism	Per Exposé XXIV title.
isogénie	isogeny	Standard.
groupe simplement connexe / adjoint	simply connected / adjoint group	Standard.
théorème d'existence	existence theorem	Per Exposé XXV title.
théorème d'unicité	uniqueness theorem	Standard.
groupes « de Tôhoku »	"Tôhoku" groups	Chevalley's groups over $\mathbb{Z}$. Italics on *Tôhoku* in journal references.
Bible (Séminaire Chevalley 1956/58)	the *Bible* (Chevalley Seminar 1956–58)	SGA 3 cites the Chevalley Seminar as "Bible". Render the same way.

French	English	Note
Bourbaki	Bourbaki	Standard.
théorie de Borel-Chevalley	Borel–Chevalley theory	Use en-dash between author names.
théorie de Demazure	Demazure's theory	Standard.

Proof movement

These follow $references/french-math-idiom.md$ of the translation skill, with SGA-specific tweaks (inherited from SGA 1/SGA 2).

French	English	Note
Soit / Soient X / X et Y	Let X / X and Y be	Match singular / plural.
Supposons (que)	Suppose (that) / Assume	"Assume" if the sentence is a hypothesis register.
Posons	Set / Put	"Put" for variable assignments; "Set" for ad-hoc definitions.
Notons	Denote / Write / Note	Choose by what follows; *Notons que* is "Note that".
On a	We have	Not "one has" except for register.
On en déduit	We deduce	"It follows" if the inference is non-trivial.
Montrons (que)	We show (that)	"It remains to show" if it closes a reduction.
Il suffit de	It suffices to	Standard.
Il reste à montrer	It remains to show	Standard.
Il en résulte (que) / Il résulte (de)	It follows (that) / It follows (from)	Choose by structure; "hence" in compact proof prose.
Par suite	Consequently / hence	Avoid "as a result of this".
D'où	Hence / whence	"Whence" only if register supports it.
En effet	Indeed	Standard introduces a justification.
Or	Now	Pivot conjunction; "but" only if the sentence really contrasts.
Donc	Thus / hence / so	Choose by rhythm.
D'une part … d'autre part	On the one hand … on the other hand	Keep both clauses.
Cela achève la démonstration	This completes the proof	Standard.
CQFD	QED	Or "This proves the claim."

French	English	Note
à savoir	namely	Standard.
compte tenu de	taking into account / in view of	Choose by clause weight.
en vertu de	by virtue of / by	“By” usually suffices.
moyennant	by means of / using	Standard.
quitte à	(possibly) by … / replacing X by Y if necessary	A standard French move.
à condition que	provided that	Standard.
dès que	as soon as	Standard.
chaque fois que	whenever	Standard.

Modality and certainty

Preserve modality exactly; do *not* upgrade certainty.

French	English
il est clair que	it is clear that
il est évident que	it is evident that
manifestement / il est manifeste que	manifestly / it is manifest that
il semble que	it seems that
on s’attend à ce que	one expects that
il est probable que	it is probable that
sans doute	doubtless / probably (context-dependent)
conjecturalement	conjecturally
il y a tout lieu de penser que	there is every reason to think that
on peut conjecturer	one may conjecture
il n’est pas exclu que	it is not excluded that
il se peut que	it may be that
vraisemblablement	presumably

Cross-volume citations

Source	English citation form
EGA I, II, III, IV (with chapter and number)	`EGA I, 1.2.3` / `EGA IV, 1.4`, etc. Preserve the chapter Roman numeral.
SGA 1 (with Exposé and number)	`SGA 1, I 5.1` / `SGA 1, VIII 1.7`, etc. Match the source’s spacing.
SGA n in general	`SGA n, X y.z` per the Introduction §7 convention.
Bible = Séminaire Chevalley 1956/58	*Bible* (italicised) or *SéminaireChevalley*1956/58. Keep the source’s wording.

Source	English citation form
TDTE	Grothendieck's *Techniques de descente et théorèmes d'existence* (Bourbaki 1959–62).
Mumford GIT	Mumford, *Geometric Invariant Theory*. Italics.
Demazure–Gabriel	*Groupes algébriques* (Demazure–Gabriel). Italics, en-dash between author names.

Common OCR repairs

The 2011 SMF re-edition is modern LaTeX, so OCR is high quality. Most math symbols survive. These items still recur:

- **Dropped sub/superscripts on big operators.** Γ, H^i, Ext^i, $R^i f_*$, `O_X`, `Lie`, $T_e G$, μ_n, G_m, G_a, α_p, $D(M)$, $w\alpha(X)$, root operators U_α, T_α may lose their indices. Recover from context.
- **`Cb` instead of `Ĉ`.** Hat on category symbol drops. Restore.
- $(*)$ / $(**)$ / $(***)$ **for original footnote markers.** Render as $[^{<}expos > - < sec > - < n >]$ slugs ($[^{I} - 1 - 1]$, $[^{I} - 1 - 2]$, ...).
- **`(N)` for N.D.E. footnote markers** (small Arabic digit in parentheses). Render as $[^{N}.D.E - < expos > - N]$ ($[^{N}.D.E - I - 3]$, etc.) and translate the body of the note.
- *ï* **or ı** (dotless i with floating diacritic) — restore to `ï`.
- **Backslash escapes** surviving from LaTeX: `\&` → `&`, `\#` → `#`.
- **Arrows.** $- \to \to \to$, $7 \to \to \mapsto$, $\sim=$ (often broken across lines) → $\cong$, `\sim\n\to` (OCR artefact) → $\xrightarrow{\sim}$ or → annotated ~.
- **Greek subscripts.** $\pi 1$, `H0`, `G m` etc. should be π_1, H^0, G_m (with backticks).
- ◦ **for opposite category** drops to o or `0`. Restore ° or ° (degree sign), matching local convention.
- **VIB, VIIA, VIIB citations.** Render as `VI_B`, VII_A, VII_B. Preserve subscript via underscore.
- **« » guillemets.** Translate to English quotes (" ⋯ ") unless the source uses them to mark a Grothendieckian coinage, in which case keep them as quotation marks in the translation too.
- **Hat across line break** for formal completions: $Et(b/X) \to \hat{E}t(\hat{X})$. Same as SGA 2.
- **Math run-ons.** Restore displayed equations as fenced `` ```text `` blocks.
- **Embedded N.D.E. footnotes.** Source convention is $(N)N.D.E. :< text >$ at page bottom. Recover the whole footnote (which may span lines) and translate as $[^{N}.D.E - < expos > - N]$.

Style anchors (non-negotiable for cross-volume consistency)

Inherited from SGA 1 / SGA 2. Deviate only if the SGA 1/2 corpus is itself inconsistent.

- **File naming.** 00-i-errata.md, 00-i-avertissement-introduction.md, 00-ii-preface.md, 00-iii-errata.md, NN-<slug>.md for each Exposé (using 06A, 06B, 07A, 07B for the split Exposés), glossary.md, README.md, zz-i-index-notations.md, zz-iii-index-notations.md, zz-iii-index-terminologique.md.
- **Heading hierarchy.** # Exposé N. <English Title> (with _A / _B suffix where applicable), then numbered ## 1. <Section Title>, then sub-sections ### 1.1. <Subsection>. The Exposé heading carries a <!-- label: III.<Roman> --> comment on the next line.
- **Statement blocks.** **Proposition.**, **Lemma.**, **Theorem.**, **Corollary.**, **Definition.**, **Remark.**, **Example.**, **Scholie.** on their own line. Statement numbers go inline: **Proposition 3.1.2.** *Statement…*. The label <!-- label: III.<Roman>.X.Y --> follows on the next line.
- **Math.** Unicode math wrapped in backticks for inline. Displayed math in fenced ```text blocks. Numbered displayed equations get <!-- label: eq:III.<Roman>.X.Y -->.
- **Footnotes.** Markdown [^{slug}] syntax. Original footnotes: [$^{<}expos > - < sec > - < n >$] (e.g. [$^{I} - 1 - 1$]). Editor footnotes: [$^{N}.D.E - < expos > - < n >$] (e.g. [$^{N}.D.E - I - 3$]).
- **Page marks.** <!-- original page N --> at page boundaries (page numbers from the 2011 re-edition).
- **Cross-references.** Source's own convention: (VI_B, 5.3), (EGA III, 4.1.5), (SGA 1, IV 3.8), (Bible, XIV).
- **Bibliographies.** Each Exposé's own bibliography section is preserved as ## Bibliography, entries as a numbered (or [Key]-keyed) list, journal titles in italics.

Cross-Exposé additions (discovered during translation)

Each translated Exposé file ends with a fenced HTML-comment block titled LEDGER DELTA — Exposé <N> recording new terminology choices that the translator encountered beyond what this glossary already covers. Those deltas are kept in place at the foot of each Exposé rather than copied here, both to preserve the audit trail and because consolidating the deltas (which span ~500–1,500 terms across the 26 Exposés) into a single table would make the glossary harder to scan than the per-Exposé tables. A future editorial pass may de-duplicate the most cross-cutting entries (e.g. *dévissage*, the prescheme/scheme split, Demazure's *épinglage* → *pinning*, Bertin's "anti-affine" envelope) into the tables above; until then, consult the deltas in the Exposé files for the authoritative per-Exposé terminology choices.

Index — Tome I

A combined index of notation and terminology for Tome I. Page numbers refer to the 2011 SMF re-edition.

Index — Tome III

A combined index of notation and terminology for Tome III. Page numbers refer to the 2011 SMF re-edition.